

Dear Professor Weil,

While trying to formulate clearly the question I was asking you before Chern's talk I was led to two more general questions. Your opinion of these questions would be appreciated. I have not had a chance to think over these questions seriously and I would not ask them except as the continuation of a casual conversation. I hope you will treat them with the tolerance they require at this stage. After I have asked them I will comment briefly on their genesis.

It will take a little discussion but I want to define some Euler products which I will call Artin-Hecke L -series because the Artin L -series, the L -series with Grössencharaktere, and the series introduced by Hecke into the theory of automorphic forms are all special cases of these series. The first question will be of course whether or not these series define meromorphic functions with functional equations. I will say a few words about the functional equation later. The other question I will formulate later. It is a generalization of the question of whether or not abelian L -series are L -series with Grössencharaktere. Since I want to formulate the question for automorphic forms on any reductive group I have to assume that certain results in the reduction theory can be pushed a little further than they have been so far.

Unfortunately I must be rather pedantic. Let k be the rational field or a completion of it. Let $\tilde{G}$ be a product of simple groups, perhaps abelian, split over k . Suppose the non-abelian factors [2] are simply connected. The case that the product is empty and $\tilde{G} = \{1\}$ is not without interest. Fix a split Cartan subgroup $\tilde{T}$ and let $\tilde{L}$ be the lattice of weights of $\tilde{T}$. $\tilde{L}$ contains the roots. I want to define "the" conjugate group to $\tilde{G}$ and "the" conjugate lattice to $\tilde{L}$. It is enough to do this for a simple group for we can then take direct products and direct sums. If $\tilde{G}$ is abelian and simple let $\tilde{L}'$ be any sublattice of $\tilde{L}$ and ${}^c\tilde{L}$, the conjugate lattice, be the dual of $\tilde{L}'$ (i.e. $\text{Hom}(\tilde{L}', \mathbf{Z})$). It contains ${}^c\tilde{L}'$, the dual of $\tilde{L}$. Let ${}^c\tilde{G}$ be a one-dimensional split torus whose lattice of weights is identified with ${}^c\tilde{L}$. If $\tilde{G}$ is simple and non-abelian let $\tilde{L}'$ be the lattice generated by the roots and let ${}^c\tilde{L}$ be the dual of $\tilde{L}'$. ${}^c\tilde{L}$ contains ${}^c\tilde{L}'$ the dual of $\tilde{L}$. Choose for each root α an element H_α in the Cartan subalgebra corresponding to $\tilde{T}$ in the usual way so that $\alpha(H_\alpha) = 2$. The linear functions ${}^c\alpha(\lambda) = \lambda(H_\alpha)$ generate $\tilde{L}'$. There is a unique simply connected group ${}^c\tilde{G}$ whose lattice of weights is isomorphic to ${}^c\tilde{L}$ in such a way that the roots of ${}^c\tilde{G}$ correspond to the elements ${}^c\alpha$. Fix simple roots $\alpha_1, \dots, \alpha_\ell$ of $\tilde{G}$; then ${}^c\alpha_1, \dots, {}^c\alpha_\ell$ can be taken as the simple roots of ${}^c\tilde{G}$. Now return to the general case.

If L is a lattice lying between $\tilde{L}'$ and $\tilde{L}$ we can associate to L in a natural way a group G covered by $\tilde{G}$. The dual lattice cL of L lies between ${}^c\tilde{L}'$ and ${}^c\tilde{L}$. It determines a group cG , covered by ${}^c\tilde{G}$, which I call the conjugate of G . Let $\mathfrak{h}$ be the Lie algebra of $\tilde{T}$ and choose for each root α a root vector X_α so that the conditions of Chevalley are satisfied. Also let ${}^c\mathfrak{h}$ be a split Cartan subalgebra of ${}^c\mathfrak{g}$ and for each root ${}^c\alpha$ choose a root vector ${}^cX_\alpha$ so that the conditions of Chevalley are satisfied. Let Ω be the group of automorphisms of $\mathfrak{g}$ which take $\mathfrak{h}$ to itself, permute $\{X_\alpha \mid \alpha \text{ simple}\}$, and take $\tilde{L}, L, \tilde{L}'$ to themselves. Define ${}^c\Omega$ in a similar fashion. ${}^c\Omega$ is the contragredient of Ω so that Ω and ${}^c\Omega$ are canonically isomorphic. Ω

thus acts as a group of [3] automorphisms of G and of cG . If K is a finite Galois extension of k and δ is a homomorphism of $\mathfrak{G} = \mathfrak{G}(K/k)$ into Ω with image Ω^δ let G^δ and ${}^cG^\delta$ be the associated forms of G and cG .

In order to define the local factors of the L -series I have to recall some facts about the Hecke algebra of G_k^δ when K is an unramified extension of the p -adic field k . If we choose a maximal compact subgroup of G_k^δ in a suitable manner then, according to Bruhat and Satake, the Hecke algebra is isomorphic to the set of elements in the group algebra of ${}^cL^\delta$, the set of elements in cL fixed by ${}^c\Omega^\delta$, which are invariant under the restricted Weyl group ${}^cW^\delta$ of ${}^cG^\delta$. (Actually we have to stretch their results a little.) Thus any homomorphism χ of the Hecke algebra into the complex numbers can be extended to a homomorphism χ' of the group algebra of cL into the complex numbers. There is at least one element g of cT so that if $f = \sum_{\lambda \in L} a_\lambda \xi_\lambda$ (ξ_λ is λ written multiplicatively) then $\chi'(f) = \sum a_\lambda \xi_\lambda(g)$. The semi-direct product $\mathfrak{G} \ltimes_\delta {}^cG$ is a complex group. Let π be a complex representation of it. If σ is the Frobenius then

$$\frac{1}{\det(1 - x\pi(\sigma \ltimes g))} \quad (x \text{ an indeterminate})$$

[4] is the local zeta function corresponding to χ and π . I have to verify that it depends only on χ and not on g . If λ is any weight let n_λ be the lowest power of σ which fixes λ and if $n_\lambda | n$ and π acts on V let $t_\lambda(n)$ be the trace of σ^n on

$$\{v \in V \mid \pi(h)v = \xi_\lambda(h)v \text{ for all } h \in {}^cT\}.$$

Then

$$\begin{aligned} \log \frac{1}{\det(1 - x\pi(\sigma \ltimes g))} &= \sum_{n=0}^{\infty} \frac{x^n}{n} \sum_{\lambda \in L} \sum_{n_\lambda | n} t_\lambda(n) \xi_\lambda(g^{\sigma^{n-1}} g^{\sigma^{n-2}} \cdots g) \\ &= \sum_{n=0}^{\infty} \frac{x^n}{n} \sum_{\lambda \in L} \sum_{n_\lambda | n} t_\lambda(n) \xi_{\frac{n}{n_\lambda}} \left(\sum_{k=0}^{n_\lambda-1} \lambda^{\sigma^k} \right) (g) \end{aligned}$$

Moreover if ω is an element of ${}^cW^\delta$ we can always choose a representative w of it which commutes with σ . Then the local zeta function does not change if g is replaced by $w^{-1}gw$ so it equals

$$\frac{1}{[{}^cW^\delta : 1]} \sum_{n=0}^{\infty} \frac{x^n}{n} \sum_{\lambda} \sum_{n_\lambda | n} t_\lambda(n) \sum_{\omega \in {}^cW^\delta} \xi_{\frac{n}{n_\lambda}} \left(\sum_{k=0}^{n_\lambda-1} \lambda^{\sigma^k} \right)^\omega (g)$$

Since

$$\sum_{\omega} \xi_{\frac{n}{n_\lambda}} \left(\sum_{k=0}^{n_\lambda-1} \lambda^{\sigma^k} \right)^\omega$$

belongs to the image of the Hecke algebra the assertion is verified.

I don't know if it is legitimate but let us assume that the characters of the complex representations separate the semi-simple conjugacy classes in $\mathfrak{G} \ltimes_\delta {}^cG$. Then by the above I can associate to each homomorphism χ of the Hecke algebra into the complex numbers the conjugacy class of the semi-simple [5] element $\sigma \ltimes g$. Conversely given a semi-simple conjugacy class in $\mathfrak{G} \ltimes_\sigma {}^cG$ it contains, by Borel-Mostow, an element in the normalizer of cT . Then it even contains an element which takes positive roots into positive roots.* Thus if the

*This is probably true and the materials for a proof are probably available in the literature. However it is not so obvious as I thought when writing the letter.

projection of the conjugacy class on $\mathfrak{G}$ (an abelian group) is σ the conjugacy class contains an element of the form $\sigma \ltimes g$, $g \in {}^cT$. As above g determines a homomorphism of the Hecke algebra into the complex numbers. If this homomorphism χ is completely determined by the local zeta factors attached to it then it is completely determined by the conjugacy class of $\sigma \ltimes g$ and we have a one-to-one correspondence between homomorphisms of the Hecke algebra into the complex numbers and semi-simple conjugacy classes in $\mathfrak{G} \ltimes {}^cG$ whose projection on $\mathfrak{G}$ is σ . It is enough to check that the value of χ on an element of the form

$$\sum_{\omega \in {}^cW^\delta} \xi\left(\sum_{k=1}^{n_\lambda} \lambda^{\sigma^k}\right)^\omega,$$

where $\sum_{k=1}^{n_\lambda} \lambda^{\sigma^k}$ belongs to the positive Weyl chamber, is determined by the local zeta functions. This can be done by the usual sort of induction, for $\sum \lambda^{\sigma^k}$ is invariant under $\mathfrak{G}$ and thus the highest weight of a representation of $\mathfrak{G} \ltimes {}^cG$ whose restriction to cG is irreducible.

Now I am going to try to define the Artin-Hecke L -series. To do this let us fix for each p an imbedding of $\overline{\mathbf{Q}}$, the algebraic closure of $\mathbf{Q}$, in $\overline{\mathbf{Q}}_p$. We will have to come back later and check that the series are independent of these choices. The choice will be implicit in the next paragraph.

Suppose we have a twisted form $\overline{G}$ of G over the rationals. The twisting can be accomplished in two steps. First for a suitable Galois extension K of $\mathbf{Q}$ take a homomorphism δ of $\mathfrak{G} = \mathfrak{G}(K/\mathbf{Q})$ into Ω to obtain G^δ . Then take an inner twisting of G^δ by means of the [6] cocycle $\{a_\tau \mid \tau \in \mathfrak{G}\}$. Let me assume the truth of the following:

- (i) Suppose G is a linear group acting on V . Let L be a Chevalley lattice in $V_{\mathbf{Q}}$. Then the intersection $G_{\mathbf{Z}_p}^\delta$ of $G_{\mathbf{Q}_p}^\delta$ with the stabilizer of $L \otimes_{\mathbf{Z}} \overline{\mathbf{Z}}_p$ ($\overline{\mathbf{Z}}_p$ is the ring of integers in $\overline{\mathbf{Q}}_p$) is, for almost all p , one of the maximal compact subgroups referred to above.
- (ii) For almost all p the restriction of $\{a_\tau\}$ to

$$\mathfrak{G}(K\mathbf{Q}_p/\mathbf{Q}_p) = \mathfrak{G}(K_p/\mathbf{Q}_p) = \mathfrak{G}_p$$

splits. Moreover for such p there is a b in the intersection of G_{K_p} with the stabilizer of $L \otimes_{\mathbf{Z}} \overline{\mathbf{Z}}_p$ so that $a_\sigma = b^{\tau\sigma} b^{-1}$, $\sigma \in \mathfrak{G}_p$.

Now take a p satisfying (i) and (ii) which does not ramify in K . Since

$$\overline{G}_{\mathbf{Q}_p} = \left\{ g \in G_{K_p}^\delta \mid g^{\sigma a_\sigma} = g, \sigma \in \mathfrak{G}_p \right\},$$

the map $g \rightarrow g^b$ is an isomorphism of $\overline{G}_{\mathbf{Q}_p}$ with $G_{\mathbf{Q}_p}^\delta$. Moreover we can take $\overline{G}_{\mathbf{Z}_p}$ to be the intersection of $\overline{G}_{\mathbf{Q}_p}$ with the stabilizer of $L \otimes_{\mathbf{Z}} \overline{\mathbf{Z}}_p$ so the map takes $\overline{G}_{\mathbf{Z}_p}$ to $G_{\mathbf{Z}_p}^\delta$. The induced isomorphism of the Hecke algebras is independent of the choice of b . Now $\overline{G}_A = \prod_p \overline{G}_{\mathbf{Q}_p}$. Suppose we have an automorphic form ϕ on $\overline{G}_{\mathbf{Q}} \backslash \overline{G}_A$ which is an eigenfunction of the Hecke algebras for almost all p . Then, for almost all p , we have a homomorphism of the Hecke algebra into the complex numbers and thus a semi-simple conjugacy class α_p in $\mathfrak{G}_p \ltimes {}^cG \subseteq \mathfrak{G} \ltimes {}^cG$. If π is a complex representation of $\mathfrak{G} \ltimes {}^cG$, I define the Artin-Hecke L -series as

$$L(s, \pi, \phi) = \prod_p \frac{1}{\det\left(1 - \frac{\pi(\alpha_p)}{p^s}\right)} \quad (\text{Product is taken over almost all } p.)$$

[7] I have to check that these series are independent of the imbeddings of $\overline{\mathbf{Q}}$ into $\overline{\mathbf{Q}}_p$.[†] For the moment fix p . We have used the original imbedding to identify $\overline{\mathbf{Q}}$ with a subfield of $\overline{\mathbf{Q}}_p$. Let us preserve this identification. Any other imbedding is obtained by sending $x \rightarrow x^\tau$ with $\tau \in \mathfrak{S}(\overline{\mathbf{Q}}/\mathbf{Q})$. If we use the original imbedding to identify $\mathfrak{S}_p$ with a subgroup of $\mathfrak{S}$ then the map of $\mathfrak{S}_p$ onto $\mathfrak{S}$ given by the new imbedding is $\sigma \rightarrow \tau\sigma\tau^{-1}$. (Identify τ with its image in $\mathfrak{S}$.) The restriction of δ to $\mathfrak{S}_p$ is replaced by δ' with $\delta'(\sigma) = \delta(\tau\sigma\tau^{-1})$. Thus $G_{\mathbf{Q}_p}^\delta$ is replaced by $G_{\mathbf{Q}_p}^{\delta'}$. The map $g \rightarrow g^{\delta(\tau)}$ is an isomorphism of $G_{\mathbf{Q}_p}^{\delta'}$ with $G_{\mathbf{Q}_p}^\delta$. If $g \in G_{\mathbf{Q}}^\delta \subset G_{\mathbf{Q}_p}^\delta$ then g is the image of g^τ so this map commutes with the imbeddings of $G_{\mathbf{Q}}^\delta$ in the two groups. The new cocycle $\{a'_\sigma\}$ is the image of

$$a_{\tau\sigma\tau^{-1}} = a_\tau^{\sigma\tau^{-1}} a_\sigma^{\tau^{-1}} a_{\tau^{-1}} = a_\tau^{\sigma\tau^{-1}} a_\sigma^{\tau^{-1}} a_\tau^{-\tau^{-1}}$$

since $a_{\sigma\tau} = a_\sigma^\tau a_\tau$ for all σ and τ . The image is $\delta(\tau)a_\tau^\sigma a_\sigma a_\tau^{-1}\delta(\tau^{-1})$. Thus

$$\overline{G}'_{\mathbf{Q}_p} = \left\{ g \in G_{K_p} \left| \begin{array}{l} g = g^{\delta(\tau)\sigma\delta(\sigma)a_\tau^\sigma a_\sigma a_\tau^{-1}\delta(\tau^{-1})} \\ = g^{\delta(\tau)a_\tau\sigma\delta(\sigma)a_\sigma a_\tau^{-1}\delta(\tau^{-1})} \text{ for } \sigma \in \mathfrak{S}_p \end{array} \right. \right\}$$

and the map $g \rightarrow g^{\delta(\tau)a_\tau}$ is an isomorphism of $\overline{G}'_{\mathbf{Q}_p}$ with $\overline{G}_{\mathbf{Q}_p}$. It commutes with the imbeddings of $\overline{G}_{\mathbf{Q}}$ in these two groups since

$$\overline{G}_{\mathbf{Q}} = \left\{ g \in G_K \mid g^{\rho\delta(\rho)a_\rho} = g \text{ for all } \rho \in \mathfrak{S} \right\}.$$

Moreover for almost all p it takes $\overline{G}'_{\mathbf{Z}_p}$ to $\overline{G}_{\mathbf{Z}_p}$. If then we choose for each p a new imbedding we get a new adèle group $\overline{G}'_A$. The above [8] maps define an isomorphism of $\overline{G}'_A$ with $\overline{G}_A$ which takes $\overline{G}_{\mathbf{Q}}$ to itself. Thus we have a map of $\overline{G}_{\mathbf{Q}} \backslash \overline{G}'_A$ to $\overline{G}_{\mathbf{Q}} \backslash \overline{G}_A$ and the automorphic form ϕ introduced above defines an automorphic form ϕ' on $\overline{G}_{\mathbf{Q}} \backslash \overline{G}'_A$ with the same properties. We have to check that

$$L(s, \phi', \pi) = L(s, \phi, \pi)$$

Fix p again. Then if $a_\sigma = b^\sigma b^{-1}$,

$$\begin{aligned} a'_\sigma &= \delta(\tau)a_\tau^\sigma b^\sigma \delta(\tau^{-1})\delta(\tau)b^{-1}a_\tau^{-1}\delta(\tau^{-1}) \\ &= [\delta'(\sigma^{-1})\sigma^{-1}\delta(\tau)a_\tau b\delta(\tau^{-1})\sigma\delta(\sigma)] [\delta(\tau)a_\tau b\delta(\tau^{-1})]^{-1} \end{aligned}$$

So a'_σ is split by $b' = \delta(\tau)a_\tau b\delta(\tau^{-1})$. For almost all p , b' lies in the stabilizer of $L \otimes_{\mathbf{Z}} \overline{\mathbf{Z}}_p$. Thus we have the following commutative diagram

$$\begin{array}{ccc} \overline{G}'_{\mathbf{Q}_p} & \xrightarrow{g \rightarrow g^{b'}} & G_{\mathbf{Q}_p}^{\delta'} \\ g \rightarrow g^{\delta(\tau)a_\tau} \downarrow & & \downarrow g \rightarrow g^{\delta(\tau)} \\ \overline{G}_{\mathbf{Q}_p} & \xrightarrow{g \rightarrow g^b} & G_{\mathbf{Q}_p}^\delta \end{array}$$

This means that if α'_p is the conjugacy class in $\tau\mathfrak{S}_p\tau^{-1} \ltimes_\delta {}^c G$ associated to ϕ' then $\alpha'_p = \tau\alpha_p\tau^{-1}$. This shows that α'_p and α_p are conjugate for almost all p and this shows that $L(s, \phi', \pi)$ and $L(s, \phi, \pi)$ differ by a finite number of factors.

[†]It should also be checked that the series converges for $\text{Re}(s)$ sufficiently large. There is a method of doing this; but I have yet to verify that it works for all groups.

The first question is whether or not these products define functions meromorphic in the entire complex plane with poles of the usual type and whether or not for each ϕ there is an automorphic form ψ so that $L(s, \phi, \pi)/L(s, \psi, \tilde{\pi})$ is an elementary function for all π † [9] $\tilde{\pi}$ is the representation contragredient to π .

Before I go into the second question let me just say that I have been making some experiments with Eisenstein series and, although the work is far from completed, it looks as though I will get some series of the above type which because of their relation to the Eisenstein series will be meromorphic in the whole plane. It might even be possible to get a functional equation in a smaller number of cases from the functional equations of the Eisenstein series. The definitions above are the result of trying to find some class of Euler products which will contain the ones coming from the Eisenstein series but which is not restricted in any artificial fashion.

Now if $G = \text{GL}(n)$, the action of $\mathfrak{G}$ is trivial, and π is the representation $g \rightarrow g$ one can perhaps use the ideas of Tamagawa to handle the above series. This leads to the second question. Suppose we have K , G , and δ as above and also K' , G' , and δ' . If $K \subset K'$ we have a homomorphism $\mathfrak{G}' \rightarrow \mathfrak{G}$. Suppose moreover that ω is a homomorphism of $\mathfrak{G} \times_{\delta'} {}^c G'$ into $\mathfrak{G} \times_{\delta} {}^c G$ so that the following diagram is commutative

$$\begin{array}{ccc} \mathfrak{G}' \times_{\delta'} {}^c G' & \longrightarrow & \mathfrak{G}' \\ \omega \downarrow & & \downarrow \\ \mathfrak{G} \times_{\delta} {}^c G & \longrightarrow & \mathfrak{G} \end{array}$$

If ϕ' is an automorphic form for some inner form of $G'^{\delta'}$ satisfying the condition we had above then for almost all p , ϕ' defines a conjugacy class α'_p in $\mathfrak{G}' \times_{\delta'} {}^c G'$. [10] Let α_p be the image of α'_p in $\mathfrak{G} \times_{\delta} {}^c G$. The second question is the following. Is there an automorphic form ϕ associated to some inner form of G^{δ} such that for almost all p the conjugacy class associated to it is α_p ?

Let me give some idea of what an affirmative answer to the question entails.

- (i) Take $\mathfrak{G}' = \mathfrak{G}$ and let G' be a split torus of rank ℓ equal to the rational rank of ${}^c G^{\delta}$ on which $\mathfrak{G}'$ acts trivially. Let A be a maximal split torus of ${}^c G^{\delta}$. Since $\mathfrak{G}$ acts trivially on A , $\mathfrak{G}' \times_{\delta'} {}^c G' \cong \mathfrak{G} \times A \subseteq \mathfrak{G} \times_{\delta} {}^c G$. Since there are ℓ parameter families of automorphic forms on $G'_{\mathbf{Q}} \backslash G'_A$ an affirmative answer implies the same is true for some inner form of G^{δ} . But this we know from the theory of Eisenstein series.
- (ii) Let $\mathfrak{G}' = \mathfrak{G}$ and let $G' = \{1\}$. Map $\mathfrak{G}' \times_{\delta'} {}^c G'$ to $\mathfrak{G} \times \{1\} \subseteq \mathfrak{G} \times_{\delta} {}^c G$. In this case it should be possible to give an affirmative answer to the question by taking ϕ to be the automorphic form obtained by setting the parameter equal to zero in a suitable Eisenstein series. ϕ' is of course a constant.
- (iii) Now let me say a few words about the relation of the question to the Artin reciprocity law. For the rational field take $\mathfrak{G}'$ abelian, $G = \{1\}$, and let χ be a character of $\mathfrak{G}'$. Let $\mathfrak{G} = \mathfrak{G}'$ and let G be a one-dimensional split torus on which $\mathfrak{G}$ acts trivially. Let ω take $\tau \times 1$ to $\tau \times \chi(\tau)$. Then an affirmative answer [11] is just the Artin reciprocity law for cyclic extensions of the rationals. Now suppose we have the following situation

[†]It appears that it will be possible to take $\psi = \phi$.

$$\begin{array}{c}
K \\
| \\
K'_1 \\
| \\
K_1 \\
| \\
\mathbf{Q}
\end{array}$$

$K/\mathbf{Q}$ is Galois and K'_1/K_1 is abelian. Let $\mathfrak{G} = \mathfrak{G}(K/\mathbf{Q})$, let $\mathfrak{G}_1$ be the elements of $\mathfrak{G}$ which fix K_1 and let $\mathfrak{G}'_1$ be the elements of $\mathfrak{G}$ which fix K'_1 . Finally suppose χ is a character of $\mathfrak{G}(K'_1/K_1) = \mathfrak{G}_1/\mathfrak{G}'_1$ and thus of $\mathfrak{G}_1$. I will take $\mathfrak{G}' = \mathfrak{G}$ and $G' = \{1\}$. Let $\ell = [\mathfrak{G}_1 : \mathfrak{G}]$ and let $\mathfrak{G} = \bigcup_{i=1}^{\ell} \mathfrak{G}_1 \tau_i$ with $\tau_1 \in \mathfrak{G}_1$. Let i_σ be such that $\tau_i \sigma \in \mathfrak{G}_1 \tau_{i_\sigma}$. Let G be the direct product $T_1 \times \cdots \times T_\ell$ of ℓ one-dimensional split tori. Define δ by $(t_1 \times \cdots \times t_\ell)^{\delta(\sigma)} = t_{1_{\sigma^{-1}}} \times \cdots \times t_{\ell_{\sigma^{-1}}}$. It is easy to check that $\delta(\sigma)\delta(\tau) = \delta(\sigma\tau)$. Moreover G^δ is just the multiplicative group of K_1 . Also ${}^c G = G$. Define $\rho_i(\sigma)$ by $\tau_i \sigma^{-1} = \rho_i^{-1}(\sigma) \tau_{i_{\sigma^{-1}}}$. Then

$$\tau_i \tau^{-1} \sigma^{-1} = \rho_i^{-1}(\tau) \tau_{i_{\tau^{-1}}} \sigma^{-1} = \rho_i^{-1}(\tau) \rho_{i_{\tau^{-1}}}(\sigma) \tau_{i_{\tau^{-1}} \sigma^{-1}};$$

so $\rho_i(\sigma\tau) = \rho_{i_{\tau^{-1}}}(\sigma) \rho_i(\tau)$. Define ω by

$$\omega(\sigma \times 1) = \sigma \times \left(\chi(\rho_1(\sigma)) \times \cdots \times \chi(\rho_\ell(\sigma)) \right).$$

Then

$$\begin{aligned}
\omega(\sigma \times 1) \omega(\tau \times 1) &= \sigma \tau \times \prod_{i=1}^{\ell} \chi(\rho_{i_{\tau^{-1}}}(\sigma)) \chi(\rho_i(\tau)) \\
&= \omega(\sigma\tau \times 1).
\end{aligned}$$

[12] By the way if the τ_i , $1 \leq i \leq \ell$ are replaced by $\tau'_i = \mu_i \tau_i$ with $\mu_i \in \mathfrak{G}_1$ then $\rho'_i(\sigma) = \mu_{i_{\sigma^{-1}}}^{-1} \rho_i(\sigma) \mu_i$ and

$$\omega'(\sigma \times 1) = (\chi(\mu_1) \times \cdots \times \chi(\mu_\ell))^{-1} \omega(\sigma \times 1) (\chi(\mu_1) \times \cdots \times \chi(\mu_\ell)).$$

So the map does not depend in an essential way on the choice of coset representatives.

I will take ϕ' to be a constant. By the Artin reciprocity law there is associated to χ a character of $K_1^\times \backslash I_{K_1}$, that is, an automorphic form ϕ on $G_{\mathbf{Q}}^\delta \backslash G_A^\delta$. I claim that ϕ is the automorphic form which provides an affirmative answer to the question.

[13] To show this we make use of the freedom we have in the choice of coset representatives. Let p be a prime which does not ramify in K . Fix an imbedding of K in $\overline{\mathbf{Q}}_p$. We identify K with its image. Let $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ be the prime divisors of p in K_1 . Choose $\mu_1, \dots, \mu_r$ in $\mathfrak{G}$ so that the map $x \rightarrow x^{\mu_j}$ of K_1 into $\overline{\mathbf{Q}}_p$ extends to a continuous map of the completion of K_1 with respect to $\mathfrak{p}_j$ into $\overline{\mathbf{Q}}_p$. Let $L_j = [K_1^{\mu_j} \mathbf{Q}_p : \mathbf{Q}_p]$, and let $n_j = [L_j : \mathbf{Q}_p]$. If σ_p is the Frobenius the automorphisms $\mu_j \sigma_p^k$, $1 \leq j \leq r$, $0 \leq k \leq n_j$ form a set of representatives for the cosets of $\mathfrak{G}_1$. If $\tau_i = \mu_j \sigma_p^k$ then

$\rho_i(\sigma_p) = 1$ unless $k = 0$ when $\rho_i(\sigma_p) = \mu_j \sigma_p^{n_j} \mu_j^{-1}$. Thus $\omega(\alpha'_p) = \alpha_p$ is the conjugacy class of

$$\sigma_p \propto \prod_{j=1}^{\ell} \left(\chi(\mu_j \sigma_p^{n_j} \mu_j^{-1}) \times 1 \times \cdots \times 1 \right).$$

$\mu_j \sigma_p^{n_j} \mu_j^{-1}$ belongs to the Frobenius conjugacy class in $\mathfrak{G}_1$ corresponding to $\mathfrak{p}_j$.

On the other hand $G_{\mathbf{Q}_p}^{\delta} \subseteq G_A^{\delta}$ is the set of elements of the form

$$\prod_{j=1}^r \prod_{k=0}^{n_j-1} x_j^{\sigma^k}$$

with x_j a non-zero element in L_j . The restriction of ϕ to such an element is, by its very definition,

$$\prod_{j=1}^{\ell} \chi(\mu_j \sigma_p^{n_j} \mu_j^{-1})^{o(x_j)}$$

if $|x_j| = p^{-o(x_j)}$. Since ${}^c G = G$ the associated conjugacy class is the one determined by any element

$$\sigma_p \propto \prod_{j=1}^r \prod_{k=0}^{n_j-1} \alpha_{jk}$$

such that

$$\prod_{j=1}^r \prod_{k=0}^{n_j-1} \alpha_{jk}^{o(x_j)} = \prod_{j=1}^{\ell} \chi(\mu_j \sigma_p^{n_j} \mu_j^{-1})^{o(x_j)}$$

[14] Looking above we see that $\omega(\alpha'_p)$ is such an element.

- (iv) Finally I want to comment on the implications an affirmative answer to the second question might have for the problem of finding a splitting law for non-abelian extensions. I had planned to discuss arbitrary ground fields but I realize now that I have to take the ground field to be $\mathbf{Q}$. However one could presumably go back and reformulate the two questions in the context of groups over a number field. The first question is not sensitive to the choice of ground field but the second is. I did not appreciate this until now; since little would be gained by rewriting the letter I content myself with taking the ground field to be $\mathbf{Q}$.

Let K be a Galois extension of $\mathbf{Q}$ and let $\mathfrak{G} = \mathfrak{G}(K/\mathbf{Q})$. We want a method of finding for almost all p the Frobenius conjugacy class $\{\sigma_p\}$ in $\mathfrak{G}$. Thus we have to find trace $\pi(\sigma_p)$ or the conjugacy class of $\pi(\sigma_p)$ in $\mathrm{GL}(m, \mathbf{C})$, if $\pi : \mathfrak{G} \rightarrow \mathrm{GL}(m, \mathbf{C})$, for all representations π of $\mathfrak{G}$. Let us fix π . As before I will take $\mathfrak{G}' = \mathfrak{G}$, $G' = \{1\}$, and ϕ' to be a constant function. I will take $G = \mathrm{GL}(m)$. Let me check that ${}^c G$ is also $\mathrm{GL}(m)$.

Take $\tilde{G} = {}^c\tilde{G} = A \times \mathrm{SL}(m)$ where A is a one-dimensional split torus. Then

$$\begin{aligned} {}^c\tilde{L} = \tilde{L} &= \left\{ (z; z_1, \dots, z_m) \left| z, z_i - z_j \in \mathbf{Z}, \sum_{i=1}^m z_i = 0 \right. \right\}, \\ {}^cL = L &= \left\{ (z; z_1, \dots, z_m) \left| z_i + \frac{z}{m} \in \mathbf{Z}, \sum_{i=1}^m z_i = 0 \right. \right\}, \\ {}^c\tilde{L}' = \tilde{L}' &= \{ (mz; z_1, z_2 - z_1, \dots, z_{n-1} - z_{n-2}, -z_{n-1}) \mid z, z_i \in \mathbf{Z} \}. \end{aligned}$$

The pairing is given by

$$\langle (z; z_1, \dots, z_m), (y; y_1, \dots, y_m) \rangle = \frac{zy}{m} + \sum_{i=1}^m z_i y_i.$$

[15] In any case $G = {}^cG = \mathrm{GL}(m)$. Define ω by

$$\omega(\sigma \ltimes 1) = \sigma \ltimes \pi(\sigma).$$

The action of $\mathfrak{G}$ on G is to be trivial. Since $\omega(\alpha'_p) = \alpha_p$ is the conjugacy class of $\sigma_p \ltimes \pi(\sigma_p)$ which of course determines the conjugacy class of $\pi(\sigma_p)$, all we need is a method of finding α_p .

Suppose there is an automorphic form ϕ on some inner form of $\mathrm{GL}(m)$ which provides an affirmative answer to the above question. To find α_p all we need do is calculate the eigenvalues of a finite number of elements of the Hecke algebra H_p corresponding to the eigenfunction ϕ . Choose a finite set S of primes containing the infinite prime so that if $\overline{G}_S = \prod_{q \in S} \overline{G}_{\mathbf{Q}_q}$ and $\overline{G}_{S'} = \prod_{q \notin S} \overline{G}_{\mathbf{Q}_q}$ then $\overline{G}_A = \overline{G}_{\mathbf{Q}} \overline{G}_S \overline{G}_{S'}$ and ϕ is a function on $\overline{G}_{\mathbf{Q}} \backslash \overline{G}_A / \overline{G}_{S'}$.

Suppose $p \notin S$ and f is the characteristic function of $\overline{G}_{\mathbf{Z}_p} a \overline{G}_{\mathbf{Z}_p}$ which is the disjoint union $\bigcup_{i=1}^n a_i \overline{G}_{\mathbf{Z}_p}$. If $g \in \overline{G}_S$

$$\begin{aligned} \lambda(f)\phi(g) &= \int_{\overline{G}_{\mathbf{Z}_p}} \phi(gh) f(h) dh \\ &= \sum_{i=1}^n \phi(ga_i) = \sum_{i=1}^n \phi(a_i g) \end{aligned}$$

since $a_i \in \overline{G}_{\mathbf{Q}_p}$. Choose $\bar{a}_1, \dots, \bar{a}_n$ in $\overline{G}_{\mathbf{Q}}$ so that $(\bar{a}_i)^{-1}a_i \in \overline{G}_{S'}$ and let b_i be the projection of $\bar{a}_i$ on $\overline{G}_S$. If $\chi(f)$ is the eigenvalue of f ,

$$\chi(f)\phi(g) = \sum \phi((\bar{a}_i)^{-1}a_i g) = \sum \phi(b_i^{-1}g).$$

[16] Now roughly speaking the elements $\bar{a}_1, \dots, \bar{a}_n$ are obtained by solving some diophantine equations involving p as a parameter. Then $\phi(b_i^{-1}g)$ depends upon the congruential properties of $\bar{a}_i$ modulo powers of the finite primes in S and the projection of $\bar{a}_i$ on $\overline{G}_{\mathbf{Q}_{\infty}} = \overline{G}_{\mathbf{R}}$. If, for each g in $\overline{G}_S$, $\phi(hg)$ as a function of h in the connected component of $\overline{G}_{\mathbf{R}}$ were rational we would get a good splitting law. It would be rather complicated but in principle no worse than the splitting law of Dedekind-Hasse for the splitting field of a cubic equation. However because of the strong approximation

$\phi(hg)$ will probably not be rational unless $m = 1$ or 2 . Thus we could only get a transcendental splitting law.

None the less if we took G to be the symplectic group in $2n$ variables and cG to be the orthogonal group in $2n + 1$ variables then strong approximation is no obstacle because G has inner forms for which $\overline{G}_{\mathbf{R}}$ is compact and we might hope to obtain laws about such things as the order of σ_p by considering imbeddings of $\mathfrak{G}$ in cG .

Yours truly,
R. Langlands

Postscript: Let me add

- (v) Let K be a quadratic extension of $\mathbf{Q}$. Let $\mathfrak{G}' = \mathfrak{G} = \mathfrak{G}(K/\mathbf{Q})$. Let $G' = {}^cG' = A_1 \times A_2$ where A_1 and A_2 are one-dimensional split tori. If σ is the non-trivial element of $\mathfrak{G}$ let $(t_1 \times t_2)^{\delta'(\sigma)} = t_2 \times t_1$. Let $G = \mathrm{GL}(2)$ and let $\mathfrak{G}$ act trivially on G . Define ω by

$$\begin{aligned}\omega(1 \times (t_1, t_2)) &= \begin{pmatrix} t_1 & 0 \\ 0 & t_2 \end{pmatrix} \\ \omega(\sigma \times (t_1, t_2)) &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} t_1 & 0 \\ 0 & t_2 \end{pmatrix} = \begin{pmatrix} 0 & t_2 \\ t_1 & 0 \end{pmatrix}\end{aligned}$$

$G_A^{\delta'}$ is just the idele group of K . Take ϕ' to be a Grössencharakter. It is not inconceivable that the work of Hecke and Maass on the relation between [17] L -series with Grössencharakter from a quadratic field and automorphic forms will provide an affirmative answer to the second question in this case.

Compiled on July 30, 2024.

PROOF OF THE GEOMETRIC LANGLANDS CONJECTURE I: CONSTRUCTION OF THE FUNCTOR

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To Sasha Beilinson

ABSTRACT. In this paper we construct the geometric Langlands functor in one direction (from the automorphic to the spectral side) in characteristic zero settings (i.e., de Rham and Betti). We prove that various forms of the conjecture (de Rham vs Betti, restricted vs. non-restricted, tempered vs. non-tempered) are equivalent. We also discuss structural properties of Hecke eigensheaves.

CONTENTS

Introduction	2
0.1. The Langlands functor in the de Rham setting	2
0.2. The Betti setting	4
0.3. Comparison of the different versions of GLC	5
0.4. Characteristic cycles of Hecke eigensheaves	7
0.5. Contents	7
0.6. Acknowledgements	9
1. Construction of the Langlands functor (de Rham context)	9
1.1. The automorphic category	9
1.2. The spectral action	10
1.3. The vacuum Poincaré sheaf	10
1.4. The coarse version of the functor	12
1.5. The case $G = T$	13
1.6. Statement of the main result	14
1.7. Compatibility with the spectral action	15
1.8. The bound on the right	16
2. Proof of Theorem 1.6.2	16
2.1. Strategy of the proof	17
2.2. Proof of Proposition 2.1.1	17
2.3. Proof of Theorem 2.1.2	19
2.4. Proof of Proposition 2.3.2	20
2.5. Proof of Proposition 1.8.2	22
3. Geometric Langlands functor in the Betti context	23
3.1. The category with nilpotent singular support	23
3.2. The Hecke action in the Betti context	23
3.3. The vacuum Poincaré sheaf in the Betti context	24
3.4. Construction of the functor	25
3.5. Comparison of de Rham and Betti versions of GLC	26
4. Proof of Theorem 3.4.4	27
4.1. Reduction steps	27
4.2. Applications of Riemann-Hilbert	29

Date: September 11, 2024.

4.3. Proof of Proposition 4.1.5	30
4.4. Proof of Lemma 4.3.2	32
5. Restricted vs full GLC	32
5.1. The de Rham context	32
5.2. Full vs tempered Langlands	35
5.3. Proof of Proposition 5.2.3	35
5.4. Proof of Proposition 5.2.4	37
5.5. The Betti context	38
6. The structure of Hecke eigensheaves	39
6.1. Statement of the main result	39
6.2. Characteristic cycles for eigensheaves	41
6.3. Proof of Theorem 6.2.2	42
References	43

INTRODUCTION

This paper is the first in a series of five, in which the¹ geometric Langlands conjecture will be proved.

The entire project is joint work of D. Arinkin, D. Beraldo, L. Chen, J. Faergeman, D. Gaitsgory, K. Lin, S. Raskin and N. Rozenblyum.

The individual papers in the five-paper series will have different subsets of the above people as authors.

0.1. The Langlands functor in the de Rham setting. We first consider the de Rham version of the geometric Langlands conjecture (GLC).

0.1.1. Let k be a field of characteristic 0. Let X be a smooth and complete curve and G a connected reductive group over k . Let $\check{G}$ be the Langlands dual of G , viewed also as a reductive group over k .

The automorphic side of GLC is the category

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

of half-twisted D-modules on the moduli stack Bun_G of principal G -bundles on X , see Sect. 1.1.

The spectral side is the category

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$$

of ind-coherent sheaves with singular support in the nilpotent cone (see [AG, Sect. 11.1.5]) on the moduli stack $\mathrm{LS}_{\check{G}}$ of de Rham $\check{G}$ -local systems on X .

0.1.2. The Langlands functor, to be constructed in this paper, goes from the automorphic to the spectral side

$$(0.1) \quad \mathbb{L}_G : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}).$$

The geometric Langlands conjecture (GLC) in the de Rham setting says that the functor (0.1) is an equivalence.

¹Global and unramified.

0.1.3. However, there is an important piece of structure that enters the game before we attempt to construct $\mathbb{L}_G$, namely, the *spectral action*.

The assertion is that the combined action of the *Hecke functors* gives rise to a uniquely defined monoidal action of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ on $\mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$.

Such an action can be viewed as a “spectral decomposition” of $\mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ along the stack $\mathrm{LS}_{\check{G}}$.

The existence of the spectral action was established by V. Drinfeld and the first author (it is recorded in [Ga2, Corollary 4.5.5]), but its idea goes back to A. Beilinson, in the form of what is called the *Beilinson projector* (see [AGKRRV, Sect. 15] for a detailed discussion).

0.1.4. The construction of $\mathbb{L}_G$ proceeds in two steps: we first construct the *coarse* version of $\mathbb{L}_G$

$$\mathbb{L}_{G,\mathrm{coarse}} : \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}}).$$

It is related to the desired functor $\mathbb{L}_G$ by

$$(0.2) \quad \Psi_{\mathrm{Nilp},\{0\}} \circ \mathbb{L}_G \simeq \mathbb{L}_{G,\mathrm{coarse}},$$

where

$$\Psi_{\mathrm{Nilp},\{0\}} : \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}})$$

is the *coarsening* functor.

0.1.5. The functor $\mathbb{L}_{G,\mathrm{coarse}}$ stems from the classical idea in the theory of automorphic functions that says that Langlands correspondence is normalized by the Whittaker model. In the geometric context this idea is incarnated by the fact that the functor $\mathbb{L}_{G,\mathrm{coarse}}$ is uniquely determined by the following two requirements:

- The functor $\mathbb{L}_{G,\mathrm{coarse}}$ is compatible with the Hecke action, i.e., it intertwines the action of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ on the two sides;
- It makes the triangle

$$\begin{array}{ccc} & \mathrm{Vect} & \\ \mathrm{coeff}^{\mathrm{Vac},\mathrm{glob}} \nearrow & & \nwarrow \Gamma(\mathrm{LS}_{\check{G}},-) \\ \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_{G,\mathrm{coarse}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \end{array}$$

commute, where $\mathrm{coeff}^{\mathrm{Vac},\mathrm{glob}}$ is the functor of *first Whittaker coefficient*, see Sect. 1.3.8.

0.1.6. Once the functor $\mathbb{L}_{G,\mathrm{coarse}}$ is constructed, we lift it to the sought-for functor $\mathbb{L}_G$ using cohomological estimates provided by Theorem 1.6.2.

Namely, Theorem 1.6.2 says that the functor $\mathbb{L}_{G,\mathrm{coarse}}$ sends compact objects of $\mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ to *eventually coconnective* objects in $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$, i.e., objects that are cohomologically bounded on the left. As $\Psi_{\mathrm{Nilp},\{0\}}$ is an equivalence on eventually coconnective subcategories, we obtain the functor $\mathbb{L}_G$, which is then uniquely characterized by (0.2) and the requirement that $\mathbb{L}_G$ send compacts to eventually coconnective objects.²

In fact, we show that the functor $\mathbb{L}_{G,\mathrm{coarse}}$ has a cohomological amplitude bounded on the left. (Note, however, that this does *not* mean that the Langlands functor $\mathbb{L}_G$ has a cohomological amplitude bounded on the left; it rather has unbounded amplitude once G is non-abelian.)

Of less immediate practical importance, we also show that the functor $\mathbb{L}_{G,\mathrm{coarse}}$ has a cohomological amplitude bounded on the *right*. This property is automatically inherited by $\mathbb{L}_G$.

²We remark that the latter requirement is necessarily satisfied by any putative geometric Langlands functor: such a functor ought to be an equivalence, so must preserve compacts; and compact objects are eventually coconnective in $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$.

0.1.7. One can thus say that the upgrade

$$(0.3) \quad \mathbb{L}_{G, \text{coarse}} \rightsquigarrow \mathbb{L}_G,$$

carried out in the present paper, is cohomological in nature. Its main ingredient, Theorem 1.6.2, uses non-trivial input: ultimately we deduce it from Theorem 2.3.8, proved in [FR] using methods developed in [AGKRRV] and [Lin].

An alternative approach to constructing $\mathbb{L}_G$ is being developed in separate work of D. Beraldo, L. Chen and K. Lin. Their work is based on “gluing” the category $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ from the categories $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\text{temp}}$ (see Sect. 5.1.2) for Levi subgroups M of G , using the *Eisenstein series* functors, see [BeLi]. Then they construct the functor using a parallel procedure on the spectral side, i.e., gluing $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$ from the categories $\mathrm{QCoh}(\mathrm{LS}_{\check{M}})$, using the *spectral Eisenstein series* functors, developed in [Be2]. Unlike the approach in the present paper, their work pursues (and better develops) the original proposal from [Ga2]³.

0.2. **The Betti setting.** We now consider GLC in the Betti setting.

0.2.1. In this case our algebraic geometry happens over the ground field $\mathbb{C}$ (so X and G are over $\mathbb{C}$), and we work with sheaves of $\mathfrak{e}$ -vector spaces, where $\mathfrak{e}$ is an arbitrary field of characteristic 0.

So, $\check{G}$ is an reductive group over $\mathfrak{e}$, and $\mathrm{LS}_{\check{G}}$ is an algebraic stack over $\mathfrak{e}$.

0.2.2. A historical and conceptual difference between the de Rham and Betti situations is that while it has “always” (i.e., since the inception of the subject by A. Beilinson and V. Drinfeld) been understood what the automorphic side in the de Rham setting was (i.e., the entire category of half-twisted D-modules), in the Betti setting it was a relatively recent discovery, due to D. Ben-Zvi and D. Nadler, see [BZN].

Namely, the entire category $\mathrm{Shv}_{\frac{1}{2}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$ of (twisted) Betti sheaves on Bun_G is too big to be equivalent to something on the spectral side. What Ben-Zvi and Nadler discovered was that there is a reasonable automorphic candidate: namely, this is the full category

$$\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G) \subset \mathrm{Shv}_{\frac{1}{2}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$$

of Betti sheaves with singular support in the global nilpotent cone.

0.2.3. The spectral side in the Betti setting must also be modified as compared to the de Rham setting, but that modification is evident: one should replace the stack $\mathrm{LS}_{\check{G}}$ of de Rham local systems on X (i.e., the stack classifying $\check{G}$ -bundles with a connection) by the stack $\mathrm{LS}_{\check{G}}^{\mathrm{Betti}}$ of Betti $\check{G}$ -local systems (i.e., if we ignore the derived structure, this is the stack of homomorphisms $\pi_1(X) \rightarrow \check{G}$, divided by the action of $\check{G}$ by conjugation).

In retrospect, knowing the spectral side, one can convince oneself that $\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$ is the right candidate for the automorphic category as follows: this is the largest subcategory of $\mathrm{Shv}_{\frac{1}{2}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$, on which the Hecke action gives rise to an action of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{Betti}})$, see [AGKRRV, Theorem 18.1.4].

0.2.4. With the candidates for the automorphic and spectral sides in place, the construction of the Langlands functor in the Betti setting

$$(0.4) \quad \mathbb{L}_G^{\mathrm{Betti}} : \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{Betti}})$$

proceeds along the same lines as in the de Rham setting, with one modification:

The functor $\mathrm{coeff}^{\mathrm{Vac}, \mathrm{glob}}$ in the de Rham setting appealed to the exponential D-module on $\mathbb{G}_a$. There is (obviously) no Betti analog of the exponential D-module, but one finds a substitute using an additional group of symmetries (given by the torus T) acting in our situation, see Sect. 3.3.

³An additional merit of this approach is that, unlike using the definition of $\mathrm{IndCoh}(-)$ from scratch, i.e., as the ind-completion of $\mathrm{Coh}(-)$, it uses the formal properties of the IndCoh category.

0.2.5. The geometric Langlands conjecture (GLC) in the Betti setting says that the functor (0.4) is an equivalence.

0.3. **Comparison of the different versions of GLC.** In addition to the de Rham (resp., Betti) versions of GLC, which we will refer to as *full* GLC, in both settings there exist other versions, which we refer to as *restricted* and *tempered*, respectively. And there exist also the restricted tempered versions.

However, it turns out that all these versions are logically equivalent in the sense that any one implies the others.

0.3.1. The restricted version of GLC, in either de Rham or Betti versions, was introduced⁴ in [AGKRRV, Conjecture 21.2.7].

On the automorphic side of the restricted GLC in the de Rham setting, we have the full subcategory

$$\mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G) \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

that consists of (twisted) D-modules with singular support in the nilpotent cone $\mathrm{Nilp} \subset T^*(\mathrm{Bun}_G)$. Objects of this category are (obviously) ind-holonomic. However, one can show that actually have regular singularities (this is a non-trivial result [AGKRRV, Corollary 16.5.6]).

On the automorphic side of the restricted GLC in the Betti setting, we have the full subcategory

$$\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G) \subset \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$$

that consists of *ind-constructible* sheaves.

In fact, both categories

$$\mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G) \text{ and } \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G)$$

fall into the paradigm of [AGKRRV, Sect. 14.1], where the restricted automorphic category is defined starting from a *constructible sheaf theory* $\mathrm{Shv}(-)$ as in [AGKRRV, Sect. 1.1.1]. The two sheaf theories in question are

$$\mathrm{Shv}(-) := \mathrm{D}\text{-mod}^{\mathrm{RS}}(-) \text{ and } \mathrm{Shv}(-) := \mathrm{Shv}^{\mathrm{Betti}, \mathrm{constr}}(-),$$

respectively.

0.3.2. On the spectral side of the restricted GLC we have the categories

$$(0.5) \quad \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{restr}}) \text{ and } \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{Betti}, \mathrm{restr}}),$$

respectively, where $\mathrm{LS}_G^{\mathrm{restr}}$ and $\mathrm{LS}_G^{\mathrm{Betti}, \mathrm{restr}}$ are the de Rham and Betti versions of the *stack of local systems with restricted variation* of [AGKRRV, Sect. 1.4].

0.3.3. Starting from the full geometric Langlands functors $\mathbb{L}_G$ and $\mathbb{L}_G^{\mathrm{Betti}}$ one produces their restricted versions

$$\mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G) \xrightarrow{\mathbb{L}_G^{\mathrm{restr}}} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{restr}})$$

and

$$\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G) \xrightarrow{\mathbb{L}_G^{\mathrm{Betti}, \mathrm{restr}}} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{Betti}, \mathrm{restr}}).$$

by applying the operations

$$- \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})} \mathrm{QCoh}(\mathrm{LS}_{\check{G}})_{\mathrm{restr}} \text{ and } - \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{Betti}})} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{Betti}})_{\mathrm{restr}},$$

where

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}})_{\mathrm{restr}} \subset \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \text{ and } \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{Betti}})_{\mathrm{restr}} \subset \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{Betti}})$$

are full subcategories that consist of objects set-theoretically supported on

$$\mathrm{LS}_G^{\mathrm{restr}} \subset \mathrm{LS}_G \text{ and } \mathrm{LS}_G^{\mathrm{Betti}, \mathrm{restr}} \subset \mathrm{LS}_G^{\mathrm{Betti}},$$

respectively.

⁴Without specifying the functor in either direction.

It is clear that if $\mathbb{L}_G$ and $\mathbb{L}_G^{\text{Betti}}$ are equivalences, then so are $\mathbb{L}_G^{\text{restr}}$ and $\mathbb{L}_G^{\text{Betti, restr}}$. However, it turns out that the converse implications also take place. This is proved in Sect. 5 of the present paper; see also Sect. 0.3.6 below. This material builds on [AGKRRV, Sect. 21.4].

0.3.4. Thus, we have:

$$\text{Full de Rham GLC} \Leftrightarrow \text{Restricted de Rham GLC}$$

and

$$\text{Full Betti GLC} \Leftrightarrow \text{Restricted Betti GLC}.$$

However, Riemann-Hilbert implies that we also have

$$\text{Restricted de Rham GLC} \Leftrightarrow \text{Restricted Betti GLC}.$$

As a result, we deduce that

$$(0.6) \quad \text{Full de Rham GLC} \Leftrightarrow \text{Full Betti GLC}.$$

0.3.5. In the rest of this series of papers we will focus on the de Rham version of GLC.

This is because it is in this context that one can use Kac-Moody localization, which provides a powerful tool in the study of $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$. It would not be an exaggeration to say that the proofs of all the key results⁵ in this series of papers ultimately appeal to this method.

However, thanks to (0.6), once we prove the de Rham version of GLC, we deduce the Betti case as well.

0.3.6. Let us now comment on how we deduce

$$\text{Restricted de Rham GLC} \Rightarrow \text{Full de Rham GLC}$$

(the Betti case is similar).

To do so, we introduce yet another version of GLC: the tempered one. Namely, the category $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ admits a localization

$$\mathbf{u} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}} \rightleftarrows \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) : \mathbf{u}^R,$$

and similarly for its the restricted version

$$\mathbf{u} : \text{D-mod}_{\frac{1}{2}, \text{Nilp}}(\text{Bun}_G)_{\text{temp}} \rightleftarrows \text{D-mod}_{\frac{1}{2}, \text{Nilp}}(\text{Bun}_G) : \mathbf{u}^R.$$

The tempered versions of the Langlands functor map

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}} \xrightarrow{\mathbb{L}_{G, \text{temp}}} \text{QCoh}(\text{LS}_{\check{G}})$$

and

$$\text{D-mod}_{\frac{1}{2}, \text{Nilp}}(\text{Bun}_G)_{\text{temp}} \xrightarrow{\mathbb{L}_{G, \text{temp}}^{\text{restr}}} \text{QCoh}(\text{LS}_{\check{G}}^{\text{restr}}),$$

respectively.

The tempered (resp., tempered restricted) version of GLC says that the functor $\mathbb{L}_{G, \text{temp}}$ (resp., $\mathbb{L}_{G, \text{temp}}^{\text{restr}}$) is an equivalence.

0.3.7. It is clear that we have the implications

$$\text{Full GLC} \Rightarrow \text{Tempered GLC} \quad \text{and} \quad \text{Restricted GLC} \Rightarrow \text{Tempered restricted GLC}$$

In Sect. 5.2 we establish an inverse implication

$$\text{Tempered GLC} \Rightarrow \text{Full GLC}.$$

(A similar argument shows that $\text{Tempered restricted GLC} \Rightarrow \text{Restricted GLC}$.)

⁵Including the existence of the spectral action.

0.3.8. It is also clear that

$$\text{Tempered GLC} \Rightarrow \text{Tempered restricted GLC}.$$

However, by mimicking the argument of [AGKRRV, Sect. 21.4] one proves that we in fact have

$$\text{Tempered GLC} \Leftrightarrow \text{Tempered restricted GLC}.$$

Remark 0.3.9. In fact, in [AGKRRV, Sect. 21.4] the equivalence

$$\text{Full GLC} \Leftrightarrow \text{Restricted GLC}$$

was proved directly, but under the assumption that the functor $\mathbb{L}_G$ admits a left adjoint.

The advantage of working with the tempered category is that the existence of the left adjoint of $\mathbb{L}_{G,\text{temp}}$ is automatic from the construction.

0.4. Characteristic cycles of Hecke eigensheaves. We prove an additional result the interaction between de Rham and Betti geometric Langlands.

Theorem 0.4.1. *Suppose that G has connected center, the genus of X is ≥ 2 , and assume the geometric Langlands conjecture. Let σ be an irreducible $\check{G}$ -local system and let $\mathcal{F}_\sigma$ be the corresponding Hecke eigensheaf.*

Then the characteristic cycle $\text{CC}(\mathcal{F}_\sigma)$ of $\mathcal{F}_\sigma$ equals the global nilpotent cone.

We prove this result via the interaction of Betti and de Rham geometric Langlands, and it is closely related to perversity properties of $\mathbb{L}_{G,\text{temp}}$ used elsewhere in the paper, which is why we consider it relevant to this paper.

We refer to Sect. 6 for more details, including a review of other known properties of $\mathcal{F}_\sigma$.

0.5. Contents. We now review the contents of the paper by section.

0.5.1. In Sect. 1 we construct the Langlands functor $\mathbb{L}_G$ in the de Rham context.

We first collect the ingredients needed for the construction of the *coarse* Langlands functor (the spectral action, the vacuum Poincaré object).

We state Theorem 1.6.2, which is the tool that allows us to upgrade $\mathbb{L}_{G,\text{coarse}}$ to the Langlands functor $\mathbb{L}_G$.

We show that $\mathbb{L}_G$ is compatible with the actions of $\text{QCoh}(\text{LS}_{\check{G}})$ on the two sides.

0.5.2. In Sect. 2 we prove Theorem 1.6.2.

We show that compact objects of $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ are bounded below (in fact, this is a general property of the category of D-modules on a *truncatable* algebraic stack with an affine diagonal), see Proposition 1.8.4. This reduces Theorem 1.6.2 to Theorem 2.1.2, which says that the functor $\mathbb{L}_{G,\text{coarse}}$ has a cohomological amplitude bounded on the left.

We express the condition on an object of $\text{QCoh}(\mathcal{Y})$ (here $\mathcal{Y}$ is an arbitrary eventually coconnective stack) to be bounded on the left in terms of its $!$ -fibers, see Proposition 2.3.2.

The latter proposition allows us to reduce Theorem 2.1.2 to its version, where instead of $\mathbb{L}_{G,\text{coarse}}$, we are dealing with its restricted version, $\mathbb{L}_{G,\text{coarse}}^{\text{restr}}$. However, in the latter case, the required assertion has been already established in [FR].

By a similar manipulation, we prove Proposition 1.8.2, which says that the functor $\mathbb{L}_{G,\text{coarse}}$ (and, hence, $\mathbb{L}_G$) has a cohomological amplitude bounded on the right.

0.5.3. In Sect. 3 we construct the geometric Langlands functor $\mathbb{L}_G$ in the Betti setting.

We first review the basics of the category $\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$ (compact generation, relation to the entire category $\mathrm{Shv}_{\frac{1}{2}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$).

We construct the Betti version of the vacuum Poincaré object, which requires a little bit of work, due to the fact that the exponential sheaf does not exist in the Betti context.

We state Theorem 3.4.4, which is a Betti counterpart of Theorem 1.6.2. Assuming this theorem, we construct $\mathbb{L}_G^{\mathrm{Betti}}$, the Betti version of the Langlands functor, by the same procedure as in the de Rham context.

We state Theorem 3.5.6, which says that the de Rham and Betti versions of GLC (along with their restricted variants) are all logically equivalent.

0.5.4. In Sect. 4, we prove Theorem 3.4.4.

We note that the proof of Theorem 3.4.4 does *not* mimic that of Theorem 1.6.2:

We do not a priori know that the compact generators of $\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$ are bounded below (although ultimately one can show that they are). Instead, we describe these compact generators explicitly and show directly that the functor $\mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{Betti}}$ sends them to bounded below objects. While doing so, we study the interactions between the following functors:

- The *left* adjoint to $\mathbf{i} : \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G) \hookrightarrow \mathrm{Shv}_{\frac{1}{2}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$;
- The *right* adjoint to $\iota^{\mathrm{Betti}} : \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G) \hookrightarrow \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$;
- The *right* adjoint to $\mathbf{i}^{\mathrm{constr}} : \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G) \hookrightarrow \mathrm{Shv}_{\frac{1}{2}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G)$.

In fact, it turns out that $(\mathbf{i}^{\mathrm{constr}})^R \simeq (\iota^{\mathrm{Betti}})^R \circ \mathbf{i} \circ \mathbf{oblv}^{\mathrm{constr}}$, as functors

$$\mathrm{Shv}_{\frac{1}{2}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G) \rightarrow \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G),$$

where $\mathbf{oblv}^{\mathrm{constr}}$ is the forgetful functor

$$\mathrm{Shv}_{\frac{1}{2}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G) \rightarrow \mathrm{Shv}_{\frac{1}{2}}^{\mathrm{Betti}}(\mathrm{Bun}_G).$$

0.5.5. In Sect. 5, we prove the implications

$$\text{Restricted de Rham GLC} \Rightarrow \text{Full de Rham GLC} \quad \text{and} \quad \text{Restricted Betti GLC} \Rightarrow \text{Full Betti GLC}$$

along the lines explained in Sect. 0.3.

We introduce the tempered GLC, in both full and restricted variants, and show, following [AGKRRV, Sect. 21.4] that these two variants are logically equivalent.

We then show that the tempered GLC is equivalent to the original full GLC.

0.5.6. In Sect. 6, we calculate characteristic cycles of Hecke eigensheaves (at least when G has connected center). We do this by proving constancy of characteristic cycles along the moduli of irreducible local systems and then appealing to [BD] to compute this constant value.

0.5.7. *Notations and conventions.* Notations and conventions in this paper largely follow those adopted in [AGKRRV].

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1. CONSTRUCTION OF THE LANGLANDS FUNCTOR (DE RHAM CONTEXT)

In this section we will first review the ingredients that are needed for the construction of the Langlands functor, and then carry out the construction in question, all in the context of D-modules.

The “coarse” version of the functor will be automatic from the spectral action (see Sect. 1.4). To upgrade it to the actual Langlands functor, we will need some cohomological estimates, which are provided by Theorem 1.6.2.

1.1. The automorphic category. In this subsection we will introduce the main player on the automorphic side: the category $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ of (half-twisted) D-modules on the moduli space Bun_G of G -bundles on X .

1.1.1. Let $\det_{\mathrm{Bun}_G}$ be the determinant line bundle on Bun_G , normalized so that it sends a G -bundle $\mathcal{P}_G$ to

$$\det(\Gamma(X, \mathfrak{g}_{\mathcal{P}_G})) \otimes \det\left(\Gamma(X, \mathfrak{g}_{\mathcal{P}_G^0})\right)^{\otimes -1},$$

where $\mathcal{P}_G^0$ is the trivial bundle.

Note also that up to the (constant) line $\det\left(\Gamma(X, \mathfrak{g}_{\mathcal{P}_G^0})\right)$, the line bundle $\det_{\mathrm{Bun}_G}$ identifies with the canonical line bundle on Bun_G .

1.1.2. According to [BD, Sect. 4], the choice of $\omega_X^{\otimes \frac{1}{2}}$ gives rise to a square root of $\det_{\mathrm{Bun}_G}$.

1.1.3. Let $\mathcal{Y}$ be a space, and let $\mathcal{G}$ be an étale μ_n -gerbe on $\mathcal{Y}$ for some $n \in \mathbb{N}$. Denote by

$$\mathrm{D}\text{-mod}_{\mathcal{G}}(\mathcal{Y})$$

the corresponding category of twisted D-modules on $\mathcal{Y}$.

1.1.4. Let

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

be the twisted category of D-modules, corresponding to the μ_2 -gerbe $\det_{\mathrm{Bun}_G}^{\frac{1}{2}}$ of square roots of $\det_{\mathrm{Bun}_G}$.

The category $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ is the de Rham incarnation of the automorphic category. It is the primary object of study in the geometric Langlands theory.

1.1.5. Note, however, that by Sect. 1.1.2, we have an identification

$$(1.1) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \simeq \mathrm{D}\text{-mod}(\mathrm{Bun}_G).$$

Such an identification can be convenient for the study of local properties of the category $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$. In particular, the results from [AGKRRV] pertaining to $\mathrm{D}\text{-mod}_{\mathrm{Nilp}}(\mathrm{Bun}_G)$ apply to the corresponding category

$$\mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G).$$

Remark 1.1.6. It is crucial, however, to consider the twisted version, when we study the Hecke functors. Otherwise, instead of the usual category $\mathrm{Rep}(\check{G})$ one has to consider its twist by a certain canonical $Z_{\check{G}}$ -gerbe, see [GLys, Sect. 6.3].

1.2. The spectral action. An ingredient, which is crucial for both for the construction of the Langlands functor and to the eventual proof that it is an equivalence, is the fact that the action of Hecke functors on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ gives rise to a monoidal action of the category $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$.

1.2.1. Let $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}$ be the de Rham version of the category $\mathrm{Rep}(\check{G})$ spread over the Ran space, see [Ga2, Sect. 4.2].

The (naive) geometric Satake functor and the Hecke action define an action of $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}$ on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$.

1.2.2. Recall now that we have a (symmetric) monoidal localization functor

$$(1.2) \quad \mathrm{Loc}_G^{\mathrm{spec}} : \mathrm{Rep}(\check{G}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}}),$$

see [Ga2, Sect. 4.3].

The category $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}$ is compactly generated, and the functor $\mathrm{Loc}_G^{\mathrm{spec}}$ preserves compactness (being a symmetric monoidal functor between rigid symmetric monoidal categories). Hence, it admits a continuous right adjoint.

It is a basic fact that this right adjoint is fully faithful. I.e., the functor $\mathrm{Loc}_G^{\mathrm{spec}}$ is a localization.

1.2.3. We have the following result (see [Ga2, Corollary 4.5.5]):

Theorem 1.2.4. *The action of $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}$ on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ factors via the functor (1.2).*

1.2.5. Thanks to Theorem 1.2.4, we obtain a canonically defined action of the monoidal category $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$.

We refer to it as the *spectral action*.

1.3. The vacuum Poincaré sheaf. The second ingredient for the construction of the Langlands functor $\mathbb{L}_G$ is the object of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$, which is the image of $\mathcal{O}_{\mathrm{LS}_{\check{G}}}$ under the would-be left adjoint $\mathbb{L}_G^L$ of $\mathbb{L}_G$.

This object is the *vacuum* Poincaré sheaf, introduced in this subsection. This particular choice for the image of $\mathcal{O}_{\mathrm{LS}_{\check{G}}}$ reflects the idea, familiar from the classical theory of automorphic functions, that the Langlands correspondence is normalized by the Whittaker model.

1.3.1. Let $\mathcal{P}_T$ be any T -bundle. Consider the stack

$$\mathrm{Bun}_{N, \mathcal{P}_T} \simeq \mathrm{Bun}_B \times_{\mathrm{Bun}_T} \mathrm{pt},$$

where the map $\mathrm{pt} \rightarrow \mathrm{Bun}_T$ corresponds the point $\mathcal{P}_T$. Denote by $\mathfrak{p}$ the map

$$\mathrm{Bun}_{N, \mathcal{P}_T} \rightarrow \mathrm{Bun}_G.$$

Note that the pullback of $\det_{\mathrm{Bun}_G}$ along this map is canonically constant. Denote the resulting line by

$$\mathbb{L}_{G, N, \mathcal{P}_T}.$$

1.3.2. The Langlands functor depends on the choice of the square root $\omega_X^{\frac{1}{2}}$ of the canonical line bundle on X , which we fix from now on. With this choice we claim:

Proposition 1.3.3. *The line $\mathfrak{l}_{G,N_{\mathcal{P}_T}}$ admits a canonical square root.*

Proof. We claim more generally that for any parabolic P with Levi quotient M and an M -bundle $\mathcal{P}_M$, line

$$\det_{\text{Bun}_G} |_{\mathcal{P}_M} \otimes \det_{\text{Bun}_M}^{\otimes -1} |_{\mathcal{P}_M}$$

admits a canonical square root.

Let P and P^- be the positive and negative parabolics corresponding to M , and let N_P and N_P^- be their respective unipotent radicals.

By definition,

$$(1.3) \quad \det_{\text{Bun}_G} |_{\mathcal{P}_M} \otimes \det_{\text{Bun}_M}^{\otimes -1} |_{\mathcal{P}_M} \simeq \\ \simeq \det(\Gamma(X, (\mathfrak{n}_P)_{\mathcal{P}_M}) \otimes \det(\Gamma(X, (\mathfrak{n}_P^-)_{\mathcal{P}_M}))^{\otimes -1} \otimes \det(\Gamma(X, \mathfrak{n}_P \otimes \mathcal{O}_X))^{\otimes -1} \otimes \det(\Gamma(X, \mathfrak{n}_P^- \otimes \mathcal{O}_X)).$$

We claim that the lines

$$(1.4) \quad \det(\Gamma(X, (\mathfrak{n}_P)_{\mathcal{P}_M} \otimes \omega_X^{\frac{1}{2}}) \otimes \det(\Gamma(X, \mathfrak{n}_P \otimes \omega_X^{\frac{1}{2}}))^{\otimes -1}$$

and

$$(1.5) \quad \det(\Gamma(X, (\mathfrak{n}_P^-)_{\mathcal{P}_M} \otimes \omega_X^{\frac{1}{2}}) \otimes \det(\Gamma(X, \mathfrak{n}_P^- \otimes \omega_X^{\frac{1}{2}}))^{\otimes -1}$$

are canonically isomorphic and their tensor product is canonically isomorphic to (1.3). This would produce the sought-for square root of (1.3).

To construct an isomorphism between (1.4) and (1.5), we identify $\mathfrak{n}_P$ with the dual of $\mathfrak{n}_P^-$ using the Killing form. Hence, we have

$$(1.6) \quad ((\mathfrak{n}_P)_{\mathcal{P}_M})^\vee \simeq (\mathfrak{n}_P^-)_{\mathcal{P}_M} \text{ and } (\mathfrak{n}_P)^* \simeq \mathfrak{n}_P^-$$

as vector bundles (resp., vector spaces).

Now the desired identification follows from the fact that for a vector bundle $\mathcal{E}$ on X , we have

$$(1.7) \quad \det(\Gamma(X, \mathcal{E})) \simeq \det(\Gamma(X, \mathcal{E}^\vee \otimes \omega_X)).$$

To construct an isomorphism between the tensor product of (1.4) and (1.5) and the right-hand side in (1.3), we recall the formula

$$(1.8) \quad \det(\Gamma(X, \mathcal{E} \otimes \mathcal{L})) \otimes \det(\Gamma(X, \mathcal{O}_X))^{\otimes \text{rk}(\mathcal{E})} \simeq \det(\Gamma(X, \mathcal{E})) \otimes \det(\Gamma(X, \mathcal{L}))^{\otimes \text{rk}(\mathcal{E})} \otimes \text{Weil}(\det(\mathcal{E}), \mathcal{L}),$$

where:

- $\mathcal{E}$ is a vector bundle on X ;
- $\mathcal{L}$ is a line bundle on X ;
- $\text{Weil}(-, -)$ is the Weil pairing.

Hence, the ratio of the tensor product of (1.4) and (1.5) and the right-hand side in (1.3) is

$$(1.9) \quad \text{Weil}(\det((\mathfrak{n}_P)_{\mathcal{P}_M}), \omega_X^{\frac{1}{2}}) \otimes \text{Weil}(\det(\mathfrak{n}_P) \otimes \mathcal{O}_X, \omega_X^{\frac{1}{2}})^{\otimes -1} \otimes \\ \otimes \text{Weil}(\det((\mathfrak{n}_P^-)_{\mathcal{P}_M}), \omega_X^{\frac{1}{2}}) \otimes \text{Weil}(\det(\mathfrak{n}_P^-) \otimes \mathcal{O}_X, \omega_X^{\frac{1}{2}})^{\otimes -1}.$$

However, the latter line is canonically trivialized thanks to (1.6). □

Corollary 1.3.4. *The line $\mathfrak{l}_{G,N_{\rho(\omega_X)}}$ admits a canonical square root.*

1.3.5. Thanks to Corollary 1.3.4, we have a well-defined functor

$$\mathfrak{p}^! : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{D-mod}(\text{Bun}_{N,\rho(\omega_X)}),$$

with a partially defined left adjoint, denoted $\mathfrak{p}_!$.

1.3.6. Note now that there is a canonical map

$$\chi : \mathrm{Bun}_{N, \rho(\omega_X)} \rightarrow \mathbb{G}_a.$$

Indeed, for every vertex i of the Dynkin diagram of G , we have a map

$$\mathrm{Bun}_{N, \rho(\omega_X)} \rightarrow \mathrm{Bun}_{N_i, \rho_i(\omega_X)},$$

where (N_i, χ_i) is the corresponding pair for the subminimal Levi subgroup attached to i .

For every i , we have a canonical map

$$\chi_i : \mathrm{Bun}_{N_i, \rho_i(\omega_X)} \rightarrow \mathbb{G}_a$$

that records the class of the extension.

We let χ be the map

$$\mathrm{Bun}_{N, \rho(\omega_X)} \rightarrow \prod_i \mathrm{Bun}_{N_i, \rho_i(\omega_X)} \xrightarrow{\prod_i \chi_i} \prod_i \mathbb{G}_a \xrightarrow{\mathrm{sum}} \mathbb{G}_a.$$

1.3.7. We let

$$\mathrm{Poinc}_{G, !}^{\mathrm{Vac}, \mathrm{glob}} \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

be the object equal to

$$\mathfrak{p}_! \circ \chi^*(\mathrm{exp}),$$

where:

- $\mathrm{exp} \in \mathrm{D}\text{-mod}(\mathbb{G}_a)$ is the exponential D-module, normalized so that

$$\mathrm{add}^*(\mathrm{exp}) \simeq \mathrm{exp} \boxtimes \mathrm{exp},$$

(i.e., $\mathrm{exp} \in \mathrm{D}\text{-mod}(\mathbb{G}_a)^{\heartsuit}[-1]$);

- The functor $\mathfrak{p}_!$ is well-defined on $\chi^*(\mathrm{exp})$, since the latter is holonomic.

1.3.8. Let

$$\mathrm{coeff}^{\mathrm{Vac}, \mathrm{glob}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{Vect}$$

denote the functor co-represented by $\mathrm{Poinc}_{G, !}^{\mathrm{Vac}, \mathrm{glob}}$.

Explicitly,

$$\mathrm{coeff}^{\mathrm{Vac}, \mathrm{glob}} \simeq \mathrm{C}(\mathrm{Bun}_{N, \rho(\omega_X)}, \mathfrak{p}^!(-) \otimes^* \mathrm{exp}_{\chi}),$$

where⁶

$$\mathrm{exp}_{\chi} := \chi^*(\mathrm{exp}) \in \mathrm{D}\text{-mod}(\mathrm{Bun}_{N, \rho(\omega_X)}).$$

1.4. The coarse version of the functor. In this subsection we will construct a *coarse* version of the Langlands functor $\mathbb{L}_G$, denoted $\mathbb{L}_{G, \mathrm{coarse}}$.

The difference between the two versions is that $\mathbb{L}_G$ is supposed to take values in the category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$, while $\mathbb{L}_{G, \mathrm{coarse}}$ takes values in the usual category $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})$.

1.4.1. Note that by Sect. 1.2.5, a choice of an object in $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ defines a $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})$ -linear functor $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$.

⁶Strictly speaking, in the formula below one should have replaces χ by $-\chi$. However, it does not matter because the situation is equivariant with respect to the action of T .

1.4.2. We define the functor

$$(1.10) \quad \mathbb{L}_{G,\text{temp}}^L : \text{QCoh}(\text{LS}_{\tilde{G}}) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G).$$

to correspond to the object

$$(1.11) \quad \text{Poinc}_{G,!}^{\text{Vac, glob}} \in \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G).$$

Since the object $\text{Poinc}_{G,!}^{\text{Vac, glob}}$ is compact and the monoidal category $\text{QCoh}(\text{LS}_{\tilde{G}})$ is compactly generated and rigid, the functor $\mathbb{L}_{G,\text{temp}}^L$ preserves compactness.

Remark 1.4.3. The notation $\mathbb{L}_{G,\text{temp}}^L$ has to do with the fact that the essential image of this functor lands in the subcategory

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}} \subset \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G),$$

see Sect. 5.1.4.

Remark 1.4.4. Once GLC (Conjecture 1.6.7) is proved, it would follow that the functor $\mathbb{L}_{G,\text{temp}}^L$ is actually an equivalence

$$\text{QCoh}(\text{LS}_{\tilde{G}}) \xrightarrow{\sim} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}},$$

see Remark 5.1.7.

1.4.5. We let

$$\mathbb{L}_{G,\text{coarse}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{QCoh}(\text{LS}_{\tilde{G}})$$

be the right adjoint of $\mathbb{L}_{G,\text{temp}}^L$.

Since the category $\text{QCoh}(\text{LS}_{\tilde{G}})$ is compactly generated and the functor $\mathbb{L}_{G,\text{temp}}^L$ preserves compactness, the functor $\mathbb{L}_{G,\text{temp}}$ is continuous. By rigidity, $\mathbb{L}_{G,\text{temp}}$ is automatically $\text{QCoh}(\text{LS}_{\tilde{G}})$ -linear.

1.4.6. One can equivalently describe $\mathbb{L}_{G,\text{coarse}}$ as follows. It is uniquely characterized by the following two pieces of structure:

- $\mathbb{L}_G$ is $\text{QCoh}(\text{LS}_{\tilde{G}})$ -linear;
- $\Gamma(\text{LS}_{\tilde{G}}, \mathbb{L}_{G,\text{coarse}}(-)) \simeq \text{coeff}^{\text{Vac, glob}}$.

1.5. **The case $G = T$.**

1.5.1. Let $G = T$ be a torus. Consider the Fourier-Mukai equivalence

$$\text{FM} : \text{QCoh}(\text{Bun}_T) \rightarrow \text{QCoh}(\text{Bun}_{\tilde{T}}),$$

given by the Poincaré line bundle

$$\mathcal{L}_{\text{Poinc}} \in \text{QCoh}(\text{Bun}_T \times \text{Bun}_{\tilde{T}}),$$

as a kernel, where $\mathcal{L}_{\text{Poinc}}$, viewed as a map $\text{Bun}_T \times \text{Bun}_{\tilde{T}} \rightarrow B\mathbb{G}_m$, is given by the Weil pairing.

1.5.2. It is known that FM can be enhanced to an equivalence

$$\text{FM}^{\text{enh}} : \text{D-mod}(\text{Bun}_T) \rightarrow \text{QCoh}(\text{LS}_{\tilde{T}}),$$

that makes the following diagram commute:

$$\begin{array}{ccc} \text{D-mod}(\text{Bun}_T) & \xrightarrow{\text{FM}^{\text{enh}}} & \text{QCoh}(\text{LS}_{\tilde{T}}) \\ \downarrow & & \downarrow \\ \text{QCoh}(\text{Bun}_T) & \xrightarrow{\text{FM}} & \text{QCoh}(\text{Bun}_{\tilde{T}}), \end{array}$$

where:

- The functor $\text{D-mod}(\text{Bun}_T) \rightarrow \text{D-mod}(\text{Bun}_T)$ is oblv^r , the forgetful functor for “right” D-modules;
- The functor $\text{QCoh}(\text{LS}_{\tilde{T}}) \rightarrow \text{QCoh}(\text{Bun}_{\tilde{T}})$ is direct image along the projection

$$\text{LS}_{\tilde{T}} \rightarrow \text{Bun}_{\tilde{T}}.$$

1.5.3. Unwinding the definitions, we obtain that the functor $\mathbb{L}_{T,\text{coarse}}$ identifies with

$$\text{FM}^{\text{enh}} \circ \tau_T,$$

where τ_T is the Cartan involution, i.e., the inversion automorphism, of T .

1.5.4. Note that in the case of $G = T$, the subset $\text{Nilp} \subset \text{Sing}(\text{LS}_{\tilde{T}})$ is the 0-section. Hence

$$\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{T}}) = \text{QCoh}(\text{LS}_{\tilde{T}}).$$

We set

$$\mathbb{L}_T := \mathbb{L}_{T,\text{coarse}}.$$

So in this case there is no difference between the coarse Langlands functor and the sought-for Langlands functor $\mathbb{L}_T$.

1.6. Statement of the main result. In this subsection we will formulate Theorem 1.6.2, which would allow us to upgrade $\mathbb{L}_{G,\text{coarse}}$ to the actual Langlands functor $\mathbb{L}_G$.

1.6.1. The main technical result of this paper pertaining to the Langlands functor in the de Rham context reads:

Theorem 1.6.2. *The functor*

$$\mathbb{L}_{G,\text{coarse}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{QCoh}(\text{LS}_{\tilde{G}})$$

sends compact objects in $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ to bounded below (i.e., eventually coconnective) objects in $\text{QCoh}(\text{LS}_{\tilde{G}})$.

The proof will be given in Sect. 2. We now proceed to the consequences of Theorem 1.6.2 that pertain to the geometric Langlands functor.

1.6.3. Let $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{G}})$ be the category of [AG, Sect. 11.1.5]. It is equipped with a forgetful functor

$$(1.12) \quad \Psi_{\text{Nilp},\{0\}} : \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{G}}) \rightarrow \text{QCoh}(\text{LS}_{\tilde{G}})$$

is t-exact and induces an equivalence

$$\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{G}})^{>-\infty} \rightarrow \text{QCoh}(\text{LS}_{\tilde{G}})^{>-\infty},$$

see [AG, Proposition 4.4.5].

1.6.4. Combining this with Theorem 1.6.2 and the fact that $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ is compactly generated (see [DG1, Theorem 0.1.2]), we obtain:

Corollary 1.6.5. *There exists a (colimit-preserving) functor*

$$\mathbb{L}_G : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{G}}),$$

uniquely characterized by the following properties:

- The functor $\mathbb{L}_G$ sends compact objects in $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ to eventually coconnective objects in $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{G}})$, i.e., to $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{G}})^{>-\infty}$.
- $\Psi_{\text{Nilp},\{0\}} \circ \mathbb{L}_G \simeq \mathbb{L}_{G,\text{coarse}}$;

1.6.6. The functor $\mathbb{L}_G$ is *the* geometric Langlands functor in the de Rham context.

We can now state the geometric Langlands conjecture (GLC) in the de Rham context:

Conjecture 1.6.7. *The functor $\mathbb{L}_G$ is an equivalence.*

Remark 1.6.8. By the same logic as in Corollary 1.6.5, using Theorem 1.6.2, we can lift $\mathbb{L}_{G,\text{coarse}}$ to a functor

$$\mathbb{L}'_G : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{IndCoh}(\text{LS}_{\check{G}}),$$

so that

$$\Psi_{\text{all},\text{Nilp}} \circ \mathbb{L}'_G \simeq \mathbb{L}_G,$$

where $\Psi_{\text{all},\text{Nilp}}$ is the right adjoint of the inclusion

$$\Xi_{\text{Nilp},\text{all}} : \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \rightarrow \text{IndCoh}(\text{LS}_{\check{G}}).$$

However, if we accept GLC, it follows from Lemma 5.4.6 below that the functor $\mathbb{L}'_G$ factors as

$$\Xi_{\text{Nilp},\text{all}} \circ \mathbb{L}_G.$$

So the refinement $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \rightsquigarrow \text{IndCoh}(\text{LS}_{\check{G}})$ does not give us anything new.

Remark 1.6.9. That said, as was suggested by D. Arinkin, one can consider a *renormalized* version⁷ $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{ren}}$ of $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$, equipped with a pair of adjoint functors

$$\text{ren} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightleftarrows \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{ren}} : \text{un-ren},$$

and one can refine $\mathbb{L}_G$ to a functor

$$\mathbb{L}_{G,\text{ren}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{ren}} \rightarrow \text{IndCoh}(\text{LS}_{\check{G}}),$$

so that the diagrams

$$\begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{ren}} & \xrightarrow{\mathbb{L}_{G,\text{ren}}} & \text{IndCoh}(\text{LS}_{\check{G}}) \\ \text{un-ren} \downarrow & & \downarrow \Psi_{\text{all},\text{Nilp}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \end{array}$$

and

$$\begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{ren}} & \xrightarrow{\mathbb{L}_{G,\text{ren}}} & \text{IndCoh}(\text{LS}_{\check{G}}) \\ \text{ren} \uparrow & & \uparrow \Xi_{\text{all},\text{Nilp}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \end{array}$$

commute.

One can then formulate a renormalized version of GLC, which says that $\mathbb{L}_{G,\text{ren}}$ is an equivalence. This will be addressed in a separate paper.

1.7. Compatibility with the spectral action. In this subsection we will assume the validity of Theorem 1.6.2, and hence of Corollary 1.6.5.

We will show that the functor $\mathbb{L}_G$ naturally upgrades to a functor between $\text{QCoh}(\text{LS}_{\check{G}})$ -module categories.

⁷The compact generators of the renormalization are !-extension of *locally compact* objects on q.c. open substacks of Bun_G .

1.7.1. Note that the category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$ is naturally a module over $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$, and the functor $\Psi_{\mathrm{Nilp},\{0\}}$ is naturally $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -linear.

Consider $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ also as a module over $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$, see Sect. 1.2. We claim:

Proposition 1.7.2. *The functor $\mathbb{L}_G$ carries a uniquely defined $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -linear structure, so that the induced $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -linear structure on*

$$\Psi_{\mathrm{Nilp},\{0\}} \circ \mathbb{L}_G \simeq \mathbb{L}_{G,\mathrm{coarse}}$$

is the natural $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -linear structure on $\mathbb{L}_{G,\mathrm{coarse}}$.

Proof. Since $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ is compactly generated and rigid, in the statement of the proposition we can replace the monoidal category $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ by its full subcategory $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})^c$.

By rigidity, the action of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})^c$ on $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ preserves compactness. Hence, it suffices to show that the functor

$$\mathbb{L}_G|_{\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$$

carries a uniquely defined $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})^c$ -linear structure, so that its composition with $\Psi_{\mathrm{Nilp},\{0\}}$ reproduces the natural $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})^c$ -linear structure on

$$\mathbb{L}_{G,\mathrm{coarse}}|_{\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}}).$$

By the construction of $\mathbb{L}_G$, the restriction $\mathbb{L}_G|_{\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c}$ factors via a full subcategory

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})^{>-\infty} \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}).$$

Hence, it suffices to show that

$$\mathbb{L}_G|_{\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})^{>-\infty}$$

carries a uniquely defined $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})^c$ -linear structure, so that its composition with $\Psi_{\mathrm{Nilp},\{0\}}$ reproduces the natural $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})^c$ -linear structure on

$$\mathbb{L}_{G,\mathrm{coarse}}|_{\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}})^{>-\infty}.$$

However, the latter is automatic since $\Psi_{\mathrm{Nilp},\{0\}}$ is an equivalence on the eventually coconnective subcategories. $\square$

1.8. **The bound on the right.** The contents of this subsection are tangential to rest of this paper (and will not be needed either for the construction of $\mathbb{L}_G$, or for the proof that it is an equivalence).

1.8.1. In Sect. 2.5 we will prove:

Proposition 1.8.2. *The functor $\mathbb{L}_{G,\mathrm{coarse}}$ has a bounded cohomological amplitude on the right, i.e., there exists an integer d so that $\mathbb{L}_{G,\mathrm{coarse}}[d]$ is right t -exact.*

1.8.3. Since the functor $\Psi_{\mathrm{Nilp},\{0\}}$ is t -exact and induces an equivalence on eventually coconnective subcategories, from Proposition 1.8.2 we obtain:

Corollary 1.8.4. *The functor $\mathbb{L}_G$ has a bounded cohomological amplitude on the right.*

2. PROOF OF THEOREM 1.6.2

In this section we will prove Theorem 1.6.2, which was used in order to bootstrap the coarse Langlands functor $\mathbb{L}_{G,\mathrm{coarse}}$ to the actual Langlands functor $\mathbb{L}_G$.

The main input is a result of [FR], stated here as Theorem 2.3.8, which says that the *restricted* version of the coarse Langlands functor is t -exact (up to a cohomological shift).

2.1. Strategy of the proof. The statement of Theorem 1.6.2 follows immediately from the combination of the following two results:

Proposition 2.1.1. *Compact objects of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ are bounded below (i.e., are eventually coconnective).*

Theorem 2.1.2. *The functor $\mathbb{L}_{G,\mathrm{coarse}}$ has a bounded cohomological amplitude on the left, i.e., there exists an integer d so that $\mathbb{L}_{G,\mathrm{coarse}}[-d]$ is left t -exact.*

Remark 2.1.3. In the course of the proof we will see that the integer d in Theorem 2.1.2 can be taken to be the dimension of the classical algebraic stack underlying $\mathrm{LS}_{\tilde{G}}$ plus $\dim(\mathrm{Bun}_{N,\rho(\omega_X)})$ minus $\dim(\mathrm{Bun}_G)$.

Note that when the genus g of our curve is > 1 and G is semi-simple, the stack $\mathrm{LS}_{\tilde{G}}$ is a classical l.c.i., so its dimension equals the virtual dimension, i.e., $(2g - 2) \cdot \dim(G)$.

When G has a connected center (but g is still > 1) the dimension of the classical prestack underlying $\mathrm{LS}_{\tilde{G}}$ is $(2g - 2) \cdot \dim(G) + \dim(Z_G)$.

2.1.4. It is likely that the integer d in Remark 2.1.3 is not the sharp bound. For example, when $G = T$ is a torus, the functor $\mathbb{L}_{G,\mathrm{coarse}}$ is left t -exact as-is. We propose:

Question 2.1.5. *What is the actual bound on the left amplitude of $\mathbb{L}_{G,\mathrm{coarse}}$?*

Remark 2.1.6. Note that Theorem 2.1.2 does *not* imply that the functor $\mathbb{L}_G$ has a bounded cohomological amplitude on the left. In fact, this amplitude *is* unbounded, unless G is a torus.

Namely, we claim the “constant sheaf” $\underline{k}_{\mathrm{Bun}_G} \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ is sent by $\mathbb{L}_G$ to an *infinitely connective* object, i.e., an object that lies in $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G)^{\leq -n}$ for any n .

Indeed, the statement is equivalent to $\Psi_{\mathrm{Nilp},\{0\}} \circ \mathbb{L}_G(\underline{k}_{\mathrm{Bun}_G}) = 0$, which is a valid statement, since $\mathbb{L}_{G,\mathrm{coarse}}(\underline{k}_{\mathrm{Bun}_G})$ is indeed zero.

2.2. Proof of Proposition 2.1.1. Before we launch the proof, let us explain what we are battling against: the stack Bun_G is not quasi-compact, and we have to estimate the cohomological amplitude of the $!$ -direct image functors for open embeddings $U \hookrightarrow U'$, for a fixed quasi-compact open substack U and variable U' .

The idea of the proof is that (since Bun_G has an affine diagonal), these amplitudes only depend on U and not on U' .

2.2.1. The proof given below applies for the category $\mathrm{D}\text{-mod}(\mathcal{Y})$, where $\mathcal{Y}$ is any *truncatable* algebraic stack (see [DG1, Sect. 0.2.3] for what this means), such that each of its quasi-compact open substacks is a global quotient⁸ (i.e., is of the form Z/H , where Z is a (separated) scheme and H as affine algebraic group).

The stack Bun_G is truncatable by [DG1, Theorem 0.2.5]. And it is a standard fact that each of its quasi-compact open substacks is a global quotient.

2.2.2. First, we remark that the assertion is clear when $\mathcal{Y}$ is quasi-compact. Indeed, in this case the category $\mathrm{D}\text{-mod}(\mathcal{Y})$ is compactly generated by objects of the form

$$\mathrm{ind}_r(\mathcal{M}), \quad \mathcal{M} \in \mathrm{Coh}(\mathcal{Y}),$$

and such objects are bounded on both sides.

In the above formula,

$$\mathrm{ind}_r : \mathrm{IndCoh}(\mathcal{Y}) \rightarrow \mathrm{D}\text{-mod}(\mathcal{Y})$$

is the left adjoint of

$$\mathrm{oblv}_r : \mathrm{D}\text{-mod}(\mathcal{Y}) \rightarrow \mathrm{IndCoh}(\mathcal{Y}),$$

see [DG2, Sect. 5.1.5].

⁸It is likely that the latter assumption is not needed, and it suffices to assume that $\mathcal{Y}$ has an affine diagonal.

2.2.3. Let now $\mathcal{Y}$ be an arbitrary truncatable algebraic stack. Write $\mathcal{Y}$ as a union of *co-truncative* open substacks

$$U \xrightarrow{j} \mathcal{Y},$$

so that the functor

$$j_! : \mathrm{D}\text{-mod}(U) \rightarrow \mathrm{D}\text{-mod}(\mathcal{Y}),$$

left adjoint to the restriction functor $j^! = j^*$ is defined.

Then compact generators of $\mathrm{D}\text{-mod}(\mathcal{Y})$ are of the form $j_!(\mathcal{F}_U)$, where $\mathcal{F}_U$ are compact generators of $\mathrm{D}\text{-mod}(U)$.

2.2.4. Write $U = Z/H$, and let f denote the map $Z \rightarrow U$. Consider the *partially defined* left adjoint $f_!$ of the functor

$$f^! : \mathrm{D}\text{-mod}(U) \rightarrow \mathrm{D}\text{-mod}(Z).$$

As in [AGKRRV, Sect. F.3.5], we can take as compact generators of $\mathrm{D}\text{-mod}(U)$ objects of the form

$$f_!(\mathcal{F}_Z), \quad \mathcal{F}_Z \in \mathrm{D}\text{-mod}(Z)^c,$$

where $\mathcal{F}_Z$ is such that the partially defined functor $f_!$ is defined on $\mathcal{F}_Z$.

It suffices to show that each of the objects

$$j_!(f_!(\mathcal{F}_Z)),$$

for $\mathcal{F}_Z$ as above, is bounded below.

Suppose that Z can be covered by n affines. We will show that if $\mathcal{F}_Z \in \mathrm{D}\text{-mod}(Z)^{\geq n}$, then $j_!(f_!(\mathcal{F}_Z)) \in \mathrm{D}\text{-mod}(\mathcal{Y})^{\geq 0}$. The assertion is equivalent to the fact that for any quasi-compact open $U' \supset U$, we have

$$j_!(f_!(\mathcal{F}_Z))|_{U'} \in \mathrm{D}\text{-mod}(U')^{\geq 0}.$$

Since the specified bound does not depend on U' , we can assume that $\mathcal{Y}$ itself is quasi-compact.

2.2.5. Denote $\tilde{f} := j \circ f$. Note that $j_!(f_!(\mathcal{F}_Z))$ is isomorphic to the value on $\mathcal{F}_Z$ of the partially defined left adjoint $\tilde{f}_!$ of $\tilde{f}^!$.

Let $g : Y' \rightarrow \mathcal{Y}$ be a smooth cover, where Y' is an affine scheme. Set

$$Z' := Z \times_{\mathcal{Y}} Y'.$$

Denote by g_Z the map $Z' \rightarrow Z$. Let $\tilde{f}'$ denote the map $Z' \rightarrow Y'$. Since g is smooth, the functors

$$g_Z^* : \mathrm{D}\text{-mod}(Z) \rightarrow \mathrm{D}\text{-mod}(Z') \text{ and } g^* : \mathrm{D}\text{-mod}(\mathcal{Y}) \rightarrow \mathrm{D}\text{-mod}(Y')$$

are well defined.

We obtain that the partially defined left adjoint $\tilde{f}'_!$ of $\tilde{f}'^!$ is well-defined on $g_Z^*(\mathcal{F}_Z)$, and we have

$$g^*(\tilde{f}'_!(\mathcal{F}_Z)) \simeq \tilde{f}'_!(g_Z^*(\mathcal{F}_Z)).$$

Hence, it suffices to show that for any

$$\mathcal{F}_{Z'} \in \mathrm{D}\text{-mod}(Z')^c \cap \mathrm{D}\text{-mod}(Z')^{\geq n},$$

on which the partially defined functor $\tilde{f}'_!$ is defined, we have

$$\tilde{f}'_!(\mathcal{F}_{Z'}) \in \mathrm{D}\text{-mod}(Y')^{\geq 0}.$$

2.2.6. Note that the functor $\widetilde{f}'_!$ is always defined as a functor

$$\mathrm{D}\text{-mod}(Z')^c \rightarrow \mathrm{Pro}(\mathrm{D}\text{-mod}(Y')^c),$$

and it is enough to show that this functor, shifted by $[-n]$, is left t-exact.

Let Z_i be the n affine schemes that cover Z , and let Z'_i denote their preimages in Z' . For a subset $\underline{i}$ of indices, let $j'_\underline{i}$ denote the open embedding

$$\bigcap_{i \in \underline{i}} Z'_i =: Z'_\underline{i} \hookrightarrow Z'.$$

By Cousin, for $\mathcal{F} \in \mathrm{D}\text{-mod}(Z')^c$, the object $\widetilde{f}'_!(\mathcal{F}) \in \mathrm{Pro}(\mathrm{D}\text{-mod}(Y')^c)$ admits a canonical filtration (of length n), whose m 's subquotient is

$$\bigoplus_{|\underline{i}|=m} (\widetilde{f}' \circ j'_\underline{i})_!(\mathcal{F}|_{Z'_\underline{i}})[m].$$

Hence, it is enough to show that each of the functors

$$(\widetilde{f}' \circ j'_\underline{i})_! : \mathrm{D}\text{-mod}(Z'_\underline{i}) \rightarrow \mathrm{Pro}(\mathrm{D}\text{-mod}(Y')^c)$$

is left t-exact.

2.2.7. Note that $\mathcal{Y}$ has an affine diagonal. Hence, the morphism $Y' \rightarrow \mathcal{Y}$ is affine, and hence so is the projection $Z' \rightarrow Z$. We obtain that each Z'_i is an affine schenme, and hence so is the morphism $\widetilde{f}' \circ j'_i$.

Now the desired left t-exactness assertion follows from [Ra, Theorem 3.4.1].

□[Proposition 2.1.1]

2.3. Proof of Theorem 2.1.2. The idea of the proof is to reduce Theorem 2.1.2 to its restricted version, namely, Theorem 2.3.8, using a Cousin-type argument. The latter is encapsulated by Proposition 2.3.2 below.

2.3.1. We have the following assertion, proved in Sect. 2.4:

Proposition 2.3.2. *Let $\mathcal{Y}$ be an eventually coconnective algebraic stack. Let $\mathcal{M}$ be an object of $\mathrm{QCoh}(\mathcal{Y})$ such that for every field extension $k \subset k'$ and a k' -point y of $\mathcal{Y}$, we have:*

$$i_y^!(\mathcal{M}') \in \mathrm{Vect}_{k'}^{\geq 0},$$

where:

- $\mathcal{M}'$ denotes the pullback of $\mathcal{M}$ along $\mathcal{Y} \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k') =: \mathcal{Y}' \rightarrow \mathcal{Y}$;
- i_y denotes the map $\mathrm{Spec}(k') \rightarrow \mathcal{Y}'$ corresponding to y ;
- $i_y^!$ denotes the (not necessarily continuous) right adjoint of $(i_y)_*$,

then $\mathcal{M} \in \mathrm{QCoh}(\mathcal{Y})^{\geq -d}$, where d is the dimension of the classical algebraic stack underlying $\mathcal{Y}$.

Remark 2.3.3. In practice, the fields k' that will appear are fields of rational functions on irreducible subschemes of an affine scheme Y that smoothly covers $\mathcal{Y}$.

2.3.4. Let us prove Theorem 2.1.2.

Let d be the integer from Proposition 2.3.2 for the stack $\mathrm{LS}_{\tilde{G}}$. We will show that the functor $\mathbb{L}_{G,\mathrm{coarse}}$ has an amplitude bounded on the left by $d - \dim(\mathrm{Bun}_G) + \dim(\mathrm{Bun}_{N,\rho(\omega_X)})$.

Let $\mathcal{F}$ be an object of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$, which is cohomologically $\geq \dim(\mathrm{Bun}_{N,\rho(\omega_X)}) - \dim(\mathrm{Bun}_G)$. Applying Proposition 2.3.2, we need to show that for every algebraically closed field extension $k \subset k'$ and a k' -point σ of

$$\mathrm{LS}'_{\tilde{G}} := \mathrm{LS}_{\tilde{G}} \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k'),$$

the object

$$i_\sigma^!(\mathbb{L}_{G,\mathrm{coarse}}(\mathcal{F})') \in \mathrm{Vect}_{k'}$$

belongs to $\mathrm{Vect}_{k'}^{\geq 0}$, where $\mathbb{L}_{G,\mathrm{coarse}}(\mathcal{F})'$ denotes the pullback of $\mathbb{L}_{G,\mathrm{coarse}}(\mathcal{F})$ to $\mathrm{LS}'_{\tilde{G}}$.

Base changing everything along $\mathrm{Spec}(k') \rightarrow \mathrm{Spec}(k)$, we can assume that $k' = k$, so that σ is a rational point of $\mathrm{LS}_{\tilde{G}}$.

2.3.5. Consider the sub prestack

$$\mathrm{LS}_{\tilde{G}}^{\mathrm{restr}} \subset \mathrm{LS}_{\tilde{G}},$$

see [AGKRRV, Sect. 4.1].

Let

$$\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})_{\mathrm{restr}} \xrightarrow{\iota^{\mathrm{spec}}} \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})$$

be the (fully faithful) embedding of the subcategory consisting of objects in with set-theoretic support on $\mathrm{LS}_{\tilde{G}}^{\mathrm{restr}}$. Let $(\iota^{\mathrm{spec}})^R$ denote the right adjoint functor.

2.3.6. Note that for a k -rational point σ of $\mathrm{LS}_{\tilde{G}}$, the morphism i_σ factors as

$$\mathrm{Spec}(k) \rightarrow \mathrm{LS}_{\tilde{G}}^{\mathrm{restr}} \rightarrow \mathrm{LS}_{\tilde{G}},$$

where the second arrow is the map of [AGKRRV, Equation (4.2)].

Hence, the functor $(i_\sigma)_*$ factors via the subcategory $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})_{\mathrm{restr}}$. Hence, the functor $i_\sigma^!$ factors as

$$(i_\sigma^!|_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})_{\mathrm{restr}}}) \circ (\iota^{\mathrm{spec}})^R.$$

Since the functor $i_\sigma^!|_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})_{\mathrm{restr}}}$ is right t-exact, it suffices to show that

$$(\iota^{\mathrm{spec}})^R(\mathbb{L}_{G, \mathrm{coarse}}(\mathcal{F})) \in \mathrm{Vect}^{\geq 0}.$$

2.3.7. Let

$$\mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G) \xrightarrow{\iota} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

be the full subcategory consisting of D-modules with singular support in the global nilpotent cone.

According to [AGKRRV, Proposition 14.5.3], we have

$$(2.1) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G) \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}^{\mathrm{restr}})} \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})_{\mathrm{restr}}.$$

Hence, the restriction of $\mathbb{L}_{G, \mathrm{coarse}}$ to $\mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G)$ is a functor

$$\mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{restr}} : \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})_{\mathrm{restr}}.$$

Furthermore, we have,

$$\mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{restr}} \circ \iota^R \simeq (\iota^{\mathrm{spec}})^R \circ \mathbb{L}_{G, \mathrm{coarse}}.$$

Since the functor ι is t-exact, the functor ι^R is left t-exact. Hence, it suffices to show that the functor $\mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{restr}}[\dim(\mathrm{Bun}_G) - \dim(\mathrm{Bun}_{N, \rho(\omega_X)})]$ is left t-exact.

However, this is the “left t-exactness half” of the following result of [FR] (see Sect. 1.6.2 in *loc. cit.*):

Theorem 2.3.8. *The functor*

$$\mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{restr}}[\dim(\mathrm{Bun}_G) - \dim(\mathrm{Bun}_{N, \rho(\omega_X)})] : \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}^{\mathrm{restr}}).$$

is t-exact.

□[Theorem 2.1.2]

Remark 2.3.9. We emphasize that, unlike $\mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{restr}}$, the functor $\mathbb{L}_{G, \mathrm{coarse}}$ (shifted cohomologically by $[\dim(\mathrm{Bun}_G) - \dim(\mathrm{Bun}_{N, \rho(\omega_X)})]$), is *not* right t-exact. One can see this already when $G = \mathbb{G}_m$, in which case $\mathbb{L}_{G, \mathrm{coarse}}$ is the Fourier-Laumon transform.

Note, however, that Proposition 1.8.2 says that its cohomological amplitude on the right is bounded.

2.4. Proof of Proposition 2.3.2.

2.4.1. Let $f : Y \rightarrow \mathcal{Y}$ be a smooth cover of $\mathcal{Y}$ by an affine scheme. Since

$$\mathcal{M} \in \mathrm{QCoh}(\mathcal{Y})^{\geq 0} \Leftrightarrow f^!(\mathcal{M})[-\dim(Y/\mathcal{Y})] \in \mathrm{QCoh}(Y)^{\geq 0},$$

we can assume that $\mathcal{Y} = Y$ is an affine scheme.

2.4.2. Let $Z \xrightarrow{i_Z} Y$ be an irreducible subvariety and let

$$\eta_Z \xrightarrow{j_Z} Z$$

be the embedding of its generic point.

Let

$$\mathrm{QCoh}(Y)_Z \xrightarrow{(\hat{i}_Z)^!} \mathrm{QCoh}(Y)$$

be the embedding of the full subcategory consisting of objects set-theoretically supported on Z . Let $(\hat{i}_Z)^!$ denote the right adjoint of $(\hat{i}_Z)_!$.

Let

$$j_Z^* : \mathrm{QCoh}(Y)_Z \rightleftarrows \mathrm{QCoh}(Y)_{\eta_Z} : (j_Z)_*$$

be the localization of $\mathrm{QCoh}(Y)_Z$ at the generic point.

Applying Cousin decomposition, we obtain that $\mathcal{M} \in \mathrm{QCoh}(Y)^{\geq -d}$ if and only if for every Z as above,

$$(2.2) \quad j_Z^* \circ (\hat{i}_Z)^!(\mathcal{M}) \in \mathrm{QCoh}(Y)_{\eta_Z}^{\geq -d}.$$

2.4.3. We can represent the formal completion of Z in Y as

$$\mathrm{colim}_n Z_n,$$

where $Z_n \xrightarrow{i_{Z_n}} Y$ are regularly embedded closed subschemes of Y (see. e.g., [GaRo1, Proposition 6.7.4]).

We have

$$(\hat{i}_Z)^!(\mathcal{M}) \simeq \mathrm{colim}_n (i_{Z_n})_* \circ (i_{Z_n})^*(\mathcal{M}).$$

Hence, in order to establish (2.2), it suffices to show that for every n above, we have

$$(2.3) \quad j_Z^* \circ (i_{Z_n})^!(\mathcal{M}) \in \mathrm{QCoh}(\eta_{Z_n})^{\geq -d},$$

where η_{Z_n} is the localization of Z_n at η_Z .

2.4.4. Let K be the field of rational functions on Z , and let $\eta_{Z_n} = \mathrm{Spec}(A_{Z_n})$. Note, however, that since Y is eventually, coconnective, A_{Z_n} is an extension of finitely many copies of K with non-positive shifts.

Hence, in order to establish (2.3), it suffices to show that

$$(2.4) \quad j_Z^* \circ i_Z^!(\mathcal{M}) \in \mathrm{Vect}_K^{\geq -d}.$$

2.4.5. Denote

$$Y' := Y \times_{\mathrm{Spec}(k)} \mathrm{Spec}(K_Z), \quad Z' := Z \times_{\mathrm{Spec}(k)} \mathrm{Spec}(K_Z);$$

let i'_Z denote the embedding $Z' \rightarrow Y'$, obtained by base-changing i_Z . Let $\mathcal{M}'$ denote the pullback of $\mathcal{M}$ to Y' .

Note that Z' has a canonical K -rational point z_{can} . Denote by i_{can} the corresponding morphism $\mathrm{Spec}(K) \rightarrow Z'$.

The morphism j_Z factors as

$$\mathrm{Spec}(K) \xrightarrow{i_{\mathrm{can}}} Z' \rightarrow Z.$$

Hence,

$$j_Z^* \circ i_Z^!(\mathcal{M}) \simeq (i_{\mathrm{can}})^* \circ (i'_Z)^!(\mathcal{M}').$$

2.4.6. Note, however, that z_{can} is a smooth point of Z' . Hence,

$$(i_{\text{can}})^* \simeq (i_{\text{can}})^! [\dim(Z)]$$

(up to a determinant line, which we ignore).

Hence, we can further rewrite

$$j_Z^* \circ i_Z^! (\mathcal{M}) \simeq (i_{\text{can}})^! \circ (i'_Z)^! (\mathcal{M}') [\dim(Z)] = (i'_Z \circ i_{\text{can}})^! (\mathcal{M}) [\dim(Z)].$$

2.4.7. Now, the condition of the proposition implies that

$$(i'_Z \circ i_{\text{can}})^! (\mathcal{M}) \in \text{Vect}_K^{\geq 0}.$$

Hence,

$$(i'_Z \circ i_{\text{can}})^! (\mathcal{M}) [\dim(Z)] \in \text{Vect}_K^{\geq -d},$$

as desired.

□[Proposition 2.3.2]

2.5. Proof of Proposition 1.8.2.

2.5.1. We have the following counterpart of Proposition 2.3.2 for right t-exactness:

Proposition 2.5.2. *Let $\mathcal{Y}$ be an eventually coconnective algebraic stack. There exists an integer d such that the following holds:*

If $\mathcal{M}$ be an object of $\text{QCoh}(\mathcal{Y})$ such that for every field extension $k \subset k'$ and a k' -point y of $\mathcal{Y}$, we have:

$$i_y^* (\mathcal{M}') \in \text{Vect}_{k'}^{\leq 0}.$$

Then $\mathcal{M} \in \text{QCoh}(\mathcal{Y})^{\leq d}$.

We take d to be the dimension of the classical scheme underlying an affine smooth cover $Y \rightarrow \mathcal{Y}$.

The proof is parallel to that of Proposition 2.3.2. The only difference is that we use the fact that

$$j_Z^* \circ (i_{Z_n})^! \simeq j_Z^* \circ (i_{Z_n})^* [-\text{codim}(Z)].$$

2.5.3. We proceed to the proof of Proposition 1.8.2. As a main ingredient we will use the fact that the functor

$$\iota^R : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{D-mod}_{\frac{1}{2}, \text{Nilp}}(\text{Bun}_G)$$

has a bounded cohomological dimension on the right, see [AGKRRV, Corollary 14.5.5 and Proposition 17.3.10]. Let d' denote this bound.

We will show that the functor $\mathbb{L}_{G, \text{coarse}}$ has a cohomological amplitude bounded on the right by $d + d' + \dim(\text{Bun}_G) - \dim(\text{Bun}_{N, \rho(\omega_X)})$, where d is the integer from Proposition 2.5.2 for the stack $\text{LS}_{\tilde{G}}$.

2.5.4. Applying Proposition 2.5.2 and arguing as in Sect. 2.3.4, it suffices to show that for a k -rational point σ of $\text{LS}_{\tilde{G}}$, the functor

$$i_\sigma^* \circ \mathbb{L}_{G, \text{coarse}} [d' + \dim(\text{Bun}_G) - \dim(\text{Bun}_{N, \rho(\omega_X)})]$$

is right t-exact.

Note that the functor i_σ^* also factors as

$$(i_\sigma^* |_{\text{QCoh}(\text{LS}_{\tilde{G}})_{\text{restr}}}) \circ (\iota^{\text{spec}})^R,$$

while the functor $i_\sigma^* |_{\text{QCoh}(\text{LS}_{\tilde{G}})_{\text{restr}}}$ is right t-exact.

Hence, it is enough to show that the functor

$$(\iota^{\text{spec}})^R \circ \mathbb{L}_{G, \text{coarse}} [d' + \dim(\text{Bun}_G) - \dim(\text{Bun}_{N, \rho(\omega_X)})]$$

is right t-exact.

2.5.5. We rewrite $(\iota^{\text{spec}})^R \circ \mathbb{L}_{G,\text{coarse}}$ as

$$\mathbb{L}_{G,\text{coarse}}^{\text{restr}} \circ \iota^R.$$

Now, the desired assertion follows from the fact that $\iota^R[d']$ is right t-exact and the fact that $\mathbb{L}_{G,\text{coarse}}^{\text{restr}}[\dim(\text{Bun}_G) - \dim(\text{Bun}_{N,\rho(\omega_X)})]$ is right t-exact (this is the “right t-exactness half” of Theorem 2.3.8).

□[Proposition 1.8.2]

3. GEOMETRIC LANGLANDS FUNCTOR IN THE BETTI CONTEXT

In this section we take our ground field k to be $\mathbb{C}$. Throughout this section we will work with the “big” category of Betti sheaves of $\mathbf{e}$ -vector spaces (see [AGKRRV, Appendix G]), where $\mathbf{e}$ is an arbitrary field of coefficients (assumed of characteristic zero).

We will construct the Langlands functor in the Betti setting.

Once the functor is constructed, we will formulate a theorem to the effect that this functor is an equivalence if and only if its de Rham counterpart is.

3.1. The category with nilpotent singular support. In this subsection we will introduce (following [BZN]) the automorphic category in the Betti context.

3.1.1. Note that since $\det_{\text{Bun}_G}^{\frac{1}{2}}$ is a gerbe with respect to $\mu_2 \simeq \mathbb{Z}/2\mathbb{Z}$, it gives rise to an étale $\mathbf{e}^\times$ -gerbe for any field $\mathbf{e}$ of characteristic different from 2.

Hence, it makes sense to consider the category $\text{Shv}_{\frac{1}{2}}^{\text{Betti}}(\text{Bun}_G)$ of sheaves of $\mathbf{e}$ -vector spaces in the classical topology on Bun_G , twisted by $\det_{\text{Bun}_G}^{\frac{1}{2}}$.

Note, however, that by Sect. 1.1.2, we can identify

$$\text{Shv}_{\frac{1}{2}}^{\text{Betti}}(\text{Bun}_G) \simeq \text{Shv}^{\text{Betti}}(\text{Bun}_G).$$

3.1.2. Let

$$\text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G) \hookrightarrow \text{Shv}_{\frac{1}{2}}^{\text{Betti}}(\text{Bun}_G)$$

be the full subcategory consisting of sheaves with singular support contained in Nilp .

According to [AGKRRV, Sect. 18.2.6], the functor $\mathbf{i}$ admits a left adjoint, to be denoted $\mathbf{i}^L$.

3.1.3. The functor $\mathbf{i}^L$ allows us to construct compact objects in $\text{Shv}_{\frac{1}{2}}^{\text{Betti}}(\text{Bun}_G)$.

For a stack $\mathcal{Y}$ and a rational point y , let $\delta_y \in \text{Shv}^{\text{Betti}}(\mathcal{Y})$ be the corresponding δ -function object, i.e., $(i_y)_!(\mathbf{e})$, where i_y be the morphism

$$\text{pt} \rightarrow \mathcal{Y}$$

corresponding to y .

According to [AGKRRV, Proposition G.3.5], objects of the form

$$\mathbf{i}^L(\delta_y), \quad y \in \text{Bun}_G$$

form a set of compact generators of $\text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G)$.

Remark 3.1.4. Note, however, that objects $\delta_y \in \text{Shv}_{\frac{1}{2}}^{\text{Betti}}(\text{Bun}_G)$ are *not* compact, and that the category $\text{Shv}_{\frac{1}{2}}^{\text{Betti}}(\text{Bun}_G)$ itself is *not* compactly generated.

3.2. The Hecke action in the Betti context.

3.2.1. Let $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}^{\mathrm{Betti}}$ be the Betti version of the category $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}$, defined as in [AGKRRV, Remark 11.1.10].

As in the de Rham setting, we have a localization functor

$$(3.1) \quad \mathrm{Loc}_G^{\mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathrm{Ran}}^{\mathrm{Betti}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_G^{\mathrm{Betti}}),$$

which admits a fully faithful continuous right adjoint.

3.2.2. We have a canonically defined monoidal action of $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}^{\mathrm{Betti}}$ on $\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$, which preserves the subcategory $\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$.

The following is a key observation from [NY] (see also [AGKRRV, Theorem 18.1.4]):

Proposition 3.2.3. *The action of $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}^{\mathrm{Betti}}$ on $\mathrm{Shv}_{\frac{1}{2}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$ factors via the localization (3.1).*

As a result, we obtain that the category $\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$ carries a canonically defined action of the monoidal category $\mathrm{QCoh}(\mathrm{LS}_G^{\mathrm{Betti}})$.

3.3. **The vacuum Poincaré sheaf in the Betti context.** Our current goal is to construct the object

$$(3.2) \quad \mathrm{Poinc}_{!, \mathrm{Nilp}}^{\mathrm{Vac, glob}} \in \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G).$$

The slight hiccup is that the exponential sheaf is not defined as an object of $\mathrm{Shv}^{\mathrm{Betti}}(\mathbb{G}_a)$. In order to circumvent this, we will first apply the procedure of $\mathbb{G}_m$ -averaging.

3.3.1. Define the object

$$\exp / \mathbb{G}_m \in \mathrm{Shv}^{\mathrm{Betti}}(\mathbb{G}_a / \mathbb{G}_m)$$

to be the $*$ -extension of

$$\mathbf{e}[-1] \in \mathrm{Vect} = \mathrm{Shv}^{\mathrm{Betti}}(\mathrm{pt})$$

along the open embedding

$$\mathrm{pt} \simeq (\mathbb{G}_a - \{0\}) / \mathbb{G}_m \hookrightarrow \mathbb{G}_a / \mathbb{G}_m.$$

Remark 3.3.2. Note that when $\mathbf{e} = \mathbb{C}$, under the Riemann-Hilbert equivalence, the object $\exp / \mathbb{G}_m$ corresponds to the $!$ -direct image of $\exp$ along the projection

$$\mathbb{G}_a \rightarrow \mathbb{G}_a / \mathbb{G}_m.$$

3.3.3. We will now consider a variant of this construction. Consider the stack

$$\prod_i \mathbb{G}_a / T,$$

where T acts on $\prod_i \mathbb{G}_a$ by means of

$$T \rightarrow T / Z_G \xrightarrow{\text{simple roots}} \prod_i \mathbb{G}_m.$$

Consider the open embedding

$$(3.3) \quad \mathrm{pt} / Z_G \simeq \left(\prod_i (\mathbb{G}_a - 0) \right) / T \hookrightarrow (\prod_i \mathbb{G}_a) / T.$$

Let

$$(\boxtimes_i \exp) / T \in \mathrm{Shv}^{\mathrm{Betti}}\left(\left(\prod_i \mathbb{G}_a\right) / T\right)$$

be the $*$ -direct image along (3.3) of

$$R_{Z_G}[-r] \in \mathrm{Shv}^{\mathrm{Betti}}(\mathrm{pt} / Z_G),$$

where:

- R_{Z_G} is the $!$ -direct image of $\mathbf{e} \in \mathrm{Vect} = \mathrm{Shv}^{\mathrm{Betti}}(\mathrm{pt})$ along $\mathrm{pt} \rightarrow \mathrm{pt} / Z_G$;
- r is the semi-simple rank of G .

Remark 3.3.4. It follows from Remark 3.3.2 and the projection formula that when $\mathbf{e} = \mathbb{C}$, under the Riemann-Hilbert equivalence, the object $(\boxtimes_i \exp)/T$ is the $!$ -direct image along the projection

$$\prod_i \mathbb{G}_a \rightarrow (\prod_i \mathbb{G}_a)/T$$

of

$$\mathrm{add}^*(\exp) \simeq \boxtimes_i \exp.$$

3.3.5. Note that the map

$$\mathbf{p} : \mathrm{Bun}_{N,\rho(\omega_X)} \rightarrow \mathrm{Bun}_G$$

factors as

$$\mathrm{Bun}_{N,\rho(\omega_X)} \rightarrow \mathrm{Bun}_{N,\rho(\omega_X)}/T \xrightarrow{\mathbf{p}/T} \mathrm{Bun}_G.$$

By construction, the map $\mathrm{Bun}_{N,\rho(\omega_X)} \xrightarrow{\underline{\chi}} \mathbb{G}_a$ factors as

$$\mathrm{Bun}_{N,\rho(\omega_X)} \xrightarrow{\underline{\chi}} \prod_i \mathbb{G}_a \xrightarrow{\mathrm{sum}} \mathbb{G}_a.$$

Consider the resulting map

$$\mathrm{Bun}_{N,\rho(\omega_X)}/T \xrightarrow{\underline{\chi}/T} (\prod_i \mathbb{G}_a)/T.$$

Let

$$\exp_\chi/T \in \mathrm{Shv}^{\mathrm{Betti}}(\mathrm{Bun}_{N,\rho(\omega_X)}/T)$$

be the pullback of the object $(\boxtimes_i \exp)/T$ along the map $\underline{\chi}/T$.

Remark 3.3.6. It follows from Remark 3.3.4 that when $\mathbf{e} = \mathbb{C}$, under the Riemann-Hilbert equivalence, the object $\exp_\chi/T$ corresponds to the $!$ -direct image along the projection

$$\mathrm{Bun}_{N,\rho(\omega_X)} \rightarrow \mathrm{Bun}_{N,\rho(\omega_X)}/T$$

of the object

$$\exp_\chi := \chi^*(\exp) \in \mathrm{D-mod}(\mathrm{Bun}_{N,\rho(\omega_X)}).$$

3.3.7. Let

$$\mathrm{Poinc}_!^{\mathrm{Vac, glob}} := (\mathbf{p}/T)_!(\exp_\chi/T) \in \mathrm{Shv}_{\frac{1}{2}}^{\mathrm{Betti}}(\mathrm{Bun}_G).$$

Finally, set

$$\mathrm{Poinc}_{!, \mathrm{Nilp}}^{\mathrm{Vac, glob}} := \mathbf{i}^L(\mathrm{Poinc}_!^{\mathrm{Vac, glob}}).$$

3.3.8. We claim:

Proposition 3.3.9. *The object $\mathrm{Poinc}_{!, \mathrm{Nilp}}^{\mathrm{Vac, glob}} \in \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$ is compact.*

Proof. Note that the object $(\boxtimes_i \exp)/T$ is a finite extension of δ -functions.

The map

$$\underline{\chi} : \mathrm{Bun}_{N,\rho(\omega_X)} \rightarrow \prod_i \mathbb{G}_a$$

is a unipotent gerbe. Hence, the map

$$\underline{\chi}/T : \mathrm{Bun}_{N,\rho(\omega_X)}/T \rightarrow (\prod_i \mathbb{G}_a)/T$$

has a similar property.

It follows that the object $\exp_\chi/T \in \mathrm{Shv}^{\mathrm{Betti}}(\mathrm{Bun}_{N,\rho(\omega_X)}/T)$ is a finite extension of δ -functions. Hence, so is $\mathrm{Poinc}_!^{\mathrm{Vac, glob}}$.

The assertion of the proposition follows now from Sect. 3.1.3. $\square$

3.4. Construction of the functor. Having the object $\mathrm{Poinc}_{!, \mathrm{Nilp}}^{\mathrm{Vac, glob}}$ at our disposal, the construction of the Langlands functor in the Betti setting mimics its de Rham counterpart.

3.4.1. As in the de Rham context, we define the functor

$$\mathbb{L}_{G,\text{temp}}^{\text{Betti},L} : \text{QCoh}(\text{LS}_G^{\text{Betti}}) \rightarrow \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G)$$

to be given by acting on the object

$$\text{Poinc}_{!,\text{Nilp}}^{\text{Vac,glob}} \in \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G).$$

Since the object $\text{Poinc}_{!,\text{Nilp}}^{\text{Vac,glob}}$ is compact and $\text{QCoh}(\text{LS}_G^{\text{Betti}})$ is compactly generated and rigid, the functor $\mathbb{L}_{G,\text{temp}}^{\text{Betti},L}$ preserves compactness.

3.4.2. Let

$$\mathbb{L}_{G,\text{coarse}}^{\text{Betti}} : \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G) \rightarrow \text{QCoh}(\text{LS}_G^{\text{Betti}})$$

denote the right adjoint of $\mathbb{L}_{G,\text{temp}}^{\text{Betti},L}$.

Since $\text{QCoh}(\text{LS}_G^{\text{Betti}})$ is compactly generated and $\mathbb{L}_{G,\text{temp}}^{\text{Betti},L}$ preserves compactness, the functor $\mathbb{L}_{G,\text{coarse}}^{\text{Betti}}$ is continuous. By rigidity, it is automatically $\text{QCoh}(\text{LS}_G^{\text{Betti}})$ -linear.

3.4.3. We have the following assertion, which is parallel to Theorem 1.6.2:

Theorem 3.4.4. *The functor*

$$\mathbb{L}_{G,\text{coarse}}^{\text{Betti}} : \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G) \rightarrow \text{QCoh}(\text{LS}_G^{\text{Betti}})$$

sends compact objects in $\text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G)$ to bounded below objects in $\text{QCoh}(\text{LS}_G^{\text{Betti}})$.

The proof will be given in Sect. 4.

3.4.5. Assuming Theorem 3.4.4, as in the de Rham context, we obtain that there exists a uniquely defined continuous functor

$$(3.4) \quad \mathbb{L}_G^{\text{Betti}} : \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G) \rightarrow \text{IndCoh}_{\text{Nilp}}(\text{LS}_G^{\text{Betti}}),$$

characterized by the requirements that:

- The functor $\mathbb{L}_G^{\text{Betti}}$ sends compact objects in $\text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G)$ to eventually coconnective objects in $\text{IndCoh}_{\text{Nilp}}(\text{LS}_G^{\text{Betti}})$;
- $\Psi_{\text{Nilp},\{0\}} \circ \mathbb{L}_G \simeq \mathbb{L}_{G,\text{coarse}}$.

Moreover, the functor $\mathbb{L}_G^{\text{Betti}}$ is automatically $\text{QCoh}(\text{LS}_G^{\text{Betti}})$ -linear.

3.4.6. The geometric Langlands conjecture in the Betti context, originally formulated by D. Ben-Zvi and D. Nadler in [BZN], says:

Conjecture 3.4.7. *The functor $\mathbb{L}_G^{\text{Betti}}$ of (3.4) is an equivalence.*

3.5. Comparson of de Rham and Betti versions of GLC. Although the Betti and de Rham versions of the automorphic category (i.e., $\text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G)$ and $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$) look very different, it turns out that the respective versions of GLC in the two contexts are logically equivalent.

The passage is realized by showing that either version is equivalent to its *restricted* variant, and the latter can be compared via Riemann-Hilbert.

The material in this subsection relies heavily on that of [AGKRRV].

3.5.1. We are going to prove:

Theorem 3.5.2. *The de Rham version of GLC implies the Betti version.*

In fact, we will prove a more precise version of Theorem 3.5.2, see Theorem 3.5.6 below.

3.5.3. Let

$$\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G)$$

be the ind-constructible category, as defined in [AGKRRV, Sect. E.5].

Recall that according to [AGKRRV, Proposition 18.3.2], the forgetful functor

$$\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G) \rightarrow \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$$

is fully faithful.

Moreover, by [AGKRRV, Theorem 18.3.6], the subcategory

$$(3.5) \quad \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G) \subset \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$$

equals

$$\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{Betti}})_{\mathrm{restr}}.$$

Note also that we have an equivalence

$$(3.6) \quad \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{Betti}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{Betti}})_{\mathrm{restr}} \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{Betti}, \mathrm{restr}}).$$

Hence, the functor $\mathbb{L}_G^{\mathrm{Betti}}$ induces a functor, to be denoted

$$\mathbb{L}_G^{\mathrm{Betti}, \mathrm{restr}} : \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{Betti}, \mathrm{restr}}).$$

3.5.4. Similarly, we have

$$(3.7) \quad \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})} \mathrm{QCoh}(\mathrm{LS}_{\check{G}})_{\mathrm{restr}} \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{restr}}).$$

Hence, by (2.1), the functor $\mathbb{L}_G$ induces a functor

$$\mathbb{L}_G^{\mathrm{restr}} : \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{restr}}).$$

3.5.5. We will prove:

Theorem 3.5.6. *The following statements are logically equivalent:*

- (i) *The functor $\mathbb{L}_G$ is an equivalence.*
- (i') *The functor $\mathbb{L}_G^{\mathrm{restr}}$ is an equivalence;*
- (ii) *The functor $\mathbb{L}_G^{\mathrm{Betti}}$ is an equivalence.*
- (ii') *The functor $\mathbb{L}_G^{\mathrm{Betti}, \mathrm{restr}}$ is an equivalence;*

Note that the implications (i) $\Rightarrow$ (i') and (ii) $\Rightarrow$ (ii') in Theorem 3.5.6 are immediate from the equivalences (3.7) and (3.6), respectively.

The equivalence of (i') and (ii') will be proved in Sect. 4.2.4.

The implications (i') $\Rightarrow$ (i) and (ii') $\Rightarrow$ (ii) will be proved in Sect. 5.

4. PROOF OF THEOREM 3.4.4

The proof of Theorem 3.4.4 follows ideas, similar to its de Rham counterpart.

The main difference is that in the Betti setting, the embedding of the category with nilpotent singular support has a *left* adjoint, while in the de Rham setting it admitted a right adjoint.

4.1. Reduction steps. The idea of the proof is to reduce the assertion of Theorem 3.4.4 to statements that concern the restricted version of the functor $\mathbb{L}_G^{\mathrm{Betti}}$. Those are formulated as Propositions 4.1.5 and 4.1.6 at the end of this subsection.

4.1.1. Recall that the left adjoint $\mathbf{i}^L$ to the tautological embedding

$$\mathbf{i} : \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G) \hookrightarrow \mathrm{Shv}_{\frac{1}{2}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$$

is well-defined.

Recall also (see Sect. 3.1.3) that objects of the form

$$\mathbf{i}^L(\delta_y), \quad y \in \mathrm{Bun}_G$$

form a set of compact generators of $\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$.

Thus, in order to prove Theorem 3.4.4, it suffices to show that the objects

$$\mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{Betti}} \circ \mathbf{i}^L(\delta_y) \in \mathrm{QCoh}(\mathrm{LS}_G^{\mathrm{Betti}})$$

are bounded below.

4.1.2. Note that for an extension of fields of coefficients $\mathbf{e} \subset \mathbf{e}'$, the base change

$$\mathrm{LS}_G^{\mathrm{Betti}} \times_{\mathrm{Spec}(\mathbf{e})} \mathrm{Spec}(\mathbf{e}')$$

identifies with the stack of Betti local systems with $\mathbf{e}'$ -coefficients.

Hence, as in Sect. 2.3.4, it suffices to show that the objects

$$i_{\sigma}^! \left(\mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{Betti}} \circ \mathbf{i}^L(\delta_y) \right) \in \mathrm{Vect}_{\mathbf{e}}$$

belong to $\mathrm{Vect}_{\mathbf{e}}^{\geq n}$, where n is an integer that only depends on y (but not on the choice of $\mathbf{e}$ or σ).

4.1.3. Let

$$\iota^{\mathrm{Betti}, \mathrm{spec}} : \mathrm{QCoh}(\mathrm{LS}_G^{\mathrm{Betti}})_{\mathrm{restr}} \rightleftarrows \mathrm{QCoh}(\mathrm{LS}_G^{\mathrm{Betti}}) : (\iota^{\mathrm{Betti}, \mathrm{spec}})^R$$

be the corresponding pair of adjoint functors.

As in Sect. 2.3.6, it suffices to show that

$$(\iota^{\mathrm{Betti}, \mathrm{spec}})^R \circ \mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{Betti}} \circ \mathbf{i}^L(\delta_y) \in \mathrm{Vect}_{\mathbf{e}}^{\geq n},$$

where n only depends on y .

4.1.4. Consider the embedding

$$\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G) \xhookrightarrow{\iota^{\mathrm{Betti}}} \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G),$$

see Sect. 3.5.3.

By (3.5), this embedding admits a continuous right adjoint, to be denoted $(\iota^{\mathrm{Betti}})^R$. Moreover, the functor $\mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{Betti}}$ induces a functor, to be denoted

$$\mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{Betti}, \mathrm{restr}} : \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{Betti}, \mathrm{restr}}),$$

and we have

$$(\iota^{\mathrm{Betti}, \mathrm{spec}})^R \circ \mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{Betti}} \simeq \mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{Betti}, \mathrm{restr}} \circ (\iota^{\mathrm{Betti}})^R.$$

Hence, it suffices to show that

$$\mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{Betti}, \mathrm{restr}} \circ (\iota^{\mathrm{Betti}})^R \circ \mathbf{i}^L(\delta_y) \in \mathrm{Vect}_{\mathbf{e}}^{\geq n}.$$

This is obtained by combining the following two assertions:

Proposition 4.1.5. *Objects of the form*

$$(\iota^{\mathrm{Betti}})^R \circ \mathbf{i}^L(\delta_y) \in \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G)$$

are bounded below.

Proposition 4.1.6. *The functor $\mathbb{L}_{G, \mathrm{coarse}}^{\mathrm{Betti}, \mathrm{restr}}[\dim(\mathrm{Bun}_G) - \dim(\mathrm{Bun}_{N, \rho(\omega_X)})]$ is t -exact.*

Remark 4.1.7. When it comes to t-exactness properties, there is a substantial difference between the de Rham and Betti settings: one can show that the entire functor

$$\mathbb{L}_{G,\text{coarse}}^{\text{Betti}}[\dim(\text{Bun}_G) - \dim(\text{Bun}_{N,\rho(\omega_X)})] : \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G) \rightarrow \text{QCoh}(\text{LS}_G^{\text{Betti}})$$

is t-exact.

4.2. Applications of Riemann-Hilbert. In this subsection we will exploit the fact that Riemann-Hilbert allows us to compare directly the categories $\text{D-mod}_{\frac{1}{2},\text{Nilp}}(\text{Bun}_G)$ and $\text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti,constr}}(\text{Bun}_G)$, and also the prestacks $\text{LS}_G^{\text{restr}}$ and $\text{LS}_G^{\text{Betti,restr}}$.

4.2.1. Proof of Proposition 4.1.6. With no restriction of generality, we can assume that $\mathbf{e} = \mathbb{C}$.

Consider the de Rham context with $k = \mathbb{C}$. Recall that according to [AGKRRV, Corollary 16.5.6], the embedding

$$\text{D-mod}_{\frac{1}{2},\text{Nilp}}^{\text{RS}}(\text{Bun}_G) \subset \text{D-mod}_{\frac{1}{2},\text{Nilp}}(\text{Bun}_G)$$

is an equality, where the superscript “RS” denotes to the sheaf-theoretic context of holonomic D-modules with regular singularities in the sense of [AGKRRV, Sect. 1.1.1]⁹.

Now, Riemann-Hilbert defines a t-exact equivalence

$$\text{D-mod}_{\frac{1}{2},\text{Nilp}}^{\text{RS}}(\text{Bun}_G) \simeq \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti,constr}}(\text{Bun}_G)$$

and an isomorphism of stacks

$$\text{LS}_G^{\text{restr}} \simeq \text{LS}_G^{\text{Betti,restr}}.$$

Indeed, the above stacks in either context are defined by Tannakian formalism, which takes as an input the corresponding symmetric monoidal category of local systems $\text{Lisse}(X)$ (see [AGKRRV, Sect. 1.2]), and these categories are identified by Riemann-Hilbert.

We claim now that Theorem 2.3.8 implies Proposition 4.1.6. Indeed, follows from the next assertion:

Lemma 4.2.2. *The diagram*

$$(4.1) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2},\text{Nilp}}^{\text{RS}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_{G,\text{coarse}}^{\text{restr}}} & \text{QCoh}(\text{LS}_G^{\text{restr}}) \\ \sim \downarrow & & \downarrow \sim \\ \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti,constr}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_{G,\text{coarse}}^{\text{Betti,restr}}} & \text{QCoh}(\text{LS}_G^{\text{Betti,restr}}) \end{array}$$

commutes.

Proof. We only have to show that the Riemann-Hilbert equivalence sends

$$\iota(\text{Poinc}_!^{\text{Vac, glob}}) \in \text{D-mod}_{\frac{1}{2},\text{Nilp}}^{\text{RS}}(\text{Bun}_G) \text{ and } \text{Poinc}_{!,\text{Nilp}}^{\text{Vac, glob}} \in \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti,constr}}(\text{Bun}_G).$$

Let us show that the functors that these objects co-represent get identified. However, in both contexts, the functor in question is

$$\mathcal{F} \mapsto \mathcal{H}om(\exp_X/T, (\mathfrak{p}/T)^1(\mathcal{F})),$$

see Sect. 3.3.5 and Remark 3.3.6

□

□[Proposition 4.1.6]

Remark 4.2.3. There was no actual need to apply to Riemann-Hilbert in order to prove Proposition 4.1.6: one could repeat the argument in [FR] verbatim in the Betti setting.

⁹Recall the formation of $\text{D-mod}^{\text{RS}}(-)$ on a stack involves the operation of in-completion on affine schemes, and the inverse limit over affine schemes mapping to the given stack.

4.2.4. In the rest of this subsection we will assume Theorem 3.4.4, so that the functor $\mathbb{L}_G^{\text{Betti}}$ is defined, and we will prove the equivalence of statements (i') and (ii') in Theorem 3.5.6.

By Lefschetz principle, we can assume that in (i') the ground field k is $\mathbb{C}$ and in (ii') the field $\mathbf{e}$ of coefficients is also $\mathbb{C}$.

Applying Riemann-Hilbert, it suffices to show that the diagram

$$(4.2) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{RS}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G^{\mathrm{restr}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{restr}}) \\ \sim \downarrow & & \downarrow \sim \\ \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G^{\mathrm{Betti}, \mathrm{restr}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{Betti}, \mathrm{restr}}) \end{array}$$

commutes.

In order to prove the commutativity of (4.2), it suffices to show that the diagram

$$(4.3) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{RS}}(\mathrm{Bun}_G)^c & \xrightarrow{\mathbb{L}_G^{\mathrm{restr}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{restr}}) \\ \sim \downarrow & & \downarrow \sim \\ \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G)^c & \xrightarrow{\mathbb{L}_G^{\mathrm{Betti}, \mathrm{restr}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{Betti}, \mathrm{restr}}) \end{array}$$

commutes.

4.2.5. Recall (see [AGKRRV, Theorem 16.1.1]) that the category

$$\mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{RS}}(\mathrm{Bun}_G) \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G)$$

is compactly generated, and since the right adjoint ι^R to the embedding

$$\mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G) \hookrightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

is continuous, compact objects of $\mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{RS}}(\mathrm{Bun}_G)$ are compact as objects of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$.

In particular, by Proposition 2.1.1, we obtain that the compact generators of $\mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{RS}}(\mathrm{Bun}_G)$ are bounded below. By Theorem 1.6.2 we obtain that the top horizontal arrow in (4.3) takes values in

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{restr}})^{>-\infty} \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{restr}}).$$

Recall (see [AGKRRV, Sect. 18.3.2]) that the embedding ι^{Betti} preserves compactness. By Theorem 3.4.4 we obtain that the bottom horizontal arrow in (4.3) takes values in

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{Betti}, \mathrm{restr}})^{>-\infty} \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{Betti}, \mathrm{restr}}).$$

Hence, we can replace (4.3) by

$$\begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{RS}}(\mathrm{Bun}_G)^c & \xrightarrow{\mathbb{L}_G^{\mathrm{restr}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{restr}})^{>-\infty} \\ \sim \downarrow & & \downarrow \sim \\ \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}, \mathrm{constr}}(\mathrm{Bun}_G)^c & \xrightarrow{\mathbb{L}_G^{\mathrm{Betti}, \mathrm{restr}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{Betti}, \mathrm{restr}})^{>-\infty}. \end{array}$$

However, the commutativity of the latter diagram follows formally from the commutativity of (4.1).

□[Equivalence of (i') and (ii')]

4.3. **Proof of Proposition 4.1.5.** The idea of the proof is to express the functor $(\iota^{\mathrm{Betti}})^R \circ \mathbf{i}^L$ via a particular Hecke functor, known as the *Beilinson projector*.

This will allow us to replace the left adjoint $\mathbf{i}^L$ by a right adjoint for some other functor, which has an evident left-exactness property,

4.3.1. Let $P_{LS_G^{Betti, restr}}$ be the Beilinson projector from [AGKRRV, Sect. 15.4.5], viewed as an endofunctor of $Shv_{\frac{1}{2}}^{Betti}(Bun_G)$.

We claim:

Lemma 4.3.2. *The functor $(\iota^{Betti})^R \circ \mathbf{i}^L$, followed by*

$$Shv_{\frac{1}{2}, Nilp}^{Betti, constr}(Bun_G) \xrightarrow{\iota^{Betti}} Shv_{\frac{1}{2}, Nilp}^{Betti}(Bun_G) \xrightarrow{\mathbf{i}} Shv_{\frac{1}{2}}^{Betti}(Bun_G)$$

identifies canonically with $P_{LS_G^{Betti, restr}}$.

Let us assume this lemma and prove Proposition 4.1.5.

4.3.3. Denote by $\mathbf{i}^{constr}$ the embedding

$$Shv_{\frac{1}{2}, Nilp}^{Betti, constr}(Bun_G) \hookrightarrow Shv_{\frac{1}{2}}^{Betti, constr}(Bun_G).$$

By [AGKRRV, Theorem 16.4.10 and Proposition 17.2.3], the functor $\mathbf{i}^{constr}$ admits a continuous right adjoint. Moreover, the comonad $\mathbf{i}^{constr} \circ (\mathbf{i}^{constr})^R$ is given by the Beilinson projector $P_{LS_G^{Betti, restr}}$, viewed as an endofunctor of $Shv_{\frac{1}{2}}^{Betti, constr}(Bun_G)$.

Note also that we have a commutative diagram

$$\begin{array}{ccc} Shv_{\frac{1}{2}}^{Betti, constr}(Bun_G) & \xrightarrow{\mathbf{oblv}^{constr}} & Shv_{\frac{1}{2}}^{Betti}(Bun_G) \\ P_{LS_G^{Betti, restr}} \downarrow & & \downarrow P_{LS_G^{Betti, restr}} \\ Shv_{\frac{1}{2}}^{Betti, constr}(Bun_G) & \xrightarrow{\mathbf{oblv}^{constr}} & Shv_{\frac{1}{2}}^{Betti}(Bun_G), \end{array}$$

in which the horizontal arrows are the tautological forgetful functor.

Remark 4.3.4. Note that the functor

$$(4.4) \quad \mathbf{oblv}^{constr} : Shv_{\frac{1}{2}}^{Betti, constr}(Bun_G) \rightarrow Shv_{\frac{1}{2}}^{Betti}(Bun_G)$$

is not fully faithful. However, the composition

$$Shv_{\frac{1}{2}, Nilp}^{Betti, constr}(Bun_G) \xrightarrow{\mathbf{i}^{constr}} Shv_{\frac{1}{2}}^{Betti, constr}(Bun_G) \xrightarrow{\mathbf{oblv}^{constr}} Shv_{\frac{1}{2}}^{Betti}(Bun_G)$$

is fully faithful, since it can be rewritten as

$$Shv_{\frac{1}{2}, Nilp}^{Betti, constr}(Bun_G) \xrightarrow{\iota^{Betti}} Shv_{\frac{1}{2}, Nilp}^{Betti}(Bun_G) \xrightarrow{\mathbf{i}} Shv_{\frac{1}{2}}^{Betti}(Bun_G).$$

4.3.5. Applying Lemma 4.3.2, we obtain that we have a canonical isomorphism

$$(\iota^{Betti})^R \circ \mathbf{i}^L \circ \mathbf{oblv}^{constr} \simeq (\mathbf{i}^{constr})^R$$

as functors

$$Shv_{\frac{1}{2}}^{Betti, constr}(Bun_G) \rightarrow Shv_{\frac{1}{2}, Nilp}^{Betti, constr}(Bun_G).$$

In particular, since the functor $(\mathbf{i}^{constr})^R$ is left t-exact, we obtain that if $\mathcal{F} \in Shv_{\frac{1}{2}}^{Betti, constr}(Bun_G)$ is eventually coconnective, then so is $(\iota^{Betti})^R \circ \mathbf{i}^L \circ \mathbf{oblv}^{constr}(\mathcal{F})$.

4.3.6. Hence, it remains to show that the objects

$$\delta_y \in Shv_{\frac{1}{2}}^{Betti, constr}(Bun_G)$$

are eventually coconnective.

This follows by Riemann-Hilbert from Proposition 2.1.1 (or can be reproved directly within $Shv_{\frac{1}{2}}^{Betti, constr}(Bun_G)$ by the same argument).

□[Proposition 4.1.5]

4.4. Proof of Lemma 4.3.2. The idea of the proof is that all functors in sight are given by the various versions of the Beilinson projector.

4.4.1. Recall (see [AGKRRV, Corollary 18.2.9(a)]) that the functor $\mathbf{i} \circ \mathbf{i}^L$ is given by the action of a *Hecke functor*, denoted $P_{LS_G}^{\text{enh}, \text{Betti}}$.

Similarly, the functor $\iota^{\text{Betti}} \circ (\iota^{\text{Betti}})^R$ is given by the restriction to

$$\text{Shv}_{\frac{1}{2}, \text{Nilp}}^{\text{Betti}}(\text{Bun}_G) \subset \text{Shv}_{\frac{1}{2}}^{\text{Betti}}(\text{Bun}_G)$$

of the Hecke functor $P_{LS_G}^{\text{Betti}, \text{restr.}}$.

Now, any two Hecke functors commute, so we have

$$\mathbf{i} \circ \iota^{\text{Betti}} \circ (\iota^{\text{Betti}})^R \circ \mathbf{i}^L \simeq P_{LS_G}^{\text{Betti}, \text{restr.}} \circ P_{LS_G}^{\text{Betti}} \simeq P_{LS_G}^{\text{enh}, \text{Betti}} \circ P_{LS_G}^{\text{Betti}, \text{restr.}}.$$

4.4.2. Hence, in order to prove Lemma 4.3.2, it remains to show that

$$P_{LS_G}^{\text{enh}, \text{Betti}} \circ P_{LS_G}^{\text{Betti}, \text{restr.}} \simeq P_{LS_G}^{\text{Betti}, \text{restr.}}.$$

I.e., we need to show that the functor $P_{LS_G}^{\text{enh}, \text{Betti}}$ acts as identity on objects lying in the essential image of the functor $P_{LS_G}^{\text{Betti}, \text{restr.}}$.

By [AGKRRV, Corollary 18.2.9(a)], it suffices to show that the essential image of the endofunctor $P_{LS_G}^{\text{Betti}, \text{restr.}}$ is contained in $\text{Shv}_{\frac{1}{2}, \text{Nilp}}^{\text{Betti}}(\text{Bun}_G)$.

4.4.3. Note that the essential image of the functor $P_{LS_G}^{\text{Betti}, \text{restr.}}$ lies in

$$\text{Shv}_{\frac{1}{2}}^{\text{Betti}}(\text{Bun}_G)^{\text{Hecke-fin.mon.}} \subset \text{Shv}_{\frac{1}{2}}^{\text{Betti}}(\text{Bun}_G)$$

(see [AGKRRV, Sect. 18.3]).

In particular, this essential image lies in

$$\text{Shv}_{\frac{1}{2}}^{\text{Betti}}(\text{Bun}_G)^{\text{Hecke-loc.const.}} \subset \text{Shv}_{\frac{1}{2}}^{\text{Betti}}(\text{Bun}_G).$$

Hence, by [AGKRRV, Theorem 18.1.6], it belongs to $\text{Shv}_{\frac{1}{2}, \text{Nilp}}^{\text{Betti}}(\text{Bun}_G)$, as required.

□[Lemma 4.3.2]

5. RESTRICTED VS FULL GLC

In this section we will prove the implications (i') $\Rightarrow$ (i) and (ii') $\Rightarrow$ (ii) in Theorem 3.5.6.

While doing so, we will encounter yet another version of GLC (in both de Rham and Betti settings), namely, the *tempered* GLC. Ultimately, the logic of the proof (in either context) will be:

$$\text{Full GLC} \Leftrightarrow \text{Full tempered GLC} \Leftrightarrow \text{Restricted tempered GLC} \Leftrightarrow \text{Restricted GLC}.$$

5.1. The de Rham context. In this subsection we will introduce the tempered version of the Langlands functor. We will show that the full tempered GLC is equivalent to its restricted version.

Remark 5.1.1. The implication (i') $\Rightarrow$ (i) was proved in [AGKRRV, Sect. 21.4], under the assumption that the functor $\mathbb{L}_G$ admits a left adjoint.

We will prove that this is the case in one of the subsequent papers in this series¹⁰. In this subsection we will give a different proof of this implication, which can then be adapted to the de Betti context.

¹⁰In fact, we will ultimately prove that $\mathbb{L}_G$ is an equivalence, so that (i) holds unconditionally.

5.1.2. Let

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}} \xhookrightarrow{\mathbf{u}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

denote the *tempered* subcategory, defined as in [AG, Sect. 17.8.4].

It follows easily from the definition that the above embedding admits a right adjoint, to be denoted $\mathbf{u}^R$. Thus, we obtain an adjunction

$$\mathbf{u} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}} \rightleftarrows \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) : \mathbf{u}^R,$$

and for most practical purposes, it is convenient to view $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}}$ as a localization of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$.

Remark 5.1.3. The definition of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}}$ in *loc. cit.* depended on the choice of a point $x \in X$, and independence of the point was conjectural. This conjecture was established by Beraldo (unpublished) and later by Faergeman-Raskin by a different method (see [FR, Sect. 2.6.2]).

However, for the purposes of this paper, one can work with $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}}$ without knowing that it is point-independent.

5.1.4. We claim that the object $\mathrm{Poinc}_!^{\mathrm{Vac, glob}}$ belongs to $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}}$. This can be shown by re-interpreting $\mathrm{Poinc}_!^{\mathrm{Vac, glob}}$ via a local-to-global construction¹¹. This will be discussed in detail in the sequel to this paper.

Since the Hecke action preserves the tempered subcategory, we obtain that the essential image of the functor $\mathbb{L}_{G, \mathrm{temp}}^L$ is contained in $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}}$.

By adjunction, we obtain that the functor $\mathbb{L}_{G, \mathrm{coarse}}$ factors as

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{\mathbf{u}^R} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}} \xrightarrow{\mathbb{L}_{G, \mathrm{temp}}} \mathrm{QCoh}(\mathrm{LS}_G)$$

for a uniquely defined functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}} \xrightarrow{\mathbb{L}_{G, \mathrm{temp}}} \mathrm{QCoh}(\mathrm{LS}_G).$$

Since

$$\Psi_{\mathrm{Nilp}, \{0\}} \circ \mathbb{L}_G \simeq \mathbb{L}_{G, \mathrm{coarse}},$$

we have a commutative diagram

$$(5.1) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}) \\ \mathbf{u}^R \downarrow & & \downarrow \Psi_{\mathrm{Nilp}, \{0\}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}} & \xrightarrow{\mathbb{L}_{G, \mathrm{temp}}} & \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}), \end{array}$$

where the functor $\mathbb{L}_{G, \mathrm{temp}}$ is the right adjoint of the functor $\mathbb{L}_{G, \mathrm{temp}}^L$ of (1.10).

5.1.5. Set

$$\mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G)_{\mathrm{temp}} := \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G) \cap \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}} \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G).$$

It follows from (2.1) that

$$\mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G)_{\mathrm{temp}} = \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}} \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})} \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})_{\mathrm{restr}},$$

as subcategories of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}}$.

It is easy to see that the functor $\mathbf{u}^R$ sends

$$\mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G)_{\mathrm{temp}}.$$

¹¹Namely, $\mathrm{Poinc}_!^{\mathrm{Vac, glob}}$ is the image of the (vacuum) object in the *local Whittaker category* $\mathrm{Whit}(G)_x$ under the Poincaré functor $\mathrm{Poinc}_{!, x} : \mathrm{Whit}(G)_x \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$, while all of $\mathrm{Whit}(G)_x$ is tempered.

Applying the functor

$$- \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})} \mathrm{QCoh}(\mathrm{LS}_{\check{G}})^{\mathrm{restr}},$$

from (5.1), we obtain a commutative diagram

$$(5.2) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G^{\mathrm{restr}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{restr}}) \\ \mathbf{u}^R \downarrow & & \downarrow \Psi_{\mathrm{Nilp}, \{0\}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G)_{\mathrm{temp}} & \xrightarrow{\mathbb{L}_{G, \mathrm{temp}}^{\mathrm{restr}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{restr}}). \end{array}$$

5.1.6. Suppose that the functor $\mathbb{L}_G^{\mathrm{restr}}$ is an equivalence. We claim that this implies that $\mathbb{L}_{G, \mathrm{temp}}^{\mathrm{restr}}$ is also an equivalence.

Indeed, the fact that $\mathbb{L}_G^{\mathrm{restr}}$ is an equivalence implies that $\mathbb{L}_{G, \mathrm{temp}}^{\mathrm{restr}}$ is a Verdier quotient. However, we also know that the functor $\mathbb{L}_{G, \mathrm{temp}}^{\mathrm{restr}}$ is conservative: this is the main result of [FR].

Remark 5.1.7. One can avoid appealing to [FR] for the proof of “ $\mathbb{L}_G^{\mathrm{restr}}$ is an equivalence” $\Rightarrow$ “ $\mathbb{L}_{G, \mathrm{temp}}^{\mathrm{restr}}$ is an equivalence”.

Indeed, it is not difficult to show that the functor $\mathbb{L}_G$ respects the action of the full Satake category (at a given point $x \in X$) on the two sides¹². Now,

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}} \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{Sat}_G} \mathrm{Sat}_{G, \mathrm{temp}}$$

and

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \otimes_{\mathrm{Sat}_G} \mathrm{Sat}_{G, \mathrm{temp}},$$

and also

$$\mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G)_{\mathrm{temp}} \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{Nilp}}(\mathrm{Bun}_G) \otimes_{\mathrm{Sat}_G} \mathrm{Sat}_{G, \mathrm{temp}}$$

and

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{restr}}) \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{restr}}) \otimes_{\mathrm{Sat}_G} \mathrm{Sat}_{G, \mathrm{temp}}.$$

By a similar logic, if $\mathbb{L}_G$ is an equivalence, then it formally follows that $\mathbb{L}_{G, \mathrm{temp}}$ is also an equivalence.

Remark 5.1.8. In Sect. 5.2 we will prove an inverse implication (which is far less trivial): the fact that $\mathbb{L}_{G, \mathrm{temp}}$ is an equivalence implies that $\mathbb{L}_G$ is an equivalence.

The same argument will show that if $\mathbb{L}_{G, \mathrm{temp}}^{\mathrm{restr}}$ is an equivalence then $\mathbb{L}_G^{\mathrm{restr}}$ is an equivalence.

5.1.9. We now claim that *if* $\mathbb{L}_{G, \mathrm{temp}}^{\mathrm{restr}}$ is an equivalence (for all extensions of the ground field $k \subset k'$), *then* $\mathbb{L}_{G, \mathrm{temp}}$ is an equivalence.

This follows by repeating verbatim the argument of [AGKRRV, Sect. 21.4].

Remark 5.1.10. Both deductions:

$$\text{Full tempered GLC} \Rightarrow \text{Restricted tempered GLC},$$

carried out above, and

$$\text{Full GLC} \Rightarrow \text{Restricted GLC},$$

carried out in [AGKRRV, Sect. 21.4], follow the same logic, and both need that the corresponding version of the Langlands functor admit an adjoint. Now, the latter is immediate from the construction for $\mathbb{L}_{G, \mathrm{temp}}$, but requires more work for the original Langlands functor $\mathbb{L}_G$.

5.1.11. Thus, we obtain that in order to prove the implication (i') $\Rightarrow$ (i) in Theorem 3.5.6, it suffices to show *if* $\mathbb{L}_{G, \mathrm{temp}}$ is an equivalence *then* $\mathbb{L}_G$ is an equivalence.

This will be done on the next section.

¹²This will be elaborated on in the subsequent paper in this series.

5.2. Full vs tempered Langlands. In this subsection we will assume that $\mathbb{L}_{G,\text{temp}}$ is an equivalence and deduce that $\mathbb{L}_G$ is an equivalence.

The idea of the proof is that the behavior of $\mathbb{L}_{G,\text{temp}}$ recovers the behavior of $\mathbb{L}_G$ on *compact objects*.

5.2.1. Recall that the category $\text{IndCoh}_{\text{Nilp}}(\text{LS}_G^{\text{restr}})$ is compactly generated (see [AG, Sect. 11.1.6]); its subcategory of compact objects is $\text{Coh}_{\text{Nilp}}(\text{LS}_G^{\text{restr}})$, the category of *coherent* sheaves with nilpotent singular support.

Note that the functor

$$\Psi : \text{IndCoh}(\text{LS}_{\tilde{G}}) \rightarrow \text{QCoh}(\text{LS}_{\tilde{G}})$$

is fully faithful, when restricted to the subcategory of compact objects. Indeed, this restriction is the natural embedding

$$\text{Coh}(\text{LS}_{\tilde{G}}) \hookrightarrow \text{QCoh}(\text{LS}_{\tilde{G}})$$

.

In particular, we obtain that the functor $\Psi_{\text{Nilp},\{0\}}$ is also fully faithful when restricted to

$$\text{Coh}_{\text{Nilp}}(\text{LS}_{\tilde{G}}^{\text{restr}}) = \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{G}}^{\text{restr}})^c.$$

5.2.2. The proof of the desired implication is a formal consequence of the combination of the following two assertions:

Proposition 5.2.3. *The functor*

$$\mathbf{u}^R|_{\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)^c} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)^c \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}}$$

is fully faithful.

Proposition 5.2.4. *Assume that $\mathbb{L}_{G,\text{temp}}$ is an equivalence (for the group G and all its Levi subgroups). Then the essential image of $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)^c$ in $\text{QCoh}(\text{LS}_{\tilde{G}})$ of $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)^c$ under*

$$\mathbb{L}_{G,\text{temp}} \circ \mathbf{u}^R \simeq \mathbb{L}_{G,\text{coarse}}$$

equals $\text{Coh}_{\text{Nilp}}(\text{LS}_{\tilde{G}})$.

5.3. Proof of Proposition 5.2.3. The proof will be obtained by combining the following two ingredients. One is the *miraculous functor* on Bun_G . The other is a result of [Be1], which says that objects $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ that are $*$ -extended from quasi-compact open substacks are tempered.

5.3.1. For an object $\mathcal{F} \in \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$, let

$$\mathcal{F}_{\text{temp}} \rightarrow \mathcal{F} \rightarrow \mathcal{F}_{\text{anti-temp}}$$

be the fiber sequence associated with

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}} \hookrightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G).$$

Explicitly,

$$\mathcal{F} \simeq \mathbf{u} \circ \mathbf{u}^R(\mathcal{F}).$$

We have to show that for a pair of compact object $\mathcal{F}_1, \mathcal{F}_2 \in \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$, the map

$$(5.3) \quad \text{Hom}_{\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)}(\mathcal{F}_1, \mathcal{F}_2) \rightarrow \text{Hom}_{\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)}(\mathcal{F}_{1,\text{temp}}, \mathcal{F}_{2,\text{temp}})$$

is an isomorphism. As we shall see, just the assumption that $\mathcal{F}_2$ be compact will suffice.

The map (5.3) is tautologically an isomorphism if $\mathcal{F}_1$ is tempered (for any $\mathcal{F}_2$). Hence, it suffices to show that if $\mathcal{F}_2$ is compact and $\mathcal{F}_1$ is *anti-tempered*, i.e., if $\mathcal{F}_{1,\text{temp}} = 0$ or, equivalently, if

$$\mathcal{F}_1 \rightarrow \mathcal{F}_{1,\text{anti-temp}}$$

is an isomorphism, then

$$\text{Hom}_{\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)}(\mathcal{F}_1, \mathcal{F}_2) = 0.$$

5.3.2. We will now use the *miraculous functor*

$$\mathrm{Mir}_{\mathrm{Bun}_G} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G),$$

see [Ga1, Sect. 3.1.1].

According to [Ga1, Theorem 3.1.5]), the functor $\mathrm{Mir}_{\mathrm{Bun}_G}$ is an equivalence. Hence, it is enough to show that for $\mathcal{F}_2$ compact and $\mathcal{F}_1$ anti-tempered,

$$(5.4) \quad \mathrm{Hom}_{\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}}(\mathrm{Mir}_{\mathrm{Bun}_G}^{-1}(\mathcal{F}_1), \mathrm{Mir}_{\mathrm{Bun}_G}^{-1}(\mathcal{F}_2)) = 0.$$

5.3.3. Recall that compact objects in $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ are of the form

$$(j_U)_!(\mathcal{F}_U),$$

where

$$U \xrightarrow{j_U} \mathrm{Bun}_G$$

is the embedding of a quasi-compact open, and $\mathcal{F}_U \in \mathrm{D}\text{-mod}(U)^c$.

With no restriction of generality, we can assume that U is *co-truncative* (see [DG1, Sect. 3.1] for what this means). In this case, the functor

$$\mathrm{Mir}_{\mathrm{Bun}_U} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(U) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(U)$$

is an equivalence (see [DG1, Lemma 4.5.7]) and we have

$$\mathrm{Mir}_{\mathrm{Bun}_G}^{-1} \circ (j_U)_! \simeq (j_U)_{*,\mathrm{co}} \circ \mathrm{Mir}_U^{-1},$$

where

$$(j_U)_{*,\mathrm{co}} : \mathrm{D}\text{-mod}(U) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}$$

is the tautological functor, see [DG1, Lemma 4.4.12].

Hence, in order to prove (5.4), it suffices to show that if $\mathcal{F} \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ is anti-tempered, then

$$(j_U)_{\mathrm{co}}^* \circ \mathrm{Mir}_{\mathrm{Bun}_G}^{-1}(\mathcal{F}) = 0,$$

where $(j_U)_{\mathrm{co}}^*$ is the left adjoint of $(j_U)_{*,\mathrm{co}}$.

5.3.4. Let

$$\mathrm{Id}^{\mathrm{nv}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

be the naive functor (see [Ga1, Sect. 2.1]). Recall that

$$(j_U)_{\mathrm{co}}^* \simeq (j_U)^* \circ \mathrm{Id}^{\mathrm{nv}}$$

(see [Ga1, Corollary 2.1.5]).

Hence, it suffices to show that the composite functor

$$\mathrm{Id}^{\mathrm{nv}} \circ \mathrm{Mir}_{\mathrm{Bun}_G}^{-1}$$

annihilates the anti-tempered subcategory.

It is easy to see that both functors $\mathrm{Id}^{\mathrm{nv}}$ and $\mathrm{Mir}_{\mathrm{Bun}_G}$ commute with the Hecke action. Hence, it suffices to show that for any $\mathcal{F} \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$, we have

$$(\mathrm{Id}^{\mathrm{nv}} \circ \mathrm{Mir}_{\mathrm{Bun}_G}^{-1}(\mathcal{F}))_{\mathrm{anti-temp}} = 0.$$

However, this follows from the fact the essential image of the functor $\mathrm{Id}^{\mathrm{nv}}$ is contained in $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}}$, see [Be1, Theorem B].

□[Proposition 5.2.3]

5.4. Proof of Proposition 5.2.4. The idea of the proof is that compact generators of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ can be assembled from tempered compact objects, and *Eisenstein series*.

The behavior of the former is controlled by the assumption that $\mathbb{L}_{G,\mathrm{temp}}$ is an equivalence. The behavior of the latter is an expression of a basic compatibility between Eisenstein series and the Langlands functor.

5.4.1. Let

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

be the full subcategory, generated by the Eisenstein functors

$$\mathrm{Eis}_! : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

for *proper* parabolic subgroups of G .

5.4.2. We claim:

Lemma 5.4.3. *The objects in*

$$\left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}}\right)^c \text{ and } \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}}\right)^c$$

generate $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$.

Proof. Let

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} := \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}}\right)^\perp.$$

Denote by $\mathbf{e}$ the embedding

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} \hookrightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G),$$

and let $\mathbf{e}^L$ denote its right adjoint.

Since the category $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}}$ is compactly generated¹³, it suffices to show that the essential image of the composite functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}} \xrightarrow{\mathbf{u}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{\mathbf{e}^L} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}$$

generates the target category. We will show that this functor is in fact a localization (i.e., its right adjoint is fully faithful).

Indeed, the right adjoint to the above functor is the composition

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}} \xleftarrow{\mathbf{u}^R} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xleftarrow{\mathbf{e}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}.$$

Now, according to [Be1, Theorem A], the essential image of the functor $\mathbf{e}$ is contained in $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}}$, i.e., $\mathbf{e}$ factors as

$$\mathbf{u} \circ \mathbf{e}_{\mathrm{temp}}$$

for a uniquely defined (fully faithful) functor

$$\mathbf{e}_{\mathrm{temp}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}}.$$

Hence, the above right adjoint identifies with

$$\mathbf{u}^R \circ \mathbf{u} \circ \mathbf{e}_{\mathrm{temp}} \simeq \mathbf{e}_{\mathrm{temp}},$$

as required. □

¹³On the one hand, since we have assumed that $\mathbb{L}_{G,\mathrm{coarse}}$ is an equivalence, the compact generation of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}}$ follows from that of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$. On the other hand, this compact generation can be proved unconditionally: in [FR] it is shown that the objects obtained by acting by $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}$ on $\mathrm{Poinc}_1^{\mathrm{Vac, glob}}$ generate $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}}$.

5.4.4. Let

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})_{\mathrm{Eis}} \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$$

be the full subcategory generated by the essential images of the *spectral Eisenstein functors*

$$\mathrm{Eis}^{\mathrm{spec}} : \mathrm{QCoh}(\mathrm{LS}_{\tilde{M}}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$$

for proper parabolic subgroups.

According to [AG, Theorem 13.3.6], the category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$ is compactly generated by

$$\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})^c \text{ and } (\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})_{\mathrm{Eis}})^c,$$

where we view $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})$ as a subcategory of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$ via the functor $\Xi_{\{0\}, \mathrm{Nilp}}$, the left adjoint of $\Psi_{\mathrm{Nilp}, \{0\}}$.

Hence, given Lemma 5.4.3, in order to prove Proposition 5.2.4, it suffices to show that:

- (1) The essential image of $(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}})^c$ in $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})$ under the functor $\mathbb{L}_{G, \mathrm{coarse}}$ equals $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})^c$;
- (2) The essential image of $(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}})^c$ under $\mathbb{L}_{G, \mathrm{coarse}}$ equals $(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})_{\mathrm{Eis}})^c$.

5.4.5. Point (1) is immediate from the fact that $\mathbb{L}_{G, \mathrm{temp}}$ is an equivalence.

Point (2) follows by induction on the semi-simple rank from the fact that $\mathbb{L}_{G, \mathrm{temp}}$ is fully faithful from the next assertion (which is well-known, and will be discussed in detail in a subsequent paper in the series):

Lemma 5.4.6. *For a given parabolic subgroup P with Levi quotient M , the diagram*

$$\begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_{M, \mathrm{coarse}}} & \mathrm{QCoh}(\mathrm{LS}_{\tilde{M}}) \\ \mathrm{Eis}_! \downarrow & & \downarrow \mathrm{Eis}_{\mathrm{coarse}}^{\mathrm{spec}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_{G, \mathrm{coarse}}} & \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}) \end{array}$$

commutes up to cohomological shifts and tensoring by line bundles, where

$$\mathrm{Eis}_{\mathrm{coarse}}^{\mathrm{spec}} := \Psi_{\mathrm{Nilp}, \{0\}} \circ \mathrm{Eis}^{\mathrm{spec}}.$$

□[Proposition 5.2.4]

5.5. The Betti context. The goal of this subsection is to prove the implication (ii') $\Rightarrow$ (ii) in Theorem 3.5.6. The argument will be parallel to the proof of the implication (i') $\Rightarrow$ (i) given above.

5.5.1. Define the full subcategory

$$\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)_{\mathrm{temp}} \subset \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$$

as in [AG, Sect. 12.8.4].

Then the material in Sects. 5.1.4-5.1.11 applies as is. I.e., it suffices to show that if the functor

$$\mathbb{L}_{G, \mathrm{temp}}^{\mathrm{Betti}} : \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)_{\mathrm{temp}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}^{\mathrm{Betti}})$$

is an equivalence, then so is $\mathbb{L}_G^{\mathrm{Betti}}$.

The latter in turn reduces to the combination of the following two propositions:

Proposition 5.5.2. *The functor*

$$\mathbf{u}^R|_{\mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)^c} : \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)^c \rightarrow \mathrm{Shv}_{\frac{1}{2}, \mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)_{\mathrm{temp}}$$

is fully faithful.

Proposition 5.5.3. *Assume that $\mathbb{L}_{G,\text{temp}}^{\text{Betti}}$ is an equivalence (for the group G and all its Levi subgroups). Then the essential image of $\text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G)^c$ in $\text{QCoh}(\text{LS}_G^{\text{Betti}})$ under*

$$\mathbb{L}_{G,\text{temp}}^{\text{Betti}} \circ \mathbf{u}^R \simeq \mathbb{L}_{G,\text{coarse}}^{\text{Betti}}$$

equals $\text{Coh}_{\text{Nilp}}(\text{LS}_G^{\text{Betti}})$.

5.5.4. Proposition 5.5.2 is proved along the same lines as Proposition 5.2.3 using the following ingredients:

- The category $\text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G)_{\text{co}}$ is well-defined (see [AGKRRV, Theorem 14.1.5]), and the functor

$$\text{Mir}_{\text{Bun}_G} : \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G)_{\text{co}} \rightarrow \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G)$$

is an equivalence;

- The essential image of the functor

$$\text{Id}^{\text{nv}} : \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G)_{\text{co}} \rightarrow \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G)$$

lies in $\text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G)_{\text{temp}}$.

Both these assertions are proved in the same way as in the de Rham setting.

5.5.5. Proposition 5.5.3 is proved along the same lines as Proposition 5.2.4 using the following ingredients:

- The subcategory

$$\text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G)_{\text{cusp}} \subset \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G)$$

is contained in the essential image of the functor Id^{nv} .

- The diagram

$$\begin{array}{ccc} \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_M) & \xrightarrow{\mathbb{L}_{M,\text{coarse}}^{\text{Betti}}} & \text{QCoh}(\text{LS}_M^{\text{Betti}}) \\ \text{Eis}_! \downarrow & & \downarrow \text{Eis}_{\text{coarse}}^{\text{spec}} \\ \text{Shv}_{\frac{1}{2},\text{Nilp}}^{\text{Betti}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_{G,\text{coarse}}^{\text{Betti}}} & \text{QCoh}(\text{LS}_G^{\text{Betti}}) \end{array}$$

commutes up to cohomological shifts and tensoring by line bundles.

Both these assertions are proved in the same way as in the de Rham setting.

6. THE STRUCTURE OF HECKE EIGENSHEAVES

In this section, we discuss the structure of Hecke eigensheaves for irreducible spectral parameters, building on recent results from [AGKRRV] and [FR].

Below, we will assume GLC in the de Rham setting, and hence in the Betti setting.

6.1. Statement of the main result.

6.1.1. *The Hecke eigensheaf.* Let $\sigma \in \text{LS}_{\tilde{G}}$ be a k -point of $\text{LS}_{\tilde{G}}$.

We define the category

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{Hecke},\sigma} := \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes_{\text{QCoh}(\text{LS}_{\tilde{G}})} \text{Vect}$$

of *Hecke eigensheaves* for σ . Here $\text{QCoh}(\text{LS}_{\tilde{G}})$ acts on Vect by pullback along $\sigma : \text{Spec}(k) \rightarrow \text{LS}_{\tilde{G}}$.

Using the pushforward functor $\sigma_* : \text{Vect} \rightarrow \text{QCoh}(\text{LS}_{\tilde{G}})$, we obtain a functor

$$\text{oblv}_{\text{Hecke},\sigma} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{Hecke},\sigma} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G).$$

Now assume that σ is an *irreducible* local system, i.e., σ does not admit a reduction to any proper parabolic $\check{P} \subsetneq \check{G}$. By [AG, Prop. 13.3.3] and our running assumption that GLC holds, the functor $\mathbb{L}_G$ induces an equivalence:

$$\mathbb{L}_{G,\sigma} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Hecke},\sigma} \simeq \mathrm{Vect}.$$

Throughout this section, we let

$$\tilde{\mathcal{F}}_\sigma \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Hecke},\sigma}$$

denote the unique object with

$$\mathbb{L}_{G,\sigma}(\tilde{\mathcal{F}}_\sigma) = k[-\dim(\mathrm{Bun}_G) + \dim(\mathrm{Bun}_{N,\rho(\omega_X)})].$$

The shift here is motivated by Theorem 2.3.8.

We then let

$$\mathcal{F}_\sigma := \mathbf{oblv}_{\mathrm{Hecke},\sigma}(\tilde{\mathcal{F}}_\sigma) \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

denote the underlying D-module of the canonical Hecke eigensheaf.

6.1.2. We will prove the following result. Most parts of it are not original to this work; we will discuss attributions after stating the result.

Theorem 6.1.3. *As above, let σ be an irreducible local system and assume the geometric Langlands conjecture.*

- (1) $\mathcal{F}_\sigma$ is holonomic and has regular singularities.
- (2) The singular support of $\mathcal{F}_\sigma$ lies in the nilpotent cone $\mathrm{Nilp} \subset T^*(\mathrm{Bun}_G)$.
- (3) $\mathcal{F}_\sigma$ lies in the heart $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^\heartsuit$ of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$.
- (4) Let S_σ denote the (non-derived) group of automorphisms of σ as a local system. Then $\mathcal{F}_\sigma$ has a decomposition:

$$\mathcal{F}_\sigma \simeq \bigoplus_{\rho \in \mathrm{Irrep}(S_\sigma)} \mathcal{F}_{\sigma,\rho}^{\dim \rho}$$

where each object

$$\mathcal{F}_{\sigma,\rho} \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^\heartsuit$$

is a simple (holonomic) D-module. Moreover, these simple objects are distinct: $\mathcal{F}_{\sigma,\rho_1} \simeq \mathcal{F}_{\sigma,\rho_2}$ if and only if $\rho_1 = \rho_2$.

In particular, $\mathcal{F}_\sigma \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^\heartsuit$ is semi-simple.

- (5) Assume the genus of X is at least two and G has connected center.

Let $[\mathrm{Nilp}]$ denote the cycle on $T^*(\mathrm{Bun}_G)$ defined by the nilpotent cone. In other words, the multiplicities of $[\mathrm{Nilp}]$ are the multiplicities of irreducible components of the closed substack¹⁴ $\mathrm{Nilp} \subset T^*(\mathrm{Bun}_G)$. Then the characteristic cycle $\mathrm{CC}(\mathcal{F}_\sigma)$ equals $[\mathrm{Nilp}]$.

In the above, (1) and (2) are due to [AGKRRV] and (3) is due to [FR]. Also, [FR] observed that its results imply (4) assuming GLC, and proved a special case of this result without GLC. The last result (5) is new.

The above results have old folklore status as conjectures. Most appear in [Laum, Conj. 6.3.2]. We expect (5) to hold without the restriction on G , and our methods below prove something in this direction in general (e.g., we prove that (5) holds for a non-trivial class of local systems σ).

Remark 6.1.4. We note that one can check directly that the multiplicity of the zero section in $[\mathrm{Nilp}]$ is $\prod_{d_i \geq 2} d_i^{\dim \Gamma(\Omega_X^{1,\otimes d_i})} = \prod d_i^{(2d_i-1)(g-1)}$ for d_i running over the exponents of G ; it follows that this is the rank of $\mathcal{F}_\sigma$ at a generic point of Bun_G .

¹⁴We recall that Nilp is defined scheme-theoretically as the preimage of 0 under the Hitchin map.

Proof of Theorem 6.1.3. (1)-(4). First, (2) is [AGKRRV, Cor. 14.4.10]. As Nilp is Lagrangian, we deduce that Hecke eigensheaves are holonomic. They have regular singularities by [AGKRRV, Cor. 16.5.7]. Then (3) is [FR, Thm. 11.2.1.2].

We now turn to (4). Recall that S_σ is an extension of a finite group by the center $Z_{\check{G}}$ of $\check{G}$; in particular, S_σ is a (disconnected) reductive group.

Let $\mathrm{LS}_{\check{G},\sigma}^{\mathrm{restr}} \subset \mathrm{LS}_{\check{G},\sigma}^{\mathrm{restr}}$ denote the connected component containing σ . Define:

$$\mathrm{D}\text{-mod}_{\mathrm{Nilp}, \frac{1}{2}}(\mathrm{Bun}_G)_\sigma := \mathrm{D}\text{-mod}_{\mathrm{Nilp}, \frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{restr}})} \mathrm{QCoh}(\mathrm{LS}_{\check{G},\sigma}^{\mathrm{restr}}).$$

As in [AGKRRV, Cor. 14.3.5], this category is a direct summand of $\mathrm{D}\text{-mod}_{\mathrm{Nilp}, \frac{1}{2}}(\mathrm{Bun}_G)$.

By Lemma 5.4.6, every object of $\mathrm{D}\text{-mod}_{\mathrm{Nilp}, \frac{1}{2}}(\mathrm{Bun}_G)_\sigma$ is cuspidal. Therefore, by [Bel], every object of $\mathrm{D}\text{-mod}_{\mathrm{Nilp}, \frac{1}{2}}(\mathrm{Bun}_G)_\sigma$ is tempered. Therefore, $\mathbb{L}_G$ induces an equivalence

$$\mathbb{L}_{G,\hat{\sigma}} : \mathrm{D}\text{-mod}_{\mathrm{Nilp}, \frac{1}{2}}(\mathrm{Bun}_G)_\sigma \simeq \mathrm{QCoh}(\mathrm{LS}_{\check{G},\sigma}^{\mathrm{restr}}).$$

We then have a diagram

$$\begin{array}{ccc} & \mathrm{Vect} & \\ \mathcal{F}_\sigma \swarrow & & \searrow \sigma_* \\ \mathrm{Shv}_{\mathrm{Nilp}, \frac{1}{2}}(\mathrm{Bun}_G)_\sigma & \xrightarrow{\mathbb{L}_{G,\hat{\sigma}}[C]} & \mathrm{QCoh}(\mathrm{LS}_{\check{G},\sigma}^{\mathrm{restr}}) \end{array}$$

that commutes by definition of $\mathcal{F}_\sigma$. Here $C := \dim(\mathrm{Bun}_G) - \dim(\mathrm{Bun}_{N,\rho(\omega_X)})$ and σ_* denotes pushforward along $\sigma : \mathrm{Spec}(k) \rightarrow \mathrm{LS}_{\check{G}}^{\mathrm{restr}}$.

The bottom arrow in the above diagram is a t-exact equivalence by GLC and Theorem 2.3.8. Therefore, Jordan-Hölder questions for $\mathcal{F}_\sigma$ are the same as for $\sigma_*(k)$.

We now remind ([AGKRRV, Prop. 4.3.5]) that σ factors as $\mathrm{Spec}(k) \rightarrow \mathrm{Spec}(k)/S_\sigma \rightarrow \mathrm{LS}_{\check{G},\sigma}^{\mathrm{restr}}$ with the latter map being a closed embedding. Therefore, there is a fully faithful embedding $\mathrm{Rep}(S_\sigma)^\vee \subset \mathrm{QCoh}(\mathrm{LS}_{\check{G},\sigma}^{\mathrm{restr}})^\vee$ with $\sigma_*(k)$ corresponding to the regular representation of S_σ .

Now the result follows from the fact that S_σ is reductive, so its (left) regular representation has the form $R_{S_\sigma} \simeq \bigoplus_{\rho \in \mathrm{Irr}(\mathrm{Rep}(S_\sigma))} \rho^{\dim \rho}$. □

6.2. Characteristic cycles for eigensheaves. We now turn to the proof of Theorem 6.1.3 (5).

6.2.1. Idea of the proof. Our main new result is the following. We always assume genus ≥ 2 in what follows, but G is allowed to be arbitrary.

Theorem 6.2.2. *For every pair of irreducible local systems $\sigma_1, \sigma_2 \in \mathrm{LS}_{\check{G}}$ lying in the same irreducible component of $\mathrm{LS}_{\check{G}}$, $\mathrm{CC}(\mathcal{F}_{\sigma_1}) = \mathrm{CC}(\mathcal{F}_{\sigma_2})$.*

The following is [BD, Prop. 5.1.2]:

Theorem 6.2.3 (Beilinson-Drinfeld). *Let $\mathrm{LS}_{\check{G},\mathrm{neut}} \subset \mathrm{LS}_{\check{G}}$ be the irreducible component of $\mathrm{LS}_{\check{G}}$ containing the trivial local system. Then there exists an irreducible local system $\sigma \in \mathrm{LS}_{\check{G},\mathrm{neut}}$ such that $\mathrm{CC}(\mathcal{F}_\sigma) = [\mathrm{Nilp}]$.*

We now recall that $\mathrm{LS}_{\check{G}}$ is irreducible when $\check{G}$ has simply-connected derived group (and the genus is ≥ 2), see [BD, Prop. 2.11.4]. Therefore, the above two results imply the claim.

Remark 6.2.4. Theorem 6.2.3 is proved by de Rham methods, but Theorem 6.2.2 will be proved by Betti methods.

6.3. Proof of Theorem 6.2.2. We will prove this result using some theory of Betti constructible sheaves.

By the Lefschetz principle, note that Theorem 6.2.2 reduces to its Betti version. Therefore, we put ourselves in this setting in what follows.

6.3.1. Review of microstalks. Let $k = \mathbb{C}$, let $\mathcal{Y}$ be a smooth algebraic stack of finite type over $\mathbb{C}$, and recall that $\mathrm{Shv}^{\mathrm{Betti}}(\mathcal{Y})$ is the category of all Betti sheaves on $\mathcal{Y}$. Let $\Lambda \subset T^*(\mathcal{Y})$ be a conical (algebraic) Lagrangian and consider $\mathrm{Shv}_{\Lambda}^{\mathrm{Betti}}(\mathcal{Y}) \subset \mathrm{Shv}^{\mathrm{Betti}}(\mathcal{Y})$, which we recall is the subcategory of sheaves with singular support in Λ and is defined in [AGKRRV, Sect. F.6] for stacks.

Let $\xi \in \Lambda$ be a point lying in the smooth locus of Λ . The following result summarizes the microstalk theory for our needs.

Theorem 6.3.2. *There is a functor $\mu_{\xi} : \mathrm{Shv}_{\Lambda}(\mathcal{Y}) \rightarrow \mathrm{Vect}$ with the following properties:*

- (1) μ_{ξ} is t -exact.
- (2) μ_{ξ} admits a left adjoint, i.e., μ_{ξ} is corepresented by an object $\mu\delta_{\xi} \in \mathrm{Shv}_{\Lambda}(\mathcal{Y})$.
- (3) For a constructible sheaf $\mathcal{F} \in \mathrm{Shv}_{\Lambda}^{\mathrm{Betti}}(\mathcal{Y})$, $\dim(\mu_{\xi}(\mathcal{F}))$ equals the order of the characteristic cycle of $\mathcal{F}$ at ξ .

Proof. First, assume $\mathcal{Y}$ is a smooth scheme, see e.g. [GPS, Sect. 4.4] for a review of the construction and the corepresentability. Exactness (up to a normalizing shift) follows from [KS, Cor. 10.3.13]. The relation to the characteristic cycle is [KS, Examples 9.5.7].

In general, choose $\pi : Y \rightarrow \mathcal{Y}$ a smooth cover by a smooth scheme, and we assume Let $y \in \mathcal{Y}$ be the point so $\xi \in T_y^*(\mathcal{Y})$; let $\tilde{y} \in \pi^{-1}(y)$ be a lift of y and let $\tilde{\xi} := \pi^*(\xi) \in T_{\tilde{y}}^*(Y)$. Define:

$$\mu_{\xi}(\mathcal{F}) := \mu_{\tilde{\xi}} \circ \pi^! (\mathcal{F})[-\dim Y + \dim \mathcal{Y}] = \mu_{\tilde{\xi}} \circ \pi^* (\mathcal{F})[\dim Y - \dim \mathcal{Y}].$$

Finally, recall from [AGKRRV, Cor. G.7.6] that the embedding $\iota : \mathrm{Shv}_{\Lambda}^{\mathrm{Betti}}(\mathcal{Y}) \rightarrow \mathrm{Shv}^{\mathrm{Betti}}(\mathcal{Y})$ admits a left adjoint ι^L . Then:

$$\mu\delta_{\xi} := \iota^L(\pi_!(\mu\delta_{\tilde{\xi}}))[\dim Y - \dim \mathcal{Y}].$$

evidently corepresents the microstalk at ξ . □

6.3.3. Microstalks and the Langlands equivalence. Let $\mathrm{Nilp}^{\alpha} \subset \mathrm{Nilp}$ be an irreducible component and let $\xi \in \mathrm{Nilp}^{\alpha}$ be a generic point lying in the smooth locus. We have a microstalk functor:

$$\mathrm{Shv}_{\mathrm{Nilp}, \frac{1}{2}}^{\mathrm{Betti}}(\mathrm{Bun}_G) \rightarrow \mathrm{Vect}$$

and a corepresenting object $\mu\delta_{\xi}$.

Define $\mathcal{E}_{\xi} \in \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{Betti}})$ as:

$$\mathcal{E}_{\xi} := \mathbb{L}_G^{\mathrm{Betti}}(\mu\delta_{\xi})[-\dim(\mathrm{Bun}_G) + \dim(\mathrm{Bun}_{N, \rho(\omega_X)})].$$

We let $\mathcal{E}_{\xi}^{\mathrm{irred}} \in \mathrm{QCoh}(\mathrm{LS}_G^{\mathrm{Betti}, \mathrm{irred}})$ be the restriction of $\mathcal{E}_{\xi}$ to the irreducible locus.

6.3.4. Observe that $\mathcal{E}_{\xi}$ is compact, i.e., it lies in $\mathrm{Coh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{Betti}})$: this follows as it corresponds to the compact object $\mu\delta_{\xi}$ under the Betti Langlands equivalence.

In particular, $\mathcal{E}_{\xi}^{\mathrm{irred}}$ is perfect, so the Euler characteristic of its fibers form a locally constant function on $\mathrm{LS}_G^{\mathrm{Betti}, \mathrm{irred}}$.

We compute the dual:

$$\mathrm{Hom}_{\mathrm{Vect}}(\sigma^*(\mathcal{E}_{\xi}), \mathbb{C})$$

to this fiber at $\sigma \in \mathrm{LS}_G^{\mathrm{Betti}, \mathrm{irred}}(\mathbb{C})$ as:

$$\mathrm{Hom}_{\mathrm{QCoh}(\mathrm{LS}_G^{\mathrm{Betti}})}(\mathcal{E}_{\xi}, \sigma_*(\mathbb{C})) \simeq \mathrm{Hom}_{\mathrm{Shv}_{\mathrm{Nilp}, \frac{1}{2}}^{\mathrm{Betti}}}(\mu\delta_{\xi}, \mathcal{F}_{\sigma}) = \mu_{\xi}(\mathcal{F}_{\sigma})$$

for $\mathcal{F}_{\sigma}$ the (Betti) eigensheaf corresponding to σ , including the shift as in Sect. 6.1.1.

Applying Theorem 6.3.2, we see (i) $\sigma^*(\mathcal{E}_\xi)$ is actually concentrated in one degree, so $\mathcal{E}_\xi^{\text{irred}}$ is actually a vector bundle, and (ii) the fiber of $\mathcal{E}_\xi^{\text{irred}}$ at σ has the same dimension as the order of the characteristic cycle of $\mathcal{F}_\sigma$ at ξ .

Therefore, the function $\sigma \mapsto \text{ord}_\xi(\text{CC}(\mathcal{F}_\sigma))$ is a locally constant function on $\text{LS}_{\tilde{G}}^{\text{Betti,irred}}$. It now suffices to observe that $\text{LS}_{\tilde{G}}^{\text{Betti,irred}}$ is smooth and dense in $\text{LS}_{\tilde{G}}^{\text{Betti}}$, so connected components of $\text{LS}_{\tilde{G}}^{\text{Betti,irred}}$ are in bijection with irreducible components of $\text{LS}_{\tilde{G}}^{\text{Betti}}$.

REFERENCES

- [AG] D. Arinkin and D. Gaitsgory, *Singular support of coherent sheaves, and the Geometric Langlands Conjecture*, Selecta Math. N.S. **21** (2015), 1–199.
- [AGKRRV] D. Arinkin, D. Gaitsgory, D. Kazhdan, S. Raskin, N. Rozenblyum and Y. Varshavsky, *The stack of local systems with restricted variation and geometric Langlands theory with nilpotent singular support*, arXiv:2010.01906.
- [BD] A. Beilinson and V. Drinfeld, *Quantization of Hitchin's integrable system and Hecke eigensheaves*, available at http://people.math.harvard.edu/~gaitsgde/grad_2009/
- [Be1] D. Beraldo, *On the geometric Ramanujan conjecture*, arXiv:2103.17211.
- [Be2] D. Beraldo, *The spectral gluing theorem revisited*, arXiv:1804.04861.
- [BeLi] D. Beraldo and L. Chen, *Automorphic Gluing*, arXiv:2204.09141.
- [BZN] D. Ben-Zvi and D. Nadler, *Betti geometric Langlands*, arXiv:1606.08523.
- [DG1] V. Drinfeld and D. Gaitsgory, *Compact generation of the category of D-modules on the stack of G-bundles on a curve* Cambridge Math Journal, **3** (2015), 19–125.
- [DG2] V. Drinfeld and D. Gaitsgory, *On some finiteness questions for algebraic stacks*, GAFA **23** (2013), 149–294.
- [FR] J. Faergeman and S. Raskin, *Non-vanishing of geometric Whittaker coefficients for reductive groups*, arXiv:2207.02955.
- [Ga1] D. Gaitsgory, *A "strange" functional equation for Eisenstein series and Verdier duality on the moduli stack of bundles*, Annales Scientifiques de l'ENS **50** (2017), 1123–1162.
- [Ga2] D. Gaitsgory, *Outline of the proof of the Geometric Langlands Conjecture for $GL(2)$* , Astérisque **370** (2015), 1–112.
- [GLys] D. Gaitsgory and S. Lysenko, *Parameters and duality for the metaplectic geometric Langlands theory*, Selecta Math. New Ser. **24** (2018), 227–301. Also arXiv: 1608.00284
- [GaRo1] D. Gaitsgory and N. Rozenblyum, *DG ind-schemes*, Contemporary Mathematics **610** (2014), 139–251.
- [GaRo2] D. Gaitsgory and N. Rozenblyum, *Crystals and D-modules*, PAMQ **10**, no. 1 (2014), 57–155.
- [GPS] S. Ganatra, J. Pardon, and V. Shende, *Microlocal Morse theory of wrapped Fukaya categories*, Ann. of Math. (2) **199** (2024), no.3, 943–1042.
- [KS] M. Kashiwara and P. Schapira, *Sheaves and Manifolds*, Grundlehren Math. Wiss., **292**, Springer-Verlag, Berlin, 1994.
- [Laum] G. Laumon, *Correspondance de Langlands géométrique pour les corps de fonctions*, Duke Math. J. **54**, no. 1 (1987): 309–359.
- [Lin] K. Lin, *Poincaré series and miraculous duality*, arXiv:2211.05282.
- [NY] D. Nadler and Z. Yun, *Spectral action in Betti geometric Langlands*, Israel Journal of Mathematics, **232** (2019), 299–349.
- [Ra] S. Raskin, *A generalization of the b-function lemma*, Comp. Math. **157** (2021), 2199–2214.

PROOF OF THE GEOMETRIC LANGLANDS CONJECTURE II: KAC-MOODY LOCALIZATION AND THE FLE

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To Joseph Bernstein

CONTENTS

Introduction	5
0.1. Overview	5
0.2. What is done in this paper?	7
0.3. Structure of this paper	10
0.4. Conventions and notation: generalities	13
0.5. Acknowledgements	13
1. Geometric Satake and Casselman-Shalika formula: recollections	15
1.1. The critical twist	16
1.2. A geometric twisting construction	18
1.3. The Whittaker category on the affine Grassmannian	18
1.4. The geometric Casselman-Shalika formula	21
1.5. The spherical category	22
1.6. The spectral spherical category	23
1.7. Geometric Satake equivalence	25
1.8. The curse of σ and τ	26
2. Kac-Moody modules and the Kazhdan-Lusztig category	27
2.1. Definition and basic properties	27
2.2. Duality	27
2.3. The functor of Drinfeld-Sokolov reduction	28
3. Ind-coherent sheaves on monodromy-free opers	30
3.1. Monodromy-free opers	30
3.2. The IndCoh^* categories	32
3.3. Properties of the $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$ functor	34
3.4. A direct product decomposition	37
3.5. The action of $\mathrm{IndCoh}^!$	38
3.6. Action of the spherical category	39
3.7. Self-duality for IndCoh on opers	40
3.8. Relation to quasi-coherent sheaves	44
4. Digression: IndCoh^* via factorization algebras	46
4.1. Factorization algebras and modules	47
4.2. Kac-Moody modules as factorization modules	48
4.3. The case of commutative factorization algebras	49
4.4. Recovering IndCoh^* of opers as factorization modules	51
4.5. Proof of Proposition 4.4.7	52
4.6. Action of the center on Kac-Moody modules	55

Date: September 11, 2024.

4.7. Action of the center and the loop group action	58
4.8. The enhanced functor of Drinfeld-Sokolov reduction at the critical level	61
5. The Feigin-Frenkel isomorphism and its applications	62
5.1. The Feigin-Frenkel isomorphism	62
5.2. The “birth” of opers	63
5.3. The Kazhdan-Lusztig category at the critical level and monodromy-free opers	64
5.4. Properties of the $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})$ -action on $\mathrm{KL}(G)_{\mathrm{crit}}$	66
6. The critical FLE	67
6.1. Construction of the critical FLE functor	67
6.2. Reduction to the pointwise version	68
6.3. The inverse of the critical FLE functor	70
6.4. Compatibility of $\mathrm{FLE}_{G,\mathrm{crit}}$ and $\mathrm{FLE}_{\check{G},\infty}$	72
6.5. The functor $\mathrm{pre-FLE}_{G,\mathrm{crit}}$	74
6.6. An alternative construction of the critical FLE functor	75
7. Proof of the pointwise version of the critical FLE	77
7.1. Temperedness	77
7.2. Proof of Theorem 6.2.2	79
7.3. Proof of Proposition 7.2.2	79
7.4. Proof of Proposition 7.2.6	81
7.5. Spherical vs Whittaker	83
8. Compatibility of the critical FLE with duality	85
8.1. Statement of the compatibility theorem	85
8.2. Feigin-Frenkel center and self-duality	87
8.3. Proof of Proposition 8.2.6	88
8.4. Self-duality on opers via Kac-Moody	89
8.5. Proof of Theorem 8.4.5	90
8.6. Proof of Theorem 8.1.4	94
9. The coefficient and Poincaré functor(s)	96
9.1. Twisted D-modules on Bun_G	96
9.2. Restricting to (twists of) Bun_N	97
9.3. The coefficient functor	98
9.4. The $!$ -Poincaré functor	100
9.5. The $*$ -Poincaré functor	102
9.6. Coefficient and Poincaré functors: global interpretation	104
10. The localization functor	105
10.1. The de Rham twisting on Bun_G corresponding to a level	105
10.2. The functor Γ_G	108
10.3. Localization functor as a left adjoint	110
10.4. The fiber of the localization functor	114
10.5. Localization functor as the dual	115
11. Digression: local-to-global functors and unitality	117
11.0. What is this section about?	117
11.1. Setup for local-to-global functors	118
11.2. The local unital structure	119
11.3. A (lax) unital structure on a local-to-global functor	122
11.4. Integrated insertion of the unit	124
11.5. Construction of the integrated functor	126
11.6. Properties of the integrated functor	128
11.7. Proofs of Lemmas 11.5.5, Lemma 11.6.3 and C.5.13	131
11.8. Unitality as a property	132
11.9. Factorization homology	133
12. Properties of the localization functor	137
12.1. Localization and the forgetful functor	137

12.2.	Proof of Proposition 10.4.5	140
12.3.	Localization and restriction	141
12.4.	Proof of Proposition 12.3.6	144
12.5.	Localization for unipotent group-schemes	146
12.6.	Proof of Theorem 12.5.13	149
12.7.	Global functions on the moduli space of bundles for a unipotent group-scheme	151
12.8.	Application: integration over (twists of) Bun_N via BRST	154
13.	Localization via the infinitesimal Hecke groupoid	156
13.1.	Another take on the functor Γ_G	156
13.2.	The functor $\mathrm{Loc}_{G,\kappa}$ and the infinitesimal Hecke groupoid	158
13.3.	Proof of Theorem 12.1.8	160
13.4.	Localization is (almost) a localization	164
13.5.	First proof of Theorem 13.4.2	166
13.6.	Second proof of Theorem 13.4.2	168
14.	The composition of localization and coefficient functors	171
14.1.	The vacuum case	171
14.2.	Composition of coefficient and localization functors: the general case	172
14.3.	Proof of Theorem 14.2.4.	174
15.	The Hecke eigen-property of critical localization	177
15.1.	Statement of the result	177
15.2.	The key local construction	179
15.3.	The expanded Hecke eigen-property	182
15.4.	Proof of Proposition 15.3.6	183
15.5.	The <i>integrated</i> Hecke eigen-property	185
15.6.	Proof of Theorem 15.1.7	188
15.7.	Construction of the algebra action	191
15.8.	Verification of the Hecke property: reduction to a local statement	193
15.9.	Verification of the Hecke property at the local level	195
15.10.	Proofs of Lemmas 15.6.4 and 15.7.3	197
16.	Proof of Theorem 15.2.8	199
16.1.	Reformulation and strategy	199
16.2.	The acting agents	201
16.3.	Construction of the action	203
16.4.	The unital structure on the action functor	204
16.5.	End of the construction	206
17.	Spectral Poincaré functor(s)	208
17.1.	Ind-coherent sheaves on local vs. global opers	209
17.2.	Interaction with self-duality	211
17.3.	Ind-coherent sheaves on local vs. global <i>monodromy-free</i> opers	213
17.4.	Two versions of the spectral Poincaré functor	216
17.5.	Action of the spectral spherical category and temperedness	218
17.6.	The spectral localization and global sections functors	219
17.7.	Composing spectral Poincaré and global sections functors	220
18.	The Langlands functor	223
18.1.	Recollections on the Langlands functor—the coarse version	223
18.2.	Compatibility with the full spherical action	225
18.3.	The actual Langlands functor	228
18.4.	Critical localization and temperedness	230
18.5.	Compatibility of the Langlands functor with critical localization	230
Appendix A.	Ind-coherent sheaves in infinite type	233
A.1.	The category $\mathrm{QCoh}_{\mathrm{co}}(-)$	233
A.2.	The category $\mathrm{QCoh}_{\mathrm{co}}$ on ind-schemes	237
A.3.	A descent property of $\mathrm{QCoh}_{\mathrm{co}}(-)$	239

A.4.	The category $\mathrm{IndCoh}^!(-)$	241
A.5.	The category $\mathrm{IndCoh}^*(-)$	244
A.6.	The multiplicative structure	245
A.7.	Further properties of IndCoh^*	247
A.8.	The t-structure on IndCoh^*	249
A.9.	Placidity	250
A.10.	The categories $\mathrm{IndCoh}^!(-)$ and $\mathrm{IndCoh}^*(-)$ in the (ind)-placid case	253
	Appendix B. Factorization patterns	256
B.1.	Factorization spaces	256
B.2.	Factorization <i>module</i> spaces	257
B.3.	Digression: formal discs	258
B.4.	Formation of (horizontal) arc and loop spaces	261
B.5.	Digression: the jet construction	263
B.6.	Digression: the notion of (almost) finite presentation in the D-sense	264
B.7.	Local systems on the formal (punctured) disc	265
B.8.	Sheaves of categories over the Ran space	269
B.9.	Factorization algebras and modules	274
B.10.	Commutative factorization algebras	279
B.11.	Factorization categories	281
B.12.	Factorization module categories	285
B.13.	Factorization categories of algebro-geometric nature	288
B.14.	Modules over Kac-Moody Lie algebras	294
B.15.	Restriction of factorization module categories, continued	301
	Appendix C. Unital structures	303
C.1.	Categorical prestacks	303
C.2.	Crystals of categories on categorical prestacks	305
C.3.	Two notions of direct image of a crystal of categories	309
C.4.	Pseudo-properness	311
C.5.	The unital Ran space	316
C.6.	Unital and counital factorization spaces	319
C.7.	Unital factorization algebras	323
C.8.	Commutative <i>unital</i> factorization algebras	326
C.9.	Factorization homology of commutative factorization algebras	329
C.10.	Unitality in correspondences	331
C.11.	Unital factorization categories	333
C.12.	Examples of unital factorization categories arising from algebraic geometry	338
C.13.	Unital structure on Kac-Moody representations	341
C.14.	Unital factorization module categories	344
C.15.	Restriction along lax unital functors	346
C.16.	Tightness	350
	Appendix D. Chiral modules	352
D.1.	A reminder: chiral vs factorization algebras	352
D.2.	The pro-projective generator for chiral modules	353
D.3.	The case of Lie-* algebras	356
D.4.	Proof of Proposition D.3.4	357
D.5.	Proof of Theorem 4.3.9	359
D.6.	Failure of Theorem 4.3.9 in the non-finitely presented case	362
D.7.	Proof of Proposition D.5.2: preparations	364
D.8.	Proof of Proposition D.5.2	366
D.9.	Proof of Lemmas D.7.8 and D.7.9	367
	Appendix E. The spectral spherical category	368
E.1.	A 1-affineness property of LS_H^{reg}	368
E.2.	Definition of $\mathrm{Sph}_H^{\mathrm{spec}}$	369

E.3.	Unital structure	373
E.4.	Duality	375
E.5.	t-structures	376
E.6.	The monoidal structure	378
E.7.	Action on $\mathrm{IndCoh}(\mathrm{LS}_H)$	381
E.8.	Action on monodromy-free opers	384
E.9.	An approach to $\mathrm{Sph}_H^{\mathrm{spec}}$ via factorization modules	386
E.10.	Compatibility of the FLE with (derived) Satake	389
E.11.	Proof of Proposition E.10.10	392
E.12.	Arc spaces of smooth D-schemes	393
E.13.	Proof of Proposition E.12.17	396
Appendix F. Horizontal sections of affine D-schemes		397
F.1.	Horizontal sections via factorization homology	398
F.2.	Variant: allowing poles	400
F.3.	Factorization with poles	402
F.4.	Sections of quasi-coherent sheaves, meromorphic version	403
F.5.	Interpretation as factorization restriction	405
Appendix G. From module categories over $\mathrm{QCoh}(\mathrm{LS}_G^{\mathrm{mer}})$ to factorization module categories over $\mathrm{Rep}(\check{G})$		406
G.1.	Creating factorization modules categories	406
G.2.	Proof of Theorem G.1.10	408
G.3.	Factorization module categories attached to affine D-schemes	410
G.4.	Proof of Proposition 7.5.7 via factorization	412
G.5.	Proof of Proposition G.4.8	413
Appendix H. Unital local-to-global functors and monoidal actions		415
H.1.	The “independent” category	415
H.2.	The calculation of the independent category in the vacuum case	419
H.3.	Non-unitality vs independence	421
H.4.	Sheaves of monoidal categories on the unital Ran space	423
H.5.	Sheaves of monoidal categories on the <i>non</i> -unital Ran space	426
H.6.	Local and integrated monoidal actions	427
H.7.	Local actions with parameters	429
Appendix I. The integration functor in the non-unital setting		431
I.1.	A left-lax unital structure on a local-to-global functor	431
I.2.	Comparison with the integration functor in the lax unital setting	432
I.3.	Proof of Proposition 11.8.8	432
I.4.	Definition of left-lax unital structures	433
I.5.	Proof of Proposition I.1.7	436
I.6.	Proof of Proposition I.2.3	437
Appendix J. A homotopical device for coaction		440
J.1.	Associative algebras via mock-simplicial sets	440
J.2.	Factorization categories attached to <i>associative</i> factorization algebras	441
J.3.	The renormalization step	442
J.4.	Adding another monoidal category	443
J.5.	Applying the paradigm	445
References		447

INTRODUCTION

0.1. **Overview.** This paper is the second in a series of five that together prove the geometric Langlands conjecture. In this paper, we study the interaction between Kac-Moody localization and the global

geometric Langlands functor of [GLC1]. We do so following the methodology of Beilinson-Drinfeld, using chiral (a.k.a, factorization) homology.

0.1.1. The main result of this paper, which appears in the main body as Theorem 18.5.2, says:

Theorem 0.1.2. *There is a commutative diagram*

$$\begin{array}{ccc} \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\ \mathrm{Loc}_{G,\mathrm{crit}} \otimes \mathbb{I} \uparrow & & \uparrow \mathrm{Poinc}_{G,*}^{\mathrm{spec}} \\ \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & \xrightarrow{\mathrm{FLE}_{G,\mathrm{crit}}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}}. \end{array}$$

The terms appearing in the above diagram warrant further discussion. We will do so at more length later in this introduction, but here is a brief synopsis:

- $\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$ is the category of *critically twisted* D-modules on Bun_G , as considered originally in [BD1];
- $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$ was defined in [AG] as the spectral category in geometric Langlands;
- The functor $\mathbb{L}_G$ is the *Langlands functor* as constructed in [GLC1];
- $\mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}}$ is the Ran space¹ version of the Kazhdan-Lusztig category at the critical level (i.e., Kac-Moody modules at the critical level integrable with respect to the arc group $\mathfrak{L}^+(G)$);
- $\mathrm{Op}_G^{\mathrm{mon-free}}$ is the factorization space parametrizing local systems on the formal disc equipped with an oper structure on the punctured disc;
- $\mathrm{FLE}_{G,\mathrm{crit}}$ is the *fundamental local equivalence* at critical level. This equivalence of factorization categories appears in Theorem 6.1.4 and extends the pointwise equivalence of [FG2]. It is the main theorem of Part I of this paper;
- $\mathrm{Loc}_{G,\mathrm{crit}}$ is the functor of critical level Kac-Moody localization;
- The Poincaré series functor, denoted functor $\mathrm{Poinc}_{G,*}^{\mathrm{spec}}$, is given at each finite set $\underline{x} \in \mathrm{Ran}$ of points in X by pull-push along the correspondence

$$\mathrm{Op}_{G,\underline{x}}^{\mathrm{mon-free}} \leftarrow \mathrm{Op}_{G,\underline{x}}^{\mathrm{mon-free, glob}} \rightarrow \mathrm{LS}_{\check{G}},$$

where the middle term parametrizes local systems on the global curve X equipped with an oper structure on $X - \underline{x}$;

- $\mathbb{I}$ is a cohomologically shifted 1-dimensional vector space that can be ignored at first approximation. Using notation defined in the paper, it is $\mathbb{I}_{G,N_{\rho(\omega_X)}}^{\otimes \frac{1}{2}} \otimes \mathbb{I}_{N_{\rho(\omega_X)}}^{\otimes -1} [-\delta_{N_{\rho(\omega_X)}}]$.

0.1.3. The above theorem has had folklore status in the subject. Its main ingredients were discussed at the 2014 conference “Towards the proof of the geometric Langlands conjecture.”

However, some of the key technical aspects have not been addressed in the existing literature. This is most notably true for the category $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}}$, i.e., the category of ind-coherent sheaves on the *Ran space* version of the space of monodromy-free opers (see Sect. 0.2.1 below).

0.1.4. The role of Theorem 0.1.2 in the geometric Langlands program is as follows:

The functor $\mathrm{Loc}_{G,\mathrm{crit}}$ is not surjective, but neither is it so far from being surjective (see [Ga1, Prop. 10.1.6]). Therefore, understanding the interaction between $\mathbb{L}_G$ and Kac-Moody localization plays a crucial role in understanding $\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$ in spectral terms. See Remark 0.2.10 for an example where this idea is applied.

0.1.5. *What goes into the proof?* As indicated above, we first need to define the various categories.

Second, we need to construct the functors appearing in the commutative diagram. Perhaps the most interesting is $\mathrm{FLE}_{G,\mathrm{crit}}$, which is the subject of Part I of this paper. We discuss it further below.

Finally, we need to prove the diagram commutes. Ultimately, we do this by expressing both circuits in terms of chiral homology for the critical level $\mathcal{W}$ -algebra and appealing to the Feigin-Frenkel isomorphism.

¹The Ran space of X parameterizes finite collection of points of X .

0.1.6. As was mentioned already, this paper builds on the ideas of Beilinson and Drinfeld.

In their seminal works [BD1] and [BD2], Beilinson and Drinfeld introduced the theory of *chiral algebras* – which are equivalent to *factorization algebras* and, suitably understood, vertex algebras – and of *chiral homology* as a tool for studying interactions between categories of *local* nature, such as sheaves on the affine Grassmannian, and categories of *global* nature, such as sheaves on Bun_G .

The functors appearing in Theorem 0.1.2 are of local-to-global nature, and may be viewed as generalizations of the functor of chiral homology. It is in this sense that one can view the present work as a continuation of [BD1, BD2].

0.1.7. In writing this text, we found that we needed to refine many foundational parts of the original work of Beilinson-Drinfeld. This ultimately accounts for the length of the present work.

A significant part of these refinements has to do with the fact that we (have to) work with ∞ -categories (whereas in [BD1, BD2] one mostly works with abelian categories).

0.2. What is done in this paper? We now highlight what we think are the most important contributions of this paper.

0.2.1. *Monodromy-free opers.* First, as an algebro-geometric object, $\mathrm{Op}_G^{\mathrm{mon-free}}$ parametrizes a point $\underline{x} \in \mathrm{Ran}$, a local system σ on the formal disc $\mathcal{D}_{\underline{x}}$ at x , and an oper structure on the restriction $\sigma|_{\mathcal{D}_{\underline{x}}^\times}$ of σ to the punctured disc.

When we work over a fixed point $x \in X$, the corresponding space $\mathrm{Op}_{\check{G},x}^{\mathrm{mon-free}}$ was introduced and studied in [FG2]. However, the Ran space version presents a host of new challenges.

This space has infinite type, so it is not immediately obvious how to define (ind-)coherent sheaves on it. We explain the relevant geometry needed to make sense of $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$ in Sect. 3.

In Sect. 4.4, we show that $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$ can *almost* be realized as a category of factorization modules. More precisely, in Sect. 4.4.4 we define a factorization algebra $R_{\check{G},\mathrm{Op}} \in \mathrm{Rep}(\check{G})$ and prove in Proposition 4.4.7 that its category of factorization modules is equivalent to $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$ modulo homological convergence issues (more precisely: the corresponding bounded below categories are equivalent).

0.2.2. *The critical FLE.* This result appears as Theorem 6.1.4. It asserts that we have a t-exact equivalence of factorization categories

$$\mathrm{FLE}_{G,\mathrm{crit}} : \mathrm{KL}(G)_{\mathrm{crit}} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}).$$

One can view this equivalence as a (substantially amplified) categorical incarnation of the Feigin-Frenkel isomorphism.

The idea of the proof is as follows:

First, we construct the functor $\mathrm{FLE}_{G,\mathrm{crit}}$. The ingredients are Feigin’s Drinfeld-Sokolov functor and Beilinson-Drinfeld’s birth of opers construction.

Second, we prove that $\mathrm{FLE}_{G,\mathrm{crit}}$ preserves compact objects (in the sense suitable for factorization categories). This expresses a finiteness property of Drinfeld-Sokolov reductions that is not immediate using classical VOA methods; we show that it is immediate from the categorical construction of $\mathcal{W}$ -algebras from [Ra2] (i.e., the *affine Skryabin theorem*).

Thanks to the preservation of compactness mentioned above, we are reduced to proving that $\mathrm{FLE}_{G,\mathrm{crit}}$ is a *pointwise* equivalence. This is a theorem of [FG2]. We actually reprove this theorem here to illustrate a more modern point of view on studying Kac-Moody representations using categorical tools.

Namely, we show that in general, for a category $\mathbf{C}$ with an $\mathfrak{L}(G)$ -action, the *tempered quotient*² $\mathrm{Sph}(\mathbf{C})_{\mathrm{temp}}$ of $\mathrm{Sph}(\mathbf{C}) := \mathbf{C}^{\mathfrak{L}^+(G)}$ can be algorithmically recovered from $\mathrm{Whit}(\mathbf{C})$, the Whittaker

²See Sect. 7.1, where this notion is defined.

model (i.e., $\mathfrak{L}(N)$ -invariants against a non-degenerate character) of $\mathbf{C}$. Heuristically, $\text{Whit}(\mathbf{C})$ should live as a sheaf over $\text{LS}_{\check{G}}(\mathcal{D}^\times)$ and its sections over $\text{LS}_{\check{G}}(\mathcal{D})$ should recover $\text{Sph}(\mathbf{C})_{\text{temp}}$; we give a precise assertion of this type in Proposition 7.5.5. The key input for this result is the pointwise version of derived Satake.

We then show that $\text{KL}(G)_{\text{crit}} = \text{Sph}(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit},x})$ equals its tempered quotient. However, by [Ra2],

$$\text{Whit}(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit},x}) \simeq \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}),$$

and hence we obtain the FLE from the previous paragraph.

To summarize: we deduce the FLE at critical level as an essentially formal consequence of derived Satake and affine Skryabin.

Remark 0.2.3. We should add that a particular case of the pointwise abelian category version of the FLE was established already in [BD1]:

Namely, in *loc. cit.* it was shown that the subcategory of $(\text{KL}(G)_{\text{crit},x})^\heartsuit$ consisting of modules with *regular central characters* is freely generated over $(\text{QCoh}(\text{Op}_{\check{G},x}^{\text{reg}}))^\heartsuit$ by the vacuum module.

Note, however, that a parallel statement would be *false* at the derived level; this observation is what led the authors of [FG2] to considering the ind-scheme of monodromy-free opers,

0.2.4. *The FLE and duality.* The category $\text{KL}(G)_{\text{crit}}$ is canonically self-dual by a construction with semi-infinite cohomology, see Sect. 2.2.4.

In Sect. 3, we show that $\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})$ is canonically self-dual, which we express as an equivalence

$$\Theta_{\text{Op}_{\check{G}}^{\text{mon-free}}} : \text{IndCoh}^!(\text{Op}_{\check{G}}^{\text{mon-free}}) \simeq \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}).$$

This equivalence comes from a similar equivalence $\Theta_{\text{Op}_{\check{G}}^{\text{mer}}}$ using all opers in place of monodromy-free opers; the latter should be thought of as a critical limit of the semi-infinite cohomology for $\mathcal{W}$ -algebras considered by Dhillon in [Dh].

In Sect. 8, we prove that these two self-duality constructions match under the FLE. As indicated above, this result should be considered as a compatibility between the FLE and two flavors of semi-infinite cohomology.

0.2.5. *The formalism of local-to-global functors.* We develop axiomatics in Sect. 11. In some part, the constructions here abstract Beilinson-Drinfeld's construction of chiral homology.

There is a separate introduction to this material in Sect. 12.0, so we describe the material briefly:

One often finds the following situation: there is a local (factorization) category $\underline{\mathbf{C}}^{\text{loc}}$, a global category $\mathbf{C}^{\text{glob}}$, and a *local-to-global* functor

$$\mathbf{F} : \mathbf{C}_{\text{Ran}}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}}.$$

Here are examples we have in mind:

- For a chiral algebra $\mathcal{A}$, take³ $\underline{\mathbf{C}}^{\text{loc}} = \mathcal{A}\text{-mod}^{\text{fact}}$, $\mathbf{C}^{\text{glob}} = \text{Vect}$, and $\mathbf{F} = \underline{\mathbf{C}}^{\text{fact}}(X, \mathcal{A}, -)$, i.e., the functor of chiral homology;
- Take $\underline{\mathbf{C}}^{\text{loc}} = \text{KL}(G)_\kappa$, $\mathbf{C}^{\text{glob}} = \text{D-mod}_\kappa(\text{Bun}_G)$, and $\mathbf{F} = \text{Loc}_{G,\kappa}$;
- Take $\underline{\mathbf{C}}^{\text{loc}} = \text{Whit}_\kappa(G)$ the category of κ -twisted Whittaker D-modules on the affine Grassmannian, $\mathbf{C}^{\text{glob}} = \text{D-mod}_\kappa(\text{Bun}_G)$, and $\mathbf{F} = \text{Poinc}_{G,!}$ (or $\text{Poinc}_{G,*}$);
- Take $\underline{\mathbf{C}}^{\text{loc}} = \text{Rep}(\check{G})$, $\mathbf{C}^{\text{glob}} = \text{QCoh}(\text{LS}_{\check{G}})$, and $\mathbf{F} = \text{Loc}^{\text{spec}}$ the spectral localization functor;
- Take $\underline{\mathbf{C}}^{\text{loc}} = \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})$, $\mathbf{C}^{\text{glob}} = \text{IndCoh}(\text{LS}_{\check{G}})$, and $\mathbf{F} = \underline{\text{Poinc}}_{\check{G},*}^{\text{spec}}$ (or $\underline{\text{Poinc}}_{\check{G},!}^{\text{spec}}$).

³Technically, $\mathcal{A}\text{-mod}^{\text{fact}}$ only forms a *lax* factorization category. In fact, the material of Sect. 11 does not assume any sort of factorization, just the existence of suitable categories over the (unital) Ran space.

The last two examples can be considered as $\kappa \rightarrow \infty$ limits of the second and third examples.

One key feature of each of the above constructions is that they are *unital*, in the sense that our factorization categories are themselves unital and F (canonically) commutes with vacuum insertion.

The following construction plays a key role:

One starts with a non-unital (but *lax* unital) functor F_0 and it turns out that there exists a procedure that canonically produces from it a *strictly unital* local-to-global functor F .

For example, for a chiral algebra $\mathcal{A}$, we could take F_0 as

$$\mathcal{A}\text{-mod}_{\text{Ran}}^{\text{fact}} \xrightarrow{\text{oblv}} \text{D-mod}(\text{Ran}) \xrightarrow{C_c^*(\text{Ran}, -)} \text{Vect}$$

with second functor that of compactly supported de Rham cochains. The resulting functor F is that of chiral homology. I.e., chiral homology can be thought of as a universal procedure that forces F_0 to commute with vacuum insertion.

We discuss this passage from lax to strict unital globalization functors at length in Sect. 11.

Remark 0.2.6. In favorable cases, Betti analogues of local-to-global functors have TQFT interpretations. Namely, given a 4d TQFT Z and a boundary condition $\mathcal{B}$ for it, we obtain $\mathbf{C}^{\text{glob}}$ as $Z(X)$; $\mathbf{C}^{\text{loc}}$ as the evaluation of Z on a closed disc $\mathcal{D}^{\text{Betti}}$, putting $\mathcal{B}$ on the boundary $\partial\mathcal{D}^{\text{Betti}}$; and for $\underline{x} \in \text{Ran}$, we suture the disc $\mathcal{D}_{\underline{x}}^{\text{Betti}}$ around $\underline{x}$ into $X \setminus (\mathcal{D}_{\underline{x}}^{\text{Betti}} \setminus \partial\mathcal{D}_{\underline{x}}^{\text{Betti}})$ to obtain F . See Remark 11.3.9 for a related discussion.

0.2.7. Localization. In Sect. 10, we construct and study the Kac-Moody localization functor, which appears in Theorem 0.1.2. We do this in a loop group equivariant way, which has the effect of making Loc_G respect the Hecke actions.

We reproduce some results from [CF]. For the purposes of Theorem 0.1.2, the most important outcome is Theorem 12.8.8, which says we have a commutative diagram

$$\begin{array}{ccc} \text{D-mod}_{\kappa}(\text{Bun}_G) & \xrightarrow{\text{coeff}_G^{\text{Vac, glob}}} & \text{Vect} \\ \text{Loc}_{G, \kappa} \otimes_{\mathcal{I}_{N_{\rho}(\omega_X)}} [\delta_{N_{\rho}(\omega_X)}] \uparrow & & \uparrow \mathbf{C}^{\text{fact}}(X, \omega, -) \\ \text{KL}(G)_{\kappa, \text{Ran}} & \xrightarrow{\text{DS}^{\text{enh}}} & \mathcal{W}_{\kappa}\text{-mod}^{\text{fact}}. \end{array}$$

Here the bottom horizontal arrow is Drinfeld-Sokolov reduction, the right vertical arrow is chiral homology, and the top horizontal arrow is the functor of *vacuum Whittaker coefficient*.

The similarity between this result and Theorem 0.1.2 is plain. In fact, the Langlands functor $\mathbb{L}_G$ is *characterized* using $\text{coeff}_G^{\text{Vac, glob}}$, so this result is quite close to Theorem 0.1.2.

0.2.8. Localization at the critical level and Hecke eigensheaves. In Sect. 15, we prove the *Hecke eigenproperty* of localization at the critical level. This result extends one of the main theorems of [BD1]; there Beilinson-Drinfeld considered the vacuum representation, but our result allows consideration of arbitrary objects of $\text{KL}(G)_{\text{crit}}$.

More precisely, Feigin-Frenkel duality (or the FLE) allows us to consider $\text{KL}(G)_{\text{crit}, \text{Ran}}$ as a module category for $\text{QCoh}(\text{Op}_{\check{G}}^{\text{mon-free}})_{\text{Ran}}$.

Now let $\text{Op}_{\check{G}}^{\text{mon-free, glob}}$ be the space over Ran parametrizing $\underline{x} \in \text{Ran}$, a $\check{G}$ -local system σ on X , and an oper structure on $\sigma|_{X \setminus \underline{x}}$. There is an evident map $\text{Op}_{\check{G}}^{\text{mon-free, glob}} \rightarrow \text{Op}_{\check{G}}^{\text{mon-free}}$.

We prove in Corollary 15.5.9 that $\text{Loc}_{G, \text{crit}}$ factors through a functor

$$\text{Loc}_{G, \text{crit}}^{\text{Op}} : \text{KL}(G)_{\text{crit}, \text{Ran}} \otimes_{\text{QCoh}(\text{Op}_{\check{G}}^{\text{mon-free}})} \text{QCoh}(\text{Op}_{\check{G}}^{\text{mon-free, glob}}) \rightarrow \text{D-mod}_{\text{crit}}(\text{Bun}_G)$$

that is $\text{Rep}(\check{G})_{\text{Ran}}$ -linear with respect to the following actions:

- $\text{Rep}(\check{G})_{\text{Ran}}$ acts on $\text{D-mod}_{\text{crit}}(\text{Bun}_G)$ through the Hecke action.

- $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}$ acts on

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}}} \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}, \mathrm{glob}})_{\mathrm{Ran}}$$

by

$$\mathrm{Rep}(\check{G})_{\mathrm{Ran}} \xrightarrow{\mathrm{Loc}_G^{\mathrm{spec}}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}})$$

and pullback along the tautological map $\mathrm{Op}_G^{\mathrm{mon-free}, \mathrm{glob}} \rightarrow \mathrm{LS}_{\check{G}}$.

Applying the FLE, we can rewrite $\mathrm{Loc}_{G, \mathrm{crit}}^{\mathrm{Op}}$ as a functor

$$\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}})} \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}, \mathrm{glob}}) \rightarrow \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G).$$

We have canonical functors

$$\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}} \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}} \xrightarrow{\Theta_{\mathrm{Op}_G^{\mathrm{mon-free}}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}}$$

that we can compose with the above to obtain a functor

$$\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}, \mathrm{glob}}) \rightarrow \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G).$$

Our Hecke property implies that this functor sends the skyscraper sheaf at an oper $\chi \in \mathrm{Op}_G^{\mathrm{mon-free}, \mathrm{glob}}$ to an eigensheaf for the local system underlying χ . When χ is a *regular* oper on X , this is the main construction of [BD1].⁴

Remark 0.2.9. Beilinson-Drinfeld show that their eigensheaves are non-zero by computing their characteristic cycles. This does not directly apply in the monodromy-free setting, but one can use Theorem 0.1.2 to verify that they are non-zero by calculating the Whittaker coefficients of these localized D-modules.

Remark 0.2.10. The above eigen-property for Kac-Moody localization was used in Drinfeld-Gaitsgory's proof of the *spectral action* of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ on $\mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$, cf. [Ga1] Theorem 4.5.2. The proof presented there is based on Kac-Moody localization, with [Ga1, Theorem 10.3.4] essentially asserting the Hecke property discussed above. In this sense, the present work fills an important gap in the literature.

0.3. Structure of this paper.

0.3.1. Overall, the paper proceeds as follows. In Part I, we formulate and prove the critical level FLE, which is the main local result of this paper. We also consider the interactions of the FLE with various duality functors.

Part II considers the vertical local-to-global functors from Theorem 0.1.2 as well as the Hecke actions. Some of our main results here reproduce results and arguments from [CF]. We put particular emphasis on the role of *unital* structures, building on ideas from [BD2], [Ra6], [Ro2] and [Ga4].

The proof of Theorem 0.1.2 relies on chiral homology and some mild variants thereof. Because [BD2] largely considered abelian categories of chiral modules, we need some extensions of their ideas to the derived setting; these appear in the appendices to this paper.

⁴Actually, even in this case, our notion of eigensheaf is somewhat more homotopically robust than the one from [BD1].

0.3.2. In more granular detail, the paper is structured as follows.

Part I deals with the local theory:

Sect. 1 reviews various factorization categories on the geometric side associated with the affine Grassmannian of G , as well as their spectral counterparts. The key points here are the geometric Casselman-Shalika formula (Theorem 1.4.2) and (derived) geometric Satake equivalence (Theorem 1.7.2). This material is largely taken from [CR].

In Sect. 2 we discuss the Kazhdan-Lusztig category and quantum Drinfeld-Sokolov reduction for its modules.

In Sect. 3, we consider ind-coherent sheaves on various spaces of local ops. We reinterpret these objects using factorization module categories in Sect. 4.4.

In Sect. 4 we explain how various factorization categories of interest can be expressed as factorization module categories over factorization algebras. We use this to define a *categorical* action of the Feigin-Frenkel center on Kac-Moody modules at the critical level.

Sect. 5 reviews the Feigin-Frenkel isomorphism at the critical level and Beilinson-Drinfeld's *birth of ops*, i.e., the local interplay of Hecke symmetries and Feigin-Frenkel duality.

In Sect. 6, we formulate the FLE at critical level. We reduce the statement to the pointwise assertion, which was proved in [FG2]. We also prove some important compatibilities for the FLE here.

As discussed above, we also provide another proof of the pointwise assertion in the present paper, by combining general considerations about derived Satake with Feigin and Frenkel's duality for $\mathcal{W}$ -algebras.

In Sect. 8, we study the interaction between the FLE and natural duality functors between the DG categories appearing in it.

0.3.3. We now turn to Part II, which deals with local-to-global constructions.

Sect. 9 reviews the definition of Whittaker coefficient functors and establishes some conventions related to them.

In Sect. 10, we introduce the Kac-Moody localization functor Loc_G .

In Sect. 11, we discuss axiomatics for local-to-global functors for factorization categories (or even just categories over Ran). The key construction produces *unital* local-to-global functors from *lax unital* such functors, abstracting the construction of chiral homology.

Sect. 12 considers the interaction between Kac-Moody localization and restriction/inflation along group homomorphisms $H \rightarrow G$. This material appeared previously in [CF].

In Sect. 13, we discuss an alternative construction of Kac-Moody localization from the eighth author's thesis. The idea is to realize D-modules on Bun_G as quasi-coherent sheaves equipped with infinitesimal Hecke equivariance structures at every point of Ran .

In Sect. 14, we apply the material from Sect. 12 to calculate Whittaker coefficients of Kac-Moody representations in terms of chiral homology for the critical level $\mathcal{W}$ -algebra.

Sect. 15 constructs a Hecke equivariance structure for Loc_G . This can be considered as an extension of the main construction of [BD1, Sect. 7], allowing more choices of characters for the Feigin-Frenkel center while adding homotopy coherence to *loc. cit.*

Sect. 16 contains the proof of Theorem 15.2.8, a technical point from Sect. 15.

Sect. 17 constructs the functor $\text{Poinc}_{G,*}^{\text{spec}}$ and its relative $\text{Poinc}_{G,!}^{\text{spec}}$.

Finally, Sect. 18 considers the interaction between the Langlands functor $\mathbb{L}_G$ and the constructions of earlier sections. Most importantly, we conclude the proof of Theorem 0.1.2 here. We also prove that $\mathbb{L}_G$ is compatible with the factorizable derived Satake equivalence here, to be used in the sequel to this paper.

0.3.4. This paper relies on a lot of foundational material, a big part (but not all) of which has not been previously written down. This material is developed in the Appendix⁵ to this paper.

In Appendix A we develop the IndCoh theory for algebro-geometric objects that are *not* of finite type. As it turns out, there are two versions, denoted $\mathrm{IndCoh}^!(-)$ and $\mathrm{IndCoh}^*(-)$, respectively, which in good situations are mutually dual. We introduce the property of schemes, called *placidity*, which guarantees that these categories behave particularly well. In addition, we introduce another category, denoted $\mathrm{QCoh}_{\mathrm{co}}(-)$, useful in many situations, and which is a *pre-dual* of $\mathrm{QCoh}_{\mathrm{co}}$.

In Appendix B we discuss the pattern of factorization. We introduce factorization spaces, and construct examples of such (e.g., loops or arcs into a given target, or various spaces attached to the formal disc). We introduce factorization algebras and modules, and various operations between them. One of the central notions in this paper is that of *factorization category*. We show how various categories of algebro-geometric or representation-theoretic nature acquire this structure (notably, $\mathrm{IndCoh}^*(-)$ of monodromy-free opers and the category of Kac-Moody representations).

In Appendix C we discuss the phenomenon of *unitality*. We introduce categorical prestacks, D-modules and sheaves of categories on them. Our main example is the unital Ran space. We introduce unital and counital factorization spaces, and their common generalization, called “unital-in-correspondences” factorization spaces; it is this latter notion that plays the most important role. We introduce unital factorization algebras and categories. We emphasize that some phenomena (such as restriction of module categories) work differently in unital and non-unital settings, and it is the former that are responsible for some of the fundamental constructions in this paper.

In Appendix D we prove one of the general fundamental theorems that describe a category of algebro-geometric nature as modules over a factorization algebra. Namely, we show that (at the level of bounded below categories), the category $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{L}_{\nabla}(\mathcal{Y}))$ (here $\mathcal{Y}$ is an affine D-scheme, and $\mathcal{L}_{\nabla}(\mathcal{Y})$ is the space of its horizontal sections on the punctured disc) identifies with factorization modules over the (commutative) factorization algebra of regular functions on $\mathcal{Y}$. The equivalence at the level of abelian categories is nearly evident. However, at the derived level, it is quite non-trivial, and requires that $\mathcal{Y}$ be of *finite presentation in the D-sense*.

In Appendix E we explain how to make sense of the *spectral spherical category*, i.e., the category of ind-coherent sheaves on the local spectral Hecke stack $\mathrm{Hecke}_G^{\mathrm{spec}, \mathrm{loc}}$, as a factorization category. The problem is that $\mathrm{Hecke}_G^{\mathrm{spec}, \mathrm{loc}}$ does not quite fit into the paradigm of Sect. A, in which we can make sense of $\mathrm{IndCoh}(-)$ by an algorithmic procedure. Yet, we give an algebro-geometric definition of $\mathrm{IndCoh}^*(\mathrm{Hecke}_G^{\mathrm{spec}, \mathrm{loc}})$, and then compare it with a representation-theoretic one of [CR].

In Appendix F we recap (essentially, following [BD2]) the relation between the scheme of horizontal sections of an affine D-scheme $\mathcal{Y}$ and the factorization homology of the factorization algebra of regular functions on $\mathcal{Y}$.

In Appendix G we describe a procedure that attaches factorization module categories over $\mathrm{Rep}(\check{G})$ to module categories over $\mathrm{QCoh}(\mathrm{LS}_G^{\mathrm{mer}})$, and show that this functor is fully faithful on a certain subcategory.

In Appendix H we recast some of the material from Sect. 11 using the notion of the “independent” category, attached to a crystal of categories on the unital Ran space. We then discuss various notions of action of a factorization monoidal category on a DG category.

Appendix I contains some complementary material to Sect. 11: we give an interpretation to the functor of the integrated insertion of the unit in terms of left-lax functors between crystals of categories over the unital Ran space.

Appendix J is homotopy-theoretic. Here we introduce a device that allows us to construct monoidal actions from Sects. 4.6 and 5.3 up to *coherent* homotopy. These monoidal actions play a key role in the definition of the FLE functor.

⁵The Appendix is coauthored by J. Campbell, L. Chen, D. Gaitsgory, K. Lin, S. Raskin and N. Rozenblyum.

0.4. Conventions and notation: generalities.

0.4.1. *The players.* Throughout the paper we work over a fixed algebraically closed field k of characteristic 0. Thus, all algebro-geometric objects are defined over k .

In particular, X is a smooth projective curve over k , G is a reductive group over k , and $\check{G}$ is the Langlands dual of G .

0.4.2. *Categories.* When we say “category”, we mean an ∞ -category. Conventions pertaining to the ∞ -categorical language are borrowed from [GaRo3, Chapter 1, Sect. 1].

0.4.3. Conventions pertaining to DG categories follow those in [GaRo3, Chapter 1, Sect. 10]. Unless explicitly stated otherwise, a DG category $\mathbf{C}$ is assumed cocomplete (i.e., to contain arbitrary direct sums). (An exception would be, e.g., the category of compact objects in a given $\mathbf{C}$, denoted $\mathbf{C}^c$.)

Unless explicitly stated otherwise, given a pair of DG categories $\mathbf{C}_1$ and $\mathbf{C}_2$, by a functor $F : \mathbf{C}_1 \rightarrow \mathbf{C}_2$ we will always understand a *continuous* functor, i.e., one that commutes with arbitrary direct sums (equivalently, colimits).

0.4.4. Given a DG category $\mathbf{C}$ with a t-structure, we will use *cohomological* conventions. I.e., $\mathbf{C}^{\leq 0}$ will denote the subcategory of *connective* objects. We will denote by $\mathbf{C}^\heartsuit$ the heart of the t-structure.

0.4.5. Conventions adopted in this paper regarding higher algebra and derived algebraic geometry follow closely those of [AGKRRV].

0.4.6. *Factorization.* Conventions and notation pertaining to the Ran space and *factorization* are explained in Sect. B.

There are several pieces of notation associated with factorization categories:

Given a factorization category $\mathbf{C}$, we will denote by $\underline{\mathbf{C}}$ the corresponding sheaf of categories over Ran , by $\mathbf{C}_{\text{Ran}}$ its category of global sections, and for $\mathcal{Z} \rightarrow \text{Ran}$ by $\mathbf{C}_{\mathcal{Z}}$ the category of sections of the pullback of $\underline{\mathbf{C}}$ to $\mathcal{Z}$. In particular, for a k -point $\underline{x} \in \text{Ran}$, we will denote by $\mathbf{C}_{\underline{x}}$ the fiber of $\underline{\mathbf{C}}$ at $\underline{x}$.

Given a pair of factorization categories $\mathbf{C}_1$ and $\mathbf{C}_2$ and a functor Φ between, we will distinguish between a property of this functor (such as admitting an adjoint or being an equivalence) taking place at the *pointwise* or *factorization* level.

The former means that the given property holds for the corresponding functor

$$\Phi : \mathbf{C}_{1,\underline{x}} \rightarrow \mathbf{C}_{2,\underline{x}}$$

for any k -point $\underline{x}$ of the Ran space. The latter means that the given property holds for

$$\Phi : \mathbf{C}_{1,\mathcal{Z}} \rightarrow \mathbf{C}_{2,\mathcal{Z}}$$

for any prestack $\mathcal{Z} \rightarrow \text{Ran}$ (equivalently, one can take $\mathcal{Z}$ to be Ran itself).

0.5. **Acknowledgements.** As should be clear from what we said above, the majority of the second part of this paper can be traced back to the ideas of A. Beilinson and V. Drinfeld recorded in [BD1] and [BD2].

The FLE as presented in Part I relies crucially on the Feigin-Frenkel isomorphism, as a passage between G and $\check{G}$, see Sect. 5.

A crucial role in local and local-to-global constructions is played by the concept of *factorization*. Its appearance in representation theory was pioneered by M. Finkelberg, I. Mirković and V. Schechtman, and it was further subsequently elucidated by A. Beilinson and J. Lurie.

Separate thanks are due to J. Lurie for enabling representation theorists to work within Higher Algebra. The mathematics developed in this paper would not be possible if one worked “up to homotopy”.

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Part I: Local Theory

This Part is mainly dedicated to the proof of a key local result: the critical FLE. It says that the Kazhdan-Lusztig category at the critical level (for G) is equivalent to the category of ind-coherent sheaves on the space of monodromy-free opers on the punctured disc (for $\check{G}$).

The FLE involves crossing the Langlands bridge. I.e., at some point, we will need to know something about the relationship between G and $\check{G}$. In fact, there are exactly two sources of such results (as long as we stay at the critical level for G and level ∞ for $\check{G}$): one is the geometric Casselman-Shalika formula (Theorem 1.4.2), and the other is the Feigin-Frenkel isomorphism (Theorem 5.1.2). The compatibility between the two is encapsulated by Theorem 5.2.5. The other results of local Langlands nature, including the FLE, are ultimately deduced from one (or a combination) of these two.

Once the FLE is proved, we will use it in Part II to establish a certain global compatibility of the Langlands functor, which will play a key role in subsequent papers in this series. This property will essentially say that the Langlands functor is compatible with the Beilinson-Drinfeld construction of eigensheaves via Kac-Moody localization and opers.

1. GEOMETRIC SATAKE AND CASSELMAN-SHALIKA FORMULA: RECOLLECTIONS

In this section we will review the constructions of categories of *geometric nature* associated, on the geometric side, to spaces of maps

$$\mathcal{D} \rightarrow G \text{ and } \mathcal{D}^\times \rightarrow G,$$

and (twisted) D-modules on these spaces, and on the spectral side to spaces of maps

$$\mathcal{D}_{\text{dR}} \rightarrow \check{G} \text{ and } \mathcal{D}_{\text{dR}}^\times \rightarrow \check{G}$$

and ind-coherent sheaves on these spaces.

Thus, the main players are:

- The category $\text{Whit}(G)$ of Whittaker D-modules on the affine Grassmannian;
- Its spectral counterpart $\text{QCoh}(\text{LS}_G^{\text{reg}}) \simeq \text{Rep}(\check{G})$;
- The equivalence $\text{Whit}(G) \simeq \text{Rep}(\check{G})$, which we call the geometric Casselman-Shalika formula (Theorem 1.4.2);
- The local spherical category Sph_G ;
- Its spectral counterpart $\text{Sph}_G^{\text{spec}}$;
- The (derived) geometric Satake equivalence $\text{Sat}_G : \text{Sph}_G \simeq \text{Sph}_G^{\text{spec}}$ (Theorem 1.7.2).

When dealing with these objects there is one major trouble and three “annoyances”, all of which will be introduced in this section, and that will plague us throughout the paper:

- (1) The trouble is that the local algebro-geometric objects on the spectral side are *not* of finite type (once we consider their factorization versions), so the $\text{IndCoh}(-)$ categories associated to them need extra work to define;
- (2) This paper is concerned with the *classical* geometric Langlands. However, “classical” for G means the *critical* level. This means that the categories on the geometric side⁶ will consist not of D-modules, but of critically or half-twisted D-modules. As a result, throughout the paper, we will have to watch carefully what happens with these twistings as we move between different spaces.
- (3) Ultimately, on the geometric side, the object we need to consider is not the constant group-scheme on X with fiber G , but rather its twist by the T -torsor $\rho(\omega_X)$. This twist is analogous to the usual ρ -shift in the representation theory of the finite-dimensional G . Thus, all spaces associated with G will undergo the corresponding twist.

⁶We avoid using the word “automorphic” in the local context, as automorphy refers to the global situation.

- (4) Both categories Sph_G and $\mathrm{Sph}_G^{\mathrm{spec}}$ are endowed with anti-involutions, denoted σ and σ^{spec} . A source of constant headache throughout this paper is that these anti-involutions are compatible under Sat_G , *up to the Chevalley involution* on G , denoted τ_G . This can be seen as a vestige (in a rather precise sense) of the fact that the square of the usual Fourier transform is not the identity, but rather is given by the action of -1 .

1.1. The critical twist.

1.1.1. We choose once and for all a square root $\omega_X^{\otimes \frac{1}{2}}$ of the canonical line bundle ω_X on X .

Warning: In this series of papers, ω_X denotes the canonical line bundle on X , and *not* the dualizing sheaf on X , which is the $[1]$ shift of that. (So, properly, we should have used Ω_X^1 , rather than ω_X .) This deviates from the convention, according to which, for a prestack $\mathcal{Y}$ we denote by $\omega_{\mathcal{Y}}$ its dualizing sheaf. So the only exception for this rule is when $\mathcal{Y}$ is the curve X itself.

1.1.2. Consider the affine Grassmannian Gr_G as a factorization space over X , equipped with an action of the (factorization) group indscheme $\mathfrak{L}(G)$.

We refer the reader to Sects. B.1.8 and B.5.5, respectively, where the definition of these objects is recalled, and to Sect. B.1.6, where the general theory of factorization spaces is set up.

1.1.3. Let $\det_{\mathrm{Gr}_G}$ denote the determinant (factorization) line bundle on Gr_G .

Remark 1.1.4. According to [BD1, Sect. 4], the choice of $\omega_X^{\otimes \frac{1}{2}}$ gives rise to a square root of $\det_{\mathrm{Gr}_G}$, as a line bundle over $\mathrm{Gr}_{G, \mathrm{Ran}}$. However, this square root is *incompatible* with factorization.⁷

1.1.5. For a line bundle $\mathcal{L}$ on a space $\mathcal{Y}$ and an integer n , let $\mathcal{L}^{\frac{1}{n}}$ denote the étale μ_n -gerbe of n th roots of $\mathcal{L}$.

Recall now that given a μ_n -gerbe $\mathcal{G}$ on a space $\mathcal{Y}$, we can consider the $\mathcal{G}$ -twisted category of D-modules on $\mathcal{Y}$, to be denoted

$$\mathrm{D}\text{-mod}_{\mathcal{G}}(\mathcal{Y}).$$

Thus, for $(\mathcal{Y}, \mathcal{L}, n)$ as above we can consider the corresponding category

$$\mathrm{D}\text{-mod}_{\mathcal{L}^{\frac{1}{n}}}(\mathcal{Y}).$$

1.1.6. Consider the μ_2 -gerbe $\det_{\mathrm{Gr}_G}^{\frac{1}{2}}$.

We will use the short-hand notation

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)$$

to denote the (factorization) category

$$\mathrm{D}\text{-mod}_{\det_{\mathrm{Gr}_G}^{\frac{1}{2}}}(\mathrm{Gr}_G)$$

of $\det_{\mathrm{Gr}_G}^{\frac{1}{2}}$ -twisted D-modules on Gr_G .

Remark 1.1.7. According to Remark 1.1.4, a choice of $\omega_X^{\otimes \frac{1}{2}}$ gives rise to a trivialization of the gerbe $\det_{\mathrm{Gr}_G}^{\frac{1}{2}}$. However, this trivialization is incompatible with factorization.

For that reason, henceforth, we will avoid using it.

⁷More precisely, this square root exists as a factorization $\mathbb{Z}/2\mathbb{Z}$ -graded line bundle, where the grading over the connected component Gr_G^λ of Gr_G (here $\lambda \in \Lambda_{G, G} = \pi_0(\mathrm{Gr}_G)$) equals $\langle \lambda, 2\check{\rho} \rangle \bmod 2$.

1.1.8. Recall that for a space $\mathcal{Y}$, we can consider *de Rham* twistings on $\mathcal{Y}$ (see, [GaRo2, Sect. 6]). These are by definition $\mathcal{O}^\times$ -gerbes on $\mathcal{Y}_{\text{dR}}$, equipped with a trivialization of their pullback to $\mathcal{Y}$.

Given a de Rham twisting $\mathcal{T}$, we can consider the corresponding twisted category of D-modules

$$\text{D-mod}_{\mathcal{T}}(\mathcal{Y}),$$

see [GaRo2, Sect. 7].

Recall also that to a line bundle $\mathcal{L}$ on $\mathcal{Y}$, we can associate a de Rham twisting, which in this paper we denote by $\text{dlog}(\mathcal{L})$ (the corresponding $\mathcal{O}^\times$ -gerbe on $\mathcal{Y}_{\text{dR}}$ is trivial, but the trivialization of its pullback to $\mathcal{Y}$ differs from the tautological one by tensoring with $\mathcal{L}$).

Note that tensoring by $\mathcal{L}$ defines an equivalence

$$(1.1) \quad \text{D-mod}(\mathcal{Y}) \rightarrow \text{D-mod}_{\text{dlog}(\mathcal{L})}(\mathcal{Y}).$$

Finally, recall (see [GaRo2, Corollary 6.4.5]) that the space of de Rham twistings on a given space $\mathcal{Y}$ carries a natural k -linear structure. Thus, for $c \in k$, we have a well-defined twisting $c \cdot \text{dlog}(\mathcal{L})$, and the corresponding category

$$\text{D-mod}_{c \cdot \text{dlog}(\mathcal{L})}(\mathcal{Y}).$$

1.1.9. Let $(\mathcal{Y}, \mathcal{L}, n)$ be as above. Note that for $c = n \in \mathbb{Z} \subset k$, we have

$$n \cdot \text{dlog}(\mathcal{L}) = \text{dlog}(\mathcal{L}^{\otimes n}).$$

In particular, we have a canonical identification of the corresponding twisted categories of D-modules:

$$(1.2) \quad \text{D-mod}_{\frac{1}{n}}(\mathcal{Y}) \xrightarrow{\sim} \text{D-mod}_{\frac{1}{n} \cdot \text{dlog}(\mathcal{L})}(\mathcal{Y}).$$

For example, when $n = 1$, the identification (1.2) is the identification of (1.1).

1.1.10. We will use the short-hand notation

$$\text{D-mod}_{\text{crit}}(\text{Gr}_G)$$

for the (factorization) category

$$\text{D-mod}_{\frac{1}{2} \cdot \text{dlog}(\det_{\text{Gr}_G})}(\text{Gr}_G).$$

1.1.11. Applying (1.2) to $\mathcal{Y} = \text{Gr}_G$ and $\mathcal{L} = \det_{\text{Gr}_G}$, we obtain a canonical equivalence of (factorization) categories

$$\text{D-mod}_{\frac{1}{2}}(\text{Gr}_G) \simeq \text{D-mod}_{\text{crit}}(\text{Gr}_G).$$

Remark 1.1.12. According to Remark 1.1.4, we can also identify

$$\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \text{Ran}}) \simeq \text{D-mod}(\text{Gr}_{G, \text{Ran}}),$$

or equivalently

$$\text{D-mod}_{\text{crit}}(\text{Gr}_{G, \text{Ran}}) \simeq \text{D-mod}(\text{Gr}_{G, \text{Ran}}),$$

as plain categories, but these identifications are incompatible with the factorization structures.

Remark 1.1.13. We distinguish $\text{D-mod}_{\text{crit}}(\text{Gr}_G)$ and $\text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)$ notationally for the following two reasons:

- (1) The étale gerbe-twisted version makes sense not just in the context of D-modules, but also in other sheaf-theoretic contexts (e.g., Betti, ℓ -adic).
- (2) The category $\text{D-mod}_{\text{crit}}(\text{Gr}_G)$ comes equipped with a natural forgetful functor to $\text{IndCoh}(\text{Gr}_G)$, while for a general étale gerbe, the gerbe-twisted category of D-modules does not carry such a functor.

Thus, the distinction between gerbes and twistings becomes relevant when discussing connections between D-modules and modules over Lie algebras, as we often do in this paper. We use the $\text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)$ (or $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$) to evoke the sheaf-theoretic geometry of these spaces, while $\text{D-mod}_{\text{crit}}(\text{Gr}_G)$ (or $\text{D-mod}_{\text{crit}}(\text{Bun}_G)$) evokes the connection to Kac-Moody representation theory at the critical level.

1.1.14. We can also consider the corresponding *multiplicative* factorization μ_2 -gerbe on $\mathfrak{L}(G)$, equipped with a multiplicative trivialization of its restriction to $\mathfrak{L}^+(G)$.

Since the group indscheme $\mathfrak{L}(N)$ is contractible, the restriction of the above gerbe to it also admits a canonical multiplicative trivialization.

In particular, if H is a factorization subgroup of either $\mathfrak{L}^+(G)$ or $\mathfrak{L}(N)$, it makes sense to consider the (factorization) category

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^H$$

of H -equivariant D-modules.

1.2. A geometric twisting construction.

1.2.1. Let H be a group mapping to G , and let $\mathcal{P}_H$ be an H -torsor over X . Taking sections over the formal disc, $\mathcal{P}_H$ gives rise to a factorization torsor over $\mathfrak{L}^+(H)$; by a slight abuse of notation, we will denote this $\mathfrak{L}^+(H)$ -factorization torsor by the same symbol $\mathcal{P}_H$.

Given a space $\mathcal{Y}$ over X , equipped with an action of $\mathfrak{L}^+(H)$, we can form a twist, to be denoted $\mathcal{Y}_{\mathcal{P}_H}$, i.e.,

$$\mathcal{Y}_{\mathcal{P}_H} := (\mathcal{P}_H \times \mathcal{Y}) / \mathfrak{L}^+(H).$$

If $\mathcal{Y}$ was endowed with a factorization structure compatible with the $\mathfrak{L}^+(H)$ -action, then so is $\mathcal{Y}_{\mathcal{P}_H}$.

1.2.2. The space $\mathcal{Y}_{\mathcal{P}_H}$ is acted on by the adjoint twist $\mathfrak{L}^+(H)_{\mathcal{P}_H}$ of $\mathfrak{L}^+(H)$.

Note that we have a canonical isomorphism

$$(1.3) \quad \mathcal{Y} / \mathfrak{L}^+(H) \simeq \mathcal{Y}_{\mathcal{P}_H} / \mathfrak{L}^+(H)_{\mathcal{P}_H}.$$

1.2.3. We will denote by the subscript $\mathcal{P}_H$ the various categories of D-modules associated with the above geometric objects, such as

$$\mathrm{D}\text{-mod}(\mathcal{Y}) \rightsquigarrow \mathrm{D}\text{-mod}(\mathcal{Y})_{\mathcal{P}_H} \text{ and } \mathrm{D}\text{-mod}(\mathcal{Y})^{\mathfrak{L}^+(H)} \rightsquigarrow (\mathrm{D}\text{-mod}(\mathcal{Y})^{\mathfrak{L}^+(H)})_{\mathcal{P}_H}.$$

Note, however, that thanks to the identification (1.3), the category $(\mathrm{D}\text{-mod}(\mathcal{Y})^{\mathfrak{L}^+(H)})_{\mathcal{P}_H}$ is canonically equivalent to the original category $\mathrm{D}\text{-mod}(\mathcal{Y})^{\mathfrak{L}^+(H)}$. We will denote this equivalence by

$$\alpha_{\mathcal{P}_H, \text{taut}} : \mathrm{D}\text{-mod}(\mathcal{Y})^{\mathfrak{L}^+(H)} \xrightarrow{\sim} (\mathrm{D}\text{-mod}(\mathcal{Y})^{\mathfrak{L}^+(H)})_{\mathcal{P}_H}.$$

1.2.4. A typical example of the above situation that we will consider is when $H = T$, and the T -bundle is $\rho(\omega_X)$, i.e., the bundle induced from $\omega_X^{\otimes \frac{1}{2}}$ by means of

$$2\rho : \mathbb{G}_m \rightarrow T.$$

1.3. The Whittaker category on the affine Grassmannian.

1.3.1. We apply the construction of Sect. 1.2.4 to $\mathcal{Y} := \mathrm{Gr}_G$, viewed as a scheme acted on by $\mathfrak{L}^+(T) \subset \mathfrak{L}^+(G)$, and the group indscheme $\mathfrak{L}(N)$.

Thus, we can form the (factorization) space $\mathrm{Gr}_{G, \rho(\omega_X)}$, which is acted on by $\mathfrak{L}(G)_{\rho(\omega_X)}$, and in particular $\mathfrak{L}(N)_{\rho(\omega_X)}$.

1.3.2. The group indscheme $\mathfrak{L}(N)_{\rho(\omega_X)}$ is equipped with a homomorphism

$$(1.4) \quad \chi : \mathfrak{L}(N)_{\rho(\omega_X)} \rightarrow \mathbb{G}_a,$$

equal to the composition

$$\mathfrak{L}(N)_{\rho(\omega_X)} \rightarrow \mathfrak{L}(N/[N, N])_{\rho(\omega_X)} \simeq \prod_I \mathfrak{L}(\mathbb{G}_a)_{\omega_X} \xrightarrow{\text{Res}} \prod_I \mathbb{G}_a \xrightarrow{\chi_0} \mathbb{G}_a,$$

where:

- I is the set of vertices of the Dynkin diagram of G ;
- $\mathfrak{L}(\mathbb{G}_a)_{\omega_X}$ is the twist formed with respect to the $\mathfrak{L}^+(\mathbb{G}_m)$ -action on $\mathfrak{L}(\mathbb{G}_a)$;
- $\text{Res} : \mathfrak{L}(\mathbb{G}_a)_{\omega_X} \rightarrow \mathbb{G}_a$ is the canonical residue map;
- χ_0 is a non-degenerate character (i.e., a character non-trivial along each factor).

1.3.3. Let $\mathbf{C}$ be a category acted on by $\mathfrak{L}(G)_{\rho(\omega_X)}$ at the critical level.⁸ Denote:

$$\text{Whit}^!(\mathbf{C}) := \mathbf{C}^{\mathfrak{L}(N)_{\rho(\omega_X)}, \chi} \text{ and } \text{Whit}_*(\mathbf{C}) := \mathbf{C}_{\mathfrak{L}(N)_{\rho(\omega_X)}, \chi},$$

where we impose equivariance against the pullback of

$$\exp \in \text{D-mod}(\mathbb{G}_a)$$

by means of χ (see [Ra2] for more details). Our normalization for $\exp$ is that it is a character sheaf in the $*$ -sense, i.e.,

$$\text{add}^*(\exp) \simeq \exp \boxtimes \exp.$$

Note that

$$(1.5) \quad (\text{Whit}_*(\mathbf{C}))^\vee \simeq \text{Whit}^!(\mathbf{C}^\vee),$$

up to replacing χ_0 by its inverse, where⁹

$$(-)^\vee := \text{Funct}((-), \text{Vect}).$$

1.3.4. Although the assignments

$$\mathbf{C} \rightsquigarrow \text{Whit}^!(\mathbf{C}) \text{ and } \mathbf{C} \rightsquigarrow \text{Whit}_*(\mathbf{C})$$

involve the group *ind*-scheme $\mathfrak{L}(N)_{\rho(\omega_X)}$, they behave nicely on the 2-category of $\mathfrak{L}(G)_{\rho(\omega_X)}$ -module categories (see [Ra2]).

Namely, they both commute with limits and colimits. Combined with (1.5), this implies that if $\mathbf{C}$ is dualizable, then so are $\text{Whit}^!(\mathbf{C})$ and $\text{Whit}_*(\mathbf{C})$.

However, more is true.

1.3.5. Let $\omega_{\mathfrak{L}(N)_{\rho(\omega_X)}}^{\text{ren}} \in \text{D-mod}(\mathfrak{L}(N)_{\rho(\omega_X)})$ be the *renormalized* dualizing sheaf on $\mathfrak{L}(N)_{\rho(\omega_X)}$, defined to be the $*$ -pullback of the dualizing sheaf along the projection

$$\mathfrak{L}(N)_{\rho(\omega_X)} \rightarrow \mathfrak{L}(N)_{\rho(\omega_X)} / \mathfrak{L}^+(N)_{\rho(\omega_X)}.$$

Consider the object

$$\omega_{\mathfrak{L}(N)_{\rho(\omega_X)}, \chi}^{\text{ren}} := \omega_{\mathfrak{L}(N)_{\rho(\omega_X)}}^{\text{ren}} \overset{*}{\otimes} \chi^*(\exp) \in \text{D-mod}(\mathfrak{L}(N)_{\rho(\omega_X)}).$$

⁸The discussion here is applicable both when we work over a fixed point $\underline{x} \in \text{Ran}$ and in the factorization setting.

⁹In the next formula $\text{Funct}(-, -)$ stands for colimit-preserving functors. We will always use this convention when talking about functors between cocomplete categories, unless explicitly specified otherwise.

1.3.6. Let $\mathbf{C}$ be as above. The operation of $*$ -convolution with $\omega_{\Sigma(N)\rho(\omega_X),\chi}^{\text{ren}}$ is an endofunctor of $\mathbf{C}$ (as a plain DG category), and this endofunctor factors as

$$\mathbf{C} \rightarrow \text{Whit}_*(\mathbf{C}) \rightarrow \text{Whit}^!(\mathbf{C}) \hookrightarrow \mathbf{C}.$$

Denote the resulting functor $\text{Whit}_*(\mathbf{C}) \rightarrow \text{Whit}^!(\mathbf{C})$ by

$$\Theta_{\text{Whit}(\mathbf{C})} : \text{Whit}_*(\mathbf{C}) \rightarrow \text{Whit}^!(\mathbf{C}).$$

The following fundamental result was established in [Ra2]:

Theorem 1.3.7. *The functor $\Theta_{\text{Whit}(\mathbf{C})}$ is an equivalence.*

Remark 1.3.8. The proof of Theorem 1.3.13, as recorded in [Ra2], is given for a fixed formal disc, but the same argument applies to prove a version of this theorem over Ran .

1.3.9. We apply the above discussion to

$$\mathbf{C} := \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X)}).$$

Thus we obtain the (factorization) categories

$$\text{Whit}^!(\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X)})) \text{ and } \text{Whit}_*(\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X)})).$$

We will use for them short-hand notations

$$\text{Whit}^!(G) \text{ and } \text{Whit}_*(G),$$

respectively.

Remark 1.3.10. The categories $\text{Whit}^!(G)$ and $\text{Whit}_*(G)$ are canonically independent of the choice of χ_0 :

Indeed, given two non-degenerate characters χ_0^1 and χ_0^2 , there exists an element $t \in T$ that conjugates χ_0^1 to χ_0^2 . Translation by t on $\text{Gr}_{G,\rho(\omega_X)}$ defines then an equivalence between the corresponding Whittaker categories.

The choice of t is unique up to an element $z \in Z_G$. However, the translation action of z on $\text{Gr}_{G,\rho(\omega_X)}$ is trivial.

1.3.11. By (1.5), the categories $\text{Whit}^!(G)$ and $\text{Whit}_*(G)$ are naturally mutually dual, up to replacing χ_0 by its inverse. Note, however, that due to Remark 1.3.10, they are actually mutually dual.

Furthermore, as is shown in [Ga6], both $\text{Whit}^!(G)$ and $\text{Whit}_*(G)$ are compactly generated (see Sect. B.11.9 for what compact generation means in the factorization setting).

1.3.12. Let

$$\Theta_{\text{Whit}(G)} : \text{Whit}_*(G) \rightarrow \text{Whit}^!(G)$$

denote the functor from Sect. 1.3.6.

As a particular case of Theorem 1.3.7, we obtain:

Theorem 1.3.13. *The functor $\Theta_{\text{Whit}(G)}$ is an equivalence (of factorization categories).*

1.3.14. The factorization categories $\text{Whit}^!(G)$ and $\text{Whit}_*(G)$ are unital (see Sect. C.11.1) for what this means. Here is the explicit description of their factorization units:

The factorization unit $\mathbf{1}_{\text{Whit}_*(G)} \in \text{Whit}_*(G)$ is the object, denoted $\text{Vac}_{\text{Whit}_*(G)}$, equal to the projection along

$$\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X)}) \rightarrow \text{Whit}_*(G)$$

of $\delta_{\text{Gr}_{G,\rho(\omega_X)}} \in \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X)})$, the latter being the factorization unit $\mathbf{1}_{\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X)})}$ for $\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X)})$ itself.

1.3.15. The factorization unit $\mathbf{1}_{\text{Whit}^!(G)} \in \text{Whit}^!(G)$ is the object, denoted $\text{Vac}_{\text{Whit}^!(G)}$, equal to the $*$ -direct image along the locally-closed embedding

$$\mathfrak{L}(N)_{\rho(\omega_X)}/\mathfrak{L}^+(N)_{\rho(\omega_X)} \hookrightarrow \text{Gr}_{G,\rho(\omega_X)}$$

of

$$\omega_{\mathfrak{L}(N)_{\rho(\omega_X)}/\mathfrak{L}^+(N)_{\rho(\omega_X)}} \otimes \chi^*(\exp) \in \text{D-mod}(\mathfrak{L}(N)_{\rho(\omega_X)}/\mathfrak{L}^+(N)_{\rho(\omega_X)})^{\mathfrak{L}(N)_{\rho(\omega_X)}, \chi}.$$

Note that the above $*$ -extension is *clean*, i.e., receives an isomorphism from the $!$ -extension.

This implies that the functor co-represented by $\text{Vac}_{\text{Whit}^!(G)}$ identifies with the functor of $!$ -fiber at the unit point $1_{\text{Gr}_{G,\rho(\omega_X)}} \in \text{Gr}_{G,\rho(\omega_X)}$, restricted to

$$\text{Whit}^!(G) \subset \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X)}).$$

1.4. The geometric Casselman-Shalika formula.

1.4.1. The following is the statement of the geometric Casselman-Shalika formula (see [Ra3, Theorem 6.36.1]¹⁰):

Theorem 1.4.2. *There exists a canonically defined equivalence of factorization categories:*

$$\text{CS}_G : \text{Whit}^!(G) \rightarrow \text{Rep}(\check{G}).$$

Remark 1.4.3. In the course of the proof of Theorem 1.4.2 one uses the *naïve* (i.e., non-derived) geometric Satake to construct a functor

$$\text{Rep}(\check{G}) \rightarrow \text{Whit}^!(G),$$

and one shows that it is an equivalence, see Remark 1.7.8.

1.4.4. The functor CS_G is normalized so that it sends the standard object

$$\Delta^\lambda \in \text{Whit}^!(G), \quad \lambda \in \Lambda_G^+,$$

corresponding to the $\mathfrak{L}(N)_{\rho(\omega_X)}$ -orbit

$$S^\lambda := \mathfrak{L}(N)_{\rho(\omega_X)} \cdot t^\lambda$$

to the highest weight module

$$V^\lambda \in \text{Rep}(\check{G}).$$

(In the above formula, t denotes the uniformizer on $\mathcal{D}$.)

Remark 1.4.5. By fixing the above normalization for CS_G we made a choice. We could have made a different choice by applying the Chevalley involution τ_G on G , or equivalently, on $\check{G}$.

The normalization for CS_G ultimately forces how we normalize the Langlands functor $\mathbb{L}_G$. Our particular choice for CS_G is dictated by the following: we want the Langlands functor $\mathbb{L}_G$ for G and its Levi subgroups to be compatible via the Eisenstein and Constant Term functors.

If we composed $\mathbb{L}_G$ with τ_G , the resulting functor would intertwine the Eisenstein/Constant Term functors on the geometric side with similar functors on the spectral side, but with respect to the *opposite parabolic*.

Note also that when G is a torus T , in our normalization, the Langlands functor $\mathbb{L}_T$ is as in [GLC1, Sect. 1.5.2].

Remark 1.4.6. For the validity of Theorem 1.4.2 at the factorization level, it is crucial that in the definition of $\text{Whit}^!(G)$ we use the twisted category $\text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)$, rather than the untwisted one, i.e., $\text{D-mod}(\text{Gr}_G)$.

¹⁰The original result in this direction is the main theorem of [FGV].

1.4.7. The following is a basic pattern of how the equivalence CS_G interacts with duality.

Let us denote by

$$\text{FLE}_{\check{G}, \infty} : \text{Rep}(\check{G}) \rightarrow \text{Whit}_*(G)$$

the functor equal to $\text{CS}_G^\vee$, with respect to the canonical dualities:

$$\text{Whit}_*(G) = (\text{Whit}^!(G))^\vee \text{ and } \text{Rep}(\check{G})^\vee \simeq \text{Rep}(\check{G}).$$

Remark 1.4.8. The notation $\text{FLE}_{\check{G}, \infty}$ stems from the fact that the above functor is indeed the limiting value of the (positive level) FLE equivalence.

1.4.9. *Example.* Note, in particular, that the functor $\text{FLE}_{\check{G}, \infty}$ sends

$$V^{-w_0(\lambda)} \in \text{Rep}(\check{G}) \mapsto \nabla^\lambda \in \text{Whit}_*(G),$$

where for $\mu \in \Lambda^+$ we denote by

$$\nabla^\mu \in \text{Whit}_*(G)$$

the object dual to $\Delta^\mu \in \text{Whit}^!(G)$, i.e.,

$$\langle \mathcal{F}, \nabla^\mu \rangle = \mathcal{H}om_{\text{Whit}^!(G)}(\Delta^\mu, \mathcal{F}), \quad \mathcal{F} \in \text{Whit}^!(G),$$

where

$$\langle -, - \rangle : \text{Whit}^!(G) \otimes \text{Whit}_*(G) \rightarrow \text{Vect}$$

is the canonical pairing.

1.4.10. Note that the Whittaker category is canonically attached to the pair (G, B) . Hence, the group of *outer automorphisms* of G (i.e., the group of automorphisms of the polarized¹¹ root datum of G) acts on both versions of the Whittaker category.

Let τ_G be the Chevalley involution, viewed as an outer automorphism of G . The corresponding automorphism of the polarized root datum acts as $\lambda \mapsto -w_0(\lambda)$.

1.4.11. We have:

Lemma 1.4.12. *The composition*

$$\text{Rep}(\check{G}) \xrightarrow{\text{FLE}_{\check{G}, \infty}} \text{Whit}_*(G) \xrightarrow{\Theta_{\text{Whit}(G)}} \text{Whit}^!(G)$$

identifies canonically with

$$\tau_G \circ (\text{CS}_G)^{-1}.$$

Remark 1.4.13. As a reality check, note that both functors in Lemma 1.4.12 send

$$V^{-w_0(\lambda)} \in \text{Rep}(\check{G}) \mapsto \Delta^\lambda \in \text{Whit}^!(G).$$

Indeed, the functor $\Theta_{\text{Whit}(G)}$ is easily seen to send Δ^λ to ∇^λ .

The proof of Lemma 1.4.12 follows easily from the construction of CS_G via naive geometric Satake.

1.5. The spherical category.

1.5.1. We denote by $\text{Sph}_G^{\text{non-ren}}$ the (factorization) monoidal category

$$\text{D-mod}_{\frac{1}{2}}(\mathfrak{L}^+(G) \backslash \mathfrak{L}(G) / \mathfrak{L}^+(G)).$$

We have a naturally defined right action of $\text{Sph}_G^{\text{non-ren}}$ on $\text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)$, compatible with the left action of $\mathfrak{L}(G)$.

¹¹By a polarization of a root datum we mean a choice of the subset of positive roots.

1.5.2. We let Sph_G denote its *renormalized* version, which is defined as the ind-completion of the full subcategory in $\mathrm{Sph}_G^{\mathrm{non-ren}}$ consisting of objects whose image under (either of) the forgetful functors

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathfrak{L}^+(G) \backslash \mathfrak{L}(G)) \leftarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathfrak{L}^+(G) \backslash \mathfrak{L}(G) / \mathfrak{L}^+(G)) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathfrak{L}(G) / \mathfrak{L}^+(G))$$

is compact (see [CR, Proposition 6.3.2] for more details).

By construction, the monoidal (and also factorization) unit

$$\mathbf{1}_{\mathrm{Sph}_G} \simeq \delta_{1_{\mathrm{Gr}_G}} \in \mathrm{Sph}_G$$

is compact.

1.5.3. We have an adjoint pair of functors

$$\mathrm{ren} : \mathrm{Sph}_G^{\mathrm{non-ren}} \rightleftarrows \mathrm{Sph}_G : \mathrm{non-ren},$$

with ren being fully faithful and $\mathrm{non-ren}$ monoidal. This makes $\mathrm{Sph}_G^{\mathrm{non-ren}}$ into a monoidal colocalization of Sph_G .

In particular, we have a right action of Sph_G on $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)$, compatible with the left action of $\mathfrak{L}(G)$ and factorization.

1.5.4. Inversion on the group $\mathfrak{L}(G)$ defines an anti-involution, denoted σ , of Sph_G . We will refer to it as the “flip” anti-involution.

Henceforth, we will use σ to pass between left and right module categories over Sph_G . In light of this, we will not necessarily distinguish between left and right actions of Sph_G .

1.5.5. The fact that Gr_G is ind-proper implies that the composition of the involution σ with Verdier duality (on compact objects) defines an equivalence

$$(1.6) \quad \mathrm{Sph}_G^\vee \simeq \mathrm{Sph}_G,$$

which identifies both with right and left monoidal dualization.

Combined with the fact that the unit in Sph_G is compact, we obtain that Sph_G is *rigid* as a monoidal category.¹²

1.5.6. Recall the setting of Sect. 1.2. For any G -bundle $\mathcal{P}_G$ on X , we can form the twisted version

$$\mathrm{Sph}_{G, \mathcal{P}_G}$$

of $\mathcal{P}_G$.

We have a naturally defined action of $\mathrm{Sph}_{G, \mathcal{P}_G}$ on $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \mathcal{P}_G})$, compatible with the left action of $\mathfrak{L}(G)_{\mathcal{P}_G}$ and factorization.

1.5.7. In particular, we have a natural action of $\mathrm{Sph}_{G, \rho(\omega_X)}$ on $\mathrm{Whit}^!(G)$ and $\mathrm{Whit}_*(G)$.

These actions are compatible both with the duality

$$(1.7) \quad (\mathrm{Whit}^!(G))^\vee \simeq \mathrm{Whit}_*(G)$$

(see Sect. 1.5.4) and the functor $\Theta_{\mathrm{Whit}(G)}$.

1.5.8. Note, however, that according to Sect. 1.2.3, we can identify¹³

$$\mathrm{Sph}_G \xrightarrow{\alpha_{\rho(\omega_X), \mathrm{taut}}} \mathrm{Sph}_{G, \rho(\omega_X)},$$

and thus we can regard $\mathrm{Whit}^!(G)$ and $\mathrm{Whit}_*(G)$ as acted on by Sph_G itself.

1.6. The spectral spherical category.

¹²Being a monoidal colocalization of a rigid category, $\mathrm{Sph}_G^{\mathrm{non-ren}}$ is semi-rigid (see [AGKRRV, Appendix C]).

¹³In the formula below we consider $\mathfrak{L}(G)$ as acted on by $\mathfrak{L}^+(G) \times \mathfrak{L}^+(G)$.

1.6.1. Consider the *local spectral Hecke stack*

$$\mathrm{Hecke}_G^{\mathrm{spec}, \mathrm{loc}} := \mathrm{LS}_G^{\mathrm{reg}} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{LS}_G^{\mathrm{reg}}.$$

as a factorization space.

In the above formula $\mathrm{LS}_G^{\mathrm{reg}}$ (resp., $\mathrm{LS}_G^{\mathrm{mer}}$) is the factorization space that attaches to $\underline{x} \in \mathrm{Ran}$ the stack $\mathrm{LS}_{\check{G}}(\mathcal{D}_{\underline{x}})$ (resp., $\mathrm{LS}_{\check{G}}(\mathcal{D}_{\underline{x}}^\times)$) of $\check{G}$ -local systems on the formal multi-disc $\mathcal{D}_{\underline{x}}$ (resp., the punctured multi-disc $\mathcal{D}_{\underline{x}}^\times := \mathcal{D}_{\underline{x}} - \underline{x}$), see Sect. B.7.1.

1.6.2. The fiber $\mathrm{Hecke}_{G, \underline{x}}^{\mathrm{spec}, \mathrm{loc}}$ of $\mathrm{Hecke}_G^{\mathrm{spec}, \mathrm{loc}}$ over a given point $\underline{x} \in \mathrm{Ran}$ is the stack

$$(1.8) \quad \mathrm{Hecke}_{G, \underline{x}}^{\mathrm{spec}, \mathrm{loc}} := \mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{reg}}.$$

The stack (1.8) is locally of finite type. In fact, it is isomorphic of the product of copies of

$$(1.9) \quad \mathrm{pt} / \check{G} \times_{\mathfrak{g}_0^\wedge / \mathrm{Ad}(\check{G})} \mathrm{pt} / \check{G}$$

for each distinct point that comprises $\underline{x}$.

1.6.3. Hence, it makes sense to consider the category

$$\mathrm{Sph}_{G, \underline{x}}^{\mathrm{spec}} := \mathrm{IndCoh}(\mathrm{Hecke}_{G, \underline{x}}^{\mathrm{spec}, \mathrm{loc}}).$$

We endow $\mathrm{Sph}_{G, \underline{x}}^{\mathrm{spec}}$ with a monoidal structure via $*$ -pull and $*$ -push along the standard convolution diagram.

1.6.4. As we let $\underline{x}$ move along Ran (or X^n for a fixed integer n), the resulting prestack is no longer locally almost of finite type, so the category of ind-coherent sheaves on it is not a priori-defined.

In fact, $\mathrm{Hecke}_G^{\mathrm{spec}, \mathrm{loc}}$ violates the condition of being (locally almost) of finite type so badly, that we do not really know how to define the corresponding category $\mathrm{IndCoh}^*(\mathrm{Hecke}_G^{\mathrm{spec}, \mathrm{loc}})$ algorithmically.

We refer the reader to Sect. E, where the definition is given (and is compared to another working definition, adopted in [CR]).

Accordingly, the proofs of all the statements that involve $\mathrm{Sph}_G^{\mathrm{spec}}$ are also delegated to Sect. E. In the main body of the text, we will supply prototypes of the corresponding proofs for the pointwise version $\mathrm{Sph}_{G, \underline{x}}^{\mathrm{spec}}$.

1.6.5. The pointwise version of the spectral Hecke category $\mathrm{Sph}_G^{\mathrm{spec}}$, i.e., $\mathrm{Sph}_{G, \underline{x}}^{\mathrm{spec}}$, is equipped with a tautological action on $\mathrm{QCoh}(\mathrm{LS}_{G, \underline{x}}^{\mathrm{reg}})$.

This construction persists in the factorization setting, i.e., we have an action of the monoidal factorization category $\mathrm{Sph}_G^{\mathrm{spec}}$ on

$$(1.10) \quad \mathrm{Rep}(\check{G}) \simeq \mathrm{QCoh}(\mathrm{LS}_G^{\mathrm{reg}}),$$

viewed as a plain¹⁴ factorization category (see Sect. B.13.5 where the equivalence (1.10) is established).

1.6.6. By construction, the category $\mathrm{Sph}_G^{\mathrm{spec}}$ receives a monoidal functor, denoted

$$\mathrm{nv} : \mathrm{Rep}(\check{G}) \rightarrow \mathrm{Sph}_G^{\mathrm{spec}},$$

to be thought of¹⁵ as the direct image functor along

$$\mathrm{LS}_G^{\mathrm{reg}} \rightarrow \mathrm{Hecke}_G^{\mathrm{spec}, \mathrm{loc}},$$

where we now view the categories appearing in (1.10) as (symmetric) *monoidal* factorization categories.

¹⁴As opposed to a (symmetric) monoidal factorization category.

¹⁵And literally so over a fixed point of Ran .

1.6.7. The flip of two factors defines an anti-involution on $\mathrm{Sph}_G^{\mathrm{spec}}$ to be denoted σ^{spec} .

We will use σ^{spec} to pass between left and right $\mathrm{Sph}_G^{\mathrm{spec}}$ -module categories.

Note that we have a commutative diagram

$$(1.11) \quad \begin{array}{ccc} \mathrm{Rep}(\check{G}) & \xrightarrow{\mathrm{nv}} & \mathrm{Sph}_G^{\mathrm{spec}} \\ \mathrm{Id} \downarrow & & \downarrow \sigma^{\mathrm{spec}} \\ \mathrm{Rep}(\check{G}) & \xrightarrow{\mathrm{nv}} & \mathrm{Sph}_G^{\mathrm{spec}}, \end{array}$$

where Id makes sense as an anti-involution of $\mathrm{Rep}(\check{G})$, since this category is symmetric monoidal.

1.7. Geometric Satake equivalence.

1.7.1. The following is the statement of the factorization version of the *derived* geometric Satake equivalence (see [CR]):

Theorem 1.7.2. *There exists a unique equivalence of monoidal factorization categories*

$$\mathrm{Sat}_G : \mathrm{Sph}_G \rightarrow \mathrm{Sph}_G^{\mathrm{spec}},$$

compatible with the actions of Sph_G on $\mathrm{Whit}(G)$ and $\mathrm{Sph}_G^{\mathrm{spec}}$ on $\mathrm{Rep}(\check{G})$ via the equivalence

$$\mathrm{CS}_G : \mathrm{Whit}^!(G) \simeq \mathrm{Rep}(\check{G}).$$

The construction of the functor Sat_G will be recalled in Sect. E.10.

Remark 1.7.3. In this series of papers we will refer to Sat_G just as “geometric Satake equivalence”, omitting the word “derived”. What is more commonly referred to as “geometric Satake” is not an equivalence, but a functor in one direction, which we will refer to as “naive Satake” and denote by $\mathrm{Sat}_G^{\mathrm{nv}}$, see Sect. 1.7.6.

1.7.4. *Example.* Unwinding the construction, we obtain that Sat_G sends the object in Sph_G corresponding to the double coset of the point t^λ (for $\lambda \in \Lambda^+$) to the object

$$\mathrm{nv}(V^{-w_0(\lambda)}) \in \mathrm{Sph}_G^{\mathrm{spec}}.$$

The above object in Sph_G is what is usually denoted by

$$\mathrm{IC}_{\overline{\mathrm{Gr}}^\lambda},$$

the intersection cohomology sheaf on the closure of the $\mathcal{L}^+(G)$ -orbit Gr^λ of t^λ .

Remark 1.7.5. As in the case of Theorem 1.4.2, for the validity of Theorem 1.7.2 at the factorization level, it is crucial that we work with the twisted category

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathcal{L}^+(G) \backslash \mathcal{L}(G) / \mathcal{L}^+(G))$$

rather than with $\mathrm{D}\text{-mod}(\mathcal{L}^+(G) \backslash \mathcal{L}(G) / \mathcal{L}^+(G))$.

1.7.6. In what follows, we will denote by $\mathrm{Sat}_G^{\mathrm{nv}}$ the functor

$$\mathrm{Rep}(\check{G}) \xrightarrow{\mathrm{nv}} \mathrm{Sph}_G^{\mathrm{spec}} \xrightarrow{\mathrm{Sat}^{-1}} \mathrm{Sph}_G.$$

Remark 1.7.7. One thing that is naive about the naive Satake functor is its direction. Our conventions are that functors in geometric Langlands go from the geometric side to the spectral side. However, the naive Satake functor produces sheaves from representations.

Remark 1.7.8. Note, for example that the functor

$$\mathrm{Rep}(\check{G}) \xrightarrow{\mathrm{Sat}_G^{\mathrm{nv}}} \mathrm{Sph}_G \xrightarrow{-\star \mathrm{Vac}_{\mathrm{Whit}^!(G)}} \mathrm{Whit}^!(G)$$

is CS_G^{-1} .

The functor

$$\mathrm{Rep}(\check{G}) \xrightarrow{\mathrm{Sat}_G^{\mathrm{nv}}} \mathrm{Sph}_G \xrightarrow{\sigma} \mathrm{Sph}_G \xrightarrow{-\star \mathrm{Vac}_{\mathrm{Whit}_*(G)}} \mathrm{Whit}_*(G)$$

is $\mathrm{FLE}_{\check{G}, \infty}$.

1.8. The curse of σ and τ .

1.8.1. The following statement results from the uniqueness assertion in Theorem 1.7.2 combined with Lemma 1.4.12:

Corollary 1.8.2. *The following diagram of anti-equivalences commutes:*

$$\begin{array}{ccc} \mathrm{Sph}_G & \xrightarrow{\mathrm{Sat}_G} & \mathrm{Sph}_{\check{G}}^{\mathrm{spec}} \\ \sigma \downarrow & & \downarrow \sigma^{\mathrm{spec}} \\ \mathrm{Sph}_G & & \\ \tau_G \downarrow & & \\ \mathrm{Sph}_G & \xrightarrow{\mathrm{Sat}_G} & \mathrm{Sph}_{\check{G}}^{\mathrm{spec}}. \end{array}$$

1.8.3. Denote by $\mathrm{Sat}_{G, \tau}$ the (factorization) equivalence

$$\mathrm{Sph}_G \xrightarrow{\tau} \mathrm{Sph}_G \xrightarrow{\mathrm{Sat}_G} \mathrm{Sph}_{\check{G}}^{\mathrm{spec}}.$$

Denote by $\mathrm{Sat}_{G, \tau}^{\mathrm{nv}}$ the functor

$$\tau_G \circ \mathrm{Sat}_G^{\mathrm{nv}}, \quad \mathrm{Rep}(\check{G}) \rightarrow \mathrm{Sph}_G.$$

1.8.4. As another corollary of Lemma 1.4.12 we obtain:

Corollary 1.8.5. *The equivalence*

$$\mathrm{Rep}(\check{G}) \xrightarrow{\mathrm{FLE}_{\check{G}, \infty}} \mathrm{Whit}_*(G)$$

is compatible with the actions of Sph_G and $\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$ via $\mathrm{Sat}_{G, \tau}$.

1.8.6. *Warning.* As has been mentioned above, we will use σ (resp., σ^{spec}) to pass between left and right module categories for Sph_G (resp., $\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$).

Note, however, that due to Corollary 1.8.2, this procedure is compatible with the geometric Satake equivalence *up to* the Chevalley involution.

In practice, this will manifest itself as follows. Let $\mathbf{C}_1$ and $\mathbf{C}_2$ (resp., $\mathbf{C}_1^{\mathrm{spec}}$ and $\mathbf{C}_2^{\mathrm{spec}}$) be left module categories over Sph_G (resp., $\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$). Thanks to the above left-right passage, we can form the tensor products

$$\mathbf{C}_1 \otimes_{\mathrm{Sph}_G} \mathbf{C}_2 \text{ and } \mathbf{C}_1^{\mathrm{spec}} \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathbf{C}_2^{\mathrm{spec}}.$$

Suppose that we have a given a functor

$$F_1 : \mathbf{C}_1 \rightarrow \mathbf{C}_1^{\mathrm{spec}},$$

which is compatible with the actions via

$$(1.12) \quad \mathrm{Sph}_G \xrightarrow{\mathrm{Sat}_G} \mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$$

and a functor

$$F_2 : \mathbf{C}_2 \rightarrow \mathbf{C}_2^{\mathrm{spec}},$$

which is compatible with the actions via

$$(1.13) \quad \mathrm{Sph}_G \xrightarrow{\mathrm{Sat}_{G, \tau}} \mathrm{Sph}_{\check{G}}^{\mathrm{spec}}.$$

In this case, we obtain a functor

$$(1.14) \quad F_1 \otimes F_2 : \mathbf{C}_1 \otimes_{\mathrm{Sph}_G} \mathbf{C}_2 \simeq \mathbf{C}_1^{\mathrm{spec}} \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathbf{C}_2^{\mathrm{spec}}.$$

1.8.7. *Warning.* Similarly, let $\mathbf{C}$ and $\mathbf{C}'$ be left module categories over Sph_G and $\mathrm{Sph}_G^{\mathrm{spec}}$, respectively. Let us view $\mathbf{C}^\vee$ (resp., $\mathbf{C}'^\vee$) again as a left module, using σ (resp., σ^{spec}).

Let $\mathbf{C} \simeq \mathbf{C}'$ be an equivalence compatible with the actions via (1.12). Then the induced equivalence

$$\mathbf{C}^\vee \simeq \mathbf{C}'^\vee$$

is compatible with the actions via (1.13).

2. KAC-MOODY MODULES AND THE KAZHDAN-LUSZTIG CATEGORY

In this section we study the local representation-theoretic category on the geometric side, which we will later connect to the global category $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ by a local-to-global procedure.

The category in question is the Kazhdan-Lusztig category at the critical level,

$$\mathrm{KL}(G)_{\mathrm{crit}} := \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}^{\mathfrak{L}^+(G)}.$$

We will need the following aspects of the theory associated with $\mathrm{KL}(G)_{\mathrm{crit}}$:

- Self-duality;
- The functor of Drinfeld-Sokolov reduction.

2.1. Definition and basic properties.

2.1.1. Let κ be a level for $\mathfrak{g}$. We consider

$$\widehat{\mathfrak{g}}\text{-mod}_\kappa,$$

the category of Kac-Moody modules at level κ . This category carries a natural action of $\mathfrak{L}(G)$ at level κ .

The definition of this category at a fixed point $\underline{x} \in \mathrm{Ran}$ is given in [Ra5]. The factorization version is defined in Sect. B.14.

2.1.2. Let

$$\mathrm{KL}(G)_\kappa := \widehat{\mathfrak{g}}\text{-mod}_\kappa^{\mathfrak{L}^+(G)},$$

denote the corresponding category of spherical objects.

We have an adjunction

$$\mathrm{oblv}_{\mathfrak{L}^+(G)} : \mathrm{KL}(G)_\kappa \rightleftarrows \widehat{\mathfrak{g}}\text{-mod}_\kappa : \mathrm{Av}_*^{\mathfrak{L}^+(G)}.$$

2.1.3. We have a monadic adjunction

$$(2.1) \quad \mathrm{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} : \mathrm{Rep}(\mathfrak{L}^+(G)) \rightleftarrows \mathrm{KL}(G)_\kappa : \mathrm{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa}.$$

In particular, $\mathrm{KL}(G)_\kappa$ is compactly generated by the image of compact generators of $\mathrm{Rep}(\mathfrak{L}^+(G))$, where the latter is *by definition* the ind-completion of the small category consisting of finite-dimensional representations.

2.1.4. Let $\mathrm{Vac}(G)_\kappa$ denote the factorization unit in $\mathrm{KL}(G)_\kappa$. By a slight abuse of notation, we will denote by the same symbol $\mathrm{Vac}(G)_\kappa$ its image under the (strictly unital) factorization functor $\mathrm{oblv}_{\mathfrak{L}^+(G)}$.

We let $\mathbb{V}_{\mathfrak{g}, \kappa}$ denote the image of $\mathrm{Vac}(G)_\kappa$ under the tautological forgetful functor

$$\widehat{\mathfrak{g}}\text{-mod}_\kappa \rightarrow \mathrm{Vect}.$$

The latter is the usual factorization (a.k.a. chiral) algebra attached to the pair $(\mathfrak{g}, \kappa)$.

2.1.5. Our primary interest in this paper is the case when $\kappa = -\frac{1}{2} \cdot \mathrm{Kil}$, where Kil is the Killing form. We will denote the corresponding level by the symbol crit .

By construction, the category $\mathrm{KL}(G)_{\mathrm{crit}}$ carries a monoidal action of Sph_G .

2.2. Duality.

2.2.1. For a given level κ , denote

$$\kappa' := -\kappa + 2 \cdot \text{crit}.$$

(In particular, $\text{crit}' = \text{crit}$.)

2.2.2. It is known that the categories

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa} \text{ and } \widehat{\mathfrak{g}}\text{-mod}_{\kappa'}$$

are canonically dual to one another, in a way compatible with the (unital) factorization structure and the $\mathfrak{L}(G)$ -action, see [Ra5, Sect. 9.16.1].

The counit of the duality is the functor

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\kappa'} \xrightarrow{\otimes} \widehat{\mathfrak{g}}\text{-mod}_{\text{Kil}} \rightarrow \text{Vect},$$

where the second arrow is the functor of semi-infinite cohomology.

By Sect. C.11.5, the above functor pairing has a structure of (lax unital¹⁶) factorization functor.

2.2.3. The above duality induces a duality between

$$(2.2) \quad (\text{KL}(G)_{\kappa})^{\vee} \simeq \text{KL}(G)_{\kappa'},$$

so that

$$(\text{oblv}_{\mathfrak{L}^+(G)})^{\vee} \simeq \text{Av}_{*}^{\mathfrak{L}^+(G)} \text{ and } \left(\text{Av}_{*}^{\mathfrak{L}^+(G)}\right)^{\vee} \simeq \text{oblv}_{\mathfrak{L}^+(G)}.$$

The unit of the duality (2.2) is the object

$$\mathfrak{CD}\mathfrak{D}(G)_{\kappa, \kappa'} \in \text{KL}(G)_{\kappa} \otimes \text{KL}(G)_{\kappa'}.$$

Under this duality and the canonical self-duality of $\text{Rep}(\mathfrak{L}^+(G))$, we have

$$\left(\text{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa}}\right)^{\vee} \simeq \text{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa}} \text{ and } \left(\text{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa}}\right)^{\vee} \simeq \text{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa}}.$$

2.2.4. Specializing to the critical level, we obtain a canonical self-duality

$$(2.3) \quad (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}})^{\vee} \simeq \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}},$$

compatible with the $\mathfrak{L}(G)$ -actions, and

$$(2.4) \quad (\text{KL}(G)_{\text{crit}})^{\vee} \simeq \text{KL}(G)_{\text{crit}},$$

compatible with the Sph_G -actions.

2.3. The functor of Drinfeld-Sokolov reduction.

2.3.1. The duality in Sect. 2.2.2 is applicable to *any* finite-dimensional Lie algebra (where the role of the level $2 \cdot \text{crit}$ is played by the Tate extension). In particular, for a unipotent Lie algebra $\mathfrak{n}'$, the corresponding category $\mathfrak{L}(\mathfrak{n}')$ is canonically self-dual, in a way compatible with the $\mathfrak{L}(N')$ -action.

This construction is functorial. Hence, if a group H acts on $\mathfrak{n}'$, and $\mathcal{P}_H$ is an H -torsor on X , we obtain a canonical self-duality on $\mathfrak{L}(\mathfrak{n}')_{\mathcal{P}_H}\text{-mod}$.

In the particular case $N' = N$, $H = T$ and $\mathcal{P}_H = \rho(\omega_X)$, we obtain a canonical duality on

$$\mathfrak{L}(\mathfrak{n})_{\rho(\omega_X)}\text{-mod}.$$

¹⁶See Sect. C.11.3 for what “lax unital” means.

2.3.2. The character χ (see Sect. 1.3.2) on the group $\mathfrak{L}(N)_{\rho(\omega_X)}$ gives rise to a character (denoted by the same symbol)

$$\mathfrak{L}(\mathfrak{n})_{\rho(\omega_X)} \rightarrow k.$$

We can regard this character as an object

$$k_\chi \in \mathfrak{L}(\mathfrak{n})_{\rho(\omega_X)}\text{-mod}.$$

The factorization property of χ equips k_χ with a structure of *factorization algebra* in $\mathfrak{L}(\mathfrak{n})_{\rho(\omega_X)}\text{-mod}$, where the latter is regarded as a lax factorization category.

The fact that $\chi|_{\mathfrak{L}(\mathfrak{n})_{\rho(\omega_X)}^+} = 0$ implies that this factorization algebra is naturally unital.

2.3.3. We define the functor

$$(2.5) \quad \text{BRST}_{\mathfrak{L}(\mathfrak{n})_{\rho(\omega_X)}, \chi} : \mathfrak{L}(\mathfrak{n})_{\rho(\omega_X)}\text{-mod} \rightarrow \text{Vect}$$

to be given by

$$\mathfrak{L}(\mathfrak{n})_{\rho(\omega_X)}\text{-mod} \xrightarrow{\text{Id} \otimes k_\chi} \mathfrak{L}(\mathfrak{n})_{\rho(\omega_X)}\text{-mod} \otimes \mathfrak{L}(\mathfrak{n})_{\rho(\omega_X)}\text{-mod} \rightarrow \text{Vect},$$

where the second arrow is the counit of the self-duality on $\mathfrak{L}(\mathfrak{n})_{\rho(\omega_X)}\text{-mod}$.

By the above, the functor (2.5) has a natural (lax unital) factorization structure.

2.3.4. Precomposing with

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa, \rho(\omega_X)} \rightarrow \mathfrak{L}(\mathfrak{n})_{\rho(\omega_X)}\text{-mod},$$

we obtain a functor of Drinfeld-Sokolov reduction, which we denote by

$$(2.6) \quad \text{DS} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa, \rho(\omega_X)} \rightarrow \text{Vect}.$$

The functor (2.6) inherits a (lax unital) factorization structure.

2.3.5. It follows from the construction that the functor DS factors as

$$(2.7) \quad \widehat{\mathfrak{g}}\text{-mod}_{\kappa, \rho(\omega_X)} \rightarrow \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\kappa, \rho(\omega_X)}) \rightarrow \text{Vect},$$

We denote the resulting functor

$$\text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\kappa, \rho(\omega_X)}) \rightarrow \text{Vect}$$

by

$$(2.8) \quad \overline{\text{DS}} : \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\kappa, \rho(\omega_X)}) \rightarrow \text{Vect}.$$

Remark 2.3.6. The category $\text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\kappa, \rho(\omega_X)})$ and the functor DS are canonically independent of the choice of the character χ_0 of $\mathfrak{n}/[\mathfrak{n}, \mathfrak{n}]$. This happens by the same mechanism as in Remark 1.3.10: the action of the center of the derived group of G on $\widehat{\mathfrak{g}}\text{-mod}_\kappa$ is trivial.

2.3.7. In the sequel, we will use the following assertion (see [FG2, Theorem 3.2.2]):

Lemma 2.3.8. *For a fixed $\underline{x} \in \text{Ran}$, the functor*

$$\text{KL}(G)_{\kappa, \rho(\omega_X), \underline{x}} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa, \underline{x}} \xrightarrow{\text{DS}} \text{Vect}$$

is t-exact.

Remark 2.3.9. One can show that the assertion of Lemma 2.3.8 holds also in the factorization setting: the proof of [Ra2, Corollary 7.2.2] given at the end of Sect. B.3 of *loc. cit.* adapts to the factorization setting.

3. IND-COHERENT SHEAVES ON MONODROMY-FREE OPERS

In this section we study the local counterpart of the Kazhdan-Lusztig category on the spectral side: this is the category

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$$

of ind-coherent sheaves on the space of monodromy-free $\check{G}$ -opers on the punctured disc. This category will be related to the global spectral category (in this case $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}(X))$) by a local-to-global procedure.

We study $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$ along with its cousins, the factorization categories

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{reg}}) \text{ and } \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}).$$

Since the geometric objects involved are *not* locally of finite type, the definition of $\mathrm{IndCoh}(-)$ on them is not automatic. However, thankfully, these objects turn out to be *(ind)-placid* (see Sect. A.9.1 for what this means), so we have well-behaved categories $\mathrm{IndCoh}^*(-)$ and $\mathrm{IndCoh}^l(-)$ attached to them.

3.1. Monodromy-free opers.

3.1.1. Let Y be a D-scheme over X . Recall the notion of $\check{G}$ -oper on Y . This is a datum of a triple

$$(\mathcal{P}_{\check{B}}, \epsilon, \nabla),$$

where

- $\mathcal{P}_{\check{B}}$ is a $\check{B}$ -bundle on $\mathcal{Y}$;
- ϵ is the identification between the induced $\check{T}$ -bundle $\mathcal{P}_{\check{T}}$ with $\check{\rho}(\omega_X)|_Y := 2\check{\rho}(\omega^{\otimes \frac{1}{2}})|_Y$;
- ∇ is a connection *along* X on the induced $\check{G}$ -bundle $\mathcal{P}_{\check{G}}$.

These data are supposed to satisfy the following compatibility condition:

The *incompatibility* of $\mathcal{P}_{\check{B}}$ and ∇ , which is an element

$$\nabla \bmod \check{\mathfrak{b}} \in (\check{\mathfrak{g}}/\check{\mathfrak{b}})_{\mathcal{P}_{\check{G}}} \otimes \omega_X|_Y$$

belongs to

$$\mathrm{Fil}_{-1}(\check{\mathfrak{g}}/\check{\mathfrak{b}})_{\mathcal{P}_{\check{G}}} \otimes \omega_X|_Y \subset (\check{\mathfrak{g}}/\check{\mathfrak{b}})_{\mathcal{P}_{\check{G}}} \otimes \omega_X|_Y$$

(here $\mathrm{Fil}_{-1}(\check{\mathfrak{g}}/\check{\mathfrak{b}}) \subset \check{\mathfrak{g}}/\check{\mathfrak{b}}$ is the bottom piece of the principal filtration), and its evaluation by means of every negative simple root $-\alpha_i$ of $\check{\mathfrak{g}}$

$$-\alpha_i(\nabla \bmod \check{\mathfrak{b}}) \in -\alpha_i(\mathcal{P}_{\check{T}}) \otimes \omega_X|_Y \xrightarrow{\epsilon} -\alpha_i(\check{\rho}(\omega_X))|_Y \otimes \omega_X|_Y \simeq \mathcal{O}_Y$$

is the unit section.

3.1.2. A priori, opers form a D-prestack over X . However, one shows (see, e.g., Sect. 3.1.7) that it is actually an affine D-scheme over X .

3.1.3. We will denote the D-scheme of $\check{G}$ -opers by $\mathrm{Op}_{\check{G}}$. Its fiber over a given point $x \in X$ is the scheme $\mathrm{Op}_{\check{G}}(\mathcal{D}_x)$ of $\check{G}$ -opers on the formal disc $\mathcal{D}_x$ around x .

We will denote by the symbol

$$\mathrm{Op}_{\check{G}}^{\mathrm{reg}} := \mathfrak{L}_{\nabla}^+(\mathrm{Op}_{\check{G}})$$

the corresponding factorization (affine) scheme (see Sect. B.4.2), i.e., its fiber $\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}}$ over a given $\underline{x} \in \mathrm{Ran}$ is the scheme $\mathrm{Op}_{\check{G}}(\mathcal{D}_{\underline{x}})$ of $\check{G}$ -opers on the formal multi-disc $\mathcal{D}_{\underline{x}}$ around $\underline{x}$.

We let

$$\mathrm{Op}_{\check{G}}^{\mathrm{mer}} := \mathfrak{L}_{\nabla}(\mathrm{Op}_{\check{G}})$$

denote the factorization ind-scheme of $\check{G}$ -opers on the formal punctured disc (see Sect. B.4.6). Its fiber $\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}$ over a given $\underline{x} \in \mathrm{Ran}$ is the ind-scheme $\mathrm{Op}_{\check{G}}(\mathcal{D}_{\underline{x}}^{\times})$ of $\check{G}$ -opers on the *punctured* multi-disc $\mathcal{D}_{\underline{x}}^{\times}$.

3.1.4. We recall the following basic fact about opers:

Once the ambient curve X is fixed, we can assume that the G -bundle underlying an oper (on X itself, a multi-disc in X , or a punctured multi-disc in X) is induced from a fixed B -bundle, to be denoted $\mathcal{P}_B^{\text{Op}}$ (see [BD1, Proposition 3.1.10(iii)]).

In what follows we will denote by $\mathcal{P}_{\tilde{G}}^{\text{Op}}$ the induced $\tilde{G}$ -bundle.

3.1.5. By construction, we have a map

$$\text{Op}_{\tilde{G}}^{\text{reg}} \rightarrow \text{LS}_{\tilde{G}}^{\text{reg}},$$

to be denoted $\mathfrak{r}^{\text{reg}}$.

Note now that thanks to Sect. 3.1.4 we also have a map¹⁷

$$\text{Op}_{\tilde{G}}^{\text{mer}} \rightarrow \text{LS}_{\tilde{G}}^{\text{mer}},$$

to be denoted $\mathfrak{r}$.

We have a commutative *but non-Cartesian* diagram

$$\begin{array}{ccc} \text{Op}_{\tilde{G}}^{\text{reg}} & \xrightarrow{\iota} & \text{Op}_{\tilde{G}}^{\text{mer}} \\ \mathfrak{r}^{\text{reg}} \downarrow & & \downarrow \mathfrak{r} \\ \text{LS}_{\tilde{G}}^{\text{reg}} & \longrightarrow & \text{LS}_{\tilde{G}}^{\text{mer}}. \end{array}$$

3.1.6. We define the factorization ind-scheme of *monodromy-free opers* as the fiber product

$$\text{Op}_{\tilde{G}}^{\text{mon-free}} := \text{LS}_{\tilde{G}}^{\text{reg}} \times_{\text{LS}_{\tilde{G}}^{\text{mer}}} \text{Op}_{\tilde{G}}^{\text{mer}}.$$

I.e., for a fixed $\underline{x} \in \text{Ran}$, the fiber $\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}}$ is the fiber product

$$\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}} := \text{LS}_{\tilde{G}, \underline{x}}^{\text{reg}} \times_{\text{LS}_{\tilde{G}, \underline{x}}^{\text{mer}}} \text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}.$$

Denote by $\iota^{\text{mon-free}}$ and $\iota^{+, \text{mon-free}}$ the resulting maps

$$\text{Op}_{\tilde{G}}^{\text{reg}} \xrightarrow{\iota^{+, \text{mon-free}}} \text{Op}_{\tilde{G}}^{\text{mon-free}} \xrightarrow{\iota^{\text{mon-free}}} \text{Op}_{\tilde{G}}^{\text{mer}}.$$

3.1.7. Recall also that the D-scheme $\text{Op}_{\tilde{G}}$ is acted on simply transitively by the D-scheme $\text{Jets}(\mathfrak{a}(\mathfrak{g})_{\omega_X})$ (see Sect. B.5.1) of jets into $\mathfrak{a}(\mathfrak{g})_{\omega_X}$, where $\mathfrak{a}(\mathfrak{g}) \subset \mathfrak{g}$ is the centralizer of a regular nilpotent element, and the twist by ω_X is performed with respect to the canonical $\mathbb{G}_m$ -action on $\mathfrak{a}(\mathfrak{g})$ (see, e.g., [BD1, Sect. 3.1.9]).

From here we obtain that $\text{Op}_{\tilde{G}}^{\text{reg}}$ (resp., $\text{Op}_{\tilde{G}}^{\text{mer}}$) is acted on simply-transitively by $\mathfrak{L}^+(\mathfrak{a}(\mathfrak{g})_{\omega_X})$ (resp., $\mathfrak{L}(\mathfrak{a}(\mathfrak{g})_{\omega_X})$).

3.1.8. Recall now the notion of *formal smoothness* of a prestack (see, e.g., [GaRo1, Sect. 8.1]). This notion has an evident relative variant.

We record the following (well-known) assertion:

Lemma 3.1.9. *The morphism $\mathfrak{r} : \text{Op}_{\tilde{G}}^{\text{mer}} \rightarrow \text{LS}_{\tilde{G}}^{\text{mer}}$ is formally smooth.*

Proof. We will show that for any classical affine scheme S and a map $\sigma : S \rightarrow \text{Op}_{\tilde{G}}^{\text{mer}}$ relative procotangent sheaf

$$(3.1) \quad T_{\sigma}^*(\text{Op}_{\tilde{G}}^{\text{mer}} / \text{LS}_{\tilde{G}}^{\text{mer}}) \in \text{Pro}(\text{QCoh}(S)^{\leq 0})$$

is a *Tate* vector bundle. This would imply the assertion of the lemma by [GaRo1, Proposition 8.2.2].

¹⁷We alert the reader to the fact that $\text{LS}_{\tilde{G}}^{\text{mer}}$ is *not* the space of loops into $\text{pt}/\tilde{G}$, see Sect. B.7.9.

In order to simplify the notation we will assume that $S = \text{pt}$ (and in particular, we work over a fixed point $\underline{x} \in \text{Ran}$). However, we will perform the analysis in such a way that it will be clear that it works in families.

Let $\mathcal{P}_B^{\text{Op}}$ be as in Sect. 3.1.4. We can represent the tangent space to $\text{Op}_{G,\underline{x}}^{\text{mer}}$ at σ is

$$\text{coFib} \left((\check{\mathfrak{n}} \otimes \mathcal{O}_{\mathcal{D}_{\underline{x}}^\times})_{\mathcal{P}_B^{\text{Op}}} \xrightarrow{\nabla_{\check{\sigma}}} (\check{\mathfrak{b}} \otimes \omega_{\mathcal{D}_{\underline{x}}^\times})_{\mathcal{P}_B^{\text{Op}}} \right),$$

where:

- $\mathcal{O}_{\mathcal{D}_{\underline{x}}^\times}$ is the ring of functions on the punctured multi-disc $\mathcal{D}_{\underline{x}}^\times$;
- $\omega_{\mathcal{D}_{\underline{x}}^\times}$ is the space of 1-forms on $\mathcal{D}_{\underline{x}}^\times$;
- The notation $(-)_{{\mathcal{P}_B^{\text{Op}}}}$ indicates the twist by $\mathcal{P}_B^{\text{Op}}$, viewed as a $\check{B}$ -bundle on $\mathcal{D}_{\underline{x}}^\times$;
- ∇_σ is the connection defined by σ .

The tangent space to $\text{LS}_{G,\underline{x}}^{\text{mer}}$ at the image of σ is

$$(3.2) \quad \text{coFib} \left((\check{\mathfrak{g}} \otimes \mathcal{O}_{\mathcal{D}_{\underline{x}}^\times})_{\mathcal{P}_G^{\text{Op}}} \xrightarrow{\nabla_{\check{\sigma}}} (\check{\mathfrak{g}} \otimes \omega_{\mathcal{D}_{\underline{x}}^\times})_{\mathcal{P}_G^{\text{Op}}} \right).$$

Hence, the relative tangent space along $\mathfrak{r}$ is

$$\text{Fib} \left((\check{\mathfrak{g}}/\check{\mathfrak{n}}) \otimes \mathcal{O}_{\mathcal{D}_{\underline{x}}^\times})_{\mathcal{P}_B^{\text{Op}}} \xrightarrow{\nabla_{\check{\sigma}}} ((\check{\mathfrak{g}}/\check{\mathfrak{b}}) \otimes \omega_{\mathcal{D}_{\underline{x}}^\times})_{\mathcal{P}_B^{\text{Op}}} \right).$$

Thus, choosing a non-degenerate invariant form on $\check{\mathfrak{g}}$, we can identify the *cotangent* space at σ again with¹⁸

$$(3.3) \quad \text{coFib} \left((\check{\mathfrak{n}} \otimes \mathcal{O}_{\mathcal{D}_{\underline{x}}^\times})_{\mathcal{P}_B^{\text{Op}}} \xrightarrow{\nabla_{\check{\sigma}}} (\check{\mathfrak{b}} \otimes \omega_{\mathcal{D}_{\underline{x}}^\times})_{\mathcal{P}_B^{\text{Op}}} \right).$$

Thus, we need to show that (3.3) is indeed a Tate vector bundle (in degree 0). Consider the principal filtration on $\check{\mathfrak{g}}$, and the induced filtrations on $\check{\mathfrak{n}}$ and $\check{\mathfrak{b}}$. The map in (3.3) sends

$$\text{Fil}_i(\check{\mathfrak{n}} \otimes \mathcal{O}_{\mathcal{D}_{\underline{x}}^\times})_{\mathcal{P}_B^{\text{Op}}} \xrightarrow{\nabla_{\check{\sigma}}} \text{Fil}_{i-1}(\check{\mathfrak{b}} \otimes \omega_{\mathcal{D}_{\underline{x}}^\times})_{\mathcal{P}_B^{\text{Op}}}.$$

It suffices to show that for every i ,

$$(3.4) \quad \text{coFib} \left(\text{gr}_i(\check{\mathfrak{n}} \otimes \mathcal{O}_{\mathcal{D}_{\underline{x}}^\times})_{\mathcal{P}_B^{\text{Op}}} \xrightarrow{\nabla_{\check{\sigma}}} \text{gr}_{i-1}(\check{\mathfrak{b}} \otimes \omega_{\mathcal{D}_{\underline{x}}^\times})_{\mathcal{P}_B^{\text{Op}}} \right)$$

is a Tate vector bundle (in degree 0).

However, the latter is evident. In fact, the maps in (3.4) are independent of σ , and the assertion follows from the fact that the maps

$$\text{gr}_i(\check{\mathfrak{n}}) \xrightarrow{\text{ad}_f} \text{gr}_{i-1}(\check{\mathfrak{b}})$$

are injective (where f is a negative principal nilpotent, fixed as an element in $\check{\mathfrak{g}}/\check{\mathfrak{b}}$). □

3.1.10. As a corollary of lemmas 3.1.9 and B.7.7, we obtain:

Corollary 3.1.11. *The ind-scheme $\text{Op}_{G,\underline{x}}^{\text{mon-free}}$ is formally smooth.*¹⁹

3.2. The IndCoh^* categories. In this subsection, for expositional purposes, we will work over a fixed point $\underline{x} \in \text{Ran}$. However, the entire discussion works when $\underline{x}$ forms a family over Ran .

¹⁸The fact that the tangent space and the relative cotangent space to opers are isomorphic is no coincidence: it reflects the interaction of the Poisson structure on $\text{Op}_{G,\underline{x}}^{\text{mer}}$ and the symplectic structure on $\text{LS}_{G,\underline{x}}^{\text{mer}}$. In fact, the morphism $\mathfrak{r}$ is Lagrangian.

¹⁹See Sect. B.1.9 for what formal smoothness means in the factorization setting.

3.2.1. First, since $\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}}$ (resp., $\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}$, $\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}$) is a scheme (resp., ind-scheme), we have a priori defined categories $\mathrm{IndCoh}^*(-)$ and $\mathrm{IndCoh}^!(-)$ attached to them, see Sects. A.4 and A.5 (the Ran space version is discussed in Sects. B.13.16 and B.13.22).

3.2.2. Using Sect. 3.1.7, we can write

$$\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}} \simeq \lim_{\mathbf{L}} \mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}} / \mathbf{L},$$

where $\mathbf{L}$ runs over the filtered poset of lattices in $\mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}}$, viewed as a Tate vector space.

This exhibits $\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}}$ as a limit of smooth schemes with smooth transition maps.

3.2.3. In particular, we obtain that $\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}}$ is *placid* (see Sect. A.9.1 for what this means), so that the categories

$$\mathrm{IndCoh}_*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}}) \text{ and } \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}})$$

are well-behaved; in particular, they are both compactly generated and are mutually dual.

Note, however, that since $\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}}$ is pro-smooth, the coarsening functor

$$\Psi_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}}) \rightarrow \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}})$$

is an equivalence, as is the functor

$$\Upsilon_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}}} : \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}}).$$

3.2.4. We can identify

$$\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}} \simeq (\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}} \times \mathfrak{L}(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}}) / \mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}}.$$

In particular, we have a pro-smooth projection

$$\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}} \rightarrow \mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}} / \mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}} \simeq \mathfrak{L}(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}} / \mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}}.$$

This exhibits $\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}$ as an *ind-placid* ind-scheme (see Sect. A.9.8 for what this means).

3.2.5. This ensures that the categories

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}) \text{ and } \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})$$

are well-behaved; in particular, they are both compactly generated and are mutually dual.

3.2.6. Note that the map

$$\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{reg}} \rightarrow \mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}}$$

is an ind-closed embedding, locally almost of finite presentation (see Lemma B.7.13).

This implies that

$$\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}} \xrightarrow{\iota^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}$$

is also an ind-closed embedding locally almost of finite presentation. In particular, since $\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}$ is ind-placid, we obtain that $\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}$ is also ind-placid (see Corollary A.9.10).

Hence, we obtain that the categories

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}) \text{ and } \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})$$

are also well-behaved; in particular, they are both compactly generated and are mutually dual.

3.2.7. We will now show how one can explicitly exhibit $\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}$ as a colimit of placid schemes.

To simplify the notation, we will assume that $\underline{x}$ consists of a single point x . Henceforth, we will omit x from the subscript, so we will write $\mathcal{D}$ instead of $\mathcal{D}_{\underline{x}}$.

Recall (see Sect. 3.1.4) that opers can be thought of as connections on a fixed $\check{G}$ -bundle $\mathcal{P}_{\check{G}}^{\mathrm{Op}}$. Trivializing this bundle on $\mathcal{D}$, we will think of opers as connection forms, to be denoted α .

Thus, we can write $\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(\mathcal{D}^\times)$ as

$$(3.5) \quad \{(\alpha \in \mathrm{Op}_{\check{G}}(\mathcal{D}^\times), g \in \mathfrak{L}(\check{G})) \mid g \cdot \alpha \in \check{\mathfrak{g}} \otimes \omega_{\mathcal{D}^\times} \} / \mathfrak{L}^+(\check{G}) \subset \mathrm{Op}_{\check{G}}(\mathcal{D}^\times) \times \mathrm{Gr}_{\check{G}},$$

Therefore, we can write $\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(\mathcal{D}^\times)$ as²⁰

$$\text{“colim”}_Y \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(\mathcal{D}^\times) \times_{\mathrm{Gr}_{\check{G}}} Y,$$

where Y runs over the filtered poset of closed subschemes of $\mathrm{Gr}_{\check{G}}$.

Let us show that each $\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(\mathcal{D}^\times) \times_{\mathrm{Gr}_{\check{G}}} Y$ is a limit of schemes almost of finite type with smooth transition maps.

Let us consider $\mathrm{Op}_{\check{G}}(\mathcal{D}^\times)$ as acted on by $\mathfrak{L}(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})$. Now, it is clear that for a fixed closed subscheme $Y \subset \mathrm{Gr}_{\check{G}}$, there is an action of *any* small enough lattice $\mathbf{L} \subset \mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})$ on

$$\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(\mathcal{D}^\times) \times_{\mathrm{Gr}_{\check{G}}} Y$$

via

$$(\alpha, g) \mapsto (\alpha + \alpha_0, g), \quad \alpha_0 \in \mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X}).$$

Further, it is easy to see that for any such $\mathbf{L}$, the quotient of (3.5) by it is locally almost of finite type.

3.3. Properties of the $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$ functor.

3.3.1. The map $\iota^{\mathrm{mon-free}}$ gives rise to the IndCoh-pushforward functor

$$(3.6) \quad (\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}).$$

The functor $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$ is t-exact with respect to the natural t-structures.

Note that since $\iota^{\mathrm{mon-free}}$ is a closed embedding locally almost of finite presentation, the right adjoint $(\iota^{\mathrm{mon-free}})^!$ of $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$ is continuous, in particular, the functor $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$ preserves compactness, see Sect. A.10.11.

3.3.2. By the same logic, the functor

$$(\iota^{\mathrm{mon-free}})^! : \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})$$

admit a left adjoint, to be denoted also by

$$(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}).$$

(We allow ourselves to use the same symbol $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$ in both instances, as the two are unlikely to be confused.)

²⁰In the next formula, the symbol “colim” (i.e., with quotes) refers to the fact that we are forming an ind-scheme, rather than taking the colimit in the category of schemes.

3.3.3. The adjoint pairs

$$(\iota^{\text{mon-free}})_*^{\text{IndCoh}} : \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}}) \rightleftarrows \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) : (\iota^{\text{mon-free}})^!$$

and

$$(\iota^{\text{mon-free}})_*^{\text{IndCoh}} : \text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}}) \rightleftarrows \text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) : (\iota^{\text{mon-free}})^!$$

are dual to one another with respect to the identifications

$$\text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}})^\vee \simeq \text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}})$$

and

$$\text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})^\vee \simeq \text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}).$$

3.3.4. We will prove:

Proposition 3.3.5.

(a) *The functor (3.6) is conservative.*

(b) *An object of $\text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}})$ is compact if (and only if) its image under $(\iota^{\text{mon-free}})_*^{\text{IndCoh}}$ is compact.*

The rest of this subsection is devoted to the proof of this proposition.

3.3.6. Let

$$(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})_{\text{mon-free}}^\wedge$$

denote the formal completion of $\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}$ along $\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}}$.

Since $\iota^{\text{mon-free}}$ is (locally) almost of finite presentation, so is the embedding

$$(\iota^{\text{mon-free}})^\wedge : (\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})_{\text{mon-free}}^\wedge \hookrightarrow \text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}.$$

In particular, $(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})_{\text{mon-free}}^\wedge$ is ind-placid, and we have a well-behaved category

$$\text{IndCoh}^*((\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})_{\text{mon-free}}^\wedge).$$

3.3.7. The functor

$$((\iota^{\text{mon-free}})^\wedge)_*^{\text{IndCoh}} : \text{IndCoh}^*((\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})_{\text{mon-free}}^\wedge) \rightarrow \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})$$

gives rise to an equivalence

$$\text{IndCoh}^*((\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})_{\text{mon-free}}^\wedge) \xrightarrow{\sim} \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})_{\text{mon-free}},$$

where

$$\text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})_{\text{mon-free}} \subset \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})$$

is the full subcategory of objects with set-theoretic support on $\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}}$.

Furthermore, the functor $((\iota^{\text{mon-free}})^\wedge)_*^{\text{IndCoh}}$ admits a continuous right adjoint, to be denoted $((\iota^{\text{mon-free}})^\wedge)^!$, so $((\iota^{\text{mon-free}})^\wedge)_*^{\text{IndCoh}}$ preserves compactness.

3.3.8. The functor $(\iota^{\text{mon-free}})_*^{\text{IndCoh}}$ of (3.6) factors as

$$\text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}}) \rightarrow \text{IndCoh}^*((\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})_{\text{mon-free}}^\wedge) \xrightarrow{((\iota^{\text{mon-free}})^\wedge)_*^{\text{IndCoh}}} \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}),$$

and in order to prove Proposition 3.3.5, it is enough to establish the corresponding properties for the above functor

$$(3.7) \quad (\iota^{\text{mon-free}})_*^{\text{IndCoh}} : \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}}) \rightarrow \text{IndCoh}^*((\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})_{\text{mon-free}}^\wedge).$$

3.3.9. Recall that according to Corollary 3.1.11, the ind-scheme $\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}$ is formally smooth. Hence, its embedding into any nilpotent thickening²¹ admits a retraction. This implies that the embedding

$$(3.8) \quad \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}} \hookrightarrow (\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}}$$

admits a retraction

$$(3.9) \quad (\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}} \rightarrow \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}.$$

3.3.10. The existence of the retraction (3.9) readily implies that $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$ is conservative:

Indeed, the functor of IndCoh-pushforward along (3.9) is a left inverse of $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$.

This proves point (a) of Proposition 3.3.5.

Remark 3.3.11. Note that the above argument implies that the functor $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$ of (3.7) is *co-monadic*. Indeed, according to what we just proved, it is conservative, and it admits a right adjoint. Hence, it remains to check that it preserves totalizations of $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$ -split cosimplicial objects.

However, since $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$ admits a left inverse, such cosimplicial objects are themselves split, and hence their totalizations are preserved by any functor.

3.3.12. Let $\mathcal{F} \in \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})$ be an object, such that $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}(\mathcal{F})$ is compact. Let us show that $\mathcal{F}$ is itself compact.

Since $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}(\mathcal{F})$ is compact, it is cohomologically bounded, i.e., there exists an integer n such that the map

$$(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}(\mathcal{F}) \rightarrow \tau^{\geq -n}((\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}(\mathcal{F}))$$

is an isomorphism.

Since the functor $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$ is t-exact, we obtain that

$$(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}(\mathcal{F}) \rightarrow (\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}(\tau^{\geq -n}(\mathcal{F}))$$

is an isomorphism.

However, since we already know that $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$ is conservative, this implies that

$$\mathcal{F} \rightarrow \tau^{\geq -n}(\mathcal{F})$$

is an isomorphism, i.e., $\mathcal{F}$ is itself cohomologically bounded.

Hence, it remains to check that the individual cohomologies $H^i(\mathcal{F})$ of $\mathcal{F}$ are coherent. However,

$$(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}(H^i(\mathcal{F})) \simeq H^i((\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}(\mathcal{F})),$$

and it is easy to see that an object $\mathcal{F}^0 \in \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})^{\heartsuit}$ is coherent if and only if

$$(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}(\mathcal{F}^0) \in \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}})^{\heartsuit}$$

is coherent (this is true for any closed embedding almost of finite presentation between ind-placid ind-schemes).

□[Proposition 3.3.5]

Remark 3.3.13. The implication “ $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}(\mathcal{F})$ is compact” $\Rightarrow$ “ $\mathcal{F}$ is compact” can also be proved using the retraction (3.9): since this map is ind-finite, the functor IndCoh-pushforward along it preserves compactness.

Remark 3.3.14. Note that the existence of a retraction implies that the ind-scheme $\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}$ is *classical*. Indeed, the ind-scheme $\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}}$ is classical, and hence so is its formal completion $(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}}$ (the latter follows from [GaRo1, Proposition 6.8.2]: we reduce to the Noetherian situation using placidity).

²¹Assuming all ind-schemes involved are $\aleph_0$.

Remark 3.3.15. The contents of this subsection apply “as-is” when $\underline{x}$ forms family over Ran . In particular, the functor $(\iota^{\text{mon-free}})_*^{\text{IndCoh}}$ has a natural factorization structure.

Moreover, when viewed as a functor between unital factorization categories, $(\iota^{\text{mon-free}})_*^{\text{IndCoh}}$ has a natural lax unital factorization structure.

We claim, however, that this unital structure is actually strict. Indeed, this follows from the fact that $(\iota^{\text{mon-free}})_*^{\text{IndCoh}}$ sends

$$\mathcal{O}_G^{\text{reg}} = \mathbf{1}_{\text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})} \rightarrow \mathbf{1}_{\text{IndCoh}^*(\text{Op}_G^{\text{mer}})} = \mathcal{O}_G^{\text{reg}},$$

see Lemma C.11.23.

3.4. A direct product decomposition.

3.4.1. We will show that we actually have a (non-canonical) isomorphism

$$(3.10) \quad (\text{Op}_{G,\underline{x}}^{\text{mer}})^{\wedge}_{\text{mon-free}} \simeq \text{Op}_{G,\underline{x}}^{\text{mon-free}} \times (\check{\mathfrak{g}}_0^{\wedge})^n, \quad |\underline{x}| = n$$

(here $\check{\mathfrak{g}}_0^{\wedge}$ is the formal completion of $\check{\mathfrak{g}}$ at 0), so that (3.8) identifies with the base change of

$$0 \rightarrow \check{\mathfrak{g}}_0^{\wedge}.$$

Remark 3.4.2. The material in this subsection is specific to the situation over a given $\underline{x} \in \text{Ran}$. I.e., we do not know what how to even formulate the corresponding statement over Ran (or even X^n for $n \geq 2$).

3.4.3. With no restriction of generality, we can assume that $\underline{x}$ consists of a single point x . Henceforth in this proof, we will drop the subscript “ x ” and simply write $\mathcal{D}$ instead of $\mathcal{D}_x$.

We have

$$\text{Op}_{\check{G}}(\mathcal{D}^{\times})^{\wedge}_{\text{mon-free}} \simeq \text{Op}_{\check{G}}(\mathcal{D}^{\times})^{\wedge}_{\text{LS}_{\check{G}}(\mathcal{D}^{\times})} \times \text{LS}_{\check{G}}(\mathcal{D}^{\times})^{\wedge}_{\text{reg}},$$

where $\text{LS}_{\check{G}}(\mathcal{D}^{\times})^{\wedge}_{\text{reg}}$ is the formal completion of $\text{LS}_{\check{G}}(\mathcal{D}^{\times})$ along $\text{LS}_{\check{G}}(\mathcal{D})$.

Note that we can identify

$$\text{LS}_{\check{G}}(\mathcal{D}^{\times})^{\wedge}_{\text{reg}} \simeq \check{\mathfrak{g}}_0^{\wedge} / \text{Ad}(\check{G}),$$

so that

$$\text{Op}_{\check{G}}^{\text{mon-free}}(\mathcal{D}) \simeq \text{Op}_{\check{G}}(\mathcal{D}^{\times})^{\wedge}_{\text{mon-free}} \times_{\check{\mathfrak{g}}_0^{\wedge} / \text{Ad}(\check{G})} \text{pt} / \check{G}.$$

3.4.4. Note that the $\check{G}$ -bundle on $\text{Op}_{\check{G}}^{\text{mon-free}}(\mathcal{D}^{\times})$ corresponding to the map

$$\text{Op}_{\check{G}}^{\text{mon-free}}(\mathcal{D}^{\times}) \xrightarrow{\tau} \text{LS}_{\check{G}}(\mathcal{D}) \simeq \text{pt} / \check{G}$$

can be (non-canonically) trivialized, see Sect. 3.1.4. I.e., the map

$$\text{Op}_{\check{G}}^{\text{mon-free}}(\mathcal{D}^{\times}) \rightarrow \text{Op}_{\check{G}}(\mathcal{D}^{\times})^{\wedge}_{\text{mon-free}} \xrightarrow{\tau} \text{LS}_{\check{G}}(\mathcal{D}^{\times})^{\wedge}_{\text{reg}} \simeq \check{\mathfrak{g}}_0^{\wedge} / \text{Ad}(\check{G}) \rightarrow \text{pt} / \check{G}$$

factors through a map

$$\text{Op}_{\check{G}}^{\text{mon-free}}(\mathcal{D}^{\times}) \rightarrow \text{pt}.$$

Hence, so does the map

$$\text{Op}_{\check{G}}(\mathcal{D}^{\times})^{\wedge}_{\text{mon-free}} \xrightarrow{\tau} \text{LS}_{\check{G}}(\mathcal{D}^{\times})^{\wedge}_{\text{reg}} \simeq \check{\mathfrak{g}}_0^{\wedge} / \text{Ad}(\check{G}) \rightarrow \text{pt} / \check{G}.$$

Hence, the map

$$\text{Op}_{\check{G}}(\mathcal{D}^{\times})^{\wedge}_{\text{mon-free}} \xrightarrow{\tau} \text{LS}_{\check{G}}(\mathcal{D}^{\times})^{\wedge}_{\text{reg}} \simeq \check{\mathfrak{g}}_0^{\wedge} / \text{Ad}(\check{G})$$

can be (non-canonically) lifted to a map

$$(3.11) \quad \text{Op}_{\check{G}}(\mathcal{D}^{\times})^{\wedge}_{\text{mon-free}} \rightarrow \check{\mathfrak{g}}_0^{\wedge}.$$

3.4.5. Combining the maps (3.9) and (3.11), we obtain a map

$$(3.12) \quad \mathrm{Op}_{\tilde{G}}(\mathcal{D}^\times)_{\mathrm{mon-free}}^\wedge \rightarrow \mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}(\mathcal{D}^\times) \times \check{\mathfrak{g}}_0^\wedge,$$

such that if we base change both sides with respect to $0 \rightarrow \check{\mathfrak{g}}_0^\wedge$, we obtain the identity map on $\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}(\mathcal{D}^\times)$.

Thus, the map (3.12) becomes an isomorphism after a base change by a nil-isomorphism. This implies that the map (3.12) is itself an isomorphism.

3.5. **The action of $\mathrm{IndCoh}^!$.** As in Sect. 3.2, for expositional purposes, we will work over a fixed point $\underline{x} \in \mathrm{Ran}$. However, the entire discussion works when $\underline{x}$ forms a family over Ran .

3.5.1. Recall that the category $\mathrm{IndCoh}^*(\mathcal{Y})$ of an ind-scheme $\mathcal{Y}$ is naturally acted on by $\mathrm{IndCoh}^!(\mathcal{Y})$. For a morphism $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$, the corresponding functor $f_* : \mathrm{IndCoh}^*(\mathcal{Y}_1) \rightarrow \mathrm{IndCoh}^*(\mathcal{Y}_2)$ is $\mathrm{IndCoh}^!(\mathcal{Y}_2)$ -linear, where $\mathrm{IndCoh}^!(\mathcal{Y}_2)$ acts on $\mathrm{IndCoh}^*(\mathcal{Y}_1)$ via $f^! : \mathrm{IndCoh}^!(\mathcal{Y}_2) \rightarrow \mathrm{IndCoh}^!(\mathcal{Y}_1)$, see Sect. A.6.6.

In particular, we obtain that the category $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$ (resp., $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mon-free}})$) is acted on by $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$ (resp., $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mon-free}})$), and the functor

$$(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$$

is linear with respect to $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$.

3.5.2. Being the right adjoint of a $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$ -linear functor, the functor

$$(\iota^{\mathrm{mon-free}})^! : \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mon-free}})$$

is right-lax $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$ -linear.

However, it is easy to see that this right-lax $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$ -linearity structure is actually strict, i.e., the adjunction

$$(3.13) \quad (\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mon-free}}) \rightleftarrows \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}}) : (\iota^{\mathrm{mon-free}})^!$$

takes place in the 2-category of $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$ -module categories.

3.5.3. Since $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$ (resp., $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mon-free}})$) is *symmetric* monoidal, we can view the dual categories

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})^\vee \text{ and } \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mon-free}})^\vee$$

as modules over $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$ and $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mon-free}})$, respectively.

Note that the identifications

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})^\vee \simeq \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$$

and

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mon-free}})^\vee \simeq \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mon-free}})$$

are compatible with the $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$ - and $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mon-free}})$ -actions, respectively.

Recall (see Sect. 3.3.2) that the dual of the adjunction (3.13) identifies with

$$(3.14) \quad (\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mon-free}}) \rightleftarrows \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}}) : (\iota^{\mathrm{mon-free}})^!$$

It is easy to see that the resulting $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$ -linear structure on (3.14) arising from the $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$ -linear structure on (3.13) is the natural $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}})$ -linear structure on

$$(\iota^{\mathrm{mon-free}})^! : \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mer}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathrm{mon-free}}).$$

3.5.4. Being a $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})$ -linear, the functor

$$(\iota^{\mathrm{mon-free}})^! : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}})$$

induces a functor

$$(3.15) \quad \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}).$$

We will prove:

Proposition 3.5.5. *The functor (3.15) is an equivalence.*

The proof will be given in Sect. 3.7.14.

3.5.6. Note that in addition to the functor (3.15), we have a tautologically defined functor

$$(3.16) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}) \rightarrow \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})}(\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}), \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})).$$

We claim:

Lemma 3.5.7. *The functor (3.16) is an equivalence.*

Proof. Recall that the category $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})$ identifies with the dual of $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})$, and this identification is compatible with the $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})$ -module structures. Therefore, for any $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})$ -module category $\mathbf{C}$, we have

$$\mathrm{Funct}_{\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})}(\mathbf{C}, \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})) \simeq \mathrm{Funct}(\mathbf{C}, \mathrm{Vect}),$$

where $\mathrm{Funct}(-, -)$ refers to colimit-preserving functors.

Applying this to $\mathbf{C} = \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}})$, we obtain that the right-hand side on (3.16) identifies with

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}})^\vee \simeq \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}).$$

It is easy to see, however, that the endomorphism of $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}})$, induced by (3.16) and the above identification, is the identity functor. $\square$

3.6. Action of the spherical category. The discussion in this subsection will be specific to the situation when $\underline{x} \in \mathrm{Ran}$ is fixed. The generalization in the factorization setting will be discussed in Sect. E.8.

3.6.1. Let us write $\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}$ as

$$(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})_{\mathrm{mon-free}}^\wedge \times_{(\mathrm{LS}_{\check{G},\underline{x}}^{\mathrm{mer}})_{\mathrm{reg}}^\wedge} \mathrm{LS}_{\check{G},\underline{x}}^{\mathrm{reg}} \simeq (\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})_{\mathrm{mon-free}}^\wedge \times_{(\check{\mathfrak{g}}_0^\wedge / \mathrm{Ad}(\check{G}))^{|\underline{x}|}} (\mathrm{pt} / \check{G})^{|\underline{x}|}.$$

From this presentation it is clear that

$$\mathrm{Sph}_{\check{G},\underline{x}}^{\mathrm{spec}} \simeq \mathrm{IndCoh}(\mathrm{LS}_{\check{G},\underline{x}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G},\underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{\check{G},\underline{x}}^{\mathrm{reg}}) \simeq \mathrm{IndCoh}(\mathrm{pt} / \check{G} \times_{\check{\mathfrak{g}}_0^\wedge / \mathrm{Ad}(\check{G})} \mathrm{pt} / \check{G})^{\otimes |\underline{x}|}$$

acts on both

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}) \text{ and } \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}).$$

Moreover, these actions are $\mathrm{IndCoh}^!((\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})_{\mathrm{mon-free}}^\wedge)$ -linear, and hence $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})$ -linear.

3.6.2. The identification

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}})^\vee \simeq \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}})$$

is compatible with the structure of $(\mathrm{Sph}_{\check{G},\underline{x}}^{\mathrm{spec}}, \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}}))$ -bimodule on the two sides.

3.6.3. It follows from the constructions that the functor (3.15) (resp., (3.16)) respects the $\mathrm{Sph}_{\tilde{G}, \underline{x}}^{\mathrm{spec}}$ -actions on the two sides, where the action on the left-hand side of (3.15) (resp., right-hand side of (3.16)) is via the $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})$ -factor.

3.6.4. Furthermore, it again follows from the construction that the functor

$$\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}, \underline{x}}^{\mathrm{reg}}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}) \xrightarrow{\otimes} \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})^{(\iota^{\mathrm{mon-free}})^{\mathrm{IndCoh}}} \rightarrow^* \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}})_{\mathrm{mon-free}}$$

canonically factors via a functor

$$(3.17) \quad \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}, \underline{x}}^{\mathrm{reg}}) \otimes_{\mathrm{Sph}_{\tilde{G}, \underline{x}}^{\mathrm{spec}}} \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}})_{\mathrm{mon-free}}.$$

We claim:

Proposition 3.6.5. *The functor (3.17) is an equivalence.*

Proof. Since the monoidal categories $\mathrm{Sph}_{\tilde{G}, \underline{x}}^{\mathrm{spec}}$ and $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}, \underline{x}}^{\mathrm{reg}})$ are rigid, the projection functor

$$(3.18) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}) \simeq \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}, \underline{x}}^{\mathrm{reg}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}, \underline{x}}^{\mathrm{reg}})} \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}) \rightarrow \\ \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}, \underline{x}}^{\mathrm{reg}}) \otimes_{\mathrm{Sph}_{\tilde{G}, \underline{x}}^{\mathrm{spec}}} \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})$$

admits a continuous right adjoint. Moreover, the corresponding adjunction is monadic.

The functor

$$((\iota^{\mathrm{mon-free}})^{\wedge})^!|_{\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}})_{\mathrm{mon-free}}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}})_{\mathrm{mon-free}} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})$$

is also monadic.

Hence, we need to show that the functor (3.17) induces an isomorphism between the two monads acting on $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})$ as plain endofunctors.

It is easy to see that the composition of (3.18) with (3.17) is the functor

$$(\iota^{\mathrm{mon-free}})^{\mathrm{IndCoh}}_* : \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}})_{\mathrm{mon-free}}.$$

This gives rise to a map between the two monads. Let us show that this map is indeed an isomorphism of the underlying endofunctors.

The monad corresponding to (3.18) is given by the action on $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})$ by the (algebra) object in $\mathrm{Sph}_{\tilde{G}, \underline{x}}^{\mathrm{spec}}$, equal to the $!$ -pullback of

$$\mathcal{O}_{\mathrm{LS}_{\tilde{G}, \underline{x}}^{\mathrm{reg}}} \in \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}, \underline{x}}^{\mathrm{reg}}) \xrightarrow{\Psi_{\mathrm{LS}_{\tilde{G}, \underline{x}}^{\mathrm{reg}}}} \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}, \underline{x}}^{\mathrm{reg}})$$

along the first projection

$$p_1 : \mathrm{LS}_{\tilde{G}, \underline{x}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\tilde{G}, \underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{\tilde{G}, \underline{x}}^{\mathrm{reg}} \rightarrow \mathrm{LS}_{\tilde{G}, \underline{x}}^{\mathrm{reg}}.$$

The monad corresponding to $(\iota^{\mathrm{mon-free}})^{\mathrm{IndCoh}}_*$ is given by $!$ -pull followed by $*$ -push along

$$\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}} \leftarrow \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}} \times_{(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}}} \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}} \rightarrow \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}.$$

However, it is easy to see that this functor is given by the action of the same object in $\mathrm{Sph}_{\tilde{G}, \underline{x}}^{\mathrm{spec}}$. $\square$

3.7. Self-duality for IndCoh on opers.

3.7.1. First, we claim that there is a canonically defined equivalence

$$(3.19) \quad \Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mer}}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}),$$

compatible with the monoidal action of $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})$ on both sides.

By $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})$ -linearity, the datum of a functor (3.19) is equivalent to a choice of an object in $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})$.

The corresponding object, to be denoted

$$\omega_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}}^{*, \mathrm{fake}} \in \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}),$$

is constructed as follows.

3.7.2. Consider $\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}$ as equipped with an action of $\mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}}$, and note that the quotient

$$\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}} / \mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}}$$

is an ind-scheme of ind-finite type. In particular, we have a well-defined category

$$\mathrm{IndCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}} / \mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}}),$$

and an object

$$\omega_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}} / \mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}}} \in \mathrm{IndCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}} / \mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}}).$$

The operation of $*$ -pullback along

$$\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}} \rightarrow \mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}} / \mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}}$$

is a well-defined functor

$$(3.20) \quad \mathrm{IndCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}} / \mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}).$$

We let $\omega_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}}^{*, \mathrm{fake}}$ be the image of $\omega_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}} / \mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}}}$ under (3.20).

3.7.3. We claim:

Lemma 3.7.4. *The functor $\Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mer}}}$ of (3.19), defined by $\omega_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}}^{*, \mathrm{fake}}$, is an equivalence.*

Proof. We can write

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}) \simeq \mathrm{colim}_{\mathbf{L}} \mathrm{IndCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}} / \mathbf{L})$$

(where the transition functors are given by $!$ -pullback) and

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}) \simeq \mathrm{colim}_{\mathbf{L}} \mathrm{IndCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}} / \mathbf{L})$$

(where the transition functors are given by $*$ -pullback), and where the $\mathbf{L}$'s run over the poset of lattices in $\mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}}$.

The functor $\Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mer}}}$ is given by the compatible family of (endo)functors

$$\mathrm{IndCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}} / \mathbf{L}) \rightarrow \mathrm{IndCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}} / \mathbf{L}),$$

each given by tensoring by the graded line

$$\det(\mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}} / \mathbf{L})[-\dim(\mathfrak{L}^+(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})_{\underline{x}} / \mathbf{L})].$$

Since all these functors are equivalences, so is their colimit.

□

Remark 3.7.5. We can combine the functor $\Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mer}}}$ of (3.19) with the identification

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\vee} \simeq \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})$$

and thus view it as the datum of self-duality on $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{reg}})$.

Remark 3.7.6. By the same token, we can define the functor

$$\Theta_{\mathrm{Op}_G^{\mathrm{reg}}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{reg}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{reg}}) \simeq \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{reg}})$$

and show that it is an equivalence.

Note that the corresponding object

$$\omega_{\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{reg}}}^{*, \mathrm{fake}} \in \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{reg}}) \simeq \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{reg}})$$

is $\mathcal{O}_{\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{reg}}}$.

3.7.7. Let

$$\omega_{\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}}^{*, \mathrm{fake}} \in \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})$$

be defined by

$$\omega_{\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}}^{*, \mathrm{fake}} := (\iota^{\mathrm{mon-free}})^!(\omega_{\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}}}^{*, \mathrm{fake}}).$$

Let

$$(3.21) \quad \Theta_{\mathrm{Op}_G^{\mathrm{mon-free}}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}),$$

be the $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})$ -linear functor, corresponding to $\omega_{\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}}^{*, \mathrm{fake}}$.

3.7.8. Note that the functor $\Theta_{\mathrm{Op}_G^{\mathrm{mon-free}}}$ is rigged so that it makes the diagram

$$(3.22) \quad \begin{array}{ccc} \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}}) & \xrightarrow{(\iota^{\mathrm{mon-free}})^!} & \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}) \\ \Theta_{\mathrm{Op}_G^{\mathrm{mer}}} \downarrow & & \downarrow \Theta_{\mathrm{Op}_G^{\mathrm{mon-free}}} \\ \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}}) & \xrightarrow{(\iota^{\mathrm{mon-free}})^!} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}) \end{array}$$

commute.

3.7.9. We claim:

Proposition 3.7.10. *The functor $\Theta_{\mathrm{Op}_G^{\mathrm{mon-free}}}$ of (3.21) is an equivalence.*

Proof. Write

$$\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}} = \text{“colim”}_i Y_i^0 \text{ and } \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}} = \text{“colim”}_i Y_i,$$

where Y_i^0 and Y_i are schemes, and the map $\iota^{\mathrm{mon-free}}$ is given by a compatible family of maps

$$Y_i^0 \xrightarrow{\iota_i^{\mathrm{mon-free}}} Y_i$$

almost of finite presentation. Moreover, we can choose Y_i so that its map to $\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}}$ is almost of finite presentation. In this case both Y_i and Y_i^0 are placid.

Set

$$\omega_{Y_i^0}^{*, \mathrm{fake}} := (\iota_i^{\mathrm{mon-free}})^!(\omega_{Y_i}^{*, \mathrm{fake}}),$$

where $\omega_{Y_i}^{*, \mathrm{fake}}$ is the $!$ -restriction of $\omega_{\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}}}^{*, \mathrm{fake}}$ along $Y_i \rightarrow \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}}$. Let

$$\Theta_{Y_i^0} : \mathrm{IndCoh}^!(Y_i^0) \rightarrow \mathrm{IndCoh}^*(Y_i^0)$$

be the $\mathrm{IndCoh}^!(Y_i^0)$ -linear functor, defined by $\omega_{Y_i^0}^{*, \mathrm{fake}}$. We will show that each $\Theta_{Y_i^0}$ is an equivalence.

Write

$$Y_i = \lim_{\alpha} Y_{i, \alpha},$$

where $Y_{i, \alpha}$ are schemes almost of finite type, with smooth transition maps.

Since $\iota_i^{\text{mon-free}}$ is almost of finite presentation (up to truncation)²², we can find an index α such that Y_i^0 fits into a Cartesian diagram

$$\begin{array}{ccc} Y_i^0 & \xrightarrow{\iota_i^{\text{mon-free}}} & Y_i \\ \pi_\alpha^0 \downarrow & & \downarrow \pi_\alpha \\ Y_{i,\alpha}^0 & \xrightarrow{(\iota^{\text{mon-free}})_i^0} & Y_{i,\alpha}. \end{array}$$

Furthermore, up to enlarging α , by the construction of $\omega_{\text{Op}_{\tilde{G},\underline{x}}^{\text{mer}}}^{*,\text{fake}}$, we can assume that

$$\omega_{Y_i}^{\text{fake},*} \simeq \pi_\alpha^*(\omega_{Y_{i,\alpha}} \otimes \mathcal{L}_{i,\alpha}),$$

where $\mathcal{L}_{i,\alpha}$ is a (comologically graded) line bundle on $Y_{i,\alpha}$. Hence,

$$\omega_{Y_i^0}^{*,\text{fake}} \simeq (\pi_\alpha^0)^*(\omega_{Y_{i,\alpha}^0} \otimes \mathcal{L}_{i,\alpha}^0),$$

where

$$\mathcal{L}_{i,\alpha}^0 := ((\iota^{\text{mon-free}})_i^0)^*(\mathcal{L}_{i,\alpha}).$$

Since the category of indices α is filtered, we can write

$$Y_i^0 \simeq \lim_{\beta \geq \alpha} Y_{i,\beta}^0, \quad Y_{i,\beta}^0 := Y_{i,\alpha}^0 \times_{Y_{i,\alpha}} Y_{i,\beta},$$

so that

$$\text{IndCoh}^*(Y_i^0) \simeq \text{colim}_{\beta \geq \alpha} \text{IndCoh}(Y_{i,\beta}^0)$$

under $*$ -pullbacks, and

$$\text{IndCoh}^!(Y_i^0) \simeq \text{colim}_{\beta \geq \alpha} \text{IndCoh}(Y_{i,\beta}^0)$$

under $!$ -pullbacks.

For each β , let $\mathcal{L}_{i,\beta}^0$ be the (canonically defined) line bundle on $Y_{i,\beta}^0$ so that

$$\omega_{Y_{i,\beta}^0} \otimes \mathcal{L}_{i,\beta}^0 \simeq (\pi_{\beta,\alpha}^0)^*(\omega_{Y_{i,\alpha}^0} \otimes \mathcal{L}_{i,\alpha}^0), \quad \pi_{\beta,\alpha}^0 : Y_{i,\beta}^0 \rightarrow Y_{i,\alpha}^0.$$

We obtain that the functor $\Theta_{Y_i^0}$ is given by the compatible system of (endo)functors

$$\text{IndCoh}(Y_{i,\beta}^0) \xrightarrow{\mathcal{L}_{i,\beta}^0 \otimes -} \text{IndCoh}(Y_{i,\beta}^0),$$

which are all equivalences. □

Remark 3.7.11. One could make the above proof more explicit by using the presentation of $\text{Op}_{\tilde{G},\underline{x}}^{\text{mon-free}}$ as in Sect. 3.2.7.

Remark 3.7.12. Note that combined with the identification

$$\text{IndCoh}^*(\text{Op}_{\tilde{G},\underline{x}}^{\text{mon-free}})^\vee \simeq \text{IndCoh}^!(\text{Op}_{\tilde{G},\underline{x}}^{\text{mon-free}}),$$

functor $\Theta_{\text{Op}_{\tilde{G}}^{\text{mon-free}}}$ of (3.21) can be viewed as the datum of self-duality on $\text{IndCoh}^*(\text{Op}_{\tilde{G},\underline{x}}^{\text{mon-free}})$.

3.7.13. As a first consequence, passing to the right adjoint functors along the horizontal arrows in (3.22), and knowing that the vertical arrows are equivalences, we obtain another commutative diagram

$$\begin{array}{ccc} \text{IndCoh}^!(\text{Op}_{\tilde{G},\underline{x}}^{\text{mon-free}}) & \xrightarrow{(\iota^{\text{mon-free}})_*^{\text{IndCoh}}} & \text{IndCoh}^!(\text{Op}_{\tilde{G},\underline{x}}^{\text{mer}}) \\ \Theta_{\text{Op}_{\tilde{G}}^{\text{mon-free}}} \downarrow & & \downarrow \Theta_{\text{Op}_{\tilde{G}}^{\text{mer}}} \\ \text{IndCoh}^*(\text{Op}_{\tilde{G},\underline{x}}^{\text{mon-free}}) & \xrightarrow{(\iota^{\text{mon-free}})_*^{\text{IndCoh}}} & \text{IndCoh}^*(\text{Op}_{\tilde{G},\underline{x}}^{\text{mer}}). \end{array}$$

²²The issue of truncation is taken over by passing to the limit.

3.7.14. *Proof of Proposition 3.5.5.* From (3.22) we obtain a commutative diagram

$$\begin{array}{ccc}
\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})} \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}}) & \xrightarrow{(\iota^{\mathrm{mon-free}})^!} & \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}) \\
\mathrm{Id} \otimes \Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mer}}} \downarrow & & \downarrow \Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}} \\
\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}}) & \xrightarrow{(\iota^{\mathrm{mon-free}})^!} & \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}).
\end{array}$$

The vertical arrows in this diagram are equivalences by Lemma 3.7.4 and Proposition 3.7.10, respectively. Since the top horizontal arrow is an equivalence, we obtain that so is the bottom horizontal arrow.

□[Proposition 3.5.5]

3.7.15. Let us observe now that once we know Proposition 3.5.5, we could view the construction of $\Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}}$ differently:

We start with the equivalence

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}}) \simeq \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}})$$

and pass to dual categories. We obtain

$$(3.23) \quad \mathrm{Func}_{\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})}(\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}), \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})) \simeq \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}).$$

Applying $\Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mer}}}$, we replace the left-hand side in (3.23) by

$$\mathrm{Func}_{\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})}(\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}), \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mer}})),$$

and applying Lemma 3.5.7, we rewrite it further as $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}})$.

Thus, we can interpret (3.23) as an equivalence

$$(3.24) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}) \simeq \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}).$$

It is easy to see, however, that (3.24) equals the (inverse of the) equivalence $\Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}}$ constructed above.

3.7.16. The following results from the definition of the $\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}$ -action on $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}})$ and $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}})$ in Sect. 3.6.1:

Lemma 3.7.17. *The equivalence (3.24) is compatible with the $\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}$ -actions.*

3.7.18. All the preceding discussion in this subsection applies also in the factorization setting.

3.8. Relation to quasi-coherent sheaves. We now consider the relationship between ind-coherent and quasi-coherent sheaves on $\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}$. We observe two pleasant categorical properties over a point and ask if they extend to the factorization setting.

3.8.1. For any prestack $\mathcal{Y}$ we have a canonically defined (symmetric monoidal) functor

$$\Upsilon_{\mathcal{Y}} : \mathrm{QCoh}(\mathcal{Y}) \rightarrow \mathrm{IndCoh}^!(\mathcal{Y}).$$

It is known that if $\mathcal{Y}$ is a formally smooth ind-scheme locally almost of finite type, then $\Upsilon_{\mathcal{Y}}$ is an equivalence (see [GaRo1, Theorem 10.1.1])²³.

²³This result was originally proved by J. Lurie.

3.8.2. Consider the functors

$$(3.25) \quad \Upsilon_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}} : \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})$$

and

$$(3.26) \quad \Upsilon_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} : \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}),$$

respectively.

3.8.3. First, we claim:

Lemma 3.8.4. *The functor $\Upsilon_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}}$ is an equivalence.*

Proof. Repeats that of Lemma 3.7.4. □

Remark 3.8.5. Both the statement and the proof of Lemma 3.8.4 carry over to the factorization setting.

3.8.6. We now claim:

Proposition 3.8.7. *For a fixed $\underline{x} \in \mathrm{Ran}$, the functor $\Upsilon_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}}$ is an equivalence.*

Proof. We will use the direct product decomposition of Sect. 3.4.

First, it is easy to see that the fact that $\Upsilon_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}}}$ is an equivalence implies that the functor

$$\Upsilon_{(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}}} : \mathrm{QCoh}((\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}})$$

is also an equivalence.

Since the category $\mathrm{QCoh}(\check{\mathfrak{g}}_0^{\wedge} / \mathrm{Ad}(\check{G}))$ is dualizable, the $\boxtimes$ functor

$$\mathrm{QCoh}(\check{\mathfrak{g}}_0^{\wedge} / \mathrm{Ad}(\check{G})) \otimes \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}) \rightarrow \mathrm{QCoh}(\check{\mathfrak{g}}_0^{\wedge} / \mathrm{Ad}(\check{G}) \times \mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})$$

is an equivalence.

The $\boxtimes$ functor

$$\mathrm{IndCoh}(\check{\mathfrak{g}}_0^{\wedge} / \mathrm{Ad}(\check{G})) \otimes \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^!(\check{\mathfrak{g}}_0^{\wedge} / \mathrm{Ad}(\check{G}) \times \mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})$$

is an equivalence tautologically.

Since the functor

$$\Upsilon_{\check{\mathfrak{g}}_0^{\wedge} / \mathrm{Ad}(\check{G})} : \mathrm{QCoh}(\check{\mathfrak{g}}_0^{\wedge} / \mathrm{Ad}(\check{G})) \rightarrow \mathrm{IndCoh}(\check{\mathfrak{g}}_0^{\wedge} / \mathrm{Ad}(\check{G}))$$

is an equivalence, in order to prove that $\Upsilon_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}}$ is an equivalence, it suffices to show that

$$\Upsilon_{\check{\mathfrak{g}}_0^{\wedge} / \mathrm{Ad}(\check{G}) \times \mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} : \mathrm{QCoh}(\check{\mathfrak{g}}_0^{\wedge} / \mathrm{Ad}(\check{G}) \times \mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^!(\check{\mathfrak{g}}_0^{\wedge} / \mathrm{Ad}(\check{G}) \times \mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})$$

is an equivalence.

However, this follows from the fact that $\Upsilon_{(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}}}$ is an equivalence, combined with the existence of an isomorphism

$$\check{\mathfrak{g}}_0^{\wedge} / \mathrm{Ad}(\check{G}) \times \mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}} \simeq (\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}}.$$

□

3.8.8. The proof of Proposition 3.8.7 given above is specific to the situation when $\underline{x} \in \mathrm{Ran}$ is fixed. Yet, we propose:

Question 3.8.9. *Is the functor*

$$\Upsilon_{\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}} : \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$$

a factorization equivalence?

3.8.10. Note that we can write

$$\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}} \simeq (\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}} \times_{(\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}}} \mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{reg}}.$$

Hence, the functor of $!$ -pullback along

$$\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}} \rightarrow (\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}}$$

gives rise to a functor

$$(3.27) \quad \mathrm{QCoh}(\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{reg}})_{\mathrm{QCoh}((\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}})} \otimes \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}).$$

We claim:

Proposition 3.8.11. *The functor (3.27) is an equivalence.*

Proof. Given that the functors

$$\Upsilon_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \text{ and } \Upsilon_{(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}}},$$

as well as

$$\Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}} \text{ and } \Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mer}}}$$

are equivalences, in order to prove Proposition 3.8.7, it suffices to show that the functor

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{reg}})_{\mathrm{QCoh}((\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}})} \otimes \mathrm{QCoh}((\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}}) \rightarrow \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})$$

is an equivalence.

However, this follows from the fact that the prestack

$$(\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}} \simeq \check{\mathfrak{g}}_0^{\wedge} / \mathrm{Ad}(\check{G})$$

is *passable* (see [GaRo3, Chapter 3, Proposition 3.5.3]). □

3.8.12. As in the case of Proposition 3.8.7, the assertion of Proposition 3.8.11 is specific to the situation when $\underline{x} \in \mathrm{Ran}$ is fixed. Parallel to Question 3.8.9, we propose:

Question 3.8.13. *Is the functor*

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}})_{\mathrm{QCoh}((\mathrm{LS}_{\check{G}}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}})} \otimes \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$$

a factorization equivalence?

4. DIGRESSION: IndCoh^* VIA FACTORIZATION ALGEBRAS

In this section we discuss the approach to factorization categories arising in the local Langlands theory, on both the geometric and spectral sides, as *factorization modules* over *factorization algebras*²⁴.

This approach is most efficient when we want to cross the Langlands bridge, i.e., map a category on the geometric side and a category on the spectral side to one another. Indeed, it is often possible to compare the corresponding factorization algebras directly (a prominent example of this is the Feigin-Frenkel isomorphism, see Theorem 5.1.2).

However, this approach comes with a caveat: typically, the given representation-theoretic or algebro-geometric category will *not* be exactly equivalent to the corresponding category of factorization modules. Rather, the two will be related by a *renormalization procedure*. Most often, this will be manifested by the fact that both sides will be endowed with t-structures, and the corresponding *eventually coconnective* subcategories will be equivalent on the nose.

²⁴The factorization algebras in question may be either plain ones (i.e., in Vect) or in some simpler or better understood factorization categories.

We apply these ideas to construct a *categorical* action of the Feigin-Frenkel center on Kac-Moody modules at the critical level; see Sect. 4.6.

4.1. Factorization algebras and modules.

4.1.1. Let $\mathcal{A}$ be a unital factorization algebra. To it we can attach a *lax* factorization category

$$\mathcal{A}\text{-mod}^{\text{fact}}$$

of unital $\mathcal{A}$ -factorization modules (see Sect. B.11.12).

It comes equipped with a conservative forgetful functor

$$\mathbf{oblv}_{\mathcal{A}} : \mathcal{A}\text{-mod}^{\text{fact}} \rightarrow \mathbf{Vect}.$$

4.1.2. By definition, the value of $\mathcal{A}\text{-mod}^{\text{fact}}$ over a given $\underline{x} \in \text{Ran}$ is the category

$$\mathcal{A}\text{-mod}_{\underline{x}}^{\text{fact}}$$

of factorization $\mathcal{A}$ -modules at $\underline{x}$.

In general, we cannot say much about homological properties of the category $\mathcal{A}\text{-mod}_{\underline{x}}^{\text{fact}}$. In particular, we do not know whether it is compactly generated.

4.1.3. This is also reflected by the following phenomenon:

For a pair of disjoint points $\underline{x}_1$ and $\underline{x}_2$, the lax factorization structure on $\mathcal{A}\text{-mod}^{\text{fact}}$ recovers the naturally defined functor

$$\mathcal{A}\text{-mod}_{\underline{x}_1}^{\text{fact}} \otimes \mathcal{A}\text{-mod}_{\underline{x}_2}^{\text{fact}} \rightarrow \mathcal{A}\text{-mod}_{\underline{x}_1 \sqcup \underline{x}_2}^{\text{fact}}.$$

However, it is not clear whether this functor is an equivalence. (If it were, and if this were true in families over $\underline{x}_1, \underline{x}_2$ moving over Ran , this would mean that the lax factorization structure on $\mathcal{A}\text{-mod}^{\text{fact}}$ is strict.)

4.1.4. Assume for a moment that $\mathcal{A}$ is connective, i.e., $\mathbf{oblv}^l(\mathcal{A}_X) \in \text{QCoh}(X)$ is connective. Then the category $\mathcal{A}\text{-mod}^{\text{fact}}$ carries a (uniquely defined) t-structure (see Sect. B.11.11 for what this means in the factorization setting), for which the functor $\mathbf{oblv}_{\mathcal{A}}$ is t-exact, see Sect. B.11.15.

In addition, it follows from the definition that $\mathcal{A}\text{-mod}^{\text{fact}}$ is *left-complete* in its t-structure.

Remark 4.1.5. The left-completeness of $\mathcal{A}\text{-mod}_{\underline{x}}^{\text{fact}}$ is an indication of its failure of compact generation:

Let $\mathbf{C}$ be a category, equipped with a t-structure, in which it is left-complete. Then every object $\mathbf{c} \in \mathbf{C}^c$ is of bounded projective dimension, i.e., the functor $\mathcal{H}om_{\mathbf{C}}(\mathbf{c}, -)$ is of bounded cohomological amplitude. However, typically, the category $\mathcal{A}\text{-mod}_{\underline{x}}^{\text{fact}}$ does not contain any objects of bounded projective dimension.

4.1.6. Here is how factorization algebras and modules will typically arise in this paper. Let $\mathbf{A}$ be a (unital) factorization category equipped with a (lax unital) factorization functor

$$F : \mathbf{A} \rightarrow \mathbf{Vect}.$$

Then $F(\mathbf{1}_{\mathbf{A}})$ is a factorization algebra (in $\mathbf{Vect}$). Moreover, the functor F naturally upgrades to a (factorization) functor, denoted

$$F^{\text{enh}} : \mathbf{A} \rightarrow F(\mathbf{1}_{\mathbf{A}})\text{-mod}^{\text{fact}},$$

see Lemma C.15.3.

Remark 4.1.7. The factorization structure on F^{enh} means for example that for disjoint points $\underline{x}_1, \underline{x}_2 \in \text{Ran}$, the diagram

$$\begin{array}{ccc} \mathbf{A}_{\underline{x}_1} \otimes \mathbf{A}_{\underline{x}_2} & \xrightarrow{\sim} & \mathbf{A}_{\underline{x}_1 \sqcup \underline{x}_2} \\ F_{\underline{x}_1} \otimes F_{\underline{x}_2} \downarrow & & \downarrow F_{\underline{x}_1 \sqcup \underline{x}_2} \\ F(\mathbf{1}_{\mathbf{A}})\text{-mod}_{\underline{x}_1}^{\text{fact}} \otimes F(\mathbf{1}_{\mathbf{A}})\text{-mod}_{\underline{x}_2}^{\text{fact}} & \longrightarrow & F(\mathbf{1}_{\mathbf{A}})\text{-mod}_{\underline{x}_1 \sqcup \underline{x}_2}^{\text{fact}} \end{array}$$

commutes (even though the bottom horizontal arrow is not in general an equivalence).

4.1.8. If in the situation of Sect. 4.1.6, the category $\mathbf{C}$ is equipped with a t-structure so that $\mathbf{1}_{\mathbf{C}}$ lies in the heart, and the functor F is t-exact, we obtain that $F(\mathbf{1}_{\mathbf{C}})$ is a *classical* factorization algebra, so that the category $F(\mathbf{1}_{\mathbf{C}})\text{-mod}^{\text{fact}}$ carries a t-structure.

In this case, the functor F^{enh} is obviously t-exact.

4.1.9. More generally, if $F : \mathbf{C}_1 \rightarrow \mathbf{C}_2$ is a lax unital factorization functor between unital factorization categories, the object

$$F(\mathbf{1}_{\mathbf{C}_1}) \in \mathbf{C}_2$$

has a natural structure of factorization algebra, and the functor F upgrades to a functor

$$F^{\text{enh}} : \mathbf{C}_1 \rightarrow F(\mathbf{1}_{\mathbf{C}_1})\text{-mod}^{\text{fact}}(\mathbf{C}_2).$$

4.2. **Kac-Moody modules as factorization modules.** Here is a typical example of the paradigm described in Sects. 4.1.6-4.1.8.

4.2.1. Consider the tautological forgetful functor

$$\mathbf{oblv}_{\widehat{\mathfrak{g}}} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa} \rightarrow \mathbf{Vect}.$$

Recall that $\mathbb{V}_{\mathfrak{g},\kappa}$ denotes the factorization algebra $\mathbf{oblv}_{\widehat{\mathfrak{g}}}(\text{Vac}(G)_{\kappa})$, so that $\mathbf{oblv}_{\widehat{\mathfrak{g}}}$ upgrades to a (t-exact) functor:

$$(4.1) \quad \mathbf{oblv}_{\widehat{\mathfrak{g}}}^{\text{enh}} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa} \rightarrow \mathbb{V}_{\mathfrak{g},\kappa}\text{-mod}^{\text{fact}}.$$

4.2.2. We have the following basic observation:

Lemma 4.2.3.

(a) *The functor $\mathbf{oblv}_{\widehat{\mathfrak{g}}}^{\text{enh}}$ of (4.1) induces an equivalence between the eventually coconnective subcategories of the two sides.*

(b) *The essential image of the subcategory of compact objects of $\widehat{\mathfrak{g}}\text{-mod}_{\kappa}$ under $\mathbf{oblv}_{\widehat{\mathfrak{g}}}^{\text{enh}}$ is contained in $(\mathbb{V}_{\mathfrak{g},\kappa}\text{-mod}^{\text{fact}})^{>-\infty}$.*

4.2.4. The rest of this subsection is devoted to the proof of Lemma 4.2.3. We will prove a pointwise version for $\underline{x} = x \in \text{Ran}$. The factorization version is just a variant of this in families.

Let $\mathbb{V}_{\mathfrak{g},\kappa}^{\text{ch}}$ be the *chiral algebra* corresponding to $\mathbb{V}_{\mathfrak{g},\kappa}$. I.e., as a D-module on X ,

$$\mathbb{V}_{\mathfrak{g},\kappa}^{\text{ch}} = \mathbb{V}_{\mathfrak{g},\kappa,X}[-1].$$

First, by [BD1, Proposition 3.4.19] (see [FraG] for the derived version), we have an equivalence

$$\mathbb{V}_{\mathfrak{g},\kappa}\text{-mod}_x^{\text{fact}} \simeq \mathbb{V}_{\mathfrak{g},\kappa}^{\text{ch}}\text{-mod}_x^{\text{ch}},$$

which commutes with the forgetful functors of both sides into $\mathbf{Vect}$, and hence preserves the t-structures on the two sides.

4.2.5. Let $L_{\mathfrak{g},\kappa}$ be the Lie-* algebra

$$\omega_X \oplus (\mathfrak{g} \otimes D_X)$$

of [BD1, Sect. 2.5.9]. Then by [BD1, Proposition 3.7.17] (which applies as-is in the derived setting), we have

$$\mathbb{V}_{\mathfrak{g},\kappa}^{\text{ch}}\text{-mod}_x^{\text{ch}} \simeq L_{\mathfrak{g},\kappa}\text{-mod}_x^{\text{ch}},$$

which commutes with the forgetful functors of both sides into $\mathbf{Vect}$, and hence preserves the t-structures on the two sides.

4.2.6. By the construction of $\widehat{\mathfrak{g}}\text{-mod}_{\kappa,x}$ in [Ra5] (or, equivalently, in [FG6, Sect. 23.1]), we have an embedding

$$(4.2) \quad (\widehat{\mathfrak{g}}\text{-mod}_{\kappa,x})^c \hookrightarrow (L_{\mathfrak{g},\kappa}\text{-mod}_x^{\text{ch}})^{>-\infty},$$

which satisfies the conditions of [FG6, Sect. 22.1.4] (see Sect. D.3.6 for an explanation of why this happens).

Hence,

$$(\widehat{\mathfrak{g}}\text{-mod}_{\kappa,x})^{>-\infty} \rightarrow (L_{\mathfrak{g},\kappa}\text{-mod}_x^{\text{ch}})^{>-\infty}$$

is an equivalence by [FG6, Proposition 22.1.5 and 22.2.1].

□[Lemma 4.2.3]

4.3. The case of commutative factorization algebras.

4.3.1. Let $\mathcal{Y}$ be an affine D-scheme over X , i.e., $\mathcal{Y} = \text{Spec}_X(A)$, where $A \in \text{ComAlg}(\text{D-mod}(X))$ with $\text{oblv}^l(A) \in \text{QCoh}(X)^{\leq 0}$.

Let $\mathcal{A} \in \text{ComAlg}(\text{FactAlg}^{\text{unl}}(X))$ denote the corresponding commutative factorization algebra, i.e., $\mathcal{A} := \text{Fact}(A)$, see Sect. B.10.2.

4.3.2. Let $\mathfrak{L}_{\nabla}^+(\mathcal{Y})$ denote the affine factorization scheme corresponding to $\mathcal{Y}$ (see Sect. B.4.2), i.e., the fiber $\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\underline{x}}$ of $\mathfrak{L}_{\nabla}^+(\mathcal{Y})$ at $\underline{x} \in \text{Ran}$ is the space

$$(4.3) \quad \text{Sect}_{\nabla}(\mathcal{D}_{\underline{x}}, \mathcal{Y}).$$

According to Sect. C.8.11, for $\mathcal{Z} \rightarrow \text{Ran}$,

$$\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\mathcal{Z}} = \text{Spec}_{\mathcal{Z}}(\mathcal{A}_{\mathcal{Z}}).$$

In particular, for $\underline{x} = \{x_1, \dots, x_n\}$, we have

$$\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\underline{x}} \simeq \prod_{i=1, \dots, n} \mathcal{Y}_{x_i}, \quad \underline{x} = \{x_1, \dots, x_n\}.$$

(Note that this is compatible with (4.3), since for a singleton $\underline{x} = \{x\}$, we have $\text{Sect}_{\nabla}(\mathcal{D}_x, \mathcal{Y}) \simeq \mathcal{Y}_x$, in agreement with (4.3)).

4.3.3. Consider the corresponding factorization category $\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}))$ (see Sect. B.13.2), so that for $\underline{x} \in \text{Ran}$, we have

$$\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}))_{\underline{x}} := \text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\underline{x}}).$$

The factorization category $\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}))$ is unital, with the structure sheaf $\mathcal{O}_{\mathfrak{L}_{\nabla}^+(\mathcal{Y})}$ being the factorization unit.

4.3.4. The functor of global sections $\Gamma(\mathfrak{L}_{\nabla}^+(\mathcal{Y}), -)$ sends

$$\mathcal{O}_{\mathfrak{L}_{\nabla}^+(\mathcal{Y})} \mapsto \mathcal{A}$$

and induces an equivalence

$$\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y})) \simeq \mathcal{A}\text{-mod}^{\text{com}}.$$

4.3.5. Let $\mathfrak{L}_{\nabla}(\mathcal{Y})$ denote the factorization D-ind-scheme that attaches to a point $\underline{x} \in \text{Ran}$ the space

$$\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}} := \text{Sect}_{\nabla}(\mathcal{D}_{\underline{x}}^{\times}, \mathcal{Y}),$$

see Sect. B.4.6.

We consider the corresponding lax factorization category $\text{QCoh}(\mathfrak{L}_{\nabla}(\mathcal{Y}))$ (see Sect. B.13.2), so that for $\underline{x} \in \text{Ran}$, we have

$$\text{QCoh}(\mathfrak{L}_{\nabla}(\mathcal{Y}))_{\underline{x}} := \text{QCoh}(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}).$$

For a general $\mathcal{Y}$, this category may be quite ill-behaved (basically, because the category of quasi-coherent sheaves on an ind-scheme may be quite unwieldy); in particular, it is not clear whether $\text{QCoh}(\mathfrak{L}_{\nabla}(\mathcal{Y}))$ is unital.

4.3.6. Recall now that we can also consider the factorization category

$$\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})),$$

see Sect. B.13.8.

Its factorization unit is the direct image of the structure sheaf on $\mathfrak{L}_{\nabla}^+(\mathcal{Y})$ along the tautological closed embedding $\mathfrak{L}_{\nabla}^+(\mathcal{Y}) \xrightarrow{\iota} \mathfrak{L}_{\nabla}(\mathcal{Y})$. By a slight abuse of notation, we will denote it by the same symbol $\mathcal{O}_{\mathfrak{L}_{\nabla}^+(\mathcal{Y})}$.

4.3.7. The operation of taking global sections is a (t-exact²⁵) factorization functor

$$\Gamma(\mathfrak{L}_{\nabla}(\mathcal{Y}), -) : \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})) \rightarrow \mathrm{Vect}.$$

Hence, by Sects. 4.1.6-4.1.8, the functor $\Gamma(\mathfrak{L}_{\nabla}(\mathcal{Y}), -)$ upgrades to a (t-exact) lax unital factorization functor

$$(4.4) \quad \Gamma(\mathfrak{L}_{\nabla}(\mathcal{Y}), -)^{\mathrm{enh}} : \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})) \rightarrow \mathcal{A}\text{-mod}^{\mathrm{fact}}.$$

4.3.8. We have the following basic assertion:

Theorem 4.3.9. *Assume that $\mathcal{Y}$ is almost finitely presented in the D -sense²⁶. Then the functor (4.4) induces an equivalence between the eventually coconnective subcategories of the two sides.*

The proof of this theorem will be given in Sect. D.5. We note that the assertion of the theorem would be *false* without the finite presentation hypothesis, see Sect. D.6.

4.3.10. Recall that if $\mathcal{Z}$ is an ind-scheme, we have a well-defined (t-exact) functor

$$\Psi_{\mathcal{Z}} : \mathrm{IndCoh}^*(\mathcal{Z}) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Z}),$$

which induces an equivalence

$$\mathrm{IndCoh}^*(\mathcal{Z})^{>-\infty} \xrightarrow{\sim} \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Z})^{>-\infty},$$

see Lemma A.8.10.

Furthermore, if $\mathcal{Z}$ is ind-placid, $\Psi_{\mathcal{Z}}$ gives rise to an equivalence between $\mathrm{IndCoh}^*(\mathcal{Z})^c$ and the subcategory of almost compact objects in $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Z})^{>-\infty}$.

Note also that the composition

$$\Gamma(\mathcal{Z}, -) \circ \Psi_{\mathcal{Z}} : \mathrm{IndCoh}^*(\mathcal{Z}) \rightarrow \mathrm{Vect}$$

is the functor $\Gamma^{\mathrm{IndCoh}}(\mathcal{Z}, -)$ of IndCoh-global sections.

4.3.11. Assume for a moment that $\mathfrak{L}_{\nabla}(\mathcal{Y})$ is ind-placid. Applying Sect. 4.3.10, we obtain a factorization functor

$$\Gamma^{\mathrm{IndCoh}}(\mathfrak{L}_{\nabla}(\mathcal{Y}), -) \simeq \Gamma(\mathfrak{L}_{\nabla}(\mathcal{Y}), -) \circ \Psi_{\mathfrak{L}_{\nabla}(\mathcal{Y})}, \quad \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathcal{Y})) \rightarrow \mathrm{Vect}$$

and its enhancement

$$(4.5) \quad \Gamma^{\mathrm{IndCoh}}(\mathfrak{L}_{\nabla}(\mathcal{Y}), -)^{\mathrm{enh}} : \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathcal{Y})) \rightarrow \mathcal{O}_{\mathcal{Y}}\text{-mod}^{\mathrm{fact}}.$$

Combining with Theorem 4.3.9 we obtain:

Corollary 4.3.12.

- (a) *The functor $\Gamma^{\mathrm{IndCoh}}(\mathfrak{L}_{\nabla}(\mathcal{Y}), -)^{\mathrm{enh}}$ of (4.5) is t-exact and induces an equivalence between the eventually coconnective subcategories of the two sides.*
- (b) *The essential image of the subcategory of compact objects in $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathcal{Y}))$ under the functor $\Gamma^{\mathrm{IndCoh}}(\mathfrak{L}_{\nabla}(\mathcal{Y}), -)^{\mathrm{enh}}$ is contained in $(\mathcal{O}_{\mathcal{Y}}\text{-mod}^{\mathrm{fact}})^{>-\infty}$.*

²⁵See Sect. A.2.8 for the definition of the t-structure on $\mathrm{QCoh}_{\mathrm{co}}$ of an ind-scheme.

²⁶See Sect. B.6.3 for what this means.

Remark 4.3.13. Point (b) in Corollary 4.3.12 can be strengthened as follows: the essential image of $\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(\mathcal{Y}))^c$ under $\Gamma^{\mathrm{IndCoh}}(\mathfrak{L}_\nabla(\mathcal{Y}), -)^{\mathrm{enh}}$ equals the category of *almost compact* objects²⁷ in $(\mathcal{O}_{\mathcal{Y}\text{-mod}}^{\mathrm{fact}})^{>-\infty}$.

Note that from Corollary 4.3.12 allows us to recover the (factorization) category $\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(\mathcal{Y}))$ from $\mathcal{O}_{\mathcal{Y}\text{-mod}}^{\mathrm{fact}}$ equipped with its t-structure:

Namely, we can identify $\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(\mathcal{Y}))$ with the ind-completion of the category of almost compact objects in $(\mathcal{O}_{\mathcal{Y}\text{-mod}}^{\mathrm{fact}})^{\geq -\infty}$.

4.4. Recovering IndCoh^* of opers as factorization modules.

4.4.1. The setup of Sect. 4.3.11 is directly applicable to the case when $\mathcal{Y} = \mathrm{Op}_{\check{G}}$ (the D-afp assumption is satisfied by Sect. 3.1.7), so that

$$\mathfrak{L}_\nabla^+(\mathcal{Y}) = \mathrm{Op}_{\check{G}}^{\mathrm{reg}} \text{ and } \mathfrak{L}_\nabla(\mathcal{Y}) = \mathrm{Op}_{\check{G}}^{\mathrm{mer}}.$$

By a slight abuse of notation, we will denote by $\mathcal{O}_{\mathrm{Op}_{\check{G}}}^{\mathrm{reg}}$ (rather than $\Gamma(\mathrm{Op}_{\check{G}}^{\mathrm{reg}}, \mathcal{O}_{\mathrm{Op}_{\check{G}}}^{\mathrm{reg}})$) the corresponding factorization algebra in Vect .

In particular, we obtain that the functor

$$\Gamma^{\mathrm{IndCoh}}(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}, -) : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}) \rightarrow \mathrm{Vect}$$

upgrades to a (t-exact) functor

$$(4.6) \quad \Gamma^{\mathrm{IndCoh}}(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}, -)^{\mathrm{enh}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}) \rightarrow \mathcal{O}_{\mathrm{Op}_{\check{G}}}^{\mathrm{reg}\text{-mod}}^{\mathrm{fact}},$$

and we have:

Corollary 4.4.2.

(a) *The functor (4.6) induces an equivalence between the corresponding eventually coconnective (a.k.a. bounded below) subcategories.*

(b) *The essential image of the subcategory of compact objects in $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mer}})$ under the functor (4.6) is contained in $(\mathcal{O}_{\mathrm{Op}_{\check{G}}}^{\mathrm{reg}\text{-mod}}^{\mathrm{fact}})^{>-\infty}$.*

We now consider the case of *monodromy-free* opers.

4.4.3. Direct image along the projection

$$\mathfrak{r}^{\mathrm{reg}} : \mathrm{Op}_{\check{G}}^{\mathrm{reg}} \rightarrow \mathrm{LS}_{\check{G}}^{\mathrm{reg}}$$

defines a t-exact (lax unital factorization) functor

$$(4.7) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{reg}}) \simeq \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{reg}}) \xrightarrow{\mathfrak{r}_*^{\mathrm{reg}}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{reg}}) \simeq \mathrm{Rep}(\check{G}).$$

Denote

$$(4.8) \quad R_{\check{G}, \mathrm{Op}} := \mathfrak{r}_*^{\mathrm{reg}}(\mathcal{O}_{\mathrm{Op}_{\check{G}}}^{\mathrm{reg}}).$$

This is naturally a commutative factorization algebra in $\mathrm{Rep}(\check{G})$.

4.4.4. Explicitly,

$$R_{\check{G}, \mathrm{Op}} := (\Gamma(\mathrm{Op}_{\check{G}}^{\mathrm{reg}}, -) \otimes \mathrm{Id}) \circ ((\mathfrak{r}^{\mathrm{reg}})^* \otimes \mathrm{Id})(R_{\check{G}}),$$

where

$$R_{\check{G}} \in \mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G})$$

is the regular representation.

²⁷Recall that an object $\mathbf{c}$ in a DG category $\mathbf{C}$ equipped with a t-structure (assumed compatible with filtered colimits) is said to be *almost compact* if the functor $\mathcal{H}\mathrm{om}_{\mathbf{C}}(\mathbf{c}, -)$ commutes with filtered colimits on $\mathbf{C}^{\geq -n}$ for all n .

4.4.5. Consider now the factorization functor

$$(4.9) \quad \mathfrak{r}_*^{\text{IndCoh}} : \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \rightarrow \text{QCoh}(\text{LS}_{\check{G}}^{\text{reg}}) \simeq \text{Rep}(\check{G}).$$

Note that we can interpret $R_{\check{G}, \text{Op}}$ also as

$$\mathfrak{r}_*^{\text{IndCoh}}(\mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}}),$$

where by a slight abuse of notation we view $\mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}}$ as an object of $\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})$ using

$$\text{QCoh}(\text{Op}_{\check{G}}^{\text{reg}}) \xrightarrow{\Psi_{\text{Op}_{\check{G}}^{\text{reg}}}} \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{reg}}) \xrightarrow{(\iota^{+, \text{mon-free}})^{\text{IndCoh}}} \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}).$$

4.4.6. The functor (4.9) naturally upgrades to a t-exact factorization functor

$$(4.10) \quad (\mathfrak{r}_*^{\text{IndCoh}})^{\text{enh}} : \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \rightarrow R_{\check{G}, \text{Op}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G})).$$

We will prove:

Proposition 4.4.7.

- (a) *The functor (4.10) induces an equivalence between the eventually coconnective subcategories of the two sides.*
- (b) *The essential image of the subcategory $\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})^c$ under (4.10) is contained in $(R_{\check{G}, \text{Op}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G})))^{>-\infty}$.*

4.5. **Proof of Proposition 4.4.7.** We will provide a general framework, of which Proposition 4.4.7 is a particular case. The assertion is local, so we can assume that X is affine.

4.5.1. Let $\mathcal{Y}$ be an affine D-scheme, and consider $\mathfrak{L}_{\nabla}^+(\mathcal{Y}) := \mathcal{T}^+$ as a factorization space. Let $\mathcal{T}$ be a factorization space, equipped with a map

$$\iota : \mathcal{T}^+ \rightarrow \mathcal{T}$$

that extends to a *unital-in-correspondences structure* on $\mathcal{T}$ relative to $\mathcal{T}^+$ (see Sect. C.10.6).

Consider the factorization category

$$\text{QCoh}_{\text{co}}(\mathcal{T}),$$

see Sect. B.13.8.

Note its factorization unit is given by

$$\iota_*(\mathcal{O}_{\mathcal{T}^+}).$$

We will assume:

- (i) $\mathcal{T}$ is an ind-placid ind-scheme.²⁸

4.5.2. Let us be given a D-prestack $\mathcal{Y}_0$, equipped with a map

$$f : \mathcal{Y} \rightarrow \mathcal{Y}_0,$$

and an extension of the map

$$\mathcal{T}^+ = \mathfrak{L}_{\nabla}^+(\mathcal{Y}) \xrightarrow{\mathfrak{L}^+(f)} \mathfrak{L}_{\nabla}^+(\mathcal{Y}_0) =: \mathcal{T}_0^+$$

to a map

$$\mathcal{T} \xrightarrow{\mathfrak{L}(f)} \mathcal{T}_0^+.$$

We will assume:

- (ii) $\mathcal{T}_0^+$ has an affine diagonal.

Note that assumptions (i) and (ii) imply in particular that the map $\mathfrak{L}(f)$ is ind-schematic.

²⁸See Sect. B.1.9 for what this means.

4.5.3. Consider the functor

$$(4.11) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathcal{T}) \xrightarrow{\mathfrak{L}(f)_*} \mathrm{QCoh}_{\mathrm{co}}(\mathcal{T}_0^+) \xrightarrow{\Omega_{\mathfrak{L}^+}(\mathcal{Y}_0)} \mathrm{QCoh}(\mathcal{T}_0^+),$$

where $\Omega_{\mathfrak{L}^+}(\mathcal{Y}_0)$ is as in (A.8). By a slight abuse of notation, we will denote the composite functor in (4.11) by the same symbol $\mathfrak{L}(f)_*$.

The functor $\mathfrak{L}(f)_*$ of (4.11) upgrades to a (factorization) functor

$$(4.12) \quad (\mathfrak{L}(f)_*)^{\mathrm{enh}} : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{T}) \rightarrow \mathfrak{L}(f)_* \circ \iota_*(\mathcal{O}_{\mathcal{T}+})\text{-mod}^{\mathrm{fact}}(\mathrm{QCoh}(\mathcal{T}_0^+)) = \mathfrak{L}^+(\mathcal{Y}_0)_*(\mathcal{O}_{\mathcal{T}+})\text{-mod}^{\mathrm{fact}}(\mathrm{QCoh}(\mathcal{T}_0^+)).$$

4.5.4. We now make an additional assumptions:

- (iii) The prestack $\mathcal{Y}_0$ admits a map

$$g : \tilde{\mathcal{Y}}_0 \rightarrow \mathcal{Y}_0,$$

where $\tilde{\mathcal{Y}}_0$ is an affine D-scheme, such that the map

$$\tilde{\mathcal{T}}_0^+ := \mathfrak{L}_{\nabla}^+(\tilde{\mathcal{Y}}_0) \xrightarrow{\mathfrak{L}^+(g)} \mathfrak{L}_{\nabla}^+(\mathcal{Y}_0) =: \mathcal{T}_0^+$$

is an fpqc cover.²⁹

- (iv) For $\tilde{\mathcal{Y}} := \mathcal{Y} \times_{\mathcal{Y}_0} \tilde{\mathcal{Y}}_0$, the resulting map

$$\tilde{\mathcal{T}}^+ := \mathcal{T}^+ \times_{\mathcal{T}_0^+} \tilde{\mathcal{T}}_0^+ \xrightarrow{\iota^{\mathrm{id}}} \mathcal{T} \times_{\mathcal{T}_0^+} \tilde{\mathcal{T}}_0^+ =: \tilde{\mathcal{T}}$$

identifies with

$$\mathfrak{L}_{\nabla}^+(\tilde{\mathcal{Y}}) \rightarrow \mathfrak{L}_{\nabla}(\tilde{\mathcal{Y}}) \times_{\mathfrak{L}_{\nabla}(\mathcal{Y}_0)} \mathfrak{L}_{\nabla}^+(\mathcal{Y}_0).$$

Note that assumption (iii) implies, in particular, that the category $\mathrm{QCoh}(\mathcal{T}_0^+)$ has a well-behaved t-structure: it is characterized uniquely by the property that the functor

$$(\mathfrak{L}^+(g))^* : \mathrm{QCoh}(\mathcal{T}_0^+) \rightarrow \mathrm{QCoh}(\tilde{\mathcal{T}}_0^+)$$

is t-exact.

It follows from assumption (iv) and base change that the functor $\mathfrak{L}(f)_*$ of (4.11) is t-exact (see Sect. B.13.8, where the t-structure on the left-hand side is defined). Hence, $(\mathfrak{L}(f)_*)^{\mathrm{enh}}$ is also t-exact.

4.5.5. Finally, we make the following assumption:

- (v) The map

$$\tilde{\mathcal{Y}} := \mathcal{Y} \times_{\mathcal{Y}_0} \tilde{\mathcal{Y}}_0 \xrightarrow{\tilde{f}} \tilde{\mathcal{Y}}_0$$

is D-afp (see Sect. B.6 for what this means).

²⁹See Sect. B.1.9 for what this means.

4.5.6. We claim:

Corollary 4.5.7. *Under the above assumptions, the functor (4.12) induces an equivalence between the eventually coconnective subcategories of the two sides.*

Proof. Let $\tilde{\mathcal{Y}}_0^\bullet$ be the Čech nerve of the map g . Denote

$$\tilde{\mathcal{Y}}^\bullet := \mathcal{Y} \times_{\mathcal{Y}_0} \tilde{\mathcal{Y}}_0^\bullet$$

and

$$\tilde{\mathcal{T}}_0^{+, \bullet} := \mathcal{L}_\nabla^+(\tilde{\mathcal{Y}}_0^\bullet), \quad \tilde{\mathcal{T}}^{+, \bullet} := \mathcal{T}^+ \times_{\mathcal{T}_0^+} \tilde{\mathcal{T}}_0^{+, \bullet} \simeq \mathcal{L}_\nabla^+(\tilde{\mathcal{Y}}^\bullet)$$

and

$$\tilde{\mathcal{T}}^\bullet := \mathcal{T}^+ \times_{\mathcal{T}_0^+} \tilde{\mathcal{T}}_0^{+, \bullet} \simeq \mathcal{L}_\nabla(\tilde{\mathcal{Y}}^\bullet) \times_{\mathcal{L}_\nabla(\tilde{\mathcal{Y}}_0^\bullet)} \mathcal{L}_\nabla^+(\tilde{\mathcal{Y}}_0^\bullet).$$

Consider the resulting maps:

$$\tilde{\mathcal{T}}^\bullet \xrightarrow{\mathcal{L}(\tilde{f})^\bullet} \tilde{\mathcal{T}}_0^{+, \bullet} \text{ and } \tilde{\mathcal{T}}^{+, \bullet} \xrightarrow{\mathcal{L}^+(\tilde{f})^\bullet} \tilde{\mathcal{T}}_0^{+, \bullet}.$$

First, by fpqc descent we have a t-exact equivalence

$$\mathrm{QCoh}(\mathcal{T}_0^+) \simeq \mathrm{Tot}(\mathrm{QCoh}(\tilde{\mathcal{T}}_0^{+, \bullet})),$$

from which we obtain a t-exact equivalence

$$\mathcal{L}^+(f)_*(\mathcal{O}_{\mathcal{T}^+})\text{-mod}^{\mathrm{fact}}(\mathrm{QCoh}(\mathcal{T}_0^+)) \simeq \mathrm{Tot} \left(\mathcal{L}^+(\tilde{f})_*(\mathcal{O}_{\tilde{\mathcal{T}}^+, \bullet})\text{-mod}^{\mathrm{fact}}(\mathrm{QCoh}(\tilde{\mathcal{T}}_0^{+, \bullet})) \right)$$

and hence

$$\mathcal{L}^+(f)_*(\mathcal{O}_{\mathcal{T}^+})\text{-mod}^{\mathrm{fact}}(\mathrm{QCoh}(\mathcal{T}_0^+))^{>-\infty} \simeq \mathrm{Tot} \left(\mathcal{L}^+(\tilde{f})_*(\mathcal{O}_{\tilde{\mathcal{T}}^+, \bullet})\text{-mod}^{\mathrm{fact}}(\mathrm{QCoh}(\tilde{\mathcal{T}}_0^{+, \bullet}))^{>-\infty} \right).$$

Next, by assumption (i) in Sect. 4.5.1 and Proposition A.3.3, the functor

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{T})^{>-\infty} \rightarrow \mathrm{Tot} \left(\mathrm{QCoh}_{\mathrm{co}}(\tilde{\mathcal{T}}^\bullet)^{>-\infty} \right)$$

is also an equivalence.

Finally, by assumption (v) and a relative version of Theorem 4.3.9, the functor

$$\begin{aligned} \mathrm{QCoh}_{\mathrm{co}}(\tilde{\mathcal{T}}^\bullet)^{>-\infty} &\simeq \mathrm{QCoh}_{\mathrm{co}} \left(\mathcal{L}_\nabla(\tilde{\mathcal{Y}}^\bullet) \times_{\mathcal{L}_\nabla(\tilde{\mathcal{Y}}_0^\bullet)} \mathcal{L}_\nabla^+(\tilde{\mathcal{Y}}_0^\bullet) \right)^{>-\infty} \rightarrow \\ &\rightarrow \mathcal{L}^+(\tilde{f})_*(\mathcal{O}_{\tilde{\mathcal{T}}^+, \bullet})\text{-mod}^{\mathrm{fact}}(\mathrm{QCoh}(\tilde{\mathcal{T}}_0^{+, \bullet}))^{>-\infty} \simeq \mathcal{L}^+(\tilde{f})_*(\mathcal{O}_{\mathcal{L}_\nabla^+(\tilde{\mathcal{Y}}^\bullet)})\text{-mod}^{\mathrm{fact}}(\mathrm{QCoh}(\mathcal{L}_\nabla^+(\tilde{\mathcal{Y}}_0^\bullet)))^{>-\infty} \end{aligned}$$

is a term-wise equivalence.

Combining, we obtain that (4.12) is also an equivalence, as required. $\square$

4.5.8. Precomposing the equivalence (4.12) with the equivalence

$$\mathrm{IndCoh}^*(\mathcal{T})^{>-\infty} \xrightarrow{\Psi_{\mathcal{T}}} \mathrm{QCoh}_{\mathrm{co}}(\mathcal{T})^{>-\infty}$$

of Lemma A.8.10, we obtain that under assumptions (i)-(iv) above, the functor

$$((\mathcal{L}(f))_*^{\mathrm{IndCoh}})^{\mathrm{enh}} := (\mathcal{L}(f)_*)^{\mathrm{enh}} \circ \Psi_{\mathcal{T}}$$

induces an equivalence

$$(4.13) \quad \mathrm{IndCoh}^*(\mathcal{T})^{>-\infty} \rightarrow ((\mathcal{L}(f))_*^{\mathrm{IndCoh}})^{\mathrm{enh}}\text{-mod}^{\mathrm{fact}}(\mathrm{QCoh}(\mathcal{T}_0^+))^{>-\infty}.$$

4.5.9. We apply the above to

$$\mathcal{Y} = \mathrm{Op}_{\check{G}}, \quad \mathcal{Y}_0 = \mathrm{pt} / \check{G} \text{ and } \mathcal{T} := \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} := \mathrm{Op}_{\check{G}}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{LS}_{\check{G}}^{\mathrm{reg}}.$$

Hence, in order to deduce the assertion of Proposition 4.4.7, we have to show that conditions (i)-v) above hold.

4.5.10. Condition (i) says that $\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}$ is placid; this has been established in Sect. 3.2.6.

We take $\widetilde{\mathcal{Y}}_0 = \mathrm{pt}$ with the tautological map $\mathrm{pt} \rightarrow \mathrm{LS}_{\check{G}}^{\mathrm{reg}}$.

Condition (iii) is the content of Lemma B.7.4. Condition (ii) also follows from Lemma B.7.4, since the property of a map being affine can be checked fpqc-locally, and

$$\mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times \mathrm{LS}_{\check{G}}^{\mathrm{reg}}} (\mathrm{pt} \times \mathrm{pt}) \simeq \mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{pt} \simeq \mathfrak{L}_{\check{V}}^+(\check{G}).$$

Condition (iv) is automatic from the construction. Finally, condition (v) is the content of the next lemma:

Lemma 4.5.11. *The affine D-scheme $\mathrm{Op}_{\check{G}} \times_{\mathrm{pt} / \check{G}} \mathrm{pt}$ is D-afp.*

Proof. We have:

$$\mathrm{Op}_{\check{G}} \times_{\mathrm{pt} / \check{G}} \mathrm{pt} \simeq \mathrm{Op}_{\check{G}} \times_{\mathrm{Jets}(\check{\mathfrak{g}} \otimes \omega_X)} \mathrm{Jets}(\check{G}),$$

where:

- The map $\mathrm{Op}_{\check{G}} \rightarrow \mathrm{Jets}(\check{\mathfrak{g}} \otimes \omega_X)$ is well-defined (Zariski-locally on X) thanks to Sect. 3.1.4;
- The map $\mathrm{Jets}(\check{G}) \rightarrow \mathrm{Jets}(\check{\mathfrak{g}} \otimes \omega_X)$ is given by the gauge action on the trivial connection.

This makes the assertion of the lemma manifest, as

$$\mathrm{Op}_{\check{G}}, \mathrm{Jets}(\check{\mathfrak{g}} \otimes \omega_X) \text{ and } \mathrm{Jets}(\check{G})$$

are all D-afp. □

□[Proposition 4.4.7]

4.6. Action of the center on Kac-Moody modules.

4.6.1. Let $\mathfrak{z}_{\mathfrak{g}}$ be the (classical) center of $\mathbb{V}_{\mathfrak{g}, \mathrm{crit}}$, viewed as a plain factorization (chiral) algebra (i.e., a factorization algebra in Vect).

By construction, $\mathfrak{z}_{\mathfrak{g}}$ is a commutative factorization algebra. It acts as such on $\mathrm{Vac}(G)_{\mathrm{crit}} \in \mathrm{KL}(G)_{\mathrm{crit}}$. In particular, we obtain a map of factorization algebras in $\mathrm{KL}(G)_{\kappa}$:

$$(4.14) \quad \mathfrak{z}_{\mathfrak{g}} \otimes \mathrm{Vac}(G)_{\mathrm{crit}} \rightarrow \mathrm{Vac}(G)_{\mathrm{crit}}.$$

4.6.2. We will denote by the symbols $\mathrm{Spec}(\mathfrak{z}_{\mathfrak{g}})$ and “Spec”($\mathfrak{z}_{\mathfrak{g}}$) the corresponding factorization scheme and ind-scheme, respectively, see Sect. 4.3.

In this subsection we will construct the an action of $\mathrm{IndCoh}^!(\text{“Spec”}(\mathfrak{z}_{\mathfrak{g}}))$ on $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}$, compatible with factorization.

Remark 4.6.3. The existence of such an action at the level of abelian categories is essentially evident: the topological algebra of global functions on “Spec”($\mathfrak{z}_{\mathfrak{g}}$) maps to the center of the completed universal enveloping algebra of $U(\widehat{\mathfrak{g}}_{\mathrm{crit}})$.

At the derived level (for a fixed point $\underline{x} \in \mathrm{Ran}$) the construction of such an action was carried out in [FG6, Sect. 23.2-23.4] and [Ra5, Sect. 11] in the language of topological associative algebras.

The methods of *loc. cit.* could be adapted to the factorization setting. However, below we present a different construction. Even though it looks more complicated (at least more abstract), its advantage

is that it is compatible with the construction of Sect. 5.3, where we do not know how to make other methods work.

The construction of the action presented below has another advantage in that it is manifestly compatible with the action of $\mathfrak{L}(G)$, see Sect. 4.7 below.

4.6.4. Let $\mathcal{Z}$ be an ind-placid ind-scheme. The categories $\mathrm{IndCoh}^!(\mathcal{Z})$ and $\mathrm{IndCoh}^*(\mathcal{Z})$ are each mutually dual, with the pairing given by

$$\mathrm{IndCoh}^!(\mathcal{Z}) \otimes \mathrm{IndCoh}^*(\mathcal{Z}) \xrightarrow{!} \mathrm{IndCoh}^*(\mathcal{Z}) \xrightarrow{\Gamma^{\mathrm{IndCoh}(\mathcal{Z}, -)}} \mathrm{Vect}.$$

Hence, we can view $\mathrm{IndCoh}^*(\mathcal{Z})$ as a comonoidal category. Moreover, the datum of an action of $\mathrm{IndCoh}^!(\mathcal{Z})$ on a (factorization) category $\mathbf{C}$ is equivalent to the datum of a coaction of $\mathrm{IndCoh}^*(\mathcal{Z})$ on $\mathbf{C}$.

4.6.5. As we shall see shortly (see Theorem 5.1.2 and Sect. 3.2.4), the factorization ind-scheme “Spec”($\mathfrak{Z}_{\mathfrak{g}}$) is ind-placid.

In particular, the category $\mathrm{IndCoh}^*(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ is well-defined (see Sect. B.13.22). Moreover, it is compactly generated and identifies with the dual of $\mathrm{IndCoh}^!(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$.

The contents of Sect. 4.6.4 apply also in the factorization context. Hence, our task will be to define a coaction of $\mathrm{IndCoh}^*(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$, viewed as a comonoidal category, on $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}$.

4.6.6. We will first explain how to construct the coaction functor

$$(4.15) \quad \mathrm{coact} : \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \rightarrow \mathrm{IndCoh}^*(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}})) \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}.$$

Since $\mathrm{Vac}(G)_{\mathrm{crit}}$ is the factorization unit in $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}$, we can write

$$\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \simeq \mathrm{Vac}(G)_{\mathrm{crit}}\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}})$$

and

$$(\mathfrak{z}_{\mathfrak{g}} \otimes \mathrm{Vac}(G)_{\mathrm{crit}})\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}) \simeq \mathfrak{z}_{\mathfrak{g}}\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}).$$

Now, restriction along the map (4.14) gives rise to a t-exact functor

$$\mathrm{Vac}(G)_{\mathrm{crit}}\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}) \rightarrow (\mathfrak{z}_{\mathfrak{g}} \otimes \mathrm{Vac}(G)_{\mathrm{crit}})\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}),$$

i.e., a functor

$$\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \rightarrow \mathfrak{z}_{\mathfrak{g}}\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}).$$

In particular, since the compact generators of $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}$ are eventually coconnective, we obtain a functor

$$(4.16) \quad (\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}})^c \rightarrow \left(\mathfrak{z}_{\mathfrak{g}}\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}) \right)^{>-\infty}.$$

4.6.7. Note now that by combining Corollary C.16.12 and Corollary 4.4.2(a), we obtain:

Corollary 4.6.8. *The functor*

$$(4.17) \quad \mathrm{IndCoh}^*(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}})) \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \rightarrow \mathfrak{z}_{\mathfrak{g}}\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}})$$

is t-exact and induces an equivalence between the eventually coconnective subcategories of the two sides.

Hence, (4.16) can be thought of as a functor

$$(4.18) \quad (\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}})^c \rightarrow (\mathrm{IndCoh}^*(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}})) \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}})^{>-\infty} \hookrightarrow \mathrm{IndCoh}^*(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}})) \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}.$$

Ind-extending, from (4.18), we obtain the desired functor (4.15).

4.6.9. Our next task is to extend the functor (4.15) to a coaction of $\mathrm{IndCoh}^*(\mathrm{Spec}(\mathfrak{Z}_{\mathfrak{g}}))$ on $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}$. In doing so we will have to overcome two hurdles:

- (i) Homological-algebraic, which has to do with inverting the functor (4.15) on the eventually coconnective subcategories.
- (ii) Homotopic-algebraic, which has to do with equipping the functor (4.15) with a homotopy-coherent associativity datum.

We will deal with (i) in the rest of this subsection, and with (ii) in Sect. J.

4.6.10. First, proceeding as in Sect. 4.6.6, for an integer n , we define an n -ry operation

$$(4.19) \quad \mathrm{coact}_n : \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \rightarrow \mathrm{IndCoh}^*(\mathrm{Spec}(\mathfrak{Z}_{\mathfrak{g}}))^{\otimes n} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}},$$

so that the composition with

$$(4.20) \quad \mathrm{IndCoh}^*(\mathrm{Spec}(\mathfrak{Z}_{\mathfrak{g}}))^{\otimes n} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \rightarrow \mathfrak{Z}_{\mathfrak{g}}^{\otimes n}\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}})$$

is the restriction functor along the action map

$$(4.21) \quad \mathfrak{Z}_{\mathfrak{g}}^{\otimes n} \otimes \mathrm{Vac}(G)_{\mathrm{crit}} \rightarrow \mathrm{Vac}(G)_{\mathrm{crit}}.$$

4.6.11. We claim:

Lemma 4.6.12. *The functor (4.19) is t-exact.*

Proof. By construction, the composition of (4.19) with (4.20) is t-exact. Since the functor (4.20) induces an equivalence on eventually eventually coconnective subcategories (see Corollary 4.6.8), it suffices to show that (4.19) has a bounded cohomological amplitude (over each X^I). By factorization, this reduces to the case when I is a singleton, and by evaluating at field-valued points of X , we reduce to the pointwise case. The latter was established in [Ra5, Sect. 11.13]. $\square$

4.6.13. From Lemma 4.6.12 we obtain:

Corollary 4.6.14. *The functor (4.15) satisfies associativity at the homotopy level.*

Proof. We need to show:

- For every $n = n_1 + n_2$, the diagram

$$\begin{array}{ccc} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} & \xrightarrow{\mathrm{coact}_{n_2}} & \mathrm{IndCoh}^*(\mathrm{Spec}(\mathfrak{Z}_{\mathfrak{g}}))^{\otimes n_2} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \\ \mathrm{coact}_n \downarrow & & \downarrow \mathrm{Id} \otimes \mathrm{coact}_{n_1} \\ \mathrm{IndCoh}^*(\mathrm{Spec}(\mathfrak{Z}_{\mathfrak{g}}))^{\otimes n} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} & \xrightarrow{\sim} & \mathrm{IndCoh}^*(\mathrm{Spec}(\mathfrak{Z}_{\mathfrak{g}}))^{\otimes n_2} \otimes \mathrm{IndCoh}^*(\mathrm{Spec}(\mathfrak{Z}_{\mathfrak{g}}))^{\otimes n_1} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \end{array}$$

commutes;

- For any n , the diagram

$$\begin{array}{ccc} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} & \xrightarrow{\mathrm{coact}} & \mathrm{IndCoh}^*(\mathrm{Spec}(\mathfrak{Z}_{\mathfrak{g}})) \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \\ \mathrm{coact}_n \downarrow & & \downarrow \otimes \mathrm{comult}_n \otimes \mathrm{Id} \\ \mathrm{IndCoh}^*(\mathrm{Spec}(\mathfrak{Z}_{\mathfrak{g}}))^{\otimes n} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} & \xrightarrow{\sim} & \mathrm{IndCoh}^*(\mathrm{Spec}(\mathfrak{Z}_{\mathfrak{g}}))^{\otimes n} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \end{array}$$

commutes.

In both cases, it suffices to show that the natural transformation in question is an isomorphism when evaluated on compact objects. In particular, it suffices to show that it is an isomorphism when evaluated on eventually coconnected subcategories.

We know that the natural transformation becomes an isomorphism after composing with the functor (4.20). Since the functor (4.20) is an equivalence on eventually coconnected subcategories (see Corollary 4.6.8), it suffices to show that all the functors involved have cohomological amplitude bounded on the left. However, this follows from Lemma 4.6.12 (for coact_n), while the functor

$$\mathrm{comult}_n : \mathrm{IndCoh}^*(\mathrm{Spec}(\mathfrak{Z}_{\mathfrak{g}})) \rightarrow \mathrm{IndCoh}^*(\mathrm{Spec}(\mathfrak{Z}_{\mathfrak{g}}))^{\otimes n} \simeq \mathrm{IndCoh}^*(\mathrm{Spec}(\mathfrak{Z}_{\mathfrak{g}}))^n$$

is t-exact (see Corollary A.8.8). □

4.7. Action of the center and the loop group action.

4.7.1. Our current goal is to show that the $\mathrm{IndCoh}^!(\mathrm{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ -module structure on $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}$ is compatible with the action of $\mathfrak{L}(G)$ on $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}$.

By the construction of the module structure, we need to show that the each of the categories

$$(\mathfrak{z}_{\mathfrak{g}}^{\otimes n} \otimes \mathrm{Vac}(G)_{\mathrm{crit}})\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}})$$

carries an action of $\mathfrak{L}(G)$, such that:

- It is compatible with the functor

$$\begin{aligned} \mathrm{IndCoh}^!(\mathrm{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))^{\otimes n} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} &\xrightarrow{(4.20)} \\ &\rightarrow \mathfrak{z}_{\mathfrak{g}}^{\otimes n}\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}) \simeq (\mathfrak{z}_{\mathfrak{g}}^{\otimes n} \otimes \mathrm{Vac}(G)_{\mathrm{crit}})\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}); \end{aligned}$$

where the $\mathfrak{L}(G)$ on the left-hand side is via the $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}$ -factor;

- The restriction functors

$$(\mathfrak{z}_{\mathfrak{g}}^{\otimes n_1} \otimes \mathrm{Vac}(G)_{\mathrm{crit}})\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}) \rightarrow (\mathfrak{z}_{\mathfrak{g}}^{\otimes n_2} \otimes \mathrm{Vac}(G)_{\mathrm{crit}})\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}})$$

along the maps

$$\mathfrak{z}_{\mathfrak{g}}^{\otimes n_2} \otimes \mathrm{Vac}(G)_{\mathrm{crit}} \rightarrow \mathfrak{z}_{\mathfrak{g}}^{\otimes n_1} \otimes \mathrm{Vac}(G)_{\mathrm{crit}}$$

that encode the $\mathfrak{z}_{\mathfrak{g}}$ -action on $\mathrm{Vac}(G)_{\mathrm{crit}}$ carry a natural $\mathfrak{L}(G)$ -equivariant structure.

4.7.2. In order to do so, it suffices to show that for *any* factorization algebra $\mathcal{A} \in \mathrm{KL}(G)_{\mathrm{crit}}$, the lax factorization category

$$\mathcal{A}\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}})$$

carries an action of $\mathfrak{L}(G)$, compatible with the forgetful functor

$$\mathbf{oblv}_{\mathcal{A}} : \mathcal{A}\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}) \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}},$$

and this construction is functorial with respect to the functors

$$\mathrm{Res}_{\phi} : \mathcal{A}_2\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}) \rightarrow \mathcal{A}_1\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}),$$

corresponding to homomorphisms

$$\phi : \mathcal{A}_1 \rightarrow \mathcal{A}_2$$

of factorization algebras in $\mathrm{KL}(G)_{\mathrm{crit}}$.

4.7.3. Let $\mathbf{oblv}_{\mathfrak{L}^+(G)}$ denote the forgetful functor. For $\mathcal{Z} \rightarrow \mathrm{Ran}$, consider

$$(4.22) \quad \mathrm{Res}_{\mathbf{oblv}_{\mathfrak{L}^+(G)}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}^{\mathrm{fact}_{\mathcal{Z}}}) \in \mathrm{KL}(G)_{\mathrm{crit}}\text{-}\mathbf{mod}_{\mathcal{Z}}^{\mathrm{fact}}.$$

By Lemma C.14.18, for $\mathcal{A} \in \mathrm{FactAlg}^{\mathrm{untl}}(X, \mathrm{KL}(G)_{\mathrm{crit}})$, we have

$$\mathbf{oblv}_{\mathfrak{L}^+(G)}(\mathcal{A})\text{-}\mathbf{mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}})_{\mathcal{Z}} \simeq \mathcal{A}\text{-}\mathbf{mod}^{\mathrm{fact}}\left(\mathrm{Res}_{\mathbf{oblv}_{\mathfrak{L}^+(G)}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}^{\mathrm{fact}_{\mathcal{Z}}})\right)_{\mathcal{Z}}.$$

Hence, it suffices to show that (4.22) carries an action of $\mathfrak{L}(G)_{\mathcal{Z}}$. We will show this in the following general framework.

4.7.4. Let $\mathbf{A}$ be a factorization category, and let $\mathbf{C}$ be a factorization module category over $\mathbf{A}$ at $\mathcal{Z} \rightarrow \text{Ran}$. We assume that $\mathbf{A}$ and $\mathbf{C}$ carry compatible actions of $\mathfrak{L}(G)$ at some level κ .

Set $\mathbf{A}_0 := \mathbf{A}^{\mathfrak{L}^+(G)}$; denote by $\mathbf{oblv}_{\mathfrak{L}^+(G)}$ the forgetful functor

$$\mathbf{A}_0 \rightarrow \mathbf{A}.$$

Consider

$$\text{Res}_{\mathbf{oblv}_{\mathfrak{L}^+(G)}}(\mathbf{C}) \in \mathbf{A}_0\text{-}\mathbf{mod}_{\mathcal{Z}}^{\text{fact}}.$$

We claim that $\text{Res}_{\mathbf{oblv}_{\mathfrak{L}^+(G)}}(\mathbf{C})$, viewed as a factorization module category over $\mathbf{A}_0$, carries an action of $\mathfrak{L}(G)_{\mathcal{Z}}$, such that the induced action on

$$\text{Res}_{\mathbf{oblv}_{\mathfrak{L}^+(G)}}(\mathbf{C})_{\mathcal{Z}} \simeq \mathbf{C}_{\mathcal{Z}}$$

is the original $\mathfrak{L}(G)_{\mathcal{Z}}$ -action on $\mathbf{C}_{\mathcal{Z}}$.

We sketch the construction of this action below; a more detailed exposition will be given in [CFGY].

4.7.5. Let $\text{Gr}_G^{\text{level}_{\mathcal{Z}}}$ be the factorization Gr_G -module space from Sect. B.2.7. It is equipped with a compatible action of $\mathfrak{L}(G)$ in the left and a commuting $\mathfrak{L}(G)_{\mathcal{Z}}$ -action on the right.

Let $\mathfrak{L}(G)^{\text{fact}_{\mathcal{Z}}}$ be the vacuum factorization module space over $\mathfrak{L}(G)$ at $\mathcal{Z}$ (see Sect. B.2.6); it carries a compatible action of $\mathfrak{L}(G) \times \mathfrak{L}(G)$.

We have a naturally defined projection

$$\pi_{\mathcal{Z}} : \mathfrak{L}(G)^{\text{fact}_{\mathcal{Z}}} \rightarrow \text{Gr}_G^{\text{level}_{\mathcal{Z}}},$$

(see Sect. B.4.10) which gives rise to the pullback functor

$$\pi_{\mathcal{Z}}^* : \text{D-mod}(\text{Gr}_G^{\text{level}_{\mathcal{Z}}}) \rightarrow \text{D-mod}(\mathfrak{L}(G)^{\text{fact}_{\mathcal{Z}}}),$$

compatible³⁰ with the factorization functor

$$\pi^* : \text{D-mod}(\text{Gr}_G) \rightarrow \text{D-mod}(\mathfrak{L}(G)),$$

given by pullback along the projection

$$\pi : \mathfrak{L}(G) \rightarrow \text{Gr}_G.$$

In particular, $\pi_{\mathcal{Z}}^*$ gives rise to a functor

$$(4.23) \quad \text{D-mod}(\text{Gr}_G^{\text{level}_{\mathcal{Z}}}) \rightarrow \text{Res}_{\pi^*}(\text{D-mod}(\mathfrak{L}(G)^{\text{fact}_{\mathcal{Z}}}),$$

as factorization module categories over $\text{D-mod}(\text{Gr}_G)$, see Sect. B.12.11.

4.7.6. We claim:

Lemma 4.7.7. *The functor (4.23) is an equivalence.*

Proof. Follows from Lemma B.15.9. □

³⁰See Sect. B.12.10 for what this means.

4.7.8. We now return to the setting of Sect. 4.7.4. Note that

$$\mathbf{C} \simeq \mathrm{D}\text{-mod}(\mathfrak{L}(G)^{\mathrm{fact}z}) \otimes_{\mathfrak{L}(G)} \mathbf{C}$$

as factorization module categories over

$$\mathbf{A} \simeq \mathrm{D}\text{-mod}(\mathfrak{L}(G)) \otimes_{\mathfrak{L}(G)} \mathbf{A}$$

at z , where we use the action of $\mathfrak{L}(G)$ on itself on the *left* to form the tensor product.

Note that the functor π_z^* and hence the equivalence (4.23) are compatible with the actions of $\mathfrak{L}(G)$ on the left. Hence, from Lemma 4.7.7 we obtain:

Corollary 4.7.9. *There is a canonical equivalence*

$$\mathrm{Res}_{\mathrm{oblv}_{\mathfrak{L}^+(G)}}(\mathbf{C}) \simeq \mathrm{D}\text{-mod}(\mathrm{Gr}_G^{\mathrm{level}z}) \otimes_{\mathfrak{L}(G)} \mathbf{C},$$

as factorization categories over

$$\mathbf{A}_0 \simeq \mathrm{D}\text{-mod}(\mathrm{Gr}_G) \otimes_{\mathfrak{L}(G)} \mathbf{A}.$$

4.7.10. Now, the action of $\mathfrak{L}(G)_z$ -action on $\mathrm{Gr}_G^{\mathrm{level}z}$ on the right gives rise to an action of $\mathfrak{L}(G)_z$ on $\mathrm{D}\text{-mod}(\mathrm{Gr}_G^{\mathrm{level}z}) \otimes_{\mathfrak{L}(G)} \mathbf{C}$, commuting with the factorization module structure over $\mathrm{D}\text{-mod}(\mathrm{Gr}_G) \otimes_{\mathfrak{L}(G)} \mathbf{A}$.

Applying Corollary 4.7.9, we produce the sought-for $\mathfrak{L}(G)_z$ -action on $\mathrm{Res}_{\mathrm{oblv}_{\mathfrak{L}^+(G)}}(\mathbf{C})$.

4.7.11. As a consequence of the compatibility of the $\mathrm{IndCoh}^1(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ and $\mathfrak{L}(G)$ -actions, we obtain:

Corollary 4.7.12.

(a) *The category $\mathrm{KL}(G)_{\mathrm{crit}}$ carries an action of $\mathrm{IndCoh}^1(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ compatible with the action of $\mathrm{IndCoh}^1(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ on $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}$ and the forgetful functor*

$$\mathrm{KL}(G)_{\mathrm{crit}} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}.$$

(b) *The action of $\mathrm{IndCoh}^1(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ on $\mathrm{KL}(G)_{\mathrm{crit}}$ is compatible with the action of Sph_G .*

Remark 4.7.13. Note that we could have equivalently defined the action of $\mathrm{IndCoh}^1(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ on $\mathrm{KL}(G)_{\mathrm{crit}}$ directly, by repeating the procedure in Sect. 4.6, replacing $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}$ by $\mathrm{KL}(G)_{\mathrm{crit}}$. An analog of Lemma 4.6.12 follows from the original variant of this lemma, since the corresponding functors

$$\mathrm{IndCoh}^*(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))^{\otimes n} \otimes \mathrm{KL}(G)_{\mathrm{crit}} \rightarrow \mathrm{IndCoh}^*(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))^{\otimes n} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}$$

are conservative.

4.7.14. Note that since $\mathfrak{Z}_{\mathfrak{g}} \subset \mathbb{V}_{\mathfrak{g},\mathrm{crit}}$ is invariant under the adjoint action, we can view it also as the (classical) center of the twisted version $\mathbb{V}_{\mathfrak{g},\mathrm{crit},\rho(\omega_X)}$.

In particular, we can regard $\mathfrak{Z}_{\mathfrak{g}}$ as acting on $\mathrm{Vac}(G)_{\mathrm{crit},\rho(\omega_X)}$ as an object of $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega_X)}$ (or $\mathrm{KL}(G)_{\mathrm{crit},\rho(\omega_X)}$).

In particular, we obtain an action of $\mathrm{IndCoh}^1(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ on $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)}$, compatible with the action of $\mathfrak{L}(G)_{\rho(\omega_X)}$. The conclusion of Lemma 4.7.12 renders automatically to the present twisted context.

In addition, we have:

Corollary 4.7.15.

(a) *The category $\mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)})$ carries a unique action of $\mathrm{IndCoh}^1(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$, compatible with the action of $\mathrm{IndCoh}^1(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ on $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)}$ and the projection*

$$\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)} \rightarrow \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)}).$$

(b) *The category $\mathrm{Whit}^!(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)})$ carries a unique action of $\mathrm{IndCoh}^!(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$, compatible with the action of $\mathrm{IndCoh}^!(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ on $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)}$ and the embedding*

$$\mathrm{Whit}^!(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)}) \hookrightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)}.$$

(c) *The functor*

$$\Theta_{\mathrm{Whit}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)})} : \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)}) \rightarrow \mathrm{Whit}^!(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)})$$

carries a natural $\mathrm{IndCoh}^!(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ -linear structure.

4.8. The enhanced functor of Drinfeld-Sokolov reduction at the critical level. In this subsection we will study the functor

$$(4.24) \quad \mathrm{DS} : \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega_X)} \rightarrow \mathrm{Vect}$$

of (2.6).

4.8.1. Consider the factorization unit

$$\mathrm{Vac}(G)_{\mathrm{crit},\rho(\omega_X)} \in \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega_X)}.$$

The functor (4.24) has a natural lax unital factorization structure (see Sect. 2.3.4). In particular, the object

$$\mathrm{DS}(\mathrm{Vac}(G)_{\mathrm{crit},\rho(\omega_X)})$$

is naturally a factorization algebra (in Vect).

4.8.2. The action map

$$\mathfrak{z}_{\mathfrak{g}} \otimes \mathrm{Vac}(G)_{\mathrm{crit},\rho(\omega_X)} \rightarrow \mathrm{Vac}(G)_{\mathrm{crit},\rho(\omega_X)}$$

gives rise to a map

$$\mathfrak{z}_{\mathfrak{g}} \otimes \mathrm{DS}(\mathrm{Vac}(G)_{\mathrm{crit},\rho(\omega_X)}) \rightarrow \mathrm{DS}(\mathrm{Vac}(G)_{\mathrm{crit},\rho(\omega_X)})$$

as factorization algebras.

Pre-composing with the unit for $\mathrm{DS}(\mathrm{Vac}(G)_{\mathrm{crit},\rho(\omega_X)})$, we obtain a map of factorization algebras

$$(4.25) \quad \mathfrak{z}_{\mathfrak{g}} \rightarrow \mathrm{DS}(\mathrm{Vac}(G)_{\mathrm{crit},\rho(\omega_X)}).$$

We have the following fundamental result, see [FF]:

Theorem 4.8.3. *The map (4.25) is an isomorphism.*

4.8.4. By Sect. 4.1.6, the functor DS of (4.24) naturally lifts to a functor

$$\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega_X)} \rightarrow \mathrm{DS}(\mathrm{Vac}(G)_{\mathrm{crit},\rho(\omega_X)})\text{-mod}^{\mathrm{fact}}.$$

Restricting along (4.25), we can view it as a functor, to be denoted

$$\mathrm{DS}^{\mathrm{enh}} : \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega_X)} \rightarrow \mathfrak{z}_{\mathfrak{g}}\text{-mod}^{\mathrm{fact}}.$$

4.8.5. Consider the functor

$$\Gamma^{\mathrm{IndCoh}}(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}), -) : \mathrm{IndCoh}^*(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}})) \rightarrow \mathfrak{z}_{\mathfrak{g}}\text{-mod}^{\mathrm{fact}}.$$

We claim:

Proposition 4.8.6. *There exists a uniquely defined (continuous) functor*

$$\mathrm{DS}^{\mathrm{enh},\mathrm{rfnd}} : \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega_X)} \rightarrow \mathrm{IndCoh}^*(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}})),$$

satisfying

- *There exists an isomorphism*

$$(4.26) \quad \mathrm{DS}^{\mathrm{enh}} \simeq \Gamma^{\mathrm{IndCoh}}(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}), -)^{\mathrm{enh}} \circ \mathrm{DS}^{\mathrm{enh},\mathrm{rfnd}};$$

- $\mathrm{DS}^{\mathrm{enh},\mathrm{rfnd}}$ *sends compact objects in $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega_X)}$ to eventually coconnective (i.e., bounded below) objects in $\mathrm{IndCoh}^*(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$.*

Furthermore, $\mathrm{DS}^{\mathrm{enh}, \mathrm{rfnd}}$ carries a uniquely defined factorization structure, so that (4.26) is an isomorphism of factorization functors.

Proof. It is enough to show that the restriction of $\mathrm{DS}^{\mathrm{enh}}$ to the subcategory

$$(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})^c \subset \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}$$

can be uniquely lifted to a functor

$$(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})^c \rightarrow \mathrm{IndCoh}^*(\mathrm{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))^{>-\infty}.$$

However, this follows from Corollary 4.3.12(a), using the fact that the initial functor DS sends

$$(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})^c \rightarrow \mathrm{Vect}^{>-\infty}.$$

□

4.8.7. Recall now that the functor DS factors via a functor

$$\overline{\mathrm{DS}} : \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \rightarrow \mathrm{Vect}.$$

It follows formally that the functor $\mathrm{DS}^{\mathrm{enh}}$ also factors via a functor, denoted

$$\overline{\mathrm{DS}}^{\mathrm{enh}} : \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \rightarrow \mathfrak{Z}_{\mathfrak{g}}\text{-mod}^{\mathrm{fact}}.$$

We now quote the following fundamental result of [Ra2]:

Theorem 4.8.8. *The functor $\mathrm{DS}^{\mathrm{enh}, \mathrm{rfnd}}$ factors via a functor*

$$(4.27) \quad \overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}} : \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \rightarrow \mathrm{IndCoh}^*(\mathrm{“Spec”}(\mathfrak{Z}_{\mathfrak{g}})),$$

and the resulting functor $\overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}}$ is an equivalence of factorization categories.

Note that by construction

$$\Gamma^{\mathrm{IndCoh}}(\mathrm{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}), -) \circ \overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}} \simeq \overline{\mathrm{DS}}^{\mathrm{enh}}.$$

4.8.9. According to Sect. 4.6 (applied to the twist $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}$ instead of the original $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}$), the category $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}$ carries an action of the monoidal category $\mathrm{IndCoh}^!(\mathrm{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$.

By construction, the functor $\mathrm{DS}^{\mathrm{enh}, \mathrm{rfnd}}$ intertwines the above $\mathrm{IndCoh}^!(\mathrm{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ -action on $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}$ and the natural $\mathrm{IndCoh}^!(\mathrm{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ -action on $\mathrm{IndCoh}^*(\mathrm{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$.

According to Corollary 4.7.15(a), the $\mathrm{IndCoh}^!(\mathrm{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ -action on $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}$ descends to an (a priori, uniquely defined) action on $\mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})$.

It follows formally that the functor $\overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}}$ intertwines this action and the $\mathrm{IndCoh}^!(\mathrm{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ -action on $\mathrm{IndCoh}^*(\mathrm{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$.

5. THE FEIGIN-FRENKEL ISOMORPHISM AND ITS APPLICATIONS

In this section we review the Feigin-Frenkel isomorphism, which provides a bridge between Kac-Moody representations and opers.

Using the Feigin-Frenkel isomorphism, we construct an action of $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})$ on $\mathrm{KL}(G)_{\mathrm{crit}}$, which is a key ingredient of the critical FLE functor, studied in the next section.

5.1. The Feigin-Frenkel isomorphism.

5.1.1. We quote the following fundamental result of Feigin and Frenkel ([FF]):

Theorem 5.1.2. *There exists a canonically defined isomorphism of factorization algebras*

$$\mathfrak{Z}_{\mathfrak{g}} \xrightarrow{\mathrm{FF}_G} \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}}}.$$

Below we will complement Theorem 5.1.2 by an assertion that describes how it interacts with geometric Satake, see Theorem 5.2.5.

5.1.3. As in Remark 1.4.5, one has a choice in the normalization of the functor FF_G ; the two different choices differ by the Chevalley involution.

However, the choices for the normalizations of CS_G and FF_G force one another via the compatibility in Theorem 6.4.5. So, once CS_G is fixed, FF_G is unambiguous.

5.1.4. *Example.* Let $G = T$ be a torus. Then $\mathfrak{z}_{\mathfrak{g}}$ is the commutative factorization algebra associated with the commutative algebra object

$$\mathrm{Sym}^!(\mathfrak{t} \otimes D_X[1]) \in \mathrm{ComAlg}(\mathrm{D}\text{-mod}(X)),$$

i.e.,

$$\mathfrak{z}_{\mathfrak{g}} = \mathrm{Fact}(\mathrm{Sym}^!(\mathfrak{t} \otimes D_X[1])),$$

see Sect. B.10.2 for the notation.

When we think of $\mathrm{D}\text{-mod}(X)$ as “left D-modules”, the above object is

$$\mathrm{Sym}_{\mathcal{O}_X}(\mathfrak{t} \otimes D_X \otimes \omega_X^{\otimes -1}) \in \mathrm{ComAlg}(\mathrm{D}\text{-mod}^l(X)).$$

The affine D-scheme $\mathrm{Op}_{\check{T}}$ identifies with the scheme of jets $\mathrm{Jets}(\check{\mathfrak{t}} \otimes \omega_X)$ (see Sect. B.5.1). Under our normalization, the isomorphism FF_T is the tautological identification

$$(5.1) \quad \mathrm{Spec}_X(\mathrm{Sym}_{\mathcal{O}_X}(\mathfrak{t} \otimes D_X \otimes \omega_X^{\otimes -1})) \simeq J(\check{\mathfrak{t}} \otimes \omega_X).$$

At the level of fibers at a given point $x \in X$, the identification (5.1) is the isomorphism

$$\mathrm{Spec}(\mathrm{Sym}(\mathfrak{t} \otimes \mathcal{K}_x / \mathcal{O}_x)) \simeq \check{\mathfrak{t}} \otimes \omega_{\mathcal{D}_x},$$

corresponding to the canonical identification of pro-finite dimensional vector spaces

$$(\mathfrak{t} \otimes \mathcal{K}_x / \mathcal{O}_x)^\vee \simeq \check{\mathfrak{t}} \otimes \omega_{\mathcal{D}_x}.$$

5.1.5. For a general G , the isomorphism FF_G is normalized so that the resulting isomorphism at the associated graded level

$$\mathrm{Jets}(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X}) \simeq \mathrm{Spec}_X(\mathrm{gr}(\mathfrak{z}_{\mathfrak{g}}, x)) \stackrel{\mathrm{FF}_G}{\simeq} \mathrm{Op}_G^{\mathrm{cl}} \simeq \mathrm{Jets}(\mathfrak{a}(\check{\mathfrak{g}})_{\omega_X}),$$

is the identity map, where $\mathrm{Op}_G^{\mathrm{cl}}$ denotes the D-scheme of *classical* $\check{G}$ -opers.

5.2. **The “birth” of opers.** In this subsection we will formulate Theorem 5.2.5, which in [BD1, Sect. 5.3] was called “the birth of opers”, that explains how the isomorphism FF_G interacts with geometric Satake.

5.2.1. Consider the (symmetric monoidal) functor

$$(5.2) \quad \mathrm{Rep}(\check{G}) \xrightarrow{\mathfrak{r}^*} \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{reg}}) \simeq \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}}\text{-mod}^{\mathrm{com}}} \stackrel{\mathrm{FF}_G}{\simeq} \mathfrak{z}_{\mathfrak{g}}\text{-mod}^{\mathrm{com}}.$$

In particular, the functor (5.2) allows us to view $\mathfrak{z}_{\mathfrak{g}}\text{-mod}^{\mathrm{com}}$ as a $\mathrm{Rep}(\check{G})$ -module category, in a way compatible with factorization.

5.2.2. Let us view $\mathrm{KL}(G)_{\mathrm{crit}}$ as a module category over $\mathrm{Rep}(\check{G})$ via

$$\mathrm{Rep}(\check{G}) \xrightarrow{\mathrm{Sat}_G^{\mathrm{nv}}} \mathrm{Sph}_G$$

and the Sph_G -action on $\mathrm{KL}(G)_{\mathrm{crit}}$. This structure is also compatible with factorization.

5.2.3. Finally, note that the action of $\mathfrak{z}_{\mathfrak{g}}$ on $\mathrm{Vac}(G)_{\mathrm{crit}}$ gives rise to a factorization functor

$$(5.3) \quad \mathfrak{z}_{\mathfrak{g}}\text{-mod}^{\mathrm{com}} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}}, \quad - \otimes_{\mathfrak{z}_{\mathfrak{g}}} \mathrm{Vac}(G)_{\mathrm{crit}}.$$

5.2.4. We claim (see [BD1, Theorem 5.5.3]):

Theorem 5.2.5. *The functor (5.3) admits a lift to a functor between $\text{Rep}(\check{G})$ -module categories. This structure is compatible with factorization.*

Remark 5.2.6. Concretely, Theorem 5.2.5 says that the object $\text{Vac}(G)_{\text{crit}}$ satisfies the *Hecke property* with respect to the action of $\text{Rep}(\check{G})$ on $\text{KL}(G)_{\text{crit}}$: i.e., we have

$$\text{Sat}_G^{\text{nv}}(V) \star \text{Vac}(G)_{\text{crit}} \simeq \text{Vac}(G)_{\text{crit}} \otimes_{\mathfrak{z}_{\mathfrak{g}}} (\text{FF}_G \circ (\mathfrak{t}^{\text{reg}})^*(V)), \quad V \in \text{Rep}(\check{G}),$$

where in the right-hand side we denoted by FF_G the equivalence

$$\mathfrak{z}_{\mathfrak{g}}\text{-mod}^{\text{com}} \simeq \mathcal{O}_{\text{Op}_G^{\text{reg}}}\text{-mod}^{\text{com}} = \text{QCoh}(\text{Op}_G^{\text{reg}}),$$

induced by the isomorphism of algebras FF_G .

The above isomorphisms are compatible with tensor products of the V 's in the natural sense.

5.2.7. From now on we will identify

$$\text{Op}_G^{\text{reg}} \simeq \text{Spec}(\mathfrak{z}_{\mathfrak{g}}) \text{ and } \text{Op}_G^{\text{mer}} \simeq \text{“Spec”}(\mathfrak{z}_{\mathfrak{g}})$$

using FF_G .

5.2.8. In particular, we will view the category $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}$ as acted on by $\text{IndCoh}^!(\text{Op}_G^{\text{mer}})$. Similarly, we will view the functors $\overline{\text{DS}}^{\text{enh,rfnd}}$ (resp., $\text{DS}^{\text{enh,rfnd}}$) as $\text{IndCoh}^!(\text{Op}_G^{\text{mer}})$ -linear functors from $\text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)})$ (resp., $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}$) to $\text{IndCoh}^*(\text{Op}_G^{\text{mer}})$.

5.2.9. Here is one particular application of Theorem 5.2.5 that will be used in the sequel. Recall the commutative algebra (factorization) object

$$R_{\check{G}, \text{Op}} \in \text{Rep}(\check{G}),$$

Sect. 4.8.

We claim that it acts on $\text{Vac}(G)_{\text{crit}} \in \text{KL}(G)_{\text{crit}}$, when we consider $\text{KL}(G)_{\text{crit}}$ as a $\text{Rep}(\check{G})$ -module category as in Sect. 5.2.2.

Indeed, by Theorem 5.2.5, in order to construct this structure, it suffices to construct an action of $R_{\check{G}, \text{Op}}$ on

$$\mathcal{O}_{\text{Op}_G^{\text{reg}}} \in \text{QCoh}(\text{Op}_G^{\text{reg}}),$$

when we consider $\text{QCoh}(\text{Op}_G^{\text{reg}})$ as a $\text{Rep}(\check{G})$ -module category via $(\mathfrak{t}^{\text{reg}})^*$. However, the latter structure comes from the map of commutative algebras in $\text{QCoh}(\text{Op}_G^{\text{reg}})$

$$(\mathfrak{t}^{\text{reg}})^*(R_{\check{G}, \text{Op}}) \rightarrow \mathcal{O}_{\text{Op}_G^{\text{reg}}},$$

given by the counit of the $((\mathfrak{t}^{\text{reg}})^*, \mathfrak{t}_*^{\text{reg}})$ -adjunction.

5.3. The Kazhdan-Lusztig category at the critical level and monodromy-free opers. In this subsection we will show that the $\text{IndCoh}^!(\text{Op}_G^{\text{mer}})$ -action on $\text{KL}(G)_{\text{crit}}$ factors through an action of $\text{IndCoh}^!(\text{Op}_G^{\text{mon-free}})$.

The construction will emulate the construction of the $\text{IndCoh}^!(\text{Op}_G^{\text{mer}})$ -action on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}$ in Sect. 4.6, with a “decoration” by $\text{Rep}(\check{G})$.

Remark 5.3.1. The construction of such an action (along with its properties discussed in Sect. 5.4) at fixed $\underline{x} \in \text{Ran}$ was the subject of the paper [FG5]. However, we do not know how to adapt the methods of *loc. cit.* to the factorization setting.

5.3.2. Using the duality between $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})$ and $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$, it suffices to construct a coaction of $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$, viewed as a comonoidal factorization category, on $\mathrm{KL}(G)_{\mathrm{crit}}$.

As in Sect. 4.6.6, we first explain how to construct the coaction functor

$$(5.4) \quad \mathrm{KL}(G)_{\mathrm{crit}} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes \mathrm{KL}(G)_{\mathrm{crit}}.$$

We will then upgrade this to the datum of coaction.

5.3.3. We start with the monoidal action of $\mathrm{Rep}(\check{G})$ on $\mathrm{KL}(G)_{\mathrm{crit}}$ as in Sect. 5.2.2. Since $\mathrm{Rep}(\check{G})$ is rigid, the right adjoint to the action is a (lax unital) factorization functor

$$(5.5) \quad \mathrm{coact}_{\mathrm{Rep}(\check{G}), \mathrm{KL}(G)_{\mathrm{crit}}} : \mathrm{KL}(G)_{\mathrm{crit}} \rightarrow \mathrm{Rep}(\check{G}) \otimes \mathrm{KL}(G)_{\mathrm{crit}}.$$

This functor upgrades to a factorization functor

$$(5.6) \quad \begin{aligned} \mathrm{coact}_{\mathrm{Rep}(\check{G}), \mathrm{KL}(G)_{\mathrm{crit}}}^{\mathrm{enh}} : \mathrm{KL}(G)_{\mathrm{crit}} &\rightarrow \\ &\rightarrow \mathrm{coact}_{\mathrm{Rep}(\check{G}), \mathrm{KL}(G)_{\mathrm{crit}}}(\mathrm{Vac}(G)_{\mathrm{crit}})\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{KL}(G)_{\mathrm{crit}}). \end{aligned}$$

5.3.4. Recall now (see Sect. 5.2.9) that $R_{\check{G}, \mathrm{Op}}$, viewed as an associative algebra object in $\mathrm{Rep}(\check{G})$ acts on $\mathrm{Vac}(G)_{\mathrm{crit}}$, compatibly with factorization. By adjunction, we obtain a map of factorization algebras

$$R_{\check{G}, \mathrm{Op}} \otimes \mathrm{Vac}(G)_{\mathrm{crit}} \rightarrow \mathrm{coact}_{\mathrm{Rep}(\check{G}), \mathrm{KL}(G)_{\mathrm{crit}}}(\mathrm{Vac}(G)_{\mathrm{crit}}).$$

Restriction along the above map defines a functor

$$(5.7) \quad \begin{aligned} \mathrm{coact}_{\mathrm{Rep}(\check{G}), \mathrm{KL}(G)_{\mathrm{crit}}}(\mathrm{Vac}(G)_{\mathrm{crit}})\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{KL}(G)_{\mathrm{crit}}) &\rightarrow \\ \rightarrow (R_{\check{G}, \mathrm{Op}} \otimes \mathrm{Vac}(G)_{\mathrm{crit}})\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{KL}(G)_{\mathrm{crit}}) &= R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{KL}(G)_{\mathrm{crit}}) \end{aligned}$$

Composing (5.6) and (5.7) we obtain a functor

$$(5.8) \quad \mathrm{KL}(G)_{\mathrm{crit}} \rightarrow R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{KL}(G)_{\mathrm{crit}}).$$

The functor (5.5) is left t-exact. Hence, since the compact generators of $\mathrm{KL}(G)_{\mathrm{crit}}$ are eventually coconnective, the functor (5.8) gives rise to a functor

$$(5.9) \quad (\mathrm{KL}(G)_{\mathrm{crit}})^c \rightarrow (R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{KL}(G)_{\mathrm{crit}}))^{\geq -\infty}.$$

5.3.5. Consider the functor

$$(5.10) \quad \begin{aligned} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes \mathrm{KL}(G)_{\mathrm{crit}} &\rightarrow R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G})) \otimes \mathrm{KL}(G)_{\mathrm{crit}} \rightarrow \\ &\rightarrow R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{KL}(G)_{\mathrm{crit}}) \end{aligned}$$

As in Corollary 4.6.8, by combining Corollary C.16.12 and Proposition 4.4.7(a), we obtain:

Lemma 5.3.6. *The functor (5.10) induces an equivalence between the eventually coconnective subcategories of the two sides.*

Hence, we can view (5.9) as a functor

$$(5.11) \quad (\mathrm{KL}(G)_{\mathrm{crit}})^c \rightarrow \left(\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes \mathrm{KL}(G)_{\mathrm{crit}} \right)^{\geq -\infty} \hookrightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes \mathrm{KL}(G)_{\mathrm{crit}}.$$

Ind-extending (5.11) we obtain the sought-for functor (5.4).

5.3.7. Our next goal is upgrade (5.4) to a datum of coaction of $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$ on $\mathrm{KL}(G)_{\mathrm{crit}}$. We do so by mimicking the strategy in Sects. 4.6.10-4.6.13.

First, we generalize the construction in Sects. 5.3.3-5.3.5 above and define the n -ry operation

$$(5.12) \quad \mathrm{KL}(G)_{\mathrm{crit}} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})^{\otimes n} \otimes \mathrm{KL}(G)_{\mathrm{crit}}.$$

5.3.8. We have the following analog of Lemma 4.6.12:

Lemma 5.3.9. *The functor (5.12) is t-exact.*

Proof. Note that the functor

$$(5.13) \quad \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})^{\otimes n} \otimes \mathrm{KL}(G)_{\mathrm{crit}} \xrightarrow{(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}} \otimes \mathrm{oblv}_{\mathfrak{L}^+(G)}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})^{\otimes n} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}$$

is t-exact and conservative.

Hence, it is enough to show that the composition of (5.12) with (5.13) is t-exact.

Note, however, that by construction, we have a commutative diagram

$$\begin{array}{ccc} \mathrm{KL}(G)_{\mathrm{crit}} & \xrightarrow{(5.12)} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})^{\otimes n} \otimes \mathrm{KL}(G)_{\mathrm{crit}} \\ \mathrm{oblv}_{\mathfrak{L}^+(G)} \downarrow & & \downarrow (\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}} \otimes \mathrm{oblv}_{\mathfrak{L}^+(G)} \\ \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} & \xrightarrow{(4.20)} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})^{\otimes n} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}. \end{array}$$

Since $\mathrm{oblv}_{\mathfrak{L}^+(G)}$ is t-exact, the assertion follows from that of Lemma 4.6.12. $\square$

5.3.10. As in Corollary 4.6.14, from Lemma 5.3.9, we obtain that the coaction functor (5.4) is associative *at the homotopy level*. We will equip it with a structure of *coherent homotopy* in Sect. J.5.

5.4. Properties of the $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})$ -action on $\mathrm{KL}(G)_{\mathrm{crit}}$. In this subsection we will discuss those properties of the $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})$ -action on $\mathrm{KL}(G)_{\mathrm{crit}}$ that will be used in the sequel.

5.4.1. First, unwinding the construction, we obtain:

Lemma 5.4.2. *The action of $\mathrm{Rep}(\check{G})$ on $\mathrm{KL}(G)_{\mathrm{crit}}$ given by*

$$(5.14) \quad \mathrm{Rep}(\check{G}) \simeq \mathrm{QCoh}(\mathrm{LS}_G^{\mathrm{reg}}) \xrightarrow{\tau^*} \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}) \xrightarrow{-\otimes \omega_{\mathrm{Op}_G^{\mathrm{mon-free}}}} \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})$$

and the above action of $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})$ on $\mathrm{KL}(G)_{\mathrm{crit}}$ identifies canonically with the action given by

$$\mathrm{Rep}(\check{G}) \xrightarrow{\mathrm{Sat}_G^{\mathrm{nv}}} \mathrm{Sph}_G$$

and the Sph_G -action on $\mathrm{KL}(G)_{\mathrm{crit}}$.

5.4.3. Further, comparing with the construction of the action of $\mathrm{IndCoh}^!(\text{“Spec”}(\mathfrak{z}_{\mathfrak{g}}))$ on $\mathrm{KL}(G)_{\mathrm{crit}}$ given by Corollary 4.7.12, we obtain that this action coincides with the precomposition of the above action of $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})$ on $\mathrm{KL}(G)_{\mathrm{crit}}$ with

$$\iota^! : \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}).$$

5.4.4. We now claim:

Corollary 5.4.5. *For a fixed $\underline{x} \in \mathrm{Ran}$, the action functor of $\mathrm{Rep}(\check{G})_{\underline{x}}$ on $\mathrm{KL}(G)_{\mathrm{crit}, \underline{x}}$ via $\mathrm{Sat}_G^{\mathrm{nv}}$ and the $\mathrm{Sph}_{G, \underline{x}}$ -action on $\mathrm{KL}(G)_{\mathrm{crit}, \underline{x}}$ is t-exact.*

Proof. By Lemma 5.4.2, we need to show that for $V \in \mathrm{Rep}(\check{G})_{\underline{x}}^{\heartsuit}$, the functor

$$\mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \xrightarrow{V \otimes \mathrm{Id}} \mathrm{Rep}(\check{G})_{\underline{x}} \otimes \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{G, \underline{x}}^{\mathrm{mon-free}}) \otimes \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}}$$

is t-exact, where $\mathrm{Rep}(\check{G}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{G, \underline{x}}^{\mathrm{mon-free}})$ is the functor (5.14).

Hence, it suffices to show that if $\mathcal{E}^{\mathrm{mon-free}} \in \mathrm{QCoh}(\mathrm{Op}_{G, \underline{x}}^{\mathrm{mon-free}})$ is a *vector bundle*, then its action on $\mathrm{KL}(G)_{\mathrm{crit}, \underline{x}}$ via

$$\mathrm{QCoh}(\mathrm{Op}_{G, \underline{x}}^{\mathrm{mon-free}}) \xrightarrow{-\otimes \omega_{\mathrm{Op}_{G, \underline{x}}^{\mathrm{mon-free}}}} \mathrm{IndCoh}^!(\mathrm{Op}_{G, \underline{x}}^{\mathrm{mon-free}})$$

and the $\mathrm{IndCoh}^!(\mathrm{Op}_{G, \underline{x}}^{\mathrm{mon-free}})$ -action on $\mathrm{KL}(G)_{\mathrm{crit}, \underline{x}}$ is t-exact.

With no restriction of generality, we can assume that $\mathcal{E}^{\text{mon-free}}$ is the restriction of a vector bundle $\mathcal{E}^{\text{mer}}$ over $\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}$. Hence, by Sect. 5.4.3, it suffices to show that for a vector bundle $\mathcal{E}^{\text{mer}}$ on $\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}$, its action on $\text{KL}(G)_{\text{crit}, \underline{x}}$ via

$$\text{QCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{-\otimes \omega_{\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}}} \text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})$$

and the action of $\text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})$ given by Corollary 4.7.12 is t-exact.

Since the forgetful functor

$$\text{KL}(G)_{\text{crit}, \underline{x}} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \underline{x}}$$

is t-exact and conservative, it suffices to show that the action of $\mathcal{E}^{\text{mer}}$ on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \underline{x}}$ is t-exact.

Unwinding the construction, it suffices to show that the composition

$$\begin{aligned} \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \underline{x}} &\rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \underline{x}} \otimes \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{\text{Id} \otimes (\mathcal{E}^{\text{mer}} \otimes -)} \\ &\rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \underline{x}} \otimes \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{\text{Id} \otimes \Gamma^{\text{IndCoh}}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}, -)} \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \underline{x}} \end{aligned}$$

is t-exact.

In the above composition, the first and the third arrows are t-exact. Hence, it suffices to show that the functor

$$\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \underline{x}} \otimes \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{\text{Id} \otimes (\mathcal{E}^{\text{mer}} \otimes -)} \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \underline{x}} \otimes \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})$$

is t-exact.

However, this easily follows from the fact that the functor

$$\text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{(\mathcal{E}^{\text{mer}} \otimes -)} \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})$$

is t-exact. □

6. THE CRITICAL FLE

In this section we prove the main result of Part I, namely, the *critical FLE*, Theorem 6.1.4, which says that there exists a canonical equivalence of factorization categories

$$(6.1) \quad \text{FLE}_{G, \text{crit}} : \text{KL}(G)_{\text{crit}} \xrightarrow{\sim} \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}}),$$

The functor in one direction in (6.1) is a variation on the theme of the functor $\overline{\text{DS}}^{\text{enh}, \text{rfnd}}$ from Sect. 4.8. Essentially $\text{FLE}_{G, \text{crit}}$ is obtained by base changing $\overline{\text{DS}}^{\text{enh}, \text{rfnd}}$ along the map from $\text{Op}_G^{\text{mon-free}}$ to Op_G^{mer} .

6.1. Construction of the critical FLE functor.

6.1.1. Let $\mathbf{C}$ be a category equipped with a $\mathfrak{L}(G)_{\rho(\omega_X)}$ -action at the critical level, in a way compatible with factorization. Consider the functor

$$(6.2) \quad \text{Sph}(\mathbf{C}) := \mathbf{C}^{\mathfrak{L}(G)_{\rho(\omega_X)}^+} \rightarrow \mathbf{C} \rightarrow \mathbf{C}_{\mathfrak{L}(N)_{\rho(\omega_X)}, X} =: \text{Whit}_*(\mathbf{C}).$$

We apply this to $\mathbf{C} = \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}$. Consider the resulting (factorization) functor

$$(6.3) \quad \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)})$$

Composing, we obtain a functor

$$(6.4) \quad \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}) \xrightarrow{\overline{\text{DS}}^{\text{enh}, \text{rfnd}}} \text{IndCoh}^*(\text{Op}_G^{\text{mer}}).$$

6.1.2. We regard $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})$ as equipped with a natural action of $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})$. We regard $\mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)}$ as acted on by $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})$.

By Sects. 5.4.3 and 4.8.9 and Corollary 4.7.12(a), the functor (6.4) is compatible with the $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})$ -actions on the two sides. Furthermore, by Sect. 5.4.3, the $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})$ -action on $\mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)}$ factors through an action of $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})$.

Hence, the functor (6.4) gives rise to a (factorization) functor

$$(6.5) \quad \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})}(\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}), \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})).$$

Finally, recall that by Lemma 3.5.7, we have a canonical identification

$$\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \simeq \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})}(\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}), \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})).$$

Thus, we can interpret (6.5) as a $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})$ -linear functor

$$(6.6) \quad \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$$

6.1.3. Precomposing (6.6) with $\mathrm{KL}(G)_{\mathrm{crit}} \xrightarrow{\alpha_{\rho(\omega_X), \mathrm{taut}}} \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)}$, we obtain a functor

$$(6.7) \quad \mathrm{FLE}_{G, \mathrm{crit}} : \mathrm{KL}(G)_{\mathrm{crit}} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$$

The functor (6.7) is the critical FLE functor. The main result of Part I of this paper reads:

Theorem 6.1.4. *The functor $\mathrm{FLE}_{G, \mathrm{crit}}$ is an equivalence of factorization categories.*

This theorem will be proved in the course of this and the next two sections.

Remark 6.1.5. A pointwise version of Theorem 6.1.4, formulated below as Theorem 6.2.2, was originally proved in [FG2].

The proof of the factorization version requires substantially new ideas.

Progress towards Theorem 6.1.4 has been made earlier in the papers [FLMM1, FLMM2].

6.1.6. Unwinding the definitions, we observe that the functor $\mathrm{FLE}_{G, \mathrm{crit}}$ carries a natural lax unital structure (as a functor between unital factorization categories). In particular, we obtain a canonical homomorphism

$$(6.8) \quad \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}}} = \mathbf{1}_{\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})} \rightarrow \mathrm{FLE}_{G, \mathrm{crit}}(\mathbf{1}_{\mathrm{KL}(G)_{\mathrm{crit}}}) = \mathrm{Vac}(G)_{\mathrm{crit}}$$

as factorization algebras in $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$.

However, we claim:

Lemma 6.1.7. *The map (6.8) is an isomorphism.*

Proof. By Proposition 3.3.5(a), it suffices to show that the map (6.8) becomes an isomorphism after applying the functor $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$. Since the latter is also a strict unital factorization functor, the resulting homomorphism identifies with a homomorphism of factorization algebras

$$\mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}}} = \mathbf{1}_{\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{reg}})} \rightarrow \mathrm{DS}^{\mathrm{enh}, \mathrm{rfnd}}(\mathrm{Vac}(G)_{\mathrm{crit}, \rho(\omega_X)}),$$

corresponding to the lax unital functor

$$\mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \widehat{\mathfrak{g}\text{-mod}}_{\mathrm{crit}, \rho(\omega_X)} \xrightarrow{\mathrm{DS}^{\mathrm{enh}, \mathrm{rfnd}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}).$$

However, the latter isomorphism is the content of Theorem 4.8.3. □

6.1.8. Combining Lemmas 6.1.7 and C.11.23, we obtain:

Corollary 6.1.9. *The functor $\mathrm{FLE}_{G, \mathrm{crit}}$ is strictly unital.*

6.2. Reduction to the pointwise version.

6.2.1. Fix a point $x \in X$. The pointwise version of Theorem 6.1.4 reads:

Theorem 6.2.2. *The functor $\mathrm{FLE}_{G,\mathrm{crit}}$ induces an equivalence*

$$\mathrm{KL}(G)_{\mathrm{crit},x} \xrightarrow{\sim} \mathrm{IndCoh}^*(\mathrm{Op}_{G,x}^{\mathrm{mon-free}}).$$

Obviously, Theorem 6.1.4 implies Theorem 6.2.2. However, in this subsection we will show that the converse implication also takes place.

In its turn, Theorem 6.2.2 is known: it is the main result of the paper [FG2]. We will, however, supply a different proof, in which we deduce it from Theorem 4.8.8, see Sect. 7.2.

6.2.3. A key step in proving the implication Theorem 6.2.2 $\Rightarrow$ Theorem 6.1.4 is the following:

Proposition 6.2.4. *The functor $\mathrm{FLE}_{G,\mathrm{crit}}$ preserves compactness.³¹*

Proof. By Proposition 3.3.5(b), it suffices to show that the composite functor

$$\mathrm{KL}(G)_{\mathrm{crit}} \xrightarrow{\mathrm{FLE}_{G,\mathrm{crit}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \xrightarrow{(\iota^{\mathrm{mon-free}})^* \mathrm{IndCoh}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})$$

preserves compactness.

I.e., it suffices to show that (6.4) preserves compactness. Since $\overline{\mathrm{DS}}^{\mathrm{enh},\mathrm{rfnd}}$ is an equivalence, it suffices to show that the functor (6.3) preserves compactness. However, we claim that this is true more generally.

Namely, we claim that the functor (6.2) admits a continuous right adjoint (and hence, preserves compactness). Indeed, the right adjoint in question is given by³² convolution with the vacuum object (i.e., the factorization unit)

$$\mathrm{Vac}_{\mathrm{Whit}^!(G)} \in \mathrm{Whit}^!(G) \simeq \mathrm{D-mod}_{\frac{1}{2}}(\mathfrak{L}(G)_{\rho(\omega_X)})^{(\mathfrak{L}(N)_{\rho(\omega_X)}, \chi), \mathfrak{L}^+(G)_{\rho(\omega_X)}}.$$

□

6.2.5. Given Proposition 6.2.4 and Theorem 6.2.2, we will deduce Theorem 6.1.4 using the following principle:

Let $F : \mathbf{C}^1 \rightarrow \mathbf{C}^2$ be a factorization functor between factorization categories. Assume that $\mathbf{C}^1$ is compactly generated, and assume that F preserves compactness.

Proposition 6.2.6. *If the induced functor $F_x : \mathbf{C}_x^1 \rightarrow \mathbf{C}_x^2$ is an equivalence for any field-valued point x , then the original functor F is also an equivalence.*

Proof. The assumption that F preserves compactness implies that its right adjoint F^R is also equipped with a factorization structure. We need to show that the unit and the counit of the (F, F^R) -adjunction are isomorphisms.

The latter assertion can be checked strata-wise on Ran . I.e., we have to show that for every n , the corresponding functor

$$F_{X(n)}^\circ : \mathbf{C}_{X(n)}^1 \rightarrow \mathbf{C}_{X(n)}^2$$

is an equivalence.

By factorization, the latter statement reduces to the case $n = 1$, i.e., we have to show that

$$F_X : \mathbf{C}_X^1 \rightarrow \mathbf{C}_X^2$$

is an equivalence.

The latter fact can be also checked after base-changing to field-valued points.

□

□[Theorem 6.1.4]

³¹See Sect. B.11.10 for what it means for a factorization functor to preserve compactness.

³²In Lemma 6.3.2 we will give another description of this right adjoint.

6.3. The inverse of the critical FLE functor. In this subsection we will assume the statement of Theorem 6.1.4, which was proved modulo Theorem 6.2.2.

6.3.1. Let $\mathbf{C}$ be a category, acted on by $\mathfrak{L}(G)_{\rho(\omega_X)}$. Note that in addition to the functor

$$(6.9) \quad \mathrm{Sph}(\mathbf{C}) \rightarrow \mathbf{C} \rightarrow \mathrm{Whit}_*(\mathbf{C}),$$

one can consider the functor

$$(6.10) \quad \mathrm{Whit}^!(\mathbf{C}) \rightarrow \mathbf{C} \xrightarrow{\mathrm{Av}_*^{\mathfrak{L}^+(G)_{\rho(\omega_X)}}} \mathrm{Sph}(\mathbf{C}).$$

We have the following elementary assertion:

Lemma 6.3.2. *The composite*

$$(6.11) \quad \mathrm{Whit}_*(\mathbf{C}) \xrightarrow{\Theta_{\mathrm{Whit}(\mathbf{C})}} \mathrm{Whit}^!(\mathbf{C}) \xrightarrow{(6.10)} \mathrm{Sph}(\mathbf{C})$$

identifies canonically with the right adjoint of (6.9).

Remark 6.3.3. This lemma is embedded into the machinery developed in [Ra2]. We supply a proof for completeness.

Proof. We need to check that for $\mathcal{F}_{\mathrm{Sph}} \in \mathrm{Sph}(\mathbf{C})$ and $\mathcal{F} \in \mathbf{C}$, we have

$$\mathcal{H}om_{\mathrm{Whit}_*(\mathbf{C})}(\overline{\mathcal{F}}_{\mathrm{Sph}}, \overline{\mathcal{F}}) \simeq \mathcal{H}om_{\mathbf{C}}(\mathcal{F}_{\mathrm{Sph}}, \Theta_{\mathrm{Whit}(\mathbf{C})}(\overline{\mathcal{F}})),$$

where $\overline{\mathcal{F}}_{\mathrm{Sph}}, \overline{\mathcal{F}}$ denotes the image of $\mathcal{F}_{\mathrm{Sph}}$ and $\mathcal{F}$ along $\mathbf{C} \rightarrow \mathrm{Whit}_*(\mathbf{C})$.

Unwinding the definitions, we reduce the assertion to the case when $\mathbf{C} := \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \rho(\omega_X)})$ and $\mathcal{F}_{\mathrm{Sph}} = \delta_{1, \mathrm{Gr}_{G, \rho(\omega_X)}}$.

Applying the definition of $\mathrm{Whit}_*(G)$, we calculate

$$(6.12) \quad \mathcal{H}om_{\mathrm{Whit}_*(G)}(\overline{\delta}_{1, \mathrm{Gr}_{G, \rho(\omega_X)}}, \overline{\mathcal{F}}) = \mathrm{colim}_{\alpha} \mathcal{H}om_{\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \rho(\omega_X)})^{N^{\alpha}, \chi}}((\delta_{1, \mathrm{Gr}_{G, \rho(\omega_X)}})^{\alpha}, \mathcal{F}^{\alpha}),$$

where:

- N^{α} is a filtered family of group subschemes that comprise $\mathfrak{L}(N)_{\rho(\omega_X)}$;
- $(\delta_{1, \mathrm{Gr}_{G, \rho(\omega_X)}})^{\alpha}$ and $\mathcal{F}^{\alpha}$ denote the projections of the corresponding objects along

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \rho(\omega_X)}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \rho(\omega_X)})^{N^{\alpha}, \chi}.$$

We have

$$\mathcal{H}om_{\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \rho(\omega_X)})^{N^{\alpha}, \chi}}((\delta_{1, \mathrm{Gr}_{G, \rho(\omega_X)}})^{\alpha}, \mathcal{F}^{\alpha}) \simeq \mathcal{H}om_{\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \rho(\omega_X)})}(\mathrm{Av}_*^{N^{\alpha}, \chi}(\delta_{1, \mathrm{Gr}_{G, \rho(\omega_X)}}), \mathcal{F}),$$

which we further rewrite as

$$\mathrm{C}^{\bullet} \left(\mathrm{Gr}_{G, \rho(\omega_X)}, \mathbb{D} \left(\mathrm{Av}_*^{N^{\alpha}, \chi}(\delta_{1, \mathrm{Gr}_{G, \rho(\omega_X)}}) \right)^{\dagger} \otimes \mathcal{F} \right) \simeq \mathrm{C}^{\bullet} \left(\mathrm{Gr}_{G, \rho(\omega_X)}, \mathrm{Av}_!^{N^{\alpha}, \chi}(\delta_{1, \mathrm{Gr}_{G, \rho(\omega_X)}})^{\dagger} \otimes \mathcal{F} \right).$$

Hence, we can rewrite (6.12) as

$$(6.13) \quad \begin{aligned} \mathrm{C}^{\bullet} \left(\mathrm{Gr}_{G, \rho(\omega_X)}, \mathrm{colim}_{\alpha} (\mathrm{Av}_!^{N^{\alpha}, \chi}(\delta_{1, \mathrm{Gr}_{G, \rho(\omega_X)}}))^{\dagger} \otimes \mathcal{F} \right) &\simeq \\ &\simeq \mathrm{C}^{\bullet} \left(\mathrm{Gr}_{G, \rho(\omega_X)}, \mathrm{Av}_!^{\mathfrak{L}(N)_{\rho(\omega_X)}, \chi}(\delta_{1, \mathrm{Gr}_{G, \rho(\omega_X)}})^{\dagger} \otimes \mathcal{F} \right). \end{aligned}$$

Now, the cleanness property from Sect. 1.3.15 implies that the natural map

$$\mathrm{Av}_!^{\mathfrak{L}(N)_{\rho(\omega_X)}, \chi}(\delta_{1, \mathrm{Gr}_{G, \rho(\omega_X)}}) \rightarrow \mathrm{Av}_{*, \mathrm{ren}}^{\mathfrak{L}(N)_{\rho(\omega_X)}, \chi}(\delta_{1, \mathrm{Gr}_{G, \rho(\omega_X)}})$$

is an isomorphism, where $\mathrm{Av}_{*, \mathrm{ren}}^{\mathfrak{L}(N)_{\rho(\omega_X)}, \chi}$ is the functor of $*$ -convolution with $\omega_{\mathfrak{L}(N)_{\rho(\omega_X)}, \chi}^{\mathrm{ren}}$.

Hence, we further rewrite (6.13) as

$$\begin{aligned} C^* \left(\mathrm{Gr}_{G, \rho(\omega_X)}, \mathrm{Av}_{*, \mathrm{ren}}^{\mathfrak{L}^{(N)} \rho(\omega_X), \chi} (\delta_{1, \mathrm{Gr}_{G, \rho(\omega_X)}}) \overset{!}{\otimes} \mathcal{F} \right) &\simeq \\ &\simeq C^* \left(\mathrm{Gr}_{G, \rho(\omega_X)}, \delta_{1, \mathrm{Gr}_{G, \rho(\omega_X)}} \overset{!}{\otimes} \mathrm{Av}_{*, \mathrm{ren}}^{\mathfrak{L}^{(N)} \rho(\omega_X), \chi} (\mathcal{F}) \right) \simeq \\ &\simeq \mathcal{H}om_{\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \rho(\omega_X)})} (\delta_{1, \mathrm{Gr}_{G, \rho(\omega_X)}}, \Theta_{\mathrm{Whit}(G)}(\overline{\mathcal{F}})), \end{aligned}$$

as desired. $\square$

6.3.4. We will now use Lemma 6.3.2 to give an explicit description of the inverse of the functor $\mathrm{FLE}_{G, \mathrm{crit}}$.

Consider the functor (6.11) for $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}$

$$(6.14) \quad \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)}.$$

By the same logic as in Sect. 6.1.2, the functor (6.14) gives rise to a functor

$$(6.15) \quad \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \overset{\otimes}{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})} \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)}.$$

Combining with the equivalence $\overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}}$, we obtain a functor

$$(6.16) \quad \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \overset{\otimes}{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)}.$$

Combining with the equivalence (3.15), from (6.16) we obtain a functor

$$(6.17) \quad \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)}.$$

We will prove:

Proposition 6.3.5. *The functor (6.17) is the inverse of (6.6).*

The rest of this subsection is devoted to the proof of Proposition 6.3.5.

6.3.6. We need to show that the composition

$$\begin{aligned} (6.18) \quad \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \overset{\otimes}{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})} \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) &\xrightarrow{\mathrm{Id} \otimes \overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}}} \\ &\rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \overset{\otimes}{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \end{aligned}$$

is isomorphic to

$$\begin{aligned} (6.19) \quad \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \overset{\otimes}{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})} \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) &\xrightarrow{(6.16)} \\ &\rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \xrightarrow{(6.6)} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}). \end{aligned}$$

Both functors are $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})$ -linear. Hence, by the

$$\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})\text{-}\mathbf{mod} \rightleftarrows \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})\text{-}\mathbf{mod}$$

adjunction, it suffices to show that the functors

$$(6.20) \quad \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \xrightarrow{\overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \xrightarrow{(\iota^{\mathrm{mon-free}})^!} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$$

and

$$\mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \xrightarrow{(6.15)} \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \xrightarrow{(6.6)} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$$

are isomorphic as $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})$ -linear functors, i.e., that the diagram

$$\begin{array}{ccc} \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) & \xrightarrow{(6.15)} & \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \\ \overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}} \downarrow & & \downarrow (6.6) \\ \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) & \xrightarrow{(\iota^{\mathrm{mon-free}})^!} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \end{array}$$

commutes, in a way compatible with the $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})$ -actions.

6.3.7. Since $\overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}}$ and (6.6) are both equivalences, it suffices to show that the diagram obtained by passing to left adjoints along the horizontal arrows, i.e.,

$$\begin{array}{ccc} \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) & \longleftarrow & \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \\ \overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}} \downarrow & & \downarrow (6.6) \\ \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) & \xleftarrow{(\iota^{\mathrm{mon-free}})_* \mathrm{IndCoh}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}), \end{array}$$

commutes, in a way compatible with the $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})$ -actions.

6.3.8. However, according to Lemma 6.3.2, the top vertical arrow in the latter diagram is the functor (6.3), and the corresponding diagram commutes by construction.

□[Proposition 6.3.5]

6.4. Compatibility of $\mathrm{FLE}_{G, \mathrm{crit}}$ and $\mathrm{FLE}_{\check{G}, \infty}$.

6.4.1. Note that by construction, the functor $\mathrm{FLE}_{G, \mathrm{crit}}$ makes the following diagram commute

$$\begin{array}{ccccc} \mathrm{KL}(G)_{\mathrm{crit}} & \xrightarrow[\sim]{\alpha_{\rho(\omega_X), \mathrm{taut}}} & \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} & \longrightarrow & \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \\ \mathrm{FLE}_{G, \mathrm{crit}} \downarrow & & & & \downarrow \overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}} \\ \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) & \xrightarrow{(\iota^{\mathrm{mon-free}})_* \mathrm{IndCoh}} & & & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}). \end{array}$$

6.4.2. Note that the functor (6.2) can be expanded to a functor

$$(6.21) \quad \mathrm{Whit}_*(G) \otimes_{\mathrm{Sph}_G} \mathrm{Sph}(\mathbf{C}) \rightarrow \mathrm{Whit}_*(\mathbf{C}).$$

Applying this to $\mathbf{C} = \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}$, we obtain a functor

$$(6.22) \quad \mathrm{Whit}_*(G) \otimes_{\mathrm{Sph}_G} \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}).$$

Composing with $\mathrm{KL}(G)_{\mathrm{crit}} \xrightarrow[\simeq]{\alpha_{\rho(\omega_X), \mathrm{taut}}} \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)}$ and $\overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}}$, we obtain a (factorization) functor

$$(6.23) \quad \mathrm{Whit}_*(G) \otimes_{\mathrm{Sph}_G} \mathrm{KL}(G)_{\mathrm{crit}} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}).$$

6.4.3. Similarly, the functor

$$(\iota^{\mathrm{mon-free}})_* \mathrm{IndCoh} : \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})$$

can be expanded to a (factorization) functor

$$(6.24) \quad \mathrm{Rep}(\check{G}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}),$$

6.4.4. We claim:

Theorem 6.4.5.

(a) *The functor $\mathrm{FLE}_{G,\mathrm{crit}}$ can be canonically endowed with the datum of compatibility with the Sph_G -action on $\mathrm{KL}(G)_{\mathrm{crit}}$ and the $\mathrm{Sph}_G^{\mathrm{spec}}$ -action on $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$, where we identify*

$$\mathrm{Sph}_G \simeq \mathrm{Sph}_G^{\mathrm{spec}}$$

via Sat_G .

(b) *Under the identification of point (a), the diagram*

$$(6.25) \quad \begin{array}{ccc} \mathrm{Whit}_*(G) \otimes_{\mathrm{Sph}_G} \mathrm{KL}(G)_{\mathrm{crit}} & \xrightarrow{(6.23)} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \\ \mathrm{FLE}_{G,\infty}^{-1} \otimes \mathrm{FLE}_{G,\mathrm{crit}} \downarrow \sim & & \sim \downarrow \mathrm{Id} \\ \mathrm{Rep}(\check{G}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) & \xrightarrow{(6.24)} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \end{array}$$

canonically commutes.

This theorem will be proved in the factorization setting in Sect. E.10.

6.4.6. Let us denote by $\mathbf{P}_G^{\mathrm{loc,enh,rfd}}$ the precomposition of (6.23) with the projection

$$\mathrm{Whit}_*(G) \otimes \mathrm{KL}(G)_{\mathrm{crit}} \rightarrow \mathrm{Whit}_*(G) \otimes_{\mathrm{Sph}_G} \mathrm{KL}(G)_{\mathrm{crit}}.$$

Explicitly, it is given by

$$(6.26) \quad \begin{aligned} & \mathrm{Whit}_*(G) \otimes \mathrm{KL}(G)_{\mathrm{crit}} \xrightarrow{\mathrm{Id} \otimes \alpha_{\rho(\omega_X), \mathrm{taut}}} \mathrm{Whit}_*(G) \otimes \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \\ & \rightarrow \mathrm{Whit}_*(G) \otimes_{\mathrm{Sph}_G} \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \xrightarrow{(6.22)} \mathrm{Whit}_*(\widehat{\mathfrak{g}\text{-mod}}_{\mathrm{crit}, \rho(\omega_X)}) \xrightarrow{\overline{\mathrm{DS}}^{\mathrm{enh,rfd}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}). \end{aligned}$$

Let $\mathbf{P}_G^{\mathrm{loc}}$ and $\mathbf{P}_G^{\mathrm{loc,enh}}$ denote the compositions of $\mathbf{P}_G^{\mathrm{loc,enh,rfd}}$ with the forgetful functors

$$(6.27) \quad \Gamma^{\mathrm{IndCoh}}(\mathrm{Op}_G^{\mathrm{mer}}, -) : \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \rightarrow \mathrm{Vect}$$

and

$$(6.28) \quad \Gamma^{\mathrm{IndCoh}}(\mathrm{Op}_G^{\mathrm{mer}}, -)^{\mathrm{enh}} : \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \rightarrow \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg-mod}} \text{fact}},$$

respectively. (These two functors are obtained by replacing the last arrow in (6.26) by $\overline{\mathrm{DS}}$ and $\overline{\mathrm{DS}}^{\mathrm{enh}}$, respectively.)

6.4.7. Let us denote by $\mathbf{P}_G^{\mathrm{loc,enh,rfd}}$ the precomposition of (6.24) with the projection

$$\mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \rightarrow \mathrm{Rep}(\check{G}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}).$$

Explicitly, it is given by

$$(6.29) \quad \begin{aligned} & \mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \xrightarrow{\tau^* \otimes \mathrm{Id}} \\ & \rightarrow \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \xrightarrow{\otimes} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \xrightarrow{(\iota^{\mathrm{mon-free}})^{\mathrm{IndCoh}}_*} \\ & \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}). \end{aligned}$$

Let $\mathbf{P}_G^{\mathrm{loc}}$ and $\mathbf{P}_G^{\mathrm{loc,enh}}$ denote the compositions of $\mathbf{P}_G^{\mathrm{loc,enh,rfd}}$ with the forgetful functors (6.27) and (6.28), respectively.

6.4.8. From Theorem 6.4.5 we immediately obtain:

Corollary 6.4.9. *The functors $P_G^{\text{loc,enh,rfd}}$ and $P_{\check{G}}^{\text{loc,enh,rfd}}$ match under the equivalences*

$$\text{KL}(G)_{\text{crit}} \xrightarrow{\text{FLE}_{G,\text{crit}}} \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \text{ and } \text{Rep}(\check{G}) \xrightarrow{\text{FLE}_{\check{G},\infty}} \text{Whit}_*(G).$$

And hence:

Corollary 6.4.10. *The functors P_G^{loc} and $P_{\check{G}}^{\text{loc}}$ (resp., $P_G^{\text{loc,enh}}$ and $P_{\check{G}}^{\text{loc,enh}}$) match under the equivalences*

$$\text{KL}(G)_{\text{crit}} \xrightarrow{\text{FLE}_{G,\text{crit}}} \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \text{ and } \text{Rep}(\check{G}) \xrightarrow{\text{FLE}_{\check{G},\infty}} \text{Whit}_*(G).$$

6.5. The functor $\text{pre-FLE}_{G,\text{crit}}$

6.5.1. By the construction of the functor $\text{FLE}_{G,\text{crit}}$ we have the following explicit descriptions of its compositions with various forgetful functors out of $\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})$:

- The composition with the functor

$$\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \xrightarrow{(\iota^{\text{mon-free}})^* \text{IndCoh}} \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}})$$

is the functor

$$\text{KL}(G)_{\text{crit}} \xrightarrow{\alpha_{\rho(\omega_X)}, \text{taut}} \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)} \xrightarrow{\text{DS}^{\text{enh, rfd}}} \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}});$$

- The composition with the functor

$$\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \xrightarrow{(\iota^{\text{mon-free}})^* \text{IndCoh}} \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \xrightarrow{\Gamma^{\text{IndCoh}}(\text{Op}_{\check{G}}^{\text{mer}}, -)^{\text{enh}}} \mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg-mod}}^{\text{fact}}}$$

is the functor

$$\text{KL}(G)_{\text{crit}} \xrightarrow{\alpha_{\rho(\omega_X)}, \text{taut}} \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)} \xrightarrow{\text{DS}^{\text{enh}}} \mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg-mod}}^{\text{fact}}};$$

- The composition with the functor

$$\Gamma^{\text{IndCoh}}(\text{Op}_{\check{G}}^{\text{mon-free}}, -) : \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \rightarrow \text{Vect}$$

is the functor

$$(6.30) \quad \text{KL}(G)_{\text{crit}} \xrightarrow{\alpha_{\rho(\omega_X)}, \text{taut}} \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)} \xrightarrow{\text{DS}} \text{Vect};$$

In this subsection we will describe explicitly the composition of $\text{FLE}_{G,\text{crit}}$ with the functor

$$\mathfrak{t}_*^{\text{IndCoh}} : \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \rightarrow \text{Rep}(\check{G}).$$

6.5.2. Define the (factorization) functor

$$(6.31) \quad \text{pre-FLE}_{G,\text{crit}} : \text{KL}(G)_{\text{crit}} \rightarrow \text{Rep}(\check{G})$$

as the composition

$$(6.32) \quad \text{KL}(G)_{\text{crit}} \xrightarrow{\alpha_{\rho(\omega_X)}, \text{taut}} \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \text{Whit}^!(G) \otimes \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}) \xrightarrow{\text{CS}_G \otimes \text{Id}} \\ \rightarrow \text{Rep}(\check{G}) \otimes \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}) \xrightarrow{\text{Id} \otimes \overline{\text{DS}}} \text{Rep}(\check{G}),$$

where the second arrow is obtained by duality from the pairing

$$\text{Whit}_*(G) \otimes \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}).$$

6.5.3. We claim:

Proposition 6.5.4. *The functor $\text{pre-FLE}_{G,\text{crit}}$ identifies canonically with $\mathfrak{t}_*^{\text{IndCoh}} \circ \text{FLE}_{G,\text{crit}}$.*

The rest of this subsection is devoted to the proof of this proposition.

6.5.5. The next assertion results from the construction of the functor $\mathrm{FLE}_{G,\mathrm{crit}}$ and Lemma 5.4.2:

Lemma 6.5.6. *The functor $\mathfrak{t}_*^{\mathrm{IndCoh}} \circ \mathrm{FLE}_{G,\mathrm{crit}}$ identifies with the composition*

$$(6.33) \quad \mathrm{KL}(G)_{\mathrm{crit}} \rightarrow \mathrm{Rep}(\check{G}) \otimes \mathrm{KL}(G)_{\mathrm{crit}} \xrightarrow{\mathrm{Id} \otimes \alpha_{\rho(\omega_X), \mathrm{taut}}} \mathrm{Rep}(\check{G}) \otimes \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{Rep}(\check{G}) \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)} \xrightarrow{\mathrm{Id} \otimes \overline{\mathrm{DS}}} \mathrm{Rep}(\check{G}),$$

where the first arrow is the functor, right adjoint to the action of $\mathrm{Rep}(\check{G})$ on $\mathrm{KL}(G)_{\mathrm{crit}}$, given by

$$\mathrm{Rep}(\check{G}) \xrightarrow{\mathrm{Sat}_G^{\mathrm{nv}}} \mathrm{Sph}_G$$

and the Sph_G -action on $\mathrm{KL}(G)_{\mathrm{crit}}$.

Hence, in order to prove Proposition 6.5.4, it suffices to establish an isomorphism between (6.32) and (6.33).

6.5.7. We rewrite the functor in (6.33) as

$$\begin{aligned} \mathrm{KL}(G)_{\mathrm{crit}} &\xrightarrow{\alpha_{\rho(\omega_X), \mathrm{taut}}} \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \\ &\rightarrow \mathrm{Rep}(\check{G}) \otimes \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{Rep}(\check{G}) \otimes \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \xrightarrow{\mathrm{Id} \otimes \overline{\mathrm{DS}}} \mathrm{Rep}(\check{G}). \end{aligned}$$

Hence, in order to prove an isomorphism between (6.32) and (6.33), it suffices to show that the functors

$$\mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{Rep}(\check{G}) \otimes \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{Rep}(\check{G}) \otimes \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})$$

and

$$\mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{Whit}^!(G) \otimes \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \xrightarrow{\mathrm{CS}_G \otimes \mathrm{Id}} \mathrm{Rep}(\check{G}) \otimes \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})$$

are canonically isomorphic.

6.5.8. By duality, this amounts to showing that the functors

$$\begin{aligned} \mathrm{Rep}(\check{G}) \otimes \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} &\xrightarrow{\mathrm{Sat}_G^{\mathrm{nv}} \otimes \mathrm{Id}} \\ &\rightarrow \mathrm{Sph}_G \otimes \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \end{aligned}$$

and

$$\mathrm{Rep}(\check{G}) \otimes \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \xrightarrow{\mathrm{FLE}_{\check{G}, \infty} \otimes \mathrm{Id}} \mathrm{Whit}_*(G) \otimes \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})$$

are canonically identified.

6.5.9. This follows by combining the following observations:

- The functor $\mathrm{Whit}_*(G) \otimes \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})$ factors as $\mathrm{Whit}_*(G) \otimes \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{Whit}_*(G) \otimes_{\mathrm{Sph}_G} \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})$;
- The functor $\mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})$ identifies with $\mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \xrightarrow{\mathrm{Vac}_{\mathrm{Whit}_*(G)} \otimes \mathrm{Id}} \mathrm{Whit}_*(G) \otimes \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})$, where $\mathbf{1}_{\mathrm{Whit}_*(G)} \in \mathrm{Whit}_*(G)$ is the vacuum object;
- The functor $\mathrm{FLE}_{\check{G}, \infty}$ identifies with

$$\mathrm{Rep}(\check{G}) \xrightarrow{\mathrm{Sat}_G^{\mathrm{nv}}} \mathrm{Sph}_G \xrightarrow{\mathrm{Vac}_{\mathrm{Whit}_*(G)} \star -} \mathrm{Whit}_*(G)$$

(see Remark 1.7.8).

□[Proposition 6.5.4]

6.6. An alternative construction of the critical FLE functor.

6.6.1. We start by observing:

Lemma 6.6.2. *There exists a canonical identification of factorization algebras in $\text{Rep}(\check{G})$*

$$(6.34) \quad \text{pre-FLE}_{G,\text{crit}}(\text{Vac}(G)_{\text{crit}}) \simeq R_{\check{G},\text{Op}}.$$

Proof. Follows from Proposition 6.5.4 and the fact that the functor $\text{FLE}_{G,\text{crit}}$ is unital (see Corollary 6.1.9). $\square$

Remark 6.6.3. One can establish the isomorphism (6.34) directly, i.e., without appealing to the functor $\text{FLE}_{G,\text{crit}}$.

6.6.4. By Sect. 4.1.9 and Lemma 6.6.2, the functor $\text{pre-FLE}_{G,\text{crit}}$ upgrades to a functor

$$\text{pre-FLE}_{G,\text{crit}}^{\text{enh}} : \text{KL}(G)_{\text{crit}} \rightarrow R_{\check{G},\text{Op}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G})).$$

Note that by construction we have a canonical isomorphism

$$\text{pre-FLE}_{G,\text{crit}}^{\text{enh}} \simeq \Gamma^{\text{IndCoh}}(\text{Op}_{\check{G}}^{\text{mon-free}}, -)^{\text{enh}} \circ \text{FLE}_{G,\text{crit}}.$$

Because of this isomorphism we will also use the notation

$$\text{FLE}_{G,\text{crit}}^{\text{coarse}} := \text{pre-FLE}_{G,\text{crit}}^{\text{enh}}.$$

6.6.5. We claim now that the functor $\text{FLE}_{G,\text{crit}}$ can be uniquely recovered from $\text{FLE}_{G,\text{crit}}^{\text{coarse}}$.

Namely, by Proposition 4.4.7(a), it suffices to show that the functor $\text{FLE}_{G,\text{crit}}^{\text{coarse}}$ sends compact objects in $\text{KL}(G)_{\text{crit}}$ to eventually coconnective objects in $R_{\check{G},\text{Op}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}))$.

6.6.6. Since the forgetful functor

$$R_{\check{G},\text{Op}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G})) \rightarrow \text{Rep}(\check{G})$$

is t-exact and conservative, it suffices to show that the functor $\text{pre-FLE}_{G,\text{crit}}$ sends compact objects in $\text{KL}(G)_{\text{crit}}$ to eventually coconnective objects in $\text{Rep}(\check{G})$.

Since the compact generators of $\text{KL}(G)_{\text{crit}}$ are eventually coconnective, it suffices to prove the following:

Lemma 6.6.7. *For a fixed $\underline{x} \in \text{Ran}$, the functor*

$$\text{pre-FLE}_{G,\text{crit}} : \text{KL}(G)_{\text{crit},\underline{x}} \rightarrow \text{Rep}(\check{G})_{\underline{x}}$$

is t-exact.

Proof. We rewrite the functor $\text{pre-FLE}_{G,\text{crit}}$ as (6.33), or equivalently

$$(6.35) \quad \begin{aligned} \text{KL}(G)_{\text{crit}} &\xrightarrow{R_{\check{G}} \otimes \text{Id}} \text{Rep}(\check{G}) \otimes \text{Rep}(\check{G}) \otimes \text{KL}(G)_{\text{crit}} \xrightarrow{\text{Id} \otimes \text{Sat}_{\check{G}}^{\text{ny}} \otimes \text{Id}} \text{Rep}(\check{G}) \otimes \text{Sph}_{\check{G}} \otimes \text{KL}(G)_{\text{crit}} \xrightarrow{\text{Id} \otimes (- \star -)} \\ &\rightarrow \text{Rep}(\check{G}) \otimes \text{KL}(G)_{\text{crit}} \xrightarrow{\text{Id} \otimes \alpha_{\rho(\omega_X), \text{taut}}} \text{Rep}(\check{G}) \otimes \text{KL}(G)_{\text{crit},\rho(\omega_X)} \xrightarrow{\text{Id} \otimes \text{DS}} \text{Rep}(\check{G}) \end{aligned}$$

In this composition, the first arrow is tautologically exact, and the second arrow is t-exact by Corollary 5.4.5. Hence, the assertion follows from Lemma 2.3.8. $\square$

6.6.8. Note that as a corollary of Lemma 6.6.7 and Proposition 6.5.4, we obtain:

Corollary 6.6.9. *For a fixed $\underline{x} \in \text{Ran}$, the functor*

$$\text{FLE}_{G,\text{crit}} : \text{KL}(G)_{\text{crit},\underline{x}} \rightarrow \text{IndCoh}^*(\text{Op}_{\check{G},\underline{x}}^{\text{mon-free}})$$

is t-exact.

7. PROOF OF THE POINTWISE VERSION OF THE CRITICAL FLE

In this section we will give a proof of the pointwise version of the critical FLE by deducing it from Theorem 4.8.8.

The idea of the proof is the that the critical FLE is essentially the *base change* of the equivalence of Theorem 4.8.8 along $LS_{\check{G},x}^{\text{reg}} \rightarrow LS_{\check{G},x}^{\text{mer}}$. In fact, such an equivalence is a general phenomenon for categories acted on by $\mathfrak{L}(G)_x$, given a *temperedness* condition (see Proposition 7.5.5 for a precise statement).

The reason this proof only works for the pointwise version is that it is only in this case that we have a good grip on the base change operation alluded to above.

We note that a completely different proof of the pointwise FLE was given in the paper [FG2].

7.1. Temperedness.

7.1.1. Let

$$(7.1) \quad \text{Sph}_{\check{G},\text{temp},x}^{\text{spec}} \hookrightarrow \text{Sph}_{\check{G},x}^{\text{spec}}$$

be the *tempered subcategory*.

By definition, this is the essential image of

$$\Xi_{\text{Hecke}_{\check{G},x}^{\text{spec,loc}}} : \text{QCoh}(\text{Hecke}_{\check{G},x}^{\text{spec,loc}}) \rightarrow \text{IndCoh}(\text{Hecke}_{\check{G},x}^{\text{spec,loc}}).$$

The embedding (7.1) admits a right adjoint, namely,

$$\Psi_{\text{Hecke}_{\check{G},x}^{\text{spec,loc}}} : \text{IndCoh}(\text{Hecke}_{\check{G},x}^{\text{spec,loc}}) \rightarrow \text{QCoh}(\text{Hecke}_{\check{G},x}^{\text{spec,loc}}),$$

whose kernel is a monoidal ideal.

This allows us to view $\text{Sph}_{\check{G},\text{temp},x}^{\text{spec}}$ as a monoidal colocalization of $\text{Sph}_{\check{G},x}^{\text{spec}}$.

Remark 7.1.2. The definition of $\text{Sph}_{\check{G},\text{temp},x}^{\text{spec}}$ is specific to the pointwise version. We do not know how to define it in the factorization setting. The reason for this is the following:

Although we can define $\text{Sph}_{\check{G},\text{temp}}^{\text{spec}} := \text{IndCoh}^*(\text{Hecke}_{\check{G}}^{\text{spec,loc}})$ in the factorization setting, we do not have the (Ξ, Ψ) -adjunction. The latter is a feature of a locally of finite type situation, which we *are* in at a fixed $\underline{x} \in \text{Ran}$, but *not* when we are allowed to vary over Ran in families.

7.1.3. Let us regard

$$\text{QCoh}(LS_{\check{G},x}^{\text{reg}}) \simeq \text{Rep}(\check{G})$$

as a bimodule with respect to $\text{QCoh}((LS_{\check{G},x}^{\text{mer}})^{\wedge}_{\text{reg}})$ and $\text{Sph}_{\check{G},x}^{\text{spec}}$.

Note, however, that the $\text{Sph}_{\check{G},x}^{\text{spec}}$ -action on $\text{QCoh}(LS_{\check{G},x}^{\text{reg}})$ factors via $\text{Sph}_{\check{G},\text{temp},x}^{\text{spec}}$: indeed, the action is given by t-exact functors and the t-structure on $\text{Rep}(\check{G})$ is separated.

7.1.4. Consider the corresponding functor

$$(7.2) \quad \text{Sph}_{\check{G},\text{temp},x}^{\text{spec}} \rightarrow \text{Funct}_{\text{QCoh}((LS_{\check{G},x}^{\text{mer}})^{\wedge}_{\text{reg}})}(\text{QCoh}(LS_{\check{G},x}^{\text{reg}}), \text{QCoh}(LS_{\check{G},x}^{\text{reg}})).$$

The following results from the definitions:

Lemma 7.1.5. *The functor (7.2) is an equivalence.*

7.1.6. Let $\mathbf{C}$ be a module category over $\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}$. Denote

$$\mathbf{C}_{\mathrm{temp}} := \mathrm{Sph}_{\check{G},\mathrm{temp},x}^{\mathrm{spec}} \otimes_{\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}} \mathbf{C}.$$

The adjunction

$$\mathrm{Sph}_{\check{G},\mathrm{temp},x}^{\mathrm{spec}} \rightleftarrows \mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}$$

gives rise to an adjunction

$$(7.3) \quad \mathbf{C}_{\mathrm{temp}} \rightleftarrows \mathbf{C},$$

making $\mathbf{C}_{\mathrm{temp}}$ into a colocalization of $\mathbf{C}$.

We let $\mathrm{temp}_{\mathbf{C}}$ denote the comonad on $\mathbf{C}$ corresponding to the adjunction (7.3)

7.1.7. Let us regard the two sides of (7.2) as right modules with respect to $\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}$, where:

- $\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}$ acts on $\mathrm{Sph}_{\check{G},\mathrm{temp},x}^{\mathrm{spec}}$ by right multiplication;
- $\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}$ acts on $\mathrm{Func}_{\mathrm{QCoh}((\mathrm{LS}_{\check{G},x}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}})}(\mathrm{QCoh}(\mathrm{LS}_{\check{G},x}^{\mathrm{reg}}), \mathrm{QCoh}(\mathrm{LS}_{\check{G},x}^{\mathrm{reg}}))$ via the target.

Tensoring (7.2) over $\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}$ with $\mathbf{C}$ we obtain a functor

$$(7.4) \quad \mathbf{C}_{\mathrm{temp}} \rightarrow \mathrm{Func}_{\mathrm{QCoh}((\mathrm{LS}_{\check{G},x}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}})} \left(\mathrm{QCoh}(\mathrm{LS}_{\check{G},x}^{\mathrm{reg}}), \mathrm{QCoh}(\mathrm{LS}_{\check{G},x}^{\mathrm{reg}}) \otimes_{\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}} \mathbf{C} \right)$$

From Lemma 7.1.5 and the fact that $\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}$ is rigid we obtain:

Corollary 7.1.8. *The functor (7.4) is an equivalence.*

7.1.9. We will say that $\mathbf{C}$ is *tempered* if the action of $\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}$ on $\mathbf{C}$ factors via $\mathrm{Sph}_{\check{G},\mathrm{temp},x}^{\mathrm{spec}}$.

This is equivalent to the condition that the functors (7.3) are mutually inverse equivalences.

Remark 7.1.10. Note that Corollary 7.1.8 says that if $\mathbf{C}$ is tempered, then it can be recovered from the $\mathrm{QCoh}((\mathrm{LS}_{\check{G},x}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}})$ -module category

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G},x}^{\mathrm{reg}}) \otimes_{\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}} \mathbf{C}$$

by applying the functor

$$\mathrm{Func}_{\mathrm{QCoh}((\mathrm{LS}_{\check{G},x}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}})} \left(\mathrm{QCoh}(\mathrm{LS}_{\check{G},x}^{\mathrm{reg}}), - \right) : \mathrm{QCoh}((\mathrm{LS}_{\check{G},x}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}})\text{-mod} \rightarrow \mathrm{Sph}_{\check{G},\mathrm{temp},x}^{\mathrm{spec}}\text{-mod} \hookrightarrow \mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}\text{-mod}.$$

7.1.11. From Corollary 7.1.8 we obtain:

Corollary 7.1.12. *The functor*

$$\mathbf{C} \mapsto \mathrm{QCoh}(\mathrm{LS}_{\check{G},x}^{\mathrm{reg}}) \otimes_{\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}} \mathbf{C}, \quad \mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}\text{-mod} \rightarrow \mathrm{DGCat}$$

is conservative, when restricted to the subcategory

$$\mathrm{Sph}_{\check{G},\mathrm{temp},x}^{\mathrm{spec}}\text{-mod} \subset \mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}\text{-mod}.$$

7.1.13. Let

$$(7.5) \quad \mathrm{Sph}_{G,x} \rightarrow \mathrm{Sph}_{G,\mathrm{temp},x}$$

be the colocalization corresponding to the colocalization

$$\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}} \rightarrow \mathrm{Sph}_{\check{G},\mathrm{temp},x}^{\mathrm{spec}}$$

(we can use either Sat_G or $\mathrm{Sat}_{G,\tau}$ to identify Sph_G with $\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$; the resulting colocalizations are the same).

The definitions and results from this subsection render automatically to the setting of $\mathrm{Sph}_{G,x}$ -module categories.

7.2. Proof of Theorem 6.2.2. We are now ready to prove the pointwise version of the FLE.

7.2.1. Recall (see Theorem 6.4.5) that the functor

$$(7.6) \quad \text{FLE}_{G,\text{crit}} : \text{KL}(G)_{\text{crit},x} \rightarrow \text{IndCoh}^*(\text{Op}_{\check{G},x}^{\text{mon-free}})$$

intertwines the actions of $\text{Sph}_{G,x}$ on the left-hand side with the $\text{Sph}_{G,x}^{\text{spec}}$ -action on the right-hand side (via Sat_G), and makes the diagram

$$(7.7) \quad \begin{array}{ccc} \text{Whit}_*(G)_x \otimes_{\text{Sph}_{G,x}} \text{KL}(G)_{\text{crit},x} & \xrightarrow{\text{FLE}_{G,\infty}^{-1} \otimes \text{FLE}_{G,\text{crit}}} & \text{Rep}(\check{G}) \otimes_{\text{Sph}_{G,x}^{\text{spec}}} \text{IndCoh}^*(\text{Op}_{\check{G},x}^{\text{mon-free}}) \\ \downarrow (6.23) & & \downarrow (6.24) \\ \text{IndCoh}^*(\text{Op}_{\check{G},x}^{\text{mer}}) & \xrightarrow{\text{Id}} & \text{IndCoh}^*(\text{Op}_{\check{G},x}^{\text{mer}}) \end{array}$$

commute.

Note that Proposition 3.6.5 says that the right vertical arrow in (7.7) is fully faithful with essential image equal to

$$\text{IndCoh}^*(\text{Op}_{\check{G},x}^{\text{mer}})_{\text{mon-free}} \subset \text{IndCoh}^*(\text{Op}_{\check{G},x}^{\text{mer}}).$$

We will prove:

Proposition 7.2.2. *The left vertical arrow in (7.7) is fully faithful with essential image equal to*

$$\text{IndCoh}^*(\text{Op}_{\check{G},x}^{\text{mer}})_{\text{mon-free}} \subset \text{IndCoh}^*(\text{Op}_{\check{G},x}^{\text{mer}}).$$

7.2.3. We now claim:

Proposition 7.2.4. *The $\text{Sph}_{\check{G},x}^{\text{spec}}$ -module category $\text{IndCoh}^*(\text{Op}_{\check{G},x}^{\text{mon-free}})$ is tempered.*

Proof. Recall the functor

$$(7.8) \quad \text{QCoh}(\text{LS}_{\check{G},\underline{x}}^{\text{reg}}) \otimes_{\text{QCoh}((\text{LS}_{\check{G},\underline{x}}^{\text{mer}})^{\wedge}_{\text{reg}})} \text{IndCoh}^*((\text{Op}_{\check{G},\underline{x}}^{\text{mer}})^{\wedge}_{\text{mon-free}}) \rightarrow \text{IndCoh}^*(\text{Op}_{\check{G},\underline{x}}^{\text{mon-free}}).$$

It intertwines the actions of $\text{Sph}_{\check{G},x}^{\text{spec}}$, where the action on the left-hand side is via the first factor.

Now, according to Proposition 3.8.11, the functor (7.8) is an equivalence. Hence, it is enough to show that the action of $\text{Sph}_{\check{G},x}^{\text{spec}}$ on the left-hand side of (7.8) factors through $\text{Sph}_{\check{G},\text{temp},x}^{\text{spec}}$.

However, this follows from the fact that the action of $\text{Sph}_{\check{G},\text{temp},x}^{\text{spec}}$ on

$$\text{QCoh}(\text{LS}_{\check{G},\underline{x}}^{\text{reg}}) \simeq \text{Rep}(\check{G})$$

factors through $\text{Sph}_{\check{G},\text{temp},x}^{\text{spec}}$. □

7.2.5. Finally, we claim:

Proposition 7.2.6. *The $\text{Sph}_{G,x}$ -module category $\text{KL}(G)_{\kappa,x}$ is tempered.*

We now observe that the combination of Propositions 7.2.2, 7.2.4 and 7.2.6, together with Corollary 7.1.12, implies that (7.6) is an equivalence. □[Pointwise FLE]

The rest of this section is devoted to the proof of Propositions 7.2.2 and 7.2.6.

7.3. Proof of Proposition 7.2.2.

7.3.1. Let $\mathbf{C}$ be a category equipped with a $\mathfrak{L}(G)_x$ -action at the critical level. Consider now the category

$$\mathrm{Sph}(\mathbf{C}) := \mathbf{C}^{\mathfrak{L}^+(G)_x}.$$

Interpreting $\mathbf{C}^{\mathfrak{L}^+(G)_x}$ as

$$\mathrm{Funct}_{\mathfrak{L}(G)_x}(\mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,x}), \mathbf{C}),$$

we obtain that $\mathrm{Sph}(\mathbf{C})$ carries a natural action of

$$\mathrm{Sph}_{G,x} \simeq \mathrm{End}_{\mathfrak{L}(G)_x} \left(\mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,x}) \right).$$

7.3.2. Denote

$$\mathbf{C}^{\mathrm{Sph}\text{-gen}} : \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,x}) \otimes_{\mathrm{Sph}_{G,x}} \mathrm{Sph}(\mathbf{C}).$$

We have a tautological functor

$$(7.9) \quad \mathbf{C}^{\mathrm{Sph}\text{-gen}} \rightarrow \mathbf{C}$$

commuting with the $\mathfrak{L}(G)_x$ -action.

The following is a standard result that results from the ind-properness of the affine Grassmannian:

Lemma 7.3.3. *The functor (7.9) is fully faithful and admits a continuous right adjoint.³³*

7.3.4. We shall say that $\mathbf{C}$ is *spherically generated* if the functor (7.9) is an equivalence.

This is equivalent to requiring that $\mathbf{C}$ is generated, as a category acted on by $\mathfrak{L}(G)_x$, by the essential image of the forgetful functor

$$\mathrm{Sph}(\mathbf{C}) \rightarrow \mathbf{C}.$$

7.3.5. The above definitions apply when we replace $\mathfrak{L}(G)_x$ by $\mathfrak{L}(G)_{\rho(\omega_X),x}$. Applying to both sides of (7.9) the functor $\mathrm{Whit}_*(-)$, we obtain a functor

$$(7.10) \quad \mathrm{Whit}_*(G) \otimes_{\mathrm{Sph}_{G,x}} \mathrm{Sph}(\mathbf{C}) \rightarrow \mathrm{Whit}_*(\mathbf{C}),$$

i.e., the functor (6.21).

From Lemma 7.3.3 we obtain:

Corollary 7.3.6. *The functor (7.10) is fully faithful.*

7.3.7. We apply Corollary 7.3.6 to

$$\mathbf{C} := \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X), x}.$$

Hence, we obtain that the left vertical arrow in (7.7) is fully faithful. Thus, to complete the proof of Proposition 7.2.2, it suffices to show that the essential image of (6.23) is contained in and generates

$$\mathrm{IndCoh}^*(\mathrm{Op}_{G,x}^{\mathrm{mer}})_{\mathrm{mon}\text{-free}} \subset \mathrm{IndCoh}^*(\mathrm{Op}_{G,x}^{\mathrm{mer}}).$$

³³This right adjoint automatically commutes with the $\mathfrak{L}(G)_x$ -action, essentially because $\mathfrak{L}(G)_x$ is a group.

7.3.8. Note that the essential image of

$$\mathrm{KL}(G)_{\mathrm{crit},x} \xrightarrow{\mathbf{1}_{\mathrm{Whit}_*(G)} \otimes \mathrm{Id}} \mathrm{Whit}_*(G)_x \otimes_{\mathrm{Sph}_{G,x}} \mathrm{KL}(G)_{\mathrm{crit},x}$$

generates the target.

Indeed, this follows by interpreting the above functor as

$$\begin{aligned} \mathrm{KL}(G)_{\mathrm{crit},x} &\simeq \mathrm{Rep}(\check{G}) \otimes_{\mathrm{Rep}(\check{G}), \mathrm{Sat}_G^{\mathrm{nv}}} \mathrm{KL}(G)_{\mathrm{crit},x} \xrightarrow{\mathrm{FLE}_{\check{G},\infty} \otimes \mathrm{Id}} \\ &\simeq \mathrm{Whit}_*(G)_x \otimes_{\mathrm{Sat}_{G,\tau}^{\mathrm{nv}}, \mathrm{Rep}(\check{G}), \mathrm{Sat}_G^{\mathrm{nv}}} \mathrm{KL}(G)_{\mathrm{crit},x} \rightarrow \mathrm{Whit}_*(G)_x \otimes_{\mathrm{Sat}_{G,\tau}^{-1}, \mathrm{Sph}_{G,x}^{\mathrm{spec}}, \mathrm{Sat}_G^{-1}} \mathrm{KL}(G)_{\mathrm{crit},x} \simeq \\ &\simeq \mathrm{Whit}_*(G)_x \otimes_{\mathrm{Sph}_{G,x}} \mathrm{KL}(G)_{\mathrm{crit},x}. \end{aligned}$$

Since the essential image of $\mathrm{FLE}_{G,x}$ is contained in $\mathrm{IndCoh}^*(\mathrm{Op}_{G,x}^{\mathrm{mer}})_{\mathrm{mon-free}}$, we obtain that so is the essential image of (6.23).

7.3.9. Hence, it remains to prove the following:

Lemma 7.3.10. *The essential image $\mathrm{KL}(G)_{\mathrm{crit},\rho(\omega_X),x}$ under*

$$\mathrm{KL}(G)_{\mathrm{crit},\rho(\omega_X),x} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega_X),x} \xrightarrow{\mathrm{DS}^{\mathrm{enh},\mathrm{rfnd}}} \mathrm{IndCoh}^*(\mathrm{Op}_{G,x}^{\mathrm{mon-free}})$$

generates $\mathrm{IndCoh}^(\mathrm{Op}_{G,x}^{\mathrm{mer}})_{\mathrm{mon-free}}$.*

Proof. We prove the lemma by matching the generators.

The compact generators of $\mathrm{KL}(G)_{\mathrm{crit},\rho(\omega_X),x}$ are the Weyl modules

$$(7.11) \quad \mathbb{V}_{\mathrm{crit}}^\lambda := \mathbf{ind}_{\mathfrak{g}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{g}^+(G))_{\mathrm{crit}}} (V^{\check{\lambda}}),$$

where $V^{\check{\lambda}} \in \mathrm{Rep}(G)$ is the irreducible representation of highest weight λ .

According to [FG3], the image of $\mathbb{V}_{\mathrm{crit}}^\lambda$ under $\mathrm{DS}^{\mathrm{enh},\mathrm{rfnd}}$ is the structure sheaf $\mathcal{O}_{\mathrm{Op}_{G,x}^{\lambda\text{-reg}}}$ of the subscheme

$$\mathrm{Op}_{G,x}^{\lambda\text{-reg}} \subset \mathrm{Op}_{G,x}^{\mathrm{mer}}$$

of $\check{\lambda}$ -opers.

We have

$$\mathrm{red}(\mathrm{Op}_{G,x}^{\mathrm{mon-free}}) = \bigsqcup_{\lambda} \mathrm{Op}_{G,x}^{\lambda\text{-reg}},$$

from which it is clear that the objects

$$\mathcal{O}_{\mathrm{Op}_{G,x}^{\lambda\text{-reg}}} \in \mathrm{IndCoh}^*(\mathrm{Op}_{G,x}^{\mathrm{mer}})$$

are the (compact) generators of $\mathrm{IndCoh}^*(\mathrm{Op}_{G,x}^{\mathrm{mer}})_{\mathrm{mon-free}}$.

□

7.4. Proof of Proposition 7.2.6.

7.4.1. We need to show that the functor

$$(7.12) \quad \mathrm{KL}(G)_{\mathrm{crit},x} := \mathrm{Sph}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},x}) \rightarrow \mathrm{Sph}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},x})_{\mathrm{temp}} =: (\mathrm{KL}(G)_{\mathrm{crit},x})_{\mathrm{temp}}$$

is an equivalence.

First, note that by combining Corollary 7.1.8 with Propositions 7.2.2 and 7.2.4 we obtain that the functor $\mathrm{FLE}_{G,\mathrm{crit}}$ factors as

$$\mathrm{KL}(G)_{\mathrm{crit},x} \xrightarrow{(7.12)} (\mathrm{KL}(G)_{\mathrm{crit},x})_{\mathrm{temp}} \xrightarrow{\mathrm{FLE}_{G,\mathrm{crit},\mathrm{temp}}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},x}^{\mathrm{mon-free}}),$$

where $\mathrm{FLE}_{G,\mathrm{crit},\mathrm{temp}}$ is an equivalence.

Combined with Proposition 6.2.4, we obtain that the functor (7.12) preserves compactness. Since (7.12) is a colocalization, we obtain that (7.12) restricts to a colocalization on compacts. Hence, it is sufficient to prove that (7.12) is conservative on $(\mathrm{KL}(G)_{\mathrm{crit},x})^c$.

To prove the latter, it is sufficient to prove that the functor $\mathrm{FLE}_{G,\mathrm{crit}}$ is conservative on $(\mathrm{KL}(G)_{\mathrm{crit},x})^c$.

7.4.2. Since $(\mathrm{KL}(G)_{\mathrm{crit},x})^c \subset (\mathrm{KL}(G)_{\mathrm{crit},x})^b$ and since $\mathrm{FLE}_{G,\mathrm{crit}}$ is t-exact (by Corollary 6.6.9), it suffices to show that $\mathrm{FLE}_{G,\mathrm{crit}}$ is conservative on $(\mathrm{KL}(G)_{\mathrm{crit},x})^\heartsuit$.

Using the fact that $\mathrm{FLE}_{G,\mathrm{crit},\mathrm{temp}}$ is an equivalence, we obtain that it is enough to show that the functor (7.12) is conservative on $(\mathrm{KL}(G)_{\mathrm{crit},x})^\heartsuit$.

7.4.3. Let $\mathrm{temp}_{\mathrm{KL}(G)_{\mathrm{crit},x}}$ denote the temperization functor (see Sect. 7.1.6). We will prove:

Lemma 7.4.4. *The functor $\mathrm{temp}_{\mathrm{KL}(G)_{\mathrm{crit},x}}$ is right t-exact, and the counit map*

$$\mathrm{temp}_{\mathrm{KL}(G)_{\mathrm{crit},x}} \rightarrow \mathrm{Id}$$

induces an isomorphism on H^0 when applied to objects in $(\mathrm{KL}(G)_{\mathrm{crit},x})^\heartsuit$.

The lemma immediately implies the conservativity of (7.12) on $(\mathrm{KL}(G)_{\mathrm{crit},x})^\heartsuit$.

□[Proposition 7.2.6]

7.4.5. *Proof of Lemma 7.4.4.* The assertion of the lemma holds more generally for a $\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}$ -module category, equipped with a t-structure, such that the action functor is t-exact. The corresponding property for $\mathrm{KL}(G)_{\mathrm{crit},x} \simeq \mathrm{KL}(G)_{\mathrm{crit},x}$ is guaranteed by Corollary 5.4.5.

Consider the temperization functor $\mathrm{temp}_{\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}}$ on $\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}$ itself, i.e., the composition of

$$\begin{aligned} \mathrm{Sph}_{\check{G},x}^{\mathrm{spec}} &= \mathrm{IndCoh}(\mathrm{Hecke}_{\check{G},x}^{\mathrm{spec,loc}}) \xrightarrow{\Psi_{\mathrm{Hecke}_{\check{G},x}^{\mathrm{spec,loc}}}} \\ &\rightarrow \mathrm{QCoh}(\mathrm{Hecke}_{\check{G},x}^{\mathrm{spec,loc}}) \xrightarrow{\Xi_{\mathrm{Hecke}_{\check{G},x}^{\mathrm{spec,loc}}}} \mathrm{IndCoh}(\mathrm{Hecke}_{\check{G},x}^{\mathrm{spec,loc}}) = \mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}. \end{aligned}$$

It suffices to show that the object

$$\mathrm{temp}_{\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}}(\mathbf{1}_{\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}})$$

lives in cohomological degrees ≤ 0 and that its 0th cohomology maps isomorphically to $\mathbf{1}_{\mathrm{Sph}_{\check{G},x}^{\mathrm{spec}}}$.

However, this is a general property of the composition $\Xi_Y \circ \Psi_Y$ on an eventually coconnective stack locally almost of finite type. Namely, for every $\mathcal{F} \in \mathrm{IndCoh}(\mathcal{Y})$, the counit of the adjunction

$$\Xi_Y \circ \Psi_Y(\mathcal{F}) \rightarrow \mathcal{F}$$

induces an isomorphism on the truncation $\tau^{\geq -n}$ for any n .

□[Lemma 7.4.4]

7.5. Spherical vs Whittaker. This subsection is not logically necessary for the sequel, but it carries an ideological significance. Here we explain how to realize the pointwise $\text{FLE}_{G,\text{crit}}$ functor as the base change of the functor $\overline{\text{DS}}^{\text{enh},\text{rfnd}}$ along $\text{LS}_{G,x}^{\text{reg}} \rightarrow \text{LS}_{G,x}^{\text{mer}}$.

7.5.1. Let $\mathbf{C}$ be a category equipped with a $\mathfrak{L}(G)_{\rho(\omega_X),x}$ -action, and assume that $\mathbf{C}$ is spherically generated.

Consider the corresponding category

$$\text{Whit}_*(\mathbf{C}) := \mathbf{C}_{\mathfrak{L}(N)_{\rho(\omega_X),x},\chi}.$$

We claim that $\text{Whit}_*(\mathbf{C})$ has a natural structure of module category over $\text{QCoh}((\text{LS}_{G,x}^{\text{mer}})^\wedge_{\text{reg}})$.

7.5.2. Indeed, we have

$$(7.13) \quad \text{Whit}_*(\mathbf{C}) \simeq \text{Whit}_*(G)_x \otimes_{\text{Sph}_{G,x}} \text{Sph}(\mathbf{C}),$$

so it is enough to show that $\text{Whit}_*(G)_x$ carries a structure of $\text{QCoh}((\text{LS}_{G,x}^{\text{mer}})^\wedge_{\text{reg}})$ -module category in a way that commutes with the $\text{Sph}_{G,x}$ -action.

We identify

$$\text{Whit}_*(G)_x \xrightarrow{\text{FLE}_{\check{G},\infty}} \text{Rep}(\check{G}) \simeq \text{QCoh}(\text{LS}_{\check{G},x}^{\text{reg}}),$$

where the $\text{Sph}_{G,x}$ -action on $\text{Whit}_*(G)_x$ corresponds to the natural $\text{Sph}_{\check{G},x}^{\text{spec}}$ -action on $\text{QCoh}(\text{LS}_{\check{G},x}^{\text{reg}})$ via $\text{Sat}_{G,\tau}$ (see Corollary 1.8.5).

The desired $\text{QCoh}((\text{LS}_{G,x}^{\text{mer}})^\wedge_{\text{reg}})$ -module structure on $\text{Whit}(G)_x$ comes from the natural action of $\text{QCoh}((\text{LS}_{\check{G},x}^{\text{mer}})^\wedge_{\text{reg}})$ -action on $\text{QCoh}(\text{LS}_{\check{G},x}^{\text{reg}})$, which naturally commutes with the $\text{Sph}_{\check{G},x}^{\text{spec}}$ -action.

7.5.3. For $\mathbf{C}$ as above, consider the category

$$(7.14) \quad \text{Funct}_{\text{QCoh}((\text{LS}_{G,x}^{\text{mer}})^\wedge_{\text{reg}})}(\text{QCoh}(\text{LS}_{\check{G},x}^{\text{reg}}), \text{Whit}_*(\mathbf{C})).$$

Recall the functor (6.2)

$$(7.15) \quad \text{Sph}(\mathbf{C}) \rightarrow \text{Whit}_*(\mathbf{C}).$$

7.5.4. We claim:

Proposition 7.5.5. *The functor (7.15) lifts to a functor*

$$(7.16) \quad \text{Sph}(\mathbf{C}) \rightarrow \text{Funct}_{\text{QCoh}((\text{LS}_{G,x}^{\text{mer}})^\wedge_{\text{reg}})}(\text{QCoh}(\text{LS}_{\check{G},x}^{\text{reg}}), \text{Whit}_*(\mathbf{C})).$$

Moreover, the functor (7.16) factors as

$$(7.17) \quad \text{Sph}(\mathbf{C}) \rightarrow \text{Sph}(\mathbf{C})_{\text{temp}} \xrightarrow{\sim} \text{Funct}_{\text{QCoh}((\text{LS}_{G,x}^{\text{mer}})^\wedge_{\text{reg}})}(\text{QCoh}(\text{LS}_{\check{G},x}^{\text{reg}}), \text{Whit}_*(\mathbf{C})),$$

where the second arrow is an equivalence.

Proof. By (7.13), we have

$$\begin{aligned} (7.18) \quad & \text{Funct}_{\text{QCoh}((\text{LS}_{G,x}^{\text{mer}})^\wedge_{\text{reg}})}(\text{QCoh}(\text{LS}_{\check{G},x}^{\text{reg}}), \text{Whit}_*(\mathbf{C})) \simeq \\ & \simeq \text{Funct}_{\text{QCoh}((\text{LS}_{G,x}^{\text{mer}})^\wedge_{\text{reg}})}(\text{QCoh}(\text{LS}_{\check{G},x}^{\text{reg}}), \text{Whit}_*(G)_x) \otimes_{\text{Sph}_{G,x}} \text{Sph}(\mathbf{C}) \xrightarrow{\text{FLE}_{G,\infty}} \\ & \simeq \text{Funct}_{\text{QCoh}((\text{LS}_{G,x}^{\text{mer}})^\wedge_{\text{reg}})}(\text{QCoh}(\text{LS}_{\check{G},x}^{\text{reg}}), \text{QCoh}(\text{LS}_{\check{G},x}^{\text{reg}})) \otimes_{\text{Sph}_{G,x}} \text{Sph}(\mathbf{C}) \simeq \text{Sph}_{\check{G},\text{temp}.x}^{\text{spec}} \otimes_{\text{Sph}_{G,x}} \text{Sph}(\mathbf{C}). \end{aligned}$$

□

7.5.6. Take $\mathbf{C} := \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X), x}^{\text{Sph-gen}}$. So we can regard

$$\text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X), x}^{\text{Sph-gen}})$$

as acted on by $\text{QCoh}((\text{LS}_{\check{G}, x}^{\text{mer}})^\wedge_{\text{reg}})$ via the recipe of Sect. 7.5.1.

Let us regard

$$\text{IndCoh}^*(\text{Op}_{\check{G}, x}^{\text{mer}})_{\text{mon-free}} \simeq \text{IndCoh}^*(\text{Op}_{\check{G}, x}^{\text{mer}})^\wedge_{\text{mon-free}}$$

as acted on by $\text{QCoh}((\text{LS}_{\check{G}, x}^{\text{mer}})^\wedge_{\text{reg}})$ via pullback along

$$\mathfrak{r} : (\text{Op}_{\check{G}, x}^{\text{mer}})^\wedge_{\text{mon-free}} \rightarrow (\text{LS}_{\check{G}, x}^{\text{mer}})^\wedge_{\text{reg}}.$$

Recall now that according to Proposition 7.2.2, the functor $\overline{\text{DS}}^{\text{enh, rfnd}}$ gives rise to an equivalence

$$(7.19) \quad \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X), x}^{\text{Sph-gen}}) \xrightarrow{\overline{\text{DS}}^{\text{enh, rfnd}}} \text{IndCoh}^*(\text{Op}_{\check{G}, x}^{\text{mer}})_{\text{mon-free}}.$$

We claim:

Proposition 7.5.7. *The equivalence (7.19) is compatible with the above actions of $\text{QCoh}((\text{LS}_{\check{G}, x}^{\text{mer}})^\wedge_{\text{reg}})$.*

Proof. Follows from the fact that the functor (7.19) can be recovered from $\text{FLE}_{G, \text{crit}}$ by applying to both sides the functor

$$\text{QCoh}(\text{LS}_{G, x}^{\text{reg}}) \otimes_{\text{Sph}_{G, x}^{\text{Spec}}} -.$$

□

Remark 7.5.8. The above proof of Proposition 7.5.7 relies on Theorem 6.4.5 as an essential ingredient.

In Sect. G we will give a different proof of Proposition 7.5.7, by showing that the functor (7.19) is compatible with the structure of *factorization module category* over $\text{Rep}(\check{G})$.

7.5.9. From Proposition 7.5.7 we obtain that we can regard the FLE functor

$$\text{FLE}_{G, \text{crit}} : \text{KL}(G)_{\text{crit}, x} \rightarrow \text{IndCoh}^*(\text{Op}_{\check{G}, x}^{\text{mon-free}})$$

as obtained from the functor (7.16) for

$$\mathbf{C} := \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X), x}^{\text{Sph-gen}}$$

by precomposing with

$$\text{KL}(G)_{\text{crit}, x} \xrightarrow{\alpha_{\rho(\omega_X), \text{taut}}} \text{KL}(G)_{\text{crit}, \rho(\omega_X), x} = \text{Sph}(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X), x})$$

and post-composing with

$$(7.20) \quad \text{Funct}_{\text{QCoh}((\text{LS}_{\check{G}, x}^{\text{mer}})^\wedge_{\text{reg}})} \left(\text{QCoh}(\text{LS}_{\check{G}, x}^{\text{reg}}), \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X), x}^{\text{Sph-gen}}) \right) \xrightarrow{\text{Proposition 7.5.7}} \simeq \\ \simeq \text{Funct}_{\text{QCoh}((\text{LS}_{\check{G}, x}^{\text{mer}})^\wedge_{\text{reg}})} \left(\text{QCoh}(\text{LS}_{\check{G}, x}^{\text{reg}}), \text{IndCoh}^*(\text{Op}_{\check{G}, x}^{\text{mer}})_{\text{mon-free}} \right) \simeq \text{IndCoh}^*(\text{Op}_{\check{G}, x}^{\text{mon-free}}),$$

where the last equivalence is obtained as follows:

$$\begin{aligned} & \text{Funct}_{\text{QCoh}((\text{LS}_{\check{G}, x}^{\text{mer}})^\wedge_{\text{reg}})} \left(\text{QCoh}(\text{LS}_{\check{G}, x}^{\text{reg}}), \text{IndCoh}^*(\text{Op}_{\check{G}, x}^{\text{mer}})_{\text{mon-free}} \right) \simeq \\ & \text{Funct}_{\text{IndCoh}^!(\text{Op}_{\check{G}, x}^{\text{mer}})_{\text{mon-free}}} \left(\text{IndCoh}^!(\text{Op}_{\check{G}, x}^{\text{mer}})_{\text{mon-free}} \otimes_{\text{QCoh}((\text{LS}_{\check{G}, x}^{\text{mer}})^\wedge_{\text{reg}})} \text{QCoh}(\text{LS}_{\check{G}, x}^{\text{reg}}), \text{IndCoh}^*(\text{Op}_{\check{G}, x}^{\text{mer}})_{\text{mon-free}} \right) \\ & \xrightarrow{\text{Proposition 3.8.11}} \text{Funct}_{\text{IndCoh}^!(\text{Op}_{\check{G}, x}^{\text{mer}})_{\text{mon-free}}} \left(\text{IndCoh}^!(\text{Op}_{\check{G}, x}^{\text{mon-free}}), \text{IndCoh}^*(\text{Op}_{\check{G}, x}^{\text{mer}})_{\text{mon-free}} \right) \simeq \\ & \simeq \text{Funct}_{\text{IndCoh}^!(\text{Op}_{\check{G}, x}^{\text{mer}})} \left(\text{IndCoh}^!(\text{Op}_{\check{G}, x}^{\text{mon-free}}), \text{IndCoh}^*(\text{Op}_{\check{G}, x}^{\text{mer}}) \right) \xrightarrow{\text{Lemma 3.5.7}} \text{IndCoh}^*(\text{Op}_{\check{G}, x}^{\text{mon-free}}). \end{aligned}$$

Remark 7.5.10. The proof of the pointwise FLE given in Sect. 7.2 relied on Proposition 7.5.7, and hence on Theorem 6.4.5 ingredient.

As was mentioned in Remark 7.5.8, we will supply a different proof of Proposition 7.5.7.

This allows us to give a proof of the pointwise FLE, avoiding Theorem 6.4.5:

Then one interprets the pointwise $\text{FLE}_{G,\text{crit}}$ functor as in Sect. 7.5.9 above. The assertion of the pointwise FLE follows now by combining Propositions 7.2.6 and 7.5.5.

7.5.11. We remark also that one can deduce the pointwise version of Theorem 6.4.5 from Proposition 7.5.7 by the following argument:

First, we note that in the context of Sect. 7.5.3, the functor

$$\text{Sph}(\mathbf{C}) \rightarrow \text{Funct}_{\text{QCoh}((\text{LS}_{G,x}^{\text{mer}})^{\wedge}_{\text{reg}})}(\text{QCoh}(\text{LS}_{G,x}^{\text{reg}}), \text{Whit}_*(\mathbf{C}))$$

intertwines the action of $\text{Sph}_{G,x}$ on $\text{Sph}(\mathbf{C})$ and the action of $\text{Sph}_{G,x}^{\text{spec}}$ on

$$\text{Funct}_{\text{QCoh}((\text{LS}_{G,x}^{\text{mer}})^{\wedge}_{\text{reg}})}(\text{QCoh}(\text{LS}_{G,x}^{\text{reg}}), \text{Whit}_*(\mathbf{C}))$$

via the source.

Unwinding the construction, we obtain that the following diagram commutes

$$\begin{array}{ccc} \text{Whit}_*(G) \otimes_{\text{Sph}_{G,x}} \text{Sph}(\mathbf{C}) & \xrightarrow[\sim]{(7.13)} & \text{Whit}_*(\mathbf{C}) \\ \text{FLE}_{G,\infty}^{-1} \otimes (7.16) \downarrow & & \downarrow \text{Id} \\ \text{QCoh}(\text{LS}_{G,x}^{\text{reg}}) \otimes_{\text{Sph}_{G,x}^{\text{spec}}} \text{Funct}_{\text{QCoh}((\text{LS}_{G,x}^{\text{mer}})^{\wedge}_{\text{reg}})}(\text{QCoh}(\text{LS}_{G,x}^{\text{reg}}), \text{Whit}_*(\mathbf{C})) & \longrightarrow & \text{Whit}_*(\mathbf{C}), \end{array}$$

where the bottom horizontal arrow is the natural evaluation functor.

This proves the desired compatibility, since the functor (7.20) is compatible with the $\text{Sph}_{G,x}^{\text{spec}}$ -actions.

8. COMPATIBILITY OF THE CRITICAL FLE WITH DUALITY

In this section we show that the FLE equivalence is compatible with the natural self-dualities of the two sides.

The proof proceeds along the following steps:

- (1) We show that the self-duality on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}$ is compatible with the $\text{IndCoh}^!(\text{"Spec"}(3_{\mathfrak{g}}))$ -action. This boils down to a particular property of the factorization algebra $\text{CDO}(G)_{\text{crit},\text{crit}}$, given by Lemma 8.3.3;
- (2) We show that the equivalence $\overline{\text{DS}}^{\text{enh},\text{rfnd}} : \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit},\rho(\omega_X)}) \rightarrow \text{IndCoh}^*(\text{Op}_G^{\text{mer}})$ is compatible with the self-dualities of the two sides. We deduce this from the $\text{IndCoh}^!(\text{Op}_G^{\text{mer}})$ -linearity (guaranteed by the previous point), combined with a general uniqueness statement;
- (3) Finally, we establish the compatibility of $\text{FLE}_{G,\text{crit}}$ with the self-dualities by essentially base-changing it from the previous point along $\text{Op}_G^{\text{mon-free}} \rightarrow \text{Op}_G^{\text{mer}}$.

As we highlight below, our methods are robust enough to show that established compatibility is automatically compatible with the actions of

$$\text{Sph}_G \simeq \text{Sph}_G^{\text{spec}}$$

on the two sides.

8.1. Statement of the compatibility theorem.

8.1.1. Recall that according to (2.4), we have a canonical identification

$$(8.1) \quad (\text{KL}(G)_{\text{crit}})^{\vee} \simeq \text{KL}(G)_{\text{crit}}.$$

By construction, the equivalence (8.1) respects the actions of Sph_G .

8.1.2. In addition, we have an equivalence

$$(8.2) \quad \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})^\vee \simeq \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \xrightarrow{\Theta_{\mathrm{Op}_G^{\mathrm{mon-free}}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}).$$

This equivalence respects the actions of $\mathrm{Sph}_G^{\mathrm{spec}}$ (see Sects. 3.6.2, 3.7.15 and Lemma 3.7.17).

8.1.3. The goal of this section is to prove the following:

Theorem 8.1.4. *With respect to the identifications (8.1) and (8.2), the (factorization) functor*

$$(\mathrm{FLE}_{G,\mathrm{crit}})^\vee : \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \rightarrow \mathrm{KL}(G)_{\mathrm{crit}}$$

identifies with

$$\tau_G \circ (\mathrm{FLE}_{G,\mathrm{crit}})^{-1}.$$

Moreover, this identification of functors respects the compatibility with the actions of

$$\mathrm{Sph}_G \xrightarrow{\mathrm{Sat}_{G,\tau}} \mathrm{Sph}_G^{\mathrm{spec}}.$$

One can rephrase Theorem 8.1.4 as a commutative diagram of factorization categories

$$(8.3) \quad \begin{array}{ccc} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})^\vee & \xrightarrow[\sim]{(8.2)} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \\ (\mathrm{FLE}_{G,\mathrm{crit}})^\vee \downarrow \sim & & \sim \uparrow \tau_G \circ \mathrm{FLE}_{G,\mathrm{crit}} \\ (\mathrm{KL}(G)_{\mathrm{crit}})^\vee & \xrightarrow[\sim]{(8.1)} & \mathrm{KL}(G)_{\mathrm{crit}}. \end{array}$$

Remark 8.1.5. Note the similarity between the statement of Theorem 8.1.4 and Lemma 1.4.12: in both cases a non-tautological self-equivalence of the Whittaker side makes the FLE inverse to its dual, up to the Chevalley involution.

Remark 8.1.6. Note again that the appearance of the Chevalley involution in Theorem 8.1.4 is in line with the curse in Sect. 1.8.7.

8.1.7. The following assertion will not be needed in the sequel, but it provides a concrete perspective on what Theorem 8.1.4 really says.

Recall that the unit of the self-duality (8.1) is the object

$$\mathfrak{CD}\mathfrak{D}_{\mathrm{crit},\mathrm{crit}} \in \mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{KL}(G)_{\mathrm{crit}}.$$

The unit of the self-duality of $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$ is

$$(\Delta_{\mathrm{Op}_G^{\mathrm{mon-free}}})_*^{\mathrm{IndCoh}}(\omega_{\mathrm{Op}_G^{\mathrm{mon-free}}}^{*,\mathrm{fake}}),$$

where

$$\omega_{\mathrm{Op}_G^{\mathrm{mon-free}}}^{*,\mathrm{fake}} \in \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$$

is as in Sect. 3.7.7, and $\Delta_{\mathrm{Op}_G^{\mathrm{mon-free}}}$ is the diagonal map of $\mathrm{Op}_G^{\mathrm{mon-free}}$.

Thus, from Theorem 8.1.4, we obtain:

Corollary 8.1.8. *We have a canonical isomorphism (of factorization algebra objects)*

$$(\mathrm{FLE}_{G,\mathrm{crit}} \otimes \mathrm{FLE}_{G,\mathrm{crit}})(\mathfrak{CD}\mathfrak{D}_{\mathrm{crit},\mathrm{crit}}) \simeq (\mathrm{Id} \otimes \tau_G) \circ (\Delta_{\mathrm{Op}_G^{\mathrm{mon-free}}})_*^{\mathrm{IndCoh}}(\omega_{\mathrm{Op}_G^{\mathrm{mon-free}}}^{*,\mathrm{fake}}).$$

Remark 8.1.9. Note that, on the one hand, the statement of Corollary 8.1.8 is actually equivalent to Theorem 8.1.4 (without the $\mathrm{Sph}_G \simeq \mathrm{Sph}_G^{\mathrm{spec}}$ compatibility).

On the other hand, the two factorization algebras appearing in Corollary 8.1.8 are *classical* (i.e., the corresponding chiral algebras lie in $\mathrm{D-mod}(X)^\heartsuit$), and one can actually prove the existence of an isomorphism between them directly.

8.1.10. Applying the forgetful functor to Vect , from Corollary 8.1.8 we obtain:

Corollary 8.1.11. *We have a canonical isomorphism of factorization algebras*

$$(8.4) \quad (\text{DS} \otimes \text{DS}) \circ (\alpha_{\rho(\omega_X), \text{taut}} \otimes \alpha_{\rho(\omega_X), \text{taut}})(\mathfrak{CD}\mathfrak{D}_{\text{crit}, \text{crit}}) \simeq \Gamma^{\text{IndCoh}}(\text{Op}_G^{\text{mon-free}}, \omega_{\text{Op}_G^{\text{mon-free}}}^{*, \text{fake}}).$$

Remark 8.1.12. We note that the factorization algebra

$$\mathcal{B}_G := (\text{DS} \otimes \text{DS}) \circ (\alpha_{\rho(\omega_X), \text{taut}} \otimes \alpha_{\rho(\omega_X), \text{taut}})(\mathfrak{CD}\mathfrak{D}_{\text{crit}, \text{crit}})$$

was studied in [FG2].

One can view (8.4) as an extension of the Feigin-Frenkel isomorphism FF_G : indeed according to Lemma 8.3.3 below, the factorization algebra $\mathcal{B}_{\mathfrak{g}}$ receives a homomorphism from

$$(\text{DS} \otimes \text{DS})(\mathbb{V}_{\mathfrak{g}, \text{crit}, \rho(\omega_X)} \otimes_{\mathfrak{z}_{\mathfrak{g}}} \mathbb{V}_{\mathfrak{g}, \text{crit}, \rho(\omega_X)}) \simeq \mathfrak{z}_{\mathfrak{g}},$$

while

$$\mathcal{B}_G^{\text{spec}} := \Gamma^{\text{IndCoh}}(\text{Op}_G^{\text{mon-free}}, \omega_{\text{Op}_G^{\text{mon-free}}}^{*, \text{fake}})$$

receives a homomorphism from

$$\Gamma(\text{Op}_G^{\text{reg}}, \mathcal{O}_{\text{Op}_G^{\text{reg}}}) =: \mathcal{O}_{\text{Op}_G^{\text{reg}}}.$$

8.2. Feigin-Frenkel center and self-duality.

8.2.1. Recall the duality identification

$$(8.5) \quad (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}})^{\vee} \simeq \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}.$$

of (2.3).

Recall also that $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}$ carries a canonical action of $\text{IndCoh}^!(\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}}))$, see Sect. 4.6. Since the category $\text{IndCoh}^!(\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}}))$ is *symmetric* monoidal, this action induces an action of $\text{IndCoh}^!(\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}}))$ on $(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}})^{\vee}$.

8.2.2. The goal of this subsection is to prove the following:

Theorem-Construction 8.2.3. *The identification (8.5) carries a canonical structure of compatibility with the $\text{IndCoh}^!(\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}}))$ -actions, up to the automorphism of $\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}})$ induced by the Chevalley involution τ_G .*

The rest of this subsection is devoted to the proof of this theorem.

8.2.4. Let

$$u_{\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}, \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}} \in \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}$$

be the unit of the self-duality. The statement of Theorem 8.2.3 is equivalent to the assertion that $u_{\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}, \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}}$ can be lifted to an object of the category

$$\text{Func}_{\text{IndCoh}^!(\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}})) \otimes \text{IndCoh}^!(\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}}))}(\text{IndCoh}^!(\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}})), \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}})$$

where $\text{IndCoh}^!(\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}})) \otimes \text{IndCoh}^!(\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}}))$ acts on $\text{IndCoh}^!(\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}}))$ via τ_G on one of the factors.

8.2.5. Consider the (factorization) category

$$\mathrm{D}\text{-mod}(\mathfrak{L}(G))_{\mathrm{crit},\mathrm{crit}}$$

of critically twisted D-modules on the loop group. We have a naturally defined (factorization) functor

$$(8.6) \quad \Gamma^{\mathrm{IndCoh}}(\mathfrak{L}(G), -) : \mathrm{D}\text{-mod}(\mathfrak{L}(G))_{\mathrm{crit},\mathrm{crit}} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}.$$

We have:

$$\mathrm{u}_{\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}, \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}} \simeq \Gamma^{\mathrm{IndCoh}}(\mathfrak{L}(G), \delta_{1, \mathfrak{L}(G)}),$$

where $\delta_{1, \mathfrak{L}(G)} \in \mathrm{D}\text{-mod}(\mathfrak{L}(G))_{\mathrm{crit},\mathrm{crit}}$ is the δ -function at the origin.

The required property of $\mathrm{u}_{\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}, \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}}$ follows from the next general assertion:

Proposition 8.2.6. *The functor (8.6) factors as*

$$\begin{aligned} \mathrm{D}\text{-mod}(\mathfrak{L}(G))_{\mathrm{crit},\mathrm{crit}} &\rightarrow \\ &\rightarrow \mathrm{Funct}_{\mathrm{IndCoh}^!(\text{"Spec"}(\mathfrak{Z}_{\mathfrak{g}})) \otimes \mathrm{IndCoh}^!(\text{"Spec"}(\mathfrak{Z}_{\mathfrak{g}}))(\mathrm{IndCoh}^!(\text{"Spec"}(\mathfrak{Z}_{\mathfrak{g}})), \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}})} \rightarrow \\ &\rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}. \end{aligned}$$

□[Theorem 8.2.3]

8.3. Proof of Proposition 8.2.6.

8.3.1. First, we record the initial input, from which we will deduce Proposition 8.2.6. Recall the (factorization algebra) object

$$\mathfrak{CD}\mathfrak{D}(G)_{\mathrm{crit},\mathrm{crit}} \in \mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{KL}(G)_{\mathrm{crit}}.$$

By a slight abuse of notation we will denote by the same symbol the image of $\mathfrak{CD}\mathfrak{D}(G)_{\mathrm{crit},\mathrm{crit}}$ under the forgetful functor

$$\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{KL}(G)_{\mathrm{crit}} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}.$$

Let us denote by $\mathrm{CDO}(G)_{\mathrm{crit},\mathrm{crit}}$ the image of $\mathfrak{CD}\mathfrak{D}(G)_{\mathrm{crit},\mathrm{crit}}$ along the further forgetful functor

$$\mathrm{oblv}_{\widehat{\mathfrak{g}} \times \widehat{\mathfrak{g}}} : \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \rightarrow \mathrm{Vect}.$$

8.3.2. The unit of $\mathfrak{CD}\mathfrak{D}(G)_{\mathrm{crit},\mathrm{crit}}$ as a factorization algebra in $\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{KL}(G)_{\mathrm{crit}}$ is a map (of factorization algebras)

$$(8.7) \quad \mathrm{Vac}(G)_{\mathrm{crit}} \otimes \mathrm{Vac}(G)_{\mathrm{crit}} \rightarrow \mathfrak{CD}\mathfrak{D}(G)_{\mathrm{crit},\mathrm{crit}},$$

which gives rise to a map

$$(8.8) \quad \mathbb{V}_{\mathfrak{g},\mathrm{crit}} \otimes \mathbb{V}_{\mathfrak{g},\mathrm{crit}} \rightarrow \mathrm{CDO}(G)_{\mathrm{crit},\mathrm{crit}}.$$

The following was established in [FG1, Theorem 5.4]:

Lemma 8.3.3. *The map (8.8) factors as*

$$\mathbb{V}_{\mathfrak{g},\mathrm{crit}} \otimes \mathbb{V}_{\mathfrak{g},\mathrm{crit}} \rightarrow \mathbb{V}_{\mathfrak{g},\mathrm{crit}} \otimes_{\mathfrak{Z}_{\mathfrak{g}}} \mathbb{V}_{\mathfrak{g},\mathrm{crit}} \rightarrow \mathrm{CDO}(G)_{\mathrm{crit},\mathrm{crit}},$$

where the $\mathfrak{Z}_{\mathfrak{g}}$ -action on one of the tensor factors is twisted by τ_G .

Since the factorization algebras involved lie in the heart of the t-structure, from Lemma 8.3.3 we obtain:

Corollary 8.3.4. *The map (8.7) factors as*

$$\mathrm{Vac}(G)_{\mathrm{crit}} \otimes \mathrm{Vac}(G)_{\mathrm{crit}} \rightarrow \mathrm{Vac}(G)_{\mathrm{crit}} \otimes_{\mathfrak{Z}_{\mathfrak{g}}} \mathrm{Vac}(G)_{\mathrm{crit}} \rightarrow \mathfrak{CD}\mathfrak{D}(G)_{\mathrm{crit},\mathrm{crit}},$$

where the $\mathfrak{Z}_{\mathfrak{g}}$ -action on the right tensor factor is twisted by τ_G .

We will now show how to use Corollary 8.3.4 to prove Proposition 8.2.6.

8.3.5. Note that $\mathrm{D}\text{-mod}(\mathfrak{L}(G))_{\mathrm{crit},\mathrm{crit}}$ is equipped with a t-structure (see Sect. B.11.11 for what this means in the factorization setting), so that the object $\delta_{\mathfrak{L}^+(G) \subset \mathfrak{L}(G)} \in \mathrm{D}\text{-mod}(\mathfrak{L}(G))_{\mathrm{crit},\mathrm{crit}}$ lies in the heart.

The category $\mathrm{D}\text{-mod}(\mathfrak{L}(G))_{\mathrm{crit},\mathrm{crit}}$ is compactly generated by objects that lie in $\mathrm{D}\text{-mod}(\mathfrak{L}(G))_{\mathrm{crit},\mathrm{crit}}^{>-\infty}$. Hence, in order to prove Proposition 8.2.6, it suffices to show that the restriction of the functor $\Gamma^{\mathrm{IndCoh}}(\mathfrak{L}(G), -)$ to $\mathrm{D}\text{-mod}(\mathfrak{L}(G))_{\mathrm{crit},\mathrm{crit}}^{>-\infty}$ factors as

$$\begin{aligned} \mathrm{D}\text{-mod}(\mathfrak{L}(G))_{\mathrm{crit},\mathrm{crit}}^{>-\infty} &\rightarrow \\ &\rightarrow \mathrm{Funct}_{\mathrm{IndCoh}^!(\text{"Spec"}(\mathfrak{Z}_{\mathfrak{g}})) \otimes \mathrm{IndCoh}^!(\text{"Spec"}(\mathfrak{Z}_{\mathfrak{g}}))}(\mathrm{IndCoh}^!(\text{"Spec"}(\mathfrak{Z}_{\mathfrak{g}})), \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}) \rightarrow \\ &\rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}. \end{aligned}$$

8.3.6. Consider the (factorization) functor $\Gamma^{\mathrm{IndCoh}}(\mathfrak{L}(G), -)$. It sends the factorization unit

$$\mathbf{1}_{\mathrm{D}\text{-mod}(\mathfrak{L}(G))_{\mathrm{crit},\mathrm{crit}}} \simeq \delta_{\mathfrak{L}^+(G) \subset \mathfrak{L}(G)} \in \Gamma^{\mathrm{IndCoh}}(\mathfrak{L}(G), -)$$

to

$$\mathfrak{C}\mathfrak{D}\mathfrak{D}(G)_{\mathrm{crit},\mathrm{crit}} \in \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}.$$

In particular, by Sect. 4.1.9, it upgrades to a functor

$$\Gamma^{\mathrm{IndCoh}}(\mathfrak{L}(G), -)^{\mathrm{enh}} : \mathrm{D}\text{-mod}(\mathfrak{L}(G))_{\mathrm{crit},\mathrm{crit}} \rightarrow \mathfrak{C}\mathfrak{D}\mathfrak{D}\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}).$$

8.3.7. Applying Corollary 8.3.4, we obtain that the functor $\Gamma^{\mathrm{IndCoh}}(\mathfrak{L}(G), -)$ factors as

$$\begin{aligned} \mathrm{D}\text{-mod}(\mathfrak{L}(G))_{\mathrm{crit},\mathrm{crit}} &\rightarrow \\ &\rightarrow (\mathrm{Vac}(G)_{\mathrm{crit}} \otimes_{\mathfrak{Z}_{\mathfrak{g}}} \mathrm{Vac}(G)_{\mathrm{crit}})\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}) \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}. \end{aligned}$$

In particular, the restriction of the functor $\Gamma^{\mathrm{IndCoh}}(\mathfrak{L}(G), -)$ to $\mathrm{D}\text{-mod}(\mathfrak{L}(G))_{\mathrm{crit},\mathrm{crit}}^{>-\infty}$ factors via the forgetful functor

$$(8.9) \quad \left((\mathrm{Vac}(G)_{\mathrm{crit}} \otimes_{\mathfrak{Z}_{\mathfrak{g}}} \mathrm{Vac}(G)_{\mathrm{crit}})\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}) \right)^{>-\infty} \rightarrow \left(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \right)^{>-\infty}.$$

8.3.8. However, unwinding the construction of the $\mathrm{IndCoh}^!(\text{"Spec"}(\mathfrak{Z}_{\mathfrak{g}}))$ -action on $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}$, we obtain that the functor (8.9) factors as

$$\begin{aligned} &\left((\mathrm{Vac}(G)_{\mathrm{crit}} \otimes_{\mathfrak{Z}_{\mathfrak{g}}} \mathrm{Vac}(G)_{\mathrm{crit}})\text{-mod}^{\mathrm{fact}}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}) \right)^{>-\infty} \rightarrow \\ &\rightarrow \mathrm{Funct}_{\mathrm{IndCoh}^!(\text{"Spec"}(\mathfrak{Z}_{\mathfrak{g}})) \otimes \mathrm{IndCoh}^!(\text{"Spec"}(\mathfrak{Z}_{\mathfrak{g}}))}(\mathrm{IndCoh}^!(\text{"Spec"}(\mathfrak{Z}_{\mathfrak{g}})), \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}) \rightarrow \\ &\rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}}. \end{aligned}$$

□[Proposition 8.2.6]

8.4. Self-duality on opers via Kac-Moody.

8.4.1. Recall the identification (1.5). Applying this to the category $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)}$ and using the identification (8.5), we obtain an identification

$$(8.10) \quad \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)})^{\vee} \simeq \mathrm{Whit}^!(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)}),$$

which fits into the commutative diagram

$$\begin{array}{ccc} \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)})^{\vee} & \xrightarrow{(8.10)} & \mathrm{Whit}^!(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)}) \\ \downarrow & & \downarrow \\ (\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)})^{\vee} & \xrightarrow{(8.5)} & \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega)}, \end{array}$$

where the left vertical arrow is the dual of the projection

$$\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega)} \rightarrow \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega)}).$$

Since the vertical arrows in the above diagram are fully faithful, from Theorem 8.2.3³⁴ (combined with Corollary 4.7.15(a,b)), we obtain:

Corollary 8.4.2. *The functor (8.10) is equipped with a natural $\text{IndCoh}^!(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ -linear structure, up to the automorphism of $\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}})$, induced by the Chevalley involution τ_G .*

8.4.3. Recall now identification of Theorem 1.3.7. Applying this to the category $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega)}$, we obtain an identification

$$(8.11) \quad \Theta_{\text{Whit}(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega)})} : \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega)}) \simeq \text{Whit}^!(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega)}).$$

Concatenating (8.10) with (8.11) we obtain an identification

$$(8.12) \quad \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega)})^\vee \simeq \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega)}).$$

Combining with Corollaries 8.4.2 and 4.7.15(c), we obtain that the functor (8.12) is also endowed with an $\text{IndCoh}^!(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}}))$ -linear structure, up to the automorphism of $\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}})$ induced by the Chevalley involution τ_G .

8.4.4. Recall now that we have an identification

$$(8.13) \quad \text{IndCoh}^*(\text{Op}_G^{\text{mer}})^\vee \simeq \text{IndCoh}^!(\text{Op}_G^{\text{mer}}) \xrightarrow{\Theta_{\text{Op}_G^{\text{mer}}}} \text{IndCoh}^*(\text{Op}_G^{\text{mer}}).$$

We will prove:

Theorem 8.4.5. *With respect to the identifications (8.12) and (8.13), the functor dual to*

$$\overline{\text{DS}}^{\text{enh, rfnd}} : \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega)}) \xrightarrow{\sim} \text{IndCoh}^*(\text{Op}_G^{\text{mer}})$$

identifies with $\tau_G \circ (\overline{\text{DS}}^{\text{enh, rfnd}})^{-1}$, compatibly with the actions of

$$\text{IndCoh}^!(\text{“Spec”}(\mathfrak{Z}_{\mathfrak{g}})) \xrightarrow{\text{FF}_G} \text{IndCoh}^!(\text{Op}_G^{\text{mer}}).$$

One can rephrase Theorem 8.1.4 as a commutative diagram

$$(8.14) \quad \begin{array}{ccc} \text{IndCoh}^*(\text{Op}_G^{\text{mer}})^\vee & \xrightarrow[\sim]{(8.13)} & \text{IndCoh}^*(\text{Op}_G^{\text{mer}}) \\ \downarrow \sim & & \uparrow \sim \\ \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega)})^\vee & \xrightarrow[\sim]{(8.12)} & \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega)}). \end{array}$$

Remark 8.4.6. As we shall see below, Theorem 8.4.5 is actually easy. However, it can be seen as a particular case of a conjecture, proposed by G. Dhillon, which says that at any level κ , the self-dualities of the (renormalized) categories of factorization modules

$$\mathcal{W}_{\mathfrak{g}, \kappa}\text{-mod}^{\text{fact}} \simeq \mathcal{W}_{\mathfrak{g}, \tilde{\kappa}}\text{-mod}^{\text{fact}}$$

that come from the identifications

$$\mathcal{W}_{\mathfrak{g}, \kappa}\text{-mod}^{\text{fact}} = \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_\kappa) \text{ and } \mathcal{W}_{\mathfrak{g}, \tilde{\kappa}}\text{-mod}^{\text{fact}} = \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\tilde{\kappa}})$$

and Theorem 1.3.7, respectively, agree.

For non-critical κ , this conjecture is completely open. What makes it tractable at the critical level is precisely the interpretation of $\mathcal{W}_{\mathfrak{g}, \text{crit}}$ as the Feigin-Frenkel center.

8.5. Proof of Theorem 8.4.5.

³⁴Here we apply Theorem 8.2.3 in the twisted setting, i.e., to $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega)}$ instead of $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}$.

8.5.1. Consider the $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})$ -linear self-equivalence of $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})$ obtained by going clockwise along the edges of (8.14). We need to show that this functor is isomorphic to the identity.

Using the equivalence $\Theta_{\mathrm{Op}_G^{\mathrm{mer}}}$, and further, the equivalence

$$\Upsilon_{\mathrm{Op}_G^{\mathrm{mer}}} : \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mer}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})$$

of Proposition 3.8.7, we can transform the above $\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})$ -linear self-equivalence of the category $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})$ into a $\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mer}})$ -linear self-equivalence of $\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mer}})$.

Such a self-equivalence is given by a (graded) line bundle, to be denoted $\mathcal{L}_{\mathrm{Op}_G^{\mathrm{mer}}}$, on $\mathrm{Op}_G^{\mathrm{mer}}$. Since all the equivalences in sight are compatible with factorization, this line bundle has a natural factorization structure.

We will now show that any such (graded) line bundle is automatically trivial.

8.5.2. The question is local, so let us choose a $\check{G}$ -oper σ on X .

The datum of σ gives rise to a section

$$\sigma_{\mathrm{Ran}} : \mathrm{Ran} \rightarrow \mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{reg}} \rightarrow \mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mer}},$$

compatible with factorization.

Set $\mathcal{L}_{\mathrm{Ran}, \sigma} := (\sigma_{\mathrm{Ran}})^*(\mathcal{L}_{\mathrm{Op}_G^{\mathrm{mer}}})$. This is a factorization line bundle on Ran . We claim that it is automatically trivial.

8.5.3. Indeed, write

$$\mathrm{Ran} \simeq \mathrm{colim}_I X_{\mathrm{dR}}^I,$$

where the index I runs over the category of non-empty finite sets and surjective maps.

Set $\mathcal{L}_{X_{\mathrm{dR}}^I, \sigma} := \mathcal{L}_{\mathrm{Ran}, \sigma}|_{X_{\mathrm{dR}}^I}$. The collection of local systems

$$(8.15) \quad I \rightsquigarrow \mathcal{L}_{X_{\mathrm{dR}}^I, \sigma}$$

is compatible with the factorization structure.

In particular,

$$\mathcal{L}_{X_{\mathrm{dR}}^I, \sigma}|_{\check{X}^I} \simeq (\mathcal{L}_{X_{\mathrm{dR}}, \sigma})^{\boxtimes I}|_{\check{X}^I},$$

where $\check{X}^I \subset X^I$ is the complement of the diagonal divisor.

Hence,

$$(8.16) \quad \mathcal{L}_{X_{\mathrm{dR}}^I, \sigma} \simeq (\mathcal{L}_{X_{\mathrm{dR}}, \sigma})^{\boxtimes I},$$

compatibly with the factorization structure.

Consider (8.16) for $I = \{1, 2\}$. Restricting to the diagonal $X \rightarrow X \times X$, we obtain

$$\mathcal{L}_{X_{\mathrm{dR}}, \sigma} \simeq (\mathcal{L}_{X_{\mathrm{dR}}, \sigma})^{\otimes 2}.$$

Hence, $\mathcal{L}_{X_{\mathrm{dR}}, \sigma}$ is trivial. By (8.16), this trivializes the system (8.15).

8.5.4. For a fixed I , consider the fiber products

$$\mathrm{Op}_{G, X_{\mathrm{dR}}^I}^{\mathrm{mer}} := X_{\mathrm{dR}}^I \times_{\mathrm{Op}_{G, \mathrm{Ran}}^{\mathrm{mer}}} \text{ and } \mathrm{Op}_{G, X^I}^{\mathrm{mer}} := X^I \times_{\mathrm{Op}_{G, \mathrm{Ran}}^{\mathrm{mer}}} \mathrm{Op}_{G, \mathrm{Ran}}^{\mathrm{mer}}$$

and the line bundles

$$\mathcal{L}_{\mathrm{Op}_{G, X_{\mathrm{dR}}^I}^{\mathrm{mer}}} := \mathcal{L}_{\mathrm{Op}_{G, \mathrm{Ran}}^{\mathrm{mer}}} |_{\mathrm{Op}_{G, X_{\mathrm{dR}}^I}^{\mathrm{reg}}} \text{ and } \mathcal{L}_{\mathrm{Op}_{G, X^I}^{\mathrm{mer}}} := \mathcal{L}_{\mathrm{Op}_{G, \mathrm{Ran}}^{\mathrm{mer}}} |_{\mathrm{Op}_{G, X^I}^{\mathrm{reg}}}.$$

The map $\mathrm{Op}_{G, X^I}^{\mathrm{mer}} \rightarrow X^I$ is a Zariski-locally trivial fibration with ind-pro-affine spaces as fibers. Since X^I is smooth, we obtain that $\mathcal{L}_{\mathrm{Op}_{G, X^I}^{\mathrm{reg}}}$ descends to a canonically defined line bundle $\mathcal{L}_{X^I}$ on X^I . Moreover, the collection

$$I \rightsquigarrow \mathcal{L}_{X^I}$$

has a natural factorization structure.

Furthermore, by construction, we have

$$\mathcal{L}_{X^I} \simeq \mathcal{L}_{X_{\mathrm{dR}, \sigma}^I} |_{X^I},$$

compatibly with factorization.

Hence, by Sect. 8.5.2, we obtain that the system

$$(8.17) \quad I \mapsto \mathcal{L}_{\mathrm{Op}_{G, X^I}^{\mathrm{mer}}}$$

is canonically trivial, compatibly with factorization.

It remains to show that this trivialization descends to a trivialization of the system

$$(8.18) \quad I \mapsto \mathcal{L}_{\mathrm{Op}_{G, X_{\mathrm{dR}}^I}^{\mathrm{mer}}}.$$

8.5.5. Note a priori, the obstruction to triviality is given by a function on $\mathcal{L}_{\mathrm{Op}_{G, X^I}^{\mathrm{mer}}}$ with values in the pullback of the sheaf of 1-forms on X^I ; denote this function by α_{X^I} .

Note that by factorization,

$$\alpha_{X^I} |_{\overset{\circ}{X}^I} \simeq \alpha_X^{\boxtimes I}.$$

Hence, it is enough to show that $\alpha_X = 0$.

8.5.6. We will first show that $\alpha_X |_{\mathrm{Op}_{G, X}^{\mathrm{reg}}}$ is trivial.

Recall that the factorization scheme $\mathrm{Op}_G^{\mathrm{reg}}$ is counital (see Sect. C.6.15 for what this means). In particular, there exist canonical projections

$$p_i : \mathrm{Op}_{G, X^2}^{\mathrm{reg}} \rightarrow \mathrm{Op}_{G, X}^{\mathrm{reg}}, \quad i = 1, 2$$

covering the two projections $p_i : X^2 \rightarrow X$.

We claim that

$$\alpha_{X^2} = p_1^*(\alpha_X) + p_2^*(\alpha_X).$$

Indeed, the equality takes place over $\overset{\circ}{X}^2$, by factorization, and hence over the entire X^2 by density.

8.5.7. Recall now that $\mathrm{Op}_G^{\mathrm{mer}}$ has a unital-in-correspondences structure relative to $\mathrm{Op}_G^{\mathrm{reg}}$ (see Sect. C.10.6 for what this means). We claim that the connection forms α_I are compatible with this structure in the following sense:

For an injection of finite sets $\phi : I_1 \rightarrow I_2$, let

$$\mathrm{Op}_{\tilde{G}, X^{I_1}}^{\mathrm{mer}} \xleftarrow{\mathrm{pr}_{\mathrm{small}}^{\mathrm{Op}}} \mathrm{Op}_{\tilde{G}, X^\phi}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}} \xrightarrow{\mathrm{pr}_{\mathrm{big}}^{\mathrm{Op}}} \mathrm{Op}_{\tilde{G}, X^{I_2}}^{\mathrm{mer}}$$

be the correspondence covering

$$X^{I_1} \xleftarrow{\Delta_\phi} X^{I_2} \xrightarrow{\mathrm{id}} X^{I_2}.$$

We claim that

$$(8.19) \quad (\mathrm{pr}_{\mathrm{small}}^{\mathrm{Op}})^*(\alpha_{X^{I_1}}) = (\mathrm{pr}_{\mathrm{big}}^{\mathrm{Op}})^*(\alpha_{X^{I_2}}).$$

Indeed, write $I_2 = I_1 \sqcup J$, and let

$$(X^{I_1} \times X^J)_{\mathrm{disj}} \subset X^{I_1} \times X^J$$

be the corresponding open subset.

It suffices that the equality (8.19) takes place over

$$\mathrm{Op}_{\tilde{G}, X^\phi}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}} \times_{X^{I_2}} (X^{I_1} \times X^J)_{\mathrm{disj}}.$$

We have

$$(8.20) \quad \mathrm{Op}_{\tilde{G}, X^\phi}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}} \times_{X^{I_2}} (X^{I_1} \times X^J)_{\mathrm{disj}} \simeq (\mathrm{Op}_{\tilde{G}, X^{I_1}}^{\mathrm{mer}} \times \mathrm{Op}_{\tilde{G}, X^J}^{\mathrm{reg}})_{X^{I_1} \times X^J} \times (X^{I_1} \times X^J)_{\mathrm{disj}}$$

and

$$\mathrm{Op}_{\tilde{G}, X^{I_2}}^{\mathrm{mer}} \times_{X^{I_2}} (X^{I_1} \times X^J)_{\mathrm{disj}} \simeq (\mathrm{Op}_{\tilde{G}, X^{I_1}}^{\mathrm{mer}} \times \mathrm{Op}_{\tilde{G}, X^J}^{\mathrm{mer}})_{X^{I_1} \times X^J} \times (X^{I_1} \times X^J)_{\mathrm{disj}},$$

where the map $\mathrm{pr}_{\mathrm{small}}^{\mathrm{Op}}$ identifies with projection on the first factor in (8.20), and the map $\mathrm{pr}_{\mathrm{big}}^{\mathrm{Op}}$ is the inclusion

$$\mathrm{Op}_{\tilde{G}, X^J}^{\mathrm{reg}} \rightarrow \mathrm{Op}_{\tilde{G}, X^J}^{\mathrm{mer}}$$

along the second factor.

By factorization, we obtain that

$$(\mathrm{pr}_{\mathrm{big}}^{\mathrm{Op}})^*(\alpha_{X^{I_2}})|_{\mathrm{Op}_{\tilde{G}, X^\phi}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}} \times_{X^{I_2}} (X^{I_1} \times X^J)_{\mathrm{disj}}}$$

equals the sum of $(\mathrm{pr}_{\mathrm{small}}^{\mathrm{Op}})^*(\alpha_{X^{I_1}})$ and the pullback of $\alpha_{X^J}|_{\mathrm{Op}_{\tilde{G}, X^J}^{\mathrm{reg}}}$ along the projection of (8.20) on the second factor.

However, the latter is zero by Sect. 8.5.6.

8.5.8. We are now ready to show that $\alpha_X = 0$. In doing so we will mimic the argument in [BD2, Proposition 3.4.7].

Write $\Omega_{X^2}^1$ as

$$\omega_X \boxtimes \mathcal{O}_X \oplus \mathcal{O}_X \boxtimes \omega_X.$$

We will show that the restriction of α_{X^2} to

$$\mathrm{Op}_{\tilde{G}, X}^{\mathrm{mer}} \simeq X \times_{X^2} \mathrm{Op}_{\tilde{G}, X^2}^{\mathrm{mer}},$$

viewed as a function on $\mathrm{Op}_{\tilde{G}, X}^{\mathrm{mer}}$ with values in the pullback of $\Omega_{X^2}^1$, takes values both in the pullback of $\omega_X \boxtimes \mathcal{O}_X$ and in the pullback of $\mathcal{O}_X \boxtimes \omega_X$. This would imply that this restriction is 0, and hence also that $\alpha_X = 0$.

We will in fact show that the restriction of α_{X^2} to

$$(8.21) \quad (X^2)^\wedge \times_{X^2} \mathrm{Op}_{\tilde{G}, X^2}^{\mathrm{mer}}$$

is 0, where $(X^2)^\wedge$ is the formal completion of the diagonal in X^2 .

8.5.9. By symmetry, it suffices to show that the restriction of α_{X^2} to takes values in the pullback of $\omega_X \boxtimes \mathcal{O}_X$.

Consider the inclusion $I_1 := \{1\} \xrightarrow{\phi} \{1, 2\} := I_2$, and the corresponding map

$$\mathrm{Op}_{\tilde{G}, X^\phi}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}} \xrightarrow{\mathrm{pt}_{\mathrm{big}}^{\mathrm{Op}}} \mathrm{Op}_{\tilde{G}, X^2}^{\mathrm{mer}}.$$

This map is an isomorphism over $X \subset X^2$, and hence, induces an isomorphism

$$(X^2)^\wedge \times_{X^2} \mathrm{Op}_{\tilde{G}, X^\phi}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}} \xrightarrow{\sim} (X^2)^\wedge \times_{X^2} \mathrm{Op}_{\tilde{G}, X^2}^{\mathrm{mer}}.$$

Hence, it is enough to prove that the pullback of α_{X^2} along target takes values in the pullback of $\omega_X \boxtimes \mathcal{O}_X$. However, this has been established in Sect. 8.5.7.

□[Theorem 8.4.5]

8.6. **Proof of Theorem 8.1.4.** By a slight abuse of notation we will use the symbol $\mathrm{FLE}_{G, \mathrm{crit}}$ for the functor (6.6).

8.6.1. Consider the following diagram

$$\begin{array}{ccccc}
 (\mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)})^\vee & & \xleftarrow{\tau_G \circ (\mathrm{FLE}_{G, \mathrm{crit}})^\vee} & & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})^\vee \\
 \downarrow \sim & & & & \\
 (\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})} \mathrm{Whit}^!(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}))^\vee & & & & \downarrow \sim \\
 \downarrow \sim & & & & \\
 (\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})} \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}))^\vee & \xleftarrow{(\mathrm{Id} \otimes \tau_G \circ \overline{\mathrm{DS}}^{\mathrm{enh}}, \mathrm{rfnd})^\vee} & (\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}))^\vee & & \\
 \sim \downarrow & & \downarrow \sim & & \\
 \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})}(\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}), \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}))^\vee & \xleftarrow{\tau_G \circ (\overline{\mathrm{DS}}^{\mathrm{enh}}, \mathrm{rfnd})^\vee} & \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})}(\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}), \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}))^\vee & & \\
 \sim \downarrow & & \downarrow \sim & & \\
 \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})}(\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}), \mathrm{Whit}^!(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})) & & \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})}(\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}), \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})) & & \\
 \sim \uparrow \Theta_{\mathrm{Whit}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})} & & \Theta_{\mathrm{Op}_G^{\mathrm{mer}}} \downarrow \sim & & \\
 \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})}(\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}), \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})) & \xrightarrow{\overline{\mathrm{DS}}^{\mathrm{enh}}, \mathrm{rfnd}} & \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})}(\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}), \mathrm{IndCoh}_*(\mathrm{Op}_G^{\mathrm{mer}})) & & \\
 \sim \uparrow & & \uparrow \sim & & \\
 \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} & \xrightarrow{\mathrm{FLE}_{G, \mathrm{crit}}} & \mathrm{IndCoh}_*(\mathrm{Op}_G^{\mathrm{mon-free}}), & &
 \end{array}$$

in which the upper vertical arrows are the duals of

$$\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})} \mathrm{Whit}^!(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \xrightarrow{(6.15)} \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)}$$

and

$$\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}),$$

respectively.

We will show:

- The left vertical composition is the identification (8.1);
- The right vertical composition is the identification (8.2);
- All inner squares commute.

This will establish the commutativity of (8.3). The compatibility of this isomorphism with the actions of

$$\mathrm{Sph}_G^{\mathrm{Sat}_{G,\tau}} \simeq \mathrm{Sph}_G^{\mathrm{spec}}.$$

is automatic from the construction.

8.6.2. *The left vertical composition.* We need to establish the commutativity of the following diagram

$$\begin{array}{ccc} (\mathrm{KL}(G)_{\mathrm{crit},\rho(\omega_X)})^\vee & \longrightarrow & (\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})} \mathrm{Whit}^!(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega_X)}))^\vee \\ & & \downarrow \sim \\ (8.1) \downarrow & & \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})}(\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}), \mathrm{Whit}^!(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega_X)}))^\vee \\ & & \downarrow \sim \\ \mathrm{KL}(G)_{\mathrm{crit},\rho(\omega_X)} & \longrightarrow & \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})}(\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}), \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega_X)})). \end{array}$$

However, this is just the fact that in the context of Sect. 6.3.1, the dual of the functor (6.9) for $\mathbf{C}$ is the functor (6.10) for $\mathbf{C}^\vee$.

8.6.3. *The right vertical composition.* The identification of the right vertical composition follows from Sect. 3.7.15.

8.6.4. *The top square.* Passing to the dual functors, we need to establish the commutativity of

$$\begin{array}{ccc} \mathrm{KL}(G)_{\mathrm{crit},\rho(\omega_X)} & \xrightarrow{\mathrm{FLE}_{G,\mathrm{crit}}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \\ \uparrow \sim & & \uparrow \sim \\ \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})} \mathrm{Whit}^!(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega_X)}) & & \\ \uparrow \sim & & \\ \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})} \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit},\rho(\omega_X)}) & \xrightarrow[\sim]{\mathrm{Id} \otimes \overline{\mathrm{DS}}^{\mathrm{enh},\mathrm{rfnd}}} & \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mer}})} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}). \end{array}$$

However, this is the content of Proposition 6.3.5.

8.6.5. *The 2nd square from the top.* This square commutes tautologically.

8.6.6. *The bottom square.* The commutation follows from the definition of the functor $\mathrm{FLE}_{G,\mathrm{crit}}$.

8.6.7. Finally, it remains to show that the 3rd square from the top commutes.³⁵

However, the required commutation is given by Theorem 8.4.5.

□[Theorem 8.1.4]

³⁵This is the only non-tautological point in the proof.

Part II: Local-to-global constructions

9. THE COEFFICIENT AND POINCARÉ FUNCTOR(S)

This section begins by introducing our main object of study: the critically twisted category of D-modules on Bun_G . In this section we will mostly think of its incarnation as $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$, see Remark 1.1.13, as the main characters in this section are sheaf-theoretic in nature.

The focus of this section is Poincaré and Whittaker coefficient functors. In fact, there are two Poincaré functors

$$\mathrm{Poinc}_{G,!} : \mathrm{Whit}^!(G)_{\mathrm{Ran}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \text{ and } \mathrm{Poinc}_{G,*} : \mathrm{Whit}_*(G)_{\mathrm{Ran}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G),$$

where $\mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G)$ is the dual category of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$. These two functors are Verdier-conjugate: the dual functor of $\mathrm{Poinc}_{G,*}$ is isomorphic to the right adjoint of $\mathrm{Poinc}_{G,!}$; this is the functor

$$\mathrm{coeff}_G : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{Whit}^!(G)_{\mathrm{Ran}}.$$

One can also give a global interpretation of the above functors, where instead of the affine Grassmannian, one uses the *twisted Drinfeld compactification*

$$\overline{\mathrm{Bun}}_{N,\rho(\omega_X)} \rightarrow \mathrm{Bun}_G.$$

This is how the global geometric Whittaker model had been mostly approached so far (see, e.g., [Ga1]). The two approaches are, however, equivalent (see [Ga6]).

For the purposes of this paper, we will only explicitly need the global interpretation of the *vacuum* cases of the above functors, see Sect. 9.6.

9.1. Twisted D-modules on Bun_G .

9.1.1. Let $\det_{\mathrm{Bun}_G}$ be the determinant line bundle on Bun_G , normalized so that it sends a G -bundle $\mathcal{P}_G$ to

$$\det(\Gamma(X, \mathfrak{g}_{\mathcal{P}_G})) \otimes \det\left(\Gamma(X, \mathfrak{g}_{\mathcal{P}_G^0})\right)^{\otimes -1},$$

where $\mathcal{P}_G^0$ is the trivial bundle.

9.1.2. Note that we have

$$\pi^*(\det_{\mathrm{Bun}_G}) \simeq \det_{\mathrm{Gr}_G, \mathrm{Ran}},$$

where π denotes the projection

$$(9.1) \quad \mathrm{Gr}_{G, \mathrm{Ran}} \rightarrow \mathrm{Bun}_G.$$

9.1.3. Note also that up to the (constant) line $\det\left(\Gamma(X, \mathfrak{g}_{\mathcal{P}_G^0})\right)$, the line bundle $\det_{\mathrm{Bun}_G}$ identifies with the canonical line bundle on Bun_G .

9.1.4. Let crit be half of the de Rham twisting defined by $\det_{\mathrm{Bun}_G}$, i.e.,

$$\mathrm{crit} = \frac{1}{2} \cdot \mathrm{dlog}(\det_{\mathrm{Bun}_G}).$$

We will denote by

$$\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$$

the corresponding category of twisted D-modules.

Note that by Sect. 9.1.3, the critical twisting on Bun_G is canonically isomorphic to the half-canonical twisting.

9.1.5. As in Sect. 1.1.11, we obtain a *canonical* identification

$$(9.2) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{\sim} \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G),$$

where $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ is the short-hand for

$$\mathrm{D}\text{-mod}_{\det_{\mathrm{Bun}_G}^{\frac{1}{2}}}(\mathrm{Gr}_G),$$

cf. Sect. 1.1.6.

Remark 9.1.6. According to [BD1, Sect. 4], the choice of $\omega_X^{\otimes \frac{1}{2}}$ gives rise to a choice of the square root of $\det_{\mathrm{Bun}_G}$ as a line bundle. This allows us to identify $\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$ (or equivalently $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$) with the untwisted category $\mathrm{D}\text{-mod}(\mathrm{Bun}_G)$.

However, we will avoid using this identification.

9.1.7. Pullback along π defines functors

$$\pi^! : \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_{G,\mathrm{Ran}})$$

and

$$\pi^! : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\mathrm{Ran}}),$$

so that the diagram

$$\begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\mathrm{Ran}}) & \longrightarrow & \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_{G,\mathrm{Ran}}) \\ \pi^! \uparrow & & \uparrow \pi^! \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \longrightarrow & \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \end{array}$$

commutes.

9.2. Restricting to (twists of) Bun_N .

9.2.1. Let $\mathcal{P}_T$ be any T -bundle. Consider the stack

$$(9.3) \quad \mathrm{Bun}_{N,\mathcal{P}_T} \simeq \mathrm{Bun}_B \times_{\mathrm{Bun}_T} \mathrm{pt},$$

where $\mathrm{pt} \rightarrow \mathrm{Bun}_T$ is the point $\mathcal{P}_T$.

Denote by $\mathfrak{p}$ the map

$$\mathrm{Bun}_{N,\mathcal{P}_T} \rightarrow \mathrm{Bun}_G.$$

Note that the pullback of $\det_{\mathrm{Bun}_G}$ along this map is canonically constant. Denote the resulting line by

$$\mathfrak{l}_{G,N\mathcal{P}_T},$$

see [GLC1, Sect. 1.3.1].

9.2.2. Note that on the one hand, we obtain an identification

$$(9.4) \quad \mathrm{D}\text{-mod}_{\frac{1}{2} \cdot \mathrm{dlog}(\mathfrak{l}_{G,N\mathcal{P}_T})}(\mathrm{Bun}_{N,\mathcal{P}_T}) \xrightarrow{\sim} \mathrm{D}\text{-mod}(\mathrm{Bun}_{N,\mathcal{P}_T}),$$

since the dlog map over pt is trivial.

On the other hand, recall (see [GLC1, Proposition 1.3.3]) that the line bundle $\mathfrak{l}_{G,N\mathcal{P}_T}$ admits a canonical square root, to be denoted $\mathfrak{l}_{G,N\mathcal{P}_T}^{\otimes \frac{1}{2}}$. Hence, we obtain *another* identification

$$(9.5) \quad \mathrm{D}\text{-mod}_{\frac{1}{2} \cdot \mathrm{dlog}(\mathfrak{l}_{G,N\mathcal{P}_T})}(\mathrm{Bun}_{N,\mathcal{P}_T}) \stackrel{(1.2)}{\simeq} \mathrm{D}\text{-mod}_{\mathfrak{l}_{G,N\mathcal{P}_T}^{\otimes \frac{1}{2}}}(\mathrm{Bun}_{N,\mathcal{P}_T}) \simeq \mathrm{D}\text{-mod}(\mathrm{Bun}_{N,\mathcal{P}_T}).$$

The discrepancy between the two trivializations is given by tensoring by the line $\mathfrak{l}_{G,N\mathcal{P}_T}^{\otimes \frac{1}{2}}$.

9.2.3. We will denote by $\mathbf{p}_{\text{crit}}^!$ the functor

$$\begin{aligned} \mathrm{D}\text{-mod}_{\text{crit}}(\mathrm{Bun}_G) &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2} \cdot \mathrm{dlog}(\det_{\mathrm{Bun}_G} |_{\mathrm{Bun}_{N, \mathcal{P}_T}})}(\mathrm{Bun}_{N, \mathcal{P}_T}) = \mathrm{D}\text{-mod}_{\frac{1}{2} \cdot \mathrm{dlog}(\mathfrak{l}_{G, N, \mathcal{P}_T})}(\mathrm{Bun}_{N, \mathcal{P}_T}) \rightarrow \\ &\xrightarrow{(9.4)} \mathrm{D}\text{-mod}(\mathrm{Bun}_{N, \mathcal{P}_T}). \end{aligned}$$

We will denote by $\mathbf{p}^!$ the functor

$$\begin{aligned} \mathrm{D}\text{-mod}_{\text{crit}}(\mathrm{Bun}_G) &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2} \cdot \mathrm{dlog}(\det_{\mathrm{Bun}_G} |_{\mathrm{Bun}_{N, \mathcal{P}_T}})}(\mathrm{Bun}_{N, \mathcal{P}_T}) = \mathrm{D}\text{-mod}_{\frac{1}{2} \cdot \mathrm{dlog}(\mathfrak{l}_{G, N, \mathcal{P}_T})}(\mathrm{Bun}_{N, \mathcal{P}_T}) \rightarrow \\ &\xrightarrow{(9.5)} \mathrm{D}\text{-mod}(\mathrm{Bun}_{N, \mathcal{P}_T}). \end{aligned}$$

Thus, we obtain a commutative diagram

$$(9.6) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}(\mathrm{Bun}_{N, \mathcal{P}_T}) & \xrightarrow{\otimes \mathfrak{l}_{G, N, \mathcal{P}_T}^{\frac{1}{2}}} & \mathrm{D}\text{-mod}(\mathrm{Bun}_{N, \mathcal{P}_T}) \\ \mathbf{p}^! \uparrow & & \uparrow \mathbf{p}_{\text{crit}}^! \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{(9.2)} & \mathrm{D}\text{-mod}_{\text{crit}}(\mathrm{Bun}_G). \end{array}$$

9.2.4. In this section we will take $\mathcal{P}_T := \rho(\omega_X)$. Consider the corresponding line

$$(9.7) \quad \mathfrak{l}_{G, N_{\rho(\omega_X)}}^{\frac{1}{2}}.$$

In this case, (9.6) reads as

$$(9.8) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}(\mathrm{Bun}_{N, \rho(\omega_X)}) & \xrightarrow{\otimes \mathfrak{l}_{G, N_{\rho(\omega_X)}}^{\frac{1}{2}}} & \mathrm{D}\text{-mod}(\mathrm{Bun}_{N, \rho(\omega_X)}) \\ \mathbf{p}^! \uparrow & & \uparrow \mathbf{p}_{\text{crit}}^! \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{(9.2)} & \mathrm{D}\text{-mod}_{\text{crit}}(\mathrm{Bun}_G). \end{array}$$

9.3. **The coefficient functor.** In this subsection we will recall the definition of the functor of Whittaker coefficient(s).

9.3.1. Consider the $\rho(\omega_X)$ -twisted version of the map (9.1)

$$\mathrm{Gr}_{G, \rho(\omega_X), \mathrm{Ran}} \rightarrow \mathrm{Bun}_G.$$

Due to the trivialization of the μ_2 -gerbe $\mathfrak{l}_{G, N_{\rho(\omega_X)}}^{\frac{1}{2}}$, we have

$$\pi^*(\det_{\mathrm{Bun}_G}^{\frac{1}{2}}) \simeq \det_{\mathrm{Gr}_{G, \rho(\omega_X)}}^{\frac{1}{2}}$$

as μ_2 -gerbes on $\mathrm{Gr}_{G, \rho(\omega_X), \mathrm{Ran}}$.

Hence, π gives rise to a well-defined functor

$$\pi^! : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \rho(\omega_X), \mathrm{Ran}}).$$

9.3.2. In this subsection we will the functor of Whittaker coefficient(s), denoted coeff_G , which maps

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{Whit}^!(G)_{\text{Ran}}.$$

and is defined as follows.

To simplify the notation, we will work over a fixed point $\underline{x} \in \text{Ran}$. So we need to define the functor

$$\text{coeff}_{G,\underline{x}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{Whit}^!(G)_{\underline{x}}.$$

Denote by

$$\pi_{\underline{x}} : \text{Gr}_{G,\rho(\omega_X),\underline{x}} \rightarrow \text{Bun}_G$$

the restriction of π along

$$(9.9) \quad \text{Gr}_{G,\rho(\omega_X),\underline{x}} \rightarrow \text{Gr}_{G,\rho(\omega_X),\text{Ran}}.$$

9.3.3. Write $\mathfrak{L}(N)_{\rho(\omega_X),\underline{x}}$ as a filtered union of subschemes N^α . For every α , consider the functor

$$\text{Av}_*^{(N^\alpha,\chi)} : \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\underline{x}}) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\underline{x}})^{N^\alpha,\chi} \hookrightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\underline{x}}).$$

For $N^\alpha \subset N^{\alpha'}$, we have a canonically defined natural transformation

$$(9.10) \quad \text{Av}_*^{(N^{\alpha'},\chi)} \rightarrow \text{Av}_*^{(N^\alpha,\chi)}.$$

We have the following (elementary) observation:

Lemma 9.3.4. *The natural transformation*

$$\text{Av}_*^{(N^{\alpha'},\chi)} \circ \pi_{\underline{x}}^! \rightarrow \text{Av}_*^{(N^\alpha,\chi)} \circ \pi_{\underline{x}}^!,$$

induced by (9.10), is an isomorphism when N^α is large enough.

Proof. Let

$$\text{Sect}(X - \underline{x}, N_{\rho(\omega_X)}) \subset \text{Sect}(X - \underline{x}, G_{\rho(\omega_X)})$$

be the group ind-schemes of sections of

$$N_{\rho(\omega_X)} \subset G_{\rho(\omega_X)}$$

over $X - \underline{x}$. Laurent expansion defines maps

$$\text{Sect}(X - \underline{x}, N_{\rho(\omega_X)}) \rightarrow \mathfrak{L}(N)_{\rho(\omega_X),\underline{x}} \text{ and } \text{Sect}(X - \underline{x}, G_{\rho(\omega_X)}) \rightarrow \mathfrak{L}(G)_{\rho(\omega_X),\underline{x}}.$$

Note that the restriction of χ to $\text{Sect}(X - \underline{x}, N_{\rho(\omega_X)})$ is trivial.

For $\mathcal{F} \in \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$, the pullback

$$\pi_{\underline{x}}^!(\mathcal{F}) \in \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\underline{x}})$$

is $\text{Sect}(X - \underline{x}, G_{\rho(\omega_X)})$ -equivariant, and hence $\text{Sect}(X - \underline{x}, N_{\rho(\omega_X)})$ -equivariant.

Hence, the map in the lemma is an isomorphism any time

$$(9.11) \quad N^\alpha \cdot \text{Sect}(X - \underline{x}, N_{\rho(\omega_X)}) = \mathfrak{L}(N)_{\rho(\omega_X),\underline{x}}.$$

□

9.3.5. By Lemma 9.10, for N^α large enough, the functor

$$(9.12) \quad \text{Av}_*^{(N^\alpha,\chi)} \circ \pi_{\underline{x}}^!$$

does not depend on the choice of N^α . In particular, its essential image is contained in

$$\bigcap_{N^\alpha} \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\underline{x}})^{N^\alpha,\chi} = \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\underline{x}})^{\mathfrak{L}(N)_{\rho(\omega_X),\underline{x}},\chi} = \text{Whit}^!(G)_{\underline{x}}.$$

Thus, we let $\text{coeff}_{G,\underline{x}}$ be the functor (9.12) for N^α large enough.

9.3.6. By construction, the functor $\text{coeff}_{G,\underline{x}}$ is compatible with the action of $\text{Sph}_{G,\underline{x}}$.

When working over the Ran space, we consider the functor, to be denoted $\text{coeff}_{G,\text{Ran}}$,

$$\begin{aligned} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G \times \text{Ran}) &\simeq \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}) \xrightarrow{\text{coeff}_G \otimes \text{Id}} \\ &\rightarrow \text{Whit}^!(G)_{\text{Ran}} \otimes \text{D-mod}(\text{Ran}) \xrightarrow{\otimes} \text{Whit}^!(G)_{\text{Ran}}. \end{aligned}$$

This functor is compatible with the natural action of $\text{Sph}_{G,\text{Ran}}$ on the two sides.

9.3.7. We let

$$\text{coeff}_G : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{Whit}^!(G)_{\text{Ran}}$$

denote the composition

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \xrightarrow{\text{Id} \otimes \omega_{\text{Ran}}} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G \times \text{Ran}) \xrightarrow{\text{coeff}_{G,\text{Ran}}} \text{Whit}^!(G)_{\text{Ran}}.$$

9.3.8. The functors $\text{coeff}_{G,\underline{x}}$ have the following property:

For $\underline{x} \subseteq \underline{x}'$ consider the natural embedding

$$\text{incl}_{\underline{x} \subseteq \underline{x}'} : \text{Gr}_{G,\rho(\omega_X),\underline{x}} \hookrightarrow \text{Gr}_{G,\rho(\omega_X),\underline{x}'}.$$

The functor $(\text{incl}_{\underline{x} \subseteq \underline{x}'})^!$ maps $\text{Whit}(G)_{\underline{x}'} \rightarrow \text{Whit}(G)_{\underline{x}}$, and we have

$$(9.13) \quad \text{coeff}_{G,\underline{x}} \simeq \text{incl}_{\underline{x} \subseteq \underline{x}'}^! \circ \text{coeff}_{G,\underline{x}'}.$$

The isomorphism (9.13) expresses the *unital* structure on the functor coeff_G , to be discussed in Sect. 11.

9.3.9. Let

$$\text{coeff}_G^{\text{Vac}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{Vect}$$

denote the composition of $\text{coeff}_{G,\underline{x}}$ with the functor

$$\text{Whit}^!(G)_{\underline{x}} \hookrightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\underline{x}}) \rightarrow \text{Vect},$$

where the second arrow is the functor of $!$ -fiber at the unit point.

By (9.13), the above definition of $\text{coeff}_G^{\text{Vac}}$ is canonically independent of the choice of $\underline{x}$.

Equivalently, $\text{coeff}_G^{\text{Vac}}$ is the unique functor $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{Vect}$ so that the diagram commutes

$$\begin{array}{ccccc} \text{Whit}^!(G)_{\text{Ran}} & \longrightarrow & \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\text{Ran}}) & \xrightarrow{(\text{Id}_{\text{Gr}_{G,\rho(\omega_X),\text{Ran}}})^!} & \text{D-mod}(\text{Ran}) \\ \text{coeff}_G \uparrow & & & & \uparrow \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & & \xrightarrow{\text{coeff}_G^{\text{Vac}}} & & \text{Vect}. \end{array}$$

(In the above diagram the right vertical arrow is the $!$ -pullback along $\text{Ran} \rightarrow \text{pt}$, which is fully faithful by the contractibility of the Ran space.)

9.3.10. As in Remark 1.3.10, both the category $\text{Whit}^!(G)_{\text{Ran}}$ and the functor coeff_G are canonically independent of the choice of a non-degenerate character $\chi_0 : N \rightarrow \mathbb{G}_a$.

9.4. The $!$ -Poincaré functor.

9.4.1. We start again by working with a fixed $\underline{x} \in \text{Ran}$. We claim:

Proposition 9.4.2. *The functor $\text{coeff}_{G,\underline{x}}$ admits a left adjoint, to be denoted $\text{Poinc}_{G,!,\underline{x}}$.*

Remark 9.4.3. In fact, as we work over a fixed point $\underline{x} \in \text{Ran}$, all objects in $\text{Whit}^!(G)_{\underline{x}}$ are ind-holonomic, which implies the assertion of the proposition. Below we give a different argument, which works also when $\underline{x}$ is allowed to move in families over Ran , see Sect. 9.4.6.

Proof of Proposition 9.4.2. Consider the *partially defined*³⁶ functor

$$(\pi_{\underline{x}})! : \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\underline{x}}) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G),$$

left adjoint to $\pi_{\underline{x}}^!$.

The assertion of the proposition is equivalent to the fact that $(\pi_{\underline{x}})!$ is defined on $\text{Whit}^!(G)_{\text{Ran}} \hookrightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\underline{x}})$.

First, it is easy to see that if $(\pi_{\underline{x}})!$ is defined on some object $\mathcal{F} \in \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\underline{x}})$ and $\mathcal{S}$ is an object of $\text{Sph}_{G,\underline{x}}$, then $(\pi_{\underline{x}})!$ is defined on $\mathcal{S} \star \mathcal{F}$, and in fact

$$(\pi_{\underline{x}})!(\mathcal{S} \star \mathcal{F}) \simeq \mathcal{S} \star (\pi_{\underline{x}})!(\mathcal{F}).$$

This follows from the properness of the convolution diagram that defines the $\text{Sph}_{G,\underline{x}}$ -action on $\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\underline{x}})$ and $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$.

Next, we observe that $(\pi_{\underline{x}})!$ is defined on the vacuum object

$$\text{Vac}_{\text{Whit}^!(G),\underline{x}} \in \text{Whit}^!(G)_{\underline{x}} \subset \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\underline{x}}).$$

Indeed, this follows from the fact that $\text{Vac}_{\text{Whit}^!(G),\underline{x}}$ is ind-holonomic, and the $!$ -direct image functor is defined on holonomic D-modules.

Finally, we claim that any object of $\text{Whit}^!(G)_{\underline{x}}$ can be obtained as a convolution of an object of $\text{Sph}_{G,\underline{x}}$ with $\text{Vac}_{\text{Whit}^!(G),\underline{x}}$. In fact, by Remark 1.7.8, the functor

$$\text{Rep}(\check{G})_{\underline{x}} \xrightarrow{\text{Sat}^{\text{nv}}} \text{Sph}_{\underline{x}} \xrightarrow{- \star \text{Vac}_{\text{Whit}^!(G),\underline{x}}} \text{Whit}^!(G)_{\underline{x}}$$

is an equivalence. □

9.4.4. The above proof shows that the functor $\text{Poinc}_{G,!,\underline{x}}$ is also compatible with the action of $\text{Sph}_{G,\underline{x}}$.

Note, however, that this also follows a priori from the compatibility of $\text{coeff}_{G,\underline{x}}$ with $\text{Sph}_{G,\underline{x}}$ -actions and the observation that $\text{Sph}_{G,\underline{x}}$ is rigid as a monoidal category.³⁷

9.4.5. For a pair of points $\underline{x}, \underline{x}'$ of Ran with $\underline{x} \subseteq \underline{x}'$, let

$$\text{ins. vac}_{\underline{x} \subseteq \underline{x}'} : \text{Whit}(G)_{\underline{x}} \rightarrow \text{Whit}(G)_{\underline{x}'}$$

be the functor left adjoint to

$$(\text{incl}_{\underline{x} \subseteq \underline{x}'})^! : \text{Whit}(G)_{\underline{x}'} \rightarrow \text{Whit}(G)_{\underline{x}}.$$

Explicitly, if $\underline{x}' = \underline{x} \sqcup \underline{x}''$, so that

$$\text{Whit}(G)_{\underline{x}'} \simeq \text{Whit}(G)_{\underline{x}} \otimes \text{Whit}(G)_{\underline{x}''},$$

then

$$\text{ins. vac}_{\underline{x} \subseteq \underline{x}'} \simeq \text{Id} \otimes \text{Vac}_{\text{Whit}^!(G),\underline{x}''}.$$

By adjunction, from (9.13), we obtain:

$$(9.14) \quad \text{Poinc}_{G,!,\underline{x}'} \circ \text{ins. vac}_{\underline{x} \subseteq \underline{x}'} \simeq \text{Poinc}_{G,!,\underline{x}}.$$

³⁶The issue here is that the “lower-!” functors are not necessarily defined on non-holonomic objects.

³⁷In fact, this was implicitly used in the proof of Proposition 9.4.2: the properness of the convolution diagram is the reason for the rigidity of $\text{Sph}_{G,\underline{x}}$.

In Sect. 11.3.7 we will formulate a version of this isomorphism when the points $\underline{x}$ and $\underline{x}'$ move in families over the Ran space.

9.4.6. By the same token, the functor

$$\text{coeff}_{G,\text{Ran}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G \times \text{Ran}) \rightarrow \text{Whit}^!(G)_{\text{Ran}}$$

admits a left adjoint, to be denoted

$$\text{Poinc}_{G,!,\text{Ran}} : \text{Whit}^!(G)_{\text{Ran}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G \times \text{Ran}).$$

The functor $\text{Poinc}_{G,!,\text{Ran}}$ is compatible with the actions of $\text{Sph}_{G,\text{Ran}}$ on the two sides.

9.4.7. The functor

$$\text{coeff}_G : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{Whit}^!(G)_{\text{Ran}}$$

admits a left adjoint, to be denoted

$$\text{Poinc}_{G,!} : \text{Whit}^!(G)_{\text{Ran}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G),$$

and given by restricting the partially defined functor

$$\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\text{Ran}}) \xrightarrow{(\pi_{\underline{x}})^!} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$$

to

$$\text{Whit}^!(G)_{\text{Ran}} \hookrightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X),\text{Ran}}).$$

Explicitly, the $\text{Poinc}_{G,!}$ identifies with the composition

$$\text{Whit}^!(G)_{\text{Ran}} \xrightarrow{\text{Poinc}_{G,!,\text{Ran}}} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G \times \text{Ran}) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G),$$

where the second arrow is the functor of $!$ -direct image.

The functor $\text{Poinc}_{G,!,\underline{x}}$ is obtained from $\text{Poinc}_{G,!}$ by restriction along (9.9).

9.4.8. It follows formally from Sect. 9.3.9 that the object

$$\text{Poinc}_{G,!,\underline{x}}(\mathbf{1}_{\text{Whit}^!(G),\underline{x}}) \in \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$$

is canonically independent of the choice of $\underline{x}$.

We will denote it by

$$\text{Poinc}_{G,!}^{\text{Vac}} \in \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G).$$

We also have

$$\text{Poinc}_{G,!}^{\text{Vac}} \simeq \text{Poinc}_{G,!}(\text{Vac}_{\text{Whit}^!(G),\text{Ran}}),$$

where

$$\text{Vac}_{\text{Whit}^!(G),\text{Ran}} \in \text{Whit}^!(G)_{\text{Ran}}$$

is the factorization unit spread over the Ran space.

9.4.9. As we saw in the proof of Proposition 9.4.2, we can recover the functor $\text{Poinc}_{G,!,\underline{x}}$ from the object $\text{Poinc}_{G,!}^{\text{Vac}}$ using the Hecke action.

By adjunction, we obtain that the functor $\text{coeff}_{G,\underline{x}}$ can be uniquely recovered from the knowledge of $\text{coeff}_G^{\text{Vac}}$ and the action of $\text{Rep}(G)_{\underline{x}}$ on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ via

$$\text{Rep}(G)_{\underline{x}} \xrightarrow{\text{Sat}_G^{\text{nv}}} \text{Sph}_{G,\underline{x}}.$$

The same applies to the functors $(\text{Poinc}_{G,!,\text{Ran}}^{\text{Vac}}, \text{coeff}_{G,\text{Ran}}^{\text{Vac}})$.

9.5. **The *-Poincaré functor.**

9.5.1. Recall that along with the category $\mathrm{D}\text{-mod}(\mathrm{Bun}_G)$, one can consider its version $\mathrm{D}\text{-mod}_{\mathrm{co}}(\mathrm{Bun}_G)$, and similarly for gerbe-twisted versions $\mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathrm{Bun}_G)$.

In the untwisted case, we have the identification

$$(\mathrm{D}\text{-mod}(\mathrm{Bun}_G))^\vee \simeq \mathrm{D}\text{-mod}_{\mathrm{co}}(\mathrm{Bun}_G).$$

In the twisted case, this becomes

$$(9.15) \quad (\mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathrm{Bun}_G))^\vee \simeq \mathrm{D}\text{-mod}_{\mathfrak{g} \otimes -1, \mathrm{co}}(\mathrm{Bun}_G).$$

For $\mathfrak{g} = \det_{\mathrm{Bun}_G}^{\frac{1}{2}}$, the identification (9.15) becomes a self-duality

$$(9.16) \quad (\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G))^\vee \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G).$$

9.5.2. Let

$$\mathrm{Poinc}_{G,*} : \mathrm{Whit}_*(G)_{\mathrm{Ran}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G)$$

be the functor *dual* to coeff_G .

Let

$$\mathrm{Poinc}_{G,*,\underline{x}} : \mathrm{Whit}_*(G)_{\underline{x}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G)$$

be the functor dual to $\mathrm{coeff}_{G,\underline{x}}$. It is easy to see that the functor $\mathrm{Poinc}_{G,*,\underline{x}}$ is obtained from $\mathrm{Poinc}_{G,*}$ by restriction along (9.9).

The functor $\mathrm{Poinc}_{G,*,\underline{x}}$ is also compatible with the action of $\mathrm{Sph}_{G,\underline{x}}$.

9.5.3. Let

$$\mathrm{Poinc}_{G,*}^{\mathrm{Vac}} \in \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G)$$

be the image under $\mathrm{Poinc}_{G,*,\underline{x}}$ of the factorization unit

$$\mathbf{1}_{\mathrm{Whit}_*(G),\underline{x}} \in \mathrm{Whit}_*(G)_{\underline{x}}$$

at some/any $\underline{x} \in \mathrm{Ran}$ or, equivalently, of

$$\mathbf{1}_{\mathrm{Whit}_*(G),\mathrm{Ran}} \in \mathrm{Whit}_*(G)_{\mathrm{Ran}}$$

under $\mathrm{Poinc}_{G,*}$.

By definition, the pairing with $\mathrm{Poinc}_{G,*}^{\mathrm{Vac}}$, viewed as a functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{Vect},$$

is the functor $\mathrm{coeff}_G^{\mathrm{Vac}}$.

9.5.4. The functor $\mathrm{Poinc}_{G,*,\underline{x}}$ can be explicitly described as follows. For N^α as in Sect. 9.3.3, consider the composition

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\rho(\omega_X),\underline{x}}) \xrightarrow{\mathrm{Av}_*^{(N^\alpha,\chi)}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\rho(\omega_X),\underline{x}}) \xrightarrow{(\pi_{\underline{x}})_*} \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G).$$

For $N^\alpha \subset N^{\alpha'}$, we have a canonically defined natural transformation

$$(9.17) \quad (\pi_{\underline{x}})_* \circ \mathrm{Av}_*^{(N^{\alpha'},\chi)} \rightarrow (\pi_{\underline{x}})_* \circ \mathrm{Av}_*^{(N^\alpha,\chi)},$$

and it follows from Lemma 9.3.4 that the maps (9.17) are isomorphisms for N^α large enough.

It follows formally that the resulting functor $(\pi_{\underline{x}})_* \circ \mathrm{Av}_*^{(N^\alpha,\chi)}$

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\rho(\omega_X),\underline{x}}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G),$$

for some/any α that is large enough, factors via the projection

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\rho(\omega_X),\underline{x}}) \rightarrow \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\rho(\omega_X),\underline{x}}) \right)_{\mathfrak{L}(N)_{\rho(\omega_X),\underline{x},\chi}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G).$$

The resulting functor

$$\mathrm{Whit}_*(G)_{\underline{x}} := \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\rho(\omega_X),\underline{x}}) \right)_{\mathfrak{L}(N)_{\rho(\omega_X),\underline{x},\chi}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G)$$

is the functor $\mathrm{Poinc}_{G,*,\underline{x}}$.

9.6. Coefficient and Poincaré functors: global interpretation.

9.6.1. Consider the stack $\mathrm{Bun}_{N,\rho(\omega_X)}$ and the map

$$\mathbf{p} : \mathrm{Bun}_{N,\rho(\omega_X)} \rightarrow \mathrm{Bun}_G.$$

Recall the functor

$$(9.18) \quad \mathbf{p}^! : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}(\mathrm{Bun}_{N,\rho(\omega_X)}),$$

see Sect. 9.2.4.

9.6.2. The character χ has a global counterpart, which is a map

$$\chi^{\mathrm{glob}} : \mathrm{Bun}_{N,\rho(\omega_X)} \rightarrow \mathbb{G}_a.$$

Namely, it is the composition

$$\mathrm{Bun}_{N,\rho(\omega_X)} \rightarrow \mathrm{Bun}_{N/[N,N],\rho(\omega_X)} \simeq \prod_i \mathrm{Bun}_{(\mathbb{G}_a)_{\omega_X}} \rightarrow \prod_i H^1(X, \omega_X) \simeq \prod_i \mathbb{G}_a \xrightarrow{\chi_0} \mathbb{G}_a,$$

where:

- $(\mathbb{G}_a)_{\omega_X}$ is the twist of the constant group-scheme with fiber $\mathbb{G}_a$ using the $\mathbb{G}_m$ -action on $\mathbb{G}_a$ and the line bundle ω_X , viewed as a $\mathbb{G}_m$ -torsor³⁸;
- $\mathrm{Bun}_{(\mathbb{G}_a)_{\omega_X}} \rightarrow H^1(X, \omega_X)$ is the map that records the class of a torsor.

9.6.3. We let

$$\mathrm{coeff}_G^{\mathrm{Vac},\mathrm{glob}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{Vect}$$

denote the functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{\mathbf{p}^!} \mathrm{D}\text{-mod}(\mathrm{Bun}_{N,\rho(\omega_X)}) \xrightarrow{-\otimes (\chi^{\mathrm{glob}})^*(\mathrm{exp})} \mathrm{D}\text{-mod}(\mathrm{Bun}_{N,\rho(\omega_X)}) \xrightarrow{\mathrm{C}_{\mathrm{dR}}^*(\mathrm{Bun}_{N,\rho(\omega_X)}, -)} \mathrm{Vect}.$$

9.6.4. Let

$$\mathrm{Poinc}_{G,!}^{\mathrm{Vac},\mathrm{glob}} \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

be the object

$$\mathbf{p}_! \circ (\chi^{\mathrm{glob}})^*(\mathrm{exp}).$$

I.e., the functor

$$\mathrm{Vect} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G), \quad k \mapsto \mathrm{Poinc}_{G,!}^{\mathrm{Vac},\mathrm{glob}} \otimes k$$

is the left adjoint of $\mathrm{coeff}_G^{\mathrm{Vac},\mathrm{glob}}$.

Let

$$\mathrm{Poinc}_{G,*}^{\mathrm{Vac},\mathrm{glob}} \in \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G)$$

be the object

$$\mathbf{p}_* \circ (\chi^{\mathrm{glob}})^*(\mathrm{exp}).$$

9.6.5. Denote

$$\delta_{N_{\rho(\omega_X)}} := \dim(\mathrm{Bun}_{N,\rho(\omega_X)}).$$

We have

$$\mathbb{D}^{\mathrm{Verdier}}(\mathrm{Poinc}_{G,!}^{\mathrm{Vac},\mathrm{glob}}) = \mathrm{Poinc}_{G,*}^{\mathrm{Vac},\mathrm{glob}}[2\delta_{N_{\rho(\omega_X)}}],$$

where $\mathbb{D}^{\mathrm{Verdier}}$ is the usual Verdier dualization functor

$$(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G))^c \rightarrow (\mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G))^c.$$

In other words, the object $\mathrm{Poinc}_{G,*}^{\mathrm{Vac},\mathrm{glob}}[2\delta_{N_{\rho(\omega_X)}}]$, viewed as a functor

$$\mathrm{Vect} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G),$$

is the dual of $\mathrm{coeff}_G^{\mathrm{Vac},\mathrm{glob}}$.

³⁸I.e., it is the total space of ω_X as a line bundle.

9.6.6. We claim:

Lemma 9.6.7.

- (i) $\text{coeff}_G^{\text{Vac, glob}} \simeq \text{coeff}_G^{\text{Vac}}[2\delta_{N_{\rho(\omega_X)}}]$;
- (ii) $\text{Poinc}_{G,!}^{\text{Vac, glob}} \simeq \text{Poinc}_{G,!}^{\text{Vac}}[-2\delta_{N_{\rho(\omega_X)}}]$;
- (iii) $\text{Poinc}_{G,*}^{\text{Vac, glob}} \simeq \text{Poinc}_{G,*}^{\text{Vac}}$.

Proof. The three statements are logically equivalent. We will prove point (iii).

Pick $\underline{x} \in \text{Ran}$, and consider the map

$$N^\alpha / \mathfrak{L}^+(N)_{\rho(\omega_X), \underline{x}} \hookrightarrow \mathfrak{L}(N)_{\rho(\omega_X), \underline{x}} / \mathfrak{L}^+(N)_{\rho(\omega_X), \underline{x}} \xrightarrow{\text{act on the unit}} \text{Gr}_{G, \rho(\omega_X), \underline{x}} \rightarrow \text{Bun}_G$$

for $N^\alpha \supset \mathfrak{L}^+(N)_{\rho(\omega_X), \underline{x}}$ as in Sect. 9.3.3.

By definition, the object $\text{Poinc}_{G,*}^{\text{Vac, glob}}$ is the direct image along this map of $(\chi|_{N^\alpha / \mathfrak{L}^+(N)_{\rho(\omega_X), \underline{x}}})^*(\text{exp})$, where:

- By a slight abuse of notation, we regard χ as a map $\mathfrak{L}(N)_{\rho(\omega_X), \underline{x}} / \mathfrak{L}^+(N)_{\rho(\omega_X), \underline{x}} \rightarrow \mathbb{G}_a$;
- The index α is taken to be large enough so that (9.11) holds.

Note that, however, the map

$$\mathfrak{L}(N)_{\rho(\omega_X), \underline{x}} / \mathfrak{L}^+(N)_{\rho(\omega_X), \underline{x}} \xrightarrow{\text{act on the unit}} \text{Gr}_{G, \rho(\omega_X), \underline{x}} \xrightarrow{\pi} \text{Bun}_G$$

factors as

$$\mathfrak{L}(N)_{\rho(\omega_X), \underline{x}} / \mathfrak{L}^+(N)_{\rho(\omega_X), \underline{x}} \rightarrow \text{Bun}_{N, \rho(\omega_X)} \rightarrow \text{Bun}_G.$$

Hence, it suffices to show that if (9.11) holds, then

$$(9.19) \quad (\pi_{N, \rho(\omega_X)}|_{N^\alpha / \mathfrak{L}^+(N)_{\rho(\omega_X), \underline{x}}})^* \circ (\chi|_{N^\alpha / \mathfrak{L}^+(N)_{\rho(\omega_X), \underline{x}}})^*(\text{exp}) \simeq (\chi^{\text{glob}})^*(\text{exp}).$$

Note that the map

$$\chi : \mathfrak{L}(N)_{\rho(\omega_X), \underline{x}} / \mathfrak{L}^+(N)_{\rho(\omega_X), \underline{x}} \rightarrow \mathbb{G}_a$$

identifies with

$$\mathfrak{L}(N)_{\rho(\omega_X), \underline{x}} / \mathfrak{L}^+(N)_{\rho(\omega_X), \underline{x}} \xrightarrow{\pi_{N, \rho(\omega_X)}} \text{Bun}_{N, \rho(\omega_X)} \xrightarrow{\chi^{\text{glob}}} \mathbb{G}_a.$$

This implies (9.19) by the projection formula, since if (9.11) holds, the map

$$\pi_{N, \rho(\omega_X)}|_{N^\alpha / \mathfrak{L}^+(N)_{\rho(\omega_X), \underline{x}}} : N^\alpha / \mathfrak{L}^+(N)_{\rho(\omega_X), \underline{x}} \rightarrow \text{Bun}_{N, \rho(\omega_X)}$$

is smooth with homologically contractible fibers; in fact, fibers are isomorphic to

$$N^\alpha \cap \text{Sect}(X - \underline{x}, N_{\rho(\omega_X)}).$$

□

10. THE LOCALIZATION FUNCTOR

A fundamental insight of Beilinson-Drinfeld in [BD1] is that the localization functor

$$(10.1) \quad \text{Loc}_{G, \kappa} : \text{KL}(G)_{\kappa, \text{Ran}} \rightarrow \text{D-mod}_\kappa(\text{Bun}_G).$$

may be used as a key local-to-global tool in geometric Langlands theory.

There are multiple (equivalent) ways to set this up, and in this section we will describe one of them.³⁹

For the sake of completeness we will define $\text{Loc}_{G, \kappa}$ for any level κ . We will specialize to the critical value of κ starting from Sect. 14.

10.1. The de Rham twisting on Bun_G corresponding to a level. In this subsection we will show how a level κ gives rise to a de Rham twisting $\mathcal{T}_\kappa^{\text{glob}}$ on Bun_G .

³⁹The reader who is willing to take the existence of $\text{Loc}_{G, \kappa}$ and its basic properties of faith may choose to skip directly to Sect. 12.

10.1.1. Let $\mathfrak{L}(G)^\wedge$ denote the formal completion of $\mathfrak{L}(G)$ along $\mathfrak{L}^+(G)$, viewed as a factorization group ind-scheme.

Note that a level κ may be thought of as a central extension of $\mathfrak{L}(G)^\wedge$ equipped with a splitting over $\mathfrak{L}^+(G)$. Equivalently, we can think of it as a factorization line bundle $\mathcal{L}_\kappa^{\text{loc}}$ on the groupoid

$$\text{pt} / \mathfrak{L}^+(G) \xleftarrow{h^{\text{loc}, \wedge}} \text{Hecke}_G^{\text{loc}, \wedge} \xrightarrow{h^{\text{loc}, \wedge}} \text{pt} / \mathfrak{L}^+(G),$$

compatible with the groupoid structure, where:

- $\text{Hecke}_G^{\text{loc}}$ is the local Hecke stack, i.e., $\mathfrak{L}^+(G) \backslash \mathfrak{L}(G) / \mathfrak{L}^+(G)$, viewed as a groupoid on $\text{pt} / \mathfrak{L}^+(G)$;
- $\text{Hecke}_G^{\text{loc}, \wedge}$ is the formal completion of $\text{Hecke}_G^{\text{loc}}$ along the unit section $\text{pt} / \mathfrak{L}^+(G) \rightarrow \text{Hecke}_G^{\text{loc}}$.

10.1.2. Let

$$\text{Bun}_G \times \text{Ran} \xleftarrow{h^{\text{glob}}} \text{Hecke}_{G, \text{Ran}}^{\text{glob}} \xrightarrow{h^{\text{glob}}} \text{Bun}_G \times \text{Ran}$$

be the global Hecke groupoid, and let $\text{Hecke}_{G, \text{Ran}}^{\text{glob}, \wedge}$ denote its formal completion along the unit section

$$\text{Bun}_G \times \text{Ran} \rightarrow \text{Hecke}_{G, \text{Ran}}^{\text{glob}}.$$

Note that we have a map of groupoids

$$(10.2) \quad \begin{array}{ccccc} \text{Bun}_G \times \text{Ran} & \xleftarrow{h^{\text{glob}}} & \text{Hecke}_{G, \text{Ran}}^{\text{glob}} & \xrightarrow{h^{\text{glob}}} & \text{Bun}_G \times \text{Ran} \\ \text{ev}_{\text{Ran}} \downarrow & & \text{ev}_{\text{Ran}} \downarrow & & \downarrow \text{ev}_{\text{Ran}} \\ (\text{pt} / \mathfrak{L}^+(G))_{\text{Ran}} & \xleftarrow{h^{\text{loc}}} & \text{Hecke}_{G, \text{Ran}}^{\text{loc}} & \xrightarrow{h^{\text{loc}}} & (\text{pt} / \mathfrak{L}^+(G))_{\text{Ran}}, \end{array}$$

in which both squares are Cartesian, where we denote by ev_{Ran} the “global-to-local” map given by restriction to the parameterized multi-disc.

Taking the formal completion along the unit sections yields the diagram

$$(10.3) \quad \begin{array}{ccccc} \text{Bun}_G \times \text{Ran} & \xleftarrow{h^{\text{glob}, \wedge}} & \text{Hecke}_{G, \text{Ran}}^{\text{glob}, \wedge} & \xrightarrow{h^{\text{glob}, \wedge}} & \text{Bun}_G \times \text{Ran} \\ \text{ev}_{\text{Ran}} \downarrow & & \text{ev}_{\text{Ran}} \downarrow & & \downarrow \text{ev}_{\text{Ran}} \\ (\text{pt} / \mathfrak{L}^+(G))_{\text{Ran}} & \xleftarrow{h^{\text{loc}, \wedge}} & \text{Hecke}_{G, \text{Ran}}^{\text{loc}, \wedge} & \xrightarrow{h^{\text{loc}, \wedge}} & (\text{pt} / \mathfrak{L}^+(G))_{\text{Ran}}. \end{array}$$

10.1.3. The pullback of the line bundle $\mathcal{L}_{\kappa, \text{Ran}}^{\text{loc}}$ on $\text{Hecke}_{G, \text{Ran}}^{\text{loc}, \wedge}$ along ev_{Ran} gives rise to a line bundle, to be denoted $\mathcal{L}_{\kappa, \text{Ran}}^{\text{glob}}$, on $\text{Hecke}_{G, \text{Ran}}^{\text{glob}, \wedge}$ that is multiplicative with respect to the groupoid structure.

Consider the prestack quotient

$$(\text{Bun}_G \times \text{Ran}) / \text{Hecke}_{G, \text{Ran}}^{\text{loc}, \wedge}.$$

The datum of $\mathcal{L}_{\kappa, \text{Ran}}^{\text{glob}}$ is equivalent to that of a $\mathcal{O}^\times$ -gerbe on $(\text{Bun}_G \times \text{Ran}) / \text{Hecke}_{G, \text{Ran}}^{\text{loc}, \wedge}$, to be denoted $\mathcal{G}_{\kappa, \text{Ran}}^{\text{Hecke}}$, equipped with a trivialization of its pullback to $\text{Bun}_G \times \text{Ran}$.

10.1.4. Note now that we have a tautological map

$$(\text{Bun}_G \times \text{Ran}) / \text{Hecke}_{G, \text{Ran}}^{\text{loc}, \wedge} \rightarrow (\text{Bun}_G)_{\text{dR}} \times \text{Ran}.$$

Consider the composite map

$$(10.4) \quad (\text{Bun}_G \times \text{Ran}) / \text{Hecke}_{G, \text{Ran}}^{\text{loc}, \wedge} \rightarrow (\text{Bun}_G)_{\text{dR}}$$

We have the following fundamental assertion (see [Ro2, Theorem 4.5.3]):

Theorem 10.1.5. *The functor of pullback along (10.4) on $\text{IndCoh}(-)$ is fully faithful.*

Corollary 10.1.6. *The functor of pullback along (10.4) is fully faithful on⁴⁰ $\text{Perf}(-)$.*

⁴⁰For a prestack $\mathcal{Y}$, we denote by $\text{Perf}(\mathcal{Y})$ the category of dualizable objects in $\text{QCoh}(\mathcal{Y})$, i.e., the objects whose pullback to any affine scheme is perfect.

10.1.7. The construction of the twisting $\mathcal{T}_\kappa$ on Bun_G corresponding to κ is provided by the following assertion:

Corollary 10.1.8. *There exists a uniquely defined de Rham twisting $\mathcal{T}_\kappa^{\text{glob}}$ on Bun_G , such that:*

- *The pullback of the underlying $\mathcal{O}^\times$ -gerbe $\mathcal{G}_\kappa^{\text{glob}}$ on $(\text{Bun}_G)_{\text{dR}}$ along (10.4) identifies with $\mathcal{G}_{\kappa, \text{Ran}}^{\text{Hecke}}$;*
- *The trivialization of $\mathcal{G}_\kappa^{\text{glob}}|_{\text{Bun}_G}$ is compatible with the trivialization of $\mathcal{G}_{\kappa, \text{Ran}}^{\text{Hecke}}|_{\text{Bun}_G \times \text{Ran}}$.*

Proof. According to [GaRo2, Sect. 6.3], using the exponential isomorphism

$$(\mathbb{G}_a)^\wedge \xrightarrow{\exp} (\mathbb{G}_m)^\wedge,$$

we can think of a twisting on a prestack $\mathcal{Y}$ as a point in $\text{Maps}(\mathcal{O}_{\mathcal{Y}_{\text{dR}}}, \mathcal{O}_{\mathcal{Y}_{\text{dR}}}[2])$ equipped with a trivialization of its pullback to $\mathcal{Y}$.

By the same logic, we can think of $\mathcal{G}_{\kappa, \text{Ran}}^{\text{Hecke}}$ as a point of

$$\text{Maps}(\mathcal{O}_{(\text{Bun}_G \times \text{Ran})/\text{Hecke}_{G, \text{Ran}}^{\text{loc}, \wedge}}, \mathcal{O}_{(\text{Bun}_G \times \text{Ran})/\text{Hecke}_{G, \text{Ran}}^{\text{loc}, \wedge}}[2])$$

equipped with a trivialization of its pullback to $\text{Bun}_G \times \text{Ran}$.

The assertion of the corollary follows now from Corollary 10.1.6, combined with the fact that the functor

$$\text{Vect} \rightarrow \text{QCoh}(\text{Ran}), \quad k \mapsto \mathcal{O}_{\text{Ran}}$$

is fully faithful. □

10.1.9. In what follows we will denote the category of $\mathcal{T}_\kappa^{\text{glob}}$ -twisted D-modules on Bun_G by

$$\text{D-mod}_\kappa(\text{Bun}_G).$$

10.1.10. Take $\kappa = 2 \cdot \text{crit}$. We claim:

Proposition 10.1.11. *The resulting twisting $\mathcal{T}_{2 \cdot \text{crit}}$ on Bun_G identifies canonically with $\text{dlog}(\det_{\text{Bun}_G})$.*

Remark 10.1.12. This proposition implies that our notations for $\text{D-mod}_{\text{crit}}(\text{Bun}_G)$ (see Sect. 9.1.4) are consistent.

Proof. Unwinding the construction, we need to show that the multiplicative line bundle $\mathcal{L}_{2 \cdot \text{crit}, \text{Ran}}^{\text{glob}}$ on $\text{Hecke}_{G, \text{Ran}}^{\text{loc}, \wedge}$ identifies with the restriction of the multiplicative line bundle on $\text{Hecke}_{G, \text{Ran}}^{\text{loc}}$ given by

$$\vec{h}^*(\det_{\text{Bun}_G}) \otimes \overleftarrow{h}^*(\det_{\text{Bun}_G})^{\otimes -1}.$$

Recall that the multiplicative line bundle $\mathcal{L}_{2 \cdot \text{crit}}^{\text{loc}}$ on $\text{Hecke}_G^{\text{loc}, \wedge}$ is itself obtained as the restriction of the *inverse* of the Tate line bundle $\mathcal{L}_{\text{Tate}}^{\text{loc}}$ on $\text{Hecke}_G^{\text{loc}}$, constructed as follows:

The line bundle $\mathcal{L}_{\text{Tate}}^{\text{loc}}$ associates to a pair of G -bundles $\mathcal{P}_G^1$ and $\mathcal{P}_G^2$ on $\mathcal{D}_{\underline{x}}$ equipped with an identification

$$\mathcal{P}_G^1|_{\mathcal{D}_{\underline{x}}} \simeq \mathcal{P}_G^2|_{\mathcal{D}_{\underline{x}}}$$

the relative determinant of the two lattices

$$\Gamma(\mathcal{D}_{\underline{x}}, \mathfrak{g}_{\mathcal{P}_G^1}) \subset \Gamma(\mathcal{D}_{\underline{x}}^\times, \mathfrak{g}_{\mathcal{P}_G^1}) = \Gamma(\mathcal{D}_{\underline{x}}^\times, \mathfrak{g}_{\mathcal{P}_G^2}) \supset \Gamma(\mathcal{D}_{\underline{x}}, \mathfrak{g}_{\mathcal{P}_G^2}),$$

i.e.,

$$(10.5) \quad \det(\Gamma(\mathcal{D}_{\underline{x}}, \mathfrak{g}_{\mathcal{P}_G^1})/\mathbf{L}) \otimes \det(\Gamma(\mathcal{D}_{\underline{x}}, \mathfrak{g}_{\mathcal{P}_G^2})/\mathbf{L})^{-1}$$

for some/any lattice $\mathbf{L}$ contained in both.

Given a pair of G -bundles $\mathcal{P}_G^1$ and $\mathcal{P}_G^2$ on X equipped with an identification

$$\mathcal{P}_G^1|_{X-\underline{x}} \simeq \mathcal{P}_G^2|_{X-\underline{x}},$$

the fiber of $\vec{h}^*(\det_{\text{Bun}_G}) \otimes \overleftarrow{h}^*(\det_{\text{Bun}_G})^{\otimes -1}$ at the corresponding point of $\text{Hecke}_{G, \text{Ran}}^{\text{loc}}$ is given by

$$(10.6) \quad \det(\Gamma(X, \mathfrak{g}_{\mathcal{P}_G^2})) \otimes \det(\Gamma(X, \mathfrak{g}_{\mathcal{P}_G^1}))^{\otimes -1}.$$

We claim that the lines (10.5) and (10.6) are indeed canonically inverse to one another.

Indeed, we can rewrite (10.5) as

$$\det \left(\text{Fib} \left(\Gamma(\mathcal{D}_{\underline{x}}, \mathfrak{g}_{\mathcal{P}_G^1}^{\text{out}} \rightarrow \mathfrak{g}^{\text{mer}}) \right) \right) \otimes \det \left(\text{Fib} \left(\Gamma(\mathcal{D}_{\underline{x}}, \mathfrak{g}_{\mathcal{P}_G^2}^{\text{out}} \rightarrow \mathfrak{g}^{\text{mer}}) \right) \right)^{\otimes -1},$$

where:

- $\Gamma(\mathcal{D}_{\underline{x}}, \mathfrak{g}_{\mathcal{P}_G^1}^{\text{out}}) =: \mathfrak{g}^{\text{mer}} := \Gamma(\mathcal{D}_{\underline{x}}, \mathfrak{g}_{\mathcal{P}_G^2}^{\text{out}})$;
- $\Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G^1}^{\text{out}}) =: \mathfrak{g}^{\text{out}} := \Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G^2}^{\text{out}})$.

However,

$$\text{Fib} \left(\Gamma(\mathcal{D}_{\underline{x}}, \mathfrak{g}_{\mathcal{P}_G^i}^{\text{out}} \rightarrow \mathfrak{g}^{\text{mer}}) \right) \simeq \Gamma(X, \mathfrak{g}_{\mathcal{P}_G^i}^{\text{out}}).$$

□

10.2. The functor Γ_G . Our approach to the construction of the localization functor is by defining it as the left adjoint of the functor Γ_G of *global sections* (not quite literally, though, see Sect. 10.3).

In this subsection we introduce the functor Γ_G .

To simplify the notation, for most of this subsection we fix a point $\underline{x} \in \text{Ran}$.

10.2.1. For an integer n , consider the stack $\text{Bun}_G^{\text{level}_{n \cdot \underline{x}}}$ of G -bundles with structure of level n at $\underline{x}$.

Consider the corresponding category

$$\text{D-mod}_{\kappa, \text{co}}(\text{Bun}_G^{\text{level}_{n \cdot \underline{x}}}),$$

see [Ra5, Sect. 1.4.25].

We endow it with the forgetful functor

$$(10.7) \quad \text{oblv}_{\kappa}^{l, \text{ren}} : \text{D-mod}_{\kappa, \text{co}}(\text{Bun}_G^{\text{level}_{n \cdot \underline{x}}}) \rightarrow \text{QCoh}_{\text{co}}(\text{Bun}_G^{\text{level}_{n \cdot \underline{x}}}),$$

which is the composition with the usual *left* forgetful functor

$$\text{oblv}_{\kappa}^l : \text{D-mod}_{\kappa, \text{co}}(\text{Bun}_G^{\text{level}_{n \cdot \underline{x}}}) \rightarrow \text{QCoh}_{\text{co}}(\text{Bun}_G^{\text{level}_{n \cdot \underline{x}}})$$

(see Corollary A.1.16), followed by the cohomological shift $[2n \dim(\mathfrak{g})]$.

Note that for $n_1 \leq n_2$, we have a commutative diagram

$$\begin{array}{ccc} \text{D-mod}_{\kappa, \text{co}}(\text{Bun}_G^{\text{level}_{n_2 \cdot \underline{x}}}) & \xrightarrow{\text{oblv}_{\kappa}^{l, \text{ren}}} & \text{QCoh}_{\text{co}}(\text{Bun}_G^{\text{level}_{n_2 \cdot \underline{x}}}) \\ \uparrow & & \uparrow \\ \text{D-mod}_{\kappa, \text{co}}(\text{Bun}_G^{\text{level}_{n_1 \cdot \underline{x}}}) & \xrightarrow{\text{oblv}_{\kappa}^{l, \text{ren}}} & \text{QCoh}_{\text{co}}(\text{Bun}_G^{\text{level}_{n_1 \cdot \underline{x}}}), \end{array}$$

where:

- The left vertical arrow is the functor of $*$ -pullback on twisted D-modules;
- The right vertical arrow is the functor of $*$ -pullback on QCoh_{co} .

Note that by definition, for $n = 0$, we have $\text{oblv}_{\kappa}^{l, \text{ren}} = \text{oblv}_{\kappa}^l$.

10.2.2. Consider the stack (in fact, a scheme)

$$\text{Bun}_G^{\text{level}_{\underline{x}}} = \lim_n \text{Bun}_G^{\text{level}_{n \cdot \underline{x}}}$$

of G -bundles with full level structure at $\underline{x}$.

Define

$$\text{D-mod}_{\kappa, \text{co}}(\text{Bun}_G^{\text{level}_{\underline{x}}}) := \text{colim}_n \text{D-mod}_{\kappa, \text{co}}(\text{Bun}_G^{\text{level}_{n \cdot \underline{x}}}),$$

where the colimit is formed using the $*$ -pullback functors

$$\text{D-mod}_{\kappa}(\text{Bun}_G^{\text{level}_{n_1 \cdot \underline{x}}}) \rightarrow \text{D-mod}_{\kappa}(\text{Bun}_G^{\text{level}_{n_2 \cdot \underline{x}}}), \quad n_2 \geq n_1.$$

The functors (10.7) combine to a functor

$$\mathbf{oblv}_\kappa^{l,\text{ren}} : \text{D-mod}_{\kappa,\text{co}}(\text{Bun}_G^{\text{level}_\underline{x}}) \rightarrow \text{QCoh}_{\text{co}}(\text{Bun}_G^{\text{level}_\underline{x}}).$$

Consider the composite functor, to be denoted $\Gamma_\kappa^{\text{ren}}(\text{Bun}_G^{\text{level}_\underline{x}}, -)$,

$$(10.8) \quad \text{D-mod}_{\kappa,\text{co}}(\text{Bun}_G^{\text{level}_\underline{x}}) \xrightarrow{\mathbf{oblv}_\kappa^{l,\text{ren}}} \text{QCoh}_{\text{co}}(\text{Bun}_G^{\text{level}_\underline{x}})^{\Gamma(\text{Bun}_G^{\text{level}_\underline{x}}, -)} \text{Vect};$$

10.2.3. According to [Ra5, Sect. 1.4.25], the category $\text{D-mod}_{\kappa,\text{co}}(\text{Bun}_G^{\text{level}_\underline{x}})$ carries a *strong* action of $\mathfrak{L}(G)_\underline{x}$ at level κ . Furthermore, the functor $\Gamma_\kappa^{\text{ren}}(\text{Bun}_G^{\text{level}_\underline{x}}, -)$ of (10.8) is *weakly* $\mathfrak{L}(G)_\underline{x}$ -equivariant.

Hence, by the *universal property of* $\widehat{\mathfrak{g}}\text{-mod}_{\kappa,\underline{x}}$ (see [Ra5, Sect. 1.4.25]), the functor $\Gamma_\kappa^{\text{ren}}(\text{Bun}_G^{\text{level}_\underline{x}}, -)$ upgrades to a functor

$$\Gamma_\kappa^{\text{ren}}(\text{Bun}_G^{\text{level}_\underline{x}}, -)^{\text{enh}} : \text{D-mod}_{\kappa,\text{co}}(\text{Bun}_G^{\text{level}_\underline{x}}) \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa,\underline{x}},$$

strongly compatible with the $\mathfrak{L}(G)_\underline{x}$ -actions.

10.2.4. In particular, the functor $\Gamma_\kappa^{\text{ren}}(\text{Bun}_G^{\text{level}_\underline{x}}, -)^{\text{enh}}$ gives rise to a functor, to be denoted $\Gamma_{G,\kappa,\underline{x}}$:

$$\text{D-mod}_{\kappa,\text{co}}(\text{Bun}_G) \simeq \left(\text{D-mod}_{\kappa,\text{co}}(\text{Bun}_G^{\text{level}_\underline{x}}) \right)^{\mathfrak{L}^+(G)_\underline{x}} \rightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\kappa,\underline{x}})^{\mathfrak{L}^+(G)_\underline{x}} = \text{KL}(G)_{\kappa,\underline{x}}.$$

10.2.5. By a similar token, letting $\underline{x}$ vary over the Ran space, we obtain a functor

$$\Gamma_{G,\kappa} : \text{D-mod}_{\kappa,\text{co}}(\text{Bun}_G) \rightarrow \text{KL}(G)_{\kappa,\text{Ran}}.$$

Furthermore, we can consider a $\text{D-mod}(\text{Ran})$ -linear functor

$$\Gamma_{G,\kappa,\text{Ran}} : \text{D-mod}_{\kappa,\text{co}}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}) \rightarrow \text{KL}(G)_{\kappa,\text{Ran}},$$

so that $\Gamma_{G,\kappa}$ is the composition

$$\text{D-mod}_{\kappa,\text{co}}(\text{Bun}_G) \xrightarrow{\text{Id} \otimes \omega_{\text{Ran}}} \text{D-mod}_{\kappa,\text{co}}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}) \xrightarrow{\Gamma_{G,\kappa,\text{Ran}}} \text{KL}(G)_{\kappa,\text{Ran}}.$$

10.2.6. Let us specialize for a moment to the case when $\kappa = \text{crit}$. In this case, both sides of

$$\Gamma_{G,\text{crit},\underline{x}} : \text{D-mod}_{\text{crit},\text{co}}(\text{Bun}_G) \rightarrow \text{KL}(G)_{\text{crit},\underline{x}}$$

are acted on by $\text{Sph}_{G,\underline{x}}$, and it follows from the construction that the functor $\Gamma_{G,\text{crit},\underline{x}}$ is compatible with these actions.

Similarly, the functor $\Gamma_{G,\kappa,\text{Ran}}$ is compatible with the action of $\text{Sph}_{G,\text{Ran}}$.

10.2.7. Note that by construction, we have a commutative diagram

$$(10.9) \quad \begin{array}{ccc} \text{D-mod}_{\kappa,\text{co}}(\text{Bun}_G) & \xrightarrow{\Gamma_{G,\kappa,\underline{x}}} & \text{KL}(G)_{\kappa,\underline{x}} \\ \mathbf{oblv}_\kappa^l \downarrow & & \downarrow \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa} \\ \text{QCoh}_{\text{co}}(\text{Bun}_G) & \xrightarrow[\Gamma_{G,\kappa,\underline{x}}^{\text{QCoh}}]{} & \text{Rep}(\mathfrak{L}^+(G)_\underline{x}), \end{array}$$

where $\Gamma_{G,\kappa,\underline{x}}^{\text{QCoh}}$ is the functor of pushforward along

$$\text{ev}_\underline{x} : \text{Bun}_G \rightarrow \text{pt}/\mathfrak{L}^+(G)_\underline{x}.$$

Remark 10.2.8. Note that $\text{Rep}(\mathfrak{L}^+(G)_\underline{x})$ is the renormalized version of $\text{QCoh}(\text{pt}/\mathfrak{L}^+(G)_\underline{x})$ (see Sect. B.14.1); however, this difference is immaterial for the definition of the functor $\Gamma_{G,\kappa,\underline{x}}^{\text{QCoh}}$.

We have:

$$\text{QCoh}_{\text{co}}(\text{Bun}_G) \simeq \text{colim}_U \text{QCoh}(U),$$

where:

- The index U runs over the filtered posets of quasi-compact open substacks of Bun_G ;
- For $U_1 \xrightarrow{j_{1,2}} U_2$, the transition functor $\text{QCoh}(U_1) \rightarrow \text{QCoh}(U_2)$ is given by $(j_{1,2})_*$.

A functor out of $\mathrm{QCoh}_{\mathrm{co}}(\mathrm{Bun}_G)$ amounts to a compatible collection of functors out of $\mathrm{QCoh}(U)$. Thus, in order to define

$$\Gamma_{G,\kappa,\underline{x}}^{\mathrm{QCoh}} : \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Bun}_G) \rightarrow \mathrm{Rep}(\mathfrak{L}^+(G)_{\underline{x}}),$$

we need to define the functors

$$\Gamma_{G,\kappa,\underline{x},U}^{\mathrm{QCoh}} : \mathrm{QCoh}(U) \rightarrow \mathrm{Rep}(\mathfrak{L}^+(G)_{\underline{x}}).$$

The sought-for functor $\Gamma_{G,\kappa,\underline{x},U}^{\mathrm{QCoh}}$ are defined as the ind-extension of the functor

$$\begin{aligned} \mathrm{QCoh}(U)^c \hookrightarrow \mathrm{QCoh}(U)^{>-\infty} &\xrightarrow{(\mathrm{ev}_{\underline{x},U})^*} \mathrm{QCoh}(\mathrm{pt} / \mathfrak{L}^+(G)_{\underline{x}})^{>-\infty} \simeq \\ &\simeq \mathrm{Rep}(\mathfrak{L}^+(G)_{\underline{x}})^{>-\infty} \hookrightarrow \mathrm{Rep}(\mathfrak{L}^+(G)_{\underline{x}}), \end{aligned}$$

where $\mathrm{ev}_{\underline{x},U}$ denotes the restriction of $\mathrm{ev}_{\underline{x}}$ to U .

10.2.9. Similarly, we have a commutative diagram

$$(10.10) \quad \begin{array}{ccc} \mathrm{D-mod}_{\kappa,\mathrm{co}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathrm{Ran}) & \xrightarrow{\Gamma_{G,\kappa,\mathrm{Ran}}} & \mathrm{KL}(G)_{\kappa,\mathrm{Ran}} \\ \mathrm{oblv}_{\kappa}^l \downarrow & & \downarrow \mathrm{oblv}_{\mathfrak{L}^+(G)}^{(\mathfrak{g}, \mathfrak{L}^+(G))_{\kappa}} \\ \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathrm{Ran}) & \xrightarrow{\Gamma_{G,\kappa,\mathrm{Ran}}^{\mathrm{QCoh}}} & \mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}}. \end{array}$$

10.2.10. Let now $\underline{x}$ and $\underline{x}'$ be two points of Ran with $\underline{x} \subseteq \underline{x}'$. Consider the functor

$$\mathrm{ins.vac}_{\underline{x} \subseteq \underline{x}'} : \mathrm{KL}(G)_{\kappa,\underline{x}} \rightarrow \mathrm{KL}(G)_{\kappa,\underline{x}'}$$

obtained by inserting the vacuum objects at the points $\underline{x}' - \underline{x}$.

Consider its right adjoint, $(\mathrm{ins.vac}_{\underline{x} \subseteq \underline{x}'})^R$. Explicitly, the functor $(\mathrm{ins.vac}_{\underline{x} \subseteq \underline{x}'})^R$ is given by

$$\begin{aligned} \mathrm{KL}(G)_{\kappa,\underline{x}'} &\simeq \mathrm{KL}(G)_{\kappa,\underline{x}} \otimes \mathrm{KL}(G)_{\kappa,\underline{x}' - \underline{x}} \xrightarrow{\mathrm{Id} \otimes \mathrm{oblv}_{\mathfrak{L}^+(G)}^{(\mathfrak{g}, \mathfrak{L}^+(G))_{\kappa}}} \\ &\rightarrow \mathrm{KL}(G)_{\kappa,\underline{x}} \otimes \mathrm{Rep}(\mathfrak{L}^+(G)_{\underline{x}' - \underline{x}}) \xrightarrow{\mathrm{Id} \otimes \mathrm{inv}_{\mathfrak{L}^+(G)_{\underline{x}' - \underline{x}}}} \mathrm{KL}(G)_{\kappa,\underline{x}}. \end{aligned}$$

It follows from the commutation of (10.9) that we have a canonical isomorphism

$$(10.11) \quad \Gamma_{G,\kappa,\underline{x}} \simeq (\mathrm{ins.vac}_{\underline{x} \subseteq \underline{x}'})^R \circ \Gamma_{G,\kappa,\underline{x}'}.$$

This isomorphism expresses the *unital* structure on the assignment

$$\underline{x} \rightsquigarrow \Gamma_{G,\mathrm{crit},\underline{x}},$$

to be discussed in Sect. 11.

10.2.11. In particular, for any $\underline{x}$, the functor

$$\mathrm{D-mod}_{\kappa,\mathrm{co}}(\mathrm{Bun}_G) \xrightarrow{\Gamma_{G,\kappa,\underline{x}}} \mathrm{KL}(G)_{\kappa,\underline{x}} \xrightarrow{\mathrm{inv}_{\mathfrak{L}^+(G)_{\underline{x}}}} \mathrm{Vect}$$

identifies with $\Gamma(\mathrm{Bun}_G, \mathrm{oblv}_{\kappa}^l(-))$.

10.3. Localization functor as a left adjoint. As was mentioned previously, we construct the localization functor $\mathrm{Loc}_{G,\kappa}$ to be *essentially* the left adjoint of Γ_G . However, there is a caveat: this adjunction takes place over quasi-compact open substacks $U \subset \mathrm{Bun}_G$, and we obtain the corresponding functors $\mathrm{Loc}_{G,\kappa,U}$. We then obtain the sought-for functor $\mathrm{Loc}_{G,\kappa}$ by passing to the limit.

10.3.1. Let

$$(10.12) \quad U \xrightarrow{j_U} \mathrm{Bun}_G$$

be a quasi-compact open substack. Consider the corresponding functor

$$j_{*,\mathrm{co}} : \mathrm{D}\text{-}\mathrm{mod}_\kappa(U) \rightarrow \mathrm{D}\text{-}\mathrm{mod}_{\kappa,\mathrm{co}}(\mathrm{Bun}_G).$$

Denote

$$\Gamma_{G,\kappa,\underline{x},U} := \Gamma_{G,\kappa,\underline{x}} \circ j_{*,\mathrm{co}}, \quad \mathrm{D}\text{-}\mathrm{mod}_\kappa(U) \rightarrow \mathrm{KL}(G)_{\kappa,\underline{x}}.$$

10.3.2. We claim:

Lemma 10.3.3. *The functor $\Gamma_{G,\kappa,\underline{x},U}$ admits a left adjoint.*

Proof. Since the essential image of

$$\mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} : \mathrm{Rep}(\mathfrak{L}^+(G)_{\underline{x}}) \rightarrow \mathrm{KL}(G)_{\kappa,\underline{x}}$$

generates the target category, it suffices to show that the composite functor

$$\mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} \circ \Gamma_{G,\kappa,\underline{x},U}, \quad \mathrm{D}\text{-}\mathrm{mod}_\kappa(U) \rightarrow \mathrm{Rep}(\mathfrak{L}^+(G)_{\underline{x}})$$

admits a left adjoint.

The above functor identifies with

$$(\mathrm{ev}_{\underline{x}} \circ j)_* \circ \mathbf{oblv}_\kappa^l.$$

In this composition, both arrows admit left adjoints: the left adjoint of $\mathbf{oblv}_\kappa^l$ is $\mathbf{ind}_\kappa^l$, and the left adjoint of $(\mathrm{ev}_{\underline{x}} \circ j)_*$ is $(\mathrm{ev}_{\underline{x}} \circ j)^*$. □

10.3.4. Let us denote the left adjoint in Lemma 10.3.3 by $\mathrm{Loc}_{G,\kappa,\underline{x},U}$.

For an inclusion between quasi-compact open substacks

$$U_1 \xrightarrow{j_{1,2}} U_2,$$

we have

$$(j_2)_{*,\mathrm{co}} \circ (j_{1,2})_* \simeq (j_1)_{*,\mathrm{co}}.$$

Hence, we obtain a canonical identification

$$\mathrm{Loc}_{G,\kappa,\underline{x},U_1} \simeq j_{1,2}^* \circ \mathrm{Loc}_{G,\kappa,\underline{x},U_2}.$$

Therefore, the system of functors

$$U \rightsquigarrow \{\mathrm{Loc}_{G,\kappa,\underline{x},U}\}$$

gives rise to a functor

$$(10.13) \quad \mathrm{Loc}_{G,\kappa,\underline{x}} : \mathrm{KL}(G)_{\kappa,\underline{x}} \rightarrow \mathrm{D}\text{-}\mathrm{mod}_\kappa(\mathrm{Bun}_G),$$

so that for every (10.12), we have

$$j^* \circ \mathrm{Loc}_{G,\kappa,\underline{x}} \simeq \mathrm{Loc}_{G,\kappa,\underline{x},U}.$$

The functor (10.13) is the sought-for localization functor.

10.3.5. The entire preceding discussion generalizes to the case when $\underline{x}$ is allowed to move in families over Ran . In particular, we obtain a functor

$$\text{Loc}_{G,\kappa,\text{Ran}} : \text{KL}(G)_{\kappa,\text{Ran}} \rightarrow \text{D-mod}_{\kappa}(\text{Bun}_G \times \text{Ran}).$$

Let $\text{Loc}_{G,\kappa}$ denote the composition

$$\text{KL}(G)_{\kappa,\text{Ran}} \xrightarrow{\text{Loc}_{G,\kappa,\text{Ran}}} \text{D-mod}_{\kappa}(\text{Bun}_G \times \text{Ran}) \rightarrow \text{D-mod}_{\kappa}(\text{Bun}_G),$$

where the second arrow is the functor of $!$ -pushforward.

Remark 10.3.6. The above construction of the localization functor is essentially equivalent to the one from [CF, Sect. 4.1].

10.3.7. Properties of the functor $\Gamma_{G,\kappa',\text{Ran}}$ induce corresponding properties of the functor $\text{Loc}_{G,\kappa,\text{Ran}}$. We will now list some of them.

10.3.8. By adjunction, for every quasi-compact open as in (10.12), from diagram (10.9) we obtain a commutative diagram:

$$(10.14) \quad \begin{array}{ccc} \text{QCoh}(U) & \xrightarrow{\text{ind}_{\kappa}^!} & \text{D-mod}_{\kappa}(U) \\ j^* \circ \text{Loc}_{G,\underline{x}}^{\text{QCoh}} \uparrow & & \uparrow \text{Loc}_{G,\kappa,\underline{x},U} \\ \text{Rep}(\mathfrak{L}^+(G)_{\underline{x}}) & \xrightarrow{\text{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa}} & \text{KL}(G)_{\kappa,\underline{x}}, \end{array}$$

where:

$$\text{Loc}_{G,\underline{x}}^{\text{QCoh}} : \text{Rep}(\mathfrak{L}^+(G)_{\underline{x}}) \rightarrow \text{QCoh}(\text{Bun}_G)$$

is the functor of pullback along $\text{ev}_{\underline{x}} : \text{Bun}_G \rightarrow \text{pt}/\mathfrak{L}^+(G)_{\underline{x}}$.

Remark 10.3.9. As in Remark 10.2.8, the difference between $\text{Rep}(\mathfrak{L}^+(G)_{\underline{x}})$ and $\text{QCoh}(\text{pt}/\mathfrak{L}^+(G)_{\underline{x}})$ does not play a role in the definition of the functor $\text{Loc}_{G,\underline{x}}^{\text{QCoh}}$.

Namely, $\text{Loc}_{G,\underline{x}}^{\text{QCoh}}$ is defined as the ind-extension of

$$\text{Rep}(\mathfrak{L}^+(G)_{\underline{x}})^c \hookrightarrow \text{QCoh}(\text{pt}/\mathfrak{L}^+(G)_{\underline{x}}) \xrightarrow{(\text{ev}_{\underline{x}})^*} \text{QCoh}(\text{Bun}_G).$$

Note also that for a quasi-compact U , the functor

$$\text{Loc}_{G,\underline{x},U}^{\text{QCoh}} := j^* \circ \text{Loc}_{G,\underline{x}}^{\text{QCoh}}$$

is the left adjoint of the functor $\Gamma_{G,\kappa,\underline{x},U}^{\text{QCoh}}$.

10.3.10. Passing to the limit over (10.12), we obtain a commutative diagram

$$(10.15) \quad \begin{array}{ccc} \text{QCoh}(\text{Bun}_G) & \xrightarrow{\text{ind}_{\kappa}^!} & \text{D-mod}_{\kappa}(\text{Bun}_G) \\ \text{Loc}_{G,\underline{x}}^{\text{QCoh}} \uparrow & & \uparrow \text{Loc}_{G,\kappa,\underline{x}} \\ \text{Rep}(\mathfrak{L}^+(G)_{\underline{x}}) & \xrightarrow{\text{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa}} & \text{KL}(G)_{\kappa,\underline{x}}. \end{array}$$

Similarly, we have

$$(10.16) \quad \begin{array}{ccc} \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}) & \xrightarrow{\text{ind}_{\kappa}^!} & \text{D-mod}_{\kappa}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}) \\ \text{Loc}_{G,\text{Ran}}^{\text{QCoh}} \uparrow & & \uparrow \text{Loc}_{G,\kappa,\text{Ran}} \\ \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}} & \xrightarrow{\text{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa}} & \text{KL}(G)_{\kappa,\text{Ran}} \end{array}$$

and

$$(10.17) \quad \begin{array}{ccc} \mathrm{QCoh}(\mathrm{Bun}_G) & \xrightarrow{\mathrm{ind}_\kappa^l} & \mathrm{D-mod}_\kappa(\mathrm{Bun}_G) \\ \mathrm{Loc}_G^{\mathrm{QCoh}} \uparrow & & \uparrow \mathrm{Loc}_{G,\kappa} \\ \mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}} & \xrightarrow{\mathrm{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa}} & \mathrm{KL}(G)_{\kappa, \mathrm{Ran}}, \end{array}$$

where $\mathrm{Loc}_G^{\mathrm{QCoh}}$ is the functor of pull-push along

$$\mathrm{Bun}_G \leftarrow \mathrm{Bun}_G \times \mathrm{Ran} \rightarrow (\mathrm{pt} / \mathfrak{L}^+(G))_{\mathrm{Ran}}.$$

10.3.11. Let now $\underline{x}$ and $\underline{x}'$ be two points of Ran with $\underline{x} \subseteq \underline{x}'$. Recall the functor

$$(10.18) \quad \mathrm{ins. vac}_{\underline{x} \subseteq \underline{x}'} : \mathrm{KL}(G)_{\kappa, \underline{x}} \rightarrow \mathrm{KL}(G)_{\kappa, \underline{x}'}.$$

For every (10.12), from (10.11) we obtain a canonical isomorphism

$$(10.19) \quad \mathrm{Loc}_{G,\kappa,\underline{x}',U} \circ \mathrm{ins. vac}_{\underline{x} \subseteq \underline{x}'} \simeq \mathrm{Loc}_{G,\kappa,\underline{x},U}.$$

Passing to the limit over (10.12), we obtain a canonical isomorphism

$$(10.20) \quad \mathrm{Loc}_{G,\kappa,\underline{x}'} \circ \mathrm{ins. vac}_{\underline{x} \subseteq \underline{x}'} \simeq \mathrm{Loc}_{G,\kappa,\underline{x}}.$$

In Sect. 11.3.7 we will formulate a version of (10.20) when the points $\underline{x}$ and $\underline{x}'$ move in families over the Ran space.

10.3.12. It follows from (10.15) that for any $\underline{x}$, we have

$$\mathrm{Loc}_{G,\kappa,\underline{x}}(\mathrm{Vac}(G)_{\kappa,\underline{x}}) \simeq \mathrm{ind}_\kappa^l(\mathcal{O}_{\mathrm{Bun}_G}).$$

Equivalently,

$$(10.21) \quad \mathrm{Loc}_{G,\kappa,\mathrm{Ran}}(\mathrm{Vac}(G)_{\kappa,\mathrm{Ran}}) \simeq \mathrm{ind}_\kappa^l(\mathcal{O}_{\mathrm{Bun}_G}) \boxtimes \omega_{\mathrm{Ran}}.$$

Note that

$$\mathrm{ind}_\kappa^l(\mathcal{O}_{\mathrm{Bun}_G}) \simeq \mathrm{D}_{\mathrm{Bun}_G,\kappa},$$

where

$$\mathrm{D}_{\mathrm{Bun}_G,\kappa} \in \mathrm{D-mod}_\kappa(\mathrm{Bun}_G)$$

is the D -module of differential operators, viewed a twisted *left* D -module.

10.3.13. Let us specialize for a moment to the case when $\kappa = \mathrm{crit}$. Then from Sect. 10.2.6 we obtain that for every U , the functor $\mathrm{Loc}_{G,\kappa,\underline{x},U}$ is compatible with the action of $\mathrm{Sph}_{G,\underline{x}}$. Hence, so is the functor $\mathrm{Loc}_{G,\kappa,\underline{x}}$.

Similarly, the functor $\mathrm{Loc}_{G,\kappa,\mathrm{Ran}}$ is compatible with the action of $\mathrm{Sph}_{G,\mathrm{Ran}}$.

10.3.14. For future reference we note that the entire discussion in this subsection applies to the case of an infinite level structure at a given $\underline{x}_0 \in \mathrm{Ran}$. I.e., for a given quasi-compact $U \subset \mathrm{Bun}_G$ and

$$U^{\mathrm{level}_{\underline{x}_0}} := U \times_{\mathrm{Bun}_G} \mathrm{Bun}^{\mathrm{level}_{\underline{x}_0}},$$

we can consider the left adjoint

$$\mathrm{Loc}_{G,\kappa,\underline{x}_0,U} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa,\underline{x}_0} \rightarrow \mathrm{D-mod}_{\kappa,\mathrm{co}}(U_G^{\mathrm{level}_{\underline{x}_0}})$$

of

$$\Gamma(U^{\mathrm{level}_{\underline{x}_0}}, -)^{\mathrm{enh}} : \mathrm{D-mod}_{\kappa,\mathrm{co}}(U_G^{\mathrm{level}_{\underline{x}_0}}) \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa,\underline{x}_0}.$$

The functors $\mathrm{Loc}_{G,\kappa,\underline{x}_0,U}$ glue to a functor

$$\mathrm{Loc}_{G,\kappa,\underline{x}_0,U} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa,\underline{x}_0} \rightarrow \mathrm{D-mod}_{\kappa,\mathrm{co}}(\mathrm{Bun}_G^{\mathrm{level}_{\underline{x}_0}}).$$

For the Ran space version, one should consider

$$\mathrm{Ran}_{\underline{x}_0} := \{\underline{x}_0\} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq},$$

where:

- $\text{Ran}^\subseteq$ is as in Sect. 11.2.2;
- The fiber product is formed using the map $\text{pr}_{\text{small}} : \text{Ran}^\subseteq \rightarrow \text{Ran}$.

10.4. The fiber of the localization functor.

10.4.1. Fix a k -point $\mathcal{P}_G \in \text{Bun}_G$. The goal of this subsection is to describe the functor

$$(10.22) \quad \text{Loc}_{G, \kappa, \underline{x}} \xrightarrow{\text{Loc}_{G, \kappa, \underline{x}}} \text{D-mod}_\kappa(\text{Bun}_G) \xrightarrow{!-\text{fiber at } \mathcal{P}_G} \text{Vect}.$$

10.4.2. Consider the Lie algebra $\Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})$. Laurent expansion defines a map

$$\Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G}) \rightarrow \mathfrak{L}(\mathfrak{g}_{\mathcal{P}_G})_{\underline{x}},$$

and recall that the Kac-Moody extension

$$0 \rightarrow k \rightarrow \widehat{\mathfrak{g}}_{\kappa, \mathcal{P}_G, \underline{x}} \rightarrow \mathfrak{L}(\mathfrak{g}_{\mathcal{P}_G})_{\underline{x}} \rightarrow 0$$

admits a canonical splitting over $\Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})$. Hence, we have a well-defined restriction functor

$$(10.23) \quad \widehat{\mathfrak{g}}\text{-mod}_{\kappa, \mathcal{P}_G, \underline{x}} \rightarrow \Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})\text{-mod}.$$

10.4.3. Composing with

$$\text{KL}(G)_{\kappa, \underline{x}} \xrightarrow{\alpha_{\mathcal{P}_G, \text{taut}}} \text{KL}(G)_{\kappa, \mathcal{P}_G, \underline{x}} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa, \mathcal{P}_G, \underline{x}}$$

we obtain a functor

$$(10.24) \quad \text{KL}(G)_{\kappa, \underline{x}} \rightarrow \Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})\text{-mod}.$$

10.4.4. We claim:

Proposition 10.4.5. *The functor (10.22) identifies canonically with the composition of (10.24) and the functor of $\Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})$ -coinvariants*

$$\Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})\text{-mod} \rightarrow \text{Vect}.$$

Remark 10.4.6. An analog of Proposition 10.4.5 for the localization functor in the finite-dimensional situation is obvious:

Let $\mathfrak{h}$ be a (discrete⁴¹) Lie algebra and let $\mathcal{Y}$ be a smooth variety equipped with an action of $\mathfrak{h}$ by vector fields. Then the corresponding localization functor

$$\text{Loc}_{\mathfrak{h}, \mathcal{Y}} : \mathfrak{h}\text{-mod} \rightarrow \text{D-mod}(\mathcal{Y}),$$

left adjoint to

$$\Gamma(\mathcal{Y}, \text{oblv}^l(-))^{\text{enh}} : \text{D-mod}(\mathcal{Y}) \rightarrow \mathfrak{h}\text{-mod},$$

is given by

$$(10.25) \quad \text{D}_{\mathcal{Y}} \otimes_{U(\mathfrak{h})} -.$$

If $y \in \mathcal{Y}$ is a point for which the action map $\mathfrak{h} \rightarrow T_y(\mathcal{Y})$ is surjective, the composition

$$\mathfrak{h}\text{-mod} \xrightarrow{\text{Loc}_{\mathfrak{h}, \mathcal{Y}}} \text{D-mod}(\mathcal{Y}) \xrightarrow{!-\text{fiber at } y} \text{Vect}$$

identifies with the functor of coinvariants with respect to

$$\text{Stab}_y(\mathfrak{h}) \subset \mathfrak{h}.$$

This follows from the fact that the $*$ -fiber at y of $\text{D}_{\mathcal{Y}}$ (as an object of $\text{QCoh}(\mathcal{Y})$ via left multiplication) identifies, as a $\mathfrak{h}$ -module, with

$$\text{ind}_{\text{Stab}_y(\mathfrak{h})}^{\mathfrak{h}}(k).$$

⁴¹As opposed to Tate.

10.4.7. One can prove Proposition 10.4.5 directly by emulating the argument in Remark 10.4.6.

In fact, such an assertion is valid for $\mathrm{Bun}_G^{\mathrm{level}_{\underline{x}}}$ replaced by a pro-scheme $\mathcal{Y}$ equipped with an action of $\mathfrak{L}(G)_{\underline{x}}^{\wedge}$ (the formal completion of $\mathfrak{L}(G)_{\underline{x}}$ along $\mathfrak{L}^+(G)_{\underline{x}}$), such that $\mathcal{Y}/\mathfrak{L}^+(G)_{\underline{x}}$ is locally of finite type, and a point $y \in \mathcal{Y}$ at which the action is infinitesimally transitive, i.e.,

$$\mathfrak{L}(\mathfrak{g})_{\underline{x}} \rightarrow T_y(\mathcal{Y})$$

is surjective.

We will, however, supply a different argument, specific to the case of Bun_G , see Sect. 12.2.

10.4.8. As an immediate corollary of Proposition 10.4.5 we obtain:

Corollary 10.4.9. *The functor*

$$\mathrm{Loc}_{G,\kappa,\underline{x}} : \mathrm{KL}(G)_{\kappa,\underline{x}} \rightarrow \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G)$$

is right t-exact, when $\mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G)$ is equipped with the left t-structure, i.e., one for which the functor oblv_{κ}^l is t-exact.

Proof. We need to show that the composite functor

$$\mathrm{KL}(G)_{\kappa,\underline{x}} \xrightarrow{\mathrm{Loc}_{G,\kappa,\underline{x}}} \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G) \xrightarrow{\mathrm{oblv}_{\kappa}^l} \mathrm{QCoh}(\mathrm{Bun}_G)$$

is right t-exact.

In order to prove that, it suffices to show that the composition of the above functor with the functor of $*$ -fiber at any field-valued point of Bun_G is right t-exact.

By base change, we can assume that the point in question is rational. In this case, the corresponding functor identifies with the functor (10.22). □

Corollary 10.4.10. *The functor $\mathrm{Loc}_{G,\kappa,\underline{x}}$ annihilates infinitely connective objects (i.e., objects that belong to $(\mathrm{KL}(G)_{\kappa,\underline{x}})^{<-n}$ for any n).*

Proof. Follows from the fact that the t-structure on $\mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G)$ is separated. □

10.5. Localization functor as the dual.

10.5.1. Let κ' be the reflected level, i.e.,

$$\kappa' := -\kappa + 2 \cdot \mathrm{crit}.$$

We claim that that we have a canonical duality

$$(10.26) \quad (\mathrm{D}\text{-mod}_{\kappa',\mathrm{co}}(\mathrm{Bun}_G))^{\vee} \simeq \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G)$$

for which the dual of the functor

$$\mathrm{oblv}_{\kappa'}^l : \mathrm{D}\text{-mod}_{\kappa',\mathrm{co}}(\mathrm{Bun}_G) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Bun}_G)$$

is the functor

$$\mathrm{ind}_{\kappa}^l : \mathrm{QCoh}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G),$$

with respect to the identification⁴²

$$\mathrm{Funct}_{\mathrm{cont}}(\mathrm{QCoh}_{\mathrm{co}}(\mathrm{Bun}_G), \mathrm{Vect}) \simeq \mathrm{QCoh}(\mathrm{Bun}_G).$$

⁴²We warn the reader that the category $\mathrm{QCoh}(\mathrm{Bun}_G)$ is not dualizable.

10.5.2. Indeed, we start with the identification

$$(\mathrm{D}\text{-mod}_{\kappa', \mathrm{co}}(\mathrm{Bun}_G))^\vee \simeq \mathrm{D}\text{-mod}_{-\kappa'}(\mathrm{Bun}_G),$$

given by Verdier duality, and compose it with the functor

$$(10.27) \quad \mathrm{D}\text{-mod}_{-\kappa'}(\mathrm{Bun}_G) \xrightarrow{\otimes K_{\mathrm{Bun}_G}} \mathrm{D}\text{-mod}_{-\kappa' + \mathrm{dlog}(K_{\mathrm{Bun}_G})}(\mathrm{Bun}_G) \simeq \mathrm{D}\text{-mod}_{-\kappa' + 2 \cdot \mathrm{crit}}(\mathrm{Bun}_G) = \\ = \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G) \xrightarrow{[\dim(\mathrm{Bun}_G)]} \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G),$$

where:

- K_{Bun_G} is the canonical line bundle on Bun_G , so that $K_{\mathrm{Bun}_G}[\dim(\mathrm{Bun}_G)] \simeq \omega_{\mathrm{Bun}_G}$;
- We have used the identification $\mathrm{dlog}(K_{\mathrm{Bun}_G}) = \mathrm{dlog}(\det_{\mathrm{Bun}_G}) = 2 \cdot \mathrm{crit}$ from Sect. 9.1.3.

10.5.3. We have the following assertion:

Proposition 10.5.4. *With respect to the identifications (10.26) and*

$$(10.28) \quad (\mathrm{KL}(G)_{\kappa, \underline{x}})^\vee \simeq \mathrm{KL}(G)_{\kappa', \underline{x}}$$

of (2.2), the functor

$$\mathrm{Loc}_{G, \kappa, \underline{x}} : \mathrm{KL}(G)_{\kappa, \underline{x}} \rightarrow \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G)$$

identifies canonically with the dual of

$$\Gamma_{G, \kappa', \underline{x}} : \mathrm{D}\text{-mod}_{\kappa', \mathrm{co}}(\mathrm{Bun}_G) \rightarrow \mathrm{KL}(G)_{\kappa', \underline{x}}.$$

The induced identification

$$\begin{aligned} \mathrm{ind}_{\kappa}^l \circ (\mathrm{ev}_{\underline{x}})^* &\simeq \mathrm{Loc}_{G, \kappa, \underline{x}} \circ \mathrm{ind}_{\mathfrak{L}^+(G)}^{(\mathfrak{g}, \mathfrak{L}^+(G))_{\kappa}} \simeq (\Gamma_{G, \kappa', \underline{x}})^\vee \circ (\mathrm{oblv}_{\mathfrak{L}^+(G)}^{(\mathfrak{g}, \mathfrak{L}^+(G))_{\kappa'}})^\vee \simeq \\ &\simeq \left(\mathrm{oblv}_{\mathfrak{L}^+(G)}^{(\mathfrak{g}, \mathfrak{L}^+(G))_{\kappa'}} \circ \Gamma_{G, \kappa', \underline{x}} \right)^\vee \simeq ((\mathrm{ev}_{\underline{x}})_* \circ \mathrm{oblv}_{\kappa'}^l)^\vee \simeq (\mathrm{oblv}_{\kappa'}^l)^\vee \circ ((\mathrm{ev}_{\underline{x}})_*)^\vee \simeq \\ &\simeq \mathrm{ind}_{\kappa}^l \circ ((\mathrm{ev}_{\underline{x}})_*)^\vee \simeq \mathrm{ind}_{\kappa}^l \circ (\mathrm{ev}_{\underline{x}})^* \end{aligned}$$

is the identity map.

This assertion is proved in [CF, Theorem 4.0.5(2)].⁴³

Remark 10.5.5. The proof of Proposition 10.5.4 in [CF] essentially emulates the following finite-dimensional phenomenon.

Let $\mathfrak{y}$ and $\mathfrak{h}$ be as in Remark 10.4.6. On the one hand, we can consider the adjoint pair

$$\mathrm{Loc}_{\mathfrak{h}, \mathfrak{y}} : \mathfrak{h}\text{-mod} \rightleftarrows \mathrm{D}\text{-mod}(\mathfrak{y}) : \Gamma(\mathfrak{y}, \mathrm{oblv}^l(-))^{\mathrm{enh}}.$$

On the other hand, consider the canonical line bundle $K_{\mathfrak{y}}$ as a line bundle acted on by $\mathfrak{h}$, and consider the corresponding functor

$$\Gamma(\mathfrak{y}, K_{\mathfrak{y}} \otimes \mathrm{oblv}^l(-))^{\mathrm{enh}} : \mathrm{D}\text{-mod}(\mathfrak{y}) \rightarrow \mathfrak{h}\text{-mod}.$$

Let

$$\mathrm{Loc}_{\mathfrak{h}, \mathfrak{y}, K_{\mathfrak{y}}} : \mathfrak{h}\text{-mod} \rightarrow \mathrm{D}\text{-mod}(\mathfrak{y})$$

denote the left adjoint of $\Gamma(\mathfrak{y}, K_{\mathfrak{y}} \otimes \mathrm{oblv}^l(-))^{\mathrm{enh}}$.

Then the functors $\mathrm{Loc}_{\mathfrak{h}, \mathfrak{y}, K_{\mathfrak{y}}}[\dim(\mathfrak{y})]$ and $\Gamma(\mathfrak{y}, \mathrm{oblv}^l(-))^{\mathrm{enh}}$ are mutually dual in terms of the Verdier duality identification

$$\mathrm{D}\text{-mod}(\mathfrak{y})^\vee \simeq \mathrm{D}\text{-mod}(\mathfrak{y}).$$

This follows from the expression for $\mathrm{Loc}_{\mathfrak{h}, \mathfrak{y}}$ given by formula (10.25), and a similar formula for $\mathrm{Loc}_{\mathfrak{h}, \mathfrak{y}, K_{\mathfrak{y}}}$.

⁴³In *loc. cit.* the dual functor to $\mathrm{Loc}_{G, \kappa}$ is denoted $\mathrm{Loc}_{\mathrm{co}}$.

10.5.6. The assertion of Proposition 10.5.4 admits an immediate generalization when $\underline{x}$ moves in families over the Ran space:

Proposition 10.5.7. *With respect to the identifications (10.26) and*

$$(\mathrm{KL}(G)_{\kappa, \mathrm{Ran}})^{\vee} \simeq \mathrm{KL}(G)_{\kappa', \mathrm{Ran}}$$

(a) *The functor*

$$\mathrm{Loc}_{G, \kappa, \mathrm{Ran}} : \mathrm{KL}(G)_{\kappa, \mathrm{Ran}} \rightarrow \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran})$$

identifies canonically with the dual of

$$\Gamma_{G, \kappa', \mathrm{Ran}} : \mathrm{D}\text{-mod}_{\kappa', \mathrm{co}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}) \rightarrow \mathrm{KL}(G)_{\kappa', \mathrm{Ran}}.$$

The induced identification

$$\begin{aligned} \mathrm{ind}_{\kappa}^l \circ (\mathrm{ev}_{\mathrm{Ran}})^* &\simeq \mathrm{Loc}_{G, \kappa, \mathrm{Ran}} \circ \mathrm{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa}} \simeq (\Gamma_{G, \kappa', \mathrm{Ran}})^{\vee} \circ (\mathrm{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}})^{\vee} \simeq \\ &\simeq \left(\mathrm{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}} \circ \Gamma_{G, \kappa', \mathrm{Ran}} \right)^{\vee} \simeq ((\mathrm{ev}_{\mathrm{Ran}})_* \circ \mathrm{oblv}_{\kappa'}^l)^{\vee} \simeq (\mathrm{oblv}_{\kappa'}^l)^{\vee} \circ ((\mathrm{ev}_{\mathrm{Ran}})_*)^{\vee} \simeq \\ &\simeq \mathrm{ind}_{\kappa}^l \circ ((\mathrm{ev}_{\mathrm{Ran}})_*)^{\vee} \simeq \mathrm{ind}_{\kappa}^l \circ (\mathrm{ev}_{\mathrm{Ran}})^* \end{aligned}$$

is the identity map.

(b) *The functor*

$$\mathrm{Loc}_{G, \kappa} : \mathrm{KL}(G)_{\kappa, \mathrm{Ran}} \rightarrow \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran})$$

identifies canonically with the dual of

$$\Gamma_{G, \kappa'} : \mathrm{D}\text{-mod}_{\kappa', \mathrm{co}}(\mathrm{Bun}_G) \rightarrow \mathrm{KL}(G)_{\kappa', \mathrm{Ran}}.$$

For the proof, see [CF, Theorem 4.0.5(2)].

10.5.8. Note that by combining Sect. 10.2.11 and Proposition 10.5.7 with the fact that the functors

$$\mathrm{KL}(G)_{\kappa'} \xrightarrow{\mathrm{inv}_{\mathfrak{L}^+(G)}} \mathrm{Vect} \quad \text{and} \quad \mathrm{Vect} \xrightarrow{k \mapsto \mathrm{Vac}(G)_{\kappa}} \mathrm{KL}(G)_{\kappa}$$

and

$$\mathrm{D}\text{-mod}_{\kappa', \mathrm{co}} \xrightarrow{\Gamma(\mathrm{Bun}_G, -) \circ \mathrm{oblv}_{\kappa}^l} \mathrm{Vect} \quad \text{and} \quad \mathrm{Vect} \xrightarrow{k \mapsto \mathrm{ind}_{\kappa}^l(\mathcal{O}_{\mathrm{Bun}_G})} \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G)$$

are mutually dual, we obtain an identification

$$(10.29) \quad \mathrm{Loc}_{G, \kappa, \mathrm{Ran}}(\mathrm{Vac}_{G, \mathrm{Ran}}) \simeq \mathrm{ind}_{\kappa}^l(\mathcal{O}_{\mathrm{Bun}_G}) \boxtimes \omega_{\mathrm{Ran}}.$$

However, it follows formally that the identification (10.29) is the same as that in (10.21).

11. DIGRESSION: LOCAL-TO-GLOBAL FUNCTORS AND UNITALITY

In this section we will introduce a general framework that formalizes the unital property of the functors

$$\mathrm{Poinc}_{G, !, \mathrm{Ran}}, \mathrm{Poinc}_{G, *, \mathrm{Ran}} \quad \text{and} \quad \mathrm{Loc}_{G, \kappa, \mathrm{Ran}}.$$

The unital property says, roughly speaking, that the insertion of the vacuum⁴⁴ does not change the value of the functor (see Sect. 11.3.3).

A key phenomenon that we will observe is the following: insertion of the vacuum along the entire Ran space improves the unital property of the functor, see Sect. 11.4. The functor of factorization homology and its generalizations are particular cases of this construction, see Sect. 11.9.

11.0. What is this section about? As this section deals with some abstract material, a general introduction is in order.

⁴⁴In the main body of this section, we use the word “unit” instead of “vacuum”.

11.0.1. In this section, we study the general formalism of local-to-global functors. The (*local*) source of such a functor is a crystal of categories $\underline{\mathbf{C}}^{\text{loc}}$ over the Ran space, while its (*global*) target is a single category $\mathbf{C}^{\text{glob}}$. Roughly, for a space $\mathcal{Z}$ equipped with a map $\underline{x} : \mathcal{Z} \rightarrow \text{Ran}$, the value of $\underline{\mathbf{C}}^{\text{loc}}$ on $\mathcal{Z}$ is a category $\mathbf{C}_{\mathcal{Z}}^{\text{loc}} = \mathbf{C}_{\mathcal{Z}, \underline{x}}^{\text{loc}}$, and a local-to-global functor $\underline{F}$ is a compatible collection of functors

$$F_{\mathcal{Z}} = F_{\mathcal{Z}, \underline{x}} : \mathbf{C}_{\mathcal{Z}, \underline{x}}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(\mathcal{Z})$$

for every $\mathcal{Z}$ and $\underline{x}$. (Here and below, the words “compatible collection” mean “collection equipped with higher coherence data”.)

11.0.2. Next, we introduce the notion of a unital crystal of categories over the Ran space (see Sect. 11.2). Informally, a unital structure on a sheaf $\underline{\mathbf{C}}^{\text{loc}}$ is a compatible collection of functors

$$\text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2} : \mathbf{C}_{\mathcal{Z}, \underline{x}_1}^{\text{loc}} \rightarrow \mathbf{C}_{\mathcal{Z}, \underline{x}_2}^{\text{loc}}$$

for every space $\mathcal{Z}$ and two maps $\underline{x}_1, \underline{x}_2 : \mathcal{Z} \rightarrow \text{Ran}$ such that $\underline{x}_1 \subseteq \underline{x}_2$. Here we view $\mathcal{Z}$ -points of the Ran space as $\mathcal{Z}$ -families of finite subsets of X .

11.0.3. Suppose now that $\underline{\mathbf{C}}^{\text{loc}}$ is a unital crystal of categories, and $\underline{F}$ is a local-to-global functor from $\underline{\mathbf{C}}^{\text{loc}}$ to a category $\mathbf{C}^{\text{glob}}$. We then introduce the notion of a unital structure on $\underline{F}$; informally, it is a compatible collection of natural transformations

$$(11.1) \quad F_{\mathcal{Z}, \underline{x}_1} \rightarrow F_{\mathcal{Z}, \underline{x}_2} \circ \text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2}.$$

for every $\mathcal{Z}$, $\underline{x}_1$, and $\underline{x}_2$ as above.

In fact, we have two notions: a (*strict*) unital structure, where the transformations (11.1) are required to be isomorphisms, and a *lax* unital structure, where (11.1) can be arbitrary transformations.

Accordingly, we obtain two categories of unital local-to-global functors: the category of (strictly) unital local-to-global functors, and the category of lax unital local-to-global functors; the former is a full subcategory of the latter. We denote the categories by

$$(11.2) \quad \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \subset \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}).$$

11.0.4. The main subject of this section is a construction on local-to-global functors, which we call *the integrated insertion of the unit*. It can be defined as the left adjoint of the embedding (11.2):

$$\int \text{ins. unit} : \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}).$$

However, the functor admits a geometric description. Remarkably, the description makes sense for all (i.e., not necessarily lax unital) local-to-global functors.

11.0.5. We will use this formalism in Sect. 12 in the context of compatibility between certain natural constructions and local-to-global functors. We will see that, in three different situations, the compatibility is only lax at the start, but composition with the functor $\int \text{ins. unit}$ makes it strict.

11.1. Setup for local-to-global functors.

11.1.1. Let $\underline{\mathbf{C}}^{\text{loc}}$ be a crystal of categories over Ran (see Sect. B.8). Let $\mathbf{C}^{\text{glob}}$ be a DG category, and let us be given a functor

$$\underline{F} : \underline{\mathbf{C}}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \underline{\text{D-mod}}(\text{Ran}),$$

where $\underline{\text{D-mod}}(\text{Ran})$ is the unit crystal of categories over Ran.

Thus, for every space $\mathcal{Z}$ mapping to Ran, we have a category $\mathbf{C}_{\mathcal{Z}}^{\text{loc}}$, tensored over $\text{D-mod}(\mathcal{Z})$ and a functor

$$(11.3) \quad F_{\mathcal{Z}} : \mathbf{C}_{\mathcal{Z}}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(\mathcal{Z}).$$

Remark 11.1.2. In the above procedure, we associate to $\mathcal{Z} \rightarrow \text{Ran}$ the category of *cristalline* sections of $\underline{\mathbf{C}}^{\text{loc}}$ over $\mathcal{Z}$, i.e., the category of sections of $\underline{\mathbf{C}}^{\text{loc}}$ over $\mathcal{Z}_{\text{dR}}$, cf. Sect. C.2.10.

11.1.3. Assume for a moment that $\mathcal{Z}$ is pseudo-proper, so that the functor

$$C_c(\mathcal{Z}, -) : \mathrm{D}\text{-mod}(\mathcal{Z}) \rightarrow \mathrm{Vect}$$

left adjoint to $k \mapsto \omega_{\mathcal{Z}}$ is defined (see Sect. C.4.12). In this case we will denote by $F_{f_{\mathcal{Z}}}$ the composition

$$\mathbf{C}_{\mathcal{Z}}^{\mathrm{loc}} \xrightarrow{F_{\mathcal{Z}}} \mathrm{D}\text{-mod}(\mathcal{Z}) \otimes \mathbf{C}^{\mathrm{glob}} \xrightarrow{C_c(\mathcal{Z}, -) \otimes \mathrm{Id}} \mathbf{C}^{\mathrm{glob}}.$$

11.1.4. In particular, for $\mathcal{Z} = \mathrm{Ran}$ and the identity map, we obtain the category $\mathbf{C}_{\mathrm{Ran}}^{\mathrm{loc}}$ and a functor

$$F_{\mathrm{Ran}} : \mathbf{C}_{\mathrm{Ran}}^{\mathrm{loc}} \rightarrow \mathbf{C}^{\mathrm{glob}} \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}).$$

We will also use the symbol $F : \mathbf{C}_{\mathrm{Ran}}^{\mathrm{loc}} \rightarrow \mathbf{C}^{\mathrm{glob}}$ for the functor $F_{f_{\mathrm{Ran}}}$.

11.1.5. Note that the datum of F recovers that of $\underline{F}$. Namely, the functor $F_{\mathcal{Z}}$ identifies with

$$\mathbf{C}_{\mathcal{Z}}^{\mathrm{loc}} \rightarrow \mathbf{C}_{\mathrm{Ran} \times \mathcal{Z}}^{\mathrm{loc}} \simeq \mathbf{C}_{\mathrm{Ran}}^{\mathrm{loc}} \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \xrightarrow{F \otimes \mathrm{Id}} \mathbf{C}^{\mathrm{glob}} \otimes \mathrm{D}\text{-mod}(\mathcal{Z}),$$

where:

- $\mathrm{Ran} \times \mathcal{Z}$ is viewed as a space over Ran via the projection on the first factor;
- The arrow $\mathbf{C}_{\mathcal{Z}}^{\mathrm{loc}} \rightarrow \mathbf{C}_{\mathrm{Ran} \times \mathcal{Z}}^{\mathrm{loc}}$ is the $!$ -pushforward along the graph $\mathcal{Z} \rightarrow \mathrm{Ran} \times \mathcal{Z}$ of the original map $\mathcal{Z} \rightarrow \mathrm{Ran}$.

11.1.6. For a general pseudo-proper $\mathcal{Z}$, the functor $F_{f_{\mathcal{Z}}}$ factors as

$$\mathbf{C}_{\mathcal{Z}}^{\mathrm{loc}} \rightarrow \mathbf{C}_{\mathrm{Ran}}^{\mathrm{loc}} \xrightarrow{F} \mathbf{C}^{\mathrm{glob}},$$

where the first arrow is the functor of $!$ -pushforward (see Corollary C.4.10).

For $\mathcal{Z} = \mathrm{pt}$ and the map $\mathcal{Z} \rightarrow \mathrm{Ran}$ given by $\underline{x} \in \mathrm{Ran}$, we obtain the category denoted $\mathbf{C}_{\underline{x}}^{\mathrm{loc}}$ and a functor

$$F_{\underline{x}} : \mathbf{C}_{\underline{x}}^{\mathrm{loc}} \rightarrow \mathbf{C}^{\mathrm{glob}}.$$

11.1.7. The main examples of the above are when $\underline{\mathbf{C}}^{\mathrm{loc}}$ is one of the factorization categories

$$\mathrm{Whit}^!(G), \mathrm{Whit}_*(G), \mathrm{KL}(G)_{\kappa}.$$

In each of these cases, the corresponding global category is

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G), \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G), \text{ and } \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G),$$

and the functor F_{Ran} is

$$\mathrm{Poinc}_{G, !, \mathrm{Ran}}, \mathrm{Poinc}_{G, *, \mathrm{Ran}} \text{ and } \mathrm{Loc}_{G, \kappa, \mathrm{Ran}},$$

respectively.

11.1.8. For given $\underline{\mathbf{C}}^{\mathrm{loc}}$ and $\mathbf{C}^{\mathrm{glob}}$, we can consider the totality of functors $\underline{F}$ as above as a category, denoted

$$\mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}).$$

By Sect. 11.1.5, this is the same as just the category

$$\mathrm{Funct}_{\mathrm{cont}}(\mathbf{C}_{\mathrm{Ran}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}).$$

11.2. The local unital structure.

11.2.1. Let $\underline{\mathbf{C}}^{\mathrm{loc}}$ be a crystal of categories over Ran . By a local unital structure on $\underline{\mathbf{C}}^{\mathrm{loc}}$: we mean an extension $\underline{\mathbf{C}}^{\mathrm{loc}, \mathrm{un}^!}$ of $\underline{\mathbf{C}}^{\mathrm{loc}}$ to a crystal of categories over $\mathrm{Ran}^{\mathrm{un}^!}$ (see Sect. C.2 for what this means).

An example of such a structure is provided by a unital lax factorization category.

Let us explain what the unital structure means in concrete terms.

11.2.2. Let $\text{Ran}^\subseteq$ be the moduli space of pairs

$$(\underline{x}, \underline{x}' \mid \underline{x} \subseteq \underline{x}'),$$

see Sect. B.2.1.

We have the maps

$$\text{pr}_{\text{small}}, \text{pr}_{\text{big}} : \text{Ran}^\subseteq \rightrightarrows \text{Ran}$$

that remember $\underline{x}$ and $\underline{x}'$, respectively.

Let diag denote the diagonal map

$$\text{Ran} \rightarrow \text{Ran}^\subseteq.$$

Note that

$$\text{pr}_{\text{small}} \circ \text{diag} \simeq \text{Id} \simeq \text{pr}_{\text{diag}} \circ \text{diag}.$$

11.2.3. Denote

$$\text{Ran}^{\subseteq^2} := \text{Ran}^\subseteq \times_{\text{pr}_{\text{small}}, \text{Ran}, \text{pr}_{\text{big}}} \text{Ran}^\subseteq.$$

In addition to the two projections

$$\text{pr}_{\text{small}^2}, \text{pr}_{\text{big}^2} : \text{Ran}^{\subseteq^2} \rightrightarrows \text{Ran}^\subseteq,$$

we have a map

$$\text{pr}_{\text{comp}} : \text{Ran}^{\subseteq^2} \rightarrow \text{Ran}^\subseteq$$

that sends

$$(\underline{x}, \underline{x}', \underline{x}'' \mid \underline{x} \subseteq \underline{x}' \subseteq \underline{x}'') \mapsto (\underline{x}, \underline{x}'').$$

11.2.4. Note that $\text{Ran}^\subseteq$ is the prestack of morphisms of Ran^{untl} . Hence, at the level of 1-morphisms, an extension of $\underline{\mathbf{C}}^{\text{loc}}$ to $\underline{\mathbf{C}}^{\text{loc}, \text{untl}}$ amounts to a functor

$$(\text{pr}_{\text{small}})^*(\underline{\mathbf{C}}^{\text{loc}}) \rightarrow (\text{pr}_{\text{big}})^*(\underline{\mathbf{C}}^{\text{loc}})$$

as crystals of categories over $\text{Ran}^\subseteq$, or equivalently, to a functor

$$(11.4) \quad \underline{\mathbf{C}}^{\text{loc}} \rightarrow (\text{pr}_{\text{small}})_* \circ (\text{pr}_{\text{big}})^*(\underline{\mathbf{C}}^{\text{loc}})$$

as crystals of categories over Ran .

In the above formula:

- $(\text{pr}_{\text{big}})^*$ (resp., $(\text{pr}_{\text{small}})^*$) is the functor of pullback along pr_{big} (resp., pr_{small}) from crystals of categories over Ran to crystals of categories over $\text{Ran}^\subseteq$;
- $(\text{pr}_{\text{small}})_*$ is the functor of pushforward along pr_{small} from sheaves of categories over $\text{Ran}^\subseteq$ to crystals of categories over Ran .

We refer the reader to Sect. C.3, where the operation of pushforward for crystals of categories is reviewed.

11.2.5. Denote the functor (11.4) by⁴⁵

$$\text{ins. unit} : \underline{\mathbf{C}}^{\text{loc}} \rightarrow (\text{pr}_{\text{small}})_* \circ (\text{pr}_{\text{big}})^*(\underline{\mathbf{C}}^{\text{loc}}).$$

The functor ins. unit has an associativity structure explained in Sect. 11.2.9. The full datum of the upgrade

$$\underline{\mathbf{C}}^{\text{loc}} \rightsquigarrow \underline{\mathbf{C}}^{\text{loc}, \text{untl}}$$

is encoded by ins. unit , together with the associativity structure satisfying a homotopy-coherent system of compatibilities.

We will now explain the concrete meaning of the functor ins. unit .

⁴⁵In the formula below ins. unit is the abbreviation of “insert unit”.

11.2.6. For $\mathcal{Z} \rightarrow \text{Ran}$, denote

$$\mathcal{Z}^{\subseteq} := \mathcal{Z} \times_{\text{Ran}} \text{Ran}^{\subseteq},$$

where in the formation of the fiber product the map $\text{Ran}^{\subseteq} \rightarrow \text{Ran}$ is pr_{small} , see Sect. B.2.2.

Denote by $\text{pr}_{\text{small}, \mathcal{Z}}$ the map

$$\mathcal{Z}^{\subseteq} \rightarrow \mathcal{Z},$$

and by pr_{big} the projection

$$\mathcal{Z}^{\subseteq} \rightarrow \text{Ran}^{\subseteq} \xrightarrow{\text{pr}_{\text{big}}} \text{Ran}.$$

We view $\mathcal{Z}^{\subseteq}$ as mapping to Ran via pr_{big} . The map diag induces a map

$$\text{diag}_{\mathcal{Z}} : \mathcal{Z} \rightarrow \mathcal{Z}^{\subseteq}.$$

11.2.7. The map $\text{diag}_{\mathcal{Z}}$ gives rise to a functor

$$(\text{diag}_{\mathcal{Z}})^! : \mathbf{C}_{\mathcal{Z}^{\subseteq}}^{\text{loc}} \rightarrow \mathbf{C}_{\mathcal{Z}}^{\text{loc}}.$$

Since $\text{diag}_{\mathcal{Z}}$ is pseudo-proper, the functor $(\text{diag}_{\mathcal{Z}})^!$ admits a left adjoint, to be denoted $(\text{diag}_{\mathcal{Z}})_!$ (see Corollary C.4.10). Thus, we have an adjoint pair:

$$(11.5) \quad (\text{diag}_{\mathcal{Z}})_! : \mathbf{C}_{\mathcal{Z}}^{\text{loc}} \rightleftarrows \mathbf{C}_{\mathcal{Z}^{\subseteq}}^{\text{loc}} : (\text{diag}_{\mathcal{Z}})^!.$$

11.2.8. The functor ins. unit assigns to $\mathcal{Z}$ a $\text{D-mod}(\mathcal{Z})$ -linear functor

$$(11.6) \quad \text{ins. unit}_{\mathcal{Z}} : \mathbf{C}_{\mathcal{Z}}^{\text{loc}} \rightarrow \mathbf{C}_{\mathcal{Z}^{\subseteq}}^{\text{loc}}.$$

Note also that

$$(11.7) \quad (\text{diag}_{\mathcal{Z}})^! \circ \text{ins. unit}_{\mathcal{Z}} \simeq \text{Id}$$

as endofunctors of $\mathbf{C}_{\mathcal{Z}}^{\text{loc}}$.

11.2.9. Denote

$$\mathcal{Z}^{\subseteq^2} := (\mathcal{Z}^{\subseteq})^{\subseteq} \simeq \mathcal{Z} \times_{\text{Ran}, \text{pr}_{\text{small}}^2} \text{Ran}^{\subseteq^2} \simeq \mathcal{Z}^{\subseteq} \times_{\text{Ran}, \text{pr}_{\text{small}}} \text{Ran}^{\subseteq}.$$

We view $\mathcal{Z}^{\subseteq^2}$ as mapping to Ran via pr_{big^2} . The map $\text{pr}_{\text{comp}} : \text{Ran}^{\subseteq^2} \rightarrow \text{Ran}^{\subseteq}$ gives rise to a map

$$\text{pr}_{\text{comp}, \mathcal{Z}} : \mathcal{Z}^{\subseteq^2} \rightarrow \mathcal{Z}^{\subseteq}$$

as spaces over Ran .

The associativity property of the functor ins. unit is encoded by the following diagram

$$\begin{array}{ccc} \mathbf{C}_{\mathcal{Z}}^{\text{loc}} & \xrightarrow{\text{ins. unit}_{\mathcal{Z}}} & \mathbf{C}_{\mathcal{Z}^{\subseteq}}^{\text{loc}} \\ \text{ins. unit}_{\mathcal{Z}} \downarrow & & \downarrow \text{pr}_{\text{comp}, \mathcal{Z}}^! \\ \mathbf{C}_{\mathcal{Z}^{\subseteq}}^{\text{loc}} & \xrightarrow{\text{ins. unit}_{\mathcal{Z}^{\subseteq}}} & \mathbf{C}_{\mathcal{Z}^{\subseteq^2}}^{\text{loc}} \end{array}$$

11.2.10. *Example.* At the pointwise level, the datum of (11.6) is a system of functors

$$\text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2} : \mathbf{C}_{\underline{x}_1}^{\text{loc}} \rightarrow \mathbf{C}_{\underline{x}_2}^{\text{loc}} \text{ for } \underline{x}_1 \subseteq \underline{x}_2.$$

When

$$\underline{x}_2 = \underline{x}_1 \sqcup \underline{x}',$$

and $\underline{\mathbf{C}}^{\text{loc}}$ is a unital lax factorization category $\mathbf{C}$, the above functor is

$$\mathbf{C}_{\underline{x}_1} \xrightarrow{\text{Id} \otimes \mathbf{1}_{\mathbf{C}, \underline{x}'}} \mathbf{C}_{\underline{x}_1} \otimes \mathbf{C}_{\underline{x}'} \rightarrow \mathbf{C}_{\underline{x}_2},$$

where the last arrow is given by the lax factorization structure.

11.2.11. Assume for a moment that $\mathcal{Z}$ is pseudo-proper. In this case the map $\mathrm{pr}_{\mathrm{big}} : \mathcal{Z}^{\subseteq} \rightarrow \mathrm{Ran}$ is pseudo-proper, and hence the functor

$$(\mathrm{pr}_{\mathrm{big}})! : \mathbf{C}_{\mathcal{Z}^{\subseteq}}^{\mathrm{loc}} \rightarrow \mathbf{C}_{\mathrm{Ran}}^{\mathrm{loc}}$$

left adjoint to $(\mathrm{pr}_{\mathrm{big}})^!$ is defined (see Corollary C.4.10).

We will consider the functor

$$\int_{\mathcal{Z}} \mathrm{ins. unit} : \mathbf{C}_{\mathcal{Z}}^{\mathrm{loc}} \rightarrow \mathbf{C}_{\mathrm{Ran}}^{\mathrm{loc}}$$

equal to the composition

$$\mathbf{C}_{\mathcal{Z}}^{\mathrm{loc}} \xrightarrow{\mathrm{ins. unit}_{\mathcal{Z}}} \mathbf{C}_{\mathcal{Z}^{\subseteq}}^{\mathrm{loc}} \xrightarrow{(\mathrm{pr}_{\mathrm{big}})!} \mathbf{C}_{\mathrm{Ran}}^{\mathrm{loc}}.$$

11.2.12. In particular, we obtain an endofunctor

$$(11.8) \quad \int_{\mathrm{Ran}} \mathrm{ins. unit} : \mathbf{C}_{\mathrm{Ran}}^{\mathrm{loc}} \rightarrow \mathbf{C}_{\mathrm{Ran}}^{\mathrm{loc}}.$$

Note that the adjunction (11.6) and the identification (11.7) give rise to a natural transformation

$$(11.9) \quad \mathrm{Id} \rightarrow \int_{\mathrm{Ran}} \mathrm{ins. unit}$$

as endofunctors of $\mathbf{C}_{\mathrm{Ran}}^{\mathrm{loc}}$. Indeed, (11.9) is given by

$$\begin{aligned} \mathrm{Id} &\simeq (\mathrm{pr}_{\mathrm{big}})! \circ (\mathrm{diag}_{\mathrm{Ran}})! \simeq (\mathrm{pr}_{\mathrm{big}})! \circ (\mathrm{diag}_{\mathrm{Ran}})! \circ (\mathrm{diag}_{\mathrm{Ran}})^! \circ \mathrm{ins. unit}_{\mathrm{Ran}} \rightarrow \\ &\rightarrow (\mathrm{pr}_{\mathrm{big}})! \circ \mathrm{ins. unit}_{\mathrm{Ran}} = \int_{\mathrm{Ran}} \mathrm{ins. unit}. \end{aligned}$$

11.2.13. *Inventory of notation.* We briefly summarize the notation related to insertion of the unit.

We denote by $\mathrm{ins. unit}_{\mathcal{Z}}$ the functor

$$\mathrm{ins. unit}_{\mathcal{Z}} : \mathbf{C}_{\mathcal{Z}}^{\mathrm{loc}} \rightarrow \mathbf{C}_{\mathcal{Z}^{\subseteq}}^{\mathrm{loc}}$$

For $\mathcal{Z}$ pseudo-proper, we denote by

$$\int_{\mathcal{Z}} \mathrm{ins. unit} : \mathbf{C}_{\mathcal{Z}}^{\mathrm{loc}} \rightarrow \mathbf{C}_{\mathrm{Ran}}^{\mathrm{loc}}$$

the composition of $\mathrm{ins. unit}_{\mathcal{Z}}$ with $(\mathrm{pr}_{\mathrm{big}})^!$.

In particular, for $\mathcal{Z} = \mathrm{Ran}$, we have $\int_{\mathrm{Ran}} \mathrm{ins. unit}$, which is an endofunctor of $\mathbf{C}_{\mathrm{Ran}}^{\mathrm{loc}}$.

11.2.14. Yet, in (11.15) we will introduce yet another symbol: just $\int \mathrm{ins. unit}$. It will be an endofunctor of the category

$$\mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}), \quad \underline{F} \mapsto \underline{F}^{\int \mathrm{ins. unit}}.$$

(see Sect. 11.1.8), defined when $\underline{\mathbf{C}}^{\mathrm{loc}}$ is equipped with a local unital structure.

We will have

$$F_{\mathcal{Z}^{\subseteq}} \circ \mathrm{ins. unit}_{\mathcal{Z}} \simeq F'_{\mathcal{Z}}$$

as functors $\mathbf{C}_{\mathcal{Z}}^{\mathrm{loc}} \rightarrow \mathbf{C}^{\mathrm{glob}}$, where:

- $\underline{F}' := F^{\int \mathrm{ins. unit}} \in \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}})$;
- The notation $F'_{\mathcal{Z}}$ is as in Sect. 11.1.3.

11.3. A (lax) unital structure on a local-to-global functor.

11.3.1. Let $(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}, \underline{\mathbf{F}})$ be as in Sect. 11.1.1. Assume now that $\underline{\mathbf{C}}^{\text{loc}}$ is equipped with a local unital structure.

A lax unital structure on $\underline{\mathbf{F}}$ is its upgrade to a *right-lax* functor

$$(11.10) \quad \underline{\mathbf{F}}^{\text{un}^{\text{tl}}} : \underline{\mathbf{C}}^{\text{loc}, \text{un}^{\text{tl}}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran}^{\text{un}^{\text{tl}}})$$

between crystals of categories over $\text{Ran}^{\text{un}^{\text{tl}}}$, see Sect. C.2.8 for what this means.

11.3.2. Concretely, a lax unital structure on $\underline{\mathbf{F}}$ means the following. Let $\mathcal{Z}$ be a space, and let

$$\underline{x}_1 \xrightarrow{\alpha} \underline{x}_2$$

be a morphism in the category $\text{Maps}(\mathcal{Z}, \text{Ran}^{\text{un}^{\text{tl}}})$.

The maps $\underline{x}_i$ give rise to categories $\mathbf{C}_{\mathcal{Z}, \underline{x}_i}^{\text{loc}}$ tensored over $\mathbf{D}\text{-mod}(\mathcal{Z})$, $i = 1, 2$. The datum of $\underline{\mathbf{F}}$ gives rise to $\mathbf{D}\text{-mod}(\mathcal{Z})$ -linear functors

$$\mathbf{F}_{\mathcal{Z}, \underline{x}_i} : \mathbf{C}_{\mathcal{Z}, \underline{x}_i}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \mathbf{D}\text{-mod}(\mathcal{Z}).$$

The local unital structure on $\underline{\mathbf{C}}^{\text{loc}}$ gives rise to a $\mathbf{D}\text{-mod}(\mathcal{Z})$ -linear functor

$$\mathbf{C}_{\mathcal{Z}, \underline{x}_1}^{\text{loc}} \xrightarrow{\alpha} \mathbf{C}_{\mathcal{Z}, \underline{x}_2}^{\text{loc}}.$$

Then the datum of $\underline{\mathbf{F}}^{\text{un}^{\text{tl}}}$ gives rise to a natural transformation

$$(11.11) \quad \mathbf{F}_{\mathcal{Z}, \underline{x}_1} \rightarrow \mathbf{F}_{\mathcal{Z}, \underline{x}_2} \circ \mathbf{C}_{\alpha}^{\text{loc}}$$

as functors

$$\mathbf{C}_{\mathcal{Z}, \underline{x}_1}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \mathbf{D}\text{-mod}(\mathcal{Z}).$$

11.3.3. *Example.* Set $\mathcal{Z} = \text{pt}$, so that $\underline{x}_1 \xrightarrow{\alpha} \underline{x}_2$ corresponds to an inclusion

$$\underline{x}_1 \subseteq \underline{x}_2.$$

Then $\underline{\mathbf{F}}^{\text{un}^{\text{tl}}}$ gives rise to a natural transformation

$$\mathbf{F}_{\underline{x}_1} \rightarrow \mathbf{F}_{\underline{x}_2} \circ \text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2}.$$

11.3.4. We can rewrite the datum of natural transformations (11.11) as follows:

Let $\underline{\mathbf{F}}$ be as in Sect. 11.1. Evaluating $\underline{\mathbf{F}}$ on $\mathcal{Z}^{\subseteq}$, we obtain a $\mathbf{D}\text{-mod}(\mathcal{Z})$ -linear functor

$$\mathbf{F}_{\mathcal{Z}^{\subseteq}} : \mathbf{C}_{\mathcal{Z}^{\subseteq}}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \mathbf{D}\text{-mod}(\mathcal{Z}^{\subseteq}).$$

The datum of $\underline{\mathbf{F}}^{\text{un}^{\text{tl}}}$ gives rise to a natural transformation

$$(11.12) \quad (\text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})^{\dagger}) \circ \mathbf{F}_{\mathcal{Z}} \rightarrow \mathbf{F}_{\mathcal{Z}^{\subseteq}} \circ \text{ins. unit}_{\mathcal{Z}}$$

as functors

$$\mathbf{C}_{\mathcal{Z}}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \mathbf{D}\text{-mod}(\mathcal{Z}^{\subseteq}).$$

The natural transformation (11.12) encodes the datum of $\underline{\mathbf{F}}^{\text{un}^{\text{tl}}}$ at the level of 1-morphisms. One can recover the full datum of $\underline{\mathbf{F}}^{\text{un}^{\text{tl}}}$ by imposing a datum of associativity that (11.12) is supposed to satisfy.

11.3.5. We shall say that a lax unital structure on $\underline{\mathbf{F}}$ is *strict* if $\underline{\mathbf{F}}^{\text{un}^{\text{tl}}}$ is a strict functor between crystals of categories over $\text{Ran}^{\text{un}^{\text{tl}}}$, see Sect. C.2.8 for what this means.

By definition, this means that the natural transformations (11.11) are isomorphisms.

In this case we will call $\underline{\mathbf{F}}^{\text{un}^{\text{tl}}}$ a *unital structure* on $\underline{\mathbf{F}}$.

11.3.6. Equivalently, a lax unital structure on $\underline{\mathbf{F}}$ is strict if the natural transformation (11.12) is an isomorphism for any $\mathcal{Z}$.

11.3.7. Each of the examples from Sect. 11.1.7 has a natural unital structure.

11.3.8. We can consider the categories

$$(11.13) \quad \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \subset \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$$

of local-to-global functors equipped with a unital or lax unital structures, respectively, with the former being a full subcategory of the latter.

Note that we have a forgetful functor

$$(11.14) \quad \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}),$$

where $\text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$ is as in Sect. 11.1.8.

Remark 11.3.9. In [HR, Sect. 1.2.6], axioms for an algebro-geometric avatar of [1,2]-extended 3d quantum field theories on X were considered, although a detailed definition was not provided. In the present setting, we can easily spell out the complete axioms:

We should have the data of a unital factorization category $\mathbf{C}$ (viewed as a crystal of categories over Ran^{untl}), a local-to-global functor $F : \mathbf{C}_{\text{Ran}} \rightarrow \text{Vect}$, and a unital structure on $\underline{F}$.

We refer to *loc. cit.* for a discussion of why these axiomatics can be geometrically interpreted in terms of 3d QFTs.

Moreover, the discussion from [HR, Sect. 1.2.13-15] suggests that local-to-global functors valued in more general global categories $\mathbf{C}^{\text{glob}}$ should generally be interpreted in terms of boundary conditions for 4d QFTs; this applies for all the examples we consider here.

11.4. Integrated insertion of the unit. The main construction in this subsection (i.e., the operation $\int \text{ins. unit}$) may be viewed as an abstraction of the definition of chiral (a.k.a. factorization) homology in [BD2, Sect. 4.2].

As we shall see, the framework introduced above allows us to reproduce this construction automatically: it amounts to the left of adjoint to the embedding

$$\text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}).$$

See also Sect. 11.9, where the specific example of the functor of factorization homology is considered.

11.4.1. Let $\underline{\mathbf{C}}^{\text{loc}}$ and $\mathbf{C}^{\text{glob}}$ be as in Sect. 11.1. Let

$$\text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$$

be the corresponding category of local-to-global functors.

Assume now that $\underline{\mathbf{C}}^{\text{loc}}$ is equipped with a local unital structure.

11.4.2. We define an endofunctor

$$(11.15) \quad \int \text{ins. unit} : \text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$$

$$\underline{F} \mapsto \underline{F}^{\int \text{ins. unit}}$$

by

$$F_z^{\int \text{ins. unit}} := (\text{Id} \otimes (\text{pr}_{\text{small}, z})!) \circ F_{z \subseteq} \circ \text{ins. unit}_z.$$

In other words, $F_z^{\int \text{ins. unit}}$ is the composition

$$\mathbf{C}_z^{\text{loc}} \xrightarrow{\text{ins. unit}_z} \mathbf{C}_{z \subseteq}^{\text{loc}} \xrightarrow{F_{z \subseteq}} \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(z \subseteq) \xrightarrow{\text{Id} \otimes (\text{pr}_{\text{small}, z})!} \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(z).$$

11.4.3. Note that we have a natural transformation

$$(11.16) \quad \text{Id} \rightarrow \int \text{ins. unit}$$

so that for a given $\mathcal{Z}$ the corresponding map

$$(11.17) \quad \mathbf{F}_{\mathcal{Z}} \rightarrow \mathbf{F}_{\mathcal{Z}}^{f \text{ ins. unit}}$$

is given by

$$(11.18) \quad \begin{aligned} \mathbf{F}_{\mathcal{Z}} &\simeq (\text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})_!) \circ (\text{Id} \otimes (\text{diag}_{\mathcal{Z}})_!) \circ \mathbf{F}_{\mathcal{Z}} \simeq (\text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})_!) \circ \mathbf{F}_{\mathcal{Z} \subseteq} \circ (\text{diag}_{\mathcal{Z}})_! \stackrel{(11.7)}{\simeq} \\ &\simeq (\text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})_!) \circ \mathbf{F}_{\mathcal{Z} \subseteq} \circ (\text{diag}_{\mathcal{Z}})_! \circ (\text{diag}_{\mathcal{Z}})^! \circ \text{ins. unit}_{\mathcal{Z}} \rightarrow \\ &\rightarrow (\text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})_!) \circ \mathbf{F}_{\mathcal{Z} \subseteq} \circ \text{ins. unit}_{\mathcal{Z}} = \underline{\mathbf{F}}_{\mathcal{Z}}^{f \text{ ins. unit}}. \end{aligned}$$

11.4.4. Assume for a moment that $\mathcal{Z}$ is pseudo-proper. Applying $\mathbf{C}_c(\mathcal{Z}, -) \otimes \text{Id}$ to both sides of (11.17), we obtain a natural transformation

$$(11.19) \quad \mathbf{F}_{f_{\mathcal{Z}}} \rightarrow \mathbf{F} \circ \int_{\mathcal{Z}} \text{ins. unit},$$

where $\int_{\mathcal{Z}} \text{ins. unit}$ is as in Sect. 11.2.11.

Take $\mathcal{Z} = \text{Ran}$. In this case, the resulting natural transformation (11.19) is

$$(11.20) \quad \mathbf{F} \rightarrow \mathbf{F} \circ \int_{\text{Ran}} \text{ins. unit},$$

where $\int_{\text{Ran}} \text{ins. unit}$ is as in Sect. 11.2.12.

It is easy to see, however, that (11.20) equals the natural transformation obtained by applying $\mathbf{F}$ to the natural transformation (11.9).

11.4.5. Suppose for a moment that $\mathbf{F}$ is equipped with a unital structure, i.e., it is the image under the forgetful functor (11.14) of an object $\underline{\mathbf{F}}^{\text{untl}} \in \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$.

We claim that in this case the map (11.17) is an isomorphism. Indeed, in this case, the *isomorphism* (11.12) identifies

$$(11.21) \quad \mathbf{F}_{\mathcal{Z}}^{f \text{ ins. unit}} \simeq (\text{pr}_{\text{small}, \mathcal{Z}})_! \circ (\text{pr}_{\text{small}, \mathcal{Z}})^! \circ \mathbf{F}_{\mathcal{Z}},$$

and the map (11.16) is the map

$$(11.22) \quad \mathbf{F}_{\mathcal{Z}} \simeq (\text{pr}_{\text{small}, \mathcal{Z}})_! \circ (\text{diag}_{\mathcal{Z}})_! \circ (\text{diag}_{\mathcal{Z}})^! \circ (\text{pr}_{\text{small}, \mathcal{Z}})^! \circ \mathbf{F}_{\mathcal{Z}} \rightarrow (\text{pr}_{\text{small}, \mathcal{Z}})_! \circ (\text{pr}_{\text{small}, \mathcal{Z}})^! \circ \mathbf{F}_{\mathcal{Z}}.$$

Now, the contractibility of the Ran space implies that the counit of the $((\text{pr}_{\text{small}, \mathcal{Z}})_!, (\text{pr}_{\text{small}, \mathcal{Z}})^!)$ -adjunction is an isomorphism. Hence, the right-hand side of (11.22) maps isomorphically to $\mathbf{F}_{\mathcal{Z}}$, and the composition

$$\mathbf{F}_{\mathcal{Z}} \xrightarrow{(11.22)} (\text{pr}_{\text{small}, \mathcal{Z}})_! \circ (\text{pr}_{\text{small}, \mathcal{Z}})^! \circ \mathbf{F}_{\mathcal{Z}} \rightarrow \mathbf{F}_{\mathcal{Z}}$$

is the identity map.

In particular, in this case the natural transformation (11.20) is an isomorphism.

11.4.6. Our next goal, carried out in Sects. 11.5-11.6, is to perform similar constructions with the same input, but in the unital context, i.e., working over Ran^{unl} rather than over Ran . Namely, we will show that, parallel to (11.15), there exists an endofunctor

$$(11.23) \quad \int \text{ins. unit} : \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-unl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-unl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$$

$$\underline{\mathbf{F}}^{\text{unl}} \mapsto \underline{\mathbf{F}}^{\text{unl}, f \text{ ins. unit}}$$

that makes the diagram

$$(11.24) \quad \begin{array}{ccc} \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-unl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) & \xrightarrow{f \text{ ins. unit}} & \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-unl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \\ \downarrow & & \downarrow \\ \text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) & \xrightarrow{f \text{ ins. unit}} & \text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \end{array}$$

commute.

11.4.7. In addition, the functor (11.23) will be equipped with a natural transformation

$$(11.25) \quad \text{Id} \rightarrow \int \text{ins. unit},$$

which is compatible with (11.16) via (11.24).

We will also show:

- The essential image of (11.23) belongs to $\text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{unl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$.
- The natural transformation (11.25) evaluates to an isomorphism on objects that belong to $\text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{unl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$;
- The two natural transformations

$$(11.26) \quad \int \text{ins. unit} \rightrightarrows \int \text{ins. unit} \circ \int \text{ins. unit},$$

arising from (11.25) coincide (it follows that they are isomorphisms).

11.4.8. The above properties combined imply that the functor (11.23) is the left adjoint of the embedding

$$\text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{unl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \hookrightarrow \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-unl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}).$$

Remark 11.4.9. The reason for discussing both versions of $\int \text{ins. unit}$, i.e., (11.23) and (11.15), is that the former has a clear categorical meaning (i.e., it is the left adjoint of the forgetful functor), while the latter is easily computable (a priori, the functor (11.23) involves taking cohomology over categorical prestacks).

However, the commutation of (11.24) implies that (11.23) is computable as well.

11.5. Construction of the integrated functor. This and the next subsection are devoted to the construction of the functor (11.23) and the verification of its properties. The reader who is willing to take this on faith may choose to skip these two subsections.

We are going to present the construction of the functor (11.23) in a hands-on manner. See, however, Sect. C.3.10 for its abstract interpretation.

11.5.1. Let $\mathcal{Y}$ be a categorical prestack and let $\mathcal{Y}^{\rightarrow}$ be the categorical prestack of 1-morphisms in $\mathcal{Y}$. I.e., for an affine scheme S , objects of $\text{Maps}(S, \mathcal{Y}^{\rightarrow})$ are

$$y_1, y_2 \in \text{Maps}(S, \mathcal{Y}), \quad y_1 \rightarrow y_2,$$

and morphisms are commutative diagrams

$$\begin{array}{ccc} y_1 & \longrightarrow & y_2 \\ \downarrow & & \downarrow \\ y'_1 & \longrightarrow & y'_2. \end{array}$$

We have the projections

$$\text{pr}_{\text{source}}, \text{pr}_{\text{target}} : \mathcal{Y}^{\rightarrow} \rightarrow \mathcal{Y},$$

with $\text{pr}_{\text{source}}$ being a Cartesian fibration.

Let $\underline{\mathbf{C}}$ be a crystal of categories on $\mathcal{Y}$. Tautologically, we have a (strict) functor

$$(11.27) \quad (\text{pr}_{\text{source}})^*(\underline{\mathbf{C}}) \rightarrow (\text{pr}_{\text{target}})^*(\underline{\mathbf{C}}),$$

as crystals of categories on $\mathcal{Y}^{\rightarrow}$.

Recall the construction of the direct image of a crystal of categories, reviewed in Sect. C.3. According to Sect. C.3.8, we have a (strict) functor

$$\underline{\mathbf{C}} \rightarrow (\text{pr}_{\text{source}})_{*, \text{strict}} \circ (\text{pr}_{\text{source}})^*(\underline{\mathbf{C}}).$$

Composing with (11.27) we obtain a (strict) functor

$$(11.28) \quad \underline{\mathbf{C}} \rightarrow (\text{pr}_{\text{source}})_{*, \text{strict}} \circ (\text{pr}_{\text{target}})^*(\underline{\mathbf{C}}).$$

11.5.2. We apply the construction in Sect. 11.5.1 to $\mathcal{Y} = \text{Ran}^{\text{untl}}$. Denote

$$\text{Ran}^{\text{untl}, \rightarrow} =: \text{Ran}^{\subseteq, \text{untl}},$$

viewed as a categorical prestacks.

Note that the prestack in groupoids underlying $\text{Ran}^{\subseteq, \text{untl}}$ is the prestack $\text{Ran}^{\subseteq}$ introduced in Sect. 11.2.2. We will use the symbols $\text{pr}_{\text{small}}^{\text{untl}}$ and $\text{pr}_{\text{big}}^{\text{untl}}$ for the corresponding maps $\text{pr}_{\text{source}}$ and $\text{pr}_{\text{target}}$.

Thus, for $\underline{\mathbf{C}}^{\text{loc}, \text{untl}}$ as in Sect. 11.2, the functor (11.28) is a functor

$$(11.29) \quad \underline{\mathbf{C}}^{\text{loc}, \text{untl}} \rightarrow (\text{pr}_{\text{small}}^{\text{untl}})_{*, \text{strict}} \circ (\text{pr}_{\text{big}}^{\text{untl}})^*(\underline{\mathbf{C}}^{\text{loc}, \text{untl}}).$$

11.5.3. Let now $\underline{\mathbf{F}}^{\text{untl}}$ be an object of $\text{Func}^{\text{loc} \rightarrow \text{glob}, \text{lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$. Applying pullback along $\text{pr}_{\text{big}}^{\text{untl}}$, we obtain a *lax* functor

$$(\text{pr}_{\text{big}}^{\text{untl}})^*(\underline{\mathbf{C}}^{\text{loc}, \text{untl}}) \xrightarrow{(\text{pr}_{\text{big}}^{\text{untl}})^*(\underline{\mathbf{F}}^{\text{untl}})} (\text{pr}_{\text{big}}^{\text{untl}})^*(\mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D-mod}}(\text{Ran}^{\text{untl}}))$$

as crystals of categories on $\text{Ran}^{\subseteq, \text{untl}}$.

Using Sect. C.3.7 we obtain a lax functor of crystal of categories on Ran^{untl}

$$(\text{pr}_{\text{small}}^{\text{untl}})_{*, \text{lax}} \circ (\text{pr}_{\text{big}}^{\text{untl}})^*(\underline{\mathbf{C}}^{\text{loc}, \text{untl}}) \xrightarrow{(\text{pr}_{\text{small}}^{\text{untl}})^* \circ (\text{pr}_{\text{big}}^{\text{untl}})^*(\underline{\mathbf{F}}^{\text{untl}})} (\text{pr}_{\text{small}}^{\text{untl}})_{*, \text{lax}} \circ (\text{pr}_{\text{big}}^{\text{untl}})^*(\mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D-mod}}(\text{Ran}^{\text{untl}})).$$

Combining, we obtain a functor

$$(11.30) \quad \begin{aligned} \underline{\mathbf{C}}^{\text{loc}, \text{untl}} &\rightarrow (\text{pr}_{\text{small}}^{\text{untl}})_{*, \text{strict}} \circ (\text{pr}_{\text{big}}^{\text{untl}})^*(\underline{\mathbf{C}}^{\text{loc}, \text{untl}}) \rightarrow (\text{pr}_{\text{small}}^{\text{untl}})_{*, \text{lax}} \circ (\text{pr}_{\text{big}}^{\text{untl}})^*(\underline{\mathbf{C}}^{\text{loc}, \text{untl}}) \rightarrow \\ &\rightarrow (\text{pr}_{\text{small}}^{\text{untl}})_{*, \text{lax}} \circ (\text{pr}_{\text{big}}^{\text{untl}})^*(\mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D-mod}}(\text{Ran}^{\text{untl}})) \simeq \mathbf{C}^{\text{glob}} \otimes (\text{pr}_{\text{small}}^{\text{untl}})_{*, \text{lax}}(\underline{\mathbf{D-mod}}(\text{Ran}^{\subseteq, \text{untl}})). \end{aligned}$$

11.5.4. Consider the (strict) functor

$$(11.31) \quad \underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\text{untl}}) \xrightarrow{(C.9)} (\mathbf{pr}_{\text{small}}^{\text{untl}})_{*,\text{strict}}(\underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\subseteq,\text{untl}})) \rightarrow (\mathbf{pr}_{\text{small}}^{\text{untl}})_{*,\text{lax}}(\underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\subseteq,\text{untl}})).$$

Lemma 11.5.5. *The functor (11.31) admits a value-wise left adjoint, to be denoted $(\mathbf{pr}_{\text{small}}^{\text{untl}})_!$. Moreover, this value-wise left adjoint, which is a priori a left-lax functor, is strict.*

The proof will be given in Sect. 11.7.2.

11.5.6. Thus, composing the lax functor (11.30) with $(\mathbf{pr}_{\text{small}}^{\text{untl}})_!$, we obtain a lax functor

$$(11.32) \quad \underline{\mathbf{C}}^{\text{loc,untl}} \rightarrow \mathbf{C}^{\text{glob}} \otimes (\mathbf{pr}_{\text{small}}^{\text{untl}})_{*,\text{lax}}(\underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\subseteq,\text{untl}})) \xrightarrow{\text{Id} \otimes (\mathbf{pr}_{\text{small}}^{\text{untl}})_!} \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}).$$

11.5.7. The functor (11.32) is the sought-for object

$$\underline{\mathbf{F}}^{\text{untl},f \text{ ins.unit}}.$$

11.6. Properties of the integrated functor. We now proceed to establish the properties of the functor (11.23).

11.6.1. Note that the construction in Sect. 11.5.1 is functorial in the following sense:

If $\Phi : \underline{\mathbf{C}}_1^{\text{loc,untl}} \rightarrow \underline{\mathbf{C}}_2^{\text{loc,untl}}$ is a *strict* functor between sheaves of categories on $\mathbf{Ran}^{\text{untl}}$, and

$$\underline{\mathbf{F}}^{\text{untl}} : \underline{\mathbf{C}}_2^{\text{loc,untl}} \rightarrow \underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\text{untl}})$$

is a lax unital functor, then we have a (tautological) isomorphism

$$(\underline{\mathbf{F}}^{\text{untl}} \circ \Phi)^{f \text{ ins.unit}} \simeq \underline{\mathbf{F}}^{\text{untl},f \text{ ins.unit}} \circ \Phi.$$

11.6.2. We first show that $f \text{ ins.unit}$ acts as identity on objects $\underline{\mathbf{F}}^{\text{untl}} \in \text{Funct}^{\text{loc} \rightarrow \text{glob,untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$.

Indeed, if the functor

$$\underline{\mathbf{F}}^{\text{untl}} : \underline{\mathbf{C}}^{\text{loc,untl}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\text{untl}})$$

is strict, then by Sect. 11.6.1 above, the functor (11.30) identifies with the composition of $\underline{\mathbf{F}}^{\text{untl}}$ with the tensor product of the identity endofunctor of $\mathbf{C}^{\text{glob}}$ with

$$(11.33) \quad \underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\text{untl}}) \xrightarrow{(11.28)} (\mathbf{pr}_{\text{small}}^{\text{untl}})_{*,\text{strict}} \circ (\mathbf{pr}_{\text{big}}^{\text{untl}})^*(\underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\text{untl}})) = \\ = (\mathbf{pr}_{\text{small}}^{\text{untl}})_{*,\text{strict}}(\underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\subseteq,\text{untl}})) \xrightarrow{(\mathbf{pr}_{\text{small}}^{\text{untl}})_!} \underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\text{untl}}).$$

We claim that (11.33) is the identity endofunctor of $\underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\text{untl}})$. Indeed, observe that for $\underline{\mathbf{C}} = \underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\text{untl}})$, the functor (11.27) is the identity endofunctor of $\underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\subseteq,\text{untl}})$, so the composition

$$\underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\text{untl}}) \xrightarrow{(11.28)} (\mathbf{pr}_{\text{small}}^{\text{untl}})_{*,\text{strict}} \circ (\mathbf{pr}_{\text{big}}^{\text{untl}})^*(\underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\text{untl}})) = \\ = (\mathbf{pr}_{\text{small}}^{\text{untl}})_{*,\text{strict}}(\underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\subseteq,\text{untl}}))$$

is the functor

$$\underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\text{untl}}) \xrightarrow{(11.28)} (\mathbf{pr}_{\text{small}}^{\text{untl}})_{*,\text{strict}} \circ (\mathbf{pr}_{\text{small}}^{\text{untl}})^*(\underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\text{untl}})) = \\ = (\mathbf{pr}_{\text{small}}^{\text{untl}})_{*,\text{strict}}(\underline{\mathbf{D}\text{-mod}}(\mathbf{Ran}^{\subseteq,\text{untl}})).$$

The assertion follows now from the next lemma:

Lemma 11.6.3. *The functor (11.31) is fully faithful.⁴⁶*

The proof will be given in Sect. 11.7.1.

⁴⁶See Sect. C.2.7 for what this means.

11.6.4. Next we show that the essential image of the functor (11.23) lies in the subcategory $\text{Func}^{\text{loc} \rightarrow \text{glob}, \text{untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$, i.e., $\underline{\mathbf{F}}^{\text{untl}, f \text{ ins. unit}}$ is strict.

Let S be an affine scheme and let $\underline{x}_1 \xrightarrow{\alpha} \underline{x}_2$ be a 1-morphism in $\text{Maps}(S, \text{Ran}^{\text{untl}})$. Consider the fiber products

$$S_{\underline{x}_i}^{\subseteq, \text{untl}} := S \times_{\underline{x}_i, \text{Ran}^{\text{untl}}} \text{Ran}^{\subseteq, \text{intl}}, \quad i = 1, 2,$$

and the resulting map

$$(11.34) \quad S_{\underline{x}_2}^{\subseteq, \text{untl}} \xrightarrow{\alpha^*} S_{\underline{x}_1}^{\subseteq, \text{untl}}.$$

Consider the diagram

$$\begin{array}{ccccc} \underline{\mathbf{C}}_{S, \underline{x}_1}^{\text{loc}, \text{untl}} & \longrightarrow & \Gamma^{\text{lax}}(S_{\underline{x}_1}^{\subseteq, \text{untl}}, \underline{\mathbf{C}}_{S, \underline{x}_1}^{\text{loc}, \text{untl}}) & \xrightarrow{\underline{\mathbf{F}}^{\text{untl}}} & \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(S_{\underline{x}_1}^{\subseteq, \text{untl}}) \\ \downarrow & & \downarrow & \nearrow & \downarrow \text{Id} \otimes (\alpha^*)^! \\ \underline{\mathbf{C}}_{S, \underline{x}_2}^{\text{loc}, \text{untl}} & \longrightarrow & \Gamma^{\text{lax}}(S_{\underline{x}_2}^{\subseteq, \text{untl}}, \underline{\mathbf{C}}_{S, \underline{x}_2}^{\text{loc}, \text{untl}}) & \xrightarrow{\underline{\mathbf{F}}^{\text{untl}}} & \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(S_{\underline{x}_2}^{\subseteq, \text{untl}}), \end{array}$$

where:

- The left vertical arrow is given by the structure of crystal of categories on $\underline{\mathbf{C}}^{\text{loc}, \text{untl}}$;
- The middle vertical arrow is (C.8);
- Both left horizontal arrows are (11.28);
- The left square commutes because the functor (C.9) is strict;
- The natural transformation in the right square is given by the lax functor structure on $\underline{\mathbf{F}}^{\text{untl}}$.

Given Lemma 11.5.5, it suffices to show that the above natural transformation is an isomorphism.

Note, however, that the middle vertical arrow in the above diagram is the functor (C.7) corresponding to the morphism α^* of (11.34). This makes the assertion manifest.

11.6.5. We now construct the natural transformation (11.25). This is done in the same way as in (11.18) using the map

$$\text{diag}^{\text{untl}} : \text{Ran}^{\text{untl}} \rightarrow \text{Ran}^{\subseteq, \text{untl}},$$

corresponding to the identity morphisms in Ran^{untl} as a categorical prestack.

11.6.6. We now show that the two maps in (11.26) coincide. This amounts to the following assertion.

Let S be an affine scheme equipped with a map to Ran^{untl} . Consider

$$S^{\subseteq, \text{untl}} := S \times_{\text{Ran}^{\text{untl}}, \text{pr}_{\text{small}}^{\text{untl}}} \text{Ran}^{\subseteq, \text{intl}} \quad \text{and} \quad S^{\subseteq^2, \text{untl}} := S^{\subseteq, \text{untl}} \times_{\text{pr}_{\text{big}}^{\text{untl}}, \text{Ran}^{\text{untl}}, \text{pr}_{\text{small}}^{\text{untl}}} \text{Ran}^{\subseteq, \text{intl}}.$$

We have the naturally defined maps

$$\text{pr}_{\text{big}^2}^{\text{untl}} : S^{\subseteq^2, \text{untl}} \rightarrow \text{Ran}^{\subseteq, \text{intl}} \quad \text{and} \quad \text{pr}_{\text{small}^2, S}^{\text{untl}} : S^{\subseteq^2, \text{untl}} \rightarrow S^{\subseteq, \text{untl}}.$$

We also have the maps

$$\text{diag}_{S, \text{big}}^{\text{untl}} \quad \text{and} \quad \text{diag}_{S, \text{small}}^{\text{untl}}, \quad S^{\subseteq, \text{untl}} \rightrightarrows S^{\subseteq^2, \text{untl}}$$

so that

$$\text{pr}_{\text{small}^2, S}^{\text{untl}} \circ \text{diag}_{S, \text{big}}^{\text{untl}} = \text{Id}, \quad \text{pr}_{\text{small}^2, S}^{\text{untl}} \circ \text{diag}_{S, \text{small}}^{\text{untl}} = \text{diag}_S^{\text{untl}} \circ \text{pr}_{\text{small}, S}^{\text{untl}}$$

and

$$\text{pr}_{\text{big}^2}^{\text{untl}} \circ \text{diag}_{S, \text{small}}^{\text{untl}} = \text{pr}_{\text{big}}^{\text{untl}}, \quad \text{pr}_{\text{big}^2}^{\text{untl}} \circ \text{diag}_{S, \text{big}}^{\text{untl}} = \text{pr}_{\text{big}}^{\text{untl}}.$$

11.6.7. Let us explain explicitly what these maps are when $S = \text{pt}$, and the map $S \rightarrow \text{Ran}^{\text{untl}}$ corresponds to a point $\underline{x} \in \text{Ran}$.

The categorical prestacks classify $S^{\subseteq, \text{untl}}$ and $S^{\subseteq^2, \text{untl}}$

$$(\underline{x} \subseteq \underline{x}_1) \text{ and } (\underline{x} \subseteq \underline{x}_1 \subseteq \underline{x}_2)$$

respectively, with the morphisms given by inclusions of the $\underline{x}_1$'s and $\underline{x}_2$.

The map $\text{pr}_{\text{small}^2, S}^{\text{untl}}$ sends

$$(\underline{x} \subseteq \underline{x}_1 \subseteq \underline{x}_2) \mapsto (\underline{x} \subseteq \underline{x}_1);$$

the map $\text{pr}_{\text{big}^2}^{\text{untl}}$ sends

$$(\underline{x} \subseteq \underline{x}_1 \subseteq \underline{x}_2) \mapsto \underline{x}_2;$$

the map $\text{diag}_{S, \text{small}}^{\text{untl}}$ sends

$$(\underline{x} \subseteq \underline{x}_1) \mapsto (\underline{x} \subseteq \underline{x} \subseteq \underline{x}_1);$$

the map $\text{diag}_{S, \text{big}}^{\text{untl}}$ sends

$$(\underline{x} \subseteq \underline{x}_1) \mapsto (\underline{x} \subseteq \underline{x}_1 \subseteq \underline{x}_1).$$

11.6.8. We obtain the natural transformations

$$(11.35) \quad (\text{pr}_{\text{small}, S}^{\text{untl}})_! \circ (\text{pr}_{\text{big}}^{\text{untl}})^! \simeq (\text{pr}_{\text{small}, S}^{\text{untl}})_! \circ (\text{pr}_{\text{small}^2, S}^{\text{untl}})_! \circ (\text{diag}_{S, \text{big}}^{\text{untl}})_! \circ (\text{pr}_{\text{big}}^{\text{untl}})^! \simeq \\ \simeq (\text{pr}_{\text{small}, S}^{\text{untl}})_! \circ (\text{pr}_{\text{small}^2, S}^{\text{untl}})_! \circ (\text{diag}_{S, \text{big}}^{\text{untl}})_! \circ (\text{diag}_{S, \text{big}}^{\text{untl}})^! \circ (\text{pr}_{\text{big}^2}^{\text{untl}})^! \rightarrow (\text{pr}_{\text{small}, S}^{\text{untl}})_! \circ (\text{pr}_{\text{small}^2, S}^{\text{untl}})_! \circ (\text{pr}_{\text{big}^2}^{\text{untl}})^!$$

and

$$(11.36) \quad (\text{pr}_{\text{small}, S}^{\text{untl}})_! \circ (\text{pr}_{\text{big}}^{\text{untl}})^! \simeq (\text{pr}_{\text{small}, S}^{\text{untl}})_! \circ (\text{diag}_S^{\text{untl}})_! \circ (\text{pr}_{\text{small}, S}^{\text{untl}})_! \circ (\text{pr}_{\text{big}}^{\text{untl}})^! \simeq \\ \simeq (\text{pr}_{\text{small}, S}^{\text{untl}})_! \circ (\text{pr}_{\text{small}^2, S}^{\text{untl}})_! \circ (\text{diag}_{S, \text{small}}^{\text{untl}})_! \circ (\text{diag}_{S, \text{small}}^{\text{untl}})^! \circ (\text{pr}_{\text{big}^2}^{\text{untl}})^! \rightarrow (\text{pr}_{\text{small}, S}^{\text{untl}})_! \circ (\text{pr}_{\text{small}^2, S}^{\text{untl}})_! \circ (\text{pr}_{\text{big}^2}^{\text{untl}})^!$$

as functors

$$\text{D-mod}(\text{Ran}^{\text{untl}}) \rightarrow \text{D-mod}(S).$$

We need to show that the natural transformations (11.35) and (11.36) coincide.

This follows from the following observation: there exists a 1-morphism between the maps

$$\text{diag}_{S, \text{small}}^{\text{untl}} \xrightarrow{\alpha} \text{diag}_{S, \text{big}}^{\text{untl}},$$

so that the induced map

$$\text{pr}_{\text{big}}^{\text{untl}} \simeq \text{pr}_{\text{big}^2}^{\text{untl}} \circ \text{diag}_{S, \text{small}}^{\text{untl}} \xrightarrow{\alpha} \text{pr}_{\text{big}^2}^{\text{untl}} \circ \text{diag}_{S, \text{big}}^{\text{untl}} \simeq \text{pr}_{\text{big}}^{\text{untl}}$$

is the identity map.

11.6.9. Finally, we prove the commutativity of (11.24).

Let $\underline{F}^{\text{untl}}$ be an object of $\text{Func}^{\text{loc} \rightarrow \text{glob}, \text{lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$, and let $\underline{F}$ be the corresponding object of $\text{Func}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$. We need to establish an isomorphism between $\underline{F}^{f, \text{ins. unit}}$ and

$$\underline{F}^{\text{untl}, f, \text{ins. unit}}|_{\underline{\mathbf{C}}^{\text{loc}}} : \underline{\mathbf{C}}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \underline{\text{D-mod}}(\text{Ran}).$$

Let S be an affine scheme and let us be given an S -point of Ran . Denote

$$S^{\subseteq, \text{untl}} := S \times_{\text{Ran}^{\text{untl}}} \text{Ran}^{\subseteq, \text{intl}} \text{ and } S^{\subseteq} := S \times_{\text{Ran}} \text{Ran}^{\subseteq},$$

so that $S^{\subseteq}$ is the prestack in groupoids underlying $S^{\subseteq, \text{untl}}$.

The value of $\underline{F}^{\text{untl}, f, \text{ins. unit}}|_{\underline{\mathbf{C}}^{\text{loc}}}$ at the above S -point of Ran is given by the composition

$$(11.37) \quad \mathbf{C}_S^{\text{loc}} = \underline{\mathbf{C}}_S^{\text{loc}, \text{untl}} \xrightarrow{(11.28)} \Gamma^{\text{lax}}(S^{\subseteq, \text{untl}}, \underline{\mathbf{C}}^{\text{loc}, \text{untl}}) \xrightarrow{\underline{F}^{\text{untl}}} \\ \rightarrow \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(S^{\subseteq, \text{untl}}) \xrightarrow{(\text{pr}_{\text{small}, S}^{\text{untl}})^!} \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(S)$$

The value of $\underline{F}^{f, \text{ins.unit}}$ at the above S -point of Ran is given by

$$(11.38) \quad \mathbf{C}_S^{\text{loc}} = \underline{\mathbf{C}}_S^{\text{loc, untl}} \xrightarrow{(11.28)} \Gamma^{\text{lax}}(S^{\subseteq, \text{untl}}, \underline{\mathbf{C}}^{\text{loc, untl}}) \xrightarrow{\underline{F}^{\text{untl}}} \\ \rightarrow \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(S^{\subseteq, \text{untl}}) \rightarrow \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(S^{\subseteq}) \xrightarrow{(\text{pr}_{\text{small}, S})^!} \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(S).$$

Let $\mathbf{t}$ denote the tautological map $S^{\subseteq} \rightarrow S^{\subseteq, \text{untl}}$. We have a natural transformation

$$(11.39) \quad (\text{pr}_{\text{small}, S})^! \circ \mathbf{t}^! \simeq (\text{pr}_{\text{small}, S}^{\text{untl}})^! \circ \mathbf{t}^! \rightarrow (\text{pr}_{\text{small}, S}^{\text{untl}})^!$$

as functors

$$\text{D-mod}(S^{\subseteq, \text{untl}}) \rightarrow \text{D-mod}(S).$$

Thus, to establish an isomorphism between (11.37) and (11.38), it suffices to prove that the natural transformation (11.39) is an isomorphism. However, this is a variant of Lemma C.5.12 (with the same proof).

11.6.10. The compatibility of (11.16) and (11.25) follows from the commutativity of the diagram

$$\begin{array}{ccccc} (\text{diag}_S)^! \circ \mathbf{t}^! & \xrightarrow{\sim} & (\text{pr}_{\text{small}, S})^! \circ (\text{diag}_S)^! \circ (\text{diag}_S)^! \circ \mathbf{t}^! & \longrightarrow & (\text{pr}_{\text{small}, S})^! \circ \mathbf{t}^! \\ \downarrow & & & & \downarrow \\ (\text{diag}_S^{\text{untl}})^! & \xrightarrow{\sim} & (\text{pr}_{\text{small}, S}^{\text{untl}})^! \circ (\text{diag}_S^{\text{untl}})^! \circ (\text{diag}_S^{\text{untl}})^! & \longrightarrow & (\text{pr}_{\text{small}, S}^{\text{untl}})^!, \end{array}$$

as functors $\text{D-mod}(S^{\subseteq, \text{untl}}) \rightarrow \text{D-mod}(S)$, where

$$\text{diag}_S : S \rightarrow S^{\subseteq} \text{ and } \text{diag}_S^{\text{untl}} : S \rightarrow S^{\subseteq, \text{untl}}$$

are the corresponding maps.

11.7. Proofs of Lemmas 11.5.5, Lemma 11.6.3 and C.5.13.

11.7.1. *Proof of Lemma 11.6.3.* We need to show that, for any affine scheme S equipped with a map $S \rightarrow \text{Ran}^{\text{untl}}$, the functor

$$\text{D-mod}(S) \xrightarrow{(\text{pr}_{\text{small}, S})^!} \text{D-mod}(S^{\subseteq, \text{untl}})^{\text{strict}} \hookrightarrow \text{D-mod}(S^{\subseteq, \text{untl}})^{\text{lax}},$$

is fully faithful.

It suffices to show that the first arrow, i.e., $(\text{pr}_{\text{small}, S})^!$ is fully faithful. However, this follows from the fact that morphism $\text{pr}_{\text{small}, S}$ admits a left adjoint. Namely, it is given by diag_S .

□[Lemma 11.6.3]

11.7.2. *Proof of Lemma 11.5.5.* Let S be an affine scheme and let $\underline{x}$ be an S -point of Ran^{untl} . Consider the map

$$S_{\underline{x}}^{\subseteq, \text{untl}} := S \times_{\underline{x}, \text{Ran}^{\text{untl}}} \text{Ran}^{\subseteq, \text{untl}} \xrightarrow{\text{pr}_{\text{small}, S, \underline{x}}^{\text{untl}}} S.$$

The first assertion of the lemma is that the functor

$$\text{D-mod}(S) \xrightarrow{(\text{pr}_{\text{small}, S, \underline{x}}^{\text{untl}})^!} \text{D-mod}(S_{\underline{x}}^{\subseteq, \text{untl}})^{\text{strict}} \rightarrow \text{D-mod}(S_{\underline{x}}^{\subseteq, \text{untl}})$$

admits a left adjoint, to be denoted $(\text{pr}_{\text{small}, S, \underline{x}}^{\text{untl}})^!$. This is a particular case of Corollary C.4.12.

The second assertion of the lemma is that for a 1-morphism $\underline{x}_1 \xrightarrow{\alpha} \underline{x}_2$ in $\text{Maps}(S, \text{Ran}^{\text{untl}})$ and the corresponding map

$$S_{\underline{x}_2}^{\subseteq, \text{untl}} \xrightarrow{\alpha^*} S_{\underline{x}_1}^{\subseteq, \text{untl}},$$

the natural transformation

$$(\text{pr}_{\text{small}, S, \underline{x}_2}^{\text{untl}})^! \circ (\alpha^*)^! \simeq (\text{pr}_{\text{small}, S, \underline{x}_1}^{\text{untl}})^! \circ (\alpha^*)^! \rightarrow (\text{pr}_{\text{small}, S, \underline{x}_1}^{\text{untl}})^!$$

is an isomorphism.

This follows from the fact that the map α^* admits a value-wise left adjoint. Namely, the left adjoint in question attaches to an affine scheme S' with a map $g_1 : S' \rightarrow S_{\underline{x}_1}^{\subseteq, \text{unl}}$ the map $g_2 : S' \rightarrow S_{\underline{x}_2}^{\subseteq, \text{unl}}$ defined as follows: the corresponding map

$$S' \xrightarrow{g_2} S_{\underline{x}_2}^{\subseteq, \text{unl}} \xrightarrow{\text{pr}_{\text{big}}^{\text{unl}}} \text{Ran}^{\text{unl}}$$

is obtained by applying the map

$$\text{union} : \text{Ran}^{\text{unl}} \times \text{Ran}^{\text{unl}} \rightarrow \text{Ran}^{\text{unl}}$$

to

$$S' \xrightarrow{g_1} S_{\underline{x}_1}^{\subseteq, \text{unl}} \xrightarrow{\text{pr}_{\text{big}}^{\text{unl}}} \text{Ran}^{\text{unl}}$$

and

$$S' \xrightarrow{g_1} S_{\underline{x}_1}^{\subseteq, \text{unl}} \xrightarrow{\text{pr}_{\text{small}, S, \underline{x}_1}^{\text{unl}}} S \xrightarrow{g_2} \text{Ran}^{\text{unl}}.$$

□[Lemma 11.5.5]

11.8. Unitality as a property. A somewhat surprising fact is that, given a local unital structure on $\underline{\mathbf{C}}^{\text{loc}}$, a *unital*⁴⁷ structure on (a non-unital local-to-global functor) $\underline{\mathbf{F}}$ is actually a *property*, and not an additional piece of structure, as we shall presently explain.

The contents of this subsection are not necessary for the sequel.

11.8.1. Let $\underline{\mathbf{F}}$ be as in Sect. 11.1. Recall the morphism (11.18)

$$(11.40) \quad \mathbf{F}_{\mathcal{Z}} \rightarrow (\text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})_!) \circ \mathbf{F}_{\mathcal{Z}^{\subseteq}} \circ \text{ins. unit}_{\mathcal{Z}}$$

as functors $\mathbf{C}_{\mathcal{Z}}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(\mathcal{Z})$.

Definition 11.8.2. We shall say that $\underline{\mathbf{F}}$ satisfies *Global Unitality Axiom 1* if the natural transformation (11.18) is an isomorphism (for any $\mathcal{Z} \rightarrow \text{Ran}$).

11.8.3. Assume that $\mathbf{F}$ satisfies Global Unitality Axiom 1. Then inverting (11.40) and applying the $((\text{pr}_{\text{small}, \mathcal{Z}})_!)^{\dagger}, (\text{pr}_{\text{small}, \mathcal{Z}})^{\dagger}$ -adjunction, we obtain a map

$$(11.41) \quad \mathbf{F}_{\mathcal{Z}^{\subseteq}} \circ \text{ins. unit}_{\mathcal{Z}} \rightarrow (\text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})^{\dagger})^{\dagger}(\mathbf{F}_{\mathcal{Z}})$$

as functors $\mathbf{C}_{\mathcal{Z}}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(\mathcal{Z}^{\subseteq})$.

Definition 11.8.4. We shall say that $\underline{\mathbf{F}}$ satisfies *Global Unitality Axiom 2* if the natural transformation (11.41) is an isomorphism (for any $\mathcal{Z} \rightarrow \text{Ran}$).

Definition 11.8.5. We shall say that $\underline{\mathbf{F}}$ has a *global unital property* if it satisfies Axioms 1 and 2.

11.8.6. It is clear that if $\underline{\mathbf{F}}$ is the image under

$$\text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{unl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$$

of

$$\underline{\mathbf{F}}^{\text{unl}} \in \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{unl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}),$$

then $\underline{\mathbf{F}}$ has a global unitality property, see Sect. 11.4.5.

⁴⁷As opposed to lax unital.

11.8.7. Vice versa, suppose that

$$\underline{F} \in \text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$$

has a global unitality property.

Then the inverse of the isomorphism (11.41) provides an isomorphism as in (11.12).

More generally, one can show that in this case $\underline{F}$ comes from a uniquely defined object $\underline{F}^{\text{unl}} \in \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{unl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$.

In other words, we claim:

Proposition 11.8.8. *The composite functor*

$$\text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{unl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-unl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$$

is fully faithful, and its essential image consists of objects that have a global unitality property.

We will give two, rather different in spirit, proofs of Proposition 11.8.8: one in Sect. H.3.3, and another in Sect. I.

11.9. Factorization homology.

11.9.1. In this section we will assume that $\underline{\mathbf{C}}^{\text{loc}, \text{unl}}$ comes from a unital lax factorization category $\mathbf{A}$, i.e., $\underline{\mathbf{C}}^{\text{loc}, \text{unl}} = \underline{\mathbf{A}}$ in the notations of Sect. B.11.1. Let $\underline{F}$ be a functor

$$\underline{\mathbf{A}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran}),$$

equipped with a lax unital structure.

Let now $\mathcal{A}$ be a unital factorization algebra in $\mathbf{A}$. We will regard $\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})$ as a lax factorization category (see Sect. C.11.9), and consider the corresponding crystal of categories $\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})$ on Ran , which naturally extends to a crystal of categories over Ran^{unl} (see Sect. C.11.13).

In the particular case of factorization algebras, we will denote the functor $\text{ins.unit}_{\mathcal{Z}}$ of Sect. 11.6 by

$$\text{ins.vac}_{\mathcal{Z}} : \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})_{\mathcal{Z}} \rightarrow \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})_{\mathcal{Z} \subseteq}, \quad \mathcal{Z} \rightarrow \text{Ran}.$$

This notation is meant to emphasize that the unit in $\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})$ is the “vacuum module”, i.e., $\mathcal{A}$, viewed as a factorization module over itself.

11.9.2. Let

$$\underline{\text{oblv}}_{\mathcal{A}} : \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}) \rightarrow \mathbf{A}$$

be the tautological forgetful functor, viewed as a functor between crystals of categories over Ran .

Note that the unital structure on $\mathcal{A}$ defines on $\underline{\text{oblv}}_{\mathcal{A}}$ a structure of *lax functor* between crystals of categories over Ran^{unl} , see Sect. C.11.14.

In particular

$$\underline{F} \circ \underline{\text{oblv}}_{\mathcal{A}} : \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}) \rightarrow \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran})$$

also acquires a lax unital structure.

11.9.3. The functor of *factorization homology*

$$\underline{C}^{\text{fact}}(X, \mathcal{A}, -)^{\mathbf{F}} : \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}) \rightarrow \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran})$$

is by definition

$$(\underline{\mathbf{F}} \circ \underline{\mathbf{oblv}}_{\mathcal{A}})^{f^{\text{ins.unit}}},$$

see (11.15).

For $\mathcal{Z} \rightarrow \text{Ran}$ we will denote the corresponding functor

$$\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})_{\mathcal{Z}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \mathbf{D}\text{-mod}(\mathcal{Z})$$

by $\underline{C}^{\text{fact}}(X, \mathcal{A}, -)^{\mathbf{F}}_{\mathcal{Z}}$.

For $\mathcal{Z}$ pseudo-proper, we will denote the composition of $\underline{C}^{\text{fact}}(X, \mathcal{A}, -)^{\mathbf{F}}_{\mathcal{Z}}$ with $\text{Id} \otimes C_c(\mathcal{Z}, -)$ by $\underline{C}^{\text{fact}}(X, \mathcal{A}, -)^{\mathbf{F}}_{\mathcal{Z}}$.

For $\mathcal{Z} = \text{Ran}$ and the identity map we will denote the resulting functor

$$\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})_{\text{Ran}} \rightarrow \mathbf{C}^{\text{glob}}$$

by $\underline{C}^{\text{fact}}(X, \mathcal{A}, -)^{\mathbf{F}}$.

11.9.4. Recall the natural transformation (11.16). In our case, this is a map

$$(11.42) \quad \underline{\mathbf{F}} \circ \underline{\mathbf{oblv}}_{\mathcal{A}} \rightarrow \underline{C}^{\text{fact}}(X, \mathcal{A}, -)^{\mathbf{F}},$$

which we will denote by $\underline{\text{Cltr}}_{\mathcal{A}}$ and refer to as the “correlator” map.

For a given $\mathcal{Z} \rightarrow \text{Ran}$, this is a map

$$\text{Cltr}_{\mathcal{A}, \mathcal{Z}} : \mathbf{F} \circ \underline{\mathbf{oblv}}_{\mathcal{A}, \mathcal{Z}} \rightarrow \underline{C}^{\text{fact}}(X, \mathcal{A}, -)^{\mathbf{F}}_{\mathcal{Z}}.$$

11.9.5. *Example.* Let $\mathbf{A} = \text{Vect}$, $\mathbf{C}^{\text{glob}} = \text{Vect}$ and $\underline{\mathbf{F}} = \text{Id}$. In this case,

$$\underline{C}^{\text{fact}}(X, \mathcal{A}, -) := \underline{C}^{\text{fact}}(X, \mathcal{A}, -)^{\mathbf{F}}, \quad \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}) \rightarrow \underline{\mathbf{D}\text{-mod}}(\text{Ran})$$

is the usual functor of factorization homology.

11.9.6. A key fact for us is that according to Sect. 11.4.6, the functor $\underline{C}^{\text{fact}}(X, \mathcal{A}, -)^{\mathbf{F}}$ acquires a natural lax unital structure. Moreover, by Sect. 11.4.7, this lax unital structure is actually *strict*.

11.9.7. Let us apply the functor $\underline{C}^{\text{fact}}(X, \mathcal{A}, -)^{\mathbf{F}}_{\text{Ran}}$ to the object

$$(\mathbf{1}_{\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})})_{\text{Ran}} = \mathcal{A}_{\text{Ran}}^{\text{fact}} \in \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})_{\text{Ran}}$$

(see Sect. C.11.15) i.e., to $\mathcal{A}_{\text{Ran}}$, viewed as a factorization module over itself at Ran .

By unitality, the above object is of the form

$$\underline{C}^{\text{fact}}(X, \mathcal{A}) \otimes \omega_{\text{Ran}},$$

for a canonically defined object

$$(11.43) \quad \underline{C}^{\text{fact}}(X, \mathcal{A}) \in \mathbf{C}^{\text{glob}}.$$

The object (11.43) is called the *vacuum factorization homology* of $\mathcal{A}$.

Explicitly,

$$\underline{C}^{\text{fact}}(X, \mathcal{A}) \simeq (\text{Id} \otimes C_c(\text{Ran}, -)) \circ \mathbf{F}(\mathcal{A}_{\text{Ran}}).$$

11.9.8. Note that again by the unitality (Sect. 11.9.6) of the functor $\underline{C}^{\text{fact}}(X, \mathcal{A}, -)$, for any $\mathcal{Z} \rightarrow \text{Ran}$ and

$$\mathcal{A}^{\text{fact}_{\mathcal{Z}}} \in \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}},$$

(see Sects. B.9.7 and B.11.16 for the notation), we have

$$(11.44) \quad \underline{C}^{\text{fact}}(X, \mathcal{A}, \mathcal{A}^{\text{fact}_{\mathcal{Z}}})_{\mathcal{Z}} \simeq \underline{C}^{\text{fact}}(X, \mathcal{A}) \otimes \omega_{\mathcal{Z}}.$$

In particular, for any $\underline{x} \in \text{Ran}$,

$$(11.45) \quad \underline{C}^{\text{fact}}(X, \mathcal{A}, \mathcal{A}^{\text{fact}_{\underline{x}}})_{\underline{x}} \simeq \underline{C}^{\text{fact}}(X, \mathcal{A}).$$

11.9.9. Let us write out explicitly the proof of the fact that the functor $\underline{\mathcal{C}}^{\text{fact}}(X, \mathcal{A}, -)^F$ is unital (we will essentially repeat the argument from Sect. 11.6.4).

Fix $\mathcal{Z} \rightarrow \text{Ran}$ and consider an object $\mathcal{M} \in \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})_{\mathcal{Z}}$. From it we produce an object

$$\text{ins. unit}_{\mathcal{Z}}(\mathcal{M}) \in \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})_{\mathcal{Z} \subseteq},$$

and further

$$\text{ins. unit}_{\mathcal{Z} \subseteq}(\text{ins. unit}_{\mathcal{Z}}(\mathcal{M})) \in \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})_{\mathcal{Z} \subseteq^2},$$

where

$$\mathcal{Z} \subseteq^2 := (\mathcal{Z} \subseteq) \subseteq.$$

Consider the object

$$\mathcal{M}' := \mathbf{F}_{\mathcal{Z} \subseteq^2} \circ \mathbf{oblv}_{\mathcal{A}, \mathcal{Z} \subseteq^2}(\text{ins. unit}_{\mathcal{Z} \subseteq}(\text{ins. unit}_{\mathcal{Z}}(\mathcal{M}))) \in \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(\mathcal{Z} \subseteq^2).$$

Consider now the map

$$\text{diag}_{\mathcal{Z}, \text{big}} : \mathcal{Z} \subseteq \rightarrow \mathcal{Z} \subseteq^2$$

(see Sect. 11.6.6).

It gives rise to a map

$$(11.46) \quad (\text{pr}_{\text{small}, \mathcal{Z}})_! \circ (\text{diag}_{\mathcal{Z}, \text{big}})^!(\mathcal{M}') \simeq (\text{pr}_{\text{small}, \mathcal{Z}})_! \circ (\text{pr}_{\text{small}, \mathcal{Z} \subseteq})_! \circ (\text{diag}_{\mathcal{Z}, \text{big}})_! \circ (\text{diag}_{\mathcal{Z}, \text{big}})^!(\mathcal{M}') \rightarrow \\ \rightarrow (\text{pr}_{\text{small}, \mathcal{Z}})_! \circ (\text{pr}_{\text{small}, \mathcal{Z} \subseteq})_!(\mathcal{M}'),$$

and we wish to show that this map is an isomorphism.

Now, the lax unital structure on $\underline{\mathbf{F}}$ implies that $\mathcal{M}'$ is the pullback of an object in

$$\mathcal{M}'' \in \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(\mathcal{Z} \subseteq^2, \text{unl}).$$

As in Lemma C.5.13, one shows that one can replace both sides in (11.46) by their unital versions, i.e., it is sufficient to show that the corresponding map

$$(\text{pr}_{\text{small}, \mathcal{Z}}^{\text{unl}})_! \circ (\text{diag}_{\mathcal{Z}, \text{big}}^{\text{unl}})^!(\mathcal{M}'') \simeq (\text{pr}_{\text{small}, \mathcal{Z}}^{\text{unl}})_! \circ (\text{pr}_{\text{small}, \mathcal{Z} \subseteq}^{\text{unl}})_! \circ (\text{diag}_{\mathcal{Z}, \text{big}}^{\text{unl}})_! \circ (\text{diag}_{\mathcal{Z}, \text{big}}^{\text{unl}})^!(\mathcal{M}'') \rightarrow \\ \rightarrow (\text{pr}_{\text{small}, \mathcal{Z}}^{\text{unl}})_! \circ (\text{pr}_{\text{small}, \mathcal{Z} \subseteq}^{\text{unl}})_!(\mathcal{M}'')$$

is an isomorphism.

However, the latter follows from the fact that the map

$$\text{diag}_{\mathcal{Z}, \text{big}}^{\text{unl}} : \mathcal{Z} \subseteq, \text{unl} \rightarrow \mathcal{Z} \subseteq^2, \text{unl}$$

is value-wise cofinal.

□

11.9.10. In what follows we will need the following variant of the isomorphism we just proved:

The natural transformation $\underline{\mathbf{F}} \rightarrow \underline{\mathbf{F}}^{f \text{ ins. unit}}$ induces a natural transformation

$$(11.47) \quad \underline{\mathcal{C}}^{\text{fact}}(X, \mathcal{A}, -)^F \rightarrow \underline{\mathcal{C}}^{\text{fact}}(X, \mathcal{A}, -)^{F^{f \text{ ins. unit}}}.$$

We claim:

Lemma 11.9.11. *The natural transformation (11.47) is an isomorphism.*

Proof. By the construction of (11.15), the unit in $\mathcal{A}$ gives rise to a natural transformation

$$(11.48) \quad \mathbf{F}^{f \text{ ins. unit}} \circ \mathbf{oblv}_{\mathcal{A}} \rightarrow (\mathbf{F} \circ \mathbf{oblv}_{\mathcal{A}})^{f \text{ ins. unit}} = \underline{\mathcal{C}}^{\text{fact}}(X, \mathcal{A}, -)^F.$$

Applying $\int \text{ins. unit}$ we obtain a natural transformation

$$\underline{\mathcal{C}}^{\text{fact}}(X, \mathcal{A}, -)^{F^{f \text{ ins. unit}}} \rightarrow \int \text{ins. unit}(\underline{\mathcal{C}}^{\text{fact}}(X, \mathcal{A}, -)^F),$$

so that the diagram

$$\begin{array}{ccc}
\underline{C}^{\text{fact}}(X, \mathcal{A}, -)^{f \text{ ins. unit}} & \xrightarrow{(11.48)} & \int \text{ins. unit} \circ \int \text{ins. unit} (F \circ \underline{\text{oblv}}_{\mathcal{A}}) \\
\uparrow & & \sim \uparrow \int \text{ins. unit}(\underline{\text{Cltr}}) \\
\underline{C}^{\text{fact}}(X, \mathcal{A}, -)^F & \xrightarrow{=} & \int \text{ins. unit} (F \circ \underline{\text{oblv}}_{\mathcal{A}})
\end{array}$$

commutes.

Hence, it suffices to check that the map (11.48) is an isomorphism. However, this is done by the same argument as in Sect. 11.9.9. $\square$

11.9.12. Here is a particular case of Lemma 11.9.11 that we will need:

Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be a pair of unital factorization algebras in a unital lax factorization category $\mathbf{A}_0$, and let $\phi : \mathcal{A}_1 \rightarrow \mathcal{A}_2$ be a unital homomorphism. Denote by res^ϕ the resulting functor

$$\mathcal{A}_2\text{-mod}(\mathbf{A}_0) \rightarrow \mathcal{A}_1\text{-mod}(\mathbf{A}_0).$$

Let us be given a lax unital functor

$$\underline{F}_0 : \mathbf{A}_0 \rightarrow \mathbf{C}^{\text{glob}} \otimes \underline{\text{D-mod}}(\text{Ran}).$$

For a given $\mathcal{Z} \rightarrow \text{Ran}$ we have a natural transformation

$$\begin{aligned}
(11.49) \quad \underline{C}^{\text{fact}}(X; \mathcal{A}_2, -)_{\mathcal{Z}}^{F_0} &= (\text{pr}_{\text{small}, \mathcal{Z}})_! \circ \underline{\text{oblv}}_{\mathcal{A}_2, \mathcal{Z} \subseteq} \circ \text{ins. vac}_{\mathcal{Z}, \mathcal{A}_2} \simeq \\
&\simeq (\text{pr}_{\text{small}, \mathcal{Z}})_! \circ \underline{\text{oblv}}_{\mathcal{A}_1, \mathcal{Z} \subseteq} \circ \text{res}^\phi \circ \text{ins. vac}_{\mathcal{Z}, \mathcal{A}_2} \simeq \\
&\simeq (\text{pr}_{\text{small}, \mathcal{Z}})_! \circ (\text{pr}_{\text{small}, \mathcal{Z} \subseteq})_! \circ (\text{diag}_{\mathcal{Z} \subseteq})_! \circ \underline{\text{oblv}}_{\mathcal{A}_1, \mathcal{Z} \subseteq} \circ \text{res}^\phi \circ \text{ins. vac}_{\mathcal{Z}, \mathcal{A}_2} \simeq \\
&\simeq (\text{pr}_{\text{small}, \mathcal{Z}})_! \circ (\text{pr}_{\text{small}, \mathcal{Z} \subseteq})_! \circ \underline{\text{oblv}}_{\mathcal{A}_1, (\mathcal{Z} \subseteq) \subseteq} \circ (\text{diag}_{\mathcal{Z} \subseteq})_! \circ \text{res}^\phi \circ \text{ins. vac}_{\mathcal{Z}, \mathcal{A}_2} \simeq \\
&\simeq (\text{pr}_{\text{small}, \mathcal{Z}})_! \circ (\text{pr}_{\text{small}, \mathcal{Z} \subseteq})_! \circ \underline{\text{oblv}}_{\mathcal{A}_1, (\mathcal{Z} \subseteq) \subseteq} \circ (\text{diag}_{\mathcal{Z} \subseteq})_! \circ (\text{diag}_{\mathcal{Z} \subseteq})^! \circ \text{ins. vac}_{\mathcal{Z} \subseteq, \mathcal{A}_1} \circ \text{res}^\phi \circ \text{ins. vac}_{\mathcal{Z}, \mathcal{A}_2} \rightarrow \\
&\rightarrow (\text{pr}_{\text{small}, \mathcal{Z}})_! \circ (\text{pr}_{\text{small}, \mathcal{Z} \subseteq})_! \circ \underline{\text{oblv}}_{\mathcal{A}_1, (\mathcal{Z} \subseteq) \subseteq} \circ \text{ins. vac}_{\mathcal{Z} \subseteq, \mathcal{A}_1} \circ \text{res}^\phi \circ \text{ins. vac}_{\mathcal{Z}, \mathcal{A}_2} \simeq \\
&\simeq (\text{pr}_{\text{small}, \mathcal{Z}})_! \circ \underline{C}^{\text{fact}}(X; \mathcal{A}_1, -)_{\mathcal{Z} \subseteq}^{F_0} \circ \text{res}^\phi \circ \text{ins. vac}_{\mathcal{Z}, \mathcal{A}_2},
\end{aligned}$$

We claim:

Corollary 11.9.13. *The natural transformation (11.49) is an isomorphism.*

Proof. Take $\mathbf{A} := \mathcal{A}_1\text{-mod}(\mathbf{A}_0)$ and $\mathcal{A} = \mathcal{A}_2$, viewed as a unital factorization algebra in $\mathcal{A}_1\text{-mod}(\mathbf{A}_0)$. Take $\underline{F} = F_0 \circ \underline{\text{oblv}}_{\mathcal{A}_1}$.

Then the assertion follows from Lemma 11.9.11, where we note that the natural transformation (11.49) is (11.47). $\square$

11.9.14. Here is another application of Lemma 11.9.11. Let $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}$ be a lax unital factorization functor between lax factorization categories. Denote by $\underline{\Phi}$ the corresponding right-lax functor

$$\underline{\mathbf{A}}_1 \rightarrow \underline{\mathbf{A}}$$

as crystals of categories on Ran^{unl} .

Let us be given a lax unital local-to-global functor

$$\underline{F} : \mathbf{A} \rightarrow \mathbf{C}^{\text{glob}} \otimes \underline{\text{D-mod}}(\text{Ran}).$$

Note that $\underline{F}_1 := \underline{F} \circ \underline{\Phi}$ also acquires a lax unital structure. By functoriality, the natural transformation

$$\underline{F}_1 = \underline{F} \circ \underline{\Phi} \rightarrow \underline{F}^{f \text{ ins. unit}} \circ \underline{\Phi}$$

gives rise to a natural transformation

$$(11.50) \quad \underline{F}_1^{f \text{ ins. unit}} \rightarrow (\underline{F}^{f \text{ ins. unit}} \circ \underline{\Phi})^{f \text{ ins. unit}}.$$

Corollary 11.9.15. *The natural transformation (11.50) is an isomorphism.*

Proof. By Sect. 4.1.6, the functor Φ factors as

$$\mathbf{A}_1 \xrightarrow{\Phi^{\text{enh}}} \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}) \xrightarrow{\text{oblv}_{\mathcal{A}}} \mathbf{A},$$

where $\mathcal{A} = \Phi(\mathbf{1}_{\mathbf{A}_1})$.

Since the functor Φ^{enh} is strictly unital, the assertion of the corollary reduces to the case when $\mathbf{A}_1 = \mathcal{A}\text{-mod}(\mathbf{A})$ and $\Phi = \text{oblv}_{\mathcal{A}}$. However, in the latter case, the map (11.50) is the map (11.47). $\square$

12. PROPERTIES OF THE LOCALIZATION FUNCTOR

In this section we study the composition of the localization functors with three constructions of global nature:

- The forgetful functor $\text{D-mod}_{\kappa}(\text{Bun}_G) \rightarrow \text{QCoh}(\text{Bun}_G)$;
- The pullback functor $\text{D-mod}_{\kappa}(\text{Bun}_G) \rightarrow \text{D-mod}_{\kappa}(\text{Bun}_{G'})$ corresponding to a group homomorphism $G' \rightarrow G$;
- For a unipotent group-scheme N' , the functor of de Rham cohomology $\text{D-mod}(\text{Bun}_{N'}) \rightarrow \text{Vect}$.

The pattern in the three composite functors mentioned above is that they can all be expressed via a local operation, followed by another localization functor:

- Restriction $\text{KL}(G)_{\kappa, \text{Ran}} \rightarrow \text{Rep}(\mathfrak{L}^+(G))$, followed by $\mathcal{O}$ -module localization $\text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}} \rightarrow \text{QCoh}(\text{Bun}_G)$;
- Restriction $\text{KL}(G)_{\kappa, \text{Ran}} \rightarrow \text{KL}(G')_{\kappa, \text{Ran}}$, followed by

$$\text{Loc}_{G', \kappa} : \text{KL}(G')_{\kappa, \text{Ran}} \rightarrow \text{D-mod}_{\kappa}(\text{Bun}_{G'});$$

- The functor of BRST reduction $\text{KL}(N')_{\text{Ran}} \rightarrow \text{Vect}$.

However, there is a caveat, common to all three of these situations: in order for the local operation to reproduce the global one, we need to precompose the former with an endofunctor of the source given by $\text{ins.vac}_{\int_{\text{Ran}}}$, see (11.8).

12.1. Localization and the forgetful functor.

12.1.1. Note that by adjunction, the commutative diagram (10.17) gives rise to a natural transformation

$$(12.1) \quad \begin{array}{ccc} \text{QCoh}(\text{Bun}_G) & \xleftarrow{\text{oblv}_{\kappa}^l} & \text{D-mod}_{\kappa}(\text{Bun}_G) \\ \uparrow \text{Loc}_G^{\text{QCoh}} & \nearrow & \uparrow \text{Loc}_{G, \kappa} \\ \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}} & \xleftarrow{\text{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa}}} & \text{KL}(G)_{\kappa, \text{Ran}} \end{array}$$

The natural transformation in (12.1) is *not* an isomorphism (unless $G = 1$): namely, evaluate both circuits on

$$\text{Vac}(G)_{\kappa, x} \in \text{KL}(G)_{\kappa, x} \hookrightarrow \text{KL}(G)_{\kappa, \text{Ran}}$$

for some $x \in X$.

We will now draw another diagram, in which a natural transformation will be an isomorphism, which encodes another basic property of the localization functor.

12.1.2. Being unital factorization categories, both $\mathrm{KL}(G)_\kappa$ and $\mathrm{Rep}(\mathfrak{L}^+(G))$, viewed as crystals of categories over Ran , carry local unital structures (see Sect. 11.2.1 for what this means). Furthermore, the local-to-global functors⁴⁸

$$(12.2) \quad \underline{\mathrm{Loc}}_{G,\kappa} : \underline{\mathrm{KL}}(G)_\kappa \rightarrow \mathrm{D}\text{-}\mathrm{mod}_\kappa(\mathrm{Bun}_G) \otimes \underline{\mathrm{D}\text{-}\mathrm{mod}}(\mathrm{Ran})$$

and

$$(12.3) \quad \underline{\mathrm{Loc}}_G^{\mathrm{QCoh}} : \mathrm{Rep}(\mathfrak{L}^+(G)) \rightarrow \mathrm{QCoh}(\mathrm{Bun}_G) \otimes \underline{\mathrm{D}\text{-}\mathrm{mod}}(\mathrm{Ran})$$

both carry naturally defined (strict) unital structures (see Sect. 11.3.1 for what this means).

Note, however, that the restriction functor

$$\underline{\mathrm{oblv}}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} : \underline{\mathrm{KL}}(G)_\kappa \rightarrow \underline{\mathrm{Rep}}(\mathfrak{L}^+(G))$$

is merely right-lax, (as is the case for any factorization functor that is lax unital as opposed to strictly unital).

12.1.3. The natural transformation

$$\underline{\mathrm{Loc}}_G^{\mathrm{QCoh}} \circ \underline{\mathrm{oblv}}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} \rightarrow \underline{\mathrm{oblv}}_\kappa^l \circ \underline{\mathrm{Loc}}_{G,\kappa}$$

from diagram (12.1) can be viewed as a natural transformation

$$(12.4) \quad \underline{\mathrm{Loc}}_G^{\mathrm{QCoh}} \circ \underline{\mathrm{oblv}}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} \rightarrow \underline{\mathrm{oblv}}_\kappa^l \circ \underline{\mathrm{Loc}}_{G,\kappa}$$

between functors

$$\underline{\mathrm{KL}}(G)_\kappa \rightrightarrows \mathrm{QCoh}(\mathrm{Bun}_G) \otimes \underline{\mathrm{D}\text{-}\mathrm{mod}}(\mathrm{Ran})$$

between crystals of categories over Ran (see Sect. 11.1.5).

The two sides in (12.4) are lax unital local-to-global functors, and the map (12.4) is compatible with the lax unital structures.

12.1.4. Note now that the right-hand side in (12.4) is strictly unital (because $\underline{\mathrm{Loc}}_{G,\kappa}$ is unital). Hence by Sect. 11.4.8, the map (12.4) gives rise to a map⁴⁹

$$(12.5) \quad \left(\underline{\mathrm{Loc}}_G^{\mathrm{QCoh}} \circ \underline{\mathrm{oblv}}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} \right)^{\int \mathrm{ins.vac}} \rightarrow \underline{\mathrm{oblv}}_\kappa^l \circ \underline{\mathrm{Loc}}_{G,\kappa}.$$

We claim:

Theorem 12.1.5. *The natural transformation (12.5) is an isomorphism.*

Remark 12.1.6. One can interpret Theorem 12.1.5 as follows: the natural transformation (12.4) fails to be an isomorphism because the right-hand side is unital (i.e., insertion of vacuum does not change the value of the functor), but the left-hand side is only lax unital. But once we correct this by applying $\int \mathrm{ins.vac}$, the corresponding map becomes an isomorphism.

12.1.7. We will now reformulate Theorem 12.1.5 in concrete terms, which do not explicitly involve categorical prestacks:

Let $\int \mathrm{ins.vac}$ be the endofunctor of $\mathrm{KL}(G)_{\kappa, \mathrm{Ran}}$ from Sect. 11.2.12. We have:

Theorem 12.1.8. *The natural transformation in (12.1) becomes an isomorphism after precomposing with $\int \mathrm{ins.vac}$.*

Note that Theorems 12.1.5 and 12.1.8 are logically equivalent. This follows from Sect. 11.1.5 and the commutativity of (11.24).

Theorem 12.1.8 will be proved in Sect. 13.3⁵⁰.

⁴⁸In the formulas below, the underline has the meaning from Sect. 11.1.1.

⁴⁹In the formula below, for the factorization category $\mathrm{KL}(G)_\kappa$ we use the notation $\mathrm{ins.vac}$ instead of $\mathrm{ins.unit}$.

⁵⁰We supply a proof for completeness. An equivalent statement appears in [CF, Lemma 4.4.16 and Variant 4.4.17]

12.1.9. Note that since $\underline{\text{Loc}}_{G,\kappa}$ is strictly unital, the map

$$\text{Loc}_{G,\kappa} \rightarrow \text{Loc}_{G,\kappa} \circ \int_{\text{Ran}} \text{ins. vac}$$

is an isomorphism (see Sect. 11.4.5). Hence, Theorem 12.1.8 implies:

Corollary 12.1.10. *We have a commutative diagram*

$$\begin{array}{ccc} \text{D-mod}_{\kappa}(\text{Bun}_G) & \xrightarrow{\text{oblv}_{\kappa}^I} & \text{QCoh}(\text{Bun}_G) \\ \text{Loc}_{G,\kappa} \uparrow & & \uparrow \text{Loc}_G^{\text{QCoh}} \\ \text{KL}(G)_{\kappa,\text{Ran}} & \xrightarrow{f \text{ ins. vac}} \text{KL}(G)_{\kappa,\text{Ran}} \xrightarrow{\text{oblv}_{\mathfrak{L}^+(G)}^{(\mathfrak{g}, \mathfrak{L}^+(G))\kappa}} & \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}}. \end{array}$$

12.1.11. Consider the functor

$$\underline{\text{Rep}}(\mathfrak{L}^+(G)) \xrightarrow{\text{Loc}_G^{\text{QCoh}}} \text{QCoh}(\text{Bun}_G) \otimes \underline{\text{D-mod}}(\text{Ran}).$$

By Sect. 11.9.3, for a factorization algebra $\mathcal{A} \in \text{Rep}(\mathfrak{L}^+(G))$, we can consider the local-to-global functor

$$\underline{\text{C}}^{\text{fact}}(X, \mathcal{A}, -)^{\text{Loc}_G^{\text{QCoh}}} : \mathcal{A}\text{-mod}^{\text{fact}}(\text{Rep}(\mathfrak{L}^+(G))) \rightarrow \text{QCoh}(\text{Bun}_G) \otimes \underline{\text{D-mod}}(\text{Ran}).$$

Moreover, by Sect. 11.9.6, the functor $\underline{\text{C}}^{\text{fact}}(X, \mathcal{A}, -)^{\text{Loc}_G^{\text{QCoh}}}$ is *strictly* unital.

12.1.12. Denote

$$\mathbb{V}_{G,\kappa} := \text{oblv}_{\mathfrak{L}^+(G)}^{(\mathfrak{g}, \mathfrak{L}^+(G))\kappa}(\text{Vac}(G)_{\kappa}),$$

viewed as a factorization algebra in $\text{Rep}(\mathfrak{L}^+(G))$.

The functor $\text{oblv}_{\mathfrak{L}^+(G)}^{(\mathfrak{g}, \mathfrak{L}^+(G))\kappa}$ upgrades to a strictly unital factorization functor

$$(\text{oblv}_{\mathfrak{L}^+(G)}^{(\mathfrak{g}, \mathfrak{L}^+(G))\kappa})^{\text{enh}} : \text{KL}(G)_{\kappa} \rightarrow \mathbb{V}_{G,\kappa}\text{-mod}^{\text{fact}}(\text{Rep}(\mathfrak{L}^+(G))).$$

Note that whereas $\text{oblv}_{\mathfrak{L}^+(G)}^{(\mathfrak{g}, \mathfrak{L}^+(G))\kappa}$ was right-lax, when viewed as a functor between sheaves of categories over Ran^{unl} , the functor $(\text{oblv}_{\mathfrak{L}^+(G)}^{(\mathfrak{g}, \mathfrak{L}^+(G))\kappa})^{\text{enh}}$ is strict.

12.1.13. With these notations, from Theorem 12.1.5, we obtain:

Corollary 12.1.14. *We have a commutative diagram*

$$(12.6) \quad \begin{array}{ccc} \text{D-mod}_{\kappa}(\text{Bun}_G) \otimes \underline{\text{D-mod}}(\text{Ran}) & \xrightarrow{\text{oblv}_{\kappa}^I \otimes \text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \underline{\text{D-mod}}(\text{Ran}) \\ \text{Loc}_{G,\kappa} \uparrow & & \uparrow \underline{\text{C}}^{\text{fact}}(X, \mathbb{V}_{G,\kappa}, -)^{\text{Loc}_G^{\text{QCoh}}} \\ \underline{\text{KL}}(G)_{\kappa} & \xrightarrow{(\text{oblv}_{\mathfrak{L}^+(G)}^{(\mathfrak{g}, \mathfrak{L}^+(G))\kappa})^{\text{enh}}} & \mathbb{V}_{G,\kappa}\text{-mod}^{\text{fact}}(\text{Rep}(\mathfrak{L}^+(G))). \end{array}$$

Integrating over Ran , we obtain:

Corollary 12.1.15. *We have a commutative diagram*

$$(12.7) \quad \begin{array}{ccc} \text{D-mod}_{\kappa}(\text{Bun}_G) & \xrightarrow{\text{oblv}_{\kappa}^I} & \text{QCoh}(\text{Bun}_G) \\ \text{Loc}_{G,\kappa} \uparrow & & \uparrow \underline{\text{C}}^{\text{fact}}(X, \mathbb{V}_{G,\kappa}, -)^{\text{Loc}_G^{\text{QCoh}}} \\ \text{KL}(G)_{\kappa,\text{Ran}} & \xrightarrow{(\text{oblv}_{\mathfrak{L}^+(G)}^{(\mathfrak{g}, \mathfrak{L}^+(G))\kappa})^{\text{enh}}} & \mathbb{V}_{G,\kappa}\text{-mod}^{\text{fact}}(\text{Rep}(\mathfrak{L}^+(G)))_{\text{Ran}}. \end{array}$$

12.2. Proof of Proposition 10.4.5. In this subsection we will use Theorem 12.1.8 (or rather Corollary 12.1.15) to deduce Proposition 10.4.5.

Let $\mathcal{P}_G$ be a k -point of Bun_G . Applying Corollary 12.1.15, we need to construct an isomorphism between

$$(12.8) \quad \text{KL}(G)_{\kappa, \underline{x}} \xrightarrow{\text{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa}} \text{Rep}(\mathfrak{L}^+(G))_{\underline{x}} \xrightarrow{\text{C}^{\text{fact}}(X, \mathbb{V}_{G, \kappa, -})_{\underline{x}}^{\text{Loc}_G^{\text{QCoh}}}} \text{QCoh}(\text{Bun}_G) \xrightarrow{*- \text{fiber at } \mathcal{P}_G} \text{Vect}$$

and

$$(12.9) \quad \text{KL}(G)_{\kappa, \underline{x}} \xrightarrow{\alpha_{\mathcal{P}_G, \text{taut}}} \text{KL}(G)_{\kappa, \mathcal{P}_G, \underline{x}} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa, \mathcal{P}_G, \underline{x}} \xrightarrow{(10.23)} \Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})\text{-mod} \xrightarrow{\text{coinv}_{\Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})}} \text{Vect}.$$

12.2.1. First, we note that the functor

$$\underline{\text{Rep}}(\mathfrak{L}^+(G)) \xrightarrow{\text{Loc}_G^{\text{QCoh}}} \text{QCoh}(\text{Bun}_G) \otimes \underline{\text{D-mod}}(\text{Ran}) \xrightarrow{(*\text{-fiber at } \mathcal{P}_G) \otimes \text{Id}} \underline{\text{D-mod}}(\text{Ran})$$

is a $\mathcal{P}_G$ -twisted version of the forgetful functor. Denote it by $\text{oblv}_{\mathfrak{L}^+(G), \mathcal{P}_G}$. We can view it as a factorization functor

$$\text{oblv}_{\mathfrak{L}^+(G), \mathcal{P}_G} : \text{Rep}(\mathfrak{L}^+(G)) \rightarrow \text{Vect}.$$

Denote

$$\mathbb{V}_{\mathfrak{g}, \kappa, \mathcal{P}_G} := \text{oblv}_{\mathfrak{L}^+(G), \mathcal{P}_G} \circ \text{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa} (\text{Vac}(G)_{\kappa}).$$

This is a factorization algebra in Vect , which is a $\mathcal{P}_G$ -twisted version of the vacuum representation $\mathbb{V}_{\mathfrak{g}, \kappa}$. We can rewrite the functor

$$\underline{\text{Rep}}(\mathfrak{L}^+(G)) \xrightarrow{\text{C}^{\text{fact}}(X, \mathbb{V}_{G, \kappa, -})_{\underline{x}}^{\text{Loc}_G^{\text{QCoh}}}} \text{QCoh}(\text{Bun}_G) \otimes \underline{\text{D-mod}}(\text{Ran}) \xrightarrow{*- \text{fiber at } \mathcal{P}_G \otimes \text{Id}} \underline{\text{D-mod}}(\text{Ran})$$

as

$$\underline{\text{Rep}}(\mathfrak{L}^+(G)) \xrightarrow{\text{C}^{\text{fact}}(X, \mathbb{V}_{\mathfrak{g}, \kappa, \mathcal{P}_G, -})_{\underline{x}}^{\text{oblv}_{\mathfrak{L}^+(G), \mathcal{P}_G}}} \underline{\text{D-mod}}(\text{Ran}).$$

12.2.2. Consider the forgetful (factorization) functor

$$\text{oblv}_{\widehat{\mathfrak{g}}, \mathcal{P}_G} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa, \mathcal{P}_G, \underline{x}} \rightarrow \text{Vect}.$$

Note that

$$\text{oblv}_{\widehat{\mathfrak{g}}, \mathcal{P}_G} \circ \text{oblv}_{\mathfrak{L}^+(G)} \circ \alpha_{\mathcal{P}_G, \text{taut}} \simeq \text{oblv}_{\mathfrak{L}^+(G), \mathcal{P}_G} \circ \text{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa}$$

as factorization functors

$$\text{KL}(G)_{\kappa} \rightarrow \text{Vect}.$$

In particular,

$$\text{oblv}_{\widehat{\mathfrak{g}}, \mathcal{P}_G} (\text{Vac}(G)_{\kappa, \mathcal{P}_G}) \simeq \mathbb{V}_{\mathfrak{g}, \kappa, \mathcal{P}_G},$$

and we obtain that the functor $\text{oblv}_{\widehat{\mathfrak{g}}, \mathcal{P}_G}$ enhances to a functor

$$(12.10) \quad \text{oblv}_{\widehat{\mathfrak{g}}, \mathcal{P}_G}^{\text{enh}} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa, \mathcal{P}_G, \underline{x}} \rightarrow \mathbb{V}_{\mathfrak{g}, \kappa, \mathcal{P}_G}\text{-mod}^{\text{fact}},$$

see Sect. 4.1.6.

12.2.3. We obtain that we can rewrite (12.8) as

$$(12.11) \quad \mathrm{KL}(G)_{\kappa, \underline{x}} \xrightarrow{\alpha_{\mathcal{P}_G, \text{taut}}} \mathrm{KL}(G)_{\kappa, \mathcal{P}_G, \underline{x}} \xrightarrow{\mathrm{oblv}_{\underline{x}^+(G)}} \widehat{\mathfrak{g}}\text{-mod}_{\kappa, \mathcal{P}_G, \underline{x}} \xrightarrow{\mathrm{oblv}_{\widehat{\mathfrak{g}}, \mathcal{P}_G}^{\mathrm{enh}}} \mathbb{V}_{\mathfrak{g}, \kappa, \mathcal{P}_G}\text{-mod}_{\underline{x}}^{\mathrm{fact}} \xrightarrow{\mathrm{C}^{\mathrm{fact}}(X, \mathbb{V}_{\mathfrak{g}, \kappa, \mathcal{P}_G}, -)} \mathrm{Vect}.$$

Thus, we obtain that it suffices to show that the composition

$$(12.12) \quad \widehat{\mathfrak{g}}\text{-mod}_{\kappa, \mathcal{P}_G, \underline{x}} \xrightarrow{\mathrm{oblv}_{\widehat{\mathfrak{g}}, \mathcal{P}_G}^{\mathrm{enh}}} \mathbb{V}_{\mathfrak{g}, \kappa, \mathcal{P}_G}\text{-mod}_{\underline{x}}^{\mathrm{fact}} \xrightarrow{\mathrm{C}^{\mathrm{fact}}(X, \mathbb{V}_{\mathfrak{g}, \kappa, \mathcal{P}_G}, -)} \mathrm{Vect}$$

identifies with

$$(12.13) \quad \widehat{\mathfrak{g}}\text{-mod}_{\kappa, \mathcal{P}_G, \underline{x}} \xrightarrow{(10.23)} \Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})\text{-mod} \xrightarrow{\mathrm{coinv}_{\Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})}} \mathrm{Vect}.$$

We will relate (12.12) to the functor of $\Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})$ -coinvariants using the calculation of factorization homology of (twisted) chiral envelopes of Lie-* algebras performed in [BD2].

12.2.4. Observe that the chiral algebra corresponding to the factorization algebra $\mathbb{V}_{\mathfrak{g}, \kappa, \mathcal{P}_G}$ identifies with $(\kappa$ -twisted) the chiral envelope

$$U^{\mathrm{ch}}(L_{\mathfrak{g}, \mathcal{P}_G})_{\kappa},$$

where $L_{\mathfrak{g}, \kappa, \mathcal{P}_G}$ is the $\mathcal{P}_G$ -twist of the Lie-algebra

$$\omega_X \oplus (\mathfrak{g} \otimes D_X).$$

Moreover, we have a canonical equivalence

$$L_{\mathfrak{g}, \kappa, \mathcal{P}_G}\text{-mod}_{\underline{x}}^{\mathrm{ch}} \simeq U^{\mathrm{ch}}(L_{\mathfrak{g}, \mathcal{P}_G})_{\kappa}\text{-mod}_{\underline{x}}^{\mathrm{ch}} \simeq \mathbb{V}_{\mathfrak{g}, \kappa, \mathcal{P}_G}\text{-mod}_{\underline{x}}^{\mathrm{fact}}$$

and a functor

$$(12.14) \quad L_{\mathfrak{g}, \kappa, \mathcal{P}_G}\text{-mod}_{\underline{x}}^{\mathrm{ch}} \rightarrow \Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})\text{-mod}.$$

Under these identifications, the functor

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa, \mathcal{P}_G, \underline{x}} \xrightarrow{\mathrm{oblv}_{\widehat{\mathfrak{g}}, \mathcal{P}_G}^{\mathrm{enh}}} \mathbb{V}_{\mathfrak{g}, \kappa, \mathcal{P}_G}\text{-mod}_{\underline{x}}^{\mathrm{fact}} \simeq L_{\mathfrak{g}, \kappa, \mathcal{P}_G}\text{-mod}_{\underline{x}}^{\mathrm{ch}} \xrightarrow{(12.14)} \Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})\text{-mod}$$

identifies with the functor (10.23).

12.2.5. Thus, it remains to show that the functor

$$L_{\mathfrak{g}, \kappa, \mathcal{P}_G}\text{-mod}_{\underline{x}}^{\mathrm{ch}} \simeq \mathbb{V}_{\mathfrak{g}, \kappa, \mathcal{P}_G}\text{-mod}_{\underline{x}}^{\mathrm{fact}} \xrightarrow{\mathrm{C}^{\mathrm{fact}}(X, \mathbb{V}_{\mathfrak{g}, \kappa, \mathcal{P}_G}, -)_{\underline{x}}} \mathrm{Vect}$$

identifies canonically with

$$L_{\mathfrak{g}, \kappa, \mathcal{P}_G}\text{-mod}_{\underline{x}}^{\mathrm{ch}} \xrightarrow{(12.14)} \Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})\text{-mod} \xrightarrow{\mathrm{coinv}_{\Gamma(X - \underline{x}, \mathfrak{g}_{\mathcal{P}_G})}} \mathrm{Vect}.$$

However, the latter is the assertion of [BD2, Proposition 4.8.2]

Remark 12.2.6. An alternative proof of the latter assertion can be found in [FraG, Corollary 6.4.4] in the special case when the coefficient module is the vacuum representation.

However, the method from [FraG] easily adapts to the present setting and can also be used to reprove [BD2, Proposition 4.8.2] in the generality in which we are using it.

□[Proposition 10.4.5]

12.3. Localization and restriction.

12.3.1. Let $\phi : G' \rightarrow G$ be a group homomorphism. We restrict the level κ to G' and consider the corresponding Kazhdan-Lusztig category

$$\mathrm{KL}(G')_{\kappa} := \widehat{\mathfrak{g}'}\text{-mod}_{\kappa}^{\mathfrak{L}^+(G')}$$

and the localization functor

$$\mathrm{Loc}_{G', \kappa} : \mathrm{KL}(G')_{\kappa, \mathrm{Ran}} \rightarrow \mathrm{D-mod}_{\kappa}(\mathrm{Bun}_{G'}).$$

12.3.2. The map ϕ gives rise to (factorization) restriction functors

$$\mathrm{Rep}(\mathfrak{L}^+(G)) \xrightarrow{\mathrm{res}^{\phi}} \mathrm{Rep}(\mathfrak{L}^+(G')) \text{ and } \mathrm{KL}(G)_{\kappa} \xrightarrow{\mathrm{res}^{\phi}} \mathrm{KL}(G')_{\kappa}$$

so that the diagram

$$\begin{array}{ccc} \mathrm{KL}(G)_{\kappa} & \xrightarrow{\mathrm{res}^{\phi}} & \mathrm{KL}(G')_{\kappa} \\ \mathrm{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa}} \downarrow & & \downarrow \mathrm{oblv}_{\mathfrak{L}^+(G')}^{(\widehat{\mathfrak{g}}', \mathfrak{L}^+(G'))_{\kappa}} \\ \mathrm{Rep}(\mathfrak{L}^+(G)) & \xrightarrow{\mathrm{res}^{\phi}} & \mathrm{Rep}(\mathfrak{L}^+(G')) \end{array}$$

commutes.

In addition, the map ϕ gives rise to a map

$$\phi^{\mathrm{glob}} : \mathrm{Bun}_{G'} \rightarrow \mathrm{Bun}_G,$$

which is compatible with the twistings and thus gives rise to a functor

$$(\phi^{\mathrm{glob}})_{\kappa}^! : \mathrm{D-mod}_{\kappa}(\mathrm{Bun}_G) \rightarrow \mathrm{D-mod}_{\kappa}(\mathrm{Bun}_{G'}),$$

which makes the diagram

$$\begin{array}{ccc} \mathrm{QCoh}(\mathrm{Bun}_G) & \xrightarrow{(\phi^{\mathrm{glob}})^*} & \mathrm{QCoh}(\mathrm{Bun}_{G'}) \\ \mathrm{oblv}_{\kappa}^! \uparrow & & \uparrow \mathrm{oblv}_{\kappa}^! \\ \mathrm{D-mod}_{\kappa}(\mathrm{Bun}_G) & \xrightarrow{(\phi^{\mathrm{glob}})_{\kappa}^!} & \mathrm{D-mod}_{\kappa}(\mathrm{Bun}_{G'}) \end{array}$$

commute.

12.3.3. Let $U \subset \mathrm{Bun}_G$ and $U' \subset \mathrm{Bun}_{G'}$ be a pair of quasi-compact substacks so that ϕ^{glob} maps $U' \rightarrow U$.

Consider the corresponding functors

$$\Gamma_{G, \kappa, U} : \mathrm{D-mod}_{\kappa}(U) \rightarrow \mathrm{KL}(G)_{\kappa, \mathrm{Ran}} \text{ and } \Gamma_{G', \kappa, U} : \mathrm{D-mod}_{\kappa}(U') \rightarrow \mathrm{KL}(G')_{\kappa, \mathrm{Ran}}.$$

By construction, we have a natural transformation

$$(12.15) \quad \begin{array}{ccc} \mathrm{D-mod}_{\kappa}(U) & \xrightarrow{(\phi^{\mathrm{glob}})_{\kappa}^!} & \mathrm{D-mod}_{\kappa}(U') \\ \Gamma_{G, \kappa, U} \downarrow & \nearrow & \downarrow \Gamma_{G', \kappa, U} \\ \mathrm{KL}(G)_{\kappa, \mathrm{Ran}} & \xrightarrow{\mathrm{res}^{\phi}} & \mathrm{KL}(G')_{\kappa, \mathrm{Ran}} \end{array}$$

By adjunction, we obtain a natural transformation

$$(12.16) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_\kappa(U) & \xrightarrow{(\phi^{\mathrm{glob}})_\kappa!} & \mathrm{D}\text{-mod}_\kappa(U') \\ \mathrm{Loc}_{G',\kappa,U} \uparrow & \swarrow & \uparrow \mathrm{Loc}_{G',\kappa,U'} \\ \mathrm{KL}(G)_{\kappa,\mathrm{Ran}} & \xrightarrow{\mathrm{res}^\phi} & \mathrm{KL}(G')_{\kappa,\mathrm{Ran}} \end{array}$$

Passing to the limit over U , from (12.16), we obtain a natural transformation

$$(12.17) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_\kappa(\mathrm{Bun}_G) & \xrightarrow{(\phi^{\mathrm{glob}})_\kappa!} & \mathrm{D}\text{-mod}_\kappa(\mathrm{Bun}_{G'}) \\ \mathrm{Loc}_{G,\kappa} \uparrow & \swarrow & \uparrow \mathrm{Loc}_{G',\kappa} \\ \mathrm{KL}(G)_{\kappa,\mathrm{Ran}} & \xrightarrow{\mathrm{res}^\phi} & \mathrm{KL}(G')_{\kappa,\mathrm{Ran}} \end{array}$$

12.3.4. The natural transformation in (12.17) is not an isomorphism (unless ϕ itself is). We will now draw another diagram, in which the natural transformation is an isomorphism, and which expresses the composition

$$(\phi^{\mathrm{glob}})_\kappa! \circ \mathrm{Loc}_{G,\kappa}$$

via $\mathrm{Loc}_{G',\kappa}$.

This will be completely parallel to Sects. 12.1.2-12.1.4.

12.3.5. The natural transformation

$$\mathrm{Loc}_{G',\kappa} \circ \mathrm{res}^\phi \rightarrow (\phi^{\mathrm{glob}})_\kappa! \circ \mathrm{Loc}_{G,\kappa}$$

in (12.17) can be viewed as a natural transformation

$$(12.18) \quad \underline{\mathrm{Loc}}_{G',\kappa} \circ \underline{\mathrm{res}}^\phi \rightarrow (\phi^{\mathrm{glob}})_\kappa! \circ \underline{\mathrm{Loc}}_{G,\kappa}$$

between functors

$$\underline{\mathrm{KL}}(G)_\kappa \rightrightarrows \mathrm{D}\text{-mod}_\kappa(\mathrm{Bun}_{G'}) \otimes \underline{\mathrm{D}\text{-mod}}(\mathrm{Ran})$$

between crystals of categories over Ran (see Sect. 11.1.5).

The two sides in (12.18) are lax unital local-to-global functors, and the map (12.4) is compatible with the lax unital structures.

Note now that the right-hand side in (12.18) is strictly unital. Hence by Sect. 11.4.8, the map (12.18) gives rise to a map

$$(12.19) \quad \left(\underline{\mathrm{Loc}}_{G',\kappa} \circ \underline{\mathrm{res}}^\phi \right)^{\int^{\mathrm{ins.vac}}} \rightarrow (\phi^{\mathrm{glob}})_\kappa! \circ \underline{\mathrm{Loc}}_{G,\kappa}$$

as unital local-to-global functors.

We claim:

Proposition 12.3.6. *The natural transformation (12.19) is an isomorphism.*

Parallel to Sect. 12.3.6, we will now reformulate Proposition 12.3.6 in several (equivalent) ways.

Proposition 12.3.7. *The natural transformation in (12.17) becomes an isomorphism after precomposing with $\int \text{ins.vac}$.*

Corollary 12.3.8. *We have a commutative diagram*

$$\begin{array}{ccccc} \text{D-mod}_\kappa(\text{Bun}_G) & & \xrightarrow{(\phi^{\text{glob}})_\kappa^!} & & \text{D-mod}_\kappa(\text{Bun}_{G'}) \\ \text{Loc}_{G,\kappa} \uparrow & & & & \uparrow \text{Loc}_{G',\kappa} \\ \text{KL}(G)_{\kappa,\text{Ran}} & \xrightarrow{\int \text{ins.vac}} & \text{KL}(G)_{\kappa,\text{Ran}} & \xrightarrow{\text{res}^\phi} & \text{KL}(G')_{\kappa,\text{Ran}} \end{array}$$

12.3.9. Consider the factorization algebra

$$\text{Vac}(G|_{G'})_\kappa := \text{res}^\phi(\text{Vac}(G)_\kappa) \in \text{KL}(G')_\kappa.$$

The functor res^ϕ upgrades to a factorization functor

$$(\text{res}^\phi)^{\text{enh}} : \text{KL}(G)_\kappa \rightarrow \text{Vac}(G|_{G'})_\kappa\text{-mod}^{\text{fact}}(\text{KL}(G')_\kappa).$$

Consider the functor

$$\underline{\mathbb{C}}^{\text{fact}}(X, \text{Vac}(G|_{G'})_\kappa, -)^{\text{Loc}_{G',\kappa}} : \text{Vac}(G|_{G'})_\kappa\text{-mod}^{\text{fact}}(\text{KL}(G')_\kappa) \rightarrow \text{D-mod}_\kappa(\text{Bun}_{G'}),$$

see Sect. 11.9.3.

Corollary 12.3.10. *We have a commutative diagram*

$$(12.20) \quad \begin{array}{ccc} \text{D-mod}_\kappa(\text{Bun}_G) \otimes \underline{\text{D-mod}}(\text{Ran}) & \xrightarrow{(\phi^{\text{glob}})_\kappa^! \otimes \text{Id}} & \text{D-mod}_\kappa(\text{Bun}_{G'}) \otimes \underline{\text{D-mod}}(\text{Ran}) \\ \text{Loc}_{G,\kappa} \uparrow & & \uparrow \underline{\mathbb{C}}^{\text{fact}}(X, \text{Vac}(G|_{G'})_\kappa, -)^{\text{Loc}_{G',\kappa}} \\ \text{KL}(G)_\kappa & \xrightarrow{(\text{res}^\phi)^{\text{enh}}} & \text{Vac}(G|_{G'})_\kappa\text{-mod}^{\text{fact}}(\text{KL}(G')_\kappa). \end{array}$$

Integrating over Ran , we obtain:

Corollary 12.3.11. *We have a commutative diagram*

$$(12.21) \quad \begin{array}{ccc} \text{D-mod}_\kappa(\text{Bun}_G) & \xrightarrow{(\phi^{\text{glob}})_\kappa^!} & \text{D-mod}_\kappa(\text{Bun}_{G'}) \\ \text{Loc}_{G,\kappa} \uparrow & & \uparrow \underline{\mathbb{C}}^{\text{fact}}(X, \text{Vac}(G|_{G'})_\kappa, -) \\ \text{KL}(G)_{\kappa,\text{Ran}} & \xrightarrow{(\text{res}^\phi)^{\text{enh}}} & \text{Vac}(G|_{G'})_\kappa\text{-mod}^{\text{fact}}(\text{KL}(G')_\kappa)_{\text{Ran}}. \end{array}$$

12.4. Proof of Proposition 12.3.6.

12.4.1. Since the functor

$$(12.22) \quad \text{oblv}_{\kappa,G'}^l : \text{D-mod}_\kappa(\text{Bun}_{G'}) \rightarrow \text{QCoh}(\text{Bun}_{G'})$$

is conservative, it is sufficient to prove that the natural transformation in (12.19) becomes an isomorphism after composing with

$$\text{oblv}_{\kappa,G'}^l \otimes \text{Id} : \text{D-mod}_\kappa(\text{Bun}_{G'}) \otimes \underline{\text{D-mod}}(\text{Ran}^{\text{untl}}) \rightarrow \text{QCoh}(\text{Bun}_{G'}) \otimes \underline{\text{D-mod}}(\text{Ran}^{\text{untl}}).$$

Remark 12.4.2. The idea of the proof is the following: the diagram

$$\begin{array}{ccc} \text{QCoh}(\text{Bun}_G) & \xrightarrow{(\phi^{\text{glob}})^*} & \text{QCoh}(\text{Bun}_{G'}) \\ \text{Loc}_G^{\text{QCoh}} \uparrow & & \uparrow \text{Loc}_{G'}^{\text{QCoh}} \\ \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}} & \xrightarrow{\text{res}^\phi} & \text{Rep}(\mathfrak{L}^+(G'))_{\text{Ran}} \end{array}$$

commutes tautologically.

Combining this observation with Corollary 12.1.10, we can express both

$$\mathbf{oblv}_\kappa^l \circ (\phi^{\text{glob}})_\kappa^! \circ \text{Loc}_{G,\kappa} \text{ and } \mathbf{oblv}_\kappa^l \circ \text{Loc}_{G',\kappa} \circ \text{res}^\phi$$

in terms of

$$\text{Loc}_{G'}^{\text{QCoh}} \circ \text{res}^\phi \circ \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa}.$$

The two sides will not match on the nose, but the difference will be accounted for by Corollary 11.9.15.

12.4.3. By construction, we have a commutative diagram of lax unital local-to-global functors

$$\begin{array}{ccc} (\mathbf{oblv}_{G',\kappa}^l \otimes \text{Id}) \circ \text{Loc}_{G',\kappa} \circ \text{res}^\phi & \xrightarrow{(\mathbf{oblv}_{\kappa,G'}^l \otimes \text{Id})(12.18)} & (\mathbf{oblv}_{G',\kappa}^l \otimes \text{Id}) \circ (\phi^{\text{glob}})_\kappa^! \circ \text{Loc}_{G,\kappa} \\ \uparrow (12.4) \circ \text{res}^\phi & & \uparrow \sim \\ \text{Loc}_{G'}^{\text{QCoh}} \circ \mathbf{oblv}_{\mathfrak{L}^+(G')}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G'))_\kappa} \circ \text{res}^\phi & & \\ \sim \uparrow & & \\ \text{Loc}_{G'}^{\text{QCoh}} \circ \text{res}^\phi \circ \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} & & \\ \sim \uparrow & & \\ (\phi^{\text{glob}})^* \circ \text{Loc}_G^{\text{QCoh}} \circ \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} & \xrightarrow{(\phi^{\text{glob}})^*(12.4)} & (\phi^{\text{glob}})^* \circ \mathbf{oblv}_{G,\kappa}^l \circ \text{Loc}_{G,\kappa}. \end{array}$$

By adjunction, we obtain a diagram

$$\begin{array}{ccc} (\mathbf{oblv}_{G',\kappa}^l \otimes \text{Id}) \circ (\text{Loc}_{G',\kappa} \circ \text{res}^\phi)^{f \text{ ins. vac}} & \xrightarrow{(\mathbf{oblv}_{\kappa,G'}^l \otimes \text{Id})(12.19)} & (\mathbf{oblv}_{G',\kappa}^l \otimes \text{Id}) \circ (\phi^{\text{glob}})_\kappa^! \circ \text{Loc}_{G,\kappa} \\ \sim \uparrow & & \uparrow = \\ ((\mathbf{oblv}_{G',\kappa}^l \otimes \text{Id}) \circ \text{Loc}_{G',\kappa} \circ \text{res}^\phi)^{f \text{ ins. vac}} & \longrightarrow & (\mathbf{oblv}_{G',\kappa}^l \otimes \text{Id}) \circ (\phi^{\text{glob}})_\kappa^! \circ \text{Loc}_{G,\kappa} \\ \uparrow ((12.4) \circ \text{res}^\phi)^{f \text{ ins. vac}} & & \\ \left(\text{Loc}_{G'}^{\text{QCoh}} \circ \mathbf{oblv}_{\mathfrak{L}^+(G')}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G'))_\kappa} \circ \text{res}^\phi \right)^{f \text{ ins. vac}} & & \\ \sim \uparrow & & \uparrow \sim \\ \left(\text{Loc}_{G'}^{\text{QCoh}} \circ \text{res}^\phi \circ \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} \right)^{f \text{ ins. vac}} & & \\ \sim \uparrow & & \\ \left((\phi^{\text{glob}})^* \circ \text{Loc}_G^{\text{QCoh}} \circ \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} \right)^{f \text{ ins. vac}} & \longrightarrow & (\phi^{\text{glob}})^* \circ \mathbf{oblv}_{G,\kappa}^l \circ \text{Loc}_{G,\kappa} \\ \sim \uparrow & & \uparrow = \\ (\phi^{\text{glob}})^* \circ \left(\text{Loc}_G^{\text{QCoh}} \circ \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} \right)^{f \text{ ins. vac}} & \xrightarrow[\sim]{(\phi^{\text{glob}})^*(12.5)} & (\phi^{\text{glob}})^* \circ \mathbf{oblv}_{G,\kappa}^l \circ \text{Loc}_{G,\kappa}, \end{array}$$

where the bottom arrow is an isomorphism by Theorem 12.1.5 (for G).

12.4.4. Hence, to prove that $(\mathbf{oblv}_{\kappa,G'}^l \otimes \text{Id})(12.19)$ is an isomorphism, it suffices to show that the map

$$\begin{aligned} (12.23) \quad \left(\text{Loc}_{G'}^{\text{QCoh}} \circ \mathbf{oblv}_{\mathfrak{L}^+(G')}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G'))_\kappa} \circ \text{res}^\phi \right)^{f \text{ ins. vac}} & \xrightarrow{((12.4) \circ \text{res}^\phi)^{f \text{ ins. vac}}} \\ & \rightarrow \left((\mathbf{oblv}_{G',\kappa}^l \otimes \text{Id}) \circ \text{Loc}_{G',\kappa} \circ \text{res}^\phi \right)^{f \text{ ins. vac}} \end{aligned}$$

induced by

$$\underline{\mathrm{Loc}}_{G'}^{\mathrm{QCoh}} \circ \underline{\mathrm{oblv}}_{\mathfrak{L}^+(G')}^{(\mathfrak{g}', \mathfrak{L}^+(G'))_\kappa} \circ \underline{\mathrm{res}}^\phi \xrightarrow{(12.4) \circ \underline{\mathrm{res}}^\phi} (\underline{\mathrm{oblv}}_{G', \kappa}^l \otimes \mathrm{Id}) \circ \underline{\mathrm{Loc}}_{G', \kappa} \circ \underline{\mathrm{res}}^\phi,$$

is an isomorphism.

Note that the map

$$\left(\underline{\mathrm{Loc}}_{G'}^{\mathrm{QCoh}} \circ \underline{\mathrm{oblv}}_{\mathfrak{L}^+(G')}^{(\mathfrak{g}', \mathfrak{L}^+(G'))_\kappa} \right)^{f \text{ ins.vac}} \xrightarrow{(12.4)^{f \text{ ins.vac}}} \left((\underline{\mathrm{oblv}}_{G', \kappa}^l \otimes \mathrm{Id}) \circ \underline{\mathrm{Loc}}_{G', \kappa} \right)^{f \text{ ins.vac}},$$

induced by

$$\underline{\mathrm{Loc}}_{G'}^{\mathrm{QCoh}} \circ \underline{\mathrm{oblv}}_{\mathfrak{L}^+(G')}^{(\mathfrak{g}', \mathfrak{L}^+(G'))_\kappa} \xrightarrow{(12.4)} (\underline{\mathrm{oblv}}_{G', \kappa}^l \otimes \mathrm{Id}) \circ \underline{\mathrm{Loc}}_{G', \kappa},$$

is an isomorphism, by Theorem 12.1.5 (for G').

This implies that (12.23) is an isomorphism by Corollary 11.9.15.

□[Proposition 12.3.6]

12.5. Localization for unipotent group-schemes. Let N' be a unipotent group-scheme over X . We will make the following technical assumption: N' admits a filtration by normal group-schemes with abelian subquotients.

12.5.1. Consider the factorization category

$$\mathrm{KL}(N') := \mathfrak{L}(\mathfrak{n}')\text{-mod}^{\mathfrak{L}^+(N')}.$$

Note that the critical level for N' is zero. In particular, we have the self-dualities

$$(12.24) \quad (\mathfrak{L}(\mathfrak{n}')\text{-mod})^\vee \simeq \mathfrak{L}(\mathfrak{n}')\text{-mod}.$$

$$(12.25) \quad \mathrm{KL}(N')^\vee \simeq \mathrm{KL}(N').$$

Both dualities take place in the sense of unital factorization categories, see Sect. C.11.5.

12.5.2. Note that since $\mathfrak{L}^+(N')$ is pro-unipotent, the forgetful functor

$$\underline{\mathrm{oblv}}_{\mathfrak{L}^+(N')} : \mathrm{KL}(N') \rightarrow \mathfrak{L}(\mathfrak{n}')\text{-mod}$$

is fully faithful.

Recall also that the right adjoint

$$\mathrm{Av}_*^{\mathfrak{L}^+(N')} : \mathfrak{L}(\mathfrak{n}')\text{-mod} \rightarrow \mathrm{KL}(N')$$

of $\underline{\mathrm{oblv}}_{\mathfrak{L}^+(N')}$ identifies also with the dual of $\underline{\mathrm{oblv}}_{\mathfrak{L}^+(N')}$ with respect to the self-dualities (12.24) and (12.25).

12.5.3. Consider the (factorization) functor of semi-infinite cohomology with respect to $\mathfrak{L}(\mathfrak{n}')$:

$$\mathrm{BRST}_{N'} : \mathfrak{L}(\mathfrak{n}')\text{-mod} \rightarrow \mathrm{Vect}.$$

In terms of the duality (12.24), the functor $\mathrm{BRST}_{N'}$ is given by

$$\langle k, - \rangle_{\mathfrak{L}(\mathfrak{n}')\text{-mod}},$$

where:

- $k \in \mathrm{KL}_{N'} \subset \mathfrak{L}(\mathfrak{n}')\text{-mod}$ is the *trivial* representation;
- $\langle -, - \rangle_{\mathfrak{L}(\mathfrak{n}')\text{-mod}}$ denotes the pairing $\mathfrak{L}(\mathfrak{n}')\text{-mod} \otimes \mathfrak{L}(\mathfrak{n}')\text{-mod} \rightarrow \mathrm{Vect}$, corresponding to (12.24).

Since k upgrades to an object of $\mathrm{FactAlg}^{\mathrm{until}}(X, \mathfrak{L}(\mathfrak{n}')\text{-mod})$, the functor $\mathrm{BRST}_{N'}$ has a natural lax unital factorization structure.

12.5.4. The value of $\text{BRST}_{\mathfrak{n}'}$ on $\text{Vac}(N')$ is

$$(12.26) \quad \langle k, \text{Vac}(N') \rangle_{\mathfrak{L}(\mathfrak{n}')\text{-mod}} \simeq \langle \text{Av}_*^{\mathfrak{L}^+(N')}(k), \text{Vac}(N') \rangle_{\text{KL}(N')} \simeq \\ \simeq \langle k, \text{Vac}(N') \rangle_{\text{KL}(N')} \simeq \langle k, \mathbf{1}_{\text{Rep}(\mathfrak{L}^+(N'))} \rangle_{\text{Rep}(\mathfrak{L}^+(N'))},$$

where:

- $\langle -, - \rangle_{\text{KL}(N')}$ is the pairing $\text{KL}(N') \otimes \text{KL}(N') \rightarrow \text{Vect}$ corresponding to (12.25);
- $\mathbf{1}_{\text{Rep}(\mathfrak{L}^+(N'))} \in \text{Rep}(\mathfrak{L}^+(N'))$ is the *trivial* representation;
- $\langle -, - \rangle_{\text{Rep}(\mathfrak{L}^+(N'))}$ denotes pairing $\text{Rep}(\mathfrak{L}^+(N')) \otimes \text{Rep}(\mathfrak{L}^+(N')) \rightarrow \text{Vect}$, corresponding to the natural self-duality of $\text{Rep}(\mathfrak{L}^+(N'))$, i.e.,

$$\langle V_1, V_2 \rangle_{\text{Rep}(\mathfrak{L}^+(N'))} = \mathcal{H}om_{\text{Rep}(\mathfrak{L}^+(N'))}(\mathbf{1}_{\text{Rep}(\mathfrak{L}^+(N'))}, V_1 \otimes V_2).$$

Denote

$$\Omega(\mathfrak{n}') := \text{inv}_{\mathfrak{L}^+(N')}(\mathbf{1}_{\text{Rep}(\mathfrak{L}^+(N'))}) \simeq C(\mathfrak{L}^+(\mathfrak{n}')) \in \text{FactAlg}^{\text{unfl}}(X).$$

Remark 12.5.5. Note also that $\Omega(\mathfrak{n}')$ is the commutative factorization algebra canonically isomorphic to

$$\text{Fact}(C_{\text{chev}}(L_{\mathfrak{n}'})),$$

where:

- $L_{\mathfrak{n}'}$ is the Lie-* algebra $\mathfrak{n}' \otimes D_X$;
- For a Lie-* algebra L , we denote by $C_{\text{chev}}(L) \in \text{ComAlg}(\text{D-mod}(X))$ its cohomological Chevalley complex, see [BD2, Sect. 1.4.10].

12.5.6. Hence, from (12.26), we obtain

$$(12.27) \quad \text{BRST}_{\mathfrak{n}'}(\text{Vac}(N')) \simeq \Omega(\mathfrak{n}'),$$

as factorization algebras (in Vect).

In particular, the functor $\text{BRST}_{\mathfrak{n}'}$ enhances to a functor

$$\text{BRST}_{\mathfrak{n}'}^{\text{enh}} : \mathfrak{L}(\mathfrak{n}')\text{-mod} \rightarrow \Omega(\mathfrak{n}')\text{-mod}^{\text{fact}}.$$

By a slight abuse of notation, we will denote by the same symbols $\text{BRST}_{\mathfrak{n}'}$ and $\text{BRST}_{\mathfrak{n}'}^{\text{enh}}$ the restrictions of the above functors along

$$\text{KL}(N') \rightarrow \mathfrak{L}(\mathfrak{n}')\text{-mod}.$$

12.5.7. Let $\delta_{N'}$ denote the integer $\dim(\text{Bun}_{N'})$.

Note that the canonical line bundle $K_{\text{Bun}_{N'}}$ of $\text{Bun}_{N'}$, i.e.,

$$\det(T^*(\text{Bun}_{N'})),$$

is canonically constant. Let $\mathbf{1}_{N'}$ denote the corresponding (ungraded) line.

The material in Sect. 10 applies as-is to the group scheme N' over X , so we can consider the localization functor

$$\text{Loc}_{N'} : \text{KL}(N')_{\text{Ran}} \rightarrow \text{D-mod}(\text{Bun}_{N'}).$$

Note also that since N' is unipotent, the stack $\text{Bun}_{N'}$ is quasi-compact. In particular, there is no difference between $\text{D-mod}(\text{Bun}_{N'})$ and $\text{D-mod}_{\text{co}}(\text{Bun}_{N'})$. Moreover, $\text{Bun}_{N'}$ is *safe* in the sense of [DG1, Sect. 10.2] so the “constant sheaf”

$$\underline{k} \in \text{D-mod}(\text{Bun}_{N'})$$

is compact, and the functor

$$C_{\text{dR}}(\text{Bun}_{N'}, -) = \mathcal{H}om_{\text{D-mod}(\text{Bun}_{N'})}(\underline{k}, -)$$

is continuous.

12.5.8. Our next goal is to construct a natural transformation from the composition

$$(12.28) \quad \mathrm{KL}(N')_{\mathrm{Ran}} \xrightarrow{\mathrm{BRST}_{n'}} \mathrm{D-mod}(\mathrm{Ran}) \xrightarrow{C_c(\mathrm{Ran}, -)} \mathrm{Vect} \xrightarrow{- \otimes \mathbb{I}_{N'}[\delta_{N'}]} \mathrm{Vect}$$

to

$$(12.29) \quad \mathrm{KL}(N')_{\mathrm{Ran}} \xrightarrow{\mathrm{Loc}_{N'}} \mathrm{D-mod}(\mathrm{Bun}_{N'}) \xrightarrow{C_{\mathrm{dR}}(\mathrm{Bun}_{N'}, -)} \mathrm{Vect},$$

i.e.,

$$(12.30) \quad (C_c(\mathrm{Ran}, -) \circ \mathrm{BRST}_{n'}) \otimes \mathbb{I}_{N'}[\delta_{N'}] \rightarrow C_{\mathrm{dR}}(\mathrm{Bun}_{N'}, -) \circ \mathrm{Loc}_{N'}.$$

12.5.9. Let us interpret $C_{\mathrm{dR}}(\mathrm{Bun}_{N'}, -) \circ \mathrm{Loc}_{N'}$ as

$$(12.31) \quad \langle \omega_{\mathrm{Bun}_{N'}}, \mathrm{Loc}_{N'}(-) \rangle_{\mathrm{Bun}_{N'}},$$

where

$$\langle -, - \rangle_{\mathrm{Bun}_{N'}} : \mathrm{D-mod}(\mathrm{Bun}_{N'}) \otimes \mathrm{D-mod}(\mathrm{Bun}_{N'}) \rightarrow \mathrm{Vect}$$

is the Verdier duality pairing.

Using Proposition 10.5.7, we rewrite (12.31) as

$$(12.32) \quad \langle \Gamma_{N', \mathrm{Ran}}(\omega_{\mathrm{Bun}_{N'}}), - \rangle_{\mathrm{KL}(N')_{\mathrm{Ran}}} \otimes \mathbb{I}_{N'}[\delta_{N'}],$$

where

$$\langle -, - \rangle_{\mathrm{KL}(N')_{\mathrm{Ran}}}$$

is the self-duality on $\mathrm{KL}_{N', \mathrm{Ran}}$ induced by (12.25).

12.5.10. Note now that for any $\underline{x} \in \mathrm{Ran}$

$$\Gamma^{\mathrm{ren}}(\mathrm{Bun}_{N'}^{\mathrm{level}_{\underline{x}}}, \omega_{\mathrm{Bun}_{N'}})$$

(see Equation (10.8) for the definition of $\Gamma^{\mathrm{ren}}(\mathrm{Bun}_{N'}^{\mathrm{level}_{\underline{x}}}, -)$) receives a map from

$$\Gamma(\mathrm{Bun}_{N'}, \mathbf{oblv}^l(\omega_{\mathrm{Bun}_{N'}})) \simeq \Gamma(\mathrm{Bun}_{N'}, \mathcal{O}_{\mathrm{Bun}_{N'}}),$$

and hence from

$$k \rightarrow \Gamma(\mathrm{Bun}_{N'}, \mathcal{O}_{\mathrm{Bun}_{N'}}).$$

Furthermore, it is easy to see that the resulting map

$$k \rightarrow \Gamma^{\mathrm{ren}}(\mathrm{Bun}_{N'}^{\mathrm{level}_{\underline{x}}}, \omega_{\mathrm{Bun}_{N'}})$$

in Vect upgrades to a map

$$k \rightarrow \Gamma^{\mathrm{ren}}(\mathrm{Bun}_{N'}^{\mathrm{level}_{\underline{x}}}, \omega_{\mathrm{Bun}_{N'}})^{\mathrm{enh}} = \Gamma_{N', \underline{x}}(\omega_{\mathrm{Bun}_{N'}}).$$

in $\mathrm{KL}(N')_{\underline{x}}$.

Making $\underline{x}$ move in families over Ran , we obtain a map

$$k_{\mathrm{Ran}} \rightarrow \Gamma_{N', \mathrm{Ran}}(\omega_{\mathrm{Bun}_{N'}}).$$

12.5.11. Thus, we obtain a map

$$(12.33) \quad \langle k_{\mathrm{Ran}}, - \rangle_{\mathrm{KL}(N'), \mathrm{Ran}} \otimes \mathbb{I}_{N'}[\delta_{N'}] \rightarrow C_{\mathrm{dR}}(\mathrm{Bun}_{N'}, -) \circ \mathrm{Loc}_{N'}.$$

Finally, we note that the functor

$$\langle k_{\mathrm{Ran}}, - \rangle_{\mathrm{KL}(N'), \mathrm{Ran}} : \mathrm{KL}(N')_{\mathrm{Ran}} \rightarrow \mathrm{Vect}$$

identifies with

$$\mathrm{KL}(N')_{\mathrm{Ran}} \xrightarrow{\mathrm{BRST}_{n'}} \mathrm{D-mod}(\mathrm{Ran}) \xrightarrow{C_c(\mathrm{Ran}, -)} \mathrm{Vect}.$$

Combining with (12.33) we obtain the desired map (12.30).

12.5.12. We will prove:⁵¹

Theorem 12.5.13. *The natural transformation (12.30) becomes an isomorphism after precomposing with the endofunctor*

$$\int \text{ins. vac} : \text{KL}(N')_{\text{Ran}} \rightarrow \text{KL}(N')_{\text{Ran}}.$$

12.5.14. Let us reformulate Theorem 12.5.13 in more concrete terms. Note that we can rewrite the precomposition of (12.28) with $\int \text{ins. vac}$ as

$$(12.34) \quad \text{KL}(N')_{\text{Ran}} \xrightarrow{\text{BRST}_{\mathfrak{n}'}^{\text{enh}}} \Omega(\mathfrak{n}')\text{-mod}_{\text{Ran}}^{\text{fact}} \xrightarrow{C_{\text{fact}}^{\text{fact}}(X, \Omega(\mathfrak{n}'), -)} \text{Vect} \xrightarrow{- \otimes \mathbb{I}_{N'}[\delta_{N'}]} \text{Vect}.$$

Further, since the functor $\text{Loc}_{N'}$ has a unital structure, the precomposition of (12.29) with $\int \text{ins. vac}$ is canonically isomorphic to (12.29) itself.

Hence, Theorem 12.5.13 implies:

Corollary 12.5.15. *There exists a canonical isomorphism between (12.34) and the functor (12.29).*

12.5.16. *Variant.* Consider the Lie algebra $\Gamma(X, \mathfrak{n}')$, and let χ^{Lie} be its character. The datum of χ^{Lie} gives rise to a factorization character

$$\chi : \mathfrak{L}(\mathfrak{n}') \rightarrow \mathbb{G}_a,$$

trivial in $\mathfrak{L}^+(\mathfrak{n}')$, and a map

$$\chi^{\text{glob}} : \text{Bun}_{N'} \rightarrow \mathbb{G}_a.$$

Let $\text{BRST}_{\mathfrak{n}', \chi}$ be the χ -twisted version of the semi-infinite cohomology functor, i.e.

$$\text{BRST}_{\mathfrak{n}', \chi}(-) = \text{BRST}_{\mathfrak{n}'}(- \otimes \chi).$$

Note that

$$\text{Vac}(N') \otimes \chi \simeq \text{Vac}(N').$$

Hence,

$$\text{BRST}_{\mathfrak{n}', \chi}(\text{Vac}(N')) \simeq \text{BRST}_{\mathfrak{n}'}(\text{Vac}(N')) \simeq \Omega(\mathfrak{n}')$$

as factorization algebras.

Let $\text{BRST}_{\mathfrak{n}', \chi}^{\text{enh}}$ be the enhancement of $\text{BRST}_{\mathfrak{n}', \chi}$

$$\text{BRST}_{\mathfrak{n}', \chi}^{\text{enh}} : \mathfrak{L}(\mathfrak{n}')\text{-mod} \rightarrow \Omega(\mathfrak{n}')\text{-mod}^{\text{fact}}.$$

12.5.17. As in (12.30) one constructs a natural transformation

$$(12.35) \quad (C_c(\text{Ran}, -) \circ \text{BRST}_{\mathfrak{n}', \chi}) \otimes \mathbb{I}_{N'}[\delta_{N'}] \rightarrow C_{\text{dR}}(\text{Bun}_{N'}, - \otimes^* \chi^*(\text{exp})) \circ \text{Loc}_{N'}.$$

And parallel to Theorem 12.5.13, we have:

Theorem 12.5.18. *The map (12.35) becomes an isomorphism after precomposing with*

$$\int \text{ins. vac} : \text{KL}(N')_{\text{Ran}} \rightarrow \text{KL}(N')_{\text{Ran}}.$$

Corollary 12.5.19. *There exists a canonical isomorphism between*

$$(12.36) \quad \text{KL}(N')_{\text{Ran}} \xrightarrow{\text{BRST}_{\mathfrak{n}', \chi}^{\text{enh}}} \Omega(\mathfrak{n}')\text{-mod}_{\text{Ran}}^{\text{fact}} \xrightarrow{C_{\text{fact}}^{\text{fact}}(X, \Omega(\mathfrak{n}'), -)} \text{Vect} \xrightarrow{- \otimes \mathbb{I}_{N'}[\delta_{N'}]} \text{Vect}$$

and

$$(12.37) \quad \text{KL}(N')_{\text{Ran}} \xrightarrow{\text{Loc}_{N'}} \text{D-mod}(\text{Bun}_{N'}) \xrightarrow{- \otimes^* \chi^*(\text{exp})} \text{D-mod}(\text{Bun}_{N'}) \xrightarrow{C_{\text{dR}}(\text{Bun}_{N'}, -)} \text{Vect},$$

12.6. Proof of Theorem 12.5.13.

⁵¹This result is established in [CF, Theorem 4.0.5(4)]. We will provide proof for completeness.

12.6.1. Since $\mathfrak{L}^+(N')$ is pro-unipotent, the category $\text{Rep}(\mathfrak{L}^+(G))$ is generated by $\mathbf{1}_{\text{Rep}(\mathfrak{L}^+(N'))}$. And hence the category $\text{KL}(N')$ is generated by $\text{Vac}(N')$. Therefore, $\text{KL}(N')_{\text{Ran}}$ is generated by $\text{Vac}(N')_{\text{Ran}}$ as a $\text{D-mod}(\text{Ran})$ -module category.

By unitality, for both sides in Theorem 12.5.13, tensoring the source by an object $\mathcal{F} \in \text{D-mod}(\text{Ran})$ has the effect of tensoring the target by $C_c(\text{Ran}, \mathcal{F})$. Hence, it is enough to show that the map in Theorem 12.5.13 evaluates to an isomorphism on $\text{Vac}(N')_{\text{Ran}}$.

12.6.2. By construction, the left-hand side is

$$C^{\text{fact}}(X; \Omega(\mathfrak{n}'))$$

(see Sect. 11.9.7 for the notation).

By (10.29), we have

$$\text{Loc}_{N'}(\text{Vac}(N)_{\text{Ran}}) \simeq \mathbf{ind}^l(\mathcal{O}_{\text{Bun}_{N'}}).$$

We rewrite

$$\mathbf{ind}^l(\mathcal{O}_{\text{Bun}_{N'}}) \simeq \mathbf{ind}^r(\omega_{\text{Bun}_{N'}}) \simeq \mathbf{ind}^r(\mathcal{O}_{\text{Bun}_{N'}}) \otimes \mathfrak{l}_{N'}[\delta_{N'}].$$

Hence, the map in Theorem 12.5.13 becomes a map

$$(12.38) \quad C^{\text{fact}}(X; \Omega(\mathfrak{n}')) \rightarrow \Gamma(\text{Bun}_{N'}, \mathcal{O}_{\text{Bun}_{N'}}).$$

The fact that (12.38) is an isomorphism is well-known. For completeness, we will supply a proof in the next subsection.

12.6.3. The material in the rest of this subsection is not logically necessary, except for the example considered in Sect. 12.6.7.

According to Remark 12.5.5, the factorization algebra $\Omega(\mathfrak{n}')$ can be thought of as the factorization algebra associated to the cohomological Chevalley complex of a Lie-* algebra. Let us consider $C^{\text{fact}}(X; \Omega(\mathfrak{n}'))$ in this paradigm.

Let L be a Lie-* algebra, whose underlying D-module is classical, finitely generated and projective (as a D-module). Consider its cohomological Chevalley complex

$$C_{\text{chev}}(L) \in \text{ComAlg}(\text{D-mod}(X)).$$

Denote

$$\Omega(L) := \text{Fact}(C_{\text{chev}}(L)) \in \text{ComAlg}(\text{FactAlg}(X)),$$

and consider

$$C^{\text{fact}}(X, \Omega(L)) \in \text{ComAlg}(\text{Vect}).$$

Unfortunately, we do not have a good grip on what $C^{\text{fact}}(X, \Omega(L))$ looks like.

12.6.4. Note that $C_{\text{chev}}(L)$ is naturally written as

$$(12.39) \quad C_{\text{chev}}(L) \simeq \varinjlim_n C_{\text{chev}}(L)_n,$$

where $C_{\text{chev}}(L)_n \in \text{ComAlg}(\text{D-mod}(X))$ is the n -step cohomological Chevalley complex.

Note, however, that the assumptions on L imply that for every n , the composite map

$$\tau^{\leq n}(C(L)) \rightarrow C(L) \rightarrow C(L)_n$$

is an isomorphism (here $\tau^{\leq n}$ refers to the left D-module structure, i.e., one for which $\mathbf{oblv}^l$ is t-exact). So, in the formation $C_{\text{ch}}(L)$ no “actual completion” is involved.

12.6.5. Denote

$$\Omega(L)_n := \text{Fact}(\mathcal{C}_{\text{chev}}(L)_n) \in \text{ComAlg}(\text{FactAlg}(X)).$$

Set:

$$\mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L))^{\wedge} := \lim_n \mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L)_n).$$

Unlike $\mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L))$, the algebra $\mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L))^{\wedge}$ can be described explicitly. Namely, according to [BD2, Proposition 7.4.1],

$$\mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L))^{\wedge} \simeq \mathcal{C}_{\text{chev}}(\mathcal{C}_{\text{dR}}(X, L)) = \mathcal{C}_{\text{chev}, \cdot}(\mathcal{C}_{\text{dR}}(X, L))^{\vee},$$

where:

- $\mathcal{C}_{\text{dR}}(X, L)$ is considered as a Lie algebra (in Vect);
- $\mathcal{C}_{\text{chev}}(-)$ and $\mathcal{C}_{\text{chev}, \cdot}(-)$ are the cohomological and homological Chevalley complexes of a Lie algebra, respectively.

12.6.6. The map $\rightarrow$ in (12.39) gives rise to a map

$$(12.40) \quad \mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L)) \rightarrow \mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L))^{\wedge}$$

but this map is in general *not* an isomorphism.

12.6.7. *Example.* Let L be abelian. Denote $L^{\vee} := \mathbb{D}(L)[-1]$. Then

$$\mathcal{C}_{\text{chev}}(L) \simeq \text{Sym}^!(\mathbb{D}(L)),$$

and hence

$$\mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L)) \simeq \text{Sym}(\mathcal{C}_{\text{dR}}(X, \mathbb{D}(L))) \simeq \text{Sym}(\mathcal{C}_{\text{dR}}(X, L)^{\vee}[-1]).$$

By contrast,

$$\mathcal{C}_{\text{chev}}(\mathcal{C}_{\text{dR}}(X, L)) \simeq \text{Sym}(\mathcal{C}_{\text{dR}}(X, L)[-1])^{\vee}.$$

So the difference between the two sides in (12.40) in this case is that between a polynomial algebra and its completion.

12.6.8. The map (12.40) is the best approximation to $\mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L))$ that we have in general. In certain situations, it allows us to recover $\mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L))$ completely.

This happens, for example, if L carries a strictly positive grading. In this case, the map (12.40) defines an isomorphism on each graded component. I.e., we have

$$\mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L))^d \simeq (\mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L))^{\wedge})^d \simeq \mathcal{C}_{\text{chev}}(\mathcal{C}_{\text{dR}}(X, L))^d \simeq (\mathcal{C}_{\text{chev}, \cdot}(\mathcal{C}_{\text{dR}}(X, L))^{-d})^{\vee}, \quad d \in \mathbb{Z}^{<0}.$$

Note, however, that $\mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L))^{\wedge}$ is *not* the direct sum of its graded pieces. Rather,

$$(\mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L))^{\wedge})^d = \lim_n \mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L)_n)^d.$$

Remark 12.6.9. Using cohomological truncations on powers of X one can show:

- $\mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L))$ is coconnective.
- Suppose that L is such that $H^0(\mathcal{C}_{\text{dR}}(X, L)) = 0$. Then $\mathcal{C}_{\cdot}^{\text{fact}}(X, \Omega(L))$ is classical.

12.7. Global functions on the moduli space of bundles for a unipotent group-scheme.

12.7.1. Let $\mathcal{Y}$ be a D-prestack over X . Let $A \in \text{ComAlg}(\text{D-mod}(X))$ be the algebra of global functions on $\mathcal{Y}$, and let $\mathcal{A} = \text{Fact}(A) \in \text{ComAlg}(\text{FactAlg}(X))$ be the corresponding factorization algebra (see Sect. B.10.2).

Recall that by (F.3), the evaluation map

$$\text{Sect}_{\nabla}(X, \mathcal{Y}) \times X \rightarrow \mathcal{Y}$$

gives rise to a map

$$(12.41) \quad C^{\text{fact}}(X, \mathcal{A}) \rightarrow \Gamma(\text{Sect}_{\nabla}(X, \mathcal{Y}), \mathcal{O}_{\text{Sect}_{\nabla}(X, \mathcal{Y})}),$$

Unwinding the construction, it is easy to see that the map (12.38) is the map (12.41) for $\mathcal{Y} = \text{Jets}(\text{pt}/N')$, where

$$\Omega(\mathfrak{n}') = \mathcal{O}_{\text{pt}/\mathfrak{L}^+(N')} \stackrel{(C.44)}{\simeq} \text{Fact}(\mathcal{O}_{\text{Jets}(\text{pt}/N')})$$

and

$$\text{Sect}_{\nabla}(X, \text{Jets}(\text{pt}/N')) \simeq \text{Sect}(X, \text{pt}/N') \simeq \text{Bun}_{N'}.$$

Thus, we need to show that (12.41) is an isomorphism in this case.

Remark 12.7.2. Recall that (12.41) is an isomorphism for $\mathcal{Y}$ that is affine over X , see Proposition F.1.4. So we want to prove that $\mathcal{Y} = \text{Jets}(\text{pt}/N')$ is not too different from the affine case.

The proof below follows closely that of Proposition F.1.4.

12.7.3. First, we will show that the map (12.41) induces an *isomorphism*

$$(12.42) \quad \begin{aligned} \text{Maps}(\text{Spec}(R), \text{Sect}_{\nabla}(X, \text{Jets}(\text{pt}/N'))) &\rightarrow \\ &\rightarrow \text{Maps}_{\text{ComAlg}(\text{Vect})}(\Gamma(\text{Sect}_{\nabla}(X, \text{Jets}(\text{pt}/N')), \mathcal{O}_{\text{Sect}_{\nabla}(X, \text{Jets}(\text{pt}/N'))}), R) \rightarrow \\ &\rightarrow \text{Maps}_{\text{ComAlg}(\text{Vect})}(C^{\text{fact}}(X, \mathcal{O}_{\text{pt}/\mathfrak{L}^+(N')}), R) \end{aligned}$$

for $R \in \text{ComAlg}(\text{Vect}^{\leq 0})$.

This essentially follows from the fact that $\text{Jets}(N')$ is pro-unipotent, and hence $\text{Jets}(\text{pt}/N')$ is *as good as affine*⁵² (see Sect. 12.7.5 below for what this means), i.e., for $\mathcal{R} \in \text{ComAlg}(\text{D-mod}(X)^{\leq 0})$, the map

$$(12.43) \quad \text{Maps}_{X, \nabla}(\text{Spec}_X(\mathcal{R}), \text{Jets}(\text{pt}/N')) \rightarrow \text{Maps}_{\text{ComAlg}(\text{D-mod}(X))}(\mathcal{O}_{\text{Jets}(\text{pt}/N')}, \mathcal{R})$$

is an isomorphism.

12.7.4. In more detail, for $R \in \text{ComAlg}(\text{Vect}^{\leq 0})$ we have:

$$(12.44) \quad \begin{aligned} \text{Maps}(\text{Spec}(R), \text{Sect}_{\nabla}(X, \text{Jets}(\text{pt}/N'))) &= \text{Maps}_{X, \nabla}(\text{Spec}(R) \times X, \text{Jets}(\text{pt}/N')) \stackrel{(12.43)}{\simeq} \\ &\simeq \text{Maps}_{\text{ComAlg}(\text{D-mod}(X))}(\mathcal{O}_{\text{Jets}(\text{pt}/N')}, R \otimes \mathcal{O}_X). \end{aligned}$$

Using Corollary C.9.5, we rewrite the expression in the right-hand side in (12.44) as

$$(12.45) \quad \text{Maps}_{\text{ComAlg}(\text{Vect})}(C^{\text{fact}}(X, \mathcal{O}_{\text{pt}/\mathfrak{L}^+(N')}), R).$$

Combining (12.44) and (12.45), we obtain an isomorphism

$$\text{Maps}(\text{Spec}(R), \text{Sect}_{\nabla}(X, \text{Jets}(\text{pt}/N'))) \simeq \text{Maps}_{\text{ComAlg}(\text{Vect})}(C^{\text{fact}}(X, \mathcal{O}_{\text{pt}/\mathfrak{L}^+(N')}), R),$$

and unwinding the definitions we obtain that this isomorphism equals the map in (12.42).

12.7.5. Let us call a prestack $\mathcal{Z}$ *as good as affine* if the functor

$$\Gamma(\mathcal{Z}, -) : \text{QCoh}(\mathcal{Z}) \rightarrow \mathcal{O}_{\mathcal{Z}\text{-mod}}$$

is an equivalence.

⁵²Up to issues of renormalization, which are irrelevant here.

12.7.6. Let R be an object of $\text{ComAlg}(\text{Vect})$. Define the prestack “ $\text{Spec}(R)$ ” by

$$\text{Maps}(\text{Spec}(R'), \text{“Spec}(R)\text{”}) := \text{Maps}_{\text{ComAlg}(\text{Vect})}(R, R'), \quad R' \in \text{ComAlg}(\text{Vect}^{\leq 0}).$$

Note the formation of “ $\text{Spec}(R)$ ” is functorial in R . In particular, we obtain a map

$$(12.46) \quad R \simeq \text{Maps}_{\text{ComAlg}(\text{Vect})}(k[t], R) \rightarrow \text{Maps}(\text{“Spec}(R)\text{”}, \text{Spec}(k[t])) \simeq \\ \simeq \text{Maps}(\text{“Spec}(R)\text{”}, \mathbb{A}^1) \simeq \Gamma(\text{“Spec}(R)\text{”}, \mathcal{O}_{\text{“Spec}(R)\text{”}}).$$

We shall say that R is “as good as connective” if:

- The prestack “ $\text{Spec}(R)$ ” is as good as affine;
- The map (12.46) is an isomorphism.

12.7.7. Thus, given that (12.42) is an isomorphism, we need to show that $C^{\text{fact}}(X, \Omega(\mathfrak{n}'))$ is as good as connective.

We will now use the assumption that N' admits a filtration by normal subgroups with abelian quotients. We will argue by induction on the length of such a filtration.

12.7.8. We first consider the base of the induction, i.e., case when N' is a vector group-scheme, i.e., is the total space of a vector bundle $\mathcal{E}$ on X . In this case, the computation of $C^{\text{fact}}(X, \Omega(\mathfrak{n}'))$ has been performed in Sect. 12.6.7.

We obtain that $C^{\text{fact}}(X, \Omega(\mathfrak{n}'))$ is (non-canonically) isomorphic to the tensor product

$$\text{Sym}(V_1) \otimes \text{Sym}(V_2[-1]),$$

where V_1 and V_2 are classical finite-dimensional vector spaces.

It is clear that the tensor product of two algebras both of which are as good as connective is itself as good as connective. Hence, it remains to see that algebras of the form $\text{Sym}(V[-1])$, where V is a classical finite-dimensional vector space, are as good as connective.

However, this is well-known: in this case

$$\text{“Spec}(\text{Sym}(V[-1]))\text{”} \simeq \text{pt}/V^\vee$$

and the assertion is manifest.

12.7.9. We now perform the induction step. Thus, we fix a short exact sequence

$$1 \rightarrow N'_2 \rightarrow N' \rightarrow N'_1 \rightarrow 1,$$

where N'_2 is a vector group-scheme.

We observe:

Lemma 12.7.10. *Let $R_1 \rightarrow R$ be a map of commutative algebras in Vect . Assume that:*

- R_1 is as good as connective;
- For any homomorphism $R_1 \rightarrow R'$ with R' connective, the base change $R' \otimes_{R_1} R$ is as good as connective.

Then R is as good as connective.

We apply this lemma to

$$R := C^{\text{fact}}(X, \Omega(\mathfrak{n}')) \text{ and } R_1 := C^{\text{fact}}(X, \Omega(\mathfrak{n}'_1)).$$

By the induction hypothesis, $C^{\text{fact}}(X, \Omega(\mathfrak{n}'_1))$ is as good as connective. Hence, it remains to show that for any connective R' and a homomorphism

$$C^{\text{fact}}(X, \Omega(\mathfrak{n}'_1)) \rightarrow R',$$

the algebra

$$R' \otimes_{C^{\text{fact}}(X, \Omega(\mathfrak{n}'_1))} C^{\text{fact}}(X, \Omega(\mathfrak{n}'))$$

is as good as connective.

12.7.11. We now apply Lemma C.9.14, and hence we can rewrite

$$R' \otimes_{C^{\text{fact}}(X, \Omega(\mathfrak{n}'_1))} C^{\text{fact}}(X, \Omega(\mathfrak{n}')) \simeq C^{\text{fact}}(X, \Omega(\mathfrak{n}')_{R'}),$$

where

$$\Omega(\mathfrak{n}')_{R'} := \Omega(\mathfrak{n}') \otimes_{\Omega(\mathfrak{n}'_1)} (R' \otimes \omega_{\text{Ran}}) \in \text{ComAlg}(\text{FactAlg}(X) \otimes R'\text{-mod}).$$

12.7.12. Recall that

$$\Omega(\mathfrak{n}') \simeq \text{Fact}(C_{\text{chev}}(L_{\mathfrak{n}'})) \text{ and } \Omega(\mathfrak{n}'_1) \simeq \text{Fact}(\text{Chev}_{\text{chev}}(L_{\mathfrak{n}'_1})),$$

where

$$L_{\mathfrak{n}'} = \mathfrak{n}' \otimes D_X, \quad L_{\mathfrak{n}'_1} = \mathfrak{n}'_1 \otimes D_X.$$

Hence, we can rewrite

$$\Omega(\mathfrak{n}')_{R'} \simeq \text{Fact} \left(C_{\text{chev}}(L_{\mathfrak{n}'}) \otimes_{C_{\text{chev}}(L_{\mathfrak{n}'_1})} (R' \otimes \mathcal{O}_X) \right).$$

12.7.13. We can interpret the datum of $C^{\text{fact}}(X, \Omega(\mathfrak{n}'_1)) \rightarrow R'$ as a map

$$\text{Spec}(R') \rightarrow \text{Bun}_{N'_1}.$$

The adjoint action of N'_1 on N'_2 gives rise to an R' -family of twisted forms of N'_2 , denoted $N'_{2,R'}$. Consider the corresponding R' -family of Lie-* algebras

$$L_{\mathfrak{n}'_{2,R'}} := \mathfrak{n}'_{2,R'} \otimes D_X.$$

We have

$$C_{\text{chev}}(L_{\mathfrak{n}'}) \otimes_{C_{\text{chev}}(L_{\mathfrak{n}'_1})} (R' \otimes \mathcal{O}_X) \simeq C_{\text{chev}}(L_{\mathfrak{n}'_{2,R'}}).$$

Thus, we can consider

$$\Omega(\mathfrak{n}'_{2,R'}) \in \text{ComAlg}(\text{FactAlg}(X) \otimes R'\text{-mod})$$

and we obtain:

$$\Omega(\mathfrak{n}')_{R'} \simeq C^{\text{fact}}(X, \Omega(\mathfrak{n}'_{2,R'})).$$

12.7.14. It remains to show that $C^{\text{fact}}(X, \Omega(\mathfrak{n}'_{2,R'}))$ is as good as connective.

Recall that $\mathfrak{n}'_2$ is abelian. Hence, $N'_{2,R'}$ is R' -family of vector group-schemes. Hence, the required assertion is a relative (over $\text{Spec}(R')$) version of the case considered in Sect. 12.7.8.

□[Theorem 12.5.13]

12.8. Application: integration over (twists of) Bun_N via BRST.

12.8.1. Let $\mathcal{P}_T$ be a T -bundle on X . Consider the unipotent group-scheme $N_{\mathcal{P}_T}$. Denote the corresponding moduli stack $\text{Bun}_{N_{\mathcal{P}_T}}$; note that it identifies with $\text{Bun}_{N, \mathcal{P}_T}$ (see (9.3)).

The resulting map

$$\text{Bun}_{N_{\mathcal{P}_T}} \simeq \text{Bun}_{N, \mathcal{P}_T} \xrightarrow{\mathfrak{p}} \text{Bun}_G$$

can be thought of as

$$\text{Bun}_{N_{\mathcal{P}_T}} \rightarrow \text{Bun}_{G_{\mathcal{P}_T}} \xrightarrow{\alpha_{\mathcal{P}_T, \text{taut}}^{-1}} \text{Bun}_G.$$

12.8.2. Since the restriction of κ to $\mathfrak{n}$ is trivial, we obtain that the restriction of the twisting $\mathcal{T}_{\kappa}$ along $\mathfrak{p}$ is canonically trivial. In particular, we have a well-defined functor

$$(12.47) \quad \mathfrak{p}_{\kappa}^! : \text{D-mod}_{\kappa}(\text{Bun}_G) \rightarrow \text{D-mod}(\text{Bun}_{N, \mathcal{P}_T}).$$

12.8.3. Note that the embedding

$$N_{\mathcal{P}_T} \rightarrow G_{\mathcal{P}_T}$$

gives rise to a map

$$\mathfrak{L}(\mathfrak{n}_{\mathcal{P}_T}) \rightarrow \widehat{\mathfrak{g}}_{\kappa, \mathcal{P}_T},$$

and this map lifts to the Kac-Moody extension.

In particular, we obtain a well-defined restriction functor

$$\mathrm{KL}(G)_{\kappa} \xrightarrow{\alpha_{\mathcal{P}_T}, \mathrm{taut}} \mathrm{KL}(G)_{\kappa, \mathcal{P}_T} \rightarrow \mathrm{KL}(N_{\mathcal{P}_T}).$$

Denote

$$\Omega(\mathfrak{n}_{\mathcal{P}_T}, \mathfrak{g})_{\kappa} := \mathrm{BRST}_{\mathfrak{n}_{\mathcal{P}_T}}(\mathrm{Vac}(G)_{\kappa, \mathcal{P}_T}).$$

This is a factorization algebra, which receives a homomorphism from $\Omega(\mathfrak{n}_{\mathcal{P}_T})$.

Thus, the composition

$$\mathrm{KL}(G)_{\kappa} \xrightarrow{\alpha_{\mathcal{P}_T}, \mathrm{taut}} \mathrm{KL}(G)_{\kappa, \mathcal{P}_T} \rightarrow \mathrm{KL}(N_{\mathcal{P}_T}) \xrightarrow{\mathrm{BRST}_{\mathfrak{n}_{\mathcal{P}_T}}^{\mathrm{enh}}} \Omega(\mathfrak{n}_{\mathcal{P}_T})\text{-mod}^{\mathrm{fact}}$$

further enhances to a (factorization) functor

$$\mathrm{BRST}_{\mathfrak{n}_{\mathcal{P}_T}}^{\mathfrak{g}\text{-enh}} : \mathrm{KL}(G)_{\kappa} \rightarrow \Omega(\mathfrak{n}_{\mathcal{P}_T}, \mathfrak{g})_{\kappa}\text{-mod}^{\mathrm{fact}}.$$

12.8.4. We are going to prove:

Theorem 12.8.5. *The composition*

$$(12.48) \quad \mathrm{KL}(G)_{\kappa, \mathrm{Ran}} \xrightarrow{\mathrm{Loc}_{G, \kappa}} \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G) \xrightarrow{\mathfrak{p}_{\kappa}^!} \mathrm{D}\text{-mod}(\mathrm{Bun}_{N, \mathcal{P}_T}) \xrightarrow{\mathrm{C}_{\mathrm{dR}}(\mathrm{Bun}_{N, \mathcal{P}_T}, -)} \mathrm{Vect}$$

identifies with the functor

$$(12.49) \quad \mathrm{KL}(G)_{\kappa, \mathrm{Ran}} \xrightarrow{\mathrm{BRST}_{\mathfrak{n}_{\mathcal{P}_T}}^{\mathfrak{g}\text{-enh}}} \Omega(\mathfrak{n}_{\mathcal{P}_T}, \mathfrak{g})\text{-mod}_{\mathrm{Ran}}^{\mathrm{fact}} \xrightarrow{\mathrm{C}_{\mathrm{dR}}^{\mathrm{fact}}(X; \Omega(\mathfrak{n}_{\mathcal{P}_T}, \mathfrak{g})_{\kappa}, -)} \mathrm{Vect} \xrightarrow{- \otimes \mathfrak{l}_{N, \mathcal{P}_T} [\delta_{N, \mathcal{P}_T}]} \mathrm{Vect},$$

where the notations $\delta_{N, \mathcal{P}_T}$ and $\mathfrak{l}_{N, \mathcal{P}_T}$ are as in Sect. 12.5.7.

12.8.6. *Proof of Theorem 12.8.5.* First, we rewrite the functor

$$\mathrm{KL}(G)_{\kappa, \mathrm{Ran}} \xrightarrow{\mathrm{Loc}_{G, \kappa}} \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G) \xrightarrow{\mathfrak{p}_{\kappa}^!} \mathrm{D}\text{-mod}(\mathrm{Bun}_{N, \mathcal{P}_T})$$

using (a $\mathcal{P}_T$ -twisted version) of Corollary 12.3.11.

We obtain that it identifies with

$$\begin{aligned} \mathrm{KL}(G)_{\kappa, \mathrm{Ran}} &\xrightarrow{\alpha_{\mathcal{P}_T}, \mathrm{taut}} \mathrm{KL}(G)_{\kappa, \mathcal{P}_T, \mathrm{Ran}} \xrightarrow{\int \mathrm{ins.vac}} \mathrm{KL}(G)_{\kappa, \mathcal{P}_T, \mathrm{Ran}} \rightarrow \\ &\rightarrow \mathrm{KL}(N_{\mathcal{P}_T})_{\mathrm{Ran}} \xrightarrow{\mathrm{Loc}_{N, \mathcal{P}_T}} \mathrm{D}\text{-mod}(\mathrm{Bun}_{N, \mathcal{P}_T}). \end{aligned}$$

By Theorem 12.5.13, the functor

$$\mathrm{KL}(N_{\mathcal{P}_T})_{\mathrm{Ran}} \xrightarrow{\mathrm{Loc}_{N, \mathcal{P}_T}} \mathrm{D}\text{-mod}(\mathrm{Bun}_{N, \mathcal{P}_T}) \xrightarrow{\mathrm{C}_{\mathrm{dR}}(\mathrm{Bun}_{N, \mathcal{P}_T}, -)} \mathrm{Vect}$$

identifies with

$$\mathrm{KL}(N_{\mathcal{P}_T})_{\mathrm{Ran}} \xrightarrow{\mathrm{BRST}_{\mathfrak{n}_{\mathcal{P}_T}}^{\mathrm{enh}}} \Omega(\mathfrak{n}_{\mathcal{P}_T})\text{-mod}_{\mathrm{Ran}}^{\mathrm{fact}} \xrightarrow{\mathrm{C}_{\mathrm{dR}}^{\mathrm{fact}}(X; \Omega(\mathfrak{n}_{\mathcal{P}_T}), -)} \mathrm{Vect} \xrightarrow{- \otimes \mathfrak{l}_{N, \mathcal{P}_T} [\delta_{N, \mathcal{P}_T}]} \mathrm{Vect}.$$

The assertion of the theorem follows now by applying Corollary 11.9.13.

□[Theorem 12.8.5]

12.8.7. Note that the same proof applies in the situation twisted by a character. Namely, χ^{Lie} be a character of $\Gamma(X, \mathfrak{n}_{\mathcal{P}_T})$ as in Sect. 12.5.16.

Denote

$$\Omega(\mathfrak{n}_{\mathcal{P}_T}, \chi, \mathfrak{g})_\kappa := \text{BRST}_{\mathfrak{n}_{\mathcal{P}_T}, \chi}(\text{Vac}(G)_{\kappa, \mathcal{P}_T}).$$

Consider the corresponding functor

$$\text{BRST}_{\mathfrak{n}_{\mathcal{P}_T}, \chi}^{\text{g-enh}} : \text{KL}(G)_\kappa \rightarrow \Omega(\mathfrak{n}_{\mathcal{P}_T}, \chi, \mathfrak{g})_\kappa\text{-mod}^{\text{fact}}.$$

Then:

Theorem 12.8.8. *The composition*

$$\begin{aligned} \text{KL}(G)_{\kappa, \text{Ran}} &\xrightarrow{\text{Loc}_{G, \kappa}} \text{D-mod}_\kappa(\text{Bun}_G) \xrightarrow{\mathfrak{p}_\kappa^!} \text{D-mod}(\text{Bun}_{N, \mathcal{P}_T}) \xrightarrow{-\otimes \chi^*(\text{exp})} \\ &\rightarrow \text{D-mod}(\text{Bun}_{N, \mathcal{P}_T}) \xrightarrow{\text{C}_{\text{dR}}(\text{Bun}_{N, \mathcal{P}_T}, -)} \text{Vect} \end{aligned}$$

identifies with the functor

$$\text{KL}(G)_{\kappa, \text{Ran}} \xrightarrow{\text{BRST}_{\mathfrak{n}_{\mathcal{P}_T}, \chi}^{\text{g-enh}}} \Omega(\mathfrak{n}_{\mathcal{P}_T}, \chi, \mathfrak{g})_\kappa\text{-mod}_{\text{Ran}}^{\text{fact}} \xrightarrow{C^{\text{fact}}(X; \Omega(\mathfrak{n}_{\mathcal{P}_T}, \chi, \mathfrak{g})_\kappa, -)} \text{Vect} \xrightarrow{-\otimes \mathfrak{l}_{N, \mathcal{P}_T}[\delta_{N, \mathcal{P}_T}]} \text{Vect}.$$

13. LOCALIZATION VIA THE INFINITESIMAL HECKE GROUPOID

This goal of this section is to prove Theorem 12.1.8. This will be based on the approach to the localization functor via the infinitesimal Hecke groupoid, which was developed in the unpublished part of the thesis of the eighth author of this paper.

We will then use these ideas to prove that the functor $\text{Loc}_{G, \kappa}$ is *almost* a localization (i.e., its right adjoint is fully faithful). Namely, it becomes a localization when we compose it with restriction to any quasi-compact open substack, see Theorem 13.4.2.

13.1. Another take on the functor Γ_G .

13.1.1. Recall the local Hecke stack completed along the diagonal, viewed as a groupoid acting on $\text{pt}/\mathfrak{L}^+(G)$:

$$\text{pt}/\mathfrak{L}^+(G) \xleftarrow{\bar{h}^{\text{loc}, \wedge}} \text{Hecke}_G^{\text{loc}, \wedge} \xrightarrow{\bar{h}^{\text{loc}, \wedge}} \text{pt}/\mathfrak{L}^+(G).$$

Recall also that the datum of a level κ gives rise to a *multiplicative* line bundle $\mathcal{L}_\kappa^{\text{loc}}$ on $\text{Hecke}_G^{\text{loc}, \wedge}$, see Sect. 10.1.1.

According to [CF, Sect. 3.3], we can identify the category $\text{KL}(G)_\kappa$ with the category

$$\text{Rep}(\mathfrak{L}^+(G))^{\text{Hecke}_G^{\text{loc}, \wedge}, \mathcal{L}_\kappa^{\text{loc}}}$$

of $\mathcal{L}_\kappa^{\text{loc}}$ -twisted $\text{Hecke}_G^{\text{loc}, \wedge}$ -equivariant objects in $\text{Rep}(\mathfrak{L}^+(G))$.

I.e., this is the category of $\mathcal{M} \in \text{Rep}(\mathfrak{L}^+(G))$ equipped with an isomorphism

$$(13.1) \quad (\bar{h}^{\text{loc}, \wedge})^*(\mathcal{M}) \simeq \mathcal{L}_\kappa^{\text{loc}} \otimes (\bar{h}^{\text{loc}, \wedge})^*(\mathcal{M})$$

(the isomorphism taking place in $\text{QCoh}(\text{Hecke}_G^{\text{loc}, \wedge})$), equipped with a homotopy-coherent system of compatibilities.

13.1.2. Let $\inf(\mathrm{Bun}_G)$ denote the infinitesimal groupoid of Bun_G , i.e.,

$$\inf(\mathrm{Bun}_G) := (\mathrm{Bun}_G \times \mathrm{Bun}_G)^\wedge,$$

where $(-)^\wedge$ means formal completion along the diagonal. Consider the corresponding diagram

$$\mathrm{Bun}_G \xleftarrow{h^{\inf}} \inf(\mathrm{Bun}_G) \xrightarrow{h^{\inf}} \mathrm{Bun}_G.$$

Note that the datum of the de Rham twisting $\mathcal{T}_\kappa$ gives rise to a multiplicative line bundle, to be denoted $\mathcal{L}_\kappa^{\inf}$ on $\inf(\mathrm{Bun}_G)$.

Let $\mathcal{Z}$ be a prestack mapping to Ran . Since Bun_G is eventually coconnective, it follows from [GaRo2, Proposition 3.4.3] that we have a canonical equivalence

$$\mathrm{D}\text{-mod}_\kappa(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \simeq (\mathrm{QCoh}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}))^{\inf(\mathrm{Bun}_G) \times \mathcal{Z}, \mathcal{L}_\kappa^{\inf}}$$

that intertwines the functor

$$(\mathbf{oblv}_\kappa^l \otimes \mathrm{Id}) : \mathrm{D}\text{-mod}_\kappa(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \rightarrow \mathrm{QCoh}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z})$$

with the tautological forgetful functor

$$(\mathrm{QCoh}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}))^{\inf(\mathrm{Bun}_G) \times \mathcal{Z}, \mathcal{L}_\kappa^{\inf}} \rightarrow \mathrm{QCoh}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}).$$

The same applies when we replace Bun_G by its open substack U .

13.1.3. Note now that we have a tautological map of groupoids

$$(13.2) \quad \mathrm{Hecke}_{G, \mathcal{Z}}^{\mathrm{glob}, \wedge} \rightarrow \inf(\mathrm{Bun}_G) \times \mathcal{Z}$$

over $\mathcal{Z}$.

By construction, the pullback of $\mathcal{L}_\kappa^{\inf}$ along (13.2) identifies canonically with $\mathcal{L}_\kappa^{\mathrm{glob}}$ as a multiplicative line bundle.

From here we obtain that $*$ -pullback along (13.2) defines a functor

$$(13.3) \quad \mathrm{D}\text{-mod}_\kappa(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \simeq (\mathrm{QCoh}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}))^{\inf(\mathrm{Bun}_G) \times \mathcal{Z}, \mathcal{L}_\kappa^{\inf}} \rightarrow \\ \rightarrow (\mathrm{QCoh}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}))^{\mathrm{Hecke}_{G, \mathcal{Z}}^{\mathrm{glob}, \wedge}, \mathcal{L}_\kappa^{\mathrm{glob}}},$$

that intertwines the forgetful functor

$$(\mathbf{oblv}_\kappa^l \otimes \mathrm{Id}) : \mathrm{D}\text{-mod}_\kappa(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \rightarrow \mathrm{QCoh}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z})$$

with the tautological forgetful functor

$$(\mathrm{QCoh}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}))^{\mathrm{Hecke}_{G, \mathcal{Z}}^{\mathrm{glob}, \wedge}, \mathcal{L}_\kappa^{\mathrm{glob}}} \rightarrow \mathrm{QCoh}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}).$$

Denote the functor (13.3) by

$$\mathbf{oblv}_{\inf \rightarrow \mathrm{Hecke}^{\wedge}, \mathcal{Z}}.$$

13.1.4. Let $U \subset \mathrm{Bun}_G$ be a quasi-compact open substack. Note that it makes sense to restrict $\mathrm{Hecke}_{G, \mathcal{Z}}^{\mathrm{glob}, \wedge}$ to U :

$$(\overleftarrow{h}^{\mathrm{glob}, \wedge})^{-1}(U \times \mathcal{Z}) =: \mathrm{Hecke}_{G, \mathcal{Z}, U}^{\mathrm{glob}, \wedge} =: (\overrightarrow{h}^{\mathrm{glob}, \wedge})^{-1}(U \times \mathcal{Z}).$$

The contents of Sect. 13.1.3 apply over U as well, and we obtain a functor, denoted

$$\mathbf{oblv}_{\inf \rightarrow \mathrm{Hecke}^{\wedge}, \mathcal{Z}, U}$$

that maps

$$(13.4) \quad \mathrm{D}\text{-mod}_\kappa(U) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \simeq (\mathrm{QCoh}(U) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}))^{\inf(\mathrm{Bun}_U) \times \mathcal{Z}, \mathcal{L}_\kappa^{\inf}} \rightarrow \\ \rightarrow (\mathrm{QCoh}(U) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}))^{\mathrm{Hecke}_{G, \mathcal{Z}, U}^{\mathrm{glob}, \wedge}, \mathcal{L}_\kappa^{\mathrm{glob}}}$$

and that intertwines the forgetful functor

$$(\mathbf{oblv}_\kappa^l \otimes \mathrm{Id}) : \mathrm{D}\text{-mod}_\kappa(U) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \rightarrow \mathrm{QCoh}(U) \otimes \mathrm{D}\text{-mod}(\mathcal{Z})$$

with the tautological forgetful functor

$$(\mathrm{QCoh}(U) \otimes \mathrm{D-mod}(\mathcal{Z}))^{\mathrm{Hecke}_{G,\mathcal{Z},U}^{\mathrm{glob},\wedge,\mathcal{L}_\kappa^{\mathrm{glob}}}} \rightarrow \mathrm{QCoh}(U) \otimes \mathrm{D-mod}(\mathcal{Z}).$$

13.1.5. Assume now that U is quasi-compact. Let j denote its embedding into Bun_G . Consider the diagram:

$$(13.5) \quad \begin{array}{ccccc} U \times \mathcal{Z} & \xleftarrow{\bar{h}^{\mathrm{glob},\wedge}} & \mathrm{Hecke}_{G,\mathcal{Z},U}^{\mathrm{glob},\wedge} & \xrightarrow{\bar{h}^{\mathrm{glob},\wedge}} & U \times \mathcal{Z} \\ \mathrm{ev}_{\mathcal{Z},U} \downarrow & & \downarrow & & \downarrow \mathrm{ev}_{\mathcal{Z},U} \\ (\mathrm{pt}/\mathfrak{L}^+(G))_{\mathcal{Z}} & \xleftarrow{\bar{h}^{\mathrm{loc},\wedge}} & \mathrm{Hecke}_{G,\mathcal{Z}}^{\mathrm{loc},\wedge} & \xrightarrow{\bar{h}^{\mathrm{loc},\wedge}} & (\mathrm{pt}/\mathfrak{L}^+(G))_{\mathcal{Z}}, \end{array}$$

where $\mathrm{ev}_{\mathcal{Z},U} := \mathrm{ev}_{\mathcal{Z}} \circ j$.

Since both squares in (13.5) are Cartesian, we obtain that the functor⁵³

$$(\mathrm{ev}_{\mathcal{Z},U})_* : \mathrm{QCoh}(U) \otimes \mathrm{D-mod}(\mathcal{Z}) \rightarrow \mathrm{Rep}(\mathrm{pt}/\mathfrak{L}^+(G))_{\mathcal{Z}}$$

gives rise to a functor

$$(13.6) \quad (\mathrm{QCoh}(U) \otimes \mathrm{D-mod}(\mathcal{Z}))^{\mathrm{Hecke}_{G,\mathcal{Z},U}^{\mathrm{glob},\wedge,\mathcal{L}_\kappa^{\mathrm{glob}}}} \rightarrow (\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathcal{Z}})^{\mathrm{Hecke}_{G,\mathcal{Z}}^{\mathrm{loc},\wedge,\mathcal{L}_\kappa^{\mathrm{loc}}}} \stackrel{\text{Sect. 13.1.1}}{\simeq} \mathrm{KL}(G)_{\kappa,\mathcal{Z}}.$$

We will denote the functor (13.6) by

$$(\mathrm{ev}_{\mathcal{Z},U})_*^{\mathrm{Hecke}^\wedge\text{-enh}}.$$

13.1.6. Composing, we obtain that the functor

$$(\mathrm{ev}_{\mathcal{Z},U})_* \circ (\mathrm{oblv}_\kappa^l \otimes \mathrm{Id}) : \mathrm{D-mod}_\kappa(U) \otimes \mathrm{D-mod}(\mathcal{Z}) \rightarrow \mathrm{Rep}(\mathfrak{L}^+(G))_{\mathcal{Z}}$$

lifts to a functor

$$(13.7) \quad \mathrm{D-mod}_\kappa(U) \otimes \mathrm{D-mod}(\mathcal{Z}) \xrightarrow{(\mathrm{ev}_{\mathcal{Z},U})_*^{\mathrm{Hecke}^\wedge\text{-enh}} \circ \mathrm{oblv}_{\mathrm{inf} \rightarrow \mathrm{Hecke}^\wedge,\mathcal{Z},U}} \mathrm{KL}(G)_{\kappa,\mathcal{Z}}.$$

13.1.7. It follows from the construction of the identification

$$(13.8) \quad (\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathcal{Z}})^{\mathrm{Hecke}_{G,\mathcal{Z}}^{\mathrm{loc},\wedge,\mathcal{L}_\kappa^{\mathrm{loc}}}} \simeq \mathrm{KL}(G)_{\kappa,\mathcal{Z}}$$

that the functor

$$(\mathrm{ev}_{\mathcal{Z},U})_*^{\mathrm{Hecke}^\wedge\text{-enh}} \circ \mathrm{oblv}_{\mathrm{inf} \rightarrow \mathrm{Hecke}^\wedge,\mathcal{Z},U}$$

of (13.7) identifies canonically with the functor

$$\Gamma_{G,\kappa,\mathcal{Z},U} := \Gamma_{G,\kappa,\mathcal{Z}} \circ j_{*,\mathrm{co}},$$

so that the diagram

$$\begin{array}{ccc} \mathrm{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} \circ (\mathrm{ev}_{\mathcal{Z},U})_*^{\mathrm{Hecke}^\wedge\text{-enh}} \circ \mathrm{oblv}_{\mathrm{inf} \rightarrow \mathrm{Hecke}^\wedge,\mathcal{Z},U} & \xrightarrow{\sim} & \mathrm{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} \circ \Gamma_{G,\kappa,\mathcal{Z},U} \\ \sim \downarrow & & \downarrow \sim \\ (\mathrm{ev}_{\mathcal{Z},U})_* \circ (\mathrm{oblv}_\kappa^l \otimes \mathrm{Id}) & \xrightarrow{\mathrm{id}} & (\mathrm{ev}_{\mathcal{Z},U})_* \circ (\mathrm{oblv}_\kappa^l \otimes \mathrm{Id}) \end{array}$$

commutes.

13.2. The functor $\mathrm{Loc}_{G,\kappa}$ and the infinitesimal Hecke groupoid.

⁵³Note that as in Remark 10.2.8, the difference between $\mathrm{Rep}(\mathfrak{L}^+(G))$ and $\mathrm{QCoh}(\mathrm{pt}/\mathfrak{L}^+(G))$ is immaterial here.

13.2.1. Let $U \subset \text{Bun}_G$ be a quasi-compact open substack, and let $\mathbb{Z}$ be a space mapping to Ran .

By the same logic as in Sect. 13.1.5, the functor

$$(\text{ev}_{\mathbb{Z}, U})^* : (\text{pt} / \mathfrak{L}^+(G))_{\text{Ran}} \rightarrow \text{QCoh}(U) \otimes \text{D-mod}(\mathbb{Z})$$

lifts to a functor

$$(13.9) \quad \text{KL}(G)_{\kappa, \mathbb{Z}} \simeq (\text{Rep}(\mathfrak{L}^+(G))_{\mathbb{Z}})^{\text{Hecke}_{G, \mathbb{Z}}^{\text{loc}, \wedge}, \mathcal{L}_{\kappa}^{\text{loc}}} \rightarrow (\text{QCoh}(U) \otimes \text{D-mod}(\mathbb{Z}))^{\text{Hecke}_{G, \mathbb{Z}, U}^{\text{glob}, \wedge}, \mathcal{L}_{\kappa}^{\text{glob}}},$$

to be denoted

$$(\text{ev}_{\mathbb{Z}, U})^*, \text{Hecke}^{\wedge} \text{-enh}.$$

Furthermore, the functors $(\text{ev}_{\mathbb{Z}, U})^*, \text{Hecke}^{\wedge} \text{-enh}$ and $(\text{ev}_{\mathbb{Z}, U})_{*}^{\text{Hecke}^{\wedge} \text{-enh}}$ are adjoint.

13.2.2. Denote by M_{κ}^{loc} the (factorization) monad

$$\mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa}} \circ \mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa}}$$

acting on $\text{Rep}(\mathfrak{L}^+(G))$.

Denote by $M_{\kappa, \mathbb{Z}, U}^{\text{glob}}$ the monad acting on $\text{QCoh}(U) \otimes \text{D-mod}(\mathbb{Z})$, corresponding to the forgetful functor

$$(\text{QCoh}(U) \otimes \text{D-mod}(\mathbb{Z}))^{\text{Hecke}_{G, \mathbb{Z}, U}^{\text{glob}, \wedge}, \mathcal{L}_{\kappa}^{\text{glob}}} \rightarrow \text{QCoh}(U) \otimes \text{D-mod}(\mathbb{Z})$$

and its left adjoint.

By adjunction, we obtain that the functor $\text{ev}_{\mathbb{Z}, U}^*$ intertwines the monads M_{κ}^{loc} and $M_{\kappa, \mathbb{Z}, U}^{\text{glob}}$, i.e., we have a commutative diagram

$$(13.10) \quad \begin{array}{ccc} \text{Rep}(\mathfrak{L}^+(G))_{\mathbb{Z}} & \xrightarrow{\text{ev}_{\mathbb{Z}, U}^*} & \text{QCoh}(U) \otimes \text{D-mod}(\mathbb{Z}) \\ M_{\kappa}^{\text{loc}} \uparrow & & \uparrow M_{\kappa, \mathbb{Z}, U}^{\text{glob}} \\ \text{Rep}(\mathfrak{L}^+(G))_{\mathbb{Z}} & \xrightarrow{\text{ev}_{\mathbb{Z}, U}^*} & \text{QCoh}(U) \otimes \text{D-mod}(\mathbb{Z}). \end{array}$$

13.2.3. The diagrams (13.10) are compatible under inclusions $U_1 \subset U_2$. Passing to the limit over U , we obtain a commutative diagram

$$(13.11) \quad \begin{array}{ccc} \text{Rep}(\mathfrak{L}^+(G))_{\mathbb{Z}} & \xrightarrow{\text{ev}_{\mathbb{Z}}^*} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) \\ M_{\kappa}^{\text{loc}} \uparrow & & \uparrow M_{\kappa, \mathbb{Z}}^{\text{glob}} \\ \text{Rep}(\mathfrak{L}^+(G))_{\mathbb{Z}} & \xrightarrow{\text{ev}_{\mathbb{Z}}^*} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}), \end{array}$$

where $M_{\kappa, \mathbb{Z}}^{\text{glob}} := M_{\kappa, \mathbb{Z}, U}^{\text{glob}}$ for $U = \text{Bun}_G$.

13.2.4. Let $M_{\kappa, \mathbb{Z}}^{\text{inf}}$ denote the monad

$$(\mathbf{oblv}_{\kappa}^l \circ \mathbf{ind}_{\kappa}^k) \otimes \text{Id}$$

acting on $\text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z})$.

The map of groupoids (13.2) gives rise to a map of monads

$$M_{\kappa, \mathbb{Z}}^{\text{glob}} \rightarrow M_{\kappa, \mathbb{Z}}^{\text{inf}}.$$

In particular, we obtain a diagram

$$(13.12) \quad \begin{array}{ccc} \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) & \xrightarrow{\text{id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) \\ M_{\kappa, \mathbb{Z}}^{\text{glob}} \uparrow & \searrow & \uparrow M_{\kappa, \mathbb{Z}}^{\text{inf}} \\ \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) & \xrightarrow{\text{id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) \end{array}$$

Concatenating diagrams (13.11) and (13.12) we obtain a diagram

$$(13.13) \quad \begin{array}{ccc} \text{Rep}(\mathfrak{L}^+(G))_{\mathbb{Z}} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\mathbb{Z}}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) \\ \uparrow M_{\kappa}^{\text{loc}} & \searrow & \uparrow M_{\kappa, \mathbb{Z}}^{\text{inf}} \\ \text{Rep}(\mathfrak{L}^+(G))_{\mathbb{Z}} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\mathbb{Z}}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) \end{array}$$

It follows from Sect. 13.1.7 that diagram (13.13) identifies with the outer diagram in

$$(13.14) \quad \begin{array}{ccc} \text{Rep}(\mathfrak{L}^+(G))_{\mathbb{Z}} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\mathbb{Z}}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) \\ \uparrow \text{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa}} & \searrow & \uparrow \text{oblv}_{\kappa}^l \\ \text{KL}(G)_{\kappa, \mathbb{Z}} & \xrightarrow{(\text{Loc}_{G, \kappa})_{\mathbb{Z}}} & \text{D-mod}_{\kappa}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) \\ \uparrow \text{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa}} & & \uparrow \text{ind}_{\kappa}^l \\ \text{Rep}(\mathfrak{L}^+(G))_{\mathbb{Z}} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\mathbb{Z}}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}), \end{array}$$

in which the lower square is (the base change along $\mathbb{Z} \rightarrow \text{Ran}$) of (10.16), and the upper square is obtained from the lower square by passing to adjoints along the vertical arrows.

13.3. Proof of Theorem 12.1.8. We are now ready to prove Theorem 12.1.8.

13.3.1. We wish to show that the natural transformation in

$$\begin{array}{ccc} \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}^{\subseteq}} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\text{Ran}^{\subseteq}}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^{\subseteq}) \\ \uparrow \text{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa}} & \searrow & \uparrow \text{oblv}_{\kappa}^l \otimes \text{Id} \\ \text{KL}(G)_{\kappa, \text{Ran}^{\subseteq}} & \xrightarrow{(\text{Loc}_{G, \kappa})_{\text{Ran}^{\subseteq}}} & \text{D-mod}_{\kappa}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^{\subseteq}) \end{array}$$

becomes an isomorphism, after we:

- We precompose with $\text{ins. vac}_{\text{Ran}} : \text{KL}(G)_{\kappa, \text{Ran}} \rightarrow \text{KL}(G)_{\kappa, \text{Ran}^{\subseteq}}$;
- Postcompose with $\text{Id} \otimes C_c(\text{Ran}^{\subseteq}, -)$.

Since the essential image of

$$\text{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa}} : \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}} \rightarrow \text{KL}(G)_{\kappa, \text{Ran}}$$

generates the target, it is sufficient to show that the natural transformation becomes an isomorphism when we further precompose with this functor.

Thus, we obtain the diagram

$$(13.15) \quad \begin{array}{ccc} \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}^\subseteq} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\text{Ran}^\subseteq}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\ \uparrow \text{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} & \searrow & \uparrow \text{oblv}_\kappa^l \otimes \text{Id} \\ \text{KL}(G)_{\kappa, \text{Ran}^\subseteq} & \xrightarrow{(\text{Loc}_{G, \kappa})_{\text{Ran}^\subseteq}} & \text{D-mod}_\kappa(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\ \uparrow \text{ins.vac}_{\text{Ran}} & & \uparrow \text{Id} \otimes (\text{pr}_{\text{small}})^! \\ \text{KL}(G)_{\kappa, \text{Ran}} & \xrightarrow{(\text{Loc}_{G, \kappa})_{\text{Ran}}} & \text{D-mod}_\kappa(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}) \\ \uparrow \text{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} & & \uparrow \text{ind}_\kappa^l \otimes \text{Id} \\ \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\text{Ran}}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}), \end{array}$$

which we then compose with

$$\text{Id} \otimes \text{C}_c(\text{Ran}^\subseteq, -) : \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \rightarrow \text{QCoh}(\text{Bun}_G),$$

and we need to show that the resulting natural transformation commutes.

13.3.2. Consider the commutative diagram obtained by concatenating the lower two squares in (13.15). It is easy to see that it identifies with the outer diagram in

$$\begin{array}{ccc} \text{KL}(G)_{\kappa, \text{Ran}^\subseteq} & \xrightarrow{(\text{Loc}_{G, \kappa})_{\text{Ran}^\subseteq}} & \text{D-mod}_\kappa(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\ \uparrow \text{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} & & \uparrow \text{ind}_\kappa^l \otimes \text{Id} \\ \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}^\subseteq} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\text{Ran}^\subseteq}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\ \uparrow \text{ins.vac}_{\text{Ran}} & & \uparrow \text{Id} \otimes (\text{pr}_{\text{small}})^! \\ \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\text{Ran}}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}). \end{array}$$

Hence, instead of (13.15), we can consider the diagram

$$(13.16) \quad \begin{array}{ccc} \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}^\subseteq} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\text{Ran}^\subseteq}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\ \uparrow \text{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa} & \searrow & \uparrow \text{oblv}_\kappa^! \otimes \text{Id} \\ \text{KL}(G)_{\kappa, \text{Ran}^\subseteq} & \xrightarrow{(\text{Loc}_G, \kappa)_{\text{Ran}^\subseteq}} & \text{D-mod}_\kappa(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\ \uparrow \text{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa} & & \uparrow \text{ind}_\kappa^! \otimes \text{Id} \\ \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}^\subseteq} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\text{Ran}^\subseteq}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\ \uparrow \text{ins.vac}_{\text{Ran}} & & \uparrow \text{Id} \otimes (\text{pr}_{\text{small}})^! \\ \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\text{Ran}}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}). \end{array}$$

13.3.3. Using Sect. 13.2.4, we can replace (13.16) by the diagram

$$(13.17) \quad \begin{array}{ccc} \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}^\subseteq} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\text{Ran}^\subseteq}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\ \uparrow M_\kappa^{\text{loc}} & \searrow & \uparrow M_{\kappa, \text{Ran}^\subseteq}^{\text{inf}} \\ \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}^\subseteq} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\text{Ran}^\subseteq}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\ \uparrow \text{ins.vac}_{\text{Ran}} & & \uparrow \text{Id} \otimes (\text{pr}_{\text{small}})^! \\ \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\text{Ran}}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}). \end{array}$$

We further rewrite (13.17) as the outer diagram in

$$(13.18) \quad \begin{array}{ccccc} \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}^\subseteq} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\text{Ran}^\subseteq}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\ \uparrow M_\kappa^{\text{loc}} & & \uparrow M_{\kappa, \text{Ran}^\subseteq}^{\text{glob}} & \searrow & \uparrow M_{\kappa, \text{Ran}^\subseteq}^{\text{inf}} \\ \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}^\subseteq} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\text{Ran}^\subseteq}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\ \uparrow \text{ins.vac}_{\text{Ran}} & & \uparrow \text{Id} \otimes (\text{pr}_{\text{small}})^! & & \uparrow \text{Id} \otimes (\text{pr}_{\text{small}})^! \\ \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}} & \xrightarrow{(\text{Loc}_G^{\text{QCoh}})_{\text{Ran}}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}). \end{array}$$

13.3.4. The left portion of (13.18) commutes. Hence, it is enough to show that the outer diagram in

$$\begin{array}{ccc}
 & \text{QCoh}(\text{Bun}_G) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \\
 & \uparrow \text{Id} \otimes C_c^*(\text{Ran}^\subseteq, -) & & \uparrow \text{Id} \otimes C_c^*(\text{Ran}^\subseteq, -) \\
 (13.19) \quad & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\
 & \uparrow M_{\kappa, \text{Ran}^\subseteq}^{\text{glob}} & \searrow & \uparrow M_{\kappa, \text{Ran}^\subseteq}^{\text{inf}} \\
 & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\
 & \uparrow \text{Id} \otimes (\text{pr}_{\text{small}})^! & & \uparrow \text{Id} \otimes (\text{pr}_{\text{small}})^! \\
 & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}).
 \end{array}$$

commutes.

We will show that already the diagram

$$\begin{array}{ccc}
 & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}) \\
 & \uparrow \text{Id} \otimes (\text{pr}_{\text{small}})^! & & \uparrow \text{Id} \otimes (\text{pr}_{\text{small}})^! \\
 (13.20) \quad & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\
 & \uparrow M_{\kappa, \text{Ran}^\subseteq}^{\text{glob}} & \searrow & \uparrow M_{\kappa, \text{Ran}^\subseteq}^{\text{inf}} \\
 & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}^\subseteq) \\
 & \uparrow \text{Id} \otimes (\text{pr}_{\text{small}})^! & & \uparrow \text{Id} \otimes (\text{pr}_{\text{small}})^! \\
 & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}).
 \end{array}$$

commutes.

This is a particular case of the next assertion:

Theorem 13.3.5. *For any $\mathcal{Z} \rightarrow \text{Ran}$, the natural transformation in the diagram*

$$\begin{array}{ccc}
 \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathcal{Z}) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathcal{Z}) \\
 \uparrow \text{Id} \otimes (\text{pr}_{\text{small}})! & & \uparrow \text{Id} \otimes (\text{pr}_{\text{small}})! \\
 \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathcal{Z}^\subseteq) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathcal{Z}^\subseteq) \\
 \uparrow M_{\kappa, \mathcal{Z}^\subseteq}^{\text{glob}} & \searrow & \uparrow M_{\kappa, \mathcal{Z}^\subseteq}^{\text{inf}} \\
 \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathcal{Z}^\subseteq) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathcal{Z}^\subseteq) \\
 \uparrow \text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})! & & \uparrow \text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})! \\
 \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathcal{Z}) & \xrightarrow{\text{Id}} & \text{QCoh}(\text{Bun}_G) \otimes \text{D-mod}(\mathcal{Z}).
 \end{array}$$

induced by the map of monads

$$M_{\kappa, \mathcal{Z}^\subseteq}^{\text{glob}} \rightarrow M_{\kappa, \mathcal{Z}^\subseteq}^{\text{inf}},$$

is an isomorphism.

This theorem is a particular case of [Ro2, Theorem 4.3.6], combined with Remark 4.5.6 in *loc. cit.* $\square$ [Theorem 12.1.8]

13.4. Localization is (almost) a localization.

13.4.1. The goal of the next few subsections is prove the following assertion:

Theorem 13.4.2. *Let $U \xrightarrow{j} \text{Bun}_G$ be a quasi-compact open substack. Then the functor*

$$\text{Loc}_{G, \kappa, U} := j^* \circ \text{Loc}_{G, \kappa}, \quad \text{KL}(G)_{\kappa, \text{Ran}} \rightarrow \text{D-mod}_{\kappa}(U)$$

is a localization.⁵⁴

13.4.3. We first consider a version of Theorem 13.4.2 when instead of $\text{D-mod}_{\kappa}(\text{Bun}_G)$, we take $\text{QCoh}(\text{Bun}_G)$.

Consider the local-to-global functor

$$\underline{\text{Loc}}_G^{\text{QCoh}} : \underline{\text{Rep}}(\mathfrak{L}^+(G)) \rightarrow \text{QCoh}(\text{Bun}_G) \otimes \underline{\text{D-mod}}(\text{Ran}).$$

For an open substack $U \subset \text{Bun}_G$, denote

$$\text{Loc}_{G, U}^{\text{QCoh}} := j^* \circ \underline{\text{Loc}}_G^{\text{QCoh}}, \quad \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}} \rightarrow \text{QCoh}(U).$$

We will prove:⁵⁵

Theorem 13.4.4. *If U is quasi-compact, then the functor $\text{Loc}_{G, U}^{\text{QCoh}}$ is a localization.*

⁵⁴I.e., its right adjoint is fully faithful.

⁵⁵A more general assertion, of which Theorem 13.4.4 is a particular case, appears in [GLC4, Proposition C.1.7].

13.4.5. Consider the following general paradigm: let

$$F : \mathbf{C}_1 \rightarrow \mathbf{C}_2$$

be a continuous functor between compactly generated categories. Assume that F preserves compactness.

Let

$$F^{\text{fake-op}} : \mathbf{C}_1^\vee \rightarrow \mathbf{C}_2^\vee$$

be the conjugate functor, i.e.,

$$F^{\text{fake-op}} = (F^R)^\vee.$$

In other words, $F^{\text{fake-op}}$ is the ind-extension of

$$(\mathbf{C}_1^\vee)^c \simeq (\mathbf{C}_1^c)^{\text{op}} \xrightarrow{F^{\text{op}}} (\mathbf{C}_2^c)^{\text{op}} \simeq (\mathbf{C}_2^\vee)^c.$$

Let $u_{\mathbf{C}_i} \in \mathbf{C}_i \otimes \mathbf{C}_i^\vee$ be the unit of the duality. Note that we have a canonically defined map

$$(13.21) \quad (F \otimes F^{\text{fake-op}})(u_{\mathbf{C}_1}) \rightarrow u_{\mathbf{C}_2}.$$

Namely,

$$\begin{aligned} (F \otimes F^{\text{fake-op}})(u_{\mathbf{C}_1}) &= (F \otimes \text{Id}) \circ (\text{Id} \otimes F^{\text{fake-op}})(u_{\mathbf{C}_1}) = (F \otimes \text{Id}) \circ (\text{Id} \otimes (F^R)^\vee)(u_{\mathbf{C}_1}) \simeq \\ &\simeq (F \otimes \text{Id}) \circ (F^R \otimes \text{Id})(u_{\mathbf{C}_1}) = ((F \circ F^R) \otimes \text{Id})(u_{\mathbf{C}_1}) \rightarrow u_{\mathbf{C}_1}. \end{aligned}$$

13.4.6. The following is elementary:

Lemma 13.4.7. *The functor F is a localization if and only if (13.21) is an isomorphism.*

13.4.8. We will prove Theorem 13.4.4 by applying Lemma 13.4.7 to $\mathbf{C}_1 := \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}}$, $\mathbf{C}_2 = \text{QCoh}(U)$ and $F := \text{Loc}_{G,U}^{\text{QCoh}}$.

First, it is easy to see that $\text{Loc}_{G,U}^{\text{QCoh}}$ preserves compactness.

13.4.9. We have a canonical self-duality

$$(13.22) \quad \text{Rep}(\mathfrak{L}^+(G))^\vee \simeq \text{Rep}(\mathfrak{L}^+(G))$$

as a factorization category. Its unit object $u_{\text{Rep}(\mathfrak{L}^+(G))}$ is the regular representation

$$R_{\mathfrak{L}^+(G)} \in \text{Rep}(\mathfrak{L}^+(G)) \otimes \text{Rep}(\mathfrak{L}^+(G)),$$

which has a natural structure of (commutative) factorization algebra.

From (13.22) we obtain a self-duality

$$(13.23) \quad (\text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}})^\vee \simeq \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}}.$$

The unit $u_{\text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}}}$ of (13.23) is given by the image of

$$R_{\mathfrak{L}^+(G), \text{Ran}} \in (\text{Rep}(\mathfrak{L}^+(G)) \otimes \text{Rep}(\mathfrak{L}^+(G)))_{\text{Ran}}$$

under

$$\begin{aligned} &(\text{Rep}(\mathfrak{L}^+(G)) \otimes \text{Rep}(\mathfrak{L}^+(G)))_{\text{Ran}} \simeq \\ &\simeq (\text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}} \otimes \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}}) \underset{\text{D-mod}(\text{Ran} \times \text{Ran})}{\otimes} \text{D-mod}(\text{Ran}) \xrightarrow{\text{Id} \otimes (\Delta_{\text{Ran}})!} \\ &\rightarrow (\text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}} \otimes \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}}) \underset{\text{D-mod}(\text{Ran} \times \text{Ran})}{\otimes} (\text{D-mod}(\text{Ran}) \otimes \text{D-mod}(\text{Ran})) = \\ &= \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}} \otimes \text{Rep}(\mathfrak{L}^+(G))_{\text{Ran}}. \end{aligned}$$

13.4.10. With respect to the canonical self-duality

$$\mathrm{QCoh}(U)^\vee \simeq \mathrm{QCoh}(U),$$

we have

$$(\mathrm{Loc}_{G,U}^{\mathrm{QCoh}})^{\mathrm{fake-op}} \simeq \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}.$$

From here we obtain

$$\begin{aligned} (\mathrm{Loc}_{G,U}^{\mathrm{QCoh}} \otimes (\mathrm{Loc}_{G,U}^{\mathrm{QCoh}}))^{\mathrm{fake-op}} (u_{\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}}}) &\simeq (\mathrm{Loc}_{G,U}^{\mathrm{QCoh}} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}) \circ (\Delta_{\mathrm{Ran}})! (R_{\mathfrak{L}^+(G), \mathrm{Ran}}) \simeq \\ &\simeq \mathrm{Loc}_{G \times G, U \times U}^{\mathrm{QCoh}} (R_{\mathfrak{L}^+(G), \mathrm{Ran}}), \end{aligned}$$

see Sect. C.5.15 for the latter isomorphism.

Thus, the map (13.21) is a map

$$(13.24) \quad \mathrm{Loc}_{G \times G, U \times U}^{\mathrm{QCoh}} (R_{\mathfrak{L}^+(G), \mathrm{Ran}}) \rightarrow (\Delta_{\mathrm{Bun}_G})_* (\mathcal{O}_{\mathrm{Bun}_G}).$$

13.4.11. Hence, by Lemma 13.4.7, in order to prove Theorem 13.4.4, we have to show that the map (13.24) is an isomorphism.

The latter is obtained by repeating verbatim the proof of [AGKRRV, Theorem 12.6.3], see also Remark 12.6.6 in *loc. cit.*

□[Theorem 13.4.4]

13.4.12. We are now ready to prove Theorem 13.4.2. We will give two proofs: one in Sect. 13.5 and another in Sect. 13.6.

13.5. First proof of Theorem 13.4.2.

13.5.1. We need to show that the functor

$$\Gamma_{G, \kappa, U} : \mathrm{D-mod}_\kappa(U) \rightarrow \mathrm{KL}(G)_{\kappa, \mathrm{Ran}}$$

is fully faithful.

For $\mathbb{Z} \rightarrow \mathrm{Ran}$, we interpret the category $\mathrm{KL}(G)_{\kappa, \mathbb{Z}}$ as

$$(\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathbb{Z}})^{\mathrm{Hecke}_{G, \mathbb{Z}}^{\mathrm{loc}, \wedge}, \mathcal{L}_\kappa^{\mathrm{loc}}}.$$

We interpret the functor $\Gamma_{G, \kappa, U}$ as a composition

$$\begin{aligned} (13.25) \quad \mathrm{D-mod}_\kappa(U) &\xrightarrow{\mathrm{Id} \otimes \omega_{\mathrm{Ran}}} \mathrm{D-mod}_\kappa(U) \otimes \mathrm{D-mod}(\mathbb{Z}) \xrightarrow{\mathrm{oblv}_{\mathrm{inf} \rightarrow \mathrm{Hecke}^\wedge, \mathrm{Ran}, U}} \\ &\rightarrow (\mathrm{QCoh}(U) \otimes \mathrm{D-mod}(\mathrm{Ran}))^{\mathrm{Hecke}_{G, \mathrm{Ran}, U}^{\mathrm{glob}, \wedge}, \mathcal{L}_\kappa^{\mathrm{glob}}} \xrightarrow{(\mathrm{ev}_{\mathrm{Ran}, U})_*^{\mathrm{Hecke}^\wedge - \mathrm{enh}}} \\ &\rightarrow (\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}})^{\mathrm{Hecke}_{G, \mathrm{Ran}}^{\mathrm{loc}, \wedge}, \mathcal{L}_\kappa^{\mathrm{loc}}}, \end{aligned}$$

where:

- $\mathrm{oblv}_{\mathrm{inf} \rightarrow \mathrm{Hecke}^\wedge, \mathrm{Ran}, U}$ is the functor (13.4);
- $(\mathrm{ev}_{\mathrm{Ran}, U})_*^{\mathrm{Hecke}^\wedge - \mathrm{enh}}$ is the functor (13.6).

We will show that (13.25) is a *retract* of a fully faithful functor⁵⁶ and is therefore itself fully faithful.

Remark 13.5.2. Here is the reason the retraction appears:

There are two copies of the Ran space that play a role: one is in Theorem 13.4.4, and the other is Theorem 10.1.5. To achieve fully faithfulness of (13.25), we need both of these copies, and that is how that Ran space becomes “doubled” in the guise of $\mathrm{Ran}^\subseteq$.

We then pass from $\mathrm{Ran}^\subseteq$ to just Ran , but at the expense of replacing the original fully faithful functor (13.27) by its retract.

⁵⁶We say that a functor $F : \mathbf{C} \rightarrow \mathbf{D}$ is a retract of a functor $F_1 : \mathbf{C} \rightarrow \mathbf{D}_1$ if there exists a retraction $\mathbf{D} \xrightarrow{\phi} \mathbf{D}_1 \xrightarrow{\psi} \mathbf{D}$ so that $F_1 \simeq \phi \circ F$ and $F \simeq F_1 \circ \psi$.

13.5.3. Note that the pullbacks of $\text{Hecke}_{G,\text{Ran}}^{\text{loc},\wedge}$ along pr_{small} and pr_{big} give rise to a well-defined groupoids on $(\text{pt}/\mathcal{L}^+(G))_{\text{Ran}\subseteq}$; we will denote them by

$$\text{Hecke}_{G,\text{Ran}\subseteq,\text{small}}^{\text{loc},\wedge} \quad \text{and} \quad \text{Hecke}_{G,\text{Ran}\subseteq,\text{big}}^{\text{loc},\wedge},$$

respectively.

The unital structure on $\text{Hecke}_{G,\text{Ran}}^{\text{loc},\wedge}$ gives rise to a map

$$(13.26) \quad \text{Hecke}_{G,\text{Ran}\subseteq,\text{small}}^{\text{loc},\wedge} \rightarrow \text{Hecke}_{G,\text{Ran}\subseteq,\text{big}}^{\text{loc},\wedge}.$$

Similar definitions apply to $\text{Hecke}_{G,\text{Ran},U}^{\text{glob},\wedge}$.

13.5.4. Consider the functor

$$(13.27) \quad \begin{aligned} \text{D-mod}_\kappa(U) &\xrightarrow{\text{Id} \otimes \omega_{\text{Ran}}} \text{D-mod}_\kappa(U) \otimes \text{D-mod}(\mathbb{Z}) \xrightarrow{\text{oblv}_{\text{inf} \rightarrow \text{Hecke}^\wedge, \text{Ran}, U}} \\ &\rightarrow (\text{QCoh}(U) \otimes \text{D-mod}(\text{Ran}))^{\text{Hecke}_{G,\text{Ran},U}^{\text{glob},\wedge}, \mathcal{L}_\kappa^{\text{glob}}} \rightarrow \\ &\xrightarrow{(\text{pr}_{\text{small}})^!} \left(\text{QCoh}(U) \otimes \text{D-mod}(\text{Ran}^\subseteq) \right)^{\text{Hecke}_{G,\text{Ran}\subseteq,\text{small},U}^{\text{glob},\wedge}, \mathcal{L}_\kappa^{\text{glob}}} \rightarrow \\ &\xrightarrow{(\text{ev}_{\text{Ran}\subseteq,U})_*^{\text{Hecke}_{\text{small}}^\wedge\text{-enh}}} \left(\text{Rep}(\mathcal{L}^+(G))_{\text{Ran}\subseteq} \right)^{\text{Hecke}_{G,\text{Ran}\subseteq,\text{small}}^{\text{loc},\wedge}, \mathcal{L}_\kappa^{\text{loc}}}, \end{aligned}$$

where $(\text{ev}_{\text{Ran}\subseteq,U})_*^{\text{Hecke}_{\text{small}}^\wedge\text{-enh}}$ is the enhancement for the corresponding groupoids of the functor $\text{ev}_{\mathbb{Z}\subseteq,U}$, cf. formula (13.4).

13.5.5. We claim that the functor (13.27) is fully faithful. In fact, we claim that it is a composition of two fully faithful functors.

Namely, the functor

$$\begin{aligned} \text{D-mod}_\kappa(U) &\xrightarrow{\text{Id} \otimes \omega_{\text{Ran}}} \text{D-mod}_\kappa(U) \otimes \text{D-mod}(\mathbb{Z}) \xrightarrow{\text{oblv}_{\text{inf} \rightarrow \text{Hecke}^\wedge, \text{Ran}, U}} \\ &\rightarrow (\text{QCoh}(U) \otimes \text{D-mod}(\text{Ran}))^{\text{Hecke}_{G,\text{Ran},U}^{\text{glob},\wedge}, \mathcal{L}_\kappa^{\text{glob}}} \end{aligned}$$

is fully faithful thanks to [Ro2, Theorem 4.5.3] (cf. Theorem 10.1.5).

The fact that

$$(13.28) \quad \begin{aligned} &(\text{QCoh}(U) \otimes \text{D-mod}(\text{Ran}))^{\text{Hecke}_{G,\text{Ran},U}^{\text{glob},\wedge}, \mathcal{L}_\kappa^{\text{glob}}} \xrightarrow{(\text{pr}_{\text{small}})^!} \\ &\rightarrow \left(\text{QCoh}(U) \otimes \text{D-mod}(\text{Ran}^\subseteq) \right)^{\text{Hecke}_{G,\text{Ran}\subseteq,\text{small},U}^{\text{glob},\wedge}, \mathcal{L}_\kappa^{\text{glob}}} \xrightarrow{(\text{ev}_{\text{Ran}\subseteq,U})_*^{\text{Hecke}_{\text{small}}^\wedge\text{-enh}}} \\ &\rightarrow \left(\text{Rep}(\mathcal{L}^+(G))_{\text{Ran}\subseteq} \right)^{\text{Hecke}_{G,\text{Ran}\subseteq,\text{small}}^{\text{loc},\wedge}, \mathcal{L}_\kappa^{\text{loc}}}, \end{aligned}$$

is fully faithful is a formal consequence of the fact that

$$(13.29) \quad \text{QCoh}(U) \otimes \text{D-mod}(\text{Ran}) \xrightarrow{(\text{pr}_{\text{small}})^!} \text{QCoh}(U) \otimes \text{D-mod}(\text{Ran}^\subseteq) \xrightarrow{\text{ev}_{\text{Ran}\subseteq,\text{small},U}} \text{Rep}(\mathcal{L}^+(G))_{\text{Ran}\subseteq}$$

is fully faithful, which is a parameterized version of Theorem 13.4.4:

Indeed, each side of (13.28) is obtained from the corresponding side of (13.29) as modules for the corresponding monad, and the functor (13.29) intertwines the actions of these monads.

13.5.6. We now claim that the functor (13.25) is a retract of (13.27).

Indeed, the corresponding functor

$$(\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}})^{\mathrm{Hecke}_{G,\mathrm{Ran}}^{\mathrm{loc},\wedge}, \mathcal{L}_\kappa^{\mathrm{loc}}} \rightarrow (\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran} \subseteq})^{\mathrm{Hecke}_{G,\mathrm{Ran} \subseteq, \mathrm{small}}^{\mathrm{loc},\wedge}, \mathcal{L}_\kappa^{\mathrm{loc}}}$$

is

$$(13.30) \quad (\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}})^{\mathrm{Hecke}_{G,\mathrm{Ran}}^{\mathrm{loc},\wedge}, \mathcal{L}_\kappa^{\mathrm{loc}}} \xrightarrow{(\mathrm{pr}_{\mathrm{big}})^!} (\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran} \subseteq})^{\mathrm{Hecke}_{G,\mathrm{Ran} \subseteq, \mathrm{big}}^{\mathrm{loc},\wedge}, \mathcal{L}_\kappa^{\mathrm{loc}}} \rightarrow (\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran} \subseteq})^{\mathrm{Hecke}_{G,\mathrm{Ran} \subseteq, \mathrm{small}}^{\mathrm{loc},\wedge}, \mathcal{L}_\kappa^{\mathrm{loc}}},$$

where the second arrow is given by restriction along (13.26).

The functor

$$(\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran} \subseteq})^{\mathrm{Hecke}_{G,\mathrm{Ran} \subseteq, \mathrm{small}}^{\mathrm{loc},\wedge}, \mathcal{L}_\kappa^{\mathrm{loc}}} \rightarrow (\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}})^{\mathrm{Hecke}_{G,\mathrm{Ran}}^{\mathrm{loc},\wedge}, \mathcal{L}_\kappa^{\mathrm{loc}}}$$

is given by

$$(13.31) \quad (\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran} \subseteq})^{\mathrm{Hecke}_{G,\mathrm{Ran} \subseteq, \mathrm{small}}^{\mathrm{loc},\wedge}, \mathcal{L}_\kappa^{\mathrm{loc}}} \xrightarrow{\mathrm{diag}^!} (\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}})^{\mathrm{Hecke}_{G,\mathrm{Ran}}^{\mathrm{loc},\wedge}, \mathcal{L}_\kappa^{\mathrm{loc}}}.$$

It is a straightforward verification that

$$(13.30) \circ (13.25) \simeq (13.27) \text{ and } (13.31) \circ (13.27) \simeq (13.25).$$

□[First proof of Theorem 13.4.2]

13.6. **Second proof of Theorem 13.4.2.** In this subsection we will apply Lemma 13.4.7 directly to the functor $\mathrm{Loc}_{G,\kappa,U}$ to deduce the assertion of Theorem 13.4.2.

13.6.1. Let $\kappa' := 2 \cdot \mathrm{crit} - \kappa$. Recall that the unit $\mathrm{u}_{\mathrm{KL}(G)_\kappa}$ of the duality

$$(\mathrm{KL}(G)_\kappa)^\vee \simeq \mathrm{KL}(G)_{\kappa'}$$

is the (factorization algebra) object $\mathfrak{CD}\mathfrak{D}(G)_{\kappa,\kappa'}$, see Sect. 2.2.3.

The dual of $\mathrm{KL}(G)_{\kappa,\mathrm{Ran}}$ identifies with $\mathrm{KL}(G)_{\kappa',\mathrm{Ran}}$ with the unit $\mathrm{u}_{\mathrm{KL}(G)_{\kappa,\mathrm{Ran}}}$ of the duality being the image of

$$(\mathfrak{CD}\mathfrak{D}(G)_{\kappa,\kappa'})_{\mathrm{Ran}} \in (\mathrm{KL}(G)_\kappa \otimes \mathrm{KL}(G)_{\kappa'})_{\mathrm{Ran}}$$

under

$$\begin{aligned} (\mathrm{KL}(G)_\kappa \otimes \mathrm{KL}(G)_{\kappa'})_{\mathrm{Ran}} &\simeq (\mathrm{KL}(G)_{\kappa,\mathrm{Ran}} \otimes \mathrm{KL}(G)_{\kappa',\mathrm{Ran}})_{\mathrm{D-mod}(\mathrm{Ran} \times \mathrm{Ran})} \otimes_{\mathrm{D-mod}(\mathrm{Ran} \times \mathrm{Ran})} \mathrm{D-mod}(\mathrm{Ran}) \xrightarrow{\mathrm{Id} \otimes (\Delta_{\mathrm{Ran}})^!} \\ &\rightarrow (\mathrm{KL}(G)_{\kappa,\mathrm{Ran}} \otimes \mathrm{KL}(G)_{\kappa',\mathrm{Ran}})_{\mathrm{D-mod}(\mathrm{Ran} \times \mathrm{Ran})} \otimes_{\mathrm{D-mod}(\mathrm{Ran} \times \mathrm{Ran})} (\mathrm{D-mod}(\mathrm{Ran}) \otimes \mathrm{D-mod}(\mathrm{Ran})) = \\ &= \mathrm{KL}(G)_{\kappa,\mathrm{Ran}} \otimes \mathrm{KL}(G)_{\kappa',\mathrm{Ran}}. \end{aligned}$$

13.6.2. We identify the category dual to $\mathrm{D-mod}_\kappa(U)$ with $\mathrm{D-mod}_{\kappa'}(U)$ by means of the functor

$$(\mathrm{D-mod}_\kappa(U))^\vee \simeq \mathrm{D-mod}_{-\kappa}(U) \stackrel{(10.27)}{\simeq} \mathrm{D-mod}_{\kappa'}(U),$$

where we apply (10.27) to U instead of Bun_G and κ instead of κ' .

In terms of this identification, the unit $\mathrm{u}_{\mathrm{D-mod}_\kappa(U)}$ of the duality is given by

$$(j_U \times j_U)^*((\Delta_{\mathrm{Bun}_G})_*(\omega_{\mathrm{Bun}_G}'))',$$

where

$$(\Delta_{\mathrm{Bun}_G})_*(\omega_{\mathrm{Bun}_G})' \in \mathrm{D-mod}_\kappa(\mathrm{Bun}_G) \otimes \mathrm{D-mod}_{\kappa'}(\mathrm{Bun}_G)$$

is the image of

$$(\Delta_{\mathrm{Bun}_G})_*(\omega_{\mathrm{Bun}_G}) \in \mathrm{D-mod}_\kappa(\mathrm{Bun}_G) \otimes \mathrm{D-mod}_{-\kappa}(\mathrm{Bun}_G)$$

under $\mathrm{Id} \otimes (10.27)$.

13.6.3. According to Proposition 10.5.7, with respect to the above dualities, the functor conjugate to $\mathrm{Loc}_{G,\kappa,U}$ identifies with $\mathrm{Loc}_{G,\kappa',U}$.

We obtain

$$\begin{aligned} (\mathrm{Loc}_{G,\kappa,U} \otimes (\mathrm{Loc}_{G,\kappa,U})^{\mathrm{fake-op}})(\mathrm{u}_{\mathrm{KL}(G)_{\kappa,\mathrm{Ran}}}) &\simeq (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Loc}_{G,\kappa',U}) \circ (\Delta_{\mathrm{Ran}})!((\mathfrak{E}\mathfrak{D}\mathfrak{D}(G)_{\kappa,\kappa'})_{\mathrm{Ran}}) \simeq \\ &\simeq (j_U \times j_U)^* (\mathrm{Loc}_{G \times G, (\kappa,\kappa')}(\mathfrak{E}\mathfrak{D}\mathfrak{D}(G)_{\kappa,\kappa'})), \end{aligned}$$

see Sect. C.5.15 for the latter isomorphism.

Thus, the map (13.21) is a map

$$(13.32) \quad (j_U \times j_U)^* (\mathrm{Loc}_{G \times G, (\kappa,\kappa')}((\mathfrak{E}\mathfrak{D}\mathfrak{D}(G)_{\kappa,\kappa'})_{\mathrm{Ran}})) \rightarrow (j_U \times j_U)^*((\Delta_{\mathrm{Bun}_G})_*(\omega_{\mathrm{Bun}_G}))'.$$

Hence, by Lemma 13.4.7, in order to prove Theorem 13.4.2, we need to show that the map (13.32) is an isomorphism.

Remark 13.6.4. The maps (13.32) are compatible under $U \subset U'$. Hence, once we establish the isomorphism (13.32), we will have established an isomorphism

$$(13.33) \quad \mathrm{Loc}_{G \times G, (\kappa,\kappa')}((\mathfrak{E}\mathfrak{D}\mathfrak{D}(G)_{\kappa,\kappa'})_{\mathrm{Ran}}) \simeq (\Delta_{\mathrm{Bun}_G})_*(\omega_{\mathrm{Bun}_G})'.$$

13.6.5. We will show that the map

$$(13.34) \quad (\mathrm{Id} \otimes \mathbf{oblv}_{\kappa'}^l) \circ (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Loc}_{G,\kappa',U})((\mathfrak{E}\mathfrak{D}\mathfrak{D}(G)_{\kappa,\kappa'})_{\mathrm{Ran}}) \rightarrow (\mathrm{Id} \otimes \mathbf{oblv}_{\kappa'}^l)((\Delta_{\mathrm{Bun}_G})_*(\omega_{\mathrm{Bun}_G}))'$$

in $\mathrm{D-mod}_{\kappa}(U) \otimes \mathrm{QCoh}(U)$, induced by the map (13.21), is an isomorphism.

This would prove that the map (13.21) is an isomorphism, since the functor $\mathrm{Id} \otimes \mathbf{oblv}_{\kappa'}^l$ is conservative.

Remark 13.6.6. The idea of the proof that (13.34) is an equivalence is that we will identify this map with the map obtained by applying $\mathbf{ind}_{\kappa}^l \otimes \mathrm{Id}$ to the map (13.24).

This is not completely automatic, since the functor the natural transformation

$$\mathrm{Loc}_{G,U}^{\mathrm{QCoh}} \circ \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}} \rightarrow \mathbf{oblv}_{\kappa'}^l \circ \mathrm{Loc}_{G,\kappa',U}$$

is *not* an isomorphism.

However, the map in question will become an isomorphism when evaluated on $\mathfrak{E}\mathfrak{D}\mathfrak{D}(G)_{\kappa,\kappa'}$, since the latter is a *unital* factorization algebra.

13.6.7. Consider the functor

$$\mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}} : \mathrm{KL}(G)_{\kappa'} \rightarrow \mathrm{Rep}(\mathfrak{L}^+(G))$$

and the corresponding functor

$$\mathrm{KL}(G)_{\kappa',\mathrm{Ran}} \rightarrow \mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}},$$

which we denote by the same character $\mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}}$.

Consider the object

$$\begin{aligned} (\mathrm{Id} \otimes \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}})(\mathrm{u}_{\mathrm{KL}(G)_{\kappa,\mathrm{Ran}}}) &\simeq \\ &\simeq (\Delta_{\mathrm{Ran}})! \circ (\mathrm{Id} \otimes \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}})((\mathfrak{E}\mathfrak{D}\mathfrak{D}(G)_{\kappa,\kappa'})_{\mathrm{Ran}}) \in \mathrm{KL}(G)_{\kappa,\mathrm{Ran}} \otimes \mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}}. \end{aligned}$$

Consider the object

$$(\mathrm{Id} \otimes \mathbf{oblv}_{\kappa'}^l)(\mathrm{u}_{\mathrm{D-mod}_{\kappa}(U)}) \simeq (\mathrm{Id} \otimes \mathbf{oblv}_{\kappa'}^l) \circ (j_U \times j_U)^*((\Delta_{\mathrm{Bun}_G})_*(\omega_{\mathrm{Bun}_G}))' \in \mathrm{D-mod}_{\kappa}(U) \otimes \mathrm{QCoh}(U).$$

We claim that there is a canonically defined map

$$(13.35) \quad (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}) \circ (\mathrm{Id} \otimes \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}})(\mathrm{u}_{\mathrm{KL}(G)_{\kappa,\mathrm{Ran}}}) \rightarrow (\mathrm{Id} \otimes \mathbf{oblv}_{\kappa'}^l)(\mathrm{u}_{\mathrm{D-mod}_{\kappa}(U)}).$$

Namely, the map (13.35) is the composition

$$\begin{aligned}
& (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}) \circ (\mathrm{Id} \otimes \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}})(\mathbf{u}_{\mathrm{KL}(G)_{\kappa, \mathrm{Ran}}}) \simeq \\
& \simeq (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Id}) \circ (\mathrm{Id} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}) \circ (\mathrm{Id} \otimes \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}})(\mathbf{u}_{\mathrm{KL}(G)_{\kappa, \mathrm{Ran}}}) \simeq \\
& \simeq (\mathrm{Id} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}) \circ (\mathrm{Id} \otimes \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}}) \circ (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Id})(\mathbf{u}_{\mathrm{KL}(G)_{\kappa, \mathrm{Ran}}}) \simeq \\
& \simeq (\mathrm{Id} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}) \circ (\mathrm{Id} \otimes \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}}) \circ (\mathrm{Id} \otimes (\mathrm{Loc}_{G,\kappa,U})^\vee)(\mathbf{u}_{\mathrm{D-mod}_\kappa(U)}) \simeq \\
& \simeq (\mathrm{Id} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}) \circ (\mathrm{Id} \otimes \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}}) \circ (\mathrm{Id} \otimes (\mathrm{Loc}_{G,\kappa',U})^R)(\mathbf{u}_{\mathrm{D-mod}_\kappa(U)}) = \\
& = \left(\mathrm{Id} \otimes (\mathrm{Loc}_{G,U}^{\mathrm{QCoh}} \circ \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}} \circ (\mathrm{Loc}_{G,\kappa',U})^R) \right) (\mathbf{u}_{\mathrm{D-mod}_\kappa(U)}) \rightarrow \\
& \rightarrow \left(\mathrm{Id} \otimes (\mathbf{oblv}_{\kappa'}^l \circ \mathrm{Loc}_{G,\kappa',U} \circ (\mathrm{Loc}_{G,\kappa',U})^R) \right) (\mathbf{u}_{\mathrm{D-mod}_\kappa(U)}) \rightarrow (\mathrm{Id} \otimes \mathbf{oblv}_{\kappa'}^l)(\mathbf{u}_{\mathrm{D-mod}_\kappa(U)}),
\end{aligned}$$

where the next-to-last arrow is induced by the natural transformation (12.5).

13.6.8. Note that by construction, the map (13.35) can be reinterpreted as

$$\begin{aligned}
(13.36) \quad & (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}) \circ (\mathrm{Id} \otimes \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}})(\mathbf{u}_{\mathrm{KL}(G)_{\kappa, \mathrm{Ran}}}) \simeq \\
& \simeq (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}) \circ (\mathrm{Id} \otimes \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa'}})((\mathfrak{C}\mathfrak{D}\mathfrak{D}(G)_{\kappa, \kappa'})_{\mathrm{Ran}}) \rightarrow \\
& \rightarrow (\mathrm{Id} \otimes \mathbf{oblv}_{\kappa'}^l) \circ (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Loc}_{G,\kappa',U})((\mathfrak{C}\mathfrak{D}\mathfrak{D}(G)_{\kappa, \kappa'})_{\mathrm{Ran}}) \xrightarrow{(13.34)} \\
& \rightarrow (\mathrm{Id} \otimes \mathbf{oblv}_{\kappa'}^l)((\Delta_{\mathrm{Bun}_G})_*(\omega_{\mathrm{Bun}_G})') \simeq (\mathrm{Id} \otimes \mathbf{oblv}_{\kappa'}^l)(\mathbf{u}_{\mathrm{D-mod}_\kappa(U)}),
\end{aligned}$$

where the second arrow is induced by (12.5).

Note, however, that the fact that $\mathfrak{C}\mathfrak{D}\mathfrak{D}(G)_{\kappa, \kappa'}$ is a *unital* factorization algebra, we obtain that the second arrow in (13.36) is an isomorphism, see Lemma 11.9.11.

Hence, we obtain that in order to show that (13.34) is an isomorphism, it suffices to show that (13.35) is an isomorphism.

13.6.9. Consider now the functor

$$\mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} : \mathrm{Rep}(\mathfrak{L}^+(G)) \rightarrow \mathrm{KL}(G)_\kappa,$$

and the corresponding functor

$$\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}} \rightarrow \mathrm{KL}(G)_{\kappa, \mathrm{Ran}},$$

which we denote by the same character $\mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa}$.

Consider the object

$$\begin{aligned}
& (\mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} \otimes \mathrm{Id})(\mathbf{u}_{\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}}}) \simeq \\
& \simeq (\Delta_{\mathrm{Ran}})! \circ (\mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} \otimes \mathrm{Id})(R_{\mathfrak{L}^+(G), \mathrm{Ran}}) \in \mathrm{KL}(G)_{\kappa, \mathrm{Ran}} \otimes \mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}}.
\end{aligned}$$

Consider the object

$$(\mathbf{ind}_\kappa^l \otimes \mathrm{Id})(\mathbf{u}_{\mathrm{QCoh}(U)}) \simeq (\mathbf{ind}_\kappa^l \otimes \mathrm{Id}) \circ (j_U \times j_U)^* \circ (\Delta_{\mathrm{Bun}_G})_*(\mathcal{O}_{\mathrm{Bun}_G}) \in \mathrm{D-mod}_\kappa(U) \otimes \mathrm{QCoh}(U).$$

We claim that there is a canonically defined map

$$(13.37) \quad (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}) \circ (\mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} \otimes \mathrm{Id})(\mathbf{u}_{\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}}}) \rightarrow (\mathbf{ind}_\kappa^l \otimes \mathrm{Id})(\mathbf{u}_{\mathrm{QCoh}(U)}).$$

Namely, the map (13.37) is the composition

$$\begin{aligned}
(13.38) \quad & (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}) \circ (\mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa} \otimes \mathrm{Id})(u_{\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}}}) \simeq \\
& \simeq (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Id}) \circ (\mathrm{Id} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}) \circ (\mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa} \otimes \mathrm{Id})(u_{\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}}}) \simeq \\
& \simeq (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Id}) \circ (\mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa} \otimes \mathrm{Id}) \circ (\mathrm{Id} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}})(u_{\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}}}) \simeq \\
& \simeq (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Id}) \circ (\mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa} \otimes \mathrm{Id})((\mathrm{Loc}_{G,U}^{\mathrm{QCoh}})^\vee \otimes \mathrm{Id})(u_{\mathrm{QCoh}(U)}) \simeq \\
& \simeq (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Id}) \circ (\mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa} \otimes \mathrm{Id}) \circ ((\mathrm{Loc}_{G,U}^{\mathrm{QCoh}})^R \otimes \mathrm{Id})(u_{\mathrm{QCoh}(U)}) \simeq \\
& \simeq (\mathbf{ind}_\kappa^l \otimes \mathrm{Id}) \circ (\mathrm{Loc}_{G,U}^{\mathrm{QCoh}} \otimes \mathrm{Id}) \circ ((\mathrm{Loc}_{G,U}^{\mathrm{QCoh}})^R \otimes \mathrm{Id})(u_{\mathrm{QCoh}(U)}) \rightarrow (\mathbf{ind}_\kappa^l \otimes \mathrm{Id})(u_{\mathrm{QCoh}(U)}).
\end{aligned}$$

Note that the last arrow in (13.38) is an isomorphism by Theorem 13.4.4. Hence, the map (13.37) is an isomorphism.

13.6.10. Note now that we have canonical isomorphisms

$$(\mathrm{Id} \otimes \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa'}) (u_{\mathrm{KL}(G)_{\kappa, \mathrm{Ran}}}) \simeq (\mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa} \otimes \mathrm{Id})(u_{\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}}}),$$

and hence

$$\begin{aligned}
& (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}) \circ (\mathrm{Id} \otimes \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa'}) (u_{\mathrm{KL}(G)_{\kappa, \mathrm{Ran}}}) \simeq \\
& \simeq (\mathrm{Loc}_{G,\kappa,U} \otimes \mathrm{Loc}_{G,U}^{\mathrm{QCoh}}) \circ (\mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa} \otimes \mathrm{Id})(u_{\mathrm{Rep}(\mathfrak{L}^+(G))_{\mathrm{Ran}}})
\end{aligned}$$

and

$$(\mathrm{Id} \otimes \mathbf{oblv}_{\kappa'}^l)(u_{\mathrm{D-mod}_\kappa(U)}) \simeq (\mathbf{ind}_\kappa^l \otimes \mathrm{Id})(u_{\mathrm{QCoh}(U)}).$$

Now, unwinding and using Proposition 10.5.7(a), we obtain that under these identifications, the map (13.35) equals the map (13.37).

Hence, we obtain that the map (13.35) is an isomorphism, as required.

□[Second proof of Theorem 13.4.2]

14. THE COMPOSITION OF LOCALIZATION AND COEFFICIENT FUNCTORS

The goal of this section is to give an expression for the composition

$$(14.1) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \xrightarrow{\mathrm{Loc}_{G, \mathrm{crit}}} \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \simeq \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{\mathrm{coeff}} \mathrm{Whit}^!(G)_{\mathrm{Ran}}$$

in terms of factorization homology.

This expression (combined with the compatibility of $\mathrm{FLE}_{G, \mathrm{crit}}$ and $\mathrm{FLE}_{\check{G}, \infty}$ expressed by Corollary 6.4.10) will be used in Sect. 18 in order to show that the Langlands functor is compatible with the critical localization and the spectral Poincaré series functors via the critical FLE.

14.1. The vacuum case.

14.1.1. Let us specialize the setting of Theorem 12.8.8 to the case $\kappa = \mathrm{crit}$ and $\mathcal{P}_T = \rho(\omega_X)$. In this case, the integer that we denoted $\delta_{N_{\mathcal{P}_T}}$ is $\delta_{N_{\rho(\omega_X)}}$. Denote the corresponding line $\mathbb{I}_{N_{\mathcal{P}_T}}$ by

$$(14.2) \quad \mathbb{I}_{N_{\rho(\omega_X)}}.$$

14.1.2. We obtain:

Theorem 14.1.3. *The composition*

$$\begin{aligned} \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} &\xrightarrow{\mathrm{Loc}_G, \mathrm{crit}} \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \xrightarrow{\mathfrak{p}_{\mathrm{crit}}^!} \mathrm{D}\text{-mod}(\mathrm{Bun}_{N, \rho(\omega_X)}) \xrightarrow{-\otimes \chi^*(\exp)} \\ &\rightarrow \mathrm{D}\text{-mod}(\mathrm{Bun}_{N, \rho(\omega_X)}) \xrightarrow{\mathrm{C}_{\mathrm{dR}}(\mathrm{Bun}_{N, \rho(\omega_X)}, -)} \mathrm{Vect} \end{aligned}$$

identifies with the functor

$$\begin{aligned} \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} &\xrightarrow{\alpha_{\rho(\omega_X), \mathrm{taut}}} \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X), \mathrm{Ran}} \xrightarrow{\mathrm{DS}^{\mathrm{enh}}} \\ &\rightarrow \mathfrak{z}_{\mathfrak{g}}\text{-mod}_{\mathrm{Ran}}^{\mathrm{fact}} \xrightarrow{\mathrm{C}_{\mathrm{fact}}^{\mathrm{fact}}(X; \mathfrak{z}_{\mathfrak{g}}, -)} \mathrm{Vect} \xrightarrow{-\otimes \mathrm{I}_{N, \rho(\omega_X)}^{[\delta_{N, \rho(\omega_X)}]}} \mathrm{Vect}. \end{aligned}$$

14.1.4. Denote by Loc_G the functor

$$(14.3) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \xrightarrow{\mathrm{Loc}_G, \mathrm{crit}} \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \xrightarrow{(9.2)} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G).$$

From (9.8) we obtain a commutative diagram

$$\begin{array}{ccc} \mathrm{Vect} & \xrightarrow{\mathrm{Id}} & \mathrm{Vect} \\ \uparrow \mathrm{C}_{\mathrm{dR}}(\mathrm{Bun}_{N, \rho(\omega_X)}, -) \circ (-\otimes \chi^*(\exp)) & & \uparrow -\otimes \mathrm{I}_{G, N, \rho(\omega_X)}^{\frac{1}{2}} \\ \mathrm{D}\text{-mod}(\mathrm{Bun}_{N, \rho(\omega_X)}) & & \mathrm{Vect} \\ \uparrow \mathfrak{p}_{\mathrm{crit}}^! & & \uparrow \mathrm{coeff}_G^{\mathrm{Vac}, \mathrm{glob}} \\ \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) & \xrightarrow{(9.2)} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \\ \uparrow \mathrm{Loc}_G, \mathrm{crit} & & \uparrow \mathrm{Loc}_G \\ \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} & \xrightarrow{\mathrm{Id}} & \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}}. \end{array}$$

Recall also that

$$\mathrm{coeff}_G^{\mathrm{Vac}, \mathrm{glob}} \simeq \mathrm{coeff}_G^{\mathrm{Vac}}[2\delta_{N, \rho(\omega_X)}].$$

14.1.5. Hence, Theorem 14.1.3 can be restated as:

Theorem 14.1.6. *The composition*

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \xrightarrow{\mathrm{Loc}_G} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{\mathrm{coeff}_G^{\mathrm{Vac}}} \mathrm{Vect} \xrightarrow{-\otimes \mathrm{I}_{G, N, \rho(\omega_X)}^{\frac{1}{2}} \otimes \mathrm{I}_{N, \rho(\omega_X)}^{\otimes -1} [\delta_{N, \rho(\omega_X)}]} \mathrm{Vect}$$

identifies with the functor

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \xrightarrow{\alpha_{\rho(\omega_X), \mathrm{taut}}} \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X), \mathrm{Ran}} \xrightarrow{\mathrm{DS}^{\mathrm{enh}}} \mathfrak{z}_{\mathfrak{g}}\text{-mod}_{\mathrm{Ran}}^{\mathrm{fact}} \xrightarrow{\mathrm{C}_{\mathrm{fact}}^{\mathrm{fact}}(X; \mathfrak{z}_{\mathfrak{g}}, -)} \mathrm{Vect}.$$

14.2. Composition of coefficient and localization functors: the general case. We are now ready to state the general theorem, describing the composition of the functors

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \xrightarrow{\mathrm{Loc}_G} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{-\otimes \mathrm{I}_{G, N, \rho(\omega_X)}^{\frac{1}{2}} \otimes \mathrm{I}_{N, \rho(\omega_X)}^{\otimes -1} [\delta_{N, \rho(\omega_X)}]} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

and

$$\mathrm{coeff}_G : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{Whit}^!(G)_{\mathrm{Ran}}.$$

14.2.1. Recall that the category $\text{Whit}^!(G)_{\text{Ran}}$ is the dual of $\text{Whit}_*(G)_{\text{Ran}}$. Hence, the description of the above composition is equivalent to describing the pairing

$$\begin{aligned}
 (14.4) \quad & \text{KL}(G)_{\text{crit,Ran}} \otimes \text{Whit}_*(G)_{\text{Ran}} \xrightarrow{\text{Loc}_G \otimes \text{Id}} \\
 & \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{Whit}_*(G)_{\text{Ran}} \xrightarrow{(-\otimes_{G,N_{\rho(\omega_X)}}^{\frac{1}{2}} \otimes_{N_{\rho(\omega_X)}}^{\otimes -1} [\delta_{N_{\rho(\omega_X)}}]) \otimes \text{Id}} \\
 & \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{Whit}_*(G)_{\text{Ran}} \xrightarrow{\text{coeff}_G \otimes \text{Id}} \text{Whit}^!(G)_{\text{Ran}} \otimes \text{Whit}_*(G)_{\text{Ran}} \rightarrow \text{Vect}.
 \end{aligned}$$

14.2.2. Recall the factorization functor $\mathbf{P}_G^{\text{loc,enh}}$, see Sect. 6.4.6. We will think of it as the functor

$$\begin{aligned}
 \text{Whit}_*(G) \otimes \text{KL}(G)_{\text{crit}} & \xrightarrow{\text{Id} \otimes \alpha_{\rho(\omega_X), \text{taut}}} \text{Whit}_*(G) \otimes \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \\
 & \rightarrow \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}) \xrightarrow{\overline{\text{DS}}^{\text{enh}}} \mathfrak{z}_{\mathfrak{g}}\text{-mod}^{\text{fact}}.
 \end{aligned}$$

Let $\mathbf{P}_G^{\text{loc}}$ be the composition of $\mathbf{P}_G^{\text{loc,enh}}$ with

$$\text{oblv}_{\mathfrak{z}_{\mathfrak{g}}} : \mathfrak{z}_{\mathfrak{g}}\text{-mod}^{\text{fact}} \rightarrow \text{Vect}.$$

14.2.3. We will prove:

Theorem 14.2.4. *The functor (14.4) identifies canonically with*

$$\begin{aligned}
 (14.5) \quad & \text{KL}(G)_{\text{crit,Ran}} \otimes \text{Whit}_*(G)_{\text{Ran}} \xrightarrow{\text{ins.vacRan} \otimes \text{ins.unitRan}} \text{KL}(G)_{\text{crit,Ran}^\subseteq} \otimes \text{Whit}_*(G)_{\text{Ran}^\subseteq} \rightarrow \\
 & \rightarrow (\text{KL}(G)_{\text{crit}} \otimes \text{Whit}_*(G))_{\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq} \xrightarrow{\mathbf{P}_G^{\text{loc,enh}}} \\
 & \rightarrow \mathfrak{z}_{\mathfrak{g}}\text{-mod}_{\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq}^{\text{fact}} \xrightarrow{C_c^{\text{fact}}(X; \mathfrak{z}_{\mathfrak{g}}, -)_{\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq}} \text{D-mod}(\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq) \rightarrow \\
 & \xrightarrow{C_c(\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq, -)} \text{Vect},
 \end{aligned}$$

where the fiber product $\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq$ is formed using the projections $\text{pr}_{\text{big}} : \text{Ran}^\subseteq \rightarrow \text{Ran}$.

Remark 14.2.5. Note that the functor (14.5), appearing in Theorem 14.2.4 can also be rewritten as

$$\begin{aligned}
 (14.6) \quad & \text{KL}(G)_{\text{crit,Ran}} \otimes \text{Whit}_*(G)_{\text{Ran}} \xrightarrow{\text{ins.vacRan} \otimes \text{ins.unitRan}} \text{KL}(G)_{\text{crit,Ran}^\subseteq} \otimes \text{Whit}_*(G)_{\text{Ran}^\subseteq} \rightarrow \\
 & \rightarrow (\text{KL}(G)_{\text{crit}} \otimes \text{Whit}_*(G))_{\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq} \xrightarrow{\mathbf{P}_G^{\text{loc}}} \text{D-mod}(\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq) \xrightarrow{C_c(\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq)} \text{Vect}
 \end{aligned}$$

i.e., instead of $C_c^{\text{fact}}(X; \mathfrak{z}_{\mathfrak{g}}, -)_{\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq}$ we can use the functor $\text{oblv}_{\mathfrak{z}_{\mathfrak{g}}, \text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq}$.

Indeed, since the functor $\mathbf{P}_G^{\text{loc,enh}}$ is strictly unital, a priori, by Theorem 14.2.4, the pairing (14.5) is

$$\begin{aligned}
 (14.7) \quad & \text{KL}(G)_{\text{crit,Ran}} \otimes \text{Whit}_*(G)_{\text{Ran}} \xrightarrow{\text{ins.vacRan} \otimes \text{ins.unitRan}} \text{KL}(G)_{\text{crit,Ran}^\subseteq} \otimes \text{Whit}_*(G)_{\text{Ran}^\subseteq} \rightarrow \\
 & \rightarrow (\text{KL}(G)_{\text{crit}} \otimes \text{Whit}_*(G))_{\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq} \xrightarrow{\text{ins.unit}_{\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq}} \\
 & \rightarrow (\text{KL}(G)_{\text{crit}} \otimes \text{Whit}_*(G))_{(\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq)^\subseteq} \xrightarrow{\mathbf{P}_G^{\text{loc}}} \text{D-mod}((\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq)^\subseteq) \rightarrow \\
 & \xrightarrow{C_c((\text{Ran}^\subseteq \times_{\text{Ran}} \text{Ran}^\subseteq)^\subseteq)} \text{Vect}.
 \end{aligned}$$

However, the composition of the first three lines in (14.7) lies in the essential image of the functor

$$\mathbf{t}^! : \mathrm{D}\text{-mod}((\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq})^{\subseteq, \mathrm{untl}}) \rightarrow \mathrm{D}\text{-mod}((\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq})^{\subseteq}).$$

Hence, by the same mechanism as in Sect. 11.9.9, the expression in (14.7) is isomorphic to that in (14.6).

14.3. Proof of Theorem 14.2.4.

14.3.1. We rewrite the functor

$$(14.8) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}} \xrightarrow{\mathrm{Loc}_G \otimes \mathrm{Id}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}} \xrightarrow{\mathrm{coeff}_G \otimes \mathrm{Id}} \mathrm{Whit}^!(G)_{\mathrm{Ran}} \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}} \rightarrow \mathrm{Vect}.$$

as

$$(14.9) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}} \xrightarrow{\mathrm{Loc}_G \otimes \mathrm{Poinc}_{G,*}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G) \xrightarrow{\langle -, - \rangle_{\mathrm{Bun}_G}} \mathrm{Vect},$$

where the last arrow is the canonical pairing

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G) \rightarrow \mathrm{Vect}.$$

14.3.2. By the unital property of Loc_G and $\mathrm{Poinc}_{G,*}$, we can rewrite the functor

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}} \xrightarrow{\mathrm{Loc}_G \otimes \mathrm{Poinc}_{G,*}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G)$$

as

$$(14.10) \quad \begin{aligned} & \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}} \xrightarrow{\mathrm{ins.vac}_{\mathrm{Ran}} \otimes \mathrm{ins.unit}_{\mathrm{Ran}}} \\ & \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}^{\subseteq}} \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}^{\subseteq}} \xrightarrow{(\mathrm{Loc}_G)_{\mathrm{Ran}^{\subseteq}} \otimes (\mathrm{Poinc}_{G,*})_{\mathrm{Ran}^{\subseteq}}} \\ & \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq}) \rightarrow \\ & \xrightarrow{\mathrm{Id} \otimes \mathrm{Id} \otimes \mathrm{C}_c'(\mathrm{Ran}^{\subseteq} \times \mathrm{Ran}^{\subseteq}, -)} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G). \end{aligned}$$

14.3.3. In this subsection we will use an analog of Sect. C.5.15, which will allow us to replace

$$\mathrm{Ran}^{\subseteq} \times \mathrm{Ran}^{\subseteq} \rightsquigarrow \mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}.$$

We note that the composition in the first three lines in (14.10) actually factors via the tensor product of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G)$ with

$$\begin{aligned} & \mathrm{D}\text{-mod}(\mathrm{Ran}_{\mathrm{Ran}, \mathrm{pr}_{\mathrm{small}}} \times \mathrm{Ran}^{\subseteq, \mathrm{untl}}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}_{\mathrm{Ran}, \mathrm{pr}_{\mathrm{small}}} \times \mathrm{Ran}^{\subseteq, \mathrm{untl}}) \xrightarrow{\mathrm{Id} \otimes \mathrm{Id} \otimes \mathbf{t}^! \otimes \mathbf{t}^!} \\ & \rightarrow \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq}) \end{aligned}$$

Hence, by the same principle as in Sect. C.5.15, its further composition with

$$\begin{aligned} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq}) \rightarrow \\ & \xrightarrow{\mathrm{Id} \otimes \mathrm{Id} \otimes \mathrm{C}_c'(\mathrm{Ran}^{\subseteq} \times \mathrm{Ran}^{\subseteq}, -)} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G) \end{aligned}$$

is isomorphic to its composition with

$$\begin{aligned}
& \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq}) \xrightarrow{\mathrm{Id} \otimes \mathrm{Id} \otimes \Delta_{\mathrm{Ran}^{\subseteq}}^!} \\
& \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}) \xrightarrow{\mathrm{Id} \otimes \mathrm{Id} \otimes \mathrm{C}_c^*(\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}, -)} \\
& \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G).
\end{aligned}$$

Hence, we can rewrite (14.10) as

$$\begin{aligned}
& \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}} \xrightarrow{\mathrm{ins.vac}_{\mathrm{Ran}} \otimes \mathrm{ins.unit}_{\mathrm{Ran}}} \\
& \rightarrow \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}^{\subseteq}} \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}^{\subseteq}} \xrightarrow{(\mathrm{Loc}_G)_{\mathrm{Ran}^{\subseteq}} \otimes (\mathrm{Poinc}_G, *)_{\mathrm{Ran}^{\subseteq}}} \\
& \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq}) \rightarrow \\
& \xrightarrow{\mathrm{Id} \otimes \mathrm{Id} \otimes (\Delta_{\mathrm{Ran}})^!} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}) \rightarrow \\
& \xrightarrow{\mathrm{Id} \otimes \mathrm{Id} \otimes \mathrm{C}_c^*(\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}, -)} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G),
\end{aligned}$$

which is the same as

$$\begin{aligned}
& \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}} \xrightarrow{\mathrm{ins.vac}_{\mathrm{Ran}} \otimes \mathrm{ins.unit}_{\mathrm{Ran}}} \\
& \rightarrow \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}^{\subseteq}} \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}^{\subseteq}} \rightarrow (\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G))_{\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}} \rightarrow \\
& \xrightarrow{(\mathrm{Loc}_G \otimes \mathrm{Poinc}_G, *)_{\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}) \rightarrow \\
& \xrightarrow{\mathrm{Id} \otimes \mathrm{Id} \otimes \mathrm{C}_c^*(\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}, -)} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G).
\end{aligned}$$

14.3.4. Thus, we obtain that we can rewrite (14.9) as

$$\begin{aligned}
(14.11) \quad & \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}} \xrightarrow{\mathrm{ins.vac} \otimes \mathrm{ins.unit}} \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}^{\subseteq}} \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}^{\subseteq}} \rightarrow \\
& \rightarrow (\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G))_{\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}} \rightarrow \\
& \xrightarrow{(\mathrm{Loc}_G \otimes \mathrm{Poinc}_G, *)_{\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}) \rightarrow \\
& \xrightarrow{\mathrm{Id} \otimes \mathrm{Id} \otimes \mathrm{C}_{\mathrm{dR}}^*(\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}, -)} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G) \xrightarrow{\langle -, - \rangle_{\mathrm{Bun}_G}} \mathrm{Vect}.
\end{aligned}$$

Hence, in order to prove the theorem, it is enough to identify the composition

$$\begin{aligned}
(14.12) \quad & (\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G))_{\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}} \rightarrow \\
& \xrightarrow{(\mathrm{Loc}_G \otimes \mathrm{Poinc}_G, *)_{\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}) \rightarrow \\
& \xrightarrow{\mathrm{Id} \otimes \mathrm{Id} \otimes \mathrm{C}_{\mathrm{dR}}^*(\mathrm{Ran}^{\subseteq} \times_{\mathrm{Ran}} \mathrm{Ran}^{\subseteq}, -)} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G) \xrightarrow{\langle -, - \rangle_{\mathrm{Bun}_G}} \\
& \rightarrow \mathrm{Vect} \xrightarrow{(- \otimes \mathrm{I}_{G,N\rho(\omega_X)}^{\otimes \frac{1}{2}}) \otimes \mathrm{I}_{N\rho(\omega_X)}^{\otimes -1}} [\delta_{N\rho(\omega_X)}] \mathrm{Vect}
\end{aligned}$$

with

$$(14.13) \quad (\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G))_{\mathrm{Ran}^\subseteq \times_{\mathrm{Ran}} \mathrm{Ran}^\subseteq} \xrightarrow{\mathbf{p}_G^{\mathrm{loc}, \mathrm{enh}}} \mathcal{C}_{\cdot}^{\mathrm{fact}}(X; \mathfrak{g}, -)_{\mathrm{Ran}^\subseteq \times_{\mathrm{Ran}} \mathrm{Ran}^\subseteq} \rightarrow \mathfrak{g}\text{-mod}_{\mathrm{Ran}^\subseteq \times_{\mathrm{Ran}} \mathrm{Ran}^\subseteq}^{\mathrm{fact}} \xrightarrow{\quad} \mathrm{D}\text{-mod}(\mathrm{Ran}^\subseteq \times_{\mathrm{Ran}} \mathrm{Ran}^\subseteq) \rightarrow \mathrm{C}_{\mathrm{dR}}(\mathrm{Ran}^\subseteq \times_{\mathrm{Ran}} \mathrm{Ran}^\subseteq, -) \xrightarrow{\quad} \mathrm{Vect}.$$

14.3.5. Applying the functor $(\mathrm{pr}_{\mathrm{big}})_!$, we obtain that it suffices to identify

$$(14.14) \quad (\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G))_{\mathrm{Ran}} \xrightarrow{(\mathrm{Loc}_G \otimes \mathrm{Poinc}_{G,*})_{\mathrm{Ran}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}) \xrightarrow{\mathrm{Id} \otimes \mathrm{Id} \otimes \mathrm{C}_{\cdot}(\mathrm{Ran}, -)} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G) \xrightarrow{\langle -, - \rangle_{\mathrm{Bun}_G}} \mathrm{Vect} \xrightarrow{(- \otimes \mathbf{l}_{G, N_{\rho}(\omega_X)}^{\otimes \frac{1}{2}} \otimes \mathbf{l}_{N_{\rho}(\omega_X)}^{\otimes -1})[\delta_{N_{\rho}(\omega_X)}]} \mathrm{Vect}$$

with

$$(14.15) \quad (\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G))_{\mathrm{Ran}} \xrightarrow{\mathbf{p}_G^{\mathrm{loc}, \mathrm{enh}}} \mathfrak{g}\text{-mod}_{\mathrm{Ran}}^{\mathrm{fact}} \xrightarrow{\mathcal{C}_{\cdot}^{\mathrm{fact}}(X; \mathfrak{g}, -)} \mathrm{Vect}.$$

14.3.6. Note that both functors (14.14) and (14.15) factor naturally via

$$(\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G))_{\mathrm{Ran}} \rightarrow (\mathrm{KL}(G)_{\mathrm{crit}} \otimes_{\mathrm{Sph}_G} \mathrm{Whit}_*(G))_{\mathrm{Ran}},$$

and in particular via

$$(\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G))_{\mathrm{Ran}} \rightarrow (\mathrm{KL}(G)_{\mathrm{crit}} \otimes_{\mathrm{Rep}(\check{G})} \mathrm{Whit}_*(G))_{\mathrm{Ran}},$$

where $\mathrm{Rep}(\check{G})$ maps to Sph_G via $\mathrm{Sat}_G^{\mathrm{nv}}$.

Hence, using the fact that the action of $\mathrm{Rep}(\check{G})$ on $\mathrm{Vac}_{\mathrm{Whit}_*(G)}$ defines an equivalence

$$\mathrm{Rep}(\check{G}) \rightarrow \mathrm{Whit}_*(G),$$

we are reduced to establishing an identification between

$$(14.16) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \xrightarrow{\mathrm{Id} \otimes \mathrm{Vac}_{\mathrm{Whit}_*(G)}} (\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G))_{\mathrm{Ran}} \rightarrow \xrightarrow{(\mathrm{Loc}_G \otimes \mathrm{Poinc}_{G,*})_{\mathrm{Ran}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}) \xrightarrow{\mathrm{Id} \otimes \mathrm{Id} \otimes \mathrm{C}_{\cdot}(\mathrm{Ran}, -)} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}, \mathrm{co}}(\mathrm{Bun}_G) \xrightarrow{\langle -, - \rangle_{\mathrm{Bun}_G}} \mathrm{Vect} \xrightarrow{(- \otimes \mathbf{l}_{G, N_{\rho}(\omega_X)}^{\otimes \frac{1}{2}} \otimes \mathbf{l}_{N_{\rho}(\omega_X)}^{\otimes -1})[\delta_{N_{\rho}(\omega_X)}]} \mathrm{Vect}$$

and

$$(14.17) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \xrightarrow{\mathrm{Id} \otimes \mathrm{Vac}_{\mathrm{Whit}_*(G)}} (\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G))_{\mathrm{Ran}} \xrightarrow{\mathbf{p}_G^{\mathrm{loc}, \mathrm{enh}}} \mathfrak{g}\text{-mod}_{\mathrm{Ran}}^{\mathrm{fact}} \xrightarrow{\mathcal{C}_{\cdot}^{\mathrm{fact}}(X; \mathfrak{g}, -)} \mathrm{Vect}.$$

14.3.7. Using the unital property of $\mathrm{Poinc}_{G,*}$ we can identify (14.16) with

$$(14.18) \quad \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} \xrightarrow{(\mathrm{Loc}_G)_{\mathrm{Ran}} \otimes \mathrm{Poinc}_{G,*}^{\mathrm{Vac}}} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathrm{Ran}) \otimes \mathrm{D-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G) \xrightarrow{\mathrm{Id} \otimes \mathrm{C}_c^*(\mathrm{Ran},-) \otimes \mathrm{Id}} \\ \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}_{\frac{1}{2},\mathrm{co}}(\mathrm{Bun}_G) \xrightarrow{\langle -, - \rangle_{\mathrm{Bun}_G}} \mathrm{Vect} \xrightarrow{(- \otimes_{G,N}^{\frac{1}{2}} \otimes_{\rho(\omega_X)}^{\otimes -1})^{[\delta_{N_{\rho(\omega_X)}}]}} \mathrm{Vect}.$$

By definition, we can identify (14.17) with

$$(14.19) \quad \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} \xrightarrow{\alpha_{\rho(\omega_X),\mathrm{taut}}} \mathrm{KL}(G)_{\mathrm{crit},\rho(\omega_X),\mathrm{Ran}} \xrightarrow{\mathrm{DS}^{\mathrm{enh}}} \mathfrak{Z}\mathfrak{g}\text{-mod}_{\mathrm{Ran}}^{\mathrm{fact}} \xrightarrow{\mathrm{C}_c^{\mathrm{fact}}(X; \mathfrak{Z}\mathfrak{g}, -)} \mathrm{Vect}.$$

So, it remains to identify (14.18) and (14.19).

14.3.8. We rewrite (14.18) as

$$(14.20) \quad \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} \xrightarrow{\mathrm{Loc}_G} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{\mathrm{coeff}_G^{\mathrm{Vac}}} \mathrm{Vect} \xrightarrow{(- \otimes_{G,N}^{\frac{1}{2}} \otimes_{\rho(\omega_X)}^{\otimes -1})^{[\delta_{N_{\rho(\omega_X)}}]}} \mathrm{Vect}.$$

Now the isomorphism between (14.20) and (14.19) is the assertion of Theorem 14.1.6.

□[Theorem 14.2.4]

15. THE HECKE EIGEN-PROPERTY OF CRITICAL LOCALIZATION

The goal of this section is to establish the Hecke eigenproperty of the functor $\mathrm{Loc}_{G,\mathrm{crit}}$, which lies in the heart of the manuscript [BD1]. What we do will essentially amount to a souped-up version of the construction in *loc. cit.*

The contents of this (and the next) section are not logically necessary either for this paper, nor for the other papers in the GLC series. However, here we fill the gap in the literature: we supply a proof of a [Ga1, Theorem 10.3.4], which is used in [Ga1, Corollary 4.5.5], while the latter is essential for the GLC series.

15.1. Statement of the result.

15.1.1. Consider the prestacks

$$\mathrm{LS}_{\tilde{G},\mathrm{Ran}}^{\mathrm{mer, glob}} \text{ and } \mathrm{Op}_{\tilde{G},\mathrm{Ran}}^{\mathrm{mer, glob}}$$

that attach to $\underline{x} \in \mathrm{Ran}$ the spaces

$$\mathrm{LS}_{\tilde{G}}(X - \underline{x}) \text{ and } \mathrm{Op}_{\tilde{G}}(X - \underline{x}),$$

respectively (see Sect. B.7.14 for what we mean by local systems on an open curve).

Set

$$\mathrm{Op}_{\tilde{G},\mathrm{Ran}}^{\mathrm{mon-free, glob}} := (\mathrm{LS}_{\tilde{G}} \times \mathrm{Ran}) \times_{\mathrm{LS}_{\tilde{G},\mathrm{Ran}}^{\mathrm{mer, glob}}} \mathrm{Op}_{\tilde{G},\mathrm{Ran}}^{\mathrm{mer, glob}}.$$

Thus, we have a Cartesian diagram

$$\begin{array}{ccc} \mathrm{Op}_{\tilde{G},\mathrm{Ran}}^{\mathrm{mon-free, glob}} & \xrightarrow{\iota^{\mathrm{mon-free, glob}}} & \mathrm{Op}_{\tilde{G},\mathrm{Ran}}^{\mathrm{mer, glob}} \\ \tau^{\mathrm{glob}} \downarrow & & \tau^{\mathrm{glob}} \downarrow \\ \mathrm{LS}_{\tilde{G}} \times \mathrm{Ran} & \longrightarrow & \mathrm{LS}_{\tilde{G},\mathrm{Ran}}^{\mathrm{mer, glob}}. \end{array}$$

All of the spaces and maps in the above diagram have natural unital structures (see Sect. C.6.1 for what this means).

As we shall see shortly, $\mathrm{Op}_{\tilde{G},\mathrm{Ran}}^{\mathrm{mon-free, glob}}$ is a relative ind-affine ind-scheme over Ran .

15.1.2. Restriction to the formal disc gives rise to the maps

$$\mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mer}, \mathrm{glob}} \rightarrow \mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mer}} \text{ and } \mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mon-free}, \mathrm{glob}} \rightarrow \mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mon-free}};$$

we will denote both by $\mathrm{ev}_{\mathrm{Ran}}$.

Note that the following commutative square is Cartesian

$$(15.1) \quad \begin{array}{ccc} \mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mon-free}, \mathrm{glob}} & \xrightarrow{\mathrm{ev}_{\mathrm{Ran}}} & \mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mon-free}} \\ \downarrow \iota^{\mathrm{mon-free}, \mathrm{glob}} & & \downarrow \iota^{\mathrm{mon-free}} \\ \mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mer}, \mathrm{glob}} & \xrightarrow{\mathrm{ev}_{\mathrm{Ran}}} & \mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mer}} \end{array}$$

15.1.3. For a given $\mathcal{Z} \rightarrow \mathrm{Ran}$, we will change the subscript

$$\mathrm{Ran} \rightsquigarrow \mathcal{Z}$$

for the corresponding base-changed spaces and maps.⁵⁷

We can consider the assignment

$$\mathcal{Z} \mapsto \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}, \mathrm{glob}})_{\mathcal{Z}} := \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \mathcal{Z}_{\mathrm{dR}}}^{\mathrm{mon-free}, \mathrm{glob}})$$

as a crystal of categories over Ran , see Sect. B.13.8.

15.1.4. Fix $\mathcal{Z} \rightarrow \mathrm{Ran}$, and recall that according to Sect. 5.3, we have a canonically defined action of the (symmetric) monoidal category $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}}$ on $\mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}$. Composing with

$$\Upsilon_{\mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mon-free}}} : \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}} \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}},$$

we obtain an action of $\mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}}$ on $\mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}$.

Consider the category

$$(15.2) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_{\check{G}}^{\mathrm{glob}}} := \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}}} \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}, \mathrm{glob}})_{\mathcal{Z}}.$$

Denote by $\mathrm{Id} \otimes (\mathrm{ev}_{\mathcal{Z}})^*$ the resulting functor

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}}} \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}, \mathrm{glob}})_{\mathcal{Z}} = \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_{\check{G}}^{\mathrm{glob}}}.$$

Note that the assignment

$$(15.3) \quad \mathcal{Z} \mapsto \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_{\check{G}}^{\mathrm{glob}}}$$

is naturally a crystal of categories over Ran .

15.1.5. We define an action of $\mathrm{Rep}(\check{G})_{\mathcal{Z}}$ on the category (15.2) as follows.

Recall that we have a symmetric monoidal functor

$$\mathrm{Loc}_{\check{G}, \mathcal{Z}}^{\mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathcal{Z}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathcal{Z}),$$

see Sect. 17.6.1.

Composing with

$$(\mathfrak{r}^{\mathrm{glob}})^* : \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathcal{Z}) \rightarrow \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}, \mathrm{glob}})_{\mathcal{Z}}$$

we obtain a symmetric monoidal functor

$$(\mathfrak{r}^{\mathrm{glob}})^* \circ \mathrm{Loc}_{\check{G}, \mathcal{Z}}^{\mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathcal{Z}} \rightarrow \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}, \mathrm{glob}})_{\mathcal{Z}}.$$

We let $\mathrm{Rep}(\check{G})_{\mathcal{Z}}$ act on (15.2) by $(\mathfrak{r}^{\mathrm{glob}})^* \circ \mathrm{Loc}_{\check{G}, \mathcal{Z}}^{\mathrm{spec}}$ via the second factor.

⁵⁷We remind that, according to our conventions in Sect. B.8.17, when discussing crystals over Ran , given $\mathcal{Z} \rightarrow \mathrm{Ran}$, by default we base change to $\mathcal{Z}_{\mathrm{dR}}$ rather than too $\mathcal{Z}$.

15.1.6. The main result of this section is the following:

Theorem-Construction 15.1.7. *There exists a canonically defined functor*

$$\mathrm{Loc}_{G,\mathrm{crit},\mathbb{Z}}^{\mathrm{Op}} : \mathrm{KL}(G)_{\mathrm{crit},\mathbb{Z}}^{\mathrm{Op}^{\mathrm{glob}}} \rightarrow \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathbb{Z})$$

such that:

(a) *The functor*

$$\mathrm{Loc}_{G,\mathrm{crit},\mathbb{Z}} : \mathrm{KL}(G)_{\mathrm{crit},\mathbb{Z}} \rightarrow \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathbb{Z})$$

factors as

$$(15.4) \quad \mathrm{KL}(G)_{\mathrm{crit},\mathbb{Z}} \xrightarrow{\mathrm{Id} \otimes (\mathrm{ev}_{\mathbb{Z}})^*} \mathrm{KL}(G)_{\mathrm{crit},\mathbb{Z}}^{\mathrm{Op}^{\mathrm{glob}}} \xrightarrow{\mathrm{Loc}_{G,\mathrm{crit},\mathbb{Z}}^{\mathrm{Op}}} \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathbb{Z}).$$

(b) *The functor $\mathrm{Loc}_{G,\mathrm{crit},\mathbb{Z}}^{\mathrm{Op}}$ intertwines the above action of $\mathrm{Rep}(\check{G})_{\mathbb{Z}}$ on (15.2) and the action of $\mathrm{Rep}(\check{G})_{\mathbb{Z}}$ on $\mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathbb{Z})$ obtained from $\mathrm{Sat}_G^{\mathrm{nv}}$ and the action of $\mathrm{Sph}_{G,\mathbb{Z}}$ on $\mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathbb{Z})$.*

Remark 15.1.8. The Hecke eigen-property of the functor $\mathrm{Loc}_{G,\mathrm{crit},\mathbb{Z}}^{\mathrm{Op}}$ formulated in point (b) of Theorem 15.1.7 is not quite the full Hecke property one wants:

For example, when $\mathbb{Z} = \mathrm{pt}$ and $\mathbb{Z} \rightarrow \mathrm{Ran}$ corresponds to $\underline{x} \in \mathrm{Ran}$, the compatibility assertion in point (b) only talks about the Hecke action at $\underline{x}$, whereas one wants the compatibility with Hecke action over the entire Ran space.

See Corollary 15.5.9 for a stronger assertion, which we will deduce from Proposition 15.3.6 below.

15.2. The key local construction. In this subsection we will formulate Theorem 15.2.8, which is a local counterpart of Theorem 15.1.7.

15.2.1. Let

$$\mathrm{Op}_{\check{G},\mathrm{Ran}^{\subseteq}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \rightarrow \mathrm{Ran}^{\subseteq}$$

be the relative indscheme that attaches to

$$(\underline{x} \subseteq \underline{x}') \in \mathrm{Ran}^{\subseteq}$$

the space

$$\mathrm{Op}_{\check{G},\underline{x} \subseteq \underline{x}'}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} := \mathrm{Op}_{\check{G}}(\mathcal{D}_{\underline{x}'} - \underline{x}) \times_{\mathrm{LS}_{\check{G}}(\mathcal{D}_{\underline{x}'} - \underline{x})} \mathrm{LS}_{\check{G}}(\mathcal{D}_{\underline{x}'}).$$

Remark 15.2.2. Note that $\mathrm{Op}_{\check{G},\mathrm{Ran}^{\subseteq}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}$ is exactly the geometric object that encodes the unital-in-correspondences structure on $\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}$, see Sect. C.10.12.

15.2.3. We have the projections

$$(15.5) \quad \mathrm{Op}_{\check{G},\mathrm{Ran}}^{\mathrm{mon-free}} \xleftarrow{\mathrm{pr}_{\mathrm{small}}^{\mathrm{Op}}} \mathrm{Op}_{\check{G},\mathrm{Ran}^{\subseteq}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \xrightarrow{\mathrm{pr}_{\mathrm{big}}^{\mathrm{Op}}} \mathrm{Op}_{\check{G},\mathrm{Ran}}^{\mathrm{mon-free}}$$

given by restrictions along

$$\mathcal{D}_{\underline{x}} - \underline{x} \hookrightarrow \mathcal{D}_{\underline{x}'} - \underline{x} \hookrightarrow \mathcal{D}_{\underline{x}'} - \underline{x}',$$

respectively.

15.2.4. *Example.* For $\underline{x}' = \underline{x} \sqcup \underline{x}''$, we have

$$\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}(\mathcal{D}_{\underline{x}'} - \underline{x}) = \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(\mathcal{D}_{\underline{x}} - \underline{x}) \times \mathrm{Op}_{\check{G}}(\mathcal{D}_{\underline{x}''}).$$

15.2.5. For $\mathcal{Z} \rightarrow \text{Ran}$, let us denote by $\text{Op}_{\mathcal{Z}, \mathcal{Z} \subseteq}^{\text{mon-free} \rightsquigarrow \text{reg}}$ the base-change

$$\mathcal{Z} \times_{\text{Ran}, \text{pr}_{\text{small}}} \text{Op}_{\mathcal{Z}, \text{Ran} \subseteq}^{\text{mon-free} \rightsquigarrow \text{reg}}.$$

The assignment

$$\mathcal{Z} \mapsto \text{QCoh}(\text{Op}_{\mathcal{Z}}^{\text{mon-free} \rightsquigarrow \text{reg}})_{\mathcal{Z} \subseteq}$$

is a crystal of symmetric monoidal categories over Ran , which we will denote by

$$\text{QCoh}(\text{Op}_{\mathcal{Z}}^{\text{mon-free} \rightsquigarrow \text{reg}}).$$

The map $\text{pr}_{\text{small}}^{\text{Op}}$ gives rise to a symmetric monoidal functor

$$(\text{pr}_{\text{small}}^{\text{Op}})^* : \text{QCoh}(\text{Op}_{\mathcal{Z}}^{\text{mon-free}}) \rightarrow \text{QCoh}(\text{Op}_{\mathcal{Z}}^{\text{mon-free} \rightsquigarrow \text{reg}}).$$

15.2.6. Denote

$$(15.6) \quad \text{KL}(G)_{\text{crit}}^{\text{Op}_{\mathcal{Z}}^{\text{loc}}} := \text{KL}(G)_{\text{crit}} \otimes_{\text{QCoh}(\text{Op}_{\mathcal{Z}}^{\text{mon-free}})} \text{QCoh}(\text{Op}_{\mathcal{Z}}^{\text{mon-free} \rightsquigarrow \text{reg}}).$$

We will regard it as a crystal of categories over Ran . Denote by $\text{Id} \otimes (\text{pr}_{\text{small}}^{\text{Op}})^*$ the functor

$$\text{KL}(G)_{\text{crit}} \rightarrow \text{KL}(G)_{\text{crit}} \otimes_{\text{QCoh}(\text{Op}_{\mathcal{Z}}^{\text{mon-free}})} \text{QCoh}(\text{Op}_{\mathcal{Z}}^{\text{mon-free} \rightsquigarrow \text{reg}}) = \text{KL}(G)_{\text{crit}}^{\text{Op}_{\mathcal{Z}}^{\text{loc}}}.$$

15.2.7. Recall that ins. vac. is a functor between crystals of categories over Ran

$$(15.7) \quad \text{KL}(G)_{\text{crit}} \rightarrow (\text{pr}_{\text{small}})^* \circ (\text{pr}_{\text{big}})^*(\text{KL}(G)_{\text{crit}}),$$

and actually a (strict) functor between crystals of categories over Ran^{untl}

$$\text{KL}(G)_{\text{crit}} \rightarrow (\text{pr}_{\text{small}}^{\text{untl}})^*, \text{strict} \circ (\text{pr}_{\text{big}}^{\text{untl}})^*(\text{KL}(G)_{\text{crit}}).$$

The key construction that we will need says the following:

Theorem-Construction 15.2.8. *There exists a canonically defined functor*

$$(15.8) \quad \text{KL}(G)_{\text{crit}}^{\text{Op}_{\mathcal{Z}}^{\text{loc}}} \xrightarrow{\text{ins. vac.}^{\text{mon-free} \rightsquigarrow \text{reg}}} (\text{pr}_{\text{small}})^* \circ (\text{pr}_{\text{big}})^*(\text{KL}(G)_{\text{crit}}),$$

such that ins. vac. factors as

$$\text{KL}(G)_{\text{crit}} \xrightarrow{\text{Id} \otimes (\text{pr}_{\text{small}}^{\text{Op}})^*} \text{KL}(G)_{\text{crit}}^{\text{Op}_{\mathcal{Z}}^{\text{loc}}} \xrightarrow{\text{ins. vac.}^{\text{mon-free} \rightsquigarrow \text{reg}}} (\text{pr}_{\text{small}})^* \circ (\text{pr}_{\text{big}})^*(\text{KL}(G)_{\text{crit}}).$$

Furthermore, the functor $\text{ins. vac.}^{\text{mon-free} \rightsquigarrow \text{reg}}$ is linear with respect to

$$(\text{pr}_{\text{small}})^* \circ (\text{pr}_{\text{big}})^*(\text{QCoh}(\text{Op}_{\mathcal{Z}}^{\text{mon-free}})),$$

where:

- $(\text{pr}_{\text{small}})^* \circ (\text{pr}_{\text{big}})^*(\text{QCoh}(\text{Op}_{\mathcal{Z}}^{\text{mon-free}}))$ acts on the left-hand side via $(\text{pr}_{\text{big}}^{\text{Op}})^*$;
- $(\text{pr}_{\text{small}})^* \circ (\text{pr}_{\text{big}})^*(\text{QCoh}(\text{Op}_{\mathcal{Z}}^{\text{mon-free}}))$ acts on the right-hand side by applying the functor $(\text{pr}_{\text{small}})^* \circ (\text{pr}_{\text{big}})^*$ to the $\text{QCoh}(\text{Op}_{\mathcal{Z}}^{\text{mon-free}})$ -action on $\text{KL}(G)_{\text{crit}}$.

The proof of Theorem 15.2.8 will be given in Sect. 16.

Remark 15.2.9. Note that Theorem 15.2.8 sounds semantically close to Theorem 15.1.7. And indeed, Theorem 15.1.7 will be deduced from Theorem 15.2.8 by a manipulation that involves factorization homology.

Yet, Theorem 15.2.8 is purely local, while Theorem 15.1.7 is “local-to-global”.

That said, we will need Theorem 15.2.8 *not only* for the proof of Theorem 15.1.7. In the next subsection, we will use it to extract a stronger Hecke property for the functor $\text{Loc}_{G, \text{crit}, \mathcal{Z}}^{\text{Op}}$.

15.2.10. Let us explain what Theorem 15.2.8 says at the pointwise level, i.e., for a fixed $(\underline{x} \subseteq \underline{x}') \in \text{Ran}^{\subseteq}$.

Write

$$\underline{x}' = \underline{x} \sqcup \underline{x}''.$$

The corresponding functor

$$\text{ins. vac}_{\underline{x} \subseteq \underline{x}'} : \text{KL}(G)_{\text{crit}, \underline{x}} \rightarrow \text{KL}(G)_{\text{crit}, \underline{x}'} \simeq \text{KL}(G)_{\text{crit}, \underline{x}} \otimes \text{KL}(G)_{\text{crit}, \underline{x}''}$$

acts as

$$\mathcal{M} \mapsto \mathcal{M} \otimes \text{Vac}(G)_{\text{crit}, \underline{x}''}.$$

The pointwise statement of Theorem 15.2.8 is that this functor can be factored as

$$\text{KL}(G)_{\text{crit}, \underline{x}} \rightarrow \text{KL}(G)_{\text{crit}, \underline{x}} \otimes \text{QCoh}(\text{Op}_{\tilde{G}, \underline{x}''}^{\text{reg}}) \rightarrow \text{KL}(G)_{\text{crit}, \underline{x}} \otimes \text{KL}(G)_{\text{crit}, \underline{x}''},$$

where the second arrow is $\text{QCoh}(\text{Op}_{\tilde{G}, \underline{x}''}^{\text{mon-free}})$ -linear.

In other words, we are saying that the object $\text{Vac}(G)_{\text{crit}, \underline{x}''} \in \text{KL}(G)_{\text{crit}, \underline{x}''}$ naturally lifts to an object of the category

$$\text{Funct}_{\text{QCoh}(\text{Op}_{\tilde{G}, \underline{x}''}^{\text{mon-free}})}(\text{QCoh}(\text{Op}_{\tilde{G}, \underline{x}''}^{\text{reg}}), \text{KL}(G)_{\text{crit}, \underline{x}''}).$$

This lift is the basic feature of the vacuum object: it says that the structure of factorization $\mathfrak{z}_{\mathfrak{g}}$ -module on $\text{Vac}(G)_{\text{crit}}$ as an object of $\text{KL}(G)_{\text{crit}}$ is obtained by restriction from a structure of *commutative* $\mathfrak{z}_{\mathfrak{g}}$ -module.

The proof of Theorem 15.2.8 will amount to spelling out the above construction in the factorization setting.

Remark 15.2.11. In the proof of Theorem 15.2.8 that we will give, we will avoid using the FLE. We do this for aesthetical reasons: the construction of the functor Theorem 15.2.8 is more or less tautological if one says the right words.

However, if we use the FLE, there would be almost nothing to prove: the FLE, combined with the equivalence $\Theta_{\text{Op}_{\tilde{G}}^{\text{mon-free}}}$, allows us to identify the two sides in (15.8) with

$$\text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free} \rightsquigarrow \text{reg}}) \text{ and } (\text{pr}_{\text{small}})_* \circ (\text{pr}_{\text{big}})^*(\text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free}})),$$

respectively, and the functor in question is obtained by taking direct image along

$$\text{Op}_{\tilde{G}, \text{Ran}^{\subseteq}}^{\text{mon-free} \rightsquigarrow \text{reg}} \xrightarrow{\text{pr}_{\text{big}}^{\text{Op}}} \text{Ran}^{\subseteq} \times_{\text{pr}_{\text{big}}, \text{Ran}} \text{Op}_{\tilde{G}, \text{Ran}}^{\text{mon-free}}.$$

That said, the functor $\text{ins. vac}^{\text{mon-free} \rightsquigarrow \text{reg}}$ that we will construct does reproduce the above functor, by the nature of the FLE.

Remark 15.2.12. Note that the space $\text{Op}_{\tilde{G}, \text{Ran}^{\subseteq}}^{\text{mon-free} \rightsquigarrow \text{reg}}$, equipped with the maps (15.5), has a structure of groupoid⁵⁸ acting on $\text{Op}_{\tilde{G}, \text{Ran}}^{\text{mon-free}}$, with the composition given by

$$\begin{aligned} & \text{Op}_{\tilde{G}}^{\text{mon-free} \rightsquigarrow \text{reg}}(\mathcal{D}_{\underline{x}'} - \underline{x}) \times_{\text{Op}_{\tilde{G}}^{\text{mon-free}}(\mathcal{D}_{\underline{x}'} - \underline{x}')} \text{Op}_{\tilde{G}}^{\text{mon-free} \rightsquigarrow \text{reg}}(\mathcal{D}_{\underline{x}''} - \underline{x}') \simeq \\ & \simeq (\text{Op}_{\tilde{G}}(\mathcal{D}_{\underline{x}'} - \underline{x}) \times_{\text{Op}_{\tilde{G}}(\mathcal{D}_{\underline{x}'} - \underline{x}')} \text{Op}_{\tilde{G}}(\mathcal{D}_{\underline{x}''} - \underline{x}')) \times_{(\text{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}'} - \underline{x}) \times_{\text{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}'} - \underline{x}')} \text{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}''} - \underline{x}'))} \text{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}''}) \rightarrow \\ & \rightarrow \text{Op}_{\tilde{G}}(\mathcal{D}_{\underline{x}''} - \underline{x}) \times_{\text{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}''} - \underline{x})} \text{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}''}) = \text{Op}_{\tilde{G}}^{\text{mon-free} \rightsquigarrow \text{reg}}(\mathcal{D}_{\underline{x}''} - \underline{x}), \end{aligned}$$

where the middle arrow is given by gluing along

$$\mathcal{D}_{\underline{x}'} - \underline{x} \sqcup_{\mathcal{D}_{\underline{x}'} - \underline{x}'} \mathcal{D}_{\underline{x}''} - \underline{x}' \simeq \mathcal{D}_{\underline{x}''} - \underline{x}.$$

⁵⁸This groupoid structure is part of the structure of being unital-in-correspondences, see Sect. C.10.

Note that one can interpret Theorem 15.2.8 as saying that the above action of $\mathrm{Op}_{\check{G}, \mathrm{Ran}^\subseteq}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}$ on $\mathrm{Op}_{\check{G}, \mathrm{Ran}^\subseteq}^{\mathrm{mon-free}}$ can be lifted to an action on $\mathrm{KL}(G)_{\mathrm{crit}}$ *at the level of 1-morphisms*.

In fact, Theorem 15.2.8 has a natural upgrade to a statement that we have a full datum of action of $\mathrm{Op}_{\check{G}, \mathrm{Ran}^\subseteq}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}$ on $\mathrm{KL}(G)_{\mathrm{crit}}$. This is again automatic if we allow ourselves to use the FLE.

15.3. The expanded Hecke eigen-property. In this subsection we will assume Theorem 15.1.7 and explain that the Hecke property of the functor $\mathrm{Loc}_{G, \mathrm{crit}, \mathbb{Z}}^{\mathrm{Op}}$ stated in point (b) of the theorem implies a stronger property (the difference between the two versions of the Hecke property is what was alluded to in Remark 15.1.8).

15.3.1. Let $\mathbb{Z} \rightarrow \mathrm{Ran}$ be as above. Consider the space

$$(15.9) \quad \mathrm{Op}_{\check{G}, \mathbb{Z}^\subseteq}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}, \mathrm{glob}} := \mathrm{Op}_{\check{G}, \mathbb{Z}}^{\mathrm{mon-free}, \mathrm{glob}} \times_{\mathbb{Z}, \mathrm{pr}_{\mathrm{small}, \mathbb{Z}}} \mathbb{Z}^\subseteq.$$

Denote by $\mathrm{pr}_{\mathrm{small}, \mathbb{Z}}^{\mathrm{Op}^{\mathrm{glob}}}$ the projection

$$\mathrm{Op}_{\check{G}, \mathbb{Z}^\subseteq}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}, \mathrm{glob}} \rightarrow \mathrm{Op}_{\check{G}, \mathbb{Z}}^{\mathrm{mon-free}, \mathrm{glob}}.$$

Note that we have a naturally defined ind-closed embedding

$$(15.10) \quad \mathrm{pr}_{\mathrm{big}, \mathbb{Z}}^{\mathrm{Op}^{\mathrm{glob}}} : \mathrm{Op}_{\check{G}, \mathbb{Z}^\subseteq}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}, \mathrm{glob}} \hookrightarrow \mathrm{Op}_{\check{G}, \mathbb{Z}^\subseteq}^{\mathrm{mon-free}, \mathrm{glob}}.$$

Denote by $\mathrm{pr}_{\mathrm{big}}^{\mathrm{Op}^{\mathrm{glob}}}$ the composition of (15.10) with the projection

$$\mathrm{Op}_{\check{G}, \mathbb{Z}^\subseteq}^{\mathrm{mon-free}, \mathrm{glob}} \xrightarrow{\mathrm{id} \times \mathrm{pr}_{\mathrm{big}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mon-free}, \mathrm{glob}}.$$

15.3.2. *Example.* Let us explain what the map (15.10) looks like for $\mathbb{Z} = \mathrm{pt}$ so that $\mathbb{Z} \rightarrow \mathrm{Ran}$ corresponds to $\underline{x} \in \mathrm{Ran}$ (in which case $\mathbb{Z}^\subseteq = \mathrm{Ran}_{\underline{x}}$, so a point on it corresponds to $\underline{x} \subseteq \underline{x}'$).

Then a point of the left-hand (resp., right-hand) side in (15.10) is a local system on X with an oper structure away from $\underline{x}$ (resp., $\underline{x}'$), and the map (15.10) is given by restriction along

$$X - \underline{x}' \subseteq X - \underline{x}.$$

15.3.3. Consider the map

$$\mathrm{Op}_{\check{G}, \mathbb{Z}^\subseteq}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}, \mathrm{glob}} = \mathrm{Op}_{\check{G}, \mathbb{Z}}^{\mathrm{mon-free}, \mathrm{glob}} \times_{\mathbb{Z}, \mathrm{pr}_{\mathrm{small}, \mathbb{Z}}} \mathbb{Z}^\subseteq \xrightarrow{\mathrm{r}^{\mathrm{glob}} \times \mathrm{id}} (\mathrm{LS}_{\check{G}} \times \mathbb{Z}) \times_{\mathbb{Z}} \mathbb{Z}^\subseteq = \mathrm{LS}_{\check{G}} \times \mathbb{Z}^\subseteq,$$

to be denoted $\mathrm{r}^{\mathrm{glob}}$.

Thus, we can consider the symmetric monoidal functor

$$(\mathrm{r}^{\mathrm{glob}})^* \circ \mathrm{Loc}_{\check{G}, \mathbb{Z}^\subseteq}^{\mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathbb{Z}^\subseteq} \rightarrow \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}, \mathrm{glob}})_{\mathbb{Z}^\subseteq}.$$

15.3.4. We can view $\mathrm{Op}_{\check{G}, \mathbb{Z}^\subseteq}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}, \mathrm{glob}}$ as mapping to $\mathrm{Op}_{\check{G}, \mathbb{Z}}^{\mathrm{mon-free}}$ by

$$\mathrm{Op}_{\check{G}, \mathbb{Z}^\subseteq}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}, \mathrm{glob}} = \mathrm{Op}_{\check{G}, \mathbb{Z}}^{\mathrm{mon-free}, \mathrm{glob}} \times_{\mathbb{Z}, \mathrm{pr}_{\mathrm{small}, \mathbb{Z}}} \mathbb{Z}^\subseteq \rightarrow \mathrm{Op}_{\check{G}, \mathbb{Z}}^{\mathrm{mon-free}, \mathrm{glob}} \xrightarrow{\mathrm{ev}_{\mathbb{Z}}} \mathrm{Op}_{\check{G}, \mathbb{Z}}^{\mathrm{mon-free}}.$$

Hence, we can form the category

$$(15.11) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathbb{Z}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathbb{Z}}} \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}, \mathrm{glob}})_{\mathbb{Z}^\subseteq}.$$

Since

$$\mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}, \mathrm{glob}})_{\mathbb{Z}} \otimes_{\mathrm{D-mod}(\mathbb{Z})} \mathrm{D-mod}(\mathbb{Z}^\subseteq) \rightarrow \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}, \mathrm{glob}})_{\mathbb{Z}^\subseteq}$$

is an equivalence, we can rewrite (15.11) as

$$(15.12) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}}} \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free, glob}})_{\mathcal{Z}} \otimes_{\mathrm{D-mod}(\mathcal{Z})} \mathrm{D-mod}(\mathcal{Z}^{\subseteq}) = \\ = \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_{\check{G}}^{\mathrm{glob}}} \otimes_{\mathrm{D-mod}(\mathcal{Z})} \mathrm{D-mod}(\mathcal{Z}^{\subseteq}).$$

Thus, we obtain that the category

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_{\check{G}}^{\mathrm{glob}}} \otimes_{\mathrm{D-mod}(\mathcal{Z})} \mathrm{D-mod}(\mathcal{Z}^{\subseteq})$$

carries a monoidal action of $\mathrm{Rep}(\check{G})_{\mathcal{Z}^{\subseteq}}$.

15.3.5. Consider the functor

$$(15.13) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_{\check{G}}^{\mathrm{glob}}} \otimes_{\mathrm{D-mod}(\mathcal{Z})} \mathrm{D-mod}(\mathcal{Z}^{\subseteq}) \xrightarrow{\mathrm{Loc}_{G, \mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}} \otimes \mathrm{Id}} \\ \rightarrow (\mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z})) \otimes_{\mathrm{D-mod}(\mathcal{Z})} \mathrm{D-mod}(\mathcal{Z}^{\subseteq}) \simeq \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z}^{\subseteq}).$$

We claim:

Proposition 15.3.6. *The functor (15.13) intertwines the actions of $\mathrm{Rep}(\check{G})_{\mathcal{Z}^{\subseteq}}$ on the two sides, where the action of $\mathrm{Rep}(\check{G})_{\mathcal{Z}^{\subseteq}}$ on the left-hand sides is the one specified in Sect. 15.3.4, and on the right-hand side it is obtained from $\mathrm{Sat}_G^{\mathrm{nv}}$ and the action of $\mathrm{Sph}_{G, \mathcal{Z}^{\subseteq}}$ on $\mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z}^{\subseteq})$.*

15.4. Proof of Proposition 15.3.6.

15.4.1. Note that we have a Cartesian square

$$\begin{array}{ccc} \mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg, glob}} & \xrightarrow{\mathrm{pr}_{\mathrm{big}, \mathcal{Z}}^{\mathrm{Op, glob}}} & \mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon-free, glob}} \\ \mathrm{ev}_{\mathcal{Z}^{\subseteq}} \downarrow & & \downarrow \mathrm{ev}_{\mathcal{Z}^{\subseteq}} \\ \mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} & \xrightarrow{\mathrm{pr}_{\mathrm{big}, \mathcal{Z}}^{\mathrm{Op}}} & \mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon-free}}, \end{array}$$

where $\mathrm{pr}_{\mathrm{big}, \mathcal{Z}}^{\mathrm{Op}}$ is the map whose composition with

$$\mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon-free}} \simeq \mathcal{Z}^{\subseteq} \times_{\mathrm{pr}_{\mathrm{big}, \mathrm{Ran}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mon-free}} \rightarrow \mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mon-free}}$$

is the map

$$\mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon-free}} \rightarrow \mathrm{Op}_{\check{G}, \mathrm{Ran}^{\subseteq}}^{\mathrm{mon-free}} \xrightarrow{\mathrm{pr}_{\mathrm{big}}^{\mathrm{Op}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mon-free}}.$$

We claim:

Lemma 15.4.2. *The functor*

$$\mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})_{\mathcal{Z}^{\subseteq}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}^{\subseteq}}} \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free, glob}})_{\mathcal{Z}^{\subseteq}} \rightarrow \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg, glob}})_{\mathcal{Z}^{\subseteq}}$$

is an equivalence.

The lemma will be proved in Sect. 15.10.

15.4.3. Consider the category

$$\begin{aligned}
 (15.14) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_G^{\mathrm{glob}}} \otimes_{\mathrm{D-mod}(\mathcal{Z})} \mathrm{D-mod}(\mathcal{Z}^{\subseteq}) &= \\
 &= \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathcal{Z}}} \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}, \mathrm{glob}})_{\mathcal{Z}} \otimes_{\mathrm{D-mod}(\mathcal{Z})} \mathrm{D-mod}(\mathcal{Z}^{\subseteq}) \simeq \\
 &\simeq \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathcal{Z}}} \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}, \mathrm{glob}})_{\mathcal{Z}^{\subseteq}}.
 \end{aligned}$$

By Lemma 15.4.2, we can rewrite it as

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathcal{Z}}} \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})_{\mathcal{Z}^{\subseteq}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathcal{Z}^{\subseteq}}} \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}, \mathrm{glob}})_{\mathcal{Z}^{\subseteq}},$$

i.e.,

$$\mathrm{KL}_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_G^{\mathrm{loc}}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathcal{Z}^{\subseteq}}} \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}, \mathrm{glob}})_{\mathcal{Z}^{\subseteq}},$$

where

$$\mathcal{Z} \mapsto \mathrm{KL}_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_G^{\mathrm{loc}}}$$

is the factorization category (15.6).

Using the functor $\mathrm{ins.vac}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}$ we obtain a functor

$$\begin{aligned}
 (15.15) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_G^{\mathrm{glob}}} \otimes_{\mathrm{D-mod}(\mathcal{Z})} \mathrm{D-mod}(\mathcal{Z}^{\subseteq}) &\simeq \\
 &\simeq \mathrm{KL}_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_G^{\mathrm{loc}}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathcal{Z}^{\subseteq}}} \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}, \mathrm{glob}})_{\mathcal{Z}^{\subseteq}} \rightarrow \\
 \mathrm{ins.vac}_{\mathcal{Z}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \otimes \mathrm{Id} &\longrightarrow \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}^{\subseteq}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathcal{Z}^{\subseteq}}} \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}, \mathrm{glob}})_{\mathcal{Z}^{\subseteq}} \simeq \\
 &\simeq \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}^{\subseteq}}^{\mathrm{Op}_G^{\mathrm{glob}}}.
 \end{aligned}$$

Remark 15.4.4. In the spirit of Remark 15.2.12, one can show that the functors (15.15) upgrade to a *local unital structure* (see Sect. 11.2.1 for what this means) on the crystal of categories over Ran given by (15.3).

15.4.5. The following property will be embedded into the construction of the assignment

$$(15.16) \quad \mathcal{Z} \mapsto \mathrm{Loc}_{G, \mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}} :$$

The composition

$$\begin{aligned}
 (15.17) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_G^{\mathrm{glob}}} &\xrightarrow{\mathrm{Id} \otimes \mathrm{pr}_{\mathrm{small}}^!} \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_G^{\mathrm{glob}}} \otimes_{\mathrm{D-mod}(\mathcal{Z})} \mathrm{D-mod}(\mathcal{Z}^{\subseteq}) \xrightarrow{(15.15)} \\
 &\rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}^{\subseteq}}^{\mathrm{Op}_G^{\mathrm{glob}}} \xrightarrow{\mathrm{Loc}_{G, \mathrm{crit}, \mathcal{Z}^{\subseteq}}^{\mathrm{Op}}} \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z}^{\subseteq})
 \end{aligned}$$

identifies with the functor

$$\begin{aligned}
 (15.18) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_G^{\mathrm{glob}}} &\xrightarrow{\mathrm{Loc}_{G, \mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}}} \\
 &\rightarrow \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z}) \xrightarrow{\mathrm{Id} \otimes \mathrm{pr}_{\mathrm{small}}^!} \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z}^{\subseteq}).
 \end{aligned}$$

Remark 15.4.6. In the spirit of Remark 15.4.4 one can show that the isomorphism between (15.17) and (15.18) upgrades to a *unital structure* (see Sect. 11.3.5 for what this means) on the assignment (15.16).

15.4.7. Since the functors

$$(15.19) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}_{\check{G}}^{\mathrm{glob}}} \otimes_{\mathrm{D-mod}(\mathcal{Z})} \mathrm{D-mod}(\mathcal{Z}^{\subseteq}) \xrightarrow{(15.15)} \\ \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}^{\subseteq}}^{\mathrm{Op}_{\check{G}}^{\mathrm{glob}}} \xrightarrow{\mathrm{Loc}_{G, \mathrm{crit}, \mathcal{Z}^{\subseteq}}^{\mathrm{Op}}} \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z}^{\subseteq})$$

and (15.13) are $\mathrm{D-mod}(\mathcal{Z}^{\subseteq})$ -linear, and the isomorphism between (15.17) and (15.18) is $\mathrm{D-mod}(\mathcal{Z})$ -linear, we obtain that the functor (15.19) identifies with (15.13).

Thus, in order to prove Proposition 15.3.6, it suffices to construct the datum of compatibility with the $\mathrm{Rep}(\check{G})_{\mathcal{Z}^{\subseteq}}$ -action for the functor (15.19).

15.4.8. We will show that each of the two arrows in (15.19) is compatible with the $\mathrm{Rep}(\check{G})_{\mathcal{Z}^{\subseteq}}$ -action.

For the second arrow, this follows from Theorem 15.1.7(b).

15.4.9. For the first arrow, unwinding the definition of the functor (15.15), we have to show that the functor

$$\mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})_{\mathcal{Z}^{\subseteq}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}^{\subseteq}}} \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free, glob}})_{\mathcal{Z}^{\subseteq}} \rightarrow \\ \rightarrow \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg, glob}})_{\mathcal{Z}^{\subseteq}}$$

is $\mathrm{Rep}(\check{G})_{\mathcal{Z}^{\subseteq}}$ -linear, where:

- $\mathrm{Rep}(\check{G})_{\mathcal{Z}^{\subseteq}}$ acts on the right-hand side via $\mathrm{Loc}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{spec}}$ followed by pullback along

$$(15.20) \quad {}'\mathfrak{t}^{\mathrm{glob}} : \mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg, glob}} \rightarrow \mathrm{LS}_{\check{G}} \times \mathcal{Z}^{\subseteq};$$

- $\mathrm{Rep}(\check{G})_{\mathcal{Z}^{\subseteq}}$ acts on the left-hand side on the second factor via $\mathrm{Loc}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{spec}}$ followed by pullback along

$$\mathfrak{t}^{\mathrm{glob}} : \mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon-free, glob}} \rightarrow \mathrm{LS}_{\check{G}} \times \mathcal{Z}^{\subseteq}.$$

The required compatibility follows from the fact that

$${}'\mathfrak{t}^{\mathrm{glob}} = \mathfrak{t}^{\mathrm{glob}} \circ \mathrm{pr}_{\mathrm{big}, \mathcal{Z}}^{\mathrm{Op}_{\check{G}}^{\mathrm{glob}}}$$

as maps

$$\mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg, glob}} \rightrightarrows \mathrm{LS}_{\check{G}} \times \mathcal{Z}^{\subseteq}.$$

15.5. The integrated Hecke eigen-property. In this subsection we will apply the paradigm of Sect. H.7, and deduce an ultimate form of compatibility of the functor $\mathrm{Loc}_{G, \mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}}$ with the Hecke action.

15.5.1. Consider $\mathrm{Rep}(\check{G})$ as a *unital* monoidal factorization category. Note that we consider the Hecke action as a local Ran -unital action of $\mathrm{Rep}(\check{G})$ on $\mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$ (see Sect. H.6.1 for what this means), i.e.,

$$\mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \in \mathrm{Rep}(\check{G})^{\mathrm{loc}}\text{-}\mathbf{mod}.$$

In particular, the action of $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}$, endowed with the *convolution* monoidal structure (i.e., the monoidal category $(\mathrm{Rep}(\check{G})_{\mathrm{Ran}})^*$, see Sect. H.5.2), on $\mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$ factors via

$$(\mathrm{Rep}(\check{G})_{\mathrm{Ran}})^* \rightarrow \mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{unl}}, \mathrm{indep}}.$$

For any $\mathcal{Z} \rightarrow \mathrm{Ran}$, we will consider

$$\mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z})$$

as a module over

$$\mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{unl}}, \mathrm{indep}} \otimes \mathrm{D-mod}(\mathcal{Z}).$$

15.5.2. Note also that the spectral localization functor can be viewed as a (strictly) unital (symmetric) monoidal functor

$$\underline{\mathrm{Loc}}_G^{\mathrm{spec}} : \underline{\mathrm{Rep}}(\check{G}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}^{\mathrm{untl}}),$$

where $\underline{\mathrm{Rep}}(\check{G})$ is the crystal of categories over $\mathrm{Ran}^{\mathrm{untl}}$ corresponding to $\mathrm{Rep}(\check{G})$, viewed as a factorization category.

In particular, the functor

$$\mathrm{Loc}_G^{\mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}})$$

factors as

$$\mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightarrow \mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{untl}}, \mathrm{indep}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}}).$$

Given $\mathcal{Z} \rightarrow \mathrm{Ran}$, we will consider the category

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}^{\mathrm{glob}}} = \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathcal{Z}}} \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free, glob}})_{\mathcal{Z}}$$

as acted on by

$$\mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{untl}}, \mathrm{indep}} \otimes \mathrm{D-mod}(\mathcal{Z})$$

via the projection

$$\mathrm{Op}_{G, \mathcal{Z}}^{\mathrm{mon-free, glob}} \rightarrow \mathrm{LS}_{\check{G}} \times \mathcal{Z}$$

and the action on the second factor.

15.5.3. We claim:

Corollary 15.5.4. *The functor*

$$\mathrm{Loc}_{G, \mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}} : \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}^{\mathrm{glob}}} \rightarrow \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z})$$

is compatible with the action of $\mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{untl}}, \mathrm{indep}} \otimes \mathrm{D-mod}(\mathcal{Z})$.

Proof. First, it is easy to see that we can assume that $\mathcal{Z}$ is an affine scheme S . By Corollary H.7.8, we can consider both sides, i.e.,

$$\mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(S) \text{ and } \mathrm{KL}(G)_{\mathrm{crit}, S}^{\mathrm{Op}^{\mathrm{glob}}},$$

as objects of

$$\mathrm{Rep}(\check{G})_S^{\mathrm{loc, untl}}\text{-}\mathbf{mod}$$

(see Sect. H.7.3 for the notation), and we have to show that $\mathrm{Loc}_{G, \mathrm{crit}, S}^{\mathrm{Op}}$ extends to a map inside this category (see Corollary H.7.8).

Now, Proposition 15.3.6 implies that the functor $\mathrm{Loc}_{G, \mathrm{crit}, S}^{\mathrm{Op}}$ gives rise to a functor between the images of these two objects under the forgetful functor

$$(15.21) \quad \mathrm{Rep}(\check{G})_S^{\mathrm{loc, untl}}\text{-}\mathbf{mod} \rightarrow \mathrm{Rep}(\check{G})_S^{\mathrm{loc}}\text{-}\mathbf{mod}.$$

Now, the assertion follows from the fact that the functor (15.21) is fully faithful, see Sect. H.7.3. $\square$

Corollary 15.5.5. *The functor*

$$\mathrm{Loc}_{G, \mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}} : \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}^{\mathrm{glob}}} \rightarrow \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z})$$

is compatible with the action of $(\mathrm{Rep}(\check{G})_{\mathrm{Ran}})^$.*

15.5.6. Let now $\mathcal{Z}$ be pseudo-proper, and consider the functor

$$(\mathrm{Id} \otimes C_c(\mathcal{Z}, -)) : \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \rightarrow \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G).$$

This functor is obviously compatible with the action of $(\mathrm{Rep}(\check{G})_{\mathrm{Ran}})^*$.

Denote

$$\mathrm{Loc}_{G, \mathrm{crit}, f_{\mathcal{Z}}}^{\mathrm{Op}} := (\mathrm{Id} \otimes C_c(\mathcal{Z}, -)) \circ \mathrm{Loc}_{G, \mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}}, \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}^{\mathrm{glob}}} \rightarrow \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G).$$

Hence, from Corollary 15.5.5 we obtain:

Corollary 15.5.7. *The functor $\mathrm{Loc}_{G, \mathrm{crit}, f_{\mathcal{Z}}}^{\mathrm{Op}}$ is compatible with the action of $(\mathrm{Rep}(\check{G})_{\mathrm{Ran}})^*$ on the two sides.*

15.5.8. Finally, we take $\mathcal{Z} = \mathrm{Ran}$. Denote the corresponding functor

$$\mathrm{Loc}_{G, \mathrm{crit}, \int_{\mathrm{Ran}}}^{\mathrm{Op}} : \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}}^{\mathrm{Op}^{\mathrm{glob}}} \rightarrow \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$$

by $\mathrm{Loc}_{G, \mathrm{crit}}^{\mathrm{Op}}$.

We obtain:

Corollary 15.5.9. *The the functor $\mathrm{Loc}_{G, \mathrm{crit}}^{\mathrm{Op}}$ is compatible with the action of $(\mathrm{Rep}(\check{G})_{\mathrm{Ran}})^*$ (with the convolution monoidal structure) on the two sides.*

This corollary is the ultimate form of the compatibility between the (globalized) critical localization functor and the (non-derived) Hecke action.

15.5.10. A variant of Corollary 15.5.5 was at the core of the construction of Hecke eigensheaves in [BD1]. In our language, this construction can be reformulated as follows.

Take $\mathcal{Z} = \mathrm{pt}$, so that $\mathcal{Z} \rightarrow \mathrm{Ran}$ corresponds to $\underline{x} \in \mathrm{Ran}$.

Recall that according to Theorem 6.1.4, Proposition 3.8.7 and Proposition 3.7.10, we can identify

$$(15.22) \quad \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \xrightarrow{\mathrm{FLE}_{G, \mathrm{crit}}} \mathrm{IndCoh}^*(\mathrm{Op}_{G, \underline{x}}^{\mathrm{mon-free}}) \xrightarrow{\Theta_{\mathrm{Op}_{G, \underline{x}}^{\mathrm{mon-free}}}} \simeq \mathrm{IndCoh}^!(\mathrm{Op}_{G, \underline{x}}^{\mathrm{mon-free}}) \xrightarrow{\Upsilon_{\mathrm{Op}_{G, \underline{x}}^{\mathrm{mon-free}}}} \mathrm{QCoh}(\mathrm{Op}_{G, \underline{x}}^{\mathrm{mon-free}})$$

as $\mathrm{QCoh}(\mathrm{Op}_{G, \underline{x}}^{\mathrm{mon-free}})$ -module categories.

Hence, we can identify the category $\mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}^{\mathrm{glob}}}$ with $\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x}))$, as a module category over $\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x}))$. Since $\mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x})$ is locally almost of finite type and formally smooth, by [GaRo1, Theorem 10.1.1], the functor

$$\Upsilon_{\mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x})} : \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x})) \rightarrow \mathrm{IndCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x}))$$

is an equivalence. Hence, we can further identify $\mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}^{\mathrm{glob}}}$ with $\mathrm{IndCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x}))$ as a $\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x}))$ -module category.

Hence, we can view $\mathrm{Loc}_{G, \mathrm{crit}, \underline{x}}^{\mathrm{Op}}$ as a functor

$$\mathrm{IndCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x})) \rightarrow \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$$

that intertwines the action of $(\mathrm{Rep}(\check{G})_{\mathrm{Ran}})^*$ on $\mathrm{IndCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x}))$, given by

$$(\mathfrak{t}^{\mathrm{glob}})^* \circ \mathrm{Loc}_G^{\mathrm{spec}} : (\mathrm{Rep}(\check{G})_{\mathrm{Ran}})^* \rightarrow \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x}))$$

and the action of $\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x}))$ on $\mathrm{IndCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x}))$, and the $(\mathrm{Rep}(\check{G})_{\mathrm{Ran}})^*$ -action on $\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$.

In particular, for a k -point $\sigma \in \mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x})$ and the corresponding sky-scraper sheaf

$$k_{\sigma} \in \mathrm{IndCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x})),$$

the object

$$\mathrm{Loc}_{G, \mathrm{crit}, \underline{x}}^{\mathrm{Op}}(k_\sigma) \in \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$$

is a Hecke eigensheaf with eigenvalue $\mathfrak{t}^{\mathrm{glob}}(\sigma) \in \mathrm{LS}_{\tilde{G}}$.

Remark 15.5.11. The case that is actually considered in [BD1] is when σ is a regular oper. In our language, this corresponds replacing $\mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{Op}^{\mathrm{glob}}}$ with

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{reg}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{reg}})_{\mathcal{Z}}} (\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}(X)) \otimes \mathrm{D}\text{-mod}(\mathcal{Z})),$$

where

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}^{\mathrm{reg}} := \mathrm{Funct}_{\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})_{\mathcal{Z}}}(\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{reg}})_{\mathcal{Z}}, \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}}).$$

15.6. Proof of Theorem 15.1.7.

15.6.1. In the proof of Theorem 15.1.7, for expositional purposes we will assume that $\mathcal{Z} = \mathrm{pt}$, so its map to Ran corresponds to $\underline{x} \in \mathrm{Ran}$. We will denote the corresponding space $\mathcal{Z}^{\subseteq}$ by $\mathrm{Ran}_{\underline{x}}$.

15.6.2. Before we launch the proof, let us explain the idea that lies behind it. Let $\mathcal{A}$ be a factorization algebra, and let $\mathcal{A}^{\mathrm{ch}}$ be the corresponding chiral algebra, see Sect. D.1.1. Let $\mathfrak{z}$ be a commutative factorization algebra, such that $\mathfrak{z}^{\mathrm{ch}}$ maps to the center of $\mathcal{A}^{\mathrm{ch}}$.

Let $\mathcal{M}$ be an object of

$$\mathcal{A}\text{-mod}_{\underline{x}}^{\mathrm{fact}} = \mathcal{A}^{\mathrm{ch}}\text{-mod}_{\underline{x}}^{\mathrm{ch}}$$

equipped with a *commutative* action of $\mathfrak{z}^{\mathrm{ch}}$, which is compatible with the action of $\mathcal{A}^{\mathrm{ch}}$.

We claim that in this case $\mathrm{C}^{\mathrm{fact}}(X, \mathcal{A}, \mathcal{M})_{\underline{x}}$ carries an action of $\mathrm{C}^{\mathrm{fact}}(X, \mathfrak{z})$. Let us construct the action morphism.

We can interpret the given structure on $\mathcal{M}$ as a map of modules

$$(15.23) \quad \mathfrak{z}_x \otimes \mathcal{M} \rightarrow \mathcal{M}$$

compatible with a map of the chiral algebras

$$\mathfrak{z}^{\mathrm{ch}} \otimes \mathcal{A}^{\mathrm{ch}} \rightarrow \mathcal{A}^{\mathrm{ch}}.$$

By the functoriality of factorization homology, we obtain a map

$$\begin{aligned} \mathrm{C}^{\mathrm{fact}}(X, \mathfrak{z}) \otimes \mathrm{C}^{\mathrm{fact}}(X, \mathcal{A}, \mathcal{M})_{\underline{x}} &\simeq \mathrm{C}^{\mathrm{fact}}(X, \mathfrak{z}, \mathfrak{z}_x) \otimes \mathrm{C}^{\mathrm{fact}}(X, \mathcal{A}, \mathcal{M})_{\underline{x}} \xrightarrow{\sim} \\ &\rightarrow \mathrm{C}^{\mathrm{fact}}(X, \mathfrak{z} \otimes \mathcal{A}, \mathfrak{z}_x \otimes \mathcal{M})_{\underline{x}} \rightarrow \mathrm{C}^{\mathrm{fact}}(X, \mathcal{A}, \mathcal{M})_{\underline{x}}, \end{aligned}$$

which is the required action map.

The resulting action of $\mathrm{C}^{\mathrm{fact}}(X, \mathfrak{z})$ on $\mathrm{C}^{\mathrm{fact}}(X, \mathcal{A}, \mathcal{M})_{\underline{x}}$ has the following property:

The action of $\mathfrak{z}_x$ on $\mathrm{C}^{\mathrm{fact}}(X, \mathcal{A}, \mathcal{M})_{\underline{x}}$ obtained from the homomorphism

$$\mathfrak{z}_x \rightarrow \mathrm{C}^{\mathrm{fact}}(X, \mathfrak{z}, \mathfrak{z}_x) \simeq \mathrm{C}^{\mathrm{fact}}(X, \mathfrak{z})$$

equals the action obtained from the $\mathfrak{z}_x$ -action on $\mathcal{M}$ by endomorphisms of the chiral $\mathcal{A}^{\mathrm{ch}}$ -module structure.

It is a souped-up version of this construction that will be used in Sect. 15.6.8 below. See also Remark 15.6.15.

15.6.3. Write

$$\mathrm{Op}_{G,\underline{x}}^{\mathrm{mon-free}} \simeq \text{“colim”}_R \mathrm{Spec}(R),$$

where $Y := \mathrm{Spec}(R) \rightarrow \mathrm{Op}_{G,\underline{x}}^{\mathrm{mon-free}}$ are closed embeddings almost of finite presentation.

We claim:

Lemma 15.6.4. *For any $\mathrm{IndCoh}^!(\mathrm{Op}_{G,\underline{x}}^{\mathrm{mon-free}})$ -module category $\mathbf{C}$, the naturally defined functor*

$$\mathrm{colim}_R \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathrm{Op}_{G,\underline{x}}^{\mathrm{mon-free}})}(\mathrm{IndCoh}^!(Y), \mathbf{C}) \rightarrow \mathbf{C}$$

is an equivalence.

This lemma will be proved in Sect. 15.10.

15.6.5. For every $Y = \mathrm{Spec}(R)$ as above, denote

$$\mathrm{KL}(G)_{\mathrm{crit},\underline{x},Y} := \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathrm{Op}_{G,\underline{x}}^{\mathrm{mon-free}})}(\mathrm{IndCoh}^!(Y), \mathrm{KL}(G)_{\mathrm{crit},\underline{x}}).$$

We consider it as a $\mathrm{QCoh}(Y)$ -linear category via

$$\Upsilon_Y : \mathrm{QCoh}(Y) \rightarrow \mathrm{IndCoh}^!(Y).$$

15.6.6. Denote

$$Y^{\mathrm{glob}} := Y \times_{\mathrm{Op}_{G,\underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_G^{\mathrm{mon-free}}(X - \underline{x}).$$

Denote by

$$(15.24) \quad \mathrm{ev}_{\underline{x}} : Y^{\mathrm{glob}} \rightarrow Y$$

the evaluation map.

Denote

$$\mathrm{KL}(G)_{\mathrm{crit},\underline{x},Y}^{\mathrm{Op}_G^{\mathrm{glob}}} := \mathrm{KL}(G)_{\mathrm{crit},\underline{x},Y} \otimes_{\mathrm{QCoh}(Y)} \mathrm{QCoh}(Y^{\mathrm{glob}}).$$

Denote by

$$\mathrm{Id} \otimes \mathrm{ev}_{\underline{x}}^* : \mathrm{KL}(G)_{\mathrm{crit},\underline{x},Y} \rightarrow \mathrm{KL}(G)_{\mathrm{crit},\underline{x},Y}^{\mathrm{Op}_G^{\mathrm{glob}}}$$

the corresponding functor.

15.6.7. By Lemma 15.6.4, in order to construct the functor $\mathrm{Loc}_{G,\mathrm{crit},\underline{x}}^{\mathrm{Op}}$, it suffices to construct a compatible family of functors

$$\mathrm{Loc}_{G,\mathrm{crit},\underline{x},Y}^{\mathrm{Op}} : \mathrm{KL}(G)_{\mathrm{crit},\underline{x},Y}^{\mathrm{Op}_G^{\mathrm{glob}}} \rightarrow \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G),$$

such that:

(a) The functor

$$(15.25) \quad \mathrm{KL}(G)_{\mathrm{crit},\underline{x},Y} \rightarrow \mathrm{KL}(G)_{\mathrm{crit},\underline{x}} \xrightarrow{\mathrm{Loc}_{G,\kappa,\underline{x}}} \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$$

factors as

$$(15.26) \quad \mathrm{KL}(G)_{\mathrm{crit},\underline{x},Y} \xrightarrow{\mathrm{Id} \otimes \mathrm{ev}_{\underline{x}}^*} \mathrm{KL}(G)_{\mathrm{crit},\underline{x},Y}^{\mathrm{Op}_G^{\mathrm{glob}}} \xrightarrow{\mathrm{Loc}_{G,\mathrm{crit},\underline{x},Y}^{\mathrm{Op}}} \mathrm{D-mod}_{\mathrm{crit}}(\mathrm{Bun}_G);$$

(b) The functor $\mathrm{Loc}_{G,\mathrm{crit},\underline{x},Y}^{\mathrm{Op}}$ is $\mathrm{Rep}(\check{G})_{\underline{x}}$ -linear.

15.6.8. Let R^{glob} denote the algebra of functions on the (affine) scheme Y^{glob} . The closed embedding (15.24) gives rise to a homomorphism

$$(15.27) \quad R \rightarrow R^{\text{glob}}.$$

The datum of $\text{Loc}_{G, \text{crit}, \underline{x}, Y}^{\text{Op}}$ together with the factorization in point (a) above is equivalent to the datum of action of R^{glob} on the functor (15.25), such that the action of R obtained by precomposing with (15.27) is identified with the action coming from the action of R on the identity endofunctor of $\text{KL}(G)_{\text{crit}, \underline{x}, Y}$.

15.6.9. By unitality, we can rewrite (15.25) as

$$(15.28) \quad \text{KL}(G)_{\text{crit}, \underline{x}, Y} \rightarrow \text{KL}(G)_{\text{crit}, \underline{x}} \xrightarrow{\text{ins.vac}_{\underline{x}}} \text{KL}(G)_{\text{crit}, \text{Ran}_{\underline{x}}} \xrightarrow{(\text{Loc}_G, \kappa)_{\text{Ran}_{\underline{x}}}} \\ \rightarrow \text{D-mod}_{\text{crit}}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}_{\underline{x}}) \xrightarrow{\text{Id} \otimes C_c^*(\text{Ran}_{\underline{x}}, -)} \text{D-mod}_{\text{crit}}(\text{Bun}_G).$$

Let $\mathcal{O}_{\text{Op}_G^{\text{reg}, Y}}$ be the object in $\text{ComAlg}(\mathcal{O}_{\text{Op}_G^{\text{reg}, Y}}\text{-mod}_{\text{fact}}^{\text{fact}})$ from Sect. F.2.6. Let $(\mathcal{O}_{\text{Op}_G^{\text{reg}, Y}})_{\text{Ran}_{\underline{x}}}$ be the underlying object in $\text{ComAlg}(\text{D-mod}(\text{Ran}_{\underline{x}}))$.

Let us view $\text{KL}(G)_{\text{crit}, \text{Ran}_{\underline{x}}}$ as a category tensored over $\text{D-mod}(\text{Ran}_{\underline{x}})$. In particular, it makes sense to talk about algebras in $\text{D-mod}(\text{Ran}_{\underline{x}})$ acting on objects in $\text{KL}(G)_{\text{crit}, \text{Ran}_{\underline{x}}}$ or on functors with values in $\text{KL}(G)_{\text{crit}, \text{Ran}_{\underline{x}}}$.

We will show that $(\mathcal{O}_{\text{Op}_G^{\text{reg}, Y}})_{\text{Ran}_{\underline{x}}}$ acts on the functor

$$(15.29) \quad \text{KL}(G)_{\text{crit}, \underline{x}, Y} \rightarrow \text{KL}(G)_{\text{crit}, \underline{x}} \xrightarrow{\text{ins.vac}_{\underline{x}}} \text{KL}(G)_{\text{crit}, \text{Ran}_{\underline{x}}}.$$

Moreover, when we evaluate the natural transformation

$$(\text{diag}_{\underline{x}})_! \simeq (\text{diag}_{\underline{x}})_! \circ (\text{diag}_{\underline{x}})^! \circ \text{ins.vac}_{\underline{x}} \rightarrow \text{ins.vac}_{\underline{x}}$$

on $\text{KL}(G)_{\text{crit}, \underline{x}, Y}$, the action of $(\mathcal{O}_{\text{Op}_G^{\text{reg}, Y}})_{\text{Ran}_{\underline{x}}}$ on the right-hand side will be compatible with the action of

$$(15.30) \quad R \simeq (\text{diag}_{\underline{x}})^!((\mathcal{O}_{\text{Op}_G^{\text{reg}, Y}})_{\text{Ran}_{\underline{x}}})$$

on the left-hand side.

15.6.10. Assume for a moment the existence of such an action of $(\mathcal{O}_{\text{Op}_G^{\text{reg}, Y}})_{\text{Ran}_{\underline{x}}}$ on (15.29), and let us produce from it an action of R^{glob} on (15.25).

First, by functoriality, we obtain an action of $(\mathcal{O}_{\text{Op}_G^{\text{reg}, Y}})_{\text{Ran}_{\underline{x}}}$ on the functor

$$(15.31) \quad \text{KL}(G)_{\text{crit}, \underline{x}, Y} \rightarrow \text{KL}(G)_{\text{crit}, \underline{x}} \xrightarrow{\text{ins.vac}_{\underline{x}}} \text{KL}(G)_{\text{crit}, \text{Ran}_{\underline{x}}} \xrightarrow{(\text{Loc}_G, \kappa)_{\text{Ran}_{\underline{x}}}} \\ \rightarrow \text{D-mod}_{\text{crit}}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}_{\underline{x}}).$$

15.6.11. Note that $(\mathcal{O}_{\text{Op}_G^{\text{reg}, Y}})_{\text{Ran}_{\underline{x}}} \in \text{D-mod}(\text{Ran}_{\underline{x}})$ belongs to the essential image of the restriction functor

$$\mathfrak{t}^! : \text{D-mod}(\text{Ran}_{\underline{x}}^{\text{untl}}) \rightarrow \text{D-mod}(\text{Ran}_{\underline{x}}).$$

In particular, it belongs to the subcategory

$$\text{D-mod}(\text{Ran}_{\underline{x}})^{\text{almost-untl}} \subset \text{D-mod}(\text{Ran}_{\underline{x}})$$

(see Sect. C.5.15).

In particular, $C_c(\text{Ran}_{\underline{x}}, (\mathcal{O}_{\text{Op}_G^{\text{reg}, Y}})_{\text{Ran}_{\underline{x}}})$ acquires a structure of (commutative) algebra.

15.6.12. The essential image of

$$\mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \xrightarrow{\mathrm{ins. vac}_{\underline{x}}} \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}_{\underline{x}}}$$

belongs to the essential image of

$$\mathbf{t}^! : \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}_{\underline{x}}^{\mathrm{untl}}} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}_{\underline{x}}}.$$

Hence, the essential image of (15.31) belongs to the essential image of

$$(\mathrm{Id} \otimes \mathbf{t}^!) : \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}_{\underline{x}}^{\mathrm{untl}}) \rightarrow \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}_{\underline{x}}).$$

In particular, it belongs to the subcategory

$$\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}_{\underline{x}})^{\mathrm{almost-untl}} \subset \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}_{\underline{x}}).$$

15.6.13. Hence, by Sect. C.5.15, the action of $(\mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}, Y}})_{\mathrm{Ran}_{\underline{x}}}$ on (15.31) gives rise to an action of the commutative algebra

$$C_c(\mathrm{Ran}_{\underline{x}}, (\mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}, Y}})_{\mathrm{Ran}_{\underline{x}}})$$

on the functor (15.28).

15.6.14. Finally, by Lemmas F.2.8 and F.3.5, we identify

$$C_c(\mathrm{Ran}_{\underline{x}}, (\mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}, Y}})_{\mathrm{Ran}_{\underline{x}}}) \simeq C_c^{\mathrm{fact}}(X, \mathcal{O}_{\mathrm{Op}^{\mathrm{reg}}}, R) \simeq R^{\mathrm{glob}}$$

as (commutative) algebras.

This gives the sought-for action of R^{glob} on (15.25). The compatibility with the R -action on the identity endofunctor of $\mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y}$ follows from the fact that the (iso)morphism

$$\begin{aligned} \mathrm{Loc}_{G, \kappa, \underline{x}} &\simeq \mathrm{Loc}_{G, \kappa, \mathrm{Ran}_{\underline{x}}} \circ (\mathrm{diag}_{\underline{x}})_! \simeq \\ &\simeq \mathrm{Loc}_{G, \kappa, \mathrm{Ran}_{\underline{x}}} \circ (\mathrm{diag}_{\underline{x}})_! \circ (\mathrm{diag}_{\underline{x}})^! \circ \mathrm{ins. vac}_{\underline{x}} \rightarrow \mathrm{Loc}_{G, \kappa, \mathrm{Ran}_{\underline{x}}} \circ \mathrm{ins. vac}_{\underline{x}} \end{aligned}$$

intertwines the R -action on the left-hand side and the $C_c^{\mathrm{fact}}(X, \mathcal{O}_{\mathrm{Op}^{\mathrm{reg}}}, R)$ -action on the right-hand side.

Remark 15.6.15. A simplified version of the above argument proves the following statement:

Let $\mathcal{A}$ be a commutative factorization algebra. Consider the factorization category $\mathcal{A}\text{-mod}^{\mathrm{com}}$ as a unital sheaf of categories on Ran . Consider the local-to-global functor

$$\mathcal{A}\text{-mod}^{\mathrm{com}} \rightarrow \underline{\mathrm{D}\text{-mod}}(\mathrm{Ran}^{\mathrm{untl}})$$

given by

$$(\mathcal{Z} \rightarrow \mathrm{Ran}) \rightsquigarrow C_c^{\mathrm{fact}}(X, \mathcal{A}, -)_{\mathcal{Z}}.$$

Then:

(a) The above functor is acted on by $C_c^{\mathrm{fact}}(X, \mathcal{A})$; in particular, upgrades to a (strictly unital) local-to-global functor

$$(15.32) \quad \mathcal{A}\text{-mod}^{\mathrm{com}} \rightarrow C_c^{\mathrm{fact}}(X, \mathcal{A})\text{-mod} \otimes \underline{\mathrm{D}\text{-mod}}(\mathrm{Ran}^{\mathrm{untl}}).$$

(b) The functor (15.32) is *universal* among strictly unital local-to-global functors.

In the language of Sect. H.1, point (b) can reformulated as saying that (15.32) induces an equivalence

$$\mathcal{A}\text{-mod}_{\mathrm{Ran}, \mathrm{indep}}^{\mathrm{com}} \rightarrow C_c^{\mathrm{fact}}(X, \mathcal{A})\text{-mod}.$$

15.7. Construction of the algebra action. In this subsection we will construct the sought-for action of $(\mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}, Y}})_{\mathrm{Ran}_{\underline{x}}}$ on the functor (15.29).

15.7.1. By Theorem 15.2.8, the functor (15.29) factors as

$$(15.33) \quad \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \xrightarrow{\mathrm{Id} \otimes (\mathrm{pr}_{\mathrm{small}}^{\mathrm{Op}})^*} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \mathrm{Ran} \underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}) \xrightarrow{\mathrm{ins. vac} \underline{x}} \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran} \underline{x}},$$

while the functor

$$\mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \xrightarrow{\mathrm{Id} \otimes (\mathrm{pr}_{\mathrm{small}}^{\mathrm{Op}})^*} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \mathrm{Ran} \underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})$$

which appears in (15.33), factors naturally as

$$\begin{aligned} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} &\rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \mathrm{Ran} \underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}) \rightarrow \\ &\rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \mathrm{Ran} \underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}). \end{aligned}$$

15.7.2. We rewrite $\mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \mathrm{Ran} \underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})$ tautologically as

$$\mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \otimes_{\mathrm{QCoh}(Y)} \left(\mathrm{QCoh}(Y) \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \mathrm{Ran} \underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}) \right).$$

We now claim:

Lemma 15.7.3. *The naturally defined functor*

$$\mathrm{QCoh}(Y) \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \mathrm{Ran} \underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}) \rightarrow \mathrm{QCoh}\left(Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran} \underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}\right)$$

is an equivalence.

The lemma will be proved in Sect. 15.10.

15.7.4. Applying Lemma 15.7.3, we obtain that the functor (15.29) can be factored as

$$(15.34) \quad \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \otimes_{\mathrm{QCoh}(Y)} \mathrm{QCoh}\left(Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran} \underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}\right) \rightarrow \\ \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \mathrm{Ran} \underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}) \xrightarrow{\mathrm{ins. vac} \underline{x}} \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran} \underline{x}}.$$

Hence, it is enough to construct an action of $(\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}, Y}})_{\mathrm{Ran} \underline{x}}$ on the composition

$$(15.35) \quad \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \otimes_{\mathrm{QCoh}(Y)} \mathrm{QCoh}\left(Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran} \underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}\right) \rightarrow \\ \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \mathrm{Ran} \underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}) \xrightarrow{\mathrm{ins. vac} \underline{x}} \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran} \underline{x}}.$$

15.7.5. Consider the category

$$(15.36) \quad \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \otimes_{\mathrm{QCoh}(Y)} \mathrm{QCoh}\left(Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}\right)$$

as tensored over

$$(15.37) \quad \mathrm{QCoh}\left(Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}\right)$$

and hence over $\mathrm{D-mod}(\mathrm{Ran}_{\underline{x}})$.

We note that, according to Lemma F.3.5,

$$Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \simeq \mathrm{Spec}_{\mathrm{Ran}_{\underline{x}}}((\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}, Y}})_{\mathrm{Ran}_{\underline{x}}}).$$

Hence, $(\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}, Y}})_{\mathrm{Ran}_{\underline{x}}}$ maps (isomorphically) to endomorphisms of the monoidal unit

$$\mathcal{O}_Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \in \mathrm{QCoh}\left(Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}\right).$$

Hence, it acts by endomorphisms of the identity functor on (15.36).

15.7.6. The functors in the composition (15.35) are $\mathrm{D-mod}(\mathrm{Ran}_{\underline{x}})$ -linear. This produces the sought-for action of $(\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}, Y}})_{\mathrm{Ran}_{\underline{x}}}$ on (15.35).

The compatibility of this action with (15.30) follows from the construction.

15.8. Verification of the Hecke property: reduction to a local statement. The goal of this and the next subsections is to verify property (b) from Sect. 15.6.7.

For expositional reasons, we will fix an object $V \in \mathrm{Rep}(\check{G})_{\underline{x}}$ and show that the functor $\mathrm{Loc}_{G, \mathrm{crit}, \underline{x}, Y}^{\mathrm{Op}}$ intertwines the actions of V on the two sides.

We will reduce the local-to-global assertion we are after to a purely local one, namely, (15.42).

15.8.1. Let $\mathcal{V}$ denote the object of $\mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})$ equal to $(\mathfrak{r}^{\mathrm{reg}})^*(V)$, where we identify

$$\mathrm{Rep}(\check{G})_{\underline{x}} \simeq \mathrm{QCoh}(\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{reg}}).$$

Let

$$\mathcal{V}^{\mathrm{glob}} := (\mathfrak{t}^{\mathrm{glob}})^* \circ \mathrm{Loc}_{\check{G}, \underline{x}}^{\mathrm{spec}}(V) \in \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(X - \underline{x})).$$

Note that we have

$$\mathcal{V}^{\mathrm{glob}} \simeq \mathrm{ev}_{\underline{x}}^*(\mathcal{V}),$$

where

$$\mathrm{ev}_{\underline{x}} : \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(X - \underline{x}) \rightarrow \mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}.$$

Denote by

$$\mathcal{V}_Y \text{ and } \mathcal{V}_Y^{\mathrm{glob}}$$

the restrictions of $\mathcal{V}$ and $\mathcal{V}^{\mathrm{glob}}$ to

$$Y \hookrightarrow \mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}} \text{ and } Y^{\mathrm{glob}} \hookrightarrow \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(X - \underline{x}),$$

respectively.

By a slight abuse of notation, we will denote by the same symbols $\mathcal{V}_Y$ and $\mathcal{V}_Y^{\mathrm{glob}}$ global sections of the corresponding vector bundles, viewed as modules over R and R^{glob} , respectively.

15.8.2. Let $\mathcal{M}$ be an object of $\mathrm{KL}(G)_{\kappa, \underline{x}, Y}$, and consider the object

$$\mathrm{Loc}_{G, \kappa, \underline{x}}(\mathcal{M}) \in \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G).$$

The construction in Sects. 15.6.10-15.6.14 endows $\mathrm{Loc}_{G, \kappa, \underline{x}}(\mathcal{M})$ with an action of R^{glob} .

Let

$$H_V : \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$$

be the Hecke endofunctor corresponding to V .

On the one hand, by functoriality, the object

$$H_V(\mathrm{Loc}_{G, \kappa, \underline{x}}(\mathcal{M})) \in \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$$

acquires an action of R^{glob} .

On the other hand, we can consider

$$\mathcal{V}_Y^{\mathrm{glob}} \otimes_{R^{\mathrm{glob}}} \mathrm{Loc}_{G, \kappa, \underline{x}}(\mathcal{M}) \in R^{\mathrm{glob}}\text{-mod}(\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)).$$

15.8.3. The statement of (b) in Sect. 15.6.7 is that we have a canonical isomorphism

$$(15.38) \quad H_V(\mathrm{Loc}_{G, \kappa, \underline{x}}(\mathcal{M})) \simeq \mathcal{V}_Y^{\mathrm{glob}} \otimes_{R^{\mathrm{glob}}} \mathrm{Loc}_{G, \kappa, \underline{x}}(\mathcal{M})$$

as objects of $R^{\mathrm{glob}}\text{-mod}(\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G))$.

Thus, our goal is to establish (15.38).

15.8.4. First, we rewrite the right-hand side in (15.38). Namely,

$$\mathcal{V}_Y^{\mathrm{glob}} \otimes_{R^{\mathrm{glob}}} \mathrm{Loc}_{G, \kappa, \underline{x}}(\mathcal{M}) \simeq \mathrm{Loc}_{G, \kappa, \underline{x}}(\mathcal{V}_Y \otimes_R \mathcal{M}),$$

where:

- We regard $\mathcal{V}_Y \otimes_R \mathcal{M}$ as an object $\mathrm{KL}(G)_{\kappa, \underline{x}, Y}$;
- $\mathrm{Loc}_{G, \kappa, \underline{x}}(-)$ acquires an action of R^{glob} via the construction in Sects. 15.6.10-15.6.14.

15.8.5. We will now rewrite the left-hand side in (15.38).

Consider the category $\mathrm{KL}(G)_{\kappa, \mathrm{Ran}_{\underline{x}}}$. It carries an action of $\mathrm{Sph}_{G, \mathrm{Ran}_{\underline{x}}}$. We will denote the action functor by

$$\mathcal{F}, \mathcal{M}' \mapsto \mathcal{F} \cdot \mathcal{M}'.$$

The unital structure on Sph_G gives rise to a (monoidal) functor

$$\mathrm{ins. unit}_{\underline{x}} : \mathrm{Sph}_{G, \underline{x}} \rightarrow \mathrm{Sph}_{G, \mathrm{Ran}_{\underline{x}}}^{\dagger}$$

(see Sect. H.5.5 for the notation).

In particular, we obtain a monoidal action of $\mathrm{Sph}_{G, \underline{x}}$ on $\mathrm{KL}(G)_{\kappa, \mathrm{Ran}_{\underline{x}}}$.

15.8.6. Recall that the object

$$\mathrm{ins. vac}_{\underline{x}}(\mathcal{M}) \in \mathrm{KL}(G)_{\kappa, \mathrm{Ran}_{\underline{x}}}$$

carries an action of the algebra object $(\mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}, Y}})_{\mathrm{Ran}_{\underline{x}}} \in \mathrm{D}\text{-mod}(\mathrm{Ran}_{\underline{x}})$. By functoriality, we obtain that

$$(15.39) \quad \mathrm{ins. unit}_{\underline{x}}(\mathrm{Sat}_G^{\mathrm{nv}}(V)) \cdot \mathrm{ins. vac}_{\underline{x}}(\mathcal{M}) \in \mathrm{KL}(G)_{\kappa, \mathrm{Ran}_{\underline{x}}}$$

also carries an action of $(\mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}, Y}})_{\mathrm{Ran}_{\underline{x}}}$.

Further, the object

$$(15.40) \quad \mathrm{Loc}_{G, \mathrm{crit}, \mathrm{Ran}_{\underline{x}}}(\mathrm{ins. unit}_{\underline{x}}(\mathrm{Sat}_G^{\mathrm{nv}}(V)) \cdot \mathrm{ins. vac}_{\underline{x}}(\mathcal{M})) \in \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}_{\underline{x}})$$

also carries an action of $(\mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}, Y}})_{\mathrm{Ran}_{\underline{x}}}$.

15.8.7. The object (15.39) belongs to the essential image of the restriction functor

$$t^! : \mathrm{KL}(G)_{\kappa, \mathrm{Ran}_{\underline{x}}^{\mathrm{untl}}} \rightarrow \mathrm{KL}(G)_{\kappa, \mathrm{Ran}_{\underline{x}}}.$$

Hence, the object (15.40) belongs to the essential image of

$$(\mathrm{Id} \otimes t^!) : \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}_{\underline{x}}^{\mathrm{untl}}) \rightarrow \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}_{\underline{x}}).$$

In particular, it belongs to

$$\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}_{\underline{x}})^{\mathrm{almost-untl}} \subset \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}_{\underline{x}}).$$

Hence, by Sect. C.5.15, we obtain that the object

$$(15.41) \quad (\mathrm{Id} \otimes C_c^*(\mathrm{Ran}_{\underline{x}}, -)) \circ \mathrm{Loc}_{G, \mathrm{crit}, \mathrm{Ran}_{\underline{x}}} \left(\mathrm{ins. unit}_{\underline{x}}(\mathrm{Sat}_G^{\mathrm{nv}}(V)) \cdot \mathrm{ins. vac}_{\underline{x}}(\mathcal{M}) \right)$$

acquires an action of

$$C_c^*(\mathrm{Ran}_{\underline{x}}, (\mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}, Y}})_{\mathrm{Ran}_{\underline{x}}}) \simeq C_c^{\mathrm{fact}}(X, \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}}, R}) \simeq R^{\mathrm{glob}}.$$

15.8.8. Recall now that the functor

$$\mathrm{Loc}_{G, \mathrm{crit}, \mathrm{Ran}_{\underline{x}}} : \mathrm{KL}(G)_{\kappa, \mathrm{Ran}_{\underline{x}}} \rightarrow \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}_{\underline{x}})$$

is $\mathrm{Sph}_{G, \mathrm{Ran}_{\underline{x}}}^{\otimes}$ -linear.

Note also that the functor

$$\mathrm{Id} \otimes C_c^*(\mathrm{Ran}_{\underline{x}}, -) : \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}_{\underline{x}}) \rightarrow \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$$

is $\mathrm{Sph}_{G, \underline{x}}$ -linear, where $\mathrm{Sph}_{G, \underline{x}}$ acts on the left-hand side via $\mathrm{ins. unit}_{\underline{x}}$.

Combining, we obtain that the functor

$$(\mathrm{Id} \otimes C_c^*(\mathrm{Ran}_{\underline{x}}, -)) \circ \mathrm{Loc}_{G, \mathrm{crit}, \mathrm{Ran}_{\underline{x}}} : \mathrm{KL}(G)_{\kappa, \mathrm{Ran}_{\underline{x}}} \rightarrow \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)$$

is $\mathrm{Sph}_{G, \underline{x}}$ -linear.

15.8.9. Combining Sects. 15.8.6-15.8.7 with Sect. 15.8.8, we obtain that

$$H_V(\mathrm{Loc}_{G, \kappa, \underline{x}}(\mathcal{M})) \in R^{\mathrm{glob}}\text{-mod}(\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G))$$

identifies with the object (15.41), with the R^{glob} -action specified in 15.8.6-15.8.7.

15.8.10. Hence, we obtain that in order to prove (15.38), it suffices to establish an isomorphism

$$(15.42) \quad \mathrm{ins. unit}_{\underline{x}}(\mathrm{Sat}_G^{\mathrm{nv}}(V)) \cdot \mathrm{ins. vac}_{\underline{x}}(\mathcal{M}) \simeq \mathrm{ins. vac}_{\underline{x}}(\mathcal{V}_Y \otimes_R \mathcal{M})$$

as $(\mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}, Y}})_{\mathrm{Ran}_{\underline{x}}}$ -modules in $\mathrm{KL}(G)_{\kappa, \mathrm{Ran}_{\underline{x}}}$, where:

- We regard $\mathcal{V}_Y \otimes_R \mathcal{M}$ as an object $\mathrm{KL}(G)_{\kappa, \underline{x}, Y}$;
- $\mathrm{ins. vac}_{\underline{x}}(-)$ on each side acquires an action of $(\mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}, Y}})_{\mathrm{Ran}_{\underline{x}}}$ via the construction in Sect. 15.7.

Remark 15.8.11. As was mentioned in the preamble to this subsection, we fixed objects $\mathcal{M} \in \mathrm{KL}(G)_{\kappa, \underline{x}, Y}$ and $V \in \mathrm{Rep}(\check{G})_{\underline{x}}$ for expositional reasons. The actual assertion behind (15.42), and one that we actually prove in Sect. 15.9, is that that $\mathrm{ins. vac}_{\underline{x}}$, viewed as a functor

$$\mathrm{KL}(G)_{\kappa, \underline{x}, Y} \rightarrow ((\mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}, Y}})_{\mathrm{Ran}_{\underline{x}}})\text{-mod}(\mathrm{KL}(G)_{\kappa, \mathrm{Ran}_{\underline{x}}}),$$

is $\mathrm{Rep}(\check{G})_{\underline{x}}$ -linear.

15.9. Verification of the Hecke property at the local level. In this subsection we will construct the identification (15.42) and thereby complete the verification of point (b) in Sect. 15.6.7.

15.9.1. Let us apply Lemma 5.4.2. It implies that we can rewrite the left-hand side in (15.42) as

$$(15.43) \quad \mathfrak{r}^*(\text{ins. unit}_{\underline{x}}(V)) \otimes \text{ins. vac}_{\underline{x}}(\mathcal{M}),$$

where:

- $\text{ins. unit}_{\underline{x}} : \text{Rep}(\check{G})_{\underline{x}} \rightarrow \text{Rep}(\check{G})_{\text{Ran}_{\underline{x}}}$ is the unital structure on $\text{Rep}(\check{G})$;
- We identify $\text{Rep}(\check{G})_{\text{Ran}_{\underline{x}}} \simeq \text{QCoh}(\text{LS}_{\check{G}, \text{Ran}_{\underline{x}}}^{\text{reg}})$;
- $\mathfrak{r}$ denotes the map $\text{Op}_{\check{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free}} \rightarrow \text{LS}_{\check{G}, \text{Ran}_{\underline{x}}}^{\text{reg}}$;
- $\otimes$ refers to the action of $\text{QCoh}(\text{Op}_{\check{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free}})$ on $\text{KL}(G)_{\kappa, \text{Ran}_{\underline{x}}}$.

The action of $(\mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}, Y}})_{\text{Ran}_{\underline{x}}}$ on (15.43) is obtained by functoriality from the $(\mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}, Y}})_{\text{Ran}_{\underline{x}}}$ -action on $\text{ins. vac}_{\underline{x}}(\mathcal{M})$.

15.9.2. We now note that the linearity with respect to

$$(\text{pr}_{\text{small}})_* \circ (\text{pr}_{\text{big}})^*(\text{QCoh}(\text{Op}_{\check{G}}^{\text{mon-free}}))$$

in Theorem 15.2.8 implies that the functor

$$(15.44) \quad \text{KL}(G)_{\text{crit}, \underline{x}, Y} \otimes_{\text{QCoh}(Y)} \text{QCoh}\left(Y \times_{\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free}}} \text{Op}_{\check{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}}\right) \rightarrow \\ \rightarrow \text{KL}(G)_{\text{crit}, \underline{x}} \otimes_{\text{QCoh}(\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free}})} \text{QCoh}(\text{Op}_{\check{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}}) \xrightarrow{\text{ins. vac}_{\underline{x}}^{\text{mon-free} \rightsquigarrow \text{reg}}} \text{KL}(G)_{\text{crit}, \text{Ran}_{\underline{x}}}$$

of (15.35) is linear with respect to $\text{QCoh}(\text{Op}_{\check{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free}})$, which acts on

$$\text{KL}(G)_{\text{crit}, \underline{x}, Y} \otimes_{\text{QCoh}(Y)} \text{QCoh}\left(Y \times_{\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free}}} \text{Op}_{\check{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}}\right)$$

via the pullback along

$$Y \times_{\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free}}} \text{Op}_{\check{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}} \xrightarrow{p_2} \text{Op}_{\check{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}} \xrightarrow{\text{pr}_{\text{big}, \underline{x}}^{\text{Op}}} \text{Op}_{\check{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free}},$$

where $\text{pr}_{\text{big}, \underline{x}}^{\text{Op}}$ is the map from Sect. 15.4.1.

15.9.3. Hence, we can rewrite (15.43) as the value on $\mathcal{M} \in \text{KL}(G)_{\kappa, \underline{x}, Y}$ of the functor

$$(15.45) \quad \text{KL}(G)_{\text{crit}, \underline{x}, Y} \rightarrow \text{KL}(G)_{\text{crit}, \underline{x}, Y} \otimes_{\text{QCoh}(Y)} \text{QCoh}\left(Y \times_{\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free}}} \text{Op}_{\check{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}}\right) \rightarrow \\ \xrightarrow{\text{Id} \otimes (- \otimes (\text{ropr}_{\text{big}, \underline{x}}^{\text{Op}} \circ p_2)^*(\text{ins. unit}_{\underline{x}}(V)))} \text{KL}(G)_{\text{crit}, \underline{x}, Y} \otimes_{\text{QCoh}(Y)} \text{QCoh}\left(Y \times_{\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free}}} \text{Op}_{\check{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}}\right) \rightarrow \\ \xrightarrow{(15.35)} \text{KL}(G)_{\kappa, \text{Ran}_{\underline{x}}}.$$

In terms of this identification, the action of $(\mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}, Y}})_{\text{Ran}_{\underline{x}}}$ on (15.43) is obtained from the action of $(\mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}, Y}})_{\text{Ran}_{\underline{x}}}$ on the functor (15.44) (which is the same as (15.35)) from Sect. 15.7.6.

15.9.4. Thus, we obtain that in order to prove (15.42), it suffices to establish an isomorphism between the value on $\mathcal{M}$ of the functor

$$(15.46) \quad \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \otimes_{\mathrm{IndCoh}^!(Y)} \mathrm{QCoh}\left(Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}\right) \rightarrow$$

$$\mathrm{Id} \otimes (- \otimes (\mathrm{r} \circ \mathrm{pr}_{\mathrm{big}, \underline{x}}^{\mathrm{Op}} \circ p_2)^* (\mathrm{ins. unit}_{\underline{x}}(V))) \longrightarrow \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \otimes_{\mathrm{QCoh}(Y)} \mathrm{QCoh}\left(Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}\right)$$

and the value on $\mathcal{V}_Y \otimes_R \mathcal{M}$ of the functor

$$(15.47) \quad \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \otimes_{\mathrm{QCoh}(Y)} \mathrm{QCoh}\left(Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}\right).$$

15.9.5. In order to do that, it suffices to establish an isomorphism between the following vector bundles on

$$Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} :$$

- The pullback of $\mathrm{ins. unit}_{\underline{x}}(V) \in \mathrm{Rep}(\check{G})_{\mathrm{Ran}_{\underline{x}}} \simeq \mathrm{QCoh}(\mathrm{LS}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{reg}})$ along

$$Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \xrightarrow{p_2} \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \xrightarrow{\mathrm{pr}_{\mathrm{big}, \underline{x}}^{\mathrm{Op}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free}} \xrightarrow{\mathrm{r}} \mathrm{LS}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{reg}} ;$$

- The pullback of $\mathcal{V}_Y$ along

$$Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \rightarrow Y.$$

15.9.6. Note that for $(\underline{x} \subseteq \underline{x}') \in \mathrm{Ran}_{\underline{x}}$, restriction along

$$\mathcal{D}_{\underline{x}} \hookrightarrow \mathcal{D}_{\underline{x}'}$$

gives rise to a map

$$(15.48) \quad \mathrm{LS}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{reg}} \rightarrow \mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{reg}},$$

so that

$$\mathrm{ins. unit}_{\underline{x}} \simeq (15.48)^*.$$

The required isomorphism of vector bundles follows from the commutative diagram

$$\begin{array}{ccccc} Y \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} & \xrightarrow{\mathrm{id} \times p_1} & Y & \longrightarrow & \mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}} \\ p_2 \downarrow & & & & \downarrow \mathrm{r} \\ \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} & & & & \\ \mathrm{pr}_{\mathrm{big}, \underline{x}}^{\mathrm{Op}} \downarrow & & & & \\ \mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free}} & \xrightarrow{\mathrm{r}} & \mathrm{LS}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{reg}} & \xrightarrow{(15.48)} & \mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{reg}}. \end{array}$$

15.10. Proofs of Lemmas 15.6.4 and 15.7.3.

15.10.1. We first prove Lemma 15.6.4. In fact, the assertion holds for $\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}(\mathcal{D}_{\underline{x}}^{\times})$ replaced by an arbitrary ind-placid ind-scheme Z .

Namely, for $Y_1 \xrightarrow{\iota_{1,2}} Y_2$, the functor

$$\mathrm{Funct}_{\mathrm{IndCoh}^!(Z)}(\mathrm{IndCoh}^!(Y_1), \mathbf{C}) \rightarrow \mathrm{Funct}_{\mathrm{IndCoh}^!(Z)}(\mathrm{IndCoh}^!(Y_2), \mathbf{C})$$

admits a right adjoint, given by precomposition with $(\iota_{1,2})_*^{\mathrm{IndCoh}}$.

Hence, we can rewrite the colimit

$$\mathrm{colim}_Y \mathrm{Funct}_{\mathrm{IndCoh}^!(Z)}(\mathrm{IndCoh}^!(Y), \mathbf{C})$$

as a limit with respect to the above right adjoint functors.

The latter limit is the same as

$$\mathrm{Funct}_{\mathrm{IndCoh}^!(Z)}\left(\mathrm{colim}_Y \mathrm{IndCoh}^!(Y), \mathbf{C}\right).$$

We now use the fact that the functor

$$\mathrm{colim}_Y \mathrm{IndCoh}^!(Y) \rightarrow \mathrm{IndCoh}(Z)$$

is an equivalence.

□[Lemma 15.6.4]

15.10.2. The rest of this subsection is devoted to the proof of Lemma 15.7.3. Consider the Cartesian diagram

$$\begin{array}{ccc} \mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} & \longrightarrow & \mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}} \\ \mathrm{pr}_{\mathrm{small}, \underline{x}}^{\mathrm{Op}} \downarrow & & \downarrow \mathrm{pr}_{\mathrm{small}, \underline{x}}^{\mathrm{Op}} \\ \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}} & \xrightarrow{\iota^{\mathrm{mon-free}}} & \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}}, \end{array}$$

where $\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}$ is as in Sect. F.3.3.

15.10.3. We will prove:

Lemma 15.10.4.

- (a) *The category $\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}})$ is dualizable as a $\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}})$ -module.*
- (b) *For any affine $Y \rightarrow \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}}$, the functor*

$$\mathrm{QCoh}(Y) \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}})} \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}) \rightarrow \mathrm{QCoh}(Y \times_{\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}}} \mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}})$$

is an equivalence.

It is easy to see that Lemma 15.10.4 implies (15.7.3) by passage to the limit.

Thus, the rest of this subsection is devoted to the proof of Lemma 15.10.4.

15.10.5. First, a standard limit-colimit procedure reduces the assertion to the case when we replace $\mathrm{Ran}_{\underline{x}}$ by

$$\mathcal{Z} := (X_{\underline{x}}^I)_{\mathrm{dR}},$$

where:

- We think of $\underline{x}$ as a point of X^J for some finite set J ;
- I is a finite set with a map $J \rightarrow I$;
- $X_{\underline{x}}^I := X^I \times_{X^J, \underline{x}} \mathrm{pt.}$

Further, we can assume that X is affine and admits an étale map to $\mathbb{A}^1$.

Second, we can replace the D-scheme $\mathrm{Op}_{\tilde{G}}$ by $\mathrm{Jets}(\mathcal{E})$, where $\mathcal{E}$ is the total space of a vector bundle on X (see Sect. 3.1.7).

15.10.6. Note that for X as above, we have (non-canonical) isomorphisms

$$\mathfrak{L}_Z^{\text{mer} \rightsquigarrow \text{reg}} \simeq \mathfrak{L}^+(\mathcal{E})_Z \times (\mathfrak{L}(\mathcal{E})_{\underline{x}} / \mathfrak{L}^+(\mathcal{E})_{\underline{x}})$$

and

$$\mathfrak{L}(\mathcal{E})_{\underline{x}} \simeq \mathfrak{L}^+(\mathcal{E})_{\underline{x}} \times (\mathfrak{L}(\mathcal{E})_{\underline{x}} / \mathfrak{L}^+(\mathcal{E})_{\underline{x}}),$$

so that the projection

$$\mathfrak{L}_{X^I}^{\text{mer} \rightsquigarrow \text{reg}} \rightarrow \mathfrak{L}(\mathcal{E})_{\underline{x}}$$

corresponds to the projection

$$\mathfrak{L}^+(\mathcal{E})_Z \rightarrow \mathfrak{L}^+(\mathcal{E})_{\underline{x}}.$$

We can therefore identify

$$\text{QCoh}(\mathfrak{L}_Z^{\text{mer} \rightsquigarrow \text{reg}}) \simeq \text{QCoh}(\mathfrak{L}^+(\mathcal{E})_Z) \otimes \text{QCoh}(\mathfrak{L}(\mathcal{E})_{\underline{x}} / \mathfrak{L}^+(\mathcal{E})_{\underline{x}})$$

and

$$\text{QCoh}(\mathfrak{L}(\mathcal{E})_{\underline{x}}) \simeq \text{QCoh}(\mathfrak{L}^+(\mathcal{E})_{\underline{x}}) \otimes \text{QCoh}(\mathfrak{L}(\mathcal{E})_{\underline{x}} / \mathfrak{L}^+(\mathcal{E})_{\underline{x}}).$$

15.10.7. To prove point (a), it suffices to show that $\text{QCoh}(\mathfrak{L}^+(\mathcal{E})_Z)$ is dualizable as a module over $\text{QCoh}(\mathfrak{L}(\mathcal{E})_{\underline{x}})$. However, this is obvious, since both geometric objects are affine schemes.

15.10.8. To prove point (b), it is sufficient to do so for a cofinal family of Y 's. Hence, we can assume that Y is invariant under translations with respect to $\mathfrak{L}^+(\mathcal{E})_{\underline{x}}$. Hence, we can identify

$$Y \times_{\mathfrak{L}(\mathcal{E})_{\underline{x}}} \mathfrak{L}_Z^{\text{mer} \rightsquigarrow \text{reg}} \simeq \mathfrak{L}^+(\mathcal{E})_{\underline{x}} \times (Y / \mathfrak{L}^+(\mathcal{E})_{\underline{x}}).$$

We have

$$\text{QCoh}(Y \times_{\mathfrak{L}(\mathcal{E})_{\underline{x}}} \mathfrak{L}_Z^{\text{mer} \rightsquigarrow \text{reg}}) \simeq \text{QCoh}(\mathfrak{L}^+(\mathcal{E})_Z) \otimes \text{QCoh}(Y / \mathfrak{L}^+(\mathcal{E})_{\underline{x}}),$$

which makes the assertion of point (b) manifest.

□[Lemma 15.10.4]

16. PROOF OF THEOREM 15.2.8

16.1. Reformulation and strategy.

16.1.1. For expositional purposes we will let $Z = \text{pt}$, so that $Z \rightarrow \text{Ran}$ corresponds to $\underline{x} \in \text{Ran}$.

Hence, our goal is to construct a functor

$$\text{QCoh}(\text{Op}_{\tilde{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}}) \otimes_{\text{QCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}})} \text{KL}(G)_{\text{crit}, \underline{x}} \xrightarrow{\text{ins. vac}_{\underline{x}}^{\text{mon-free} \rightsquigarrow \text{reg}}} \text{KL}(G)_{\kappa, \text{Ran}_{\underline{x}}}$$

such that

$$\text{ins. vac}_{\underline{x}} : \text{KL}(G)_{\text{crit}, \underline{x}} \rightarrow \text{KL}(G)_{\kappa, \text{Ran}_{\underline{x}}}$$

factors as

$$(16.1) \quad \text{KL}(G)_{\text{crit}, \underline{x}} \xrightarrow{(\text{pr}_{\text{small}, \underline{x}}^{\text{Op}})^* \otimes \text{Id}} \text{QCoh}(\text{Op}_{\tilde{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}}) \otimes_{\text{QCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}})} \text{KL}(G)_{\text{crit}, \underline{x}} \xrightarrow{\text{ins. vac}_{\underline{x}}^{\text{mon-free} \rightsquigarrow \text{reg}}} \text{KL}(G)_{\kappa, \text{Ran}_{\underline{x}}},$$

and such that the functor $\text{ins. vac}_{\underline{x}}^{\text{mon-free} \rightsquigarrow \text{reg}}$ is $\text{QCoh}(\text{Op}_{\tilde{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free}})$ -linear via

$$(\text{pr}_{\text{big}, \underline{x}}^{\text{Op}})^* : \text{QCoh}(\text{Op}_{\tilde{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free}}) \rightarrow \text{QCoh}(\text{Op}_{\tilde{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}}).$$

16.1.2. Precomposing with the functor

$$(16.2) \quad \Upsilon_{\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}} : \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}),$$

we obtain that it suffices to carry out the construction in Sect. 16.1.1 above for $\mathrm{QCoh}(-)$ replaced⁵⁹ by $\mathrm{IndCoh}^!(-)$.

I.e., from now on our goal will be to construct a functor

$$(16.3) \quad \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \xrightarrow{\mathrm{ins.vac}_{\underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}} \mathrm{KL}(G)_{\kappa, \mathrm{Ran}_{\underline{x}}}$$

such that $\mathrm{ins.vac}_{\underline{x}}$ factors as

$$(16.4) \quad \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \xrightarrow{(\mathrm{pr}_{\mathrm{small}, \underline{x}}^{\mathrm{Op}})^! \otimes \mathrm{Id}} \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \xrightarrow{\mathrm{ins.vac}_{\underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}} \mathrm{KL}(G)_{\kappa, \mathrm{Ran}_{\underline{x}}},$$

and such that the functor $\mathrm{ins.vac}_{\underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}$ is $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free}})$ -linear via

$$(\mathrm{pr}_{\mathrm{big}, \underline{x}}^{\mathrm{Op}})^! : \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}).$$

16.1.3. Consider the factorization functor

$$\mathrm{Vac}(G)_{\mathrm{crit}} : \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{reg}}) \simeq \mathfrak{z}\text{-mod}^{\mathrm{cl}} \rightarrow \mathrm{KL}(G)_{\kappa},$$

given by

$$\mathcal{F} \mapsto \mathcal{F} \otimes_{\mathfrak{z}} \mathrm{Vac}(G)_{\mathrm{crit}}.$$

16.1.4. Recall that according to the conventions in Sect. B.12.4, for a factorization category $\mathbf{A}$, we denote by $\mathbf{A}^{\mathrm{fact}_{\underline{x}}}$ the vacuum object of $\mathbf{A}\text{-mod}_{\underline{x}}^{\mathrm{fact}}$, i.e., the object whose underlying category is $\mathbf{A}_{\underline{x}}$.

Thus, we can consider

$$(\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}} \in \mathrm{KL}(G)_{\kappa}\text{-mod}_{\underline{x}}^{\mathrm{fact}}.$$

Consider the object

$$\mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}}) \in \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{reg}})\text{-mod}_{\underline{x}}^{\mathrm{fact}}.$$

16.1.5. *Example.* For $\underline{x} \sqcup \underline{x}'' = \underline{x}' \in \mathrm{Ran}_{\underline{x}'}$, the fiber of $\mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}})$ at $\underline{x}'$ is

$$\mathrm{KL}(G)_{\kappa, \underline{x}} \otimes \mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}''}^{\mathrm{reg}}).$$

16.1.6. Consider the assignment

$$(\mathcal{Z} \rightarrow \mathrm{Ran}_{\underline{x}}) \mapsto \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \times_{\mathrm{Ran}_{\underline{x}}} \mathcal{Z}_{\mathrm{dR}})$$

as a crystal of (symmetric) monoidal categories over $\mathrm{Ran}_{\underline{x}}$, denote it by $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}}$.

Note that it extends naturally to a crystal of (symmetric) monoidal categories over $\mathrm{Ran}_{\underline{x}}^{\mathrm{unfl}}$, in which the monoidal structure is given by *strict* functors between sheaves of categories.

16.1.7. The key step will be to construct an action of

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}}$$

on

$$\mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}})$$

(as crystals of categories over $\mathrm{Ran}_{\underline{x}}$). Furthermore, we will show this action extends to a *strict* action as crystals of categories over $\mathrm{Ran}_{\underline{x}}^{\mathrm{unfl}}$.

⁵⁹One can show (using Lemma 15.10.4 combined with a parallel statement for $\mathrm{IndCoh}^!$) that the functor (16.2) is actually an equivalence.

16.1.8. The above strict compatibility means in particular that the functor

$$\text{ins. vac}_{\underline{x}} : \text{KL}(G)_{\kappa, \underline{x}} \rightarrow \text{Res}_{\text{vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\text{Ran}_{\underline{x}}}$$

intertwines the $\text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}})$ -action on $\text{KL}(G)_{\kappa, \underline{x}}$ and the $\text{IndCoh}^!(\text{Op}_{\tilde{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}})$ -action on $\text{Res}_{\text{vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\text{Ran}_{\underline{x}}}$ via the functor

$$\text{ins. unit}_{\underline{x}} : \text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}}) \rightarrow \text{IndCoh}^!(\text{Op}_{\tilde{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}}),$$

while the latter is the functor of $!$ -pullback along

$$\text{pr}_{\text{small}, \underline{x}}^{\text{Op}} : \text{Op}_{\tilde{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}} \rightarrow \text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}}.$$

Hence, we obtain a functor

$$(16.5) \quad \text{IndCoh}^!(\text{Op}_{\tilde{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}}) \otimes_{\text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}})} \text{KL}(G)_{\kappa, \underline{x}} \rightarrow \text{Res}_{\text{vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\text{Ran}_{\underline{x}}}.$$

16.1.9. Composing with the tautological functor

$$\text{Res}_{\text{vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\text{Ran}_{\underline{x}}} \rightarrow \text{KL}(G)_{\kappa, \text{Ran}_{\underline{x}}},$$

we obtain the sought-for functor

$$\text{ins. vac}_{\underline{x}}^{\text{mon-free} \rightsquigarrow \text{reg}} : \text{IndCoh}^!(\text{Op}_{\tilde{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}}) \otimes_{\text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}})} \text{KL}(G)_{\kappa, \underline{x}} \rightarrow \text{KL}(G)_{\kappa, \text{Ran}_{\underline{x}}}$$

of (16.3).

16.2. **The acting agents.** In this subsection we will interpret the category $\text{IndCoh}^!(\text{Op}_{\tilde{G}, \text{Ran}_{\underline{x}}}^{\text{mon-free} \rightsquigarrow \text{reg}})$, in terms of *factorization restriction*.

16.2.1. Recall the map of factorization spaces

$$\iota^{+, \text{mon-free}} : \text{Op}_{\tilde{G}}^{\text{reg}} \rightarrow \text{Op}_{\tilde{G}}^{\text{mon-free}}.$$

Consider the corresponding factorization functors

$$(16.6) \quad \text{QCoh}(\text{Op}_{\tilde{G}}^{\text{reg}}) \simeq \text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{reg}}) \xrightarrow{(\iota^{+, \text{mon-free}})_* \text{IndCoh}} \text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}})$$

and

$$(16.7) \quad \text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{reg}}) \xrightarrow{(\iota^{+, \text{mon-free}})^* \text{IndCoh}} \text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free}}).$$

16.2.2. We start with

$$\text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}})^{\text{fact}_{\underline{x}}} \in \text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}})\text{-}\mathbf{mod}_{\underline{x}}^{\text{fact}}$$

and

$$\text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free}})^{\text{fact}_{\underline{x}}} \in \text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free}})\text{-}\mathbf{mod}_{\underline{x}}^{\text{fact}}$$

and consider the resulting objects

$$(16.8) \quad \text{Res}_{(\iota^{+, \text{mon-free}})_* \text{IndCoh}}(\text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}})^{\text{fact}_{\underline{x}}}) \in \text{QCoh}(\text{Op}_{\tilde{G}}^{\text{reg}})\text{-}\mathbf{mod}_{\underline{x}}^{\text{fact}}$$

and

$$(16.9) \quad \text{Res}_{(\iota^{+, \text{mon-free}})^* \text{IndCoh}}(\text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free}})^{\text{fact}_{\underline{x}}}) \in \text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{reg}})\text{-}\mathbf{mod}_{\underline{x}}^{\text{fact}}.$$

Since the factorization functors (16.6) and (16.7) are unital, the module categories (16.8) and (16.9) have natural unital structures, see Sect. C.14.15.

16.2.3. Let us consider the assignments

$$(\mathcal{Z} \rightarrow \mathrm{Ran}_{\underline{x}}) \mapsto \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \times_{\mathrm{Ran}_{\underline{x}}} \mathcal{Z}_{\mathrm{dR}})$$

and

$$(\mathcal{Z} \rightarrow \mathrm{Ran}_{\underline{x}}) \mapsto \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \times_{\mathrm{Ran}_{\underline{x}}} \mathcal{Z}_{\mathrm{dR}})$$

as crystals of categories over $\mathrm{Ran}_{\underline{x}}$ (the latter is the crystal of categories that we have introduced in Sect. 16.1.6).

They have natural structures of unital module categories (at $\underline{x}$) over

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{reg}}) \text{ and } \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{reg}}),$$

respectively. We will denote them by

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}} \text{ and } \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}},$$

respectively.

16.2.4. We will regard $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}}$ (resp., $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}}$) as equipped with a comonoidal (resp., monoidal) structure given by IndCoh^* -pushforward (resp., $!$ -pullback) along the diagonal morphism.

We note that when we view $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}}$ as a crystal of categories over $\mathrm{Ran}_{\underline{x}}^{\mathrm{unl}}$, its comonoidal structure is given by right-lax functors.

By contrast, when we view $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}}$ as a crystal of categories over $\mathrm{Ran}_{\underline{x}}^{\mathrm{unl}}$, its monoidal structure is given by *strict* functors.

16.2.5. The map $\iota^{+, \mathrm{mon-free}}$ gives rise to a map

$$\iota^{+, \mathrm{mon-free} \rightarrow \mathrm{reg}} : \mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \rightarrow \mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free}}.$$

(Note that $\iota^{+, \mathrm{mon-free}}$ is the same as the map $\mathrm{pr}_{\mathrm{big}, \underline{x}}^{\mathrm{Op}}$).

We obtain functors of (unital) module categories

$$(16.10) \quad (\iota^{+, \mathrm{mon-free} \rightarrow \mathrm{reg}})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}}$$

and

$$(16.11) \quad (\iota^{+, \mathrm{mon-free} \rightarrow \mathrm{reg}})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}} \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}},$$

compatible with the functors (16.6) and (16.7), respectively.

16.2.6. By Sect. B.12.11, the functors (16.10) and (16.11) give rise to maps

$$(16.12) \quad (\iota^{+, \mathrm{mon-free}})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}} \rightarrow \mathrm{Res}_{(\iota^{+, \mathrm{mon-free}})_*^{\mathrm{IndCoh}}}(\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}})$$

and

$$(16.13) \quad (\iota^{+, \mathrm{mon-free}})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}} \rightarrow \mathrm{Res}_{(\iota^{+, \mathrm{mon-free}})_*^{\mathrm{IndCoh}}}(\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}})$$

in

$$\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{reg}})\text{-}\mathbf{mod}_{\underline{x}}^{\mathrm{fact}} \text{ and } \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{reg}})\text{-}\mathbf{mod}_{\underline{x}}^{\mathrm{fact}},$$

respectively. Moreover, the maps (16.12) and (16.13) are compatible with the unital structures, see Lemma C.14.16.

16.2.7. The following is a variant of Lemma F.5.7:

Lemma 16.2.8. *The functors (16.12) and (16.13) are equivalences.*

Proof. We prove the assertion for IndCoh^* . The case of $\mathrm{IndCoh}^!$ is analogous.

By Lemma B.15.9, it suffices to check that:

- (i) The functor (16.6) admits a right adjoint (as a functor between sheaves of categories);
- (ii) The functor (16.10) admits a right adjoint (as a functor between sheaves of categories);
- (iii) The functor (16.10) induces an equivalence between the fibers of the two sides at $\underline{x} \in \mathrm{Ran}_{\underline{x}}$.

We note that point (iii) holds tautologically.

Points (i) and (ii) are also automatic: the right adjoints in question are given by the functors $(\iota^{+, \mathrm{mon-free}})^!$ and $(\iota^{+, \mathrm{mon-free} \rightarrow \mathrm{reg}})^!$, respectively. $\square$

16.3. Construction of the action.

16.3.1. Recall the object

$$\mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}}) \in \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{reg}})\text{-}\mathbf{mod}_{\underline{x}}^{\mathrm{fact}}.$$

We claim that $\mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}})$, viewed as a sheaf of categories over $\mathrm{Ran}_{\underline{x}}$, carries a canonically defined action of $\mathrm{Res}_{(\iota^{+, \mathrm{mon-free}})_{*} \mathrm{IndCoh}}(\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}})$.

16.3.2. By duality, a datum of such an action is equivalent to the datum of a coaction of the sheaf of comonoidal categories $\mathrm{Res}_{(\iota^{+, \mathrm{mon-free}})_{*} \mathrm{IndCoh}}(\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}})$.

We will construct the corresponding coaction functor

$$(16.14) \quad \begin{aligned} \mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}}) &\rightarrow \\ &\rightarrow \mathrm{Res}_{(\iota^{+, \mathrm{mon-free}})_{*} \mathrm{IndCoh}}(\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}}) \otimes \mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}}). \end{aligned}$$

The full datum of coaction is defined similar to Sect. 5.3, using the device from Sect. J.

16.3.3. We interpret the right-hand side in (16.14) as the restriction of

$$(\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}) \otimes \mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}} \in (\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}) \otimes \mathrm{KL}(G)_{\kappa})\text{-}\mathbf{mod}_{\underline{x}}^{\mathrm{fact}}$$

along the factorization functor

$$((\iota^{+, \mathrm{mon-free}})_{*}^{\mathrm{IndCoh}} \otimes \mathrm{Vac}(G)_{\mathrm{crit}}) : \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{reg}}) \otimes \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{reg}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}) \otimes \mathrm{KL}(G)_{\kappa}.$$

The functor (16.14) is given by the procedure of restriction from Sect. B.12.15 along the diagram

$$\begin{array}{ccc} \mathrm{KL}(G)_{\kappa} & \longrightarrow & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}) \otimes \mathrm{KL}(G)_{\kappa} \\ \mathrm{Vac}(G)_{\mathrm{crit}} \uparrow & & \uparrow (\iota^{+, \mathrm{mon-free}})_{*}^{\mathrm{IndCoh}} \otimes \mathrm{Vac}(G)_{\mathrm{crit}} \\ \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{reg}}) & \longrightarrow & \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{reg}}) \otimes \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{reg}}), \end{array}$$

where:

- The top horizontal arrow is the coaction of $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})$ on $\mathrm{KL}(G)_{\kappa}$;
- The bottom horizontal arrow is the comonoidal operation, i.e., the functor of direct image along the diagonal map.

16.3.4. *Example.* Here is what the above action (resp., coaction) does at the pointwise level. Write

$$\underline{x}' = \underline{x} \sqcup \underline{x}'',$$

so that

$$\begin{aligned} \mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_\kappa)^{\mathrm{fact}_{\underline{x}}})_{\underline{x}'} &\simeq \mathrm{KL}(G)_{\kappa, \underline{x}} \otimes \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}}), \\ \mathrm{Res}_{(\ell^+, \mathrm{mon-free})_*} \mathrm{IndCoh}(\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}})_{\underline{x}'} &\simeq \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}) \otimes \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}}), \\ \mathrm{Res}_{(\ell^+, \mathrm{mon-free})_*} \mathrm{IndCoh}(\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}})_{\underline{x}'} &\simeq \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}) \otimes \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}}). \end{aligned}$$

The coaction of $\mathrm{Res}_{(\ell^+, \mathrm{mon-free})_*} \mathrm{IndCoh}(\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}})_{\underline{x}'}$ on $\mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_\kappa)^{\mathrm{fact}_{\underline{x}}})_{\underline{x}'}$ acts as the tensor product of

- The coaction of $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})$ on $\mathrm{KL}(G)_{\kappa, \underline{x}}$, and
- The functor of direct image along the diagonal map $\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}}) \rightarrow \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}}) \otimes \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}})$.

The action of $\mathrm{Res}_{(\ell^+, \mathrm{mon-free})_*} \mathrm{IndCoh}(\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}})_{\underline{x}'}$ on $\mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_\kappa)^{\mathrm{fact}_{\underline{x}}})_{\underline{x}'}$ acts as the tensor product of

- The action of $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})$ on $\mathrm{KL}(G)_{\kappa, \underline{x}}$, and
- The canonical action of $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}})$ on $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}}) \simeq \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}})$.

16.4. The unital structure on the action functor.

16.4.1. Let us regard

$$\mathrm{KL}(G)_\kappa \text{ and } \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})$$

as crystals of categories on Ran , equipped with a *unital* structure (see Sect. 11.2.1 for what this means).

The coaction of $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})$ on $\mathrm{KL}(G)_\kappa$ has the following feature: it extends to a coaction in the 2-category of crystals of categories on $\mathrm{Ran}^{\mathrm{un}^{\mathrm{t}}}$ with *right-lax* functors.

This follows from the construction of this coaction in Sect. 5.3, using the following observation:

For a map of factorization algebras $\mathcal{A}_1 \rightarrow \mathcal{A}_2$ in a given factorization category $\mathbf{A}$, the restriction functor

$$\mathcal{A}_2\text{-mod}(\mathbf{A}) \rightarrow \mathcal{A}_1\text{-mod}(\mathbf{A}),$$

viewed as a functor between crystals of categories on Ran , admits a natural extension to a right-lax functor between crystals of categories on $\mathrm{Ran}^{\mathrm{un}^{\mathrm{t}}}$.

16.4.2. It follows from the construction in Sect. 16.3.3 and Sect. C.14.20 that the functor (16.14) extends to a right-lax functor between crystals of categories on $\mathrm{Ran}_{\underline{x}}^{\mathrm{un}^{\mathrm{t}}}$.

By a similar token, we obtain that the full datum of coaction of the comonoidal category $\mathrm{Res}_{(\ell^+, \mathrm{mon-free})_*} \mathrm{IndCoh}(\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}})$ on $\mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_\kappa)^{\mathrm{fact}_{\underline{x}}})$ extends to a coaction in the 2-category of crystals of categories on $\mathrm{Ran}_{\underline{x}}^{\mathrm{un}^{\mathrm{t}}}$ with right-lax functors.

16.4.3. Combining with Lemma 16.2.8, we obtain that $\mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_\kappa)^{\mathrm{fact}_{\underline{x}}})$ carries a coaction of $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}}$.

Furthermore, this coaction extends to a coaction in the 2-category of crystals of categories on $\mathrm{Ran}_{\underline{x}}^{\mathrm{un}^{\mathrm{t}}}$ with right-lax functors.

16.4.4. We note now that the unital structures on

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}} \text{ and } \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}}$$

have the following feature:

The counit of the duality

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}} \otimes \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}} \rightarrow \underline{\mathrm{D-mod}}(\mathrm{Ran}_{\underline{x}})$$

extends to a right-lax functor between crystals of categories on $\mathrm{Ran}_{\underline{x}}^{\mathrm{un}^{\mathrm{t}}}$.

16.4.5. The coaction of $\text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free} \rightsquigarrow \text{reg}})^{\text{fact}_{\underline{x}}}$ on $\text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})$ gives rise to an action of $\text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free} \rightsquigarrow \text{reg}})^{\text{fact}_{\underline{x}}}$ on $\text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})$.

Moreover, by Sect. 16.4.4, this action extends to an action in the 2-category of sheaves of categories on Ran^{untl} with *right-lax* functors.

16.4.6. We now claim:

Lemma 16.4.7. *The right-lax functors that define the action of $\text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free} \rightsquigarrow \text{reg}})^{\text{fact}_{\underline{x}}}$ on $\text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})$ as crystals of categories on Ran^{untl} are strict.*

Proof. We need to show the following: for $\underline{x} \subseteq \underline{x}_1 \subseteq \underline{x}_2$, the natural transformation from

$$(16.15) \quad \text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x} \subseteq \underline{x}_1}^{\text{mon-free} \rightsquigarrow \text{reg}}) \otimes \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_1} \xrightarrow{\text{action}} \\ \rightarrow \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_1} \xrightarrow{\text{ins.unit}_{\underline{x}_1 \subseteq \underline{x}_2}} \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_2}$$

to

$$(16.16) \quad \text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x} \subseteq \underline{x}_1}^{\text{mon-free} \rightsquigarrow \text{reg}}) \otimes \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_1} \xrightarrow{\text{ins.unit}_{\underline{x}_1 \subseteq \underline{x}_2} \otimes \text{ins.unit}_{\underline{x}_1 \subseteq \underline{x}_2}} \\ \rightarrow \text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x} \subseteq \underline{x}_2}^{\text{mon-free} \rightsquigarrow \text{reg}}) \otimes \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_2} \xrightarrow{\text{action}} \\ \rightarrow \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_2}$$

is an isomorphism.

The question of a functor between crystals of categories (in this case, over $\text{Ran}_{\underline{x}}$) being an isomorphism can be checked strata-wise. So we can assume that

$$\underline{x}_2 = \underline{x}_1 \sqcup \underline{x}'.$$

We identify

$$(16.17) \quad \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_2} \simeq \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_1} \otimes \text{QCoh}(\text{Op}_{\tilde{G}, \underline{x}''}^{\text{reg}})$$

and

$$(16.18) \quad \text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x} \subseteq \underline{x}_2}^{\text{mon-free} \rightsquigarrow \text{reg}}) \otimes \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_2} \simeq \\ \simeq \text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x} \subseteq \underline{x}_1}^{\text{mon-free} \rightsquigarrow \text{reg}}) \otimes \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_1} \bigotimes \\ \bigotimes \text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x}''}^{\text{reg}}) \otimes \text{QCoh}(\text{Op}_{\tilde{G}, \underline{x}''}^{\text{reg}}),$$

so that the functor

$$\text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_1} \xrightarrow{\text{ins.unit}_{\underline{x}_1 \subseteq \underline{x}_2}} \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_2}$$

identifies with

$$\text{Id} \otimes \mathcal{O}_{\text{Op}_{\tilde{G}, \underline{x}''}^{\text{reg}}}$$

and the functor

$$\text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x} \subseteq \underline{x}_1}^{\text{mon-free} \rightsquigarrow \text{reg}}) \otimes \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_1} \xrightarrow{\text{ins.unit}_{\underline{x}_1 \subseteq \underline{x}_2} \otimes \text{ins.unit}_{\underline{x}_1 \subseteq \underline{x}_2}} \\ \rightarrow \text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x} \subseteq \underline{x}_2}^{\text{mon-free} \rightsquigarrow \text{reg}}) \otimes \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_2}$$

identifies with

$$\text{Id} \otimes \omega_{\text{Op}_{\tilde{G}, \underline{x}''}^{\text{reg}}} \otimes \mathcal{O}_{\text{Op}_{\tilde{G}, \underline{x}''}^{\text{reg}}}.$$

In terms of (16.17) and (16.18), the functor

$$\text{IndCoh}^!(\text{Op}_{\tilde{G}, \underline{x} \subseteq \underline{x}_2}^{\text{mon-free} \rightsquigarrow \text{reg}}) \otimes \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_2} \xrightarrow{\text{action}} \text{Res}_{\text{Vac}(G)_{\text{crit}}}((\text{KL}(G)_{\kappa})^{\text{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_2}$$

is the tensor product of

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x} \subseteq \underline{x}_1}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}) \otimes \mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_\kappa)^{\mathrm{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_1} \xrightarrow{\mathrm{action}} \mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_\kappa)^{\mathrm{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_1}$$

along the first factor and

$$\begin{aligned} \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}}) \otimes \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}}) &\simeq \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}}) \xrightarrow{!} \\ &\rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}}) \simeq \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}}) \end{aligned}$$

along the second factor.

Hence, we obtain that, in terms of (16.17) and (16.18), both functors (16.15) and (16.16) are identified with

$$\begin{aligned} (16.19) \quad \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x} \subseteq \underline{x}_1}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}) \otimes \mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_\kappa)^{\mathrm{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_1} &\xrightarrow{\mathrm{action}} \\ &\rightarrow \mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_\kappa)^{\mathrm{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_1} \xrightarrow{\mathrm{Id} \otimes \omega_{\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}}}} \\ &\rightarrow \mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_\kappa)^{\mathrm{fact}_{\underline{x}}})_{\underline{x} \subseteq \underline{x}_1} \otimes \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}''}^{\mathrm{reg}}). \end{aligned}$$

Unwinding the construction, we obtain that the endomorphism of (16.19) defined by the structure of right-lax functor on the action map is the identity map. $\square$

16.5. End of the construction.

16.5.1. Thus, we have carried out the construction announced in Sect. 16.1.7. In particular, we obtain a functor

$$(16.20) \quad \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \rightarrow \mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_\kappa)^{\mathrm{fact}_{\underline{x}}})_{\mathrm{Ran}_{\underline{x}}},$$

and its composition of the functor (16.5) with

$$(16.21) \quad \mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_\kappa)^{\mathrm{fact}_{\underline{x}}})_{\mathrm{Ran}_{\underline{x}}} \rightarrow \mathrm{KL}(G)_{\kappa, \mathrm{Ran}_{\underline{x}}}$$

produces the sought-for functor

$$(16.22) \quad \mathrm{ins. vac}_{\underline{x}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}) \otimes_{\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{KL}(G)_{\kappa, \underline{x}} \rightarrow \mathrm{KL}(G)_{\kappa, \mathrm{Ran}_{\underline{x}}}.$$

16.5.2. It remains to show that the functor (16.22) is $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})$ -linear.

The functor (16.20) is $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})$ -linear by construction. So it remains to show that the functor (16.21) is also $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})$ -linear.

16.5.3. Unwinding the construction of the coaction of $\mathrm{Res}_{(\ell^+, \mathrm{mon-free})_*} \mathrm{IndCoh}(\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}})$ on $\mathrm{Res}_{\mathrm{Vac}(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_\kappa)^{\mathrm{fact}_{\underline{x}}})$ in Sect. 16.3.3, we obtain that the functor (16.21) intertwines the coaction of $\mathrm{Res}_{(\ell^+, \mathrm{mon-free})_*} \mathrm{IndCoh}(\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}})$ on the left-hand side with the coaction of $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})$ on the right hand side via the tautological forgetful functor

$$\mathrm{Res}_{(\ell^+, \mathrm{mon-free})_*} \mathrm{IndCoh}(\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{\mathrm{fact}_{\underline{x}}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}).$$

Hence, the functor (16.21) intertwines the coaction of $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free}})$ on the left-hand side of (16.21) with the action of $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})$ on the right-hand side of (16.21) via $(\mathrm{pr}_{\mathrm{big}, \underline{x}}^{\mathrm{Op}})_* \mathrm{IndCoh}$.

Passing to the dual of the acting agents, we obtain that functor (16.21) intertwines the action of $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free}})$ on the left-hand side of (16.21) with the action of $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})$ on the right-hand side of (16.21) via $(\mathrm{pr}_{\mathrm{big}, \underline{x}}^{\mathrm{Op}})^!$, as desired.

□[Theorem 15.2.8]

16.5.4. The next assertion is not needed for the sequel; we mention it for the sake of completeness:

Proposition 16.5.5. *The functor*

$$(16.23) \quad \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}) \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \rightarrow \mathrm{Res}_{\mathcal{V}ac(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}})_{\mathrm{Ran}_{\underline{x}}},$$

induced by (16.20), is an equivalence.

The rest of this subsection is devoted to the proof of this proposition.

16.5.6. The functor (16.23) comes from a morphism in the 2-category $\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{reg}})\text{-}\mathbf{mod}_{\underline{x}}^{\mathrm{fact}}$:

$$(16.24) \quad \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \rightarrow \mathrm{Res}_{\mathcal{V}ac(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}})$$

Hence, by Lemma B.15.9, in order to check that (16.23) is an equivalence, it suffices to check that:

(i) The composition of (16.24) with the tautological functor

$$\mathrm{Res}_{\mathcal{V}ac(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}}) \rightarrow (\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}}$$

admits a right adjoint (as a functor between sheaves of categories);

(ii) The functor $\mathcal{V}ac(G)_{\mathrm{crit}}$ admits a right adjoint (as a functor between sheaves of categories);

(iii) The functor (16.24) induces an equivalence between the fibers of the two sides at $\underline{x} \in \mathrm{Ran}_{\underline{x}}$.

16.5.7. Point (iii) above is immediate. Point (ii) follows from the fact that the functor $\mathcal{V}ac(G)_{\mathrm{crit}}$ preserves compactness.

Hence, it remains to show that the composition in point (i) preserves compactness (and the left-hand side is compactly generated).

16.5.8. Let Y be as in Sect. 15.6.3.

It is easy to see that the corresponding category $\mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y}$ is compactly generated. By Lemma 15.6.4, it suffices to show that the composition

$$(16.25) \quad \begin{aligned} & \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \rightarrow \\ & \rightarrow \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \rightarrow \\ & \rightarrow \mathrm{Res}_{\mathcal{V}ac(G)_{\mathrm{crit}}}((\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}}) \rightarrow (\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}} \end{aligned}$$

preserves compactness.

We rewrite the left-hand side in (16.25) as

$$\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_{\underline{x}}} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{QCoh}(Y) \otimes_{\mathrm{QCoh}(Y)} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y},$$

and further, by Lemma 15.7.3 as

$$(16.26) \quad \mathrm{QCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \times_{\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}} Y)^{\mathrm{fact}_{\underline{x}}} \otimes_{\mathrm{QCoh}(Y)} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y}.$$

16.5.9. Now, since $\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} Y$ is a relative affine *scheme* (as opposed to ind-scheme) over $\mathrm{Ran}_{\underline{x}}$, the category (16.26) is compactly generated by the essential image of compact objects along the functor

$$\mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \rightarrow \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \times_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}} Y)^{\mathrm{fact}_x} \otimes_{\mathrm{QCoh}(Y)} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y},$$

given by tensoring with the structure sheaf along the first factor.

Hence, it suffices to show that the functor

$$\begin{aligned} (16.27) \quad \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} &\rightarrow \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_x} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \rightarrow \\ &\rightarrow \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_x} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \rightarrow \\ &\rightarrow \mathrm{Res}_{\mathcal{V}_{ac}(G)_{\mathrm{crit}}} ((\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}}) \rightarrow (\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}} \end{aligned}$$

preserves compactness.

16.5.10. The forgetful functor

$$\mathrm{KL}(G)_{\mathrm{crit}, \underline{x}, Y} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}}$$

preserves compactness (indeed, it admits a continuous right adjoint). Hence, it suffices to show that the functor

$$\mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \rightarrow \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_x} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}})} \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \rightarrow (\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}}$$

preserves compactness.

However, by construction, the latter functor is

$$\mathrm{ins. vac}_{\underline{x}} : \mathrm{KL}(G)_{\mathrm{crit}, \underline{x}} \rightarrow (\mathrm{KL}(G)_{\kappa})^{\mathrm{fact}_{\underline{x}}},$$

and the assertion follows. $\square$ [Proposition 16.5.5]

17. SPECTRAL POINCARÉ FUNCTOR(S)

In this section we start dealing with the local-to-global constructions on the spectral side, i.e., when the recipient category is $\mathrm{IndCoh}(\mathrm{LS}_{\check{G}})$.

We introduce two versions of the spectral Poincaré functor:

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \xrightarrow{\mathrm{Poinc}_{\check{G}, !}^{\mathrm{spec}}} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}(X))$$

and

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \xrightarrow{\mathrm{Poinc}_{\check{G}, *}^{\mathrm{spec}}} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}(X)).$$

However, we show (Theorem 17.4.7) that they are intertwined by the “self-duality” functor

$$\Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}),$$

up to tensoring by a graded line.

Next we recall the definition of the *spectral localization and global sections functors*

$$\mathrm{Loc}_{\check{G}}^{\mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightleftarrows \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) : \Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}}.$$

Finally, we give the expression for the composition

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \xrightarrow{\mathrm{Poinc}_{\check{G}, *}^{\mathrm{spec}}} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \xrightarrow{\Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}}} \mathrm{Rep}(\check{G})_{\mathrm{Ran}}$$

via factorization homology, which exactly matches the composition (14.1) under $\mathrm{FLE}_{G, \mathrm{crit}}$ and $\mathrm{FLE}_{\check{G}, \infty}$.

17.1. Ind-coherent sheaves on local vs. global opers.

17.1.1. For $\mathcal{Z} \rightarrow \text{Ran}$ consider the morphism

$$\text{Op}_{\tilde{G}, \mathcal{Z}}^{\text{mer, glob}} \xrightarrow{\text{ev}_{\mathcal{Z}}} \text{Op}_{\tilde{G}, \mathcal{Z}}^{\text{mer}}.$$

Note that the prestack $\text{Op}_{\tilde{G}, \mathcal{Z}}^{\text{mer, glob}}$ is locally almost of finite type, so we have a well-defined category $\text{IndCoh}(\text{Op}_{\tilde{G}}^{\text{mer, glob}})_{\mathcal{Z}} := \text{IndCoh}(\text{Op}_{\tilde{G}, \mathcal{Z}}^{\text{mer, glob}})$.

Consider the pair of *mutually dual* functors

$$(17.1) \quad (\text{ev}_{\mathcal{Z}})_*^{\text{IndCoh}} : \text{IndCoh}(\text{Op}_{\tilde{G}}^{\text{mer, glob}})_{\mathcal{Z}} \rightarrow \text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mer}})_{\mathcal{Z}}$$

and

$$(17.2) \quad (\text{ev}_{\mathcal{Z}})^! : \text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mer}})_{\mathcal{Z}} \rightarrow \text{IndCoh}(\text{Op}_{\tilde{G}}^{\text{mer, glob}})_{\mathcal{Z}}.$$

17.1.2. We claim:

Lemma 17.1.3. *The functor (17.2) preserves compactness.*

17.1.4. Before we prove Lemma 17.1.3, we need to introduce some notation. For expositional purposes, we will assume that $\mathcal{Z} = \text{pt}$, so that $\mathcal{Z} \rightarrow \text{Ran}$ corresponds to $\underline{x} \in \text{Ran}$.

Let $\mathbf{V}$ denote the Tate vector space

$$\Gamma(\mathcal{D}_{\underline{x}}^{\times}, \mathfrak{a}(\check{\mathfrak{g}})_{\omega_X}).$$

Let $\mathbf{L}_0 \subset \mathbf{V}$ denote the standard lattice, i.e.,

$$\mathbf{L}_0 := \Gamma(\mathcal{D}_{\underline{x}}, \mathfrak{a}(\check{\mathfrak{g}})_{\omega_X}).$$

Recall that, according to Sect. 3.1.7, we have a simply-transitive action of $\mathbf{L}_0$ on $\text{Op}_{\tilde{G}, \underline{x}}^{\text{reg}}$, so that

$$\mathbf{V}^{\mathbf{L}_0} \times \text{Op}_{\tilde{G}, \underline{x}}^{\text{reg}} \simeq \text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}.$$

For a lattice $\mathbf{L} \supset \mathbf{L}_0$ denote

$$\mathbf{L} \times \text{Op}_{\tilde{G}, \underline{x}}^{\text{reg}} =: \text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}} \subset \text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}.$$

Denote

$$\mathbf{V}^{\text{glob}} := \Gamma(X - \underline{x}, \mathfrak{a}(\check{\mathfrak{g}})_{\omega_X}).$$

We have a simply-transitive action of $\mathbf{V}^{\text{glob}}$ on $\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer, glob}}$, compatible with the embedding $\text{ev}_{\underline{x}}$ via

$$\mathbf{V}^{\text{glob}} \hookrightarrow \mathbf{V}.$$

For $\mathbf{L} \supset \mathbf{L}_0$ set

$$\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \text{glob}} := \text{Op}_{\tilde{G}, \underline{x}}^{\text{mer, glob}} \times_{\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}} \text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}.$$

17.1.5. *Proof of Lemma 17.1.3.* We have

$$(17.3) \quad \text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mer}})_{\mathcal{Z}} \simeq \text{colim}_{\mathbf{L} \subset \mathbf{L}_0} \text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\mathbf{L}})_{\mathcal{Z}},$$

where the colimit is taken with respect to the IndCoh -pushforward functors.

Hence, it is enough to show that the composition

$$\text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\mathbf{L}})_{\mathcal{Z}} \rightarrow \text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mer}})_{\mathcal{Z}} \xrightarrow{\text{ev}_{\mathcal{Z}}^!} \text{IndCoh}(\text{Op}_{\tilde{G}}^{\text{mer, glob}})_{\mathcal{Z}}$$

preserves compactness.

The Cartesian diagram

$$(17.4) \quad \begin{array}{ccc} \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \mathrm{glob}} & \xrightarrow{\mathrm{ev}_{\underline{x}}} & \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}} \\ \downarrow & & \downarrow \\ \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}, \mathrm{glob}} & \xrightarrow{\mathrm{ev}_{\underline{x}}} & \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}} \end{array}$$

gives rise to a commutative diagram

$$\begin{array}{ccc} \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \mathrm{glob}}) & \xleftarrow{(\mathrm{ev}_{\underline{x}})^!} & \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}) \\ \text{pushforward} \downarrow & & \downarrow \text{pushforward} \\ \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}, \mathrm{glob}}) & \xleftarrow{(\mathrm{ev}_{\underline{x}})^!} & \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}}), \end{array}$$

see Sect. A.10.12.

Hence, it suffices to show that the functor

$$(\mathrm{ev}_{\underline{x}})^! : \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \mathrm{glob}})$$

preserves compactness.

Write

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}) \simeq \mathrm{colim}_{\mathbf{L}' \subset \mathbf{L}_0} \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}} / \mathbf{L}'),$$

where the colimit is taken with respect to the $!$ -pullback functors.

Hence, it suffices to show that the $!$ -pullback functors along

$$\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}} \rightarrow \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \mathrm{glob}} \rightarrow \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}} / \mathbf{L}'$$

preserve compactness.

However, the latter is obvious, since the above morphism goes between two smooth schemes.

□[Lemma 17.1.3]

17.1.6. As an immediate corollary of Lemma 17.1.3, we obtain:

Corollary 17.1.7. *The functor (17.1) admits a left adjoint, to be denoted $\mathrm{ev}_z^{*, \mathrm{IndCoh}}$.*

17.1.8. Note that we have a tautological commutative diagram

$$(17.5) \quad \begin{array}{ccc} \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mer}, \mathrm{glob}})_z & \xrightarrow{(\mathrm{ev}_z)_*^{\mathrm{IndCoh}}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mer}})_z \\ \Psi_{\mathrm{Op}_{\tilde{G}, z}^{\mathrm{mer}, \mathrm{glob}}} \downarrow & & \downarrow \Psi_{\mathrm{Op}_{\tilde{G}, z}^{\mathrm{mer}}} \\ \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mer}, \mathrm{glob}})_z & \xrightarrow{(\mathrm{ev}_z)_*} & \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mer}})_z, \end{array}$$

see Sect. A.7.3.

Since the morphism ev_z is schematic, the functor

$$(\mathrm{ev}_z)_* : \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mer}, \mathrm{glob}})_z \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mer}})_z$$

admits a left adjoint, denoted $(\mathrm{ev}_z)^*$, see Sect. A.1.4.

Passing to left adjoints along the horizontal arrows in (17.5), we obtain a diagram

$$(17.6) \quad \begin{array}{ccc} \mathrm{IndCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mer},\mathrm{glob}})_Z & \xleftarrow{(\mathrm{ev}_Z)^*, \mathrm{IndCoh}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}})_Z \\ \Psi_{\mathrm{Op}_{\check{G},Z}^{\mathrm{mer},\mathrm{glob}}} \downarrow & \swarrow & \downarrow \Psi_{\mathrm{Op}_{\check{G},Z}^{\mathrm{mer}}} \\ \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Op}_{\check{G}}^{\mathrm{mer},\mathrm{glob}})_Z & \xleftarrow{(\mathrm{ev}_Z)^*} & \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Op}_{\check{G}}^{\mathrm{mer}})_Z. \end{array}$$

We claim:

Lemma 17.1.9. *The natural transformation in (17.6) is an isomorphism.*

Proof. With no restriction of generality, we can assume that $Z = X^I$; in particular, it is smooth.

We claim that the vertical arrows in (17.5) are in fact equivalences. Indeed, we write

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}})_Z \simeq \mathrm{colim}_{\mathbf{L} \subset \mathbf{L}_0} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathbf{L}})_Z$$

and

$$\mathrm{IndCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mer},\mathrm{glob}})_Z \simeq \mathrm{colim}_{\mathbf{L} \subset \mathbf{L}_0} \mathrm{IndCoh}(\mathrm{Op}_{\check{G}}^{\mathbf{L},\mathrm{glob}})_Z$$

where both colimits are formed with respect to the pushforward functors.

Hence it enough to show that the functors

$$\Psi_{\mathrm{Op}_{\check{G},Z}^{\mathbf{L}}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathbf{L}})_Z \rightarrow \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathbf{L}})_Z$$

and

$$\Psi_{\mathrm{Op}_{\check{G},Z}^{\mathbf{L},\mathrm{glob}}} : \mathrm{IndCoh}(\mathrm{Op}_{\check{G}}^{\mathbf{L},\mathrm{glob}})_Z \rightarrow \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathbf{L},\mathrm{glob}})_Z$$

are equivalences.

However, this follows from the fact that $\mathrm{Op}_{\check{G},Z}^{\mathbf{L},\mathrm{glob}}$ (resp., $\mathrm{Op}_{\check{G},Z}^{\mathbf{L}}$) is smooth (resp., pro-smooth). $\square$

17.2. Interaction with self-duality.

17.2.1. Recall now the functor

$$\Theta_{\mathrm{Op}_{\check{G},Z}^{\mathrm{mer}}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mer}})_Z \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}})_Z,$$

see Sect. 3.7.1.

17.2.2. Denote by $\mathbf{l}_{\mathrm{Kost}(\check{G})}$ the (non-graded) line

$$\det(\Gamma(X, \mathfrak{a}(\check{\mathfrak{g}})_{\omega_X})).$$

Set

$$\delta_G := \dim(\mathrm{Bun}_G) = (g-1) \cdot \dim(G).$$

17.2.3. We claim:

Proposition 17.2.4. *There exists a commutative diagram*

$$\begin{array}{ccc} \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mer}})_Z & \xrightarrow{\Theta_{\mathrm{Op}_{\check{G},Z}^{\mathrm{mer}}}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}})_Z \\ (\mathrm{ev}_Z)^! \downarrow & & \downarrow (\mathrm{ev}_Z)^*, \mathrm{IndCoh} \\ \mathrm{IndCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mer},\mathrm{glob}})_Z & \xrightarrow{-\otimes \mathbf{l}_{\mathrm{Kost}(\check{G})}[-\delta_G]} & \mathrm{IndCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mer},\mathrm{glob}})_Z. \end{array}$$

The rest of this subsection is devoted to proof of Proposition 17.2.4.

17.2.5. For expositional purposes, we will assume that $\mathcal{Z} = \text{pt}$, so that $\mathcal{Z} \rightarrow \text{Ran}$ corresponds to $\underline{x} \in \text{Ran}$. We will use the notation from Sect. 17.1.4.

17.2.6. By the definition of the functor $\Theta_{\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}}$, we need to establish an isomorphism of the following two objects in $\text{IndCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer, glob}})$:

$$\omega_{\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer, glob}}} \otimes \mathbb{I}_{\text{Kost}(\tilde{G})}[-\delta_G] \simeq (\text{ev}_{\underline{x}})^*, \text{IndCoh}(\omega_{\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}}^{*, \text{fake}}),$$

where

$$\omega_{\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}}^{*, \text{fake}} \in \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}})$$

is as in Sect. 3.7.2.

17.2.7. In terms of the presentation

$$\text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \simeq \text{colim}_{\mathbf{L} \subset \mathbf{L}_0} \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}) \xrightarrow{\Psi_{\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}}} \text{QCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}),$$

the object $\omega_{\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}}^{*, \text{fake}}$ is, by construction, the colimit of the images of

$$\mathcal{O}_{\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}} \otimes \det(\mathbf{L}/\mathbf{L}_0)^{\otimes -1} [\dim(\mathbf{L}/\mathbf{L}_0)].$$

In terms of the presentation

$$\text{IndCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{glob}}) \simeq \text{colim}_{\mathbf{L} \subset \mathbf{L}_0} \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \text{glob}}) \xrightarrow{\Psi_{\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \text{glob}}}} \text{QCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \text{glob}}),$$

the object $\omega_{\text{Op}_{\tilde{G}, \underline{x}}^{\text{glob}}}$ is, tautologically, the colimit of the images of

$$\Psi_{\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \text{glob}}}(\omega_{\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \text{glob}}}).$$

17.2.8. The Cartesian diagram (17.4) gives rise to a commutative diagram

$$\begin{array}{ccc} \text{IndCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \text{glob}}) & \xrightarrow{(\text{ev}_{\underline{x}})_*^{\text{IndCoh}}} & \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}) \\ \uparrow \text{!-pullback} & & \uparrow \text{!-pullback} \\ \text{IndCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{glob}}) & \xrightarrow{(\text{ev}_{\underline{x}})_*^{\text{IndCoh}}} & \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}), \end{array}$$

see Sect. A.10.12.

Passing to the left adjoints, we obtain a commutative diagram

$$(17.7) \quad \begin{array}{ccc} \text{IndCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \text{glob}}) & \xleftarrow{(\text{ev}_{\underline{x}})^*, \text{IndCoh}} & \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}) \\ \downarrow *- \text{pushforward} & & \downarrow *- \text{pushforward} \\ \text{IndCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{glob}}) & \xleftarrow{(\text{ev}_{\underline{x}})^*, \text{IndCoh}} & \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}). \end{array}$$

As in Lemma 17.1.9, we also have a commutative diagram

$$\begin{array}{ccc} \text{QCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \text{glob}}) & \xleftarrow{(\text{ev}_{\underline{x}})^*} & \text{QCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}) \\ \uparrow \Psi_{\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \text{glob}}} & & \uparrow \Psi_{\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}} \\ \text{IndCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \text{glob}}) & \xleftarrow{(\text{ev}_{\underline{x}})^*, \text{IndCoh}} & \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}). \end{array}$$

Hence, it is enough to construct a compatible collection of identifications

$$\Psi_{\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \text{glob}}}(\omega_{\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \text{glob}}}) \otimes \mathbb{I}_{\text{Kost}(\tilde{G})}[-\delta_G] \simeq \mathcal{O}_{\text{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \text{glob}}} \otimes \det(\mathbf{L}/\mathbf{L}_0)^{\otimes -1} [\dim(\mathbf{L}/\mathbf{L}_0)],$$

taking place in $\mathrm{QCoh}(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathbf{L}, \mathrm{glob}})$.

17.2.9. Note now that $\mathrm{Op}_{\check{G}, \underline{x}}^{\mathbf{L}, \mathrm{glob}}$ is an affine space with respect to

$$\mathbf{L}^{\mathrm{glob}} := \mathbf{V}^{\mathrm{glob}} \cap \mathbf{L}.$$

Hence,

$$\Psi_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathbf{L}, \mathrm{glob}}}(\omega_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathbf{L}, \mathrm{glob}}}) \simeq \mathcal{O}_{\mathrm{Op}_{\check{G}, \underline{x}}^{\mathbf{L}, \mathrm{glob}}} \otimes \det(\mathbf{L}^{\mathrm{glob}})^{\otimes -1} [\dim(\mathbf{L}^{\mathrm{glob}})].$$

17.2.10. Thus, it remains to establish a compatible collection of isomorphisms between the lines

$$(17.8) \quad \det(\mathbf{L}^{\mathrm{glob}})^{\otimes -1} \otimes \mathfrak{l}_{\mathrm{Kost}(\check{G})}[\dim(\mathbf{L}^{\mathrm{glob}}) - \delta_G] \simeq \det(\mathbf{L}/\mathbf{L}_0)^{\otimes -1} [\dim(\mathbf{L}/\mathbf{L}_0)].$$

However, this follows from the fact that

$$\mathfrak{l}_{\mathrm{Kost}(\check{G})} \simeq \det(\mathbf{V}^{\mathrm{glob}} \cap \mathbf{L}_0) \text{ and } \delta_G = \dim(\Gamma(X, \mathfrak{a}(\mathfrak{g})_{\omega_X})) = \dim(\mathbf{V}^{\mathrm{glob}} \cap \mathbf{L}_0).$$

17.3. Ind-coherent sheaves on local vs. global *monodromy-free* opers.

17.3.1. For $\mathcal{Z} \rightarrow \mathrm{Ran}$ recall the (relative ind-scheme) $\mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mon-free}, \mathrm{glob}}$, which fits into the Cartesian square

$$(17.9) \quad \begin{array}{ccc} \mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mon-free}, \mathrm{glob}} & \xrightarrow{\mathrm{ev}_{\mathcal{Z}}} & \mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mon-free}} \\ \downarrow \iota^{\mathrm{mon-free}, \mathrm{glob}} & & \downarrow \iota^{\mathrm{mon-free}} \\ \mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}, \mathrm{glob}} & \xrightarrow{\mathrm{ev}_{\mathcal{Z}}} & \mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}} \end{array}$$

Consider the morphism:

$$\mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mon-free}, \mathrm{glob}} \xrightarrow{\mathrm{ev}_{\mathcal{Z}}} \mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mon-free}}$$

and the resulting pair of mutually dual functors

$$(17.10) \quad (\mathrm{ev}_{\mathcal{Z}})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}, \mathrm{glob}})_{\mathcal{Z}} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}}$$

and

$$(17.11) \quad (\mathrm{ev}_{\mathcal{Z}})^! : \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}} \rightarrow \mathrm{IndCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}, \mathrm{glob}})_{\mathcal{Z}}.$$

17.3.2. We claim:

Lemma 17.3.3. *The functor (17.11) preserves compactness.*

Proof. From (17.9) we obtain a commutative square

$$(17.12) \quad \begin{array}{ccc} \mathrm{IndCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}, \mathrm{glob}})_{\mathcal{Z}} & \xleftarrow{(\mathrm{ev}_{\mathcal{Z}})^!} & \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}} \\ \downarrow (\iota^{\mathrm{mon-free}, \mathrm{glob}})_*^{\mathrm{IndCoh}} & & \downarrow (\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}} \\ \mathrm{IndCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mer}, \mathrm{glob}})_{\mathcal{Z}} & \xleftarrow{(\mathrm{ev}_{\mathcal{Z}})^!} & \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mer}})_{\mathcal{Z}}, \end{array}$$

see Sect. A.10.12.

As in Proposition 3.3.5(b), one shows that an object in $\mathrm{IndCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}, \mathrm{glob}})_{\mathcal{Z}}$ is compact if and only if its image in $\mathrm{IndCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mer}, \mathrm{glob}})_{\mathcal{Z}}$ under $(\iota^{\mathrm{glob}})_*^{\mathrm{IndCoh}}$ is compact.

Hence, it suffices to show that the clockwise circuit in (17.12) preserves compactness.

For the functor $(\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}}$ this is evident (since the morphism $\iota^{\mathrm{mon-free}}$ is of finite presentation). For the bottom horizontal arrow in (17.12) this follows from Lemma 17.1.3. $\square$

17.3.4. As a corollary of Lemma 17.3.3 we obtain:

Corollary 17.3.5. *The functor (17.10) admits a left adjoint, to be denoted $\mathrm{ev}_{\mathcal{Z}}^{*, \mathrm{IndCoh}}$.*

17.3.6. The fact that (17.9) is Cartesian implies that the diagram

$$(17.13) \quad \begin{array}{ccc} \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}, \mathrm{glob}})_Z & \xrightarrow{(\mathrm{ev}_Z)_*^{\mathrm{IndCoh}}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})_Z \\ (\iota^{\mathrm{mon-free}, \mathrm{glob}})^! \uparrow & & \uparrow (\iota^{\mathrm{mon-free}})^! \\ \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mer}, \mathrm{glob}})_Z & \xrightarrow{(\mathrm{ev}_Z)_*^{\mathrm{IndCoh}}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mer}})_Z \end{array}$$

commutes, see Sect. A.10.12.

By passing to left adjoints along all arrows in (17.13) we obtain that the diagram

$$(17.14) \quad \begin{array}{ccc} \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}, \mathrm{glob}})_Z & \xleftarrow{(\mathrm{ev}_Z)^*, \mathrm{IndCoh}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})_Z \\ (\iota^{\mathrm{mon-free}, \mathrm{glob}})_*^{\mathrm{IndCoh}} \downarrow & & \downarrow (\iota^{\mathrm{mon-free}})_*^{\mathrm{IndCoh}} \\ \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mer}, \mathrm{glob}})_Z & \xleftarrow{(\mathrm{ev}_Z)^*, \mathrm{IndCoh}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mer}})_Z \end{array}$$

commutes as well.

However, passing left adjoints only along the horizontal arrows in (17.13) we obtain a diagram

$$(17.15) \quad \begin{array}{ccc} \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}, \mathrm{glob}})_Z & \xleftarrow{(\mathrm{ev}_Z)^*, \mathrm{IndCoh}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})_Z \\ (\iota^{\mathrm{mon-free}, \mathrm{glob}})^! \uparrow & \swarrow & \uparrow (\iota^{\mathrm{mon-free}})^! \\ \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}}^{\mathrm{mer}, \mathrm{glob}})_Z & \xleftarrow{(\mathrm{ev}_Z)^*, \mathrm{IndCoh}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mer}})_Z. \end{array}$$

We claim:

Lemma 17.3.7. *The natural transformation in (17.15) is an isomorphism.*

Proof. For expositional purposes we will assume that $Z = \mathrm{pt}$, so that $Z \rightarrow \mathrm{Ran}$ corresponds to $\underline{x} \in \mathrm{Ran}$. We will use the notations from Sect. 17.1.4.

For $\mathbf{L} \supset \mathbf{L}_0$ denote

$$\begin{aligned} \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L} \cap \mathrm{mon-free}} &:= \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}} \times_{\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}}} \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}, \\ \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L} \cap \mathrm{mon-free}, \mathrm{glob}} &:= \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}, \mathrm{glob}} \times_{\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mer}}} \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}, \end{aligned}$$

Using (17.7) and a similar diagram for “mer” replaced by “mon-free”, it suffices to show that the natural transformation in the diagram

$$\begin{array}{ccc} \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L} \cap \mathrm{mon-free}, \mathrm{glob}}) & \xleftarrow{(\mathrm{ev}_{\underline{x}})^*, \mathrm{IndCoh}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L} \cap \mathrm{mon-free}}) \\ (\iota^{\mathrm{mon-free}, \mathrm{glob}})^! \uparrow & \swarrow & \uparrow (\iota^{\mathrm{mon-free}})^! \\ \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}, \mathrm{glob}}) & \xleftarrow{(\mathrm{ev}_{\underline{x}})^*, \mathrm{IndCoh}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L}}). \end{array}$$

is an isomorphism.

According to Sect. 3.2.7, for a small enough lattice $\mathbf{L}' \subset \mathbf{L}$, we have a well-defined action of $\mathbf{L}'$ on $\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L} \cap \mathrm{mon-free}}$ by translations, and the quotient $\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathbf{L} \cap \mathrm{mon-free}} / \mathbf{L}'$ is a prestack locally almost of finite type.

We have a commutative diagram

$$\begin{array}{ccc} \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathbf{L} \cap \text{mon-free}}) & \xleftarrow{* \text{-pullback}} & \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathbf{L} \cap \text{mon-free}}/\mathbf{L}') \\ \uparrow (\iota^{\text{mon-free}})^! & & \uparrow (\iota^{\text{mon-free}}/\mathbf{L}')^! \\ \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathbf{L}}) & \xleftarrow{* \text{-pullback}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathbf{L}}/\mathbf{L}'). \end{array}$$

Hence, it suffices to show that the natural transformation in the diagram

$$\begin{array}{ccc} \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathbf{L} \cap \text{mon-free, glob}}) & \xleftarrow{* \text{-pullback}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathbf{L} \cap \text{mon-free}}/\mathbf{L}') \\ \uparrow (\iota^{\text{mon-free, glob}})^! & \nearrow & \uparrow (\iota^{\text{mon-free}}/\mathbf{L}')^! \\ \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathbf{L, glob}}) & \xleftarrow{* \text{-pullback}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathbf{L}}/\mathbf{L}'). \end{array}$$

is an isomorphism.

However, this follows from the fact that the diagram

$$\begin{array}{ccc} \mathrm{Op}_{\tilde{G},\underline{x}}^{\mathbf{L} \cap \text{mon-free, glob}} & \longrightarrow & \mathrm{Op}_{\tilde{G},\underline{x}}^{\mathbf{L} \cap \text{mon-free}}/\mathbf{L}' \\ \downarrow & & \downarrow \\ \mathrm{Op}_{\tilde{G},\underline{x}}^{\mathbf{L, glob}} & \longrightarrow & \mathrm{Op}_{\tilde{G},\underline{x}}^{\mathbf{L}}/\mathbf{L}' \end{array}$$

is Cartesian, combined with the fact that $\mathrm{Op}_{\tilde{G},\underline{x}}^{\mathbf{L}}/\mathbf{L}'$ is a smooth scheme. $\square$

17.3.8. Recall the functor

$$\Theta_{\mathrm{Op}_{\tilde{G},z}^{\text{mon-free}}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\text{mon-free}})_z \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\text{mon-free}})_z,$$

see Sect. 3.7.7.

We claim:

Proposition 17.3.9. *There exists a commutative diagram*

$$\begin{array}{ccc} \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\text{mon-free}})_z & \xrightarrow{\Theta_{\mathrm{Op}_{\tilde{G},z}^{\text{mon-free}}}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\text{mon-free}})_z \\ (\mathrm{ev}_z)^! \downarrow & & \downarrow (\mathrm{ev}_z)^*, \mathrm{IndCoh} \\ \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}}^{\text{mon-free, glob}})_z & \xrightarrow{-\otimes \mathbb{I}_{\mathrm{Kost}(\tilde{G})}[-\delta_G]} & \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}}^{\text{mon-free, glob}})_z. \end{array}$$

Proof. Both circuits of the diagram are $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\text{mon-free}})_z$ -linear functors. Hence, it suffices to identify the objects that correspond to the image of the unit.

I.e., we wish to identify

$$(17.16) \quad \omega_{\mathrm{Op}_{\tilde{G},z}^{\text{mon-free, glob}}} \otimes \mathbb{I}_{\mathrm{Kost}(\tilde{G})}[-\delta_G] \simeq (\mathrm{ev}_z)^*, \mathrm{IndCoh}(\omega_{\mathrm{Op}_{\tilde{G},z}^{\text{mer}}}^{*, \text{fake}}).$$

We start with

$$\omega_{\mathrm{Op}_{\tilde{G},z}^{\text{mer, glob}}} \otimes \mathbb{I}_{\mathrm{Kost}(\tilde{G})}[-\delta_G] \simeq (\mathrm{ev}_z)^*, \mathrm{IndCoh}(\omega_{\mathrm{Op}_{\tilde{G},z}^{\text{mer}}}^{*, \text{fake}}),$$

given by Proposition 17.2.4 and apply the functor $(\iota^{\text{mon-free, glob}})^!$.

The left-hand side gives the left-hand side of (17.16). The right-hand side gives the right-hand side of (17.16) thanks to Lemma 17.3.7. $\square$

17.4. Two versions of the spectral Poincaré functor.

17.4.1. For $\mathcal{Z} \rightarrow \text{Ran}$, we define the spectral !-Poincaré functor

$$\text{Poinc}_{\tilde{G},!,\mathcal{Z}}^{\text{spec}} : \text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free}})_{\mathcal{Z}} \rightarrow \text{IndCoh}(\text{LS}_{\tilde{G}}) \otimes \text{D-mod}(\mathcal{Z})$$

as

$$\text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free}})_{\mathcal{Z}} \xrightarrow{\text{ev}_{\mathcal{Z}}^!} \text{IndCoh}(\text{Op}_{\tilde{G}}^{\text{mon-free,glob}})_{\mathcal{Z}} \xrightarrow{(\text{r}^{\text{glob}})^{\text{IndCoh}}} \text{IndCoh}(\text{LS}_{\tilde{G}}) \otimes \text{D-mod}(\mathcal{Z}).$$

17.4.2. We define the spectral *-Poincaré functor

$$\text{Poinc}_{\tilde{G},*,\mathcal{Z}}^{\text{spec}} : \text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}})_{\mathcal{Z}} \rightarrow \text{IndCoh}(\text{LS}_{\tilde{G}}) \otimes \text{D-mod}(\mathcal{Z})$$

as

$$\text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}})_{\mathcal{Z}} \xrightarrow{\text{ev}_{\mathcal{Z}}^{*,\text{IndCoh}}} \text{IndCoh}(\text{Op}_{\tilde{G}}^{\text{mon-free,glob}})_{\mathcal{Z}} \xrightarrow{(\text{r}^{\text{glob}})^{\text{IndCoh}}} \text{IndCoh}(\text{LS}_{\tilde{G}}) \otimes \text{D-mod}(\mathcal{Z}).$$

17.4.3. Both

$$(17.17) \quad \mathcal{Z} \rightsquigarrow \text{Poinc}_{\tilde{G},!,\mathcal{Z}}^{\text{spec}} \text{ and } \mathcal{Z} \rightsquigarrow \text{Poinc}_{\tilde{G},*,\mathcal{Z}}^{\text{spec}}$$

are naturally local-to-global functors in the sense of Sect. 11.1.1, to be denoted

$$\underline{\text{Poinc}}_{\tilde{G},!}^{\text{spec}} \text{ and } \underline{\text{Poinc}}_{\tilde{G},*}^{\text{spec}},$$

respectively.

17.4.4. Furthermore, the assignments (17.17) have natural *unital* structures, in the sense of Sect. 11.3.5. Let us spell it out explicitly for the *-version (!-version is analogous).

The local unital structure on the source crystal of categories, i.e., $\text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}})$ assigns to

$$(\underline{x} \subseteq \underline{x}') \in \text{Ran}$$

the functor

$$\text{IndCoh}^*(\text{Op}_{\tilde{G},\underline{x}}^{\text{mon-free}}) \rightarrow \text{IndCoh}^*(\text{Op}_{\tilde{G},\underline{x}'}^{\text{mon-free}})$$

given by *-pull followed by *-push along the diagram

$$(17.18) \quad \begin{array}{ccc} & \text{Op}_{\tilde{G}}(\mathcal{D}_{\underline{x}' - \underline{x}}) \times_{\text{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}' - \underline{x}})} \text{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}'}) & \\ & \swarrow \quad \searrow & \\ \text{Op}_{\tilde{G}}(\mathcal{D}_{\underline{x} - \underline{x}}) \times_{\text{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x} - \underline{x}})} \text{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}}) & & \text{Op}_{\tilde{G}}(\mathcal{D}_{\underline{x}' - \underline{x}'} \times_{\text{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}' - \underline{x}'})} \text{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}'})) \\ \parallel & & \parallel \\ \text{Op}_{\tilde{G},\underline{x}}^{\text{mon-free}} & & \text{Op}_{\tilde{G},\underline{x}'}^{\text{mon-free}}, \end{array}$$

in which the slanted arrows are given by restriction along the inclusions

$$(\mathcal{D}_{\underline{x} - \underline{x}}) \rightarrow (\mathcal{D}_{\underline{x}' - \underline{x}}) \leftarrow (\mathcal{D}_{\underline{x}' - \underline{x}'}),$$

respectively.

17.4.5. Consider the diagram

$$\begin{array}{ccccc}
 \mathrm{Op}_{\tilde{G}, \underline{x}'}^{\mathrm{mon-free}} & & \xleftarrow{\mathrm{ev}_{\underline{x}'}} & \mathrm{Op}_{\tilde{G}, \underline{x}'}^{\mathrm{mon-free, glob}} & \xrightarrow{\tau_{\underline{x}'}^{\mathrm{glob}}} \mathrm{LS}_{\tilde{G}} \\
 \uparrow & & & \uparrow & \uparrow \mathrm{id} \\
 \mathrm{Op}_{\tilde{G}}(\mathcal{D}_{\underline{x}'} - \underline{x}) \times_{\mathrm{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x} - \underline{x}})} \mathrm{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}}) & \xleftarrow{\quad} & & \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free, glob}} & \xrightarrow{\tau_{\underline{x}}^{\mathrm{glob}}} \mathrm{LS}_{\tilde{G}} \\
 \downarrow & & & \downarrow \mathrm{ev}_{\underline{x}} & \\
 \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}} & \xleftarrow{\mathrm{id}} & & \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}, &
 \end{array}$$

in which the square

$$\begin{array}{ccc}
 \mathrm{Op}_{\tilde{G}, \underline{x}'}^{\mathrm{mon-free}} & \xleftarrow{\mathrm{ev}_{\underline{x}'}} & \mathrm{Op}_{\tilde{G}, \underline{x}'}^{\mathrm{mon-free, glob}} \\
 \uparrow & & \uparrow \\
 \mathrm{Op}_{\tilde{G}}(\mathcal{D}_{\underline{x}'} - \underline{x}) \times_{\mathrm{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x} - \underline{x}})} \mathrm{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}}) & \xleftarrow{\quad} & \mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free, glob}}
 \end{array}$$

is Cartesian.

By Sect. A.10.12, we have a commutative diagram

$$\begin{array}{ccc}
 \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}'}^{\mathrm{mon-free}}) & \xrightarrow{* \text{-pushforward}} & \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}'}^{\mathrm{mon-free, glob}}) \\
 \downarrow \text{!-pullback} & & \downarrow \text{!-pullback} \\
 \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}(\mathcal{D}_{\underline{x}'} - \underline{x}) \times_{\mathrm{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x} - \underline{x}})} \mathrm{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}})) & \xrightarrow{* \text{-pushforward}} & \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free, glob}}).
 \end{array}$$

Passing to left adjoints, we obtain a commutative diagram

$$\begin{array}{ccc}
 \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}'}^{\mathrm{mon-free}}) & \xrightarrow{* \text{-pullback}} & \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}'}^{\mathrm{mon-free, glob}}) \\
 \uparrow * \text{-pushforward} & & \uparrow * \text{-pushforward} \\
 \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}(\mathcal{D}_{\underline{x}'} - \underline{x}) \times_{\mathrm{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x} - \underline{x}})} \mathrm{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}})) & \xrightarrow{* \text{-pullback}} & \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free, glob}}).
 \end{array}$$

Now, the unital structure on $\mathrm{Poinc}_{\tilde{G}, *}^{\mathrm{spec}}$ is encoded by the following diagram:

$$\begin{array}{ccccc}
 \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}'}^{\mathrm{mon-free}}) & \xrightarrow{\mathrm{ev}_{\underline{x}'}^*, \mathrm{IndCoh}} & \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}'}^{\mathrm{mon-free, glob}}) & \xrightarrow{(\tau_{\underline{x}'}^{\mathrm{glob}})^* \mathrm{IndCoh}} & \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}) \\
 \uparrow * \text{-pushforward} & & \uparrow * \text{-pushforward} & & \uparrow \mathrm{id} \\
 \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}(\mathcal{D}_{\underline{x}'} - \underline{x}) \times_{\mathrm{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x} - \underline{x}})} \mathrm{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}})) & \xrightarrow{* \text{-pullback}} & \mathrm{IndCoh}(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free, glob}}) & \xrightarrow{(\tau_{\underline{x}}^{\mathrm{glob}})^* \mathrm{IndCoh}} & \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}) \\
 \uparrow * \text{-pullback} & & \uparrow \mathrm{ev}_{\underline{x}}^* & & \\
 \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}) & \xrightarrow{\mathrm{Id}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \underline{x}}^{\mathrm{mon-free}}) & &
 \end{array}$$

17.4.6. Note that from Proposition 17.3.9 we obtain:

Theorem 17.4.7. *There is an isomorphism of local-to-global functors*

$$\mathrm{Poinc}_{\tilde{G}, !}^{\mathrm{spec}} \otimes \mathfrak{l}_{\mathrm{Kost}(\tilde{G})}[-\delta_G] \simeq \mathrm{Poinc}_{\tilde{G}, *}^{\mathrm{spec}} \circ \Theta_{\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}}.$$

Remark 17.4.8. The reason that we discuss both $\underline{\text{Poinc}}_{\check{G},!}^{\text{spec}}$ and $\underline{\text{Poinc}}_{\check{G},*}^{\text{spec}}$ (despite the fact that, thanks to Theorem 17.4.7, they are easily expressible one through another) is that the $!$ -version is naturally compatible with Eisenstein series (which we will exploit in the sequel to this paper), and the $*$ -version is naturally compatible with the functor

$$\Gamma^{\text{IndCoh}}(\text{LS}_{\check{G}}, -) : \text{IndCoh}(\text{LS}_{\check{G}}) \rightarrow \text{Vect},$$

which we will use in the next section.

17.5. Action of the spectral spherical category and temperedness. In this subsection we will work with $\underline{\text{Poinc}}_{\check{G},*}^{\text{spec}}$, but a parallel discussion is applicable to $\underline{\text{Poinc}}_{\check{G},!}^{\text{spec}}$.

17.5.1. For a fixed $\underline{x} \in \text{Ran}$ we have a naturally defined action of $\text{Sph}_{\check{G},\underline{x}}^{\text{spec}}$ on $\text{IndCoh}(\text{LS}_{\check{G}})$. Namely, it is given by $*$ -pull followed by $*$ -push along the following diagram

$$\begin{array}{ccccc} \text{LS}_{\check{G}} & \longleftarrow & \text{Hecke}_{\check{G},\underline{x}}^{\text{spec, glob}} & \longrightarrow & \text{LS}_{\check{G}} \\ \text{ev}_{\underline{x}} \downarrow & & \text{ev}_{\underline{x}} \downarrow & & \downarrow \text{ev}_{\underline{x}} \\ \text{LS}_{\check{G},\underline{x}}^{\text{reg}} & \longleftarrow & \text{Hecke}_{\check{G},\underline{x}}^{\text{spec, loc}} & \longrightarrow & \text{LS}_{\check{G},\underline{x}}^{\text{reg}}, \end{array}$$

in which both squares are Cartesian.

Remark 17.5.2. In Sect. 18.2.1 we will consider a Ran version of this action. This involves some technical difficulties, inherent to the definition of $\text{Sph}_{\check{G}}^{\text{spec}}$ as a factorization category, see Sect. 1.6.4.

17.5.3. We have a natural action of $\text{Sph}_{\check{G},\underline{x}}^{\text{spec}}$ on $\text{IndCoh}^*(\text{Op}_{\check{G},\text{Ran}_{\underline{x}}}^{\text{mon-free}})$ given by pull-push along the following diagram

$$\begin{array}{ccccc} \text{Op}_{\check{G},\underline{x}'}^{\text{mon-free}} & \longleftarrow & \text{Hecke}_{\check{G},\underline{x}}^{\text{spec, Op}_{\check{G},\underline{x}'}^{\text{mon-free}}} & \longrightarrow & \text{Op}_{\check{G},\underline{x}'}^{\text{mon-free}} \\ \downarrow & & \downarrow & & \downarrow \\ \text{LS}_{\check{G},\underline{x}}^{\text{reg}} & \longleftarrow & \text{Hecke}_{\check{G},\underline{x}}^{\text{spec, loc}} & \longrightarrow & \text{LS}_{\check{G},\underline{x}}^{\text{reg}}, \end{array}$$

in which both squares are Cartesian, where:

- $\underline{x} \subseteq \underline{x}'$;
- $\text{Hecke}_{\check{G},\underline{x}}^{\text{spec, Op}_{\check{G},\underline{x}'}^{\text{mon-free}}} := \text{Op}_{\check{G},\underline{x}'}^{\text{mer}} \times_{\text{LS}_{\check{G},\underline{x}'}^{\text{mer}}} (\text{LS}_{\check{G},\underline{x}'}^{\text{reg}} \times_{\text{LS}_{\check{G},\underline{x} \subseteq \underline{x}'}^{\text{mer} \rightsquigarrow \text{reg}}} \text{LS}_{\check{G},\underline{x}}^{\text{reg}})$;
- $\text{LS}_{\check{G},\underline{x} \subseteq \underline{x}'}^{\text{mer} \rightsquigarrow \text{reg}} := \text{LS}_{\check{G}}(\mathcal{D}_{\underline{x}'} - \underline{x})$.

17.5.4. Consider the functor

$$\text{Poinc}_{\check{G},*,\text{Ran}_{\underline{x}}}^{\text{spec}} : \text{IndCoh}^*(\text{Op}_{\check{G},\text{Ran}_{\underline{x}}}^{\text{mon-free}}) \rightarrow \text{IndCoh}(\text{LS}_{\check{G}}) \otimes \text{D-mod}(\text{Ran}_{\underline{x}}).$$

The following results by unwinding the constructions:

Lemma 17.5.5. *The functor $\text{Poinc}_{\check{G},*,\text{Ran}_{\underline{x}}}^{\text{spec}}$ intertwines the actions of $\text{Sph}_{\check{G},\underline{x}}^{\text{spec}}$ on the two sides.*

Remark 17.5.6. One can define an action of $\text{Sph}_{\check{G}}^{\text{spec}}$ on $\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})$ as a factorization category. Furthermore, one can show that the functor $\underline{\text{Poinc}}_{\check{G},*}^{\text{spec}}$ is compatible with the action of $\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})$, where the latter is thought of as a crystal of monoidal categories over Ran.

Moreover, the above action and compatibility are in turn compatible with the unital structures. This will be performed in Sect. E.8.

17.5.7. Recall (see [AG, Sect. 12.8.2]) that the subcategory

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \subset \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})$$

can be singled out by the *temperedness* condition:

Namely, it is the maximal subcategory on which for some/any $\underline{x} \in \mathrm{Ran}$, the action of $\mathrm{Sph}_{\check{G}, \underline{x}}^{\mathrm{spec}}$ on $\mathrm{IndCoh}(\mathrm{LS}_{\check{G}})$ factors via

$$\mathrm{Sph}_{\check{G}, \underline{x}}^{\mathrm{spec}} \rightarrow \mathrm{Sph}_{\check{G}, \mathrm{temp}, \underline{x}}^{\mathrm{spec}},$$

see Sect. 7.1.1.

17.5.8. A basic property of the spectral Poincaré functor is the following:

Proposition 17.5.9. *The essential image of the functor*

$$\mathrm{Poinc}_{\check{G}, * }^{\mathrm{spec}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})$$

lies in

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \subset \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}).$$

Proof. Choose some/any $\underline{x} \in \mathrm{Ran}$. By the unital property of $\mathrm{Poinc}_{\check{G}, * }^{\mathrm{spec}}$, it suffices to show that the functor

$$\mathrm{Poinc}_{\check{G}, *, \mathrm{Ran}_{\underline{x}}}^{\mathrm{spec}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathrm{Ran}_{\underline{x}})$$

takes values in

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathrm{Ran}_{\underline{x}}) \subset \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathrm{Ran}_{\underline{x}}).$$

By Sect. 17.5.7 and Lemma 17.5.5, it suffices to show that the $\mathrm{Sph}_{\check{G}, \underline{x}}^{\mathrm{spec}}$ -action on $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free}})$ factors via $\mathrm{Sph}_{\check{G}, \mathrm{temp}, \underline{x}}^{\mathrm{spec}}$.

The latter assertion can be checked strata-wise, so it is enough to show that for a fixed $\underline{x} \subseteq \underline{x}'$, the action of $\mathrm{Sph}_{\check{G}, \underline{x}}^{\mathrm{spec}}$ on $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}'}}^{\mathrm{mon-free}})$ factors via $\mathrm{Sph}_{\check{G}, \mathrm{temp}, \underline{x}}^{\mathrm{spec}}$.

Write $\underline{x}' = \underline{x} \sqcup \underline{x}''$. In terms of the factorization

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}'}}^{\mathrm{mon-free}}) \simeq \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}}}^{\mathrm{mon-free}}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}''}}^{\mathrm{mon-free}}),$$

the action of $\mathrm{Sph}_{\check{G}, \underline{x}}^{\mathrm{spec}}$ on $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \mathrm{Ran}_{\underline{x}'}}^{\mathrm{mon-free}})$ is via the first factor.

The required assertion follows now from Proposition 7.2.4. □

Remark 17.5.10. Note that thanks to Theorem 17.4.7, we obtain that an assertion parallel to Proposition 17.5.9 holds for the functor $\mathrm{Poinc}_{\check{G}, ! }^{\mathrm{spec}}$. (Alternatively, one can prove it by the same argument.)

17.6. The spectral localization and global sections functors.

17.6.1. The spectral localization functor $\mathrm{Loc}_{\check{G}}^{\mathrm{spec}}$, i.e., the collection functors

$$\mathrm{Loc}_{\check{G}, \mathbb{Z}}^{\mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathbb{Z}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z})$$

for $\mathbb{Z} \rightarrow \mathrm{Ran}$, is defined as pullback along

$$(17.19) \quad \mathrm{LS}_{\check{G}} \times \mathbb{Z} \rightarrow \mathrm{LS}_{\check{G}, \mathbb{Z}}^{\mathrm{reg}},$$

where we identify

$$\mathrm{Rep}(\check{G})_{\mathbb{Z}} \simeq \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{reg}})_{\mathbb{Z}}.$$

The functor $\mathrm{Loc}_{\check{G}}^{\mathrm{spec}}$ possesses a natural unital structure (see Sect. 11.3.5 for what this means).

17.6.2. The functor

$$\mathrm{Loc}_{\check{G}}^{\mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}})$$

admits a right adjoint, denoted

$$\Gamma_{\check{G}}^{\mathrm{spec}} : \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \rightarrow \mathrm{Rep}(\check{G})_{\mathrm{Ran}}.$$

Explicitly, for a given $\underline{x} \in \mathrm{Ran}$, the corresponding functor

$$\Gamma_{\check{G}, \underline{x}}^{\mathrm{spec}} : \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \rightarrow \mathrm{Rep}(\check{G})_{\underline{x}}$$

is given by $*$ -direct image along

$$\mathrm{LS}_{\check{G}} \rightarrow \mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{reg}}.$$

17.6.3. Note also that the categories $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ and $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}$ are both canonically self-dual, and with respect to these dualities, we have

$$(\mathrm{Loc}_{\check{G}}^{\mathrm{spec}})^{\vee} \simeq \Gamma_{\check{G}}^{\mathrm{spec}}.$$

17.6.4. By a slight abuse of notation we will denote by the same symbol $\mathrm{Loc}_{\check{G}}^{\mathrm{spec}}$ the composite functor

$$\mathrm{Rep}(\check{G})_{\mathrm{Ran}} \xrightarrow{\mathrm{Loc}_{\check{G}}^{\mathrm{spec}}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{\Xi_{\mathrm{LS}_{\check{G}}}} \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}).$$

We will denote by $\Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}}$ the functor

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{\Psi_{\mathrm{LS}_{\check{G}}}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{\Gamma_{\check{G}}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_{\mathrm{Ran}}.$$

The functors

$$\mathrm{Loc}_{\check{G}}^{\mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightleftarrows \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) : \Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}}$$

also form an adjoint pair.

17.6.5. Note that the category $\mathrm{IndCoh}(\mathrm{LS}_{\check{G}})$ is also self-dual by means of *Serre* duality. Under this duality and the standard self-duality of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$, we have

$$\Psi_{\mathrm{LS}_{\check{G}}}^{\vee} \simeq \Upsilon_{\mathrm{LS}_{\check{G}}}.$$

However, note that $\mathrm{LS}_{\check{G}}$ is quasi-smooth and Calabi-Yau:

$$\omega_{\mathrm{LS}_{\check{G}}} = \mathcal{O}_{\mathrm{LS}_{\check{G}}} \otimes \det(\mathrm{Lie}(Z_G))^{\otimes (2-2g)} [2\delta_G].$$

Hence, we have

$$\Upsilon_{\mathrm{LS}_{\check{G}}} \simeq \Xi_{\mathrm{LS}_{\check{G}}} \otimes \det(\mathrm{Lie}(Z_G))^{\otimes (2-2g)} [2\delta_G].$$

17.6.6. Hence, we obtain that with respect to the self-duality of $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}$ and the Serre duality of $\mathrm{IndCoh}(\mathrm{LS}_{\check{G}})$, we have

$$(\mathrm{Loc}_{\check{G}}^{\mathrm{spec}})^{\vee} \simeq \Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}} \otimes \det(\mathrm{Lie}(Z_G))^{\otimes (2-2g)} [2\delta_G].$$

17.7. Composing spectral Poincaré and global sections functors.

17.7.1. Our current goal is to study the composite functor

$$(17.20) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \xrightarrow{\mathrm{Poinc}_{\check{G},*}^{\mathrm{spec}}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{\Gamma_{\check{G}}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_{\mathrm{Ran}}.$$

Applying the canonical self-duality of $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}$, the datum of the functor (17.20) is equivalent to the datum of the pairing

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \otimes \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightarrow \mathrm{Vect},$$

given by

$$(17.21) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \otimes \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \xrightarrow{\mathrm{Poinc}_{\check{G},*}^{\mathrm{spec}} \otimes \mathrm{Id}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightarrow \\ \xrightarrow{\Gamma_{\check{G}}^{\mathrm{spec}} \otimes \mathrm{Id}} \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \otimes \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightarrow \mathrm{Vect}.$$

We will prove (cf. Theorem 14.2.4):

Theorem 17.7.2. *The functor (17.21) identifies canonically with*

$$(17.22) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \otimes \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \xrightarrow{\mathrm{ins.unit}_{\mathrm{Ran}} \otimes \mathrm{ins.unit}_{\mathrm{Ran}}} \\ \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}^{\subseteq}} \otimes \mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\subseteq}} \rightarrow \\ \rightarrow \left(\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes \mathrm{Rep}(\check{G}) \right)_{\mathrm{Ran}^{\subseteq} \times \mathrm{Ran}^{\subseteq}} \xrightarrow{\mathrm{P}_{\check{G}}^{\mathrm{loc,enh}}} \\ \rightarrow \mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg-mod}} \times \mathrm{Ran}^{\subseteq}}^{\mathrm{fact}} \xrightarrow{\mathrm{C}_{\check{G}}^{\mathrm{fact}}(X; \mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}, -)_{\mathrm{Ran}^{\subseteq}} \times \mathrm{Ran}^{\subseteq}} \mathrm{D-mod}(\mathrm{Ran}^{\subseteq} \times \mathrm{Ran}^{\subseteq}) \rightarrow \\ \xrightarrow{\mathrm{C}_{\check{G}}(\mathrm{Ran}^{\subseteq} \times \mathrm{Ran}^{\subseteq}, -)} \mathrm{Vect},$$

where $\mathrm{P}_{\check{G}}^{\mathrm{loc,enh}}$ is the functor introduced in Sect. 6.4.7.

Remark 17.7.3. Note that the functor (17.22), appearing in Theorem 17.7.2 can also be rewritten as

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \otimes \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \xrightarrow{\mathrm{ins.unit}_{\mathrm{Ran}} \otimes \mathrm{ins.unit}_{\mathrm{Ran}}} \\ \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}^{\subseteq}} \otimes \mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\subseteq}} \rightarrow \\ \left(\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes \mathrm{Rep}(\check{G}) \right)_{\mathrm{Ran}^{\subseteq} \times \mathrm{Ran}^{\subseteq}} \xrightarrow{\mathrm{P}_{\check{G}}^{\mathrm{loc}}} \mathrm{D-mod}(\mathrm{Ran}^{\subseteq} \times \mathrm{Ran}^{\subseteq}) \xrightarrow{\mathrm{C}_{\check{G}}(\mathrm{Ran}^{\subseteq} \times \mathrm{Ran}^{\subseteq})} \mathrm{Vect},$$

i.e., instead of $\mathrm{C}_{\check{G}}^{\mathrm{fact}}(X; \mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}, -)_{\mathrm{Ran}^{\subseteq} \times \mathrm{Ran}^{\subseteq}}$ we can use the functor $\mathrm{oblv}_{\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}, \mathrm{Ran}^{\subseteq} \times \mathrm{Ran}^{\subseteq}}$. This follows by the same manipulation as in Remark 14.2.5

17.7.4. The rest of this subsection is devoted to the proof of Theorem 17.7.2.

First, using the (non-derived) Satake action, as in the proof of Theorem 14.2.4, we obtain that the assertion of the theorem is equivalent to that of the following:

Theorem 17.7.5. *The functor*

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \xrightarrow{\mathrm{Poinc}_{\check{G},*}^{\mathrm{spec}}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{\Gamma(\mathrm{LS}_{\check{G}}, -)} \mathrm{Vect}$$

identifies canonically with

$$(17.23) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \xrightarrow{(\iota^{\mathrm{mon-free}})^* \mathrm{IndCoh}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}})_{\mathrm{Ran}} \xrightarrow{\Gamma^{\mathrm{IndCoh}}(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}, -)^{\mathrm{enh}}} \\ \rightarrow \mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg-mod}} \times \mathrm{Ran}^{\subseteq}}^{\mathrm{fact}} \xrightarrow{\mathrm{C}_{\check{G}}^{\mathrm{fact}}(X; \mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}, -)} \mathrm{Vect}.$$

17.7.6. For expositional purposes, will replace the situation over Ran by one with a fixed $\underline{x} \in \text{Ran}$. So, we want to show that the composition

$$(17.24) \quad \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}}) \xrightarrow{\text{Poinc}_{\tilde{G}, *, \underline{x}}^{\text{spec}}} \text{QCoh}(\text{LS}_{\tilde{G}}) \xrightarrow{\Gamma(\text{LS}_{\tilde{G}}, -)} \text{Vect}$$

identifies canonically with

$$(17.25) \quad \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}}) \xrightarrow{(\iota^{\text{mon-free}})^* \text{IndCoh}} \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{\Gamma \text{IndCoh}(\text{Op}_{\tilde{G}}^{\text{mer}}, -)^{\text{enh}}} \\ \rightarrow \mathcal{O}_{\text{Op}_{\tilde{G}}^{\text{reg-mod}} \underline{x}}^{\text{fact}} \xrightarrow{C^{\text{ch}}(X, \mathcal{O}_{\text{Op}_{\tilde{G}}^{\text{reg}}, -})_{\underline{x}}} \text{Vect}.$$

17.7.7. The functor (17.24) can be tautologically rewritten as the composition

$$\text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}}) \xrightarrow{\text{ev}_{\underline{x}}^*, \text{IndCoh}} \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free, glob}}) \xrightarrow{\Gamma \text{IndCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free, glob}}, -)} \text{Vect},$$

and further as

$$(17.26) \quad \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}}) \xrightarrow{\text{ev}_{\underline{x}}^*, \text{IndCoh}} \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free, glob}}) \xrightarrow{(\iota^{\text{mon-free, glob}})^* \text{IndCoh}} \\ \rightarrow \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer, glob}}) \xrightarrow{\Gamma \text{IndCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer, glob}}, -)} \text{Vect}.$$

Applying (17.14), we rewrite (17.26) as

$$(17.27) \quad \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mon-free}}) \xrightarrow{(\iota^{\text{mon-free}})^* \text{IndCoh}} \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{\text{ev}_{\underline{x}}^*, \text{IndCoh}} \\ \rightarrow \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer, glob}}) \xrightarrow{\Gamma \text{IndCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer, glob}}, -)} \text{Vect}.$$

Thus, it suffices to establish an isomorphism between

$$(17.28) \quad \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{\text{ev}_{\underline{x}}^*, \text{IndCoh}} \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer, glob}}) \xrightarrow{\Gamma \text{IndCoh}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer, glob}}, -)} \text{Vect}$$

and

$$(17.29) \quad \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{\Gamma \text{IndCoh}(\text{Op}_{\tilde{G}}^{\text{mer}}, -)^{\text{enh}}} \mathcal{O}_{\text{Op}_{\tilde{G}}^{\text{reg-mod}} \underline{x}}^{\text{fact}} \xrightarrow{C^{\text{ch}}(X, \mathcal{O}_{\text{Op}_{\tilde{G}}^{\text{reg}}, -})_{\underline{x}}} \text{Vect}.$$

17.7.8. By Lemma 17.1.9, we can rewrite (17.28) as

$$(17.30) \quad \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{\Psi_{\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}}^*} \text{QCoh}_{\text{co}}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{\text{ev}_{\underline{x}}^*} \text{QCoh}_{\text{co}}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer, glob}}) \xrightarrow{\Gamma(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer, glob}}, -)} \text{Vect},$$

while (17.29) is by definition

$$(17.31) \quad \text{IndCoh}^*(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{\Psi_{\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}}^*} \text{QCoh}_{\text{co}}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{\Gamma(\text{Op}_{\tilde{G}}^{\text{mer}}, -)^{\text{enh}}} \mathcal{O}_{\text{Op}_{\tilde{G}}^{\text{reg-mod}} \underline{x}}^{\text{fact}} \xrightarrow{C^{\text{ch}}(X, \mathcal{O}_{\text{Op}_{\tilde{G}}^{\text{reg}}, -})_{\underline{x}}} \text{Vect}.$$

Hence, it suffices to establish an isomorphism between

$$(17.32) \quad \text{QCoh}_{\text{co}}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{\text{ev}_{\underline{x}}^*} \text{QCoh}_{\text{co}}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer, glob}}) \xrightarrow{\Gamma(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer, glob}}, -)} \text{Vect}$$

and

$$(17.33) \quad \text{QCoh}_{\text{co}}(\text{Op}_{\tilde{G}, \underline{x}}^{\text{mer}}) \xrightarrow{\Gamma(\text{Op}_{\tilde{G}}^{\text{mer}}, -)^{\text{enh}}} \mathcal{O}_{\text{Op}_{\tilde{G}}^{\text{reg-mod}} \underline{x}}^{\text{fact}} \xrightarrow{C^{\text{ch}}(X, \mathcal{O}_{\text{Op}_{\tilde{G}}^{\text{reg}}, -})_{\underline{x}}} \text{Vect}.$$

However, the latter is the statement of Proposition F.4.4.

18. THE LANGLANDS FUNCTOR

In this section we recall the construction of the Langlands functor, and establish the following of its properties:

- Compatibility with the functors coeff_G and $\Gamma_{\check{G}}^{\text{spec}}$;
- Compatibility with the actions of Sph_G and $\text{Sph}_{\check{G}}^{\text{spec}}$;
- Compatibility with the functors Loc_G and $\text{Poinc}_{\check{G},*}^{\text{spec}}$.

18.1. Recollections on the Langlands functor—the coarse version. In this and the next subsections we recall the construction of the *coarse version of the* Langlands functor

$$(18.1) \quad \mathbb{L}_{G,\text{coarse}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{QCoh}(\text{LS}_{\check{G}}).$$

18.1.1. We consider $(\text{Rep}(\check{G})_{\text{Ran}})^*$ as a monoidal category (see Sect. H.5.2), and

$$(18.2) \quad \text{Loc}_{\check{G}}^{\text{spec}} : (\text{Rep}(\check{G})_{\text{Ran}})^* \rightarrow \text{QCoh}(\text{LS}_{\check{G}})$$

as a monoidal functor. Recall that $\text{Loc}_{\check{G}}^{\text{spec}}$ is a *localization*, i.e., its right adjoint is fully faithful (the proof is given, e.g., in [GLC4, Corollary C.1.8 and Sect. C.1.9]).

18.1.2. We consider $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ as acted on by $(\text{Rep}(\check{G})_{\text{Ran}})^*$ via the action of $(\text{Sph}_{G,\text{Ran}})^*$ on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ (see Sect. H.6.8) and

$$\text{Sat}_G^{\text{nv}} : (\text{Rep}(\check{G})_{\text{Ran}})^* \rightarrow (\text{Sph}_{G,\text{Ran}})^*.$$

According to [Ga1, Corollary 4.5.5], the action of $(\text{Rep}(\check{G})_{\text{Ran}})^*$ on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ factors through (18.2), so we obtain an action on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ of $\text{QCoh}(\text{LS}_{\check{G}})$.

18.1.3. The coarse Langlands functor

$$\mathbb{L}_{G,\text{coarse}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{QCoh}(\text{LS}_{\check{G}}),$$

as constructed in [GLC1, Sect. 1.4], is uniquely characterized by the following two properties:

- The functor $\mathbb{L}_{G,\text{coarse}}$ is $\text{QCoh}(\text{LS}_{\check{G}})$ -linear;
- The diagram

$$(18.3) \quad \begin{array}{ccc} \text{Vect} & \xrightarrow{\text{Id}} & \text{Vect} \\ \text{coeff}_G^{\text{vac, glob}} \uparrow & & \uparrow \Gamma(\text{LS}_{\check{G}}, -) \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_{G,\text{coarse}}} & \text{QCoh}(\text{LS}_{\check{G}}) \end{array}$$

commutes.

18.1.4. Note that since $\text{Loc}_{\check{G}}^{\text{spec}}$ is a localization, the second property can be equivalently formulated as linearity with respect to $(\text{Rep}(\check{G})_{\text{Ran}})^*$.

By Corollary H.6.11, we obtain that the functor

$$\mathbb{L}_{G,\text{coarse}} \otimes \text{Id} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}) \rightarrow \text{QCoh}(\text{LS}_{\check{G}}) \otimes \text{D-mod}(\text{Ran})$$

intertwines the actions of $(\text{Rep}(\check{G})_{\text{Ran}})^{\otimes}$ on both sides.

In fact, Corollary H.6.11 implies that the functor

$$\mathbb{L}_{G,\text{coarse}} \otimes \text{Id} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \underline{\text{D-mod}}(\text{Ran}^{\text{untl}}) \rightarrow \text{QCoh}(\text{LS}_{\check{G}}) \otimes \underline{\text{D-mod}}(\text{Ran}^{\text{untl}})$$

intertwines the actions of $\text{Rep}(\check{G})$, viewed as a crystal of monoidal categories over Ran^{untl} .

18.1.5. We now claim:

Proposition 18.1.6. *The following diagram commutes:*

$$(18.4) \quad \begin{array}{ccc} \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xrightarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{coeff}_G[2\delta_{N_{\rho(\omega_X)}}]^\dagger \uparrow & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_{G,\mathrm{coarse}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}), \end{array}$$

where $\delta_{N_{\rho(\omega_X)}}$ is as in Sect. 9.6.5.

Proof. It suffices to construct the datum of commutativity for the diagram

$$(18.5) \quad \begin{array}{ccc} \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xrightarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{coeff}_{G,\mathrm{Ran}}[2\delta_{N_{\rho(\omega_X)}}]^\dagger \uparrow & & \uparrow \Gamma_{\check{G},\mathrm{Ran}}^{\mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathrm{Ran}) & \xrightarrow{\mathbb{L}_{G,\mathrm{coarse}} \otimes \mathrm{Id}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathrm{Ran}). \end{array}$$

Consider the categories appearing in (18.5) as equipped with an action of $(\mathrm{Rep}(\check{G})_{\mathrm{Ran}})^\dagger$. We will construct a datum of commutativity of (18.5) as $(\mathrm{Rep}(\check{G})_{\mathrm{Ran}})^\dagger$ -module categories.

Note, however, that the upper right corner, i.e.,

$$\mathrm{Rep}(\check{G})_{\mathrm{Ran}} = (\mathrm{Rep}(\check{G})_{\mathrm{Ran}})^\dagger,$$

is *co-free*, when viewed as a module over itself. Hence, the datum of commutativity of (18.5) as $(\mathrm{Rep}(\check{G})_{\mathrm{Ran}})^\dagger$ -module categories, is equivalent to the datum of commutativity of the *outer diagram* in

$$(18.6) \quad \begin{array}{ccc} \mathrm{D-mod}(\mathrm{Ran}) & \xrightarrow{\mathrm{Id}} & \mathrm{D-mod}(\mathrm{Ran}) \\ \uparrow & & \uparrow \\ \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xrightarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{coeff}_{G,\mathrm{Ran}}[2\delta_{N_{\rho(\omega_X)}}]^\dagger \uparrow & & \uparrow \Gamma_{\check{G},\mathrm{Ran}}^{\mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathrm{Ran}) & \xrightarrow{\mathbb{L}_{G,\mathrm{coarse}} \otimes \mathrm{Id}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathrm{Ran}), \end{array}$$

as $\mathrm{D-mod}(\mathrm{Ran})$ -linear categories, where:

- The upper right vertical arrow is the factorization functor $\mathrm{inv}_{\check{G}} : \mathrm{Rep}(\check{G}) \rightarrow \mathrm{Vect}$;
- The upper left vertical arrow is the factorization functor

$$\mathrm{Whit}^!(G)_{\mathrm{Ran}} \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\rho(\omega_X)}) \rightarrow \mathrm{Vect},$$

where the second arrow is the functor of $!$ -fiber at the unit.

In its turn, the datum of commutativity of the outer diagram in (18.6) is equivalent to the datum of commutativity of

$$(18.7) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}(\mathrm{Ran}) & \xrightarrow{\mathrm{Id}} & \mathrm{D}\text{-mod}(\mathrm{Ran}) \\ \uparrow & & \uparrow \\ \mathrm{Whit}^!(G)_{\mathrm{Ran}} & & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{coeff}_{G,\mathrm{Ran}}[2\delta_{N_{\rho(\omega_X)}}] \uparrow & & \uparrow \Gamma_{\check{G},\mathrm{Ran}}^{\mathrm{spec}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}) & & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}) \\ \mathrm{Id} \otimes \omega_{\mathrm{Ran}} \uparrow & & \uparrow \mathrm{Id} \otimes \omega_{\mathrm{Ran}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_{G,\mathrm{coarse}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \end{array}$$

just as DG categories.

Note, however, that the composite left vertical arrow in (18.7) is the functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{\mathrm{coeff}_G^{\mathrm{Vac}}[2\delta_{N_{\rho(\omega_X)}}]} \mathrm{Vect} \xrightarrow{\omega_{\mathrm{Ran}}} \mathrm{D}\text{-mod}(\mathrm{Ran})$$

and the composite right vertical arrow in (18.7) is the functor

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{\Gamma(\mathrm{LS}_{\check{G}}, -)} \mathrm{Vect} \xrightarrow{\omega_{\mathrm{Ran}}} \mathrm{D}\text{-mod}(\mathrm{Ran}).$$

Now, the required commutativity is supplied by (18.3), combined with Lemma 9.6.7. $\square$

18.2. Compatibility with the full spherical action.

18.2.1. As was mentioned in Sect. 17.5.1, for a fixed $\underline{x} \in \mathrm{Ran}$, the category $\mathrm{IndCoh}(\mathrm{LS}_{\check{G}})$ carries an action of $\mathrm{Sph}_{\check{G},\underline{x}}^{\mathrm{spec}}$.

In Sect. E.7.1 we will extend this to an action of $(\mathrm{Sph}_{\check{G},\mathrm{Ran}}^{\mathrm{spec}})^{\dagger}$ on $\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran})$. In fact, we have an action of $\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$, viewed as a crystal of monoidal categories over $\mathrm{Ran}^{\mathrm{unl}}$ on $\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \underline{\mathrm{D}\text{-mod}}(\mathrm{Ran}^{\mathrm{unl}})$.

By Sect. H.6.8, this gives rise to an action of

$$(\mathrm{Sph}_{\check{G},\mathrm{Ran}}^{\mathrm{spec}})^{\star} \rightarrow \mathrm{Sph}_{\check{G},\mathrm{Ran}^{\mathrm{unl}},\mathrm{indep}}^{\mathrm{spec}}$$

on $\mathrm{IndCoh}(\mathrm{LS}_{\check{G}})$.

18.2.2. A basic feature of the above action is that the functor

$$\Gamma_{\check{G},\mathrm{Ran}}^{\mathrm{spec},\mathrm{IndCoh}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}) \rightarrow \mathrm{Rep}(\check{G})_{\mathrm{Ran}}$$

is compatible with the actions of $(\mathrm{Sph}_{\check{G},\mathrm{Ran}}^{\mathrm{spec}})^{\dagger}$ on the two sides.

18.2.3. We claim:

Lemma 18.2.4.

(a) *The action of $(\mathrm{Sph}_{\check{G},\mathrm{Ran}}^{\mathrm{spec}})^{\dagger}$ on $\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran})$ preserves the subcategory*

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}) \xrightarrow{\Xi_{\mathrm{LS}_{\check{G}}} \otimes \mathrm{Id}} \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}).$$

(b) *The resulting action of $(\mathrm{Sph}_{\check{G},\mathrm{Ran}}^{\mathrm{spec}})^{\dagger}$ on $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran})$ is compatible with the projection*

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}) \xrightarrow{\Psi_{\mathrm{LS}_{\check{G}}} \otimes \mathrm{Id}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}).$$

Proof. To prove point (a), it suffices to show that the generators of $(\mathrm{Sph}_{\check{G}, \mathrm{Ran}}^{\mathrm{spec}})^{\dagger}$ preserve the subcategory $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathrm{Ran})$. We take these generators to be the essential image of the factorization functor

$$\mathrm{nv} : \mathrm{Rep}(\check{G}) \rightarrow \mathrm{Sph}_{\check{G}}^{\mathrm{spec}}.$$

This makes the assertion evident: the resulting action is the natural action of $(\mathrm{Rep}(\check{G})_{\mathrm{Ran}})^{\dagger}$ on $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathrm{Ran})$.

Point (b) of the lemma follows similarly. $\square$

Remark 18.2.5. Note that for a fixed $\underline{x} \in \mathrm{Ran}$, the action of $\mathrm{Sph}_{\check{G}, \underline{x}}^{\mathrm{spec}}$ on $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ factors through the quotient

$$\mathrm{Sph}_{\check{G}, \underline{x}}^{\mathrm{spec}} \twoheadrightarrow \mathrm{Sph}_{\check{G}, \mathrm{temp}, \underline{x}}^{\mathrm{spec}},$$

see Sect. 7.1.1. This follows from the fact that the action of $\mathrm{Sph}_{\check{G}, \underline{x}}^{\mathrm{spec}}$ on $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ is given by t-exact functors, combined with the fact that the t-structure on $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ is separated.

We do *not* know how to formulate a parallel property for the action of $(\mathrm{Sph}_{\check{G}, \mathrm{Ran}}^{\mathrm{spec}})^{\dagger}$ on the category $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathrm{Ran})$, see Remark 7.1.2.

18.2.6. We now claim:

Proposition 18.2.7. *The functor*

$$\mathbb{L}_{G, \mathrm{coarse}} \otimes \mathrm{Id} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathrm{Ran}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathrm{Ran})$$

intertwines the $(\mathrm{Sph}_{G, \mathrm{Ran}})^{\dagger}$ action on the left-hand side with the $(\mathrm{Sph}_{\check{G}, \mathrm{Ran}}^{\mathrm{spec}})^{\dagger}$ -action on the right-hand side via the functor

$$(\mathrm{Sph}_{G, \mathrm{Ran}})^{\dagger} \rightarrow (\mathrm{Sph}_{\check{G}, \mathrm{Ran}}^{\mathrm{spec}})^{\dagger},$$

induced by the factorization functor

$$\mathrm{Sat}_G : \mathrm{Sph}_G \rightarrow \mathrm{Sph}_{\check{G}}^{\mathrm{spec}}.$$

The rest of this subsection is devoted to the proof of Proposition 18.2.7.

18.2.8. By Corollaries H.6.4 and H.6.7, we can reformulate the assertion of the proposition as follows: the functor

$$\mathbb{L}_{G, \mathrm{coarse}} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}})$$

intertwines the actions of

$$\mathrm{Sph}_{G, \mathrm{Ran}^{\mathrm{untl}}, \mathrm{indep}}$$

on the left-hand side with the action of

$$\mathrm{Sph}_{\check{G}, \mathrm{Ran}^{\mathrm{untl}}, \mathrm{indep}}^{\mathrm{spec}}$$

on the right-hand side via the functor

$$\mathrm{Sph}_{G, \mathrm{Ran}^{\mathrm{untl}}, \mathrm{indep}} \rightarrow \mathrm{Sph}_{\check{G}, \mathrm{Ran}^{\mathrm{untl}}, \mathrm{indep}}^{\mathrm{spec}},$$

induced by the factorization functor

$$\mathrm{Sat}_G : \mathrm{Sph}_G \rightarrow \mathrm{Sph}_{\check{G}}^{\mathrm{spec}}.$$

18.2.9. We start with the commutative diagram (18.4)

$$(18.8) \quad \begin{array}{ccc} \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xrightarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{coeff}_G[2\delta_{N_{\rho(\omega_X)}}] \uparrow & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_{G,\mathrm{coarse}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}), \end{array}$$

and note that the vertical arrows factor as

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{Whit}^!(G)_{\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}} \hookrightarrow \mathrm{Whit}^!(G)_{\mathrm{Ran}}$$

and

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \rightarrow \mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}} \hookrightarrow \mathrm{Rep}(\check{G})_{\mathrm{Ran}},$$

respectively, so that we obtain a commutative diagram

$$(18.9) \quad \begin{array}{ccc} \mathrm{Whit}^!(G)_{\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}} & \xrightarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}} \\ \mathrm{coeff}_{G,\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}}[2\delta_{N_{\rho(\omega_X)}}] \uparrow & & \uparrow \Gamma_{\check{G},\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}}^{\mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_{G,\mathrm{coarse}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}). \end{array}$$

18.2.10. Since the functor

$$\mathrm{coeff}_{G,\mathrm{Ran}} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathrm{Ran}) \rightarrow \mathrm{Whit}^!(G)_{\mathrm{Ran}}$$

is compatible with the action of $(\mathrm{Sph}_{G,\mathrm{Ran}})^{\otimes}$, from Corollaries H.6.4 and H.6.7 we obtain that the functor

$$(18.10) \quad \begin{aligned} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{-\otimes \omega_{\mathrm{Ran}^{\mathrm{untl}}}} \simeq \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathrm{Ran}^{\mathrm{untl}})_{\mathrm{indep}} \xrightarrow{\mathrm{coeff}_{G,\mathrm{Ran}^{\mathrm{untl}}}} \mathrm{Whit}^!(G)_{\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}}, \end{aligned}$$

appearing as the left vertical arrow in (18.9), is compatible with the action of $\mathrm{Sph}_{G,\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}}$.

Similarly, since the functor

$$\Gamma_{\check{G},\mathrm{Ran}}^{\mathrm{spec}} : \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathrm{Ran}) \rightarrow \mathrm{Rep}(\check{G})_{\mathrm{Ran}}$$

is compatible with the actions of $(\mathrm{Sph}_{G,\mathrm{Ran}}^{\mathrm{spec}})^{\otimes}$, we obtain that the functor

$$(18.11) \quad \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{-\otimes \omega_{\mathrm{Ran}^{\mathrm{untl}}}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathrm{Ran}^{\mathrm{untl}})_{\mathrm{indep}} \xrightarrow{\Gamma_{\check{G},\mathrm{Ran}^{\mathrm{untl}}}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}},$$

appearing as the right vertical arrow in (18.9), is compatible with the action of $\mathrm{Sph}_{G,\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}}^{\mathrm{spec}}$.

18.2.11. Recall now that the functor

$$\Gamma_{\check{G}}^{\mathrm{spec}} : \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \rightarrow \mathrm{Rep}(\check{G})_{\mathrm{Ran}}$$

is fully faithful. By Proposition H.3.2, this implies that the functor $\Gamma_{\check{G},\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}}^{\mathrm{spec}}$ is fully faithful.

Hence, in order to equip $\mathbb{L}_{G,\mathrm{coarse}}$ with a datum of compatibility with respect to

$$\mathrm{Sph}_{G,\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}} \xrightarrow{\mathrm{Sat}_G} \mathrm{Sph}_{\check{G},\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}}^{\mathrm{spec}}$$

it suffices to do so for the counter-clockwise composition in (18.9).

By the commutativity of (18.9), this is equivalent to endowing the clockwise composition in (18.9) with a datum of compatibility with the above action.

However, this follows from the compatibility for (18.10) mentioned above, combined with the compatibility of CS_G with Sat_G .

□[Proposition 18.2.7]

18.3. The actual Langlands functor.

18.3.1. We now quote the following result established in [GLC1, Corollary 1.6.5]:

Theorem 18.3.2. *There exists a uniquely defined functor*

$$\mathbb{L}_G : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}),$$

subject to the following conditions:

- $\Psi_{\mathrm{LS}_{\check{G}}} \circ \mathbb{L}_G \simeq \mathbb{L}_{G, \mathrm{coarse}};$
- The functor $\mathbb{L}_G$ sends compact objects in $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ to

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})^{>-\infty} \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}).$$

18.3.3. Let

$$\Xi_{0, \mathrm{Nilp}} : \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \rightleftarrows \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) : \Psi_{\mathrm{Nilp}, 0}$$

and

$$\Xi_{\mathrm{Nilp}, \mathrm{all}} : \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \rightleftarrows \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) : \Psi_{\mathrm{all}, \mathrm{Nilp}}$$

denote the resulting pairs of adjoint functors.

By a slight abuse of notation we will denote by $\Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}}$ the functor

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \xrightarrow{\Xi_{\mathrm{Nilp}, \mathrm{all}}} \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{\Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}}} \mathrm{Rep}(\check{G})_{\mathrm{Ran}},$$

which is the same as

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \xrightarrow{\Psi_{\mathrm{Nilp}, 0}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{\Gamma_{\check{G}}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_{\mathrm{Ran}}.$$

18.3.4. From Proposition 18.1.6 we formally obtain:

Corollary 18.3.5. *The following diagram commutes:*

$$(18.12) \quad \begin{array}{ccc} \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xrightarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{coeff}_G[2\delta_{N_{\rho(\omega_X)}}] \uparrow & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}). \end{array}$$

18.3.6. Consider again the action of $(\mathrm{Sph}_{\check{G}, \mathrm{Ran}}^{\mathrm{spec}})^{\otimes}$ on $\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran})$. By the same mechanism as in Lemma 18.2.4, this action gives rise to an action of $(\mathrm{Sph}_{\check{G}, \mathrm{Ran}}^{\mathrm{spec}})^{\otimes}$ on the category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran})$, which is compatible with the functors

$$(\Xi_{0, \mathrm{Nilp}}, \Psi_{\mathrm{Nilp}, 0}) \text{ and } (\Xi_{\mathrm{Nilp}, \mathrm{all}}, \Psi_{\mathrm{all}, \mathrm{Nilp}}).$$

In particular, we obtain an action of

$$(\mathrm{Sph}_{\check{G}, \mathrm{Ran}}^{\mathrm{spec}})^{\star} \rightarrow \mathrm{Sph}_{\check{G}, \mathrm{Ran}}^{\mathrm{spec}, \mathrm{untl}, \mathrm{indep}}$$

on $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$.

18.3.7. We now claim:

Proposition 18.3.8. *The functor $\mathbb{L}_G$ intertwines the action of $(\mathrm{Sph}_{G, \mathrm{Ran}})^{\star}$ on the left-hand side and the action of $(\mathrm{Sph}_{\check{G}, \mathrm{Ran}}^{\mathrm{spec}})^{\star}$ on the right-hand side via*

$$\mathrm{Sat}_G : \mathrm{Sph}_G \rightarrow \mathrm{Sph}_{\check{G}}^{\mathrm{spec}}.$$

Proof. Note that $(\mathrm{Sph}_{G,\mathrm{Ran}})^*$ is compactly generated, and the subcategory

$$((\mathrm{Sph}_{G,\mathrm{Ran}})^*)^c \subset (\mathrm{Sph}_{G,\mathrm{Ran}})^*$$

is closed under the monoidal operation, and its action on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ preserves the subcategory

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G).$$

Hence, in order to prove the proposition, it suffices to equip the functor

$$\mathbb{L}_G|_{\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$$

with a datum of compatibility with respect to the action of $((\mathrm{Sph}_{G,\mathrm{Ran}})^*)^c$.

By the definition of the functor $\mathbb{L}_G$, the restriction $\mathbb{L}_G|_{\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c}$ factors as

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c \xrightarrow{\mathbb{L}_G} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})^{>-\infty} \hookrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}).$$

where

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})^{>-\infty} \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$$

is also preserved by the action of

$$((\mathrm{Sph}_{G,\mathrm{Ran}})^*)^c \simeq ((\mathrm{Sph}_{\tilde{G},\mathrm{Ran}}^{\mathrm{spec}})^*)^c.$$

Hence, it suffices to endow the functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c \xrightarrow{\mathbb{L}_G} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})^{>-\infty}$$

with a datum of compatibility with respect to the action of $((\mathrm{Sph}_{G,\mathrm{Ran}})^*)^c$.

Next, we note that the functor

$$\Psi_{\mathrm{Nilp},0}|_{\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})^{>-\infty}} : \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})^{>-\infty} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})$$

is compatible with the action of $((\mathrm{Sph}_{\tilde{G},\mathrm{Ran}}^{\mathrm{spec}})^*)^c$ and is fully faithful.

Hence, it suffices to endow the composition

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c \xrightarrow{\mathbb{L}_G} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})^{>-\infty} \xrightarrow{\Psi_{\mathrm{Nilp},0}} \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})$$

with a datum of compatibility with respect to the action of $((\mathrm{Sph}_{G,\mathrm{Ran}})^*)^c$.

However, the latter composition is the functor

$$\mathbb{L}_{G,\mathrm{coarse}}|_{\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^c},$$

and the required datum is supplied by Proposition 18.2.7. □

18.3.9. Combining Proposition 18.3.8 with Corollaries H.6.4, H.6.7 and H.6.11, we obtain:

Corollary 18.3.10.

(a) *The functor $\mathbb{L}_G$ intertwines the action of $\mathrm{Sph}_{G,\mathrm{Ran},\mathrm{indep}}$ on the left-hand side and the action of $\mathrm{Sph}_{\tilde{G},\mathrm{Ran}^{\mathrm{untl}},\mathrm{indep}}^{\mathrm{spec}}$ on the right-hand side.*

(b) *The functor*

$$\mathbb{L}_G \otimes \mathrm{Id} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran})$$

intertwines the action of $(\mathrm{Sph}_{G,\mathrm{Ran}})^{\frac{!}{\otimes}}$ on the left-hand side and the action of $(\mathrm{Sph}_{\tilde{G},\mathrm{Ran}}^{\mathrm{spec}})^{\frac{!}{\otimes}}$ on the right-hand side.

(c) *The functor*

$$\mathbb{L}_G \otimes \mathrm{Id} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \underline{\mathrm{D}\text{-mod}}(\mathrm{Ran}^{\mathrm{untl}}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}) \otimes \underline{\mathrm{D}\text{-mod}}(\mathrm{Ran}^{\mathrm{untl}})$$

intertwines the actions of Sph_G and $\mathrm{Sph}_{\tilde{G}}^{\mathrm{spec}}$, viewed as crystals of monoidal categories over $\mathrm{Ran}^{\mathrm{untl}}$.

18.4. Critical localization and temperedness.

18.4.1. Choose $\underline{x} \in \text{Ran}$, and let

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}, \underline{x}} := \text{Sph}_{G, \text{temp}, \underline{x}} \otimes_{\text{Sph}_{G, \underline{x}}} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G).$$

The pair of adjoint functors

$$\text{Sph}_{G, \text{temp}, \underline{x}} \rightleftarrows \text{Sph}_{G, \underline{x}}$$

allows us to view $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}, \underline{x}}$ as a colocalization of $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$.

According to [FR, Sect. 2.6.2], this colocalization is actually independent of the choice of $\underline{x}$. So from now on we will omit the subscript and denote the corresponding sub/quotient category by $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}}$. Denote by

$$\mathbf{u} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}} \rightleftarrows \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) : \mathbf{u}^R$$

the corresponding pair of adjoint functors.

18.4.2. From Proposition 18.3.8 we obtain:

Corollary 18.4.3. *There exists a uniquely defined functor*

$$\mathbb{L}_{G, \text{temp}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}} \rightarrow \text{QCoh}(\text{LS}_{\check{G}}),$$

which makes both squares in the next diagram commute:

$$\begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \\ \mathbf{u}^R \downarrow & & \downarrow \Psi_{\text{Nilp}, 0} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}} & \xrightarrow{\mathbb{L}_{G, \text{temp}}} & \text{QCoh}(\text{LS}_{\check{G}}) \\ \mathbf{u} \downarrow & & \downarrow \Xi_{0, \text{Nilp}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}). \end{array}$$

Furthermore,

$$\mathbb{L}_{G, \text{temp}} \simeq \mathbb{L}_{G, \text{coarse}} \circ \mathbf{u}.$$

18.4.4. Let

$$\text{Loc}_G : \text{KL}(G)_{\text{crit}, \text{Ran}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$$

be as in Sect. 14.1.4.

The following assertion is a counterpart of Proposition 17.5.9:

Proposition 18.4.5. *The essential image of the functor*

$$\text{Loc}_G : \text{KL}(G)_{\text{crit}, \text{Ran}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$$

lies in

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}} \subset \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G).$$

Proof. Repeats the proof of Proposition 17.5.9 using Proposition 7.2.6. □

18.5. Compatibility of the Langlands functor with critical localization.

18.5.1. The following theorem expresses the compatibility of the Langlands functor with critical localization:

Theorem 18.5.2. *The diagram*

$$\begin{array}{ccc}
 \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\
 \mathrm{Loc}_G \otimes \mathbb{I}_{G,N_{\rho(\omega_X)}}^{\otimes \frac{1}{2}} \otimes \mathbb{I}_{N_{\rho(\omega_X)}}^{\otimes -1} [-\delta_{N_{\rho(\omega_X)}}] \uparrow & & \uparrow \mathrm{Poinc}_{\check{G},*}^{\mathrm{spec}} \\
 \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & \xrightarrow{\mathrm{FLE}_{G,\mathrm{crit}}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}}
 \end{array}$$

commutes, where the lines $\mathbb{I}_{G,N_{\rho(\omega_X)}}^{\otimes \frac{1}{2}}$ and $\mathbb{I}_{N_{\rho(\omega_X)}}$ are as in (14.2) and (9.7), respectively.

The rest of the subsection is devoted to the proof of Theorem 18.5.2.

18.5.3. First, by Propositions 18.4.5 and 17.5.9, the commutativity of the diagram in Theorem 18.5.2 is equivalent to the commutativity of the following one:

$$\begin{array}{ccc}
 \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}} & \xrightarrow{\mathbb{L}_{G,\mathrm{temp}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \\
 \uparrow & & \uparrow \mathrm{Poinc}_{\check{G},*}^{\mathrm{spec}} \\
 \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & \xrightarrow{\mathrm{FLE}_{G,\mathrm{crit}}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}},
 \end{array}$$

and is further equivalent to the commutativity of

$$(18.13) \quad \begin{array}{ccc}
 \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_{G,\mathrm{coarse}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \\
 \uparrow & & \uparrow \mathrm{Poinc}_{\check{G},*}^{\mathrm{spec}} \\
 \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & \xrightarrow{\mathrm{FLE}_{G,\mathrm{crit}}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}}.
 \end{array}$$

18.5.4. Since the right vertical arrow in (18.4) is fully faithful, it suffices to show that the two circuits in (18.13) become isomorphic after composing with the functor $\Gamma_{\check{G}}^{\mathrm{spec}}$.

Since the diagram (18.4) is commutative, we obtain that it suffices to establish the commutativity of the diagram

$$\begin{array}{ccc}
 \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xrightarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\
 \mathrm{coeff}_G[2\delta_{N_{\rho(\omega_X)}}] \uparrow & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}} \\
 \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \\
 \mathrm{Loc}_G \otimes \mathbb{I}_{G,N_{\rho(\omega_X)}}^{\otimes \frac{1}{2}} \otimes \mathbb{I}_{N_{\rho(\omega_X)}}^{\otimes -1} [-\delta_{N_{\rho(\omega_X)}}] \uparrow & & \uparrow \mathrm{Poinc}_{\check{G},*}^{\mathrm{spec}} \\
 \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & \xrightarrow{\mathrm{FLE}_{G,\mathrm{crit}}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}},
 \end{array}$$

or which is the same

$$(18.14) \quad \begin{array}{ccc}
 \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xrightarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\
 \mathrm{coeff}_G \uparrow & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}} \\
 \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \\
 \mathrm{Loc}_G \otimes \mathbb{I}_{G,N_{\rho(\omega_X)}}^{\otimes \frac{1}{2}} \otimes \mathbb{I}_{N_{\rho(\omega_X)}}^{\otimes -1} [\delta_{N_{\rho(\omega_X)}}] \uparrow & & \uparrow \mathrm{Poinc}_{\check{G},*}^{\mathrm{spec}} \\
 \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & \xrightarrow{\mathrm{FLE}_{G,\mathrm{crit}}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}}.
 \end{array}$$

18.5.5. Applying duality, we obtain that it suffices to show that the pairing

$$(18.15) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}} \xrightarrow{\mathrm{Loc}_G \otimes \mathrm{Id}} \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}} \xrightarrow{\mathrm{coeff}_G \otimes \mathrm{Id}} \\ \rightarrow \mathrm{Whit}^!(G)_{\mathrm{Ran}} \otimes \mathrm{Whit}_*(G)_{\mathrm{Ran}} \rightarrow \mathrm{Vect} \xrightarrow{(- \otimes_{G, N_{\rho(\omega_X)}}^{\otimes \frac{1}{2}}) \otimes \mathrm{I}_{N_{\rho(\omega_X)}}^{\otimes -1}} [\delta_{N_{\rho(\omega_X)}}] \mathrm{Vect}$$

agrees under the FLE equivalences

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \xrightarrow{\mathrm{FLE}_{G, \mathrm{crit}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}} \text{ and } \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \xrightarrow{\mathrm{FLE}_{\check{G}, \infty}} \mathrm{Whit}_*(G)$$

with

$$(18.16) \quad \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}} \otimes \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \xrightarrow{\mathrm{Poinc}_{\check{G}, *}}^{\mathrm{spec} \otimes \mathrm{Id}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightarrow \\ \xrightarrow{\Gamma_{\check{G}}^{\mathrm{spec} \otimes \mathrm{Id}}} \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \otimes \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightarrow \mathrm{Vect}.$$

18.5.6. By Theorem 14.2.4, the functor (18.15) identifies canonically with (14.5). By Theorem 17.7.2, the functor (18.16) identifies canonically with (17.22).

The desired assertion follows now from Corollary 6.4.10.

□[Theorem 18.5.2]

Part III: Appendix

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The main body of the paper relies on a lot of foundational material, which is developed in this Appendix. The main points are:

- Ind-coherent sheaves on algebro-geometric objects of infinite type (our main, but *by far, not only* example is $\mathrm{Op}_G^{\mathrm{mon-free}}$). This is developed out in Sect. A;
- The notion of *factorization category*, and associated objects (factorization module categories, factorization algebras within a factorization category, etc.). This is developed in Sect. B.
- The notion *unitarity* in the factorization setting. This is developed in C;
- A result connecting the (pre dual of the) category of quasi-coherent sheaves on the loop space and the category of factorization modules over the corresponding factorization algebra D;
- The definition of the *spectral spherical category* in the factorization setting (the underlying algebro-geometric object is so unwieldy that one cannot algorithmically apply a procedure from Sect. A). This is developed in Sect. E.

The majority of this material is of *local* nature, i.e., it is needed to set up the local Langlands theory. That said, some sections in this Appendix (notably, Sects. F , H and I) consider local-to-global constructions.

APPENDIX A. IND-COHERENT SHEAVES IN INFINITE TYPE

This section is devoted to the development of the theory of ind-coherent sheaves on algebro-geometric objects of infinite-type.

Prior to doing so, we introduce another player, which in some sense lies in between $\mathrm{QCoh}(-)$ and $\mathrm{IndCoh}(-)$. This object is denoted by

$$\mathcal{Y} \in \mathrm{PreStk} \rightsquigarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}),$$

and it is defined by a *colimit* procedure (unlike $\mathrm{QCoh}(\mathcal{Y})$, which is defined as a *limit*). The category $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$ is a *predual* of $\mathrm{QCoh}(\mathcal{Y})$.

We now turn to the IndCoh theory. A priori, $\mathrm{IndCoh}(-)$ is defined for affine schemes almost of finite type, and by the process of right Kan extension on all prestacks that are locally almost of finite type. When S is an affine scheme that is not of finite type, one can approximate it by affine schemes of finite type S_α , but then one faces a choice: one can define $\mathrm{IndCoh}(S)$ either as the colimit of $\mathrm{IndCoh}(S_\alpha)$ with respect to $!$ -pullbacks and as a limit of $\mathrm{IndCoh}(S_\alpha)$ with respect to $*$ -pullbacks. This leads to two different categories, denoted $\mathrm{IndCoh}^!(S)$ and $\mathrm{IndCoh}^*(S)$, respectively. In good cases (technically, when S is *placid*), the Serre duality in finite type gives rise to a duality between $\mathrm{IndCoh}^!(S)$ and $\mathrm{IndCoh}^*(S)$.

The majority of this section is devoted to developing the $\mathrm{IndCoh}^!(-)$ and $\mathrm{IndCoh}^*(-)$ theories, and their interactions with other actors.

A.1. The category $\mathrm{QCoh}_{\mathrm{co}}(-)$.

A.1.1. In section we will work with *all* affine schemes (i.e., ones not necessarily almost of finite type). We denote the corresponding category by $\mathrm{Sch}^{\mathrm{aff}}$. We denote by PreStk the category of all functors

$$(\mathrm{Sch}^{\mathrm{aff}})^{\mathrm{op}} \rightarrow \mathrm{Spc}.$$

A.1.2. Consider the functor

$$(A.1) \quad \mathrm{Sch}^{\mathrm{aff}} \rightarrow \mathrm{DGCat}, \quad S \mapsto \mathrm{QCoh}(S), \quad (S_1 \xrightarrow{f} S_2) \rightsquigarrow \mathrm{QCoh}(S_1) \xrightarrow{f_*} \mathrm{QCoh}(S_2).$$

Consider the left Kan extension of (A.1) along the fully faithful embedding

$$\mathrm{Sch}^{\mathrm{aff}} \hookrightarrow \mathrm{PreStk};$$

this yields a functor

$$(A.2) \quad \mathrm{PreStk} \rightarrow \mathrm{DGCat}.$$

We will denote the value of (A.2) on a given prestack $\mathcal{Y}$ by

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}).$$

Explicitly,

$$(A.3) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}) \simeq \operatorname{colim}_{S \rightarrow \mathcal{Y}, S \in \mathrm{Sch}^{\mathrm{aff}}} \mathrm{QCoh}(S),$$

where the colimit is formed using the pushforward functors.

Remark A.1.3. The above definition turns out to be useful in many contexts, but in the present paper its main application is the following (see Sect. A.2):

Let $\mathcal{Y}$ be an ind-affine ind-scheme

$$\mathcal{Y} = \operatorname{"colim"}_i \operatorname{Spec}(R_i).$$

Let R denote the topological ring $\lim_i R_i$.

Then

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}) \simeq \operatorname{colim}_i R_i\text{-mod},$$

i.e., this formalizes the notion of “the category of discrete R -modules”.

Note that the above definition is close, but not the same, as $\operatorname{IndCoh}(\mathcal{Y})$. The difference is two-fold:

- $\operatorname{IndCoh}(\mathcal{Y})$ is a priori defined only when $\mathcal{Y}$ is locally almost of finite type (however, we will generalize that in Sect. A.5.6 below), while $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$ does not require this assumption;
- When $\mathcal{Y}$ is an affine scheme Y , we have $\mathrm{QCoh}_{\mathrm{co}}(Y) = \mathrm{QCoh}(Y)$, i.e., there is no renormalization procedure involved. (The price we will have to pay for this is that, even for ind-schemes locally almost of finite type, the category $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$ is not necessarily dualizable.)

A.1.4. By construction, the assignment

$$\mathcal{Y} \mapsto \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$$

has a functoriality with respect to pushforwards, i.e., for a map $\mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ we have the functor

$$f_* : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_1) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_2).$$

A.1.5. The construction

$$\mathcal{Y} \rightsquigarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$$

has a natural multiplicativity property. Namely, for a pair of prestacks $\mathcal{Y}_1$ and $\mathcal{Y}_2$, we have a naturally defined equivalence

$$(A.4) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_1) \otimes \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_2) \simeq \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_1 \times \mathcal{Y}_2).$$

Namely, we have, by definition:

$$\begin{aligned} \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_1) \otimes \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_2) &\simeq \operatorname{colim}_{S_1 \in \mathrm{Sch}^{\mathrm{aff}}/\mathcal{Y}_1, S_2 \in \mathrm{Sch}^{\mathrm{aff}}/\mathcal{Y}_2} \mathrm{QCoh}(S_1) \otimes \mathrm{QCoh}(S_2) \simeq \\ &\simeq \operatorname{colim}_{S_1 \in \mathrm{Sch}^{\mathrm{aff}}/\mathcal{Y}_1, S_2 \in \mathrm{Sch}^{\mathrm{aff}}/\mathcal{Y}_2} \mathrm{QCoh}(S_1 \times S_2), \end{aligned}$$

and

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_1 \times \mathcal{Y}_2) \simeq \operatorname{colim}_{S \in \mathrm{Sch}^{\mathrm{aff}}_{\mathcal{Y}_1 \times \mathcal{Y}_2}} \mathrm{QCoh}(S).$$

Now, the functor

$$\mathrm{Sch}^{\mathrm{aff}}_{\mathcal{Y}_1} \times \mathrm{Sch}^{\mathrm{aff}}_{\mathcal{Y}_2} \rightarrow \mathrm{Sch}^{\mathrm{aff}}_{\mathcal{Y}_1 \times \mathcal{Y}_2}, \quad S_1, S_2 \mapsto S_1 \times S_2$$

is cofinal.

A.1.6. Note that there is no reason for the category $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$ to be dualizable. However, we claim that $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$ is a *pre-dual* of $\mathrm{QCoh}(\mathcal{Y})$, i.e., we have a canonical identification

$$(A.5) \quad \mathrm{Funct}_{\mathrm{cont}}(\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}), \mathrm{Vect}) =: \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})^\vee \simeq \mathrm{QCoh}(\mathcal{Y}).$$

Indeed, using (A.3), we have

$$\begin{aligned} \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})^\vee &\simeq \left(\operatorname{colim}_{S \rightarrow \mathcal{Y}, S \in \mathrm{Sch}^{\mathrm{aff}}} \mathrm{QCoh}(S) \right)^\vee \simeq \\ &\simeq \lim_{S \rightarrow \mathcal{Y}, S \in \mathrm{Sch}^{\mathrm{aff}}} \mathrm{QCoh}(S)^\vee \simeq \lim_{S \rightarrow \mathcal{Y}, S \in \mathrm{Sch}^{\mathrm{aff}}} \mathrm{QCoh}(S) =: \mathrm{QCoh}(\mathcal{Y}), \end{aligned}$$

where we recall that for $f : S_1 \rightarrow S_2$, with respect to the self-dualities

$$\mathrm{QCoh}(S_i)^\vee \simeq \mathrm{QCoh}(S_i), \quad i = 1, 2,$$

the dual of f_* is f^* .

A.1.7. We claim that there is a natural action of $\mathrm{QCoh}(\mathcal{Y})$ on $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$. Namely, in terms of (A.3), an object $\mathcal{F} \in \mathrm{QCoh}(\mathcal{Y})$ gives rise to a compatible family of endofunctors of $\mathrm{QCoh}(S)$ for $y : S \rightarrow \mathcal{Y}$, namely

$$S \rightsquigarrow y^*(\mathcal{F}) \otimes (-).$$

This action is compatible with the identification (A.5).

Furthermore, it satisfies the projection formula: for $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ we have

$$f_*(f^*(\mathcal{F}_2) \otimes \mathcal{F}_1) \simeq \mathcal{F}_2 \otimes f_*(\mathcal{F}_1), \quad \mathcal{F}_1 \in \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_1), \quad \mathcal{F}_2 \in \mathrm{QCoh}(\mathcal{Y}_2).$$

A.1.8. We can rewrite the canonical pairing

$$(A.6) \quad \mathrm{QCoh}(\mathcal{Y}) \otimes \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}) \rightarrow \mathrm{Vect}$$

in terms of the above action of $\mathrm{QCoh}(\mathcal{Y})$ on $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$.

Namely, it is given by

$$\mathrm{QCoh}(\mathcal{Y}) \otimes \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}) \xrightarrow{\text{action}} \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}) \xrightarrow{\Gamma(\mathcal{Y}, -)} \mathrm{Vect}.$$

A.1.9. Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be affine. In this case we claim that the functor

$$f_* : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_1) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_2)$$

admits a left adjoint, to be denoted f^* .

Indeed, the functor

$$\mathrm{Sch}^{\mathrm{aff}}_{\mathcal{Y}_2} \rightarrow \mathrm{Sch}^{\mathrm{aff}}_{\mathcal{Y}_1}, \quad S \mapsto S \times_{\mathcal{Y}_2} \mathcal{Y}_1$$

is cofinal, so the functor

$$\operatorname{colim}_{S \in \mathcal{Y}_2, S \in \mathrm{Sch}^{\mathrm{aff}}} \mathrm{QCoh}(S \times_{\mathcal{Y}_2} \mathcal{Y}_1) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_1).$$

is an equivalence.

In terms of this identification, the functor f^* is given by the compatible family of the pullback functors

$$\mathrm{QCoh}(S) \rightarrow \mathrm{QCoh}(S \times_{\mathcal{Y}_2} \mathcal{Y}_1).$$

A.1.10. Let $\mathcal{Y}$ be a prestack over an affine scheme S . Let $f : S' \rightarrow S$ be a map between affine schemes; set $\mathcal{Y}' := S' \times_S \mathcal{Y}$. By a slight abuse of notation we will denote by the same character f the resulting map $\mathcal{Y}' \rightarrow \mathcal{Y}$.

The category $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$ (resp., $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}')$) is naturally tensored over $\mathrm{QCoh}(S)$ (resp., $\mathrm{QCoh}(S')$), and the functor $f_* : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}') \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$ is $\mathrm{QCoh}(S)$ -linear. Hence, so is its right adjoint f^* . From here we obtain a functor

$$(A.7) \quad \mathrm{QCoh}(S') \otimes_{\mathrm{QCoh}(S)} \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}').$$

We claim:

Lemma A.1.11. *The functor (A.7) is an equivalence.*

Proof. Follows from the fact that the functor

$$\tilde{S} \in \mathrm{Sch}_{/\mathcal{Y}}^{\mathrm{aff}} \mapsto S' \times_S \tilde{S} \in \mathrm{Sch}_{/\mathcal{Y}'}^{\mathrm{aff}}$$

is cofinal. □

A.1.12. Let $\mathcal{Y}$ have an affine diagonal. In this case we claim that there is a naturally defined functor

$$(A.8) \quad \Omega_{\mathcal{Y}} : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}) \rightarrow \mathrm{QCoh}(\mathcal{Y}).$$

Namely, in terms of (A.3), the functor (A.8) is given by the (compatible family) of direct image functors

$$\mathrm{QCoh}(S) \rightarrow \mathrm{QCoh}(\mathcal{Y}),$$

which are well-defined, since the morphisms $S \rightarrow \mathcal{Y}$ are affine.

Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a schematic map. Note that, by construction, the following diagram commutes:

$$\begin{array}{ccc} \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_1) & \xrightarrow{f_*} & \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_2) \\ \Omega_{\mathcal{Y}_1} \downarrow & & \downarrow \Omega_{\mathcal{Y}_2} \\ \mathrm{QCoh}(\mathcal{Y}_1) & \xrightarrow{f_*} & \mathrm{QCoh}(\mathcal{Y}_2). \end{array}$$

A.1.13. The following assertion is established in [Ga5, Theorems 2.2.4 or 2.2.6 and Proposition 6.3.8]:

Theorem A.1.14. *Let $\mathcal{Y}$ be an quasi-compact algebraic stack with an affine diagonal. Suppose that one of the following conditions holds:*

- (i) $\mathcal{Y}$ can be realized as a quotient of an algebraic space by an action of a (finite-dimensional) algebraic group;
- (ii) $\mathcal{Y}$ is eventually coconnective algebraic stack almost of finite type.

Then the functor $\Omega_{\mathcal{Y}}$ of (A.8) is an equivalence.

A.1.15. Let $\mathcal{Y}$ be a (not necessarily quasi-compact) algebraic stack. Suppose that $\mathcal{Y}$ can be written as a union of quasi-compact open substacks $\mathcal{Y}_i$ that satisfy one of the conditions in Theorem A.1.14.

Corollary A.1.16. *Under the above circumstances, we have a canonical equivalence*

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}) \simeq \mathrm{colim}_i \mathrm{QCoh}(\mathcal{Y}_i),$$

where the colimit is taken with respect to the pushforward functors.

Proof. Note that the map

$$\operatorname{colim}_i \mathcal{Y}_i \rightarrow \mathcal{Y}$$

is an isomorphism in PreStk .

Hence, the functor

$$\operatorname{colim}_i \operatorname{QCoh}_{\operatorname{co}}(\mathcal{Y}_i) \rightarrow \operatorname{QCoh}_{\operatorname{co}}(\mathcal{Y})$$

is an equivalence.

Now the assertion follows from Theorem A.1.14. $\square$

A.2. The category $\operatorname{QCoh}_{\operatorname{co}}$ on ind-schemes.

A.2.1. Let $\mathcal{Y}$ be an ind-affine ind-scheme (see [GaRo1]). According to Corollary 1.6.6 in *loc. cit.*, the map

$$\operatorname{colim}_{S \text{ closed in } \mathcal{Y}} S \rightarrow \mathcal{Y}$$

is an isomorphism in PreStk , where the index category is that of affine schemes, equipped with a closed embedding to $\mathcal{Y}$.

Hence, in this case, we have

$$(A.9) \quad \operatorname{QCoh}_{\operatorname{co}}(\mathcal{Y}) \simeq \operatorname{colim}_{S \text{ closed in } \mathcal{Y}} \operatorname{QCoh}(S).$$

A.2.2. Recall the following general paradigm:

Let

$$i \mapsto \mathbf{C}_i, \quad i \in I$$

be a diagram in DGCat . Denote

$$\mathbf{C} := \operatorname{colim}_{i \in I} \mathbf{C}_i,$$

where, per our conventions, the colimit is taken in DGCat (i.e., the category of cocomplete DG categories and continuous functors).

Let $\operatorname{ins}_i : \mathbf{C}_i \rightarrow \mathbf{C}$ denote the tautological functors.

A.2.3. Suppose that each $\mathbf{C}_i$ is equipped with a t-structure, compatible with filtered colimits, i.e., $\mathbf{C}_i^{\geq 0}$ is closed under filtered colimits. And suppose that the transition functors

$$\mathbf{C}_i \xrightarrow{F_{i,j}} \mathbf{C}_j$$

are t-exact.

We equip with $\mathbf{C}$ with a t-structure by declaring that $\mathbf{C}^{\leq 0}$ is generated under colimits by the essential images of $\operatorname{ins}_i(\mathbf{C}_i^{\leq 0})$.

So by construction, the functors ins_i are right t-exact.

A.2.4. We claim:

Lemma A.2.5. *Assume that I is filtered. Then:*

- (a) *The t-structure on $\mathbf{C}$ is compatible with filtered colimits.*
- (b) *The functors ins_i are t-exact.*
- (c) *The category $\mathbf{C}^{\geq 0}$ is generated under filtered colimits by the essential images of $\mathbf{C}_i^{\geq 0}$ along the functors ins_i .*

Proof. The first two points are proved in [Lu3, Proposition C.3.3.5].

Namely, in the notation of loc. cit., the category $\mathrm{Groth}^{\mathrm{lex}}$ is equivalent to the category of presentable stable ∞ -categories with right-complete t-structures and colimit preserving functors which are t-exact. The first two points are equivalent to the assertion that $\mathrm{Groth}^{\mathrm{lex}}$ admits filtered colimits and the forgetful functor

$$\mathrm{Groth}^{\mathrm{lex}} \rightarrow \mathrm{Pr}^{\mathrm{L}}$$

given by $(\mathbf{C}, \mathbf{C}^{\leq 0}) \mapsto \mathbf{C}$ preserves filtered colimits. By [Lu3, Proposition C.3.3.5], $\mathrm{Groth}^{\mathrm{lex}}$ admits filtered colimits and the functor $(\mathbf{C}, \mathbf{C}^{\leq 0}) \mapsto \mathbf{C}^{\leq 0}$ preserves filtered colimits. The result now follows from the fact that for any $(\mathbf{C}, \mathbf{C}^{\leq 0}) \in \mathrm{Groth}^{\mathrm{lex}}$, we have

$$\mathbf{C} \simeq \mathrm{Stab}(\mathbf{C}^{\leq 0})$$

is given by the stabilization, and the functor $\mathrm{Stab} : \mathrm{Pr}^{\mathrm{L}} \rightarrow \mathrm{Pr}^{\mathrm{L}}$ preserves colimits.

To prove the third point we note that any $\mathbf{c} \in \mathbf{C}$ is canonically isomorphic to

$$\mathrm{colim}_{i \in I} \mathrm{ins}_i \circ \mathrm{ins}_i^R(\mathbf{c}).$$

If $\mathbf{c} \in \mathbf{C}^{\geq 0}$, then so are all $\mathrm{ins}_i^R(\mathbf{c})$ (since the functors ins_i^R are left t-exact, being right adjoints of (right) t-exact functors. □

Remark A.2.6. Note that in the situation of Lemma A.2.5, we have

$$\mathbf{C} \simeq \lim_{iiI^{\mathrm{op}}} \mathbf{C}_i,$$

where:

- The limit is taken in the category of ∞ -categories;
- The functor $\rightarrow$ is given by the (compatible collection of) the functors ins_i^R .

Since the functors ins_i^R send $\mathbf{C}^{\geq 0}$ to $\mathbf{C}_i^{\geq 0}$, we obtain they also induce an equivalence

$$(A.10) \quad \mathbf{C}^{\geq 0} \simeq \lim_{iiI^{\mathrm{op}}} \mathbf{C}_i^{\geq 0}.$$

Corollary A.2.7. *Let $\Phi : \mathbf{C} \rightarrow \mathbf{D}$ be a continuous functor, where $\mathbf{D}$ is also equipped with a t-structure.*

- (a) *The functor Φ is right t-exact if and only if each $\Phi \circ \mathrm{ins}_i =: \Phi_i : \mathbf{C}_i \rightarrow \mathbf{D}$ is right t-exact.*
- (b) *If Φ is left t-exact, then so is each Φ_i .*
- (b') *Suppose that the t-structure on $\mathbf{D}$ is compatible with filtered colimits. Then the assertion in (b) is “if and only if”.*

A.2.8. Let $\mathcal{Y}$ be an ind-affine ind-scheme. We use the presentation (A.9) and the construction in Sect. A.2.3 to equip the category $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$ with a t-structure:

By definition, $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})^{\leq 0}$ is generated under colimits by the essential images of $\mathrm{QCoh}(S)^{\leq 0}$ for $S \in \mathrm{Sch}_{/\mathcal{Y}}^{\mathrm{aff}}$.

A.2.9. From Lemma A.2.5, we obtain:

Corollary A.2.10.

- (a) *The t-structure on $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$ is compatible with filtered colimits.*
- (b) *For every $S \in \mathrm{Sch}_{/\mathcal{Y}}^{\mathrm{aff}}$, the direct image functor $\mathrm{QCoh}(S) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$ is t-exact.*

A.2.11. Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a map between ind-affine ind-schemes. It follows by definition that the functor

$$f_* : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_1) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_2)$$

is right t-exact.

However, from Corollaries A.2.10(b) and A.2.7 we obtain:

Corollary A.2.12. *The functor f_* is t-exact.*

A.2.13. Assume that f is affine, in which case we have a well-defined functor

$$f^* : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_2) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_1).$$

From Corollary A.2.12 we obtain:

Corollary A.2.14. *The functor f^* is right t -exact.*

Finally, assume that f is flat. In this case, unwinding the construction of f^* in Sect. A.1.9 and using Corollaries A.2.10(b) and A.2.7, we obtain:

Lemma A.2.15. *For a flat map f between ind-affine ind-schemes, the functor $f^* : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_2) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}_1)$ is t -exact.*

A.3. A descent property of $\mathrm{QCoh}_{\mathrm{co}}(-)$. In this subsection we will prove a certain technical assertion used in the main body of the text.

A.3.1. Let $\mathcal{Y}$ be an ind-affine ind-scheme. Let $g : \tilde{\mathcal{Y}} \rightarrow \mathcal{Y}$ be a map of prestacks that is an affine fpqc cover, and let $\tilde{\mathcal{Y}}^\bullet$ denote its Čech nerve.

Consider $\mathrm{QCoh}_{\mathrm{co}}(\tilde{\mathcal{Y}}^\bullet)$ as a cosimplicial category, equipped with an augmentation by $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$ using $*$ -pullbacks (they are well-defined since the maps involved are affine, see Sect. A.1.9).

Thus, we obtain a functor

$$(A.11) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}) \rightarrow \mathrm{Tot}(\mathrm{QCoh}_{\mathrm{co}}(\tilde{\mathcal{Y}}^\bullet)).$$

A.3.2. From now on we will perceive $\mathrm{QCoh}_{\mathrm{co}}(\tilde{\mathcal{Y}}^\bullet)$ as a *semi*-cosimplicial category, so that transition functors involved are t -exact (by the flatness assumption on g , see Lemma A.2.15). Hence, the functor (A.11) induces a functor

$$(A.12) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})^{>-\infty} \rightarrow \mathrm{Tot}(\mathrm{QCoh}_{\mathrm{co}}(\tilde{\mathcal{Y}}^\bullet)^{>-\infty}).$$

We will prove:

Proposition A.3.3. *Suppose that $\mathcal{Y}$ can be exhibited as a filtered colimit in PreStk of affine schemes with transition maps that are almost finitely presented.⁶⁰ Then the functor (A.12) is an equivalence.*

The rest of this subsection is devoted to the proof of this proposition.

A.3.4. It suffices to show that

$$(A.13) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})^{\geq 0} \rightarrow \mathrm{Tot}(\mathrm{QCoh}_{\mathrm{co}}(\tilde{\mathcal{Y}}^\bullet)^{\geq 0})$$

is an equivalence.

A.3.5. Let

$$\mathcal{Y} = \mathrm{colim}_{i \in I} Y_i$$

be the presentation of $\mathcal{Y}$ as in the statement of the proposition. Denote the map $Y_i \rightarrow Y_j$ for $(i \rightarrow j) \in I$ by $f_{i,j}$.

Set

$$\tilde{Y}_i^\bullet := \tilde{\mathcal{Y}}^\bullet \times_{\mathcal{Y}} Y_i.$$

Then for every m , we also have

$$\tilde{\mathcal{Y}}^m \simeq \mathrm{colim}_i \tilde{Y}_i^m.$$

Denote the corresponding maps $\tilde{Y}_i^m \rightarrow \tilde{Y}_j^m$ by $f_{i,j}^m$.

⁶⁰I.e., finitely presented after each coconnective truncation.

A.3.6. We have

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}) \simeq \operatorname{colim}_{i \in I} \mathrm{QCoh}(Y_i)$$

and

$$\mathrm{QCoh}_{\mathrm{co}}(\tilde{\mathcal{Y}}^m) \simeq \operatorname{colim}_{i \in I} \mathrm{QCoh}(\tilde{Y}_i^m),$$

where in both cases the colimit is taken with respect to the pushforward functors (recall that by default, colimits are taken in DGCat , i.e., in the ∞ -category of cocomplete DG categories and continuous functors).

We can rewrite the above colimits as *limits* (in the category of DG categories with not necessarily continuous functors)

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}) \simeq \lim_{i \in I^{\mathrm{op}}} \mathrm{QCoh}(Y_i)$$

and

$$\mathrm{QCoh}_{\mathrm{co}}(\tilde{\mathcal{Y}}^m) \simeq \lim_{i \in I^{\mathrm{op}}} \mathrm{QCoh}(\tilde{Y}_i^m),$$

where the transition functors are $f_{i,j}^! := (f_{i,j})_*^R$ and $(f_{i,j}^m)^! := (f_{i,j}^m)_*^R$, respectively (note that these right adjoints are indeed in general *discontinuous*).

From here we obtain:

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})^{\geq 0} \simeq \lim_{i \in I^{\mathrm{op}}} \mathrm{QCoh}(Y_i)^{\geq 0}$$

and

$$\mathrm{QCoh}_{\mathrm{co}}(\tilde{\mathcal{Y}}^m)^{\geq 0} \simeq \lim_{i \in I^{\mathrm{op}}} \mathrm{QCoh}(\tilde{Y}_i^m)^{\geq 0},$$

where limits are taken in the category of ∞ -categories and all functors, see (A.10).

A.3.7. For every $\phi : [m'] \rightarrow [m'']$ in⁶¹ Δ denote by g^ϕ the corresponding map

$$\tilde{\mathcal{Y}}^{m''} \rightarrow \tilde{\mathcal{Y}}^{m'}.$$

For every index i , let g_i^ϕ denote the resulting map $\tilde{Y}_i^{m''} \rightarrow \tilde{Y}_i^{m'}$.

For every arrow $(i \rightarrow j) \in I$, we have a Cartesian diagram of schemes

$$\begin{array}{ccc} \tilde{Y}_i^{m''} & \xrightarrow{f_{i,j}^{m''}} & \tilde{Y}_j^{m''} \\ g_i^\phi \downarrow & & \downarrow g_j^\phi \\ \tilde{Y}_i^{m'} & \xrightarrow{f_{i,j}^{m'}} & \tilde{Y}_j^{m'}, \end{array}$$

which gives rise to a commutative diagram

$$\begin{array}{ccc} \mathrm{QCoh}(\tilde{Y}_i^{m''}) & \xrightarrow{(f_{i,j}^{m''})_*} & \mathrm{QCoh}(\tilde{Y}_j^{m''}) \\ (g_i^\phi)^* \uparrow & & \uparrow (g_j^\phi)^* \\ \mathrm{QCoh}(\tilde{Y}_i^{m'}) & \xrightarrow{(f_{i,j}^{m'})_*} & \mathrm{QCoh}(\tilde{Y}_j^{m'}). \end{array}$$

From here we obtain a natural transformation

$$(A.14) \quad (g_i^\phi)^* \circ (f_{i,j}^{m'})^! \rightarrow (f_{i,j}^{m''})^! \circ (g_j^\phi)^*.$$

Now, the assumption that the maps $f_{i,j}$ are almost of finite presentation and the maps g^ϕ are flat implies that the natural transformations (A.14) are isomorphisms when evaluated on $\mathrm{QCoh}(\tilde{Y}_j^{m'})^{\geq 0}$.

⁶¹We remind that we only consider injective maps ϕ .

Hence, we obtain a family of commutative diagrams

$$\begin{array}{ccc} \mathrm{QCoh}(\tilde{Y}_i^{m''})^{\geq 0} & \xleftarrow{(f_{i,j}^{m''})^!} & \mathrm{QCoh}(\tilde{Y}_j^{m''})^{\geq 0} \\ (g_i^\phi)^* \uparrow & & \uparrow (g_j^\phi)^* \\ \mathrm{QCoh}(\tilde{Y}_i^{m'})^{\geq 0} & \xleftarrow{(f_{i,j}^{m'})^!} & \mathrm{QCoh}(\tilde{Y}_j^{m'})^{\geq 0}. \end{array}$$

A.3.8. Thus, we obtain a well-defined functor from $\Delta \times I$ to the category of ∞ -categories

$$m, i \mapsto \mathrm{QCoh}(\tilde{Y}_i^m)^{\geq 0},$$

and we can rewrite

$$\mathrm{Tot}(\mathrm{QCoh}_{\mathrm{co}}(\tilde{\mathcal{Y}}^\bullet)^{\geq 0}) := \lim_{m \in \Delta} \lim_{i \in I^{\mathrm{op}}} \mathrm{QCoh}(\tilde{Y}_i^m)^{\geq 0}$$

as

$$\lim_{(m,i) \in \Delta \times I^{\mathrm{op}}} \mathrm{QCoh}(\tilde{Y}_i^m)^{\geq 0}$$

and further as

$$\lim_{i \in I^{\mathrm{op}}} \lim_{m \in \Delta} \mathrm{QCoh}(\tilde{Y}_i^m)^{\geq 0}.$$

Unwinding the construction, we obtain that the following diagram commutes

$$(A.15) \quad \begin{array}{ccc} \mathrm{Tot}(\mathrm{QCoh}_{\mathrm{co}}(\tilde{\mathcal{Y}}^\bullet)^{\geq 0}) & \xrightarrow{\sim} & \lim_{i \in I^{\mathrm{op}}} \lim_{m \in \Delta} \mathrm{QCoh}(\tilde{Y}_i^m)^{\geq 0} \\ \uparrow & & \uparrow \\ \mathrm{QCoh}_{\mathrm{co}}(\tilde{\mathcal{Y}})^{\geq 0} & \xrightarrow{\sim} & \lim_{i \in I^{\mathrm{op}}} \mathrm{QCoh}(Y_i)^{\geq 0}, \end{array}$$

where the right vertical arrow is comprised of the functors

$$(A.16) \quad \mathrm{QCoh}(Y_i)^{\geq 0} \rightarrow \mathrm{Tot}(\mathrm{QCoh}_{\mathrm{co}}(\tilde{Y}_i^\bullet)^{\geq 0}).$$

Now, the functors (A.16) are equivalences by the usual fpqc descent. Hence, the right vertical arrow in (A.15) is an equivalence.

Hence, the left vertical arrow is also an equivalence, as required.

□[Proposition A.3.3]

A.4. The category $\mathrm{IndCoh}^!(-)$.

A.4.1. Let $\leq^n \mathrm{Sch}^{\mathrm{aff}}$ denote the category of n -coconnective affine schemes, i.e.,

$$\leq^n \mathrm{Sch}^{\mathrm{aff}} = (\mathrm{ComAlg}(\mathrm{Vect}^{\geq -n, \leq 0}))^{\mathrm{op}}.$$

We have

$$\mathrm{Sch}^{\mathrm{aff}} \simeq \lim_n (\leq^n \mathrm{Sch}^{\mathrm{aff}}).$$

A.4.2. Let

$$\leq^n \mathrm{Sch}_{\mathrm{ft}}^{\mathrm{aff}} \subset \leq^n \mathrm{Sch}^{\mathrm{aff}}$$

be the full subcategory consisting of n -coconnective affine schemes of *finite type* (see [GaRo3, Chapter 1, Sect. 1.5]).

Note that

$$\leq^n \mathrm{Sch}^{\mathrm{aff}} \simeq \mathrm{Pro}(\leq^n \mathrm{Sch}_{\mathrm{ft}}^{\mathrm{aff}}).$$

Let $\mathrm{Sch}_{\mathrm{aft}}^{\mathrm{aff}}$ denote the full subcategory of $\mathrm{Sch}^{\mathrm{aff}}$ consisting of affine schemes almost of finite type, which is by definition

$$\lim_n (\leq^n \mathrm{Sch}_{\mathrm{ft}}^{\mathrm{aff}}).$$

Let $\mathrm{PreStk}_{\mathrm{aft}} \subset \mathrm{PreStk}$ the full subcategory consisting of prestacks locally almost of finite type. We have

$$\mathrm{PreStk}_{\mathrm{aft}} \simeq \lim_n (\leq^n \mathrm{PreStk}_{\mathrm{lft}}),$$

where

$$\leq^n \text{PreStk}_{\text{lft}} \simeq \text{Func}((\leq^n \text{Sch}_{\text{ft}}^{\text{aff}})^{\text{op}}, \infty\text{-Grpd}).$$

A.4.3. We define the functor

$$\leq^n \text{IndCoh}^! : (\leq^n \text{Sch}^{\text{aff}})^{\text{op}} \rightarrow \text{DGCat},$$

to be the left Kan extension of the functor

$$\text{IndCoh} : (\leq^n \text{Sch}_{\text{ft}}^{\text{aff}})^{\text{op}} \rightarrow \text{DGCat}$$

along

$$(\leq^n \text{Sch}_{\text{ft}}^{\text{aff}})^{\text{op}} \hookrightarrow (\leq^n \text{Sch}^{\text{aff}})^{\text{op}}.$$

Explicitly, for $S \in \leq^n \text{Sch}^{\text{aff}}$, we have

$$(A.17) \quad \text{IndCoh}^!(S) = \text{colim}_{S \rightarrow S_0, S_0 \in \leq^n \text{Sch}_{\text{ft}}^{\text{aff}}} \text{IndCoh}(S_0),$$

where the transition functors are given by

$$(S \rightarrow S'_0 \xrightarrow{f} S''_0) \rightsquigarrow \text{IndCoh}(S''_0) \xrightarrow{f^!} \text{IndCoh}(S'_0).$$

A.4.4. It is easy to see that the natural transformation from

$$(\leq^n \text{Sch}^{\text{aff}})^{\text{op}} \xrightarrow{\leq^n \text{IndCoh}^!} \text{DGCat}$$

to

$$(\leq^n \text{Sch}^{\text{aff}})^{\text{op}} \hookrightarrow (\leq^{n+1} \text{Sch}^{\text{aff}})^{\text{op}} \xrightarrow{\leq^{n+1} \text{IndCoh}^!} \text{DGCat}$$

is an isomorphism. Hence, we obtain a well-defined functor

$$(A.18) \quad \text{IndCoh}^! : (<^\infty \text{Sch}^{\text{aff}})^{\text{op}} \rightarrow \text{DGCat},$$

where

$$<^\infty \text{Sch}^{\text{aff}} = \text{colim}_n \leq^n \text{Sch}^{\text{aff}}.$$

A.4.5. We define the functor

$$\text{IndCoh}^! : (\text{PreStk})^{\text{op}} \rightarrow \text{DGCat}$$

to be the right Kan extension of (A.18) along the embedding

$$(<^\infty \text{Sch}^{\text{aff}})^{\text{op}} \hookrightarrow (\text{PreStk})^{\text{op}}.$$

Explicitly, for $\mathcal{Y} \in \text{PreStk}$, we have

$$\text{IndCoh}^!(\mathcal{Y}) = \lim_{S \rightarrow \mathcal{Y}, S \in <^\infty \text{Sch}^{\text{aff}}} \text{IndCoh}^!(S),$$

where the transition functors are given by

$$(S' \xrightarrow{f} S'' \rightarrow \mathcal{Y}) \rightsquigarrow \text{IndCoh}^!(S'') \xrightarrow{f^!} \text{IndCoh}^!(S').$$

A.4.6. Thus, by definition, for a map $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ in PreStk , we have a well-defined functor

$$f^! : \text{IndCoh}^!(\mathcal{Y}_2) \rightarrow \text{IndCoh}^!(\mathcal{Y}_1).$$

In particular, taking the projection $\mathcal{Y} \rightarrow \text{pt}$, we obtain that for any $\mathcal{Y}$, we have a well-defined object

$$\omega_{\mathcal{Y}} \in \text{IndCoh}^!(\mathcal{Y}).$$

A.4.7. It follows from the convergence property of the usual IndCoh functor

$$\mathrm{IndCoh} : (\mathrm{PreStk}_{\mathrm{laft}})^{\mathrm{op}} \rightarrow \mathrm{DGCat}$$

(see [GaRo3, Chapter 4, Prop. 6.4.3]) that the natural transformation

$$\mathrm{IndCoh}^!|_{\mathrm{PreStk}_{\mathrm{laft}}} \rightarrow \mathrm{IndCoh}$$

is an isomorphism.

I.e., the value of $\mathrm{IndCoh}^!(\mathcal{Y})$ on a prestack locally almost of finite type recovers the usual $\mathrm{IndCoh}(\mathcal{Y})$.

Remark A.4.8. The above construction gives a definition of the functor $\mathrm{IndCoh}^!(\mathcal{Y})$ for a general prestack $\mathcal{Y}$. But unless some conditions on $\mathcal{Y}$ are imposed, we will not be able to say much about the properties of this category.

For example, it is not even clear (and, probably, not true) whether for $S \in \mathrm{Sch}^{\mathrm{aff}}$, the category $\mathrm{IndCoh}(S)$ is dualizable.

A condition on $\mathcal{Y}$ that makes $\mathrm{IndCoh}^!(\mathcal{Y})$ manageable is called “placidity”, to be discussed in Sect. A.9.

A.4.9. In the sequel we will need the following property of $\mathrm{IndCoh}^!$:

Let $\mathcal{Y}$ be a prestack mapping to a *smooth* affine scheme S of finite type. Let $f : S' \rightarrow S$ be a map, where $S' \in \mathrm{Sch}_{\mathrm{aff}}^{\mathrm{aff}}$. Denote

$$\mathcal{Y}' := S' \times_S \mathcal{Y}.$$

By a slight abuse of notation, we will denote by the same symbol f the resulting map $\mathcal{Y}' \rightarrow \mathcal{Y}$.

The functor $f^! : \mathrm{IndCoh}^!(\mathcal{Y}) \rightarrow \mathrm{IndCoh}^!(\mathcal{Y}')$ extends to a functor

$$(A.19) \quad \mathrm{QCoh}(S') \otimes_{\mathrm{QCoh}(S)} \mathrm{IndCoh}^!(\mathcal{Y}) \rightarrow \mathrm{IndCoh}^!(\mathcal{Y}').$$

We claim:

Lemma A.4.10. *The functor (A.19) is fully faithful. If f is smooth, it is an equivalence.*

Proof. Follows from [Ga7, Propositions 4.4.2 and 7.5.7]. □

A.4.11. In the sequel we will need the following assertion about the behavior of $\mathrm{IndCoh}^!(-)$. Let

$$\begin{array}{ccc} \mathcal{Y}_1 & \xrightarrow{f_Y} & \mathcal{Y}_2 \\ \downarrow & & \downarrow \\ S_1 & \xrightarrow{f_S} & S_2 \end{array}$$

be a fiber square, where S_i are affine schemes almost of finite type, and f_S is a closed embedding of finite Tor-dimension. Unwinding the definitions, we obtain:

Lemma A.4.12. *Under the above circumstances, the functor $f_Y^! : \mathrm{IndCoh}^!(\mathcal{Y}_2) \rightarrow \mathrm{IndCoh}^!(\mathcal{Y}_1)$ admits a left adjoint, to be denoted $(f_Y)_*^{\mathrm{IndCoh}}$. Furthermore, the functor $(f_Y)_*^{\mathrm{IndCoh}}$ satisfies base change for any fiber square*

$$\begin{array}{ccc} \mathcal{Y}'_1 & \xrightarrow{f'_Y} & \mathcal{Y}'_2 \\ \downarrow & & \downarrow \\ \mathcal{Y}_1 & \xrightarrow{f_Y} & \mathcal{Y}_2. \end{array}$$

A.4.13. Let S be an affine scheme almost of finite type, and let $S' \subset S$ be a Zariski-closed subset. Let $S^\wedge$ denote the formal completion of S along S' . Let $\mathcal{Y}$ be a prestack over S , and set $\mathcal{Y}^\wedge := S^\wedge \times_S \mathcal{Y}$:

$$\begin{array}{ccc} \mathcal{Y}^\wedge & \xrightarrow{i_{\mathcal{Y}}} & \mathcal{Y} \\ \downarrow & & \downarrow \\ S^\wedge & \xrightarrow{i_S} & S. \end{array}$$

We now claim:

Proposition A.4.14. *Under the above circumstances, the functor $i_{\mathcal{Y}}^! : \mathrm{IndCoh}^!(\mathcal{Y}) \rightarrow \mathrm{IndCoh}^!(\mathcal{Y}^\wedge)$ is a colocalization, i.e., it admits a fully faithful left adjoint. The essential image of this left adjoint is the full subcategory of $\mathrm{IndCoh}^!(\mathcal{Y})$ consisting of objects with set-theoretic support on $\mathcal{Y}' := S' \times_S \mathcal{Y}$. The formation of the left adjoint satisfies base change for any fiber square*

$$\begin{array}{ccc} \tilde{\mathcal{Y}}_1^\wedge & \longrightarrow & \tilde{\mathcal{Y}} \\ \downarrow & & \downarrow \\ \tilde{\mathcal{Y}} & \longrightarrow & \mathcal{Y}. \end{array}$$

Proof. By [GaRo1, Proposition 6.7.4], we can write $S^\wedge$ as “colim” S_i , where S_i are closed subschemes of S , and the maps $S_i \rightarrow S$ are of finite Tor-dimension.

Unwinding the definition of $\mathrm{IndCoh}^!(-)$, we reduce the assertion to the case when $\mathcal{Y}$ is an eventually connective affine scheme $\tilde{S}$, so that

$$\tilde{S}^\wedge \simeq \text{“colim”}_i \tilde{S}_i, \quad \tilde{S}_i := S_i \times_S \tilde{S};$$

note that all $\tilde{S}_i$ are eventually coconnective.

Unwinding further, we reduce the assertion to the case when $\tilde{S}$ is of finite type; in this case the assertion follows from [GaRo1, Proposition 7.4.5]. $\square$

A.5. The category $\mathrm{IndCoh}^*(-)$.

A.5.1. We define the functor

$$\leq^n \mathrm{IndCoh}^* : \leq^n \mathrm{Sch}^{\mathrm{aff}} \rightarrow \mathrm{DGCat},$$

to be

$$\leq^n \mathrm{Sch}^{\mathrm{aff}} \xrightarrow{(\leq^n \mathrm{IndCoh}^!)^{\mathrm{op}}} (\mathrm{DGCat})^{\mathrm{op}} \rightarrow \mathrm{DGCat},$$

where:

- The first arrow is the opposite of the functor

$$S \mapsto \mathrm{IndCoh}^!(S), \quad (S_1 \xrightarrow{f} S_2) \rightsquigarrow (\mathrm{IndCoh}^!(S_2) \xrightarrow{f^!} \mathrm{IndCoh}^!(S_1));$$

- The second arrow is

$$\mathbf{D} \mapsto \mathbf{D}^\vee := \mathrm{Funct}_{\mathrm{cont}}(\mathbf{D}, \mathrm{Vect})$$

(note that in the above formula, the DG category $\mathbf{D}$ is not assumed dualizable).

A.5.2. Explicitly, for $S \in \leq^n \mathrm{Sch}^{\mathrm{aff}}$, we have:

$$(A.20) \quad \leq^n \mathrm{IndCoh}^*(S) = \lim_{S \rightarrow S_0, S_0 \in \leq^n \mathrm{Sch}_{\mathrm{ft}}^{\mathrm{aff}}} \mathrm{IndCoh}(S_0),$$

where the transition functors are given by

$$(S \rightarrow S'_0 \xrightarrow{f} S''_0) \rightsquigarrow \mathrm{IndCoh}(S'_0) \xrightarrow{f_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}(S''_0).$$

A.5.3. As in Sect. A.4.4, it is easy to see that the collection

$$n \rightsquigarrow {}^{\leq n} \text{IndCoh}^*$$

gives rise to a well-defined functor

$$(A.21) \quad \text{IndCoh}^* : {}^{<\infty} \text{Sch}^{\text{aff}} \rightarrow \text{DGCat}.$$

A.5.4. Note that, *by construction*, for $S \in {}^{<\infty} \text{Sch}^{\text{aff}}$, the category $\text{IndCoh}^!(S)$ is naturally a pre-dual of $\text{IndCoh}^*(S)$.

This will be a perfect duality if S is placid, see Sect. A.10.2 below.

A.5.5. Let $f : S_1 \rightarrow S_2$ be a morphism between eventually coconnective affine schemes. Unwinding the definitions, we obtain that with respect to the identifications

$$\text{IndCoh}^!(S_i)^\vee \simeq \text{IndCoh}^*(S_i), \quad i = 1, 2,$$

we have

$$(f^!)^\vee \simeq f_*^{\text{IndCoh}}.$$

A.5.6. Unlike $\text{IndCoh}^!$, we do not even attempt to define IndCoh^* on all prestacks. Rather, we define it on *ind-affine ind-schemes* (see [GaRo4, Chap. 3.1]).

Namely, we let the functor

$$\text{IndCoh}^* : \text{indSch} \rightarrow \text{DGCat}$$

to be the left Kan extension of (A.21) along the embedding

$${}^{<\infty} \text{Sch}^{\text{aff}} \hookrightarrow \text{indSch}^{\text{ind-aff}}.$$

A.5.7. Explicitly, for $\mathcal{Y} \in \text{indSch}^{\text{ind-aff}}$, we have

$$(A.22) \quad \text{IndCoh}^*(\mathcal{Y}) = \text{colim}_S \text{IndCoh}^*(S),$$

where:

- The index category is that of $S \in {}^{<\infty} \text{Sch}^{\text{aff}}$ equipped with a closed embedding $S \rightarrow \mathcal{Y}$;
- The transition functors are given by

$$(S' \xrightarrow{f} S'' \rightarrow \mathcal{Y}) \rightsquigarrow \text{IndCoh}^*(S') \xrightarrow{f_*^{\text{IndCoh}}} \text{IndCoh}^*(S'').$$

A.5.8. Thus, by definition, for a map $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ in $\text{indSch}^{\text{ind-aff}}$, we have a well-defined functor

$$f_*^{\text{IndCoh}} : \text{IndCoh}^*(\mathcal{Y}_1) \rightarrow \text{IndCoh}^*(\mathcal{Y}_2).$$

In particular, taking the projection $\mathcal{Y} \rightarrow \text{pt}$, we obtain that for any $\mathcal{Y} \in \text{indSch}^{\text{ind-aff}}$ there is a well-defined functor

$$\Gamma^{\text{IndCoh}}(\mathcal{Y}, -) : \text{IndCoh}^*(\mathcal{Y}) \rightarrow \text{Vect}.$$

A.5.9. It follows from [GaRo1, Sect. 2.4.2] that if $\mathcal{Y} \in \text{indSch}_{\text{aft}}^{\text{ind-aff}}$, the naturally defined functor

$$\text{IndCoh}(\mathcal{Y}) \rightarrow \text{IndCoh}^*(\mathcal{Y})$$

is an equivalence.

A.6. The multiplicative structure.

A.6.1. Note that since the index category in (A.17) is filtered (and, in particular, sifted), for $S \in {}^{\leq n}\text{Sch}^{\text{aff}}$, the category $\text{IndCoh}^!(S)$ carries a naturally defined monoidal structure.

Explicitly, the corresponding binary operation is given by

$$\text{IndCoh}^!(S) \otimes \text{IndCoh}^!(S) \rightarrow \text{IndCoh}^!(S \times S) \xrightarrow{\Delta_S^!} \text{IndCoh}^!(S).$$

In other words, the functor

$$\text{IndCoh}^! : (<^\infty \text{Sch}^{\text{aff}})^{\text{op}} \rightarrow \text{DGCat}$$

lifts to a functor

$$(<^\infty \text{Sch}^{\text{aff}})^{\text{op}} \rightarrow \text{ComAlg}(\text{DGCat}) = \text{DGCat}^{\text{SymMon}}.$$

A.6.2. By construction, we obtain that the functor

$$\text{IndCoh}^! : (\text{PreStk})^{\text{op}} \rightarrow \text{DGCat}$$

also lifts to a functor

$$(\text{PreStk})^{\text{op}} \rightarrow \text{ComAlg}(\text{DGCat}) = \text{DGCat}^{\text{SymMon}},$$

i.e., for every $\mathcal{Y} \in \text{PreStk}$, the category $\text{IndCoh}^!(\mathcal{Y})$ has a naturally defined symmetric monoidal structure.

Namely, the corresponding binary operation is given by

$$\text{IndCoh}^!(\mathcal{Y}) \otimes \text{IndCoh}^!(\mathcal{Y}) \rightarrow \text{IndCoh}^!(\mathcal{Y} \times \mathcal{Y}) \xrightarrow{\Delta_{\mathcal{Y}}^!} \text{IndCoh}^!(\mathcal{Y}).$$

The unit for this symmetric monoidal structure is the object $\omega_{\mathcal{Y}}$.

A.6.3. Let $\mathcal{Y}_1$ and $\mathcal{Y}_2$ be a pair of prestacks. The operation of pullback and tensor product gives rise to a functor

$$(A.23) \quad \text{IndCoh}^!(\mathcal{Y}_1) \otimes \text{IndCoh}^!(\mathcal{Y}_2) \rightarrow \text{IndCoh}^!(\mathcal{Y}_1 \times \mathcal{Y}_2).$$

For general prestacks there is no reason for (A.23) to be an equivalence.

A.6.4. Let S_1 and S_2 be a pair of eventually coconnective schemes. Given an eventually coconnective scheme S of finite type and a map $S_1 \times S_2 \rightarrow S$, the category of factorizations of this map as

$$S_1 \times S_2 \rightarrow S_{1,0} \times S_{2,0} \rightarrow S$$

is contractible, where:

- $S_{i,0}$ are eventually coconnective;
- The first arrow comes from a pair of maps $S_i \rightarrow S_{i,0}$.

This implies that we have a well-defined functor

$$\text{IndCoh}^*(S_1) \otimes \text{IndCoh}^*(S_2) \rightarrow \text{IndCoh}^*(S).$$

Passing to the limit over S , we obtain a functor

$$(A.24) \quad \text{IndCoh}^*(S_1) \otimes \text{IndCoh}^*(S_2) \rightarrow \text{IndCoh}^*(S_1 \times S_2).$$

Ind-extending, we obtain a functor

$$(A.25) \quad \text{IndCoh}^*(\mathcal{Y}_1) \otimes \text{IndCoh}^*(\mathcal{Y}_2) \rightarrow \text{IndCoh}^*(\mathcal{Y}_1 \times \mathcal{Y}_2),$$

where $\mathcal{Y}_1$ and $\mathcal{Y}_2$ are ind-schemes.

For general ind-schemes there is no reason for (A.25) to be an equivalence.

A.6.5. By a similar principle, we obtain a symmetric monoidal functor

$$\Upsilon_{\mathcal{Y}} : \text{QCoh}(\mathcal{Y}) \rightarrow \text{IndCoh}^!(\mathcal{Y}).$$

A.6.6. Note also that by the definition of IndCoh^* , for $S \in {}^{<\infty}\mathrm{Sch}^{\mathrm{aff}}$ we have a naturally defined action of $\mathrm{IndCoh}^!(S)$ on $\mathrm{IndCoh}^*(S)$.

For a map $f : S_1 \rightarrow S_2$, this action satisfies the projection formula

$$\mathcal{F}_2 \otimes f_*^{\mathrm{IndCoh}}(\mathcal{F}_1) \simeq f_*^{\mathrm{IndCoh}}(f^!(\mathcal{F}_2) \otimes \mathcal{F}_1), \quad \mathcal{F}_1 \in \mathrm{IndCoh}^*(S_1), \mathcal{F}_2 \in \mathrm{IndCoh}^!(S_2).$$

This implies that for $\mathcal{Y} \in \mathrm{indSch}_{\mathrm{laft}}^{\mathrm{ind-aff}}$, we also have a natural action of $\mathrm{IndCoh}^!(\mathcal{Y})$ on $\mathrm{IndCoh}^*(\mathcal{Y})$, and the projection formula holds.

A.6.7. For an ind-scheme $\mathcal{Y}$ we have a canonically defined pairing:

$$(A.26) \quad \mathrm{IndCoh}^*(\mathcal{Y}) \otimes \mathrm{IndCoh}^!(\mathcal{Y}) \xrightarrow{\text{action}} \mathrm{IndCoh}^*(\mathcal{Y}) \xrightarrow{\Gamma^{\mathrm{IndCoh}(\mathcal{Y}, -)}} \mathrm{Vect}.$$

Note, however, that unlike the case of schemes, we do not claim that the above pairing realizes $\mathrm{IndCoh}^!(\mathcal{Y})$ as the predual of $\mathrm{IndCoh}^*(\mathcal{Y})$. (It will, however, be a perfect duality, under the placidity assumption.)

A.7. Further properties of IndCoh^* .

A.7.1. *The functor Ψ .* Let S be an eventually coconnective scheme. The presentation in (A.20) shows that we have a canonically defined functor

$$\Psi_S : \mathrm{IndCoh}^*(S) \rightarrow \mathrm{QCoh}(S).$$

Indeed, it is given by the compatible family of functors

$$\Psi_{S_0} : \mathrm{IndCoh}(S_0) \rightarrow \mathrm{QCoh}(S_0),$$

where we use the fact that for any affine scheme S , written as a limit of other affine schemes

$$S \simeq \lim_{\alpha} S_{\alpha},$$

the functor

$$\mathrm{QCoh}(S) \rightarrow \lim_{\alpha} \mathrm{QCoh}(S_{\alpha}),$$

given by taking direct images along $S \rightarrow S_{\alpha}$, is an equivalence.

For a map $f : S_1 \rightarrow S_2$, we have a commutative diagram

$$\begin{array}{ccc} \mathrm{IndCoh}^*(S_1) & \xrightarrow{\Psi_{S_1}} & \mathrm{QCoh}(S_1) \\ f_*^{\mathrm{IndCoh}} \downarrow & & \downarrow f_* \\ \mathrm{IndCoh}^*(S_2) & \xrightarrow{\Psi_{S_2}} & \mathrm{QCoh}(S_2). \end{array}$$

A.7.2. Unwinding, we obtain that with respect to the identification

$$\mathrm{IndCoh}^*(S) \simeq \mathrm{IndCoh}^!(S)^{\vee}$$

of Sect. A.5.4 and the canonical self-duality on $\mathrm{QCoh}(S)$, we have

$$\Psi_S \simeq (\Upsilon_S)^{\vee}$$

A.7.3. For an ind-affine ind-scheme $\mathcal{Y}$, we have the functor

$$\Psi_{\mathcal{Y}} : \mathrm{IndCoh}^*(\mathcal{Y}) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$$

defined in terms of the presentation (A.22) by

$$\mathrm{IndCoh}^*(\mathcal{Y}) \simeq \mathrm{colim}_{S \rightarrow \mathcal{Y}} \mathrm{IndCoh}^*(S) \xrightarrow{\{\Psi_S\}} \mathrm{colim}_{S \rightarrow \mathcal{Y}} \mathrm{QCoh}(S) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}),$$

where the colimits are taken over the index category of eventually coconnective affine schemes equipped with a closed embedding into $\mathcal{Y}$.

Note that the functors $\Psi_{\mathcal{Y}}$ and $\Upsilon_{\mathcal{Y}}$ are mutually dual in the sense that the following diagram commutes

$$\begin{array}{ccc} \mathrm{IndCoh}^*(\mathcal{Y}) \otimes \mathrm{QCoh}(\mathcal{Y}) & \xrightarrow{\mathrm{Id} \otimes \Upsilon_{\mathcal{Y}}} & \mathrm{IndCoh}^*(\mathcal{Y}) \otimes \mathrm{IndCoh}^!(\mathcal{Y}) \\ \Psi_{\mathcal{Y}} \otimes \mathrm{Id} \downarrow & & \downarrow \text{(A.26)} \\ \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}) \otimes \mathrm{QCoh}(\mathcal{Y}) & \xrightarrow{\text{(A.6)}} & \mathrm{Vect}. \end{array}$$

A.7.4. Let $f : S_1 \rightarrow S_2$ be a map between eventually connective affine schemes. Assume that S_1 is finitely presented over S_2 (inside the category of n -coconnective schemes for some n) and that f is of finite Tor-dimension.

We claim that in this case the functor

$$f_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(S_1) \rightarrow \mathrm{IndCoh}^*(S_2)$$

admits a left adjoint, to be denoted $f^{*, \mathrm{IndCoh}}$.

By Noetherian approximation (see [Lu3, Sect. 4.4.1]), the assumption on f implies that we can write S_2 as $\lim_{\alpha} S_{2,\alpha}$, where $S_{2,\alpha} \in \leq^n \mathrm{Sch}^{\mathrm{aff}}$, so that there exists a compatible family of Cartesian diagrams

$$\begin{array}{ccc} S_1 & \xrightarrow{f} & S_2 \\ \downarrow & & \downarrow \\ S_{1,\alpha} & \xrightarrow{f_{\alpha}} & S_{2,\alpha} \end{array}$$

with $S_{1,\alpha} \in \leq^n \mathrm{Sch}^{\mathrm{aff}}$ and $S_1 \simeq \lim_{\alpha} S_{1,\alpha}$.

In this case we have

$$\mathrm{IndCoh}^*(S_2) \simeq \lim_{\alpha} \mathrm{IndCoh}(S_{2,\alpha}) \text{ and } \mathrm{IndCoh}^*(S_1) \simeq \lim_{\alpha} \mathrm{IndCoh}(S_{1,\alpha}),$$

and the functor $f^{*, \mathrm{IndCoh}}$ is given by the compatible family of functors

$$\mathrm{IndCoh}(S_{2,\alpha}) \xrightarrow{f_{\alpha}^{*, \mathrm{IndCoh}}} \mathrm{IndCoh}(S_{1,\alpha}),$$

which exists thanks to [Ga7, Lemma 3.5.8].

A.7.5. Let

$$\begin{array}{ccc} S'_1 & \xrightarrow{f'} & S'_2 \\ g_1 \downarrow & & \downarrow g_2 \\ S_1 & \xrightarrow{f} & S_2 \end{array}$$

be a Cartesian diagram of eventually connective affine schemes, where the horizontal arrows are of finite presentation and finite Tor-dimension.

We have the tautological isomorphism

$$(g_2)_*^{\mathrm{IndCoh}} \circ f'^{\mathrm{IndCoh}}_* \simeq f_*^{\mathrm{IndCoh}} \circ (g_1)_*^{\mathrm{IndCoh}},$$

from which we obtain a natural transformation

$$f^{*, \mathrm{IndCoh}} \circ (g_2)_*^{\mathrm{IndCoh}} \rightarrow (g_1)_*^{\mathrm{IndCoh}} \circ f'^{*, \mathrm{IndCoh}}.$$

However, it is easy to see that the above natural transformation is an isomorphism, see [Ga7, Lemma 3.6.9].

A.7.6. Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a map between ind-affine ind-schemes, and assume that f is (i) affine, (ii) of finite presentation, (iii) of finite Tor-dimension (i.e., the above properties hold after base change of f by an affine scheme).

We claim that in this case the functor

$$f_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathcal{Y}_1) \rightarrow \mathrm{IndCoh}^*(\mathcal{Y}_2)$$

admits a left adjoint (to be denoted $f^{*,\mathrm{IndCoh}}$).

Indeed, write

$$\mathrm{IndCoh}^*(\mathcal{Y}_2) \simeq \operatorname{colim}_{S_{2,\alpha} \rightarrow \mathcal{Y}_2} \mathrm{IndCoh}^*(S_{2,\alpha}),$$

where the index category is that of $S_{2,\alpha} \in {}^{<\infty}\mathrm{Sch}^{\mathrm{aff}}$ equipped with a closed embedding $S_{2,\alpha} \rightarrow \mathcal{Y}_2$.

For $S_{2,\alpha}$ as above, set

$$S_{1,\alpha} := \mathcal{Y}_1 \times_{\mathcal{Y}_2} S_{2,\alpha}.$$

Then the family

$$S_{1,\alpha} \rightarrow \mathcal{Y}_1$$

is cofinal in the category of eventually coconnective affine schemes mapping to $\mathcal{Y}_1$, and hence we have

$$\mathrm{IndCoh}^*(\mathcal{Y}_1) \simeq \operatorname{colim}_{S_{1,\alpha} \rightarrow \mathcal{Y}_1} \mathrm{IndCoh}^*(S_{1,\alpha}).$$

In terms of this presentation, the functor $f^{*,\mathrm{IndCoh}}$ is given by the (compatible) family of functors

$$f_\alpha^{*,\mathrm{IndCoh}} : \mathrm{IndCoh}^*(S_{2,\alpha}) \rightarrow \mathrm{IndCoh}^*(S_{1,\alpha}),$$

see Sect. A.7.4.

A.7.7. The following is a counterpart of Sect. A.4.9 for IndCoh^* . Let us be in the situation of *loc. cit.*, but let us assume that $\mathcal{Y}$ is an ind-scheme.

Since f is finite Tor-dimension, we can consider the functor $f^{*,\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathcal{Y}) \rightarrow \mathrm{IndCoh}^*(\mathcal{Y}')$, and it extends to a functor

$$(A.27) \quad \mathrm{QCoh}(S') \otimes_{\mathrm{QCoh}(S)} \mathrm{IndCoh}^*(\mathcal{Y}) \rightarrow \mathrm{IndCoh}^*(\mathcal{Y}').$$

We have:

Lemma A.7.8. *The functor (A.27) is fully faithful. If f is smooth, it is an equivalence.*

A.8. The t-structure on IndCoh^* .

A.8.1. Let S be an eventually coconnective affine scheme. The presentation in (A.20) endows the category $\mathrm{IndCoh}^*(S)$ with a t-structure. It is uniquely characterized by the property that for a map

$$f : S \rightarrow S_0, \quad S_0 \in {}^{<\infty}\mathrm{Sch}_{\mathrm{ft}}^{\mathrm{aff}}$$

the functor

$$f_* : \mathrm{IndCoh}^*(S) \rightarrow \mathrm{IndCoh}(S_0)$$

is t-exact.

This t-structure is compatible with filtered colimits, by construction.

A.8.2. For a map $S_1 \rightarrow S_2$ between eventually coconnective affine schemes, the corresponding functor

$$f_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(S_1) \rightarrow \mathrm{IndCoh}^*(S_2)$$

is t-exact.

A.8.3. By construction, the functor Ψ_S is t-exact and induces an equivalence

$$\mathrm{IndCoh}^*(S)^{>-\infty} \xrightarrow{\sim} \mathrm{QCoh}(S)^{>-\infty}.$$

A.8.4. Let $\mathcal{Y}$ be an ind-affine ind-scheme. We use the presentation (A.22) and the construction in Sect. A.2.3 to equip $\mathrm{IndCoh}^*(\mathcal{Y})$ with a t -structure:

By definition, $\mathrm{IndCoh}^*(\mathcal{Y})^{\leq 0}$ is generated under colimits by the essential images of $\mathrm{IndCoh}^*(S)^{\leq 0}$ for $S \in {}^{<\infty}\mathrm{Sch}^{\mathrm{aff}}$ equipped with a closed embedding $S \rightarrow \mathcal{Y}$.

A.8.5. From Lemma A.2.5, we obtain:

Corollary A.8.6.

- (a) *The t -structure on $\mathrm{IndCoh}^*(\mathcal{Y})$ is compatible with filtered colimits.*
- (b) *For every $S \in {}^{<\infty}\mathrm{Sch}^{\mathrm{aff}}$ equipped with a closed embedding $S \rightarrow \mathcal{Y}$, the direct image functor $\mathrm{IndCoh}^*(S) \rightarrow \mathrm{IndCoh}^*(\mathcal{Y})$ is t -exact.*

A.8.7. Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a map between ind-affine ind-schemes. As in Corollary A.2.12, we have:

Corollary A.8.8. *The functor*

$$f_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathcal{Y}_1) \rightarrow \mathrm{IndCoh}^*(\mathcal{Y}_2)$$

is t -exact.

A.8.9. Recall the functor

$$\Psi_{\mathcal{Y}} : \mathrm{IndCoh}^*(\mathcal{Y}) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}),$$

see Sect. A.7.3. We claim:

Lemma A.8.10. *The functor $\Psi_{\mathcal{Y}}$ is t -exact and induces an equivalence:*

$$\mathrm{IndCoh}^*(\mathcal{Y})^{>-\infty} \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})^{>-\infty}.$$

Proof. The fact that Ψ is t -exact follows from Corollaries A.2.10(b) and A.2.7.

To prove the equivalence statement, it suffices to show that for any n , the corresponding functor

$$\mathrm{IndCoh}^*(\mathcal{Y})^{\geq 0, \leq n} \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})^{\geq 0, \leq n}$$

is an equivalence.

As in the proof of Lemma A.2.5, we can write

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})^{\geq 0, \leq n} \simeq \mathrm{colim}_{S \rightarrow \mathcal{Y}} \mathrm{QCoh}(S)^{\geq 0, \leq n}$$

and

$$\mathrm{IndCoh}^*(\mathcal{Y})^{\geq 0, \leq n} \simeq \mathrm{colim}_{S' \rightarrow \mathcal{Y}} \mathrm{IndCoh}^*(S')^{\geq 0, \leq n},$$

where:

- The index S runs over the category affine schemes equipped with a closed embedding into $\mathcal{Y}$;
- The index S' runs over the category of eventually coconnective affine schemes equipped with a closed embedding into $\mathcal{Y}$.
- Both colimits are taken in the ∞ -category of categories closed under filtered colimits and functors that preserve filtered colimits.

Now, the assertion follows from the fact that for $m \geq n$, the direct image functor

$$\mathrm{QCoh}(\leq^m S) \rightarrow \mathrm{QCoh}(S)$$

induces an equivalence

$$\mathrm{QCoh}(\leq^m S)^{\geq 0, \leq n} \rightarrow \mathrm{QCoh}(S)^{\geq 0, \leq n}.$$

□

A.9. Placidity.

A.9.1. An affine scheme S is said to be *placid* if it can be written as a limit

$$(A.28) \quad S \simeq \lim_{\alpha \in A} S_\alpha,$$

where:

- $S_\alpha \in \text{Sch}_{\text{aff}}^{\text{aff}}$;
- The transition maps $f_{\beta,\alpha} : S_\beta \rightarrow S_\alpha$ are flat.
- The category A of indices is co-filtered (i.e., the opposite category is filtered).

Remark A.9.2. There are in fact two variants of the definition of placidity. The less restricted one is what we just gave above. In the more restrictive one, one requires that the maps $f_{\beta,\alpha}$ be smooth.

The flatness condition is sufficient for our purposes, which are to ensure the compact generation of the categories IndCoh^* and $\text{IndCoh}^!$ (see Sect. A.10.1). One needs smoothness when one works with D-modules.

That said, in most examples of placid schemes that we will encounter, the smoothness condition is satisfied as well.

A.9.3. Note that if S is placid, then so is any of its truncations: for a presentation (A.28), we have

$$\leq^n S \simeq \lim_{\alpha} \leq^n S_\alpha,$$

and the truncated maps $\leq^n S_\beta \rightarrow \leq^n S_\alpha$ are also flat; in fact the flatness of $f_{\beta,\alpha}$ implies that

$$\leq^n S_\beta \simeq S_\beta \times_{S_\alpha} \leq^n S_\alpha.$$

A.9.4. Fix an integer n . Let $R \rightarrow R'$ be a map in $\text{ComAlg}(\text{Vect}^{\leq 0, \geq -n})$. Recall that R' is said to be *finitely presented* as an R -algebra if R' is compact as an object of $\text{ComAlg}((R\text{-mod})^{\leq 0, \geq -n})$.

Let $R \rightarrow R'$ be a map in $\text{ComAlg}(\text{Vect}^{\leq 0})$. We shall say that R' *almost finitely presented* as an R -algebra if for every n , the map $\tau^{\geq -n}(R) \rightarrow \tau^{\geq -n}(R')$ realizes $\tau^{\geq -n}(R')$ as a finitely presented $\tau^{\geq -n}(R)$ -algebra.

We shall say that a morphism of n -coconnective affine schemes (resp., affine schemes) $\text{Spec}(R') = S' \rightarrow S = \text{Spec}(R)$ is of finite presentation (resp., almost of finite presentation) if R' is finitely presented (resp., almost finitely presented) as an R -algebra.

A.9.5. Let $\mathcal{Y}$ be an ind-affine ind-scheme mapping to an affine scheme S . We shall say that $\mathcal{Y}$ is locally almost of finite presentation over S if for every n , the truncation

$$\leq^n \mathcal{Y} \in \leq^n \text{PreStk}$$

can be exhibited as a filtered colimit

$$\leq^n \mathcal{Y} \simeq \text{colim}_{i \in I} S_i, \quad S_i \in \leq^n \text{Sch}^{\text{aff}},$$

such that the maps $S_i \rightarrow S$ are of finite presentation.

Let $\mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a map between ind-affine ind-schemes. We shall say that f is locally almost of finite presentation if the base change of f by any affine scheme S yields an ind-affine ind-scheme locally almost of finite presentation over S .

A.9.6. We have the following hereditary property of placidity:

Lemma A.9.7. *Let $S' \rightarrow S$ be a map almost of finite presentation between affine schemes. Suppose that every coconnective truncation of S is placid. Then the same is true for S' .*

Proof. Fix n and consider the corresponding map $\leq^n S' \rightarrow \leq^n S$. Write $\leq^n S$ as

$$\lim_{\alpha \in A} S_\alpha, \quad S_\alpha \in \leq^n \text{Sch}_{\text{ft}}^{\text{aff}}$$

as in (A.28).

Then for some index α , we have an affine scheme $S'_\alpha \in \leq^n \text{Sch}_{\text{ft}}^{\text{aff}}$ and a Cartesian square

$$\begin{array}{ccc} \leq^n S' & \longrightarrow & \leq^n S \\ \downarrow & & \downarrow \\ S'_\alpha & \longrightarrow & S_\alpha. \end{array}$$

Consider the category $A_{/\alpha}$. Since A is cofiltered, the category $A_{/\alpha}$ is also cofiltered and the opposite of the inclusion functor $A_{/\alpha} \rightarrow A$ is cofinal.

For any $\beta \in A_{/\alpha}$ denote

$$S'_\beta := S'_\alpha \times_{S_\alpha} S_\beta.$$

Then we have

$$\leq^n S' \simeq \lim_{\beta \in A_{/\alpha}} S'_\beta,$$

and the maps $S'_{\beta_1} \rightarrow S'_{\beta_2}$ are flat. □

A.9.8. Let $\mathcal{Y}$ be an ind-affine ind-scheme. We shall say that $\mathcal{Y}$ is *ind-placid* if for every n , the truncation

$$\leq^n \mathcal{Y} \in \leq^n \text{PreStk}$$

can be exhibited as a filtered colimit

$$(A.29) \quad \leq^n \mathcal{Y} \simeq \text{colim}_{i \in I} S_i, \quad S_i \in \leq^n \text{Sch}^{\text{aff}},$$

where:

- The affine schemes S_i are placid;
- The transition maps $S_i \rightarrow S_j$ are of finite presentation.

A.9.9. From Lemma A.9.7 we obtain:

Corollary A.9.10. *Let $\mathcal{Y}$ be an ind-placid ind-scheme and let $f : \mathcal{Y}' \rightarrow \mathcal{Y}$ be a map locally almost of finite presentation. Then $\mathcal{Y}'$ is also an ind-placid ind-scheme.*

Proof. Fix an integer n and a presentation of $\leq^n \mathcal{Y}$ as in (A.29). For every index i , set

$$\mathcal{Y}'_i := \leq^n (\mathcal{Y}' \times_{\mathcal{Y}} S_i).$$

By assumption, this is an ind-affine ind-scheme of ind-finite presentation over S_i . Consider the category

$$\mathbf{F}_i := \{S'_i \in \leq^n \text{Sch}^{\text{aff}}, S'_i \xrightarrow{\text{closed emb}} \mathcal{Y}'_i\}.$$

By assumption, this subcategory contains a full cofinal subcategory, denoted $\mathbf{F}'_i$ consisting of those objects for which the map $S'_i \rightarrow S_i$ is of finite presentation.

The assignment $i \mapsto \mathbf{F}_i$ extends to a co-Cartesian fibration

$$\mathbf{F} \rightarrow I.$$

The assumption that the maps $S_i \rightarrow S_j$ are of finite presentation implies that the assignment $i \mapsto \mathbf{F}'_i$ corresponds to a full cofinal subcategory $\mathbf{F}' \subset \mathbf{F}$, and

$$\mathbf{F}' \rightarrow I$$

is also a co-Cartesian fibration. By cofinality, we obtain that the map

$$(A.30) \quad \text{colim}_{(i, S'_i) \in \mathbf{F}} S'_i \rightarrow \mathcal{Y}'$$

is an isomorphism in $\leq^n \text{PreStk}$. Moreover, for a map $(i, S'_i) \rightarrow (j, S'_j)$ in $\mathbf{F}'$, the corresponding map $S'_i \rightarrow S'_j$ is of finite presentation (because its composition with $S'_j \rightarrow S_j$ is).

Since I is filtered and each $\mathbf{F}'_i$ is filtered, we obtain that $\mathbf{F}'$ is filtered.

Finally, by Lemma A.9.7, the affine schemes S'_i are placid. Hence, (A.30) gives the desired presentation of $\mathcal{Y}'$. $\square$

A.10. The categories $\text{IndCoh}^!(-)$ and $\text{IndCoh}^*(-)$ in the (ind)-placid case.

A.10.1. Let $S \in \leq^n \text{Sch}^{\text{aff}}$ be placid. We claim that in this case, the category $\text{IndCoh}^!(S)$ is compactly generated.

Indeed, write S as in (A.28). Since the maps $f_{\beta,\alpha}$ are flat, the functors $f_{\beta,\alpha}^!$ preserves coherence, and hence compactness. We obtain

$$\text{IndCoh}^!(S) \simeq \text{colim}_{\alpha} \text{IndCoh}(S_{\alpha}),$$

where the terms are compactly generated, and the transition functors preserve compactness.

Hence, the images of $\text{Coh}(S_{\alpha}) \subset \text{IndCoh}(S_{\alpha})$ under the $!$ -pullback functors

$$\text{IndCoh}(S_{\alpha}) \rightarrow \text{IndCoh}^!(S)$$

provide a set of compact generators of $\text{IndCoh}^!(S)$.

A.10.2. From the identification

$$(A.31) \quad \text{IndCoh}^*(S) \simeq \text{IndCoh}^!(S)^{\vee}$$

of Sect. A.5.4, we obtain that $\text{IndCoh}^*(S)$ is also compactly generated, and (A.31) is a duality in DGCat .

A.10.3. Explicitly, the presentation

$$\text{IndCoh}^*(S) \simeq \lim_{\alpha} \text{IndCoh}(S_{\alpha})$$

(with respect to the $*$ -pushforward functors) implies that

$$\text{IndCoh}^*(S) \simeq \text{colim}_{\alpha} \text{IndCoh}(S_{\alpha}),$$

with respect to the $*$ -pullback functors (which are well-defined, due to the flatness assumption).

Thus, the compact generators of $\text{IndCoh}^*(S)$ are the images of $\text{Coh}(S_{\alpha}) \subset \text{IndCoh}(S_{\alpha})$ along the $*$ -pullback functors

$$\text{IndCoh}(S_{\alpha}) \rightarrow \text{IndCoh}^*(S).$$

A.10.4. Let $f : S_1 \rightarrow S_2$ be a morphism almost of finite presentation between placid eventually coconnective affine schemes. Assume now that f is a closed embedding. We claim that in this case the functor

$$f_*^{\text{IndCoh}} : \text{IndCoh}^*(S_1) \rightarrow \text{IndCoh}^*(S_2)$$

admits a continuous right adjoint (to be denoted $f^!$).

Indeed, we claim that f_*^{IndCoh} preserves compactness. This follows from the manipulation in the proof of Lemma A.9.7 using the fact that for a Cartesian diagram of affine schemes almost of finite type

$$\begin{array}{ccc} S'_1 & \xrightarrow{f'} & S'_2 \\ g_1 \downarrow & & \downarrow g_2 \\ S''_1 & \xrightarrow{f''} & S''_2 \end{array}$$

with the maps g_1, g_2 flat, the natural transformation

$$g_2^{\text{IndCoh},*} \circ f''_*^{\text{IndCoh}} \rightarrow f'_*^{\text{IndCoh}} \circ g_1^{\text{IndCoh},*}$$

is an isomorphism.

A.10.5. Moreover, in the above situation, for a Cartesian diagram

$$\begin{array}{ccc} \tilde{S}_1 & \xrightarrow{\tilde{f}} & \tilde{S}_2 \\ g_1 \downarrow & & \downarrow g_2 \\ S_1 & \xrightarrow{f} & S_2, \end{array}$$

where:

- All affine schemes involved are eventually coconnective and placid;
- The maps f and $\tilde{f}$ are closed embeddings of finite presentation,

the natural transformation

$$(g_1)_*^{\mathrm{IndCoh}} \circ \tilde{f}^! \rightarrow f^! \circ (g_2)_*^{\mathrm{IndCoh}}, \quad \mathrm{IndCoh}^*(\tilde{S}_2) \rightrightarrows \mathrm{IndCoh}^*(S_1),$$

obtained by adjunction from

$$f_*^{\mathrm{IndCoh}} \circ (g_1)_*^{\mathrm{IndCoh}} \rightarrow (g_2)_*^{\mathrm{IndCoh}} \circ \tilde{f}_*^{\mathrm{IndCoh}},$$

is an isomorphism.

A.10.6. Recall (see Sect. A.5.5) that that with respect to the dualities

$$\mathrm{IndCoh}^*(S_i) \simeq \mathrm{IndCoh}^!(S_i)^\vee, \quad i = 1, 2,$$

we have

$$f_*^{\mathrm{IndCoh}} \simeq (f^!)^\vee.$$

Hence, the existence of a continuous right adjoint of

$$f_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(S_1) \rightarrow \mathrm{IndCoh}^*(S_2)$$

implies the existence of a *left* adjoint of the functor

$$f^! : \mathrm{IndCoh}^!(S_2) \rightarrow \mathrm{IndCoh}^!(S_1).$$

We will denote this left adjoint by

$$f_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^!(S_1) \rightarrow \mathrm{IndCoh}^!(S_2).$$

A.10.7. Let now $\mathcal{Y}$ be an ind-placid ind-scheme. We claim that in this case the category $\mathrm{IndCoh}^*(\mathcal{Y})$ is compactly generated.

Indeed, writing

$$\mathrm{IndCoh}^*(\mathcal{Y}) \simeq \mathrm{colim}_i \mathrm{IndCoh}^*(S_i),$$

where S_i are eventually coconnective placid affine schemes and the transition maps $S_i \xrightarrow{f_{i,j}} S_j$ are closed embeddings almost of finite presentation, we obtain that the transition functors

$$\mathrm{IndCoh}^*(S_i) \rightarrow \mathrm{IndCoh}^*(S_j)$$

preserve compactness (see Sect. A.10.4 above).

Similarly, the category $\mathrm{IndCoh}^!(\mathcal{Y})$ can be written as

$$\mathrm{colim}_i \mathrm{IndCoh}^!(S_i),$$

where the transition functors are $(f_{i,j})_*^{\mathrm{IndCoh}}$. This implies that $\mathrm{IndCoh}^!(\mathcal{Y})$ is also compactly generated by the essential images of $\mathrm{IndCoh}^!(S_i)^c$.

A.10.8. Note also that it follows that if $\mathcal{Y}$ is placid, the pairing (A.26) is a perfect duality.

A.10.9. By a similar logic we obtain:

Lemma A.10.10. *Let $\mathcal{Y}_1$ and $\mathcal{Y}_2$ be a pair of ind-placid ind-schemes. Then the functors (A.23) and (A.25) are equivalences.*

A.10.11. Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a morphism between ind-placid ind-affine ind-schemes that is locally almost of finite presentation. Assume that f is an ind-closed embedding (i.e., for every closed embedding $S \rightarrow \mathcal{Y}_1$, the composite map $S \rightarrow \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ is also a closed embedding).

Similarly to the above, we obtain that in this case, the functor

$$f_*^{\text{IndCoh}} : \text{IndCoh}^*(\mathcal{Y}_1) \rightarrow \text{IndCoh}^*(\mathcal{Y}_2)$$

preserves compactness, and hence admits a continuous right adjoint, to be denoted $f^!$.

By duality, the functor

$$f^! : \text{IndCoh}^!(\mathcal{Y}_2) \rightarrow \text{IndCoh}^!(\mathcal{Y}_1)$$

admits a left adjoint, to be denoted

$$\tilde{f}_*^{\text{IndCoh}} : \text{IndCoh}^!(\mathcal{Y}_1) \rightarrow \text{IndCoh}^!(\mathcal{Y}_2).$$

A.10.12. Let

$$\begin{array}{ccc} \tilde{\mathcal{Y}}_1 & \xrightarrow{\tilde{f}} & \tilde{\mathcal{Y}}_2 \\ g_1 \downarrow & & \downarrow g_2 \\ \mathcal{Y}_1 & \xrightarrow{f} & \mathcal{Y}_2 \end{array}$$

be a Cartesian diagram, where:

- All objects are involved are ind-placid affine ind-schemes;
- The maps f and $\tilde{f}$ are ind-closed embeddings of finite presentation.

Unwinding, it follows from Sect. A.10.5 that in this case, the natural transformation

$$(g_1^{\text{IndCoh}})_* \circ \tilde{f}^! \rightarrow f^! \circ (g_2^{\text{IndCoh}})_*, \quad \text{IndCoh}^*(\tilde{\mathcal{Y}}_2) \rightrightarrows \text{IndCoh}^*(\mathcal{Y}_1), \quad \text{IndCoh}^*(\tilde{\mathcal{Y}}_2) \rightrightarrows \text{IndCoh}^*(\mathcal{Y}_1),$$

obtained by adjunction from

$$f_*^{\text{IndCoh}} \circ (g_1^{\text{IndCoh}})_* \rightarrow (g_2^{\text{IndCoh}})_* \circ \tilde{f}_*^{\text{IndCoh}},$$

is an isomorphism.

By duality, the natural transformation

$$\tilde{f}_*^{\text{IndCoh}} \circ g_1^! \rightarrow g_2^! \circ f_*^{\text{IndCoh}}, \quad \text{IndCoh}^*(\mathcal{Y}_1) \rightrightarrows \text{IndCoh}^*(\tilde{\mathcal{Y}}_2),$$

obtained by adjunction from

$$g_1^! \circ f^! \simeq \tilde{f}^! \circ g_2^!,$$

is also an isomorphism.

A.10.13. Let now $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a morphism between ind-placid ind-schemes. Assume that f is affine and of finite Tor-dimension (but we are not assuming that f be of finite presentation).

We claim, generalizing Sect. A.7.4, that in this case the functor

$$f^{*, \text{IndCoh}} : \text{IndCoh}^*(\mathcal{Y}_2) \rightarrow \text{IndCoh}^*(\mathcal{Y}_1),$$

left adjoint to f_*^{IndCoh} , exists, and satisfies base change against functors g_*^{IndCoh} for $g : \mathcal{Y}'_1 \rightarrow \mathcal{Y}_1$, where g is another ind-placid ind-scheme.

Indeed, by Sect. A.10.11, the question reduces to the case when $\mathcal{Y}_2$ is an eventually coconnective affine scheme, to be denoted S_2 . In this case, $\mathcal{Y}_1$ is also an eventually coconnective affine scheme, to be denoted S_1 .

Furthermore, in this case we can further reduce to the case when S_2 is of finite type. Since S_1 is placid, we can factor the morphism f as

$$S_1 \xrightarrow{h} S_{1,0} \xrightarrow{f_0} S_2,$$

where:

- $S_{1,0}$ is an eventually coconnective scheme of finite type,
- The morphism h is flat;

- The morphism f_0 is of finite Tor-dimension.

This reduces the assertion to the case of eventually coconnective affine schemes of finite type, where it follows from [Ga7, Lemma 3.5.8].

APPENDIX B. FACTORIZATION PATTERNS

The local Langlands theory considers various representation-theoretic categories $\mathbf{A}_x$ attached to the group G (or its dual $\check{G}$) and the formal disc $\mathcal{D}_x$, attached to a point $x \in X$. More generally, one is led to consider the multi-disc $\mathcal{D}_{\underline{x}}$, $\underline{x} = x_1, \dots, x_n$; moreover the points $x_1, \dots, x_n$ are allowed to move in families over X and collide. In this case, we shall say that $\underline{x}$ is a (scheme-theoretic) point of the *Ran* space of X .

The datum of a *factorization category* attached to such $\underline{x}$ a category $\mathbf{A}_{\underline{x}}$, such that if $\underline{x}_1$ and $\underline{x}_2$ are disjoint, we are given an isomorphism

$$\mathbf{A}_{\underline{x}_1 \sqcup \underline{x}_2} \simeq \mathbf{A}_{\underline{x}_1} \otimes \mathbf{A}_{\underline{x}_2}.$$

We develop the theory of factorization categories in this section, along with various adjoining notions (factorization spaces, factorization algebras). Can can view this section as a natural development of the theory of chiral algebras, initiated in [BD2]. The main difference with *loc. cit.* is that all our constructions take place in the world of ∞ -categories, whereas in [BD2] one mainly worked at the abelian level.

In order to produce examples of factorization categories one often uses geometric objects associated to the formal (resp., formal punctured) disc, such as arcs and loop spaces. Some of the work in this section is devoted to the study of the relevant geometries.

B.1. Factorization spaces.

B.1.1. The Ran space of X , denoted Ran , is the prestack that assigns to an affine test scheme S the set of finite-non-empty subsets of $\mathrm{Hom}(S_{\mathrm{red}}, X)$.

Note that, by definition, the map $\mathrm{Ran} \rightarrow \mathrm{Ran}_{\mathrm{dR}}$ is an isomorphism.

We denote k -points of Ran by $\underline{x}$. By definition, these are finite non-empty collections

$$(B.1) \quad \underline{x} = \{x_1, \dots, x_n\}$$

of k -points of X .

B.1.2. In what follows we will use the following notations. Let $\underline{x} : S \rightarrow \mathrm{Ran}$ be a map corresponding to $I \subset \mathrm{Hom}(S_{\mathrm{red}}, X)$.

- For $i \in I$, we will denote by x_i the corresponding map $S_{\mathrm{red}} \rightarrow X$;
- We will denote by $\mathrm{Graph}_{x_i} \subset S \times X$ the graph of x_i (viewed as a closed *subset*, i.e., we ignore its scheme-theoretic structure);
- We will denote by $\mathrm{Graph}_{\underline{x}}$ the Zariski-closed subset of $S \times X$ equal to $\bigcup_i \mathrm{Graph}_{x_i}$.

B.1.3. One can exhibit Ran explicitly as a colimit of de Rham spaces of schemes. Namely,

$$\mathrm{Ran} \simeq \operatorname{colim}_{I \in (\mathrm{fSet}^{\mathrm{surj}})^{\mathrm{op}}} X_{\mathrm{dR}}^I,$$

where $\mathrm{fSet}^{\mathrm{surj}}$ is the category of non-empty finite sets and surjective maps (see [Ro2, Sect. 2] for a detailed discussion).

B.1.4. The presentation (B.1.3) implies, in particular, that X_{dR} is locally almost of finite type as a prestack. Hence, it is sufficient to probe it by eventually connective affine schemes of finite type.

Hence, in the discussion below, we will be tacitly assuming that schemes and prestacks mapping to Ran are laft (locally almost of finite type).

B.1.5. Here are the two basic features of Ran that will be used in the sequel:

(i) There is a canonically defined map

$$\text{union} : \text{Ran} \times \text{Ran} \rightarrow \text{Ran},$$

given by the operation of *union* of finite subsets.

(ii) There exists an open subspace

$$(\text{Ran} \times \text{Ran})_{\text{disj}} \subset \text{Ran} \times \text{Ran},$$

corresponding to the condition that the two subsets are disjoint. Namely, for a affine test scheme S , a pair of S -points $\underline{x}_1, \underline{x}_2$ of Ran maps to $(\text{Ran} \times \text{Ran})_{\text{disj}}$ if

$$\text{Graph}_{\underline{x}_1} \cap \text{Graph}_{\underline{x}_2} = \emptyset.$$

Note that the restriction of the map union to $(\text{Ran} \times \text{Ran})_{\text{disj}}$ is étale.

B.1.6. By a factorization space $\mathcal{T}$ over X we will mean a prestack

$$\mathcal{T}_{\text{Ran}} \rightarrow \text{Ran},$$

equipped with a *factorization structure*, which is by definition the datum of an isomorphism

$$(B.2) \quad \mathcal{T}_{\text{Ran}} \times_{\text{Ran}, \text{union}} (\text{Ran} \times \text{Ran})_{\text{disj}} \simeq (\mathcal{T}_{\text{Ran}} \times \mathcal{T}_{\text{Ran}}) \times_{\text{Ran} \times \text{Ran}} (\text{Ran} \times \text{Ran})_{\text{disj}},$$

equipped with a homotopy-coherent data of associativity and commutativity (see [Ra6, Sect. 6], where this is spelled out in detail).

B.1.7. Given a map $\mathcal{Z} \rightarrow \text{Ran}$, we will denote by $\mathcal{T}_{\mathcal{Z}}$ the base change

$$\mathcal{Z} \times_{\text{Ran}} \mathcal{T}_{\text{Ran}}.$$

For $\mathcal{Z} = \text{pt}$ so that $\mathcal{Z} \rightarrow \text{Ran}$ corresponds to $\underline{x} \in \text{Ran}$, we will write $\mathcal{T}_{\underline{x}}$ for the corresponding $\mathcal{Z}$. The factorization structure on $\mathcal{T}$ implies that for $\underline{x}$ as in (B.1), we have

$$\mathcal{T}_{\underline{x}} \simeq \prod_i \mathcal{T}_{x_i}.$$

B.1.8. A basic example of a factorization space is the affine Grassmannian Gr_G . Namely, for an affine test scheme S and a map $\underline{x} : S \rightarrow \text{Ran}$, its lift to $\text{Gr}_{G, \text{Ran}}$ is a datum of

$$(\mathcal{P}_G, \alpha),$$

where $\mathcal{P}_G$ is a G -bundle on $S \times X$, and α is a trivialization of $\mathcal{P}_G$ over the open

$$S \times X - \text{Graph}_{\underline{x}}.$$

B.1.9. Let $\mathcal{T}$ be a factorization space. We can talk about its local properties, such as being a scheme, being an ind-scheme, being (ind)-placid, being formally smooth, etc.

By definition, this means that these properties hold for $\mathcal{T}_S$ relatively to S for every $S \in \text{Sch}_{\text{Ran}}^{\text{aff}}$.

In a similar way, we can talk about local properties of a map between factorization spaces (e.g., being flat or an fpqc cover).

B.2. Factorization *module* spaces.

B.2.1. Let $\text{Ran}^{\subseteq}$ be the subfunctor of $\text{Ran} \times \text{Ran}$, such that $\text{Maps}(S, \text{Ran}^{\subseteq})$ corresponds to pairs $\underline{x} \subseteq \underline{x}'$, as subsets of $\text{Hom}(S_{\text{red}}, X)$.

Denote by pr_{small} and pr_{big} the two projections $\text{Ran}^{\subseteq} \rightrightarrows \text{Ran}$ that send a pair $(\underline{x}, \underline{x}')$ to $\underline{x}$ and $\underline{x}'$, respectively.

B.2.2. Let $\mathcal{Z}$ be a prestack equipped with a map to Ran . Denote

$$\mathcal{Z}^\subseteq := \mathcal{Z} \times_{\mathrm{Ran}, \mathrm{pr}_{\mathrm{small}}} \mathrm{Ran}^\subseteq.$$

For $\mathcal{Z} = \mathrm{pt}$, so that $\mathcal{Z} \rightarrow \mathrm{Ran}$ corresponds to $\underline{x} \in \mathrm{Ran}$, we will write $\mathrm{Ran}_{\underline{x}}$ for the corresponding space $\mathcal{Z}^\subseteq$.

B.2.3. Note that we have a variant of the map union:

$$\mathrm{union} : \mathrm{Ran} \times \mathcal{Z}^\subseteq \rightarrow \mathcal{Z}^\subseteq, \quad (\underline{x}, (z, \underline{x}')) \mapsto (z, \underline{x} \cup \underline{x}').$$

Denote by

$$(\mathrm{Ran} \times \mathcal{Z}^\subseteq)_{\mathrm{disj}} \subset \mathrm{Ran} \times \mathcal{Z}^\subseteq$$

the open subfunctor equal to the preimage of $(\mathrm{Ran} \times \mathrm{Ran})_{\mathrm{disj}}$ under

$$\mathrm{Ran} \times \mathcal{Z}^\subseteq \rightarrow \mathrm{Ran} \times \mathrm{Ran}^\subseteq \xrightarrow{\mathrm{id} \times \mathrm{pr}_{\mathrm{big}}} \mathrm{Ran} \times \mathrm{Ran}.$$

B.2.4. Given a factorization space $\mathcal{T}$, a factorization module space $\mathcal{T}_m$ over $\mathcal{T}$ at $\mathcal{Z}$ is a prestack

$$(\mathcal{T}_m)_{\mathcal{Z}^\subseteq} \rightarrow \mathcal{Z}^\subseteq,$$

equipped with a datum of factorization *against* $\mathcal{T}$:

$$(B.3) \quad (\mathcal{T}_m)_{\mathcal{Z}^\subseteq} \times_{\mathcal{Z}^\subseteq, \mathrm{union}} (\mathrm{Ran} \times \mathcal{Z}^\subseteq)_{\mathrm{disj}} \simeq (\mathcal{T}_{\mathrm{Ran}} \times (\mathcal{T}_m)_{\mathcal{Z}^\subseteq}) \times_{\mathrm{Ran} \times \mathcal{Z}^\subseteq} (\mathrm{Ran} \times \mathcal{Z}^\subseteq)_{\mathrm{disj}},$$

equipped with a homotopy-coherent data of associativity; see [Ra6, Sect. 6] for complete details.

B.2.5. For a factorization module space $\mathcal{T}_m$ at $\mathcal{Z}$, denote

$$(\mathcal{T}_m)_{\mathcal{Z}} := \mathcal{Z} \times_{\mathcal{Z}^\subseteq} (\mathcal{T}_m)_{\mathcal{Z}^\subseteq},$$

where $\mathcal{Z} \rightarrow \mathcal{Z}^\subseteq$ is the map $\mathrm{diag}_{\mathcal{Z}}$ of Sect. 11.2.6.

We will refer to $(\mathcal{T}_m)_{\mathcal{Z}}$ as the *prestack underlying* the factorization module space $\mathcal{T}_m$.

B.2.6. A basic example of a factorization module space over $\mathcal{T}$, defined for any $\mathcal{Z} \rightarrow \mathrm{Ran}$, denoted $\mathcal{T}^{\mathrm{fact}_{\mathcal{Z}}}$, is constructed as follows:

$$\mathcal{T}^{\mathrm{fact}_{\mathcal{Z}}} := \mathcal{T}_{\mathcal{Z}^\subseteq},$$

where $\mathcal{Z}^\subseteq \rightarrow \mathrm{Ran}$ is the map $\mathrm{pr}_{\mathrm{big}, \mathcal{Z}}$, with the datum of (B.3) being provided by the factorization structure on $\mathcal{T}$ itself.

We refer to $\mathcal{T}^{\mathrm{fact}_{\mathcal{Z}}}$ as the *vacuum* factorization module space over $\mathcal{T}$ at $\mathcal{Z}$.

B.2.7. Here is an example of a factorization module space over Gr_G , also defined for any $\mathcal{Z} \rightarrow \mathrm{Ran}$, to be denoted

$$\mathrm{Gr}_G^{\mathrm{level}_{\mathcal{Z}}}.$$

For $(z, \underline{x}) : S \rightarrow \mathcal{Z}^\subseteq$, a lift of this point to $\mathrm{Gr}_G^{\mathrm{level}_{\mathcal{Z}}}$ is a lift of $\underline{x}$ to a point of $\mathrm{Gr}_{G, \mathrm{Ran}}$, and the trivialization of the restriction of the resulting G -bundle $\mathcal{P}_G$ over $S \times X$ to $\widehat{\mathcal{D}}_{\underline{x}}$ (see Sect. B.3.1 below).

B.3. Digression: formal discs.

B.3.1. Fix a map $\underline{x} : S \rightarrow \text{Ran}$. Let $\widehat{\mathcal{D}}_{\underline{x}}$ be the formal scheme equal to the formal completion of $S \times X$ along the closed subset

$$\text{Graph}_{\underline{x}} \subset S \times X,$$

i.e.,

$$\widehat{\mathcal{D}}_{\underline{x}} := (S \times X) \times_{(S \times X)_{\text{dR}}} (\text{Graph}_{\underline{x}})_{\text{dR}},$$

where $(\text{Graph}_{\underline{x}})_{\text{dR}} \rightarrow (S \times X)_{\text{dR}}$ is given by the embedding

$$(\text{Graph}_{\underline{x}})_{\text{red}} \hookrightarrow S \times X.$$

Note that $\widehat{\mathcal{D}}_{\underline{x}}$ is naturally the pullback of a relative formal scheme, denoted $\widehat{\mathcal{D}}_{\underline{x}, \nabla}$, over $S \times X_{\text{dR}}$. Namely,

$$\widehat{\mathcal{D}}_{\underline{x}, \nabla} := (S \times X_{\text{dR}}) \times_{(S \times X)_{\text{dR}}} (\text{Graph}_{\underline{x}})_{\text{dR}}.$$

B.3.2. Let $\text{PreStk}_{\text{disj-loc}} \subset \text{PreStk}$ be the full subcategory of disjoint-union-local prestacks, i.e., prestacks $\mathcal{Z}$, for which for a disjoint union of affine schemes

$$S = S_1 \sqcup S_2,$$

the map

$$(B.4) \quad \text{Maps}(S, \mathcal{Z}) \rightarrow \text{Maps}(S_1, \mathcal{Z}) \times \text{Maps}(S_2, \mathcal{Z})$$

is an isomorphism.

Most prestacks that one encounters in practice satisfy this condition. E.g., note that if $\mathcal{Z}$ satisfies Zariski descent, then it is disjoint-union-local.

Note that the Yoneda embedding

$$\text{Sch}^{\text{aff}} \rightarrow \text{PreStk}_{\text{disj-loc}}$$

commutes with coproducts (this would not be true for the original Yoneda embedding into PreStk).

B.3.3. Let $\underline{x}_1$ and $\underline{x}_2$ be a pair of S -points of Ran , such that $(\underline{x}_1, \underline{x}_2)$ lands in $(\text{Ran} \times \text{Ran})_{\text{disj}}$. Set $\underline{x} := \text{union}(\underline{x}_1, \underline{x}_2)$. In this case we have

$$(B.5) \quad \widehat{\mathcal{D}}_{\underline{x}} \simeq \widehat{\mathcal{D}}_{\underline{x}_1} \sqcup \widehat{\mathcal{D}}_{\underline{x}_2} \text{ and } \widehat{\mathcal{D}}_{\underline{x}, \nabla} \simeq \widehat{\mathcal{D}}_{\underline{x}_1, \nabla} \sqcup \widehat{\mathcal{D}}_{\underline{x}_2, \nabla},$$

where $\sqcup$ is the coproduct taken in $\text{PreStk}_{\text{disj-loc}}$.

B.3.4. Note that, when viewed as an ind-scheme, $\widehat{\mathcal{D}}_{\underline{x}}$ is ind-affine, i.e., of the form

$$\text{“colim”}_{\alpha} \text{Spec}(R_{\alpha}).$$

Let $\mathcal{D}_{\underline{x}}$ denote the affine *scheme* equal to

$$\text{colim}_{\alpha} \text{Spec}(R_{\alpha}),$$

where the colimit is taken in Sch^{aff} . I.e.,

$$\mathcal{D}_{\underline{x}} = \text{Spec}(R), \quad R = \lim_{\alpha} R_{\alpha}.$$

Now, by [Bh, Theorem 1.1], the map

$$(B.6) \quad \widehat{\mathcal{D}}_{\underline{x}} \rightarrow X$$

canonically extends to a map

$$(B.7) \quad \mathcal{D}_{\underline{x}} \rightarrow X.$$

Remark B.3.5. Here we have used [Bh, Theorem 1.1] in a very elementary situation, in which the required assertion can be handled explicitly:

The observation is that maps from an affine scheme $\tilde{S}$ to X can be described as symmetric monoidal functors

$$\mathrm{Perf}(X) \rightarrow \mathrm{Perf}(\tilde{S}),$$

and hence this is true for $\tilde{S}$ replaced by any prestack.

Now, the required assertion is that the restriction functor

$$\mathrm{Perf}(\mathcal{D}_{\underline{x}}) \rightarrow \mathrm{Perf}(\widehat{\mathcal{D}}_{\underline{x}})$$

is an equivalence, so the restriction map

$$\mathrm{Maps}(\mathcal{D}_{\underline{x}}, X) \rightarrow \mathrm{Maps}(\widehat{\mathcal{D}}_{\underline{x}}, X)$$

is an isomorphism.

Remark B.3.6. Note that while the assignment

$$\underline{x} \rightsquigarrow \widehat{\mathcal{D}}_{\underline{x}}$$

is compatible with Zariski localization along S , the formation of $\mathcal{D}_{\underline{x}}$ is *not*.

I.e., for an open $S' \subset S$ and $\underline{x}' := \underline{x}|_{S'}$, the square

$$\begin{array}{ccc} \mathcal{D}_{\underline{x}'} & \longrightarrow & \mathcal{D}_{\underline{x}} \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

is *not* Cartesian.

B.3.7. We have an ind-closed embedding

$$\widehat{\mathcal{D}}_{\underline{x}} \rightarrow \mathcal{D}_{\underline{x}}.$$

In particular, $\mathrm{Graph}_{\underline{x}}$ is a Zariski-closed subset of $\mathcal{D}_{\underline{x}}$. Set

$$\mathcal{D}_{\underline{x}}^{\times} := \mathcal{D}_{\underline{x}} - \mathrm{Graph}_{\underline{x}}.$$

Composing with (B.7), we obtain a map

$$\mathcal{D}_{\underline{x}}^{\times} \rightarrow X.$$

B.3.8. The following material reproduces [Bogd, Construction A.1.3].

We claim that $\mathcal{D}_{\underline{x}}$ descends to a relative affine scheme $\mathcal{D}_{\underline{x}, \nabla}$ over X_{dR} . In order to construct this descent, it is enough to construct a version of $\mathcal{D}_{\underline{x}}$ over the Čech nerve of the infinitesimal groupoid

$$(X \times X)^{\wedge}$$

of X .

I.e., we have to construct a *compatible* family of affine schemes

$$\mathcal{D}'_{\underline{x}} \rightarrow X'$$

over each infinitesimal thickening X' of the main diagonal in $X^{\times n}$ for all n .

We let

$$\widehat{\mathcal{D}}'_{\underline{x}} := X' \times_{X_{\mathrm{dR}}} \widehat{\mathcal{D}}_{\underline{x}, \nabla},$$

viewed as an ind-affine ind-scheme, and we let $\mathcal{D}'_{\underline{x}}$ be the colimit of $\widehat{\mathcal{D}}'_{\underline{x}}$, taken in the category of affine schemes.

The map $\widehat{\mathcal{D}}'_{\underline{x}} \rightarrow X'$ extends to a map $\mathcal{D}'_{\underline{x}} \rightarrow X'$ extends by the same principle as in the case of $\mathcal{D}_{\underline{x}}$.

B.3.9. This material repeats [Bogd, Lemma A.1.5].

In order to establish the compatibility of the above construction, it suffices to show that (with respect to any of the projections $X' \rightarrow X$), the map

$$(B.8) \quad \mathcal{D}'_{\underline{x}} \rightarrow X' \times_X \mathcal{D}_{\underline{x}}$$

is an isomorphism.

We note that $X' \rightarrow X$ has the form $\mathrm{Spec}_X(A_0)$, where is given by $A_0 \in \mathrm{Perf}(X)$, and hence $X' \times_X \mathcal{D}_{\underline{x}}$ has the form $\mathrm{Spec}_{\mathcal{D}_{\underline{x}}}(A)$, where $A \in \mathrm{Perf}(\mathcal{D}_{\underline{x}})$.

In other words, if R is as in Sect. B.3.4, then $X' \times_X \mathcal{D}_{\underline{x}} \simeq \mathrm{Spec}(R')$, where R' is compact as an object of $R\text{-mod}$.

Let R_α be as in Sect. B.3.4. Then

$$\widehat{\mathcal{D}}'_{\underline{x}} = \text{“colim”}_{\alpha} \mathrm{Spec}(R' \otimes_R R_\alpha).$$

The assertion that (B.8) is an isomorphism is equivalent to saying that the map

$$R' \simeq R' \otimes_R (\lim_{\alpha} R_\alpha) \rightarrow \lim_{\alpha} (R' \otimes_R R_\alpha)$$

is an isomorphism. However, this follows from the fact that R' is compact as an R -module.

B.3.10. We let $\mathcal{D}_{\underline{x}, \nabla}^\times$ be the open sub-functor of $\mathcal{D}_{\underline{x}, \nabla}$, equal to

$$\mathcal{D}_{\underline{x}, \nabla} - \mathrm{Graph}_{\underline{x}}.$$

It is easy to see that the prestacks

$$(B.9) \quad \mathcal{D}_{\underline{x}, \nabla} \text{ and } \mathcal{D}_{\underline{x}, \nabla}^\times,$$

when viewed as relative affine schemes over X_{dR} , are independent of the choice of X° .

B.3.11. The prestacks (B.9) satisfy a splitting property parallel to (B.5).

B.4. Formation of (horizontal) arc and loop spaces. In this subsection we will discuss two ubiquitous sources of examples of factorization spaces.

B.4.1. Let $\mathcal{Y} \rightarrow X$ be a D-prestack⁶² over X , i.e., $\mathcal{Y}$ is the pullback along $X \rightarrow X_{\mathrm{dR}}$ of a prestack $\mathcal{Y}_{\nabla} \rightarrow X_{\mathrm{dR}}$.

We will assume that $\mathcal{Y}_{\nabla}$ is disjoint-union-local (see Sect. B.3.2).

B.4.2. We define the factorization prestack $\mathfrak{L}_{\nabla}^+(\mathcal{Y})$ as follows. For an affine test scheme S and a map $\underline{x} : S \rightarrow \mathrm{Ran}$, a lift of this map to $\mathfrak{L}_{\nabla}^+(\mathcal{Y})$ is an X_{dR} -map

$$\widehat{\mathcal{D}}_{\underline{x}, \nabla} \rightarrow \mathcal{Y}_{\nabla}.$$

The factorization structure on $\mathfrak{L}_{\nabla}^+(\mathcal{Y})$ follows from (B.5) and (B.4).

Remark B.4.3. Note that the restriction $\mathfrak{L}_{\nabla}^+(\mathcal{Y})|_X$ recovers the original $\mathcal{Y}$. In Sect. C.6.10 we will upgrade the assignment

$$\mathcal{Y} \mapsto \mathfrak{L}_{\nabla}^+(\mathcal{Y})$$

to an equivalence of categories.

⁶²A.k.a., crystal of prestacks.

B.4.4. For the rest of this subsection we will assume that $\mathcal{Y}$ is affine over X . I.e., $\mathcal{Y}$ is an affine D-scheme.

There is an (obvious) equivalence between the category of D-schemes and that of commutative algebras A in $\mathrm{D}\text{-mod}(X)$ so that $\mathbf{oblv}^l(A)$ is connective.

The plain affine scheme underlying a given affine D-scheme is

$$\mathrm{Spec}_X(\mathbf{oblv}^l(A)).$$

We will write

$$\mathrm{Spec}_X(A)$$

when we want to emphasize the D-structure.

B.4.5. Note that the assumption that $\mathcal{Y}$ is affine over X allows us to interpret the datum of a map

$$\widehat{\mathcal{D}}_{\underline{x}, \nabla} \rightarrow \mathcal{Y}_{\nabla}$$

as

$$\mathcal{D}_{\underline{x}, \nabla} \rightarrow \mathcal{Y}_{\nabla},$$

where $\mathcal{D}_{\underline{x}, \nabla}$ is an in Sect. B.3.8.

B.4.6. We will now define another factorization space, denoted $\mathfrak{L}_{\nabla}(\mathcal{Y})$. A lift of this map to $\mathfrak{L}_{\nabla}(\mathcal{Y})_{\mathrm{Ran}}$ is by definition a X_{dR} map

$$\mathcal{D}_{\underline{x}, \nabla}^{\times} \rightarrow \mathcal{Y}_{\nabla}.$$

The factorization structure on $\mathfrak{L}_{\nabla}(\mathcal{Y})_{\mathrm{Ran}}$ follows from the splitting property in Sect. B.3.11.

B.4.7. Using Sect. B.4.5 and the open embeddings

$$\mathcal{D}_{\underline{x}, \nabla}^{\times} \hookrightarrow \mathcal{D}_{\underline{x}, \nabla}$$

we obtain a map of factorization spaces:

$$\iota : \mathfrak{L}_{\nabla}^+(\mathcal{Y}) \rightarrow \mathfrak{L}_{\nabla}(\mathcal{Y}).$$

We claim:

Lemma B.4.8.

- (a) $\mathfrak{L}_{\nabla}^+(\mathcal{Y})$ is an factorization affine scheme.
- (b) $\mathfrak{L}_{\nabla}(\mathcal{Y})$ is a factorization ind-affine ind-scheme.
- (c) The map ι is a closed embedding.⁶³

B.4.9. *Proof of Lemma B.4.8.* At any level of coconnective truncation, we can write $\mathcal{Y}$ as a finite product of affine D-schemes that are (potentially infinite) products of affine D-schemes of the form

$$(B.10) \quad \mathrm{Spec}_X(A), \quad A = \mathrm{Sym}^l(\mathbf{ind}^l(\mathcal{E})),$$

where $\mathcal{E}$ is a vector bundle on X .

Since (ind-)affine (ind-)schemes are closed under finite limits and products, and the functors

$$\mathcal{Y} \mapsto \mathfrak{L}_{\nabla}^+(\mathcal{Y}) \text{ and } \mathcal{Y} \mapsto \mathfrak{L}_{\nabla}(\mathcal{Y})$$

map products to products, we are reduced to considering $\mathcal{Y}$ of the form specified in (B.10).

In the latter case, the assertions of the lemma can be (easily) checked directly (see Sect. B.5.6 below). $\square$

⁶³By a slight abuse of terminology, we call a map $Z \rightarrow \mathcal{Z}$ from a scheme Z to an ind-scheme $\mathcal{Z}$ a “closed embedding” if the map from Z to some/any of the schemes that comprise $\mathcal{Z}$ is a closed embedding. Note that such a map is *not* a closed embedding in the DAG sense, but rather an *ind-closed embedding*.

B.4.10. *Example.* Recall the factorization space Gr_G . By Beauville-Laszlo theorem, we can rewrite it as follows: for $\underline{x} : S \rightarrow \mathrm{Ran}$, a lift of this point to a point of $\mathrm{Gr}_{G, \mathrm{Ran}}$ is the datum of G -bundle on $\mathcal{D}_{\underline{x}}$ (which is equivalent to that of a G -bundle on $\widehat{\mathcal{D}}_{\underline{x}}$) and the trivialization of its restriction to $\mathcal{D}_{\underline{x}}^\times$.

From this description, we obtain a canonical projection

$$\mathfrak{L}(G) \rightarrow \mathrm{Gr}_G.$$

We claim that this projection identifies Gr_G with the étale quotient $\mathfrak{L}(G)/\mathfrak{L}^+(G)$. Indeed, this follows from the fact that a G -bundle on $\widehat{\mathcal{D}}_{\underline{x}}$ can be trivialized after an étale sheafification along S (see Sect. B.7.2).

A similar description applies to factorization Gr_G -module space $\mathrm{Gr}_G^{\mathrm{level}_Z}$, see Sect. B.2.7.

B.5. Digression: the jet construction.

B.5.1. Let $\mathcal{Y}$ be a prestack over X . Its jets construction, denoted

$$\mathrm{Jets}(\mathcal{Y}) \rightarrow X_{\mathrm{dR}}$$

is by definition restriction of scalars à la Weil of $\mathcal{Y}$ along the projection

$$X \rightarrow X_{\mathrm{dR}}.$$

B.5.2. Explicitly for $x : S \rightarrow X_{\mathrm{dR}}$, a lift of this map to a map $S \rightarrow \mathrm{Jets}(\mathcal{Y})$ is an X -map

$$\widehat{\mathcal{D}}_x \rightarrow \mathcal{Y}.$$

B.5.3. Suppose for a moment that $\mathcal{Y}$ is affine over X , i.e.,

$$\mathcal{Y} = \mathrm{Spec}_X(A_0), \quad A_0 \in \mathrm{ComAlg}(\mathrm{QCoh}(X)^{\leq 0}).$$

In this case

$$\mathrm{Jets}(\mathcal{Y}) = \mathrm{Spec}_X(A),$$

where A is the D-algebra obtained by applying to A_0 the left adjoint of the forgetful functor

$$\mathbf{oblv}^! : \mathrm{ComAlg}(\mathrm{D-mod}(X)) \rightarrow \mathrm{ComAlg}(\mathrm{QCoh}(X)).$$

For example, when

$$A_0 = \mathrm{Sym}(\mathcal{E}), \quad \mathcal{E} \in \mathrm{QCoh}(X),$$

the above left adjoint produces

$$\mathrm{Sym}^!(\mathbf{ind}^!(\mathcal{E})).$$

B.5.4. Denote

$$\mathfrak{L}^+(\mathcal{Y}) := \mathfrak{L}_\nabla^+(\mathrm{Jets}(\mathcal{Y})).$$

One can tautologically rewrite the definition of $\mathfrak{L}^+(\mathcal{Y})$ as follows. For $\underline{x} : S \rightarrow \mathrm{Ran}$, its lift to $\mathfrak{L}^+(\mathcal{Y})_{\mathrm{Ran}}$ is an X -map

$$\widehat{\mathcal{D}}_{\underline{x}} \rightarrow \mathcal{Y}.$$

B.5.5. Assume again that $\mathcal{Y}$ is affine over X . Then $\mathrm{Jets}(\mathcal{Y})$ is affine over X_{dR} (see Sect. B.5.3), and one can consider

$$\mathfrak{L}(\mathcal{Y}) := \mathfrak{L}_\nabla(\mathrm{Jets}(\mathcal{Y})).$$

Explicitly, for an affine test scheme S and $\underline{x} : S \rightarrow \mathrm{Ran}$, its lift to a map $\mathfrak{L}(\mathcal{Y})_{\mathrm{Ran}}$ is an X -map

$$\mathcal{D}_{\underline{x}}^\times \rightarrow \mathcal{Y}.$$

Note that in this case a lift to $\mathfrak{L}^+(\mathcal{Y})_{\mathrm{Ran}}$ can also be described as an X -map

$$\mathcal{D}_{\underline{x}} \rightarrow \mathcal{Y}.$$

B.5.6. *Example.* Let $\mathcal{Y} = \mathrm{Spec}_X(\mathrm{Sym}(\mathcal{E}))$, where $\mathcal{E}$ is a vector bundle on X . Then the spaces

$$\mathfrak{L}^+(\mathcal{Y}) \text{ and } \mathfrak{L}(\mathcal{Y})$$

can be described explicitly as follows.

Fix a map $\underline{x} : S \rightarrow \mathrm{Ran}$. Fix lifts of the maps $x_i : S_{\mathrm{red}} \rightarrow X$ to maps $\tilde{x}_i : S \rightarrow X$. Let D be the divisor on $S \times X$ equal

$$\sum_i n_i \cdot \mathrm{Graph}_{\tilde{x}_i}$$

for some/any choice of $n_i \geq 1$.

Then a lift of $\underline{x}$ to a point of $\mathfrak{L}^+(\mathcal{Y})$ is a point of

$$\lim_n \Gamma(S \times X, \mathcal{O}_X \otimes \mathcal{E}^\vee / \mathcal{O}_X \otimes \mathcal{E}^\vee(-n \cdot D)).$$

A lift of $\underline{x}$ to a point of $\mathfrak{L}(\mathcal{Y})$ is a point of

$$\mathrm{colim}_m \lim_n \Gamma(S \times X, \mathcal{O}_X \otimes \mathcal{E}^\vee(m \cdot D) / \mathcal{O}_X \otimes \mathcal{E}^\vee(-n \cdot D)).$$

B.5.7. *Example.* Let H be a smooth group-scheme over X . On the one hand, we can consider the algebraic stack⁶⁴ pt/H over X , and consider the corresponding D-prestack

$$\mathrm{Jets}(\mathrm{pt}/H).$$

On the other hand, we can consider

$$\mathrm{pt} / \mathrm{Jets}(H),$$

i.e., the étale sheafification of $B(\mathrm{Jets}(H))$.

We have a tautological map

$$(B.11) \quad \mathrm{pt} / \mathrm{Jets}(H) \rightarrow \mathrm{Jets}(\mathrm{pt}/H).$$

We claim that (B.11) is an isomorphism. Indeed, this follows from the fact that if an H -bundle on $\widehat{\mathcal{D}}_{\underline{x}}$ is such that its restriction to S is trivial, then it is itself trivial.

B.6. Digression: the notion of (almost) finite presentation in the D-sense.

B.6.1. Fix a natural number n , and consider the category $\mathrm{ComAlg}(\mathrm{D-mod}(X))^{\leq 0, \geq -n}$ of connective n -coconnective commutative algebras in $\mathrm{D-mod}(X)$, which is by definition

$$\mathrm{ComAlg}(\mathrm{D-mod}(X)) \times_{\mathrm{QCoh}(X)} \mathrm{QCoh}(X)^{\leq 0, \geq -n},$$

where the functor $\mathrm{ComAlg}(\mathrm{D-mod}(X)) \rightarrow \mathrm{QCoh}(X)$ is

$$\mathrm{ComAlg}(\mathrm{D-mod}(X)) \xrightarrow{\mathrm{oblv}_{\mathrm{ComAlg}}} \mathrm{D-mod}(X) \xrightarrow{\mathrm{oblv}^l} \mathrm{QCoh}(X).$$

We shall say that $A \in \mathrm{ComAlg}(\mathrm{D-mod}(X))^{\leq 0, \geq -n}$ is n -D-afp if it is *compact* as an object of this category.

B.6.2. Suppose that $A \in \mathrm{ComAlg}(\mathrm{D-mod}(X))^{\leq 0, \geq -n}$ is isomorphic to the geometric realization of a simplicial object $A_\bullet$ in $\mathrm{ComAlg}(\mathrm{D-mod}(X))^{\leq 0, \geq -n}$ with terms of the form

$$A_n = \mathrm{Sym}^!(\mathcal{M}_n), \quad \mathcal{M}_n \in \mathrm{D-mod}(X), \quad \mathrm{oblv}^l(\mathcal{M}_n) \in \mathrm{D-mod}^{\heartsuit, \mathrm{f.g.}}.$$

It is clear that such A is compact.

Furthermore, it is easy to see that objects of this form generate $\mathrm{ComAlg}(\mathrm{D-mod}(X))^{\leq 0, \geq -n}$ under filtered colimits. Furthermore, for the generation statement, we can take $\mathcal{M}_n$ to be locally free, i.e., of the form $\mathrm{ind}^l(\mathcal{E}_n)$, where $\mathcal{E}_n$ is a vector bundle on X .

From here it follows that every n -D-afp algebra can be written as a retract of such $|A_\bullet|$.

⁶⁴Throughout the paper we write pt/H , where we mean the étale sheafification of $B(H)$, where the latter is the quotient of the base (in this case, X) by the trivial action of H .

B.6.3. Consider now the category $\mathrm{ComAlg}(\mathrm{D}\text{-mod}(X))^{\leq 0}$ of connective commutative algebras in $\mathrm{D}\text{-mod}(X)$, i.e.,

$$\mathrm{ComAlg}(\mathrm{D}\text{-mod}(X)) \times_{\mathrm{QCoh}(X)} \mathrm{QCoh}(X)^{\leq 0}.$$

We shall say that A is D-afp if for every n , the truncation $\tau^{\geq -n}(A) \in \mathrm{ComAlg}(\mathrm{D}\text{-mod}(X))^{\leq 0, \geq -n}$ is n -D-afp.

We shall say that $\mathcal{Y} = \mathrm{Spec}_X(A)$ is D-afp if A is.

Here are some examples of D-afp algebras.

B.6.4. Let $\mathcal{Y}_0 \rightarrow X$ be an affine scheme almost of finite type, and take $\mathcal{Y} := \mathrm{Jets}(\mathcal{Y}_0)$. Then $\mathcal{Y}$ is D-afp.

Indeed, let $\mathcal{Y}_0 = \mathrm{Spec}_X(A_0)$, where $A_0 \in \mathrm{ComAlg}(\mathrm{QCoh}(X)^{\leq 0})$. We can write

$$A_0 = |A_{0,\bullet}|, \quad A_{0,n} = \mathrm{Sym}_{\mathcal{O}_X}(\mathcal{E}_n), \quad \mathcal{E}_n \in \mathrm{Perf}(X)^\vee.$$

Then $\mathcal{Y} = \mathrm{Spec}_X(A)$ for

$$A = |A_\bullet|, \quad A_n = \mathrm{Sym}^!(\mathbf{ind}^l(\mathcal{E}_n)).$$

B.6.5. Let now $\mathcal{Y}$ be the constant affine D-scheme with fiber $\mathcal{Y}_0$, where $\mathcal{Y}_0$ is almost of finite type. We claim that it is D-afp.

Indeed, write $\mathcal{Y}_0 = \mathrm{Spec}(A_0)$, where

$$A_0 = |A_{0,\bullet}|, \quad A_{0,\bullet} = \mathrm{Sym}(V_n), \quad V_n \in \mathrm{Vect}^{\vee, \mathrm{f.d.}}.$$

Then $\mathcal{Y} = \mathrm{Spec}(A)$ where

$$A = |A_\bullet|, \quad A_n = \mathrm{Sym}(V_n) \otimes \omega_X[1] = \mathrm{Sym}^!(V_n \otimes \omega_X[1])$$

(recall that according to our conventions in Sect. 1.1.1, the object $\omega_X[1] \in \mathrm{D}\text{-mod}(X)$ is the dualizing sheaf on X).

B.6.6. Let $\mathcal{Y} \rightarrow \mathcal{Y}_0$ be a map of affine D-schemes. The notion of being D-afp has a straightforward analog in this situation:

If $\mathcal{Y}_0 = \mathrm{Spec}_X(A_0)$ and $\mathcal{Y} = \mathrm{Spec}_X(A)$, so that $\mathcal{Y} \rightarrow \mathcal{Y}_0$ corresponds to a map $A_0 \rightarrow A$, it makes sense to talk about A being finitely presented over A_0 in the D-sense.

The notion of being D-afp is transitive: if

$$\mathcal{Y}'' \rightarrow \mathcal{Y}' \rightarrow \mathcal{Y}$$

are maps of affine D-schemes, with $\mathcal{Y}' \rightarrow \mathcal{Y}$ and $\mathcal{Y}'' \rightarrow \mathcal{Y}'$ D-afp, then so is $\mathcal{Y}'' \rightarrow \mathcal{Y}$.

B.7. Local systems on the formal (punctured) disc.

B.7.1. Let H be a (finite-dimensional) algebraic group. Consider the algebraic stack pt/H . We regard it as a constant D-prestack over X , i.e.,

$$\mathcal{Y}_\nabla := \mathrm{pt}/H \times X_{\mathrm{dR}}.$$

We define the factorization space

$$\mathrm{LS}_H^{\mathrm{reg}} := \mathcal{L}_\nabla^+(\mathrm{pt}/H).$$

B.7.2. Note that the natural map

$$(B.12) \quad \mathrm{pt}/\mathcal{L}_\nabla^+(H) \rightarrow \mathcal{L}_\nabla^+(\mathrm{pt}/H) =: \mathrm{LS}_H^{\mathrm{reg}}$$

is an isomorphism (as was mentioned earlier, the notation $\mathrm{pt}/\mathcal{L}_\nabla^+(H)$ means the étale sheafification of $B(\mathcal{L}_\nabla^+(H))$, the quotient of the base by the trivial action of the corresponding group-scheme).

Indeed, this follows from the fact that for $\underline{x}: S \rightarrow \mathrm{Ran}$, the étale topology on S generates the étale topology on $\widehat{\mathcal{D}}_{\underline{x}}$.

B.7.3. First, we claim:

Lemma B.7.4. *The map $\mathrm{pt} \rightarrow \mathrm{LS}_H^{\mathrm{reg}}$ is étale-locally surjective and is an fpqc cover.*

Proof. The étale surjectivity follows from the isomorphism (B.12).

Hence, to prove the lemma it remains to show that

$$\mathrm{pt} \times_{\mathrm{LS}_H^{\mathrm{reg}}} \mathrm{pt} \simeq \mathfrak{L}_{\nabla}^+(H)$$

is flat. I.e., we need to show that for every finite set I , the scheme $\mathfrak{L}_{\nabla}^+(H)_{X^I}$ is flat over X^I .

The latter is the assertion that for any flat D-algebra A on X , the factorization algebra $\mathrm{Fact}(A)$ (see (C.45)) has the property that for every I , the restriction $\mathbf{oblv}^I(\mathrm{Fact}(A)_{X^I})$ is flat over X^I (this is essentially [BD2, Lemma 3.4.12]). $\square$

B.7.5. One of the key facts about $\mathrm{LS}_H^{\mathrm{reg}}$ is that it can be accessed via *gauge* forms. Namely, consider the tautological map

$$\mathrm{pt}/H \rightarrow \mathrm{Jets}(\mathrm{pt}/H) \simeq \mathrm{pt}/\mathrm{Jets}(H)$$

Note that the fiber product

$$\mathrm{pt}/H \times_{\mathrm{pt}/\mathrm{Jets}(H)} \mathrm{pt}$$

identifies with the D-scheme

$$\mathrm{Jets}(\mathfrak{h} \otimes \omega_X) =: \mathrm{Conn}(\mathfrak{h})$$

of jets into $\mathfrak{h} \otimes \omega_X$ (i.e., the total space of the corresponding vector bundle, viewed as an affine scheme over X).⁶⁵ The resulting $\mathrm{Jets}(H)$ -action on $\mathrm{Conn}(\mathfrak{h})$ is called *the gauge action*.

Hence, we can identify

$$\mathrm{pt}/H \simeq \mathrm{Conn}(\mathfrak{h})/\mathrm{Jets}(H)$$

as D-prestacks over X .

In particular, we obtain an action of $\mathfrak{L}^+(H)$ on $\mathfrak{L}_{\nabla}^+(\mathrm{Conn}(\mathfrak{h}))$ as factorization spaces, and an identification:

$$(B.13) \quad \mathrm{LS}_H^{\mathrm{reg}} \simeq \mathfrak{L}_{\nabla}^+(\mathrm{Conn}(\mathfrak{h}))/\mathfrak{L}^+(H).$$

B.7.6. We note:

Lemma B.7.7. *The factorization space $\mathrm{LS}_H^{\mathrm{reg}}$ is formally smooth.*

Proof. This follows from (B.13). $\square$

Remark B.7.8. Note that the map

$$\mathrm{pt} \rightarrow \mathrm{LS}_H^{\mathrm{reg}}$$

is *not* formally smooth. Indeed, the fiber product

$$\mathrm{pt} \times_{\mathrm{LS}_H^{\mathrm{reg}}} \mathrm{pt}$$

is the affine factorization scheme $\mathfrak{L}_{\nabla}^+(H)$, associated to the constant D-scheme H (see Sect. 4.3.2), and one can show that $\mathfrak{L}_{\nabla}^+(H)_{X^2}$ is *not* formally smooth over X^2 .

Namely, the cotangent space to $\mathfrak{L}_{\nabla}^+(H)_{X^2}$ at the unit section is the object of $\mathrm{QCoh}(X^2)$ underlying the *left* D-module

$$(B.14) \quad \mathrm{Fib}(j_*((\mathfrak{h}^* \oplus \mathfrak{h}^*) \otimes \mathcal{O}_{X^2 - \Delta(X)}) \rightarrow \Delta_*(\mathfrak{h}^* \otimes \mathcal{O}_X)),$$

where the map in (B.14) is the composition

$$\mathrm{Fib}(j_*((\mathfrak{h}^* \oplus \mathfrak{h}^*) \otimes \mathcal{O}_{X^2 - \Delta(X)}) \rightarrow \Delta_*((\mathfrak{h}^* \oplus \mathfrak{h}^*) \otimes \mathcal{O}_X) \rightarrow \Delta_*(\mathfrak{h}^* \otimes \mathcal{O}_X),$$

⁶⁵Indeed, the map $\mathrm{dlog} : \mathrm{Jets}(H) \rightarrow \mathrm{Jets}(\mathfrak{h} \otimes \omega_X)$ identifies the target with $\mathrm{Jets}(H)/H$.

where the second arrow is induced by the addition map $\mathfrak{h}^* \oplus \mathfrak{h}^* \rightarrow \mathfrak{h}^*$.

The object (B.14) of $\mathrm{QCoh}(X^2)$ is flat, but not projective, implying that $\mathfrak{L}_\nabla^+(H)$ is not formally smooth.

Note that this is not in contradiction with the conclusion of Corollary 3.1.11. Let us see that $T_{X^2}^*(\mathrm{LS}_H^{\mathrm{reg}})$ does satisfy the infinitesimal condition for formal smoothness, i.e., that

$$\mathrm{Hom}(T_{X^2}^*(\mathrm{LS}_H^{\mathrm{reg}}), \mathcal{F}) = 0 \text{ if } \mathcal{F} \in \mathrm{QCoh}(X^2)^{\leq -1}.$$

We have

$$T_{X^2}^*(\mathrm{LS}_H^{\mathrm{reg}}) \simeq T_{X^2}^*(\mathfrak{L}_\nabla^+(H))[-1].$$

Now, we claim for any flat *countably generated* object $\mathcal{E} \in \mathrm{QCoh}(X^2)$, we have

$$\mathrm{Hom}(\mathcal{E}[-1], \mathcal{F}) = 0 \text{ if } \mathcal{F} \in \mathrm{QCoh}(X^2)^{\leq -1}.$$

Indeed, any such $\mathcal{E}$ can be written as a countable filtered colimit of projective modules $\mathcal{E}_i$, and

$$\mathcal{H}om(\mathcal{E}[-1], \mathcal{F}) = \lim. \mathrm{proj.} \mathcal{H}om(\mathcal{E}_i[-1], \mathcal{F}).$$

Here all $\mathcal{H}om(\mathcal{E}_i[-1], \mathcal{F})$ live in cohomological degrees ≤ -2 , while $\lim. \mathrm{proj.}$ has amplitude 1, due to the countability assumption.

B.7.9. We now proceed to defining the factorization space $\mathrm{LS}_H^{\mathrm{mer}}$. Naively, one would want to apply the functor $\mathfrak{L}_\nabla(-)$ to the constant D-prestack pt/H . But we cannot quite do this, and that is for two reasons, which already occur for $\mathrm{pt}/\mathrm{Jets}(H)$:

- (i) When considering $\mathfrak{L}_\nabla(-)$, we only allow affine targets.
- (ii) Even over a fixed point $x = \underline{x} \in \mathrm{Ran}$ (but an arbitrary affine scheme $S = \mathrm{Spec}(R)$ of parameters), the space of maps

$$\mathcal{D}_x^\times \rightarrow \mathrm{pt}/H$$

is the space of étale H -torsors over the affine scheme $\mathrm{Spec}(R((t)))$, where t is a local coordinate near x . However, we do not want to consider the étale topology on $\mathrm{Spec}(R((t)))$. Rather, we want to étale-localize with respect to $S = \mathrm{Spec}(R)$ itself, i.e., we only want to consider covers of $\mathrm{Spec}(R((t)))$ of the form $\mathrm{Spec}(\tilde{R}((t)))$, where $\mathrm{Spec}(\tilde{R}) \rightarrow \mathrm{Spec}(R)$ is étale.

B.7.10. The gauge action of $\mathrm{Jets}(H)$ on $\mathrm{Conn}(\mathfrak{h})$ gives rise to an action of $\mathfrak{L}(H)$ as a factorization group ind-scheme on $\mathfrak{L}_\nabla(\mathrm{Conn}(\mathfrak{h}))$.

We define $\mathrm{LS}_H^{\mathrm{mer}}$ to be

$$\mathfrak{L}_\nabla(\mathrm{Conn}(\mathfrak{h}))/\mathfrak{L}(H),$$

the étale sheafification of the non-sheafified quotient of $\mathfrak{L}_\nabla(\mathrm{Conn}(\mathfrak{h}))$ by the gauge action of $\mathfrak{L}(H)$.

B.7.11. We can rephrase the definition of $\mathrm{LS}_H^{\mathrm{mer}}$ as follows:

Consider the étale quotient

$$\mathfrak{L}_\nabla(\mathrm{Conn}(\mathfrak{h}))/\mathfrak{L}^+(H).$$

It carries an action of the groupoid

$$\mathfrak{L}^+(H) \backslash \mathfrak{L}(H) / \mathfrak{L}^+(H).$$

Then $\mathrm{LS}_H^{\mathrm{mer}}$ is the quotient of $\mathfrak{L}_\nabla(\mathrm{Conn}(\mathfrak{h}))/\mathfrak{L}^+(H)$ by $\mathfrak{L}^+(H) \backslash \mathfrak{L}(H) / \mathfrak{L}^+(H)$, subsequent sheafified in the étale topology.

B.7.12. By construction, we have a naturally defined map

$$(B.15) \quad \mathrm{LS}_H^{\mathrm{reg}} \rightarrow \mathrm{LS}_H^{\mathrm{mer}}.$$

We claim:

Lemma B.7.13. *The map (B.15) is ind-affine locally almost of finite presentation⁶⁶.*

Proof. By construction, it suffices to show that the fiber product

$$(B.16) \quad \mathrm{LS}_H^{\mathrm{reg}} \times_{\mathrm{LS}_H^{\mathrm{mer}}} \mathfrak{L}_{\nabla}(\mathrm{Conn}(\mathfrak{h})) \rightarrow \mathfrak{L}_{\nabla}(\mathrm{Conn}(\mathfrak{h}))$$

is ind-schematic locally almost of finite presentation.

The left-hand side in (B.16) is the space

$$\{g \in \mathfrak{L}(H), \alpha \in \mathfrak{L}_{\nabla}(\mathrm{Conn}(\mathfrak{h})) \mid g \cdot \alpha \in \mathfrak{L}_{\nabla}^+(\mathrm{Conn}(\mathfrak{h}))\} / \mathfrak{L}^+(H).$$

In other words, we can rewrite it as

$$(\mathrm{Gr}_H \times \mathfrak{L}_{\nabla}(\mathrm{Conn}(\mathfrak{h}))) \times_{\mathfrak{L}_{\nabla}(\mathrm{Conn}(\mathfrak{h})) / \mathfrak{L}^+(H)} \mathfrak{L}_{\nabla}^+(\mathrm{Conn}(\mathfrak{h})) / \mathfrak{L}^+(H)$$

and its map to the right-hand side of (B.16) is the composition

$$(B.17) \quad (\mathrm{Gr}_H \times \mathfrak{L}_{\nabla}(\mathrm{Conn}(\mathfrak{h}))) \times_{\mathfrak{L}_{\nabla}(\mathrm{Conn}(\mathfrak{h})) / \mathfrak{L}^+(H)} \mathfrak{L}_{\nabla}^+(\mathrm{Conn}(\mathfrak{h})) / \mathfrak{L}^+(H) \rightarrow \mathrm{Gr}_H \times \mathfrak{L}_{\nabla}(\mathrm{Conn}(\mathfrak{h})) \rightarrow \mathfrak{L}_{\nabla}(\mathrm{Conn}(\mathfrak{h})).$$

Since the map $\mathfrak{L}_{\nabla}^+(\mathrm{Conn}(\mathfrak{h})) \rightarrow \mathfrak{L}_{\nabla}(\mathrm{Conn}(\mathfrak{h}))$ is an ind-closed embedding locally almost of finite presentation, we obtain that the first arrow in (B.17) has this property.

The second arrow in (B.17) is ind-schematic and locally almost of finite presentation since Gr_H is an ind-scheme locally almost of finite type. Hence (B.16) is ind-schematic and locally almost of finite presentation.

Finally, let us show that (B.15) is ind-affine. Let H' denote the reductive quotient of H . Factor the map (B.15) as

$$(B.18) \quad \mathrm{LS}_H^{\mathrm{reg}} \rightarrow \mathrm{LS}_H^{\mathrm{mer}} \times_{\mathrm{LS}_{H'}^{\mathrm{mer}}} \mathrm{LS}_{H'}^{\mathrm{reg}} \rightarrow \mathrm{LS}_H^{\mathrm{mer}},$$

and it is enough to show that both arrows in (B.18) are ind-affine.

For the first arrow in (B.18), after base-changing to the fpqc cover $\mathrm{pt} \rightarrow \mathrm{LS}_{H'}^{\mathrm{reg}}$, it suffices to show that the map

$$\mathrm{LS}_{H''}^{\mathrm{reg}} \rightarrow \mathrm{LS}_{H''}^{\mathrm{mer}}$$

is ind-affine, where H'' is the unipotent radical of H . This follows from the fact that in this case the second arrow (B.17) is ind-affine, since the affine Grassmannian for a unipotent group is ind-affine.

For the second arrow in (B.18), it suffices to show that the original map (B.15) is ind-affine when H is reductive. Note that in this case Gr_H is ind-proper, so the second arrow in (B.17) is ind-proper. Hence, the map (B.15) is ind-proper. However, since it is also injective at the level of k -points, we obtain that it is ind-affine (indeed, a proper map between schemes that is injective at k -points is affine). $\square$

⁶⁶See Sect. A.9.5 for what this means.

B.7.14. We now discuss a global version of the factorization space $\mathrm{LS}_H^{\mathrm{mer}}$.

We define the (non-factorization!) space

$$\mathrm{LS}_{H,\mathrm{Ran}}^{\mathrm{mer},\mathrm{glob}} \rightarrow \mathrm{Ran}$$

as follows:

For an affine test scheme S and a map $\underline{x} \rightarrow \mathrm{Ran}$, its lift to $\mathrm{LS}_{H,\mathrm{Ran}}^{\mathrm{mer},\mathrm{glob}}$ is a datum of a map

$$(B.19) \quad (S \times X_{\mathrm{dR}} - \mathrm{Graph}_{\underline{x}}) \rightarrow \mathrm{pt} / H$$

such that étale-locally on S , the composite map

$$(S \times X - \mathrm{Graph}_{\underline{x}}) \rightarrow (S \times X_{\mathrm{dR}} - \mathrm{Graph}_{\underline{x}}) \rightarrow \mathrm{pt} / H$$

admits an extension to a map

$$(B.20) \quad S \times X \rightarrow \mathrm{pt} / H.$$

B.7.15. We claim that we have a naturally defined evaluation map

$$\mathrm{ev}_{\mathrm{Ran}} : \mathrm{LS}_{H,\mathrm{Ran}}^{\mathrm{mer},\mathrm{glob}} \rightarrow \mathrm{LS}_{H,\mathrm{Ran}}^{\mathrm{mer}}.$$

Indeed, for $\underline{x} : S \rightarrow \mathrm{Ran}$, given a map (B.19) and a lift (B.20), the restriction of the connection form to $\mathcal{D}_{\underline{x}}$ gives rise to a section of

$$\mathcal{L}_{\nabla}(\mathrm{Conn}(\mathfrak{h}))_S / \mathcal{L}^+(H)_S.$$

A modification of (B.20) results in an action of the groupoid

$$\mathcal{L}^+(H)_S \backslash \mathcal{L}(H)_S / \mathcal{L}^+(H)_S.$$

B.8. Sheaves of categories over the Ran space.

B.8.1. Let $\mathcal{Y}$ be a prestack. When discussing sheaves of categories over $\mathcal{Y}$, we will assume that $\mathcal{Y}$ is locally almost of finite type. In this context, when considering affine schemes S or general prestacks $\mathcal{Z}$ mapping to $\mathcal{Y}$, we will assume that they are also locally almost of finite type.

B.8.2. A sheaf of categories $\underline{\mathbf{C}}$ on $\mathcal{Y}$ is an assignment

$$(S, y) \in \mathrm{Sch}_{\mathcal{Y}}^{\mathrm{aff}} \rightsquigarrow \mathbf{C}_{S,y} \in \mathrm{QCoh}(S)\text{-}\mathbf{mod},$$

equipped with identifications:

$$S' \xrightarrow{f} S, \quad \mathbf{C}_{S',y \circ f} \simeq \mathrm{QCoh}(S') \otimes_{\mathrm{QCoh}(S)} \mathbf{C}_{S,y},$$

satisfying a homotopy-coherent system of compatibilities; we refer the reader to [Ga5] for details.

Let $\mathrm{ShvCat}(\mathcal{Y})$ denote the (2-)category of sheaves of categories over $\mathcal{Y}$. It has a natural symmetric monoidal structure given by

$$(\mathbf{C}_1 \otimes \mathbf{C}_2)_{S,y} := \mathbf{C}_{1,S,y} \otimes_{\mathrm{QCoh}(S)} \mathbf{C}_{2,S,y}.$$

B.8.3. A basic example of a sheaf of categories over $\mathcal{Y}$ is

$$S \rightsquigarrow \mathrm{QCoh}(S);$$

we denote it by $\underline{\mathrm{QCoh}}(\mathcal{Y})$.

This is the unit for the above symmetric monoidal structure on $\mathrm{ShvCat}(\mathcal{Y})$.

B.8.4. Given a sheaf of categories $\underline{\mathbf{C}}$ on $\mathcal{Y}$, the category of its global sections, denoted $\Gamma(\mathcal{Y}, \underline{\mathbf{C}})$ is defined as

$$\lim_{(S, y) \in \mathrm{Sch}_{/\mathcal{Y}}^{\mathrm{aff}}} \mathbf{C}_{S, y}.$$

This category is naturally a module over

$$\lim_{(S, y) \in \mathrm{Sch}_{/\mathcal{Y}}^{\mathrm{aff}}} \mathrm{QCoh}(S) =: \mathrm{QCoh}(\mathcal{Y}).$$

Thus, the functor

$$\Gamma(\mathcal{Y}, -) : \mathrm{ShvCat}(\mathcal{Y}) \rightarrow \mathrm{DGCat}$$

upgrades to a functor

$$\Gamma(\mathcal{Y}, -)^{\mathrm{enh}} : \mathrm{ShvCat}(\mathcal{Y}) \rightarrow \mathrm{QCoh}(\mathcal{Y})\text{-}\mathbf{mod}.$$

B.8.5. *Example.* We have

$$\Gamma(\mathcal{Y}, \underline{\mathrm{QCoh}}(\mathcal{Y})) \simeq \mathrm{QCoh}(\mathcal{Y}),$$

as a module over itself.

B.8.6. A prestack $\mathcal{Y}$ is said to be *1-affine* if the functor $\Gamma(\mathcal{Y}, -)^{\mathrm{enh}}$ is an equivalence.

In the paper [Ga5] a number of results was proved, showing that various classes of prestacks are 1-affine.

The most relevant for us are:

- The stack pt/H , where H is a (finite-dimensional) algebraic group, is 1-affine.
- The stack Z_{dR} , where Z is a scheme of finite type, is 1-affine.

B.8.7. Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a map between prestacks. We define a functor

$$f^* : \mathrm{ShvCat}(\mathcal{Y}_2) \rightarrow \mathrm{ShvCat}(\mathcal{Y}_1)$$

by sending $\underline{\mathbf{C}} \in \mathrm{ShvCat}(\mathcal{Y}_2)$ to $f^*(\underline{\mathbf{C}}) \in \mathrm{ShvCat}(\mathcal{Y}_1)$ that assigns

$$(S, y) \in \mathrm{Sch}_{/\mathcal{Y}_1}^{\mathrm{aff}} \rightsquigarrow \mathbf{C}_{S, f \circ y} \in \mathrm{QCoh}(S)\text{-}\mathbf{mod}.$$

The above functor f^* has a right adjoint, denoted f_* . Explicitly, f_* sends $\underline{\mathbf{C}} \in \mathrm{ShvCat}(\mathcal{Y}_1)$ to $f_*(\underline{\mathbf{C}}) \in \mathrm{ShvCat}(\mathcal{Y}_2)$ that assigns

$$(S, y) \in \mathrm{Sch}_{/\mathcal{Y}_2}^{\mathrm{aff}} \rightsquigarrow \Gamma(S \times_{\mathcal{Y}_1} \mathcal{Y}_2, y'^*(\underline{\mathbf{C}})) \in \mathrm{QCoh}(S \times_{\mathcal{Y}_1} \mathcal{Y}_2)\text{-}\mathbf{mod} \xrightarrow{f'^*} \mathrm{QCoh}(S)\text{-}\mathbf{mod},$$

where y' denotes the map

$$S \times_{\mathcal{Y}_1} \mathcal{Y}_2 \rightarrow \mathcal{Y}_1$$

and f' denotes the map

$$S \times_{\mathcal{Y}_1} \mathcal{Y}_2 \rightarrow S.$$

B.8.8. Let $\underline{\mathbf{C}}$ be a sheaf of categories over $\mathcal{Y}$. We shall say that $\underline{\mathbf{C}}$ is compactly generated if for every $S \in \mathrm{Sch}_{/\mathcal{Y}}^{\mathrm{aff}}$, the category $\mathbf{C}_S$ is compactly generated.

B.8.9. Suppose for a moment that $\underline{\mathbf{C}}$ is pulled back from a sheaf of categories on $\mathcal{Y}_{\mathrm{dR}}$. Hence, for $S \in \mathrm{Sch}_{/\mathcal{Y}}^{\mathrm{aff}}$ we have a well-defined category $\mathbf{C}_{S_{\mathrm{dR}}}$. Note that if $\mathbf{C}_S$ is compactly generated, then so is $\mathbf{C}_{S_{\mathrm{dR}}}$:

Indeed, with no restriction of generality we can assume that S is eventually coconnective. By the 1-affineness of S_{dR} , we have

$$\mathbf{C}_S \simeq \mathrm{QCoh}(S) \underset{\mathrm{D-mod}(S)}{\otimes} \mathbf{C}_{S_{\mathrm{dR}}},$$

and the pair of (D-mod(S))-linear adjoint functors

$$\mathbf{ind}^l : \mathrm{QCoh}(S) \rightleftarrows \mathrm{D-mod}(S) : \mathbf{oblv}^l$$

induces an adjoint pair

$$(\mathbf{ind}^l \otimes \mathrm{Id}) : \mathbf{C}_S \rightleftarrows \mathbf{C}_{S_{\mathrm{dR}}} : (\mathbf{oblv}^l \otimes \mathrm{Id}),$$

where the essential image of the left adjoint generates the target category.

B.8.10. Let $F : \underline{\mathbf{C}}_1 \rightarrow \underline{\mathbf{C}}_2$ be a functor between sheaves of categories. Suppose that $\underline{\mathbf{C}}_1$ is compactly generated (in the above sense).

We shall say that F preserves compactness if for every $S \in \mathrm{Sch}_{/\mathrm{Ran}}^{\mathrm{aff}}$, the corresponding functor

$$F_S : \mathbf{C}_{1,S} \rightarrow \mathbf{C}_{2,S}$$

preserves compactness.

Then the usual argument (using the fact that for an affine schemes S , the category $\mathrm{QCoh}(S)$ is rigid) shows that in this case the functor F admits a right adjoint, to be denoted F^R , as a functor between sheaves of categories over Ran .

B.8.11. Let $\underline{\mathbf{C}}$ be a sheaf of categories over $\mathcal{Y}$. A t-structure on $\underline{\mathbf{C}}$ is a collection of t-structures on $\mathbf{C}_S$ for any $S \in \mathrm{Sch}_{/\mathcal{Y}}^{\mathrm{aff}}$ such that:

For any $f : S' \rightarrow S$, the functor

$$f_* : \mathbf{C}_{S'} \rightarrow \mathbf{C}_S$$

is t-exact.

Note that this condition can be rewritten as saying that with respect to the identification

$$\mathbf{C}_{S'} \simeq \mathrm{QCoh}(S') \underset{\mathrm{QCoh}(S)}{\otimes} \mathbf{C}_S,$$

the t-structure on $\mathbf{C}_{S'}$ is the *tensor product* t-structure, i.e., $\left(\mathrm{QCoh}(S') \underset{\mathrm{QCoh}(S)}{\otimes} \mathbf{C}_S \right)^{\leq 0}$ is generated under colimits by the essential image of

$$\mathrm{QCoh}(S')^{\leq 0} \times \mathbf{C}_S^{\leq 0} \rightarrow \mathrm{QCoh}(S') \underset{\mathrm{QCoh}(S)}{\otimes} \mathbf{C}_S.$$

B.8.12. Let $\underline{\mathbf{C}}$ be equipped with a t-structure. For $S \in \mathrm{Sch}_{/\mathcal{Y}}^{\mathrm{aff}}$, consider the category

$$\mathrm{IndCoh}(S) \underset{\mathrm{QCoh}(S)}{\otimes} \mathbf{C}_S.$$

We equip it with the tensor product t-structure.

Assume now that the t-structure on $\mathbf{C}_S$ is right-complete and compatible with filtered colimits. Then a standard argument shows that the functor

$$\mathrm{IndCoh}(S) \underset{\mathrm{QCoh}(S)}{\otimes} \mathbf{C}_S \xrightarrow{\Psi_S \otimes \mathrm{Id}} \mathrm{QCoh}(S) \underset{\mathrm{QCoh}(S)}{\otimes} \mathbf{C}_S \simeq \mathbf{C}_S$$

is t-exact and induces an equivalence of the eventually coconnective subcategories of the two sides (see, e.g., [Lu3, Proposition C.4.6.1]).

B.8.13. Suppose again that $\underline{\mathbf{C}}$ is the pull back of a sheaf of categories on $\mathcal{Y}_{\mathrm{dR}}$. For S as above consider the category $\mathbf{C}_{S_{\mathrm{dR}}}$. Note that we can identify

$$\mathrm{IndCoh}(S) \otimes_{\mathrm{QCoh}(S)} \mathbf{C}_S \simeq \mathrm{IndCoh}(S) \otimes_{\mathrm{QCoh}(S)} \mathrm{QCoh}(S) \otimes_{\mathrm{D-mod}(S)} \mathbf{C}_{S_{\mathrm{dR}}} \simeq \mathrm{IndCoh}(S) \otimes_{\mathrm{D-mod}(S)} \mathbf{C}_{S_{\mathrm{dR}}}.$$

The pair of $(\mathrm{D-mod}(S)$ -linear) adjoint functors

$$\mathbf{ind}^r : \mathrm{IndCoh}(S) \rightleftarrows \mathrm{D-mod}(S) : \mathbf{oblv}^r$$

gives rise to an adjunction

$$(\mathbf{ind}^r \otimes \mathrm{Id}) : \mathrm{IndCoh}(S) \otimes_{\mathrm{QCoh}(S)} \mathbf{C}_S \rightleftarrows \mathbf{C}_{S_{\mathrm{dR}}} : (\mathbf{oblv}^r \otimes \mathrm{Id}).$$

We define a t-structure on $\mathbf{C}_{S_{\mathrm{dR}}}$ by letting $\mathbf{C}_{S_{\mathrm{dR}}}^{\leq 0}$ be generated under colimits by the essential image under $(\mathbf{ind}^r \otimes \mathrm{Id})$ of $(\mathrm{IndCoh}(S) \otimes_{\mathrm{QCoh}(S)} \mathbf{C}_S)^{\leq 0}$.

Assume for a moment that S is smooth. In this case the functor $\mathbf{ind}^r$ is t-exact, and hence the endofunctor $\mathbf{oblv}^r \circ \mathbf{ind}^r$ of $\mathrm{IndCoh}(S)$ is t-exact. This implies that the functor

$$(\mathbf{oblv}^r \otimes \mathrm{Id}) : \mathbf{C}_{S_{\mathrm{dR}}} \rightarrow \mathrm{IndCoh}(S) \otimes_{\mathrm{QCoh}(S)} \mathbf{C}_S$$

is t-exact.

B.8.14. We now specialize the case of $\mathcal{Y} = \mathrm{Ran}$. We claim:

Lemma B.8.15. *The prestack Ran is 1-affine.*

Proof. The proof that we will give applies to any prestack $\mathcal{Y}$ that can be written as a colimit

$$\mathrm{colim}_{i \in I} (Z_i)_{\mathrm{dR}},$$

where Z_i are schemes and the transition maps are proper, and for which the diagonal morphism

$$\mathcal{Y} \rightarrow \mathcal{Y} \times \mathcal{Y}$$

is closed (at the reduced level⁶⁷).

We can think of an object of $\mathrm{ShvCat}(\mathcal{Y})$ as a compatible collection of categories $\{\mathbf{C}_i\}$

$$\mathbf{C}_i \in \mathrm{D-mod}(Z_i), \quad (Z_i \xrightarrow{f_{i,j}} Z_j) \rightsquigarrow \mathbf{C}_i \simeq \mathrm{D-mod}(Z_i) \otimes_{\mathrm{D-mod}(Z_j)} \mathbf{C}_j.$$

The functor $\Gamma(\mathcal{Y}, -)$ sends such a collection to

$$\lim_{i \in I^{\mathrm{op}}} \mathbf{C}_i,$$

viewed as a module over

$$\mathrm{QCoh}(\mathcal{Y}) = \mathrm{D-mod}(\mathcal{Y}) \simeq \lim_i \mathrm{D-mod}(Z_i).$$

Given a $\mathrm{D-mod}(\mathcal{Y})$ -module category $\mathbf{D}$, we attach to it an object of $\mathrm{ShvCat}(\mathcal{Y})$ by setting

$$\mathbf{C}_i := \mathrm{Funct}_{\mathrm{D-mod}(\mathcal{Y})}(\mathrm{D-mod}(Z_i), \mathbf{D}).$$

For $(i \rightarrow j) \in I$, the corresponding transition functor

$$\mathrm{Funct}_{\mathrm{D-mod}(\mathcal{Y})}(\mathrm{D-mod}(Z_j), \mathbf{D}) \rightarrow \mathrm{Funct}_{\mathrm{D-mod}(\mathcal{Y})}(\mathrm{D-mod}(Z_i), \mathbf{D})$$

is given by precomposition with $(f_{i,j})_!$.

For $(i \rightarrow j) \in I$, we have, tautologically,

$$\mathrm{Funct}_{\mathrm{D-mod}(Z_j)}(\mathrm{D-mod}(Z_i), \mathrm{Funct}_{\mathrm{D-mod}(\mathcal{Y})}(\mathrm{D-mod}(Z_j), \mathbf{D})) \simeq \mathrm{Funct}_{\mathrm{D-mod}(\mathcal{Y})}(\mathrm{D-mod}(Z_i), \mathbf{D}).$$

However, this implies that we also have

$$\mathrm{D-mod}(Z_i) \otimes_{\mathrm{D-mod}(Z_j)} \mathrm{Funct}_{\mathrm{D-mod}(\mathcal{Y})}(\mathrm{D-mod}(Z_j), \mathbf{D}) \simeq \mathrm{Funct}_{\mathrm{D-mod}(\mathcal{Y})}(\mathrm{D-mod}(Z_i), \mathbf{D}),$$

⁶⁷I.e., the base change of this morphism by an affine scheme yields a morphism of prestacks, such that the morphism of the underlying reduced prestacks is a closed embedding.

since $\mathrm{D}\text{-mod}(Z_i)$ is self-dual as a $\mathrm{D}\text{-mod}(Z_j)$ -module.

Let us establish the equivalence

$$(B.21) \quad \lim_{i \in I^{\mathrm{op}}} \mathrm{Funct}_{\mathrm{D}\text{-mod}(\mathcal{Y})}(\mathrm{D}\text{-mod}(Z_i), \mathbf{D}) \simeq \mathbf{D}.$$

Indeed, we can rewrite the left-hand side as

$$\mathrm{Funct}_{\mathrm{D}\text{-mod}(\mathcal{Y})}(\mathrm{colim}_{i \in I} \mathrm{D}\text{-mod}(Z_i), \mathbf{D}),$$

where the colimit is taken with respect to the $(f_{i,j})_!$ -functors.

Now, we can rewrite

$$\mathrm{colim}_{i \in I} \mathrm{D}\text{-mod}(Z_i) \simeq \lim_{i \in I^{\mathrm{op}}} \mathrm{D}\text{-mod}(Z_i),$$

where in the right-hand side the limit is taken with respect to the $f_{i,j}^!$ functors. Finally,

$$\mathrm{colim}_{i \in I^{\mathrm{op}}} \mathrm{D}\text{-mod}(Z_i) \simeq \mathrm{D}\text{-mod}(\mathcal{Y}),$$

by definition. Hence, the left-hand side in (B.21) identifies with

$$\mathrm{Funct}_{\mathrm{D}\text{-mod}(\mathcal{Y})}(\mathrm{D}\text{-mod}(\mathcal{Y}), \mathbf{D}) \simeq \mathbf{D},$$

as required.

It is easy to see that in order to show that the above two functors are mutually inverse, it remains to show that for $i_1, i_2 \in I$, and the corresponding maps

$$Z_{i_1} \xrightarrow{f_{i_1}} \mathcal{Y} \xleftarrow{f_{i_2}} Z_{i_2}$$

the naturally defined functor

$$(B.22) \quad \mathrm{D}\text{-mod}(Z_{i_1} \times_{\mathcal{Y}} Z_{i_2}) \rightarrow \mathrm{Funct}_{\mathrm{D}\text{-mod}(\mathcal{Y})}(\mathrm{D}\text{-mod}(Z_{i_1}), \mathrm{D}\text{-mod}(Z_{i_2}))$$

is an equivalence.

We calculate the right-hand side in (B.22) as the totalization of the cosimplicial category with terms

$$\mathrm{D}\text{-mod}(Z_{i_1} \times \mathcal{Y}^m \times Z_{i_2}),$$

with terms given by $!$ -pushforward functors along the maps in the corresponding cosimplicial prestack.

However, due to the assumption that $\mathcal{Y}$ has a closed diagonal, the face maps in this cosimplicial prestack are closed embeddings, and hence the corresponding $!$ -pushforward functors are fully faithful. Hence, the above totalization is the equalizer of

$$\mathrm{D}\text{-mod}(Z_{i_1} \times Z_{i_2}) \rightrightarrows \mathrm{D}\text{-mod}(Z_{i_1} \times \mathcal{Y} \times Z_{i_2}).$$

Objects of this equalizer are supported on

$$(Z_{i_1} \times Z_{i_2}) \times_{\mathrm{Graph}_{f_{i_1}} \times \mathrm{id}, Z_{i_1} \times \mathcal{Y} \times Z_{i_2}, \mathrm{id} \times \mathrm{Graph}_{f_{i_2}}} (Z_{i_1} \times Z_{i_2}).$$

However, the above fiber product identifies with

$$Z_{i_1} \times_{\mathcal{Y}} Z_{i_2}.$$

□

Remark B.8.16. Let $\underline{\mathbf{C}}$ be a sheaf of categories over Ran . In the course of the proof of Lemma B.8.15, we have encountered another way of how one may think of the category $\Gamma(\mathrm{Ran}, \underline{\mathbf{C}})$. Namely,

$$\Gamma(\mathrm{Ran}, \underline{\mathbf{C}}) \simeq \mathrm{colim}_{I \in (\mathrm{fSet}^{\mathrm{surj}})^{\mathrm{op}}} \mathbf{C}_{X_{\mathrm{dR}}}^I,$$

where the colimit is formed using the $!$ -pushforward functors, i.e., for $(I \xrightarrow{\phi} J) \in \mathrm{fSet}^{\mathrm{surj}}$ and the corresponding map

$$\Delta_{\phi} : X^J \rightarrow X^I,$$

the functor in question is

$$\mathbf{C}_{X_{\mathrm{dR}}^J} \simeq \mathrm{D}\text{-mod}(X^J) \underset{\mathrm{D}\text{-mod}(X^I)}{\otimes} \mathbf{C}_{X_{\mathrm{dR}}^I} \xrightarrow{(\Delta_\phi)_I \otimes \mathrm{Id}} \mathrm{D}\text{-mod}(X^I) \underset{\mathrm{D}\text{-mod}(X^I)}{\otimes} \mathbf{C}_{X_{\mathrm{dR}}^I} \simeq \mathbf{C}_{X_{\mathrm{dR}}^I}.$$

B.8.17. Notational convention. We use the term *crystal of categories* for sheaves of categories on $\mathcal{Y}_{\mathrm{dR}}$ and denote

$$\mathrm{CrystCat}(\mathcal{Y}) := \mathrm{ShvCat}(\mathcal{Y}_{\mathrm{dR}}).$$

From this perspective, we refer to crystalline objects of $\underline{\mathbf{C}}$ in $\mathrm{CrystCat}(\mathcal{Y})$ to mean the objects of the category of global sections of $\underline{\mathbf{C}}$ as a sheaf of categories on $\mathcal{Y}_{\mathrm{dR}}$.

When we use such “crystalline” terminology and are given $f : Z \rightarrow \mathcal{Y}$, we use the symbol $\mathbf{C}_Z$ to denote the crystalline objects of $\underline{\mathbf{C}}$ restricted to Z , that is,

$$\mathbf{C}_Z := \Gamma(\mathcal{Z}_{\mathrm{dR}}, (f_{\mathrm{dR}})^*(\underline{\mathbf{C}})).$$

B.9. Factorization algebras and modules.

B.9.1. A factorization algebra $\mathcal{A}$ on X is an object

$$\mathcal{A}_{\mathrm{Ran}} \in \mathrm{D}\text{-mod}(\mathrm{Ran}),$$

equipped with a datum of factorization

$$(B.23) \quad \mathrm{union}^!(\mathcal{A}_{\mathrm{Ran}})|_{(\mathrm{Ran} \times \mathrm{Ran})_{\mathrm{disj}}} \simeq \mathcal{A}_{\mathrm{Ran}} \boxtimes \mathcal{A}_{\mathrm{Ran}}|_{(\mathrm{Ran} \times \mathrm{Ran})_{\mathrm{disj}}}$$

and with a homotopy-coherent data of associativity and commutativity, see [Ra6, Sect. 6] for details.

Let $\mathrm{FactAlg}(X)$ denote the category of factorization algebras on X .

For $\mathcal{A} \in \mathrm{FactAlg}(X)$ and $\mathcal{Z} \rightarrow \mathcal{A}$, we will denote by $\mathcal{A}_{\mathcal{Z}} \in \mathrm{D}\text{-mod}(\mathcal{Z})$ the pullback of $\mathcal{A}$ to $\mathcal{Z}$.

B.9.2. Example. We let k denote the unit factorization algebra. I.e., the underlying object

$$k_{\mathrm{Ran}} \in \mathrm{D}\text{-mod}(\mathrm{Ran})$$

is ω_{Ran} , equipped with the natural factorization structure.

B.9.3. Let $\mathcal{A}$ be a factorization algebra on X . Let $\mathcal{Z}$ be a prestack equipped with a map $\mathcal{Z} \rightarrow \mathrm{Ran}$. Recall the space $\mathcal{Z}^\subseteq$, see Sect. B.2.2.

A factorization $\mathcal{A}$ -module $\mathcal{M}$ at $\mathcal{Z}$ is an object

$$\mathcal{M}_{\mathcal{Z}^\subseteq} \in \mathrm{QCoh}(\mathcal{Z}^\subseteq)$$

equipped with a datum of factorization against $\mathcal{A}$:

$$(B.24) \quad \mathcal{M}_{\mathcal{Z}^\subseteq}|_{(\mathrm{Ran} \times \mathcal{Z}^\subseteq)_{\mathrm{disj}}} \simeq (\mathcal{A}_{\mathrm{Ran}} \boxtimes \mathcal{M}_{\mathcal{Z}^\subseteq})|_{(\mathrm{Ran} \times \mathcal{Z}^\subseteq)_{\mathrm{disj}}}$$

and a homotopy-coherent data of associativity.

We denote the category of factorization $\mathcal{A}$ -modules at $\mathcal{Z}$ by

$$\mathcal{A}\text{-mod}_{\mathcal{Z}}^{\mathrm{fact}}.$$

This category is naturally tensored over $\mathrm{QCoh}(\mathcal{Z})$ via the projection

$$\mathcal{Z}^\subseteq \xrightarrow{\mathrm{Pr}_{\mathrm{small}, \mathcal{Z}}} \mathcal{Z}.$$

Remark B.9.4. For $\mathcal{Z} = X^I$, one can described the category $\mathcal{A}\text{-mod}_{X^I}^{\mathrm{fact}}$ explicitly as chiral modules, see [Ro1, Sect. 3].

B.9.5. Let $\text{diag}_{\mathcal{Z}} : \mathcal{Z} \rightarrow \mathcal{Z}^{\subseteq}$ be the diagonal map (see Sect. 11.2.6). For $\mathcal{M} \in \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}}$ we will denote by $\mathcal{M}_{\mathcal{Z}}$ the object

$$\text{diag}_{\mathcal{Z}}^*(\mathcal{M}_{\mathcal{Z}^{\subseteq}}) \in \text{QCoh}(\mathcal{Z}).$$

We will refer to $\mathcal{M}_{\mathcal{Z}}$ as the *quasi-coherent sheaf on $\mathcal{Z}$ underlying $\mathcal{M}$* .

The resulting functor

$$(B.25) \quad \mathbf{oblv}_{\mathcal{A}} : \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}} \rightarrow \text{QCoh}(\mathcal{Z})$$

is conservative and compatible with colimits. (Sometimes instead of $\mathbf{oblv}_{\mathcal{A}}$ we write $\mathbf{oblv}_{\mathcal{A}, \mathcal{Z}}$ in order to emphasize the dependence on $\mathcal{Z}$.)

We will think of the datum of $\mathcal{M}$ as $\mathcal{M}_{\mathcal{Z}} \in \text{QCoh}(\mathcal{Z})$, equipped with an additional datum of factorization against $\mathcal{A}$.

B.9.6. Take $\mathcal{A} = k$ from Sect. B.9.2. We claim that there is a naturally defined functor

$$(B.26) \quad \text{QCoh}(\mathcal{Z}) \rightarrow k\text{-mod}_{\mathcal{Z}}^{\text{fact}}.$$

Namely, the functor (B.26) is given by pullback along the projection

$$\mathcal{Z}^{\subseteq} \rightarrow \mathcal{Z}.$$

In Sect. C.7.7, we will upgrade the functor (B.26) to an equivalence of categories, once we replace the right-hand side by the category of *unital* factorization modules.

B.9.7. A basic example of a factorization $\mathcal{A}$ -module is the *vacuum* module, denoted

$$\mathcal{A}^{\text{fact}_{\mathcal{Z}}} \in \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}}.$$

Namely, the corresponding object

$$\mathcal{A}_{\mathcal{Z}^{\subseteq}}^{\text{fact}_{\mathcal{Z}}} \in \text{QCoh}(\mathcal{Z}^{\subseteq})$$

is the pullback of $\mathcal{A}_{\text{Ran}}$ along the map

$$\mathcal{Z}^{\subseteq} \xrightarrow{\text{Pr}_{\text{big}}} \text{Ran}.$$

Note that

$$\mathcal{A}_{\mathcal{Z}}^{\text{fact}_{\mathcal{Z}}} \simeq \mathcal{A}_{\mathcal{Z}}.$$

For that reason, we will sometimes abuse the notation and write $\mathcal{A}_{\mathcal{Z}}$ instead of $\mathcal{A}^{\text{fact}_{\mathcal{Z}}}$. (This is similar to denoting the free object over an associative algebra A by the same symbol A as the underlying vector space.)

B.9.8. Generalizing the above construction, given $\mathcal{Z} \rightarrow \text{Ran}$ and $f : \mathcal{Z}' \rightarrow \mathcal{Z}^{\subseteq}$, which we perceive as mapping to Ran by means of

$$\mathcal{Z}' \rightarrow \mathcal{Z}^{\subseteq} \xrightarrow{\text{Pr}_{\text{big}}} \text{Ran},$$

there is a naturally defined functor⁶⁸

$$(B.27) \quad \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}} \rightarrow \mathcal{A}\text{-mod}_{\mathcal{Z}'}^{\text{fact}}$$

that makes the following diagram commute

$$\begin{array}{ccc} \mathcal{A}\text{-mod}_{\mathcal{Z}'}^{\text{fact}} & \xrightarrow{\mathbf{oblv}_{\mathcal{A}, \mathcal{Z}'}} & \text{QCoh}(\mathcal{Z}') \\ \uparrow & & \uparrow f^! \\ \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}} & \longrightarrow & \text{QCoh}(\mathcal{Z}^{\subseteq}). \end{array}$$

We will denote the functor (B.27) by

$$\mathcal{M} \mapsto \mathcal{M}|_{\mathcal{Z}'}.$$

⁶⁸See Sect. C.11.13 for more details.

Remark B.9.9. The functor (B.27) reflects the *unital structure* on the assignment

$$\mathcal{Z} \mapsto \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}},$$

see Sect. C.11.13.

B.9.10. Let S be an affine scheme mapping to Ran , and let $S' \xrightarrow{f} S$ be a map of affine schemes. Pullback along

$$f^{\subseteq} : S'^{\subseteq} \rightarrow S^{\subseteq}$$

defines a functor

$$(B.28) \quad f^* : \mathcal{A}\text{-mod}_S^{\text{fact}} \rightarrow \mathcal{A}\text{-mod}_{S'}^{\text{fact}}.$$

This functor is $\text{QCoh}(S)$ -linear, and hence gives rise to a functor

$$(B.29) \quad \text{QCoh}(S') \otimes_{\text{QCoh}(S)} \mathcal{A}\text{-mod}_S^{\text{fact}} \rightarrow \mathcal{A}\text{-mod}_{S'}^{\text{fact}}.$$

We claim:

Lemma B.9.11. *The functor (B.29) is an equivalence.*

Proof. Pushforward along $f^{\subseteq}$ gives rise to a functor

$$f_* : \mathcal{A}\text{-mod}_{S'}^{\text{fact}} \rightarrow \mathcal{A}\text{-mod}_S^{\text{fact}},$$

which is a right adjoint of (B.28). It is easy to see that it is monadic.

The functor (B.29) gives rise to a map of monads, and in order to prove the lemma, it suffices to show that this map of monads induces an isomorphism of the underlying endofunctors.

However, the latter is obvious: both endofunctors are given by tensoring by $f_*(\mathcal{O}_{S'})$, viewed as an algebra on $\text{QCoh}(S)$. □

B.9.12. From Lemma B.9.11 we obtain that the assignment

$$(S \rightarrow \text{Ran}) \rightsquigarrow \mathcal{A}\text{-mod}_S^{\text{fact}}$$

forms a sheaf of categories over Ran .

We will denote this sheaf of categories by $\mathcal{A}\text{-mod}^{\text{fact}}$. For any $f : \mathcal{Z} \rightarrow \text{Ran}$, we have

$$\mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}} \simeq \Gamma(\mathcal{Z}, f^*(\mathcal{A}\text{-mod}^{\text{fact}})).$$

B.9.13. We shall say that a factorization algebra $\mathcal{A}$ is *connective* if

$$\text{oblv}^I(\mathcal{A}_X) \in \text{QCoh}(X)$$

is connective, where $X \rightarrow \text{Ran}$ is the tautological map.

B.9.14. Recall that for any finite non-empty set, we have a tautological map

$$X_{\text{dR}}^I \rightarrow \text{Ran}.$$

We have (by the argument in [BD2, Lemma 3.4.12]):

Lemma B.9.15. *Suppose that $\mathcal{A}$ is connective in the above sense, and unital (see Sect. C.7.1) for what this means. Then for any I , the object*

$$\text{oblv}^I(\mathcal{A}_{X^I}) \in \text{QCoh}(X^I)$$

is connective.

Corollary B.9.16. *Under the assumptions of Lemma B.9.15, for any scheme Z equipped with a map to Ran , the object*

$$\mathcal{A}_Z \in \text{QCoh}(Z)$$

is connective.

B.9.17. We now claim:

Proposition B.9.18. *Let $\mathcal{A}$ be connective.⁶⁹ Then for any scheme Z mapping to Ran , the category $\mathcal{A}\text{-mod}_Z^{\mathrm{fact}}$ carries a t -structure, uniquely characterized by the property that the forgetful functor (B.25) is t -exact. Furthermore, $\mathcal{A}\text{-mod}_Z^{\mathrm{fact}}$ is left-complete in its t -structure.*

Proof. This follows by interpreting factorization modules as chiral modules, see [FraG]. The connectivity assumption translates into the fact that the corresponding chiral algebra is connective (for the *right* t -structure on $\mathrm{D}\text{-mod}(X)$):

Namely, we consider the category $\mathrm{QCoh}(Z^\subseteq)$ as acted on via the map union by $\mathrm{D}\text{-mod}(\mathrm{Ran})$, equipped with the *chiral* symmetric monoidal structure (see [BD2, Sect. 3.4.10] and/or [FraG, Sect 2.2]).

Let

$$\mathcal{A}^{\mathrm{ch}} \in \mathrm{D}\text{-mod}(X) \subset \mathrm{D}\text{-mod}(\mathrm{Ran})$$

be the chiral algebra, corresponding to $\mathcal{A}$ (see Sect. D.1.1); it has a structure of Lie algebra, viewed as an object of $\mathrm{D}\text{-mod}(\mathrm{Ran})$ in the chiral symmetric monoidal structure.

We can interpret $\mathcal{A}\text{-mod}_Z^{\mathrm{fact}}$ as the full subcategory of

$$\mathcal{A}^{\mathrm{ch}}\text{-mod}(\mathrm{QCoh}(Z^\subseteq)),$$

consisting of objects with set-theoretic support on $Z \xrightarrow{\mathrm{diag}_Z} Z^\subseteq$ (note that the map diag_Z identifies Z with its formal completion inside $Z^\subseteq$).

Now the assertion of the proposition follows from Lemma B.9.20 below. □

B.9.19. Let $\mathbf{C}$ be a DG category with a t -structure compatible with filtered colimits, and suppose we are given an embedding $\mathbf{C} \subset \mathbf{C}'$. Let $\mathbf{A}$ be a symmetric monoidal (resp., monoidal) category acting on $\mathbf{C}'$; denote the action functor by $\star$.

Let L (resp., A) be a Lie algebra (resp., associative) in $\mathbf{A}$ with the property that for every $\mathbf{c}_1 \in \mathbf{C}^{\leq 0}$ and every $\mathbf{c}_2 \in \mathbf{C}^{> 0}$, $\mathrm{Maps}(L \star \mathbf{c}_1, \mathbf{c}_2) = 0$ (resp., $\mathrm{Maps}(A \star \mathbf{c}_1, \mathbf{c}_2) = 0$).

Lemma B.9.20. *Under the above circumstances, the category $L\text{-mod}(\mathbf{C}) := L\text{-mod}(\mathbf{C}') \times_{\mathbf{C}'} \mathbf{C}$ (resp., $A\text{-mod}(\mathbf{C}) := A\text{-mod}(\mathbf{C}') \times_{\mathbf{C}'} \mathbf{C}$) admits a unique t -structure for which the forgetful functor to $\mathbf{C}$ is t -exact.*

Remark B.9.21. In the terminology of Sect. B.8.11, the assertion of Proposition B.9.18 is that if $\mathcal{A}$ is connective, the sheaf of categories $\mathcal{A}\text{-mod}^{\mathrm{fact}}$ carries a t -structure, for which the forgetful functor $\mathbf{oblv}_C \mathcal{A}$ is conservative.

Remark B.9.22. An assertion parallel to Proposition B.9.18 holds for the category

$$\mathcal{A}\text{-mod}_{Z_{\mathrm{dR}}}^{\mathrm{fact}},$$

i.e., for crystalline factorization $\mathcal{A}$ -modules at Z . In this case, we consider $\mathrm{QCoh}(Z_{\mathrm{dR}}) \simeq \mathrm{D}\text{-mod}(Z)$ equipped with the *right* t -structure, i.e., one for which the functor $\mathbf{ind}^r$ is t -exact.

Remark B.9.23. Assume for a moment that $\mathcal{A}$ is such that $\mathbf{oblv}^l(\mathcal{A}_X)$ belongs to $\mathrm{QCoh}(X)^\vee$ and is flat. Then [BD2, Lemma 3.4.12] implies that the same is true for $\mathbf{oblv}^l(\mathcal{A}_{X^I})$ for all finite sets I .

This in turn implies that for any $S \rightarrow \mathrm{Ran}$ with $S \in \mathrm{Sch}^{\mathrm{aff}}$, the object $\mathcal{A}_S \in \mathrm{QCoh}(S)$ also belongs to $\mathrm{QCoh}(S)^\vee$ and is flat.

⁶⁹The unitality assumption on $\mathcal{A}$ here is irrelevant, as one can add a unit to it (and then consider the corresponding category of unital modules).

B.9.24. Let now $\phi : \mathcal{A}_1 \rightarrow \mathcal{A}_2$ be a homomorphism of factorization algebras. Let

$$\mathcal{M}_i \in \mathcal{A}_i\text{-mod}_{\mathcal{Z}}^{\text{fact}}.$$

There is a natural notion of map $\mathcal{M}_1 \rightarrow \mathcal{M}_2$, compatible with factorization. Namely, this is a map

$$\phi_m : \mathcal{M}_{1,\mathcal{Z} \subseteq} \rightarrow \mathcal{M}_{2,\mathcal{Z} \subseteq}$$

in $\text{QCoh}(\mathcal{Z}^{\subseteq})$, which makes the following diagram commute,

$$\begin{array}{ccc} \mathcal{M}_{1,\mathcal{Z} \subseteq} |_{(\text{Ran} \subset \mathcal{Z} \subseteq)} & \xrightarrow{\sim} & (\mathcal{A}_{1,\text{Ran}} \boxtimes \mathcal{M}_{1,\mathcal{Z} \subseteq}) |_{(\text{Ran} \subset \mathcal{Z} \subseteq)} \\ \phi_m \downarrow & & \downarrow \phi \boxtimes \phi_m \\ \mathcal{M}_{2,\mathcal{Z} \subseteq} |_{(\text{Ran} \subset \mathcal{Z} \subseteq)} & \xrightarrow{\sim} & (\mathcal{A}_{2,\text{Ran}} \boxtimes \mathcal{M}_{2,\mathcal{Z} \subseteq}) |_{(\text{Ran} \subset \mathcal{Z} \subseteq)}, \end{array}$$

along with a homotopy-coherent system of higher compatibilities.

Denote the corresponding space of maps by

$$(B.30) \quad \text{Maps}_{\mathcal{A}_1 \rightarrow \mathcal{A}_2}(\mathcal{M}_1, \mathcal{M}_2).$$

B.9.25. For a fixed $\mathcal{M}_2 \in \mathcal{A}_2\text{-mod}_{\mathcal{Z}}^{\text{fact}}$, we can consider the functor

$$(\mathcal{A}_1\text{-mod}_{\mathcal{Z}}^{\text{fact}})^{\text{op}} \rightarrow \text{Spc},$$

given by

$$(B.31) \quad \mathcal{M}_1 \mapsto \text{Maps}_{\mathcal{A}_1 \rightarrow \mathcal{A}_2}(\mathcal{M}_1, \mathcal{M}_2).$$

This functor sends colimits to limits, and hence is representable. We denote the representing object by

$$\text{Res}_{\phi}(\mathcal{M}_2) \in \mathcal{A}_1\text{-mod}_{\mathcal{Z}}^{\text{fact}}.$$

We have a tautologically defined object in $\text{Maps}_{\mathcal{A}_1 \rightarrow \mathcal{A}_2}(\text{Res}_{\phi}(\mathcal{M}_2), \mathcal{M}_2)$, i.e., a map

$$(B.32) \quad \text{Res}_{\phi}(\mathcal{M}_2) \rightarrow \mathcal{M}_2,$$

compatible with factorization.

We claim:

Lemma B.9.26. *The map (B.32) induces an isomorphism*

$$\text{Res}_{\phi}(\mathcal{M}_2)_{\mathcal{Z}} \rightarrow \mathcal{M}_{2,\mathcal{Z}}.$$

Proof. This also follows by interpreting factorization modules as chiral modules. Under this equivalence, restriction corresponds to restriction of modules along a homomorphism of chiral algebras. $\square$

Remark B.9.27. Let us emphasize that Lemma B.9.26 says that the operation of restriction acts as identity on the underlying object $\text{QCoh}(\mathcal{Z})$, and its content is that one can restrict the factorization action on it of $\mathcal{A}_2$ to obtain a factorization action of $\mathcal{A}_1$.

B.9.28. *Example.* One can describe the object $\text{Res}_{\phi}(\mathcal{M}_2)_{\mathcal{Z} \subseteq}$ explicitly. Let us consider the example of $\mathcal{Z} = \text{pt}$, where $\mathcal{Z} \rightarrow \text{Ran}$ corresponds to a singleton $x \in X$, so that $\mathcal{Z}^{\subseteq} = \text{Ran}_x$.

We have a natural map

$$X \rightarrow \text{Ran}_x, \quad x' \mapsto x \cup x',$$

and let us describe the restriction of $\text{Res}_{\phi}(\mathcal{M}_2)$ along this map.

Let j denote the embedding $X - x \hookrightarrow X$. Then

$$\text{Res}_{\phi}(\mathcal{M}_2)_X \simeq \mathcal{M}_{2,X} \times_{j_* \circ j^*(\mathcal{A}_2 \boxtimes \mathcal{M}_{2,x})} j_* \circ j^*(\mathcal{A}_1 \boxtimes \mathcal{M}_{2,x}),$$

where the map

$$\mathcal{M}_{2,X} \rightarrow j_* \circ j^*(\mathcal{A}_2 \boxtimes \mathcal{M}_{2,x})$$

is

$$\mathcal{M}_{2,X} \xrightarrow{\text{adjunction}} j_* \circ j^*(\mathcal{M}_{2,X}) \xrightarrow{\text{factorization}} j_* \circ j^*(\mathcal{A}_2 \boxtimes \mathcal{M}_{2,x}).$$

Using a description of $\text{Res}_\phi(\mathcal{M}_2)_{\mathcal{Z} \subseteq}$ along these lines, one can prove Lemma B.9.26 without resorting to chiral algebras.

B.9.29. *Example.* Let $\phi : \mathcal{A}_1 \rightarrow \mathcal{A}_2$ be a map of factorization algebras. For $\mathcal{Z} \rightarrow \text{Ran}$, consider the resulting map

$$\text{pr}_{\text{big}}^*(\mathcal{A}_{1,\text{Ran}}) \rightarrow \text{pr}_{\text{big}}^*(\mathcal{A}_{2,\text{Ran}})$$

in $\text{QCoh}(\mathcal{Z}^\subseteq)$.

It is easy to see that this map is compatible with factorization, i.e., gives rise to a map

$$\phi : \mathcal{A}_1^{\text{fact}\mathcal{Z}} \rightarrow \mathcal{A}_2^{\text{fact}\mathcal{Z}},$$

compatible with factorization.

In particular, we obtain a map

$$\mathcal{A}_1^{\text{fact}\mathcal{Z}} \rightarrow \text{Res}_\phi(\mathcal{A}_2^{\text{fact}\mathcal{Z}})$$

in $\mathcal{A}_1\text{-mod}_{\mathcal{Z}}^{\text{fact}}$.

B.10. Commutative factorization algebras.

B.10.1. The category $\text{FactAlg}(X)$ has an evident (pointwise) symmetric monoidal structure. Consider the category

$$\text{ComAlg}(\text{FactAlg}(X)).$$

We will refer to its objects as *commutative factorization algebras*.

We have a tautological forgetful functor

$$(B.33) \quad \text{ComAlg}(\text{FactAlg}(X)) \rightarrow \text{ComAlg}(\text{D-mod}(\text{Ran}))$$

B.10.2. Restriction along $X \rightarrow \text{Ran}$ defines a functor

$$(B.34) \quad \text{ComAlg}(\text{FactAlg}(X)) \rightarrow \text{ComAlg}(\text{D-mod}(\text{Ran})) \rightarrow \text{ComAlg}(\text{D-mod}(X)).$$

This functor has a right inverse, to be denoted

$$A \mapsto \text{Fact}(A)$$

.

Remark B.10.3. In Sect. C.8 we will upgrade the functor $\text{Fact}(-)$ to an equivalence of categories.

B.10.4. The object

$$\text{Fact}(A)_{\text{Ran}} \in \text{D-mod}(\text{Ran})$$

can be described by an explicit colimit procedure:

Let $\text{TwArr}(\text{fSet}^{\text{surj}})$ be the twisted arrows category of $\text{fSet}^{\text{surj}}$. I.e., its objects are

$$I \xrightarrow{\phi} J$$

and morphisms $(I_1 \xrightarrow{\phi_1} J_1) \rightarrow (I_2 \xrightarrow{\phi_2} J_2)$ are diagrams

$$(B.35) \quad \begin{array}{ccc} I_1 & \xrightarrow{\phi_1} & J_1 \\ \psi_I \downarrow & & \uparrow \psi_J \\ I_2 & \xrightarrow{\phi_2} & J_2. \end{array}$$

We define a functor

$$A_{\text{TwArr}} : \text{TwArr}(\text{fSet}^{\text{surj}}) \rightarrow \text{D-mod}(\text{Ran})$$

be sending $I \xrightarrow{\phi} J$ to the direct image along

$$\Delta_{X^J, \text{Ran}} : X_{\text{dR}}^J \rightarrow \text{Ran}$$

of

$$\boxtimes_{j \in J} A^{\otimes \phi^{-1}(j)}.$$

For a morphism (B.35), the corresponding map

$$(\Delta_{X^{J_1}, \text{Ran}})! \left(\boxtimes_{j_1 \in J_1} A^{\otimes \phi_1^{-1}(j_1)} \right) \rightarrow (\Delta_{X^{J_2}, \text{Ran}})! \left(\boxtimes_{j_2 \in J_2} A^{\otimes \phi_2^{-1}(j_2)} \right)$$

is

$$\begin{aligned} (\Delta_{X^{J_1}, \text{Ran}})! \left(\boxtimes_{j_1 \in J_1} A^{\otimes \phi_1^{-1}(j_1)} \right) &\simeq (\Delta_{X^{J_2}, \text{Ran}})! \circ (\Delta_{\psi_J})! \left(\boxtimes_{j_1 \in J_1} A^{\otimes \phi_1^{-1}(j_1)} \right) \simeq \\ &\simeq (\Delta_{X^{J_2}, \text{Ran}})! \circ (\Delta_{\psi_J})! \circ (\Delta_{\psi_J})! \left(\boxtimes_{j_2 \in J_2} A^{\otimes (\phi_2 \circ \psi_I)^{-1}(j_2)} \right) \rightarrow (\Delta_{X^{J_2}, \text{Ran}})! \left(\boxtimes_{j_2 \in J_2} A^{\otimes (\phi_2 \circ \psi_I)^{-1}(j_2)} \right) \simeq \\ &\simeq (\Delta_{X^{J_2}, \text{Ran}})! \left(\boxtimes_{j_2 \in J_2} \bigotimes_{i_2 \in \phi_2^{-1}(j_2)} A^{\otimes \psi_I^{-1}(i_2)} \right) \rightarrow (\Delta_{X^{J_2}, \text{Ran}})! \left(\boxtimes_{j_2 \in J_2} \bigotimes_{i_2 \in \phi_2^{-1}(j_2)} A \right) \simeq \\ &\simeq (\Delta_{X^{J_2}, \text{Ran}})! \left(\boxtimes_{j_2 \in J_2} A^{\otimes (\phi_2)^{-1}(j_2)} \right), \end{aligned}$$

where the arrow in the third line is the tensor product of the maps

$$A^{\otimes \psi_I^{-1}(i_2)} \rightarrow A,$$

given by the commutative algebra structure on A .

Then

$$\text{Fact}(A)_{\text{Ran}} \simeq \text{colim}_{\text{TwArr}(\text{fSet}^{\text{surj}})} A_{\text{TwArr}}.$$

B.10.5. Let $\mathcal{A}$ be a commutative factorization algebra on X . Let $\mathcal{Z}$ be a prestack mapping to Ran . Denote

$$\mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{com}} := \mathcal{A}_{\mathcal{Z}}\text{-mod}(\text{QCoh}(\mathcal{Z})),$$

where we regard $\mathcal{A}_{\mathcal{Z}}$ as an object of $\text{ComAlg}(\text{QCoh}(\mathcal{Z}))$.

We will view the assignment

$$\mathcal{Z} \rightsquigarrow \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{com}}$$

as a sheaf of categories over Ran , which we will denote by $\mathcal{A}\text{-mod}^{\text{com}}$.

B.10.6. Let $\mathcal{A}$ be of the form $\text{Fact}(A)$, for $A \in \text{ComAlg}(\text{D-mod}(X))$. We claim that in this case there is a canonically defined functor of sheaves of categories,

$$\mathcal{A}\text{-mod}^{\text{com}} \rightarrow \mathcal{A}\text{-mod}^{\text{fact}},$$

i.e., a compatible collection of functors

$$(B.36) \quad \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{com}} \rightarrow \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}}.$$

In order to construct (B.36), it suffices to exhibit an object of $\mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}}$, equipped with an action of $\mathcal{A}_{\mathcal{Z}} \in \text{ComAlg}(\text{QCoh}(\mathcal{Z}))$, where we view $\mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}}$ as tensored over $\text{QCoh}(\mathcal{Z})$.

The object in question is the vacuum module $\mathcal{A}^{\text{fact}, \mathcal{Z}}$, see Sect. B.9.7. Thus, we need to define an action of $\mathcal{A}_{\mathcal{Z}}$ on $\mathcal{A}^{\text{fact}, \mathcal{Z}}$. To do so, we can consider the universal case, i.e., $\mathcal{Z} = \text{Ran}$.

Thus, we need to define an action of $\text{pr}_{\text{small}}^*(\mathcal{A}_{\text{Ran}})$ on $\text{pr}_{\text{big}}^*(\mathcal{A}_{\text{Ran}})$, compatible with the factorization. This map comes from the commutative algebra structure on $\mathcal{A}_{\text{Ran}}$, and a homomorphism

$$\text{pr}_{\text{small}}^*(\mathcal{A}_{\text{Ran}}) \rightarrow \text{pr}_{\text{big}}^*(\mathcal{A}_{\text{Ran}}),$$

given by the *unital structure* on $\mathcal{A}$, see Sect. C.8.4 below.

Note that since

$$\mathcal{A}_{\mathcal{Z}}^{\text{fact}, \mathcal{Z}} \simeq \mathcal{A}_{\mathcal{Z}},$$

the functor (B.36) is compatible with the forgetful functors to $\text{QCoh}(\mathcal{Z})$.

B.10.7. Let $\mathcal{A}$ be a commutative factorization algebra. Restriction along the binary operation

$$\mathcal{A} \overset{!}{\otimes} \mathcal{A} \rightarrow \mathcal{A}$$

in the sense of Sect. B.9.24 defines on $\mathcal{A}\text{-mod}_Z^{\text{fact}}$ a structure of (symmetric) *pseudo-monoidal* category (see [BD2, Sect. 1.1]) for what this means.

Concretely, for $\mathcal{M}_1, \mathcal{M}_2, \mathcal{N} \in \mathcal{A}\text{-mod}_Z^{\text{fact}}$ this means that we know what it means to map

$$\mathcal{M}_1 \text{ “} \otimes \text{” } \mathcal{M}_2 \rightarrow \mathcal{N}.$$

Namely, by definition, the space of such maps is

$$\text{Maps}_{\mathcal{A} \overset{!}{\otimes} \mathcal{A} \rightarrow \mathcal{A}}(\mathcal{M}_1 \otimes \mathcal{M}_2, \mathcal{N})$$

in the notation of (B.30), where we regard $\mathcal{M}_1 \otimes \mathcal{M}_2$ as an object of $(\mathcal{A} \overset{!}{\otimes} \mathcal{A})\text{-mod}_Z^{\text{fact}}$.

In particular, it makes sense to talk about (commutative) algebra objects in $\mathcal{A}\text{-mod}_Z^{\text{fact}}$.

Note that by construction, the functor (B.36) is right-lax (pseudo)-monoidal. In particular, it maps (commutative) algebras to (commutative) algebras. In addition, the forgetful functor

$$\mathcal{A}\text{-mod}_Z^{\text{fact}} \rightarrow \text{QCoh}(Z^{\subseteq})$$

is also right-lax (pseudo)-monoidal; in particular, it maps (commutative) algebras to (commutative) algebras.

B.11. Factorization categories.

B.11.1. A factorization category $\mathbf{A}$ on X is a sheaf/crystal of categories $\underline{\mathbf{A}}$ over Ran , equipped with a datum of factorization

$$(B.37) \quad \text{union}^*(\underline{\mathbf{A}})|_{(\text{Ran} \times \text{Ran})_{\text{disj}}} \simeq \underline{\mathbf{A}} \boxtimes \underline{\mathbf{A}}|_{(\text{Ran} \times \text{Ran})_{\text{disj}}}$$

(here $-|_{-}$ denotes pullback of sheaves of categories along an open embedding) and with a homotopy-coherent datum of associativity and commutativity; see [Ra6, Sect. 6], where the definition is written out in detail.

For $f : Z \rightarrow \text{Ran}$, we will denote

$$\mathbf{A}_Z := \Gamma(Z, f^*(\underline{\mathbf{A}})).$$

In particular, we denote

$$\mathbf{A}_{\text{Ran}} := \Gamma(\text{Ran}, \underline{\mathbf{A}});$$

this is a category tensored over $\text{D-mod}(\text{Ran})$.

Let $\text{FactCat}(X)$ denote the (2-)category of factorization categories over X . This category carries a naturally defined symmetric monoidal structure.

B.11.2. The unit object in this symmetric monoidal category is the sheaf of categories $\underline{\text{QCoh}}(\text{Ran})$.

By a slight abuse of notation, we will denote this factorization category by Vect (i.e., its fiber at any $\underline{x} \in \text{Ran}$ is $\text{Vect} \in \text{DGCat}$).

B.11.3. Note that since Ran is 1-affine (by Lemma B.8.15), the datum of the sheaf of categories $\underline{\mathbf{A}}$ is equivalent to that of the category $\mathbf{A}_{\text{Ran}}$, equipped with an action of $\text{D-mod}(\text{Ran})$.

Furthermore, the datum of factorization is equivalent to

$$(B.38) \quad \mathbf{A}_{\text{Ran}} \underset{\text{D-mod}(\text{Ran}), \text{union}^!}{\otimes} \text{D-mod}((\text{Ran} \times \text{Ran})_{\text{disj}}) \simeq (\mathbf{A}_{\text{Ran}} \otimes \mathbf{A}_{\text{Ran}}) \underset{\text{D-mod}(\text{Ran}) \otimes \text{D-mod}(\text{Ran})}{\otimes} \text{D-mod}((\text{Ran} \times \text{Ran})_{\text{disj}}),$$

equipped with a homotopy-coherent datum of associativity and commutativity.

B.11.4. Let $\mathbf{A}$ be a factorization category. A factorization algebra $\mathcal{A} \in \mathbf{A}$ is an object $\mathcal{A}_{\text{Ran}} \in \mathbf{A}_{\text{Ran}}$, equipped with a datum of factorization

$$\text{union}^!(\mathcal{A}_{\text{Ran}})|_{(\text{Ran} \times \text{Ran})_{\text{disj}}} \simeq (\mathcal{A}_{\text{Ran}} \boxtimes \mathcal{A}_{\text{Ran}})|_{(\text{Ran} \times \text{Ran})_{\text{disj}}},$$

as objects in the two sides of (B.37), and with a homotopy-coherent datum of associativity and commutativity.

Let $\text{FactAlg}(X, \mathbf{A})$ denote the category of factorization algebras in $\mathbf{A}$.

A factorization functor $\mathbf{A}_1 \rightarrow \mathbf{A}_2$ induces a functor

$$\text{FactAlg}(X, \mathbf{A}_1) \rightarrow \text{FactAlg}(X, \mathbf{A}_2).$$

B.11.5. Given $\mathcal{A} \in \text{FactAlg}(X, \mathbf{A})$, parallel to Sect. B.9.3, given $\mathcal{Z} \rightarrow \text{Ran}$, one defines the category of factorization $\mathcal{A}$ -modules in $\mathbf{A}$ at $\mathcal{Z}$, denoted

$$\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})_{\mathcal{Z}}.$$

The assignment

$$(B.39) \quad \mathcal{Z} \rightarrow \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})_{\mathcal{Z}}$$

forms a sheaf of categories over Ran that we denote by

$$\underline{\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})}.$$

B.11.6. We will see many examples of factorization categories in the sequel. However, one family of examples we can produce right away:

Let $\underline{\mathbf{A}}_X$ be a crystal of symmetric monoidal categories over X . To it we attach a (symmetric monoidal) factorization category, denoted $\text{Fact}(\underline{\mathbf{A}}_X)$, see [GLys, Sect. 8.1].

Equivalently, let $\mathbf{A}_X$ be a symmetric monoidal category tensored over $\text{D-mod}(X)$. To it we can attach a DG category, denoted $\text{Fact}(\mathbf{A}_X)$, tensored over $\text{D-mod}(\text{Ran})$ (in fact, a commutative algebra object in $\text{D-mod}(\text{Ran})\text{-mod}$), and equipped with a factorization structure as in Sect. B.11.3.

This can be done by directly mimicking the procedure in Sect. B.10.4.

Similarly, if $A \in \mathbf{A}_X$ is a commutative algebra object, it gives rise to a commutative algebra object

$$\text{Fact}(A) \in \text{FactAlg}(X, \text{Fact}(\underline{\mathbf{A}}_X)).$$

B.11.7. Let A be an object in $\text{ComAlg}(\text{D-mod}(X))$. Denote $\mathcal{A} := \text{Fact}(A) \in \text{FactAlg}(X)$. It is easy to see that the sheaf of categories $\underline{\mathcal{A}\text{-mod}^{\text{com}}}$ (see Sect. B.10.5) has a natural factorization structure. We denote the resulting factorization category by $\mathcal{A}\text{-mod}^{\text{com}}$.

Furthermore, we have:

$$\mathcal{A}\text{-mod}^{\text{com}} \simeq \text{Fact}(A\text{-mod}(\text{D-mod}(X))).$$

B.11.8. *Notational convention.* In the special case when $\mathbf{A}_X$ is constant, i.e., of the form $\mathbf{A} \otimes \text{D-mod}(X)$, where $\mathbf{A}$ is a symmetric monoidal category, we will denote the factorization category $\text{Fact}(\mathbf{A}_X)$ simply by $\mathbf{A}$.

For example, when $\mathbf{A} = \text{Rep}(\check{G})$, we will use the symbol $\text{Rep}(\check{G})$ to denote the corresponding factorization category.

Note that this is in line with the notation for Vect in Sect. B.11.2.

B.11.9. Let $\mathbf{A}$ be a factorization category over X . We shall say that $\mathbf{A}$ is dualizable if for every $S \in \text{Sch}_{\text{Ran}}^{\text{aff}}$, the category $\mathbf{A}_S$ is dualizable. This is equivalent to $\underline{\mathbf{A}}$ being a dualizable object in $\text{ShvCat}(\text{Ran})$.

If $\mathbf{A}$ is dualizable, then the dual $\underline{\mathbf{A}}^\vee$ of $\underline{\mathbf{A}}$ as a sheaf of categories admits a natural factorization structure; we will denote the resulting factorization category by $\mathbf{A}^\vee$.

The datum of duality between two factorization categories $\mathbf{A}_1$ and $\mathbf{A}_2$ is equivalent to that of factorization functors

$$\text{Vect} \rightarrow \mathbf{A}_1 \otimes \mathbf{A}_2 \text{ and } \mathbf{A}_1 \otimes \mathbf{A}_2 \rightarrow \text{Vect},$$

satisfying the usual axioms.

B.11.10. Let $\mathbf{A}$ be a factorization category over X . We shall say that $\mathbf{A}$ is compactly generated if $\underline{\mathbf{A}}$ is compactly generated as a sheaf of categories over Ran .

Let $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$ be a factorization functor between factorization categories. Suppose that $\mathbf{A}_1$ is compactly generated. Suppose also that Φ preserves compactness, so that Φ admits a right adjoint Φ^R , as a functor between sheaves of categories.

It follows automatically that Φ^R carries a structure of compatibility with factorization.

B.11.11. Let $\mathbf{A}$ be a factorization category over X . A t-structure on $\mathbf{A}$ is a t-structure on $\underline{\mathbf{A}}$ as a sheaf of categories that is compatible with factorization in the following sense:

Let us be given a map

$$(\underline{x}_1, \underline{x}_2) : S \rightarrow (\text{Ran} \times \text{Ran})_{\text{disj}}, \quad S \in \text{Sch}^{\text{aff}}.$$

We need that the factorization equivalence

$$\mathbf{A}_{S, \text{union} \circ (\underline{x}_1, \underline{x}_2)} \simeq \mathbf{A}_{S, \underline{x}_1} \otimes_{\text{QCoh}(S)} \mathbf{A}_{S, \underline{x}_2}$$

be t-exact, where the right-hand side is equipped with the tensor product t-structure.

B.11.12. A lax factorization category $\mathbf{A}$ is a sheaf of categories $\mathbf{A}_{\text{Ran}}$ over Ran , equipped with *functors*

$$(B.40) \quad \underline{\mathbf{A}} \boxtimes \underline{\mathbf{A}}|_{(\text{Ran} \times \text{Ran})_{\text{disj}}} \rightarrow \text{union}^*(\underline{\mathbf{A}})|_{(\text{Ran} \times \text{Ran})_{\text{disj}}},$$

equipped with a homotopy-coherent datum of associativity and commutativity.

The entire discussion above equally applies to lax factorization categories.

B.11.13. In particular, we can talk about factorization algebras in a lax factorization category:

We require that the corresponding functor

$$(B.41) \quad (\mathbf{A}_{\text{Ran}} \otimes \mathbf{A}_{\text{Ran}})_{\text{D-mod}(\text{Ran}) \otimes \text{D-mod}(\text{Ran})} \otimes \text{D-mod}((\text{Ran} \times \text{Ran})_{\text{disj}}) \rightarrow \mathbf{A}_{\text{Ran}} \otimes_{\text{D-mod}(\text{Ran}), \text{union}^!} \text{D-mod}((\text{Ran} \times \text{Ran})_{\text{disj}}),$$

maps the object

$$(\mathcal{A}_{\text{Ran}} \boxtimes \mathcal{A}_{\text{Ran}})|_{(\text{Ran} \times \text{Ran})_{\text{disj}}} \in (\mathbf{A}_{\text{Ran}} \otimes \mathbf{A}_{\text{Ran}})_{\text{D-mod}(\text{Ran}) \otimes \text{D-mod}(\text{Ran})} \otimes \text{D-mod}((\text{Ran} \times \text{Ran})_{\text{disj}})$$

to the object

$$\text{union}^!(\mathcal{A}_{\text{Ran}})|_{(\text{Ran} \times \text{Ran})_{\text{disj}}} \in \mathbf{A}_{\text{Ran}} \otimes_{\text{D-mod}(\text{Ran}), \text{union}^!} \text{D-mod}((\text{Ran} \times \text{Ran})_{\text{disj}}).$$

B.11.14. *Example.* Let $\mathcal{A}$ be a factorization algebra on X . Note that the sheaf of categories $\mathcal{A}\text{-}\underline{\text{mod}}^{\text{fact}}$ carries a natural lax factorization structure:

Let us be given a pair of maps $\underline{x}_i : S_i \rightarrow \text{Ran}$, $i = 1, 2$ such that

$$S_1 \times S_2 \xrightarrow{\underline{x}_1 \times \underline{x}_2} \text{Ran} \times \text{Ran}$$

lands in $(\text{Ran} \times \text{Ran})_{\text{disj}}$. Let us be given a pair of objects $\mathcal{M}_i \in \mathcal{A}\text{-}\text{mod}_{S_i}^{\text{fact}}$. We define the corresponding object

$$\mathcal{M}_1 \boxtimes \mathcal{M}_2 \in \mathcal{A}\text{-}\text{mod}_{S_1 \times S_2}^{\text{fact}}$$

as follows.

Consider the prestack

$$(S_1 \times S_2)^{\subseteq} = \{\underline{x}' \in \text{Ran}, | \underline{x}_1 \subseteq \underline{x}', \underline{x}_2 \subseteq \underline{x}'\}.$$

For $i = 1, 2$, let $U_i \subset (S_1 \times S_2)^{\subseteq}$ be the open sub-prestack corresponding to the condition that $(\underline{x}_i, \underline{x}')$ belongs to $(\text{Ran} \times \text{Ran})_{\text{disj}}$. The condition that $\underline{x}_1$ and $\underline{x}_2$ are disjoint implies that

$$U_1 \cup U_2 = (S_1 \times S_2)^{\subseteq}.$$

Denote $U_{1,2} := U_1 \cap U_2$.

Let pr_1 (resp., pr_2) denote the map $U_1 \rightarrow S_1 \times S_2^{\subseteq}$ (resp., $U_2 \rightarrow S_1^{\subseteq} \times S_2$) whose second (resp., first) component remembers the data of $\underline{x}$. Let $\text{pr}_{1,2}$ denote the map $U_{1,2} \rightarrow S_1 \times S_2 \times \text{Ran}$, whose last component remembers $\underline{x}'$.

We let

$$(\mathcal{M}_1 \boxtimes \mathcal{M}_2)_{(S_1 \times S_2)^{\subseteq}}|_{U_1} \simeq \text{pr}_1^*(\mathcal{M}_{1,S_1} \boxtimes \mathcal{M}_{2,S_2^{\subseteq}})$$

and

$$(\mathcal{M}_1 \boxtimes \mathcal{M}_2)_{(S_1 \times S_2)^{\subseteq}}|_{U_2} \simeq \text{pr}_2^*(\mathcal{M}_{1,S_1^{\subseteq}} \boxtimes \mathcal{M}_{2,S_2}).$$

The factorization structures on $\mathcal{M}_1$ and $\mathcal{M}_2$ imply that

$$(\mathcal{M}_1 \boxtimes \mathcal{M}_2)_{(S_1 \times S_2)^{\subseteq}}|_{U_{1,2}} \simeq \text{pr}_{1,2}^*(\mathcal{M}_{1,S_1} \boxtimes \mathcal{M}_{2,S_2} \boxtimes \mathcal{A}_{\text{Ran}}) \simeq (\mathcal{M}_1 \boxtimes \mathcal{M}_2)_{(S_1 \times S_2)^{\subseteq}}|_{U_2}|_{U_{1,2}}.$$

This defines the object

$$(\mathcal{M}_1 \boxtimes \mathcal{M}_2)_{(S_1 \times S_2)^{\subseteq}} \in \text{QCoh}((S_1 \times S_2)^{\subseteq}).$$

The factorization structure on it against $\mathcal{A}$ follows from the construction.

B.11.15. Let us denote the resulting lax factorization category by $\mathcal{A}\text{-}\text{mod}^{\text{fact}}$. For a map $\phi : \mathcal{A}_1 \rightarrow \mathcal{A}_2$ between the factorization algebras, the restriction functor

$$\text{Res}_{\phi} : \mathcal{A}_2\text{-}\underline{\text{mod}}^{\text{fact}} \rightarrow \mathcal{A}_1\text{-}\underline{\text{mod}}^{\text{fact}}$$

upgrades to a factorization functor, denoted

$$\text{Res}_{\phi} : \mathcal{A}_2\text{-}\text{mod}^{\text{fact}} \rightarrow \mathcal{A}_1\text{-}\text{mod}^{\text{fact}}.$$

Denote by $\text{oblv}_{\mathcal{A}}$ the forgetful functor $\mathcal{A}\text{-}\text{mod}^{\text{fact}} \rightarrow \text{Vect}$ (see Sect. B.11.8 above for the convention regarding Vect).

Note that the assignment

$$\mathcal{Z} \mapsto \mathcal{A}^{\text{fact}\mathcal{Z}}$$

(see Sect. B.9.7) gives rise to a factorization algebra object in $\mathcal{A}\text{-}\text{mod}^{\text{fact}}$, to be denoted $\mathcal{A}^{\text{fact}}$, so that

$$\text{oblv}_{\mathcal{A}}(\mathcal{A}^{\text{fact}}) = \mathcal{A}.$$

That said, we will sometimes abuse the notation and instead of $\mathcal{A}^{\text{fact}}$ simply write $\mathcal{A}$.

Suppose for moment that $\mathcal{A}$ is connective. Then the construction in Sect. B.9.17 equips $\mathcal{A}\text{-}\text{mod}^{\text{fact}}$ with a t-structure in the sense of Sect. B.11.11.⁷⁰

⁷⁰In the case of lax factorization categories, we require that the functor (B.40) be t-exact.

B.11.16. The example in Sect. B.11.14 generalizes to the situation when $\mathcal{A}$ is a factorization algebra in a given lax factorization category $\mathbf{A}$. We obtain that the sheaf of categories

$$\mathcal{A}\text{-}\mathbf{mod}^{\text{fact}}(\mathbf{A})$$

carries a structure of lax factorization category. We denote it by $\mathcal{A}\text{-}\mathbf{mod}^{\text{fact}}(\mathbf{A})$.

As in Sect. B.11.15, we can consider $\mathcal{A}^{\text{fact}}$ as an object of $\text{FactAlg}(X, \mathcal{A}\text{-}\mathbf{mod}^{\text{fact}}(\mathbf{A}))$. By a slight abuse of notation, we will sometimes denote this factorization algebra simply by $\mathcal{A}$.

By definition, for any $\mathcal{Z} \rightarrow \text{Ran}$

$$(\mathcal{A}^{\text{fact}})_{\mathcal{Z}} \simeq \mathcal{A}^{\text{fact}_{\mathcal{Z}}}$$

as objects of $\mathcal{A}\text{-}\mathbf{mod}^{\text{fact}}(\mathbf{A})_{\mathcal{Z}}$.

If $\mathbf{A}$ carries a t-structure (in the sense of Sect. B.11.11) and $\mathcal{A}$ is connective (i.e., $\mathbf{oblv}^l(\mathcal{A}_X) \in \mathbf{A}_X$ is connective), then the construction in Sect. B.9.17 equips $\mathcal{A}\text{-}\mathbf{mod}^{\text{fact}}(\mathbf{A})$ with a t-structure.

B.11.17. Let $\mathbf{A}$ be a lax factorization category, and let $\phi : \mathcal{A} \rightarrow \mathcal{A}'$ be a homomorphism between factorization algebras in $\mathbf{A}$. Restriction along ϕ denotes a factorization functor

$$\text{Res}_{\phi} : \mathcal{A}'\text{-}\mathbf{mod}^{\text{fact}}(\mathbf{A}) \rightarrow \mathcal{A}\text{-}\mathbf{mod}^{\text{fact}}(\mathbf{A}).$$

In particular, we can consider

$$\text{Res}_{\phi}(\mathcal{A}'^{\text{fact}}) \in \text{FactAlg}(X, \mathcal{A}\text{-}\mathbf{mod}^{\text{fact}}(\mathbf{A})).$$

We have

$$\mathbf{oblv}_{\mathcal{A}}(\text{Res}_{\phi}(\mathcal{A}'^{\text{fact}})) = \mathcal{A}'.$$

We will sometimes abuse the notation, and instead of $\text{Res}_{\phi}(\mathcal{A}'^{\text{fact}})$ simply write $\mathcal{A}'$.

B.12. Factorization module categories.

B.12.1. Let $\mathbf{A}$ be a factorization category over X . Let $\mathcal{Z}$ be a prestack mapping to Ran . A factorization module category $\mathbf{C}$ over $\mathbf{A}$ at $\mathcal{Z}$ is a sheaf of categories $\underline{\mathbf{C}}$ on $\mathcal{Z}^{\subseteq}$, equipped with a factorization structure:

$$(B.42) \quad \text{union}^*(\underline{\mathbf{C}})|_{(\text{Ran} \times \mathcal{Z}^{\subseteq})_{\text{disj}}} \simeq \underline{\mathbf{A}} \boxtimes \underline{\mathbf{C}}|_{(\text{Ran} \times \mathcal{Z}^{\subseteq})_{\text{disj}}}$$

and with a homotopy-coherent datum of associativity; see [Ra6, Sect. 6] for details.

Let $\mathbf{A}\text{-}\mathbf{mod}_{\mathcal{Z}}^{\text{fact}}$ denote the (2-)category of factorization module categories over $\mathbf{A}$ at $\mathcal{Z}$.

B.12.2. For an object $\mathbf{C} \in \mathbf{A}\text{-}\mathbf{mod}_{\mathcal{Z}}^{\text{fact}}$ and $f : \mathcal{Z}' \rightarrow \mathcal{Z}^{\subseteq}$, we denote

$$\mathbf{C}_{\mathcal{Z}'} := \Gamma(\mathcal{Z}', f^*(\underline{\mathbf{C}})).$$

Taking $\mathcal{Z}' = \mathcal{Z}$ and $f = \text{diag}_{\mathcal{Z}}$, we obtain the category $\mathbf{C}_{\mathcal{Z}}$, tensored over $\text{QCoh}(\mathcal{Z})$. We will refer to $\mathbf{C}_{\mathcal{Z}}$ as the category underlying $\mathbf{C}$.

B.12.3. As in Sect. B.9.8, the above category $\mathbf{C}_{\mathcal{Z}'}$ is in fact the category underlying an $\mathbf{A}$ -module category at $\mathcal{Z}'$, to be denoted $\mathbf{C}|_{\mathcal{Z}'}$.

B.12.4. *Example.* Repeating the construction in Sect. B.9.7, we obtain that for any $\mathcal{Z}$ there exists a distinguished object

$$\mathbf{A}^{\text{fact}_{\mathcal{Z}}} \in \mathbf{A}\text{-}\mathbf{mod}_{\mathcal{Z}}^{\text{fact}},$$

whose underlying category is $\mathbf{A}_{\mathcal{Z}}$.

We will refer to $\mathbf{A}^{\text{fact}_{\mathcal{Z}}}$ as the *vacuum* factorization module category at $\mathcal{Z}$.

B.12.5. *Example.* Take $\mathbf{A} = \text{Vect}$, and let $\underline{\mathbf{C}}_0$ be a sheaf of categories over $\mathcal{Z}$. We claim that it gives rise to a factorization module category over Vect at $\mathcal{Z}$, to be denoted $\mathbf{C}$ (cf. Sect. B.9.6).

Namely, the corresponding sheaf of categories $\underline{\mathbf{C}}$ over $\mathcal{Z}^{\subseteq}$ is the pullback of $\underline{\mathbf{C}}_0$ along the projection $\text{pr}_{\text{small}} : \mathcal{Z}^{\subseteq} \rightarrow \mathcal{Z}$.

B.12.6. Let $\mathcal{A}$ be a factorization algebra in $\mathbf{A}$, and let $\mathbf{C}$ be an object of $\mathbf{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}}$.

A factorization $\mathcal{A}$ -module $\mathcal{M}$ in $\mathbf{C}$ is an object

$$\mathcal{M}_{\mathcal{Z} \subseteq} \in \mathbf{C}_{\mathcal{Z} \subseteq},$$

equipped with an isomorphism

$$(B.43) \quad \text{union}^*(\mathcal{M}_{\mathcal{Z} \subseteq})|_{(\text{Ran} \times \mathcal{Z} \subseteq)_{\text{disj}}} \simeq (\mathcal{A}_{\text{Ran}} \boxtimes \mathcal{M}_{\mathcal{Z} \subseteq})|_{(\text{Ran} \times \mathcal{Z} \subseteq)_{\text{disj}}},$$

where:

- $\text{union}^*(\mathcal{M}_{\mathcal{Z} \subseteq})$ is an object in

$$\Gamma\left((\text{Ran} \times \mathcal{Z} \subseteq)_{\text{disj}}, \text{union}^*(\underline{\mathbf{C}})|_{(\text{Ran} \times \mathcal{Z} \subseteq)_{\text{disj}}}\right);$$

- $\mathcal{A}_{\text{Ran}} \boxtimes \mathcal{M}_{\mathcal{Z} \subseteq}$ is an object in

$$\Gamma\left((\text{Ran} \times \mathcal{Z} \subseteq)_{\text{disj}}, \underline{\mathbf{A}} \boxtimes \underline{\mathbf{C}}|_{(\text{Ran} \times \mathcal{Z} \subseteq)_{\text{disj}}}\right);$$

- The isomorphisms between the two sides is understood in the sense of the identification (B.42).

The isomorphism (B.43) is required to be equipped with a homotopy-coherent datum of associativity.

B.12.7. We denote the category of factorization $\mathcal{A}$ -modules in $\mathbf{C}$ by

$$\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{C})_{\mathcal{Z}}.$$

Note that when $\mathbf{C} := \mathbf{A}^{\text{fact}_{\mathcal{Z}}}$, recover the category (B.39).

B.12.8. Let $\underline{\mathbf{C}}_0$ be as in Sect. B.12.5. Denote

$$\mathbf{C}_0 := \Gamma(\mathcal{Z}, \underline{\mathbf{C}}_0).$$

Let $\mathcal{A}$ be a factorization algebra (i.e., a factorization algebra in Vect , viewed as a factorization category). Then it makes sense to consider the category

$$\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{C})_{\mathcal{Z}}.$$

Note that we have a tautologically defined functor

$$(B.44) \quad \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}} \otimes_{\text{QCoh}(\mathcal{Z})} \mathbf{C}_0 \rightarrow \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{C})_{\mathcal{Z}}.$$

We claim:

Lemma B.12.9. *Assume that $\mathcal{Z} = S$ is an affine scheme, and assume that $\underline{\mathbf{C}}_0$ dualizable as a sheaf of categories. Then the functor (B.44) is an equivalence.*

Proof. Let $\underline{\mathbf{C}}_0^{\vee}$ be the dual sheaf of categories; note that $\Gamma(\mathcal{Z}, \underline{\mathbf{C}}_0^{\vee})$ identifies with the dual of $\mathbf{C}_0$ as a $\text{QCoh}(S)$ -linear category, and hence also as a plain DG category.

We have a naturally defined functor

$$\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{C})_S \otimes_{\text{QCoh}(S)} \mathbf{C}_0^{\vee} \rightarrow \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{C} \otimes \mathbf{C}^{\vee})_S,$$

where $\mathbf{C}^{\vee} \in \text{Vect}\text{-mod}_{\mathcal{Z}}^{\text{fact}}$ is attached to $\underline{\mathbf{C}}_0^{\vee}$ by the procedure of Sect. B.12.5.

Composing with the evaluation map

$$\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{C} \otimes \mathbf{C}^{\vee})_S \rightarrow \mathcal{A}\text{-mod}^{\text{fact}}(\text{Vect}_S^{\text{fact}})_S \simeq \mathcal{A}\text{-mod}_S^{\text{fact}},$$

we obtain a functor

$$(B.45) \quad \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{C})_S \otimes_{\text{QCoh}(S)} \mathbf{C}_0^{\vee} \rightarrow \mathcal{A}\text{-mod}_S^{\text{fact}}.$$

The sought-for inverse functor to (B.44) is given by

$$\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{C})_{\mathcal{Z}} \xrightarrow{\text{Id} \otimes \text{co-eval}} \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{C})_{\mathcal{Z}} \otimes_{\text{QCoh}(S)} \underline{\mathbf{C}}_0^{\vee} \otimes_{\text{QCoh}(S)} \underline{\mathbf{C}}_0 \xrightarrow{(B.45)} \mathcal{A}\text{-mod}_S^{\text{fact}} \otimes_{\text{QCoh}(S)} \underline{\mathbf{C}}_0.$$

□

B.12.10. Let $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$ be a factorization functor between factorization categories. Let $\mathbf{C}_1$ and $\mathbf{C}_2$ be objects in $\mathbf{A}_1\text{-mod}_{\mathcal{Z}}^{\text{fact}}$ and $\mathbf{A}_2\text{-mod}_{\mathcal{Z}}^{\text{fact}}$, respectively.

A functor $\Phi_m : \mathbf{C}_1 \rightarrow \mathbf{C}_2$ compatible with factorization is a functor

$$\Phi_m : \underline{\mathbf{C}}_1 \rightarrow \underline{\mathbf{C}}_2$$

between sheaves of categories on $\mathcal{Z}^{\subseteq}$ that makes the following diagram commute

$$\begin{array}{ccc} \text{union}^*(\underline{\mathbf{C}}_1)|_{(\text{Ran} \times \mathcal{Z}^{\subseteq})_{\text{disj}}} & \xrightarrow{\sim} & \underline{\mathbf{A}}_1 \boxtimes \underline{\mathbf{C}}_1|_{(\text{Ran} \times \mathcal{Z}^{\subseteq})_{\text{disj}}} \\ \Phi_m \downarrow & & \downarrow \Phi \boxtimes \Phi_m \\ \text{union}^*(\underline{\mathbf{C}}_2)|_{(\text{Ran} \times \mathcal{Z}^{\subseteq})_{\text{disj}}} & \xrightarrow{\sim} & \underline{\mathbf{A}}_2 \boxtimes \underline{\mathbf{C}}_2|_{(\text{Ran} \times \mathcal{Z}^{\subseteq})_{\text{disj}}} \end{array}$$

along with a homotopy-coherent system of higher compatibilities.

Let

$$\text{Funct}_{\mathbf{A}_1 \rightarrow \mathbf{A}_2}(\mathbf{C}_1, \mathbf{C}_2)$$

denote the category of such functors. When an ambiguity is likely to occur, we will use the notation $\text{Funct}_{\Phi: \mathbf{A}_1 \rightarrow \mathbf{A}_2}(\mathbf{C}_1, \mathbf{C}_2)$, i.e., we will insert Φ in the subscript.

B.12.11. Given $\mathbf{C}_2 \in \mathbf{A}_2\text{-mod}_{\mathcal{Z}}^{\text{fact}}$, one defines its restriction,

$$\text{Res}_{\Phi}(\mathbf{C}_2) \in \mathbf{A}_1\text{-mod}_{\mathcal{Z}}^{\text{fact}}$$

by the universal property as in Sect. B.9.25, i.e.,

$$\text{Funct}_{\mathbf{A}_1\text{-mod}_{\mathcal{Z}}^{\text{fact}}}(\mathbf{C}_1, \text{Res}_{\Phi}(\mathbf{C}_2)) \simeq \text{Funct}_{\mathbf{A}_1 \rightarrow \mathbf{A}_2}(\mathbf{C}_1, \mathbf{C}_2).$$

One can explicitly describe $\text{Res}_{\Phi}(\mathbf{C}_2)$ by a limit procedure as in Sect. B.9.28. ⁷¹

We have a tautologically defined functor

$$(B.46) \quad \text{Res}_{\Phi}(\mathbf{C}_2) \rightarrow \mathbf{C}_2$$

compatible with factorization.

Parallel to Lemma B.9.26, we have:

Lemma B.12.12. *The functor (B.46) induces an equivalence of the underlying categories*

$$\text{Res}_{\Phi}(\mathbf{C}_2)_{\mathcal{Z}} \rightarrow \mathbf{C}_{2, \mathcal{Z}}.$$

B.12.13. Let $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$ be a factorization functor, and let $\mathcal{A}_1$ be a factorization algebra in $\mathbf{A}_1$. Note that $\Phi(\mathcal{A}_1)$ has a natural structure of factorization algebra in $\mathbf{A}_2$.

Let $\mathbf{C}_1$ and $\mathbf{C}_2$ as in Sect. B.12.10. Given a functor $\Phi_m : \mathbf{C}_1 \rightarrow \mathbf{C}_2$ compatible with factorization, we obtain a naturally defined functor

$$(B.47) \quad \mathcal{A}_1\text{-mod}^{\text{fact}}(\mathbf{C}_1)_{\mathcal{Z}} \rightarrow \Phi(\mathcal{A}_1)\text{-mod}^{\text{fact}}(\mathbf{C}_2)_{\mathcal{Z}}.$$

Parallel with Lemma B.12.12 one proves:

Lemma B.12.14. *Assume that the functor $\Phi_m : \mathbf{C}_1 \rightarrow \mathbf{C}_2$ induces an equivalence*

$$\mathbf{C}_1 \rightarrow \text{Res}_{\Phi}(\mathbf{C}_2)$$

as factorization module categories over $\mathbf{A}_1$. Then the functor (B.47) is an equivalence.

⁷¹This will be written out in detail in [CFGY].

B.12.15. As in Sect. B.9.29, given a factorization functor $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$, for any $\mathcal{Z} \rightarrow \text{Ran}$, we have a canonically defined functor

$$\mathbf{A}_1^{\text{fact}_{\mathcal{Z}}} \rightarrow \text{Res}_{\Phi}(\mathbf{A}_2^{\text{fact}_{\mathcal{Z}}}).$$

In particular, given a commutative diagram of factorization categories

$$\begin{array}{ccc} \mathbf{A}'_1 & \longrightarrow & \mathbf{A}'_2 \\ \Psi_1 \uparrow & & \uparrow \Psi_2 \\ \mathbf{A}_1 & \longrightarrow & \mathbf{A}_2. \end{array}$$

we obtain a functor

$$\text{Res}_{\Psi_1}(\mathbf{A}'_1{}^{\text{fact}_{\mathcal{Z}}}) \rightarrow \text{Res}_{\Psi_2}(\mathbf{A}'_2{}^{\text{fact}_{\mathcal{Z}}}),$$

compatible with factorization.

B.13. Factorization categories of algebro-geometric nature.

B.13.1. Let $\mathcal{Z}_{\text{Ran}} \rightarrow \text{Ran}$ be a prestack. We attach to it a sheaf of categories $\underline{\text{QCoh}}(\mathcal{Z})$ over Ran , namely,

$$\underline{\text{QCoh}}(\mathcal{Z}) := \pi_*(\underline{\text{QCoh}}(\mathcal{Z}_{\text{Ran}})),$$

where π denotes the projection $\mathcal{Z}_{\text{Ran}} \rightarrow \text{Ran}$.

Explicitly, for $S \in \text{Sch}_{/\text{Ran}}^{\text{aff}}$, we have

$$\text{QCoh}(\mathcal{Z})_S = \text{QCoh}(S \times_{\text{Ran}} \mathcal{Z}_{\text{Ran}}).$$

B.13.2. Let $\mathcal{T}$ be a factorization space over X . Consider the corresponding sheaf of categories $\underline{\text{QCoh}}(\mathcal{T})$, i.e.,

$$\text{QCoh}(\mathcal{T})_S = \text{QCoh}(\mathcal{T}_S).$$

The factorization structure on $\mathcal{T}$ equips $\underline{\text{QCoh}}(\mathcal{T})$ with a lax factorization structure. We denote the corresponding lax factorization category by $\text{QCoh}(\mathcal{T})$.

Remark B.13.3. The reason that $\mathcal{T}$ is a priori only lax is that for a pair of prestacks $\mathcal{Y}_1$ and $\mathcal{Y}_2$, the naturally defined functor

$$\text{QCoh}(\mathcal{Y}_1) \otimes \text{QCoh}(\mathcal{Y}_2) \rightarrow \text{QCoh}(\mathcal{Y}_1 \otimes \mathcal{Y}_2)$$

is not necessarily an equivalence.

This lax structure is strict, e.g., if $\underline{\text{QCoh}}(\mathcal{T})$ is dualizable (as a sheaf of categories over Ran , which is equivalent to each $\text{QCoh}(\mathcal{T})_S$ being dualizable). This happens, e.g., if $\mathcal{T}$ is a factorization *affine scheme*.

B.13.4. *Example.* Let $\mathcal{Y}$ be an affine D-scheme over X . On the one hand, we can consider the factorization scheme $\mathfrak{L}_{\nabla}^+(\mathcal{Y})$ and the corresponding factorization category $\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}))$. As such, it is equipped with a symmetric monoidal structure.

On the other hand, we can consider the symmetric monoidal factorization category

$$\text{Fact}(\text{QCoh}(\mathcal{Y})).$$

We claim that there is a canonical equivalence:

$$\text{Fact}(\text{QCoh}(\mathcal{Y})) \simeq \text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y})).$$

Indeed, both sides identify, as factorization categories with $\mathcal{A}\text{-mod}^{\text{com}}$ (see Sect. B.11.7), where

$$\mathcal{A} = \text{Fact}(A), \quad \mathcal{Y} = \text{Spec}_X(A),$$

see (C.44) below.

B.13.5. Let now $\mathcal{T} := \mathrm{LS}_H^{\mathrm{reg}}$. We claim that the lax factorization category $\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}})$ identifies with $\mathrm{Rep}(H)$ (see Sect. B.11.8 for the notational conventions); in particular $\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}})$ is a *factorization* category.

Indeed, on the one hand, unwinding the definitions, we obtain that $\mathrm{Rep}(H)$, viewed as a factorization category is the (factorization) category of comodules with respect to $\mathrm{Fact}(\mathcal{O}_H)$, viewed as a factorization coalgebra.

We can rewrite it as the totalization of the cosimplicial factorization category with terms

$$\mathrm{Fact}(\mathcal{O}_{H^\bullet})\text{-mod}^{\mathrm{com}},$$

where $H^\bullet$ is the Čech nerve of $\mathrm{pt} \rightarrow \mathrm{pt}/H$.

On the other hand since $\mathrm{pt} \rightarrow \mathrm{LS}_H^{\mathrm{reg}}$ is an fpqc cover (see Lemma B.7.4), $\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}})$ identifies with the totalization of the cosimplicial factorization category with terms $\mathrm{QCoh}(-)$ of the Čech nerve of $\mathrm{pt} \rightarrow \mathrm{LS}_H^{\mathrm{reg}}$, the latter being $\mathcal{L}_{\nabla}^+(H^\bullet)$.

How the desired equivalence follows from the fact that

$$\mathcal{O}_{\mathcal{L}_{\nabla}^+(H^\bullet)} \simeq \mathrm{Fact}(\mathcal{O}_{H^\bullet}),$$

see (C.44) below.

Remark B.13.6. For completeness, we remark on the following comparison with [Ra3].

In *loc. cit.*, a different construction of the factorization category associated to a symmetric monoidal category was used; in particular, $\mathrm{Rep}(H)_{\mathrm{Ran}}$ has an a priori different meaning than how it is used in this paper. One can directly compare the two constructions, but rather than doing so here, we note that the above material combined with [Ra4, Lemma 9.8.1] allows us to indirectly deduce that the two constructions coincide.

B.13.7. Let $\mathcal{Z}_{\mathrm{Ran}} \rightarrow \mathrm{Ran}$ be a prestack. We attach to it a sheaf of categories $\underline{\mathrm{QCoh}}_{\mathrm{co}}(\mathcal{Z})$ over Ran by setting for $S \in \mathrm{Sch}_{\mathrm{Ran}}^{\mathrm{aff}}$

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Z})_S := \mathrm{QCoh}_{\mathrm{co}}(S \times_{\mathrm{Ran}} \mathcal{Z}_{\mathrm{Ran}}).$$

The sheaf of categories structure holds thanks to Lemma A.1.11.

B.13.8. Let $\mathcal{T}$ be factorization space over X . Consider the corresponding sheaf of categories $\underline{\mathrm{QCoh}}_{\mathrm{co}}(\mathcal{T})$, i.e.,

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{T})_S := \mathrm{QCoh}_{\mathrm{co}}(\mathcal{T}_S).$$

The factorization structure on $\mathcal{T}$ induces a *factorization* structure on $\underline{\mathrm{QCoh}}_{\mathrm{co}}(\mathcal{T})$, see Lemma A.1.5. We will denote the resulting factorization category by $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{T})$.

By Sect. A.2.8, if $\mathcal{T}$ is a factorization ind-scheme, the factorization category $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{T})$ carries a naturally defined t-structure.

B.13.9. Let $\mathcal{Z}_{\mathrm{Ran}} \rightarrow \mathrm{Ran}$ be a prestack. We will attach to it a sheaf of categories over Ran , denoted $\underline{\mathrm{IndCoh}}^!(\mathcal{Z})$. Unlike the cases of $\mathrm{QCoh}(-)$ and $\mathrm{QCoh}_{\mathrm{co}}(-)$, this will use some special features of Ran .

Namely, we will use the fact that Ran can be exhibited as a colimit of prestacks $S_{\alpha, \mathrm{dR}}$, where S_α are *smooth* schemes with transition $f_{\alpha, \beta} : S_\alpha \rightarrow S_\beta$ maps being closed embeddings. In practice, $S_\alpha = X^I$ for finite non-empty sets I , see Sect. B.1.3.

B.13.10. For S as above set

$$\mathrm{IndCoh}^!(\mathcal{Z})_{S_\alpha} := \mathrm{IndCoh}(\mathcal{Z}_{S_\alpha}).$$

For a map $S_\alpha \xrightarrow{f_{\alpha,\beta}} S_\beta$ denote by the same symbol $f_{\alpha,\beta}$ the corresponding map

$$\mathcal{Z}_{S_\alpha} \rightarrow \mathcal{Z}_{S_\beta}.$$

The functor

$$f_{\alpha,\beta}^! : \mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta}) \rightarrow \mathrm{IndCoh}^!(\mathcal{Z}_{S_\alpha})$$

gives rise to a functor

$$(B.48) \quad \mathrm{QCoh}(S_\alpha) \otimes_{\mathrm{QCoh}(S_\beta)} \mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta}) \rightarrow \mathrm{IndCoh}^!(\mathcal{Z}_{S_\alpha}).$$

We claim:

Lemma B.13.11. *The functor (B.48) is an equivalence.*

Proof. The question is Zariski-local, so we can assume that S_α and S_β are affine. By Lemma A.4.10, the functor (B.48) is fully faithful. So we only have to show that it is essentially surjective.

Let $S_\beta^\wedge$ be the formal completion of S_β along S_α . Consider the corresponding functor

$$(B.49) \quad \mathrm{QCoh}(S_\beta^\wedge) \otimes_{\mathrm{QCoh}(S_\beta)} \mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta}) \rightarrow \mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta^\wedge}).$$

We have a commutative diagram

$$\begin{array}{ccc} \mathrm{QCoh}(S_\beta^\wedge) \otimes_{\mathrm{QCoh}(S_\beta)} \mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta}) & \longrightarrow & \mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta^\wedge}) \\ \uparrow & & \uparrow \\ \mathrm{QCoh}(S_\beta) \otimes_{\mathrm{QCoh}(S_\beta)} \mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta}) & \xrightarrow{=} & \mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta}). \end{array}$$

The left vertical arrow is a colocalization. We claim that the right vertical arrow is also a colocalization: indeed, this follows from Proposition A.4.14. Hence, we obtain that the functor (B.49) is also a colocalization.⁷² In particular, (B.49) is essentially surjective.

We have a commutative diagram

$$(B.50) \quad \begin{array}{ccc} \mathrm{QCoh}(S_\alpha) \otimes_{\mathrm{QCoh}(S_\beta)} \mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta}) & \longrightarrow & \mathrm{IndCoh}^!(\mathcal{Z}_{S_\alpha}) \\ \uparrow & & \uparrow \\ \mathrm{QCoh}(S_\beta^\wedge) \otimes_{\mathrm{QCoh}(S_\beta)} \mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta}) & \longrightarrow & \mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta^\wedge}), \end{array}$$

where:

- The right vertical arrow is given by $!$ -pullback along $\mathcal{Z}_{S_\alpha} \rightarrow \mathcal{Z}_{S_\beta^\wedge}$;
- The left vertical arrows is given by the $*$ -pullback functor along $S_\alpha \rightarrow S_\beta^\wedge$ along the $\mathrm{QCoh}(-)$ factors.

We wish to show that the top horizontal arrow in (B.50) is essentially surjective. By the above, the bottom horizontal arrow is essentially surjective. Hence, it suffices to show that the functor

$$\mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta^\wedge}) \rightarrow \mathrm{IndCoh}^!(\mathcal{Z}_{S_\alpha})$$

is essentially surjective.

We claim, however, that the map $\mathcal{Z}_{S_\alpha} \rightarrow \mathcal{Z}_{S_\beta^\wedge}$ admits a retraction. Indeed, since S_α is smooth, the embedding

$$S_\alpha \rightarrow S_\beta^\wedge$$

⁷²In fact, the above argument shows that (B.49) is an equivalence.

admits a retraction, denote it by $g_{\beta,\alpha}$. Note that the two maps

$$S_\beta^\wedge \rightarrow \text{Ran} \text{ and } S_\beta^\wedge \xrightarrow{g_{\beta,\alpha}} S_\alpha \rightarrow \text{Ran}$$

agree on $(S_\beta^\wedge)_{\text{red}} \simeq S_\alpha$. Hence, since $\text{Ran} \rightarrow \text{Ran}_{\text{dR}}$ is an isomorphism, we can identify

$$\mathcal{Z}_{S_\beta^\wedge} \simeq S_\beta^\wedge \times_{g_{\beta,\alpha}, S_\alpha} \mathcal{Z}_{S_\alpha}.$$

In terms of this identification, the projection

$$S_\beta^\wedge \times_{g_{\beta,\alpha}, S_\alpha} \mathcal{Z}_{S_\alpha} \rightarrow \mathcal{Z}_{S_\alpha}$$

provides the sought-for retraction. $\square$

B.13.12. Let S be an affine scheme mapping to Ran . This map factors as

$$S \xrightarrow{f} S_\alpha \rightarrow \text{Ran}$$

for some α .

Set

$$\text{IndCoh}^!(\mathcal{Z})_{S,f} := \text{QCoh}(S) \otimes_{\text{QCoh}(S_\alpha)} \text{IndCoh}^!(\mathcal{Z}_{S_\alpha}).$$

In order to show that the assignment

$$S \in \text{Sch}_{\text{Ran}}^{\text{aff}} \rightsquigarrow \text{IndCoh}^!(\mathcal{Z})_{S,f}$$

gives a well-defined sheaf of categories over Ran , it remains to show that for two maps f_1 and f_2 as above, for which $f_1|_{S_{\text{red}}} \simeq f_2|_{S_{\text{red}}}$, we have a canonical identification

$$\text{IndCoh}^!(\mathcal{Z})_{S,f_1} \simeq \text{IndCoh}^!(\mathcal{Z})_{S,f_2},$$

i.e.,

$$(B.51) \quad \text{QCoh}(S) \otimes_{f_1^*, \text{QCoh}(S_\alpha)} \text{IndCoh}^!(\mathcal{Z}_{S_\alpha}) \simeq \text{QCoh}(S) \otimes_{f_2^*, \text{QCoh}(S_\alpha)} \text{IndCoh}^!(\mathcal{Z}_{S_\alpha}).$$

(In addition, one needs to show that these identifications satisfy a homotopy-coherent system of compatibilities for multi-fold comparisons $f_1|_{S_{\text{red}}} \simeq f_2|_{S_{\text{red}}} \simeq \dots \simeq f_n|_{S_{\text{red}}}$, but this will be automatic from the construction explained below.)

B.13.13. Let $(S_\alpha \times S_\alpha)^\wedge$ be the formal completion of the diagonal in $S_\alpha \times S_\alpha$. Note that we have a well-defined map

$$(S_\alpha \times S_\alpha)^\wedge \rightarrow \text{Ran},$$

Consider the corresponding prestack $\mathcal{Z}_{(S_\alpha \times S_\alpha)^\wedge}$ and the category

$$\text{IndCoh}^!(\mathcal{Z}_{(S_\alpha \times S_\alpha)^\wedge}).$$

For $i = 1, 2$, let

$$p_i : (S_\alpha \times S_\alpha)^\wedge \rightarrow S_\alpha$$

denote the corresponding projection. We will denote by the same symbol p_i the corresponding map

$$\mathcal{Z}_{(S_\alpha \times S_\alpha)^\wedge} \rightarrow \mathcal{Z}_{S_\alpha}.$$

The functor

$$p_i^! : \text{IndCoh}^!(\mathcal{Z}_{S_\alpha}) \rightarrow \text{IndCoh}^!(\mathcal{Z}_{(S_\alpha \times S_\alpha)^\wedge})$$

gives rise to a functor

$$(B.52) \quad \text{QCoh}((S_\alpha \times S_\alpha)^\wedge) \otimes_{p_i^*, \text{QCoh}(S_\alpha)} \text{IndCoh}^!(\mathcal{Z}_{S_\alpha}) \rightarrow \text{IndCoh}^!(\mathcal{Z}_{(S_\alpha \times S_\alpha)^\wedge}).$$

We claim:

Lemma B.13.14. *The functor (B.52) is an equivalence.*

Proof. Proceeds along the same lines as the proof of Lemma B.13.11. $\square$

B.13.15. Using Lemma B.13.14, we obtain

$$\begin{aligned} \mathrm{QCoh}(S) & \underset{f_i^*, \mathrm{QCoh}(S_\alpha)}{\otimes} \mathrm{IndCoh}^!(\mathcal{Z}_{S_\alpha}) = \\ & = \mathrm{QCoh}(S) \underset{(f_1 \times f_2)^*, \mathrm{QCoh}((S_\alpha \times S_\alpha)^\wedge)}{\otimes} \mathrm{QCoh}((S_\alpha \times S_\alpha)^\wedge) \underset{p_i^*, \mathrm{QCoh}(S_\alpha)}{\otimes} \mathrm{IndCoh}^!(\mathcal{Z}_{S_\alpha}) \simeq \\ & \simeq \mathrm{QCoh}(S) \underset{(f_1 \times f_2)^*, \mathrm{QCoh}((S_\alpha \times S_\alpha)^\wedge)}{\otimes} \mathrm{IndCoh}^!(\mathcal{Z}_{(S_\alpha \times S_\alpha)^\wedge}), \end{aligned}$$

thereby establishing (B.51).

This completes the construction of $\underline{\mathrm{IndCoh}}^!(\mathcal{Z})$ as a sheaf of categories over Ran .

B.13.16. Let $\mathcal{T}$ be a factorization space over X . The multiplicative structure in Sect. A.6.3 equips the corresponding sheaf of categories $\underline{\mathrm{IndCoh}}^!(\mathcal{T})$ with a structure of lax factorization category; we will denote it by $\mathrm{IndCoh}^!(\mathcal{T})$.

Suppose now that $\mathcal{T}$ is ind-placid. In this case, from Lemma A.10.10, we obtain that the lax factorization structure on $\mathrm{IndCoh}^!(\mathcal{T})$ is a *factorization* structure.

B.13.17. Let now $\mathcal{Z}_{\mathrm{Ran}} \rightarrow \mathrm{Ran}$ be a relative ind-placid ind-scheme. In this case, we are going to define the sheaf of categories $\underline{\mathrm{IndCoh}}^*(\mathcal{Z})$.

We proceed with the same recipe as in the case of $\underline{\mathrm{IndCoh}}^!(\mathcal{Z})$. For an index α , set

$$\mathrm{IndCoh}^*(\mathcal{Z})_{S_\alpha} := \mathrm{IndCoh}^*(\mathcal{Z}_{S_\alpha}).$$

For a map $f_{\alpha,\beta}$, we have a well-defined functor

$$f_{\alpha,\beta}^{*, \mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathcal{Z}_{S_\beta}) \rightarrow \mathrm{IndCoh}^*(\mathcal{Z}_{S_\alpha})$$

(see Sect. A.7.4). Consider the resulting functor

$$(B.53) \quad \mathrm{QCoh}(S_\alpha) \underset{\mathrm{QCoh}(S_\beta)}{\otimes} \mathrm{IndCoh}^*(\mathcal{Z}_{S_\beta}) \rightarrow \mathrm{IndCoh}^*(\mathcal{Z}_{S_\alpha}).$$

We claim:

Lemma B.13.18. *The functor (B.53) is an equivalence.*

Proof. We can reformulate the assertion of the lemma as saying that the right adjoint of $f_{\alpha,\beta}^{*, \mathrm{IndCoh}}$, i.e., the functor

$$(f_{\alpha,\beta})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathcal{Z}_{S_\alpha}) \rightarrow \mathrm{IndCoh}^*(\mathcal{Z}_{S_\beta})$$

gives rise to an equivalence

$$(B.54) \quad \mathrm{IndCoh}^*(\mathcal{Z}_{S_\alpha}) \rightarrow (f_{\alpha,\beta})_*(\mathcal{O}_{S_\alpha})\text{-mod}(\mathrm{IndCoh}^*(\mathcal{Z}_{S_\beta})).$$

Note that Lemma B.13.11 can be reformulated as saying that the functor

$$f_{\alpha,\beta}^! : \mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta}) \rightarrow \mathrm{IndCoh}^!(\mathcal{Z}_{S_\alpha})$$

admits a right adjoint, and this right adjoint identifies

$$(f_{\alpha,\beta})_*(\mathcal{O}_{S_\alpha})\text{-mod}(\mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta})) \simeq \mathrm{IndCoh}^!(\mathcal{Z}_{S_\alpha}).$$

Passing to the duals, we obtain that the dual of $f_{\alpha,\beta}^!$ identifies

$$(B.55) \quad (f_{\alpha,\beta})_*(\mathcal{O}_{S_\alpha})\text{-mod}(\mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta})^\vee) \simeq \mathrm{IndCoh}^!(\mathcal{Z}_{S_\alpha})^\vee.$$

We will now use the identifications

$$\mathrm{IndCoh}^*(\mathcal{Z}_{S_\alpha}) \simeq \mathrm{IndCoh}^!(\mathcal{Z}_{S_\alpha})^\vee \text{ and } \mathrm{IndCoh}^*(\mathcal{Z}_{S_\beta}) \simeq \mathrm{IndCoh}^!(\mathcal{Z}_{S_\beta})^\vee,$$

see Sect. A.10.8.

Under these identifications

$$(f_{\alpha,\beta}^!)^\vee \simeq (f_{\alpha,\beta})_*^{\mathrm{IndCoh}}.$$

Unwinding the definitions, it is easy to see that the resulting identification (B.55) is the same as (B.54). $\square$

B.13.19. Let S be an affine scheme mapping to Ran . This map factors as

$$S \xrightarrow{f} S_\alpha \rightarrow \text{Ran}$$

for some α .

Set

$$\text{IndCoh}^*(\mathcal{Z})_{S,f} := \text{QCoh}(S) \underset{\text{QCoh}(S_\alpha)}{\otimes} \text{IndCoh}^*(\mathcal{Z}_{S_\alpha}).$$

As in the case of $\text{IndCoh}^!$, in order to complete the construction of $\underline{\text{IndCoh}}^*(\mathcal{Z})$ as a sheaf of categories over Ran , it suffices to show that for a pair of maps f_1 and f_2 as above, for which $f_1|_{S_{\text{red}}} \simeq f_2|_{S_{\text{red}}}$, we have a canonical identification

$$\text{IndCoh}^*(\mathcal{Z})_{S,f_1} \simeq \text{IndCoh}^*(\mathcal{Z})_{S,f_2},$$

i.e.,

$$(B.56) \quad \text{QCoh}(S) \underset{f_1^*, \text{QCoh}(S_\alpha)}{\otimes} \text{IndCoh}^*(\mathcal{Z}_{S_\alpha}) \simeq \text{QCoh}(S) \underset{f_2^*, \text{QCoh}(S_\alpha)}{\otimes} \text{IndCoh}^*(\mathcal{Z}_{S_\alpha}).$$

This follows from the following:

Lemma B.13.20. *In the notations of Lemma B.13.14, the functor*

$$\text{QCoh}((S_\alpha \times S_\alpha)^\wedge) \underset{p_i^*, \text{QCoh}(S_\alpha)}{\otimes} \text{IndCoh}^*(\mathcal{Z}_{S_\alpha}) \rightarrow \text{IndCoh}^*(\mathcal{Z}_{(S_\alpha \times S_\alpha)^\wedge}).$$

is an equivalence.

The lemma follows by duality from Lemma B.13.20.

B.13.21. By construction, the sheaves of categories $\underline{\text{IndCoh}}^*(\mathcal{Z})$ and $\underline{\text{IndCoh}}^!(\mathcal{Z})$ are mutually dual.

By Sect. A.8, the sheaf of categories $\underline{\text{IndCoh}}^*(\mathcal{Z})$ is equipped with a t-structure.

B.13.22. Let $\mathcal{T}$ be a factorization ind-placid ind-scheme over X . The multiplicative structure in Sect. A.6.4 and Lemma A.10.10 imply that in this case the sheaf of categories $\underline{\text{IndCoh}}^*(\mathcal{T})$ carries a factorization structure. Denote the resulting factorization category by $\text{IndCoh}^*(\mathcal{T})$.

By the construction of $\text{IndCoh}^*(\mathcal{T})$ in Sect. B.13.17 and Sect. A.8, the factorization category $\text{IndCoh}^*(\mathcal{T})$ carries a naturally defined t-structure.

By construction, $\text{IndCoh}^*(\mathcal{T})$ is dual to $\text{IndCoh}^!(\mathcal{T})$ as a factorization category.

B.13.23. Let $\mathcal{T}$ be a factorization space over X , and let $\mathcal{T}_m$ be a factorization module space over $\mathcal{T}$ at some $\mathcal{Z} \rightarrow \text{Ran}$.

Suppose that $\mathcal{T}$ is such that the categories $\text{QCoh}(\mathcal{T}_S)$ for $S \in \text{Sch}_{/\text{Ran}}^{\text{aff}}$ are dualizable. Then the sheaf of categories

$$\pi_*(\text{QCoh}((\mathcal{T}_m)_{\mathcal{Z}^\subseteq}))$$

on $\mathcal{Z}^\subseteq$ (here π denotes the structural map $\mathcal{T}_{\mathcal{Z}^\subseteq} \rightarrow \mathcal{Z}^\subseteq$) admits a natural structure of factorization module category over $\text{QCoh}(\mathcal{T})$ at $\mathcal{Z}$. We will denote it by $\text{QCoh}(\mathcal{T}_m)$.

For general $\mathcal{T}$ and $\mathcal{T}_m$, the assignment

$$S \rightarrow \mathcal{Z}^\subseteq, \rightsquigarrow \text{QCoh}_{\text{co}}((\mathcal{T}_m)_S)$$

is a sheaf of categories over $\mathcal{Z}^\subseteq$, to be denoted $\text{QCoh}_{\text{co}}(\mathcal{T}_m)$. It has a natural structure of factorization module category over $\text{QCoh}_{\text{co}}(\mathcal{T})$ at $\mathcal{Z}$. We will denote it by $\text{QCoh}_{\text{co}}(\mathcal{T}_m)$.

Assume now that $\mathcal{T}_m$ is an ind-placid ind-scheme relative to $\mathcal{Z}^\subseteq$. In this case, we can consider the sheaves of categories

$$\underline{\text{IndCoh}}^!(\mathcal{T}_m) \text{ and } \underline{\text{IndCoh}}^*(\mathcal{T}_m)$$

and they have natural factorization module structures over $\mathrm{IndCoh}^!(\mathcal{T})$ and $\mathrm{IndCoh}^*(\mathcal{T})$, respectively. We will denote the resulting factorization module categories by $\mathrm{IndCoh}^!(\mathcal{T}_m)$ and $\mathrm{IndCoh}^*(\mathcal{T}_m)$, respectively.

B.14. Modules over Kac-Moody Lie algebras. In this subsection we will show how to adapt the theory developed in [Ra5] to the factorization setting. We start by defining the factorization category $\mathrm{Rep}(\mathfrak{L}^+(G))$.

In this subsection G can be arbitrary an algebraic group (i.e., not necessarily reductive).

B.14.1. For $S \in \mathrm{Sch}_{\mathrm{Ran}}^{\mathrm{aff}}$ consider the group scheme $\mathfrak{L}^+(G)_S$ over S . It is pro-smooth,

$$\mathfrak{L}^+(G)_S \simeq \lim_{\alpha \in A^{\mathrm{op}}} G_S^\alpha,$$

where G_S^α are smooth group-schemes over S of finite type, and the transition maps are smooth and surjective.

For every α we can consider the algebraic stack pt/G_S^α . Set

$$\mathrm{Rep}(G_S^\alpha) := \mathrm{QCoh}(\mathrm{pt}/G_S^\alpha).$$

Set

$$(B.57) \quad \mathrm{Rep}(\mathfrak{L}^+(G))_S := \mathrm{colim}_{\alpha \in A} \mathrm{Rep}(G_S^\alpha),$$

where the transition functors are given by restriction:

$$G_S^\beta \rightarrow G_S^\alpha \rightsquigarrow \mathrm{Rep}(G_S^\alpha) \rightarrow \mathrm{Rep}(G_S^\beta).$$

Note that the above transition functors admit (continuous) right adjoints, given by

$$\mathrm{inv}_{\ker(G_S^\beta \rightarrow G_S^\alpha)}.$$

Hence, we can rewrite $\mathrm{Rep}(\mathfrak{L}^+(G))_S$ also as the limit

$$\mathrm{Rep}(\mathfrak{L}^+(G))_S := \lim_{\alpha \in A^{\mathrm{op}}} \mathrm{Rep}(G_S^\alpha),$$

with respect to the above right adjoints.

B.14.2. We claim that $\mathrm{Rep}(\mathfrak{L}^+(G))_S$ is compactly generated. In order to prove that, it suffices to show that each $\mathrm{Rep}(G_S^\alpha)$ is compactly generated. This can be proved on general grounds (the category of quasi-coherent sheaves on a smooth algebraic stack is compactly generated). What follows below is an explicit construction of compact generators.

First, it is easy to see that if an object $\mathcal{V} \in \mathrm{Rep}(G_S^\alpha)$ is such that $\mathrm{oblv}_{G_S^\alpha}(\mathcal{V}) \in \mathrm{QCoh}(S)$ is compact, then $\mathcal{V}$ itself is compact. Hence, in order to prove the compact generation, it suffices to exhibit a generating collection of compact objects. We do that as follows.

We can assume that the category of indices A has an initial element α_0 for which $G_S^{\alpha_0}$ is the following explicit group-scheme:

Let I be a finite set such that the map $S \rightarrow \mathrm{Ran}$ factors as

$$S \rightarrow X^I \rightarrow \mathrm{Ran}.$$

Let $\mathrm{Graph}_I \subset X^I \times X$ be the incidence divisor. Let G_{X^I} be the group-scheme over X^I equal to the restriction of scalars à la Weil along

$$\mathrm{Graph}_I \subset X^I \rightarrow X^I$$

of the pullback of the constant group-scheme with fiber G along

$$\mathrm{Graph}_I \subset X^I \rightarrow X.$$

Note that Graph_I receives a map from the disjoint union of I many copies of X^I (i.e., the pairwise diagonals of the i th and the last coordinate in $X^I \times X$). In particular, we obtain a map

$$(B.58) \quad G_{X^I} \rightarrow (G \times X)^I$$

as group-schemes over X^I .

We take $G_S^{\alpha_0}$ to be the pullback of G_{X^I} along $S \rightarrow X^I$.

Note that for every α , the kernel of the projection

$$(B.59) \quad G_S^\alpha \rightarrow G_S^{\alpha_0}$$

is unipotent (in fact, admits a filtration with subquotients isomorphic to the constant group-scheme with fiber $\mathbb{G}_a$).

Hence, the essential image of the forgetful functor

$$\mathrm{Rep}(G_S^{\alpha_0}) \rightarrow \mathrm{Rep}(G_S^\alpha)$$

generates $\mathrm{Rep}(G_S^\alpha)$.

The map (B.58) induces a map

$$G_S^\alpha \rightarrow G^I.$$

In particular, we obtain a functor

$$(B.60) \quad \mathrm{Rep}(G)^{\otimes I} \rightarrow \mathrm{Rep}(G_S^{\alpha_0}).$$

It is easy to see that the essential image of (B.60) generates $\mathrm{Rep}(G_S^{\alpha_0})$. Hence, we obtain that the images of the compact objects in $\mathrm{Rep}(G)^{\otimes I}$ under the composition

$$(B.61) \quad \mathrm{Rep}(G)^{\otimes I} \rightarrow \mathrm{Rep}(G_S^{\alpha_0}) \rightarrow \mathrm{Rep}(G_S^\alpha)$$

provide a set of compact generators of $\mathrm{Rep}(G_S^\alpha)$.

Remark B.14.3. Note that the category $\mathrm{Rep}(\mathfrak{L}^+(G))_S$ is *not* the same as representations of the group-scheme $\mathfrak{L}^+(G)_S$, i.e.,

$$\mathcal{O}_{\mathfrak{L}^+(G)_S}\text{-comod} \simeq \mathrm{QCoh}(\mathrm{pt}/\mathfrak{L}^+(G)_S).$$

Rather, it is its renormalized version, in which we declare the compacts to be the images of the compacts in $\mathrm{Rep}(G_S^\alpha)$ under the restriction along $\mathfrak{L}^+(G)_S \rightarrow G_S^\alpha$.

In particular, the forgetful functor

$$\mathbf{oblv}_{\mathfrak{L}^+(G)_S} : \mathrm{Rep}(\mathfrak{L}^+(G))_S \rightarrow \mathrm{QCoh}(S)$$

is *not* conservative.

B.14.4. The presentation (B.57) equips $\mathrm{Rep}(\mathfrak{L}^+(G))_S$ with a t-structure, for which the forgetful functor $\mathbf{oblv}_{\mathfrak{L}^+(G)_S}$ is t-exact.

B.14.5. Let $f : S' \rightarrow S$ be a map of affine schemes. Pullback along f gives rise to a functor

$$f^* : \mathrm{Rep}(\mathfrak{L}^+(G))_S \rightarrow \mathrm{Rep}(\mathfrak{L}^+(G))_{S'},$$

which in turn gives rise to a functor

$$(B.62) \quad \mathrm{QCoh}(S') \otimes_{\mathrm{QCoh}(S)} \mathrm{Rep}(\mathfrak{L}^+(G))_S \rightarrow \mathrm{Rep}(\mathfrak{L}^+(G))_{S'}.$$

We claim that the functor (B.62) is an equivalence. Indeed, this follows from the fact that for every α , the corresponding functor

$$\mathrm{QCoh}(S') \otimes_{\mathrm{QCoh}(S)} \mathrm{Rep}(G_S^\alpha) \rightarrow \mathrm{Rep}(G_{S'}^\alpha)$$

is an equivalence.

This defines on the assignment

$$S \in \mathrm{Sch}_{\mathrm{Ran}}^{\mathrm{aff}} \rightsquigarrow \mathrm{Rep}(\mathfrak{L}^+(G))_S$$

a structure of sheaf of categories over Ran .

We will denote this sheaf of categories by $\underline{\mathrm{Rep}}(\mathfrak{L}^+(G))$.

B.14.6. Let $\underline{x}_i : S_i \rightarrow \text{Ran}$, $i = 1, 2$ be points such that

$$S_1 \times S_2 \xrightarrow{\underline{x}_1, \underline{x}_2} \text{Ran} \times \text{Ran}$$

lands in $(\text{Ran} \times \text{Ran})_{\text{disj}}$.

Note that we have

$$\mathfrak{L}^+(G)_{S_1 \times S_2} \simeq \mathfrak{L}^+(G)_{S_1} \times \mathfrak{L}^+(G)_{S_2}.$$

Tensor product of representations defines a functor

$$(B.63) \quad \text{Rep}(\mathfrak{L}^+(G)_{S_1}) \otimes \text{Rep}(\mathfrak{L}^+(G)_{S_2}) \rightarrow \text{Rep}(\mathfrak{L}^+(G)_{S_1 \times S_2}).$$

We claim that (B.63) is an equivalence.⁷³ Indeed, we can write $\mathfrak{L}^+(G)_{S_1 \times S_2}$ as

$$\lim_{(\alpha_1, \alpha_2) \in A_1^{\text{op}} \times A_2^{\text{op}}} G_{S_1}^{\alpha_1} \times G_{S_2}^{\alpha_2},$$

and for each pair of indices α_1, α_2 , the functor

$$\text{Rep}(G_{S_1}^{\alpha_1}) \otimes \text{Rep}(G_{S_2}^{\alpha_2}) \rightarrow \text{Rep}(G_{S_1}^{\alpha_1} \times G_{S_2}^{\alpha_2})$$

is an equivalence (e.g., by [Ga7, Cor. 10.3.6]).

This endows the sheaf of categories $\text{Rep}(\mathfrak{L}^+(G))$ with a factorization structure. We denote the resulting factorization category by $\text{Rep}(\mathfrak{L}^+(G))$.

The t-structures in Sect. B.14.4 define a t-structure on $\text{Rep}(\mathfrak{L}^+(G))$ as a factorization category.

B.14.7. Let $S \in \text{Sch}_{/\text{Ran}}^{\text{aff}}$ be as above. Following [Ra5], one defines the (2-)category

$$\mathfrak{L}^+(G)_S\text{-}\mathbf{mod}^{\text{weak}}$$

of $\text{QCoh}(S)$ -linear categories equipped with a *weak* action of $\mathfrak{L}^+(G)_S$ to be equivalent to

$$\text{Rep}(\mathfrak{L}^+(G))_S\text{-}\mathbf{mod}.$$

We consider the forgetful functor

$$(B.64) \quad \mathbf{oblv}_{\mathfrak{L}^+(G)_S, \text{weak}} : \text{Rep}(\mathfrak{L}^+(G))_S\text{-}\mathbf{mod} \rightarrow \text{QCoh}(S)\text{-}\mathbf{mod}$$

given by

$$\mathbf{C} \mapsto \mathbf{C} \otimes_{\text{Rep}(\mathfrak{L}^+(G))_S, \mathbf{oblv}_{\mathfrak{L}^+(G)_S}} \text{QCoh}(S).$$

Remark B.14.8. As in Remark B.14.3, the 2-category $\text{Rep}(\mathfrak{L}^+(G))_S\text{-}\mathbf{mod}$ is *not* the same as $\text{QCoh}(S)$ -linear categories equipped with a co-action of $\text{QCoh}(\mathfrak{L}^+(G)_S)$. Indeed, for the unit object

$$\text{Rep}(\mathfrak{L}^+(G))_S \in \text{Rep}(\mathfrak{L}^+(G))_S\text{-}\mathbf{mod},$$

its category of endofunctors in $\text{Rep}(\mathfrak{L}^+(G))_S\text{-}\mathbf{mod}$ is $\text{Rep}(\mathfrak{L}^+(G))_S$, while for the unit object

$$\text{QCoh}(S) \in \text{QCoh}(\mathfrak{L}^+(G)_S)\text{-}\mathbf{comod},$$

its category of endofunctors is the non-renormalized category

$$\mathcal{O}_{\mathfrak{L}^+(G)_S}\text{-comod} \simeq \text{QCoh}(\text{pt}/\mathfrak{L}^+(G)_S).$$

That said, for each individual α , the stack pt/G_S^α is 1-affine, and hence the functor

$$(B.65) \quad \text{Rep}(G_S^\alpha)\text{-}\mathbf{mod} \rightarrow \text{QCoh}(G_S^\alpha)\text{-}\mathbf{comod}, \quad \mathbf{C} \mapsto \mathbf{C} \otimes_{\text{Rep}(G_S^\alpha), \mathbf{oblv}_{G_S^\alpha}} \text{QCoh}(S)$$

is an equivalence

⁷³This would not be the case (at least, not obviously so) if instead of $\text{Rep}(\mathfrak{L}^+(G))_{S_i}$ we used their naive versions, i.e., the categories of representations of the group-schemes $\mathfrak{L}^+(G)_{S_i}$.

B.14.9. Parallel to [Ra5], one defines the $(2-)$ category

$$\mathfrak{L}^+(G)_{S\text{-mod}}$$

of $\mathrm{QCoh}(S)$ -linear categories equipped with a *strong* action of $\mathfrak{L}^+(G)_S$. Namely, this is the category of comodules (inside $\mathrm{QCoh}(S)\text{-mod}$) for

$$\mathrm{D}\text{-mod}_{\mathrm{rel}/S}(\mathfrak{L}^+(G)_S),$$

where:

- $\mathrm{D}\text{-mod}_{\mathrm{rel}/S}(\mathfrak{L}^+(G)_S) := \mathrm{colim}_{\alpha \in A} \mathrm{D}\text{-mod}_{\mathrm{rel}/S}(G_S^\alpha)$;
- $\mathrm{D}\text{-mod}_{\mathrm{rel}/S}(G_S^\alpha) := \mathrm{QCoh}(G_{S_{\mathrm{dR}}}^\alpha \times_{S_{\mathrm{dR}}} S)$.

One shows that any object

$$\mathbf{C} \in \mathfrak{L}^+(G)_{S\text{-mod}}$$

can be *canonically written* as

$$\mathrm{colim}_{\alpha \in A} \mathbf{C}^{\ker(\mathfrak{L}^+(G)_S \rightarrow G_S^\alpha)},$$

where

$$\mathbf{C}^{\ker(\mathfrak{L}^+(G)_S \rightarrow G_S^\alpha)} \subset \mathbf{C}$$

is the full subcategory of strong invariants with respect to the (pro-unipotent) group-scheme

$$\ker(\mathfrak{L}^+(G)_S \rightarrow G_S^\alpha).$$

B.14.10. We have a forgetful functor

$$\mathbf{oblv}_{\mathrm{weak}}^{\mathrm{strong}} : \mathfrak{L}^+(G)_{S\text{-mod}} \rightarrow \mathfrak{L}^+(G)_{S\text{-mod}}^{\mathrm{weak}}$$

that sends an object $\mathbf{C}$ to

$$\mathrm{colim}_{\alpha \in A} \mathbf{oblv}_{\mathrm{weak}}^{\mathrm{strong}}(\mathbf{C}^{\ker(\mathfrak{L}^+(G)_S \rightarrow G_S^\alpha)}),$$

where in the right-hand side $\mathbf{oblv}_{\mathrm{weak}}^{\mathrm{strong}}$ denotes the family of forgetful functors

$$\mathrm{D}\text{-mod}_{\mathrm{rel}/S}(G_S^\alpha)\text{-comod} \xrightarrow{\mathbf{oblv}^l} \mathrm{QCoh}(G_S^\alpha)\text{-comod} \stackrel{(\mathrm{B.65})}{\simeq} \mathrm{Rep}(G_S^\alpha)\text{-mod}.$$

This functor intertwines the natural forgetful functor

$$\mathbf{oblv}_{\mathfrak{L}^+(G)_S, \mathrm{weak}} : \mathfrak{L}^+(G)_{S\text{-mod}} \rightarrow \mathrm{QCoh}(S)\text{-mod}$$

with the functor $\mathbf{oblv}_{\mathfrak{L}^+(G)_S, \mathrm{weak}}$ of (B.64).

B.14.11. Our next goal is to define the category $\mathfrak{L}^+(\mathfrak{g})\text{-mod}$ of modules for the arc Lie algebra $\mathfrak{L}^+(\mathfrak{g})$. The definition that we are about to give mimics the following finite-dimensional situation:

Let H be a finite-dimensional algebraic group. Then the category $\mathfrak{h}\text{-mod}$ of modules over its Lie algebra has the following structures:

- It carries a (strong) action of H ;
- It is equipped with a forgetful functor $\mathbf{oblv}_{\mathfrak{h}} : \mathfrak{h}\text{-mod} \rightarrow \mathrm{Vect}$;
- The functor $\mathbf{oblv}_{\mathfrak{h}}$ is equipped with a structure of compatibility with the induced *weak* action of H .

Moreover, the category $\mathfrak{h}\text{-mod}$ is universal with respect to the above pieces of structure. I.e., for a category $\mathbf{C}$, equipped with a (strong) action of H , compositing with $\mathbf{oblv}_{\mathfrak{h}}$ defines an equivalence between:

- Functors $\mathbf{C} \rightarrow \mathfrak{h}\text{-mod}$, compatible with (strong) actions of H ;
- Functors $\mathbf{C} \rightarrow \mathrm{Vect}$, compatible with weak actions of H .

The above universal property can be established by realizing $\mathfrak{h}\text{-mod}$ as follows

$$(B.66) \quad \mathfrak{h}\text{-mod} \simeq \mathrm{D}\text{-mod}(H)^{H\text{-weak}}.$$

B.14.12. We define the category

$$\mathfrak{L}^+(\mathfrak{g})\text{-mod}_S$$

by the universal property as in Sect. B.14.11. I.e., this is a category, equipped with:

- A (strong) action of $\mathfrak{L}^+(G)_S$;
- A forgetful functor $\mathbf{oblv}_{\mathfrak{L}^+(\mathfrak{g})} : \mathfrak{L}^+(\mathfrak{g})\text{-mod}_S \rightarrow \text{QCoh}(S)$;
- A datum of compatibility on $\mathbf{oblv}_{\mathfrak{L}^+(\mathfrak{g})}$ with the weak action of $\mathfrak{L}^+(G)_S$.

Moreover, $\mathfrak{L}^+(\mathfrak{g})\text{-mod}_S$ is universal with respect to the above pieces of structure.

As in (B.66), we can explicitly realize $\mathfrak{L}^+(\mathfrak{g})\text{-mod}_S$ as follows:

$$(B.67) \quad \mathfrak{L}^+(\mathfrak{g})\text{-mod}_S \simeq \text{D-mod}_{\text{rel}/S}(\mathfrak{L}^+(G)_S)^{\mathfrak{L}^+(G)_S\text{-weak}}.$$

Remark B.14.13. A feature of $\mathfrak{L}^+(\mathfrak{g})\text{-mod}_S$ that one has to keep in mind, and which distinguishes it from the finite-dimensional situation, is that the functor

$$\mathbf{oblv}_{\mathfrak{L}^+(\mathfrak{g})} : \mathfrak{L}^+(\mathfrak{g})\text{-mod}_S \rightarrow \text{QCoh}(S)$$

is *not* conservative.

B.14.14. Note that, by definition, we have

$$(B.68) \quad \mathfrak{L}^+(\mathfrak{g})\text{-mod}_S^{\mathfrak{L}^+(G)_S} \simeq \text{Funct}_{\mathfrak{L}^+(G)_S\text{-mod}}(\text{QCoh}(S), \mathfrak{L}^+(\mathfrak{g})\text{-mod}_S) \simeq \\ \simeq \text{Funct}_{\mathfrak{L}^+(G)_S\text{-mod}^{\text{weak}}}(\text{QCoh}(S), \text{QCoh}(S)) \simeq \text{Rep}(\mathfrak{L}^+(G))_S.$$

B.14.15. By the universal property of $\mathfrak{L}^+(\mathfrak{g})\text{-mod}_S$, we have naturally defined restriction functors

$$\text{Lie}(G_S^\alpha)\text{-mod} \rightarrow \mathfrak{L}^+(\mathfrak{g})\text{-mod}_S.$$

Furthermore, the functor

$$(B.69) \quad \text{colim}_{\alpha \in A} \text{Lie}(G_S^\alpha)\text{-mod} \rightarrow \mathfrak{L}^+(\mathfrak{g})\text{-mod}_S$$

is an equivalence.

Since the transition functors in the left-hand side of (B.67) preserve compactness, we obtain that $\mathfrak{L}^+(\mathfrak{g})\text{-mod}_S$ is compactly generated.

B.14.16. The presentation of $\mathfrak{L}^+(\mathfrak{g})\text{-mod}_S$ as in (B.69) equips it with a t-structure, for which the forgetful functor $\mathbf{oblv}_{\mathfrak{L}^+(\mathfrak{g})}$ is t-exact.

B.14.17. For $S' \rightarrow S$, the universal property of $\mathfrak{L}^+(\mathfrak{g})\text{-mod}_S$ gives rise to a functor

$$\text{QCoh}(S') \otimes_{\text{QCoh}(S)} \mathfrak{L}^+(\mathfrak{g})\text{-mod}_S \rightarrow \mathfrak{L}^+(\mathfrak{g})\text{-mod}_{S'}.$$

One shows (e.g., using (B.67)) that the above functor is equivalence.

This endows the assignment

$$S \rightsquigarrow \mathfrak{L}^+(\mathfrak{g})\text{-mod}_S$$

with a structure of sheaf of categories over Ran . We will denote it by $\mathfrak{L}^+(\mathfrak{g})\text{-mod}$.

B.14.18. Let $\underline{x}_i : S_i \rightarrow \text{Ran}$ be as in Sect. B.14.6. By the universal property of $\mathfrak{L}^+(\mathfrak{g})\text{-mod}_{S_1 \times S_2}$, we obtain a functor

$$\mathfrak{L}^+(\mathfrak{g})\text{-mod}_{S_1} \otimes \mathfrak{L}^+(\mathfrak{g})\text{-mod}_{S_2} \rightarrow \mathfrak{L}^+(\mathfrak{g})\text{-mod}_{S_1 \times S_2}.$$

One shows (e.g., using (B.67)) that this functor is an equivalence. This endows the sheaf of categories $\mathfrak{L}^+(\mathfrak{g})\text{-mod}$ with a factorization structure.

We denote the resulting factorization category by $\mathfrak{L}^+(\mathfrak{g})\text{-mod}$. The t-structures from Sect. B.14.16 give rise to a t-structure on $\mathfrak{L}^+(\mathfrak{g})\text{-mod}$ as a factorization category.

B.14.19. We finally consider representations of Kac-Moody algebras. First, fix $S \in \text{Sch}_{/\text{Ran}}^{\text{aff}}$. Having the (2-)categories

$$\mathfrak{L}^+(G)_S\text{-}\mathbf{mod}^{\text{weak}} \text{ and } \mathfrak{L}^+(G)_S\text{-}\mathbf{mod},$$

proceeding as in [Ra5, Sect. 7-8], we define the (2-)categories

$$\mathfrak{L}(G)_S\text{-}\mathbf{mod}^{\text{weak}} \text{ and } \mathfrak{L}(G)_{\kappa,S}\text{-}\mathbf{mod}$$

of $\text{QCoh}(S)$ -linear categories, equipped with weak (resp., strong at level κ) actions of $\mathfrak{L}(G)_S$.

We define the category $\widehat{\mathfrak{g}}\text{-mod}_{\kappa,S}$ as the category, equipped with and universal with respect to the following pieces of structure:

- A *strong* action of $\mathfrak{L}(G)_S$ at level κ ;
- A functor to $\text{QCoh}(S)$;
- A datum of compatibility on the above functor with respect to the *weak* action of $\mathfrak{L}(G)_S$.

We can explicitly realize the category $\widehat{\mathfrak{g}}\text{-mod}_{\kappa,S}$ as

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa,S} \simeq \text{D-mod}_{\kappa,\text{rel}/S}(\mathfrak{L}(G)_S)^{\mathfrak{L}(G)_S\text{-weak}}.$$

B.14.20. By the universal property of $\widehat{\mathfrak{g}}\text{-mod}_{\kappa,S}$, it comes equipped with a (conservative) forgetful functor

$$\mathbf{oblv}_{\mathfrak{L}^+(\mathfrak{g})}^{\widehat{\mathfrak{g}}\kappa} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa,S} \rightarrow \mathfrak{L}^+(\mathfrak{g})\text{-mod}_S.$$

As in [Ra5, §9.12, §11.10], one shows that this functor admits a left adjoint, to be denoted

$$\mathbf{ind}_{\mathfrak{L}^+(\mathfrak{g})}^{\widehat{\mathfrak{g}}\kappa} : \mathfrak{L}^+(\mathfrak{g})\text{-mod}_S \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa,S}.$$

In particular, the fact that $\mathfrak{L}^+(\mathfrak{g})\text{-mod}_S$ is compactly generated implies that so is $\widehat{\mathfrak{g}}\text{-mod}_{\kappa,S}$.

B.14.21. We note that the endofunctor of $\mathfrak{L}^+(\mathfrak{g})\text{-mod}_S$ underlying the monad

$$\mathbf{oblv}_{\mathfrak{L}^+(\mathfrak{g})}^{\widehat{\mathfrak{g}}\kappa} \circ \mathbf{ind}_{\mathfrak{L}^+(\mathfrak{g})}^{\widehat{\mathfrak{g}}\kappa}$$

is t-exact.

This allows us to equip $\widehat{\mathfrak{g}}\text{-mod}_{\kappa,S}$ with a t-structure for which both functors $\mathbf{oblv}_{\mathfrak{L}^+(\mathfrak{g})}^{\widehat{\mathfrak{g}}\kappa}$ and $\mathbf{ind}_{\mathfrak{L}^+(\mathfrak{g})}^{\widehat{\mathfrak{g}}\kappa}$ are t-exact.

B.14.22. As in Sects. Sect. B.14.17 one endows the assignment

$$S \rightsquigarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa,S}$$

with a structure of sheaf of categories over Ran . We denote it by $\widehat{\mathfrak{g}}\text{-mod}_{\kappa}$.

As in Sect. B.14.18, one endows $\widehat{\mathfrak{g}}\text{-mod}_{\kappa}$ with a factorization structure. We denote the resulting factorization category by $\widehat{\mathfrak{g}}\text{-mod}_{\kappa}$.

The t-structures in Sect. B.14.21 combine to a t-structure on $\widehat{\mathfrak{g}}\text{-mod}_{\kappa}$ as a factorization category.

B.14.23. For $S \in \text{Sch}_{/\text{Ran}}^{\text{aff}}$ set

$$\text{KL}(G)_{\kappa,S} := (\widehat{\mathfrak{g}}\text{-mod}_{\kappa,S})^{\mathfrak{L}^+(G)_S}.$$

The adjoint functors

$$\mathbf{ind}_{\mathfrak{L}^+(\mathfrak{g})}^{\widehat{\mathfrak{g}}\kappa} : \mathfrak{L}^+(\mathfrak{g})\text{-mod}_S \rightleftarrows \widehat{\mathfrak{g}}\text{-mod}_{\kappa,S} : \mathbf{oblv}_{\mathfrak{L}^+(\mathfrak{g})}^{\widehat{\mathfrak{g}}\kappa}$$

induce a pair of adjoint functors

$$(B.70) \quad \mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa} : \text{Rep}(\mathfrak{L}^+(G))_S \rightleftarrows \text{KL}(G)_{\kappa,S} : \mathbf{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))\kappa}.$$

In particular, we obtain that $\text{KL}(G)_{\kappa,S}$ is compactly generated.

B.14.24. Let us describe in concrete terms a set of compact generators of $\mathrm{KL}(G)_{\kappa,S}$. Let I be a finite set as in Sect. B.14.2.

For a compact object $V \in \mathrm{Rep}(G)^{\otimes I}$, let us denote by a light abuse of notation by the same character V its image under (B.61).

The objects

$$\mathbf{ind}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_{\kappa}}(V) \in \mathrm{KL}(G)_{\kappa,S}$$

are called Weyl modules, and they compactly generate $\mathrm{KL}(G)_{\kappa,S}$.

Remark B.14.25. Let $\underline{\mathbf{C}}$ be a sheaf of categories on S_{dR} , where S is an affine scheme of finite type. There is a stronger notion than compact generation, called *ULA generation*:

An object $\mathbf{c} \in \Gamma(S_{\mathrm{dR}}, \underline{\mathbf{C}})$ is said to be ULA if its image in $\Gamma(S, \underline{\mathbf{C}})$ is compact.

We say that $\underline{\mathbf{C}}$ is ULA-generated if it contains a collection of ULA objects, whose images in $\Gamma(S, \underline{\mathbf{C}})$ generate this category.

We will say that a factorization category $\mathbf{C}$ is ULA-generated if for every $S \rightarrow \mathrm{Ran}$, the corresponding category $\mathbf{C}_{S_{\mathrm{dR}}}$ is. This property is enough to check for $S = X^I$.

Many factorization categories that appear in geometric representation theory have this property, e.g., $\mathbf{C} = \mathrm{Sph}_G$. However, the category $\mathrm{KL}(G)_{\kappa}$ does not: namely, $\mathrm{KL}(G)_{\kappa, X^I}$ is not ULA-generated for $|I| \geq 2$.

B.14.26. As in Sect. B.14.21, we obtain that $\mathrm{KL}(G)_{\kappa,S}$ carries a t-structure, for which both functors (B.70) are t-exact.

B.14.27. For $S' \rightarrow S$, the equivalence

$$\mathrm{QCoh}(S') \otimes_{\mathrm{QCoh}(S)} \widehat{\mathfrak{g}}\text{-mod}_{\kappa,S} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa,S'}$$

induces a functor

$$\begin{aligned} \text{(B.71)} \quad \mathrm{QCoh}(S') \otimes_{\mathrm{QCoh}(S)} \mathrm{KL}(G)_{\kappa,S} &= \mathrm{QCoh}(S') \otimes_{\mathrm{QCoh}(S)} \widehat{\mathfrak{g}}\text{-mod}_{\kappa,S}^{\mathfrak{L}^+(G)_S} \rightarrow \\ &\rightarrow (\mathrm{QCoh}(S') \otimes_{\mathrm{QCoh}(S)} \mathrm{KL}(G)_{\kappa,S})^{\mathfrak{L}^+(G)_S} \simeq (\mathrm{QCoh}(S') \otimes_{\mathrm{QCoh}(S)} \mathrm{KL}(G)_{\kappa,S})^{\mathfrak{L}^+(G)_{S'}} \simeq \mathrm{KL}(G)_{\kappa,S'}. \end{aligned}$$

However, the functor

$$\mathbf{C} \rightarrow \mathbf{C}^{\mathfrak{L}^+(G)_S}, \quad \mathfrak{L}^+(G)_S\text{-mod} \rightarrow \mathrm{QCoh}(S)\text{-mod}$$

is known to commute with colimits. Hence, the functor (B.71) is an equivalence.

This endows the assignment

$$S \in \mathrm{Sch}_{\mathrm{Ran}}^{\mathrm{aff}}, \rightsquigarrow \mathrm{KL}(G)_{\kappa,S}$$

with a structure of sheaf of categories. We denote it by $\underline{\mathrm{KL}}(G)_{\kappa}$.

B.14.28. Similarly, in the situation Sect. B.14.6, the equivalence

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa,S_1} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\kappa,S_2} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa,S_1 \times S_2}$$

induces an equivalence

$$\begin{aligned} \mathrm{KL}(G)_{\kappa,S_1} \otimes \mathrm{KL}(G)_{\kappa,S_2} &= (\widehat{\mathfrak{g}}\text{-mod}_{\kappa,S_1})^{\mathfrak{L}^+(G)_{S_1}} \otimes (\widehat{\mathfrak{g}}\text{-mod}_{\kappa,S_1})^{\mathfrak{L}^+(G)_{S_2}} \simeq \\ &\simeq (\widehat{\mathfrak{g}}\text{-mod}_{\kappa,S_1} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\kappa,S_2})^{\mathfrak{L}^+(G)_{S_1} \times \mathfrak{L}^+(G)_{S_2}} = \mathrm{KL}(G)_{\kappa,S_1 \times S_2}. \end{aligned}$$

This endows $\underline{\mathrm{KL}}(G)_{\kappa}$ with a factorization structure. The resulting factorization category, denoted $\mathrm{KL}(G)_{\kappa}$ is the sought-for factorization incarnation of the Kazhdan-Lusztig category.

The t-structures in Sect. B.14.26 define a t-structure on $\mathrm{KL}(G)_{\kappa}$ as a factorization category.

B.15. Restriction of factorization module categories, continued. In this subsection we will discuss some additional aspects of the operation of restriction of factorization module categories, introduced in Sect. B.12.11.

We will omit most proofs (they are elaborations of the limit construction described in Sect. B.9.28); a more detailed discussion will appear in [CFGY].

B.15.1. Fix $\mathcal{Z} \rightarrow \text{Ran}$. Consider the totality of pairs $(\mathbf{A}, \mathbf{C})$, where $\mathbf{A}$ is a factorization category, and $\mathbf{C} \in \mathbf{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}}$. We view it as a 2-category, to be denoted $\text{FactCat-and-Mod}(X)$ with the above objects, and the categories of morphisms defined as follows:

For a pair of objects $(\mathbf{A}_1, \mathbf{C}_1)$ and $(\mathbf{A}_2, \mathbf{C}_2)$ of $\text{FactCat-and-Mod}(X)$, the category

$$\text{Maps}_{\text{FactCat-and-Mod}(X)}((\mathbf{A}_1, \mathbf{C}_1), (\mathbf{A}_2, \mathbf{C}_2))$$

consists of pairs of morphisms $(\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2, \Phi_m : \mathbf{C}_1 \rightarrow \mathbf{C}_2)$ as in Sect. B.12.11.

Consider the natural projection

$$(B.72) \quad \text{FactCat-and-Mod}(X) \rightarrow \text{FactCat}(X).$$

The following assertion encodes the functoriality of the assignment

$$\mathbf{A} \rightsquigarrow \mathbf{C} \in \mathbf{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}}.$$

Theorem B.15.2. *The functor*

$$\text{FactCat-and-Mod}(X)^{2\text{-op}} \rightarrow \text{FactCat}(X)^{2\text{-op}},$$

induced by (B.72) is a 2-Cartesian fibration, where:

- The symbol $(-)^{2\text{-op}}$ refers to reversing 2-morphisms in a given 2-category;
- The notion of 2-Cartesian fibration is as in [GaRo3, Chapter 11, Sect. 1.1.].

Remark B.15.3. Note that the situation in Theorem B.15.2 is parallel to the following more familiar paradigm, when instead of $\text{FactCat-and-Mod}(X)$ we consider the category of pairs $(\mathbf{A}, \mathbf{C})$, where $\mathbf{A}$ is a monoidal category, and $\mathbf{C} \in \mathbf{A}\text{-mod}$, and where the category of functors

$$(\mathbf{A}_1, \mathbf{C}_1) \rightarrow (\mathbf{A}_2, \mathbf{C}_2)$$

consists of right-lax monoidal functors between the monoidal categories and compatible right-lax functors between the module categories.

B.15.4. The concrete meaning of this theorem is the following. It says that:

- For

$$\mathbf{A}_1 \xrightarrow{\Phi_{1,2}} \mathbf{A}_2 \xrightarrow{\Phi_{2,3}} \mathbf{A}_3, \quad \Phi_{1,3} = \Phi_{2,3} \circ \Phi_{1,2},$$

and $\mathbf{C}_3 \in \mathbf{A}_3\text{-mod}_{\mathcal{Z}}^{\text{fact}}$, the tautological functor

$$\text{Res}_{\Phi_{1,2}} \circ \text{Res}_{\Phi_{2,3}}(\mathbf{C}_3) \rightarrow \text{Res}_{\Phi_{1,3}}$$

is an equivalence;

- For $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$, $\mathbf{C}_i \in \mathbf{A}_i\text{-mod}_{\mathcal{Z}}^{\text{fact}}$, $i = 1, 2$, $\Phi_m \in \text{Funct}_{\Phi: \mathbf{A}_1 \rightarrow \mathbf{A}_2}(\mathbf{C}_1, \mathbf{C}_2)$ and a natural transformation $\Phi \rightarrow \Phi'$, there exists an object $\Phi'_m \in \text{Funct}_{\Phi': \mathbf{A}_1 \rightarrow \mathbf{A}_2}(\mathbf{C}_1, \mathbf{C}_2)$ together with a compatible natural transformation $\Phi'_m \rightarrow \Phi_m$, which is universal with respect to this property.

B.15.5. According to [GaRo3, Chapter 11, Theorem 1.1.8], we can interpret Theorem B.15.2 as saying that the assignment

$$\mathbf{A} \mapsto \mathbf{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}}$$

extends to a functor between 2-categories

$$\text{FactCat}(X)^{1\text{-op}, 2\text{-op}} \rightarrow 2\text{-Cat},$$

where:

- The symbol $(-)^{1\text{-op}, 2\text{-op}}$ refers to reversing both 1-morphisms and 2-morphisms in a given 2-category;
- 2-Cat is the totality of $(\infty, 2)$ -categories, viewed as a 2-category.

B.15.6. We will need the following corollary of Theorem B.15.2. Let $\mathbf{A}_1$ and $\mathbf{A}_2$ be a pair of factorization categories, and let

$$\Phi : \mathbf{A}_1 \rightrightarrows \mathbf{A}_2 : \Phi^R$$

be a pair of factorization functors.

We claim:

Corollary B.15.7. *For $\mathbf{C}_i \in \mathbf{A}_i\text{-mod}_{\mathbb{Z}}^{\text{fact}}$, $i = 1, 2$ there is a canonical equivalence*

$$\text{Funct}_{\mathbf{A}_1\text{-mod}_{\mathbb{Z}}^{\text{fact}}}(\text{Res}_{\Phi}(\mathbf{C}_2), \mathbf{C}_1) \simeq \text{Funct}_{\mathbf{A}_2\text{-mod}_{\mathbb{Z}}^{\text{fact}}}(\mathbf{C}_2, \text{Res}_{\Phi^R}(\mathbf{C}_1)).$$

This proposition can be reformulated as saying that the functor Res_{Φ} is the *left* adjoint of the functor Res_{Φ^R} .

Proof. This is a formal corollary of having a 2-Cartesian fibration. To be explicit, let us exhibit the unit and the counit of the adjunction.

The unit is given by

$$\text{Id} = \text{Res}_{\text{Id}} \xrightarrow{\Phi \circ \Phi^R \rightarrow \text{Id}} \text{Res}_{\Phi \circ \Phi^R} \simeq \text{Res}_{\Phi^R} \circ \text{Res}_{\Phi}.$$

The counit is given by

$$\text{Res}_{\Phi} \circ \text{Res}_{\Phi^R} \simeq \text{Res}_{\Phi^R \circ \Phi} \xrightarrow{\text{Id} \rightarrow \Phi^R \circ \Phi} \text{Res}_{\text{Id}} = \text{Id}.$$

□

B.15.8. We will now use Corollary B.15.7 to prove the following partial converse to Lemma B.12.12:

Lemma B.15.9. *Let $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$ be a factorization functor, and let $\mathbf{C}_1 \rightarrow \mathbf{C}_2$ be a functor between objects of $\mathbf{A}_i\text{-mod}_{\mathbb{Z}}^{\text{fact}}$, $i = 1, 2$, compatible with factorization. Assume that:*

- (i) *The functor $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$ between sheaves of categories on Ran admits a right adjoint;*
- (ii) *The functor $\Phi_m : \mathbf{C}_1 \rightarrow \mathbf{C}_2$ between sheaves of categories on $\mathbb{Z}^{\subseteq}$ admits a right adjoint;*
- (iii) *The induced functor $\Phi_m : \mathbf{C}_{1,\mathbb{Z}} \rightarrow \mathbf{C}_{2,\mathbb{Z}}$ is an equivalence.*

Then the resulting functor

$$(B.73) \quad \mathbf{C}_1 \rightarrow \text{Res}_{\Phi}(\mathbf{C}_2)$$

as module categories over $\mathbf{A}_1$, is an equivalence.

Proof. We claim that the functor (B.73) admits a right adjoint. Once we prove this, the lemma will follow, because a functor between sheaves of categories that admits a right adjoint is an equivalence if and only if it is an equivalence strata-wise.

The right adjoint of $\Phi_m : \mathbf{C}_1 \rightarrow \mathbf{C}_2$ is a functor compatible with factorization against $\Phi^R : \mathbf{A}_1 \rightarrow \mathbf{A}_2$. Hence, it gives rise to an object

$$\Phi_m^R \in \text{Funct}_{\Phi^R : \mathbf{A}_2 \rightarrow \mathbf{A}_1}(\mathbf{C}_2, \mathbf{C}_1) \simeq \text{Funct}_{\mathbf{A}_2\text{-mod}_{\mathbb{Z}}^{\text{fact}}}(\mathbf{C}_2, \text{Res}_{\Phi^R}(\mathbf{C}_1)).$$

(In fact, the pair (Φ^R, Φ_m^R) is the right adjoint of (Φ, Φ_m) as a 1-morphism in $\text{FactCat-and-Mod}(X)$.)

Using Corollary B.15.7, we identify the latter category with

$$\text{Funct}_{\mathbf{A}_1\text{-mod}_{\mathbb{Z}}^{\text{fact}}}(\text{Res}_{\Phi}(\mathbf{C}_2), \mathbf{C}_1).$$

Unwinding the constructions, we obtain that the resulting functor

$$\text{Res}_{\Phi}(\mathbf{C}_2) \rightarrow \mathbf{C}_1$$

is indeed the right adjoint of (B.73).

□

APPENDIX C. UNITAL STRUCTURES

In the previous section we introduced the notion of factorization category. In this section we will describe an extra structure that factorization categories often carry: the *unital* structure.

The role that the unital structure plays is two-fold. For one thing, it enables various local-to-global constructions (see Sect. 11). But it also leads to purely local constructions, which play a key role in this paper: given a lax-unital factorization functor $F : \mathbf{A}_1 \rightarrow \mathbf{A}_2$ between unital factorization categories, the image $F(\mathbf{1}_{\mathbf{A}_1})$ of the unite $\mathbf{1}_{\mathbf{A}_1} \in \mathbf{A}_2$ is a factorization algebra in $\mathbf{A}_2$, and the functor F *enhances* to a functor

$$F^{\text{enh}} : \mathbf{A}_1 \rightarrow F(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\mathbf{A}_2).$$

Now the functor is often, if not an equivalence, but is close to be such⁷⁴, and that allows to understand the more complicated category $\mathbf{A}_1$ in terms of $\mathbf{A}_1$.

In order to talk about unitality we have to enlarge our world of algebro-geometric objects. Namely, we normally work with prestacks (i.e., spaces in algebro-geometric sense, whose functor of points takes place in (∞) -groupoids). But in order to talk about unitality, we need to work with *categorical prestacks*; whose functors of points take place in (∞) -categories. The rudiments of categorical prestacks (D-modules and sheaves of categories on them) are also developed in this section.

C.1. Categorical prestacks.

C.1.1. When discussing categorical prestacks, we will work in the locally almost of finite type (lft) category. Accordingly, when we write Sch^{aff} (resp., PreStk), we will mean affine schemes (resp., prestacks) locally almost of finite type.

By a categorical prestack we shall mean a functor

$$(\text{Sch}^{\text{aff}})^{\text{op}} \rightarrow \infty\text{-Cat}.$$

Given a categorical prestack $\mathcal{Y}$ we will denote by

$$\mathcal{Y}(S) \text{ or } \text{Maps}(S, \mathcal{Y})$$

the category of its values on $S \in \text{Sch}^{\text{aff}}$.

We let CatPreStk denote the category of categorical prestacks.

C.1.2. Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a map between categorical prestacks. We shall say that some categorical property of f (such as being Cartesian/co-Cartesian, cofinal, admitting an adjoint) holds *value-wise* if this property holds for the corresponding functor

$$\mathcal{Y}_1(S) \rightarrow \mathcal{Y}_2(S)$$

for every $S \in \text{Sch}^{\text{aff}}$.

C.1.3. Let $\mathcal{Y}$ be a categorical prestack. To it we will associate two prestacks in groupoids, by applying the right and left adjoints to the embedding

$$\infty\text{-Grpd} \hookrightarrow \infty\text{-Cat},$$

respectively.

We let $\mathcal{Y}^{\text{grpd}}$ denote the prestack whose value on $S \in \text{Sch}^{\text{aff}}$ is the groupoid underlying the category $\mathcal{Y}(S)$.

We will denote by $\mathbf{t}$ the tautological map

$$\mathcal{Y}^{\text{grpd}} \rightarrow \mathcal{Y}.$$

We will denote by $\mathcal{Y}^{\text{strict}}$ the prestack in groupoids, whose value on $S \in \text{Sch}^{\text{aff}}$ is the groupoid obtained from $\mathcal{Y}(S)$ by inverting all 1-morphisms.

⁷⁴E.g., in multiple instances, in the presence of a $\mathbf{t}$ -structure, it is an equivalence on the bounded below categories.

We will denote the tautological projection by

$$\text{strict} : \mathcal{Y} \rightarrow \mathcal{Y}^{\text{strict}}.$$

C.1.4. Given a categorical prestack $\mathcal{Y}$, we associate to it several DG categories of algebro-geometric nature.

Let $\mathcal{Y}_{\text{Sch}^{\text{aff}}}$ denote the Cartesian fibration over Sch^{aff} that attaches to $S \in \text{Sch}^{\text{aff}}$ the category $\mathcal{Y}(S)$.

Let

$$\text{QCoh}_{\text{Sch}^{\text{aff}}}, \text{IndCoh}_{\text{Sch}^{\text{aff}}} \text{ and } \text{D-mod}_{\text{Sch}^{\text{aff}}}$$

denote the Cartesian fibrations over Sch^{aff} that attach to S the categories

$$(C.1) \quad \text{QCoh}(S), \text{IndCoh}(S) \text{ and } \text{D-mod}(S),$$

respectively.

We define the categories

$$(C.2) \quad \text{QCoh}(\mathcal{Y}), \text{IndCoh}(\mathcal{Y}) \text{ and } \text{D-mod}(\mathcal{Y})$$

as functors from $\mathcal{Y}_{\text{Sch}^{\text{aff}}}$ to the categories in (C.2) that map arrows that are Cartesian over Sch^{aff} to arrows with a similar property. See [Ro2, Sect. C.3] for more details.

C.1.5. For a map of prestacks $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$, precomposition with

$$(\mathcal{Y}_1)_{\text{Sch}^{\text{aff}}} \rightarrow (\mathcal{Y}_2)_{\text{Sch}^{\text{aff}}}$$

gives rise to a functor

$$f^! : \text{D-mod}(\mathcal{Y}_2) \rightarrow \text{D-mod}(\mathcal{Y}_1),$$

and similarly for $\text{IndCoh}(\mathcal{Y})$ and $\text{D-mod}(\mathcal{Y})$.

C.1.6. Categorical prestacks form an $(\infty, 2)$ -category, so there is a natural notion of adjunction between morphisms. Explicitly, a morphism

$$f : \mathcal{Y}_1 \rightrightarrows \mathcal{Y}_2$$

admits a right adjoint, if for every $S \in \text{Sch}^{\text{aff}}$, the corresponding functor

$$f : \mathcal{Y}_1(S) \rightarrow \mathcal{Y}_2(S)$$

admits a right adjoint, and for every $S' \rightarrow S$, the natural transformation

$$\begin{array}{ccc} \mathcal{Y}_1(S') & \xleftarrow{f^R} & \mathcal{Y}_2(S') \\ \uparrow & \nearrow & \uparrow \\ \mathcal{Y}_1(S) & \xleftarrow{f^R} & \mathcal{Y}_2(S) \end{array}$$

arising by adjunction from the commutative diagram

$$\begin{array}{ccc} \mathcal{Y}_1(S') & \xrightarrow{f} & \mathcal{Y}_2(S') \\ \uparrow & & \uparrow \\ \mathcal{Y}_1(S) & \xrightarrow{f} & \mathcal{Y}_2(S), \end{array}$$

is an isomorphism.

We have the following useful observation:

Lemma C.1.7. *Let*

$$f : \mathcal{Y}_1 \rightleftarrows \mathcal{Y}_2 : g$$

be mutually adjoint maps. Then the functors $(g^!, f^!)$ form an adjoint pair.

C.1.8. For a categorical prestack $\mathcal{Y}$, we let $\mathcal{Y}_{\text{dR}}$ denote the categorical prestack defined by

$$\mathcal{Y}_{\text{dR}}(S) := \mathcal{Y}(S_{\text{red}}).$$

A standard manipulation shows that

$$\text{IndCoh}(\mathcal{Y}_{\text{dR}}) \simeq \text{D-mod}(\mathcal{Y}).$$

C.1.9. One can describe the categories (C.2) explicitly as follows. We will do this for $\text{D-mod}(\mathcal{Y})$, while the other two cases are similar.

An object $\mathcal{F} \in \text{D-mod}(\mathcal{Y})$ is an assignment:

- For every $y : S \rightarrow \mathcal{Y}$ of an object $\mathcal{F}_{S,y} \in \text{D-mod}(S)$;
- For a map $(y_1 \xrightarrow{\alpha} y_2) \in \mathcal{Y}(S)$ of a map $\mathcal{F}_{S,y_1} \xrightarrow{\mathcal{F}_{\alpha}} \mathcal{F}_{S,y_2}$ in $\text{D-mod}(S)$;
- For $f : S' \rightarrow S$ and $y' = y \circ f$ of an isomorphism $\mathcal{F}_{S',y'} \simeq f^!(\mathcal{F}_{S,y})$ in $\text{D-mod}(S')$;
- The datum of commutativity for the diagram

$$\begin{array}{ccc} \mathcal{F}_{S',y'_1} & \xrightarrow{\mathcal{F}_{f^*(\alpha)}} & \mathcal{F}_{S',y'_2} \\ \simeq \downarrow & & \downarrow \simeq \\ f^!(\mathcal{F}_{S,y_1}) & \xrightarrow{f^!(\mathcal{F}_{\alpha})} & f^!(\mathcal{F}_{S,y_2}); \end{array}$$

- A homotopy-coherent system of compatibilities for compositions.

C.1.10. In addition to the categories (C.2), one can consider their strict versions:

$$\text{QCoh}(\mathcal{Y})^{\text{strict}} := \text{QCoh}(\mathcal{Y}^{\text{strict}}), \quad \text{IndCoh}(\mathcal{Y})^{\text{strict}} := \text{IndCoh}(\mathcal{Y}^{\text{strict}})$$

and

$$\text{D-mod}(\mathcal{Y})^{\text{strict}} := \text{D-mod}(\mathcal{Y}^{\text{strict}}),$$

respectively.

Unwinding the definitions, we obtain that pullback along

$$\text{strict} : \mathcal{Y} \rightarrow \mathcal{Y}^{\text{strict}}$$

is a fully faithful embedding, with essential image described as follows:

It consists of those objects $\mathcal{F}$ in Sect. C.1.9, for which the maps

$$\mathcal{F}_{S,y_1} \xrightarrow{\mathcal{F}_{\alpha}} \mathcal{F}_{S,y_2}$$

are isomorphisms.

Note that we can describe $\text{D-mod}(\mathcal{Y})^{\text{strict}}$ also as

$$\text{D-mod}(\mathcal{Y})^{\text{strict}} = \lim_{S \in \text{Sch}_{/\mathcal{Y}}^{\text{aff}}} \text{D-mod}(S).$$

The same applies also to $\text{QCoh}(\mathcal{Y})^{\text{strict}}$ and $\text{IndCoh}(\mathcal{Y})^{\text{strict}}$.

C.2. Crystals of categories on categorical prestacks.

C.2.1. Let $\mathcal{Y}$ be a categorical prestack. Combining the ideas of Sects. B.8.2 and C.1.4 we obtain the notion of *sheaf of categories* over $\mathcal{Y}$.

Thus, a sheaf of categories $\underline{\mathbf{C}}$ over $\mathcal{Y}$ is an assignment:

- For every affine scheme S and a map $y : S \rightarrow \mathcal{Y}$ of a category $\mathbf{C}_{S,y}$ tensored over $\mathrm{QCoh}(S)$;
- For a map $y_1 \xrightarrow{\alpha} y_2$ in $\mathrm{Maps}(S, \mathcal{Y})$ of a $\mathrm{D}\text{-mod}(S)$ -linear functor $\mathbf{C}_{S,y_1} \xrightarrow{\mathbf{C}_{S,\alpha}} \mathbf{C}_{S,y_2}$;
- For $S' \xrightarrow{f} S$ and $y' = y \circ f$ of an identification $\mathbf{C}_{S',y'} \simeq \mathrm{QCoh}(S') \otimes_{\mathrm{QCoh}(S)} \mathbf{C}_{S,y}$;
- For $\alpha' = \alpha \circ f$ of a datum of commutativity for

$$\begin{array}{ccc} \mathbf{C}_{S,y_1} & \xrightarrow{\mathbf{C}_{S,\alpha}} & \mathbf{C}_{S,y_2} \\ f^! \downarrow & & \downarrow f^! \\ \mathbf{C}_{S',y'_1} & \xrightarrow{\mathbf{C}_{S',\alpha'}} & \mathbf{C}_{S',y'_2}. \end{array}$$

- A homotopy-coherent system of compatibilities for compositions.

C.2.2. We shall say that $\underline{\mathbf{C}}$ is strict if the functors $\mathbf{C}_{S,\alpha}$ are equivalences. Note that $\underline{\mathbf{C}}$ is strict if it is the pullback⁷⁵ along

$$\mathcal{Y} \rightarrow \mathcal{Y}^{\mathrm{strict}}$$

of a crystal of categories on $\mathcal{Y}^{\mathrm{strict}}$.

C.2.3. One can assign to $\underline{\mathbf{C}}$ two categories, denoted

$$\Gamma^{\mathrm{lax}}(\mathcal{Y}, \underline{\mathbf{C}}) \text{ and } \Gamma^{\mathrm{strict}}(\mathcal{Y}, \underline{\mathbf{C}}),$$

respectively, defined as follows, the latter being a full subcategory of the former.

An object of $\Gamma^{\mathrm{lax}}(\mathcal{Y}, \underline{\mathbf{C}})$ assigns to every affine scheme S and a map $y : S \rightarrow \mathcal{Y}$ an object $\mathbf{c}_{S,y} \in \mathbf{C}_{S,y}$ together with the following data:

- For a map $y_1 \xrightarrow{\alpha} y_2$ in $\mathrm{Maps}(S, \mathcal{Y})$ a morphism

$$(C.3) \quad \mathbf{C}_{S,\alpha}(\mathbf{c}_{S,y_1}) \rightarrow \mathbf{c}_{S,y_2};$$

- For a map $f : S' \rightarrow S$ and $y' = y \circ f$ an isomorphism

$$f^!(\mathbf{c}_{S,y}) \simeq \mathbf{c}_{S',y'}$$

as objects in $\mathbf{C}_{S',y'}$.

- A homotopy-coherent datum of compatibility for the above pieces of data.

The subcategory $\Gamma^{\mathrm{strict}}(\mathcal{Y}, \underline{\mathbf{C}})$ consists of those assignments for which the maps (C.3) are isomorphisms.

Remark C.2.4. We alert the reader to the discrepancy between the notations

$$\Gamma^{\mathrm{lax}}(-, -) \text{ and } \Gamma^{\mathrm{strict}}(-, -)$$

introduced above and those used in [Ra6, Sect. 4].

Namely, what we denote $\Gamma^{\mathrm{lax}}(-, -)$ is denoted $\Gamma(-, -)$ in *loc. cit.*, and what we denote $\Gamma^{\mathrm{strict}}(-, -)$ is denoted $\Gamma^{\mathrm{naive}}(-, -)$ in *loc. cit.*

Similarly, the notion of *functor* of sheaves of categories considered in [Ra6] corresponds to the notion of *right-lax functor* considered below.

⁷⁵In the sense of Sect. C.2.17.

C.2.5. *Example.* Let $\underline{\mathbf{C}}$ be $\underline{\mathrm{QCoh}}(\mathcal{Y})$, the unit crystal of categories, i.e., its value for $(S, y) \in \mathrm{Sch}_{/\mathcal{Y}}^{\mathrm{aff}}$ is $\mathrm{QCoh}(S)$.

Then

$$\Gamma^{\mathrm{lax}}(\mathcal{Y}, \underline{\mathbf{C}}) = \mathrm{QCoh}(\mathcal{Y}),$$

see Sect. C.1.4 and

$$\Gamma^{\mathrm{strict}}(\mathcal{Y}, \underline{\mathbf{C}}) = \mathrm{QCoh}(\mathcal{Y})^{\mathrm{strict}} \simeq \mathrm{QCoh}(\mathcal{Y}^{\mathrm{strict}})$$

(see Sect. C.1.10).

C.2.6. Let $\underline{\mathbf{C}}'$ and $\underline{\mathbf{C}}''$ be two crystals of categories on $\mathcal{Y}$. In this case there is an (evident) notion of functor

$$\Phi : \underline{\mathbf{C}}' \rightarrow \underline{\mathbf{C}}''.$$

When an ambiguity is likely to occur, we will call such functors *strict*.

C.2.7. For future reference, a (strict) functor Φ is said to be fully faithful if for every $y : S \rightarrow \mathcal{Y}$, the resulting functor

$$\mathbf{C}'_{S,y} \rightarrow \mathbf{C}''_{S,y}$$

is fully faithful.

C.2.8. In addition, there is a notion of *right-lax* functor. A right-lax functor $\Phi : \underline{\mathbf{C}}' \rightarrow \underline{\mathbf{C}}''$ is an assignment:

- For every (S, y) of a functor $\mathbf{C}'_{S,y} \xrightarrow{\Phi_{S,y}} \mathbf{C}''_{S,y}$;
 - For every map $y_1 \xrightarrow{\alpha} y_2$ in $\mathrm{Maps}(S, \mathcal{Y})$ we have a *natural transformation*
- (C.4)
$$\mathbf{C}''_{S,\alpha} \circ \Phi_{S,y_1} \rightarrow \Phi_{S,y_2} \circ \mathbf{C}'_{S,\alpha}.$$
- For $f : \tilde{S} \rightarrow S$ and $\tilde{y} = y \circ f$ of an isomorphism $f^! \circ \Phi_{S,y} \simeq \Phi_{\tilde{S},\tilde{y}} \circ f^!$;
 - A homotopy-coherent system of compatibilities for the above data.

By definition, a right-lax functor is *strict* if the natural transformations (C.4) are isomorphisms.

We denote the categories of right-lax and strict functors by

$$\mathrm{Funct}_{\mathrm{CrystCat}(\mathcal{Y})}^{\mathrm{lax}}(\underline{\mathbf{C}}', \underline{\mathbf{C}}'') \text{ and } \mathrm{Funct}_{\mathrm{CrystCat}(\mathcal{Y})}^{\mathrm{strict}}(\underline{\mathbf{C}}', \underline{\mathbf{C}}''),$$

respectively.

C.2.9. We will denote the (2-)category of sheaves of categories on $\mathcal{Y}$, with 1-morphisms being strict functors by $\mathrm{ShvCat}^{\mathrm{strict}}(\mathcal{Y})$.

We will denote the (2-)category of sheaves of categories on $\mathcal{Y}$, with 1-morphisms being right-lax functors by $\mathrm{ShvCat}^{\mathrm{lax}}(\mathcal{Y})$.

Sometimes we will simply write $\mathrm{ShvCat}(\mathcal{Y})$, when the discussion is applicable in both contexts.

Both $\mathrm{ShvCat}^{\mathrm{strict}}(\mathcal{Y})$ and $\mathrm{ShvCat}^{\mathrm{lax}}(\mathcal{Y})$ carry a natural symmetric monoidal structure with the unit being $\underline{\mathrm{QCoh}}(\mathcal{Y})$.

C.2.10. We set

$$\mathrm{CrystCat}^{\mathrm{strict}}(\mathcal{Y}) := \mathrm{ShvCat}^{\mathrm{strict}}(\mathcal{Y}_{\mathrm{dR}}) \text{ and } \mathrm{CrystCat}^{\mathrm{lax}}(\mathcal{Y}) := \mathrm{ShvCat}^{\mathrm{lax}}(\mathcal{Y}_{\mathrm{dR}})$$

Terminologically, when we talk about $\underline{\mathbf{C}}$ being a *crystal of categories* over $\mathcal{Y}$, for $(S, y) \in \mathrm{Sch}_{/\mathcal{Y}}^{\mathrm{aff}}$, we will denote by

$$\mathbf{C}_{S,y} \in \mathrm{D-mod}(S)\text{-}\mathbf{mod}$$

the corresponding category of crystalline sections.

We let $\underline{\mathrm{D-mod}}(\mathcal{Y})$ denote the unit crystal of categories on $\mathcal{Y}$, i.e., its value for $(S, y) \in \mathrm{Sch}_{/\mathcal{Y}}^{\mathrm{aff}}$ is $\mathrm{D-mod}(S)$.

C.2.11. Let $\underline{\mathbf{C}}_1$ and $\underline{\mathbf{C}}_2$ be two crystals of categories on $\mathcal{Y}$, and let

$$\Phi : \underline{\mathbf{C}}_1 \rightarrow \underline{\mathbf{C}}_2$$

be a strict functor.

Assume that the induced functor

$$\mathbf{t}^!(\underline{\mathbf{C}}_1) \xrightarrow{\mathbf{t}^!(\Phi)} \mathbf{t}^!(\underline{\mathbf{C}}_2)$$

admits a right adjoint⁷⁶, to be denoted $(\mathbf{t}^!(\Phi))^R$.

In this case $(\mathbf{t}^!(\Phi))^R$ admits a natural extension, to be denoted $\underline{\Phi}^R$ to a right-lax functor

$$\underline{\mathbf{C}}_2 \rightarrow \underline{\mathbf{C}}_1,$$

see, e.g., [AMR, Lemma B.5.9].

C.2.12. Let $\underline{\mathbf{C}}$ be a crystal of categories over $\mathcal{Y}$. Assume that $\mathbf{t}^!(\underline{\mathbf{C}})$ is dualizable. Assume moreover that for every $(y_1 \xrightarrow{\alpha} y_2) \in \text{Maps}(S, \mathcal{Y})$, the functor

$$(C.5) \quad \mathbf{C}_{S, y_1} \xrightarrow{\mathbf{C}_{S, \alpha}} \mathbf{C}_{S, y_2}$$

admits a right adjoint.

In this case, we can extend the dual $(\mathbf{t}^!(\underline{\mathbf{C}}))^\vee$ to a crystal of categories $\underline{\mathbf{C}}^\vee$ over $\mathcal{Y}$ by letting

$$\mathbf{C}_{S, y_1}^\vee \xrightarrow{\mathbf{C}_{S, \alpha}^\vee} \mathbf{C}_{S, y_2}^\vee$$

be the dual of the right adjoint of (C.5).

Under the above circumstances we will say that $\underline{\mathbf{C}}$ is dualizable, and we will refer to the above crystal of categories $\underline{\mathbf{C}}^\vee$ as the dual of $\underline{\mathbf{C}}$.

Note that we have the natural evaluation and coevaluation that are *right-lax* functors

$$(C.6) \quad \underline{\mathbf{D}}\text{-mod}(\mathcal{Y}) \xrightarrow{\text{co-eval}} \underline{\mathbf{C}} \otimes \underline{\mathbf{D}} \text{ and } \underline{\mathbf{D}} \otimes \underline{\mathbf{C}} \xrightarrow{\text{eval}} \underline{\mathbf{D}}\text{-mod}(\mathcal{Y}),$$

i.e., the duality between $\underline{\mathbf{C}}$ and $\underline{\mathbf{C}}^\vee$ takes place in the symmetric monoidal category $\text{ShvCat}^{\text{lax}}(\mathcal{Y})$.

C.2.13. Vice versa, let us be given two crystal of categories $\underline{\mathbf{C}}$ and $\underline{\mathbf{D}}$ and right-lax functors as in (C.6). Suppose that the following conditions hold:

- For every $S \xrightarrow{y} \mathcal{Y}$, the functors

$$\text{D-mod}(S) \rightarrow \mathbf{C}_{S, y} \underset{\text{D-mod}(S)}{\otimes} \mathbf{D}_{S, y} \text{ and } \mathbf{D}_{S, y} \underset{\text{D-mod}(S)}{\otimes} \mathbf{C}_{S, y} \rightarrow \text{D-mod}(S)$$

define a perfect pairing;

- The identification of the pullback of the composition

$$\underline{\mathbf{C}} \xrightarrow{\text{co-eval} \otimes \text{Id}} \underline{\mathbf{C}} \otimes \underline{\mathbf{D}} \otimes \underline{\mathbf{C}} \xrightarrow{\text{Id} \otimes \text{eval}} \underline{\mathbf{C}}$$

along $\mathbf{t}$ with the identity endofunctor of $\underline{\mathbf{C}}|_{\mathcal{Y}^{\text{grp d}}}$ extends to $\mathcal{Y}$;

- The identification of the pullback of the composition

$$\underline{\mathbf{D}} \xrightarrow{\text{Id} \otimes \text{eval}} \underline{\mathbf{D}} \otimes \underline{\mathbf{C}} \otimes \underline{\mathbf{D}} \xrightarrow{\text{co-eval} \otimes \text{Id}} \underline{\mathbf{D}}$$

along $\mathbf{t}$ with the identity functor of $\underline{\mathbf{D}}|_{\mathcal{Y}^{\text{grp d}}}$ extends to $\mathcal{Y}$.

Then $\underline{\mathbf{C}}$ is dualizable, and $\underline{\mathbf{D}}$ identifies canonically with the dual of $\underline{\mathbf{C}}$ in the sense of Sect. C.2.12, see [CF, Remark 11.11.17].

⁷⁶As a functor between sheaves of categories on $\mathcal{Y}^{\text{grp d}}$, i.e., it admits a continuous right adjoint value-wise.

C.2.14. Still equivalently, let us be given a pair of crystals of categories $\underline{\mathbf{C}}$ and $\underline{\mathbf{D}}$ and *either* a right-lax functor

$$\underline{\mathbf{D}} \otimes \underline{\mathbf{C}} \xrightarrow{\text{eval}} \underline{\mathbf{D-mod}}(\mathcal{Y})$$

or a right-lax functor

$$\underline{\mathbf{D-mod}}(\mathcal{Y}) \xrightarrow{\text{co-eval}} \underline{\mathbf{C}} \otimes \underline{\mathbf{D}}.$$

Suppose that:

- The pullback of eval (resp., co-eval) along $\mathbf{t}^!$ is a perfect pairing;
- For every $(y_1 \xrightarrow{\alpha} y_2) \in \text{Maps}(S, \mathcal{Y})$, the resulting natural transformation $\text{Id} \rightarrow \mathbf{D}_{S,\alpha}^\vee \circ \mathbf{C}_{S,\alpha}$ (resp., $\mathbf{C}_{S,\alpha} \circ \mathbf{D}_{S,\alpha}^\vee \rightarrow \text{Id}$) is the unit (resp., counit) of an adjunction.

Then this datum extends uniquely to a datum of duality between $\underline{\mathbf{C}}$ and $\underline{\mathbf{D}}$ as crystals of categories on $\mathcal{Y}$.

C.2.15. For $\underline{\mathbf{C}}'$ and $\underline{\mathbf{C}}''$ as above, we observe that a right-lax functor $\Phi : \underline{\mathbf{C}}' \rightarrow \underline{\mathbf{C}}''$ gives rise to a functor

$$\Phi : \Gamma^{\text{lax}}(\mathcal{Y}, \underline{\mathbf{C}}') \rightarrow \Gamma^{\text{lax}}(\mathcal{Y}, \underline{\mathbf{C}}'').$$

If Φ is strict, then Φ induces a functor

$$\Gamma^{\text{strict}}(\mathcal{Y}, \underline{\mathbf{C}}') \rightarrow \Gamma^{\text{strict}}(\mathcal{Y}, \underline{\mathbf{C}}'').$$

C.2.16. Note that the functors

$$\underline{\mathbf{C}} \mapsto \Gamma^{\text{lax}}(\mathcal{Y}, \underline{\mathbf{C}}) \text{ and } \underline{\mathbf{C}} \mapsto \Gamma^{\text{strict}}(\mathcal{Y}, \underline{\mathbf{C}})$$

can be recovered as adjoints:

For $\mathbf{D} \in \text{DGCat}$, we have

$$\text{Funct}_{\text{cont}}(\mathbf{D}, \Gamma^{\text{lax}}(\mathcal{Y}, \underline{\mathbf{C}})) \simeq \text{Funct}_{\text{CrystCat}(\mathcal{Y})}^{\text{lax}}(\mathbf{D} \otimes \underline{\mathbf{D-mod}}(\mathcal{Y}), \underline{\mathbf{C}})$$

and

$$\text{Funct}_{\text{cont}}(\mathbf{D}, \Gamma^{\text{strict}}(\mathcal{Y}, \underline{\mathbf{C}})) \simeq \text{Funct}_{\text{CrystCat}(\mathcal{Y})}^{\text{strict}}(\mathbf{D} \otimes \underline{\mathbf{D-mod}}(\mathcal{Y}), \underline{\mathbf{C}}),$$

respectively.

C.2.17. Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a map between categorical prestacks. As in Sects. B.8.7 and C.1.5, there is a naturally defined functor

$$f^* : \text{CrystCat}(\mathcal{Y}_1) \rightarrow \text{CrystCat}(\mathcal{Y}_2).$$

C.2.18. For a map $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ between categorical prestacks and a crystal of categories $\underline{\mathbf{C}}$ on $\mathcal{Y}_2$, we have a naturally defined functor

$$(C.7) \quad f^! : \Gamma^{\text{lax}}(\mathcal{Y}_2, \underline{\mathbf{C}}) \rightarrow \Gamma^{\text{lax}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}})),$$

which induces a functor

$$f^! : \Gamma^{\text{strict}}(\mathcal{Y}_2, \underline{\mathbf{C}}) \rightarrow \Gamma^{\text{strict}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}})).$$

C.3. Two notions of direct image of a crystal of categories.

C.3.1. Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a map between categorical prestacks. We shall say that f is a (co)Cartesian fibration, i.e.:

- f is a value-wise (co)Cartesian fibration;
- For $S' \rightarrow S$, the functor $\mathcal{Y}_1(S) \rightarrow \mathcal{Y}_1(S')$ sends arrows that are (co)Cartesian over $\mathcal{Y}_2(S)$ to arrows in $\mathcal{Y}_1(S')$ with the same property.

Note that the second condition is automatic if f is a value-wise (co)Cartesian fibration *in groupoids*.

C.3.2. Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a Cartesian fibration.

Let $\underline{\mathbf{C}}_1$ be a crystal of categories over $\mathcal{Y}_1$. In this case one can form two sheaves of categories, denoted

$$f_*(\underline{\mathbf{C}}_1) \text{ and } f_{*,\text{lax}}(\underline{\mathbf{C}}_1)$$

on $\mathcal{Y}_2$, as follows.

C.3.3. For $y : S \rightarrow \mathcal{Y}_2$, the value of $f_{*,\text{lax}}(\underline{\mathbf{C}}_1)$ on (S, y_2) is

$$\Gamma^{\text{lax}}(S \times_{\mathcal{Y}_2} \mathcal{Y}_1, \underline{\mathbf{C}}_1|_{S \times_{\mathcal{Y}_2} \mathcal{Y}_1})$$

and the value of $f_{*,\text{strict}}(\underline{\mathbf{C}}_1)$ is

$$\Gamma^{\text{strict}}(S \times_{\mathcal{Y}_2} \mathcal{Y}_1, \underline{\mathbf{C}}_1|_{S \times_{\mathcal{Y}_2} \mathcal{Y}_1}).$$

The data of crystal of categories on $f_{*,\text{lax}}(\underline{\mathbf{C}}_1)$ is defined as follows.

C.3.4. For a map $y' \xrightarrow{\alpha} y''$ in $\text{Maps}(S, \mathcal{Y}_2)$ we have a map

$$S \times_{y'', \mathcal{Y}_2} \mathcal{Y}_1 \xrightarrow{\alpha_{y_2}^*} S \times_{y', \mathcal{Y}_2} \mathcal{Y}_1.$$

The structure on $\underline{\mathbf{C}}_1$ of crystal of categories gives rise to a (strict) functor between crystals of categories on $S \times_{y'', \mathcal{Y}_2} \mathcal{Y}_1$

$$(\alpha_{y_2}^*)^*(\underline{\mathbf{C}}_1|_{S \times_{y', \mathcal{Y}_2} \mathcal{Y}_1}) \rightarrow \underline{\mathbf{C}}_1|_{S \times_{y'', \mathcal{Y}_2} \mathcal{Y}_1}.$$

Hence, the constructions of Sects. C.2.15 and C.2.18 combine to gives rise to a functor

$$\begin{aligned} \text{(C.8)} \quad f_{*,\text{lax}}(\underline{\mathbf{C}}_1)_{S, y'} &:= \Gamma^{\text{lax}}(S \times_{y', \mathcal{Y}_2} \mathcal{Y}_1, \underline{\mathbf{C}}_1|_{S \times_{y', \mathcal{Y}_2} \mathcal{Y}_1}) \xrightarrow{(\alpha_{y_2}^*)^!} \\ &\rightarrow \Gamma^{\text{lax}}(S \times_{y'', \mathcal{Y}_2} \mathcal{Y}_1, (\alpha_{y_2}^*)^*(\underline{\mathbf{C}}_1|_{S \times_{y', \mathcal{Y}_2} \mathcal{Y}_1})) \rightarrow \Gamma^{\text{lax}}(S \times_{y'', \mathcal{Y}_2} \mathcal{Y}_1, \underline{\mathbf{C}}_1|_{S \times_{y'', \mathcal{Y}_2} \mathcal{Y}_1}) =: f_{*,\text{lax}}(\underline{\mathbf{C}}_1)_{S, y''}. \end{aligned}$$

This functor induces a functor

$$\begin{aligned} f_{*,\text{strict}}(\underline{\mathbf{C}}_1)_{S, y'} &:= \Gamma^{\text{strict}}(S \times_{y', \mathcal{Y}_2} \mathcal{Y}_1, \underline{\mathbf{C}}_1|_{S \times_{y', \mathcal{Y}_2} \mathcal{Y}_1}) \xrightarrow{(\alpha_{y_2}^*)^!} \\ &\rightarrow \Gamma^{\text{strict}}(S \times_{y'', \mathcal{Y}_2} \mathcal{Y}_1, (\alpha_{y_2}^*)^*(\underline{\mathbf{C}}_1|_{S \times_{y', \mathcal{Y}_2} \mathcal{Y}_1})) \rightarrow \Gamma^{\text{strict}}(S \times_{y'', \mathcal{Y}_2} \mathcal{Y}_1, \underline{\mathbf{C}}_1|_{S \times_{y'', \mathcal{Y}_2} \mathcal{Y}_1}) =: f_{*,\text{strict}}(\underline{\mathbf{C}}_1)_{S, y''}. \end{aligned}$$

C.3.5. For $g : \tilde{S} \rightarrow S$ and $\tilde{y} = f \circ g$, we have a map

$$g : \tilde{S} \times_{\mathcal{Y}_2} \mathcal{Y}_1 \rightarrow S \times_{\mathcal{Y}_2} \mathcal{Y}_1,$$

and the construction of Sect. C.2.18 gives rise to a functor

$$f_{*,\text{lax}}(\underline{\mathbf{C}}_1)_{S, y} := \Gamma^{\text{lax}}(S \times_{\mathcal{Y}_2} \mathcal{Y}_1, \underline{\mathbf{C}}_1|_{S \times_{\mathcal{Y}_2} \mathcal{Y}_1}) \xrightarrow{g^!} \Gamma^{\text{lax}}(\tilde{S} \times_{\mathcal{Y}_2} \mathcal{Y}_1, \underline{\mathbf{C}}_1|_{\tilde{S} \times_{\mathcal{Y}_2} \mathcal{Y}_1}) =: f_{*,\text{lax}}(\underline{\mathbf{C}}_1)_{\tilde{S}, \tilde{y}}.$$

This functor induces a functor

$$f_{*,\text{strict}}(\underline{\mathbf{C}}_1)_{S, y} := \Gamma^{\text{strict}}(S \times_{\mathcal{Y}_2} \mathcal{Y}_1, \underline{\mathbf{C}}_1|_{S \times_{\mathcal{Y}_2} \mathcal{Y}_1}) \xrightarrow{g^!} \Gamma^{\text{strict}}(\tilde{S} \times_{\mathcal{Y}_2} \mathcal{Y}_1, \underline{\mathbf{C}}_1|_{\tilde{S} \times_{\mathcal{Y}_2} \mathcal{Y}_1}) =: f_{*,\text{strict}}(\underline{\mathbf{C}}_1)_{\tilde{S}, \tilde{y}}.$$

C.3.6. By construction, we have a strict functor

$$f_{*,\text{strict}}(\underline{\mathbf{C}}_1) \rightarrow f_{*,\text{lax}}(\underline{\mathbf{C}}_1),$$

which is a value-wise fully faithful embedding.

C.3.7. The above construction is functorial in the following sense: for a lax functor $\underline{\Phi} : \underline{\mathbf{C}}'_1 \rightarrow \underline{\mathbf{C}}''_1$ between crystals of categories on $\mathcal{Y}_1$ we obtain a lax functor

$$f_*(\underline{\Phi}) : f_{*,\text{lax}}(\underline{\mathbf{C}}'_1) \rightarrow f_{*,\text{lax}}(\underline{\mathbf{C}}''_1).$$

If $\underline{\Phi}$ is strict, then so is $f_*(\underline{\Phi})$, and it also induces a strict functor

$$f_*(\underline{\Phi}) : f_{*,\text{strict}}(\underline{\mathbf{C}}'_1) \rightarrow f_{*,\text{strict}}(\underline{\mathbf{C}}''_1).$$

C.3.8. The operations

$$\underline{\mathbf{C}}_1 \mapsto f_{*,\text{strict}}(\underline{\mathbf{C}}_1) \text{ and } \underline{\mathbf{C}}_1 \mapsto f_{*,\text{lax}}(\underline{\mathbf{C}}_1)$$

can also be realized as right adjoints. Namely, for $\underline{\mathbf{C}}_2 \in \text{CrystCat}(\mathcal{Y}_2)$, we have

$$\text{Funct}_{\text{CrystCat}(\mathcal{Y}_1)}^{\text{strict}}(f^*(\underline{\mathbf{C}}_2), \underline{\mathbf{C}}_1) \simeq \text{Funct}_{\text{CrystCat}(\mathcal{Y}_2)}^{\text{strict}}(\underline{\mathbf{C}}_2, f_{*,\text{strict}}(\underline{\mathbf{C}}_1))$$

and

$$\text{Funct}_{\text{CrystCat}(\mathcal{Y}_1)}^{\text{lax}}(f^*(\underline{\mathbf{C}}_2), \underline{\mathbf{C}}_1) \simeq \text{Funct}_{\text{CrystCat}(\mathcal{Y}_2)}^{\text{lax}}(\underline{\mathbf{C}}_2, f_{*,\text{lax}}(\underline{\mathbf{C}}_1)).$$

In particular, we have a canonically defined (strict) functor

$$(C.9) \quad \underline{\mathbf{C}}_2 \rightarrow f_{*,\text{strict}} \circ f^*(\underline{\mathbf{C}}_2).$$

C.3.9. Consider the forgetful functor

$$\text{CrystCat}^{\text{strict}}(\mathcal{Y}) \rightarrow \text{CrystCat}^{\text{lax}}(\mathcal{Y}).$$

We claim that it admits a right adjoint. Namely, consider the map

$$\text{pr}_{\text{source}} : \mathcal{Y}^{\rightarrow} \rightarrow \mathcal{Y}$$

is a Cartesian fibration.

The above right adjoint is given by

$$(\text{pr}_{\text{source}})_{*,\text{lax}} \circ (\text{pr}_{\text{target}})^*.$$

In particular, for $\underline{\mathbf{C}}_1, \underline{\mathbf{C}}_2 \in \text{CrystCat}^{\text{strict}}(\mathcal{Y})$ we have a canonical identification

$$(C.10) \quad \text{Funct}^{\text{lax}}(\underline{\mathbf{C}}_1, \underline{\mathbf{C}}_2) \simeq \text{Funct}^{\text{strict}}(\underline{\mathbf{C}}_1, (\text{pr}_{\text{source}})_{*,\text{lax}} \circ (\text{pr}_{\text{target}})^*(\underline{\mathbf{C}}_2)),$$

with the inclusion

$$\text{Funct}^{\text{lax}}(\underline{\mathbf{C}}_1, \underline{\mathbf{C}}_2) \hookrightarrow \text{Funct}^{\text{strict}}(\underline{\mathbf{C}}_1, \underline{\mathbf{C}}_2)$$

corresponding to the strict functor

$$\underline{\mathbf{C}}_2 \rightarrow (\text{pr}_{\text{source}})_{*,\text{strict}} \circ (\text{pr}_{\text{target}})^*(\underline{\mathbf{C}}_2) \rightarrow (\text{pr}_{\text{source}})_{*,\text{lax}} \circ (\text{pr}_{\text{strict}})^*(\underline{\mathbf{C}}_2),$$

where the first arrow corresponds by adjunction to the functor

$$(\text{pr}_{\text{source}})^*(\underline{\mathbf{C}}_2) \rightarrow (\text{pr}_{\text{target}})^*(\underline{\mathbf{C}}_2),$$

of (11.27).

C.3.10. We now explain an abstract framework for the construction in Sect. 11.5.

Let $\mathcal{Y}$ be a categorical prestack, and let $\underline{\mathbf{C}}_1, \underline{\mathbf{C}}_2$ be crystals of categories on it.

Suppose that the functor

$$(C.11) \quad \underline{\mathbf{C}}_2 \rightarrow (\text{pr}_{\text{source}})_{*,\text{strict}}(\underline{\mathbf{C}}_2) \rightarrow (\text{pr}_{\text{source}})_{*,\text{lax}}(\underline{\mathbf{C}}_2)$$

admits a left adjoint that is a strict functor.

Then the functor

$$\text{Funct}^{\text{strict}}(\underline{\mathbf{C}}_1, \underline{\mathbf{C}}_2) \rightarrow \text{Funct}^{\text{lax}}(\underline{\mathbf{C}}_1, \underline{\mathbf{C}}_2)$$

admits a left adjoint, given, in terms of (C.10), by composing with the left adjoint of (C.11).

C.4. Pseudo-properness.

C.4.1. We introduce the category of *pseudo-proper* categorical prestacks as

$$\mathrm{CatPreStk}_{\mathrm{ps-proper}} := \mathrm{Funct}((\mathrm{Sch}^{\mathrm{proper}})^{\mathrm{op}}, \infty\text{-Cat}).$$

The embedding

$$\mathrm{Sch}^{\mathrm{proper}} \hookrightarrow \mathrm{PreStk} \hookrightarrow \mathrm{CatPreStk}$$

uniquely extends to a colimit-preserving functor

$$(C.12) \quad \mathrm{CatPreStk}_{\mathrm{ps-proper}} \rightarrow \mathrm{CatPreStk}.$$

Similarly, we define the category

$$\mathrm{PreStk}_{\mathrm{ps-proper}} := \mathrm{Funct}((\mathrm{Sch}^{\mathrm{proper}})^{\mathrm{op}}, \infty\text{-Grpd})$$

and the functor

$$(C.13) \quad \mathrm{PreStk}_{\mathrm{ps-proper}} \rightarrow \mathrm{PreStk}.$$

Remark C.4.2. Note that the functors (C.12) and (C.13) are *not* fully faithful. I.e., with the above definition, pseudo-properness is not a property, but extra structure.

However, since the functor $\mathrm{Sch}^{\mathrm{proper}} \hookrightarrow \mathrm{PreStk}$ preserves fiber products, so do the functors (C.12) and (C.13).

C.4.3. Concretely, pseudo-properness means the following:

A prestack $\mathcal{Y}$ is pseud-proper when it can be written as

$$\mathrm{colim}_{i \in I} Z_i,$$

where:

- Z_i are proper schemes;
- The colimit is taken in PreStk .

As morphisms

$$\mathrm{colim}_{i \in I} Z_i \rightarrow \mathrm{colim}_{i' \in I'} Z'_{i'}$$

we take

$$\lim_{i \in I^{\mathrm{op}}} \mathrm{colim}_{i' \in I'} \mathrm{Maps}(Z_i, Z'_{i'}).$$

A categorical prestack $\mathcal{Y}$ is pseudo-proper if the prestacks $\mathrm{Mor}^n(\mathcal{Y})$ classifying n -fold composition of morphisms in $\mathcal{Y}$ are pseudo-proper, and for the maps $[n_1] \rightarrow [n_2]$ in $\mathbf{\Delta}$, the corresponding maps

$$\mathrm{Mor}^{[n_2]}(\mathcal{Y}) \rightarrow \mathrm{Mor}^{[n_1]}(\mathcal{Y})$$

take place in $\mathrm{PreStk}_{\mathrm{ps-proper}}$.

C.4.4. *Example.* The prestack Ran and the categorical prestacks $\mathrm{Ran}^{\mathrm{untl}}$ and $\mathrm{Ran}^{\mathrm{untl},*}$ (see Sect. C.5.6) are pseudo-proper.

C.4.5. We define the functors $\mathrm{D-mod}(-)$ and $\mathrm{CrystCat}(-)$ on $\mathrm{CatPreStk}_{\mathrm{ps-proper}}$ precomposing the same-named functors out of $\mathrm{CatPreStk}$ with (C.12).

C.4.6. For $\mathcal{Y} \in \mathrm{CatPreStk}_{\mathrm{ps-proper}}$, we can describe $\mathrm{D-mod}(\mathcal{Y})$, $\mathrm{CrystCat}(\mathcal{Y})$ and

$$\Gamma^{\mathrm{lax}}(\mathcal{Y}, \underline{\mathbf{C}}), \quad \mathbf{C} \in \mathrm{CrystCat}(\mathcal{Y})$$

in terms of proper schemes mapping to $\mathcal{Y}$.

I.e., in the appropriate definitions, we can replace

$$\mathrm{Sch}_{/\mathcal{Y}}^{\mathrm{aff}} \rightsquigarrow \mathrm{Sch}_{/\mathcal{Y}}^{\mathrm{proper}},$$

where in the right-hand side the morphisms take place in $\mathrm{CatPreStk}_{\mathrm{ps-proper}}$,

C.4.7. Let $\underline{\mathbf{C}}$ be a crystal of categories over $\mathcal{Y}$, where $\mathcal{Y}$ is pseudo-proper. We claim:

Lemma C.4.8. *For $Z \xrightarrow{y} \mathcal{Y}$ with Z proper and y taking place in $\text{CatPreStk}_{\text{ps-proper}}$, the functor of evaluation*

$$\Gamma^{\text{la}}(\mathcal{Y}, \underline{\mathbf{C}}) \rightarrow \Gamma(Z, y^*(\underline{\mathbf{C}})) =: \mathbf{C}_{Z,y}$$

commutes with limits.

Proof. We can describe the category

$$\Gamma^{\text{la}}(\mathcal{Y}, \underline{\mathbf{C}})$$

as a family of assignments

$$(Z \xrightarrow{y} \mathcal{Y}) \in \text{CatPreStk}_{\text{ps-proper}} \rightsquigarrow \mathbf{c}_{Z,y} \in \mathbf{C}_{Z,y}, \quad Z \text{ is proper,}$$

compatible under pullbacks:

For $f : Z' \rightarrow Z$ we are given an isomorphism

$$f^!(\mathbf{c}_{Z,y}) \simeq \mathbf{c}_{Z',y \circ f}$$

in $\mathbf{C}_{Z',y \circ f}$.

To prove the lemma, it suffices to show that the functors

$$f^! : \mathbf{C}_{Z,y} \rightarrow \mathbf{C}_{Z',y \circ f}$$

commute with limits. Indeed, this would imply that limits in $\Gamma^{\text{la}}(\mathcal{Y}, \underline{\mathbf{C}})$ are computed component-wise in terms of $\{\mathbf{c}_{Z,y}\}$.

We claim that for any proper map $f : Z' \rightarrow Z$ and a crystal of categories $\underline{\mathbf{D}}$ on Z , the functor

$$f^! : \Gamma(Z, \underline{\mathbf{D}}) \rightarrow \Gamma(Z', f^*(\underline{\mathbf{D}}))$$

commutes with limits.

Indeed, since for a scheme its de Rham space is 1-affine, we can think of $\underline{\mathbf{D}}$ as a $\text{D-mod}(Z)$ -linear category $\mathbf{D}$, so that the functor $f^!$ is

$$(C.14) \quad \Gamma(Z, \underline{\mathbf{D}}) = \mathbf{D} \simeq \text{D-mod}(Z) \underset{\text{D-mod}(Z)}{\otimes} \mathbf{D} \xrightarrow{f^! \otimes \text{Id}} \text{D-mod}(Z') \underset{\text{D-mod}(Z)}{\otimes} \mathbf{D} = \Gamma(Z', f^*(\underline{\mathbf{D}})).$$

Now, since f is proper, the functor $f^! : \text{D-mod}(Z) \rightarrow \text{D-mod}(Z')$ admits a left adjoint, namely, $f_!$, which is automatically $\text{D-mod}(Z)$ -linear. Hence, the functor (C.14) admits a left adjoint, namely,

$$\text{D-mod}(Z') \underset{\text{D-mod}(Z)}{\otimes} \mathbf{D} \xrightarrow{f_! \otimes \text{Id}} \text{D-mod}(Z) \underset{\text{D-mod}(Z)}{\otimes} \mathbf{D} \simeq \mathbf{D}.$$

This implies that $f^!$ commutes with limits. □

C.4.9. As a consequence of Lemma C.4.8 we obtain:

Corollary C.4.10. *Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a map between pseudo-proper categorical prestacks. Then for $\underline{\mathbf{C}} \in \text{CrystCat}(\mathcal{Y}_2)$, the functor*

$$f^! : \Gamma^{\text{la}}(\mathcal{Y}_2, \underline{\mathbf{C}}) \rightarrow \Gamma^{\text{la}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}}))$$

admits a left adjoint (to be denoted $f_!$).

Proof. It suffices to check that the functor $f^!$ commutes with limits. The latter follows from Lemma C.4.8. □

Remark C.4.11. We warn the reader that although for a map f between pseudo-proper prestacks, the functor $f_!$ exists, it does not in general satisfy base change. (It does, however, if f is a value-wise co-Cartesian fibration, see Lemma C.4.17 below.)

As a particular case of Corollary C.4.14 we have:

Corollary C.4.12. *Let $\mathcal{Y}$ be pseudo-proper. Then the functor*

$$\mathbf{C}_c(\mathcal{Y}, -) : \mathbf{D}\text{-mod}(\mathcal{Y}) \rightarrow \mathbf{Vect},$$

left adjoint to

$$\mathbf{Vect} \xrightarrow{\omega_{\mathcal{Y}}} \mathbf{D}\text{-mod}(\mathcal{Y})$$

is well-defined.

C.4.13. Let $\mathcal{Y}$ be pseudo-proper. It follows formally that the prestack in groupoids $\mathcal{Y}^{\text{strict}}$ is also pseudo-proper. Hence, from Corollary C.4.10 we obtain:

Corollary C.4.14. *Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a map between pseudo-proper categorical prestacks. Then for a strict $\underline{\mathbf{C}}$, the functor*

$$f^! : \Gamma^{\text{strict}}(\mathcal{Y}_2, \underline{\mathbf{C}}) \rightarrow \Gamma^{\text{strict}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}}))$$

admits a left adjoint (to be denoted $f_!$).

Remark C.4.15. We warn the reader that the functors

$$f_! : \Gamma^{\text{lax}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}})) \rightarrow \Gamma^{\text{lax}}(\mathcal{Y}_2, \underline{\mathbf{C}})$$

and

$$f_! : \Gamma^{\text{strict}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}})) \rightarrow \Gamma^{\text{strict}}(\mathcal{Y}_2, \underline{\mathbf{C}})$$

are in general *incompatible* with the embeddings

$$\Gamma^{\text{strict}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}})) \hookrightarrow \Gamma^{\text{lax}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}})) \text{ and } \Gamma^{\text{strict}}(\mathcal{Y}_2, \underline{\mathbf{C}}) \hookrightarrow \Gamma^{\text{lax}}(\mathcal{Y}_2, \underline{\mathbf{C}}),$$

respectively.

C.4.16. We will now show how to compute the functor $f_!$ more explicitly. First, unwinding the definitions, we obtain:

Lemma C.4.17. *Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a map in $\mathbf{CatPreStk}_{\text{ps-proper}}$ that is a co-Cartesian fibration.⁷⁷ Then for $\underline{\mathbf{C}} \in \mathbf{CrystCat}(\mathcal{Y}_2)$, the functor $f_!$ satisfies base change, i.e., for a pullback diagram in $\mathbf{CatPreStk}_{\text{ps-proper}}$*

$$\begin{array}{ccc} \mathcal{Y}'_1 & \xrightarrow{g_1} & \mathcal{Y}_1 \\ f' \downarrow & & \downarrow f \\ \mathcal{Y}'_2 & \xrightarrow{g_2} & \mathcal{Y}_2, \end{array}$$

the natural transformation

$$f'_! \circ g_1^! \rightarrow g_2^! \circ f_!, \quad \Gamma^{\text{lax}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}})) \rightrightarrows \Gamma^{\text{lax}}(\mathcal{Y}'_2, g_2^*(\underline{\mathbf{C}})),$$

obtained by adjunction from

$$g_1^! \circ f^! \simeq f'^! \circ g_2^!.$$

is an isomorphism.

Corollary C.4.18. *Under the assumptions of Lemma C.4.17, for a proper scheme Z equipped with a map $Z \xrightarrow{y_1} \mathcal{Y}_1$ in $\mathbf{CatPreStk}_{\text{ps-proper}}$, the composition*

$$y_1^! \circ f_! : \Gamma^{\text{lax}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}})) \rightarrow \mathbf{C}_{Z, y_1}$$

identifies canonically with

$$\Gamma^{\text{lax}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}})) \xrightarrow{\text{pullback}} \Gamma^{\text{lax}}(\mathcal{Y}_{1,Z}, \underline{\mathbf{C}}|_{\mathcal{Y}_{1,Z}}) \xrightarrow{(f_Z)^!} \mathbf{C}_{Z, y_1},$$

where:

- $\mathcal{Y}_{1,Z} := Z \times_{\mathcal{Y}_2} \mathcal{Y}_1$;
- f_Z is the map $\mathcal{Y}_{1,Z} \rightarrow Z$.

⁷⁷As in Sect. C.3.1, but in the category $\mathbf{Funct}((\mathbf{Sch}^{\text{proper}})^{\text{op}}, \infty\text{-Cat})$.

C.4.19. Let now f be an arbitrary map in $\text{CatPreStk}_{\text{ps-proper}}$. Denote $\mathcal{Y}_{2,f/}$ be the (pseudo-proper) categorical prestack given by the slice construction, i.e.,

$$\mathcal{Y}_{2,f/}(S) = \{y_1 \in \mathcal{Y}_1(S), y_2 \in \mathcal{Y}_2(S), f(y_1) \rightarrow y_2\}.$$

Let $\tilde{f}$ and pr_f denote the projections

$$\mathcal{Y}_{2,f/} \rightarrow \mathcal{Y}_2, \quad (y_1, y_2, f(y_1) \rightarrow y_2) \mapsto y_2, \quad \mathcal{Y}_{2,f/} \rightarrow \mathcal{Y}_1, \quad (y_1, y_2, f(y_1) \rightarrow y_2) \mapsto y_1,$$

respectively.

Assume now that $\underline{\mathbf{C}}$ is strict. In this case we have a canonical equivalence

$$\tilde{f}^*(\underline{\mathbf{C}}) \simeq \text{pr}_f^* \circ f^*(\underline{\mathbf{C}}).$$

We claim:

Lemma C.4.20. *Assume that $\underline{\mathbf{C}}$ is strict. Then the functor $f_!$ identifies canonically with*

$$\Gamma^{\text{lax}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}})) \xrightarrow{\text{pr}_f^!} \Gamma^{\text{lax}}(\mathcal{Y}_{2,f/}, \text{pr}_f^* \circ f^*(\underline{\mathbf{C}})) \simeq \Gamma^{\text{lax}}(\mathcal{Y}_{2,f/}, \tilde{f}^*(\underline{\mathbf{C}})) \xrightarrow{\tilde{f}_!} \Gamma^{\text{lax}}(\mathcal{Y}_2, \underline{\mathbf{C}}),$$

Proof. Note that f factors as

$$\text{diag}_f \circ \tilde{f},$$

where

$$\text{diag}_f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_{2,f/}, \quad y_1 \mapsto (y_1, f(y_1), f(y_1) \xrightarrow{\text{id}} f(y_1)).$$

Hence, we obtain

$$f_! \simeq \tilde{f}_! \circ (\text{diag}_f)_!.$$

Now we claim that

$$(\text{diag}_f)_! \simeq \text{pr}_f^!, \quad \Gamma^{\text{lax}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}})) \rightrightarrows \Gamma^{\text{lax}}(\mathcal{Y}_{2,f/}, \text{pr}_f^* \circ f^*(\underline{\mathbf{C}})).$$

Indeed, this follows from the fact that the morphisms $(\text{diag}_f, \text{pr}_f)$ form an adjoint pair, cf. Lemma C.1.7. $\square$

Corollary C.4.21. *In the setting of Lemma C.4.20, for a proper scheme Z equipped with a map $Z \xrightarrow{y_1} \mathcal{Y}_1$ in $\text{CatPreStk}_{\text{ps-proper}}$, the composition*

$$y_1^! \circ f_! : \Gamma^{\text{lax}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}})) \rightarrow \mathbf{C}_{Z, y_1}$$

identifies canonically with

$$\Gamma^{\text{lax}}(\mathcal{Y}_1, f^*(\underline{\mathbf{C}})) \xrightarrow{\text{pullback}} \Gamma^{\text{lax}}(\mathcal{Y}_{2,f/,Z}, \underline{\mathbf{C}}|_{\mathcal{Y}_{2,f/,Z}}) \xrightarrow{(\tilde{f}_Z)_!} \mathbf{C}_{Z, y_1},$$

where:

- $\mathcal{Y}_{2,f/,Z} := Z \times_{\mathcal{Y}_2} \mathcal{Y}_{2,f/}$;
- $\tilde{f}_Z$ is the map $\mathcal{Y}_{2,f/,Z} \rightarrow Z$.

C.4.22. *Example.* Let $\mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be the map

$$\mathbf{t} : \mathcal{Y}^{\text{grpd}} \rightarrow \mathcal{Y}.$$

The functor $\mathbf{t}_!$ is the left adjoint of the forgetful functor, and the formula for it, given by Proposition C.4.20, coincides with that of [Ga4, Proposition 4.4.2].

C.4.23. Consider now the commutative square

$$(C.15) \quad \begin{array}{ccc} \mathcal{Y}^{\text{grp d}} & \xrightarrow{t} & \mathcal{Y} \\ \mathfrak{t}^{\text{op}} \downarrow & & \downarrow \text{strict}_{\mathcal{Y}} \\ \mathcal{Y}^{\text{op}} & \xrightarrow{\text{strict}_{\mathcal{Y}^{\text{op}}}} & \mathcal{Y}^{\text{strict}}. \end{array}$$

We obtain a natural transformation

$$(C.16) \quad (\mathfrak{t}^{\text{op}})_! \circ \mathfrak{t}^! \rightarrow (\text{strict}_{\mathcal{Y}^{\text{op}}})^! \circ (\text{strict}_{\mathcal{Y}})_!.$$

Suppose now that $\mathcal{Y}$ has the property that (C.15) is Cartesian. In particular, since the right vertical arrow is a co-Cartesian fibration, then so is the left vertical arrow. By Lemma C.4.17, we obtain that in this case (C.16) is an isomorphism.

Applying this in the case $\mathcal{Y} = \text{Ran}^{\text{untl}}$, this gives a conceptual explanation of the commutativity of (11.24), at least in the particular case when $\underline{\mathbf{C}}^{\text{loc}} = \underline{\mathbf{D}\text{-mod}}(\text{Ran}^{\text{untl}})$, $\mathbf{C}^{\text{glob}} = \text{Vect}$.

C.4.24. The above definitions and assertions admit a variant when we consider prestacks over a given affine base scheme S . In this case, one can talk about pseudo-properness relative to S , and the entire discussion applies.

C.5. The unital Ran space.

C.5.1. There are two versions of the unital Ran space that we will consider: Ran^{untl} and $\text{Ran}^{\text{untl},*}$.

For $S \in \text{Sch}^{\text{aff}}$, the category $\text{Ran}^{\text{untl}}(S)$ is that of finite *non-empty* subsets in $\text{Maps}(S_{\text{dR}}, X)$, with the morphisms defined as follows:

$$\text{Maps}_{\text{Maps}(S_{\text{dR}}, X)}(\underline{x}_1, \underline{x}_2) = \begin{cases} \{*\} & \text{if } \underline{x}_1 \subseteq \underline{x}_2, \\ \emptyset & \text{otherwise.} \end{cases}$$

In the above formula, $\underline{x}_i$ denotes the finite subset of $\text{Hom}(S_{\text{red}}, X)$ corresponding to the same-named $S \rightarrow \text{Ran}$.

In the case of $\text{Ran}^{\text{untl},*}$, we allow $\underline{x}$ to be empty, i.e., we add to Ran^{untl} a point $\{\emptyset\}$, which is value-wise initial, corresponding to the empty set.

Remark C.5.2. There is a variant of the above definition, where the morphisms are defined by

$$\text{Maps}_{\text{Maps}(S_{\text{dR}}, X)}(\underline{x}_1, \underline{x}_2) = \begin{cases} \{*\} & \text{if } \text{Graph}_{\underline{x}_1} \subseteq \text{Graph}_{\underline{x}_2}, \\ \emptyset & \text{otherwise.} \end{cases}$$

where in the formula $\subseteq$ means containment as closed subsets of $S \times X$. Denote the resulting categorical prestack by $'\text{Ran}^{\text{untl}}$.

The two versions are equivalent for most practical purposes. Namely, we have a naturally defined map $\text{Ran}^{\text{untl}} \rightarrow '\text{Ran}^{\text{untl}}$, and we claim that it induces an equivalence between the corresponding (2-)categories of crystals of categories.

This follows from the fact that the corresponding map

$$\text{Mor}^1(\text{Ran}^{\text{untl}}) \rightarrow \text{Mor}^1(' \text{Ran}^{\text{untl}})$$

becomes an isomorphism after sheafification in the Grothendieck topology generated by finite surjective maps, while D-modules satisfy descent for this topology.

C.5.3. The above two versions of the unital Ran space, i.e., Ran^{unl} and $\text{Ran}^{\text{unl},*}$, are convenient in slightly different situations: the $\text{Ran}^{\text{unl},*}$ version is more convenient for discussing factorization, while the Ran^{unl} version is more convenient for the discussion of local-to-global functors. Yet, the next assertion says that we could use $\text{Ran}^{\text{unl},*}$ for the latter too.

Proposition C.5.4. *Let $\underline{\mathbf{C}}$ be a sheaf of categories over $\text{Ran}^{\text{unl},*}$. Then for a DG category $\mathbf{D}$, pullback along $\text{Ran}^{\text{unl}} \rightarrow \text{Ran}^{\text{unl},*}$ gives rise to an equivalence*

$$\begin{aligned} \text{Funct}_{\text{CrystCat}(\text{Ran}^{\text{unl},*})}^{\text{strict}}(\underline{\mathbf{C}}, \mathbf{D} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran}^{\text{unl},*})) &\xrightarrow{\sim} \text{Funct}_{\text{CrystCat}(\text{Ran}^{\text{unl}})}^{\text{strict}}(\underline{\mathbf{C}}|_{\text{Ran}^{\text{unl}}}, \mathbf{D} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran}^{\text{unl}})). \end{aligned}$$

Proof. The assertion follows from the fact that the inclusion

$$\text{Ran}^{\text{unl}} \rightarrow \text{Ran}^{\text{unl},*}$$

is value-wise cofinal, and the following general claim:

Lemma C.5.5. *Let $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$ be a value-wise cofinal morphism between categorical prestacks. Then for any pair of crystals of categories $\underline{\mathbf{C}}', \underline{\mathbf{C}}''$ on $\mathcal{Y}_2$ with $\underline{\mathbf{C}}'$ strict, the functor*

$$\text{Funct}_{\mathcal{Y}_2}^{\text{strict}}(\underline{\mathbf{C}}', \underline{\mathbf{C}}'') \rightarrow \text{Funct}_{\mathcal{Y}_1}^{\text{strict}}(f^*(\underline{\mathbf{C}}'), f^*(\underline{\mathbf{C}}'')).$$

□

C.5.6. Note that the presentation of Ran as in Sect. B.1.3 shows that it is pseudo-proper.

We claim that Ran^{unl} is also pseudo-proper. We can write

$$\text{Mor}^1(\text{Ran}^{\text{unl}}) = ((\text{Ran}^{\text{unl}})^{\rightarrow})^{\text{grpd}} \simeq \text{Ran}^{\subseteq}$$

as the colimit

$$\text{colim}_{I_{\text{small}} \subseteq I_{\text{big}}} X^{I_{\text{big}}},$$

where the colimit is taken over the (opposite of the) category whose objects are pairs of non-empty finite sets $I_1 \rightarrow I_2$ and whose morphisms are commutative squares

$$\begin{array}{ccc} I_{\text{small}} & \longrightarrow & I_{\text{big}} \\ \downarrow & & \downarrow \\ I'_{\text{small}} & \longrightarrow & I'_{\text{big}} \end{array}$$

with the vertical arrows surjective.

The maps pr_{big} and pr_{small} send the term corresponding to $I_{\text{small}} \subseteq I_{\text{big}}$ to

$$X^{I_{\text{big}}} \rightarrow \text{Ran} \text{ and } X^{I_{\text{big}}} \rightarrow X^{I_{\text{small}}} \rightarrow \text{Ran},$$

respectively.

The prestacks of higher-order compositions $\text{Mor}^n(\text{Ran}^{\text{unl}})$ are described similarly.

C.5.7. By Corollary C.4.12, the functors

$$\text{C}_c(\text{Ran}^{\text{unl}}, -) : \mathbf{D}\text{-mod}(\text{Ran}^{\text{unl}}) \rightarrow \text{Vect} \text{ and } \text{C}_c(\text{Ran}, -) : \mathbf{D}\text{-mod}(\text{Ran}^{\text{unl}}) \rightarrow \text{Vect},$$

left adjoint to

$$k \mapsto \omega_{\text{Ran}^{\text{unl}}} \text{ and } k \mapsto \omega_{\text{Ran}},$$

respectively, are well-defined.

The same applies to $\text{Ran}^{\text{unl},*}$.

C.5.8. Being the left adjoint of a symmetric monoidal structure, the functor $C_c(\mathrm{Ran}^{\mathrm{untl}}, -)$ carries a naturally defined left-lax symmetric monoidal structure.

We claim:

Lemma C.5.9. *The left-lax monoidal structure on $C_c(\mathrm{Ran}^{\mathrm{untl}}, -)$ is strict.*

Proof. We need to show that the natural transformation

$$C_c(\mathrm{Ran}^{\mathrm{untl}}, -) \circ (\Delta_{\mathrm{Ran}^{\mathrm{untl}}})^! \rightarrow C_c(\mathrm{Ran}^{\mathrm{untl}} \times \mathrm{Ran}^{\mathrm{untl}}, -),$$

induced by the $((\Delta_{\mathrm{Ran}^{\mathrm{untl}}})_!, (\Delta_{\mathrm{Ran}^{\mathrm{untl}}})^!)$ -adjunction, is an isomorphism.

This follows, however, from the fact that the morphism

$$\Delta_{\mathrm{Ran}^{\mathrm{untl}}} : \mathrm{Ran}^{\mathrm{untl}} \rightarrow \mathrm{Ran}^{\mathrm{untl}} \times \mathrm{Ran}^{\mathrm{untl}}$$

is value-wise cofinal. □

C.5.10. We now consider the relation between the functors

$$C_c(\mathrm{Ran}^{\mathrm{untl}}, -) \text{ and } C_c(\mathrm{Ran}, -).$$

We have the natural transformation

$$(C.17) \quad C_c(\mathrm{Ran}, -) \circ \mathbf{t}^! \simeq C_c(\mathrm{Ran}^{\mathrm{untl}}, -) \circ \mathbf{t}_! \circ \mathbf{t}^! \rightarrow C_c(\mathrm{Ran}^{\mathrm{untl}}, -)$$

as functors $\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl}}) \rightarrow \mathrm{Vect}$.

C.5.11. We claim:

Lemma C.5.12. *The natural transformation transformation (C.17) is an isomorphism.*

This assertion is proved in [Ga4, Theorem 4.6.2]. We include the proof for completeness:

Recall what it means for a morphism between categorical prestacks to be *universally homologically cofinal*, see [Ga4, Sect. 3.5.1]. Using [Ga4, Corollary 3.5.12], the assertion of Lemma C.5.12 follows from the next one:

Lemma C.5.13. *The map $\mathbf{t}$ is universally homologically cofinal.*

Proof of Lemma C.5.13. Let S be an affine scheme and let us be given an S -point $\underline{x}$ of Ran . Consider the corresponding prestack

$$\mathrm{Ran}_{\underline{x}/},$$

see [Ga4, Sect. 3.5.1]. We need to show that it is *universally homologically contractible* over S (see [Ga4, Sect. 2.5.1] for what this means).

Note, however, that $\mathrm{Ran}_{\underline{x}/}$ is a prestack in groupoids isomorphic to $S_{\underline{x}}^{\subseteq}$, and its projection to S is pseudo-proper. Hence, it is enough to show that the fibers of the map

$$S_{\underline{x}}^{\subseteq} \rightarrow S$$

have trivial homology. The latter follows by the usual argument for the contractibility of the Ran space. □

Remark C.5.14. Statements parallel to Lemmas C.5.9 and C.5.12 hold for the $\mathrm{Ran}^{\mathrm{untl},*}$ version of the unital Ran space, see [Ro2, Sect. 2.5].

C.5.15. An analog of the assertion of Lemma C.5.9 would of course fail for the usual (i.e., non-unital) Ran space. I.e., the natural transformation

$$(C.18) \quad C_c(\text{Ran}, -) \circ (\Delta_{\text{Ran}})^! \simeq C_c(\text{Ran} \times \text{Ran}, -) \circ (\Delta_{\text{Ran}})! \circ (\Delta_{\text{Ran}})^! \rightarrow C_c(\text{Ran} \times \text{Ran}, -)$$

is not an isomorphism.

However, it admits the following variant:

Let us denote by

$$\text{D-mod}(\text{Ran})^{\text{almost-unital}} \subset \text{D-mod}(\text{Ran})$$

the full subcategory generated by the essential image of the forgetful functor

$$t^! : \text{D-mod}(\text{Ran}^{\text{unital}}) \rightarrow \text{D-mod}(\text{Ran}).$$

Note this subcategory is preserved by the monoidal operation.

We claim the left-lax monoidal structure on $C_c(\text{Ran}, -)$, given by (C.18), becomes strict when restricted to $\text{D-mod}(\text{Ran})^{\text{almost-unital}}$. Indeed, this follows from Lemma C.5.12.

C.5.16. By a similar token, for $\mathcal{Z} \rightarrow \text{Ran}$ one defines a unital version

$$\mathcal{Z}^{\subseteq, \text{unital}} := \mathcal{Z} \times_{\text{Ran}^{\text{unital}}} (\text{Ran}^{\text{unital}})^{\rightarrow}$$

of $\mathcal{Z}^{\subseteq}$.

The (categorical) prestacks $\mathcal{Z}^{\subseteq}$ and $\mathcal{Z}^{\subseteq, \text{unital}}$ are pseudo-proper relative to $\mathcal{Z}$.

The assertion of Lemma C.5.12 renders to the present context, when instead of the functors

$$C_c(\text{Ran}, -) \text{ and } C_c(\text{Ran}^{\text{unital}}, -)$$

we use the functors

$$(\text{pr}_{\text{small}, \mathcal{Z}})^! : \text{D-mod}(\mathcal{Z}^{\subseteq}) \rightarrow \text{D-mod}(\mathcal{Z})$$

and

$$(\text{pr}_{\text{small}, \mathcal{Z}}^{\text{unital}})^! : \text{D-mod}(\mathcal{Z}^{\subseteq, \text{unital}}) \rightarrow \text{D-mod}(\mathcal{Z}),$$

respectively.

C.6. Unital and counital factorization spaces.

C.6.1. Let $\mathcal{Z}_{\text{Ran}} \rightarrow \text{Ran}$ be a prestack. A unital structure on $\mathcal{Z}_{\text{Ran}}$ is its extension to a categorical prestack

$$(C.19) \quad \mathcal{Z}_{\text{Ran}^{\text{unital}}, *} \rightarrow \text{Ran}^{\text{unital}}, *$$

such that (C.19) is a value-wise *co-Cartesian fibration in groupoids*.

Let $\text{PreStk}_{\text{Ran}}^{\text{unital}}$ denote the category of prestacks over Ran , equipped with a unital structure.

C.6.2. In concrete terms, an upgrade

$$\mathcal{Z}_{\text{Ran}} \rightsquigarrow \mathcal{Z}_{\text{Ran}^{\text{unital}}, *}$$

means that for every $\underline{x} \subseteq \underline{x}'$ we give ourselves a map

$$\text{ins. unit}_{\underline{x} \subseteq \underline{x}'} : \mathcal{Z}_{\underline{x}} \rightarrow \mathcal{Z}_{\underline{x}'},$$

in a way compatible with compositions.

In addition, we give ourselves a space $\mathcal{Z}_{\emptyset}$ and a system of maps

$$\text{ins. unit}_{\emptyset \subseteq \underline{x}} : \mathcal{Z}_{\emptyset} \rightarrow \mathcal{Z}_{\underline{x}}$$

equipped with identifications

$$\text{ins. unit}_{\underline{x} \subseteq \underline{x}'} \circ \text{ins. unit}_{\emptyset \subseteq \underline{x}} \simeq \text{ins. unit}_{\emptyset \subseteq \underline{x}'}.$$

C.6.3. Here is an example of a prestack equipped with a unital structure (see [Ro2, Sect. 3.3]). Let $\mathcal{Y}$ be an affine D-scheme. Let

$$\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Y})_{\mathrm{Ran}}$$

be the space over Ran that attaches to $\underline{x} \in \mathrm{Ran}$ the space

$$\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Y})_{\underline{x}} := \mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}).$$

This prestack has a natural unital structure: namely for $\underline{x} \subseteq \underline{x}'$, the corresponding map

$$\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}) \rightarrow \mathrm{Sect}_{\nabla}(X - \underline{x}', \mathcal{Y})$$

is given by restriction.

Note that

$$\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Y})_{\emptyset} \simeq \mathrm{Sect}_{\nabla}(X, \mathcal{Y}).$$

C.6.4. Let $\mathcal{T}$ be a factorization space over X . A unital structure on it is a unital structure on $\mathcal{T}_{\mathrm{Ran}}$ (in the sense of Sect. C.6.1) and an extension of (B.2) to an isomorphism

$$(C.20) \quad \mathcal{T}_{\mathrm{Ran}^{\mathrm{untl},*}} \times_{\mathrm{Ran}^{\mathrm{untl},*}, \mathrm{union}} (\mathrm{Ran}^{\mathrm{untl},*} \times \mathrm{Ran}^{\mathrm{untl},*})_{\mathrm{disj}} \simeq (\mathcal{T}_{\mathrm{Ran}^{\mathrm{untl},*}} \times \mathcal{T}_{\mathrm{Ran}^{\mathrm{untl},*}})_{\mathrm{Ran}^{\mathrm{untl},*} \times \mathrm{Ran}^{\mathrm{untl},*}} \times (\mathrm{Ran}^{\mathrm{untl},*} \times \mathrm{Ran}^{\mathrm{untl},*})_{\mathrm{disj}},$$

equipped with a homotopy-coherent data of associativity and commutativity, where

$$(\mathrm{Ran}^{\mathrm{untl},*} \times \mathrm{Ran}^{\mathrm{untl},*})_{\mathrm{disj}} \subset \mathrm{Ran}^{\mathrm{untl},*} \times \mathrm{Ran}^{\mathrm{untl},*}.$$

is the corresponding open subfunctor.

In addition, we stipulate that

$$(C.21) \quad \mathcal{T}_{\emptyset} \simeq \mathrm{pt},$$

and this identification behaves (homotopically coherently) as a unit for the isomorphisms (C.20), i.e., the map

$$\mathcal{T}_{\mathrm{Ran}^{\mathrm{untl},*}} \rightarrow \mathcal{T}_{\emptyset} \times \mathcal{T}_{\mathrm{Ran}^{\mathrm{untl},*}},$$

obtained by base-changing (C.20) with respect to

$$\{\emptyset\} \times \mathrm{Ran}^{\mathrm{untl},*} \rightarrow (\mathrm{Ran}^{\mathrm{untl},*} \times \mathrm{Ran}^{\mathrm{untl},*})_{\mathrm{disj}}$$

identifies via (C.21) with the identity map.

C.6.5. A typical example of a unital factorization space is Gr_G . Namely, for $\underline{x} \subseteq \underline{x}'$ the corresponding map

$$\mathrm{Gr}_{G,\underline{x}} \rightarrow \mathrm{Gr}_{G,\underline{x}'}$$

is defined as follows.

Recall (see Sect. B.4.10) that $\mathrm{Gr}_{G,\underline{x}}$ can be described as the space of G -bundles on $\mathcal{D}_{\underline{x}}$ equipped with a trivialization on $\mathcal{D}_{\underline{x}} - \underline{x}$. We have:

Lemma C.6.6. *The map*

$$\mathcal{D}_{\underline{x}} \sqcup_{\mathcal{D}_{\underline{x}} - \underline{x}} (\mathcal{D}_{\underline{x}'} - \underline{x}) \rightarrow \mathcal{D}_{\underline{x}'}$$

is an isomorphism when the pushout is taken in the category of affine schemes.

Using this lemma, we can interpret $\mathrm{Gr}_{G,\underline{x}}$ as the space of G -bundles on $\mathcal{D}_{\underline{x}'}$ with a trivialization on $\mathcal{D}_{\underline{x}'} - \underline{x}$.

The desired map

$$\mathrm{ins. unit}_{\underline{x} \subseteq \underline{x}'} : \mathrm{Gr}_{G,\underline{x}} \rightarrow \mathrm{Gr}_{G,\underline{x}'}$$

is given by restricting the trivialization from $\mathcal{D}_{\underline{x}'} - \underline{x}$ to $\mathcal{D}_{\underline{x}'} - \underline{x}'$.

C.6.7. Let $\mathcal{Z}_{\text{Ran}} \rightarrow \text{Ran}$ be a prestack. A *counital* structure on $\mathcal{Z}_{\text{Ran}}$ is its extension to a categorical prestack

$$(C.22) \quad \mathcal{Z}_{\text{Ran}^{\text{untl},*}} \rightarrow \text{Ran}^{\text{untl},*},$$

such that (C.22) is a value-wise *Cartesian fibration in groupoids*.

Let $\text{PreStk}_{/\text{Ran}}^{\text{co-untl}}$ denote the category of prestacks over Ran , equipped with a counital structure.

C.6.8. In concrete terms, an upgrade

$$\mathcal{Z}_{\text{Ran}} \rightsquigarrow \mathcal{Z}_{\text{Ran}^{\text{untl},*}}$$

means that for every $\underline{x} \subseteq \underline{x}'$ we give ourselves a map

$$\text{proj. counit}_{\underline{x} \subseteq \underline{x}'} : \mathcal{Z}_{\underline{x}'} \rightarrow \mathcal{Z}_{\underline{x}},$$

in a way compatible with compositions.

In addition, we give ourselves a space $\mathcal{Z}_{\emptyset}$ and a system of maps

$$\text{proj. counit}_{\emptyset \subseteq \underline{x}} : \mathcal{Z}_{\underline{x}} \rightarrow \mathcal{Z}_{\emptyset}$$

equipped with identifications

$$\text{proj. counit}_{\emptyset \subseteq \underline{x}} \circ \text{proj. counit}_{\underline{x} \subseteq \underline{x}'} \simeq \text{proj. counit}_{\emptyset \subseteq \underline{x}'}.$$

C.6.9. Let $\mathcal{Y}$ be a D-prestack over X . Note that the arc space $\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\text{Ran}}$ has a natural counital structure:

For $\underline{x} \subseteq \underline{x}'$, the corresponding maps

$$\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\underline{x}'} \rightarrow \mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\underline{x}}$$

are given by restriction along $\widehat{\mathcal{D}}_{\underline{x}} \rightarrow \widehat{\mathcal{D}}_{\underline{x}'}$.

C.6.10. We claim:

Proposition C.6.11. *The functor $\mathcal{Y} \mapsto \mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\text{Ran}}$ is the right adjoint to the functor*

$$(C.23) \quad \text{PreStk}_{/\text{Ran}}^{\text{co-untl}} \rightarrow \text{PreStk}_{/\text{Ran}} \rightarrow \text{PreStk}_{X_{\text{dR}}},$$

where the last arrow is given by pullback along $X_{\text{dR}} \rightarrow \text{Ran}$.

Proof. Let us construct the unit and counit map for the adjunction. The counit is easy, and it is actually an isomorphism: we have

$$\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X_{\text{dR}}} \simeq \mathcal{Y}_{\nabla}.$$

To construct the unit, for any $\underline{x} : S \rightarrow \text{Ran}$ and $Z_{\text{Ran}} \in \text{PreStk}_{/\text{Ran}}^{\text{co-untl}}$, we need to define the map

$$\mathcal{Z}_S \rightarrow \text{Weil-Res}_S^{\widehat{\mathcal{D}}_{\underline{x}, \nabla}}(\widehat{\mathcal{D}}_{\underline{x}, \nabla} \times_{X_{\text{dR}}} \mathcal{Z}_{X_{\text{dR}}}),$$

where Weil-Res is the functor of restriction of scalars à la Weil.

By adjunction, the datum of the latter map is equivalent to that of a map

$$(C.24) \quad \widehat{\mathcal{D}}_{\underline{x}, \nabla} \times_S \mathcal{Z}_S \rightarrow \widehat{\mathcal{D}}_{\underline{x}, \nabla} \times_{X_{\text{dR}}} \mathcal{Z}_{X_{\text{dR}}}.$$

Note that the two sides in (C.24) are the pullbacks of $\mathcal{Z}_{\text{Ran}}$ along the following two maps

$$\widehat{\mathcal{D}}_{\underline{x}, \nabla} \rightrightarrows \text{Ran}.$$

One is

$$\widehat{\mathcal{D}}_{\underline{x}, \nabla} \rightarrow S \xrightarrow{\underline{x}} \text{Ran},$$

and the other is

$$\widehat{\mathcal{D}}_{\underline{x}, \nabla} \rightarrow X_{\text{dR}} \rightarrow \text{Ran}.$$

Now, by the definition of $\widehat{\mathcal{D}}_{\underline{x}, \nabla}$, there is a natural map from latter map to the former map inside the category

$$\text{Maps}(\widehat{\mathcal{D}}_{\underline{x}, \nabla}, \text{Ran}^{\text{untl}, *}).$$

Hence, the required map is provided by the unital structure on $\mathcal{Z}_{\text{Ran}}$.

The fact that the unit and counit maps constructed above satisfy the adjunction axioms is a straightforward verification. $\square$

C.6.12. Let $\mathcal{Z}_{\text{Ran}}$ be equipped with a counital structure. Note that this structure gives rise to a map

$$(C.25) \quad \mathcal{Z}_{\text{Ran}} \times_{\text{Ran}, \text{union}} (\text{Ran} \times \text{Ran}) \rightarrow \mathcal{Z}_{\text{Ran}} \times \mathcal{Z}_{\text{Ran}}.$$

Base changing along

$$(\text{Ran} \times \text{Ran})_{\text{disj}} \rightarrow \text{Ran} \times \text{Ran},$$

we obtain a map

$$(C.26) \quad \mathcal{Z}_{\text{Ran}} \times_{\text{Ran}, \text{union}} (\text{Ran} \times \text{Ran})_{\text{disj}} \rightarrow (\mathcal{Z}_{\text{Ran}} \times \mathcal{Z}_{\text{Ran}}) \times_{\text{Ran} \times \text{Ran}} (\text{Ran} \times \text{Ran})_{\text{disj}}.$$

We shall say that the counital structure is *factorizable* if (C.26) is an isomorphism and the diagonal map

$$\mathcal{Z}_{\emptyset} \rightarrow \mathcal{Z}_{\emptyset} \times \mathcal{Z}_{\emptyset}$$

is also an isomorphism (implying that $\mathcal{Z}_{\emptyset} \simeq \text{pt}$).

Note that the fact that the maps (C.26) are isomorphisms implies that the maps

$$(C.27) \quad \mathcal{Z}_{\text{Ran}^{\text{untl}, *}} \times_{\text{Ran}^{\text{untl}, *}, \text{union}} (\text{Ran}^{\text{untl}, *} \times \text{Ran}^{\text{untl}, *})_{\text{disj}} \rightarrow (\mathcal{Z}_{\text{Ran}^{\text{untl}, *}} \times \mathcal{Z}_{\text{Ran}^{\text{untl}, *}}) \times_{\text{Ran}^{\text{untl}, *} \times \text{Ran}^{\text{untl}, *}} (\text{Ran}^{\text{untl}, *} \times \text{Ran}^{\text{untl}, *})_{\text{disj}},$$

are also isomorphisms.

Let

$$(\text{PreStk}_{/\text{Ran}}^{\text{co-untl}})^{\text{factzbl}} \subset \text{PreStk}_{/\text{Ran}}^{\text{co-untl}}$$

denote the full subcategory that consists of factorizable objects.

C.6.13. We have the following more precise version of Proposition C.6.11:

Proposition C.6.14. *The functor*

$$(C.28) \quad \mathcal{Y} \mapsto \mathcal{Z}_{\nabla}^+(\mathcal{Y})_{\text{Ran}}$$

has an essential image in $(\text{PreStk}_{/\text{Ran}}^{\text{co-untl}})^{\text{factzbl}}$. The resulting functor

$$\text{PreStk}_{/X_{\text{dR}}} \rightarrow (\text{PreStk}_{/\text{Ran}}^{\text{co-untl}})^{\text{factzbl}}$$

is fully faithful and defines an equivalence between the full subcategories consisting of objects that are affine over X_{dR} in the left-hand side and over Ran in the right-hand side.

Proof. The fact that the essential image of the functor (C.28) lands in $(\text{PreStk}_{/\text{Ran}}^{\text{co-untl}})^{\text{factzbl}}$ has been established in Sect. B.4.2.

We have seen that the counit of the adjunction in Proposition C.6.11 is an isomorphism. Hence, the functor (C.28) is fully faithful.

In order to prove the proposition, it remains to show that the functor (C.23) is conservative on objects in $(\text{PreStk}_{/\text{Ran}}^{\text{co-untl}})^{\text{factzbl}}$ that are affine over Ran .

Let $\phi : \mathcal{Z}_{1, \text{Ran}} \rightarrow \mathcal{Z}_{2, \text{Ran}}$ be a map between two objects in $(\text{PreStk}_{/\text{Ran}}^{\text{co-untl}})^{\text{factzbl}}$, such that the map

$$\phi|_{X_{\text{dR}}} : \mathcal{Z}_{1, X_{\text{dR}}} \rightarrow \mathcal{Z}_{2, X_{\text{dR}}}$$

is an isomorphism.

In order to check that ϕ is an isomorphism, it suffices to show that it is such when restricted to X_{dR}^I for every finite non-empty set I :

$$\mathcal{Z}_{1, X_{\mathrm{dR}}^I} \rightarrow \mathcal{Z}_{2, X_{\mathrm{dR}}^I}.$$

Since both prestacks are affine over X_{dR}^I , the question of a map being an isomorphism can be checked strata-wise. Using the diagonal stratification of X^I , it suffices to show that the further restriction to

$$\overset{\circ}{X}_{\mathrm{dR}}^I \subset X_{\mathrm{dR}}^I$$

is an isomorphism. Since $\mathcal{Z}_{1, \mathrm{Ran}}$ and $\mathcal{Z}_{2, \mathrm{Ran}}$ are both factorizable, the latter map is the direct product of I copies of the map $\phi|_{X_{\mathrm{dR}}}$. $\square$

C.6.15. As in Sect. C.6.4, given a factorization space, we can talk about a counital structure on it.

It is easy to see, however, that if $\mathcal{T}$ is a counital factorization space, then the corresponding prestack $\mathcal{T}_{\mathrm{Ran}^{\mathrm{untl}, *}}$ is factorizable (in the sense of Sect. C.6.12). Vice versa, given an object

$$\mathcal{Z}_{\mathrm{Ran}^{\mathrm{untl}, *}} \in (\mathrm{PreStk}_{\mathrm{Ran}}^{\mathrm{co-untl}})^{\mathrm{factzbl}},$$

the isomorphism (C.27) defines on it a factorization structure.

So the categories of counital factorization spaces and factorizable counital prestacks over Ran are tautologically equivalent.

C.7. Unital factorization algebras.

C.7.1. Let $\mathcal{A}$ be a factorization algebra on X . A unital structure on $\mathcal{A}$ is an extension of $\mathcal{A}_{\mathrm{Ran}}$ to an object

$$\mathcal{A}_{\mathrm{Ran}^{\mathrm{untl}, *}} \in \mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl}, *}),$$

and an extension of the isomorphism (B.23) to an isomorphism

$$(C.29) \quad \mathrm{union}^!(\mathcal{A}_{\mathrm{Ran}^{\mathrm{untl}, *}})|_{(\mathrm{Ran}^{\mathrm{untl}, *} \times \mathrm{Ran}^{\mathrm{untl}, *})_{\mathrm{disj}}} \simeq \mathcal{A}_{\mathrm{Ran}^{\mathrm{untl}, *}} \boxtimes \mathcal{A}_{\mathrm{Ran}^{\mathrm{untl}, *}}|_{(\mathrm{Ran}^{\mathrm{untl}, *} \times \mathrm{Ran}^{\mathrm{untl}, *})_{\mathrm{disj}}},$$

equipped with a homotopy-coherent data of associativity and commutativity.

In addition, we stipulate that

$$(C.30) \quad \mathcal{A}_{\emptyset} \simeq k$$

and this isomorphism behaves (homotopically coherently) as a unit for the identifications (C.29), i.e., the map

$$\mathcal{A}_{\mathrm{Ran}^{\mathrm{untl}, *}} \rightarrow \mathcal{A}_{\emptyset} \otimes \mathcal{A}_{\mathrm{Ran}^{\mathrm{untl}, *}},$$

obtained by restricting (C.29) to

$$\{\emptyset\} \times \mathrm{Ran}^{\mathrm{untl}, *} \rightarrow (\mathrm{Ran}^{\mathrm{untl}, *} \times \mathrm{Ran}^{\mathrm{untl}, *})_{\mathrm{disj}}$$

identifies via (C.30) with the identity map.

Let $\mathrm{FactAlg}^{\mathrm{untl}}(X)$ denote the category of unital factorization algebras on X .

Remark C.7.2. Pullback along

$$\mathfrak{t} : \mathrm{Ran} \rightarrow \mathrm{Ran}^{\mathrm{untl}, *}$$

gives rise to a functor

$$(C.31) \quad \mathrm{FactAlg}^{\mathrm{untl}}(X) \rightarrow \mathrm{FactAlg}(X).$$

This functor is *not* fully faithful. However, by analogy with the topological situation, we expect that:

- The functor (C.31) induces a *monomorphism* on the mapping spaces;
- The functor (C.31) induces an isomorphism on the union of the components of the mapping spaces that correspond to *isomorphisms*.

The second property can be phrased as saying that for a factorization algebra, being unital is a property and not a structure.

We will not prove this in this paper. However, we will prove a result in this direction, see Proposition C.7.13.

C.7.3. Let us denote by k the unit factorization algebra, see Sect. B.9.2. Note that it naturally upgrades to a unital factorization algebra: namely, the corresponding object in $\mathrm{D-mod}(\mathrm{Ran}^{\mathrm{unl},*})$ is $\omega_{\mathrm{Ran}^{\mathrm{unl},*}}$.

Let $\mathcal{A}$ be a unital factorization algebra on X . Note that the initial point $\{\emptyset\} \in \mathrm{Ran}^{\mathrm{unl},*}$ gives rise to a map

$$\omega_{\mathrm{Ran}^{\mathrm{unl},*}} \rightarrow \mathcal{A}_{\mathrm{Ran}^{\mathrm{unl},*}}$$

in $\mathrm{D-mod}(\mathrm{Ran}^{\mathrm{unl},*})$. It follows from the axioms that this map is compatible with factorization.

I.e., we obtain a map of unital factorization algebras

$$(C.32) \quad \mathrm{vac}_{\mathcal{A}} : k \rightarrow \mathcal{A},$$

which we will refer to it as the *vacuum* map for $\mathcal{A}$.

C.7.4. Let $\mathcal{Z}$ be a prestack mapping to Ran . Consider the corresponding categorical prestack $\mathcal{Z}^{\subseteq, \mathrm{unl}}$, see Sect. C.5.16.

Let $\mathcal{A}$ be a factorization algebra on X , and let $\mathcal{M}$ be a factorization module $\mathcal{M}$ over $\mathcal{A}$ at $\mathcal{Z}$. Let $\mathcal{A}$ be equipped with a unital structure. A unital structure on $\mathcal{M}$ is an extension of $\mathcal{M}_{\mathcal{Z}^{\subseteq}}$ to an object

$$\mathcal{M}_{\mathcal{Z}^{\subseteq}} \in \mathrm{D-mod}(\mathcal{Z}^{\subseteq, \mathrm{unl}})$$

and an extension of the isomorphism (B.24) to an isomorphism

$$(C.33) \quad \mathcal{M}_{\mathcal{Z}^{\subseteq, \mathrm{unl}}} |_{(\mathrm{Ran}^{\mathrm{unl},*} \times \mathcal{Z}^{\subseteq, \mathrm{unl}})_{\mathrm{disj}}} \simeq (\mathcal{A}_{\mathrm{Ran}^{\mathrm{unl},*}} \boxtimes \mathcal{M}_{\mathcal{Z}^{\subseteq, \mathrm{unl}}}) |_{(\mathrm{Ran}^{\mathrm{unl},*} \times \mathcal{Z}^{\subseteq, \mathrm{unl}})_{\mathrm{disj}}}.$$

In addition, we stipulate that the isomorphism

$$\mathcal{M}_{\mathcal{Z}^{\subseteq, \mathrm{unl}}} \rightarrow \mathcal{A}_{\emptyset} \otimes \mathcal{M}_{\mathcal{Z}^{\subseteq, \mathrm{unl}}},$$

obtained by restricting (C.33) along

$$\{\emptyset\} \times \mathcal{Z}^{\subseteq, \mathrm{unl}} \rightarrow (\mathrm{Ran}^{\mathrm{unl},*} \times \mathcal{Z}^{\subseteq, \mathrm{unl}})_{\mathrm{disj}},$$

identifies via (C.30) with the identity map.

The contents of Sect. B.9 apply to unital factorization modules.

C.7.5. *Notational convention.* When $\mathcal{A}$ is a unital factorization algebra, we will denote by

$$\mathcal{A}\text{-mod}_{\mathcal{Z}}^{\mathrm{fact}}$$

the category of *unital* factorization modules over $\mathcal{A}$ at $\mathcal{Z}$.

The category of modules over $\mathcal{A}$ as a plain⁷⁸ factorization algebra will be denoted by

$$\mathcal{A}\text{-mod}_{\mathcal{Z}}^{\mathrm{fact-n.u.}}.$$

C.7.6. *Example.* Let $\mathcal{A}$ be a unital factorization algebra. For $\mathcal{Z} \rightarrow \mathrm{Ran}$ consider the factorization module $\mathcal{A}^{\mathrm{fact}_{\mathcal{Z}}}$ from Sect. B.9.7.

Unwinding the definitions, we obtain that $\mathcal{A}^{\mathrm{fact}_{\mathcal{Z}}}$ carries a natural unital structure.

⁷⁸I.e., non-unital.

C.7.7. *Example.* Take $\mathcal{A} = k$ from Sect. C.7.3. Note that pullback along

$$\mathrm{pr}_{\mathrm{small}, \mathcal{Z}}^{\mathrm{unl}} : \mathcal{Z}^{\subseteq, \mathrm{unl}} \rightarrow \mathcal{Z}$$

gives rise to a functor

$$(C.34) \quad \mathrm{D}\text{-mod}(\mathcal{Z}) \rightarrow k\text{-mod}_{\mathcal{Z}}^{\mathrm{fact}},$$

cf. Sect. B.9.6.

In other words, the functor (C.34) is given by tensoring (over $\mathrm{D}\text{-mod}(\mathcal{Z})$) with the object

$$k^{\mathrm{fact}, \mathcal{Z}} \in k\text{-mod}_{\mathcal{Z}}^{\mathrm{fact}}.$$

We claim that the functor (C.34) is an equivalence, with the inverse functor being

$$\mathcal{M} \mapsto \mathcal{M}_{\mathcal{Z}} := (\mathrm{diag}_{\mathcal{Z}}^{\mathrm{unl}})^!(\mathcal{M}).$$

Indeed, the fact that the map

$$\mathrm{diag}_{\mathcal{Z}}^{\mathrm{unl}} \rightarrow \mathcal{Z}^{\subseteq, \mathrm{unl}}$$

is initial relative to $\mathcal{Z}$, implies that for any $\mathcal{M} \in k\text{-mod}_{\mathcal{Z}}^{\mathrm{fact}}$, we have a canonically defined map

$$(\mathrm{pr}_{\mathrm{small}, \mathcal{Z}}^{\mathrm{unl}})^! \circ (\mathrm{diag}_{\mathcal{Z}}^{\mathrm{unl}})^!(\mathcal{M}) \rightarrow \mathcal{M}.$$

Now, the factorization condition implies that this map is actually an isomorphism.

C.7.8. Let $\mathcal{A}$ be a unital factorization algebra on X .

Restriction along

$$\mathbf{t} : \mathcal{Z}^{\subseteq} \rightarrow \mathcal{Z}^{\subseteq, \mathrm{unl}}$$

gives rise to a functor

$$(C.35) \quad \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\mathrm{fact}} \rightarrow \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\mathrm{fact-n.u.}}.$$

We have the following assertion, proved in [CR, Proposition 3.8.4]:

Proposition C.7.9. *The functor (C.35) is fully faithful with essential image*

$$\mathcal{A}\text{-mod}_{\mathcal{Z}}^{\mathrm{fact-n.u.}} \times_{k\text{-mod}_{\mathcal{Z}}^{\mathrm{fact-n.u.}}} k\text{-mod}_{\mathcal{Z}}^{\mathrm{fact}} \simeq \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\mathrm{fact-n.u.}} \times_{k\text{-mod}_{\mathcal{Z}}^{\mathrm{fact-n.u.}}} \mathrm{D}\text{-mod}(\mathcal{Z}),$$

where

$$\mathcal{A}\text{-mod}_{\mathcal{Z}}^{\mathrm{fact-n.u.}} \rightarrow k\text{-mod}_{\mathcal{Z}}^{\mathrm{fact-n.u.}}$$

is the functor of restriction along (C.32).

C.7.10. Let $\phi : \mathcal{A}_1 \rightarrow \mathcal{A}_2$ be a unital map between unital factorization algebras. From Proposition C.7.9 we obtain:

Corollary C.7.11. *The restriction functor*

$$\mathrm{Res}_{\phi} : \mathcal{A}_2\text{-mod}_{\mathcal{Z}}^{\mathrm{fact-n.u.}} \rightarrow \mathcal{A}_1\text{-mod}_{\mathcal{Z}}^{\mathrm{fact-n.u.}}$$

sends

$$\mathcal{A}_2\text{-mod}_{\mathcal{Z}}^{\mathrm{fact}} \rightarrow \mathcal{A}_1\text{-mod}_{\mathcal{Z}}^{\mathrm{fact}}.$$

C.7.12. Let $\text{FactAlg}^{\text{q-untl}}(X)$ be the category of pairs $(\mathcal{A}, \text{vac}_{\mathcal{A}})$, where $\mathcal{A}$ is a non-unital factorization algebra, and $\text{vac}_{\mathcal{A}}$ is a homomorphism $k \rightarrow \mathcal{A}$, such that the object

$$\text{Res}_{\text{vac}_{\mathcal{A}}}(\mathcal{A}^{\text{fact}_{\text{Ran}}}) \in k\text{-mod}_{\text{Ran}}^{\text{fact-n.u.}}$$

belongs to

$$\text{D-mod}(\text{Ran}) \simeq k\text{-mod}_{\text{Ran}}^{\text{fact}} \subset k\text{-mod}_{\text{Ran}}^{\text{fact-n.u.}}.$$

We will call objects of $\text{FactAlg}^{\text{q-untl}}(X)$ “quasi-unital factorization algebras”.

We have a tautological functor

$$(C.36) \quad \text{FactAlg}^{\text{untl}}(X) \rightarrow \text{FactAlg}^{\text{q-untl}}(X).$$

We claim:

Proposition C.7.13. *The functor (C.36) is an equivalence.*

The proof will be given in Sect. C.11.20.

Note that the second assertion of the proposition says that a quasi-unital factorization algebra

$$k \xrightarrow{\text{vac}_{\mathcal{A}}} \mathcal{A}$$

carries a canonical unital structure, for which $\text{vac}_{\mathcal{A}}$ is the unit.

C.8. Commutative *unital* factorization algebras.

C.8.1. By the same token as in Sect. B.10.1, one can consider the category

$$\text{ComAlg}(\text{FactAlg}^{\text{untl}}(X)).$$

It is equipped with a tautological forgetful functor

$$(C.37) \quad \text{ComAlg}(\text{FactAlg}^{\text{untl}}(X)) \rightarrow \text{ComAlg}(\text{D-mod}(\text{Ran}^{\text{untl},*}))$$

and also with a functor

$$(C.38) \quad \mathfrak{t}^! : \text{ComAlg}(\text{FactAlg}^{\text{untl}}(X)) \rightarrow \text{ComAlg}(\text{FactAlg}(X)).$$

C.8.2. Let $\mathcal{A}_{\text{Ran}^{\text{untl},*}}$ be an object of $\text{ComAlg}(\text{D-mod}(\text{Ran}^{\text{untl},*}))$. Note that the unital structure on $\mathcal{A}$ gives rise to the maps

$$(p_i)^!(\mathcal{A}_{\text{Ran}^{\text{untl},*}}) \rightarrow \text{union}^!(\mathcal{A}_{\text{Ran}^{\text{untl},*}}), \quad i = 1, 2$$

where p_1 and p_2 are the two projections $\text{Ran}^{\text{untl},*} \times \text{Ran}^{\text{untl},*} \rightarrow \text{Ran}^{\text{untl},*}$.

Since the coproduct in ComAlg is the tensor product, we obtain a map

$$(C.39) \quad \mathcal{A}_{\text{Ran}^{\text{untl},*}} \boxtimes \mathcal{A}_{\text{Ran}^{\text{untl},*}} \rightarrow \text{union}^!(\mathcal{A}_{\text{Ran}^{\text{untl},*}}).$$

The map (C.39) gives rise to a map

$$(C.40) \quad \mathcal{A}_{\text{Ran}^{\text{untl},*}} \boxtimes \mathcal{A}_{\text{Ran}^{\text{untl},*}}|_{(\text{Ran}^{\text{untl},*} \times \text{Ran}^{\text{untl},*})_{\text{disj}}} \rightarrow \text{union}^!(\mathcal{A}_{\text{Ran}^{\text{untl},*}})|_{(\text{Ran}^{\text{untl},*} \times \text{Ran}^{\text{untl},*})_{\text{disj}}}.$$

We shall say that $\mathcal{A}_{\text{Ran}^{\text{untl},*}}$ is *factorizable* if the map (C.40) is an isomorphism. Note that this automatically implies that $\mathcal{A}_{\emptyset} \simeq k$.

C.8.3. Let

$$\mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl},*}))^{\mathrm{factzble}} \subset \mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl},*}))$$

denote the full subcategory consisting of factorizable objects.

It follows from the axioms that the essential image of the functor (C.37) lands in

$$\mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl},*}))^{\mathrm{factzble}}.$$

Vice versa, for an object

$$\mathcal{A}_{\mathrm{Ran}^{\mathrm{untl},*}} \in \mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl},*}))^{\mathrm{factzble}},$$

the isomorphism (C.40) defines on $\mathcal{A}_{\mathrm{Ran}^{\mathrm{untl},*}}$ a factorization structure.

It is easy to see that the resulting two functors

$$(C.41) \quad \mathrm{ComAlg}(\mathrm{FactAlg}^{\mathrm{untl}}(X)) \leftrightarrow \mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl},*}))^{\mathrm{factzble}}$$

are mutually inverse.

C.8.4. Recall the functor

$$A \mapsto \mathrm{Fact}(A),$$

see Sect. B.10.2.

Unwinding the construction, we obtain that $\mathrm{Fact}(-)$ upgrades to a functor

$$\mathrm{ComAlg}(\mathrm{D}\text{-mod}(X)) \rightarrow \mathrm{ComAlg}(\mathrm{FactAlg}^{\mathrm{untl}}(X)).$$

By a slight abuse of notation, we will use the same symbol $\mathrm{Fact}(-)$ to denote the latter functor.

C.8.5. Consider now the functor

$$(C.42) \quad \mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl},*})) \rightarrow \mathrm{ComAlg}(\mathrm{D}\text{-mod}(X)),$$

given by restriction along $\Delta_{X, \mathrm{Ran}^{\mathrm{untl},*}} : X_{\mathrm{dR}} \rightarrow \mathrm{Ran}^{\mathrm{untl},*}$.

We claim:

Proposition C.8.6. *The functor (C.42) admits a left adjoint. Moreover, this left adjoint is fully faithful and lands in $\mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl},*}))^{\mathrm{factzble}}$.*

Proof. It is enough to prove the assertion of the proposition on objects of $\mathrm{ComAlg}(\mathrm{D}\text{-mod}(X))$ of the form

$$\mathrm{Sym}^!(\mathcal{M}), \quad \mathcal{M} \in \mathrm{D}\text{-mod}(X).$$

The value of the left adjoint on such an object is

$$(C.43) \quad \mathrm{Sym}^!((\Delta_{X, \mathrm{Ran}^{\mathrm{untl},*}})_!(\mathcal{M})).$$

To prove that this left adjoint is fully faithful, it is enough to show that the unit of the adjunction

$$\mathcal{M} \rightarrow (\Delta_{X, \mathrm{Ran}^{\mathrm{untl},*}})^! \circ (\Delta_{X, \mathrm{Ran}^{\mathrm{untl},*}})_!(\mathcal{M})$$

is an isomorphism. However, this follows from Corollary C.4.21: indeed, the categorical prestack

$$X_{\mathrm{dR}} \times_{\mathrm{Ran}^{\mathrm{untl},*}} (\mathrm{Ran}^{\mathrm{untl},*})_{\Delta_{X, \mathrm{Ran}^{\mathrm{untl},*}}} /$$

identifies with X_{dR} .

In order to show that (C.43) belongs to $\mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl},*}))^{\mathrm{factzble}}$, it suffices to show that the canonical map

$$p_1^!((\Delta_{X, \mathrm{Ran}^{\mathrm{untl},*}})_!(\mathcal{M})) \oplus p_2^!((\Delta_{X, \mathrm{Ran}^{\mathrm{untl},*}})_!(\mathcal{M})) \rightarrow \mathrm{union}^!((\Delta_{X, \mathrm{Ran}^{\mathrm{untl},*}})_!(\mathcal{M}))$$

becomes an isomorphism after restricting to $(\mathrm{Ran}^{\mathrm{untl},*} \times \mathrm{Ran}^{\mathrm{untl},*})_{\mathrm{disj}}$. This is the content of [Ro2, Theorem 2.7.6]. We include the argument for completeness.

However, this follows again from Corollary C.4.21: we have a canonical isomorphism

$$\begin{aligned} & \left((\text{Ran}^{\text{untl},*})_{\Delta_{X, \text{Ran}^{\text{untl},*}/}} \times \text{Ran}^{\text{untl},*} \sqcup \text{Ran}^{\text{untl},*} \times (\text{Ran}^{\text{untl},*})_{\Delta_{X, \text{Ran}^{\text{untl},*}/}} \right)_{\text{Ran}^{\text{untl},*} \times \text{Ran}^{\text{untl},*}} \times \\ & \quad \times_{\text{Ran}^{\text{untl},*} \times \text{Ran}^{\text{untl},*}} (\text{Ran}^{\text{untl},*} \times \text{Ran}^{\text{untl},*})_{\text{disj}} \simeq \\ & \simeq (\text{Ran}^{\text{untl},*})_{\Delta_{X, \text{Ran}^{\text{untl},*}/}} \times_{\text{Ran}^{\text{untl},*}, \text{union}} (\text{Ran}^{\text{untl},*} \times \text{Ran}^{\text{untl},*})_{\text{disj}}. \end{aligned}$$

Indeed, this is just the fact that for a disjoint pair $\underline{x}_1, \underline{x}_2$ of points of Ran , and a singleton x ,

$$x \subseteq \underline{x}_1 \cup \underline{x}_2 \Leftrightarrow x \subseteq \underline{x}_1 \text{ or } x \subseteq \underline{x}_2.$$

□

C.8.7. As a corollary, we obtain:

Corollary C.8.8. *The composite functor*

$$\text{ComAlg}(\text{FactAlg}^{\text{untl}}(X)) \rightarrow \text{ComAlg}(\text{D-mod}(\text{Ran}^{\text{untl},*})) \xrightarrow{\Delta_{X, \text{Ran}^{\text{untl},*}}^!} \text{ComAlg}(\text{D-mod}(X))$$

is an equivalence, with the inverse given by $\text{Fact}(-)$.

Proof. Given Proposition C.8.6, we only need to prove that the functor in Corollary C.8.8 is conservative. But this is immediate from the factorization. □

Corollary C.8.9. *The functor*

$$\text{ComAlg}(\text{D-mod}(X)) \xrightarrow{\text{Fact}(-)} \text{ComAlg}(\text{FactAlg}^{\text{untl}}(X)) \rightarrow \text{ComAlg}(\text{D-mod}(\text{Ran}^{\text{untl},*}))$$

is the left adjoint of

$$\text{ComAlg}(\text{D-mod}(\text{Ran}^{\text{untl},*})) \xrightarrow{\Delta_{X, \text{Ran}^{\text{untl},*}}^!} \text{ComAlg}(\text{D-mod}(X)).$$

C.8.10. Let $\mathcal{Y}$ be a D-prestack over X . Suppose that the prestack $\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\text{Ran}} \rightarrow \text{Ran}$ is such that the formation of direct image of the structure sheaf along

$$\mathfrak{L}_{\nabla}^+(\mathcal{Y})_S \rightarrow S, \quad S \in \text{Sch}_{\text{Ran}}^{\text{aff}}$$

is compatible with base change, and satisfies Kunneth formula.

This happens, e.g., when $\mathcal{Y}$ is affine over X , and hence $\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\text{Ran}}$ is affine over Ran .

Taking the direct image of the structure sheaf along

$$\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\text{Ran}^{\text{untl},*}} \rightarrow \text{Ran}^{\text{untl},*},$$

we obtain an object in $\text{ComAlg}(\text{D-mod}(\text{Ran}^{\text{untl},*}))$, which by a slight abuse of notation we denote by $\mathcal{O}_{\mathfrak{L}_{\nabla}^+(\mathcal{Y}), \text{Ran}^{\text{untl},*}}$. By Kunneth formula and factorization, $\mathcal{O}_{\mathfrak{L}_{\nabla}^+(\mathcal{Y}), \text{Ran}^{\text{untl},*}}$ has a natural structure of factorization algebra.

Denote the resulting object by

$$\mathcal{O}_{\mathfrak{L}_{\nabla}^+(\mathcal{Y})} \in \text{FactAlg}(X).$$

The value of $\mathcal{O}_{\mathfrak{L}_{\nabla}^+(\mathcal{Y})}$ on X , i.e., the restriction of $\mathcal{O}_{\mathfrak{L}_{\nabla}^+(\mathcal{Y}), \text{Ran}^{\text{untl},*}}$ along $X \rightarrow \text{Ran}^{\text{untl},*}$, is the direct image of the structure sheaf along $\mathcal{Y} \rightarrow X$, which by a slight abuse of notation we denote by $\mathcal{O}_{\mathcal{Y}}$.

Hence, by the equivalence of Corollary C.8.8, we obtain that

$$(C.44) \quad \mathcal{O}_{\mathfrak{L}_{\nabla}^+(\mathcal{Y})} \simeq \text{Fact}(\mathcal{O}_{\mathcal{Y}}).$$

C.8.11. Assume that $\mathcal{Y}$ is affine over X , i.e., $\mathcal{Y} = \mathrm{Spec}_X(A)$ for $A \in \mathrm{ComAlg}(\mathrm{D}\text{-mod}(X))$ with $\mathbf{obl}^I(A)$ is connective.

Let $\mathcal{A} := \mathrm{Fact}(A)$. Then (C.44) says that for $S \rightarrow \mathrm{Ran}$,

$$(C.45) \quad \mathfrak{L}_{\nabla}^+(\mathcal{Y})_S \simeq \mathrm{Spec}_S(\mathcal{A}_S).$$

C.8.12. Note that the identification (C.45) is in agreement with Proposition C.6.11.

Namely, let us be given a counital prestack $\mathcal{Z}_{\mathrm{Ran}}$ over Ran , such that the direct image of its structure sheaf satisfies base change. Let

$$\mathcal{B}_{\mathrm{Ran}^{\mathrm{untl},*}} \in \mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl},*}))$$

denote the corresponding object. Set $\mathcal{B}_X := \mathcal{B}_{\mathrm{Ran}^{\mathrm{untl},*}}|_{X_{\mathrm{dR}}}$.

Then the following diagram commutes:

$$\begin{array}{ccc} \mathrm{Maps}_{\mathrm{PreStk}^{\mathrm{co-untl}}/\mathrm{Ran}}(\mathcal{Z}_{\mathrm{Ran}^{\mathrm{untl},*}}, \mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\mathrm{Ran}^{\mathrm{untl},*}}) & \xrightarrow[\sim]{\text{Proposition C.6.11}} & \mathrm{Maps}_{X_{\mathrm{dR}}}(X_{\mathrm{dR}} \times_{\mathrm{Ran}^{\mathrm{untl},*}} \mathcal{Z}_{\mathrm{Ran}^{\mathrm{untl},*}}, \mathcal{Y}_{\nabla}) \\ (C.45) \downarrow \sim & & \downarrow \sim \\ \mathrm{Maps}_{\mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl},*}))}(\mathcal{A}, \mathcal{B}_{\mathrm{Ran}^{\mathrm{untl},*}}) & \xrightarrow{\text{Proposition C.8.6}} & \mathrm{Maps}_{\mathrm{ComAlg}(\mathrm{D}\text{-mod}(X))}(A, \mathcal{B}_X). \end{array}$$

C.9. Factorization homology of commutative factorization algebras.

C.9.1. Consider the functor

$$\mathrm{ComAlg}(\mathrm{Vect}) \rightarrow \mathrm{ComAlg}(\mathrm{D}\text{-mod}(X)), \quad R \mapsto R \otimes \mathcal{O}_X,$$

where $\mathcal{O}_X$ is perceived as a *left* D-module.

In this subsection, we will describe, following [BD2, Sect. 4.6.1], its left adjoint.

C.9.2. From Lemma C.5.9 we obtain that the functor $\mathrm{C}_c(\mathrm{Ran}^{\mathrm{untl}}, -)$ gives rise to a functor

$$\mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl}})) \rightarrow \mathrm{ComAlg}(\mathrm{Vect}),$$

left adjoint to

$$\mathrm{ComAlg}(\mathrm{Vect}) \rightarrow \mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl}})), \quad R \mapsto R \otimes \omega_{\mathrm{Ran}^{\mathrm{untl}}}.$$

Remark C.9.3. Note that by Sect. C.5.15, the restriction of $\mathrm{C}_c(\mathrm{Ran}, -)$ to the subcategory

$$\mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran})^{\mathrm{almost-untl}}) \subset \mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}))$$

defines a left adjoint to the functor

$$\mathrm{ComAlg}(\mathrm{Vect}) \rightarrow \mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran})^{\mathrm{almost-untl}}), \quad R \mapsto R \otimes \omega_{\mathrm{Ran}}.$$

C.9.4. Recall now that according to Corollary C.8.9, the functor

$$\mathrm{ComAlg}(\mathrm{D}\text{-mod}(X)) \xrightarrow{\mathrm{Fact}} \mathrm{ComAlg}(\mathrm{FactAlg}^{\mathrm{untl}}(X)) \rightarrow \mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl},*}))$$

provides a left adjoint to the restriction functor

$$\mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl},*})) \rightarrow \mathrm{ComAlg}(\mathrm{D}\text{-mod}(X)).$$

Combined with Sect. C.9.2, we obtain:

Corollary C.9.5. *The functor*

$$\begin{aligned} \mathrm{ComAlg}(\mathrm{D}\text{-mod}(X)) &\xrightarrow{\mathrm{Fact}} \mathrm{ComAlg}(\mathrm{FactAlg}^{\mathrm{untl}}(X)) \rightarrow \\ &\rightarrow \mathrm{ComAlg}(\mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl},*})) \xrightarrow{\mathrm{C}_c(\mathrm{Ran}^{\mathrm{untl},*}, -)} \mathrm{ComAlg}(\mathrm{Vect}) \end{aligned}$$

is the left adjoint of

$$R \mapsto R \otimes \mathcal{O}_X, \quad \mathrm{ComAlg}(\mathrm{Vect}) \rightarrow \mathrm{ComAlg}(\mathrm{D}\text{-mod}(X)).$$

C.9.6. For $\mathcal{A} \in \text{FactAlg}^{\text{unfl}}(X)$ recall the object

$$C^{\text{fact}}(X, \mathcal{A}) = C_c(\text{Ran}, \mathcal{A}_{\text{Ran}}),$$

see Sect. 11.9.7.

Note that by Lemma C.5.12, we can rewrite this also as

$$C_c(\text{Ran}^{\text{unfl}}, \mathcal{A}_{\text{Ran}^{\text{unfl}}}).$$

C.9.7. Thus, Corollary C.9.5 says that the functor

$$A \mapsto C^{\text{fact}}(X, \text{Fact}(A)), \quad \text{ComAlg}(\text{D-mod}(X)) \rightarrow \text{ComAlg}(\text{Vect})$$

is the left adjoint of

$$R \mapsto R \otimes \mathcal{O}_X, \quad \text{ComAlg}(\text{Vect}) \rightarrow \text{ComAlg}(\text{D-mod}(X)).$$

Remark C.9.8. Note that when we think of $C^{\text{fact}}(X, \mathcal{A})$ as $C_c(\text{Ran}, \mathcal{A}_{\text{Ran}})$, the commutative algebra structure on it follows from Remark C.9.3, since

$$\mathcal{A}_{\text{Ran}} \in \text{D-mod}(\text{D-mod}_{\text{Ran}})^{\text{almost-unfl}}.$$

C.9.9. Let $A \rightarrow B$ be a map in $\text{ComAlg}(\text{D-mod}(X))$. Denote

$$\mathcal{A} := \text{Fact}(A), \quad \mathcal{B} := \text{Fact}(B).$$

Let R be an object of $\text{ComAlg}(\text{Vect})$, and fix a map

$$A \rightarrow R \otimes \mathcal{O}_X,$$

or, equivalently by Corollary C.9.5, a map

$$C^{\text{fact}}(X, \mathcal{A}) \rightarrow R.$$

C.9.10. Denote

$$B_R := B \otimes_A (R \otimes \mathcal{O}_X).$$

We can view B_R as an object of

$$\text{ComAlg}(\text{D-mod}(X) \otimes R\text{-mod}),$$

i.e., as an R -linear object in $\text{ComAlg}(\text{D-mod}(X))$.

C.9.11. Consider the corresponding object

$$\mathcal{B}_R \in \text{ComAlg}(\text{FactAlg}(X) \otimes R\text{-mod}).$$

We can apply the construction of factorization homology in the R -linear context, and thus form

$$C^{\text{fact}}(X, \mathcal{B}_R) \in \text{ComAlg}(R\text{-mod}).$$

C.9.12. We have the naturally defined maps in $\text{ComAlg}(\text{Vect})$

$$R \rightarrow C^{\text{fact}}(X, \mathcal{B}_R) \leftarrow C^{\text{fact}}(X, \mathcal{B}),$$

which fit into the commutative diagram

$$\begin{array}{ccc} C^{\text{fact}}(X, \mathcal{A}) & \longrightarrow & R \\ \downarrow & & \downarrow \\ C^{\text{fact}}(X, \mathcal{B}) & \longrightarrow & C^{\text{fact}}(X, \mathcal{B}_R). \end{array}$$

In particular, we obtain a map

$$(C.46) \quad R \otimes_{C^{\text{fact}}(X, \mathcal{A})} C^{\text{fact}}(X, \mathcal{B}) \rightarrow C^{\text{fact}}(X, \mathcal{B}_R).$$

C.9.13. We claim:

Lemma C.9.14. *The map (C.46) is an isomorphism.*

Proof. Follows immediately from Corollary C.9.5. □

C.10. Unitality in correspondences.

C.10.1. Let $\Phi : \mathbf{C} \rightarrow \mathbf{D}$ be a functor between ∞ -categories. We shall say that Φ is a *fibration-in-correspondences*⁷⁹ if the following two conditions hold:

- For every $\mathbf{d} \in \mathbf{D}$, the fiber $\mathbf{C}_{\mathbf{d}}$ is a groupoid;
- For every composable pair of arrows $\mathbf{d}_0 \xrightarrow{\alpha_{0,1}} \mathbf{d}_1 \xrightarrow{\alpha_{1,2}} \mathbf{d}_2$ in $\mathbf{D}$, the map

$$\mathbf{C}_{\alpha_{0,1}} \times_{\mathbf{C}_{\mathbf{d}_1}} \mathbf{C}_{\alpha_{1,2}} \rightarrow \mathbf{C}_{\alpha_{1,2} \circ \alpha_{0,1}}$$

is an isomorphism, where for an arrow $\mathbf{d}' \xrightarrow{\alpha} \mathbf{d}''$, viewed as a functor $[0, 1] \rightarrow \mathbf{D}$, we denote by $\mathbf{C}_{\alpha}$ the category of lifts of $\alpha : [0, 1] \rightarrow \mathbf{D}$ to a functor $[0, 1] \rightarrow \mathbf{C}$, i.e.,

$$\text{Funct}_{/\mathbf{D}}([0, 1], \mathbf{C}).$$

Note that the second condition can be reformulated as follows: given an arrow $\mathbf{c}_0 \xrightarrow{\beta_{0,2}} \mathbf{c}_2$ in $\mathbf{C}$ and a factorization of its image $\Phi(\beta_{0,2})$ as

$$\Phi(\mathbf{d}_0) =: \mathbf{d}_0 \xrightarrow{\alpha_{0,1}} \mathbf{d}_1 \xrightarrow{\alpha_{1,2}} \mathbf{d}_2 := \Phi(\mathbf{d}_2)$$

the space of the factorizations

$$\beta_{0,2} = \beta_{1,2} \circ \beta_{0,1}, \quad \Phi(\beta_{0,1}) = \alpha_{0,1}, \quad \Phi(\beta_{1,2}) = \alpha_{1,2}$$

is contractible.

Remark C.10.2. There is a version of straightening construction that attaches to a fibration-in-correspondences $\Phi : \mathbf{C} \rightarrow \mathbf{D}$ a functor from $\mathbf{D}$ to $\text{Corr}(\text{Grpd})$, i.e., the category, whose objects are groupoids and the morphisms are correspondences between groupoids, see [AF, Theorem 0.8(2) and Theorem 0.10(1)].

C.10.3. Note that if Φ is either a Cartesian or a co-Cartesian fibration in groupoids, then it is a fibration-in-correspondences.

C.10.4. Let $\Phi : \mathbf{C} \rightarrow \mathbf{D}$ be a fibration-in-correspondences. Suppose for a moment that $\mathbf{D}$ contains an initial object $\{\emptyset\}$. Let $\mathbf{C}^+$ denote the category

$$\{\mathbf{c}_{\emptyset} \in \mathbf{C}_{\{\emptyset\}}, \mathbf{c} \in \mathbf{C}, \mathbf{c}_{\emptyset} \xrightarrow{\beta} \mathbf{c}\}.$$

Note the functor

$$\Phi^+ : \mathbf{C}^+ \rightarrow \mathbf{D}, \quad (\mathbf{c}_{\emptyset}, \mathbf{c}, \beta) \mapsto \Phi(\mathbf{c})$$

is a Cartesian fibration in groupoids.

Indeed, given

$$\mathbf{c}_{\emptyset} \xrightarrow{\beta} \mathbf{c} \text{ and } \mathbf{d}' \rightarrow \Phi(\mathbf{c}),$$

we let

$$\mathbf{c}_{\emptyset} \xrightarrow{\beta'} \mathbf{c}' \rightarrow \mathbf{c}$$

be its unique factorization covering the canonical factorization

$$\{\emptyset\} \rightarrow \mathbf{d}' \rightarrow \Phi(\mathbf{c}).$$

Note that if $\mathbf{C} \rightarrow \mathbf{D}$ is co-Cartesian fibration in groupoids, then $\mathbf{C}^+ \simeq \mathbf{C}_{\emptyset} \times \mathbf{D}$. If $\mathbf{C} \rightarrow \mathbf{D}$ is Cartesian fibration in groupoids, then $\mathbf{C}^+ \rightarrow \mathbf{C}$ is an equivalence.

⁷⁹Another name for this is “conservative exponentiable fibration”, see [AF].

C.10.5. Let $\mathcal{Z}_{\text{Ran}} \rightarrow \text{Ran}$ be a prestack. A unital-in-correspondences structure on it is an extension of $\mathcal{Z}_{\text{Ran}}$ to a categorical prestack

$$\mathcal{Z}_{\text{Ran}^{\text{untl},*}} \rightarrow \text{Ran}^{\text{untl},*},$$

which is a value-wise fibration-in-correspondences

C.10.6. Note that the construction in Sect. C.10.4 associates to such $\mathcal{Z}_{\text{Ran}}$ a prestack $\mathcal{Z}_{\text{Ran}}^+$, equipped with a counital structure.

In what follows we will say that $\mathcal{Z}_{\text{Ran}}$ admits a unital-in-correspondences structure *relative* to $\mathcal{Z}^+$. And we will refer to $\mathcal{Z}_{\text{Ran}}^+$ as the counital prestack *underlying* $\mathcal{Z}_{\text{Ran}}$.

For an arrow in $\text{Ran}^{\text{untl},*}$ given by $\underline{x} \subseteq \underline{x}'$ we will denote by

$$\mathcal{Z}_{\underline{x} \subseteq \underline{x}'}^{\text{all} \rightsquigarrow +}$$

the prestack of its lifts to $\mathcal{Z}_{\text{Ran}}$ (see Sect. C.10.1 above). It is equipped with maps

$$\mathcal{Z}_{\underline{x}} \xleftarrow{\text{pr}_{\text{small}}^{\mathcal{Z}}} \mathcal{Z}_{\underline{x} \subseteq \underline{x}'}^{\text{all} \rightsquigarrow +} \xrightarrow{\text{pr}_{\text{big}}^{\mathcal{Z}}} \mathcal{Z}_{\underline{x}'}.$$

C.10.7. Note that by Sect. C.10.4, a unital structure on $\mathcal{Z}_{\text{Ran}}$ gives rise to a unital-in-correspondences structure, with $\mathcal{Z}^+ \simeq \mathcal{Z}_{\emptyset} \times \text{Ran}$.

A counital structure on $\mathcal{Z}_{\text{Ran}}$ gives rise to a unital-in-correspondences structure with $\mathcal{Z}_{\text{Ran}}^+ \rightarrow \mathcal{Z}_{\text{Ran}}$ being an isomorphism.

C.10.8. Let $\mathcal{T}$ be factorization space over X . There is a natural notion of unital-in-correspondences structure on $\mathcal{T}$ (i.e., $\mathcal{T}_{\text{Ran}}$ has a unital-in-correspondences structure, compatible with factorization, and we stipulate $\mathcal{T}_{\emptyset} \simeq \text{pt}$).

Let $\mathcal{T}^+$ be the corresponding counital factorization space (see Sect. C.10.4); in this case we will say that $\mathcal{T}$ has a unital-in-correspondences structure *relative* to $\mathcal{T}^+$. We will refer to $\mathcal{T}^+$ as the counital factorization space *underlying* $\mathcal{T}$.

C.10.9. Let $\mathcal{T}$ be a unital factorization space. Then $\mathcal{T}$ acquires a natural unital-in-correspondences structure, for which $\mathcal{T}^+ \rightarrow \text{pt}$ is an isomorphism.

According to Sect. C.10.4, if $\mathcal{T}$ is counital as a factorization space, it acquires a natural structure of unitality-in-correspondences with $\mathcal{T}^+ \rightarrow \mathcal{T}$ being an isomorphism.

C.10.10. Let $\mathcal{Y}$ be an affine D-scheme over X . We claim that the factorization space $\mathfrak{L}_{\nabla}(\mathcal{Y})$ (see Sect. B.4.6) possesses a natural unital-in-correspondences structure relative to $\mathfrak{L}_{\nabla}^+(\mathcal{Y})$.

Namely, let $\underline{x} \subseteq \underline{x}' : S \rightarrow \text{Ran}$ be a pair of S -points of Ran . Consider the corresponding prestack $\mathcal{D}_{\underline{x}', \nabla}$. It contains $\text{Graph}_{\underline{x}}$ as a closed subset.

By definition, a lift of $\underline{x} \subseteq \underline{x}'$ to an S -point of $\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x} \subseteq \underline{x}'}^{\text{all} \rightsquigarrow +}$ is a X_{dR} -map

$$(\mathcal{D}_{\underline{x}', \nabla} - \text{Graph}_{\underline{x}}) \rightarrow \mathcal{Y}_{\nabla}.$$

NB: in the particular case of $\mathcal{T} = \mathfrak{L}_{\nabla}(\mathcal{Y})$, we use the notation $\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\underline{x} \subseteq \underline{x}'}$ instead of $\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x} \subseteq \underline{x}'}^{\text{all} \rightsquigarrow +}$.

In order to define the composition of morphisms, we need to establish an isomorphism

$$\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\underline{x}_1 \subseteq \underline{x}_2} \times_{\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}_2}^{\text{mer}}} \mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\underline{x}_2 \subseteq \underline{x}_3} \simeq \mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\underline{x}_1 \subseteq \underline{x}_3}$$

for $\underline{x}_1 \subseteq \underline{x}_2 \subseteq \underline{x}_3$. This follows from the isomorphism

$$(\mathcal{D}_{\underline{x}_2, \nabla} - \underline{x}_1) \sqcup_{\mathcal{D}_{\underline{x}_2, \nabla} - \underline{x}_2} (\mathcal{D}_{\underline{x}_3, \nabla} - \underline{x}_2) \simeq (\mathcal{D}_{\underline{x}_3, \nabla} - \underline{x}_1),$$

cf. Lemma C.6.6.

C.10.11. We will now define a unital-in-correspondences structure on the factorization space $\mathrm{LS}_H^{\mathrm{mer}}$ from Sect. B.7.10.

Let $\underline{x} \subseteq \underline{x}' : S \rightarrow \mathrm{Ran}$ be as above. We consider $\mathfrak{L}_{\nabla}(\mathrm{Conn}(\mathfrak{h}))_{\underline{x}'}$ as acted on by $\mathfrak{L}(H)_{\underline{x}'}$. Similarly, $\mathfrak{L}_{\nabla}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}(\mathrm{Conn}(\mathfrak{h}))_{\underline{x} \subseteq \underline{x}'}$ is acted on by

$$\mathfrak{L}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}(H)_{\underline{x} \subseteq \underline{x}'}$$

Let $\mathrm{LS}_{H, \underline{x} \subseteq \underline{x}'}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}$ be the prestack equal to the étale sheafification of the (non-sheafified) quotient of $\mathfrak{L}_{\nabla}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}(\mathrm{Conn}(\mathfrak{h}))_{\underline{x} \subseteq \underline{x}'}$ by $\mathfrak{L}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}(H)_{\underline{x} \subseteq \underline{x}'}$.

The composition of morphisms is defined as in Sect. C.10.10.

Note that the underlying counital factorization space of $\mathrm{LS}_H^{\mathrm{mer}}$ is $\mathrm{LS}_H^{\mathrm{reg}}$.

C.10.12. We now define a unital-in-correspondences structure on $\mathrm{Op}_G^{\mathrm{mon-free}}$. This is, however, automatic since

$$\mathrm{Op}_G^{\mathrm{mon-free}} := \mathrm{Op}_G^{\mathrm{mer}} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{LS}_G^{\mathrm{reg}},$$

so the unital-in-correspondences structure on $\mathrm{Op}_G^{\mathrm{mer}}$, $\mathrm{LS}_G^{\mathrm{mer}}$ and $\mathrm{LS}_G^{\mathrm{reg}}$ induces one on $\mathrm{Op}_G^{\mathrm{mon-free}}$.

Note that by construction, the underlying counital factorization space of $\mathrm{Op}_G^{\mathrm{mon-free}}$ is $\mathrm{Op}_G^{\mathrm{reg}}$.

C.11. Unital factorization categories.

C.11.1. Let $\mathbf{A}$ be a factorization category on X . A unital structure on $\mathbf{A}$ is an extension of the crystal of categories $\underline{\mathbf{A}}$ over Ran to a crystal of categories over $\mathrm{Ran}^{\mathrm{untl},*}$ in a way compatible with factorization, i.e., we extend the identifications (B.37) to

$$(C.47) \quad \mathrm{union}^*(\underline{\mathbf{A}})|_{(\mathrm{Ran}^{\mathrm{untl},*} \times \mathrm{Ran}^{\mathrm{untl},*})_{\mathrm{disj}}} \simeq \underline{\mathbf{A}} \boxtimes \underline{\mathbf{A}}|_{(\mathrm{Ran}^{\mathrm{untl},*} \times \mathrm{Ran}^{\mathrm{untl},*})_{\mathrm{disj}}}.$$

In addition, we stipulate

$$(C.48) \quad \mathbf{A}_{\emptyset} \simeq \mathrm{Vect},$$

so that this identification behaves (homotopically coherently) as a unit for the identifications (C.47), i.e., the functor

$$\underline{\mathbf{A}} \rightarrow \mathbf{A}_{\emptyset} \otimes \underline{\mathbf{A}},$$

obtained by restricting (C.47) to

$$\{\emptyset\} \times \mathrm{Ran}^{\mathrm{untl},*} \rightarrow (\mathrm{Ran}^{\mathrm{untl},*} \times \mathrm{Ran}^{\mathrm{untl},*})_{\mathrm{disj}}$$

identifies via (C.48) with the identity functor.

We let

$$\mathbf{1}_{\mathbf{A}} \in \Gamma^{\mathrm{strict}}(\mathrm{Ran}^{\mathrm{untl},*}, \underline{\mathbf{A}})$$

the canonical object, whose value at any $\underline{x} \in \mathrm{Ran}^{\mathrm{untl},*}$ is

$$\mathrm{ins. unit}_{\emptyset \subseteq \underline{x}}(k),$$

where:

- $k \in \mathrm{Vect} \simeq \mathbf{A}_{\emptyset}$;
- $\mathrm{ins. unit}_{\emptyset \subseteq \underline{x}}$ is the functor $\mathbf{A}_{\emptyset} \rightarrow \mathbf{A}_{\underline{x}}$ corresponding to the unique morphism $\{\emptyset\} \rightarrow \underline{x}$.

C.11.2. Note that the factorization category Vect from Sect. B.11.2 admits a tautological unital structure. Namely, the underlying crystal of categories on $\mathrm{Ran}^{\mathrm{untl},*}$ is $\underline{\mathrm{D-mod}}(\mathrm{Ran}^{\mathrm{untl},*})$.

C.11.3. Given a pair of unital factorization categories, we can talk about lax unital or strictly unital functors between them, compatible with factorization, see Sect. C.2.8.

We denote the resulting (2-)categories by

$$\mathrm{FactCat}^{\mathrm{untl}, \mathrm{lax}}(X) \text{ and } \mathrm{FactCat}^{\mathrm{untl}, \mathrm{strict}}(X),$$

respectively.

C.11.4. Pointwise tensor product defines a symmetric monoidal structure on the category of unital factorization categories (for both variants: lax or strictly unital functors).

The unit for the above symmetric monoidal structure is $\mathbf{Vect}$.

C.11.5. Let $\mathbf{A}$ be a unital factorization category. We shall say that $\mathbf{A}$ is dualizable if it is a dualizable as an object of the above category, with *lax* unital functors as morphisms.

In this case, the evaluation and the coevaluation functors

$$\mathbf{A} \otimes \mathbf{A}^\vee \rightarrow \mathbf{Vect} \text{ and } \mathbf{Vect} \rightarrow \mathbf{A}^\vee \otimes \mathbf{A}$$

carry lax unital structures.

C.11.6. *Example.* Let $\mathbf{A}_X$ be as in Sect. B.11.6. The explicit construction of the (symmetric monoidal) factorization category $\mathbf{Fact}(\mathbf{A}_X)$ shows that it admits a natural unital structure.

We will denote by the same symbol $\mathbf{Fact}(\mathbf{A}_X)$ the resulting (symmetric monoidal) unital factorization category.

C.11.7. Given a unital factorization category $\mathbf{A}$, we can talk about unital factorization algebras in it: by definition, a unital factorization algebra $\mathcal{A}$ in $\mathbf{A}$ is a lax unital factorization functor $\mathbf{Vect} \rightarrow \mathbf{A}$.

Explicitly, the datum of $\mathcal{A}$ is an object

$$\mathcal{A}_{\mathbf{Ran}^{\text{unl},*}} \in \Gamma^{\text{lax}}(\mathbf{Ran}^{\text{unl},*}, \mathbf{A}),$$

which is compatible with factorization in the natural sense (i.e., combine the ideas from Sects. C.7.1 and B.11.4).

We denote by

$$\mathbf{FactAlg}^{\text{unl}}(X, \mathbf{A})$$

the category of unital factorization algebras in $\mathbf{A}$.

The object $\mathbf{1}_{\mathbf{A}}$ has a natural structure of unital factorization algebra in $\mathbf{A}$. Furthermore, for any $\mathcal{A} \in \mathbf{FactAlg}^{\text{unl}}(X, \mathbf{A})$ we have a canonically defined map

$$\text{vac}_{\mathcal{A}} : \mathbf{1}_{\mathbf{A}} \rightarrow \mathcal{A},$$

which we will refer to as the *unit* or *vacuum* for $\mathcal{A}$.

C.11.8. If $\mathbf{A}$ is a unital factorization category, it admits a canonically defined (strictly) unital functor

$$\mathbf{Vac}_{\mathbf{A}} : \mathbf{Vect} \rightarrow \mathbf{A}.$$

By a slight abuse of notation, we will denote this functor by $\mathbf{1}_{\mathbf{A}}$; the image of

$$k \in \mathbf{FactAlg}^{\text{unl}}(X, \mathbf{Vect})$$

under $\mathbf{Vac}_{\mathbf{A}}$ is $\mathbf{1}_{\mathbf{A}}$.

C.11.9. Given $\mathcal{A} \in \mathbf{FactAlg}^{\text{unl}}(X, \mathbf{A})$ and $\mathcal{Z} \rightarrow \mathbf{Ran}$, we can talk about unital factorization $\mathcal{A}$ -modules at $\mathcal{Z}$. Denote this category by

$$\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})_{\mathcal{Z}}.$$

When we talk about *non-unital* factorization $\mathcal{A}$ -modules, we will denote the corresponding category by

$$\mathcal{A}\text{-mod}^{\text{fact-n.u.}}(\mathbf{A})_{\mathcal{Z}}.$$

We have a forgetful functor

$$\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})_{\mathcal{Z}} \rightarrow \mathcal{A}\text{-mod}^{\text{fact-n.u.}}(\mathbf{A})_{\mathcal{Z}}$$

and Proposition C.7.9 applies in the present context as well.

The assignment

$$\mathcal{Z} \rightsquigarrow \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})_{\mathcal{Z}}$$

is a crystal of categories over $\mathbf{Ran}$ that we will denote by $\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})$. This crystal of categories has a natural lax factorization structure.

We will denote the resulting lax factorization category by $\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})$.

C.11.10. Let $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$ be a lax unital functor between unital factorization categories. Then it naturally gives rise to a functor

$$\Phi : \text{FactAlg}^{\text{unital}}(X, \mathbf{A}_1) \rightarrow \text{FactAlg}^{\text{unital}}(X, \mathbf{A}_2).$$

In particular, $\Phi(\mathbf{1}_{\mathbf{A}_1})$ has a natural structure of factorization algebra in $\mathbf{A}_2$, and we have a map of factorization algebras

$$(C.49) \quad \mathbf{1}_{\mathbf{A}_2} \rightarrow \Phi(\mathbf{1}_{\mathbf{A}_1}).$$

C.11.11. Let $\underline{\mathbf{A}}_X$ be as in Sect. C.11.6. Let $A \in \mathbf{A}_X$ be a commutative algebra object. Then the object

$$\text{Fact}(A) \in \text{ComAlg}(\text{FactAlg}(X, \text{Fact}(\underline{\mathbf{A}}_X)))$$

from Sect. B.11.6 naturally lifts to an object of

$$\text{ComAlg}(\text{FactAlg}^{\text{unital}}(X, \text{Fact}(\underline{\mathbf{A}}_X))).$$

By a slight abuse of notation, we will denote it by the same symbol $\text{Fact}(A)$.

C.11.12. By a similar token, one defines the notion of unital structure on a *lax* factorization category (see Sect. B.11.12). The entire preceding discussion equally applies to unital lax factorization categories.

C.11.13. *Example.* Let $\mathcal{A}$ be a (not necessarily unital) factorization algebra. Recall the lax factorization category $\mathcal{A}\text{-mod}^{\text{fact}}$ (see Sect. B.11.15). We claim that it acquires a natural unital structure.

In order to define it, we need to provide the following data: for any $\mathcal{Z} \rightarrow \mathbf{Ran}$, we need to define a functor

$$(C.50) \quad \text{ins. unit}_{\mathcal{Z}} : \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}} \rightarrow \mathcal{A}\text{-mod}_{\mathcal{Z}^{\subseteq}}^{\text{fact}},$$

equipped with an appropriate associativity structure. This construction was already mentioned in Sect. B.9.8:

Let $\mathcal{M}$ be an object of $\mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}}$. The corresponding object $\text{ins. unit}_{\mathcal{Z}}(\mathcal{M}) \in \mathcal{A}\text{-mod}_{\mathcal{Z}^{\subseteq}}^{\text{fact}}$ is constructed as follows.

Note that there is a canonical projection

$$\mathcal{Z}^{\subseteq^2} := (\mathcal{Z}^{\subseteq})^{\subseteq} \xrightarrow{\text{pr}_{\text{comp}, \mathcal{Z}}} \mathcal{Z}^{\subseteq}, \quad (z, \underline{x}_1 \subseteq \underline{x}_2) \mapsto (z, \underline{x}_2).$$

We let

$$(\text{ins. unit}_{\mathcal{Z}}(\mathcal{M}))_{\mathcal{Z}^{\subseteq^2}} := (\text{pr}_{\text{comp}, \mathcal{Z}})^!(\mathcal{M}_{\mathcal{Z}^{\subseteq}}).$$

The factorization structure on $(\text{ins. unit}_{\mathcal{Z}}(\mathcal{M}))_{\mathcal{Z}^{\subseteq^2}}$ against $\mathcal{A}$ is induced by that on $\mathcal{M}_{\mathcal{Z}^{\subseteq}}$.

The factorization unit in $\mathcal{A}\text{-mod}^{\text{fact}}$ is the object $\mathcal{A}^{\text{fact}}$ from Sect. B.11.15.

Let $\phi : \mathcal{A}_1 \rightarrow \mathcal{A}_2$ be a map of (non-unital) factorization algebras. Then the functor

$$\text{Res}_{\phi} : \mathcal{A}_2\text{-mod}^{\text{fact}} \rightarrow \mathcal{A}_1\text{-mod}^{\text{fact}}$$

carries a natural lax unital structure.

C.11.14. Let now $\mathcal{A}$ be a unital factorization algebra. The above construction applies verbatim to the category of unital factorization $\mathcal{A}$ -modules.

Consider the forgetful functor

$$\mathbf{oblv}_{\mathcal{A}} : \mathcal{A}\text{-mod}^{\text{fact}} \rightarrow \mathbf{Vect}$$

(see (B.25)) as a factorization functor (where the left-hand side is a lax factorization category). We claim that it carries a naturally defined lax unital structure.

In order to define it, we need to provide the following data: for any $\mathcal{Z} \rightarrow \mathbf{Ran}$ and $\mathcal{M} \in \mathcal{A}\text{-mod}_{\mathcal{Z}}^{\text{fact}}$ we need to define a map

$$\mathrm{pr}_{\text{small}, \mathcal{Z}}^! (\mathcal{M}_{\mathcal{Z}}) \rightarrow (\mathrm{ins. unit}_{\mathcal{Z}}(\mathcal{M}))_{\mathcal{Z} \subseteq}.$$

We note, however, that by construction

$$(\mathrm{ins. unit}_{\mathcal{Z}}(\mathcal{M}))_{\mathcal{Z} \subseteq} := \mathcal{M}_{\mathcal{Z} \subseteq}.$$

Now, the sought-for map

$$\mathrm{pr}_{\text{small}, \mathcal{Z}}^! (\mathcal{M}_{\mathcal{Z}}) \rightarrow \mathcal{M}_{\mathcal{Z} \subseteq}$$

is exactly provided by the structure on $\mathcal{M}$ of unital factorization $\mathcal{A}$ -module.

C.11.15. By a similar token, given a (non-unital) lax factorization category $\mathbf{A}$ and a factorization algebra $\mathcal{A}$ in it, the lax factorization category $\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})$ (see Sect. B.11.16) acquires a naturally defined unital structure. The object

$$\mathcal{A}^{\text{fact}} \in \mathbf{FactAlg}(X, \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}))$$

from Sect. B.11.16 extends to an object of $\mathbf{FactAlg}^{\text{unital}}(X, \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}))$, and equals in fact the unit in $\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})$, i.e., the map

$$\mathbf{1}_{\mathcal{A}\text{-mod}^{\text{fact}}} \rightarrow \mathcal{A}^{\text{fact}}$$

is an isomorphism.

For a factorization functor $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$ between non-unital lax factorization categories and $\mathcal{A}_1 \in \mathbf{FactAlg}(\mathbf{A}_1, X)$, the resulting functor

$$(C.51) \quad \Phi : \mathcal{A}_1\text{-mod}^{\text{fact}}(\mathbf{A}_1) \rightarrow \Phi(\mathcal{A}_1)\text{-mod}^{\text{fact}}(\mathbf{A}_2)$$

has a natural (strictly) unital structure.

Similarly, if $\mathbf{A}$ is unital and $\mathcal{A}$ is unital, then the lax factorization category $\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})$ (of *unital* $\mathcal{A}$ -modules) acquires a naturally defined unital structure. Furthermore, in this case the forgetful functor

$$\mathbf{oblv}_{\mathcal{A}} : \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}) \rightarrow \mathbf{A}$$

carries a naturally defined lax unital structure.

For a lax unital functor $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$ between unital lax factorization categories and $\mathcal{A}_1 \in \mathbf{FactAlg}^{\text{unital}}(\mathbf{A}_1, X)$, the resulting functor

$$(C.52) \quad \Phi : \mathcal{A}_1\text{-mod}^{\text{fact}}(\mathbf{A}_1) \rightarrow \Phi(\mathcal{A}_1)\text{-mod}^{\text{fact}}(\mathbf{A}_2)$$

has a natural (strictly) unital structure.

C.11.16. Let $\mathbf{A}$ be a (non-unital) lax factorization category, and let $\phi : \mathcal{A} \rightarrow \mathcal{A}'$ be a homomorphism between factorization (non-unital) algebras in it.

Since the functor

$$\mathrm{Res}_{\phi} : \mathcal{A}'\text{-mod}^{\text{fact}}(\mathbf{A}) \rightarrow \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})$$

is lax unital and $\mathcal{A}'^{\text{fact}}$ is the factorization unit in $\mathcal{A}'\text{-mod}^{\text{fact}}(\mathbf{A})$, we obtain that the factorization algebra $\mathrm{Res}_{\phi}(\mathcal{A}'^{\text{fact}}) \in \mathbf{FactAlg}(X, \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}))$ acquires a natural unital structure.

Moreover, the functor (C.52) applied to $\mathbf{A}_1 = \mathcal{A}'\text{-mod}^{\text{fact}}(\mathbf{A})$, $\mathbf{A}_2 = \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})$ and $\Phi = \mathrm{Res}_{\phi}$ gives rise to a unital factorization functor

$$(C.53) \quad \mathcal{A}'\text{-mod}^{\text{fact}}(\mathbf{A}) \rightarrow \mathrm{Res}_{\phi}(\mathcal{A}'^{\text{fact}})\text{-mod}^{\text{fact-unital}}(\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})).$$

Unwinding the definitions, we obtain:

Lemma C.11.17. *The functor (C.53) is an equivalence.*

C.11.18. Let $\mathbf{A}$ be a unital factorization category, and let $\phi : \mathcal{A} \rightarrow \mathcal{A}'$ be a homomorphism between unital factorization algebras in $\mathbf{A}$.

Consider the unital lax factorization categories $\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})$ and $\mathcal{A}'\text{-mod}^{\text{fact}}(\mathbf{A})$, and the restriction functor

$$\text{Res}_\phi : \mathcal{A}'\text{-mod}^{\text{fact}}(\mathbf{A}) \rightarrow \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}).$$

This functor carries a natural lax unital structure. In particular, the object

$$\text{Res}_\phi(\mathcal{A}'^{\text{fact}}) \in \text{FactAlg}(\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}))$$

from Sect. B.11.17 lifts to an object of $\text{FactAlg}^{\text{untl}}(\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}))$.

Note that the forgetful functor $\text{oblv}_{\mathcal{A}} : \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}) \rightarrow \mathbf{A}$ sends $\text{Res}_\phi(\mathcal{A}'^{\text{fact}}) \rightarrow \mathcal{A}'$. In particular, it induces a functor

$$(C.54) \quad \text{Res}_\phi(\mathcal{A}'^{\text{fact}})\text{-mod}^{\text{fact}}(\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})) \rightarrow \mathcal{A}'\text{-mod}^{\text{fact}}(\mathbf{A}).$$

The following is obtained by unwinding the definitions:

Lemma C.11.19. *The functor (C.54) is an equivalence.*

In fact, the inverse of the functor (C.54) is given by the functor (C.52) for $\mathbf{A}_1 = \mathcal{A}'\text{-mod}^{\text{fact}}(\mathbf{A})$, $\mathbf{A}_2 = \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A})$, $\Phi = \text{Res}_\phi$ and $\mathcal{A}_1 = \mathcal{A}'^{\text{fact}}$.

C.11.20. *Proof of Proposition C.7.13.* We will explicitly construct an inverse functor.

Let $k \xrightarrow{\text{vac}_{\mathcal{A}}} \mathcal{A}$ be a quasi-unital factorization algebra. Consider the unital lax factorization category⁸⁰ $\mathcal{A}\text{-mod}^{\text{fact-n.u.}}$. It is equipped with a lax unital factorization functor

$$\text{Res}_{\text{vac}_{\mathcal{A}}} : \mathcal{A}\text{-mod}^{\text{fact-n.u.}} \rightarrow k\text{-mod}^{\text{fact-n.u.}}.$$

Consider the fiber product

$$\mathcal{A}\text{-mod}^{\text{fact-q.u.}} := \text{Vect} \times_{k\text{-mod}^{\text{fact-n.u.}}} \mathcal{A}\text{-mod}^{\text{fact-n.u.}} \subset C\mathcal{A}\text{-mod}^{\text{fact-n.u.}}.$$

It has a natural factorization structure, and the fact that

$$\text{Res}_{\text{vac}_{\mathcal{A}}}(\mathcal{A}^{\text{fact}_{\text{Ran}}}) \in k\text{-mod}_{\text{Ran}}^{\text{fact}} \subset k\text{-mod}_{\text{Ran}}^{\text{fact-n.u.}}$$

implies that the unital structure on $\mathcal{A}\text{-mod}^{\text{fact-n.u.}}$ induces one on $\mathcal{A}\text{-mod}^{\text{fact-q.u.}}$. The factorization unit $\mathbf{1}_{\mathcal{A}\text{-mod}^{\text{fact-q.u.}}}$ in $\mathcal{A}\text{-mod}^{\text{fact-q.u.}}$ is $\mathcal{A}^{\text{fact}}$ (see Sect. C.11.13).

Restricting $\text{Res}_{\text{vac}_{\mathcal{A}}}$ to $\mathcal{A}\text{-mod}^{\text{fact-q.u.}}$, we obtain a lax unital factorization functor

$$(C.55) \quad \mathcal{A}\text{-mod}^{\text{fact-q.u.}} \rightarrow \text{Vect}.$$

The image of the factorization unit $\mathbf{1}_{\mathcal{A}\text{-mod}^{\text{fact-q.u.}}} = \mathcal{A}^{\text{fact}}$ under the above functor is a unital factorization algebra (whose underlying plain factorization algebra is $\mathcal{A}$ itself). This defines a functor

$$(C.56) \quad \text{FactAlg}^{\text{q-untl}}(X) \rightarrow \text{FactAlg}^{\text{untl}}(X),$$

which commutes with a forgetful functor to $\text{FactAlg}(X)$.

Let us show that the functors (C.36) and (C.56) are mutually inverse. We first show that the composition (C.56) $\circ$ (C.36) is isomorphic to the identity functor.

Indeed, when $\mathcal{A}$ is unital, by Proposition C.7.9, the unital factorization category $\mathcal{A}\text{-mod}^{\text{fact-q.u.}}$ identifies with $\mathcal{A}\text{-mod}^{\text{fact}}$, and the functor (C.56) identifies with $\text{oblv}_{\mathcal{A}}$, equipped with its natural lax unital structure. Hence, in this case, the image of $\mathcal{A}^{\text{fact}}$ under (C.56) identifies with $\mathcal{A}$ as a unital factorization algebra.

⁸⁰The appearance of “n.u.” in the superscript in the next formula is meant to emphasize that we are dealing with non-unital factorization modules, even though since $\mathcal{A}$ is non-unital, we cannot even talk about unital modules.

Vice versa, let us start with a quasi-unital factorization algebra $k \xrightarrow{\text{vac}, \mathcal{A}} \mathcal{A}$. We have a commutative diagram of unital lax factorization categories and lax unital functors, with the horizontal arrows being strict:

$$\begin{array}{ccc} \mathcal{A}\text{-mod}^{\text{fact-q.u.}} & \longrightarrow & \mathcal{A}\text{-mod}^{\text{fact-n.u.}} \\ \downarrow & & \downarrow \\ \text{Vect} & \longrightarrow & k\text{-mod}^{\text{fact-n.u.}} \end{array}$$

Applying each circuit to $\mathbf{1}_{\mathcal{A}\text{-mod}^{\text{fact-q.u.}}}$ we obtain a factorization algebra in $k\text{-mod}^{\text{fact-n.u.}}$, equipped with a homomorphism from k . For the clockwise circuit, we obtain the original $k \xrightarrow{\text{vac}, \mathcal{A}} \mathcal{A}$. For the anti-clockwise circuit, we obtain (C.56)($k \xrightarrow{\text{vac}, \mathcal{A}} \mathcal{A}$), equipped with a map from k to it, given by its unital structure.

□[Proposition C.7.13]

C.11.21. In the rest of this subsection we will focus on strict (i.e., non-lax) factorization categories.

Let $\mathbf{A}_1$ and $\mathbf{A}_2$ be a pair of unital factorization categories, and let $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$ be a strictly unital functor. Assume that Φ admits a right adjoint Φ^R as a functor between the underlying crystals of categories on Ran .

Then Φ^R , viewed as a factorization functor between factorization categories admits a natural extension to a lax unital functor between unital factorization categories, see Sect. C.2.11.

C.11.22. We claim:

Lemma C.11.23. *Let $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$ be a lax unital factorization functor between unital factorization categories. Then Φ is strictly unital if and only if the map (C.49) is an isomorphism.*

Proof. The “only if” direction is tautological. Let us prove the “if” direction, so let us assume that (C.49) is an isomorphism.

Let $\underline{x} \subseteq \underline{x}'$ be an arrow in $\text{Ran}^{\text{untl},*}(S)$ for $S \in \text{Sch}_{\text{Ran}}^{\text{aff}}$. We need to show that the natural transformation

$$(C.57) \quad \text{ins. unit}_{\mathbf{A}_2, \underline{x} \subseteq \underline{x}'} \circ \Phi_{\underline{x}} \rightarrow \Phi_{\underline{x}'} \circ \text{ins. unit}_{\mathbf{A}_1, \underline{x} \subseteq \underline{x}'},$$

given by the lax unital structure on Φ , is an isomorphism.

This assertion can be checked strata-wise, so we can assume that $\underline{x}$ and $\underline{x}'$ are field-valued points. Write

$$\underline{x}' = \underline{x} \sqcup \underline{x}''.$$

We have

$$\mathbf{A}_{i, \underline{x}'} \simeq \mathbf{A}_{i, \underline{x}} \otimes \mathbf{A}_{i, \underline{x}''}, \quad i = 1, 2$$

and the natural transformation (C.57) is the tensor product of the identity endomorphism of the functor

$$\Phi_{\underline{x}} : \mathbf{A}_{1, \underline{x}} \rightarrow \mathbf{A}_{2, \underline{x}}$$

and the natural transformation (C.57) for $\emptyset \subset \underline{x}$.

However, the latter is exactly the map

$$\mathbf{1}_{\mathbf{A}_{2, \underline{x}}} \rightarrow \Phi_{\underline{x}}(\mathbf{1}_{\mathbf{A}_{1, \underline{x}}}).$$

□

C.12. Examples of unital factorization categories arising from algebraic geometry.

C.12.1. Let $\mathcal{T}$ be a counital factorization space. We claim that the lax factorization category $\mathrm{QCoh}(\mathcal{T})$ admits a natural unital structure.

Indeed, for a pair of S -points $\underline{x} \subseteq \underline{x}'$ of $\mathrm{Ran}^{\mathrm{untl},*}$, the corresponding functor

$$\mathrm{QCoh}(\mathcal{T}_{S,\underline{x}}) \rightarrow \mathrm{QCoh}(\mathcal{T}_{S,\underline{x}'})$$

is given by pullback along

$$(C.58) \quad \mathcal{T}_{S,\underline{x}'} \rightarrow \mathcal{T}_{S,\underline{x}}.$$

In particular, the unit $\mathbf{1}_{\mathrm{QCoh}(\mathcal{T})}$ is the structure sheaf

$$\mathcal{O}_{\mathcal{T}} \in \mathrm{QCoh}(\mathcal{T}).$$

C.12.2. By a similar token, using Sect. B.13.16 the (lax) factorization category $\mathrm{IndCoh}^!(\mathcal{T})$ admits a natural unital structure.

Assume now that $\mathcal{T}$ is affine and placid. Recall that according to Sect. B.13.22, we can consider the factorization category $\mathrm{IndCoh}^*(\mathcal{T})$.

Assume now that the maps

$$\mathcal{T}_{S,X^{I_2}} \rightarrow \mathcal{T}_{S,X^{I_1}}$$

for inclusions of finite sets $I_1 \hookrightarrow I_2$ are of finite Tor-dimension (e.g., they are flat). In this case, by Sect. A.10.13, the functors of $*$ -pullback along the maps (C.58) are defined for $\mathrm{IndCoh}^*(-)$

$$\mathrm{IndCoh}^*(\mathcal{T}_{S,\underline{x}}) \rightarrow \mathrm{IndCoh}^*(\mathcal{T}_{S,\underline{x}'}).$$

Hence, we obtain that in this case, $\mathrm{IndCoh}^*(\mathcal{T})$ also admits a natural unital structure.

C.12.3. Let now $\mathcal{T}$ be a unital factorization space. We claim that in this case the factorization category $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{T})$ has a natural unital structure.

Indeed, for a pair of S -points $\underline{x} \subseteq \underline{x}'$ of $\mathrm{Ran}^{\mathrm{untl},*}$, the corresponding functor

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{T}_{S,\underline{x}}) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{T}_{S,\underline{x}'})$$

is given by pushforward along

$$(C.59) \quad \mathcal{T}_{S,\underline{x}} \rightarrow \mathcal{T}_{S,\underline{x}'}.$$

Assume now that $\mathcal{T}$ is an ind-placid factorization ind-scheme, so that we can consider the factorization category $\mathrm{IndCoh}^*(\mathcal{T})$.

Taking the IndCoh -pushforwards along (C.59) we obtain that $\mathrm{IndCoh}^*(\mathcal{T})$ acquires a natural unital structure.

C.12.4. *Example.* We obtain that the factorization categories $\mathrm{QCoh}_{\mathrm{co}}(\mathrm{Gr}_G)$ and $\mathrm{IndCoh}(\mathrm{Gr}_G)$ acquire a natural unital structure.

By a similar mechanism, the factorization category $\mathrm{D-mod}(\mathrm{Gr}_G)$ also acquires a unital structure.

C.12.5. Let us now in addition assume that for an injection of finite sets $I_1 \hookrightarrow I_2$, the corresponding map

$$X^{I_2} \times_{X^{I_1}} \mathcal{T}_{X^{I_1}} \rightarrow \mathcal{T}_{X^{I_2}}$$

is an ind-closed embedding locally almost of finite-presentation.

In this case, by Sect. A.10.11, the IndCoh -pushforward functors along (C.59) are defined on $\mathrm{IndCoh}^!(-)$:

$$\mathrm{IndCoh}^!(\mathcal{T}_{S,\underline{x}}) \rightarrow \mathrm{IndCoh}^!(\mathcal{T}_{S,\underline{x}'}).$$

Hence, we obtain that in this case, $\mathrm{IndCoh}^!(\mathcal{T})$ also acquires a unital structure.

C.12.6. Let $\mathcal{T}$ be a factorization space over X , equipped with a unital-in-correspondences structure. Assume that for a pair of S -points $\underline{x} \subseteq \underline{x}'$ of $\text{Ran}^{\text{untl},*}$, the map

$$(C.60) \quad \mathcal{T}_{\underline{x} \subseteq \underline{x}'}^{\text{all} \rightsquigarrow +} \xrightarrow{\text{Pr}_{\text{small}}^{\mathcal{T}}} \mathcal{T}_{\underline{x}}$$

is affine.

We claim that in this case, the factorization category $\text{QCoh}_{\text{co}}(\mathcal{T})$ acquires a unital structure, and the functor

$$\text{QCoh}(\mathcal{T}^+) = \text{QCoh}_{\text{co}}(\mathcal{T}^+) \rightarrow \text{QCoh}_{\text{co}}(\mathcal{T}),$$

given by direct image along

$$\iota : \mathcal{T}^+ \rightarrow \mathcal{T}$$

is strictly unital. In particular, the unit $\mathbf{1}_{\text{QCoh}_{\text{co}}(\mathcal{T})}$ is the direct image of $\mathcal{O}_{\mathcal{T}^+}$ along ι .

Namely, for a pair of S -points $\underline{x} \subseteq \underline{x}'$ of $\text{Ran}^{\text{untl},*}$, the corresponding functor

$$\text{QCoh}_{\text{co}}(\mathcal{T}_{S,\underline{x}}) \rightarrow \text{QCoh}_{\text{co}}(\mathcal{T}_{S,\underline{x}'})$$

is given by pull-push along

$$(C.61) \quad \mathcal{T}_{\underline{x}} \xleftarrow{\text{Pr}_{\text{small}}^{\mathcal{T}}} \mathcal{T}_{\underline{x} \subseteq \underline{x}'}^{\text{all} \rightsquigarrow +} \xrightarrow{\text{Pr}_{\text{big}}^{\mathcal{T}}} \mathcal{T}_{\underline{x}'}$$

C.12.7. *Example.* Thus, we obtain that for an affine D-scheme $\mathcal{Y}$, the factorization category

$$\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y}))$$

acquires a unital structure, with the unit $\mathbf{1}_{\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y}))}$ being the direct image of $\mathcal{O}_{\mathfrak{L}_{\nabla}^+(\mathcal{Y})}$ along

$$\mathfrak{L}_{\nabla}^+(\mathcal{Y}) \xrightarrow{\iota} \mathfrak{L}_{\nabla}(\mathcal{Y}).$$

As another example, we can take $\mathcal{T} = \text{Op}_G^{\text{mon-free}}$, and we obtain that $\text{QCoh}_{\text{co}}(\text{Op}_G^{\text{mon-free}})$ acquires a unital structure. The unit $\mathbf{1}_{\text{QCoh}_{\text{co}}(\text{Op}_G^{\text{mon-free}})}$ is the direct image of $\mathcal{O}_{\text{Op}_G^{\text{reg}}}$ along

$$\text{Op}_G^{\text{reg}} \xrightarrow{\iota^{+, \text{mon-free}}} \text{Op}_G^{\text{mon-free}}.$$

C.12.8. Assume now that $\mathcal{T}$ is an ind-placid ind-affine factorization ind-scheme. Assume that the maps

$$\mathcal{T}_{I_1 \subseteq I_2}^{\text{all} \rightsquigarrow +} \xrightarrow{\text{Pr}_{\text{small}}^{\mathcal{T}}} \mathcal{T}_{X^{I_1}}$$

for an injection of finite sets $I_1 \subseteq I_2$ are affine and of finite Tor-dimension.

In this case, by Sect. A.10.13, the functors of $*$ -pullback along the maps (C.60) are well-defined on $\text{IndCoh}^*(-)$ and satisfy base change.

We define a unital structure on the factorization category $\text{IndCoh}^*(\mathcal{T})$ by $(\text{IndCoh}, *)$ -pull followed by IndCoh -pushforward along (C.61).

Note that by construction, the functor of IndCoh -pushforward along ι

$$\text{IndCoh}^*(\mathcal{T}^+) \rightarrow \text{IndCoh}^*(\mathcal{T})$$

is (strictly) unital.

In particular, the unit in $\text{IndCoh}^*(\mathcal{T})$ is the direct image of $\mathcal{O}_{\mathcal{T}^+} \in \text{IndCoh}^*(\mathcal{T}^+)$ along ι .

C.12.9. Let us continue to assume that $\mathcal{T}$ is an ind-placid ind-affine factorization ind-scheme. Assume now that for an inclusion of finite sets $I_1 \subseteq I_2$, the map

$$\mathcal{T}_{I_1 \subseteq I_2}^{\text{all} \rightsquigarrow +} \xrightarrow{\text{pr}_{\text{big}}^{\mathcal{T}}} \mathcal{T}_{X^{I_2}}$$

is an ind-closed embedding locally almost of finite-presentation.

In this case, Sect. A.10.11, the IndCoh -pushforward functors along

$$\mathcal{T}_{\underline{x} \subseteq \underline{x}'}^{\text{all} \rightsquigarrow +} \xrightarrow{\text{pr}_{\text{big}}^{\mathcal{T}}} \mathcal{T}_{\underline{x}'}$$

are defined on $\text{IndCoh}^1(-)$ and satisfy base change.

We define a unital structure on the factorization category $\text{IndCoh}^1(\mathcal{T})$ by $!$ -pull followed by IndCoh -pushforward along (C.61).

Note that by construction, the functor of IndCoh -pushforward along ι

$$\text{IndCoh}^1(\mathcal{T}^+) \rightarrow \text{IndCoh}^1(\mathcal{T})$$

is (strictly) unital.

In particular, the unit in $\text{IndCoh}^1(\mathcal{T})$ is the direct image of $\omega_{\mathcal{T}^+} \in \text{IndCoh}^1(\mathcal{T}^+)$ along ι .

C.12.10. *Example.* Let $\mathcal{Y}$ be an affine D-scheme, such that:

- $\mathfrak{L}_{\nabla}(\mathcal{Y})$ is ind-placid;
- The maps

$$\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{I_1 \subseteq I_2} \xrightarrow{\text{pr}_{\text{small}}^{\mathcal{Y}}} \mathfrak{L}_{\nabla}(\mathcal{Y})_{X^{I_1}}$$

for $I_1 \subseteq I_2$ are flat;

- The maps

$$\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{I_1 \subseteq I_2} \xrightarrow{\text{pr}_{\text{big}}^{\mathcal{Y}}} \mathfrak{L}_{\nabla}(\mathcal{Y})_{X^{I_2}}$$

for $I_1 \subseteq I_2$ are locally almost of finite-presentation.

This happens, e.g., for $\mathcal{Y} = \text{Jets}(H)$, where H is a smooth group-scheme over X .

We obtain that the factorization categories $\text{IndCoh}^1(\mathfrak{L}_{\nabla}(\mathcal{Y}))$ and $\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathcal{Y}))$ acquire unital structures.

C.12.11. By a similar procedure, the factorization category $\text{D-mod}(\mathfrak{L}(H))$ acquires a unital structure.

C.12.12. As yet another example, we obtain that the categories

$$\text{IndCoh}^1(\text{Op}_{\mathcal{G}}^{\text{mon-free}}) \text{ and } \text{IndCoh}^*(\text{Op}_{\mathcal{G}}^{\text{mon-free}})$$

acquire unital structures.

C.12.13. Let $\mathcal{T}$ be an ind-placid ind-affine factorization ind-scheme. Assume that both additional conditions in Sects. C.12.8 and C.12.9 are satisfied, so both $\text{IndCoh}^*(\mathcal{T})$ and $\text{IndCoh}^1(\mathcal{T})$ acquire unital structures.

Note that the canonical pairing

$$(C.62) \quad \text{IndCoh}^1(\mathcal{T}) \otimes \text{IndCoh}^*(\mathcal{T}) \rightarrow \text{Vect}$$

(see Sect. A.10.8) as factorization categories, admits a natural lax unital structure as a functor between unital factorization categories.

Moreover, unwinding the definitions we obtain that the condition from Sect. C.2.14 is satisfied.

Hence, we obtain that (C.62) realizes $\text{IndCoh}^1(\mathcal{T})$ and $\text{IndCoh}^*(\mathcal{T})$ as each other duals as unital factorization categories.

C.13. Unital structure on Kac-Moody representations.

C.13.1. Our current goal is to construct a unital structure on the factorization category $\widehat{\mathfrak{g}}\text{-mod}_{\kappa}$. Let us be given a pair of S -points $\underline{x}, \underline{x}'$ of Ran with $\underline{x} \subseteq \underline{x}'$. We need to construct a functor

$$(C.63) \quad \widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, \underline{x}} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, \underline{x}'}.$$

The unital-in-correspondences structure on $\mathfrak{L}(G)$ gives rise to the following diagram of group ind-schemes over S :

$$\mathfrak{L}(G)_{\underline{x}} \xleftarrow{\text{pr}_{\text{small}}^G} \mathfrak{L}^{\text{mer} \rightsquigarrow \text{reg}}(G)_{\underline{x} \subseteq \underline{x}'} \xrightarrow{\text{pr}_{\text{small}}^G} \mathfrak{L}(G)_{\underline{x}'}$$

Proceeding as in Sect. B.14.19, we consider the corresponding categories

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, \underline{x}}, \widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, \underline{x}'} \text{ and } \widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, \underline{x} \subseteq \underline{x}'}$$

The universal properties of these categories give rise to restriction functors

$$(C.64) \quad \text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x} \subseteq \underline{x}'}}^{\widehat{\mathfrak{g}}_{\underline{x}'}} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, \underline{x}'} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, \underline{x} \subseteq \underline{x}'}$$

and

$$(C.65) \quad \text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x} \subseteq \underline{x}}}^{\widehat{\mathfrak{g}}_{\underline{x}}} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, \underline{x}} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, \underline{x} \subseteq \underline{x}'},$$

both strongly compatible with the forgetful functors to $\text{D-mod}(S)$ and the actions of $\mathfrak{L}^{\text{mer} \rightsquigarrow \text{reg}}(G)_{\underline{x} \subseteq \underline{x}'}$ at level κ .

C.13.2. One checks directly that the functor $\text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x} \subseteq \underline{x}'}}^{\widehat{\mathfrak{g}}_{\underline{x}}}$ admits a left adjoint.

We define the functor

$$\text{ins. vac}_{\underline{x} \subseteq \underline{x}'} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, \underline{x}} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, \underline{x}'}$$

as the composition

$$(C.66) \quad (\text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x} \subseteq \underline{x}'}}^{\widehat{\mathfrak{g}}_{\underline{x}'}})^L \circ \text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x} \subseteq \underline{x}}}^{\widehat{\mathfrak{g}}_{\underline{x}}}.$$

C.13.3. In order to upgrade the collection of functors $\text{ins. vac}_{\underline{x} \subseteq \underline{x}'}$ to a unital structure, we need to construct associativity isomorphisms

$$(C.67) \quad \text{ins. vac}_{\underline{x}_3 \subseteq \underline{x}_2} \circ \text{ins. vac}_{\underline{x}_1 \subseteq \underline{x}_2} \simeq \text{ins. vac}_{\underline{x}_1 \subseteq \underline{x}_3}$$

for $\underline{x}_1 \subseteq \underline{x}_2 \subseteq \underline{x}_3$.

Note that the inclusions

$$(\mathcal{D}_{\underline{x}_2} - \underline{x}_1) \hookrightarrow (\mathcal{D}_{\underline{x}_3} - \underline{x}_1) \hookrightarrow (\mathcal{D}_{\underline{x}_3} - \underline{x}_2)$$

give rise to maps

$$\mathfrak{L}^{\text{mer} \rightsquigarrow \text{reg}}(G)_{\underline{x}_1 \subseteq \underline{x}_2} \leftarrow \mathfrak{L}^{\text{mer} \rightsquigarrow \text{reg}}(G)_{\underline{x}_1 \subseteq \underline{x}_3} \rightarrow \mathfrak{L}^{\text{mer} \rightsquigarrow \text{reg}}(G)_{\underline{x}_3 \subseteq \underline{x}_2}.$$

We have the corresponding functors

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, \underline{x}_1 \subseteq \underline{x}_2} \xrightarrow{\text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x}_1 \subseteq \underline{x}_3}}^{\widehat{\mathfrak{g}}_{\underline{x}_1 \subseteq \underline{x}_2}}} \widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, \underline{x}_1 \subseteq \underline{x}_3} \xleftarrow{\text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x}_1 \subseteq \underline{x}_3}}^{\widehat{\mathfrak{g}}_{\underline{x}_2 \subseteq \underline{x}_3}}} \widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, \underline{x}_2 \subseteq \underline{x}_2}$$

and an isomorphism

$$\text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x}_1 \subseteq \underline{x}_3}}^{\widehat{\mathfrak{g}}_{\underline{x}_1 \subseteq \underline{x}_2}} \circ \text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x}_1 \subseteq \underline{x}_2}}^{\widehat{\mathfrak{g}}_{\underline{x}_2}} \simeq \text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x}_1 \subseteq \underline{x}_3}}^{\widehat{\mathfrak{g}}_{\underline{x}_2 \subseteq \underline{x}_3}} \circ \text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x}_2 \subseteq \underline{x}_3}}^{\widehat{\mathfrak{g}}_{\underline{x}_2}}.$$

From here we obtain a natural transformation

$$(C.68) \quad (\text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x}_1 \subseteq \underline{x}_3}}^{\widehat{\mathfrak{g}}_{\underline{x}_2 \subseteq \underline{x}_3}})^L \circ \text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x}_1 \subseteq \underline{x}_2}}^{\widehat{\mathfrak{g}}_{\underline{x}_1 \subseteq \underline{x}_2}} \rightarrow \text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x}_2 \subseteq \underline{x}_3}}^{\widehat{\mathfrak{g}}_{\underline{x}_2}} \circ (\text{oblv}_{\widehat{\mathfrak{g}}_{\underline{x}_1 \subseteq \underline{x}_2}}^{\widehat{\mathfrak{g}}_{\underline{x}_2}})^L.$$

We claim that (C.68) is an isomorphism. Indeed, this follows from the fact that the diagram of Lie algebras

$$\begin{array}{ccc} \mathfrak{L}^{\text{mer} \rightsquigarrow \text{reg}}(\mathfrak{g})_{\underline{x}_1 \subseteq \underline{x}_3} & \longrightarrow & \mathfrak{L}^{\text{mer} \rightsquigarrow \text{reg}}(\mathfrak{g})_{\underline{x}_2 \subseteq \underline{x}_3} \\ \downarrow & & \downarrow \\ \mathfrak{L}^{\text{mer} \rightsquigarrow \text{reg}}(\mathfrak{g})_{\underline{x}_1 \subseteq \underline{x}_2} & \longrightarrow & \mathfrak{L}(\mathfrak{g})_{\underline{x}_2} \end{array}$$

is Cartesian.

Now, (C.67) follows by precomposing both sides of (C.68) with $\mathbf{oblv}_{\widehat{\mathfrak{g}}_{x_1} \subseteq x_2}^{\widehat{\mathfrak{g}}_{x_1}}$ and post-composing with $(\mathbf{oblv}_{\widehat{\mathfrak{g}}_{x_2} \subseteq x_3}^{\widehat{\mathfrak{g}}_{x_3}})^L$.

C.13.4. Recall that the functor $\mathbf{oblv}_{\widehat{\mathfrak{g}}_{x \subseteq x'}}^{\widehat{\mathfrak{g}}_x}$ is compatible with the action of $\mathfrak{L}^{\text{mer} \rightsquigarrow \text{reg}}(G)_{x \subseteq x'}$. In particular, it is compatible with the action of $\mathfrak{L}^+(G)_{x'}$, which acts on $\widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, x}$ via $\mathfrak{L}^+(G)_{x'} \rightarrow \mathfrak{L}^+(G)_x$. Hence, it induces a functor

$$\text{KL}(G)_{\kappa, S, x} = (\widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, x})^{\mathfrak{L}^+(G)_x} \rightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, x \subseteq x'})^{\mathfrak{L}^+(G)_{x'}}.$$

Since the functor $\mathbf{oblv}_{\widehat{\mathfrak{g}}_{x \subseteq x'}}^{\widehat{\mathfrak{g}}_{x'}}$ is compatible with the action of $\mathfrak{L}^{\text{mer} \rightsquigarrow \text{reg}}(G)_{x \subseteq x'}$, then so is its left adjoint. In particular, we obtain that $(\mathbf{oblv}_{\widehat{\mathfrak{g}}_{x \subseteq x'}}^{\widehat{\mathfrak{g}}_{x'}})^L$ is compatible with the action of $\mathfrak{L}^+(G)_{x'}$, and hence induces a functor

$$(\widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, x \subseteq x'})^{\mathfrak{L}^+(G)_{x'}} \rightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, x'})^{\mathfrak{L}^+(G)_{x'}} = \text{KL}(G)_{\kappa, S, x'}.$$

Hence, we obtain that the functor (C.66) induces a functor

$$(C.69) \quad \text{ins. vac}_{x \subseteq x'} : \text{KL}(G)_{\kappa, S, x} \rightarrow \text{KL}(G)_{\kappa, S, x'}.$$

The functors (C.69) define a unital structure on $\text{KL}(G)_{\kappa}$.

C.13.5. By construction, the unit object $\mathbf{1}_{\widehat{\mathfrak{g}}\text{-mod}_{\kappa}}$ is the vacuum module $\text{Vac}(G)_{\kappa}$, i.e.,

$$\text{Vac}(G)_{\kappa, x} := (\mathbf{oblv}_{\mathfrak{L}^+(G)_x}^{\widehat{\mathfrak{g}}_x})^L \circ \mathbf{oblv}_{\mathfrak{L}^+(G)_x}^0(k).$$

Furthermore, $\text{Vac}(G)_{\kappa}$ naturally upgrades to an object of $\text{KL}(G)_{\kappa}$ and coincides with its unit.

C.13.6. Let κ' be as in Sect. 2.2.

Recall that according to [Ra5, Sect. 9.16.11], we have canonical pairings

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa, x} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\kappa', x} \rightarrow \text{D-mod}(S)$$

making

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa} \text{ and } \widehat{\mathfrak{g}}\text{-mod}_{\kappa'}$$

mutually dual as factorization categories.

We claim that this duality extends to a duality between $\widehat{\mathfrak{g}}\text{-mod}_{\kappa}$ and $\widehat{\mathfrak{g}}\text{-mod}_{\kappa'}$ as unital factorization categories (see Sect. C.2.12 for what this means).

Namely, imitating the construction in [Ra5, Sect. 9.16] we obtain a duality between

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, x \subseteq x'} \text{ and } \widehat{\mathfrak{g}}\text{-mod}_{\kappa', S, x \subseteq x'}.$$

Under this identification, the dual of the functor $\mathbf{oblv}_{\widehat{\mathfrak{g}}_{x \subseteq x'}}^{\widehat{\mathfrak{g}}_x}$ of (C.65) is the *right adjoint* of

$$\mathbf{oblv}_{\widehat{\mathfrak{g}}_{x \subseteq x'}}^{\widehat{\mathfrak{g}}_x} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa', S, x} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa', S, x \subseteq x'},$$

and the dual of the functor $\mathbf{oblv}_{\widehat{\mathfrak{g}}_{x \subseteq x'}}^{\widehat{\mathfrak{g}}_{x'}}$ of (C.64) is the left adjoint of the functor

$$\mathbf{oblv}_{\widehat{\mathfrak{g}}_{x \subseteq x'}}^{\widehat{\mathfrak{g}}_x} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa', S, x'} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa', S, x \subseteq x'}.$$

From here we obtain the desired identification between the dual of

$$\text{ins. vac}_{x \subseteq x'} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, x} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa, S, x'}$$

and the right adjoint of

$$\text{ins. vac}_{x \subseteq x'} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa', S, x} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\kappa', S, x'}.$$

C.13.7. In a similar fashion we obtain that the duality between

$$\mathrm{KL}(G)_\kappa \text{ and } \mathrm{KL}(G)_{\kappa'}$$

as factorization categories extends to a duality as unital factorization categories.

C.14. Unital factorization module categories.

C.14.1. Let $\mathbf{A}$ be a factorization category. Let $\mathcal{Z}$ be a prestack mapping to Ran , and let $\mathbf{C}$ be a factorization module category over $\mathbf{A}$ at $\mathcal{Z}$.

Suppose that $\mathbf{A}$ is equipped with a unital structure. Combining the ideas of Sects. C.11.1, B.12.1 and C.7.4, we define the notion of unital structure on $\mathbf{C}$.

C.14.2. Concretely, the unital structure on $\mathbf{C}$ amounts to the following: given a pair of points $(z, \underline{x})$ and $(z, \underline{x}')$ of $\mathcal{Z}^\subseteq$ with $\underline{x} \subseteq \underline{x}'$, we must be given a functor

$$\mathrm{ins. unit}_{\underline{x} \subseteq \underline{x}'} : \mathbf{C}_{(z, \underline{x})} \rightarrow \mathbf{C}_{(z, \underline{x}')},$$

compatible with factorization.

The latter compatibility means the following: for $\underline{x}' = \underline{x} \sqcup \underline{x}''$, with respect to the identification

$$\mathbf{C}_{(z, \underline{x}')} \simeq \mathbf{A}_{\underline{x}'} \otimes \mathbf{C}_{(z, \underline{x})},$$

the functor $\mathrm{ins. unit}_{\underline{x} \subseteq \underline{x}'}$ is

$$1_{\mathbf{A}_{\underline{x}''}} \otimes \mathrm{Id}.$$

In what follows we will denote by $\mathrm{ins. unit}_{\mathcal{Z}}$ the corresponding functor

$$(C.70) \quad \mathbf{C}_{\mathcal{Z}} \rightarrow \mathbf{C}_{\mathcal{Z}^\subseteq, \mathrm{unl}}.$$

C.14.3. For a unital factorization category $\mathbf{A}$ and any $\mathcal{Z} \rightarrow \mathrm{Ran}$, consider the factorization module category $\mathbf{A}^{\mathrm{fact}_{\mathcal{Z}}}$ from Sect. B.12.4.

Unwinding the definitions, we obtain that $\mathbf{A}^{\mathrm{fact}_{\mathcal{Z}}}$ carries a natural unital structure (cf. Sect. C.7.6).

C.14.4. Given a pair of unital factorization module categories $\mathbf{C}_1$ and $\mathbf{C}_2$ over $\mathbf{A}$ at $\mathcal{Z}$ one can a priori talk about strictly unital or lax unital functors between them, compatible with factorization. However, as in Lemma C.11.23 one shows that any lax unital functor between them is automatically strictly unital.

Thus, we can unambiguously talk about the (2-) category of unital factorization module categories over $\mathbf{A}$ at $\mathcal{Z}$.

C.14.5. *Notational convention.* When $\mathbf{A}$ is unital, we will denote the (2-) category of unital factorization module categories over $\mathbf{A}$ at $\mathcal{Z}$ by

$$\mathbf{A}\text{-}\mathbf{mod}_{\mathcal{Z}}^{\mathrm{fact}}.$$

We will denote the category of plain (i.e., non-unital) factorization module categories over $\mathbf{A}$ at $\mathcal{Z}$ by

$$\mathbf{A}\text{-}\mathbf{mod}_{\mathcal{Z}}^{\mathrm{fact-n.u.}}.$$

Remark C.14.6. Unlike the case of modules over factorization algebras, the forgetful functor

$$(C.71) \quad \mathbf{A}\text{-}\mathbf{mod}_{\mathcal{Z}}^{\mathrm{fact}} \rightarrow \mathbf{A}\text{-}\mathbf{mod}_{\mathcal{Z}}^{\mathrm{fact-n.u.}}$$

is *not* fully faithful.

Indeed, take $\mathbf{A} = \mathrm{Vect}$ and $\mathcal{Z} = \mathrm{pt}$ corresponding to a singleton $\{x\} = \underline{x} \in \mathrm{Ran}$. Take

$$\mathbf{C} = \mathrm{Vect}^{\mathrm{fact}_x} \in \mathrm{Vect}\text{-}\mathbf{mod}_x^{\mathrm{fact}}.$$

Then the category of endofunctors of $\mathbf{C}$ as an object of $\mathrm{Vect}\text{-}\mathbf{mod}_x^{\mathrm{fact}}$ is

$$k\text{-}\mathbf{mod}_x^{\mathrm{fact}}$$

(where k is the unit factorization algebra), and the latter identifies with Vect , see Sect. C.7.7.

By contrast, the category of endofunctors of $\mathbf{C}$ as an object of $\mathbf{Vect}\text{-}\mathbf{mod}_x^{\text{fact-n.u.}}$ is

$$k\text{-mod}_x^{\text{fact-n.u.}}.$$

So, at the level of endofunctors of the above object, the forgetful functor (C.71) is the forgetful functor

$$k\text{-mod}_x^{\text{fact}} \rightarrow k\text{-mod}_x^{\text{fact-n.u.}},$$

which is fully faithful, but not an equivalence.

C.14.7. *Example.* Recall the construction from Sect. B.12.5. It is easy to see that the resulting Vect-factorization module category $\mathbf{C}$ is unital.

In Sect. C.16.6 we will show that the functor

$$\text{CrystCat}(\mathcal{Z}_0) \rightarrow \mathbf{Vect}\text{-}\mathbf{mod}_{\mathcal{Z}_0}^{\text{fact}}$$

is fully faithful, and we will characterize its essential image.

Note, however, that one categorical level down, the corresponding functor was an equivalence, see Sect. C.7.7.

C.14.8. Let $\mathcal{A}$ be a unital factorization algebra in a unital factorization category $\mathbf{A}$. Let $\mathbf{C}$ be a unital factorization module category over $\mathbf{A}$ at some $\mathcal{Z} \rightarrow \text{Ran}$.

Parallel to Sects. B.12.6 and C.7.4, one defines the notion of unital factorization modules *over* $\mathcal{A}$ in $\mathbf{C}$.

We will denote the corresponding category by

$$\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{C})_{\mathcal{Z}}.$$

By contrast, we will denote the category of plain (i.e., non-unital) $\mathcal{A}$ -modules in $\mathbf{C}$ by

$$\mathcal{A}\text{-mod}^{\text{fact-n.u.}}(\mathbf{C})_{\mathcal{Z}}.$$

C.14.9. We claim:

Lemma C.14.10. $\mathbf{1}_{\mathbf{A}}\text{-mod}^{\text{fact}}(\mathbf{C})_{\mathcal{Z}} \simeq \mathbf{C}_{\mathcal{Z}}.$

Proof. The proof essentially repeats the contents of Sect. C.7.7:

Starting from an object $\mathcal{M}' \in \mathbf{C}_{\mathcal{Z}}$, consider

$$\text{ins. unit}_{\mathcal{Z}}(\mathcal{M}') \in \mathbf{C}_{\mathcal{Z} \subseteq, \text{unl}}.$$

This object has a tautological factorization structure against $\mathbf{1}_{\mathbf{A}}$.

Vice versa, starting from $\mathcal{M} \in \mathbf{1}_{\mathbf{A}}\text{-mod}^{\text{fact}}(\mathbf{C})_{\mathcal{Z}}$, the unital structure on $\mathcal{M}$ gives rise to a map

$$\text{ins. unit}_{\mathcal{Z}}(\mathcal{M}) \rightarrow \mathcal{M}_{\mathcal{Z} \subseteq, \text{unl}}$$

in $\mathbf{C}_{\mathcal{Z} \subseteq, \text{unl}}$, compatible with factorization.

It suffices to show that the latter map is an isomorphism. This can be checked stratawise, in which case it follows from factorization. $\square$

C.14.11. Let $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$ be a lax unital factorization functor between unital factorization categories. Let $\mathbf{C}_1$ and $\mathbf{C}_2$ be unital factorization module categories over $\mathbf{A}_1$ and $\mathbf{A}_2$, respectively, at some $\mathcal{Z} \rightarrow \text{Ran}$.

Mimicking Sect. B.12.10 we have the notion of lax unital functor

$$\Phi_m : \mathbf{C}_1 \rightarrow \mathbf{C}_2,$$

compatible with factorization. Denote the category of such functors

$$\text{Funct}_{\mathbf{A}_1 \rightarrow \mathbf{A}_2}(\mathbf{C}_1, \mathbf{C}_2).$$

C.14.12. Let $(\Phi, \Phi_m) : (\mathbf{A}_1, \mathbf{C}_1) \rightarrow (\mathbf{A}_2, \mathbf{C}_2)$ be as above. Let $\mathcal{A}_1 \in \mathbf{A}_1$ be a unital factorization algebra, and consider its image $\Phi(\mathcal{A}_1) \in \text{FactAlg}^{\text{unital}}(X, \mathbf{A}_2)$.

Then the functor Φ_m induces a functor

$$(C.72) \quad \Phi_m : \mathcal{A}_1\text{-mod}^{\text{fact}}(\mathbf{C}_1)_Z \rightarrow \Phi(\mathcal{A}_1)\text{-mod}^{\text{fact}}(\mathbf{C}_2)_Z.$$

C.14.13. As in Lemma C.11.23, we have:

Lemma C.14.14. *Suppose that Φ is strictly unital. Then the functor between crystals of categories on $\mathcal{Z}^{\subseteq, \text{unital}}$ underlying every $\Phi_m \in \text{Funct}_{\mathbf{A}_1 \rightarrow \mathbf{A}_2}(\mathbf{C}_1, \mathbf{C}_2)$*

$$\underline{\mathbf{C}}_1 \rightarrow \underline{\mathbf{C}}_2$$

is strict.

C.14.15. Let $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$ be a strictly unital functor between unital factorization categories. Let $\mathbf{C}_2$ be a unital module category over $\mathbf{A}_2$ at some $\mathcal{Z} \rightarrow \text{Ran}$.

In this case, it follows from the construction of the restriction functor Res_Φ that $\text{Res}_\Phi(\mathbf{C}_2)$ possesses a natural unital structure, and the tautological functor

$$(C.73) \quad \text{Res}_\Phi(\mathbf{C}_2) \rightarrow \mathbf{C}_2$$

admits a natural lax unital structure compatible with factorization.

Furthermore, the resulting object

$$\text{Res}_\Phi(\mathbf{C}_2) \in \mathbf{A}_1\text{-mod}_{\mathcal{Z}}^{\text{fact}}$$

has a universal property parallel to that in the non-unital case:

Lemma C.14.16. *For $\mathbf{C}_1 \in \mathbf{A}_1\text{-mod}_{\mathcal{Z}}^{\text{fact}}$, composition with (C.73) defines an equivalence*

$$\text{Funct}_{\mathbf{A}_1\text{-mod}_{\mathcal{Z}}^{\text{fact}}}(\mathbf{C}_1, \text{Res}_\Phi(\mathbf{C}_2)) \xrightarrow{\sim} \text{Funct}_{\mathbf{A}_1 \rightarrow \mathbf{A}_2}(\mathbf{C}_1, \mathbf{C}_2).$$

C.14.17. Let $\mathcal{A}_1 \in \mathbf{A}_1$ be a unital factorization algebra. As in Lemma B.12.14, we have:

Lemma C.14.18. *The functor (C.73) induces an equivalence*

$$\mathcal{A}_1\text{-mod}^{\text{fact}}(\text{Res}_\Phi(\mathbf{C}_2))_Z \rightarrow \Phi(\mathcal{A}_1)\text{-mod}^{\text{fact}}(\mathbf{C}_2)_Z.$$

Remark C.14.19. The material in Sect. B.15 is applicable in the context of unital factorization categories and strictly unital factorization functors between them.

C.14.20. Let us place ourselves momentarily in the context of Sect. B.12.15, where the factorization categories in the diagram

$$\begin{array}{ccc} \mathbf{A}'_1 & \xrightarrow{\Phi'} & \mathbf{A}'_2 \\ \Psi_1 \uparrow & & \uparrow \Psi_2 \\ \mathbf{A}_1 & \xrightarrow{\Phi} & \mathbf{A}_2. \end{array}$$

are unital, all functors are lax unital, and the vertical arrows are strictly unital.

Unwinding the constructions, one obtains that in this case the resulting functor

$$\text{Res}_{\Psi_1}(\mathbf{A}'_1{}^{\text{fact}}_{\mathcal{Z}}) \rightarrow \text{Res}_{\Psi_2}(\mathbf{A}'_2{}^{\text{fact}}_{\mathcal{Z}})$$

viewed as a functor between unital module categories over $\mathbf{A}_1$ and $\mathbf{A}_2$, respectively, possesses a natural lax unital structure compatible with factorization.

C.15. Restriction along lax unital functors. In this subsection we will study the phenomenon of restriction with respect to a functor

$$\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2,$$

which is only lax unital.

C.15.1. Consider the (unital) factorization algebra

$$\Phi(\mathbf{1}_{\mathbf{A}_1}) \in \text{FactAlg}^{\text{unl}}(X, \mathbf{A}_2),$$

see Sect. C.14.11.

Let $\mathcal{Z}$ be a prestack mapping to Ran , and let

$$\Phi_m : \mathbf{C}_1 \rightarrow \mathbf{C}_2$$

be an object of $\text{Func}_{\mathbf{A}_1 \rightarrow \mathbf{A}_2}(\mathbf{C}_1, \mathbf{C}_2)$.

Consider the induced functor

$$(C.74) \quad \Phi_m : \mathbf{C}_{1,\mathcal{Z}} \rightarrow \mathbf{C}_{2,\mathcal{Z}}$$

between the underlying DG categories.

C.15.2. We claim:

Lemma-Construction C.15.3. *The functor (C.74) naturally enhances to a functor*

$$\Phi_m^{\text{enh}} : \mathbf{C}_{1,\mathcal{Z}} \rightarrow \Phi(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\mathbf{C}_2)_{\mathcal{Z}}.$$

Proof. We rewrite

$$\mathbf{C}_{1,\mathcal{Z}} \simeq \mathbf{1}_{\mathbf{A}_1}\text{-mod}^{\text{fact}}(\mathbf{C}_1)_{\mathcal{Z}},$$

and now the required functor is a particular case of (C.72). □

C.15.4. *Variant.* Let us return for a moment to the setting of Sect. C.14.20. By Lemma C.15.3 for every $\underline{x} : S \rightarrow \text{Ran}$ we obtain that the functor

$$\Phi' : \text{Res}_{\Psi_1}(\mathbf{A}'_1{}^{\text{fact}_{\underline{x}}}) \rightarrow \text{Res}_{\Psi_2}(\mathbf{A}'_2{}^{\text{fact}_{\underline{x}}})$$

gives rise to a functor

$$(C.75) \quad \Phi'^{\text{enh}} : \mathbf{A}'_{1,\underline{x}} \rightarrow \Phi(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\text{Res}_{\Psi_2}(\mathbf{A}'_2{}^{\text{fact}_{\underline{x}}}))_{\underline{x}} \stackrel{\text{Lemma C.14.18}}{\simeq} (\Psi_2 \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\mathbf{A}'_2{}^{\text{fact}_{\underline{x}}})_{\underline{x}}.$$

Note that the left-hand side in (C.75) is the value at $\underline{x}$ of a unital factorization category, namely, $\mathbf{A}'_1$ itself.

The right-hand side in (C.75) is the value at $\underline{x}$ of a unital *lax* factorization category, namely,

$$(\Psi_2 \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\mathbf{A}'_2),$$

see Sect. C.11.15.

Unwinding the constructions, we obtain that (C.75) upgrades to a *unital* factorization functor

$$(C.76) \quad \Phi'^{\text{enh}} : \mathbf{A}'_1 \rightarrow (\Psi_2 \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\mathbf{A}'_2).$$

Remark C.15.5. Note that in the setting of Sect. C.15.4, we have

$$(\Psi_2 \circ \Phi)(\mathbf{1}_{\mathbf{A}_1}) \simeq (\Phi' \circ \Psi_1)(\mathbf{1}_{\mathbf{A}_1}) \simeq \Phi'(\mathbf{1}_{\mathbf{A}'_1}).$$

So, the information contained by the functor (C.76) is completely captured by the case when Ψ_1 and Ψ_2 are the identity functors. I.e., the claim is that the lax unital factorization functor

$$\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$$

upgrades to a unital factorization functor

$$\Phi^{\text{enh}} : \mathbf{A}_1 \rightarrow \Phi(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\mathbf{A}_2),$$

with the caveat that the right-hand side is a lax factorization category.

C.15.6. Let $\mathcal{Z}$ and $\mathbf{C}_2$ be as in Sect. C.15.1. Consider the contravariant functor on $\mathbf{A}_1\text{-mod}_{\mathcal{Z}}^{\text{fact}}$ that assigns to $\mathbf{C}_1$ the category $\text{Func}_{\mathbf{A}_1 \rightarrow \mathbf{A}_2}(\mathbf{C}_1, \mathbf{C}_2)$. One shows that this functor is representable, and let

$$\text{Res}_{\Phi}^{\text{unl}}(\mathbf{C}_2) \in \mathbf{A}_1\text{-mod}_{\mathcal{Z}}^{\text{fact}}$$

denote the representing object.

By Lemma C.15.3, the tautological functor

$$(C.77) \quad \text{Res}_{\Phi}^{\text{unl}}(\mathbf{C}_2) \rightarrow \mathbf{C}_2$$

upgrades to a functor

$$(C.78) \quad \text{Res}_{\Phi}^{\text{unl}}(\mathbf{C}_2)_{\mathcal{Z}} \rightarrow \Phi(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\mathbf{C}_2)_{\mathcal{Z}}.$$

C.15.7. We have the following generalization of Lemma B.12.12:

Lemma C.15.8. *The functor (C.78) is an equivalence.*

Remark C.15.9. Note that when Φ is strictly unital, then by Lemma C.14.16

$$\text{Res}_{\Phi}^{\text{unl}}(\mathbf{C}_2) \simeq \text{Res}_{\Phi}(\mathbf{C}_2),$$

and the assertion of Lemma C.15.8 coincides with that of Lemma B.12.12.

C.15.10. *Example.* Let $\mathcal{Z} = \text{pt}$ and let $\mathcal{Z} \rightarrow \text{Ran}$ correspond to a singleton $\{x\} \in \text{Ran}$. Recall the notations of Sect. B.9.28. Let us give an explicit description of the category

$$\text{Res}_{\Phi}^{\text{unl}}(\mathbf{C}_2)_X.$$

Namely,

$$\begin{aligned} \text{Res}_{\Phi}^{\text{unl}}(\mathbf{C}_2)_X &\simeq \\ &\simeq \Phi(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}((\mathbf{C}_2)|_X)_X \times_{\Phi(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}((\mathbf{C}_2)|_{X-x})_{X-x}} \left(\mathbf{A}_{1,X-x} \otimes \Phi(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\mathbf{C}_2)_x \right), \end{aligned}$$

where:

- The notation $(\mathbf{C}_2)|_X$ is as in Sect. B.12.3, i.e., we regard the pullback of $\underline{\mathbf{C}}_2$ along $X \rightarrow \text{Ran}_x$ as a module category over $\mathbf{A}_2$ at X ;
- The functor

$$\mathbf{A}_{1,X-x} \otimes \Phi(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\mathbf{C}_2)_x \rightarrow \Phi(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}((\mathbf{C}_2)|_{X-x})_{X-x}$$

is the composition

$$\begin{aligned} \mathbf{A}_{1,X-x} \otimes \Phi(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\mathbf{C}_2)_x &\xrightarrow{\Phi^{\text{enh}} \otimes \text{Id}} \Phi(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\mathbf{A}_2)_{X-x} \otimes \Phi(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\mathbf{C}_2)_x \rightarrow \\ &\xrightarrow{\text{factorization of } \mathbf{C}_2} \Phi(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}((\mathbf{C}_2)|_{X-x})_{X-x}, \end{aligned}$$

where the second arrow is defined as in Sect. B.11.14.

C.15.11. We have the following analog of Lemma C.14.18

Lemma C.15.12. *The functor (C.77) induces an equivalence*

$$\mathcal{A}_1\text{-mod}^{\text{fact}}(\text{Res}_{\Phi}^{\text{unl}}(\mathbf{C}_2))_{\mathcal{Z}} \rightarrow \Phi(\mathcal{A}_1)\text{-mod}^{\text{fact}}(\mathbf{C}_2)_{\mathcal{Z}}.$$

C.15.13. The material in Sect. B.15 is applicable in the context of unital factorization categories and lax unital factorization functors between them. We will not need it in the full generality, except for an analog of Corollary B.15.7, formulated as Lemma C.15.16 below.

C.15.14. Let

$$\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$$

be a unital functor between unital factorization categories.

Suppose that Φ admits a right adjoint as a functor between the underlying crystals of categories over Ran . According to Sect. C.11.21, the right adjoint Φ^R of Φ admits a natural extension to a lax unital functor between unital factorization categories.

Let $\mathbf{C}_1$ (resp., $\mathbf{C}_2$) be a unital module category over $\mathbf{A}_1$ (resp., $\mathbf{A}_2$) at some $\mathcal{Z} \rightarrow \text{Ran}$.

C.15.15. We claim:

Lemma C.15.16. *There is a canonical equivalence*

$$\text{Funct}_{\mathbf{A}_1\text{-mod}_{\mathcal{Z}}^{\text{fact}}}(\text{Res}_{\Phi}(\mathbf{C}_2), \mathbf{C}_1) \simeq \text{Funct}_{\mathbf{A}_2\text{-mod}_{\mathcal{Z}}^{\text{fact}}}(\mathbf{C}_2, \text{Res}_{\Phi^R}^{\text{unl}}(\mathbf{C}_1)).$$

C.15.17. We will not give a full proof of this lemma; rather we will sketch the construction of the maps in both directions in the framework of Sect. C.15.10.

Namely, for $\mathbf{C}_2 \in \mathbf{A}_2\text{-mod}_{\mathcal{Z}}^{\text{fact}}$ we will construct a functor

$$(C.79) \quad \mathbf{C}_2 \rightarrow \text{Res}_{\Phi^R}^{\text{unl}} \circ \text{Res}_{\Phi}(\mathbf{C}_2)$$

and for $\mathbf{C}_1 \in \mathbf{A}_1\text{-mod}_{\mathcal{Z}}^{\text{fact}}$ we will construct a functor

$$(C.80) \quad \text{Res}_{\Phi} \circ \text{Res}_{\Phi^R}^{\text{unl}}(\mathbf{C}_1) \rightarrow \mathbf{C}_1.$$

C.15.18. By the universal property of $\text{Res}_{\Phi^R}^{\text{unl}}$, the datum of (C.79) is equivalent to the datum of a functor

$$(C.81) \quad \mathbf{C}_2 \rightarrow \text{Res}_{\Phi}(\mathbf{C}_2)$$

as module categories over $\mathbf{A}_2$ and $\mathbf{A}_1$, respectively, compatible with factorization against the functor Φ^R .

We now specialize to the context of Sect. C.15.10 and construct the corresponding functor

$$(C.82) \quad \mathbf{C}_{2,X} \rightarrow \text{Res}_{\Phi}(\mathbf{C}_2)_X.$$

We write

$$(C.83) \quad \text{Res}_{\Phi}(\mathbf{C}_2)_X \simeq \mathbf{C}_{2,X} \times_{\mathbf{C}_{2,X-x}} (\mathbf{A}_{1,X-x} \otimes \mathbf{C}_{2,x}),$$

where

$$\mathbf{A}_{1,X-x} \otimes \mathbf{C}_{2,x} \rightarrow \mathbf{C}_{2,X-x}$$

is the functor

$$\mathbf{A}_{1,X-x} \otimes \mathbf{C}_{2,x} \xrightarrow{\Phi \otimes \text{Id}} \mathbf{A}_{2,X-x} \otimes \mathbf{C}_{2,x} \xrightarrow{\text{fact}} \mathbf{C}_{2,X-x}.$$

We write

$$(C.84) \quad \mathbf{C}_{2,X} \simeq \mathbf{C}_{2,X} \times_{\mathbf{C}_{2,X-x}} (\mathbf{A}_{2,X-x} \otimes \mathbf{C}_{2,x}),$$

where

$$\mathbf{A}_{2,X-x} \otimes \mathbf{C}_{2,x} \xrightarrow{\sim} \mathbf{C}_{2,X-x}$$

is the factorization equivalence.

We define the functor (C.82) by sending an object $\mathbf{c}_2 \in \mathbf{C}_{2,X}$ to

$$\left(\mathbf{c}_2 \times_{j_* \circ j^*(\mathbf{c}_2)} j_*(((\Phi \circ \Phi^R) \otimes \text{Id})(j^*(\mathbf{c}_2))), (\Phi^R \otimes \text{Id})(j^*(\mathbf{c}_2)) \right) \in \mathbf{C}_{2,X} \times_{\mathbf{C}_{2,X-x}} (\mathbf{A}_{1,X-x} \otimes \mathbf{C}_{2,x}),$$

where the map

$$((\Phi \circ \Phi^R) \otimes \text{Id})(j^*(\mathbf{c}_2)) \rightarrow j^*(\mathbf{c}_2)$$

is the counit of the adjunction.

C.15.19. Write

$$(C.85) \quad \text{Res}_\Phi \circ \text{Res}_{\Phi^R}^{\text{untl}}(\mathbf{C}_1)_X \simeq (\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}((\mathbf{C}_1|_X)_X)_{(\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}((\mathbf{C}_1|_{X-x})_{X-x})} \times_{(\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}((\mathbf{C}_1|_{X-x})_{X-x})} \left(\mathbf{A}_{1,X-x} \otimes ((\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\mathbf{C}_1)_x) \right).$$

The sought-for functor

$$\text{Res}_\Phi \circ \text{Res}_{\Phi^R}^{\text{untl}}(\mathbf{C}_1)_X \rightarrow \mathbf{C}_{1,X}$$

sends an object $(\mathbf{c}'_1, \mathbf{c}''_1)$ in the right-hand side of (C.85), i.e.,

$$\mathbf{c}'_1 \in (\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}((\mathbf{C}_1|_X)_X)$$

and

$$\mathbf{c}''_1 \in \mathbf{A}_{1,X-x} \otimes ((\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})\text{-mod}^{\text{fact}}(\mathbf{C}_1)_x)$$

to the object

$$\text{oblv}_{(\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})}(\mathbf{c}'_1) \times_{j_* \circ j^* \circ \text{oblv}_{(\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})}(\mathbf{c}'_1)} \left(j_* \circ (\text{Id} \otimes \text{oblv}_{(\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})})(\mathbf{c}''_1) \right),$$

where the map

$$(\text{Id} \otimes \text{oblv}_{(\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})})(\mathbf{c}''_1) \rightarrow j^* \circ \text{oblv}_{(\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})}(\mathbf{c}'_1)$$

is obtained by applying the functor $\text{oblv}_{(\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})}$ to the isomorphism

$$((\Phi^R \circ \Phi)^{\text{enh}} \otimes \text{Id})(\mathbf{c}''_1) \simeq j^*(\mathbf{c}'_1),$$

precomposed with the unit of the (Φ, Φ^R) -adjunction

$$\begin{aligned} (\text{Id} \otimes \text{oblv}_{(\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})})(\mathbf{c}''_1) &\rightarrow ((\Phi^R \circ \Phi) \otimes \text{oblv}_{(\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})})(\mathbf{c}''_1) \simeq \\ &\simeq \text{oblv}_{(\Phi^R \circ \Phi)(\mathbf{1}_{\mathbf{A}_1})} \left(((\Phi^R \circ \Phi)^{\text{enh}} \otimes \text{Id})(\mathbf{c}''_1) \right). \end{aligned}$$

C.16. Tightness.

C.16.1. Let $\mathbf{A}$ be a unital factorization category. We will say that $\mathbf{A}$ is *tight* if for every $\underline{x}, \underline{x}' : S \rightarrow \text{Ran}$ with $\underline{x} \subseteq \underline{x}'$, the corresponding functor

$$\text{ins}_{\underline{x} \subseteq \underline{x}'} : \mathbf{A}_{S, \underline{x}} \rightarrow \mathbf{A}_{S, \underline{x}'}$$

admits a continuous right adjoint.

C.16.2. Many of the unital factorization categories we have introduced satisfy this property. This includes representation-theoretic examples, e.g.,

$$\widehat{\mathfrak{g}}\text{-mod}_\kappa, \text{KL}(G)_\kappa$$

and algebro-geometric examples:

$$\text{QCoh}(\mathcal{T}),$$

where $\mathcal{T}$ is an affine counital factorization space, and

$$\text{IndCoh}^*(\mathcal{T}) \text{ and } \text{IndCoh}^!(\mathcal{T}),$$

where $\mathcal{T}$ is a unital-in-correspondences ind-placid factorization ind-schemes, satisfying the conditions from Sects. C.12.8 and C.12.9, respectively.

C.16.3. Let $\mathbf{A}$ be a tight unital factorization category, and let $\mathbf{C}$ be a unital factorization module category over $\mathbf{C}$ at some $\mathcal{Z}$.

We shall say that $\mathbf{C}$ is *tight* if for every $(z, \underline{x}), (z, \underline{x}') : S \rightarrow \mathcal{Z}^\subseteq$ with $\underline{x} \subseteq \underline{x}'$, the corresponding functor

$$\text{ins}_{\underline{x} \subseteq \underline{x}'} : \mathbf{C}_{S, \underline{x}} \rightarrow \mathbf{C}_{S, \underline{x}'}$$

admits a continuous right adjoint.

C.16.4. The following is immediate:

Lemma C.16.5. *Suppose that $\mathbf{A}$ is tight. Then for any $\mathcal{Z} \rightarrow \text{Ran}$, the factorization module category $\mathbf{A}^{\text{fact}_{\mathcal{Z}}}$ is tight.*

C.16.6. Take $\mathbf{A} = \text{Vect}$, and recall the construction from Sect. C.14.7

$$(C.86) \quad \underline{\mathbf{C}}_0 \in \text{CrysCat}(\mathcal{Z}_0) \rightsquigarrow \mathbf{C} \in \text{Vect-}\mathbf{mod}_{\mathcal{Z}}^{\text{fact}}.$$

It is clear that the essential image of (C.86) is contained in the full subcategory of $\text{Vect-}\mathbf{mod}_{\mathcal{Z}}^{\text{fact}}$ consisting of tight unital factorization module categories.

We claim:

Lemma C.16.7. *The functor (C.86) is an equivalence onto the full subcategory of $\text{Vect-}\mathbf{mod}_{\mathcal{Z}}^{\text{fact}}$ consisting of tight objects.*

Proof. The functor (C.86) admits a retraction (i.e., a left inverse), given by restricting the crystal of categories from $\mathcal{Z}^\subseteq$ to $\mathcal{Z}$ along $\text{diag}_{\mathcal{Z}}$. We claim that this left inverse is an actual inverse when applied to tight objects.

Indeed, let $\mathbf{C}'$ be a tight unital factorization module category at $\mathcal{Z}$. Let $\underline{\mathbf{C}}'$ be the corresponding crystal of categories over $\mathcal{Z}^\subseteq$, and let $\underline{\mathbf{C}}'_0$ denote the restriction of $\underline{\mathbf{C}}'$ along $\text{diag}_{\mathcal{Z}}$.

The unital structure on $\mathbf{C}'$ gives rise to a functor

$$(C.87) \quad \text{pr}_{\text{small}}^*(\underline{\mathbf{C}}'_0) \rightarrow \underline{\mathbf{C}}.$$

We have to show that (C.87) is an equivalence.

By the tightness assumption, the functor (C.87) admits a right adjoint. Hence, in order to show that it is an equivalence, it suffices to show that it is an equivalence strata-wise. However, this follows from factorization. □

C.16.8. Consider the following situation. Let $\mathbf{A}$ be a tight unital factorization category, and let $\mathbf{C}$ be a tight unital factorization module category over it at some $\mathcal{Z}$.

Consider the (strictly) unital factorization functor

$$\text{Vac}_{\mathbf{A}} : \text{Vect} \rightarrow \mathbf{A},$$

see Sect. C.11.8.

C.16.9. We claim:

Lemma C.16.10. *The unital factorization module category*

$$\text{Res}_{\text{Vac}_{\mathbf{A}}}(\mathbf{C}) \in \text{Vect-}\mathbf{mod}_{\mathcal{Z}}^{\text{fact}}$$

is tight.

Proof. Set $\underline{\mathbf{C}}'_0 := \underline{\mathbf{C}}_{\mathcal{Z}}$ be the sheaf of categories on $\mathcal{Z}$ underlying $\mathbf{C}$. Let $\mathbf{C}'$ be the (tight, unital) factorization module over $\mathbf{Vect}$, attached $\underline{\mathbf{C}}'_0$ by the functor (C.86).

The unital structure on $\mathbf{C}$ gives to a (strictly unital) functor

$$\mathbf{C}' \rightarrow \mathbf{C},$$

compatible with factorization (in the sense of Sect. C.14.11). Hence, we obtain a functor

$$(C.88) \quad \mathbf{C}' \rightarrow \mathbf{Res}_{\mathbf{Vac}_{\mathbf{A}}}(\mathbf{C}).$$

We claim that the functor (C.88) is an equivalence. Indeed, this follows from the assumptions by applying Lemma B.15.9. □

C.16.11. We will use Lemma C.16.10 as follows. Let $\mathbf{A}$ be a tight unital factorization category, and let R be a factorization algebra (in $\mathbf{Vect}$).

Using the functor $\mathbf{Vac}_{\mathbf{A}}$, we can consider the factorization algebra

$$\mathbf{Vac}_{\mathbf{A}}(R) \simeq R \otimes \mathbf{1}_{\mathbf{A}}$$

in $\mathbf{A}$, and consider the corresponding lax factorization category

$$R\text{-mod}^{\text{fact}}(\mathbf{A}) := (R \otimes \mathbf{1}_{\mathbf{A}})\text{-mod}^{\text{fact}}(\mathbf{A}).$$

We have a naturally defined functor between lax factorization categories

$$(C.89) \quad R\text{-mod}^{\text{fact}} \otimes \mathbf{A} \rightarrow R\text{-mod}^{\text{fact}}(\mathbf{A}),$$

see (B.44).

Combining Lemmas C.16.10 and B.12.9, we obtain:

Corollary C.16.12. *Assume that $\mathbf{A}$ is dualizable as a factorization category. Then the functor (C.89) is an equivalence.*

APPENDIX D. CHIRAL MODULES

The main purpose of this section is to prove Theorem 4.3.9, which gives a geometric description of modules over commutative factorization/chiral algebras.

To do so, we first develop a general theory describing modules over chiral algebras in terms of modules over topological algebras (although we do not write in these exact terms), with an especially explicit understanding for “nice” Lie- $\ast$ algebras.

This material largely consists of transporting [BD2, Sect. 3.6] into the derived setting. However, we will encounter a surprise: a certain equivalence that always takes place at an abelian level, in order to hold at the derived level requires a finiteness condition (see Sect. D.6).

D.1. A reminder: chiral vs factorization algebras.

D.1.1. Recall that, according to [BD2, Proposition 3.4.19] (see [FraG] for the derived version), factorization algebras are the same as chiral algebras. Given a factorization algebra $\mathcal{A}$, the corresponding chiral algebra, thought of as a D-module on X , is given by

$$\mathcal{A}^{\text{ch}} := \mathcal{A}_X[-1].$$

For example, the unit factorization algebra corresponds to the chiral algebra ω_X (see our conventions in Sect. 1.1.1⁸¹).

D.1.2. Generalizing [BD2, Proposition 3.4.19], we have

$$(D.1) \quad \mathcal{A}\text{-mod}^{\text{fact}} \simeq \mathcal{A}^{\text{ch}}\text{-mod}^{\text{ch}},$$

where modules on both sides can be taken on any space that maps to $\mathbf{Ran}$.

⁸¹This was one of the main reasons for this choice of conventions, i.e., in order to be in line with [BD2].

D.1.3. Let us recall also how the bijection between between chiral and factorization algebras plays out in the commutative case.

Let A be a commutative algebra object in $\mathrm{D}\text{-mod}(X)$. Then the corresponding factorization algebra $\mathcal{A} := \mathrm{Fact}(A)$ is such that

$$\mathcal{A}_X = A.$$

And the corresponding chiral algebra $\mathcal{A}^{\mathrm{ch}}$ is $A[-1]$. Note that

$$\mathbf{oblv}^r(\mathcal{A}^{\mathrm{ch}}) = \mathbf{oblv}^l(A) \otimes \omega_X.$$

D.1.4. For example, for the free commutative algebra $A = \mathrm{Sym}^!(\mathcal{M}[1])$ for $\mathcal{M} \in \mathrm{D}\text{-mod}(X)$, we have

$$\mathcal{A}^{\mathrm{ch}} = \mathrm{Sym}^!(\mathcal{M}[1])[-1] \simeq U^{\mathrm{ch}}(\mathcal{M}),$$

where:

- In the left-hand side $\mathcal{M}$ is considered as an abelian Lie-* algebra;
- U^{ch} is the functor of chiral envelope.

Note also that in this case.

$$\mathbf{oblv}^l(A) = \mathrm{Sym}_{\mathcal{O}_X}(\mathbf{oblv}^l(\mathcal{M}[1])) \simeq \mathrm{Sym}_{\mathcal{O}_X}(\mathbf{oblv}^r(\mathcal{M}) \otimes \omega_X^{\otimes -1}).$$

So, if $\mathcal{M} = \mathbf{ind}^r(\mathcal{E}) = \mathcal{E} \otimes D_X$ for a classical locally free sheaf $\mathcal{E}$ on X , then the corresponding D-scheme

$$\mathrm{Spec}_X(A)$$

is the scheme $\mathrm{Jets}(\mathcal{E}^\vee \otimes \omega_X)$ of jets into the vector bundle $\mathcal{E}^\vee \otimes \omega_X$.

D.1.5. *Convention.* In what follows, by a slight abuse of notation, for a given factorization algebra $\mathcal{A}$, we will use the same symbol $\mathcal{A}$ to denote the corresponding chiral algebra (i.e., we will not write $\mathcal{A}^{\mathrm{ch}}$).

D.2. The pro-projective generator for chiral modules.

D.2.1. Let $\mathcal{A}$ be a unital chiral algebra on X . Let $\mathcal{A}\text{-mod}_x^{\mathrm{ch}}$ denote the category of *unital* chiral $\mathcal{A}$ -modules at x . We let $\mathbf{oblv}_{\mathcal{A}}$ denote the forgetful functor $\mathcal{A}\text{-mod}_x^{\mathrm{ch}} \rightarrow \mathrm{Vect}$.

For $\mathcal{M} \in \mathcal{A}\text{-mod}_x^{\mathrm{ch}}$, we will consider the action map

$$j^*(\mathcal{A}) \otimes \mathcal{M} \xrightarrow{\mathrm{action}} i_*(\mathcal{M}),$$

as a map of chiral $\mathcal{A}$ -modules on X , where:

- j denotes the open embedding $X - x \hookrightarrow X$;
- i denotes the embedding of the point x into X .

D.2.2. We will also use a short-hand notation

$$\underline{\mathcal{M}} := \mathbf{oblv}_{\mathcal{A}}(\mathcal{M}), \quad \mathcal{M} \in \mathcal{A}\text{-mod}_x^{\mathrm{ch}}.$$

In what follows we will take about “elements” of $\underline{\mathcal{M}}$:

For $V \in \mathrm{Vect}$, by an element $v \in V$ we mean a point of the space $\mathrm{Maps}_{\mathrm{Vect}}(k, V)$.

D.2.3. In what follows we will denote by $\mathcal{A}_x$ is the $[1]$ -shifted $!$ -fiber of $\mathcal{A}$ at x (this is the same as the $!$ -fiber at $x \in \mathrm{Ran}$ of the factorization algebra corresponding to $\mathcal{A}$), viewed as an object⁸² of $\mathcal{A}\text{-mod}_x^{\mathrm{ch}}$.

We let

$$1_{\mathcal{A},x} \in \mathcal{A}_x$$

denote the vacuum vector, i.e., the image of $1 \in k \simeq (\omega_X)_x$ under the unit map

$$\mathrm{vac}_{\mathcal{A}} : \omega_X \rightarrow \mathcal{A}.$$

⁸²In fact, $\mathcal{A}_x$ should more properly be denoted $\mathcal{A}^{\mathrm{fact},x}$, see Sect. B.9.7.

D.2.4. Consider the category $\text{Modif}(\mathcal{A})$ of unital chiral algebras $\mathcal{A}'$ equipped with an isomorphism

$$\mathcal{A}'|_{X-x} \simeq \mathcal{A}|_{X-x}.$$

This category has fiber products, and hence is cofiltered.

Note that the category $\mathcal{A}\text{-mod}_x^{\text{ch}}$ only depends on $\mathcal{A}|_{X-x}$, so for any $\mathcal{A}' \in \text{Modif}(\mathcal{A})$ we have a canonical identification

$$\mathcal{A}\text{-mod}_x^{\text{ch}} \simeq \mathcal{A}'\text{-mod}_x^{\text{ch}}.$$

D.2.5. Consider the functor

$$\text{Modif}(\mathcal{A}) \rightarrow \mathcal{A}\text{-mod}_x^{\text{ch}}, \quad \mathcal{A}' \mapsto \mathcal{A}'_x \in \mathcal{A}'\text{-mod}_x^{\text{ch}} \simeq \mathcal{A}\text{-mod}_x^{\text{ch}}.$$

Set

$$\mathbf{P}_{\mathcal{A},x} := \varprojlim_{\mathcal{A}' \in \text{Modif}(\mathcal{A})} \mathcal{A}'_x \in \text{Pro}(\mathcal{A}\text{-mod}_x^{\text{ch}}).$$

The object

$$\mathbf{oblv}_{\mathcal{A}}(\mathbf{P}_{\mathcal{A},x}) \in \text{Pro}(\text{Vect})$$

is equipped with a canonical vector $1_{\mathbf{P},x}$ comprised of the vacuum vectors $1_{\mathcal{A}',x} \in \mathcal{A}'_x$.

D.2.6. Evaluation on $1_{\mathbf{P},x}$ gives rise to a natural transformation

$$(D.2) \quad \mathcal{H}om(\mathbf{P}_{\mathcal{A},x}, -) \rightarrow \mathbf{oblv}_{\mathcal{A}}, \quad \mathcal{A}\text{-mod}_x^{\text{ch}} \rightarrow \text{Vect}.$$

The following assertion is a derived version of [BD2, Proposition 3.6.16]:

Proposition D.2.7. *The natural transformation is an isomorphism.*

D.2.8. The proof of Proposition D.2.7 is based on the following observation: we claim that the category $\text{Modif}(\mathcal{A})$ is equivalent to the category of pairs

$$(\mathcal{M} \in \mathcal{A}\text{-mod}_x^{\text{ch}}, m \in \underline{\mathcal{M}}).$$

Namely, in one direction, to $\mathcal{A}' \in \text{Modif}(\mathcal{A})$ we attach the pair $(\mathcal{A}'_x, 1_{\mathcal{A}',x})$.

Vice versa, given $(\mathcal{M}, m)$ we let $\mathcal{A}'$ be the fiber of the map

$$(D.3) \quad j_* \circ j^*(\mathcal{A}) \xrightarrow{\text{id} \otimes m} j_* \circ j^*(\mathcal{A}) \otimes \mathcal{M} \xrightarrow{\text{action}} i_*(\mathcal{M}).$$

We claim that $\mathcal{A}'$ has a natural structure of unital chiral algebra. This fits into the following general paradigm:

Let L be a Lie algebra in a symmetric pseudo-monoidal monoidal category $\mathbf{A}$, and let M be a module over it. Consider $L \oplus M$ as a split square-zero extension of M . For an element $m \in M$ (i.e., a map $\mathbf{1}_{\mathbf{A}} \rightarrow M$), the action of L on M gives rise to an automorphism ϕ_m of $L \oplus M$. Then we can identify

$$\text{Fib}(L \xrightarrow{\text{act on } m} M)$$

with

$$L \times_{L \oplus M} L,$$

where the two maps $L \rightrightarrows L \oplus M$ are the compositions of the tautological embedding with: (1) the identity map, and (2) ϕ_m .

We apply this to $\mathbf{A} := \text{D-mod}(X)$, equipped with the chiral pseudo-monoidal monoidal structure, so that $\mathbf{1}_{\mathbf{A}}$ is the “constant sheaf” on X . We take $L = j_* \circ j^*(\mathcal{A})$ and $M = i_*(\mathcal{M})$. This endows $\mathcal{A}'$ with a chiral algebra structure, i.e.,

$$\mathcal{A}' = j_* \circ j^*(\mathcal{A}) \times_{j_* \circ j^*(\mathcal{A}) \oplus i_*(\mathcal{M})} j_* \circ j^*(\mathcal{A}).$$

In order to show that $\mathcal{A}'$ is unital, by Proposition C.7.13, it is enough to equip it with a quasi-unital structure (see Sect. C.7.12 for what this means). The above fiber product presentation defines this structure on the nose.

Proof of Proposition D.2.7. This is tautological from Sect. D.2.8: the assertion of the proposition is just the fact that the map

$$\operatorname{colim}_{(\mathcal{M}, m \in \underline{\mathcal{M}})} \mathcal{H}om(\mathcal{M}, \mathcal{M}') \rightarrow \underline{\mathcal{M}'}$$

is an isomorphism. $\square$

D.2.9. Assume now that $j^*(\mathcal{A})$ is connective. Let

$$(D.4) \quad \operatorname{Modif}_{\operatorname{conn}}(\mathcal{A}) \subset \operatorname{Modif}(\mathcal{A})$$

be the full subcategory consisting of those $\mathcal{A}'$ that are connective.

Truncation ≤ 0 on chiral algebras defines a right adjoint to the above inclusion. Hence, the opposite of (D.4) is cofinal.

In particular, the object $P_{\mathcal{A},x}$ maps isomorphically to

$$\operatorname{"lim"}_{\mathcal{A}' \in \operatorname{Modif}_{\operatorname{conn}}(\mathcal{A})} \mathcal{A}'_x \in \operatorname{Pro}(\mathcal{A}\text{-mod}_x^{\operatorname{ch}}).$$

D.2.10. For an integer m , let

$$\operatorname{Modif}_{\operatorname{conn}, \geq -m}(\mathcal{A}) \subset \operatorname{Modif}_{\operatorname{conn}}(\mathcal{A})$$

be the full subcategory consisting of those objects $\mathcal{A}'$, for which $\mathcal{A}'_x \in \operatorname{Vect}^{\geq -m, \leq 0}$.

Let $\operatorname{Modif}_{\operatorname{conn}, \operatorname{ev-c}}(\mathcal{A})$ be the full subcategory of $\operatorname{Modif}_{\operatorname{conn}}(\mathcal{A})$ consisting of those objects $\mathcal{A}'$, for which $\mathcal{A}'_x$ is eventually coconnective (as an object of Vect). I.e.,

$$\operatorname{Modif}_{\operatorname{conn}, \operatorname{ev-c}}(\mathcal{A}) = \operatorname{colim}_m \operatorname{Modif}_{\operatorname{conn}, \geq -m}(\mathcal{A}).$$

Note that if $j^*(\mathcal{A})$ is itself eventually coconnective, the above condition on $\mathcal{A}'_x$ is equivalent to $\mathcal{A}'$ being eventually coconnective.

D.2.11. Set

$$P_{\mathcal{A},x,\operatorname{ev-c}} := \operatorname{"lim"}_{\mathcal{A}' \in \operatorname{Modif}_{\operatorname{conn}, \operatorname{ev-c}}(\mathcal{A})} \mathcal{A}'_x \in \operatorname{Pro}(\mathcal{A}\text{-mod}_x^{\operatorname{ch}}).$$

We have a tautological map

$$(D.5) \quad P_{\mathcal{A},x} \rightarrow P_{\mathcal{A},x,\operatorname{ev-c}}$$

in $\operatorname{Pro}(\mathcal{A}\text{-mod}_x^{\operatorname{ch}})$.

Lemma D.2.12. *The map*

$$\mathcal{H}om(P_{\mathcal{A},x,\operatorname{ev-c}}, -) \rightarrow \mathcal{H}om(P_{\mathcal{A},x}, -),$$

defined by (D.5) is an isomorphism, when evaluated on $(\mathcal{A}\text{-mod}_x^{\operatorname{ch}})^{>-\infty}$.

Proof. It suffices to show that if $\mathcal{M} \in (\mathcal{A}\text{-mod}_x^{\operatorname{ch}})^{\geq -m}$ for some m , then

$$\operatorname{colim}_{\mathcal{A}' \in \operatorname{Modif}_{\operatorname{conn}, \operatorname{ev-c}}(\mathcal{A})} \operatorname{Maps}(\mathcal{A}'_x, \mathcal{M}) \rightarrow \operatorname{colim}_{\mathcal{A}' \in \operatorname{Modif}_{\operatorname{conn}}(\mathcal{A})} \operatorname{Maps}(\mathcal{A}'_x, \mathcal{M})$$

is an isomorphism.

However, it is clear that the map

$$\operatorname{colim}_{\mathcal{A}' \in \operatorname{Modif}_{\operatorname{conn}, \geq -m'}(\mathcal{A})} \operatorname{Maps}(\mathcal{A}'_x, \mathcal{M}) \rightarrow \operatorname{colim}_{\mathcal{A}' \in \operatorname{Modif}_{\operatorname{conn}}(\mathcal{A})} \operatorname{Maps}(\mathcal{A}'_x, \mathcal{M})$$

is an isomorphism for every $m' \geq m$. $\square$

Corollary D.2.13. *The map*

$$\mathcal{H}om(P_{\mathcal{A},x,\operatorname{ev-c}}, -) \rightarrow \operatorname{oblv}_{\mathcal{A}}$$

is an isomorphism, when evaluated on $(\mathcal{A}\text{-mod}_x^{\operatorname{ch}})^{>-\infty}$.

D.3. The case of Lie-* algebras. In this subsection we will assume that the chiral algebra $\mathcal{A}$ is the chiral universal envelope $U^{\text{ch}}(L)$ of a (connective) Lie-* algebra L . Recall that we can identify

$$\mathcal{A}\text{-mod}_x^{\text{ch}} \simeq L\text{-mod}_x^{\text{ch}}.$$

We will study how the object $P_{\mathcal{A},x}$ looks like in this case.

D.3.1. Consider the categories

$$\text{Modif}(L) \text{ and } \text{Modif}_{\text{conn}}(L)$$

defined as in the case of chiral algebras, i.e., these are Lie-* algebras equipped with an isomorphism with L over $X - x$.

We have two pairs of adjoint functors

$$U^{\text{ch}} : \text{Modif}(L) \rightleftarrows \text{Modif}(\mathcal{A}) : \mathbf{oblv}^{\text{ch} \rightarrow \text{Lie-*}}$$

and

$$U^{\text{ch}} : \text{Modif}_{\text{conn}}(L) \rightleftarrows \text{Modif}_{\text{conn}}(\mathcal{A}) : \mathbf{oblv}^{\text{ch} \rightarrow \text{Lie-*}}.$$

In particular, the corresponding functors

$$\text{Modif}(L)^{\text{op}} \rightarrow \text{Modif}(\mathcal{A})^{\text{op}} \text{ and } \text{Modif}_{\text{conn}}(L)^{\text{op}} \rightarrow \text{Modif}_{\text{conn}}(\mathcal{A})^{\text{op}}$$

are cofinal.

D.3.2. In particular, we obtain that we can write

$$P_{\mathcal{A},x} \simeq \varinjlim_{L' \in \text{Modif}(L)} \mathbf{ind}_{L'\text{-mod}_x^{\text{Lie-*}}}^{L\text{-mod}_x^{\text{ch}}}(k)$$

and when L is connective also as

$$P_{\mathcal{A},x} \simeq \varinjlim_{L' \in \text{Modif}_{\text{conn}}(L)} \mathbf{ind}_{L'\text{-mod}_x^{\text{Lie-*}}}^{L\text{-mod}_x^{\text{ch}}}(k),$$

where

- $\mathbf{ind}_{L'\text{-mod}_x^{\text{Lie-*}}}^{L\text{-mod}_x^{\text{ch}}} : L'\text{-mod}_x^{\text{Lie-*}} \rightarrow L'\text{-mod}_x^{\text{ch}} \simeq L\text{-mod}_x^{\text{ch}}$ is the left adjoint of the restriction functor.

D.3.3. Assume now that $j^*(L)$ is classical (i.e., is in cohomological degree 0 as a right D-module), and let

$$\text{Modif}_{\text{cl,flat}}(L) \subset \text{Modif}_{\text{conn}}(L)$$

be the full subcategory, consisting of those modifications L' that are classical and flat (as $\mathcal{O}_X$ -modules).

Note that the functor of chiral universal envelope maps

$$(D.6) \quad \text{Modif}_{\text{cl,flat}}(L) \rightarrow \text{Modif}_{\text{conn, ev-c}}(\mathcal{A}).$$

Proposition D.3.4. *Assume that $j^*(L)$ is finitely generated and locally free as a D-module. Then the (opposite of the) functor (D.6) is cofinal.*

We note that the assertion of the proposition would be false without the finite generation assumption, see Sect. D.6.

The proposition will be proved in Sect. D.4. We will now consider some applications.

Corollary D.3.5. *Under the assumptions of Proposition D.3.4, we have*

$$P_{\mathcal{A},x,\text{ev-c}} \simeq \varinjlim_{L' \in \text{Modif}_{\text{cl,flat}}(L)} \mathbf{ind}_{L'\text{-mod}_x^{\text{Lie-*}}}^{L\text{-mod}_x^{\text{ch}}}(k).$$

D.3.6. Take

$$L = L_{\mathfrak{g}} = \mathfrak{g} \otimes D_X,$$

where $\mathfrak{g}$ is a classical finite-dimensional Lie algebra (or a central extension of $L_{\mathfrak{g}}$).

In this case, the category $\text{Modif}_{\text{cl,flat}}(L)^{\text{op}}$ contains a cofinal family of objects of the form

$$\mathfrak{g} \otimes \mathcal{O}_X(-n \cdot x) \otimes_{\mathcal{O}_X} D_X.$$

Note that the corresponding objects $\mathbf{ind}_{L' \text{-mod}_{L_{\text{Lie}}^*}}^{L \text{-mod}_x^{\text{ch}}}(k)$ are the images under (4.2) of

$$\mathbf{ind}_{\mathfrak{g}_n}^{\widehat{\mathfrak{g}}}(k),$$

where $\mathfrak{g}_n \subset \widehat{\mathfrak{g}}$ is the n th congruence subalgebra.

Hence, combining Corollaries D.3.5 and D.2.13, we obtain:

Corollary D.3.7. *The natural transformation*

$$\text{colim}_n \mathcal{H}om(\mathbf{ind}_{\mathfrak{g}_n}^{\widehat{\mathfrak{g}}}(k), -) \rightarrow \mathbf{oblv}_L, \quad (L \text{-mod}_x^{\text{ch}})^{>-\infty} \rightarrow \text{Vect}$$

is an isomorphism.

D.3.8. Assume now that $\mathcal{M}$ is abelian (and finitely generated and locally free as a D-module). Note that in this case

$$\mathcal{A} = U^{\text{ch}}(\mathcal{M}) \simeq \text{Sym}^!(\mathcal{M}[1])[1]$$

is a commutative chiral algebra (see Sect. D.1.4).

In this case we can talk about *commutative* chiral $\mathcal{A}$ -modules: this is by definition the category of modules over the commutative algebra $\text{Sym}(\mathcal{M}_x)$, and it has a natural forgetful functor to

$$\mathcal{A}\text{-mod}_x^{\text{ch}} \simeq \mathcal{M}\text{-mod}_x^{\text{ch}}.$$

Denote

$$\text{Modif}_{\text{cl,flat,f.g.}}(\mathcal{M}) := \{\mathcal{M}' \in \text{D-mod}(X)^{\nabla, \text{flat, f.g.}}, \mathcal{M}'|_{X-x} \simeq j^*(\mathcal{M})\}.$$

From Corollaries D.3.5 and D.2.13, we obtain:

Corollary D.3.9. *The natural transformation*

$$\text{colim}_{\mathcal{M}' \in \text{Modif}_{\text{cl,flat,f.g.}}(\mathcal{M})} \mathcal{H}om(\text{Sym}(\mathcal{M}'_x), -) \rightarrow \mathbf{oblv}_{\mathcal{M}}, \quad (\mathcal{M}\text{-mod}_x^{\text{ch}})^{>-\infty} \rightarrow \text{Vect}$$

is an isomorphism, where

$$\text{Sym}(\mathcal{M}'_x) \in \text{Sym}(\mathcal{M}'_x)\text{-mod} \rightarrow \mathcal{M}'\text{-mod}_x^{\text{ch}} \simeq \mathcal{M}\text{-mod}_x^{\text{ch}}.$$

D.4. Proof of Proposition D.3.4.

D.4.1. Let $\widetilde{L}$ be an eventually coconnective Lie- $*$ algebra on X , and let us be given a map

$$j^*(L) \xrightarrow{\alpha} j^*(\widetilde{L}).$$

Consider the category

$$\mathbf{C}_{\widetilde{L}} := \{L' \in \text{Modif}_{\text{cl,flat}}(L), L' \xrightarrow{\alpha'} \widetilde{L} | \alpha'|_{X-x} = \alpha\}.$$

We will prove:

Proposition D.4.2. *Assume that $j^*(L)$ is finitely generated and locally free as a D-module. Then the category $\mathbf{C}_{\widetilde{L}}$ is non-empty.*

Let us show how Proposition D.4.2 implies Proposition D.3.4.

Proof of Proposition D.3.4. By adjunction, it suffices to prove that in the setup of Proposition D.4.2, the category $\mathbf{C}_{\tilde{L}}$ is contractible. We will show that it is cofiltered.

Let

$$L'_I : I \rightarrow \mathbf{C}_{\tilde{L}}$$

be a finite diagram. We need to show that it can be extended to a diagram

$$L'_{I^\triangleleft} : I^\triangleleft \rightarrow \mathbf{C}_{\tilde{L}},$$

where $I^\triangleleft$ is a left cone over I .

Set

$$\tilde{L}_I := \lim_I L'_I,$$

where the limit is taken in the category of Lie-* algebras over $\tilde{L}$. Note that $\tilde{L}_I$ is eventually coconnective. By construction, we have a map

$$\alpha_I : j^*(L) \rightarrow j^*(\tilde{L}_I).$$

The datum of $L'_{I^\triangleleft}$ is equivalent to finding $L' \in \text{Modif}_{\text{cl,flat}}(L)$ and a map

$$\alpha'_I : L' \rightarrow \tilde{L}_I,$$

extending α_I .

However, the existence of (L', α'_I) is guaranteed by Proposition D.4.2

□

[Proposition D.3.4]

D.4.3. The rest of this subsection is devoted to the proof of Proposition D.4.2. The starting point is the following observation ([BD2, Lemma 2.5.13]):

Lemma D.4.4. *Assume that $j^*(L)$ is classical, and the underlying D -module is finitely generated and locally free. Then the tautological map*

$$j_! \circ j^*(\mathbf{oblv}_{\text{Lie-*}}(L)) \rightarrow \lim_{L' \in \text{Modif}_{\text{cl,flat}}(L)} \mathbf{oblv}_{\text{Lie-*}}(L')$$

is an isomorphism in $\text{Pro}(\text{D-mod}(X))$, where:

- $\mathbf{oblv}_{\text{Lie-*}}$ is the forgetful functor from the category of Lie-* algebras to the category of D -modules on X ;
- $j_! : \text{D-mod}(X - x) \rightarrow \text{Pro}(\text{D-mod}(X))$ is the pro-left adjoint of j^* .

D.4.5. Let $\tilde{L}$ be concentrated in degrees $[-n, 0]$. We will argue by induction on n .

Consider first the case $n = 0$.

By Lemma D.4.4, we can find $L' \in \text{Modif}_{\text{cl,flat}}(L)$, so that α extends to a map

$$\alpha' : L' \rightarrow \tilde{L}$$

as plain D -modules.

The obstruction to α' being a map of Lie-* algebras is a map

$$L' \boxtimes L' \rightarrow \Delta_*(\tilde{L}),$$

which vanishes on $(X - x) \times (X - x)$.

The assumption that $j^*(L)$ is finitely generated implies that the naturally defined map

$$(j \times j)_! \circ (j \times j)^*(L \boxtimes L) \rightarrow j_! \circ j^*(L) \boxtimes j_! \circ j^*(L)$$

in $\text{Pro}(\text{D-mod}(X \times X))$ is an isomorphism. Hence, again by Lemma D.4.4, there exists an arrow $L'' \rightarrow L'$ in $\text{Modif}_{\text{cl,flat}}(L)$ such that the composition

$$L'' \boxtimes L'' \rightarrow L' \boxtimes L' \rightarrow \Delta_*(\tilde{L})$$

vanishes.

Hence, $L'' \rightarrow L' \rightarrow \tilde{L}$ provides the desired object of $\mathbf{C}_{\tilde{L}}$.

D.4.6. We now perform the induction step. Suppose the assertion is valid for $\tau^{\geq n-1}(\tilde{L})$, i.e., that can find an object $L' \in \text{Modif}_{\text{cl,flat}}(L)$ and a lift of α to a map of Lie-* algebras

$$L' \rightarrow \tau^{\geq n-1}(\tilde{L}).$$

Fix this map, and consider the fiber product

$$\tilde{L} \times_{\tau^{\geq n-1}(\tilde{L})} L' =: \tilde{L}'.$$

We wish to find an arrow $L'' \rightarrow L'$ in $\text{Modif}_{\text{cl,flat}}(L)$, so that the map

$$\tilde{L}' \times_{L'} L'' =: \tilde{L}'' \rightarrow L''$$

admits a left inverse.

By Lemma D.4.4, after replacing L' , the extension

$$\tilde{L}' \rightarrow L'$$

is given by an $(n+2)$ -cocycle, which is a map

$$(L')^{\boxtimes(n+2)} \rightarrow \Delta_*^n(\tilde{L}),$$

which vanishes on $(X-x)^n$, where Δ^n denotes the main diagonal $X \rightarrow X^n$.

Now, by the same argument as above, using the fact that L is finitely generated, we can find an arrow $L'' \rightarrow L'$ in $\text{Modif}_{\text{cl,flat}}(L)$ such that the composition

$$(L'')^{\boxtimes n} \rightarrow (L')^{\boxtimes n} \rightarrow \Delta_*^n(\tilde{L})$$

vanishes.

Hence, the extension $\tilde{L}'' \rightarrow L''$ admits a splitting.

□[Proposition D.4.2]

D.5. **Proof of Theorem 4.3.9.** We will prove the variant of the lemma with a fixed $\underline{x} = x \in \text{Ran}$. The factorization version is a variant of this in families.

D.5.1. We start with the following observation: let us regard the assignment

$$(D.7) \quad \mathcal{Y} \rightsquigarrow \text{QCoh}_{\text{co}}(\mathcal{Y})^{>-\infty}, \quad (\mathcal{Y}_1 \xrightarrow{f} \mathcal{Y}_2) \rightsquigarrow f_*$$

as a functor from the category of ind-affine ind-schemes to ∞ -categories.

We claim:

Proposition D.5.2. *The functor (D.7) commutes with totalizations, in the sense that if $\mathcal{Y}^\bullet$ is a cosimplicial ind-affine ind-scheme and*

$$\mathcal{Y} \simeq \text{Tot}(\mathcal{F}^\bullet),$$

where the limit is taken in PreStk , then the functor

$$\text{QCoh}_{\text{co}}(\mathcal{Y})^{>-\infty} \rightarrow \text{Tot}(\text{QCoh}_{\text{co}}(\mathcal{Y}^\bullet)^{>-\infty})$$

is an equivalence.

The proposition will be proved in Sect. D.8. We now proceed with the proof of Theorem 4.3.9.

D.5.3. For an affine D-scheme $\mathcal{Y}$, let A denote the corresponding commutative algebra in $\mathrm{D}\text{-mod}(X)$ so that $\mathbf{oblv}^l(A)$ is connective and $\mathcal{Y} = \mathrm{Spec}_X(A)$.

Let $\mathcal{A}$ denote the corresponding (commutative) chiral algebra, see Sect. D.1.3, so that $\mathcal{A}$ corresponds to the factorization algebra $\mathcal{O}_{\mathcal{Y}}$, and

$$\mathcal{O}_{\mathcal{Y}}\text{-mod}_x^{\mathrm{fact}} \simeq \mathcal{A}\text{-mod}_x^{\mathrm{ch}}.$$

Over the next few subsections we will reduce the statement of Theorem 4.3.9 to the case when $A = \mathrm{Sym}^l(\mathcal{M}[1])$ for $\mathcal{M}$ a classical locally free finitely generated D-module.

D.5.4. We interpret the functor $\Gamma(\mathfrak{L}_{\nabla}(\mathcal{Y}), -)^{\mathrm{enh}}$ of (4.4) as

$$\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})) \rightarrow \mathcal{A}\text{-mod}_x^{\mathrm{ch}}.$$

This functor is t-exact and both categories are right complete in their respective t-structures. Hence, it is enough to show that the functor $\Gamma(\mathfrak{L}_{\nabla}(\mathcal{Y}), -)^{\mathrm{enh}}$ induces an equivalence

$$\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y}))^{\geq 0, \leq m} \rightarrow (\mathcal{A}\text{-mod}_x^{\mathrm{ch}})^{\geq 0, \leq m}$$

for every m .

Note now that if $A_1 \rightarrow A_2$ is a map in $\mathrm{ComAlg}(\mathrm{D}\text{-mod}(X))$, such that the induced map $\tau^{\geq -m}(A_1) \rightarrow \tau^{\geq -m}(A_2)$ is an isomorphism, the corresponding functors

$$\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y}_2)) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y}_1))$$

and

$$\mathcal{A}_2\text{-mod}_x^{\mathrm{ch}} \rightarrow \mathcal{A}_1\text{-mod}_x^{\mathrm{ch}}$$

induces equivalences on the corresponding $(-)^{\geq 0, \leq m}$ categories.

In particular, we obtain that it is enough to show that the functor

$$\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}({}^m\mathcal{Y}))^{\geq 0, \leq m} \rightarrow ({}^m\mathcal{A}\text{-mod}_x^{\mathrm{ch}})^{\geq 0, \leq m}$$

is an equivalence for ${}^mA := \tau^{\geq -m}(A)$, and the corresponding ${}^m\mathcal{Y}$ and mA .

D.5.5. By the assumption that A is D-afp, we can find a simplicial object $A_{\bullet}$ in $\mathrm{ComAlg}(\mathrm{D}\text{-mod}(X))^{\leq 0}$ with terms $A_n = \mathrm{Sym}^l(\mathcal{M}_n[1])$, where $\mathcal{M}_n$ is a classical⁸³ locally free finitely generated D-module, such that mA is a retract of $\tau^{\geq -m}(|A_{\bullet}|)$, see Sect. B.6.2.

Hence, it is enough to prove that

$$\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y}'))^{\geq 0, \leq m} \rightarrow (\mathcal{A}'\text{-mod}_x^{\mathrm{ch}})^{\geq 0, \leq m}$$

is an equivalence for $\mathcal{A}' := \tau^{\geq -m}(|A_{\bullet}|)$ for $A_{\bullet}$ as above.

Applying Sect. D.5.4 again, we obtain that it is enough to prove that

$$\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y}''))^{\geq 0, \leq m} \rightarrow (\mathcal{A}''\text{-mod}_x^{\mathrm{ch}})^{\geq 0, \leq m}$$

is an equivalence for $\mathcal{A}'' := |A_{\bullet}|$ for $A_{\bullet}$ as above.

Hence, we can obtain that it is enough to prove Theorem 4.3.9 for A is of the form $|A_{\bullet}|$ for $A_{\bullet}$ as above.

⁸³Recall that according to our conventions, this means that $\mathbf{oblv}^r(\mathcal{M}_n)$ is classical, i.e., $\mathbf{oblv}^l(\mathcal{M}_n)[1]$ is classical.

D.5.6. For $A_\bullet$ as above set $\mathcal{Y}^n := \mathrm{Spec}_X(A_n)$, and consider the corresponding simplicial affine D-scheme $\mathcal{Y}^\bullet$, so that

$$\mathcal{Y} \simeq \mathrm{Tot}(\mathcal{Y}^\bullet).$$

It is clear that the functor

$$\mathcal{Y} \mapsto \mathfrak{L}_\nabla(\mathcal{Y})$$

preserves limits, so that

$$\mathfrak{L}_\nabla(\mathcal{Y}) \simeq |\mathfrak{L}_\nabla(\mathcal{Y}^\bullet)|.$$

Hence, by Proposition D.5.2, the functor

$$\mathrm{QCoh}(\mathfrak{L}_\nabla(\mathcal{Y}))^{>-\infty} \rightarrow \mathrm{Tot}(\mathrm{QCoh}(\mathfrak{L}_\nabla(\mathcal{Y}^\bullet)))^{>-\infty}$$

is an equivalence.

The functor

$$\mathcal{A}\text{-mod}_x^{\mathrm{ch}} \rightarrow \mathrm{Tot}(\mathcal{A}_n\text{-mod}_x^{\mathrm{ch}})$$

is also an equivalence: indeed, this is obvious for non-unital modules (this is a general property of categories of modules over operad algebras), and this property is inherited by unital modules by [CR, Proposition 3.8.4].

Hence, it is enough to show that the functors

$$\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_\nabla(\mathcal{Y}^n))^{>-\infty} \rightarrow (\mathcal{A}\text{-mod}_x^{\mathrm{ch}})^{>-\infty}$$

are equivalences for every n .

This reduces the assertion of Theorem 4.3.9 to the case when $A = \mathrm{Sym}^!(\mathcal{M}[1])$ for $\mathcal{M}$ a classical locally free finitely generated D-module.

D.5.7. Let $\mathcal{Y}$ be general an affine D-scheme. Consider the functor

$$\Gamma(\mathfrak{L}_\nabla(\mathcal{Y}), -)^{\mathrm{enh}} : \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_\nabla(\mathcal{Y})) \rightarrow \mathcal{A}\text{-mod}_x^{\mathrm{ch}}$$

and its (a priori discontinuous) right adjoint. Tautologically, we have

$$\Gamma(\mathfrak{L}_\nabla(\mathcal{Y}), -) \simeq \mathbf{oblv}_{\mathcal{A}} \circ \Gamma(\mathfrak{L}_\nabla(\mathcal{Y}), -)^{\mathrm{enh}},$$

hence we obtain a natural transformation

$$(D.8) \quad \Gamma(\mathfrak{L}_\nabla(\mathcal{Y}), -) \circ (\Gamma(\mathfrak{L}_\nabla(\mathcal{Y}), -)^{\mathrm{enh}})^R \rightarrow \mathbf{oblv}_{\mathcal{A}}.$$

We claim that it suffices to show that (D.8) is an isomorphism when evaluated on $(\mathcal{A}\text{-mod}_x^{\mathrm{ch}})^{>-\infty}$.

Indeed, this follows from the fact that both $\mathbf{oblv}_{\mathcal{A}}$ and $\Gamma(\mathfrak{L}_\nabla(\mathcal{Y}), -)$ are conservative on the eventually coconnective subcategories (see Sect. D.7.1).

D.5.8. We will now specialize to the case when $\mathcal{A} = \mathrm{Sym}^!(\mathcal{M}[1])$ for $\mathcal{M} \in \mathrm{D-mod}^l(X)^{\heartsuit, \mathrm{loc. free}, \mathrm{f.g.}}$, and prove that (D.8) is an isomorphism on $(\mathcal{A}\text{-mod}_x^{\mathrm{ch}})^{>-\infty}$ by an explicit calculation.

D.5.9. Let $\mathrm{Modif}_{\mathrm{cl}, \mathrm{flat}, \mathrm{f.g.}}(\mathcal{M})$ be the category

$$\{\mathcal{M}' \in \mathrm{D-mod}^l(X)^{\heartsuit, \mathrm{flat}, \mathrm{f.g.}}, \mathcal{M}'|_{X-x} \simeq \mathcal{M}|_{X-x}\}.$$

We claim that the ind-scheme $\mathfrak{L}_\nabla(\mathcal{Y})$ identifies in this case with

$$\begin{array}{c} \text{“colim”} \\ \mathcal{M}' \in \mathrm{Modif}_{\mathrm{cl}, \mathrm{f.g.}, \mathrm{flat}} \end{array} \mathrm{Spec}(\mathrm{Sym}(\mathcal{M}'_x)).$$

Indeed, since the question is local around x , with no restriction of generality we can assume that X is affine. Let t be a uniformizer at x . Then for a connective commutative algebra R , we have by definition

$$\mathrm{Maps}(\mathrm{Spec}(R), \mathfrak{L}_\nabla(\mathcal{Y})) = \mathrm{Maps}_{\mathrm{ComAlg}(\mathrm{D-mod}(X))}(\mathrm{Sym}_{\odot_X}(\mathcal{M}), R((t))) \simeq \mathrm{Maps}_{\mathrm{D-mod}(X)}(\mathcal{M}, R((t)))$$

Since the map

$$j_! \circ j^*(\mathcal{M}) \rightarrow \begin{array}{c} \text{“lim”} \\ \mathcal{M}' \in \mathrm{Modif}_{\mathrm{cl}, \mathrm{flat}, \mathrm{f.g.}} \end{array} \mathcal{M}'$$

is an isomorphism in $\mathrm{Pro}(\mathrm{D}\text{-}\mathrm{mod}(X))$, we have

$$\begin{aligned} \mathrm{Maps}_{\mathrm{D}\text{-}\mathrm{mod}(X)}(\mathcal{M}, R((t))) &\simeq \mathrm{colim}_{\mathcal{M}'} \mathrm{Maps}_{\mathrm{D}\text{-}\mathrm{mod}(X)}(\mathcal{M}', R[[t]]) \simeq \mathrm{colim}_{\mathcal{M}'} \mathrm{Maps}_{\mathrm{Vect}}(\mathcal{M}'_x, R) \simeq \\ &\simeq \mathrm{colim}_{\mathcal{M}'} \mathrm{Maps}_{\mathrm{ComAlg}(\mathrm{Vect})}(\mathrm{Sym}(\mathcal{M}'_x), R) = \mathrm{colim}_{\mathcal{M}'} \mathrm{Maps}(\mathrm{Spec}(R), \mathrm{Spec}(\mathrm{Sym}(\mathcal{M}'_x))) = \\ &=: \mathrm{Maps}\left(\mathrm{Spec}(R), \text{“colim}_{\mathcal{M}'} \mathrm{Spec}(\mathrm{Sym}(\mathcal{M}'_x))\right), \end{aligned}$$

as desired.

D.5.10. For $\mathcal{M}'$ as above, let us denote by

$$\mathcal{O}_{(\mathcal{M}'_x)^\vee} \in \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_\nabla(\mathcal{Y}))$$

the direct images of the structure sheaf along the map,

$$\mathrm{Spec}(\mathrm{Sym}(\mathcal{M}'_x)) \rightarrow \mathfrak{L}_\nabla(\mathcal{Y}).$$

The above description of $\mathfrak{L}_\nabla(\mathcal{Y})$ implies that the functor $\Gamma(\mathfrak{L}_\nabla(\mathcal{Y}), -)$ is isomorphic to

$$\mathrm{colim}_{\mathcal{M}'} \mathcal{H}om(\mathcal{O}_{(\mathcal{M}'_x)^\vee}, -).$$

The functor

$$\Gamma(\mathfrak{L}_\nabla(\mathcal{Y}), -)^{\mathrm{enh}} : \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_\nabla(\mathcal{Y})) \rightarrow \mathcal{A}\text{-mod}_x^{\mathrm{ch}}$$

sends $\mathcal{O}_{(\mathcal{M}'_x)^\vee}$ to

$$\mathrm{Sym}(\mathcal{M}'_x) \in \mathcal{A}\text{-mod}_x^{\mathrm{ch}}.$$

Now, the required isomorphism follows from Corollary D.3.9.

□[Theorem 4.3.9]

D.6. Failure of Theorem 4.3.9 in the non-finitely presented case. In this subsection we will explain why Theorem 4.3.9 does *not* hold when $\mathcal{Y}$ is *not* almost finitely presented (in the D-sense).

Remark D.6.1. One can show that the functor

$$(D.9) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_\nabla(\mathcal{Y}))^{>-\infty} \rightarrow (\mathcal{A}\text{-mod}_x^{\mathrm{ch}})^{>-\infty}$$

induces an equivalence of the abelian categories

$$\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_\nabla(\mathcal{Y}))^\heartsuit \rightarrow (\mathcal{A}\text{-mod}_x^{\mathrm{ch}})^\heartsuit.$$

So, the failure of (D.9) to be an equivalence occurs at the derived level.

D.6.2. We will take $\mathcal{A} = \mathrm{Sym}_{\mathcal{O}_X}(\mathcal{M})$, where $\mathcal{M}$ is an infinitely-generated D-module (i.e., the direct sum of countably many copies of D_X). We will show that (D.9) fails to be an equivalence in this case.

Namely, we will construct two objects $\mathcal{F}_1, \mathcal{F}_2 \in (\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_\nabla(\mathcal{Y}))^\heartsuit)$ with images in $(\mathcal{A}\text{-mod}_x^{\mathrm{ch}})^\heartsuit$ denoted $\mathcal{M}_1, \mathcal{M}_2$, respectively, and an element in $\mathrm{Ext}_{\mathcal{A}\text{-mod}_x^{\mathrm{ch}}}^2(\mathcal{M}_1, \mathcal{M}_2)$ that does not come from an element in $\mathrm{Ext}_{\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_\nabla(\mathcal{Y}))}^2(\mathcal{F}_1, \mathcal{F}_2)$.

Remark D.6.3. It follows from the description of the category $\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_\nabla(\mathcal{Y}))^\heartsuit$ in Sect. D.7.5 that the category $\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_\nabla(\mathcal{Y}))^b$ is the bounded derived category of its heart. So the above inequality of the Ext^2 spaces means that $(\mathcal{A}\text{-mod}_x^{\mathrm{ch}})^b$ is *not* the bounded derived category of its heart.

D.6.4. Write

$$\mathcal{M} = \operatorname{colim}_i \mathcal{M}_i,$$

where $\mathcal{M}_i$ are finitely generated.

Let us call a modification $\mathcal{M}'$ “quasi-finitely generated” if all the intersections $\mathcal{M}' \cap \mathcal{M}_i$ are finitely generated. As in Sect. D.5.9, we have

$$\mathfrak{L}_{\nabla}(\mathcal{Y}) \simeq \operatorname{“colim”}_{\mathcal{M}' \in \operatorname{Mod}_{\operatorname{cl}, \operatorname{flat}, \operatorname{q-f.g.}}} \operatorname{Spec}(\operatorname{Sym}(\mathcal{M}'_x)).$$

By the same logic as in Sect. D.5.10, the functor

$$\mathcal{F} \mapsto \operatorname{colim}_{\mathcal{M}' \in \operatorname{Mod}_{\operatorname{cl}, \operatorname{flat}, \operatorname{q-f.g.}}} \operatorname{Hom}(\mathcal{O}_{(\mathcal{M}'_x)^\vee}, -), \quad \operatorname{QCoh}_{\operatorname{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})) \rightarrow \operatorname{Vect}$$

identifies with $\Gamma(\mathfrak{L}_{\nabla}(\mathcal{Y}), -)$. In particular, it is t-exact.

Hence, if for some $\mathcal{M}'$ and $\mathcal{F} \in \operatorname{QCoh}_{\operatorname{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y}))^\vee$ we have a class

$$\alpha' \in \operatorname{Ext}_{\operatorname{QCoh}_{\operatorname{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y}))}^2(\mathcal{O}_{(\mathcal{M}'_x)^\vee}, \mathcal{F}),$$

we can find $\mathcal{M}'' \subset \mathcal{M}'$ such that the image α'' of α' in

$$\operatorname{Ext}_{\operatorname{QCoh}_{\operatorname{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y}))}^2(\mathcal{O}_{(\mathcal{M}''_x)^\vee}, \mathcal{F})$$

vanishes.

D.6.5. As in Sect. D.5.10, the image of $\mathcal{O}_{(\mathcal{M}'_x)^\vee}$ in $\mathcal{A}\text{-mod}_x^{\operatorname{ch}}$ is $\operatorname{Sym}(\mathcal{M}'_x) \in \mathcal{A}\text{-mod}_x^{\operatorname{ch}}$.

By adjunction for any $\mathcal{M} \in \mathcal{A}\text{-mod}_x^{\operatorname{ch}}$, we have

$$\operatorname{Hom}_{\mathcal{A}\text{-mod}_x^{\operatorname{ch}}}(\operatorname{Sym}(\mathcal{M}'_x), \mathcal{M}) \simeq \operatorname{Hom}_{\mathcal{M}'\text{-mod}_x^{\operatorname{Lie}^*}}(k, \mathcal{M}).$$

D.6.6. Hence, it suffices to find $\mathcal{F} \in \operatorname{QCoh}_{\operatorname{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y}))^\vee$ with image $\mathcal{M} \in (\mathcal{A}\text{-mod}_x^{\operatorname{ch}})^\vee$ and a class

$$\beta' \in \operatorname{Ext}_{\mathcal{M}'\text{-mod}_x^{\operatorname{Lie}^*}}^2(k, \mathcal{M})$$

such that for any $\mathcal{M}'' \subset \mathcal{M}'$, the image β'' of β' in

$$\operatorname{Ext}_{\mathcal{M}''\text{-mod}_x^{\operatorname{Lie}^*}}^2(k, \mathcal{M})$$

is non-zero.

We will take $\mathcal{F}$ to be the sky-scraper at the origin of $\mathcal{Y}$, so that $\mathcal{M} = k$, with the trivial chiral action of $\mathcal{M}$.

D.6.7. We calculate

$$\operatorname{Hom}_{\mathcal{M}'\text{-mod}_x^{\operatorname{Lie}^*}}(\mathcal{M}_1, \mathcal{M}_2) \simeq \lim_i \operatorname{Hom}_{(\mathcal{M}' \cap \mathcal{M}_i)\text{-mod}_x^{\operatorname{Lie}^*}}(\mathcal{M}_1, \mathcal{M}_2).$$

For $\mathcal{M}_1 = \mathcal{M}_2 = k$, we obtain

$$\begin{aligned} \operatorname{Ext}_{\mathcal{M}'\text{-mod}_x^{\operatorname{Lie}^*}}^2(k, k) &\simeq \lim_i \Lambda^2(\mathbb{D}(\mathcal{M}' \cap \mathcal{M}_i)_x) \simeq \\ &\simeq \lim_i \operatorname{Hom}_{\operatorname{D-mod}(X \times X)^\vee}^{\operatorname{antisym}}((\mathcal{M}' \cap \mathcal{M}_i) \boxtimes (\mathcal{M}' \cap \mathcal{M}_i), \delta_{x,x}) \simeq \operatorname{Hom}_{\operatorname{D-mod}(X \times X)^\vee}^{\operatorname{antisym}}(\mathcal{M}' \boxtimes \mathcal{M}', \delta_{x,x}). \end{aligned}$$

Now, it is clear that since $\mathcal{M}'$ is infinitely generated, we can find an element

$$\beta' \in \operatorname{Hom}_{\operatorname{D-mod}(X \times X)^\vee}^{\operatorname{antisym}}(\mathcal{M}' \boxtimes \mathcal{M}', \delta_{x,x}),$$

such that for any $\mathcal{M}''$, the restriction β'' of β' to

$$\operatorname{Hom}_{\operatorname{D-mod}(X \times X)^\vee}^{\operatorname{antisym}}(\mathcal{M}'' \boxtimes \mathcal{M}'', \delta_{x,x})$$

is non-zero.

Remark D.6.8. Note that the above counterexample does not work for Ext^1 instead of Ext^2 (as must be the case, since (D.9) is an equivalence at the abelian level). Indeed, for any

$$\gamma' \in \mathrm{Hom}_{\mathrm{D-mod}(X)^\heartsuit}(\mathcal{M}', \delta_x),$$

there exists $\mathcal{M}'' \subset \mathcal{M}'$, such that the restriction γ'' of γ' to

$$\mathrm{Hom}_{\mathrm{D-mod}(X)^\heartsuit}(\mathcal{M}'', \delta_x)$$

vanishes.

D.7. Proof of Proposition D.5.2: preparations.

D.7.1. First, we claim that if $\mathcal{Y}$ is an ind-affine ind-scheme, then the functor

$$\Gamma(\mathcal{Y}, -) : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y}) \rightarrow \mathrm{Vect}$$

is conservative on $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})^{>-\infty}$.

Indeed, since $\Gamma(\mathcal{Y}, -)$ is t-exact and the t-structure on $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})$ is right-complete, it suffices to show that $\Gamma(\mathcal{Y}, -)$ does not annihilate objects from $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})^\heartsuit$.

Write $\mathcal{Y}$ is a filtered colimit of schemes Y_α under closed embeddings

$$f_{\alpha,\beta} : Y_\alpha \rightarrow Y_\beta.$$

An object $\mathcal{F} \in \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})^\heartsuit$ amounts to a collection

$$\{\mathcal{F}_\alpha \in \mathrm{QCoh}(Y_\alpha)^\heartsuit, \mathcal{F}_\alpha \simeq H^0(f_{\alpha,\beta}^!(\mathcal{F}_\beta))\}.$$

In particular, the maps

$$\Gamma(\mathcal{Y}_\alpha, \mathcal{F}_\alpha) \rightarrow \Gamma(\mathcal{Y}_\beta, \mathcal{F}_\beta)$$

are injective.

We have

$$\Gamma(\mathcal{Y}, \mathcal{F}) \simeq \mathrm{colim}_\alpha \Gamma(\mathcal{Y}_\alpha, \mathcal{F}_\alpha)$$

and the statement is manifest.

D.7.2. We claim:

Proposition D.7.3. *The functor*

$$\Gamma(\mathcal{Y}, -) : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})^{\geq 0} \rightarrow \mathrm{Vect}^{\geq 0}$$

is comonadic.

Given the conservativity, the assertion of the proposition follows from the next general observation:

Lemma D.7.4. *Let $\mathbf{C}, \mathbf{D}$ be cocomplete DG categories, equipped with t-structures, compatible with filtered colimits. Assume that $\mathbf{D}$ is right-complete in its t-structure. Let $F : \mathbf{C} \rightarrow \mathbf{D}$ be a t-exact continuous functor. Assume that F is conservative on $\mathbf{C}^{>-\infty}$. Then the induced functor*

$$\mathbf{C}^{\geq 0} \rightarrow \mathbf{D}^{\geq 0}$$

is comonadic.

Proof. By Barr-Beck-Lurie, suffices to show that F preserves totalizations of cosimplicial objects in $\mathbf{C}^{\geq 0}$. Thus, let $\mathbf{c}^\bullet$ be a cosimplicial object in $\mathbf{C}$. We have to show that the map

$$F(\mathrm{Tot}(\mathbf{c}^\bullet)) \rightarrow \mathrm{Tot}(F(\mathbf{c}^\bullet))$$

is an isomorphism.

Since $\mathbf{D}$ is right-complete in its t-structure, it suffices to show that for every n ,

$$\tau^{\leq n}(F(\mathrm{Tot}(\mathbf{c}^\bullet))) \rightarrow \tau^{\leq n}(\mathrm{Tot}(F(\mathbf{c}^\bullet)))$$

is an isomorphism.

Let

$$\mathrm{Tot}^{\leq n}(\mathbf{c}^\bullet) \text{ and } \mathrm{Tot}^{\leq n}(F(\mathbf{c}^\bullet))$$

be the totalizations of the corresponding $(n+1)$ -skeleta.

We have natural maps

$$\mathrm{Tot}(\mathbf{c}^\bullet) \rightarrow \mathrm{Tot}^{\leq n}(\mathbf{c}^\bullet) \text{ and } \mathrm{Tot}(F(\mathbf{c}^\bullet)) \rightarrow \mathrm{Tot}^{\leq n}(F(\mathbf{c}^\bullet)).$$

Since the terms of $\mathbf{c}^\bullet$ and $F(\mathbf{c}^\bullet)$ are in $\mathbf{C}^{\geq 0}$, the above maps induce isomorphisms between the $\tau^{\leq n}$ truncations.

Hence, it suffices to show that

$$F(\mathrm{Tot}^{\leq n}(\mathbf{c}^\bullet)) \rightarrow \mathrm{Tot}^{\leq n}(F(\mathbf{c}^\bullet))$$

is an isomorphism.

However, this is obvious, since the limit over $\Delta_{\leq n}$ is a finite limit. $\square$

D.7.5. Note that we have a canonical equivalence

$$\mathrm{Pro}(\mathrm{Vect})^{\mathrm{op}} \simeq \mathrm{Funct}_{\mathrm{discont}}(\mathrm{Vect}, \mathrm{Vect}), \quad \mathbf{V} \mapsto \mathcal{H}om(\mathbf{V}, -).$$

where $\mathrm{Funct}_{\mathrm{discont}}(-, -)$ denotes the category of exact k -linear functors that are not necessarily continuous.

Under this equivalence, the monoidal structure on $\mathrm{Funct}_{\mathrm{discont}}(-, -)$ given by composition corresponds to the $\vec{\otimes}$ monoidal structure on $\mathrm{Pro}(\mathrm{Vect})$ (see [Bei]):

For

$$\mathbf{V} = \varprojlim_{i \in I} V_i \text{ and } \mathbf{W} = \varprojlim_{j \in J} W_j,$$

we have

$$\mathbf{V} \vec{\otimes} \mathbf{W} = \varprojlim_{j \in J} \left(\mathrm{colim}_{V_j^f \subset V_j} \left((\varprojlim_{i \in I} V_i) \otimes V_j^f \right) \right),$$

where:

- V_j^f runs the category of compact objects mapping to V_j ;
- In the right-hand side, the outer limit and the inner colimit are taken in $\mathrm{Pro}(\mathrm{Vect})$.

Thus, comonads on Vect correspond to algebra objects in $\mathrm{Pro}(\mathrm{Vect})$ with respect to $\vec{\otimes}$. Comonads that are left t-exact correspond to algebra objects in $\mathrm{Pro}(\mathrm{Vect}^{\leq 0})$.

D.7.6. Let $\mathbf{M}_{\mathcal{Y}}$ denote the comonad on Vect corresponding to the functor $\Gamma(\mathcal{Y}, -)$, so that

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{Y})^{\geq 0} \simeq \mathbf{M}_{\mathcal{Y}}\text{-comod}(\mathrm{Vect}^{\geq 0}).$$

The object of $\mathrm{Pro}(\mathrm{Vect}^{\leq 0})$ that corresponds to $\mathbf{M}_{\mathcal{Y}}$ is described as follows.

Let $\mathcal{O}_{\mathcal{Y}}$ be the object of $\mathrm{Pro}(\mathrm{ComAlg}(\mathrm{Vect}^{\leq 0}))$ associated to the ind-affine ind-scheme $\mathcal{Y}$. I.e., if

$$\mathcal{Y} = \varinjlim_i Y_i,$$

then

$$\mathcal{O}_{\mathcal{Y}} := \varinjlim_i \mathcal{O}_{Y_i}.$$

Let

$$\mathbf{oblv}_{\mathrm{ComAlg}}^{\mathrm{Pro}} : \mathrm{Pro}(\mathrm{ComAlg}(\mathrm{Vect})) \rightarrow \mathrm{Pro}(\mathrm{Vect})$$

be the pro-extension of the functor

$$\mathbf{oblv}_{\mathrm{ComAlg}} : \mathrm{ComAlg}(\mathrm{Vect}) \rightarrow \mathrm{Vect}.$$

The endofunctor of Vect underlying the comonad $\mathbf{M}_{\mathcal{Y}}$ is given by

$$V \mapsto \mathcal{H}om(\mathbf{oblv}_{\mathrm{ComAlg}}^{\mathrm{Pro}}(\mathcal{O}_{\mathcal{Y}}), V).$$

D.7.7. The proof of Proposition D.5.2 will be based on the following two observations:

Lemma D.7.8. *For every natural number n , the functor $\mathbf{oblv}_{\mathbf{ComAlg}}^{\mathbf{Pro}} : \mathbf{Pro}(\mathbf{ComAlg}(\mathbf{Vect}^{\leq 0})) \rightarrow \mathbf{Pro}(\mathbf{Vect}^{\leq 0})$, followed by the truncation*

$$\mathbf{Pro}(\mathbf{Vect}^{\leq 0}) \rightarrow \mathbf{Pro}(\mathbf{Vect}^{\leq 0, \geq -n})$$

commutes with geometric realizations.

Lemma D.7.9. *For every natural number n and for every natural number m , the functor*

$$\mathbf{V} \mapsto \mathbf{V}^{\vec{\otimes} m}, \quad \mathbf{Pro}(\mathbf{Vect}^{\leq 0}) \rightarrow \mathbf{Pro}(\mathbf{Vect}^{\leq 0}),$$

followed by the truncation

$$\mathbf{Pro}(\mathbf{Vect}^{\leq 0}) \rightarrow \mathbf{Pro}(\mathbf{Vect}^{\leq 0, \geq -n})$$

commutes with geometric realizations.

Let us temporarily assume these lemmas and prove Proposition D.5.2.

D.8. Proof of Proposition D.5.2.

D.8.1. It is enough to show that for every n , the functor

$$(D.10) \quad \mathbf{QCoh}_{\mathbf{co}}(\mathcal{Y})^{\geq 0, \leq n} \rightarrow \mathbf{Tot}(\mathbf{QCoh}_{\mathbf{co}}(\mathcal{Y}^\bullet)^{\geq 0, \leq n})$$

is an equivalence.

D.8.2. Note that $\mathbf{M}$ is a left-exact endofunctor $\mathbf{Vect}$, we can create its truncation $\mathbf{M}^{\leq n}$ that acts on $\mathbf{Vect}^{\geq 0, \leq n}$, namely,

$$\mathbf{M}^{\leq n}(V) := \tau^{\leq n}(\mathbf{M}(V)).$$

This assignment is monoidal, so if $\mathbf{M}$ is a comonad, then $\mathbf{M}^{\leq n}$ inherits a natural comonad structure, and we have

$$\mathbf{M}\text{-comod}(\mathbf{Vect}^{\geq 0}) \times_{\mathbf{Vect}^{\geq 0}} \mathbf{Vect}^{\geq 0, \leq n} \simeq \mathbf{M}^{\leq n}\text{-comod}(\mathbf{Vect}^{\geq 0, \leq n}).$$

D.8.3. According to Proposition D.7.3, we have:

$$\mathbf{QCoh}_{\mathbf{co}}(\mathcal{Y})^{\geq 0} \simeq \mathbf{M}_{\mathcal{Y}}\text{-comod}(\mathbf{Vect}^{\geq 0}) \text{ and } \mathbf{QCoh}_{\mathbf{co}}(\mathcal{Y}^\bullet)^{\geq 0} \simeq \mathbf{M}_{\mathcal{Y}^\bullet}\text{-comod}(\mathbf{Vect}^{\geq 0}).$$

Hence,

$$\mathbf{QCoh}_{\mathbf{co}}(\mathcal{Y})^{\geq 0, \leq n} \simeq \mathbf{M}_{\mathcal{Y}}^{\leq n}\text{-comod}(\mathbf{Vect}^{\geq 0, \leq n}) \text{ and } \mathbf{QCoh}_{\mathbf{co}}(\mathcal{Y}^\bullet)^{\geq 0, \leq n} \simeq \mathbf{M}_{\mathcal{Y}^\bullet}^{\leq n}\text{-comod}(\mathbf{Vect}^{\geq 0, \leq n}).$$

D.8.4. We now observe:

Lemma D.8.5. *Let I be an index category and let*

$$\mathbf{M}_I : I \rightarrow \mathbf{Comonad}(\mathbf{C}), \quad i \mapsto \mathbf{M}_i$$

be an I -diagram of comonads on a category $\mathbf{C}$. Let $\mathbf{M}$ be another comonad, equipped with a compatible collection of maps $\mathbf{M} \rightarrow \mathbf{M}_i$. Suppose that for every natural number m , the map

$$\mathbf{M}^{\times m} \rightarrow \lim_i \mathbf{M}_i^{\times m}$$

is an isomorphism, where:

- The notation $\mathbf{M}^{\times m}$ means an m -fold composition of $\mathbf{M}$, and similarly for $\mathbf{M}_i$;
- The limit in the right-hand side is taken in the category of endofunctors of $\mathbf{C}$ (i.e., is computed value-wise).

Then the induced functor

$$\mathbf{M}\text{-comod}(\mathbf{C}) \rightarrow \lim_i (\mathbf{M}_i\text{-comod}(\mathbf{C}))$$

is an equivalence.

D.8.6. Hence, in order to prove that (D.10) is an equivalence, it suffices to show that for any m , the map

$$(D.11) \quad (M_{\vec{y}}^{\leq n})^{\times m} \rightarrow \text{Tot} \left((M_{\vec{y}\bullet}^{\leq n})^{\times m} \right)$$

is an equivalence.

D.8.7. By Sect. D.7.6, we have

$$M_{\vec{y}}\text{-comod}(\text{Vect}^{\leq 0}) \simeq \mathcal{O}_{\vec{y}}\text{-mod}(\text{Vect}^{\leq 0})$$

and

$$M_{\vec{y}\bullet}\text{-comod}(\text{Vect}^{\leq 0}) \simeq \mathcal{O}_{\vec{y}\bullet}\text{-mod}(\text{Vect}^{\leq 0}),$$

where we regard $\mathcal{O}_{\vec{y}}$ as an algebra object in $(\text{Pro}(\text{Vect}), \overset{\rightarrow}{\otimes})$ via the natural forgetful functor

$$\text{Pro}(\text{ComAlg}(\text{Vect})) \rightarrow \text{AssocAlg}(\text{Pro}(\text{Vect}), \overset{\rightarrow}{\otimes}).$$

Note now that in the situation of Proposition D.5.2, the map

$$\mathcal{O}_{\vec{y}} \rightarrow |\mathcal{O}_{\vec{y}\bullet}|$$

is an isomorphism, where the geometric realization is taken in $\text{Pro}(\text{ComAlg}(\text{Vect}))$.

Hence, D.7.8 and D.7.9 guarantee that the maps (D.11) are isomorphisms, as required.

□[Proposition D.5.2]

D.9. Proof of Lemmas D.7.8 and D.7.9.

D.9.1. The key input is the following:

Lemma D.9.2. *Let $\mathbf{C}$ be an n -truncated category (i.e., the mapping spaces have homotopy groups π_m vanish for $m > n$). Then the map*

$$|-|_{\leq n+1} \rightarrow |-|, \quad \text{Funct}(\Delta^{\text{op}}, \mathbf{C}) \rightarrow \mathbf{C}$$

is an isomorphism, where the left-hand side is the colimit over $\Delta_{\leq n+1}^{\text{op}}$.

D.9.3. *Proof Lemma D.7.9.* Since Δ^{op} is sifted, it suffices to show that the binary operation

$$(D.12) \quad \text{Pro}(\text{Vect}^{\leq 0, \geq -n}) \times \text{Pro}(\text{Vect}^{\leq 0, \geq -n}) \rightarrow \text{Pro}(\text{Vect}^{\leq 0, \geq -n}), \quad \mathbf{V}, \mathbf{W} \mapsto \mathbf{V} \overset{\rightarrow}{\otimes} \mathbf{W}$$

commutes with geometric realizations in each variable.

However, by Lemma D.9.2, the functor of geometric realization in $\text{Vect}^{\leq 0, \geq -n}$ is a *finite colimit*, and it is clear that (D.12) commutes with finite colimits in each variable.

□[Lemma D.7.9]

D.9.4. *Proof of Lemma D.7.8.* By Lemma D.9.2, it suffices to show that the functor

$$\mathbf{oblv}_{\text{ComAlg}}^{\text{Pro}} : \text{Pro}(\text{ComAlg}(\text{Vect}^{\leq 0, \geq -n})) \rightarrow \text{Pro}(\text{Vect}^{\leq 0, \geq -n})$$

commutes with colimits over $\Delta_{\leq n+1}^{\text{op}}$.

The rest of the argument essentially reproduces [Lu2, Proposition 6.1.5.3]:

Note that for any finite index category I and a category $\mathbf{C}$, the naturally defined functor

$$\text{Pro Funct}(I, \mathbf{C}) \rightarrow \text{Funct}(I, \text{Pro}(\mathbf{C}))$$

is an equivalence.

Furthermore, for an object

$$“\lim_{\alpha}” (i \mapsto \mathbf{c}_{i,\alpha}) \in \text{Pro}(\text{Funct}(I, \mathbf{C}))$$

and the resulting object

$$i \mapsto “\lim_{\alpha}” \mathbf{c}_{i,\alpha} \in \text{Funct}(I, \text{Pro}(\mathbf{C})),$$

we have

$$\operatorname{colim}_{i \in I} \left(\lim_{\alpha} \mathbf{c}_{i,\alpha} \right) \simeq \lim_{\alpha} \left(\operatorname{colim}_{i \in I} \mathbf{c}_{i,\alpha} \right).$$

We apply this observation to $I = \Delta_{\leq n+1}^{\operatorname{op}}$ and $\mathbf{C}$ being

$$\operatorname{ComAlg}(\operatorname{Vect}^{\leq 0, \geq -n}) \text{ and } \operatorname{Vect}^{\leq 0, \geq -n}.$$

Hence, in order to prove Lemma D.7.8, it suffices to show that the functor

$$\mathbf{oblv}_{\operatorname{ComAlg}} : \operatorname{ComAlg}(\operatorname{Vect}^{\leq 0, \geq -n}) \rightarrow \operatorname{Vect}^{\leq 0, \geq -n}$$

commutes with colimits over $\Delta_{\leq n+1}^{\operatorname{op}}$.

However, this follows from Lemma D.9.2 combined with the fact that the usual geometric realization functor commutes with $\mathbf{oblv}_{\operatorname{ComAlg}}$.

□[Lemma D.7.8]

APPENDIX E. THE SPECTRAL SPHERICAL CATEGORY

Throughout this section we let H be an arbitrary finite-dimensional algebraic group. Our goal is to define the factorization category

$$\operatorname{IndCoh}^*(\operatorname{Hecke}_H^{\operatorname{spec}, \operatorname{loc}}).$$

When $H = \check{G}$, the category $\operatorname{IndCoh}^*(\operatorname{Hecke}_{\check{G}}^{\operatorname{spec}, \operatorname{loc}}) =: \operatorname{Spc}_{\check{G}}^{\operatorname{spec}}$ is the spectral counterpart of Sph_G , and it acts by Hecke functors on the global spectral category. This action will play a key role in the sequel to this paper.

The difficulty we face is that we have not found a way to plug $\operatorname{Hecke}_H^{\operatorname{spec}, \operatorname{loc}}$ into one of the previously discussed constructions, i.e.,

$$\operatorname{QCoh}(-), \operatorname{QCoh}_{\operatorname{co}}(-), \text{ or } \operatorname{IndCoh}^*(-)$$

to obtain the desired category.

Instead, we will define it as bi-coinvariants with respect to $\mathfrak{L}_{\nabla}^+(H)$ inside $\operatorname{IndCoh}^*(\mathfrak{L}_{\nabla}(H))$, where the latter also requires some care, as the factorization scheme $\mathfrak{L}_{\nabla}(H)$ is *not* ind-placid.

E.1. A 1-affineness property of $\operatorname{LS}_H^{\operatorname{reg}}$. Throughout this subsection we fix an affine scheme S and a map $\underline{x} : S \rightarrow \operatorname{Ran}$.

E.1.1. Consider $\operatorname{QCoh}(\mathfrak{L}_{\nabla}^+(H))_S$ as an $\operatorname{QCoh}(S)$ -linear monoidal category with respect to convolution.

Note that we have:

$$\operatorname{Funct}_{\operatorname{QCoh}(\mathfrak{L}_{\nabla}^+(H))_S\text{-}\mathbf{mod}}(\operatorname{QCoh}(S), \operatorname{QCoh}(S)) \simeq \operatorname{QCoh}(\operatorname{LS}_{H,S}^{\operatorname{reg}})$$

as monoidal categories.

This gives rise to a pair of adjoint functors

$$(E.1) \quad \operatorname{QCoh}(\operatorname{LS}_{H,S}^{\operatorname{reg}})\text{-}\mathbf{mod} \rightleftarrows \operatorname{QCoh}(\mathfrak{L}_{\nabla}^+(H))_S\text{-}\mathbf{mod},$$

$$\mathbf{C} \mapsto \mathbf{C} \otimes_{\operatorname{QCoh}(\operatorname{LS}_{H,S}^{\operatorname{reg}})} \operatorname{QCoh}(S), \quad \mathbf{C}' \mapsto (\mathbf{C}')^{\mathfrak{L}_{\nabla}^+(H)_S}.$$

Remark E.1.2. Throughout this section, the symbols $(-)^{\mathfrak{L}_{\nabla}^+(H)_S}$ and $(-)_{\mathfrak{L}_{\nabla}^+(H)_S}$ indicate *weak*⁸⁴ invariants/coinvariants of the group-scheme $\mathfrak{L}_{\nabla}^+(H)_S$ acting on a $\operatorname{QCoh}(S)$ -linear category.

E.1.3. We have the following assertion ([Ra4, Lemma 9.8.1]):

Proposition E.1.4. *The functors (E.1) are mutually inverse equivalences.*

⁸⁴Here “weak” is as opposed to “strong”. Note that we could not even talk about strong invariants/coinvariants, because we are talking about weak actions.

E.1.5. Let $\mathbf{C}$ be a module category over $\mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(H))_S$. The functor of $\mathfrak{L}_{\nabla}^+(H)_S$ -averaging

$$\mathrm{Av}_*^{\mathfrak{L}_{\nabla}^+(H)_S} : \mathbf{C} \rightarrow \mathbf{C}^{\mathfrak{L}_{\nabla}^+(H)_S}$$

naturally factors via a functor

$$(E.2) \quad \mathbf{C}_{\mathfrak{L}_{\nabla}^+(H)_S} \rightarrow \mathbf{C}^{\mathfrak{L}_{\nabla}^+(H)_S}.$$

We claim:

Corollary E.1.6. *The functor (E.2) is an equivalence.*

Proof. Proposition E.1.4 implies that the functor

$$\mathbf{C} \mapsto \mathbf{C}^{\mathfrak{L}_{\nabla}^+(H)_S}$$

commutes with colimits.

Hence, both sides in (E.2) commute with colimits. Any object in $\mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(H))_S\text{-}\mathbf{mod}$ can be written as a colimit of objects of the form $\mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(H))_S \otimes \mathbf{D}$, where the module structure comes from the first factor. Hence, we obtain that it is sufficient to prove that (E.2) is an equivalence for such objects.

However, the latter is obvious: the corresponding functor is the identity functor

$$\mathrm{QCoh}(S) \otimes \mathbf{D} \simeq (\mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(H))_S \otimes \mathbf{D})_{\mathfrak{L}_{\nabla}^+(H)_S} \rightarrow (\mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(H))_S \otimes \mathbf{D})^{\mathfrak{L}_{\nabla}^+(H)_S} \simeq \mathrm{QCoh}(S) \otimes \mathbf{D}.$$

□

E.1.7. We now claim:

Corollary E.1.8. *The functor*

$$\Omega_{\mathrm{LS}_{H,S}^{\mathrm{reg}}} : \mathrm{QCoh}_{\mathrm{co}}(\mathrm{LS}_{H,S}^{\mathrm{reg}}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{H,S}^{\mathrm{reg}})$$

is an equivalence.

Proof. The fact that $S \rightarrow \mathrm{LS}_{H,S}^{\mathrm{reg}}$ is an fpqc cover implies that the functor

$$\mathrm{QCoh}(S) \otimes_{\mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(H))_S} \mathrm{QCoh}(S) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathrm{LS}_{H,S}^{\mathrm{reg}})$$

is an equivalence (cf. [Ga5, Proposition 6.2.7]).

Hence, it remains to show that the functor

$$\mathrm{QCoh}(S) \otimes_{\mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(H))_S} \mathrm{QCoh}(S) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{H,S}^{\mathrm{reg}})$$

is an equivalence.

However, the latter functor is the functor (E.2) for $\mathbf{C} = \mathrm{QCoh}(S)$.

□

E.2. Definition of $\mathrm{Sph}_H^{\mathrm{spec}}$.

E.2.1. Recall that the local spectral Hecke stack is by definition

$$\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}} := \mathrm{LS}_H^{\mathrm{reg}} \times_{\mathrm{LS}_H^{\mathrm{mer}}} \mathrm{LS}_H^{\mathrm{reg}}.$$

Our approach to the definition of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ is based on the following observation:

Lemma E.2.2. *The factorization prestack $\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}$ identifies canonically with the double quotient*

$$\mathcal{L}_\nabla^+(H) \backslash \mathcal{L}_\nabla(H) / \mathcal{L}_\nabla^+(H).$$

Proof. By definition, the fiber product $\mathrm{LS}_H^{\mathrm{reg}} \times_{\mathrm{LS}_H^{\mathrm{mer}}} \mathrm{LS}_H^{\mathrm{reg}}$ identifies with

$$\mathcal{L}_\nabla^+(H) \backslash \mathrm{Stab}_{\mathcal{L}_\nabla(\mathrm{Jets}(H))}(0) / \mathcal{L}_\nabla^+(H),$$

where:

- $0 \in \mathcal{L}_\nabla(\mathrm{Conn}(\mathfrak{h}))$ is the trivial connection;
- $\mathrm{Stab}_{\mathcal{L}_\nabla(\mathrm{Jets}(H))}(0)$ denotes the stabilizer of 0 with respect to the gauge action of $\mathcal{L}_\nabla(\mathrm{Jets}(H)) \simeq \mathcal{L}(H)$.

However,

$$\mathrm{Stab}_{\mathcal{L}_\nabla(\mathrm{Jets}(H))}(0) = \mathcal{L}_\nabla(\mathrm{Stab}_{\mathrm{Jets}(H)}(0)),$$

while

$$\mathrm{Stab}_{\mathrm{Jets}(H)}(0) \simeq H,$$

as a group D-scheme. □

E.2.3. We also note:

Lemma E.2.4. *For $S \in \mathrm{Sch}_{\mathrm{aff}}^{\mathrm{aff}}$, the quotient $(\mathcal{L}_\nabla(H) / \mathcal{L}_\nabla^+(H))_S$ is locally almost of finite type.*

Proof. First, we note that the unit section

$$S \rightarrow (\mathcal{L}_\nabla(H) / \mathcal{L}_\nabla^+(H))_S$$

is an isomorphism at the classical level. (Indeed, for any affine Y , the map $\mathcal{L}_\nabla^+(Y) \rightarrow \mathcal{L}_\nabla(Y)$ is an isomorphism at the classical level.)

Hence, by [GaRo4, Chapter 1, Theorem 9.1.2], it suffices to show that the cotangent space to $(\mathcal{L}_\nabla(H) / \mathcal{L}_\nabla^+(H))_S$ at the unit section is left (see [GaRo4, Chapter 1, Sect. 3.4.1] for what this means). However, this cotangent space is the dual of

$$(\mathcal{L}_\nabla(\mathfrak{h}) / \mathcal{L}_\nabla^+(\mathfrak{h}))_S,$$

which makes the assertion manifest. □

E.2.5. In what follows we will define the (monoidal) factorization category $\mathrm{IndCoh}^*(\mathcal{L}_\nabla(H))$, equipped with an action of the monoidal category $\mathrm{QCoh}(\mathcal{L}_\nabla^+(H))$ on the two sides. We will then set

$$(E.3) \quad \mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}) := (\mathrm{IndCoh}^*(\mathcal{L}_\nabla(H)))_{\mathcal{L}_\nabla^+(H) \times \mathcal{L}_\nabla^+(H)}.$$

The caveat here is that $\mathcal{L}_\nabla^+(H)$ is *not* placid (and hence, $\mathcal{L}_\nabla(H)$ is not ind-placid). Yet, we will show that the construction of $\mathrm{IndCoh}^*(-)$ in B.13.17-B.13.22 is applicable in this particular case.

E.2.6. Let S_α be as in Sect. B.13.9. Consider the relative affine scheme $\mathfrak{L}_\nabla^+(H)_{S_\alpha}$ and the relative ind-affine ind-scheme $\mathfrak{L}_\nabla(H)_{S_\alpha}$.

First, we claim:

Lemma E.2.7. *The functor*

$$\Psi_{\mathfrak{L}_\nabla^+(H)_{S_\alpha}} : \mathrm{IndCoh}^*(\mathfrak{L}_\nabla^+(H))_{S_\alpha} \rightarrow \mathrm{QCoh}(\mathfrak{L}_\nabla^+(H))_{S_\alpha}$$

is an equivalence.

Proof. The assertion holds for any smooth target scheme Y . Indeed, one shows that for $S_\alpha = X^I$, the relative affine scheme $\mathfrak{L}_\nabla^+(Y)_{X^I}$ is isomorphic to the limit of a sequence of affine blow-ups with smooth centers, starting with $Y^I \rightarrow X^I$, see Sect. E.12.

In particular, $\mathfrak{L}_\nabla^+(Y)_{X^I}$ is isomorphic to a filtered limit of relative affine schemes $Y_n \rightarrow S_\alpha$ that are *smooth*.

We have:

$$\mathrm{IndCoh}^*(\mathfrak{L}_\nabla^+(Y))_{S_\alpha} \simeq \lim_n \mathrm{IndCoh}(Y_n)$$

(with respect to push-forwards) and

$$\mathrm{QCoh}(\mathfrak{L}_\nabla^+(Y))_{S_\alpha} \simeq \lim_n \mathrm{QCoh}(Y_n),$$

and the functor $\Psi_{\mathfrak{L}_\nabla^+(H)_{S_\alpha}}$ corresponds to the compatible family of functors

$$\Psi_{Y_n} : \mathrm{IndCoh}(Y_n) \rightarrow \mathrm{QCoh}(Y_n),$$

all of which are equivalences, since Y_n are smooth. □

E.2.8. According to Sect. A.5, since $\mathfrak{L}_\nabla(H)_{S_\alpha}$ is an ind-affine ind-scheme, we have a well-defined category $\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H))_{S_\alpha}$.

The action of $\mathfrak{L}_\nabla^+(H)_{S_\alpha} \times \mathfrak{L}_\nabla^+(H)_{S_\alpha}$ on $\mathfrak{L}_\nabla(H)_{S_\alpha}$ defines on $\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H))_{S_\alpha}$ a structure of bimodule with respect to $\mathrm{IndCoh}^*(\mathfrak{L}_\nabla^+(H))_{S_\alpha}$.

Hence, thanks to Lemma E.2.7, we can think of $\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H))_{S_\alpha}$ as a bimodule with respect to $\mathrm{QCoh}(\mathfrak{L}_\nabla^+(H))_{S_\alpha}$.

E.2.9. Consider $\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H))_{S_\alpha}$ as a module over $\mathrm{QCoh}(\mathfrak{L}_\nabla^+(H))_{S_\alpha}$ with respect to the action on the right.

Direct image with respect to the projection

$$(E.4) \quad \mathfrak{L}_\nabla(H)_{S_\alpha} \rightarrow (\mathfrak{L}_\nabla(H)/\mathfrak{L}_\nabla^+(H))_{S_\alpha}$$

gives rise to a functor

$$(E.5) \quad (\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H))_{S_\alpha})_{\mathfrak{L}_\nabla^+(H)_{S_\alpha}} \rightarrow \mathrm{IndCoh}^*((\mathfrak{L}_\nabla(H)/\mathfrak{L}_\nabla^+(H))_{S_\alpha}) \simeq \underset{\text{Lemma E.2.4}}{\simeq} \mathrm{IndCoh}((\mathfrak{L}_\nabla(H)/\mathfrak{L}_\nabla^+(H))_{S_\alpha}).$$

We claim:

Lemma E.2.10. *The functor (E.5) is an equivalence.*

Proof. Since $S_\alpha \rightarrow (\mathfrak{L}_\nabla(H)/\mathfrak{L}_\nabla^+(H))_{S_\alpha}$ is an isomorphism at the reduced level, Zariski-locally on S_α , the map (E.4) splits as a product: indeed, the restriction of the étale $\mathfrak{L}_\nabla^+(H)$ -torsor (E.4) to S_α is trivial, and hence over any open affine of $S_\alpha^\circ \subset S_\alpha$, the $\mathfrak{L}_\nabla^+(H)$ -torsor (E.4) itself is trivial.

Hence, by Zariski descent, it suffices to show that for $Z_\alpha \rightarrow S_\alpha$, where Z_α is an ind-affine ind-scheme locally almost of finite type, the functor

$$\left(\mathrm{IndCoh}^*(Z_\alpha \times_{S_\alpha} \mathfrak{L}_\nabla^+(H)_{S_\alpha}) \right)_{\mathfrak{L}_\nabla^+(H)_{S_\alpha}} \rightarrow \mathrm{IndCoh}(Z_\alpha)$$

is an equivalence.

To prove this, it suffices to show that the functor

$$(E.6) \quad \mathrm{IndCoh}(Z_\alpha) \otimes_{\mathrm{QCoh}(S_\alpha)} \mathrm{QCoh}(\mathfrak{L}_\nabla^+(H))_{S_\alpha} \simeq \mathrm{IndCoh}^*(Z_\alpha \times_{S_\alpha} \mathfrak{L}_\nabla^+(H)_{S_\alpha})$$

is an equivalence.

To prove (E.6), we can assume that Z_α is an affine scheme. Writing $\mathfrak{L}_\nabla^+(H)_{S_\alpha}$ as a limit of relative affine schemes Y_n *smooth* over S_α as in the proof of Lemma E.2.7, it suffices to show that each of the functors

$$(E.7) \quad \mathrm{IndCoh}(Z_\alpha) \otimes_{\mathrm{QCoh}(S_\alpha)} \mathrm{QCoh}(Y_n) \simeq \mathrm{IndCoh}^*(Z_\alpha \times_{S_\alpha} Y_n)$$

is an equivalence.

However, this follows from Lemma A.4.10. □

E.2.11. We are finally ready to define $\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H))$ as a factorization category. Proceeding as in B.13.17-B.13.22, we need to show that Lemmas B.13.18 and B.13.20 hold for $\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H))$.

We will prove Lemma B.13.18; Lemma B.13.20 is proved similarly.

We need to show that the functor

$$(E.8) \quad \mathrm{QCoh}(S_\alpha) \otimes_{\mathrm{QCoh}(S_\beta)} \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H))_{S_\beta} \rightarrow \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H))_{S_\alpha}.$$

is an equivalence.

We consider both sides as modules over

$$\mathrm{QCoh}(S_\alpha) \otimes_{\mathrm{QCoh}(S_\beta)} \mathrm{QCoh}(\mathfrak{L}_\nabla^+(H))_{S_\beta} \simeq \mathrm{QCoh}(\mathfrak{L}_\nabla^+(H))_{S_\alpha}.$$

By Proposition E.1.4, it suffices to show that (E.8) becomes an equivalence after taking $\mathfrak{L}_\nabla^+(H)_{S_\alpha}$ -invariants, or, equivalently, thanks to Lemma E.1.6, $\mathfrak{L}_\nabla^+(H)_{S_\alpha}$ -coinvariants.

However, by Lemma E.2.10, when we take $\mathfrak{L}_\nabla^+(H)_{S_\alpha}$ -coinvariants in (E.8), the resulting functor identifies with

$$(E.9) \quad \mathrm{QCoh}(S_\alpha) \otimes_{\mathrm{QCoh}(S_\beta)} \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H)/\mathfrak{L}_\nabla^+(H))_{S_\beta} \rightarrow \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H)/\mathfrak{L}_\nabla^+(H))_{S_\alpha}.$$

Now, (E.9) is an equivalence by Lemma B.13.18, since $\mathfrak{L}_\nabla(H)/\mathfrak{L}_\nabla^+(H)$ is locally almost of finite type (by Lemma E.2.4) and in particular is placid.

E.2.12. By construction, $\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H))$ is equipped, as a factorization category, with an action of $\mathrm{QCoh}(\mathfrak{L}_\nabla^+(H)) \otimes \mathrm{QCoh}(\mathfrak{L}_\nabla^+(H))$.

We define $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ by formula (E.3).

By Proposition E.1.4, we have

$$\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}}) \otimes \mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}})} \mathrm{Vect} \simeq \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H))$$

and

$$\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}})} \mathrm{Vect} \simeq \mathrm{IndCoh}(\mathfrak{L}_\nabla(H)/\mathfrak{L}_\nabla^+(H)).$$

E.2.13. The pair of adjoint functors

$$(E.10) \quad \iota_*^{\text{IndCoh}} : \text{IndCoh}^*(\mathfrak{L}_{\nabla}^+(H)) \rightleftarrows \text{IndCoh}^*(\mathfrak{L}_{\nabla}(H)) : \iota^!$$

gives rise via

$$\begin{aligned} \text{QCoh}(\text{LS}_H^{\text{reg}}) &\stackrel{\text{Corollary E.1.8}}{\simeq} \text{QCoh}_{\text{co}}(\text{LS}_H^{\text{reg}}) \simeq \text{Vect}_{\mathfrak{L}_{\nabla}^+(H)} \simeq \\ &\simeq \text{QCoh}(\mathfrak{L}_{\nabla}^+(H))_{\mathfrak{L}_{\nabla}^+(H) \times \mathfrak{L}_{\nabla}^+(H)} \stackrel{\Psi_{\mathfrak{L}_{\nabla}^+(H)}}{\simeq} \text{IndCoh}^*(\mathfrak{L}_{\nabla}^+(H))_{\mathfrak{L}_{\nabla}^+(H) \times \mathfrak{L}_{\nabla}^+(H)} \end{aligned}$$

to an adjoint pair

$$(E.11) \quad \iota_*^{\text{IndCoh}} : \text{QCoh}(\text{LS}_H^{\text{reg}}) \rightleftarrows \text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}}) : \iota^!.$$

Since the essential image of the left adjoint in (E.10) generates the essential image, the same is true for (E.11).

In particular, images of compact objects in $\text{QCoh}(\text{LS}_H^{\text{reg}})$ under ι_*^{IndCoh} provide compact generators of $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$.

E.3. Unital structure.

E.3.1. We claim that the factorization categories we defined above, namely,

$$\text{IndCoh}^*(\mathfrak{L}_{\nabla}(H)) \text{ and } \text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$$

carry naturally defined unital structures.

Let us carry out the construction for $\text{IndCoh}^*(\mathfrak{L}_{\nabla}(H))$; the case of $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$ will follow by taking $\mathfrak{L}_{\nabla}^+(H) \times \mathfrak{L}_{\nabla}^+(H)$ -coinvariants.

E.3.2. By Sect. C.10.10, the factorization space $\mathfrak{L}_{\nabla}(H)$ carries a natural unital-in-correspondences structure. (Note, however, that we cannot deduce from there the unital structure on $\text{IndCoh}^*(\mathfrak{L}_{\nabla}(H))$ by applying Sect. C.12.8 directly because we are not in an ind-placid situation.)

For an injection of finite sets $I_1 \subseteq I_2$ consider the corresponding diagram

$$(E.12) \quad \mathfrak{L}_{\nabla}(H)_{X^{I_1}} \xleftarrow{\text{pr}_{\text{small}}^H} \mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(H)_{I_1 \subseteq I_2} \xrightarrow{\text{pr}_{\text{big}}^H} \mathfrak{L}_{\nabla}(H)_{X^{I_2}}.$$

E.3.3. We claim that the functor

$$(\text{pr}_{\text{small}}^H)_*^{\text{IndCoh}} : \text{IndCoh}^*(\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(H))_{I_1 \subseteq I_2} \rightarrow \text{IndCoh}^*(\mathfrak{L}_{\nabla}(H))_{X^{I_1}}$$

admits a left adjoint, to be denoted $(\text{pr}_{\text{small}}^H)^*, \text{IndCoh}$.

Factor the map $\text{pr}_{\text{small}}^H$ as

$$\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(H)_{I_1 \subseteq I_2} \xrightarrow{\text{pr}_{\text{small}}^H} \mathfrak{L}_{\nabla}(H)_{X^{I_1}} \times_{X^{I_1}} X^{I_2} \rightarrow \mathfrak{L}_{\nabla}(H)_{X^{I_1}},$$

and it is sufficient to prove the existence of the left adjoint for the first arrow, i.e., that the functor

$$(E.13) \quad (\text{pr}_{\text{small}}^H)_*^{\text{IndCoh}} : \text{IndCoh}^*(\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(H))_{I_1 \subseteq I_2} \rightarrow \text{IndCoh}^*(\mathfrak{L}_{\nabla}(H)_{X^{I_1}} \times_{X^{I_1}} X^{I_2})$$

admits a left adjoint.

E.3.4. We consider the two sides of (E.13) as acted on by

$$\mathfrak{L}_{\nabla}^+(H)_{X^{I_2}} \text{ and } \mathfrak{L}_{\nabla}^+(H)_{X^{I_1}} \times_{X^{I_1}} X^{I_2},$$

respectively. These actions are compatible via the map

$$\mathfrak{L}_{\nabla}^+(H)_{X^{I_2}} \rightarrow \mathfrak{L}_{\nabla}^+(H)_{X^{I_1}} \times_{X^{I_1}} X^{I_2},$$

corresponding to the counital structure on $\mathfrak{L}_{\nabla}^+(H)$.

By Proposition E.1.4, it suffices to show that the functor

$$(E.14) \quad (\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}(H)_{I_1 \subseteq I_2}))_{\mathfrak{L}_{\nabla}^+(H)_{X^{I_2}}} \rightarrow \left(\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H)_{X^{I_1}} \times_{X^{I_1}} X^{I_2}) \right)_{\mathfrak{L}_{\nabla}^+(H)_{X^{I_1}} \times_{X^{I_1}} X^{I_2}},$$

induced by (E.13), admits a left adjoint.

However, by Lemma E.2.10, the latter functor is the identity endofunctor of

$$\mathrm{IndCoh} \left((\mathfrak{L}_{\nabla}(H)/\mathfrak{L}_{\nabla}^+(H))_{X^{I_1}} \times_{X^{I_1}} X^{I_2} \right).$$

E.3.5. We define the functor

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H))_{X^{I_1}} \rightarrow \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H))_{X^{I_2}},$$

to be denoted $\mathrm{ins.unit}_{I_1 \subseteq I_2}$, to be

$$(\mathrm{pr}_{\mathrm{big}}^H)^{\mathrm{IndCoh}} \circ (\mathrm{pr}_{\mathrm{small}}^H)^{*, \mathrm{IndCoh}}.$$

E.3.6. In order to promote this to a unital structure on $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H))$, we need to construct isomorphisms

$$\mathrm{ins.unit}_{I_2 \subseteq I_2} \circ \mathrm{ins.unit}_{I_1 \subseteq I_2} \simeq \mathrm{ins.unit}_{I_1 \subseteq I_3}$$

for $I_1 \subseteq I_2 \subseteq I_3$.

Denote the maps in (E.12) by

$$\mathrm{pr}_{\mathrm{small}, I_1 \subseteq I_2}^H \text{ and } \mathrm{pr}_{\mathrm{big}, I_1 \subseteq I_2}^H$$

to indicate the dependence on the finite sets involved.

Thus, we need to construct an isomorphism

$$(E.15) \quad (\mathrm{pr}_{\mathrm{big}, I_2 \subseteq I_3}^H)^{\mathrm{IndCoh}} \circ (\mathrm{pr}_{\mathrm{small}, I_2 \subseteq I_3}^H)^{*, \mathrm{IndCoh}} \circ (\mathrm{pr}_{\mathrm{big}, I_1 \subseteq I_2}^H)^{\mathrm{IndCoh}} \circ (\mathrm{pr}_{\mathrm{small}, I_1 \subseteq I_1}^H)^{*, \mathrm{IndCoh}} \xrightarrow{\sim} (\mathrm{pr}_{\mathrm{big}, I_1 \subseteq I_3}^H)^{\mathrm{IndCoh}} \circ (\mathrm{pr}_{\mathrm{small}, I_1 \subseteq I_3}^H)^{*, \mathrm{IndCoh}}.$$

E.3.7. Note that we have a commutative diagram,

$$\begin{array}{ccc} \mathfrak{L}_{\nabla}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}(H)_{I_1 \subseteq I_3} & \xrightarrow{(\mathrm{pr}_{\mathrm{big}, I_1 \subseteq I_2}^H)^*} & \mathfrak{L}_{\nabla}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}(H)_{I_2 \subseteq I_3} \xrightarrow{\mathrm{pr}_{\mathrm{big}, I_2 \subseteq I_3}^H} \mathfrak{L}_{\nabla}(H)_{X^{I_3}} \\ \downarrow (\mathrm{pr}_{\mathrm{small}, I_2 \subseteq I_3}^H)^* & & \downarrow \mathrm{pr}_{\mathrm{small}, I_2 \subseteq I_3}^H \\ \mathfrak{L}_{\nabla}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}(H)_{I_1 \subseteq I_2} & \xrightarrow{(\mathrm{pr}_{\mathrm{big}, I_1 \subseteq I_2}^H)^*} & \mathfrak{L}_{\nabla}(H)_{X^{I_2}} \\ \downarrow \mathrm{pr}_{\mathrm{small}, I_1 \subseteq I_2}^H & & \\ \mathfrak{L}_{\nabla}(H)_{X^{I_1}}, & & \end{array}$$

in which the inner square is Cartesian.

We rewrite the right-hand side in (E.15) as

$$(\mathrm{pr}_{\mathrm{big}, I_2 \subseteq I_3}^H)^{\mathrm{IndCoh}} \circ (\mathrm{pr}_{\mathrm{big}, I_1 \subseteq I_2}^H)^{\mathrm{IndCoh}} \circ (\mathrm{pr}_{\mathrm{small}, I_2 \subseteq I_3}^H)^{*, \mathrm{IndCoh}} \circ (\mathrm{pr}_{\mathrm{small}, I_1 \subseteq I_2}^H)^{*, \mathrm{IndCoh}}.$$

The isomorphism

$$(\mathrm{pr}_{\mathrm{big}, I_1 \subseteq I_2}^H)_*^{\mathrm{IndCoh}} \circ (\mathrm{pr}_{\mathrm{small}, I_2 \subseteq I_3}^H)_*^{\mathrm{IndCoh}} \simeq (\mathrm{pr}_{\mathrm{small}, I_2 \subseteq I_3}^H)_*^{\mathrm{IndCoh}} \circ (\mathrm{pr}_{\mathrm{big}, I_1 \subseteq I_2}^H)_*^{\mathrm{IndCoh}}$$

induces a natural transformation

$$(E.16) \quad (\mathrm{pr}_{\mathrm{small}, I_2 \subseteq I_3}^H)_*^{\mathrm{IndCoh}} \circ (\mathrm{pr}_{\mathrm{big}, I_1 \subseteq I_2}^H)_*^{\mathrm{IndCoh}} \rightarrow (\mathrm{pr}_{\mathrm{big}, I_1 \subseteq I_2}^H)_*^{\mathrm{IndCoh}} \circ (\mathrm{pr}_{\mathrm{small}, I_2 \subseteq I_3}^H)_*^{\mathrm{IndCoh}}.$$

E.3.8. We claim that (E.16) is an isomorphism. Indeed, this follows by the same argument as that proving the existence of $(\mathrm{pr}_{\mathrm{small}}^H)_*^{\mathrm{IndCoh}}$ in Sects. E.3.3-E.3.4.

E.3.9. Finally, we define the natural isomorphism in (E.15) by precomposing the isomorphism (E.16) with $(\mathrm{pr}_{\mathrm{small}, I_1 \subseteq I_2}^H)_*^{\mathrm{IndCoh}}$ and post-composing with $(\mathrm{pr}_{\mathrm{big}, I_2 \subseteq I_3}^H)_*^{\mathrm{IndCoh}}$.

The higher compatibilities are constructed by a similar procedure.

E.3.10. By construction, the functor

$$\mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(H)) \xrightarrow{\Psi_{\mathfrak{L}_{\nabla}^+(H)}} \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(H)) \xrightarrow{\iota_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H))$$

is unital.

In particular, the object

$$\iota_*^{\mathrm{IndCoh}}(\mathcal{O}_{\mathfrak{L}_{\nabla}^+(H)}) \in \mathrm{FactAlg}(X, \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H)))$$

is the factorization unit in $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H))$.

Similarly, the functor

$$\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}}) \xrightarrow{\iota_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$$

is unital, and

$$\iota_*^{\mathrm{IndCoh}}(\mathcal{O}_{\mathrm{LS}_H^{\mathrm{reg}}}) \in \mathrm{FactAlg}(X, \mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}))$$

is the factorization unit in $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$.

E.4. Duality.

E.4.1. Let S_α be as in Sect. B.13.9.

Note that the same argument as in Lemma E.2.7 shows that the functor

$$\Upsilon_{\mathfrak{L}_{\nabla}^+(H))_{S_\alpha}} : \mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(H))_{S_\alpha} \rightarrow \mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}^+(H))_{S_\alpha}$$

is an equivalence.

As a consequence, we obtain that the pairing

$$\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}^+(H))_{S_\alpha} \otimes \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(H))_{S_\alpha} \rightarrow \mathrm{Vect}$$

of (A.26) is perfect.

E.4.2. Similarly, an argument parallel to that in Lemma E.2.10 shows that the $!$ -pullback functor along

$$\mathrm{IndCoh}(\mathfrak{L}_{\nabla}(H)/\mathfrak{L}_{\nabla}^+(H))_{S_\alpha} \rightarrow \mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(H))_{S_\alpha}$$

gives rise to an equivalence

$$\mathrm{IndCoh}(\mathfrak{L}_{\nabla}(H)/\mathfrak{L}_{\nabla}^+(H))_{S_\alpha} \simeq \left(\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(H))_{S_\alpha} \right)^{\mathfrak{L}_{\nabla}^+(H)_{S_\alpha}}.$$

Combining with Proposition E.1.4, we obtain that the pairing

$$\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(H))_{S_\alpha} \otimes \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H))_{S_\alpha} \rightarrow \mathrm{Vect}$$

of (A.26) is perfect.

E.4.3. In particular, we obtain that Lemmas B.13.11 and B.13.14 hold for $\mathrm{IndCoh}^!(\mathfrak{L}_\nabla(H))$. I.e., the recipe in Sects. B.13.10-B.13.15 gives rise to a well-defined factorization category $\mathrm{IndCoh}^!(\mathfrak{L}_\nabla(H))$.

Moreover, we obtain that (A.26) defines a perfect pairing between

$$\mathrm{IndCoh}^!(\mathfrak{L}_\nabla(H)) \text{ and } \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H))$$

as factorization categories.

E.4.4. By a similar logic as in Sect. E.2.12, we obtain that the assignment

$$S_\alpha \rightsquigarrow \mathrm{IndCoh}^!(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})_{S_\alpha}$$

extends to a well-defined factorization category $\mathrm{IndCoh}^!(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$.

Moreover, we have

$$\mathrm{IndCoh}^!(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}) \simeq \mathrm{IndCoh}^!(\mathfrak{L}_\nabla(H))^{\mathfrak{L}_\nabla^+(H) \times \mathfrak{L}_\nabla^+(H)}$$

and

$$\mathrm{IndCoh}^!(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}})} \mathrm{Vect} \simeq \mathrm{IndCoh}^!(\mathfrak{L}_\nabla(H)/\mathfrak{L}_\nabla^+(H)).$$

We obtain that

$$\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}) \text{ and } \mathrm{IndCoh}^!(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$$

are mutually dual as factorization categories.

E.4.5. Finally, a procedure dual to that in Sect. E.3 defines on

$$\mathrm{IndCoh}^!(\mathfrak{L}_\nabla(H)) \text{ and } \mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$$

unital structures, and the identifications

$$\mathrm{IndCoh}^!(\mathfrak{L}_\nabla(H))^\vee \simeq \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H)) \text{ and } \mathrm{IndCoh}^!(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})^\vee \simeq \mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$$

extend to identifications of the corresponding unital factorization categories.

E.5. t-structures. In this section we will discuss an alternative approach to the definition of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$.

Namely, we can start with (the more elementary) $\mathrm{QCoh}_{\mathrm{co}}(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$, and obtain from it $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ by a *renormalization procedure* (i.e., ind-completion of a specified small subcategory), see Sect. E.5.6 below.

E.5.1. As in Sect. B.13.22, the factorization category

$$\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H))$$

carries a naturally defined t-structure.

By construction, the functor

$$\Gamma^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H)) \rightarrow \mathrm{Vect}$$

is t-exact, and conservative when restricted to the eventually coconnective subcategory.

Moreover, the functor

$$\ell_*^{\mathrm{IndCoh}} : \mathrm{QCoh}(\mathfrak{L}_\nabla^+(H)) \rightarrow \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(H))$$

is t-exact.

E.5.2. We now define a t-structure on $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$. Namely, by Proposition E.1.4 the projection

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H)) \rightarrow \mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$$

admits a left adjoint, which is comonadic.

Moreover, the resulting comonad on $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H))$ is t-exact. This implies that the category $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ acquires a unique t-structure for which both functors

$$\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}) \rightleftarrows \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H))$$

are t-exact.

E.5.3. Consider the untal factorization categories

$$\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(H)), \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(H)/\mathfrak{L}_{\nabla}^+(H)) \text{ and } \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}).$$

The category $\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(H))$ carries a natural action of $\mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(H))$, and it follows formally that the functors

$$(E.17) \quad (\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(H)))_{\mathfrak{L}_{\nabla}^+(H) \times \mathfrak{L}_{\nabla}^+(H)} \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$$

and

$$(E.18) \quad (\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(H)))_{\mathfrak{L}_{\nabla}^+(H)} \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(H)/\mathfrak{L}_{\nabla}^+(H)),$$

induced by the direct image functors

$$\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(H)) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}) \text{ and } \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(H)) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(H)/\mathfrak{L}_{\nabla}^+(H)),$$

respectively, are equivalences.

In addition, we have the unital factorization functors

$$\mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(H)) \xrightarrow{\iota_*} \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(H)) \text{ and } \mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}}) \simeq \mathrm{QCoh}_{\mathrm{co}}(\mathrm{LS}_H^{\mathrm{reg}}) \xrightarrow{\iota_*} \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}).$$

E.5.4. Let S_{α} be as in Sect. B.13.9. Recall (see Lemma A.8.10) that the functor

$$\Psi_{\mathfrak{L}_{\nabla}(H)_{S_{\alpha}}} : \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H))_{S_{\alpha}} \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(H))_{S_{\alpha}}$$

is t-exact, and induces an equivalence between the eventually coconnective subcategories on both sides.

It follows from the definition of $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H))$ as a factorization category that the functors $\Psi_{\mathfrak{L}_{\nabla}(H)_{S_{\alpha}}}$ combine to give rise to a factorization functor

$$\Psi_{\mathfrak{L}_{\nabla}(H)} : \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H)) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(H)).$$

Moreover, the functor $\Psi_{\mathfrak{L}_{\nabla}(H)}$ is t-exact and induces an equivalences between the eventually coconnective subcategories on both sides.

Furthermore, the functor $\Psi_{\mathfrak{L}_{\nabla}(H)}$ has a naturally defined unital structure.

E.5.5. Note that the contents of Sect. E.5.4 allows us to recover $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H))$, as a unital factorization category, from $\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(H))$ with its t-structure.

Namely, for $S \rightarrow \mathrm{Ran}$, the category $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(H))_S$ identifies with the ind-completion of the full subcategory of $\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(H))_S$, generated by *finite* colimits by the essential image of $\mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(H))_S$ along ι_* .

E.5.6. It follows formally from Sect. E.5.2 that we have a naturally defined t-exact unital factorization functor

$$\Psi_{\text{Hecke}_H^{\text{spec}, \text{loc}}} : \text{IndCoh}^*(\text{Hecke}_H^{\text{spec}, \text{loc}}) \rightarrow \text{QCoh}_{\text{co}}(\text{Hecke}_H^{\text{spec}, \text{loc}}),$$

which induces an equivalences between the eventually coconnective subcategories on both sides.

Furthermore, as in Sect. E.5.5, we can recover $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec}, \text{loc}})$ (as a unital factorization category) from $\text{QCoh}_{\text{co}}(\text{Hecke}_H^{\text{spec}, \text{loc}})$ with its t-structure.

Namely, for $S \rightarrow \text{Ran}$, the category $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec}, \text{loc}})_S$ identifies with the ind-completion of the full subcategory of $(\text{QCoh}_{\text{co}}(\text{Hecke}_H^{\text{spec}, \text{loc}})_S)^{>-\infty}$, generated by *finite* colimits by the essential image of $\text{QCoh}(\text{LS}_H^{\text{reg}})_S$ along ι_* .

E.6. The monoidal structure.

E.6.1. Let S_α be as in Sect. B.13.9. The group-scheme structure on $\mathcal{L}_\nabla(H)_{S_\alpha}$ induces on the category $\text{IndCoh}^*(\mathcal{L}_\nabla(H))_{S_\alpha}$ a structure of monoidal category (under convolution).

By the construction of $\text{IndCoh}^*(\mathcal{L}_\nabla(H))$ as a factorization category, we obtain that $\text{IndCoh}^*(\mathcal{L}_\nabla(H))$ acquires a structure of monoidal factorization category.

This structure is compatible with the unital structure on $\text{IndCoh}^*(\mathcal{L}_\nabla(H))$ in the sense that the monoidal operation has a natural lax unital structure. I.e., $\text{IndCoh}^*(\mathcal{L}_\nabla(H))$ is an associative algebra object in the symmetric monoidal category of unital factorization categories with lax unital functors as morphisms.

E.6.2. Note that for S_α as above, the monoidal operation on $\text{IndCoh}^*(\mathcal{L}_\nabla(H))_{S_\alpha}$, viewed as a functor

$$\text{IndCoh}^*(\mathcal{L}_\nabla(H))_{S_\alpha} \otimes_{\text{QCoh}(S_\alpha)} \text{IndCoh}^*(\mathcal{L}_\nabla(H))_{S_\alpha} \rightarrow \text{IndCoh}^*(\mathcal{L}_\nabla(H))_{S_\alpha}$$

is t-exact.

Hence, the monoidal operation on $\text{IndCoh}^*(\mathcal{L}_\nabla(H))$ is t-exact.

E.6.3. We will now descend the above monoidal structure to one on $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec}, \text{loc}})$.

Consider the correspondence

$$(E.19) \quad (\text{LS}_H^{\text{reg}} \times_{\text{LS}_H^{\text{mer}}} \text{LS}_H^{\text{reg}}) \times (\text{LS}_H^{\text{reg}} \times_{\text{LS}_H^{\text{mer}}} \text{LS}_H^{\text{reg}}) \xleftarrow{\Delta_{\text{LS}_H^{\text{reg}}}} \text{LS}_H^{\text{reg}} \times_{\text{LS}_H^{\text{mer}}} \text{LS}_H^{\text{reg}} \xrightarrow{\text{mult}} \text{LS}_H^{\text{reg}} \times_{\text{LS}_H^{\text{mer}}} \text{LS}_H^{\text{reg}}.$$

Note that by Lemma E.2.2 we can think of this diagram also as

$$(E.20) \quad (\mathcal{L}_\nabla^+(H) \backslash \mathcal{L}_\nabla(H) / \mathcal{L}_\nabla^+(H)) \times (\mathcal{L}_\nabla^+(H) \backslash \mathcal{L}_\nabla(H) / \mathcal{L}_\nabla^+(H)) \leftarrow \mathcal{L}_\nabla^+(H) \backslash \left(\frac{\mathcal{L}_\nabla(H) \times \mathcal{L}_\nabla(H)}{\mathcal{L}_\nabla^+(H)} \right) / \mathcal{L}_\nabla^+(H) \rightarrow \mathcal{L}_\nabla^+(H) \backslash \mathcal{L}_\nabla(H) / \mathcal{L}_\nabla^+(H),$$

where:

- $\frac{\mathcal{L}_\nabla(H) \times \mathcal{L}_\nabla(H)}{\mathcal{L}_\nabla^+(H)}$ denotes the quotient with respect to the diagonal action by right multiplication along the left factor and the left multiplication along the right factor;
- The arrow $\rightarrow$ is induced by the product map

$$\frac{\mathcal{L}_\nabla(H) \times \mathcal{L}_\nabla(H)}{\mathcal{L}_\nabla^+(H)} \rightarrow \mathcal{L}_\nabla(H).$$

E.6.4. As in Sect. E.2, we obtain a well-defined factorization category

$$\mathrm{IndCoh}^*(\mathrm{LS}_H^{\mathrm{reg}} \times_{\mathrm{LS}_H^{\mathrm{mer}}} \mathrm{LS}_H^{\mathrm{reg}} \times_{\mathrm{LS}_H^{\mathrm{mer}}} \mathrm{LS}_H^{\mathrm{reg}}),$$

equipped with factorization functors

$$(E.21) \quad \mathrm{IndCoh}^*(\mathrm{LS}_H^{\mathrm{reg}} \times_{\mathrm{LS}_H^{\mathrm{mer}}} \mathrm{LS}_H^{\mathrm{reg}}) \otimes \mathrm{IndCoh}^*(\mathrm{LS}_H^{\mathrm{reg}} \times_{\mathrm{LS}_H^{\mathrm{mer}}} \mathrm{LS}_H^{\mathrm{reg}}) \xrightarrow{(\Delta_{\mathrm{LS}_H^{\mathrm{reg}}})_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*(\mathrm{LS}_H^{\mathrm{reg}} \times_{\mathrm{LS}_H^{\mathrm{mer}}} \mathrm{LS}_H^{\mathrm{reg}}) \xleftarrow{(\mathrm{mult})_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*(\mathrm{LS}_H^{\mathrm{reg}} \times_{\mathrm{LS}_H^{\mathrm{mer}}} \mathrm{LS}_H^{\mathrm{reg}}).$$

It is easy to see that the functor $(\Delta_{\mathrm{LS}_H^{\mathrm{reg}}})_*^{\mathrm{IndCoh}}$ admits a left adjoint, to be denoted $(\Delta_{\mathrm{LS}_H^{\mathrm{reg}}})^*, \mathrm{IndCoh}$. We define the monoidal structure on $\mathrm{IndCoh}^*(\mathrm{LS}_H^{\mathrm{reg}} \times_{\mathrm{LS}_H^{\mathrm{mer}}} \mathrm{LS}_H^{\mathrm{reg}})$ with the binary operation given by

$$(\mathrm{mult})_*^{\mathrm{IndCoh}} \circ (\Delta_{\mathrm{LS}_H^{\mathrm{reg}}})^*, \mathrm{IndCoh}.$$

One defines similarly n -fold compositions, and they form a compatible system thanks to the fact that the functors $(\Delta_{\mathrm{LS}_H^{\mathrm{reg}}})^*, \mathrm{IndCoh}$ satisfy base change against IndCoh -pushforwards.

This defines on

$$\mathrm{IndCoh}^*(\mathrm{LS}_H^{\mathrm{reg}} \times_{\mathrm{LS}_H^{\mathrm{mer}}} \mathrm{LS}_H^{\mathrm{reg}}) =: \mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$$

a structure of monoidal factorization category.

As in the case of $\mathrm{IndCoh}^*(\mathcal{L}_{\nabla}(H))$, it is easy to see that the monoidal operation on the category $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ is t-exact.

E.6.5. A similar procedure gives rise to an action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ on $\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}})$.

E.6.6. By construction, the monoidal structure on $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ is compatible with the unital structure, in the sense that $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ is an associative algebra object in the symmetric monoidal category of unital factorization categories with lax unital functors as morphisms.

However, we claim that, unlike $\mathrm{IndCoh}^*(\mathcal{L}_{\nabla}(H))$, in the case of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ more is true: namely, the monoidal operation is strictly unital.

Indeed, this follows from Lemma C.11.23, since the factorization unit for $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$, namely, $\iota_*^{\mathrm{IndCoh}}(\mathcal{O}_{\mathrm{LS}_H^{\mathrm{reg}}})$ is also the monoidal unit.

The same observation applies to the action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ on $\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}})$.

E.6.7. A similar procedure defines the monoidal factorization categories

$$(E.22) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathcal{L}_{\nabla}(H)) \text{ and } \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}),$$

with t-exact monoidal operations.

We note that the procedure Sect. E.5.6 allows us to recover

$$(E.23) \quad \mathrm{IndCoh}^*(\mathcal{L}_{\nabla}(H)) \text{ and } \mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$$

as *monoidal factorization categories* from those in (E.22).

Namely, these monoidal structures are uniquely determined by the requirement that the functors

$$\Psi_{\mathcal{L}_{\nabla}(H)} : \mathrm{IndCoh}^*(\mathcal{L}_{\nabla}(H)) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{L}_{\nabla}(H))$$

and

$$\Psi_{\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}} : \mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$$

are monoidal.

E.6.8. Note that the functor $(\text{mult})_*^{\text{IndCoh}}$, involved in the definition of the monoidal structure on $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$ admits a (continuous) right adjoint, to be denoted $(\text{mult})^!$.

To prove this, it suffices to show that the corresponding functor

$$\text{IndCoh}^*\left(\frac{\mathfrak{L}_\nabla(H) \times \mathfrak{L}_\nabla(H)}{\mathfrak{L}_\nabla^+(H)}\right) \xrightarrow{(\text{mult})_*^{\text{IndCoh}}} \text{IndCoh}^*(\mathfrak{L}_\nabla(H))$$

admits a (continuous) right adjoint.

However, we can isomorph the projection

$$\frac{\mathfrak{L}_\nabla(H) \times \mathfrak{L}_\nabla(H)}{\mathfrak{L}_\nabla^+(H)} \xrightarrow{\text{mult}} \mathfrak{L}_\nabla(H)$$

to the projection

$$\mathfrak{L}_\nabla(H) \times (\mathfrak{L}_\nabla(H)/\mathfrak{L}_\nabla^+(H)) \rightarrow \mathfrak{L}_\nabla(H),$$

and the assertion follows from the fact that $\mathfrak{L}_\nabla(H)/\mathfrak{L}_\nabla^+(H)$ is ind-proper.

E.6.9. In particular, we obtain that the monoidal operation on $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$ admits a continuous right adjoint, namely,

$$(\Delta_{LS_H^{\text{reg}}})_*^{\text{IndCoh}} \circ (\text{mult})^!.$$

Furthermore, it easy to see that this right adjoint

$$\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}}) \rightarrow \text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}}) \otimes \text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$$

is compatible with the $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$ -bimodule structure.

Since the monoidal unit in $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$ is compact, we obtain that $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$ is *rigid* (see [GaRo3, Chapter 1, Definition 9.1.2] for what this means), i.e., for any $S \rightarrow \text{Ran}$, the monoidal category

$$\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})_S$$

is rigid.

E.6.10. Passing to duals, the monoidal structure on $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$ induces a comonoidal structure on its dual, i.e., $\text{IndCoh}^!(\text{Hecke}_H^{\text{spec,loc}})$. Since $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$ is rigid, the comonoidal operation on $\text{IndCoh}^!(\text{Hecke}_H^{\text{spec,loc}})$ admits a left adjoint, i.e., $\text{IndCoh}^!(\text{Hecke}_H^{\text{spec,loc}})$ is naturally a monoidal category.

Let us describe the monoidal operation on $\text{IndCoh}^!(\text{Hecke}_H^{\text{spec,loc}})$ explicitly. In terms of (E.19), it is given by

$$(\text{mult})_*^{\text{IndCoh}} \circ (\Delta_{LS_H^{\text{reg}}})^!,$$

where $(\text{mult})_*^{\text{IndCoh}}$ is the left adjoint of

$$\text{mult}^! : \text{IndCoh}^!(LS_H^{\text{reg}} \times_{LS_H^{\text{mer}}} LS_H^{\text{reg}}) \rightarrow \text{IndCoh}^!(LS_H^{\text{reg}} \times_{LS_H^{\text{mer}}} LS_H^{\text{reg}} \times_{LS_H^{\text{mer}}} LS_H^{\text{reg}}),$$

or, which is the same as the dual of

$$(\text{mult})^! : \text{IndCoh}^*(LS_H^{\text{reg}} \times_{LS_H^{\text{mer}}} LS_H^{\text{reg}}) \rightarrow \text{IndCoh}^*(LS_H^{\text{reg}} \times_{LS_H^{\text{mer}}} LS_H^{\text{reg}} \times_{LS_H^{\text{mer}}} LS_H^{\text{reg}}),$$

whose existence was proved above.

E.6.11. Since $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$ is rigid, by [GaRo3, Sect. 9.2.1], a choice of “right” or “left” determines an equivalence

$$(E.24) \quad \text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}}) \simeq \text{IndCoh}^!(\text{Hecke}_H^{\text{spec,loc}})$$

as monoidal categories.

E.7. Action on $\text{IndCoh}(\text{LS}_H)$. In this subsection we will define a (local action) of the monoidal factorization category $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$ on $\text{IndCoh}(\text{LS}_H)$ in the sense of Sect. H.6.1.

E.7.1. Our goal is to define an action of the monoidal category $(\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}}))_{\text{Ran}}^!$ (see Sect. H.5.5 for the notation) on $\text{IndCoh}(\text{LS}_H) \otimes \underline{\text{D-mod}}(\text{Ran})$. In other words, we need to define an action of $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})_S$ on $\text{IndCoh}(\text{LS}_H) \otimes \text{QCoh}(S)$ for any $S \rightarrow \text{Ran}$.

Let $\text{Hecke}_{G,S}^{\text{spec,glob}}$ denote the fiber product

$$(\text{LS}_H \times S) \times_{\text{LS}_{H,S}^{\text{mer,glob}}} (\text{LS}_H \times S),$$

where

$$\text{LS}_{H,S}^{\text{mer,glob}} := S \times_{\text{Ran}} \text{LS}_{H,\text{Ran}}^{\text{mer,glob}}$$

for $\text{LS}_{H,\text{Ran}}^{\text{mer,glob}}$ as in Sect. B.7.14.

Restriction to the formal disc gives rise to vertical arrows in the following diagram, see Sect. B.7.15:

$$(E.25) \quad \begin{array}{ccccc} \text{LS}_H \times S & \xleftarrow{\bar{h}^{\text{spec,glob}}} & \text{Hecke}_{H,S}^{\text{spec,glob}} & \xrightarrow{\bar{h}^{\text{spec,glob}}} & \text{LS}_H \times S \\ \text{ev}_S \downarrow & & \text{ev}_S \downarrow & & \downarrow \text{ev}_S \\ \text{LS}_{H,S}^{\text{reg}} & \xleftarrow{\bar{h}^{\text{spec,loc}}} & \text{Hecke}_{H,S}^{\text{spec,loc}} & \xrightarrow{\bar{h}^{\text{spec,loc}}} & \text{LS}_{H,S}^{\text{reg}}. \end{array}$$

Lemma E.7.2. *Both squares in (E.25) are Cartesian.*

Proof. First, we claim that both squares are Cartesian at the classical level. Indeed, classically, the horizontal arrows in (E.25) are isomorphisms.

Hence, in order to prove the lemma, it remains to check the Cartesian property of the tangent spaces on the unit section of $\text{Hecke}_{H,S}^{\text{spec,glob}}$.

Let σ be a point of LS_H . Then for any $s \in S$, the relative tangent space of

$$\text{Hecke}_{H,S}^{\text{spec,glob}} \rightarrow \text{LS}_H \times S$$

at (the image along the unit section of) (σ, s) identifies with

$$\text{Fib}(C^*(X, \mathfrak{h}_\sigma)[1] \rightarrow C^*(X - \underline{x}, \mathfrak{h}_\sigma)[1]) \simeq \text{coFib}(C^*(X, \mathfrak{h}_\sigma) \rightarrow C^*(X - \underline{x}, \mathfrak{h}_\sigma)),$$

where $\underline{x} \in \text{Ran}$ is the image of s .

The relative tangent space of

$$\text{Hecke}_{H,S}^{\text{spec,loc}} \rightarrow \text{LS}_{H,S}^{\text{reg}}$$

at the image of this point identifies with

$$(\mathfrak{L}_{\nabla}^+(\mathfrak{h}_\sigma)/\mathfrak{L}_{\nabla}^+(\mathfrak{h}_\sigma))_{\underline{x}}.$$

The required Cartesian property of the tangent spaces follows from the fact that diagram

$$\begin{array}{ccc} C^*(X, \mathfrak{h}_\sigma) & \longrightarrow & C^*(X - \underline{x}, \mathfrak{h}_\sigma) \\ \downarrow & & \downarrow \\ \mathfrak{L}_{\nabla}^+(\mathfrak{h}_\sigma)_{\underline{x}} & \longrightarrow & \mathfrak{L}_{\nabla}(\mathfrak{h}_\sigma)_{\underline{x}} \end{array}$$

is Cartesian. □

E.7.3. For $S \rightarrow \text{Ran}$ as above, denote

$$(\text{LS}_H \times S)^{\text{level}} := (\text{LS}_H \times S) \times_{\text{LS}_{H,S}^{\text{reg}}} S,$$

where $S \rightarrow \text{LS}_{H,S}^{\text{reg}}$ is the unit point.

By construction, $(\text{LS}_H \times S)^{\text{level}}$ is acted on by $\mathfrak{L}_{\nabla}^+(H)_S$, so that

$$(\text{LS}_H \times S)^{\text{level}} / \mathfrak{L}_{\nabla}^+(H)_S \simeq \text{LS}_H \times S.$$

Note that Lemma E.7.2 can be reformulated as saying that the above action of $\mathfrak{L}^+(H)_S$ on $(\text{LS}_H \times S)^{\text{level}}$ extends to an action of $\mathfrak{L}(H)_S$.

E.7.4. Let S_α be as in Sect. B.13.9. Consider the category

$$\text{IndCoh}^*((\text{LS}_H \times S_\alpha)^{\text{level}}).$$

The action of $\mathfrak{L}_{\nabla}(H)_{S_\alpha}$ on $(\text{LS}_H \times S_\alpha)^{\text{level}}$ gives rise to an action of the monoidal category $\text{IndCoh}^*(\mathfrak{L}_{\nabla}(H)_{S_\alpha})$ on $\text{IndCoh}^*((\text{LS}_H \times S_\alpha)^{\text{level}})$.

As in Lemma E.2.7, the IndCoh -pushforward functor

$$\text{IndCoh}^*((\text{LS}_H \times S_\alpha)^{\text{level}}) \rightarrow \text{IndCoh}(\text{LS}_H \times S_\alpha)$$

gives rise to an equivalence

$$\left(\text{IndCoh}^*((\text{LS}_H \times S_\alpha)^{\text{level}}) \right)_{\mathfrak{L}_{\nabla}^+(H)_{S_\alpha}} \xrightarrow{\sim} \text{IndCoh}(\text{LS}_H \times S_\alpha).$$

Then by the same mechanism as in Sect. E.6.4, we obtain an action of the monoidal category $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})_{S_\alpha}$ on $\text{IndCoh}(\text{LS}_H \times S_\alpha)$.

E.7.5. Note that we can explicitly describe the action functor as follows:

$$\begin{aligned} \text{(E.26)} \quad \text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})_{S_\alpha} \otimes_{\text{QCoh}(S_\alpha)} \text{IndCoh}(\text{LS}_H \times S_\alpha) &\xrightarrow{(\text{ev}_S \times \vec{h}^{\text{spec,glob}})^*, \text{IndCoh}} \\ &\rightarrow \text{IndCoh}((\text{Hecke}_H^{\text{spec,glob}})_{S_\alpha}) \xrightarrow{(\vec{h}^{\text{spec,glob}})^{\text{IndCoh}}_*} \text{IndCoh}(\text{LS}_H \times S_\alpha), \end{aligned}$$

where the first arrow is obtained by identifying

$$\begin{aligned} \text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})_{S_\alpha} \otimes_{\text{QCoh}(S_\alpha)} \text{IndCoh}(\text{LS}_H \times S_\alpha) &\simeq \\ &\simeq \left(\text{IndCoh}^*(\mathfrak{L}_{\nabla}(H) \times_{S_\alpha} (\text{LS}_H \times S_\alpha)^{\text{level}}) \right)_{(\mathfrak{L}^+(H) \times \mathfrak{L}^+(H) \times \mathfrak{L}^+(H))_{S_\alpha}} \end{aligned}$$

and

$$\text{IndCoh}((\text{Hecke}_H^{\text{spec,glob}})_{S_\alpha}) \simeq \left(\text{IndCoh}^*(\mathfrak{L}_{\nabla}(H) \times_{S_\alpha} (\text{LS}_H \times S_\alpha)^{\text{level}}) \right)_{(\mathfrak{L}^+(H) \times \mathfrak{L}^+(H))_{S_\alpha}},$$

and the functor $(\text{ev}_S \times \vec{h}^{\text{spec,glob}})^*, \text{IndCoh}$ is the left adjoint to the projection from $(\mathfrak{L}^+(H) \times \mathfrak{L}^+(H))_{S_\alpha}$ -coinvariants to $(\mathfrak{L}^+(H) \times \mathfrak{L}^+(H) \times \mathfrak{L}^+(H))_{S_\alpha}$ -coinvariants.

E.7.6. Having defined the action of $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})_{S_\alpha}$ on $\text{IndCoh}(\text{LS}_H \times S_\alpha)$, the procedure in Sects. B.13.17-B.13.19 defines an action of $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})_S$ on $\text{IndCoh}(\text{LS}_H \times S)$ for any $S \rightarrow \text{Ran}$.

Thus, we obtain the sought-for local action of $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$ on $\text{IndCoh}(\text{LS}_H)$. Furthermore, unwinding the construction, we obtain that this action has a natural Ran -unital structure (see Sect. H.6.1 for what this means).

E.7.7. Recall the functor

$$\mathrm{Loc}_{H,\mathrm{Ran}}^{\mathrm{spec}} : \mathrm{Rep}(H)_{\mathrm{Ran}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_H) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}).$$

We claim:

Proposition E.7.8. *The functor $\mathrm{Loc}_{H,\mathrm{Ran}}^{\mathrm{spec}}$ intertwines the actions of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})_{\mathrm{Ran}}^!$ on the two sides.*

Proof. Unwinding the construction, we need to construct the datum of compatibility for the functor

$$\mathrm{Loc}_{H,S_\alpha}^{\mathrm{spec}} : \mathrm{Rep}(H)_{S_\alpha} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_H) \otimes \mathrm{QCoh}(S_\alpha)$$

for S_α as in Sect. B.13.9.

We identify

$$\mathrm{Rep}(H)_{S_\alpha} \simeq \mathrm{QCoh}(\mathrm{LS}_{H,S_\alpha}^{\mathrm{reg}}),$$

so that the functor $\mathrm{Loc}_{H,S_\alpha}^{\mathrm{spec}}$ identifies with the functor

$$\mathrm{QCoh}(\mathrm{LS}_{H,S_\alpha}^{\mathrm{reg}}) \xrightarrow{(\mathrm{ev}_{S_\alpha})^*, \mathrm{IndCoh}} \mathrm{IndCoh}(\mathrm{LS}_H \otimes S_\alpha) \simeq \mathrm{IndCoh}(\mathrm{LS}_H) \otimes \mathrm{QCoh}(S_\alpha).$$

Now the assertion of the proposition follows by unwinding the constructions, using the fact that diagram (E.25) is Cartesian (see Lemma E.7.2). $\square$

E.7.9. Consider now the functor

$$\Gamma_{H,\mathrm{Ran}}^{\mathrm{spec},\mathrm{IndCoh}} : \mathrm{IndCoh}(\mathrm{LS}_H) \otimes \mathrm{D-mod}(\mathrm{Ran}) \rightarrow \mathrm{Rep}(H)_{\mathrm{Ran}},$$

right adjoint to $\mathrm{Loc}_{H,\mathrm{Ran}}^{\mathrm{spec}}$.

Since $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})$ is rigid, from Proposition E.7.8, we obtain:

Corollary E.7.10. *The functor $\Gamma_{H,\mathrm{Ran}}^{\mathrm{spec},\mathrm{IndCoh}}$ intertwines the actions of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})_{\mathrm{Ran}}^!$ on the two sides.*

E.7.11. The (local) action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})$ on $\mathrm{IndCoh}(\mathrm{LS}_H)$ gives rise to a (local) *right* action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})$ on the dual $\mathrm{IndCoh}(\mathrm{LS}_H)^\vee$ of $\mathrm{IndCoh}(\mathrm{LS}_H)$.

We identify

$$\mathrm{IndCoh}(\mathrm{LS}_H)^\vee \simeq \mathrm{IndCoh}(\mathrm{LS}_H)$$

by Serre duality. Thus, we obtain a new (local) right action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})$ on $\mathrm{IndCoh}(\mathrm{LS}_H)$.

Note now that we can pass between right and left modules over $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})$ using the anti-involution σ^{spec} , induced by the inversion operation of $\mathfrak{L}_\nabla(H)$.

Unwinding the construction, we obtain that the resulting new (local) action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})$ on $\mathrm{IndCoh}(\mathrm{LS}_H)$ coincides with the original one.

E.7.12. The (local) action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})$ on $\mathrm{IndCoh}(\mathrm{LS}_H)$ gives rise to a (local) *coaction* of the factorization comonoidal category $\mathrm{IndCoh}^!(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})$ on $\mathrm{IndCoh}(\mathrm{LS}_H)$.

Since $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})$ is rigid, the coaction functor admits a left adjoint, so we obtain a (local) *action* of $\mathrm{IndCoh}^!(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})$, viewed as a factorization monoidal category, on $\mathrm{IndCoh}(\mathrm{LS}_H)$.

The corresponding monoidal operation is described explicitly as follows.

For S_α as in Sect. B.13.9, the action functor

$$\mathrm{IndCoh}^!(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})_{S_\alpha} \otimes_{\mathrm{QCoh}(S_\alpha)} \mathrm{IndCoh}(\mathrm{LS}_H \times S_\alpha) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_H \times S_\alpha)$$

is given by

$$\begin{aligned}
 (E.27) \quad \text{IndCoh}(\text{LS}_H \times S_\alpha) &\otimes_{\text{QCoh}(S_\alpha)} \text{IndCoh}^!(\text{Hecke}_H^{\text{spec,loc}})_{S_\alpha} \simeq \\
 &\simeq \text{IndCoh}^! \left((\text{LS}_H \times S_\alpha) \times_{S_\alpha} (\text{Hecke}_H^{\text{spec,loc}})_{S_\alpha} \right) \xrightarrow{(\overleftarrow{h}^{\text{spec,glob}} \times \text{ev})^!} \\
 &\rightarrow \text{IndCoh}((\text{Hecke}_H^{\text{spec,glob}})_{S_\alpha}) \xrightarrow{(\overrightarrow{h}^{\text{spec,glob}})^{\text{IndCoh}}_*} \text{IndCoh}(\text{LS}_H \times S_\alpha).
 \end{aligned}$$

For an arbitrary $S \rightarrow \text{Ran}$, this action is extended by the mechanism of Sects. B.13.10-B.13.15.

E.7.13. Recall now that according to (E.24), we can identify $\text{IndCoh}^!(\text{Hecke}_H^{\text{spec,loc}})$ as a monoidal category with $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$.

It is a formal property of actions of rigid categories that with respect to this identification, the above action of $\text{IndCoh}^!(\text{Hecke}_H^{\text{spec,loc}})$ on $\text{IndCoh}(\text{LS}_H)$ identifies with the right action of $\text{IndCoh}^*(\text{Hecke}_H^{\text{spec,loc}})$ on $\text{IndCoh}(\text{LS}_H)$ from Sect. E.7.11.

E.8. Action on monodromy-free opers. In this subsection we take $H = \check{G}$, the Langlands dual of a reductive group G . We will construct a factorization version of the action of $\text{IndCoh}^*(\text{Hecke}_{\check{G}}^{\text{spec,loc}})$ on $\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})$.

E.8.1. Let $\text{Hecke}_{\check{G}}^{\text{spec,Op}_{\check{G}}^{\text{mon-free}}}$ be the factorization ind-scheme, defined as

$$\text{Hecke}_{\check{G}}^{\text{spec,Op}_{\check{G}}^{\text{mon-free}}} := \text{Op}_{\check{G}}^{\text{mer}} \times_{\text{LS}_{\check{G}}^{\text{mer}}} \text{Hecke}_{\check{G}}^{\text{spec,loc}}.$$

Note that we have a commutative diagram

$$\begin{array}{ccccc}
 \text{Op}_{\check{G}}^{\text{mon-free}} & \xleftarrow{\overleftarrow{h}^{\text{spec,Op}}} & \text{Hecke}_{\check{G}}^{\text{spec,Op}_{\check{G}}^{\text{mon-free}}} & \xrightarrow{\overrightarrow{h}^{\text{spec,Op}}} & \text{Op}_{\check{G}}^{\text{mon-free}} \\
 \downarrow \tau & & \downarrow \tau & & \downarrow \tau \\
 \text{LS}_{\check{G}}^{\text{reg}} & \xleftarrow{\overleftarrow{h}^{\text{spec,loc}}} & \text{Hecke}_{\check{G}}^{\text{spec,loc}} & \xrightarrow{\overrightarrow{h}^{\text{spec,loc}}} & \text{LS}_{\check{G}}^{\text{reg}},
 \end{array}
 \quad (E.28)$$

in which both arrows are Cartesian.

From here, by the same mechanism as in Sect. E.6.4, we obtain an action of the monoidal category $\text{IndCoh}^*(\text{Hecke}_{\check{G}}^{\text{spec,loc}})$ on $\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})$.

We write the action functor symbolically as

$$\begin{aligned}
 \text{IndCoh}^*(\text{Hecke}_{\check{G}}^{\text{spec,loc}}) \otimes \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) &\xrightarrow{(\tau \times \overrightarrow{h}^{\text{spec,Op}})^*, \text{IndCoh}} \\
 &\rightarrow \text{IndCoh}^*(\text{Hecke}_{\check{G}}^{\text{spec,Op}_{\check{G}}^{\text{mon-free}}}) \xrightarrow{(\overleftarrow{h}^{\text{spec,Op}})^{\text{IndCoh}}_*} \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}),
 \end{aligned}$$

where the functor $(\tau \times \overrightarrow{h}^{\text{spec,Op}})^*, \text{IndCoh}$ is assigned a meaning as in Sect. E.7.12.

Remark E.8.2. Note that $\text{Hecke}_{\check{G}}^{\text{spec,Op}_{\check{G}}^{\text{mon-free}}}$ is ind-placid. Indeed, this follows from the fact that $\text{Op}_{\check{G}}^{\text{mon-free}}$ is placid, combined with the fact that the map $\overleftarrow{h}^{\text{spec,loc}}$ is locally almost of finite presentation (see Lemma E.2.4).

E.8.3. Recall the functor

$$\mathrm{Poinc}_{\tilde{G},*,\mathrm{Ran}}^{\mathrm{spec}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G},\mathrm{Ran}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}) \otimes \mathrm{D-mod}(\mathrm{Ran}),$$

see Sect. 17.4.2.

We claim:

Proposition E.8.4. *The functor $\mathrm{Poinc}_{\tilde{G},*,\mathrm{Ran}}^{\mathrm{spec}}$ intertwines the actions of $\mathrm{IndCoh}^*(\mathrm{Hecke}_{\tilde{G}}^{\mathrm{spec,loc}})$ on the two sides.*

Proof. Unwinding the construction, we need to show that the functor

$$\mathrm{Poinc}_{\tilde{G},*,S_\alpha}^{\mathrm{spec}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})_S \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}) \otimes \mathrm{QCoh}(S_\alpha)$$

is compatible with the action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_{\tilde{G}}^{\mathrm{spec,loc}})_{S_\alpha}$ for S_α as in Sect. B.13.9.

Recall that the functor $\mathrm{Poinc}_{\tilde{G},*,S_\alpha}^{\mathrm{spec}}$ is the composition of:

- *-pullback along

$$\mathrm{Op}_{\tilde{G},S_\alpha}^{\mathrm{mon-free,glob}} \xrightarrow{\mathrm{ev}_{S_\alpha}} \mathrm{Op}_{\tilde{G},S_\alpha}^{\mathrm{mon-free}};$$

- IndCoh -pushforward along $\mathrm{Op}_{\tilde{G},S_\alpha}^{\mathrm{mon-free,glob}} \xrightarrow{\mathrm{r}^{\mathrm{glob}}} \mathrm{LS}_{\tilde{G}} \times S_\alpha$.

Denote

$$\mathrm{Hecke}_{\tilde{G},S_\alpha}^{\mathrm{spec,glob,Op}_{\tilde{G}}^{\mathrm{mon-free}}} := \mathrm{Op}_{\tilde{G},S_\alpha}^{\mathrm{mer,glob}} \times_{\mathrm{LS}_{\tilde{G},S_\alpha}^{\mathrm{mer,glob}}} \mathrm{Hecke}_{H,S_\alpha}^{\mathrm{spec,glob}}.$$

The assertion of the proposition holds by unwinding the constructions from the fact that in the following diagrams both square are Cartesian

$$\begin{array}{ccccc} \mathrm{Op}_{\tilde{G},S_\alpha}^{\mathrm{mon-free,glob}} & \xleftarrow{h^{\mathrm{spec,Op}}} & \mathrm{Hecke}_{\tilde{G},S_\alpha}^{\mathrm{spec,glob,Op}_{\tilde{G}}^{\mathrm{mon-free}}} & \xleftarrow{h^{\mathrm{spec,Op}}} & \mathrm{Op}_{\tilde{G},S_\alpha}^{\mathrm{mon-free,glob}} \\ \mathrm{ev} \downarrow & & \mathrm{ev} \downarrow & & \downarrow \mathrm{ev} \\ \mathrm{Op}_{\tilde{G},S_\alpha}^{\mathrm{mon-free}} & \xleftarrow{h^{\mathrm{spec,glob,Op}}} & \mathrm{Hecke}_{\tilde{G},S_\alpha}^{\mathrm{spec,Op}_{\tilde{G}}^{\mathrm{mon-free}}} & \xrightarrow{h^{\mathrm{spec,glob,Op}}} & \mathrm{Op}_{\tilde{G},S_\alpha}^{\mathrm{mon-free}} \end{array}$$

and

$$\begin{array}{ccccc} \mathrm{Op}_{\tilde{G},S_\alpha}^{\mathrm{mon-free,glob}} & \xleftarrow{h^{\mathrm{spec,Op}}} & \mathrm{Hecke}_{\tilde{G},S_\alpha}^{\mathrm{spec,glob,Op}_{\tilde{G}}^{\mathrm{mon-free}}} & \xleftarrow{h^{\mathrm{spec,Op}}} & \mathrm{Op}_{\tilde{G},S_\alpha}^{\mathrm{mon-free,glob}} \\ \mathrm{r}^{\mathrm{glob}} \downarrow & & \mathrm{r}^{\mathrm{glob}} \downarrow & & \downarrow \mathrm{r}^{\mathrm{glob}} \\ \mathrm{LS}_{\tilde{G}} & \xleftarrow{h^{\mathrm{spec,loc}}} & \mathrm{Hecke}_{\tilde{G}}^{\mathrm{spec,glob}} & \xrightarrow{h^{\mathrm{spec,loc}}} & \mathrm{LS}_{\tilde{G}}. \end{array}$$

□

E.8.5. The action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_{\tilde{G}}^{\mathrm{spec,loc}})$ on $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})$ gives rise to a *right* action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_{\tilde{G}}^{\mathrm{spec,loc}})$ on the dual of $\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})$. Identifying

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^\vee \simeq \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}),$$

we thus obtain a right action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_{\tilde{G}}^{\mathrm{spec,loc}})$ on $\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})$.

Applying the anti-involution σ^{spec} , we can turn this right action into a left action.

E.8.6. Passing to dual functors, from the action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_{\check{G}}^{\mathrm{spec}, \mathrm{loc}})$ on $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$, we obtain a coaction of $\mathrm{IndCoh}^!(\mathrm{Hecke}_{\check{G}}^{\mathrm{spec}, \mathrm{loc}})$ on $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$.

By rigidity, the above coaction functor admits a left adjoint, i.e., we obtain an action of $\mathrm{IndCoh}^!(\mathrm{Hecke}_{\check{G}}^{\mathrm{spec}, \mathrm{loc}})$, viewed as a monoidal category, on $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$.

The corresponding action functor is explicitly given by

$$\begin{aligned} \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes \mathrm{IndCoh}^!(\mathrm{Hecke}_{\check{G}}^{\mathrm{spec}, \mathrm{loc}}) &\xrightarrow{(\check{h}^{\mathrm{spec}, \mathrm{Op}} \times \mathrm{r})^*, \mathrm{IndCoh}} \\ &\rightarrow \mathrm{IndCoh}^!(\mathrm{Hecke}_{\check{G}}^{\mathrm{spec}, \mathrm{Op}} \times \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \xrightarrow{(\check{h}^{\mathrm{spec}, \mathrm{Op}})^{\mathrm{IndCoh}}} \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}). \end{aligned}$$

As in Sect. E.7.13, it follows formally that with respect to the identification (E.24), the above action of $\mathrm{IndCoh}^!(\mathrm{Hecke}_{\check{G}}^{\mathrm{spec}, \mathrm{loc}})$ on $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$ identifies with the action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_{\check{G}}^{\mathrm{spec}, \mathrm{loc}})$ on $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$ from Sect. E.8.5.

E.8.7. In a way analogous to Proposition E.8.4, one proves:

Proposition E.8.8. *The functor $\mathrm{Poinc}_{\check{G}, !, \mathrm{Ran}}^{\mathrm{spec}}$ intertwines the coactions of $\mathrm{IndCoh}^!(\mathrm{Hecke}_{\check{G}}^{\mathrm{spec}, \mathrm{loc}})$ on the two sides.*

Corollary E.8.9. *The functor $\mathrm{Poinc}_{\check{G}, !, \mathrm{Ran}}^{\mathrm{spec}}$ intertwines the actions of $\mathrm{IndCoh}^*(\mathrm{Hecke}_{\check{G}}^{\mathrm{spec}, \mathrm{loc}})$ on the two sides, where the action on the left-hand side is as in Sect. E.8.5.*

E.8.10. Note that Lemma 3.7.17 adapts to the factorization setting as follows:

Lemma E.8.11. *The equivalence*

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mon-free}}) \xrightarrow{\Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mon-free}})$$

is compatible with the actions of $\mathrm{IndCoh}^(\mathrm{Hecke}_{\check{G}, \mathrm{Ran}}^{\mathrm{spec}, \mathrm{loc}})$ on the two sides, where: the action on the left-hand side is one from Sect. E.8.5.*

E.8.12. Recall now that according to Theorem 17.4.7, we have a canonical isomorphism:

$$(E.29) \quad \mathrm{Poinc}_{\check{G}, !, \mathrm{Ran}}^{\mathrm{spec}} \otimes \mathbf{k}_{\mathrm{Kost}(\check{G})}[-\delta_G] \simeq \mathrm{Poinc}_{\check{G}, *, \mathrm{Ran}}^{\mathrm{spec}} \circ \Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}}.$$

Unwinding the constructions, we obtain:

Lemma E.8.13. *The commutative diagram*

$$\begin{array}{ccc} \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mon-free}}) & \xrightarrow{\Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \mathrm{Ran}}^{\mathrm{mon-free}}) \\ \mathrm{Poinc}_{\check{G}, !, \mathrm{Ran}}^{\mathrm{spec}} \downarrow & & \downarrow \mathrm{Poinc}_{\check{G}, *, \mathrm{Ran}}^{\mathrm{spec}} \\ \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathrm{Ran}) & \xrightarrow{\mathrm{Id}} & \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathrm{Ran}) \end{array}$$

upgrades to a commutative diagram of categories equipped with actions of $\mathrm{IndCoh}^(\mathrm{Hecke}_{\check{G}, \mathrm{Ran}}^{\mathrm{spec}, \mathrm{loc}})$, where:*

- The compatibility for the left vertical arrow is given by Corollary E.8.9;
- The compatibility for the right vertical arrow is given by Proposition E.8.4;
- The compatibility for the top horizontal arrow with (E.24) is given by Lemma E.8.11;

E.9. An approach to $\mathrm{Sph}_H^{\mathrm{spec}}$ via factorization modules. In this subsection we will review the connection between the definition of $\mathrm{IndCoh}^*(\mathrm{Hecke}_{H, \mathrm{Ran}}^{\mathrm{spec}, \mathrm{loc}})$ developed above, and one given in [CR].

E.9.1. The projection

$$\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}} := \mathrm{LS}_H^{\mathrm{reg}} \times_{\mathrm{LS}_H^{\mathrm{mer}}} \mathrm{LS}_H^{\mathrm{reg}} \xrightarrow{f} \mathrm{LS}_H^{\mathrm{reg}} \times \mathrm{LS}_H^{\mathrm{reg}}$$

gives rise to a lax unital functor

$$\begin{aligned} f_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}) &\rightarrow \mathrm{IndCoh}^*(\mathrm{LS}_H^{\mathrm{reg}} \times \mathrm{LS}_H^{\mathrm{reg}}) \xrightarrow{\Psi_{\mathrm{LS}_H^{\mathrm{reg}} \times \mathrm{LS}_H^{\mathrm{reg}}}} \\ &\simeq \mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}} \times \mathrm{LS}_H^{\mathrm{reg}}) \simeq \mathrm{Rep}_{H \times H} \simeq \mathrm{Rep}_H \otimes \mathrm{Rep}_H, \end{aligned}$$

where the first arrow is obtained by applying the functor of $\mathfrak{L}_\nabla^+(H) \times \mathfrak{L}_\nabla^+(H)$ -coinvariants to

$$\Gamma^{\mathrm{IndCoh}}(\mathfrak{L}_\nabla(H), -) \rightarrow \mathrm{Vect}.$$

E.9.2. Note that the image of the factorization unit

$$\mathbf{1}_{\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})} \simeq \iota_*^{\mathrm{IndCoh}}(\mathcal{O}_{\mathrm{LS}_H^{\mathrm{reg}}})$$

identifies with

$$(\Delta_{\mathrm{LS}_H^{\mathrm{reg}}})_*(\mathcal{O}_{\mathrm{LS}_H^{\mathrm{reg}}}) \in \mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}} \times \mathrm{LS}_H^{\mathrm{reg}}) \Leftrightarrow R_H \in \mathrm{Rep}_{H \times H},$$

where R_H denotes the regular representation, viewed as a commutative factorization algebra in $\mathrm{Rep}_{H \times H}$

By Lemma C.15.3, the functor f_* enhances to a unital functor

$$(E.30) \quad (f_*^{\mathrm{IndCoh}})^{\mathrm{enh}} : \mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}) \rightarrow R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}_H \otimes \mathrm{Rep}_H),$$

where the right-hand side is viewed as a unital lax factorization category.

E.9.3. We claim:

Proposition E.9.4.

(a) The functor (E.30) induces an equivalence between the eventually coconnective subcategories of the two sides.

(b) The essential image of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})^c \subset \mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ under the functor (E.30) is contained in $(R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}_H \otimes \mathrm{Rep}_H))^{>-\infty}$.

Proof. The proof proceeds along the same lines as that of Proposition 4.4.7, with the following difference:

Instead of appealing to Proposition A.3.3, we claim that $\mathrm{QCoh}_{\mathrm{co}}(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ identifies with the totalization of the cosimplicial category

$$\mathrm{QCoh}_{\mathrm{co}}(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}} \times_{\mathrm{LS}_H^{\mathrm{reg}} \times \mathrm{LS}_H^{\mathrm{reg}}} \mathrm{pt}^\bullet),$$

where $\mathrm{pt}^\bullet$ is the Čech nerve of the projection

$$\mathrm{pt} \rightarrow \mathrm{LS}_H^{\mathrm{reg}} \times \mathrm{LS}_H^{\mathrm{reg}},$$

i.e., we claim that the functor

$$(E.31) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}}) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_\nabla(H))^{\mathfrak{L}_\nabla^+(H) \times \mathfrak{L}_\nabla^+(H)}$$

is an equivalence.

Indeed, the precomposition of (E.31) with (E.17) is the functor

$$\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_\nabla(H))^{\mathfrak{L}_\nabla^+(H) \times \mathfrak{L}_\nabla^+(H)} \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_\nabla(H))^{\mathfrak{L}_\nabla^+(H) \times \mathfrak{L}_\nabla^+(H)}$$

of (E.2), which is an equivalence by Corollary E.1.6.

□

E.9.5. Note that we have a commutative diagram

$$\begin{array}{ccc}
\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}}) & \xrightarrow{\iota_*^{\mathrm{IndCoh}}} & \mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}}) \\
\sim \downarrow & & \downarrow (f_*^{\mathrm{IndCoh}})^{\mathrm{enh}} \\
\mathrm{Rep}(H) & & \\
\sim \downarrow & & \\
R_H\text{-mod}^{\mathrm{com}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H)) & \longrightarrow & R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H)).
\end{array}$$

Recall now that $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})^c$ is generated under finite colimits by the essential image of $\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}})$ along $\iota_*^{\mathrm{IndCoh}}$.

Hence, from Proposition E.9.4, we obtain that the essential image of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})^c$ under (E.30) is the full subcategory of $R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H))$ generated under finite colimits by the essential image of $\mathrm{Rep}(H)^c$ under the functor

$$(E.32) \quad \mathrm{Rep}(H) \simeq R_H\text{-mod}^{\mathrm{com}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H)) \rightarrow R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H)).$$

This allows us to recover $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})$ from $R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H))$ by an explicit procedure:

Namely, $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})$ identifies with the ind-completion of the full subcategory of $R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H))$ generated under finite colimits by the essential image of $\mathrm{Rep}(H)^c$ under the functor (E.32).

In the rest of this subsection we will show how to recover various pieces of structure on $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})$ from those on $R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H))$.

E.9.6. Note that $\mathrm{Rep}(H) \otimes \mathrm{Rep}(H)$ is naturally a factorization monoidal category under *convolution*, and R_H is the monoidal unit.

Hence, the monoidal operation

$$(E.33) \quad (\mathrm{Rep}(H) \otimes \mathrm{Rep}(H)) \otimes (\mathrm{Rep}(H) \otimes \mathrm{Rep}(H)) \xrightarrow{*} \mathrm{Rep}(H) \otimes \mathrm{Rep}(H)$$

sends the factorization algebra

$$R_H \otimes R_H \in \mathrm{FactAlg}^{\mathrm{unfl}}(X, (\mathrm{Rep}(H) \otimes \mathrm{Rep}(H)) \otimes (\mathrm{Rep}(H) \otimes \mathrm{Rep}(H)))$$

to

$$R_H \in \mathrm{FactAlg}^{\mathrm{unfl}}(X, \mathrm{Rep}(H) \otimes \mathrm{Rep}(H)).$$

From here we obtain that the functor (E.33) induces a functor

$$(E.34) \quad (R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H))) \otimes (R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H))) \xrightarrow{*} \\ \rightarrow R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H)),$$

which naturally extends to a monoidal structure on $R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H))$.

E.9.7. Unwinding the constructions, we obtain that the functor (E.30) is monoidal.

The monoidal operation (E.34) is t-exact. Hence, the procedure of recovering $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec},\mathrm{loc}})$ from $R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H))$ described in Sect. E.9.5 allows us also to recover its monoidal structure.

E.9.8. Let $\mathbf{A}$ be a (unital) factorization category, equipped with a monoidal action of $\mathrm{Rep}(H) \otimes \mathrm{Rep}(H)$. Let $\mathcal{A}$ be *any* (unital) factorization algebra in $\mathbf{A}$.

The action functor

$$(E.35) \quad (\mathrm{Rep}(H) \otimes \mathrm{Rep}(H)) \otimes \mathbf{A} \xrightarrow{*} \mathbf{A}$$

automatically sends the factorization algebra

$$R_H \otimes \mathcal{A} \in \mathrm{FactAlg}^{(\mathrm{untl})}(X, (\mathrm{Rep}(H) \otimes \mathrm{Rep}(H)) \otimes \mathbf{A})$$

to

$$\mathcal{A} \in \mathrm{FactAlg}^{(\mathrm{untl})}(X, \mathrm{Rep}(H) \otimes \mathrm{Rep}(H)).$$

Hence, we obtain that the functor (E.35) gives rise to a functor

$$(E.36) \quad (R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H))) \otimes (\mathcal{A}\text{-mod}^{\mathrm{fact}}\text{-mod}(\mathbf{A})) \xrightarrow{*} \mathcal{A}\text{-mod}^{\mathrm{fact}}\text{-mod}(\mathbf{A}),$$

which extends to a monoidal action of $R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H))$ on $\mathcal{A}\text{-mod}^{\mathrm{fact}}\text{-mod}(\mathbf{A})$.

E.9.9. Take $\mathbf{A} = \mathrm{Rep}(H)$ and $\mathcal{A} = \mathbf{1}_{\mathrm{Rep}(H)} = \mathcal{O}_{\mathrm{Rep}(H)}$, so that

$$\mathcal{A}\text{-mod}^{\mathrm{fact}}\text{-mod}(\mathbf{A}) = \mathrm{Rep}(H).$$

Hence, we obtain an action of $R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H))$ on $\mathrm{Rep}(H)$.

Unwinding the definitions, we obtain that the functor (E.30) intertwines the above action with the action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ on $\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}})$ from Sect. E.6.5.

Furthermore, as in Sect. E.9.7, this allows us to recover the latter action from the action of $R_H\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(H) \otimes \mathrm{Rep}(H))$ on $\mathrm{Rep}(H)$.

E.9.10. Let us now take $H = \check{G}$. Let us take again $\mathbf{A} = \mathrm{Rep}(\check{G})$, but let us take $\mathcal{A} := R_{\check{G}, \mathrm{Op}}$ from (4.8).

We obtain an action of $R_{\check{G}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G}))$ on $R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}))$.

Unwinding the constructions, we obtain that the functors (E.30) and $(\tau_*^{\mathrm{IndCoh}})^{\mathrm{enh}}$ intertwine the above action with the action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ on $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$ from Sect. E.8.1.

Recall now that according to Proposition 4.4.7 the functor

$$(\tau_*^{\mathrm{IndCoh}})^{\mathrm{enh}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \rightarrow R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}))$$

induces an equivalence between the eventually coconnective subcategories of the two sides and sends compact objects to eventually coconnective ones. Combined with Proposition E.9.4, this allows us to recover the action of $\mathrm{IndCoh}^*(\mathrm{Hecke}_H^{\mathrm{spec}, \mathrm{loc}})$ on $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$ from the above action of $R_{\check{G}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G}))$ on $R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}))$.

E.10. Compatibility of the FLE with (derived) Satake. In this subsection we continue to take $H = \check{G}$. Our goal is to prove Theorem 6.4.5 in the factorization setting.

E.10.1. As a first step, we recall the construction of the geometric equivalence functor $\mathrm{Sat}_{\check{G}}$. Consider the factorization category

$$\mathrm{Whit}^!(G) \otimes \mathrm{Whit}_*(G)$$

and note that it is naturally a bimodule⁸⁵ for the factorization monoidal category $\mathrm{Sph}_{\check{G}}$.

We identify

$$\mathrm{Whit}^!(G) \xrightarrow{\mathrm{CS}_{\check{G}}} \mathrm{Rep}(\check{G}),$$

and we identify

$$\mathrm{Whit}_*(G) \xrightarrow{\mathrm{FLE}_{\check{G}, \infty}} \mathrm{Rep}(\check{G}).$$

⁸⁵Recall that according to Sect. 1.5.4 we freely pass between left and right modules for $\mathrm{Sph}_{\check{G}}$ using the anti-involution σ .

Thus, we obtain that $\text{Rep}(\check{G}) \otimes \text{Rep}(\check{G})$ acquires a bimodule structure with respect to Sph_G . The action on the factorization algebra object $R_{\check{G}} \in \text{Rep}(G) \otimes \text{Rep}(\check{G})$ gives rise to a factorization functor

$$\text{pre-Sat}_G : \text{Sph}_G \rightarrow \text{Rep}(\check{G}) \otimes \text{Rep}(\check{G}).$$

Since the monoidal unit $\delta_{1, \text{Gr}_G} \in \text{Sph}_G$ equals the factorization unit, the functor pre-Sat_G sends $\mathbf{1}_{\text{Sph}_G}$ to $R_{\check{G}}$.

The above operations are compatible with unital structures. Hence, by Lemma C.15.3, the functor pre-Sat_G upgrades to a functor

$$(\text{pre-Sat}_G)^{\text{enh}} : \text{Sph}_G \rightarrow R_{\check{G}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{G})).$$

Unwinding the constructions, we obtain that the above functor $(\text{pre-Sat}_G)^{\text{enh}}$ carries a naturally defined monoidal structure.

E.10.2. We observe:

Lemma E.10.3. *The functor $(\text{pre-Sat}_G)^{\text{enh}}$ sends the subcategory $(\text{Sph}_G)^c \subset \text{Sph}_G$ to the full subcategory of $R_{\check{G}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{G}))$ generated under finite colimits by the essential image of $\text{Rep}(\check{G})^c$ under the functor*

$$R_{\check{G}}\text{-mod}^{\text{com}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{G})) \rightarrow R_{\check{G}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{G})).$$

Thus, combining with Sect. E.9.5, we obtain that the functor $(\text{pre-Sat}_G)^{\text{enh}}$ can be uniquely factored as

$$\text{Sph}_G \rightarrow \text{Sph}_G^{\text{spec}} := \text{IndCoh}^*(\text{Hecke}_G^{\text{spec, loc}}) \xrightarrow{(\text{E.30})} R_{\check{G}}\text{-mod}^{\text{fact}}(\text{Rep}_{\check{G}} \otimes \text{Rep}_{\check{G}}),$$

where the first arrow preserves compactness.

The resulting functor

$$\text{Sph}_G \rightarrow \text{Sph}_G^{\text{spec}}$$

is the functor Sat_G , as it was constructed in [CR].

E.10.4. For the proof of Theorem 6.4.5, we will need the following output of the above construction:

Consider the functor

$$(E.37) \quad \text{Sph}_G \xrightarrow{\text{pre-Sat}_G} \text{Rep}(\check{G}) \otimes \text{Rep}(\check{G}) \xrightarrow{\text{Id} \otimes \text{FLE}_{\check{G}, \infty}} \text{Rep}(\check{G}) \otimes \text{Whit}_*(G).$$

By construction, it has the following properties:

- It intertwines the action of Sph_G on itself by right multiplication with the natural action of Sph_G on $\text{Whit}_*(G)$;
- It intertwines the action of Sph_G on itself by left multiplication with the action of $\text{Sph}_G^{\text{spec}}$ on $\text{Rep}(\check{G})$ via Sat_G .

E.10.5. Let $\mathbf{A}$ be a factorization category, equipped with a factorization action of $\mathfrak{L}(G)_{\rho(\omega_X)}$ at the critical level. We have a naturally defined factorization functor

$$\text{D-mod}_{\text{crit}}(\text{Gr}_{G, \rho(\omega_X)}) \underset{\text{Sph}_G}{\otimes} \text{Sph}(\mathbf{A}) \rightarrow \mathbf{A},$$

which gives rise to a factorization functor

$$(E.38) \quad \text{Whit}_*(G) \underset{\text{Sph}_G}{\otimes} \text{Sph}(\mathbf{A}) \rightarrow \text{Whit}_*(\mathbf{A}).$$

By duality, the functor (E.38) gives rise to a functor

$$(E.39) \quad \text{Sph}(\mathbf{A}) \rightarrow \text{Whit}^!(G) \otimes \text{Whit}_*(\mathbf{A}),$$

compatible with the action of Sph_G .

E.10.6. Composing with CS_G along the first factor, from (E.39) we obtain a functor

$$(E.40) \quad \mathrm{Sph}(\mathbf{A}) \rightarrow \mathrm{Rep}(\check{G}) \otimes \mathrm{Whit}_*(\mathbf{A}).$$

We claim:

Lemma E.10.7. *The functor (E.40) intertwines the action of Sph_G on $\mathrm{Sph}(\mathbf{A})$ with the action of $\mathrm{Sph}_G^{\mathrm{spec}}$ on $\mathrm{Rep}(\check{G})$ via Sat_G .*

Proof. We can interpret the functor (E.40) as follows:

$$(E.41) \quad \begin{aligned} \mathrm{Sph}(\mathbf{A}) &\simeq \mathrm{Sph}_G \otimes_{\mathrm{Sph}_G} \mathrm{Sph}(\mathbf{A}) \xrightarrow{(E.37)} (\mathrm{Rep}(\check{G}) \otimes \mathrm{Whit}_*(G)) \otimes_{\mathrm{Sph}_G} \mathrm{Sph}(\mathbf{A}) \simeq \\ &\simeq \mathrm{Rep}(\check{G}) \otimes (\mathrm{Whit}_*(G) \otimes_{\mathrm{Sph}_G} \mathrm{Sph}(\mathbf{A})) \xrightarrow{\mathrm{Id} \otimes (E.38)} \mathrm{Rep}(\check{G}) \otimes \mathrm{Whit}_*(\mathbf{A}). \end{aligned}$$

Now the assertion follows from the second bullet point in Sect. E.10.4. $\square$

E.10.8. Consider the morphism

$$\mathrm{Op}_G^{\mathrm{mon-free}} \xrightarrow{\tau \times \iota^{\mathrm{mon-free}}} \mathrm{LS}_G^{\mathrm{reg}} \times \mathrm{Op}_G^{\mathrm{mer}},$$

and the corresponding factorization functor

$$(E.42) \quad \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \xrightarrow{(\tau \times \iota^{\mathrm{mon-free}})^{\mathrm{IndCoh}}_*} \mathrm{IndCoh}^*(\mathrm{LS}_G^{\mathrm{reg}} \times \mathrm{Op}_G^{\mathrm{mer}}) \simeq \mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}).$$

Unwinding the construction, we obtain that (E.42) is compatible with the actions of $\mathrm{Sph}_G^{\mathrm{spec}}$ on the two sides, where $\mathrm{Sph}_G^{\mathrm{spec}}$ acts on the right-hand side via the $\mathrm{LS}_G^{\mathrm{reg}}$ -factor.

Denote

$$R_{G, \mathrm{Op}}^{\mathrm{Rep}} := ((\tau \times \iota^{\mathrm{mon-free}})^{\mathrm{IndCoh}}_*)(\mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}}}) \in \mathrm{FactAlg}^{\mathrm{untl}}(X, \mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})).$$

By Lemma C.15.3, the functor (E.42) enhances to a functor

$$(E.43) \quad ((\tau \times \iota^{\mathrm{mon-free}})^{\mathrm{IndCoh}}_*)^{\mathrm{enh}} : \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \rightarrow R_{G, \mathrm{Op}}^{\mathrm{Rep}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})).$$

The functor (E.43) intertwines the $\mathrm{Sph}_G^{\mathrm{spec}}$ -action on the left-hand side and the action of $R_{G, \mathrm{Op}}^{\mathrm{Rep}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G}))$ from Sect. E.9.8 on the right-hand side via the functor (E.30).

E.10.9. We will prove:

Proposition E.10.10.

(a) *The functor (E.43) induces an equivalence between the eventually coconnective subcategories of the two sides.*

(b) *The essential image of the subcategory of compact objects in $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$ under the functor (E.43) is contained in $(R_{G, \mathrm{Op}}^{\mathrm{Rep}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})))^{>-\infty}$.*

The proof will be given in Sect. E.11. Let us accept this proposition temporarily and proceed with the proof of Theorem 6.4.5.

E.10.11. We now launch the proof of Theorem 6.4.5 proper.

Consider the functor

$$(E.44) \quad \mathrm{KL}(G)_{\mathrm{crit}} \xrightarrow{\mathrm{FLE}_{G, \mathrm{crit}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \xrightarrow{(E.42)} \mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}).$$

We consider its enhancement

$$(E.45) \quad \mathrm{KL}(G)_{\mathrm{crit}} \xrightarrow{\mathrm{FLE}_{G, \mathrm{crit}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \xrightarrow{(E.43)} R_{G, \mathrm{Op}}^{\mathrm{Rep}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})).$$

By Propositions E.9.4 and E.10.10, in order to show that the functor $\mathrm{FLE}_{G, \mathrm{crit}}$ intertwines the action of Sph_G on $\mathrm{KL}(G)_{\mathrm{crit}}$ with the action of $\mathrm{Sph}_G^{\mathrm{spec}}$ on $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$, it suffices to show that the

composite functor in (E.45) intertwines the action of Sph_G on the left-hand side with the action of $R_{\check{G}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G}))$ from Sect. E.9.8 on the right-hand side via the functor $(\mathrm{pre}\text{-}\mathrm{Sat}_G)^{\mathrm{enh}}$.

E.10.12. By the construction of the action of $R_{\check{G}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G}))$ on the right-hand side of (E.45) in Sect. E.9.8, it suffices to show that the original functor (E.44) intertwines the action of Sph_G on the left-hand side with the action of $R_{\check{G}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G}))$ on the right-hand side via the functor $(\mathrm{pre}\text{-}\mathrm{Sat}_G)^{\mathrm{enh}}$.

We will show that the functor (E.44) intertwines the action of Sph_G on the left-hand side with the action of $\mathrm{SpC}_{\check{G}}^{\mathrm{spec}}$ on the right-hand side via Sat_G .

E.10.13. We claim that the functor (E.44) identifies canonically with the functor

$$(E.46) \quad \mathrm{KL}(G)_{\mathrm{crit}} \xrightarrow{\alpha_{\rho(\omega_X), \mathrm{taut}}} \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} = \mathrm{Sph}(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \xrightarrow{(E.40)} \\ \rightarrow \mathrm{Rep}(\check{G}) \otimes \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \xrightarrow{\mathrm{Id} \otimes \overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}}} \mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}).$$

Indeed, by construction, both functors are $\mathrm{Rep}(\check{G})$ -linear. Hence, since $\mathrm{Rep}(\check{G})$ is rigid as a monoidal category, it suffices to identify the compositions of (E.44) and (E.46) with the functor

$$(\mathrm{inv}_{\check{G}} \otimes \mathrm{Id}) : \mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}).$$

The composition involving (E.44) becomes

$$\mathrm{KL}(G)_{\mathrm{crit}} \xrightarrow{\mathrm{FLE}_{G, \mathrm{crit}}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \xrightarrow{(t^{\mathrm{mon-free}})^* \mathrm{IndCoh}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}),$$

which is by construction

$$(E.47) \quad \mathrm{KL}(G)_{\mathrm{crit}} \xrightarrow{\alpha_{\rho(\omega_X), \mathrm{taut}}} \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \\ \rightarrow \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \xrightarrow{\overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}).$$

Unwinding, we obtain that the composition involving (E.46) also identifies with (E.47).

E.10.14. Thus, we have to show that the functor (E.46) intertwines the action of Sph_G on the left-hand side with the action of $\mathrm{SpC}_{\check{G}}^{\mathrm{spec}}$ on the right-hand side via Sat_G .

However, this follows from Lemma E.10.7.

E.10.15. It remains to establish the commutativity of (6.25). This is equivalent to the commutativity of the diagram

$$\begin{array}{ccc} \mathrm{KL}(G)_{\mathrm{crit}} & \xrightarrow{\alpha_{\rho(\omega_X), \mathrm{taut}}} \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} & \longrightarrow \mathrm{Whit}^!(G) \otimes \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}) \\ \mathrm{FLE}_{G, \mathrm{crit}} \downarrow & & \downarrow \mathrm{CS}_G \otimes \overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}} \\ \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) & \xrightarrow{(E.42)} & \mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}), \end{array}$$

compatibly with the action of Sph_G on the top row and the action of $\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$ on the bottom row via Sat_G .

However, this amounts to the identification between (E.44) and (E.46) established above.

□[Theorem 6.4.5]

E.11. Proof of Proposition E.10.10.

E.11.1. Consider the functor

$$(E.48) \quad \mathrm{Id} \otimes \Gamma^{\mathrm{IndCoh}}(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}, -) : \mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}) \rightarrow \mathrm{Rep}(\check{G}).$$

It sends $R_{\check{G}, \mathrm{Op}}^{\mathrm{Rep}} \mapsto R_{\check{G}, \mathrm{Op}}$ and hence induces a functor

$$(E.49) \quad R_{\check{G}, \mathrm{Op}}^{\mathrm{Rep}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}})) \rightarrow R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}))$$

The composition of the functor (E.43) with (E.49) is the functor (4.10). Hence, by Proposition 4.4.7, it suffices to show that the functor (E.49) induces an equivalence between the eventually coconnective subcategories of the two sides.

E.11.2. The functor $\mathrm{Id} \otimes \Gamma^{\mathrm{IndCoh}}(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}, -)$ enhances to a functor

$$(E.50) \quad \mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}) \rightarrow \mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}))$$

where $\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}$ is regarded as a constant (commutative) factorization algebra in $\mathrm{Rep}(\check{G})$.

The functor (E.50) sends $R_{\check{G}, \mathrm{Op}}^{\mathrm{Rep}} \in \mathrm{FactAlg}^{\mathrm{untl}}(X, \mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}))$ to $R_{\check{G}, \mathrm{Op}}$, regarded as a unital factorization algebra in $\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}))$ via the homomorphism $\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}} \rightarrow R_{\check{G}, \mathrm{Op}}$, see Sect. C.11.18.

Hence, (E.50) induces a functor

$$(E.51) \quad R_{\check{G}, \mathrm{Op}}^{\mathrm{Rep}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}})) \rightarrow R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}\left(\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}))\right).$$

The functor (E.49) is the composition of (E.51) with the forgetful functor

$$(E.52) \quad \mathrm{oblv}_{\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}} : R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}\left(\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}))\right) \rightarrow R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G})).$$

E.11.3. Note now that by (a relative version of) Corollary 4.4.2, the functor (E.50) induces an equivalence

$$(E.53) \quad (\mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}))^{\geq -\infty} \xrightarrow{\sim} \left(\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}))\right)^{\geq -\infty}.$$

Note also that $R_{\check{G}, \mathrm{Op}} \in \mathrm{FactAlg}^{\mathrm{untl}}(X, \mathrm{Rep}(\check{G}))$ belongs to

$$\mathrm{Rep}(\check{G})^{\heartsuit} \subset \mathrm{Rep}(\check{G})^{\geq -\infty}.$$

Hence, it makes sense to consider the full subcategory

$$\left(R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}\left(\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}))\right)\right)^{\geq -\infty} \subset R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}\left(\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}))\right).$$

It follows formally from the equivalence (E.53) that the functor (E.51) induces an equivalence

$$(E.54) \quad \left(R_{\check{G}, \mathrm{Op}}^{\mathrm{Rep}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}))\right)^{\geq -\infty} \rightarrow \left(R_{\check{G}, \mathrm{Op}}\text{-mod}^{\mathrm{fact}}\left(\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}))\right)\right)^{\geq -\infty}.$$

E.11.4. Hence, it suffices to show that the functor (E.52) induces an equivalence between the eventually coconnective subcategories of the two sides.

However, we claim that the functor (E.52) is itself an equivalence. Indeed, this is a particular case of Lemma C.11.19.

□[Proposition E.10.10]

E.12. Arc spaces of smooth D-schemes. In this subsection, we will prove the following result.

Proposition E.12.1. *Let $\mathcal{Y} \rightarrow X$ be a smooth affine D-scheme over X . Then for any test scheme S and a map $\underline{x} : S \rightarrow \mathrm{Ran}$, the S -scheme $\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\underline{x}}$ is isomorphic to the limit of a sequence of smooth affine S -schemes.*

E.12.2. To prove Proposition E.12.1, it is enough to treat the case for the canonical map $X^I \rightarrow \text{Ran}$. So our goal will be to prove the following:

Proposition E.12.3. *Let $\mathcal{Y} \rightarrow X$ be a smooth D -scheme over X . Then $\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^I}$ is isomorphic to the limit of a sequence of relative smooth affine schemes over X^I .*

We will now describe $\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^I}$ as a limit of affine blow-ups $\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^I}^n$ (see below) for *any* $\mathcal{Y}$. When $\mathcal{Y}$ is smooth, we will show that all $\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^I}^n$ are smooth.

E.12.4. We first give a brief review of the classical theory of affine blow-ups (a.k.a. dilations). From now on, we only work in *classical* algebraic geometry, i.e., schemes means classical schemes, and fiber products of schemes mean non-derived fiber products, etc.

Let S be a smooth scheme and E be an effective Cartier divisor on S . Let Z be a E -regular S -scheme, i.e., the closed subscheme $Z_E := Z \times_S E$ is an effective Cartier divisor on Z . Let V be a closed subscheme of Z_E . The *affine blow-up* of Z with center V (with respect to E) is defined to be

$$\text{Dil}_V^E(Z) := \text{Bl}_V(Z) - \widetilde{Z}_E,$$

where $\text{Bl}_V(Z)$ is the blow-up of Z with center V , and $\widetilde{Z}_E$ is the strict transform of Z_E .

E.12.5. More explicitly, let $\mathcal{O}_S(-E) \subset \mathcal{O}_S$ be the ideal sheaf defining E , and $\mathcal{I} \subset \mathcal{O}_Z$ be the ideal sheaf defining V . Note that $\mathcal{O}_Z(-E) := \mathcal{O}_Z \otimes_{\mathcal{O}_S} \mathcal{O}_S(-E)$ is a subsheaf of $\mathcal{I}$. Consider the inductive colimit

$$\text{Dil}_{\mathcal{I}}^E(\mathcal{O}_Z) := \text{colim}(\mathcal{O}_Z \rightarrow \mathcal{I}(E) \rightarrow \mathcal{I}^2(2E) \rightarrow \cdots).$$

Note that the connecting morphisms are injective because Z is E -regular. This is a quasi-coherent $\mathcal{O}_Z$ -algebra with multiplication defined in the obvious way. More explicitly, if E is locally cut out by a function f of $\mathcal{O}_S$, then $\text{Dil}_{\mathcal{I}}^E(\mathcal{O}_Z)$ is obtained from $\mathcal{O}_Z$ by adding local sections

$$f^{-n} \cdot a_1 \cdots a_n, \quad a_k \in \mathcal{I}.$$

We have

$$\text{Dil}_V^E(Z) \simeq \text{Spec}_Z(\text{Dil}_{\mathcal{I}}^E(\mathcal{O}_Z)).$$

E.12.6. We observe:

Lemma E.12.7. *In the setting of Sect. E.12.4, let $\text{Sch}_{S,E-\text{reg}}$ be the category of E -regular S -schemes. Then $\text{Dil}_V^E(Z)$ represents the functor*

$$\text{Sch}_{S,E-\text{reg}}^{\text{op}} \rightarrow \text{Set}, \quad W \mapsto \{f : W \rightarrow Z \mid f(W_E) \subseteq V\}.$$

Here $f(W_E) \subseteq V$ means the restriction of $f|_{W_E}$ factors through V .

Proof. Follows by unwinding the definitions. □

E.12.8. Let $S^\circ \subset S$ be an open, whose complement has codimension ≥ 2 . Denote $Z^\circ := Z \times_S S^\circ$. Observe:

Lemma E.12.9. *Assume that Z equals the affine closure of Z° . Assume moreover that V is the closure of $V^\circ := V \times_S S^\circ$ in Z . Then $\text{Dil}_V^E(Z)$ equals the affine closure of*

$$\text{Dil}_V^E(Z)^\circ := \text{Dil}_V^E(Z) \times_S S^\circ.$$

E.12.10. We first construct the schemes $\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^I}^n$ for $I = \{1, 2\}$.

Let Δ_X be the diagonal of X^2 and consider the divisor $\Delta_{X,n} := n \cdot \Delta_X$. Note that the connection on $\mathcal{Y}$ relative to X provides a closed subscheme $\Delta_{\mathcal{Y},n}$ of $\mathcal{Y}^2$:

$$(E.55) \quad \Delta_{\mathcal{Y},n} := \mathcal{Y}_{\nabla} \times_{X_{\text{dR}}} \Delta_{X,n},$$

viewed as a closed subscheme of $\mathcal{Y}_{\nabla}^2 \times_{X_{\text{dR}}^2} X^2$.

Set

$$\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^2}^n := \text{Dil}_{\Delta_{\mathcal{Y},n}}^{n \cdot \Delta_X}(\mathcal{Y}^2).$$

We claim that we have a canonical isomorphism

$$\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^2} \simeq \lim_n \mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^2}^n.$$

First, by the universal property of Lemma E.12.7, we have a canonically defined map

$$(E.56) \quad \mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^2} \rightarrow \lim_n \mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^2}^n.$$

Let us show that this map is an isomorphism. Let A be the commutative algebra in left D-modules on X corresponding to $\mathcal{Y}$. Choose a local coordinate t on X , so that X^2 has coordinates t_1, t_2 .

By Sect. E.12.5, the algebra of functions on the right-hand side in (E.56) is the submodule of

$$j_*(A \boxtimes A)$$

that consists of local sections of the form

$$(t_1 - t_2)^{-n} \cdot a, \quad a|_{n \cdot \Delta} \in \ker(A \otimes_{\mathcal{O}_X} A|_{n \cdot \Delta_X} \rightarrow A|_{n \cdot \Delta_X}),$$

where:

- The connection on A allows us to extend it and also $A \otimes_{\mathcal{O}_X} A$ to quasi-coherent sheaves on $n \cdot \Delta_X$, denoted $A \otimes_{\mathcal{O}_X} A|_{n \cdot \Delta_X}$ and $A|_{n \cdot \Delta_X}$, respectively.
- We identify $A \boxtimes A|_{n \cdot \Delta_X} \simeq A \otimes_{\mathcal{O}_X} A|_{n \cdot \Delta_X}$.

However, this description coincides with the description of

$$\mathcal{A}_{X^2} \subset j_*(A \boxtimes A), \quad \mathcal{A} = \text{Fact}(\mathcal{A}).$$

E.12.11. We now construct $\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^I}^n$ for an arbitrary finite set I . Let $\Delta_{X,I}$ be the diagonal divisor in X^I , i.e., the sum of the pairwise diagonals. We define the subscheme

$$\Delta_{\mathcal{Y},I,n} \subset \mathcal{Y}^I$$

over $n \cdot \Delta_{X,I}$ as follows:

Let

$$X^{I,\circ} \xrightarrow{j_{I,\circ}} X^I$$

be the open corresponding to the condition that not more than two coordinates coincide (i.e., we remove diagonals of codimension ≥ 2). Denote

$$\mathcal{Y}^{I,\circ} := \mathcal{Y}^I \times_{X^I} X^{I,\circ}.$$

Formula (E.55) defines a subscheme

$$\Delta_{\mathcal{Y},I,n}^{\circ} \subset \mathcal{Y}^{I,\circ}.$$

We let $\Delta_{\mathcal{Y},I,n}$ be the closure of $\Delta_{\mathcal{Y},I,n}^{\circ}$ in $\mathcal{Y}^I$.

E.12.12. Set

$$\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^I}^n := \mathrm{Dil}_{\Delta_{\mathcal{Y}, I, n}}^{n \cdot \Delta_{X, I}}(\mathcal{Y}^I).$$

By the the universal property of Lemma E.12.7, we have a map

$$(E.57) \quad \mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^I} \rightarrow \lim_n \mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^I}^n.$$

We claim that (E.57) is an isomorphism.

Indeed, the fact that (E.57) is an isomorphism over $X^{I, \circ}$ follows from the case $I = \{1, 2\}$, considered above. We now claim that both sides in (E.57) are affine closures of their respective restrictions to $X^{I, \circ}$.

Indeed, for the right-hand side, this follows from Lemma E.12.9. For the left-hand side, this follows from the fact that the map

$$\mathcal{A}_{X^I} \rightarrow (j_{I, \circ})_* \circ (j_{I, \circ})^*(\mathcal{A}_{X^I})$$

induces an isomorphism on H^0 .

E.12.13. We will now show that if $\mathcal{Y}$ is smooth, then the schemes $\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{X^I}^n$ are smooth. Note that the assertion is invariant under *formal isomorphisms*. This allows us to replace $\mathcal{Y}$ by $\mathbb{A}^k$.

To prove the smoothness, we can assume $k = 1$. Thus, from now on, we will consider the schemes $\mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_{X^I}^n$.

E.12.14. Let $H_I \subset X^I \times X$ be the incidence divisor. Consider the correspondence

$$\begin{array}{ccc} n \cdot H_I & \longrightarrow & X \\ \downarrow & & \\ X^I & & \end{array}$$

For an X -scheme $\mathcal{Y}$, let

$$\mathrm{Jets}_I^n(\mathcal{Y})$$

be the scheme of n -truncated jets into $\mathcal{Y}$, i.e., the restriction of scalars à la Weil of $(n \cdot H_I) \times_X \mathcal{Y}$ along the map $n \cdot H_I \rightarrow X^I$.

E.12.15. Consider the schemes

$$\mathrm{Jets}_I^n(\mathbf{A}^1) \text{ and } \mathrm{Jets}_I^{n-1}((\mathbf{A}^1)_{\omega_X}),$$

where $(\mathbf{A}^1)_{\omega_X}$ is the total space of the line bundle of 1-forms on X .

Both schemes are vector bundles over X^I of ranks $|I| \cdot n$ and $|I| \cdot (n - 1)$, respectively. De Rham differential defines a map

$$(E.58) \quad d_{\mathrm{dR}} : \mathrm{Jets}_I^n(\mathbf{A}^1) \rightarrow \mathrm{Jets}_I^{n-1}((\mathbf{A}^1)_{\omega_X}).$$

Set

$$' \mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_{X^I}^n := \ker(d_{\mathrm{dR}}).$$

Since d_{dR} is fiber-wise surjective as a map of vector bundles, $' \mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_{X^I}^n$ is also a vector bundle. In particular, it is smooth over X^I .

E.12.16. We are going to prove:

Proposition E.12.17. *There exists a canonical isomorphism*

$$' \mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_{X^I}^n \simeq \mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_{X^I}^n.$$

E.13. **Proof of Proposition E.12.17.**

E.13.1. First, we note that when $I = \{*\}$, there is an obvious isomorphism

$$\begin{aligned} ' \mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_X^n &= \ker(\mathrm{Jets}^n(\mathbf{A}^1) \rightarrow \mathrm{Jets}^{n-1}((\mathbf{A}^1)_{\omega_X})) \xleftarrow{\sim} \ker(\mathrm{Jets}(\mathbf{A}^1) \rightarrow \mathrm{Jets}((\mathbf{A}^1)_{\omega_X})) \simeq \\ &\simeq \mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_X = \mathbb{A}^1 \times X. \end{aligned}$$

E.13.2. Note that for a map of finite sets $I' \rightarrow I$, there is a canonically defined map

$$\mathrm{Jets}_I^n(\mathcal{Y}) \rightarrow \mathrm{Jets}_{I'}^n(\mathcal{Y}),$$

covering $X^I \rightarrow X^{I'}$.

In particular, we obtain a map

$$(E.59) \quad {}'\mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_{X^I}^n \rightarrow ({}'\mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_X^n)^I \simeq (\mathbb{A}^1 \times X)^I$$

over X^I . This map is an isomorphism away from the diagonal divisor.

E.13.3. We will show that the map (E.59) lifts to and defines an isomorphism

$$(E.60) \quad {}'\mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_{X^I}^n \rightarrow \mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_{X^I}^n$$

over $X^{I,\circ}$.

This would imply Proposition E.12.17, since both schemes in question are affine closures of their respective restrictions to $X^{I,\circ}$.

E.13.4. In order to prove (E.60), it suffices to consider the case $I = \{1, 2\}$. The following is obtained by a straightforward calculation:

Lemma E.13.5. *The map*

$$(E.61) \quad {}'\mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_{X^2}^n \rightarrow (\mathbb{A}^1 \times X)^2$$

has the following properties:

- (i) *Its restriction to $n \cdot \Delta_X$ maps to $\Delta_{\mathbb{A}^1, n}$;*
- (ii) *Its restriction to $(n+1) \cdot \Delta_X$ does not map to $\Delta_{\mathbb{A}^1, n+1}$.*

E.13.6. From Lemmas E.13.5(i) and (E.12.7), we obtain that (E.61) lifts to a map

$$(E.62) \quad {}'\mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_{X^2}^n \rightarrow \mathfrak{L}_{\nabla}^+(\mathbb{A}^2)_{X^I}^n.$$

It remains to show that (E.62) is an isomorphism.

E.13.7. The map (E.61) respects the vector bundle structures. Hence, so does the map (E.62). It is an isomorphism away from the diagonal, so it remains to show that it is surjective over the diagonal.

E.13.8. It is clear that the image of

$${}'\mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_{X^2}^n|_{\Delta_X} \rightarrow \mathfrak{L}_{\nabla}^+(\mathbb{A}^2)_{X^I}^n|_{\Delta_X} \simeq \mathbb{A}^1 \times \mathbb{A}^1$$

contains the diagonal copy of $\mathbb{A}^1$.

Furthermore, if the entire image *had been* contained in the diagonal copy of $\mathbb{A}^1$, it would have meant that the map (E.62) lifts further to a map

$${}'\mathfrak{L}_{\nabla}^+(\mathbb{A}^1)_{X^2}^n \rightarrow \mathfrak{L}_{\nabla}^+(\mathbb{A}^2)_{X^I}^{n+1}.$$

However, the latter contradicts point (ii) of Lemma E.13.5.

□[Proposition E.12.17]

APPENDIX F. HORIZONTAL SECTIONS OF AFFINE D-SCHEMES

Let $\mathcal{Y}$ be an affine D-prestack over X . Let $\mathrm{Sect}_{\nabla}(X, \mathcal{Y})$ denote the prestack of its horizontal sections, i.e.,

$$\mathrm{Maps}(\mathrm{Spec}(R), \mathrm{Sect}_{\nabla}(X, \mathcal{Y})) := \mathrm{Maps}_{X, \nabla}(\mathrm{Spec}(R) \times X, \mathcal{Y}).$$

In this section we will describe, following [BD2], $\mathrm{Sect}_{\nabla}(X, \mathcal{Y})$ explicitly in terms of (vacuum) factorization homology of the algebra of functions $\mathcal{O}_{\mathcal{Y}}$ of $\mathcal{Y}$. We also describe spaces of sections of quasi-coherent sheaves on $\mathrm{Sect}_{\nabla}(X, \mathcal{Y})$, in terms of factorization homology of $\mathcal{O}_{\mathcal{Y}}$ with coefficients in corresponding modules.

We then generalize this discussion, when instead of $\mathrm{Sect}_{\nabla}(X, \mathcal{Y})$ we consider the space $\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y})$ of sections, where we allow punctures at $\underline{x} \subset X$.

F.1. Horizontal sections via factorization homology.

F.1.1. We start with an arbitrary D-prestack $\mathcal{Y} \rightarrow X$ (later on in this section, we will assume that $\mathcal{Y}$ is affine over X).

Let $A \in \text{ComAlg}(\text{D-mod}(X))$ be the direct image of the structure sheaf of $\mathcal{Y}$, and let $\mathcal{A}$ be the corresponding object in $\text{ComAlg}(\text{FactAlg}(X))$, i.e., $\mathcal{A} = \text{Fact}(A)$.

Consider the evaluation map

$$\text{Sect}_{\nabla}(X, \mathcal{Y}) \times X \rightarrow \mathcal{Y}.$$

It gives rise to a map

$$(F.1) \quad A \rightarrow \mathcal{O}_{\text{Sect}_{\nabla}(X, \mathcal{Y})} \otimes \mathcal{O}_X$$

in $\text{ComAlg}(\text{D-mod}(X))$, where by a slight abuse of notation we denote by $\mathcal{O}_{\text{Sect}_{\nabla}(X, \mathcal{Y})}$ the algebra of global functions on $\text{Sect}_{\nabla}(X, \mathcal{Y})$.

By the adjunction of Corollary C.8.9, from (F.1) we obtain a map

$$(F.2) \quad \mathcal{A} \rightarrow \mathcal{O}_{\text{Sect}_{\nabla}(X, \mathcal{Y})} \otimes \omega_{\text{Ran}^{\text{untl}, *}}$$

in $\text{ComAlg}(\text{D-mod}(\text{Ran}^{\text{untl}, *}))$, and further, by Sect. C.9.2, a map

$$(F.3) \quad C^{\text{fact}}(X, \mathcal{A}) \rightarrow \mathcal{O}_{\text{Sect}_{\nabla}(X, \mathcal{Y})}$$

in $\text{ComAlg}(\text{Vect})$.

F.1.2. Suppose that the prestack $\mathcal{Y}$ satisfies the assumption of Sect. C.8.10 (e.g., $\mathcal{Y} \rightarrow X$ is affine). In this case, the map (F.2) can also be interpreted as follows:

Consider the map

$$\text{Sect}_{\nabla}(X, \mathcal{Y}) \times \text{Ran} \xrightarrow{\text{ev}} \mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\text{Ran}}.$$

Pullback at the level of functions defines a map

$$(F.4) \quad \mathcal{A} \xrightarrow{(C.44)} \mathcal{O}_{\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\text{Ran}}} \rightarrow \mathcal{O}_{\text{Sect}_{\nabla}(X, \mathcal{Y})} \otimes \omega_{\text{Ran}}$$

in $\text{ComAlg}(\text{D-mod}(\text{Ran}))$.

It follows by unwinding the definition that the map (F.2) is the same as (F.4).

F.1.3. Suppose now that $\mathcal{Y} \rightarrow X$ is affine. In this case we claim:

Proposition F.1.4. *The prestack $\text{Sect}_{\nabla}(X, \mathcal{Y})$ is an affine scheme, and the map (F.3) is an isomorphism.*

Proof. The key fact is that for any factorization algebra $\mathcal{A}$ such that $\mathbf{oblv}^l(\mathcal{A}_X) \in \text{QCoh}(X)^{\leq 0}$, we have

$$C^{\text{fact}}(X, \mathcal{A}) \in \text{Vect}^{\leq 0}.$$

This follows from the fact that

$$C^{\text{fact}}(X, \mathcal{A}) \simeq C_c(\text{Ran}, \mathcal{A}_{\text{Ran}})$$

can be written as a colimit with terms

$$C(X^I, \mathcal{A}_{X^I})$$

for (non-empty) finite sets I , while each $\mathcal{A}_{X^I}$ satisfies $\mathbf{oblv}^l(\mathcal{A}_{X^I}) \in \text{QCoh}(X^I)^{\leq 0}$ (see [BD2, Sect. 3.4.11]) and hence $C(X^I, \mathcal{A}_{X^I}) \in \text{Vect}^{\leq 0}$.

In particular, in our case

$$C^{\text{fact}}(X, \mathcal{A}) \in \text{ComAlg}(\text{Vect}^{\leq 0}).$$

Now, the assertion follows immediately from Corollary C.9.5: for $R \in \text{ComAlg}(\text{Vect}^{\leq 0})$ we have

$$(F.5) \quad \begin{aligned} \text{Maps}(\text{Spec}(R), \text{Sect}_{\nabla}(X, \mathcal{Y})) &= \text{Maps}_{X, \nabla}(\text{Spec}(R) \times X, \mathcal{Y}) \xrightarrow{\text{affineness of } \mathcal{Y}} \\ &\simeq \text{Maps}_{\text{ComAlg}(\text{D-mod}(X))}(A, R \otimes \mathcal{O}_X) \simeq \text{Maps}_{\text{ComAlg}(\text{Vect})}(C^{\text{fact}}(X, \mathcal{A}), R), \end{aligned}$$

so $\text{Sect}_\nabla(X, \mathcal{Y})$ is the affine scheme $\text{Spec}(C^{\text{fact}}(X, \mathcal{A}))$, and by construction the map (F.3) is the map

$$C^{\text{fact}}(X, \mathcal{A}) \rightarrow \Gamma(\text{Spec}(C^{\text{fact}}(X, \mathcal{A})), \mathcal{O}_{\text{Spec}(C^{\text{fact}}(X, \mathcal{A}))})$$

resulting from (F.5).

□

F.1.5. Recall now that the functor

$$\Gamma(\mathfrak{L}_\nabla^+(\mathcal{Y})_{\underline{x}}, -) : \text{QCoh}(\mathfrak{L}_\nabla^+(\mathcal{Y})_{\underline{x}}) \rightarrow \text{Vect}$$

factors via an equivalence

$$\Gamma(\mathfrak{L}_\nabla^+(\mathcal{Y})_{\underline{x}}, -)^{\text{enh}} : \text{QCoh}(\mathfrak{L}_\nabla^+(\mathcal{Y})_{\underline{x}}) \rightarrow \mathcal{A}_{\underline{x}}\text{-mod} \simeq \mathcal{A}\text{-mod}_x^{\text{com}},$$

followed by the forgetful functor

$$\text{oblv}_{\mathcal{A}, \underline{x}} : \mathcal{A}\text{-mod}_x^{\text{com}} \rightarrow \text{Vect}.$$

We claim:

Proposition F.1.6. *There exists a canonical isomorphism between*

$$(F.6) \quad \text{QCoh}(\mathfrak{L}_\nabla^+(\mathcal{Y})_{\underline{x}}) \xrightarrow{\text{ev}_x^*} \text{QCoh}(\text{Sect}_\nabla(X, \mathcal{Y})) \xrightarrow{\Gamma(\text{Sect}_\nabla(X, \mathcal{Y}), -)} \text{Vect}$$

and

$$(F.7) \quad \text{QCoh}(\mathfrak{L}_\nabla^+(\mathcal{Y})_{\underline{x}}) \xrightarrow{\Gamma(\mathfrak{L}_\nabla^+(\mathcal{Y})_{\underline{x}}, -)^{\text{enh}}} \mathcal{A}\text{-mod}_x^{\text{com}} \rightarrow \mathcal{A}\text{-mod}_x^{\text{fact}} \xrightarrow{C^{\text{fact}}(X, \mathcal{A}, -)_{\underline{x}}} \text{Vect}.$$

Proof. Since

$$\text{QCoh}(\mathfrak{L}_\nabla^+(\mathcal{Y})_{\underline{x}}) \simeq \mathcal{A}_{\underline{x}}\text{-mod} \simeq \mathcal{A}\text{-mod}_x^{\text{com}},$$

the assertion of the proposition amounts to the following:

We have a canonical identification

$$\Gamma(\text{Sect}_\nabla(X, \mathcal{Y}), \mathcal{O}_{\text{Sect}_\nabla(X, \mathcal{Y})}) \simeq C^{\text{fact}}(X, \mathcal{A}, \mathcal{A}_{\underline{x}})_{\underline{x}}$$

as $\mathcal{A}_{\underline{x}}$ -modules, where:

- $\mathcal{A}_{\underline{x}}$ acts on the left-hand side via

$$\mathcal{A}_x \xrightarrow{\sim} \text{End}(\mathcal{O}_{\mathfrak{L}_\nabla^+(\mathcal{Y})_{\underline{x}}}) \xrightarrow{\text{ev}_x^*} \text{End}(\text{ev}_x^*(\mathcal{O}_{\mathfrak{L}_\nabla^+(\mathcal{Y})_{\underline{x}}})) \simeq \text{End}(\mathcal{O}_{\text{Sect}_\nabla(X, \mathcal{Y})});$$

- $\mathcal{A}_{\underline{x}}$ acts on the right-hand side via

$$\mathcal{A}_x \xrightarrow{\sim} \text{End}_{\mathcal{A}\text{-mod}_x^{\text{com}}}(\mathcal{A}_x) \rightarrow \text{End}_{\mathcal{A}\text{-mod}_x^{\text{fact}}}(\mathcal{A}_{\underline{x}}).$$

However, this follows from the fact that the isomorphism

$$\Gamma(\text{Sect}_\nabla(X, \mathcal{Y}), \mathcal{O}_{\text{Sect}_\nabla(X, \mathcal{Y})}) \xrightarrow{(F.3)} C^{\text{fact}}(X, \mathcal{A}) \xrightarrow{\text{unitality}} C^{\text{fact}}(X, \mathcal{A}, \mathcal{A}_{\underline{x}})_{\underline{x}}$$

(as commutative algebras) is, by construction, compatible with the homomorphisms

$$\mathcal{A}_{\underline{x}} \simeq \Gamma(\mathfrak{L}_\nabla^+(\mathcal{Y})_{\underline{x}}, \mathcal{O}_{\mathfrak{L}_\nabla^+(\mathcal{Y})_{\underline{x}}}) \rightarrow \Gamma(\text{Sect}_\nabla(X, \mathcal{Y}), \mathcal{O}_{\text{Sect}_\nabla(X, \mathcal{Y})})$$

and

$$\mathcal{A}_{\underline{x}} \rightarrow C_c(\text{Ran}_{\underline{x}}^{\text{untl}}, \mathcal{A}_{\text{Ran}_{\underline{x}}^{\text{untl}}}) = C^{\text{fact}}(X, \mathcal{A}, \mathcal{A}_{\underline{x}})_{\underline{x}}.$$

□

Remark F.1.7. Recall that Remark 15.6.15 says that for a commutative factorization algebra $\mathcal{A}$, we have a canonical identification

$$\mathcal{A}\text{-mod}_{\text{Ran}, \text{indep}}^{\text{com}} \simeq \text{C}(X, \mathcal{A})\text{-mod}.$$

Hence, by Proposition F.1.4 we obtain:

$$\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}))_{\text{Ran}, \text{indep}} \simeq \text{QCoh}(\text{Sect}_{\nabla}(X, \mathcal{Y})).$$

We can interpret Proposition F.1.6 as saying that under the above equivalence, the corresponding functor

$$\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y})) \rightarrow \text{QCoh}(\text{Sect}_{\nabla}(X, \mathcal{Y})) \otimes \underline{\text{D-mod}}(\text{Ran}^{\text{unfl}})$$

sends $\underline{x} \in \text{Ran}$ to the functor

$$\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\underline{x}}) \xrightarrow{\text{ev}_{\underline{x}}^*} \text{QCoh}(\text{Sect}_{\nabla}(X, \mathcal{Y})).$$

Remark F.1.8. The above remark applies to D-prestacks that are “as good as affine”, see Sect. 12.7. For example, for a unipotent group-scheme N' over X and $\mathcal{Y} = \text{pt} / \mathfrak{L}^+(N')$, we obtain an equivalence

$$(F.8) \quad \text{Rep}(\text{pt} / \mathfrak{L}^+(N'))_{\text{Ran}, \text{indep}} \simeq \text{QCoh}(\text{Bun}_{N'}),$$

where the composite functor

$$\text{Rep}(\text{pt} / \mathfrak{L}^+(N'))_{\text{Ran}} \twoheadrightarrow \text{Rep}(\text{pt} / \mathfrak{L}^+(N'))_{\text{Ran}, \text{indep}} \simeq \text{QCoh}(\text{Bun}_{N'})$$

is $\text{Loc}_{N'}^{\text{QCoh}}$.

Similarly, if N' is equipped with a connection (e.g., N' is constant), we have

$$(F.9) \quad \text{Rep}(N')_{\text{Ran}, \text{indep}} \simeq \text{QCoh}(\text{LS}_{N'}),$$

where the composite functor

$$\text{Rep}(N')_{\text{Ran}} \twoheadrightarrow \text{Rep}(N')_{\text{Ran}, \text{indep}} \simeq \text{QCoh}(\text{LS}_{N'})$$

is $\text{Loc}_{N'}^{\text{spec}}$.

Remark F.1.9. We warn the reader that the equivalences (F.8) and (F.9) are a feature of unipotent group schemes. For a reductive G (e.g., for $G = \mathbb{G}_m$), the corresponding functors are far from being equivalences.

F.2. Variant: allowing poles. Let A and $\mathcal{Y}$ be as in the previous subsection. For $\underline{x} \in \text{Ran}$, consider the prestack

$$\text{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}).$$

We will now give an explicit description of $\text{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y})$ in terms of factorization homology. In particular, we will show that it is an ind-affine ind-scheme.

F.2.1. Write

$$\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}} \simeq \text{“colim”}_R \text{Spec}(R'), \quad R' \in \text{ComAlg}(\text{Vect}^{\leq 0}),$$

where each

$$(F.10) \quad \text{Spec}(R') \rightarrow \mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}$$

is a closed embedding.

Consider the fiber product

$$\text{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y})' := \text{Spec}(R') \times_{\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}} \text{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}).$$

We will show that $\text{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y})'$ is a scheme and describe the algebra of functions on it in terms of factorization homology.

F.2.2. First, we note that as in Sect. D.2.8 the datum of a closed embedding

$$(F.11) \quad \mathrm{Spec}(R') \rightarrow \mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}$$

is equivalent to the datum of a modification A' of A at $\underline{x}$, i.e., A' is an object of $\mathrm{ComAlg}(\mathrm{D-mod}(X)^{\leq 0})$ equipped with an isomorphism

$$A'|_{X-\underline{x}} \simeq A|_{X-\underline{x}}.$$

Indeed, given A' , set

$$(F.12) \quad \mathcal{Y}' := \mathrm{Spec}_X(A'),$$

and we recover R' as $\mathcal{A}'_{\underline{x}}$, where $\mathcal{A}' := \mathrm{Fact}(A')$, and (F.11) as

$$\mathrm{Spec}(R') \simeq \mathfrak{L}_{\nabla}^+(\mathcal{Y}') \hookrightarrow \mathfrak{L}_{\nabla}(\mathcal{Y}') \simeq \mathfrak{L}_{\nabla}(\mathcal{Y}').$$

F.2.3. Vice versa, given (F.11), we interpret it as a map of commutative D-algebras

$$j_* \circ j^*(A) \rightarrow R'((t))$$

(here t is a coordinate on the multidisc $\mathcal{D}_{\underline{x}}$ and j is the open embedding $X - \underline{x} \hookrightarrow X$), where the condition that (F.11) is a closed embedding corresponds to the condition that the map of D-modules

$$(F.13) \quad j_* \circ j^*(A) \rightarrow R'((t)) \rightarrow R'((t))/R'[[t]]$$

is surjective on H^0 .

We recover A' as the fiber product

$$j_* \circ j^*(A) \times_{R'((t))} R[[t]],$$

and the surjectivity of (F.13) is equivalent to the condition that A' is connective.

F.2.4. Let R' be as in (F.11), and let A' be the corresponding modification of A . Unwinding the definitions, we obtain:

$$\mathrm{Spec}(R') \times_{\mathfrak{L}(\mathcal{Y})_{\underline{x}}} \mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}) \simeq \mathrm{Sect}_{\nabla}(X - x, \mathcal{Y}').$$

Hence, by Proposition F.1.4, we obtain:

Corollary F.2.5. *The prestack $\mathrm{Spec}(R') \times_{\mathfrak{L}(\mathcal{Y})_{\underline{x}}} \mathrm{Sect}_{\nabla}(X - x, \mathcal{Y})$ is affine and the naturally defined map*

$$C^{\mathrm{fact}}(X, \mathcal{A}') \rightarrow \mathcal{O}_{\mathrm{Spec}(R')} \times_{\mathfrak{L}(\mathcal{Y})_{\underline{x}}} \mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y})$$

is an isomorphism, where $\mathcal{A}' := \mathrm{Fact}(A')$.

F.2.6. We will now rewrite $C^{\mathrm{fact}}(X, \mathcal{A}')$ in terms of factorization homology of $\mathcal{A}$ itself.

We can view R' as an object of $\mathcal{A}\text{-mod}_{\underline{x}}^{\mathrm{fact}}$ via

$$R' \in \mathcal{A}'\text{-mod}_{\underline{x}}^{\mathrm{com}} \rightarrow \mathcal{A}'\text{-mod}_{\underline{x}}^{\mathrm{fact}} \simeq \mathcal{A}\text{-mod}_{\underline{x}}^{\mathrm{fact}}.$$

When viewed as such, we will denote it by $R'_{\mathcal{A}}$. Let $R'_{\mathcal{A}, \mathrm{Ran}_{\underline{x}}^{\mathrm{untl}}}$ be the corresponding object of $\mathrm{D-mod}(\mathrm{Ran}_{\underline{x}}^{\mathrm{untl}})$.

Recall (see Sect. B.10.7) that since $\mathcal{A}$ is a *commutative* factorization algebra, the category $\mathcal{A}\text{-mod}_{\underline{x}}^{\mathrm{fact}}$ has a natural symmetric pseudo-monoidal structure. The commutative algebra structure on R' endows $R'_{\mathcal{A}}$ with a structure of commutative algebra object in $\mathcal{A}\text{-mod}_{\underline{x}}^{\mathrm{fact}}$. In particular, $R'_{\mathcal{A}, \mathrm{Ran}_{\underline{x}}^{\mathrm{untl}}}$ acquires a structure of commutative algebra object in $\mathrm{D-mod}(\mathrm{Ran}_{\underline{x}}^{\mathrm{untl}})$.

F.2.7. The assertion of Lemma C.5.9 is valid for $\text{Ran}_{\underline{x}}^{\text{unl}}$, hence

$$C^{\text{fact}}(X, \mathcal{A}, R'_{\mathcal{A}})_{\underline{x}} := C_c(\text{Ran}_{\underline{x}}^{\text{unl}}, R'_{\mathcal{A}, \text{Ran}_{\underline{x}}^{\text{unl}}})$$

acquires a structure of commutative algebra in Vect .

We claim:

Lemma F.2.8. *There is a canonical isomorphism*

$$(F.14) \quad C^{\text{fact}}(X, \mathcal{A}, R'_{\mathcal{A}})_{\underline{x}} \simeq C^{\text{fact}}(X, \mathcal{A}')$$

as objects of $\text{ComAlg}(\text{Vect})$.

Proof. Note that with respect to the (symmetric monoidal) equivalence

$$\mathcal{A}\text{-mod}_{\underline{x}}^{\text{fact}} \simeq \mathcal{A}'\text{-mod}_{\underline{x}}^{\text{fact}},$$

the object

$$R'_{\mathcal{A}} \in \mathcal{A}\text{-mod}_{\underline{x}}^{\text{fact}}$$

corresponds to the vacuum object

$$(\mathcal{A}')^{\text{fact}_{\underline{x}}} \in \mathcal{A}'\text{-mod}_{\underline{x}}^{\text{fact}},$$

where we recall that $(\mathcal{A}')^{\text{fact}_{\underline{x}}}$ denotes the vacuum factorization module at $\underline{x}$.

Moreover, this isomorphism respects the structure of commutative algebra object on both sides. In particular,

$$R'_{\mathcal{A}, \text{Ran}_{\underline{x}}^{\text{unl}}} \simeq \mathcal{A}'_{\text{Ran}_{\underline{x}}^{\text{unl}}}$$

as objects of $\text{ComAlg}(\text{D-mod}(\text{Ran}_{\underline{x}}^{\text{unl}}))$.

Now, (F.14) follows by concatenating

$$C^{\text{fact}}(X, \mathcal{A}, R'_{\mathcal{A}})_{\underline{x}} \simeq C^{\text{fact}}(X, \mathcal{A}', (\mathcal{A}')^{\text{fact}_{\underline{x}}})_{\underline{x}},$$

and

$$C^{\text{fact}}(X, \mathcal{A}', (\mathcal{A}')^{\text{fact}_{\underline{x}}})_{\underline{x}} \xrightarrow{\text{unitality}} C^{\text{fact}}(X, \mathcal{A}').$$

□

F.2.9. Thus, $\text{Spec}(C^{\text{fact}}(X, \mathcal{A}, R'_{\mathcal{A}})_{\underline{x}})$ gives the desired expression for

$$\text{Spec}(R') \times_{\mathcal{L}(\mathcal{Y})_{\underline{x}}} \text{Sect}_{\nabla}(X - x, \mathcal{Y})$$

in terms of factorization homology of $\mathcal{A}$.

F.2.10. Let

$$R'_{\mathcal{A}, \text{Ran}_{\underline{x}}} \in \text{ComAlg}(\text{D-mod}(\text{Ran}_{\underline{x}}))$$

be the restriction of $R'_{\mathcal{A}, \text{Ran}_{\underline{x}}^{\text{unl}}}$ to the non-unital Ran space.

By construction, it belongs to $\text{ComAlg}(\text{D-mod}(\text{Ran}_{\underline{x}})^{\text{almost-unl}})$. Hence, by Sect. C.5.15,

$$C_c(\text{Ran}_{\underline{x}}, R'_{\mathcal{A}, \text{Ran}_{\underline{x}}})$$

acquires a commutative algebra structure.

By Lemma C.5.12 we have an isomorphism

$$C_c(\text{Ran}_{\underline{x}}, R'_{\mathcal{A}, \text{Ran}_{\underline{x}}}) \simeq C_c(\text{Ran}_{\underline{x}}^{\text{unl}}, R'_{\mathcal{A}, \text{Ran}_{\underline{x}}^{\text{unl}}}) = C^{\text{fact}}(X, \mathcal{A}, R')_{\underline{x}}$$

as commutative algebras.

F.3. Factorization with poles. In this subsection we will add a slightly different spin to the discussion in Sect. F.2.

F.3.1. As in [Bogd, Definition A.2.1], let $\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{Ran} \subseteq}$ be the factorization ind-scheme over $\text{Ran} \subseteq$ that attaches to

$$(\underline{x} \subseteq \underline{x}')$$

the space

$$\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\underline{x} \subseteq \underline{x}'} := \text{Sect}_{\nabla}(\mathcal{D}_{\underline{x}'} - \underline{x}, \mathcal{Y}).$$

We have the projections

$$(F.15) \quad \mathfrak{L}_{\nabla}(\mathcal{Y}) \xleftarrow{\text{pr}_{\text{small}}^{\mathcal{Y}}} \mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{Ran} \subseteq} \xrightarrow{\text{pr}_{\text{big}}^{\mathcal{Y}}} \mathfrak{L}_{\nabla}(\mathcal{Y})$$

given by restrictions along

$$\mathcal{D}_{\underline{x}} - \underline{x} \hookrightarrow \mathcal{D}_{\underline{x}'} - \underline{x} \hookleftarrow \mathcal{D}_{\underline{x}'} - \underline{x}',$$

respectively.

F.3.2. *Example.* When

$$\underline{x}' = \underline{x} \sqcup \underline{x}'',$$

we have

$$\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\underline{x} \subseteq \underline{x}'} \simeq \mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}} \times \mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\underline{x}''}.$$

F.3.3. Fix $\underline{x}$, and consider the space

$$\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{Ran} \subseteq} := \mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{Ran} \subseteq} \times_{\text{Ran} \subseteq} \text{Ran}_{\underline{x}}.$$

It has a natural structure of factorization module space with respect to $\mathfrak{L}_{\nabla}^+(\mathcal{Y})$, with the underlying space being $\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}$.

Moreover, viewed as such, $\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{Ran} \subseteq}$ has a natural *counital* structure (see Sect. C.6.7 for what this means).

In particular, we have the map

$$\text{pr}_{\text{small}, \underline{x}}^{\mathcal{Y}} : \mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{Ran} \subseteq} \rightarrow \mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}.$$

F.3.4. Fix a closed embedding $\text{Spec}(R') \rightarrow \mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}$, and consider the fiber product

$$\text{Spec}(R') \times_{\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}} \mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{Ran} \subseteq}.$$

Unwinding the definitions, we obtain:

Lemma F.3.5. *We have a canonical isomorphism*

$$\text{Spec}(R') \times_{\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}} \mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{Ran} \subseteq} \simeq \mathfrak{L}_{\nabla}^+(\mathcal{Y}')_{\text{Ran} \subseteq},$$

where $\mathcal{Y}'$ is as in (F.12).

F.4. Sections of quasi-coherent sheaves, meromorphic version.

F.4.1. Recall (see Sect. 4.3.7) that the functor

$$\Gamma(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}, -) : \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}) \rightarrow \text{Vect}$$

admits an enhancement to a functor

$$\Gamma(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}, -)^{\text{enh}} : \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}) \rightarrow \mathcal{A}\text{-mod}_{\underline{x}}^{\text{fact}}.$$

F.4.2. Note also that since the morphism

$$\text{ev}_{\underline{x}} : \text{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}) \rightarrow \mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}$$

is schematic, we have a well-defined functor

$$\text{ev}_{\underline{x}}^* : \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}) \rightarrow \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y})).$$

F.4.3. The goal of this subsection is to prove the following:

Proposition F.4.4. *There exists a canonical isomorphism between*

$$(F.16) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}) \xrightarrow{\mathrm{ev}_{\underline{x}}^*} \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}))^{\Gamma(\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}), -)} \mathrm{Vect}$$

and

$$(F.17) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}})^{\Gamma(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}, -)^{\mathrm{enh}}} \mathcal{A}\text{-mod}_{\underline{x}}^{\mathrm{fact}} \xrightarrow{C^{\mathrm{fact}}(X, \mathcal{A}, -)_{\underline{x}}} \mathrm{Vect}.$$

The rest of this subsection is devoted to the proof of Proposition F.4.4.

F.4.5. First, we construct a natural transformation

$$(F.17) \rightarrow (F.16).$$

Note that the functor

$$\mathrm{ins.unit}_{\underline{x}} : \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\mathrm{Ran}_{\underline{x}}})$$

is given by

$$(\mathrm{pr}_{\mathrm{big}, \underline{x}}^{\mathcal{Y}})^* \circ (\mathrm{pr}_{\mathrm{small}, \underline{x}}^{\mathcal{Y}})^*,$$

where

$$\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}} \xleftarrow{\mathrm{pr}_{\mathrm{small}, \underline{x}}^{\mathcal{Y}}} \mathfrak{L}_{\nabla}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}(\mathcal{Y})_{\mathrm{Ran}_{\underline{x}}} \xrightarrow{\mathrm{pr}_{\mathrm{big}, \underline{x}}^{\mathcal{Y}}} \mathfrak{L}_{\nabla}(\mathcal{Y})_{\mathrm{Ran}_{\underline{x}}}.$$

Hence, we obtain that the functor

$$\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}})^{\Gamma(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}, -)^{\mathrm{enh}}} \mathcal{A}\text{-mod}_{\underline{x}}^{\mathrm{fact}} \rightarrow \mathrm{D-mod}(\mathrm{Ran}_{\underline{x}})$$

is given by

$$(p_{\mathrm{Ran}_{\underline{x}}})^* \circ (\mathrm{pr}_{\mathrm{small}, \underline{x}}^{\mathcal{Y}})^*,$$

where

$$p : \mathfrak{L}_{\nabla}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}(\mathcal{Y})_{\mathrm{Ran}_{\underline{x}}} \rightarrow \mathrm{Ran}_{\underline{x}}.$$

From here we obtain a natural transformation from

$$(F.18) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}})^{\Gamma(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}, -)^{\mathrm{enh}}} \mathcal{A}\text{-mod}_{\underline{x}}^{\mathrm{fact}} \xrightarrow{\mathrm{ins.vac}_{\underline{x}}} \mathcal{A}\text{-mod}_{\mathrm{Ran}_{\underline{x}}}^{\mathrm{fact}} \xrightarrow{\mathrm{oblv}_{\mathcal{A}, \mathrm{Ran}_{\underline{x}}}} \mathrm{D-mod}(\mathrm{Ran}_{\underline{x}})$$

to

$$(F.19) \quad \begin{aligned} & \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}})^{(\mathrm{pr}_{\mathrm{small}, \underline{x}}^{\mathcal{Y}})^*} \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}(\mathcal{Y})_{\mathrm{Ran}_{\underline{x}}})^{(\mathrm{ev}_{\mathrm{Ran}_{\underline{x}}})^*} \\ & \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y})) \otimes \mathrm{D-mod}(\mathrm{Ran}_{\underline{x}})^{\Gamma(\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}), -) \otimes \mathrm{Id}} \mathrm{D-mod}(\mathrm{Ran}_{\underline{x}}). \end{aligned}$$

F.4.6. Note, however, that the map

$$\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}) \times \mathrm{Ran}_{\underline{x}} \xrightarrow{\mathrm{ev}_{\mathrm{Ran}_{\underline{x}}}} \mathfrak{L}_{\nabla}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}(\mathcal{Y})_{\mathrm{Ran}_{\underline{x}}} \xrightarrow{\mathrm{pr}_{\mathrm{small}, \underline{x}}^{\mathcal{Y}}} \mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}$$

equals

$$\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}) \times \mathrm{Ran}_{\underline{x}} \rightarrow \mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}) \xrightarrow{\mathrm{ev}_{\underline{x}}} \mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}.$$

Hence, we obtain that (F.19) can be rewritten as

$$(F.20) \quad \begin{aligned} & \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}) \xrightarrow{\mathrm{ev}_{\underline{x}}^*} \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y})) \xrightarrow{\mathrm{Id} \otimes \omega_{\mathrm{Ran}_{\underline{x}}}} \\ & \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y})) \otimes \mathrm{D-mod}(\mathrm{Ran}_{\underline{x}})^{\Gamma(\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}), -) \otimes \mathrm{Id}} \mathrm{D-mod}(\mathrm{Ran}_{\underline{x}}), \end{aligned}$$

which is the same as

$$(F.21) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}) \xrightarrow{\mathrm{ev}_{\underline{x}}^*} \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}))^{\Gamma(\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}), -)} \mathrm{Vect} \xrightarrow{\omega_{\mathrm{Ran}_{\underline{x}}}} \mathrm{D-mod}(\mathrm{Ran}_{\underline{x}}).$$

Thus, we obtain a natural transformation from (F.18) to (F.21). By adjunction, this gives rise to the desired natural transformation from (F.17) to (F.16).

F.4.7. We will now prove that the above natural transformation (F.17) $\rightarrow$ (F.16) is an isomorphism.

Write

$$\mathfrak{L}_{\nabla}(\mathcal{Y}) = \text{“colim”}_{R'} \text{Spec}(R') = \mathfrak{L}_{\nabla}^+(\mathcal{Y}')$$

as in Sect. F.2.1 and $\mathcal{Y}'$ as an (F.12), so that

$$\text{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}) = \text{“colim”}_{R'} \text{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}').$$

Thus, we obtain that

$$\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})) \simeq \text{colim}_{R'} \text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}')),$$

and we obtain in order to show that (F.17) $\rightarrow$ (F.16) is an isomorphism, it suffices to show that it becomes such if we precompose both functors with

$$\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}')) \xrightarrow{\iota'_*} \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y}))$$

for every R' as above, where ι' denotes the corresponding map $\iota' : \mathfrak{L}_{\nabla}^+(\mathcal{Y}') \rightarrow \mathfrak{L}_{\nabla}(\mathcal{Y})$.

F.4.8. Note that for every R' as above, the functor

$$(F.22) \quad \text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}')) \xrightarrow{\iota'_*} \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})) \xrightarrow{\Gamma(\mathfrak{L}_{\nabla}(\mathcal{Y})_{\underline{x}}, -)^{\text{enh}}} \mathcal{A}\text{-mod}_{\underline{x}}^{\text{fact}}$$

identifies with

$$(F.23) \quad \text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}')) \xrightarrow{\Gamma(\mathfrak{L}_{\nabla}^+(\mathcal{Y}')_{\underline{x}}, -)} \mathcal{A}'\text{-mod}_{\underline{x}}^{\text{com}} \rightarrow \mathcal{A}'\text{-mod}_{\underline{x}}^{\text{fact}} \simeq \mathcal{A}\text{-mod}_{\underline{x}}^{\text{fact}},$$

and the functor

$$(F.24) \quad \text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}')) \xrightarrow{\iota'_*} \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})) \xrightarrow{\text{ev}_{\underline{x}}^*} \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y})) \xrightarrow{\Gamma(\text{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y}), -)} \text{Vect}$$

identifies with

$$(F.25) \quad \text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}')) \xrightarrow{\text{ev}_{\underline{x}}^*} \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X, \mathcal{Y}')) \xrightarrow{\Gamma(\text{Sect}_{\nabla}(X, \mathcal{Y}'), -)} \text{Vect}.$$

Composing (F.22) and (F.23) with the functor $C_{\cdot}^{\text{fact}}(X, \mathcal{A}, -)_{\underline{x}}$, we obtain that the natural transformation (F.17) $\rightarrow$ (F.16), constructed above, gives rise to a natural transformation from

$$\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}')) \xrightarrow{\Gamma(\mathfrak{L}_{\nabla}^+(\mathcal{Y}')_{\underline{x}}, -)} \mathcal{A}'\text{-mod}_{\underline{x}}^{\text{com}} \rightarrow \mathcal{A}'\text{-mod}_{\underline{x}}^{\text{fact}} \simeq \mathcal{A}\text{-mod}_{\underline{x}}^{\text{fact}} \xrightarrow{C_{\cdot}^{\text{fact}}(X, \mathcal{A}, -)_{\underline{x}}} \text{Vect},$$

which is the same as

$$(F.26) \quad \text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}')) \xrightarrow{\Gamma(\mathfrak{L}_{\nabla}^+(\mathcal{Y}')_{\underline{x}}, -)} \mathcal{A}'\text{-mod}_{\underline{x}}^{\text{com}} \rightarrow \mathcal{A}'\text{-mod}_{\underline{x}}^{\text{fact}} \xrightarrow{C_{\cdot}^{\text{fact}}(X, \mathcal{A}', -)_{\underline{x}}} \text{Vect},$$

to (F.25).

However, unwinding the definitions, we obtain that the resulting natural transformation

$$(F.26) \rightarrow (F.25),$$

coincides with the natural isomorphism of Proposition F.1.6.

□[Proposition F.4.4]

F.5. Interpretation as factorization restriction.

F.5.1. In this section we will assume that $\mathfrak{L}_{\nabla}(\mathcal{Y})$ is ind-placid, and that the embeddings

$$\iota : \mathfrak{L}_{\nabla}^+(\mathcal{Y}) \rightarrow \mathfrak{L}_{\nabla}(\mathcal{Y}) \text{ and } \mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{Ran}_{\underline{x}}} \xrightarrow{\iota^{\text{mer} \rightsquigarrow \text{reg}}} \mathfrak{L}_{\nabla}(\mathcal{Y})_{\text{Ran}_{\underline{x}}}$$

are locally almost of finite presentation, where

$$\iota^{\text{mer} \rightsquigarrow \text{reg}} = \text{pr}_{\text{big}, \underline{x}}^{\mathcal{Y}}.$$

This assumption implies that $\mathfrak{L}_{\nabla}^+(\mathcal{Y})$ (resp., $\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{Ran}_{\underline{x}}}$) is placid (resp., ind-placid).

F.5.2. *Example.* The above assumptions hold for $\mathcal{Y} = \text{Jets}(\mathcal{E})$, where $\mathcal{E}$ is the total space of a vector bundle on X .

In particular, they hold for $\mathcal{Y} = \text{Op}_{\tilde{G}}$.

F.5.3. We can consider the factorization categories

$$\text{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\mathcal{Y})) \text{ and } \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathcal{Y}))$$

and the factorization functor

$$(F.27) \quad \iota_*^{\text{IndCoh}} : \text{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\mathcal{Y})) \rightarrow \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathcal{Y})).$$

F.5.4. Consider the assignment

$$(\mathcal{Z} \rightarrow \text{Ran}_{\underline{x}}) \mapsto \text{IndCoh}^*(\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{Ran}_{\underline{x}}} \times_{\text{Ran}_{\underline{x}}} \mathcal{Z})$$

as a crystal of categories over $\text{Ran}_{\underline{x}}$.

It has a natural structure of factorization module category with respect to $\text{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\mathcal{Y}))$; when viewed in this capacity we will denote it by $\text{IndCoh}^*(\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y}))^{\text{fact}_{\underline{x}}}$.

F.5.5. The map $\iota^{\text{mer} \rightsquigarrow \text{reg}}$ gives rise to a functor

$$(F.28) \quad (\iota^{\text{mer} \rightsquigarrow \text{reg}})_*^{\text{IndCoh}} : \text{IndCoh}^*(\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y}))^{\text{fact}_{\underline{x}}} \rightarrow \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathcal{Y}))^{\text{fact}_{\underline{x}}}$$

as factorization module categories, compatible with the factorization functor (F.27).

Hence, by Sect. B.9.25, we obtain a functor

$$(F.29) \quad \text{IndCoh}^*(\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y}))^{\text{fact}_{\underline{x}}} \rightarrow \text{Res}_{\iota_*^{\text{IndCoh}}}(\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathcal{Y}))^{\text{fact}_{\underline{x}}}).$$

F.5.6. We claim:

Lemma F.5.7. *The functor (F.29) is an equivalence.*

Proof. Repeats that of Lemma 16.2.8. □

APPENDIX G. FROM MODULE CATEGORIES OVER $\text{QCoh}(\text{LS}_{\tilde{G}}^{\text{mer}})$ TO FACTORIZATION MODULE CATEGORIES OVER $\text{Rep}(\tilde{G})$

This section is not logically necessary for the rest of the paper. Here we explain a procedure that associates to a module category over $\text{QCoh}(\text{LS}_{\tilde{G}}^{\text{mer}})$ a factorization module category over $\text{Rep}(\tilde{G})$, and show that this functor is fully faithful (on a certain subcategory).

Using this, we deduce an alternative proof of Proposition 7.5.7.

G.1. Creating factorization modules categories. Throughout this section we will work at a fixed point $x \in X$. In this subsection we let H be an arbitrary algebraic group.

G.1.1. Consider the space $\text{LS}_{H,x}^{\text{mer}}$ and the monoidal category $\text{QCoh}(\text{LS}_{H,x}^{\text{mer}})$. Let us recall the construction of a functor

$$(G.1) \quad \text{QCoh}(\text{LS}_{H,x}^{\text{mer}})\text{-mod} \rightarrow \text{Rep}(H)\text{-mod}_x^{\text{fact}}, \quad \mathbf{C} \mapsto \mathbf{C}^{\text{fact}_{x, \text{Rep}(H)}}.$$

G.1.2. Namely, we will construct an object

$$\text{QCoh}(\text{LS}_H^{\text{mer} \rightsquigarrow \text{reg}})^{\text{fact}_{x, \text{Rep}(H)}} \in \text{Rep}(H)\text{-mod}_x^{\text{fact}}$$

that carries a commuting action of $\text{QCoh}(\text{LS}_{H,x}^{\text{mer}})$. The functor (G.1) will then be given by

$$\text{QCoh}(\text{LS}_H^{\text{mer} \rightsquigarrow \text{reg}})^{\text{fact}_{x, \text{Rep}(H)}} \otimes_{\text{QCoh}(\text{LS}_{H,x}^{\text{mer}})} -.$$

G.1.3. The object $\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{mer} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_x, \mathrm{Rep}(H)}$ will have the feature that its underlying DG category, equipped with an action of $\mathrm{QCoh}(\mathrm{LS}_{H,x}^{\mathrm{mer}})$, identifies with $\mathrm{QCoh}(\mathrm{LS}_{H,x}^{\mathrm{mer}})$ itself.

This will imply that the functor (G.1) has the feature that for $\mathbf{C} \in \mathrm{QCoh}(\mathrm{LS}_{H,x}^{\mathrm{mer}})\text{-}\mathbf{mod}$, the category underlying $\mathbf{C}^{\mathrm{fact}_x, \mathrm{Rep}(H)}$ identifies with the original $\mathbf{C}$.

G.1.4. The object $\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{mer} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_x, \mathrm{Rep}(H)}$ is constructed as follows.

For our fixed point $x \in X$ and a finite subset $x \in \underline{x} \subset X$, consider the multi-disc $\mathcal{D}_{\underline{x}}$, and set

$$(\mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{mer} \rightsquigarrow \mathrm{reg}})^{\mathrm{fact}_x, \mathrm{Rep}(H)})_{\underline{x}} := \mathrm{QCoh}(\mathrm{LS}_{H,x \subseteq \underline{x}}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}),$$

see Sect. C.10.11.

Remark G.1.5. The construction above is a cousin of the following construction for affine D-schemes: starting from a module category over $\mathrm{QCoh}(\mathfrak{L}_{\nabla}(\mathcal{Y}))$ we can associate to it a factorization module category over $\mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}))$ by the operation of tensoring with

$$\mathrm{QCoh}_{\mathrm{co}}(\mathfrak{L}_{\nabla}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}(\mathcal{Y}))^{\mathrm{fact}_{\underline{x}}},$$

see Sect. F.5.4.

Remark G.1.6. This construction also be viewed as an analog of a construction in [CFGY] that associates to a $\mathfrak{L}(G)_x$ -module category a factorization module category over $\mathrm{D-mod}(\mathrm{Gr}_G)$; in fact a version of the latter construction has appeared in Sect. 4.7.8.

G.1.7. In what follows we will need a variant of the above construction, when instead of $\mathrm{LS}_{H,x}^{\mathrm{mer}}$ we use its formal completion $(\mathrm{LS}_{H,x}^{\mathrm{mer}})_{\mathrm{reg}}^{\wedge}$ around $\mathrm{LS}_{H,x}^{\mathrm{reg}}$.

The corresponding functor

$$(G.2) \quad \mathrm{QCoh}((\mathrm{LS}_{H,x}^{\mathrm{mer}})_{\mathrm{reg}}^{\wedge})\text{-}\mathbf{mod} \rightarrow \mathrm{Rep}(H)\text{-}\mathbf{mod}_x^{\mathrm{fact}}, \quad \mathbf{C} \mapsto \mathbf{C}^{\mathrm{fact}_x, \mathrm{Rep}(H)}$$

is constructed using the prestack

$$(x \subseteq \underline{x}) \rightsquigarrow (\mathrm{LS}_{H,x \subseteq \underline{x}}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}})_{\mathrm{reg}}^{\wedge},$$

i.e., the formal completion of $\mathrm{LS}_{H,x \subseteq \underline{x}}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}}$ along $(\mathrm{LS}_H^{\mathrm{reg}})_{\underline{x}}^{\mathrm{fact}_x}$.

G.1.8. *Example.* Consider $\mathrm{QCoh}(\mathrm{LS}_{H,x}^{\mathrm{reg}}) \simeq \mathrm{Rep}(H)$ as an object of $\mathrm{QCoh}((\mathrm{LS}_{H,x}^{\mathrm{mer}})_{\mathrm{reg}}^{\wedge})\text{-}\mathbf{mod}$, via the restriction functor

$$\mathrm{QCoh}((\mathrm{LS}_{H,x}^{\mathrm{mer}})_{\mathrm{reg}}^{\wedge}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{H,x}^{\mathrm{reg}}).$$

We claim that

$$(G.3) \quad \mathrm{QCoh}(\mathrm{LS}_{H,x}^{\mathrm{reg}})^{\mathrm{fact}_x, \mathrm{Rep}(H)} \simeq \mathrm{Rep}(H)^{\mathrm{fact}_x},$$

i.e., the vacuum object of $\mathrm{Rep}(H)\text{-}\mathbf{mod}_x^{\mathrm{fact}}$.

Indeed, this follows from [GaRo3, Chapter 3, Proposition 3.5.3] using the fact that

$$(\mathrm{LS}_{H,x \subseteq \underline{x}'}^{\mathrm{mer} \rightsquigarrow \mathrm{reg}})_{\mathrm{reg}}^{\wedge} \times_{(\mathrm{LS}_{H,x'}^{\mathrm{mer}})_{\mathrm{reg}}^{\wedge}} \mathrm{LS}_{H,x}^{\mathrm{reg}} \simeq (\mathrm{LS}_H^{\mathrm{reg}})_{\underline{x}}^{\mathrm{fact}_x}.$$

When applying [GaRo3, Chapter 3, Proposition 3.5.3] we use the fact that we can identify

$$(\mathrm{LS}_{H,x}^{\mathrm{mer}})_{\mathrm{reg}}^{\wedge} \simeq \mathfrak{h}_0^{\wedge} / \mathrm{Ad}(H),$$

so this prestack is passable.

G.1.9. We claim:

Theorem G.1.10. *The functor (G.2) is fully faithful.*

This theorem is proved in [Ra4, Theorem 9.13.1]. We will supply a proof for completeness.

Remark G.1.11. We conjecture that the functor (G.1) is itself fully faithful. A partial result in this direction has recently been established in [Bogd]: the composition (G.1) with the restriction functor

$$\mathrm{QCoh}(\mathrm{LS}_{H,x}^{\mathrm{restr}})\text{-}\mathbf{mod} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{H,x}^{\mathrm{mer}})\text{-}\mathbf{mod}$$

is fully faithful, where

$$\mathrm{LS}_{H,x}^{\mathrm{restr}} \subset \mathrm{LS}_{H,x}^{\mathrm{mer}}$$

is the stack of local systems with restricted variation (see [AGKRRV, Sect. 1.4]).

Remark G.1.12. The reason the discussion in this section is for a fixed point in the Ran space is that we do not know how to prove the analog of Theorem G.1.10 in the factorization setting (and are not confident in its validity).

G.2. Proof of Theorem G.1.10. The proof is a “baby version” of the argument proving the corresponding assertion in [CFGY], see Remark G.1.6.

G.2.1. *Reduction steps.* First, we note that the functor (G.2) preserves both limits and colimits.

Second, the (symmetric monoidal) restriction functor

$$\mathrm{QCoh}((\mathrm{LS}_{H,x}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{H,x}^{\mathrm{reg}})$$

is comonadic, and its right adjoint is $\mathrm{QCoh}((\mathrm{LS}_{H,x}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}})$ -linear.

It follows that the 2-category $\mathrm{QCoh}((\mathrm{LS}_{H,x}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}})\text{-}\mathbf{mod}$ is generated under colimits by the essential image of

$$(G.4) \quad \mathrm{QCoh}(\mathrm{LS}_{H,x}^{\mathrm{reg}})\text{-}\mathbf{mod} \rightarrow \mathrm{QCoh}((\mathrm{LS}_{H,x}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}})\text{-}\mathbf{mod},$$

given by restriction along

$$\mathrm{LS}_{H,x}^{\mathrm{reg}} \rightarrow (\mathrm{LS}_{H,x}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}}.$$

Moreover, by passing to right adjoints, the 2-category $\mathrm{QCoh}((\mathrm{LS}_{H,x}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}})\text{-}\mathbf{mod}$ is also generated under *limits* by the essential image of (G.4).

Third, $\mathrm{QCoh}(\mathrm{LS}_{H,x}^{\mathrm{reg}})\text{-}\mathbf{mod} \simeq \mathrm{Rep}(H)\text{-}\mathbf{mod}$ is generated under colimits (and separately, limits) by objects of the form $\mathbf{C} \otimes \mathrm{Rep}(H)$, for $\mathbf{C} \in \mathrm{DGCat}$.

And fourth, since

$$\mathrm{Rep}(H) \simeq \mathrm{Vect} \underset{\mathrm{QCoh}(H)}{\otimes} \mathrm{Vect},$$

we obtain that the object $\mathrm{Rep}(H) \in \mathrm{Rep}(H)\text{-}\mathbf{mod}$ is a colimit of objects on which the action of $\mathrm{Rep}(H)$ is trivial, i.e., factors via the augmentation functor

$$\mathrm{Rep}(H) \rightarrow \mathrm{Vect},$$

corresponding to

$$\mathrm{pt} \rightarrow \mathrm{LS}_{H,x}^{\mathrm{reg}}.$$

Combining these observations, we obtain that the 2-category $\mathrm{QCoh}((\mathrm{LS}_{H,x}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}})\text{-}\mathbf{mod}$ is generated under colimits by trivial modules and under limits by objects of the form $\mathbf{C}_0 \otimes \mathrm{Rep}(H)$ for a DG category $\mathbf{C}_0$ with a trivial action of $\mathrm{Rep}(H)$.

Hence, it is enough to show that the functor

$$(G.5) \quad \mathrm{Funct}_{\mathrm{QCoh}((\mathrm{LS}_{H,x}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}})\text{-}\mathbf{mod}}(\mathrm{Vect}, \mathbf{C}_0 \otimes \mathrm{Rep}(H)) \rightarrow \mathrm{Funct}_{\mathrm{Rep}(H)\text{-}\mathbf{mod}^{\mathrm{fact}_x}}(\mathrm{Vect}^{\mathrm{fact}_x, \mathrm{Rep}(H)}, \mathbf{C}_0 \otimes \mathrm{Rep}(H)^{\mathrm{fact}_x})$$

is an equivalence.

G.2.2. Recall that for a (unital) factorization functor $\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$ and an object $\mathbf{C}_2 \in \mathbf{A}_2\text{-}\mathbf{mod}_x^{\text{fact}}$, we denote by $\text{Res}_\Phi(\mathbf{C}_2)$ its restriction along Φ .

Consider the tautological object $\text{Vect}^{\text{fact}_x} \in \text{Vect}\text{-}\mathbf{mod}_x^{\text{fact}}$, and the corresponding object

$$\text{Res}_{\mathbf{oblv}_H}(\text{Vect}^{\text{fact}_x}) \in \text{Rep}(H)\text{-}\mathbf{mod}_x^{\text{fact}},$$

where $\mathbf{oblv}_H : \text{Rep}(H) \rightarrow \text{Vect}$ is the forgetful functor, viewed as a (strictly unital) factorization functor.

Pullback along the unit section of $(\text{LS}_{H,x \subseteq x'}^{\text{mer} \rightsquigarrow \text{reg}})_{\text{reg}}^\wedge$ gives rise to a functor

$$(G.6) \quad \text{Vect}^{\text{fact}_x, \text{Rep}(H)} \rightarrow \text{Vect}^{\text{fact}_x}$$

compatible with factorization (in the sense of Sect. C.14.11). Hence, we obtain a morphism

$$(G.7) \quad \text{Vect}^{\text{fact}_x, \text{Rep}(H)} \rightarrow \text{Res}_{\mathbf{oblv}_H}(\text{Vect}^{\text{fact}_x})$$

as unital factorization modules over $\text{Rep}(H)$.

G.2.3. We claim:

Lemma G.2.4. *The functor (G.7) is an equivalence.*

Proof. Follows from Lemma B.15.9. □

G.2.5. Recall now the paradigm of Lemma C.15.16. We apply it to $\mathbf{oblv}_H : \text{Rep}(H) \rightarrow \text{Vect}$. Combined with Lemma G.2.4 above, we obtain that for $\mathbf{C} \in \text{Rep}(H)\text{-}\mathbf{mod}_x^{\text{fact}}$ we have a canonical equivalence

$$(G.8) \quad \text{Funct}_{\text{Rep}(H)\text{-}\mathbf{mod}_x^{\text{fact}}}(\text{Vect}^{\text{fact}_x, \text{Rep}(H)}, \mathbf{C}) \simeq R_H\text{-mod}^{\text{fact}}(\mathbf{C}_x).$$

G.2.6. Taking $\mathbf{C} := \text{Rep}(H)^{\text{fact}_x} \otimes \mathbf{C}_0$ in (G.8) and using (G.3), we obtain that (G.5) reduces to showing that the resulting functor

$$(G.9) \quad \text{Funct}_{\text{QCoh}((\text{LS}_{H,x}^{\text{mer}})_{\text{reg}}^\wedge)\text{-mod}}(\text{Vect}, \text{Rep}(H) \otimes \mathbf{C}_0) \rightarrow R_H\text{-mod}^{\text{fact}}(\text{Rep}(H) \otimes \mathbf{C}_0)$$

is an equivalence.

It is clear that the functor

$$\text{Funct}_{\text{QCoh}((\text{LS}_{H,x}^{\text{mer}})_{\text{reg}}^\wedge)\text{-mod}}(\text{Vect}, \text{Rep}(H)) \otimes \mathbf{C}_0 \rightarrow \text{Funct}_{\text{QCoh}((\text{LS}_{H,x}^{\text{mer}})_{\text{reg}}^\wedge)\text{-mod}}(\text{Vect}, \text{Rep}(H) \otimes \mathbf{C}_0)$$

is an equivalence (since $\text{QCoh}((\text{LS}_{H,x}^{\text{mer}})_{\text{reg}}^\wedge)$ is a semi-rigid monoidal category, see [AGKRRV, Sect. C]).

Now the fact that R_H is holonomic implies that

$$R_H\text{-mod}^{\text{fact}}(\text{Rep}(H)) \otimes \mathbf{C}_0 \rightarrow R_H\text{-mod}^{\text{fact}}(\text{Rep}(H) \otimes \mathbf{C}_0)$$

is also an equivalence (see [Ra4, Theorem 8.13.1]).

Hence, it is enough to show that the functor

$$(G.10) \quad \text{Funct}_{\text{QCoh}((\text{LS}_{H,x}^{\text{mer}})_{\text{reg}}^\wedge)\text{-mod}}(\text{Vect}, \text{Rep}(H)) \rightarrow R_H\text{-mod}^{\text{fact}}(\text{Rep}(H))$$

is an equivalence.

G.2.7. We rewrite the left-hand side in (G.10) as

$$\text{QCoh}(\text{pt} \times_{(\text{LS}_{H,x}^{\text{mer}})_{\text{reg}}^\wedge} \text{LS}_{H,x}^{\text{reg}}) \simeq \text{QCoh}(\text{pt} \times_{\text{LS}_{H,x}^{\text{mer}}} \text{LS}_{H,x}^{\text{reg}}).$$

G.2.8. By Sect. 4.5.4, the category $(R_H\text{-mod}^{\text{fact}}(\text{Rep}(H)))^{>-\infty}$ identifies with

$$\text{QCoh}_{\text{co}}\left(\text{pt} \times_{\text{LS}_{H,x}^{\text{mer}}} \text{LS}_{H,x}^{\text{reg}}\right)^{>-\infty}.$$

Since $R_H\text{-mod}^{\text{fact}}(\text{Rep}(H))$ is left-complete (see Proposition B.9.18), we obtain that it identifies with the left completion of

$$\text{QCoh}_{\text{co}}\left(\text{pt} \times_{\text{LS}_{H,x}^{\text{mer}}} \text{LS}_{H,x}^{\text{reg}}\right).$$

Since $\text{pt} \times_{\text{LS}_{H,x}^{\text{mer}}} \text{LS}_{H,x}^{\text{reg}}$ is almost of finite type, we have an equivalence

$$\Psi_{\text{pt} \times_{\text{LS}_{H,x}^{\text{mer}}} \text{LS}_{H,x}^{\text{reg}}} : \text{QCoh}_{\text{co}}\left(\text{pt} \times_{\text{LS}_{H,x}^{\text{mer}}} \text{LS}_{H,x}^{\text{reg}}\right)^{>-\infty} \xrightarrow{\sim} \text{IndCoh}\left(\text{pt} \times_{\text{LS}_{H,x}^{\text{mer}}} \text{LS}_{H,x}^{\text{reg}}\right)^{>-\infty}.$$

Hence, the above left-completion identifies with

$$\text{QCoh}\left(\text{pt} \times_{\text{LS}_{H,x}^{\text{mer}}} \text{LS}_{H,x}^{\text{reg}}\right).$$

G.2.9. Unwinding the constructions, we obtain that the endo-functor of $\text{QCoh}\left(\text{pt} \times_{\text{LS}_{H,x}^{\text{mer}}} \text{LS}_{H,x}^{\text{reg}}\right)$, induced by (G.10) and the above two identifications is the identity.

□[Theorem G.1.10]

G.3. Factorization module categories attached to affine D-schemes.

G.3.1. Let $\mathcal{Y}$ be an affine D-scheme over X , equipped with a map

$$\mathcal{Y} \rightarrow \text{pt}/H.$$

Consider the corresponding factorization spaces

$$(G.11) \quad \mathfrak{L}_{\nabla}^+(\mathcal{Y}) \xhookrightarrow{\iota} \mathfrak{L}_{\nabla}(\mathcal{Y})$$

and the commutative (but *not necessarily* Cartesian) diagram

$$\begin{array}{ccc} \mathfrak{L}_{\nabla}^+(\mathcal{Y}) & \longrightarrow & \mathfrak{L}_{\nabla}(\mathcal{Y}) \\ \tau^{\text{reg}} \downarrow & & \downarrow \tau \\ \text{LS}_H^{\text{reg}} & \longrightarrow & \text{LS}_H^{\text{mer}} \end{array}$$

G.3.2. On the one hand, we can consider $\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_x)$ as an object of $\text{QCoh}(\text{LS}_{H,x}^{\text{mer}})\text{-mod}$. Consider the resulting object

$$(G.12) \quad \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_x)^{\text{fact}_x, \text{Rep}(H)} \in \text{Rep}(H)\text{-mod}^{\text{fact}}.$$

G.3.3. On the other hand, consider the $\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y}))$ -factorization category

$$\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y}))^{\text{fact}_x},$$

defined as in Sect. F.5.4.

Pullback along

$$\tau^{\text{reg}} : \mathfrak{L}_{\nabla}^+(\mathcal{Y}) \rightarrow \text{LS}_H^{\text{reg}}$$

defines a (strictly unital) factorization functor

$$\text{Rep}(H) \simeq \text{QCoh}(\text{LS}_H^{\text{reg}}) \xrightarrow{(\tau^{\text{reg}})^*} \text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Y})).$$

Consider the resulting object

$$(G.13) \quad \text{Res}_{(\tau^{\text{reg}})^*} \left(\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y}))^{\text{fact}_x} \right) \in \text{Rep}(H)\text{-mod}_x^{\text{fact}}.$$

G.3.4. Note that the natural morphism

$$(G.14) \quad \mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{x \subseteq \underline{x}} \rightarrow \text{LS}_{H, x \subseteq \underline{x}}^{\text{mer} \rightsquigarrow \text{reg}} \times_{\text{LS}_{H, x}^{\text{mer}}} \mathfrak{L}_{\nabla}(\mathcal{Y})_x$$

is affine.

Hence, pullback along (G.14) gives rise to a functor

$$(G.15) \quad \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_x)^{\text{fact}_x, \text{Rep}(H)} \rightarrow \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y}))^{\text{fact}_x}$$

compatible with factorization.

Hence, we obtain a functor

$$(G.16) \quad \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Y})_x)^{\text{fact}_x, \text{Rep}(H)} \rightarrow \text{Res}_{(\text{rreg})^*} \left(\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y}))^{\text{fact}_x} \right)$$

in $\text{Rep}(H)\text{-}\mathbf{mod}^{\text{fact}}$.

G.3.5. *Variant.* Let $\mathfrak{L}_{\nabla}(\mathcal{Y})_{\text{mon-free}}^{\wedge}$ denote the formal completion of $\mathfrak{L}_{\nabla}(\mathcal{Y})$ along

$$\mathfrak{L}_{\nabla}^{\text{mon-free}}(\mathcal{Y}) := \mathfrak{L}_{\nabla}(\mathcal{Y}) \times_{\text{LS}_H^{\text{mer}}} \text{LS}_H^{\text{reg}} \rightarrow \mathfrak{L}_{\nabla}(\mathcal{Y}),$$

or which is the same

$$\mathfrak{L}_{\nabla}(\mathcal{Y}) \times_{\text{LS}_H^{\text{mer}}} (\text{LS}_H^{\text{mer}})^{\wedge}_{\text{reg}}.$$

Similar to the above, we can consider

$$\text{QCoh}_{\text{co}}((\mathfrak{L}_{\nabla}(\mathcal{Y})_{\text{mon-free}}^{\wedge})_x)^{\text{fact}_x, \text{Rep}(H)} \in \text{Rep}(H)\text{-}\mathbf{mod}^{\text{fact}}$$

and

$$\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{mon-free}}^{\wedge})^{\text{fact}_x},$$

and we obtain a functor

$$(G.17) \quad \text{QCoh}_{\text{co}}((\mathfrak{L}_{\nabla}(\mathcal{Y})_{\text{mon-free}}^{\wedge})_x)^{\text{fact}_x, \text{Rep}(H)} \rightarrow \text{Res}_{(\text{rreg})^*} \left(\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{mon-free}}^{\wedge})^{\text{fact}_x} \right)$$

in $\text{Rep}(H)\text{-}\mathbf{mod}^{\text{fact}}$.

G.3.6. *Variant.* Let us assume now that $\mathfrak{L}_{\nabla}(\mathcal{Y})$ and also $\text{pt} \times_{\text{LS}_H^{\text{mer}}} \mathfrak{L}_{\nabla}(\mathcal{Y})$ satisfy the assumptions of Sect. C.12.10.

This implies that the morphism

$$\text{LS}_{H, x \subseteq \underline{x}}^{\text{fact}_x} \times_{\text{LS}_{H, x \subseteq \underline{x}}^{\text{mer} \rightsquigarrow \text{reg}}} \mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{x \subseteq \underline{x}} \rightarrow \text{LS}_{H, x \subseteq \underline{x}}^{\text{fact}_x} \times_{\text{LS}_{H, x}^{\text{reg}}} \text{LS}_{H, x}^{\text{reg}} \times_{\text{LS}_{H, x}^{\text{mer}}} \mathfrak{L}_{\nabla}(\mathcal{Y})_x = \text{LS}_{H, x \subseteq \underline{x}}^{\text{fact}_x} \times_{\text{LS}_{H, x}^{\text{mer}}} \mathfrak{L}_{\nabla}(\mathcal{Y})_x$$

is of finite Tor-dimension.

Hence, so is the morphism

$$(G.18) \quad (\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{x \subseteq \underline{x}})^{\wedge}_{\text{mon-free}} \rightarrow \left(\text{LS}_{H, x \subseteq \underline{x}}^{\text{mer} \rightsquigarrow \text{reg}} \right)^{\wedge}_{\text{mon-free}} \times_{(\text{LS}_{H, x}^{\text{mer}})^{\wedge}_{\text{reg}}} (\mathfrak{L}_{\nabla}(\mathcal{Y})_{\text{mon-free}}^{\wedge})_x.$$

Hence, by Sect. A.10.13, the $(\text{IndCoh}, *)$ -pullback along (G.18) gives rise to a functor

$$(G.19) \quad \text{IndCoh}^* ((\mathfrak{L}_{\nabla}(\mathcal{Y})_{\text{mon-free}}^{\wedge})_x)^{\text{fact}_x, \text{Rep}(H)} \rightarrow \text{IndCoh}^* ((\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{mon-free}}^{\wedge})^{\text{fact}_x})$$

compatible with factorization.

Hence, we obtain a functor

$$(G.20) \quad \text{IndCoh}^* ((\mathfrak{L}_{\nabla}(\mathcal{Y})_{\text{mon-free}}^{\wedge})_x)^{\text{fact}_x, \text{Rep}(H)} \rightarrow \text{Res}_{(\text{rreg})^*} \left(\text{IndCoh}^* ((\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y})_{\text{mon-free}}^{\wedge})^{\text{fact}_x}) \right)$$

in $\text{Rep}(H)\text{-}\mathbf{mod}^{\text{fact}}$.

G.3.7. We claim:

Lemma G.3.8. *The functor (G.20) is an equivalence.*

Proof. Follows from Lemma B.15.9. □

G.3.9. Let $\iota^{+, \text{mon-free}}$ denote the morphism

$$\mathfrak{L}_{\nabla}^+(\mathcal{Y}) \rightarrow \mathfrak{L}_{\nabla}(\mathcal{Y})_{\text{mon-free}}^{\wedge}.$$

The operation of IndCoh-pushforward along $\iota^{+, \text{mon-free}}$ gives rise to a (strictly) unital factorization functor

$$(\iota^{+, \text{mon-free}})_*^{\text{IndCoh}} : \text{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\mathcal{Y})) \rightarrow \text{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\text{mon-free}}^{\wedge}).$$

It follows from Lemma F.5.7 that the naturally defined functor

$$\text{IndCoh}^*((\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Y}))_{\text{mon-free}}^{\wedge})^{\text{fact}_x} \rightarrow \text{Res}_{(\iota^{+, \text{mon-free}})_*^{\text{IndCoh}}} \left((\text{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\text{mon-free}}^{\wedge}))^{\text{fact}_x} \right)$$

is an equivalence.

Hence, combining with Lemma G.3.8, we obtain:

Corollary G.3.10. *There is a canonical equivalence*

$$\text{IndCoh}^*((\mathfrak{L}_{\nabla}(\mathcal{Y})_{\text{mon-free}}^{\wedge})^{\text{fact}_x, \text{Rep}(H)}) \simeq \text{Res}_{(\iota^{+, \text{mon-free}})_*^{\text{IndCoh}} \circ (\mathfrak{r}^{\text{reg}})^*} \left((\text{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\text{mon-free}}^{\wedge}))^{\text{fact}_x} \right)$$

in $\text{Rep}(H)\text{-mod}^{\text{fact}}$.

G.4. Proof of Proposition 7.5.7 via factorization.

G.4.1. Consider the following (strictly unital) factorization functor, to be denoted Φ_1

$$(G.21) \quad \text{Rep}(\check{G}) \xrightarrow{\text{FLE}_{\check{G}, \infty}} \text{Whit}_*(G) \xrightarrow{\text{Id} \otimes \text{Vac}(G)_{\text{crit}, \rho(\omega_X)}} \text{Whit}_*(G) \otimes \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}^{\text{Sph-gen}}).$$

Restricting the vacuum factorization module category

$$\text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}^{\text{Sph-gen}})^{\text{fact}_x} \in \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}^{\text{Sph-gen}})\text{-mod}_x^{\text{fact}}$$

along (G.21), we obtain an object

$$(G.22) \quad \text{Res}_{\Phi_1}(\text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}^{\text{Sph-gen}})^{\text{fact}_x}) \in \text{Rep}(\check{G})\text{-mod}_x^{\text{fact}}.$$

G.4.2. We claim:

Lemma G.4.3. *The object (G.22) identifies canonically with*

$$\text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}^{\text{Sph-gen}})^{\text{fact}_x, \text{Rep}(\check{G})},$$

where we regard $\text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}^{\text{Sph-gen}})$ as an object of $\text{QCoh}((\text{LS}_{G, x}^{\text{mer}})^{\wedge})\text{-mod}$ by the recipe of Sect. 7.5.1.

Proof. We rewrite Φ_1 as

$$(G.23) \quad \text{Rep}(\check{G}) \xrightarrow{\text{FLE}_{\check{G}, \infty}} \text{Whit}_*(G) \xrightarrow{\text{Id} \otimes \text{Vac}(G)_{\text{crit}, \rho(\omega_X)}} \text{Whit}_*(G) \otimes \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \text{Whit}_*(G) \otimes_{\text{Sph}_G} \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \simeq \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}^{\text{Sph-gen}}).$$

Therefore, we can reinterpret the object (G.22) as follows. Consider

$$\text{Res}_{\text{FLE}_{\check{G}, \infty}}(\text{Whit}_*(G)^{\text{fact}_x}) \in \text{Rep}(\check{G})\text{-mod}_x^{\text{fact}}$$

as an object equipped with a commuting action of $\text{Sph}_{G, x}$. Then (G.22) identifies with

$$\text{Res}_{\text{FLE}_{\check{G}, \infty}}(\text{Whit}_*(G)^{\text{fact}_x}) \otimes_{\text{Sph}_{G, x}} \text{KL}(G)_{\text{crit}, \rho(\omega_X), x}.$$

Comparing with the definition of

$$\mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}^{\mathrm{Sph-gen}}) \in \mathrm{QCoh}((\mathrm{LS}_{G,x}^{\mathrm{mer}})^\wedge_{\mathrm{reg}})\text{-}\mathbf{mod}$$

(see Sect. 7.5.1), the assertion of the lemma follows from the identification (G.3). $\square$

G.4.4. Consider the following (strictly unital) factorization functor, to be denoted Φ_2

$$(G.24) \quad \mathrm{Rep}(\check{G}) \xrightarrow{(\mathfrak{r}^{\mathrm{reg}})^*} \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{reg}}) \xrightarrow{(\iota^{\mathrm{mon-free}})^* \mathrm{IndCoh}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}}.$$

Restricting the vacuum factorization module category

$$(\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}})^{\mathrm{fact}_x} \in \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}}\text{-}\mathbf{mod}_x^{\mathrm{fact}}$$

along (G.24), we obtain an object

$$(G.25) \quad \mathrm{Res}_{\Phi_2}((\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}})^{\mathrm{fact}_x}) \in \mathrm{Rep}(\check{G})\text{-}\mathbf{mod}_x^{\mathrm{fact}}.$$

G.4.5. We claim:

Lemma G.4.6. *The object (G.25) identifies canonically with*

$$(\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}})^{\mathrm{fact}_x, \mathrm{Rep}(\check{G})},$$

where we regard $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}}$ as an object of $\mathrm{QCoh}((\mathrm{LS}_{G,x}^{\mathrm{mer}})^\wedge_{\mathrm{reg}})\text{-}\mathbf{mod}$ via

$$\mathfrak{r} : (\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}}^\wedge \rightarrow (\mathrm{LS}_{G,x}^{\mathrm{mer}})_{\mathrm{reg}}^\wedge.$$

Proof. This is a particular case of Lemma G.3.8. $\square$

G.4.7. We now claim:

Proposition G.4.8. *We have a canonical isomorphism of (strictly unital) factorization functors*

$$\overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}} \circ \Phi_1 \simeq \Phi_2.$$

Let us accept this proposition for a moment and finish the proof of Proposition 7.5.7.

G.4.9. By Theorem G.1.10, combined with Lemmas G.4.3 and G.4.6, it suffices to show that the functor $\overline{\mathrm{DS}}^{\mathrm{enh}, \mathrm{rfnd}}$ appearing in Proposition 7.5.7 can be realized as the fiber at x of a functor between

$$\mathrm{Res}_{\Phi_1}(\mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)}^{\mathrm{Sph-gen}})^{\mathrm{fact}_x}) \text{ and } \mathrm{Res}_{\Phi_2}((\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}})^{\mathrm{fact}_x}),$$

viewed as objects of $\mathrm{Rep}(\check{G})\text{-}\mathbf{mod}_x^{\mathrm{fact}}$.

The latter structure is supplied by Proposition G.4.8.

$\square$ [Proposition 7.5.7]

G.5. Proof of Proposition G.4.8.

G.5.1. It is enough to show that the two functors match after we compose them with the fully faithful embedding

$$\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}} \hookrightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}).$$

We will first establish an isomorphism between the compositions of the two functors in question with

$$\Gamma^{\mathrm{IndCoh}}(\mathrm{Op}_G^{\mathrm{mer}}, -)^{\mathrm{enh}} : \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \rightarrow \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}}\text{-mod}}^{\mathrm{fact}}.$$

G.5.2. We start by rewriting the corresponding composition for Φ_1 . It is equal to

$$(G.26) \quad \text{Rep}(\check{G}) \xrightarrow{\text{FLE}_{\check{G}, \infty}} \text{Whit}_*(G) \xrightarrow{\text{Id} \otimes \text{Vac}(G)_{\text{crit}, \rho(\omega_X)}} \text{Whit}_*(G) \otimes \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \\ \rightarrow \text{Whit}_*(G) \otimes_{\text{Sph}_G} \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}) \xrightarrow{\overline{\text{DS}}^{\text{enh}}} \mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}\text{-mod}}^{\text{fact}}.$$

By Remark 1.7.8, we rewrite this as

$$(G.27) \quad \text{Rep}(\check{G}) \xrightarrow{\text{Sat}_{\check{G}}^{\text{nv}}} \text{Sph}_G \xrightarrow{\sigma} \text{Sph}_G \xrightarrow{-\star \mathbf{1}_{\text{Whit}_*(G)}} \text{Whit}_*(G) \xrightarrow{\text{Id} \otimes \text{Vac}(G)_{\text{crit}, \rho(\omega_X)}} \\ \rightarrow \text{Whit}_*(G) \otimes \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \text{Whit}_*(G) \otimes_{\text{Sph}_G} \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \\ \rightarrow \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)}) \xrightarrow{\overline{\text{DS}}^{\text{enh}}} \mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}\text{-mod}}^{\text{fact}}.$$

We can rewrite the composition in the first two lines in (G.27) as

$$(G.28) \quad \text{Rep}(\check{G}) \xrightarrow{\text{Sat}_{\check{G}}^{\text{nv}}} \text{Sph}_G \xrightarrow{-\star \text{Vac}(G)_{\text{crit}, \rho(\omega_X)}} \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \xrightarrow{\mathbf{1}_{\text{Whit}_*(G)} \otimes \text{Id}} \\ \rightarrow \text{Whit}_*(G) \otimes \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \text{Whit}_*(G) \otimes_{\text{Sph}_G} \text{KL}(G)_{\text{crit}, \rho(\omega_X)}.$$

Hence, we can rewrite (G.27) as

$$(G.29) \quad \text{Rep}(\check{G}) \xrightarrow{\text{Sat}_{\check{G}}^{\text{nv}}} \text{Sph}_G \xrightarrow{-\star \text{Vac}(G)_{\text{crit}, \rho(\omega_X)}} \\ \rightarrow \text{KL}(G)_{\text{crit}, \rho(\omega_X)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}, \rho(\omega_X)} \xrightarrow{\text{DS}^{\text{enh}}} \mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}\text{-mod}}^{\text{fact}}.$$

G.5.3. We now apply Theorem 5.2.5. It says that the functor

$$\text{Rep}(\check{G}) \xrightarrow{\text{Sat}_{\check{G}}^{\text{nv}}} \text{Sph}_G \xrightarrow{-\star \text{Vac}(G)_{\text{crit}, \rho(\omega_X)}} \text{KL}(G)_{\text{crit}, \rho(\omega_X)}$$

is isomorphic to

$$(G.30) \quad \text{Rep}(\check{G}) \xrightarrow{(\tau^{\text{reg}})^*} \text{QCoh}(\text{Op}_{\check{G}}^{\text{reg}}) \xrightarrow{\Gamma(\text{Op}_{\check{G}}^{\text{reg}}, -)} \mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}\text{-mod}}^{\text{com}} \xrightarrow{- \otimes_{\mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}}} \text{Vac}(G)_{\text{crit}, \rho(\omega_X)}} \text{KL}(G)_{\text{crit}, \rho(\omega_X)}.$$

Hence, we can rewrite (G.27) as

$$(G.31) \quad \text{Rep}(\check{G}) \xrightarrow{(\tau^{\text{reg}})^*} \text{QCoh}(\text{Op}_{\check{G}}^{\text{reg}}) \xrightarrow{\Gamma(\text{Op}_{\check{G}}^{\text{reg}}, -)} \mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}\text{-mod}}^{\text{com}} \xrightarrow{- \otimes_{\mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}}} \text{DS}^{\text{enh}}(\text{Vac}(G)_{\text{crit}, \rho(\omega_X)})} \mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}\text{-mod}}^{\text{fact}}.$$

G.5.4. We now use Theorem 4.8.3, which says that the natural map

$$\mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}} \rightarrow \text{DS}^{\text{enh}}(\text{Vac}(G)_{\text{crit}, \rho(\omega_X)})$$

is an isomorphism.

This allows us to rewrite (G.31) as

$$(G.32) \quad \text{Rep}(\check{G}) \xrightarrow{(\tau^{\text{reg}})^*} \text{QCoh}(\text{Op}_{\check{G}}^{\text{reg}}) \xrightarrow{\Gamma(\text{Op}_{\check{G}}^{\text{reg}}, -)} \mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}\text{-mod}}^{\text{com}} \rightarrow \mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}\text{-mod}}^{\text{fact}}.$$

G.5.5. We now consider the composition involving Φ_2 :

$$(G.33) \quad \text{Rep}(\check{G}) \xrightarrow{(\tau^{\text{reg}})^*} \text{QCoh}(\text{Op}_{\check{G}}^{\text{reg}}) \xrightarrow{(\iota^{\text{mon-free}})^*} \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \xrightarrow{\Gamma^{\text{IndCoh}}(\text{Op}_{\check{G}}^{\text{mer}}, -)^{\text{enh}}} \mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}\text{-mod}}^{\text{fact}}.$$

This functor identifies with (G.32) on the nose.

G.5.6. Thus, we have identified the compositions of the two functors

$$(G.34) \quad \mathrm{Rep}(\check{G}) \rightrightarrows \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})$$

with

$$\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \xrightarrow{\Gamma^{\mathrm{IndCoh}}(\mathrm{Op}_G^{\mathrm{mer}}, -)^{\mathrm{enh}}} \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg-mod}}^{\mathrm{fact}}}.$$

Let us show how to upgrade this isomorphism to an isomorphism between the two functors in (G.34).

G.5.7. It is enough to establish the isomorphism between the two functors in question on the compact generators of $\mathrm{Rep}(\check{G})$. These generators can be taken to be eventually coconnective. Hence, by Corollary 4.4.2(a), it is enough to show that both functors are t-exact (in fact, it is sufficient to know that they are t-exact on $\mathrm{Rep}(\check{G})^c$).

The t-exactness of the composition involving Φ_2 is clear. For Φ_1 , we interpret it as

$$(G.35) \quad \begin{aligned} \mathrm{Rep}(\check{G}) &\xrightarrow{(\mathrm{r}^{\mathrm{reg}})^*} \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{reg}}) \xrightarrow{\Gamma(\mathrm{Op}_G^{\mathrm{reg}}, -)} \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg-mod}}^{\mathrm{com}}} \xrightarrow{- \otimes_{\mathrm{Op}_G^{\mathrm{reg}}} \mathrm{Vac}(G)_{\mathrm{crit}, \rho(\omega_X)}} \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \rightarrow \\ &\rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)} \xrightarrow{\mathrm{DS}^{\mathrm{enh}, \mathrm{rfnd}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}). \end{aligned}$$

As was remarked already, it is enough to show that this functor is t-exact on $\mathrm{Rep}(\check{G})^c$. The composition in the first line of (G.35) is t-exact. Hence, by the construction of $\mathrm{DS}^{\mathrm{enh}, \mathrm{rfnd}}$, the corresponding functor

$$\mathrm{Rep}(\check{G})^c \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})$$

maps to $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})^{>-\infty}$, and its t-exactness follows from the t-exactness of its composition with $\Gamma^{\mathrm{IndCoh}}(\mathrm{Op}_G^{\mathrm{mer}}, -)^{\mathrm{enh}}$, while the latter is the functor (G.32), which is evidently t-exact.

APPENDIX H. UNITAL LOCAL-TO-GLOBAL FUNCTORS AND MONOIDAL ACTIONS

This section serves as a complement to Sect. 11. Here we express the notion of a (strictly) unital local-to-global functor as a functor of what we call the *independent* category.

This will allow us to study the interaction between various global monoidal categories attached to a local unital monoidal categories. These various variants are handy when studying the pattern of the Hecke action.

H.1. The “independent” category.

H.1.1. Let $\underline{\mathbf{C}}^{\mathrm{loc}, \mathrm{unl}}$ be a crystal of categories over $\mathrm{Ran}^{\mathrm{unl}}$. Denote

$$\mathbf{C}_{\mathrm{Ran}^{\mathrm{unl}}}^{\mathrm{loc}} := \Gamma^{\mathrm{lax}}(\mathrm{Ran}^{\mathrm{unl}}, \underline{\mathbf{C}}^{\mathrm{loc}, \mathrm{unl}}).$$

For example, when $\underline{\mathbf{C}}^{\mathrm{loc}, \mathrm{unl}}$ is the unit sheaf of categories, i.e., $\underline{\mathbf{D-mod}}(\mathrm{Ran}^{\mathrm{unl}})$, the above category is $\mathbf{D-mod}(\mathrm{Ran}^{\mathrm{unl}})$.

H.1.2. Let $\mathbf{C}^{\mathrm{glob}}$ be a target DG category. On the one hand we can consider the category $\mathrm{Func}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{lax-unl}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}})$ of lax unital local-to-global functors, i.e., right-lax functors

$$\mathbf{F}^{\mathrm{unl}} : \underline{\mathbf{C}}^{\mathrm{loc}, \mathrm{unl}} \rightarrow \mathbf{C}^{\mathrm{glob}} \otimes \underline{\mathbf{D-mod}}(\mathrm{Ran}^{\mathrm{unl}})$$

between sheaves of categories, see Sect. 11.3.8.

On the other hand, we can consider the category $\mathrm{Func}(\mathbf{C}_{\mathrm{Ran}^{\mathrm{unl}}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}})$ of (continuous) functors

$$\mathbf{F}^{\mathrm{unl}} : \mathbf{C}_{\mathrm{Ran}^{\mathrm{unl}}}^{\mathrm{loc}} \rightarrow \mathbf{C}^{\mathrm{glob}}.$$

There is a naturally defined functor

$$(H.1) \quad \mathrm{Func}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{lax-unl}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}) \rightarrow \mathrm{Func}(\mathbf{C}_{\mathrm{Ran}^{\mathrm{unl}}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}).$$

Namely, given $\underline{F}^{\text{untl}}$, we construct $\underline{F}^{\text{untl}}$ by applying the functor $\Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, -)$, followed by

$$\mathbf{C}^{\text{glob}} \otimes \text{D-mod}(\text{Ran}^{\text{untl}}) \xrightarrow{\text{Id} \otimes C_c^*(\text{Ran}^{\text{untl}}, -)} \mathbf{C}^{\text{glob}}.$$

Remark H.1.3. Note that unlike the case of the usual Ran space, the functor (H.1) is *not* an equivalence.

For example, for

$$\underline{\mathbf{C}}^{\text{loc, untl}} = \underline{\text{D-mod}}(\text{Ran}^{\text{untl}}) \text{ and } \mathbf{C}^{\text{glob}} = \text{Vect},$$

we have

$$\text{Funct}^{\text{loc} \rightarrow \text{glob, lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \simeq \text{D-mod}(\text{Ran}^{\text{untl}}),$$

while $\text{Funct}(\mathbf{C}_{\text{Ran}^{\text{untl}}}, \mathbf{C}^{\text{glob}})$ identifies with $\text{D-mod}(\text{Ran}^{\text{untl}})^{\vee}$. In terms of this identification, the functor (H.2) is the pairing

$$\text{D-mod}(\text{Ran}^{\text{untl}}) \otimes \text{D-mod}(\text{Ran}^{\text{untl}}) \rightarrow \text{Vect}, \quad \mathcal{F}_1, \mathcal{F}_2 \mapsto C_c^*(\text{Ran}^{\text{untl}}, \mathcal{F}_1 \overset{!}{\otimes} \mathcal{F}_2).$$

However, this pairing is not perfect. The reason for this is that, although the functor $(\Delta_{\text{Ran}^{\text{untl}}})^!$ is defined, it does *not* satisfy the projection formula.

More generally, the category $\text{Funct}^{\text{loc} \rightarrow \text{glob, lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$ can be described explicitly as lax sections of another crystal of categories on Ran^{untl} , see Remark H.1.8.

H.1.4. From now on we will make the following assumption on $\underline{\mathbf{C}}^{\text{loc, untl}}$:

For every $S \in \text{Sch}^{\text{aff}}$ and a map $\underline{x}_1 \xrightarrow{\alpha} \underline{x}_2$ in $\text{Maps}(S, \text{Ran}^{\text{untl}})$, the corresponding functor

$$\mathbf{C}_{S, \underline{x}_1}^{\text{loc}} \xrightarrow{\text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2}} \mathbf{C}_{S, \underline{x}_2}^{\text{loc}}$$

admits a *continuous* right adjoint.

Note that in this case, this right adjoint is automatically $\text{D-mod}(S)$ -linear.

Remark H.1.5. This assumption is made in order to simplify the exposition. One can make do without it, but in what follows one will have to describe $\underline{\mathbf{C}}^{\text{loc, untl}}$ using *proper* (rather than affine) schemes mapping to Ran , see Sect. C.4.6.

The properness assumption would guarantee that for map $f : Z' \rightarrow Z$ (of proper schemes) and

$$(\underline{x}_1 \rightarrow \underline{x}_2) \in \text{Maps}(Z, \text{Ran}^{\text{untl}}),$$

the diagram

$$\begin{array}{ccc} \mathbf{C}_{Z', \underline{x}_1 \circ f} & \xleftarrow{\text{ins. unit}_{\underline{x}_1 \circ f \subseteq \underline{x}_2 \circ f}^R} & \mathbf{C}_{Z', \underline{x}_2 \circ f} \\ f^! \uparrow & & \uparrow f^! \\ \mathbf{C}_{Z, \underline{x}_1} & \xleftarrow{\text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2}^R} & \mathbf{C}_{Z, \underline{x}_2} \end{array}$$

obtained by passing to right adjoints in the commutative diagram

$$\begin{array}{ccc} \mathbf{C}_{Z', \underline{x}_1 \circ f} & \xleftarrow{\text{ins. unit}_{\underline{x}_1 \circ f \subseteq \underline{x}_2 \circ f}} & \mathbf{C}_{Z', \underline{x}_2 \circ f} \\ f_! \uparrow & & \uparrow f_! \\ \mathbf{C}_{Z, \underline{x}_1} & \xleftarrow{\text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2}} & \mathbf{C}_{Z, \underline{x}_2}. \end{array}$$

commutes.

H.1.6. Under the above assumption, passage to right adjoints defines a sheaf of categories

$$(\underline{\mathbf{C}}^{\text{loc, untl}})^{\text{op}}$$

over $(\text{Ran}^{\text{untl}})^{\text{op}}$.

Note that we can tautologically identify $\mathbf{C}_{\text{Ran}^{\text{untl}}}^{\text{loc}} := \Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \underline{\mathbf{C}}^{\text{loc, untl}})$ with

$$\Gamma^{\text{lax}}((\text{Ran}^{\text{untl}})^{\text{op}}, (\underline{\mathbf{C}}^{\text{loc, untl}})^{\text{op}}).$$

Set

$$\mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}} := \Gamma^{\text{strict}}((\text{Ran}^{\text{untl}})^{\text{op}}, (\underline{\mathbf{C}}^{\text{loc, untl}})^{\text{op}}) \subset \Gamma^{\text{lax}}((\text{Ran}^{\text{untl}})^{\text{op}}, (\underline{\mathbf{C}}^{\text{loc, untl}})^{\text{op}}).$$

H.1.7. Let us describe $\mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}}$ explicitly as a full subcategory of $\mathbf{C}_{\text{Ran}^{\text{untl}}}^{\text{loc}}$:

An object

$$(S \xrightarrow{x} \text{Ran}) \mapsto \mathbf{c}_{S, x} \in \mathbf{C}_{S, x}^{\text{loc}}$$

belongs to $\mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}}$ if for every

$$(x_1 \xrightarrow{\alpha} x_2) \in \text{Maps}(S, \text{Ran}^{\text{untl}})$$

the map

$$\mathbf{c}_{S, x_1} \rightarrow (\text{ins. unit}_{x_1 \subseteq x_2})^R(\mathbf{c}_{S, x_2}),$$

obtained by adjunction from

$$\text{ins. unit}_{x_1 \subseteq x_2}(x_1) \rightarrow x_2,$$

is an isomorphism.

Remark H.1.8. Let assume in addition that $\underline{\mathbf{C}}^{\text{loc, untl}}$ is value-wise dualizable. Then passing to duals in $(\underline{\mathbf{C}}^{\text{loc, untl}})^{\text{op}}$, we obtain a crystal of categories $((\underline{\mathbf{C}}^{\text{loc, untl}})^{\text{op}})^{\vee}$ on Ran^{untl} .

It is easy to see that the category

$$\text{Func}^{\text{loc} \rightarrow \text{glob, lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$$

identifies with

$$((\mathbf{C}^{\text{op}})^{\vee})_{\text{Ran}^{\text{untl}}} := \Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, ((\underline{\mathbf{C}}^{\text{loc, untl}})^{\text{op}})^{\vee} \otimes \mathbf{C}^{\text{glob}}).$$

H.1.9. We claim:

Lemma H.1.10. *The embedding*

$$\text{emb. indep} : \mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}} \hookrightarrow \mathbf{C}_{\text{Ran}^{\text{untl}}}^{\text{loc}}$$

admits a left adjoint.

Proof. It is sufficient to show that

$$\mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}} \subset \mathbf{C}_{\text{Ran}^{\text{untl}}}^{\text{loc}}$$

is closed under limits.

However, this follows from the fact that limits in $\mathbf{C}_{\text{Ran}^{\text{untl}}}$ exists and have the property that they commute with evaluation on every *proper* Z mapping to Ran^{untl} , see Remark H.1.5. $\square$

H.1.11. Let emb. indep^L denote the left adjoint of emb. indep . Thus, we can view $\mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}}$ as a localization of $\mathbf{C}_{\text{Ran}^{\text{untl}}}^{\text{loc}}$.

Let us view the category $\text{Func}(\mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$ as a full subcategory of $\text{Func}(\mathbf{C}_{\text{Ran}^{\text{untl}}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$ via precomposition with emb. indep^L .

H.1.12. We claim:

Lemma H.1.13. *The functor (H.1) sends*

$$\mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{unl}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}) \subset \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{lax-unl}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}})$$

to

$$\mathrm{Funct}(\mathbf{C}_{\mathrm{Ran}^{\mathrm{unl}}, \mathrm{indep}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}) \subset \mathrm{Funct}(\mathbf{C}_{\mathrm{Ran}^{\mathrm{unl}}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}),$$

and the resulting functor

$$(H.2) \quad \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{unl}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}) \rightarrow \mathrm{Funct}(\mathbf{C}_{\mathrm{Ran}^{\mathrm{unl}}, \mathrm{indep}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}})$$

is an equivalence.

Proof. We will access $\mathrm{Ran}^{\mathrm{unl}}$ via proper schemes mapping to it, see Sect. C.4.6.

Let us be given an object $\underline{F} \in \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}})$. For every proper Z equipped with a map $\underline{x} : Z \rightarrow \mathrm{Ran}$, consider the corresponding functor

$$F_{f_Z, \underline{x}} : \mathbf{C}_{Z, \underline{x}}^{\mathrm{loc}} \rightarrow \mathbf{C}^{\mathrm{glob}}$$

and its (not necessarily continuous) right adjoint

$$(F_{f_Z, \underline{x}})^R : \mathbf{C}^{\mathrm{glob}} \rightarrow \mathbf{C}_{Z, \underline{x}}^{\mathrm{loc}}.$$

For $Z' \xrightarrow{g} Z$, the diagram

$$(H.3) \quad \begin{array}{ccc} \mathbf{C}_{Z', \underline{x} \circ g}^{\mathrm{loc}} & \xrightarrow{F_{Z', \underline{x} \circ g}} & \mathbf{C}^{\mathrm{glob}} \otimes \mathrm{D-mod}(Z') \\ g^! \uparrow & & \uparrow \mathrm{id} \otimes g^! \\ \mathbf{C}_{Z, \underline{x}}^{\mathrm{loc}} & \xrightarrow{F_{Z, \underline{x}}} & \mathbf{C}^{\mathrm{glob}} \otimes \mathrm{D-mod}(Z) \end{array}$$

is equipped with a datum of commutativity. Since g is proper, the diagram

$$(H.4) \quad \begin{array}{ccc} \mathbf{C}_{Z', \underline{x} \circ g}^{\mathrm{loc}} & \xrightarrow{F_{Z', \underline{x} \circ g}} & \mathbf{C}^{\mathrm{glob}} \otimes \mathrm{D-mod}(Z') \\ g^! \downarrow & & \downarrow \mathrm{id} \otimes g^! \\ \mathbf{C}_{Z, \underline{x}}^{\mathrm{loc}} & \xrightarrow{F_{Z, \underline{x}}} & \mathbf{C}^{\mathrm{glob}} \otimes \mathrm{D-mod}(Z'), \end{array}$$

obtained from (H.3) by passing to left adjoints along the vertical arrows, also commutes.

From (H.4), we obtain a datum of commutativity for the diagram

$$(H.5) \quad \begin{array}{ccc} \mathbf{C}_{Z', \underline{x} \circ g}^{\mathrm{loc}} & \xrightarrow{F_{f_{Z'}, \underline{x} \circ g}} & \mathbf{C}^{\mathrm{glob}} \\ g^! \downarrow & & \downarrow \mathrm{id} \\ \mathbf{C}_{Z, \underline{x}}^{\mathrm{loc}} & \xrightarrow{F_{f_Z, \underline{x}}} & \mathbf{C}^{\mathrm{glob}}. \end{array}$$

Finally, by passing to right adjoints in (H.5), we obtain a datum of commutativity for the diagram

$$(H.6) \quad \begin{array}{ccc} \mathbf{C}_{Z', \underline{x} \circ g}^{\mathrm{loc}} & \xleftarrow{(F_{f_{Z'}, \underline{x} \circ g})^R} & \mathbf{C}^{\mathrm{glob}} \\ g^! \uparrow & & \uparrow \mathrm{id} \\ \mathbf{C}_{Z, \underline{x}}^{\mathrm{loc}} & \xleftarrow{(F_{f_Z, \underline{x}})^R} & \mathbf{C}^{\mathrm{glob}}, \end{array}$$

where the functors $(F_{f_Z, \underline{x}})^R$ are not necessarily continuous, but limit-preserving.

The functor $\underline{F}$ is determined by the data of $F_{f_Z, \underline{x}}$ plus the data of commutativity for the diagrams (H.5), which is equivalent to having the data of limit-preserving functors $(F_{f_Z, \underline{x}})^R$ plus the data of commutativity for the diagrams (H.6).

Suppose now that $\underline{F}$ is upgraded to an object $\underline{F}^{\text{untl}}$ of $\text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$.

Let us be given a map $\underline{x}_1 \rightarrow \underline{x}_2$ in $\text{Maps}(Z, \text{Ran}^{\text{untl}})$. The unital structure on $\underline{\mathbf{C}}^{\text{loc}}$ gives rise to a functor

$$\text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2} : \mathbf{C}_{Z, \underline{x}_1}^{\text{loc}} \rightarrow \mathbf{C}_{Z, \underline{x}_2}^{\text{loc}},$$

and the lax unital structure $\underline{F}^{\text{untl}}$ on $\underline{F}$ gives rise to a natural transformation

$$F_{f_Z, \underline{x}_1} \rightarrow F_{f_Z, \underline{x}_2} \circ \text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2}.$$

By definition, this natural transformation is an isomorphism if and only if the above lax unital structure on $\underline{F}$ is strict. By adjunction, we obtain a natural transformation

$$(\text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2})^R \circ (F_{f_Z, \underline{x}_2})^R \rightarrow (F_{f_Z, \underline{x}_1})^R,$$

which is an isomorphism if and only if $\underline{F}^{\text{untl}}$ is strictly unital.

Since for every $(Z, \underline{x})$, the evaluation functor

$$\mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}} \hookrightarrow \mathbf{C}_{\text{Ran}^{\text{untl}}}^{\text{loc}} \rightarrow \mathbf{C}_{Z, \underline{x}}^{\text{loc}}$$

commutes with limits, we obtain that the datum of a strictly unital object $\underline{F}^{\text{untl}}$ is equivalent to that of a limit-preserving functor

$$\mathbf{C}^{\text{glob}} \rightarrow \mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}},$$

which is equivalent to that of a continuous functor

$$\mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}}.$$

Thus, we have constructed an equivalence

$$\text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \simeq \text{Funct}(\mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}}, \mathbf{C}^{\text{glob}}).$$

Unwinding the construction, it is easy to see that the diagram

$$\begin{array}{ccc} \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) & \longrightarrow & \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \\ \sim \downarrow & & \downarrow \\ \text{Funct}(\mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) & \longrightarrow & \text{Funct}(\mathbf{C}_{\text{Ran}^{\text{untl}}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \end{array}$$

commutes.

□

Remark H.1.14. The discussion in this subsection is not specific to Ran^{untl} . It applies to any pseudo-proper categorical prestack.

H.2. The calculation of the independent category in the vacuum case.

H.2.1. Here is a sample calculation of the category $\mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}}$. Take $\underline{\mathbf{C}}^{\text{loc}, \text{untl}} := \underline{\text{D-mod}}(\text{Ran}^{\text{untl}})$. Denote the corresponding independent category by

$$\text{Vect}_{\text{Ran}^{\text{untl}}, \text{indep}}.$$

H.2.2. Take $\mathbf{C}^{\text{glob}} = \text{Vect}$. Using Lemma H.1.13, from the identity functor

$$\underline{\text{D-mod}}(\text{Ran}^{\text{untl}}) \rightarrow \underline{\text{D-mod}}(\text{Ran}^{\text{untl}}),$$

we obtain a functor

$$(H.7) \quad \text{Vect}_{\text{Ran}^{\text{untl}}, \text{indep}} \rightarrow \text{Vect}.$$

We claim:

Proposition H.2.3. *The functor (H.7) is an equivalence.*

Remark H.2.4. Note that by Sect. H.1.6 can be reformulated as saying that the functor

$$(H.8) \quad \text{Vect} \xrightarrow{k \mapsto \omega_{\text{Ran}^{\text{unl}}}} \Gamma^{\text{strict}}(\text{Ran}^{\text{unl}}, \underline{\text{D-mod}}(\text{Ran}^{\text{unl}}))$$

is an equivalence.

Remark H.2.5. Note that, unlike the fact that Ran is contractible, which requires X to be *connected*, the assertion of Proposition H.2.3 is valid for any X that is non-empty.

Proof of Proposition H.2.3. We will show that the functor (H.8) is an equivalence. The right adjoint of this functor is given by the restriction of $C_c(\text{Ran}^{\text{unl}}, -)$ to

$$\Gamma^{\text{strict}}(\text{Ran}^{\text{unl}}, \underline{\text{D-mod}}(\text{Ran}^{\text{unl}})) \subset \Gamma^{\text{lax}}(\text{Ran}^{\text{unl}}, \underline{\text{D-mod}}(\text{Ran}^{\text{unl}})) = \text{D-mod}(\text{Ran}^{\text{unl}}).$$

It suffices to show that for $\mathcal{F} \in \Gamma^{\text{strict}}(\text{Ran}^{\text{unl}}, \underline{\text{D-mod}}(\text{Ran}^{\text{unl}}))$, the map

$$\mathcal{F} \rightarrow C_c(\text{Ran}^{\text{unl}}, \mathcal{F}) \otimes_{\omega_{\text{Ran}^{\text{unl}}}}$$

is an isomorphism.

For that it suffices to show that for any $\underline{x} \in \text{Ran}$, the map

$$(H.9) \quad \mathcal{F}_{\underline{x}} \rightarrow C_c(\text{Ran}^{\text{unl}}, \mathcal{F})$$

corresponding to

$$(H.10) \quad \text{pt} \xrightarrow{\underline{x}} \text{Ran}^{\text{unl}},$$

is an isomorphism.

We factor (H.9) as

$$(H.11) \quad \text{pt} \rightarrow \text{Ran}_{\underline{x}}^{\text{unl}} \rightarrow \text{Ran}^{\text{unl}},$$

and hence (H.10) as

$$(H.12) \quad \mathcal{F}_{\underline{x}} \rightarrow C_c(\text{Ran}_{\underline{x}}^{\text{unl}}, \mathcal{F}|_{\text{Ran}_{\underline{x}}^{\text{unl}}}) \rightarrow C_c(\text{Ran}^{\text{unl}}, \mathcal{F}).$$

We claim that both maps in (H.12) are isomorphisms. Indeed, the first map is an isomorphism, because

$$\mathcal{F}|_{\text{Ran}_{\underline{x}}^{\text{unl}}} \in \Gamma^{\text{strict}}(\text{Ran}^{\text{unl}}, \underline{\text{D-mod}}(\text{Ran}_{\underline{x}}^{\text{unl}})),$$

and $\underline{x}$ is the (value-wise) initial point of $\text{Ran}_{\underline{x}}^{\text{unl}}$.

The second arrow in (H.12) is an isomorphism because the map

$$\text{Ran}_{\underline{x}}^{\text{unl}} \rightarrow \text{Ran}^{\text{unl}}$$

is value-wise cofinal: its value-wise left adjoint is given by $\underline{x}' \mapsto \underline{x} \cup \underline{x}'$.

□

H.2.6. Let us return to Proposition C.5.4, and explain its meaning in terms of local-to-global functors.

Let us be given a strictly unital functor

$$\underline{F}^{\text{unl}} : \underline{C}^{\text{loc,unl}} \rightarrow C^{\text{glob}} \otimes \underline{\text{D-mod}}(\text{Ran}^{\text{unl}}).$$

Assume now that $\underline{C}^{\text{loc,unl}}$ is the restriction of a sheaf of categories $C^{\text{loc,unl},*}$ on $\text{Ran}^{\text{unl},*}$ (as is the case in all our examples).

Let $C_{\emptyset}^{\text{loc}}$ be the value of this extension on the initial point. The entire datum of the extension

$$\underline{C}^{\text{loc,unl}} \rightsquigarrow C^{\text{loc,unl},*}$$

is equivalent to the datum of a functor

$$(H.13) \quad C_{\emptyset}^{\text{loc}} \xrightarrow{\text{ins.unl}_{\emptyset}} \Gamma^{\text{strict}}(\text{Ran}^{\text{unl}}, \underline{C}^{\text{loc,unl}}) \hookrightarrow \Gamma^{\text{lax}}(\text{Ran}^{\text{unl}}, \underline{C}^{\text{loc,unl}}) = C_{\text{Ran}^{\text{unl}}}^{\text{loc}}.$$

Remark H.2.7. Note that in most examples, $\mathbf{C}_\emptyset^{\text{loc}} \simeq \text{Vect}$, so that datum of (H.13) is that of an object

$$\mathbf{1}_{\mathbf{C}^{\text{loc}}} \in \Gamma^{\text{strict}}(\text{Ran}^{\text{untl}}, \mathbf{C}^{\text{loc, untl}}).$$

I.e., for every $\underline{x} \in \text{Ran}$, we have an object $\mathbf{1}_{\mathbf{C}^{\text{loc}}, \underline{x}} \in \mathbf{C}_{\underline{x}}^{\text{loc}}$ and for every $\underline{x} \subseteq \underline{x}'$ we have an isomorphism

$$\text{ins. unit}_{\underline{x} \subseteq \underline{x}'}(\mathbf{1}_{\mathbf{C}^{\text{loc}}, \underline{x}} \simeq \mathbf{1}_{\mathbf{C}^{\text{loc}}, \underline{x}'}).$$

H.2.8. Consider the composition

$$\mathbf{F}^{\text{untl}} \circ \text{ins. unit}_\emptyset : \mathbf{C}_\emptyset^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(\text{Ran}^{\text{untl}}).$$

The claim is that it factors canonically as

$$\mathbf{C}_\emptyset^{\text{loc}} \xrightarrow{\mathbf{F}_\emptyset} \mathbf{C}^{\text{glob}} \xrightarrow{\text{Id} \otimes \omega_{\text{Ran}^{\text{untl}}}} \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(\text{Ran}^{\text{untl}}).$$

Indeed, the functor $\mathbf{F}^{\text{untl}} \circ \text{ins. unit}_\emptyset$ factors as

$$\begin{aligned} \mathbf{C}_\emptyset^{\text{loc}} &\rightarrow \mathbf{C}^{\text{glob}} \otimes \Gamma^{\text{strict}}(\text{Ran}^{\text{untl}}, \underline{\text{D-mod}}(\text{Ran}^{\text{untl}})) \rightarrow \\ &\rightarrow \mathbf{C}^{\text{glob}} \otimes \Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \underline{\text{D-mod}}(\text{Ran}^{\text{lax}})) = \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(\text{Ran}^{\text{untl}}), \end{aligned}$$

and according to Remark H.2.4, the functor

$$\mathbf{C}^{\text{glob}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \Gamma^{\text{strict}}(\text{Ran}^{\text{untl}}, \underline{\text{D-mod}}(\text{Ran}^{\text{untl}}))$$

is an equivalence.

H.3. Non-unitality vs independence. Let $\mathbf{C}^{\text{loc, untl}}$ be as in the previous subsection.

In this subsection we will utilize the contractibility of the Ran space to explain the relation between the “independent” category $\mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}}$ and the non-unital version $\mathbf{C}_{\text{Ran}}^{\text{loc}}$.

H.3.1. We claim:

Proposition H.3.2. *The composite functor*

$$\mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}} \xrightarrow{\text{emb, indep}} \mathbf{C}_{\text{Ran}^{\text{untl}}}^{\text{loc}} \xrightarrow{\mathbf{t}^!} \mathbf{C}_{\text{Ran}}^{\text{loc}}$$

is fully faithful.

Proof. The assertion of the proposition amounts to the following. Let us be given two objects

$$\mathbf{c}', \mathbf{c}'' \in \mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}} \subset \mathbf{C}_{\text{Ran}^{\text{untl}}}^{\text{loc}},$$

and a map

$$\mathbf{t}^!(\mathbf{c}') \xrightarrow{\phi} \mathbf{t}^!(\mathbf{c}'').$$

We need to show that this map can be uniquely upgraded to a map

$$\mathbf{c}' \xrightarrow{\phi_{\text{untl}}} \mathbf{c}''.$$

This amounts to the following. Fix $S \in \text{Sch}^{\text{aff}}$ and let us be given a map

$$(\underline{x}_1 \rightarrow \underline{x}_2) \in \text{Maps}(S, \text{Ran}^{\text{untl}}).$$

We need to equip the following diagram (taking place in $\mathbf{C}_{S, \underline{x}_2}^{\text{loc}}$)

$$\begin{array}{ccc} \mathbf{c}'_{S, \underline{x}_2} & \xrightarrow{\phi_{S, \underline{x}_2}} & \mathbf{c}''_{S, \underline{x}_2} \\ \uparrow & & \uparrow \\ \text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2}(\mathbf{c}'_{S, \underline{x}_1}) & \xrightarrow{\text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2}(\phi_{S, \underline{x}_1})} & \text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2}(\mathbf{c}''_{S, \underline{x}_1}) \end{array}$$

(where the vertical arrows are given by the structure of objects of $\mathbf{C}_{\text{Ran}^{\text{untl}}}^{\text{loc}}$ on $\mathbf{c}'$ and $\mathbf{c}''$, respectively) with a datum of commutativity.

The datum of commutativity of the above diagram is equivalent to the datum of commutativity of the following diagram (taking place in $\mathbf{C}_{S, \underline{x}_1}$):

$$(H.14) \quad \begin{array}{ccc} \text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2}^R(\mathbf{c}'_{S, \underline{x}_2}) & \xrightarrow{\text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2}^R(\phi_{S, \underline{x}_2})} & \text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2}^R(\mathbf{c}''_{S, \underline{x}_2}) \\ \uparrow & \xrightarrow{\phi_{S, \underline{x}_1}} & \uparrow \\ \mathbf{c}'_{S, \underline{x}_1} & & \mathbf{c}''_{S, \underline{x}_1} \end{array}$$

Note the vertical arrows in (H.14) are isomorphisms, by the assumption that $\mathbf{c}', \mathbf{c}'' \in \mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\text{loc}}$. Thus, we can view both circuits in (H.14) as an object of $\mathcal{H}om_{\mathbf{C}_{S, \underline{x}_1}^{\text{loc}}}(\mathbf{c}'_{S, \underline{x}_1}, \mathbf{c}''_{S, \underline{x}_1})$.

Thus, letting $\underline{x}_2$ vary, we can view the clockwise circuit in (H.14) as a map

$$(H.15) \quad (\text{Id} \otimes \text{pr}_{\text{small}, S}^!)(\mathbf{c}'_{S, \underline{x}_1}) \rightarrow (\text{Id} \otimes \text{pr}_{\text{small}, S}^!)(\mathbf{c}''_{S, \underline{x}_1})$$

in

$$\mathbf{C}_{S, \underline{x}_1}^{\text{loc}} \otimes_{\text{D-mod}(S)} \text{D-mod}(S_{\underline{x}_1}^{\subseteq}).$$

Now, the universal homological contractibility of the map

$$\text{pr}_{\text{small}, S} : S_{\underline{x}_1}^{\subseteq} \rightarrow S$$

implies that the functor

$$\text{pr}_{\text{small}, S}^! : \text{D-mod}(S) \rightarrow \text{D-mod}(S_{\underline{x}_1}^{\subseteq})$$

is fully faithful, and hence so is

$$\text{Id} \otimes \text{pr}_{\text{small}, S}^! : \mathbf{C}_{S, \underline{x}_1}^{\text{loc}} \rightarrow \mathbf{C}_{S, \underline{x}_1}^{\text{loc}} \otimes_{\text{D-mod}(S)} \text{D-mod}(S_{\underline{x}_1}^{\subseteq}).$$

Hence, the space

$$\mathcal{H}om_{\mathbf{C}_{S, \underline{x}_1}^{\text{loc}} \otimes_{\text{D-mod}(S)} \text{D-mod}(S_{\underline{x}_1}^{\subseteq})}((\text{Id} \otimes \text{pr}_{\text{small}, S}^!)(\mathbf{c}'_{S, \underline{x}_1}), (\text{Id} \otimes \text{pr}_{\text{small}, S}^!)(\mathbf{c}''_{S, \underline{x}_1}))$$

is isomorphic to $\mathcal{H}om_{\mathbf{C}_{S, \underline{x}_1}^{\text{loc}}}(\mathbf{c}'_{S, \underline{x}_1}, \mathbf{c}''_{S, \underline{x}_1})$, with the mutually inverse isomorphisms given by the functors $\text{Id} \otimes \text{pr}_{\text{small}, S}^!$ and $\text{Id} \otimes \text{diag}_S^!$, respectively.

This equips the family of diagrams (H.14) with a unique datum of commutativity as $\underline{x}_2$ varies over $S_{\underline{x}_1}^{\subseteq}$. □

H.3.3. Combining Proposition H.3.2 and Lemma H.1.13, we obtain:

Corollary H.3.4. *The functor from the category of strict functors of crystals of categories*

$$\underline{\mathbf{F}}^{\text{untl}} : \underline{\mathbf{C}}^{\text{loc}, \text{untl}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \underline{\text{D-mod}}(\text{Ran}^{\text{untl}})$$

to the category of functors

$$\underline{\mathbf{F}} : \underline{\mathbf{C}}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \underline{\text{D-mod}}(\text{Ran}),$$

given by restriction along $\text{Ran} \rightarrow \text{Ran}^{\text{untl}}$, is fully faithful.

Unwinding the definitions, we obtain that the essential image of the above fully faithful functor

$$\text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$$

consists of objects that have a global unitality property.

I.e., this proves Proposition 11.8.8.

H.3.5. As another immediate corollary of Proposition H.3.2 we obtain:

Corollary H.3.6. *The natural transformation*

$$\text{emb.indep}^L \circ \text{ot}_! \circ \text{t}^! \rightarrow \text{emb.indep}^L$$

is an isomorphism.

H.3.7. Finally, we record:

Corollary H.3.8. *The composite functor*

$$\mathbf{C}_{\text{Ran}} \xrightarrow{\text{t}_!} \mathbf{C}_{\text{Ran}^{\text{untl}}} \xrightarrow{\text{emb.indep}^L} \mathbf{C}_{\text{Ran}^{\text{untl}}, \text{indep}}$$

is a localization.

H.4. Sheaves of monoidal categories on the unital Ran space. Let $\mathbf{A}^{\text{loc}, \text{untl}}$ be a sheaf of unital monoidal categories over Ran^{untl} . We will assume that $\mathbf{A}^{\text{loc}, \text{untl}}$ satisfies the condition from Sect. H.1.4.

We will also assume that the monoidal operation admits a right adjoint, which is a strict functor between sheaves of categories.

In this subsection we will study the categories

$$(H.16) \quad \mathbf{A}_{\text{Ran}^{\text{untl}}} := \Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc}, \text{untl}}) \text{ and } \mathbf{A}_{\text{Ran}^{\text{untl}}, \text{indep}}.$$

We will equip them with monoidal structures, and study the interactions between them.

H.4.1. *Example.* An example of such $\mathbf{A}^{\text{loc}, \text{untl}}$ is the sheaf of unital monoidal categories over Ran^{untl} attached to a unital monoidal factorization category $\mathbf{A}$.

An important example of such an $\mathbf{A}$ is Sph_G .

H.4.2. Another example is the *symmetric* monoidal factorization category $\mathbf{A}$ associated to a crystal of symmetric monoidal categories over X (this is a categorical counterpart of the procedure from Sect. C.8).

An example of this is the constant crystal of symmetric monoidal categories over X with fiber $\text{Rep}(\check{G})$.

Remark H.4.3. Note that being a sheaf of *unital* monoidal categories over Ran^{untl} automatically imposes a condition on the compatibility between the monoidal unit and a structure of sheaf of categories:

The monoidal unit

$$\mathbf{1}_{\mathbf{A}, \text{Ran}} \in \Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc}, \text{untl}}) = \mathbf{A}_{\text{Ran}^{\text{untl}}}$$

belongs to

$$\Gamma^{\text{strict}}(\text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc}, \text{untl}}) \subset \Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc}, \text{untl}}).$$

H.4.4. Consider the map

$$\text{union} : \text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}} \rightarrow \text{Ran}^{\text{untl}}.$$

We have the 1-morphisms

$$p_1 \rightarrow \text{union} \leftarrow p_2$$

in the category of maps from $\text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}$ to Ran^{untl} . From here we obtain functors

$$p_1^*(\mathbf{A}^{\text{loc}, \text{untl}}) \rightarrow \text{union}^*(\mathbf{A}^{\text{loc}, \text{untl}}) \leftarrow p_2^*(\mathbf{A}^{\text{loc}, \text{untl}})$$

as crystals of categories over $\text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}$, and hence a functor

$$(H.17) \quad \mathbf{A}^{\text{loc}, \text{untl}} \boxtimes \mathbf{A}^{\text{loc}, \text{untl}} \rightarrow \text{union}^*(\mathbf{A}^{\text{loc}, \text{untl}} \otimes \mathbf{A}^{\text{loc}, \text{untl}}).$$

Combining with the monoidal operation on $\mathbf{A}^{\text{loc}, \text{untl}}$, we obtain a functor

$$(H.18) \quad \mathbf{A}^{\text{loc}, \text{untl}} \boxtimes \mathbf{A}^{\text{loc}, \text{untl}} \rightarrow \text{union}^*(\mathbf{A}^{\text{loc}, \text{untl}}).$$

Applying $\Gamma^{\text{lax}}(\text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}, -)$, from (H.18) we obtain a functor

$$(H.19) \quad \mathbf{A}_{\text{Ran}^{\text{untl}}} \otimes \mathbf{A}_{\text{Ran}^{\text{untl}}} \rightarrow \Gamma^{\text{lax}}(\text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}, \text{union}^*(\mathbf{A}^{\text{loc}, \text{untl}})).$$

Finally, composing (H.19) with the functor

$$\text{union}^! : \Gamma^{\text{lax}}(\text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}, \text{union}^*(\mathbf{A}^{\text{loc, untl}})) \rightarrow \Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc, untl}}),$$

left adjoint to $\text{union}^!$ (it exists thanks to Corollary C.4.10), we obtain a functor

$$(H.20) \quad \mathbf{A}_{\text{Ran}^{\text{untl}}} \otimes \mathbf{A}_{\text{Ran}^{\text{untl}}} \rightarrow \mathbf{A}_{\text{Ran}^{\text{untl}}}.$$

We will denote the resulting binary operation on $\mathbf{A}_{\text{Ran}^{\text{untl}}}$ by $\star$, and will refer to it as “convolution”.

H.4.5. The above binary operation extends to a monoidal structure on $\mathbf{A}_{\text{Ran}^{\text{untl}}}$, which we will refer to us the *convolution* monoidal structure. We will denote $\mathbf{A}_{\text{Ran}^{\text{untl}}}$, viewed as a monoidal category equipped with the convolution structure by $\mathbf{A}_{\text{Ran}^{\text{untl}}}^*$.

H.4.6. Note now that the monoidal operation on $\mathbf{A}^{\text{loc, untl}}$ defines a *pointwise* monoidal structure on $\mathbf{A}_{\text{Ran}^{\text{untl}}}$.

We will denote $\mathbf{A}_{\text{Ran}^{\text{untl}}}$, viewed as a monoidal category equipped with the pointwise structure by $\mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes}$.

Note that $\mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes}$ is unital: its unit is the object $\mathbf{1}_{\text{Ran}^{\text{untl}}}$.

H.4.7. Unwinding the definitions, one obtains:

Lemma H.4.8. *The pointwise monoidal structure on $\mathbf{A}_{\text{Ran}^{\text{untl}}}$ descends to the quotient*

$$\mathbf{A}_{\text{Ran}^{\text{untl}}} \twoheadrightarrow \mathbf{A}_{\text{Ran}^{\text{untl}}, \text{indep}}.$$

In what follows we will consider $\mathbf{A}_{\text{Ran}^{\text{untl}}, \text{indep}}$ as a monoidal category, with the monoidal structure furnished by Lemma H.4.8.

Since $\mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes}$ is unital, we obtain that so is $\mathbf{A}_{\text{Ran}^{\text{untl}}, \text{indep}}$.

H.4.9. Note now that the natural transformation

$$(\Delta_{\text{Ran}^{\text{untl}}})^! \simeq \text{union}^! \circ (\Delta_{\text{Ran}^{\text{untl}}})^! \circ (\Delta_{\text{Ran}^{\text{untl}}})^! \rightarrow \text{union}^!$$

defines on the identity functor on $\mathbf{A}_{\text{Ran}^{\text{untl}}}$ a structure of *right-lax* monoidal functor

$$(H.21) \quad \mathbf{A}_{\text{Ran}^{\text{untl}}}^* \rightarrow \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes}.$$

H.4.10. However, we claim:

Lemma H.4.11. *The right-lax monoidal structure on the functor (H.21) is strict.*

Proof. Let $\underline{\mathbf{C}}$ denote the crystal of categories

$$\text{union}^*(\mathbf{A}^{\text{loc, untl}} \otimes \mathbf{A}^{\text{loc, untl}})$$

on $\text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}$.

Let ins. union denote the functor (H.17); we will use the same notation for the induced functor

$$\mathbf{A}_{\text{Ran}^{\text{untl}}} \otimes \mathbf{A}_{\text{Ran}^{\text{untl}}} \simeq \Gamma^{\text{lax}}(\text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc, untl}} \boxtimes \mathbf{A}^{\text{loc, untl}}) \rightarrow \Gamma^{\text{lax}}(\text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}, \underline{\mathbf{C}}).$$

We have to show that for $\mathcal{F}_1, \mathcal{F}_2, \mathcal{F} \in \mathbf{A}_{\text{Ran}^{\text{untl}}}^*$, the map

$$(H.22) \quad \mathcal{H}om_{\Gamma^{\text{lax}}(\text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}, \underline{\mathbf{C}})}(\text{ins. union}(\mathcal{F}_1 \boxtimes \mathcal{F}_2), \text{union}^!(\text{mult}^R(\mathcal{F}))) \rightarrow \\ \rightarrow \mathcal{H}om_{\Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc, untl}})}(\text{mult}(\mathcal{F}_1 \otimes \mathcal{F}_2), \mathcal{F}),$$

is an isomorphism, where:

- mult denotes the functor

$$\Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc,untl}} \otimes \mathbf{A}^{\text{loc,untl}}) \rightarrow \Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc,untl}})$$

induced by the monoidal operation

$$\mathbf{A}^{\text{loc,untl}} \otimes \mathbf{A}^{\text{loc,untl}} \xrightarrow{\text{mult}} \mathbf{A}^{\text{loc,untl}},$$

- mult^R denotes the functor

$$\Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc,untl}}) \rightarrow \Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc,untl}} \otimes \mathbf{A}^{\text{loc,untl}})$$

induced by the functor

$$\mathbf{A}^{\text{loc,untl}} \xrightarrow{\text{mult}^R} \mathbf{A}^{\text{loc,untl}} \otimes \mathbf{A}^{\text{loc,untl}},$$

right adjoint to the monoidal operation;

- $\text{union}^!$ denotes the functor

$$\Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc,untl}} \otimes \mathbf{A}^{\text{loc,untl}}) \rightarrow \Gamma^{\text{lax}}(\text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}, \underline{\mathbf{C}});$$

- The map (H.22) is given by the composition

$$\begin{aligned} & \text{Hom}_{\Gamma^{\text{lax}}(\text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}, \underline{\mathbf{C}})}(\text{ins. union}(\mathcal{F}_1 \boxtimes \mathcal{F}_2), \text{union}^!(\text{mult}^R(\mathcal{F}))) \xrightarrow{\Delta_{\text{Ran}^{\text{untl}}}^!} \\ & \text{Hom}_{\Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \Delta_{\text{Ran}^{\text{untl}}}^*(\underline{\mathbf{C}}))}(\Delta_{\text{Ran}^{\text{untl}}}^!(\text{ins. union}(\mathcal{F}_1 \boxtimes \mathcal{F}_2)), \Delta_{\text{Ran}^{\text{untl}}}^!(\text{union}^!(\text{mult}^R(\mathcal{F})))) \simeq \\ & \simeq \text{Hom}_{\Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc,untl}} \otimes \mathbf{A}^{\text{loc,untl}})}(\mathcal{F}_1 \overset{!}{\otimes} \mathcal{F}_2, \text{mult}^R(\mathcal{F})) \simeq \\ & \simeq \text{Hom}_{\Gamma^{\text{lax}}(\text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc,untl}})}(\text{mult}(\mathcal{F}_1 \overset{!}{\otimes} \mathcal{F}_2), \mathcal{F}). \end{aligned}$$

Now the isomorphism assertion holds for any crystal of categories $\underline{\mathbf{C}}$ on $\text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}$, since the map

$$\Delta_{\text{Ran}^{\text{untl}}} : \text{Ran}^{\text{untl}} \rightarrow \text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}$$

is cofinal. □

H.4.12. Thanks to Lemma H.4.11, we do not need to distinguish between the two monoidal structures on $\mathbf{A}_{\text{Ran}^{\text{untl}}}$. Thus we will use the symbol $\mathbf{A}_{\text{Ran}^{\text{untl}}}$ unambiguously to refer to a tensor structure on $\mathbf{A}_{\text{Ran}^{\text{untl}}}$.

Yet we will sometimes use the symbols $\mathbf{A}_{\text{Ran}^{\text{untl}}}^*$ or $\mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes}$ to emphasize that we are thinking of the monoidal structure as convolution or pointwise tensor product, respectively.

H.4.13. Let $\mathbf{A}^{\text{loc,untl}}$ be as in Sect. H.4.2, i.e., it is associated to a crystal $\underline{\mathbf{A}}_X$ of symmetric monoidal categories on X . In this case, one can describe the corresponding (symmetric) monoidal category $\mathbf{A}_{\text{Ran}^{\text{untl}}, \text{indep}}$ explicitly.

Namely, as a DG category it is isomorphic to the colimit over the twisted arrows category of the category of finite non-empty sets (and arbitrary maps) of the functor that associates to

$$I_1 \xrightarrow{\phi} I_2$$

the category

$$(H.23) \quad \Gamma(X^{I_2}, \boxtimes_{i \in I_2} \underline{\mathbf{A}}_X^{\otimes \phi^{-1}(i_2)}).$$

The symmetric monoidal structure is given by the operation of disjoint union of finite sets, see [FraG, Sect. 2.2.1].

H.5. Sheaves of monoidal categories on the *non-unital* Ran space. We now consider the usual (i.e., non-unital) Ran space. For $\mathbf{A}^{\text{loc}, \text{unl}}$ as above, let $\mathbf{A}^{\text{loc}}$ denote its restriction along the map $\mathbf{t} : \text{Ran} \rightarrow \text{Ran}^{\text{unl}}$.

Denote

$$\mathbf{A}_{\text{Ran}} := \Gamma(\text{Ran}, \mathbf{A}^{\text{loc}}).$$

In this subsection we will endow $\mathbf{A}_{\text{Ran}}$ with monoidal structure(s) and study its interactions with the unital counterparts.

H.5.1. By a slight abuse of notation, we will use the same symbol union to denote the corresponding map

$$\text{Ran} \times \text{Ran} \rightarrow \text{Ran}.$$

Restricting (H.18) along

$$\mathbf{t} : \text{Ran} \rightarrow \text{Ran}_{\text{unl}}$$

we obtain a map of crystals of categories

$$(H.24) \quad \mathbf{A}^{\text{loc}} \boxtimes \mathbf{A}^{\text{loc}} \rightarrow \text{union}^*(\mathbf{A}^{\text{loc}})$$

on $\text{Ran} \times \text{Ran}$.

Since the map union is pseudo-proper, the functor (H.24) induces a functor

$$\mathbf{A}_{\text{Ran}} \otimes \mathbf{A}_{\text{Ran}} \xrightarrow{*} \mathbf{A}_{\text{Ran}}.$$

H.5.2. The above binary operation extends to a monoidal structure on $\mathbf{A}_{\text{Ran}}$, which we will refer to as the *convolution* monoidal structure. We will denote $\mathbf{A}_{\text{Ran}}$, viewed as a monoidal category equipped with the convolution monoidal structure by $\mathbf{A}_{\text{Ran}}^*$.

H.5.3. By construction, the functor

$$\mathbf{t}_! : \mathbf{A}_{\text{Ran}} \rightarrow \mathbf{A}_{\text{Ran}^{\text{unl}}}$$

has a monoidal structure, when we consider both as equipped with the convolution monoidal structure.

In particular, we obtain that the functor

$$(H.25) \quad \text{emb. indep}^L \circ \mathbf{t}_! : \mathbf{A}_{\text{Ran}}^* \rightarrow \mathbf{A}_{\text{Ran}^{\text{unl}}, \text{indep}}$$

acquires a monoidal structure.

Combining with Corollary H.3.8, we obtain that the functor (H.25) is a monoidal localization.

H.5.4. Consider now the functor

$$\mathbf{t}^! : \mathbf{A}_{\text{Ran}^{\text{unl}}} \rightarrow \mathbf{A}_{\text{Ran}}.$$

Being the right adjoint of a monoidal functor, the functor $\mathbf{t}^!$ acquires a right-lax monoidal structure as a functor

$$\mathbf{A}_{\text{Ran}^{\text{unl}}}^* \rightarrow \mathbf{A}_{\text{Ran}}^*.$$

Note that the natural transformation

$$(H.26) \quad \text{emb. indep}^L \circ \mathbf{t}_! \circ \mathbf{t}^! \rightarrow \text{emb. indep}^L$$

has a natural right-lax monoidal structure. However, from Corollary H.3.6 we obtain that the natural transformation (as right-lax monoidal functors) is an isomorphism.

In particular, we obtain that the right-lax monoidal structure on (H.26) is strict.

H.5.5. As in Sect. H.4.6, we can also consider the $\overset{!}{\otimes}$ -monoidal structure on $\mathbf{A}_{\text{Ran}}$. We will denote $\mathbf{A}_{\text{Ran}}$, viewed as a monoidal category equipped with the pointwise structure by $\mathbf{A}_{\text{Ran}}^{\overset{!}{\otimes}}$.

As in Sect. H.4.6, the identity functor on $\mathbf{A}_{\text{Ran}}$ has a right-lax monoidal structure, when viewed as a functor

$$\mathbf{A}_{\text{Ran}}^* \rightarrow \mathbf{A}_{\text{Ran}}^{\overset{!}{\otimes}}.$$

H.5.6. The functor $t^!$ is monoidal, when viewed as a functor

$$\mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes} \rightarrow \mathbf{A}_{\text{Ran}}^{\otimes}.$$

Hence, the functor

$$t_! : \mathbf{A}_{\text{Ran}}^{\otimes} \rightarrow \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes}$$

acquires a left-lax monoidal structure, when viewed as a functor

$$\mathbf{A}_{\text{Ran}}^{\otimes} \rightarrow \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes}.$$

H.5.7. Let

$$\mathbf{A}_{\text{Ran}}^{\text{almost-untl}} \subset \mathbf{A}_{\text{Ran}}$$

be the full subcategory generated by the essential image of the functor $t^!$.

It is easy to see that it is preserved by the $\otimes$ monoidal operation, and hence it acquires a monoidal structure.

H.5.8. We claim:

Lemma H.5.9. *The left-lax monoidal structure on the functor*

$$\mathbf{A}_{\text{Ran}}^{\otimes} \xrightarrow{t_!} \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes} \xrightarrow{\text{emb.indep}^L} \mathbf{A}_{\text{Ran}^{\text{untl}}, \text{indep}}$$

becomes strict when restricted to $\mathbf{A}_{\text{Ran}}^{\text{almost-untl}}$.

Proof. It suffices to show that the left-lax monoidal structure on the functor

$$\mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes} \xrightarrow{t^!} \mathbf{A}_{\text{Ran}}^{\otimes} \xrightarrow{t_!} \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes} \xrightarrow{\text{emb.indep}^L} \mathbf{A}_{\text{Ran}^{\text{untl}}, \text{indep}}$$

is strict.

The assertion follows now from Corollary H.3.8, which implies that the above composition is isomorphic to

$$\mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes} \xrightarrow{\text{emb.indep}^L} \mathbf{A}_{\text{Ran}^{\text{untl}}, \text{indep}}$$

as a left-lax monoidal functor. □

H.5.10. Let $\mathbf{A}$ be as in Sect. H.4.13. In this case, we can also describe the category $\mathbf{A}_{\text{Ran}}^*$ explicitly.

It is given by the colimit of the functor (H.23), with the only difference that we take the twisted arrows category of the category of finite non-empty sets and *surjective* maps.

The functor

$$t_! : \mathbf{A}_{\text{Ran}}^* \rightarrow \mathbf{A}_{\text{Ran}^{\text{untl}}}^*$$

is given by embedding the index categories one into the other.

H.6. Local and integrated monoidal actions.

H.6.1. Let $\mathbf{A}^{\text{loc}, \text{untl}}$ be as above. Let $\mathbf{A}^{\text{loc}}$ denote the restriction of $\mathbf{A}^{\text{loc}, \text{untl}}$ along $t : \text{Ran} \rightarrow \text{Ran}^{\text{untl}}$.

Let $\mathbf{D}$ be a DG category. We give the following definitions:

- A local action of $\mathbf{A}^{\text{loc}}$ on $\mathbf{D}$ is a (unital) action of the crystal of monoidal categories $\mathbf{A}^{\text{loc}}$ on $\mathbf{D} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran})$;
- A local lax-Ran-unital action of $\mathbf{A}^{\text{loc}, \text{untl}}$ on $\mathbf{D}$ is a (unital) action of the crystal of monoidal categories $\mathbf{A}^{\text{loc}, \text{untl}}$ on $\mathbf{D} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran}^{\text{untl}})$, in the 2-category of crystals of categories and right-lax functors between them;
- A local Ran-unital action of $\mathbf{A}^{\text{loc}, \text{untl}}$ on $\mathbf{D}$ is a (unital) action of the crystal of monoidal categories $\mathbf{A}^{\text{loc}, \text{untl}}$ on $\mathbf{D} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran}^{\text{untl}})$, in the 2-category of crystals of categories and strict functors between them.

H.6.2. At the pointwise level, a local action of $\mathbf{A}^{\text{loc}}$ on $\mathbf{D}$ yields an action of the monoidal category $\mathbf{A}_{\underline{x}}$ on $\mathbf{D}$ for every $\underline{x} \in \text{Ran}$, denoted

$$\mathbf{a}_{\underline{x}}, \mathbf{d} \mapsto \mathbf{a} \cdot \mathbf{d}, \quad \mathbf{a}_{\underline{x}} \in \mathbf{A}_{\underline{x}}, \mathbf{d} \in \mathbf{D}.$$

A lax-Ran-unital structure on such an action is a natural transformation

$$\mathbf{a}_{\underline{x}_1} \cdot \mathbf{d} \rightarrow \text{ins. unit}_{\underline{x}_1 \subseteq \underline{x}_2}(\mathbf{a}_{\underline{x}_1}) \cdot \mathbf{d}, \quad \mathbf{a}_{\underline{x}_1} \in \mathbf{A}_{\underline{x}_1}, \mathbf{d} \in \mathbf{D}.$$

A lax-Ran-unital structure is strict if the above natural transformation is an isomorphism.

H.6.3. Thus, we obtain the 2-categories

$$\mathbf{A}^{\text{loc}}\text{-mod}, \mathbf{A}^{\text{loc, untl}}\text{-mod}^{\text{lax}} \text{ and } \mathbf{A}^{\text{loc, untl}}\text{-mod}.$$

We have a fully faithful functor

$$\mathbf{A}^{\text{loc, untl}}\text{-mod} \hookrightarrow \mathbf{A}^{\text{loc, untl}}\text{-mod}^{\text{lax}}$$

and a forgetful functor

$$\mathbf{A}^{\text{loc, untl}}\text{-mod}^{\text{lax}} \rightarrow \mathbf{A}^{\text{loc}}\text{-mod}.$$

Note, however, that from Corollary H.3.4, we obtain:

Corollary H.6.4. *The composite functor*

$$(H.27) \quad \mathbf{A}^{\text{loc, untl}}\text{-mod} \hookrightarrow \mathbf{A}^{\text{loc, untl}}\text{-mod}^{\text{lax}} \rightarrow \mathbf{A}^{\text{loc}}\text{-mod}$$

is fully faithful.

H.6.5. Consider now the (unital) monoidal category $\mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes}$. We claim that given a local lax-Ran-unital action of $\mathbf{A}^{\text{loc, untl}}$ on $\mathbf{D}$, we can construct a (unital) action of $\mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes}$ on $\mathbf{D}$.

Indeed, this follows from the fact that the functor

$$C_c(\text{Ran}^{\text{untl}}, -) : \mathbf{D}\text{-mod}(\text{Ran}^{\text{untl}}) \rightarrow \mathbf{Vect}$$

is symmetric monoidal.

Explicitly, given $\mathbf{a} \in \mathbf{A}_{\text{Ran}^{\text{untl}}}$, the action is given by

$$\begin{aligned} \mathbf{D} &\xrightarrow{- \otimes \omega_{\text{Ran}^{\text{untl}}}} \mathbf{D} \otimes \mathbf{D}\text{-mod}(\text{Ran}^{\text{untl}})^{\text{strict}} \hookrightarrow \\ &\rightarrow \mathbf{D} \otimes \mathbf{D}\text{-mod}(\text{Ran}^{\text{untl}}) \xrightarrow{\mathbf{a} \cdot -} \mathbf{D} \otimes \mathbf{D}\text{-mod}(\text{Ran}^{\text{untl}}) \xrightarrow{\text{Id} \otimes C_c(\text{Ran}^{\text{untl}}, -)} \mathbf{D}. \end{aligned}$$

H.6.6. From Lemma H.1.13 we obtain:

Corollary H.6.7. *The composite functor*

$$\mathbf{A}^{\text{loc, untl}}\text{-mod} \hookrightarrow \mathbf{A}^{\text{loc, untl}}\text{-mod}^{\text{lax}} \rightarrow \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes}\text{-mod}$$

is fully faithful with essential image being

$$\mathbf{A}_{\text{Ran}^{\text{untl}}, \text{indep}}\text{-mod} \subset \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes}\text{-mod}.$$

H.6.8. Precomposing the construction in Sect. H.6.5 with the monoidal functor

$$\mathbf{A}_{\text{Ran}}^* \xrightarrow{t_!} \mathbf{A}_{\text{Ran}^{\text{untl}}}^* \simeq \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes \dagger}$$

we obtain that for

$$\mathbf{D} \in \mathbf{A}^{\text{loc,untl}}\text{-mod}^{\text{lax}}$$

we have an action of $\mathbf{A}_{\text{Ran}}^*$ on $\mathbf{D}$.

Remark H.6.9. Unwinding the definitions, for $\mathbf{a} \in \mathbf{A}_{\text{Ran}}$, its action on $\mathbf{D}$ is given by the composition

$$\mathbf{D} \xrightarrow{- \otimes \omega_{\text{Ran}}} \mathbf{D} \otimes \text{D-mod}(\text{Ran}) \xrightarrow{\mathbf{a}^-} \mathbf{D} \otimes \text{D-mod}(\text{Ran}) \xrightarrow{\text{Id} \otimes C_c^*(\text{Ran}, -)} \mathbf{D}.$$

I.e., the binary operation only depends on structure on $\mathbf{D}$ of object of $\mathbf{A}^{\text{loc}}\text{-mod}$.

However, one would not be able to define either left-lax or right-lax structure on this binary operation without $\mathbf{D}$ being extended to an object of $\mathbf{A}^{\text{loc,untl}}\text{-mod}^{\text{lax}}$.

H.6.10. From Sect. H.5.3 and Corollary H.6.7 we obtain:

Corollary H.6.11. *The composite functor*

$$\mathbf{A}^{\text{loc,untl}}\text{-mod} \hookrightarrow \mathbf{A}^{\text{loc,untl}}\text{-mod}^{\text{lax}} \rightarrow \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes \dagger}\text{-mod} \simeq \mathbf{A}_{\text{Ran}^{\text{untl}}}^*\text{-mod} \rightarrow \mathbf{A}_{\text{Ran}}^*\text{-mod}$$

is fully faithful with the essential image being

$$\mathbf{A}_{\text{Ran}^{\text{untl}}, \text{indep}}\text{-mod} \subset \mathbf{A}_{\text{Ran}}^*\text{-mod}.$$

H.7. Local actions with parameters.

H.7.1. Let now S be an affine scheme mapping to Ran . The discussion in Sects. H.4-H.6 applies when we replace Ran (resp., Ran^{untl}) by $S^{\subseteq}$ (resp., $S^{\subseteq, \text{untl}}$).

Note that $S^{\subseteq}$ and $S^{\subseteq, \text{untl}}$ are pseudo-proper *relative* to S , so Sect. C.4.24 applies.

H.7.2. In particular, one can consider the $\text{D-mod}(S)$ -linear monoidal categories

$$\mathbf{A}_{S^{\subseteq, \text{untl}}}^*, \mathbf{A}_{S^{\subseteq, \text{untl}}}^{\otimes \dagger}, \mathbf{A}_{S^{\subseteq}}^*, \mathbf{A}_{S^{\subseteq}}^{\otimes \dagger} \text{ and } \mathbf{A}_{S^{\subseteq, \text{untl}}, \text{indep}},$$

equipped with the monoidal functors

$$\begin{aligned} \mathbf{A}_{S^{\subseteq, \text{untl}}}^* &\xrightarrow{\sim} \mathbf{A}_{S^{\subseteq, \text{untl}}}^{\otimes \dagger}, \\ \mathbf{A}_{S^{\subseteq}}^* &\xrightarrow{t_!} \mathbf{A}_{S^{\subseteq, \text{untl}}}^*, \\ \mathbf{A}_{S^{\subseteq, \text{untl}}}^{\otimes \dagger} &\xrightarrow{t^!} \mathbf{A}_{S^{\subseteq, \text{untl}}}^{\otimes \dagger}, \\ \mathbf{A}_{S^{\subseteq, \text{untl}}}^* &\xrightarrow{\text{embed.indep}^L} \mathbf{A}_{S^{\subseteq, \text{untl}}, \text{indep}}, \\ \mathbf{A}_{S^{\subseteq}}^* &\xrightarrow{\text{embed.indep}^L \circ t_!} \mathbf{A}_{S^{\subseteq, \text{untl}}, \text{indep}}, \end{aligned}$$

where the last two functors are monoidal localizations.

H.7.3. We also have the corresponding notions of local action, so we have the 2-categories

$$\mathbf{A}_S^{\text{loc}}\text{-mod}, \mathbf{A}_S^{\text{loc,untl}}\text{-mod}^{\text{lax}} \text{ and } \mathbf{A}_S^{\text{loc,untl}}\text{-mod},$$

the fully faithful embedding

$$\mathbf{A}_S^{\text{loc,untl}}\text{-mod} \hookrightarrow \mathbf{A}_S^{\text{loc,untl}}\text{-mod}^{\text{lax}}$$

and a forgetful functor

$$\mathbf{A}_S^{\text{loc,untl}}\text{-mod}^{\text{lax}} \rightarrow \mathbf{A}_S^{\text{loc}}\text{-mod},$$

so that composition

$$\mathbf{A}_S^{\text{loc,untl}}\text{-mod} \hookrightarrow \mathbf{A}_S^{\text{loc,untl}}\text{-mod}^{\text{lax}} \rightarrow \mathbf{A}_S^{\text{loc}}\text{-mod}$$

is fully faithful.

H.7.4. In addition, we have the commutative diagrams

$$\begin{array}{ccc} \mathbf{A}_S^{\text{loc, untl}}\text{-mod} & \longrightarrow & \mathbf{A}_S^{\text{loc, untl}}\text{-mod}^{\text{lax}} \\ \sim \downarrow & & \downarrow \\ \mathbf{A}_{S^{\subseteq, \text{untl}}, \text{indep}}\text{-mod} & \longrightarrow & \mathbf{A}_{S^{\subseteq, \text{untl}}}^*\text{-mod} \end{array}$$

H.7.5. We now claim:

Proposition H.7.6. *The functor*

$$\text{D-mod}(S) \otimes \mathbf{A}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\otimes} \xrightarrow{(\text{pr}_{\text{small}, S}^{\text{untl}})^! \otimes (\text{pr}_{\text{big}}^{\text{untl}})^!} \mathbf{A}_{S^{\subseteq, \text{untl}}, \text{indep}}^{\otimes}$$

is an equivalence.

Proof. Note that operation of union defines a map

$$S \times \text{Ran}^{\text{untl}} \xrightarrow{\text{union}_S} S^{\subseteq, \text{untl}}.$$

Hence, we obtain a functor

$$\mathbf{A}_{S^{\subseteq, \text{untl}}, \text{indep}}^{\otimes} = \Gamma^{\text{lax}}(S^{\subseteq, \text{untl}}, \mathbf{A}^{\text{loc, untl}}) \xrightarrow{\text{union}_S^!} \Gamma^{\text{lax}}(S \times \text{Ran}^{\text{untl}}, \mathbf{A}^{\text{loc, untl}}) \simeq \text{D-mod}(S) \otimes \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes},$$

and it is easy to see that it sends

$$\mathbf{A}_{S^{\subseteq, \text{untl}}, \text{indep}}^{\otimes} \rightarrow \text{D-mod}(S) \otimes \mathbf{A}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\otimes}.$$

The composition

$$S^{\subseteq, \text{untl}} \xrightarrow{\text{pr}_{\text{small}, S}^{\text{untl}} \times \text{pr}_{\text{big}}^{\text{untl}}} S \times \text{Ran}^{\text{untl}} \xrightarrow{\text{union}_S} S^{\subseteq, \text{untl}}$$

is the identity map. Hence, the composition

$$\mathbf{A}_{S^{\subseteq, \text{untl}}, \text{indep}}^{\otimes} \xrightarrow{\text{union}_S^!} \text{D-mod}(S) \otimes \mathbf{A}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\otimes} \xrightarrow{(\text{pr}_{\text{small}, S}^{\text{untl}})^! \otimes (\text{pr}_{\text{big}}^{\text{untl}})^!} \mathbf{A}_{S^{\subseteq, \text{untl}}, \text{indep}}^{\otimes}$$

is the identity functor.

The composition

$$S \times \text{Ran}^{\text{untl}} \xrightarrow{\text{union}_S} S^{\subseteq, \text{untl}} \xrightarrow{\text{pr}_{\text{small}, S}^{\text{untl}} \times \text{pr}_{\text{big}}^{\text{untl}}} S \times \text{Ran}^{\text{untl}}$$

receives a 1-morphism from the identity map. This 1-morphism defines a natural transformation from the identity endofunctor on $\text{D-mod}(S) \otimes \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes}$ to the composition

$$\text{D-mod}(S) \otimes \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes} \xrightarrow{(\text{pr}_{\text{small}, S}^{\text{untl}} \times \text{pr}_{\text{big}}^{\text{untl}})^!} \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes} \xrightarrow{\text{union}_S^!} \text{D-mod}(S) \otimes \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes}.$$

However, this natural transformation is an isomorphism when restricted to $\text{D-mod}(S) \otimes \mathbf{A}_{\text{Ran}^{\text{untl}}}^{\otimes}$, by the definition of this subcategory. $\square$

H.7.7. From Proposition H.7.6 we obtain:

Corollary H.7.8. *For a $\text{D-mod}(S)$ -module category $\mathbf{D}$, pullback along $\text{pr}_{\text{big}}^{\text{untl}} : S^{\subseteq, \text{untl}} \rightarrow \text{Ran}^{\text{untl}}$ defines an equivalence between the following data:*

- (i) *A local Ran-unital action of $\mathbf{A}_S^{\text{loc, untl}}$ on $\underline{\text{D-mod}}(S^{\subseteq, \text{untl}}) \otimes_{\text{D-mod}(S)} \mathbf{D}$;*
- (ii) *A local Ran-unital action of $\mathbf{A}^{\text{loc, untl}}$ on $\mathbf{D}$, compatible with the $\text{D-mod}(S)$ -action.*
- (ii') *An action on $\mathbf{D}$ of the monoidal category $\mathbf{A}_{\text{Ran}^{\text{untl}}, \text{indep}}^{\otimes}$, compatible with the $\text{D-mod}(S)$ -action.*

APPENDIX I. THE INTEGRATION FUNCTOR IN THE NON-UNITAL SETTING

In this appendix, we give a categorical meaning to the functor (11.15)

$$\int \text{ins. unit} : \text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}).$$

using the notion of *left-lax* (a.k.a. left-lax) unital structure on a local-to-global functor $\underline{\mathbf{F}}$. As a byproduct, we provide a proof to Proposition 11.8.8.

I.1. A left-lax unital structure on a local-to-global functor.

I.1.1. In this subsection, we describe the notion of a left-lax unital structure on a local-to-global functor $\underline{\mathbf{F}}$ in concrete words. The precise definition will be given in Sect. I.4. Also, we explain why left-lax unital structures provide categorical meaning to the functor $\int \text{ins. unit}$.

I.1.2. Let $\mathcal{Z}$ be a space, and let $\underline{x}_1 \xrightarrow{\alpha} \underline{x}_2$ be a morphism in the category $\text{Maps}(\mathcal{Z}, \text{Ran}^{\text{un}^{\text{t}}})$. Recall (see Sect. 11.3.2) that a *lax* unital structure on $\underline{\mathbf{F}}$ means there is a natural transformation

$$\mathbf{F}_{\mathcal{Z}, \underline{x}_1} \rightarrow \mathbf{F}_{\mathcal{Z}, \underline{x}_2} \circ \mathbf{C}_{\alpha}^{\text{loc}}$$

as functors $\mathbf{C}_{\mathcal{Z}, \underline{x}_1}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(\mathcal{Z})$. Then a *left-lax* unital structure on $\underline{\mathbf{F}}$ means there is a natural transformation of the opposite direction, i.e.,

$$\mathbf{F}_{\mathcal{Z}, \underline{x}_2} \circ \mathbf{C}_{\alpha}^{\text{loc}} \rightarrow \mathbf{F}_{\mathcal{Z}, \underline{x}_1}.$$

I.1.3. As in Sect. 11.3.4, using the prestack $\mathcal{Z}^{\subseteq}$, we can rewrite the datum of these natural transformations as a natural transformation

$$(I.1) \quad \mathbf{F}_{\mathcal{Z}^{\subseteq}} \circ \text{ins. unit}_{\mathcal{Z}} \rightarrow (\text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})^{\dagger}) \circ \mathbf{F}_{\mathcal{Z}}$$

as functors

$$\mathbf{C}_{\mathcal{Z}^{\subseteq}}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \text{D-mod}(\mathcal{Z}^{\subseteq})$$

which is supposed to be equipped with a datum of associativity.

I.1.4. Using the adjunction $((\text{pr}_{\text{small}, \mathcal{Z}})^{\dagger}, (\text{pr}_{\text{small}, \mathcal{Z}})^{\dagger})$, knowing (I.1) is equivalent to knowing a natural transformation

$$(\text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})^{\dagger}) \circ \mathbf{F}_{\mathcal{Z}^{\subseteq}} \circ \text{ins. unit}_{\mathcal{Z}} \rightarrow \mathbf{F}_{\mathcal{Z}}.$$

Note that the LHS is exactly the functor $\mathbf{F}_{\mathcal{Z}}^{f \text{ ins. unit}}$ (see (11.15)). Hence a left-lax unital structure on $\underline{\mathbf{F}}$ means a natural transformation

$$\underline{\mathbf{F}}^{f \text{ ins. unit}} \rightarrow \underline{\mathbf{F}}$$

equipped with a datum of associativity. As we will see in the proof of Proposition I.1.7, the endofunctor

$$\int \text{ins. unit} : \text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \text{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$$

has a natural monad structure, and this datum of associativity says exactly that $\underline{\mathbf{F}}$ is a module for this monad.

I.1.5. In particular, for *any* local-to-global functor $\underline{\mathbf{F}}$, the functor $\underline{\mathbf{F}}^{f \text{ ins. unit}}$ is an induced (a.k.a. free) module for this monad. Therefore we obtain a left-lax unital structure on $\underline{\mathbf{F}}^{f \text{ ins. unit}}$, and this left-lax unital structure satisfies the following universal property. For any local-to-global functor $\underline{\mathbf{G}}$ equipped with a left-lax unital structure, knowing a (plain) natural transformation $\underline{\mathbf{F}} \rightarrow \underline{\mathbf{G}}$ is equivalent to knowing a natural transformation $\underline{\mathbf{F}}^{f \text{ ins. unit}} \rightarrow \underline{\mathbf{G}}$ compatible with the left-lax unital structure.

I.1.6. Let

$$\mathbf{Funct}^{\text{loc} \rightarrow \text{glob}, \text{left-lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$$

be the category of left-lax unital local-to-global functors. The above observation suggests the following result, which will be proved in I.5 (after we give a precise definition to left-lax unital functors).

Proposition I.1.7. *The forgetful functor*

$$\iota_{\text{left-lax} \rightarrow \text{all}} : \mathbf{Funct}^{\text{loc} \rightarrow \text{glob}, \text{left-lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \mathbf{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$$

has a left adjoint

$$\iota_{\text{left-lax} \rightarrow \text{all}}^L : \mathbf{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \mathbf{Funct}^{\text{loc} \rightarrow \text{glob}, \text{left-lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$$

that sends

$$\underline{F} \mapsto \underline{F}^{\text{ins.unit}}.$$

In particular, the latter has a natural left-lax unital structure.

I.2. Comparison with the integration functor in the lax unital setting.

I.2.1. Tautologically, there is a commutative square

$$(I.2) \quad \begin{array}{ccc} \mathbf{Funct}^{\text{loc} \rightarrow \text{glob}, \text{untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) & \xrightarrow[\iota_{\text{st} \rightarrow \text{lax}}]{\subset} & \mathbf{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \\ \downarrow \iota_{\text{st} \rightarrow \text{left-lax}} \subset & & \downarrow \iota_{\text{lax} \rightarrow \text{all}} \\ \mathbf{Funct}^{\text{loc} \rightarrow \text{glob}, \text{left-lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) & \xrightarrow[\iota_{\text{left-lax} \rightarrow \text{all}}]{} & \mathbf{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \end{array}$$

where each $\iota_?$ is a forgetful functor from a category of local-to-global functors equipped with certain unital structures to a category of such functors equipped with coarser structures. Here only $\iota_{\text{st} \rightarrow \text{lax}}$ and $\iota_{\text{st} \rightarrow \text{left-lax}}$ are fully faithful.

I.2.2. Recall in Sect. 11.5, we also constructed an adjoint pair:

$$\iota_{\text{st} \rightarrow \text{lax}}^L : \mathbf{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightleftarrows \mathbf{Funct}^{\text{loc} \rightarrow \text{glob}, \text{untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) : \iota_{\text{st} \rightarrow \text{lax}}$$

such that the left adjoint sends

$$\underline{F}^{\text{untl}} \mapsto \underline{F}^{\text{untl}, f \text{ ins.unit}}.$$

The following result, which will be proved in I.6, says this adjoint pair is compatible with that in Proposition I.1.7.

Proposition I.2.3. *The commutative square (I.2) is left adjointable along the horizontal direction, i.e., the Bech–Chevalley natural transformation from the clockwise circuit in the diagram below to the counter-clockwise circuit is invertible:*

$$(I.3) \quad \begin{array}{ccc} \mathbf{Funct}^{\text{loc} \rightarrow \text{glob}, \text{untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) & \xleftarrow[\iota_{\text{st} \rightarrow \text{lax}}^L]{} & \mathbf{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \\ \downarrow \iota_{\text{st} \rightarrow \text{left-lax}} \subset & & \downarrow \iota_{\text{lax} \rightarrow \text{all}} \\ \mathbf{Funct}^{\text{loc} \rightarrow \text{glob}, \text{left-lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) & \xleftarrow[\iota_{\text{left-lax} \rightarrow \text{all}}^L]{} & \mathbf{Funct}^{\text{loc} \rightarrow \text{glob}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}). \end{array}$$

Moreover,

- The monad $\iota_{\text{st} \rightarrow \text{lax}} \circ \iota_{\text{st} \rightarrow \text{lax}}^L$ can be identified with (11.23).
- The underlying endofunctor of the monad $\iota_{\text{left-lax} \rightarrow \text{all}} \circ \iota_{\text{left-lax} \rightarrow \text{all}}^L$ can be identified with (11.15), and the unit of the monad can be identified with (11.16).
- The combination of (I.3) and (I.2) gives the commutative diagram (11.24).

I.3. Proof of Proposition 11.8.8. In this subsection, we deduce Proposition 11.8.8 from Proposition I.1.7. This is essentially a formal diagram chase.

I.3.1. We need to show the functor

$$\iota_{\text{st} \rightarrow \text{all}} := \iota_{\text{lax} \rightarrow \text{all}} \circ \iota_{\text{st} \rightarrow \text{lax}} \simeq \iota_{\text{left-lax} \rightarrow \text{all}} \circ \iota_{\text{st} \rightarrow \text{left-lax}}$$

is fully faithful and identify its essential image.

Given

$$\underline{\mathbf{F}}^{\text{untl}}, \underline{\mathbf{G}}^{\text{untl}} \in \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{untl}}(\underline{\mathbf{C}}^{\text{loc}}, \underline{\mathbf{C}}^{\text{glob}}),$$

we have

$$\begin{aligned} & \text{Maps}(\iota_{\text{st} \rightarrow \text{all}}(\underline{\mathbf{F}}^{\text{untl}}), \iota_{\text{st} \rightarrow \text{all}}(\underline{\mathbf{G}}^{\text{untl}})) \simeq \\ & \simeq \text{Maps}(\iota_{\text{st} \rightarrow \text{all}}(\underline{\mathbf{F}}^{\text{untl}}), \iota_{\text{left-lax} \rightarrow \text{all}} \circ \iota_{\text{st} \rightarrow \text{left-lax}}(\underline{\mathbf{G}}^{\text{untl}})) \simeq \\ & \simeq \text{Maps}(\iota_{\text{left-lax} \rightarrow \text{all}}^L \circ \iota_{\text{st} \rightarrow \text{all}}(\underline{\mathbf{F}}^{\text{untl}}), \iota_{\text{st} \rightarrow \text{left-lax}}(\underline{\mathbf{G}}^{\text{untl}})) \simeq \\ & \simeq \text{Maps}(\iota_{\text{left-lax} \rightarrow \text{all}}^L \circ \iota_{\text{lax} \rightarrow \text{all}} \circ \iota_{\text{st} \rightarrow \text{lax}}(\underline{\mathbf{F}}^{\text{untl}}), \iota_{\text{st} \rightarrow \text{left-lax}}(\underline{\mathbf{G}}^{\text{untl}})) \simeq \\ & \simeq \text{Maps}(\iota_{\text{st} \rightarrow \text{left-lax}} \circ \iota_{\text{st} \rightarrow \text{lax}}^L \circ \iota_{\text{st} \rightarrow \text{lax}}(\underline{\mathbf{F}}^{\text{untl}}), \iota_{\text{st} \rightarrow \text{left-lax}}(\underline{\mathbf{G}}^{\text{untl}})) \simeq \\ & \simeq \text{Maps}(\iota_{\text{st} \rightarrow \text{lax}}^L \circ \iota_{\text{st} \rightarrow \text{lax}}(\underline{\mathbf{F}}^{\text{untl}}), (\underline{\mathbf{G}}^{\text{untl}})) \simeq \\ & \simeq \text{Maps}(\iota_{\text{st} \rightarrow \text{left-lax}}(\underline{\mathbf{F}}^{\text{untl}}), \iota_{\text{st} \rightarrow \text{left-lax}}(\underline{\mathbf{G}}^{\text{untl}})) \simeq \\ & \simeq \text{Maps}(\underline{\mathbf{F}}^{\text{untl}}, \underline{\mathbf{G}}^{\text{untl}}), \end{aligned}$$

where

- The fourth equivalence is due to Proposition I.1.7;
- The sixth equivalence is because $\iota_{\text{st} \rightarrow \text{lax}}$ is fully faithful;
- The seventh equivalence is because $\iota_{\text{st} \rightarrow \text{left-lax}}$ is fully faithful.

We leave it to the reader to check the resulting composition is the inverse to the obvious map from the RHS to the LHS. This proves $\iota_{\text{st} \rightarrow \text{all}}$ is fully faithful.

I.3.2. Let $\underline{\mathbf{F}}$ be a local-to-global functor. By definition, the Global Unitality Axiom (i) means exactly the unit adjunction

$$\underline{\mathbf{F}} \rightarrow \iota_{\text{left-lax} \rightarrow \text{all}} \circ \iota_{\text{left-lax} \rightarrow \text{all}}^L(\underline{\mathbf{F}})$$

is invertible. In particular, $\underline{\mathbf{F}}$ can be naturally lifted to the object

$$\iota_{\text{left-lax} \rightarrow \text{all}}^L(\underline{\mathbf{F}}) \in \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{left-lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \underline{\mathbf{C}}^{\text{glob}}).$$

By the proof of Proposition I.1.7, the Global Unitality Axiom (ii) means exactly the left-lax structure on $\iota_{\text{left-lax} \rightarrow \text{all}}^L(\underline{\mathbf{F}})$ is strict (see Remark I.5.8 below). It follows that $\underline{\mathbf{F}}$ is contained in the essential image of $\iota_{\text{st} \rightarrow \text{all}}$ if and only if it satisfies the Global Unitality Axioms.

□[Proposition 11.8.8]

I.4. Definition of left-lax unital structures. To give a homotopy-coherent definition of a left-lax unital structure on a local-to-global functor, we need some higher algebra. This will also complete the omitted higher datum in Sect. 11.2.5, e.t.c..

I.4.1. For any $[n] \in \Delta^{\text{op}}$, let $\text{Ran}^{\subseteq, [n]}$ be the moduli space of chains $\underline{x}_0 \subseteq \underline{x}_1 \subseteq \cdots \subseteq \underline{x}_n$. In particular, we have

$$\text{Ran}^{\subseteq, [0]} = \text{Ran}, \text{Ran}^{\subseteq, [1]} = \text{Ran}^{\subseteq}.$$

We obtain a simplicial prestack $\text{Ran}^{\subseteq, \bullet}$ which is a $(\infty\text{-})\text{categorical object}$ in PreStk . Indeed, $\text{Ran}^{\subseteq, [0]} = \text{Ran}$ is the “prestack of objects” and $\text{Ran}^{\subseteq, [1]} = \text{Ran}^{\subseteq}$ is the “prestack of 1-morphisms”. The projections

$$\text{pr}_{\text{small}}, \text{pr}_{\text{big}} : \text{Ran}^{\subseteq} \rightarrow \text{Ran}$$

remember respectively the source and the target of a 1-morphism, while

$$\text{diag} : \text{Ran} \rightarrow \text{Ran}^{\subseteq}$$

sends an object to the identity 1-morphism at it. The higher categorical structure on $\text{Ran}^{\subseteq, \bullet}$ is provided by its simplicial structure.

I.4.2. Let $\mathcal{Y}^\bullet$ be any categorical object in $\mathbf{PreStk}$. Write $\mathcal{Obj} := \mathcal{Y}^0$, $\mathcal{Mor}_1 := \mathcal{Y}^1$ and $\mathcal{Mor}_2 := \mathcal{Y}^2$. In general, there is a *comonad* on the 2-category

$$\mathbf{CrystCat}(\mathcal{Obj})$$

of sheaves of categories on $\mathcal{Obj}$, whose underlying endofunctor is the composition

$$\mathbf{CrystCat}(\mathcal{Obj}) \xrightarrow{p_t^*} \mathbf{CrystCat}(\mathcal{Mor}_1) \xrightarrow{p_{s,*}} \mathbf{CrystCat}(\mathcal{Obj}),$$

where $p_s, p_t : \mathcal{Mor}_1 \rightarrow \mathcal{Obj}$ are the projections that remember respectively the sources and the targets.

I.4.3. The operation of composition on $p_{s,*} \circ p_t^*$ can be obtained as follows. View

$$(I.4) \quad \mathcal{Obj} \xleftarrow{p_s} \mathcal{Mor}_1 \xrightarrow{p_t} \mathcal{Obj}$$

as an endomorphism ϕ on $\mathcal{Obj}$ in the 2-category $\mathbf{Corr}(\mathbf{PreStk})_{\text{all,all}}^{\text{all}}$ of correspondences. The categorical structure gives a monad structure on this endomorphism. Namely, $\phi \circ \phi$ can be identified with the correspondence

$$(I.5) \quad \mathcal{Obj} \xleftarrow{p_s} \mathcal{Mor}_2 \xrightarrow{p_t} \mathcal{Obj}$$

Then $\phi \circ \phi \rightarrow \phi$ is induced by the 2-morphism from (I.5) to (I.4) given by the projection

$$p_{02} : \mathcal{Mor}_2 \rightarrow \mathcal{Mor}_1.$$

that corresponds to the map $[1] \rightarrow [2]$, $0 \mapsto 0, 1 \mapsto 2$. We have a functor between 2-categories⁸⁶

$$\mathbf{CrystCat} : \mathbf{Corr}(\mathbf{PreStk})_{\text{all,all}}^{\text{all}, 2-\text{op}} \rightarrow \mathbf{2} - \mathbf{Cat}$$

that sends the endomorphism ϕ to the endomorphism $p_{s,*} \circ p_t^*$. It follows that the monad structure on ϕ gives a comonad structure on $p_{s,*} \circ p_t^*$.

I.4.4. Applying to $\mathbf{Ran}^{\subseteq, \bullet}$, we obtain a comonad

$$\mathbf{P} : \mathbf{CrystCat}(\mathbf{Ran}) \rightarrow \mathbf{CrystCat}(\mathbf{Ran})$$

whose underlying endofunctor is $(\text{pr}_{\text{small}})_* \circ (\text{pr}_{\text{big}})^*$. In other words, we have (see Sect. 11.2.5)

$$\mathbf{P}(\underline{\mathbf{C}}^{\text{loc}}) \simeq \underline{\mathbf{C}}^{\text{loc}, \subseteq}.$$

As explained in *loc.cit.*, we have

Lemma I.4.5. *A local unital structure on $\underline{\mathbf{C}}^{\text{loc}}$ is the same as a $\mathbf{P}$ -comodule structure on it, where*

$$\mathbf{P} = (\text{pr}_{\text{small}})_* \circ (\text{pr}_{\text{big}})^* : \mathbf{CrystCat}(\mathbf{Ran}) \rightarrow \mathbf{CrystCat}(\mathbf{Ran})$$

is a comonad acting on $\mathbf{CrystCat}(\mathbf{Ran})$.

I.4.6. From now on, whenever $\underline{\mathbf{C}}^{\text{loc}}$ is equipped with a local unital structure, we view it as a $\mathbf{P}$ -comodule via the above lemma. Note that the coaction functor fits into the following commutative diagram

$$\begin{array}{ccc} \underline{\mathbf{C}}^{\text{loc}} & \xrightarrow{\text{coact}} & \mathbf{P}(\underline{\mathbf{C}}^{\text{loc}}) \\ \text{Id} \downarrow & & \downarrow \simeq \\ \underline{\mathbf{C}}^{\text{loc}} & \xrightarrow{\text{ins.unitt}} & \underline{\mathbf{C}}^{\text{loc}, \subseteq}. \end{array}$$

⁸⁶See [GaRo3, Chapter 7] for the notation.

I.4.7. *Example.* The constant sheaf of categories

$$\mathbf{D}^{\text{cons}} := \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran}) \in \text{CrystCat}(\text{Ran})$$

has an obvious local unital structure. Note that

$$\mathbf{P}(\mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran})) \simeq \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran}^{\subseteq}),$$

where the RHS is viewed as a sheaf of categories over Ran via the small projection $\text{pr}_{\text{small}} : \text{Ran}^{\subseteq} \rightarrow \text{Ran}$. The corresponding $\mathbf{P}$ -comodule structure is given by the functor

$$\text{Id} \otimes (\text{pr}_{\text{small}})^! : \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran}) \rightarrow \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran}^{\subseteq}).$$

Note that we have an adjunction

$$\text{coact}^L : \mathbf{P}(\mathbf{D}^{\text{cons}}) \rightleftarrows \mathbf{D}^{\text{cons}} : \text{coact}$$

in the 2-category $\text{CrystCat}(\text{Ran})$, where the left adjoint coact^L is given by the functor $\text{Id} \otimes (\text{pr}_{\text{small}})^!$. Also, the right adjoint coact is fully faithful, i.e.

$$\text{coact}^L \circ \text{coact} \xrightarrow{\sim} \text{Id}.$$

Indeed, this follows from the contractibility of the map $\text{pr}_{\text{small}} : \text{Ran}^{\subseteq} \rightarrow \text{Ran}$.

I.4.8. *A general paradigm.* Let $\mathbf{P}$ be a comonad acting on a 2-category $\mathbb{S}$. Let $\mathbf{c}, \mathbf{d}$ be two $\mathbf{P}$ -comodules. For any morphism $f : \mathbf{c} \rightarrow \mathbf{d}$ in $\mathbb{S}$, we can talk about (co)lax $\mathbf{P}$ -linear structures on f . Namely, a lax $\mathbf{P}$ -linear structure on f is a 2-morphism

$$\begin{array}{ccc} \mathbf{c} & \xrightarrow{\text{coact}} & \mathbf{P}(\mathbf{c}) \\ f \downarrow & \nearrow & \downarrow \mathbf{P}(f) \\ \mathbf{d} & \xrightarrow{\text{coact}} & \mathbf{P}(\mathbf{d}), \end{array}$$

i.e.,

$$\alpha : \text{coact} \circ f \rightarrow \mathbf{P}(f) \circ \text{coact},$$

equipped with a datum of associativity.

Recall any comonad $\mathbf{P}$ has a counit natural transformation $\epsilon : \mathbf{P} \rightarrow \text{Id}$, and for any comodule $\mathbf{c}$, the composition $\mathbf{c} \xrightarrow{\text{coact}} \mathbf{P}(\mathbf{c}) \xrightarrow{\epsilon} \mathbf{c}$ is isomorphic to the identity morphism. Then the above datum of associativity in particular says the outer square in the following diagram commutes:

$$\begin{array}{ccccc} \mathbf{c} & \xrightarrow{\text{coact}} & \mathbf{P}(\mathbf{c}) & \xrightarrow{\epsilon} & \mathbf{c} \\ f \downarrow & \nearrow & \downarrow \mathbf{P}(f) & & \downarrow f \\ \mathbf{d} & \xrightarrow{\text{coact}} & \mathbf{P}(\mathbf{d}) & \xrightarrow{\epsilon} & \mathbf{d}. \end{array}$$

In other words, α becomes invertible after composing with the counit $\mathbf{P}(\mathbf{d}) \xrightarrow{\epsilon} \mathbf{d}$.

Dually, a colax $\mathbf{P}$ -linear structure on f is a 2-morphism

$$\beta : \mathbf{P}(f) \circ \text{coact} \rightarrow \text{coact} \circ f$$

equipped with a datum of associativity. Also, β becomes invertible after composing with ϵ .

Given a (co)lax $\mathbf{P}$ -linear structure on f , we say it is *strict*, or equivalently f is $\mathbf{P}$ -linear, if the above 2-morphism α (resp. β) is invertible.

I.4.9. Now for $\mathbb{S} := \text{CrystCat}(\text{Ran})$, $\mathbf{c} := \underline{\mathbf{C}}^{\text{loc}}$ and $\mathbf{d} := \mathbf{D}^{\text{const}} := \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran})$, a morphism $f : \mathbf{c} \rightarrow \mathbf{d}$ in $\mathbb{S}$ is just a local-to-global functor

$$\underline{f} : \underline{\mathbf{C}}^{\text{loc}} \rightarrow \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D}\text{-mod}}(\text{Ran}).$$

Suppose $\underline{\mathbf{C}}^{\text{loc}}$ is equipped with a local unital structure, i.e., $\mathbf{c}$ is equipped with a $\mathbf{P}$ -comodule structure, where recall $\mathbf{P}(\underline{\mathbf{C}}^{\text{loc}}) \simeq \mathbf{C}^{\text{loc}, \subseteq}$. As explained in Sect. 11.3.4, we have

Lemma I.4.10. *In the above notations, knowing a lax unital structure on $\underline{f}$ is equivalent to knowing a lax $\mathbf{P}$ -linear structure on f . Via this correspondence, the natural transformation*

$$\alpha : \text{coact} \circ f \rightarrow \mathbf{P}(f) \circ \text{coact},$$

is given by (11.12).

I.4.11. Now we *define* a left-lax unital structure on $\underline{f}$ to be a left-lax $\mathbf{P}$ -linear structure on f . This is the homotopically sound definition promised in Sect. I.1.

I.5. Proof of Proposition I.1.7.

I.5.1. Using the notations in Sect. I.4.8, the forgetful functor $\iota_{\text{left-lax} \rightarrow \text{all}}$ is given by

$$\iota_{\text{left-lax} \rightarrow \text{all}} : \text{Funct}_{\text{left-lax-}\mathbf{P}}(\mathbf{c}, \mathbf{d}) \rightarrow \text{Funct}(\mathbf{c}, \mathbf{d})$$

which sends a left-lax $\mathbf{P}$ -linear morphism $f : \mathbf{c} \rightarrow \mathbf{d}$ to its underlying morphism. Now Proposition I.1.7 is a particular case of the following general result.

Lemma I.5.2. *Let $(\mathbb{S}, \mathbf{P}, \mathbf{c}, \mathbf{d})$ be as in Sect. I.4.8. Suppose:*

(*) *The morphism $\text{coact} : \mathbf{d} \rightarrow \mathbf{P}(\mathbf{d})$ has a left adjoint $\text{coact}^L : \mathbf{P}(\mathbf{d}) \rightarrow \mathbf{d}$.*

Then the forgetful functor

$$(I.6) \quad \text{Funct}_{\text{left-lax-}\mathbf{P}}(\mathbf{c}, \mathbf{d}) \rightarrow \text{Funct}(\mathbf{c}, \mathbf{d})$$

has a left adjoint given by

$$(I.7) \quad \text{Funct}(\mathbf{c}, \mathbf{d}) \rightarrow \text{Funct}_{\text{left-lax-}\mathbf{P}}(\mathbf{c}, \mathbf{d}), \quad f \mapsto \text{coact}^L \circ \mathbf{P}(f) \circ \text{coact}.$$

I.5.3. *Proof.* The rest of this subsection is devoted to the proof of the lemma. We first define the desired left-lax $\mathbf{P}$ -linear structure on the composition

$$(I.8) \quad \mathbf{c} \xrightarrow{\text{coact}} \mathbf{P}(\mathbf{c}) \xrightarrow{\mathbf{P}(f)} \mathbf{P}(\mathbf{d}) \xrightarrow{\text{coact}^L} \mathbf{d}.$$

It is the composition of the following three left-lax $\mathbf{P}$ -linear structures:

- The coaction morphism $\mathbf{c} \xrightarrow{\text{coact}} \mathbf{P}(\mathbf{c})$ always has a natural strict $\mathbf{P}$ -linear structure;
- The morphism $\mathbf{P}(f) : \mathbf{P}(\mathbf{c}) \rightarrow \mathbf{P}(\mathbf{d})$ is a strictly $\mathbf{P}$ -linear morphism between cofree $\mathbf{P}$ -comodules;
- As the left adjoint of the strictly $\mathbf{P}$ -linear morphism $\mathbf{d} \xrightarrow{\text{coact}} \mathbf{P}(\mathbf{d})$, the morphism $\mathbf{P}(\mathbf{d}) \xrightarrow{\text{coact}^L} \mathbf{d}$ has a natural left-lax $\mathbf{P}$ -linear structure.

I.5.4. To show (I.7) is indeed left adjoint to (I.6), we provide the unit and counit natural transformations for this adjunction.

I.5.5. *Unit natural transformation.* Given $f \in \text{Funct}(\mathbf{c}, \mathbf{d})$, we have the following commutative diagram

$$\begin{array}{ccccc} \mathbf{c} & \xrightarrow{\text{coact}} & \mathbf{P}(\mathbf{c}) & \xrightarrow{\epsilon} & \mathbf{c} \\ & & \downarrow \mathbf{P}(f) & & \downarrow f \\ \mathbf{d} & \xrightarrow{\text{coact}} & \mathbf{P}(\mathbf{d}) & \xrightarrow{\epsilon} & \mathbf{d}. \end{array}$$

Then the desired morphism $f \rightarrow \text{coact}^L \circ \mathbf{P}(f) \circ \text{coact}$ is obtained by passing to left adjoints along the bottom line and using the fact that $\epsilon \circ \text{coact} \simeq \text{Id}$. More precisely, we have a morphism

$$\epsilon \rightarrow \epsilon \circ \text{coact} \circ \text{coact}^L \simeq \text{coact}^L,$$

where the first morphism is given by the unit adjunction of $(\text{coact}^L, \text{coact})$. Hence we obtain a morphism (I.9)

$$f \simeq f \circ \epsilon \circ \text{coact} \simeq \epsilon \circ P(f) \circ \text{coact} \rightarrow \text{coact}^L \circ P(f) \circ \text{coact}.$$

which is the value of the desired unit natural transformation at f .

I.5.6. Counit natural transformation. Given $g \in \text{Funct}_{\text{left-lax-P}}(\mathbf{c}, \mathbf{d})$, the left-lax P -linear structure on it provides a morphism in $\text{Funct}_{\text{left-lax-P}}(\mathbf{c}, \mathbf{d})$:

$$P(g) \circ \text{coact} \rightarrow \text{coact} \circ g.$$

Using the adjunction $(\text{coact}^L, \text{coact})$, we obtain a morphism

$$\text{coact}^L \circ P(g) \circ \text{coact} \rightarrow g$$

which is the value of the desired counit natural transformation at g .

I.5.7. It is a routine exercise to verify these natural transformations indeed satisfy the axioms of an adjunction. We leave it to the readers. $\square$

I.5.8. Remark. Note that Proposition I.1.7 is indeed a particular case of Lemma I.5.2 because of Sect. I.4.7. Also, unwinding the definitions, for a local-to-global functor $\underline{F}$ and the corresponding $f : \mathbf{c} \rightarrow \mathbf{d}$, the unit adjunction (I.9) is exactly (11.16).

Moreover, by definition, $\underline{F}$ satisfies the Global Unitality Axioms iff the corresponding f satisfies

- (i) The unit adjunction (I.9) is invertible;
- (ii) The natural transformation

$$(I.10) \quad P(f) \circ \text{coact} \rightarrow \text{coact} \circ f$$

obtained from the inverse of (I.9) is invertible.

Now (i) implies f has a natural left-lax P -linear structure given by that of $\text{coact}^L \circ P(f) \circ \text{coact}$. A direct diagram chasing shows this left-lax P -linear structure is exhibited by the natural transformation (I.10). Hence (ii) says this left-lax P -linear structure on f is strict. This was used in the proof of Proposition 11.8.8 in Sect. I.3.2.

I.6. Proof of Proposition I.2.3.

I.6.1. Unlike Proposition I.1.7, the proof of Proposition I.2.3 is *not* completely formal. Instead, we need the following particular feature of the setting of local-to-global functors.

Lemma I.6.2. *The composition*

$$\iota_{\text{left-lax} \rightarrow \text{all}}^L \circ \iota_{\text{lax} \rightarrow \text{all}} : \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}}) \rightarrow \text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{left-lax-untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$$

takes image in $\text{Funct}^{\text{loc} \rightarrow \text{glob}, \text{untl}}(\underline{\mathbf{C}}^{\text{loc}}, \mathbf{C}^{\text{glob}})$.

I.6.3. Remark. The claim of the lemma is a priori *stronger* than the results listed in Sect. 11.5. Namely, given a local-to-global functor $\underline{F}$ equipped with a lax unital structure, Sect. 11.5 says there is a strictly unital structure on $\underline{F}^{\text{ins.unit}}$, while the lemma says the natural left-lax unital structure on it (provided by Proposition I.1.7) is strict. Nevertheless, it is easy to see the proof in loc.cit. actually implies this stronger claim. For completeness, we repeat this proof.

Proof. We will deduce the claim from Lemma 11.5.5. Let $\underline{F}$ be a local-to-global functor equipped with a lax unital structure, and $f : \mathbf{c} \rightarrow \mathbf{d}$ be the corresponding lax P -linear morphism. We need to show that the left-lax P -linear structure on $\text{coact}^L \circ P(f) \circ \text{coact}$ is strict. Recall that $P(f) \circ \text{coact}$, i.e.,

$$(I.11) \quad \mathbf{C}_{\text{Ran}}^{\text{loc}} \xrightarrow{\text{ins.unit}} \mathbf{C}_{\text{Ran}}^{\text{loc}, \subseteq} \xrightarrow{F \subseteq} \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D-mod}}(\text{Ran}^{\subseteq})$$

is naturally P -linear. On the other hand, $\text{coact}^L : P(\mathbf{d}) \rightarrow \mathbf{d}$, i.e.,

$$(I.12) \quad \text{Id} \otimes (\text{pr}_{\text{small}})! : \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D-mod}}(\text{Ran}^{\subseteq}) \rightarrow \mathbf{C}^{\text{glob}} \otimes \underline{\mathbf{D-mod}}(\text{Ran}).$$

has a natural left-lax P -linear structure because its right adjoint is naturally P -linear.

Consider the categorical prestack

$$\mathrm{Ran} \times_{\mathrm{Ran}^{\mathrm{untl}}} \mathrm{Ran}^{\subseteq, \mathrm{untl}}$$

where $\mathrm{Ran}^{\subseteq, \mathrm{untl}} \rightarrow \mathrm{Ran}^{\mathrm{untl}}$ is the small projection $\mathrm{pr}_{\mathrm{small}}^{\mathrm{untl}}$. Viewing it as a categorical prestack over Ran , we obtain a crystal of categories over Ran

$$\underline{\mathrm{D-mod}}(\mathrm{Ran} \times_{\mathrm{Ran}^{\mathrm{untl}}} \mathrm{Ran}^{\subseteq, \mathrm{untl}})$$

and a functor

$$(I.13) \quad \mathbf{t}^! : \underline{\mathrm{D-mod}}(\mathrm{Ran} \times_{\mathrm{Ran}^{\mathrm{untl}}} \mathrm{Ran}^{\subseteq, \mathrm{untl}}) \rightarrow \underline{\mathrm{D-mod}}(\mathrm{Ran}^{\subseteq})$$

given by pullback along $\mathbf{t} : \mathrm{Ran}^{\subseteq} \rightarrow \mathrm{Ran} \times_{\mathrm{Ran}^{\mathrm{untl}}} \mathrm{Ran}^{\subseteq, \mathrm{untl}}$. Note that both sides of (I.13) have natural local unital structures and the functor is obviously unital. Now Lemma 11.5.5, combined with a variant of Lemma C.5.12, says the composition

$$\underline{\mathrm{D-mod}}(\mathrm{Ran} \times_{\mathrm{Ran}^{\mathrm{untl}}} \mathrm{Ran}^{\subseteq, \mathrm{untl}}) \rightarrow \underline{\mathrm{D-mod}}(\mathrm{Ran}^{\subseteq}) \xrightarrow{(\mathrm{pr}_{\mathrm{small}})^!} \underline{\mathrm{D-mod}}(\mathrm{Ran}),$$

which is a priori left-lax unital, is strictly unital. The statement remains true if we tensor it with the DG category $\mathbf{C}^{\mathrm{glob}}$. Then we finish the proof because the composition (I.11) factors through (I.13) by the lax unital structure on $\underline{\mathbf{F}}$. $\square$

I.6.4. By Lemma I.6.2, there is a unique functor $\tau_{\mathrm{lax} \rightarrow \mathrm{st}}$ making the following diagram commute

$$(I.14) \quad \begin{array}{ccc} \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{untl}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}) & \xleftarrow{\tau_{\mathrm{lax} \rightarrow \mathrm{st}}} & \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{lax-untl}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}) \\ \downarrow \iota_{\mathrm{st} \rightarrow \mathrm{left-lax}} \downarrow \subset & & \downarrow \iota_{\mathrm{lax} \rightarrow \mathrm{all}} \\ \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{left-lax-untl}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}) & \xleftarrow{\iota_{\mathrm{left-lax} \rightarrow \mathrm{all}}^L} & \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}). \end{array}$$

In particular, we obtained:

- A lifting of the endofunctor

$$\iota_{\mathrm{left-lax} \rightarrow \mathrm{all}}^L \circ \iota_{\mathrm{left-lax} \rightarrow \mathrm{all}}^L : \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}) \rightarrow \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}})$$

to an endofunctor

$$\iota_{\mathrm{st} \rightarrow \mathrm{lax}} \circ \tau_{\mathrm{lax} \rightarrow \mathrm{st}} : \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{lax-untl}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}) \rightarrow \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{lax-untl}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}});$$

- A lifting of the endofunctor

$$\iota_{\mathrm{left-lax} \rightarrow \mathrm{all}}^L \circ \iota_{\mathrm{left-lax} \rightarrow \mathrm{all}}^L : \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{left-lax-untl}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}) \rightarrow \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{left-lax-untl}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}})$$

to an endofunctor

$$\tau_{\mathrm{lax} \rightarrow \mathrm{st}} \circ \iota_{\mathrm{st} \rightarrow \mathrm{lax}} : \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{untl}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}) \rightarrow \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{untl}}(\underline{\mathbf{C}}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}});$$

I.6.5. To finish the proof, we only need to lift the unit and counit adjunctions

$$\mathrm{Id} \rightarrow \iota_{\mathrm{left-lax} \rightarrow \mathrm{all}}^L \circ \iota_{\mathrm{left-lax} \rightarrow \mathrm{all}}^L, \quad \iota_{\mathrm{left-lax} \rightarrow \mathrm{all}}^L \circ \iota_{\mathrm{left-lax} \rightarrow \mathrm{all}}^L \rightarrow \mathrm{Id}$$

to

$$\mathrm{Id} \rightarrow \iota_{\mathrm{st} \rightarrow \mathrm{lax}} \circ \tau_{\mathrm{lax} \rightarrow \mathrm{st}}, \quad \tau_{\mathrm{lax} \rightarrow \mathrm{st}} \circ \iota_{\mathrm{st} \rightarrow \mathrm{lax}} \rightarrow \mathrm{Id}$$

and verify they satisfy the axioms of an adjunction. Indeed, these will induce an equivalence $\tau_{\mathrm{lax} \rightarrow \mathrm{st}} \simeq \iota_{\mathrm{st} \rightarrow \mathrm{lax}}^L$ such that (I.14) can be identified with the Bech–Chevalley natural transformation (I.3). Then the other claims in Proposition I.2.3 follow from definitions.

I.6.6. We will lift the counit and unit natural transformations, and leave it to the readers to verify they satisfy the axioms of an adjunction.

I.6.7. *Lift the counit.* The lifting of the counit is obvious because the forgetful functor

$$\mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{untl}}(\mathbf{C}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}}) \rightarrow \mathrm{Funct}^{\mathrm{loc} \rightarrow \mathrm{glob}, \mathrm{left-lax-untl}}(\mathbf{C}^{\mathrm{loc}}, \mathbf{C}^{\mathrm{glob}})$$

is fully faithful and the any morphism in the RHS (such as the counit $\iota_{\mathrm{left-lax} \rightarrow \mathrm{all}}^L \circ \iota_{\mathrm{left-lax} \rightarrow \mathrm{all}} \rightarrow \mathrm{Id}$) has a unique lifting to the LHS as long as its source and target are contained in the LHS.

I.6.8. *Lift the unit.* The rest of this subsection is devoted to lift the unit. Recall its definition in Sect. I.5.5. By definition, the morphism $\epsilon : \mathbf{P}(\mathbf{d}) \rightarrow \mathbf{d}$ is

$$\mathrm{Id} \otimes \mathrm{diag}^! : \mathbf{C}^{\mathrm{glob}} \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}^{\subseteq}) \rightarrow \mathbf{C}^{\mathrm{glob}} \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran})$$

and the morphism

$$\epsilon \rightarrow \mathrm{coact}^L$$

is induced by the natural transformation

$$(I.15) \quad \mathrm{diag}^! \rightarrow \mathrm{diag}^! \circ \mathrm{pr}_{\mathrm{small}}^! \circ (\mathrm{pr}_{\mathrm{small}})^! \simeq (\mathrm{pr}_{\mathrm{small}})^!$$

provided by the isomorphism $\mathrm{pr}_{\mathrm{small}} \circ \mathrm{diag} \simeq \mathrm{Id}$.

I.6.9. Recall the restriction functor

$$\mathbf{t}^! : \underline{\mathrm{D-mod}}(\mathrm{Ran} \times_{\mathrm{Ran}^{\mathrm{untl}}} \mathrm{Ran}^{\subseteq, \mathrm{untl}}) \rightarrow \underline{\mathrm{D-mod}}(\mathrm{Ran}^{\subseteq}).$$

We claim the (horizontal) composition of (I.15) with $\mathbf{t}^!$, i.e.,

$$(I.16) \quad \mathrm{diag}^! \circ \mathbf{t}^! \rightarrow (\mathrm{pr}_{\mathrm{small}})^! \circ \mathbf{t}^!$$

can be naturally lifted to a natural transformation between lax unital functors

$$\underline{\mathrm{D-mod}}(\mathrm{Ran} \times_{\mathrm{Ran}^{\mathrm{untl}}} \mathrm{Ran}^{\subseteq, \mathrm{untl}}) \rightarrow \underline{\mathrm{D-mod}}(\mathrm{Ran}).$$

Here $\underline{\mathrm{D-mod}}(\mathrm{Ran} \times_{\mathrm{Ran}^{\mathrm{untl}}} \mathrm{Ran}^{\subseteq, \mathrm{untl}})$ and $\underline{\mathrm{D-mod}}(\mathrm{Ran})$ are equipped with the local unital structures given respectively by $\mathrm{Ran}^{\subseteq, \mathrm{untl}}$ and $\mathrm{Ran}^{\mathrm{untl}}$.

I.6.10. Consider the map

$$\mathrm{diag}^{\mathrm{untl}} : \mathrm{Ran}^{\mathrm{untl}} \rightarrow \mathrm{Ran}^{\subseteq, \mathrm{untl}}$$

and its left inverse

$$\mathrm{pr}_{\mathrm{small}}^{\mathrm{untl}} : \mathrm{Ran}^{\subseteq, \mathrm{untl}} \rightarrow \mathrm{Ran}^{\mathrm{untl}}.$$

Similar to (I.15), we have a natural transformation

$$(I.17) \quad (\mathrm{diag}^{\mathrm{untl}})^! \rightarrow (\mathrm{diag}^{\mathrm{untl}})^! \circ (\mathrm{pr}_{\mathrm{small}}^{\mathrm{untl}})^! \circ (\mathrm{pr}_{\mathrm{small}}^{\mathrm{untl}})^! \simeq (\mathrm{pr}_{\mathrm{small}}^{\mathrm{untl}})^!.$$

Note that $(\mathrm{diag}^{\mathrm{untl}})^!$ has an obvious lax unital structure, i.e., is a lax functor between sheaves of categories on $\mathrm{Ran}^{\mathrm{untl}}$, while $(\mathrm{pr}_{\mathrm{small}}^{\mathrm{untl}})^!$ has a strictly unital structure by Lemma 11.5.5. The composition (I.17) is compatible with the lax unital structures on both sides because each natural transformation is.

By (a variant of) Lemma C.5.12, when restricted along $\mathrm{Ran} \rightarrow \mathrm{Ran}^{\mathrm{untl}}$, (I.17) gives exactly (I.16). In other words, we have proved the claim in Sect. I.6.9.

I.6.11. Recall that $\mathbf{P}(f) \circ \mathrm{coact}$ (which is just (I.11)) factors through $\mathrm{Id} \otimes \mathbf{t}^!$. It follows that (I.16) induces a natural transformation

$$\epsilon \circ \mathbf{P}(f) \circ \mathrm{coact} \rightarrow \mathrm{coact}^L \circ \mathbf{P}(f) \circ \mathrm{coact}$$

compatible with the lax unital structures on both sides.

I.6.12. On the other hand, we have

$$f \simeq f \circ \epsilon \circ \mathrm{coact} \simeq \epsilon \circ \mathbf{P}(f) \circ \mathrm{coact}$$

because ϵ is the counit of the comonad $\mathbf{P}$. Moreover, this isomorphism is obviously compatible with the lax unital structures on both sides.

I.6.13. Combining the above two subsections, we obtain that

$$f \rightarrow \text{coact}^L \circ P(f) \circ \text{coact}$$

is naturally compatible with the lax unital structures on both sides. In other words, we have found the desired lifting of the unit adjunction.

I.6.14. We leave it to the readers to check the above liftings indeed satisfy the axioms of an adjunction. $\square$ [Proposition I.1.7]

APPENDIX J. A HOMOTOPICAL DEVICE FOR COACTION

The goal of this section is to introduce a homotopical device that will help us carry out the constructions in Sects. 4.6 and 5.3 up to *coherent* homotopy.

J.1. Associative algebras via mock-simplicial sets.

J.1.1. Let us recall the following device of encoding the structure of associative algebra (resp., module over a given associative algebra) in a monoidal category (see [Lu2, Sect. 2.2.4]).

Let $\Delta^{\text{op}, \text{mock}}$ be the category of (possibly empty) finite ordered sets. The operation of (ordered) union defines on $\Delta^{\text{op}, \text{mock}}$ a structure of monoidal category. Its monoidal unit is $\emptyset$.

In what follows we will denote⁸⁷

$$\Delta^{\text{mock}} := (\Delta^{\text{op}, \text{mock}})^{\text{op}}.$$

J.1.2. The datum of a unital associative algebra in a monoidal category $\mathfrak{A}$ is equivalent to that of a monoidal functor

$$\Delta^{\text{op}, \text{mock}} \rightarrow \mathfrak{A}.$$

Under this correspondence, for a given functor $F : \Delta^{\text{op}, \text{mock}} \rightarrow \mathfrak{A}$, the corresponding algebra object $\mathfrak{a} \in \mathfrak{A}$ is $F(\{*\})$. The unit in $\mathfrak{a}$ is given by the map

$$1_{\mathfrak{A}} = F(\emptyset) \rightarrow F(\{*\}) = \mathfrak{a},$$

corresponding to the (unique) map $\emptyset \rightarrow \{*\}$.

The binary operation on $\mathfrak{a}$ corresponds to the map

$$\mathfrak{a} \otimes \mathfrak{a} = F(\{*\}) \otimes F(\{*\}) \simeq F(\{1, 2\}) \rightarrow F(\{*\}) = \mathfrak{a},$$

where the arrows is given by the (unique) map $\{1, 2\} \rightarrow \{*\}$ in $\Delta^{\text{op}, \text{mock}}$.

J.1.3. Let $\Delta_*^{\text{op}, \text{mock}}$ be the category of non-empty finite ordered sets, pointed by their maximal element. The category $\Delta_*^{\text{op}, \text{mock}}$ is naturally a module over $\Delta^{\text{op}, \text{mock}}$.

J.1.4. Let us be given a monoidal category $\mathfrak{A}$. Let $\mathfrak{a}$ be an associative algebra in $\mathfrak{A}$, thought of as a monoidal functor

$$F_{\mathfrak{a}} : \Delta^{\text{op}, \text{mock}} \rightarrow \mathfrak{A}.$$

Then the category

$$\mathfrak{a}\text{-mod}(\mathfrak{A})$$

of $\mathfrak{a}$ -modules in $\mathfrak{A}$ is equivalent to that of functors of left $\Delta^{\text{op}, \text{mock}}$ -module categories

$$\Delta_*^{\text{op}, \text{mock}} \rightarrow \mathfrak{A},$$

where $\mathfrak{A}$ is a left $\Delta^{\text{op}, \text{mock}}$ -module via $F_{\mathfrak{a}}$.

⁸⁷The category Δ^{mock} is equivalent to the subcategory of Δ consisting of *active* morphisms, i.e. those that preserve the maximal and minimal element. For our purposes, the nonstandard “mock” terminology will be more convenient. We caution the reader that the usual simplex category, denoted Δ , is that of non-empty finite ordered sets. Note that arrows in Δ and Δ^{mock} go in opposite directions.

J.1.5. Under this correspondence, given a functor $F_m : \Delta_*^{\text{op}, \text{mock}} \rightarrow \mathfrak{A}$, the object of $\mathfrak{A}$ underlying the corresponding $\mathfrak{a}$ -module is

$$\mathfrak{m} := F_m(\{*\}).$$

The action map $\mathfrak{a} \otimes M \rightarrow M$ is given by

$$\mathfrak{a} \otimes \mathfrak{m} \simeq F_a(\{0\}) \otimes F_m(\{*\}) \simeq F_m(\{0, *\}) \rightarrow F_m(\{*\}) \simeq \mathfrak{m}.$$

J.1.6. As above, in what follows we will denote

$$\Delta_*^{\text{mock}} := (\Delta_*^{\text{op}, \text{mock}})^{\text{op}}.$$

We will refer to functors $\Delta^{\text{op}, \text{mock}} \rightarrow \mathfrak{A}$ (resp., $\Delta_*^{\text{op}, \text{mock}} \rightarrow \mathfrak{A}$, $\Delta^{\text{mock}} \rightarrow \mathfrak{A}$, $\Delta_*^{\text{mock}} \rightarrow \mathfrak{A}$) as *mock-simplicial* (resp., pointed mock-simplicial, mock cosimplicial, pointed mock cosimplicial) objects of $\mathfrak{A}$.

J.2. Factorization categories attached to associative factorization algebras. Recall that our goal is to carry out the constructions in Sects. 4.6 and 5.3. We will achieve this by introducing appropriate objects to feed into the machine in Sect. J.1.

J.2.1. Let $\mathfrak{A}_1$ be a monoidal category and let $\mathfrak{C}_1$ be a module category over it. Note that given two such pairs, we can talk about *strictly* and *right-lax* monoidal functors of pairs

$$(F_a, F_c) : (\mathfrak{A}_1, \mathfrak{C}_1) \rightarrow (\mathfrak{A}_2, \mathfrak{C}_2),$$

i.e., $F_a : \mathfrak{A}_1 \rightarrow \mathfrak{A}_2$ is right-lax monoidal, and $F_c : \mathfrak{C}_1 \rightarrow \mathfrak{C}_2$ is right-lax compatible with the actions.

We introduce several such pairs now.

J.2.2. Take

$$\mathfrak{A}_1 = \Delta^{\text{mock}}, \mathfrak{A}_2 = (\text{FactAlg}^{\text{untl}}(X))^{\text{op}}, \mathfrak{A}_3 = \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X),$$

where the subscript “lax” means that we are considering lax factorization categories.

J.2.3. By Sect. J.1, an object

$$(J.1) \quad \mathcal{R} \in \text{AssAlg}(\text{FactAlg}^{\text{untl}}(X))$$

can be viewed as a *strictly* monoidal functor

$$(J.2) \quad \Delta^{\text{op}, \text{mock}} \rightarrow \text{FactAlg}^{\text{untl}}(X),$$

and hence also as a *strictly* monoidal functor

$$(J.3) \quad \Delta^{\text{mock}} \rightarrow (\text{FactAlg}^{\text{untl}}(X))^{\text{op}}.$$

J.2.4. In addition, we have a naturally defined *right-lax* monoidal functor

$$(J.4) \quad (\text{FactAlg}^{\text{untl}})^{\text{op}} \rightarrow \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X), \quad \mathcal{R} \rightarrow \mathcal{R}\text{-mod}^{\text{fact}}, \quad (\mathcal{R}_1 \xrightarrow{\phi} \mathcal{R}_2) \rightsquigarrow \text{Res}_{\phi}.$$

Composing, for $\mathcal{R}$ as above, we obtain a *right-lax* monoidal functor

$$(J.5) \quad \Delta^{\text{mock}} \rightarrow \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X), \quad n \mapsto \mathcal{R}^{\otimes n}\text{-mod}^{\text{fact}}.$$

J.2.5. Take

$$\mathfrak{C}_1 = \Delta_*^{\text{mock}}, \mathfrak{C}_2 = \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X), \mathfrak{C}_3 = \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X),$$

where:

- Δ^{mock} acts on Δ_*^{mock} as in Sect. J.1.3;
- $\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)$ acts on itself via the (symmetric) monoidal structure on $\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)$;
- $(\text{FactAlg}^{\text{untl}}(X))^{\text{op}}$ acts on $\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)$ by

$$\mathcal{R}, \mathbf{A} \mapsto \mathcal{R}\text{-mod}^{\text{fact}}(\mathbf{A}) \simeq (\mathcal{R} \otimes \mathbf{1}_{\mathbf{A}})\text{-mod}^{\text{fact}}(\mathbf{A}).$$

We note that the above action of $(\text{FactAlg}^{\text{untl}})^{\text{op}}$ on $\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)$ is monoidal thanks to Lemma C.11.19.

J.2.6. Let $\mathcal{R}$ be as in (J.1) and suppose that it acts on the unit $\mathbf{1}_{\mathbf{C}} \in \mathbf{C}$ for some $\mathbf{C} \in \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)$.

Then the functor (J.2) extends to a *strictly* monoidal functor of pairs

$$(J.6) \quad (\Delta^{\text{op}, \text{mock}}, \Delta_*^{\text{op}, \text{mock}}) \rightarrow (\text{FactAlg}^{\text{untl}}(X), (\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X))^{\text{op}}), \quad * \mapsto \mathbf{C}.$$

Hence, we obtain a *strictly* monoidal functor of pairs

$$(J.7) \quad (\Delta^{\text{mock}}, \Delta_*^{\text{mock}}) \rightarrow ((\text{FactAlg}^{\text{untl}}(X))^{\text{op}}, \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)), \quad * \mapsto \mathbf{C}.$$

J.2.7. In addition, the functor (J.4) extends to a *right-lax* monoidal functor

$$(J.8) \quad ((\text{FactAlg}^{\text{untl}}(X))^{\text{op}}, \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)) \rightarrow (\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X), \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)),$$

which acts as identity on the module component.

J.2.8. Combining, we obtain that for $(\mathcal{R}, \mathbf{C})$ as in Sect. J.2.6 we obtain that (J.5) extends to a *right-lax* monoidal functor of pairs

$$(J.9) \quad (\Delta_*^{\text{mock}}, \Delta_*^{\text{mock}}) \rightarrow (\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X), \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X))$$

that sends

$$(* \in \Delta_*^{\text{mock}}) \rightsquigarrow \mathbf{C}.$$

J.3. The renormalization step.

J.3.1. We apply Sect. J.2.8 to

$$\mathcal{R} = \mathfrak{z}_{\mathfrak{g}} \text{ and } \mathbf{C} = \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}},$$

and consider the resulting right-lax monoidal functor of pairs (J.9); denote it

$$(J.10) \quad (F_{\mathfrak{z}_{\mathfrak{g}}}, F_{\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}}).$$

J.3.2. We now introduce several more (symmetric) monoidal categories. Let

$$\mathfrak{A} = \mathfrak{A}_3 := \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)$$

be as above.

We let $\mathfrak{A}'$ be the following 1-full subcategory of $\mathfrak{A}$, to be denoted

$$\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)^{\text{t-str}}.$$

Its objects are lax unital factorization categories, equipped with a t-structure. For a pair of objects $\mathbf{C}_1, \mathbf{C}_2$, we let

$$\text{Maps}_{\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)^{\text{t-str}}}(\mathbf{C}_1, \mathbf{C}_2) \subset \text{Maps}_{\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)}(\mathbf{C}_1, \mathbf{C}_2)$$

be the full subcategory consisting of *left t-exact* functors.

J.3.3. We let $\mathfrak{A}''$ be the full subcategory of the category of arrows in $\mathfrak{A}'$, whose objects are those pairs

$$\Phi : \mathbf{C}^{\text{ren}} \rightarrow \mathbf{C},$$

for which:

- Φ induces an equivalence between the eventually coconnective subcategories;
- $\mathbf{C}^{\text{ren}}$ is compactly generated by objects that are eventually coconnective.

Note that the tautological forgetful functor

$$\mathfrak{A}'' \rightarrow \mathfrak{A}', \quad (\Phi : \mathbf{C}^{\text{ren}} \rightarrow \mathbf{C}) \mapsto \mathbf{C}$$

is 1-fully faithful, i.e., induces a fully faithful functor on spaces of morphisms.

Explicitly, given two pairs $\mathbf{C}_1^{\text{ren}} \xrightarrow{\Phi_1} \mathbf{C}_1$ and $\mathbf{C}_2^{\text{ren}} \xrightarrow{\Phi_2} \mathbf{C}_2$, a functor $\Psi : \mathbf{C}_1 \rightarrow \mathbf{C}_2$ in $\mathfrak{A}'$ lifts to $\mathfrak{A}''$ if and only if the ind-extension of

$$(\mathbf{C}_1^{\text{ren}})^c \subset (\mathbf{C}_1^{\text{ren}})^{>-\infty} \simeq \mathbf{C}_1^{>-\infty} \xrightarrow{\Psi} \mathbf{C}_2^{>-\infty} \simeq (\mathbf{C}_2^{\text{ren}})^{>-\infty}$$

is left t-exact (equivalently, has a bounded cohomological amplitude on the left).

J.3.4. Finally, we let $\mathfrak{A}'''$ be again $\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)$, and we consider the forgetful functor

$$(J.11) \quad \mathfrak{A}'' \rightarrow \mathfrak{A}''', \quad (\Phi : \mathbf{C}^{\text{ren}} \rightarrow \mathbf{C}) \mapsto \mathbf{C}^{\text{ren}}.$$

J.3.5. Note now that given a lax monoidal functor

$$(\Delta^{\text{mock}}, \Delta_*^{\text{mock}}) \rightarrow (\mathfrak{A}, \mathfrak{A}),$$

in order to lift to a lax monoidal functor

$$(\Delta^{\text{mock}}, \Delta_*^{\text{mock}}) \rightarrow (\mathfrak{A}'', \mathfrak{A}''),$$

it suffices to do so at the level of objects and 1-morphisms, i.e., it is sufficient to do so *at the homotopy level* (moreover, the lift at the level of objects defines it completely).

We start with the functor $(F_{\mathfrak{z}_{\mathfrak{g}}}, F_{\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}})$ of (J.10). We lift to a functor

$$(F_{\mathfrak{z}_{\mathfrak{g}}}, F_{\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}})'' : (\Delta^{\text{mock}}, \Delta_*^{\text{mock}}) \rightarrow (\mathfrak{A}'', \mathfrak{A}'')$$

at the level of objects by sending

$$(n \in \Delta^{\text{mock}}) \rightsquigarrow \left(\text{IndCoh}^*(\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}}))^{\otimes n} \rightarrow \mathfrak{z}_{\mathfrak{g}}^{\otimes n}\text{-mod}^{\text{fact}} \right)$$

$$(\{n, *\} \in \Delta_*^{\text{mock}}) \rightsquigarrow \left(\text{IndCoh}^*(\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}}))^{\otimes n} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}} \rightarrow \mathfrak{z}_{\mathfrak{g}}^{\otimes n}\text{-mod}(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}) \right).$$

The existence of the lift at the level of 1-morphisms is guaranteed by Lemma 4.6.12.

J.3.6. We compose $(F_{\mathfrak{z}_{\mathfrak{g}}}, F_{\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}})''$ with (J.11) to obtain a right-lax monoidal functor of pairs

$$(F_{\mathfrak{z}_{\mathfrak{g}}}, F_{\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}})''' : (\Delta^{\text{mock}}, \Delta_*^{\text{mock}}) \rightarrow (\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X), \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)).$$

However, by construction, the latter functor is strictly monoidal. I.e., the functor

$$F_{\mathfrak{z}_{\mathfrak{g}}}''' : \Delta^{\text{mock}} \rightarrow \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)$$

is strictly monoidal, and

$$F_{\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}}''' : \Delta_*^{\text{mock}} \rightarrow \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)$$

is a functor between Δ^{mock} -module categories.

Applying Sect. J.1 to $(F_{\mathfrak{z}_{\mathfrak{g}}}, F_{\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}})'''$ we obtain the desired coaction of $\text{IndCoh}^*(\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}}))$ on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}$.

This completes the construction from Sect. 4.6.

J.4. Adding another monoidal category. In order to carry out the construction in Sect. 5.3, we need to enhance the setting of Sect. J.2.

J.4.1. We modify the setting of Sect. J.2.1, and we now take $\mathfrak{A}_2$ to be the category, denoted

$$(\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X),$$

whose objects are pairs $(\mathbf{A}, \mathcal{R})$, where:

- $\mathbf{A} \in \text{FactCat}_{\text{lax}}^{\text{untl}}(X)$;
- $\mathcal{R} \in \text{FactAlg}^{\text{untl}}(X, \mathbf{A})$.

The space of morphisms

$$\text{Maps}_{(\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)}((\mathbf{A}_1, \mathcal{R}_1), (\mathbf{A}_2, \mathcal{R}_2))$$

consists of pairs: (Φ, ϕ) , where:

- Φ is a lax unital functor $\mathbf{A}_1 \rightarrow \mathbf{A}_2$;
- ϕ is a map of unital factorization algebras in $\mathbf{A}_2$

$$\mathcal{R}_2 \rightarrow \Phi(\mathcal{R}_1).$$

Remark J.4.2. Note that we can interpret $(\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)$, equipped with the forgetful functor to $\text{FactCat}_{\text{lax}}^{\text{untl}}(X)$ as the co-Cartesian fibration corresponding to the functor

$$\text{FactCat}_{\text{lax}}^{\text{untl}}(X) \rightarrow \infty\text{-Cat}, \quad \mathbf{A} \mapsto (\text{FactAlg}^{\text{untl}}(X, \mathbf{A}))^{\text{op}}.$$

The monoidal structure on $(\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)$ corresponds to the right-lax monoidal structure on the above functor.

J.4.3. We let $\mathfrak{A}_3$ be the same as in Sect. J.2.5, i.e., $\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)$.

Note that we have a lax monoidal functor

$$(J.12) \quad (\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X) \rightarrow \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X), \quad (\mathbf{A}, \mathcal{R}) \mapsto \mathcal{R}\text{-mod}^{\text{fact}}(\mathbf{A}).$$

J.4.4. We take $\mathfrak{C}_2 = \mathfrak{A}_2$ and $\mathfrak{C}_3 = \mathfrak{A}_3$. The functor (J.12) gives rise to a right-lax monoidal functor of pairs

$$(J.13) \quad \left((\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X), (\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X) \right) \rightarrow \\ \rightarrow (\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X), \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)).$$

J.4.5. We now explain a procedure that gives rise to strictly monoidal functors

$$\Delta^{\text{mock}} \rightarrow (\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)$$

and strictly monoidal functors of pairs

$$(\Delta^{\text{mock}}, \Delta_*^{\text{mock}}) \rightarrow \left((\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X), (\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X) \right).$$

J.4.6. Let $(\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)$ denote the category defined as follows. Its objects are pairs $(\mathbf{A}, \mathcal{R})$, where:

- $\mathbf{A} \in \text{FactCat}_{\text{lax}}^{\text{untl}}(X)$;
- $\mathcal{R} \in \text{FactAlg}^{\text{untl}}(X, \mathbf{A})$.

The space of morphisms

$$\text{Maps}_{(\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)}((\mathbf{A}_1, \mathcal{R}_1), (\mathbf{A}_2, \mathcal{R}_2))$$

consists of pairs: (Φ, ϕ) , where:

- Φ is a strictly unital functor $\mathbf{A}_1 \rightarrow \mathbf{A}_2$;
- ϕ is a map of unital factorization algebras in $\mathbf{A}_2$

$$\Phi(\mathcal{R}_1) \rightarrow \mathcal{R}_2.$$

Remark J.4.7. As in Remark J.4.2, the category $(\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)$, equipped with the forgetful functor to $\text{FactCat}_{\text{lax}}^{\text{untl}}(X)$, is the co-Cartesian fibration corresponding to the functor

$$\text{FactCat}_{\text{lax}}^{\text{untl}}(X) \rightarrow \infty\text{-Cat}, \quad \mathbf{A} \mapsto \text{FactAlg}^{\text{untl}}(X, \mathbf{A}).$$

J.4.8. Let $\mathbf{A}$ a monoidal unital lax factorization category $\mathbf{A}$ and let $\mathcal{R} \in \text{FactAlg}^{\text{untl}}(X, \mathbf{A})$ be an object, equipped with a structure of *associative algebra*, in the sense of the monoidal structure on $\mathbf{A}$.

We can think of $(\mathbf{A}, \mathcal{R})$ as an associative algebra object in $(\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)$. Hence, by Sect. J.1, it gives rise to a (strictly) monoidal functor

$$F_{\mathbf{A}, \mathcal{R}} : \Delta^{\text{op}, \text{mock}} \rightarrow (\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X).$$

Let $\mathbf{C}$ be a unital lax factorization category, equipped with an action of $\mathbf{A}$ as a monoidal factorization category. Suppose, moreover, that $\mathcal{R}$ acts on $\mathbf{1}_{\mathbf{C}}$ in the sense of the action of $\mathbf{A}$ on $\mathbf{C}$.

Then we can consider $(\mathbf{C}, \mathbf{1}_{\mathbf{C}})$ as a module over $(\mathbf{A}, \mathcal{R})$ in $(\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)$. Hence, by Sect. J.1, the functor $F_{\mathbf{A}, \mathcal{R}}$ extends to a (strictly) monoidal functor of pairs

$$(F_{\mathbf{A}, \mathcal{R}}, F_{\mathbf{C}}) : (\Delta^{\text{op}, \text{mock}}, \Delta_*^{\text{op}, \text{mock}}) \rightarrow \left((\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X), (\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X) \right).$$

J.4.9. Let

$$((\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X))_{\text{adj}} \subset (\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)$$

be a 1-full subcategory, where we take the same objects, but as 1-morphisms we let

$$\begin{aligned} \text{Maps}_{((\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X))_{\text{adj}}}((\mathbf{A}_1, \mathcal{R}_1), (\mathbf{A}_2, \mathcal{R}_2)) \subset \\ \subset \text{Maps}_{(\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)}((\mathbf{A}_1, \mathcal{R}_1), (\mathbf{A}_2, \mathcal{R}_2)) \end{aligned}$$

be the full subcategory consisting of those pairs (Φ, ϕ) , for which Φ admits a factorization right adjoint, which then automatically acquires a lax unital structure (see Sect. C.11.21).

J.4.10. Note that the operation of passage to the right adjoint functor defines a (1-fully faithful, symmetric) monoidal functor

$$(J.14) \quad ((\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X))_{\text{adj}} \rightarrow ((\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X))^{\text{op}},$$

which acts as identity on objects, and sends

$$(\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2, \phi(\mathcal{R}_1) \rightarrow \mathcal{R}_2) \rightsquigarrow (\Phi^R : \mathbf{A}_2 \rightarrow \mathbf{A}_1, \psi : \mathcal{R}_1 \rightarrow \Phi^R(\mathcal{R}_2)),$$

where ψ is obtained from ϕ by adjunction.

J.4.11. Therefore, given a (strictly) monoidal functor of pairs

$$(\Delta^{\text{op, mock}}, \Delta_*^{\text{op, mock}}) \rightarrow ((\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X), (\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)),$$

which at the level of 1-morphisms lands in

$$(J.15) \quad (((\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X))_{\text{adj}}, ((\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X))_{\text{adj}}),$$

by passing to right adjoints, we can create from it a (strictly) monoidal functor of pairs

$$(\Delta^{\text{mock}}, \Delta_*^{\text{mock}}) \rightarrow ((\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X), (\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)).$$

J.5. **Applying the paradigm.** We are finally ready to complete the construction from Sect. 5.3.

J.5.1. In the context of Sect. J.4.8, we take

$$\mathbf{A} := \text{Rep}(\check{G}), \quad \mathbf{C} := \text{KL}(G)_{\text{crit}}, \quad \mathcal{R} = R_{\check{G}, \text{Op}}.$$

The data of action from Sect. J.4.8 is provided by Sect. 5.2.9.

Denote the resulting monoidal functor of pairs by

$$\begin{aligned} (F_{\text{Rep}(\check{G}), R_{\check{G}, \text{Op}}}, F_{\text{KL}(G)_{\text{crit}}}) : (\Delta^{\text{op, mock}}, \Delta_*^{\text{op, mock}}) \rightarrow \\ \rightarrow ((\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X), (\text{FactAlg} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)). \end{aligned}$$

J.5.2. Note now that since $\text{Rep}(\check{G})$ is rigid, the above functor $(F_{\text{Rep}(\check{G}), R_{\check{G}, \text{Op}}}, F_{\text{KL}(G)_{\text{crit}}})$ lands in the subcategory (J.15).

Hence, by Sect. J.4.11, we can produce from it a (strictly) monoidal functor of pairs

$$\begin{aligned} (F_{\text{Rep}(\check{G}), R_{\check{G}, \text{Op}}}, F_{\text{KL}(G)_{\text{crit}}})_{\text{adj}} : (\Delta^{\text{mock}}, \Delta_*^{\text{mock}}) \rightarrow \\ \rightarrow ((\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X), (\text{FactAlg}^{\text{op}} \in \text{FactCat}_{\text{lax}})^{\text{untl}}(X)). \end{aligned}$$

J.5.3. Composing with (J.13), we obtain a right-lax monoidal functor of pairs

$$(F_{\text{Rep}(\check{G}), R_{\check{G}, \text{Op}}}, F_{\text{KL}(G)_{\text{crit}}})_{\text{adj}}^{\sim} : (\Delta^{\text{mock}}, \Delta_*^{\text{mock}}) \rightarrow (\text{FactCat}_{\text{lax}}^{\text{untl, lax}}(X), \text{FactCat}_{\text{lax}}^{\text{untl, lax}}(X)).$$

J.5.4. The final step consists of lifting the functor $(F_{\text{Rep}(\check{G}), R_{\check{G}, \text{Op}}}, F_{\text{KL}(G)_{\text{crit}}})_{\text{adj}}^{\sim}$ to a (strictly) monoidal functor of pairs

$$(F_{\text{Rep}(\check{G}), R_{\check{G}, \text{Op}}}, F_{\text{KL}(G)_{\text{crit}}})'_{\text{adj}} : (\Delta^{\text{mock}}, \Delta_*^{\text{mock}}) \rightarrow (\text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X), \text{FactCat}_{\text{lax}}^{\text{untl}, \text{lax}}(X)),$$

which at the level of objects sends

$$n \mapsto \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})^{\otimes n}, \quad (n \sqcup *) \mapsto \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})^{\otimes n} \otimes \text{KL}(G)_{\text{crit}}.$$

This is achieved by repeating the procedure in Sect. J.3.6 using Lemma 5.3.9.

REFERENCES

- [AF] D. Ayala and J. Francis, *Fibrations of ∞ -categories*, Higher Structures **4** (2020), 168–265.
- [AG] D. Arinkin and D. Gaitsgory, *Singular support of coherent sheaves, and the geometric Langlands conjecture*, Selecta Math. N.S. **21** (2015), 1–199.
- [AGKRRV] D. Arinkin, D. Gaitsgory, D. Kazhdan, S. Raskin, N. Rozenblyum and Y. Varshavsky, *The stack of local systems with restricted variation and geometric Langlands theory with nilpotent singular support*, arXiv:2010.01906.
- [AMR] D. Ayala, A. Mazel-Gee and N. Rozenblyum, *Stratified non-commutative geometry*, arXiv:1910.14602.
- [BD1] A. Beilinson and V. Drinfeld, *Quantization of Hitchin's integrable system and Hecke eigensheaves*, available at http://people.math.harvard.edu/~gaitsgde/grad_2009/
- [BD2] A. Beilinson and V. Drinfeld, *Chiral algebras*, American Mathematical Soc. **51** (2004).
- [Be1] D. Beraldo, *On the extended Whittaker category*, Selecta Mathematica **25** (2019).
- [Be2] D. Beraldo, *On the geometric Ramanujan conjecture*, arXiv:2103.17211.
- [Bei] A. Beilinson, *Remarks on topological algebras*, arXiv:0711.2527.
- [Bh] B. Bhatt, *Algebraization and Tannaka duality*, Cambridge Journal of Math **4** (2016), 403–461.
- [Bogd] E. Bogdabova, *in preparation*.
- [CR] J. Campbell and S. Raskin, *Langlands duality on the Beilinson-Drinfeld Grassmannian*, arXiv:2310.19734.
- [Ch1] L. Chen, *Nearby cycles on Drinfeld-Gaitsgory-Vinberg interpolation Grassmannian and long intertwining functor*, Duke Math. Jour. **172** (2023), 447–553.
- [Ch2] L. Chen, *Nearby cycles and dualities in geometric Langlands program*, PhD Thesis (Harvard University 2021), available at: <https://windshower.github.io/linchen/papers/dissertation.pdf>
- [CF] L. Chen and Y. Fu, *An extension of Kazhdan-Lusztig equivalence*, arXiv:2111.14606
- [CFGY] L. Chen, Y. Fu, D. Gaitsgory and D. Yang, *Loop group actions via factorization*, preprint.
- [Dh] G. Dhillon, *Semi-infinite cohomology and the linkage principle for W -algebras*, Advances in Mathematics **381** (2021).
- [Dr] V. Drinfeld, *Two-dimensional ℓ -adic representations of the fundamental group of a curve over a finite field and automorphic forms on $GL(2)$* , Amer. Jour. of Math. **105** (1983), 85–114.
- [DG1] V. Drinfeld and D. Gaitsgory, *On some finiteness questions for algebraic stacks*, Geometric and Functional Analysis. **23** (2013), 149–294.
- [DrGa2] V. Drinfeld and D. Gaitsgory, *Compact generation of the category of D -modules on the stack of G -bundles on a curve* Cambridge Math Journal, **3** (2015), 19–125.
- [FR] J. Faergeman and S. Raskin, *Non-vanishing of geometric Whittaker coefficients for reductive groups*, arXiv:2207.02955.
- [FraG] J. Francis and D. Gaitsgory, *Chiral Koszul duality*, Selecta Math. (N.S.) **18** (2012), 27–87.
- [FF] B. Feigin and E. Frenkel, *Affine Kac-Moody algebras at the critical level and Gelfand-Dikii algebras*, International Journal of Modern Physics A. **7** (1992), 197–215.
- [FG1] E. Frenkel and D. Gaitsgory, *D -modules on the affine Grassmannian and representations of affine Kac-Moody algebras*, Duke Math. Jour. **152** (2004), 279–327.
- [FG2] E. Frenkel and D. Gaitsgory, *Local geometric Langlands correspondence: the spherical case*, Algebraic Analysis and around, **54** (2009), 167–186.
- [FG3] E. Frenkel and D. Gaitsgory, *Weyl modules and opers without monodromy*, Arithmetic and geometry around quantization, (2010), 101–121.
- [FG4] E. Frenkel and D. Gaitsgory, *Local geometric Langlands correspondence and affine Kac-Moody algebras*, in: Algebraic Geometry and Number Theory: In Honor of Vladimir Drinfeld's 50th birthday, (2006), 69–260.
- [FG5] E. Frenkel and D. Gaitsgory, *Fusion and convolution: applications to affine Kac-Moody algebras at the critical level*, Pure Appl. Math. Q. **2** (2006), 1255–1312.
- [FG6] E. Frenkel and D. Gaitsgory, *D -modules on the affine flag variety and representations of affine Kac-Moody algebras*, Represent. Theory **13** (2009), 470–608.
- [FGV] E. Frenkel, D. Gaitsgory and K. Vilonen, *Whittaker patterns in the geometry of moduli spaces of bundles on curves*, Annals of Math. **153** (2001), 699–748.
- [FLMM1] G. Fortuna, D. Lombardo, A. Maffei, V. Melani, *Local opers with two singularities: the case of $\mathfrak{sl}_2$* , Comm.Math.Phys **394**, (2022), 303–1360.
- [FLMM2] G. Fortuna, D. Lombardo, A. Maffei, V. Melani, *The semi-infinite cohomology of Weyl modules with two singular points*, PAMQ **20** (2024), 1251–1284.
- [Ga1] D. Gaitsgory, *Outline of the proof of the geometric Langlands conjecture for $GL(2)$* , Astérisque **370** (2015), 1–112.
- [Ga2] D. Gaitsgory, *A conjectural extension of the Kazhdan-Lusztig equivalence*, Publications of RIMS **57** (2021), 1227–1376.
- [Ga3] D. Gaitsgory, *A "strange" functional equation for Eisenstein series and miraculous duality on the moduli stack of bundles*, Annales Scientifiques de l'ENS **50** (2017), 1123–1162.
- [Ga4] D. Gaitsgory, *The Atiyah-Bott formula for the cohomology of the moduli space of bundles on a curve*, arXiv:1505.02331 (2015).
- [Ga5] D. Gaitsgory, *Sheaves of categories and the notion of 1-affineness*, Stacks and categories in geometry, topology, and algebra, **643** (2015), 127–225.
- [Ga6] D. Gaitsgory, *The local and global versions of the Whittaker category*, PAMQ **16**, (2020), 775–904.
- [Ga7] D. Gaitsgory, *Ind-coherent sheaves*, Moscow Mathematical Journal, **13** (2013), 399–528.
- [GaLu] D. Gaitsgory and J. Lurie, *Weil's conjecture for function fields. Volume 1*, Annals of Mathematics Studies (2014).
- [GLys] D. Gaitsgory and S. Lysenko, *Parameters and duality for the metaplectic geometric Langlands theory*, Selecta Math. New Ser. **24** (2018), 227–301. Also arXiv: 1608.00284
- [GaRo1] D. Gaitsgory and N. Rozenblyum, *DG indschemes*, Perspectives in representation theory, **610** (2014), 139–251.

- [GaRo2] D. Gaitsgory and N. Rozenblyum, *Crystals and D-modules*, Pure and Applied Mathematics Quarterly, **10** (2014), 57–154.
- [GaRo3] D. Gaitsgory and N. Rozenblyum, *A study in derived algebraic geometry, Vol. 1: Correspondences and Duality*, Mathematical surveys and monographs **221** (2017), AMS, Providence, RI.
- [GaRo4] D. Gaitsgory and N. Rozenblyum, *A study in derived algebraic geometry, Vol. 2: Deformations, Lie theory and formal geometry*, Mathematical surveys and monographs **221** (2017), AMS, Providence, RI.
- [GLC1] D. Gaitsgory and S. Raskin, *Proof of the geometric Langlands conjecture I: construction of the functor*, arXiv:2405.03599
- [GLC4] D. Arinkin, D. Beraldo, L. Chen, J. Faergeman, D. Gaitsgory, K. Lin, S. Raskin and N. Rozenblyum, *Proof of the geometric Langlands conjecture IV: ambidexterity*, available at: <https://people.mpim-bonn.mpg.de/gaitsgde/GLC/>
- [HR] J. Hilburn and S. Raskin, *Tate's thesis in the de Rham setting*, Journal of the American Mathematical Society, **36** (2023), 917–1001.
- [Lin] K. Lin, *Poincaré series and miraculous duality*, arXiv:2211.05282.
- [Lu1] J. Lurie, *Higher topos theory*, Princeton University Press (2009).
- [Lu2] J. Lurie, *Higher Algebra* (2017). Available at: <https://www.math.ias.edu/lurie/papers/HA.pdf>.
- [Lu3] J. Lurie, *Spectral algebraic geometry* (2018). Available at: <https://www.math.ias.edu/lurie/papers/SAG-rootfile.pdf>.
- [Ra1] S. Raskin, *A geometric proof of the Feigin-Frenkel theorem*, Representation Theory **16** (2012), 489–512.
- [Ra2] S. Raskin, *W-algebras and Whittaker categories*, Selecta Mathematica **27** (2021), 46.
- [Ra3] S. Raskin, *Chiral principal series categories I: finite dimensional calculations*, Adv. in Math. **388** (2021).
- [Ra4] S. Raskin, *Chiral principal series categories II: the factorizable Whittaker category*, available at: <https://gauss.math.yale.edu/sr2532/cpsii.pdf>.
- [Ra5] S. Raskin, *Homological methods in semi-infinite contexts*, arXiv:2002.01395 (2020).
- [Ra6] S. Raskin, *Chiral categories* (2015), available at: <https://gauss.math.yale.edu/sr2532/chiralcats.pdf>.
- [Ro1] N. Rozenblyum, *Modules over a chiral algebra*, arXiv:1010.1998 (2010).
- [Ro2] N. Rozenblyum, *Connections on moduli spaces and infinitesimal Hecke modifications*, arXiv:2108.07745 (2021).

PROOF OF THE GEOMETRIC LANGLANDS CONJECTURE III: COMPATIBILITY WITH PARABOLIC INDUCTION

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ABSTRACT. We establish the compatibility of the Langlands functor $\mathbb{L}_G$ with the operations of Eisenstein series constant term, and deduce that $\mathbb{L}_G$ induces an equivalence on Eisenstein-generated subcategories.

CONTENTS

Introduction	5
0.1. What is done in this paper?	5
0.2. The logical structure: compatibility with Eisenstein series and constant terms	5
0.3. The logical structure: equivalence on Eisenstein parts	7
0.4. All this enhancement business	9
0.5. The local FLE/Jacquet compatibility	11
0.6. Description of the actual contents	13
0.7. Conventions and notation	15
0.8. Acknowledgements	15
1. The local semi-infinite category	17
1.1. The corrected Jacquet functor	17
1.2. Digression: renormalized equivariant categories	18
1.3. The local spherical semi-infinite category	20
1.4. Digression: Sph_G as a rigid category	22
1.5. The dual of the spherical semi-infinite category	25
1.6. A twisting construction: recollections	27
1.7. Twisting with respect to the center	29
1.8. The enhanced Jacquet functor	30
2. Spectral semi-infinite category/ies	34
2.1. The spherical spectral semi-infinite category	34
2.2. The (factorization) algebras $\tilde{\Omega}^{\text{spec}}$ and Ω^{spec}	37
2.3. Spectral Jacquet functors	39
2.4. Semi-infinite geometric Satake	40
2.5. Consequences for Jacquet functors	41
2.6. Self-duality on the spectral side	43
2.7. The semi-infinite Casselman-Shalika equivalence	45
3. Jacquet and Wakimoto functors for Kac-Moody modules	46
3.1. Twisting by Z_G^0 -torsors	47
3.2. Jacquet functor(s) for Kac-Moody representations	48
3.3. The enhanced Jacquet functor for Kac-Moody representations	50
3.4. The Wakimoto functor(s)	51
3.5. Restriction of Wakimoto modules	52

¹With an appendix coauthored additionally by J. Faergeman, K. Lin and N. Rozenblyum.
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3.6.	The “usual” Wakimoto functor	53
3.7.	The enhanced Wakimoto functor	54
3.8.	Twisting by Z_G -torsors	56
3.9.	The twisted Jacquet and Wakimoto functors	58
3.10.	Drinfeld-Sokolov reduction of Wakimoto modules	61
3.11.	Proof of Proposition 3.6.10	62
4.	Jacquet functor(s) for opers	63
4.1.	(Parabolic) Miura opers	63
4.2.	Definition of the Jacquet functor for opers	65
4.3.	The <i>semi-infinite</i> Jacquet functor for opers, pointwise version	66
4.4.	The semi-infinite category for opers	69
4.5.	A formal completion of Miura opers	73
4.6.	The semi-infinite Jacquet functor for opers, factorization version	76
4.7.	The semi-infinite Jacquet functor for opers, enhanced version	79
5.	Compatibility of the FLE with the Jacquet functors	80
5.1.	The twisted critical FLE	80
5.2.	The initial formulation	82
5.3.	The enhanced version	83
5.4.	The dual formulation	85
6.	Proof of Theorem 5.4.5	86
6.1.	Structure of the proof	86
6.2.	The composition on the spectral side	90
6.3.	The composition on the geometric side	92
6.4.	Identification of the compositions	95
6.5.	Upgrade to factorization algebras I	96
6.6.	Interlude: Miura transform and the Wakimoto functor	98
6.7.	Proof of Proposition 6.3.9	99
6.8.	Upgrade to an isomorphism with values in $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})$	100
6.9.	Upgrade to factorization algebras II	102
7.	Proof of Theorem 5.4.5, continuation	104
7.1.	Structure of the argument	104
7.2.	The t-structure on the spectral side	105
7.3.	The left t-exactness of the co-Jacquet functor	105
7.4.	Fully faithfulness on the spectral side	107
8.	Eisenstein and constant term functors on the geometric side	111
8.1.	The usual Eisenstein and constant term functors	111
8.2.	The geometric (co-)enhanced category	113
8.3.	The <i>co</i> -enhanced constant term functor	115
8.4.	Proof of Lemma 8.3.9	118
8.5.	The enhanced Eisenstein and constant term functors	118
8.6.	Enhanced functors and duality	120
8.7.	Enhanced functors and miraculous duality	122
8.8.	The ρ -shift	124
9.	Global interpretation of the enhanced functors	127
9.1.	Drinfeld’s compactifications	127
9.2.	Local-to-global construction for Drinfeld’s compactification, the $*$ -version	130
9.3.	The strong approximation argument	132
9.4.	The global enhanced functors, the $*$ -version	134
9.5.	Proof of Lemma 9.4.5	137
9.6.	Local-to-global construction for Drinfeld’s compactification, the $!$ -version	138
9.7.	The global enhanced functors, the $!$ -version	143
9.8.	Relation to compactified Eisenstein series	145
10.	Interaction of parabolic induction with the Whittaker model	146

10.1.	Whittaker coefficients of Eisenstein series	147
10.2.	The global model for the Whittaker category	149
10.3.	The space of <i>generic</i> reductions	152
10.4.	Closures of semi-infinite orbits	153
10.5.	Zastava spaces	155
10.6.	Proof of Theorem 10.1.2	157
10.7.	Proof of Proposition 10.6.8	161
10.8.	Whittaker coefficients of Eisenstein series: the enhanced version	163
10.9.	Proof of Theorem 10.8.2	164
10.10.	Proof of Proposition 10.9.5	165
11.	Constant term of Kac-Moody localization	167
11.1.	Linear twistings on Bun_G	167
11.2.	Integration along $\mathrm{Bun}_{P^-} \rightarrow \mathrm{Bun}_M$ and BRST	168
11.3.	Identification of line bundles	169
11.4.	Proof of Theorem 11.2.3	171
11.5.	Constant term of localization: the enhanced version	171
11.6.	Construction of the natural transformation for the enhanced functor	174
11.7.	Addendum: Whittaker coefficients of localization, take II	177
11.8.	Localization and translation by Z_G -torsors	179
11.9.	Constant term of localization: the translated version	180
12.	Spectral Eisenstein and constant term functors	182
12.1.	The usual spectral Eisenstein and constant term functors	182
12.2.	Spectral Eisenstein functor and localization	185
12.3.	Proof of Theorem 12.2.10	188
12.4.	Global (co-)enhanced categories on the spectral side	190
12.5.	The enhanced spectral Eisenstein functor	191
12.6.	Partial enhancement	194
12.7.	Partial enhancement, continued	196
12.8.	The “compactified” spectral Eisenstein functor	198
12.9.	Spectral Eisenstein functor and localization: the enhanced version	200
12.10.	The (co-)enhanced spectral constant term functor	202
12.11.	Partial enhancement for the constant term functor	205
12.12.	The “compactified” spectral constant term functor	207
12.13.	Enhancement and duality on the spectral side	208
13.	Spectral constant term functor and opers	210
13.1.	Statement of the result—the unenhanced version	210
13.2.	Proof of Theorem 13.1.13	214
13.3.	How do Miura opers appear?	216
13.4.	Opers force transversality	218
13.5.	The enhanced version of spectral Poincaré vs constant term compatibility	218
13.6.	Addendum: spectral global sections and the spectral Poincaré functor	221
14.	Langlands functor and Eisenstein series	224
14.1.	The Langlands functor: recollections	224
14.2.	Compatibility of the Langlands functor with Eisenstein series	226
14.3.	Compatibility of the Langlands functor with Eisenstein series—the enhanced version	229
14.4.	Proof of Theorem 14.1.6	230
15.	Langlands functor and constant terms	233
15.1.	Statement of the compatibility result	233
15.2.	Proof of Theorem 15.1.13 for $P = B$	237
15.3.	Proof of Proposition 15.2.10	240
15.4.	Proof of Theorem 15.1.13 for a general Levi	243
15.5.	Proof of Proposition 15.4.10	246
15.6.	Constant term compatibility: the enhanced version	248

16.	The left adjoint of the Langlands functor	254
16.1.	The existence of the left adjoint	254
16.2.	The left adjoint as a dual	255
16.3.	Proof of Lemma 16.2.12	258
16.4.	The composition $\mathbb{L}_G \circ \mathbb{L}_G^L$	259
16.5.	Proof of Proposition 16.4.10	262
17.	The Langlands functor is an equivalence on the Eisenstein part(s)	263
17.1.	Statement of the main result	263
17.2.	Constant terms and the left adjoint of the Langlands functor	263
17.3.	Proof of Theorem 17.1.2	265
17.4.	Proof of Proposition 17.3.6	267
17.5.	Langlands functor and the “compactified” constant term functor(s)	269
Appendix A.	The spectral semi-infinite category/ies over the Ran space	272
A.1.	Horizontal loops into the (parabolic) base affine space	272
A.2.	Ind-coherent sheaves on horizontal loop and arc spaces	273
A.3.	Ind-coherent sheaves on $\mathfrak{L}_\nabla(\tilde{G}/\tilde{N}_P^-)$	274
A.4.	Compact generation and duality	276
A.5.	Construction of the spherical spectral semi-infinite category	277
A.6.	Construction of the category $\mathrm{IndCoh}^*((\mathrm{LS}_{\tilde{P}^-}^{\mathrm{mon-free} \tilde{M}})_{\mathrm{mon-free}}^\wedge)$	281
A.7.	t-structures on the spectral semi-infinite categories	283
A.8.	The global counterpart	284
Appendix B.	Proof of semi-infinite Casselman-Shalika	286
B.1.	Orientation session	286
B.2.	The semi-infinite Whittaker category	287
B.3.	Digression: monoidal actions of factorization categories	289
B.4.	The spectral counterpart	290
B.5.	Identification of factorization algebras	291
B.6.	Finiteness properties of Ω^{spec}	292
B.7.	Finiteness properties of Ω	293
B.8.	Proof of Theorem 2.7.6	294
B.9.	Proof of Theorem B.2.12(a)	294
Appendix C.	Sheaves on the (pointwise) $\mathrm{LS}_H^{\mathrm{mer}}$ for a unipotent group	295
C.1.	The statement for unipotent groups	295
C.2.	Method of proof	295
C.3.	Digression: factorization modules over holonomic factorization algebras	296
C.4.	Proof of Proposition C.2.4	297
C.5.	Proof of Proposition C.4.5	298
C.6.	Proof of Proposition C.4.6	298
Appendix D.	Compactness of Ω^{spec}	299
D.1.	The shearing trick	299
D.2.	From <i>almost</i> compactness to compactness	300
D.3.	Proof of Proposition D.2.5	303
Appendix E.	Proof of semi-infinite geometric Satake	304
E.1.	Another orientation session	305
E.2.	The factorization algebra $\Omega(R)^{\mathrm{spec}}$	305
E.3.	Fully faithfulness on the eventually coconnective part	307
E.4.	Proof of Proposition E.3.6	308
E.5.	The functor $\mathbf{F}^{\mathrm{geom}}$	311
E.6.	Connecting the geometric and spectral sides	312
E.7.	Construction of the functor	313
E.8.	Proof of the pointwise equivalence	314
E.9.	Identification of the semi-infinite IC sheaf	316
Appendix F.	Proof of Theorem 6.9.3	318

F.1. Regular Kac-Moody representations	318
F.2. Transverse Kac-Moody representations	319
F.3. Wakimoto modules as objects of the transverse subcategory	320
F.4. Proof of Theorem F.3.3	322
F.5. Proof of Theorem F.3.6	323
F.6. Proof of Lemma F.5.2	324
F.7. Proof of Lemma F.5.3	327
References	329

INTRODUCTION

0.1. **What is done in this paper?** This paper is the third in the series of five, in the course of which a proof of the geometric Langlands conjecture (stated as [GLC1, Conjecture 1.6.7]) will be given.

0.1.1. As far as the program of proving the geometric Langlands conjecture is concerned, in this paper the following two steps toward the proof are performed:

- It is shown that the geometric Langlands functor

$$(0.1) \quad \mathbb{L}_G : \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$$

is compatible with the functors of *Eisenstein series* and *constant term*, see Theorems 14.2.2 and 15.1.13, respectively.

- Assuming the geometric Langlands conjecture for proper Levi subgroups, it is shown (Theorem 17.1.2) that $\mathbb{L}_G$ induces an equivalence on Eisenstein-generated subcategories

$$\mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \xrightarrow{\sim} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}.$$

Another result that concerns the global geometric Langlands program, proved here, and which is of independent interest, is Theorem 16.2.3, which says that:

- The left adjoint functor of $\mathbb{L}_G$ can be obtained from the functor *dual* to $\mathbb{L}_G$, by composing with the *Miraculous functor* and Chevalley involution.

0.1.2. Other results established in this paper concern the *local Langlands theory*. Apart from being of independent interest, these results provide local ingredients for the proofs of global theorems mentioned above.

The main local result is Theorem 5.2.3, which establishes the compatibility of the critical FLE with the Jacquet functors on the geometric and spectral sides.

0.2. **The logical structure: compatibility with Eisenstein series and constant terms.** We will now describe the logical structure of the paper from the point of view of the geometric Langlands conjecture.

0.2.1. The Langlands functor $\mathbb{L}_G$ is constructed so that it makes the following diagram commute¹

$$(0.2) \quad \begin{array}{ccc} \mathrm{Whit}^!(\mathrm{Gr}_{G,\mathrm{Ran}}) & \xrightarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{coeff}_G \uparrow & & \uparrow \Gamma_G^{\mathrm{IndCoh},\mathrm{spec}} \\ \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}). \end{array}$$

Here:

- CS_G is the geometric Casselman-Shalika equivalence;
- coeff_G is the functor of Whittaker coefficient(s);

¹Up to a cohomological shift, which we omit in the Introduction.

- $\Gamma_{\check{G}}^{\text{IndCoh, spec}}$ is the functor right adjoint to the localization functor

$$\text{Rep}(\check{G})_{\text{Ran}} \xrightarrow{\text{Loc}_{\check{G}}^{\text{spec}}} \text{QCoh}(\text{LS}_{\check{G}}) \hookrightarrow \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}).$$

The geometric Langlands conjecture ([GLC2, Conjecture 1.6.7]) says that the functor $\mathbb{L}_G$ is an equivalence.

0.2.2. The compatibility of the geometric Casselman-Shalika equivalence with Jacquet functors implies that $\mathbb{L}_G$ is compatible with the *Eisenstein functors*, i.e., for a standard (negative) parabolic P^- with Levi quotient M , the following diagram commutes (again, up to a cohomological shift):

$$(0.3) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{M}}) \\ \text{Eis}_{*, \rho_P(\omega_X)}^- \downarrow & & \downarrow \text{Eis}^-, \text{spec} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}). \end{array}$$

In the above formula, $\text{Eis}_{*, \rho_P(\omega_X)}^-$ is the *translated* Eisenstein series functor, see Sect. 8.8.

0.2.3. Given diagram (0.3), by passing to right adjoint functors along the vertical arrows, we obtain a diagram

$$(0.4) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{M}}), \\ \text{CT}_{*, \rho_P(\omega_X)}^- \uparrow & \searrow & \uparrow \text{CT}^-, \text{spec} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}). \end{array}$$

that commutes *up to a natural transformation*.

However, it is entirely not obvious that the natural transformation in (0.4) is an isomorphism. Ultimately, we establish that it is an isomorphism (Theorem 15.1.2), but this comes after we prove our main result, Theorem 17.1.2.

0.2.4. What we prove *a priori* is that there exists *some* natural isomorphism that makes the diagram

$$(0.5) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{M}}) \\ \text{CT}_{*, \rho_P(\omega_X)}^- \uparrow & & \uparrow \text{CT}^-, \text{spec} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \end{array}$$

commute².

The existence of the commutative diagram (0.5) is one of the main results of this paper (Theorem 15.1.13). It uses the commutative square³

$$(0.6) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \\ \text{Loc}_G \uparrow & & \uparrow \text{Poinc}_G^{\text{spec}} \\ \text{KL}(G)_{\text{crit, Ran}} & \xrightarrow{\text{FLE}_{G, \text{crit}}} & \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}}(\mathcal{D}^\times))_{\text{Ran}}, \end{array}$$

constructed in [GLC2, Theorem 18.5.2].

²We do not know, and are not sure that it is true, that the natural isomorphism in (0.5) equals one of (0.4). One can show, however, that the two differ by a (non-zero) scalar.

³Up to a graded line, omitted in the Introduction.

0.2.5. We prove the existence of (0.5) by constructing a commutative cube (see diagram (15.2)) that relates the diagram (0.6) for G with a similar diagram for M , with the crucial ingredient being the compatibility of the critical FLE with Jacquet functors, given by Theorem 5.2.3 mentioned above.

0.2.6. Of course, the compatibility of $\mathbb{L}_G$ with the critical localization functor, expressed by diagram (0.6) plays a much bigger role in this project than just proving the existence of (0.5).

In the next paper, it will be used to show that the functor $\mathbb{L}_G$ is *ambidextrous* (at least on the cuspidal part), which would be another crucial step towards the proof of the geometric Langlands conjecture.

Remark 0.2.7. One can say that our approach to the proof of the geometric Langlands conjecture consists of playing diagrams (0.2) and (0.6) one against the other.

Note that the approach to geometric Langlands via (0.2) was essentially the idea behind Drinfeld's founding work [Dr] (taken up by [Laum] and developed further in [FGV]), and the approach via (0.6) was the idea behind the work of Beilinson-Drinfeld [BD1].

0.3. The logical structure: equivalence on Eisenstein parts.

0.3.1. The existence of diagram (0.5) immediately implies that the functor $\mathbb{L}_G$ admits a left adjoint, to be denoted $\mathbb{L}_G^L$, which makes the diagram

$$(0.7) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xleftarrow{\mathbb{L}_M^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}}) \\ \mathrm{Eis}_{\tilde{M}, \rho_F(\omega_X)}^- \downarrow & & \downarrow \mathrm{Eis}^-, \mathrm{spec} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xleftarrow{\mathbb{L}_G^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}). \end{array}$$

commute.

Having both diagrams (0.3) and (0.7) implies that the functors $\mathbb{L}_G$ and $\mathbb{L}_G^L$ send the subcategories

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

and

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})_{\mathrm{red}} \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$$

to one another.

The main result of this paper, Theorem 17.1.2, says that the resulting adjoint functors

$$(0.8) \quad \mathbb{L}_G : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \rightleftarrows \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})_{\mathrm{red}} : \mathbb{L}_G^L$$

are mutually inverse equivalences, provided that we know that the geometric Langlands conjecture holds for all proper Levi subgroups of G .

We will now explain the logic of the proof of this theorem.

0.3.2. Given diagrams (0.3) and (0.7), and using the inductive hypothesis that GLC holds for Levi quotients of proper parabolic subgroups, it suffices to show that $\mathbb{L}_G^L$ is fully faithful.

0.3.3. Consider the composition

$$\mathbb{L}_G \circ \mathbb{L}_G^L,$$

viewed as a monad acting on $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$.

We show (Theorem 16.4.2) that this monad is given by the action of an associative algebra object

$$\mathcal{A}_G \in \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})$$

(we view $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$ as a module category over $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})$).

0.3.4. The assertion that the functor $\mathbb{L}_G^L$ is fully faithful is equivalent to the assertion that the unit

$$(0.9) \quad \mathcal{O}_{\mathrm{LS}_{\check{G}}} \rightarrow \mathcal{A}_G$$

is an isomorphism.⁴

The assertion that $\mathbb{L}_G^L|_{\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}}$ is fully faithful is equivalent to the assertion that the map

$$(0.10) \quad \mathcal{O}_{\mathrm{LS}_{\check{G}}^{\mathrm{red}}} \rightarrow \mathcal{A}_G|_{\mathrm{LS}_{\check{G}}^{\mathrm{red}}},$$

induced by (0.9), is an isomorphism, where $\mathrm{LS}_{\check{G}}^{\mathrm{red}} \subset \mathrm{LS}_{\check{G}}$ is any closed substack whose underlying subset consists of reducible local systems.

0.3.5. The latter assertion is equivalent to the map

$$(0.11) \quad \mathcal{O}_{\mathrm{LS}_{\check{P}^-}} \rightarrow (\mathfrak{q}^{-, \mathrm{spec}})^*(\mathcal{A}_G)$$

being an isomorphism for any *proper* standard (negative) parabolic $\check{P}^- \subset \check{G}$, where

$$\mathfrak{q}^{-, \mathrm{spec}} : \mathrm{LS}_{\check{P}^-} \rightarrow \mathrm{LS}_{\check{G}}$$

is the natural projection.

Now, a simple but crucial observation is given by Lemma 17.3.7, which says that in order to prove that (0.11) is an isomorphism, it is enough to show that the object

$$(\mathfrak{q}^{-, \mathrm{spec}})^*(\mathcal{A}_G) \in \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-})$$

is a *line bundle*.

Further, we show that in order to prove that $(\mathfrak{q}^{-, \mathrm{spec}})^*(\mathcal{A}_G)$ is a line bundle, it suffices to show that its *restriction* along the map

$$\mathrm{LS}_{\check{M}} \rightarrow \mathrm{LS}_{\check{P}^-}$$

is a line bundle on $\mathrm{LS}_{\check{M}}$.

0.3.6. Now, the functor of pullback along

$$\mathrm{LS}_{\check{M}} \rightarrow \mathrm{LS}_{\check{P}^-} \rightarrow \mathrm{LS}_{\check{G}},$$

is (up to tensoring by a line bundle) the functor of “compactified constant term”,⁵ denoted $\mathrm{CT}_{!*}^{-, \mathrm{spec}}$, introduced in Sect. 12.12.

Note that

$$\mathcal{A}_G \simeq \mathbb{L}_G \circ \mathbb{L}_G^L(\mathcal{O}_{\mathrm{LS}_{\check{G}}}).$$

We will prove that there exists a canonical isomorphism

$$(0.12) \quad \mathrm{CT}_{!*}^{-, \mathrm{spec}} \circ \mathbb{L}_G \circ \mathbb{L}_G^L \simeq \mathrm{CT}_{!*}^{-, \mathrm{spec}},$$

see Proposition 17.5.2, so that

$$\mathrm{CT}_{!*}^{-, \mathrm{spec}}(\mathcal{A}_G) \simeq \mathrm{CT}_{!*}^{-, \mathrm{spec}}(\mathcal{O}_{\mathrm{LS}_{\check{G}}}).$$

This will establish the desired property of $(\mathfrak{q}^{-, \mathrm{spec}})^*(\mathcal{A}_G)$, since this property clearly holds for $\mathcal{O}_{\mathrm{LS}_{\check{G}}}$.

⁴In fact, combining the main theorem of [FR] with the commutation of (0.5) and the inductive hypothesis that GLC holds for proper Levis, we obtain that the fully-faithfulness of $\mathbb{L}_G^L$ is equivalent to the geometric Langlands conjecture.

⁵This is so on the subcategory $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \subset \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})$, which is what is relevant for us here. The actual definition of $\mathrm{CT}_{!*}^{-, \mathrm{spec}}$ is more involved, see Sect. 12.12.

0.3.7. In order to prove (0.12), we break it into two isomorphisms

$$(0.13) \quad \mathrm{CT}_{!*}^{-,\mathrm{spec}} \circ \mathbb{L}_G \simeq \mathbb{L}_M \circ \mathrm{CT}_{!*}^{-}(\omega_X)$$

and

$$(0.14) \quad \mathrm{CT}_{!*}^{-}(\omega_X) \circ \mathbb{L}_G^L \simeq \mathbb{L}_M^L \circ \mathrm{CT}_{!*}^{-,\mathrm{spec}},$$

for a suitably defined functor

$$\mathrm{CT}_{!*}^{-}(\omega_X) : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M).$$

Then (0.12) would follow by the induction hypothesis that the geometric Langlands conjecture holds for M , and hence $\mathbb{L}_M \circ \mathbb{L}_M^L \simeq \mathrm{Id}$.

0.3.8. The upshot of the above discussion is that in addition to the commutative diagram (0.5), one needs to have a commutative diagram that involves the pair $(\mathrm{CT}_{!*}^{-}(\omega_X), \mathrm{CT}_{!*}^{-,\mathrm{spec}})$. This would imply the isomorphism (0.13).

It will turn out by duality considerations that (0.14) is equivalent to a cousin of (0.3), where instead of $(\mathrm{Eis}_{!*}^{-}(\omega_X), \mathrm{Eis}_{!*}^{-,\mathrm{spec}})$ one uses suitably defined functors $(\mathrm{Eis}_{!*}^{-}(\omega_X), \mathrm{Eis}_{!*}^{-,\mathrm{spec}})$.

This brings us to the idea of *enhancement*.

0.4. All this enhancement business.

0.4.1. Informally speaking, the enhancement procedure(s)

$$\mathrm{Eis}_{!*}^{-} \rightsquigarrow \mathrm{Eis}_{!*}^{-}, \quad \mathrm{Eis}_{!*}^{-,\mathrm{spec}} \rightsquigarrow \mathrm{Eis}_{!*}^{-,\mathrm{spec}}$$

and

$$\mathrm{CT}_{!*}^{-} \rightsquigarrow \mathrm{CT}_{!*}^{-}, \quad \mathrm{CT}_{!*}^{-,\mathrm{spec}} \rightsquigarrow \mathrm{CT}_{!*}^{-,\mathrm{spec}}$$

is a mechanism to encode the (pre)compositions of the original functors with the Hecke functors for G and M (resp., $\check{G}$ and $\check{M}$).

0.4.2. Technically, this idea is realized as follows. We introduce the *local semi-infinite categories*, denoted

$$(0.15) \quad \mathrm{I}(G, P^-)^{\mathrm{loc}} \text{ and } \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec,loc}},$$

respectively.

The category on the geometric side, i.e., $\mathrm{I}(G, P^-)^{\mathrm{loc}}$, is the category of (half-twisted) D-modules on Gr_G , satisfying the parabolic semi-infinite equivariance condition, i.e., equivariance with respect to $\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)$, up to a small renormalization procedure.

On the spectral side, the category $\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec,loc}}$ is that of (appropriately defined) ind-coherent sheaves on the fiber product

$$(0.16) \quad \mathrm{LS}_{\check{M}}(\mathcal{D}) \times_{\mathrm{LS}_{\check{M}}(\mathcal{D}^\times)} \mathrm{LS}_{\check{P}^-}(\mathcal{D}^\times) \times_{\mathrm{LS}_{\check{G}}(\mathcal{D}^\times)} \mathrm{LS}_{\check{G}}(\mathcal{D}),$$

where $\mathcal{D}$ (resp., $\mathcal{D}^\times$) is the formal (resp., formal punctured) disc.

The categories (0.15) are acted on by

$$\mathrm{Sph}_G \otimes \mathrm{Sph}_M \text{ and } \mathrm{Sph}_G^{\mathrm{spec}} \otimes \mathrm{Sph}_M^{\mathrm{spec}},$$

respectively. So they give rise to functors between

$$\mathrm{Sph}_G\text{-}\mathbf{mod} \leftrightarrow \mathrm{Sph}_M\text{-}\mathbf{mod} \text{ and } \mathrm{Sph}_G^{\mathrm{spec}}\text{-}\mathbf{mod} \leftrightarrow \mathrm{Sph}_M^{\mathrm{spec}}\text{-}\mathbf{mod},$$

respectively.

One forms the global enhanced categories⁶

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^{-,\mathrm{enh}} := \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)^{\mathrm{loc}},$$

⁶The tensor products below should be spread over the Ran space, a point which we elide in the Introduction.

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)^{-,\mathrm{enh}} := \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes_{\mathrm{Sph}_M} \mathrm{I}(G, P^-)^{\mathrm{loc}}$$

and

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})^{-,\mathrm{enh}} := \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}},$$

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}})^{-,\mathrm{enh}} := \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \otimes_{\mathrm{Sph}_{\check{M}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}},$$

and we show that the enhanced Eisenstein and constant term functors naturally extend to functors between the enhanced categories.

0.4.3. We show that the (derived) geometric Satake equivalences

$$\mathrm{Sph}_G \xrightarrow{\mathrm{Sat}_G} \mathrm{Sph}_G^{\mathrm{spec}} \text{ and } \mathrm{Sph}_M \xrightarrow{\mathrm{Sat}_M} \mathrm{Sph}_M^{\mathrm{spec}}$$

extend to an equivalence

$$\mathrm{I}(G, P^-)^{\mathrm{loc}} \xrightarrow{\mathrm{Sat}^-, \infty} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}},$$

and we show that the enhanced Langlands functor(s)

$$\mathbb{L}_G^{-,\mathrm{enh}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^{-,\mathrm{enh}} \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})^{-,\mathrm{enh}}$$

and

$$\mathbb{L}_M^{-,\mathrm{enh}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)^{-,\mathrm{enh}} \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}})^{-,\mathrm{enh}}$$

are compatible with the enhanced versions of the Eisenstein and constant term functors.

0.4.4. Now, the point of all this is that one can express the “compactified” functors

$$\mathrm{Eis}_{!*}^-, \mathrm{Eis}_{!*}^{-,\mathrm{spec}}, \mathrm{CT}_{!*}^-, \mathrm{CT}_{!*}^{-,\mathrm{spec}}$$

in terms⁷ of the enhanced functors

$$\mathrm{Eis}_!^{-,\mathrm{enh}}, \mathrm{Eis}^{-,\mathrm{spec}, \mathrm{enh}}, \mathrm{CT}_*^{-,\mathrm{enh}}, \mathrm{CT}^{-,\mathrm{spec}, \mathrm{enh}},$$

respectively, and the enhanced compatibility of the Langlands functor with the Eisenstein and constant term functors implies the compatibility with the “compactified” functors, alluded to in Sect. 0.3.8.

0.4.5. The above explains why we need the enhancements for the proof of Theorem 17.1.2.

However, there is one more aspect that makes these enhancements necessary: they are used in the proof in the main local result, Theorem 5.2.3, which expresses the compatibility of the FLE with Jacquet functors, see Sect. 0.5 below.

0.4.6. We conclude this subsection with the following remark. The enhancement

$$\mathrm{Eis}_!^- \rightsquigarrow \mathrm{Eis}_!^{-,\mathrm{enh}}$$

is in fact something natural when one traces back the path from sheaves to functions.

Recall that in the classical theory of automorphic functions, the Eisenstein operator is defined as pull-push along the following diagram

$$(0.17) \quad \begin{array}{ccc} & G(\mathbb{A})/P^-(K) & \\ \swarrow & & \searrow \\ G(\mathbb{A})/G(K) & & G(\mathbb{A})/N_P^-(\mathbb{A}) \cdot M(K). \end{array}$$

⁷This expression is provided using the *semi-infinite IC sheaf* constructed in [Ga4, Ga5].

This is a map of $G(\mathbb{A})$ -representations. When one takes spherical vectors everywhere, one obtains the diagram

$$(0.18) \quad \begin{array}{ccc} & G(\mathbb{O}) \backslash G(\mathbb{A}) / P^-(K) & \\ \swarrow & & \searrow \\ G(\mathbb{O}) \backslash G(\mathbb{A}) / G(K) & & G(\mathbb{O}) \backslash G(\mathbb{A}) / N_P^-(\mathbb{A}) \cdot M(K). \end{array}$$

Now, since for a local field $\mathcal{K}$ with local ring $\mathcal{O}$, we have

$$(0.19) \quad G(\mathcal{O}) \backslash G(\mathcal{K}) \simeq P^-(\mathcal{O}) \backslash P^-(\mathcal{K}),$$

we have

$$G(\mathbb{O}) \backslash G(\mathbb{A}) \simeq P^-(\mathbb{O}) \backslash P^-(\mathbb{A}),$$

and hence we can rewrite (0.18) as

$$(0.20) \quad \begin{array}{ccc} & P^-(\mathbb{O}) \backslash P^-(\mathbb{A}) / P^-(K) & \\ \swarrow & & \searrow \\ G(\mathbb{O}) \backslash G(\mathbb{A}) / G(K) & & M(\mathbb{O}) \backslash M(\mathbb{A}) / M(K). \end{array}$$

Now, if we take the geometric counterpart of (0.20), we obtain the diagram

$$\begin{array}{ccc} & \text{Bun}_{P^-} & \\ \swarrow & & \searrow \\ \text{Bun}_G & & \text{Bun}_M, \end{array}$$

which is the diagram that defines the functor $\text{Eis}_!^-$.

By contrast, the geometric counterpart of (0.18) gives rise to the functor $\text{Eis}_!^{-, \text{enh}}$.

The point is that the geometric counterpart of the set-theoretic isomorphism

$$G(\mathcal{O}) \backslash G(\mathcal{K}) / N_P^-(\mathcal{K}) \cdot M(\mathcal{O}) \simeq P^-(\mathcal{O}) \backslash P^-(\mathcal{K}) / N_P^-(\mathcal{K}) \cdot M(\mathcal{O}) = M(\mathcal{O}) \backslash M(\mathcal{K}) / M(\mathcal{O})$$

is a *stratification* of Gr_G by orbits of $\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)$, so that the category $\text{I}(G, P^-)^{\text{loc}}$ is related to Sph_M by a monadic adjunction, which is not at all an equivalence.

0.5. The local FLE/Jacquet compatibility. In this subsection we describe the main local result, Theorem 5.2.3, whose proof occupies most of Part I of the paper.

0.5.1. In [GLC2] we constructed the factorization category $\text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})$, and we established the *Fundamental Local Equivalence* (the FLE)

$$\text{FLE}_{G, \text{crit}} : \text{KL}(G)_{\text{crit}} \simeq \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}}).$$

There is a similar equivalence for M , and also one that incorporates a “ ρ ”-shift (see (5.9)):

$$(0.21) \quad \text{FLE}_{M, \text{crit}_M + \check{\rho}_P} : \text{KL}(M)_{\text{crit}_M + \check{\rho}_P} \simeq \text{IndCoh}^*(\text{Op}_{M, \check{\rho}_P}^{\text{mon-free}}).$$

0.5.2. In Sect. 3.9.7 we construct a Jacquet functor

$$J_{\text{KM}, \rho_P(\omega_X)}^{-, \text{Sph}} : \text{KL}(G)_{\text{crit}_G} \rightarrow \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P}.$$

In Sect. 4.2.4 we construct a Jacquet functor

$$J_{\text{Op}, \Theta}^{-, \text{spec}} : \text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}}) \rightarrow \text{IndCoh}^*(\text{Op}_{\tilde{M}, \tilde{\rho}_P}^{\text{mon-free}}).$$

Theorem 5.2.3 says that the diagram of factorization categories

$$(0.22) \quad \begin{array}{ccc} \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P} & \xrightarrow{\text{FLE}_{M, \text{crit}_G + \tilde{\rho}_P}} & \text{IndCoh}^*(\text{Op}_{\tilde{M}, \tilde{\rho}_P}^{\text{mon-free}}) \\ J_{\text{KM}, \rho_P(\omega_X)}^{-, \text{Sph}} \uparrow & & \uparrow J_{\text{Op}, \Theta}^{-, \text{spec}} \\ \text{KL}(G)_{\text{crit}_G} & \xrightarrow{\text{FLE}_{G, \text{crit}_G}} & \text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}}). \end{array}$$

commutes.

0.5.3. Theorem 5.2.3 is the most difficult among the local results in this paper and its prequel ([GLC2]) and it digs deep into the original works of Feigin and Frenkel on Wakimoto modules (with some developments of the latter in [FG1]).

In order to prove Theorem 5.2.3, we replace (0.22) by a dual diagram

$$(0.23) \quad \begin{array}{ccc} \text{KL}(M)_{\text{crit}_G - \tilde{\rho}_P} & \xrightarrow{\text{FLE}_{M, \text{crit}_G + \tilde{\rho}_P} \circ \tau_M} & \text{IndCoh}^*(\text{Op}_{\tilde{M}, \tilde{\rho}_P}^{\text{mon-free}}) \\ \text{Wak}_{\rho_P(\omega_X)}^{-, \text{Sph}} \downarrow & & \downarrow \text{co}J_{\text{Op}}^{-, \text{spec}} \\ \text{KL}(G)_{\text{crit}_G} & \xrightarrow{\text{FLE}_{G, \text{crit}_G} \circ \tau_G} & \text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}}), \end{array}$$

where $\text{Wak}_{\rho_P(\omega_X)}^{-, \text{Sph}}$ is the *spherical* Wakimoto functor.

Further, in order to establish the commutation of (0.23), we form its enhancement

$$(0.24) \quad \begin{array}{ccc} \text{KL}(M)_{\text{crit}_G - \tilde{\rho}_P} & \xrightarrow{\text{FLE}_{M, \text{crit}_G + \tilde{\rho}_P} \circ \tau_M} & \text{IndCoh}^*(\text{Op}_{\tilde{M}, \tilde{\rho}_P}^{\text{mon-free}}) \\ \text{Wak}_{\rho_P(\omega_X)}^{-, \text{enh}} \downarrow & & \downarrow \text{co}J_{\text{Op}}^{-, \text{spec}, \text{enh}} \\ \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} & \xrightarrow{\text{FLE}_{G, \text{crit}}^{-, \text{enh}}} & \text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}}) \otimes_{\text{Sph}_G^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}, \end{array}$$

where the lower horizontal arrow is the *enhancement* of the functor $\text{FLE}_{G, \text{crit}}$. This is the second essential usage of the enhancement procedure in this paper.⁸

It turns out that, although the terms of (0.24) look more complicated than those in (0.23), this diagram is more manageable since the functor $\text{Wak}_{\rho_P(\omega_X)}^{-, \text{enh}}$ is more directly related to the classical Wakimoto functor.

⁸We also need (the dual of) the enhanced diagram (0.24) to prove the compatibility of the *enhanced* Langlands functor with constant terms.

0.5.4. The proof of the commutation of (0.24) uses techniques from the theory of factorization algebras.

We map the lower right corner, i.e., the category $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec,loc}}$ to the category of factorization modules over a certain factorization algebra

$$\Omega(R)^{\mathrm{Op}} \in \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M}).$$

This functor turns out to be *almost an equivalence* (technically, an equivalence on eventually coconnective subcategories). The bulk of the proof of the commutation of (0.24) consists of showing that the two circuits in (0.24) become isomorphic when composed with the above functor.

At the end of the day, this comes down to the computation of the Drinfeld-Sokolov reduction of Wakimoto modules, together with an action of the Feigin-Frenkel center (which is given by Miura transform).

0.5.5. One can say that the above calculation is the computational core of this paper, hence the dedication to B. Feigin.

0.6. **Description of the actual contents.** This paper consists of three Parts.

0.6.1. *Contents of Part I.* This Part is devoted to the study of Jacquet functors (and their duals) on the two sides of Langlands correspondence. And on each side there are two pairs of such functors: Whittaker-style and Kac-Moody style.

In Sect. 1 we introduce and study the *local semi-infinite category* on the geometric side, $\mathrm{I}(G, P^-)^{\mathrm{loc}}$. It serves as a connector between categories associated with G and M .

In Sect. 2 we introduce the spectral counterpart of $\mathrm{I}(G, P^-)^{\mathrm{loc}}$, denoted $\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec,loc}}$.

Over a fixed formal disc, this is the category of ind-coherent sheaves on (0.16); it is well-defined because this fiber product is a (derived) algebraic stack locally of finite type. However, when we work in families over the Ran space, we no longer have the locally of finite type property, and special care is needed in order to define the category of ind-coherent sheaves (in fact, there are two such categories that one can consider: $\mathrm{IndCoh}^!(-)$ and $\mathrm{IndCoh}^*(-)$). These issues are dealt with in Sect. A.

Coming back to Sect. 2, we state the existence of an equivalence (0.15) (Theorem 2.4.3), and a closely related assertion, Theorem 2.7.6. The proofs of these theorems is delegated to the Appendix, Sects. B-E.

In Sect. 3 we study various versions of the Jacquet functors between the categories of Kac-Moody modules for $\mathfrak{g}$ and $\mathfrak{m}$. These Jacquet functors are obtained essentially by decorating the functor of BRST reduction with respect to $\mathfrak{L}(\mathfrak{n}_P^-)$. We also study the dual functors, which yield various versions of the Wakimoto module construction.

In Sect. 4 we study Jacquet functors between the various categories of ind-coherent sheaves on opers for $\check{G}$ and $\check{M}$. The central tool is a geometric object called *parabolic Miura opers*, which appeared prominently in the works of Feigin and Frenkel.

In Sect. 5 we state our main local result, Theorem 5.2.3. We reformulate it in dual terms, and also state the enhanced version, Theorem 5.4.5. The proof of Theorem 5.4.5 spans Sects. 6 and 7.

In Sect. 6 we supply the computational/representation-theoretic part of the proof, with one statement, Theorem 6.9.3, delegated to the Appendix, Sect. F.

In Sect. 7 we deal with convergence issues.

0.6.2. *Contents of Part II.* This Part is thematically close to Part II of [GLC2] in that it deals with the interactions between Eisenstein and constant term functors (on either side of Langlands correspondence) with local-to-global functors studied in *loc. cit.* (there are two such on each side).

We note that *none* of the results from Part II cross the Langlands bridge, i.e., we work on the geometric and spectral sides separately. We first deal with the geometric side (Sects. 8-11) and then with the spectral side (Sects. 12-13).

In Sect. 8 we introduce the geometric Eisenstein and constant term functors, as well as their enhancements.

In Sect. 9 we interpret the enhanced Eisenstein and constant term functors via Drinfeld's compactifications. This interpretation is needed for the next section, where we compute constant terms of *Poincaré functors*.

In Sect. 10 we perform the first calculation of the type: “what do we get when we compose a local-to-global functor with the constant term functor”. Namely, we compute the composition of the Poincaré functor with the constant term functor. The result of Theorem 10.1.2 says that the resulting functor equals the composition of a local Jacquet functor (from the Whittaker category of G to that of M), followed by the Poincaré functor for M .

However, there is a caveat: we need to precompose the local functor with the functor of “insertion of the unit along the marked Ran space.” This feature will be common to all four instances of such calculations, performed in Part II.

Additionally, in Sect. 10 we perform a similar calculation, but for the enhanced functors.

In Sect. 11 we calculate the composition of the localization functor

$$\mathrm{Loc}_G : \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \rightarrow \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

with the constant term functor. The answer is given again as a composition of a local Jacquet functor (from $\mathrm{KL}(G)_{\mathrm{crit}}$ to $\mathrm{KL}(M)_{\mathrm{crit}}$), followed by the localization functor for M , modulo the same caveat as was mentioned above. We also perform an analogous computation for the enhanced version of the functors involved.

In Sect. 12 we introduce the spectral Eisenstein and constant term functors, as well as their enhancements. We calculate the composition of the spectral localization functor

$$\mathrm{Loc}_G^{\mathrm{IndCoh}, \mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})$$

with the spectral constant term functor (and also for the enhanced versions of these functors).

In Sect. 13 we calculate the composition of the *spectral Poincaré functor* and the spectral constant term functor.

0.6.3. *Contents of Part III.* In this Part, we finally turn to problems of Langlands duality in the global setting, by combining the results of Parts I and II.

0.6.4. In Sect. 14 we prove that the Langlands functor is compatible with the Eisenstein functors, i.e., we establish the commutation of the diagram (0.3), as well as its enhanced version.

In Sect. 15 we prove that the Langlands functor is compatible with the constant term functors, i.e., we establish the commutation of the diagram (0.5), as well as its enhanced version.

In Sect. 16, we prove the following three results:

- The Langlands functor $\mathbb{L}_G$ admits a left adjoint (denoted $\mathbb{L}_G^L$; moreover, the functor $\mathbb{L}_G^L$ is compatible with the Eisenstein functors);
- The left adjoint of the Langlands functor is isomorphic to the dual of the Langlands functor, up to Miraculous duality and the Chevalley involution;

- The composition $\mathbb{L}_G \circ \mathbb{L}_G^L$, viewed as an endofunctor of $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}})$ is given by tensor product with an object⁹ of $\text{QCoh}(\text{LS}_{\check{G}})$.

Finally, in Sect. 17 we state and prove the main result of this paper, Theorem 17.1.2, which says that the Langlands functor $\mathbb{L}_G$ induces an equivalence between the Eisenstein-generated subcategories of the two sides.

0.7. Conventions and notation.

0.7.1. Notations and conventions in this paper follow those adopted in [GLC2]. We only need to introduce those that have to do with parabolic subgroups of G .

To simplify the discussion, we choose a Cartan subgroup $T \subset G$.

0.7.2. For a standard parabolic subgroup P we denote by P^- its negative, so that $M := P \cap P^-$ is a Levi subgroup in both. We let N_P (resp., N_{P^-}) the unipotent radical of P (resp., P^-).

We will identify the Levi *quotients* of P and P^- by means of

$$P/N_P \simeq P \cap P^- \simeq P^-/N_{P^-}.$$

0.7.3. We let Λ denote the *coweight* lattice of G . We denote by $\Lambda_G^+ \subset \Lambda$ the submonoid of dominant coweights.

We let $\check{\Lambda}$ denote the weight lattice of G , and we let $\check{\Lambda}_G^+$ the submonoid of dominant weights.

0.7.4. We let $\check{P}$ (resp., $\check{P}^-$) the positive (resp., negative) parabolic in $\check{G}$, corresponding to P (resp., P^-), so that $\check{M}$ is the Langlands dual of M .

We let $\check{N}_P$ (resp., $\check{N}_{P^-}$) denote the unipotent radical of $\check{P}$ (resp., $\check{P}^-$). We denote by $\check{\mathfrak{n}}_P$ and $\check{\mathfrak{n}}_{P^-}$ the corresponding Lie algebras.

We let $B(M), N(M), B^-(M), N^-(M)$ (resp., $\check{B}(M), \check{N}(M), \check{B}^-(M), \check{N}^-(M)$) the corresponding subgroups of M (resp., $\check{M}$).

0.7.5. *Why this minus sign?* The reason that in most of the paper we work with the *negative* parabolic P^- is that when consider its interaction with the Whittaker model (which is taken with respect to the *positive* N), the open Bruhat cell corresponds to the unit element of the Weyl group.

In other words, if we worked with P , we would have the longest element $w_0 \in W$ enter a lot of formulas.

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In addition, we wish to acknowledge the contributions of the following mathematicians (in the order in which their ideas appear in the paper):

The Langlands-type equivalences of local semi-infinite categories given by Theorems 2.4.3 and 2.7.6, at the pointwise level, reproduce results from Bezrukavnikov's theory. His results was the reason that we thought these equivalences were possible in the first place.

The theory of Wakimoto and BRST functors, as well as the notion of Miura opers, which play a crucial role in Part I of the paper, was pioneered and developed by B. Feigin and E. Frenkel.

The study of Eisenstein series in the geometric context was pioneered by G. Laumon.

The constant term/Whittaker calculations are crucially based on the idea of Zastava spaces, introduced by M. Finkelberg and I. Mirković.

Drinfeld's compactifications, which are extensively used in Sects. 10 were (obviously) introduced by V. Drinfeld.

⁹A priori, this could have been an object of $\text{IndCoh}_{\text{Nilp} \times \text{Nilp}}(\text{LS}_{\check{G}} \times \text{LS}_{\check{G}})_{\text{QCoh}(\text{LS}_{\check{G}}) \otimes \text{QCoh}(\text{LS}_{\check{G}})}^{\otimes} \text{QCoh}(\text{LS}_{\check{G}})$.

The idea that factorization homology appears in local-to-global calculations for the localization functor goes back to A. Beilinson.

The idea of enhancement of Eisenstein series originates in the joint work of the third-named author with A. Braverman, and is strongly influenced by ideas taught to him by J. Bernstein.

The third- and fourth-named authors wish to thank IHES, where a significant part of this paper was written.

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Part I: Jacquet functors in the local theory

The key players in this Part are the *local semi-infinite category*, denoted $I(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}}$, and its spectral counterpart, denoted $I(\check{G}, \check{P}^-)^{\text{spec, loc}}$.

Objects of $I(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}}$ can be used as *generalized Jacquet functors*

$$\text{Whit}^!(G) \rightarrow \text{Whit}^!(M)$$

and objects of $I(\check{G}, \check{P}^-)^{\text{spec, loc}}$ can be used as *generalized spectral Jacquet functors*

$$\text{Rep}(\check{G}) \rightarrow \text{Rep}(\check{M}).$$

The first main result of this Part is Theorem 2.4.3, which says that there is a canonically defined equivalence of factorization categories

$$\text{Sat}^{-, \frac{\infty}{2}} : I(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \simeq I(\check{G}, \check{P}^-)^{\text{spec, loc}},$$

essentially determined by the fact that it intertwines the Jacquet functors on the two sides via the Casselman-Shalika equivalences for G and M , respectively. The proof of this theorem is delegated to the Appendices, Sects. B-E.

In addition, objects of (the dual of) $I(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}}$ can be used as *generalized Jacquet functors*

$$\text{KL}(G)_{\text{crit}} \rightarrow \text{KL}(M)_{\text{crit}}$$

and objects of (the dual of) $I(\check{G}, \check{P}^-)^{\text{spec, loc}}$ can be used as *generalized Jacquet functors* for opers

$$\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \rightarrow \text{IndCoh}^*(\text{Op}_M^{\text{mon-free}}).$$

The second main result of this Part, Theorem 5.3.6, says that the equivalence $\text{Sat}^{-, \frac{\infty}{2}}$ intertwines the corresponding Jacquet functors via the FLE equivalences for G and M , respectively.

Both of the above compatibilities would be used in a crucial way in Part II in order to establish the compatibility of the Langlands functor with the functors of Eisenstein series and constant term.

1. THE LOCAL SEMI-INFINITE CATEGORY

In this section we study the category that ultimately allows one to connect categories (on the local geometric side) associated with the group G and corresponding categories for its Levi subgroups.

The category in question is the *local spherical semi-infinite category*, which is a *twisted, translated and renormalized version* of

$$(1.1) \quad \text{D-mod}(\text{Gr}_G)^{-, \frac{\infty}{2}} := \text{D-mod}(\text{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)}.$$

The twist and the translation referred to above are familiar from [GLC2]. The twist has to do with the fact that we deal not with plain D-modules, but with critically twisted ones. The translation has to do with the T -bundle $\rho(\omega_X)$, see [GLC2, Sect. 1.2.4]. The renormalization has also appeared in [GLC2, Sect. 1.5.2], but will be reviewed in more detail in Sect. 1.2.

1.1. The corrected Jacquet functor. In addition to the twisting, translation and renormalization mentioned above, there is one tweak that is inherent to the passage between G and M on the geometric side: a certain cohomological shift built into Jacquet functors. The need for this shift appears via the requirement of compatibility with factorization.

This shift will be introduced in this subsection.

1.1.1. Let P^- be the (negative) standard parabolic of G with Levi quotient M . Consider the restrictions of the line bundles

$$\det_{\mathrm{Gr}_G} \text{ and } \det_{\mathrm{Gr}_M}$$

along the maps

$$(1.2) \quad \mathrm{Gr}_G \xleftarrow{\mathfrak{p}^-} \mathrm{Gr}_{P^-} \xrightarrow{\mathfrak{q}^-} \mathrm{Gr}_M,$$

respectively.

Denote by $\det_{\mathrm{Gr}_{G,M}}$ their ratio:

$$\det_{\mathrm{Gr}_{G,M}} = \det_{\mathrm{Gr}_G} |_{\mathrm{Gr}_{P^-}} \otimes (\det_{\mathrm{Gr}_M} |_{\mathrm{Gr}_{P^-}})^{\otimes -1};$$

it naturally descends (as a factorization line bundle) to Gr_M . By a slight abuse of notation, we will denote the resulting line bundle on Gr_M by the same symbol $\det_{\mathrm{Gr}_{G,M}}$.

1.1.2. We consider $\det_{\mathrm{Gr}_{G,M}}$ as an (evenly) graded line bundle on Gr_M , so that its portion over the connected component Gr_M^λ has grading

$$2\langle \lambda, 2\check{\rho}_P \rangle,$$

where $2\check{\rho}_P$ is the character of M equal to the determinant of its action on $\mathfrak{n}(P)$.

It was shown in [GLys, Sect. 5.2] that $\det_{\mathrm{Gr}_{G,M}}$ admits a canonical square root,¹⁰ to be denoted $\det_{\mathrm{Gr}_{G,M}}^{\otimes \frac{1}{2}}$, viewed as a $\mathbb{Z}$ -graded and hence $\mathbb{Z}/2\mathbb{Z}$ -graded (=super) factorization line bundle, so that its portion over the connected component Gr_M^λ has grading

$$\langle \lambda, 2\check{\rho}_P \rangle.$$

1.1.3. Ignoring the grading, The line bundle $\det_{\mathrm{Gr}_{G,M}}^{\otimes \frac{1}{2}}$ gives rise to an identification of the pullbacks of the gerbes

$$\det_{\mathrm{Gr}_G}^{\frac{1}{2}} \text{ and } \det_{\mathrm{Gr}_M}^{\frac{1}{2}}$$

to Gr_{P^-} .

Due to the above identification of gerbes, we have a well-defined functor

$$J_{\mathrm{Gr}}^{-,*,\mathrm{nv}} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_M),$$

given by !-pull and *-push along (1.2).

We will refer to $J_{\mathrm{Gr}}^{-,*,\mathrm{nv}}$ as the “naive” Jacquet functor.

1.1.4. However, due to the fact that $\det_{\mathrm{Gr}_{G,M}}^{\otimes \frac{1}{2}}$ does not factorize as a line bundle, but only as a super line bundle, the above identification of gerbes is *incompatible* with factorization.

Hence, the functor $J_{\mathrm{Gr}}^{-,*,\mathrm{nv}}$ is *not* compatible with factorization either.

1.1.5. We introduce the corrected Jacquet functor

$$J_{\mathrm{Gr}}^{-,*} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_M),$$

as follows:

Over a connected component Gr_M^λ , we set

$$J_{\mathrm{Gr}}^{-,*}(-) := J_{\mathrm{Gr}}^{-,*,\mathrm{nv}}(-)[\langle \lambda, 2\check{\rho}_P \rangle].$$

The functor $J_{\mathrm{Gr}}^{-,*}$ is naturally compatible with the factorization structures, due to the sign rules.

1.2. Digression: renormalized equivariant categories.

¹⁰Which depends on the choice of a square root of ω_X .

1.2.1. Let $\mathbf{C}$ be a category acted on by an algebraic group H . Assume that the category $\mathbf{C}^H$ is compactly generated.

We let $\mathbf{C}^{H,\text{ren}}$ denote the following renormalized version of the category $\mathbf{C}^H$:

By definition, $\mathbf{C}^{H,\text{ren}}$ is the ind-completion of the full subcategory of $\mathbf{C}^H$, consisting of objects, whose images under the forgetful functor

$$\mathbf{C}^H \rightarrow \mathbf{C}$$

are compact.

The ind-extension of the inclusion

$$(\mathbf{C}^{H,\text{ren}})^c \hookrightarrow \mathbf{C}^H$$

is a functor

$$(1.3) \quad \mathbf{C}^{H,\text{ren}} \rightarrow \mathbf{C}^H.$$

1.2.2. Note that when H is unipotent, an object of $\mathbf{C}^H$ is compact *if* (and only if) if its image in $\mathbf{C}$ is compact, so in this case, (1.3) is an equivalence.

1.2.3. We have a tautological inclusion

$$(\mathbf{C}^H)^c \subset (\mathbf{C}^{H,\text{ren}})^c,$$

whose ind-extension is a (fully faithful) functor

$$\mathbf{C}^H \rightarrow \mathbf{C}^{H,\text{ren}},$$

left adjoint to (1.3).

I.e., we obtain $\mathbf{C}^H$ is a colocalization of $\mathbf{C}^{H,\text{ren}}$:

$$(1.4) \quad \mathbf{C}^H \rightleftarrows \mathbf{C}^{H,\text{ren}}.$$

1.2.4. *Example.* Take $\mathbf{C} = \text{Vect}$, equipped with the trivial action of H . Note that

$$\text{Vect}^H \simeq \text{D-mod}(\text{pt}/H).$$

Then $\text{Vect}^{H,\text{ren}}$ is compactly generated by $k \in \mathbf{C}^H$, and so

$$\text{Vect}^{H,\text{ren}} \simeq \text{C}^*(\text{pt}/H)\text{-mod}.$$

The full subcategory

$$\text{Vect}^H \subset \text{Vect}^{H,\text{ren}}$$

corresponds to the full subcategory

$$\text{C}^*(\text{pt}/H)\text{-mod}_0 \subset \text{C}^*(\text{pt}/H)\text{-mod},$$

generated by the augmentation module.

1.2.5. The category $\mathbf{C}^H$ is acted on tautologically by the symmetric monoidal category Vect^H .

Similarly, $\text{Vect}^{H,\text{ren}}$ acquires a symmetric monoidal structure, and $\mathbf{C}^{H,\text{ren}}$ is a module over it.

Moreover, the functor (1.3) is symmetric monoidal, and we have

$$\mathbf{C}^H \simeq \text{Vect}^H \underset{\text{Vect}^{H,\text{ren}}}{\otimes} \mathbf{C}^{H,\text{ren}}.$$

1.2.6. *Example.* Take $\mathbf{C} = \mathfrak{h}\text{-mod}$, so that

$$\mathbf{C}^H \simeq \text{Rep}(H).$$

In this case, it is easy to see that the adjoint functors (1.4) are mutually inverse equivalences.

1.2.7. The renormalization construction works also for group-schemes. We will apply it in the case when $\underline{\mathbf{C}}$ is a sheaf of categories over Ran , with the group-scheme in question being $\mathfrak{L}^+(H)$ for an algebraic group H . We refer the reader to [CR, Sect. 6.11] for details.

1.3. The local spherical semi-infinite category.

1.3.1. Consider the factorization category

$$I(G, P^-)^{\text{loc}} := \text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M), \text{ren}},$$

where the renormalization procedure is taken with respect to $\mathfrak{L}^+(M)$, viewed as acting on the category $\text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)^{\mathfrak{L}(N_P^-)}$ (the corresponding compact generation property is established in [Ga5, Sect. 1.5] for $P = B$ and in [FH] for an arbitrary parabolic).

Convolution equips it with a natural action of Sph_G . (Recall that according to the convention in [GLC2, Sect. 1.5.4], we identify left and right actions of Sph_G via the involution σ .)

1.3.2. The trivialization of the gerbe in Sect. 1.1.3 gives rise to an action of Sph_M on $I(G, P^-)^{\text{loc}}$. However, as in Sect. 1.1.4, this action is *incompatible* with factorization.

We introduce a corrected version of Sph_M -action on $I(G, P^-)^{\text{loc}}$ by precomposing the one above with the automorphism of Sph_M that acts as cohomological shift $[\langle \lambda, 2\check{\rho}_P \rangle]$ on the connected component Gr_M^λ .

From now on, unless explicitly mentioned otherwise, we will only consider this corrected version of the Sph_M -action on $I(G, P^-)^{\text{loc}}$.

The resulting Sph_M -action on $I(G, P^-)^{\text{loc}}$ is compatible with factorization, and commutes with the Sph_G -action, making $I(G, P^-)^{\text{loc}}$ into a $(\text{Sph}_G, \text{Sph}_M)$ -bimodule.

1.3.3. By the same logic, we have a factorization functor

$$(1.5) \quad I(G, P^-)^{\text{loc}} = \text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M), \text{ren}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{P^-})^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M), \text{ren}} \simeq \\ \simeq \text{D-mod}_{\frac{1}{2}}(\text{Gr}_M)^{\mathfrak{L}^+(M), \text{ren}} =: \text{Sph}_M^{[\text{shift}]} \text{Sph}_M,$$

compatible with the Sph_M -actions, where:

- The second arrow is $(\mathfrak{p}^-)^\dagger$;
- The third arrow is the equivalence, induced by

$$\text{D-mod}_{\frac{1}{2}}(\text{Gr}_M)^{(\mathfrak{q}^-)^\dagger} \simeq \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{P^-})^{\mathfrak{L}(N_P^-)};$$

- The the last arrow is the cohomological shift $[\langle \lambda, 2\check{\rho}_P \rangle]$ on Gr_M^λ ;

We denote this functor by $\mathbf{oblv}_{\frac{\infty}{2} \rightarrow \text{Sph}_M}$.

1.3.4. According to [Ga5, Proposition 1.5.3], the functor (1.5) admits a (factorization) left adjoint.¹¹

Explicitly, this left adjoint is given by

$$(1.6) \quad \text{Sph}_M^{[-\text{shift}]} \text{Sph}_M := \text{D-mod}_{\frac{1}{2}}(\text{Gr}_M)^{\mathfrak{L}^+(M), \text{ren}} (\mathfrak{q}^-)^\dagger \\ \simeq \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{P^-})^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M), \text{ren}} (\mathfrak{p}^-)^\dagger \text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M), \text{ren}} = I(G, P^-)^{\text{loc}}.$$

In the above formula, $(\mathfrak{p}^-)^\dagger$ is the functor, induced by the same-named functor

$$(\mathfrak{p}^-)^\dagger : \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{P^-})^{\mathfrak{L}(N_P^-)} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)^{\mathfrak{L}(N_P^-)},$$

left adjoint to $(\mathfrak{p}^-)^\dagger$.

The content of [Ga5, Proposition 1.5.3] is that the partially defined functor $(\mathfrak{p}^-)^\dagger$, left adjoint to

$$(\mathfrak{p}^-)^\dagger : \text{D-mod}_{\frac{1}{2}}(\text{Gr}_G) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{P^-}),$$

¹¹In [Ga5] only the case $P = B$ is considered. The case of an arbitrary parabolic is considered in the forthcoming paper [FH].

is defined on

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{P^-})^{\mathfrak{L}(N_P^-)} \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{P^-}),$$

and has a natural factorization structure.

1.3.5. We will denote the above left adjoint by $\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}$.

Denote

$$\Delta^{-, \frac{\infty}{2}} := \mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}(\delta_{1, \mathrm{Gr}_M}).$$

Explicitly, $\Delta^{-, \frac{\infty}{2}}$ is given by the $!$ -extension of the dualizing sheaf on the $\mathfrak{L}(N_P^-)$ -orbit through the origin in Gr_G . (Again, it is the assertion of [Ga5, Proposition 1.5.3] that this $!$ -extension is well-defined over the Ran space and its formation commutes with $!$ -pullbacks.)

1.3.6. Since Sph_M is rigid, the functor $\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}$ is also compatible with the Sph_M -actions.¹² In particular, we have:

$$\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}(-) \simeq (-) \star_M \Delta^{-, \frac{\infty}{2}},$$

where $\star_M$ refers to the Sph_M -action on $\mathrm{I}(G, P^-)^{\mathrm{loc}}$.

Since the functor (1.5) is conservative, we obtain a monadic adjunction

$$(1.7) \quad \mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}} : \mathrm{Sph}_M \rightleftarrows \mathrm{I}(G, P^-)^{\mathrm{loc}} : \mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M}$$

as Sph_M -module categories.

1.3.7. Consider the above adjunction (1.7). Due to the monadicity, there exists a canonically defined (factorization) associative algebra object

$$\tilde{\Omega} \in \mathrm{Sph}_M$$

(where Sph_M is viewed as a monoidal factorization category) so that

$$(1.8) \quad \mathrm{I}(G, P^-)^{\mathrm{loc}} \simeq \tilde{\Omega}\text{-mod}^r(\mathrm{Sph}_M)$$

as Sph_M -module categories, and the adjunction (1.7) identifies with

$$\mathbf{ind}_{\tilde{\Omega}} : \mathrm{Sph}_M \rightleftarrows \tilde{\Omega}\text{-mod}^r(\mathrm{Sph}_M) : \mathbf{oblv}_{\tilde{\Omega}}.$$

1.3.8. The factorization category $\mathrm{I}(G, P^-)^{\mathrm{loc}}$ carries a natural *unital structure* (see [GLC2, Sect. C.11] for what this means), for which the functor $\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}$ is (strictly) unital.

The factorization unit of $\mathbf{1}_{\mathrm{I}(G, P^-)^{\mathrm{loc}}} \in \mathrm{I}(G, P^-)^{\mathrm{loc}}$ is the above object $\Delta^{-, \frac{\infty}{2}}$.

1.3.9. Note that we also have:

$$\Delta^{-, \frac{\infty}{2}} \simeq \mathrm{Av}_!^{\mathfrak{L}(N_P^-)}(\delta_{1, \mathrm{Gr}_G}),$$

where $\mathrm{Av}_!^{\mathfrak{L}(N_P^-)}$ is the left adjoint¹³ to the forgetful functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M), \mathrm{ren}} \hookrightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}^+(M), \mathrm{ren}}.$$

Hence, we obtain an adjunction of factorization functors

$$(1.9) \quad \mathrm{Av}_!^{\mathfrak{L}(N_P^-)} : \mathrm{Sph}_G \rightleftarrows \mathrm{I}(G, P^-)^{\mathrm{loc}} : \mathrm{Av}_*^{\mathfrak{L}^+(G)/\mathfrak{L}^+(M)},$$

with the left adjoint being unital.

The functors in (1.9) are compatible with the Sph_G -actions. Hence, we have

$$\mathrm{Av}_!^{\mathfrak{L}(N_P^-)}(-) \simeq (-) \star_G \Delta^{-, \frac{\infty}{2}},$$

where $\star_G$ refers to the Sph_G -action on $\mathrm{I}(G, P^-)^{\mathrm{loc}}$.

¹²Again, we do not distinguish between left and right actions of Sph_M .

¹³This left adjoint is defined by an argument similar to one in [Ch1, Lemma 2.3.4(2,4)].

1.3.10. In addition to the functor $\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}$, there exists a more simply-minded functor, denoted $\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}^*$, given by

$$(1.10) \quad \mathrm{Sph}_M \xrightarrow{[-\text{shift}]} \mathrm{Sph}_M = \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_M)^{\mathfrak{L}^+(M), \mathrm{ren}} (\mathfrak{q}^-)^! \\ \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{P^-})^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M), \mathrm{ren}} (\mathfrak{p}^-)_* \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M), \mathrm{ren}} = \mathrm{I}(G, P^-)^{\mathrm{loc}}.$$

I.e., the difference between $\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}^*$ and $\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}$ is that we use $(\mathfrak{p}^-)_*$ instead of $(\mathfrak{p}^-)!$.

Denote by $\nabla^{-, \frac{\infty}{2}}$ the image of $\delta_{1, \mathrm{Gr}_M} = \mathbf{1}_{\mathrm{Sph}_M}$ under (1.10).

Note that the functor $\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}^*$ is also compatible with the actions of Sph_M . Hence, we have:

$$\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}^* \simeq (-) \star \nabla^{-, \frac{\infty}{2}}.$$

Note also that the composition

$$\mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M} \circ \mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}^*$$

is canonically isomorphic to the identity endofunctor of Sph_M .

In particular,

$$(1.11) \quad \mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M}(\nabla^{-, \frac{\infty}{2}}) \simeq \delta_{1, \mathrm{Gr}_M}.$$

1.3.11. Note that the category Sph_M is naturally graded by

$$\Lambda_M / \Lambda_{[M, M]_{\mathrm{sc}}} \simeq \pi_{1, \mathrm{alg}}(M)$$

and in terms of this grading, the algebra $\tilde{\Omega}$ is graded by the sub-monoid

$$\Lambda_{G, P}^{\mathrm{pos}} \subset \Lambda_M / \Lambda_{[M, M]_{\mathrm{sc}}}$$

spanned by the images of the simple coroots, with the 0-weight component being $\mathbf{1}_{\mathrm{Sph}_M}$.

In particular, the algebra $\tilde{\Omega}$ admits a unique augmentation, and hence we have a well-defined object

$$(1.12) \quad \delta_{1, \mathrm{Gr}_M} = \mathbf{1}_{\mathrm{Sph}_M} \in \tilde{\Omega}\text{-mod}^r(\mathrm{Sph}_M).$$

It follows from (1.11) that under the equivalence (1.8) the object $\nabla^{-, \frac{\infty}{2}} \in \mathrm{I}(G, P^-)^{\mathrm{loc}}$ corresponds to (1.12).

1.3.12. We now recall that in addition to the factorization algebras

$$\Delta^{-, \frac{\infty}{2}} \text{ and } \nabla^{-, \frac{\infty}{2}},$$

there exists another factorization algebra in $\mathrm{I}(G, P^-)^{\mathrm{loc}}$, denoted

$$\mathrm{IC}^{-, \frac{\infty}{2}},$$

constructed in [Ga4, Ga5].¹⁴

It is unital and is equipped with homomorphisms of unital factorization algebras

$$\Delta^{-, \frac{\infty}{2}} \rightarrow \mathrm{IC}^{-, \frac{\infty}{2}} \rightarrow \nabla^{-, \frac{\infty}{2}}.$$

1.4. Digression: Sph_G as a rigid category.

¹⁴In *loc.cit.* this object was defined for $P = B$. The case of a general parabolic is similar, and will be considered in detail in [FH].

1.4.1. Let $\mathbf{A}$ be a rigid monoidal category (see [GaRo3, Chapter 1, Definition 9.1.2] for what this means). Let $\text{mult} : \mathbf{A} \otimes \mathbf{A} \rightarrow \mathbf{A}$ denote the binary operation, and let mult^R denote its right adjoint. Let $\mathbf{1}_{\mathbf{A}} \in \mathbf{A}$ denote the monoidal unit.

Recall (see [GaRo3, Chapter 1, Sect. 9.2.1]) that the object

$$\text{mult}^R(\mathbf{1}_{\mathbf{A}}) \in \mathbf{A} \otimes \mathbf{A}$$

provides the unit for a self-duality on $\mathbf{A}$.

1.4.2. Note, however, that there are *two* such self-dualities:

On the one hand, we have the identification

$$\mathbf{A} \xrightarrow{\psi^l} \mathbf{A}^\vee,$$

for which

$$(\psi^l \otimes \text{id})(\text{mult}^R(\mathbf{1}_{\mathbf{A}})) \in \mathbf{A}^\vee \otimes \mathbf{A}$$

is the unit.

On the other hand, we have the identification

$$\mathbf{A} \xrightarrow{\psi^r} \mathbf{A}^\vee,$$

for which

$$(\text{id} \otimes \psi^r)(\text{mult}^R(\mathbf{1}_{\mathbf{A}})) \in \mathbf{A} \otimes \mathbf{A}^\vee$$

is the unit.

1.4.3. Both ψ^l and ψ^r are monoidal equivalences, where we equip $\mathbf{A}^\vee$ with a monoidal structure dual to the comonoidal structure on $\mathbf{A}$, where the latter is given by passing to right adjoints from the original monoidal structure.

Let $\psi^{l \rightarrow r}$ be the endofunctor of $\mathbf{A}$, given by

$$(\psi^r)^{-1} \circ \psi^l.$$

It is a monoidal self-equivalence of $\mathbf{A}$.

A *pivotal structure* on $\mathbf{A}$ is a trivialization of $\psi^{l \rightarrow r}$ (as a monoidal self-equivalence).

Given a pivotal structure on $\mathbf{A}$, we will use notation

$$\psi^l =: \psi := \psi^r.$$

1.4.4. *Example.* Suppose that $\mathbf{A}$ is compactly generated. Then ψ^l (resp., ψ^r) is given on compact objects by the operation of left (resp., right) dual.

Hence, $\psi^{l \rightarrow r}$ is given on compact objects as the square of functor of left duality.

1.4.5. Let $\mathbf{M}^l$ (resp., $\mathbf{M}^r$) be a left (resp., right) $\mathbf{A}$ -module category. Consider the action functors

$$\mathbf{A} \otimes \mathbf{M}^l \xrightarrow{\text{act}} \mathbf{M}^l \text{ and } \mathbf{M}^r \otimes \mathbf{A} \xrightarrow{\text{act}} \mathbf{M}^r.$$

Consider their duals

$$\mathbf{M}^l \xrightarrow{\text{act}^\vee} \mathbf{A}^\vee \otimes \mathbf{M}^l \text{ and } \mathbf{M}^r \xrightarrow{\text{act}^\vee} \mathbf{M}^r \otimes \mathbf{A}^\vee.$$

We have the commutative diagrams

$$\begin{array}{ccc} \mathbf{M}^l & \xrightarrow{\text{act}^\vee} & \mathbf{A}^\vee \otimes \mathbf{M}^l \\ \text{Id} \uparrow & & \sim \uparrow \psi^l \otimes \text{Id} \\ \mathbf{M}^l & \xrightarrow{\text{act}^R} & \mathbf{A} \otimes \mathbf{M}^l \end{array}$$

and

$$\begin{array}{ccc} \mathbf{M}^r & \xrightarrow{\text{act}^\vee} & \mathbf{M}^r \otimes \mathbf{A}^\vee \\ \text{Id} \uparrow & & \sim \uparrow \text{Id} \otimes \psi^r \\ \mathbf{M}^r & \xrightarrow{\text{act}^R} & \mathbf{M}^r \otimes \mathbf{A}. \end{array}$$

In particular, we have a commutative diagram

$$\begin{array}{ccc} \mathbf{M}^r & \xrightarrow{\text{act}^\vee} & \mathbf{M}^r \otimes \mathbf{A}^\vee \\ \text{Id} \uparrow & & \sim \uparrow \text{Id} \otimes \psi^l \\ \mathbf{M}_{\psi^l \rightarrow r}^r & \xrightarrow{\text{act}^R} & \mathbf{M}_{\psi^l \rightarrow r}^r \otimes \mathbf{A}, \end{array}$$

where $\mathbf{M}_{\psi^l \rightarrow r}^r$ the right $\mathbf{A}$ -module category, obtained by precomposing the action functor with the automorphism $\psi^{l \rightarrow r}$ of $\mathbf{A}$.

From here it follows that we have a canonical identification

$$(1.13) \quad \mathbf{M}^l \otimes_{\mathbf{A}} \mathbf{M}^r \simeq \mathbf{M}_{\psi^l \rightarrow r}^r \otimes_{\mathbf{A}} \mathbf{M}^l,$$

where

$$\mathbf{M}^l \otimes_{\mathbf{A}} \mathbf{M}^r := \text{Funct}_{\mathbf{A} \otimes \mathbf{A}^o}(\mathbf{A}, \mathbf{M}^l \otimes \mathbf{M}^r),$$

see¹⁵ [GaRo3, Chapter 1, Proposition 9.5.6].

In particular, we obtain that if $\mathbf{A}$ is equipped with a pivotal structure, we have a canonical identification

$$(1.14) \quad \mathbf{M}^r \otimes_{\mathbf{A}} \mathbf{M}^l \simeq \mathbf{M}^r \otimes_{\mathbf{A}} \mathbf{M}^l.$$

Further, we obtain that if $\mathbf{A}$ is equipped with a pivotal structure, and $\mathbf{M}^l$ (resp., $\mathbf{M}^r$) is dualizable as a plain DG category, then its dualizable as an $\mathbf{A}$ -module category, and its $\mathbf{A}$ -module dual identifies with $(\mathbf{M}^l)^\vee$ (resp., $(\mathbf{M}^r)^\vee$) as a right (resp., left) $\mathbf{A}$ -module category.

1.4.6. The above discussion applies verbatim to the case when $\mathbf{A}$ is a monoidal factorization category.

1.4.7. We take $\mathbf{A} = \text{Sph}_G$. We claim that $\mathbf{A}$ is equipped with a naturally defined pivotal structure.

Indeed, in this case we have a well-defined *Verdier duality* identification

$$(1.15) \quad \text{Sph}_G \simeq \text{Sph}_G^\vee$$

as monoidal factorization categories (see Sect. 1.4.3), and ψ^l (resp., ψ^r) is obtained from (1.15) by composing (resp., precomposing) with the anti-involution σ of Sph_G .

The identification $\psi^l \simeq \psi^r$ follows now from the fact that σ commutes with Verdier duality on Sph_G .

1.4.8. Thus, thanks to Sect. 1.4.5, we will freely identify duals of Sph_G -modules as plain categories with their duals as Sph_G -modules.

Furthermore, given a left and right module categories $\mathbf{M}^l$ and $\mathbf{M}^r$ over Sph_G we will freely identify

$$(1.16) \quad \mathbf{M}^r \otimes_{\text{Sph}_G} \mathbf{M}^l \simeq \mathbf{M}^r \otimes_{\text{Sph}_G} \mathbf{M}^l.$$

¹⁵The notation $\mathbf{A} \rightsquigarrow \mathbf{A}^o$ means reversing the monoidal structure.

1.4.9. We now consider the case of $\mathbf{A} = \text{Sph}_G^{\text{spec}}$.

On the one hand, the geometric Satake equivalence

$$\text{Sat}_G : \text{Sph}_G \simeq \text{Sph}_G^{\text{spec}}$$

allows us to transport the pivotal structure from Sph_G to $\text{Sph}_G^{\text{spec}}$

However, on the other hand, one would like to understand the geometric meaning of this pivotal structure on $\text{Sph}_G^{\text{spec}}$ purely spectral side. This will be done in a future publication, and in the meantime we will avoid using it in main the narrative in this paper. This means that some categories that could ultimately be identified will *not* be identified in this paper.

Thus, given a left and right module categories $\mathbf{M}^l$ and $\mathbf{M}^r$ over Sph_G , we have

$$(1.17) \quad \mathbf{M}^r \otimes_{\text{Sph}_G^{\text{spec}}} \mathbf{M}^l \simeq \mathbf{M}_{\psi_G^{l \rightarrow r}}^r \otimes_{\text{Sph}_G^{\text{spec}}} \mathbf{M}^l,$$

where $\psi_G^{l \rightarrow r}$ is the corresponding monoidal self-equivalence of $\text{Sph}_G^{\text{spec}}$.

1.4.10. *An observation.* Let B be an coassociative coalgebra in a symmetric monoidal category $\mathbf{O}$. and let M^l and M^r be left and right B -comodules in $\mathbf{O}$, respectively. In this case we can consider the object

$$M^r \otimes^B M^l \in \mathbf{O}.$$

Let now A be an associative algebra in $\mathbf{O}$, dualizable as an object. Let M^l and M^r be left and right A -modules in $\mathbf{O}$, respectively. Set $B := A^\vee$. Passing to the duals of the action maps, we can regard M^l and M^r as B -comodules.

Unwinding the definitions, we obtain a canonical isomorphism

$$M^r \otimes^B M^l \simeq M^l \otimes^A M^r.$$

We will mostly apply this for $\mathbf{O} = \text{DGCat}$.

1.5. The dual of the spherical semi-infinite category.

1.5.1. Consider the category

$$\text{D-mod}(\text{Gr}_G)_{\mathfrak{L}(N_P^-)}.$$

It is proved in [Ch1] that it is dualizable as a factorization category. Once the dualizability is established, it follows formally that Verdier duality on Gr_G gives rise to an identification

$$(\text{D-mod}(\text{Gr}_G)^{\mathfrak{L}(N_P^-)})^\vee \simeq \text{D-mod}(\text{Gr}_G)_{\mathfrak{L}(N_P^-)}.$$

1.5.2. Denote

$$\text{I}(G, P^-)_{\text{co}}^{\text{loc}} := \text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)_{\mathfrak{L}(N_P^-)}^{\mathfrak{L}^+(M), \text{ren}}.$$

By similar logic, we have a canonical identification

$$(1.18) \quad \left(\text{I}(G, P^-)^{\text{loc}} \right)^\vee \simeq \text{I}(G, P^-)_{\text{co}}^{\text{loc}}.$$

1.5.3. Note that the identification (1.18) is compatible with the Sph_M -actions in the following sense:

- The action on $\left(\text{I}(G, P^-)^{\text{loc}} \right)^\vee$ is induced by the Sph_M -action on $\text{I}(G, P^-)^{\text{loc}}$ specified in Sect. 1.3.2 (as always, we pass from right to left Sph_M -modules using σ).
- The action on $\text{I}(G, P^-)_{\text{co}}^{\text{loc}}$ is obtained from the natural geometric action, by applying the inverse cohomological shift to the one from Sect. 1.3.2.

Since Sph_M is rigid, the duality (1.18) realizes $\text{I}(G, P^-)_{\text{co}}^{\text{loc}}$ as a Sph_M -module dual of $\text{I}(G, P^-)^{\text{loc}}$.

1.5.4. Note that the functor dual to

$$\mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M} : \mathrm{I}(G, P^-)^{\mathrm{loc}} \rightarrow \mathrm{Sph}_M$$

is a functor, to be denoted

$$(1.19) \quad \mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}, \mathrm{co}} : \mathrm{Sph}_M \rightarrow \mathrm{I}(G, P^-)_{\mathrm{co}}^{\mathrm{loc}},$$

given by

$$(1.20) \quad \begin{aligned} \mathrm{Sph}_M &\xrightarrow{[\mathrm{shift}]} \mathrm{Sph}_M = \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_M)^{\mathfrak{L}^+(M), \mathrm{ren}} \simeq \\ &\simeq \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_{P^-})_{\mathfrak{L}(N_P^-)}^{\mathfrak{L}^+(M), \mathrm{ren}} \xrightarrow{(\mathfrak{p}^-)^*} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)_{\mathfrak{L}(N_P^-)}^{\mathfrak{L}^+(M), \mathrm{ren}} = \mathrm{I}(G, P^-)_{\mathrm{co}}^{\mathrm{loc}}, \end{aligned}$$

where:

- The first arrow is the cohomological shift $[\langle \lambda, 2\check{\rho}_P \rangle]$ on Gr_M^λ ;
- The third arrow is induced by the equivalence

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_{P^-})_{\mathfrak{L}(N_P^-)} \xrightarrow{(\mathfrak{q}^-)^*} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_M).$$

1.5.5. The factorization category $\mathrm{I}(G, P^-)_{\mathrm{co}}^{\mathrm{loc}}$ has a natural unital structure, and the functor $\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}, \mathrm{co}}$ is (strictly) unital.

The factorization unit in $\mathrm{I}(G, P^-)_{\mathrm{co}}^{\mathrm{loc}}$ is the object, to be denoted $\Delta_{\mathrm{co}}^{-, \frac{\infty}{2}}$, equal to the image of $\delta_{1, \mathrm{Gr}} \in \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}^+(P^-), \mathrm{ren}}$ under the natural projection

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}^+(P^-), \mathrm{ren}} \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)_{\mathfrak{L}(N_P^-)}^{\mathfrak{L}^+(M), \mathrm{ren}} = \Delta_{\mathrm{co}}^{-, \frac{\infty}{2}}.$$

1.5.6. By duality, we obtain that the functor (1.19) admits a right adjoint, to be denoted $\mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M, \mathrm{co}}$, namely, the dual of $\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}, \mathrm{co}}$, so that we have a monadic adjunction

$$(1.21) \quad \mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}, \mathrm{co}} : \mathrm{Sph}_M \rightleftarrows \mathrm{I}(G, P^-)_{\mathrm{co}}^{\mathrm{loc}} : \mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M, \mathrm{co}}.$$

1.5.7. Note also that we can also consider the functor

$$(1.22) \quad \begin{aligned} \mathrm{I}(G, P^-)_{\mathrm{co}}^{\mathrm{loc}} &= \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)_{\mathfrak{L}(N_P^-)}^{\mathfrak{L}^+(M), \mathrm{ren}} \xrightarrow{(\mathfrak{p}^-)^!} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_{P^-})_{\mathfrak{L}(N_P^-)}^{\mathfrak{L}^+(M), \mathrm{ren}} \simeq \\ &\simeq \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_M)^{\mathfrak{L}^+(M), \mathrm{ren}} = \mathrm{Sph}_M^{[\mathrm{shift}]} \simeq \mathrm{Sph}_M. \end{aligned}$$

Denote this functor by $\mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M, \mathrm{co}}^*$. By construction, we have

$$\mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M, \mathrm{co}}^* \simeq (\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}^*)^\vee.$$

1.5.8. Let $\mathbf{A}$ be a monoidal category and $A \in \mathbf{A}$ an associative algebra. We consider $A\text{-mod}(\mathbf{A})$ as a right $\mathbf{A}$ -module category, and $A\text{-mod}^r(\mathbf{A})$ as a left $\mathbf{A}$ -module category.

Tautologically, $A\text{-mod}(\mathbf{A})$ and $A\text{-mod}^r(\mathbf{A})$ are each other's duals (as right and left $\mathbf{A}$ -module categories, respectively).

Note also that

$$A\text{-mod}^r(\mathbf{A}) \simeq A^\circ\text{-mod}(\mathbf{A}^\circ),$$

where:

- For a monoidal category $\mathbf{A}$, we denote by $\mathbf{A}^\circ$ denotes the monoidal category category obtained by reversing the monoidal operation;
- For an associative algebra A in a monoidal category $\mathbf{A}$, we denote by A° the corresponding associative algebra in $\mathbf{A}^\circ$.

1.5.9. Let $\tilde{\Omega}_{\text{co}}$ be the associative (factorization) algebra object in Sph_M , so that (1.21) identifies with

$$\mathbf{ind}_{\tilde{\Omega}_{\text{co}}} : \text{Sph}_M \rightleftarrows \tilde{\Omega}_{\text{co}}\text{-mod}^r(\text{Sph}_M) : \mathbf{oblv}_{\tilde{\Omega}_{\text{co}}}.$$

From the duality between

$$\mathbf{I}(G, P^-)^{\text{loc}} \text{ and } \mathbf{I}(G, P^-)_{\text{co}}^{\text{loc}}$$

as Sph_M -module categories, we obtain an identification

$$(1.23) \quad \tilde{\Omega}_{\text{co}} \simeq \sigma((\tilde{\Omega})^o).$$

1.5.10. We now quote the following result of [Ch1]:

Theorem 1.5.11. *The composite functor*

$$\text{D-mod}(\text{Gr}_G)^{\mathfrak{L}(N(P))} \hookrightarrow \text{D-mod}(\text{Gr}_G) \twoheadrightarrow \text{D-mod}(\text{Gr}_G)_{\mathfrak{L}(N_P^-)}$$

is an equivalence (as factorization categories).

From Theorem 1.5.11 we formally obtain:

Corollary 1.5.12. *The functor*

$$(1.24) \quad \mathbf{I}(G, P)^{\text{loc}} := \text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)^{\mathfrak{L}(N(P)) \cdot \mathfrak{L}^+(M), \text{ren}} \hookrightarrow \\ \hookrightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)^{\mathfrak{L}^+(M), \text{ren}} \twoheadrightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)_{\mathfrak{L}(N_P^-)}^{\mathfrak{L}^+(M), \text{ren}} = \mathbf{I}(G, P^-)_{\text{co}}^{\text{loc}}$$

is an equivalence (as factorization categories).

In what follows we will denote by $\Theta_{\mathbf{I}(G, P^-)^{\text{loc}}}$ the functor inverse to the equivalence of (1.24).

1.5.13. Applying the Chevalley involution on G (normalized so that it swaps P and P^- , and thus is compatible with the Chevalley involution τ_M on M), we obtain an equivalence

$$\mathbf{I}(G, P)^{\text{loc}} \xrightarrow{\tau_G} \mathbf{I}(G, P^-)^{\text{loc}}.$$

Thus, composing (1.24) with (1.18) and τ_G , we obtain a self-duality

$$(1.25) \quad (\mathbf{I}(G, P^-)^{\text{loc}})^{\vee} \simeq \mathbf{I}(G, P^-)^{\text{loc}}.$$

By construction, the equivalence (1.25) is compatible with the actions of Sph_M , where:

- The action on $(\mathbf{I}(G, P^-)^{\text{loc}})^{\vee}$ is the one specified in Sect. 1.5.3;
- The action on $\mathbf{I}(G, P^-)^{\text{loc}}$ is precomposition of the action specified in Sect. 1.3.2 with the automorphism τ_M of Sph_M .

1.5.14. As in Sect. 1.5.9, from the equivalence (1.25) we obtain an isomorphism

$$(1.26) \quad \tau_M(\tilde{\Omega}) \simeq \sigma((\tilde{\Omega})^o),$$

as associative (factorization) algebras in Sph_M .

1.6. A twisting construction: recollections. In this and the next subsections we will (re)introduce the translation by $\rho(\omega_X)$, mentioned in the preamble to this section.

The reader may choose to skip this material on the first pass and returned to when necessary.

1.6.1. Recall the twisting construction from [GLC2, Sect. 1.2]. Let $\mathcal{P}_M$ be an M -torsor on X . We can consider $\mathcal{P}_M$ -twisted versions of all objects in sight, i.e.,

$$\mathrm{Gr}_{M,\mathcal{P}_M}, \mathrm{Gr}_{G,\mathcal{P}_M}, \mathfrak{L}(N_P^-)_{\mathcal{P}_M},$$

see [GLC2, Sect. 1.2].

We will denote by subscript $\mathcal{P}_M$ the categories associated with the corresponding twisted geometric objects, i.e.,

$$\mathrm{Sph}_{M,\mathcal{P}_M}, \mathrm{Sph}_{G,\mathcal{P}_M}, \mathrm{I}(G, P^-)_{\mathcal{P}_M}^{\mathrm{loc}}.$$

In particular, we have a monadic adjunction

$$\mathrm{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}} : \mathrm{Sph}_{M,\mathcal{P}_M} \rightleftarrows \mathrm{I}(G, P^-)_{\mathcal{P}_M}^{\mathrm{loc}} : \mathrm{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M},$$

and the corresponding associative (factorization) algebra object

$$\tilde{\Omega}_{\mathcal{P}_M} \in \mathrm{Sph}_{M,\mathcal{P}_M}.$$

We will denote by the same symbol $\Delta^{-, \frac{\infty}{2}}$ the factorization unit in $\mathrm{I}(G, P^-)_{\mathcal{P}_M}^{\mathrm{loc}}$, and by

$$\mathrm{IC}^{-, \frac{\infty}{2}} \in \mathrm{I}(G, P^-)_{\mathcal{P}_M}^{\mathrm{loc}}$$

the corresponding “semi-infinite IC sheaf”.

1.6.2. Note, however, that the local Hecke stacks for M (or G), i.e.,

$$\mathrm{Hecke}_M^{\mathrm{loc}} := \mathfrak{L}^+(M) \backslash \mathfrak{L}(M) / \mathfrak{L}^+(M) \text{ and } \mathrm{Hecke}_G^{\mathrm{loc}} := \mathfrak{L}^+(G) \backslash \mathfrak{L}(G) / \mathfrak{L}^+(G)$$

are canonically isomorphic to their twisted versions, see [GLC2, Equation (1.3)].

So we have canonical identifications of monoidal (factorization) categories

$$\mathrm{Sph}_M \xrightarrow{\alpha_{\mathcal{P}_M, \mathrm{taut}}} \mathrm{Sph}_{M,\mathcal{P}_M} \text{ and } \mathrm{Sph}_G \xrightarrow{\alpha_{\mathcal{P}_M, \mathrm{taut}}} \mathrm{Sph}_{G,\mathcal{P}_M}.$$

Similarly, we have a canonical equivalence

$$(1.27) \quad \alpha_{\mathcal{P}_M, \mathrm{taut}} : \mathrm{I}(G, P^-)_{\mathcal{P}_M}^{\mathrm{loc}} \simeq \mathrm{I}(G, P^-)_{\mathcal{P}_M}^{\mathrm{loc}}.$$

Tautologically, we have:

$$\alpha_{\mathcal{P}_M, \mathrm{taut}}(\tilde{\Omega}) \simeq \tilde{\Omega}_{\mathcal{P}_M}$$

and

$$\alpha_{\mathcal{P}_M, \mathrm{taut}}(\Delta^{-, \frac{\infty}{2}}) \simeq \Delta^{-, \frac{\infty}{2}}.$$

1.6.3. As in Sect. 1.5.2, we can consider the category $\mathrm{I}(G, P^-)_{\mathrm{co}, \mathcal{P}_M}^{\mathrm{loc}}$.

We still have an equivalence

$$(1.28) \quad \left(\mathrm{I}(G, P^-)_{\mathcal{P}_M}^{\mathrm{loc}} \right)^{\vee} \simeq \mathrm{I}(G, P^-)_{\mathrm{co}, \mathcal{P}_M}^{\mathrm{loc}},$$

compatible with Sph_M -actions.

Similarly, we can consider the associative (factorization) algebra object

$$\tilde{\Omega}_{\mathrm{co}, \mathcal{P}_M} \in \mathrm{Sph}_M,$$

and as in (1.23) we have

$$(1.29) \quad \tilde{\Omega}_{\mathrm{co}, \mathcal{P}_M} \simeq \sigma((\tilde{\Omega}_{\mathcal{P}_M})^o).$$

1.6.4. The assertion of Theorem 1.5.11 renders automatically to the twisted context, so that the functor

$$(1.30) \quad \mathrm{I}(G, P)_{\mathcal{P}_M}^{\mathrm{loc}} \rightarrow \mathrm{I}(G, P^-)_{\mathrm{co}, \mathcal{P}_M}^{\mathrm{loc}}$$

is an equivalence.

The Chevalley involution induces an equivalence

$$\mathrm{I}(G, P)_{\mathcal{P}_M}^{\mathrm{loc}} \xrightarrow{\tau_{\mathcal{G}}} \mathrm{I}(G, P^-)_{\tau_M(\mathcal{P}_M)}^{\mathrm{loc}}.$$

One can compose it with the equivalence

$$\mathrm{I}(G, P^-)_{\tau_M(\mathcal{P}_M)}^{\mathrm{loc}} \xrightarrow{(\alpha_{\mathcal{P}_M, \mathrm{taut}}) \circ (\alpha_{\tau_M(\mathcal{P}_M), \mathrm{taut}})^{-1}} \mathrm{I}(G, P)_{\mathcal{P}_M}^{\mathrm{loc}},$$

and thus obtain again a self-duality

$$(1.31) \quad (\mathrm{I}(G, P^-)_{\mathcal{P}_M}^{\mathrm{loc}})^{\vee} \simeq \mathrm{I}(G, P^-)_{\mathcal{P}_M}^{\mathrm{loc}}.$$

1.6.5. Similarly to (1.26), the equivalence (1.31) yields an isomorphism of associative (factorization) algebras in Sph_M :

$$(1.32) \quad (\alpha_{\tau_M(\mathcal{P}_M), \mathrm{taut}})^{-1}(\tau_M(\tilde{\Omega}_{\mathcal{P}_M})) \simeq (\alpha_{\mathcal{P}_M, \mathrm{taut}})^{-1} \circ \sigma((\tilde{\Omega}_{\mathcal{P}_M})^o).$$

1.7. **Twisting with respect to the center.** This subsection is a continuation of the preceding one. Here we will assume that the M -bundle $\mathcal{P}_M$ is induced from a Z_M -bundle $\mathcal{P}_{Z_M}$.

By contrast with the previous section, in which “not much was going on”, here there is something to pay attention to: the non-triviality of the automorphism (1.34) of Sph_M .

That said, this subsection can also be skipped on the first pass, and returned to when necessary.

1.7.1. Assume now that in the context of [GLC2, Sect. 1.2], the group H is abelian and the action of $\mathfrak{L}^+(H)$ on $\mathcal{Y}$ is trivial. In this case, we have canonical isomorphisms

$$\mathcal{Y}_{\mathcal{P}_H} \simeq \mathcal{Y} \text{ and } \mathfrak{L}(H)_{\mathcal{P}_H} \simeq \mathfrak{L}(H).$$

In particular, we obtain an equivalence

$$\alpha_{\mathcal{P}_H, \mathrm{cent}} : \mathrm{D}\text{-mod}(\mathcal{Y})_{\mathcal{P}_H} \xrightarrow{\sim} \mathrm{D}\text{-mod}(\mathcal{Y})$$

and hence *another* identification

$$\alpha_{\mathcal{P}_H, \mathrm{cent}} : (\mathrm{D}\text{-mod}(\mathcal{Y})^{\mathfrak{L}^+(H)})_{\mathcal{P}_H} \xrightarrow{\sim} \mathrm{D}\text{-mod}(\mathcal{Y})^{\mathfrak{L}^+(H)}.$$

We will denote by

$$(\mathrm{transl}_{\mathcal{P}_{Z_H}})^* := \alpha_{\mathcal{P}_H, \mathrm{cent}} \circ \alpha_{\mathcal{P}_H, \mathrm{taut}}$$

the resulting auto-equivalence of $\mathrm{D}\text{-mod}(\mathcal{Y})^{\mathfrak{L}^+(H)}$.

The functor $(\mathrm{transl}_{\mathcal{P}_{Z_H}})^*$ is the pullback along the automorphism, denoted $\mathrm{transl}_{\mathcal{P}_{Z_H}}$, of the stack $\mathcal{Y}/\mathfrak{L}^+(H)$ given by translation by the point $\mathcal{P}_H \in \mathrm{pt}/\mathfrak{L}^+(H)$ via the action map

$$\mathrm{pt}/\mathfrak{L}^+(H) \times \mathcal{Y}/\mathfrak{L}^+(H) \rightarrow \mathcal{Y}/\mathfrak{L}^+(H).$$

1.7.2. A typical example of the situation of Sect. 1.7.1 is when $\mathcal{Y} = \mathrm{Gr}_M$, so that $\mathcal{Y}/\mathfrak{L}^+(M)$ is the local Hecke stack

$$\mathrm{Hecke}_M^{\mathrm{loc}} := \mathfrak{L}^+(M) \backslash \mathfrak{L}(M) / \mathfrak{L}^+(M).$$

Let H map to the center of M . In this case, the action of $\mathfrak{L}^+(H)$ on Gr_M is trivial.

The above automorphism $\mathrm{transl}_{\mathcal{P}_{Z_H}}$ of

$$\mathrm{Hecke}_M^{\mathrm{loc}} = \{\mathcal{P}'_M, \mathcal{P}''_M, \mathcal{P}'_M|_{\mathcal{D}^\times} \sim \mathcal{P}''_M|_{\mathcal{D}^\times}\}$$

is given by tensoring the M -bundles involved by $\mathcal{P}_H$, using the canonical map

$$\mathrm{pt}/H \times \mathrm{pt}/M \rightarrow \mathrm{pt}/M.$$

1.7.3. Assume now that in the context of Sect. 1.6.1, the M -torsor $\mathcal{P}_M$ is induced by a Z_M -torsor $\mathcal{P}_{Z_M}$. Hence, we obtain *another* identification

$$(1.33) \quad \mathrm{Sph}_{M, \mathcal{P}_{Z_M}} \xrightarrow{\alpha_{\mathcal{P}_{Z_M}, \mathrm{cent}}} \mathrm{Sph}_M$$

as monoidal (factorization) categories.

The composite

$$(1.34) \quad (\mathrm{transl}_{\mathcal{P}_{Z_M}})^* := \alpha_{\mathcal{P}_{Z_M}, \mathrm{cent}} \circ \alpha_{\mathcal{P}_{Z_M}, \mathrm{taut}}$$

is a monoidal (factorization) automorphism of Sph_M .

Remark 1.7.4. The automorphism $(\mathrm{transl}_{\mathcal{P}_{Z_M}})^*$ of Sph_M is *non-trivial*. However, its composition with the naive Satake functor

$$\mathrm{Sat}_M^{\mathrm{nv}} : \mathrm{Rep}(\check{M}) \rightarrow \mathrm{Sph}_M$$

is *canonically* trivial.

1.7.5. *Convention.* Henceforth, unless explicitly specified otherwise, when we consider $\mathrm{I}(G, P^-)_{\mathcal{P}_{Z_M}}^{\mathrm{loc}}$ as acted on by Sph_M , we will do so using the identification $\alpha_{\mathcal{P}_{Z_M}, \mathrm{cent}}$ of (1.33).

In particular, we will view $\tilde{\Omega}_{\mathcal{P}_{Z_M}}$ as an associative (factorization) algebra object in Sph_M via $\alpha_{\mathcal{P}_{Z_M}, \mathrm{cent}}$.

Tautologically, we have

$$\tilde{\Omega}_{\mathcal{P}_{Z_M}} \simeq (\mathrm{transl}_{\mathcal{P}_{Z_M}})^*(\tilde{\Omega}),$$

as associative (factorization) algebras in Sph_M .

1.7.6. Note that the equivalence

$$(1.35) \quad (\mathrm{I}(G, P^-)_{\mathcal{P}_{Z_M}}^{\mathrm{loc}})^{\vee} \simeq \mathrm{I}(G, P^-)_{\mathcal{P}_{Z_M}}^{\mathrm{loc}}$$

of (1.31) is compatible with the Sph_M -actions, where

- The action on $(\mathrm{I}(G, P^-)_{\mathcal{P}_{Z_M}}^{\mathrm{loc}})^{\vee}$ is induced by the Sph_M -action on $\mathrm{I}(G, P^-)_{\mathcal{P}_{Z_M}}^{\mathrm{loc}}$ specified in Sect. 1.7.5 (as always, we pass from right to left Sph_M -modules using σ);
- The action on $\mathrm{I}(G, P^-)_{\mathcal{P}_{Z_M}}^{\mathrm{loc}}$ is precomposition of the action specified in Sect. 1.3.2 with the automorphisms τ_M and $(\mathrm{transl}_{\mathcal{P}_{Z_M} \otimes \tau(\mathcal{P}_{Z_M}^{-1})})^*$ of Sph_M .

1.7.7. The equivalence (1.35) translates into the following isomorphism of associative (factorization) algebras in Sph_M :

$$(1.36) \quad \tau_M(\tilde{\Omega}_{\mathcal{P}_{Z_M}}) \simeq (\mathrm{transl}_{\mathcal{P}_{Z_M}^{\otimes -1} \otimes \tau(\mathcal{P}_{Z_M})})^* \left(\sigma((\tilde{\Omega}_{\mathcal{P}_{Z_M}})^o) \right).$$

Note, however, that (1.36) can be equivalently obtained by applying $(\mathrm{transl}_{\mathcal{P}_{Z_M}^{\otimes -1}})^*$ to (1.26).

1.7.8. In practice we will take

$$\mathcal{P}_{Z_M} := \rho_P(\omega_X) := 2\rho_P(\omega_X^{\otimes \frac{1}{2}}).$$

1.7.9. Note that

$$\tau(\rho_P(\omega_X)) = -\rho_P(\omega_X).$$

Hence, in this case, (1.36) specializes to

$$(1.37) \quad \tau_M(\tilde{\Omega}_{\rho_P(\omega_X)}) \simeq (\mathrm{transl}_{-2\rho_P(\omega_X)})^* \left(\sigma((\tilde{\Omega}_{\rho_P(\omega_X)})^o) \right).$$

1.8. The enhanced Jacquet functor. In this subsection we introduce the *raison d'être* of $\mathrm{I}(G, P^-)^{\mathrm{loc}}$, namely, how it can be used to define generalized Jacquet functors.

1.8.1. Note that we can identify

$$\mathrm{D}\text{-mod}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)} \simeq \left(\mathrm{D}\text{-mod}(\mathrm{Gr}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Gr}_M) \right)^{\mathfrak{L}(P^-)}.$$

This observation allows us to view objects of $\mathrm{D}\text{-mod}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)}$ *kernels of functors* between $\mathrm{D}\text{-mod}(\mathrm{Gr}_G)$ and $\mathrm{D}\text{-mod}(\mathrm{Gr}_M)$.

Namely, we consider the factorization functor

$$\begin{aligned} (1.38) \quad & \mathrm{D}\text{-mod}(\mathrm{Gr}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)} \simeq \\ & \simeq \mathrm{D}\text{-mod}(\mathrm{Gr}_G) \otimes \left(\mathrm{D}\text{-mod}(\mathrm{Gr}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Gr}_M) \right)^{\mathfrak{L}(P^-)} \rightarrow \\ & \rightarrow \mathrm{D}\text{-mod}(\mathrm{Gr}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Gr}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Gr}_M) \xrightarrow{(\overset{!}{\otimes}) \otimes (\mathrm{Id})} \mathrm{D}\text{-mod}(\mathrm{Gr}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Gr}_M)^{\Gamma_{\mathrm{dR}}(\mathrm{Gr}_{G_1^-}) \otimes \mathrm{Id}} \\ & \rightarrow \mathrm{D}\text{-mod}(\mathrm{Gr}_M). \end{aligned}$$

This functor is equivariant with respect to the $\mathfrak{L}(P^-)$ -actions on $\mathrm{D}\text{-mod}(\mathrm{Gr}_G)$ (via $\mathfrak{L}(P^-) \rightarrow \mathfrak{L}(G)$) and on $\mathrm{D}\text{-mod}(\mathrm{Gr}_M)$ (via $\mathfrak{L}(P^-) \rightarrow \mathfrak{L}(M)$), respectively.

1.8.2. The functor (1.38) has a variant, given by

$$\begin{aligned} (1.39) \quad & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \otimes \mathrm{I}(G, P^-)^{\mathrm{loc}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)} \simeq \\ & \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \otimes \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_M) \right)^{\mathfrak{L}(P^-)} \rightarrow \\ & \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_M) \xrightarrow{(\overset{!}{\otimes}) \otimes (\mathrm{Id})} \\ & \rightarrow \mathrm{D}\text{-mod}(\mathrm{Gr}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_M)^{\Gamma_{\mathrm{dR}}(\mathrm{Gr}_{G_1^-}) \otimes \mathrm{Id}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_M) \xrightarrow{[\mathrm{shift}]} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_M), \end{aligned}$$

where:

- The category $\left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_M) \right)^{\mathfrak{L}(P^-)}$ makes sense due to the identification of the two multiplicative μ_2 -gerbes on $\mathfrak{L}(P^-)$ (one obtained by restriction from $\mathfrak{L}(G)$, and another from $\mathfrak{L}(M)$) that results from the existence of the square root $\det_{\mathrm{Gr}_{G,M}}^{\otimes \frac{1}{2}}$;
- The $\overset{!}{\otimes}$ -tensor product functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \rightarrow \mathrm{D}\text{-mod}(\mathrm{Gr}_G)$$

makes sense due to the fact that the square of the gerbe $\det_{\mathrm{Gr}_G}^{\otimes \frac{1}{2}}$ is canonically trivial.

- The last arrow is the cohomological shift by $\langle \lambda, 2\check{\rho}_P \rangle$ on Gr_M^λ .

1.8.3. Denote the functor (1.39) by $J_{\mathrm{Gr}}^{-, \mathrm{pre-enh}}$. It has the following properties:

- (1) It has an equivariance property with respect to $\mathfrak{L}(P^-)$, similar to that of (1.38);
- (2) It carries a natural factorization structure;
- (3) It is compatible with the Sph_M -actions.

Furthermore, $J_{\mathrm{Gr}}^{-, \mathrm{pre-enh}}$ factors as

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \otimes \mathrm{I}(G, P^-)^{\mathrm{loc}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)^{\mathrm{loc}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_M),$$

and the resulting functor

$$(1.40) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)^{\mathrm{loc}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_M)$$

also has the properties (1)-(3) above.

We denote the functor (1.40) by $J_{\text{Gr}}^{-, \text{enh}}$.

1.8.4. We will denote by

$$J_{\text{Gr}}^{-, !} : \text{D-mod}_{\frac{1}{2}}(\text{Gr}_G) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_M)$$

the functor obtained from $J_{\text{Gr}}^{-, \text{pre-enh}}$ by inserting $\Delta^{-, \frac{\infty}{2}}$ along the second factor.

Note that the functor $\text{D-mod}_{\frac{1}{2}}(\text{Gr}_G) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_M)$, obtained from $J_{\text{Gr}}^{-, \text{pre-enh}}$ by inserting $\nabla^{-, \frac{\infty}{2}}$ along the second factor, identifies with the functor $J_{\text{Gr}}^{-, *}$ from Sect. 1.1.5.

We will denote the functor $\text{D-mod}_{\frac{1}{2}}(\text{Gr}_G) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_M)$, obtained from $J_{\text{Gr}}^{-, \text{pre-enh}}$ by inserting $\text{IC}^{-, \frac{\infty}{2}}$ along the second factor by $J_{\text{Gr}}^{-, !*}$.

1.8.5. Let $\mathcal{P}_M$ be an M -torsor on X . Note also that the functor (1.39) admits a twisted version

$$(1.41) \quad \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \mathcal{P}_M}) \otimes \text{I}(G, P^-)_{\mathcal{P}_M}^{\text{loc}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{M, \mathcal{P}_M}).$$

Let us now be given an M -bundle $\mathcal{P}'_M$ and a Z_M -bundle $\mathcal{P}''_{Z_M}$ on X , so that

$$\mathcal{P}_M \simeq \mathcal{P}'_M \otimes \mathcal{P}''_{Z_M}.$$

We have the following version of the functor (1.41):

$$(1.42) \quad \begin{aligned} \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \mathcal{P}_M}) \otimes \text{I}(G, P^-)_{\mathcal{P}'_M}^{\text{loc}} &\xrightarrow{\text{Id} \otimes \alpha_{\mathcal{P}'_M, \text{taut}}} \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \mathcal{P}_M}) \otimes \text{I}(G, P^-)_{\mathcal{P}_M}^{\text{loc}} \xrightarrow{(1.41)} \\ &\rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{M, \mathcal{P}_M}) \xrightarrow{\alpha_{\mathcal{P}''_{Z_M}, \text{cent}}} \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{M, \mathcal{P}'_M}). \end{aligned}$$

1.8.6. The functor (1.42) inherits the following properties:

- (1) It is equivariant with respect to the natural action of $\mathfrak{L}(P^-)_{\mathcal{P}_M}$ on $\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \mathcal{P}_M})$ and the action of

$$\mathfrak{L}(P^-)_{\mathcal{P}_M} \twoheadrightarrow \mathfrak{L}(M)_{\mathcal{P}_M} \xrightarrow{\alpha_{\mathcal{P}''_{Z_M}, \text{cent}}} \mathfrak{L}(M)_{\mathcal{P}'_M}$$

on $\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{M, \mathcal{P}'_M})$.

- (2) It carries a natural factorization structure;
- (3) It is compatible with the Sph_M -action on $\text{I}(G, P^-)_{\mathcal{P}'_M}^{\text{loc}}$ from Sect. 1.7.5, and the action of Sph_M on $\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{M, \mathcal{P}'_M})$ obtained from the identification

$$\text{Sph}_M \xrightarrow{\alpha_{\mathcal{P}'_M, \text{taut}}} \text{Sph}_{M, \mathcal{P}'_M},$$

(see Sect. 1.6.2).

1.8.7. In practice, we will take

$$\mathcal{P}'_M := \rho_M(\omega_X), \quad \mathcal{P}'_{Z_M} := \rho_P(\omega_X),$$

so that

$$\mathcal{P}_M = \rho(\omega_X).$$

We will consider the resulting functor

$$(1.43) \quad \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X)}) \otimes \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{M, \rho_M(\omega_X)}).$$

We will denote the functor (1.43) by $J_{\text{Gr}}^{-, \text{pre-enh}}$, i.e., the same symbol as in the untwisted situation.

1.8.8. The functor $J_{\text{Gr}}^{-, \text{pre-enh}}$ naturally factors as

$$\begin{aligned} \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X)}) \otimes \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} &\rightarrow \\ &\rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X)}) \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{M, \rho_M(\omega_X)}). \end{aligned}$$

We will denote the resulting functor

$$\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X)}) \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{M, \rho_M(\omega_X)})$$

by $J_{\text{Gr}}^{-, \text{enh}}$. It inherits properties (1)-(3) from Sect. 1.8.6. We will refer to it as the *enhanced Jacquet functor*.

1.8.9. We will denote by the same symbols

$$(1.44) \quad J_{\text{Gr}}^{-, !}, J_{\text{Gr}}^{-, !*} \text{ and } J_{\text{Gr}}^{-, *} \text{ and } J_{\text{Gr}}^{-, \infty}$$

the functors

$$\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X)}) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{M, \rho_M(\omega_X)})$$

obtained from $J_{\text{Gr}}^{-, \text{pre-enh}}$ by inserting the objects

$$\Delta^{-, \frac{\infty}{2}}, \text{IC}^{-, \frac{\infty}{2}} \text{ and } \nabla^{-, \frac{\infty}{2}} \in \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}},$$

respectively, along the second factor.

1.8.10. The equivariance property of (1.43) with respect¹⁶ to $\mathfrak{L}(N(M)) \subset \mathfrak{L}(P^-)$ implies that the functors $J_{\text{Gr}}^{-, \text{pre-enh}}$ and $J_{\text{Gr}}^{-, \text{enh}}$ restrict to functors

$$(1.45) \quad \text{Whit}^!(G) \otimes \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \rightarrow \text{Whit}^!(M) \text{ and } \text{Whit}^!(G) \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \rightarrow \text{Whit}^!(M).$$

respectively.

We will denote the resulting functors (1.45) by

$$(1.46) \quad J_{\text{Whit}}^{-, \text{pre-enh}} \text{ and } J_{\text{Whit}}^{-, \text{enh}},$$

respectively.

We will denote by

$$J_{\text{Whit}}^{-, !}, J_{\text{Whit}}^{-, !*} \text{ and } J_{\text{Whit}}^{-, *}$$

the functors

$$\text{Whit}^!(G) \rightarrow \text{Whit}^!(M)$$

obtained from the functors (1.44) by restriction.

1.8.11. Since the functor $J_{\text{Gr}}^{-, \text{enh}}$ respects the Sph_M -actions on the two sides, by duality (see Sect. 1.4.8), it gives rise to a Sph_G -linear functor

$$(1.47) \quad \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X)}) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{M, \rho_M(\omega_X)}) \otimes_{\text{Sph}_M} \text{I}(G, P^-)_{\text{co}, \rho_P(\omega_X)}^{\text{loc}},$$

which we will denote by $J_{\text{Gr}}^{-, \text{enhco}}$.

The functor (1.47) gives rise to a Sph_G -linear functor

$$J_{\text{Whit}}^{-, \text{enhco}} : \text{Whit}^!(G) \rightarrow \text{Whit}^!(M) \otimes_{\text{Sph}_M} \text{I}(G, P^-)_{\text{co}, \rho_P(\omega_X)}^{\text{loc}}.$$

¹⁶Here $N(M)$ is the maximal unipotent subgroup of the Levi M .

1.8.12. In what follows we will use the notations

$$\mathrm{Whit}^!(M)^{-,\mathrm{enh}} := \mathrm{Whit}^!(M) \otimes_{\mathrm{Sph}_M} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}}$$

and

$$\mathrm{Whit}^!(M)^{-,\mathrm{enh}_{\mathrm{co}}} := \mathrm{Whit}^!(M) \otimes_{\mathrm{Sph}_M} \mathrm{I}(G, P^-)_{\mathrm{co}, \rho_P(\omega_X)}^{\mathrm{loc}}.$$

So, $J_{\mathrm{Whit}}^{-,\mathrm{enh}_{\mathrm{co}}}$ is a functor

$$\mathrm{Whit}^!(G) \rightarrow \mathrm{Whit}^!(M)^{-,\mathrm{enh}_{\mathrm{co}}}.$$

We recover the functor $J_{\mathrm{Whit}}^{-,!}$ from $J_{\mathrm{Whit}}^{-,\mathrm{enh}_{\mathrm{co}}}$ as the composite

$$\mathrm{Whit}^!(G) \xrightarrow{J_{\mathrm{Whit}}^{-,\mathrm{enh}_{\mathrm{co}}}} \mathrm{Whit}^!(M)^{-,\mathrm{enh}_{\mathrm{co}}} \xrightarrow{\mathrm{Id} \otimes \mathrm{oblv} \xrightarrow{\frac{\infty}{2}} \mathrm{Sph}_M, \mathrm{co}} \mathrm{Whit}^!(M) \otimes_{\mathrm{Sph}_M} \mathrm{Sph}_M = \mathrm{Whit}^!(M).$$

2. SPECTRAL SEMI-INFINITE CATEGORY/IES

In this section we introduce the spectral counterpart of $\mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}}$, which is the category

$$\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}},$$

thought of as ind-coherent sheaves on the (factorization) space

$$(2.1) \quad \mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{M}}^{\mathrm{mer}}} \mathrm{LS}_{\check{M}}^{\mathrm{reg}}.$$

When doing so, we will have to overcome difficulties having to do with the fact that the Ran version of (2.1) is not locally (almost) of finite type. This discussion is mostly delegated to the Appendix, Sect. A.

The main result stated in this section is Theorem 2.4.3, which says that there is a canonical equivalence

$$\mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} \xrightarrow{\mathrm{Sat}^-, \frac{\infty}{2}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}};$$

the proof of this theorem is also delegated to the Appendix and occupies Sects. B-E.

2.1. The spherical spectral semi-infinite category.

2.1.1. Denote by

$$\mathrm{Hecke}_{\check{G}, \check{P}^-}^{\mathrm{spec}, \mathrm{loc}}$$

the factorization space (2.1).

2.1.2. For a fixed $\underline{x} \in \mathrm{Ran}$, the corresponding space

$$\mathrm{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{spec}, \mathrm{loc}}$$

is an algebraic stack locally almost of finite type. Indeed, for $\underline{x}$ being a singleton x , we have¹⁷

$$\mathrm{Hecke}_{\check{G}, \check{P}^-, x}^{\mathrm{spec}, \mathrm{loc}} \simeq \mathrm{pt} / \check{G} \times_{\check{\mathfrak{g}} / \mathrm{Ad}(\check{G})} \check{\mathfrak{p}}^- / \mathrm{Ad}(\check{P}^-) \times_{\check{\mathfrak{m}} / \mathrm{Ad}(\check{M})} \mathrm{pt} / \check{M}.$$

So the category $\mathrm{IndCoh}(\mathrm{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{spec}, \mathrm{loc}})$ is well-defined.

However, when $\underline{x}$ moves in families over Ran , as an algebro-geometric object, $\mathrm{Hecke}_{\check{G}, \check{P}^-}^{\mathrm{spec}, \mathrm{loc}}$ does not fit into the framework developed in [GLC2, Appendix A.5], in which the category $\mathrm{IndCoh}^*(-)$ is defined.

Rather, we refer the reader to Sect. A.5, where we define

$$\mathrm{IndCoh}^*(\mathrm{Hecke}_{\check{G}, \check{P}^-}^{\mathrm{spec}, \mathrm{loc}})$$

as a (unital) factorization category.

¹⁷The isomorphism below follows from the fact that the formal completion $(\mathrm{LS}_{\check{H}, x}^{\mathrm{mer}})_{\mathrm{reg}}^{\wedge}$ of $\mathrm{LS}_{\check{H}, x}^{\mathrm{mer}}$ along $\mathrm{LS}_{\check{H}, x}^{\mathrm{reg}}$ identifies with $\mathfrak{h}^{\wedge} / \mathrm{Ad}(\mathfrak{h})$, where $\mathfrak{h}^{\wedge}$ is the formal completion of $\mathfrak{h}$ at 0.

In addition, in Sect. A.5.6, we will define the factorization category

$$\mathrm{IndCoh}^!(\mathrm{Hecke}_{\check{G}, \check{P}^-}^{\mathrm{spec}, \mathrm{loc}}),$$

which will turn out to be the dual of $\mathrm{IndCoh}^*(\mathrm{Hecke}_{\check{G}, \check{P}^-}^{\mathrm{spec}, \mathrm{loc}})$.

2.1.3. We will use the notations:

$$I(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} := \mathrm{IndCoh}^*(\mathrm{Hecke}_{\check{G}, \check{P}^-}^{\mathrm{spec}, \mathrm{loc}})$$

and

$$I(\check{G}, \check{P}^-)_{\mathrm{co}}^{\mathrm{spec}, \mathrm{loc}} := \mathrm{IndCoh}^!(\mathrm{Hecke}_{\check{G}, \check{P}^-}^{\mathrm{spec}, \mathrm{loc}}).$$

2.1.4. At a fixed point $\underline{x} \in \mathrm{Ran}$, the category $I(\check{G}, \check{P}^-)_{\underline{x}}^{\mathrm{spec}, \mathrm{loc}} = \mathrm{IndCoh}^*(\mathrm{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{spec}, \mathrm{loc}})$ carries a natural structure of bimodule category with respect to

$$\mathrm{Sph}_{M, \underline{x}}^{\mathrm{spec}} := \mathrm{IndCoh}^*(\mathrm{Hecke}_{M, \underline{x}}^{\mathrm{spec}, \mathrm{loc}}) \text{ and } \mathrm{Sph}_{\check{G}, \underline{x}}^{\mathrm{spec}} := \mathrm{IndCoh}^*(\mathrm{Hecke}_{\check{G}, \underline{x}}^{\mathrm{spec}, \mathrm{loc}}).$$

In Sect. A.5.9, we will construct a bimodule structure on $I(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$ with respect to

$$\mathrm{Sph}_M^{\mathrm{spec}} \text{ and } \mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$$

in the factorization setting.

2.1.5. We have a correspondence

$$\begin{array}{ccc} \mathrm{LS}_{\check{P}^-}^{\mathrm{reg}} & \xrightarrow{\mathfrak{p}^-, \mathrm{spec} \times \iota} & \mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \\ \downarrow \mathfrak{q}^-, \mathrm{spec} & & \\ \mathrm{LS}_M^{\mathrm{reg}} & & \end{array}$$

and its base change

$$(2.2) \quad \begin{array}{ccc} \mathrm{LS}_{\check{P}^-}^{\mathrm{reg}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{reg}} & \longrightarrow & \mathrm{Hecke}_{\check{G}, \check{P}^-}^{\mathrm{spec}, \mathrm{loc}} \\ \downarrow & & \\ \mathrm{Hecke}_M^{\mathrm{spec}, \mathrm{loc}} & & \end{array}$$

2.1.6. For a fixed $\underline{x} \in \mathrm{Ran}$, the functors of $!$ -pull and $*$ -push along (2.2) define a forgetful functor

$$\mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M}^{\mathrm{spec}} : I(\check{G}, \check{P}^-)_{\underline{x}}^{\mathrm{spec}, \mathrm{loc}} \rightarrow \mathrm{Sph}_{M, \underline{x}}^{\mathrm{spec}},$$

which admits a left adjoint, denoted $\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}^{\mathrm{spec}}$, given by $*$ -pull followed by $*$ -push.

Thus, we obtain an adjoint pair

$$(2.3) \quad \mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}^{\mathrm{spec}} : \mathrm{Sph}_{M, \underline{x}}^{\mathrm{spec}} \rightleftarrows I(\check{G}, \check{P}^-)_{\underline{x}}^{\mathrm{spec}, \mathrm{loc}} : \mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M}^{\mathrm{spec}}.$$

Moreover, it is easy to see that the adjunction (2.4) is monadic and respects the $\mathrm{Sph}_{M, \underline{x}}^{\mathrm{spec}}$ -actions.

2.1.7. In Sect. A.5.3, we will construct the functor

$$\mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M}^{\mathrm{spec}} : I(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \rightarrow \mathrm{Sph}_M^{\mathrm{spec}}$$

in the factorization setting.

Moreover, we will show that $\mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M}^{\mathrm{spec}}$ admits a left adjoint, denoted $\mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}^{\mathrm{spec}}$, which is a (strictly unital) factorization functor.

As in Sect. 1.3.7, we can view the adjunction

$$(2.4) \quad \mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}^{\mathrm{spec}} : \mathrm{Sph}_M^{\mathrm{spec}} \rightleftarrows I(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} : \mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M}^{\mathrm{spec}}$$

as between $\mathrm{Sph}_M^{\mathrm{spec}}$ -module categories.

2.1.8. The factorization unit

$$\mathbf{1}_{I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}} \in I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}$$

is the $(\text{IndCoh}, *)$ -direct image of

$$\mathcal{O}_{\text{LS}_{\check{P}^-}^{\text{reg}}} \in \text{QCoh}(\text{LS}_{\check{P}^-}^{\text{reg}}) \simeq \text{IndCoh}^*(\text{LS}_{\check{P}^-}^{\text{reg}})$$

along

$$(2.5) \quad \text{LS}_{\check{P}^-}^{\text{reg}} \xrightarrow{\iota} \text{LS}_{\check{G}}^{\text{reg}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_{\check{P}^-}^{\text{reg}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_M^{\text{reg}} = \text{Hecke}_{\check{G}, \check{P}^-}^{\text{spec}, \text{loc}}.$$

We will also use the notation

$$\Delta^{-, \text{spec}, \frac{\infty}{2}} := \mathbf{1}_{I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}}.$$

Since the functor $\mathbf{ind}_{\text{Sph}_M \rightarrow \frac{\infty}{2}}^{\text{spec}}$ is $\text{Sph}_M^{\text{spec}}$ -linear, we have

$$\mathbf{ind}_{\text{Sph}_M \rightarrow \frac{\infty}{2}}^{\text{spec}}(-) \simeq (-) \star_M \Delta^{-, \text{spec}, \frac{\infty}{2}},$$

where $\star_M$ denotes the $\text{Sph}_M^{\text{spec}}$ -action on $I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}$.

2.1.9. The $\text{Sph}_{\check{G}}^{\text{spec}}$ -action on $\Delta^{-, \text{spec}, \frac{\infty}{2}}$ is a $\text{Sph}_{\check{G}}^{\text{spec}}$ -linear (strictly unital) factorization functor

$$(2.6) \quad \text{Sph}_{\check{G}}^{\text{spec}} \rightarrow I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}, \quad (-) \star_{\check{G}} \Delta^{-, \text{spec}, \frac{\infty}{2}},$$

where $\star_{\check{G}}$ denotes the $\text{Sph}_{\check{G}}^{\text{spec}}$ -action on $I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}$.

Explicitly, the functor (2.6) is given by $(\text{IndCoh}, *)$ -pullback, followed by $(\text{IndCoh}, *)$ -direct image along the diagram

$$\begin{array}{ccc} \text{LS}_{\check{G}}^{\text{reg}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_{\check{P}^-}^{\text{reg}} & \longrightarrow & \text{LS}_{\check{G}}^{\text{reg}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_{\check{P}^-}^{\text{mer}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_M^{\text{reg}} \\ \downarrow & & \\ \text{LS}_{\check{G}}^{\text{reg}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_{\check{G}}^{\text{reg}} & & \end{array}$$

The functor (2.6) admits a factorization right adjoint, given by $!$ -pullback, followed by $(\text{IndCoh}, *)$ -direct image *in the opposite direction* along the above diagram.

2.1.10. In addition to the factorization functor $\mathbf{ind}_{\text{Sph}_M \rightarrow \frac{\infty}{2}}^{\text{spec}}$, one can consider another functor, denoted

$$\mathbf{ind}_{\text{Sph}_M \rightarrow \frac{\infty}{2}}^{\text{spec}, *} : \text{Sph}_M^{\text{spec}} \rightarrow I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}},$$

defined as $!$ -pullback, followed by $(\text{IndCoh}, *)$ -direct image along the diagram¹⁸

$$\begin{array}{ccc} \text{LS}_{\check{P}^-}^{\text{reg}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_M^{\text{reg}} & \xrightarrow{=} & \text{LS}_{\check{P}^-}^{\text{reg}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_M^{\text{mer}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_M^{\text{reg}} \longrightarrow \text{LS}_{\check{G}}^{\text{reg}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_{\check{P}^-}^{\text{mer}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_M^{\text{reg}} \\ \downarrow & & \downarrow = \\ \text{LS}_M^{\text{reg}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_M^{\text{reg}} & & \text{Hecke}_{\check{G}, \check{P}^-}^{\text{spec}, \text{loc}} \end{array}$$

Note that the composition

$$\mathbf{oblv}_{\frac{\infty}{2} \rightarrow \text{Sph}_M}^{\text{spec}} \circ \mathbf{ind}_{\text{Sph}_M \rightarrow \frac{\infty}{2}}^{\text{spec}, *}$$

¹⁸Note the appearance of the *positive* parabolic $\check{P}$ in this diagram.

is isomorphic to the identity endofunctor of $\mathrm{Sph}_M^{\mathrm{spec}}$. This follows by base change from the fact that the diagram

$$\begin{array}{ccc} \mathrm{LS}_M^{\mathrm{reg}} & \longrightarrow & \mathrm{LS}_{\check{P}^-}^{\mathrm{reg}} \\ \downarrow & & \downarrow \\ \mathrm{LS}_{\check{P}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{P}}^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{mer}} & \longrightarrow & \mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \end{array}$$

is Cartesian.

We denote

$$\nabla^{-, \mathrm{spec}, \frac{\infty}{2}} := \mathbf{ind}_{\mathrm{Sph}_M \rightarrow \frac{\infty}{2}}^{\mathrm{spec}, *} (\mathbf{1}_{\mathrm{Sph}_M^{\mathrm{spec}}}) \in \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}.$$

We have

$$(2.7) \quad \mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}_M}^{\mathrm{spec}} (\nabla^{-, \mathrm{spec}, \frac{\infty}{2}}) \simeq \mathbf{1}_{\mathrm{Sph}_M^{\mathrm{spec}}}.$$

2.1.11. In addition to the factorization algebras

$$\Delta^{-, \mathrm{spec}, \frac{\infty}{2}} \text{ and } \nabla^{-, \mathrm{spec}, \frac{\infty}{2}},$$

we will consider the following factorization algebra object in $\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$, denoted

$$\mathrm{IC}^{-, \mathrm{spec}, \frac{\infty}{2}}.$$

Namely, it is defined as the mage of $\mathcal{O}_{\mathrm{LS}_M^{\mathrm{reg}}} \in \mathrm{QCoh}(\mathrm{LS}_M^{\mathrm{reg}})$ along

$$\mathrm{QCoh}(\mathrm{LS}_M^{\mathrm{reg}}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) \simeq \mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) \xrightarrow{\iota_*^{\mathrm{IndCoh}}} \mathrm{Hecke}_{\check{G}, \check{P}^-}^{\mathrm{spec}, \mathrm{loc}},$$

where the first arrow is *direct image* along the map

$$(2.8) \quad \mathrm{LS}_M^{\mathrm{reg}} \rightarrow \mathrm{LS}_{\check{P}^-}^{\mathrm{reg}},$$

corresponding to the canonical map¹⁹ $\mathrm{pt}/\check{M} \rightarrow \mathrm{pt}/\check{P}^-$.

2.2. The (factorization) algebras $\tilde{\Omega}^{\mathrm{spec}}$ and Ω^{spec} . The definition of $\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$ involves some complicated algebraic geometry in infinite type. Yet, the datum of this category can be equivalently recorded by a more elementary object: an associative factorization algebra in $\mathrm{Sph}_M^{\mathrm{spec}}$, denoted $\tilde{\Omega}^{\mathrm{spec}}$.

We introduce it in the subsection.

2.2.1. Let

$$\tilde{\Omega}^{\mathrm{spec}} \in \mathrm{Sph}_M^{\mathrm{spec}}$$

denote the associative (factorization) algebra object so that (2.4) identifies with the adjunction

$$(2.9) \quad \mathbf{ind}_{\tilde{\Omega}^{\mathrm{spec}}}^{\mathrm{spec}} : \mathrm{Sph}_M^{\mathrm{spec}} \rightleftarrows \tilde{\Omega}^{\mathrm{spec}}\text{-mod}^r(\mathrm{Sph}_M^{\mathrm{spec}}) : \mathbf{oblv}_{\tilde{\Omega}^{\mathrm{spec}}}.$$

2.2.2. In terms of the identification

$$(2.10) \quad \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \simeq \tilde{\Omega}^{\mathrm{spec}}\text{-mod}^r(\mathrm{Sph}_M^{\mathrm{spec}}),$$

the object

$$\Delta^{-, \mathrm{spec}, \frac{\infty}{2}} \in \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$$

corresponds to

$$\tilde{\Omega}^{\mathrm{spec}} \in \tilde{\Omega}^{\mathrm{spec}}\text{-mod}^r(\mathrm{Sph}_M^{\mathrm{spec}}).$$

¹⁹A choice of a simplitting $\check{M} \rightarrow \check{P}^-$ gives rise to a map $\mathrm{pt}/\check{M} \rightarrow \mathrm{pt}/\check{P}^-$; every two splittings differ by conjugation by a *uniquely* defined element of $\check{N}_{\check{P}^-}$, and the resulting two maps $\mathrm{pt}/\check{M} \rightrightarrows \mathrm{pt}/\check{P}^-$ coincide.

2.2.3. Note that the adjunction

$$(\mathfrak{q}^{-, \text{spec}})^* : \text{QCoh}(\text{LS}_M^{\text{reg}}) \rightleftarrows \text{QCoh}(\text{LS}_{\check{P}^-}^{\text{reg}}) : (\mathfrak{q}^{-, \text{spec}})_*$$

identifies with

$$(2.11) \quad \text{Res}_{\check{P}^-}^{\check{M}} : \text{Rep}(\check{M}) \rightleftarrows \text{Rep}(\check{P}^-) : \text{inv}_{\check{N}_P^-},$$

where $\text{inv}_{\check{N}_P^-}$ is the functor of $\check{\mathfrak{n}}_P^-$ -invariants (a.k.a. cohomological Chevalley complex).

2.2.4. Let

$$\Omega^{\text{spec}} \in \text{FactAlg}^{\text{unl}}(\text{Rep}(\check{M}))$$

denote the (commutative) factorization algebra corresponding (via [GLC2, Corollary C.8.8]) to

$$\text{inv}_{\check{N}_P^-}(k) \in \text{ComAlg}(\text{Rep}(\check{M}));$$

note that the latter is the same as the cohomological Chevalley complex

$$\text{C}_{\text{chev}}^*(\check{\mathfrak{n}}_P^-),$$

of $\check{\mathfrak{n}}_P^-$ with coefficients in the trivial module.

The (monadic) adjunction (2.11) can therefore be rewritten as

$$(2.12) \quad \text{ind}_{\Omega^{\text{spec}}} : \text{Rep}(\check{M}) \rightleftarrows \Omega^{\text{spec}}\text{-mod}^r(\text{Rep}(\check{M})) : \text{oblv}_{\Omega^{\text{spec}}}.$$

2.2.5. By a slight abuse of notation, we will denote by the same symbol Ω^{spec} its image along the functor

$$\text{Rep}(\check{M}) \xrightarrow{\text{nv}} \text{Sph}_M^{\text{spec}}.$$

From diagram (2.2) we obtain that there exists a canonically defined map of associative algebra objects

$$(2.13) \quad \Omega^{\text{spec}} \rightarrow \tilde{\Omega}^{\text{spec}},$$

so that the adjunction (2.9) factors as a composition of

$$(2.14) \quad \text{ind}_{\Omega^{\text{spec}}} : \text{Sph}_M^{\text{spec}} \rightleftarrows \Omega^{\text{spec}}\text{-mod}^r(\text{Sph}_M^{\text{spec}}) : \text{oblv}_{\Omega^{\text{spec}}}.$$

and

$$(2.15) \quad \text{ind}_{\Omega^{\text{spec}} \rightarrow \tilde{\Omega}^{\text{spec}}} : \Omega^{\text{spec}}\text{-mod}^r(\text{Sph}_M^{\text{spec}}) \rightleftarrows \tilde{\Omega}^{\text{spec}}\text{-mod}^r(\text{Sph}_M^{\text{spec}}) : \text{oblv}_{\tilde{\Omega}^{\text{spec}} \rightarrow \Omega^{\text{spec}}}.$$

2.2.6. Note that the category $\text{Sph}_M^{\text{spec}}$ is naturally graded by

$$\Lambda_M / \Lambda_{[M, M]_{\text{sc}}} \simeq \pi_{1, \text{alg}}(M) \simeq \text{Hom}(Z(\check{M}), \mathbb{G}_m),$$

and in terms of this grading, the algebras

$$\tilde{\Omega}^{\text{spec}} \text{ and } \Omega^{\text{spec}}$$

are graded by the sub-monoid

$$\Lambda_{G, P}^{\text{pos}} \subset \Lambda_M / \Lambda_{[M, M]_{\text{sc}}}$$

spanned by the images of the positive simple coroots, with the 0-weight component being $\mathbf{1}_{\text{Sph}_M^{\text{spec}}}$.

In particular, both $\tilde{\Omega}^{\text{spec}}$ and Ω^{spec} are naturally augmented. In particular, we have the well-defined objects

$$\mathbf{1}_{\text{Sph}_M^{\text{spec}}} \in \tilde{\Omega}^{\text{spec}}\text{-mod}^r(\text{Sph}_M^{\text{spec}}) \text{ and } \mathbf{1}_{\text{Sph}_M^{\text{spec}}} \in \Omega^{\text{spec}}\text{-mod}^r(\text{Sph}_M^{\text{spec}}).$$

2.2.7. By (2.7), in terms of the equivalence (2.10), the object

$$\nabla^{-, \text{spec}, \frac{\infty}{2}} \in \text{I}(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}$$

corresponds to

$$\mathbf{1}_{\text{Sph}_M^{\text{spec}}} \in \tilde{\Omega}^{\text{spec}}\text{-mod}^r(\text{Sph}_M^{\text{spec}}).$$

2.2.8. By construction, the object

$$\mathrm{IC}^{-,\mathrm{spec},\frac{\infty}{2}} \in \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}$$

corresponds to

$$\mathrm{ind}_{\Omega^{\mathrm{spec}} \rightarrow \tilde{\Omega}^{\mathrm{spec}}}(\mathbf{1}_{\mathrm{Sph}_M^{\mathrm{spec}}}) \in \tilde{\Omega}^{\mathrm{spec}}\text{-mod}^r(\mathrm{Sph}_M^{\mathrm{spec}}).$$

2.3. Spectral Jacquet functors. This subsection is a spectral counterpart of Sect. 1.8: we will show how objects of $\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}$ give rise to generalized Jacquet functors $\mathrm{Rep}(\check{G}) \rightarrow \mathrm{Rep}(\check{M})$.

2.3.1. Consider the diagram

$$(2.16) \quad \begin{array}{ccc} \mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{reg}} & & \\ \swarrow p_1 & & \searrow p_2 \\ \mathrm{LS}_{\check{G}}^{\mathrm{reg}} & & \mathrm{LS}_M^{\mathrm{reg}}. \end{array}$$

At a fixed point $\underline{x} \in \mathrm{Ran}$, the corresponding diagram

$$\begin{array}{ccc} \mathrm{LS}_{\check{G},\underline{x}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G},\underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mer}} \times_{\mathrm{LS}_{M,\underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{M,\underline{x}}^{\mathrm{reg}} & & \\ \swarrow p_1 & & \searrow p_2 \\ \mathrm{LS}_{\check{G},\underline{x}}^{\mathrm{reg}} & & \mathrm{LS}_{M,\underline{x}}^{\mathrm{reg}} \end{array}$$

consists of stacks locally almost of finite type. Hence, we have a well-defined functor

$$\begin{aligned} \mathrm{Rep}(\check{G})_{\underline{x}} \otimes \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}_{\underline{x}} &= \mathrm{QCoh}(\mathrm{LS}_{\check{G},\underline{x}}^{\mathrm{reg}}) \otimes \mathrm{IndCoh}^*(\mathrm{LS}_{\check{G},\underline{x}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G},\underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mer}} \times_{\mathrm{LS}_{M,\underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{M,\underline{x}}^{\mathrm{reg}}) \xrightarrow{p_1^*(-) \otimes (-)} \\ &\rightarrow \mathrm{IndCoh}^*(\mathrm{LS}_{\check{G},\underline{x}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G},\underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mer}} \times_{\mathrm{LS}_{M,\underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{M,\underline{x}}^{\mathrm{reg}}) \xrightarrow{(p_2)_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*(\mathrm{LS}_{M,\underline{x}}^{\mathrm{reg}}) \simeq \mathrm{Rep}(\check{M})_{\underline{x}}. \end{aligned}$$

2.3.2. In Sect. A.5.7 we will show that the functor $(p_2)_*^{\mathrm{IndCoh}}$ in the composition

$$(2.17) \quad \begin{aligned} \mathrm{Rep}(\check{G}) \otimes \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}} &= \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{reg}}) \otimes \mathrm{IndCoh}^*(\mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{reg}}) \xrightarrow{p_1^*(-) \otimes (-)} \\ &\rightarrow \mathrm{IndCoh}^*(\mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{reg}}) \xrightarrow{(p_2)_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*(\mathrm{LS}_M^{\mathrm{reg}}) \simeq \mathrm{Rep}(\check{M}) \end{aligned}$$

is well-defined also in the factorization setting.

We will denote the functor (2.17) by $J^{-,\mathrm{spec},\mathrm{pre}\text{-}\mathrm{enh}}$.

2.3.3. For a fixed point $\underline{x} \in \mathrm{Ran}$, the functor $J^{-,\mathrm{spec},\mathrm{pre}\text{-}\mathrm{enh}}$ naturally factors as

$$\mathrm{Rep}(\check{G})_{\underline{x}} \otimes \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}_{\underline{x}} \rightarrow \mathrm{Rep}(\check{G})_{\underline{x}} \otimes_{\mathrm{Sph}_{\check{G},\underline{x}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}_{\underline{x}} \rightarrow \mathrm{Rep}(\check{M})_{\underline{x}}.$$

In Sect. A.5.10 we will show that a similar factorization takes place when $\underline{x}$ is allowed to move in families over Ran , thereby giving rise to a factorization functor

$$\mathrm{Rep}(\check{G}) \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}} \rightarrow \mathrm{Rep}(\check{M}),$$

which we denote by $J^{-,\mathrm{spec},\mathrm{enh}}$. We will refer to it as the *spectral enhanced Jacquet functor*.

2.3.4. We will denote by

$$J^{-,\text{spec},!}, J^{-,\text{spec},!*} \text{ and } J^{-,\text{spec},*}$$

the functors

$$\text{Rep}(\check{G}) \rightarrow \text{Rep}(\check{M}),$$

obtained from $J^{-,\text{spec},\text{pre-enh}}$ by inserting the objects

$$\Delta^{-,\text{spec},\frac{\infty}{2}}, \text{IC}^{-,\text{spec},\frac{\infty}{2}} \text{ and } \nabla^{-,\text{spec},\frac{\infty}{2}} \in \text{I}(\check{G}, \check{P}^-)^{\text{spec},\text{loc}},$$

respectively, along the second factor.

2.3.5. Unwinding the definitions, we obtain that the functor $J^{-,\text{spec},!}$ identifies canonically with the $(\text{IndCoh}, *)$ -pullback, followed by $(\text{IndCoh}, *)$ -pushforward along the diagram

$$(2.18) \quad \begin{array}{ccc} & \text{LS}_{\check{P}^-}^{\text{reg}} & \\ \text{p}^{-,\text{spec}} \swarrow & & \searrow \text{q}^{-,\text{spec}} \\ \text{LS}_{\check{G}}^{\text{reg}} & & \text{LS}_{\check{M}}^{\text{reg}}. \end{array}$$

I.e., $J^{-,\text{spec},!}$ is the functor

$$\text{Rep}(\check{G}) \xrightarrow{\text{Res}_{\check{P}^-}^{\check{G}}} \text{Rep}(\check{P}^-) \xrightarrow{\text{inv}_{\check{M}^-}} \text{Rep}(\check{M}).$$

2.3.6. Similarly, the functor $J^{-,\text{spec},!*}$ is simply the restriction functor

$$\text{Rep}(\check{G}) \xrightarrow{\text{Res}_{\check{M}}^{\check{G}}} \text{Rep}(\check{M})$$

along $\check{M} \rightarrow \check{G}$.

2.3.7. Finally, the functor $J^{-,\text{spec},*}$ is described as follows. Consider the fiber product

$$\mathcal{Y} := \text{LS}_M^{\text{reg}} \times_{\text{LS}_M^{\text{reg}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_M^{\text{reg}}} (\text{LS}_{\check{P}}^{\text{reg}} \times_{\text{LS}_{\check{P}}^{\text{mer}}} \text{LS}_{\check{M}}^{\text{reg}}) \simeq \text{LS}_{\check{P}}^{\text{reg}} \times_{\text{LS}_{\check{P}}^{\text{mer}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_M^{\text{reg}}} \text{LS}_M^{\text{reg}}.$$

It is equipped with maps

$$p_1 : \mathcal{Y} \rightarrow \text{LS}_{\check{G}}^{\text{reg}} \text{ and } p_2 : \mathcal{Y} \rightarrow \text{LS}_{\check{M}}^{\text{reg}}.$$

Set

$$\mathcal{F}_{\mathcal{Y}} := p_2^!(\mathcal{O}_{\text{LS}_M^{\text{reg}}}) \in \text{IndCoh}^*(\mathcal{Y}).$$

Then $J^{-,\text{spec},*}$ is given by

$$\text{Rep}(\check{G}) \simeq \text{QCoh}(\text{LS}_{\check{G}}^{\text{reg}}) \xrightarrow{p_1^*(-) \otimes \mathcal{F}_{\mathcal{Y}}} \text{IndCoh}^*(\mathcal{Y}) \xrightarrow{(p_2)_*^{\text{IndCoh}}} \text{IndCoh}^*(\text{LS}_{\check{M}}^{\text{reg}}) \simeq \text{Rep}(\check{M}).$$

2.3.8. The functor $J^{-,\text{spec},\text{enh}}$ is $\text{Sph}_M^{\text{spec}}$ -linear, and hence, by duality, it gives rise to a $\text{Sph}_{\check{G}}^{\text{spec}}$ -linear functor

$$\text{Rep}(\check{G}) \rightarrow \text{Rep}(\check{M}) \otimes_{\text{Sph}_M^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)^{\text{spec},\text{loc}},$$

which we denote by $J^{-,\text{spec},\text{enh}_{\text{co}}}$.

2.4. **Semi-infinite geometric Satake.** In this subsection we state one of the two main theorems in Part I of the paper, namely, Theorem 2.4.3.

2.4.1. Recall (see [GLC2, Sect. 1.8]) that the anti-involutions σ and σ^{spec} of Sph_G and $\text{Sph}_G^{\text{spec}}$, respectively, are compatible with the geometric Satake equivalence

$$(2.19) \quad \text{Sph}_G \xrightarrow{\text{Sat}_G} \text{Sph}_{\check{G}}^{\text{spec}}$$

up to the Chevalley involution τ_G .

Hence, τ_G comes up every time when we compare the passage between left and right modules categories on the two sides of geometric Satake.

2.4.2. Denote

$$\text{CS}_{G,\tau} := \text{CS}_G \circ \tau_G \text{ and } \text{CS}_{M,\tau} := \text{CS}_M \circ \tau_M.$$

We are now ready to state the *semi-infinite geometric Satake* equivalence:

Theorem 2.4.3. *There exists an equivalence of unital factorization categories*

$$\text{Sat}^{-, \frac{\infty}{2}} : \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \rightarrow \text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}},$$

compatible with the actions of

$$(2.20) \quad \text{Sph}_G \xrightarrow{\text{Sat}_G} \text{Sph}_{\check{G}}^{\text{spec}},$$

which makes the diagram

$$(2.21) \quad \begin{array}{ccc} \text{Whit}^!(G) \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} & \xrightarrow{(\text{CS}_{G,\tau}) \otimes \text{Sat}^{-, \frac{\infty}{2}}} & \text{Rep}(\check{G}) \otimes_{\text{Sph}_{\check{G}}^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}} \\ J_{\text{Whit}}^{-, \text{enh}} \downarrow & & \downarrow J^{-, \text{spec, enh}} \\ \text{Whit}^!(M) & \xrightarrow{\text{CS}_{M,\tau}} & \text{Rep}(\check{M}) \end{array}$$

commute. Moreover, the functor $\text{Sat}^{-, \frac{\infty}{2}}$ and the functors in (2.21) are compatible with the actions of

$$(2.22) \quad \text{Sph}_M \xrightarrow{\text{Sat}_{M,\tau}} \text{Sph}_{\check{M}}^{\text{spec}}.$$

We will supply a proof in Sect. E.

Remark 2.4.4. The existence of an equivalence

$$\text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \simeq \text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}}$$

was established in [CR].²⁰ However, instead of the commutativity of (2.21), the construction in *loc. cit.* makes a different diagram commute (see Remark 2.5.4 below).

So, the assertion of Theorem 2.4.3 does not quite follow from [CR]. Our proof of Theorem 2.4.3 follows ideas similar to those of [CR], but is formally independent of it.²¹

That said, one can deduce Theorem 2.4.3 from the results of [CR] by an argument involving duality; this will be addressed in a future publication.

2.5. Consequences for Jacquet functors.

²⁰At the pointwise level, the existence of an equivalence between $\text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}}$ and $\text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}}$ had been established in [ABG], which was the reason Theorem 2.4.3 was conjectured.

²¹Except that we use [CR, Theorem 4.6.1] as a key technical point, see the proof of Proposition D.1.8.

2.5.1. Being an equivalence of unital factorization categories, the functor $\text{Sat}^{-, \frac{\infty}{2}}$ satisfies

$$\text{Sat}^{-, \frac{\infty}{2}}(\Delta^{-, \frac{\infty}{2}}) \simeq \Delta^{-, \text{spec}, \frac{\infty}{2}}.$$

From here, we obtain:

Corollary 2.5.2. *The following diagram of factorization categories and functors commutes:*

$$(2.23) \quad \begin{array}{ccc} \text{Whit}^!(G) & \xrightarrow{\text{CS}_{G, \tau}} & \text{Rep}(\check{G}) \\ J_{\text{Whit}}^{-, !} \downarrow & & \downarrow \text{inv}_{\check{N}_P^-} = J^{-, \text{spec}, !} \\ \text{Whit}^!(M) & \xrightarrow{\text{CS}_{M, \tau}} & \text{Rep}(\check{M}). \end{array}$$

Remark 2.5.3. Although we stated the commutativity of (2.23) as a corollary of Theorem 2.4.3, when it comes to proofs, the actual logic of the proofs is the other way around: one establishes Corollary 2.5.2 first, and this is as one of the initial steps in the proof of Theorem 2.4.3.

Remark 2.5.4. What follows from [CR] is that the diagram

$$(2.24) \quad \begin{array}{ccc} \text{Whit}^!(G) & \xrightarrow{\text{CS}_{G, \tau}} & \text{Rep}(\check{G}) \\ J_{\text{Whit}}^{-, *} \downarrow & & \downarrow J^{-, \text{spec}, *} \\ \text{Whit}^!(M) & \xrightarrow{\text{CS}_{M, \tau}} & \text{Rep}(\check{M}) \end{array}$$

commutes.

2.5.5. For future reference, we introduce the functor

$$(2.25) \quad J_{\text{Whit}, \Theta}^{-, !} : \text{Whit}_*(G) \rightarrow \text{Whit}_*(M)$$

so that the diagram

$$\begin{array}{ccc} \text{Whit}_*(G) & \xrightarrow{J_{\text{Whit}, \Theta}^{-, !}} & \text{Whit}_*(M) \\ \Theta_{\text{Whit}(G)} \downarrow \sim & & \sim \downarrow \Theta_{\text{Whit}(M)} \\ \text{Whit}^!(G) & \xrightarrow{J_{\text{Whit}}^{-, !}} & \text{Whit}^!(M) \end{array}$$

commutes, where the vertical arrows are as in [GLC2, Sect. 1.3.12].

From Corollary 2.5.2 we obtain a commutative diagram

$$(2.26) \quad \begin{array}{ccc} \text{Whit}^*(G) & \xleftarrow{\text{FLE}_{\check{G}, \infty}} & \text{Rep}(\check{G}) \\ J_{\text{Whit}, \Theta}^{-, !} \downarrow & & \downarrow \text{inv}_{\check{N}_P^-} \\ \text{Whit}^*(M) & \xleftarrow{\text{FLE}_{\check{M}, \infty}} & \text{Rep}(\check{M}). \end{array}$$

2.5.6. The compatibility of the functor $\text{Sat}^{-, \frac{\infty}{2}}$ with (2.22) yields a commutative diagram

$$\begin{array}{ccc} \text{Sph}_M & \xrightarrow{\text{Sat}_{M, \tau}} & \text{Sph}_M^{\text{spec}} \\ \text{ind}_{\text{Sph}_M \rightarrow \frac{\infty}{2}} \downarrow & & \downarrow \text{ind}_{\text{Sph}_M^{\text{spec}} \rightarrow \frac{\infty}{2}} \\ \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} & \xrightarrow{\text{Sat}^{-, \frac{\infty}{2}}} & \text{I}(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}. \end{array}$$

Passing to right adjoints, we obtain a commutative diagram

$$\begin{array}{ccc} \text{Sph}_M & \xrightarrow{\text{Sat}_M} & \text{Sph}_M^{\text{spec}} \\ \text{oblv}_{\frac{\infty}{2} \rightarrow \text{Sph}_M} \uparrow & & \uparrow \text{oblv}_{\frac{\infty}{2} \rightarrow \text{Sph}_M^{\text{spec}}} \\ \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} & \xrightarrow{\text{Sat}^{-, \frac{\infty}{2}}} & \text{I}(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}. \end{array}$$

Hence, we can view the equivalence $\text{Sat}^{-, \frac{\infty}{2}}$ as the statement that we have an isomorphism of associative (factorization) algebra objects

$$(2.27) \quad \text{Sat}_{M, \tau}(\tilde{\Omega}_{\rho_P(\omega_X)}) \simeq \tilde{\Omega}^{\text{spec}}.$$

2.5.7. Note that for reasons of augmentation, we have

$$\text{Sat}^{-, \frac{\infty}{2}}(\nabla^{-, \frac{\infty}{2}}) \simeq \nabla^{-, \text{spec}, \frac{\infty}{2}}.$$

Note that this isomorphism immediately implies the commutativity of (2.24).

2.5.8. The next assertion will be proved in Sect. E.9:

Proposition 2.5.9. *We have a canonical isomorphism of unital factorization algebras*

$$\text{Sat}^{-, \frac{\infty}{2}}(\text{IC}^{-, \frac{\infty}{2}}) \simeq \text{IC}^{-, \text{spec}, \frac{\infty}{2}}.$$

Corollary 2.5.10. *The following diagram of factorization categories and functors commutes:*

$$(2.28) \quad \begin{array}{ccc} \text{Whit}^!(G) & \xrightarrow{\text{CS}_{G, \tau}} & \text{Rep}(\check{G}) \\ J_{\text{Whit}}^{-, !*} \downarrow & & \downarrow \text{Res}_{\check{M}}^{\check{G}} \simeq J^{-, \text{spec}, !*} \\ \text{Whit}^!(M) & \xrightarrow{\text{CS}_{M, \tau}} & \text{Rep}(\check{M}). \end{array}$$

Remark 2.5.11. The commutativity of (2.28) for $P = B$ (essentially) follows from [Ra4, Theorem 4.15.1]. A direct proof of Corollary 2.5.10 (without referring to Proposition 2.5.9) for an arbitrary parabolic will be given in [FH].

2.6. Self-duality on the spectral side.

2.6.1. Recall that according to (1.31), we have a canonical equivalence

$$(\text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}})^{\vee} \simeq \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}}.$$

Combining with Theorem 2.4.3, we obtain an equivalence

$$(2.29) \quad \text{I}(\check{G}, \check{P}^-)_{\text{co}}^{\text{spec}, \text{loc}} \simeq \text{I}(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}.$$

Remark 2.6.2. Note according to Sect. 1.7.6, the identification (2.29) is compatible with:

- The natural actions of $\text{Sph}_{\check{G}}^{\text{spec}}$ on the two sides;
- The natural actions of $\text{Sph}_{\check{M}}^{\text{spec}}$ on the two sides, *up to* the automorphism of $\text{Sph}_{\check{M}}^{\text{spec}}$ that corresponds under Sat_M to the automorphism $\text{transl}_{2\rho_P(\omega_X)}^*$ of Sph_M .

2.6.3. One can describe the equivalence (2.29) purely in spectral terms. However, this is non-trivial, and will be addressed in a separate publication.

For example, at a fixed $\underline{x} \in \text{Ran}$, in which case $\text{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\text{spec}, \text{loc}}$ is a stack locally almost of finite type and so

$$\text{I}(\check{G}, \check{P}^-)_{\text{co}}^{\text{spec}, \text{loc}} \simeq \text{IndCoh}(\text{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\text{spec}, \text{loc}}) \simeq \text{I}(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}},$$

the identification (2.29) corresponds to the usual (i.e., Serre) self-duality of $\text{IndCoh}(\text{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\text{spec}, \text{loc}})$

For the purposes of this paper, we will need a particular property of (2.29), stated as Proposition 2.6.8 below.

2.6.4. Let $\Delta_{\text{co}}^{-, \text{spec}, \frac{\infty}{2}}$ be the factorization unit of $I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}_{\text{co}}$. Explicitly,

$$\Delta_{\text{co}}^{-, \text{spec}, \frac{\infty}{2}} = \iota_*^{\text{IndCoh}}(\omega_{\text{LS}_{\check{P}^-}^{\text{reg}}}),$$

where:

- The morphism ι is as in (2.5);
- The functor

$$\text{IndCoh}^!(\text{LS}_{\check{P}^-}^{\text{reg}}) \xrightarrow{\iota_*^{\text{IndCoh}}} \text{IndCoh}^!(\text{Hecke}_{\check{G}, \check{P}^-}^{\text{spec}, \text{loc}}) = I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}_{\text{co}}$$

is the dual of the functor $\iota^!$ from (A.11), and so is the left adjoint of the tautologically defined functor

$$I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}_{\text{co}} = \text{IndCoh}^!(\text{Hecke}_{\check{G}, \check{P}^-}^{\text{spec}, \text{loc}}) \xrightarrow{\iota^!} \text{IndCoh}^!(\text{LS}_{\check{P}^-}^{\text{reg}}).$$

Since (2.29) is an equivalence of unital factorization categories, we obtain that the functor (2.29) send

$$\Delta_{\text{co}}^{-, \text{spec}, \frac{\infty}{2}} \in I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}_{\text{co}} \mapsto \Delta^{-, \text{spec}, \frac{\infty}{2}} \in I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}.$$

2.6.5. Let

$$\text{IC}_{\text{co}}^{-, \text{spec}, \frac{\infty}{2}} \in I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}_{\text{co}}$$

be the factorization algebra object equal to the image of

$$\omega_{\text{LS}_M^{\text{reg}}} \in \text{IndCoh}^!(\text{LS}_M^{\text{reg}})$$

along the functor

$$\text{IndCoh}^!(\text{LS}_M^{\text{reg}}) \rightarrow \text{IndCoh}^!(\text{LS}_{\check{P}^-}^{\text{reg}}) \xrightarrow{\iota_*^{\text{IndCoh}}} \text{IndCoh}^!(\text{Hecke}_{\check{G}, \check{P}^-}^{\text{spec}, \text{loc}}),$$

where:

- The first arrow is the dual of

$$\text{IndCoh}^*(\text{LS}_{\check{P}^-}^{\text{reg}}) \simeq \text{QCoh}(\text{LS}_{\check{P}^-}^{\text{reg}}) \rightarrow \text{QCoh}(\text{LS}_M^{\text{reg}}) \simeq \text{IndCoh}^*(\text{LS}_M^{\text{reg}}),$$

with the middle arrow being the functor of $*$ -pullback along (2.8);

- The second arrow is the functor from Sect. 2.6.4 above.

2.6.6. Note that the (factorization) functor

$$I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}} \rightarrow \text{Vect}$$

corresponding to the (factorization algebra) object $\text{IC}_{\text{co}}^{-, \text{spec}, \frac{\infty}{2}}$ is the composition

$$\begin{aligned} I(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}} &= \text{IndCoh}^*(\text{Hecke}_{\check{G}, \check{P}^-}^{\text{spec}, \text{loc}}) \xrightarrow{\iota^!} \text{IndCoh}^*(\text{LS}_{\check{P}^-}^{\text{reg}}) \simeq \\ &\simeq \text{QCoh}(\text{LS}_{\check{P}^-}^{\text{reg}}) \rightarrow \text{QCoh}(\text{LS}_M^{\text{reg}}) \xrightarrow{\Gamma(\text{LS}_M^{\text{reg}}, -)} \text{Vect}, \end{aligned}$$

where the functor $\text{QCoh}(\text{LS}_{\check{P}^-}^{\text{reg}}) \rightarrow \text{QCoh}(\text{LS}_M^{\text{reg}})$ is $*$ -pullback with respect to (2.8).

2.6.7. We claim:

Proposition 2.6.8. *The factorization algebra $\text{IC}_{\text{co}}^{-, \text{spec}, \frac{\infty}{2}}$ corresponds under the identification (2.29) to the factorization algebra $\text{IC}^{-, \text{spec}, \frac{\infty}{2}}$.*

The proof will be given in Sect. E.9.8.

2.7. The semi-infinite Casselman-Shalika equivalence. Just like the proof of the (derived) geometric Satake equivalence

$$\mathrm{Sph}_G^{\mathrm{Sat}_G} \simeq \mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$$

in [CR] is bootstrapped from the geometric Casselman-Shalika equivalence

$$\mathrm{CS}_G : \mathrm{Whit}^!(G) \simeq \mathrm{Rep}(\check{G}),$$

the proof of Theorem 2.4.3 is based on another local Langlands-type equivalence, Theorem 2.7.6.

We introduce the actors in the latter equivalence in this subsection, and the proof is delegated to Sect. B.

2.7.1. Consider the factorization prestack

$$(2.30) \quad \mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}_{\check{M}}} := \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{M}}^{\mathrm{mer}}} \mathrm{LS}_{\check{M}}^{\mathrm{reg}}.$$

When we work over a fixed point $\underline{x} \in \mathrm{Ran}$, we have

$$(2.31) \quad \mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free}_{\check{M}}} \simeq (\check{\mathfrak{n}}_{\check{P}^-} / \mathrm{Ad}(\check{P}^-))^{\times |\underline{x}|}.$$

Hence, the prestack (2.31) is an algebraic stack locally of finite type, and thus the category

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free}_{\check{M}}})$$

is well-defined. Moreover, we have map

$$(2.32) \quad \iota : \mathrm{LS}_{\check{P}^-}^{\mathrm{reg}} \rightarrow \mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}_{\check{M}}},$$

which is a *closed embedding* over $\underline{x}$, and which gives rise to a pair of adjoint functors

$$(2.33) \quad \iota_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{reg}}) \rightleftarrows \mathrm{IndCoh}(\mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free}_{\check{M}}}) : \iota^!.$$

However, we do not know how to make sense of $\mathrm{IndCoh}^*(-)$ or $\mathrm{IndCoh}^!(-)$ of (2.30) as a factorization category.

Remark 2.7.2. That said, one can consider the factorization category

$$(2.34) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}_{\check{M}}}),$$

see [GLC2, Sect. B.13.8].

Over a fixed point $\underline{x} \in \mathrm{Ran}$, the functor

$$\Psi_{\mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free}_{\check{M}}}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free}_{\check{M}}}) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free}_{\check{M}}})$$

is an equivalence, since (2.31) is smooth.

So, it may be that (2.34) is a reasonable candidate for

$$\mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}_{\check{M}}}).$$

However, for example, the basic thing we do not know is whether with this definition we have an adjunction

$$\iota_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) \rightleftarrows \mathrm{IndCoh}(\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{M}}^{\mathrm{mer}}} \mathrm{LS}_{\check{M}}^{\mathrm{reg}}) : \iota^!$$

of factorization functors, extending the pointwise adjunction (2.33).

2.7.3. Let

$$(\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}_{\check{M}}})_{\mathrm{mon-free}}^{\wedge}$$

be the formal completion of (2.30) along (2.32).

In Sect. A.6 we will show how to construct

$$\mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}_{\check{M}}})_{\mathrm{mon-free}}^{\wedge})$$

as a factorization category.

2.7.4. The category $\mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}-}^{\mathrm{mon-free}\check{M}})_{\mathrm{mon-free}}^\wedge)$ will come equipped with the following identification

$$(2.35) \quad \mathrm{Rep}(\check{G}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \simeq \mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}-}^{\mathrm{mon-free}\check{M}})_{\mathrm{mon-free}}^\wedge),$$

see Sect. A.6.3.

With respect to this equivalence, the functor

$$J^{-, \mathrm{spec}, \mathrm{enh}} : \mathrm{Rep}(\check{G}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \rightarrow \mathrm{Rep}(\check{M})$$

corresponds to the functor

$$\mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}-}^{\mathrm{mon-free}\check{M}})_{\mathrm{mon-free}}^\wedge) \xrightarrow{((\mathfrak{q}^{-, \mathrm{spec}})_{\mathrm{mon-free}}^\wedge)_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*(\mathrm{LS}_{\check{M}}^{\mathrm{reg}}) \simeq \mathrm{Rep}(\check{M}),$$

where $(\mathfrak{q}^{-, \mathrm{spec}})_{\mathrm{mon-free}}^\wedge$ denotes the map

$$(\mathrm{LS}_{\check{P}-}^{\mathrm{mon-free}\check{M}})_{\mathrm{mon-free}}^\wedge \rightarrow \mathrm{LS}_{\check{M}}^{\mathrm{reg}}.$$

2.7.5. Combing (2.35) with Theorem 2.4.3 we obtain:

Theorem 2.7.6. *There exists an equivalence of unital factorization categories*

$$(2.36) \quad \mathrm{CS}_\tau^{-, \frac{\infty}{2}} : \mathrm{Whit}^!(G) \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)^{\mathrm{loc}}_{\rho_P(\omega_X)} \simeq \mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}-}^{\mathrm{mon-free}\check{M}})_{\mathrm{mon-free}}^\wedge),$$

which makes the following diagram commute

$$(2.37) \quad \begin{array}{ccc} \mathrm{Whit}^!(G) \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)^{\mathrm{loc}}_{\rho_P(\omega_X)} & \xrightarrow{\mathrm{CS}_\tau^{-, \frac{\infty}{2}}} & \mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}-}^{\mathrm{mon-free}\check{M}})_{\mathrm{mon-free}}^\wedge) \\ J_{\mathrm{Whit}}^{-, \mathrm{enh}} \downarrow & & \downarrow ((\mathfrak{q}^{-, \mathrm{spec}})_{\mathrm{mon-free}}^\wedge)_*^{\mathrm{IndCoh}} \\ \mathrm{Whit}^!(M) & \xrightarrow{\mathrm{CS}_{M, \tau}} & \mathrm{Rep}(\check{M}). \end{array}$$

Furthermore, the functor $\mathrm{CS}_\tau^{-, \frac{\infty}{2}}$ is compatible with the actions of

$$\mathrm{Sph}_M^{\mathrm{Sat}_{M, \tau}} \simeq \mathrm{Sph}_M^{\mathrm{spec}}$$

on the two sides.

We view Theorem 2.7.6 as a semi-infinite version of the Casselman-Shalika equivalence.

Remark 2.7.7. As far as the actual logic of the proofs is concerned, we will prove Theorem 2.7.6 first, and then use it to deduce Theorem 2.4.3.

3. JACQUET AND WAKIMOTO FUNCTORS FOR KAC-MOODY MODULES

In this subsection we introduce generalized Jacquet functors between categories of Kac-Moody representations for G and M . The input for the construction is the functor of BRST reduction with respect to the loops Lie algebra into $\mathfrak{n}_P^-$.

We also study the *dual* functors, which go from Kac-Moody representations of M to those of G , called the *Wakimoto* functors.

We import the definitions pertaining to Kac-Moody modules and the Kazhdan-Lusztig category from [GLC2, Sect. 2].

For most of this section we will work at an arbitrary level κ , but at some point we will specialize to the case $\kappa = \mathrm{crit}_G$.

3.1. Twisting by Z_G^0 -torsors. In this subsection we will introduce a new kind of twist, specific to Kac-Moody extensions: it turns out that one can twist the Kac-Moody extension for G by a torsor with respect to the (connected) center of $\check{G}$.

In the bulk of the paper, the observations from this subsection will be applied when the reductive group in question is the Levi subgroup M of the original G . The specific twist that we will need is the Langlands dual of the *Miura shift* for opers, see Sect. 4.1.2.

3.1.1. Let $[\mathfrak{g}, \mathfrak{g}]$ be the Lie algebra of the derived group of G , so that $\mathfrak{g}_{\text{ab}} := \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ is the cocenter of $\mathfrak{g}$.

Consider the vector space $\mathfrak{L}(\mathfrak{g}_{\text{ab}})/\mathfrak{L}^+(\mathfrak{g}_{\text{ab}})$. Its dual, viewed as a group, acts by automorphisms of the Kac-Moody extension, and hence also by automorphisms of the categories $\widehat{\mathfrak{g}}\text{-mod}_\kappa$ and $\text{KL}(G)_\kappa$.

In particular, given a $(\mathfrak{L}(\mathfrak{g}_{\text{ab}})/\mathfrak{L}^+(\mathfrak{g}_{\text{ab}}))^*$ -torsor $\mathcal{P}$, we can form a twist $\widehat{\mathfrak{g}}_{\kappa+\mathcal{P}}$ of $\widehat{\mathfrak{g}}$ and consider the twisted versions of $\widehat{\mathfrak{g}}\text{-mod}_\kappa$ and $\text{KL}(G)_\kappa$, denoted

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa+\mathcal{P}} \text{ and } \text{KL}(G)_{\kappa+\mathcal{P}},$$

respectively.

3.1.2. Note that we have a canonical duality

$$\mathfrak{g}_{\text{ab}} \simeq (z_{\check{\mathfrak{g}}})^*.$$

In particular, we can identify

$$(\mathfrak{L}(\mathfrak{g}_{\text{ab}})/\mathfrak{L}^+(\mathfrak{g}_{\text{ab}}))^* \simeq \mathfrak{L}^+(z_{\check{\mathfrak{g}}} \otimes \omega_X),$$

where the right-hand side is the space of sections of $z_{\check{\mathfrak{g}}}$ -valued 1-forms on the formal disc.

3.1.3. Let Z_G^0 denote the neutral connected component of the center of $\check{G}$. Consider the homomorphism

$$\text{dlog} : \mathfrak{L}^+(Z_G^0) \rightarrow \mathfrak{L}^+(z_{\check{\mathfrak{g}}} \otimes \omega_X).$$

Thus, starting from a Z_G^0 -torsor $\mathcal{P}_{Z_G^0}$ on X , we can:

- Produce a $\mathfrak{L}^+(Z_G^0)$ -torsor (by restricting to the formal disc);
- Induce a $\mathfrak{L}^+(z_{\check{\mathfrak{g}}} \otimes \omega_X)$ -torsor using dlog ;
- Think of the latter as a $(\mathfrak{L}(\mathfrak{g}_{\text{ab}})/\mathfrak{L}^+(\mathfrak{g}_{\text{ab}}))^*$ -torsor, which by a slight abuse of notation we denote by

$$\text{dlog}(\mathcal{P}_{Z_G^0}).$$

The above construction is naturally compatible with factorization.

We denote the resulting (factorization) categories of Kac-Moody modules by

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa+\text{dlog}(\mathcal{P}_{Z_G^0})} \text{ and } \text{KL}(G)_{\kappa+\text{dlog}(\mathcal{P}_{Z_G^0})},$$

respectively.

3.1.4. Denote $G_{\text{ab}} = G/[G, G]$, so that it is a torus dual to Z_G^0 . The Contou-Carrère symbol defines a bilinear pairing

$$(3.1) \quad \langle -, - \rangle_{\text{CtCr}} : \mathfrak{L}(Z_G^0) \times \mathfrak{L}(G_{\text{ab}}) \rightarrow \mathbb{G}_m$$

We normalize (3.1) so that for an element $g \in \mathfrak{L}(Z_G^0)$, the resulting homomorphism

$$\langle g, - \rangle_{\text{CtCr}} : \mathfrak{L}(G_{\text{ab}}) \rightarrow \mathbb{G}_m$$

is such that its differential, which is an element of

$$\mathfrak{L}(\mathfrak{g}_{\text{ab}})^* \simeq \mathfrak{L}(z_{\check{\mathfrak{g}}} \otimes \omega_X)$$

is given by $\text{dlog}(g) \in \mathfrak{L}(z_{\check{\mathfrak{g}}} \otimes \omega_X)$.

3.1.5. Restricting $\langle -, - \rangle_{\text{CtCr}}$ along

$$\mathfrak{L}^+(Z_G^0) \times \mathfrak{L}(G) \rightarrow \mathfrak{L}^+(Z_G^0) \times \mathfrak{L}(G_{\text{ab}}),$$

we obtain that a Z_G^0 -torsor $\mathcal{P}_{Z_G^0}$ on X gives rise to a multiplicative line bundle $\mathcal{L}_{\mathfrak{L}(G), \mathcal{P}_{Z_G^0}}^{\text{loc}}$ on $\mathfrak{L}(G)$, i.e., a central extension by means of $\mathbb{G}_m$, compatible with the factorization structure.

The induced central extension of $\mathfrak{L}(\mathfrak{g})$ is $\widehat{\mathfrak{g}}\text{-mod}_{\text{dlog}(\mathcal{P}_{Z_G^0})}$.

3.1.6. Hence, tensoring by $\mathcal{L}_{\mathfrak{L}(G), \mathcal{P}_{Z_G^0}}^{\text{loc}}$ defines a bijection between categories acted on (strongly) by $\mathfrak{L}(G)$ at levels κ and $\kappa + \text{dlog}(\mathcal{P}_{Z_G^0})$, respectively.

In particular, we obtain an action of $\mathfrak{L}(G)$ at level κ on the category $\widehat{\mathfrak{g}}\text{-mod}_{\kappa + \text{dlog}(\mathcal{P}_{Z_G^0})}$, and we have the convolution functor

$$\text{D-mod}_{\kappa}(\text{Gr}_G) \otimes \text{KL}(G)_{\kappa + \text{dlog}(\mathcal{P}_{Z_G^0})} \xrightarrow{*} \widehat{\mathfrak{g}}\text{-mod}_{\kappa + \text{dlog}(\mathcal{P}_{Z_G^0})}.$$

3.1.7. *Notational convention.* Assume for a moment that the above Z_G^0 -torsor is of the form $\check{\lambda}(\omega_X)$ where $\check{\lambda} : \mathbb{G}_m \rightarrow Z_G^0$.

In this case, we will use a shorthand notation

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa + \check{\lambda}} := \widehat{\mathfrak{g}}\text{-mod}_{\kappa + \text{dlog}(\check{\lambda}(\omega_X))} \text{ and } \text{KL}(G)_{\kappa + \check{\lambda}} := \text{KL}(G)_{\kappa + \text{dlog}(\check{\lambda}(\omega_X))}$$

for the corresponding twisted categories.

Note that the assignment

$$\check{\lambda} \rightsquigarrow \text{dlog}(\check{\lambda}(\omega_X))$$

is linear. So the resulting torsor with respect to

$$\mathfrak{L}^+(z_{\check{\mathfrak{g}}} \otimes \omega_X) \simeq (\mathfrak{L}(\mathfrak{g}_{\text{ab}})/\mathfrak{L}^+(\mathfrak{g}_{\text{ab}}))^*$$

makes sense for any $\check{\lambda} \in z_{\check{\mathfrak{g}}}$.

Thus, the twisted categories

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa + \check{\lambda}} \text{ and } \text{KL}(G)_{\kappa + \check{\lambda}}$$

are defined for any $\check{\lambda} \in z_{\check{\mathfrak{g}}}$.

3.2. Jacquet functor(s) for Kac-Moody representations. In this subsection we recall the construction of the (usual) BRST reduction functor from Kac-Moody representations of G to those of M .

3.2.1. Note that that $\kappa|_{\mathfrak{p}^-}$ factors through $\mathfrak{p}^- \rightarrow \mathfrak{m}$. We will use the same character κ to denote the resulting level for $\mathfrak{m}$.

Note that the central extensions of $\mathfrak{L}(\mathfrak{p}^-)$ induced from $\widehat{\mathfrak{g}}_{\kappa}$ and $\widehat{\mathfrak{m}}_{\kappa}$ via the maps

$$\mathfrak{L}(\mathfrak{g}) \leftarrow \mathfrak{L}(\mathfrak{p}^-) \rightarrow \mathfrak{L}(\mathfrak{m}),$$

respectively, are canonically isomorphic.

We denote the resulting central extension of $\mathfrak{L}(\mathfrak{p}^-)$ by $\widehat{\mathfrak{p}}_{\kappa}^-$.

3.2.2. It is a basic fact (see [BD2, Proof of Theorem 2.8.17, point (c) on page 142]) that the Tate extension of $\mathfrak{L}(\mathfrak{p}^-)$ identifies with

$$\widehat{\mathfrak{p}}_{-\text{crit}_G - \text{crit}_M + \check{\rho}_P}^-$$

where:

- The subscripts crit_G and crit_M are the critical levels for G and M , respectively.
- The subscript $\check{\rho}_P$ is as in Sect. 3.1.7.

In particular, for any κ we have a canonical duality

$$(3.2) \quad (\widehat{\mathfrak{p}}^- \text{-mod}_{\text{crit}_G + \kappa})^\vee \simeq \widehat{\mathfrak{p}}^- \text{-mod}_{\text{crit}_M - \kappa - \check{\rho}_P}$$

as unital factorization categories, compatible with the $\mathfrak{L}(G)$ -action, where:

- On the left-hand side, $\mathfrak{L}(P^-)$ acts at level $-(\text{crit}_G + \kappa)$ restricted along $P^- \rightarrow G$;
- On the right-hand side, $\mathfrak{L}(P^-)$ acts at level $\text{crit}_M - \kappa - \check{\rho}_P$ restricted along $P^- \rightarrow M$;
- We pass between actions at these levels as in Sect. 3.1.6.

3.2.3. The projection

$$\widehat{\mathfrak{p}}^- \text{-mod}_{\text{crit}_M - \kappa - \check{\rho}_P} \rightarrow \widehat{\mathfrak{m}} \text{-mod}_{\text{crit}_M - \kappa - \check{\rho}_P}$$

gives rise to a (strictly) unital factorization functor

$$\text{Res}_{\widehat{\mathfrak{p}}^-}^{\widehat{\mathfrak{m}}} : \widehat{\mathfrak{m}} \text{-mod}_{\text{crit}_M - \kappa - \check{\rho}_P} \rightarrow \widehat{\mathfrak{p}}^- \text{-mod}_{\text{crit}_M - \kappa - \check{\rho}_P}.$$

Its dual, which is a functor

$$(3.3) \quad \text{BRST}^- : \widehat{\mathfrak{p}}^- \text{-mod}_{\text{crit}_G + \kappa} \rightarrow \widehat{\mathfrak{m}} \text{-mod}_{\text{crit}_M + \kappa + \check{\rho}_P},$$

is called the functor of BRST reduction with respect to $\mathfrak{L}(\mathfrak{n}_P^-)$.

Being the dual of a (strictly) unital factorization functor, the functor BRST^- acquires a lax unital factorization structure.

3.2.4. Note that the functor $\text{Res}_{\widehat{\mathfrak{p}}^-}^{\widehat{\mathfrak{m}}}$ respects the actions of $\mathfrak{L}(P^-)$ on the two sides. Hence, the functor BRST^- acquires a similar structure, where:

- On the left-hand side, $\mathfrak{L}(P^-)$ acts at level $\text{crit}_G + \kappa$ restricted along $P^- \rightarrow G$;
- On the left-hand side, $\mathfrak{L}(P^-)$ acts at level $\text{crit}_M + \kappa + \check{\rho}_P$ restricted along $P^- \rightarrow M$;
- We pass between actions at these levels as in Sect. 3.1.6.

In particular, the functor BRST^- is $\mathfrak{L}(N_P^-)$ -invariant, and hence canonically factors as

$$(3.4) \quad \widehat{\mathfrak{p}}^- \text{-mod}_{\text{crit}_G + \kappa} \rightarrow (\widehat{\mathfrak{p}}^- \text{-mod}_{\text{crit}_G + \kappa})_{\mathfrak{L}(N_P^-)} \xrightarrow{\overline{\text{BRST}}^-} \widehat{\mathfrak{m}} \text{-mod}_{\text{crit}_M + \kappa + \check{\rho}_P},$$

where $\overline{\text{BRST}}^-$ acquires a $\mathfrak{L}(M)$ -equivariant structure.

In particular, the functor BRST^- induces a functor

$$\widehat{\mathfrak{p}}^- \text{-mod}_{\text{crit}_G + \kappa}^{\mathfrak{L}^+(P^-)} \rightarrow \widehat{\mathfrak{m}} \text{-mod}_{\text{crit}_M + \kappa + \check{\rho}_P}^{\mathfrak{L}^+(P^-)} \simeq \widehat{\mathfrak{m}} \text{-mod}_{\text{crit}_M + \kappa + \check{\rho}_P}^{\mathfrak{L}^+(M)} = \text{KL}(M)_{\text{crit}_M + \kappa + \check{\rho}_P},$$

denoted by the same symbol BRST^- , and the functor $\overline{\text{BRST}}^-$ induces a functor

$$(\widehat{\mathfrak{p}}^- \text{-mod}_{\text{crit}_G + \kappa})_{\mathfrak{L}(N_P^-)}^{\mathfrak{L}^+(M)} \rightarrow \widehat{\mathfrak{m}} \text{-mod}_{\text{crit}_M + \kappa + \check{\rho}_P}^{\mathfrak{L}^+(M)} = \text{KL}(M)_{\text{crit}_M + \kappa + \check{\rho}_P},$$

denoted by the same symbol $\overline{\text{BRST}}^-$.

3.2.5. We let

$$J_{\text{KM}}^- : \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa} \rightarrow \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa + \check{\rho}_P}$$

denote the composition

$$\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa} \rightarrow \widehat{\mathfrak{p}}^-\text{-mod}_{\text{crit}_G + \kappa} \xrightarrow{\text{BRST}^-} \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa + \check{\rho}_P}.$$

We will denote by the same symbol J_{KM}^- the induced functor

$$\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa}^{\mathfrak{g}^+(P^-)} \rightarrow \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa + \check{\rho}_P}^{\mathfrak{g}^+(P^-)} \simeq \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa + \check{\rho}_P}^{\mathfrak{g}^+(M)} = \text{KL}(M)_{\text{crit}_M + \kappa + \check{\rho}_P}.$$

3.2.6. Let

$$\overline{J}_{\text{KM}}^- : (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa})_{\mathfrak{L}(N_P^-)} \rightarrow \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa + \check{\rho}_P}$$

denote the composition

$$(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa})_{\mathfrak{L}(N_P^-)} \rightarrow (\widehat{\mathfrak{p}}^-\text{-mod}_{\text{crit}_G + \kappa})_{\mathfrak{L}(N_P^-)} \xrightarrow{\overline{\text{BRST}}^-} \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa + \check{\rho}_P}.$$

We will denote by the same symbol $\overline{J}_{\text{KM}}^-$ the induced functor

$$(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa})_{\mathfrak{L}(N_P^-)}^{\mathfrak{g}^+(M)} \rightarrow \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa + \check{\rho}_P}^{\mathfrak{g}^+(M)} = \text{KL}(M)_{\text{crit}_M + \kappa + \check{\rho}_P}.$$

3.2.7. We will denote by $J_{\text{KM}}^{-, \text{Sph}}$ the composition

$$(3.5) \quad \text{KL}(G)_{\text{crit}_G + \kappa} = \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa}^{\mathfrak{g}^+(G)} \xrightarrow{\text{oblv}_{\mathfrak{g}^+(G) \rightarrow \mathfrak{g}^+(P^-)}} \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa}^{\mathfrak{g}^+(P^-)} \xrightarrow{J_{\text{KM}}^-} \text{KL}(M)_{\text{crit}_M + \kappa + \check{\rho}_P}.$$

3.3. The enhanced Jacquet functor for Kac-Moody representations. In this subsection we couple the Jacquet functor $\overline{J}_{\text{KM}}^-$ from the previous subsection with objects $\text{I}(G, P^-)_{\text{co}}^{\text{loc}}$ to obtain generalized Jacquet functors that map $\text{KL}(G)_{\text{crit}_G}$ to $\text{KL}(M)_{\text{crit}_M}$.

3.3.1. We now specialize to the critical level. In this case we will introduce one more version of the Jacquet functor, which will play a fundamental role in the sequel.

Note that we have a fully faithful functor

$$\text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)_{\mathfrak{L}(N_P^-)}^{\mathfrak{g}^+(M)} \hookrightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})_{\mathfrak{L}(N_P^-)}^{\mathfrak{g}^+(M)}.$$

Pre-composing with

$$\text{I}(G, P^-)_{\text{co}}^{\text{loc}} := \text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)_{\mathfrak{L}(N_P^-)}^{\mathfrak{g}^+(M), \text{ren}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)_{\mathfrak{L}(N_P^-)}^{\mathfrak{g}^+(M)}$$

and composing with $\overline{J}_{\text{KM}}^-$, we obtain a functor

$$(3.6) \quad \text{KL}(G)_{\text{crit}_G}^{-, \text{enh}_{\text{co}}} := \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\text{co}}^{\text{loc}} \rightarrow \text{KL}(M)_{\text{crit}_M + \check{\rho}_P},$$

which we denote by

$$J_{\text{KM}}^{-, \text{enh}_{\text{co}}}.$$

3.3.2. The functor $J_{\text{KM}}^{-, \text{enh}_{\text{co}}}$ is Sph_M -linear. Hence, by Sect. 1.4.8, the datum of $J_{\text{KM}}^{-, \text{enh}_{\text{co}}}$ is equivalent to the datum of a Sph_G -linear functor

$$(3.7) \quad \text{KL}(G)_{\text{crit}_G} \rightarrow \text{KL}(M)_{\text{crit}_M + \check{\rho}_P} \otimes_{\text{Sph}_M} \text{I}(G, P^-)^{\text{loc}} =: \text{KL}(M)_{\text{crit}_M + \check{\rho}_P}^{-, \text{enh}}.$$

We denote the functor (3.7) by

$$J_{\text{KM}}^{-, \text{enh}}.$$

3.3.3. Note that the functor $J_{\text{KM}}^{-, \text{Sph}}$ of (3.5) is recovered from $J_{\text{KM}}^{-, \text{enh}}$ as

$$\text{KL}(G)_{\text{crit}_G} \xrightarrow{J_{\text{KM}}^{-, \text{enh}}} \text{KL}(M)_{\text{crit}_M + \bar{\rho}_P}^{-, \text{enh}} \xrightarrow{\text{oblv}_{\text{enh}}} \text{KL}(M)_{\text{crit}_M + \bar{\rho}_P},$$

where oblv_{enh} is the functor

$$\text{KL}(M)_{\text{crit}_M + \bar{\rho}_P} \otimes_{\text{Sph}_M} \text{I}(G, P^-)^{\text{loc}} \xrightarrow{\text{Id} \otimes \text{oblv}_{\frac{\infty}{2}} \rightarrow \text{Sph}} \text{KL}(M)_{\text{crit}_M + \bar{\rho}_P} \otimes_{\text{Sph}_M} \text{Sph}_M = \text{KL}(M)_{\text{crit}_M + \bar{\rho}_P}.$$

3.4. **The Wakimoto functor(s).** In this subsection we define the *Wakimoto* functor, which is a functor dual to J_{KM}^- .

Note, however, that our definition of the Wakimoto functor (or the values of this functor on particular objects) is different from what is called *Wakimoto modules* in most of the literature: our Wakimoto modules are semi-infinite in nature. The usual Wakimoto modules can be obtained from ours by a subsequent averaging procedure, see Sect. 3.6.

3.4.1. Let

$$(3.8) \quad \text{Wak}^{-, \frac{\infty}{2}} : \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M - \kappa - \bar{\rho}_P} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa}$$

be the functor dual to the functor J_{KM}^- .

Since J_{KM}^- factors as

$$\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa} \rightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa})_{\mathfrak{L}(N_P^-)} \xrightarrow{\bar{J}_{\text{KM}}} \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa + \bar{\rho}_P},$$

the functor $\text{Wak}^{-, \frac{\infty}{2}}$ takes values in

$$(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa})^{\mathfrak{L}(N_P^-)} \subset \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa}.$$

The resulting functor

$$\widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M - \kappa - \bar{\rho}_P} \rightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa})^{\mathfrak{L}(N_P^-)}$$

is the dual of the functor $\bar{J}_{\text{KM}}^-$.

We will keep the same symbol $\text{Wak}^{-, \frac{\infty}{2}}$ for the functor

$$(3.9) \quad \text{KL}(M)_{\text{crit}_M - \kappa - \bar{\rho}_P} := \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M - \kappa - \bar{\rho}_P}^{\mathfrak{L}^+(M)} \rightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa})^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)} =: \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa}^{-, \frac{\infty}{2}}$$

Remark 3.4.2. Recall (see [GLC2, Sect. B.14]) that $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa}$ has a natural structure of (unital) factorization category, and this structure is inherited by $(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa})_{\mathfrak{L}(N_P^-)}$.

However, we do not know whether the operation of $\mathfrak{L}(N_P^-)$ -invariants commutes with colimits or tensor products. Hence, a priori

$$\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa}^{\mathfrak{L}^+(N_P^-)} \text{ and } \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa}^{-, \frac{\infty}{2}}$$

are only *lax* factorization categories (see [GLC2, Sect. B.11.12] for what this means), equipped with a unital structure.²²

The functor $\text{Wak}^{-, \frac{\infty}{2}}$ (in both variants) has a natural structure of (lax unital) factorization functor.

Remark 3.4.3. The version of the Wakimoto functor given by (3.9) may seem somewhat exotic. For example, it produces objects that are not seen by the t-structure on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa}$ (i.e., all of their cohomologies are 0).

One recovers from it the usual Wakimoto functor by composing $\text{Wak}^{-, \frac{\infty}{2}}$ with the functor

$$\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa} \xrightarrow{\text{Av}_*^{\mathfrak{L}^+(N)}} \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa}^{\mathfrak{L}^+(N)} \hookrightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa},$$

see Sect. 3.6.1.

²²For a unital factorization category $\mathbf{C}$, equipped with an action of $\mathfrak{L}(G)$, the category $\mathbf{C}^{\mathfrak{L}^+(N_P^-)}$ acquires a unital structure, in which the factorization unit is $\Delta^{-, \frac{\infty}{2}} \star \mathbf{1}_{\mathbf{C}}$.

3.4.4. Denote:

$$(3.10) \quad \text{Wak}^{-, \text{Sph}} := \text{Av}_*^{\mathfrak{L}^+(G)/\mathfrak{L}^+(M)} \circ \text{Wak}^{-, \frac{\infty}{2}}, \quad \text{KL}(M)_{\text{crit}_M - \kappa - \check{\rho}_P} \rightarrow \text{KL}(G)_{\text{crit}_G - \kappa}.$$

This is the functor dual to $J_{\text{KM}}^{-, \text{Sph}}$. It has a natural (lax unital) factorization structure.

3.4.5. The functor (3.10) can be explicitly described in terms of the duality

$$(3.11) \quad (\text{KL}(M)_{\text{crit}_M - \kappa - \check{\rho}_P})^\vee \simeq \text{KL}(M)_{\text{crit}_G + \kappa - \check{\rho}_P}.$$

Namely, the corresponding object of

$$(\text{KL}(M)_{\text{crit}_M - \kappa - \check{\rho}_P})^\vee \otimes \text{KL}(G)_{\text{crit}_G - \kappa} \simeq \text{KL}(M)_{\text{crit}_G + \kappa - \check{\rho}_P} \otimes \text{KL}(G)_{\text{crit}_G - \kappa}$$

is given by

$$(J_{\text{KM}}^{-, \text{Sph}} \otimes \text{Id})(\mathfrak{C}\mathfrak{D}\mathfrak{D}(G)_{\text{crit}_G + \kappa, \text{crit}_G - \kappa}),$$

where

$$\mathfrak{C}\mathfrak{D}\mathfrak{D}(G)_{\text{crit}_G + \kappa, \text{crit}_G - \kappa} \in \text{KL}(G)_{\text{crit}_G + \kappa} \otimes \text{KL}(G)_{\text{crit}_G - \kappa}$$

is as in [GLC2, Sect. 2.2.3].

3.5. Restriction of Wakimoto modules.

3.5.1. We now consider the relationship between the Wakimoto construction and restriction to the opposite Borel.

Let $\mathfrak{b}(M) = \mathfrak{p}^- \cap \mathfrak{b}$ (resp., $\mathfrak{n}(M) = \mathfrak{p}^- \cap \mathfrak{n}$) be the Lie algebra of the positive Borel (resp., its unipotent radical) in M .

Note that as in Sect. 3.2.2, the Tate extension for $\mathfrak{L}(\mathfrak{b}(M))$ is $-\text{crit}_M - \check{\rho}_M$ while the Tate extension for $\mathfrak{L}(\mathfrak{b})$ itself is $-\text{crit}_G - \check{\rho}_G$, so the difference between these two Tate extensions is the same as the difference between the Tate extensions for $\mathfrak{L}(\mathfrak{p})$ and $\mathfrak{L}(\mathfrak{g})$. Therefore, the following result makes sense:

Proposition 3.5.2. *The following diagram of (lax unital) factorization functors commutes:*

$$\begin{array}{ccc} \widehat{\mathfrak{p}}^- \text{-mod}_{\text{crit}_M + \kappa - \check{\rho}_P} & \xrightarrow{\text{oblv}^\vee} & \widehat{\mathfrak{g}} \text{-mod}_{\text{crit}_G + \kappa} \\ \downarrow \text{oblv} & & \downarrow \text{oblv} \\ \widehat{\mathfrak{b}}(M) \text{-mod}_{\text{crit}_M + \kappa - \check{\rho}_P} & \xrightarrow{\text{oblv}^\vee} & \widehat{\mathfrak{b}} \text{-mod}_{\text{crit}_G + \kappa} \end{array}$$

Here the functors $\text{oblv}^\vee$ are dual to forgetful functors oblv under the equivalences

$$\begin{aligned} (\widehat{\mathfrak{p}}^- \text{-mod}_{\text{crit}_M + \kappa - \check{\rho}_P})^\vee &\simeq \widehat{\mathfrak{p}}^- \text{-mod}_{\text{crit}_G - \kappa}, & (\widehat{\mathfrak{g}} \text{-mod}_{\text{crit}_G + \kappa})^\vee &\simeq \widehat{\mathfrak{g}} \text{-mod}_{\text{crit}_G - \kappa}, \\ (\widehat{\mathfrak{b}}(M) \text{-mod}_{\text{crit}_M + \kappa - \check{\rho}_P})^\vee &\simeq \widehat{\mathfrak{b}}(M) \text{-mod}_{-\kappa + \check{\rho}_G}, & (\widehat{\mathfrak{b}} \text{-mod}_{\text{crit}_G + \kappa})^\vee &\simeq \widehat{\mathfrak{b}} \text{-mod}_{-\kappa + \check{\rho}_G} \end{aligned}$$

given by semi-infinite cohomology.

Remark 3.5.3. Unwinding the abstraction, Proposition 3.5.2 amounts to the identity

$$(3.12) \quad \Gamma^{\text{IndCoh}}(\mathfrak{L}(G), \delta_1) \simeq \Gamma^{\text{IndCoh}}(\mathfrak{L}(P^-), \delta_1) \underset{\widehat{\mathfrak{b}}(M)}{\overset{\frac{\infty}{2}}{\otimes}} \Gamma^{\text{IndCoh}}(\mathfrak{L}(B), \delta_1),$$

where $\Gamma^{\text{IndCoh}}(\mathfrak{L}(H), \delta_1)$ is the unit of the duality between $\widehat{\mathfrak{h}} \text{-mod}_?$ and $\widehat{\mathfrak{h}} \text{-mod}_{??}$ for $H = G, P^-, B, B(M)$ and $?$ and $??$ are appropriate levels.

So, (3.12) expresses the triangular decomposition of $\mathfrak{g}$.

Remark 3.5.4. A similar result holds for any parabolic $\mathfrak{p}_2$ in place of the Borel, or more generally, any subalgebra $\mathfrak{h} \subset \mathfrak{g}$ with the property that $\mathfrak{p}^- + \mathfrak{h} = \mathfrak{g}$. In particular, we can apply it to $\mathfrak{h} = \mathfrak{n}$, see Lemma 3.5.8 below.

3.5.5. As a formal consequence of Proposition 3.5.2, we obtain:

Corollary 3.5.6. *The following diagram of (lax unital) factorization functors commutes:*

$$\begin{array}{ccc} \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa - \bar{\rho}_P} & \xrightarrow{\text{Wak}^-, \frac{\infty}{2}} & \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa} \\ \downarrow \text{oblv} & & \downarrow \text{oblv} \\ \widehat{\mathfrak{b}}(M)\text{-mod}_{\text{crit}_M + \kappa - \bar{\rho}_P} & \xrightarrow{\text{oblv}^\vee} & \widehat{\mathfrak{b}}\text{-mod}_{\text{crit}_G + \kappa} \end{array}$$

3.5.7. For future reference, we note (see Remark 3.5.4):

Lemma 3.5.8. *The following diagram of (lax unital) factorization functors commutes*

$$\begin{array}{ccc} \widehat{\mathfrak{p}}^-\text{-mod}_{\text{crit}_M + \kappa - \bar{\rho}_P} & \xrightarrow{\text{oblv}^\vee} & \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa} \\ \downarrow \text{oblv} & & \downarrow \text{oblv} \\ \mathfrak{L}(\mathfrak{n}(M))\text{-mod} & \xrightarrow{\text{oblv}^\vee} & \mathfrak{L}(\mathfrak{n})\text{-mod}, \end{array}$$

where the lower horizontal arrow is the dual of the natural restriction functor.

Corollary 3.5.9. *The following diagram of (lax unital) factorization functors commutes:*

$$\begin{array}{ccc} \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa - \bar{\rho}_P} & \xrightarrow{\text{Wak}^-, \frac{\infty}{2}} & \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa} \\ \downarrow \text{oblv} & & \downarrow \text{oblv} \\ \mathfrak{L}(\mathfrak{n}(M))\text{-mod} & \xrightarrow{\text{oblv}^\vee} & \mathfrak{L}(\mathfrak{n})\text{-mod}, \end{array}$$

where the lower horizontal arrow is the dual of the natural restriction functor.

3.6. The “usual” Wakimoto functor.

3.6.1. The “usual” Wakimoto functor, to be denoted

$$\text{Wak} : \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa - \bar{\rho}_P} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa}$$

is defined as the composition

$$\widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa - \bar{\rho}_P} \xrightarrow{\text{Wak}^-, \frac{\infty}{2}} (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa})^{\mathfrak{L}(N_P^-)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa} \xrightarrow{\text{Av}_*^{\mathfrak{L}^+(N_P)}} (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa})^{\mathfrak{L}^+(N_P)}.$$

This is the version of the Wakimoto functor that was studied in the early papers of Feigin and Frenkel (see [Ga2, Sect. 2.4]).

The functor Wak induces a functor

$$(3.13) \quad \text{KL}(M)_{\text{crit}_M + \kappa - \bar{\rho}_P} \rightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa})^{\mathfrak{L}^+(P)},$$

which we will denote by the same character Wak.

3.6.2. Here is an explicit description of the functor Wak in the spirit of Sect. 3.4.5.

Let $\overset{\circ}{G} := P \cdot P^- \subset G$ be the big Bruhat cell. We can consider the factorization algebra

$$\mathfrak{CD}\mathfrak{D}(\overset{\circ}{G})_{\text{crit}_G + \kappa, \text{crit}_G - \kappa} \in \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa}.$$

Consider the factorization algebra

$$(3.14) \quad (\text{Id} \otimes J_{\text{KM}}^-)(\mathfrak{CD}\mathfrak{D}(\overset{\circ}{G})_{\text{crit}_G + \kappa, \text{crit}_G - \kappa}) \in (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa})^{\mathfrak{L}^+(P)} \otimes \text{KL}(M)_{\text{crit}_G - \kappa + \bar{\rho}_P}.$$

Then the object (3.14) corresponds to the functor Wak of (3.13) in terms of the duality (3.11).

3.6.3. We will denote by $\mathbb{W}_{\mathfrak{g}, \text{crit}_G + \kappa}$ the image of $\text{Wak}(\text{Vac}(M)_{\text{crit}_M + \kappa})$ under the forgetful functor

$$(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa})^{\mathfrak{L}^+(P)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa} \rightarrow \text{Vect}.$$

This is a *classical* factorization algebra. The unital structure on $\text{Wak}^{-, \frac{\infty}{2}}$ (and hence on Wak) equips $\mathbb{W}_{\mathfrak{g}, \text{crit}_G + \kappa}$ with a homomorphism

$$\mathbb{V}_{\mathfrak{g}, \text{crit}_G + \kappa} \rightarrow \mathbb{W}_{\mathfrak{g}, \text{crit}_G + \kappa}.$$

3.6.4. The following is immediate from Corollary 3.5.6:

Corollary 3.6.5. *The following diagram of (lax unital) factorization functors commutes:*

$$\begin{array}{ccc} \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa - \bar{\rho}_P} & \xrightarrow{\text{Wak}} & \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa}^{\mathfrak{L}^+(N_P)} \\ \downarrow \text{oblv} & & \downarrow \text{oblv} \\ \widehat{\mathfrak{b}}(M)\text{-mod}_{\text{crit}_M + \kappa - \bar{\rho}_P} & \longrightarrow & \widehat{\mathfrak{b}}\text{-mod}_{\text{crit}_G + \kappa}^{\mathfrak{L}^+(N_P)} \end{array}$$

where the bottom horizontal arrow is the dual of the forgetful functor.

3.6.6. From Corollary 3.6.5, we obtain:

Corollary 3.6.7. *The following diagram of (lax unital) factorization functors commutes:*

$$\begin{array}{ccc} \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa - \bar{\rho}_P} & \xrightarrow{\text{Wak}} & \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa}^{\mathfrak{L}^+(N_P)} \\ \downarrow \text{oblv} & & \downarrow \text{oblv} \\ \widehat{\mathfrak{b}}(M)\text{-mod}_{\text{crit}_M + \kappa - \bar{\rho}_P} & \xrightarrow{\text{oblv}(-) \otimes \mathfrak{L}(N_P)} & \mathfrak{L}(N_P)\text{-mod}^{\mathfrak{L}^+(N_P)}, \end{array}$$

where in the bottom horizontal arrow, the functor **oblv** is the forgetful functor

$$\widehat{\mathfrak{b}}(M)\text{-mod}_{\text{crit}_M + \kappa - \bar{\rho}_P} \rightarrow \text{Vect}.$$

Remark 3.6.8. Corollary 3.6.7 expresses the following idea: the functor Wak produces objects that are “semi-infinite free” with respect to $\mathfrak{L}(\mathfrak{n}_P)$, see [Ga2, Sect. 2.2].

3.6.9. We have the following key result.

Proposition 3.6.10. *At the pointwise level, the functor Wak is t-exact.*

Remark 3.6.11. Note that Corollary 3.6.7 implies that the t-exactness property in Proposition 3.6.10 holds after applying the forgetful functor $\text{oblv} : \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa}^{\mathfrak{L}^+(N_P)} \rightarrow \text{Vect}$.

However, since the t-structure on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa}^{\mathfrak{L}^+(N_P)}$ is not left-separated, this is not enough to deduce the assertion of proposition. We fill in the details in Sect. 3.11.

3.7. The enhanced Wakimoto functor.

3.7.1. We now specialize the critical level. In this case, we will encounter one more version of the Wakimoto functor.

Namely, let

$$\text{Wak}^{-, \text{enh}} : \text{KL}(M)_{\text{crit}_M - \bar{\rho}_P} \rightarrow \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)^{\text{loc}}$$

be the functor dual to the functor $J_{\text{KM}}^{-, \text{enhco}}$ of (3.6), see Sect. 1.4.8.

3.7.2. Explicitly, the functor $\mathrm{Wak}^{-,\mathrm{enh}}$ is the composition

$$\mathrm{KL}(M)_{\mathrm{crit}_M - \bar{\rho}_P} \xrightarrow{\mathrm{Wak}^{-,\frac{\infty}{2}}} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{-,\frac{\infty}{2}} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)^{\mathrm{loc}},$$

where the second arrow, i.e.,

$$(3.15) \quad \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{-,\frac{\infty}{2}} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)^{\mathrm{loc}}$$

is the functor dual to

$$(3.16) \quad \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)^{\mathrm{loc}} = \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}^+(M), \mathrm{ren}}_{\mathfrak{L}(N_P^-)} \rightarrow \\ \rightarrow \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}^+(M)}_{\mathfrak{L}(N_P^-)} \hookrightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G})^{\mathfrak{L}^+(M)}_{\mathfrak{L}(N_P^-)}.$$

3.7.3. We record:

Lemma 3.7.4. *The functors*

$$\mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}^+(M), \mathrm{ren}}_{\mathfrak{L}(N_P^-)} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}^+(M)}_{\mathfrak{L}(N_P^-)}$$

and

$$\mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M), \mathrm{ren}} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)}$$

are equivalences. In particular, the functors (3.16) and

$$(3.17) \quad \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)^{\mathrm{loc}} = \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M), \mathrm{ren}} \rightarrow \\ \rightarrow \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)} \hookrightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G})^{\mathfrak{L}^+(M) \cdot \mathfrak{L}(N_P^-)} =: \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{-,\frac{\infty}{2}}$$

are fully faithful, and the functor (3.15) identifies with the right adjoint of (3.17).

Proof. It is enough to establish the corresponding pointwise assertion with respect to Ran . In this case, by (the parabolic version of) [Ra4, Theorem 6.2.1 and Corollary 6.2.3], at the pointwise level, we can replace both categories

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}^+(M)}_{\mathfrak{L}(N_P^-)} \text{ and } \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)}$$

by

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{I_P^-},$$

where I_P^- is the corresponding (negative) parahoric subgroup.

Similarly,

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)_{\mathfrak{L}(N_P^-)} \text{ and } \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-)}$$

can both be identified with

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{I_P^{-,0}},$$

where $I_P^{-,0}$ is the unipotent radical of I_P^- .

We can rewrite $(\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{I_P^{-,0}})^{M, \mathrm{ren}}$ as

$$\mathrm{D-mod}_{\frac{1}{2}}(I_P^{-,0} \backslash \mathfrak{L}(G)/K)^{M \times G, \mathrm{ren}},$$

where $K \subset \mathfrak{L}^+(G)$ is the first congruence subgroup, and the renormalization construction is taken with respect to the action of the group $M \times G$.

We can further rewrite this as

$$\mathrm{D-mod}_{\frac{1}{2}}(I_P^- \backslash \mathfrak{L}(G)/K)^{G, \mathrm{ren}},$$

and where the renormalization is taken with respect to G .

Let now be $\mathbf{C}_1, \mathbf{C}_2$ be an arbitrary pair of categories, acted on by $\mathfrak{L}(G)$. We claim that the functor

$$(3.18) \quad \mathbf{C}_1^{\mathfrak{L}^+(G)} \otimes_{\mathrm{Sph}_G} ((\mathbf{C}_2)^K)^{G, \mathrm{ren}} \rightarrow \mathbf{C}_1^{\mathfrak{L}^+(G)} \otimes_{\mathrm{Sph}_G} ((\mathbf{C}_2)^K)^G$$

is an equivalence.

Note that

$$\mathrm{C}(\mathrm{pt}/G)\text{-mod} \simeq \mathrm{D}\text{-mod}(\mathrm{pt}/G) \simeq \mathrm{D}\text{-mod}(\mathrm{pt}/\mathfrak{L}^+(G))$$

maps to the center of Sph_G . It suffices to show that the action of $\mathrm{C}(\mathrm{pt}/G)$ on the left-hand side of (3.18) (via the $((\mathbf{C}_2)^K)^{G, \mathrm{ren}}$ term) factors via an action of $\mathrm{C}(\mathrm{pt}/G)\text{-mod}_0$.

However, the above action equals the action via the $\mathbf{C}_1^{\mathfrak{L}^+(G)}$ term, and the required factorization is manifest. $\square$

3.7.5. Will denote the (fully faithful) functor

$$\mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)^{\mathrm{loc}} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{-, \frac{\infty}{2}}$$

of (3.17) by Accs .

Will denote its essential image by

$$\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{-, \frac{\infty}{2} - \mathrm{Accs}} \subset \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{-, \frac{\infty}{2}},$$

and we will refer it as *the spherically accessible subcategory* of $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{-, \frac{\infty}{2}}$.

3.7.6. From Lemma 3.7.4 we obtain:

Corollary 3.7.7. *The functor (3.15) identifies with the right adjoint of Accs .*

3.7.8. From Corollary 3.7.7 we obtain:

Corollary 3.7.9. *The functor $\mathrm{Wak}^{-, \mathrm{Sph}}$ is recovered from $\mathrm{Wak}^{-, \mathrm{enh}}$ as*

$$\begin{aligned} \mathrm{KL}(M)_{\mathrm{crit}_M - \bar{\rho}_P} &\xrightarrow{\mathrm{Wak}^{-, \mathrm{enh}}} \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)^{\mathrm{loc}} \xrightarrow{\mathrm{Id} \otimes ((-) \star_G \Delta^{-, \frac{\infty}{2}})^R} \\ &\rightarrow \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{Sph}_G = \mathrm{KL}(G)_{\mathrm{crit}_G}. \end{aligned}$$

Note that the functor $(\Delta^{-, \frac{\infty}{2}})^R$ appearing in the above formula is just

$$\begin{aligned} \mathrm{I}(G, P^-)^{\mathrm{loc}} &\simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M), \mathrm{ren}} \rightarrow \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}^+(M), \mathrm{ren}} \xrightarrow{\mathrm{Av}_*^{\mathfrak{L}^+(G)/\mathfrak{L}^+(M)}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}^+(G), \mathrm{ren}} =: \mathrm{Sph}_G. \end{aligned}$$

3.8. **Twisting by Z_G -torsors.** In this subsection we introduce the twists by central torsors (see Sect. 1.7) in the context of Kac-Moody representations.

As in Sect. 3.1, in the bulk of the paper, the observations from this subsection will be applied when the reductive group in question is not the original G , but rather one of its Levi subgroups.

3.8.1. Let $\mathcal{P}_G$ be a G -bundle on X . As in [GLC2, Sect. 1.2], we can consider the $\mathcal{P}_G$ -twists of $\widehat{\mathfrak{g}}\text{-mod}_\kappa$ and $\mathrm{KL}(G)_\kappa$, denoted

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa, \mathcal{P}_G} \text{ and } \mathrm{KL}(G)_{\kappa, \mathcal{P}_G},$$

respectively.

3.8.2. Note, however, that as in [GLC2, Sect. 1.2.3], we have a canonical equivalence

$$(3.19) \quad \alpha_{\mathcal{P}_G, \text{taut}} : \text{KL}(G)_\kappa \rightarrow \text{KL}(G)_{\kappa, \mathcal{P}_G}.$$

This equivalence fits into the commutative diagram

$$\begin{array}{ccc} \text{KL}(G)_\kappa & \xrightarrow{\alpha_{\mathcal{P}_G, \text{taut}}} & \text{KL}(G)_{\kappa, \mathcal{P}_G} \\ \text{Res}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} \downarrow & & \downarrow \text{oblv}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))_\kappa} \\ \text{Rep}(\mathfrak{L}^+(G)) & \xrightarrow{\alpha_{\mathcal{P}_G, \text{taut}}} & \text{Rep}(\mathfrak{L}^+(G)_{\mathcal{P}_G}). \end{array}$$

3.8.3. Assume now that $\mathcal{P}_G$ is induced from a Z_G -torsor $\mathcal{P}_{Z_G}$. In this case, we have a canonical identification

$$(3.20) \quad \alpha_{\mathcal{P}_{Z_G}, \text{cent}} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa, \mathcal{P}_{Z_G}} \xrightarrow{\sim} \widehat{\mathfrak{g}}\text{-mod}_{\kappa + \kappa(\text{dlog}(\mathcal{P}_{Z_G}), -)},$$

where:

- $\text{dlog}(\mathcal{P}_{Z_G})$ is the $\mathfrak{L}^+(z_{\mathfrak{g}} \otimes \omega_X)$ -torsor, induced by means of the map

$$\text{dlog} : \mathfrak{L}^+(Z_G) \rightarrow \mathfrak{L}^+(z_{\mathfrak{g}} \otimes \omega_X).$$

- The level $\kappa(-, -)$ is viewed as defining a map

$$\kappa(-, -) : z_{\mathfrak{g}} \rightarrow z_{\widehat{\mathfrak{g}}},$$

and so we can view

$$\kappa(\text{dlog}(\mathcal{P}_{Z_G}), -)$$

as a torsor with respect to

$$\mathfrak{L}^+(z_{\widehat{\mathfrak{g}}} \otimes \omega_X) \simeq (\mathfrak{L}(\mathfrak{g}_{\text{ab}})/\mathfrak{L}^+(\mathfrak{g}_{\text{ab}}))^*.$$

3.8.4. The equivalence (3.20) induces an equivalence

$$(3.21) \quad \alpha_{\mathcal{P}_{Z_G}, \text{cent}} : \text{KL}(G)_{\kappa, \mathcal{P}_{Z_G}} \xrightarrow{\sim} \text{KL}(G)_{\kappa + \kappa(\text{dlog}(\mathcal{P}_{Z_G}), -)}.$$

Composing with (3.19), we obtain an equivalence, denoted

$$(3.22) \quad \text{KL}(G)_\kappa \xrightarrow{(\text{transl}_{\mathcal{P}_{Z_G}})^*} \text{KL}(G)_{\kappa + \kappa(\text{dlog}(\mathcal{P}_{Z_G}), -)}.$$

This equivalence fits into the commutative diagram

$$\begin{array}{ccc} \text{KL}(G)_\kappa & \xrightarrow{(\text{transl}_{\mathcal{P}_{Z_G}})^*} & \text{KL}(G)_{\kappa + \kappa(\text{dlog}(\mathcal{P}_{Z_G}), -)} \\ \text{Res}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))} \downarrow & & \downarrow \text{Res}_{\mathfrak{L}^+(G)}^{(\widehat{\mathfrak{g}}, \mathfrak{L}^+(G))} \\ \text{Rep}(\mathfrak{L}^+(G)) & & \text{Rep}(\mathfrak{L}^+(G)) \\ = \downarrow & & \downarrow = \\ \text{QCoh}(\text{pt}/\mathfrak{L}^+(G)) & \xrightarrow{(\text{transl}_{\mathcal{P}_{Z_G}})^*} & \text{QCoh}(\text{pt}/\mathfrak{L}^+(G)) \end{array}$$

where the bottom horizontal arrow is the functor of pullback with respect to the automorphism $\text{transl}_{\mathcal{P}_{Z_G}}$ of $\text{pt}/\mathfrak{L}^+(G)$, given by tensoring with $\mathcal{P}_{Z_G}|_{\mathcal{D}}$.

3.8.5. Note that when $\kappa|_{z_{\mathfrak{g}}} = 0$, (e.g., when we are at the critical level), we have

$$\kappa(\text{dlog}(\mathcal{P}_{Z_G}), -) = 0.$$

So in this case, $(\text{transl}_{\mathcal{P}_{Z_G}})^*$ is an endofunctor of $\text{KL}(G)_\kappa$. It can be explicitly described as follows:

For every trivialization of $\mathcal{P}_{Z_G}$, we have

$$(\text{transl}_{\mathcal{P}_{Z_G}})^* \simeq \text{Id}.$$

A change of trivialization given by an element $g \in \mathfrak{L}^+(Z_G)$ corresponds to the automorphism of the identity endofunctor of $\text{KL}(G)_\kappa$ given by the action of g on modules.

3.8.6. *Notational convention.* Assume for a moment that the above Z_G -torsor is of the form $\lambda(\omega_X)$, where $\lambda : \mathbb{G}_m \rightarrow Z_G$.

In this case, we will use a shorthand notation

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa+\kappa(\lambda,-)} := \widehat{\mathfrak{g}}\text{-mod}_{\kappa+\kappa(\mathrm{dlog}(\lambda(\omega_X)), -)} \text{ and } \mathrm{KL}(G)_{\kappa+\kappa(\lambda,-)} := \mathrm{KL}(G)_{\kappa+\kappa(\mathrm{dlog}(\lambda(\omega_X)), -)}.$$

Note that this agrees with the notation introduced in Sect. 3.1.7, where we regard $\kappa(\lambda, -)$ as an element of $z_{\mathfrak{g}}$.

As in *loc.cit.*, the notations

$$\widehat{\mathfrak{g}}\text{-mod}_{\kappa+\kappa(\lambda,-)} \text{ and } \mathrm{KL}(G)_{\kappa+\kappa(\lambda,-)}$$

make sense for an arbitrary element $\lambda \in z_{\mathfrak{g}}$.

3.8.7. Similar constructions apply if instead of $\mathrm{KL}(G)_{\kappa}$ (resp., $\widehat{\mathfrak{g}}\text{-mod}_{\kappa}$), we start with the category of the form $\mathrm{KL}(G)_{\kappa+\mathrm{dlog}(\mathcal{P}_{Z_G^0})}$ (resp., $\widehat{\mathfrak{g}}\text{-mod}_{\kappa+\mathrm{dlog}(\mathcal{P}_{Z_G^0})}$) for a Z_G^0 -torsor $\mathcal{P}_{Z_G^0}$.

We denote the resulting categories by

$$\mathrm{KL}(G)_{\kappa+\mathrm{dlog}(\mathcal{P}_{Z_G^0})+\kappa(\mathrm{dlog}(\mathcal{P}_{Z_G}), -)} \text{ and } \widehat{\mathfrak{g}}\text{-mod}_{\kappa+\mathrm{dlog}(\mathcal{P}_{Z_G^0})+\kappa(\mathrm{dlog}(\mathcal{P}_{Z_G}), -)}$$

or

$$\mathrm{KL}(G)_{\kappa+\bar{\lambda}+\kappa(\lambda,-)} \text{ and } \widehat{\mathfrak{g}}\text{-mod}_{\kappa+\bar{\lambda}+\kappa(\lambda,-)},$$

respectively.

3.9. **The twisted Jacquet and Wakimoto functors.** In this subsection we will adapt the (generalized) Jacquet and Wakimoto functors to the twisted situation.

3.9.1. Let $\mathcal{P}_M$ be an M -torsor. Consider the $\mathcal{P}_M$ -twists of the objects appearing in Sect. 3.2, see Sect. 1.6.

In particular, we obtain the functor

$$J_{\mathrm{KM}, \mathcal{P}_M}^- : \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G + \kappa, \mathcal{P}_M} \rightarrow \widehat{\mathfrak{m}}\text{-mod}_{\mathrm{crit}_M + \kappa + \bar{\rho}_P, \mathcal{P}_M}.$$

3.9.2. Note, however, that by Sect. 3.8.2, we have the equivalences

$$\alpha_{\mathcal{P}_M, \mathrm{taut}} : \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G + \kappa}^{\mathfrak{g}^+(M)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G + \kappa, \mathcal{P}_M}^{\mathfrak{g}^+(M)}$$

and

$$\alpha_{\mathcal{P}_M, \mathrm{taut}} : \mathrm{KL}(M)_{\mathrm{crit}_M + \kappa + \bar{\rho}_P} \rightarrow \mathrm{KL}(M)_{\mathrm{crit}_M + \kappa + \bar{\rho}_P, \mathcal{P}_M}$$

so that the diagram

$$(3.23) \quad \begin{array}{ccc} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G + \kappa}^{\mathfrak{g}^+(M)} & \xrightarrow{\alpha_{\mathcal{P}_M, \mathrm{taut}}} & \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G + \kappa, \mathcal{P}_M}^{\mathfrak{g}^+(M)} \\ J_{\mathrm{KM}}^- \downarrow & & \downarrow J_{\mathrm{KM}, \mathcal{P}_M}^- \\ \mathrm{KL}(M)_{\mathrm{crit}_M + \kappa + \bar{\rho}_P} & \xrightarrow{\alpha_{\mathcal{P}_M, \mathrm{taut}}} & \mathrm{KL}(M)_{\mathrm{crit}_M + \kappa + \bar{\rho}_P, \mathcal{P}_M} \end{array}$$

commutes.

3.9.3. Assume now that $\mathcal{P}_M$ is induced by a Z_M -bundle $\mathcal{P}_{Z_M}$. In this case, we can identify

$$\widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa + \bar{\rho}_P, \mathcal{P}_{Z_M}} \xrightarrow{\alpha_{\mathcal{P}_{Z_M}, \text{cent}}} \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa + \bar{\rho}_P + \kappa(\text{dlog}(\mathcal{P}_{Z_M}), -)}$$

and

$$\text{KL}(M)_{\text{crit}_M + \kappa + \bar{\rho}_P, \mathcal{P}_{Z_M}} \xrightarrow{\alpha_{\mathcal{P}_{Z_M}, \text{cent}}} \text{KL}(M)_{\text{crit}_M + \kappa + \bar{\rho}_P + \kappa(\text{dlog}(\mathcal{P}_{Z_M}), -)}$$

see (3.20) and (3.21), respectively.

Using the above identifications, we will view $J_{\text{KM}, \mathcal{P}_{Z_M}}^-$ as a functor

$$\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa}^{\mathfrak{s}^+(M)} \rightarrow \text{KL}(M)_{\text{crit}_M + \kappa + \bar{\rho}_P + \kappa(\text{dlog}(\mathcal{P}_{Z_M}), -)}$$

equal to

$$(3.24) \quad \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa}^{\mathfrak{s}^+(M)} \xrightarrow{\alpha_{\mathcal{P}_{Z_M}, \text{taut}}} \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa, \mathcal{P}_{Z_M}}^{\mathfrak{s}^+(M)} \xrightarrow{J_{\text{KM}, \mathcal{P}_{Z_M}}^-} \text{KL}(M)_{\text{crit}_M + \kappa + \bar{\rho}_P, \mathcal{P}_{Z_M}} \xrightarrow{\alpha_{\mathcal{P}_{Z_M}, \text{cent}}} \text{KL}(M)_{\text{crit}_M + \kappa + \bar{\rho}_P + \kappa(\text{dlog}(\mathcal{P}_{Z_M}), -),}$$

or, which is the same,

$$(3.25) \quad \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa}^{\mathfrak{s}^+(M)} \xrightarrow{J_{\text{KM}}^-} \text{KL}(M)_{\text{crit}_M + \kappa + \bar{\rho}_P} \xrightarrow{(\text{transl}_{\mathcal{P}_{Z_M}})^*} \text{KL}(M)_{\text{crit}_M + \kappa + \bar{\rho}_P + \kappa(\text{dlog}(\mathcal{P}_{Z_M}), -)}.$$

3.9.4. *Warning.* Note that when $\kappa = 0$ (i.e., we are at the critical level), both functors

$$J_{\text{KM}}^- \text{ and } J_{\text{KM}, \mathcal{P}_{Z_M}}^-$$

can be viewed as mapping $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G}^{\mathfrak{s}^+(M)}$ to the *same* category $\text{KL}(M)_{\text{crit}_M + \bar{\rho}_P}$.

Yet, these two functors are different: namely, they differ by the automorphism of $\text{KL}(M)_{\text{crit}_M + \bar{\rho}_P}$ given by $(\text{transl}_{\mathcal{P}_{Z_M}})^*$, see Sect. 3.8.5.

In particular, they have different compatibilities with respect to the actions of Sph_M . Indeed, in the twisted version, while Sph_M acts naturally on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G}^{\mathfrak{s}^+(M)}$, its action on $\text{KL}(M)_{\text{crit}_M + \bar{\rho}_P}$ is twisted by the automorphism $(\text{transl}_{\mathcal{P}_{Z_M}})^*$ of Sph_M , see Remark 1.7.4

3.9.5. We will denote by $J_{\text{KM}, \mathcal{P}_{Z_M}}^{-, \text{Sph}}$ the restriction of (3.24) along

$$\text{KL}(G)_{\text{crit}_G + \kappa} := \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa}^{\mathfrak{s}^+(G)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa}^{\mathfrak{s}^+(M)},$$

so this is a functor

$$(3.26) \quad J_{\text{KM}, \mathcal{P}_{Z_M}}^{-, \text{Sph}} : \text{KL}(G)_{\text{crit}_G + \kappa} \rightarrow \text{KL}(M)_{\text{crit}_M + \kappa + \bar{\rho}_P - \kappa(\text{dlog}(\mathcal{P}_{Z_M}), -)}.$$

From (3.23) we obtain a commutative diagram

$$(3.27) \quad \begin{array}{ccc} \text{KL}(G)_{\text{crit}_G + \kappa} & \xrightarrow{\text{Id}} & \text{KL}(G)_{\text{crit}_G + \kappa} \\ J_{\text{KM}}^{-, \text{Sph}} \downarrow & & \downarrow J_{\text{KM}, \mathcal{P}_{Z_M}}^{-, \text{Sph}} \\ \text{KL}(M)_{\text{crit}_M + \kappa + \bar{\rho}_P} & \xrightarrow{(\text{transl}_{\mathcal{P}_{Z_M}})^*} & \text{KL}(M)_{\text{crit}_M + \kappa + \bar{\rho}_P + \kappa(\text{dlog}(\mathcal{P}_{Z_M}), -)}. \end{array}$$

3.9.6. Similar conventions apply to the Wakimoto functors. In particular, we obtain the functors

$$\text{Wak}_{\mathcal{P}_{Z_M}}^{-, \frac{\infty}{2}} : \text{KL}(M)_{\text{crit}_M - \kappa - \bar{\rho}_P - \kappa(\text{dlog}(\mathcal{P}_{Z_M}), -)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa}^{-, \frac{\infty}{2}}$$

and

$$\text{Wak}_{\mathcal{P}_{Z_M}}^{-, \text{Sph}} : \text{KL}(M)_{\text{crit}_M - \kappa - \bar{\rho}_P - \kappa(\text{dlog}(\mathcal{P}_{Z_M}), -)} \rightarrow \text{KL}(G)_{\text{crit}_G - \kappa}.$$

3.9.7. We will apply the above constructions mostly in the case when $\mathcal{P}_{Z_M} = \rho_P(\omega_X)$. So we obtain the functors

$$\begin{aligned} \overline{J}_{\text{KM}, \rho_P(\omega_X)}^- : (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa})_{\mathfrak{L}(N_P^-)}^{\mathfrak{L}^+(M)} &\rightarrow \text{KL}(M)_{\text{crit}_M + \kappa + \tilde{\rho}_P + \kappa(\rho_P, -)}, \\ \text{Wak}_{\rho_P(\omega_X)}^{-, \frac{\infty}{2}} : \text{KL}(M)_{\text{crit}_M - \kappa - \tilde{\rho}_P - \kappa(\rho_P, -)} &\rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa}^{-, \frac{\infty}{2}}, \\ \text{Wak}_{\rho_P(\omega_X)} : \text{KL}(M)_{\text{crit}_M - \kappa - \tilde{\rho}_P - \kappa(\rho_P, -)} &\rightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \kappa})^{\mathfrak{L}^+(P)}, \end{aligned}$$

as well as

$$J_{\rho_P(\omega_X)}^{-, \text{Sph}} : \text{KL}(G)_{\text{crit}_G + \kappa} \rightarrow \text{KL}(M)_{\text{crit}_M + \kappa + \tilde{\rho}_P + \kappa(\rho_P, -)}$$

and

$$\text{Wak}_{\rho_P(\omega_X)}^{-, \text{Sph}} : \text{KL}(M)_{\text{crit}_M - \kappa - \tilde{\rho}_P - \kappa(\rho_P, -)} \rightarrow \text{KL}(G)_{\text{crit}_G - \kappa}.$$

3.9.8. We now specialize to the critical level. In this case the functors in Sect. 3.9.7 map

$$\begin{aligned} \overline{J}_{\text{KM}, \rho_P(\omega_X)}^- : (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})_{\mathfrak{L}(N_P^-)}^{\mathfrak{L}^+(M)} &\rightarrow \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P}, \\ \text{Wak}_{\rho_P(\omega_X)}^{-, \frac{\infty}{2}} : \text{KL}(M)_{\text{crit}_M - \tilde{\rho}_P} &\rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G}^{-, \frac{\infty}{2}}, \\ \text{Wak}_{\rho_P(\omega_X)} : \text{KL}(M)_{\text{crit}_M - \tilde{\rho}_P} &\rightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})^{\mathfrak{L}^+(P)}, \end{aligned}$$

as well as

$$J_{\rho_P(\omega_X)}^{-, \text{Sph}} : \text{KL}(G)_{\text{crit}_G} \rightarrow \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P}$$

and

$$\text{Wak}_{\rho_P(\omega_X)}^{-, \text{Sph}} : \text{KL}(M)_{\text{crit}_M - \tilde{\rho}_P} \rightarrow \text{KL}(G)_{\text{crit}_G}.$$

3.9.9. In what follows we will also need twisted versions of the functors $J_{\text{KM}}^{-, \text{enhco}}$, $J_{\text{KM}}^{-, \text{enh}}$ and $\text{Wak}^{-, \text{enh}}$, respectively.

We let

$$J_{\text{KM}, \rho_P(\omega_X)}^{-, \text{enhco}} : \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\text{co}, \rho_P(\omega_X)}^{\text{loc}} \rightarrow \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P}$$

be the functor

$$\begin{aligned} \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\text{co}, \rho_P(\omega_X)}^{\text{loc}} &\xrightarrow{\text{Id} \otimes \alpha_{\text{taut}, \rho_P(\omega_X)}^{-1}} \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\text{co}}^{\text{loc}} \xrightarrow{J_{\text{KM}}^{-, \text{enhco}}} \\ &\rightarrow \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P} \xrightarrow{(\text{transl}_{\rho_P(\omega_X)})^*} \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P}, \end{aligned}$$

which is the same as

$$\begin{aligned} \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\text{co}, \rho_P(\omega_X)}^{\text{loc}} &\xrightarrow{\alpha_{\text{taut}, \rho_P(\omega_X)} \otimes \text{Id}} \text{KL}(G)_{\text{crit}_G, \rho_P(\omega_X)} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\text{co}, \rho_P(\omega_X)}^{\text{loc}} \rightarrow \\ &\rightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \rho_P(\omega_X)})_{\mathfrak{L}(N_P^-)}^{\mathfrak{L}^+(M)} \xrightarrow{\overline{J}_{\text{KM}, \rho_P(\omega_X)}^-} \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P, \rho_P(\omega_X)} \rightarrow \\ &\xrightarrow{\alpha_{\text{cent}, \rho_P(\omega_X)}} \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P}. \end{aligned}$$

This functor intertwines the natural actions of Sph_M on the two sides (see Sect. 1.7.5 for the conventions regarding the action of Sph_M on $\text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}}$).

The composition

$$\text{KL}(G)_{\text{crit}_G} \xrightarrow{\text{Id} \otimes \mathbf{1}_{\text{I}(G, P^-)_{\text{co}, \rho_P(\omega_X)}^{\text{loc}}}} \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\text{co}, \rho_P(\omega_X)}^{\text{loc}} \xrightarrow{J_{\text{KM}, \rho_P(\omega_X)}^{-, \text{enhco}}} \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P}$$

recovers the functor $J_{\rho_P(\omega_X)}^{-, \text{Sph}}$.

3.9.10. We let

$$J_{\text{KM}, \rho_P(\omega_X)}^{-, \text{enh}} : \text{KL}(G)_{\text{crit}_G} \rightarrow \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P} \otimes_{\text{Sph}_M} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}}$$

be the functor obtained from $J_{\text{KM}, \rho_P(\omega_X)}^{-, \text{enh}_{\text{co}}}$ by duality.

This functor intertwines the natural actions of Sph_G on the two sides.

The composition

$$\text{KL}(G)_{\text{crit}_G} \xrightarrow{J_{\text{KM}, \rho_P(\omega_X)}^{-, \text{enh}}} \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P} \otimes_{\text{Sph}_M} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \xrightarrow{\text{oblv}_{\text{enh}}} \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P},$$

recovers the functor $J_{\rho_P(\omega_X)}^{-, \text{Sph}}$, where oblv_{enh} denotes the composition

$$\text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P} \otimes_{\text{Sph}_M} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \xrightarrow{\text{Id} \otimes \text{oblv}_{\frac{\infty}{2}} \rightarrow \text{Sph}} \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P} \otimes_{\text{Sph}_M} \text{Sph}_M = \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P},$$

3.9.11. In the sequel we will also use the notations

$$\text{KL}(G)_{\text{crit}_G, \rho_P(\omega_X)}^{-, \text{enh}_{\text{co}}} := \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\text{co}, \rho_P(\omega_X)}^{\text{loc}}$$

and

$$\text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P, \rho_P(\omega_X)}^{-, \text{enh}} := \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}},$$

so that we can view $J_{\text{KM}, \rho_P(\omega_X)}^{-, \text{enh}_{\text{co}}}$ as a functor

$$\text{KL}(G)_{\text{crit}_G, \rho_P(\omega_X)}^{-, \text{enh}_{\text{co}}} \rightarrow \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P}$$

and $J_{\text{KM}, \rho_P(\omega_X)}^{-, \text{enh}}$ as a functor

$$\text{KL}(G)_{\text{crit}_G} \rightarrow \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P, \rho_P(\omega_X)}^{-, \text{enh}}.$$

3.9.12. We let

$$\text{Wak}_{\rho_P(\omega_X)}^{-, \text{enh}} : \text{KL}(M)_{\text{crit}_M - \tilde{\rho}_P} \rightarrow \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}}$$

be the functor dual to $J_{\text{KM}, \rho_P(\omega_X)}^{-, \text{enh}_{\text{co}}}$.

I.e., $\text{Wak}_{\rho_P(\omega_X)}^{-, \text{enh}}$ is the composition

$$\begin{aligned} & \text{KL}(M)_{\text{crit}_M - \tilde{\rho}_P} \xrightarrow{(\text{transl}_{-\rho_P(\omega_X)})^*} \text{KL}(M)_{\text{crit}_M - \tilde{\rho}_P} \xrightarrow{\text{Wak}_{\rho_P(\omega_X)}^{-, \text{enh}}} \\ & \rightarrow \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \xrightarrow{\text{Id} \otimes \alpha_{\text{taut}, \rho_P(\omega_X)}} \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} := \text{KL}(G)_{\text{crit}_G, \rho_P(\omega_X)}^{-, \text{enh}}. \end{aligned}$$

3.10. Drinfeld-Sokolov reduction of Wakimoto modules. In this subsection we compute the composition of the Drinfeld-Sokolov reduction and Wakimoto functors. This computation lies at the core of the proof of Theorem 5.4.5.

At its essence, this computation goes back to Feigin and Frenkel, and is lies also in the core of their work on duality for W-algebras.

3.10.1. We have the following basic assertion, which follows by duality from Corollary 3.5.9:

Lemma 3.10.2. *The following diagram of (lax unital) factorization functors commutes:*

$$(3.28) \quad \begin{array}{ccc} \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa - \tilde{\rho}_P, \rho(\omega_X)} & \xrightarrow{\alpha_{\rho_P(\omega), \text{cent}}} & \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \kappa - \tilde{\rho}_P + \kappa(\rho_P, -), \rho_M(\omega_X)} \\ \text{Wak}_{\rho(\omega_X)}^{-, \frac{\infty}{2}} \downarrow & & \downarrow \text{DS}_M \\ \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \kappa, \rho(\omega_X)} & \xrightarrow{\text{DS}_G} & \text{Vect} . \end{array}$$

3.10.3. From Lemma 3.10.2 we obtain:

Corollary 3.10.4. *The following diagram of (lax unital) factorization functors commutes:*

$$\begin{array}{ccc}
\mathrm{KL}(M)_{\mathrm{crit}_M + \kappa - \tilde{\rho}_P + \kappa(\rho_P, -)} & \xrightarrow{\alpha_{\rho_M(\omega), \mathrm{taut}}} & \mathrm{KL}(M)_{\mathrm{crit}_M + \kappa - \tilde{\rho}_P + \kappa(\rho_P, -), \rho_M(\omega_X)} \\
\mathrm{Wak}_{\rho_P(\omega_X)}^{-, \frac{\infty}{2}} \downarrow & & \downarrow \mathrm{DS}_M \\
\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G + \kappa}^{\mathfrak{L}^+(M)} & \xrightarrow{\alpha_{\rho(\omega), \mathrm{taut}}} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G + \kappa, \rho(\omega_X)}^{\mathfrak{L}^+(M)} \xrightarrow{\mathrm{DS}_G} & \mathrm{Vect}.
\end{array}$$

3.10.5. For future reference we observe that the functors

$$\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)} \xrightarrow{\mathrm{DS}_G} \mathrm{Vect}$$

and

$$\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)} \xrightarrow{\mathrm{Av}_*^{\mathfrak{L}^+(N_P)\rho(\omega_X)}} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)}^{\mathfrak{L}^+(N_P)\rho(\omega_X)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)} \xrightarrow{\mathrm{DS}_G} \mathrm{Vect}$$

are canonically isomorphic.

In particular, from Corollary 3.10.4 we obtain:

Corollary 3.10.6. *The following diagram of (lax unital) factorization functors commutes:*

$$\begin{array}{ccc}
\mathrm{KL}(M)_{\mathrm{crit}_M + \kappa - \tilde{\rho}_P + \kappa(\rho_P, -)} & \xrightarrow{\alpha_{\rho_M(\omega), \mathrm{taut}}} & \mathrm{KL}(M)_{\mathrm{crit}_M + \kappa - \tilde{\rho}_P + \kappa(\rho_P, -), \rho_M(\omega_X)} \\
\mathrm{Wak}_{\rho_P(\omega_X)} \downarrow & & \downarrow \mathrm{DS}_M \\
\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G + \kappa}^{\mathfrak{L}^+(P)} & \xrightarrow{\alpha_{\rho(\omega), \mathrm{taut}}} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G + \kappa, \rho(\omega_X)}^{\mathfrak{L}^+(P)} \xrightarrow{\mathrm{DS}_G} & \mathrm{Vect}.
\end{array}$$

3.10.7. Specializing to the critical level, from Corollary 3.10.4 we obtain:

Corollary 3.10.8. *The following diagrams of (lax unital) factorization functors commute:*

$$\begin{array}{ccc}
\mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P} & \xrightarrow{\alpha_{\rho_M(\omega), \mathrm{taut}}} & \mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P, \rho_M(\omega_X)} \\
\mathrm{Wak}_{\rho_P(\omega_X)}^{-, \frac{\infty}{2}} \downarrow & & \downarrow \mathrm{DS}_M \\
\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{\mathfrak{L}^+(M)} & \xrightarrow{\alpha_{\rho(\omega), \mathrm{taut}}} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)}^{\mathfrak{L}^+(M)} \xrightarrow{\mathrm{DS}_G} & \mathrm{Vect}
\end{array}$$

and

$$\begin{array}{ccc}
\mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P} & \xrightarrow{\alpha_{\rho_M(\omega), \mathrm{taut}}} & \mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P, \rho_M(\omega_X)} \\
\mathrm{Wak}_{\rho_P(\omega_X)} \downarrow & & \downarrow \mathrm{DS}_M \\
\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{\mathfrak{L}^+(M)} & \xrightarrow{\alpha_{\rho(\omega), \mathrm{taut}}} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)}^{\mathfrak{L}^+(M)} \xrightarrow{\mathrm{DS}_G} & \mathrm{Vect}.
\end{array}$$

3.11. Proof of Proposition 3.6.10.

3.11.1. As was explained in Remark 3.6.11, it is enough to prove that Wak is left t-exact.

Suppose

$$\mathcal{M} \in \widehat{\mathfrak{m}}\text{-mod}_{\mathrm{crit}_M + \kappa - \tilde{\rho}_P}^{\geq 0}$$

is coconnective. We wish to show that $\mathrm{Wak}(\mathcal{M})$ is coconnective as well. It suffices to do this in the case that

$$\mathcal{M} \in \widehat{\mathfrak{m}}\text{-mod}_{\mathrm{crit}_M + \kappa - \tilde{\rho}_P}^{\mathfrak{L}_n^+(M)}$$

for $\mathfrak{L}_n^+(M) \subset \mathfrak{L}^+(M)$ the n th congruence subgroup as every coconnective $\mathcal{M}$ is a filtered colimit of coconnective $\mathfrak{L}_n^+(M)$ -integrable modules. Therefore, we assume $\mathcal{M}$ is $\mathfrak{L}_n^+(M)$ -integrable to start.

Now for n fixed, it suffices to show that $\mathrm{Wak}(\mathcal{M})$ is eventually coconnective, since Corollary 3.6.7 will then force it to be coconnective as is.

3.11.2. Note that $\mathrm{Wak}(\mathcal{M}) \in \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G + \kappa}^{\mathfrak{L}_n^+(G) \cdot \mathfrak{L}^+(N_P)}$ under our assumption.

Therefore, showing that $\mathrm{Wak}(\mathcal{M})$ is eventually coconnective is equivalent to showing that

$$(3.29) \quad \mathcal{H}om(\mathbb{V}_{\mathrm{crit}_G + \kappa, n}, \mathrm{Wak}(\mathcal{M})) \in \mathrm{Vect}$$

is eventually coconnective, where $\mathbb{V}_{\mathrm{crit}_G + \kappa, n}$ is induced from the trivial module for $\mathfrak{L}_n^+(\mathfrak{g}) = t^n \mathfrak{g}[[t]]$.

3.11.3. We can rewrite (3.29) as

$$(3.30) \quad \langle \mathbb{V}_{\mathrm{crit}_G - \kappa, n}[n \cdot \dim \mathfrak{g}], \mathrm{Wak}(\mathcal{M}) \rangle_{\widehat{\mathfrak{g}}}$$

where $\langle -, \rangle_{\widehat{\mathfrak{g}}}$ is the pairing

$$\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G - \kappa} \otimes \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G + \kappa} \rightarrow \mathrm{Vect}.$$

By the definition of Wak , we can rewrite (3.30) up to a shift – which we can ignore by Sect. 3.11.1 – as

$$(3.31) \quad \langle J_{\mathrm{KM}}^-(\mathbb{V}_{\mathrm{crit}_G - \kappa, n}), \mathcal{M} \rangle_{\widehat{\mathfrak{m}}}.$$

3.11.4. We now make the following elementary observation:

Lemma 3.11.5. *Suppose $\mathfrak{h}$ is a Tate Lie algebra, $\mathfrak{k} \subset \mathfrak{h}$ is a subalgebra that is a lattice, and $\mathfrak{h}_0 \subset \mathfrak{h}$ is a Tate Lie subalgebra. Then for any $\mathcal{N} \in \mathfrak{k}\text{-mod}$, $\mathrm{ind}_{\mathfrak{k}}^{\mathfrak{h}}(\mathcal{N})$ has a canonical $\mathbf{Z}^{\geq 0}$ -indexed filtration*

$$\mathrm{fil}_{\bullet}(\mathrm{ind}_{\mathfrak{k}}^{\mathfrak{h}}(\mathcal{N}))$$

as an $\mathfrak{h}_0$ -module so that

$$\mathrm{gr}_i(\mathrm{ind}_{\mathfrak{k}}^{\mathfrak{h}}(\mathcal{N})) \simeq \mathrm{ind}_{\mathfrak{k} \cap \mathfrak{h}_0}^{\mathfrak{h}_0} \left(\mathrm{Sym}^i(\mathfrak{h}/(\mathfrak{k} + \mathfrak{h}_0)) \otimes \mathcal{N} \right) \in \mathfrak{h}_0\text{-mod}.$$

3.11.6. We apply the lemma to $\mathfrak{h} = \widehat{\mathfrak{g}}$, $\mathfrak{h}_0 = \widehat{\mathfrak{p}}^-$, $\mathfrak{k} = \mathfrak{L}_n^+(\mathfrak{g})$, and $\mathcal{N}$ being the trivial module. It follows that $\mathbb{V}_{\mathrm{crit}_G - \kappa, n}$ carries a filtration as a $\widehat{\mathfrak{p}}^-$ -module so that

$$\mathrm{gr}_{\bullet}(\mathbb{V}_{\mathrm{crit}_G - \kappa, n}) = \mathrm{ind}_{\mathfrak{L}_n^+(\mathfrak{p}^-)}^{\widehat{\mathfrak{p}}_{\mathrm{crit}_G - \kappa}} \left(\mathrm{Sym}(\mathfrak{L}(\mathfrak{g})/(\mathfrak{L}_n^+(\mathfrak{g}) + \mathfrak{L}(\mathfrak{p}^-))) \right).$$

Therefore, it suffices to prove that

$$(3.32) \quad \langle \mathrm{BRST}^-(\mathrm{ind}_{\mathfrak{L}_n^+(\mathfrak{p}^-)}^{\widehat{\mathfrak{p}}_{\mathrm{crit}_G - \kappa}}(\mathrm{Sym}(\mathfrak{L}(\mathfrak{g})/(\mathfrak{L}_n^+(\mathfrak{g}) + \mathfrak{L}(\mathfrak{p}^-))))), \mathcal{M} \rangle_{\widehat{\mathfrak{m}}}$$

is eventually coconnective.

3.11.7. By general properties of semi-infinite cohomology, we can compute (3.32) as

$$\mathrm{inv}_{\mathfrak{L}_n^+(\mathfrak{p}^-)} \left(\mathrm{Sym}(\mathfrak{L}(\mathfrak{g})/(\mathfrak{L}_n^+(\mathfrak{g}) + \mathfrak{L}(\mathfrak{p}^-))) \otimes \mathcal{M} \otimes \det(\mathfrak{L}^+(\mathfrak{p}^-)/\mathfrak{L}_n^+(\mathfrak{p}^-))^{\vee} \right)[- \dim \mathfrak{L}^+(\mathfrak{p}^-)/\mathfrak{L}_n^+(\mathfrak{p}^-)].$$

Because invariants is left t-exact, this complex is evidently bounded from below.

4. JACQUET FUNCTOR(S) FOR OPERS

In the present section we introduce generalized Jacquet functors (as well as their duals) that connect the $\mathrm{IndCoh}^*(-)$ categories on the spaces of monodromy-free opers for $\check{G}$ and $\check{M}$ to one another. Note, however, that the corresponding space of opers for $\check{M}$ is translated by the *Miura shift*, see Sect. 4.1.2.

The definition of these functors uses another geometric object that we call (parabolic) *Miura opers*.

4.1. (Parabolic) Miura opers.

4.1.1. *Translated opers.* Let $\check{G}$ be a reductive group, and let $\mathcal{P}_{Z_{\check{G}}^0}$ be a $Z_{\check{G}}^0$ -torsor on X . We let

$$\mathrm{Op}_{\check{G}, \mathcal{P}_{Z_{\check{G}}^0}}$$

be the following variant of the D-scheme $\mathrm{Op}_{\check{G}}$:

In the definition of opers (see [GLC2, Sect. 3.1.1]), instead of requiring that the induced $\check{T}$ -bundle be $\check{\rho}(\omega_X)$, we require that it be

$$(4.1) \quad \check{\rho}(\omega_X) \otimes \mathcal{P}_{Z_{\check{G}}^0}.$$

We let

$$\mathrm{Op}_{\check{G}, \mathcal{P}_{Z_{\check{G}}^0}}^{\mathrm{reg}} \subset \mathrm{Op}_{\check{G}, \mathcal{P}_{Z_{\check{G}}^0}}^{\mathrm{mon-free}} \subset \mathrm{Op}_{\check{G}, \mathcal{P}_{Z_{\check{G}}^0}}^{\mathrm{mer}}$$

denote the corresponding factorization spaces.²³

The material from [GLC2, Sect. 3.1] transfers verbatim to the present context.

4.1.2. We now take the reductive group in question to be $\check{M}$, the Levi subgroup of a standard parabolic $\check{P}$. We take

$$\mathcal{P}_{Z_{\check{M}}^0} := \check{\rho}_P(\omega_X).$$

Note that in this case, the $\check{T}$ -bundle (4.1) is

$$\check{\rho}_M(\omega_X) \otimes \check{\rho}_P(\omega_X) = \check{\rho}(\omega_X),$$

where $\check{\rho}$ in the right-hand side is the $\check{\rho}$ for $\check{G}$.

We will use a shorthand notation

$$\mathrm{Op}_{\check{M}, \check{\rho}_P} := \mathrm{Op}_{\check{M}, \check{\rho}_P(\omega_X)}.$$

4.1.3. *Example.* When $P = B$, we obtain that the scheme

$$\mathrm{Op}_{\check{T}, \check{\rho}}$$

that classifies connections on the $\check{T}$ -bundle $\check{\rho}(\omega_X)$.

4.1.4. Let $\mathrm{MOp}_{\check{G}, \check{P}^-}$ denote the D-scheme of $\check{P}^-$ -Miura opers, i.e.,

$$\mathrm{MOp}_{\check{G}, \check{P}^-} := (\mathrm{Op}_{\check{G}} \times_{\mathrm{LS}_{\check{G}}} \mathrm{LS}_{\check{P}^-})^{\mathrm{trans}},$$

where the superscript “trans” refers to the condition that the $\check{B}$ -reduction of the $\check{G}$ -bundles involved in the oper structure is transversal to the $\check{P}^-$ -reduction.²⁴

We have the natural forgetful map

$$(4.2) \quad \mathbf{p}^{\mathrm{Miu}} : \mathrm{MOp}_{\check{G}, \check{P}^-} \rightarrow \mathrm{Op}_{\check{G}}$$

4.1.5. Note also that we have a map

$$(4.3) \quad \mathbf{q}^{\mathrm{Miu}} : \mathrm{MOp}_{\check{G}, \check{P}^-} \rightarrow \mathrm{Op}_{\check{M}, \check{\rho}_P},$$

constructed as follows:

The $\check{M}$ -bundle and the connection are induced from the $\check{P}^-$ -bundle with its connection. The reduction to $\check{B}(M)$ comes from the reduction of the original $\check{G}$ -bundle to $\check{B}$.

²³As in [GLC2, Sect. 3.1.3], we distinguish notationally between the D-scheme $\mathrm{Op}_{\check{G}, \mathcal{P}_{Z_{\check{G}}^0}}$ and the corresponding factorization space $\mathrm{Op}_{\check{G}, \mathcal{P}_{Z_{\check{G}}^0}}^{\mathrm{reg}}$.

²⁴Elsewhere in the literature, what we call “Miura opers” is sometimes referred to as “generic Miura opers”.

4.1.6. The following assertion is fundamental, albeit immediate:

Lemma 4.1.7. *The map (4.3) is an isomorphism.*

Remark 4.1.8. The composition

$$\mathfrak{p}^{\text{Miu}} \circ (\mathfrak{q}^{\text{Miu}})^{-1}$$

is a map

$$\text{Op}_{\tilde{M}, \tilde{\rho}_P} \rightarrow \text{Op}_{\tilde{G}},$$

called the (parabolic) Miura transform.

Remark 4.1.9. The assertion of Lemma 4.1.7 has the following classical counterpart: let

$$\mathfrak{b} = \text{Fil}_0(\mathfrak{g}) \subset \text{Fil}_{-1}(\mathfrak{g}) \subset \mathfrak{g}$$

be the corresponding piece of the principal filtration. Then

$$(\text{Fil}_{-1}(\mathfrak{g})/\text{Ad}(B) \times_{\mathfrak{g}/\text{Ad}(G)} \mathfrak{p}^-/\text{Ad}(P^-)) \times_{\text{pt}/B \times_{\text{pt}/G} \text{pt}/P^-} (\text{pt}/B \times_{\text{pt}/G} \text{pt}/P^-)^{\text{trans}},$$

projects isomorphically to

$$\text{Fil}_{-1}(\mathfrak{m})/\text{Ad}(B_M),$$

where

$$(\text{pt}/B \times_{\text{pt}/G} \text{pt}/P^-)^{\text{trans}} \subset \text{pt}/B \times_{\text{pt}/G} \text{pt}/P^-$$

is the transversal locus, i.e., one corresponding to the open $B \times P^-$ -orbit acting on G .

4.2. Definition of the Jacquet functor for opers. In this subsection we introduce the unenhanced version of the Jacquet functor for opers. The definition will be simple in that it does not require $\text{IndCoh}^*(-)$ of algebro-geometric objects of semi-infinite nature. The latter will appear, after some preparations, in the definition of the enhanced version, in Sect. 4.6.

4.2.1. Consider the fiber product

$$(4.4) \quad \text{MOP}_{\tilde{G}, \tilde{P}^-}^{\text{mon-free } \tilde{P}^-} := \text{MOP}_{\tilde{G}, \tilde{P}^-}^{\text{mer}} \times_{\text{LS}_{\tilde{P}^-}^{\text{mer}}} \text{LS}_{\tilde{P}^-}^{\text{reg}}.$$

Note that the maps

$$\text{Op}_{\tilde{G}}^{\text{mer}} \leftarrow \text{MOP}_{\tilde{G}, \tilde{P}^-}^{\text{mon-free } \tilde{P}^-} \rightarrow \text{Op}_{\tilde{M}, \tilde{\rho}_P}^{\text{mer}},$$

induced by $\mathfrak{p}^{\text{Miu}}$ and $\mathfrak{q}^{\text{Miu}}$, respectively, naturally factor via maps

$$(4.5) \quad \text{Op}_{\tilde{G}}^{\text{mon-free}} \xleftarrow{\mathfrak{p}^{\text{Miu, mon-free}}} \text{MOP}_{\tilde{G}, \tilde{P}^-}^{\text{mon-free } \tilde{P}^-} \xrightarrow{\mathfrak{q}^{\text{Miu, mon-free}}} \text{Op}_{\tilde{M}, \tilde{\rho}_P}^{\text{mon-free}}.$$

4.2.2. Recall that according to [GLC2, Sect. 3.2.6], $\text{MOP}_{\tilde{G}, \tilde{P}^-}^{\text{mer}} \simeq \text{Op}_{\tilde{M}, \tilde{\rho}_P}^{\text{mer}}$ is an ind-placid ind-affine factorization ind-scheme.

In addition, the map

$$\text{LS}_{\tilde{P}^-}^{\text{reg}} \rightarrow \text{LS}_{\tilde{P}^-}^{\text{mer}}$$

is ind-affine, locally almost of finite presentation (see [GLC2, Lemma B.7.13]). Hence, $\text{MOP}_{\tilde{G}, \tilde{P}^-}^{\text{mon-free } \tilde{P}^-}$ is also an ind-placid ind-affine factorization ind-scheme.

In particular, we have well-defined mutually dual unital factorization categories

$$\text{IndCoh}^!(\text{MOP}_{\tilde{G}, \tilde{P}^-}^{\text{mon-free } \tilde{P}^-}) \text{ and } \text{IndCoh}^*(\text{MOP}_{\tilde{G}, \tilde{P}^-}^{\text{mon-free } \tilde{P}^-}),$$

see [GLC2, Sect. B.13.16-21]. And we have a pair of mutually dual (lax unital) factorization functors

$$(\mathfrak{p}^{\text{Miu, mon-free}})^! : \text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free}}) \rightarrow \text{IndCoh}^!(\text{MOP}_{\tilde{G}, \tilde{P}^-}^{\text{mon-free } \tilde{P}^-})$$

and

$$(\mathfrak{p}^{\text{Miu, mon-free}})_*^{\text{IndCoh}} : \text{IndCoh}^*(\text{MOP}_{\tilde{G}, \tilde{P}^-}^{\text{mon-free } \tilde{P}^-}) \rightarrow \text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}}).$$

Furthermore, the morphism $\mathbf{q}^{\text{Miu, mon-free}}$ is an ind-closed embedding,²⁵ locally almost of finite presentation. Hence, by [GLC2, Sect. A.10.11], we have a pair of mutually dual (lax unital) factorization functors

$$(\mathbf{q}^{\text{Miu, mon-free}})_*^{\text{IndCoh}} : \text{IndCoh}^*(\text{MOp}_{\tilde{G}, \tilde{P}^-}^{\text{mon-free}}) \rightarrow \text{IndCoh}^*(\text{Op}_{M, \tilde{P}}^{\text{mon-free}})$$

and

$$(\mathbf{q}^{\text{Miu, mon-free}})^! : \text{IndCoh}^!(\text{Op}_{M, \tilde{P}}^{\text{mon-free}}) \rightarrow \text{IndCoh}^!(\text{MOp}_{\tilde{G}, \tilde{P}^-}^{\text{mon-free}}).$$

4.2.3. Let

$$J_{\text{Op}}^{-, \text{spec}} : \text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free}}) \rightarrow \text{IndCoh}^!(\text{Op}_{M, \tilde{P}}^{\text{mon-free}})$$

denote the functor

$$(\mathbf{q}^{\text{Miu, mon-free}})_* \circ (\mathbf{p}^{\text{Miu, mon-free}})^!.$$

4.2.4. Denote by $J_{\text{Op}, \Theta}^{-, \text{spec}}$ the functor

$$\text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}}) \rightarrow \text{IndCoh}^*(\text{Op}_{M, \tilde{P}}^{\text{mon-free}}),$$

so that we have a commutative diagram

$$\begin{array}{ccc} \text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free}}) & \xrightarrow{J_{\text{Op}}^{-, \text{spec}}} & \text{IndCoh}^!(\text{Op}_{M, \tilde{P}}^{\text{mon-free}}) \\ \Theta_{\text{Op}_{\tilde{G}}^{\text{mon-free}}} \downarrow \sim & & \sim \downarrow \Theta_{\text{Op}_M^{\text{mon-free}}} \\ \text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}}) & \xrightarrow{J_{\text{Op}, \Theta}^{-, \text{spec}}} & \text{IndCoh}^*(\text{Op}_{M, \tilde{P}}^{\text{mon-free}}), \end{array}$$

where $\Theta_{\text{Op}_{\tilde{G}}^{\text{mon-free}}}$ and $\Theta_{\text{Op}_M^{\text{mon-free}}}$ are the identifications of [GLC2, Sect. 3.7.7].

4.2.5. We will refer to $J_{\text{Op}}^{-, \text{spec}}$ (and also $J_{\text{Op}, \Theta}^{-, \text{spec}}$) as the *spectral Jacquet functor*. The main theorem in Part I of the paper will establish its relationship with the functor J_{KM}^- at the critical level.

4.2.6. We let

$$\text{co}J_{\text{Op}}^{-, \text{spec}} : \text{IndCoh}^*(\text{Op}_{M, \tilde{P}}^{\text{mon-free}}) \rightarrow \text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}})$$

the functor dual to $J_{\text{Op}}^{-, \text{spec}}$.

Explicitly, it is given by

$$\text{co}J_{\text{Op}}^{-, \text{spec}} = (\mathbf{p}^{\text{Miu, mon-free}})_*^{\text{IndCoh}} \circ (\mathbf{q}^{\text{Miu, mon-free}})^!.$$

4.3. **The *semi-infinite* Jacquet functor for opers, pointwise version.** In this subsection we start exploring the semi-infinite Jacquet functor for opers.

Ultimately, we will need a factorization version of this functor. However, by way of motivation, in this subsection we will work over a fixed point $\underline{x} \in \text{Ran}$, and we will be able to define a version of the Jacquet functor more general than in the factorization situation.

²⁵Namely, the morphism in [GLC2, Lemma B.7.13] is an ind-closed embedding if $\mathbf{H}$ is unipotent

4.3.1. Consider the (non-affine) D-scheme

$$\mathrm{Op}_{\check{G}, \check{P}^-} := \mathrm{Op}_{\check{G}} \times_{\mathrm{pt}/\check{G}} \mathrm{pt}/\check{P}^-$$

over X .

Consider the corresponding factorization spaces:

$$\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{reg}} := \mathrm{Op}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{reg}};$$

$$\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mer}} := \mathrm{Op}_{\check{G}}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}};$$

$$\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-} := \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{reg}};$$

$$\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}} := \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{M}}^{\mathrm{mer}}} \mathrm{LS}_{\check{M}}^{\mathrm{reg}};$$

$$\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{G}} := \mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mer}}$$

and

$$\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{G}, \check{M}} := \mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{M}}^{\mathrm{mer}}} \mathrm{LS}_{\check{M}}^{\mathrm{reg}}.$$

Note that the latter space can also be written as

$$\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Hecke}_{\check{G}, \check{P}^-}^{\mathrm{spec}, \mathrm{loc}}.$$

4.3.2. Consider the following maps:

- Let j denote the map

$$\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mer}} \rightarrow \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mer}},$$

and also the maps

$$\mathrm{Op}_{\check{M}, \check{P}^-}^{\mathrm{mon-free} \check{q}^{\mathrm{Miu}}} \simeq \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}} := \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{M}}^{\mathrm{mer}}} \mathrm{LS}_{\check{M}}^{\mathrm{reg}} \rightarrow \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}$$

and

$$\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-} := \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{reg}} \rightarrow \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}$$

obtained by base change;

- Let $\mathrm{p}^{-, \mathrm{spec}}$ denote the map

$$\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}} \rightarrow \mathrm{LS}_{\check{G}}^{\mathrm{reg}},$$

and also its base change by means of $\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} \rightarrow \mathrm{LS}_{\check{G}}^{\mathrm{reg}}$, which is the map

$$\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-} \rightarrow \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}};$$

- ι denote the map

$$\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}} \rightarrow \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{M}}^{\mathrm{mer}}} \mathrm{LS}_{\check{M}}^{\mathrm{reg}} =: \mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free} \check{M}},$$

and also the maps

$$(4.6) \quad \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-} \rightarrow \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}.$$

and

$$\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-} \rightarrow \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}},$$

obtained by base change.

Remark 4.3.3. Note that although the map

$$\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{reg}} \rightarrow \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{reg}}$$

is an open embedding, this is *no longer the case* for the three maps denoted j above.

4.3.4. Note that we have a Cartesian diagram

$$(4.7) \quad \begin{array}{ccccc} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-} & \xrightarrow{\iota} & \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}} & \xrightarrow[\sim]{q^{\mathrm{Miu}, \mathrm{mon-free}}} & \mathrm{Op}_{\check{M}, \check{\rho}_P}^{\mathrm{mon-free}} \\ j \downarrow & & \downarrow j & & \\ \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-} & \xrightarrow{\iota} & \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}} & & \\ \mathfrak{p}^-, \mathrm{spec} \downarrow & & & & \\ \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} & & & & \end{array}$$

The composite vertical arrow in (4.7) is the map $\mathfrak{p}^{\mathrm{Miu}, \mathrm{mon-free}}$.

4.3.5. Since

$$\mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}} \simeq (\check{\mathfrak{n}}^- / \mathrm{Ad}(\check{P}^-))^{\times |\underline{x}|},$$

it is easy to see that $\mathrm{Op}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}}$ is an ind-scheme that admits a finite open cover by ind-affine ind-placid ind-schemes.

Hence, by [GLC2, Sect. A.10.12], the functor

$$\iota_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{P}^-}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}})$$

is well-defined, and we have an isomorphism

$$(4.8) \quad (q^{\mathrm{Miu}, \mathrm{mon-free}})_*^{\mathrm{IndCoh}} \circ j^! \simeq j^! \circ \iota_*^{\mathrm{IndCoh}},$$

as functors

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{P}^-}) \rightrightarrows \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}}).$$

4.3.6. We let

$$J_{\mathrm{Op}}^{-, \mathrm{spec}, \frac{\infty}{2}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M}, \check{\rho}_P, \underline{x}}^{\mathrm{mon-free}})$$

be the functor

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}}) \xrightarrow{j^!} \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M}, \check{\rho}_P, \underline{x}}^{\mathrm{mon-free}}) \xrightarrow{(q^{\mathrm{Miu}, \mathrm{mon-free}})_*^{\mathrm{IndCoh}} \simeq} \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M}, \check{\rho}_P, \underline{x}}^{\mathrm{mon-free}}),$$

and refer to it as the *semi-infinite* Jacquet functor for opers.

By (4.8), we have:

$$(4.9) \quad J_{\mathrm{Op}}^{-, \mathrm{spec}} \simeq J_{\mathrm{Op}}^{-, \mathrm{spec}, \frac{\infty}{2}} \circ \iota_*^{\mathrm{IndCoh}} \circ (\mathfrak{p}^-, \mathrm{spec})^!.$$

Remark 4.3.7. Generalizing [FG2], we conjecture²⁶ that the category $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}})$ is equivalent to the category

$$((\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G})_{\mathfrak{L}(N_P^-)}^{\mathfrak{L}^+(M)})_{\underline{x}},$$

so that the following diagram commutes:

$$\begin{array}{ccc} ((\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G})_{\mathfrak{L}(N_P^-)}^{\mathfrak{L}^+(M)})_{\underline{x}} & \longrightarrow & \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}}) \\ \overline{J}_{\mathrm{KM}} \downarrow & & \downarrow J_{\mathrm{Op}}^{-, \mathrm{spec}, \frac{\infty}{2}} \\ \mathrm{KL}(M)_{\mathrm{crit}_M + \check{\rho}_P, \underline{x}} & \xrightarrow{\Theta_{\mathrm{Op}_{\check{M}}^{\mathrm{mon-free}} \circ \mathrm{FLE}_{M, \mathrm{crit}_M}}^{-1}} & \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M}, \check{\rho}_P, \underline{x}}^{\mathrm{mon-free}}). \end{array}$$

²⁶In fact, we believe that this conjecture can be proved by the methods of *loc. cit.*

4.3.8. However, we do *not* know how to make sense of

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}})$$

in the factorization setting.

What we will do instead is replace $\mathrm{Op}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}}$ by its formal completion along the map ι of (4.6). This will place us in a context where we can make sense of the categories involved in the factorization setting.

4.4. **The semi-infinite category for opers.** Let

$$(\mathrm{Op}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}}$$

be the formal completion of $\mathrm{Op}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}}$ along the map ι of (4.6).

The goal of this subsection is to define unital factorization categories

$$\mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}}) \text{ and } \mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}}).$$

We will do so by mimicking the procedure in Sect. A.6.

4.4.1. Consider the fiber product

$$(4.10) \quad \mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}.$$

This is a factorization ind-affine ind-scheme. It is *not* ind-placid. Yet, we claim that the factorization categories

$$(4.11) \quad \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \text{ and } \mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$$

are well-defined and mutually dual.

Indeed, this follows by the same principle as in [GLC2, Sect. E.2.11]. Namely, by construction, (4.10) carries an action of $\mathfrak{L}_{\check{\nabla}}^+(\check{G})$, so that

$$\mathfrak{L}_{\check{\nabla}}^+(\check{G}) \backslash (\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \simeq \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}},$$

while

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \text{ and } \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$$

are well-defined and mutually dual, since $\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}$ is placid.

4.4.2. Note now that we can rewrite (4.10) also as

$$\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}}^{\mathrm{mer}}.$$

Hence, we obtain that the above $\mathfrak{L}_{\check{\nabla}}^+(\check{G})$ -action on (4.10) extends to an action of $\mathfrak{L}_{\check{\nabla}}(\check{G})$. From here we obtain an action of $\mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}(\check{G}))$ on $\mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$ and a coaction of $\mathrm{IndCoh}^!(\mathfrak{L}_{\check{\nabla}}(\check{G}))$ on $\mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$.

In this way we obtain an action of

$$\mathrm{Sph}_{\check{G}}^{\mathrm{spec}} := \mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}^+(\check{G}) \backslash \mathfrak{L}_{\check{\nabla}}(\check{G}) / \mathfrak{L}_{\check{\nabla}}^+(\check{G}))$$

on $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$ and a coaction of

$$(\mathrm{Sph}_{\check{G}}^{\mathrm{spec}})^{\vee} \simeq \mathrm{IndCoh}^!(\mathfrak{L}_{\check{\nabla}}^+(\check{G}) \backslash \mathfrak{L}_{\check{\nabla}}(\check{G}) / \mathfrak{L}_{\check{\nabla}}^+(\check{G}))$$

on $\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$; this is what the construction from [GLC2, Sect. E.8] amounts to.

The above structures on the $\mathrm{IndCoh}^*(-)$ and $\mathrm{IndCoh}^!(-)$ versions are mutually dual in the natural sense.

4.4.3. Recall the factorization space

$$\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{G}, \check{M}}} = \mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}},$$

and consider the fiber product

$$\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{G}, \check{M}}} \simeq \mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}};$$

this is a factorization space acted on by $\mathfrak{L}_{\nabla}(\check{G})$.

Note that we can identify

$$\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}} \simeq \left(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} \right) \times \left(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_{\check{P}}^-)/\mathfrak{L}_{\nabla}^+(\check{M}) \right).$$

Hence, we can identify $(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}})_{\mathrm{mon-free}}^{\wedge}$ with the étale quotient

$$\left(\left(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} \right) \times \left(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_{\check{P}}^-)/\mathfrak{L}_{\nabla}^+(\check{M}) \right) \right) / \mathfrak{L}_{\nabla}(\check{G}).$$

We will use this presentation to access the corresponding $\mathrm{IndCoh}^*(-)$ and $\mathrm{IndCoh}^!(-)$ categories, cf. Sect. A.6.1

4.4.4. By the above, we have a well-defined (unital) factorization category

$$\begin{aligned} \mathrm{IndCoh}^* \left(\left(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} \right) \times \left(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_{\check{P}}^-)/\mathfrak{L}_{\nabla}^+(\check{M}) \right) \right) &:= \\ &:= \mathrm{IndCoh}^* \left(\left(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} \right) \right) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{IndCoh}^* \left(\left(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_{\check{P}}^-)/\mathfrak{L}_{\nabla}^+(\check{M}) \right) \right), \end{aligned}$$

equipped with the (diagonal) action of the monoidal factorization category $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))$.

We set

$$\begin{aligned} (4.12) \quad \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}})_{\mathrm{mon-free}}^{\wedge}) &:= \\ &:= \mathrm{IndCoh}^* \left(\left(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} \right) \times \left(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_{\check{P}}^-)/\mathfrak{L}_{\nabla}^+(\check{M}) \right) \right) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect}. \end{aligned}$$

Similarly, we have a well-defined (unital factorization) category

$$\begin{aligned} \mathrm{IndCoh}^! \left(\left(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} \right) \times \left(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_{\check{P}}^-)/\mathfrak{L}_{\nabla}^+(\check{M}) \right) \right) &:= \\ &:= \mathrm{IndCoh}^! \left(\left(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} \right) \right) \otimes_{\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{IndCoh}^! \left(\left(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_{\check{P}}^-)/\mathfrak{L}_{\nabla}^+(\check{M}) \right) \right), \end{aligned}$$

equipped with a coaction of the comonoidal factorization category $\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}))$.

We set

$$\begin{aligned} (4.13) \quad \mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}})_{\mathrm{mon-free}}^{\wedge}) &:= \\ &:= \mathrm{IndCoh}^! \left(\left(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} \right) \times \left(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_{\check{P}}^-)/\mathfrak{L}_{\nabla}^+(\check{M}) \right) \right) \otimes_{\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect} \end{aligned}$$

(see Sect. 1.4.10 for the $\otimes$ notation).

By construction, $\mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge)$ and $\mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge)$ are mutually dual unitary factorization categories.

4.4.5. Note that we can also write

$$(4.14) \quad \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge) \simeq \\ \simeq \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{reg}}} \mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(\check{G}))} \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(\check{G}/\check{N}_P^-))_{\mathfrak{L}_\nabla^+(\check{M})}$$

and

$$(4.15) \quad \mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge) \simeq \\ \simeq \mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{reg}}} \mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^!(\mathfrak{L}_\nabla(\check{G}))} \mathrm{IndCoh}^!(\mathfrak{L}_\nabla(\check{G}/\check{N}_P^-))_{\mathfrak{L}_\nabla^+(\check{M})}.$$

And we can rewrite further:

$$(4.16) \quad \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge) \simeq \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$$

and

$$(4.17) \quad \mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge) \simeq \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}}^{\mathrm{Sph}_G^{\mathrm{spec}} \vee} \mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}}^{\mathrm{spec}, \mathrm{loc}},$$

see Sect. A.6.3.

4.4.6. In particular, we have the (strictly unital) factorization functor

$$(4.18) \quad \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \xrightarrow{\mathrm{Id} \otimes \mathbf{1}_{\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \rightarrow \\ \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \simeq \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge)$$

which admits a right adjoint as a factorization functor, namely

$$(4.19) \quad \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge) = \\ = \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \xrightarrow{\mathrm{Id} \otimes (- \star_G^{\Delta^-, \mathrm{spec}, \frac{\infty}{2}})^R} \\ \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{Sph}_G^{\mathrm{spec}} \simeq \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}),$$

where $(- \star_G^{\Delta^-, \mathrm{spec}, \frac{\infty}{2}})^R$ is the right adjoint of the functor

$$\mathrm{Sph}_G^{\mathrm{spec}} \xrightarrow{- \star_G^{\Delta^-, \mathrm{spec}, \frac{\infty}{2}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}.$$

4.4.7. The dual of (4.19) is a (strictly unital) factorization functor

$$(4.20) \quad \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge).$$

Note that the functor (4.20) can also be interpreted as

$$(4.21) \quad \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \xrightarrow{\mathrm{Id} \otimes \mathbf{1}_{\mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}}^{\mathrm{spec}, \mathrm{loc}}}} \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes \mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}}^{\mathrm{spec}, \mathrm{loc}} \rightarrow \\ \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}}^{\mathrm{Sph}_G^{\mathrm{spec}} \vee} \mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}}^{\mathrm{spec}, \mathrm{loc}} \simeq \mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge),$$

where the second arrow is the left adjoint of the forgetful functor

$$\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})^{(\mathrm{Sph}_G^{\mathrm{spec}})^\vee} \otimes \mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}}^{\mathrm{spec}, \mathrm{loc}} \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes \mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}}^{\mathrm{spec}, \mathrm{loc}},$$

which is well-defined as a factorization functor, since $\mathrm{Sph}_G^{\mathrm{spec}}$ is rigid.

4.4.8. Consider now the fiber product

$$\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-} \simeq \mathrm{Op}_G^{\mathrm{mon-free}} \times_{\mathrm{LS}_G^{\mathrm{reg}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}.$$

By the same principle as in Sect. 4.4.1, we have well-defined unital factorization categories

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) \text{ and } \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}).$$

Namely,

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) := \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{reg}}} \mathrm{Op}_G^{\mathrm{mon-free}})_{\mathfrak{L}_\nabla^+(\check{P}^-)}$$

and

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) := \mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{reg}}} \mathrm{Op}_G^{\mathrm{mon-free}})_{\mathfrak{L}_\nabla^+(\check{P}^-)}$$

We have the naturally defined factorization functors

$$(\mathfrak{p}^{-, \mathrm{spec}})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$$

and

$$(\mathfrak{p}^{-, \mathrm{spec}})^! : \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}).$$

Moreover, the functor $(\mathfrak{p}^{-, \mathrm{spec}})_*^{\mathrm{IndCoh}}$ admits a left adjoint as a factorization functor

$$(\mathfrak{p}^{-, \mathrm{spec}})^* : \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}).$$

The above functors carry lax unital structures, and for $(\mathfrak{p}^{-, \mathrm{spec}})^*$ and $(\mathfrak{p}^{-, \mathrm{spec}})^!$ this structure is strict.

4.4.9. Let ι denote the map

$$\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-} \rightarrow (\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge.$$

We claim that we have adjoint pairs

$$(4.22) \quad (\iota)_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) \rightleftarrows \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge) : (\iota)^!$$

and

$$(4.23) \quad (\iota)_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) \rightleftarrows \mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge) : (\iota)^!$$

Namely, we interpret the functor $(\iota)_*^{\mathrm{IndCoh}}$ in (4.22) as

$$\begin{aligned} \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{reg}}} \mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{IndCoh}^*(\mathfrak{L}_\nabla^+(\check{P}^-))} &\otimes \mathrm{Vect} \simeq \\ &\simeq \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{reg}}} \mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{IndCoh}^*(\mathfrak{L}_\nabla^+(\check{G}))} \otimes \mathrm{IndCoh}^*(\mathfrak{L}_\nabla^+(G/\check{N}_P^-))_{\mathfrak{L}_\nabla^+(\check{M})} \rightarrow \\ &\rightarrow \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{reg}}} \mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{IndCoh}^*(\mathfrak{L}_\nabla^+(\check{G}))} \otimes \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(G/\check{N}_P^-))_{\mathfrak{L}_\nabla^+(\check{M})} \rightarrow \\ &\rightarrow \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{reg}}} \mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{IndCoh}^*(\mathfrak{L}_\nabla^+(\check{G}))} \otimes \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(G/\check{N}_P^-))_{\mathfrak{L}_\nabla^+(\check{M})}, \end{aligned}$$

and it admits a right adjoint as a factorization functor by Sect. A.6.5.

The functors in (4.23) are set to be the duals of ones in (4.22).

4.4.10. Unwinding the definitions, we obtain that the functor (4.18) identifies with

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{(\mathrm{p}^-, \mathrm{spec})^*} \xrightarrow{\quad} \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}})^{(\iota)_*^{\mathrm{IndCoh}}} \xrightarrow{\quad} \mathrm{IndCoh}^*((\mathrm{Op}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}}),$$

and hence its right adjoint, i.e., the functor (4.19), identifies with

$$\mathrm{IndCoh}^*((\mathrm{Op}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}})^{(\iota)^!} \xrightarrow{\quad} \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}})^{(\mathrm{p}^-, \mathrm{spec})^*} \xrightarrow{\quad} \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}).$$

Furthermore, the functor (4.20) identifies with

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{(\mathrm{p}^-, \mathrm{spec})^!} \xrightarrow{\quad} \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}})^{(\iota)_*^{\mathrm{IndCoh}}} \xrightarrow{\quad} \mathrm{IndCoh}^!((\mathrm{Op}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}}).$$

4.5. A formal completion of Miura opers. The semi-infinite Jacquet functor for opers goes via the category of ind-coherent sheaves on the formal completion of the space of Miura opers along the monodromy-free locus.

We study this category in the present subsection.

4.5.1. Let

$$(\mathrm{MOp}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}}$$

denote the formal completion of

$$\mathrm{Op}_{\tilde{M}, \tilde{\rho}_P}^{\mathrm{mon-free}} \simeq \mathrm{MOp}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}} := \mathrm{MOp}_{\tilde{G}, \tilde{P}-}^{\mathrm{mer}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{reg}}$$

along the map

$$(4.24) \quad \mathrm{MOp}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}} \xrightarrow{\mathrm{q}^{\mathrm{Miu}, \mathrm{mon-free}}} \mathrm{Op}_{\tilde{M}, \tilde{\rho}_P}^{\mathrm{mon-free}},$$

i.e., the map

$$\mathrm{MOp}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}} \rightarrow \mathrm{MOp}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}}.$$

Note that the image of the above map coincides set-theoretically with the preimage of

$$(4.25) \quad \mathrm{LS}_G^{\mathrm{reg}} \subset \mathrm{LS}_G^{\mathrm{mer}}$$

along the map

$$(4.26) \quad \mathrm{MOp}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}} \rightarrow \mathrm{MOp}_{\tilde{G}, \tilde{P}-}^{\mathrm{mer}} \xrightarrow{\mathrm{p}^{\mathrm{Miu}}} \mathrm{Op}_{\tilde{G}}^{\mathrm{mer}} \rightarrow \mathrm{LS}_G^{\mathrm{mer}}.$$

4.5.2. Let $(\mathrm{q}^{\mathrm{Miu}, \mathrm{mon-free}})^{\wedge}$ denote the ind-closed embedding

$$(\mathrm{MOp}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}} \rightarrow \mathrm{Op}_{\tilde{M}, \tilde{\rho}_P}^{\mathrm{mon-free}}.$$

Since (4.25) is locally almost of finite presentation (see [GLC2, Lemma B.7.13]), we obtain that the map $(\mathrm{q}^{\mathrm{Miu}, \mathrm{mon-free}})^{\wedge}$ is also locally almost of finite presentation.

In particular, since $\mathrm{Op}_{\tilde{M}, \tilde{\rho}_P}^{\mathrm{mon-free}}$ is ind-placid, we obtain that $(\mathrm{MOp}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}}$ is also ind-placid (see [GLC2, Corollary A.9.10]). Hence, we have well-defined unital factorization categories

$$(4.27) \quad \mathrm{IndCoh}^*((\mathrm{MOp}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}}) \text{ and } \mathrm{IndCoh}^!((\mathrm{MOp}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}})$$

and pairs of adjoint functors:

$$(4.28) \quad ((\mathrm{q}^{\mathrm{Miu}, \mathrm{mon-free}})^{\wedge})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*((\mathrm{MOp}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}}) \rightleftarrows \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{M}, \tilde{\rho}_P}^{\mathrm{mon-free}}) : ((\mathrm{q}^{\mathrm{Miu}, \mathrm{mon-free}})^{\wedge})^!$$

and

$$(4.29) \quad ((\mathrm{q}^{\mathrm{Miu}, \mathrm{mon-free}})^{\wedge})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^!((\mathrm{MOp}_{\tilde{G}, \tilde{P}-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}}) \rightleftarrows \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{M}, \tilde{\rho}_P}^{\mathrm{mon-free}}) : ((\mathrm{q}^{\mathrm{Miu}, \mathrm{mon-free}})^{\wedge})^!$$

with the left adjoints being strictly unital.

4.5.3. Let ι denote the map

$$\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-} \rightarrow (\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^{\wedge}.$$

We obtain that the adjunctions

$$(\mathbf{q}^{\mathrm{Miu}, \mathrm{mon-free}})_{*}^{\mathrm{IndCoh}} : \mathrm{IndCoh}^{*}(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) \rightleftarrows \mathrm{IndCoh}^{*}(\mathrm{Op}_{\check{M}, \check{P}^-}^{\mathrm{mon-free}}) : (\mathbf{q}^{\mathrm{Miu}, \mathrm{mon-free}})^{!}$$

and

$$(\mathbf{q}^{\mathrm{Miu}, \mathrm{mon-free}})^{!}_{*}{}^{\mathrm{IndCoh}} : \mathrm{IndCoh}^{!}(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) \rightleftarrows \mathrm{IndCoh}^{!}(\mathrm{Op}_{\check{M}, \check{P}^-}^{\mathrm{mon-free}}) : (\mathbf{q}^{\mathrm{Miu}, \mathrm{mon-free}})^{!}$$

factor as compositions of

$$(4.30) \quad \iota_{*}^{\mathrm{IndCoh}} : \mathrm{IndCoh}^{*}(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) \rightleftarrows \mathrm{IndCoh}^{*}((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^{\wedge}) : \iota^{!}$$

and

$$(4.31) \quad \iota_{*}^{\mathrm{IndCoh}} : \mathrm{IndCoh}^{!}(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) \rightleftarrows \mathrm{IndCoh}^{!}((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^{\wedge}) : \iota^{!}$$

with the adjunctions (4.28) and (4.29), respectively.

4.5.4. We now claim that the categories (4.27) admit descriptions similar to those in (4.14) and (4.15), respectively.

Namely, recall the map (4.26), and consider the fiber products

$$\mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}} \quad \text{and} \quad \mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}.$$

Both are ind-affine factorization ind-schemes, with $\mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}$ being placid, so that

$$\mathrm{IndCoh}^{*}(\mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}) \quad \text{and} \quad \mathrm{IndCoh}^{!}(\mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})$$

are well-defined as (mutually dual) unital factorization categories.

The ind-scheme $\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}$ carries an action of $\mathfrak{L}_{\nabla}(\check{G})$ and

$$\mathfrak{L}_{\nabla}^{+}(\check{G}) \backslash (\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}) \simeq \mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}.$$

Hence, by the same mechanism as in Sect. 4.4.1, the categories

$$\mathrm{IndCoh}^{*}(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}) \quad \text{and} \quad \mathrm{IndCoh}^{!}(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})$$

are well-defined as (mutually dual) unital factorization categories.

Moreover, $\mathrm{IndCoh}^{*}(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})$ carries an action of the monoidal factorization category $\mathrm{IndCoh}^{*}(\mathfrak{L}_{\nabla}(\check{G}))$, and $\mathrm{IndCoh}^{!}(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})$ carries a coaction of the comonoidal factorization category $\mathrm{IndCoh}^{!}(\mathfrak{L}_{\nabla}(\check{G}))$.

4.5.5. Let $''l$ denote the map

$$\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{G}, \check{M}}} := \mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}} \rightarrow (\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}})_{\mathrm{mon-free}}^{\wedge}.$$

We have the adjunctions

$$(''l)_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{G}, \check{M}}}) \rightleftarrows \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}})_{\mathrm{mon-free}}^{\wedge}) : (''l)^!$$

and

$$(''l)_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^!(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{G}, \check{M}}}) \rightleftarrows \mathrm{IndCoh}^!((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}})_{\mathrm{mon-free}}^{\wedge}) : (''l)^!.$$

Note now that the functor

$$(4.32) \quad \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}}) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\check{G}))} \mathrm{Vect} \simeq \\ \simeq \mathrm{IndCoh}^*(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{G}, \check{M}}}) \xrightarrow{(''l)_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}})_{\mathrm{mon-free}}^{\wedge})$$

naturally factors as

$$\mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}}) \rightarrow \\ \rightarrow \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}}) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect} \rightarrow \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}})_{\mathrm{mon-free}}^{\wedge}).$$

4.5.6. We claim:

Proposition 4.5.7. *The functor*

$$(4.33) \quad \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}}) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect} \rightarrow \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}})_{\mathrm{mon-free}}^{\wedge})$$

is an equivalence.

Proof. First, we claim that the functor (4.33) preserves compactness. To prove this, we note that its precomposition with

$$(4.34) \quad \mathrm{IndCoh}^*(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{G}, \check{M}}}) \simeq \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}}) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\check{G}))} \mathrm{Vect} \rightarrow \\ \rightarrow \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}}) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect},$$

which is the functor

$$(''l)_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{G}, \check{M}}}) \rightarrow \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}_{\check{M}}})_{\mathrm{mon-free}}^{\wedge})$$

preserves compactness.

However, the functor (4.34) itself also preserves compactness, and its essential image generates the target.

Hence, by [GLC2, Proposition 6.2.6], it suffices to show that the functor (4.33) is a pointwise equivalence (i.e., induces an equivalence over each $\underline{x} \in \mathrm{Ran}$).

We rewrite the resulting functor

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{\rho}_P, \underline{x}}^{\mathrm{mon-free}} \times_{\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}}} \mathrm{pt}) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G})_{\underline{x}})} \mathrm{Vect} \rightarrow \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{mon-free}})_{\mathrm{mon-free}}^{\wedge})$$

as

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{\rho}_P, \underline{x}}^{\mathrm{mon-free}} \times_{\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{reg}}) \otimes_{\mathrm{Sph}_{\check{G}, \underline{x}}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_{\underline{x}} \rightarrow \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{mon-free}})_{\mathrm{mon-free}}^{\wedge}),$$

and the required assertion follows by the same mechanism as [GLC2, Proposition 3.6.5]. $\square$

4.5.8. Similarly to (4.32), the functor

$$\begin{aligned} ("l)^! : \mathrm{IndCoh}^!((\mathrm{MOP}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^{\wedge}) &\xrightarrow{("l)^!} \mathrm{IndCoh}^!(\mathrm{MOP}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{G}, \check{M}}) \simeq \\ &\simeq \mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOP}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}}) \otimes^{\mathrm{IndCoh}^! (\mathfrak{L}_{\nabla}^+(\check{G}))} \mathrm{Vect} \rightarrow \mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOP}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}}) \end{aligned}$$

naturally factors as

$$\begin{aligned} \mathrm{IndCoh}^!((\mathrm{MOP}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^{\wedge}) &\rightarrow \mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOP}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}}) \otimes^{\mathrm{IndCoh}^! (\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect} \rightarrow \\ &\rightarrow \mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOP}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}}). \end{aligned}$$

As a formal consequence of Proposition 4.5.7, we obtain:

Corollary 4.5.9. *The functor*

$$(4.35) \quad \mathrm{IndCoh}^!((\mathrm{MOP}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^{\wedge}) \rightarrow \mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOP}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}}) \otimes^{\mathrm{IndCoh}^! (\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect}$$

is an equivalence.

4.6. The semi-infinite Jacquet functor for opers, factorization version. Recall the functor $J_{\mathrm{Op}}^{-, \mathrm{spec}, \frac{\infty}{2}}$ from Sect. 4.3.6. The goal of this subsection is to introduce its variant, denoted

$$J_{\mathrm{Op}}^{-, \mathrm{spec}, \frac{\infty}{2} \mathrm{mon-free}},$$

which will be a (lax unital) factorization functor

$$\mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^{\wedge}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M}, \check{P}}^{\mathrm{mon-free}}),$$

as well as its dual, denoted

$$\mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \frac{\infty}{2} \mathrm{mon-free}},$$

which will be (lax unital) factorization functor

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{P}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^{\wedge}).$$

Intuitively, the functor $J_{\mathrm{Op}}^{-, \mathrm{spec}, \frac{\infty}{2} \mathrm{mon-free}}$ is pullback along the map

$$(\mathrm{MOP}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^{\wedge} \rightarrow (\mathrm{Op}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^{\wedge},$$

induced by the map

$$(4.36) \quad j : \mathrm{MOP}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}} \rightarrow \mathrm{Op}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}},$$

followed by pushforward along

$$(\mathrm{MOP}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^{\wedge} \rightarrow \mathrm{MOP}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}} \stackrel{\mathrm{qMiu}}{\simeq} \mathrm{Op}_{\check{M}, \check{P}}^{\mathrm{mon-free}}.$$

4.6.1. Note that the map j of (4.36) induces a map, denoted by the same symbol j ,

$$(4.37) \quad j : \mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{MOP}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}} \rightarrow \mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}}$$

as factorization prestacks, compatible with the actions of $\mathfrak{L}_{\nabla}(\check{G})$.

Remark 4.6.2. Note that the map (4.37) is *not* an open embedding, see Remark 4.3.3 above.

4.6.3. We have a pair of mutually dual (lax unital) factorization functors

$$j_* : \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})$$

and

$$j^! : \mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}).$$

They induce a pair of (lax unital) factorization functors

$$(4.38) \quad j_* : \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect} \rightarrow \\ \rightarrow \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect}$$

$$(4.39) \quad j^! : \mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}) \otimes_{\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect} \rightarrow \\ \rightarrow \mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}) \otimes_{\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect},$$

respectively.

4.6.4. Using the identifications

$$\mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect} =: \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}})$$

and

$$\mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}) \otimes_{\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect} =: \mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}})$$

and

$$\mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect} \stackrel{\text{Proposition 4.5.7}}{\simeq} \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}})$$

and

$$\mathrm{IndCoh}^!(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}}) \otimes_{\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect} \stackrel{\text{Corollary 4.5.9}}{\simeq} \mathrm{IndCoh}^!((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}}),$$

we can interpret the functors (4.38) and (4.39) as a pair of mutually dual (lax unital) factorization functors

$$(4.40) \quad j_* : \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}}),$$

and

$$(4.41) \quad j^! : \mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^!((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}}),$$

respectively.

4.6.5. We define the functor

$$J_{\mathrm{Op}}^{-, \mathrm{spec}, \frac{\infty}{2} \mathrm{mon-free}} : \mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M}, \check{P}^-}^{\mathrm{mon-free}})$$

to be the composition

$$(4.42) \quad \mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}}) \xrightarrow{j^!} \\ \rightarrow \mathrm{IndCoh}^!((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}}) \xrightarrow{((q^{\mathrm{Miu}, \mathrm{mon-free}})^{\wedge})_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M}, \check{P}^-}^{\mathrm{mon-free}}).$$

We define the functor

$$\mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \frac{\infty}{2} \mathrm{mon-free}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{P}^-}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}})$$

to be

$$(4.43) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{P}^-}^{\mathrm{mon-free}})^{((q^{\mathrm{Miu}, \mathrm{mon-free}})^\wedge)^\dagger} \rightarrow \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^\wedge_{\mathrm{mon-free}})^{J_*^{\mathrm{IndCoh}}} \rightarrow \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^\wedge_{\mathrm{mon-free}}).$$

4.6.6. We have a commutative diagram of functors

$$(4.44) \quad \begin{array}{ccc} \mathrm{IndCoh}^!(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) & \xleftarrow{(\iota')^\dagger} & \mathrm{IndCoh}^!((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^\wedge_{\mathrm{mon-free}}) \\ j^! \uparrow & & \uparrow j^! \\ \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) & \xleftarrow{(\iota')^\dagger} & \mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^\wedge_{\mathrm{mon-free}}). \end{array}$$

By adjunction, it gives rise to a natural transformation

$$(4.45) \quad (\iota)_*^{\mathrm{IndCoh}} \circ j^! \rightarrow j^! \circ (\iota)_*^{\mathrm{IndCoh}}$$

as functors

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) \rightrightarrows \mathrm{IndCoh}^!((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^\wedge_{\mathrm{mon-free}}).$$

Similarly, we have a commutative diagram of functors

$$(4.46) \quad \begin{array}{ccc} \mathrm{IndCoh}^*(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) & \xrightarrow{(\iota)_*^{\mathrm{IndCoh}}} & \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^\wedge_{\mathrm{mon-free}}) \\ j_*^{\mathrm{IndCoh}} \downarrow & & \downarrow j_*^{\mathrm{IndCoh}} \\ \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) & \xrightarrow{(\iota)_*^{\mathrm{IndCoh}}} & \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^\wedge_{\mathrm{mon-free}}). \end{array}$$

By adjunction, it gives rise to a natural transformation

$$(4.47) \quad j_*^{\mathrm{IndCoh}} \circ (\iota)^\dagger \rightarrow (\iota)^\dagger \circ j_*^{\mathrm{IndCoh}}$$

as functors

$$\mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})^\wedge_{\mathrm{mon-free}}) \rightrightarrows \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}).$$

We claim:

Lemma 4.6.7. *The natural transformations (4.45) and (4.47) are isomorphisms.*

Proof. We will prove the assertion for (4.47); the one for (4.45) would then follow by duality.

Consider the fiber products

$$\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-} \simeq \mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mer}}$$

and

$$\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-} \simeq \mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mer}}.$$

Both are factorization ind-affine ind-schemes equipped with an action of $\mathfrak{L}_{\nabla}(\check{P}^-)$. We have a commutative diagram

$$(4.48) \quad \begin{array}{ccc} (\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mer}}) \times \mathfrak{L}_{\nabla}(\check{P}^-)/\mathfrak{L}_{\nabla}^+(\check{P}^-) & \xrightarrow{\mathrm{id} \times q^-, \mathrm{spec}} & (\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mer}}) \times \mathfrak{L}_{\nabla}(\check{M})/\mathfrak{L}_{\nabla}^+(\check{M}) \\ j \times \mathrm{id} \downarrow & & \downarrow j \times \mathrm{id} \\ (\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mer}}) \times \mathfrak{L}_{\nabla}(\check{P}^-)/\mathfrak{L}_{\nabla}^+(\check{P}^-) & \xrightarrow{\mathrm{id} \times q} & (\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mer}}) \times \mathfrak{L}_{\nabla}(\check{M})/\mathfrak{L}_{\nabla}^+(\check{M}), \end{array}$$

where q^-, spec denotes the map $\mathfrak{L}_{\nabla}(\check{P}^-)/\mathfrak{L}_{\nabla}^+(\check{P}^-) \rightarrow \mathfrak{L}_{\nabla}(\check{M})/\mathfrak{L}_{\nabla}^+(\check{M})$.

We can interpret (4.46) as obtained by the operation

$$- \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{P}^-))} \mathrm{Vect}$$

from the diagram

$$\begin{array}{ccc} \mathrm{IndCoh}^*((\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{MOP}_{\check{G}, \check{P}^-}^{\mathrm{mer}}) \times \mathfrak{L}_{\nabla}(\check{P}^-)/\mathfrak{L}_{\nabla}^+(\check{P}^-)) & \xrightarrow{(\mathrm{id} \times \mathfrak{q}^-, \mathrm{spec})_*^{\mathrm{IndCoh}}} & \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{MOP}_{\check{G}, \check{P}^-}^{\mathrm{mer}}) \times \mathfrak{L}_{\nabla}(\check{M})/\mathfrak{L}_{\nabla}^+(\check{M})) \\ \downarrow (j \times \mathrm{id})_*^{\mathrm{IndCoh}} & & \downarrow (j \times \mathrm{id})_*^{\mathrm{IndCoh}} \\ \mathrm{IndCoh}^*((\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mer}}) \times \mathfrak{L}_{\nabla}(\check{P}^-)/\mathfrak{L}_{\nabla}^+(\check{P}^-)) & \xrightarrow{(\mathrm{id} \times \mathfrak{q}^-, \mathrm{spec})_*^{\mathrm{IndCoh}}} & \mathrm{IndCoh}^*((\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{MOP}_{\check{G}, \check{P}^-}^{\mathrm{mer}}) \times \mathfrak{L}_{\nabla}(\check{M})/\mathfrak{L}_{\nabla}^+(\check{M})). \end{array}$$

Hence, it is enough to show that the natural transformation

$$(j \times \mathrm{id})_*^{\mathrm{IndCoh}} \circ (\mathrm{id} \times \mathfrak{q}^-, \mathrm{spec})^! \rightarrow (\mathrm{id} \times \mathfrak{q}^-, \mathrm{spec})^! \circ j_*^{\mathrm{IndCoh}}$$

is an isomorphism.

However, the latter diagram splits as a tensor product of

$$j_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*((\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{MOP}_{\check{G}, \check{P}^-}^{\mathrm{mer}})) \rightarrow \mathrm{IndCoh}^*((\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mer}}))$$

along the vertical arrows and

$$(\mathfrak{q}^-, \mathrm{spec})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}(\mathfrak{L}_{\nabla}(\check{P}^-)/\mathfrak{L}_{\nabla}^+(\check{P}^-)) \rightarrow \mathrm{IndCoh}(\mathfrak{L}_{\nabla}(\check{M})/\mathfrak{L}_{\nabla}^+(\check{M}))$$

along the horizontal arrows, and the assertion follows. $\square$

4.6.8. Combining Lemma 4.6.7 with Sect. 4.4.10 we obtain:

Corollary 4.6.9.

(a) *The functor $J_{\mathrm{Op}}^{-, \mathrm{spec}}$ identifies canonically with*

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \xrightarrow{(4.20)} \mathrm{IndCoh}^!((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}}) \xrightarrow{J_{\mathrm{Op}}^{-, \mathrm{spec}, \frac{\infty}{2}} \mathrm{mon-free}} \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M}, \check{P}^-}^{\mathrm{mon-free}}).$$

(b) *The functor $\mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}}$ identifies canonically with*

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{P}^-}^{\mathrm{mon-free}}) \xrightarrow{\mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \frac{\infty}{2}} \mathrm{mon-free}} \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}}) \xrightarrow{(4.19)} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}).$$

4.7. **The semi-infinite Jacquet functor for opers, enhanced version.** In this subsection we will give slightly different interpretations of the functors $J_{\mathrm{Op}}^{-, \mathrm{spec}, \frac{\infty}{2}} \mathrm{mon-free}$ and $\mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \frac{\infty}{2}} \mathrm{mon-free}$.

These interpretations play a crucial role in the statements of Theorems 5.3.6 and 5.4.5.

4.7.1. Recall the identification

$$\mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}}) \simeq \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$$

of (4.16).

Thus, we can interpret the functor $\mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \frac{\infty}{2}} \mathrm{mon-free}$ as a functor

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{P}^-}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$$

When viewed as such, we will denote it by

$$\mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \mathrm{enh}}.$$

Unwinding the constructions, we obtain that the functor $\mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \mathrm{enh}}$ respects the actions of $\mathrm{Sph}_{\check{M}}^{\mathrm{spec}}$ on the two sides.

Note that Corollary 4.6.9(b) says that the original functor $\mathrm{co}J_{\mathrm{Op}}^{-,\mathrm{spec}}$ identifies with

$$\begin{aligned} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M},\check{\rho}_P}^{\mathrm{mon-free}}) &\xrightarrow{\mathrm{co}J_{\mathrm{Op}}^{-,\mathrm{spec},\mathrm{enh}}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}} \xrightarrow{\mathrm{Id} \otimes (-\star \Delta^{-,\mathrm{spec},\frac{\infty}{2}})^R} \\ &\rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{Sph}_{\check{G}}^{\mathrm{spec}} \simeq \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}). \end{aligned}$$

We will also use the notation

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})^{-,\mathrm{enh}} := \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}.$$

4.7.2. Recall now the identification

$$\mathrm{IndCoh}^!((\mathrm{Op}_{\check{G},\check{P}^-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}}) \simeq \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})^{(\mathrm{Sph}_{\check{G}}^{\mathrm{spec}})^{\vee}} \otimes_{\mathrm{co}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}$$

of (4.17).

Thus, we can view the functor $J_{\mathrm{Op}}^{-,\mathrm{spec},\frac{\infty}{2},\mathrm{mon-free}}$ as a functor

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})^{(\mathrm{Sph}_{\check{G}}^{\mathrm{spec}})^{\vee}} \otimes_{\mathrm{co}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}} \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M},\check{\rho}_P}^{\mathrm{mon-free}}).$$

When viewed as such, we will denote it by

$$J_{\mathrm{Op}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}}.$$

Unwinding the construction, we obtain that the functor $J_{\mathrm{Op}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}}$ respects the coactions of $(\mathrm{Sph}_{\check{M}}^{\mathrm{spec}})^{\vee}$ on the two sides.

In what follows we will also use the notation

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})^{-,\mathrm{enh}_{\mathrm{co}}} := \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})^{(\mathrm{Sph}_{\check{G}}^{\mathrm{spec}})^{\vee}} \otimes_{\mathrm{co}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}.$$

4.7.3. It follows from Corollary 4.6.9(a) that the original functor $J_{\mathrm{Op}}^{-,\mathrm{spec}}$ identifies with

$$\begin{aligned} \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) &\xrightarrow{\mathrm{Id} \otimes \mathrm{I}_{\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}}_{\mathrm{co}}} \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}_{\mathrm{co}} \rightarrow \\ &\rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})^{(\mathrm{Sph}_{\check{G}}^{\mathrm{spec}})^{\vee}} \otimes_{\mathrm{co}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}} \xrightarrow{J_{\mathrm{Op}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}}} \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M},\check{\rho}_P}^{\mathrm{mon-free}}), \end{aligned}$$

where the second arrow is the left adjoint of the forgetful functor

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})^{(\mathrm{Sph}_{\check{G}}^{\mathrm{spec}})^{\vee}} \otimes_{\mathrm{co}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}} \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}_{\mathrm{co}},$$

which exists thanks to rigidity.

5. COMPATIBILITY OF THE FLE WITH THE JACQUET FUNCTORS

In this section we first formulate the theorem that expresses the compatibility of the critical FLE with the Jacquet functors (Theorem 5.2.3), as well as its enhanced version (Theorem 5.3.6).

However, in order to prove both these theorems, we will reformulate them in dual terms. Thus, we will formulate Theorems 5.4.2 and 5.4.5, which are equivalent to Theorems 5.2.3 and 5.3.6, respectively.

A feature of the present situation is that although Theorem 5.4.2 looks simpler than its enhanced version, namely, Theorem 5.4.5, we will have to prove the latter in order to prove the former. I.e., the enhanced statement ends up being more accessible than the unenhanced one. The reason for this lies in the computational core of the proof, namely, our knowledge of the Drinfeld-Sokolov reduction of Wakimoto modules, see Sect. 3.10.

5.1. The twisted critical FLE. In this subsection we will adapt the setting the critical FLE to the situation twisted by a $Z_{\check{G}}^0$ -torsor.

5.1.1. Let us return to the setting of Sect. 3.1. Let $\mathcal{P}_{Z_G^0}$ be a Z_G^0 -torsor on X .

Recall the *critical FLE*, i.e., the equivalence

$$(5.1) \quad \text{FLE}_{G, \text{crit}_G} : \text{KL}(G)_{\text{crit}_G} \rightarrow \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})$$

of [GLC2, Theorem 6.1.4].

We will consider twists of the two sides of (5.1) by $\mathcal{P}_{Z_G^0}$.

5.1.2. On the Kac-Moody side, we consider the factorization categories

$$\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0})} \text{ and } \text{KL}(G)_{\text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0})},$$

see Sect. 3.1.1.

On the oper side, we consider the factorization category

$$\text{IndCoh}^*(\text{Op}_{G, \mathcal{P}_{Z_G^0}}^{\text{mon-free}}),$$

see Sect. 4.1.1.

We claim that the equivalence (5.1) admits a twisted version, namely,

$$(5.2) \quad \text{FLE}_{G, \text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0})} : \text{KL}(G)_{\text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0})} \rightarrow \text{IndCoh}^*(\text{Op}_{G, \mathcal{P}_{Z_G^0}}^{\text{mon-free}}).$$

5.1.3. Indeed, in order to construct (5.2), one repeats the construction from [GLC2, Sect. 6.1], using the following version of the Feigin-Frenkel isomorphism [GLC2, Theorem 5.1.2]:

$$(5.3) \quad \mathfrak{z}_{\mathfrak{g}, \mathcal{P}_{Z_G^0}} \xrightarrow{\text{FF}_G} \mathcal{O}_{\text{Op}_{G, \mathcal{P}_{Z_G^0}}^{\text{reg}}},$$

where $\mathfrak{z}_{\mathfrak{g}, \mathcal{P}_{Z_G^0}}$ is the classical center of $\mathbb{V}_{\mathfrak{g}, \text{crit}_G, \mathcal{P}_{Z_G^0}}$.

The identification (5.3) is obtained as the tensor product of the untwisted Feigin-Frenkel isomorphism for $\mathfrak{g}_{\text{der}} := [\mathfrak{g}, \mathfrak{g}]$ with the twisted version for the torus $\mathfrak{g}_{\text{ab}}$, see [GLC2, Sect. 5.1.3].

5.1.4. The following commutative diagram is the twisted version of [GLC2, Theorem 6.4.5]:

$$(5.4) \quad \begin{array}{ccc} \text{Whit}_*(G) \otimes_{\text{Sph}_G} \text{KL}(G)_{\text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0})} & \xrightarrow{\text{Id} \otimes \alpha_{\rho(\omega_X), \text{taut}}} & \text{Whit}_*(G) \otimes_{\text{Sph}_G} \text{KL}(G)_{\text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0}), \rho(\omega_X)} \\ \downarrow \text{FLE}_{G, \infty}^{-1} \otimes \text{FLE}_{G, \text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0})} & & \downarrow \\ \text{Rep}(\check{G}) \otimes_{\text{Sph}_G^{\text{spec}}} \text{IndCoh}^*(\text{Op}_{G, \mathcal{P}_{Z_G^0}}^{\text{mon-free}}) & & \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0}), \rho(\omega_X)}) \\ \downarrow & & \downarrow \overline{\text{DS}}^{\text{enh}, \text{rfnd}} \\ \text{IndCoh}^*(\text{Op}_{G, \mathcal{P}_{Z_G^0}}^{\text{mer}}) & \xleftarrow{\text{FF}_G} & \text{IndCoh}^*(\text{"Spec"}(\mathfrak{z}_{\mathfrak{g}, \mathcal{P}_{Z_G^0}})), \end{array}$$

(see [GLC2, Sect. 4.8] where the notation $\overline{\text{DS}}^{\text{enh}, \text{rfnd}}$ is explained).

5.1.5. Note the dualities of [GLC2, Equations (2.3) and (2.4)] induce dualities

$$(5.5) \quad (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \text{dlog}(\mathcal{P}_{Z_G^0})})^\vee \simeq \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0})}$$

and

$$(5.6) \quad (\text{KL}(G)_{\text{crit}_G, -\text{dlog}(\mathcal{P}_{Z_G^0})})^\vee \simeq \text{KL}(G)_{\text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0})}.$$

Note also that the Chevalley involution τ_G induces equivalences

$$\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0})} \simeq \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G - \text{dlog}(\mathcal{P}_{Z_G^0})}$$

and

$$\text{KL}(G)_{\text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0})} \simeq \text{KL}(G)_{\text{crit}_G - \text{dlog}(\mathcal{P}_{Z_G^0})}.$$

5.1.6. The equivalence of [GLC2, Equation (3.21)] induces an equivalence

$$\Theta_{\text{Op}_G^{\text{mon-free}}} : \text{IndCoh}^!(\text{Op}_{G, \mathcal{P}_{Z_G^0}}^{\text{mon-free}}) \rightarrow \text{IndCoh}^*(\text{Op}_{G, \mathcal{P}_{Z_G^0}}^{\text{mon-free}}).$$

In particular, we obtain a self-duality

$$(5.7) \quad \left(\text{IndCoh}^*(\text{Op}_{G, \mathcal{P}_{Z_G^0}}^{\text{mon-free}}) \right)^\vee \simeq \text{IndCoh}^!(\text{Op}_{G, \mathcal{P}_{Z_G^0}}^{\text{mon-free}})^{\Theta_{\text{Op}_G^{\text{mon-free}}} \simeq} \text{IndCoh}^*(\text{Op}_{G, \mathcal{P}_{Z_G^0}}^{\text{mon-free}}).$$

It follows formally from [GLC2, Theorem 8.1.4] that we have a commutative diagram

$$(5.8) \quad \begin{array}{ccc} \left(\text{IndCoh}^*(\text{Op}_{G, \mathcal{P}_{Z_G^0}}^{\text{mon-free}}) \right)^\vee & \xrightarrow[\sim]{(5.7)} & \text{IndCoh}^*(\text{Op}_{G, \mathcal{P}_{Z_G^0}}^{\text{mon-free}}) \\ \downarrow (\text{FLE}_{G, \text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0})})^\vee & & \uparrow \text{FLE}_{G, \text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0})} \circ \tau_G \\ (\text{KL}(G)_{\text{crit}_G + \text{dlog}(\mathcal{P}_{Z_G^0})})^\vee & \xrightarrow[\sim]{(5.6)} & \text{KL}(G)_{\text{crit}_G - \text{dlog}(\mathcal{P}_{Z_G^0})} \end{array}$$

5.1.7. In practice, we will take the reductive group in question to be the Levi subgroup M of a standard parabolic P , and $\mathcal{P}_{Z_M^0} := \check{\rho}_P(\omega_X)$.

So in this case, the equivalence (5.2) specializes to

$$(5.9) \quad \text{FLE}_{M, \text{crit}_M + \check{\rho}_P} : \text{KL}(M)_{\text{crit}_M + \check{\rho}_P} \simeq \text{IndCoh}^*(\text{Op}_{M, \check{\rho}_P}^{\text{mon-free}}),$$

see Sect. 3.1.7 for the notational conventions.

5.2. The initial formulation.

5.2.1. Recall the functor

$$J_{\text{KM}, \rho_P(\omega_X)}^{-, \text{Sph}} : \text{KL}(G)_{\text{crit}_G} \rightarrow \text{KL}(M)_{\text{crit}_M - \check{\rho}_P},$$

see Sect. 3.9.7.

Recall also the functor

$$J_{\text{Op}, \Theta}^{-, \text{spec}} : \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}}) \rightarrow \text{IndCoh}^*(\text{Op}_{M, \check{\rho}_P}^{\text{mon-free}}),$$

see Sect. 4.2.4.

5.2.2. The following theorem, which is one of the main results of this paper, expresses the compatibility of the FLE with the Jacquet functors:

Theorem 5.2.3. *The following diagram of (lax unital) factorization functors commutes*

$$(5.10) \quad \begin{array}{ccc} \mathrm{KL}(M)_{\mathrm{crit}_M + \tilde{\rho}_P} & \xrightarrow{\mathrm{FLE}_{M, \mathrm{crit}_G + \tilde{\rho}_P}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{M}, \tilde{\rho}_P}^{\mathrm{mon-free}}) \\ J_{\mathrm{KM}, \rho_P(\omega_X)}^{-, \mathrm{Sph}} \uparrow & & \uparrow J_{\mathrm{Op}, \Theta}^{-, \mathrm{spec}} \\ \mathrm{KL}(G)_{\mathrm{crit}_G} & \xrightarrow{\mathrm{FLE}_{G, \mathrm{crit}_G}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}). \end{array}$$

Remark 5.2.4. Note that the statement of Theorem 5.2.3 bears a similarity with that of Corollary 2.5.2.

5.3. The enhanced version.

5.3.1. Recall that the equivalence

$$\mathrm{FLE}_{G, \mathrm{crit}_G} : \mathrm{KL}(G)_{\mathrm{crit}_G} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})$$

is compatible with the actions of

$$\mathrm{Sph}_G^{\mathrm{Sat}_G} \simeq \mathrm{Sph}_{\tilde{G}}^{\mathrm{spec}},$$

(see [GLC2, Theorem 6.4.5(a)]).

Consider the equivalence

$$(5.11) \quad \mathrm{KL}(G)_{\mathrm{crit}_G} \xrightarrow{\mathrm{FLE}_{G, \mathrm{crit}_G}} \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}) \xrightarrow{\Theta_{\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}}^{-1}} \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}).$$

This equivalence is compatible with the actions of

$$\mathrm{Sph}_G^{\mathrm{Sat}_G} \simeq \mathrm{Sph}_{\tilde{G}}^{\mathrm{spec}},$$

and hence with the coactions of

$$(5.12) \quad \mathrm{Sph}_G^{\vee} \xrightarrow{(\mathrm{Sat}_G^{\vee})^{-1}} (\mathrm{Sph}_{\tilde{G}}^{\mathrm{spec}})^{\vee}.$$

Consider also a similar equivalence for M :

$$(5.13) \quad \mathrm{KL}(M)_{\mathrm{crit}_M + \tilde{\rho}_P} \xrightarrow{\mathrm{FLE}_{M, \mathrm{crit}_G + \tilde{\rho}_P}} \mathrm{IndCoh}^*(\mathrm{Op}_{\tilde{M}, \tilde{\rho}_P}^{\mathrm{mon-free}}) \xrightarrow{\Theta_{\mathrm{Op}_{\tilde{M}, \tilde{\rho}_P}^{\mathrm{mon-free}}}^{-1}} \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{M}, \tilde{\rho}_P}^{\mathrm{mon-free}}).$$

5.3.2. Consider the factorization category $\mathrm{I}(G, P^-)_{\mathrm{co}, \rho_P(\omega_X)}^{\mathrm{loc}}$ as a comodule over the comonoidal factorization category $\mathrm{Sph}_G^{\vee}$.

Consider the factorization category $\mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}}^{\mathrm{spec}, \mathrm{loc}}$ as a comodule over the comonoidal factorization category $(\mathrm{Sph}_{\tilde{G}}^{\mathrm{spec}})^{\vee}$.

The equivalence $\mathrm{Sat}^{-, \frac{\infty}{2}}$ induces an equivalence

$$(5.14) \quad \mathrm{I}(G, P^-)_{\mathrm{co}, \rho_P(\omega_X)}^{\mathrm{loc}} \xrightarrow{((\mathrm{Sat}^{-, \frac{\infty}{2}})^{\vee})^{-1}} \mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}}^{\mathrm{spec}, \mathrm{loc}},$$

This equivalence is compatible with the coactions of

$$(5.15) \quad \mathrm{Sph}_G^{\vee} \xrightarrow{(\mathrm{Sat}_{G, \tau}^{\vee})^{-1}} (\mathrm{Sph}_{\tilde{G}}^{\mathrm{spec}})^{\vee}.$$

Hence, by Sect. 2.4.1, the equivalences (5.11) and (5.14) combine to an equivalence

$$(5.16) \quad \mathrm{KL}(G)_{\mathrm{crit}_G, \rho_P(\omega_X)}^{-, \mathrm{enh}_{\mathrm{co}}} \simeq \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})^{-, \mathrm{enh}_{\mathrm{co}}},$$

where

$$\mathrm{KL}(G)_{\mathrm{crit}_G, \rho_P(\omega_X)}^{-, \mathrm{enh}_{\mathrm{co}}} := \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\mathrm{co}, \rho_P(\omega_X)}^{\mathrm{loc}} \simeq \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G^{\vee}} \mathrm{I}(G, P^-)_{\mathrm{co}, \rho_P(\omega_X)}^{\mathrm{loc}}$$

(see Sects. 3.9.11 and 1.4.8) and

$$\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})^{-, \mathrm{enh}_{\mathrm{co}}} := \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})^{(\mathrm{Sph}_G^{\mathrm{spec}})^\vee} \mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}}^{\mathrm{spec}, \mathrm{loc}},$$

(see Sect. 4.7.2).

5.3.3. Consider the functor

$$J_{\mathrm{KM}, \rho_P(\omega_X)}^{-, \mathrm{enh}_{\mathrm{co}}} : \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\mathrm{co}, \rho_P(\omega_X)}^{\mathrm{loc}} \rightarrow \mathrm{KL}(M)_{\mathrm{crit}_M + \check{\rho}_P}.$$

(see Sects. 1.4.8 and 1.4.10).

We will think of it as a functor

$$\mathrm{KL}(G)_{\mathrm{crit}_G, \rho_P(\omega_X)}^{-, \mathrm{enh}_{\mathrm{co}}} \rightarrow \mathrm{KL}(M)_{\mathrm{crit}_M + \check{\rho}_P}.$$

5.3.4. Consider the functor

$$J_{\mathrm{Op}}^{-, \mathrm{spec}, \mathrm{enh}_{\mathrm{co}}} : \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})^{-, \mathrm{enh}_{\mathrm{co}}} \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M}, \check{\rho}_P}^{\mathrm{mon-free}}),$$

see Sect. 4.7.2.

5.3.5. The following is an enhancement of Theorem 5.2.3:

Theorem 5.3.6. *The following diagram of (lax unital) factorization functors commutes*

$$\begin{array}{ccc} \mathrm{KL}(M)_{\mathrm{crit}_M + \check{\rho}_P} & \xrightarrow{(5.13)} & \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M}, \check{\rho}_P}^{\mathrm{mon-free}}) \\ J_{\mathrm{KM}, \rho_P(\omega_X)}^{-, \mathrm{enh}_{\mathrm{co}}} \uparrow & & \uparrow J_{\mathrm{Op}}^{-, \mathrm{spec}, \mathrm{enh}_{\mathrm{co}}} \\ \mathrm{KL}(G)_{\mathrm{crit}_G, \rho_P(\omega_X)}^{-, \mathrm{enh}_{\mathrm{co}}} & \xrightarrow{(5.16)} & \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})^{-, \mathrm{enh}_{\mathrm{co}}}, \end{array}$$

in a way compatible with the actions of

$$\mathrm{Sph}_M^{\mathrm{Sat}_M} \simeq \mathrm{Sph}_M^{\mathrm{spec}}.$$

Remark 5.3.7. Note the formal similarity between the statement of Theorem 5.4.5 and that of Theorem 2.4.3.

5.3.8. Note that the statement of Theorem 5.2.3 can be obtained from that of Theorem 5.3.6 by concatenating with the commutative diagram

$$\begin{array}{ccc} \mathrm{KL}(G)_{\mathrm{crit}_G, \rho_P(\omega_X)}^{-, \mathrm{enh}_{\mathrm{co}}} & \xrightarrow{(5.16)} & \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})^{-, \mathrm{enh}_{\mathrm{co}}} \\ \uparrow = & & \uparrow = \\ \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G^\vee} \mathrm{I}(G, P^-)_{\mathrm{co}, \rho_P(\omega_X)}^{\mathrm{loc}} & & \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})^{(\mathrm{Sph}_G^{\mathrm{spec}})^\vee} \mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}}^{\mathrm{spec}, \mathrm{loc}} \\ \uparrow & & \uparrow \\ \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes \mathrm{I}(G, P^-)_{\mathrm{co}, \rho_P(\omega_X)}^{\mathrm{loc}} & & \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes \mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}}^{\mathrm{spec}, \mathrm{loc}} \\ \uparrow \mathrm{Id} \otimes \mathbf{1}_{\mathrm{I}(G, P^-)_{\mathrm{co}, \rho_P(\omega_X)}^{\mathrm{loc}}} & & \uparrow \mathrm{Id} \otimes \mathbf{1}_{\mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}}^{\mathrm{spec}, \mathrm{loc}}} \\ \mathrm{KL}(G)_{\mathrm{crit}_G} & \xrightarrow{(5.11)} & \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}), \end{array}$$

where the middle vertical arrows are the left adjoints to the natural forgetful functors.

Indeed, after concatenation, we obtain the diagram

$$\begin{array}{ccc} \mathrm{KL}(M)_{\mathrm{crit}_M + \check{\rho}_P} & \xrightarrow{(5.13)} & \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M}, \check{\rho}_P}^{\mathrm{mon-free}}) \\ J_{\mathrm{KM}, \rho_P(\omega_X)}^{-, \mathrm{Sph}} \uparrow & & \uparrow J_{\mathrm{Op}}^{-, \mathrm{spec}} \\ \mathrm{KL}(G)_{\mathrm{crit}_G} & \xrightarrow{(5.11)} & \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}}), \end{array}$$

which identifies with diagram (5.10) via (5.8).

5.4. The dual formulation. In order to prove Theorems 5.2.3 and 5.3.6, we will reformulate them in dual terms.

5.4.1. Dualizing the arrows in (5.10), and taking into account (5.8), we obtain that the following result is equivalent to Theorem 5.2.3:

Theorem 5.4.2. *The following diagram of (lax unital) factorization functors commutes:*

$$(5.17) \quad \begin{array}{ccc} \mathrm{KL}(M)_{\mathrm{crit}_G - \bar{\rho}_P} & \xrightarrow{\mathrm{FLE}_{M, \mathrm{crit}_G + \bar{\rho}_P} \circ \tau_M} & \mathrm{IndCoh}^*(\mathrm{Op}_{M, \bar{\rho}_P}^{\mathrm{mon-free}}) \\ \mathrm{Wak}_{\rho_P(\omega_X)}^{-, \mathrm{Sph}} \downarrow & & \downarrow \mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}} \\ \mathrm{KL}(G)_{\mathrm{crit}_G} & \xrightarrow{\mathrm{FLE}_{G, \mathrm{crit}_G} \circ \tau_G} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}). \end{array}$$

We now proceed to formulating the statement dual to that of Theorem 5.3.6.

5.4.3. Combining the functors $\mathrm{FLE}_{G, \mathrm{crit}_G} \circ \tau_G$ and $\mathrm{Sat}^{-, \frac{\infty}{2}}$ we obtain an equivalence

$$(5.18) \quad \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} \simeq \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}.$$

Note that the functor (5.16) is the dual of (5.18).

5.4.4. Thus, we obtain that the statement dual to Theorem 5.3.6 reads as follows:

Theorem 5.4.5. *The following diagram of (lax unital) factorization functors commutes:*

$$(5.19) \quad \begin{array}{ccc} \mathrm{KL}(M)_{\mathrm{crit}_G - \bar{\rho}_P} & \xrightarrow{\mathrm{FLE}_{M, \mathrm{crit}_G + \bar{\rho}_P} \circ \tau_M} & \mathrm{IndCoh}^*(\mathrm{Op}_{M, \bar{\rho}_P}^{\mathrm{mon-free}}) \\ \mathrm{Wak}_{\rho_P(\omega_X)}^{-, \mathrm{enh}} \downarrow & & \downarrow \mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \mathrm{enh}} \\ \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} & \xrightarrow{(5.18)} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}, \end{array}$$

in a way compatible with the actions of

$$\mathrm{Sph}_M^{\mathrm{Sat}_M, \tau} \simeq \mathrm{Sph}_M^{\mathrm{spec}}.$$

5.4.6. Note that the statement of Theorem 5.4.2 is obtained from that of Theorem 5.4.5 by concatenating (5.19) with the diagram

$$\begin{array}{ccc} \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} & \xrightarrow{(5.18)} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \\ \downarrow & & \downarrow \\ \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{Sph}_G & \xrightarrow{(\mathrm{FLE}_{G, \mathrm{crit}_G} \circ \tau_G) \otimes \mathrm{Sat}_{G, \tau}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{Sph}_G^{\mathrm{spec}} \\ \sim \downarrow & & \downarrow \sim \\ \mathrm{KL}(G)_{\mathrm{crit}_G} & \xrightarrow{\mathrm{FLE}_{G, \mathrm{crit}_G} \circ \tau_G} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}), \end{array}$$

where the upper vertical arrows are

$$((-) \star_G \Delta^{-, \frac{\infty}{2}})^R \text{ and } ((-) \star_{\check{G}} \Delta^{-, \mathrm{spec}, \frac{\infty}{2}})^R,$$

respectively.

Remark 5.4.7. When working at the pointwise level, the statement of Theorem 5.4.5 is close to what is proved in [FG1], and in fact one can generalize the results of *loc. cit.* to deduce the full statement of Theorem 5.4.5 at the pointwise level.

However, the methods of *loc. cit.* do not allow to treat families over the Ran space; the reason for this is that these methods are based on considering the Iwahori subgroup, and the latter is a special feature of the pointwise set-up.

6. PROOF OF THEOREM 5.4.5

6.1. Structure of the proof.

6.1.1. The proof of Theorem 5.4.5 will be roughly modeled on that of Theorem 2.4.3. It will be structured as follows:

In this subsection, we will define a (lax) factorization category $\mathbf{C}$, equipped with a unital structure and (strictly unital) factorization functors

$$\mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} \xrightarrow{\mathrm{F}_{\mathrm{KM}(G)}} \mathbf{C} \xleftarrow{\mathrm{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$$

such that:

- Both functors $\mathrm{F}_{\mathrm{KM}(G)}$ and $\mathrm{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}}$ are fully faithful when restricted to the *eventually coconnective subcategories*, with respect to the t-structures to be introduced;
- The functors $\mathrm{Wak}_{\rho_P(\omega_X)}^{-, \mathrm{enh}}$ and $\mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \mathrm{enh}}$ send compact objects to eventually coconnective objects;

The above two bullet points will be carried out in Sect. 7.

- There exists a canonical identification $\mathrm{F}_{\mathrm{KM}(G)} \simeq \mathrm{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}}$ as functors from

$$\mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} \stackrel{(5.18)}{\simeq} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$$

to $\mathbf{C}$;

This bullet point will be carried out at the end of this subsection.

- There is a canonical identification between the compositions

$$\mathrm{F}_{\mathrm{KM}(G)} \circ \mathrm{Wak}_{\rho_P(\omega_X)}^{-, \mathrm{enh}} \simeq \mathrm{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}} \circ \mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \mathrm{enh}}$$

as factorization functors from

$$\mathrm{KL}(M)_{\mathrm{crit}_G - \check{\rho}_P} \stackrel{\mathrm{FLE}_{M, \mathrm{crit}_G + \check{\rho}_P} \circ \tau_M}{\simeq} \mathrm{IndCoh}^*(\mathrm{Op}_{M, \check{\rho}_P}^{\mathrm{mon-free}})$$

to $\mathbf{C}$.

The proof of this property will occupy the majority of this section.

We depict the third property above as a commutative triangle:

$$(6.1) \quad \begin{array}{ccc} \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} & \xrightarrow{(5.18)} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \\ & \searrow \mathrm{F}_{\mathrm{KM}(G)} & \swarrow \mathrm{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}} \\ & \mathbf{C} & \end{array}$$

We depict the fourth property above as a commutative pentagon:

$$\begin{array}{ccc}
 \mathrm{KL}(M)_{\mathrm{crit}_G - \check{\rho}_P} & \xrightarrow{\mathrm{FLE}_{M, \mathrm{crit}_G + \check{\rho}_P} \circ \tau_M} & \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{\rho}_P}^{\mathrm{mon-free}}) \\
 \downarrow \mathrm{Wak}_{\rho_P(\omega_X)}^{-, \mathrm{enh}} & & \downarrow \mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \mathrm{enh}} \\
 \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} & & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \\
 & \searrow \mathrm{F}_{\mathrm{KM}(G)} \quad \swarrow \mathrm{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}} & \\
 & \mathbf{C} &
 \end{array}
 \tag{6.2}$$

The above properties clearly imply the assertion of Theorem 5.4.5.

6.1.2. The compatibility with the actions of

$$\mathrm{Sph}_M^{\mathrm{Sat}_{M, \tau}} \cong \mathrm{Sph}_M^{\mathrm{spec}}$$

will be embedded into the constructions, see Sect. B.3.

6.1.3. We proceed to the definition of the category $\mathbf{C}$. Recall the factorization algebra

$$\Omega(R)^{\mathrm{spec}} \in \mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M}),$$

see Sect. E.2.2.

Let $\Omega(R)^{\mathrm{Op}}$ denote the image of $\Omega(R)^{\mathrm{spec}}$ under the functor

$$((\mathbf{r}^{\mathrm{reg}})^* \otimes \mathrm{Id}) : \mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M}) \rightarrow \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{reg}}) \otimes \mathrm{Rep}(\check{M}),$$

viewed as a factorization algebra in

$$\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{reg}}) \otimes \mathrm{Rep}(\check{M}),$$

where $\mathbf{r}^{\mathrm{reg}}$ is the projection $\mathrm{Op}_G^{\mathrm{reg}} \rightarrow \mathrm{LS}_G^{\mathrm{reg}}$, see [GLC2, Sect. 3.1.5].

Using the functor

$$\mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{reg}}) \simeq \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{reg}}) \xrightarrow{\iota_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}),$$

we can regard $\Omega(R)^{\mathrm{Op}}$ as a factorization algebra in

$$\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M}).$$

6.1.4. We define the (lax) factorization category $\mathbf{C}$ to be

$$(6.3) \quad \Omega(R)^{\mathrm{Op-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M})).$$

We now proceed to the definition of the functors

$$(6.4) \quad \mathrm{F}_{\mathrm{KM}(G)} \text{ and } \mathrm{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}}.$$

Remark 6.1.5. Informally, one should think of $\mathbf{C}$ as a *partial left completion* of the *would-be* category

$$(6.5) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}} \check{M}).$$

The problem is, however, that we do not know how to define (6.5) as a factorization category (cf. Sect. 4.3.8). So we have to work directly with its surrogate provided by (6.3).

6.1.6. Consider the full subcategory

$$\mathrm{IndCoh}^*((\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}}^\wedge) \subset \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}).$$

As in Proposition 4.5.7, we have a canonical identification

$$(6.6) \quad \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{reg}}} \mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))} \mathrm{Vect} \simeq \mathrm{IndCoh}^*((\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}}^\wedge).$$

From here we obtain an identification

$$(6.7) \quad \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{Rep}(\check{G}) \simeq \mathrm{IndCoh}^*((\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}}^\wedge).$$

6.1.7. Note that the lax unital factorization functor

$$(6.8) \quad \mathrm{Rep}(\check{G}) \xrightarrow{\mathbf{1}_{\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})} \otimes \mathrm{Id}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{Rep}(\check{G}) \simeq \mathrm{IndCoh}^*((\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}}^\wedge) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})$$

identifies with

$$\mathrm{Rep}(\check{G}) \xrightarrow{(\mathrm{r}_{\check{G}}^{\mathrm{reg}})^*} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{reg}}) \xrightarrow{\iota_{\mathrm{IndCoh}}^*} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}).$$

In particular, the image of $\Omega(R)^{\mathrm{spec}}$ along the functor

$$\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M}) \xrightarrow{(6.8) \otimes \mathrm{Id}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M})$$

identifies with $\Omega(R)^{\mathrm{Op}}$.

6.1.8. Consider the full subcategory

$$(6.9) \quad \Omega(R)^{\mathrm{Op-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*((\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}}^\wedge) \otimes \mathrm{Rep}(\check{M})) \subset \Omega(R)^{\mathrm{Op-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M})).$$

From the fact that $\mathrm{Sph}_G^{\mathrm{spec}}$ is rigid and $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})$ is dualizable, we obtain that the functor

$$(6.10) \quad \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \left(\Omega(R)^{\mathrm{spec-mod}^{\mathrm{fact}}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M})) \right) \rightarrow \Omega(R)^{\mathrm{spec-mod}^{\mathrm{fact}}} \left(\left(\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{Rep}(\check{G}) \right) \otimes \mathrm{Rep}(\check{M}) \right)$$

is an equivalence, cf. [GLC2, Lemma B.12.9].

6.1.9. Combining (6.10) with (6.7) and Sect. 6.1.7, we obtain an identification

$$(6.11) \quad \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \left(\Omega(R)^{\mathrm{spec-mod}^{\mathrm{fact}}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M})) \right) \simeq \Omega(R)^{\mathrm{Op-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*((\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}}^\wedge) \otimes \mathrm{Rep}(\check{M})).$$

Remark 6.1.10. Recall that although we could not define (6.5) as a factorization category, we could do so for $\mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge)$.

In the spirit of Remark 6.1.5, the category

$$\Omega(R)^{\mathrm{Op-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*((\mathrm{Op}_G^{\mathrm{mer}})_{\mathrm{mon-free}}^\wedge) \otimes \mathrm{Rep}(\check{M}))$$

can indeed be thought as a partial left completion of $\mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge)$:

6.1.11. Recall now from Sect. E.7.2 that we have a commutative diagram of (strictly unital) $\mathrm{Sph}_G^{\mathrm{spec}}$ -linear factorization functors

$$(6.12) \quad \begin{array}{ccc} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} & \xrightarrow{\mathrm{Sat}^-, \frac{\infty}{2}} & \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \\ & \searrow \mathrm{F} & \swarrow \mathrm{F}^{\mathrm{spec}} \\ & \Omega(R)^{\mathrm{spec}\text{-mod}^{\mathrm{fact}}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M})) & \end{array}$$

6.1.12. Tensoring the functors F and $\mathrm{F}^{\mathrm{spec}}$ with

$$\mathrm{KL}(G)_{\mathrm{crit}_G} \xrightarrow{\mathrm{FLE}_G, \mathrm{crit}_G \circ \tau_G} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon}\text{-free}}) \xleftarrow{\mathrm{Id}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon}\text{-free}}),$$

respectively, we obtain functors

$$\mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon}\text{-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \left(\Omega(R)^{\mathrm{spec}\text{-mod}^{\mathrm{fact}}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M})) \right)$$

and

$$\begin{aligned} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon}\text{-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \rightarrow \\ & \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon}\text{-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \left(\Omega(R)^{\mathrm{spec}\text{-mod}^{\mathrm{fact}}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M})) \right). \end{aligned}$$

Composing with (6.11) and with (6.9) we obtain the sought-for functors (6.4). These functors make the diagram (6.1) commute by construction.

Remark 6.1.13. Recall (see (4.16)) that we can think of

$$\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon}\text{-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$$

as $\mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon}\text{-free} \check{M}})_{\mathrm{mon}\text{-free}}^{\wedge})$.

In terms of this identification and Remark 6.1.5, the functor $\mathrm{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}}$ should informally²⁷ be thought of as the composition of the inclusion

$$\mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon}\text{-free} \check{M}})_{\mathrm{mon}\text{-free}}^{\wedge}) \hookrightarrow \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon}\text{-free} \check{M}}))$$

and the tautological functor

$$\mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon}\text{-free} \check{M}})) \rightarrow \mathbf{C}.$$

More rigorously, in terms of Remark 6.1.10, the functor $\mathrm{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}}$ maps $\mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon}\text{-free} \check{M}})_{\mathrm{mon}\text{-free}}^{\wedge})$ to its partial left completion.²⁸

6.1.14. The rest of this section is devoted to the verification of the commutativity of the pentagon (6.2).

As was mentioned earlier, the first two bullet points in Sect. 6.1.1 will be carried out in Sect. 7.

²⁷Informally: because the target category in the next formula is not defined.

²⁸By this we only mean that the functor in question induces an equivalence on the eventually coconnective subcategories.

6.2. The composition on the spectral side. Denote

$$\mathbf{F}_{\mathrm{MOp}(\check{M})}^{\mathrm{spec}} := \mathbf{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}} \circ \mathrm{co}J_{\mathrm{Op}}^{-,\mathrm{spec},\mathrm{enh}}, \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M},\check{\rho}_P}^{\mathrm{mon-free}}) \rightarrow \mathbf{C}.$$

As a first step towards establishing the commutativity of the pentagon (6.2), in this subsection, we will break down this functor into a composition of more elementary operations.

Remark 6.2.1. In terms of Remarks 6.1.5 and 6.1.10, the functor $\mathbf{F}_{\mathrm{MOp}(\check{M})}^{\mathrm{spec}}$ is the composition

$$\begin{aligned} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M},\check{\rho}_P}^{\mathrm{mon-free}}) &\xrightarrow{((q^{\mathrm{Miu},\mathrm{mon-free}})^\wedge)^!} \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G},\check{P}-}^{\mathrm{mon-free}})_{\mathrm{mon-free}}^\wedge) \xrightarrow{J_*^{\mathrm{IndCoh}}} \\ &\rightarrow \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G},\check{P}-}^{\mathrm{mon-free}})_{\mathrm{mon-free}}^\wedge) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\check{P}-}^{\mathrm{mon-free}}) \rightarrow \mathbf{C}. \end{aligned}$$

The idea of this section is to (informally) rewrite this functor as the composition of the endofunctor

$$(6.13) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M},\check{\rho}_P}^{\mathrm{mon-free}}) \xrightarrow{((q^{\mathrm{Miu},\mathrm{mon-free}})^\wedge)^!} \rightarrow \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G},\check{P}-}^{\mathrm{mon-free}})_{\mathrm{mon-free}}^\wedge) \xrightarrow{((q^{\mathrm{Miu},\mathrm{mon-free}})^\wedge)_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M},\check{\rho}_P}^{\mathrm{mon-free}})$$

of $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{M},\check{\rho}_P}^{\mathrm{mon-free}})$, followed by

$$(6.14) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M},\check{\rho}_P}^{\mathrm{mon-free}}) \xrightarrow{J_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\check{P}-}^{\mathrm{mon-free}}) \rightarrow \mathbf{C}.$$

The point is that although we cannot define the individual functors in (6.14) (because we do not know how to make sense of $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\check{P}-}^{\mathrm{mon-free}})$), we *can* define their composition. The resulting functor

$$(6.15) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M},\check{\rho}_P}^{\mathrm{mon-free}}) \rightarrow \mathbf{C}$$

is the functor $\mathbf{F}_{\mathrm{MOp}(\check{M})}^{\mathrm{spec},\mathrm{ext}}$ defined at the end of this section.

The reason for doing so is that one can express the functor (6.15) in terms of factorization algebras, and thus compare it more directly with its counterpart on the geometric side, thereby establishing the commutativity of (6.2).

6.2.2. We will first describe the functor $\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{MOp}(\check{M})}^{\mathrm{spec}}$, which is the composition of $\mathbf{F}_{\mathrm{MOp}(\check{M})}^{\mathrm{spec}}$ with the forgetful functor

$$(6.16) \quad \mathrm{oblv}_{\Omega(R) \circ_P} : \Omega(R)^{\mathrm{Op-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M})) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M}).$$

Prior to that, we will describe the functor $\mathrm{pre}\text{-}\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{MOp}(\check{M})}^{\mathrm{spec}}$, equal to the composition of $\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{MOp}(\check{M})}^{\mathrm{spec}}$ with

$$(6.17) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M}) \xrightarrow{\mathrm{Id} \otimes \mathrm{inv}_{\check{M}}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}),$$

cf. Sect. E.2.4 and Remark E.2.6.

6.2.3. Namely, unwinding the definitions, we obtain that the functor $\mathrm{pre}\text{-}\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{MOp}(\check{M})}^{\mathrm{spec}}$ identifies with the composition of the endofunctor (6.13) of $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{M},\check{\rho}_P}^{\mathrm{mon-free}})$ with the functor

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{M},\check{\rho}_P}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}})$$

equal to

$$(6.18) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M},\check{\rho}_P}^{\mathrm{mon-free}}) \xrightarrow{q^{\mathrm{Miu}}} \mathrm{IndCoh}^*(\mathrm{MOp}_{\check{G},\check{P}-}^{\mathrm{mon-free}}) \xrightarrow{(p^{\mathrm{Miu}})_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}).$$

6.2.4. We recover the functor $\text{pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec}}$ from $\text{pre-pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec}}$ using the self-duality of $\text{Rep}(\check{M})$.

Namely, this is the unique $\text{Rep}(\check{M})$ -linear functor, whose composition with (6.17) is $\text{pre-pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec}}$.

Explicitly, $\text{pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec}}$ is the composition

$$\begin{aligned} \text{IndCoh}^*(\text{Op}_{\check{M}, \check{\rho}_P}^{\text{mon-free}}) &\xrightarrow{\text{Id} \otimes \mathbf{u}_{\text{Rep}(\check{M})}} \text{IndCoh}^*(\text{Op}_{\check{M}, \check{\rho}_P}^{\text{mon-free}}) \otimes \text{Rep}(\check{M}) \otimes \text{Rep}(\check{M}) \xrightarrow{\text{act} \otimes \text{Id}} \\ &\rightarrow \text{IndCoh}^*(\text{Op}_{\check{M}, \check{\rho}_P}^{\text{mon-free}}) \otimes \text{Rep}(\check{M}) \xrightarrow{\text{pre-pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec}} \otimes \text{Id}} \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \otimes \text{Rep}(\check{M}), \end{aligned}$$

where $\mathbf{u}_{\text{Rep}(\check{M})} \in \text{Rep}(\check{M}) \otimes \text{Rep}(\check{M})$ is the unit of the self-duality.

6.2.5. Denote by

$$\mathcal{A}^{\text{spec}} \in \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \otimes \text{Rep}(\check{M})$$

the image of the factorization unit, i.e.,

$$\mathcal{O}_{\text{Op}_{\check{M}, \check{\rho}_P}^{\text{reg}}} \in \text{IndCoh}^*(\text{Op}_{\check{M}, \check{\rho}_P}^{\text{mon-free}})$$

under the functor $\text{pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec}}$.

The lax unital structure on the functor $\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec}}$ gives rise to a map

$$\phi^{\text{spec}} : \Omega(R)^{\text{Op}} \rightarrow \mathcal{A}^{\text{spec}}$$

as unital factorization algebras in $\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \otimes \text{Rep}(\check{M})$.

Vice versa, given $\text{pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec}}$, the datum of ϕ^{spec} recovers $\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec}}$ as a lax unital factorization functor.

Namely, $\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec}}$ is the composition of the functor

$$\text{IndCoh}^*(\text{Op}_{\check{M}, \check{\rho}_P}^{\text{mon-free}}) \rightarrow \mathcal{A}^{\text{spec}}\text{-mod}^{\text{fact}}(\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \otimes \text{Rep}(\check{M}))$$

induced by $\text{pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec}}$, followed by the functor

$$\mathcal{A}^{\text{spec}}\text{-mod}^{\text{fact}}(\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \otimes \text{Rep}(\check{M})) \rightarrow \Omega(R)^{\text{Op}}\text{-mod}^{\text{fact}}(\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \otimes \text{Rep}(\check{M})),$$

given by restriction along ϕ^{spec} .

6.2.6. Let us denote by $\text{pre-pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec, ext}}$ the functor (6.18). Note that by the same mechanism as above, the functor $\text{pre-pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec, ext}}$ gives rise to a (lax unital) factorization functor

$$\text{pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec, ext}} : \text{IndCoh}^*(\text{Op}_{\check{M}, \check{\rho}_P}^{\text{mon-free}}) \rightarrow \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \otimes \text{Rep}(\check{M}),$$

so that $\text{pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec}}$ is the precomposition of $\text{pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec, ext}}$ with (6.13).

6.2.7. Furthermore, since the functor (6.13) is *strictly unital*, the image of the factorization unit under $\text{pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec, ext}}$ is the same factorization algebra $\mathcal{A}^{\text{spec}}$.

Hence, the homomorphism ϕ^{spec} gives rise to a lift of $\text{pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec, ext}}$ to a lax unital factorization functor

$$\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec, ext}} : \text{IndCoh}^*(\text{Op}_{\check{M}, \check{\rho}_P}^{\text{mon-free}}) \rightarrow \Omega(R)^{\text{Op}}\text{-mod}^{\text{fact}}(\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \otimes \text{Rep}(\check{M})) =: \mathbf{C},$$

so that $\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec}}$ is the precomposition of $\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec, ext}}$ with (6.13).

Remark 6.2.8. See Remark 6.2.1 for the geometric meaning of the functor $\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec, ext}}$.

6.3. The composition on the geometric side. Denote

$$\mathbf{F}_{\mathrm{KL}(M)} := \mathbf{F}_{\mathrm{KM}(G)} \circ \mathrm{Wak}_{\rho_P(\omega_X)}^{-, \mathrm{enh}}.$$

In this subsection we will rewrite this functor, along the lines of how the functor $\mathbf{F}_{\mathrm{MOp}(\check{M})}^{\mathrm{spec}}$ was described in Sect. 6.2.

Namely, we will express it as a composition of an endofunctor of $\mathrm{KL}(M)_{\mathrm{crit}_G - \check{\rho}_P}$ parallel to (6.13), followed by a certain functor

$$\mathbf{F}_{\mathrm{KL}(M)}^{\mathrm{ext}} : \mathrm{KL}(M)_{\mathrm{crit}_G - \check{\rho}_P} \rightarrow \Omega(R)^{\mathrm{Op}\text{-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M})) =: \mathbf{C},$$

which we will be able to access explicitly in terms of factorization algebras.

6.3.1. First, by the same mechanism as in Sect. 6.2, we can consider the functors

$$\mathrm{pre}\text{-}\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KL}(M)} : \mathrm{KL}(M)_{\mathrm{crit}_G - \check{\rho}_P} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}),$$

$$\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KL}(M)} : \mathrm{KL}(M)_{\mathrm{crit}_G - \check{\rho}_P} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M})$$

and the factorization algebra

$$\mathcal{A} \in \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M}),$$

equipped with a homomorphism

$$\phi : \Omega(R)^{\mathrm{Op}} \rightarrow \mathcal{A}.$$

The datum of $\mathrm{pre}\text{-}\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KL}(M)}$ recovers $\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KL}(M)}$ by the same mechanism as in Sect. 6.2.4.

In its turn, we recover $\mathbf{F}_{\mathrm{KL}(M)}$ from $\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KL}(M)}$ and the homomorphism ϕ , see Sect. 6.2.5.

6.3.2. Our current goal is to construct the counterparts on the geometric side of the functors $\mathrm{pre}\text{-}\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{MOp}(\check{M})}^{\mathrm{spec}, \mathrm{ext}}$ (resp., $\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{MOp}(\check{M})}^{\mathrm{spec}, \mathrm{ext}}, \mathbf{F}_{\mathrm{MOp}(\check{M})}^{\mathrm{spec}, \mathrm{ext}}$), to be denoted

$$\mathrm{pre}\text{-}\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KL}(M)}^{\mathrm{ext}}, \mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KL}(M)}^{\mathrm{ext}} \text{ and } \mathbf{F}_{\mathrm{KL}(M)}^{\mathrm{ext}},$$

respectively.

6.3.3. Recall the (lax) factorization category

$$\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{-, \frac{\infty}{2}} := (\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G})^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)},$$

see Sect. 3.4.1.

Define the functor

$$\mathrm{pre}\text{-}\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KM}(G)}^{\mathrm{ext}} : \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{-, \frac{\infty}{2}} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})$$

to be the composition

$$\begin{aligned} (\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G})^{-, \frac{\infty}{2}} &\rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{\mathfrak{L}^+(M)} \xrightarrow{\alpha_{\rho(\omega_X)}, \mathrm{taut}} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)}^{\mathfrak{L}^+(M)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)}^{\mathrm{DS}^{\mathrm{enh}, \mathrm{rfnd}}} \\ &\rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \xrightarrow{\tau_{\check{G}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}). \end{aligned}$$

where:

- The first and the third arrows are the forgetful functors;
- The functor $\mathrm{DS}^{\mathrm{enh}, \mathrm{rfnd}}$ is as in [GLC2, Sect. 5.2.8];

From $\mathrm{pre}\text{-}\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KM}(G)}^{\mathrm{ext}}$ we produce a $\mathrm{Rep}(\check{M})$ -linear functor

$$\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KM}(G)}^{\mathrm{ext}} : (\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G})^{-, \frac{\infty}{2}} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M})$$

by the same principle as in Sect. 6.2.4.

6.3.4. Let $\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KM}(G)}$ and $\mathrm{pre}\text{-}\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KM}(G)}$ denote the compositions of $\mathbf{F}_{\mathrm{KM}(G)}$ with the functors

$$\mathbf{oblv}_{\Omega(R)^{\mathrm{Op}}} \text{ and } (\mathrm{Id} \otimes \mathrm{inv}_{\check{M}}) \circ \mathbf{oblv}_{\Omega(R)^{\mathrm{Op}}},$$

respectively.

Note that the functor $\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KM}(G)}$ is recovered from $\mathrm{pre}\text{-}\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KM}(G)}$ by the same principle as in Sect. 6.2.4.

6.3.5. We now claim:

Proposition 6.3.6. *The functor $\text{pre-pre-F}_{\text{KM}(G)}$ identifies canonically with the composition*

$$(6.19) \quad \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \xrightarrow{\text{Id} \otimes \alpha_{\rho_P(\omega_X)}^{-1}, \text{taut}} \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)^{\text{loc}} \xrightarrow{\text{Accs}} \\ \rightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})^{-, \frac{\infty}{2}} \xrightarrow{\text{pre-pre-F}_{\text{KM}(G)}^{\text{ext}}} \text{IndCoh}^*(\text{Op}_G^{\text{mer}}),$$

where Accs is the functor from (3.17).

Proof. Let

$$\text{pre-pre-F} : \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \rightarrow \text{Rep}(\check{G})$$

be as in Sect. E.6.3. We need to identify the composition (6.19) with the functor obtained by tensoring pre-pre-F with

$$\text{KL}(G)_{\text{crit}_G} \xrightarrow{\tau_G \circ \text{FLE}_G^{\text{crit}}} \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})$$

over

$$\text{Sph}_G \xrightarrow{\text{Sat}_G, \tau} \text{Sph}_G^{\text{spec}}.$$

By construction, the functor (6.19) can be rewritten as

$$\begin{aligned} \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} &\rightarrow \text{KL}(G)_{\text{crit}_G} \otimes_{\text{Sph}_G} \text{Whit}_*(G) \xrightarrow{\alpha_{\rho(\omega_X)}, \text{taut} \otimes \text{Id}} \\ &\rightarrow \text{KL}(G)_{\text{crit}_G, \rho(\omega_X)} \otimes_{\text{Sph}_G} \text{Whit}_*(G) \rightarrow \text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \rho(\omega_X)}) \xrightarrow{\overline{\text{DS}}^{\text{enh}, \text{rfnd}}} \\ &\rightarrow \text{IndCoh}^*(\text{Op}_G^{\text{mer}}) \xrightarrow{\tau_G} \text{IndCoh}^*(\text{Op}_G^{\text{mer}}), \end{aligned}$$

where the first arrow is the tensor product of the identity functor on $\text{KL}(G)_{\text{crit}_G}$ with

$$(6.20) \quad \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \xrightarrow{\alpha_{\rho_M(\omega_X)}, \text{taut}} \text{I}(G, P^-)_{\rho(\omega_X)}^{\text{loc}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X)}) \rightarrow \text{Whit}_*(G).$$

Note, however, that the composition of (6.20) with (the inverse of)

$$\text{Rep}(\check{G}) \xrightarrow{\text{FLE}_G^{\text{crit}, \infty}} \text{Whit}_*(G)$$

is exactly the functor pre-pre-F .

The required identification follows now from [GLC2, Theorem 6.4.5]. $\square$

6.3.7. Recall (see [GLC2, Sect. 5.3]) that the category $\text{KL}(M)_{\text{crit}_M + \check{\rho}_P}$ is acted on by the monoidal category $\text{QCoh}(\text{Op}_{M, \check{\rho}_P}^{\text{mon-free}})$. The Chevalley involution on M allows us to identify

$$\text{KL}(M)_{\text{crit}_M + \check{\rho}_P} \simeq \text{KL}(M)_{\text{crit}_M - \check{\rho}_P}.$$

Hence, we can view $\text{KL}(M)_{\text{crit}_M - \check{\rho}_P}$ also as acted on by $\text{QCoh}(\text{Op}_{M, \check{\rho}_P}^{\text{mon-free}})$.

Let

$$(\text{KL}(M)_{\text{crit}_M - \check{\rho}_P})_{\text{mon-free}_G} \subset \text{KL}(M)_{\text{crit}_M - \check{\rho}_P}$$

be the full subcategory set-theoretically supported over the image of the map $\mathfrak{q}^{\text{Miu}, \text{mon-free}}$ of (4.24).

Let

$$((\mathfrak{q}^{\text{Miu}, \text{mon-free}})^{\wedge})_*^{\text{IndCoh}} : (\text{KL}(M)_{\text{crit}_M - \check{\rho}_P})_{\text{mon-free}_G} \rightleftarrows \text{KL}(M)_{\text{crit}_M - \check{\rho}_P} : ((\mathfrak{q}^{\text{Miu}, \text{mon-free}})^{\wedge})^!$$

denote the resulting pair of adjoint functors.

6.3.8. Recall now that according to Lemma 3.7.4, the functor

$$(6.21) \quad \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} \xrightarrow{\mathrm{Id} \otimes \alpha_{\rho_P(\omega_X), \mathrm{taut}}^{-1}} \simeq \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)^{\mathrm{loc}} \xrightarrow{\mathrm{Accs}} (\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G})^{-, \frac{\infty}{2}}$$

is fully faithful.

Proposition 6.3.9.

(a) *The functor*

$$\mathrm{Wak}_{\rho_P(\omega_X)}^{-, \frac{\infty}{2}} : \mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P} \rightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G})^{-, \frac{\infty}{2}}$$

sends $(\mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P})_{\mathrm{mon-free}_{\tilde{G}}} \subset \mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P}$ to essential image of (6.21). I.e., we have a commutative diagram

$$(6.22) \quad \begin{array}{ccc} (\mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P})_{\mathrm{mon-free}_{\tilde{G}}} & \xrightarrow{((\mathfrak{q}^{\mathrm{Miu}, \mathrm{mon-free}})^{\wedge})_*^{\mathrm{IndCoh}}} & \mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P} \\ \mathrm{Wak}_{\rho_P(\omega_X)}^{-, \frac{\infty}{2}} \downarrow & & \downarrow \mathrm{Wak}_{\rho_P(\omega_X)}^{-, \frac{\infty}{2}} \\ \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} & \xrightarrow{(6.21)} & (\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G})^{-, \frac{\infty}{2}}. \end{array}$$

(b) *The natural transformation in the diagram*

$$(6.23) \quad \begin{array}{ccc} (\mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P})_{\mathrm{mon-free}_{\tilde{G}}} & \xleftarrow{((\mathfrak{q}^{\mathrm{Miu}, \mathrm{mon-free}})^{\wedge})^!} & \mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P} \\ \mathrm{Wak}_{\rho_P(\omega_X)}^{-, \frac{\infty}{2}} \downarrow & \searrow & \downarrow \mathrm{Wak}_{\rho_P(\omega_X)}^{-, \frac{\infty}{2}} \\ \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} & \xleftarrow{\quad} & (\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G})^{-, \frac{\infty}{2}} \end{array}$$

obtained from (6.22) by passing to right adjoints along the horizontal arrows, is an isomorphism.

The proof of the proposition will be given in Sect. 6.7.

6.3.10. Combining Proposition 6.3.9 with Corollary 3.7.7, we obtain:

Corollary 6.3.11. *We have a canonical isomorphism*

$$\mathrm{Wak}_{\rho_P(\omega_X)}^{-, \frac{\infty}{2}}|_{(\mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P})_{\mathrm{mon-free}_{\tilde{G}}}} \circ ((\mathfrak{q}^{\mathrm{Miu}, \mathrm{mon-free}})^{\wedge})^! \simeq \mathrm{Wak}_{\rho_P(\omega_X)}^{-, \mathrm{enh}}$$

as functors

$$\mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}}.$$

6.3.12. Set

$$\mathrm{pre-pre-F}_{\mathrm{KL}(M)}^{\mathrm{ext}} := \mathrm{pre-pre-F}_{\mathrm{KM}(G)}^{\mathrm{ext}} \circ \mathrm{Wak}_{\rho_P(\omega_X)}^{-, \frac{\infty}{2}}$$

as a factorization functor

$$\mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}).$$

By combining Proposition 6.3.6, Proposition 6.3.9(a) and Corollary 6.3.11 we obtain an isomorphism

$$\mathrm{pre-pre-F}_{\mathrm{KL}(M)} \simeq \mathrm{pre-pre-F}_{\mathrm{KL}(M)}^{\mathrm{ext}} \circ ((\mathfrak{q}^{\mathrm{Miu}, \mathrm{mon-free}})^{\wedge})_*^{\mathrm{IndCoh}} \circ ((\mathfrak{q}^{\mathrm{Miu}, \mathrm{mon-free}})^{\wedge})^!$$

as factorization functors

$$\mathrm{KL}(M)_{\mathrm{crit}_M - \tilde{\rho}_P} \rightrightarrows \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}).$$

6.3.13. Let $\text{pre-}\mathbf{F}_{\text{KL}(M)}^{\text{ext}}$ be the functor obtained from $\text{pre-pre-}\mathbf{F}_{\text{KL}(M)}^{\text{ext}}$ by the procedure in Sect. 6.2.4.

It follows formally that we have an isomorphism

$$\text{pre-}\mathbf{F}_{\text{KL}(M)} \simeq \text{pre-}\mathbf{F}_{\text{KL}(M)}^{\text{ext}} \circ ((\mathbf{q}^{\text{Miu, mon-free}})^{\wedge})_*^{\text{IndCoh}} \circ ((\mathbf{q}^{\text{Miu, mon-free}})^{\wedge})^!$$

as factorization functors

$$\text{KL}(M)_{\text{crit}_M - \check{\rho}_P} \rightrightarrows \text{IndCoh}^*(\text{Op}_G^{\text{mer}}) \otimes \text{Rep}(\check{M}).$$

6.3.14. From the fact that the functors Accs and $((\mathbf{q}^{\text{Miu, mon-free}})^{\wedge})_*^{\text{IndCoh}}$ are strictly unital, it follows that the factorization algebra $\mathcal{A}$ identifies with the image of the factorization unit under the functor $\text{pre-}\mathbf{F}_{\text{KL}(M)}^{\text{ext}}$.

Hence, the homomorphism

$$\phi : \Omega(R)^{\text{Op}} \rightarrow \mathcal{A}.$$

can be interpreted as a homomorphism

$$\Omega(R)^{\text{Op}} \rightarrow \text{pre-}\mathbf{F}_{\text{KL}(M)}^{\text{ext}}(\mathbf{1}_{\text{KL}(M)_{\text{crit}_M - \check{\rho}_P}}).$$

Hence, by the same mechanism as in Sect. 6.2.5, the functor $\text{pre-}\mathbf{F}_{\text{KL}(M)}^{\text{ext}}$ upgrades to a functor

$$\mathbf{F}_{\text{KL}(M)}^{\text{ext}} : \text{KL}(M)_{\text{crit}_M - \check{\rho}_P} \rightarrow \Omega(R)^{\text{Op}}\text{-mod}^{\text{fact}}(\text{IndCoh}^*(\text{Op}_G^{\text{mer}}) \otimes \text{Rep}(\check{M})) =: \mathbf{C}.$$

It follows formally that we have an isomorphism

$$\mathbf{F}_{\text{KL}(M)} \simeq \mathbf{F}_{\text{KL}(M)}^{\text{ext}} \circ ((\mathbf{q}^{\text{Miu, mon-free}})^{\wedge})_*^{\text{IndCoh}} \circ ((\mathbf{q}^{\text{Miu, mon-free}})^{\wedge})^!$$

as factorization functors

$$\text{KL}(M)_{\text{crit}_M - \check{\rho}_P} \rightrightarrows \mathbf{C}.$$

6.4. Identification of the compositions.

6.4.1. As was mentioned above, our goal is to establish the commutativity of the pentagon (6.2). By Sects. 6.2.7 and 6.3.14, this amounts to:

- Establishing an identification

$$(6.24) \quad \text{pre-}\mathbf{F}_{\text{KL}(M)}^{\text{ext}} \simeq \text{pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec, ext}}$$

as factorization functors from

$$\text{KL}(M)_{\text{crit}_G - \check{\rho}_P} \xrightarrow{\text{FLE}_{M, \text{crit}_G + \check{\rho}_P} \circ \tau_M} \text{IndCoh}^*(\text{Op}_{\check{M}, \check{\rho}_P}^{\text{mon-free}})$$

to $\text{IndCoh}^*(\text{Op}_G^{\text{mer}}) \otimes \text{Rep}(\check{M})$.

- Showing that with respect to the resulting isomorphism of factorization algebras

$$\mathcal{A} = \text{pre-}\mathbf{F}_{\text{KL}(M)}^{\text{ext}}(\mathbf{1}_{\text{KL}(M)_{\text{crit}_G - \check{\rho}_P}}) \simeq \text{pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec, ext}}(\mathbf{1}_{\text{IndCoh}^*(\text{Op}_{\check{M}, \check{\rho}_P}^{\text{mon-free}})}) = \mathcal{A}^{\text{spec}},$$

the homomorphisms ϕ and ϕ^{spec} are canonically identified.

6.4.2. We start by constructing the isomorphism (6.24).

Note, however, that according to Sect. 6.2.4, in order to establish (6.24), it suffices to establish an isomorphism

$$(6.25) \quad \text{pre-pre-}\mathbf{F}_{\text{KL}(M)}^{\text{ext}} \simeq \text{pre-pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec, ext}}$$

as factorization functors with values in $\text{IndCoh}^*(\text{Op}_G^{\text{mer}})$.

In this subsection we will construct an isomorphism between the compositions of the two functors in (6.25) with the forgetful functor

$$\Gamma^{\text{IndCoh}}(\text{Op}_G^{\text{mer}}, -) : \text{IndCoh}^*(\text{Op}_G^{\text{mer}}) \rightarrow \text{Vect},$$

i.e.,

$$(6.26) \quad \Gamma^{\text{IndCoh}}(\text{Op}_G^{\text{mer}}, -) \circ \text{pre-pre-}\mathbf{F}_{\text{KL}(M)}^{\text{ext}} \simeq \Gamma^{\text{IndCoh}}(\text{Op}_G^{\text{mer}}, -) \circ \text{pre-pre-}\mathbf{F}_{\text{MOp}(\check{M})}^{\text{spec, ext}}.$$

6.4.3. By construction, the composition

$$\Gamma^{\text{IndCoh}}(\text{Op}_G^{\text{mer}}, -) \circ \text{pre-pre-F}_{\text{KL}(M)}^{\text{ext}}$$

identifies with

$$\begin{aligned} \text{KL}(M)_{\text{crit}_M - \tilde{\rho}_P} &\xrightarrow{\text{Wak}_{\rho_P(\omega_X)}^{-, \frac{\infty}{2}}} (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})^{-, \frac{\infty}{2}} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G}^{\mathfrak{g}^+(M)} \xrightarrow{\alpha_{\rho(\omega_X), \text{taut}}} \\ &\rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \rho(\omega_X)}^{\mathfrak{g}^+(M)_{\rho(\omega_X)}} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \rho(\omega_X)} \xrightarrow{\text{DS}_G} \text{Vect}. \end{aligned}$$

The composition

$$\Gamma^{\text{IndCoh}}(\text{Op}_G^{\text{mer}}, -) \circ \text{pre-pre-F}_{\text{MOp}(\tilde{M})}^{\text{spec, ext}} \circ (\tau_M \circ \text{FLE}_{M, \text{crit}})$$

identifies with

$$\Gamma^{\text{IndCoh}}(\text{Op}_M^{\text{mer}}, -) \circ (\tau_M \circ \text{FLE}_{M, \text{crit}}),$$

and hence with

$$\text{KL}(M)_{\text{crit}_M - \tilde{\rho}_P} \xrightarrow{\alpha_{\rho_M(\omega_X), \text{taut}}} \text{KL}(M)_{\text{crit}_M - \tilde{\rho}_P, \rho_M(\omega_X)} \rightarrow \mathfrak{m}\text{-mod}_{\text{crit}_M - \tilde{\rho}_P, \rho_M(\omega_X)} \xrightarrow{\text{DS}_M} \text{Vect}.$$

Now, the two functors match by Corollary 3.10.8.

Remark 6.4.4. Note that in computing the above compositions, the Chevalley involutions on either G or M played no role. They will, however, play a role for the next step in constructing (6.25), see Sect. 6.5.7 below.

6.5. Upgrade to factorization algebras I.

6.5.1. Consider now the functor

$$(6.27) \quad (\Gamma^{\text{IndCoh}}(\text{Op}_G^{\text{mer}}, -))^{\text{enh}} : \text{IndCoh}^*(\text{Op}_G^{\text{mer}}) \rightarrow \mathcal{O}_{\text{Op}_G^{\text{reg}}}\text{-mod}^{\text{fact}},$$

see [GLC2, Formula (4.6)].²⁹ We will now refine (6.26) to an isomorphism

$$(6.28) \quad (\Gamma^{\text{IndCoh}}(\text{Op}_G^{\text{mer}}, -))^{\text{enh}} \circ \text{pre-pre-F}_{\text{KL}(M)}^{\text{ext}} \simeq (\Gamma^{\text{IndCoh}}(\text{Op}_G^{\text{mer}}, -))^{\text{enh}} \circ \text{pre-pre-F}_{\text{MOp}(\tilde{M})}^{\text{spec, ext}}.$$

Let $\mathcal{B}$ and $\mathcal{B}^{\text{spec}}$ be the factorization algebras (in Vect), obtained by applying the functors in (6.26) to $\text{Vac}(M)_{\text{crit}_M - \tilde{\rho}_P}$, respectively.

6.5.2. The isomorphism (6.26) gives rise to an identification

$$\mathcal{B} \simeq \mathcal{B}^{\text{spec}}$$

as factorization algebras.

Now, each side in (6.28) gives rise to maps

$$\psi : \mathcal{O}_{\text{Op}_G^{\text{reg}}} \rightarrow \mathcal{B} \text{ and } \psi^{\text{spec}} : \mathcal{O}_{\text{Op}_G^{\text{reg}}} \rightarrow \mathcal{B}^{\text{spec}}.$$

The datum of (6.28) amounts to an identification

$$(6.29) \quad \psi = \psi^{\text{spec}}.$$

Note that $\mathcal{B}^{\text{spec}}$ (and hence $\mathcal{B}$) is a *classical* factorization algebra. Hence, it is enough to establish (6.29) at the pointwise level.

²⁹Note that the subscript “enh” here refers to the procedure in [GLC2, Sect. 4.1.6], rather than tensoring by $\text{I}(G, P^-)_{\rho_P(\omega_X)}$.

6.5.3. Denote

$$(6.30) \quad \mathbb{W}_{\mathfrak{g}, \text{crit}_G, \rho_P(\omega_X)} := \mathbf{oblv}_{\widehat{\mathfrak{g}}} \circ \text{Wak}_{\rho_P(\omega_X)}(\text{Vac}(M)_{\text{crit}_M - \bar{\rho}_P}),$$

cf. Sect. 3.6.3.

Note that we have a canonical identification

$$\mathbb{W}_{\mathfrak{g}, \text{crit}_G, \rho_P(\omega_X)} \simeq \mathbb{W}_{\mathfrak{g}, \text{crit}_G} := \mathbf{oblv}_{\widehat{\mathfrak{g}}} \circ \text{Wak}(\text{Vac}(M)_{\text{crit}_M - \bar{\rho}_P})$$

(indeed, the twisted by the Z_M -torsor $\rho_P(\omega)$ acts as identity on vacuum objects).

6.5.4. Consider the corresponding homomorphism

$$\mathbb{V}_{\mathfrak{g}, \text{crit}_G} \rightarrow \mathbb{W}_{\mathfrak{g}, \text{crit}_G, \rho_P(\omega_X)}.$$

By construction, the homomorphism ψ equals

$$\mathcal{O}_{\text{Op}_G^{\text{reg}}} \xrightarrow{\tau_G^*} \mathcal{O}_{\text{Op}_G^{\text{reg}}} \xrightarrow{\text{FF}_G} \mathfrak{z}_{\mathfrak{g}} \rightarrow \text{DS}_G \circ \alpha_{\rho(\omega_X), \text{taut}}(\mathbb{V}_{\mathfrak{g}, \text{crit}}) \rightarrow \text{DS}_G \circ \alpha_{\rho(\omega_X), \text{taut}}(\mathbb{W}_{\mathfrak{g}, \text{crit}_G, \rho_P(\omega_X)}).$$

6.5.5. Note, however, that the map

$$(6.31) \quad \mathfrak{z}_{\mathfrak{g}} \rightarrow \text{DS}_G \circ \alpha_{\rho(\omega_X), \text{taut}}(\mathbb{V}_{\mathfrak{g}, \text{crit}}) \rightarrow \text{DS}_G \circ \alpha_{\rho(\omega_X), \text{taut}}(\mathbb{W}_{\mathfrak{g}, \text{crit}_G, \rho_P(\omega_X)}),$$

which appears in the above composition, can be rewritten as follows:

It is known that when we view $\mathbb{W}_{\mathfrak{g}, \text{crit}_G, \rho_P(\omega_X)} \simeq \mathbb{W}_{\mathfrak{g}, \text{crit}_G}$ as a factorization module over $\mathfrak{z}_{\mathfrak{g}}$, it is *commutative*³⁰ (see, e.g., [FG2, Sect. 10]). In particular, we have an action map

$$(6.32) \quad \mathfrak{z}_{\mathfrak{g}} \otimes \mathbb{W}_{\mathfrak{g}, \text{crit}_G, \rho_P(\omega_X)} \rightarrow \mathbb{W}_{\mathfrak{g}, \text{crit}_G, \rho_P(\omega_X)}.$$

By transport of structure, we obtain a map

$$(6.33) \quad \mathfrak{z}_{\mathfrak{g}} \otimes \text{DS}_G \circ \alpha_{\rho(\omega_X), \text{taut}}(\mathbb{W}_{\mathfrak{g}, \text{crit}_G, \rho_P(\omega_X)}) \rightarrow \text{DS}_G \circ \alpha_{\rho(\omega_X), \text{taut}}(\mathbb{W}_{\mathfrak{g}, \text{crit}_G, \rho_P(\omega_X)}).$$

The map (6.31) is obtained from (6.33) by acting on the vacuum vector.

6.5.6. The map ψ^{spec} is described as follows. We identify

$$\mathcal{B}^{\text{spec}} \simeq \text{DS}_M \circ \alpha_{\rho_M(\omega_X), \text{taut}}(\text{Vac}(M)_{\text{crit}_M - \bar{\rho}_P}),$$

and in terms of this identification ψ^{spec} equals

$$\begin{aligned} \mathcal{O}_{\text{Op}_G^{\text{reg}}} &\xrightarrow{(\rho^{\text{Miu}})^*} \mathcal{O}_{\text{MOp}_{G, \bar{P}}^{\text{reg}}} \xrightarrow{\mathfrak{q}^{\text{Miura}}} \simeq \\ &\simeq \mathcal{O}_{\text{Op}_{M, \bar{\rho}_P}^{\text{reg}}} \xrightarrow{\text{FF}_M} \mathfrak{z}_{\mathfrak{m}, \bar{\rho}_M(\omega_X)} \xrightarrow{\tau_M} \mathfrak{z}_{\mathfrak{m}, -\bar{\rho}_M(\omega_X)} \simeq \text{DS}_M \circ \alpha_{\rho_M(\omega_X), \text{taut}}(\text{Vac}(M)_{\text{crit}_M - \bar{\rho}_P}). \end{aligned}$$

Note the last map in the above composition, i.e.,

$$(6.34) \quad \mathfrak{z}_{\mathfrak{m}, -\bar{\rho}_M(\omega_X)} \simeq \text{DS}_M \circ \alpha_{\rho_M(\omega_X), \text{taut}}(\text{Vac}(M)_{\text{crit}_M - \bar{\rho}_P})$$

is by definition constructed as follows (see [GLC2, Sect. 4.8.2]):

We consider $\text{Vac}(M)_{\text{crit}_M - \bar{\rho}_P}$ as acted on by $\mathfrak{z}_{\mathfrak{m}, -\bar{\rho}_M(\omega_X)}$. Hence, by transport of structure, $\text{DS}_M \circ \alpha_{\rho_M(\omega_X), \text{taut}}(\text{Vac}(M)_{\text{crit}_M - \bar{\rho}_P})$ also acquires an action of $\mathfrak{z}_{\mathfrak{m}, -\bar{\rho}_M(\omega_X)}$. The map (6.34) is given by the action of $\mathfrak{z}_{\mathfrak{m}, -\bar{\rho}_M(\omega_X)}$ on the vacuum vector of $\text{DS}_M \circ \alpha_{\rho_M(\omega_X), \text{taut}}(\text{Vac}(M)_{\text{crit}_M - \bar{\rho}_P})$.

³⁰See [GLC2, Sect. B.10.5-6] what this means.

6.5.7. Note that by the construction of the isomorphism (6.26), the isomorphism $\mathcal{B} \simeq \mathcal{B}^{\text{spec}}$ comes from the identification

$$\text{DS}_G \circ \alpha_{\rho(\omega_X), \text{taut}} \circ \text{Wak}_{\rho_P(\omega_X)}(\text{Vac}(M)_{\text{crit}_M - \bar{\rho}_P}) \simeq \text{DS}_M \circ \alpha_{\rho_M(\omega_X), \text{taut}}(\text{Vac}(M)_{\text{crit}_M - \bar{\rho}_P}).$$

Hence, in order to construct the identification (6.29), it suffices to prove the following:

The action of $\mathfrak{z}_{\mathfrak{g}}$ on $\mathbb{W}_{\mathfrak{g}, \text{crit}_G}$ in (6.32) equals the action obtained via

$$\mathfrak{z}_{\mathfrak{g}} \xrightarrow{\tau_G} \mathfrak{z}_{\mathfrak{g}} \xrightarrow{\text{FF}_G} \mathcal{O}_{\text{Op}_G^{\text{reg}}} \xrightarrow{(\text{p}^{\text{Miu}})^*} \mathcal{O}_{\text{MOp}_{\bar{G}, \bar{P}}^{\text{reg}}} \xrightarrow{\text{q}^{\text{Miura}}} \mathcal{O}_{\text{Op}_{\bar{M}, \bar{\rho}_P}^{\text{reg}}} \xrightarrow{\text{FF}_M} \mathfrak{z}_{\mathfrak{m}, \bar{\rho}_M(\omega_X)} \xrightarrow{\tau_M} \mathfrak{z}_{\mathfrak{m}, -\bar{\rho}_M(\omega_X)}$$

and the action of $\mathfrak{z}_{\mathfrak{m}, -\bar{\rho}_M(\omega_X)}$ on

$$\mathbb{W}_{\mathfrak{g}, \text{crit}_G} = \text{oblv}_{\bar{\mathfrak{g}}} \circ \text{Wak}(\text{Vac}(M)_{\text{crit}_M - \bar{\rho}_P})$$

by transport of structure.

However, the latter is the basic feature of the Feigin-Frenkel isomorphism³¹, see [Fr, Theorem 11.3] or [FG2, Sect. 10].³²

6.6. Interlude: Miura transform and the Wakimoto functor.

6.6.1. Recall that according to [GLC2, Sect. 4.6], we have an action of the monoidal factorization category

$$\text{QCoh}(\text{Op}_G^{\text{mer}}) \xrightarrow{\Upsilon_{\text{Op}_G^{\text{mer}}}} \text{IndCoh}^!(\text{Op}_G^{\text{mer}})$$

on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G}$. In what follows we will twist this action by the Chevalley involution τ_G .

Consider the induced action on the full subcategory

$$(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})^{\mathfrak{L}^+(N_P)} \subset \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G}.$$

Consider also a similar action of

$$\text{QCoh}(\text{Op}_{\bar{M}, \bar{\rho}_P}^{\text{mer}}) \xrightarrow{\Upsilon_{\text{Op}_{\bar{M}, \bar{\rho}_P}^{\text{mer}}}} \text{IndCoh}^!(\text{Op}_{\bar{M}, \bar{\rho}_P}^{\text{mer}})$$

on $\widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M - \bar{\rho}_P}$, i.e., we compose the action from [GLC2, Sect. 4.6] with τ_M .

6.6.2. Unwinding the construction in [GLC2, Sect. 4.6], from the identification of the $\mathfrak{z}_{\mathfrak{g}}$ -action on $\mathbb{W}_{\mathfrak{g}, \text{crit}_G}$ in Sect. 6.5.7 above, we obtain:

Corollary 6.6.3. *The functor*

$$\text{Wak} : \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M - \bar{\rho}_P} \rightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})^{\mathfrak{L}^+(N_P)}$$

is compatible with the above actions via the monoidal functor

$$(6.35) \quad \text{IndCoh}^!(\text{Op}_G^{\text{mer}}) \xrightarrow{(\text{p}^{\text{Miu}})^!} \text{IndCoh}^!(\text{MOp}_{\bar{G}, \bar{P}}^{\text{mer}}) \xrightarrow{\text{q}^{\text{Miu}}} \text{IndCoh}^!(\text{Op}_{\bar{M}, \bar{\rho}_P}^{\text{mer}}).$$

³¹See Corollary 6.6.5 for the explanation of the appearance of the Chevalley involution in the above formula.

³²These references treat the case of $P = B$. The case of the general parabolic follows formally, since the vacuum module for a given Levi *embeds* into the corresponding principal Wakimoto module.

6.6.4. Note now that the self-duality

$$(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}})^\vee \simeq \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}}$$

is linear with respect to the action of $\text{IndCoh}^!(\text{Op}_G^{\text{mer}})$, *up to the Chevalley involution*, see [GLC2, Theorem 8.2.3]. A similar assertion applies to G replaced by M .

Hence, from Corollary 6.6.3, we obtain:

Corollary 6.6.5. *The composite functor*

$$(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})^{\mathfrak{L}^+(N_P)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G} \xrightarrow{J_{\text{KM}}^-} \widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \bar{\rho}_P}$$

is $\text{IndCoh}^!(\text{Op}_G^{\text{mer}})$ -linear via (6.35), where:

- $\text{IndCoh}^!(\text{Op}_G^{\text{mer}})$ acts on $(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})^{\mathfrak{L}^+(N_P)}$ as in [GLC2, Sect. 4.6] (without τ_G);
- $\text{IndCoh}^!(\text{Op}_{M, \bar{\rho}_P}^{\text{mer}})$ acts on $\widehat{\mathfrak{m}}\text{-mod}_{\text{crit}_M + \bar{\rho}_P}$ by a twisted version of [GLC2, Sect. 4.6] (without τ_M).

6.7. Proof of Proposition 6.3.9.

6.7.1. First, we note that both points of the proposition are enough to check strata-wise over Ran . We can further assume that we work over a fixed point $\underline{x} \in \text{Ran}$.

6.7.2. Let $I_{P, \underline{x}} \subset \mathfrak{L}^+(G)_{\underline{x}}$ be the parahoric subgroup corresponding to P . Recall that the functor $\text{Av}_*^{\mathfrak{L}^+(N_P)}$ defines equivalences

$$(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_{G, \underline{x}}})^{-, \frac{\infty}{2}} \rightarrow (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_{G, \underline{x}}})^{I_{P, \underline{x}}}$$

and

$$\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(N_P^-)_{\underline{x}} \cdot \mathfrak{L}^+(M)_{\underline{x}, \text{ren}}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \underline{x}})^{I_{P, \underline{x}}, \text{ren}}.$$

Hence, in the statement of Proposition 6.3.9 we can replace

$$(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_{G, \underline{x}}})^{-, \frac{\infty}{2}} \rightsquigarrow (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_{G, \underline{x}}})^{I_{P, \underline{x}}};$$

$$\text{Wak}_{\rho_P(\omega_X)}^{-, \frac{\infty}{2}} \rightsquigarrow \text{Wak};$$

$$\text{I}(G, P^-)_{\rho_P(\omega_X)} \rightsquigarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \underline{x}})^{I_{P, \underline{x}}, \text{ren}}.$$

6.7.3. We regard the category $(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})^{I_{P, \underline{x}}}$ as acted on by

$$\text{QCoh}(\text{Op}_{G, \underline{x}}^{\text{mer}}) \xrightarrow{\Upsilon_{\text{Op}_{G, \underline{x}}^{\text{mer}}}} \text{IndCoh}^!(\text{Op}_{G, \underline{x}}^{\text{mer}}).$$

Let

$$(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})_{\text{mon-free}}^{I_{P, \underline{x}}} \subset (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})^{I_{P, \underline{x}}}$$

be the full subcategory consisting of objects that are set-theoretically supported over $\text{Op}_{G, \underline{x}}^{\text{mon-free}} \subset \text{Op}_{G, \underline{x}}^{\text{mer}}$.

Denote the above fully faithful embedding by $((\ell^{\text{mon-free}})^\wedge)_*^{\text{IndCoh}}$, and by $((\ell^{\text{mon-free}})^\wedge)^!$ its right adjoint.

6.7.4. It follows from Corollary 6.6.3 that the functor Wak sends $(\text{KL}(M)_{\text{crit}_M - \bar{\rho}_P})_{\text{mon-free}_{\check{G}}}$ to $(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})_{\text{mon-free}}^{I_{P,\underline{x}}}$, i.e., we have a commutative diagram

$$(6.36) \quad \begin{array}{ccc} (\text{KL}(M)_{\text{crit}_M - \bar{\rho}_P})_{\text{mon-free}_{\check{G}}} & \xrightarrow{((q^{\text{Miu, mon-free}})^{\wedge})_*^{\text{IndCoh}}} & \text{KL}(M)_{\text{crit}_M - \bar{\rho}_P} \\ \text{Wak} \downarrow & & \downarrow \text{Wak} \\ (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})_{\text{mon-free}}^{I_{P,\underline{x}}} & \xrightarrow{((\iota^{\text{mon-free}})^{\wedge})_*^{\text{IndCoh}}} & (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})^{I_{P,\underline{x}}}. \end{array}$$

Moreover, the natural transformation in the diagram

$$(6.37) \quad \begin{array}{ccc} (\text{KL}(M)_{\text{crit}_M - \bar{\rho}_P})_{\text{mon-free}_{\check{G}}} & \xleftarrow{((q^{\text{Miu, mon-free}})^{\wedge})^!} & \text{KL}(M)_{\text{crit}_M - \bar{\rho}_P} \\ \text{Wak} \downarrow & & \downarrow \text{Wak} \\ (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})_{\text{mon-free}}^{I_{P,\underline{x}}} & \xleftarrow{((\iota^{\text{mon-free}})^{\wedge})^!} & (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})^{I_{P,\underline{x}}}, \end{array}$$

obtained by passing to right adjoints along the horizontal arrows in (6.36), is an isomorphism.

6.7.5. Note now that we have an inclusion of subcategories of $(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G})^{I_{P,\underline{x}}}$:

$$(6.38) \quad \text{KL}(G)_{\text{crit}} \otimes_{\text{Sph}_G} \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\underline{x}})^{I_{P,\underline{x}}, \text{ren}} \subset (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}})_{\text{mon-free}}^{I_{P,\underline{x}}}.$$

Hence, in order to prove Proposition 6.3.9, it suffices to show that the inclusion (6.38) is an equality. In fact, we claim:

Proposition 6.7.6. *The inclusion*

$$\text{KL}(G)_{\text{crit}} \otimes_{\text{Sph}_G} \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\underline{x}}) \subset (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}})_{\text{mon-free}}$$

is an equality.

Proof. The assertion of the proposition is equivalent to the statement that $(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}})_{\text{mon-free}}$ is generated, as a category acted on by $\mathfrak{L}(G)$, by its $\mathfrak{L}(G)^+$ -equivariant objects.

A version of this statement for $\overset{\circ}{I}$ (the unipotent radical of the Iwahori subgroup I) is [Ya-D, Theorem 4.3]. One passes from $\overset{\circ}{I}$ to $\mathfrak{L}(G)^+$ using the same methods as in [FG3]:

In *loc. cit.* it is shown that the category

$$(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}})_{\text{reg}}^{\overset{\circ}{I}} \subset (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}})^{\overset{\circ}{I}},$$

consisting of objects that are set-theoretically supported over $\text{Op}_G^{\text{reg}} \subset \text{Op}_G^{\text{mer}}$, is spherically generated.

The same arguments apply to all other connected components of $\text{Op}_G^{\text{mon-free}}$. □

6.8. Upgrade to an isomorphism with values in $\text{IndCoh}^*(\text{Op}_G^{\text{mer}})$. Recall that the functor (6.27) is “almost” an equivalence. More precisely, it induces an equivalence between the eventually coconnective parts of the two sides with respect to the natural t-structures, see [GLC2, Lemma 4.2.3].

Thus, the isomorphism (6.28) constructed above implies that (6.25) holds *up to renormalization*. In this subsection, we will carry out the construction of (6.25) “as-is”.

6.8.1. Since the category $\mathrm{KL}(M)_{\mathrm{crit}_M - \bar{\rho}_P}$ is compactly generated (see [GLC2, Sect. B.11.10] for what this means), it suffices to construct an identification between the restrictions of the two functors in (6.25) to the subcategory of compact objects.

By [GLC2, Corollary 4.4.2(a)], it suffices to show that both functors in (6.25) send compact objects in $\mathrm{KL}(M)_{\mathrm{crit}_M - \bar{\rho}_P}$ to eventually coconnective objects in $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})$.

Since compact objects in $\mathrm{KL}(M)_{\mathrm{crit}_M - \bar{\rho}_P}$ are eventually coconnective (over any X^I), it suffices to show that both functors in (6.25) have a cohomological amplitude on the left (over any X^I). The latter property can be checked at the pointwise on X^I .

Thus, for the duration of this subsection we will work at a fixed point $\underline{x} \in \mathrm{Ran}$. Moreover, by factorization, we can assume that $\underline{x}$ is a singleton $\underline{x} = \{x\}$.

6.8.2. We first consider the functor $\mathrm{pre}\text{-}\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{MOp}(\bar{M})}^{\mathrm{spec}, \mathrm{ext}}$.

We claim that the functor in question is in fact t-exact. Indeed, $\mathrm{pre}\text{-}\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{MOp}(\bar{M})}^{\mathrm{spec}, \mathrm{ext}}$ is the functor (6.18), and the t-exactness follows from [GLC2, Corollary A.8.8].

6.8.3. We now show that the functor $\mathrm{pre}\text{-}\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KL}(M)}^{\mathrm{ext}}$ has a bounded cohomological amplitude.

As in Sect. 3.10.5, since the functor $\mathrm{DS}^{\mathrm{enh}, \mathrm{rfnd}}$ factors as

$$\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)} \rightarrow \mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)}) \xrightarrow{\mathrm{DS}^{\mathrm{enh}, \mathrm{rfnd}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})$$

(see [GLC2, Theorem 4.8.8]), the functor $\mathrm{pre}\text{-}\mathrm{pre}\text{-}\mathbf{F}_{\mathrm{KL}(M)}^{\mathrm{ext}}$ identifies with the composition

$$\begin{aligned} \mathrm{KL}(M)_{\mathrm{crit}_M - \bar{\rho}_P} &\xrightarrow{\mathrm{Wak}_{\rho_P(\omega_X)}} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{\mathfrak{L}^+(P)} \xrightarrow{\alpha_{\rho(\omega_X), \mathrm{taut}}} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)}^{\mathfrak{L}^+(P)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)} \xrightarrow{\mathrm{DS}^{\mathrm{enh}, \mathrm{rfnd}}} \\ &\rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \xrightarrow{\tau_G} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}), \end{aligned}$$

i.e., we can replace the semi-infinite Wakimoto functor $\mathrm{Wak}_{\rho_P(\omega_X)}^{-, \frac{\infty}{2}}$ by the “usual” Wakimoto functor $\mathrm{Wak}_{\rho_P(\omega_X)}$.

Thus, it suffices to show that the composition

$$\mathrm{KL}(M)_{\mathrm{crit}_M - \bar{\rho}_P} \xrightarrow{\mathrm{Wak}_{\rho_P(\omega_X)}} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{\mathfrak{L}^+(P)} \xrightarrow{\alpha_{\rho(\omega_X), \mathrm{taut}}} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)}^{\mathfrak{L}^+(P)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)} \xrightarrow{\mathrm{DS}^{\mathrm{enh}, \mathrm{rfnd}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})$$

has a bounded cohomological amplitude.

6.8.4. It follows from the explicit shape of the functor $\mathrm{Wak}_{\rho_P(\omega_X)}$ that it is t-exact (at the pointwise level). In addition, from the description of $\mathrm{Wak}_{\rho_P(\omega_X)}$ in Sect. 3.6.2, it follows that it factors via

$$\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{I_P} \subset \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, x}^{\mathfrak{L}^+(P)_x},$$

where $I_P \subset \mathfrak{L}^+(G)_x$ is the parahoric subgroup, corresponding to P .

Hence, it suffices to show that the functor

$$\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{I_P} \xrightarrow{\alpha_{\rho(\omega_X), \mathrm{taut}}} \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)}^{I_P, \rho(\omega_X)} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)} \xrightarrow{\mathrm{DS}^{\mathrm{enh}, \mathrm{rfnd}}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}})$$

has a bounded cohomological amplitude.

However, this follows from [Ra2, Lemma 5.3.1].³³

³³More precisely, we first *-average from $\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G}^{I_P}$ to the “baby Whittaker category”, which has a bounded amplitude, then apply [Ra2, Lemma 5.3.1], which says that the projection from the baby Whittaker category to $\mathrm{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \rho(\omega_X)})$ is t-exact, up to a shift.

6.9. Upgrade to factorization algebras II. So far, we have carried out the construction of the isomorphism of functors in the first bullet point in Sect. 6.4.1. In particular, we obtain an isomorphism of factorization algebras

$$(6.39) \quad \mathcal{A} \xrightarrow{\nu} \mathcal{A}^{\text{spec}}$$

in $\text{IndCoh}^*(\text{Op}_G^{\text{mer}}) \otimes \text{Rep}(\check{M})$.

Thus, it remains to construct an identification of the second bullet point, i.e., an identification

$$\phi \simeq \phi^{\text{spec}}$$

as maps from $\Omega(R)^{\text{Op}}$ to the two sides of (6.39).

Our strategy will be as follows: we will construct *another* identification

$$(6.40) \quad \mathcal{A} \xrightarrow{\nu'} \mathcal{A}^{\text{spec}}$$

as factorization algebras in $\text{IndCoh}^*(\text{Op}_G^{\text{mer}}) \otimes \text{Rep}(\check{M})$, which is *a priori compatible* with the maps ψ and ψ^{spec} . We will then show that ν' equals the already existing isomorphism ν .

6.9.1. Note that the isomorphism (6.24) established above implies that the pentagon (6.2) becomes commutative after we apply the forgetful functor

$$\mathbf{C} \rightarrow \text{IndCoh}^*(\text{Op}_G^{\text{mer}}) \otimes \text{Rep}(\check{M}),$$

and further the functor

$$(6.41) \quad \Gamma^{\text{IndCoh}}(\text{Op}_G^{\text{mer}}, -) \otimes \text{Id} : \text{IndCoh}^*(\text{Op}_G^{\text{mer}}) \otimes \text{Rep}(\check{M}) \rightarrow \text{Rep}(\check{M}).$$

In other words, the diagram (5.19) becomes commutative after we compose it with

$$(6.42) \quad \begin{aligned} & \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}}) \otimes_{\text{Sph}_G^{\text{spec}}} \mathbf{I}(\check{G}, \check{P}^-)^{\text{spec, loc}} \xrightarrow{\text{Id} \otimes \text{pre-F}^{\text{spec}}} \\ & \rightarrow \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}}) \otimes_{\text{Sph}_G^{\text{spec}}} \text{Rep}(\check{G}) \otimes \text{Rep}(\check{M}) \rightarrow \\ & \rightarrow \text{IndCoh}^*(\text{Op}_G^{\text{mer}}) \otimes \text{Rep}(\check{M}) \xrightarrow{\Gamma^{\text{IndCoh}}(\text{Op}_G^{\text{mer}}, -) \otimes \text{Id}} \text{Rep}(\check{M}), \end{aligned}$$

where $\text{pre-F}^{\text{spec}}$ is as in Sect. E.2.4.

6.9.2. We will take the following input from the paper [FG1, Theorem 4.11], adapted to the parahoric setting:

Theorem 6.9.3. *At the pointwise level, the two circuits of the diagram (5.19) commute when evaluated on the object $\text{Vac}(M)_{\text{crit}_M - \check{\rho}_P} \in \text{KL}(M)_{\text{crit}_M - \check{\rho}_P}$. Moreover:*

(i) *This isomorphism is compatible with the maps to each side from*

$$\mathbf{1}_{\text{IndCoh}^*(\text{Op}_G^{\text{mon-free}}) \otimes_{\text{Sph}_G^{\text{spec}}} \mathbf{I}(\check{G}, \check{P}^-)^{\text{spec, loc}}} \in \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}}) \otimes_{\text{Sph}_G^{\text{spec}}} \mathbf{I}(\check{G}, \check{P}^-)^{\text{spec, loc}};$$

(ii) *This isomorphism is compatible with the identification obtained by applying the functor (6.42).*

We will supply a proof in Sect. F.

6.9.4. Denote the two unital factorization algebras obtained by applying the two circuits of the diagram (5.19) to $\text{Vac}(M)_{\text{crit}_M - \check{P}_P}$ by

$$\tilde{\mathcal{A}} \text{ and } \tilde{\mathcal{A}}^{\text{spec}},$$

respectively.

We will use Theorem 6.9.3 to deduce the following:

Corollary 6.9.5. *The unital factorization algebras $\tilde{\mathcal{A}}$ and $\tilde{\mathcal{A}}^{\text{spec}}$ are canonically isomorphic.*

Proof. Note that for an object

$$\mathcal{F} \in \text{IndCoh}^*((\text{Op}_{\check{G}, \check{P}^-}^{\text{mon-free } \check{M}})_{\text{mon-free}}^\wedge) \simeq \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \otimes_{\text{Sph}_{\check{G}}^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}}$$

and an open subscheme

$$U \subset \text{Op}_{\check{G}, \check{P}^-}^{\text{mon-free}},$$

it makes sense to talk about the *localization* of $\mathcal{F}$ on U ; to be denoted $\mathcal{F}_U$. It comes equipped with a universal map

$$\mathcal{F} \rightarrow \mathcal{F}_U.$$

Take $U := \text{MOp}_{\check{G}, \check{P}^-}^{\text{reg}}$. Tautologically, the map

$$\mathcal{O}_{\text{Op}_{\check{G}, \check{P}^-}^{\text{mon-free}}} \xrightarrow{\sim} \mathbf{1}_{\text{IndCoh}^*((\text{Op}_{\check{G}, \check{P}^-}^{\text{mon-free } \check{M}})_{\text{mon-free}}^\wedge)} \rightarrow \tilde{\mathcal{A}}^{\text{spec}}$$

identifies $\tilde{\mathcal{A}}^{\text{spec}}$ with the localization of $\mathcal{O}_{\text{Op}_{\check{G}, \check{P}^-}^{\text{mon-free}}}$ on U .

In order to construct an isomorphism $\tilde{\mathcal{A}} \simeq \tilde{\mathcal{A}}^{\text{spec}}$, it suffices to show that the map

$$(6.43) \quad \mathbf{1}_{\text{IndCoh}^*((\text{Op}_{\check{G}, \check{P}^-}^{\text{mon-free } \check{M}})_{\text{mon-free}}^\wedge)} \rightarrow \tilde{\mathcal{A}}^{\text{spec}}$$

also identifies $\tilde{\mathcal{A}}$ with the localization of $\mathbf{1}_{\text{IndCoh}^*((\text{Op}_{\check{G}, \check{P}^-}^{\text{mon-free } \check{M}})_{\text{mon-free}}^\wedge)}$ on U .

The property of a map to be a localization on a given open can be checked strata-wise. Hence, since the map (6.43) is compatible with factorization, this property can be checked at the pointwise level.

Hence, the required property holds follows from the isomorphism stated in Theorem 6.9.3 and the compatibility in point (i). $\square$

6.9.6. Applying the functor

$$\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \otimes_{\text{Sph}_{\check{G}}^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}} \xrightarrow{\text{Op}(\check{G})^{\text{spec}}} \Omega(R)^{\text{Op-mod}^{\text{fact}}}(\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \otimes \text{Rep}(\check{M})) =: \mathbf{C},$$

to the isomorphism

$$\tilde{\mathcal{A}} \xrightarrow{\nu'} \tilde{\mathcal{A}}^{\text{spec}}$$

of Corollary 6.9.5, we obtain the sought-for isomorphism ν' in (6.40).

To finish the proof, we need to show that ν' equals the isomorphism ν of (6.39).

6.9.7. Note that both objects in (6.39) are factorization algebras and as such they are *classical* (i.e., their values on X_{dR} shifted by $[-1]$ belong to the heart of the t-structure). Hence, in order to show that two given morphisms between them are equal, it is enough to do so at the pointwise level.

At the pointwise level, in order to show that two given maps between objects of

$$(\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \otimes \text{Rep}(\check{M}))^\heartsuit$$

are equal, it is enough to show that this is the case after applying the functor (6.41).

The required assertion follows now the compatibility of point (ii) in Theorem 6.9.3.

7. PROOF OF THEOREM 5.4.5, CONTINUATION

In the previous section, we introduced the (lax) factorization category $\mathbf{C}$ and constructed the commutative diagrams (6.1) and (6.2).

In this section we will introduce the relevant t-structures and prove the first two bullet points in Sect. 6.1.1.

7.1. Structure of the argument.

7.1.1. Our proof will consist of the following steps:

- We will show that the functor

$$\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \xrightarrow{\mathbf{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}}} \mathbf{C}$$

is fully-faithful “as-is” at the pointwise level over Ran (i.e. over each $\underline{x} \in \mathrm{Ran}$);

- We will introduce a t-structure on $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$ and show that the functor $\mathbf{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}}$ is fully faithful when restricted to the eventually coconnective subcategory;
- We will show that the functor

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{\rho}_P}^{\mathrm{mon-free}}) \xrightarrow{\mathrm{co}J_{\mathrm{Op}}^-, \mathrm{spec}, \mathrm{enh}} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$$

has a cohomological amplitude bounded on the left.

Let us show how the above three properties, together with the commutativity of (6.1) and (6.2), which has been already established, imply the first two bullet points in Sect. 6.1.1, thereby completing the proof of Theorem 5.4.5.

7.1.2. First, we note that the first bullet point in Sect. 7.1.1 combined with the commutativity of (6.2) implies that the diagram

$$(7.1) \quad \begin{array}{ccc} \mathrm{KL}(M)_{\mathrm{crit}_G - \check{\rho}_P, \underline{x}} & \xrightarrow{\mathrm{FLE}_{M, \mathrm{crit}_G + \check{\rho}_P} \circ \tau_M} & \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{\rho}_P}^{\mathrm{mon-free}})_{\underline{x}} \\ \mathrm{Wak}_{\rho_P(\omega_X)}^-, \mathrm{enh} \downarrow & & \downarrow \mathrm{co}J_{\mathrm{Op}}^-, \mathrm{spec}, \mathrm{enh} \\ \mathrm{KL}(G)_{\mathrm{crit}_G, \underline{x}} \otimes_{\mathrm{Sph}_{G, \underline{x}}} \mathrm{I}(G, P^-)_{\rho_P(\omega_X), \underline{x}}^{\mathrm{loc}} & \xrightarrow{(5.18)} & \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \underline{x}}^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_{G, \underline{x}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)_{\underline{x}}^{\mathrm{spec}, \mathrm{loc}}. \end{array}$$

commutes for every $\underline{x} \in \mathrm{Ran}$.

I.e., we obtain that Theorem 5.4.5 holds at the pointwise level.

7.1.3. The t-structure on

$$\mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}}$$

is defined by transferring the structure on

$$\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$$

via the equivalence (5.18).

The fact that the functor $\mathbf{F}_{\mathrm{KM}(G)}$ is fully faithful when restricted to the eventually coconnective subcategory follows tautologically from the commutativity of (6.1) and the corresponding property of $\mathbf{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}}$.

7.1.4. Since compact objects in $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{P}}^{\mathrm{mon-free}})$ are eventually coconnective, the fact that $\mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \mathrm{enh}}$ has a cohomological amplitude bounded on the left implies that it sends compact objects to objects that are eventually coconnective.

Note that compact objects in $\mathrm{KL}(M)_{\mathrm{crit}_G - \check{P}}^{\mathrm{mon-free}}$ are also eventually coconnective. Hence, it remains to show that the functor $\mathrm{Wak}_{\rho_P(\omega_X)}^{-, \mathrm{enh}}$ has a cohomological amplitude bounded on the left.

Note, however, that this property can be checked pointwise on Ran . Recall that the functor $\mathrm{FLE}_{M, \mathrm{crit}_G + \check{P}}^{\mathrm{mon-free}}$ is t-exact at the pointwise level (see [GLC2, Corollary 6.6.9]). Hence, the required property holds from the commutativity of (7.1) and the corresponding property of $\mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \mathrm{enh}}$.

7.1.5. Thus, the rest of this section is devoted to carrying out the steps outlined in Sect. 7.1.1.

7.2. **The t-structure on the spectral side.** We will define a t-structure on

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$$

by the same principle as in Sect. A.7.1.

7.2.1. Consider the factorization space

$$\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}} \simeq \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{reg}},$$

and the factorization category $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}})$, which we rewrite as

$$(7.2) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes_{\mathrm{Rep}(\check{G})} \mathrm{Rep}(\check{P}^-).$$

We endow (7.2) with a tensor product t-structure.

7.2.2. Recall the equivalence

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \simeq \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}})$$

of (4.16), and recall the adjunction

$$(\iota)_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}}) \rightleftarrows \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}}) : (\iota)^!$$

of (4.22).

As in Proposition A.7.2 one shows that the monad $(\iota)^! \circ (\iota)_*^{\mathrm{IndCoh}}$ acting on $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}})$ is left t-exact. Hence, we obtain that

$$\begin{aligned} \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}}) &\stackrel{(4.14)}{\simeq} \\ &\simeq \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{reg}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\check{V}}(\check{G}))} \mathrm{IndCoh}^*(\mathfrak{L}_{\check{V}}(\check{G}/\check{N}_{\check{P}}^-))_{\mathfrak{L}_{\check{V}}^+(\check{M})} \end{aligned}$$

acquires a t-structure, uniquely characterized by the condition that the functor $(\iota)_*^{\mathrm{IndCoh}}$ is left t-exact.

7.2.3. By construction, an object of $\mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}})^{\wedge}_{\mathrm{mon-free}})$ is coconnective if and only if its image under $(\iota)^!$ is coconnective.

7.3. **The left t-exactness of the co-Jacquet functor.** In this subsection we show that the functor $\mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \mathrm{enh}}$ has a bounded cohomological amplitude on the left. In fact, we will show that it is left t-exact.

7.3.1. First, we note that since $\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}$ and $(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge$ are (factorization) ind-placid ind-schemes, the categories

$$\mathrm{IndCoh}^*(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) \text{ and } \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge)$$

have naturally defined t-structures, so that the functor $(\iota)_*^{\mathrm{IndCoh}}$ of (4.30) is t-exact, and its right adjoint $(\iota)^!$ is left t-exact, see [GLC2, Sects. A.8 and B.13.22].

7.3.2. Recall that the functor $\mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \mathrm{enh}}$ is the composition (4.43), where the first arrow, i.e., $((\mathbf{q}^{\mathrm{Miu}, \mathrm{mon-free}})^\wedge)^!$ is left t-exact. So it remains to show that the second arrow, i.e.,

$$(7.3) \quad \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge) \xrightarrow{j_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge)$$

is also left t-exact.

In order to show that (7.3) is left t-exact, by Sect. 7.2.3 and because the natural transformation (4.47) is an isomorphism (by Lemma 4.6.7), it suffices to show that the functor

$$\mathrm{IndCoh}^*(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) \xrightarrow{j_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-})$$

is left t-exact.

We will show that this functor is actually t-exact.

7.3.3. We have a commutative diagram

$$\begin{array}{ccc} \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mer}}) & \longleftarrow & \mathrm{IndCoh}^*(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) \\ j_*^{\mathrm{IndCoh}} \downarrow & & \downarrow j_*^{\mathrm{IndCoh}} \\ \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mer}}) & \longleftarrow & \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}), \end{array}$$

in which the horizontal arrows are given by $*$ -pullback. These horizontal arrows are t-exact and conservative.

Hence, it is enough to show that the functor

$$\mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mer}}) \xrightarrow{j_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*(\mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mer}})$$

is t-exact.

However, this holds by [GLC2, Corollary A.8.8], since

$$j : \mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mer}} \rightarrow \mathrm{pt} \times_{\mathrm{LS}_{\check{P}^-}^{\mathrm{mer}}} \mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mer}}$$

is a morphism between (factorization) ind-affine ind-schemes.

Remark 7.3.4. The above argument shows that the restriction of the functor $\mathrm{co}J_{\mathrm{Op}}^{-, \mathrm{spec}, \mathrm{enh}}$ to

$$\mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge) \subset \mathrm{IndCoh}^*(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})$$

is actually t-exact.

Indeed, the connective subcategory of $\mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge)$ is generated under colimits by the essential image of the connective subcategory of $\mathrm{IndCoh}^*(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-})$ along the functor

$$(\iota)_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) \rightarrow \mathrm{IndCoh}^*((\mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge).$$

7.4. Fully faithfulness on the spectral side. In order to complete the steps in Sect. 7.1.1, it remains to show that the functor $F_{\mathrm{Op}(\check{G})}^{\mathrm{spec}}$ is fully faithful at the pointwise level, and also when restricted to the eventually coconnective subcategory in the factorization setting.

We will prove this in the present subsection.

7.4.1. Let $R_{\check{G}}$ be the regular representation of $\check{G}$, viewed as a commutative algebra, and hence also as a factorization algebra, in $\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G})$.

By a slight abuse of notation, we will denote by the same symbol the image of $R_{\check{G}}$ under the restriction functor

$$\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G}) \rightarrow \mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{P}^-).$$

Let $R_{\check{G}}^{\mathrm{Op}}$ denote the image of $R_{\check{G}}$ under the functors

$$\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G}) \xrightarrow{(\mathrm{r}^{\mathrm{reg}})^*} \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{reg}}) \otimes \mathrm{Rep}(\check{G}) \text{ or } \mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{P}^-) \xrightarrow{(\mathrm{r}^{\mathrm{reg}})^*} \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{reg}}) \otimes \mathrm{Rep}(\check{P}^-),$$

respectively.

Remark 7.4.2. The object

$$R_{\check{G}}^{\mathrm{Op}} \in \mathrm{FactAlg}^{\mathrm{untl}}(\mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{reg}}) \otimes \mathrm{Rep}(\check{G}))$$

introduced above is closely related to the object

$$R_{\check{G}, \mathrm{Op}} \in \mathrm{FactAlg}^{\mathrm{untl}}(\mathrm{Rep}(\check{G}))$$

from [GLC2, Equation (4.8)].

Namely,

$$(\Gamma(\mathrm{Op}_{\check{G}}^{\mathrm{reg}}, -) \otimes \mathrm{Id})(R_{\check{G}}^{\mathrm{Op}}) \simeq R_{\check{G}, \mathrm{Op}}$$

as objects of $\mathrm{FactAlg}^{\mathrm{untl}}(\mathrm{Rep}(\check{G}))$.

7.4.3. Using the functor

$$\iota_*^{\mathrm{IndCoh}} : \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{reg}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}),$$

we can view $R_{\check{G}}^{\mathrm{Op}}$ as a factorization algebra in

$$\mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes \mathrm{Rep}(\check{G}) \text{ or } \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes \mathrm{Rep}(\check{P}^-).$$

Consider the resulting (lax) factorization categories

$$R_{\check{G}}^{\mathrm{Op}}\text{-mod}^{\mathrm{fact}}(\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{G})) \text{ and } R_{\check{G}}^{\mathrm{Op}}\text{-mod}^{\mathrm{fact}}(\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{P}^-)),$$

and also their full subcategories

$$R_{\check{G}}^{\mathrm{Op}}\text{-mod}^{\mathrm{fact}}(\mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}}^{\mathrm{mer}})_{\mathrm{mon-free}}^{\wedge}) \otimes \mathrm{Rep}(\check{G}))$$

and

$$R_{\check{G}}^{\mathrm{Op}}\text{-mod}^{\mathrm{fact}}(\mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}}^{\mathrm{mer}})_{\mathrm{mon-free}}^{\wedge}) \otimes \mathrm{Rep}(\check{P}^-))$$

respectively.

7.4.4. The operation of $(\mathrm{IndCoh}, *)$ -pushforward along

$$\iota^{\mathrm{mon-free}} \times \mathfrak{r} : \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} \rightarrow \mathrm{Op}_{\check{G}}^{\mathrm{mer}} \times \mathrm{LS}_{\check{G}}^{\mathrm{reg}}$$

gives rise to a (lax unital) factorization functor

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{G}).$$

This functor sends the factorization unit, i.e., $\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}} \in \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$ to $R_{\check{G}}^{\mathrm{Op}}$. Hence, it upgrades to a functor

$$(7.4) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \rightarrow R_{\check{G}}^{\mathrm{Op-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{G})),$$

see [GLC2, Sect. 4.1.6].

Tensoring over $\mathrm{Rep}(\check{G})$ with $\mathrm{Rep}(\check{P}^-)$, we obtain a functor

$$(7.5) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free}}) \rightarrow R_{\check{G}}^{\mathrm{Op-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{P}^-)).$$

7.4.5. We claim:

Proposition 7.4.6.

- (a) *The functors (7.4) and (7.5) are fully faithful at the pointwise level.*
- (b) *In the factorization setting, the functors (7.4) and (7.5) induce equivalences between the eventually coconnective subcategories of the two sides. In particular, the restrictions of the functors (7.4) and (7.5) to the eventually coconnective subcategory of the source is fully faithful.*

Proof. We first prove point (a). We will show that at the pointwise level, the functor (7.4) gives rise to an equivalence

$$(7.6) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \rightarrow R_{\check{G}}^{\mathrm{Op-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}}) \otimes \mathrm{Rep}(\check{G})).$$

The assertion for (7.5) would then follow by base change.

Recall the functor

$$(7.7) \quad \mathrm{Sph}_{\check{G}}^{\mathrm{spec}} \rightarrow R_{\check{G}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G}))$$

of (E.4). This functor is linear with respect to $\mathrm{Sph}_{\check{G}}^{\mathrm{spec}} \otimes \mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$, where:

- $\mathrm{Sph}_{\check{G}}^{\mathrm{spec}} \otimes \mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$ acts on $\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$ naturally;
- $\mathrm{Sph}_{\check{G}}^{\mathrm{spec}} \otimes \mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$ acts on $R_{\check{G}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G}))$ by acting on $\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G})$ naturally, see Sect. B.3.2.

Note that the functor (7.6) factors as

$$\begin{aligned} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) &= \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{Sph}_{\check{G}}^{\mathrm{spec}} \xrightarrow{\mathrm{Id} \otimes (7.7)} \\ &\rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \left(R_{\check{G}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G})) \right) \xrightarrow{\sim} \\ &\xrightarrow{\sim} R_{\check{G}}\text{-mod}^{\mathrm{fact}} \left(\left(\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{Rep}(\check{G}) \right) \otimes \mathrm{Rep}(\check{G}) \right) \simeq \\ &\simeq R_{\check{G}}^{\mathrm{Op-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}}^{\mathrm{mer}})^{\wedge}_{\mathrm{mon-free}}) \otimes \mathrm{Rep}(\check{G})), \end{aligned}$$

where:

- The third arrow is an equivalence due to the fact that $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})$ is dualizable and $\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$ is rigid;
- The last equivalence is [GLC2, Proposition 3.6.5].

Hence, in order to prove that (7.6) is an equivalence at the pointwise level, it suffices to show that the functor

$$(7.8) \quad \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \otimes_{\text{Sph}_{\check{G}}^{\text{spec}}} \text{Sph}_{\check{G}}^{\text{spec}} \xrightarrow{\text{Id} \otimes (7.7)} \\ \rightarrow \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \otimes_{\text{Sph}_{\check{G}}^{\text{spec}}} \left(R_{\check{G}\text{-mod}^{\text{fact}}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{G})) \right)$$

has this property.

Fix $\underline{x} \in \text{Ran}$. Note that at the level of fibers at $\underline{x}$, the functor (7.7) factors as

$$\text{Sph}_{\check{G}, \underline{x}}^{\text{spec}} \rightarrow \text{Sph}_{\check{G}, \text{temp}, \underline{x}}^{\text{spec}} \rightarrow R_{\check{G}\text{-mod}^{\text{fact}}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{G}))_{\underline{x}},$$

where the second arrow is an equivalence. Indeed, both

$$\text{Sph}_{\check{G}, \text{temp}, \underline{x}}^{\text{spec}} \text{ and } R_{\check{G}\text{-mod}^{\text{fact}}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{G}))_{\underline{x}}$$

identify with the left completion of $\text{Sph}_{\check{G}, \underline{x}}^{\text{spec}}$ with respect to the natural t-structure.

Hence, in order to prove that (7.8) is a pointwise equivalence, it suffices to show that

$$\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})_{\underline{x}} = \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})_{\underline{x}} \otimes_{\text{Sph}_{\check{G}, \underline{x}}^{\text{spec}}} \text{Sph}_{\check{G}, \underline{x}}^{\text{spec}} \rightarrow \\ \rightarrow \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})_{\underline{x}} \otimes_{\text{Sph}_{\check{G}, \underline{x}}^{\text{spec}}} \text{Sph}_{\check{G}, \text{temp}, \underline{x}}^{\text{spec}}$$

is an equivalence.

However, this is the content of [GLC2, Proposition 7.2.4].

We now prove point (b). Again, by base change, it is enough to prove the assertion about (7.4). Let $\mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}}$ be the factorization algebra in Vect from [GLC2, Sect.4.4.1].

By [GLC2, Corollary 4.4.2], the functor

$$(\Gamma^{\text{IndCoh}}(\text{Op}_{\check{G}}^{\text{mer}}, -) \otimes \text{Id}) : \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \otimes \text{Rep}(\check{G}) \rightarrow (\mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}} \otimes \mathbf{1}_{\text{Rep}(\check{G})})\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}))$$

induces an equivalence on eventually coconnective subcategories.

Note that we have a tautological map

$$\mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}} \otimes \mathbf{1}_{\text{Rep}(\check{G})} \rightarrow (\Gamma^{\text{IndCoh}}(\text{Op}_{\check{G}}^{\text{mer}}, -) \otimes \text{Id})(R_{\check{G}}^{\text{Op}})$$

as factorization algebras in $\text{Rep}(\check{G})$. Hence, the functor $\Gamma^{\text{IndCoh}}(\text{Op}_{\check{G}}^{\text{mer}}, -) \otimes \text{Id}$ induces an equivalence between the eventually coconnective subcategory of $R_{\check{G}}^{\text{Op}}\text{-mod}^{\text{fact}}(\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \otimes \text{Rep}(\check{G}))$ and the eventually coconnective subcategory of

$$(\Gamma^{\text{IndCoh}}(\text{Op}_{\check{G}}^{\text{mer}}, -) \otimes \text{Id})(R_{\check{G}}^{\text{Op}})\text{-mod}^{\text{fact}}(\text{Rep}(\check{G})).$$

Note, however, that we have a tautological identification

$$(\Gamma^{\text{IndCoh}}(\text{Op}_{\check{G}}^{\text{mer}}, -) \otimes \text{Id})(R_{\check{G}}^{\text{Op}}) \simeq R_{\check{G}, \text{Op}},$$

where $R_{\check{G}, \text{Op}}$ is as in [GLC2, Equation (4.8)].

With respect to this identification, the composite functor

$$\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \xrightarrow{(7.4)} \\ \rightarrow R_{\check{G}}^{\text{Op}}\text{-mod}^{\text{fact}}(\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) \otimes \text{Rep}(\check{G})) \xrightarrow{\Gamma^{\text{IndCoh}}(\text{Op}_{\check{G}}^{\text{mer}}, -) \otimes \text{Id}} R_{\check{G}, \text{Op}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}))$$

is the functor of [GLC2, Equation (4.10)].

The required assertion follows now by [GLC2, Proposition 4.4.7].

□

7.4.7. Note that the functor

$$\mathrm{inv}_{\check{N}^-} : \mathrm{Rep}(\check{P}^-) \rightarrow \mathrm{Rep}(\check{M})$$

induces a functor

$$\mathrm{Rep}(\check{P}^-) \rightarrow \Omega^{\mathrm{spec}\text{-mod}^{\mathrm{fact}}}(\mathrm{Rep}(\check{M})).$$

From here we obtain a functor

$$(7.9) \quad R_G^{\mathrm{Op}\text{-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{P}^-)) \rightarrow \Omega(R)^{\mathrm{Op}\text{-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M})).$$

Unwinding, we obtain a commutative diagram

$$(7.10) \quad \begin{array}{ccc} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) & \xrightarrow{(\iota)_*^{\mathrm{IndCoh}}} & \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge) \\ (7.5) \downarrow & & \downarrow \mathbf{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}} \\ R_G^{\mathrm{Op}\text{-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{P}^-)) & \xrightarrow{(7.9)} & \Omega(R)^{\mathrm{Op}\text{-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M})). \end{array}$$

As in Proposition E.3.6 one shows:

Proposition 7.4.8. *The functor (7.9) admits a continuous right adjoint. Moreover, the natural transformation in the diagram*

$$\begin{array}{ccc} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) & \xleftarrow{(\iota)^\dagger} & \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge) \\ \downarrow (7.5) & \searrow & \downarrow \mathbf{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}} \\ R_G^{\mathrm{Op}\text{-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{P}^-)) & \xleftarrow{\quad} & \Omega(R)^{\mathrm{Op}\text{-mod}^{\mathrm{fact}}}(\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}}) \otimes \mathrm{Rep}(\check{M})), \end{array}$$

obtained by passing to right adjoints along the horizontal arrows in (7.10), is an isomorphism.

7.4.9. We are now ready to prove the desired properties of the functor $\mathbf{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}}$. The adjunction (4.22) is monadic and induces a monadic adjunction on the eventually coconnective categories.

Hence, in order to prove that $\mathbf{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}}$ is fully faithful (“as-is” at the pointwise level, and on the eventually coconnective subcategories in the factorization context), it suffices to prove that the monad corresponding to the functor

$$(\iota)_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{P}^-}) \rightarrow \mathrm{IndCoh}^*((\mathrm{Op}_{\check{G}, \check{P}^-}^{\mathrm{mon-free} \check{M}})_{\mathrm{mon-free}}^\wedge)$$

maps isomorphically to the monad corresponding to the composition

$$\mathbf{F}_{\mathrm{Op}(\check{G})}^{\mathrm{spec}} \circ (\iota)_*^{\mathrm{IndCoh}}.$$

However, this follows by combining Proposition 7.4.6 and Proposition 7.4.8.

7.4.10. This concludes the proof of the properties announced in Sect. 6.1.1, and hence of Theorem 5.4.5.

Part II: Eisenstein and constant term functors

In Part II of the paper [GLC2] we introduced the basic local-to-global constructions that play a role in the geometric Langlands theory.

On the geometric side these were the functors of Whittaker coefficient

$$\text{coeff}_G : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{Whit}^!(G)_{\text{Ran}},$$

and the localization functor

$$\text{Loc}_G : \text{KL}(G)_{\text{crit,Ran}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G).$$

On the spectral side these were the spectral localization functor

$$\text{Loc}_G^{\text{spec}} : \text{Rep}(\check{G})_{\text{Ran}} \rightarrow \text{IndCoh}(\text{LS}_{\check{G}})$$

and the spectral Poincaré functor

$$\text{Poinc}_G^{\text{spec}} : \text{IndCoh}(\text{Op}_G^{\text{mon-free}})_{\text{Ran}} \rightarrow \text{IndCoh}(\text{LS}_{\check{G}}).$$

The main theme of the current paper is the interaction between the global (geometric and spectral) categories for G with their counterparts for Levi subgroups. We introduce the basic functors that connect these categories.

On the geometric side, this is the adjoint pair

$$\text{Eis}_!^- : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \rightleftarrows \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) : \text{CT}_*^-$$

and on the spectral side, this is the adjoint pair

$$\text{Eis}^{-,\text{spec}} : \text{IndCoh}(\text{LS}_{\check{M}}) \rightleftarrows \text{IndCoh}(\text{LS}_{\check{G}}) : \text{CT}^{-,\text{spec}}.$$

The goal of Part II of the current paper is to study the compositions of these functors with the local-to-global functors mentioned above. In each case, we will express a given composition as a local-to-global functor for the Levi and a functor of local nature.

The ultimate goal of these calculations is to show that the Langlands functor is compatible with the Eis and CT functors, as will be established in Part III of the present paper.

8. EISENSTEIN AND CONSTANT TERM FUNCTORS ON THE GEOMETRIC SIDE

In this section we develop the functors that connect the automorphic category for G with the automorphic categories for its Levi subgroups.

We start by introducing the most fundamental pair of mutually adjoint functors

$$\text{Eis}_!^- : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \rightleftarrows \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) : \text{CT}_*^-.$$

These functors are a direct geometric counterpart of the operators of Eisenstein series and constant term in the theory of automorphic functions.

However, by contrast with the classical theory, in the geometric context, we can enhance these functors, by (locally) tensoring up with $\text{I}(G, P^-)^{\text{loc}}$ (or its dual). This way, we obtain various versions of enhanced Eisenstein and constant term functors. These enhanced functors will be used in the proof of the main theorem, Theorem 17.1.2, in Part III of the paper.

8.1. The usual Eisenstein and constant term functors.

8.1.1. Consider the diagram

$$(8.1) \quad \text{Bun}_G \xleftarrow{p_-^-} \text{Bun}_{P^-} \xrightarrow{q_-^-} \text{Bun}_M.$$

The functors of Eisenstein and constant term are defined by pull-push along the above diagram, followed by various corrections explained in this subsection.

8.1.2. Consider the restrictions of the line bundles

$$\det_{\mathrm{Bun}_G} \text{ and } \det_{\mathrm{Bun}_M}$$

along the maps in (8.1).

Denote by $\det_{\mathrm{Bun}_{G,M}}$ their ratio:

$$\det_{\mathrm{Bun}_{G,M}} = \det_{\mathrm{Bun}_G}|_{\mathrm{Bun}_{P-}} \otimes (\det_{\mathrm{Bun}_M}|_{\mathrm{Bun}_{P-}})^{\otimes -1};$$

it naturally descends to Bun_M . By a slight abuse of notation, we will denote the resulting line bundle on Bun_M by the same symbol $\det_{\mathrm{Bun}_{G,M}}$.

According to [GLC1, Proposition 1.3.3], the line bundle $\det_{\mathrm{Bun}_{G,M}}$ admits a canonical square root, to be denoted $\det_{\mathrm{Bun}_{G,M}}^{\otimes \frac{1}{2}}$ (see Sect. 11.3.2 for the explicit formula).

In particular, the pullback of the μ_2 -gerbes

$$\det_{\mathrm{Bun}_G}^{\frac{1}{2}} \text{ and } \det_{\mathrm{Bun}_M}^{\frac{1}{2}}$$

along the maps (8.1) are canonically identified.

8.1.3. We will denote the resulting category of twisted D-modules by

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{P-}).$$

The morphisms (8.1) give rise to functors

$$(\mathfrak{p}^-)^! : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{P-})$$

and

$$(\mathfrak{q}^-)_* : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{P-}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M),$$

where the latter is continuous because the morphism $\mathfrak{q}^-$ is *safe* (see [DG1, Defn. 10.2.2] for what this means).

8.1.4. *The naive constant term functor.* We define the functor

$$\mathrm{CT}_*^{-,\mathrm{nv}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)$$

to be

$$(\mathfrak{q}^-)_* \circ (\mathfrak{p}^-)^!.$$

8.1.5. The morphism $\mathfrak{q}^-$ is smooth. Hence, the functor

$$(\mathfrak{q}^-)^* : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{P-}),$$

left adjoint to $(\mathfrak{q}^-)_*$ is well-defined.

According to [DG3, Proposition 1.1.2], the (partially defined) functor $(\mathfrak{p}^-)^!$, left adjoint to $(\mathfrak{p}^-)^!$ is defined on the essential image of $(\mathfrak{q}^-)^!$. Hence, the functor $\mathrm{CT}_*^{-,\mathrm{nv}}$ admits a left adjoint, explicitly given by

$$\mathrm{Eis}_!^{-,\mathrm{nv}} := (\mathfrak{p}^-)^! \circ (\mathfrak{q}^-)^*.$$

8.1.6. We define the corrected constant term and Eisenstein functors

$$\mathrm{Eis}_!^- : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightleftarrows \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) : \mathrm{CT}_*^-$$

by

$$(8.2) \quad \mathrm{Eis}_!^- := \mathrm{Eis}_!^{-,\mathrm{nv}}[\mathrm{shift}]$$

and

$$(8.3) \quad \mathrm{CT}_*^- := \mathrm{CT}_*^{-,\mathrm{nv}}[-\mathrm{shift}],$$

where

$$(8.4) \quad \mathrm{shift} := \dim. \mathrm{rel.}(\mathrm{Bun}_{P^-} / \mathrm{Bun}_M).$$

NB: the integer $\dim. \mathrm{rel.}(\mathrm{Bun}_{P^-} / \mathrm{Bun}_M)$ depends on the connected component of Bun_M .

8.1.7. *The ρ -shift.* Let $\mathcal{P}_{Z_M}$ be a Z_M -torsor on X . Translation by $\mathcal{P}_{Z_M}$ is an automorphism

$$\mathrm{transl}_{\mathcal{P}_{Z_M}} : \mathrm{Bun}_M \rightarrow \mathrm{Bun}_M.$$

By the construction of the line bundle $\det_{\mathrm{Bun}_M}$, it is equivariant with respect to $\mathrm{transl}_{\mathcal{P}_{Z_M}}$.

In particular, we obtain mutually inverse self-equivalences

$$(\mathrm{transl}_{\mathcal{P}_{Z_M}})_* : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightleftarrows \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) : (\mathrm{transl}_{\mathcal{P}_{Z_M}})^*.$$

We define the translated constant term and Eisenstein functors

$$\mathrm{Eis}_{!,\mathcal{P}_{Z_M}}^- : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightleftarrows \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) : \mathrm{CT}_{*,\mathcal{P}_{Z_M}}^-$$

by

$$\mathrm{Eis}_{!,\mathcal{P}_{Z_M}}^- := \mathrm{Eis}_!^- \circ (\mathrm{transl}_{\mathcal{P}_{Z_M}})_* \text{ and } \mathrm{CT}_{*,\mathcal{P}_{Z_M}}^- := (\mathrm{transl}_{\mathcal{P}_{Z_M}})^* \circ \mathrm{CT}_*^-.$$

We will mostly apply the above construction when $\mathcal{P}_{Z_M} = \rho_P(\omega_X)$.

8.1.8. *The constant term functor for a general level.* Let κ be a level for $\mathfrak{g}$, and recall (see Sect. 3.2.1) that we use the same character κ to denote the corresponding level for $\mathfrak{m}$.

We let

$$\mathrm{CT}_*^- : \mathrm{D}\text{-mod}_{\mathrm{crit}_G + \kappa}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\mathrm{crit}_M + \kappa}(\mathrm{Bun}_M)$$

denote the composition

$$\begin{aligned} \mathrm{D}\text{-mod}_{\mathrm{crit}_G + \kappa}(\mathrm{Bun}_G) &= \\ &= \mathrm{D}\text{-mod}_{\frac{1}{2} \mathrm{dlog}(\det(\mathrm{Bun}_G)) + \kappa}(\mathrm{Bun}_G) \xrightarrow{(\mathfrak{p}^-)^!} \mathrm{D}\text{-mod}_{\frac{1}{2} \mathrm{dlog}(\det(\mathrm{Bun}_G)|_{\mathrm{Bun}_{P^-}}) + \kappa}(\mathrm{Bun}_{P^-}) \simeq \\ &\simeq \mathrm{D}\text{-mod}_{\frac{1}{2} \mathrm{dlog}(\det(\mathrm{Bun}_M)|_{\mathrm{Bun}_{P^-}}) + \mathrm{dlog}(\det_{\mathrm{Bun}_G,M}^{\otimes \frac{1}{2}}) + \kappa}(\mathrm{Bun}_{P^-}) \xrightarrow{\otimes \det_{\mathrm{Bun}_G,M}^{\otimes -\frac{1}{2}}} \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2} \mathrm{dlog}(\det(\mathrm{Bun}_M)|_{\mathrm{Bun}_{P^-}}) + \kappa}(\mathrm{Bun}_{P^-}) \xrightarrow{(\mathfrak{q}^-)^*} \mathrm{D}\text{-mod}_{\frac{1}{2} \mathrm{dlog}(\det(\mathrm{Bun}_M)) + \kappa}(\mathrm{Bun}_M) = \\ &= \mathrm{D}\text{-mod}_{\mathrm{crit}_M + \kappa}(\mathrm{Bun}_M) \xrightarrow{[-\mathrm{shift}]} \mathrm{D}\text{-mod}_{\mathrm{crit}_M + \kappa}(\mathrm{Bun}_M). \end{aligned}$$

8.2. The geometric (co-)enhanced category.

8.2.1. Let $\mathcal{Z}$ be a prestack mapping to Ran . Consider the category

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{Z}}^{-,\mathrm{enh}} := \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \right)_{\mathrm{Sph}_M, \mathcal{Z}} \otimes \mathrm{I}(G, P^-)_{\mathcal{Z}}^{\mathrm{loc}}.$$

The assignment

$$\mathcal{Z} \rightsquigarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{Z}}^{-,\mathrm{enh}}$$

is a unital crystal of categories over Ran (see [GLC2, Sect. 11.2.1] for the terminology), to be denoted $\underline{\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)}^{-,\mathrm{enh}}$.

8.2.2. The monadic adjunction

$$\mathbf{ind}_{\mathrm{Sph} \rightarrow \frac{\infty}{2}} : \mathrm{Sph}_M \rightleftarrows \mathrm{I}(G, P^-)^{\mathrm{loc}} : \mathbf{oblv}_{\frac{\infty}{2} \rightarrow \mathrm{Sph}}$$

gives rise to a monadic adjunction

$$(8.5) \quad \mathbf{ind}_{\mathrm{enh}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathbb{Z}) \rightleftarrows \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathbb{Z}}^{-, \mathrm{enh}} : \mathbf{oblv}_{\mathrm{enh}}.$$

8.2.3. Here is another way to think about the category $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathbb{Z}}^{-, \mathrm{enh}}$.

Let $\tilde{\Omega}$ be as in Sect. 1.3.7. We can consider

$$\tilde{\Omega}_{\mathbb{Z}} \in \mathrm{AssocAlg}(\mathrm{Sph}_{M, \mathbb{Z}}),$$

where we consider $\mathrm{Sph}_{M, \mathbb{Z}}$ as a monoidal category.

Consider $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathbb{Z})$ as a right module category over $\mathrm{Sph}_{M, \mathbb{Z}}$. Then we have

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathbb{Z}}^{-, \mathrm{enh}} \simeq \tilde{\Omega}_{\mathbb{Z}}\text{-mod}^r(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathbb{Z})),$$

so that the adjunction (8.5) corresponds to

$$\mathbf{ind}_{\tilde{\Omega}_{\mathbb{Z}}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightleftarrows \tilde{\Omega}_{\mathbb{Z}}\text{-mod}^r(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathbb{Z})) : \mathbf{oblv}_{\tilde{\Omega}_{\mathbb{Z}}}.$$

8.2.4. Let

$$(8.6) \quad \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)^{-, \mathrm{enh}} \right)_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}}$$

be the corresponding “independent” category (see [GLC2, Sect. H.1]).

Note that it fits into the pullback square

$$(8.7) \quad \begin{array}{ccc} \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)^{-, \mathrm{enh}} \right)_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}} & \longrightarrow & \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{Ran}}^{-, \mathrm{enh}} \right) \\ \mathbf{oblv}_{\mathrm{enh}} \downarrow & & \downarrow \mathbf{oblv}_{\mathrm{enh}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}^{\mathrm{untl}})_{\mathrm{indep}} & \longrightarrow & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran}) \\ \sim \downarrow & & \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M). & & \end{array}$$

Remark 8.2.5. The contents of this remark will not be used in the sequel:

One can show that the category $\left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)^{-, \mathrm{enh}} \right)_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}}$ is equivalent to the category $\mathrm{I}(G, P^-)^{\mathrm{glob}}$ from [Ga1, Sect. 6.1], defined as follows:

For a group H , let $\mathrm{Bun}_{H, \mathrm{gen}}$ denote the prestack of H -bundles defined generically on X . There exists a tautological map

$$\mathrm{Bun}_H \rightarrow \mathrm{Bun}_{H, \mathrm{gen}}.$$

Consider the prestack

$$\mathrm{Bun}_{P^-, \mathrm{gen}} \times_{\mathrm{Bun}_{G, \mathrm{gen}}} \mathrm{Bun}_G.$$

The category $\mathrm{I}(G, P^-)^{\mathrm{glob}}$ is by definition the fiber product

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{P^-, \mathrm{gen}} \times_{\mathrm{Bun}_{G, \mathrm{gen}}} \mathrm{Bun}_G) \times_{\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{P^-})} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M),$$

where:

- The twisting on $\mathrm{Bun}_{P^-, \mathrm{gen}} \times_{\mathrm{Bun}_{G, \mathrm{gen}}} \mathrm{Bun}_G$ is pulled back from Bun_G ;

- The functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{P^-})$$

is $(q^-)^*$;

- The functor $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{P^-, \mathrm{gen}} \times_{\mathrm{Bun}_{G, \mathrm{gen}}} \mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{P^-})$ is $!$ -pullback with respect to

$$\mathrm{Bun}_{P^-} \rightarrow \mathrm{Bun}_{P^-, \mathrm{gen}} \times_{\mathrm{Bun}_{G, \mathrm{gen}}} \mathrm{Bun}_G.$$

In other words, $I(G, P^-)^{\mathrm{glob}}$ is the full subcategory of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{P^-, \mathrm{gen}} \times_{\mathrm{Bun}_{G, \mathrm{gen}}} \mathrm{Bun}_G)$, consisting of objects, whose $!$ -pullback to Bun_{P^-} lies in the essential image of the (fully faithful) functor $(q^-)^*$:

$$(8.8) \quad \begin{array}{ccc} I(G, P^-)^{\mathrm{glob}} & \longrightarrow & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{P^-, \mathrm{gen}} \times_{\mathrm{Bun}_{G, \mathrm{gen}}} \mathrm{Bun}_G) \\ \downarrow & & \downarrow \text{!-pullback} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{(q^-)^*} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{P^-, \mathrm{gen}}). \end{array}$$

Under the equivalence

$$\left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)^{-, \mathrm{enh}} \right)_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}} \simeq I(G, P^-)^{\mathrm{glob}},$$

the left vertical arrow in (8.7) corresponds to the left vertical arrow in (8.8).

8.2.6. We define a unital crystal of categories $\underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}(\mathrm{Bun}_G)^{-, \mathrm{enhco}}$ over Ran by

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathcal{Z}}^{-, \mathrm{enhco}} := \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \right)_{\mathrm{Sph}_{G, \mathcal{Z}}} \otimes I(G, P^-)_{\mathrm{co}, \mathcal{Z}}^{\mathrm{loc}}.$$

8.3. The *co-enhanced constant term* functor.

8.3.1. Our first goal is to define the co-enhanced constant term functor

$$(8.9) \quad \underline{\mathrm{CT}}_*^{-, \mathrm{enhco}} : \underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}(\mathrm{Bun}_G)^{-, \mathrm{enhco}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \underline{\mathrm{D}\text{-mod}}(\mathrm{Ran}),$$

which fits into the formalism of “local-to-global” functors of [GLC2, Sect. 11.1].

I.e. we wish to define a family of functors

$$(8.10) \quad \mathrm{CT}_{*, \mathcal{Z}}^{-, \mathrm{enhco}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathcal{Z}}^{-, \mathrm{enhco}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}), \quad \mathcal{Z} \rightarrow \mathrm{Ran}.$$

Following the convention of [GLC2, Sect. 11.1.4], we will denote by $\mathrm{CT}_*^{-, \mathrm{enhco}}$ the resulting functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Ran}}^{-, \mathrm{enhco}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M).$$

8.3.2. In order to simplify the notation, we will consider the case when $\mathcal{Z} = \mathrm{pt}$, and $\mathcal{Z} \rightarrow \mathrm{Ran}$ corresponds to a point $\underline{x} \in \mathrm{Ran}$. Thus, we will define a functor

$$(8.11) \quad \mathrm{CT}_{*, \underline{x}}^{-, \mathrm{enhco}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\underline{x}}^{-, \mathrm{enhco}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M).$$

8.3.3. Consider the Hecke global stack

$$\mathrm{Bun}_G \xleftarrow{\bar{h}_G} \mathrm{Hecke}_{G,\underline{x}}^{\mathrm{glob}} \xrightarrow{\bar{h}_G} \mathrm{Bun}_G.$$

Denote by $\mathbf{s}_{\underline{x}}$ the projection

$$\mathrm{Hecke}_{G,\underline{x}}^{\mathrm{glob}} \rightarrow \mathrm{Hecke}_{G,\underline{x}}^{\mathrm{loc}} := \mathfrak{L}^+(G)_{\underline{x}} \backslash \mathrm{Gr}_{G,\underline{x}}.$$

Note that we have a canonical identification of line bundles on $\mathrm{Hecke}_{G,\underline{x}}^{\mathrm{glob}}$

$$\bar{h}_G^*(\det_{\mathrm{Bun}_G}) \simeq \bar{h}_G^*(\det_{\mathrm{Bun}_G}) \otimes \mathbf{s}^*(\det_{\mathrm{Gr}_{G,\underline{x}}}).$$

In particular, we have a canonical identification of μ_2 -gerbes

$$(8.12) \quad \bar{h}_G^*(\det_{\mathrm{Bun}_G}^{\frac{1}{2}}) \simeq \bar{h}_G^*(\det_{\mathrm{Bun}_G}^{\frac{1}{2}}) \otimes \mathbf{s}^*(\det_{\mathrm{Gr}_{G,\underline{x}}}^{\frac{1}{2}}).$$

8.3.4. Consider the fiber product

$$(8.13) \quad \mathrm{Hecke}_{G,P^-, \underline{x}}^{\mathrm{glob}} := \mathrm{Hecke}_{G,\underline{x}}^{\mathrm{glob}} \xrightarrow{\bar{h}_{G,\mathrm{Bun}_G}} \times \mathrm{Bun}_{P^-}.$$

Denote the resulting maps by

$$\mathrm{Bun}_G \xleftarrow{\bar{h}_{G,P^-}} \mathrm{Hecke}_{G,P^-, \underline{x}}^{\mathrm{glob}} \xrightarrow{\bar{h}_{G,P^-}} \mathrm{Bun}_{P^-}.$$

We have a natural map

$$(8.14) \quad \mathrm{Hecke}_{G,P^-, \underline{x}}^{\mathrm{glob}} \rightarrow \mathfrak{L}^+(P^-)_{\underline{x}} \backslash \mathrm{Gr}_{G,\underline{x}},$$

which we denote by the same symbol $\mathbf{s}_{\underline{x}}$.

By Sect. 8.1.2, the identification (8.12) gives rise to an identification of μ_2 -gerbes

$$(8.15) \quad \bar{h}_{G,P^-}^*(\det_{\mathrm{Bun}_G}^{\frac{1}{2}}) \otimes \mathbf{s}^*(\det_{\mathrm{Gr}_{G,\underline{x}}}^{\frac{1}{2}}) \simeq (\bar{h}_{G,P^-})^* \circ (\mathbf{q}^-)^*(\det_{\mathrm{Bun}_M}^{\frac{1}{2}}).$$

8.3.5. Note that we can think of $\mathrm{Hecke}_{G,P^-, \underline{x}}^{\mathrm{glob}}$ as a twisted product

$$\mathrm{Gr}_{G,\underline{x}} \widetilde{\times} \mathrm{Bun}_{P^-}.$$

We will denote by $\mathbf{s}_{\underline{x}}^!$ the naturally defined functor

$$\mathrm{D}\text{-mod}(\mathrm{Gr}_{G,\underline{x}})^{\mathfrak{L}^+(P^-)_{\underline{x}}} \rightarrow \mathrm{D}\text{-mod}(\mathrm{Hecke}_{G,P^-, \underline{x}}^{\mathrm{glob}}), \quad \mathcal{F} \mapsto \mathcal{F} \widetilde{\boxtimes} \omega_{\mathrm{Bun}_{P^-}}.$$

We will use the same symbol $\mathbf{s}_{\underline{x}}^!$ for the corresponding functor in the context of twisted D-modules.

8.3.6. Thus, we have a well-defined functor

$$(8.16) \quad (\bar{h}_{G,P^-})_* \left((\bar{h}_{G,P^-})^!(-) \otimes \mathbf{s}_{\underline{x}}^!(-) \right),$$

which maps

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\underline{x}})^{\mathfrak{L}^+(P^-)_{\underline{x}}} \rightarrow \mathrm{D}\text{-mod}_{(\mathbf{q}^-)^*(\det_{\mathrm{Bun}_M}^{\frac{1}{2}})}(\mathrm{Bun}_{P^-}).$$

8.3.7. Consider the functor

$$(8.17) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\underline{x}})^{\mathfrak{L}^+(P^-)_{\underline{x}}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)$$

equal to the composition of (8.16) with:

- The functor

$$\mathbf{q}_*^- : \mathrm{D}\text{-mod}_{(\mathbf{q}^-)^*(\det_{\mathrm{Bun}_M}^{\frac{1}{2}})}(\mathrm{Bun}_{P^-}) \rightarrow \mathrm{D}\text{-mod}_{\det_{\mathrm{Bun}_M}^{\frac{1}{2}}}(\mathrm{Bun}_M) =: \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M);$$

- The functor of cohomological shift to the right by (8.4) over a given connected component of Bun_M .

8.3.8. We claim:

Lemma 8.3.9. *The functor (8.17) factors via the quotient*

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\underline{x}})^{\mathfrak{L}^+(P^-)_{\underline{x}}} \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\underline{x}})_{\mathfrak{L}^+(N_P^-)_{\underline{x}}}^{\mathfrak{L}^+(M)_{\underline{x}}} \twoheadrightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\underline{x}})_{\mathfrak{L}(N_P^-)_{\underline{x}}}^{\mathfrak{L}^+(M)_{\underline{x}}}$$

along the second factor.

The proof will be given in Sect. 8.4.

8.3.10. Composing with

$$\mathrm{I}(G, P^-)_{\mathrm{co},\underline{x}}^{\mathrm{loc}} := \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\underline{x}})_{\mathfrak{L}(N_P^-)_{\underline{x}}}^{\mathfrak{L}^+(M)_{\underline{x}},\mathrm{ren}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\underline{x}})_{\mathfrak{L}(N_P^-)_{\underline{x}}}^{\mathfrak{L}^+(M)_{\underline{x}}},$$

we obtain a functor

$$(8.18) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{I}(G, P^-)_{\mathrm{co},\underline{x}}^{\mathrm{loc}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M),$$

which we denote by $\mathrm{pre}\text{-CT}_{*,\underline{x}}^{-,\mathrm{enhco}}$

8.3.11. Unwinding the construction, we obtain that the functor $\mathrm{pre}\text{-CT}_{*,\underline{x}}^{-,\mathrm{enhco}}$ factors as

$$(8.19) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{I}(G, P^-)_{\mathrm{co},\underline{x}}^{\mathrm{loc}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{Sph}_{G,\underline{x}}} \mathrm{I}(G, P^-)_{\mathrm{co},\underline{x}}^{\mathrm{loc}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M).$$

The second arrow in (8.19) is the sought-for functor $\mathrm{CT}_{*,\underline{x}}^{-,\mathrm{enhco}}$ of (8.11).

Remark 8.3.12. Note that by the same mechanism as in Lemma 3.7.4, the functor

$$\begin{aligned} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\underline{x}}^{-,\mathrm{enhco}} &= \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{Sph}_{G,\underline{x}}} \mathrm{I}(G, P^-)_{\mathrm{co},\underline{x}}^{\mathrm{loc}} \rightarrow \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{Sph}_{G,\underline{x}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\underline{x}})_{\mathfrak{L}(N_P^-)_{\underline{x}}}^{\mathfrak{L}^+(M)_{\underline{x}}} \end{aligned}$$

is an equivalence.

8.3.13. By construction, the composition

$$\begin{aligned} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) &\xrightarrow{\mathrm{Id} \otimes \mathbf{1}_{\mathrm{I}(G, P^-)_{\mathrm{co},\underline{x}}^{\mathrm{loc}}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{I}(G, P^-)_{\mathrm{co},\underline{x}}^{\mathrm{loc}} \rightarrow \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{Sph}_{G,\underline{x}}} \mathrm{I}(G, P^-)_{\mathrm{co},\underline{x}}^{\mathrm{loc}} = \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\underline{x}}^{-,\mathrm{enhco}} \xrightarrow{\mathrm{CT}_{*,\underline{x}}^{-,\mathrm{enhco}}} \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \end{aligned}$$

recovers the functor CT_*^- .

In families, the composition

$$\begin{aligned} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) &\xrightarrow{\mathrm{Id} \otimes \mathbf{1}_{\mathrm{I}(G, P^-)_{\mathrm{co},\mathcal{Z}}^{\mathrm{loc}}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{I}(G, P^-)_{\mathrm{co},\mathcal{Z}}^{\mathrm{loc}} \rightarrow \\ &\rightarrow (\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z})) \otimes_{\mathrm{Sph}_{G,\mathcal{Z}}} \mathrm{I}(G, P^-)_{\mathrm{co},\mathcal{Z}}^{\mathrm{loc}} = \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathcal{Z}}^{-,\mathrm{enhco}} \xrightarrow{\mathrm{CT}_{*,\mathcal{Z}}^{-,\mathrm{enhco}}} \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \end{aligned}$$

identifies with $\mathrm{CT}_*^- \otimes \mathrm{Id}_{\mathrm{D}\text{-mod}(\mathcal{Z})}$.

8.3.14. By construction, the functor (8.11) respects the actions of $\mathrm{Sph}_{M,x}$ on the two sides.

In families, the functor (8.10) respects the actions of $\mathrm{Sph}_{M,\mathcal{Z}}$ on the two sides.

8.3.15. Unwinding the construction, we obtain that the resulting functor (8.9) has a natural unital structure, in the sense of [GLC2, Sect. 11.3].

8.4. Proof of Lemma 8.3.9.

8.4.1. In order to simplify the exposition, we prove the assertion at the level of $!$ -fibers at the trivial M -bundle $\mathcal{P}_M^0 \in \text{Bun}_M$.

Denote:

$$\text{Hecke}_{G, N_P^-, \underline{x}}^{\text{glob}} := \text{pt}_{\mathcal{P}_M^0, \text{Bun}_M} \times \text{Hecke}_{G, P^-, \underline{x}}^{\text{glob}}$$

and

$$\text{Bun}_G \xleftarrow{h_{G, N_P^-}} \text{Hecke}_{G, N_P^-, \underline{x}}^{\text{glob}} \xrightarrow{h_{G, N_P^-}} \text{Bun}_{N_P^-}.$$

We denote by the same symbol $\mathbf{s}_{\underline{x}}$ the resulting map

$$\text{Hecke}_{G, N_P^-, \underline{x}}^{\text{glob}} \rightarrow \mathfrak{L}^+(N_P^-)_{\underline{x}} \backslash \text{Gr}_{G, \underline{x}}.$$

Consider the resulting functor

$$(8.20) \quad (\vec{h}_{G, N_P^-})_* \left((\overleftarrow{h}_{G, N_P^-})^!(-) \otimes \mathbf{s}_{\underline{x}}^!(-) \right),$$

which maps

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(N_P^-)_{\underline{x}}} \rightarrow \text{D-mod}(\text{Bun}_{N_P^-}).$$

We will show that the composite functor

$$(8.21) \quad \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(N_P^-)_{\underline{x}}} \xrightarrow{(8.20)} \text{D-mod}(\text{Bun}_{N_P^-}) \xrightarrow{C^*(\text{Bun}_{N_P^-}, -)} \text{Vect}$$

factors via

$$\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(N_P^-)_{\underline{x}}} \simeq \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \underline{x}})_{\mathfrak{L}^+(N_P^-)_{\underline{x}}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \underline{x}})_{\mathfrak{L}(N_P^-)_{\underline{x}}}$$

along the second factor.

8.4.2. Let $\text{Bun}_{N_P^-}^{\text{level}_{\underline{x}}}$ be the moduli stack of N_P^- -bundles on X with a full level structure at $\underline{x}$. Parallel to (8.20) we have a functor

$$(8.22) \quad \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \underline{x}}) \rightarrow \text{D-mod}(\text{Bun}_{N_P^-}^{\text{level}_{\underline{x}}}),$$

which is $\mathfrak{L}(N_P^-)_{\underline{x}}$ -equivariant, so that the functor (8.20) is obtained from (8.22) by taking $\mathfrak{L}^+(N_P^-)_{\underline{x}}$ -invariants/co-invariants on both sides.

Consider the functor

$$(8.23) \quad \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \underline{x}}) \xrightarrow{(8.20)} \text{D-mod}(\text{Bun}_{N_P^-}^{\text{level}_{\underline{x}}}) \xrightarrow{C^*(\text{Bun}_{N_P^-}^{\text{level}_{\underline{x}}}, -)} \text{Vect}$$

It naturally factors via $\mathfrak{L}^+(N_P^-)_{\underline{x}}$ -coinvariants along the second factor, and the resulting functor

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \underline{x}})_{\mathfrak{L}^+(N_P^-)_{\underline{x}}} \rightarrow \text{Vect}$$

is the functor (8.21).

8.4.3. Now the desired assertion follows from the fact that the functor (8.23) is $\mathfrak{L}(N_P^-)_{\underline{x}}$ -equivariant, since both arrows that comprise it have this property.

8.5. **The enhanced Eisenstein and constant term functors.** In this subsection, we will use duality and adjunction manipulations to produce from $\underline{\text{CT}}_*^{\text{enhco}}$ other enhanced functors.

8.5.1. By duality (see Sect. 1.4.8), the functor (8.10) gives rise to a $\mathrm{Sph}_{G,\mathbb{Z}}$ -linear functor

$$(8.24) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathbb{Z}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathbb{Z}}^{-,\mathrm{enh}}.$$

We denote the functor (8.24) by $\mathrm{CT}_{*,\mathbb{Z}}^{-,\mathrm{enh}}$ and refer to it as the *enhanced constant term* functor.

We will view (8.24) as a map between crystals of categories on Ran , to be denoted $\underline{\mathrm{CT}}_*^{-,\mathrm{enh}}$.

8.5.2. By construction, the composition

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathbb{Z}) \xrightarrow{\mathrm{CT}_{*,\mathbb{Z}}^{-,\mathrm{enh}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathbb{Z}}^{-,\mathrm{enh}} \xrightarrow{\mathrm{oblv}_{\mathrm{enh}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathbb{Z})$$

identifies with $\mathrm{CT}_*^{-} \otimes \mathrm{Id}_{\mathrm{D}\text{-mod}(\mathbb{Z})}$.

8.5.3. Note that the datum of $\underline{\mathrm{CT}}_*^{-,\mathrm{enh}}$ (without the action of Sph_G) is recovered from the datum of the functor

$$(8.25) \quad \mathrm{CT}_{*,\mathrm{Ran}}^{-,\mathrm{enh}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{Ran}}^{-,\mathrm{enh}}.$$

The fact that (8.9) is unital implies that the essential image of (8.25) is contained in the full subcategory

$$\left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}}^{-,\mathrm{enh}} \right) \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{Ran}}^{-,\mathrm{enh}},$$

see Remark 8.2.5.

Remark 8.5.4. In terms of the equivalence

$$\left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}}^{-,\mathrm{enh}} \right) \simeq \mathrm{I}(G, P^-)^{\mathrm{glob}},$$

the resulting functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{I}(G, P^-)^{\mathrm{glob}}$$

identifies with the enhanced constant term functor of [Gal, Sect. 6.3.1].

8.5.5. We claim now that the functor $\mathrm{CT}_{*,\mathbb{Z}}^{-,\mathrm{enh}}$ admits a left adjoint, to be denoted $\mathrm{Eis}_{!,\mathbb{Z}}^{-,\mathrm{enh}}$.

Indeed, it suffices to show that $\mathrm{Eis}_{!,\mathbb{Z}}^{-,\mathrm{enh}}$ is defined on the generators of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathbb{Z}}^{-,\mathrm{enh}}$. Hence, it is enough to show that the functor

$$\mathrm{Eis}_{!,\mathbb{Z}}^{-,\mathrm{enh}} \circ \mathbf{ind}_{\mathrm{enh}},$$

left adjoint to $\mathbf{oblv}_{\mathrm{enh}} \circ \mathrm{CT}_{*,\mathbb{Z}}^{-,\mathrm{enh}}$ is well-defined.

However, by Sect. 8.5.2,

$$\mathbf{oblv}_{\mathrm{enh}} \circ \mathrm{CT}_{*,\mathbb{Z}}^{-,\mathrm{enh}} \simeq \mathrm{CT}_*^{-} \otimes \mathrm{Id}_{\mathrm{D}\text{-mod}(\mathbb{Z})},$$

and hence

$$\mathrm{Eis}_{!,\mathbb{Z}}^{-,\mathrm{enh}} \circ \mathbf{ind}_{\mathrm{enh}} \simeq \mathrm{Eis}_!^{-} \otimes \mathrm{Id}_{\mathrm{D}\text{-mod}(\mathbb{Z})}.$$

Since $\mathrm{Sph}_{G,\mathbb{Z}}$ is rigid (over $\mathbb{Z}$), the fact that $\mathrm{CT}_{*,\mathbb{Z}}^{-,\mathrm{enh}}$ is $\mathrm{Sph}_{G,\mathbb{Z}}$ -linear implies that $\mathrm{Eis}_{!,\mathbb{Z}}^{-,\mathrm{enh}}$ inherits this structure.

8.5.6. The assignment

$$\mathcal{Z} \rightsquigarrow \mathrm{Eis}_{!,\mathcal{Z}}^{-,\mathrm{enh}}$$

is a “local-to-global” functor in the sense of [GLC2, Sect. 11.1]. We denote it by

$$\underline{\mathrm{Eis}}_{!}^{-,\mathrm{enh}} : \underline{\mathrm{D-mod}}_{\frac{1}{2}}(\mathrm{Bun}_M)^{-,\mathrm{enh}} \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}).$$

Following the conventions of [GLC2, Sect. 11.1.4], we will denote by $\mathrm{Eis}_{!}^{-,\mathrm{enh}}$ the resulting functor

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{Ran}}^{-,\mathrm{enh}} \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G).$$

The fact that the essential image of $\mathrm{CT}_{*,\mathrm{Ran}}^{-,\mathrm{enh}}$ is contained in

$$\left(\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)^{-,\mathrm{enh}} \right)_{\mathrm{Ran}^{\mathrm{un}1}, \mathrm{indep}}$$

implies that the functor $\underline{\mathrm{Eis}}_{!}^{-,\mathrm{enh}}$ carries a unital structure.

8.5.7. Let $\mathcal{A}$ be a factorization algebra in $\mathrm{I}(G, P^{-})^{\mathrm{loc}}$. We define the functor

$$\mathrm{Eis}_{!,\mathcal{A}}^{-} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

to be the composition

$$\begin{aligned} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) &\xrightarrow{\mathrm{Id} \otimes \mathcal{A}_{\mathrm{Ran}}} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{I}(G, P^{-})_{\mathrm{Ran}}^{\mathrm{loc}} \rightarrow \\ &\rightarrow (\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}(\mathrm{Ran})) \otimes_{\mathrm{Sph}_{M,\mathrm{Ran}}} \mathrm{I}(G, P^{-})_{\mathrm{Ran}}^{\mathrm{loc}} = \\ &= \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{Ran}}^{-,\mathrm{enh}} \xrightarrow{\mathrm{Eis}_{!}^{-,\mathrm{enh}}} \mathrm{D-mod}(\mathrm{Bun}_G). \end{aligned}$$

Taking $\mathcal{A}$ to be

$$\Delta^{-,\frac{\infty}{2}}, \nabla^{-,\frac{\infty}{2}} \text{ and } \mathrm{IC}^{-,\frac{\infty}{2}},$$

respectively, we obtain the functors

$$(8.26) \quad \mathrm{Eis}_{!,\Delta}^{-} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G),$$

$$(8.27) \quad \mathrm{Eis}_{!,\nabla}^{-} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

and

$$(8.28) \quad \mathrm{Eis}_{!,\mathrm{IC}}^{-} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G).$$

Since $\Delta^{-,\frac{\infty}{2}}$ is the factorization unit in $\mathrm{I}(G, P^{-})^{\mathrm{loc}}$, by unitality we have

$$\mathrm{Eis}_{!,\Delta}^{-} \simeq \mathrm{Eis}_{!}^{-}.$$

We will shortly see that the functor (8.28) also identifies with an a priori familiar functor, see Proposition 9.8.5.

A similar (but simpler) argument shows that the functor (8.27) identifies with the functor Eis_{*}^{-} from [Ga3, Sect. 1.1.6] (up to the cohomological shift by (8.4)).

8.5.8. Similarly, for a factorization algebra $\mathcal{B}$ in $\mathrm{I}(G, P^{-})_{\mathrm{co}}^{\mathrm{loc}}$, we obtain a functor

$$\mathrm{CT}_{*,\mathcal{B}}^{-} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M).$$

For $\mathcal{B} := \Delta_{\mathrm{co}}^{-,\frac{\infty}{2}} = \mathbf{1}_{\mathrm{I}(G, P^{-})_{\mathrm{co}}^{\mathrm{loc}}}$, we recover the functor CT_{*}^{-} .

8.6. **Enhanced functors and duality.**

8.6.1. Consider now the categories

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} \text{ and } \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}},$$

see [DG2, Sect. 4.3.3].

We introduce the following (unital) crystals of categories over the Ran space:

$$\underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}^{-,\mathrm{enh}} :$$

$$\mathcal{Z} \mapsto \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co},\mathcal{Z}}^{-,\mathrm{enh}} := \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \right)_{\mathrm{Sph}_{G,\mathcal{Z}}} \otimes I(G, P^-)_{\mathcal{Z}}^{\mathrm{loc}},$$

and

$$\underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}}^{-,\mathrm{enh}_{\mathrm{co}}} :$$

$$\mathcal{Z} \mapsto \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co},\mathcal{Z}}^{-,\mathrm{enh}_{\mathrm{co}}} := \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \right)_{\mathrm{Sph}_{M,\mathcal{Z}}} \otimes I(G, P^-)_{\mathrm{co},\mathcal{Z}}^{\mathrm{loc}}.$$

8.6.2. Passing to dual functors, we obtain the functors

$$\underline{\mathrm{Eis}}_{\mathrm{co},*}^{-,\mathrm{enh}_{\mathrm{co}}} := \left(\underline{\mathrm{CT}}_*^{-,\mathrm{enh}} \right)^\vee, \quad \underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}}^{-,\mathrm{enh}_{\mathrm{co}}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \otimes \underline{\mathrm{D}\text{-mod}}(\mathrm{Ran}),$$

$$\underline{\mathrm{CT}}_{\mathrm{co},?}^{-,\mathrm{enh}_{\mathrm{co}}} := \left(\underline{\mathrm{Eis}}_{\mathrm{co},*}^{-,\mathrm{enh}_{\mathrm{co}}} \right)^\vee, \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \otimes \underline{\mathrm{D}\text{-mod}}(\mathrm{Ran}) \rightarrow \underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}}^{-,\mathrm{enh}_{\mathrm{co}}}.$$

Note that the functors $(\underline{\mathrm{Eis}}_{\mathrm{co},*}^{-,\mathrm{enh}_{\mathrm{co}}}, \underline{\mathrm{CT}}_{\mathrm{co},?}^{-,\mathrm{enh}_{\mathrm{co}}})$ form adjoint pair.

Using Sect. 1.4.8, from the functor $\underline{\mathrm{CT}}_{\mathrm{co},?}^{-,\mathrm{enh}_{\mathrm{co}}}$ we produce the functor

$$\underline{\mathrm{CT}}_{\mathrm{co},?}^{-,\mathrm{enh}} : \underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}^{-,\mathrm{enh}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} \otimes \underline{\mathrm{D}\text{-mod}}(\mathrm{Ran}).$$

8.6.3. For any $\mathcal{Z} \rightarrow \mathrm{Ran}$, the composition

$$\begin{aligned} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} &\xrightarrow{\mathrm{Id} \otimes \mathbf{1}_{I(G, P^-)_{\mathrm{co},\mathcal{Z}}^{\mathrm{loc}}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} \otimes I(G, P^-)_{\mathrm{co},\mathcal{Z}}^{\mathrm{loc}} \rightarrow \\ &\rightarrow \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \right)_{\mathrm{Sph}_{M,\mathcal{Z}}} \otimes I(G, P^-)_{\mathrm{co},\mathcal{Z}}^{\mathrm{loc}} = \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co},\mathcal{Z}}^{-,\mathrm{enh}_{\mathrm{co}}} \xrightarrow{\underline{\mathrm{Eis}}_{\mathrm{co},*}^{-,\mathrm{enh}_{\mathrm{co}}}} \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \end{aligned}$$

identifies with

$$\underline{\mathrm{Eis}}_{\mathrm{co},*} \otimes \mathrm{Id}_{\mathrm{D}\text{-mod}(\mathcal{Z})},$$

where $\underline{\mathrm{Eis}}_{\mathrm{co},*}$ is the functor from [Ga3, Sect. 1.4.1] (up to the cohomological shift by (8.4)).

The composition

$$\begin{aligned} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} &\xrightarrow{\mathrm{Id} \otimes \mathbf{1}_{I(G, P^-)_{\mathcal{Z}}^{\mathrm{loc}}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \otimes I(G, P^-)_{\mathcal{Z}}^{\mathrm{loc}} \rightarrow \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \otimes_{\mathrm{Sph}_{G,\mathcal{Z}}} I(G, P^-)_{\mathcal{Z}}^{\mathrm{loc}} = \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co},\mathcal{Z}}^{-,\mathrm{enh}} \xrightarrow{\underline{\mathrm{CT}}_{\mathrm{co},?}^{-,\mathrm{enh}}} \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \end{aligned}$$

identifies with

$$\underline{\mathrm{CT}}_{\mathrm{co},?}^{-,\mathrm{enh}} \otimes \mathrm{Id}_{\mathrm{D}\text{-mod}(\mathcal{Z})},$$

where

$$(8.29) \quad \underline{\mathrm{CT}}_{\mathrm{co},?}^{-,\mathrm{enh}} := (\underline{\mathrm{Eis}}_{\mathrm{co},*}^{-,\mathrm{enh}})^\vee$$

is the functor from [Ga3, Sect. 1.4.5] (up to the cohomological shift by (8.4)).

8.6.4. The functors $\underline{\text{Eis}}_{\text{co},*}^{-,\text{enh}_{\text{co}}}$ and $\underline{\text{CT}}_{\text{co},?}^{-,\text{enh}}$ are unital “local-to-global” functors in the sense of [GLC2, Sect. 11.1 and 11.3]. We will denote by

$$\text{Eis}_{\text{co},*}^{-,\text{enh}_{\text{co}}} \quad \text{and} \quad \text{CT}_{\text{co},?}^{-,\text{enh}}$$

the resulting functors

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co,Ran}}^{-,\text{enh}_{\text{co}}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}}$$

and

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co,Ran}}^{-,\text{enh}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}},$$

respectively.

8.6.5. By the same recipe as in Sect. 8.5.7, for a factorization algebra $\mathcal{A}$ in $\text{I}(G, P^-)^{\text{loc}}$, we obtain a functor

$$\text{CT}_{\text{co},?,\mathcal{A}}^{-} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}}.$$

Taking $\mathcal{A}$ to be

$$\Delta^{-,\frac{\infty}{2}}, \nabla^{-,\frac{\infty}{2}} \quad \text{and} \quad \text{IC}^{-,\frac{\infty}{2}},$$

respectively, we obtain the functors

$$(8.30) \quad \text{CT}_{\text{co},?,\Delta}^{-} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}}.$$

$$(8.31) \quad \text{CT}_{\text{co},?,\nabla}^{-} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}}$$

and

$$(8.32) \quad \text{CT}_{\text{co},?,\text{IC}}^{-} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}}.$$

Since $\Delta^{-,\frac{\infty}{2}}$ is the factorization unit in $\text{I}(G, P^-)^{\text{loc}}$, we have

$$\text{CT}_{\text{co},?,\Delta}^{-} \simeq \text{CT}_{\text{co},?}^{-}.$$

Additionally, it is not difficult to show that

$$\text{CT}_{\text{co},?,\nabla}^{-} \simeq \text{CT}_{\text{co},*}^{-},$$

where $\text{CT}_{\text{co},*}^{-} \simeq (\text{Eis}_{\text{co},*}^{-})^{\vee}$ is as in [Ga3, Sect. 1.4.3] (up to the cohomological shift by (8.4)).

8.6.6. Similarly, for a factorization algebra $\mathcal{B}$ in $\text{I}(G, P^-)^{\text{loc}}$, we obtain a functor

$$\text{Eis}_{\text{co},*,\mathcal{B}}^{-} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}}.$$

For $\mathcal{B} = \Delta_{\text{co}}^{-,\frac{\infty}{2}}$ we recover the functor $\text{Eis}_{\text{co},*}^{-}$.

8.7. Enhanced functors and miraculous duality.

8.7.1. Recall now the miraculous functor

$$\text{Mir}_{\text{Bun}_G} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G),$$

see [DG2, Sect. 4.4.8].³⁴

Remark 8.7.2. In [Ga3], the functor $\text{Mir}_{\text{Bun}_G}$ was defined in the untwisted setting, i.e., for $\text{D-mod}(\text{Bun}_G)$ rather than for $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$. However, the same procedure defines is also for $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$, and the resulting functor has the same properties: the two setting are actually equivalent, see [GLC2, Remark 9.1.6].

³⁴In *loc. cit.*, this functor is denoted $\text{Ps-Id}_{\text{Bun}_G,!}$.

8.7.3. By construction, $\text{Mir}_{\text{Bun}_G}$ commutes with the natural actions of Sph_G on the two sides.

According to [Ga3, Theorem 3.1.5], the functor $\text{Mir}_{\text{Bun}_G}$ is an equivalence. Consider also the corresponding functor for M .

$$\text{Mir}_{\text{Bun}_M} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M).$$

Denote

$$\text{Mir}_{\text{Bun}_G, \tau} := \tau_G \circ \text{Mir}_{\text{Bun}_G} \quad \text{and} \quad \text{Mir}_{\text{Bun}_M, \tau} := \tau_M \circ \text{Mir}_{\text{Bun}_M},$$

where τ_G and τ_M are the Chevalley involutions in G and M respectively.

8.7.4. Recall now the equivalence

$$\Theta_{\text{I}(G, P^-)^{\text{loc}}} : \text{I}(G, P^-)^{\text{loc}}_{\text{co}} \rightarrow \text{I}(G, P)^{\text{loc}},$$

see Corollary 1.5.12.

Composing with the Chevalley involution τ_G , we obtain an equivalence that we will denote

$$(8.33) \quad \Theta_{\text{I}(G, P^-)^{\text{loc}}, \tau} : \text{I}(G, P^-)^{\text{loc}}_{\text{co}} \rightarrow \text{I}(G, P^-)^{\text{loc}}.$$

It commutes with the actions of Sph_G and Sph_M , up to the Chevalley involutions.

Tensoring $\Theta_{\text{I}(G, P^-)^{\text{loc}}, \tau}$ with $\text{Mir}_{\text{Bun}_M, \tau}$ we obtain an equivalence that we denote

$$\text{Mir}_{\text{Bun}_M, \tau}^{-, \text{enh}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}}^{-, \text{enhco}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)^{-, \text{enh}}.$$

8.7.5. We now quote the following result (see [Ch2, Theorem 5.3.5(b)]):

Theorem 8.7.6. *The following diagram commutes*

$$(8.34) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}, \text{Ran}}^{-, \text{enhco}} & \xrightarrow{\text{Mir}_{\text{Bun}_M, \tau}^{-, \text{enh}}} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{Ran}}^{-, \text{enh}} \\ \text{Eis}_{\text{co}, *}^{-, \text{enhco}}[\delta_{N_P^-}] \downarrow & & \downarrow \text{Eis}_1^{-, \text{enh}}[-\delta_{N_P^-}] \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}} & \xrightarrow{\text{Mir}_{\text{Bun}_G, \tau}} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G), \end{array}$$

where

$$\delta_{N_P^-} = \dim(\text{Bun}_{N_P^-}).$$

Remark 8.7.7. More precisely, Theorem 8.7.6 differs from [Ch2, Theorem 5.3.5(b)] in the following technical points:

- In *loc.cit.*, the theorem was about the corresponding “independent” categories (see Remark 8.2.5). Note that the functors $\text{Eis}_{\text{co}, *}^{-, \text{enhco}}$ and $\text{Eis}_1^{-, \text{enh}}$ indeed factor through these “independent” categories.
- In *loc.cit.*, the equivalence $\text{Mir}_{\text{Bun}_M, \tau}^{-, \text{enh}}$, referred as the *global inv-inv duality* (up to the Chevalley involution), was constructed by a different method, which can be shown to be equivalent to the current construction by combining [Ch2, Corollary 3.2.4] and [Ch2, Theorem 5.1.7].

Remark 8.7.8. Theorem 8.7.6 generalizes the result of [Ga3, Theorem 4.1.2], which says that the diagram

$$(8.35) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}} & \xrightarrow{\text{Mir}_{\text{Bun}_M, \tau}} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \\ \text{Eis}_{\text{co}, *}^{-, [\delta_{N_P^-}]} \downarrow & & \downarrow \text{Eis}_1^{-, [-\delta_{N_P^-}]} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}} & \xrightarrow{\text{Mir}_{\text{Bun}_G, \tau}} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G), \end{array}$$

commutes.

8.7.9. Tensoring $\mathrm{Mir}_{\mathrm{Bun}_G, \tau}$ with $(\Theta_{I(G, P^-)^{\mathrm{loc}}, \tau})^{-1}$ we obtain an equivalence that we denote

$$\mathrm{Mir}_{\mathrm{Bun}_G, \tau}^{-, \mathrm{enh}} : \underline{\mathrm{D-mod}}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}^{-, \mathrm{enh}} \rightarrow \underline{\mathrm{D-mod}}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Ran}}^{-, \mathrm{enhco}}.$$

By duality, from Theorem 8.7.6 we obtain:

Corollary 8.7.10. *The following diagram commutes:*

$$\begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{\mathrm{Bun}_M, \tau}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \\ \mathrm{CT}_{\mathrm{co}, ?}^{-, \mathrm{enh}}[-\delta_{N_P^-}] \uparrow & & \uparrow \mathrm{CT}_*^{-, \mathrm{enhco}}[\delta_{N_P^-}] \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}, \mathrm{Ran}}^{-, \mathrm{enh}} & \xrightarrow{\mathrm{Mir}_{\mathrm{Bun}_G, \tau}^{-, \mathrm{enh}}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Ran}}^{-, \mathrm{enhco}}. \end{array}$$

In particular, we have:

Corollary 8.7.11. *The following diagram commutes:*

$$\begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{\mathrm{Bun}_M, \tau}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \\ \mathrm{CT}_{\mathrm{co}, ?}^{-, \mathrm{enh}}[-\delta_{N_P^-}] \uparrow & & \uparrow \mathrm{CT}_*^{-, \mathrm{enh}}[\delta_{N_P^-}] \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{\mathrm{Bun}_G, \tau}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G). \end{array}$$

(Note, however, that Corollary 8.7.11 can be directly obtained by duality from (8.35)).

8.8. The ρ -shift.

8.8.1. Let $\mathcal{P}_{Z_M}$ be a Z_M -torsor on X , and consider again the auto-equivalence

$$(8.36) \quad (\mathrm{transl}_{\mathcal{P}_{Z_M}})^* : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M).$$

Recall now the automorphism $(\mathrm{transl}_{\mathcal{P}_{Z_M}})^*$ of Sph_M , see (1.34). These two automorphisms are compatible with the action of $\mathrm{Sph}_{M, \mathbb{Z}}$ on $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}(\mathbb{Z})$, for $\mathbb{Z} \rightarrow \mathrm{Ran}$.

8.8.2. Recall the identification

$$(8.37) \quad I(G, P^-)^{\mathrm{loc}} \xrightarrow{\alpha_{\mathcal{P}_{Z_M}, \mathrm{taut}}} I(G, P^-)_{\mathcal{P}_{Z_M}}^{\mathrm{loc}}$$

as categories acted on by Sph_G and Sph_M (see Sect. 1.7.5 regarding our conventions for the latter action).

The tensor product of (8.36) and (8.37) defines an identification

$$(8.38) \quad \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathbb{Z}}^{-, \mathrm{enh}} \simeq \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{P}_{Z_M}, \mathbb{Z}}^{-, \mathrm{enh}},$$

as categories acted on by $\mathrm{Sph}_{G, \mathbb{Z}}$, where

$$(8.39) \quad \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{P}_{Z_M}, \mathbb{Z}}^{-, \mathrm{enh}} := \left(\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}(\mathbb{Z}) \right)_{\mathrm{Sph}_{M, \mathbb{Z}}} \otimes I(G, P^-)_{\mathcal{P}_{Z_M}, \mathbb{Z}}^{\mathrm{loc}}.$$

8.8.3. We have an adjoint pair

$$\mathbf{ind}_{\mathrm{enh}} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}(\mathbb{Z}) \rightleftarrows \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{P}_{Z_M}, \mathbb{Z}}^{-, \mathrm{enh}} : \mathbf{oblv}_{\mathrm{enh}}$$

so that we have a commutative diagram

$$\begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathbb{Z}}^{-, \mathrm{enh}} & \xrightarrow{(8.38)} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{P}_{Z_M}, \mathbb{Z}}^{-, \mathrm{enh}} \\ \mathbf{oblv}_{\mathrm{enh}} \downarrow & & \downarrow \mathbf{oblv}_{\mathrm{enh}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}(\mathbb{Z}) & \xrightarrow{(\mathrm{transl}_{\mathcal{P}_{Z_M}})^* \otimes \mathrm{Id}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}(\mathbb{Z}). \end{array}$$

8.8.4. We let

$$\mathrm{CT}_{*,\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}}$$

denote the composition

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \xrightarrow{\mathrm{CT}_{*,\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{Z}}^{-,\mathrm{enh}} \xrightarrow{(8.38)} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}}.$$

This functor is compatible with the actions of $\mathrm{Sph}_{G,\mathrm{Ran}}$ on the two sides.

By construction, the composition

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \xrightarrow{\mathrm{CT}_{*,\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}} \xrightarrow{\mathrm{obl}_{\mathrm{enh}} \otimes \mathrm{Id}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z})$$

identifies with $\mathrm{CT}_{*,\mathcal{P}_{Z_M}} \otimes \mathrm{Id}$.

8.8.5. We let

$$\mathrm{Eis}_{!,\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z})$$

be the left adjoint of $\mathrm{CT}_{*,\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}}$.

By construction, we have a commutative diagram

$$\begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{Z}}^{-,\mathrm{enh}} & \xrightarrow{(8.38)} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}} \\ \mathrm{Eis}_{!,\mathcal{Z}}^{-,\mathrm{enh}} \downarrow & & \downarrow \mathrm{Eis}_{!,\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) & \xrightarrow{\mathrm{Id}} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}). \end{array}$$

Also, by construction, the composition

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \xrightarrow{\mathrm{ind}_{\mathrm{enh}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}} \xrightarrow{\mathrm{Eis}_{!,\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z})$$

identifies with $\mathrm{Eis}_{!,\mathcal{P}_{Z_M}} \otimes \mathrm{Id}$.

8.8.6. Denote

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}_{\mathrm{co}}} := \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \right)_{\mathrm{Sph}_{G,\mathcal{Z}}} \otimes \mathrm{I}(G, P^-)_{\mathrm{co},\mathcal{P}_{Z_M},\mathcal{Z}}^{\mathrm{loc}}.$$

Applying (8.37) along the second factor we obtain an identification

$$(8.40) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathcal{Z}}^{-,\mathrm{enh}_{\mathrm{co}}} \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}_{\mathrm{co}}}.$$

Let

$$\mathrm{CT}_{*,\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}_{\mathrm{co}}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}_{\mathrm{co}}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z})$$

be the functor obtained from $\mathrm{CT}_{*,\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}}$ by duality.

By construction, this functor is compatible with the actions of $\mathrm{Sph}_{M,\mathrm{Ran}}$ on the two sides.

The composition

$$\begin{aligned} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) & \xrightarrow{\mathrm{Id} \otimes \mathrm{I}(G, P^-)_{\mathrm{co},\mathcal{P}_{Z_M},\mathcal{Z}}^{\mathrm{loc}}} \\ & \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}_{\mathrm{co}}} \xrightarrow{\mathrm{CT}_{*,\mathcal{P}_{Z_M},\mathcal{Z}}^{-,\mathrm{enh}_{\mathrm{co}}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \end{aligned}$$

identifies with $\mathrm{CT}_{*,\mathcal{P}_{Z_M}} \otimes \mathrm{Id}$.

Unwinding, we obtain that the functor $\mathrm{CT}_{*, \mathcal{P}_{Z_M}, \mathcal{Z}}^{-, \mathrm{enh}_{\mathrm{co}}}$ identifies with the composition

$$\begin{aligned} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathcal{P}_{Z_M}, \mathcal{Z}}^{-, \mathrm{enh}_{\mathrm{co}}} &\stackrel{(8.40)}{\simeq} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathcal{Z}}^{-, \mathrm{enh}_{\mathrm{co}}} \xrightarrow{\mathrm{CT}_{*, \mathcal{Z}}^{-, \mathrm{enh}_{\mathrm{co}}}} \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \xrightarrow{(\mathrm{transl}_{\mathcal{P}_{Z_M}})^* \otimes \mathrm{Id}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}). \end{aligned}$$

8.8.7. By a similar principle, we define the categories

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}, \mathcal{P}_{Z_M}, \mathcal{Z}}^{-, \mathrm{enh}} \text{ and } \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}, \mathcal{P}_{Z_M}, \mathcal{Z}}^{-, \mathrm{enh}_{\mathrm{co}}}$$

and the functors

$$\mathrm{Eis}_{\mathrm{co}, *, \mathcal{P}_{Z_M}, \mathcal{Z}}^{-, \mathrm{enh}_{\mathrm{co}}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}, \mathcal{P}_{Z_M}, \mathcal{Z}}^{-, \mathrm{enh}_{\mathrm{co}}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(\mathcal{Z}),$$

$$\mathrm{CT}_{\mathrm{co}, ?, \mathcal{P}_{Z_M}, \mathcal{Z}}^{-, \mathrm{enh}_{\mathrm{co}}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}, \mathcal{P}_{Z_M}, \mathcal{Z}}^{-, \mathrm{enh}_{\mathrm{co}}}$$

and

$$\mathrm{CT}_{\mathrm{co}, ?, \mathcal{P}_{Z_M}, \mathcal{Z}}^{-, \mathrm{enh}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}, \mathcal{P}_{Z_M}, \mathcal{Z}}^{-, \mathrm{enh}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(\mathcal{Z}).$$

8.8.8. Let $\mathcal{A}$ be a factorization algebra in $\mathrm{I}(G, P^-)^{\mathrm{loc}}$. Using the identification (8.37), we produce from it a factorization algebra in $\mathrm{I}(G, P^-)^{\mathrm{loc}}_{\mathcal{P}_{Z_M}}$, to be denoted $\mathcal{A}_{\mathcal{P}_{Z_M}}$.

The recipe in Sects. 8.5.7 and 8.6.6 gives rise to functors

$$\mathrm{Eis}_{\mathrm{I}, \mathcal{P}_{Z_M}, \mathcal{A}}^{-} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

and

$$\mathrm{CT}_{\mathrm{co}, ?, \mathcal{P}_{Z_M}, \mathcal{A}}^{-} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}}.$$

We have

$$\mathrm{Eis}_{\mathrm{I}, \mathcal{P}_{Z_M}, \mathcal{A}}^{-} \simeq \mathrm{Eis}_{\mathrm{I}, \mathcal{A}}^{-} \circ (\mathrm{transl}_{\mathcal{P}_{Z_M}})^* \text{ and } \mathrm{CT}_{\mathrm{co}, ?, \mathcal{P}_{Z_M}, \mathcal{A}}^{-} \simeq (\mathrm{transl}_{\mathcal{P}_{Z_M}})^* \circ \mathrm{CT}_{\mathrm{co}, ?, \mathcal{A}}^{-, \mathrm{enh}}.$$

8.8.9. Similarly, if $\mathcal{B}$ is a factorization algebra in $\mathrm{I}(G, P^-)^{\mathrm{loc}}_{\mathrm{co}}$, we construct functors

$$\mathrm{Eis}_{\mathrm{co}, *, \mathcal{P}_{Z_M}, \mathcal{B}}^{-} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}$$

and

$$\mathrm{CT}_{*, \mathcal{P}_{Z_M}, \mathcal{B}}^{-} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)$$

and we have

$$\mathrm{Eis}_{\mathrm{co}, *, \mathcal{P}_{Z_M}, \mathcal{B}}^{-} \simeq \mathrm{Eis}_{\mathrm{co}, *, \mathcal{B}}^{-, \mathrm{enh}_{\mathrm{co}}} \circ (\mathrm{transl}_{\mathcal{P}_{Z_M}})^*,$$

$$\mathrm{CT}_{*, \mathcal{P}_{Z_M}, \mathcal{B}}^{-} \simeq (\mathrm{transl}_{\mathcal{P}_{Z_M}})^* \circ \mathrm{CT}_{*, \mathcal{B}}^{-}.$$

8.8.10. Recall the equivalence $\Theta_{\mathrm{I}(G, P^-)^{\mathrm{loc}}, \tau}$ of (8.33), and note that in the translated context it gives rise to an equivalence

$$\Theta_{\mathrm{I}(G, P^-)^{\mathrm{loc}}_{\mathcal{P}_{Z_M}}, \tau} : \mathrm{I}(G, P^-)^{\mathrm{loc}}_{\mathcal{P}_{Z_M}, \mathrm{co}} \rightarrow \mathrm{I}(G, P^-)^{\mathrm{loc}}_{\tau_M(\mathcal{P}_{Z_M})}.$$

Let

$$\mathrm{Mir}_{\mathrm{Bun}_M, \tau, \mathcal{P}_{Z_M}}^{-, \mathrm{enh}} : \underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}, \mathcal{P}_{Z_M}}^{-, \mathrm{enh}_{\mathrm{co}}} \rightarrow \underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\tau_M(\mathcal{P}_{Z_M})}^{-, \mathrm{enh}}$$

denote the tensor product of $\mathrm{Mir}_{\mathrm{Bun}_M, \tau}^{-, \mathrm{enh}}$ and $\Theta_{\mathrm{I}(G, P^-)^{\mathrm{loc}}_{\mathcal{P}_{Z_M}}, \tau}$.

From Theorem 8.7.6 we obtain:

Corollary 8.8.11. *The following diagram commutes*

$$(8.41) \quad \begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}, \mathcal{P}_{Z_M}, \mathrm{Ran}}^{-, \mathrm{enhco}} & \xrightarrow{\mathrm{Mir}_{\mathrm{Bun}_M, \tau, \mathcal{P}_{Z_M}}^{-, \mathrm{enh}}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\tau_M(\mathcal{P}_{Z_M}), \mathrm{Ran}}^{-, \mathrm{enh}} \\ \mathrm{Eis}_{\mathrm{co}, *, \mathcal{P}_{Z_M}}^{-, \mathrm{enhco}} [\delta_{N_P}^-] \downarrow & & \downarrow \mathrm{Eis}_{!, \tau_M(\mathcal{P}_{Z_M})}^{-, \mathrm{enhco}} [-\delta_{N_P}^-] \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{\mathrm{Bun}_G, \tau}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G). \end{array}$$

8.8.12. Let

$$\mathrm{Mir}_{\mathrm{Bun}_G, \tau, \mathcal{P}_{Z_M}}^{-, \mathrm{enh}} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}, \mathcal{P}_{Z_M}, \mathrm{Ran}}^{-, \mathrm{enh}} \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\tau_M(\mathcal{P}_{Z_M}), \mathrm{Ran}}^{-, \mathrm{enhco}}$$

be the equivalence obtained by tensoring $\mathrm{Mir}_{\mathrm{Bun}_G, \tau}^{-, \mathrm{enh}}$ with $\Theta_{\mathrm{I}(G, P^-)_{\mathcal{P}_{Z_M}}^{\mathrm{loc}}}, \tau$.

By duality, from Corollary 8.8.11 we obtain:

Corollary 8.8.13. *The following diagram commutes:*

$$\begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{\mathrm{Bun}_M, \tau}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \\ \mathrm{CT}_{\mathrm{co}, ?, \mathcal{P}_{Z_M}}^{-, \mathrm{enh}} [-\delta_{N_P}^-] \uparrow & & \uparrow \mathrm{CT}_{*, \tau_M(\mathcal{P}_{Z_M})}^{-, \mathrm{enhco}} [\delta_{N_P}^-] \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}, \mathcal{P}_{Z_M}, \mathrm{Ran}}^{-, \mathrm{enh}} & \xrightarrow{\mathrm{Mir}_{\mathrm{Bun}_G, \tau, \mathcal{P}_{Z_M}}^{-, \mathrm{enh}}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\tau_M(\mathcal{P}_{Z_M}), \mathrm{Ran}}^{-, \mathrm{enhco}}. \end{array}$$

8.8.14. We will mostly apply the above discussion in the case when $\mathcal{P}_{Z_M} = \rho_P(\omega_X)$. Note that in this case $\tau_M(\mathcal{P}_{Z_M}) = -\rho_P(\omega_X)$.

9. GLOBAL INTERPRETATION OF THE ENHANCED FUNCTORS

In this section we will give an interpretation of the functors

$$\mathrm{Eis}_{!}^{-, \mathrm{enh}} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{Ran}}^{-, \mathrm{enh}} \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G),$$

$$\mathrm{Eis}_{\mathrm{co}, *}^{-, \mathrm{enhco}} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}, \mathrm{Ran}}^{-, \mathrm{enhco}} \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}},$$

$$\mathrm{CT}_{*}^{-, \mathrm{enhco}} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Ran}}^{-, \mathrm{enhco}} \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M),$$

and

$$\mathrm{CT}_{\mathrm{co}, ?}^{-, \mathrm{enh}} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}, \mathrm{Ran}}^{-, \mathrm{enh}} \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}}$$

via Drinfeld's compactifications.

This interpretation will be crucial for the computation of Whittaker coefficients of enhanced Eisenstein series.

9.1. Drinfeld's compactifications.

9.1.1. Let

$$\mathrm{Bun}_G \xleftarrow{\widetilde{\mathfrak{p}}^-} \widetilde{\mathrm{Bun}}_{P^-} \xrightarrow{\widetilde{\mathfrak{q}}^-} \mathrm{Bun}_M$$

be as in [BG1, Sect. 1.3.6]. Let

$$j : \mathrm{Bun}_{P^-} \hookrightarrow \widetilde{\mathrm{Bun}}_{P^-}$$

denote the tautological open embedding.

9.1.2. Let

$$(9.1) \quad \widetilde{\mathrm{Bun}}_{P^-, \mathrm{Ran}} \rightarrow \mathrm{Ran}$$

be the polar version of this space, where we allow the generalized P^- -reduction to have poles at the specified locus of the curve, see [Ga5, Sect. 6.2.1] (in the case $P = B$) or [BG1, Sect. 4.1.1] (pointwise version for any P).

We denote by the same symbols the corresponding projections

$$\mathrm{Bun}_G \xleftarrow{\widetilde{\mathfrak{p}}^-} \widetilde{\mathrm{Bun}}_{P^-, \mathrm{Ran}} \xrightarrow{\widetilde{\mathfrak{q}}^-} \mathrm{Bun}_M.$$

For $\mathcal{Z} \rightarrow \mathrm{Ran}$ denote

$$\widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}} := \widetilde{\mathrm{Bun}}_{P^-, \mathrm{Ran}} \times_{\mathrm{Ran}} \mathcal{Z};$$

and let

$$\mathrm{Bun}_G \times \mathcal{Z} \xleftarrow{\widetilde{\mathfrak{p}}_{\mathcal{Z}}^-} \widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}} \xrightarrow{\widetilde{\mathfrak{q}}_{\mathcal{Z}}^-} \mathrm{Bun}_M \times \mathcal{Z}$$

be the corresponding maps.

9.1.3. We will consider three μ_2 -gerbes on $\widetilde{\mathrm{Bun}}_{P^-, \mathrm{Ran}}$:

$$\det_{\frac{1}{2}}^{\frac{1}{2}} := (\widetilde{\mathfrak{p}}^-)^*(\det_{\frac{1}{2}}^{\frac{1}{2}}_{\mathrm{Bun}_G}), \quad \det_{\frac{1}{2}}^{\frac{1}{2}} := (\widetilde{\mathfrak{q}}^-)^*(\det_{\frac{1}{2}}^{\frac{1}{2}}_{\mathrm{Bun}_M})$$

and

$$\det_{\frac{1}{2}}^{\frac{1}{2}}_{G, M} := \det_{\frac{1}{2}}^{\frac{1}{2}}_G \otimes \det_{\frac{1}{2}}^{-\frac{1}{2}}_M \simeq \det_{\frac{1}{2}}^{\frac{1}{2}}_G \otimes \det_{\frac{1}{2}}^{\frac{1}{2}}_M.$$

Note that according to Sect. 8.1.2, the restriction $\det_{\frac{1}{2}}^{\frac{1}{2}}_{G, M}|_{\mathrm{Bun}_{P^-}}$ is canonically trivialized.

9.1.4. Let $\mathcal{G}$ be any of the above μ_2 -gerbes. For $\mathcal{Z} \rightarrow \mathrm{Ran}$, we will consider the following categories of (twisted) D-modules on $\widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}}$

$$\mathrm{D}\text{-mod}_{\mathcal{G}}(\widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}}), \quad \mathrm{D}\text{-mod}_{\mathcal{G}}(\widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}})_{\mathrm{co}} \quad \text{and} \quad \mathrm{D}\text{-mod}_{\mathcal{G}}(\widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}})_{\mathrm{co}} / \mathrm{Bun}_M,$$

defined as follows.

First, we note that (9.1) is a relative ind-algebraic stack. For a scheme $S \rightarrow \mathrm{Ran}$, write

$$\widetilde{\mathrm{Bun}}_{P^-, S} \simeq \mathrm{colim}_i \mathcal{Y}_{S, i},$$

where for $i \rightarrow j$, the maps

$$f_{i, j} : \mathcal{Y}_{S, i} \rightarrow \mathcal{Y}_{S, j}$$

are closed embeddings. (Namely, the closed substacks $\mathcal{Y}_{S, i} \subset \widetilde{\mathrm{Bun}}_{P^-, S}$ are obtained by bounding the order of poles of the generalized P^- -reduction.)

Note that for every i , the map

$$\widetilde{\mathfrak{p}}^-|_{\mathcal{Y}_{S, i}} : \mathcal{Y}_{S, i} \rightarrow \mathrm{Bun}_G \times S_i$$

is schematic.

9.1.5. For every S and i as above, we let $\mathrm{D}\text{-mod}_{\mathcal{G}}(\mathcal{Y}_{S, i})$ be the usual category of (twisted) D-modules.

Let

$$\mathrm{D}\text{-mod}_{\mathcal{G}}(\mathcal{Y}_{S, i})_{\mathrm{co}} := \mathrm{colim}_{U_{\mathcal{Y}}} \mathrm{D}\text{-mod}_{\mathcal{G}}(U_{\mathcal{Y}})_{\mathrm{co}},$$

where:

- $U_{\mathcal{Y}}$ runs over the poset of quasi-compact open substacks of $\mathcal{Y}_{S, i}$;
- For an inclusion $U_{\mathcal{Y}} \xrightarrow{j} U'_{\mathcal{Y}}$, the functor $\mathrm{D}\text{-mod}_{\mathcal{G}}(U_{\mathcal{Y}})_{\mathrm{co}} \rightarrow \mathrm{D}\text{-mod}_{\mathcal{G}}(U'_{\mathcal{Y}})_{\mathrm{co}}$ is j_* .

9.1.6. We let

$$\mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathcal{Y}_{S,i})_{\mathrm{co}} / \mathrm{Bun}_M := \lim_{U_M} \mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathcal{Y}_{S,i} \times_{\mathrm{Bun}_M} U_M)_{\mathrm{co}},$$

where:

- U_M runs over the poset of quasi-compact open substacks of Bun_M ;
- For an inclusion $U_M \xrightarrow{j} U'_M$, the functor

$$\mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathcal{Y}_{S,i} \times_{\mathrm{Bun}_M} U'_M)_{\mathrm{co}} \rightarrow \mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathcal{Y}_{S,i} \times_{\mathrm{Bun}_M} U_M)_{\mathrm{co}}$$

is given by restriction.

9.1.7. We let $\mathrm{D}\text{-mod}_{\mathfrak{g}}(\widetilde{\mathrm{Bun}}_{P^-,S})$ be the usual category of twisted D-modules, i.e.,

$$\mathrm{D}\text{-mod}_{\mathfrak{g}}(\widetilde{\mathrm{Bun}}_{P^-,S}) = \lim_i \mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathcal{Y}_{S,i}),$$

and we let

$$\begin{aligned} \mathrm{D}\text{-mod}_{\mathfrak{g}}(\widetilde{\mathrm{Bun}}_{P^-,S})_{\mathrm{co}} &:= \lim_i \mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathcal{Y}_{S,i})_{\mathrm{co}}, \\ \mathrm{D}\text{-mod}_{\mathfrak{g}}(\widetilde{\mathrm{Bun}}_{P^-,S})_{\mathrm{co}} / \mathrm{Bun}_M &:= \lim_i \mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathcal{Y}_{S,i})_{\mathrm{co}} / \mathrm{Bun}_M, \end{aligned}$$

where in all three cases the transition functors are given by $f_{i,j}^!$.

Note that in all three cases we can rewrite the limits as colimits

$$\begin{aligned} \mathrm{colim}_i \mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathcal{Y}_{S,i}), \\ \mathrm{colim}_i \mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathcal{Y}_{S,i})_{\mathrm{co}} \end{aligned}$$

and

$$\mathrm{colim}_i \mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathcal{Y}_{S,i})_{\mathrm{co}} / \mathrm{Bun}_M,$$

respectively, where the transition functors are given by $(f_{i,j})_*$.

9.1.8. We let $\mathrm{D}\text{-mod}_{\mathfrak{g}}(\widetilde{\mathrm{Bun}}_{P^-,Z})$ be the usual category of twisted D-modules. I.e.,

$$\mathrm{D}\text{-mod}_{\mathfrak{g}}(\widetilde{\mathrm{Bun}}_{P^-,Z}) \simeq \lim_{S \rightarrow Z} \mathrm{D}\text{-mod}_{\mathfrak{g}}(\widetilde{\mathrm{Bun}}_{P^-,S}),$$

and we let

$$\begin{aligned} \mathrm{D}\text{-mod}_{\mathfrak{g}}(\widetilde{\mathrm{Bun}}_{P^-,Z})_{\mathrm{co}} &\simeq \lim_{S \rightarrow Z} \mathrm{D}\text{-mod}_{\mathfrak{g}}(\widetilde{\mathrm{Bun}}_{P^-,S})_{\mathrm{co}}, \\ \mathrm{D}\text{-mod}_{\mathfrak{g}}(\widetilde{\mathrm{Bun}}_{P^-,Z})_{\mathrm{co}} / \mathrm{Bun}_M &\simeq \lim_{S \rightarrow Z} \mathrm{D}\text{-mod}_{\mathfrak{g}}(\widetilde{\mathrm{Bun}}_{P^-,S})_{\mathrm{co}} / \mathrm{Bun}_M, \end{aligned}$$

where the transition functors for $g : S' \rightarrow S$ are given by $g^!$.

If Z is pseudo-proper (see [GLC2, Sect. C.4] for what this means), e.g., $Z = \mathrm{Ran}$ itself, we can choose S to be proper schemes. In this case the above limits can also be written as colimits

$$\begin{aligned} \mathrm{colim}_{S \rightarrow Z} \mathrm{D}\text{-mod}_{\mathfrak{g}}(\widetilde{\mathrm{Bun}}_{P^-,S}), \\ \mathrm{colim}_{S \rightarrow Z} \mathrm{D}\text{-mod}_{\mathfrak{g}}(\widetilde{\mathrm{Bun}}_{P^-,S})_{\mathrm{co}} \end{aligned}$$

and

$$\mathrm{colim}_{S \rightarrow Z} \mathrm{D}\text{-mod}_{\mathfrak{g}}(\widetilde{\mathrm{Bun}}_{P^-,S})_{\mathrm{co}} / \mathrm{Bun}_M,$$

where the transition functors are given by g_* .

Write also

$$\mathrm{D}\text{-mod}_{\mathfrak{g}_G}(\mathrm{Bun}_G \times Z)_{\mathrm{co}} := \lim_{S \rightarrow Z} \mathrm{D}\text{-mod}_{\mathfrak{g}_G}(\mathrm{Bun}_G \times S)_{\mathrm{co}} \simeq \mathrm{D}\text{-mod}_{\mathfrak{g}_G}(\mathrm{Bun}_G)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(Z)$$

(for a gerbe $\mathfrak{g}_G$ on Bun_G) and

$$\mathrm{D}\text{-mod}_{\mathfrak{g}_M}(\mathrm{Bun}_M \times Z)_{\mathrm{co}} := \lim_{S \rightarrow Z} \mathrm{D}\text{-mod}_{\mathfrak{g}_M}(\mathrm{Bun}_M \times S)_{\mathrm{co}} \simeq \mathrm{D}\text{-mod}_{\mathfrak{g}_M}(\mathrm{Bun}_M)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(Z)$$

(for a gerbe $\mathfrak{g}_M$ on Bun_M).

9.1.9. For a gerbe $\mathcal{G}_G$ on Bun_G , we have well-defined functors

$$(9.2) \quad (\mathbf{p}_Z^-)^\dagger : \text{D-mod}_{\mathcal{G}_G}(\text{Bun}_G \times \mathbb{Z}) \rightarrow \text{D-mod}_{(\mathbf{p}^-)^*(\mathcal{G}_G)}(\widetilde{\text{Bun}}_{P^-, \mathbb{Z}}),$$

$$(9.3) \quad (\mathbf{p}_Z^-)^\dagger : \text{D-mod}_{\mathcal{G}_G}(\text{Bun}_G \times \mathbb{Z})_{\text{co}} \rightarrow \text{D-mod}_{(\mathbf{p}^-)^*(\mathcal{G}_G)}(\widetilde{\text{Bun}}_{P^-, \mathbb{Z}})_{\text{co}},$$

and

$$(9.4) \quad (\mathbf{p}_Z^-)_* : \text{D-mod}_{(\mathbf{p}^-)^*\mathcal{G}_G}(\widetilde{\text{Bun}}_{P^-, \mathbb{Z}})_{\text{co}} \rightarrow \text{D-mod}_{\mathcal{G}_G}(\text{Bun}_G \times \mathbb{Z})_{\text{co}}.$$

For a gerbe $\mathcal{G}_M$ on Bun_M , we have well-defined functors

$$(9.5) \quad (\mathbf{q}_Z^-)_* : \text{D-mod}_{(\mathbf{q}^-)^*(\mathcal{G}_M)}(\widetilde{\text{Bun}}_{P^-, \text{Ran}})_{\text{co}} \rightarrow \text{D-mod}_{\mathcal{G}_M}(\text{Bun}_M \times \mathbb{Z})_{\text{co}}$$

and

$$(9.6) \quad (\mathbf{q}_Z^-)_* : \text{D-mod}_{(\mathbf{q}^-)^*(\mathcal{G}_M)}(\widetilde{\text{Bun}}_{P^-, \text{Ran}})_{\text{co}} / \text{Bun}_M \rightarrow \text{D-mod}_{\mathcal{G}_M}(\text{Bun}_M \times \mathbb{Z}).$$

9.1.10. For a pair of gerbes $\mathcal{G}_1$ and $\mathcal{G}_2$ with

$$\mathcal{G}_{1,2} := \mathcal{G}_1 \otimes \mathcal{G}_2,$$

the operation of $\overset{!}{\otimes}$ product of twisted D-modules gives rise to functors

$$(9.7) \quad \text{D-mod}_{\mathcal{G}_1}(\widetilde{\text{Bun}}_{P^-, \mathbb{Z}}) \otimes \text{D-mod}_{\mathcal{G}_2}(\widetilde{\text{Bun}}_{P^-, \mathbb{Z}})_{\text{co}} \rightarrow \text{D-mod}_{\mathcal{G}_{1,2}}(\widetilde{\text{Bun}}_{P^-, \mathbb{Z}})_{\text{co}}$$

and

$$(9.8) \quad \text{D-mod}_{\mathcal{G}_1}(\widetilde{\text{Bun}}_{P^-, \mathbb{Z}}) \otimes \text{D-mod}_{\mathcal{G}_2}(\widetilde{\text{Bun}}_{P^-, \mathbb{Z}})_{\text{co}} / \text{Bun}_M \rightarrow \text{D-mod}_{\mathcal{G}_{1,2}}(\widetilde{\text{Bun}}_{P^-, \mathbb{Z}})_{\text{co}} / \text{Bun}_M.$$

9.1.11. In addition, for a gerbe $\mathcal{G}_M$ on Bun_M we have a functor

$$(9.9) \quad (\mathbf{q}_Z^-)^\dagger(-) \overset{!}{\otimes} (-) : \text{D-mod}_{\mathcal{G}_M}(\text{Bun}_M \times \mathbb{Z})_{\text{co}} \otimes \text{D-mod}_{\mathcal{G}}(\widetilde{\text{Bun}}_{P^-, \mathbb{Z}})_{\text{co}} / \text{Bun}_M \rightarrow \\ \rightarrow \text{D-mod}_{(\mathbf{q}^-)^*(\mathcal{G}_M) \otimes \mathcal{G}}(\widetilde{\text{Bun}}_{P^-, \mathbb{Z}})_{\text{co}}.$$

9.2. Local-to-global construction for Drinfeld's compactification, the *-version.

9.2.1. Let

$$\text{Gr}_{G, \text{Bun}_M, \text{Ran}} := \text{Gr}_{G, \text{Ran}} \widetilde{\times} \text{Bun}_M$$

be the fibration over $\text{Bun}_M \times \text{Ran}$ that associates to a point

$$(\mathcal{P}_M, \underline{x}) \in \text{Bun}_M \times \text{Ran}$$

the twist $\text{Gr}_{G, \mathcal{P}_M, \underline{x}}$, i.e., we twist $\text{Gr}_{G, \underline{x}}$ by the $\mathfrak{L}^+(G)_{\underline{x}}$ -torsor induced by the *canonical*³⁵ map

$$(9.10) \quad \text{pt} / M \rightarrow \text{pt} / P^- \rightarrow \text{pt} / G$$

from the $\mathfrak{L}^+(M)_{\underline{x}}$ -torsor $\mathcal{P}_M|_{\mathcal{D}_{\underline{x}}}$.

9.2.2. Let $\widetilde{\mathfrak{s}}_{\text{Ran}}$ denote the tautological projection

$$\text{Gr}_{G, \text{Bun}_M, \text{Ran}} \rightarrow \mathfrak{L}^+(P^-)_{\text{Ran}} \setminus \text{Gr}_{G, \text{Ran}}.$$

We will denote by $\widetilde{\mathfrak{s}}_{\text{Ran}}^\dagger$ the functor

$$\mathcal{F} \mapsto \mathcal{F} \widetilde{\otimes}_{\omega_{\text{Bun}_M}}, \quad \text{D-mod}(\text{Gr}_{G, \text{Ran}})^{\mathfrak{L}^+(P^-)_{\text{Ran}}} \rightarrow \text{D-mod}(\text{Gr}_{G, \text{Bun}_M, \text{Ran}}).$$

We will use the same symbol $\widetilde{\mathfrak{s}}_{\text{Ran}}^\dagger$ for the corresponding functor in the context of twisted D-modules.

³⁵The splitting $M \rightarrow P^-$ depends on the choice of P , and hence so does the map $\text{pt} / M \rightarrow \text{pt} / P^-$ as *pointed* stacks. However, the map $\text{pt} / M \rightarrow \text{pt} / P^-$ as just stacks is canonically independent of the choice of P .

9.2.3. By the Beauville-Laszlo theorem, we can interpret $\mathrm{Gr}_{G, \mathrm{Bun}_M, \mathrm{Ran}}$ as the moduli space of quadruples

$$(\underline{x}, \mathcal{P}_G, \mathcal{P}_M, \alpha),$$

where:

- $\underline{x} \in \mathrm{Ran}$;
- $\mathcal{P}_M$ is an M -bundle on X (we will denote by $\mathcal{P}_{G, M}^0$ the induced G -bundle by means of (9.10));
- $\mathcal{P}_G$ is a G -bundle on X ;
- α is an isomorphism between $\mathcal{P}_G$ and $\mathcal{P}_{G, M}^0$ over $X - \underline{x}$.

9.2.4. From the above interpretation of $\mathrm{Gr}_{G, \mathrm{Bun}_M, \mathrm{Ran}}$, and the fact that $\mathcal{P}_{G, M}^0$ has a tautological reduction to P^- , we obtain that there exists a canonically defined map

$$\pi_{\mathrm{Ran}} : \mathrm{Gr}_{G, \mathrm{Bun}_M, \mathrm{Ran}} \rightarrow \widetilde{\mathrm{Bun}}_{P^-, \mathrm{Ran}},$$

which commutes with the projections of both spaces to Ran , Bun_G and Bun_M , respectively.

Note also that as in (8.15), we have a canonical identification of μ_2 -gerbes on $\mathrm{Gr}_{G, \mathrm{Bun}_M, \mathrm{Ran}}$

$$\pi_{\mathrm{Ran}}^*(\det_{G, M}^{\frac{1}{2}}) \simeq \widetilde{\mathfrak{s}}_{\mathrm{Ran}}^*(\det_{\mathrm{Gr}_{G, \mathrm{Ran}}}^{\frac{1}{2}}).$$

9.2.5. Let $\mathcal{Z} \rightarrow \mathrm{Ran}$. We will denote by $\pi_{\mathcal{Z}}$ the corresponding map

$$\mathrm{Gr}_{G, \mathrm{Bun}_M, \mathcal{Z}} \rightarrow \widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}}.$$

We claim that there is a canonically defined functor

$$(9.11) \quad \mathrm{I}(G, P^-)_{\mathrm{co}, \mathcal{Z}}^{\mathrm{loc}} \rightarrow \mathrm{D-mod}_{\det_{G, M}^{\frac{1}{2}}}(\widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}})_{\mathrm{co}} / \mathrm{Bun}_M,$$

to be denoted $(\pi_* \circ \widetilde{\mathfrak{s}}^!)_{\mathcal{Z}}^{\mathrm{Av}}$.

To simplify the notation, instead of (9.11), we will describe its version when $\mathcal{Z} = \mathrm{pt}$ corresponding to $\underline{x} \in \mathrm{Ran}$:

$$(9.12) \quad (\pi_* \circ \widetilde{\mathfrak{s}}^!)_{\underline{x}}^{\mathrm{Av}} : \mathrm{I}(G, P^-)_{\mathrm{co}, \underline{x}}^{\mathrm{loc}} \rightarrow \mathrm{D-mod}_{\det_{G, M}^{\frac{1}{2}}}(\widetilde{\mathrm{Bun}}_{P^-, \underline{x}})_{\mathrm{co}} / \mathrm{Bun}_M.$$

Warning: For the construction below, it is essential that $\underline{x} \neq \emptyset$.

Remark 9.2.6. The construction below could be coarsened and considered to take values in be given with values in $\mathrm{D-mod}_{\det_{G, M}^{\frac{1}{2}}}(\widetilde{\mathrm{Bun}}_{P^-, \underline{x}})$, instead of $\mathrm{D-mod}_{\det_{G, M}^{\frac{1}{2}}}(\widetilde{\mathrm{Bun}}_{P^-, \underline{x}})_{\mathrm{co}} / \mathrm{Bun}_M$. However, we need the finer versions (i.e., with values in $\mathrm{D-mod}_{\det_{G, M}^{\frac{1}{2}}}(\widetilde{\mathrm{Bun}}_{P^-, \underline{x}})_{\mathrm{co}} / \mathrm{Bun}_M$) in what follows.

Note also that we could *not* further refine it to take values in $\mathrm{D-mod}_{\det_{G, M}^{\frac{1}{2}}}(\widetilde{\mathrm{Bun}}_{P^-, \underline{x}})_{\mathrm{co}}$: indeed objects that we construct exhibit a $\mathrm{D-mod}(-)$ -behavior (rather than a $\mathrm{D-mod}(-)_{\mathrm{co}}$ -behavior) along Bun_M .

9.2.7. We start with the functor

$$\mathrm{D-mod}_{\det_{\mathrm{Gr}_{G, \mathrm{Ran}}}^{\frac{1}{2}}}(\mathrm{Gr}_{G, \underline{x}})^{\mathfrak{S}^+(P^-)} \xrightarrow{\widetilde{\mathfrak{s}}_{\underline{x}}^!} \mathrm{D-mod}_{\widetilde{\mathfrak{s}}^*(\det_{\mathrm{Gr}_{G, \mathrm{Ran}}}^{\frac{1}{2}})}(\mathrm{Gr}_{G, \mathrm{Bun}_M, \underline{x}}) = \mathrm{D-mod}_{\pi^*(\det_{G, M}^{\frac{1}{2}})}(\mathrm{Gr}_{G, \mathrm{Bun}_M, \underline{x}}).$$

Composing with the functor

$$(\pi_{\underline{x}})_* : \mathrm{D-mod}_{\widetilde{\mathfrak{s}}^*(\det_{\mathrm{Gr}_{G, \mathrm{Ran}}}^{\frac{1}{2}})}(\mathrm{Gr}_{G, \mathrm{Bun}_M, \underline{x}}) \rightarrow \mathrm{D-mod}_{\det_{G, M}^{\frac{1}{2}}}(\widetilde{\mathrm{Bun}}_{P^-, \underline{x}})_{\mathrm{co}} / \mathrm{Bun}_M$$

we obtain a functor

$$(9.13) \quad (\pi_{\underline{x}})_* \circ \widetilde{\mathfrak{s}}_{\underline{x}}^! : \mathrm{D-mod}_{\det_{\mathrm{Gr}_{G, \mathrm{Ran}}}^{\frac{1}{2}}}(\mathrm{Gr}_{G, \underline{x}})^{\mathfrak{S}^+(P^-)} \rightarrow \mathrm{D-mod}_{\det_{G, M}^{\frac{1}{2}}}(\widetilde{\mathrm{Bun}}_{P^-, \underline{x}})_{\mathrm{co}} / \mathrm{Bun}_M.$$

9.2.8. We now apply to (9.13) the procedure analogous to one in [GLC2, Sect. 9.5.4]. Namely, let N^α be sequence of group subschemes of $\mathfrak{L}(N_P^-)$, normalized by $\mathfrak{L}^+(P^-)$.

For a pair of indices $\alpha' \geq \alpha$ we have a natural transformation

$$(9.14) \quad (\pi_{\underline{x}})_* \circ \widetilde{\mathfrak{s}}_{\underline{x}}^! \circ \mathrm{Av}_*^{N^{\alpha'}} \rightarrow (\pi_{\underline{x}})_* \circ \widetilde{\mathfrak{s}}_{\underline{x}}^! \circ \mathrm{Av}_*^{N^\alpha}.$$

As in [GLC2, Sect. 9.5.4] one shows that (9.14) is an isomorphism for α large enough.³⁶

9.2.9. This implies that for any α that is large enough, the functor

$$(\pi_{\underline{x}})_* \circ \widetilde{\mathfrak{s}}_{\underline{x}}^! \circ \mathrm{Av}_*^{N^\alpha} : \mathrm{D}\text{-mod}_{\det \frac{1}{2} \mathrm{Gr}_{G, \mathrm{Ran}}} (\mathrm{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(P^-)} \rightarrow \mathrm{D}\text{-mod}_{\det \frac{1}{2} \widetilde{G}, M} (\widetilde{\mathrm{Bun}}_{P^-, \underline{x}})_{\mathrm{co}} / \mathrm{Bun}_M$$

factors as

$$(9.15) \quad \begin{aligned} \mathrm{D}\text{-mod}_{\det \frac{1}{2} \mathrm{Gr}_{G, \mathrm{Ran}}} (\mathrm{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(P^-)} &\simeq \mathrm{D}\text{-mod}_{\det \frac{1}{2} \mathrm{Gr}_{G, \mathrm{Ran}}} (\mathrm{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(M)} \rightarrow \\ &\rightarrow \mathrm{D}\text{-mod}_{\det \frac{1}{2} \mathrm{Gr}_{G, \mathrm{Ran}}} (\mathrm{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(M)}_{\mathfrak{L}(N_P^-)} \rightarrow \mathrm{D}\text{-mod}_{\det \frac{1}{2} \widetilde{G}, M} (\widetilde{\mathrm{Bun}}_{P^-, \underline{x}})_{\mathrm{co}} / \mathrm{Bun}_M. \end{aligned}$$

Precomposing the last arrow in (9.15) with

$$\mathrm{I}(G, P^-)_{\mathrm{co}, \underline{x}}^{\mathrm{loc}} \rightarrow \mathrm{D}\text{-mod}_{\det \frac{1}{2} \mathrm{Gr}_{G, \mathrm{Ran}}} (\mathrm{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(M)}_{\mathfrak{L}(N_P^-)},$$

we obtain the sought-for functor $(\pi_* \circ \widetilde{\mathfrak{s}}^!)_{\underline{x}}^{\mathrm{Av}}$ of (9.12).

9.2.10. Note that we can view the assignment

$$\mathcal{Z} \rightsquigarrow \mathrm{D}\text{-mod}_{\det \frac{1}{2} \widetilde{G}, M} (\widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}})_{\mathrm{co}} / \mathrm{Bun}_M$$

as a unital crystal of categories over Ran .

Namely, for $\underline{x}_1 \subset \underline{x}_2$, the corresponding functor

$$\mathrm{D}\text{-mod}_{\det \frac{1}{2} \widetilde{G}, M} (\widetilde{\mathrm{Bun}}_{P^-, \underline{x}_1})_{\mathrm{co}} / \mathrm{Bun}_M \rightarrow \mathrm{D}\text{-mod}_{\det \frac{1}{2} \widetilde{G}, M} (\widetilde{\mathrm{Bun}}_{P^-, \underline{x}_2})_{\mathrm{co}} / \mathrm{Bun}_M$$

is given by direct image along the closed embedding

$$\widetilde{\mathrm{Bun}}_{P^-, \underline{x}_1} \rightarrow \widetilde{\mathrm{Bun}}_{P^-, \underline{x}_2}.$$

9.2.11. By construction, the assignment

$$\mathcal{Z} \rightsquigarrow (\pi_* \circ \widetilde{\mathfrak{s}}^!)_{\mathcal{Z}}^{\mathrm{Av}}$$

has a structure of *left-lax* unital structure³⁷ as a functor between crystals of categories over Ran . Denote this functor by $(\pi_* \circ \widetilde{\mathfrak{s}}^!)^{\mathrm{Av}}$.

However, we claim:

Proposition 9.2.12. *The left-lax unital structure on $(\pi_* \circ \widetilde{\mathfrak{s}}^!)^{\mathrm{Av}}$ is strict.*

9.3. **The strong approximation argument.** In this subsection we will prove Proposition 9.2.12.

As we shall see, the proof essentially amounts to the strong approximation theorem applied to the *unipotent* group N_P^- , cf. [Ga6, Sect. 5.2.7 and Proposition 5.2.8].

³⁶The index α for which the stabilization occurs depends on the connected component of Bun_M .

³⁷See [GLC2, Sect. I.1.2] what this means.

9.3.1. To simplify the notation, we will prove the proposition for a fixed pair of points $\underline{x} \subseteq \underline{x}'$ of Ran .

Further, let us work over $1 \in \text{Bun}_M$, so that instead of $\widetilde{\text{Bun}}_{P^-, \underline{x}}$ we are dealing with

$$\widetilde{\text{Bun}}_{N_P^-, \underline{x}} := \widetilde{\text{Bun}}_{P^-, \underline{x}} \times_{\text{Bun}_M} \text{pt}$$

and instead of $\text{Gr}_{G, \text{Bun}_M, \underline{x}}$ just with $\text{Gr}_{G, \underline{x}}$, and similarly for $\underline{x}'$.

9.3.2. Let $N_{P, \text{out}, \underline{x}}^-$ be the group ind-scheme $\mathbf{Maps}(X - \underline{x}, N_P^-)$. Let

$$\mathfrak{L}^+(N_P^-)_{\underline{x}} \subset N_{\underline{x}}^\alpha \subset \mathfrak{L}(N_P^-)_{\underline{x}}$$

be a group subscheme large enough so that

$$N_{P, \text{out}, \underline{x}}^- \cdot N^\alpha = \mathfrak{L}(N_P^-)_{\underline{x}}.$$

By definition, the functor $(\pi_* \circ \widetilde{\mathfrak{S}}^!_{\underline{x}})^{\text{Av}}$ equals

$$(\pi_{\underline{x}})_* \circ \text{Av}_*^{N_{\underline{x}}^\alpha},$$

Let $N_{\underline{x}}^\alpha \subset \mathfrak{L}(N_P^-)_{\underline{x}'}$ be a suitable subgroup for $\underline{x}'$.

So,

$$(\pi_* \circ \widetilde{\mathfrak{S}}^!_{\underline{x}'})^{\text{Av}} = (\pi_{\underline{x}'})_* \circ \text{Av}_*^{N_{\underline{x}'}^\alpha}.$$

9.3.3. Let

$$N_{\underline{x} \subseteq \underline{x}'}^\alpha := N_{\underline{x}'}^\alpha \cap \mathbf{Maps}(\mathcal{D}_{\underline{x}'} - \underline{x}, N_P^-) \subset N_{\underline{x}'}^\alpha.$$

With no restriction of generality, we can assume that the subgroups $N_{\underline{x}}^\alpha$ and $N_{\underline{x}'}^\alpha$ are chosen so that $N_{\underline{x} \subseteq \underline{x}'}^\alpha$ equals the preimage of $N_{\underline{x}}^\alpha$ under the projection

$$\mathbf{Maps}(\mathcal{D}_{\underline{x}'} - \underline{x}, N_P^-) \rightarrow \mathbf{Maps}(\mathcal{D}_{\underline{x}} - \underline{x}, N_P^-) = \mathfrak{L}(N_P^-)_{\underline{x}}.$$

9.3.4. The left-unital structure on $(\pi_* \circ \widetilde{\mathfrak{S}}^!_{\underline{x}})^{\text{Av}}$ is given by the system of maps

$$\begin{aligned} (9.16) \quad & (\pi_* \circ \widetilde{\mathfrak{S}}^!_{\underline{x}'})^{\text{Av}} \circ (\text{ins. unit}_{\underline{x} \subseteq \underline{x}'})_* = \\ & = (\pi_{\underline{x}'})_* \circ \text{Av}_*^{N_{\underline{x}'}^\alpha} \circ (\text{ins. unit}_{\underline{x} \subseteq \underline{x}'})_* \rightarrow (\pi_{\underline{x}'})_* \circ \text{Av}_*^{N_{\underline{x} \subseteq \underline{x}'}^\alpha} \circ (\text{ins. unit}_{\underline{x} \subseteq \underline{x}'})_* \simeq \\ & \simeq (\pi_{\underline{x}'})_* \circ (\text{ins. unit}_{\underline{x} \subseteq \underline{x}'})_* \circ \text{Av}_*^{N_{\underline{x}}^\alpha} \simeq (\pi_{\underline{x}})_* \circ \text{Av}_*^{N_{\underline{x}}^\alpha} = (\pi_* \circ \widetilde{\mathfrak{S}}^!_{\underline{x}})^{\text{Av}}, \end{aligned}$$

where:

- $\text{ins. unit}_{\underline{x} \subseteq \underline{x}'} : \text{Gr}_{G, \underline{x}} \rightarrow \text{Gr}_{G, \underline{x}'}$ is the unital structure on Gr_G ;
- The next-to-last isomorphism is thanks to the choice of subgroups in Sect. 9.3.3.

We wish to show that the maps (9.16) are isomorphisms. I.e., we wish to show that the second arrow in (9.16) is an isomorphism.

9.3.5. Denote

$$N_{\text{out}, \underline{x}}^\alpha := N_{P, \text{out}, \underline{x}}^- \cap N^\alpha,$$

and similarly for $\underline{x}'$.

We factor the second arrow in (9.16) as

$$\begin{aligned} (9.17) \quad & (\pi_{\underline{x}'})_* \circ \text{Av}_*^{N_{\underline{x}'}^\alpha} \circ (\text{ins. unit}_{\underline{x} \subseteq \underline{x}'})_* \rightarrow (\pi_{\underline{x}'})_* \circ \text{Av}_*^{N_{\text{out}, \underline{x}'}^\alpha} \circ \text{Av}_*^{N_{\underline{x} \subseteq \underline{x}'}^\alpha} \circ (\text{ins. unit}_{\underline{x} \subseteq \underline{x}'})_* \rightarrow \\ & \rightarrow (\pi_{\underline{x}'})_* \circ \text{Av}_*^{N_{\underline{x} \subseteq \underline{x}'}^\alpha} \circ (\text{ins. unit}_{\underline{x} \subseteq \underline{x}'})_*. \end{aligned}$$

We claim that both arrows in (9.17) are isomorphisms when the subgroups $N_{\underline{x}}^\alpha$ and $N_{\underline{x}'}^\alpha$ are large enough.

9.3.6. Up to enlarging $N_{\underline{x}}^\alpha$ and $N_{\underline{x}'}^\alpha$, we can assume that

$$(9.18) \quad N_{\text{out}, \underline{x}'}^\alpha \cdot N_{\underline{x} \subseteq \underline{x}'}^\alpha = N_{\underline{x}'}^\alpha,$$

which is a form of *strong approximation* for N_P^- .

The equality (9.18) implies that the first arrow in (9.17) is an isomorphism.

9.3.7. We now prove that the second arrow in (9.17) is an isomorphism. We rewrite it equivalently as

$$(\pi_{\underline{x}'})_* \circ \text{Av}_*^{N_{\text{out}, \underline{x}'}^\alpha} \circ (\text{ins. unit}_{\underline{x} \subseteq \underline{x}'})_* \circ \text{Av}_*^{N_{\underline{x}}^\alpha} \rightarrow (\pi_{\underline{x}'})_* \circ (\text{ins. unit}_{\underline{x} \subseteq \underline{x}'})_* \circ \text{Av}_*^{N_{\underline{x}}^\alpha}.$$

In fact, we claim that the map

$$(9.19) \quad (\pi_{\underline{x}'})_* \circ \text{Av}_*^{N_{\text{out}, \underline{x}'}^\alpha} \circ (\text{ins. unit}_{\underline{x} \subseteq \underline{x}'})_* \rightarrow (\pi_{\underline{x}'})_* \circ (\text{ins. unit}_{\underline{x} \subseteq \underline{x}'})_*$$

is an isomorphism when evaluated on $N_{\text{out}, \underline{x}}^\alpha$ -equivariant objects.

9.3.8. Indeed, we note that the map

$$\pi_{\underline{x}} : \text{Gr}_{G, \underline{x}} \rightarrow \widetilde{\text{Bun}}_{N_P^-, \underline{x}}$$

factors as

$$(9.20) \quad \text{Gr}_{G, \underline{x}} \rightarrow N_{P, \text{out}, \underline{x}}^- \backslash \text{Gr}_{G, \underline{x}} \xrightarrow{\pi_{\underline{x}}} \widetilde{\text{Bun}}_{N_P^-, \underline{x}},$$

and similarly for $\underline{x}'$.

The isomorphism (9.19) follows now from the next commutative diagram:

$$(9.21) \quad \begin{array}{ccc} N_{\text{out}, \underline{x}'}^\alpha \backslash \left(\frac{N_{P, \text{out}, \underline{x}'}^\alpha \times \text{Gr}_{G, \underline{x}}}{N_{\text{out}, \underline{x}}^\alpha} \right) & \xrightarrow{\text{action} \circ \text{ins. unit}_{\underline{x} \subseteq \underline{x}'}} & N_{\text{out}, \underline{x}'}^\alpha \backslash \text{Gr}_{G, \underline{x}'} \\ \sim \downarrow & & \\ N_{\text{out}, \underline{x}}^\alpha \backslash \text{Gr}_{G, \underline{x}} & & \downarrow \pi_{\underline{x}'} \\ \pi_{\underline{x}} \downarrow & & \\ \widetilde{\text{Bun}}_{N_P^-, \underline{x}} & \longrightarrow & \widetilde{\text{Bun}}_{N_P^-, \underline{x}'} \end{array}$$

Indeed, the left- (resp. right-) hand side in (9.19) is given by pushforward along the clockwise (resp., counterclockwise) circuit in (9.21).

9.4. The global enhanced functors, the *-version.

9.4.1. For $\mathbb{Z} \rightarrow \text{Ran}$, define the functor

$$\begin{aligned} \text{pre-CT}_{*, \mathbb{Z}}^{-, \text{enhco, glob}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{I}(G, P^-)_{\text{co}, \mathbb{Z}}^{\text{loc}} &\rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M \times \mathbb{Z}) \simeq \\ &\simeq \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathbb{Z}) \end{aligned}$$

to be

$$(9.22) \quad (\widetilde{\mathfrak{q}}_{\mathbb{Z}}^-)_* \left((\widetilde{\mathfrak{p}}^-)^!(-) \overset{!}{\otimes} (\pi_* \circ \widetilde{\mathfrak{s}}^!_{\mathbb{Z}})^{\text{Av}}(-) \right) [\text{shift}],$$

where:

- The functor $(\widetilde{\mathfrak{p}}^-)^!$ is understood in the sense (9.2);
- The functor $\overset{!}{\otimes}$ is understood in the sense (9.8);
- The functor $(\widetilde{\mathfrak{q}}_{\mathbb{Z}}^-)_*$ is understood in the sense (9.6);
- [shift] is as in (8.4).

NB: the sign of the shift is opposite to that in (8.3).

9.4.2. Unwinding the constructions, we obtain that the functor $\text{pre-CT}_{*,\mathbb{Z}}^{-,\text{enhco, glob}}$ factors canonically as

$$\begin{aligned}
 (9.23) \quad & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{I}(G, P^-)_{\text{co}, \mathbb{Z}}^{\text{loc}} \simeq \\
 & \simeq \left(\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) \right)_{\text{D-mod}(\mathbb{Z})} \otimes \text{I}(G, P^-)_{\text{co}, \mathbb{Z}}^{\text{loc}} \rightarrow \\
 & \rightarrow \left(\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) \right)_{\text{Sph}_{G, \mathbb{Z}}} \otimes \text{I}(G, P^-)_{\text{co}, \mathbb{Z}}^{\text{loc}} = \\
 & = \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\mathbb{Z}}^{-, \text{enhco}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathbb{Z}).
 \end{aligned}$$

Denote the last arrow in (9.23) by

$$\text{CT}_{*, \mathbb{Z}}^{-, \text{enhco, glob}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\mathbb{Z}}^{-, \text{enhco}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathbb{Z}).$$

By construction, the functor $\text{CT}_{*, \mathbb{Z}}^{-, \text{enhco, glob}}$ intertwines the actions of $\text{Sph}_{M, \mathbb{Z}}$ on the two sides.

Denote also

$$\text{CT}_*^{-, \text{enhco, glob}} := (\text{Id} \otimes \text{C}(\text{Ran}, -)) \circ \text{CT}_{*, \text{Ran}}^{-, \text{enhco, glob}}, \quad \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{Ran}}^{-, \text{enhco}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M).$$

9.4.3. We claim:

Proposition 9.4.4. *There is a canonical isomorphism*

$$\text{CT}_{*, \mathbb{Z}}^{-, \text{enhco, glob}} \simeq \text{CT}_{*, \mathbb{Z}}^{-, \text{enhco}},$$

as $\text{Sph}_{M, \mathbb{Z}}$ -linear functors

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\mathbb{Z}}^{-, \text{enhco}} \rightrightarrows \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathbb{Z}).$$

Proof. In order to simplify the notation we will establish the required isomorphism when $\mathbb{Z} = \text{pt}$, i.e.,

$$(9.24) \quad \text{CT}_{*, \underline{x}}^{-, \text{enhco, glob}} \simeq \text{CT}_{*, \underline{x}}^{-, \text{enhco}}$$

as $\text{Sph}_{M, \underline{x}}$ -linear functors

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\underline{x}}^{-, \text{enhco}} \rightrightarrows \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)$$

for $\underline{x} \in \text{Ran}$.

Note that there is a naturally defined map

$$\mathbf{f}_{\underline{x}} : \text{Hecke}_{G, P^-, \underline{x}}^{\text{glob}} \rightarrow \widetilde{\text{Bun}}_{P^-, \underline{x}},$$

which respects the projections of both spaces to Bun_G and Bun_M , respectively. In addition, we have

$$\mathbf{f}_{\underline{x}}^*(\det_{G, M}^{\frac{1}{2}}) \simeq \mathbf{s}_{\underline{x}}^*(\det_{\text{Gr}_{G, \underline{x}}}^{\frac{1}{2}}),$$

where $\mathbf{s}_{\underline{x}}$ is the map from (8.14).

Consider the functor

$$(\mathbf{f}_{\underline{x}})_* \circ \mathbf{s}_{\underline{x}}^! : \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \underline{x}})^{\mathfrak{S}^+(P^-)_{\underline{x}}} \rightarrow \text{D-mod}_{\det_{G, M}^{\frac{1}{2}}}(\widetilde{\text{Bun}}_{P^-, \underline{x}})_{\text{co}} / \text{Bun}_M.$$

As in Sects. 9.2.8-9.2.9, from $(\mathbf{f}_{\underline{x}})_* \circ \mathbf{s}_{\underline{x}}^!$ we obtain a functor

$$((\mathbf{f}_{\underline{x}})_* \circ \mathbf{s}_{\underline{x}}^!)^{\text{Av}} : \text{I}(G, P^-)_{\text{co}, \underline{x}}^{\text{loc}} \rightarrow \text{D-mod}_{\det_{G, M}^{\frac{1}{2}}}(\widetilde{\text{Bun}}_{P^-, \underline{x}})_{\text{co}} / \text{Bun}_M.$$

We claim:

Lemma 9.4.5. *There is a canonical isomorphism*

$$((\mathbf{f}_{\underline{x}})_* \circ \mathbf{s}_{\underline{x}}^!)^{\text{Av}} \simeq (\pi_* \circ \widetilde{\mathbf{s}}^!)_{\underline{x}}^{\text{Av}}[2 \cdot \text{shift}].$$

The proof will be given in Sect. 9.5.

The isomorphism (9.24) follows from Lemma 9.4.5 by the projection formula. $\square$

Corollary 9.4.6. *There is a canonical isomorphism*

$$\mathrm{CT}_*^{-,\mathrm{enh}_{\mathrm{co}},\mathrm{glob}} \simeq \mathrm{CT}_*^{-,\mathrm{enh}_{\mathrm{co}}},$$

as functors

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^{-,\mathrm{enh}_{\mathrm{co}}} \Rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M).$$

9.4.7. Define the functor

$$\begin{aligned} \mathrm{pre}\text{-}\mathrm{Eis}_{\mathrm{co},*,\mathbb{Z}}^{-,\mathrm{enh}_{\mathrm{co}},\mathrm{glob}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} \otimes \mathrm{I}(G, P^-)_{\mathrm{co},\mathbb{Z}}^{\mathrm{loc}} &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G \times \mathbb{Z})_{\mathrm{co}} \simeq \\ &\simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(\mathbb{Z}) \end{aligned}$$

to be

$$(9.25) \quad (\tilde{\mathbf{p}}_{\mathbb{Z}}^-)_* \left((\tilde{\mathbf{q}}^-)^!(-) \otimes (\pi_* \circ \tilde{\mathbf{s}}^!)_{\mathbb{Z}}^{\mathrm{Av}}(-) \right) [\mathrm{shift}],$$

where:

- The functor $(\tilde{\mathbf{q}}^-)^!(-) \otimes (-)$ is understood in the sense of (9.9);
- The functor $(\tilde{\mathbf{p}}_{\mathbb{Z}}^-)_*$ is understood in the sense of (9.4);
- $[\mathrm{shift}]$ is as in (8.4).

9.4.8. Unwinding the constructions, we obtain that the functor $\mathrm{pre}\text{-}\mathrm{Eis}_{\mathrm{co},*,\mathbb{Z}}^{-,\mathrm{enh}_{\mathrm{co}},\mathrm{glob}}$ factors canonically as

$$\begin{aligned} (9.26) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} \otimes \mathrm{I}(G, P^-)_{\mathrm{co},\mathbb{Z}}^{\mathrm{loc}} &\simeq \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(\mathbb{Z}) \right)_{\mathrm{D}\text{-mod}(\mathbb{Z})} \otimes \mathrm{I}(G, P^-)_{\mathrm{co},\mathbb{Z}}^{\mathrm{loc}} \rightarrow \\ &\rightarrow \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(\mathbb{Z}) \right)_{\mathrm{Sph}_{M,\mathbb{Z}}} \otimes \mathrm{I}(G, P^-)_{\mathrm{co},\mathbb{Z}}^{\mathrm{loc}} = \\ &= \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co},\mathbb{Z}}^{-,\mathrm{enh}_{\mathrm{co}}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(\mathbb{Z}). \end{aligned}$$

Denote the last arrow in (9.23) by

$$\mathrm{Eis}_{\mathrm{co},*,\mathbb{Z}}^{-,\mathrm{enh}_{\mathrm{co}},\mathrm{glob}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co},\mathbb{Z}}^{-,\mathrm{enh}_{\mathrm{co}}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(\mathbb{Z}).$$

By construction, the functor $\mathrm{Eis}_{\mathrm{co},*,\mathbb{Z}}^{-,\mathrm{enh}_{\mathrm{co}}}$ intertwines the actions of $\mathrm{Sph}_{G,\mathbb{Z}}$ on the two sides.

9.4.9. Finally, denote

$$\mathrm{Eis}_{\mathrm{co},*}^{-,\mathrm{enh}_{\mathrm{co}},\mathrm{glob}} := (\mathrm{Id} \otimes \mathbb{C}(\mathrm{Ran}, -)) \circ \mathrm{Eis}_{\mathrm{co},*,\mathrm{Ran}}^{-,\mathrm{enh}_{\mathrm{co}},\mathrm{glob}}, \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co},\mathrm{Ran}}^{-,\mathrm{enh}_{\mathrm{co}}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}.$$

By duality, from Proposition 9.4.4 we obtain:

Corollary 9.4.10.

(a) *There is a canonical isomorphism*

$$\mathrm{Eis}_{\mathrm{co},*,\mathbb{Z}}^{-,\mathrm{enh}_{\mathrm{co}},\mathrm{glob}} \simeq \mathrm{Eis}_{\mathrm{co},*}^{-,\mathrm{enh}_{\mathrm{co}}}$$

as $\mathrm{Sph}_{G,\mathbb{Z}}$ -linear functors

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co},\mathbb{Z}}^{-,\mathrm{enh}_{\mathrm{co}}} \Rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \otimes \mathrm{D}\text{-mod}(\mathbb{Z}).$$

(b) *There is a canonical isomorphism*

$$\mathrm{Eis}_{\mathrm{co},*}^{-,\mathrm{enh}_{\mathrm{co}},\mathrm{glob}} \simeq \mathrm{Eis}_{\mathrm{co},*}^{-,\mathrm{enh}_{\mathrm{co}}}$$

as functors

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)^{-,\mathrm{enh}_{\mathrm{co}}}_{\mathrm{co},\mathrm{Ran}} \rightrightarrows \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}.$$

9.5. Proof of Lemma 9.4.5.

9.5.1. In order to simplify the exposition, we will prove the required isomorphism after restriction to the preimage of the trivial bundle $\mathcal{P}_M^0 \in \mathrm{Bun}_M$.

Denote:

$$\widetilde{\mathrm{Bun}}_{N_P^-, \underline{x}} := \mathrm{pt}_{\mathcal{P}_M^0, \mathrm{Bun}_M} \times \widetilde{\mathrm{Bun}}_{P^-, \underline{x}} \text{ and } \mathrm{Hecke}_{G, N_P^-, \underline{x}}^{\mathrm{glob}} := \mathrm{pt}_{\mathcal{P}_M^0, \mathrm{Bun}_M} \times \mathrm{Hecke}_{G, P^-, \underline{x}}^{\mathrm{glob}}.$$

We keep the same symbols for the corresponding maps

$$\mathbf{f}_{\underline{x}} : \mathrm{Hecke}_{G, N_P^-, \underline{x}}^{\mathrm{glob}} \rightarrow \widetilde{\mathrm{Bun}}_{N_P^-, \underline{x}},$$

$$\pi_{\underline{x}} : \mathrm{Gr}_{G, \underline{x}} \rightarrow \widetilde{\mathrm{Bun}}_{N_P^-, \underline{x}},$$

$$\mathbf{s}_{\underline{x}} : \mathrm{Hecke}_{G, N_P^-, \underline{x}}^{\mathrm{glob}} \rightarrow \mathfrak{L}^+(N_P^-)_{\underline{x}} \backslash \mathrm{Gr}_{G, \underline{x}},$$

and

$$\widetilde{\mathbf{s}}_{\underline{x}} : \mathrm{Gr}_{G, \underline{x}} \rightarrow \mathfrak{L}^+(N_P^-)_{\underline{x}} \backslash \mathrm{Gr}_{G, \underline{x}}.$$

We will prove an isomorphism

$$((\mathbf{f}_{\underline{x}})_* \circ \mathbf{s}_{\underline{x}}^!)^{\mathrm{Av}} \simeq (\pi_* \circ \widetilde{\mathbf{s}}_{\underline{x}}^!)^{\mathrm{Av}} [2\delta_{N_P^-}]$$

as functors

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(N_P^-)_{\underline{x}}} \rightrightarrows \mathrm{D}\text{-mod}_{\frac{1}{2}}(\widetilde{\mathrm{Bun}}_{N_P^-, \underline{x}}).$$

9.5.2. Let $\mathrm{Bun}_{N_P^-}^{\mathrm{level}_{\underline{x}}}$ denote the moduli space of N_P^- -bundles on X with a full level structure at $\underline{x}$. We have

$$\mathrm{Hecke}_{G, N_P^-, \underline{x}}^{\mathrm{glob}} \simeq (\mathrm{Bun}_{N_P^-}^{\mathrm{level}_{\underline{x}}} \times \mathrm{Gr}_{G, \underline{x}}) / \mathfrak{L}^+(N_P^-)_{\underline{x}}.$$

For a group subscheme

$$\mathfrak{L}^+(N_P^-)_{\underline{x}} \subseteq N^\alpha \subset \mathfrak{L}(N_P^-)_{\underline{x}}$$

denote

$$\mathcal{Y}^\alpha := (\mathrm{Bun}_{N_P^-}^{\mathrm{level}_{\underline{x}}} \times \mathrm{Gr}_{G, \underline{x}}) / N^\alpha.$$

We can regard $\mathcal{Y}^\alpha$ as a fibration with typical fiber $\mathrm{Gr}_{G, \underline{x}}$ over

$$\mathrm{Bun}_{N_P^-}^\alpha := \mathrm{Bun}_{N_P^-}^{\mathrm{level}_{\underline{x}}} / N^\alpha,$$

i.e.,

$$\mathcal{Y}^\alpha \simeq \mathrm{Gr}_{G, \underline{x}} \widetilde{\times} \mathrm{Bun}_{N_P^-}^\alpha.$$

Denote by $\mathbf{s}^\alpha$ the resulting map

$$\mathcal{Y}^\alpha \rightarrow N^\alpha \backslash \mathrm{Gr}_{G, \underline{x}}.$$

Let $(\mathbf{s}^\alpha)^!$ denote the functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \underline{x}})^{N^\alpha} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathcal{Y}^\alpha), \quad \mathcal{F} \mapsto \mathcal{F} \boxtimes_{\mathrm{Bun}_{N_P^-}^\alpha}.$$

9.5.3. Note that the map $f_{\underline{x}}$ factors as

$$\mathrm{Hecke}_{G, N_P^-, \underline{x}}^{\mathrm{glob}} \rightarrow \mathcal{Y}^\alpha \xrightarrow{f^\alpha} \widetilde{\mathrm{Bun}}_{N_P^-, \underline{x}},$$

where the first arrow is a fibration with contractible fibers of relative dimension

$$d_\alpha := \dim(N^\alpha / \mathcal{L}^+(N_P^-)_{\underline{x}}).$$

From here, we obtain that

$$(9.27) \quad f_*^\alpha \circ (s^\alpha)^\dagger \simeq (f_{\underline{x}})_* \circ s_{\underline{x}}^\dagger[-2d_\alpha],$$

as functors

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \underline{x}})^{N_\alpha} \rightrightarrows \mathrm{D}\text{-mod}_{\frac{1}{2}}(\widetilde{\mathrm{Bun}}_{N_P^-, \underline{x}}).$$

9.5.4. Note that we have a canonically defined map

$$g_{\underline{x}} : \mathrm{Gr}_{G, \underline{x}} \rightarrow \mathrm{Hecke}_{G, N_P^-, \underline{x}}^{\mathrm{glob}}$$

so that

$$f_{\underline{x}} \circ g_{\underline{x}} = \pi_{\underline{x}}, \quad \mathrm{Gr}_{G, \underline{x}} \rightrightarrows \widetilde{\mathrm{Bun}}_{P^-, \underline{x}}$$

and

$$s_{\underline{x}} \circ g_{\underline{x}} = \widetilde{s}_{\underline{x}}, \quad \mathrm{Gr}_{G, \underline{x}} \rightrightarrows \mathcal{L}^+(P^-)_{\underline{x}} \backslash \mathrm{Gr}_{G, \underline{x}}.$$

9.5.5. Let g^α denote the composition

$$\mathrm{Gr}_{G, \underline{x}} \xrightarrow{g_{\underline{x}}} \mathrm{Hecke}_{G, N_P^-, \underline{x}}^{\mathrm{glob}} \rightarrow \mathcal{Y}_\alpha.$$

We have

$$(9.28) \quad f^\alpha \circ g^\alpha = \pi_{\underline{x}} \text{ and } s^\alpha \circ g^\alpha = \widetilde{s}_{\underline{x}}.$$

9.5.6. The map g^α is the base change of the map

$$\mathrm{pt} \rightarrow \mathrm{Bun}_{N_P^-}^\alpha.$$

When α is large enough, the latter map is a fibration with contractible fibers of relative dimension $d_\alpha - \delta_{N_P^-}$. Hence, the same is true for g^α .

Hence, from (9.28) we obtain

$$(9.29) \quad (\pi_{\underline{x}})_* \circ \widetilde{s}_{\underline{x}}^\dagger \simeq (f^\alpha)_* \circ (s^\alpha)^\dagger[2(d_\alpha - \delta_{N_P^-})]$$

as functors

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \underline{x}})^{N_\alpha} \rightrightarrows \mathrm{D}\text{-mod}_{\frac{1}{2}}(\widetilde{\mathrm{Bun}}_{N_P^-, \underline{x}}).$$

9.5.7. Comparing (9.27) and (9.29), we obtain

$$(\pi_{\underline{x}})_* \circ \widetilde{s}^\dagger \simeq (f_{\underline{x}})_* \circ s_{\underline{x}}^\dagger[-2\delta_{N_P^-}],$$

as desired.

□[Lemma 9.4.5]

9.6. Local-to-global construction for Drinfeld's compactification, the !-version.

9.6.1. Consider the (partially defined) left adjoint $(\pi_{\mathcal{Z}})_!$ to

$$(9.30) \quad \begin{aligned} \pi_{\mathcal{Z}}^! : \mathrm{D}\text{-mod}_{\det_{G, M}^{\frac{1}{2}}}(\widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}}) &\rightarrow \\ &\rightarrow \mathrm{D}\text{-mod}_{\pi^*(\det_{G, M}^{\frac{1}{2}})}(\mathrm{Gr}_{G, \mathrm{Bun}_M, \mathcal{Z}}) \simeq \mathrm{D}\text{-mod}_{\widetilde{s}^*(\det_{\mathrm{Gr}_G, \mathrm{Ran}}^{\frac{1}{2}})}(\mathrm{Gr}_{G, \mathrm{Bun}_M, \mathcal{Z}}). \end{aligned}$$

9.6.2. We claim:

Proposition 9.6.3.

(a) The functor $(\pi_{\mathcal{Z}})_!$ is well-defined and is $(\mathrm{Sph}_G \otimes \mathrm{Sph}_M)_{\mathcal{Z}}$ -linear on the essential image of

$$(9.31) \quad \begin{aligned} \mathrm{I}(G, P^-)_{\mathcal{Z}}^{\mathrm{loc}} &\rightarrow \mathrm{D}\text{-mod}_{\det_{\mathrm{Gr}_{G,\mathcal{Z}}}^{\frac{1}{2}}} (\mathrm{Gr}_{G,\mathcal{Z}})^{\mathfrak{L}(N_P^-)_{\mathcal{Z}} \cdot \mathfrak{L}^+(M)_{\mathcal{Z}}} \hookrightarrow \\ &\rightarrow \mathrm{D}\text{-mod}_{\det_{\mathrm{Gr}_{G,\mathcal{Z}}}^{\frac{1}{2}}} (\mathrm{Gr}_{G,\mathcal{Z}})^{\mathfrak{L}^+(P^-)_{\mathcal{Z}}} \xrightarrow{\mathfrak{S}^!} \mathrm{D}\text{-mod}_{\mathfrak{S}^*(\det_{\mathrm{Gr}_{G,\mathcal{Z}}}^{\frac{1}{2}})} (\mathrm{Gr}_{G,\mathrm{Bun}_M,\mathcal{Z}}). \end{aligned}$$

Denote the resulting functor

$$\mathrm{I}(G, P^-)_{\mathcal{Z}}^{\mathrm{loc}} \rightarrow \mathrm{D}\text{-mod}_{\det_{G,M}^{\frac{1}{2}}} (\widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}})$$

by $(\pi_! \circ \mathfrak{S}^!)_{\mathcal{Z}}$.

(b) The assignment $\mathcal{Z} \rightsquigarrow (\pi_! \circ \mathfrak{S}^!)_{\mathcal{Z}}$ commutes with base change, i.e., for a map $f : \mathcal{Z}' \rightarrow \mathcal{Z}$, the natural transformation

$$(\pi_! \circ \mathfrak{S}^!)_{\mathcal{Z}'} \circ f^! \rightarrow f^! \circ (\pi_! \circ \mathfrak{S}^!)_{\mathcal{Z}}$$

is an isomorphism as functors

$$\mathrm{I}(G, P^-)_{\mathcal{Z}'}^{\mathrm{loc}} \rightrightarrows \mathrm{D}\text{-mod}_{\det_{G,M}^{\frac{1}{2}}} (\widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}'}).$$

(c) Let $\mathcal{Z} = S$ be a scheme. Then the functor $(\pi_! \circ \mathfrak{S}^!)_S$ sends compact objects in $\mathrm{I}(G, P^-)_S^{\mathrm{loc}}$ to objects in $\mathrm{D}\text{-mod}_{\det_{G,M}^{\frac{1}{2}}} (\widetilde{\mathrm{Bun}}_{P^-, S})$ that are ULA with respect to $\mathfrak{q}^- : \widetilde{\mathrm{Bun}}_{P^-, S} \rightarrow \mathrm{Bun}_M$.

The proof Proposition 9.6.3 will occupy the majority of this subsection.

Note that to prove points (a) and (b) we can assume that $\mathcal{Z}$ (resp., $\mathcal{Z}$ and $\mathcal{Z}'$) are schemes, to be denoted S and S' , respectively.

9.6.4. We will first show that $(\pi_! \circ \mathfrak{S}^!)_S^!$ is defined on $\Delta_S^{-, \frac{\infty}{2}} \in \mathrm{I}(G, P^-)_S^{\mathrm{loc}}$ and

$$(\pi_! \circ \mathfrak{S}^!)_S^! (\Delta_S^{-, \frac{\infty}{2}}) \simeq (j_S)_! (\omega_{\mathrm{Bun}_{P^-} \times S}),$$

where:

- j_S denotes the corresponding locally closed embedding $\mathrm{Bun}_{P^-} \times S \rightarrow \widetilde{\mathrm{Bun}}_{P^-, S}$;
- The object $\omega_{\mathrm{Bun}_{P^-} \times S} \in \mathrm{D}\text{-mod}_{\det_{G,M}^{\frac{1}{2}}} (\mathrm{Bun}_{P^-} \times S)$ makes sense due to the trivialization of the gerbe $\det_{G,M}^{\frac{1}{2}}|_{\mathrm{Bun}_{P^-}}$.

9.6.5. Consider the unit orbit of $\mathfrak{L}(N_P^-)$ acting on Gr_G :

$$\mathbf{i}_S : (\mathfrak{L}(N_P^-)/\mathfrak{L}^+(N_P^-))_S \rightarrow \mathrm{Gr}_{G,S}.$$

Recall that

$$(9.32) \quad \Delta_S^{-, \frac{\infty}{2}} = (\mathbf{i}_S)_! (\omega_{(\mathfrak{L}(N_P^-)/\mathfrak{L}^+(N_P^-))_S}).$$

Consider the corresponding locally closed embedding

$$(\mathfrak{L}(N_P^-)/\mathfrak{L}^+(N_P^-))_{\mathrm{Bun}_M, S} \xrightarrow{\mathbf{i}_{\mathrm{Bun}_M, S}} \mathrm{Gr}_{G, \mathrm{Bun}_M, S}.$$

From (9.32), we obtain

$$\mathfrak{S}_S^! (\Delta_S^{-, \frac{\infty}{2}}) \simeq (\mathbf{i}_{\mathrm{Bun}_M, S})_! (\omega_{(\mathfrak{L}(N_P^-)/\mathfrak{L}^+(N_P^-))_{\mathrm{Bun}_M, S}}).$$

In particular, the object $\mathfrak{S}_S^! (\Delta_S^{-, \frac{\infty}{2}})$ is *ind-holonomic*, so the partially defined functor $(\pi_S)_!$ is automatically defined on it.

9.6.6. Note that the map

$$(\mathfrak{L}(N_P^-)/\mathfrak{L}^+(N_P^-))_{\mathrm{Bun}_M, S} \xrightarrow{\mathbf{i}_{\mathrm{Bun}_M, S}} \mathrm{Gr}_{G, \mathrm{Bun}_M, S} \xrightarrow{\pi_S} \widetilde{\mathrm{Bun}}_{P^-, S}$$

factors as

$$(\mathfrak{L}(N_P^-)/\mathfrak{L}^+(N_P^-))_{\mathrm{Bun}_M, S} \xrightarrow{\pi_S^0} \mathrm{Bun}_{P^-} \times S \xrightarrow{j_S} \widetilde{\mathrm{Bun}}_{P^-, S}.$$

Moreover, the above map

$$(\mathfrak{L}(N_P^-)/\mathfrak{L}^+(N_P^-))_{\mathrm{Bun}_M, S} \xrightarrow{\pi_S^0} \mathrm{Bun}_{P^-} \times S$$

is a locally trivial fibration, whose fibers are ind-contractible.³⁸

Hence, the naturally defined map

$$(\pi_S^0)_!(\omega_{(\mathfrak{L}(N_P^-)/\mathfrak{L}^+(N_P^-))_{\mathrm{Bun}_M, S}}) \rightarrow \omega_{\mathrm{Bun}_{P^-} \times S}$$

is an isomorphism.

Therefore.

$$\begin{aligned} (\pi_! \circ \widetilde{\mathfrak{s}})_S(\Delta_S^{-, \frac{\infty}{2}}) &\simeq (\pi_S)_! \circ (\mathbf{i}_{\mathrm{Bun}_M, S})_!(\omega_{(\mathfrak{L}(N_P^-)/\mathfrak{L}^+(N_P^-))_{\mathrm{Bun}_M, S}}) \simeq \\ &\simeq (j_S)_! \circ (\pi_S^0)_!(\omega_{(\mathfrak{L}(N_P^-)/\mathfrak{L}^+(N_P^-))_{\mathrm{Bun}_M, S}}) \simeq (j_S)_!(\omega_{\mathrm{Bun}_{P^-} \times S}), \end{aligned}$$

as desired.

9.6.7. Let

$$p_{\mathrm{Gr}, S} : \mathrm{Gr}_{G, S} \rightarrow S, \quad p_{\mathrm{Gr}, \mathrm{Bun}_M, S} : \mathrm{Gr}_{G, \mathrm{Bun}_M, S} \rightarrow S \text{ and } p_{\widetilde{\mathrm{Bun}}_{P^-, S}} : \widetilde{\mathrm{Bun}}_{P^-, S} \rightarrow S$$

denote the tautological projections.

Next we will show that the functor $(\pi_! \circ \widetilde{\mathfrak{s}})_S$ is defined on objects of the form

$$\Delta_S^{-, \frac{\infty}{2}} \stackrel{!}{\otimes} p_S^!(\mathcal{F}_S), \quad \mathcal{F}_S \in \mathrm{D}\text{-mod}(S).$$

Moreover, we will show that the naturally defined map

$$(9.33) \quad (\pi_! \circ \widetilde{\mathfrak{s}})_S(\Delta_S^{-, \frac{\infty}{2}} \stackrel{!}{\otimes} p_{\mathrm{Gr}, S}^!(\mathcal{F}_S)) \rightarrow (\pi_! \circ \widetilde{\mathfrak{s}})_S(\Delta_S^{-, \frac{\infty}{2}}) \stackrel{!}{\otimes} p_{\widetilde{\mathrm{Bun}}_{P^-, S}}^!(\mathcal{F}_S)$$

is an isomorphism.

9.6.8. We have

$$\begin{aligned} \widetilde{\mathfrak{s}}(\Delta_S^{-, \frac{\infty}{2}} \stackrel{!}{\otimes} p_{\mathrm{Gr}, S}^!(\mathcal{F}_S)) &\simeq \widetilde{\mathfrak{s}}(\Delta_S^{-, \frac{\infty}{2}}) \stackrel{!}{\otimes} p_{\mathrm{Gr}, \mathrm{Bun}_M, S}^!(\mathcal{F}_S) \simeq \\ &\simeq (\mathbf{i}_{\mathrm{Bun}_M, S})_!(\omega_{(\mathfrak{L}(N_P^-)/\mathfrak{L}^+(N_P^-))_{\mathrm{Bun}_M, S}}) \stackrel{!}{\otimes} p_{\mathrm{Gr}, \mathrm{Bun}_M, S}^!(\mathcal{F}_S). \end{aligned}$$

Now, it follows from [Ga5, Proposition 1.6.3] that the naturally defined map

$$(9.34) \quad (\mathbf{i}_{\mathrm{Bun}_M, S})_! \circ (\mathbf{i}_{\mathrm{Bun}_M, S})^! \circ p_{\mathrm{Gr}, \mathrm{Bun}_M, S}^!(\mathcal{F}_S) \rightarrow (\mathbf{i}_{\mathrm{Bun}_M, S})_!(\omega_{(\mathfrak{L}(N_P^-)/\mathfrak{L}^+(N_P^-))_{\mathrm{Bun}_M, S}}) \stackrel{!}{\otimes} p_{\mathrm{Gr}, \mathrm{Bun}_M, S}^!(\mathcal{F}_S)$$

is an isomorphism; in particular, the left-hand side in (9.34) is defined.

Hence, in order to prove (9.33), it suffices to show that the naturally defined map

$$(9.35) \quad (\pi_S)_! \circ (\mathbf{i}_{\mathrm{Bun}_M, S})_! \circ (\mathbf{i}_{\mathrm{Bun}_M, S})^! \circ p_{\mathrm{Gr}, \mathrm{Bun}_M, S}^!(\mathcal{F}_S) \rightarrow (\pi_S)_! \circ (\mathbf{i}_{\mathrm{Bun}_M, S})_!(\omega_{(\mathfrak{L}(N_P^-)/\mathfrak{L}^+(N_P^-))_{\mathrm{Bun}_M, S}}) \stackrel{!}{\otimes} p_{\widetilde{\mathrm{Bun}}_{P^-, S}}^!(\mathcal{F}_S) \simeq (j_S)_!(\omega_{\mathrm{Bun}_{P^-} \times S}) \stackrel{!}{\otimes} p_{\widetilde{\mathrm{Bun}}_{P^-, S}}^!(\mathcal{F}_S)$$

is an isomorphism.

³⁸It is for this assertion to hold that we need S to be non-empty.

9.6.9. We rewrite the left-hand side in (9.35) as

$$(j_S)! \circ (\pi_S^0)! \circ (p_{\text{Gr}, \text{Bun}_M, S} \circ \mathbf{i}_{\text{Bun}_M, S})^!(\mathcal{F}_S) \simeq (j_S)! \circ (\pi_S^0)! \circ (\pi_S^0)^! \circ j_S^! \circ p_{\widetilde{\text{Bun}_{P^-}, S}}^!(\mathcal{F}_S).$$

By the contractibility result mentioned above, the above expression maps isomorphically to

$$(j_S)! \left(\omega_{\text{Bun}_{P^-} \times S} \overset{!}{\otimes} \left(j_S^! \circ p_{\widetilde{\text{Bun}_{P^-}, S}}^!(\mathcal{F}_S) \right) \right).$$

Thus, in order to prove (9.33), we have to show that the naturally defined map

$$(j_S)! \left(\omega_{\text{Bun}_{P^-} \times S} \overset{!}{\otimes} \left(j_S^! \circ p_{\widetilde{\text{Bun}_{P^-}, S}}^!(\mathcal{F}_S) \right) \right) \rightarrow (j_S)! (\omega_{\text{Bun}_{P^-} \times S}) \overset{!}{\otimes} p_{\widetilde{\text{Bun}_{P^-}, S}}^!(\mathcal{F}_S)$$

is an isomorphism. However, the latter is straightforward:

The map j_S factors as a composition

$$\text{Bun}_{P^-} \times S \xrightarrow{j \times \text{id}} \widetilde{\text{Bun}_{P^-}} \times S \xrightarrow{i_S} \widetilde{\text{Bun}_{P^-, S}},$$

where i_S is a closed embedding. Hence, it is enough to show that

$$(j \times \text{id})! (\omega_{\text{Bun}_{P^-}} \boxtimes \mathcal{F}_S) \rightarrow j! (\omega_{\text{Bun}_{P^-}}) \boxtimes \mathcal{F}_S$$

is an isomorphism, which is obvious.

9.6.10. Thus, we have shown that the functor $(\pi_! \circ \mathfrak{S}^!)_S$ is defined and $\text{D-mod}(S)$ -linear on the full subcategory of

$$\text{I}(G, P^-)_S^{0, \text{loc}} \subset \text{I}(G, P^-)_S^{\text{loc}}$$

spanned by objects of the form

$$\Delta_S^{-, \frac{\infty}{2}} \overset{!}{\otimes} p_S^!(\mathcal{F}_S).$$

9.6.11. We now prove point (a) of Proposition 9.6.3 on all of $\text{I}(G, P^-)_S^{\text{loc}}$. Note that the functor (9.30) (for $\mathcal{Z} = S$) is $\text{Sph}_{M, S}$ -linear with respect to the natural actions of $\text{Sph}_{M, S}$ on the two sides.

Recall that $\text{I}(G, P^-)_S^{\text{loc}}$ is generated by $\text{I}(G, P^-)_S^{0, \text{loc}}$ under the action of $\text{Sph}_{M, S}$.

Finally, recall that $\text{Sph}_{M, S}$ is rigid relative to $\text{D-mod}(S)$. Now, the assertion of Proposition 9.6.3(a) follows from the following general paradigm:

Let $\mathbf{A}^0$ and $\mathbf{A}$ be a pair of monoidal categories, with $\mathbf{A}^0$ symmetric monoidal and $\mathbf{A}$ rigid over $\mathbf{A}^0$. Let $F : \mathbf{C} \rightarrow \mathbf{D}$ be a functor between $\mathbf{A}$ -module categories. Suppose that the (partially defined) left adjoint F^L to F is defined and is $\mathbf{A}^0$ -linear on a full subcategory $\mathbf{D}^0 \subset \mathbf{D}$.

We claim:

Lemma 9.6.12. *Under the above circumstances, F^L is defined and is $\mathbf{A}$ -linear on the full subcategory of $\mathbf{D}$ generated by $\mathbf{D}^0$ under the action of $\mathbf{A}$.*

9.6.13. Point (b) of Proposition 9.6.3 on all of $\text{I}(G, P^-)_S^{\text{loc}}$ follows by a similar rigidity argument.

9.6.14. Let us now prove point (c) of Proposition 9.6.3. The starting point is that the object

$$j! (\omega_{\text{Bun}_{P^-}}) \in \text{D-mod}_{\det_{G, M}^{\frac{1}{2}}} (\widetilde{\text{Bun}_{P^-}})$$

is ULA over Bun_M . This is the assertion of [BG1, Theorem 5.1.5].

Now, it is easy to see that whenever $\mathcal{F} \in \text{D-mod}_{\det_{G, M}^{\frac{1}{2}}} (\widetilde{\text{Bun}_{P^-}})$ is ULA over Bun_M , then so is any object of the form

$$\mathcal{F}_M \star_M (i_S)_* (\mathcal{F} \boxtimes \omega_S), \quad \mathcal{F}_M \in (\text{Sph}_{M, S})^c.$$

Now, the assertion of Proposition 9.6.3(c) follows from the fact that every compact object in $I(G, P^-)_S^{\text{loc}}$ is a retract of an object of the form

$$\mathcal{F}_M \star_M \Delta_S^{-, \frac{\infty}{2}}, \quad \mathcal{F}_M \in (\text{Sph}_{M,S})^c$$

and that the functor $(\pi_! \circ \tilde{\mathfrak{f}}^!)_S$ is $\text{Sph}_{M,S}$ -linear.

□[Proposition 9.6.3]

9.6.15. Let $\mathcal{Z} = S$ be a scheme, and let $U \subset \text{Bun}_M$ be a quasi-compact open substack.

Note that the functor $(\pi_! \circ \tilde{\mathfrak{f}}^!)_S$ sends compact objects to objects that are *locally compact* relative to Bun_M , i.e., they become compact after a pullback $U \rightarrow \text{Bun}_M$, whenever U is a scheme. Note that such objects are $!$ -extended from a quasi-compact substack of $\widetilde{\text{Bun}}_{P^-,S} \times_{\text{Bun}_M} U$, see [DG2, Proposition 2.3.7].

9.6.16. We claim:

Lemma 9.6.17. *For $\mathcal{F} \in (I(G, P^-)_S^{\text{loc}})^c$ we have*

$$(9.36) \quad \mathbb{D}^{\text{glob}}((\pi_! \circ \tilde{\mathfrak{f}}^!)_S(\mathcal{F})) \simeq (\pi_* \circ \tilde{\mathfrak{f}}^!)_Z^{\text{Av}}(\mathbb{D}^{\text{loc}}(\mathcal{F}))[-2 \dim(\text{Bun}_M)],$$

where:

- $\mathbb{D}^{\text{loc}}$ denote the canonical equivalence $((I(G, P^-)_S^{\text{loc}})^c)^{\text{op}} \rightarrow (I(G, P^-)_{\text{co},S}^{\text{loc}})^c$;
- $\mathbb{D}^{\text{glob}}$ denotes the Verdier duality functor

$$(\text{D-mod}_{\det_{G,M}^{\frac{1}{2}}}(\widetilde{\text{Bun}}_{P^-,Z})^{\text{loc.c} / \text{Bun}_M})^{\text{op}} \rightarrow \text{D-mod}_{\det_{G,M}^{\frac{1}{2}}}(\widetilde{\text{Bun}}_{P^-,Z})_{\text{co} / \text{Bun}_M},$$

which make sense due to the $!$ -extension property mentioned above.

Proof. To simplify the exposition, we will take $S = \text{pt}$, so that $S \rightarrow \text{Ran}$ corresponds to $\underline{x} \in \text{Ran}$.

Any compact object $\mathcal{F} \in I(G, P^-)_{\underline{x}}^{\text{loc}}$ is of the form

$$\text{Av}_!^{\mathfrak{L}(N_P^-)_S}(\mathcal{F}'), \quad \mathcal{F}' \in \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\underline{x}})^{N^\alpha \cdot \mathfrak{L}^+(M)_{\underline{x},\text{ren}}},$$

where $N^\alpha \subset \mathfrak{L}(N_P^-)_{\underline{x}}$ is a large enough subgroup, normalized by $\mathfrak{L}^+(M)_{\underline{x}}$, and $\mathcal{F}'$ is compact as an object of $\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\underline{x}})^{N^\alpha}$.

Taking N^α to be as Sect. 9.2.9, we have

$$(\pi_! \circ \tilde{\mathfrak{f}}^!)_\underline{x}(\mathcal{F}) \simeq (\pi_\underline{x})_!(\mathcal{F}' \tilde{\boxtimes} \omega_{\text{Bun}_M}).$$

Note that $\mathbb{D}^{\text{loc}}(\mathcal{F})$ is the image of

$$\mathbb{D}(\mathcal{F}') \in \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\underline{x}})^{N^\alpha \cdot \mathfrak{L}^+(M)_{\underline{x},\text{ren}}}$$

under

$$\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\underline{x}})^{N^\alpha \cdot \mathfrak{L}^+(M)_{\underline{x},\text{ren}}} \simeq \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\underline{x}})^{\mathfrak{L}^+(M)_{\underline{x},\text{ren}}} \twoheadrightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\underline{x}})^{\mathfrak{L}^+(M)_{\underline{x},\text{ren}} / \mathfrak{L}(N_P^-)_{\underline{x}}},$$

where $\mathbb{D}$ is Verdier duality on $\text{Gr}_{G,\underline{x}}$.

Now, the assertion of the lemma follows from the fact that Verdier duality swaps $*$ - and $!$ - direct images, whenever the latter are defined, and taking into account that

$$\mathbb{D}(\mathcal{F}' \tilde{\boxtimes} \omega_{\text{Bun}_M}) \simeq \mathbb{D}(\mathcal{F}') \tilde{\boxtimes} \omega_{\text{Bun}_M}[-2 \dim(\text{Bun}_M)].$$

□

9.6.18. In particular, for $\mathcal{F}$ as above and

$$\mathcal{F}_G \in \mathrm{D}\text{-mod}_{\det_G^{\frac{1}{2}}}(\widetilde{\mathrm{Bun}}_{P^-, S}), \quad \mathcal{F}_M \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M \times S)$$

we have a canonical isomorphism

$$(9.37) \quad \mathcal{H}om_{\mathrm{D}\text{-mod}_{\det_G^{\frac{1}{2}}}(\widetilde{\mathrm{Bun}}_{P^-, S})} \left((\widetilde{\mathbf{q}}_S^-)^* (\mathcal{F}_M) \otimes^* (\pi_! \circ \widetilde{\mathbf{s}}^!)_S (\mathcal{F}), \mathcal{F}_G \right) \simeq \\ \simeq \mathcal{H}om_{\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M \times S)} \left(\mathcal{F}_M, (\widetilde{\mathbf{q}}_S^-)_* \left(\mathcal{F}_G \otimes^! (\pi_* \circ \widetilde{\mathbf{s}}^!)_{\mathbb{Z}}^{\mathrm{Av}} (\mathbb{D}^{\mathrm{loc}}(\mathcal{F})) \right) \right) [-2 \dim(\mathrm{Bun}_M)],$$

where the operation $\otimes^*$ is well-defined thanks to Lemma 9.6.3(c).

9.6.19. As in Sects. 9.2.10-9.2.11, we can view the assignment

$$\mathbb{Z} \rightsquigarrow \mathrm{D}\text{-mod}_{\det_{G,M}^{\frac{1}{2}}}(\widetilde{\mathrm{Bun}}_{P^-, \mathbb{Z}})$$

as a unital crystal of categories over Ran and the assignment

$$\mathbb{Z} \rightsquigarrow (\pi_! \circ \widetilde{\mathbf{s}})_{\mathbb{Z}}$$

as a *right-lax* unital functor, to be denoted $(\pi_! \circ \widetilde{\mathbf{s}})$.

9.6.20. We claim:

Corollary 9.6.21. *The right-lax unital structure on $(\pi_! \circ \widetilde{\mathbf{s}})$ is strict.*

Proof. Follows from Proposition 9.2.12 using Lemma 9.6.17. □

9.7. The global enhanced functors, the !-version.

9.7.1. For $\mathbb{Z} \rightarrow \mathrm{Ran}$, define the functor

$$\mathrm{pre}\text{-}\mathrm{Eis}_{!, \mathbb{Z}}^{-, \mathrm{enh}, \mathrm{glob}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{I}(G, P^-)_{\mathbb{Z}}^{\mathrm{loc}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G \times \mathbb{Z}) \simeq \\ \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathrm{Ran})$$

to be

$$(9.38) \quad (\widetilde{\mathbf{p}}_{\mathbb{Z}}^-)_* \left((\widetilde{\mathbf{q}}^-)^!(-) \otimes^! (\pi_! \circ \widetilde{\mathbf{s}}^!)_{\mathbb{Z}}(-) \right) [-\mathrm{shift}],$$

where the functors $(\widetilde{\mathbf{q}}^-)^!$, $(\widetilde{\mathbf{p}}_{\mathbb{Z}}^-)_*$ and $\otimes^!$ are the corresponding functors on the usual categories of (twisted) D -modules.

9.7.2. Recall the functor $\mathrm{pre}\text{-}\mathrm{CT}_{*, \mathbb{Z}}^{-, \mathrm{enh}, \mathrm{co}, \mathrm{glob}}$ from Sect. 9.4.1. By duality, it gives rise to a functor

$$(9.39) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G \times \mathbb{Z}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{I}(G, P^-)_{\mathbb{Z}}^{\mathrm{loc}}.$$

We claim:

Lemma 9.7.3. *The functor $\mathrm{pre}\text{-}\mathrm{Eis}_{!, \mathbb{Z}}^{-, \mathrm{enh}, \mathrm{glob}}$ is the left adjoint of the functor (9.39).*

Proof. With no restriction of generality, we can assume that $\mathbb{Z} = S$ is a scheme. In order to simplify the notation, we will assume that $S = \mathrm{pt}$, corresponding to $\underline{x} \in \mathrm{Ran}$.

Fix objects

$$\mathcal{F}_G \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G), \quad \mathcal{F}_M \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M), \quad \mathcal{F}_{G,M} \in (\mathrm{I}(G, P^-)_{\underline{x}}^{\mathrm{loc}})^c.$$

Let $\mathbb{D}^{\mathrm{loc}}(\mathcal{F}_{G,M}) \in \mathrm{I}(G, P^-)_{\mathrm{co}, \underline{x}}^{\mathrm{loc}}$ be the “formal dual” of $\mathcal{F}_{G,M}$, see Sect. 9.6.15.

We need to establish a canonical isomorphism

$$(9.40) \quad \mathcal{H}om_{\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)} \left((\tilde{\mathfrak{p}}^-)_* \left((\tilde{\mathfrak{q}}^-)^! (\mathcal{F}_M) \otimes (\pi_! \circ \tilde{\mathfrak{s}}^!)_{\mathcal{Z}} (\mathcal{F}_{G,M}) \right) [-\text{shift}], \mathcal{F}_G \right) \simeq \\ \simeq \mathcal{H}om_{\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)} \left(\mathcal{F}_M, (\tilde{\mathfrak{q}}_S^-)_* \left((\tilde{\mathfrak{p}}^-)^! (\mathcal{F}_G) \otimes (\pi_* \circ \tilde{\mathfrak{s}}^!)_{\mathcal{Z}}^{\mathrm{Av}} (\mathbb{D}^{\mathrm{loc}}(\mathcal{F}_{G,M})) \right) [\text{shift}] \right).$$

Since the morphism $(\tilde{\mathfrak{p}}_S^-)_*$ is ind-proper, we can rewrite the left-hand side of (9.40) as

$$(9.41) \quad \mathcal{H}om_{\mathrm{D-mod}_{\frac{1}{2}}(\widetilde{\mathrm{Bun}}_{P^-, \underline{x}})} \left((\tilde{\mathfrak{q}}^-)^! (\mathcal{F}_M) \otimes (\pi_! \circ \tilde{\mathfrak{s}}^!)_{\mathcal{Z}} (\mathcal{F}_{G,M}) [-\text{shift}], (\tilde{\mathfrak{p}}_S^-)^! (\mathcal{F}_G) \right).$$

By Proposition 9.6.3(c), we can rewrite (9.41) further as

$$(9.42) \quad \mathcal{H}om_{\mathrm{D-mod}_{\frac{1}{2}}(\widetilde{\mathrm{Bun}}_{P^-, \underline{x}})} \left((\tilde{\mathfrak{q}}^-)^* (\mathcal{F}_M) \otimes (\pi_! \circ \tilde{\mathfrak{s}}^!)_{\mathcal{Z}} (\mathcal{F}_{G,M}) [-\text{shift} - 2 \dim(\mathrm{Bun}_M)], (\tilde{\mathfrak{p}}_S^-)^! (\mathcal{F}_G) \right).$$

Applying (9.37), we obtain that (9.42) identifies with the right-hand side in (9.40), as required. $\square$

9.7.4. Unwinding the constructions, we obtain that the functor $\mathrm{pre-Eis}_{!, \mathcal{Z}}^{-, \mathrm{enh}, \mathrm{glob}}$ factors canonically as

$$(9.43) \quad \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{I}(G, P^-)_{\mathcal{Z}}^{\mathrm{loc}} \simeq \\ \simeq \left(\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}(\mathcal{Z}) \right)_{\mathrm{D-mod}(\mathcal{Z})} \otimes \mathrm{I}(G, P^-)_{\mathcal{Z}}^{\mathrm{loc}} \rightarrow \\ \rightarrow \left(\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}(\mathcal{Z}) \right)_{\mathrm{Sph}_{M, \mathcal{Z}}} \otimes \mathrm{I}(G, P^-)_{\mathcal{Z}}^{\mathrm{loc}} = \\ = \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{Z}}^{-, \mathrm{enh}} \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z}).$$

Denote the last arrow in (9.43) by

$$\mathrm{Eis}_{!, \mathcal{Z}}^{-, \mathrm{enh}, \mathrm{glob}} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{Z}}^{-, \mathrm{enh}} \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z}).$$

By construction, the functor $\mathrm{Eis}_{!, \mathcal{Z}}^{-, \mathrm{enh}, \mathrm{glob}}$ intertwines the actions of $\mathrm{Sph}_{G, \mathcal{Z}}$ on the two sides.

Denote also

$$\mathrm{Eis}_{!}^{-, \mathrm{enh}, \mathrm{glob}} := (\mathrm{Id} \otimes \mathrm{C}(\mathrm{Ran}, -)) \circ \mathrm{Eis}_{!, \mathrm{Ran}}^{-, \mathrm{enh}, \mathrm{glob}}, \quad \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{Ran}}^{-, \mathrm{enh}} \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G).$$

9.7.5. Note that the identification of Lemma 9.7.3 is compatible with the actions of $\mathrm{Sph}_{M, \mathcal{Z}}$ and $\mathrm{Sph}_{G, \mathcal{Z}}$. Hence, combining with Proposition 9.4.4, we obtain:

Corollary 9.7.6.

(a) *There is a canonical isomorphism*

$$\mathrm{Eis}_{!, \mathcal{Z}}^{-, \mathrm{enh}, \mathrm{glob}} \simeq \mathrm{Eis}_{!}^{-, \mathrm{enh}}$$

as $\mathrm{Sph}_{G, \mathcal{Z}}$ -linear functors

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathcal{Z}}^{-, \mathrm{enh}} \rightrightarrows \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z}).$$

(b) *There is a canonical isomorphism*

$$\mathrm{Eis}_{!}^{-, \mathrm{enh}, \mathrm{glob}} \simeq \mathrm{Eis}_{!}^{-, \mathrm{enh}}$$

as functors

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{Ran}}^{-, \mathrm{enh}} \rightrightarrows \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G).$$

9.7.7. Define the functor

$$\begin{aligned} \text{pre-CT}_{\text{co},?,\mathbb{Z}}^{-,\text{enh},\text{glob}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}} \otimes \text{I}(G, P^-)_{\mathbb{Z}}^{\text{loc}} &\rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M \times \mathbb{Z})_{\text{co}} \simeq \\ &\simeq \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}} \otimes \text{D-mod}(\text{Ran}) \end{aligned}$$

to be

$$(9.44) \quad (\tilde{\mathfrak{q}}_{\mathbb{Z}}^-)_* \left((\tilde{\mathfrak{p}}^-)^!(-) \overset{!}{\otimes} (\pi_! \circ \mathfrak{S}^!)_{\mathbb{Z}}(-) \right) [-\text{shift}],$$

where:

- The functor $(\tilde{\mathfrak{p}}^-)^!$ is understood in the sense (9.3);
- The functor $\overset{!}{\otimes}$ is understood in the sense (9.7);
- The functor $(\tilde{\mathfrak{q}}_{\mathbb{Z}}^-)_*$ is understood in the sense (9.5);
- $[\text{shift}]$ is as in (8.4).

9.7.8. Unwinding the constructions, we obtain that the functor $\text{pre-CT}_{\text{co},?,\mathbb{Z}}^{-,\text{enh},\text{glob}}$ factors canonically as

$$\begin{aligned} (9.45) \quad \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}} \otimes \text{I}(G, P^-)_{\mathbb{Z}}^{\text{loc}} &\simeq \\ &\simeq \left(\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}} \otimes \text{D-mod}(\mathbb{Z}) \right)_{\text{D-mod}(\mathbb{Z})} \otimes \text{I}(G, P^-)_{\mathbb{Z}}^{\text{loc}} \rightarrow \\ &\rightarrow \left(\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}} \otimes \text{D-mod}(\mathbb{Z}) \right)_{\text{Sph}_{G,\mathbb{Z}}} \otimes \text{I}(G, P^-)_{\mathbb{Z}}^{\text{loc}} = \\ &= \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co},\mathbb{Z}}^{-,\text{enh}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}} \otimes \text{D-mod}(\mathbb{Z}). \end{aligned}$$

Denote the last arrow in (9.45) by

$$\text{CT}_{\text{co},?,\mathbb{Z}}^{-,\text{enh},\text{glob}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co},\mathbb{Z}}^{-,\text{enh}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}} \otimes \text{D-mod}(\mathbb{Z}).$$

By construction, the functor $\text{CT}_{\text{co},?,\mathbb{Z}}^{-,\text{enh},\text{glob}}$ intertwines the actions of $\text{Sph}_{M,\mathbb{Z}}$ on the two sides.

9.7.9. Denote

$$\text{CT}_{\text{co},?}^{-,\text{enh},\text{glob}} := (\text{Id} \otimes \text{C}(\text{Ran}, -)) \circ \text{CT}_{\text{co},?,\text{Ran}}^{-,\text{enh},\text{glob}}, \quad \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co},\text{Ran}}^{-,\text{enh}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}}.$$

By duality, from Proposition 9.4.4 we obtain:

Corollary 9.7.10.

(a) *There is a canonical isomorphism*

$$\text{CT}_{\text{co},?,\mathbb{Z}}^{-,\text{enh},\text{glob}} \simeq \text{CT}_{\text{co},?,\mathbb{Z}}^{-,\text{enh}}$$

as $\text{Sph}_{M,\mathbb{Z}}$ -linear functors

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co},\mathbb{Z}}^{-,\text{enh}} \rightrightarrows \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}} \otimes \text{D-mod}(\mathbb{Z}).$$

(b) *There is a canonical isomorphism*

$$\text{CT}_{\text{co},?}^{-,\text{enh},\text{glob}} \simeq \text{CT}_{\text{co},?}^{-,\text{enh}}$$

as functors

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co},\text{Ran}}^{-,\text{enh}} \rightrightarrows \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}}.$$

9.8. **Relation to compactified Eisenstein series.**

9.8.1. Let $\mathrm{IC}^{-, \frac{\infty}{2}, \mathrm{glob}}$ be the (shifted) half-twisted intersection cohomology sheaf on $\widetilde{\mathrm{Bun}}_{P-}$, i.e.,

$$\mathrm{IC}^{-, \frac{\infty}{2}, \mathrm{glob}} := j_{!*}(\mathrm{IC}_{\mathrm{Bun}_{P-}}[\dim(\mathrm{Bun}_M)]) \in \mathrm{D-mod}_{\det \frac{1}{2}_{G,M}}(\widetilde{\mathrm{Bun}}_{P-}),$$

where:

- j denotes the embedding $\mathrm{Bun}_{P-} \hookrightarrow \widetilde{\mathrm{Bun}}_{P-}$;
- $\mathrm{IC}_{\mathrm{Bun}_{P-}}[\dim(\mathrm{Bun}_M)] \simeq \omega_{\mathrm{Bun}_{P-}}[-\text{shift}]$, where $[\text{shift}]$ is as in (8.4);
- We regard $\omega_{\mathrm{Bun}_{P-}}$ as an object of $\mathrm{D-mod}_{\det \frac{1}{2}_{G,M}}(\mathrm{Bun}_{P-})$ thanks to the trivialization of $\det \frac{1}{2}_{G,M} |_{\mathrm{Bun}_{P-}}$.

9.8.2. Following [BG1], define the functor

$$\mathrm{Eis}_{!*}^{-} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

to be

$$(\widetilde{\mathfrak{p}}^{-})_*((\widetilde{\mathfrak{q}}^{-})^!(-) \otimes \mathrm{IC}^{-, \frac{\infty}{2}, \mathrm{glob}}).$$

9.8.3. Define the functor

$$\mathrm{CT}_{\mathrm{co},!*}^{-} := (\mathrm{Eis}_{!*}^{-})^{\vee}, \quad \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}}.$$

I.e.,

$$\mathrm{CT}_{\mathrm{co},!*}^{-} \simeq (\widetilde{\mathfrak{q}}^{-})_*((\widetilde{\mathfrak{p}}^{-})^!(-) \otimes \mathrm{IC}^{-, \frac{\infty}{2}, \mathrm{glob}}).$$

9.8.4. Recall the functor $\mathrm{Eis}_{!,\mathrm{IC}}^{-}$, see Sect. 8.5.7. We are going to prove:

Proposition 9.8.5. *The functor $\mathrm{Eis}_{!*}^{-}$ identifies canonically with $\mathrm{Eis}_{!,\mathrm{IC}}^{-}$.*

Note that as an immediate corollary we obtain:

Corollary 9.8.6. *The functor $\mathrm{CT}_{\mathrm{co},!*}^{-}$ identifies canonically with $\mathrm{CT}_{\mathrm{co},?,\mathrm{IC}}^{-}$.*

9.8.7. *Proof of Proposition 9.8.5.* First, by the definition of $\mathrm{IC}^{-, \frac{\infty}{2}}$, its support on Gr_G is contained in the closure of the unit orbit of $\mathfrak{L}(N_P^-)$. This implies that the support of $\widetilde{\mathfrak{s}}^!(\mathrm{IC}_{\mathrm{Ran}}^{-, \frac{\infty}{2}})$ on $\mathrm{Gr}_{G, \mathrm{Bun}_M, \mathrm{Ran}}$ is contained in $\pi^{-1}(\widetilde{\mathrm{Bun}}_{P-} \times \mathrm{Ran})$.

In particular, the object

$$(9.46) \quad (\pi_! \circ \widetilde{\mathfrak{s}}^!)_{\mathrm{Ran}}(\mathrm{IC}_{\mathrm{Ran}}^{-, \frac{\infty}{2}}) \in \mathrm{D-mod}_{\det \frac{1}{2}_{G,M}}(\widetilde{\mathrm{Bun}}_{P-, \mathrm{Ran}})$$

is supported on $\widetilde{\mathrm{Bun}}_{P-} \times \mathrm{Ran} \subset \widetilde{\mathrm{Bun}}_{P-, \mathrm{Ran}}$.

Hence, in order to prove the proposition, it suffices to show that if we apply

$$(\mathrm{Id} \otimes C^*(\mathrm{Ran}, -)) : \mathrm{D-mod}_{\det \frac{1}{2}_{G,M}}(\widetilde{\mathrm{Bun}}_{P-} \times \mathrm{Ran}) \rightarrow \mathrm{D-mod}_{\det \frac{1}{2}_{G,M}}(\widetilde{\mathrm{Bun}}_{P-})$$

to (9.46), viewed as an object of $\mathrm{D-mod}_{\det \frac{1}{2}_{G,M}}(\widetilde{\mathrm{Bun}}_{P-} \times \mathrm{Ran})$, the result is isomorphic to $\mathrm{IC}^{-, \frac{\infty}{2}, \mathrm{glob}}$.

However, the latter is the assertion of [Ga5, Theorems 6.3.2 and 3.4.4] (adapted to the case of an arbitrary parabolic).

□[Proposition 9.8.5]

10. INTERACTION OF PARABOLIC INDUCTION WITH THE WHITTAKER MODEL

In this section we will perform the computation of Whittaker coefficients of Eisenstein series. When combined with its spectral counterpart, this will ensure the compatibility of the Langlands functor with the Eisenstein functors.

10.1. Whittaker coefficients of Eisenstein series. In this subsection we will state several assertions, all equivalent to the calculation of Whittaker coefficients of (usual, i.e., unenhanced) Eisenstein series.

10.1.1. We will prove:

Theorem 10.1.2. *For $\mathcal{Z} \rightarrow \text{Ran}$, the following diagram of functors commutes:*

$$(10.1) \quad \begin{array}{ccc} \text{Whit}_*(G)_{\mathcal{Z}} & \xrightarrow{\text{ins.unit}_{\mathcal{Z}}} & \text{Whit}_*(G)_{\mathcal{Z}^{\subseteq}} \\ \downarrow \text{Poinc}_{G,*,\mathcal{Z}}[2\delta_{(N_P)_{\rho_P(\omega_X)}} + \delta_{(N_P^-)_{\rho_P(\omega_X)}}] & & \downarrow J_{\text{Whit},\Theta}^{-,!} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}} \otimes \text{D-mod}(\mathcal{Z}) & & \text{Whit}_*(M)_{\mathcal{Z}^{\subseteq}} \\ \downarrow \text{CT}_{\text{co},?,\rho_P(\omega_X)}^{-} \otimes \text{Id} & & \downarrow \text{Poinc}_{M,*,\mathcal{Z}^{\subseteq}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}} \otimes \text{D-mod}(\mathcal{Z}) & \xleftarrow{\text{Id} \otimes (\text{pr}_{\text{small},\mathcal{Z}})^!} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}} \otimes \text{D-mod}(\mathcal{Z}^{\subseteq}), \end{array}$$

where:

- The notation $\text{ins.unit}_{\text{Ran}}$ refers to the unital structure on $\text{Whit}_*(G)$, see [GLC2, Sect. 11.2.8];
- When forming $\text{Poinc}_{M,*,\mathcal{Z}^{\subseteq}}$, we regard $\mathcal{Z}^{\subseteq}$ as mapping to Ran by pr_{big} , see [GLC2, Sect. 11.2.6];
- The functor $J_{\text{Whit},\Theta}^{-,!}$ is one from (2.25);
- $\text{CT}_{\text{co},?,\rho_P(\omega_X)}^{-} := (\text{Eis}_{!,\rho_P(\omega_X)}^{-})^{\vee}$;
- $\delta_{(N_P)_{\rho_P(\omega_X)}} := \dim(\text{Bun}_{(N_P)_{\rho_P(\omega_X)}})$ and $\delta_{(N_P^-)_{\rho_P(\omega_X)}} := \dim(\text{Bun}_{(N_P^-)_{\rho_P(\omega_X)}})$.

10.1.3. Note that the unital property of the functor $\text{Poinc}_{G,!}$ implies that the essential image of the functor

$$\text{coeff}_G : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{Whit}^!(G)_{\text{Ran}}$$

belongs to the full subcategory

$$\text{Whit}^!(G)_{\text{Ran}^{\text{untl},\text{indep}}} \subset \text{Whit}^!(G)_{\text{Ran}},$$

see [GLC2, Sect. H.1].

When viewed as such, we will denote it, as well as its composition with the embedding

$$\text{Whit}^!(G)_{\text{Ran}^{\text{untl},\text{indep}}} \hookrightarrow \text{Whit}^!(G)_{\text{Ran}^{\text{untl}}},$$

by $\text{coeff}_{G,\text{untl}}$.

10.1.4. Recall now that the functor

$$\text{emb.indep}_{\text{Whit}^!(G)} : \text{Whit}^!(G)_{\text{Ran}^{\text{untl},\text{indep}}} \xrightarrow{\text{embed.indep}_{\text{Whit}^!(G)}} \text{Whit}^!(G)_{\text{Ran}^{\text{untl}}},$$

admits a left adjoint, see [GLC2, Lemma H.1.10].

We claim:

Corollary 10.1.5. *The following diagram of functors commutes:*

$$\begin{array}{ccccc} \text{Whit}^!(G)_{\text{Ran}^{\text{untl},\text{indep}}} & \xleftarrow{\text{emb.indep}_{\text{Whit}^!(G)}^L} & \text{Whit}^!(G)_{\text{Ran}^{\text{untl}}} & \xleftarrow{(J_{\text{Whit},\Theta}^{-,!})^{\vee}} & \text{Whit}^!(M)_{\text{Ran}^{\text{untl}}} \\ \uparrow \text{coeff}_{G,\text{untl}}[2\delta_{(N_P)_{\rho_P(\omega_X)}} + \delta_{(N_P^-)_{\rho_P(\omega_X)}}] & & & & \uparrow \text{coeff}_{M,\text{untl}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & & \xleftarrow{\text{Eis}_{!,\rho_P(\omega_X)}^{-}} & & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M). \end{array}$$

Proof. Passing to the dual functors in (10.1) with $\mathcal{Z} = \text{Ran}$, we obtain that it suffices to show that the functor

$$(10.2) \quad \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \xrightarrow{\text{Id} \otimes \omega_{\text{Ran} \subseteq}} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\text{Ran} \subseteq) \xrightarrow{\text{coeff}_{M, \text{Ran} \subseteq}} \\ \rightarrow \text{Whit}^!(M)_{\text{Ran} \subseteq} \xrightarrow{(J_{\text{Whit}, \Theta}^-)^{\vee}} \text{Whit}^!(G)_{\text{Ran} \subseteq} \xrightarrow{(\text{ins. unit}_{\text{Ran}})^{\vee}} \text{Whit}^!(G)_{\text{Ran}}$$

takes values in $\text{Whit}^!(G)_{\text{Ran}^{\text{untl}, \text{indep}}}$, and as such identifies with

$$(10.3) \quad \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \xrightarrow{\text{coeff}_M} \text{Whit}^!(M)_{\text{Ran}^{\text{untl}}} \xrightarrow{(J_{\text{Whit}, \Theta}^-)^{\vee}} \\ \rightarrow \text{Whit}^!(G)_{\text{Ran}^{\text{untl}}} \xrightarrow{\text{emb. indep}_{\text{Whit}^!(G)}^L} \text{Whit}^!(G)_{\text{Ran}^{\text{untl}, \text{indep}}}.$$

Note that the composition of the first two arrows in (10.2) can be rewritten as

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \xrightarrow{\text{coeff}_M} \text{Whit}^!(M)_{\text{Ran}^{\text{untl}}} \rightarrow \text{Whit}^!(M)_{\text{Ran}} \xrightarrow{(\text{pr}_{\text{big}})^!} \text{Whit}^!(M)_{\text{Ran} \subseteq},$$

where the second arrow is the forgetful functor.

Hence, it suffices to establish an isomorphism between

$$\text{Whit}^!(M)_{\text{Ran}^{\text{untl}}} \rightarrow \text{Whit}^!(M)_{\text{Ran}} \xrightarrow{(\text{pr}_{\text{big}})^!} \\ \rightarrow \text{Whit}^!(M)_{\text{Ran} \subseteq} \xrightarrow{(J_{\text{Whit}, \Theta}^-)^{\vee}} \text{Whit}^!(G)_{\text{Ran} \subseteq} \xrightarrow{(\text{ins. unit}_{\text{Ran}})^{\vee}} \text{Whit}^!(G)_{\text{Ran}},$$

which is the same as

$$\text{Whit}^!(M)_{\text{Ran}^{\text{untl}}} \xrightarrow{(J_{\text{Whit}, \Theta}^-)^{\vee}} \text{Whit}^!(G)_{\text{Ran}^{\text{untl}}} \rightarrow \text{Whit}^!(G)_{\text{Ran}} \xrightarrow{(\text{pr}_{\text{big}})^!} \\ \rightarrow \text{Whit}^!(G)_{\text{Ran} \subseteq} \xrightarrow{(\text{ins. unit}_{\text{Ran}})^{\vee}} \text{Whit}^!(G)_{\text{Ran}},$$

and

$$\text{Whit}^!(M)_{\text{Ran}^{\text{untl}}} \xrightarrow{(J_{\text{Whit}, \Theta}^-)^{\vee}} \text{Whit}^!(G)_{\text{Ran}^{\text{untl}}} \xrightarrow{\text{emb. indep}_{\text{Whit}^!(G)}^L} \text{Whit}^!(G)_{\text{Ran}^{\text{untl}, \text{indep}}}.$$

Note that the functor $\text{Whit}^!(G)_{\text{Ran} \subseteq} \xrightarrow{(\text{ins. unit}_{\text{Ran}})^{\vee}} \text{Whit}^!(G)_{\text{Ran}}$ is the composition

$$\text{Whit}^!(G)_{\text{Ran} \subseteq} := \text{Whit}^!(G)_{\text{Ran} \subseteq, \text{pr}_{\text{big}}} \rightarrow \text{Whit}^!(G)_{\text{Ran} \subseteq, \text{pr}_{\text{small}}} \xrightarrow{(\text{pr}_{\text{small}})^!} \text{Whit}^!(G)_{\text{Ran}},$$

where the second arrow is the dual of to the functor

$$(10.4) \quad \text{Whit}_*(G)_{\text{Ran} \subseteq, \text{pr}_{\text{small}}} \rightarrow \text{Whit}_*(G)_{\text{Ran} \subseteq, \text{pr}_{\text{big}}}$$

defined by the unital structure. Note also that the dual of (10.4) identifies canonically with the right adjoint of

$$(10.5) \quad \text{Whit}^!(G)_{\text{Ran} \subseteq, \text{pr}_{\text{small}}} \rightarrow \text{Whit}^!(G)_{\text{Ran} \subseteq, \text{pr}_{\text{big}}}.$$

Now, the required isomorphism follows from Lemma 10.1.6 below. $\square$

Lemma 10.1.6. *In the context of [GLC2, Lemma H.1.10] assume that $\underline{\mathbf{C}}$ is dualizable as a crystal of categories over Ran^{untl} . Then the functor left adjoint to*

$$\text{emb. indep} : \mathbf{C}_{\text{Ran}^{\text{untl}, \text{indep}}}^{\text{loc}} \hookrightarrow \mathbf{C}_{\text{Ran}^{\text{untl}}}^{\text{loc}},$$

composed with the embedding $\mathbf{C}_{\text{Ran}^{\text{untl}, \text{indep}}}^{\text{loc}} \hookrightarrow \mathbf{C}_{\text{Ran}}^{\text{loc}}$, is given by

$$\mathbf{C}_{\text{Ran}^{\text{untl}}}^{\text{loc}} \rightarrow \mathbf{C}_{\text{Ran}}^{\text{loc}} \xrightarrow{(\text{pr}_{\text{big}})^!} \mathbf{C}_{\text{Ran} \subseteq, \text{pr}_{\text{big}}}^{\text{loc}} \rightarrow \mathbf{C}_{\text{Ran} \subseteq, \text{pr}_{\text{small}}}^{\text{loc}} \xrightarrow{(\text{pr}_{\text{small}})^!} \mathbf{C}_{\text{Ran}}^{\text{loc}},$$

where the third arrow is the functor right adjoint to the functor

$$\mathbf{C}_{\text{Ran} \subseteq, \text{pr}_{\text{small}}}^{\text{loc}} \rightarrow \mathbf{C}_{\text{Ran} \subseteq, \text{pr}_{\text{big}}}^{\text{loc}},$$

defined by the unital structure on $\underline{\mathbf{C}}^{\text{loc}}$.

Proof. Follows from [GLC2, Proposition I.2.3] applied to the dual of $\mathbf{C}^{\text{loc}}$. $\square$

10.1.7. We now deduce another consequence of Theorem 10.1.2, this time for the computation of the constant term of the $!$ -Poincaré functor:

Corollary 10.1.8. *For $\mathcal{Z} \rightarrow \text{Ran}$, the following diagram of functors commutes:*

$$\begin{array}{ccc} \text{Whit}^!(G)_{\mathcal{Z}} & \xrightarrow{\text{ins.unit}_{\mathcal{Z}}} & \text{Whit}^!(G)_{\mathcal{Z} \subseteq} \\ \downarrow \text{Poinc}_{G,!,\mathcal{Z}}[d] & & \downarrow J_{\text{Whit},\tau}^{-,!} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathcal{Z}) & & \text{Whit}^!(M)_{\mathcal{Z} \subseteq} \\ \downarrow \text{CT}_{*,\rho_P(\omega_X)} \otimes \text{Id} & & \downarrow \text{Poinc}_{M,!,\mathcal{Z} \subseteq} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathcal{Z}) & \xleftarrow{\text{Id} \otimes (\text{pr}_{\text{small},\mathcal{Z}})!} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathcal{Z}^{\subseteq}), \end{array}$$

where:

- $\text{CT}_{*,\rho_P(\omega_X)} \simeq (\text{transl}_{2\rho_P(\omega_X)})^* \circ \tau_M \circ \text{CT}_{*,\rho_P(\omega_X)}^- \circ \tau_G \simeq \tau_M \circ (\text{transl}_{-2\rho_P(\omega_X)})^* \circ \text{CT}_{*,\rho_P(\omega_X)}^- \circ \tau_G$;
- $J_{\text{Whit},\tau}^{-,!} = \tau_M \circ J_{\text{Whit}}^{-,!} \circ \tau_G$;
- $d = -2\delta_{(N_P)_{\rho_P(\omega_X)}} - \delta_{(N_P^-)_{\rho_P(\omega_X)}} + 2\delta_{N_P^-}$.

10.1.9. Before we prove Corollary 10.1.8, we record the following result from [Lin, Theorem 1.1.6]³⁹:

Theorem 10.1.10. *The diagram*

$$(10.6) \quad \begin{array}{ccc} \text{Whit}^!(G)_{\text{Ran}} & \xleftarrow{\Theta_{\text{Whit}(G)}} & \text{Whit}_*(\text{Gr}_G)_{\text{Ran}} \\ \text{Poinc}_{G,!} \downarrow & & \downarrow \text{Poinc}_{G,*}[2\delta_{N_{\rho(\omega_X)}}] \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xleftarrow{\text{Mir}_{\text{Bun}_G}[2\delta_G]} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}}, \end{array}$$

commutes, where

$$\delta_G = \dim(\text{Bun}_G).$$

Proof of Corollary 10.1.8. Obtained juxtaposing Theorem 10.1.2, Corollary 8.7.11 and diagrams (10.6) for G and M . $\square$

10.1.11. In the next several subsections we will recall the definitions of several geometric tools needed for the proof of Theorem 10.1.2.

10.2. The global model for the Whittaker category.

³⁹In order to apply the result from *loc. cit.* to deduce Theorem 10.1.10, one needs to combine it with Theorem 10.2.8(c) below.

10.2.1. Let

$$\overline{\mathrm{Bun}}_{N,\rho(\omega_X)} \xrightarrow{\bar{p}} \mathrm{Bun}_G.$$

be as in [Ga6, Sect. 4.1].

For $\mathcal{Z} \rightarrow \mathrm{Ran}$, we let $\overline{\mathrm{Bun}}_{N,\rho(\omega_X),\mathcal{Z}}$ be a version of $\overline{\mathrm{Bun}}_{N,\rho(\omega_X)}$, where we allow poles at the specified points of the curve. When $\mathcal{Z} = S$ is a scheme, the prestack $\overline{\mathrm{Bun}}_{N,\rho(\omega_X),S}$ is an ind-algebraic stack.

We will denote by the same symbol $\bar{p}$ the projection

$$\overline{\mathrm{Bun}}_{N,\rho(\omega_X),\mathcal{Z}} \rightarrow \mathrm{Bun}_G,$$

and by $\bar{p}_{\mathcal{Z}}$ the projection

$$\overline{\mathrm{Bun}}_{N,\rho(\omega_X),\mathcal{Z}} \rightarrow \mathrm{Bun}_G \times \mathcal{Z}.$$

We will denote by

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\overline{\mathrm{Bun}}_{N,\rho(\omega_X),\mathcal{Z}}) \text{ and } \mathrm{D}\text{-mod}_{\frac{1}{2}}(\overline{\mathrm{Bun}}_{N,\rho(\omega_X),\mathcal{Z}})_{\mathrm{co}}$$

the corresponding categories of twisted D-modules, by the means of the μ_2 -gerbe $\pi^*(\det_{\mathrm{Bun}_G}^{\frac{1}{2}})$, see Sects. 9.1.4-9.1.8.

10.2.2. Following [Ga6, Sect. 4.7], one singles out a full subcategory

$$\mathrm{Whit}^{\mathrm{glob}}(G)_{\mathcal{Z}} \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\overline{\mathrm{Bun}}_{N,\rho(\omega_X),\mathcal{Z}}).$$

10.2.3. Let $\mathcal{Z} = S$ be a scheme. Then as in Sect. 9.1.4, we can write $\overline{\mathrm{Bun}}_{N,\rho(\omega_X),S}$ as a union of algebraic stacks $\mathcal{Y}_{S,i}$. We can write each $\mathcal{Y}_{S,i}$ as a union of opens $U \subset \mathcal{Y}_{S,i}$ stable under the loop groupoid from [Ga6, Sect. 4.4].

Hence, we can make sense of

$$\mathrm{Whit}^{\mathrm{glob}}(G)_{S,U} \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(U),$$

by imposing the Whittaker equivariance condition as in [Ga6, Sect. 4.7].

Then for

$$\mathrm{Whit}^{\mathrm{glob}}(G)_i := \mathrm{Whit}^{\mathrm{glob}}(G)_S \cap \mathrm{D}\text{-mod}_{\frac{1}{2}}(CY_{S,i}),$$

we have

$$\mathrm{Whit}^{\mathrm{glob}}(G)_i \simeq \lim_i \mathrm{Whit}^{\mathrm{glob}}(G)_{S,U},$$

where the limit is formed using the restriction functors.

We have

$$\mathrm{Whit}^{\mathrm{glob}}(G)_S \simeq \lim_i \mathrm{Whit}^{\mathrm{glob}}(G)_i \simeq \mathrm{colim}_i \mathrm{Whit}^{\mathrm{glob}}(G)_i,$$

where:

- The limit is formed using $!$ -pullback functors;
- The colimit is formed using the $*$ -pushforward functors.

10.2.4. Define

$$\mathrm{Whit}^{\mathrm{glob}}(G)_{\mathrm{co},i} = \mathrm{colim}_U \mathrm{Whit}^{\mathrm{glob}}(G)_{S,U},$$

where the colimit is formed using $*$ -pushforward functors, and the index runs over open substacks U as above.

Define

$$\mathrm{Whit}^{\mathrm{glob}}(G)_{\mathrm{co},S} = \mathrm{colim}_i \mathrm{Whit}^{\mathrm{glob}}(G)_{\mathrm{co},i},$$

where the colimit is formed using $*$ -pushforward functors (this time over closed embeddings).

We can rewrite

$$\mathrm{Whit}^{\mathrm{glob}}(G)_{\mathrm{co},S} \simeq \lim_i \mathrm{Whit}^{\mathrm{glob}}(G)_{\mathrm{co},i},$$

where the limit is formed using $!$ -pullback functors.

The functor⁴⁰ $\mathrm{Ps}\text{-}\mathrm{Id}_{CY_{S,i}}^{\mathrm{nv}}$ gives rise to a functor

$$(10.7) \quad \mathrm{Whit}^{\mathrm{glob}}(G)_{\mathrm{co},i} \rightarrow \mathrm{Whit}^{\mathrm{glob}}(G)_i.$$

The functors (10.7) combine to a functor

$$\Theta_{\mathrm{Whit}(G)}^{\mathrm{glob}} : \mathrm{Whit}^{\mathrm{glob}}(G)_{\mathrm{co},S} \xrightarrow{\sim} \mathrm{Whit}^{\mathrm{glob}}(G)_S.$$

We define the category $\mathrm{Whit}^{\mathrm{glob}}(G)_{\mathrm{co},Z}$ and the functor $\Theta_{\mathrm{Whit}(G)}^{\mathrm{glob}}$ for a general prestack Z mapping to Ran by right Kan extending from the case of schemes.

10.2.5. We now note the following property of the Whittaker subcategory (see [Ga6, Corollary 4.7.9(a)]):

For every i , there exists a quasi-compact open $U_0 \subset \mathcal{Y}_{S,i}$, such for any $U \subset U_0$, any object of $\mathrm{Whit}^{\mathrm{glob}}(G)_{S,U}$ is a *clean*⁴¹ extension of its restriction to U_0 along $U_0 \hookrightarrow U$.

In particular, we obtain that the $*$ -extension functor

$$\mathrm{Whit}^{\mathrm{glob}}(G)_{S,U_0} \rightarrow \mathrm{Whit}^{\mathrm{glob}}(G)_{\mathrm{co},i}$$

is already an equivalence.

It follows formally that the functors (10.7), and hence $\Theta_{\mathrm{Whit}(G)}^{\mathrm{glob}}$, are equivalences.

10.2.6. As in Sect. 9.2.4, we have a naturally defined map

$$\pi_Z : \mathrm{Gr}_{G,\rho(\omega_X),Z} \rightarrow \overline{\mathrm{Bun}}_{N,\rho(\omega_X),Z}.$$

Thanks to the existence of a canonical square root of the line $\mathfrak{l}_{G,N\rho(\omega_X)}$ (see [GLC1, Sect. 9.1.1]), the pullback of the μ_2 -gerbe $\det_{\mathrm{Bun}_G}^{\frac{1}{2}}$ along π_Z identifies with the μ_2 -gerbe $\det_{\mathrm{Gr}_{G,\rho(\omega_X)}}^{\frac{1}{2}}$ on $\mathrm{Gr}_{G,\rho(\omega_X),Z}$, see [GLC2, Sect. 9.3.1].

Consider the resulting functor

$$(\pi_Z)_* : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\rho(\omega_X),Z}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\overline{\mathrm{Bun}}_{N,\rho(\omega_X),Z})_{\mathrm{co}}.$$

As in Sects. 9.2.8-9.2.9, from $(\pi_Z)_*$ we produce a functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\rho(\omega_X),Z})_{\mathfrak{L}(N)_{\rho(\omega),\chi}} =: \mathrm{Whit}_*(G)_Z \xrightarrow{((\pi_Z)_*)^{\mathrm{Av},\chi}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\overline{\mathrm{Bun}}_{N,\rho(\omega_X),Z})_{\mathrm{co}}.$$

Note that by the definition of the functor $\mathrm{Poinc}_{G,*,Z}$, we have

$$\mathrm{Poinc}_{G,*,Z} \simeq (\overline{\mathfrak{p}}_Z)_* \circ ((\pi_Z)_*)^{\mathrm{Av},\chi}.$$

⁴⁰See [DG2, Sect. 4.4.2].

⁴¹Clean=both $!$ and $*$

10.2.7. The following is [Ga6, Theorem 5.1.4, Proposition 5.4.2, Corollary 5.4.5]:

Theorem 10.2.8.

(a) The functor $((\pi_Z)_*)^{\text{Av}, \chi}$ takes values in $\text{Whit}^{\text{glob}}(G)_{\text{co}, Z}$, and the resulting functor

$$\text{Whit}_*(G)_Z \xrightarrow{((\pi_Z)_*)^{\text{Av}, \chi}} \text{Whit}^{\text{glob}}(G)_{\text{co}, Z}$$

is an equivalence.

(b) The restriction of the functor $\pi_Z^! : \text{D-mod}_{\frac{1}{2}}(\overline{\text{Bun}}_{N, \rho(\omega_X), Z}) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X), Z})$ to

$$\text{Whit}^{\text{glob}}(G)_Z \subset \text{D-mod}_{\frac{1}{2}}(\overline{\text{Bun}}_{N, \rho(\omega_X), Z})$$

takes values in

$$(10.8) \quad \text{Whit}^!(G)_Z := \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X), Z})^{\mathfrak{d}(N)_{\rho(\omega_X), Z, \chi}} \subset \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X), Z}),$$

and the resulting functor

$$(10.9) \quad \pi_Z^! : \text{Whit}^{\text{glob}}(G)_Z \rightarrow \text{Whit}^!(G)_Z$$

is an equivalence.

(b') The inverse to the functor (10.9) is given by the restriction of the (partially defined) left adjoint $(\pi_Z)_!$ of $\pi_Z^!$ to (10.8).

(c) The composition

$$\pi_Z^! \circ \Theta_{\text{Whit}(G)}^{\text{glob}} \circ ((\pi_Z)_*)^{\text{Av}, \chi} : \text{Whit}_*(G)_Z \rightarrow \text{Whit}^!(G)_Z$$

is canonically isomorphic to $\Theta_{\text{Whit}(G)}[-2\delta_{N, \rho(\omega_X)}]$.

10.2.9. As in Sects. 9.2.10-9.2.11 and 9.6.19 the assignments

$$Z \rightsquigarrow \text{D-mod}_{\frac{1}{2}}(\overline{\text{Bun}}_{N, \rho(\omega_X), Z}) \text{ and } Z \rightsquigarrow \text{D-mod}_{\frac{1}{2}}(\overline{\text{Bun}}_{N, \rho(\omega_X), Z})_{\text{co}}$$

have natural structures of unital crystals of categories over Ran , and the assignments

$$Z \rightsquigarrow ((\pi_Z)_*)^{\text{Av}, \chi} \text{ and } Z \rightsquigarrow (\pi_Z)_!$$

are left-lax and right-lax unital functors, to be denoted,

$$\underline{\pi}_*^{\text{Av}, \chi} \text{ and } \underline{\pi}_!,$$

respectively.

However, as in Proposition 9.2.12 (resp., Corollary 9.6.21) one shows that these structures are strictly unital.

10.3. The space of *generic* reductions.

10.3.1. Let

$$\overline{\text{Bun}}_{N, \rho(\omega_X), \text{gen}} := \text{Bun}_{N, \rho(\omega_X), \text{gen}} \times_{\text{Bun}_{G, \text{gen}}} \text{Bun}_G$$

be the space of G -bundles equipped with a *generic* reduction to $N_{\rho(\omega_X)}$ (see [Bar] or [Ga5, Appendix A], where the definition in families is given). Denote by $\bar{\mathbf{p}}_{\text{gen}}$ the projection

$$\overline{\text{Bun}}_{N, \rho(\omega_X), \text{gen}} \rightarrow \text{Bun}_G.$$

Note that we have a naturally defined map

$$(10.10) \quad \mathbf{f} : \overline{\text{Bun}}_{N, \rho(\omega_X), \text{Ran}} \rightarrow \overline{\text{Bun}}_{N, \rho(\omega_X), \text{gen}}.$$

The following is established in [Ga5, Theorem A.1.10]:

Lemma 10.3.2. *The map $\mathbf{f}$ of (10.10) is pseudo-proper and universally homologically acyclic.*⁴²

⁴²See [Ga5, Sect. A.1.8] for what this means.

10.3.3. Similarly, for $\mathcal{Z} \rightarrow \text{Ran}$ we have a map

$$(10.11) \quad f_{\mathcal{Z}} : \overline{\text{Bun}}_{N, \rho(\omega_X), \mathcal{Z} \subseteq} \rightarrow \overline{\text{Bun}}_{N, \rho(\omega_X), \text{gen}} \times \mathcal{Z},$$

and parallel to Lemma 10.3.2, one shows that the map (10.11) is also universally homologically acyclic.

10.4. Closures of semi-infinite orbits.

10.4.1. Let $\text{Gr}_G^{\text{pos}} \subset \text{Gr}_G$ be the factorization subspace equal to the closure of the unit orbit of $\mathfrak{L}(N_P)$, see [Ga6, Sect. 6.1.5].

We can regard Gr_G as a factorization *module* space over Gr_G^{pos} . For $\mathcal{Z} \rightarrow \text{Ran}$, denote by $\text{Gr}_{G, \mathcal{Z} \subseteq}^{\text{pos}}$ the corresponding prestack over $\mathcal{Z} \subseteq$, i.e.,

$$\text{Gr}_{G, \mathcal{Z} \subseteq}^{\text{pos}} \times_{\mathcal{Z} \subseteq} \mathcal{Z} = \text{Gr}_{G, \mathcal{Z}}$$

and

$$\text{Gr}_{G, \mathcal{Z} \subseteq}^{\text{pos}} \times_{\mathcal{Z} \subseteq} (\text{Ran} \times \mathcal{Z} \subseteq)_{\text{disj}} \simeq (\text{Gr}_{G, \text{Ran}}^{\text{pos}} \times \text{Gr}_{G, \mathcal{Z} \subseteq}^{\text{pos}})_{\text{Ran} \times \mathcal{Z} \subseteq} \times (\text{Ran} \times \mathcal{Z} \subseteq)_{\text{disj}}.$$

10.4.2. Note that we have a Cartesian square

$$\begin{array}{ccc} \text{Gr}_{G, \rho(\omega)_X, \text{Ran}}^{\text{pos}} & \longrightarrow & \text{Gr}_{G, \rho(\omega)_X, \text{Ran}} \\ \pi_{\text{Ran}}^{\text{pos}} \downarrow & & \downarrow \pi_{\text{Ran}} \\ \overline{\text{Bun}}_{N, \rho(\omega_X)} \times \text{Ran} & \longrightarrow & \overline{\text{Bun}}_{N, \rho(\omega_X), \text{Ran}}, \end{array}$$

and, more generally, for $\mathcal{Z} \rightarrow \text{Ran}$:

$$\begin{array}{ccc} \text{Gr}_{G, \rho(\omega)_X, \mathcal{Z} \subseteq}^{\text{pos}} & \longrightarrow & \text{Gr}_{G, \rho(\omega)_X, \mathcal{Z} \subseteq} \\ \pi_{\mathcal{Z} \subseteq}^{\text{pos}} \downarrow & & \downarrow \pi_{\mathcal{Z} \subseteq} \\ \overline{\text{Bun}}_{N, \rho(\omega_X), \mathcal{Z}} \times_{\mathcal{Z}} \mathcal{Z} \subseteq & \longrightarrow & \overline{\text{Bun}}_{\rho(\omega)_X, \mathcal{Z} \subseteq} \\ \text{id} \times \text{pr}_{\text{small}, \mathcal{Z}} \downarrow & & \\ \overline{\text{Bun}}_{N, \rho(\omega_X), \mathcal{Z}} & & \end{array}$$

10.4.3. We claim:

Lemma 10.4.4. *For $\mathcal{F} \in \text{Whit}^{\text{glob}}(G)_{\mathcal{Z}}$, the object*

$$(\pi_{\mathcal{Z} \subseteq}^{\text{pos}})^! \circ (\text{id} \times \text{pr}_{\text{small}, \mathcal{Z}})^!(\mathcal{F}) \in \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega)_X, \mathcal{Z} \subseteq}^{\text{pos}}) \subset \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega)_X, \mathcal{Z} \subseteq}),$$

viewed as an object of $\text{Whit}^!(G)_{\mathcal{Z} \subseteq}$, identifies canonically with

$$\text{ins. unit}_{\mathcal{Z}}(\pi_{\mathcal{Z}}^!(\mathcal{F})).$$

Proof. To simplify the notation, we will assume that $\mathcal{Z} = \text{pt}$, corresponding to $\underline{x} \in \text{Ran}$, so that

$$\mathcal{Z} \subseteq =: \text{Ran}_{\underline{x}}.$$

The unital structure on Gr_G as a factorization space gives rise to a map

$$\text{ins. unit}_{\text{Gr}, \underline{x}} : \text{Gr}_{G, \rho(\omega)_X, \underline{x}} \times \text{Ran}_{\underline{x}} \rightarrow \text{Gr}_{G, \rho(\omega)_X, \text{Ran}_{\underline{x}}}.$$

It follows from [Ga6, Theorem 6.2.5] that the functor of $!$ -pullback along $\text{ins. unit}_{\text{Gr}, \underline{x}}$ gives rise to an equivalence between the full subcategory of $\text{Whit}^!(G)_{\text{Ran}_{\underline{x}}}$ consisting of objects supported on

$$\text{Gr}_{G, \rho(\omega)_X, \text{Ran}_{\underline{x}}}^{\text{pos}} \subset \text{Gr}_{G, \rho(\omega)_X, \text{Ran}_{\underline{x}}}$$

and

$$\text{Whit}^!(G)_{\underline{x}} \otimes \text{D-mod}(\text{Ran}_{\underline{x}}).$$

Hence, it suffices to construct an isomorphism between

$$(\text{ins. unit}_{\text{Gr}, \underline{x}})^! \circ (\pi_{\text{Ran}, \underline{x}}^{\text{pos}})^! \circ (\text{id} \times \text{pr}_{\text{small}, \underline{x}})^!(\mathcal{F}) \text{ and } (\text{ins. unit}_{\text{Gr}, \underline{x}})^! \circ \text{ins. unit}_{\underline{x}}(\pi_{\mathcal{Z}}^!(\mathcal{F})).$$

First, we tautologically have:

$$(\text{ins. unit}_{\text{Gr}, \underline{x}})^! \circ (\pi_{\text{Ran}, \underline{x}}^{\text{pos}})^! \circ (\text{id} \times \text{pr}_{\text{small}, \underline{x}})^!(\mathcal{F}) \simeq \pi_{\underline{x}}^!(\mathcal{F}) \boxtimes \omega_{\text{Ran}, \underline{x}}.$$

By the definition of the unital structure on $\text{Whit}^!(G)$, the functor

$$\text{ins. unit}_{\underline{x}} : \text{Whit}^!(G)_{\underline{x}} \rightarrow \text{Whit}^!(G)_{\text{Ran}, \underline{x}}$$

is given by

$$\text{Av}_!^{\Omega(N)_{\rho(\omega_X), \text{Ran}, \underline{x}}, \chi} \circ (\text{ins. unit}_{\text{Gr}, \underline{x}})_* ((-) \boxtimes \omega_{\text{Ran}, \underline{x}}),$$

where $\text{Av}_!^{\Omega(N)_{\rho(\omega_X), \text{Ran}, \underline{x}}, \chi}$ is the (partially defined) left adjoint to the embedding

$$\text{Whit}^!(G)_{\text{Ran}, \underline{x}} \hookrightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega)_X, \text{Ran}, \underline{x}}).$$

Now, by [Ga6, Theorem 6.4.8], the functor

$$\text{Av}_!^{\Omega(N)_{\rho(\omega_X), \text{Ran}, \underline{x}}, \chi} \circ (\text{ins. unit}_{\text{Gr}, \underline{x}})_* : \text{Whit}^!(G)_{\underline{x}} \otimes \text{D-mod}(\text{Ran}, \underline{x}) \rightarrow \text{Whit}^!(G)_{\text{Ran}, \underline{x}}$$

is a right inverse of $(\text{ins. unit}_{\text{Gr}, \underline{x}})^!$.

Hence, we obtain

$$(\text{ins. unit}_{\text{Gr}, \underline{x}})^!(\text{ins. unit}_{\mathcal{Z}}(\pi_{\underline{x}}^!(\mathcal{F}))) \simeq \pi_{\underline{x}}^!(\mathcal{F}) \boxtimes \omega_{\text{Ran}, \underline{x}},$$

as desired. □

10.4.5. Let

$$(\text{Gr}_G \times \text{Gr}_M)^{\text{neg}} \subset \text{Gr}_G \times \text{Gr}_M$$

be the factorization space equal to the closure of the unit orbit of $\mathfrak{L}(P^-)$.

As in Sect. 10.4.1, for $\mathcal{Z} \rightarrow \text{Ran}$, we can form the space

$$(\text{Gr}_G \times \text{Gr}_M)_{\mathcal{Z} \subseteq}^{\text{neg}}.$$

Furthermore, as in Sect. 10.4.2, we have a natural projection

$$\pi_{\text{Ran}} : (\text{Gr}_G \times \text{Gr}_M)_{\text{Ran}} \rightarrow \widetilde{\text{Bun}}_{P^-, \text{Ran}},$$

so that we have Cartesian squares

$$\begin{array}{ccc} (\text{Gr}_G \times \text{Gr}_M)_{\text{Ran}}^{\text{neg}} & \longrightarrow & (\text{Gr}_G \times \text{Gr}_M)_{\text{Ran}} \\ \pi_{\text{Ran}}^{\text{neg}} \downarrow & & \downarrow \pi_{\text{Ran}} \\ \widetilde{\text{Bun}}_{P^-, \text{Ran}} \times \text{Ran} & \longrightarrow & \widetilde{\text{Bun}}_{P^-, \text{Ran}} \end{array}$$

and

$$\begin{array}{ccc} (\text{Gr}_G \times \text{Gr}_M)_{\mathcal{Z} \subseteq}^{\text{neg}} & \longrightarrow & (\text{Gr}_G \times \text{Gr}_M)_{\mathcal{Z} \subseteq} \\ \pi_{\mathcal{Z} \subseteq}^{\text{neg}} \downarrow & & \downarrow \pi_{\mathcal{Z} \subseteq} \\ \widetilde{\text{Bun}}_{P^-, \mathcal{Z}} \times_{\mathcal{Z}} \mathcal{Z} \subseteq & \longrightarrow & \widetilde{\text{Bun}}_{P^-, \mathcal{Z} \subseteq}. \end{array}$$

10.4.6. Denote by π^{neg} the composition

$$(\text{Gr}_G \times \text{Gr}_M)_{\text{Ran}}^{\text{neg}} \xrightarrow{\pi_{\text{Ran}}^{\text{neg}}} \widetilde{\text{Bun}}_{P^-, \text{Ran}} \times \text{Ran} \rightarrow \widetilde{\text{Bun}}_{P^-}.$$

Note also that we have a canonical isomorphism of μ_2 -gerbes on $(\text{Gr}_G \times \text{Gr}_M)_{\text{Ran}}^{\text{neg}}$

$$(\pi^{\text{neg}})^*(\det_{G, M}^{\frac{1}{2}}) \simeq \det_{\text{Gr}_G}^{\frac{1}{2}} \otimes \det_{\text{Gr}_M}^{\frac{1}{2}},$$

and similarly for the map $\pi_{\mathcal{Z} \subseteq}^{\text{neg}}$.

10.4.7. Let $\tilde{\Delta}^{-, \frac{\infty}{2}}$ denote the image of $\Delta^{-, \frac{\infty}{2}}$ under

$$\begin{aligned} \mathrm{I}(G, P^-)^{\mathrm{loc}} &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \rho(\omega_X)})^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)} \simeq \\ &\simeq \mathrm{D}\text{-mod}_{\det_{\mathrm{Gr}_G}^{\frac{1}{2}} \otimes \det_{\mathrm{Gr}_M}^{\frac{1}{2}}}(\mathrm{Gr}_G \times \mathrm{Gr}_M)^{\mathfrak{L}(P^-)} \rightarrow \mathrm{D}\text{-mod}_{\det_{\mathrm{Gr}_G}^{\frac{1}{2}} \otimes \det_{\mathrm{Gr}_M}^{\frac{1}{2}}}(\mathrm{Gr}_G \times \mathrm{Gr}_M). \end{aligned}$$

Note that $\tilde{\Delta}_{\mathrm{Ran}}^{-, \frac{\infty}{2}}$ is supported on

$$(\mathrm{Gr}_G \times \mathrm{Gr}_M)_{\mathrm{Ran}}^{\mathrm{neg}} \subset (\mathrm{Gr}_G \times \mathrm{Gr}_M)_{\mathrm{Ran}}.$$

We have a tautologically defined map⁴³

$$(\pi^{\mathrm{neg}})_!(\tilde{\Delta}_{\mathrm{Ran}}^{-, \frac{\infty}{2}}) \rightarrow j_!(\omega_{\mathrm{Bun}_{P^-}}).$$

Hence, we obtain a map

$$(10.12) \quad \tilde{\Delta}_{\mathrm{Ran}}^{-, \frac{\infty}{2}} \rightarrow (\pi^{\mathrm{neg}})^! \circ j_!(\omega_{\mathrm{Bun}_{P^-}}).$$

The following assertion is a variant of [Ga5, Corollary 3.6.6]:

Lemma 10.4.8. *The map (10.12) is an isomorphism.*

10.5. Zastava spaces.

10.5.1. Let **Zas** be the *Zastava space*, defined to be the open substack of

$$\overline{\mathrm{Bun}}_{N, \rho(\omega_X)} \times_{\mathrm{Bun}_G} \widetilde{\mathrm{Bun}}_{P^-},$$

corresponding to the condition that the generalized reductions of the given G -bundle to $N_{\rho(\omega_X)}$ and P^- are transversal *at the generic point* of X .

For $\mathbb{Z} \rightarrow \mathrm{Ran}$, we let **Zas_ℤ** denote the version of **Zas**, where we allow both generalized reductions to have poles at the specified point of the curve, i.e., **Zas_ℤ** is the open substack of

$$\overline{\mathrm{Bun}}_{N, \rho(\omega_X), \mathbb{Z}} \times_{\mathrm{Bun}_G \times \mathbb{Z}} \widetilde{\mathrm{Bun}}_{P^-, \mathbb{Z}},$$

given by the generic transversality condition.

We will denote by

$$' \tilde{\mathbf{p}}_{\mathbb{Z}}^- : \mathbf{Zas}_{\mathbb{Z}} \rightarrow \overline{\mathrm{Bun}}_{N, \rho(\omega_X), \mathbb{Z}} \text{ and } ' \bar{\mathbf{p}}_{\mathbb{Z}} : \mathbf{Zas}_{\mathbb{Z}} \rightarrow \widetilde{\mathrm{Bun}}_{P^-, \mathbb{Z}}$$

the corresponding maps.

10.5.2. We will consider the μ_2 -gerbes on **Zas_ℤ**

$$\det_G^{\frac{1}{2}}, \det_M^{\frac{1}{2}} \text{ and } \det_{G, M}^{\frac{1}{2}},$$

pulled back by means of $' \bar{\mathbf{p}}_{\mathbb{Z}}$ from the corresponding gerbes on $\widetilde{\mathrm{Bun}}_{P^-, \mathbb{Z}}$.

We will consider the corresponding categories of twisted D-modules

$$\mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathbf{Zas}_{\mathbb{Z}}), \mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathbf{Zas}_{\mathbb{Z}})_{\mathrm{co}}, \mathrm{D}\text{-mod}_{\mathfrak{g}}(\mathbf{Zas}_{\mathbb{Z}})_{\mathrm{co}} / \mathrm{Bun}_M,$$

defined by the recipe of Sects. 9.1.4-9.1.8.

⁴³In fact, one can show that this map is an isomorphism.

10.5.3. Note now that we have a canonically defined map⁴⁴

$$\mathfrak{s} : \mathbf{Zas} \rightarrow \overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \text{gen}}$$

(see Sect. 10.3.1), so that there is a commutative diagram

$$\begin{array}{ccc} \mathbf{Zas} & \xrightarrow{\mathfrak{s}} & \overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \text{gen}} \\ \downarrow \text{'}\bar{\mathfrak{p}} & & \downarrow \bar{\mathfrak{p}}_{M, \text{gen}} \\ \widetilde{\mathbf{Bun}}_{P^-} & \xrightarrow{\bar{\mathfrak{q}}^-} & \mathbf{Bun}_M. \end{array}$$

Similarly, for $\mathcal{Z} \rightarrow \text{Ran}$, we have a map

$$\mathfrak{s}_{\mathcal{Z}} : \mathbf{Zas}_{\mathcal{Z}} \rightarrow \overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \text{gen}} \times \mathcal{Z}$$

and a commutative diagram

$$(10.13) \quad \begin{array}{ccc} \mathbf{Zas}_{\mathcal{Z}} & \xrightarrow{\mathfrak{s}_{\mathcal{Z}}} & \overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \text{gen}} \times \mathcal{Z} \\ \downarrow \text{'}\bar{\mathfrak{p}}_{\mathcal{Z}} & & \downarrow \bar{\mathfrak{p}}_{M, \text{gen}} \times \text{id}_{\mathcal{Z}} \\ \widetilde{\mathbf{Bun}}_{P^-, \mathcal{Z}} & \xrightarrow{\bar{\mathfrak{q}}_{\mathcal{Z}}} & \mathbf{Bun}_M \times \mathcal{Z} \end{array}$$

10.5.4. Denote

$$\mathbf{Zas}^{\text{Ran}} := \mathbf{Zas} \times_{\overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \text{gen}}} \overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \text{Ran}}$$

and

$$\mathbf{Zas}_{\mathcal{Z}}^{\text{Ran}} := \mathbf{Zas}_{\mathcal{Z}} \times_{\overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \text{gen}} \times \mathcal{Z}} \overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \mathcal{Z} \subseteq},$$

where

$$\overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \text{Ran}} \rightarrow \overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \text{gen}} \text{ and } \overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \mathcal{Z} \subseteq} \rightarrow \overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \text{gen}} \times \mathcal{Z}$$

are the maps (10.10) and (10.11), respectively.

Remark 10.5.5. The superscript Ran in the notations $\mathbf{Zas}^{\text{Ran}}$ and $\mathbf{Zas}_{\mathcal{Z}}^{\text{Ran}}$ is meant to distinguish these spaces from $\mathbf{Zas}_{\text{Ran}}$ and $\text{Ran}_{\mathcal{Z} \subseteq}$, respectively.

10.5.6. Consider the fiber products

$$\begin{array}{ccc} \mathbf{Zas}^{\text{Ran}} \times_{\overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \text{Ran}}} \text{Gr}_{M, \rho_M(\omega_X), \text{Ran}} & \longrightarrow & \text{Gr}_{M, \rho(\omega_X), \text{Ran}} \\ \downarrow & & \downarrow \pi_{\text{Ran}} \\ \mathbf{Zas}^{\text{Ran}} & \xrightarrow{\text{'}\bar{\mathfrak{s}}} & \overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \text{Ran}} \end{array}$$

and

$$(10.14) \quad \begin{array}{ccc} \mathbf{Zas}_{\mathcal{Z}}^{\text{Ran}} \times_{\overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \mathcal{Z} \subseteq}} \text{Gr}_{M, \rho_M(\omega_X), \mathcal{Z} \subseteq} & \longrightarrow & \text{Gr}_{M, \rho(\omega_X), \mathcal{Z} \subseteq} \\ \downarrow & & \downarrow \pi_{\mathcal{Z} \subseteq} \\ \mathbf{Zas}_{\mathcal{Z}}^{\text{Ran}} & \xrightarrow{\text{'}\bar{\mathfrak{s}}_{\mathcal{Z}}} & \overline{\mathbf{Bun}}_{N(M), \rho(\omega_X), \mathcal{Z} \subseteq}. \end{array}$$

⁴⁴Here $N(M)$ is the maximal unipotent subgroup of the Levi M .

10.5.7. Recall the space $(\mathrm{Gr}_G \times \mathrm{Gr}_M)_{\mathcal{Z} \subseteq}^{\mathrm{neg}}$, see Sect. 10.4.5. The key geometric input in the proof of Theorem 10.1.2 is the following observation:

Lemma 10.5.8. *There exists a canonical isomorphism*

$$\mathrm{Zas}_{\mathcal{Z}}^{\mathrm{Ran}} \times_{\overline{\mathrm{Bun}}_{N(M), \rho(\omega_X), \mathcal{Z} \subseteq}} \mathrm{Gr}_{M, \rho_M(\omega_X), \mathcal{Z} \subseteq} \simeq \mathrm{Gr}_{G, \rho(\omega_X), \mathcal{Z} \subseteq}^{\mathrm{pos}} \times_{\mathrm{Gr}_{G, \rho(\omega_X), \mathcal{Z} \subseteq}} (\mathrm{Gr}_{G, \rho(\omega_X)} \times \mathrm{Gr}_{M, \rho(\omega_X)})_{\mathcal{Z} \subseteq}^{\mathrm{neg}},$$

so that:

- The map

$$\mathrm{Zas}_{\mathcal{Z}}^{\mathrm{Ran}} \times_{\overline{\mathrm{Bun}}_{N(M), \rho(\omega_X), \mathcal{Z} \subseteq}} \mathrm{Gr}_{M, \rho_M(\omega_X), \mathcal{Z} \subseteq} \rightarrow \mathrm{Zas}_{\mathcal{Z}}^{\mathrm{Ran}} \rightarrow \mathrm{Zas}_{\mathcal{Z}} \xrightarrow{\bar{q}_{\mathcal{Z}}} \overline{\mathrm{Bun}}_{N, \rho(\omega_X), \mathcal{Z}}$$

identifies with

$$\begin{aligned} \mathrm{Gr}_{G, \rho(\omega_X), \mathcal{Z} \subseteq}^{\mathrm{pos}} \times_{\mathrm{Gr}_{G, \rho(\omega_X), \mathcal{Z} \subseteq}} (\mathrm{Gr}_{G, \rho(\omega_X)} \times \mathrm{Gr}_{M, \rho(\omega_X)})_{\mathcal{Z} \subseteq}^{\mathrm{neg}} &\rightarrow \mathrm{Gr}_{G, \rho(\omega_X), \mathcal{Z} \subseteq}^{\mathrm{pos}} \xrightarrow{\pi_{\mathcal{Z} \subseteq}^{\mathrm{pos}}} \\ &\rightarrow \overline{\mathrm{Bun}}_{N, \rho(\omega_X), \mathcal{Z}} \times_{\mathcal{Z}} \mathcal{Z} \subseteq \rightarrow \overline{\mathrm{Bun}}_{N, \rho(\omega_X), \mathcal{Z}}; \end{aligned}$$

- The map

$$\mathrm{Zas}_{\mathcal{Z}}^{\mathrm{Ran}} \times_{\overline{\mathrm{Bun}}_{N(M), \rho(\omega_X), \mathcal{Z} \subseteq}} \mathrm{Gr}_{M, \rho_M(\omega_X), \mathcal{Z} \subseteq} \rightarrow \mathrm{Zas}_{\mathcal{Z}}^{\mathrm{Ran}} \rightarrow \mathrm{Zas}_{\mathcal{Z}} \xrightarrow{\bar{p}_{\mathcal{Z}}} \widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}}$$

identifies with

$$\begin{aligned} \mathrm{Gr}_{G, \rho(\omega_X), \mathcal{Z} \subseteq}^{\mathrm{pos}} \times_{\mathrm{Gr}_{G, \rho(\omega_X), \mathcal{Z} \subseteq}} (\mathrm{Gr}_{G, \rho(\omega_X)} \times \mathrm{Gr}_{M, \rho(\omega_X)})_{\mathcal{Z} \subseteq}^{\mathrm{neg}} &\rightarrow \\ &\rightarrow (\mathrm{Gr}_{G, \rho(\omega_X)} \times \mathrm{Gr}_{M, \rho(\omega_X)})_{\mathcal{Z} \subseteq}^{\mathrm{neg}} \xrightarrow{\pi_{\mathcal{Z} \subseteq}^{\mathrm{neg}}} \widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}} \times_{\mathcal{Z}} \mathcal{Z} \subseteq \xrightarrow{\mathrm{id} \times \mathrm{pr}_{\mathrm{small}, \mathcal{Z}}} \widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}}. \end{aligned}$$

10.6. Proof of Theorem 10.1.2.

10.6.1. Recall that $j_{\mathcal{Z}}$ denotes the locally closed embedding

$$\mathrm{Bun}_{P^-} \times \mathcal{Z} \hookrightarrow \widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}}.$$

Note that as in Sect. 9.8.1, we can regard $\omega_{\mathrm{Bun}_{P^-} \times \mathcal{Z}}$ as an object of $\mathrm{D}\text{-mod}_{\det \frac{1}{2}, M}(\mathrm{Bun}_{P^-} \times \mathcal{Z})$.

Hence, we can regard $(j_{\mathcal{Z}})_!(\omega_{\mathrm{Bun}_{P^-} \times \mathcal{Z}})$ as an object of $\mathrm{D}\text{-mod}_{\det \frac{1}{2}, M}(\widetilde{\mathrm{Bun}}_{P^-, \mathcal{Z}})$.

The functor

$$(\mathrm{CT}_{\mathrm{co}, ?, \rho_P(\omega_X)}^- \otimes \mathrm{Id}) \circ \mathrm{Poinc}_{G, *, \mathcal{Z}}$$

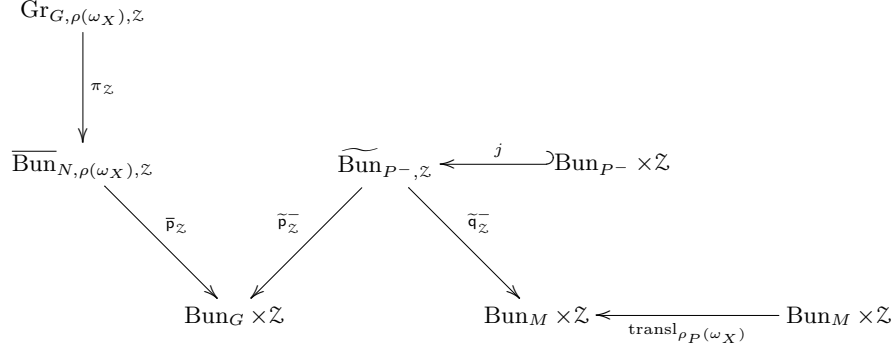
is by definition

$$(\mathrm{transl}_{\rho_P(\omega_X)})^* \circ (\widetilde{q}_{\mathcal{Z}})_* \circ \left(\left((\widetilde{p}_{\mathcal{Z}}^-)^! \circ \mathrm{Poinc}_{G, *, \mathcal{Z}}(-) \right)^! \otimes (j_{\mathcal{Z}})_!(\omega_{\mathrm{Bun}_{P^-} \times \mathcal{Z}}) \right) [-\mathrm{shift}],$$

which we rewrite as

$$(10.15) \quad (\mathrm{transl}_{\rho_P(\omega_X)})^* \circ (\widetilde{q}_{\mathcal{Z}})_* \circ \left(\left((\widetilde{p}_{\mathcal{Z}}^-)^! \circ (\bar{p}_{\mathcal{Z}})_* \circ ((\pi_{\mathcal{Z}})_*)^{\mathrm{Av}, \chi}(-) \right)^! \otimes (j_{\mathcal{Z}})_!(\omega_{\mathrm{Bun}_{P^-} \times \mathcal{Z}}) \right) [-\mathrm{shift}].$$

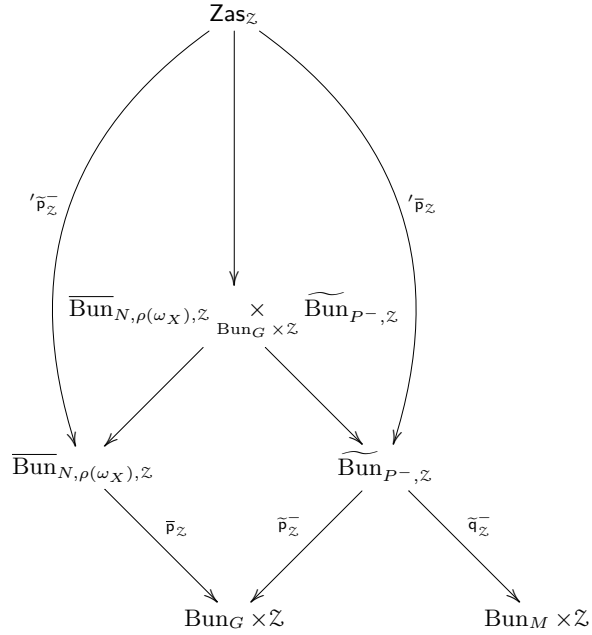
We depict this diagrammatically as follows:



10.6.2. By base change for any $\mathcal{F} \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\overline{\mathrm{Bun}}_{N, \rho(\omega_X), \mathbb{Z}})$, we obtain a morphism

$$\begin{aligned}
(10.16) \quad (\tilde{q}_{\mathbb{Z}}^-)_* \circ \left((\tilde{p}_{\mathbb{Z}}^-)^! \circ (\bar{p}_{\mathbb{Z}})_*(\mathcal{F}) \otimes (j_{\mathbb{Z}})^!(\omega_{\mathrm{Bun}_{P^-} \times \mathbb{Z}}) \right) \rightarrow \\
\rightarrow (\tilde{q}_{\mathbb{Z}}^-)_* \circ (\tilde{p}_{\mathbb{Z}}^-)_* \left((\tilde{p}_{\mathbb{Z}}^-)^!(\mathcal{F}) \otimes (\tilde{p}_{\mathbb{Z}}^-)^! \circ (j_{\mathbb{Z}})^!(\omega_{\mathrm{Bun}_{P^-} \times \mathbb{Z}}) \right),
\end{aligned}$$

see the maps in the next diagram⁴⁵:



The following is established as in [Lin, Proposition 4.1.8]:

Lemma 10.6.3. *The map (10.16) is an isomorphism if $\mathcal{F} \in \mathrm{Whit}^{\mathrm{glob}}(G)_{\mathrm{co}, \mathbb{Z}}$.*

⁴⁵Note that the long vertical arrow is an open embedding.

10.6.4. Applying Lemma 10.6.3, we rewrite the expression in (10.15) (without the ρ_P -translation and the cohomological shift) as

$$(10.17) \quad (\tilde{q}_Z^-)_* \circ (\tilde{p}_Z^-)_* \left(\left((\tilde{p}_Z^-)^! \circ ((\pi_Z)_*)^{A_{V,X}}(-) \right)^! \otimes (\tilde{p}_Z^-)^! \circ (j_Z)_!(\omega_{\text{Bun}_{P^-} \times Z}) \right).$$

We depict this by the following diagram:

$$(10.18) \quad \begin{array}{ccc} \text{Gr}_{G,\rho(\omega_X),Z} & & \text{Zas}_Z \\ \pi_Z \searrow & \swarrow \tilde{p}_Z^- & \searrow \tilde{p}_Z^- \\ \overline{\text{Bun}}_{N,\rho(\omega_X),Z} & & \widetilde{\text{Bun}}_{P^-,Z} \\ & & \searrow \tilde{q}_Z^- \\ & & \text{Bun}_M \times Z \end{array}$$

10.6.5. Using (10.13), we rewrite (10.17) as

$$(10.19) \quad (\bar{p}_{M,\text{gen}} \times \text{id}_Z)_* \circ (s_Z)_* \left(\left((\tilde{p}_Z^-)^! \circ ((\pi_Z)_*)^{A_{V,X}}(-) \right)^! \otimes (\tilde{p}_Z^-)^! \circ (j_Z)_!(\omega_{\text{Bun}_{P^-} \times Z}) \right),$$

where $\bar{p}_{M,\text{gen}}$ denotes the map

$$\overline{\text{Bun}}_{N(M),\rho(\omega_X),\text{gen}} \rightarrow \text{Bun}_M.$$

I.e., we replace (10.18) by

$$(10.20) \quad \begin{array}{ccc} \text{Gr}_{G,\rho(\omega_X),Z} & & \text{Zas}_Z \\ \pi_Z \searrow & \swarrow \tilde{p}_Z^- & \searrow s_Z \\ \overline{\text{Bun}}_{N,\rho(\omega_X),Z} & & \overline{\text{Bun}}_{N(M),\rho(\omega_X),\text{gen}} \times Z \\ & & \searrow \bar{p}_{M,\text{gen}} \times \text{id}_Z \\ & & \text{Bun}_M \times Z \end{array}$$

10.6.6. Note that according to Lemma 10.3.2, the map

$$(f_{M,Z})_* \circ (f_{M,Z})^! \rightarrow \text{Id}$$

is an isomorphism, where $f_{M,Z}$ denotes the map

$$f_{M,Z} : \overline{\text{Bun}}_{N(M),\rho(\omega_X),Z \subseteq} \rightarrow \overline{\text{Bun}}_{N(M),\rho(\omega_X),\text{gen}} \times Z$$

and also its base change

$$\text{Zas}_Z^{\text{Ran}} \rightarrow \text{Zas}_Z.$$

Hence, we can rewrite the expression in (10.17) as

$$\begin{aligned} & (\text{id} \times \text{pr}_{\text{small},Z})_* \circ (\bar{p}_{M,Z \subseteq})_* \circ (s_{Z \subseteq})_* \circ f_{M,Z}^! \left(\left((\tilde{p}_Z^-)^! \circ ((\pi_Z)_*)^{A_{V,X}}(-) \right)^! \otimes (\tilde{p}_Z^-)^! \circ (j_Z)_!(\omega_{\text{Bun}_{P^-} \times Z}) \right) \simeq \\ & \simeq (\text{id} \times \text{pr}_{\text{small},Z})_* \circ (\bar{p}_{M,Z \subseteq})_* \circ (s_{Z \subseteq})_* \circ f_{M,Z}^! \left(\left((\tilde{p}_Z^-)^! \circ ((\pi_Z)_*)^{A_{V,X}}(-) \right)^! \otimes (\tilde{p}_Z^-)^! \circ (j_Z)_!(\omega_{\text{Bun}_{P^-} \times Z}) \right), \end{aligned}$$

where the isomorphism is due to the fact that $\mathrm{pr}_{\mathrm{small}, \mathbb{Z}}$ is pseudo-proper, and where the maps are as in the diagram below:

(10.21)

$$\begin{array}{ccccc}
 & & \mathrm{Zas}_{\mathbb{Z}}^{\mathrm{Ran}} & & \\
 & \swarrow \mathrm{f}_{M, \mathbb{Z}} & & \searrow \mathfrak{s}_{\mathbb{Z} \subseteq} & \\
 \mathrm{Gr}_{G, \rho(\omega_X), \mathbb{Z}} & & \mathrm{Zas}_{\mathbb{Z}} & & \overline{\mathrm{Bun}}_{N(M), \rho(\omega_X), \mathbb{Z} \subseteq} \\
 \searrow \pi_{\mathbb{Z}} & \swarrow \mathrm{i}' \bar{\mathfrak{p}}_{\mathbb{Z}} & \searrow \mathrm{i}' \bar{\mathfrak{p}}_{\mathbb{Z}} & \swarrow \bar{\mathfrak{p}}_{M, \mathbb{Z} \subseteq} & \\
 & \overline{\mathrm{Bun}}_{N, \rho(\omega_X), \mathbb{Z}} & & \mathrm{Bun}_M \times \mathbb{Z}^{\subseteq} & \\
 & & \mathrm{Bun}_{P^-} \times \mathbb{Z}^{\subseteq} \xrightarrow{j} \widetilde{\mathrm{Bun}}_{P^-, \mathbb{Z}} \xrightarrow{\bar{\mathfrak{q}}_{\mathbb{Z}}} \mathrm{Bun}_M \times \mathbb{Z} & & \\
 & & \uparrow \mathrm{id} \times \mathrm{pr}_{\mathrm{small}, \mathbb{Z}} & &
 \end{array}$$

10.6.7. Note now that since ρ_P is central in M , we have a well-defined map

$$\mathrm{transl}_{\rho_P(\omega_X)} : \overline{\mathrm{Bun}}_{N(M), \rho_M(\omega_X), \mathbb{Z} \subseteq} \rightarrow \overline{\mathrm{Bun}}_{N(M), \rho(\omega_X), \mathbb{Z} \subseteq}.$$

The next assertion is the main point in the proof of Theorem 10.1.2:

Proposition 10.6.8. *There is a canonical isomorphism*

$$\begin{aligned}
 (10.22) \quad & (\mathrm{transl}_{\rho_P(\omega_X)})^* \circ (\mathfrak{s}_{\mathbb{Z} \subseteq})_* \circ \mathrm{f}_{M, \mathbb{Z}}^! \left(\left((\mathrm{i}' \bar{\mathfrak{p}}_{\mathbb{Z}})^! \circ ((\pi_{\mathbb{Z}})_*)^{\mathrm{Av}, \chi}(-) \right)^! \otimes (\mathrm{i}' \bar{\mathfrak{p}}_{\mathbb{Z}})^! \circ (j_{\mathbb{Z}})_!(\omega_{\mathrm{Bun}_{P^-} \times \mathbb{Z}}) \right) \simeq \\
 & \simeq ((\pi_{M, \mathbb{Z} \subseteq})_*)^{\mathrm{Av}, \chi} \circ J_{\mathrm{Whit}, \Theta}^{-, !} \circ \mathrm{ins. unit}_{\mathbb{Z}}(-) [\mathrm{shift} - 2\delta_{(N_P)_{\rho_P(\omega_X)}} - \delta_{(N_P^-)_{\rho_P(\omega_X)}}],
 \end{aligned}$$

as functors

$$\mathrm{Whit}_*(G)_{\mathbb{Z}} \rightrightarrows \mathrm{D-mod}_{\frac{1}{2}}(\overline{\mathrm{Bun}}_{N(M), \rho_M(\omega_X), \mathbb{Z} \subseteq})_{\mathrm{co}},$$

where $\pi_{M, \mathbb{Z} \subseteq}$ denotes the map

$$\mathrm{Gr}_{M, \rho_M(\omega_X), \mathbb{Z} \subseteq} \rightarrow \overline{\mathrm{Bun}}_{N(M), \rho_M(\omega_X), \mathbb{Z} \subseteq}.$$

10.6.9. Proposition 10.6.8 combined with Sect. 10.6.6 implies that the expression in (10.15) identifies with

$$\begin{aligned}
 & (\mathrm{id} \times \mathrm{pr}_{\mathrm{small}, \mathbb{Z}})_! \circ (\bar{\mathfrak{p}}_{M, \mathbb{Z} \subseteq})_* \circ ((\pi_{M, \mathbb{Z} \subseteq})_*)^{\mathrm{Av}, \chi} \circ J_{\mathrm{Whit}, \Theta}^{-, !} \circ \mathrm{ins. unit}_{\mathbb{Z}}[-2\delta_{(N_P)_{\rho_P(\omega_X)}} - \delta_{(N_P^-)_{\rho_P(\omega_X)}}] \simeq \\
 & \simeq (\mathrm{id} \times \mathrm{pr}_{\mathrm{small}, \mathbb{Z}})_! \circ \mathrm{Poinc}_{M, *, \mathbb{Z} \subseteq} \circ J_{\mathrm{Whit}, \Theta}^{-, !} \circ \mathrm{ins. unit}_{\mathbb{Z}}[-2\delta_{(N_P)_{\rho_P(\omega_X)}} - \delta_{(N_P^-)_{\rho_P(\omega_X)}}],
 \end{aligned}$$

establishing Theorem 10.1.2.

□[Theorem 10.1.2]

Remark 10.6.10. Note that one can regard Proposition 10.6.8 as giving an expression for $J_{\mathrm{Whit}}^{-, !}$ in “finite-dimensional” terms, i.e., in terms of sheaves with finite-dimensional support. Namely, it says that in terms of the equivalences

$$\mathrm{Whit}^!(G)_{\mathbb{Z}} \xrightarrow{\Theta_{\mathrm{Whit}(G)}} \mathrm{Whit}_*(G)_{\mathbb{Z}} \xrightarrow{((\pi_{\mathbb{Z}})_*)^{\mathrm{Av}, \chi}} \mathrm{Whit}^{\mathrm{glob}}(G)_{\mathrm{co}, \mathbb{Z}}$$

and

$$\mathrm{Whit}^!(M)_{\mathbb{Z} \subseteq} \xrightarrow{\Theta_{\mathrm{Whit}(M)}} \mathrm{Whit}_*(M)_{\mathbb{Z} \subseteq} \xrightarrow{((\pi_{M, \mathbb{Z} \subseteq})_*)^{\mathrm{Av}, \chi}} \mathrm{Whit}^{\mathrm{glob}}(M)_{\mathrm{co}, \mathbb{Z} \subseteq},$$

the functor

$$\mathrm{Whit}^!(G)_Z \xrightarrow{\mathrm{ins.unit}_Z} \mathrm{Whit}^!(G)_{Z \subseteq} \xrightarrow{J_{\mathrm{Whit}}^{-,!}} \mathrm{Whit}^!(M)_{Z \subseteq}$$

corresponds to

$$(\mathrm{transl}_{\rho_P(\omega_X)})^* \circ (\mathfrak{s}_{Z \subseteq})^* \circ \mathbf{f}_{M,Z}^! \left(\left((' \widetilde{\mathbf{p}}_Z^-)^!(-) \right) \overset{!}{\otimes} (' \widetilde{\mathbf{p}}_Z)^! \circ (j_Z)! (\omega_{\mathrm{Bun}_{P-} \times Z}) \right).$$

10.7. Proof of Proposition 10.6.8.

10.7.1. First, we notice that translation by $\rho_P(\omega_X)$ allows us to think about $\mathrm{Whit}^!(M)_Z$ as a full subcategory in $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_{M,\rho(\omega_X)})$ (rather than $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_{M,\rho_M(\omega_X)})$).

A similar remark applies also to $\mathrm{Whit}_*(M)_Z$, $\mathrm{Whit}^{\mathrm{glob}}(M)_Z$ and $\mathrm{Whit}^{\mathrm{glob}}(M)_{co,Z}$.

Thus, our goal is to establish an isomorphism

$$(10.23) \quad (\mathfrak{s}_{Z \subseteq})^* \circ \mathbf{f}_{M,Z}^! \left(\left((' \widetilde{\mathbf{p}}_Z^-)^! \circ ((\pi_Z)_*)^{\mathrm{Av},\chi}(-) \right) \overset{!}{\otimes} (' \widetilde{\mathbf{p}}_Z)^! \circ (j_Z)! (\omega_{\mathrm{Bun}_{P-} \times Z}) \right) \simeq \\ \simeq ((\pi_{M,Z \subseteq})^*)^{\mathrm{Av},\chi} \circ J_{\mathrm{Whit},\Theta,\rho_P}^{-,!} \circ \mathrm{ins.unit}_Z(-) [\mathrm{shift} - 2\delta_{(N_P)_{\rho_P(\omega_X)}} - \delta_{(N_P^-)_{\rho_P(\omega_X)}}],$$

as functors

$$\mathrm{Whit}_*(G)_Z \rightrightarrows \mathrm{D-mod}_{\frac{1}{2}}(\overline{\mathrm{Bun}}_{N(M),\rho(\omega_X),Z \subseteq})_{co},$$

where:

- $\pi_{M,Z \subseteq}$ denotes now the map $\mathrm{Gr}_{M,\rho(\omega_X),Z \subseteq} \rightarrow \overline{\mathrm{Bun}}_{N(M),\rho(\omega_X),Z \subseteq}$;
- $J_{\mathrm{Whit},\Theta,\rho_P}^{-,!} := (\mathrm{transl}_{\rho_P(\omega_X)})^* \circ J_{\mathrm{Whit},\Theta}^{-,!}$;

10.7.2. First, it is straightforward to check⁴⁶ that the left-hand side in (10.23) belongs to

$$\mathrm{Whit}^{\mathrm{glob}}(M)_{co,Z \subseteq} \subset \mathrm{D-mod}_{\frac{1}{2}}(\overline{\mathrm{Bun}}_{N(M),\rho(\omega_X),Z \subseteq})_{co}.$$

Hence, by Theorem 10.2.8(b,c), the isomorphism (10.23) is equivalent to

$$(10.24) \quad (\pi_{M,Z \subseteq})^! \circ \Theta_{\mathrm{Whit}(M)}^{\mathrm{glob}} \circ (\mathfrak{s}_{Z \subseteq})^* \circ \mathbf{f}_{M,Z}^! \left(\left((' \widetilde{\mathbf{p}}_Z^-)^! \circ ((\pi_Z)_*)^{\mathrm{Av},\chi}(-) \right) \overset{!}{\otimes} (' \widetilde{\mathbf{p}}_Z)^! \circ (j_Z)! (\omega_{\mathrm{Bun}_{P-} \times Z}) \right) \simeq \\ \simeq J_{\mathrm{Whit},\rho_P}^{-,!} \circ \mathrm{ins.unit}_Z \circ \Theta_{\mathrm{Whit}(G)}(-) [\mathrm{shift} - 2\delta_{N_{\rho(\omega_X)}} - \delta_{(N_P^-)_{\rho_P(\omega_X)}}]$$

as functors

$$\mathrm{Whit}_*(G)_Z \rightrightarrows \mathrm{Whit}_*(M)_{Z \subseteq} \subset \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Gr}_{M,\rho(\omega_X),Z \subseteq}).$$

Note that the left-hand side in (10.24) identifies with

$$(10.25) \quad (\pi_{M,Z \subseteq})^! \circ (\mathfrak{s}_{Z \subseteq})^* \circ \mathbf{f}_{M,Z}^! \left(\left((' \widetilde{\mathbf{p}}_Z^-)^! \circ \Theta_{\mathrm{Whit}(G)}^{\mathrm{glob}} \circ ((\pi_Z)_*)^{\mathrm{Av},\chi}(-) \right) \overset{!}{\otimes} (' \widetilde{\mathbf{p}}_Z)^! \circ (j_Z)! (\omega_{\mathrm{Bun}_{P-} \times Z}) \right).$$

⁴⁶One inspects the actions of the corresponding groupoids, cf. Sect. 1.8.10.

10.7.3. Note that by Lemma 10.5.8 we have a diagram in which the square is Cartesian:

$$\begin{array}{ccccc}
 & & \mathrm{Gr}_{G,\rho(\omega_X),Z\subseteq}^{\mathrm{pos}} \times \mathrm{Gr}_{G,\rho(\omega_X),Z\subseteq} & \times & (\mathrm{Gr}_{G,\rho(\omega_X)} \times \mathrm{Gr}_{M,\rho(\omega_X)})_{Z\subseteq}^{\mathrm{neg}} \\
 & & \downarrow \scriptstyle ' \pi_{M,Z\subseteq} & & \downarrow \scriptstyle ' s_{Z\subseteq} \\
 & & \mathrm{Zas}_Z^{\mathrm{Ran}} & & \mathrm{Gr}_{M,\rho(\omega_X),Z\subseteq} \\
 & \swarrow \scriptstyle f_{M,Z} & \searrow \scriptstyle s_{Z\subseteq} & & \downarrow \scriptstyle \pi_{M,Z\subseteq} \\
 (10.26) \quad \mathrm{Gr}_{G,\rho(\omega_X),Z} & & \mathrm{Zas}_Z & & \overline{\mathrm{Bun}}_{N(M),\rho(\omega_X),Z\subseteq} \\
 & \searrow \scriptstyle \pi_Z & \swarrow \scriptstyle ' \bar{p}_Z^- & & \downarrow \scriptstyle ' \bar{p}_Z \\
 & \overline{\mathrm{Bun}}_{N,\rho(\omega_X),Z} & & & \mathrm{Bun}_{P^-} \times^{\mathcal{Z}} \mathrm{Z} \xrightarrow{j} \widetilde{\mathrm{Bun}}_{P^-,Z}
 \end{array}$$

Hence, by base change, the expression in (10.25) identifies with $('s_{Z\subseteq})_*$ applied to

$$(10.27) \quad (' \bar{p}_Z^- \circ f_{M,Z} \circ ' \pi_{M,Z\subseteq})^! \left(\Theta_{\mathrm{Whit}(G)}^{\mathrm{glob}} \circ ((\pi_Z)_*)^{\mathrm{Av},X}(-) \right)^! \otimes (' \bar{p}_Z^- \circ f_{M,Z} \circ ' \pi_{M,Z\subseteq})^! \left((j_Z)_! (\omega_{\mathrm{Bun}_{P^-} \times Z}) \right).$$

10.7.4. We claim:

Proposition 10.7.5.

(a) *The first factor in (10.27), viewed as an object of*

$$\mathrm{D}\text{-mod}_{\det_G^{\frac{1}{2}}}(\mathrm{Gr}_{G,\rho(\omega_X)} \times \mathrm{Gr}_{M,\rho(\omega_X)})_{Z\subseteq},$$

identifies with

$$p_1^! \circ \mathrm{ins. unit}_Z \circ \pi_Z^! \circ \Theta_{\mathrm{Whit}(G)}^{\mathrm{glob}} \circ ((\pi_Z)_*)^{\mathrm{Av},X}(-),$$

where

$$p_1 : (\mathrm{Gr}_{G,\rho(\omega_X)} \times \mathrm{Gr}_{M,\rho(\omega_X)})_{Z\subseteq} \rightarrow \mathrm{Gr}_{G,\rho(\omega_X)}.$$

(b) *The second factor in (10.27), viewed as an object of*

$$\mathrm{D}\text{-mod}_{\det_{G,M}^{\frac{1}{2}}}(\mathrm{Gr}_{G,\rho(\omega_X)} \times \mathrm{Gr}_{M,\rho(\omega_X)})_{Z\subseteq},$$

identifies with the object⁴⁷ $\widetilde{\Delta}_{Z\subseteq}^{-,\infty}$.

⁴⁷See Sect. 10.4.7 for the notation.

10.7.6. Let us show how Proposition 10.7.5 implies Proposition 10.6.8. Indeed, combining Proposition 10.7.5(a) and Theorem 10.2.8(c), we obtain that the first factor in (10.27) identifies with

$$p_1^! \circ \text{ins.unit}_Z(-)[-2\delta_{N_{\rho_P(\omega_X)}}].$$

Now, the assertion of Proposition 10.6.8 follows from the definition of the functor $J_{\text{Whit}}^{-,!}$; there is a built-in cohomological shift⁴⁸ and the additional shift by $\delta_{(N_P^-)_{\rho_P(\omega_X)}}$ comes from the translation by $\rho_P(\omega_X)$.

□[Proposition 10.7.5]

10.7.7. *Proof of Proposition 10.7.5(a).* By Lemma 10.5.8, the composition

$$\text{Gr}_{G,\rho(\omega_X),Z \subseteq}^{\text{pos}} \times_{\text{Gr}_{G,\rho(\omega_X),Z \subseteq}} (\text{Gr}_{G,\rho(\omega_X)} \times \text{Gr}_{M,\rho(\omega_X)})_{Z \subseteq}^{\text{neg}} \xrightarrow{\pi_{M,Z \subseteq}^{\text{pos}}} \text{Zas}_Z^{\text{Ran}} \xrightarrow{f_{M,Z}} \text{Zas}_Z \xrightarrow{\bar{p}_Z^-} \overline{\text{Bun}}_{N,\rho(\omega_X),Z}$$

is the same as

$$\begin{aligned} & \text{Gr}_{G,\rho(\omega_X),Z \subseteq}^{\text{pos}} \times_{\text{Gr}_{G,\rho(\omega_X),Z \subseteq}} (\text{Gr}_{G,\rho(\omega_X)} \times \text{Gr}_{M,\rho(\omega_X)})_{Z \subseteq}^{\text{neg}} \xrightarrow{p_1^!} \\ & \rightarrow \text{Gr}_{G,\rho(\omega_X),Z \subseteq}^{\text{pos}} \xrightarrow{\pi_{Z \subseteq}^{\text{pos}}} \overline{\text{Bun}}_{N,\rho(\omega_X),Z} \times_{Z \subseteq} \text{Zas}_Z \xrightarrow{\text{id} \times \text{pr}_{\text{small},Z}} \overline{\text{Bun}}_{N,\rho(\omega_X),Z}. \end{aligned}$$

Now, the assertion of Proposition 10.7.5(a) follows from Lemma 10.4.4.

10.7.8. *Proof of Proposition 10.7.5(b).* By Lemma 10.5.8, the composition

$$\text{Gr}_{G,\rho(\omega_X),Z \subseteq}^{\text{pos}} \times_{\text{Gr}_{G,\rho(\omega_X),Z \subseteq}} (\text{Gr}_{G,\rho(\omega_X)} \times \text{Gr}_{M,\rho(\omega_X)})_{Z \subseteq}^{\text{neg}} \xrightarrow{\pi_{M,Z \subseteq}^{\text{pos}}} \text{Zas}_Z^{\text{Ran}} \xrightarrow{f_{M,Z}} \text{Zas}_Z \xrightarrow{\bar{p}_Z^-} \widetilde{\text{Bun}}_{P^-,Z}$$

is the same as

$$\begin{aligned} & \text{Gr}_{G,\rho(\omega_X),Z \subseteq}^{\text{pos}} \times_{\text{Gr}_{G,\rho(\omega_X),Z \subseteq}} (\text{Gr}_{G,\rho(\omega_X)} \times \text{Gr}_{M,\rho(\omega_X)})_{Z \subseteq}^{\text{neg}} \rightarrow \\ & \rightarrow (\text{Gr}_{G,\rho(\omega_X)} \times \text{Gr}_{M,\rho(\omega_X)})_{Z \subseteq}^{\text{neg}} \xrightarrow{\pi_{Z \subseteq}^{\text{neg}}} \widetilde{\text{Bun}}_{P^-,Z} \times_{Z \subseteq} \text{Zas}_Z \xrightarrow{\text{id} \times \text{pr}_{\text{small},Z}} \widetilde{\text{Bun}}_{P^-,Z}. \end{aligned}$$

The assertion follows now from Lemma 10.4.8.

10.8. Whittaker coefficients of Eisenstein series: the enhanced version.

10.8.1. The next result is an enhanced version of Theorem 10.1.2

Theorem 10.8.2. *For $Z \rightarrow \text{Ran}$, the following diagram commutes*

$$(10.28) \quad \begin{array}{ccc} (\text{Whit}_*(G) \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}})_Z & \xrightarrow{\text{ins.unit}_Z} & (\text{Whit}_*(G) \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}})_{Z \subseteq} \\ \downarrow \text{Poinc}_{G,*,Z}^{-,\text{enh}}[d] & & \downarrow J_{\text{Whit},\Theta}^{-,\text{enh}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co},\rho_P(\omega_X),Z}^{-,\text{enh}} & & \text{Whit}_*(M)_{Z \subseteq} \\ \downarrow \text{CT}_{\text{co},?,\rho_P(\omega_X),Z}^{-,\text{enh}} \otimes \text{Id} & & \downarrow \text{Poinc}_{M,*,Z} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}} \otimes \text{D-mod}(Z) & \xleftarrow{\text{Id} \otimes (\text{pr}_{\text{small},Z})^!} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}} \otimes \text{D-mod}(Z \subseteq), \end{array}$$

where:

⁴⁸This shift is designed to be compatible with [shift] in the definition of CT_*^- .

- $\text{Poinc}_{G,*,Z}^{-,\text{enh}}$ is the functor

$$\begin{aligned} & (\text{Whit}_*(G) \otimes_{\text{Sph}_G} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}})_{\mathbb{Z}} \xrightarrow{\text{Poinc}_{G,*,Z} \otimes \text{Id}_{\text{I}(G, P^-)_{\rho_P(\omega_X)}}} \\ & \rightarrow \left(\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}} \otimes \text{D-mod}(\mathbb{Z}) \right) \otimes_{\text{Sph}_{G,Z}} \text{I}(G, P^-)_{\rho_P(\omega_X),Z}^{\text{loc}} =: \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co},\rho_P(\omega_X),Z}^{-,\text{enh}}; \end{aligned}$$

- $J_{\text{Whit},\Theta}^{-,\text{enh}} := \Theta_{\text{Whit}(M)}^{-1} \circ J_{\text{Whit}}^{-,\text{enh}} \circ (\Theta_{\text{Whit}(G)} \otimes \text{Id}_{\text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}}})$;
- $d = 2\delta_{(N_P)_{\rho_P(\omega_X)}} + \delta_{(N_P^-)_{\rho_P(\omega_X)}}$.

Moreover, the isomorphism of functors in (10.28) respects the actions of $\text{Sph}_{M,Z}$.

10.8.3. Note that unital property of the local-to-global functor $\underline{\text{Poinc}}_{G,!}$ implies that the functor

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) \xrightarrow{\text{Id} \otimes (\text{pr}_{\text{small},Z})^!} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}_{\subseteq}) \xrightarrow{\text{coeff}_{G,Z_{\subseteq}}} \text{Whit}^!(G)_{Z_{\subseteq}}$$

takes values in

$$\text{Whit}^!(G)_{Z_{\subseteq}, \text{unl}, \text{indep}} \subset \text{Whit}^!(G)_{Z_{\subseteq}}.$$

We will denote the corresponding functor

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) \rightarrow \text{Whit}^!(G)_{Z_{\subseteq}, \text{unl}, \text{indep}},$$

as well as its composition with the embedding

$$\text{Whit}^!(G)_{Z_{\subseteq}, \text{unl}, \text{indep}} \hookrightarrow \text{Whit}^!(G)_{Z_{\subseteq}, \text{unl}}$$

by $\text{coeff}_{G,Z,\text{unl}}$.

10.8.4. Passing to dual functors as in Corollary 10.1.5, from Theorem 10.8.2 we obtain:

Corollary 10.8.5. *For $Z \rightarrow \text{Ran}$, the following diagram commutes:*

$$\begin{array}{ccc} \text{Whit}^!(G)_{Z_{\subseteq}, \text{unl}, \text{indep}} & \xleftarrow{\text{emb.indep}_{\text{Whit}^!(G)}^L} & \text{Whit}^!(G)_{Z_{\subseteq}, \text{unl}} \\ \text{coeff}_{G,Z,\text{unl}}[d] \uparrow & & \uparrow \text{co}J_{\text{Whit},\Theta}^{-,\text{enh}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) & & (\text{Whit}^!(M) \otimes_{\text{Sph}_M} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}})_{Z_{\subseteq}, \text{unl}} \\ & & \uparrow \\ & & \text{Whit}^!(M)_{Z_{\subseteq}, \text{unl}} \otimes_{\text{Sph}_{M,Z}} \text{I}(G, P^-)_{\rho_P(\omega_X),Z_{\subseteq}, \text{unl}}^{\text{loc}} \\ & & \uparrow \text{coeff}_{M,Z,\text{unl}} \otimes \text{ins.unit}_Z \\ \text{Eis}_{!,\rho_P(\omega_X),Z}^{-,\text{enh}} \uparrow & & \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\rho_P(\omega_X),Z}^{-,\text{enh}} & \xrightarrow{=} & (\text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathbb{Z})) \otimes_{\text{Sph}_{M,Z}} \text{I}(G, P^-)_{\rho_P(\omega_X),Z}^{\text{loc}} \end{array}$$

where:

- $\text{co}J_{\text{Whit},\Theta}^{-,\text{enh}}$ is the functor obtained from $J_{\text{Whit},\Theta}^{-,\text{enh}}$ by duality;
- $d = 2\delta_{(N_P)_{\rho_P(\omega_X)}} + \delta_{(N_P^-)_{\rho_P(\omega_X)}}$.

Moreover, the isomorphism of functors in the above diagram respects the actions of $\text{Sph}_{G,Z}$.

10.9. Proof of Theorem 10.8.2.

10.9.1. We will show that the two functors

$$\text{Whit}_*(G)_Z \otimes \text{I}(G, P^-)_{\rho_P(\omega_X),Z}^{\text{loc}} \rightrightarrows \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\text{co}} \otimes \text{D-mod}(\mathbb{Z})$$

are canonically isomorphic. The compatibility with the actions of $\text{Sph}_{G,Z}$ and $\text{Sph}_{M,Z}$ will follow from the construction.

10.9.2. By Corollary 9.7.10, for $\mathcal{F}_W \in \text{Whit}_*(G)_Z$ and $\mathcal{F}_I \in \text{I}(G, P^-)_{\rho_P(\omega_X), Z}^{\text{loc}}$, the object

$$(10.29) \quad (\text{CT}_{\text{co}, ?, \rho_P(\omega_X), Z}^{-, \text{enh}} \otimes \text{Id}) \circ (\text{Poinc}_{G, *, Z} \otimes \text{Id}_{\text{I}(G, P^-)_{\rho_P(\omega_X)}})(\mathcal{F}_W \otimes \mathcal{F}_I)$$

is

$$(10.30) \quad (\text{transl}_{\rho_P(\omega_X)})^* \circ (\tilde{\mathbf{q}}_Z^-)^* \circ \left(\left((\tilde{\mathbf{p}}_Z^-)^! \circ (\bar{\mathbf{p}}_Z)^* \circ ((\pi_Z)_*)^{\text{Av}, \chi}(\mathcal{F}_W) \right)^! \otimes (\pi_! \circ \tilde{\mathbf{s}}^!)_Z(\mathcal{F}_{I, -\rho_P}) \right) [-\text{shift}],$$

where:

- The functor

$$(\pi_! \circ \tilde{\mathbf{s}}^!)_Z : \text{I}(G, P^-)_{\rho_P(\omega_X), Z}^{\text{loc}} \rightarrow \text{D-mod}_{\det_{\tilde{G}, M}^{\frac{1}{2}}}(\widetilde{\text{Bun}}_{P^-, Z})$$

is as in Proposition 9.6.3(a).

- $\mathcal{F}_{I, -\rho_P}$ denotes the image of $\mathcal{F}_I$ under $\text{I}(G, P^-)_{\rho_P(\omega_X), Z}^{\text{loc}} \xrightarrow{\alpha_{\text{taut}, \rho_P(\omega_X)}^{-1}} \text{I}(G, P^-)_Z^{\text{loc}}$.

10.9.3. An analog of Lemma 10.6.3 is applicable in the current situation, and we rewrite (10.30) as

$$(10.31) \quad (\text{transl}_{\rho_P(\omega_X)})^* \circ (\tilde{\mathbf{q}}_Z^-)^* \circ (\bar{\mathbf{p}}_Z)^* \circ \left(\left((\tilde{\mathbf{p}}_Z^-)^! \circ ((\pi_Z)_*)^{\text{Av}, \chi}(\mathcal{F}_W) \right)^! \otimes (\bar{\mathbf{p}}_Z)^! \circ (\pi_! \circ \tilde{\mathbf{s}}^!)_Z(\mathcal{F}_{I, -\rho_P}) \right) [-\text{shift}].$$

Further, as in the proof of Theorem 10.1.2 given in Sect. 10.6.9, it suffices to construct an isomorphism

$$(10.32) \quad (\text{transl}_{\rho_P(\omega_X)})^* \circ (\mathfrak{s}_{Z \subseteq})^* \circ \mathbf{f}_{M, Z}^! \circ \left(\left((\tilde{\mathbf{p}}_Z^-)^! \circ ((\pi_Z)_*)^{\text{Av}, \chi}(\mathcal{F}_W) \right)^! \otimes (\bar{\mathbf{p}}_Z)^! \circ (\pi_! \circ \tilde{\mathbf{s}}^!)_Z(\mathcal{F}_{I, -\rho_P}) \right) \simeq \\ \simeq ((\pi_{M, Z \subseteq})^*)^{\text{Av}, \chi} \circ \text{ins. unit}_Z \left(J_{\text{Whit}, \Theta}^{\text{enh}}(\mathcal{F}_W \otimes \mathcal{F}_I) \right) [\text{shift} - 2\delta_{(N_P)_{\rho_P(\omega_X)}} - \delta_{(N_P^-)_{\rho_P(\omega_X)}}].$$

10.9.4. Finally, as in the proof of Proposition 10.6.8, in order to prove (10.32), it suffices to establish the following extension of Lemma 10.4.8:

Proposition 10.9.5. *For $\mathcal{F}_I \in \text{I}(G, P^-)_Z^{\text{loc}}$, the object*

$$(\pi_{Z \subseteq}^{\text{neg}})^! \circ (\text{id} \times \text{pr}_{\text{small}, Z})^! \circ (\pi_! \circ \tilde{\mathbf{s}}^!)_Z(\mathcal{F}_I) \in \\ \in \text{D-mod}_{(\pi^{\text{neg}})^*(\det_{\tilde{G}, M}^{\frac{1}{2}})}((\text{Gr}_G \times \text{Gr}_M)_{Z \subseteq}^{\text{neg}}) \subset \text{D-mod}_{\det_{\tilde{G}_G}^{\frac{1}{2}} \otimes \det_{\tilde{G}_M}^{\frac{1}{2}}}(\text{Gr}_G \times \text{Gr}_M)_{Z \subseteq}$$

identifies canonically with the image of

$$\text{ins. unit}_Z(\mathcal{F}_I) \in \text{I}(G, P^-)_{Z \subseteq}^{\text{loc}}$$

under

$$(10.33) \quad \text{I}(G, P^-)^{\text{loc}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)} \simeq \\ \simeq \text{D-mod}_{\det_{\tilde{G}_G}^{\frac{1}{2}} \otimes \det_{\tilde{G}_M}^{\frac{1}{2}}}(\text{Gr}_G \times \text{Gr}_M)^{\mathfrak{L}(P^-)} \rightarrow \text{D-mod}_{\det_{\tilde{G}_G}^{\frac{1}{2}} \otimes \det_{\tilde{G}_M}^{\frac{1}{2}}}(\text{Gr}_G \times \text{Gr}_M).$$

□[Theorem 10.8.2]

Remark 10.9.6. As the proof of Proposition 10.9.5 will show, the isomorphism of functors

$$\text{I}(G, P^-)_Z^{\text{loc}} \rightrightarrows \text{D-mod}_{\det_{\tilde{G}_G}^{\frac{1}{2}} \otimes \det_{\tilde{G}_M}^{\frac{1}{2}}}(\text{Gr}_G \times \text{Gr}_M)_{Z \subseteq}$$

stated in Proposition 10.9.5 is compatible with the actions of $(\text{Sph}_G \otimes \text{Sph}_M)_Z$ on the two sides.

10.10. Proof of Proposition 10.9.5.

10.10.1. For $\mathcal{F}_I \in I(G, P^-)_{\mathbb{Z}}^{\text{loc}}$, let $\text{Av}_!^{\mathfrak{L}(M)/\mathfrak{L}^+(M)}(\mathcal{F}_I)$ denote⁴⁹ its image along the functor (10.33).

We first construct a map in one direction:

$$(10.34) \quad \text{Av}_!^{\mathfrak{L}(M)/\mathfrak{L}^+(M)}(\text{ins. unit}_{\mathbb{Z}}(\mathcal{F}_I)) \rightarrow (\pi_{\mathbb{Z} \subseteq}^{\text{neg}})^! \circ (\text{id} \times \text{pr}_{\text{small}, \mathbb{Z}})^! \left((\pi_! \circ \widetilde{\mathfrak{s}}^!)_{\mathbb{Z}}(\mathcal{F}_I) \right).$$

10.10.2. Let $\widetilde{\pi}_{\mathbb{Z}}$ denote the map

$$(\text{Gr}_G \times \text{Gr}_M)_{\mathbb{Z}} \rightarrow \widetilde{\text{Bun}}_{P^-, \mathbb{Z}}.$$

Note also that we have a naturally defined map

$$\mathbf{g}_{\mathbb{Z}} : (\text{Gr}_G \times \text{Gr}_M)_{\mathbb{Z}} \rightarrow \text{Gr}_{G, \text{Bun}_M, \mathbb{Z}}$$

so that

$$\widetilde{\pi}_{\mathbb{Z}} = \pi_{\mathbb{Z}} \circ \mathbf{g}_{\mathbb{Z}}.$$

By construction

$$\text{Av}_!^{\mathfrak{L}(M)/\mathfrak{L}^+(M)}(\mathcal{F}_I) \simeq \mathbf{g}_{\mathbb{Z}}^! \circ \widetilde{\mathfrak{s}}_{\mathbb{Z}}^!(\mathcal{F}_I).$$

From here we obtain a map

$$(\widetilde{\pi}_{\mathbb{Z}})_! \circ \text{Av}_!^{\mathfrak{L}(M)/\mathfrak{L}^+(M)}(\mathcal{F}_I) \rightarrow (\pi_! \circ \widetilde{\mathfrak{s}}^!)_{\mathbb{Z}}(\mathcal{F}_I).$$

In particular, we obtain a map

$$(10.35) \quad (\pi_{\mathbb{Z} \subseteq}^{\text{neg}})_! \circ \text{Av}_!^{\mathfrak{L}(M)/\mathfrak{L}^+(M)}(\text{ins. unit}_{\mathbb{Z}}(\mathcal{F}_I)) = \\ = (\widetilde{\pi}_{\mathbb{Z} \subseteq})_! \circ \text{Av}_!^{\mathfrak{L}(M)/\mathfrak{L}^+(M)}(\text{ins. unit}_{\mathbb{Z}}(\mathcal{F}_I)) \rightarrow (\pi_! \circ \widetilde{\mathfrak{s}}^!)_{\mathbb{Z} \subseteq}(\text{ins. unit}_{\mathbb{Z}}(\mathcal{F}_I)).$$

10.10.3. Using the fact that the functor $(\pi_! \circ \widetilde{\mathfrak{s}}^!)$ is strictly unital, we obtain that the right-hand side in (10.35) identifies with

$$(\text{id} \times \text{pr}_{\text{small}, \mathbb{Z}})^! \circ (\pi_! \circ \widetilde{\mathfrak{s}}^!)_{\mathbb{Z}}(\mathcal{F}_I).$$

To summarize we obtain a map

$$(10.36) \quad (\pi_{\mathbb{Z} \subseteq}^{\text{neg}})_! \circ \text{Av}_!^{\mathfrak{L}(M)/\mathfrak{L}^+(M)}(\text{ins. unit}_{\mathbb{Z}}(\mathcal{F}_I)) \rightarrow (\text{id} \times \text{pr}_{\text{small}, \mathbb{Z}})^! \circ (\pi_! \circ \widetilde{\mathfrak{s}}^!)_{\mathbb{Z}}(\mathcal{F}_I)$$

By adjunction, from (10.36), we obtain the sought-for map (10.34).

Remark 10.10.4. One can show that the map

$$(\text{id} \times \text{pr}_{\text{small}, \mathbb{Z}})_! \circ (\pi_{\mathbb{Z} \subseteq}^{\text{neg}})_! \circ \text{Av}_!^{\mathfrak{L}(M)/\mathfrak{L}^+(M)}(\text{ins. unit}_{\mathbb{Z}}(\mathcal{F}_I)) \rightarrow (\pi_! \circ \widetilde{\mathfrak{s}}^!)_{\mathbb{Z}}(\mathcal{F}_I),$$

obtained by adjunction from (10.36), is an isomorphism.

10.10.5. We now show that the map (10.34) is an isomorphism.

The categories

$$I(G, P^-)_{\mathbb{Z}}^{\text{loc}} \text{ and } \text{D-mod}_{\det \frac{1}{2}_{\text{Gr}_G} \otimes \det \frac{1}{2}_{\text{Gr}_M}} (\text{Gr}_G \times \text{Gr}_M)_{\mathbb{Z} \subseteq}$$

are acted on by $\text{Sph}_{M, \mathbb{Z}}$, and the two functors in the lemma, as well as the natural transformation between them constructed above, respect these actions.

Hence, in order to show that (10.34) is an isomorphism, it suffices to do so for $\mathcal{F}_I = \Delta_{\mathbb{Z}}^{-, \frac{\infty}{2}}$, since this object generates $I(G, P^-)_{\mathbb{Z}}^{\text{loc}}$ under the $\text{Sph}_{M, \mathbb{Z}}$ -action.

However, in the latter case, the map (10.34) equals the map (10.12), and the assertion follows from Lemma 10.4.8.

□[Proposition 10.9.5]

⁴⁹We use the symbol $\text{Av}_!^{\mathfrak{L}(M)/\mathfrak{L}^+(M)}$ because the functor $\text{D-mod}(\text{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)} \rightarrow \text{D-mod}(\text{Gr}_G \times \text{Gr}_M)^{\mathfrak{L}(P^-)}$ is given embedding along $\text{Gr}_G \xrightarrow{\text{id} \times 1_{\text{Gr}_M}} \text{Gr}_G \times \text{Gr}_M$, followed by $!$ -averaging along $\mathfrak{L}(M)/\mathfrak{L}^+(M)$.

11. CONSTANT TERM OF KAC-MOODY LOCALIZATION

The goal of this subsection is to express the composition of constant term and localization functors in terms of the Jacquet functor for Kac-Moody modules.

Once combined with its spectral counterpart, this will be used in Part III of the paper in order to establish the compatibility of the Langlands functor with constant terms.

11.1. Linear twistings on Bun_G .

11.1.1. Consider the torus $G_{\text{ab}} := G/[G, G]$. Since $\text{Bun}_{G_{\text{ab}}}$ is an (abelian) group-stack, it make sense to talk about (commutative) multiplicative de Rham twistings on it. Such twistings are in bijection with (abelian) central extensions of the Lie algebra of $\text{Bun}_{G_{\text{ab}}}$, i.e., $\Gamma(X, \mathcal{O}_X) \otimes \mathfrak{g}_{\text{ab}}[1]$.

In particular, points in

$$(\Gamma(X, \mathcal{O}_X) \otimes \mathfrak{g}_{\text{ab}})^* \simeq \Gamma(X, \omega_X \otimes \mathfrak{g}_{\text{ab}}^*[1]),$$

i.e., torsors on X with respect to $\mathfrak{g}_{\text{ab}}^* \otimes \omega_X$, give rise to commutative multiplicative de Rham twistings on $\text{Bun}_{G_{\text{ab}}}$, (cf. Sect. 3.1.1).

11.1.2. Given such a torsor $\mathcal{P}$, we consider the de Rham twisting on Bun_G obtained by pullback from the corresponding commutative multiplicative de Rham twisting on $\text{Bun}_{G_{\text{ab}}}$; we will denote it by the same symbol $\mathcal{P}$.

For a point $\underline{x} \in \text{Ran}$, the resulting twisting on $\text{Bun}_G^{\text{level}_{\underline{x}}}$ is compatible with the *inverse* of the multiplicative twisting on $\mathcal{L}(G)_{\underline{x}}$ corresponding to the central extension of $\widehat{\mathfrak{g}}$ from Sect. 3.1.1.

Thus, for $\mathcal{P}$ as above and a level κ , we can consider the corresponding category $\text{D-mod}_{\kappa+\mathcal{P}}(\text{Bun}_G)$ and the localization functor

$$\underline{\text{Loc}}_{G, \kappa+\mathcal{P}} : \underline{\text{KL}}(G)_{\kappa+\mathcal{P}} \rightarrow \text{D-mod}_{\kappa-\mathcal{P}}(\text{Bun}_G) \otimes \underline{\text{D-mod}}(\text{Ran}).$$

11.1.3. Let now $\mathcal{P}_{Z_G^0}$ be a Z_G^0 -torsor. Weil pairing with $\mathcal{P}_{Z_G^0}$ gives rise to a line bundle on $\text{Bun}_{G_{\text{ab}}}$, to be denoted $\mathcal{L}_{\mathcal{P}_{Z_G^0}}$; by a slight abuse of notation, we will denote by the same symbol $\mathcal{L}_{\mathcal{P}_{Z_G^0}}$ its pullback along

$$\text{Bun}_G \rightarrow \text{Bun}_{G_{\text{ab}}}.$$

11.1.4. Note that as in Sect. 3.1.3 we can regard $\text{dlog}(\mathcal{P}_{Z_G^0})$ as a torsor on X with respect to $\mathfrak{g}_{\text{ab}}^* \otimes \omega_X$.

It is easy to see, however, that

$$\text{dlog}(\mathcal{P}_{Z_G^0}) \simeq \text{dlog}(\mathcal{L}_{\mathcal{P}_{Z_G^0}})$$

as de Rham twistings on Bun_G .

11.1.5. Thus, for a level κ we have a naturally defined local-to-global functor

$$\underline{\text{Loc}}_{G, \kappa+\text{dlog}(\mathcal{P}_{Z_G^0})} : \underline{\text{KL}}(G)_{\kappa+\text{dlog}(\mathcal{P}_{Z_G^0})} \rightarrow \text{D-mod}_{\kappa-\text{dlog}(\mathcal{L}_{\mathcal{P}_{Z_G^0}})}(\text{Bun}_G) \otimes \underline{\text{D-mod}}(\text{Ran}).$$

For $\kappa = \text{crit}$ we will denote the resulting functor

$$\begin{aligned} \underline{\text{KL}}(G)_{\text{crit}+\text{dlog}(\mathcal{P}_{Z_G^0})} &\xrightarrow{\underline{\text{Loc}}_{G, \text{crit}+\text{dlog}(\mathcal{P}_{Z_G^0})}} \text{D-mod}_{\text{crit}-\text{dlog}(\mathcal{L}_{\mathcal{P}_{Z_G^0}})}(\text{Bun}_G) \otimes \underline{\text{D-mod}}(\text{Ran}) \xrightarrow{(-) \otimes \mathcal{L}_{\mathcal{P}_{Z_G^0}}} \\ &\rightarrow \text{D-mod}_{\text{crit}}(\text{Bun}_G) \otimes \underline{\text{D-mod}}(\text{Ran}) \simeq \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \underline{\text{D-mod}}(\text{Ran}) \end{aligned}$$

by $\underline{\text{Loc}}_{G, \mathcal{P}_{Z_G^0}}$.

11.1.6. When $\mathcal{P}_{Z_G^0} = \check{\lambda}(\omega_X)$ for a coweight $\check{\lambda}$ of Z_G^0 (i.e., a character of G), we will use a shorthand notation

$$\mathrm{D}\text{-mod}_{\kappa+\check{\lambda}}(\mathrm{Bun}_G) := \mathrm{D}\text{-mod}_{\kappa+\mathrm{dlog}(\check{\lambda}(\omega_X))}(\mathrm{Bun}_G) \otimes \underline{\mathrm{D}\text{-mod}}(\mathrm{Ran}),$$

and also

$$\underline{\mathrm{Loc}}_{G,\kappa+\check{\lambda}} : \underline{\mathrm{KL}}(G)_{\kappa+\check{\lambda}} \rightarrow \mathrm{D}\text{-mod}_{\kappa-\check{\lambda}}(\mathrm{Bun}_G) \otimes \underline{\mathrm{D}\text{-mod}}(\mathrm{Ran})$$

and

$$\underline{\mathrm{Loc}}_{G,\check{\lambda}} : \underline{\mathrm{KL}}(G)_{\mathrm{crit}+\check{\lambda}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \underline{\mathrm{D}\text{-mod}}(\mathrm{Ran})$$

for the corresponding functors.

Note that as Sect. 3.1.7, the above notions are well-defined for any element $\lambda \in z_G := \mathrm{Lie}(Z_G^0)$.

11.1.7. Contents of this subsection will mostly be applied when the reductive group in question is M , a Levi subgroup of the original G .

11.2. Integration along $\mathrm{Bun}_{P^-} \rightarrow \mathrm{Bun}_M$ and BRST.

11.2.1. For a level κ for M , let us denote by $\mathbf{q}_*^{-,\mathrm{shift}}$ the functor

$$\begin{aligned} \mathrm{D}\text{-mod}_{\mathrm{crit}_G+\kappa}(\mathrm{Bun}_{P^-}) &= \mathrm{D}\text{-mod}_{\frac{1}{2} \mathrm{dlog}(\det(\mathrm{Bun}_G)|_{\mathrm{Bun}_{P^-}})+\kappa}(\mathrm{Bun}_{P^-}) \simeq \\ &\simeq \mathrm{D}\text{-mod}_{\frac{1}{2} \mathrm{dlog}(\det(\mathrm{Bun}_M)|_{\mathrm{Bun}_{P^-}})+\mathrm{dlog}(\det_{\mathrm{Bun}_{G,M}}^{\otimes \frac{1}{2}})+\kappa}(\mathrm{Bun}_{P^-}) \xrightarrow{\otimes \det_{\mathrm{Bun}_{G,M}}^{\otimes -\frac{1}{2}}} \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2} \mathrm{dlog}(\det(\mathrm{Bun}_M)|_{\mathrm{Bun}_{P^-}})+\kappa}(\mathrm{Bun}_{P^-}) \xrightarrow{(\mathbf{q}^-)^*} \mathrm{D}\text{-mod}_{\frac{1}{2} \mathrm{dlog}(\det(\mathrm{Bun}_M))+\kappa}(\mathrm{Bun}_M) \simeq \\ &\simeq \mathrm{D}\text{-mod}_{\mathrm{crit}_M+\kappa}(\mathrm{Bun}_M) \xrightarrow{[-\mathrm{shift}]} \mathrm{D}\text{-mod}_{\mathrm{crit}_M+\kappa}(\mathrm{Bun}_M), \end{aligned}$$

where $[\mathrm{shift}]$ is as in (8.4).

11.2.2. Let $\mathbf{l}_{N_P^-}$ be the (ungraded) line from [GLC2, Sect. 12.5.7]. I.e.,

$$\mathbf{l}_{N_P^-} = \det(\mathfrak{n}_P^- \otimes \Gamma(X, \mathcal{O}_X)) \simeq \det(\Gamma(X, \mathcal{O}_X))^{\otimes \dim(\mathfrak{n}_P^-)} \otimes \det(\mathfrak{n}_P^-)^{\otimes (1-g)}.$$

Let $\mathcal{L}_{\check{\rho}_P(\omega_X)}$ be the line bundle on Bun_M from Sect. 11.1.3.

We will prove:

Theorem 11.2.3. *For $\mathcal{Z} \rightarrow \mathrm{Ran}$, the following diagram of functors canonically commutes*

$$(11.1) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\mathrm{crit}_G+\kappa}(\mathrm{Bun}_{P^-}) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) & \xrightarrow{\mathbf{q}_*^{-,\mathrm{shift}} \otimes \mathrm{Id}} & \mathrm{D}\text{-mod}_{\mathrm{crit}_M+\kappa}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \\ \mathrm{Loc}_{P^-, \mathrm{crit}_G+\kappa, \mathcal{Z}} \uparrow & & \uparrow \mathbf{l}_{N_P^-} \otimes ((-) \otimes \mathcal{L}_{\check{\rho}_P(\omega_X)}) \otimes (\mathrm{pr}_{\mathrm{small}, \mathcal{Z}})! \\ \mathrm{KL}(P^-)_{\mathrm{crit}_G+\kappa, \mathcal{Z}} & & \mathrm{D}\text{-mod}_{\mathrm{crit}_M+\kappa-\check{\rho}_P}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}^{\subseteq}) \\ \mathrm{ins.vac.}_{\mathcal{Z}} \downarrow & & \uparrow \mathrm{Loc}_{M, \mathrm{crit}_M+\kappa+\check{\rho}_P, \mathcal{Z}^{\subseteq}} \\ \mathrm{KL}(P^-)_{\mathrm{crit}_G+\kappa, \mathcal{Z}^{\subseteq}} & \xrightarrow{\mathrm{BRST}^-} & \mathrm{KL}(M)_{\mathrm{crit}_M+\kappa+\check{\rho}_P, \mathcal{Z}^{\subseteq}}. \end{array}$$

11.2.4. Let CT_*^- be as in Sect. 8.1.8. We claim:

Corollary 11.2.5. *For $\mathcal{Z} \rightarrow \mathrm{Ran}$, the following diagram of functors canonically commutes*

$$(11.2) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\mathrm{crit}_G + \kappa}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) & \xrightarrow{\mathrm{CT}_*^- \otimes \mathrm{Id}} & \mathrm{D}\text{-mod}_{\mathrm{crit}_M + \kappa}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \\ \mathrm{Loc}_{G, \mathrm{crit}_G + \kappa, \mathcal{Z}} \uparrow & & \uparrow \mathfrak{l}_{N_P^-} \otimes ((-) \otimes \mathcal{L}_{\check{\rho}_P(\omega_X)}) \otimes (\mathrm{pr}_{\mathrm{small}, \mathcal{Z}})_! \\ \mathrm{KL}(G)_{\mathrm{crit}_G + \kappa, \mathcal{Z}} & & \mathrm{D}\text{-mod}_{\mathrm{crit}_M + \kappa - \check{\rho}_P}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \\ \mathrm{ins.vac.}_{\mathcal{Z}} \downarrow & & \uparrow \mathrm{Loc}_{M, \mathrm{crit}_M + \kappa + \check{\rho}_P, \mathcal{Z} \subseteq} \\ \mathrm{KL}(G)_{\mathrm{crit}_G + \kappa, \mathcal{Z} \subseteq} & \xrightarrow{J_{\mathrm{KM}}^-, \mathrm{Sph}} & \mathrm{KL}(M)_{\mathrm{crit}_M + \kappa + \check{\rho}_P, \mathcal{Z} \subseteq} \end{array}$$

Proof. Follows from Theorem 11.2.3 in the same way as [GLC2, Theorem 12.8.5] follows from the combination of [GLC2, Corollary 13.3.11 and Theorem 12.5.13]. $\square$

11.2.6. In particular, specializing to the critical level, we obtain:

Corollary 11.2.7. *For $\mathcal{Z} \rightarrow \mathrm{Ran}$, the following diagram of functors canonically commutes*

$$(11.3) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) & \xrightarrow{\mathrm{CT}_*^- \otimes \mathrm{Id}} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z}) \\ \mathrm{Loc}_{G, \mathcal{Z}} \uparrow & & \uparrow \mathfrak{l}_{N_P^-} \otimes (\mathrm{Id} \otimes (\mathrm{pr}_{\mathrm{small}, \mathcal{Z}})_!) \\ \mathrm{KL}(G)_{\mathrm{crit}_G, \mathcal{Z}} & & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathcal{Z} \subseteq) \\ \mathrm{ins.vac.}_{\mathcal{Z}} \downarrow & & \uparrow \mathrm{Loc}_{M, \check{\rho}_P, \mathcal{Z} \subseteq} \\ \mathrm{KL}(G)_{\mathrm{crit}_G, \mathcal{Z} \subseteq} & \xrightarrow{J_{\mathrm{KM}}^-, \mathrm{Sph}} & \mathrm{KL}(M)_{\mathrm{crit}_M + \check{\rho}_P, \mathcal{Z} \subseteq} \end{array}$$

11.3. Identification of line bundles. As a first step towards the proof of Theorem 11.2.3, we will explain the appearance of the line $\mathfrak{l}_{N_P^-}$.

11.3.1. Consider the relative cotangent bundle $T^*(\mathrm{Bun}_{P^-} / \mathrm{Bun}_M)$. It is easy to see that its determinant (viewed as an ungraded line bundle) canonically descends to a line bundle on Bun_M , to be denoted

$$\det(T^*(\mathrm{Bun}_{P^-} / \mathrm{Bun}_M)).$$

11.3.2. Recall now (see [GLC1, Proposition 1.3.3]) that the line bundle $\det_{\mathrm{Bun}_{G,M}}^{\otimes \frac{1}{2}}$ on Bun_M attaches to $\mathcal{P}_M \in \mathrm{Bun}_M$ the line

$$\det(\Gamma(X, (\mathfrak{n}_P^-)_{\mathcal{P}_M} \otimes \omega_X^{\otimes \frac{1}{2}})) \otimes \det(\Gamma(X, \mathfrak{n}_P^- \otimes \omega_X^{\otimes \frac{1}{2}}))^{\otimes -1}.$$

Note that this is the same

$$\det(\Gamma(X, (\mathfrak{n}_P^-)_{\mathcal{P}_{P^-}} \otimes \omega_X^{\otimes \frac{1}{2}})) \otimes \det(\Gamma(X, \mathfrak{n}_P^- \otimes \omega_X^{\otimes \frac{1}{2}}))^{\otimes -1},$$

where $\mathcal{P}_{P^-}$ is any P^- -bundle that projects to $\mathcal{P}_M$.

11.3.3. We claim:

Proposition 11.3.4. *There is a canonical isomorphism of line bundles on Bun_M :*

$$\det(T^*(\mathrm{Bun}_{P^-} / \mathrm{Bun}_M)) \simeq \det_{\mathrm{Bun}_{G,M}}^{\otimes \frac{1}{2}} \otimes \mathcal{L}_{\check{\rho}_P(\omega_X)} \otimes \mathfrak{l}_{N_P^-},$$

where

$$\check{\rho}_P(\omega_X) := 2\check{\rho}_P(\omega_X^{\otimes \frac{1}{2}}).$$

Proof. The tangent space to Bun_M at $\mathcal{P}_M$ is

$$\Gamma(X, (\mathfrak{n}_P^-)_{\mathcal{P}_M})[1].$$

Since

$$\det((V[1])^*) \simeq \det(V),$$

the line bundle $\det(T^*(\text{Bun}_{P-} / \text{Bun}_M))$ attaches to $\mathcal{P}_M \in \text{Bun}_M$ the line

$$\det(\Gamma(X, (\mathfrak{n}_P^-)_{\mathcal{P}_M})).$$

Thus, we need to establish an isomorphism

$$(11.4) \quad \det(\Gamma(X, (\mathfrak{n}_P^-)_{\mathcal{P}_M} \otimes \omega_X^{\otimes \frac{1}{2}})) \otimes \det(\Gamma(X, (\mathfrak{n}_P^-)_{\mathcal{P}_M}))^{\otimes -1} \simeq \\ \simeq \det(\Gamma(X, \mathfrak{n}_P^- \otimes \omega_X^{\otimes \frac{1}{2}})) \otimes \det(\Gamma(X, \mathfrak{n}_P^- \otimes \mathcal{O}_X))^{\otimes -1} \otimes \text{Weil}(-2\check{\rho}_P(\mathcal{P}_M), \omega^{\otimes \frac{1}{2}}).$$

Recall that for a vector bundle $\mathcal{E}$ of rank n and a line bundle $\mathcal{L}$ on X we have

$$(11.5) \quad \det(\Gamma(X, \mathcal{E} \otimes \mathcal{L})) \simeq \det(\Gamma(X, \mathcal{E})) \otimes \det(\Gamma(X, \mathcal{L}))^{\otimes n} \otimes \det(\Gamma(X, \mathcal{O}_X))^{\otimes -n} \otimes \text{Weil}(\det(\mathcal{E}), \mathcal{L}).$$

Hence, (11.4) is equivalent to

$$\text{Weil}(\det((\mathfrak{n}_P^-)_{\mathcal{P}_M}), \omega^{\otimes \frac{1}{2}}) \simeq \text{Weil}(\det(\mathfrak{n}_P^- \otimes \mathcal{O}_X), \omega^{\otimes \frac{1}{2}}) \otimes \text{Weil}(-2\check{\rho}_P(\mathcal{P}_M), \omega^{\otimes \frac{1}{2}}),$$

where $\text{Weil}(\det(\mathfrak{n}_P^- \otimes \mathcal{O}_X), \omega^{\otimes \frac{1}{2}})$ is a constant line.⁵⁰

□

11.3.5. Thanks to Proposition 11.3.4, the operation of tensoring by $\det(T^*(\text{Bun}_{P-} / \text{Bun}_M))$ gives rise to a functor

$$\text{D-mod}_{\text{crit}_M + \kappa}(\text{Bun}_{P-}) \rightarrow \text{D-mod}_{\text{crit}_G + \kappa + \check{\rho}_P}(\text{Bun}_{P-}).$$

Hence, the operation of tensoring by $\det(T^*(\text{Bun}_{P-}))$ defines a functor

$$\text{D-mod}_{\text{crit}_M + \kappa}(\text{Bun}_{P-}) \rightarrow \text{D-mod}_{\text{crit}_G + 2\text{crit}_M + \kappa + \check{\rho}_P}(\text{Bun}_{P-}).$$

11.3.6. We record the following particular case of [GLC2, Proposition 10.5.7]:

Proposition 11.3.7. *With respect to the identification*

$$(\text{KL}(P^-)_{\text{crit}_M - \kappa - \check{\rho}_P})^\vee \simeq \text{KL}(P^-)_{\text{crit}_G + \kappa}$$

induced by (3.2), and

$$(\text{D-mod}_{\text{crit}_M - \kappa + \check{\rho}_P}(\text{Bun}_{P-})_{\text{co}})^\vee \simeq \text{D-mod}_{-\text{crit}_M + \kappa - \check{\rho}_P}(\text{Bun}_{P-}),$$

the dual of the functor⁵¹

$$\Gamma_{P-, \text{crit}_M - \kappa - \check{\rho}_P, \mathcal{Z}} : \text{D-mod}_{\text{crit}_M - \kappa + \check{\rho}_P}(\text{Bun}_{P-})_{\text{co}} \otimes \text{D-mod}(\mathcal{Z}) \rightarrow \text{KL}(P^-)_{\text{crit}_M - \kappa - \check{\rho}_P, \mathcal{Z}}$$

identifies with the functor

$$\text{KL}(P^-)_{\text{crit}_G + \kappa, \mathcal{Z}} \xrightarrow{\text{Loc}_{P-, \text{crit}_G + \kappa, \mathcal{Z}}} \text{D-mod}_{\text{crit}_G + \kappa}(\text{Bun}_{P-}) \otimes \text{D-mod}(\mathcal{Z}) \xrightarrow{((-) \otimes \det(T^*(\text{Bun}_{P-}))^{\otimes -1}) \otimes \text{Id}} \\ \rightarrow \text{D-mod}_{-\text{crit}_M + \kappa - \check{\rho}_P}(\text{Bun}_{P-}) \otimes \text{D-mod}(\mathcal{Z}) \xrightarrow{[-\dim(\text{Bun}_{P-})]} \text{D-mod}_{-\text{crit}_M + \kappa - \check{\rho}_P}(\text{Bun}_{P-}) \otimes \text{D-mod}(\mathcal{Z}),$$

where $\dim(\text{Bun}_{P-})$ refers to the dimension of a given connected component of Bun_{P-} .

Remark 11.3.8. In the statement of Proposition 11.3.7, it is not claimed that the category

$$\text{D-mod}_{\text{crit}_M - \kappa + \check{\rho}_P}(\text{Bun}_{P-})_{\text{co}}$$

is dualizable, so the symbol $(\text{D-mod}_{\text{crit}_M - \kappa + \check{\rho}_P}(\text{Bun}_{P-})_{\text{co}})^\vee$ is understood as

$$\text{Funct}_{\text{cont}}(\text{D-mod}_{\text{crit}_M - \kappa + \check{\rho}_P}(\text{Bun}_{P-})_{\text{co}}, \text{Vect}),$$

and similarly for the notion of the dual functor.

⁵⁰Note that for a line $\mathfrak{l}$ and a line bundle $\mathcal{L}$, we have $\text{Weil}(\mathfrak{l} \otimes \mathcal{O}_X, \mathcal{L}) \simeq \mathfrak{l}^{\deg(\mathcal{L})}$.

⁵¹See Sect. 11.1.2 for the sign flip.

11.4. Proof of Theorem 11.2.3.

11.4.1. First, as in [GLC2, Sect. 12.5.10], we observe that there exists a natural transformation

$$(11.6) \quad \text{Res}_{\mathfrak{p}^-}^{\widehat{\mathfrak{m}}} \circ \Gamma_{M, \text{crit}_M - \kappa - \check{\rho}_P, \mathbb{Z}} \rightarrow \Gamma_{P^-, \text{crit}_M - \kappa - \check{\rho}_P, \mathbb{Z}} \circ ((\mathfrak{q}^-)^! \otimes \text{Id})$$

as functors

$$\text{D-mod}_{\text{crit}_M - \kappa + \check{\rho}_P}(\text{Bun}_M)_{\text{co}} \otimes \text{D-mod}(\mathbb{Z}) \rightrightarrows \text{KL}(P^-)_{\text{crit}_M - \kappa - \check{\rho}_P, \mathbb{Z}}.$$

11.4.2. Passing to dual functors and using Proposition 11.3.7 and [GLC2, Proposition 10.5.7], from (11.6) we obtain a natural transformation

$$(11.7) \quad \begin{array}{ccc} \text{D-mod}_{\text{crit}_G + \kappa}(\text{Bun}_{P^-}) \otimes \text{D-mod}(\mathbb{Z}) & \xrightarrow{\mathfrak{q}_*^{-, \text{shift}} \otimes \text{Id}} & \text{D-mod}_{\text{crit}_M + \kappa}(\text{Bun}_M) \otimes \text{D-mod}(\mathbb{Z}) \\ \uparrow \text{Loc}_{P^-, \text{crit}_G + \kappa, \mathbb{Z}} & \swarrow \text{BRST}^- & \uparrow \mathfrak{l}_{N_P^-} \otimes ((-) \otimes \mathcal{L}_{\check{\rho}_P}(\omega_X)) \circ \text{Loc}_{M, \text{crit}_M + \kappa + \check{\rho}_P, \mathbb{Z}} \\ \text{KL}(P^-)_{\text{crit}_G + \kappa, \mathbb{Z}} & \xrightarrow{\text{BRST}^-} & \text{KL}(M)_{\text{crit}_M + \kappa + \check{\rho}_P, \mathbb{Z}} \end{array}$$

Thus, we obtain a natural transformation

$$(11.8) \quad ((-) \otimes \mathcal{L}_{\check{\rho}_P}(\omega_X) \otimes \mathfrak{l}_{N_P^-}) \circ \underline{\text{Loc}}_{M, \text{crit}_M + \kappa + \check{\rho}_P} \circ \underline{\text{BRST}}^- \rightarrow \mathfrak{p}_*^{-, \text{shift}} \circ \underline{\text{Loc}}_{P^-, \text{crit}_G + \kappa}$$

as local-to-global functors

$$\underline{\text{KL}}(P^-)_{\text{crit}_G + \kappa} \rightrightarrows \text{D-mod}_{\text{crit}_M + \kappa}(\text{Bun}_M) \otimes \underline{\text{D-mod}}(\text{Ran}).$$

11.4.3. Both sides in (11.8) have natural lax unital structures and the map (11.8) is compatible with these structures. Furthermore, the lax unital structure on the right-hand side is strict.

Hence, by [GLC2, Sect. 11.4.8], the natural transformation (11.8) extends to a natural transformation

$$(11.9) \quad ((-) \otimes \mathcal{L}_{\check{\rho}_P}(\omega_X) \otimes \mathfrak{l}_{N_P^-}) \circ \underline{\text{Loc}}_{M, \text{crit}_M + \kappa + \check{\rho}_P} \circ (\underline{\text{BRST}}^-)^{\int \text{ins. vac}} \rightarrow \mathfrak{p}_*^{-, \text{shift}} \circ \underline{\text{Loc}}_{P^-, \text{crit}_G + \kappa},$$

see [GLC2, Sect. 11.4] for the notation $\int \text{ins. vac}$.

In other words, we obtain a natural transformation from the counterclockwise circuit to the clockwise circuit in (11.1).

11.4.4. We now claim that (11.9) is an isomorphism. Indeed, this reduces to the statement of [GLC2, Theorem 12.5.13] by the same mechanism as [GLC2, Theorem 12.8.15].

□[Theorem 11.2.3]

11.5. Constant term of localization: the enhanced version. In this section we specialize to the critical level.

11.5.1. Let $\text{Loc}_{G, \mathbb{Z}}^{-, \text{enhco}}$ denote the functor

$$\begin{aligned} \text{KL}(G)_{\text{crit}, \mathbb{Z}}^{-, \text{enhco}} &:= \text{KL}(G)_{\text{crit}, \mathbb{Z}} \otimes_{\text{Sph}_{G, \mathbb{Z}}} \text{I}(G, P^-)_{\text{co}, \mathbb{Z}}^{\text{loc}} \xrightarrow{\text{Loc}_{G, \mathbb{Z}} \otimes \text{Id}} \\ &\rightarrow \left(\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) \right) \otimes_{\text{Sph}_{G, \mathbb{Z}}} \text{I}(G, P^-)_{\text{co}, \mathbb{Z}}^{\text{loc}} =: \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\mathbb{Z}}^{-, \text{enhco}}. \end{aligned}$$

11.5.2. We will prove the following enhanced version of Corollary 11.2.7:

Theorem 11.5.3. *For $\mathcal{Z} \rightarrow \text{Ran}$, the following diagram of functors canonically commutes, in a way compatible with the $\text{Sph}_{M,\mathcal{Z}}$ -actions:*

$$\begin{array}{ccc}
 \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\mathcal{Z}}^{-,\text{enhco}} & \xrightarrow{\text{CT}_{*,\mathcal{Z}}^{-,\text{enhco}}} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathcal{Z}) \\
 \text{Loc}_{G,\mathcal{Z}}^{-,\text{enhco}} \uparrow & & \uparrow \mathfrak{l}_{N_P}^- \otimes (\text{Id} \otimes (\text{pr}_{\text{small},\mathcal{Z}})!) \\
 \text{KL}(G)_{\text{crit}_G,\mathcal{Z}}^{-,\text{enhco}} & & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathcal{Z}^{\subseteq}) \\
 \text{ins.vac.}_{\mathcal{Z}} \downarrow & & \uparrow \text{Loc}_{M,\tilde{\rho}_P,\mathcal{Z}^{\subseteq}} \\
 \text{KL}(G)_{\text{crit}_G,\mathcal{Z}^{\subseteq}}^{-,\text{enhco}} & \xrightarrow{J_{\text{KM}}^{-,\text{enhco}}} & \text{KL}(M)_{\text{crit}_M+\tilde{\rho}_P,\mathcal{Z}^{\subseteq}},
 \end{array}$$

where $J_{\text{KM}}^{-,\text{enhco}}$ is the functor from Sect. 3.3.1.

11.5.4. *Strategy of proof.* We will construct a natural transformation

$$(11.10) \quad \begin{array}{ccc}
 \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\mathcal{Z}}^{-,\text{enhco}} & \xrightarrow{\text{CT}_{*,\mathcal{Z}}^{-,\text{enhco}}} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathcal{Z}) \\
 \uparrow \text{Loc}_{G,\mathcal{Z}}^{-,\text{enhco}} & \swarrow & \uparrow \mathfrak{l}_{N_P}^- \otimes \text{Loc}_{M,\tilde{\rho}_P,\mathcal{Z}} \\
 \text{KL}(G)_{\text{crit}_G,\mathcal{Z}}^{-,\text{enhco}} & \xrightarrow{J_{\text{KM}}^{-,\text{enhco}}} & \text{KL}(M)_{\text{crit}_M+\tilde{\rho}_P,\mathcal{Z}},
 \end{array}$$

compatible with the $\text{Sph}_{M,\mathcal{Z}}$ -actions, such that its concatenation with the commutative diagram

$$\begin{array}{ccc}
 \text{KL}(G)_{\text{crit}_G,\mathcal{Z}}^{-,\text{enhco}} & \xrightarrow{J_{\text{KM}}^{-,\text{enhco}}} & \text{KL}(M)_{\text{crit}_M+\tilde{\rho}_P,\mathcal{Z}} \\
 \text{Id} \otimes \mathbf{1}_{\text{I}(G,P^-)^{\text{loc}}_{\text{co},\mathcal{Z}}} \uparrow & & \uparrow \text{Id} \\
 \text{KL}(G)_{\text{crit}_G,\mathcal{Z}} & \xrightarrow{J_{\text{KM}}^{-,\text{Sph}}} & \text{KL}(M)_{\text{crit}_M+\tilde{\rho}_P,\mathcal{Z}},
 \end{array}$$

which is a natural transformation

(11.11)

$$\begin{array}{ccc}
 \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\mathcal{Z}}^{-,\text{enhco}} & \xrightarrow{\text{CT}_{*,\mathcal{Z}}^{-,\text{enhco}}} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathcal{Z}) \\
 \uparrow \text{Loc}_{G,\mathcal{Z}}^{-,\text{enhco}} \circ (\text{id} \otimes \mathbf{1}_{\text{I}(G,P^-)^{\text{loc}}_{\text{co},\mathcal{Z}}}) & \swarrow & \uparrow \mathfrak{l}_{N_P}^- \otimes \text{Loc}_{M,\tilde{\rho}_P,\mathcal{Z}} \\
 \text{KL}(G)_{\text{crit}_G,\mathcal{Z}} & \xrightarrow{J_{\text{KM}}^{-,\text{Sph}}} & \text{KL}(M)_{\text{crit}_M+\tilde{\rho}_P,\mathcal{Z}},
 \end{array}$$

is obtained by concatenating the commutative diagram

$$(11.12) \quad \begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathbb{Z}}^{-, \mathrm{enh}_{\mathrm{co}}} & \xrightarrow{\mathrm{CT}_{*, \mathbb{Z}}^{-, \mathrm{enh}_{\mathrm{co}}}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}(\mathbb{Z}) \\ \mathrm{Id} \otimes \mathbf{1}_{\mathrm{I}(G, P^-)_{\mathrm{co}, \mathbb{Z}}^{\mathrm{loc}}} \uparrow & & \uparrow \mathrm{Id} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathbb{Z}) & \xrightarrow{\mathrm{CT}_{*, \mathbb{Z}}^{-}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}(\mathbb{Z}) \end{array}$$

with the natural transformation

$$(11.13) \quad \begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathbb{Z}) & \xrightarrow{\mathrm{CT}_*^{-} \otimes \mathrm{Id}_{\mathrm{D-mod}(\mathbb{Z})}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}(\mathbb{Z}) \\ \uparrow \mathrm{Loc}_{G, \mathbb{Z}} & \swarrow & \uparrow \mathbf{l}_{N_P^-} \otimes \mathrm{Loc}_{M, \tilde{\rho}_P, \mathbb{Z}} \\ \mathrm{KL}(G)_{\mathrm{crit}_G, \mathbb{Z}} & \xrightarrow{J_{\mathrm{KM}}^{-, \mathrm{Sph}}} & \mathrm{KL}(M)_{\mathrm{crit}_M + \tilde{\rho}_P, \mathbb{Z}} \end{array}$$

obtained from (11.3).

11.5.5. Let us show how the existence of a diagram (11.10) with the above properties implies the assertion of Theorem 11.5.3.

Namely, let us view the family of diagrams (11.10) as a natural transformation

$$(11.14) \quad \mathbf{l}_{N_P^-} \otimes \underline{\mathrm{Loc}}_{M, \tilde{\rho}_P} \circ J_{\mathrm{KM}}^{-, \mathrm{enh}_{\mathrm{co}}} \rightarrow \underline{\mathrm{CT}}_*^{-, \mathrm{enh}_{\mathrm{co}}} \circ \underline{\mathrm{Loc}}_G^{-, \mathrm{enh}_{\mathrm{co}}}$$

as lax-unital local-to-global functors

$$(11.15) \quad \underline{\mathrm{KL}}(G)_{\mathrm{crit}_G}^{-, \mathrm{enh}_{\mathrm{co}}} \rightrightarrows \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}).$$

The right-hand side in (11.15) is strictly unital, and hence (11.15) induces a natural transformation

$$(11.16) \quad \mathbf{l}_{N_P^-} \otimes \underline{\mathrm{Loc}}_{M, \tilde{\rho}_P} \circ (J_{\mathrm{KM}}^{-, \mathrm{enh}_{\mathrm{co}}})^f \mathrm{ins.vac} \rightarrow \underline{\mathrm{CT}}_*^{-, \mathrm{enh}_{\mathrm{co}}} \circ \underline{\mathrm{Loc}}_G^{-, \mathrm{enh}_{\mathrm{co}}}.$$

We claim that (11.16) is an isomorphism.

11.5.6. Indeed, since $\mathbf{1}_{\mathrm{I}(G, P^-)_{\mathrm{co}}^{\mathrm{loc}}}$ generates $\mathrm{I}(G, P^-)_{\mathrm{co}}^{\mathrm{loc}}$ as a Sph_M -module category and since the natural transformation (11.16) is compatible with the Sph_M -actions, in order to show that (11.16) is an isomorphism, it suffices to show that the induced natural transformation

$$(11.17) \quad \mathbf{l}_{N_P^-} \otimes \underline{\mathrm{Loc}}_{M, \tilde{\rho}_P} \circ \left(J_{\mathrm{KM}}^{-, \mathrm{enh}_{\mathrm{co}}} \circ (\mathrm{Id} \otimes \mathbf{1}_{\mathrm{I}(G, P^-)_{\mathrm{co}}^{\mathrm{loc}}}) \right)^f \mathrm{ins.vac} \rightarrow \underline{\mathrm{CT}}_*^{-, \mathrm{enh}_{\mathrm{co}}} \circ \underline{\mathrm{Loc}}_G^{-, \mathrm{enh}_{\mathrm{co}}} \circ (\mathrm{Id} \otimes \mathbf{1}_{\mathrm{I}(G, P^-)_{\mathrm{co}}^{\mathrm{loc}}})$$

is an isomorphism as local-to-global functors

$$\underline{\mathrm{KL}}(G)_{\mathrm{crit}_G} \rightrightarrows \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran})$$

is an isomorphism.

However, by Sect. 11.5.4, the natural transformation is obtained by concatenating (11.12) with the natural transformation

$$(11.18) \quad \mathbf{l}_{N_P^-} \otimes \underline{\mathrm{Loc}}_{M, \tilde{\rho}_P} \circ (J_{\mathrm{KM}}^{-, \mathrm{Sph}})^f \mathrm{ins.vac} \rightarrow (\mathrm{CT}_*^{-} \otimes \mathrm{Id}_{\underline{\mathrm{D-mod}}(\mathrm{Ran})}) \circ \underline{\mathrm{Loc}}_G.$$

Hence, it is enough to show that (11.18) is an isomorphism. However, the latter follows by construction from the commutativity of (11.3).

□[Theorem 11.5.3]

11.6. Construction of the natural transformation for the enhanced functor. To simplify the notation, we will assume that $\mathcal{Z} = \text{pt}$ so that $\mathcal{Z} \rightarrow \text{Ran}$ corresponds to a point $\underline{x} \in \text{Ran}$.

Thus, our goal will be to construct a natural transformation in

$$(11.19) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes_{\text{Sph}_{G,\underline{x}}} \text{I}(G, P^-)_{\text{co},\underline{x}}^{\text{loc}} & \xrightarrow{\text{CT}_{*,\underline{x}}^{\text{enhco}}} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \\ \uparrow \text{Loc}_{G,\underline{x}} \otimes \text{Id} & \swarrow & \uparrow \text{I}_{N_P^-} \otimes \text{Loc}_{M,\tilde{\rho}_P,\underline{x}} \\ \text{KL}(G)_{\text{crit},\underline{x}} \otimes_{\text{Sph}_{G,\underline{x}}} \text{I}(G, P^-)_{\text{co},\underline{x}}^{\text{loc}} & \xrightarrow{J_{\text{KM}}^{\text{enhco}}} & \text{KL}(M)_{\text{crit}_M + \tilde{\rho}_P,\underline{x}} \end{array}$$

11.6.1. Let κ be an arbitrary level. Parallel to the functor

$$\text{Loc}_{G,\kappa,\underline{x}} : \text{KL}(G)_{\kappa,\underline{x}} \rightarrow \text{D-mod}_{\kappa}(\text{Bun}_G),$$

the construction in [GLC2, Sect. 10.2 and 10.3] naturally produces a functor

$$\text{Loc}_{G,\kappa,\underline{x}} : \widehat{\mathfrak{g}}\text{-mod}_{\kappa,\underline{x}} \rightarrow \text{D-mod}_{\kappa}(\text{Bun}_G^{\text{level}_{\underline{x}}}),$$

compatible with the actions of $\mathfrak{L}(G)_{\underline{x}}$.

In particular, for a subgroup $H \subset \mathfrak{L}^+(G)_{\underline{x}}$, we have a commutative diagram

$$(11.20) \quad \begin{array}{ccc} \text{D-mod}_{\kappa}(\text{Bun}_G) \otimes \text{D-mod}_{\kappa}(\text{Gr}_{G,\underline{x}})^H & \xrightarrow{*} & \text{D-mod}_{\kappa}(H \backslash \text{Bun}_G^{\text{level}_{\underline{x}}}) \\ \uparrow \text{Loc}_{G,\kappa,\underline{x}} \otimes \text{Id} & & \uparrow \text{Loc}_{G,\kappa,\underline{x}} \\ \text{KL}(G)_{\kappa,\underline{x}} \otimes \text{D-mod}_{\kappa}(\text{Gr}_{G,\underline{x}})^H & \xrightarrow{*} & \widehat{\mathfrak{g}}\text{-mod}_{\kappa,\underline{x}}^H \end{array}$$

Consider the particular case of (11.20) when $H = \mathfrak{L}^+(P^-)_{\underline{x}}$. We obtain a commutative diagram:

$$(11.21) \quad \begin{array}{ccc} \text{D-mod}_{\kappa}(\text{Bun}_G) \otimes \text{D-mod}_{\kappa}(\text{Gr}_{G,\underline{x}})^{\mathfrak{L}^+(P^-)_{\underline{x}}} & \xrightarrow{*} & \text{D-mod}_{\kappa}(\text{Bun}_G \times_{\text{pt}/\mathfrak{L}^+(G)_{\underline{x}}} \text{pt}/\mathfrak{L}^+(P^-)_{\underline{x}}) \\ \uparrow \text{Loc}_{G,\kappa,\underline{x}} \otimes \text{Id} & & \uparrow \text{Loc}_{G,\kappa,\underline{x}} \\ \text{KL}(G)_{\kappa,\underline{x}} \otimes \text{D-mod}_{\kappa}(\text{Gr}_{G,\underline{x}})^{\mathfrak{L}^+(P^-)_{\underline{x}}} & \xrightarrow{*} & \widehat{\mathfrak{g}}\text{-mod}_{\kappa,\underline{x}}^{\mathfrak{L}^+(P^-)_{\underline{x}}} \end{array}$$

11.6.2. Note now that the map

$$\mathfrak{p}^- : \text{Bun}_{P^-} \rightarrow \text{Bun}_G$$

naturally factors via a map

$$\mathfrak{p}^- : \text{Bun}_{P^-} \rightarrow \text{Bun}_G \times_{\text{pt}/\mathfrak{L}^+(G)_{\underline{x}}} \text{pt}/\mathfrak{L}^+(P^-)_{\underline{x}}.$$

As in [GLC2, Equation (12.17)], we obtain a natural transformation

$$(11.22) \quad \begin{array}{ccc} \text{D-mod}_{\kappa}(\text{Bun}_G \times_{\text{pt}/\mathfrak{L}^+(G)_{\underline{x}}} \text{pt}/\mathfrak{L}^+(P^-)_{\underline{x}}) & \xrightarrow{(\mathfrak{p}^-)^!} & \text{D-mod}_{\kappa}(\text{Bun}_{P^-}) \\ \uparrow \text{Loc}_{G,\kappa,\underline{x}} & \swarrow & \uparrow \text{Loc}_{P^-, \kappa, \underline{x}} \\ \widehat{\mathfrak{g}}\text{-mod}_{\kappa,\underline{x}}^{\mathfrak{L}^+(P^-)_{\underline{x}}} & \xrightarrow{\text{Res}_{\mathfrak{p}^-}^{\widehat{\mathfrak{g}}}} & \text{KL}(P^-)_{\kappa,\underline{x}} \end{array}$$

11.6.3. Recall also the diagram

$$\begin{array}{ccc}
 \mathrm{D}\text{-mod}_{\mathrm{crit}_G+\kappa}(\mathrm{Bun}_{P^-}) & \xrightarrow{\mathbf{q}_*^-, \text{shift}} & \mathrm{D}\text{-mod}_{\mathrm{crit}_M+\kappa}(\mathrm{Bun}_M) \\
 \uparrow \mathrm{Loc}_{P^-, \mathrm{crit}_G+\kappa, \underline{x}} & \nwarrow & \uparrow \mathbf{l}_{N_P^-} \otimes ((-) \otimes \mathcal{L}_{\tilde{\rho}_P(\omega_X)}) \circ \mathrm{Loc}_{M, \mathrm{crit}_M+\kappa+\tilde{\rho}_P, \underline{x}} \\
 \mathrm{KL}(P^-)_{\mathrm{crit}_G+\kappa, \underline{x}} & \xrightarrow{\mathrm{BRST}^-} & \mathrm{KL}(M)_{\mathrm{crit}_M+\kappa+\tilde{\rho}_P, \underline{x}},
 \end{array}$$

see (11.7).

Concatenating it with diagrams (11.22) and (11.21) we obtain a natural transformation

$$\begin{array}{ccc}
 \mathrm{D}\text{-mod}_{\mathrm{crit}_G+\kappa}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\mathrm{crit}_G+\kappa}(\mathrm{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(P^-)_{\underline{x}}} & \longrightarrow & \mathrm{D}\text{-mod}_{\mathrm{crit}_M+\kappa}(\mathrm{Bun}_M) \\
 \uparrow \mathrm{Loc}_{G, \mathrm{crit}_G+\kappa, \underline{x}} \otimes \mathrm{Id} & \nwarrow & \uparrow \mathbf{l}_{N_P^-} \otimes ((-) \otimes \mathcal{L}_{\tilde{\rho}_P(\omega_X)}) \circ \mathrm{Loc}_{M, \mathrm{crit}_M+\kappa+\tilde{\rho}_P, \underline{x}} \\
 \mathrm{KL}(G)_{\mathrm{crit}_G+\kappa, \underline{x}} \otimes \mathrm{D}\text{-mod}_{\mathrm{crit}_G+\kappa}(\mathrm{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(P^-)_{\underline{x}}} & \longrightarrow & \mathrm{KL}(M)_{\mathrm{crit}_M+\kappa+\tilde{\rho}_P, \underline{x}}.
 \end{array}$$

11.6.4. We now specialize to the critical level, so that the latter diagram becomes

$$\begin{array}{ccc}
 \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(P^-)_{\underline{x}}} & \longrightarrow & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \\
 \uparrow \mathrm{Loc}_{G, \underline{x}} \otimes \mathrm{Id} & \nwarrow & \uparrow \mathbf{l}_{N_P^-} \otimes \mathrm{Loc}_{M, \tilde{\rho}_P, \underline{x}} \\
 \mathrm{KL}(G)_{\mathrm{crit}_G, \underline{x}} \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(P^-)_{\underline{x}}} & \longrightarrow & \mathrm{KL}(M)_{\mathrm{crit}_M+\tilde{\rho}_P, \underline{x}}.
 \end{array}
 \tag{11.23}$$

Note that the functor

$$\begin{aligned}
 \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \underline{x}})^{\mathfrak{L}^+(P^-)_{\underline{x}}} & \xrightarrow{\star} \\
 & \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G \times_{\mathrm{pt}/\mathfrak{L}^+(G)_{\underline{x}}} \mathrm{pt}/\mathfrak{L}^+(P^-)_{\underline{x}}) \xrightarrow{(\mathbf{p}^-)^!} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{P^-})
 \end{aligned}$$

identifies with the functor (8.16).

Hence, the upper horizontal arrow in (11.23) is the functor (8.17). In particular, (11.23) factors via

$$(11.24) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\underline{x}})^{\mathfrak{L}^+(M)_{\underline{x}}} & \longrightarrow & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \\ \uparrow \mathrm{Loc}_{G,\underline{x}} \otimes \mathrm{Id} & \swarrow & \uparrow \mathfrak{l}_{N_P^-} \otimes \mathrm{Loc}_{M,\tilde{\rho}_P,\underline{x}} \\ \mathrm{KL}(G)_{\mathrm{crit}_{G,\underline{x}}} \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\underline{x}})^{\mathfrak{L}^+(M)_{\underline{x}}} & \longrightarrow & \mathrm{KL}(M)_{\mathrm{crit}_M + \tilde{\rho}_P,\underline{x}} \end{array}$$

11.6.5. Precomposing (11.25) with

$$\mathrm{I}(G, P^-)_{\mathrm{co},\underline{x}}^{\mathrm{loc}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\underline{x}})^{\mathfrak{L}^+(M)_{\underline{x}}},$$

we obtain a natural transformation

$$(11.25) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{I}(G, P^-)_{\mathrm{co},\underline{x}}^{\mathrm{loc}} & \xrightarrow{\mathrm{pre}\text{-}\mathrm{CT}_{*,\underline{x}}^{\mathrm{enhco}}} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \\ \uparrow \mathrm{Loc}_{G,\underline{x}} \otimes \mathrm{Id} & \swarrow & \uparrow \mathfrak{l}_{N_P^-} \otimes \mathrm{Loc}_{M,\tilde{\rho}_P,\underline{x}} \\ \mathrm{KL}(G)_{\mathrm{crit}_{G,\underline{x}}} \otimes \mathrm{I}(G, P^-)_{\mathrm{co},\underline{x}}^{\mathrm{loc}} & \longrightarrow & \mathrm{KL}(M)_{\mathrm{crit}_M + \tilde{\rho}_P,\underline{x}} \end{array}$$

The functors and the natural transformation in (11.25) are compatible with⁵² the action of $\mathrm{Sph}_{G,x}$. Hence, it further factors via

$$(11.26) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{Sph}_{G,\underline{x}}} \mathrm{I}(G, P^-)_{\mathrm{co},\underline{x}}^{\mathrm{loc}} & \xrightarrow{\mathrm{CT}_{*,\underline{x}}^{\mathrm{enhco}}} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \\ \uparrow \mathrm{Loc}_{G,\underline{x}} \otimes \mathrm{Id} & \swarrow & \uparrow \mathfrak{l}_{N_P^-} \otimes \mathrm{Loc}_{M,\tilde{\rho}_P,\underline{x}} \\ \mathrm{KL}(G)_{\mathrm{crit}_{G,\underline{x}}} \otimes_{\mathrm{Sph}_{G,\underline{x}}} \mathrm{I}(G, P^-)_{\mathrm{co},\underline{x}}^{\mathrm{loc}} & \xrightarrow{J_{\mathrm{KM}}^{\mathrm{enhco}}} & \mathrm{KL}(M)_{\mathrm{crit}_M + \tilde{\rho}_P,\underline{x}} \end{array}$$

as desired.

The compatibility with the $\mathrm{Sph}_{M,x}$ -action follows from the construction.

⁵²I.e., the horizontal arrows coequalize the Sph_G actions, and the above diagram is compatible with this structure.

11.7. Addendum: Whittaker coefficients of localization, take II. In this subsection we use the methods developed above in order to sharpen a theorem from [GLC2, Sect. 14], which computes Whittaker coefficients of Kac-Moody localization.

Namely, we will show that the computation in *loc. cit.* is compatible with the action of Sph_G .

11.7.1. Let us denote by $\mathcal{P}_G^{\mathrm{loc}, \mathrm{true}}$ the (lax unital) factorization functor

$$\begin{aligned} \mathrm{KL}(G)_{\mathrm{crit}} \otimes_{\mathrm{Sph}_G} \mathrm{Whit}_*(G) &\xrightarrow{\alpha_{\mathrm{taut}, \rho(\omega_X)} \otimes \mathrm{Id}} \\ &\rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \rho(\omega_X)} \otimes_{\mathrm{Sph}_G} \mathrm{Whit}_*(G) \xrightarrow{*} (\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}, \rho(\omega_X)})_{\mathfrak{L}(N)_{\rho(\omega_X), X}} \xrightarrow{\overline{\mathrm{DS}}} \mathrm{Vect}, \end{aligned}$$

where the last arrow is [GLC2, Equation (2.8)].

Recall that the functor $\mathcal{P}_G^{\mathrm{loc}}$ is defined to be the composition

$$\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G) \rightarrow \mathrm{KL}(G)_{\mathrm{crit}} \otimes_{\mathrm{Sph}_G} \mathrm{Whit}_*(G) \xrightarrow{\mathcal{P}_G^{\mathrm{loc}, \mathrm{true}}} \mathrm{Vect},$$

see [GLC2, Sect. 6.4.6].

11.7.2. In [GLC2, Theorem 14.1.4] it was shown that the functor

$$(11.27) \quad (\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G))_{\mathbb{Z}} \xrightarrow{\mathrm{Loc}_{G, \mathbb{Z}} \otimes \mathrm{Id}} (\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathbb{Z})) \otimes_{\mathrm{D-mod}(\mathbb{Z})} \mathrm{Whit}_*(G)_{\mathbb{Z}} \rightarrow$$

$$\xrightarrow{\mathrm{coeff}_{G, \mathbb{Z}} \otimes \mathrm{id}} (\mathrm{Whit}^!(G) \otimes \mathrm{Whit}_*(G))_{\mathbb{Z}} \rightarrow \mathrm{D-mod}(\mathbb{Z})$$

is canonically isomorphic to

$$(11.28) \quad (\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G))_{\mathbb{Z}} \xrightarrow{\mathrm{ins.unit}_{\mathbb{Z}}} (\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G))_{\mathbb{Z} \subseteq} \xrightarrow{\mathcal{P}_G^{\mathrm{loc}}} \\ \rightarrow \mathrm{D-mod}(\mathbb{Z} \subseteq) \xrightarrow{(\mathrm{pr}_{\mathrm{small}, \mathbb{Z}})^!} \mathrm{D-mod}(\mathbb{Z}) \xrightarrow{\mathfrak{l}_{G, N_{\rho(\omega_X)}}^{\otimes -\frac{1}{2}} \otimes \mathfrak{l}_{N_{\rho(\omega_X)}}^{\otimes \delta_{N_{\rho(\omega_X)}}}} \mathrm{D-mod}(\mathbb{Z}),$$

where:

- $\delta_{N_{\rho(\omega_X)}} := \dim(\mathrm{Bun}_{N_{\rho(\omega_X)}})$;
- $\mathfrak{l}_{N_{\rho(\omega_X)}}$ is the line from [GLC2, Sect. 14.1.1], i.e.,

$$\det(T^*(\mathrm{Bun}_{N_{\rho(\omega_X)}})) \simeq \mathcal{O}_{\mathrm{Bun}_{N_{\rho(\omega_X)}}} \otimes \mathfrak{l}_{N_{\rho(\omega_X)}}[\delta_{N_{\rho(\omega_X)}}];$$

- $\mathfrak{l}_{G, N_{\rho(\omega_X)}}^{\otimes \frac{1}{2}}$ is the line from [GLC2, Sect. 9.2.4].⁵³

11.7.3. Note now that both (11.27) and (11.28) factor via

$$(\mathrm{KL}(G)_{\mathrm{crit}} \otimes \mathrm{Whit}_*(G))_{\mathbb{Z}} \rightarrow (\mathrm{KL}(G)_{\mathrm{crit}} \otimes_{\mathrm{Sph}_G} \mathrm{Whit}_*(G))_{\mathbb{Z}} \rightarrow \mathrm{D-mod}(\mathbb{Z}).$$

Indeed, for (11.27) this factorization is provided by

$$(11.29) \quad (\mathrm{KL}(G)_{\mathrm{crit}} \otimes_{\mathrm{Sph}_G} \mathrm{Whit}_*(G))_{\mathbb{Z}} \xrightarrow{\mathrm{Loc}_{G, \mathbb{Z}} \otimes \mathrm{Id}} (\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathbb{Z})) \otimes_{\mathrm{Sph}_{G, \mathbb{Z}}} \mathrm{Whit}_*(G)_{\mathbb{Z}} \rightarrow$$

$$\xrightarrow{\mathrm{coeff}_{G, \mathbb{Z}} \otimes \mathrm{Id}} (\mathrm{Whit}^!(G) \otimes_{\mathrm{Sph}_{G, \mathbb{Z}}} \mathrm{Whit}_*(G))_{\mathbb{Z}} \rightarrow \mathrm{D-mod}(\mathbb{Z}).$$

⁵³See [GLC2, Sect. 14.1.4] for why the twist by this line appears.

For (11.28) this factorization is provided by

$$(11.30) \quad (\mathrm{KL}(G)_{\mathrm{crit}} \otimes_{\mathrm{Sph}_G} \mathrm{Whit}_*(G))_{\mathcal{Z}} \xrightarrow{\mathrm{ins.unit}_{\mathcal{Z}}} (\mathrm{KL}(G)_{\mathrm{crit}} \otimes_{\mathrm{Sph}_G} \mathrm{Whit}_*(G))_{\mathcal{Z} \subseteq} \xrightarrow{\mathbf{p}_G^{\mathrm{loc}, \mathrm{true}}} \\ \rightarrow \mathrm{D-mod}(\mathcal{Z} \subseteq) \xrightarrow{(\mathrm{pr}_{\mathrm{small}, \mathcal{Z}})^!} \mathrm{D-mod}(\mathcal{Z}) \xrightarrow{\mathfrak{l}_{G, N_{\rho}(\omega_X)}^{\otimes -\frac{1}{2}} \otimes \mathfrak{l}_{N_{\rho}(\omega_X)} [\delta_{N_{\rho}(\omega_X)}]}} \mathrm{D-mod}(\mathcal{Z}).$$

11.7.4. Denote the functor (11.27) by $\mathbf{P}_{G, \mathcal{Z}}^{\mathrm{glob}}$ and the functor (11.29) by $\mathbf{P}_{G, \mathcal{Z}}^{\mathrm{glob}, \mathrm{true}}$.

We can view the assignments

$$\mathcal{Z} \rightsquigarrow \mathbf{P}_{G, \mathcal{Z}}^{\mathrm{glob}} \text{ and } \mathcal{Z} \rightsquigarrow \mathbf{P}_{G, \mathcal{Z}}^{\mathrm{glob}, \mathrm{true}}$$

as (unital) local-to-global functors, to be denoted

$$\underline{\mathbf{P}}_G^{\mathrm{glob}} \text{ and } \underline{\mathbf{P}}_G^{\mathrm{glob}, \mathrm{true}},$$

respectively.

Note that we can think of the functor (11.30), up to tensoring by a line and a cohomological shift, as

$$(\underline{\mathbf{P}}_G^{\mathrm{loc}, \mathrm{true}})^f \xrightarrow{\mathrm{ins.unit}}.$$

11.7.5. In this subsection we will prove the following sharpening of [GLC2, Theorem 14.1.4]:

Theorem 11.7.6. *We have a canonical isomorphism*

$$(\underline{\mathbf{P}}_G^{\mathrm{loc}, \mathrm{true}})^f \xrightarrow{\mathrm{ins.unit}} \otimes \mathfrak{l}_{G, N_{\rho}(\omega_X)}^{\otimes -\frac{1}{2}} \otimes \mathfrak{l}_{N_{\rho}(\omega_X)} [\delta_{N_{\rho}(\omega_X)}] \simeq \underline{\mathbf{P}}_G^{\mathrm{glob}, \mathrm{true}}$$

as local-to-global functors

$$\underline{\mathrm{KL}}(G)_{\mathrm{crit}} \otimes_{\underline{\mathrm{Sph}}_G} \underline{\mathrm{Whit}}_*(G) \Rightarrow \mathrm{Vect}.$$

The rest of this subsection is devoted to the proof of this theorem.

11.7.7. Parallel to the functor $\mathrm{CT}_{*, \mathcal{Z}}^{-, \mathrm{enhco}}$ constructed in Sect. 8.3, one can consider the functor

$$(11.31) \quad (\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z})) \otimes_{\mathrm{Sph}_{G, \mathcal{Z}}} \mathrm{Whit}_*(G)_{\mathcal{Z}} \rightarrow \mathrm{D-mod}(\mathcal{Z}),$$

so that the composition

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z}) \xrightarrow{\mathrm{Id} \otimes \mathbf{1}_{\mathrm{Whit}_*(G)_{\mathcal{Z}}}} \\ \rightarrow (\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z})) \otimes_{\mathrm{Sph}_{G, \mathcal{Z}}} \mathrm{Whit}_*(G)_{\mathcal{Z}} \rightarrow \mathrm{D-mod}(\mathcal{Z})$$

identifies with

$$\mathrm{coeff}_G^{\mathrm{Vac}, \mathrm{glob}} \otimes \mathrm{Id}_{\mathrm{D-mod}(\mathcal{Z})},$$

where $\mathrm{coeff}_G^{\mathrm{Vac}, \mathrm{glob}}$ is the functor from [GLC2, Sect. 9.3.6].

11.7.8. A construction parallel to that in Sect. 11.6, combined with [GLC2, Sect. 12.5.16-12.5.17] gives rise to a natural transformation

$$\begin{array}{ccc} (\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z})) \otimes_{\mathrm{Sph}_{G, \mathcal{Z}}} \mathrm{Whit}_*(G)_{\mathcal{Z}} & \xrightarrow{(11.31)} & \mathrm{D-mod}(\mathcal{Z}) \\ \uparrow \mathrm{Loc}_{G, \mathcal{Z}} \otimes \mathrm{Id} & \swarrow & \uparrow \mathfrak{l}_{G, N_{\rho}(\omega_X)}^{\otimes -\frac{1}{2}} \otimes \mathfrak{l}_{N_{\rho}(\omega_X)} [\delta_{N_{\rho}(\omega_X)}] \\ \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}} \otimes_{\mathrm{Sph}_{G, \mathcal{Z}}} \mathrm{Whit}_*(G)_{\mathcal{Z}} & \xrightarrow{\mathbf{p}_G^{\mathrm{loc}, \mathrm{true}}} & \mathrm{D-mod}(\mathcal{Z}). \end{array}$$

As in Sects. 11.5.5-11.5.6, one shows that the induced natural transformation

$$\begin{array}{ccc}
 (\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathbb{Z})) \otimes_{\mathrm{Sph}_{G,z}} \mathrm{Whit}_*(G)_z & \xrightarrow{(11.31)} & \mathrm{D}\text{-mod}(\mathbb{Z}) \\
 \uparrow \mathrm{Loc}_{G,z} \otimes \mathrm{Id} & \nearrow & \uparrow \mathfrak{l}_{G,N_{\rho(\omega_X)}^{-\frac{1}{2}}} \otimes \mathfrak{l}_{N_{\rho(\omega_X)}}[\delta_{N_{\rho(\omega_X)}}] \\
 \mathrm{KL}(G)_{\mathrm{crit},z} \otimes_{\mathrm{Sph}_{G,z}} \mathrm{Whit}_*(G)_z & & \mathrm{D}\text{-mod}(\mathbb{Z}) \\
 \downarrow \mathrm{ins.unit}_z & \searrow & \downarrow (\mathrm{Pr}_{\mathrm{small},z})! \\
 \mathrm{KL}(G)_{\mathrm{crit},z \subseteq} \otimes_{\mathrm{Sph}_{G,z \subseteq}} \mathrm{Whit}_*(G)_{z \subseteq} & \xrightarrow{\mathfrak{p}_G^{\mathrm{loc},\mathrm{true}}} & \mathrm{D}\text{-mod}(\mathbb{Z}^{\subseteq})
 \end{array}$$

is an isomorphism.

11.7.9. However, now, as in Proposition 9.4.4 one shows that the functor (11.31) identifies with

$$\begin{aligned}
 & (\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathbb{Z})) \otimes_{\mathrm{Sph}_{G,z}} \mathrm{Whit}_*(G)_z \rightarrow \\
 & \xrightarrow{\mathrm{coeff}_{G,z} \otimes \mathrm{id}} (\mathrm{Whit}^!(G) \otimes_{\mathrm{Sph}_{G,z}} \mathrm{Whit}_*(G))_z \rightarrow \mathrm{D}\text{-mod}(\mathbb{Z}),
 \end{aligned}$$

as required.

□[Theorem 11.7.6]

11.8. Localization and translation by Z_G -torsors.

11.8.1. Let $\mathcal{P}_G$ be a G -torsor on X . Parallel to [GLC2, Sect. 10], we can consider the localization functor

$$\mathrm{Loc}_{G,\kappa,\mathcal{P}_G,z} : \mathrm{KL}(G)_{\kappa,\mathcal{P}_G,z} \rightarrow \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_{G,\mathcal{P}_G}) \otimes \mathrm{D}\text{-mod}(\mathbb{Z}),$$

which makes the diagram

$$\begin{array}{ccc}
 \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathbb{Z}) & \xrightarrow{\alpha_{\mathcal{P}_G,\mathrm{taut}} \otimes \mathrm{Id}} & \mathrm{D}\text{-mod}_{\kappa}(\mathrm{Bun}_{G,\mathcal{P}_G}) \otimes \mathrm{D}\text{-mod}(\mathbb{Z}) \\
 \uparrow \mathrm{Loc}_{G,\kappa,z} & & \uparrow \mathrm{Loc}_{G,\kappa,\mathcal{P}_G,z} \\
 \mathrm{KL}(G)_{\kappa,z} & \xrightarrow{\alpha_{\mathcal{P}_G,\mathrm{taut}}} & \mathrm{KL}(G)_{\kappa,\mathcal{P}_G,z},
 \end{array}$$

commute, where in the top row, $\alpha_{\mathcal{P}_G,\mathrm{taut}}$ is the tautological identification

$$\mathrm{Bun}_G \xrightarrow{\alpha_{\mathcal{P}_G,\mathrm{taut}}} \mathrm{Bun}_{G,\mathcal{P}_G}.$$

11.8.2. Assume now that $\mathcal{P}_G$ is induced from a Z_G -torsor $\mathcal{P}_{Z_G}$. In this case, we have a canonical identification

$$\mathrm{Bun}_{G,\mathcal{P}_{Z_G}} \xrightarrow{\alpha_{\mathcal{P}_{Z_G},\mathrm{cent}}} \mathrm{Bun}_G.$$

Recall the notation $\kappa(\mathrm{dlog}(\mathcal{P}_{Z_G}), -)$ for the corresponding torsor over $\mathfrak{g}_{\mathrm{ab}}^* \otimes \omega_X$, see Sect. 3.8.3. We claim:

Lemma 11.8.3. *The isomorphism $\alpha_{\mathcal{P}_{Z_G}, \text{cent}}$ identifies the twisting κ on $\text{Bun}_{G, \mathcal{P}_{Z_G}}$ with the twisting $\kappa - \kappa(\text{dlog}(\mathcal{P}_{Z_G}, -))$*

on Bun_G . Furthermore, the diagram

$$(11.33) \quad \begin{array}{ccc} \text{D-mod}_{\kappa}(\text{Bun}_{G, \mathcal{P}_{Z_G}}) \otimes \text{D-mod}(\mathbb{Z}) & \xrightarrow{\alpha_{\mathcal{P}_{Z_G}, \text{cent}}} & \text{D-mod}_{\kappa - \kappa(\text{dlog}(\mathcal{P}_{Z_G}, -))}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) \\ \uparrow \text{Loc}_{G, \kappa, \mathcal{P}_{Z_G}, \mathbb{Z}} & & \uparrow \text{Loc}_{G, \kappa + \kappa(\text{dlog}(\mathcal{P}_{Z_G}, -), \mathbb{Z})} \\ \text{KL}(G)_{\kappa, \mathcal{P}_{Z_G}, \mathbb{Z}} & \xrightarrow{\alpha_{\mathcal{P}_{Z_G}, \text{cent}}} & \text{KL}(G)_{\kappa + \kappa(\text{dlog}(\mathcal{P}_{Z_G}, -), \mathbb{Z})} \end{array}$$

11.8.4. Note that the composite map

$$\text{Bun}_G \xrightarrow{\alpha_{\mathcal{P}_{Z_G}, \text{taut}}} \text{Bun}_{G, \mathcal{P}_{Z_G}} \xrightarrow{\alpha_{\mathcal{P}_{Z_G}, \text{cent}}} \text{Bun}_G$$

is the inverse of the automorphism

$$\text{transl}_{\mathcal{P}_{Z_G}} : \text{Bun}_G \rightarrow \text{Bun}_G,$$

given by tensoring with $\mathcal{P}_{Z_G}$.

11.8.5. From Lemma 11.8.3 we obtain:

Corollary 11.8.6. *The pullback of the twisting of the twisting κ on Bun_G with respect to $\text{transl}_{\mathcal{P}_{Z_G}}$ identifies canonically with the twisting $\kappa - \kappa(\text{dlog}(\mathcal{P}_{Z_G}, -))$.*

11.8.7. Concatenating diagrams (11.32) and (11.33), we obtain a commutative diagram

$$(11.34) \quad \begin{array}{ccc} \text{D-mod}_{\kappa}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) & \xrightarrow{(\text{transl}_{\mathcal{P}_{Z_G}})^*} & \text{D-mod}_{\kappa - \kappa(\text{dlog}(\mathcal{P}_{Z_G}, -))}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) \\ \uparrow \text{Loc}_{G, \kappa, \mathbb{Z}} & & \uparrow \text{Loc}_{G, \kappa - \kappa(\text{dlog}(\mathcal{P}_{Z_G}, -), \mathbb{Z})} \\ \text{KL}(G)_{\kappa, \mathbb{Z}} & \xrightarrow{(\text{transl}_{\mathcal{P}_{Z_G}})^*} & \text{KL}(G)_{\kappa + \kappa(\text{dlog}(\mathcal{P}_{Z_G}, -), \mathbb{Z})} \end{array}$$

where the functor $(\text{transl}_{\mathcal{P}_{Z_G}})^*$ in the bottom line is as in Sect. 3.22.

11.8.8. Parallel to Sect. 3.8.6 for $\mathcal{P}_{Z_G}$ of the form $\lambda(\omega_X)$ for $\lambda : \mathbb{G}_m \rightarrow Z_G$, we will use a shorthand-notation

$$\kappa - \kappa(\lambda, -) := \kappa - \kappa(\text{dlog}(\lambda(\omega_X), -)).$$

Note that this notation is consistent with one in Sect. 11.1.6, i.e., we can treat $\kappa(\lambda, -)$ as a bona fide element of $z_{\mathfrak{g}}$.

11.9. Constant term of localization: the translated version.

11.9.1. Let $\mathcal{P}_{Z_M}$ be a Z_M -torsor. Let $\text{CT}_{*, \mathcal{P}_{Z_M}}^-$ be the functor

$$\text{D-mod}_{\text{crit}_G + \kappa}(\text{Bun}_G) \xrightarrow{\text{CT}_{*, \mathcal{P}_{Z_M}}^-} \text{D-mod}_{\text{crit}_M + \kappa}(\text{Bun}_M) \xrightarrow{(\text{transl}_{\mathcal{P}_{Z_M}})^*} \text{D-mod}_{\text{crit}_M + \kappa - \kappa(\text{dlog}(\mathcal{P}_{Z_M}, -))}(\text{Bun}_M).$$

11.9.2. Concatenating the diagrams (11.34), (11.2) and (3.27). we obtain:

Corollary 11.9.3. *For $\mathbb{Z} \rightarrow \text{Ran}$, the following diagram of functors canonically commutes*

$$\begin{array}{ccc} \text{D-mod}_{\text{crit}_G + \kappa}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) & \xrightarrow{\text{CT}_{*, \mathcal{P}_{Z_M}}^- \otimes \text{Id}} & \text{D-mod}_{\text{crit}_M + \kappa - \kappa(\text{dlog}(\mathcal{P}_{Z_M}, -))}(\text{Bun}_M) \otimes \text{D-mod}(\mathbb{Z}) \\ \uparrow \text{Loc}_{G, \text{crit}_G + \kappa, \mathbb{Z}} & & \uparrow \text{I} \otimes ((-) \otimes \mathcal{L}_{\tilde{\rho}_P}(\omega_X)) \otimes (\text{Pr}_{\text{small}}, \mathbb{Z})! \\ \text{KL}(G)_{\text{crit}_G + \kappa, \mathbb{Z}} & & \text{D-mod}_{\text{crit}_M + \kappa + \tilde{\rho}_P - \kappa(\text{dlog}(\mathcal{P}_{Z_M}, -))}(\text{Bun}_M) \otimes \text{D-mod}(\mathbb{Z}) \\ \downarrow \text{ins.vac.} \mathbb{Z} & & \uparrow \text{Loc}_{M, \text{crit}_M + \kappa + \tilde{\rho}_P + \kappa(\text{dlog}(\mathcal{P}_{Z_M}, -), \mathbb{Z}) \subseteq \\ \text{KL}(G)_{\text{crit}_G + \kappa, \mathbb{Z} \subseteq} & \xrightarrow{J_{\text{KM}, \mathcal{P}_{Z_M}}^-, \text{Sph}} & \text{KL}(M)_{\text{crit}_M + \kappa + \tilde{\rho}_P + \kappa(\text{dlog}(\mathcal{P}_{Z_M}, -), \mathbb{Z} \subseteq} \end{array}$$

where

$$\mathfrak{I} := \mathfrak{I}_{N_P^-} \otimes \text{Weil}(\mathcal{P}_{Z_M}, \check{\rho}_P(\omega_X)).$$

Specializing to the critical level, we obtain:

Corollary 11.9.4. *For $\mathcal{Z} \rightarrow \text{Ran}$, the following diagram of functors canonically commutes*

$$\begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathcal{Z}) & \xrightarrow{\text{CT}_{*, \mathcal{P}_{Z_M}}^- \otimes \text{Id}} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathcal{Z}) \\ \uparrow \text{Loc}_{G, \mathcal{Z}} & & \uparrow \mathfrak{I} \otimes (\text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})_!) \\ \text{KL}(G)_{\text{crit}_G, \mathcal{Z}} & & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathcal{Z}^\subseteq) \\ \downarrow \text{ins.vac.}_{\mathcal{Z}} & & \uparrow \text{Loc}_{M, \check{\rho}_P, \mathcal{Z}^\subseteq} \\ \text{KL}(G)_{\text{crit}_G, \mathcal{Z}^\subseteq} & \xrightarrow{J_{\text{KM}, \mathcal{P}_{Z_M}}^-, \text{Sph}} & \text{KL}(M)_{\text{crit}_M + \check{\rho}_P, \mathcal{Z}^\subseteq}, \end{array}$$

where

$$\mathfrak{I} := \mathfrak{I}_{N_P^-} \otimes \text{Weil}(\mathcal{P}_{Z_M}, \check{\rho}_P(\omega_X)).$$

In the specific case of $\mathcal{P}_{Z_M} = \rho_P(\omega_X)$ we obtain:

Corollary 11.9.5. *For $\mathcal{Z} \rightarrow \text{Ran}$, the following diagram of functors canonically commutes*

$$\begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathcal{Z}) & \xrightarrow{\text{CT}_{*, \rho_P(\omega_X)}^- \otimes \text{Id}} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathcal{Z}) \\ \uparrow \text{Loc}_{G, \mathcal{Z}} & & \uparrow \mathfrak{I} \otimes (\text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})_!) \\ \text{KL}(G)_{\text{crit}_G, \mathcal{Z}} & & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathcal{Z}^\subseteq) \\ \downarrow \text{ins.vac.}_{\mathcal{Z}} & & \uparrow \text{Loc}_{M, \check{\rho}_P, \mathcal{Z}^\subseteq} \\ \text{KL}(G)_{\text{crit}_G, \mathcal{Z}^\subseteq} & \xrightarrow{J_{\text{KM}, \rho_P(\omega_X)}^-, \text{Sph}} & \text{KL}(M)_{\text{crit}_M + \check{\rho}_P, \mathcal{Z}^\subseteq}, \end{array}$$

where

$$\mathfrak{I} := \mathfrak{I}_{N_P^-} \otimes \text{Weil}(\mathcal{P}_{Z_M}, \check{\rho}_P(\omega_X)).$$

11.9.6. Similarly, from Theorem 11.5.3, we obtain:

Corollary 11.9.7. *For $\mathcal{Z} \rightarrow \text{Ran}$, the following diagram of functors canonically commutes*

$$\begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\mathcal{Z}}^{-, \text{enhco}} & \xrightarrow{\text{CT}_{*, \mathcal{P}_{Z_M}, \mathcal{Z}}^{-, \text{enhco}}} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathcal{Z}) \\ \uparrow \text{Loc}_{G, \mathcal{Z}}^{-, \text{enhco}} & & \uparrow \mathfrak{I} \otimes (\text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})_!) \\ \text{KL}(G)_{\text{crit}_G, \mathcal{Z}}^{-, \text{enhco}} & & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathcal{Z}^\subseteq) \\ \downarrow \text{ins.vac.}_{\mathcal{Z}} & & \uparrow \text{Loc}_{M, \check{\rho}_P, \mathcal{Z}^\subseteq} \\ \text{KL}(G)_{\text{crit}_G, \mathcal{Z}^\subseteq}^{-, \text{enhco}} & \xrightarrow{J_{\text{KM}, \mathcal{P}_{Z_M}}^{-, \text{enhco}} \otimes \mathfrak{I}_{N_P^-} \otimes \text{Weil}(\mathcal{P}_{Z_M}, \check{\rho}_P(\omega_X))} & \text{KL}(M)_{\text{crit}_M + \check{\rho}_P, \mathcal{Z}^\subseteq}, \end{array}$$

where

$$\mathfrak{I} := \mathfrak{I}_{N_P^-} \otimes \text{Weil}(\mathcal{P}_{Z_M}, \check{\rho}_P(\omega_X)).$$

In the specific case of $\mathcal{P}_{Z_M} = \rho_P(\omega_X)$ we obtain:

Corollary 11.9.8. *For $\mathcal{Z} \rightarrow \text{Ran}$, the following diagram of functors canonically commutes*

$$\begin{array}{ccc}
\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\mathcal{Z}}^{-, \text{enhco}} & \xrightarrow{\text{CT}_{*, \rho_P(\omega_X), \mathcal{Z}}^{-, \text{enhco}}} & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathcal{Z}) \\
\text{Loc}_{G, \mathcal{Z}}^{-, \text{enhco}} \uparrow & & \uparrow \mathfrak{l} \otimes (\text{Id} \otimes (\text{pr}_{\text{small}, \mathcal{Z}})_!) \\
\text{KL}(G)_{\text{crit}_G, \mathcal{Z}}^{-, \text{enhco}} & & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(\mathcal{Z}^{\subseteq}) \\
\text{ins.vac.}_{\mathcal{Z}} \downarrow & & \uparrow \text{Loc}_{M, \check{\rho}_P, \mathcal{Z}^{\subseteq}} \\
\text{KL}(G)_{\text{crit}_G, \mathcal{Z}^{\subseteq}}^{-, \text{enhco}} & \xrightarrow{J_{\text{KM}, \rho_P(\omega_X)}^{-, \text{enhco}}} & \text{KL}(M)_{\text{crit}_M + \check{\rho}_P, \mathcal{Z}^{\subseteq}},
\end{array}$$

where

$$\mathfrak{l} := \mathfrak{l}_{N_P}^- \otimes \text{Weil}(\rho_P(\omega_X), \check{\rho}_P(\omega_X)).$$

11.9.9. For future reference, we record the following observation, which is a particular case of Proposition 11.3.4:

Corollary 11.9.10. *There is a canonical isomorphism of lines*

$$\mathfrak{l}_{N_P}^- \otimes \text{Weil}(\rho_P(\omega_X), \check{\rho}_P(\omega_X)) \simeq \mathfrak{l}_{G, N_{\rho(\omega_X)}}^{\otimes -\frac{1}{2}} \otimes \mathfrak{l}_{N_{\rho(\omega_X)}} \otimes \mathfrak{l}_{M, N(M)_{\rho_M(\omega_X)}}^{\otimes \frac{1}{2}} \otimes \mathfrak{l}_{N(M)_{\rho_M(\omega_X)}}^{\otimes -1},$$

where:

- $\mathfrak{l}_{G, N_{\rho(\omega_X)}}^{\otimes \frac{1}{2}}$ is the square root of the line $\mathfrak{l}_{G, N_{\rho(\omega_X)}}$, given by [GLC1, Proposition 1.3.3], and
- $\mathfrak{l}_{M, N(M)_{\rho_M(\omega_X)}}^{\otimes \frac{1}{2}}$ is the corresponding line for M ;
- $\mathfrak{l}_{N_{\rho(\omega_X)}}$ is the line from [GLC2, Sect. 14.1.1], and $\mathfrak{l}_{N(M)_{\rho_M(\omega_X)}}$ is the corresponding line for M .

12. SPECTRAL EISENSTEIN AND CONSTANT TERM FUNCTORS

In this section start dealing with the global spectral category, i.e., $\text{IndCoh}(\text{LS}_{\tilde{G}})$, where $\text{LS}_{\tilde{G}}$ is the stack of $\tilde{G}$ -local systems on X .

We develop the spectral counterparts of the constructions in Sects. 8-9, i.e., the spectral Eisenstein and constant term functors, as well as their enhancements.

We will perform a spectral analog of the “localization and constant term” calculation from 11. The analog of the “Whittaker vs Eisenstein” calculation⁵⁴ from Sect. 10 will be dealt with in Sect. 13.

12.1. The usual spectral Eisenstein and constant term functors.

12.1.1. Consider the diagram

$$\text{LS}_{\tilde{G}} \xleftarrow{\mathfrak{p}^{-, \text{spec}}} \text{LS}_{\tilde{P}} \xrightarrow{\mathfrak{q}^{-, \text{spec}}} \text{LS}_{\tilde{M}}.$$

The (usual) spectral Eisenstein and constant term functors are the adjoint pair

$$\text{Eis}^{-, \text{spec}} : \text{IndCoh}(\text{LS}_{\tilde{M}}) \rightleftarrows \text{IndCoh}(\text{LS}_{\tilde{G}}) : \text{CT}^{-, \text{spec}}$$

defined by

$$\text{Eis}^{-, \text{spec}} := (\mathfrak{p}^{-, \text{spec}})_*^{\text{IndCoh}} \circ (\mathfrak{q}^{-, \text{spec}})^{\text{IndCoh}, *}$$

and

$$\text{CT}^{-, \text{spec}} := (\mathfrak{q}^{-, \text{spec}})_*^{\text{IndCoh}} \circ (\mathfrak{p}^{-, \text{spec}})^!,$$

where the functor $(\mathfrak{q}^{-, \text{spec}})^{\text{IndCoh}, *}$ is well-defined since the morphism $\mathfrak{q}^{-, \text{spec}}$ has a finite Tor-dimension (in fact, the morphism $\mathfrak{q}^{-, \text{spec}}$ is quasi-smooth).

⁵⁴Note, however, that the Langlands functor takes the localization functor on the geometric side to the Poincaré functor on the spectral side.

12.1.2. It is shown in [AG, Proposition 13.2.6] that the functor $\mathrm{Eis}^{-,\mathrm{spec}}$ sends

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}).$$

Similarly, it follows from [AG, Theorem 7.1.3] that the functor $\mathrm{CT}^{-,\mathrm{spec}}$ also sends

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}).$$

Hence, the functors $(\mathrm{Eis}^{-,\mathrm{spec}}, \mathrm{CT}^{-,\mathrm{spec}})$ induce an adjoint pair

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \rightleftarrows \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}).$$

12.1.3. In addition to the functor $\mathrm{Eis}^{-,\mathrm{spec}}$ we will consider the functor

$$\mathrm{Eis}_{\mathrm{QCoh}}^{-,\mathrm{spec}} : \mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}})$$

defined by

$$\mathrm{Eis}_{\mathrm{QCoh}}^{-,\mathrm{spec}} := (\mathfrak{p}^{-,\mathrm{spec}})_* \circ (\mathfrak{q}^{-,\mathrm{spec}})^*.$$

We have a commutative diagram

$$(12.1) \quad \begin{array}{ccc} \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) & \xrightarrow{\Psi_{\mathrm{LS}_{\check{M}}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \\ \mathrm{Eis}^{-,\mathrm{spec}} \downarrow & & \downarrow \mathrm{Eis}_{\mathrm{QCoh}}^{-,\mathrm{spec}} \\ \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) & \xrightarrow{\Psi_{\mathrm{LS}_{\check{G}}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}). \end{array}$$

12.1.4. Let

$$\mathrm{CT}_{\mathrm{QCoh}}^{-,\mathrm{spec}} : \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{M}})$$

be the functor defined by

$$\mathrm{CT}_{\mathrm{QCoh}}^{-,\mathrm{spec}} := (\mathfrak{q}^{-,\mathrm{spec}})_* \circ (\mathfrak{p}^{-,\mathrm{spec}})^*.$$

Note that we have

$$(12.2) \quad (\mathrm{Eis}_{\mathrm{QCoh}}^{-,\mathrm{spec}})^\vee \simeq \mathrm{CT}_{\mathrm{QCoh}}^{-,\mathrm{spec}}$$

with respect to the canonical self-duality

$$(12.3) \quad \mathrm{QCoh}(\mathrm{LS}_{\check{G}})^\vee \simeq \mathrm{QCoh}(\mathrm{LS}_{\check{G}}),$$

and a similar self-duality for $\check{M}$.

12.1.5. Consider the (ungraded) line bundle $\det(T^*(\mathrm{LS}_{\check{P}-} / \mathrm{LS}_{\check{G}}))$ on $\mathrm{LS}_{\check{P}-}$. It is easy to see that it canonically descends to a line bundle on $\mathrm{LS}_{\check{M}}$; by a slight abuse of notation, we will denote it by the same character $\det(T^*(\mathrm{LS}_{\check{P}-} / \mathrm{LS}_{\check{G}}))$.

We claim:

Lemma 12.1.6. *There is a commutative diagram*

$$\begin{array}{ccc} \mathrm{QCoh}(\mathrm{LS}_{\check{M}}) & \xrightarrow{((-) \otimes \det(T^*(\mathrm{LS}_{\check{P}-} / \mathrm{LS}_{\check{G}}))) \circ \Xi_{\mathrm{LS}_{\check{M}}}[-(2g-2) \cdot \dim(\check{\mathfrak{n}}_P)]} & \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \\ \mathrm{CT}_{\mathrm{QCoh}}^{-,\mathrm{spec}} \uparrow & & \uparrow \mathrm{CT}^{-,\mathrm{spec}} \\ \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) & \xrightarrow{\Xi_{\mathrm{LS}_{\check{G}}}} & \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}). \end{array}$$

Proof. Since both $\mathrm{LS}_{\check{G}}$ and $\mathrm{LS}_{\check{P}-}$ are quasi-smooth, we have a commutative diagram

$$\begin{array}{ccc} \mathrm{QCoh}(\mathrm{LS}_{\check{P}-}) & \xrightarrow{((-) \otimes \det(T^*(\mathrm{LS}_{\check{P}-} / \mathrm{LS}_{\check{G}}))) \circ \Xi_{\mathrm{LS}_{\check{P}-}}[-(2g-2) \cdot \dim(\check{\mathfrak{n}}_P)]} & \mathrm{IndCoh}(\mathrm{LS}_{\check{P}-}) \\ (\mathfrak{p}^{-,\mathrm{spec}})^* \uparrow & & \uparrow (\mathfrak{p}^{-,\mathrm{spec}})! \\ \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) & \xrightarrow{\Xi_{\mathrm{LS}_{\check{G}}}} & \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}), \end{array}$$

where

$$(2g-2) \cdot \dim(\check{\mathfrak{n}}_P) = \dim(\mathrm{LS}_{\check{G}}) - \dim(\mathrm{LS}_{\check{P}-}).$$

Since $\mathbf{q}^{-,\text{spec}}$ is quasi-smooth, the diagram

$$\begin{array}{ccc} \mathrm{QCoh}(\mathrm{LS}_{\tilde{P}-}) & \xrightarrow{\Xi_{\mathrm{LS}_{\tilde{P}-}}} & \mathrm{IndCoh}(\mathrm{LS}_{\tilde{P}-}) \\ (\mathbf{q}^{-,\text{spec}})_* \downarrow & & \downarrow (\mathbf{q}^{-,\text{spec}})_*^{\mathrm{IndCoh}} \\ \mathrm{QCoh}(\mathrm{LS}_{\tilde{M}}) & \xrightarrow{\Xi_{\mathrm{LS}_{\tilde{M}}}} & \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \end{array}$$

also commutes.

Concatenating the two diagrams, we obtain the assertion of the lemma. $\square$

12.1.7. Recall that Serre duality defines an identification

$$(12.4) \quad \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}})^\vee \simeq \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}),$$

and similarly for $\tilde{M}$.

Since $\mathbf{q}^{-,\text{spec}}$ is quasi-smooth, with respect to these identifications, we have

$$(12.5) \quad (\mathrm{CT}^{-,\text{spec}})^\vee \simeq \mathrm{Eis}^{-,\text{spec}} \circ \left((-) \otimes \det(T^*(\mathrm{LS}_{\tilde{P}-} / \mathrm{LS}_{\tilde{M}})) \right) [(2g-2) \cdot \dim(\mathfrak{n}_{\tilde{P}}^-)]$$

as functors

$$\mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \rightrightarrows \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}),$$

where:

- By a slight abuse of notation we denote by $\det(T^*(\mathrm{LS}_{\tilde{P}-} / \mathrm{LS}_{\tilde{M}}))$ the natural descent of the same-named line bundle from $\mathrm{LS}_{\tilde{P}-}$ to $\mathrm{LS}_{\tilde{M}}$;
- The integer $(2g-2) \cdot \dim(\mathfrak{n}_{\tilde{P}}^-)$ appears as $\dim(\mathrm{LS}_{\tilde{P}-}) - \dim(\mathrm{LS}_{\tilde{M}})$.

12.1.8. For future reference we note:

Lemma 12.1.9. *There is a canonical isomorphism of (ungraded) line bundles*

$$\det(T^*(\mathrm{LS}_{\tilde{P}-} / \mathrm{LS}_{\tilde{M}})) \simeq \mathcal{L}_{2\rho_P(\omega_X)}$$

and

$$\det(T^*(\mathrm{LS}_{\tilde{P}-} / \mathrm{LS}_{\tilde{G}})) \simeq \mathcal{L}_{-2\rho_P(\omega_X)}$$

where by a slight abuse of notation we denote by the same symbol $\mathcal{L}_{2\rho_P(\omega_X)}$ the pullback of the corresponding line bundle along

$$\mathrm{LS}_{\tilde{M}} \rightarrow \mathrm{Bun}_{\tilde{M}}.$$

Proof. Let $\mathcal{E}$ be a vector bundle with a connection on X . First, we claim that there is a canonical isomorphism

$$(12.6) \quad \det(C(X, \mathcal{E})) \simeq \mathrm{Weil}(\det(\mathcal{E}), \omega_X)^{\otimes -1}.$$

Indeed, we calculate $C(X, \mathcal{E})$ as

$$\mathrm{Fib} \left(\Gamma(X, \mathcal{E}) \xrightarrow{\nabla} \Gamma(X, \mathcal{E} \otimes \omega_X) \right),$$

and hence

$$(12.7) \quad \det(C(X, \mathcal{E}))^{\otimes -1} \simeq \det(\Gamma(X, \mathcal{E} \otimes \omega_X)) \otimes \det(\Gamma(X, \mathcal{E}))^{\otimes -1}.$$

Applying (11.5), we rewrite the right-hand side in (12.7) as

$$(12.8) \quad \det(\Gamma(X, \omega_X))^{\otimes \mathrm{rk}(\mathcal{E})} \otimes \det(\Gamma(X, \mathcal{O}_X))^{\otimes -\mathrm{rk}(\mathcal{E})} \otimes \mathrm{Weil}(\det(\mathcal{E}), \omega_X).$$

Applying (12.7) to $\mathcal{E} = \mathcal{O}_X$, we obtain

$$\det(\Gamma(X, \omega_X)) \otimes \det(\Gamma(X, \mathcal{O}_X))^{\otimes -1} \simeq \det(C(X))^{\otimes -1}.$$

Note, however, the line $\det(C(X))$ is canonically trivialized by the symplectic structure on $C(X)[1]$. Thus, the expression in (12.8) identifies with $\mathrm{Weil}(\det(\mathcal{E}), \omega_X)$, as desired.

We now prove

$$\det(T^*(\mathrm{LS}_{\check{P}-} / \mathrm{LS}_{\check{M}})) \simeq \mathcal{L}_{2\rho_P(\omega_X)};$$

the second isomorphism is proved similarly.

For $\sigma = (\mathcal{P}_{\check{P}-}, \nabla) \in \mathrm{LS}_{\check{P}-}$, the relative tangent space $T_\sigma(\mathrm{LS}_{\check{P}-} / \mathrm{LS}_{\check{M}})$ identifies with

$$C(X, (\check{\mathfrak{n}}_P^-)_\sigma)[1].$$

Hence, by (12.6),

$$(12.9) \quad \det(T^*(\mathrm{LS}_{\check{P}-} / \mathrm{LS}_{\check{M}}))_\sigma \simeq \mathrm{Weil}(\det((\check{\mathfrak{n}}_P^-)_{\mathcal{P}_{\check{P}-}}), \omega_X)^{\otimes -1}.$$

We have

$$\begin{aligned} \mathrm{Weil}(\det((\check{\mathfrak{n}}_P^-)_{\mathcal{P}_{\check{P}-}}), \omega_X) &\simeq \mathrm{Weil}(-2\rho_P(\mathcal{P}_{\check{P}-}), \omega_X) \otimes \mathrm{Weil}(\det(\check{\mathfrak{n}}_P^-) \otimes \mathcal{O}_X, \omega_X) \simeq \\ &\simeq \mathrm{Weil}(-2\rho_P(\mathcal{P}_{\check{P}-}), \omega_X) \otimes \det(\check{\mathfrak{n}}_P^-)^{\otimes (2g-2)} = (\mathcal{L}_{-2\rho_P(\omega_X)})_{\mathcal{P}_{\check{P}-}} \otimes \det(\check{\mathfrak{n}}_P^-)^{\otimes (2g-2)}. \end{aligned}$$

Finally, we claim that the line $\det(\check{\mathfrak{n}}_P^-)^{\otimes 2}$ is canonically trivial. Indeed, the Killing form on $\check{\mathfrak{g}}$ gives rise to an identification

$$\det(\check{\mathfrak{n}}_P^-) \simeq \det(\check{\mathfrak{n}}_P)^{\otimes -1},$$

while the Chevalley involution on $\check{\mathfrak{g}}$ gives rise to an identification

$$\check{\mathfrak{n}}_P^- \simeq \check{\mathfrak{n}}_P.$$

□

12.2. Spectral Eisenstein functor and localization. The goal of this section is to express the composition

$$\Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}} \circ \mathrm{Eis}^{-, \mathrm{spec}}$$

via $\Gamma_{\check{M}}^{\mathrm{spec}, \mathrm{IndCoh}}$ and a functor of local nature.

The main result is given by Corollary 12.2.6.

12.2.1. For $\mathcal{Z} \rightarrow \mathrm{Ran}$, recall the functor

$$\Gamma_{\check{G}, \mathrm{Ran}}^{\mathrm{spec}} : \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathcal{Z}) \rightarrow \mathrm{Rep}(\check{G})_{\mathcal{Z}},$$

see [GLC2, Sect. 17.6.2].

Explicitly, it is given by direct image along

$$\mathrm{LS}_{\check{G}} \times \mathcal{Z} \xrightarrow{\mathrm{ev}_{\mathcal{Z}}} \mathrm{LS}_{\check{G}, \mathcal{Z}}^{\mathrm{reg}}.$$

12.2.2. From the commutative diagram

$$(12.10) \quad \begin{array}{ccccc} \mathrm{LS}_{\check{G}, \mathcal{Z}}^{\mathrm{reg}} & \xleftarrow{\mathfrak{p}^{-, \mathrm{spec}}} & \mathrm{LS}_{\check{P}-, \mathcal{Z}}^{\mathrm{reg}} & \xrightarrow{\mathfrak{q}^{-, \mathrm{spec}}} & \mathrm{LS}_{\check{M}, \mathcal{Z}}^{\mathrm{reg}} \\ \mathrm{ev}_{\mathcal{Z}} \uparrow & & \mathrm{ev}_{\mathcal{Z}} \uparrow & & \uparrow \mathrm{ev}_{\mathcal{Z}} \\ \mathrm{LS}_{\check{G}} \times \mathcal{Z} & \xleftarrow{\mathfrak{p}^{-, \mathrm{spec}}} & \mathrm{LS}_{\check{P}-} \times \mathcal{Z} & \xrightarrow{\mathfrak{q}^{-, \mathrm{spec}}} & \mathrm{LS}_{\check{M}} \times \mathcal{Z} \end{array}$$

we obtain a natural transformation

$$(12.11) \quad \begin{array}{ccc} \mathrm{Rep}(\check{G})_{\mathcal{Z}} & \xleftarrow{\mathrm{co}J^{-, \mathrm{spec}, !}} & \mathrm{Rep}(\check{M})_{\mathcal{Z}} \\ \uparrow \Gamma_{\check{G}, \mathcal{Z}}^{\mathrm{spec}} & & \uparrow \Gamma_{\check{M}, \mathcal{Z}}^{\mathrm{spec}} \\ \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathcal{Z}) & \xleftarrow{\mathrm{Eis}_{\mathrm{QCoh}}^{\mathrm{spec}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathcal{Z}), \end{array}$$

where

$$\mathrm{co}J^{-, \mathrm{spec}, !} : \mathrm{Rep}(\check{M}) \rightarrow \mathrm{Rep}(\check{G})$$

is the functor dual to $J^{-,\text{spec},!}$, i.e., this is the functor

$$\text{Rep}(\check{M}) \xrightarrow{\text{Res}_{\check{P}^-}^{\check{M}}} \text{Rep}(\check{P}^-) \xrightarrow{\text{coind}_{\check{P}^-}^{\check{G}}} \text{Rep}(\check{G}).$$

12.2.3. Applying (12.11) to $\mathcal{Z} = \text{Ran}$ and taking into account that the essential image of the functor

$$\Gamma_{\check{G}} : \text{QCoh}(\text{LS}_{\check{G}}) \rightarrow \text{Rep}(\check{G})_{\text{Ran}}$$

lands in the full subcategory

$$\text{Rep}(\check{G})_{\text{Ran}^{\text{unl}}, \text{indep}} \subset \text{Rep}(\check{G})_{\text{Ran}},$$

we obtain a natural transformation

$$(12.12) \quad \begin{array}{ccccc} \text{Rep}(\check{G})_{\text{Ran}^{\text{unl}}, \text{indep}} & \xleftarrow{\text{emb.indep}_{\text{Rep}(\check{G})}^L} & \text{Rep}(\check{G})_{\text{Ran}^{\text{unl}}} & \xleftarrow{\text{co}J^{-,\text{spec},!}} & \text{Rep}(\check{M})_{\text{Ran}^{\text{unl}}} \\ \uparrow \Gamma_{\check{G}}^{\text{spec}} & & \nearrow & & \uparrow \Gamma_{\check{M}, \text{unl}}^{\text{spec}} \\ \text{QCoh}(\text{LS}_{\check{G}}) & \xleftarrow{\text{Eis}_{\text{QCoh}}^{\text{spec}}} & & & \text{QCoh}(\text{LS}_{\check{M}}). \end{array}$$

We will prove:

Theorem 12.2.4. *The natural transformation (12.12) is an isomorphism.*

This theorem is a spectral counterpart of Corollary 10.1.5. We will deduce it from Theorem 12.2.10, which is a spectral analog of Theorem 10.1.2.

12.2.5. Precomposing (12.11) with the functors $\Psi_{\text{LS}_{\check{G}}}$ and $\Psi_{\text{LS}_{\check{M}}}$ and using (12.1), we obtain a natural transformation

$$(12.13) \quad \begin{array}{ccc} \text{Rep}(\check{G})_{\mathcal{Z}} & \xleftarrow{\text{co}J^{-,\text{spec},!}} & \text{Rep}(\check{M})_{\mathcal{Z}} \\ \uparrow \Gamma_{\check{G}, \mathcal{Z}}^{\text{spec}, \text{IndCoh}} & & \uparrow \Gamma_{\check{M}, \mathcal{Z}}^{\text{spec}, \text{IndCoh}} \\ \text{IndCoh}(\text{LS}_{\check{G}}) \otimes \text{D-mod}(\mathcal{Z}) & \xleftarrow{\text{Eis}^{-,\text{spec}}} & \text{IndCoh}(\text{LS}_{\check{M}}) \otimes \text{D-mod}(\mathcal{Z}), \end{array}$$

and further

$$(12.14) \quad \begin{array}{ccccc} \text{Rep}(\check{G})_{\text{Ran}^{\text{unl}}, \text{indep}} & \xleftarrow{\text{emb.indep}_{\text{Rep}(\check{G})}^L} & \text{Rep}(\check{G})_{\text{Ran}^{\text{unl}}} & \xleftarrow{\text{co}J^{-,\text{spec},!}} & \text{Rep}(\check{M})_{\text{Ran}^{\text{unl}}} \\ \uparrow \Gamma_{\check{G}}^{\text{spec}, \text{IndCoh}} & & \nearrow & & \uparrow \Gamma_{\check{M}, \text{unl}}^{\text{spec}, \text{IndCoh}} \\ \text{IndCoh}(\text{LS}_{\check{G}}) & \xleftarrow{\text{Eis}^{-,\text{spec}}} & & & \text{IndCoh}(\text{LS}_{\check{M}}). \end{array}$$

From Theorem 12.2.4 we obtain:

Corollary 12.2.6. *The natural transformation (12.14) is an isomorphism.*

12.2.7. In order to prove Theorem 12.2.4 we will reformulate it in dual terms. Recall the functor

$$\mathrm{Loc}_{\check{G}, \mathbb{Z}}^{\mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathbb{Z}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z}),$$

given by pullback along $\mathrm{ev}_{\mathbb{Z}}$, see [GLC2, Sect. 17.6.1].

Diagram (12.10) gives rise to a natural transformation

$$(12.15) \quad \begin{array}{ccc} \mathrm{Rep}(\check{G})_{\mathbb{Z}} & \xrightarrow{J^{-, \mathrm{spec}, !}} & \mathrm{Rep}(\check{M})_{\mathbb{Z}} \\ \mathrm{Loc}_{\check{G}, \mathbb{Z}}^{\mathrm{spec}} \downarrow & \nearrow & \downarrow \mathrm{Loc}_{\check{M}, \mathbb{Z}}^{\mathrm{spec}} \\ \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z}) & \xrightarrow{\mathrm{CT}_{\mathrm{QCoh}}^{-, \mathrm{spec}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathbb{Z}), \end{array}$$

which can be viewed as obtained by passing to dual functors in (12.11).

12.2.8. We can view the system of natural transformations (12.15) as a map

$$(12.16) \quad \underline{\mathrm{Loc}}_{\check{M}}^{\mathrm{spec}} \circ \underline{J}^{-, \mathrm{spec}, !} \rightarrow (\mathrm{CT}_{\mathrm{QCoh}}^{\mathrm{spec}} \otimes \mathrm{Id}) \circ \underline{\mathrm{Loc}}_{\check{G}}^{\mathrm{spec}}$$

as lax unital local-to-global functors

$$\underline{\mathrm{Rep}}(\check{G}) \rightrightarrows \mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}),$$

with the right-hand sides being strictly unital.

By [GLC2, Sect. 11.4.8], the natural transformation (12.16) gives rise to a natural transformation

$$(12.17) \quad \left(\underline{\mathrm{Loc}}_{\check{M}}^{\mathrm{spec}} \circ \underline{J}^{-, \mathrm{spec}, !} \right)^f \circ \mathrm{ins.unit} \rightarrow (\mathrm{CT}_{\mathrm{QCoh}}^{\mathrm{spec}} \otimes \mathrm{Id}) \circ \underline{\mathrm{Loc}}_{\check{G}}^{\mathrm{spec}}.$$

In other words, we obtain a system of natural transformations

$$(12.18) \quad \begin{array}{ccccc} \mathrm{Rep}(\check{G})_{\mathbb{Z}} & \xrightarrow{\mathrm{ins.unit}_{\mathbb{Z}}} & \mathrm{Rep}(\check{G})_{\mathbb{Z} \subseteq} & \xrightarrow{J^{-, \mathrm{spec}, !}} & \mathrm{Rep}(\check{M})_{\mathbb{Z} \subseteq} \\ \mathrm{Loc}_{\check{G}, \mathbb{Z}}^{\mathrm{spec}} \downarrow & & & \nearrow & \downarrow \mathrm{Loc}_{\check{M}, \mathbb{Z} \subseteq}^{\mathrm{spec}} \\ \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z}) & \xrightarrow{\mathrm{CT}_{\mathrm{QCoh}}^{-, \mathrm{spec}} \otimes \mathrm{Id}} & \mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathbb{Z}) & \xleftarrow{\mathrm{Id} \otimes (\mathrm{pr}_{\mathrm{small}, \mathbb{Z}})_!} & \mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathbb{Z}^{\subseteq}). \end{array}$$

12.2.9. We will prove:

Theorem 12.2.10. *The natural transformation (12.17) is an isomorphism.*

12.2.11. We claim that Theorem 12.2.10 formally implies Theorem 12.2.4. Indeed, this is a duality manipulation parallel to one the proof of Corollary 10.1.5.

12.2.12. Recall (see [GLC2, Sect. 17.6.4]) that by a slight abuse of notation, we use the same symbol $\mathrm{Loc}_{\check{G}, \mathbb{Z}}^{\mathrm{spec}}$ to denote the composition:

$$\mathrm{Rep}(\check{G})_{\mathbb{Z}} \xrightarrow{\mathrm{Loc}_{\check{G}, \mathbb{Z}}^{-, \mathrm{spec}}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z}) \xrightarrow{\Xi_{\mathrm{LS}_{\check{G}}} \otimes \mathrm{Id}} \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z}).$$

12.2.13. From Theorem 12.2.10 and Lemmas 12.1.6 and 12.1.9 we obtain:

Corollary 12.2.14. *The following diagram of functors commutes:*

$$\begin{array}{ccccc}
 \mathrm{Rep}(\check{G})_{\mathbb{Z}} & \xrightarrow{\mathrm{ins.unit}_{\mathbb{Z}}} & \mathrm{Rep}(\check{G})_{\mathbb{Z} \subseteq} & \xrightarrow{J^{-,\mathrm{spec},!}} & \mathrm{Rep}(\check{M})_{\mathbb{Z} \subseteq} \\
 \mathrm{Loc}_{\check{G},\mathbb{Z}}^{-,\mathrm{spec}} \downarrow & & & & \downarrow \mathrm{Loc}_{\check{M},\mathbb{Z} \subseteq}^{-,\mathrm{spec}} \\
 \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z}) & \longrightarrow & \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathbb{Z}) & \xleftarrow{\mathrm{Id} \otimes (\mathrm{pr}_{\mathrm{small},\mathbb{Z}})^!} & \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathbb{Z}^{\subseteq}),
 \end{array}$$

where the left bottom horizontal arrow is the tensor product of $\mathrm{Id}_{\mathrm{D-mod}(\mathbb{Z})}$ with

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{\mathrm{CT}^{-,\mathrm{spec}}} \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \xrightarrow{(-) \otimes \mathcal{L}_{2\rho_P(\omega_X)}[(2g-2) \cdot \dim(\check{n}_P)]} \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}).$$

Remark 12.2.15. Corollary 12.2.14 is the Langlands-dual counterpart of Corollary 10.1.8. Note that the functor

$$(\mathrm{transl}_{2\rho_P(\omega_X)})^* : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M),$$

appearing in Corollary 10.1.8, is the Langlands-dual of

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{(-) \otimes \mathcal{L}_{2\rho_P(\omega_X)}} \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}).$$

12.3. Proof of Theorem 12.2.10.

12.3.1. The natural transformation (12.15) is obtained by concatenating the commutative diagram

$$\begin{array}{ccc}
 \mathrm{Rep}(\check{G})_{\mathbb{Z}} & \xrightarrow{\mathrm{Res}_{\check{P}-}^{\check{G}}} & \mathrm{Rep}(\check{P}^-)_{\mathbb{Z}} \\
 \mathrm{Loc}_{\check{G},\mathbb{Z}}^{\mathrm{spec}} \downarrow & & \downarrow \mathrm{Loc}_{\check{P}-,\mathbb{Z}}^{\mathrm{spec}} \\
 \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z}) & \xrightarrow{(\mathfrak{p}^{-,\mathrm{spec}})^* \otimes \mathrm{Id}} & \mathrm{QCoh}(\mathrm{LS}_{\check{P}-}) \otimes \mathrm{D-mod}(\mathbb{Z})
 \end{array}$$

with the natural transformation

$$\begin{array}{ccc}
 \mathrm{Rep}(\check{P}^-)_{\mathbb{Z}} & \xrightarrow{\mathrm{inv}_{\check{N}_P^-}} & \mathrm{Rep}(\check{M})_{\mathbb{Z}} \\
 \mathrm{Loc}_{\check{P}-,\mathbb{Z}}^{\mathrm{spec}} \downarrow & \nearrow & \downarrow \mathrm{Loc}_{\check{M},\mathbb{Z}}^{\mathrm{spec}} \\
 \mathrm{QCoh}(\mathrm{LS}_{\check{P}-}) \otimes \mathrm{D-mod}(\mathbb{Z}) & \xrightarrow{(\mathfrak{q}^{-,\mathrm{spec}})_* \otimes \mathrm{Id}} & \mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathbb{Z}).
 \end{array} \tag{12.19}$$

Hence, it is enough to show that the resulting map

$$\left(\mathrm{Loc}_{\check{M}}^{\mathrm{spec}} \circ \mathrm{inv}_{\check{N}_P^-} \right)^f \mathrm{ins.unit} \rightarrow ((\mathfrak{q}^{-,\mathrm{spec}})_* \otimes \mathrm{Id}) \circ \mathrm{Loc}_{\check{P}-}^{\mathrm{spec}} \tag{12.20}$$

is an isomorphism.

12.3.2. To prove that (12.20) is an isomorphism, it suffices to show that it becomes such after taking $*$ -fibers at all field-valued points. Let σ be a k' -valued point of $\mathrm{LS}_{\check{M}}$.

Up to base changing the situation, we can assume that $k' = k$.

12.3.3. Let $\check{N}_{P,\sigma}^-$ be the twist of the constant group-scheme with a connection with fiber $\check{N}_P^-$ by σ (via the adjoint action of $\check{M}$ on $\check{N}_P^-$). Note that

$$\mathrm{LS}_{\check{N}_{P,\sigma}^-} \simeq \mathrm{pt} \times_{\sigma, \mathrm{LS}_{\check{M}}} \mathrm{LS}_{\check{P}^-}.$$

For $\mathcal{Z} \rightarrow \mathrm{Ran}$, the map

$$\mathrm{ev}_{\mathcal{Z}} : \mathrm{LS}_{\check{N}_{P,\sigma}^-} \times \mathcal{Z} \rightarrow \mathrm{LS}_{\check{N}_{P,\sigma}^-, \mathcal{Z}}^{\mathrm{reg}}$$

gives rise to a natural transformation

$$(12.21) \quad \begin{array}{ccc} \mathrm{Rep}(\check{N}_{P,\sigma}^-)_{\mathcal{Z}} & \xrightarrow{\mathrm{inv}_{\check{N}_{P,\sigma}^-}} & \mathrm{D-mod}(\mathcal{Z}) \\ \downarrow \mathrm{Loc}_{\check{N}_{P,\sigma}^-, \mathcal{Z}}^{\mathrm{spec}} & \nearrow & \downarrow \mathrm{Id} \\ \mathrm{QCoh}(\mathrm{LS}_{\check{N}_{P,\sigma}^-}) \otimes \mathrm{D-mod}(\mathcal{Z}) & \xrightarrow{\Gamma(\mathrm{LS}_{\check{N}_{P,\sigma}^-}, -) \otimes \mathrm{Id}} & \mathrm{D-mod}(\mathcal{Z}). \end{array}$$

The concatenation of (12.19) with the functor of taking the $*$ -fiber at σ identifies with the concatenation of the commutative diagram

$$\begin{array}{ccc} \mathrm{Rep}(\check{P}^-)_{\mathcal{Z}} & \longrightarrow & \mathrm{Rep}(\check{N}_{P,\sigma}^-)_{\mathcal{Z}} \\ \downarrow \mathrm{Loc}_{\check{P}^-, \mathcal{Z}}^{\mathrm{spec}} & & \downarrow \mathrm{Loc}_{\check{N}_{P,\sigma}^-, \mathcal{Z}}^{\mathrm{spec}} \\ \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}) \otimes \mathrm{D-mod}(\mathcal{Z}) & \longrightarrow & \mathrm{QCoh}(\mathrm{LS}_{\check{N}_{P,\sigma}^-}) \otimes \mathrm{D-mod}(\mathcal{Z}) \end{array}$$

(where the horizontal arrow are given by $*$ -restriction) with the natural transformation (12.21).

Hence, it suffices to show that the map

$$(12.22) \quad (\mathrm{inv}_{\check{N}_{P,\sigma}^-})^f \circ \mathrm{ins.unit} \rightarrow (\Gamma(\mathrm{LS}_{\check{N}_{P,\sigma}^-}, -) \otimes \mathrm{Id}) \circ \mathrm{Loc}_{\check{N}_{P,\sigma}^-, \mathcal{Z}}^{\mathrm{spec}}$$

as functors

$$\underline{\mathrm{Rep}}(\check{N}_{P,\sigma}^-) \rightrightarrows \underline{\mathrm{D-mod}}(\mathrm{Ran})$$

is an isomorphism.

12.3.4. Since $\check{N}_{P,\sigma}^-$ is unipotent, the category $\mathrm{Rep}(\check{N}_{P,\sigma}^-)$ is generated by the trivial representation. Hence, in order to prove that (12.22) is an isomorphism, it is enough to show that the map

$$(12.23) \quad \mathrm{C}^{\mathrm{fact}}(X, \mathrm{inv}_{\check{N}_{P,\sigma}^-}(k)) \rightarrow \Gamma\left(\mathrm{LS}_{\check{N}_{P,\sigma}^-}, \mathcal{O}_{\mathrm{LS}_{\check{N}_{P,\sigma}^-}}\right),$$

induced by (12.22) is an isomorphism.

12.3.5. Recall the construction

$$R \mapsto \text{“Spec}(R)\text{”}$$

from [GLC2, Sect. 12.7.6], and reviewed in Sect. C.4.3.

Since $\check{N}_P^-$ is unipotent, the classifying stack $\mathrm{pt}/\check{N}_{P,\sigma}^-$ identifies with

$$\text{“Spec}_X(\mathrm{inv}_{\check{N}_{P,\sigma}^-}(k))\text{”}.$$

From here it formally follows that

$$\mathrm{LS}_{\check{N}_{P,\sigma}^-} \simeq \text{“Spec}(\mathrm{C}^{\mathrm{fact}}(X, \mathrm{inv}_{\check{N}_{P,\sigma}^-}(k)))\text{”},$$

cf. [GLC2, last formula in Sect. 12.7.4].

In terms of this identification, the map (12.23) identifies with the map [GLC2, Equation (12.46)], which is reproduced as (C.7).

Hence, in order to show that (12.23) is an isomorphism, we have to show that

$$C^{\text{fact}}(X, \text{inv}_{\tilde{N}_{P,\sigma}^-}(k)) \in \text{ComAlg}(\text{Vect})$$

is *as good as connective* (see [GLC2, Sect. 12.7.6] or Sect. C.4.3 for what this means).

However, this is done by the same argument as Proposition C.4.6.

12.4. Global (co-)enhanced categories on the spectral side.

12.4.1. Let $\mathcal{Z}$ be a prestack mapping to Ran . We define the category $\text{IndCoh}(\text{LS}_{\tilde{M}})_{\mathcal{Z}}^{-,\text{enh}}$ in a way parallel to Sect. 8.2. Namely,

$$\text{IndCoh}(\text{LS}_{\tilde{M}})_{\mathcal{Z}}^{-,\text{enh}} := (\text{IndCoh}(\text{LS}_{\tilde{M}}) \otimes \text{D-mod}(\mathcal{Z})) \otimes_{\text{Sph}_{\tilde{M},\mathcal{Z}}^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)_{\mathcal{Z}}^{\text{spec,loc}}.$$

The assignment

$$(12.24) \quad \mathcal{Z} \rightsquigarrow \text{IndCoh}(\text{LS}_{\tilde{M}})_{\mathcal{Z}}^{-,\text{enh}}$$

is a crystal of categories over Ran , which we will denote by

$$\underline{\text{IndCoh}}(\text{LS}_{\tilde{M}})^{-,\text{enh}}.$$

12.4.2. The monadic adjunction

$$\mathbf{ind}_{\text{Sph} \rightarrow \frac{\infty}{2}}^{\text{spec}} : \text{Sph}_{\tilde{M}}^{\text{spec}} \rightleftarrows \text{I}(\check{G}, \check{P}^-)^{\text{spec,loc}} : \mathbf{oblv}_{\frac{\infty}{2} \rightarrow \text{Sph}}^{\text{spec}}$$

gives rise to an adjunction

$$(12.25) \quad \mathbf{ind}_{\text{enh}}^{\text{spec}} : \text{IndCoh}(\text{LS}_{\tilde{M}}) \otimes \text{D-mod}(\mathcal{Z}) \rightleftarrows \text{IndCoh}(\text{LS}_{\tilde{M}})_{\mathcal{Z}}^{-,\text{enh}} : \mathbf{oblv}_{\text{enh}}^{\text{spec}}.$$

12.4.3. Here is another way to think about the categories (12.24).

Let $\tilde{\Omega}_{\mathcal{Z}}^{\text{spec}}$ be as in Sect. 2.2.1. Consider

$$\tilde{\Omega}_{\mathcal{Z}}^{\text{spec}} \in \text{AssocAlg}(\text{Sph}_{\tilde{M},\mathcal{Z}}^{\text{spec}}),$$

where we consider $\text{Sph}_{\tilde{M},\mathcal{Z}}^{\text{spec}}$ as a monoidal category.

Consider $\text{IndCoh}(\text{LS}_{\tilde{M}}) \otimes \text{D-mod}(\mathcal{Z})$ as a right module category over $\text{Sph}_{\tilde{M},\mathcal{Z}}^{\text{spec}}$. Then

$$\text{IndCoh}(\text{LS}_{\tilde{M}})_{\mathcal{Z}}^{-,\text{enh}} \simeq \tilde{\Omega}_{\mathcal{Z}}^{\text{spec}}\text{-mod}^r(\text{IndCoh}(\text{LS}_{\tilde{M}}) \otimes \text{D-mod}(\mathcal{Z})),$$

so that (12.25) is the adjunction

$$\mathbf{ind}_{\tilde{\Omega}_{\mathcal{Z}}} : \text{IndCoh}(\text{LS}_{\tilde{M}}) \otimes \text{D-mod}(\mathcal{Z}) \rightleftarrows \tilde{\Omega}_{\mathcal{Z}}^{\text{spec}}\text{-mod}^r(\text{IndCoh}(\text{LS}_{\tilde{M}}) \otimes \text{D-mod}(\mathcal{Z})) : \mathbf{oblv}_{\tilde{\Omega}_{\mathcal{Z}}}.$$

12.4.4. Consider the full category

$$\text{IndCoh}(\text{LS}_{\tilde{M}})_{\text{Ran}^{\text{untl},\text{indep}}}^{-,\text{enh}} \subset \text{IndCoh}(\text{LS}_{\tilde{M}})_{\text{Ran}}^{-,\text{enh}}.$$

As in (8.7), it fits into the pullback square

$$(12.26) \quad \begin{array}{ccc} \text{IndCoh}(\text{LS}_{\tilde{M}})_{\text{Ran}^{\text{untl},\text{indep}}}^{-,\text{enh}} & \longrightarrow & \text{IndCoh}(\text{LS}_{\tilde{M}})_{\text{Ran}}^{-,\text{enh}} \\ \mathbf{oblv}_{\text{enh}}^{\text{spec}} \downarrow & & \downarrow \mathbf{oblv}_{\text{enh}}^{\text{spec}} \\ \text{IndCoh}(\text{LS}_{\tilde{M}}) \otimes \text{D-mod}(\text{Ran}^{\text{untl}})_{\text{indep}} & \longrightarrow & \text{IndCoh}(\text{LS}_{\tilde{M}}) \otimes \text{D-mod}(\text{Ran}) \\ \sim \downarrow & & \\ \text{IndCoh}(\text{LS}_{\tilde{M}}) & & \end{array}$$

12.4.5. In a way parallel to Sect. 8.2.6, we define

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{G}})^{-, \mathrm{enh}_{\mathrm{co}}} := (\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathcal{Z})) \otimes_{\mathrm{Sph}_{\check{G}, \mathcal{Z}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}_{\mathrm{co}, \mathcal{Z}}.$$

We regard the assignment

$$\mathcal{Z} \rightsquigarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})^{-, \mathrm{enh}_{\mathrm{co}}}$$

as a crystal of categories over Ran , to be denoted

$$\underline{\mathrm{IndCoh}}(\mathrm{LS}_{\check{G}})^{-, \mathrm{enh}_{\mathrm{co}}}.$$

12.5. The enhanced spectral Eisenstein functor. In this subsection we define the *enhanced* spectral Eisenstein functor, which is a unital local-to-global functor

$$\underline{\mathrm{Eis}}^{-, \mathrm{spec}, \mathrm{enh}} : \underline{\mathrm{IndCoh}}(\mathrm{LS}_{\check{M}})^{-, \mathrm{enh}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}).$$

12.5.1. Let $\mathrm{LS}_{\check{P}^-, \mathrm{Ran}}^{\mathrm{mer}, \mathrm{glob}}$ be the space of meromorphic $\check{P}^-$ -local systems on X with poles at specified points, defined as in [GLC2, Sect. B.7.15].

Denote

$$\mathrm{LS}_{\check{P}^-, \mathrm{Ran}}^{\mathrm{mon-free}_{\check{G}}, \mathrm{glob}} := (\mathrm{LS}_{\check{G}} \times \mathrm{Ran}) \times_{\mathrm{LS}_{\check{G}, \mathrm{Ran}}^{\mathrm{mer}, \mathrm{glob}}} \mathrm{LS}_{\check{P}^-, \mathrm{Ran}}^{\mathrm{mer}, \mathrm{glob}}.$$

We will denote by $\mathfrak{p}_{\mathrm{Ran}}^{-, \mathrm{spec}}$ the map

$$\mathrm{LS}_{\check{P}^-, \mathrm{Ran}}^{\mathrm{mon-free}_{\check{G}}, \mathrm{glob}} \rightarrow \mathrm{LS}_{\check{G}} \times \mathrm{Ran}.$$

Given $\mathcal{Z} \rightarrow \mathrm{Ran}$, we will substitute $\mathrm{Ran} \rightsquigarrow \mathcal{Z}$ in the subscript to denote the corresponding base-changed objects.

12.5.2. Let $\mathfrak{s}_{\mathrm{Ran}}^{\mathrm{spec}, \check{G}}$ denote the naturally defined map

$$(12.27) \quad \mathrm{LS}_{\check{P}^-, \mathrm{Ran}}^{\mathrm{mon-free}_{\check{G}}, \mathrm{glob}} \rightarrow \mathrm{LS}_{\check{G}, \mathrm{Ran}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}, \mathrm{Ran}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \mathrm{Ran}}^{\mathrm{mer}} =: \mathrm{LS}_{\check{P}^-, \mathrm{Ran}}^{\mathrm{mon-free}_{\check{G}}}.$$

Note that we have a commutative (but not Cartesian) diagram

$$\begin{array}{ccc} \mathrm{LS}_{\check{P}^-, \mathrm{Ran}}^{\mathrm{mon-free}_{\check{G}}, \mathrm{glob}} & \xrightarrow{\mathfrak{p}_{\mathrm{Ran}}^{-, \mathrm{spec}}} & \mathrm{LS}_{\check{G}} \times \mathrm{Ran} \\ \mathfrak{s}_{\mathrm{Ran}}^{\mathrm{spec}, \check{G}} \downarrow & & \downarrow \mathrm{ev}_{\mathrm{Ran}}^{\check{G}} \\ \mathrm{LS}_{\check{P}^-, \mathrm{Ran}}^{\mathrm{mon-free}_{\check{G}}} & \xrightarrow[\mathfrak{p}_{\mathrm{Ran}}^{-, \mathrm{spec}, \mathrm{loc}}]{} & \mathrm{LS}_{\check{G}, \mathrm{Ran}}^{\mathrm{reg}}, \end{array}$$

where

$$\mathfrak{p}_{\mathrm{Ran}}^{-, \mathrm{spec}, \mathrm{loc}} : \mathrm{LS}_{\check{P}^-, \mathrm{Ran}}^{\mathrm{mon-free}_{\check{G}}} \rightarrow \mathrm{LS}_{\check{G}, \mathrm{Ran}}^{\mathrm{reg}}$$

is the projection on the first factor.

12.5.3. We have a naturally defined map

$$(12.28) \quad \iota^{\mathrm{glob}, \check{G}} : \mathrm{LS}_{\check{P}^-} \times \mathrm{Ran} \rightarrow \mathrm{LS}_{\check{P}^-, \mathrm{Ran}}^{\mathrm{mon-free}_{\check{G}}, \mathrm{glob}},$$

so that the diagram

$$\begin{array}{ccc} \mathrm{LS}_{\check{P}^-} \times \mathrm{Ran} & \xrightarrow{\iota^{\mathrm{glob}, \check{G}}} & \mathrm{LS}_{\check{P}^-, \mathrm{Ran}}^{\mathrm{mon-free}_{\check{G}}, \mathrm{glob}} \\ \mathrm{ev}_{\mathrm{Ran}}^{\check{P}^-} \downarrow & & \downarrow \mathfrak{s}_{\mathrm{Ran}}^{\mathrm{spec}, \check{G}} \\ \mathrm{LS}_{\check{P}^-, \mathrm{Ran}}^{\mathrm{reg}} & \xrightarrow[\iota^{\check{G}}]{} & \mathrm{LS}_{\check{P}^-, \mathrm{Ran}}^{\mathrm{mon-free}_{\check{G}}} \end{array}$$

is Cartesian.

The map $\iota^{\text{glob}, \check{G}}$ is an isomorphism at the classical level. This observation, combined with the calculation of cotangent spaces implies that $\text{LS}_{\check{P}^-, \text{Ran}}^{\text{mon-free}_{\check{G}}, \text{glob}}$ is a (derived) algebraic stack locally almost of finite type (see [GaRo4, Chapter 1, Theorem 9.1.2]).

In particular, we obtain that the category $\text{IndCoh}(\text{LS}_{\check{P}^-, \text{Ran}}^{\text{mon-free}_{\check{G}}, \text{glob}})$ is well-defined.

12.5.4. We claim that there is a well-defined functor

$$\text{IndCoh}(\text{LS}_{\check{M}}^-)^{-, \text{enh}} \rightarrow \text{IndCoh}(\text{LS}_{\check{P}^-, \check{Z}}^{\text{mon-free}_{\check{G}}, \text{glob}}),$$

to be denoted $(\mathbf{q}_{\check{Z}}^{-, \text{spec}})^*, \text{IndCoh, enh}$.

We will perform this construction in the present subsection when $\check{Z} = \underline{x}$, and the case of a general $\check{Z}$ will be considered in Sects. A.8.2-A.8.7.

12.5.5. Consider the map

$$\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}_{\check{G}}} \rightarrow \text{LS}_{\check{P}^-, \underline{x}}^{\text{mer}} \rightarrow \text{LS}_{\check{M}, \underline{x}}^{\text{mer}},$$

and the fiber product

$$(12.29) \quad \text{LS}_{\check{M}, \underline{x}}^{\text{mer, glob}} \times_{\text{LS}_{\check{M}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}_{\check{G}}}.$$

We claim that when $\underline{x}$ is fixed, (12.29) is an algebraic stack locally almost of finite type. This follows from the calculation of cotangent spaces combined with the fact that the map

$$(12.30) \quad \text{LS}_{\check{M}} \times_{\text{LS}_{\check{M}, \underline{x}}^{\text{reg}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{reg}} \rightarrow \text{LS}_{\check{M}, \underline{x}}^{\text{mer, glob}} \times_{\text{LS}_{\check{M}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}_{\check{G}}}$$

is an isomorphism at the classical level.

In particular, the category $\text{IndCoh}(\text{LS}_{\check{M}, \underline{x}}^{\text{mer, glob}} \times_{\text{LS}_{\check{M}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}_{\check{G}}})$ is well-defined. Moreover, as in (A.19), we have a canonical isomorphism

$$(12.31) \quad \text{IndCoh}(\text{LS}_{\check{M}}^-)^{-, \text{enh}} := \text{IndCoh}(\text{LS}_{\check{M}}^-) \otimes_{\text{Sph}_{\check{M}, \underline{x}}^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}} \simeq \text{IndCoh}(\text{LS}_{\check{M}, \underline{x}}^{\text{mer, glob}} \times_{\text{LS}_{\check{M}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}_{\check{G}}}).$$

12.5.6. Combining the maps

$$\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}_{\check{G}}, \text{glob}} \rightarrow \text{LS}_{\check{P}^-, \underline{x}}^{\text{mer, glob}} \rightarrow \text{LS}_{\check{M}, \underline{x}}^{\text{mer, glob}}$$

and $\mathbf{s}_{\underline{x}}^{\text{spec}, \check{G}}$, we obtain a map

$$\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}_{\check{G}}, \text{glob}} \rightarrow \text{LS}_{\check{M}, \underline{x}}^{\text{mer, glob}} \times_{\text{LS}_{\check{M}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}_{\check{G}}},$$

which we denote by $\mathbf{q}_{\underline{x}}^{-, \text{spec, enh}}$.

We claim that $\mathbf{q}_{\underline{x}}^{-, \text{spec, enh}}$ is schematic of finite Tor-dimension. Indeed, its base change by the nil-isomorphism (12.30) is the map

$$\text{LS}_{\check{P}^-} \xrightarrow{\mathbf{q}_{\underline{x}}^{-, \text{spec}} \times \text{ev}_{\underline{x}}^{\check{P}^-}} \text{LS}_{\check{M}} \times_{\text{LS}_{\check{M}, \underline{x}}^{\text{reg}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{reg}},$$

which has the required property.

12.5.7. Hence, we obtain a well-defined functor

$$(\mathbf{q}_{\underline{x}}^{-,\text{spec},\text{enh}})^{\text{IndCoh},*} : \text{IndCoh}(\text{LS}_{\check{M},\underline{x}}^{\text{mer},\text{glob}} \times_{\text{LS}_{\check{M},\underline{x}}^{\text{mer}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}\check{G}}) \rightarrow \text{IndCoh}(\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}\check{G},\text{glob}}).$$

Combining with (12.31), we obtain the sought-for functor

$$(\mathbf{q}_{\underline{x}}^{-,\text{spec}})^{*,\text{IndCoh},\text{enh}} : \text{IndCoh}(\text{LS}_{\check{M}}^{-,\text{enh}})_{\underline{x}} \rightarrow \text{IndCoh}(\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}\check{G},\text{glob}}).$$

Remark 12.5.8. Here is another way to think about the composition

$$(12.32) \quad \text{IndCoh}(\text{LS}_{\check{M}}) \otimes \text{I}(\check{G}, \check{P}^-)_{\underline{x}}^{\text{spec},\text{loc}} \rightarrow \text{IndCoh}(\text{LS}_{\check{M}}^{-,\text{enh}})_{\underline{x}} \xrightarrow{(\mathbf{q}_{\underline{x}}^{-,\text{spec}})^{*,\text{IndCoh},\text{enh}}} \text{IndCoh}(\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}\check{G},\text{glob}}).$$

Denote:

$$\text{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\text{spec},\text{glob}} := \text{LS}_{\check{G}} \times_{\text{LS}_{\check{G}, \underline{x}}^{\text{mer},\text{glob}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{mer},\text{glob}} \times_{\text{LS}_{\check{M}, \underline{x}}^{\text{mer},\text{glob}}} \text{LS}_{\check{M}}.$$

This is an algebraic stack, locally almost of finite type, equipped with maps

$$\text{LS}_{\check{G}} \xleftarrow{\mathbf{p}_{\underline{x}}^{-,\text{spec},\text{mer}}} \text{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\text{spec},\text{glob}} \xrightarrow{\mathbf{q}_{\underline{x}}^{-,\text{spec},\text{mer}}} \text{LS}_{\check{M}}.$$

and

$$\mathbf{s}_{\underline{x}}^{\text{spec}} : \text{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\text{spec},\text{glob}} \rightarrow \text{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\text{spec},\text{loc}},$$

where $\text{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\text{spec},\text{loc}}$ is as in Sect. 2.1.1.

We claim that the functor

$$(12.33) \quad (\mathbf{q}_{\underline{x}}^{-,\text{spec},\text{mer}})^*(-) \otimes (\mathbf{s}_{\underline{x}}^{\text{spec}})^*(-),$$

which maps

$$\text{IndCoh}(\text{LS}_{\check{M}}) \otimes \text{I}(\check{G}, \check{P}^-)_{\underline{x}}^{\text{spec},\text{loc}} = \text{IndCoh}(\text{LS}_{\check{M}}) \otimes \text{IndCoh}(\text{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\text{spec},\text{loc}}) \rightarrow \text{IndCoh}(\text{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\text{spec},\text{glob}}),$$

is well-defined. Indeed, this follows from the fact that the map

$$\text{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\text{spec},\text{glob}} \xrightarrow{\mathbf{q}_{\underline{x}}^{-,\text{spec},\text{mer}} \times \mathbf{s}_{\underline{x}}^{\text{spec}}} \text{LS}_{\check{M}} \times \text{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\text{spec},\text{loc}}$$

is schematic of finite Tor-dimension. (The former follows by deformation theory using, e.g., [AG, Theorem B.2.14]; for the latter, we observe that the base change of the above morphism with respect to the nil-embedding

$$\text{LS}_{\check{P}^-, \underline{x}}^{\text{reg}} \rightarrow \text{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\text{spec},\text{loc}}$$

is the morphism

$$\text{LS}_{\check{P}^-} \xrightarrow{\mathbf{q}^{-,\text{spec}} \times \text{ev}_{\underline{x}}} \text{LS}_{\check{M}} \times \text{LS}_{\check{P}^-, \underline{x}}^{\text{reg}},$$

which is of finite Tor-dimension.)

Now, the functor (12.32) is isomorphic to the composition of (12.33) with the $(\text{IndCoh}, *)$ -pushforward functor along

$$\text{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\text{spec},\text{glob}} \rightarrow \text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}\check{G},\text{glob}}.$$

12.5.9. The spectral Hecke groupoid $\text{Hecke}_{\check{G}, \underline{x}}^{\text{spec},\text{loc}}$ acts naturally on

$$\text{LS}_{\check{M}, \underline{x}}^{\text{mer},\text{glob}} \times_{\text{LS}_{\check{M}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}\check{G}} \text{ and } \text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}\check{G},\text{glob}},$$

and the map $\mathbf{q}_{\underline{x}}^{-,\text{spec},\text{enh}}$ is compatible with this action.

From here we obtain that the functor $(\mathbf{q}_{\underline{x}}^{-,\text{spec}})^{*,\text{IndCoh},\text{enh}}$ is compatible with the actions of $\text{Sph}_{\check{G}, \underline{x}}^{\text{spec}}$ on the two sides. The same will be true when $\underline{x} \in \text{Ran}$ is replaced by an arbitrary $\mathcal{Z} \rightarrow \text{Ran}$.

12.5.10. We define the functor

$$\mathrm{Eis}_Z^{-,\mathrm{spec},\mathrm{enh}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_Z^{-,\mathrm{enh}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z})$$

as the composition

$$(\mathfrak{p}_Z^{-,\mathrm{spec}})_*^{\mathrm{IndCoh}} \circ (\mathfrak{q}_Z^{-,\mathrm{spec}})^*, \mathrm{IndCoh}, \mathrm{enh}.$$

Since both functors $(\mathfrak{q}_Z^{-,\mathrm{spec}})^*, \mathrm{IndCoh}, \mathrm{enh}$ and $(\mathfrak{p}_Z^{-,\mathrm{spec}})_*^{\mathrm{IndCoh}}$ respect the actions of $\mathrm{Sph}_{\check{G},Z}^{\mathrm{spec}}$, we obtain that $\mathrm{Eis}_Z^{-,\mathrm{spec},\mathrm{enh}}$ also respects the actions of $\mathrm{Sph}_{\check{G},Z}^{\mathrm{spec}}$.

12.5.11. Unwinding the construction, we obtain that the composition

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathbb{Z}) \xrightarrow{\mathrm{ind}_{\mathrm{enh}}^{\mathrm{spec}}} \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_Z^{-,\mathrm{enh}} \xrightarrow{\mathrm{Eis}_Z^{-,\mathrm{spec},\mathrm{enh}}} \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z})$$

identifies canonically with $\mathrm{Eis}^{-,\mathrm{spec}} \otimes \mathrm{Id}$.

12.5.12. We will view the assignment

$$\mathbb{Z} \rightsquigarrow \mathrm{Eis}_Z^{-,\mathrm{spec},\mathrm{enh}}$$

as a local-to-global functor

$$\underline{\mathrm{IndCoh}}(\mathrm{LS}_{\check{M}})^{-,\mathrm{enh}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}),$$

to be denoted $\underline{\mathrm{Eis}}^{-,\mathrm{spec},\mathrm{enh}}$.

It has a natural lax unital structure. However, the isomorphism

$$\mathrm{Eis}_Z^{-,\mathrm{spec},\mathrm{enh}} \circ \mathbf{ind}_{\mathrm{enh}} \simeq \mathrm{Eis}^{-,\mathrm{spec}} \otimes \mathrm{Id}$$

implies that this lax unital structure is strict.

12.5.13. Let $\mathcal{A}$ be a factorization algebra in $\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}$. Parallel to Sect. 8.5.7 we can consider the functor

$$\mathrm{Eis}_{\mathcal{A}}^{-,\mathrm{spec}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}).$$

12.6. **Partial enhancement.** This subsection serves as a preparation to Sect. 12.8.

12.6.1. Set

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_Z^{-,\mathrm{part},\mathrm{enh}} := (\mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathbb{Z})) \otimes_{\mathrm{Rep}(\check{M})_Z} \mathrm{Rep}(\check{P}^-)_Z.$$

The monadic adjunction

$$\mathrm{Res}_{\check{P}^-}^{\check{M}} : \mathrm{Rep}(\check{M}) \rightleftarrows \mathrm{Rep}(\check{P}^-) : \mathrm{inv}_{\check{N}_{\check{P}}^-}$$

gives rise to a monadic adjunction

$$\mathbf{ind}_{\mathrm{part},\mathrm{enh}}^{\mathrm{spec}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathbb{Z}) \rightleftarrows \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_Z^{-,\mathrm{part},\mathrm{enh}} : \mathbf{obl}_{\mathrm{part},\mathrm{enh}}^{\mathrm{spec}}.$$

We will consider the assignment

$$\mathbb{Z} \rightsquigarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_Z^{-,\mathrm{part},\mathrm{enh}}$$

as a crystal of categories over Ran , to be denoted

$$\underline{\mathrm{IndCoh}}(\mathrm{LS}_{\check{M}})^{-,\mathrm{part},\mathrm{enh}}.$$

12.6.2. Consider the full category

$$\mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}}^{-, \mathrm{part. enh}} \subset \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}})_{\mathrm{Ran}}^{-, \mathrm{part. enh}}.$$

As in (12.26), it fits into the pullback square

$$(12.34) \quad \begin{array}{ccc} \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}}^{-, \mathrm{part. enh}} & \longrightarrow & \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}})_{\mathrm{Ran}}^{-, \mathrm{part. enh}} \\ \downarrow & & \downarrow \mathrm{oblv}_{\mathrm{part. enh}}^{\mathrm{spec}} \\ \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \otimes \mathrm{D-mod}(\mathrm{Ran}^{\mathrm{untl}})_{\mathrm{indep}} & \longrightarrow & \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \otimes \mathrm{D-mod}(\mathrm{Ran}) \\ \sim \downarrow & & \\ \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}). & & \end{array}$$

12.6.3. Consider the category

$$\mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{M}})} \mathrm{QCoh}(\mathrm{LS}_{\tilde{P}-}).$$

The operation

$$(-) \otimes (\mathrm{ev}_{\mathcal{Z}}^{\tilde{P}-})^*(-)$$

gives rise to a functor

$$\mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}})_{\mathcal{Z}}^{-, \mathrm{part. enh}} \rightarrow \left(\mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{M}})} \mathrm{QCoh}(\mathrm{LS}_{\tilde{P}-}) \right) \otimes \mathrm{D-mod}(\mathcal{Z}),$$

which we denote by $(\mathbf{q}_{\mathcal{Z}}^{-, \mathrm{spec}})^*, \mathrm{part. enh}$.

12.6.4. Note now that the functor

$$(\mathbf{q}^{-, \mathrm{spec}})^{\mathrm{IndCoh}, *} : \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{P}-})$$

gives rise to a functor

$$(12.35) \quad (\mathbf{q}^{-, \mathrm{spec}})^{\mathrm{IndCoh}, *} \otimes \mathrm{Id} : \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{M}})} \mathrm{QCoh}(\mathrm{LS}_{\tilde{P}-}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{P}-}).$$

12.6.5. Consider the composition

$$(\mathbf{q}_{\mathcal{Z}}^{-, \mathrm{spec}})^{\mathrm{IndCoh}, *, \mathrm{part. enh}} := ((\mathbf{q}^{-, \mathrm{spec}})^{\mathrm{IndCoh}, *} \otimes \mathrm{Id}) \circ (\mathbf{q}_{\mathcal{Z}}^{-, \mathrm{spec}})^*, \mathrm{part. enh}.$$

Composing further with $(\mathbf{p}^{-, \mathrm{spec}})^{\mathrm{IndCoh}, *}$, we obtain a functor

$$(\mathbf{p}^{-, \mathrm{spec}})^{\mathrm{IndCoh}, *} \circ (\mathbf{q}_{\mathcal{Z}}^{-, \mathrm{spec}})^{\mathrm{IndCoh}, *, \mathrm{part. enh}} : \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}})_{\mathcal{Z}}^{-, \mathrm{part. enh}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}) \otimes \mathrm{D-mod}(\mathcal{Z}),$$

which we denote by $\mathrm{Eis}_{\mathcal{Z}}^{-, \mathrm{spec}, \mathrm{part. enh}}$.

Note that the composition $\mathrm{Eis}_{\mathcal{Z}}^{-, \mathrm{spec}, \mathrm{part. enh}} \circ \mathrm{ind}_{\mathrm{part. enh}}^{\mathrm{spec}}$ identifies canonically with $\mathrm{Eis}^{-, \mathrm{spec}} \otimes \mathrm{Id}$.

12.6.6. We will view the assignment

$$\mathcal{Z} \rightsquigarrow \mathrm{Eis}_{\mathcal{Z}}^{-, \mathrm{spec}, \mathrm{part. enh}}$$

as a local-to-global functor

$$\underline{\mathrm{IndCoh}}(\mathrm{LS}_{\tilde{M}})^{-, \mathrm{part. enh}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}),$$

to be denoted $\underline{\mathrm{Eis}}^{-, \mathrm{spec}, \mathrm{part. enh}}$.

It has a natural lax unital structure. However, the isomorphism

$$\mathrm{Eis}_{\mathcal{Z}}^{-, \mathrm{spec}, \mathrm{part. enh}} \circ \mathrm{ind}_{\mathrm{part. enh}}^{\mathrm{spec}} \simeq \mathrm{Eis}^{-, \mathrm{spec}} \otimes \mathrm{Id}$$

implies that this lax unital structure is strict.

12.6.7. Let $\mathcal{A}$ be a factorization algebra in $\mathrm{Rep}(\tilde{P}^-)$. Parallel to Sect. 8.5.7 we can consider the functor

$$\mathrm{Eis}_{\mathcal{A}}^{-, \mathrm{spec}, \mathrm{enh}} : \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}).$$

12.6.8. Recall now that we have a pair of mutually adjoint factorization functors

$$\iota_* : \mathrm{Rep}(\check{P}^-) \rightleftarrows \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} : \iota^!$$

It induces a pair of adjoint functors

$$(\mathrm{Id} \otimes \iota_*) : \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_{\mathcal{Z}}^{-, \mathrm{part}, \mathrm{enh}} \rightleftarrows \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_{\mathcal{Z}}^{-, \mathrm{enh}} : (\mathrm{Id} \otimes \iota^!).$$

Unwinding the constructions, we obtain that there is a canonical isomorphism

$$(\mathbf{q}_{\mathcal{Z}}^{-, \mathrm{spec}})^{\mathrm{IndCoh}, *, \mathrm{part}, \mathrm{enh}} \simeq (\mathbf{q}_{\mathcal{Z}}^{-, \mathrm{spec}})^{*, \mathrm{IndCoh}, \mathrm{enh}} \circ (\mathrm{Id} \otimes \iota_*).$$

From here, we obtain an isomorphism

$$(12.36) \quad \underline{\mathrm{Eis}}^{-, \mathrm{spec}, \mathrm{part}, \mathrm{enh}} \simeq \underline{\mathrm{Eis}}^{-, \mathrm{spec}, \mathrm{enh}} \circ (\mathrm{Id} \otimes \iota_*).$$

12.7. Partial enhancement, continued. The contents of this subsection are optional, in that they will not be necessary in the sequel.

12.7.1. Consider the assignment

$$\mathcal{Z} \rightsquigarrow (\mathbf{q}_{\mathcal{Z}}^{-, \mathrm{spec}})^{*, \mathrm{part}, \mathrm{enh}}$$

as a local-to-global functor, to be denoted $(\mathbf{q}^{-, \mathrm{spec}})^{*, \mathrm{part}, \mathrm{enh}}$.

It has a natural unital structure, and hence gives rise to a functor

$$(12.37) \quad \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}}^{-, \mathrm{part}, \mathrm{enh}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{M}})} \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}).$$

12.7.2. We claim:

Proposition 12.7.3. *The functor (12.37) is an equivalence.*

Proof. Denote

$$\mathrm{QCoh}(\mathrm{LS}_{\check{M}})_{\mathcal{Z}}^{-, \mathrm{part}, \mathrm{enh}} := (\mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathcal{Z})) \otimes_{\mathrm{Rep}(\check{M})_{\mathcal{Z}}} \mathrm{Rep}(\check{P}^-)_{\mathcal{Z}}.$$

The operation

$$(-) \otimes (\mathrm{ev}_{\mathcal{Z}}^{\check{P}^-})^*(-)$$

gives rise to a functor

$$\mathrm{QCoh}(\mathrm{LS}_{\check{M}})_{\mathcal{Z}}^{-, \mathrm{part}, \mathrm{enh}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}) \otimes \mathrm{D-mod}(\mathcal{Z}),$$

which we denote by $\mathbf{q}^{-, \mathrm{spec}} \otimes \mathrm{Loc}_{\check{P}^-, \mathcal{Z}}^{\mathrm{spec}}$.

The assignment

$$\mathcal{Z} \rightsquigarrow \mathbf{q}^{-, \mathrm{spec}} \otimes \mathrm{Loc}_{\check{P}^-, \mathcal{Z}}^{\mathrm{spec}}$$

is a unital local to global functor, to be denoted

$$(12.38) \quad \mathbf{q}^{-, \mathrm{spec}} \otimes \underline{\mathrm{Loc}}_{\check{P}^-}^{\mathrm{spec}} : \mathrm{QCoh}(\mathrm{LS}_{\check{M}})^{-, \mathrm{part}, \mathrm{enh}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}).$$

It suffices to show that the resulting functor

$$(12.39) \quad \mathrm{QCoh}(\mathrm{LS}_{\check{M}})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}}^{-, \mathrm{part}, \mathrm{enh}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-})$$

is an equivalence.

To simplify the exposition, we will show that the functor (12.39) becomes an equivalence after applying the operation

$$\mathrm{Vect} \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{M}})} -,$$

where $\mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \rightarrow \mathrm{Vect}$ corresponds to the trivial local system.

The resulting functor is the functor

$$(12.40) \quad \mathrm{Rep}(\check{N}_{\check{P}}^-)_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{N}_{\check{P}}^-}),$$

induced by

$$\underline{\mathrm{Loc}}_{\tilde{N}_P^-}^{\mathrm{spec}} : \underline{\mathrm{Rep}}(\tilde{N}_P^-) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\tilde{N}_P^-}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}).$$

We rewrite

$$\mathrm{Rep}(\tilde{N}_P^-) \simeq \mathcal{A}\text{-mod}^{\mathrm{com}},$$

where $\mathcal{A} = \mathrm{inv}_{\tilde{N}_P^-}(k)$.

By Sect. 12.3.5, we have

$$\mathrm{QCoh}(\mathrm{LS}_{\tilde{N}_P^-}) \simeq \mathrm{C}^{\mathrm{fact}}(X, \mathcal{A})\text{-mod}.$$

With respect to these identifications, the functor (12.40) is the functor

$$(12.41) \quad (\mathcal{A}\text{-mod}^{\mathrm{com}})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}} \rightarrow \mathrm{C}^{\mathrm{fact}}(X, \mathcal{A})\text{-mod}$$

of [GLC2, Equation (15.32)].

The fact that (12.41) is an equivalence is the content of [GLC2, Remark 15.6.16].

□

12.7.4. Next we observe:

Lemma 12.7.5. *The functor (12.35) defines an equivalence*

$$\mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{M}})} \mathrm{QCoh}(\mathrm{LS}_{\tilde{P}-}) \simeq \mathrm{IndCoh}_{\tilde{M}}(\mathrm{LS}_{\tilde{P}-}),$$

where

$$\mathrm{IndCoh}_{\tilde{M}}(\mathrm{LS}_{\tilde{P}-}) \subset \mathrm{IndCoh}(\mathrm{LS}_{\tilde{P}-})$$

is the full subcategory, consisting of objects with singular support contained in

$$\mathrm{Sing}(\mathrm{LS}_{\tilde{M}}) \times_{\mathrm{LS}_{\tilde{M}}} \mathrm{LS}_{\tilde{P}-} \subset \mathrm{Sing}(\mathrm{LS}_{\tilde{P}-}).$$

Proof. This is a particular case of [AG, Corollary 7.6.2].

□

12.7.6. Combining Proposition 12.7.3 and Lemma 12.7.5, we obtain:

Corollary 12.7.7. *The functor*

$$\mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}}^{-, \mathrm{part}, \mathrm{enh}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{P}-}),$$

defined by the (unital) local-to-global functor

$$\mathcal{Z} \rightsquigarrow (\mathfrak{q}_{\mathcal{Z}}^{-, \mathrm{spec}})^{\mathrm{IndCoh}, *, \mathrm{part}, \mathrm{enh}},$$

is an equivalence onto $\mathrm{IndCoh}_{\tilde{M}}(\mathrm{LS}_{\tilde{P}-})$.

12.7.8. To summarize, we obtain that the datum of the functor $\underline{\mathrm{Eis}}^{-, \mathrm{spec}, \mathrm{part}, \mathrm{enh}}$ is equivalent to the datum of the functor

$$\mathrm{IndCoh}_{\tilde{M}}(\mathrm{LS}_{\tilde{P}-}) \hookrightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{P}-}) \xrightarrow{(\mathfrak{p}^{-, \mathrm{spec}})^{\mathrm{IndCoh}}} \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}).$$

Next, we will describe a similar interpretation of the functor $\underline{\mathrm{Eis}}^{-, \mathrm{spec}, \mathrm{enh}}$.

12.7.9. Consider the fiber product

$$(\mathrm{LS}_{\check{P}-})_{\mathrm{dR}} \times_{(\mathrm{LS}_{\check{G}})_{\mathrm{dR}}} \mathrm{LS}_{\check{G}}.$$

It comes equipped with a natural map

$$\mathrm{LS}_{\check{P}-} \rightarrow (\mathrm{LS}_{\check{P}-})_{\mathrm{dR}} \times_{(\mathrm{LS}_{\check{G}})_{\mathrm{dR}}} \mathrm{LS}_{\check{G}}.$$

Let

$$\mathrm{IndCoh}_{\check{M}} \left((\mathrm{LS}_{\check{P}-})_{\mathrm{dR}} \times_{(\mathrm{LS}_{\check{G}})_{\mathrm{dR}}} \mathrm{LS}_{\check{G}} \right) \subset \mathrm{IndCoh} \left((\mathrm{LS}_{\check{P}-})_{\mathrm{dR}} \times_{(\mathrm{LS}_{\check{G}})_{\mathrm{dR}}} \mathrm{LS}_{\check{G}} \right)$$

be the full subcategory consisting of objects, whose image under the !-pullback functor

$$(12.42) \quad \mathrm{IndCoh} \left((\mathrm{LS}_{\check{P}-})_{\mathrm{dR}} \times_{(\mathrm{LS}_{\check{G}})_{\mathrm{dR}}} \mathrm{LS}_{\check{G}} \right) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{P}-})$$

is contained in

$$\mathrm{IndCoh}_{\check{M}}(\mathrm{LS}_{\check{P}-}) \subset \mathrm{IndCoh}(\mathrm{LS}_{\check{P}-}).$$

12.7.10. The following theorem can be deduced from the results of [Ro]:

Theorem 12.7.11. *The category $\mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}}^{-, \mathrm{enh}}$ identifies canonically with*

$$\mathrm{I}(\check{G}, \check{P}^{-})^{\mathrm{spec}, \mathrm{glob}} := \mathrm{IndCoh}_{\check{M}} \left((\mathrm{LS}_{\check{P}-})_{\mathrm{dR}} \times_{(\mathrm{LS}_{\check{G}})_{\mathrm{dR}}} \mathrm{LS}_{\check{G}} \right).$$

Under this identification, the forgetful functor

$$(\mathrm{Id} \otimes \iota^!) : \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}}^{-, \mathrm{enh}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}}^{-, \mathrm{part}, \mathrm{enh}}$$

corresponds to the functor

$$\mathrm{IndCoh}_{\check{M}} \left((\mathrm{LS}_{\check{P}-})_{\mathrm{dR}} \times_{(\mathrm{LS}_{\check{G}})_{\mathrm{dR}}} \mathrm{LS}_{\check{G}} \right) \rightarrow \mathrm{IndCoh}_{\check{M}}(\mathrm{LS}_{\check{P}-}),$$

induced by (12.42), and the functor

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}}^{-, \mathrm{enh}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}),$$

induced by $\underline{\mathrm{Eis}}^{-, \mathrm{spec}, \mathrm{enh}}$ identifies with

$$\mathrm{IndCoh}_{\check{M}} \left((\mathrm{LS}_{\check{P}-})_{\mathrm{dR}} \times_{(\mathrm{LS}_{\check{G}})_{\mathrm{dR}}} \mathrm{LS}_{\check{G}} \right) \hookrightarrow \mathrm{IndCoh} \left((\mathrm{LS}_{\check{P}-})_{\mathrm{dR}} \times_{(\mathrm{LS}_{\check{G}})_{\mathrm{dR}}} \mathrm{LS}_{\check{G}} \right) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}),$$

*where the second arrow is the $(\mathrm{IndCoh}, *)$ -pushforward functor.*

12.8. The “compactified” spectral Eisenstein functor. In this subsection we will perform a calculation (formulated in Proposition 12.8.4) that will be used in the proof of the main theorem of this paper, Theorem 17.1.2.

12.8.1. Consider the construction from Sect. 12.5.13 applied to

$$\mathcal{A} := \mathrm{IC}^{-, \mathrm{spec}, \frac{\infty}{2}}.$$

Denote the resulting functor $\mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})$ by $\mathrm{Eis}_{\mathrm{IC}}^{-, \mathrm{spec}}$.

12.8.2. Consider now the functor

$$\mathrm{Eis}_{!*}^{-,\mathrm{spec}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}),$$

equal to the composition

$$\begin{aligned} \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) &= \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \underset{\mathrm{QCoh}(\mathrm{LS}_{\check{M}})}{\otimes} \mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \underset{\mathrm{QCoh}(\mathrm{LS}_{\check{M}})}{\otimes} \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}) \rightarrow \\ &\xrightarrow{(\mathbf{q}^{-,\mathrm{spec}})^{\mathrm{IndCoh},*} \otimes \mathrm{Id}} \mathrm{IndCoh}(\mathrm{LS}_{\check{P}^-}) \xrightarrow{(\mathbf{p}^{-,\mathrm{spec}})^{\mathrm{IndCoh}}} \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}), \end{aligned}$$

where the second arrow is the tensor product of $\mathrm{Id}_{\mathrm{IndCoh}(\mathrm{LS}_{\check{M}})}$ with the functor

$$(12.43) \quad \mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}),$$

given by *direct image* along the canonical map $\mathrm{LS}_{\check{M}} \rightarrow \mathrm{LS}_{\check{P}^-}$.

12.8.3. We are going to prove:

Proposition 12.8.4. *The functors $\mathrm{Eis}_{\mathrm{IC}}^{-,\mathrm{spec}}$ and $\mathrm{Eis}_{!*}^{-,\mathrm{spec}}$ are canonically isomorphic.*

The rest of this subsection is devoted to the proof of Proposition 12.8.4.

12.8.5. Note that using the functor $\mathbf{q}^{-,\mathrm{spec}} \otimes \underline{\mathrm{Loc}}_{\check{P}^-}^{\mathrm{spec}}$ of (12.38), we can associate to

$$\mathcal{B} \in \mathrm{FactAlg}(\mathrm{Rep}(\check{P}^-))$$

a $\mathrm{QCoh}(\mathrm{LS}_{\check{M}})$ -linear functor

$$(\mathbf{q}^{-,\mathrm{spec}} \otimes \underline{\mathrm{Loc}}_{\check{P}^-}^{\mathrm{spec}})_{\mathcal{B}} : \mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}),$$

where $\mathrm{QCoh}(\mathrm{LS}_{\check{P}^-})$ is regarded as a $\mathrm{QCoh}(\mathrm{LS}_{\check{M}})$ -module category via $(\mathbf{q}^{-,\mathrm{spec}})^*$.

Take $\mathcal{B}$ to be the direct image of $\mathbf{1}_{\mathrm{Rep}(\check{M})}$ under the factorization functor

$$\mathrm{Rep}(\check{M}) = \mathrm{QCoh}(\mathrm{pt} / \check{M}) \rightarrow \mathrm{QCoh}(\mathrm{pt} / \check{P}^-) = \mathrm{Rep}(\check{P}^-),$$

where the arrow is the functor of direct image along

$$\mathrm{pt} / \check{M} \rightarrow \mathrm{pt} / \check{P}^-.$$

By (12.36), it suffices to show that the resulting functor $(\mathbf{q}^{-,\mathrm{spec}} \otimes \underline{\mathrm{Loc}}_{\check{P}^-}^{\mathrm{spec}})_{\mathcal{B}}$ identifies with (12.43).

12.8.6. To simplify the exposition, we will perform the corresponding calculation after applying the operation

$$\mathrm{Vect} \underset{\mathrm{QCoh}(\mathrm{LS}_{\check{M}})}{\otimes} -,$$

where $\mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \rightarrow \mathrm{Vect}$ corresponds to the trivial local system.

12.8.7. Thus, we need to show that the functor $\mathrm{Vect} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{N}_P^-})$ corresponding to the local-to-global functor

$$\underline{\mathrm{Loc}}_{\check{N}_P^-}^{\mathrm{spec}} : \underline{\mathrm{Rep}}(\check{N}_P^-) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{N}_P^-}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran})$$

and

$$\mathcal{B} \in \mathrm{FactAlg}(\mathrm{Rep}(\check{N}_P^-))$$

equal to the regular representation $R_{\check{N}_P^-}$, identifies canonically with the functor of direct image along

$$(12.44) \quad \mathrm{pt} \rightarrow \mathrm{LS}_{\check{N}_P^-},$$

corresponding to the trivial $\check{N}_P^-$ -local system.

12.8.8. The latter is a standard calculation, valid for any algebraic group H , cf. [AGKRRV, Theorem 12.6.3].

Let us, however, give a different argument, specific to the unipotent case.

12.8.9. Recall that according to Sect. 12.3.5, we have

$$\mathrm{QCoh}(\mathrm{LS}_{\check{N}_P^-}) = \mathrm{C}^{\mathrm{fact}}(X, \mathcal{A})\text{-mod},$$

where $\mathcal{A} := \mathrm{inv}_{\check{N}_P^-}(k)$.

Under the equivalence

$$\mathrm{Rep}(\check{N}_P^-) \simeq \mathcal{A}\text{-mod}^{\mathrm{fact}},$$

the object

$$\mathcal{B} \in \mathrm{ComAlg}(\mathrm{FactAlg}(\mathrm{Rep}(\check{N}_P^-)))$$

corresponds to the augmentation

$$(12.45) \quad \mathcal{A} \rightarrow k.$$

Therefore, the functor

$$\mathrm{Loc}_{\check{N}_P^-, \mathcal{B}}^{\mathrm{spec}} : \mathrm{Vect} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{N}_P^-})$$

corresponds to the homomorphism

$$(12.46) \quad \mathrm{C}^{\mathrm{fact}}(X, \mathcal{A}) \xrightarrow{(12.45)} \mathrm{C}^{\mathrm{fact}}(X, k) \simeq k,$$

i.e., to the natural augmentation on $\mathrm{C}^{\mathrm{fact}}(X, \mathcal{A})$.

However, the homomorphism (12.46) corresponds under

$$\mathrm{LS}_{\check{N}_P^-} \simeq \text{“Spec}(\mathrm{C}^{\mathrm{fact}}(X, \mathcal{A}))\text{”}$$

to the map (12.44).

□[Proposition 12.8.4]

12.9. Spectral Eisenstein functor and localization: the enhanced version.

12.9.1. In this section we will provide an enhancement of Corollary 12.2.6. Namely, we will express the composition

$$\Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}} \circ \mathrm{Eis}^{-, \mathrm{spec}, \mathrm{enh}},$$

via $\Gamma_{\check{M}}^{\mathrm{spec}, \mathrm{IndCoh}}$ and a functor of local nature.

This calculation is parallel to one in Sect. 11.

12.9.2. Recall the factorization functor

$$J^{-, \mathrm{spec}, \mathrm{pre-enh}} : \mathrm{Rep}(\check{G}) \otimes \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \rightarrow \mathrm{Rep}(\check{M})$$

of (2.17).

Let us denote by

$$\mathrm{co}J^{-, \mathrm{spec}, \mathrm{pre-enh}} : \mathrm{Rep}(\check{M}) \otimes \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \rightarrow \mathrm{Rep}(\check{G})$$

the functor obtained from $J^{-, \mathrm{spec}, \mathrm{pre-enh}}$ via the canonical self-dualities on $\mathrm{Rep}(\check{G})$ and $\mathrm{Rep}(\check{M})$, respectively.

Explicitly, $\mathrm{co}J^{-, \mathrm{spec}, \mathrm{pre-enh}}$ is given by

$$\mathcal{F}_M, \mathcal{F}_{G, P^-} \mapsto (p_1)_*^{\mathrm{IndCoh}} \circ (p_2 \times \mathrm{Id})^{\mathrm{IndCoh}, *}(\mathcal{F}_M \otimes \mathcal{F}_{G, P^-}),$$

where

$$\mathcal{F}_M \in \mathrm{Rep}(\check{M}) \simeq \mathrm{QCoh}(\mathrm{LS}_M^{\mathrm{reg}}) =: \mathrm{IndCoh}^*(\mathrm{LS}_M^{\mathrm{reg}}) \text{ and } \mathcal{F}_{G, P^-} \in \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}},$$

and p_1 and p_2 are the two maps in (2.16).

The functor $\mathrm{co}J^{-, \mathrm{spec}, \mathrm{pre-enh}}$ factors naturally as

$$(12.47) \quad \mathrm{co}J^{-, \mathrm{spec}, \mathrm{pre-enh}} : \mathrm{Rep}(\check{M}) \otimes \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \rightarrow \mathrm{Rep}(\check{M}) \otimes_{\mathrm{Sph}_M^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \rightarrow \mathrm{Rep}(\check{G}).$$

We denote the second arrow in (12.47) by $\mathrm{co}J^{-, \mathrm{spec}, \mathrm{enh}}$.

12.9.3. The functor $\mathrm{co}J^{-,\mathrm{spec},\mathrm{enh}}$ can also be interpreted as follows. Recall that according to Sect. A.8.1, we have

$$(12.48) \quad \mathrm{Rep}(\check{M}) \otimes_{\mathrm{Sph}_M^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}} \simeq \mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}\check{G}}).$$

In terms of this identification, the functor $\mathrm{co}J^{-,\mathrm{spec},\mathrm{enh}}$ corresponds to the functor

$$\mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}\check{G}}) \xrightarrow{(\mathrm{p}_{\mathrm{Ran}}^{-,\mathrm{spec},\mathrm{loc}})^* \mathrm{IndCoh}} \mathrm{IndCoh}^*(\mathrm{LS}_{\check{G}}^{\mathrm{reg}}) \simeq \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{reg}}) \simeq \mathrm{Rep}(\check{G}).$$

12.9.4. From the commutative (but not Cartesian!) diagram

$$\begin{array}{ccc} \mathrm{LS}_{\check{P}^-,z}^{\mathrm{mon-free}\check{G},\mathrm{glob}} & \xrightarrow{\mathrm{q}_z^{-,\mathrm{spec},\mathrm{enh}}} & \mathrm{LS}_{\check{M},z}^{\mathrm{mer},\mathrm{glob}} \times_{\mathrm{LS}_{\check{M},z}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-,z}^{\mathrm{mon-free}\check{G}} \\ \mathrm{p}_z^{-,\mathrm{spec}} \downarrow & & \downarrow \mathrm{ev}_z^{\check{M}} \times \mathrm{id} \\ \mathrm{LS}_{\check{G}} & & \mathrm{LS}_{\check{M},z}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{M},z}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-,z}^{\mathrm{mon-free}\check{G}} \simeq \mathrm{LS}_{\check{P}^-,z}^{\mathrm{mon-free}\check{G}} \\ \mathrm{ev}_z^{\check{G}} \downarrow & \xrightarrow{\mathrm{id}} & \downarrow \mathrm{p}_z^{-,\mathrm{spec},\mathrm{loc}} \\ \mathrm{LS}_{\check{G},z}^{\mathrm{reg}} & & \mathrm{LS}_{\check{G},z}^{\mathrm{reg}} \end{array}$$

we obtain a natural transformation

$$(12.49) \quad (\mathrm{p}_z^{-,\mathrm{spec},\mathrm{loc}})^* \mathrm{IndCoh} \circ (\mathrm{ev}_z^{\check{M}} \times \mathrm{id})^* \mathrm{IndCoh} \rightarrow (\mathrm{ev}_z^{\check{G}})^* \mathrm{IndCoh} \circ (\mathrm{p}_z^{-,\mathrm{spec}})^* \mathrm{IndCoh} \circ (\mathrm{q}_z^{-,\mathrm{spec},\mathrm{enh}})^* \mathrm{IndCoh},$$

as functors

$$\mathrm{IndCoh}^*(\mathrm{LS}_{\check{M},z}^{\mathrm{mer},\mathrm{glob}} \times_{\mathrm{LS}_{\check{M},z}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-,z}^{\mathrm{mon-free}\check{G}}) \Rightarrow \mathrm{IndCoh}^*(\mathrm{LS}_{\check{G}}^{\mathrm{reg}})_z.$$

12.9.5. Using the identification (12.48), we interpret (12.49) as a natural transformation

$$(12.50) \quad \mathrm{co}J^{-,\mathrm{spec},\mathrm{enh}} \circ (\Gamma_M^{\mathrm{spec},\mathrm{IndCoh}} \otimes \mathrm{Id}) \rightarrow \Gamma_{\check{G},z}^{\mathrm{spec},\mathrm{IndCoh}} \circ \mathrm{Eis}_z^{-,\mathrm{spec},\mathrm{enh}}$$

as functors

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_z^{-,\mathrm{enh}} \Rightarrow \mathrm{Rep}(\check{G})_z,$$

i.e.,

$$(12.51) \quad \begin{array}{ccc} \mathrm{Rep}(\check{G})_z & \xleftarrow{\mathrm{co}J^{-,\mathrm{spec},\mathrm{enh}}} & (\mathrm{Rep}(\check{M}) \otimes_{\mathrm{Sph}_M^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}})_z \\ \uparrow \Gamma_{\check{G},z}^{\mathrm{spec},\mathrm{IndCoh}} & \nearrow & \uparrow \Gamma_M^{\mathrm{spec},\mathrm{IndCoh}} \otimes \mathrm{Id} \\ \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z}) & \xleftarrow{\mathrm{Eis}^{-,\mathrm{spec},\mathrm{enh}}} & \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_z^{-,\mathrm{enh}} \end{array}$$

12.9.6. Note that when we precompose both sides of (12.50) with

$$\mathbf{ind}_{\mathrm{enh}}^{\mathrm{spec}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathbb{Z}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_z^{-,\mathrm{enh}},$$

we recover the natural transformation (12.13).

12.9.7. From (12.51) we obtain a natural transformation from the counterclockwise to the clockwise composition in the diagram

$$(12.52) \quad \begin{array}{ccc} \text{Rep}(\check{G})_{Z \subseteq, \text{untl}, \text{indep}} & \xleftarrow{\text{emb.indep}_{\text{Rep}(\check{G})}^L} & \text{Rep}(\check{G})_{Z \subseteq, \text{untl}} \\ \uparrow \Gamma_{\check{G}, Z, \text{untl}}^{\text{spec}, \text{IndCoh}} & & \uparrow \text{co}J^{-, \text{spec}, \text{enh}} \\ \text{IndCoh}(\text{LS}_{\check{G}}) \otimes \text{D-mod}(Z) & & (\text{Rep}(\check{M}) \otimes_{\text{Sph}_M^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}})_{Z \subseteq, \text{untl}} \\ \uparrow \text{Eis}^{-, \text{spec}, \text{enh}} & & \uparrow \\ \text{IndCoh}(\text{LS}_{\check{M}})_Z^{-, \text{enh}} & \xrightarrow{=} & (\text{IndCoh}(\text{LS}_{\check{M}}) \otimes \text{D-mod}(Z)) \otimes_{\text{Sph}_{\check{M}, Z}^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}_Z \\ & & \uparrow \Gamma_{\check{M}, Z, \text{untl}}^{\text{spec}, \text{IndCoh}} \otimes \text{ins.unit}_Z \\ & & \text{Rep}(\check{M})_{Z \subseteq, \text{untl}} \otimes_{\text{Sph}_{\check{M}, Z}^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}_{Z \subseteq, \text{untl}} \end{array}$$

where the notations $\Gamma_{\check{G}, Z, \text{untl}}^{\text{spec}, \text{IndCoh}}$ and $\Gamma_{\check{M}, Z, \text{untl}}^{\text{spec}, \text{IndCoh}}$ have the same meaning as in Sect. 10.8.3.

12.9.8. We claim:

Theorem 12.9.9. *The natural transformation in (12.52) is an isomorphism.*

Proof. It is enough to show that the natural transformation in (12.52) becomes an isomorphism after precomposing with

$$\text{ind}_{\text{enh}}^{\text{spec}} : \text{IndCoh}(\text{LS}_{\check{M}}) \otimes \text{D-mod}(Z) \rightarrow \text{IndCoh}(\text{LS}_{\check{M}})_Z^{-, \text{enh}}.$$

However, by Sect. 12.9.6, up to tensoring with $\text{D-mod}(Z)$, the resulting natural transformation equals one in (12.14). Now, the latter natural transformation is an isomorphism by Corollary 12.2.6. $\square$

12.10. The (co-)enhanced spectral constant term functor. The goal of this section is to define the co-enhanced spectral constant term functor

$$\underline{\text{CT}}^{-, \text{spec}, \text{enh}_{\text{co}}} : \underline{\text{IndCoh}}(\text{LS}_{\check{G}})^{-, \text{enh}_{\text{co}}} \rightarrow \underline{\text{IndCoh}}(\text{LS}_{\check{M}}) \otimes \underline{\text{D-mod}}(\text{Ran}),$$

where the left-hand side is the category defined in Sect. 12.4.5.

12.10.1. Denote

$$\text{LS}_{\check{P}^-, \text{Ran}}^{\text{mon-free}_{\check{M}}, \text{glob}} := (\text{LS}_{\check{M}} \times \text{Ran}) \times_{\text{LS}_{\check{M}, \text{Ran}}^{\text{mer}, \text{glob}}} \text{LS}_{\check{P}^-, \text{Ran}}^{\text{mer}, \text{glob}}.$$

We will denote by $\mathfrak{q}_{\text{Ran}}^{-, \text{spec}}$ the map

$$\text{LS}_{\check{P}^-, \text{Ran}}^{\text{mon-free}_{\check{M}}, \text{glob}} \rightarrow \text{LS}_{\check{M}} \times \text{Ran}.$$

12.10.2. Let $s_{\text{Ran}}^{\text{spec}, \tilde{M}}$ denote the naturally defined map

$$(12.53) \quad \text{LS}_{\tilde{P}^-, \text{Ran}}^{\text{mon-free}, \tilde{M}, \text{glob}} \rightarrow \text{LS}_{\tilde{M}, \text{Ran}}^{\text{reg}} \times_{\text{LS}_{\tilde{M}, \text{Ran}}^{\text{mer}}} \text{LS}_{\tilde{P}^-, \text{Ran}}^{\text{mer}} =: \text{LS}_{\tilde{P}^-, \text{Ran}}^{\text{mon-free}, \tilde{M}}.$$

Note that we have a commutative (but not Cartesian) diagram

$$\begin{array}{ccc} \text{LS}_{\tilde{P}^-, \text{Ran}}^{\text{mon-free}, \tilde{M}, \text{glob}} & \xrightarrow{q_{\text{Ran}}^-, \text{spec}} & \text{LS}_{\tilde{M}} \times \text{Ran} \\ s_{\text{Ran}}^{\text{spec}, \tilde{M}} \downarrow & & \downarrow \text{ev}_{\text{Ran}}^{\tilde{M}} \\ \text{LS}_{\tilde{P}^-, \text{Ran}}^{\text{mon-free}, \tilde{M}} & \xrightarrow{q_{\text{Ran}}^-, \text{spec}, \text{loc}} & \text{LS}_{\tilde{M}, \text{Ran}}^{\text{reg}}, \end{array}$$

where

$$q_{\text{Ran}}^-, \text{spec}, \text{loc} : \text{LS}_{\tilde{P}^-, \text{Ran}}^{\text{mon-free}, \tilde{M}} \rightarrow \text{LS}_{\tilde{M}, \text{Ran}}^{\text{reg}}$$

is the projection on the first factor.

12.10.3. We have a naturally defined map

$$(12.54) \quad \iota^{\text{glob}, \tilde{M}} : \text{LS}_{\tilde{P}^-} \times \text{Ran} \rightarrow \text{LS}_{\tilde{P}^-, \text{Ran}}^{\text{mon-free}, \tilde{M}, \text{glob}},$$

so that the diagram

$$\begin{array}{ccc} \text{LS}_{\tilde{P}^-} \times \text{Ran} & \xrightarrow{\iota^{\text{glob}, \tilde{M}}} & \text{LS}_{\tilde{P}^-, \text{Ran}}^{\text{mon-free}, \tilde{M}, \text{glob}} \\ \text{ev}_{\text{Ran}}^{\tilde{P}^-} \downarrow & & \downarrow s_{\text{Ran}}^{\text{spec}, \tilde{M}} \\ \text{LS}_{\tilde{P}^-, \text{Ran}}^{\text{reg}} & \xrightarrow{\iota^{\tilde{M}}} & \text{LS}_{\tilde{P}^-, \text{Ran}}^{\text{mon-free}, \tilde{M}} \end{array}$$

is Cartesian.

Note, however, that unlike the case of (12.28), the map (12.54) is *not* a nil-isomorphism.

12.10.4. We let

$$(12.55) \quad \left(\text{LS}_{\tilde{P}^-, \text{Ran}}^{\text{mon-free}, \tilde{M}, \text{glob}} \right)_{\text{mon-free}}^{\wedge}$$

denote the formal completion of $\text{LS}_{\tilde{P}^-, \text{Ran}}^{\text{mon-free}, \tilde{M}, \text{glob}}$ along the map (12.54).

By the same mechanism as in Sect. 12.5.3, we obtain that (12.55) is a formal algebraic stack, locally almost of finite type. In particular, the category

$$\text{IndCoh} \left(\left(\text{LS}_{\tilde{P}^-, \text{Ran}}^{\text{mon-free}, \tilde{M}, \text{glob}} \right)_{\text{mon-free}}^{\wedge} \right)$$

is well-defined.

Remark 12.10.5. For a fixed $\underline{x}$, the prestack $\text{LS}_{\tilde{P}^-, \underline{x}}^{\text{mon-free}, \tilde{M}, \text{glob}}$ is actually an algebraic stack, locally almost of finite type, so we could work directly with the IndCoh category on it.

However, we do not know how to define IndCoh of $\text{LS}_{\tilde{P}^-, \mathcal{Z}}^{\text{mon-free}, \tilde{M}, \text{glob}}$ for general $\mathcal{Z} \rightarrow \text{Ran}$, cf. Remark 2.7.2.

12.10.6. We claim that there is a well-defined functor

$$\text{IndCoh}(\text{LS}_{\tilde{\mathcal{Z}}}^-)^{\text{enh}, \text{co}} \rightarrow \text{IndCoh} \left(\left(\text{LS}_{\tilde{P}^-, \mathcal{Z}}^{\text{mon-free}, \tilde{M}, \text{glob}} \right)_{\text{mon-free}}^{\wedge} \right),$$

to be denoted $(\mathfrak{p}_{\mathcal{Z}}^-, \text{spec})^{\text{!}, \text{enh}}$.

We will perform this construction in this subsection when $\mathcal{Z} = \underline{x}$, and the case of a general $\mathcal{Z}$ will be considered in Sect. A.8.8.

12.10.7. Let $(\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}}$ be the formal completion of $\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}, \mathrm{glob}}$ along

$$\mathrm{LS}_{\check{G}} \rightarrow \mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}, \mathrm{glob}}.$$

Consider the map

$$\mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}} \rightarrow \mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mer}} \rightarrow \mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}},$$

and the fiber product

$$(12.56) \quad (\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}} \times_{\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}}.$$

We claim that when $\underline{x}$ is fixed, (12.56) is an algebraic stack locally almost of finite type. This follows from deformation theory combined with the fact that the map

$$(12.57) \quad \mathrm{LS}_{\check{G}} \times_{\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{reg}}} \mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{reg}} \rightarrow (\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}} \times_{\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}}$$

is a nil-isomorphism.

In particular, the category

$$\mathrm{IndCoh} \left((\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}} \times_{\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}} \right)$$

is well-defined. Moreover, as in (A.20), we have a canonical isomorphism

$$(12.58) \quad \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})^{-, \mathrm{enh}_{\mathrm{co}}} := \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes^{\mathrm{Sph}_{\check{G}, \underline{x}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}_{\mathrm{co}, \underline{x}} \simeq \\ \simeq \mathrm{IndCoh} \left((\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}} \times_{\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}} \right).$$

12.10.8. Combining the maps

$$\mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}, \mathrm{glob}} \rightarrow \mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mer}, \mathrm{glob}} \rightarrow \mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}, \mathrm{glob}}$$

and $\mathfrak{s}_{\underline{x}}^{\mathrm{spec}, \check{M}}$, we obtain a map

$$(12.59) \quad (\mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}} \rightarrow (\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}} \times_{\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}}$$

which we denote by $\mathfrak{p}_{\underline{x}}^{-, \mathrm{spec}, \mathrm{enh}}$.

Thus, we can consider the functor

$$(\mathfrak{p}_{\underline{x}}^{-, \mathrm{spec}, \mathrm{enh}})^! : \mathrm{IndCoh} \left((\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}} \times_{\mathrm{LS}_{\check{G}, \underline{x}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}} \right) \rightarrow \\ \rightarrow \mathrm{IndCoh} \left((\mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}} \right).$$

Combining with (12.58), we obtain the sought-for functor

$$(\mathfrak{p}_{\underline{x}}^{-, \mathrm{spec}})^!, \mathrm{enh} : \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})^{-, \mathrm{enh}_{\mathrm{co}}} \rightarrow \mathrm{IndCoh} \left((\mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}} \right).$$

Remark 12.10.9. Parallel to Remark 12.5.8, the composition⁵⁵

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}_{\mathrm{co}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})^{-, \mathrm{enh}_{\mathrm{co}}} \xrightarrow{(\mathfrak{p}_{\underline{x}}^{-, \mathrm{spec}})^!, \mathrm{enh}} \\ \rightarrow \mathrm{IndCoh} \left((\mathrm{LS}_{\check{P}^-, \underline{x}}^{\mathrm{mon-free} \check{M}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}} \right)$$

⁵⁵In the formula below, the first arrow is the left adjoint of the forgetful functor.

can be described using the stack $\mathrm{Hecke}_{\check{G}, \check{P}^-, \underline{x}}^{\mathrm{spec}, \mathrm{glob}}$.

12.10.10. The spectral Hecke groupoid $\mathrm{Hecke}_{\check{M}, \underline{x}}^{\mathrm{spec}, \mathrm{loc}}$ acts naturally on both sides of (12.59) and the map $\mathfrak{p}_{\underline{x}}^{-, \mathrm{spec}, \mathrm{enh}}$ is compatible with this action.

From here we obtain that the functor $(\mathfrak{p}_{\underline{x}}^{-, \mathrm{spec}})^{!, \mathrm{enh}}$ is compatible with the actions of $\mathrm{Sph}_{\check{M}, \underline{x}}^{\mathrm{spec}}$ on the two sides. The same will be true when $\underline{x} \in \mathrm{Ran}$ is replaced by an arbitrary $\mathcal{Z} \rightarrow \mathrm{Ran}$.

12.10.11. We define the functor

$$\mathrm{CT}_{\mathcal{Z}}^{-, \mathrm{spec}, \mathrm{enh}_{\mathrm{co}}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})_{\mathcal{Z}}^{-, \mathrm{enh}_{\mathrm{co}}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathcal{Z})$$

as the composition

$$(\mathfrak{q}_{\mathcal{Z}}^{-, \mathrm{spec}})^{\mathrm{IndCoh}}_* \circ (\mathfrak{p}_{\mathcal{Z}}^{-, \mathrm{spec}})^{!, \mathrm{enh}}.$$

Since both functors $(\mathfrak{p}_{\mathcal{Z}}^{-, \mathrm{spec}})^{!, \mathrm{enh}}$ and $(\mathfrak{q}_{\mathcal{Z}}^{-, \mathrm{spec}})^{\mathrm{IndCoh}}_*$ respect the actions of $\mathrm{Sph}_{\check{M}, \underline{x}}^{\mathrm{spec}}$, we obtain that $\mathrm{CT}_{\mathcal{Z}}^{-, \mathrm{spec}, \mathrm{enh}_{\mathrm{co}}}$ also respects the actions of $\mathrm{Sph}_{\check{M}, \mathcal{Z}}^{\mathrm{spec}}$.

12.10.12. Unwinding the construction, we obtain that the composition

$$(12.60) \quad \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathcal{Z}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})_{\mathcal{Z}}^{-, \mathrm{enh}_{\mathrm{co}}} \xrightarrow{\mathrm{CT}_{\mathcal{Z}}^{-, \mathrm{spec}, \mathrm{enh}_{\mathrm{co}}}} \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathcal{Z})$$

identifies canonically with $\mathrm{CT}^{-, \mathrm{spec}} \otimes \mathrm{Id}$.

12.10.13. We will view the assignment

$$\mathcal{Z} \rightsquigarrow \mathrm{CT}_{\mathcal{Z}}^{-, \mathrm{spec}, \mathrm{enh}_{\mathrm{co}}}$$

as a local-to-global functor

$$\underline{\mathrm{IndCoh}}(\mathrm{LS}_{\check{G}})^{-, \mathrm{enh}_{\mathrm{co}}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}),$$

to be denoted $\underline{\mathrm{CT}}^{-, \mathrm{spec}, \mathrm{enh}_{\mathrm{co}}}$.

It has a natural lax unital structure. However, the isomorphism between (12.60) and $\mathrm{CT}^{-, \mathrm{spec}} \otimes \mathrm{Id}$ implies that this lax unital structure is strict.

12.10.14. Let $\mathcal{A}$ be a factorization algebra in $\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}_{\mathrm{co}}$. Parallel to Sect. 8.5.7 we can consider the functor

$$\mathrm{CT}_{\mathcal{A}}^{-, \mathrm{spec}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}).$$

12.11. Partial enhancement for the constant term functor.

12.11.1. Set

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{G}})_{\mathcal{Z}}^{-, \mathrm{part}, \mathrm{enh}_{\mathrm{co}}} := (\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathcal{Z})) \otimes_{\mathrm{Rep}(\check{G})_{\mathcal{Z}}} \mathrm{Rep}(\check{P}^-)_{\mathcal{Z}}.$$

We will consider the assignment

$$\mathcal{Z} \rightsquigarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})_{\mathcal{Z}}^{-, \mathrm{part}, \mathrm{enh}_{\mathrm{co}}}$$

as a crystal of categories over Ran , to be denoted

$$\underline{\mathrm{IndCoh}}(\mathrm{LS}_{\check{G}})^{-, \mathrm{part}, \mathrm{enh}_{\mathrm{co}}}.$$

12.11.2. Consider the functor

$$(\mathfrak{p}_Z^{-,\text{spec}})^{!,\text{part.enh}_{\text{co}}} : \text{IndCoh}(\text{LS}_{\check{G}})_Z^{-,\text{part.enh}_{\text{co}}} \rightarrow \text{IndCoh}(\text{LS}_{\check{P}-}) \otimes \text{D-mod}(\mathbb{Z})$$

defined as

$$(\mathfrak{p}^{-,\text{spec}})^{!}(-) \otimes (\text{ev}_Z^{\check{P}-})^*(-).$$

Set

$$\text{CT}_Z^{-,\text{spec,enh}_{\text{co}}} := ((\mathfrak{q}^{-,\text{spec}})^{\text{IndCoh}}_* \otimes \text{Id}) \circ (\mathfrak{p}_Z^{-,\text{spec}})^{!,\text{part.enh}_{\text{co}}};$$

this is a functor

$$\text{IndCoh}(\text{LS}_{\check{G}})_Z^{-,\text{part.enh}_{\text{co}}} \rightarrow \text{IndCoh}(\text{LS}_{\check{M}}) \otimes \text{D-mod}(\mathbb{Z}).$$

The composition

$$(12.61) \quad \text{IndCoh}(\text{LS}_{\check{G}}) \otimes \text{D-mod}(\mathbb{Z}) \rightarrow \text{IndCoh}(\text{LS}_{\check{G}})_Z^{-,\text{part.enh}_{\text{co}}} \xrightarrow{\text{CT}_Z^{-,\text{spec,enh}_{\text{co}}}} \text{IndCoh}(\text{LS}_{\check{M}}) \otimes \text{D-mod}(\mathbb{Z})$$

identifies canonically with $\text{CT}^{-,\text{spec}} \otimes \text{Id}$.

12.11.3. We will view the assignment

$$Z \rightsquigarrow \text{CT}_Z^{-,\text{spec,part.enh}_{\text{co}}}$$

as a local-to-global functor

$$\text{IndCoh}(\text{LS}_{\check{G}})^{-,\text{part.enh}_{\text{co}}} \rightarrow \text{IndCoh}(\text{LS}_{\check{M}}) \otimes \underline{\text{D-mod}}(\text{Ran}),$$

to be denoted $\underline{\text{CT}}^{-,\text{spec,part.enh}_{\text{co}}}$.

It has a natural lax unital structure. However, the isomorphism between (12.61) and $\text{CT}^{-,\text{spec}} \otimes \text{Id}$ implies that this lax unital structure is strict.

12.11.4. Let $\mathcal{B}$ be a factorization algebra in $\text{Rep}(\check{G}^-)$. Parallel to Sect. 8.5.7 we can consider the functor

$$\text{CT}_{\mathcal{B}}^{-,\text{spec}} : \text{IndCoh}(\text{LS}_{\check{G}}) \rightarrow \text{IndCoh}(\text{LS}_{\check{M}}).$$

12.11.5. Recall the factorization functor

$$\iota_*^{\text{IndCoh}} : \text{IndCoh}^!(\text{LS}_{\check{P}-}^{\text{reg}}) \rightarrow \text{I}(\check{G}, \check{P}^-)_{\text{co}}^{\text{spec,loc}}$$

introduced in Sect. 2.6.4.

Since $\text{Rep}(\check{G})$ is rigid, we can interpret $\text{IndCoh}(\text{LS}_{\check{G}})_Z^{-,\text{part.enh}_{\text{co}}}$ as

$$(\text{IndCoh}(\text{LS}_{\check{G}}) \otimes \text{D-mod}(\mathbb{Z}))^{\text{Rep}(\check{G})_Z} \otimes \text{IndCoh}^!(\text{LS}_{\check{P}-}^{\text{reg}})_Z.$$

Tensoring ι_* with the identity functor on $\text{IndCoh}(\text{LS}_{\check{G}})$, we obtain a functor

$$(\text{Id} \otimes \iota_*) : \text{IndCoh}(\text{LS}_{\check{G}})_Z^{-,\text{part.enh}_{\text{co}}} \rightarrow \text{IndCoh}(\text{LS}_{\check{G}})_Z^{-,\text{part.enh}}.$$

12.11.6. Unwinding the constructions, we obtain a canonical isomorphism

$$(12.62) \quad (\mathfrak{p}_{\underline{Z}}^{-,\text{spec}})^{!,\text{enh}} \circ (\text{Id} \otimes \iota_*) \simeq (\iota^{\text{glob},\check{M}})_*^{\text{IndCoh}} \circ (\mathfrak{p}_Z^{-,\text{spec}})^{!,\text{part.enh}_{\text{co}}},$$

as functors

$$\text{IndCoh}(\text{LS}_{\check{G}})_Z^{-,\text{part.enh}_{\text{co}}} \rightrightarrows \text{IndCoh} \left(\left(\text{LS}_{\check{P}-,Z}^{\text{mon-free},\check{M},\text{glob}} \right)_{\text{mon-free}}^{\wedge} \right),$$

where we view $(\iota^{\text{glob},\check{M}})_*^{\text{IndCoh}}$ as a functor

$$\text{IndCoh}(\text{LS}_{\check{P}-}) \otimes \text{D-mod}(\mathbb{Z}) \rightarrow \text{IndCoh} \left(\left(\text{LS}_{\check{P}-,Z}^{\text{mon-free},\check{M},\text{glob}} \right)_{\text{mon-free}}^{\wedge} \right).$$

From here we obtain an isomorphism

$$(12.63) \quad \underline{\text{CT}}^{-,\text{spec,enh}_{\text{co}}} \circ (\text{Id} \otimes \iota_*) \simeq \underline{\text{CT}}^{-,\text{spec,part.enh}_{\text{co}}}$$

as local-to-global functors

$$\text{IndCoh}(\text{LS}_{\check{G}})^{-,\text{part.enh}_{\text{co}}} \rightrightarrows \text{IndCoh}(\text{LS}_{\check{M}}) \otimes \underline{\text{D-mod}}(\text{Ran}).$$

12.11.7. Note that, thanks to (12.63), for a factorization algebra $\mathcal{B}$ in $\text{Rep}(\check{B}^-)$, we have

$$\text{CT}_{\mathcal{B}}^{-,\text{spec}} \simeq \text{CT}_{\iota_*^{\text{IndCoh}(\mathcal{B})}}^{-,\text{spec}},$$

where:

- The left-hand side is understood in the sense of Sect. 12.11.4;
- The right-hand side is understood in the sense of Sect. 12.5.13.

12.12. The “compactified” spectral constant term functor.

12.12.1. In the framework of Sect. 12.10.14, take $\mathcal{A}$ to be the object

$$\text{IC}_{\text{co}}^{-,\text{spec},\frac{\infty}{2}} \in \text{I}(\check{G}, \check{P}^-)^{\text{spec},\text{loc}},$$

introduced in Sect. 2.6.5.

Denote the resulting functor $\text{CT}_{\mathcal{A}}^{-,\text{spec}}$ by

$$\text{CT}_{\text{IC}}^{-,\text{spec}} : \text{IndCoh}(\text{LS}_{\check{G}}) \rightarrow \text{IndCoh}(\text{LS}_{\check{M}}).$$

12.12.2. Consider now the functor

$$\text{CT}_{!_*}^{-,\text{spec}} : \text{IndCoh}(\text{LS}_{\check{G}}) \rightarrow \text{IndCoh}(\text{LS}_{\check{M}}),$$

defined as follows.

Note that since the functor

$$(\mathfrak{q}^{-,\text{spec}})_*^{\text{IndCoh}} : \text{IndCoh}(\text{LS}_{\check{P}^-}) \rightarrow \text{IndCoh}(\text{LS}_{\check{M}})$$

is $\text{QCoh}(\text{LS}_{\check{M}})$ -linear, it upgrades to a functor

$$(\mathfrak{q}^{-,\text{spec}})_*^{\text{IndCoh},\text{enh}} : \text{IndCoh}(\text{LS}_{\check{P}^-}) \rightarrow \text{IndCoh}(\text{LS}_{\check{M}}) \otimes_{\text{QCoh}(\text{LS}_{\check{M}})} \text{QCoh}(\text{LS}_{\check{P}^-}),$$

so that the composition

$$\begin{aligned} \text{IndCoh}(\text{LS}_{\check{P}^-}) &\xrightarrow{(\mathfrak{q}^{-,\text{spec}})_*^{\text{IndCoh},\text{enh}}} \text{IndCoh}(\text{LS}_{\check{M}}) \otimes_{\text{QCoh}(\text{LS}_{\check{M}})} \text{QCoh}(\text{LS}_{\check{P}^-}) \xrightarrow{\text{Id} \otimes \mathfrak{q}_*^{-,\text{spec}}} \\ &\rightarrow \text{IndCoh}(\text{LS}_{\check{M}}) \otimes_{\text{QCoh}(\text{LS}_{\check{M}})} \text{QCoh}(\text{LS}_{\check{M}}) = \text{IndCoh}(\text{LS}_{\check{M}}) \end{aligned}$$

is the original functor $(\mathfrak{q}^{-,\text{spec}})_*^{\text{IndCoh}}$.

We let $\text{CT}_{!_*}^{-,\text{spec}}$ be the composition

$$\begin{aligned} (12.64) \quad \text{IndCoh}(\text{LS}_{\check{G}}) &\xrightarrow{(\mathfrak{p}^{-,\text{spec}})^!} \text{IndCoh}(\text{LS}_{\check{P}^-}) \xrightarrow{(\mathfrak{q}^{-,\text{spec}})_*^{\text{IndCoh},\text{enh}}} \\ &\rightarrow \text{IndCoh}(\text{LS}_{\check{M}}) \otimes_{\text{QCoh}(\text{LS}_{\check{M}})} \text{QCoh}(\text{LS}_{\check{P}^-}) \rightarrow \\ &\rightarrow \text{IndCoh}(\text{LS}_{\check{M}}) \otimes_{\text{QCoh}(\text{LS}_{\check{M}})} \text{QCoh}(\text{LS}_{\check{M}}) = \text{IndCoh}(\text{LS}_{\check{M}}), \end{aligned}$$

where the third arrow is the tensor product of $\text{Id}_{\text{IndCoh}(\text{LS}_{\check{M}})}$ with the functor

$$\text{QCoh}(\text{LS}_{\check{P}^-}) \rightarrow \text{QCoh}(\text{LS}_{\check{M}}),$$

given by $*$ -pullback along

$$\text{LS}_{\check{M}} \rightarrow \text{LS}_{\check{P}^-},$$

cf. Sect. 12.8.2.

12.12.3. From Lemma 12.1.9, we obtain:

Lemma 12.12.4. *With respect to the Serre duality identifications*

$$\text{IndCoh}(\text{LS}_{\check{G}})^{\vee} \simeq \text{IndCoh}(\text{LS}_{\check{G}}) \text{ and } \text{IndCoh}(\text{LS}_{\check{M}})^{\vee} \simeq \text{IndCoh}(\text{LS}_{\check{M}}),$$

there is a canonical isomorphism

$$(\text{CT}_{!_*}^{-,\text{spec}})^{\vee} \simeq \text{Eis}_{!_*}^{-,\text{spec}} \circ \left((-) \otimes \mathcal{L}_{2\rho_P(\omega_X)} \right) [(2g-2) \cdot \dim(\check{\mathfrak{n}}_{\check{P}})].$$

12.12.5. We are going to prove:

Proposition 12.12.6. *The functors $\mathrm{CT}_{\mathrm{IC}}^{-,\mathrm{spec}}$ and $\mathrm{CT}_{!_*}^{-,\mathrm{spec}}$ are canonically isomorphic.*

The rest of this subsection is devoted to the proof of this proposition.

12.12.7. Let $\mathcal{B}$ be a factorization algebra in $\mathrm{Rep}(\check{P}^-)$. By (12.63), it suffices to establish an isomorphism

$$\mathrm{CT}_{!_*}^{-,\mathrm{spec}} \simeq \mathrm{CT}_{\mathcal{B}}^{-,\mathrm{spec}}$$

for $\mathrm{CT}_{\mathcal{B}}^{-,\mathrm{spec}}$ as in Sect. 12.11.4 when $\mathcal{B}$ is as in Sect. 12.8.5.

12.12.8. By construction, for any $\mathcal{B} \in \mathrm{FactAlg}(\mathrm{Rep}(\check{P}^-))$, we have

$$\mathrm{CT}_{\mathcal{B}}^{-,\mathrm{spec}} \simeq (\mathfrak{q}^{-,\mathrm{spec}})_*^{\mathrm{IndCoh}} \left((\mathfrak{p}^{-,\mathrm{spec}})^!(-) \otimes \mathrm{Loc}_{\check{P}^-, \mathcal{B}}^{\mathrm{spec}}(k) \right).$$

In Sects. 12.8.5-12.8.9 it was shown that for $\mathcal{B}$ being the image of $\mathbf{1}_{\mathrm{Rep}(\check{M})}$ along

$$\mathrm{Rep}(\check{M}) \xrightarrow{\mathrm{coind}_{\check{M}}^{\check{P}^-}} \mathrm{Rep}(\check{P}^-),$$

the object

$$\mathrm{Loc}_{\check{P}^-, \mathcal{B}}^{\mathrm{spec}}(k) \in \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-})$$

identifies with the direct image of $\mathcal{O}_{\mathrm{LS}_{\check{M}}}$ along

$$\mathrm{LS}_{\check{M}} \rightarrow \mathrm{LS}_{\check{P}^-};$$

by a slight abuse of notation, we will denote this object by the same symbol $\mathcal{O}_{\mathrm{LS}_{\check{M}}}$.

12.12.9. Thus, it remains to show that the functor

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{P}^-}) \xrightarrow{(-) \otimes^{\mathcal{O}_{\mathrm{LS}_{\check{M}}}} \mathrm{IndCoh}(\mathrm{LS}_{\check{P}^-})} (\mathfrak{q}^{-,\mathrm{spec}})_*^{\mathrm{IndCoh}} \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})$$

identifies with the composition of the last three arrows in (12.64).

However, this follows from the fact that both functors in question can be identified with

$$\begin{aligned} \mathrm{IndCoh}(\mathrm{LS}_{\check{P}^-}) &\xrightarrow{(\mathfrak{q}^{-,\mathrm{spec}})_*^{\mathrm{IndCoh}, \mathrm{enh}}} \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{M}})} \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}) \xrightarrow{\mathrm{Id} \otimes ((-) \otimes^{\mathcal{O}_{\mathrm{LS}_{\check{M}}}})} \\ &\rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{M}})} \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}) \xrightarrow{\mathrm{Id} \otimes (\mathfrak{q}^{-,\mathrm{spec}})_*} \\ &\rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{M}})} \mathrm{QCoh}(\mathrm{LS}_{\check{M}}) = \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}). \end{aligned}$$

□[Proposition 12.12.6]

12.13. Enhancement and duality on the spectral side. The contents of this subsection will not be necessary for the sequel.

12.13.1. Unwinding the construction of the functor $\mathrm{CT}_{\mathcal{Z}}^{-,\mathrm{spec}, \mathrm{enhco}}$, we obtain that it can be obtained from the functor $\mathrm{Eis}_{\mathcal{Z}}^{-,\mathrm{spec}, \mathrm{enh}}$ as follows:

The functor $\mathrm{Eis}_{\mathcal{Z}}^{-,\mathrm{spec}, \mathrm{enh}}$ preserves compactness; hence admits a right adjoint, which is a functor

$$(12.65) \quad \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathcal{Z}) \rightarrow (\mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathcal{Z})) \otimes_{\mathrm{Sph}_{\check{M}, \mathcal{Z}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)_{\mathcal{Z}}^{\mathrm{spec}, \mathrm{loc}} =: \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_{\mathcal{Z}}^{-,\mathrm{enh}}.$$

Denote the above functor (12.65) by $\mathrm{CT}_{\mathcal{Z}}^{-,\mathrm{spec}, \mathrm{enh}}$.

Using the relative rigidity of $\mathrm{Sph}_{\check{M}, \mathcal{Z}}^{\mathrm{spec}}$ and $\mathrm{Sph}_{\check{G}, \mathcal{Z}}^{\mathrm{spec}}$ over $\mathcal{Z}$, from $\mathrm{CT}_{\mathcal{Z}}^{-,\mathrm{spec}, \mathrm{enh}}$, we obtain a functor

$$(\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathcal{Z})) \otimes_{\mathrm{Sph}_{\check{M}, \mathcal{Z}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}, \mathcal{Z}}^{\mathrm{spec}, \mathrm{loc}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathcal{Z}),$$

which is our functor $\mathrm{CT}_{\mathbb{Z}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}}$.

12.13.2. Note now that there is another variant for enhancement: we can take the *dual* of $\mathrm{CT}_{*,\mathbb{Z}}^{-,\mathrm{enh}}$ and thus obtain a functor

$$\begin{aligned} \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})^{-,\mathrm{enh}_{\mathrm{co}}} &:= (\mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathbb{Z})) \otimes^{\mathrm{Sph}_{\check{M},\mathbb{Z}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}_{\mathrm{co},\mathbb{Z}} \rightarrow \\ &\rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z}), \end{aligned}$$

to be denoted $\mathrm{Eis}_{\mathbb{Z}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}}$.

The following functors are each others counterparts on the spectral and geometric sides, respectively:

$$\mathrm{CT}_{\mathbb{Z}}^{-,\mathrm{spec},\mathrm{enh}} \leftrightarrow \mathrm{CT}_{*,\mathbb{Z}}^{-,\mathrm{enh}}$$

$$\mathrm{CT}_{\mathbb{Z}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}} \leftrightarrow \mathrm{CT}_{*,\mathbb{Z}}^{-,\mathrm{enh}_{\mathrm{co}}}$$

$$\mathrm{Eis}_{\mathbb{Z}}^{-,\mathrm{spec},\mathrm{enh}} \leftrightarrow \mathrm{Eis}_{!,\mathbb{Z}}^{-,\mathrm{enh}}$$

and

$$\mathrm{Eis}_{\mathbb{Z}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}} \leftrightarrow \mathrm{Eis}_{\mathrm{co},*,\mathbb{Z}}^{-,\mathrm{enh}_{\mathrm{co}}}.$$

Recall now that there was another kind of relationship between the functors $\mathrm{Eis}_{!}^{-,\mathrm{enh}}$ and $\mathrm{Eis}_{\mathrm{co},*}^{-,\mathrm{enh}_{\mathrm{co}}}$, namely, one given by Theorem 8.7.6.

Hence, one expects a similar relationship between the functors $\mathrm{Eis}_{\mathbb{Z}}^{-,\mathrm{spec},\mathrm{enh}}$ and $\mathrm{Eis}_{\mathbb{Z}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}}$.

12.13.3. Recall the equivalence

$$\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}_{\mathrm{co},\mathbb{Z}} \simeq \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}_{\mathbb{Z}}$$

of (2.29); denote it by $\Theta_{\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}}$.

Recall also that the monoidal category $\mathrm{Sph}_{\check{M},\mathbb{Z}}^{\mathrm{spec}}$ is rigid relative to $\mathbb{Z}$, so we can identify

$$\otimes^{\mathrm{Sph}_{\check{M},\mathbb{Z}}^{\mathrm{spec}}} \quad \text{and} \quad \otimes^{\mathrm{Sph}_{\check{M},\mathbb{Z}}^{\mathrm{spec}}}.$$

Let $\Theta_{\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{glob}}}$ denote the equivalence

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{M}})^{-,\mathrm{enh}_{\mathrm{co}}}_{\mathbb{Z}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})^{-,\mathrm{enh}}_{\mathbb{Z}}$$

obtained by tensoring $\Theta_{\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}}}$ with the functor

$$((-) \otimes \mathcal{L}_{2\rho_P(\omega_X)})[(2g-2) \cdot \dim(\check{\mathfrak{n}}_P^-)] : \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})$$

over⁵⁶ $\mathrm{Sph}_{\check{M},\mathbb{Z}}^{\mathrm{spec}}$.

⁵⁶Note that according to Remark 2.6.2 it is the latter functor that makes the actions of $\mathrm{Sph}_{\check{M},\mathbb{Z}}^{\mathrm{spec}}$ on the two sides compatible.

12.13.4. We propose:

Conjecture 12.13.5. *The following diagram of functors commutes:*

$$\begin{array}{ccc} \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}})_{\mathbb{Z}}^{-, \mathrm{enh}_{\mathrm{co}}} & \xrightarrow{\Theta_{\mathrm{I}(\tilde{G}, \tilde{P}-)^{\mathrm{spec}, \mathrm{glob}}}} & \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}})_{\mathbb{Z}}^{-, \mathrm{enh}} \\ \mathrm{Eis}_{\mathbb{Z}}^{-, \mathrm{spec}, \mathrm{enh}_{\mathrm{co}}} \downarrow & & \downarrow \mathrm{Eis}_{\mathbb{Z}}^{-, \mathrm{spec}, \mathrm{enh}} \\ \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}) \otimes \mathrm{D-mod}(\mathbb{Z}) & \xrightarrow{\mathrm{id}} & \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}) \otimes \mathrm{D-mod}(\mathbb{Z}), \end{array}$$

in a way compatible with the $\mathrm{Sph}_{\tilde{G}, \mathbb{Z}}^{\mathrm{spec}}$ -actions.

Remark 12.13.6. As a reality check, note, that according to (12.5), the diagram in Conjecture 12.13.5 does become commutative when juxtaposed with the commutative square

$$\begin{array}{ccc} \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \otimes \mathrm{D-mod}(\mathbb{Z}) & \xrightarrow{((-) \otimes \mathcal{L}_{2\rho_P(\omega_X)})[(2g-2) \cdot \dim(\tilde{\mathfrak{n}}_{\tilde{P}})]} & \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \otimes \mathrm{D-mod}(\mathbb{Z}) \\ \mathrm{ind}_{\mathrm{enh}_{\mathrm{co}}} \downarrow & & \downarrow \mathrm{ind}_{\mathrm{enh}} \\ \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}})_{\mathbb{Z}}^{-, \mathrm{enh}_{\mathrm{co}}} & \xrightarrow{\Theta_{\mathrm{I}(\tilde{G}, \tilde{P}-)^{\mathrm{spec}, \mathrm{glob}}}} & \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}})_{\mathbb{Z}}^{-, \mathrm{enh}}. \end{array}$$

12.13.7. Note that the functor $\Theta_{\mathrm{I}(\tilde{G}, \tilde{P}-)^{\mathrm{spec}, \mathrm{glob}}}$ induces a self-duality on $\mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}})_{\mathrm{Ran}}^{-, \mathrm{enh}}$, which in turn induces a self-duality on $\mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}}^{-, \mathrm{enh}}$.

Recall now that according to Theorem 12.7.11, we have a canonical equivalence

$$\mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}}^{-, \mathrm{enh}} \simeq \mathrm{IndCoh}_{\tilde{M}} \left((\mathrm{LS}_{\tilde{P}-})_{\mathrm{dR}} \times_{(\mathrm{LS}_{\tilde{G}})_{\mathrm{dR}}} \mathrm{LS}_{\tilde{G}} \right).$$

Thus, $\Theta_{\mathrm{I}(\tilde{G}, \tilde{P}-)^{\mathrm{spec}, \mathrm{glob}}}$ induces a self-duality on $\mathrm{IndCoh}_{\tilde{M}} \left((\mathrm{LS}_{\tilde{P}-})_{\mathrm{dR}} \times_{(\mathrm{LS}_{\tilde{G}})_{\mathrm{dR}}} \mathrm{LS}_{\tilde{G}} \right)$.

12.13.8. We propose:

Conjecture 12.13.9. *The self-duality on $\mathrm{IndCoh}_{\tilde{M}} \left((\mathrm{LS}_{\tilde{P}-})_{\mathrm{dR}} \times_{(\mathrm{LS}_{\tilde{G}})_{\mathrm{dR}}} \mathrm{LS}_{\tilde{G}} \right)$ induced by the functor $\Theta_{\mathrm{I}(\tilde{G}, \tilde{P}-)^{\mathrm{spec}, \mathrm{glob}}}$ equals the canonical (Serre) self-duality defined on IndCoh of a prestack locally almost of finite type.*

Remark 12.13.10. Note that Conjecture 12.13.9 implies Conjecture 12.13.5.

13. SPECTRAL CONSTANT TERM FUNCTOR AND OPERS

The goal of this section is to express the composition

$$\mathrm{CT}^{-, \mathrm{spec}} \circ \mathrm{Poinc}_{\tilde{G}, !}^{\mathrm{spec}}$$

via the (local) functor $J_{\mathrm{Op}}^{-, \mathrm{spec}}$ followed by $\mathrm{Poinc}_{\tilde{M}, !}^{\mathrm{spec}}$, and similarly for the enhanced versions of the functors involved.

This will be *parallel* to the “Whittaker vs Eisenstein” calculation in Sect. 10. That said, this calculation is the *Langlands dual* of the “localization vs constant term” calculation from Sect. 11.

13.1. Statement of the result—the unenhanced version. In this subsection we will state a theorem that relates $\mathrm{CT}^{-, \mathrm{spec}} \circ \mathrm{Poinc}_{\tilde{G}, !}^{\mathrm{spec}}$ and $\mathrm{Poinc}_{\tilde{M}, !}^{\mathrm{spec}} \circ J_{\mathrm{Op}}^{-, \mathrm{spec}}$.

13.1.1. Recall the (unital) local-to-global functor

$$\underline{\mathrm{Poinc}}_{\tilde{G},!}^{\mathrm{spec}} : \underline{\mathrm{IndCoh}}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}),$$

see [GLC2, Sect. 17.4.1] or Sect. 13.1.3 below.

For a $Z_{\tilde{G}}^0$ -torsor $\mathcal{P}_{Z_{\tilde{G}}^0}$, recall the space of translated opers, see Sect. 4.1.1. By a similar token, we obtain a functor

$$\underline{\mathrm{Poinc}}_{\tilde{G},!,\mathcal{P}_{Z_{\tilde{G}}^0}}^{\mathrm{spec}} : \underline{\mathrm{IndCoh}}^!(\mathrm{Op}_{\tilde{G},\mathcal{P}_{Z_{\tilde{G}}^0}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}).$$

In particular, we can consider the functor

$$\underline{\mathrm{Poinc}}_{\tilde{M},!,\tilde{\rho}_P}^{\mathrm{spec}} : \underline{\mathrm{IndCoh}}^!(\mathrm{Op}_{\tilde{M},\tilde{\rho}_P}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}).$$

13.1.2. On the one hand, we consider the (unital) local-to-global functor

$$\mathrm{CT}^{-,\mathrm{spec}} \circ \underline{\mathrm{Poinc}}_{\tilde{G},!}^{\mathrm{spec}} : \underline{\mathrm{IndCoh}}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}).$$

On the other hand, we consider the *lax* unital local-to-global functor

$$\underline{\mathrm{Poinc}}_{\tilde{M},!,\tilde{\rho}_P}^{\mathrm{spec}} \circ J_{\mathrm{Op}}^{-,\mathrm{spec}} : \underline{\mathrm{IndCoh}}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}),$$

where

$$J_{\mathrm{Op}}^{-,\mathrm{spec}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}}) \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{M},\tilde{\rho}_P}^{\mathrm{mon-free}})$$

is the factorization functor from Sect. 4.2.3.

Our first goal will be to construct a natural transformation

$$(13.1) \quad \underline{\mathrm{Poinc}}_{\tilde{M},!,\tilde{\rho}_P}^{\mathrm{spec}} \circ J_{\mathrm{Op}}^{-,\mathrm{spec}} \rightarrow \mathrm{CT}^{-,\mathrm{spec}} \circ \underline{\mathrm{Poinc}}_{\tilde{G},!}^{\mathrm{spec}}.$$

Fix $\mathcal{Z} \rightarrow \mathrm{Ran}$. Thus, we wish to construct a natural transformation

$$(13.2) \quad \underline{\mathrm{Poinc}}_{\tilde{M},!,\tilde{\rho}_P,\mathcal{Z}}^{\mathrm{spec}} \circ J_{\mathrm{Op}}^{-,\mathrm{spec}} \rightarrow \mathrm{CT}^{-,\mathrm{spec}} \circ \underline{\mathrm{Poinc}}_{\tilde{G},!,\mathcal{Z}}^{\mathrm{spec}}$$

as functors

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G},\mathcal{Z}}^{\mathrm{mon-free}}) \rightrightarrows \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \otimes \underline{\mathrm{D-mod}}(\mathcal{Z}).$$

13.1.3. We begin by rewriting the right-hand side in (13.2). Recall the space

$$\mathrm{Op}_{\tilde{G},\mathcal{Z}}^{\mathrm{mon-free,glob}} := \mathrm{Op}_{\tilde{G},\mathcal{Z}}^{\mathrm{mer,glob}} \times_{\mathrm{LS}_{\tilde{G},\mathcal{Z}}^{\mathrm{mer,glob}}} (\mathrm{LS}_{\tilde{G}} \times \mathcal{Z})$$

(see [GLC2, Sect. 15.1.1] for the notation).

Recall (see [GLC2, Sect. 17.4.1]) that the functor $\underline{\mathrm{Poinc}}_{\tilde{G},!,\mathcal{Z}}^{\mathrm{spec}}$ is given by $!$ -pull followed by $(\mathrm{IndCoh}, *)$ -push along the diagram

$$\begin{array}{ccc} & & \mathrm{LS}_{\tilde{G}} \times \mathcal{Z} \\ & & \uparrow \tau_{\tilde{G}}^{\mathrm{glob}} \\ \mathrm{Op}_{\tilde{G},\mathcal{Z}}^{\mathrm{mon-free}} & \xleftarrow{\mathrm{ev}_{\mathcal{Z}}^{\mathrm{Op}_{\tilde{G}}}} & \mathrm{Op}_{\tilde{G},\mathcal{Z}}^{\mathrm{mon-free,glob}}. \end{array}$$

Then the right-hand side in (13.2) is given by the diagram

$$(13.3) \quad \begin{array}{ccccc} & & & & \mathrm{LS}_{\tilde{M}} \times \mathcal{Z} \\ & & & & \uparrow \mathfrak{q}^{-,\mathrm{spec}} \\ & & \mathrm{LS}_{\tilde{G}} \times \mathcal{Z} & \xleftarrow{\mathfrak{p}^{-,\mathrm{spec}}} & \mathrm{LS}_{\tilde{P}} \times \mathcal{Z} \\ & & \uparrow \tau_{\tilde{G}}^{\mathrm{glob}} & & \\ \mathrm{Op}_{\tilde{G},\mathcal{Z}}^{\mathrm{mon-free}} & \xleftarrow{\mathrm{ev}_{\mathcal{Z}}^{\mathrm{Op}_{\tilde{G}}}} & \mathrm{Op}_{\tilde{G},\mathcal{Z}}^{\mathrm{mon-free,glob}}, & & \end{array}$$

in which we apply $!$ -pullback along the horizontal arrows and $(\text{IndCoh}, *)$ -pushforward along the vertical ones.

By base change, we can replace (13.3) by the diagram

$$(13.4) \quad \begin{array}{c} \text{LS}_{\tilde{M}} \times \mathbb{Z} \\ \uparrow \mathbf{q}^{-, \text{spec}} \\ \text{LS}_{\tilde{P}^-} \times \mathbb{Z} \\ \uparrow \mathbf{r}_G^{\text{glob}} \\ \text{Op}_{\tilde{G}, \mathbb{Z}}^{\text{mon-free}} \xleftarrow[\text{ev}_Z]{\text{Op}_{\tilde{G}}} \text{Op}_{\tilde{G}, \mathbb{Z}}^{\text{mon-free, glob}} \xleftarrow[\mathbf{p}^{-, \text{spec}}]{} \text{Op}_{\tilde{G}, \mathbb{Z}}^{\text{mon-free, glob}} \times_{\text{LS}_{\tilde{G}} \times \mathbb{Z}} (\text{LS}_{\tilde{P}^-} \times \mathbb{Z}) \end{array}$$

13.1.4. Recall the affine D-scheme $\text{MOp}_{\tilde{G}, \tilde{P}^-}$ (see Sect. 4.1.4). Consider the corresponding ind-scheme

$$\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\text{mer, glob}}$$

and the fiber product

$$\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\text{mer, mon-free } \tilde{P}^-, \text{glob}} := \text{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\text{mer, glob}} \times_{\text{LS}_{\tilde{P}^-, \mathbb{Z}}^{\text{mer, glob}}} (\text{LS}_{\tilde{P}^-} \times \mathbb{Z}).$$

We have a naturally defined map

$$(13.5) \quad j^{\text{glob}} : \text{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\text{mer, mon-free } \tilde{P}^-, \text{glob}} \rightarrow \text{Op}_{\tilde{G}, \mathbb{Z}}^{\text{mon-free, glob}} \times_{\text{LS}_{\tilde{G}} \times \mathbb{Z}} (\text{LS}_{\tilde{P}^-} \times \mathbb{Z}).$$

13.1.5. The following observation is key for our construction:

Proposition 13.1.6. *The map j^{glob} of (13.5) is ind-proper.⁵⁷*

In a sense, this proposition explains the mechanism of the appearance of Miura ops.

The proof of Proposition 13.1.6 will be given in Sect. 13.3.7. We assume it temporarily and proceed with the construction of the natural transformation (13.2).

13.1.7. From Proposition 13.1.6 we obtain an adjoint pair $((j^{\text{glob}})_*^{\text{IndCoh}}, (j^{\text{glob}})^!)$, and hence a natural transformation

$$(j^{\text{glob}})_*^{\text{IndCoh}} \circ (j^{\text{glob}})^! \rightarrow \text{Id}.$$

Hence, the right-hand side in (13.2) receives a map from the functor

$$(13.6) \quad (\mathbf{q}^{-, \text{spec}})_*^{\text{IndCoh}} \circ (\mathbf{r}_G^{\text{glob}})_*^{\text{IndCoh}} \circ (j^{\text{glob}})_*^{\text{IndCoh}} \circ (j^{\text{glob}})^! \circ (\mathbf{p}^{-, \text{spec}})^! \circ (\text{ev}_Z^{\text{Op}_{\tilde{G}}})^!.$$

We will now show that the functor (13.6) identifies with the left-hand side in (13.2).

13.1.8. Let $\mathbf{p}^{\text{Miura, glob}}$ denote the composition

$$\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\text{mer, mon-free } \tilde{P}^-, \text{glob}} \xrightarrow{j^{\text{glob}}} \text{Op}_{\tilde{G}, \mathbb{Z}}^{\text{mon-free, glob}} \times_{\text{LS}_{\tilde{G}} \times \mathbb{Z}} (\text{LS}_{\tilde{P}^-} \times \mathbb{Z}) \xrightarrow[\mathbf{p}^{-, \text{spec}}]{} \text{Op}_{\tilde{G}, \mathbb{Z}}^{\text{mon-free, glob}}.$$

Note also that we have a naturally defined map

$$\mathbf{q}^{\text{Miura, glob}} : \text{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\text{mer, mon-free } \tilde{P}^-, \text{glob}} \rightarrow \text{Op}_{\tilde{M}, \tilde{P}^-, \mathbb{Z}}^{\text{mon-free, glob}}$$

so that the composition

$$\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\text{mer, mon-free } \tilde{P}^-, \text{glob}} \xrightarrow{j^{\text{glob}}} \text{Op}_{\tilde{G}, \mathbb{Z}}^{\text{mon-free, glob}} \times_{\text{LS}_{\tilde{G}} \times \mathbb{Z}} (\text{LS}_{\tilde{P}^-} \times \mathbb{Z}) \xrightarrow[\mathbf{r}_G^{\text{glob}}]{} \text{LS}_{\tilde{P}^-} \times \mathbb{Z} \xrightarrow[\mathbf{p}^{-, \text{spec}} \times \text{id}]{} \text{LS}_{\tilde{M}} \times \mathbb{Z}$$

equals

$$\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\text{mer, mon-free } \tilde{P}^-, \text{glob}} \xrightarrow{\mathbf{q}^{\text{Miura, glob}}} \text{Op}_{\tilde{M}, \tilde{P}^-, \mathbb{Z}}^{\text{mon-free, glob}} \xrightarrow[\mathbf{r}_M^{\text{glob}}]{} \text{LS}_{\tilde{M}} \times \mathbb{Z}.$$

⁵⁷In fact, as the proof will show, this map is an ind-closed embedding.

With these notations, the functor (13.6) is given by pull-push along the diagram

$$(13.7) \quad \begin{array}{ccccc} & & & & \mathrm{LS}_{\tilde{M}} \times \mathbb{Z} \\ & & & \uparrow \tau_M^{\mathrm{glob}} & \\ & & & \mathrm{Op}_{\tilde{M}, \tilde{\rho}_P, \mathbb{Z}}^{\mathrm{mon-free, glob}} & \\ & & \uparrow q^{\mathrm{Miura, glob}} & & \\ \mathrm{Op}_{\tilde{G}, \mathbb{Z}}^{\mathrm{mon-free}} & \xleftarrow{\mathrm{Op}_{\tilde{G}}} & \mathrm{Op}_{\tilde{G}, \mathbb{Z}}^{\mathrm{mon-free, glob}} & \xleftarrow{\mathfrak{p}^{\mathrm{Miura, glob}}} & \mathrm{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\mathrm{mer, mon-free}_{\tilde{P}^-, \mathrm{glob}}} \end{array}.$$

13.1.9. Note that the evaluation map

$$\mathrm{ev}_Z^{\mathrm{MOp}} : \mathrm{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\mathrm{mer, glob}} \rightarrow \mathrm{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\mathrm{mer}}$$

gives rise to a map

$$\mathrm{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\mathrm{mer, mon-free}_{\tilde{P}^-, \mathrm{glob}}} \rightarrow \mathrm{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\mathrm{mer, mon-free}_{\tilde{P}^-}},$$

where

$$\mathrm{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\mathrm{mer, mon-free}_{\tilde{P}^-}} := \mathrm{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\mathrm{mer}} \times_{\mathrm{LS}_{\tilde{P}^-, \mathbb{Z}}^{\mathrm{mer}}} \mathrm{LS}_{\tilde{P}^-, \mathbb{Z}}^{\mathrm{reg}},$$

see Sect. 4.2.1. Denote resulting map by the same symbol $\mathrm{ev}_Z^{\mathrm{MOp}}$.

Then the horizontal composition in (13.7) equals

$$\mathrm{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\mathrm{mer, mon-free}_{\tilde{P}^-, \mathrm{glob}}} \xrightarrow{\mathrm{ev}_Z^{\mathrm{MOp}}} \mathrm{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\mathrm{mer, mon-free}_{\tilde{P}^-}} \xrightarrow{\mathfrak{p}^{\mathrm{Miura, mon-free}}} \mathrm{Op}_{\tilde{G}, \mathbb{Z}}^{\mathrm{mon-free}},$$

where $\mathfrak{p}^{\mathrm{Miura, mon-free}}$ is as in Sect. 4.2.2.

Hence, we can replace (13.7) by

$$(13.8) \quad \begin{array}{ccccc} & & & & \mathrm{LS}_{\tilde{M}} \times \mathbb{Z} \\ & & & \uparrow \tau_M^{\mathrm{glob}} & \\ & & & \mathrm{Op}_{\tilde{M}, \tilde{\rho}_P, \mathbb{Z}}^{\mathrm{mon-free, glob}} & \\ & & \uparrow q^{\mathrm{Miura, glob}} & & \\ \mathrm{Op}_{\tilde{G}, \mathbb{Z}}^{\mathrm{mon-free}} & \xleftarrow{\mathfrak{p}^{\mathrm{Miura, mon-free}}} & \mathrm{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\mathrm{mer, mon-free}_{\tilde{P}^-}} & \xleftarrow{\mathrm{MOp}_{\tilde{G}}} & \mathrm{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\mathrm{mer, mon-free}_{\tilde{P}^-, \mathrm{glob}}} \end{array}.$$

13.1.10. Note now that we have a *Cartesian* square

$$\begin{array}{ccc} \mathrm{Op}_{\tilde{M}, \tilde{\rho}_P, \mathbb{Z}}^{\mathrm{mon-free}} & \xleftarrow{\mathrm{Op}_{\tilde{M}}} & \mathrm{Op}_{\tilde{M}, \tilde{\rho}_P, \mathbb{Z}}^{\mathrm{mon-free, glob}} \\ \uparrow q^{\mathrm{Miura, mon-free}} & & \uparrow q^{\mathrm{Miura, glob}} \\ \mathrm{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\mathrm{mer, mon-free}_{\tilde{P}^-}} & \xleftarrow{\mathrm{MOp}_{\tilde{G}}} & \mathrm{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\mathrm{mer, mon-free}_{\tilde{P}^-, \mathrm{glob}}} \end{array}.$$

Hence, we can replace (13.8) by

$$(13.9) \quad \begin{array}{ccc} & & \mathrm{LS}_{\tilde{M}} \times \mathbb{Z} \\ & & \uparrow \tau_M^{\mathrm{glob}} \\ & & \mathrm{Op}_{\tilde{M}, \tilde{\rho}_P, \mathbb{Z}}^{\mathrm{mon-free, glob}} \\ & \xleftarrow{\mathrm{Op}_{\tilde{M}}} & \\ \mathrm{Op}_{\tilde{M}, \tilde{\rho}_P, \mathbb{Z}}^{\mathrm{mon-free}} & & \\ \uparrow q^{\mathrm{Miura, mon-free}} & & \\ \mathrm{Op}_{\tilde{G}, \mathbb{Z}}^{\mathrm{mon-free}} & \xleftarrow{\mathfrak{p}^{\mathrm{Miura, mon-free}}} & \mathrm{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\mathrm{mer, mon-free}_{\tilde{P}^-}} \end{array}.$$

Finally, the functor defined by (13.9) is by definition the left-hand side in (13.2).

13.1.11. Thus, we have completed the construction of the natural transformation (13.1).

By [GLC2, Sect. 11.4.8], the natural transformation (13.1) gives rise to a natural transformation

$$(13.10) \quad \left(\underline{\mathrm{Poinc}}_{\tilde{M}, !, \tilde{\rho}_P}^{\mathrm{spec}} \circ J_{\mathrm{Op}}^{-, \mathrm{spec}} \right)^{f, \mathrm{ins.unit}} \rightarrow \mathrm{CT}^{-, \mathrm{spec}} \circ \underline{\mathrm{Poinc}}_{\tilde{G}, !}^{\mathrm{spec}}.$$

13.1.12. The main result of this section reads:

Theorem 13.1.13. *The natural transformation (13.10) is an isomorphism.*

13.2. Proof of Theorem 13.1.13.

13.2.1. Fix $\mathcal{Z} \rightarrow \mathrm{Ran}$. We wish to show that the natural transformation from the counterclockwise to the clockwise circuit in the diagram

$$(13.11) \quad \begin{array}{ccc} \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}) \otimes \mathrm{D-mod}(\mathcal{Z}) & \xrightarrow{\mathrm{CT}^{-, \mathrm{spec}} \otimes \mathrm{Id}} & \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \otimes \mathrm{D-mod}(\mathcal{Z}) \\ \uparrow \mathrm{Poinc}_{\tilde{G}, !, \mathcal{Z}}^{\mathrm{spec}} & & \uparrow \mathrm{Id} \otimes (\mathrm{pr}_{\mathrm{small}, \mathcal{Z}})_! \\ \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})_{\mathcal{Z}} & & \mathrm{IndCoh}(\mathrm{LS}_{\tilde{M}}) \otimes \mathrm{D-mod}(\mathcal{Z}^{\subseteq}) \\ \downarrow \mathrm{ins.unit}_{\mathcal{Z}} & & \uparrow \mathrm{Poinc}_{\tilde{M}, !, \tilde{\rho}_P, \mathcal{Z}^{\subseteq}}^{\mathrm{spec}} \\ \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})_{\mathcal{Z}^{\subseteq}} & \xrightarrow{J_{\mathrm{Op}}^{-, \mathrm{spec}}} & \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{M}, \tilde{\rho}_P}^{\mathrm{mon-free}})_{\mathcal{Z}^{\subseteq}} \end{array}$$

arising from (13.1) is an isomorphism.

13.2.2. Let $\mathrm{Op}_{\tilde{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}}$ be as in [GLC2, Sect. 15.2.5]. I.e., it is the space that associates to a point of $\mathcal{Z}^{\subseteq}$ mapping to $(\underline{x} \subseteq \underline{x}') \in \mathrm{Ran}^{\subseteq}$ the space of opers on $\mathcal{D}_{\underline{x}'} - \underline{x}$ that are monodromy-free around $\underline{x}$, i.e.,

$$\mathrm{Op}_{\tilde{G}}(\mathcal{D}_{\underline{x}'} - \underline{x})_{\mathrm{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}'} - \underline{x})} \times \mathrm{LS}_{\tilde{G}}(\mathcal{D}_{\underline{x}'}).$$

Consider the corresponding maps

$$\mathrm{Op}_{\tilde{G}, \mathcal{Z}}^{\mathrm{mon-free}} \xleftarrow{\mathrm{pr}_{\mathrm{small}, \mathcal{Z}}^{\mathrm{Op}_{\tilde{G}}}} \mathrm{Op}_{\tilde{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \xrightarrow{\mathrm{pr}_{\mathrm{big}, \mathcal{Z}}^{\mathrm{Op}_{\tilde{G}}}} \mathrm{Op}_{\tilde{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon-free}}.$$

Note that the map $\mathrm{pr}_{\mathrm{big}, \mathcal{Z}}^{\mathrm{Op}_{\tilde{G}}}$ is an ind-closed embedding.

13.2.3. Recall that the functor

$$\mathrm{ins.unit}_{\mathcal{Z}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})_{\mathcal{Z}} \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\tilde{G}}^{\mathrm{mon-free}})_{\mathcal{Z}^{\subseteq}}$$

is given by pull-push along the diagram

$$\begin{array}{ccc} & \mathrm{Op}_{\tilde{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon-free}} & \\ & \uparrow \mathrm{pr}_{\mathrm{big}, \mathcal{Z}}^{\mathrm{Op}_{\tilde{G}}} & \\ \mathrm{Op}_{\tilde{G}, \mathcal{Z}}^{\mathrm{mon-free}} & \xleftarrow[\mathrm{pr}_{\mathrm{small}, \mathcal{Z}}^{\mathrm{Op}_{\tilde{G}}}]{} & \mathrm{Op}_{\tilde{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon-free} \rightsquigarrow \mathrm{reg}} \end{array}$$

13.2.4. By Sects. 13.1.8-13.1.10, the counterclockwise composition in (13.11) is given by pull-push along the following diagram

$$(13.12) \quad \begin{array}{ccccc} & & & & \mathrm{LS}_{\check{M}} \times \mathcal{Z} \\ & & & & \uparrow \mathrm{Id} \otimes (\mathrm{pr}_{\mathrm{small}, \mathcal{Z}})! \\ & & & & \mathrm{LS}_{\check{M}} \times \mathcal{Z}^{\subseteq} \\ & & & \uparrow & \\ & & & \mathrm{MOP}_{\check{G}, \check{P}^-, \mathcal{Z}^{\subseteq}}^{\mathrm{mer}, \mathrm{mon}\text{-}\mathrm{free}_{\check{P}^-, \mathrm{glob}}} & \xleftarrow{\mathrm{ev}_Z^{\mathrm{MOP}}} \mathrm{MOP}_{\check{G}, \check{P}^-, \mathcal{Z}^{\subseteq}}^{\mathrm{mer}, \mathrm{mon}\text{-}\mathrm{free}_{\check{P}^-}} \\ & & \xleftarrow{\mathrm{pMiura}, \mathrm{mon}\text{-}\mathrm{free}} & \mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon}\text{-}\mathrm{free}} & \\ & \uparrow \mathrm{pr}_{\mathrm{big}, \mathcal{Z}}^{\mathrm{Op}_{\check{G}}} & & & \\ \mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mon}\text{-}\mathrm{free}} & \xleftarrow{\mathrm{pr}_{\mathrm{small}, \mathcal{Z}}^{\mathrm{Op}_{\check{G}}}} & \mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon}\text{-}\mathrm{free} \rightsquigarrow \mathrm{reg}} & & \end{array}.$$

Note that the fiber product

$$\mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon}\text{-}\mathrm{free} \rightsquigarrow \mathrm{reg}} \times_{\mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon}\text{-}\mathrm{free}}} \mathrm{MOP}_{\check{G}, \check{P}^-, \mathcal{Z}^{\subseteq}}^{\mathrm{mer}, \mathrm{mon}\text{-}\mathrm{free}_{\check{P}^-, \mathrm{glob}}}$$

identifies with

$$(13.13) \quad \widetilde{\mathrm{MOP}}_{\check{G}, \check{P}^-, \mathcal{Z}^{\subseteq}}^{\mathrm{mer}, \mathrm{mon}\text{-}\mathrm{free}_{\check{P}^-, \mathrm{glob}}} := \left(\mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mon}\text{-}\mathrm{free}, \mathrm{glob}} \times_{\mathcal{Z}, \mathrm{pr}_{\mathrm{small}, \mathcal{Z}}} \mathcal{Z}^{\subseteq} \right) \times_{\mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon}\text{-}\mathrm{free}, \mathrm{glob}}} \mathrm{MOP}_{\check{G}, \check{P}^-, \mathcal{Z}^{\subseteq}}^{\mathrm{mer}, \mathrm{mon}\text{-}\mathrm{free}_{\check{P}^-, \mathrm{glob}}},$$

where

$$\mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mon}\text{-}\mathrm{free}, \mathrm{glob}} \times_{\mathcal{Z}, \mathrm{pr}_{\mathrm{small}, \mathcal{Z}}} \mathcal{Z}^{\subseteq} \rightarrow \mathrm{Op}_{\check{G}, \mathcal{Z}^{\subseteq}}^{\mathrm{mon}\text{-}\mathrm{free}, \mathrm{glob}}$$

is the natural restriction map.

Hence, we can replace the diagram (13.12) by

$$(13.14) \quad \begin{array}{ccccc} & & & & \mathrm{LS}_{\check{M}} \times \mathcal{Z} \\ & & & & \uparrow \mathrm{Id} \otimes (\mathrm{pr}_{\mathrm{small}, \mathcal{Z}})! \\ & & & & \mathrm{LS}_{\check{M}} \times \mathcal{Z}^{\subseteq} \\ & & & \uparrow & \\ & & & \widetilde{\mathrm{MOP}}_{\check{G}, \check{P}^-, \mathcal{Z}^{\subseteq}}^{\mathrm{mer}, \mathrm{mon}\text{-}\mathrm{free}_{\check{P}^-, \mathrm{glob}}} & \xleftarrow{\mathrm{ev}_Z^{\mathrm{Op}_{\check{G}}}} \mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mon}\text{-}\mathrm{free}, \mathrm{glob}} \\ & \xleftarrow{\mathrm{ev}_Z^{\mathrm{Op}_{\check{G}}}} & \mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mon}\text{-}\mathrm{free}, \mathrm{glob}} & \xleftarrow{\quad} \mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mon}\text{-}\mathrm{free}, \mathrm{glob}} \times_{\mathcal{Z}, \mathrm{pr}_{\mathrm{small}, \mathcal{Z}}} \mathcal{Z}^{\subseteq} & \end{array}.$$

13.2.5. Note that we have a naturally defined map

$$\tilde{\mathcal{J}}^{\mathrm{glob}} : \widetilde{\mathrm{MOP}}_{\check{G}, \check{P}^-, \mathcal{Z}^{\subseteq}}^{\mathrm{mer}, \mathrm{mon}\text{-}\mathrm{free}_{\check{P}^-, \mathrm{glob}}} \rightarrow \mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mon}\text{-}\mathrm{free}, \mathrm{glob}} \times_{\mathrm{LS}_{\check{G}} \times \mathcal{Z}} (\mathrm{LS}_{\check{P}^-} \times \mathcal{Z}),$$

such that:

- The horizontal composition in (13.14) equals the composition of $\tilde{\mathcal{J}}^{\mathrm{glob}}$ with the horizontal composition in (13.4);
- The vertical composition in (13.14) equals the composition of $\tilde{\mathcal{J}}^{\mathrm{glob}}$ with the vertical composition in (13.4).

It follows formally from Proposition 13.1.6 that the map $\tilde{\mathcal{J}}^{\text{glob}}$ is pseudo-proper.

Unwinding, we obtain that the natural transformation in (13.11) equals

$$\begin{aligned} (\mathbf{q}^{-, \text{spec}})_*^{\text{IndCoh}} \circ (\mathbf{r}_G^{\text{glob}})_*^{\text{IndCoh}} \circ (\tilde{\mathcal{J}}^{\text{glob}})_*^{\text{IndCoh}} \circ (\tilde{\mathcal{J}}^{\text{glob}})^! \circ (\mathbf{p}^{-, \text{spec}})^! \circ (\text{ev}_Z^{\text{Op}_G})^! \rightarrow \\ \rightarrow (\mathbf{q}^{-, \text{spec}})_*^{\text{IndCoh}} \circ (\mathbf{r}_G^{\text{glob}})_*^{\text{IndCoh}} \circ (\mathbf{p}^{-, \text{spec}})^! \circ (\text{ev}_Z^{\text{Op}_G})^!, \end{aligned}$$

induced by the counit of the adjunction

$$(13.15) \quad (\tilde{\mathcal{J}}^{\text{glob}})_*^{\text{IndCoh}} \circ (\tilde{\mathcal{J}}^{\text{glob}})^! \rightarrow \text{Id}.$$

13.2.6. We will prove:

Proposition 13.2.7. *The natural transformation (13.15) is an isomorphism.*

The proof will be given in Sect. 13.3.8.

13.2.8. From Proposition 13.2.7, we obtain that the natural transformation in (13.11) is an isomorphism as well, as desired.

□[Theorem 13.1.13]

13.3. How do Miura opers appear?

13.3.1. In Sect. 10.5.1 we introduced the Zastava space, $\mathbf{Zas}$. Here we will use a variant of this construction, defined as follows.

For $Z \rightarrow \text{Ran}$, let $\text{Bun}_{\check{G}, Z}^{\text{mer}, \text{glob}}$ (resp., $\text{Bun}_{\check{B}, Z}^{\text{mer}, \text{glob}}$) be the space of $\check{G}$ - (resp., $\check{B}$ -) bundles on X defined away from the marked point.

Consider the fiber product

$$\text{Bun}_{\check{B}, Z}^{\text{mer}, \text{glob}} \times_{\text{Bun}_{\check{G}, Z}^{\text{mer}, \text{glob}}} (\text{Bun}_{\check{P}^-} \times Z).$$

Let Zast_Z be its open subfunctor corresponding to the condition that the $\check{B}$ -reduction and the $\check{P}^-$ -reduction are transversal at the generic point of the curve.

13.3.2. We have a tautological forgetful map

$$(13.16) \quad \text{Op}_{\check{G}, Z}^{\text{mon-free}, \text{glob}} \times_{\text{LS}_{\check{G}} \times Z} (\text{LS}_{\check{P}^-} \times Z) \rightarrow \text{Bun}_{\check{B}, Z}^{\text{mer}, \text{glob}} \times_{\text{Bun}_{\check{G}, Z}^{\text{mer}, \text{glob}}} (\text{Bun}_{\check{P}^-} \times Z).$$

The following observation is the reason Miura opers make an appearance:

Proposition 13.3.3. *The image of (13.16) lies in Zast_Z .*

The proof will be given in Sect. 13.4.

13.3.4. Let

$$\text{Zast}_{Z \subseteq}^\circ \subset \text{Zast}_Z \times_{Z, \text{pr}_{\text{small}, Z}} Z \subseteq$$

be the subfunctor given by the following condition: for $z \in Z \subseteq$ mapping to $(\underline{x} \subseteq \underline{x}') \in \text{Ran}^\subseteq$, we require that the $\check{B}$ -reduction and the $\check{P}^-$ -reduction be transversal on $X - \underline{x}'$.

We have a forgetful map

$$\tilde{\mathcal{J}}_{\text{Zast}} : \text{Zast}_{Z \subseteq}^\circ \rightarrow \text{Zast}_Z,$$

which fits into the *Cartesian* square

$$(13.17) \quad \begin{array}{ccc} \widetilde{\text{MOp}}_{\check{G}, \check{P}^-, Z \subseteq}^{\text{mer}, \text{mon-free}, \text{glob}} & \longrightarrow & \text{Zast}_{Z \subseteq}^\circ \\ \tilde{\mathcal{J}}^{\text{glob}} \downarrow & & \downarrow \tilde{\mathcal{J}}_{\text{Zast}} \\ \text{Op}_{\check{G}, Z}^{\text{mon-free}, \text{glob}} \times_{\text{LS}_{\check{G}} \times Z} (\text{LS}_{\check{P}^-} \times Z) & \longrightarrow & \text{Zast}_Z, \end{array}$$

where $\widetilde{\text{MOp}}_{\check{G}, \check{P}^-, Z \subseteq}^{\text{mer}, \text{mon-free}, \text{glob}}$ is as in (13.13).

13.3.5. We will prove:

Lemma 13.3.6.

- (a) *The map $\tilde{j}_{\text{Zast}}$ is pseudo-proper;*
- (b) *The fibers of $\tilde{j}_{\text{Zast}}$ over geometric points of $\text{Zast}_{\mathbb{Z}}$ are $\mathcal{O}$ -contractible.*

Let us show how Lemma 13.3.6 implies Propositions 13.1.6 and 13.2.7.

13.3.7. *Proof of Proposition 13.1.6.* The map j^{glob} is ind-schematic (and injective). Therefore, in order to show that it is ind-proper (in fact, an ind-closed embedding), it suffices to show that it is pseudo-proper.

By Lemma 13.3.6(a) and (13.17), the map $\tilde{j}^{\text{glob}}$ is pseudo-proper. Now, the map j^{glob} is the composition

$$\begin{aligned} \text{MOp}_{\tilde{G}, \tilde{P}^-, \mathbb{Z}}^{\text{mer, mon-free } \tilde{P}^-, \text{glob}} &\simeq \mathbb{Z} \times_{\mathbb{Z} \subseteq} \widetilde{\text{MOp}}_{\tilde{G}, \tilde{P}^-, \mathbb{Z} \subseteq}^{\text{mer, mon-free } \tilde{P}^-, \text{glob}} \rightarrow \\ &\rightarrow \widetilde{\text{MOp}}_{\tilde{G}, \tilde{P}^-, \mathbb{Z} \subseteq}^{\text{mer, mon-free } \tilde{P}^-, \text{glob}} \xrightarrow{\tilde{j}^{\text{glob}}} \text{Op}_{\tilde{G}, \mathbb{Z}}^{\text{mon-free, glob}} \times_{\text{LS}_{\tilde{G}}} (\text{LS}_{\tilde{P}^-} \times \mathbb{Z}), \end{aligned}$$

where the second arrow is the base change of

$$\mathbb{Z} \xrightarrow{\text{diag}_{\mathbb{Z}}} \mathbb{Z} \subseteq$$

and thus is a closed embedding.

□[Proposition 13.1.6]

13.3.8. *Proof of Proposition 13.2.7.* As was remarked above, the map $\tilde{j}^{\text{glob}}$ is pseudo-proper; hence, the functor $(\tilde{j}^{\text{glob}})_*^{\text{IndCoh}}$ satisfies base change. Hence, in order to show that (13.15) is an isomorphism, it is enough to do so at the level of fibers over field-valued points.

By (13.17), the assertion follows now from Lemma 13.3.6(b).

□[Proposition 13.2.7]

13.3.9. *Proof of Lemma 13.3.6(a).* It suffices to show that the inclusion

$$\text{Zast}_{\mathbb{Z} \subseteq}^{\circ} \hookrightarrow \text{Zast}_{\mathbb{Z}} \times_{\mathbb{Z}, \text{Pr}_{\text{small}, \mathbb{Z}}} \mathbb{Z} \subseteq$$

is an ind-closed embedding.

Let us be given $\underline{x} \in X$ and

- A $\tilde{B}$ -bundle $\mathcal{P}_{\tilde{B}}$ on $X - \underline{x}$;
- A $\tilde{P}^-$ -bundle $\mathcal{P}_{\tilde{P}^-}$ on $X - \underline{x}$;
- An isomorphism $\tilde{G} \times^{\tilde{B}} \mathcal{P}_{\tilde{B}} \xrightarrow{\beta} \tilde{G} \times^{\tilde{P}^-} \mathcal{P}_{\tilde{P}^-}$ over $X - \underline{x}$ that is *generically transversal*.

We have to show that the locus of $\underline{x}' \supseteq \underline{x}$ such that β be transversal over $X - \underline{x}'$ is an ind-closed subfunctor of $\text{Ran}_{\underline{x}}$.

This reduces to the following situation: let $\gamma : \mathcal{L}_1 \rightarrow \mathcal{L}_2$ be a *non-zero* map between line bundles on X with poles at $\underline{x}$. Then the locus of $\underline{x}' \supseteq \underline{x}$ such that γ is an isomorphism on $X - \underline{x}'$ is an ind-closed subfunctor of $\text{Ran}_{\underline{x}}$.

Up to modifying $\mathcal{L}_2$ at $\underline{x}$, we can assume that γ is a regular map. In this case the above locus in $\text{Ran}_{\underline{x}}$ is the support of $\text{coker}(\gamma)$.

□[Lemma 13.3.6(a)]

13.3.10. *Proof of Lemma 13.3.6(b).* Fix a geometric point $(\underline{x}, \mathcal{P}_{\tilde{B}}, \mathcal{P}_{\tilde{P}^-}, \beta) \in \text{Zast}_{\mathbb{Z}}$ as above. Let $U \subset X - \underline{x}$ be the locus over which β is transversal. Write $U = X - \underline{y}$.

Then the fiber of $\tilde{j}^{\text{Zast}}$ over the above point is $\text{Ran}_{\underline{y}}$. Now the assertion follows from the homological contractibility of $\text{Ran}_{\underline{y}}$, since

$$\text{Ran}_{\underline{y}} \simeq (\text{Ran}_{\underline{y}})_{\text{dR}}.$$

□[Lemma 13.3.6(b)]

13.4. **Opers force transversality.** In this subsection we prove Proposition 13.3.3.

13.4.1. The assertion of Proposition 13.3.3 can be checked at the level of geometric points. Let

$$(\mathcal{P}_{\tilde{G}}, \nabla, \mathcal{P}_{\tilde{B}}, \mathcal{P}_{\tilde{P}^-})$$

be an oper equipped with a reduction to $\tilde{P}^-$ of the underlying local system over an open curve $U \subset X$.

Assume by contradiction that $\mathcal{P}_{\tilde{B}}$ and $\mathcal{P}_{\tilde{P}^-}$ are not generically transversal. Then, up to shrinking U , we can assume that the relative position of $\mathcal{P}_{\tilde{B}}$ and $\mathcal{P}_{\tilde{P}^-}$ corresponds to a non-unit element $w \in W/W_M$, where W is the Weyl group.

13.4.2. Consider the fiber square of vector bundles on U :

$$\begin{array}{ccc} \check{\mathfrak{g}}^{\mathcal{P}_{\tilde{G}}}/\check{\mathfrak{b}}^{\mathcal{P}_{\tilde{B}}} \cap \check{\mathfrak{p}}_{\mathcal{P}_{\tilde{P}^-}}^- & \longrightarrow & \check{\mathfrak{g}}^{\mathcal{P}_{\tilde{G}}}/\check{\mathfrak{b}}^{\mathcal{P}_{\tilde{B}}} \\ \downarrow & & \downarrow \\ \check{\mathfrak{g}}^{\mathcal{P}_{\tilde{G}}}/\check{\mathfrak{p}}_{\mathcal{P}_{\tilde{P}^-}}^- & \longrightarrow & \check{\mathfrak{g}}^{\mathcal{P}_{\tilde{G}}}/\check{\mathfrak{b}}^{\mathcal{P}_{\tilde{B}}} + \check{\mathfrak{p}}_{\mathcal{P}_{\tilde{P}^-}}^- . \end{array}$$

The data of the connection ∇ and that of reductions of $\mathcal{P}_{\tilde{G}}$ to $\mathcal{P}_{\tilde{B}}$ and $\mathcal{P}_{\tilde{P}^-}$ gives rise to a section

$$(13.18) \quad \nabla \bmod \check{\mathfrak{b}}^{\mathcal{P}_{\tilde{B}}} \cap \check{\mathfrak{p}}_{\mathcal{P}_{\tilde{P}^-}}^-$$

of $(\check{\mathfrak{g}}^{\mathcal{P}_{\tilde{G}}}/\check{\mathfrak{b}}^{\mathcal{P}_{\tilde{B}}} \cap \check{\mathfrak{p}}_{\mathcal{P}_{\tilde{P}^-}}^-) \otimes \omega_X$.

By assumption, the image of (13.18) in $(\check{\mathfrak{g}}^{\mathcal{P}_{\tilde{G}}}/\check{\mathfrak{p}}_{\mathcal{P}_{\tilde{P}^-}}^-) \otimes \omega_X$ vanishes, and hence, so does its further image on $(\check{\mathfrak{g}}^{\mathcal{P}_{\tilde{G}}}/\check{\mathfrak{b}}^{\mathcal{P}_{\tilde{B}}} + \check{\mathfrak{p}}_{\mathcal{P}_{\tilde{P}^-}}^-) \otimes \omega_X$.

13.4.3. The image of (13.18) in $(\check{\mathfrak{g}}^{\mathcal{P}_{\tilde{G}}}/\check{\mathfrak{b}}^{\mathcal{P}_{\tilde{B}}}) \otimes \omega_X$ is given by the oper condition.

However, it is easy to see that for $w \neq 1$ the map

$$(\check{\mathfrak{g}}^{\mathcal{P}_{\tilde{G}}}/\check{\mathfrak{b}}^{\mathcal{P}_{\tilde{B}}}) \otimes \omega_X \rightarrow (\check{\mathfrak{g}}^{\mathcal{P}_{\tilde{G}}}/\check{\mathfrak{b}}^{\mathcal{P}_{\tilde{B}}} + \check{\mathfrak{p}}_{\mathcal{P}_{\tilde{P}^-}}^-) \otimes \omega_X$$

is non-zero on oper elements.

□[Proposition 13.3.3]

13.5. **The enhanced version of spectral Poincaré vs constant term compatibility.**

13.5.1. Since the functor

$$\text{Poinc}_{\tilde{G}, !, \mathbb{Z}}^{\text{spec}} : \text{IndCoh}^!(\text{Op}_{\tilde{G}, \mathbb{Z}}^{\text{mon-free}}) \rightarrow \text{IndCoh}(\text{LS}_{\tilde{G}}) \otimes \text{D-mod}(\mathbb{Z})$$

respects the actions of $\text{Sph}_{\tilde{G}, \mathbb{Z}}^{\text{spec}}$, it induces a functor

$$\text{Poinc}_{\tilde{G}, !, \mathbb{Z}}^{-, \text{spec}, \text{enhco}} : \text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free}})^{-, \text{enhco}}_{\mathbb{Z}} \rightarrow \text{IndCoh}(\text{LS}_{\tilde{G}})^{-, \text{enhco}}_{\mathbb{Z}},$$

where:

- $\text{IndCoh}^!(\text{Op}_{\tilde{G}}^{\text{mon-free}})^{-, \text{enhco}}$ is as in Sect. 4.7.2.
- $\text{IndCoh}(\text{LS}_{\tilde{G}})^{-, \text{enhco}}_{\mathbb{Z}}$ is as in Sect. 12.4.5.

Denote by

$$\underline{\mathrm{Poinc}}_{\check{G},!}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}} : \underline{\mathrm{IndCoh}}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})^{-,\mathrm{enh}_{\mathrm{co}}} \rightarrow \underline{\mathrm{IndCoh}}(\mathrm{LS}_{\check{G}})^{-,\mathrm{enh}_{\mathrm{co}}}$$

the resulting functor between crystals of categories.

The unital structure on $\underline{\mathrm{Poinc}}_{\check{G},!}^{-,\mathrm{spec}}$ induces one on $\underline{\mathrm{Poinc}}_{\check{G},!}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}}$.

13.5.2. Let

$$\underline{\mathrm{CT}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}} : \underline{\mathrm{IndCoh}}(\mathrm{LS}_{\check{G}})^{-,\mathrm{enh}_{\mathrm{co}}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran})$$

be as in Sect. 12.10.

Let

$$J_{\mathrm{Op}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})^{-,\mathrm{enh}_{\mathrm{co}}} \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M},\check{P}}^{\mathrm{mon-free}})$$

be as in Sect. 4.7.2.

Our current goal is to construct a natural transformation

$$(13.19) \quad \underline{\mathrm{Poinc}}_{\check{M},!,\check{P}}^{\mathrm{spec}} \circ J_{\mathrm{Op}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}} \rightarrow \underline{\mathrm{CT}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}} \circ \underline{\mathrm{Poinc}}_{\check{G},!}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}}$$

as local-to-global functors

$$\underline{\mathrm{IndCoh}}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})^{-,\mathrm{enh}_{\mathrm{co}}} \rightrightarrows \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \underline{\mathrm{D-mod}}(\mathrm{Ran}),$$

compatible with the lax unital structure.

13.5.3. Given $\mathcal{Z}$, we wish to construct a natural transformation

$$(13.20) \quad \underline{\mathrm{Poinc}}_{\check{M},!,\check{P},\mathcal{Z}}^{\mathrm{spec}} \circ J_{\mathrm{Op}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}} \rightarrow \underline{\mathrm{CT}}_{\mathcal{Z}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}} \circ \underline{\mathrm{Poinc}}_{\check{G},!,\mathcal{Z}}^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}}$$

as functors

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})^{-,\mathrm{enh}_{\mathrm{co}}} \rightrightarrows \mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \underline{\mathrm{D-mod}}(\mathcal{Z}).$$

By construction, the right-hand side in (13.20) is given by pull-push along the diagram

$$\begin{array}{ccc} & & \mathrm{LS}_{\check{M}} \times \mathcal{Z} \\ & & \uparrow \mathbf{q}_{\mathcal{Z}}^{-,\mathrm{spec}} \\ & & (\mathrm{LS}_{\check{P}^-, \mathcal{Z}}^{\mathrm{mon-free}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}} \\ & \xleftarrow{\rho_{\mathcal{Z}}^{-,\mathrm{spec}, \mathrm{enh}}} & (\mathrm{LS}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}} \times_{\mathrm{LS}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \mathcal{Z}}^{\mathrm{mon-free}, \check{M}} \\ & \uparrow & \\ (\mathrm{Op}_{\check{G}, \check{P}^-, \mathcal{Z}}^{\mathrm{mon-free}, \check{M}})^{\wedge}_{\mathrm{mon-free}} & \xleftarrow{\quad} & (\mathrm{Op}_{\check{G}, \check{P}^-, \mathcal{Z}}^{\mathrm{mon-free}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}}, \end{array}$$

where:

- $(\mathrm{Op}_{\check{G}, \check{P}^-, \mathcal{Z}}^{\mathrm{mon-free}, \check{M}})^{\wedge}_{\mathrm{mon-free}}$ is as in Sect. 4.4, i.e., is the formal completion of

$$\mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \mathcal{Z}}^{\mathrm{mon-free}, \check{M}}$$

along the natural map from $\mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \mathcal{Z}}^{\mathrm{reg}}$;

- $(\mathrm{Op}_{\check{G}, \check{P}^-, \mathcal{Z}}^{\mathrm{mon-free}, \mathrm{glob}})^{\wedge}_{\mathrm{mon-free}}$ denotes the formal completion of

$$\mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}, \mathrm{glob}} \times_{\mathrm{LS}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \mathcal{Z}}^{\mathrm{mon-free}, \check{M}}$$

along the natural map from $\mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}, \mathrm{glob}} \times_{\mathrm{LS}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \mathcal{Z}}^{\mathrm{reg}}$.

- The left horizontal arrow is induced by the restriction map $\mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}, \mathrm{glob}} \rightarrow \mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}}$;
- The middle vertical arrow is induced by the map $\mathrm{Op}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}, \mathrm{glob}} \rightarrow \mathrm{LS}_{\check{G}, \mathcal{Z}}^{\mathrm{mer}, \mathrm{glob}}$.

13.5.4. As in Sect. 13.1.7, the above functor receives a natural transformation from the functor given by pull-push along the diagram

$$(13.21) \quad \begin{array}{c} \text{LS}_{\tilde{M}} \times \mathcal{Z} \\ \uparrow \mathfrak{q}_{\mathcal{Z}}^-, \text{spec} \\ \left(\text{LS}_{\tilde{P}^-, \mathcal{Z}}^{\text{mon-free}, \text{glob}} \right)_{\text{mon-free}}^{\wedge} \\ \uparrow \\ (\text{Op}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mon-free}, \text{glob}})_{\text{mon-free}}^{\wedge} \longleftarrow (\text{Op}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mon-free}, \text{glob}})_{\text{mon-free}}^{\wedge} \longleftarrow (\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mon-free}, \text{glob}})_{\text{mon-free}}^{\wedge}, \end{array}$$

where $(\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mon-free}, \text{glob}})_{\text{mon-free}}^{\wedge}$ is the completion of

$$\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mer}, \text{glob}} \times_{\text{LS}_{\tilde{M}, \mathcal{Z}}^{\text{mer}}} \text{LS}_{\tilde{M}, \mathcal{Z}}^{\text{reg}}$$

along the natural map from

$$\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mer}, \text{mon-free}, \text{glob}} := \text{MOp}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mer}, \text{glob}} \times_{\text{LS}_{\tilde{P}^-, \mathcal{Z}}^{\text{mer}, \text{glob}}} (\text{LS}_{\tilde{P}^-} \times \mathcal{Z}) \simeq \text{MOp}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mer}, \text{glob}} \times_{\text{LS}_{\tilde{P}^-, \mathcal{Z}}^{\text{mer}}} \text{LS}_{\tilde{P}^-, \mathcal{Z}}^{\text{reg}}.$$

13.5.5. We now show that the latter functor equals the left-hand side in (13.20). Indeed, we can replace the diagram (13.21) by

$$(13.22) \quad \begin{array}{c} \text{LS}_{\tilde{M}} \times \mathcal{Z} \\ \uparrow \\ \text{Op}_{\tilde{M}, \tilde{\rho}_P, \mathcal{Z}}^{\text{mon-free}, \text{glob}} \\ \uparrow \\ (\text{Op}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mon-free}, \text{glob}})_{\text{mon-free}}^{\wedge} \longleftarrow (\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mon-free}, \text{glob}})_{\text{mon-free}}^{\wedge} \longleftarrow (\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mon-free}, \text{glob}})_{\text{mon-free}}^{\wedge}, \end{array}$$

where $(\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mon-free}, \text{glob}})_{\text{mon-free}}^{\wedge}$ is as Sect. 4.5.1, i.e., is the completion of

$$\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mer}} \times_{\text{LS}_{\tilde{M}, \mathcal{Z}}^{\text{mer}}} \text{LS}_{\tilde{M}, \mathcal{Z}}^{\text{reg}}$$

along the natural map from

$$\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mer}} \times_{\text{LS}_{\tilde{P}^-, \mathcal{Z}}^{\text{mer}}} \text{LS}_{\tilde{P}^-, \mathcal{Z}}^{\text{reg}}.$$

13.5.6. We note now that we have a Cartesian square

$$\begin{array}{ccc} \text{Op}_{\tilde{M}, \tilde{\rho}_P, \mathcal{Z}}^{\text{mon-free}} & \longleftarrow & \text{Op}_{\tilde{M}, \tilde{\rho}_P, \mathcal{Z}}^{\text{mon-free}, \text{glob}} \\ \uparrow & & \uparrow \\ (\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mon-free}, \text{glob}})_{\text{mon-free}}^{\wedge} & \longleftarrow & (\text{MOp}_{\tilde{G}, \tilde{P}^-, \mathcal{Z}}^{\text{mon-free}, \text{glob}})_{\text{mon-free}}^{\wedge} \end{array}$$

This diagram satisfies base change for $!$ -pullbacks and $*$ -pushforwards. Hence, we can further replace (13.22) by

$$\begin{array}{ccc}
 & & \text{LS}_{\check{M}} \times \mathbb{Z} \\
 & & \uparrow \\
 & \text{Op}_{\check{M}, \check{\rho}_P, \mathbb{Z}}^{\text{mon-free}} & \longleftarrow \text{Op}_{\check{M}, \check{\rho}_P, \mathbb{Z}}^{\text{mon-free, glob}} \\
 & \uparrow & \\
 (\text{MOp}_{\check{G}, \check{P}^-, \mathbb{Z}}^{\text{mon-free}})^{\wedge}_{\text{mon-free}} & \longleftarrow & (\text{MOp}_{\check{G}, \check{P}^-, \mathbb{Z}}^{\text{mon-free, glob}})^{\wedge}_{\text{mon-free}}
 \end{array}$$

The functor define by the latter diagram is by definition the left-hand side in (13.20).

13.5.7. Thus, we have completed the construction of the natural transformation (13.19).

By construction, if we precompose (13.19) on both sides with the functor

$$\begin{aligned}
 (13.23) \quad \text{IndCoh}^!(\text{Op}_{\check{G}}^{\text{mon-free}}) &\xrightarrow{\text{Id} \otimes \mathbf{1}_{\text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}}_{\text{co}}}} \text{IndCoh}^!(\text{Op}_{\check{G}}^{\text{mon-free}}) \otimes \text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}}_{\text{co}} \rightarrow \\
 &\rightarrow \text{IndCoh}^!(\text{Op}_{\check{G}}^{\text{mon-free}})^{\otimes (\text{Sph}_{\check{G}}^{\text{spec}})^{\vee}} \text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}}_{\text{co}} = \text{IndCoh}^!(\text{Op}_{\check{G}}^{\text{mon-free}})^{-, \text{enh}_{\text{co}}},
 \end{aligned}$$

we recover the natural transformation (13.1).

13.5.8. Since the right-hand side in (13.19) is a (strictly) unital local-to-global functor, by [GLC2, Sect. 11.4.8], from (13.19) we obtain a natural transformation

$$(13.24) \quad \left(\underline{\text{Poinc}}_{\check{M}, !, \check{\rho}_P}^{\text{spec}} \circ \underline{J}_{\text{Op}}^{-, \text{spec, enh}_{\text{co}}} \right)^f \text{ins. unit} \rightarrow \underline{\text{CT}}^{-, \text{spec, enh}_{\text{co}}} \circ \underline{\text{Poinc}}_{\check{G}, !}^{-, \text{spec, enh}_{\text{co}}}.$$

We claim:

Theorem 13.5.9. *The natural transformation (13.24) is an isomorphism.*

Proof. By construction, the categories involved carry an action of $\text{Sph}_{\check{M}}^{\text{spec}}$, and the two functors, as well as the natural transformation (13.24) are compatible with the action.

Hence, since $\text{IndCoh}(\text{Op}_{\check{G}}^{\text{mon-free}})^{-, \text{enh}_{\text{co}}}$ is generated under the action of $\text{Sph}_{\check{M}}^{\text{spec}}$ by the image of (13.23), it suffices to show that (13.24) becomes an isomorphism after precomposing with (13.23).

However, by Sect. 13.5.7, the resulting natural transformation identifies with (13.1). Hence, the required assertion follows from Theorem 13.1.13. $\square$

13.6. Addendum: spectral global sections and the spectral Poincaré functor. In this subsection we will prove a spectral counterpart of Theorem 11.7.6.

13.6.1. Let us denote by $\mathbf{P}_{\check{G}}^{\text{spec, loc, true}}$ the (lax unital) factorization functor

$$\begin{aligned}
 \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \otimes_{\text{Sph}_{\check{G}}^{\text{spec}}} \text{Rep}(\check{G}) &\simeq \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}})_{\text{mon-free}} \hookrightarrow \\
 &\hookrightarrow \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}})^{\Gamma^{\text{IndCoh}(\text{Op}_{\check{G}}^{\text{mer}}, -)}} \text{Vect},
 \end{aligned}$$

where the first arrow is [GLC2, Equation (3.17)].

Recall that the functor $\mathbf{P}_{\check{G}}^{\text{spec, loc}}$ is defined to be the composition

$$\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \otimes \text{Rep}(\check{G}) \rightarrow \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \otimes_{\text{Sph}_{\check{G}}^{\text{spec}}} \text{Rep}(\check{G}) \xrightarrow{\mathbf{P}_{\check{G}}^{\text{spec, loc, true}}} \text{Vect},$$

see [GLC2, Sect. 6.4.7].

13.6.2. In [GLC2, Theorem 17.7.2] it was shown that the functor

$$(13.25) \quad (\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes \mathrm{Rep}(\check{G}))_Z \xrightarrow{\mathrm{Poinc}_{\check{G},*}^Z \otimes \mathrm{Id}} (\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z})) \otimes_{\mathrm{D-mod}(\mathbb{Z})} \mathrm{Rep}(\check{G})_Z \rightarrow \\ \xrightarrow{\Gamma_Z^{\mathrm{spec}, \mathrm{IndCoh}} \otimes \mathrm{Id}} (\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G}))_Z \rightarrow \mathrm{D-mod}(\mathbb{Z})$$

is canonically isomorphic to

$$(13.26) \quad (\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes \mathrm{Rep}(\check{G}))_Z \xrightarrow{\mathrm{ins.unit}_Z} (\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes \mathrm{Rep}(\check{G}))_{Z \subseteq} \xrightarrow{\mathbf{P}_G^{\mathrm{spec}, \mathrm{loc}}} \\ \rightarrow \mathrm{D-mod}(\mathbb{Z} \subseteq) \xrightarrow{(\mathrm{pr}_{\mathrm{small}, Z})!} \mathrm{D-mod}(\mathbb{Z}).$$

13.6.3. Note now that both (13.25) and (13.26) factor via

$$(\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes \mathrm{Rep}(\check{G}))_Z \rightarrow (\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{Rep}(\check{G}))_Z \rightarrow \mathrm{D-mod}(\mathbb{Z}).$$

Indeed, for (13.25) this factorization is provided by

$$(13.27) \quad (\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{Rep}(\check{G}))_Z \xrightarrow{\mathrm{Poinc}_{\check{G},*}^Z \otimes \mathrm{Id}} \\ \rightarrow (\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z})) \otimes_{\mathrm{Sph}_{\check{G}, Z}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_Z \xrightarrow{\Gamma_Z^{\mathrm{spec}, \mathrm{IndCoh}} \otimes \mathrm{Id}} \\ \rightarrow (\mathrm{Rep}(\check{G}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{Rep}(\check{G}))_Z \rightarrow \mathrm{D-mod}(\mathbb{Z}).$$

For (13.26) this factorization is provided by

$$(13.28) \quad (\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{Rep}(\check{G}))_Z \xrightarrow{\mathrm{ins.unit}_Z} \\ \rightarrow (\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{Rep}(\check{G}))_{Z \subseteq} \xrightarrow{\mathbf{P}_G^{\mathrm{spec}, \mathrm{loc}, \mathrm{true}}} \mathrm{D-mod}(\mathbb{Z} \subseteq) \xrightarrow{(\mathrm{pr}_{\mathrm{small}, Z})!} \mathrm{D-mod}(\mathbb{Z}).$$

13.6.4. Denote the functor (13.25) by $\mathbf{P}_{\check{G}, Z}^{\mathrm{spec}, \mathrm{glob}}$ and the functor (13.27) by $\mathbf{P}_{\check{G}, Z}^{\mathrm{spec}, \mathrm{glob}, \mathrm{true}}$.

Note that we can think of the functor (13.28) as

$$(\mathbf{P}_{\check{G}}^{\mathrm{spec}, \mathrm{loc}, \mathrm{true}})_Z \int_Z^{\mathrm{ins.unit}}.$$

13.6.5. In this subsection we will prove the following sharpening of [GLC2, Theorem 14.1.4]:

Theorem 13.6.6. *There is a canonical isomorphism*

$$\mathbf{P}_{\check{G}, Z}^{\mathrm{spec}, \mathrm{glob}, \mathrm{true}} \simeq (\mathbf{P}_{\check{G}}^{\mathrm{spec}, \mathrm{loc}, \mathrm{true}})_Z \int_Z^{\mathrm{ins.unit}}$$

as local-to-global functors

$$\underline{\mathrm{IndCoh}}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\underline{\mathrm{Sph}}_G^{\mathrm{spec}}} \underline{\mathrm{Rep}}(\check{G}) \rightarrow \underline{\mathrm{D-mod}}(\mathrm{Ran}).$$

The rest of this subsection is devoted to the proof of this theorem.

13.6.7. Fix $\mathcal{Z} \rightarrow \text{Ran}$. Unwinding the definitions, we can rewrite the functor $\mathbf{P}_{\check{G}, \mathcal{Z}}^{\text{spec, glob, true}}$ as $*$ -pull followed by $*$ -push along the following diagram

$$\begin{array}{ccc} & & \mathcal{Z} \\ & & \uparrow \\ & & (\text{LS}_{\check{G}, \mathcal{Z}}^{\text{mer, glob}})^{\wedge}_{\text{mon-free}} \\ & & \uparrow \\ (\text{Op}_{\check{G}, \mathcal{Z}}^{\text{mer}})^{\wedge}_{\text{mon-free}} & \longleftarrow & (\text{Op}_{\check{G}, \mathcal{Z}}^{\text{mer, glob}})^{\wedge}_{\text{mon-free}} \end{array}$$

while $\mathbf{P}_{\check{G}, \mathcal{Z}}^{\text{spec, loc, true}}$ is just the $*$ -pushforward along

$$(\text{Op}_{\check{G}, \mathcal{Z}}^{\text{mer}})^{\wedge}_{\text{mon-free}} \rightarrow \mathcal{Z}.$$

13.6.8. This gives rise to a natural transformation

$$(13.29) \quad \underline{\mathbf{P}}_{\check{G}}^{\text{spec, loc, true}} \rightarrow \underline{\mathbf{P}}_{\check{G}}^{\text{spec, glob, true}},$$

as lax unital local-to-global functors

$$\underline{\text{IndCoh}}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \otimes_{\underline{\text{Sph}}_{\check{G}}^{\text{spec}}} \underline{\text{Rep}}(\check{G}) \rightrightarrows \text{Vect},$$

where the right-hand side in (13.29) is strictly unital.

We claim that the induced natural transformation

$$(13.30) \quad (\underline{\mathbf{P}}_{\check{G}}^{\text{spec, loc, true}})^f \text{ ins. unital} \rightarrow \underline{\mathbf{P}}_{\check{G}}^{\text{spec, glob, true}}$$

is an isomorphism.

13.6.9. Indeed, when we precompose the functors on the two sides of (13.29) with

$$\underline{\text{IndCoh}}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \otimes \underline{\text{Rep}}(\check{G}) \rightarrow \underline{\text{IndCoh}}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \otimes_{\underline{\text{Sph}}_{\check{G}}^{\text{spec}}} \underline{\text{Rep}}(\check{G}),$$

we obtain a natural transformation

$$\underline{\mathbf{P}}_{\check{G}}^{\text{spec, loc}} \rightarrow \underline{\mathbf{P}}_{\check{G}}^{\text{spec, glob}},$$

(between lax unital functors, with the right-hand side strictly unital), such that the induced natural transformation

$$(\underline{\mathbf{P}}_{\check{G}}^{\text{spec, loc}})^f \text{ ins. unit} \rightarrow \underline{\mathbf{P}}_{\check{G}}^{\text{spec, glob}}$$

is the isomorphism of [GLC2, Theorem 17.7.2].

This formally implies that (13.30) is also an isomorphism.

□[Theorem 13.6.6]

Part III: Applications to the Langlands functor

In this Part we use the material developed in Part II of the paper to establish properties of the Langlands functor

$$\mathbb{L}_G :: \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$$

vis-a-vis functors that connect G to Levi quotients of its parabolic subgroups.

We prove the following:

- It intertwines the $\mathrm{Eis}_!$ functor on the geometric side with the $\mathrm{Eis}^{\mathrm{spec}}$ functor on the spectral side (up to some twists and shifts), see Theorem 14.2.2, and similarly for the enhanced versions;
- It intertwines the CT_* functor on the geometric side with the $\mathrm{CT}^{\mathrm{spec}}$ functor on the spectral side (up to some twists and shifts), see Theorem 15.1.13, and similarly for the enhanced versions;
- The Langlands functor admits a left adjoint, see Theorem 16.1.2;
- The left adjoint of the Langlands functor identifies with its dual (up to some twists and shifts), see Theorem 16.2.3.

The main result of this Part and of the entire paper, Theorem 17.1.2, says that the Langlands functor $\mathbb{L}_G$ and its left adjoint $\mathbb{L}_G^L$ define mutually inverse equivalences between the following full subcategories on the geometric and spectral sides, respectively:

- On the geometric side, this is the subcategory

$$\mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \subset \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G),$$

generated by the essential images of the Eisenstein functors $\mathrm{Eis}_!$ for all *proper* parabolics;

- On the spectral side, this is the full subcategory

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}),$$

consisting of objects, set-theoretically supported on the locus $\mathrm{LS}_{\check{G}}^{\mathrm{red}} \subset \mathrm{LS}_{\check{G}}$, consisting of *reducible* local systems.

We will reduce the assertion of Theorem 17.1.2 to Proposition 17.3.9, which says that the functors

$$\mathrm{CT}_{!*}^{-,\mathrm{spec}} \quad \text{and} \quad \mathrm{CT}_{!*}^{-,\mathrm{spec}} \circ \mathbb{L}_G \circ \mathbb{L}_G^L,$$

are canonically isomorphic, where $\mathrm{CT}_{!*}^{-,\mathrm{spec}}$ is the “compactified” spectral constant term functor from Sect. 12.12.2.

14. LANGLANDS FUNCTOR AND EISENSTEIN SERIES

In this section we establish the compatibility of the Langlands functor with the functors $\mathrm{Eis}_!$ and $\mathrm{Eis}^{\mathrm{spec}}$, as well as their enhancements.

14.1. The Langlands functor: recollections. In this subsection we recall some material from [GLC2, Sect. 18].

14.1.1. The Langlands functor

$$\mathbb{L}_G : \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$$

was introduced in [GLC1, Sect. 1.6.6].

14.1.2. *Whittaker compatibility.* Combining [GLC1, Corollary 1.6.5] with [GLC2, Proposition 18.1.6] (and the fully faithfulness of the functor $\Gamma_{\check{G}}^{\text{spec}}$), we obtain that the functor $\mathbb{L}_G$ is characterized uniquely by the following two properties:

- It makes the diagram

$$(14.1) \quad \begin{array}{ccc} \text{Whit}^!(G)_{\text{Ran}} & \xrightarrow{\text{CS}_G} & \text{Rep}(\check{G})_{\text{Ran}} \\ \text{coeff}_G[2\delta_{N_{\rho(\omega_X)}}] \uparrow & & \uparrow \Gamma_{\check{G}}^{\text{spec}, \text{IndCoh}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \end{array}$$

commute;

- It sends compact objects in $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ to $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}})^{>-\infty}$.

14.1.3. *Compatibility with the Hecke action.* According to [GLC2, Corollary 18.3.10], the functor $\mathbb{L}_G$ is compatible with the spherical Hecke action in the sense that the functor

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}) \xrightarrow{\mathbb{L}_G \otimes \text{Id}} \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \otimes \text{D-mod}(\text{Ran})$$

is compatible with the actions of

$$(14.2) \quad \text{Sph}_{G, \text{Ran}} \xrightarrow{\text{Sat}_G} \text{Sph}_{G, \text{Ran}}^{\text{spec}}$$

on each side, respectively.

14.1.4. *Compatibility with Kac-Moody localization.* According to [GLC2, Theorem 18.5.2], we have the following commutative diagram

$$(14.3) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \\ \text{Loc}_G \otimes \mathbb{I}[-d] \uparrow & & \uparrow \text{Poinc}_{G, *}^{\text{spec}} \\ \text{KL}(G)_{\text{crit}, \text{Ran}} & \xrightarrow{\text{FLE}_{G, \text{crit}}} & \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})_{\text{Ran}}, \end{array}$$

where

$$\mathbb{I} := \mathbb{I}_{G, N_{\rho(\omega_X)}}^{\otimes \frac{1}{2}} \otimes \mathbb{I}_{N_{\rho(\omega_X)}}^{\otimes -1} \quad \text{and} \quad d = \delta_{N_{\rho(\omega_X)}}$$

for the lines $\mathbb{I}_{G, N_{\rho(\omega_X)}}^{\otimes \frac{1}{2}}$ and $\mathbb{I}_{N_{\rho(\omega_X)}}$ from Sect. 11.7.2.

14.1.5. Let us rewrite (14.3) equivalently as

$$(14.4) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\text{Ran}) & \xrightarrow{\mathbb{L}_G \otimes \text{Id}} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \otimes \text{D-mod}(\text{Ran}) \\ \text{Loc}_{G, \text{Ran}} \otimes \mathbb{I}[-d] \uparrow & & \uparrow \text{Poinc}_{G, *, \text{Ran}}^{\text{spec}} \\ \text{KL}(G)_{\text{crit}, \text{Ran}} & \xrightarrow{\text{FLE}_{G, \text{crit}}} & \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})_{\text{Ran}}, \end{array}$$

and note that left column is a functor between $\text{Sph}_{G, \text{Ran}}$ -module categories, and the right column is a functor between $\text{Sph}_{G, \text{Ran}}^{\text{spec}}$ -module categories, while the horizontal arrows intertwine these actions via (14.2).

We claim now that the statement of [GLC2, Theorem 18.5.2] can be complemented as follows:

Theorem 14.1.6. *The datum of commutativity in (14.4) is compatible with the Hecke actions via (14.2).*

The proof will be given in Sect. 14.4.

14.1.7. *The twisted version.* Let $\mathcal{P}_{Z_G^0}$ be a Z_G^0 -torsor on X . Recall the twisted FLE:

$$\mathrm{FLE}_{G, \mathrm{crit} + \mathrm{dlog}(\mathcal{P}_{Z_G^0})} : \mathrm{KL}(G)_{\mathrm{crit} + \mathrm{dlog}(\mathcal{P}_{Z_G^0})} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{G, \mathcal{P}_{Z_G^0}}^{\mathrm{mon-free}}),$$

see Sect. 5.2.

Using (5.4) (instead of the untwisted version, i.e., [GLC2, Theorem 6.4.5]), one proves the following variant of [GLC2, Theorem 18.5.2] :

Theorem 14.1.8. *We have a commutative diagram*

$$\begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\ \mathrm{Loc}_{G, \mathcal{P}_{Z_G^0}} \otimes \mathbb{I}[-d] \uparrow & & \uparrow \mathrm{Poinc}_{G, *, \mathcal{P}_{Z_G^0}}^{\mathrm{spec}} \\ \mathrm{KL}(G)_{\mathrm{crit} + \mathrm{dlog}(\mathcal{P}_{Z_G^0}), \mathrm{Ran}} & \xrightarrow{\mathrm{FLE}_{G, \mathrm{crit} + \mathrm{dlog}(\mathcal{P}_{Z_G^0})}} & \mathrm{IndCoh}^*(\mathrm{Op}_{G, \mathcal{P}_{Z_G^0}}^{\mathrm{mon-free}})_{\mathrm{Ran}}. \end{array}$$

14.1.9. We will apply Theorem 14.1.8 when the group in question is the Levi subgroup M of the original G and $\mathcal{P}_{Z_G^0} = \check{\rho}_P(\omega_X)$. In this case, the commutative diagram in Theorem 14.1.8 reads as

$$(14.5) \quad \begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ \mathrm{Loc}_{M, \check{\rho}_P} \otimes \mathbb{I}[-d] \uparrow & & \uparrow \mathrm{Poinc}_{G, *, \check{\rho}_P}^{\mathrm{spec}} \\ \mathrm{KL}(M)_{\mathrm{crit}_M + \check{\rho}_P, \mathrm{Ran}} & \xrightarrow{\mathrm{FLE}_{M, \mathrm{crit} + \check{\rho}_P}} & \mathrm{IndCoh}^*(\mathrm{Op}_{M, \check{\rho}_P}^{\mathrm{mon-free}})_{\mathrm{Ran}}, \end{array}$$

where

$$\mathbb{I} := \mathbb{I}_{M, N(M)_{\rho_M(\omega_X)}}^{\otimes \frac{1}{2}} \otimes \mathbb{I}_{N(M)_{\rho_M(\omega_X)}}^{\otimes -1}$$

and

$$d := \delta_{N(M)_{\rho_M(\omega_X)}}.$$

14.2. Compatibility of the Langlands functor with Eisenstein series. In this subsection we establish the compatibility of the Langlands functor with (usual) Eisenstein series.

14.2.1. We are going to prove:

Theorem 14.2.2. *The following diagram commutes*

$$(14.6) \quad \begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\ \mathrm{Eis}_{\mathbb{I}, \rho_P(\omega_X)}^- [\delta_{(N_P^-)_{\rho_P(\omega_X)}}] \uparrow & & \uparrow \mathrm{Eis}^-, \mathrm{spec} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}). \end{array}$$

The rest of this subsection is devoted to the proof of this theorem.

14.2.3. We claim that in order to prove the commutativity of (14.6), it is sufficient to prove the commutativity of

$$(14.7) \quad \begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\Psi_{\mathrm{LS}_{\check{G}}} \circ \mathbb{L}_G} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \\ \mathrm{Eis}_{\mathbb{I}, \rho_P(\omega_X)}^- [\delta_{(N_P^-)_{\rho_P(\omega_X)}}] \uparrow & & \uparrow \Psi_{\mathrm{LS}_{\check{G}}} \circ \mathrm{Eis}^-, \mathrm{spec} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}). \end{array}$$

Indeed, it is sufficient to show that (14.6) commutes, when evaluated on compact objects of $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)$.

We claim that both circuits in (14.6) send $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)^c$ to $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})^{>-\infty}$. This would prove the required assertion, since the functor $\Psi_{\mathrm{LS}_{\check{G}}}$ is fully faithful when restricted to $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})^{>-\infty}$.

For the clockwise circuit, the claim follows from the fact that the functor $\mathrm{Eis}_{!,\rho_P(\omega_X)}^-$ preserves compactness (it admits a right adjoint, namely, $\mathrm{CT}_{*,\rho_P(\omega_X)}^-$) and the fact that $\mathbb{L}_G$ sends compact objects to $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})^{>-\infty}$.

For the counterclockwise circuit, this follows from the fact that the functor $\mathbb{L}_M$ sends compact objects to $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}})^{>-\infty}$ and the fact that the functor $\mathrm{Eis}^{-,\mathrm{spec}}$ is left t-exact (up to a cohomological shift).

14.2.4. Next, since the functor

$$\Gamma_{\check{G}}^{\mathrm{spec}} : \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \rightarrow \mathrm{Rep}(\check{G})_{\mathrm{Ran}}$$

is fully faithful, in order to prove the commutativity of (14.7), it suffices to prove the commutativity of

$$\begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\Gamma_{\check{G}}^{\mathrm{spec}} \circ \Psi_{\mathrm{LS}_{\check{G}}} \circ \mathbb{L}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{Eis}_{!,\rho_P(\omega_X)}^- [\delta_{(N_P^-)_{\rho_P(\omega_X)}}] \uparrow & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}} \circ \Psi_{\mathrm{LS}_{\check{G}}} \circ \mathrm{Eis}^{-,\mathrm{spec}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}), \end{array}$$

which is the same as

$$(14.8) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}} \circ \mathbb{L}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{Eis}_{!,\rho_P(\omega_X)}^- [\delta_{(N_P^-)_{\rho_P(\omega_X)}}] \uparrow & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}} \circ \mathrm{Eis}^{-,\mathrm{spec}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}), \end{array}$$

Hence, given the commutation of (14.1), we obtain that it suffices to prove the commutation of

$$(14.9) \quad \begin{array}{ccc} \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xrightarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{coeff}_G[2\delta_{N_{\rho_P(\omega_X)}}] \uparrow & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\ \mathrm{Eis}_{!,\rho_P(\omega_X)}^- [\delta_{(N_P^-)_{\rho_P(\omega_X)}}] \uparrow & & \uparrow \mathrm{Eis}^{-,\mathrm{spec}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}). \end{array}$$

14.2.5. We now use the material from Part II of the paper. Namely, by Corollary 10.1.5, the left vertical composition in (14.9) identifies with

$$(14.10) \quad \begin{aligned} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathrm{coeff}_M^{\mathrm{untl}}} \mathrm{Whit}^!(M)_{\mathrm{Ran}^{\mathrm{untl}}} \xrightarrow{(J_{\mathrm{Whit},\Theta}^-)^{\vee}} \\ & \rightarrow \mathrm{Whit}^!(G)_{\mathrm{Ran}^{\mathrm{untl}}} \xrightarrow{\mathrm{emb.indep}_L^{\mathrm{Whit}^!(G)}} \mathrm{Whit}^!(G)_{\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}} \rightarrow \mathrm{Whit}^!(G)_{\mathrm{Ran}}. \end{aligned}$$

Similarly, by Corollary 12.2.6, the right vertical composition in (14.9) identifies with

$$(14.11) \quad \begin{aligned} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) & \xrightarrow{\Gamma_{\check{M}}^{\mathrm{spec}, \mathrm{IndCoh}}} \mathrm{Rep}(\check{M})_{\mathrm{Ran}^{\mathrm{untl}}} \xrightarrow{\mathrm{co}J_{\mathrm{Rep}(\check{G})}^{\mathrm{spec},!}} \\ & \rightarrow \mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{untl}}} \xrightarrow{\mathrm{emb.indep}_L^{\mathrm{Rep}(\check{G})}} \mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{untl},\mathrm{indep}}} \rightarrow \mathrm{Rep}(\check{G})_{\mathrm{Ran}}. \end{aligned}$$

Thus, it suffices to establish the commutation of each of the three squares in

$$(14.12) \quad \begin{array}{ccc} \mathrm{Whit}^!(G)_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}} & \xrightarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}} \\ \mathrm{emb.indep}^L_{\mathrm{Whit}^!(G)} \uparrow & & \uparrow \mathrm{emb.indep}^L_{\mathrm{Rep}(\check{G})} \\ \mathrm{Whit}^!(G)_{\mathrm{Ran}^{\mathrm{untl}}} & \xrightarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{untl}}} \\ (J_{\mathrm{Whit}, \Theta}^{-, !})^\vee \uparrow & & \uparrow \mathrm{co}J^{-, \mathrm{spec}, !} \\ \mathrm{Whit}^!(M)_{\mathrm{Ran}^{\mathrm{untl}}} & \xrightarrow{\mathrm{CS}_M} & \mathrm{Rep}(\check{M})_{\mathrm{Ran}^{\mathrm{untl}}} \\ \mathrm{coeff}_{M, \mathrm{untl}} \uparrow & & \uparrow \Gamma_{M, \mathrm{untl}}^{\mathrm{spec}, \mathrm{IndCoh}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}). \end{array}$$

14.2.6. The upper square in (14.12) commutes tautologically (since CS_G is an equivalence of unital factorization categories).

The lower square is (14.1) for M .

14.2.7. Finally, we use the input from Part I of the paper. Namely, the commutative of the middle square in (14.12) is obtained by duality from (2.26).

□[Theorem 14.2.2]

Remark 14.2.8. Note that one can depict the proof of Theorem 14.2.2 given above as a cube:

$$(14.13) \quad \begin{array}{ccccc} & & \mathrm{Whit}^!(G)_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}} & \xrightarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}} \\ & \nearrow & \uparrow & & \nearrow \\ \mathrm{Whit}^!(M)_{\mathrm{Ran}^{\mathrm{untl}}} & \xrightarrow{\mathrm{CS}_M} & \mathrm{Rep}(\check{M})_{\mathrm{Ran}^{\mathrm{untl}}} & & \\ \uparrow \mathrm{coeff}_{M, \mathrm{untl}} & & \uparrow \mathrm{coeff}_{G, \mathrm{untl}} & & \uparrow \Gamma_{\check{G}, \mathrm{untl}}^{\mathrm{spec}, \mathrm{IndCoh}} \\ & & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}(X)) \\ & \nearrow \mathrm{Eis}^{-, !, \rho_P(\omega_X)} & \uparrow \Gamma_{\check{M}, \mathrm{untl}}^{\mathrm{spec}, \mathrm{IndCoh}} & & \nearrow \mathrm{Eis}^{-, \mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_M(X)), & & \end{array}$$

where the upper slanted arrows are

$$\mathrm{Whit}^!(M)_{\mathrm{Ran}^{\mathrm{untl}}} \xrightarrow{(J_{\mathrm{Whit}, \Theta}^{-, !})^\vee} \mathrm{Whit}^!(G)_{\mathrm{Ran}^{\mathrm{untl}}} \xrightarrow{\mathrm{emb.indep}^L_{\mathrm{Whit}^!(G)}} \mathrm{Whit}^!(G)_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}}$$

and

$$\mathrm{Rep}(\check{M})_{\mathrm{Ran}^{\mathrm{untl}}} \xrightarrow{\mathrm{co}J^{-, \mathrm{spec}, !}} \mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{untl}}} \xrightarrow{\mathrm{emb.indep}^L_{\mathrm{Rep}(\check{G})}} \mathrm{Rep}(\check{G})_{\mathrm{Ran}^{\mathrm{untl}, \mathrm{indep}}},$$

respectively.

Theorem 14.2.2 asserts that the bottom lid of (14.13) commutes. The proof consists of showing that all the other faces commute:

- The front and back face encode the compatibility of the Langlands functor with Whittaker coefficients;
- The side faces encode the compatibility of the Eisenstein functor with the corresponding local-to-global functor on each side of Langlands correspondence;
- The top lid encodes the compatibility of the Casselman-Shalika equivalence with Jacquet functors.

14.3. Compatibility of the Langlands functor with Eisenstein series—the enhanced version. In this subsection we will prove a generalization of Theorem 14.2.2, which establishes the compatibility of the Langlands functor with *enhanced* Eisenstein series.

14.3.1. Fix $\mathcal{Z} \rightarrow \text{Ran}$. Since the Langlands functor $\mathbb{L}_M$ is compatible with the Hecke actions, we can tensor it with $\text{Sat}^{\frac{\infty}{2}}$ and thus obtain a functor

$$\mathbb{L}_{M,\mathcal{Z}}^{-,\text{enh}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\mathcal{Z}}^{-,\text{enh}} \rightarrow \text{IndCohNilp}(\text{LS}_{\check{M}})_{\mathcal{Z}}^{-,\text{enh}}.$$

The fact that the functor $\text{Eis}^{-,\text{spec,enh}}$ sends $\text{IndCohNilp}(\text{LS}_{\check{M}})$ to $\text{IndCohNilp}(\text{LS}_{\check{G}})$ implies that the functor $\text{Eis}_{\mathcal{Z}}^{-,\text{spec,enh}}$ sends $\text{IndCohNilp}(\text{LS}_{\check{M}})_{\mathcal{Z}}^{-,\text{enh}}$ to $\text{IndCohNilp}(\text{LS}_{\check{G}}) \otimes \text{D-mod}(\mathcal{Z})$ (indeed, the essential image if $\mathbf{ind}_{\text{enh}}^{\text{spec}}$ generates the target).

14.3.2. We are going to prove the following:

Theorem 14.3.3. *The following diagram commutes*

$$(14.14) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathcal{Z}) & \xrightarrow{\mathbb{L}_G \otimes \text{Id}_{\mathcal{Z}}} & \text{IndCohNilp}(\text{LS}_{\check{G}}) \otimes \text{D-mod}(\mathcal{Z}) \\ \text{Eis}_{1,\rho_P(\omega_X),\mathcal{Z}}^{-,\text{enh}}[\delta_{(N_P^-)_{\rho_P(\omega_X)}}] \uparrow & & \uparrow \text{Eis}_{\mathcal{Z}}^{-,\text{spec,enh}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\mathcal{Z}}^{-,\text{enh}} & \xrightarrow{\mathbb{L}_{M,\mathcal{Z}}^{-,\text{enh}}} & \text{IndCohNilp}(\text{LS}_{\check{M}})_{\mathcal{Z}}^{-,\text{enh}}, \end{array}$$

in a way compatible with the Hecke actions via (14.2).

Proof. The proof follows closely that of Theorem 14.2.2. The fact that both circuits in (14.6) send compact objects in $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)$ to eventually coconnective objects in $\text{IndCohNilp}(\text{LS}_{\check{G}})$ formally implies that the same is true for (14.14),

Hence, we obtain that it is sufficient to show that the diagram

$$(14.15) \quad \begin{array}{ccc} \text{Whit}^!(G)_{\mathcal{Z} \subseteq, \text{untl}, \text{indep}} & \xrightarrow{\text{CS}_G} & \text{Rep}(\check{G})_{\mathcal{Z} \subseteq, \text{untl}, \text{indep}} \\ \text{coeff}_{G,\mathcal{Z},\text{untl}}[2\delta_{N_P(\omega_X)}] \uparrow & & \uparrow \Gamma_{\check{G},\text{untl}}^{\text{spec,IndCoh}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathcal{Z}) & & \text{IndCohNilp}(\text{LS}_{\check{G}}) \otimes \text{D-mod}(\mathcal{Z}) \\ \text{Eis}_{1,\rho_P(\omega_X),\mathcal{Z}}^{-,\text{enh}}[\delta_{(N_P^-)_{\rho_P(\omega_X)}}] \uparrow & & \uparrow \text{Eis}_{\mathcal{Z}}^{-,\text{spec,enh}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\mathcal{Z}}^{-,\text{enh}} & \xrightarrow{\mathbb{L}_{M,\mathcal{Z}}^{-,\text{enh}}} & \text{IndCohNilp}(\text{LS}_{\check{M}})_{\mathcal{Z}}^{-,\text{enh}} \end{array}$$

commutes, in a way compatible with the Hecke actions via (14.2).

By Corollary 10.8.5 and Theorem 12.9.9, we replace (14.15) by

$$\begin{array}{ccc}
\text{Whit}^!(G)_{Z \subseteq, \text{untl}, \text{indep}} & \xrightarrow{\text{CS}_G} & \text{Rep}(\check{G})_{Z \subseteq, \text{untl}, \text{indep}} \\
\uparrow \text{emb.indep}^L_{\text{Whit}^!(G)} & & \uparrow \text{emb.indep}^L_{\text{Rep}(\check{G})} \\
\text{Whit}^!(G)_{Z \subseteq, \text{untl}} & \xrightarrow{\text{CS}_G} & \text{Rep}(\check{G})_{Z \subseteq, \text{untl}} \\
\uparrow \text{co}J_{\text{Whit}, \Theta}^{-, \text{enh}} & & \uparrow A_{\text{co}J^{-, \text{spec}, \text{enh}}} \\
(\text{Whit}^!(M) \otimes_{\text{Sph}_M} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}})_{Z \subseteq, \text{untl}} & \xrightarrow{\text{CS}_M \otimes \text{Sat}^{\frac{\infty}{2}}} & (\text{Rep}(\check{M}) \otimes_{\text{Sph}_M^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}})_{Z \subseteq, \text{untl}} \\
\uparrow & & \uparrow \\
\text{Whit}^!(M)_{Z \subseteq, \text{untl}} \otimes_{\text{Sph}_{M, Z}} \text{I}(G, P^-)_{\rho_P(\omega_X), Z \subseteq, \text{untl}}^{\text{loc}} & \xrightarrow{\text{CS}_M \otimes \text{Sat}^{\frac{\infty}{2}}} & \text{Rep}(\check{M})_{Z \subseteq, \text{untl}} \otimes_{\text{Sph}_{M, Z}^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}_{Z \subseteq, \text{untl}} \\
\uparrow \text{coeff}_{M, Z, \text{untl}} \otimes \text{ins.unit}_Z & & \uparrow \Gamma_{M, Z, \text{untl}}^{\text{spec}, \text{IndCoh}} \otimes \text{ins.unit}_Z \\
(\text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \otimes \text{D-mod}(Z)) \otimes_{\text{Sph}_{M, Z}} \text{I}(G, P^-)_{\rho_P(\omega_X), Z}^{\text{loc}} & \xrightarrow{\text{L}_M \otimes \text{Sat}^{\frac{\infty}{2}}} & (\text{IndCoh}(\text{LS}_{\check{M}}) \otimes \text{D-mod}(Z)) \otimes_{\text{Sph}_{M, Z}^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}}_Z \\
\uparrow = & & \uparrow = \\
\text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)_{\rho_P(\omega_X), Z}^{-, \text{enh}} & \xrightarrow{\text{L}_{M, Z}^{-, \text{enh}}} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{M}})_Z^{-, \text{enh}}.
\end{array}$$

We claim that all the inner squares in (14.16) commute. This is tautological for the top, middle and bottom squares. For the second square from the bottom, the commutation follows from the fact that diagram (14.1) for M commutes, compatibly with Hecke actions.

For the second square from the top, the commutation follows from the commutation of

$$\begin{array}{ccc}
\text{Whit}^!(G) & \xrightarrow{\text{CS}_G} & \text{Rep}(\check{G}) \\
\uparrow \text{co}J_{\text{Whit}, \Theta}^{-, \text{enh}} & & \uparrow \text{co}J^{-, \text{spec}, \text{enh}} \\
\text{Whit}^!(M) \otimes_{\text{Sph}_M} \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} & \xrightarrow{\text{CS}_M \otimes \text{Sat}^{\frac{\infty}{2}}} & \text{Rep}(\check{M}) \otimes_{\text{Sph}_M^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)^{\text{spec}, \text{loc}},
\end{array}$$

which is obtained by duality from (2.37). □

14.3.4. Combining Theorem 14.3.3 with Propositions 2.5.9, 9.8.5 and 12.8.4, we obtain:

Corollary 14.3.5. *The following diagram commutes*

$$\begin{array}{ccc}
\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\text{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \\
\uparrow \text{Eis}_{!*, \rho_P(\omega_X)}^{-, \text{loc}} \text{I}(\delta_{(N_P^-)_{\rho_P(\omega_X)}}) & & \uparrow \text{Eis}_{!*, \text{spec}} \\
\text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) & \xrightarrow{\text{L}_M} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{M}}).
\end{array}$$

14.4. **Proof of Theorem 14.1.6.**

14.4.1. We need to show that for $\mathbb{Z} \rightarrow \text{Ran}$, the diagram

$$\begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) & \xrightarrow{\text{L}_G \otimes \text{Id}} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \otimes \text{D-mod}(\mathbb{Z}) \\ \uparrow \text{Loc}_{G,\mathbb{Z}} \otimes \mathbb{I}[-d] & & \uparrow \text{Poinc}_{G,*,\mathbb{Z}}^{\text{spec}} \\ \text{KL}(G)_{\text{crit},\mathbb{Z}} & \xrightarrow{\text{FLE}_{G,\text{crit}}} & \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})_{\mathbb{Z}}, \end{array}$$

commutes, in a way compatible with actions of

$$(14.18) \quad \text{Sph}_{G,\mathbb{Z}} \xrightarrow{\text{Sat}} \text{Sph}_{\check{G},\mathbb{Z}}^{\text{spec}}.$$

By the same logic as in the proof of [GLC2, Theorem 18.5.2], it is enough to show that the diagram

$$(14.19) \quad \begin{array}{ccc} \text{Whit}^!(G)_{\mathbb{Z} \subseteq} & \longrightarrow & \text{Rep}(\check{G})_{\mathbb{Z} \subseteq} \\ \uparrow \text{coeff}_{G,\mathbb{Z},\text{untl}} & & \uparrow \Gamma_{G,\mathbb{Z},\text{untl}}^{\text{spec,IndCoh}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z}) & & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \otimes \text{D-mod}(\mathbb{Z}) \\ \uparrow \text{Loc}_{G,\mathbb{Z}} \otimes \mathbb{I}[d] & & \uparrow \text{Poinc}_{G,*,\mathbb{Z}}^{\text{spec}} \\ \text{KL}(G)_{\text{crit},\mathbb{Z}} & \xrightarrow{\text{FLE}_{G,\text{crit}}} & \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})_{\mathbb{Z}} \end{array}$$

commutes in a way compatible with the actions of (14.18), where $\text{coeff}_{G,\mathbb{Z},\text{untl}}$ is as in Sect. 10.8.3, and similarly for $\Gamma_{\check{G},\mathbb{Z},\text{untl}}^{\text{spec,IndCoh}}$.

14.4.2. By duality, the datum of the left column in (14.19) as a functor equipped with an action of $\text{Sph}_{G,\mathbb{Z}}$ is equivalent to the datum of the $\text{D-mod}(\mathbb{Z})$ -linear functor

$$(14.20) \quad \begin{aligned} & \text{KL}(G)_{\text{crit},\mathbb{Z}} \otimes_{\text{Sph}_{G,\mathbb{Z}}} \text{Whit}_*(G)_{\mathbb{Z} \subseteq} \xrightarrow{(\text{Loc}_{G,\mathbb{Z}} \otimes \mathbb{I}[d]) \otimes \text{Id}} \\ & \rightarrow (\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes \text{D-mod}(\mathbb{Z})) \otimes_{\text{Sph}_{G,\mathbb{Z}}} \text{Whit}_*(G)_{\mathbb{Z} \subseteq} \rightarrow \\ & \xrightarrow{\text{coeff}_{G,\mathbb{Z},\text{untl}} \otimes \text{Id}} \text{Whit}^!(G)_{\mathbb{Z} \subseteq} \otimes_{\text{Sph}_{G,\mathbb{Z}}} \text{Whit}_*(G)_{\mathbb{Z} \subseteq} \rightarrow \text{Whit}^!(G)_{\mathbb{Z} \subseteq} \otimes_{\text{Sph}_{G,\mathbb{Z} \subseteq}} \text{Whit}_*(G)_{\mathbb{Z} \subseteq} \rightarrow \\ & \rightarrow \text{D-mod}(\mathbb{Z} \subseteq) \xrightarrow{(\text{pr}_{\text{small},\mathbb{Z}})^!} \text{D-mod}(\mathbb{Z}), \end{aligned}$$

where the second-to-last functor is the canonical pairing.

Similarly, the datum of the right column in (14.19) as a functor equipped with an action of $\text{Sph}_{\check{G},\mathbb{Z}}^{\text{spec}}$ is equivalent to the datum of the $\text{D-mod}(\mathbb{Z})$ -linear functor

$$(14.21) \quad \begin{aligned} & \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})_{\mathbb{Z}} \otimes_{\text{Sph}_{\check{G},\mathbb{Z}}^{\text{spec}}} \text{Rep}(\check{G})_{\mathbb{Z} \subseteq} \xrightarrow{\text{Poinc}_{G,*,\mathbb{Z}}^{\text{spec}} \otimes \text{Id}} \\ & \rightarrow (\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \otimes \text{D-mod}(\mathbb{Z})) \otimes_{\text{Sph}_{\check{G},\mathbb{Z}}^{\text{spec}}} \text{Rep}(\check{G})_{\mathbb{Z} \subseteq} \rightarrow \\ & \xrightarrow{\Gamma_{\check{G},\mathbb{Z},\text{untl}}^{\text{spec,IndCoh}} \otimes \text{Id}} \text{Rep}(\check{G})_{\mathbb{Z} \subseteq} \otimes_{\text{Sph}_{\check{G},\mathbb{Z}}^{\text{spec}}} \text{Rep}(\check{G})_{\mathbb{Z} \subseteq} \rightarrow \text{Rep}(\check{G})_{\mathbb{Z} \subseteq} \otimes_{\text{Sph}_{\check{G},\mathbb{Z} \subseteq}^{\text{spec}}} \text{Rep}(\check{G})_{\mathbb{Z} \subseteq} \rightarrow \\ & \rightarrow \text{D-mod}(\mathbb{Z} \subseteq) \xrightarrow{(\text{pr}_{\text{small},\mathbb{Z}})^!} \text{D-mod}(\mathbb{Z}), \end{aligned}$$

where the second-to-last functor is the canonical pairing, corresponding to the self-duality of $\text{Rep}(\check{G})$.

Hence, in order to establish the commutativity of (14.19), it suffices to show that the functors (14.20) and (14.21) match up under

$$(14.22) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}} \otimes_{\mathrm{Sph}_{G, \mathcal{Z}}} \mathrm{Whit}_*(G)_{\mathcal{Z} \subseteq} \xrightarrow{\mathrm{FLE}_{G, \mathrm{crit}} \otimes \mathrm{FLE}_{G, \infty}^{-1}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}} \otimes_{\mathrm{Sph}_{\check{G}, \mathcal{Z}}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_{\mathcal{Z} \subseteq}.$$

14.4.3. We can rewrite (14.20) as

$$(14.23) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}} \otimes_{\mathrm{Sph}_{G, \mathcal{Z}}} \mathrm{Whit}_*(G)_{\mathcal{Z} \subseteq} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z} \subseteq} \otimes_{\mathrm{Sph}_{G, \mathcal{Z} \subseteq}} \mathrm{Whit}_*(G)_{\mathcal{Z} \subseteq} \rightarrow \\ \xrightarrow{\mathrm{P}_{G, \mathcal{Z} \subseteq}^{\mathrm{glob}, \mathrm{true}} \otimes \mathbb{I}[d]} \mathrm{D-mod}(\mathcal{Z} \subseteq) \xrightarrow{(\mathrm{pr}_{\mathrm{small}, \mathcal{Z}})^!} \mathrm{D-mod}(\mathcal{Z}),$$

where for $\mathcal{Z}' \rightarrow \mathrm{Ran}$ we denote by $\mathrm{P}_{G, \mathcal{Z}'}^{\mathrm{glob}, \mathrm{true}}$ the functor

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}'} \otimes_{\mathrm{Sph}_{G, \mathcal{Z}'}} \mathrm{Whit}_*(G)_{\mathcal{Z}'} \xrightarrow{\mathrm{Loc}_{G, \mathcal{Z}'} \otimes \mathrm{Id}} (\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathcal{Z}')) \otimes_{\mathrm{Sph}_{G, \mathcal{Z}'}} \mathrm{Whit}_*(G)_{\mathcal{Z}'} \rightarrow \\ \xrightarrow{\mathrm{coeff}_{G, \mathcal{Z}'} \otimes \mathrm{Id}} \mathrm{Rep}(\check{G})_{\mathcal{Z}'} \otimes_{\mathrm{Sph}_{G, \mathcal{Z}'}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_{\mathcal{Z}'} \rightarrow \mathrm{D-mod}(\mathcal{Z}'),$$

see Sect. 11.7.4.

Similarly, we can rewrite

$$(14.24) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}} \otimes_{\mathrm{Sph}_{\check{G}, \mathcal{Z}}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_{\mathcal{Z} \subseteq} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z} \subseteq} \otimes_{\mathrm{Sph}_{\check{G}, \mathcal{Z} \subseteq}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_{\mathcal{Z} \subseteq} \rightarrow \\ \xrightarrow{\mathrm{P}_{\check{G}, \mathcal{Z} \subseteq}^{\mathrm{spec}, \mathrm{glob}, \mathrm{true}}} \mathrm{D-mod}(\mathcal{Z} \subseteq) \xrightarrow{(\mathrm{pr}_{\mathrm{small}, \mathcal{Z}})^!} \mathrm{D-mod}(\mathcal{Z}),$$

where for $\mathcal{Z}' \rightarrow \mathrm{Ran}$ we denote by $\mathrm{P}_{G, \mathcal{Z}'}^{\mathrm{spec}, \mathrm{glob}}$ is the functor

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}'} \otimes_{\mathrm{Sph}_{\check{G}, \mathcal{Z}'}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_{\mathcal{Z}'} \rightarrow \\ \xrightarrow{\mathrm{Poinc}_{G, *, \mathcal{Z}'}^{\mathrm{spec}} \otimes \mathrm{Id}} (\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathcal{Z}')) \otimes_{\mathrm{Sph}_{\check{G}, \mathcal{Z}'}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_{\mathcal{Z}'} \xrightarrow{\Gamma_{\check{G}, \mathcal{Z}'}^{\mathrm{spec}, \mathrm{IndCoh}} \otimes \mathrm{Id}} \\ \rightarrow \mathrm{Rep}(\check{G})_{\mathcal{Z}'} \otimes_{\mathrm{Sph}_{\check{G}, \mathcal{Z}'}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_{\mathcal{Z}'} \rightarrow \mathrm{D-mod}(\mathcal{Z}'),$$

see Sect. 13.6.4.

14.4.4. Hence, in order to establish the commutativity of (14.19), it suffices to show that the functors

$$\mathrm{P}_{G, \mathcal{Z}'}^{\mathrm{glob}, \mathrm{true}} \otimes \mathbb{I}[d] \text{ and } \mathrm{P}_{G, \mathcal{Z}'}^{\mathrm{spec}, \mathrm{glob}, \mathrm{true}}$$

match up under

$$(14.25) \quad \mathrm{KL}(G)_{\mathrm{crit}, \mathcal{Z}'} \otimes_{\mathrm{Sph}_{G, \mathcal{Z}'}} \mathrm{Whit}_*(G)_{\mathcal{Z}'} \xrightarrow{\mathrm{FLE}_{G, \mathrm{crit}} \otimes \mathrm{FLE}_{G, \infty}^{-1}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathcal{Z}'} \otimes_{\mathrm{Sph}_{\check{G}, \mathcal{Z}'}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_{\mathcal{Z}'}.$$

for any $\mathcal{Z}' \rightarrow \mathrm{Ran}$.

14.4.5. We now use Theorems 11.7.6 and 13.6.6 to identify

$$\mathbf{P}_{G,Z'}^{\text{glob,true}} \otimes \mathbb{I}[d] \simeq (\mathbf{P}_{G,Z'}^{\text{loc,true}})^f \text{ins.unit}$$

and

$$\mathbf{P}_{\tilde{G},Z'}^{\text{spec,glob,true}} \simeq (\mathbf{P}_{\tilde{G},Z'}^{\text{spec,loc,true}})^f \text{ins.unit},$$

respectively.

Hence, it suffices to show that under (14.25), the functors

$$\mathbf{P}_{\tilde{G}}^{\text{loc,true}} \text{ and } \mathbf{P}_{\tilde{G}}^{\text{spec,loc,true}}$$

match up.

However, this follows from [GLC2, Theorem 6.4.5(b)] by applying the functor $\Gamma^{\text{IndCoh}}(\text{Op}_{\tilde{G}}^{\text{mer}}, -)$.
 $\square$ [Theorem 14.1.6]

15. LANGLANDS FUNCTOR AND CONSTANT TERMS

Starting from this section, we will assume the validity of the geometric Langlands conjecture (as stated in [GLC1, Conjecture 1.6.7]) for the Levi quotients of all proper parabolic subgroups of G .

We will establish a result pertaining to compatibility between the Langlands functor and the operation of constant term. This result will be central to our proof that the Langlands functor induces an equivalence on Eisenstein-generated categories.

15.1. Statement of the compatibility result.

15.1.1. Passing to right adjoints along the vertical arrows in the commutative diagram (14.6), we obtain a natural transformation

$$(15.1) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{G}}) \\ \downarrow \text{CT}_{*,\rho_P(\omega_X)}^{-}[-\delta_{(N_P^-)_{\rho_P(\omega_X)}}] & \nearrow & \downarrow \text{CT}^{-,\text{spec}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}}) \end{array}$$

At the end of this paper (see Sect. 17.1.3) we will prove:

Theorem 15.1.2. *The natural transformation (15.1) is an isomorphism.*

15.1.3. However, the proof of Theorem 15.1.2 that we will give will not be direct: we will deduce Theorem 15.1.2 once we prove that the functor $\mathbb{L}_G$ is an equivalence on Eisenstein-generated categories (Theorem 17.1.2), and this will rely on the inductive assumption that GLC holds for proper Levi subgroups of G .

In this section we will prove that the two circuits in (15.1) are *abstractly* isomorphic, see Theorem 15.1.13. Although we will give a canonical construction of the isomorphism, its relation to the natural transformation (15.1) is not evident. We will use the existence of an isomorphism (and its enhancement, Theorem 15.6.5) in the proof of Theorem 17.1.2.

Remark 15.1.4. We repeat that we do not know whether the isomorphisms of Theorems 15.1.13 and 15.1.2 are the same. However, one can show that they differ at most by a (non-zero) scalar.

The potential discrepancy between (analogues of) isomorphisms of functors in Theorems 15.1.2 and 15.1.13 is present also in the quantum geometric Langlands program.

15.1.5. We form the following cube whose commutativity we analyze in what follows:

(15.2)

$$\begin{array}{ccccc}
 & & \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\
 & \nearrow \mathrm{CT}_{*,\rho_P(\omega_X)}^{-}[-\delta_{(N_P^-)\rho_P(\omega_X)}] & & & \nearrow \mathrm{CT}^{-,\mathrm{spec}} \\
 \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) & & \\
 \uparrow \mathrm{Loc}_G & & \uparrow \mathrm{Loc}_M & & \uparrow \mathrm{Poinc}_{M,*,\check{\rho}_P}^{\mathrm{spec}} \\
 & \nearrow \mathrm{KL}(M)_{\mathrm{crit}_M+\check{\rho}_P,\mathrm{Ran}\subseteq} & & & \nearrow \mathrm{IndCoh}^*(\mathrm{Op}_{M,\check{\rho}_P}^{\mathrm{mon-free}})_{\mathrm{Ran}\subseteq} \\
 & \nearrow \mathcal{J}_{\mathrm{KM},\rho_P(\omega_X)}^{\mathrm{Sph}} \circ \mathrm{oins.vac} & \xrightarrow{\mathrm{FLE}_{M,\mathrm{crit}+\check{\rho}_P}} & & \nearrow \mathcal{J}_{\mathrm{Op},\Theta}^{-,\mathrm{spec}} \circ \mathrm{oins.unit} \\
 \mathrm{KL}(G)_{\mathrm{crit}_G,\mathrm{Ran}} & \xrightarrow{\mathrm{FLE}_{G,\mathrm{crit}}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} & &
 \end{array}$$

where:

- $\mathrm{Loc}_M := (\mathrm{Id} \otimes \mathrm{C}^*(\mathrm{Ran}^{\subseteq}, -)) \circ \mathrm{Loc}_{M,\mathrm{Ran}\subseteq}$;
- $\mathrm{Poinc}_{M,*,\check{\rho}_P}^{\mathrm{spec}} := (\mathrm{Id} \otimes \mathrm{C}^*(\mathrm{Ran}^{\subseteq}, -)) \circ \mathrm{Poinc}_{M,*,\check{\rho}_P,\mathrm{Ran}\subseteq}^{\mathrm{spec}}$.

The commutation of its upper lid in (15.2) is what we eventually want to establish. We will see that various results from Part II of this paper amount to commutativity of the other five faces up to various shifts and lines. We will use these five faces to establish the commutation of its upper lid, see Theorem 15.1.13 for the precise statement.

15.1.6. We now analyze its 2-skeleton:

- By Theorem 5.2.3, the bottom lid commutes;
- By Corollary 11.9.5, the left face commutes after tensoring the clockwise circuit with

$$\mathbb{I}_{N_P^-}^{\otimes -1} \otimes \mathrm{Weil}(\rho_P(\omega_X), \check{\rho}_P(\omega_X))^{\otimes -1} [\delta_{(N_P^-)\rho_P(\omega_X)}];$$

- By (14.3), the front face commutes after tensoring the clockwise circuit with

$$\mathbb{I}_{G,N_{\rho}(\omega_X)}^{\otimes \frac{1}{2}} \otimes \mathbb{I}_{N_{\rho}(\omega_X)}^{\otimes -1} [-\delta_{N_{\rho}(\omega_X)}];$$

- By (14.5), the back face commutes after tensoring the clockwise circuit with

$$\mathbb{I}_{M,N(M)\rho_M(\omega_X)}^{\otimes \frac{1}{2}} \otimes \mathbb{I}_{N(M)\rho_M(\omega_X)}^{\otimes -1} [-\delta_{N(M)\rho_M(\omega_X)}];$$

- By Theorem 13.1.13 and [GLC2, Theorem 17.4.7], the right face commutes after tensoring the clockwise circuit with

$$\mathbb{I}_{\mathrm{Kost}(\check{G})}^{\otimes -1} [\delta_{\check{G}}]$$

and the counterclockwise circuit with

$$\mathbb{I}_{\mathrm{Kost}(\check{M})}^{\otimes -1} [\delta_{\check{M}}],$$

where the line $\mathbb{I}_{\mathrm{Kost}(\check{G})}$ is as in [GLC2, Sect.17.2.2], and similarly for $\check{M}$.

15.1.7. From Sect. 15.1.6 above, we obtain that the two circuits in the upper lid in (15.2) commute *after* we precompose them with the functor Loc_G , *up to* a cohomological shifts and twist by a constant line.

We will now show that this cohomological shift and line are actually trivial.

15.1.8. The cancelation of cohomological shifts this amounts to the equality

$$-\delta_{(N_P^-)_{\rho_P(\omega_X)}} - \delta_{N_{\rho(\omega_X)}} + \delta_G = -\delta_{N(M)_{\rho_M(\omega_X)}} + \delta_M$$

which is equivalent to the valid equality

$$\begin{aligned} \delta_G - \delta_M &= \\ &= -\dim(\Gamma(X, \mathfrak{g}_{\rho_P(\omega_X)})) + \dim(\Gamma(X, \mathfrak{m}_{\rho_P(\omega_X)})) = -\dim(\Gamma(X, (\mathfrak{n}_P)_{\rho_P(\omega_X)})) - \dim(\Gamma(X, (\mathfrak{n}_P^-)_{\rho_P(\omega_X)})) = \\ &= \delta_{(N_P)_{\rho_P(\omega_X)}} + \delta_{(N_P^-)_{\rho_P(\omega_X)}} = \delta_{(N_P^-)_{\rho_P(\omega_X)}} + \delta_{N_{\rho(\omega_X)}} - \delta_{N(M)_{\rho_M(\omega_X)}}. \end{aligned}$$

15.1.9. Let us now show that twists by the constant lines specified in Sect. 15.1.6 cancel out. This follows by combining Corollary 11.9.10 with the following assertion (and its counterpart for M):⁵⁸

Proposition 15.1.10. *There exists a canonical isomorphism of lines*

$$\mathfrak{l}_{\text{Kost}(\tilde{G})} \simeq \mathfrak{l}_{G, N_{\rho(\omega_X)}} \otimes \mathfrak{l}_{N_{\rho(\omega_X)}}^{\otimes 2} \otimes \det(\Gamma(X, \mathcal{O}_X) \otimes \mathfrak{g}) \otimes ((z_{\mathfrak{g}})^{\otimes 2})^{\otimes (1-g)}.$$

Proof. First, using the fact that $z_{\mathfrak{g}}$ is the dual of $z_{\tilde{\mathfrak{g}}}$ and

$$\mathfrak{l}_{\text{Kost}(\tilde{G})} \simeq \mathfrak{l}_{\text{Kost}(\tilde{G}_{\text{ad}})} \otimes \det(\Gamma(X, \mathcal{O}_X) \otimes z_{\tilde{\mathfrak{g}}})$$

and

$$\det(\Gamma(X, \mathcal{O}_X) \otimes \mathfrak{g}) \simeq \det(\Gamma(X, \mathcal{O}_X) \otimes \mathfrak{g}_{\text{ad}}) \otimes \det(\Gamma(X, \mathcal{O}_X) \otimes z_{\mathfrak{g}}),$$

in order to prove the proposition, we can replace G by G_{ad} . Thus, we can assume that G is semi-simple. In this case, using the Killing form, we identify $\mathfrak{a}(\tilde{\mathfrak{g}})$ with $\mathfrak{a}(\mathfrak{g})$.

We have:

$$\mathfrak{l}_{G, N_{\rho(\omega_X)}} \otimes \det(\Gamma(X, \mathcal{O}_X) \otimes \mathfrak{g}) = \det(\Gamma(X, \mathfrak{g}_{\rho(\omega_X)}))$$

and

$$\mathfrak{l}_{N_{\rho(\omega_X)}} = \det(\Gamma(X, \mathfrak{n}_{\rho(\omega_X)})).$$

Thus, we need to establish an isomorphism

$$(15.3) \quad \det(\Gamma(X, \mathfrak{a}(\mathfrak{g})_{\omega_X})) \otimes \det(\Gamma(X, \mathfrak{n}_{\rho(\omega_X)})) \simeq \det(\Gamma(X, (\mathfrak{g}/\mathfrak{n})_{\rho(\omega_X)})).$$

Decompose

$$\mathfrak{g} \simeq \bigoplus_{e \in \mathbb{Z}_{\geq 0}} V^e.$$

with respect to the action of the principal SL_2 .⁵⁹ Decompose each V^e into its weight spaces

$$V^e = \bigoplus_n V^e(n).$$

Note that we can identify

$$\mathfrak{a}(\mathfrak{g}) \simeq \bigoplus_e V^e(e),$$

Note, however, that canonical $\mathbb{G}_m$ -action on $\mathfrak{a}(\mathfrak{g})$ is shifted by 1 relative to⁶⁰ the action of $\mathbb{G}_m \hookrightarrow SL_2$ on $V^e(e)$, so that

$$\det(\Gamma(X, \mathfrak{a}(\mathfrak{g})_{\omega_X})) \simeq \bigotimes_e \det(\Gamma(X, V^e(e)_{\rho(\omega_X)} \otimes \omega_X)).$$

⁵⁸Note that the Killing form on $\mathfrak{g}$ restricts to a non-degenerate form on $z_{\mathfrak{m}}/z_{\mathfrak{g}}$, thereby trivializing the line $\det(z_{\mathfrak{m}}/z_{\mathfrak{g}})^{\otimes 2} \simeq \det(z_{\mathfrak{m}})^{\otimes 2} \otimes \det(z_{\mathfrak{g}})^{\otimes -2}$.

⁵⁹We emphasize that in the above formula, e is a non-negative integer, and not the “ e ” element in the principal SL_2 .

⁶⁰See [BD1, Sect. 3.1.9].

To prove (15.3), it suffices to show that for each e , we have:

$$(15.4) \quad \det(\Gamma(X, V^e(e)_{\rho(\omega_X)} \otimes \omega_X)) \otimes \det(\Gamma(X, \bigoplus_{n>0} V^e(n)_{\rho(\omega_X)})) \simeq \det(\Gamma(X, \bigoplus_{n \leq 0} V^e(n)_{\rho(\omega_X)})),$$

where we note that

$$V^e(n)_{\rho(\omega_X)} \simeq V^e(n) \otimes \omega_X^{\otimes n}.$$

We will show that

$$(15.5) \quad \det(\Gamma(X, V^e(e)_{\rho(\omega_X)} \otimes \omega_X)) \simeq \det(\Gamma(X, V^e(-e)_{\rho(\omega_X)}))$$

and for every $n > 0$

$$(15.6) \quad \det(\Gamma(X, V^e(n)_{\rho(\omega_X)})) \simeq \det(\Gamma(X, V^e(-n+2)_{\rho(\omega_X)})).$$

The Killing form identifies $V^e(e)$ with the dual of $V^e(-e)$, so that by Serre duality

$$\Gamma(X, V^e(e)_{\rho(\omega_X)} \otimes \omega_X)^\vee \simeq \Gamma(X, V^e(-e)_{\rho(\omega_X)})[1].$$

This implies (15.5).

Similarly, for $n > 0$, the Killing form and the action of the positive generator of $\mathfrak{sl}_2$ identifies $V^e(n)$ with the dual vector space of $V^e(-n+2)$, and hence by Serre duality

$$\Gamma(X, V^e(n)_{\rho(\omega_X)})^\vee \simeq \Gamma(X, V^e(-n+2)_{\rho(\omega_X)})[1].$$

This implies (15.6). □

Remark 15.1.11. The isomorphism constructed above was canonical. However, it relied on certain conventions. For example, we used weight lowering operators f^i at one point, but sometimes one finds weight lowering operators $\frac{f^i}{i!}$ are better choices. These sorts of conventions do not affect our main results, but they do affect the scalar discussed in Remark 15.1.4, and are one reason we are not confident that that scalar is one.

15.1.12. We are now ready to state the main result of this section:

Theorem 15.1.13. *Assuming the validity of GLC for M , there exists a unique datum of commutation for the upper lid in (15.2), i.e., the square*

$$(15.7) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\ \mathrm{CT}_{*, \rho_P(\omega_X)}^{-} \downarrow [-\delta_{(N_P^-)_{\rho_P(\omega_X)}}] & & \downarrow \mathrm{CT}^{-, \mathrm{spec}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}), \end{array}$$

that makes (15.2) commute (with the specified cohomological shifts and twists by constant lines along the edges).

15.1.14. We now commence the proof of Theorem 15.1.13.

The commutation of the five of the faces of the cube (15.2) established above implies that the *outer* diagram in

$$\begin{array}{ccccc}
 & & \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\
 & \nearrow & \uparrow \mathrm{CT}_{*,\rho_P(\omega_X)}^{-}[-\delta_{(N_P^-)_{\rho_P(\omega_X)}}] & & \uparrow \mathrm{CT}^{-,\mathrm{spec}} \\
 (15.8) \quad \mathrm{CT}_{*,\rho_P(\omega_X)}^{-} \circ \mathrm{Loc}_G[-\delta_{N(P^-)_{\rho_P(\omega_X)}}] & & \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\
 & \nearrow \mathrm{Loc}_G & \nearrow \mathbb{L}_G \circ \mathrm{Loc}_G & & \\
 & \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & & &
 \end{array}$$

is endowed with a commutation datum.

The statement of Theorem 15.1.13 is equivalent to the fact that this datum comes from a uniquely defined commutation datum of the inner square in (15.8).

For expositional purposes, we first consider the case when $P = B$.

15.2. Proof of Theorem 15.1.13 for $P = B$.

15.2.1. The category $\mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_T) \simeq \mathrm{D}\text{-}\mathrm{mod}(\mathrm{Bun}_T)$ splits as a direct sum according to connected components of Bun_T , which are indexed by the coweight lattice of T . For each coweight μ , let $\mathrm{CT}_{*,\rho_P(\omega_X)}^{-,\mu}$ denote the corresponding direct summand of $\mathrm{CT}_{*,\rho_P(\omega_X)}^{-}$.

Let $\mathrm{CT}^{-,\mathrm{spec},\mu}$ denote the corresponding direct summand of $\mathrm{CT}^{-,\mathrm{spec}}$. It corresponds to the direct summand $\mathrm{QCoh}(\mathrm{LS}_{\check{T}})^\mu$ of

$$\mathrm{QCoh}(\mathrm{LS}_{\check{T}}) = \mathrm{IndCoh}_{\{0\}}(\mathrm{LS}_{\check{T}}) = \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{T}})$$

consisting of objects, on which the action of $\tilde{T}$ by 1-automorphisms⁶¹ of $\mathrm{LS}_{\check{T}}$ has character $-\mu$ (here we regard μ as a weight of $\tilde{T}$), see [GLC1, Sect. 1.5.3]

Thus, in proving Theorem 15.1.13, instead of the diagram (15.8), we can consider

⁶¹I.e., Here and below “1-automorphisms” of a given object means the group of automorphisms of the identity map on this object.

(15.9)

$$\begin{array}{ccccc}
& & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M^\mu) & \xrightarrow{\mathbb{L}_M^\mu} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}}(X))^\mu \\
& \nearrow & \uparrow \mathrm{CT}_{*,\rho_P(\omega_X)}^{-,\mu}[-\delta_{N(P^-)}\rho_P(\omega_X)] & & \uparrow \mathrm{CT}^{-,\mathrm{spec},\mu} \\
\mathrm{CT}_{*,\rho_P(\omega_X)}^{-,\mu} \circ \mathrm{Loc}_G[-\delta_{N(P^-)}\rho_P(\omega_X)] & & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}(X)) \\
& \nearrow \mathrm{Loc}_G & \nearrow \mathbb{L}_G \circ \mathrm{Loc}_G & & \\
\mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & & & &
\end{array}$$

for a fixed μ .

15.2.2. For a fixed μ , let $\lambda \in \Lambda_G^{+,\mathbb{Q}}$ be large enough so that the image of the map

$$\mathrm{Bun}_B^\mu \rightarrow \mathrm{Bun}_G$$

is contained in the open union of Harder-Narasimhan strata $\mathrm{Bun}_G^{(<\lambda)}$ (see [DG2, Sect. 7.4] for our conventions regarding the parameterization of the Harder-Narasimhan strata).

By construction, we have:

Lemma 15.2.3. *The functor $\mathrm{CT}_{*,\rho_P(\omega_X)}^{-,\mu}$ factors as*

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G^{(<\lambda)}) \xrightarrow{(\mathrm{CT}_{*,\rho_P(\omega_X)}^{-,\mu})^{(<\lambda)}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_T),$$

where the first arrow is the restriction functor.

15.2.4. Let P' be a standard parabolic in G with Levi quotient M' . Recall that $\Lambda_{G,P'}$ denotes the quotient of Λ by the root lattice of M' , i.e.,

$$\Lambda_{G,P'} \simeq \pi_{1,\mathrm{alg}}(M') \simeq \pi_0(\mathrm{Bun}_{M'}).$$

Recall (see [DG2, Sects. 7.1.3-7.1.5]) that we can view $\Lambda_{G,P'}$ as a subset of $\Lambda_G^{\mathbb{Q}}$. Denote

$$\Lambda_{G,P'}^+ := \Lambda_{G,P'} \cap \Lambda_G^{+,\mathbb{Q}}.$$

15.2.5. Let

$$\mathrm{Bun}_{M'}^{\not\prec \lambda} \subset \mathrm{Bun}_{M'}$$

be the union of connected components, indexed by coweights $\lambda' \in \Lambda_{G,P'}^+$ with

$$\lambda' \not\prec_G \lambda.$$

Note that the functor

$$\mathrm{Eis}_! : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{M'}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

induces a functor

$$(15.10) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{M'}^{\not\prec \lambda}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G^{(<\lambda)}),$$

where

$$\mathrm{Bun}_G^{(<\lambda)} \subset \mathrm{Bun}_G$$

is the closed substack equal to the complement of $\mathrm{Bun}_G^{(<\lambda)}$.

15.2.6. We claim:

Lemma 15.2.7. *For λ large enough, the essential image of (15.10) generates the target.*

Proof. Let λ satisfy

$$\langle \lambda, \check{\alpha}_i \rangle \geq \max(0, 2g - 2)$$

for all simple roots $\check{\alpha}_i$.

Then according to [DG2, Sect. 3], we can partition the set $\Lambda_G^{(<\lambda)}$ into locally closed subsets⁶² S , such that for

$$\mathrm{Bun}_G^{(S)} := \bigcup_{\mu \in S} \mathrm{Bun}_G^{(\mu)}$$

the following holds:

For every S there exists a parabolic P' such that for

$$\mathrm{Bun}_{P'}^{(S)} := \mathrm{Bun}_{P'} \times_{\mathrm{Bun}_{M'}} \mathrm{Bun}_{M'}^{(S)},$$

- The map $\mathbf{p} : \mathrm{Bun}_{P'}^{(S)} \rightarrow \mathrm{Bun}_G^{(S)}$ is an isomorphism;
- The map $\mathbf{q} : \mathrm{Bun}_{P'}^{(S)} \rightarrow \mathrm{Bun}_M^{(S)}$ is a unipotent gerbe.⁶³

It is clear that for S as above, the functor $\mathrm{Eis}_!$ induces an equivalence

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{M'}^{(S)}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G^{(S)}).$$

□

15.2.8. Let λ be sufficiently large. Applying Lemma 15.2.7, we can identify $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G^{(<\lambda)})$ with the quotient of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ by the full subcategory generated by the essential images of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{M'}^{(<\lambda)})$ along the Eisenstein functors

$$\mathrm{Eis}_! : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{M'}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

for all standard proper parabolics P' .

15.2.9. For $\lambda' \in \Lambda_{G,P'}$, let $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}'}^{\lambda'})$ be the direct summand of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}'})$ consisting of objects on which the action of $Z_{\check{M}'}$ by 1-automorphisms⁶⁴ of $\mathrm{LS}_{\check{M}'}$ has character $-\lambda'$.

Let $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})^{(<\lambda)}$ denote the quotient of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$ by the full subcategory generated by the essential images of⁶⁵

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{M'})^{\lambda'}, \quad \lambda' \not\leq \lambda + 2(g-1) \cdot \rho_{P'}.$$

We will prove:

Proposition 15.2.10. *For a fixed μ , and λ large enough, for every standard parabolic P' and $\lambda' \in \Lambda_{G,P'}$ satisfying $\lambda' \not\leq \lambda$, the functor*

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}'}^{\lambda'}) \xrightarrow{\mathrm{Eis}^{\mathrm{spec}}} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \xrightarrow{\mathrm{CT}^-, \mathrm{spec}, \mu} \mathrm{QCoh}(\mathrm{LS}_{\check{T}})^{\mu}$$

vanishes.

15.2.11. Assuming Proposition 15.2.10 temporarily, we obtain that for λ sufficiently large, the functor $\mathrm{CT}^-, \mathrm{spec}, \mu$ factors as

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \twoheadrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})^{(<\lambda)} \xrightarrow{(\mathrm{CT}^-, \mathrm{spec}, \mu)^{(<\lambda)}} \mathrm{QCoh}(\mathrm{LS}_{\check{T}})^{\mu}.$$

⁶²Here “locally closed” refers to the order relation $\leq$ on Λ_G , see [DG2, Appendix A].

⁶³See [DG1, Definition 10.3.5] for what this means.

⁶⁴See footnote in Sect. 15.2.1 for the meaning of “1-automorphisms”.

⁶⁵The shift by $2(g-1) \cdot \rho_{P'}$ in the formula below is due to the fact that on the geometric side in Theorem 14.2.2, we are dealing with the functor $\mathrm{Eis}_{1, \rho_P(\omega_X)}^-$ rather than just $\mathrm{Eis}_!^-$.

15.2.12. The compatibility of the Langlands functor with the Eisenstein functors given by Theorem 14.2.2 implies that the functor $\mathbb{L}_G$ descends to a well-defined functor

$$\mathbb{L}_G^{(<\lambda)} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G^{(<\lambda)}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})^{(<\lambda)}.$$

We obtain that the commutativity datum for (15.9) is equivalent to that of the commutativity datum for the inner square in

(15.11)

$$\begin{array}{ccccc}
 & & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M^\mu) & \xrightarrow{\mathbb{L}_M^\mu} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}})^\mu, \\
 & & \uparrow & & \uparrow \\
 & & (\mathrm{CT}^-, \mu_{*, \rho_P(\omega_X)})^{(<\lambda)} [-\delta_{N(P^-), \rho_P(\omega_X)}] & & (\mathrm{CT}^-, \mathrm{spec}, \mu)^{(<\lambda)} \\
 & & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G^{(<\lambda)}) & \xrightarrow{\mathbb{L}_G^{(<\lambda)}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})^{(<\lambda)} \\
 & \nearrow & & \nearrow & \\
 & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & & & \\
 \nearrow & & & & \\
 \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} & & & &
 \end{array}$$

Loc_G

compatible with the existing commutativity datum for the outer diagram.

Furthermore, we can replace $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G^{(<\lambda)})$ by its connected component $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G^{(<\lambda), \bar{\mu}})$ corresponding to the image $\bar{\mu}$ of μ under

$$\Lambda \rightarrow \Lambda/\Lambda_{\mathrm{s.c.}} \simeq \pi_{1, \mathrm{alg}}(G) \simeq \pi_0(\mathrm{Bun}_G).$$

Now, the required assertion follows now from the next observation:

Lemma 15.2.13. *The functor*

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} \xrightarrow{\mathrm{Loc}_G} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \twoheadrightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G^{(<\lambda), \bar{\mu}})$$

is a Verdier quotient.

Proof. This is a particular case of [GLC2, Theorem 13.4.2]. □

15.3. Proof of Proposition 15.2.10. For the duration of the proof, we will change the notation from P' to P .

15.3.1. By base change, the functor

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \xrightarrow{\mathrm{Eis}^{\mathrm{spec}}} \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{\mathrm{CT}^-, \mathrm{spec}} \mathrm{IndCoh}(\mathrm{LS}_{\check{T}})$$

can be rewritten as the composition of:

- $*$ -pullback along $\mathrm{LS}_{\check{P}} \rightarrow \mathrm{LS}_{\check{M}}$;
- $!$ -pullback along $\mathrm{LS}_{\check{P}} \times_{\mathrm{LS}_{\check{G}}} \mathrm{LS}_{\check{B}^-} \rightarrow \mathrm{LS}_{\check{P}}$;

- $*$ -pushforward along $LS_{\tilde{P}} \times_{LS_{\tilde{G}}} LS_{\tilde{B}^-} \rightarrow LS_{\tilde{B}^-} \rightarrow LS_{\tilde{T}}$.

However, since the morphism $LS_{\tilde{P}} \rightarrow LS_{\tilde{M}}$ is quasi-smooth, up to shifting the degree, we can replace the $*$ -pullback by the $!$ -pullback. So the functor in question becomes $!$ -pullback followed by $*$ -pushforward along the diagram

$$\begin{array}{ccc} & LS_{\tilde{P}} \times_{LS_{\tilde{G}}} LS_{\tilde{B}^-} & \\ \swarrow & & \searrow \\ LS_{\tilde{M}} & & LS_{\tilde{T}} \end{array}$$

15.3.2. We decompose $LS_{\tilde{P}} \times_{LS_{\tilde{G}}} LS_{\tilde{B}^-}$ according to relative positions of the two reductions, which are indexed by the elements of W/W_M .

For each $w \in W/W_M$, let

$$(LS_{\tilde{P}} \times_{LS_{\tilde{G}}} LS_{\tilde{B}^-})_w \subset LS_{\tilde{P}} \times_{LS_{\tilde{G}}} LS_{\tilde{B}^-}$$

denote the corresponding locally closed substack, and let

$$(LS_{\tilde{P}} \times_{LS_{\tilde{G}}} LS_{\tilde{B}^-})_w^\wedge$$

denote its formal completion inside $LS_{\tilde{P}} \times_{LS_{\tilde{G}}} LS_{\tilde{B}^-}$.

We will show that for every w and λ large enough, the pull-push functor along

$$(15.12) \quad \begin{array}{ccc} & (LS_{\tilde{P}} \times_{LS_{\tilde{G}}} LS_{\tilde{B}^-})_w^\wedge & \\ \swarrow & & \searrow \\ LS_{\tilde{M}} & & LS_{\tilde{T}} \end{array}$$

has the property that its (λ', μ) component vanishes for

$$(15.13) \quad \lambda' \not\leq \lambda, \quad \lambda' \in \Lambda_{G,P}^+.$$

We will first establish the corresponding fact for the diagram

$$(15.14) \quad \begin{array}{ccc} & (LS_{\tilde{P}} \times_{LS_{\tilde{G}}} LS_{\tilde{B}^-})_w & \\ \swarrow & & \searrow \\ LS_{\tilde{M}} & & LS_{\tilde{T}} \end{array}$$

15.3.3. Let $w \in W$ the shortest representative of the given coset. Consider the subgroup

$$H_w := \tilde{P} \cap w^{-1}(\tilde{B}^-) \stackrel{w}{\simeq} w(\tilde{P}) \cap \tilde{B}^-.$$

We can identify

$$(LS_{\tilde{P}} \times_{LS_{\tilde{G}}} LS_{\tilde{B}^-})_w \simeq LS_{H_w},$$

so that (15.14) identifies with

$$(15.15) \quad \begin{array}{ccc} & \text{LS}_{H_w} & \\ & \swarrow \quad \searrow & \\ \text{LS}_{\check{M}} & & \text{LS}_{\check{T}}, \end{array}$$

where the two projections correspond to

$$(15.16) \quad H_w = \check{P} \cap w^{-1}(\check{B}^-) \rightarrow \check{P} \rightarrow \check{M}$$

$$(15.17) \quad H_w = \check{P} \cap w^{-1}(\check{B}^-) \xrightarrow{w} w(\check{P}) \cap \check{B}^- \rightarrow \check{B}^- \rightarrow \check{T},$$

respectively.

15.3.4. Note that diagram (15.15) naturally factors as

$$(15.18) \quad \begin{array}{ccc} & \text{LS}_{H_w} & \\ & \downarrow h & \\ & \text{LS}_{\check{B}^-(M)} & \\ \text{LS}_{\check{M}} & \swarrow \text{p}_M^{-,\text{spec}} \quad \searrow w \circ \text{q}_M^{-,\text{spec}} & \text{LS}_{\check{T}}. \end{array}$$

Pull-push along (15.15) identifies with

$$(15.19) \quad (w \circ \text{q}_M^{-,\text{spec}})_*^{\text{IndCoh}} \left((\text{p}_M^{-,\text{spec}})^!(-) \otimes^! h_*^{\text{IndCoh}}(\omega_{\text{LS}_{H_w}}) \right).$$

15.3.5. Note that the stack LS_{H_w} and the map h are quasi-smooth (the latter because $H_w \rightarrow \check{B}^-(M)$ is surjective⁶⁶). Hence, up to shift by a fixed weight, we can replace (15.19) by

$$(15.20) \quad (w \circ \text{q}_M^{-,\text{spec}})_*^{\text{IndCoh}} \left((\text{p}_M^{-,\text{spec}})^!(-) \otimes h_*(\mathcal{O}_{\text{LS}_{H_w}}) \right),$$

where:

- $h_*(\mathcal{O}_{\text{LS}_{H_w}})$ is viewed as an object of $\text{QCoh}(\text{LS}_{\check{B}^-(M)})$;
- $\otimes$ denotes the tensor product functor $\text{IndCoh}(-) \otimes \text{QCoh}(-) \rightarrow \text{IndCoh}(-)$.

15.3.6. Consider the action of $Z_{\check{M}}$ by 1-automorphisms on the stacks in

$$\begin{array}{ccc} & \text{LS}_{\check{B}^-(M)} & \\ & \swarrow \text{p}_M^{-,\text{spec}} \quad \searrow w \circ \text{q}_M^{-,\text{spec}} & \\ \text{LS}_{\check{M}} & & \text{LS}_{\check{T}}. \end{array}$$

The characters of $Z_{\check{M}}$ that appear in $h_*(\mathcal{O}_{\text{LS}_{H_w}})$ are sums of characters of the form

$$(15.21) \quad -\alpha, \quad \alpha \text{ is a root in } \check{\mathfrak{n}}_P \cap w^{-1}(\check{\mathfrak{n}}^-).$$

⁶⁶Indeed, $H_w \cap \check{M} = \check{B}^-(M)$.

Hence, for $\mathcal{F} \in \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{M}})^{\lambda'}$, the characters of $Z_{\check{M}}$ on the pull-push of $Z_{\check{M}}$ along (15.14) are of the form

$$-w(\lambda' + \sum_{\alpha} n_{\alpha} \cdot \alpha),$$

for α in (15.21) and $n_{\alpha} \in \mathbb{Z}^{\geq 0}$.

Hence, for the μ -direct summand of the above pull-push to be non-zero, we must have

$$-w(\lambda' + \sum_{\alpha} n_{\alpha} \cdot \alpha) = \mu|_{Z_{\check{M}}}.$$

I.e.,

$$(15.22) \quad \lambda' + \sum_{\alpha} n_{\alpha} \cdot \alpha = -w^{-1}(\mu)|_{Z_{\check{M}}}.$$

15.3.7. However, (15.22) is impossible once (15.13) is satisfied with λ large enough.

Indeed, choose λ so that

$$(15.23) \quad \langle \lambda, \check{\omega}_i \rangle > \langle -w^{-1}(\mu), \check{\omega}_i \rangle$$

for all fundamental coweights $\check{\omega}_i$.

If (15.13) is satisfied, there exists a fundamental coweight $\check{\omega}_i$ central in $\check{M}$, such that

$$\langle \lambda', \check{\omega}_i \rangle \geq \langle \lambda, \check{\omega}_i \rangle.$$

Pairing the left hand side of (15.22) with this $\check{\omega}_i$, we obtain

$$\langle \lambda', \check{\omega}_i \rangle + \sum_{\alpha} n_{\alpha} \cdot \langle \alpha, \check{\omega}_i \rangle \geq \langle \lambda', \check{\omega}_i \rangle \geq \langle \lambda, \check{\omega}_i \rangle > \langle -w^{-1}(\mu), \check{\omega}_i \rangle,$$

contradicting (15.22).

15.3.8. We now prove the assertion for the pull-push along (15.12). This functor admits a filtration with subquotients of the form

$$(w \circ \mathbf{q}_M^{-, \text{spec}})_*^{\text{IndCoh}} \left((\mathbf{p}_M^{-, \text{spec}})^!(-) \otimes^{\dagger} h_*^{\text{IndCoh}} (\omega_{\text{LS}_{\mathbf{H}_w}} \otimes \text{Sym}(\text{Norm}_w)) \right),$$

where Norm_w is the normal bundle to $\text{LS}_{\mathbf{H}_w}$ inside $\text{LS}_{\check{P}} \times_{\text{LS}_{\check{G}}} \text{LS}_{\check{B}^-}$.

The fiber of this normal at a given $\sigma \in \text{LS}_{\mathbf{H}_w}$ is

$$\text{Cofib} \left(\text{C}^*(X, (\mathbf{h}_w)_{\sigma})[1] \rightarrow \text{C}^*(X, (\check{\mathfrak{p}} \times w^{-1}(\check{\mathfrak{b}}^-))_{\sigma})[1] \right) \simeq \text{C}^* \left(X, \left(\check{\mathfrak{g}}/(\check{\mathfrak{p}} + w^{-1}(\check{\mathfrak{b}}^-)) \right)_{\sigma} \right)[1].$$

Hence, the characters of $Z_{\check{M}}$ on

$$h_*^{\text{IndCoh}} (\omega_{\text{LS}_{\mathbf{H}_w}} \otimes \text{Sym}(\text{Norm}_w))$$

are sums of characters of the form (15.21) and also those of the form

$$(15.24) \quad \beta, \quad \beta \text{ is a root in } \check{\mathfrak{g}}/(\check{\mathfrak{p}} + w^{-1}(\check{\mathfrak{b}}^-)).$$

The proof is concluded by the same argument as in Sects. 15.3.6-15.3.7.

□[Proposition 15.2.10]

15.4. Proof of Theorem 15.1.13 for a general Levi.

15.4.1. Recall that we are assuming the validity of the geometric Langlands conjecture for M .

Remark 15.4.2. What we really need to assume for the proof to go through is a certain property of the category of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}})$, see Sect. 15.4.7. This property takes place purely on the spectral side, and it follows from GLC.

Let us formulate this property for $\tilde{G}$. For $\lambda \in \Lambda_G^{+, \mathbb{Q}}$, consider the Verdier quotient category

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}) \twoheadrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})^{(<\lambda)}, \quad \lambda \in \Lambda_G^{+, \mathbb{Q}},$$

where we kill the subcategory generated by the essential images of the functors

$$\mathrm{Eis}^{\mathrm{spec}} : \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}'}^{\lambda'}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}), \quad \lambda' \in \Lambda_{G, P'}^+, \quad \lambda' \not\leq_G \lambda - 2(g-1) \cdot \rho_{P'}.$$

for all standard parabolics P' of G .

What we need is that the functor

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}) \rightarrow \lim_{\lambda, \leq_G} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})^{(<\lambda)}$$

is an equivalence.

With this property at hand, we will imitate the argument from Sect. 15.2 essentially word for word.

Remark 15.4.3. For $\lambda \in \Lambda_G^{+, \mathbb{Q}}$, denote by $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})^{(<\lambda)}$ the corresponding quotient of $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})$ so that we have a commutative diagram.

$$(15.25) \quad \begin{array}{ccc} \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}) & \xleftarrow{\Xi_{0, \mathrm{Nilp}}^R} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}) \\ \downarrow & & \downarrow \\ \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})^{(<\lambda)} & \xleftarrow{\Xi_{0, \mathrm{Nilp}}^R} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})^{(<\lambda)}, \end{array}$$

Assuming GLC for G , it follows from the localization argument given below that the category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})^{(<\lambda)}$ is generated by the essential image of

$$\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}) \xrightarrow{\Xi_{0, \mathrm{Nilp}}^R} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})^{(<\lambda)}.$$

This implies that the bottom horizontal arrow in (15.25) is actually an equivalence.

In particular, we obtain that the category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$ can be recovered from the usual $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})$ also as

$$\lim_{\lambda, \leq_G} \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})^{(<\lambda)}.$$

15.4.4. Fix $\mu \in \Lambda_M^+$, and let

$$\mathrm{Bun}_M^{(<\mu)} \subset \mathrm{Bun}_M$$

be the quasi-compact open equal to the union of Harder-Narasimhan strata $\mathrm{Bun}_M^{(\mu')}$ with

$$\mu' \leq_M \mu.$$

We consider $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M^{(<\mu)})$ as a quotient of $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)$.

Assuming that μ is dominant enough (as a coweight of M), by Lemma 15.2.7, the kernel of the projection

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M^{(<\mu)})$$

is generated by the essential images of

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{M'}^{\mu'}), \quad \mu' \in \Lambda_{G, P'}^+, \quad \mu' \not\leq_M \mu + 2(g-1) \cdot \rho_{P'}$$

along the functors

$$\mathrm{Eis}_{!, -\rho_{P'}(\omega_X)} : \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{M'}) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M),$$

where M' is the Levi of a standard parabolic P' of M .

15.4.5. Let $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})^{(<\mu)}$ denote the quotient of $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})$ by the full subcategory generated by the essential images of

$$\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}'})^{\mu'}, \quad \mu' \in \Lambda_{G,P'}^+, \quad \mu' \not\leq_M \mu + 2(g-1) \cdot \rho_{P'}$$

along the functors

$$\text{Eis}^{\text{spec}} : \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}'}) \rightarrow \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}}).$$

15.4.6. The compatibility of the Langlands functor $\mathbb{L}_M$ with Eisenstein series implies that there exists a commutative diagram

$$\begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M^{(<\mu)}) & \xrightarrow[\sim]{\mathbb{L}_M^{(<\mu)}} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})^{(<\mu)} \\ \uparrow & & \uparrow \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) & \xrightarrow[\sim]{\mathbb{L}_M} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}}). \end{array}$$

15.4.7. Since

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \rightarrow \lim_{\mu} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M^{(<\mu)})$$

is an equivalence, we obtain that

$$\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}}) \rightarrow \lim_{\mu} \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})^{(<\mu)}$$

is also an equivalence.

Hence, in order to construct a datum of commutativity for (15.8), it enough to construct a compatible data of commutativity for the diagrams (for varying μ)

$$(15.26) \quad \begin{array}{ccccc} & & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M^{(<\mu)}) & \xrightarrow{\mathbb{L}_M^{(<\mu)}} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})^{(<\mu)} \\ & & \uparrow & & \uparrow \\ & & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) & & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}}) \\ & & \uparrow \text{CT}_{*,\rho_P(\omega_X)}^-[-\delta_{N(P^-)}\rho_P(\omega_X)] & & \uparrow \text{CT}^-, \text{spec} \\ & & \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{G}}) \\ & \nearrow & & \nearrow & \\ & \text{KL}(G)_{\text{crit,Ran},} & & & \end{array}$$

compatible with the given data of commutativity for the outer diagram.

15.4.8. Let $\text{Bun}_G^{(<\lambda)} \subset \text{Bun}_G$ be a quasi-compact open union of Harder-Narasimhan strata, such that the functor

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \xrightarrow{\text{CT}_{*,\rho_P(\omega_X)}^-} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M^{(<\mu)})$$

factors via the quotient

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \twoheadrightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G^{(<\lambda)}).$$

Denote the resulting functor

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G^{(<\lambda)}) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M^{(<\mu)})$$

by $(\mathrm{CT}_{*, \rho_P(\omega_X)}^-)^{(<\mu), (<\lambda)}$.

15.4.9. Let

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})^{(<\lambda)}$$

denote the corresponding quotient, see Sect. 15.2.9.

The following is a generalization of Proposition 15.2.10:

Proposition 15.4.10. *For a fixed μ and λ large enough, for every standard parabolic P' and $\lambda' \in \Lambda_{G, P'}$ satisfying $\lambda' \not\leq \lambda$, the functor*

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}'}^{\lambda'}) \xrightarrow{\mathrm{Eis}^{\mathrm{spec}}} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}) \xrightarrow{\mathrm{CT}^-, \mathrm{spec}} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}})^{(<\mu)}$$

vanishes.

Let us assume this proposition for a moment and finish the proof of Theorem 15.1.13.

15.4.11. From Proposition 15.4.10, combined with Theorem 14.2.2 and Lemma 15.2.7, we obtain:

Corollary 15.4.12. *For λ large enough, the composite functor*

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{\mathbb{L}_G} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}) \xrightarrow{\mathrm{CT}^-, \mathrm{spec}} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}})^{(<\mu)}$$

also factors via the quotient

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G^{(<\lambda)}).$$

15.4.13. Denote the resulting functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G^{(<\lambda)}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}})^{(<\mu)}$$

by

$$(\mathrm{CT}^-, \mathrm{spec} \circ \mathbb{L}_G)^{(<\mu), (<\lambda)}.$$

We obtain that a datum of commutativity for (15.26) is equivalent to that for the diagram

$$(15.27) \quad \begin{array}{ccc} & & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}})^{(<\mu)} \\ & \nearrow & \uparrow \\ & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G^{(<\lambda)}) & \xrightarrow{(\mathrm{CT}^-, \mathrm{spec} \circ \mathbb{L}_G)^{(<\mu), (<\lambda)}} \\ & \nearrow & \\ & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \\ \nearrow \mathrm{Loc}_G & & \\ \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} & & \end{array}$$

However, this follows again from Lemma 15.2.13.

15.5. **Proof of Proposition 15.4.10.** The proof proceeds along the same lines as that of Proposition 15.2.10, using a generalization of diagram (15.18), explained below.

15.5.1. Let P_1 and P_2 be a pair of standard parabolics of G with Levi quotients M_1 and M_2 , respectively. For an element

$$w \in W_1 \backslash W / W_2,$$

let

$$(\mathrm{LS}_{\check{P}_1} \times_{\mathrm{LS}_{\check{G}}} \mathrm{LS}_{\check{P}_2})_w \subset \mathrm{LS}_{\check{P}_1} \times_{\mathrm{LS}_{\check{G}}} \mathrm{LS}_{\check{P}_2}$$

be the corresponding locally closed substack.

Let w be the shortest representative of the given double coset. Consider the subgroup

$$\mathbf{H}_w := \check{P}_1 \cap w^{-1}(\check{P}_2) \stackrel{w}{\simeq} w(\check{P}_1) \cap \check{P}_2,$$

so that we can identify

$$(\mathrm{LS}_{\check{P}_1} \times_{\mathrm{LS}_{\check{G}}} \mathrm{LS}_{\check{P}_2})_w \simeq \mathrm{LS}_{\mathbf{H}_w}.$$

15.5.2. Set

$$\check{P}'_1 := \check{P}_1 \cap w^{-1}(\check{P}_2) / (\check{N}_{P_1} \cap w^{-1}(\check{P}_2)) \text{ and } \check{P}'_2 := w(\check{P}_1) \cap \check{P}_2 / (w(\check{P}_1) \cap \check{N}_{P_2}).$$

The projections

$$\check{P}'_1 \rightarrow \check{P}_1 / \check{N}_{P_1} = \check{M}_1 \text{ and } \check{P}'_2 \rightarrow \check{P}_2 / \check{N}_{P_2} = \check{M}_2$$

realize $\check{P}'_1$ and $\check{P}'_2$ as standard parabolics in $\check{M}_1$ and $\check{M}_2$, respectively.

The element w gives rise to an identification of the Levi quotients

$$\check{M}'_1 \stackrel{w}{\simeq} \check{M}'_2$$

of $\check{P}'_1$ and $\check{P}'_2$, respectively.

Finally, note also that we have a Cartesian diagram

$$\begin{array}{ccc} \mathrm{LS}_{\check{P}_1 \cap w^{-1}(\check{P}_2) / (N(\check{P}_1) \cap w^{-1}(N(\check{P}_2)))} & \stackrel{w}{\sim} & \mathrm{LS}_{w(\check{P}_1) \cap \check{P}_2 / (w(N(\check{P}_1)) \cap N(\check{P}_2))} \\ \swarrow & & \searrow \\ \mathrm{LS}_{\check{P}'_1} & & \mathrm{LS}_{\check{P}'_2} \\ \searrow & & \swarrow \\ \mathrm{LS}_{\check{M}'_1} & \stackrel{w}{\sim} & \mathrm{LS}_{\check{M}'_2} \end{array}$$

15.5.3. Then the diagram

$$\begin{array}{ccc} & (\mathrm{LS}_{\check{P}_1} \times_{\mathrm{LS}_{\check{G}}} \mathrm{LS}_{\check{P}_2})_w & \\ \swarrow & & \searrow \\ \mathrm{LS}_{\check{M}_1} & & \mathrm{LS}_{\check{M}_2} \end{array}$$

can be factored as

$$\begin{array}{c}
 (\mathrm{LS}_{\tilde{P}_1} \times_{\mathrm{LS}_{\tilde{G}}} \mathrm{LS}_{\tilde{P}_2})_w \\
 \downarrow \sim \\
 \mathrm{LS}_{H_w} \\
 \downarrow h \\
 \mathrm{LS}_{\tilde{P}_1 \cap w^{-1}(\tilde{P}_2)/(N(\tilde{P}_1) \cap w^{-1}(N(\tilde{P}_2)))} \stackrel{w}{\sim} \mathrm{LS}_{w(\tilde{P}_1) \cap \tilde{P}_2/(w(N(\tilde{P}_1)) \cap N(\tilde{P}_2))} \\
 \swarrow \quad \searrow \\
 \mathrm{LS}_{\tilde{P}'_1} \quad \mathrm{LS}_{\tilde{P}'_2} \\
 \swarrow \quad \searrow \quad \swarrow \quad \searrow \\
 \mathrm{LS}_{\tilde{M}_1} \quad \mathrm{LS}_{\tilde{M}'_1} \stackrel{w}{\sim} \mathrm{LS}_{\tilde{M}'_2} \quad \mathrm{LS}_{\tilde{M}_2} .
 \end{array}$$

15.5.4. From this point, the argument proceeds as in Sects. 15.3.4-15.3.8.

15.6. Constant term compatibility: the enhanced version.

15.6.1. Given $\mathcal{Z} \rightarrow \mathrm{Ran}$, consider the 1-skeleton of the following cube:

(15.28)

$$\begin{array}{ccccc}
& & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D}\text{-mod}(\mathbb{Z}) & \xrightarrow{C_M^{\mathrm{glob}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D}\text{-mod}(\mathbb{Z}) \\
& \nearrow A_{G,M}^{\mathrm{glob}} & \uparrow C_G^{\mathrm{glob}} & & \nearrow A_{G,M}^{\mathrm{spec,glob}} \\
\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathbb{Z}}^{-,\mathrm{enhco}} & \xrightarrow{\quad} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathbb{Z}}^{-,\mathrm{enhco}} & & \\
\uparrow B_G & & \uparrow B_M & & \uparrow B_M^{\mathrm{spec}} \\
& \nearrow A_{G,M}^{\mathrm{loc}} & \mathrm{KL}(M)_{\mathrm{crit}_M + \check{\rho}_P, (\mathbb{Z} \subseteq)} & \xrightarrow{\quad} & \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{\rho}_P}^{\mathrm{mon-free}})_{(\mathbb{Z} \subseteq)} \\
& & \uparrow C_M^{\mathrm{loc}} & & \nearrow A_{G,M}^{\mathrm{spec,loc}} \\
\mathrm{KL}(G)_{\mathrm{crit}_G, \mathbb{Z} \subseteq} \otimes^{\mathrm{Sph}_{G, \mathbb{Z}}} \mathrm{I}(G, P^-)_{\mathrm{co}, \rho_P(\omega_X), \mathbb{Z}}^{\mathrm{loc}} & \xrightarrow{C_G^{\mathrm{loc}}} & \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathbb{Z} \subseteq} \otimes^{\mathrm{Sph}_{\check{G}, \mathbb{Z}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}, \mathbb{Z}}^{\mathrm{spec,loc}}, & &
\end{array}$$

where:

- $A_{G,M}^{\mathrm{glob}} := \mathrm{CT}_{*, \rho_P(\omega_X), \mathbb{Z}}^{-, \mathrm{enhco}}[-\delta_{(N_P^-)_{\rho_P(\omega_X)}}]$
- $A_{G,M}^{\mathrm{spec,glob}} := \mathrm{CT}_{\mathbb{Z}}^{-, \mathrm{spec,enhco}};$
- $A_{G,M}^{\mathrm{loc}}$ is the composition

$$\begin{aligned}
& \mathrm{KL}(G)_{\mathrm{crit}_G, \mathbb{Z} \subseteq} \otimes^{\mathrm{Sph}_{G, \mathbb{Z}}} \mathrm{I}(G, P^-)_{\mathrm{co}, \rho_P(\omega_X), \mathbb{Z}}^{\mathrm{loc}} \xrightarrow{\mathrm{Id} \otimes \mathrm{ins}, \mathrm{unit}} \mathrm{KL}(G)_{\mathrm{crit}_G, \mathbb{Z} \subseteq} \otimes^{\mathrm{Sph}_{G, \mathbb{Z} \subseteq}} \mathrm{I}(G, P^-)_{\mathrm{co}, \rho_P(\omega_X), \mathbb{Z} \subseteq}^{\mathrm{loc}} = \\
& = (\mathrm{KL}(G)_{\mathrm{crit}_G} \otimes^{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\mathrm{co}, \rho_P(\omega_X)}^{\mathrm{loc}})_{\mathbb{Z} \subseteq} \xrightarrow{\mathrm{ins}, \mathrm{unit}} (\mathrm{KL}(G)_{\mathrm{crit}_G} \otimes^{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\mathrm{co}, \rho_P(\omega_X)}^{\mathrm{loc}})_{(\mathbb{Z} \subseteq) \subseteq} = \\
& = \mathrm{KL}(G)_{\mathrm{crit}_G, (\mathbb{Z} \subseteq) \subseteq}^{-, \mathrm{enhco}} \xrightarrow{J_{\mathrm{KM}, \rho_P(\omega_X)}^{-, \mathrm{enhco}}} \mathrm{KL}(M)_{\mathrm{crit}_M + \check{\rho}_P, (\mathbb{Z} \subseteq) \subseteq};
\end{aligned}$$

- $A_{G,M}^{\mathrm{spec,loc}}$ is the composition

$$\begin{aligned}
& \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathbb{Z} \subseteq} \otimes^{\mathrm{Sph}_{\check{G}, \mathbb{Z}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}, \mathbb{Z}}^{\mathrm{spec,loc}} \xrightarrow{\mathrm{Id} \otimes \mathrm{ins}, \mathrm{unit}} \\
& \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathbb{Z} \subseteq} \otimes^{\mathrm{Sph}_{\check{G}, \mathbb{Z} \subseteq}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)_{\mathbb{Z} \subseteq}^{\mathrm{spec,loc}} = \\
& = (\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes^{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}, \mathbb{Z}}^{\mathrm{spec,loc}})_{\mathbb{Z} \subseteq} \xrightarrow{\mathrm{ins}, \mathrm{unit}} \\
& \rightarrow (\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes^{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)_{\mathrm{co}, \mathbb{Z}}^{\mathrm{spec,loc}})_{(\mathbb{Z} \subseteq) \subseteq} = \\
& =: \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{(\mathbb{Z} \subseteq) \subseteq}^{-, \mathrm{enhco}} \xrightarrow{J_{\mathrm{Op}, \Theta}^{-, \mathrm{spec,enhco}}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{\rho}_P}^{\mathrm{mon-free}})_{(\mathbb{Z} \subseteq) \subseteq},
\end{aligned}$$

where $J_{\mathrm{Op}, \Theta}^{-, \mathrm{spec}, \mathrm{enh}_{\mathrm{co}}}$ is the composition

$$\begin{aligned} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})^{-, \mathrm{enh}_{\mathrm{co}}} &:= \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}) \otimes^{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \otimes^{\Theta_{\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}} \otimes \mathrm{Id}} \\ &\simeq \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})^{-, \mathrm{enh}_{\mathrm{co}}} \otimes^{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} := \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})^{-, \mathrm{enh}_{\mathrm{co}}} \xrightarrow{J_{\mathrm{Op}}^{-, \mathrm{spec}, \mathrm{enh}_{\mathrm{co}}}} \\ &\rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{M}, \check{\rho}_P}^{\mathrm{mon-free}}) \otimes^{\Theta_{\mathrm{Op}_{\check{M}, \check{\rho}_P}^{\mathrm{mon-free}}}^{-1}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{\rho}_P}^{\mathrm{mon-free}}); \end{aligned}$$

- B_G is the composition

$$\begin{aligned} \mathrm{KL}(G)_{\mathrm{crit}_G, \mathbb{Z} \subseteq} &\otimes^{\mathrm{Sph}_{G, \mathbb{Z}}} \mathrm{I}(G, P^-)^{\mathrm{loc}}_{\mathrm{co}, \rho_P(\omega_X), \mathbb{Z}} \xrightarrow{\mathrm{Loc}_{G, \mathbb{Z} \subseteq} \otimes \mathrm{Id}} \\ &\rightarrow (\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathbb{Z}^{\subseteq})) \otimes^{\mathrm{Sph}_{G, \mathbb{Z}}} \mathrm{I}(G, P^-)^{\mathrm{loc}}_{\mathrm{co}, \rho_P(\omega_X), \mathbb{Z}} \rightarrow \\ \mathrm{Id} \otimes (\mathrm{pr}_{\mathrm{small}, \mathbb{Z}})_! \otimes \mathrm{Id} &(\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D-mod}(\mathbb{Z})) \otimes^{\mathrm{Sph}_{G, \mathbb{Z}}} \mathrm{I}(G, P^-)^{\mathrm{loc}}_{\mathrm{co}, \rho_P(\omega_X), \mathbb{Z}} = \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathbb{Z}}^{-, \mathrm{enh}_{\mathrm{co}}}; \end{aligned}$$

- B_M is the composition

$$\begin{aligned} \mathrm{KL}(M)_{\mathrm{crit}_M + \check{\rho}_P, (\mathbb{Z} \subseteq) \subseteq} &\xrightarrow{\mathrm{Loc}_{M, (\mathbb{Z} \subseteq) \subseteq}} \\ &\rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}((\mathbb{Z}^{\subseteq}) \subseteq) \xrightarrow{\mathrm{Id} \otimes (\mathrm{pr}_{\mathrm{small}^2, \mathbb{Z}})_!} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}(\mathbb{Z}); \end{aligned}$$

where $\mathrm{pr}_{\mathrm{small}^2, \mathbb{Z}}$ is the projection $(\mathbb{Z}^{\subseteq}) \subseteq \rightarrow \mathbb{Z}$;

- $B_{\check{G}}^{\mathrm{spec}}$ is the composition

$$\begin{aligned} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathbb{Z} \subseteq} &\otimes^{\mathrm{Sph}_{\check{G}, \mathbb{Z}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}_{\mathrm{co}, \mathbb{Z}} \xrightarrow{\mathrm{Poinc}_{\check{G}, *, \mathbb{Z} \subseteq} \otimes \mathrm{Id}} \\ &\rightarrow (\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z}^{\subseteq})) \otimes^{\mathrm{Sph}_{\check{G}, \mathbb{Z}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}_{\mathrm{co}, \mathbb{Z}} \rightarrow \\ \mathrm{Id} \otimes (\mathrm{pr}_{\mathrm{small}, \mathbb{Z}})_! \otimes \mathrm{Id} &(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \otimes \mathrm{D-mod}(\mathbb{Z})) \otimes^{\mathrm{Sph}_{\check{G}, \mathbb{Z}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}_{\mathrm{co}, \mathbb{Z}} = \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathbb{Z}}^{-, \mathrm{enh}_{\mathrm{co}}} \end{aligned}$$

- B_M^{spec} is the composition

$$\begin{aligned} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{M}, \check{\rho}_P}^{\mathrm{mon-free}})_{(\mathbb{Z} \subseteq) \subseteq} &\xrightarrow{\mathrm{Poinc}_{\check{M}, *, \check{\rho}_P, (\mathbb{Z} \subseteq) \subseteq}} \\ &\rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}((\mathbb{Z}^{\subseteq}) \subseteq) \xrightarrow{\mathrm{Id} \otimes (\mathrm{pr}_{\mathrm{small}^2, \mathbb{Z}})_!} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathbb{Z}); \end{aligned}$$

- C_G^{loc} is the tensor product of $\mathrm{FLE}_{G, \mathrm{crit}}^{-, \mathrm{enh}_{\mathrm{co}}}$ and $((\mathrm{Sat}^{\frac{\infty}{2}})^{\vee})^{-1}$, cf. (5.16);
- $C_M^{\mathrm{loc}} := \mathrm{FLE}_{M, \mathrm{crit} + \check{\rho}_P}$;
- C_G^{glob} is the functor $\mathbb{L}_{G, \mathbb{Z}}^{-, \mathrm{enh}_{\mathrm{co}}}$, defined as the tensor product of $\mathbb{L}_G$ and $((\mathrm{Sat}^{\frac{\infty}{2}})^{\vee})^{-1}$, cf. Sect. 14.3.1;
- $C_M^{\mathrm{glob}} := \mathbb{L}_M \otimes \mathrm{Id}$;

15.6.2. We claim that the five faces of this diagram commute (up to cohomological shifts and tensor products by constant lines) in the same way as in diagram (15.2).

The only points of difference are:

- For the commutation of the bottom lid, we use Theorem 5.3.6 instead of Theorem 5.2.3;
- For the commutation of the left face, we use Corollary 11.9.8 instead of Corollary 11.9.5 ;
- For the commutation of the right face we use Theorem 13.5.9 instead of Theorem 13.1.13;
- The front face commutes thanks to Theorem 14.1.6.

15.6.3. By construction, all the above data are compatible with the actions of

$$(15.29) \quad \mathrm{Sph}_{M,Z} \xrightarrow{\mathrm{Sat}_M} \mathrm{Sph}_{M,Z}^{\mathrm{spec}}$$

15.6.4. We will prove the following extension of Theorem 15.1.13:

Theorem 15.6.5. *Assuming the validity of GLC for M , there exists a unique datum of commutativity for the upper lid in (15.28), i.e., the square*

$$(15.30) \quad \begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_Z^{-,\mathrm{enhco}} & \xrightarrow{\mathbb{L}_{G,Z}^{-,\mathrm{enhco}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})_Z^{-,\mathrm{enhco}} \\ \mathrm{CT}_{*,\rho_P(\omega_X),Z}^{-,\mathrm{enhco}}[-\delta_{(N_P^-)_{\rho_P(\omega_X)}}] \downarrow & & \downarrow \mathrm{CT}_Z^{-,\mathrm{spec,enhco}} \otimes \mathrm{Id} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}(Z) & \xrightarrow{\mathbb{L}_M \otimes \mathrm{Id}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}}) \otimes \mathrm{D-mod}(Z), \end{array}$$

that makes (15.28) commute (with the specified cohomological shifts and twists by constant lines along the edges), in a way compatible with the actions of (15.29).

The rest of this subsection will be devoted to the proof of Theorem 15.6.5.

15.6.6. Consider the diagram

$$(15.31) \quad \begin{array}{ccccc} & & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes \mathrm{D-mod}(Z) & \xrightarrow{\mathbb{L}_M \otimes \mathrm{Id}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}}) \otimes \mathrm{D-mod}(Z) \\ & \nearrow & \uparrow \mathrm{CT}_{*,\rho_P(\omega_X),Z}^{-,\mathrm{enhco}}[-\delta_{(N_P^-)_{\rho_P(\omega_X)}}] & & \uparrow \mathrm{CT}_Z^{-,\mathrm{spec,enhco}} \\ & & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_Z^{-,\mathrm{enhco}} & \xrightarrow{\mathbb{L}_{G,Z}^{-,\mathrm{enhco}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})_Z^{-,\mathrm{enhco}} \\ & \nearrow B_G & \nearrow & \nearrow & \\ \mathrm{KL}(G)_{\mathrm{crit}_G,Z} \subseteq \mathrm{Sph}_{G,Z} \otimes \mathrm{I}(G,P^-)_{\mathrm{co},\rho_P(\omega_X),Z}^{\mathrm{loc}} & & & & \end{array}$$

where:

- The arrow marked as B_G is as in (15.28);
- The two triangles with slanted arrows commute.

The commutation of the five faces in (15.28) implies that the outer diagram in (15.31) commutes. The statement of Theorem 15.6.5 is that this datum comes from a uniquely defined commutativity datum of the inner square in (15.8).

We consider (15.31) as a diagram of categories acted on by

$$\mathrm{Sph}_M \xrightarrow{\mathrm{Sat}_M} \mathrm{Sph}_M^{\mathrm{spec}}.$$

15.6.7. By duality, the required property of (15.31) is equivalent to a parallel property of the following diagram of categories acted on by

$$\mathrm{Sph}_G \xrightarrow{\mathrm{Sat}_G} \mathrm{Sph}_G^{\mathrm{spec}}:$$

$$\begin{array}{ccccc}
& & \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_Z^{-,\mathrm{enh}} & \xrightarrow{\mathbb{L}_{M,Z}^{-,\mathrm{enh}}} & \mathrm{IndCohNilp}(\mathrm{LS}_{\tilde{M}})_Z^{-,\mathrm{enh}}, \\
& \nearrow & \uparrow \mathrm{CT}_{*,\rho_P(\omega_X)}^{-,\mathrm{enh},Z}[-\delta_{(N_P^-)_{\rho_P(\omega_X)}}] & & \uparrow \mathrm{CT}_Z^{-,\mathrm{spec},\mathrm{enh}} \\
& & \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-}\mathrm{mod}(\mathbb{Z}) & \xrightarrow{\mathbb{L}_G \otimes \mathrm{Id}} & \mathrm{IndCohNilp}(\mathrm{LS}_{\tilde{G}}) \otimes \mathrm{D}\text{-}\mathrm{mod}(\mathbb{Z}) \\
& \nearrow \mathrm{Loc}'_{G,Z \subseteq} & & \nearrow & \\
\mathrm{KL}(G)_{\mathrm{crit},Z \subseteq} & & & &
\end{array}$$

where:

- $\mathrm{Loc}'_{G,Z \subseteq} := (\mathrm{pr}_{\mathrm{small},Z})! \circ \mathrm{Loc}_{G,Z \subseteq}$;
- The functor $\mathrm{CT}_Z^{-,\mathrm{spec},\mathrm{enh}}$ is obtained from $\mathrm{CT}_Z^{-,\mathrm{spec},\mathrm{enh}_{\mathrm{co}}}$ by duality.

15.6.8. In other words, we obtain a diagram

$$\begin{array}{ccc}
& & \mathrm{IndCohNilp}(\mathrm{LS}_{\tilde{M}})_Z^{-,\mathrm{enh}} \\
& \nearrow & \uparrow \\
(15.32) \quad & \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-}\mathrm{mod}(\mathbb{Z}) & \\
& \nearrow \mathrm{Loc}'_{G,Z \subseteq} & \\
& \mathrm{KL}(G)_{\mathrm{crit},Z \subseteq} &
\end{array}$$

in which the two curved arrows become isomorphic after precomposing with the functor $\mathrm{Loc}'_{G,Z \subseteq}$. We wish to show that this isomorphism comes from a uniquely defined isomorphism between the curved arrows themselves.

15.6.9. Recall that the forgetful functor

$$\mathrm{IndCohNilp}(\mathrm{LS}_{\tilde{M}})_Z^{-,\mathrm{enh}} \rightarrow \mathrm{IndCohNilp}(\mathrm{LS}_{\tilde{M}}) \otimes \mathrm{D}\text{-}\mathrm{mod}(\mathbb{Z})$$

is monadic, and that the corresponding monad is given by the action of the associative algebra object

$$\tilde{\Omega}_Z^{\mathrm{spec}} \in \mathrm{Sph}_{M,Z}^{\mathrm{spec}},$$

see Sect. 12.4.3.

The endofunctor F of $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}}) \otimes \text{D-mod}(\mathbb{Z})$ underlying the above monad is naturally filtered⁶⁷ by the poset $\Lambda_{G,P}^{\text{pos}}$ (see Sect. 2.2.6, where the notation is introduced). Write

$$F \simeq \text{colim}_{\lambda \in \Lambda_{G,P}^{\text{pos}}} F_{\lambda}.$$

15.6.10. Let $F_{\underline{\lambda}}$ be a composite of a finite collection of the functors F_{λ} . Let

$$F_{\underline{\lambda}}\text{-mod}$$

denote the category of

$$\{x \in \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}}) \otimes \text{D-mod}(\mathbb{Z}), \alpha : F_{\underline{\lambda}}(x) \rightarrow x\}.$$

The category $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})_{\mathbb{Z}}^{-,\text{enh}}$ can be identified with a limit of categories of the form $F_{\underline{\lambda}}\text{-mod}$.

15.6.11. We will show that for each $\underline{\lambda}$ *separately*, there exists a *unique* isomorphism between the composition of the two curved arrows in (15.32) with the projection

$$\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})_{\mathbb{Z}}^{-,\text{enh}} \rightarrow F_{\underline{\lambda}}\text{-mod},$$

so that after pre-composing with $\text{Loc}'_{G,\mathbb{Z} \subseteq}$ we obtain the already existing isomorphism.

The uniqueness assertion implies that these isomorphisms give rise to a uniquely defined isomorphism for the diagram (15.32) itself.

15.6.12. For a sufficiently large element $\mu \in \Lambda_M^+$, let $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})^{(<\mu)}$ be corresponding the quotient of $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})$ (see Sect. 15.4.5), so that

$$\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}}) \rightarrow \lim_{\mu} \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})^{(<\mu)}$$

is an equivalence, see Sect. 15.4.7.

The functors F_{λ} (and, hence, their compositions $F_{\underline{\lambda}}$) have the following property:

For every μ , the composition

$$\begin{aligned} \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}}) \otimes \text{D-mod}(\mathbb{Z}) &\xrightarrow{F_{\underline{\lambda}}} \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}}) \otimes \text{D-mod}(\mathbb{Z}) \rightarrow \\ &\rightarrow \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})^{(<\mu)} \otimes \text{D-mod}(\mathbb{Z}) \end{aligned}$$

factors as

$$\begin{aligned} \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}}) \otimes \text{D-mod}(\mathbb{Z}) &\twoheadrightarrow \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})^{(<\mu')} \otimes \text{D-mod}(\mathbb{Z}) \xrightarrow{F_{\underline{\lambda}}^{\mu',\mu}} \\ &\rightarrow \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})^{(<\mu)} \otimes \text{D-mod}(\mathbb{Z}) \end{aligned}$$

for some sufficiently large μ' .

15.6.13. For $\underline{\lambda}$ and $\mu \leq \mu'$ as above, let

$$F_{\underline{\lambda}}^{\mu',\mu}\text{-mod}$$

denote the category of

$$\{x_{\mu'} \in \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})^{(<\mu')} \otimes \text{D-mod}(\mathbb{Z}), \alpha : F_{\underline{\lambda}}^{\mu',\mu}(x_{\mu'}) \rightarrow x_{\mu}\},$$

where x_{μ} denotes the image of $x_{\mu'}$ along the projection

$$\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})^{(<\mu')} \otimes \text{D-mod}(\mathbb{Z}) \rightarrow \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}})^{(<\mu)} \otimes \text{D-mod}(\mathbb{Z}).$$

We have:

$$F_{\underline{\lambda}}\text{-mod} \simeq \lim_{\mu \leq \mu'} F_{\underline{\lambda}}^{\mu',\mu}\text{-mod}.$$

⁶⁷In fact, this filtration comes from a grading.

Hence, it is enough to show that for each $\mu \leq \mu'$, there exists a *unique* isomorphism between the composition of the two curved arrows in (15.32) with the projection

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}})_{\mathbb{Z}}^{-, \mathrm{enh}} \rightarrow \mathbf{F}_{\tilde{\Delta}}^{\mu', \mu}\text{-mod},$$

so that after pre-composing with $\mathrm{Loc}'_{G, \mathbb{Z} \subseteq}$ we obtain the already existing isomorphism.

15.6.14. We now note that for a fixed $\mu \leq \mu'$, the resulting two arrows

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightrightarrows \mathbf{F}_{\tilde{\Delta}}^{\mu', \mu}\text{-mod}$$

both factor as

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^{(<\nu), \bar{\mu}'} \rightrightarrows \mathbf{F}_{\tilde{\Delta}}^{\mu', \mu}\text{-mod}$$

for some $\nu \in \Lambda_G^+$.

Indeed, since the functor

$$\mathbf{F}_{\tilde{\Delta}}^{\mu', \mu}\text{-mod} \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}})^{(<\mu')} \otimes \mathrm{D}\text{-mod}(\mathbb{Z})$$

is conservative, this follows from the corresponding property of the two curved arrows

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightrightarrows \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}})^{(<\mu')}$$

in (15.27) established in the course of the proof of Theorem 15.1.13.

15.6.15. Now, the required assertion follows the fact that the functor

$$\mathrm{KL}(G)_{\mathrm{crit}, \mathbb{Z} \subseteq} \xrightarrow{\mathrm{Loc}'_{G, \mathbb{Z} \subseteq}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes \mathrm{D}\text{-mod}(\mathbb{Z}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^{(<\nu), \bar{\mu}'} \otimes \mathrm{D}\text{-mod}(\mathbb{Z})$$

is a Verdier quotient, which follows from [GLC2, Theorem 13.4.2].

□[Theorem 15.6.5]

16. THE LEFT ADJOINT OF THE LANGLANDS FUNCTOR

In this section we start reaping the benefits from the work done until this point.

- We show that the functor $\mathbb{L}_G$ admits a left adjoint (to be denoted $\mathbb{L}_G^L$), which is also compatible with the geometric and spectral Eisenstein series functors;
- We show that, up to a cohomological shift, the functor $\mathbb{L}_G^L$ identifies with the composition

$$(16.1) \quad \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}) \xrightarrow{\mathrm{Serre}} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})^{\vee} \xrightarrow{\mathbb{L}_G^{\vee}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^{\vee} \xrightarrow{\mathrm{Verdier}} \\ \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \xrightarrow{\mathrm{Mir}_{\mathrm{Bun}_G}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{\tau_G} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G),$$

where $\mathrm{Mir}_{\mathrm{Bun}_G}$ is the *Miraculous functor*, and τ_G is the Chevalley involution;

- We show that the composition $\mathbb{L}_G \circ \mathbb{L}_G^L$, which is an endofunctor of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$ is given by tensor product by an associative algebra object $\mathcal{A}_G \in \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})$.

16.1. The existence of the left adjoint.

16.1.1. The goal of this subsection is to prove the following statement:

Theorem 16.1.2. *The functor $\mathbb{L}_G$ admits a left adjoint. Moreover, for every standard parabolic P , we have a commutative diagram*

$$(16.2) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xleftarrow{\mathbb{L}_M^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}}) \\ \mathrm{Eis}_{1, \rho_P(\omega_X)}^{-} [\delta_{N(P^-)}]_{\rho_P(\omega_X)} \Big\downarrow & & \Big\downarrow \mathrm{Eis}^{-, \mathrm{spec}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xleftarrow{\mathbb{L}_G^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}). \end{array}$$

The rest of the subsection is devoted to the proof of this result.

16.1.3. As was shown in [AG, Theorem 13.3.6], the category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$ is generated by the essential images of $\mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}})$ along the functors

$$\mathrm{Eis}^{-, \mathrm{spec}} : \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}).$$

In Theorem 15.1.13, we have constructed a commutative square

$$\begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ \mathrm{CT}_{*, \rho_P(\omega_X)}^{-}[-\delta_{N(P^-)}]_{\rho_P(\omega_X)} \uparrow & & \uparrow \mathrm{CT}^{-, \mathrm{spec}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}), \end{array}$$

in which the vertical arrows are the right adjoints to the ones in (16.2).

It follows formally that in order to prove that $\mathbb{L}_G^L$ exists and makes (16.2) commute, it suffices to show that for every M , the (a priori partially defined) left adjoint $\mathbb{L}_M^L$, is actually defined on $\mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}})$.

16.1.4. Up to changing the notation, we can assume that $M = G$. However, then the existence of $\mathbb{L}_G^L|_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})}$ was built into the construction of $\mathbb{L}_G$: this is the functor $\mathbb{L}_{G, \mathrm{temp}}^L$ of [GLC1, Formula (1.10)].

Namely, $\mathbb{L}_{G, \mathrm{temp}}^L$ is obtained by the spectral action of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ on the object

$$\mathrm{Poinc}_{G, !}^{\mathrm{Vac}, \mathrm{glob}} \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G).$$

□[Theorem 16.1.2]

Remark 16.1.5. Note that by passing to left adjoints along the horizontal arrows in (14.6), we obtain a natural transformation

$$(16.3) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xleftarrow{\mathbb{L}_G^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\ \mathrm{Eis}_{!, \rho_P(\omega_X)}^{-}[\delta_{(N_F^-)}]_{\rho_P(\omega_X)} \uparrow & & \uparrow \mathrm{Eis}^{-, \mathrm{spec}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xleftarrow{\mathbb{L}_M^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \end{array}$$

Alternatively, (16.3) can be viewed as obtained by passing to left adjoints along all the arrows in (15.1).

Once we prove Theorem 15.1.2, we will have also proved that the natural transformation in (16.3) is an isomorphism.

However, we do not know whether this isomorphism equals one constructed in Theorem 16.2.

16.2. The left adjoint as a dual.

16.2.1. Recall that Verdier duality allows us to identify

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^\vee \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}.$$

Consider the functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \xleftarrow{\mathbb{L}_G^\vee} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}(X)),$$

dual to $\mathbb{L}_G^\vee$, where we identify $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}(X))$ with its own dual via Serre duality.

Define the functor

$$\Phi_G : \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}(X)) \longrightarrow \mathrm{D}\text{-mod}(\mathrm{Bun}_G)$$

to be

$$\Phi_G := \tau_G \circ \text{Mir}_{\text{Bun}_G} \circ \mathbb{L}_G^\vee \otimes \mathbb{I}_{z_{\check{G}}} [4\delta_G - 2\delta_{N_{\rho}(\omega_X)}],$$

where

$$\mathbb{I}_{z_{\check{G}}} := \det(T_1^*(\text{LS}_{Z_G^0})) \simeq \det(z_{\check{G}})^{\otimes (2-2g)}.$$

16.2.2. We are going to prove:

Theorem 16.2.3. *There exists a canonical isomorphism*

$$\Phi_G \simeq \mathbb{L}_G^L.$$

The rest of this subsection is devoted to the proof of Theorem 16.2.3.

16.2.4. Both functors appearing in Theorem 16.2.3 are (automatically) compatible with the Hecke actions via

$$\text{Sph}_G \xrightarrow{\text{Sat}_G} \text{Sph}_{\check{G}}^{\text{spec}},$$

see [GLC2, Sect. 1.8].

Hence, they both give rise to functors

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}} \xrightleftharpoons[\Phi_{G,\text{temp}}]{\mathbb{L}_{G,\text{temp}}^L} \text{QCoh}(\text{LS}_{\check{G}})$$

that make both diagrams

$$\begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightleftharpoons[\Phi_G]{\mathbb{L}_G^L} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \\ \uparrow \mathbf{u} & & \uparrow \Xi_{0,\text{Nilp}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}} & \xrightleftharpoons[\Phi_{G,\text{temp}}]{\mathbb{L}_{G,\text{temp}}^L} & \text{QCoh}(\text{LS}_{\check{G}}) \end{array}$$

and

$$(16.4) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightleftharpoons[\Phi_G]{\mathbb{L}_G^L} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \\ \downarrow \mathbf{u}^R & & \downarrow (\Xi_{0,\text{Nilp}})^R \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}} & \xrightleftharpoons[\Phi_{G,\text{temp}}]{\mathbb{L}_{G,\text{temp}}^L} & \text{QCoh}(\text{LS}_{\check{G}}) \end{array}$$

commute, where $(\mathbf{u}, \mathbf{u}^R)$ are as in [GLC2, Sect. 18.4.1].

16.2.5. We will first show that the functors $\mathbb{L}_{G,\text{temp}}^L$ and $\Phi_{G,\text{temp}}$ are (canonically) isomorphic.

Since the functor

$$\text{Loc}_{\check{G}}^{\text{spec}} : \text{Rep}(\check{G})_{\text{Ran}} \rightarrow \text{QCoh}(\text{LS}_{\check{G}})$$

is a Verdier quotient, it suffices to establish an equivalence

$$(16.5) \quad \mathbb{L}_{G,\text{temp}}^L \circ \text{Loc}_{\check{G}}^{\text{spec}} \simeq \Phi_{G,\text{temp}} \circ \text{Loc}_{\check{G}}^{\text{spec}}.$$

By construction, the functor $\mathbb{L}_{G,\text{temp}}^L$ makes the diagram

$$\begin{array}{ccc} \text{Whit}^!(G)_{\text{Ran}} & \xleftarrow{(CS_G)^{-1}} & \text{Rep}(\check{G})_{\text{Ran}} \\ \text{Poinc}_{G,!}[-2\delta_{N_{\rho}(\omega_X)}] \downarrow & & \downarrow \text{Loc}_{\check{G}}^{\text{spec}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{temp}} & \xleftarrow{\mathbb{L}_{G,\text{temp}}^L} & \text{QCoh}(\text{LS}_{\check{G}}) \end{array}$$

commute, see [GLC1, Sect. 1.4.2].

Hence, it suffices to check that the diagram

$$\begin{array}{ccc}
 \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xleftarrow{(\mathrm{CS}_G)^{-1}} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\
 \mathrm{Poinc}_{G,!}[-2\delta_{N_{\rho(\omega_X)}}] \downarrow & & \downarrow \mathrm{Loc}_G^{\mathrm{spec}} \\
 \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}} & \xleftarrow{\tau_G \circ \mathrm{Mir}_{\mathrm{Bun}_G} \circ \mathbb{L}_G^\vee|_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})} \otimes \mathbb{I}_{z_{\check{\mathfrak{g}}}}[4\delta_G - 2\delta_{N_{\rho(\omega_X)}}]} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}})
 \end{array}$$

commutes as well.

16.2.6. Recall (see [GLC2, Lemma 1.4.12]) that

$$\tau_G \circ (\mathrm{CS}_G)^{-1} \simeq \Theta_{\mathrm{Whit}(G)} \circ \mathrm{FLE}_{\check{G},\infty}.$$

Thus, we need to establish the commutativity of

$$(16.6) \quad \begin{array}{ccc}
 \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xleftarrow{\Theta_{\mathrm{Whit}(G)} \circ \mathrm{FLE}_{\check{G},\infty}} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\
 \mathrm{Poinc}_{G,!} \downarrow & & \downarrow \mathrm{Loc}_G^{\mathrm{spec}} \\
 \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}} & \xleftarrow{\mathrm{Mir}_{\mathrm{Bun}_G} \circ \mathbb{L}_G^\vee|_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})} \otimes \mathbb{I}_{z_{\check{\mathfrak{g}}}}[4\delta_G]} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}).
 \end{array}$$

16.2.7. Note that $\mathrm{LS}_{\check{G}}$ is quasi-smooth and *almost*⁶⁸ canonically derived-symplectic, so that

$$(16.7) \quad \omega_{\mathrm{LS}_{\check{G}}} \simeq \mathcal{O}_{\mathrm{LS}_{\check{G}}} \otimes \mathbb{I}_{z_{\check{\mathfrak{g}}}}[\dim(\mathrm{LS}_{\check{G}})].$$

Indeed, consider the short exact sequence

$$1 \rightarrow Z_G^0 \rightarrow \check{G} \rightarrow \check{G}_1 \rightarrow 1,$$

so that

$$\mathrm{LS}_{\check{G}} / \mathrm{LS}_{Z_G^0} \simeq \mathrm{LS}_{\check{G}_1}.$$

Note that $T^*(\mathrm{LS}_{Z_G^0})$ is a constant bundle with fiber $T_1^*(\mathrm{LS}_{Z_G^0})$, while $\mathrm{LS}_{\check{G}_1}$ is *canonically* derived-symplectic: for $\sigma \in \check{\mathrm{LS}}_{\check{G}_1}$ we have

$$T_\sigma(\mathrm{LS}_{\check{G}_1}) \simeq C(X, (\check{\mathfrak{g}}_1)_\sigma)[1],$$

so

$$(T_\sigma(\mathrm{LS}_{\check{G}_1}))^* \simeq (C(X, (\check{\mathfrak{g}}_1)_\sigma)[1])^* \stackrel{\mathrm{Verdier}}{\simeq} C(X, (\check{\mathfrak{g}}_1)_\sigma^*)[1] \stackrel{\mathrm{Killing Form}}{\simeq} C(X, (\check{\mathfrak{g}}_1)_\sigma)[1] \simeq T_\sigma(\mathrm{LS}_{\check{G}_1}).$$

16.2.8. Recall that the functors

$$\mathrm{Loc}_G^{\mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightleftarrows \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) : \Gamma_G^{\mathrm{spec}}$$

are mutually dual, when we identify $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ with its own dual via the usual duality (i.e., usual dualization on perfect complexes).

Hence, when we use the identification

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}})^\vee \simeq \mathrm{QCoh}(\mathrm{LS}_{\check{G}}),$$

induced by the Serre duality on $\mathrm{IndCoh}(\mathrm{LS}_{\check{G}})$, the dual of the functor Γ_G^{spec} becomes identified with

$$\mathrm{Loc}_G^{\mathrm{spec}} \otimes \mathbb{I}_{z_{\check{\mathfrak{g}}}}[\dim(\mathrm{LS}_{\check{G}})],$$

where $\dim(\mathrm{LS}_{\check{G}}) = 2 \dim(\mathrm{Bun}_G) = 2\delta_G$, and $\mathbb{I}_{z_{\check{\mathfrak{g}}}}$ appears via (16.7).

⁶⁸The symplectic structure depends on the choice of a non-degenerate symmetric bilinear form on $z_{\check{\mathfrak{g}}}$.

16.2.9. Hence, by taking the duals in the commutative diagram

$$\begin{array}{ccc}
 \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xrightarrow{\mathrm{CS}_G = (\mathrm{FLE}_{\check{G}, \infty})^\vee} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\
 \mathrm{coeff}_G[2\delta_{N_{\rho(\omega_X)}}]^\dagger \uparrow & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}} \\
 \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}} & \xrightarrow{\mathbb{L}_{G, \mathrm{temp}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}),
 \end{array}$$

we obtain a commutative diagram

$$\begin{array}{ccc}
 \mathrm{Whit}_*(\mathrm{Gr}_G)_{\mathrm{Ran}} & \xleftarrow{\mathrm{FLE}_{\check{G}, \infty}} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\
 \mathrm{Poinc}_{G, *} [2\delta_{N_{\rho(\omega_X)}}]^\dagger \downarrow & & \downarrow \mathrm{Loc}_{\check{G}}^{\mathrm{spec}} \otimes \mathbb{I}_{z_{\check{G}}} [2\delta_G] \\
 \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}, \mathrm{temp}} & \xleftarrow{\mathbb{L}_G^\vee|_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}).
 \end{array}
 \tag{16.8}$$

16.2.10. We now apply Theorem 10.1.10. Concatenating the diagrams (16.8) and (10.6) horizontally, we obtain the desired commutative diagram (16.6).

Thus, we have established the isomorphism

$$\mathbb{L}_{G, \mathrm{temp}}^L \simeq \Phi_{G, \mathrm{temp}}.
 \tag{16.9}$$

16.2.11. We will now deduce

$$\mathbb{L}_{G, \mathrm{temp}}^L \simeq \Phi_{G, \mathrm{temp}} \Rightarrow \mathbb{L}_G^L \simeq \Phi_G.$$

The functor $\mathbb{L}_G^L$ being a left adjoint, preserves compactness. We will prove:

Lemma 16.2.12. *The functor Φ_G preserves compactness.*

16.2.13. Recall now that according to [GLC1, Proposition 5.2.3], the functor $\mathbf{u}^R$ is fully faithful on compacts.

Hence, the assertion of Theorem 16.2.3 follows from Lemma 16.2.12 and (16.9) via the commutative diagram (16.4).

16.3. Proof of Lemma 16.2.12.

16.3.1. It is enough to show that Φ_G sends objects of the form $\mathrm{Eis}^{-, \mathrm{spec}}(\mathcal{F})$, for $\mathcal{F} \in \mathrm{QCoh}(\mathrm{LS}_{\check{M}})^c$ to compacts.

16.3.2. Recall that the functors

$$\mathrm{Eis}^{-, \mathrm{spec}} : \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \rightleftarrows \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) : \mathrm{CT}^{-, \mathrm{spec}}$$

are mutually dual, up to tensoring by a line bundle on $\mathrm{LS}_{\check{M}}$, see (12.5).

Hence

$$\mathbb{L}_G^\vee \circ \mathrm{Eis}^{-, \mathrm{spec}}$$

is isomorphic to

$$(\mathrm{CT}^{-, \mathrm{spec}} \circ \mathbb{L}_G)^\vee,$$

up to tensoring by a line bundle.

Hence, combining with Theorem 15.1.13, we obtain that $\mathbb{L}_G^\vee \circ \mathrm{Eis}^{-, \mathrm{spec}}$ is isomorphic to

$$(\mathbb{L}_M \circ \mathrm{CT}_*^-)^\vee \simeq \mathrm{Eis}_{\mathrm{co}, *}^- \circ \mathbb{L}_M^\vee,$$

again up to tensoring by a line bundle.

16.3.3. Now, by (8.35),

$$\tau_G \circ \text{Mir}_{\text{Bun}_G} \circ \text{Eis}_{\text{co},*}^- \simeq \text{Eis}_!^- \circ \text{Mir}_{\text{Bun}_M} \circ \tau_M.$$

Combining, we obtain that

$$\Phi_G \circ \text{Eis}^{-,\text{spec}} \simeq \text{Eis}_!^- \circ \Phi_M,$$

again up to tensoring by a line bundle.

Now, the assertion follows from the fact that the functors Φ_M send compacts in $\text{QCoh}(\text{LS}_{\bar{M}})$ to compacts in $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)$, which follows from the isomorphism $\mathbb{L}_{M,\text{temp}}^L \simeq \Phi_{M,\text{temp}}$.

□[Lemma 16.2.12]

16.4. The composition $\mathbb{L}_G \circ \mathbb{L}_G^L$. Recall that the geometric Langlands conjecture says that the functor $\mathbb{L}_G$ is an equivalence. We can now reformulate this as saying that the unit of the adjunction

$$\text{Id} \rightarrow \mathbb{L}_G \circ \mathbb{L}_G^L$$

is an isomorphism as endofunctors of $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}})$, combined with the fact that the functor $\mathbb{L}_G$ is conservative.

In this subsection we commence the study of the composition $\mathbb{L}_G \circ \mathbb{L}_G^L$.

16.4.1. Note that $\mathbb{L}_G \circ \mathbb{L}_G^L$, viewed as an endofunctor of $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}})$, is $\text{QCoh}(\text{LS}_{\check{G}})$ -linear. Hence, it is a priori given by an object in

$$\mathcal{A}_G \in \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \otimes_{\text{QCoh}(\text{LS}_{\check{G}})} \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}).$$

The goal of this subsection is to prove the following assertion:

Theorem 16.4.2. *The object $\mathcal{A}_G$ belongs to*

$$\begin{aligned} \text{QCoh}(\text{LS}_{\check{G}}) &\simeq \text{QCoh}(\text{LS}_{\check{G}}) \otimes_{\text{QCoh}(\text{LS}_{\check{G}})} \text{QCoh}(\text{LS}_{\check{G}}) \subset \\ &\subset \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \otimes_{\text{QCoh}(\text{LS}_{\check{G}})} \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}). \end{aligned}$$

In other words, this theorem implies that we have an isomorphism

$$\mathbb{L}_G \circ \mathbb{L}_G^L \simeq \mathcal{A}_G \otimes_{\mathcal{O}_{\text{LS}_{\check{G}}}} -, \quad \mathcal{A}_G \in \text{QCoh}(\text{LS}_{\check{G}}).$$

Remark 16.4.3. Once Theorem 16.4.2 is proved, and given the fact that the functor $\mathbb{L}_G$ is conservative, we will have interpreted GLC (i.e., [GLC1, Conjecture 1.6.7]) as the statement that the unit of the adjunction

$$(16.10) \quad \mathcal{O}_{\text{LS}_{\check{G}}} \rightarrow \mathcal{A}_G$$

is an isomorphism in $\text{QCoh}(\text{LS}_{\check{G}})$.

In the next section, we will show that the map (16.10) becomes an isomorphism when restricted to the *reducible* locus of $\text{LS}_{\check{G}}$. This will reduce GLC to the study of the restriction of $\mathcal{A}_G$ to the *irreducible* locus $\text{LS}_{\check{G}}^{\text{irred}}$.

In the subsequent paper of this series ([GLC4]), we will show that

$$\mathcal{A}_{G,\text{irred}} := \mathcal{A}_G|_{\text{LS}_{\check{G}}^{\text{irred}}}$$

is a *classical* vector bundle, equipped with a flat connection.

In the last paper of the series ([GLC5]), we will deduce from this that (16.10) is an isomorphism also over $\text{LS}_{\check{G}}^{\text{irred}}$, thereby concluding the proof of GLC.

16.4.4. We now begin the proof of Theorem 16.4.2.

Let $j : \mathrm{LS}_{\check{G}}^{\mathrm{irred}} \subset \mathrm{LS}_{\check{G}}$ denote the embedding of the locus of irreducible local systems. We denote by $i : \mathrm{LS}_{\check{G}}^{\mathrm{red}} \rightarrow \mathrm{LS}_{\check{G}}$ be the embedding of its complement (with any scheme-theoretic structure).

Let

$$(16.11) \quad \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$$

be the full subcategory of objects set-theoretically supported on $\mathrm{LS}_{\check{G}}^{\mathrm{red}}$.

In other words,

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} = \ker(j^* : \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})).$$

Denote by $\widehat{i}_!$ the tautological embedding,

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} \hookrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$$

and by $\widehat{i}^!$ its right adjoint.

$$\widehat{i}_! : \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} \rightleftarrows \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) : \widehat{i}^!.$$

16.4.5. In order to prove that $\mathcal{A}_G$ belongs to $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$, it suffices to show that

$$(j^* \otimes j^*)(\mathcal{A}_G) \in \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$$

and

$$(16.12) \quad (\widehat{i}^! \otimes \widehat{i}^!)(\mathcal{A}_G) \in \mathrm{QCoh}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}.$$

The former is automatic, since over $\mathrm{LS}_{\check{G}}^{\mathrm{irred}}$, the nilpotent cone vanishes, so the embedding

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \subseteq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$$

is an equality.

Remark 16.4.6. Alternatively, it is easy to reduce the assertion of Theorem 16.4.2 to the case when G is semi-simple. In this case $\mathrm{LS}_{\check{G}}^{\mathrm{irred}}$ is smooth, so the inclusion

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \subseteq \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$$

is also an equality.

16.4.7. We now proceed to proving (16.12).

Since both functors $\mathbb{L}_G$ and $\mathbb{L}_G^L$ are compatible with the inclusions and projections

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \hookrightarrow \mathrm{QCoh}(\mathrm{LS}_G) \text{ and } \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \hookrightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{temp}},$$

so is the composition $\mathbb{L}_G \circ \mathbb{L}_G^L$.

Denote the induced endofunctor of $\mathrm{QCoh}(\mathrm{LS}_G)$ by $\mathcal{A}_{G,\mathrm{temp}}$. In other words,

$$\mathbb{L}_{G,\mathrm{temp}} \circ \mathbb{L}_{G,\mathrm{temp}}^L =: \mathcal{A}_{G,\mathrm{temp}} \otimes_{\mathcal{O}_{\mathrm{LS}_{\check{G}}}} -, \quad \mathcal{A}_{G,\mathrm{temp}} \in \mathrm{QCoh}(\mathrm{LS}_{\check{G}}).$$

In order to prove (16.12), it suffices to show that

$$(\mathbb{L}_G \circ \mathbb{L}_G^L)|_{\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}} \simeq \mathcal{A}_{G,\mathrm{temp}} \otimes_{\mathcal{O}_{\mathrm{LS}_{\check{G}}}} -$$

as endofunctors of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}$.

To that end, it suffices to establish a functorial isomorphism

$$(16.13) \quad \mathbb{L}_G \circ \mathbb{L}_G^L(\mathcal{M}) \simeq \mathcal{A}_{G,\mathrm{temp}} \otimes_{\mathcal{O}_{\mathrm{LS}_{\check{G}}}} \mathcal{M}, \quad \mathcal{M} \in \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}^c.$$

16.4.8. For $\mathcal{M}$ as above, let

$$\mathcal{M}_{\text{temp}} \text{ and } (\mathbb{L}_G \circ \mathbb{L}_G^L(\mathcal{M}))_{\text{temp}}$$

denote the projections of $\mathcal{M}$ and $\mathbb{L}_G \circ \mathbb{L}_G^L(\mathcal{M})$, respectively, along

$$(16.14) \quad \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}})_{\text{red}} \rightarrow \text{QCoh}(\text{LS}_{\check{G}})_{\text{red}},$$

where

$$(16.15) \quad \text{QCoh}(\text{LS}_{\check{G}})_{\text{red}} := \ker \left(j^* : \text{QCoh}(\text{LS}_{\check{G}}) \rightarrow \text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}}) \right).$$

By construction

$$(16.16) \quad (\mathbb{L}_G \circ \mathbb{L}_G^L(\mathcal{M}))_{\text{temp}} \simeq \mathcal{A}_{G,\text{temp}} \otimes_{\mathcal{O}_{\text{LS}_{\check{G}}}} \mathcal{M}_{\text{temp}}.$$

Thus, we wish to show that (16.16) implies (16.13).

16.4.9. We claim:

Proposition 16.4.10. *The functor $(\mathbb{L}_G \circ \mathbb{L}_G^L)|_{\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}})_{\text{red}}}$ preserves compactness.*

Let us assume this proposition temporarily and finish the proof of Theorem 16.4.2.

16.4.11. We claim that Proposition 16.4.10 implies the following:

Corollary 16.4.12. *The restriction of $\mathcal{A}_{G,\text{temp}}$ to the formal completion $(\text{LS}_{\check{G}})_{\text{LS}_{\check{G}}^{\text{red}}}^{\wedge}$ is perfect as an object of $\text{QCoh}((\text{LS}_{\check{G}})_{\text{LS}_{\check{G}}^{\text{red}}}^{\wedge})$.*

Proof of Corollary 16.4.12. The assertion of Corollary 16.4.12 can be reformulated as saying that

$$\mathcal{A}_{G,\text{temp}}|_{(\text{LS}_{\check{G}})_{\text{LS}_{\check{G}}^{\text{red}}}^{\wedge}}$$

is dualizable as an object of $\text{QCoh}((\text{LS}_{\check{G}})_{\text{LS}_{\check{G}}^{\text{red}}}^{\wedge})$, and equivalently, as saying that the functor

$$\mathcal{A}_{G,\text{temp}} \otimes_{\mathcal{O}_{\text{LS}_{\check{G}}}} - : \text{QCoh}((\text{LS}_{\check{G}})_{\text{LS}_{\check{G}}^{\text{red}}}^{\wedge}) \rightarrow \text{QCoh}((\text{LS}_{\check{G}})_{\text{LS}_{\check{G}}^{\text{red}}}^{\wedge})$$

admits a continuous right adjoint.

Note that restriction along

$$(\text{LS}_{\check{G}})_{\text{LS}_{\check{G}}^{\text{red}}}^{\wedge} \hookrightarrow \text{LS}_{\check{G}}$$

defines an equivalence

$$\text{QCoh}(\text{LS}_{\check{G}})_{\text{red}} \rightarrow \text{QCoh}((\text{LS}_{\check{G}})_{\text{LS}_{\check{G}}^{\text{red}}}^{\wedge}),$$

where

$$\text{QCoh}(\text{LS}_{\check{G}})_{\text{red}} \subset \text{QCoh}(\text{LS}_{\check{G}})$$

is as in (16.15).

Hence, it suffices to show that the functor

$$(16.17) \quad \mathcal{A}_{G,\text{temp}} \otimes_{\mathcal{O}_{\text{LS}_{\check{G}}}} - : \text{QCoh}(\text{LS}_{\check{G}})_{\text{red}} \rightarrow \text{QCoh}(\text{LS}_{\check{G}})_{\text{red}}$$

admits a continuous right adjoint.

It is easy to see, however, that (16.17) admits a continuous right adjoint if and only if its composition with

$$\Xi_{\text{LS}_{\check{G}}} : \text{QCoh}(\text{LS}_{\check{G}})_{\text{red}} \rightarrow \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}})_{\text{red}}$$

does.

However, the latter follows from the commutative diagram

$$\begin{array}{ccc}
\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} & \xrightarrow{\mathbb{L}_G \circ \mathbb{L}_{\check{G}}^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} \\
\Xi_{\mathrm{LS}_{\check{G}}} |_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}} \uparrow & & \uparrow \Xi_{\mathrm{LS}_{\check{G}}} |_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}} \\
\mathrm{QCoh}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} & \xrightarrow{\mathcal{A}_{G, \mathrm{temp}} \otimes_{\mathcal{O}_{\mathrm{LS}_{\check{G}}}} -} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}.
\end{array}$$

Indeed, the top horizontal arrow admits a continuous right adjoint by Proposition 16.4.10, since the category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}$ is compactly generated. And the right adjoint of the left vertical arrow is the (continuous) functor $\Psi_{\mathrm{LS}_{\check{G}}} |_{\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}}$. $\square$

16.4.13. Let $\mathcal{M}$ be as in (16.13). By Proposition 16.4.10,

$$\mathbb{L}_G \circ \mathbb{L}_{\check{G}}^L(\mathcal{M}) \in \mathrm{Coh}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} := \mathrm{Coh}(\mathrm{LS}_{\check{G}}) \cap \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})_{\mathrm{red}},$$

and by Corollary 16.4.12, the object $\mathcal{A}_{G, \mathrm{temp}} \otimes_{\mathcal{O}_{\mathrm{LS}_{\check{G}}}} \mathcal{M}$ also belongs to this subcategory.

The projections of these objects along (16.14) are identified by (16.16). The isomorphism (16.13) now follows, since the restriction of (16.14) to

$$\mathrm{Coh}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} \subset \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}$$

is fully faithful.

$\square$ [Theorem 16.4.2]

16.5. Proof of Proposition 16.4.10.

16.5.1. Let

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

denote the full subcategory generated by the essential images of the functors

$$\mathrm{Eis}_! : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

for Levi quotients M of *proper* parabolics $P \subset G$.

16.5.2. We claim that the functors

$$\mathbb{L}_G : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \hookrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) : \mathbb{L}_G^L$$

send the full subcategories

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \text{ and } \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$$

to one another, and the resulting functors

$$\mathbb{L}_G : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \hookrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} : \mathbb{L}_G^L$$

preserve compactness.

This would imply that $(\mathbb{L}_G \circ \mathbb{L}_{\check{G}}^L)|_{\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}}$ also preserves compactness.

16.5.3. For the functor $\mathbb{L}_G$, this follows from its compatibility with the Eisenstein procedure, expressed by Theorem 14.2.2, and the assumption that GLC holds for M .

16.5.4. To prove the assertion for $\mathbb{L}_G^L$, we note that the subcategory (16.11) is generated by the essential images of the functors

$$\mathrm{Eis}^{\mathrm{spec}} : \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$$

for proper parabolics (indeed, the collection of their right adjoints (i.e., the functors $\mathrm{CT}^{\mathrm{spec}}$) is conservative on $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}$).

Now, the assertion concerning $\mathbb{L}_G^L$ follows from the commutative diagram (16.2).

$\square$ [Proposition 16.4.10]

17. THE LANGLANDS FUNCTOR IS AN EQUIVALENCE ON THE EISENSTEIN PART(S)

In this section we state and prove the main result of this paper, Theorem 17.1.2, which says that the Langlands functor $\mathbb{L}_G$ induces an equivalence between the Eisenstein-generated categories on the two sides.

17.1. Statement of the main result.

17.1.1. In Sect. 16.5.1, we have considered the pair of (mutually adjoint) functors

$$(17.1) \quad \mathbb{L}_G : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \rightleftarrows \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} : \mathbb{L}_G^L.$$

We will prove:

Main Theorem 17.1.2. *Let us assume that GLC is valid for Levi subgroups of all proper parabolics of G . Then the adjoint functors in (17.1) are mutually inverse equivalences.*

17.1.3. *Proof of Theorem 15.1.2.* Let us assume Theorem 17.1.2 for a moment and deduce from it Theorem 15.1.2.

By passing to the right adjoints along the vertical arrows in

$$\begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow[\sim]{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ \mathrm{Eis}_{\dagger, \rho_P(\omega_X)}^{-}[\delta_{N(P^-)}]_{\rho_P(\omega_X)} \downarrow & & \downarrow \mathrm{Eis}^{-, \mathrm{spec}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} & \xrightarrow[\sim]{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}}, \end{array}$$

we obtain that the natural transformation in question is an equivalence, once both circuits in the diagram (15.1) are restricted to

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G).$$

Hence, in order to prove the corollary, it suffices to show that both circuits vanish when restricted to

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} := (\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}})^{\perp}.$$

The vanishing is tautological for the counterclockwise circuit. For the vanishing of the clockwise circuit, it suffices to show that there exists *some* isomorphism

$$\mathrm{CT}^{-, \mathrm{spec}} \circ \mathbb{L}_G \simeq \mathbb{L}_M \circ \mathrm{CT}_{*, \rho_P(\omega_X)}^{-}[-\delta_{N(P^-)}]_{\rho_P(\omega_X)}.$$

However, the latter is given by Theorem 15.1.13.

□[Theorem 15.1.2]

Remark 17.1.4. As was already mentioned, we do not know whether the isomorphism

$$\mathrm{CT}^{-, \mathrm{spec}} \circ \mathbb{L}_G \simeq \mathbb{L}_M \circ \mathrm{CT}_{*, \rho_P(\omega_X)}^{-}[-\delta_{N(P^-)}]_{\rho_P(\omega_X)},$$

constructed in Theorem 15.1.13 above equals the one given by Theorem 15.1.2.

Remark 17.1.5. Note that Theorem 17.1.2 implies that GLC is equivalent to the statement that the functors $(\mathbb{L}_G^L, \mathbb{L}_G)$ define mutually inverse equivalences

$$(17.2) \quad \mathbb{L}_{G, \mathrm{irred}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} \rightleftarrows \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \simeq \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) : \mathbb{L}_{G, \mathrm{irred}}^L.$$

As was mentioned in Remark 16.4.3, this is equivalent to showing that the map

$$(17.3) \quad \mathcal{O}_{\mathrm{LS}_{\check{G}}^{\mathrm{irred}}} \rightarrow \mathcal{A}_{G, \mathrm{irred}},$$

induced by (16.10), is an isomorphism.

17.2. Constant terms and the left adjoint of the Langlands functor.

17.2.1. As a preparation to the proof of Theorem 17.1.2, we will need the following:

Proposition 17.2.2. *The following diagram of functors commutes*

$$\begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xleftarrow{\mathbb{L}_M^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ \mathrm{CT}_{*,\rho_P(\omega_X)}^{-}[-\delta_{(N_P^-)_{\rho_P(\omega_X)}}] \uparrow & & \uparrow_{\mathrm{CT}^{-,\mathrm{spec}}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xleftarrow{\mathbb{L}_G^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}). \end{array}$$

The rest of this subsection is devoted to the proof of this proposition.

17.2.3. Applying Theorem 16.2.3 for G and M , we need to show that the following diagram commutes

$$\begin{array}{ccccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xleftarrow{\mathrm{Mir}_{M,\tau}} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} & \xleftarrow{\mathbb{L}_M^\vee} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ \mathrm{CT}_{*,\rho_P(\omega_X)}^{-} \uparrow & & & & \uparrow_{\mathrm{CT}^{-,\mathrm{spec}}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xleftarrow{\mathrm{Mir}_{G,\tau}} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} & \xleftarrow{\mathbb{L}_G^\vee} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}), \end{array}$$

up to a cohomological shift by

$$4\delta_G - 2\delta_{N_{\rho(\omega_X)}} - 4\delta_M + 2\delta_{N(M)_{\rho_M(\omega_X)}} - \delta_{(N_P^-)_{\rho_P(\omega_X)}} = 3(\delta_G - \delta_M) - \delta_{(N_P)_{\rho_P(\omega_X)}}.$$

NB: the (constant) lines $\mathbb{L}_{z_{\check{\mathfrak{g}}}}$ and $\mathbb{L}_{z_{\check{\mathfrak{m}}}}$ cancel out, since the Killing form of $\check{\mathfrak{g}}$ is non-degenerate on $z_{\check{\mathfrak{m}}}/z_{\check{\mathfrak{g}}}$.

Passing to dual functors, we obtain that it suffices to show that the following diagram commutes

$$\begin{array}{ccccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{M,\tau}} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ (\mathrm{CT}_{*,\rho_P(\omega_X)}^{-})^\vee \downarrow & & & & \downarrow_{(\mathrm{CT}^{-,\mathrm{spec}})^\vee} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{G,\tau}} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}), \end{array}$$

up to the same cohomological shift.

17.2.4. Recall that according to (12.5) and Lemma 12.1.9, with respect to the Serre duality identifications

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})^\vee \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \text{ and } \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}})^\vee \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}),$$

we have a canonical identification

$$(\mathrm{CT}^{-,\mathrm{spec}})^\vee \simeq \mathrm{Eis}^{-,\mathrm{spec}} \circ (\mathcal{L}_{2\rho_P(\omega_X)} \otimes (-)),$$

up to a cohomological shift by

$$\dim(\check{\mathfrak{n}}_P^-) \cdot (2g - 2).$$

Also, by construction, we have

$$(\mathrm{CT}_{*,\rho_P(\omega_X)}^{-})^\vee \simeq \mathrm{Eis}_{\mathrm{co},*,\rho_P(\omega_X)}^{-}.$$

Hence, we need to show that the following diagram commutes

$$(17.4) \quad \begin{array}{ccccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{M,\tau}} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ \mathrm{Eis}_{\mathrm{co},*,\rho_P(\omega_X)}^{-} \downarrow & & & & \downarrow_{\mathrm{Eis}^{-,\mathrm{spec}} \circ (\mathcal{L}_{2\rho_P(\omega_X)} \otimes (-))} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{G,\tau}} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}), \end{array}$$

up to a cohomological shift by

$$3(\delta_G - \delta_M) - \delta_{(N_P)_{\rho_P(\omega_X)}} - \dim(\check{\mathfrak{n}}_P^-) \cdot (2g - 2) = 2(\delta_G - \delta_M) - \delta_{(N_P)_{\rho_P(\omega_X)}}.$$

17.2.5. Note that by (8.35), we have the following diagram

$$(17.5) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{M,\tau}} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \\ \mathrm{Eis}_{\mathrm{co},*,\rho_P(\omega_X)}^- \downarrow & & \downarrow \mathrm{Eis}_{!,-\rho_P(\omega_X)}^- \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{G,\tau}} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G), \end{array}$$

which commutes up to a cohomological shift by $2\delta_{N_P^-}$.

Hence, we obtain that it suffices to show that the following diagram commutes

$$(17.6) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ \mathrm{Eis}_{!,-\rho_P(\omega_X)}^- \downarrow & & \downarrow \mathrm{Eis}^-, \mathrm{spec} \circ (\mathcal{L}_{2\rho_P(\omega_X)} \otimes (-)) \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}), \end{array}$$

up to a cohomological shift by

$$2(\delta_G - \delta_M) - \delta_{(N_P)_{\rho_P(\omega)_X}} - 2\delta_{N_P^-} = (\delta_G - \delta_M) - \delta_{(N_P)_{\rho_P(\omega)_X}} = \delta_{(N_P^-)_{\rho_P(\omega)_X}}.$$

17.2.6. Note that the following diagram commutes

$$(17.7) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ (\mathrm{transl}_{-2\rho_P(\omega_X)})_* \downarrow & & \downarrow \mathcal{L}_{2\rho_P(\omega_X)} \otimes (-) \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}), \end{array}$$

see [GLC1, Sect. 1.5.3].

We also note that

$$\mathrm{Eis}_{!,-\rho_P(\omega_X)}^- \simeq \mathrm{Eis}_{!,\rho_P(\omega_X)}^- \circ (\mathrm{transl}_{-2\rho_P(\omega_X)})_*.$$

Hence, the commutation of (17.17) is equivalent to the commutation of

$$(17.8) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ \mathrm{Eis}_{!,\rho_P(\omega_X)}^- [\delta_{(N_P^-)_{\rho_P(\omega_X)}}] \downarrow & & \downarrow \mathrm{Eis}^-, \mathrm{spec} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}). \end{array}$$

However, the latter was established in Theorem 14.2.2.

□[Proposition 17.2.2]

17.2.7. Combining with Theorem 15.1.13 we obtain:

Corollary 17.2.8. *We have a canonical isomorphism*

$$\mathrm{CT}^-, \mathrm{spec} \simeq \mathrm{CT}^-, \mathrm{spec} \circ \mathbb{L}_G \circ \mathbb{L}_G^L$$

as functors

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \rightrightarrows \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}).$$

In particular, we have:

Corollary 17.2.9. *We have an isomorphism in $\mathrm{QCoh}(\mathrm{LS}_{\check{M}}) \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}})$*

$$\mathrm{CT}^-, \mathrm{spec}(\mathcal{A}_G) \simeq \mathrm{CT}^-, \mathrm{spec}(\mathcal{O}_{\mathrm{LS}_{\check{G}}}).$$

17.3. **Proof of Theorem 17.1.2.**

17.3.1. In the course of the proof of Proposition 16.4.10, we have seen that the essential image of each of the functors in (17.1) generates the target category.

In particular, we obtain that the functor $\mathbb{L}_G|_{\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}}}$ is conservative.

Hence, in order to prove Theorem 17.1.2, it suffices to show that the functor $\mathbb{L}_G^L|_{\mathrm{IndCohNilp}(\mathrm{LS}_{\tilde{G}})_{\mathrm{red}}}$ is fully faithful.

I.e., we need to show that the unit of the adjunction

$$(17.9) \quad \mathrm{Id}_{\mathrm{IndCohNilp}(\mathrm{LS}_{\tilde{G}})_{\mathrm{red}}} \rightarrow (\mathbb{L}_G \circ \mathbb{L}_G^L)|_{\mathrm{IndCohNilp}(\mathrm{LS}_{\tilde{G}})_{\mathrm{red}}}$$

is an isomorphism.

17.3.2. By Theorem 16.4.2, the endofunctor $\mathbb{L}_G \circ \mathbb{L}_G^L$ is given by tensor product with an object

$$\mathcal{A}_G \in \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}).$$

Moreover, the structure of monad on $\mathbb{L}_G \circ \mathbb{L}_G^L$ corresponds to a structure of associative algebra on $\mathcal{A}_G$ as an object of the (symmetric) monoidal category $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})$.

Under this identification, the unit of the adjunction

$$\mathrm{Id}_{\mathrm{IndCohNilp}(\mathrm{LS}_{\tilde{G}})} \rightarrow \mathbb{L}_G \circ \mathbb{L}_G^L$$

corresponds to the map of associative algebras

$$(17.10) \quad \mathcal{O}_{\mathrm{LS}_{\tilde{G}}} \rightarrow \mathcal{A}_G.$$

17.3.3. In order to show that (17.9) is an isomorphism, it suffices to show that the $*$ -restriction of the map (17.10) to the formal completion

$$(\mathrm{LS}_{\tilde{G}})_{\mathrm{LS}_{\tilde{G}}^{\mathrm{red}}}^{\wedge} \subset \mathrm{LS}_{\tilde{G}}$$

is an isomorphism.

17.3.4. Note that the composition

$$\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})_{\mathrm{red}} \hookrightarrow \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}) \xrightarrow{\hat{i}^*} \mathrm{QCoh}((\mathrm{LS}_{\tilde{G}})_{\mathrm{LS}_{\tilde{G}}^{\mathrm{red}}}^{\wedge})$$

is an equivalence.

Note also that the collection of functors

$$(\mathfrak{p}^{-, \mathrm{spec}})^* : \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\tilde{P}-}),$$

for proper standard parabolics of G , is conservative on $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})_{\mathrm{red}}$.

Hence, it suffices to show that for each proper parabolic as above, the induced map

$$(17.11) \quad \mathcal{O}_{\mathrm{LS}_{\tilde{P}-}} \rightarrow (\mathfrak{p}^{-, \mathrm{spec}})^*(\mathcal{A}_G)$$

is an isomorphism.

17.3.5. We will deduce this from the following statement:

Proposition 17.3.6. *For every proper parabolic, the object*

$$(\mathfrak{p}^{-, \mathrm{spec}})^*(\mathcal{A}_G) \in \mathrm{QCoh}(\mathrm{LS}_{\tilde{P}-})$$

is a line bundle.

Indeed, the fact that Proposition 17.3.6 implies that (17.11) is an isomorphism follows from the next observation:

Lemma 17.3.7. *Let $\mathcal{Y}$ be an algebraic stack, and let $\mathcal{A}_{\mathcal{Y}}$ be a unital associative algebra object in $\mathrm{QCoh}(\mathcal{Y})$. Assume that $\mathcal{A}_{\mathcal{Y}}$ is a line bundle. Then the unit map $\mathcal{O}_{\mathcal{Y}} \rightarrow \mathcal{A}_{\mathcal{Y}}$ is an isomorphism.*

Proof 1. It suffices to show that the unit map is an isomorphism at the level of fibers at field-valued points. This reduces the assertion to the case when $\mathcal{Y}$ is the spectrum of a field, in which case it is obvious. $\square$

Proof 2. Let $\mathcal{A}$ be a unital associative algebra object in a symmetric monoidal stable ∞ -category $\mathbf{A}$. Suppose that, as a plain object, $\mathcal{A}$ is invertible. We claim that in this case, the unit map

$$u : \mathbf{1}_{\mathbf{A}} \rightarrow \mathcal{A}$$

is an isomorphism.

Indeed, tensoring

$$m : \mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A}$$

on the left with $\mathcal{A}^{\otimes -1}$, we obtain a map

$$v : \mathcal{A} \rightarrow \mathbf{1}_{\mathbf{A}}.$$

Let us show that the maps u and v are each other's inverses. The fact that $v \circ u = \text{id}$ is obvious. Hence, we obtain that $\mathbf{1}_{\mathbf{A}}$ is a retract of $\mathcal{A}$. Write

$$\mathcal{A} = \mathbf{1}_{\mathbf{A}} \oplus \mathcal{B}.$$

We wish to show that $\mathcal{B} = 0$. It suffices to show that the tautological map $\alpha : \mathcal{B} \rightarrow \mathcal{A}$ is null-homotopic. Consider the composition:

$$\mathcal{B} \xrightarrow{u \otimes \text{id}} \mathcal{A} \otimes \mathcal{B} \xrightarrow{\text{id} \otimes \alpha} \mathcal{A} \otimes \mathcal{A} \xrightarrow{m} \mathcal{A}$$

On the one hand, it equals α . On the other hand, since $m = \text{id} \otimes v$ and $v \circ \alpha = 0$, we obtained that the above composition vanishes. $\square$

17.3.8. Consider now the map

$$(17.12) \quad \text{LS}_{\check{M}} \rightarrow \text{LS}_{\check{P}^-}$$

induced by the canonical map

$$(17.13) \quad \text{pt} / \check{M} \rightarrow \text{pt} / \check{P}^-.$$

We will deduce Proposition 17.3.6 from the following assertion:

Proposition 17.3.9. *The $*$ -pullback of $(\mathfrak{p}^{-, \text{spec}})^*(\mathcal{A}_G)$ along (17.12) is a line bundle on $\text{LS}_{\check{M}}$.*

We will prove Proposition 17.3.9 in Sect. 17.5. The implication Proposition 17.3.9 $\Rightarrow$ Proposition 17.3.6 will be carried out in Sect. 17.4 below.

17.3.10. Once Propositions 17.3.6 and 17.3.9 are proved, the proof of Theorem 17.1.2 would be complete.

17.4. Proof of Proposition 17.3.6.

17.4.1. Choose a coweight

$$\check{\theta} : \mathbb{G}_m \rightarrow Z_{\check{M}},$$

which is *anti-dominant and regular*, i.e., its action on any $\alpha \in \check{\mathfrak{n}}_{\check{P}}^-$ is a strictly positive integer.

The adjoint action of $\mathbb{G}_m$ on $\check{P}^-$ gives rise to an action of $\mathbb{G}_m$ on $\text{LS}_{\check{P}^-}$. This action has the following properties (cf. [DG2, Sect. 11.2]):

- (a) It respects the projection $\mathfrak{q}^{-, \text{spec}} : \text{LS}_{\check{P}^-} \rightarrow \text{LS}_{\check{M}}$ (because the adjoint action of $\check{\theta}$ on $\check{M}$ is trivial);
- (b) It is *contracting*, i.e., the action of $\mathbb{G}_m$ extends to an action of the monoid $\mathbb{A}^1$, with the fixed-point locus being the section of $\mathfrak{q}^{-, \text{spec}}$ given by (17.12);
- (c) It is isomorphic to the trivial action (because the $\mathbb{G}_m$ -action on $\check{P}^-$ is inner).

17.4.2. Note that the discrepancy between the trivialization of the $\mathbb{G}_m$ -action on $\mathrm{LS}_{\tilde{P}-}$ and the data of $\mathbb{G}_m$ -equivariance on the projection $\mathfrak{q}^{-,\mathrm{spec}}$ is given by the action of $\mathbb{G}_m$ by 1-automorphisms on $\mathrm{LS}_{\tilde{M}}$.

In particular, for $\mathcal{F} \in \mathrm{QCoh}(\mathrm{LS}_{\tilde{P}-})$, the $\mathbb{G}_m$ -action on $(\mathfrak{q}^{-,\mathrm{spec}})_*(\mathcal{F})$ coming from:

- the $\mathbb{G}_m$ -equivariant structure on $\mathcal{F}$ given by point (c) above, and
- the structure of $\mathbb{G}_m$ -invariance on $\mathfrak{q}^{-,\mathrm{spec}}$ from point (a)

equals the $\mathbb{G}_m$ -action on $(\mathfrak{q}^{-,\mathrm{spec}})_*(\mathcal{F})$ as an object of $\mathrm{QCoh}(\mathrm{LS}_{\tilde{M}})$, coming from the action of $\mathbb{G}_m$ by 1-automorphisms on $\mathrm{LS}_{\tilde{M}}$.

17.4.3. Consider

$$(\mathfrak{q}^{-,\mathrm{spec}})_*(\mathcal{O}_{\mathrm{LS}_{\tilde{P}-}})$$

as a graded connective commutative algebra in $\mathrm{QCoh}(\mathrm{LS}_{\tilde{M}})$.

Note also that $(\mathfrak{q}^{-,\mathrm{spec}})_*(\mathcal{O}_{\mathrm{LS}_{\tilde{P}-}})$ is *as good as connective* (see [GLC2, Sect. 12.7.6] or Sect. C.4.3 for what this means). In particular, the functor

$$(\mathfrak{q}^{-,\mathrm{spec}})_* : \mathrm{QCoh}(\mathrm{LS}_{\tilde{P}-}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\tilde{M}})$$

upgrades to an equivalence

$$(17.14) \quad \mathrm{QCoh}(\mathrm{LS}_{\tilde{P}-}) \rightarrow (\mathfrak{q}^{-,\mathrm{spec}})_*(\mathcal{O}_{\mathrm{LS}_{\tilde{P}-}})\text{-mod}(\mathrm{QCoh}(\mathrm{LS}_{\tilde{M}})).$$

By property (b) in Sect. 17.4.1, the grading on $(\mathfrak{q}^{-,\mathrm{spec}})_*(\mathcal{O}_{\mathrm{LS}_{\tilde{P}-}})$ is non-negative, with the degree 0 part isomorphic to $\mathcal{O}_{\mathrm{LS}_{\tilde{M}}}$, which is a retract of $(\mathfrak{q}^{-,\mathrm{spec}})_*(\mathcal{O}_{\mathrm{LS}_{\tilde{P}-}})$ via (17.12).

In particular, the functor of pullback along (17.12) corresponds under the equivalence (17.14) to the functor

$$\mathcal{O}_{\mathrm{LS}_{\tilde{M}}} \otimes_{(\mathfrak{q}^{-,\mathrm{spec}})_*(\mathcal{O}_{\mathrm{LS}_{\tilde{P}-}})} (-).$$

17.4.4. We now claim:

Lemma 17.4.5. *The grading on $(\mathfrak{q}^{-,\mathrm{spec}})_* \circ (\mathfrak{p}^{-,\mathrm{spec}})^*(\mathcal{A}_G)$ is non-negative.*

Assuming the lemma for a moment, the fact that $(\mathfrak{p}^{-,\mathrm{spec}})^*(\mathcal{A}_G)$ is a line bundle follows from Proposition 17.3.9 using the next version of the graded Nakayama lemma:

Lemma 17.4.6. *Let A be a non-negatively graded commutative algebra. Denote by A_0 its degree 0 component. Let M be a non-negatively graded A -module. Suppose that the A_0 -module*

$$M_0 := A_0 \otimes_A M$$

is invertible. Then M is itself invertible.

□[Proposition 17.3.6]

17.4.7. *Proof of Lemma 17.4.5.* Recall that, according to Sect. 17.4.2, this grading equals the grading on $(\mathfrak{q}^{-,\mathrm{spec}})_* \circ (\mathfrak{p}^{-,\mathrm{spec}})^*(\mathcal{A}_G)$ as an object of $\mathrm{QCoh}(\mathrm{LS}_{\tilde{M}})$ coming from the action of $\mathbb{G}_m$ by 1-automorphisms on $\mathrm{LS}_{\tilde{M}}$.

Note that

$$(\mathfrak{q}^{-,\mathrm{spec}})_* \circ (\mathfrak{p}^{-,\mathrm{spec}})^*(\mathcal{A}_G) \simeq ((\mathfrak{q}^{-,\mathrm{spec}})_* \circ (\mathfrak{p}^{-,\mathrm{spec}})^!(\mathcal{A}_G)) \otimes \mathcal{L}$$

for a particular (cohomologically shifted) line bundle on $\mathrm{LS}_{\tilde{M}}$.

We also have

$$(\mathfrak{q}^{-,\mathrm{spec}})_* \circ (\mathfrak{p}^{-,\mathrm{spec}})^*(\mathcal{O}_{\mathrm{LS}_{\tilde{G}}}) \simeq ((\mathfrak{q}^{-,\mathrm{spec}})_* \circ (\mathfrak{p}^{-,\mathrm{spec}})^!(\mathcal{O}_{\mathrm{LS}_{\tilde{G}}})) \otimes \mathcal{L}$$

for the same line bundle $\mathcal{L}$.

Now, according to Corollary 17.2.9, we have an isomorphism

$$(\mathfrak{q}^{-,\mathrm{spec}})_* \circ (\mathfrak{p}^{-,\mathrm{spec}})^!(\mathcal{A}_G) \simeq (\mathfrak{q}^{-,\mathrm{spec}})_* \circ (\mathfrak{p}^{-,\mathrm{spec}})^!(\mathcal{O}_{\mathrm{LS}_{\tilde{G}}}).$$

Hence, we also have

$$(\mathfrak{q}^{-, \text{spec}})_* \circ (\mathfrak{p}^{-, \text{spec}})^*(\mathcal{A}_G) \simeq (\mathfrak{q}^{-, \text{spec}})_* \circ (\mathfrak{p}^{-, \text{spec}})^*(\mathcal{O}_{\text{LS}_{\tilde{G}}}).$$

Hence, the required assertion follows from the fact that the grading on

$$(\mathfrak{q}^{-, \text{spec}})_* \circ (\mathfrak{p}^{-, \text{spec}})^*(\mathcal{O}_{\text{LS}_{\tilde{G}}}) \simeq (\mathfrak{q}^{-, \text{spec}})_*(\mathcal{O}_{\text{LS}_{\tilde{P}-}})$$

is non-negative.

□[Lemma 17.4.5]

17.5. Langlands functor and the “compactified” constant term functor(s). The goal of this subsection is to prove Proposition 17.3.9. We will reduce it to a statement about the compatibility of the Langlands functor and its left adjoint with the “compactified” constant term functors.

17.5.1. Since both $\text{LS}_{\tilde{G}}$ and $\text{LS}_{\tilde{P}-}$ are quasi-smooth, and

$$\mathcal{A}_G \in \text{QCoh}(\text{LS}_{\tilde{G}}) \subset \text{IndCoh}(\text{LS}_{\tilde{G}}),$$

when proving Proposition 17.3.9, instead of $(\mathfrak{p}^{-, \text{spec}})^*(\mathcal{A}_G)$, we can consider $(\mathfrak{p}^{-, \text{spec}})^!(\mathcal{A}_G)$ (the two differ by tensoring by a line bundle).

Note that the resulting object of

$$\text{QCoh}(\text{LS}_{\tilde{M}}) \subset \text{IndCoh}(\text{LS}_{\tilde{M}})$$

is by definition

$$\text{CT}_{!*}^{-, \text{spec}}(\mathcal{A}_G),$$

see Sect. 12.12.2.

Note also that by definition,

$$\mathcal{A}_G \simeq \mathbb{L}_G \circ \mathbb{L}_G^L(\mathcal{O}_{\text{LS}_{\tilde{G}}}).$$

Hence, in order to prove Proposition 17.3.9, it suffices to show the following:

Proposition 17.5.2. *There is a canonical isomorphism*

$$\text{CT}_{!*}^{-, \text{spec}} \circ \mathbb{L}_G \circ \mathbb{L}_G^L \simeq \text{CT}_{!*}^{-, \text{spec}}$$

as functors

$$\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{G}}) \rightrightarrows \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\tilde{M}}).$$

17.5.3. Recall the equivalence

$$\Theta_{\text{I}(G, P^-)^{\text{loc}, \tau}} : \text{I}(G, P^-)_{\text{co}}^{\text{loc}} \simeq \text{I}(G, P^-)^{\text{loc}},$$

see (8.33). Let

$$\text{IC}_{\text{co}}^{-, \frac{\infty}{2}} \in \text{I}(G, P^-)_{\text{co}}^{\text{loc}}$$

be the factorization algebra object equal to the image of $\text{IC}_{\text{co}}^{-, \frac{\infty}{2}}$ under $\Theta_{\text{I}(G, P^-)^{\text{loc}, \tau}}$.

Let

$$\text{CT}_{!*, \rho_P(\omega_X)}^{-} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_M)$$

be the functor as in Sect. 8.8.9 with $\mathcal{B} := \text{IC}_{\text{co}}^{-}$.

17.5.4. Proposition 17.5.2 is obtained by combining the following two assertions:

Proposition 17.5.5. *The following diagram of functors commutes*

$$\begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ \mathrm{CT}_{!*, \rho_P(\omega_X)}^{-}[-\delta_{(N_P^-)_{\rho_P(\omega_X)}}] \uparrow & & \uparrow \mathrm{CT}_{!*, \mathrm{spec}}^{-} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}). \end{array}$$

Proposition 17.5.6. *The following diagram of functors commutes*

$$\begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xleftarrow{\mathbb{L}_M^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ \mathrm{CT}_{!*, \rho_P(\omega_X)}^{-}[-\delta_{(N_P^-)_{\rho_P(\omega_X)}}] \uparrow & & \uparrow \mathrm{CT}_{!*, \mathrm{spec}}^{-} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xleftarrow{\mathbb{L}_G^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}). \end{array}$$

17.5.7. *Proof of Proposition 17.5.5.* By Proposition 12.12.6, we rewrite $\mathrm{CT}_{!*, \mathrm{spec}}^{-}$ as $\mathrm{CT}_{\mathrm{IC}}^{-, \mathrm{spec}}$.

Note now that the object $\mathrm{IC}_{\mathrm{co}}^{-, \frac{\infty}{2}}$ can be also described as the image of

$$\mathrm{IC}_{\mathrm{co}}^{-, \mathrm{spec}, \frac{\infty}{2}} \in \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}_{\mathrm{co}}$$

under the equivalence $((\mathrm{Sat}^{\frac{\infty}{2}})^{\vee})^{-1}$.

The assertion of the proposition follows now from Theorem 15.6.5.

□[Proposition 17.5.5]

17.5.8. The rest of this subsection is devoted to the proof of Proposition 17.5.6. In fact, it will follow line-by-line the proof of Proposition 17.2.2.

NB: in the argument below we will not specify the exact amounts of cohomological shifts, because they are the same as in the proof of Proposition 17.2.2.

Applying Theorem 16.2.3 for G and M , we need to show that the following diagram commutes

$$\begin{array}{ccccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xleftarrow{\mathrm{Mir}_M, \tau} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} & \xleftarrow{\mathbb{L}_M^{\vee}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ \mathrm{CT}_{!*, \rho_P(\omega_X)}^{-} \uparrow & & & & \uparrow \mathrm{CT}_{!*, \mathrm{spec}}^{-} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xleftarrow{\mathrm{Mir}_G, \tau} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} & \xleftarrow{\mathbb{L}_G^{\vee}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}), \end{array}$$

up to a cohomological shift.

Passing to dual functors, we obtain that it suffices to show that the following diagram commutes

$$\begin{array}{ccccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_M, \tau} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ (\mathrm{CT}_{!*, \rho_P(\omega_X)}^{-})^{\vee} \downarrow & & & & \downarrow (\mathrm{CT}_{!*, \mathrm{spec}}^{-})^{\vee} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_G, \tau} & \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}), \end{array}$$

up to the same cohomological shift.

17.5.9. Recall that according to Lemma 12.12.4, with respect to the Serre duality identifications

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})^\vee \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \text{ and } \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}})^\vee \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}),$$

we have a canonical identification

$$(\mathrm{CT}_{!*}^{-,\mathrm{spec}})^\vee \simeq \mathrm{Eis}_{!*}^{-,\mathrm{spec}} \circ (\mathcal{L}_{2\rho_P(\omega_X)} \otimes (-)),$$

up to a cohomological shift.

Also, by construction, we have

$$(\mathrm{CT}_{!*,\rho_P(\omega_X)}^-)^\vee \simeq \mathrm{Eis}_{\mathrm{co},*,\rho_P(\omega_X),\mathrm{IC}}^-,$$

where $\mathrm{Eis}_{\mathrm{co},*,\rho_P(\omega_X),\mathrm{IC}}^-$ is the functor from Sect. 8.8.9 with $\mathcal{B} := \mathrm{IC}_{\mathrm{co}}^-$.

Hence, we need to show that the following diagram commutes

$$(17.15) \quad \begin{array}{ccccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{M,\tau}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ \mathrm{Eis}_{\mathrm{co},*,\rho_P(\omega_X),\mathrm{IC}}^- \downarrow & & & & \downarrow \mathrm{Eis}_{!*}^{-,\mathrm{spec}} \circ (\mathcal{L}_{2\rho_P(\omega_X)} \otimes (-)) \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{G,\tau}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}), \end{array}$$

up to a cohomological shift.

17.5.10. Note that by Corollary 8.8.11, we have the following diagram

$$(17.16) \quad \begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{M,\tau}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \\ \mathrm{Eis}_{\mathrm{co},*,\rho_P(\omega_X),\mathrm{IC}}^- \downarrow & & \downarrow \mathrm{Eis}_{!,-\rho_P(\omega_X),\mathrm{IC}}^- \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} & \xrightarrow{\mathrm{Mir}_{G,\tau}} & \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G), \end{array}$$

which commutes up to a cohomological shift.

Hence, we obtain that it suffices to show that the following diagram commutes

$$(17.17) \quad \begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ \mathrm{Eis}_{!,-\rho_P(\omega_X),\mathrm{IC}}^- \downarrow & & \downarrow \mathrm{Eis}_{!*}^{-,\mathrm{spec}} \circ (\mathcal{L}_{2\rho_P(\omega_X)} \otimes (-)) \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}), \end{array}$$

up to a cohomological shift.

17.5.11. Recall the commutative diagram (17.7). We also note that

$$\mathrm{Eis}_{!,-\rho_P(\omega_X),\mathrm{IC}}^- \simeq \mathrm{Eis}_{!,\rho_P(\omega_X),\mathrm{IC}}^- \circ (\mathrm{transl}_{-2\rho_P(\omega_X)})_*.$$

Hence, the commutation of (17.17) is equivalent to the commutation of

$$(17.18) \quad \begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{M}}) \\ \mathrm{Eis}_{!,\rho_P(\omega_X),\mathrm{IC}}^- [\delta_{(N_P^-)\rho_P(\omega_X)}] \downarrow & & \downarrow \mathrm{Eis}_{!*}^{-,\mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}). \end{array}$$

However, the latter was established in Corollary 14.3.5.

□[Proposition 17.5.6]

Appendices

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APPENDIX A. THE SPECTRAL SEMI-INFINITE CATEGORY/IES OVER THE RAN SPACE

We will construct the category

$$\mathrm{IndCoh}^*(\mathrm{LS}_G^{\mathrm{reg}} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{LS}_{P^-}^{\mathrm{mer}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{reg}})$$

by emulating the recipe of [GLC2, Sect. E.2] for the construction of

$$\mathrm{IndCoh}^*(\mathrm{LS}_H^{\mathrm{reg}} \times_{\mathrm{LS}_H^{\mathrm{mer}}} \mathrm{LS}_H^{\mathrm{reg}})$$

for an algebraic group H .

Namely, we will first define the factorization category

$$(A.1) \quad \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)),$$

equipped with an action of the factorization group-scheme $\mathfrak{L}_{\nabla}^+(\check{G}) \times \mathfrak{L}_{\nabla}^+(\check{M})$, and set

$$(A.2) \quad \mathrm{IndCoh}^*(\mathrm{LS}_G^{\mathrm{reg}} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{LS}_{P^-}^{\mathrm{mer}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{reg}}) := \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))_{\mathfrak{L}_{\nabla}^+(\check{G}) \times \mathfrak{L}_{\nabla}^+(\check{M})}$$

Note, however, that the definition of (A.1) (even over an individual X^I) is not immediate, as it does not fit into the paradigm developed in [GLC2, Sect. A.5], since $\check{G}/\check{N}_P^-$ is not affine.

A.1. Horizontal loops into the (parabolic) base affine space.

A.1.1. Let Y be a quasi-affine scheme almost of finite type. Even though it is not affine, we can apply the definition from [GLC2, Sect. B.4.6] and consider the factorization space $\mathfrak{L}_{\nabla}(Y)$.

We claim:

Proposition A.1.2. *The prestack $\mathfrak{L}_{\nabla}(Y)_{\mathrm{Ran}} \rightarrow \mathrm{Ran}$ is a relative ind-scheme over Ran .*

Proof. The map

$$(A.3) \quad \mathfrak{L}_{\nabla}^+(Y)_{\mathrm{Ran}} \rightarrow \mathfrak{L}_{\nabla}(Y)_{\mathrm{Ran}}$$

induces an isomorphism of the underlying classical prestacks,⁶⁹ while

$$\mathfrak{L}_{\nabla}^+(Y)_{\mathrm{Ran}} \rightarrow \mathrm{Ran}$$

is a relative scheme.

Hence, in order to show that $\mathfrak{L}_{\nabla}(Y)$ is an ind-scheme, it suffices to show that it admits a connective deformation theory (see [GaRo4, Chapter 2, Theorem 1.3.12]). However, the latter is a straightforward verification. $\square$

A.1.3. Note that since (A.3) is an isomorphism at the classical level, $\mathfrak{L}_{\nabla}(Y)_{\mathrm{Ran}} \rightarrow \mathrm{Ran}$ is a relative *formal* scheme⁷⁰ over Ran .

Additionally, we obtain that if $U \subset Y$ is an open subscheme (resp., a Zariski cover of Y), then

$$\mathfrak{L}_{\nabla}(U)_{\mathrm{Ran}} \rightarrow \mathfrak{L}_{\nabla}(Y)_{\mathrm{Ran}}$$

is an open embedding (resp., Zariski cover).

⁶⁹Indeed, this is true for $Y = \mathbb{A}^1$, hence for any affine Y , and hence for any quasi-affine Y as well.

⁷⁰In the sense of [GaRo4, Chapter 2, Sect. 0.2.3].

A.1.4. We apply the above discussion to Y being the (parabolic) base affine space $\check{G}/\check{N}_P^-$. Thus, we obtain a (factorization) ind-scheme $\mathfrak{L}_\nabla(\check{G}/\check{N}_P^-)$.

Remark A.1.5. Here is another way to see that $\mathfrak{L}_\nabla(\check{G}/\check{N}_P^-)$ is a relative ind-scheme. Let $\overline{\check{G}/\check{N}_P^-}$ denote the affine closure of $\check{G}/\check{N}_P^-$.

Then, by Sect. A.1.3, $\mathfrak{L}_\nabla(\check{G}/\check{N}_P^-)$ is the open subscheme of $\mathfrak{L}_\nabla(\overline{\check{G}/\check{N}_P^-})$, whose underlying classical scheme equals that of

$$\mathfrak{L}_\nabla^+(\check{G}/\check{N}_P^-) \subset \mathfrak{L}_\nabla^+(\overline{\check{G}/\check{N}_P^-}).$$

A.1.6. The relevance of $\mathfrak{L}_\nabla(\check{G}/\check{N}_P^-)$ for us is explained by the following assertion:

Lemma A.1.7. *There is a canonical isomorphism*

$$\mathfrak{L}_\nabla(\check{G}/\check{N}_P^-) \simeq \mathrm{pt} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{pt}.$$

Proof. Parallel to that of [GLC2, Lemma E.2.2]. □

Corollary A.1.8. *We have a canonical isomorphism*

$$\mathfrak{L}_\nabla^+(\check{G}) \backslash \mathfrak{L}_\nabla(\check{G}/\check{N}_P^-) / \mathfrak{L}_\nabla^+(\check{M}) \simeq \mathrm{LS}_G^{\mathrm{reg}} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{reg}}.$$

A.2. Ind-coherent sheaves on horizontal loop and arc spaces.

A.2.1. Recall the construction from [GLC2, Sect. A.5] that attaches to an ind-affine ind-scheme $\mathcal{Y}$ the category $\mathrm{IndCoh}^*(\mathcal{Y})$.

A.2.2. Note that if $\mathcal{Y}^0 \xrightarrow{j} \mathcal{Y}$ is an open embedding, then the functor

$$j_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathcal{Y}^0) \rightarrow \mathrm{IndCoh}^*(\mathcal{Y})$$

admits a left adjoint, denoted $j^{\mathrm{IndCoh},*}$ (see [GLC2, Sect. A.7.6]). Moreover, it is easy to see that j_*^{IndCoh} is fully faithful, making $j^{\mathrm{IndCoh},*}$ a localization.⁷¹

A.2.3. We now record the following observation to the effect that the construction

$$\mathcal{Y} \rightsquigarrow \mathrm{IndCoh}^*(\mathcal{Y})$$

satisfies Zariski descent.

Let $\mathcal{U} \rightarrow \mathcal{Y}$ is a Zariski cover (i.e., $\mathcal{U}$ is a finite disjoint union of open subfunctors of $\mathcal{Y}$, each of which is itself an ind-affine ind-scheme, that cover it set-theoretically).

Lemma A.2.4.

(a) *The functor*

$$\mathrm{IndCoh}^*(\mathcal{Y}) \rightarrow \mathrm{Tot}(\mathrm{IndCoh}^*(\mathcal{U}^\bullet)),$$

*given by $(\mathrm{IndCoh}, *)$ -restriction, is an equivalence.*

(b) *The functor*

$$|\mathrm{IndCoh}^*(\mathcal{U}^\bullet)| \rightarrow \mathrm{IndCoh}^*(\mathcal{Y}),$$

*given by $(\mathrm{IndCoh}, *)$ -pushforward, is also an equivalence.*

A.2.5. Note that Lemma A.2.4 allows us to extend the assignment

$$\mathcal{Y} \rightsquigarrow \mathrm{IndCoh}^*(\mathcal{Y})$$

to (not necessarily ind-affine) ind-schemes that admit finite open covers by ind-affine ones.

⁷¹The fact that the counit of the adjunction $\mathrm{Id} \rightarrow j^{\mathrm{IndCoh},*} \circ j_{\mathrm{IndCoh},*}$ is an isomorphism follows by base change.

A.2.6. Let I be a finite set and consider the scheme X^I . Let Y be a quasi-affine scheme of finite type. By Lemma A.2.4, we have a well-defined category

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(Y)_{X^I}).$$

However, we do not know whether for a general Y , the assignment

$$I \rightsquigarrow \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(Y))_{X^I}$$

can be extended to a datum of factorization category. Namely, we do not know how to establish the analogs of [GLC2, Lemmas B.13.18 and B.13.20].

A.2.7. We claim, however:

Lemma A.2.8. *Assume that Y is smooth. The functor*

$$\Psi_{\mathfrak{L}_{\nabla}^+(Y)} : \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(Y)) \rightarrow \mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(Y))$$

is an equivalence.

Proof. Follows from [GLC2, Proposition E.12.1]. □

The above lemma allows us to define $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(Y))$ as a (unital) factorization category for a smooth Y . Namely, [GLC2, Lemmas B.13.18 and B.13.20] hold automatically for $\mathrm{QCoh}(-)$.

Note that by Lemma A.2.8

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(Y)) \simeq \mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(Y)),$$

as factorization categories.

A.2.9. Let Y be a smooth quasi-affine scheme of finite type. We now wish to define the (factorization) category $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(Y))$.

Combining Lemma A.2.4 and Proposition A.1.2, we obtain that for a fixed finite set I , we have a well-defined category

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(Y)_{X^I}).$$

However, we do not know whether for a general Y , the assignment

$$I \rightsquigarrow \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(Y))_{X^I}$$

can be extended to a datum of factorization category (for the same reason as in Sect. A.2.6 above).

That said, we *can* establish the required assertions in the following cases:

- (i) When Y is an algebraic group H .
- (ii) When Y is an open subset of $\mathbb{A}^n$.

Indeed, case (i) was considered in [GLC2, Sect. E.2]. For case (ii), if $Y = \mathbb{A}^n$, this is a particular case of (i), and in general the required properties are inherited under open embeddings, see Sect. A.2.2.

A.2.10. In particular, we obtain that if Y is a quasi-affine scheme almost of finite type *that admits a cover by open subsets isomorphic to open subsets of $\mathbb{A}^n$* , then we have a well-defined unital factorization category $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(Y))$.

A.3. Ind-coherent sheaves on $\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)$. In this subsection we define

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))$$

as a factorization category. We will give two different constructions, which will ultimately be equivalent.

A.3.1. For the first definition of $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))$, we start with the unital factorization category $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))$.

It is equipped with an action of the monoidal unital factorization category

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G} \times \check{G})) \simeq \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G})) \otimes \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G})),$$

and the action morphism has a naturally defined lax unital structure.

In particular, we can restrict this action to $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{N}_P^-))$, which we consider as mapping to the second factor.

A.3.2. We set

$$(A.4) \quad {}'\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)) := \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G})) \underset{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{N}_P^-))}{\otimes} \mathrm{Vect}.$$

Note that by construction, ${}'\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))$ carries an action of

$$(A.5) \quad \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G})) \otimes \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M})).$$

A.3.3. For the second definition, we apply the paradigm of Sect. A.2.10 directly to $Y = \check{G}/\check{N}_P^-$, where the open cover of $\check{G}/\check{N}_P^-$ is provided by translates of the big Schubert cell.

Thus, we obtain a well-defined unital factorization category

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)).$$

Again, by construction, this category carries a monoidal action of (A.5).

A.3.4. It follows that $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))$ as constructed in Sect. A.3.3 comes equipped with a (strictly unital) factorization functor

$$(A.6) \quad \iota_{*}^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\check{G}/\check{N}_P^-)) \rightarrow \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)),$$

given by $(\mathrm{IndCoh}, *)$ -pushforward.

In particular, the factorization unit in $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))$ is the image of

$$\mathcal{O}_{\mathfrak{L}_{\nabla}^+(\check{G}/\check{N}_P^-)} \in \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\check{G}/\check{N}_P^-))$$

under the above functor.

A.3.5. Note that the operation of $(\mathrm{IndCoh}, *)$ -pushforward gives rise to a functor

$$(A.7) \quad {}'\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)) \rightarrow \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)).$$

Proposition A.3.6. *The functor (A.7) is an equivalence.*

Proof. Let U be the cover of $\check{G}/\check{N}_P^-$ by translates of the big Schubert cell. Set

$$\tilde{U} := U \underset{\check{G}/\check{N}_P^-}{\times} \check{G}.$$

By construction,

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)) \simeq |\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(U^{\bullet}))|.$$

By Lemma A.2.4(b), we have:

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G})) \simeq |\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\tilde{U}^{\bullet}))|.$$

Hence,

$${}'\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)) \simeq |\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\tilde{U}^{\bullet})) \underset{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{N}_P^-))}{\otimes} \mathrm{Vect}|.$$

Therefore, it suffices to show that for every connected component U' of U^n (for every n), the functor

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\tilde{U}')) \underset{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{N}_P^-))}{\otimes} \mathrm{Vect} \rightarrow \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(U'))$$

is an equivalence.

Note that for every U' as above, the $\tilde{N}_P^-$ -torsor $\tilde{U}' \rightarrow U'$ splits as a product

$$U' \times \tilde{N}_P^-.$$

Hence, it suffices to show that

$$\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(U') \times \mathfrak{L}_\nabla(\tilde{N}_P^-)) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(\tilde{N}_P^-))} \mathrm{Vect} \rightarrow \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(U'))$$

is an equivalence.

However, from the fact that $\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(\tilde{N}_P^-))$ is dualizable (see Sect. A.4.6 below), we obtain that the functor

$$\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(U')) \otimes \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(\tilde{N}_P^-)) \rightarrow \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(U') \times \mathfrak{L}_\nabla(\tilde{N}_P^-))$$

is an equivalence,⁷² and the assertion follows. $\square$

A.4. Compact generation and duality.

A.4.1. Recall the notion of compact generation for a factorization category, see [GLC2, Sect. B.11.10].

Let Y be a smooth scheme. Consider the factorization category $\mathrm{IndCoh}^*(\mathfrak{L}_\nabla^+(Y))$. It follows from Lemma A.2.8 that it is compactly generated.

A.4.2. We now claim:

Proposition A.4.3. *The functor (A.6) admits a (conservative) right adjoint as a factorization functor, to be denoted $\iota^!$.*

Proof. It suffices to show that if U is an open affine in $\mathbb{A}^n$, then the functor

$$(A.8) \quad \iota_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathfrak{L}_\nabla^+(U)) \rightarrow \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(U))$$

admits a conservative right adjoint as a factorization functor and that the formation of these right adjoints is compatible with $(\mathrm{IndCoh}, *)$ -pushforwards along further open embeddings $U' \hookrightarrow U$.

We first consider the case when $U = \mathbb{A}^n$. In the latter case, taking $\mathfrak{L}_\nabla^+(\mathbb{A}^n)$ -invariants on both sides (cf. [GLC2, Lemma E.2.10]), we reduce the assertion to that of the functor

$$(A.9) \quad \mathrm{Vect} \simeq \mathrm{IndCoh}(\mathrm{pt}) \rightarrow \mathfrak{L}_\nabla(\mathbb{A}^n)/\mathfrak{L}_\nabla^+(\mathbb{A}^n).$$

However, the functor (A.9) is the $(\mathrm{IndCoh}, *)$ -pushforward in the laft category, and the assertion follows.

We now claim that if the functor (A.8) admits a right adjoint (as a factorization functor), then the same will be true for any open $U' \subset U$, and that the formation of these right adjoints is compatible with $(\mathrm{IndCoh}, *)$ -pushforwards along open embeddings $j : U' \hookrightarrow U$. Indeed, this follows from Sect. A.2.2 by passing to right adjoints in the commutative diagram

$$\begin{array}{ccc} \mathrm{IndCoh}^*(\mathfrak{L}_\nabla^+(U)) & \xrightarrow{j^{\mathrm{IndCoh},*}} & \mathrm{IndCoh}^*(\mathfrak{L}_\nabla^+(U')) \\ \iota_*^{\mathrm{IndCoh}} \downarrow & & \downarrow \iota_*^{\mathrm{IndCoh}} \\ \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(U)) & \xrightarrow{j^{\mathrm{IndCoh},*}} & \mathrm{IndCoh}^*(\mathfrak{L}_\nabla(U')). \end{array}$$

$\square$

A.4.4. Combining Proposition A.4.3 with Lemma A.2.8, we obtain:

Corollary A.4.5. *The factorization category $\mathrm{IndCoh}^*(\mathfrak{L}_\nabla(\tilde{G}/\tilde{N}_P^-))$ is compactly generated.*

⁷²The dualizability is used in commuting limits involved in the definition of $\mathrm{IndCoh}^*(-)$ with colimits and tensor products, cf. [GaRo3, Chapter 3, Proposition 3.1.7.]

A.4.6. Being compactly generated, the category $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))$ is dualizable as a factorization category (see [GLC2, Sect. B.11.9] for what this means).

Moreover, it is easy to see that it is dualizable as a *unital* factorization category (see [GLC2, C.11.5] for what this means).

We will now describe explicitly the dual of $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))$.

A.4.7. Recall (see [GLC2, Sect. A.4]) that to any prestack $\mathcal{Y}$ we can attach a category $\mathrm{IndCoh}^!(\mathcal{Y})$.

We take

$$\mathcal{Y} := \mathfrak{L}_{\nabla}(Y),$$

viewed as a factorization space, where Y is a scheme of finite type. We would like to extend the assignment

$$I \in \mathrm{fSet} \rightsquigarrow \mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(Y))_I$$

to a datum of factorization category.

In order to do so, we need to prove assertions parallel to [GLC2, Lemmas B.13.11 and B.13.14]. We do not know how to do this in general.

However, these assertions do hold in the cases considered in Sect. A.2.9, see [GLC2, Sect. 4].

A.4.8. An assertion parallel to Lemma A.2.4 holds also for $\mathrm{IndCoh}^!(-)$. Hence, by the same mechanism as in Sect. A.3.3, we obtain a well-defined factorization category

$$\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)).$$

We now claim:

Proposition A.4.9. *The factorization categories*

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)) \text{ and } \mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))$$

are naturally dual.

Proof. By construction, we need to establish a duality between

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))_{X^I} \text{ and } \mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))_{X^I}$$

for finite sets I , compatibly with maps between finite sets.

Note that, up to Zariski sheafification, $\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)_{X^I}$ can be identified with the quotient of $\mathfrak{L}_{\nabla}(\check{G})_{X^I}$ by $\mathfrak{L}_{\nabla}(\check{N}_P^-)_{X^I}$. Hence, by the $\mathrm{IndCoh}^!(-)$ -version of Lemma A.2.4, we can identify

$$\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)_{X^I}) \simeq \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{N}_P^-))_{X^I}}(\mathrm{QCoh}(X^I), \mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}))_{X^I}),$$

where $\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{N}_P^-))_{X^I}$ is viewed as a comonoidal category via the group-structure on $\mathfrak{L}_{\nabla}(\check{N}_P^-)_{X^I}$.

Now the assertion of the proposition follows from the identifications

$$\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{N}_P^-))_{X^I} \simeq (\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{N}_P^-))_{X^I})^{\vee}$$

as comonoidal categories and

$$\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}))_{X^I} \simeq (\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))_{X^I})^{\vee}$$

as modules over them, see [GLC2, Sect. E.4.2].

□

A.5. Construction of the spherical spectral semi-infinite category.

A.5.1. We regard $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))$ as a module category for the monoidal factorization category

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\check{G})) \otimes \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\check{M})).$$

Moreover, the action functor has a natural lax unital structure.

Recall now that for an algebraic group H , the functor

$$\Psi_{\mathfrak{L}_{\nabla}^+(H)} : \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(H)) \rightarrow \mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(H))$$

is an equivalence (see Lemma A.2.8). Hence, we can regard $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))$ as acted on by the group-scheme $\mathfrak{L}_{\nabla}^+(\check{G}) \times \mathfrak{L}_{\nabla}^+(\check{M})$.

A.5.2. Given Corollary A.1.8, we define the category

$$(A.10) \quad \mathrm{IndCoh}^*(\mathrm{LS}_G^{\mathrm{reg}} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{reg}})$$

by formula (A.2).

We now proceed to establishing its properties.

A.5.3. Note that we can identify

$$\begin{aligned} & \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\check{G}/\check{N}_P^-))_{\mathfrak{L}_{\nabla}^+(\check{G}) \times \mathfrak{L}_{\nabla}^+(\check{M})} \simeq \\ & \simeq \mathrm{QCoh}(\mathfrak{L}_{\nabla}^+(\check{G}/\check{N}_P^-))_{\mathfrak{L}_{\nabla}^+(\check{G}) \times \mathfrak{L}_{\nabla}^+(\check{M})} \simeq \mathrm{QCoh}(\mathrm{pt}/\mathfrak{L}_{\nabla}^+(\check{P}^-)) = \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) =: \mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}). \end{aligned}$$

Hence, the (monadic) adjunction

$$\iota_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\check{G}/\check{N}_P^-)) \rightleftarrows \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)) : \iota^!$$

gives rise to a monadic adjunction

$$(A.11) \quad \iota_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) \rightleftarrows \mathrm{IndCoh}^*(\mathrm{LS}_G^{\mathrm{reg}} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{reg}}) : \iota^!.$$

A.5.4. A straightforward verification shows that the functor $\iota_*^{\mathrm{IndCoh}}$ in (A.11) is strictly unital.

In particular, the factorization unit in $\mathrm{IndCoh}^*(\mathrm{LS}_G^{\mathrm{reg}} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{reg}})$ is given by

$$\iota_*^{\mathrm{IndCoh}}(\mathcal{O}_{\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}}).$$

A.5.5. The fact that the factorization category $\mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}})$ is compactly generated, combined with the monadic adjunction (A.11), implies that the category (A.10) is compactly generated.

Namely, compact generators of (A.10) are provided as the images under $\iota_*^{\mathrm{IndCoh}}$ of compact objects in $\mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) \simeq \mathrm{Rep}(\check{P}^-)$.

A.5.6. Note that the operation of coinvariants with respect to $\mathfrak{L}_{\nabla}^+(\check{G}) \times \mathfrak{L}_{\nabla}^+(\check{M})$ preserves dualizability (see [GLC2, Proposition E.1.4]). Hence, we obtain that the category (A.10) is dualizable.

Furthermore, it follows formally from Proposition A.4.9 that its dual identifies with

$$\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))_{\mathfrak{L}_{\nabla}^+(\check{G}) \times \mathfrak{L}_{\nabla}^+(\check{M})} \simeq \mathrm{IndCoh}^!(\mathrm{LS}_G^{\mathrm{reg}} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{reg}}).$$

A.5.7. Note that the functor

$$\Gamma^{\text{IndCoh}}(\mathfrak{L}_{\nabla}(\check{G}), -) : \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G})) \rightarrow \text{Vect}$$

is $\mathfrak{L}_{\nabla}(\check{G}) \times \mathfrak{L}_{\nabla}(\check{G})$ -invariant, and hence gives rise to a functor

$$\Gamma^{\text{IndCoh}}(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-), -) : \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)) \rightarrow \text{Vect},$$

which is in turn $\mathfrak{L}_{\nabla}(\check{G}) \times \mathfrak{L}_{\nabla}(\check{M})$ -invariant.

In particular, the functor $\Gamma^{\text{IndCoh}}(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-), -)$ gives rise to a functor

$$\begin{aligned} \text{IndCoh}^*(\text{LS}_{\check{G}}^{\text{reg}} \times_{\text{LS}_{\check{G}}^{\text{mer}}} \text{LS}_{\check{P}^-}^{\text{mer}} \times_{\text{LS}_{\check{M}}^{\text{mer}}} \text{LS}_{\check{M}}^{\text{reg}}) &:= \\ &:= \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))_{\mathfrak{L}_{\nabla}(\check{G}) \times \mathfrak{L}_{\nabla}(\check{M})} \rightarrow \text{Vect}_{\mathfrak{L}_{\nabla}(\check{M})} = \text{Rep}(\check{M}). \end{aligned}$$

This is the sought-for functor $(p_2)_*^{\text{IndCoh}}$ in formula (2.17).

A.5.8. Recall the construction from [GLC2, Sect. E.6]. It can be reformulated as follows: given a (factorization) category $\mathbf{C}$ equipped with an action of the monoidal (factorization) category $\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathbf{H}))$, then the (factorization) category

$$\mathbf{C}_{\mathfrak{L}_{\nabla}(\mathbf{H})} \simeq \mathbf{C} \otimes_{\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathbf{H}))} \text{Vect}$$

acquires an action of

$$\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathbf{H}))_{\mathfrak{L}_{\nabla}(\mathbf{H}) \times \mathfrak{L}_{\nabla}(\mathbf{H})} =: \text{Sph}_{\mathbf{H}}^{\text{spec}}.$$

Moreover, if $\mathbf{C}$ was unital and the action of $\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathbf{H}))$ on $\mathbf{C}$ had a lax unital structure, then the resulting action of $\text{Sph}_{\mathbf{H}}^{\text{spec}}$ on $\mathbf{C}_{\mathfrak{L}_{\nabla}(\mathbf{H})}$ is strictly unital.

A.5.9. In particular, the action of

$$\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G})) \otimes \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))$$

on $\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))$ gives rise to an action of

$$\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))_{\mathfrak{L}_{\nabla}(\check{G}) \times \mathfrak{L}_{\nabla}(\check{G})} \otimes \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))_{\mathfrak{L}_{\nabla}(\check{M}) \times \mathfrak{L}_{\nabla}(\check{M})} = \text{Sph}_{\check{G}}^{\text{spec}} \otimes \text{Sph}_{\check{M}}^{\text{spec}}$$

on

$$\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))_{\mathfrak{L}_{\nabla}(\check{G}) \times \mathfrak{L}_{\nabla}(\check{M})} =: \text{IndCoh}^*(\text{LS}_{\check{G}}^{\text{reg}} \times_{\text{LS}_{\check{G}}^{\text{mer}}} \text{LS}_{\check{P}^-}^{\text{mer}} \times_{\text{LS}_{\check{M}}^{\text{mer}}} \text{LS}_{\check{M}}^{\text{reg}}).$$

This provides the action of the spherical categories promised in Sect. 2.1.4.

A.5.10. Note that the functor

$$(A.12) \quad \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))_{\mathfrak{L}_{\nabla}(\check{M})} \rightarrow \text{Vect}_{\mathfrak{L}_{\nabla}(\check{M})}$$

considered in Sect. A.5.7 is compatible with the actions of $\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))_{\mathfrak{L}_{\nabla}(\check{M}) \times \mathfrak{L}_{\nabla}(\check{M})}$ on the two sides, and it is invariant with respect to the action of $\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))$ on the left-hand side.

This implies that the functor

$$\begin{aligned} \text{Vect}_{\mathfrak{L}_{\nabla}(\check{G})} \otimes \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))_{\mathfrak{L}_{\nabla}(\check{G}) \times \mathfrak{L}_{\nabla}(\check{M})} &\xrightarrow{\otimes} \\ &\rightarrow \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))_{\mathfrak{L}_{\nabla}(\check{G}) \times \mathfrak{L}_{\nabla}(\check{M})} \rightarrow \text{Vect}_{\mathfrak{L}_{\nabla}(\check{M})} \end{aligned}$$

factors as

$$\begin{aligned} \text{Vect}_{\mathfrak{L}_{\nabla}(\check{G})} \otimes \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))_{\mathfrak{L}_{\nabla}(\check{G}) \times \mathfrak{L}_{\nabla}(\check{M})} &\rightarrow \\ &\rightarrow \text{Vect}_{\mathfrak{L}_{\nabla}(\check{G})} \otimes_{\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))_{\mathfrak{L}_{\nabla}(\check{G}) \times \mathfrak{L}_{\nabla}(\check{G})}} \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))_{\mathfrak{L}_{\nabla}(\check{G}) \times \mathfrak{L}_{\nabla}(\check{M})} \rightarrow \text{Vect}_{\mathfrak{L}_{\nabla}(\check{M})}, \end{aligned}$$

where the second arrow is still compatible with the actions of $\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))_{\mathfrak{L}_{\nabla}(\check{M}) \times \mathfrak{L}_{\nabla}(\check{M})}$.

In other words, we obtain the desired factorization of the functor

$$(A.13) \quad J^{-, \text{spec, pre-enh}} : \text{Rep}(\check{G}) \otimes \text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}} \rightarrow \text{Rep}(\check{M})$$

via a functor

$$J^{-, \text{spec, enh}} : \text{Rep}(\check{G}) \otimes_{\text{Sph}_{\check{G}}^{\text{spec}}} \text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}} \rightarrow \text{Rep}(\check{M}),$$

which is $\text{Sph}_{\check{M}}^{\text{spec}}$ -linear.

A.5.11. In the rest of this subsection we will construct the pair of adjoint functors from Sect. 2.1.7. Consider the fiber product $\text{LS}_{\check{P}^-}^{\text{reg}} \times_{\text{LS}_{\check{M}}^{\text{mer}}} \text{LS}_{\check{M}}^{\text{reg}}$.

As in [GLC2, Sect. E.2], we *define*

$$\text{IndCoh}^*(\text{LS}_{\check{P}^-}^{\text{reg}} \times_{\text{LS}_{\check{M}}^{\text{mer}}} \text{LS}_{\check{M}}^{\text{reg}}) := \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))_{\mathfrak{L}_{\nabla}^+(\check{P}^-) \times \mathfrak{L}_{\nabla}^+(\check{M})}.$$

Let us first construct the $\text{Sph}_{\check{M}}^{\text{spec}}$ -linear monadic adjunction

$$\text{IndCoh}^*(\text{Hecke}_{\check{M}}^{\text{spec, loc}}) := \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))_{\mathfrak{L}_{\nabla}^+(\check{M}) \times \mathfrak{L}_{\nabla}^+(\check{M})} \rightleftarrows \text{IndCoh}^*(\text{LS}_{\check{P}^-}^{\text{reg}} \times_{\text{LS}_{\check{M}}^{\text{mer}}} \text{LS}_{\check{M}}^{\text{reg}}).$$

A.5.12. We have the tautological functor

$$(A.14) \quad \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))_{\mathfrak{L}_{\nabla}^+(\check{P}^-)} \rightarrow \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))_{\mathfrak{L}_{\nabla}^+(\check{M})},$$

where the coinvariants are taken with respect to the action by left translations.

It is compatible with the actions of $\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))$ by right translations, and hence induces a functor

$$(A.15) \quad \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))_{\mathfrak{L}_{\nabla}^+(\check{P}^-) \times \mathfrak{L}_{\nabla}^+(\check{M})} \rightarrow \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))_{\mathfrak{L}_{\nabla}^+(\check{M}) \times \mathfrak{L}_{\nabla}^+(\check{M})},$$

compatible with the actions of $\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))_{\mathfrak{L}_{\nabla}^+(\check{M}) \times \mathfrak{L}_{\nabla}^+(\check{M})} =: \text{Sph}_{\check{M}}^{\text{spec}}$.

It is easy to see that the functor (A.15) is conservative.⁷³

A.5.13. We claim that the functor (A.15) admits a left adjoint. To prove this, it suffices to show that the functor (A.14) admits a left adjoint, and, equivalently, that the functor

$$(A.16) \quad \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))_{\mathfrak{L}_{\nabla}^+(\check{N}_{\check{P}}^-)} \rightarrow \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))$$

admits a left adjoint.

However, the latter is obvious from the identification

$$\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))_{\mathfrak{L}_{\nabla}^+(\check{N}_{\check{P}}^-)} \simeq \text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M})) \otimes \text{Rep}(\check{N}_{\check{P}}^-),$$

so that the functor (A.16) is induced by the functor of $\check{N}_{\check{P}}^-$ -invariants

$$\text{Rep}(\check{N}_{\check{P}}^-) \rightarrow \text{Vect},$$

whose left adjoint is the functor of the trivial representation.

⁷³E.g., conservativity can be checked pointwise.

A.5.14. We now construct the $\mathrm{Sph}_M^{\mathrm{spec}}$ -linear monadic adjunction

$$\mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{reg}}) \rightleftarrows \mathrm{IndCoh}^*(\mathrm{Hecke}_{\check{G}, \check{P}^-}^{\mathrm{spec}, \mathrm{loc}}).$$

The map $\check{M} \rightarrow \check{G}/\check{N}_{\check{P}}^-$ gives rise to the $(\mathrm{IndCoh}, *)$ -pushforward functor

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}(\check{M})) \rightarrow \mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}(\check{G}/\check{N}_{\check{P}}^-)),$$

which commutes with the actions of $\mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}(\check{M}))$ on the right, and further to a functor

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}(\check{M}))_{\mathfrak{L}_{\check{\nabla}}^+(\check{P}^-)} \rightarrow \mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}(\check{G}/\check{N}_{\check{P}}^-))_{\mathfrak{L}_{\check{\nabla}}^+(\check{G})},$$

which is still compatible with the actions of $\mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}(\check{M}))$ on the right.

Passing to $\mathfrak{L}_{\check{\nabla}}^+(\check{M})$ -coinvariants, we obtain a functor

$$(A.17) \quad \mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}(\check{M}))_{\mathfrak{L}_{\check{\nabla}}^+(\check{P}^-) \times \mathfrak{L}_{\check{\nabla}}^+(\check{M})} \rightarrow \mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}(\check{G}/\check{N}_{\check{P}}^-))_{\mathfrak{L}_{\check{\nabla}}^+(\check{G}) \times \mathfrak{L}_{\check{\nabla}}^+(\check{M})}$$

compatible with the actions of $\mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}(\check{M}))_{\mathfrak{L}_{\check{\nabla}}^+(\check{M}) \times \mathfrak{L}_{\check{\nabla}}^+(\check{M})}$.

A.5.15. We claim that the functor (A.17) admits a (conservative) right adjoint as a factorization functor. To prove this, it suffices to show that the functor (A.17) preserves compactness.

For that we note that the category

$$\mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}(\check{M}))_{\mathfrak{L}_{\check{\nabla}}^+(\check{P}^-) \times \mathfrak{L}_{\check{\nabla}}^+(\check{M})}$$

is compactly generated by the essential image of the functor

$$(A.18) \quad \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) \simeq \mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}(\check{M}))_{\mathfrak{L}_{\check{\nabla}}^+(\check{P}^-) \times \mathfrak{L}_{\check{\nabla}}^+(\check{M})} \xrightarrow{\iota_{\mathrm{IndCoh}}^*} \mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}(\check{M}))_{\mathfrak{L}_{\check{\nabla}}^+(\check{P}^-) \times \mathfrak{L}_{\check{\nabla}}^+(\check{M})}.$$

Now, the composition of (A.18) and (A.17) is the left adjoint in (A.11), which preserves compactness.

A.6. Construction of the category $\mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}, \check{M}})^{\wedge}_{\mathrm{mon-free}})$. In this subsection we define another semi-infinite spectral category, namely, $\mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}, \check{M}})^{\wedge}_{\mathrm{mon-free}})$, as was promised in Sect. 2.7.

A.6.1. Let $(\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}, \check{M}})^{\wedge}_{\mathrm{mon-free}}$ be as in Sect. 2.7.3.

Our current goal is to define

$$\mathrm{IndCoh}^*\left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}, \check{M}})^{\wedge}_{\mathrm{mon-free}}\right) \text{ and } \mathrm{IndCoh}^!\left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}, \check{M}})^{\wedge}_{\mathrm{mon-free}}\right)$$

as unital factorization categories.

The idea of the definition is as follows: we have

$$\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}} \simeq \mathrm{pt} / \mathfrak{L}_{\check{\nabla}}^+(\check{P}^-) = \mathrm{pt} / \mathfrak{L}_{\check{\nabla}}^+(\check{N}_{\check{P}}^-) \cdot \mathfrak{L}_{\check{\nabla}}^+(\check{M}).$$

Hence, we can identify $(\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}, \check{M}})^{\wedge}_{\mathrm{mon-free}}$ with the étale quotient

$$\mathrm{pt} / \mathfrak{L}_{\check{\nabla}}(\check{N}_{\check{P}}^-) \cdot \mathfrak{L}_{\check{\nabla}}^+(\check{M}),$$

and we will use this presentation to define $\mathrm{IndCoh}^*(-)$ and $\mathrm{IndCoh}^!(-)$ of $(\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}, \check{M}})^{\wedge}_{\mathrm{mon-free}}$.

A.6.2. We set

$$\mathrm{IndCoh}^*\left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}, \check{M}})^{\wedge}_{\mathrm{mon-free}}\right) := \mathrm{Vect} \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\check{\nabla}}(\check{N}_{\check{P}}^-) \cdot \mathfrak{L}_{\check{\nabla}}^+(\check{M}))} \mathrm{Vect}$$

and

$$\mathrm{IndCoh}^!\left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}, \check{M}})^{\wedge}_{\mathrm{mon-free}}\right) := \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathfrak{L}_{\check{\nabla}}(\check{N}_{\check{P}}^-) \cdot \mathfrak{L}_{\check{\nabla}}^+(\check{M}))}(\mathrm{Vect}, \mathrm{Vect}).$$

A.6.3. For future reference, we note that we have

$$\begin{aligned} \mathrm{Vect} \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{N}_P^-) \cdot \mathfrak{L}_{\nabla}^+(\check{M}))} \mathrm{Vect} &\simeq \\ &\simeq \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G})) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}) \times \mathfrak{L}_{\nabla}(\check{N}_P^-) \cdot \mathfrak{L}_{\nabla}^+(\check{M}))} \mathrm{Vect} \simeq \\ &\simeq \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}) \times \mathfrak{L}_{\nabla}^+(\check{M}))} \mathrm{Vect}, \end{aligned}$$

and so

$$\mathrm{IndCoh}^* \left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}} \right) \simeq \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-)) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}) \times \mathfrak{L}_{\nabla}^+(\check{M}))} \mathrm{Vect}.$$

Similarly,

$$\mathrm{IndCoh}^! \left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}} \right) \simeq \mathrm{Funct}_{\mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}) \times \mathfrak{L}_{\nabla}^+(\check{M}))}(\mathrm{Vect}, \mathrm{IndCoh}^!(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))).$$

In particular, we have

$$(A.19) \quad \mathrm{IndCoh}^* \left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}} \right) \simeq \mathrm{Rep}(\check{G}) \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$$

and

$$(A.20) \quad \mathrm{IndCoh}^! \left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}} \right) \simeq \mathrm{Rep}(\check{G}) \otimes_{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}}^{\mathrm{Sph}_{\check{G}}^{\mathrm{spec}} \vee} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}},$$

where:

- $(\mathrm{Sph}_{\check{G}}^{\mathrm{spec}})^{\vee}$ is the dual of $\mathrm{Sph}_{\check{G}}^{\mathrm{spec}}$, viewed as a comonoidal category;
- The notation $\mathbf{C}_1 \otimes^{\mathbf{B}} \mathbf{C}_2$ is as in Sect. 1.4.10.

A.6.4. Let ι denote the map

$$\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}} \rightarrow (\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}}.$$

We define the (strictly unital) factorization functor

$$(\iota)_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) \rightarrow \mathrm{IndCoh}^* \left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}} \right)$$

to be the composition

$$(A.21) \quad \begin{aligned} \mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) &\simeq \mathrm{Vect} \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\check{P}^-))} \mathrm{Vect} = \mathrm{Vect} \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\check{N}_P^-) \cdot \mathfrak{L}_{\nabla}^+(\check{M}))} \mathrm{Vect} \rightarrow \\ &\rightarrow \mathrm{Vect} \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{N}_P^-) \cdot \mathfrak{L}_{\nabla}^+(\check{M}))} \mathrm{Vect} = \mathrm{IndCoh}^* \left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free} \check{M}})^{\wedge}_{\mathrm{mon-free}} \right). \end{aligned}$$

We claim that the functor (A.21), admits a right adjoint as a factorization functor, to be denoted $(\iota)^!$.

A.6.5. More generally, we claim that if $\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathbf{H}))$ acts on a sheaf of categories $\underline{\mathbf{C}}$ over Ran , then the functor

$$\underline{\mathbf{C}} \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\mathbf{H}))} \mathrm{Vect} \rightarrow \underline{\mathbf{C}} \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathbf{H}))} \mathrm{Vect}$$

admits a right adjoint as a factorization functor.

Indeed, we write the above functor as

$$(A.22) \quad \begin{aligned} \underline{\mathbf{C}} \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\mathbf{H}))} \mathrm{Vect} &\simeq \underline{\mathbf{C}} \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\mathbf{H}) \times \mathfrak{L}_{\nabla}(\mathbf{H}))} \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathbf{H})) \simeq \\ &\simeq \underline{\mathbf{C}} \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathbf{H}))} \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathbf{H})/\mathfrak{L}_{\nabla}^+(\mathbf{H})) \xrightarrow{\mathrm{Id} \otimes \Gamma^{\mathrm{IndCoh}}(\mathfrak{L}_{\nabla}(\mathbf{H})/\mathfrak{L}_{\nabla}^+(\mathbf{H}), -)} \underline{\mathbf{C}} \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathbf{H}))} \mathrm{Vect}, \end{aligned}$$

and the assertion follows from the fact that the functor

$$\Gamma^{\mathrm{IndCoh}}(\mathfrak{L}_{\nabla}(\mathbf{H})/\mathfrak{L}_{\nabla}^+(\mathbf{H}), -) : \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\mathbf{H})/\mathfrak{L}_{\nabla}^+(\mathbf{H})) \rightarrow \mathrm{Vect}$$

admits a $\mathfrak{L}_{\nabla}(\mathbf{H})$ -equivariant right adjoint.

A.6.6. Thus, we obtain that the (factorization) category

$$\mathrm{IndCoh}^*\left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}\check{M}})^{\wedge}_{\mathrm{mon-free}}\right)$$

is compactly generated, and in particular, is dualizable.

By the same mechanism as in Sect. A.5.6, we obtain an identification

$$\mathrm{IndCoh}^*\left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}\check{M}})^{\wedge}_{\mathrm{mon-free}}\right)^{\vee} \simeq \mathrm{IndCoh}^!\left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}\check{M}})^{\wedge}_{\mathrm{mon-free}}\right).$$

A.7. **t-structures on the spectral semi-infinite categories.**

A.7.1. We claim that the factorization category $\mathrm{IndCoh}^*\left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}\check{M}})^{\wedge}_{\mathrm{mon-free}}\right)$ carries a (unique) t-structure, so that the functor

$$\mathrm{Rep}(\check{P}^-) \simeq \mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) \xrightarrow{(\iota)_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}^*\left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}\check{M}})^{\wedge}_{\mathrm{mon-free}}\right)$$

is t-exact.

Since the adjunction

$$(\iota)_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) \rightleftarrows \mathrm{IndCoh}^*\left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}\check{M}})^{\wedge}_{\mathrm{mon-free}}\right) : (\iota)^!$$

monadic, this assertion is equivalent to the following:

Proposition A.7.2. *The endofunctor of $\mathrm{Rep}(\check{P}^-) \simeq \mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}})$ underlying the monad*

$$(\iota)^! \circ (\iota)_*^{\mathrm{IndCoh}}$$

is left t-exact.

Proof. We have a commutative diagram

$$(A.23) \quad \begin{array}{ccc} \mathrm{IndCoh}(\mathfrak{L}_{\nabla}(\check{P}^-)/\mathfrak{L}_{\nabla}^+(\check{P}^-)) & \xrightarrow{(\mathfrak{q}^-, \mathrm{spec})_*^{\mathrm{IndCoh}}} & \mathrm{IndCoh}(\mathfrak{L}_{\nabla}(\check{M})/\mathfrak{L}_{\nabla}^+(\check{M})) \\ \downarrow & & \downarrow \\ \mathrm{Vect}_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{P}^-))} \otimes \mathrm{IndCoh}(\mathfrak{L}_{\nabla}(\check{P}^-)/\mathfrak{L}_{\nabla}^+(\check{P}^-)) & \longrightarrow & \mathrm{Vect}_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))} \otimes \mathrm{IndCoh}(\mathfrak{L}_{\nabla}(\check{P}^-)/\mathfrak{L}_{\nabla}^+(\check{M})) \\ \sim \downarrow & & \downarrow \sim \\ \mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) & \xrightarrow{(\iota)_*^{\mathrm{IndCoh}}} & \mathrm{IndCoh}^*\left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}\check{M}})^{\wedge}_{\mathrm{mon-free}}\right), \end{array}$$

such that the natural transformation in diagram

$$\begin{array}{ccc} \mathrm{IndCoh}(\mathfrak{L}_{\nabla}(\check{P}^-)/\mathfrak{L}_{\nabla}^+(\check{P}^-)) & \xleftarrow{(\mathfrak{q}^-, \mathrm{spec})^!} & \mathrm{IndCoh}(\mathfrak{L}_{\nabla}(\check{M})/\mathfrak{L}_{\nabla}^+(\check{M})) \\ \downarrow & \searrow & \downarrow \\ \mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) & \xleftarrow{(\iota)^!} & \mathrm{IndCoh}^*\left((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}\check{M}})^{\wedge}_{\mathrm{mon-free}}\right), \end{array}$$

obtained from (A.23) by passing to right adjoints along the horizontal arrows, is an isomorphism.

We claim that the left vertical arrow in diagram (A.23) sends the coconnective subcategory in $\mathrm{IndCoh}(\mathfrak{L}_{\nabla}(\check{P}^-)/\mathfrak{L}_{\nabla}^+(\check{P}^-))$ surjectively onto the coconnective subcategory in

$$\mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) \simeq \mathrm{QCoh}(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) \simeq \mathrm{Rep}(\check{P}^-).$$

Indeed, this functor admits a right inverse given by direct image along

$$\mathrm{pt}/\mathfrak{L}_{\nabla}^+(\check{P}^-) \rightarrow \mathfrak{L}_{\nabla}(\check{P}^-)/\mathfrak{L}_{\nabla}^+(\check{P}^-).$$

Hence, in order to prove that $(\iota)^! \circ (\iota)_*^{\mathrm{IndCoh}}$ is left t-exact, it suffices to show that the endofunctor of $\mathrm{IndCoh}(\mathfrak{L}_{\nabla}(\check{P}^-)/\mathfrak{L}_{\nabla}^+(\check{P}^-))$ underlying the monad

$$(\mathfrak{q}^{-,\mathrm{spec}})^! \circ (\mathfrak{q}^{-,\mathrm{spec}})_*^{\mathrm{IndCoh}}$$

is left t-exact.

However, this follows from the fact that the functor

$$(\mathfrak{q}^{-,\mathrm{spec}})_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}(\mathfrak{L}_{\nabla}(\check{P}^-)/\mathfrak{L}_{\nabla}^+(\check{P}^-)) \rightarrow \mathrm{IndCoh}(\mathfrak{L}_{\nabla}(\check{M})/\mathfrak{L}_{\nabla}^+(\check{M}))$$

is left t-exact. □

A.7.3. Similarly, we claim that the category $\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$ carries a (unique) t-structure, so that the functor

$$\iota_*^{\mathrm{IndCoh}} : \mathrm{Rep}(\check{P}^-) \rightarrow \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$$

is t-exact, where ι denotes here the map

$$\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}} \rightarrow \mathrm{LS}_{\check{G}}^{\mathrm{reg}} \times_{\mathrm{LS}_{\check{G}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \times_{\mathrm{LS}_{\check{M}}^{\mathrm{mer}}} \mathrm{LS}_{\check{M}}^{\mathrm{reg}}.$$

Since the adjunction in (A.11) is monadic, the statement is equivalent to the fact that the endofunctor of $\mathrm{Rep}(\check{P}^-)$ underlying the monad $\iota^! \circ \iota_*^{\mathrm{IndCoh}}$ is left t-exact. This is proved in a way similar to Proposition A.7.2.

A.8. The global counterpart. The goal of this subsection is to construct the functors from Sects. 12.5 and 12.10 in families over Ran .

A.8.1. First, let us explain how to use the techniques developed earlier in this section to define the factorization category $\mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}_{\check{G}}})$.

Namely,

$$\mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}_{\check{G}}}) := \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G})) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\check{G})) \otimes \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{P}^-))} \mathrm{Vect}.$$

Note that by the same principle as in (A.19), we have:

$$\mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}_{\check{G}}}) \simeq \mathrm{Rep}(\check{M}) \otimes_{\mathrm{Sph}_M^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}.$$

A.8.2. Next, we proceed to the construction of the functor

$$(\mathfrak{q}^{-,\mathrm{spec}})_Z^{\mathrm{IndCoh}, *, \mathrm{enh}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_Z^{-, \mathrm{enh}} \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{P}^-, Z}^{\mathrm{mon-free}_{\check{G}}, \mathrm{glob}})$$

from Sect. 12.5.4.

Consider the fiber product

$$\mathrm{LS}_{\check{M}, Z}^{\mathrm{mer}, \mathrm{glob}} \times_{\mathrm{LS}_{\check{M}, Z}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, Z}^{\mathrm{mon-free}_{\check{G}}}.$$

Since

$$(\mathrm{LS}_{\check{M}} \times Z) \times_{\mathrm{LS}_{\check{M}, Z}^{\mathrm{reg}}} \mathrm{LS}_{\check{P}^-, Z}^{\mathrm{reg}} \rightarrow \mathrm{LS}_{\check{M}, Z}^{\mathrm{mer}, \mathrm{glob}} \times_{\mathrm{LS}_{\check{M}, Z}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, Z}^{\mathrm{mon-free}_{\check{G}}}$$

is an isomorphism at the classical level, as in Sect. A.6.1, we can interpret it as

$$\left((\mathrm{LS}_{\check{M}} \times \mathbb{Z})^{\mathrm{level}} \times_{\mathbb{Z}} (\mathfrak{L}_{\nabla}^+(\check{G}) \backslash \mathfrak{L}_{\nabla}(\check{G}/\check{N}_{\check{P}}^-)_{\mathbb{Z}}) \right) / \mathfrak{L}_{\nabla}(\check{M})_{\mathbb{Z}},$$

where $(\mathrm{LS}_{\check{M}} \times \mathbb{Z})^{\mathrm{level}}$ is as in [GLC2, Sect. E.7.3].

A.8.3. Recall that

$$\mathbb{Z} \rightsquigarrow \mathrm{IndCoh}^*((\mathrm{LS}_{\check{M}} \times \mathbb{Z})^{\mathrm{level}})$$

is well-defined as a crystal of categories over Ran , equipped with action of

$$\mathbb{Z} \rightsquigarrow \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{M}))_{\mathbb{Z}}.$$

Explicitly

$$\mathrm{IndCoh}^*((\mathrm{LS}_{\check{M}} \times \mathbb{Z})^{\mathrm{level}}) \simeq (\mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathbb{Z})) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\check{M}))_{\mathbb{Z}}} \mathrm{D-mod}(\mathbb{Z}).$$

A.8.4. Set

$$\begin{aligned} \text{(A.24)} \quad & \mathrm{IndCoh}^* \left(\mathrm{LS}_{\check{M}, \mathbb{Z}}^{\mathrm{mer}, \mathrm{glob}} \times_{\mathrm{LS}_{\check{M}, \mathbb{Z}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \mathbb{Z}}^{\mathrm{mon-free}_{\check{G}}} \right) := \\ & = \left(\mathrm{IndCoh}^*((\mathrm{LS}_{\check{M}} \times \mathbb{Z})^{\mathrm{level}}) \otimes_{\mathrm{D-mod}(\mathbb{Z})} \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_{\check{P}}^-))_{\mathbb{Z}} \right) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(G) \times \mathfrak{L}_{\nabla}(\check{P}^-))_{\mathbb{Z}}} \mathrm{D-mod}(\mathbb{Z}). \end{aligned}$$

We can rewrite the above expression also as

$$\text{(A.25)} \quad \left(\mathrm{IndCoh}^*((\mathrm{LS}_{\check{M}} \times \mathbb{Z})^{\mathrm{level}}) \otimes_{\mathrm{D-mod}(\mathbb{Z})} \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))_{\mathbb{Z}} \right) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(G) \times \mathfrak{L}_{\nabla}(\check{P}^-))_{\mathbb{Z}}} \mathrm{D-mod}(\mathbb{Z}).$$

A.8.5. Note, however, that by the same principle as in (A.19), we have

$$\begin{aligned} & \mathrm{IndCoh}^* \left(\mathrm{LS}_{\check{M}, \mathbb{Z}}^{\mathrm{mer}, \mathrm{glob}} \times_{\mathrm{LS}_{\check{M}, \mathbb{Z}}^{\mathrm{mer}}} \mathrm{LS}_{\check{P}^-, \mathbb{Z}}^{\mathrm{mon-free}_{\check{G}}} \right) \simeq \\ & \simeq (\mathrm{IndCoh}(\mathrm{LS}_{\check{M}}) \otimes \mathrm{D-mod}(\mathbb{Z})) \otimes_{\mathrm{Sph}_{\check{M}, \mathbb{Z}}^{\mathrm{spec}}} \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(G) \backslash \mathfrak{L}_{\nabla}(\check{G}/\check{N}_{\check{P}}^-) / \mathfrak{L}_{\nabla}^+(\check{M}))_{\mathbb{Z}} = \\ & =: \mathrm{IndCoh}(\mathrm{LS}_{\check{M}})_{\mathbb{Z}}^{-, \mathrm{enh}}. \end{aligned}$$

A.8.6. By the same principle as in Sect. A.6.1, we also have

$$\mathrm{LS}_{\check{P}^-, \mathbb{Z}}^{\mathrm{mon-free}_{\check{G}}, \mathrm{glob}} \simeq \left((\mathrm{LS}_{\check{P}^-} \times \mathbb{Z})^{\mathrm{level}} \times_{\mathbb{Z}} (\mathfrak{L}_{\nabla}^+(\check{G}) \backslash \mathfrak{L}_{\nabla}(\check{G}))_{\mathbb{Z}} \right) / \mathfrak{L}_{\nabla}(\check{P})_{\mathbb{Z}}.$$

Hence, as in Proposition A.3.6, we have

$$\begin{aligned} \text{(A.26)} \quad & \mathrm{IndCoh}(\mathrm{LS}_{\check{P}^-, \mathbb{Z}}^{\mathrm{mon-free}_{\check{G}}, \mathrm{glob}}) \simeq \\ & \simeq \left(\mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}^-} \times \mathbb{Z})^{\mathrm{level}}) \otimes_{\mathrm{D-mod}(\mathbb{Z})} \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))_{\mathbb{Z}} \right) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(G) \times \mathfrak{L}_{\nabla}(\check{P}^-))_{\mathbb{Z}}} \mathrm{D-mod}(\mathbb{Z}). \end{aligned}$$

A.8.7. Hence, we can interpret the sought-for functor $(\mathfrak{q}_z^{-, \mathrm{spec}})^{\mathrm{IndCoh}, *, \mathrm{enh}}$ as a functor

$$\begin{aligned} \text{(A.27)} \quad & \left(\mathrm{IndCoh}^*((\mathrm{LS}_{\check{M}} \times \mathbb{Z})^{\mathrm{level}}) \otimes_{\mathrm{D-mod}(\mathbb{Z})} \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))_{\mathbb{Z}} \right) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(G) \times \mathfrak{L}_{\nabla}(\check{P}^-))_{\mathbb{Z}}} \mathrm{D-mod}(\mathbb{Z}) \rightarrow \\ & \rightarrow \left(\mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}^-} \times \mathbb{Z})^{\mathrm{level}}) \otimes_{\mathrm{D-mod}(\mathbb{Z})} \mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}))_{\mathbb{Z}} \right) \otimes_{\mathrm{IndCoh}^*(\mathfrak{L}_{\nabla}^+(G) \times \mathfrak{L}_{\nabla}(\check{P}^-))_{\mathbb{Z}}} \mathrm{D-mod}(\mathbb{Z}). \end{aligned}$$

The latter functor is induced by the functor

$$(\mathfrak{q}_z^{-, \mathrm{spec}, \mathrm{level}})^{\mathrm{IndCoh}, *} : \mathrm{IndCoh}^*((\mathrm{LS}_{\check{M}} \times \mathbb{Z})^{\mathrm{level}}) \rightarrow \mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}^-} \times \mathbb{Z})^{\mathrm{level}})$$

left adjoint to the

$$(q_z^{-,\text{spec},\text{level}})_*^{\text{IndCoh}} : \text{IndCoh}^*((\text{LS}_{\bar{P}-} \times \mathbb{Z})^{\text{level}}) \rightarrow \text{IndCoh}^*((\text{LS}_{\bar{M}} \times \mathbb{Z})^{\text{level}}).$$

Explicitly, the functor $(q_z^{-,\text{spec},\text{level}})^{\text{IndCoh},*}$ equals

$$\begin{aligned} \text{IndCoh}^*((\text{LS}_{\bar{M}} \times \mathbb{Z})^{\text{level}}) &\simeq (\text{IndCoh}^*(\text{LS}_{\bar{M}}) \otimes \text{D-mod}(\mathbb{Z})) \otimes_{\text{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\bar{M}))} \text{D-mod}(\mathbb{Z}) \xrightarrow{(q_z^{-,\text{spec}})^* \otimes \text{Id}} \\ &\rightarrow (\text{IndCoh}^*(\text{LS}_{\bar{P}-}) \otimes \text{D-mod}(\mathbb{Z})) \otimes_{\text{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\bar{M}))} \text{D-mod}(\mathbb{Z}) \rightarrow \\ &\rightarrow (\text{IndCoh}^*(\text{LS}_{\bar{P}-}) \otimes \text{D-mod}(\mathbb{Z})) \otimes_{\text{IndCoh}^*(\mathfrak{L}_{\nabla}^+(\bar{P}-))} \text{D-mod}(\mathbb{Z}) \simeq \text{IndCoh}^*((\text{LS}_{\bar{P}-} \times \mathbb{Z})^{\text{level}}). \end{aligned}$$

A.8.8. We now proceed to the definition of the functor

$$(p_z^{-,\text{spec}})^{!,\text{enh}} : \text{IndCoh}(\text{LS}_{\bar{G}})_z^{-,\text{enhco}} \rightarrow \text{IndCoh}(\text{LS}_{\bar{P}-,z}^{\text{mon-free},\bar{M},\text{glob}}).$$

First, as above, we identify

$$\begin{aligned} \text{IndCoh}(\text{LS}_{\bar{G}})_z^{-,\text{enhco}} &\simeq \text{IndCoh}^! \left(\left(\text{LS}_{\bar{G},z}^{\text{mer},\text{glob}} \right)^{\wedge}_{\text{mon-free}} \times_{\text{LS}_{\bar{G},z}^{\text{mer}}} \text{LS}_{\bar{P}-,z}^{\text{mon-free},\bar{M}} \right) := \\ &= \text{Funct}_{\text{IndCoh}^! (\mathfrak{L}_{\nabla}(\bar{N}_{\bar{P}})_z \cdot \mathfrak{L}_{\nabla}^+(\bar{M})_z)} \left(\text{D-mod}(\mathbb{Z}), \text{IndCoh}^!((\text{LS}_{\bar{G}} \times \mathbb{Z})^{\text{level}}) \right). \end{aligned}$$

Furthermore, we identify

$$\text{IndCoh}(\text{LS}_{\bar{P}-,z}^{\text{mon-free},\bar{M},\text{glob}}) \simeq \text{Funct}_{\text{IndCoh}^! (\mathfrak{L}_{\nabla}(\bar{N}_{\bar{P}})_z \cdot \mathfrak{L}_{\nabla}^+(\bar{M})_z)} \left(\text{D-mod}(\mathbb{Z}), \text{IndCoh}^!((\text{LS}_{\bar{P}-} \times \mathbb{Z})^{\text{level}}) \right).$$

In terms of these identifications, the sought-for functor $(p_z^{-,\text{spec}})^{!,\text{enh}}$ is induced by the $!$ -pullback functor

$$\text{IndCoh}^!((\text{LS}_{\bar{P}-} \times \mathbb{Z})^{\text{level}}) \rightarrow \text{IndCoh}^!((\text{LS}_{\bar{G}} \times \mathbb{Z})^{\text{level}}).$$

APPENDIX B. PROOF OF SEMI-INFINITE CASSELMAN-SHALIKA

The goal of this section is to prove Theorem 2.7.6, which serves as a preparation for the proof of Theorem 2.4.3.

Note that Theorem 2.7.6 “crosses the Langlands bridge”, i.e., we need to find a way to connect a category on the geometric side to a category on the spectral side. This is done by interpreting⁷⁴ both sides as the category of factorization modules over the *same* factorization algebra.

B.1. Orientation session. The material in this section is closely related what has already appeared in the existing literature, To orient the reader, we highlight these connections and the points of departure.

B.1.1. First, the origin of these ideas is [AB], which proves a version of these results in the pointwise/Iwahori setting.

B.1.2. Second, [Ra4] considers a close variant of the main result here. There are two key differences.

One is that [Ra4] considers $\mathfrak{L}(N^-) \cdot \mathfrak{L}^+(T)$ -*coinvariants*, where here we consider *invariants* (and a general parabolic).

The other is that [Ra4] considers the factorization algebra Υ (cf. *loc. cit.*), where we consider the factorization algebra Ω . We encourage the reader to think that Ω is fundamentally connected to invariants while Υ is fundamentally connected to coinvariants; we hope to study this idea further in future works.

B.1.3. Finally, [Ya-R] proves a pointwise (and quantum!) version of our results, i.e., recovers this work recovers the main result of [AB] but using factorization algebras in the main constructions.

⁷⁴Here “interpreting” = “mapping *almost* fully faithfully.”

B.1.4. Compared to the previous works, the main novelty of the present text is a ULA-ness result for Ω , Theorem B.6.2 below, which builds on finiteness results from [CR]. A parallel result for Υ was given in [Ra3] and plays a key technical role in [Ra4]. Using this result, we are able to reduce Theorem 2.7.6 to the pointwise theorem of [Ya-R].

B.2. The semi-infinite Whittaker category. In this section we will interpret the geometric side of Theorem 2.7.6 as the category of factorization modules over a factorization algebra.

While doing so, we will provide a more conceptual point of view on the category in the geometric side of Theorem 2.7.6.

B.2.1. Consider the factorization category

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G)).$$

Imposing the condition of $\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)$ -equivariance with respect to the action by right translations, we obtain a unital lax^{75} factorization category

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G))^{-, \frac{\infty}{2}} := \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G))^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)}.$$

B.2.2. The category $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G))^{-, \frac{\infty}{2}}$ is equipped with a (strictly unital) factorization functor

$$\begin{aligned} \text{(B.1)} \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-) &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \otimes_{\mathrm{Sph}_G} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)} \rightarrow \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G))^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)} =: \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G))^{-, \frac{\infty}{2}}, \end{aligned}$$

to be denoted Accs .

By the same principle as in Lemma 3.7.4, one shows that the functor Accs is fully faithful. We will denote its essential image by

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G))^{-, \frac{\infty}{2} - \mathrm{Accs}} \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G))^{-, \frac{\infty}{2}};$$

and refer to it as the *spherically accessible* subcategory.

The lax factorization structure on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G))^{-, \frac{\infty}{2}}$ induces a genuine factorization structure on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G))^{-, \frac{\infty}{2} - \mathrm{Accs}}$.

B.2.3. We have a naturally defined functor

$$J^{-, \frac{\infty}{2}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G))^{-, \frac{\infty}{2}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_M),$$

given by

$$\begin{aligned} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G))^{-, \frac{\infty}{2}} &\xrightarrow{\sim} \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G)) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_M) \right)^{\mathfrak{L}(P^-)} \rightarrow \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G)) \otimes \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_M) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_M) \xrightarrow{\text{shift}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_M), \end{aligned}$$

where:

- The second arrow is the functor of forgetting the $\mathfrak{L}(P^-)$ -equivariance;
- The third arrow is given by $!$ -restriction along $1_{\mathfrak{L}(G)} \rightarrow \mathfrak{L}(G)$;
- The last arrow is the cohomological shift by $\langle \lambda, 2\check{\rho}_P \rangle$ on Gr_M^λ .

We observe:

Lemma B.2.4. *The precomposition of the functor $J^{-, \frac{\infty}{2}}$ with the functor (B.1) is the functor $J_{\mathrm{Gr}}^{-, \mathrm{enh}}$ of (1.40).*

⁷⁵It is possible that the lax factorization structure is strict, but we do not know how to prove this.

Proof. Unwinding the definitions, the assertion boils down to the fact that composition

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \otimes_{\mathrm{Sph}_G} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G) \xrightarrow{*} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G)) \xrightarrow{\text{l-fiber at } 1_{\mathfrak{L}(G)}} \mathrm{Vect}$$

identifies with the Verdier duality pairing. $\square$

B.2.5. By a similar token, we can consider the twisted versions

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G)_{\rho(\omega_X)}) \text{ and } \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G)_{\rho(\omega_X)})^{-, \frac{\infty}{2}},$$

respectively, and the corresponding functors

$$(B.2) \quad \mathrm{Accs} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G, \rho(\omega_X)}) \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G)_{\rho(\omega_X)})$$

and

$$(B.3) \quad J^{-, \frac{\infty}{2}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G)_{\rho(\omega_X)})^{-, \frac{\infty}{2}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{M, \rho_M(\omega_X)}),$$

see Sect. 1.8.5.

B.2.6. Set

$$\mathrm{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2}} := \mathrm{Whit}^!(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathfrak{L}(G)_{\rho(\omega_X)})^{-, \frac{\infty}{2}}),$$

see [GLC2, Sect. 1.3.3] for the notation $\mathrm{Whit}^!(-)$.

Remark B.2.7. One could equivalently define $\mathrm{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2}}$ as

$$\left(\mathrm{Whit}^!(\mathrm{D}\text{-mod}(\mathfrak{L}(G)_{\rho(\omega_X)})) \right)^{-, \frac{\infty}{2}} := \left(\mathrm{Whit}^!(\mathrm{D}\text{-mod}(\mathfrak{L}(G)_{\rho(\omega_X)})) \right)^{\mathfrak{L}(N_P^-)_{\rho(\omega_X)} \cdot \mathfrak{L}^+(M)_{\rho(\omega_X)}}.$$

B.2.8. The functor (B.2) induces a fully faithful (strictly unital) factorization functor

$$(B.4) \quad \mathrm{Accs} : \mathrm{Whit}^!(G) \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)} \rightarrow \mathrm{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2}}.$$

We will denote its essential image by

$$\mathrm{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2} - \mathrm{Accs}} \subset \mathrm{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2}},$$

and refer to it as the *spherically accessible* subcategory.

The category $\mathrm{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2}}$ carries a lax factorization structure, and it induces a genuine factorization structure on $\mathrm{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2} - \mathrm{Accs}}$.

The functor $J^{-, \frac{\infty}{2}}$ induces a functor

$$\mathrm{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2}} \rightarrow \mathrm{Whit}^!(M),$$

to be denoted $J_{\mathrm{Whit}}^{-, \frac{\infty}{2}}$.

By Lemma B.2.4, the composition of the functors Accs and $J_{\mathrm{Whit}}^{-, \frac{\infty}{2}}$ is the functor $J_{\mathrm{Whit}}^{-, \mathrm{enh}}$ of (1.46).

B.2.9. Let Ω be the (unital) factorization algebra in $\mathrm{Whit}^!(M)$ equal to the image of the factorization unit under the functor $J_{\mathrm{Whit}}^{-, \frac{\infty}{2}}$.

Note that this is the same as the image of the factorization unit in $\mathrm{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2} - \mathrm{Accs}}$ along the functor $J_{\mathrm{Whit}}^{-, \mathrm{enh}}$.

Still equivalently, this is the same as the image of the unit in $\mathrm{Whit}^!(G)$ along the functor⁷⁶ $J_{\mathrm{Whit}}^{-, !}$.

B.2.10. Thus, the functor $J_{\mathrm{Whit}}^{-, \frac{\infty}{2}}$ gives rise to a (strictly unital) factorization functor

$$(B.5) \quad \mathrm{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2}} \rightarrow \Omega\text{-mod}^{\mathrm{fact}}(\mathrm{Whit}^!(M)),$$

to be denoted $J_{\mathrm{Whit}, \Omega}^{-, \frac{\infty}{2}}$.

⁷⁶Because the averaging functor $\mathrm{Whit}^!(G) \rightarrow \mathrm{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2} - \mathrm{Accs}}$ is strictly unital.

B.2.11. We will prove:

Theorem B.2.12.

(a) The functor $J_{\text{Whit}, \Omega}^{-, \frac{\infty}{2}}$ is a pointwise equivalence, i.e., it induces an equivalence

$$\text{Whit}^!(\mathfrak{L}(G))_{\underline{x}}^{-, \frac{\infty}{2}} \rightarrow \Omega\text{-mod}^{\text{fact}}(\text{Whit}^!(M))_{\underline{x}}$$

for every $\underline{x} \in \text{Ran}$.

(b) The restriction of $J_{\text{Whit}, \Omega}^{-, \frac{\infty}{2}}$ to $\text{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2}-\text{Accs}}$ is fully faithful, i.e., for every affine scheme $S \rightarrow \text{Ran}$, the corresponding functor

$$\text{Whit}^!(\mathfrak{L}(G))_S^{-, \frac{\infty}{2}-\text{Accs}} \rightarrow \Omega\text{-mod}^{\text{fact}}(\text{Whit}^!(M))_S$$

is fully faithful.

Point (a) of Theorem B.2.12 will be proved in Sect. B.9; as we shall explain in *loc. cit.*, Theorem B.2.12(a) is essentially a paraphrase of the (parabolic version of the) *classical*⁷⁷ case of the main result of [Ya-R], namely Theorem 6.4.8 in *loc. cit.*.

Point (b) Theorem B.2.12 will be proved in Sect. B.7.7.

Remark B.2.13. In Theorem B.7.6 we will provide a more precise version of Theorem B.2.12(b). Namely, we will describe explicitly the essential image of the restriction of $J_{\text{Whit}, \Omega}^{-, \frac{\infty}{2}}$ to $\text{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2}-\text{Accs}}$.

B.3. Digression: monoidal actions of factorization categories.

B.3.1. Let $\mathbf{A}$ be a factorization category, and let $\mathbf{C}$ be a monoidal factorization category that acts on $\mathbf{A}$.

We will assume that both $\mathbf{C}$ and $\mathbf{A}$ are unital as factorization categories. Moreover, we will assume that the monoidal structure on $\mathbf{C}$ and the action of $\mathbf{C}$ on $\mathbf{A}$ are strictly unital.

In particular, the monoidal unit in $\mathbf{C}$, thought of as a factorization functor $\text{Vect} \rightarrow \mathbf{C}$, is unital, i.e., the monoidal unit in $\mathbf{C}$ equals the factorization unit.

B.3.2. Let $\mathcal{A}$ be a factorization algebra in $\mathbf{A}$. Note that in this case the action of $\mathbf{C}$ canonically extends to an action of $\mathbf{C}$ on the (lax factorization) category

$$\mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}),$$

in a way compatible with the forgetful functor.

Indeed, the corresponding binary operation is given by

$$\mathbf{C} \otimes \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}) \simeq \mathbf{1}_{\mathbf{C}}\text{-mod}^{\text{fact}}(\mathbf{C}) \otimes \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}) \rightarrow \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}),$$

where the second arrow is induced by action $\mathbf{C} \otimes \mathbf{A} \rightarrow \mathbf{A}$, and it sends

$$\mathbf{1}_{\mathbf{C}} \otimes \mathcal{A} \mapsto \mathcal{A},$$

by assumption.

B.3.3. Let now $\mathbf{A}_1$ and $\mathbf{A}_2$ be a pair of factorization categories, both acted on by $\mathbf{C}$. Let

$$\Phi : \mathbf{A}_1 \rightarrow \mathbf{A}_2$$

be a factorization functor, compatible with the action of $\mathbf{C}$.

Let $\mathcal{A}$ be a factorization algebra in $\mathbf{A}_1$, so that Φ extends to a functor

$$\Phi_{\mathcal{A}} : \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{A}_1) \rightarrow \Phi(\mathcal{A})\text{-mod}^{\text{fact}}(\mathbf{A}_2).$$

Then the functor $\Phi_{\mathcal{A}}$ is compatible with the actions of $\mathbf{C}$ on the two sides.

B.3.4. We apply the paradigm of Sect. B.3.2 to $\mathbf{A} := \text{Whit}^!(M)$ and $\mathbf{C} := \text{Sph}_M$. From Sect. B.3.2 we obtain that the (lax factorization) category $\Omega\text{-mod}^{\text{fact}}(\text{Whit}^!(M))$ carries a canonical Sph_M -action.

⁷⁷Trivial=no twisting

B.3.5. We apply the paradigm of Sect. B.3.3 to $\Phi := J_{\text{Whit}}^{-, \frac{\infty}{2}}$ and $\mathcal{A}$ being the factorization unit. We obtain that the functor $J_{\text{Whit}, \Omega}^{-, \frac{\infty}{2}}$ is compatible with the actions of Sph_M on the two sides.

B.4. The spectral counterpart. In this subsection we will interpret the spectral side in Theorem 2.7.6 as the category of modules over a factorization algebra.

B.4.1. Recall the (unital) factorization category $\text{IndCoh}^*((\text{LS}_{\check{P}^-}^{\text{mon-free } \check{M}})_{\text{mon-free}}^\wedge)$, which is equipped with a (strictly unital) factorization functor

$$(B.6) \quad \text{Rep}(\check{P}^-) \simeq \text{QCoh}(\text{LS}_{\check{P}^-}^{\text{reg}}) \simeq \text{IndCoh}^*(\text{LS}_{\check{P}^-}^{\text{reg}}) \xrightarrow{(\iota)_*^{\text{IndCoh}}} \text{IndCoh}^*((\text{LS}_{\check{P}^-}^{\text{mon-free } \check{M}})_{\text{mon-free}}^\wedge).$$

Recall the (lax unital) factorization functor

$$\text{IndCoh}^*((\text{LS}_{\check{P}^-}^{\text{mon-free } \check{M}})_{\text{mon-free}}^\wedge) \xrightarrow{((q^-, \text{spec})_{\text{mon-free}}^\wedge)_*^{\text{IndCoh}}} \text{IndCoh}^*(\text{LS}_{\check{M}}^{\text{reg}}) \simeq \text{QCoh}(\text{LS}_{\check{M}}^{\text{reg}}) \simeq \text{Rep}(\check{M}),$$

see Sect. 2.7.4.

By construction, the composition $((q^-, \text{spec})_{\text{mon-free}}^\wedge)_*^{\text{IndCoh}} \circ (\iota)_*^{\text{IndCoh}}$ is the functor of direct image

$$(B.7) \quad (q^-, \text{spec})_*^{\text{IndCoh}} : \text{IndCoh}^*(\text{LS}_{\check{P}^-}^{\text{reg}}) \rightarrow \text{IndCoh}^*(\text{LS}_{\check{M}}^{\text{reg}}),$$

i.e., this is the functor

$$\text{inv}_{\check{N}_P^-} : \text{Rep}(\check{P}^-) \rightarrow \text{Rep}(\check{M}).$$

B.4.2. In particular, we obtain that the functor $((q^-, \text{spec})_{\text{mon-free}}^\wedge)_*^{\text{IndCoh}}$ sends the factorization unit to the (commutative) factorization algebra $\Omega^{\text{spec}} \in \text{Rep}(\check{M})$, see Sect. 2.2.4.

Thus, the functor $((q^-, \text{spec})_{\text{mon-free}}^\wedge)_*^{\text{IndCoh}}$ induces a (strictly unital) factorization functor

$$\text{IndCoh}^*((\text{LS}_{\check{P}^-}^{\text{mon-free } \check{M}})_{\text{mon-free}}^\wedge) \rightarrow \Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M})),$$

to be denoted by $\mathbf{Q}_{\text{mon-free}}^\wedge$.

B.4.3. Recall the action of $\text{Sph}_M^{\text{spec}}$ on $\text{Rep}(\check{M})$ (see [GLC2, Sect. E.6.5]). As in Sect. B.3.4, we obtain that the category $\Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))$ carries a canonical action of $\text{Sph}_M^{\text{spec}}$.

As in Sect. B.3.5, we obtain that the functor $\mathbf{Q}_{\text{mon-free}}^\wedge$ respects the actions of $\text{Sph}_M^{\text{spec}}$ on the two sides.

B.4.4. We will prove:

Theorem B.4.5. *The functor $\mathbf{Q}_{\text{mon-free}}^\wedge$ is fully faithful.*

The proof will be given in Sect. B.6.8.

Remark B.4.6. One should view Theorem B.4.5 as a spectral counterpart of point (b) of Theorem B.2.12. We will now proceed to formulating a spectral counterpart of Theorem B.2.12(a).

In addition, in Theorem B.6.7 we will describe explicitly the essential image of the functor Theorem B.4.5, making this a spectral counterpart of Theorem B.7.6.

B.4.7. Fix a point $\underline{x} \in \text{Ran}$, and note that we can regard

$$\text{IndCoh}(\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}_{\check{M}}}) \simeq \text{QCoh}(\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}_{\check{M}}})$$

as a factorization module category over

$$\text{IndCoh}^*(\text{LS}_{\check{P}^-}^{\text{reg}}) \simeq \text{QCoh}(\text{LS}_{\check{P}^-}^{\text{reg}}) \simeq \text{Rep}(\check{P}^-)$$

at $\underline{x}$.

The map

$$\mathbf{q}^{-, \text{spec}} : \text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}_{\check{M}}} \rightarrow \text{LS}_{\check{M}, \underline{x}}^{\text{reg}}$$

gives rise to a functor

$$(B.8) \quad \text{IndCoh}(\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}_{\check{M}}}) \xrightarrow{(\mathbf{q}^{-, \text{spec}})^{\text{IndCoh}}} \text{IndCoh}^*(\text{LS}_{\check{M}, \underline{x}}^{\text{reg}}) \simeq \text{QCoh}(\text{LS}_{\check{M}, \underline{x}}^{\text{reg}}) \simeq \text{Rep}(\check{M})_{\underline{x}}.$$

Moreover, this functor is compatible with the factorization module structure on the two sides with respect to $\text{IndCoh}^*(\text{LS}_{\check{P}^-}^{\text{reg}})$ and $\text{IndCoh}^*(\text{LS}_{\check{M}}^{\text{reg}})$, respectively, via the functor (B.7), see [GLC2, Sect. B.12.10] for what this means.

In particular, by [GLC2, Sect. C.14.12], we obtain that the functor (B.8) gives rise to a functor

$$(B.9) \quad \text{IndCoh}(\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}_{\check{M}}}) \rightarrow \Omega^{\text{spec-mod}^{\text{fact}}}(\text{Rep}(\check{M}))_{\underline{x}},$$

to be denoted $\mathbf{Q}$.

B.4.8. The spectral counterpart of Theorem B.2.12(a) reads:

Theorem B.4.9. *The functor $\mathbf{Q}$ of (B.9) is an equivalence.*

The proof will be given in Sect. C.

B.5. Identification of factorization algebras.

B.5.1. The computational core for the proof of Theorem 2.7.6 is provided by the following assertion:

Theorem B.5.2. *The equivalence*

$$\text{CS}_{\check{M}, \tau}^* : \text{Whit}^!(M) \rightarrow \text{Rep}(\check{M})$$

maps the factorization algebras Ω and Ω^{spec} to one another.

This theorem is a particular case of Theorem B.5.4 just below.

B.5.3. We will now discuss an extension of Theorem B.5.2, which will be used for the proof of Theorem 2.4.3:

Theorem B.5.4. *The following diagram of (lax unital) factorization functors commutes:*

$$(B.10) \quad \begin{array}{ccc} \text{Whit}^!(G) & \xrightarrow{\text{CS}_{G, \tau}} & \text{Rep}(\check{G}) \\ J_{\text{Whit}}^{-, !} \downarrow & & \downarrow \text{inv}_{\check{N}_P^-} \\ \text{Whit}^!(M) & \xrightarrow{\text{CS}_{M, \tau}} & \text{Rep}(\check{M}). \end{array}$$

Note that what we have just stated as Theorem B.5.4 is identical to Corollary 2.5.2, see, Remark 2.5.3.

Remark B.5.5. Note that the assertion of Theorem B.5.2 is a particular case of that of Theorem B.5.4. Namely, the two sides in Theorem B.5.2 are obtained by evaluating the two circuits in (B.10) on the factorization unit.

Remark B.5.6. A detailed proof of Theorem B.5.4 will appear in [FH]. Here is the main idea:

In the setting of coinvariants and Υ (see Sect. B.1.2), as analogue of Theorem B.5.4 was proved in [Ra4, Theorem 4.15.1] by reducing to a Zastava space calculation, performed [Ra3, Theorem 7.9.1]. This was done in the $P = B$ case, and the case of a general parabolic follows the same ideas.

The invariants/ Ω -version follows formally from the coinvariants/ Υ case by Verdier duality, once we combine with the *cleanness* result [Lin, Lemma 4.1.10].⁷⁸

B.6. Finiteness properties of Ω^{spec} . In this subsection we will state a crucial result, Theorem B.6.2, pertaining to finiteness properties of Ω^{spec} , viewed as a factorization module over itself.

B.6.1. In Sect. D we will prove:

Theorem B.6.2. *The factorization algebra Ω^{spec} is ULA as⁷⁹ an object of $\Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))$, i.e., for every affine scheme $S \rightarrow \text{Ran}$, the object*

$$\Omega_S^{\text{spec}} \in \Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))_S$$

is compact.

B.6.3. Recall the action of $\text{Sph}_M^{\text{spec}}$ on $\Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))$ (see Sect. B.4.3); denote the corresponding binary operation by $-\star_M-$.

For every $S \rightarrow \text{Ran}$, denote by

$$\Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))_S^{\text{Accs}} \subset \Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))_S$$

the full subcategory generated by objects of the form

$$(B.11) \quad \mathcal{F} \star_M \Omega_S^{\text{spec}} \in \Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))_S, \quad \mathcal{F} \in (\text{Sph}_M^{\text{spec}})_S^c.$$

Equivalently, one can take $\mathcal{F}$ to lie in the essential image of the functor $\text{nv} : \text{Rep}(\check{M})^c \rightarrow \text{Sph}_M^{\text{spec}}$.

B.6.4. Note that Theorem B.6.2 (combined with the fact that $(\text{Sph}_M^{\text{spec}})_S$ is rigid) implies that the objects (B.11) are compact.

The assignment

$$S \rightsquigarrow \Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))_S^{\text{Accs}}$$

extends to a (lax) factorization category, which we will denote by

$$\Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))^{\text{Accs}}.$$

B.6.5. Recall that the functor $\mathbf{Q}_{\text{mon-free}}^\wedge$ is compatible with the actions of $\text{Sph}_M^{\text{spec}}$ on the two sides (see Sect. B.4.3).

Now, it is easy to see that the category $\text{IndCoh}^*((\text{LS}_{\check{P}-}^{\text{mon-free}})^{\wedge}_{\text{mon-free}})$ is generated by objects of the form

$$\mathcal{F} \star_M (\iota)_*^{\text{IndCoh}}(\mathbf{1}_{\text{LS}_{\check{P}-}^{\text{reg}}}), \quad \mathcal{F} \in \text{Sph}_M^{\text{spec}},$$

where $(\iota)_*^{\text{IndCoh}}$ is as in (B.6).

Hence, we obtain that the essential image of the functor $\mathbf{Q}_{\text{mon-free}}^\wedge$ is contained in the subcategory $\Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))^{\text{Accs}}$.

B.6.6. We have the following more precise version of Theorem B.4.5:

Theorem B.6.7. *The functor $\mathbf{Q}_{\text{mon-free}}^\wedge$ induces an equivalence*

$$\text{IndCoh}^*((\text{LS}_{\check{P}-}^{\text{mon-free}})^{\wedge}_{\text{mon-free}}) \rightarrow \Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))^{\text{Accs}}.$$

⁷⁸It is referred to as “purity” in *loc.cit.*

⁷⁹See [Ra3, Appendix B] or [CR, Sect. 2] for a detailed discussion of ULA objects in factorization categories.

B.6.8. *Proof of Theorem B.6.7.* Before we launch into the proof, note that Theorem B.6.7 tautologically implies Theorem B.4.5.

We will prove Theorem B.6.7 assuming Theorem B.4.9, which will be proved independently (see Sect. C).

We need to show that for every affine scheme $S \rightarrow \text{Ran}$, the corresponding functor

$$(B.12) \quad \text{IndCoh}^*((\text{LS}_{\tilde{P}^-}^{\text{mon-free}, \check{M}})_{\text{mon-free}}^\wedge)_S \rightarrow \Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))_S^{\text{Accs}}$$

is an equivalence. With no restriction of generality we can assume that S is eventually coconnective (in fact, it is enough to take S to be smooth).

By construction, the essential image of (B.12) generates the target. Hence, it is enough to show that (B.12) is fully faithful.

Since the objects (B.11) are compact, we obtain that the functor (B.12) preserves compactness. Hence, it admits a continuous right adjoint, which is automatically $\text{QCoh}(S)$ -linear.

We need to show that the unit of the adjunction is an isomorphism. In order to do so, since S was assumed eventually coconnective, it is enough to do so at the level of fibers at geometric points of S . I.e., we obtain that it is enough to show that for every $\underline{x} \in \text{Ran}$, the corresponding functor

$$\mathbf{Q}_{\text{mon-free}}^\wedge : \text{IndCoh}^*((\text{LS}_{\tilde{P}^-}^{\text{mon-free}, \check{M}})_{\text{mon-free}}^\wedge)_{\underline{x}} \rightarrow \Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))_{\underline{x}}^{\text{Accs}}$$

is fully faithful.

Note, however, that we have a commutative diagram

$$\begin{array}{ccc} \text{IndCoh}^*((\text{LS}_{\tilde{P}^-}^{\text{mon-free}, \check{M}})_{\text{mon-free}}^\wedge)_{\underline{x}} & \xrightarrow{\mathbf{Q}_{\text{mon-free}}^\wedge} & \Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))_{\underline{x}}^{\text{Accs}} \\ \downarrow & & \downarrow \\ \text{IndCoh}(\text{LS}_{\tilde{P}^-, \underline{x}}^{\text{mon-free}, \check{M}}) & \xrightarrow{\mathbf{Q}} & \Omega\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))_{\underline{x}}, \end{array}$$

where the vertical arrows are the tautological fully faithful embeddings.

Hence, we obtain that the fully faithfulness of $\mathbf{Q}_{\text{mon-free}}^\wedge$ at the pointwise level follows from Theorem B.4.9.

□[Theorem B.6.7]

B.7. Finiteness properties of Ω . In this subsection we will establish results parallel to those in the previous subsection, but now on the geometric side.

B.7.1. First, combining Theorems B.5.2 and B.6.2 we obtain:

Corollary B.7.2. *The factorization algebra Ω is ULA as an object of $\Omega\text{-mod}^{\text{fact}}(\text{Whit}^!(M))$.*

B.7.3. We define the (lax) factorization category

$$\Omega\text{-mod}^{\text{fact}}(\text{Whit}^!(M))^{\text{Accs}} \subset \Omega\text{-mod}^{\text{fact}}(\text{Whit}^!(M))$$

by the procedure parallel to that of Sect. B.6.3.

B.7.4. Since the essential image of the functor

$$\mathbf{ind}_{\text{Sph}_M \rightarrow \frac{\infty}{2}} : \text{Sph}_M \rightarrow \mathbf{I}(G, P^-)_{\rho_P(\omega_X)}$$

generates the target, we obtain that the functor $J_{\text{Whit}, \Omega}^{-, \frac{\infty}{2}}$ maps

$$\text{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2}-\text{Accs}} \subset \text{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2}}$$

to the full subcategory

$$\Omega\text{-mod}^{\text{fact}}(\text{Whit}^!(M))^{\text{Accs}} \subset \Omega\text{-mod}^{\text{fact}}(\text{Whit}^!(M)).$$

B.7.5. We claim:

Theorem B.7.6. *The resulting functor*

$$J_{\text{Whit}, \Omega}^{-, \frac{\infty}{2}} : \text{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2} - \text{Accs}} \rightarrow \Omega\text{-mod}^{\text{fact}}(\text{Whit}^!(M))^{\text{Accs}}$$

is an equivalence.

Proof. Parallel to the proof of Theorem B.6.7 given in Sect. B.6.8, using Theorem B.2.12(a), which will be proved independently. $\square$

B.7.7. Note that the assertion of Theorem B.7.6 tautologically implies that of Theorem B.2.12(b).

B.8. Proof of Theorem 2.7.6. We are now ready to prove Theorem 2.7.6.

In fact, this is an immediate consequence of the combination of Theorems B.5.2, B.7.6 and B.6.7.

Namely, $\text{CS}_{\tau}^{-, \frac{\infty}{2}}$ is the unique functor that makes the following diagram commute

$$\begin{array}{ccc} \text{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2} - \text{Accs}} & \xrightarrow{\text{CS}_{\tau}^{-, \frac{\infty}{2}}} & \text{IndCoh}^*((\text{LS}_{\check{P}-}^{\text{mon-free}})_{\text{mon-free}}^{\wedge}) \\ J_{\text{Whit}, \Omega}^{-, \frac{\infty}{2}} \downarrow \sim & & \sim \downarrow \mathbb{Q}_{\text{mon-free}}^{\wedge} \\ \Omega\text{-mod}^{\text{fact}}(\text{Whit}^!(M))^{\text{Accs}} & \xrightarrow[\text{CS}_{M, \tau}]{} & \Omega^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M}))^{\text{Accs}}. \end{array}$$

The functor $\text{CS}_{\tau}^{-, \frac{\infty}{2}}$ is compatible with the actions of

$$\text{Sph}_M^{\text{Sat}_{M, \tau}} \simeq^{\tau} \text{Sph}_M^{\text{spec}},$$

because the functor $\text{CS}_{M, \tau}$ is.

$\square$ [Theorem 2.7.6]

B.9. Proof of Theorem B.2.12(a).

B.9.1. As in the proof of Lemma 3.7.4, we can replace the category $\text{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2}}$ by

$$\text{Whit}^!(\text{Fl}_G^{\text{aff}, I_P^-}) := \text{Whit}^!\left(\text{D-mod}_{\frac{1}{2}}((\text{Fl}_G^{\text{aff}, I_P^-})_{\rho(\omega_X)})\right),$$

where $\text{Fl}_G^{\text{aff}, I_P^-}$ is the partial affine flag scheme

$$\text{Fl}_G^{\text{aff}, I_P^-} := \mathfrak{L}(G)/I_P^-$$

at $\underline{x}$.

Under this identification, the functor $J_{\text{Whit}, \Omega}^{-, \frac{\infty}{2}}$ corresponds to (the parahoric version of) the functor

$$(B.13) \quad \text{Whit}^!(\text{Fl}_G^{\text{aff}, I_P^-}) \rightarrow \Omega\text{-mod}^{\text{fact}}(\text{Whit}^!(M))_{\underline{x}}$$

of [Ya-R, Equation (6.4.7)].

Hence, the assertion follows from (the parahoric version of) [Ya-R, Theorem 6.4.8].

$\square$ [Theorem B.2.12(a)]

Remark B.9.2. When $P = B$, the existence of an equivalence between $\text{Whit}^!(\text{Fl}_G^{\text{aff}, I_P^-})$ and $\Omega\text{-mod}^{\text{fact}}(\text{Whit}^!(M))_{\underline{x}}$, interpreted as $\text{IndCoh}(\text{LS}_{\check{P}-, \underline{x}}^{\text{mon-free}})$, was established in [AB].

One can show, however, that their equivalence coincides with one provided by (B.13).

APPENDIX C. SHEAVES ON THE (POINTWISE) $\mathrm{LS}_H^{\mathrm{mer}}$ FOR A UNIPOTENT GROUP

The goal of this section is to prove Theorem B.4.9.

While doing so, we will prove a general statement concerning the category of ind/quasi-coherent sheaves on the stack $\mathrm{LS}_{H,\underline{x}}^{\mathrm{mer}}$ for a unipotent group H .

C.1. The statement for unipotent groups.

C.1.1. Applying base change with respect to $\mathrm{pt} \rightarrow \mathrm{LS}_M^{\mathrm{reg}}$, we reduce Theorem B.4.9 to the following assertion, see Theorem C.1.5 below, valid for any unipotent group, to be denoted H , where in our case

$$H := \check{N}_P^-.$$

C.1.2. Consider $\mathrm{IndCoh}(\mathrm{LS}_{H,\underline{x}}^{\mathrm{mer}})$ as a factorization module category with respect to

$$\mathrm{IndCoh}^*(\mathrm{LS}_H^{\mathrm{reg}}) \simeq \mathrm{QCoh}(\mathrm{LS}_H^{\mathrm{reg}}) \simeq \mathrm{Rep}(H).$$

The functor

$$\Gamma(\mathrm{LS}_{H,\underline{x}}^{\mathrm{mer}}, -)^{\mathrm{IndCoh}} : \mathrm{IndCoh}(\mathrm{LS}_{H,\underline{x}}^{\mathrm{mer}}) \rightarrow \mathrm{Vect}$$

is compatible with factorization module structure with respect to the functor

$$(C.1) \quad \Gamma(\mathrm{LS}_H^{\mathrm{reg}}, -)^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{LS}_H^{\mathrm{reg}}) \rightarrow \mathrm{Vect}.$$

C.1.3. The functor (C.1) sends the factorization unit to the (commutative) unital factorization algebra Ω_H^{spec} (in Vect) corresponding to the commutative algebra

$$\mathrm{inv}_H(k) \in \mathrm{ComAlg}(\mathrm{Vect}),$$

where the latter is the same as the cohomological Chevalley complex $C_{\mathrm{chev}}(\mathfrak{h})$ of $\mathfrak{h}$ with trivial coefficients.

Hence, the functor $\Gamma(\mathrm{LS}_{H,\underline{x}}^{\mathrm{mer}}, -)^{\mathrm{IndCoh}}$ gives rise to a functor

$$\mathrm{IndCoh}(\mathrm{LS}_{H,\underline{x}}^{\mathrm{mer}}) \rightarrow \Omega_H^{\mathrm{spec}}\text{-mod}^{\mathrm{fact}}(\mathrm{Vect})_{\underline{x}} =: \Omega_H^{\mathrm{spec}}\text{-mod}_{\underline{x}}^{\mathrm{fact}},$$

to be denoted $\mathcal{Q}$.

C.1.4. We will prove:

Theorem C.1.5. *The functor $\mathcal{Q}$ is an equivalence.*

The proof occupies the rest of this section.

C.2. Method of proof.

C.2.1. In order to simplify the notation, we will assume that $\underline{x}$ is a singleton $\{x\}$. However, the entire discussion is equally applicable to a general $\underline{x} \in \mathrm{Ran}$.

Recall that $\mathrm{LS}_{H,x}^{\mathrm{mer}}$ is isomorphic to the algebraic stack $\mathfrak{h}/\mathrm{Ad}(H)$. In particular, it is smooth, so the functor

$$\Psi_{\mathrm{LS}_{H,x}^{\mathrm{mer}}} : \mathrm{IndCoh}(\mathrm{LS}_{H,\underline{x}}^{\mathrm{mer}}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{H,\underline{x}}^{\mathrm{mer}})$$

is an equivalence.

Since H is unipotent, the functor

$$\Gamma(\mathrm{LS}_{H,\underline{x}}^{\mathrm{mer}}, -) : \mathrm{QCoh}(\mathrm{LS}_{H,\underline{x}}^{\mathrm{mer}}) \rightarrow \mathrm{Vect}$$

is conservative.

Hence, $\mathcal{O}_{\mathrm{LS}_{H,\underline{x}}^{\mathrm{mer}}}$ is a compact generator of $\mathrm{IndCoh}(\mathrm{LS}_{H,\underline{x}}^{\mathrm{mer}})$.

C.2.2. Let

$$P_{\Omega_H^{\text{spec}}, x} \in \text{Pro}(\Omega_H^{\text{spec}}\text{-mod}_{\underline{x}}^{\text{fact}})$$

be as in [GLC2, Sect. D.2.5]. I.e., this is the (pro)-object that (pro)-corepresents the forgetful functor

$$\mathbf{oblv}_{\Omega_H^{\text{spec}}} : \Omega_H^{\text{spec}}\text{-mod}_{\underline{x}}^{\text{fact}} \rightarrow \mathbf{Vect},$$

see [GLC2, Proposition D.2.7].

We have a tautologically defined map:

$$(C.2) \quad P_{\Omega_H^{\text{spec}}, x} \rightarrow Q(\mathcal{O}_{\text{LSmer}_{H, \underline{x}}}).$$

C.2.3. We will prove:

Proposition C.2.4. *The map (C.2) is an isomorphism.*

Clearly, Proposition C.2.4 implies Theorem C.1.5.

Remark C.2.5. Note that Proposition C.2.4 in particular says that $P_{\Omega_H^{\text{spec}}, x}$ is an *object* of $\Omega_H^{\text{spec}}\text{-mod}_{\underline{x}}^{\text{fact}}$ (rather than a pro-object).

But we will establish this fact *a priori*.

C.3. Digression: factorization modules over holonomic factorization algebras.

C.3.1. Let $\mathcal{A}$ be a factorization algebra on X . Recall the category $\text{Modif}(\mathcal{A})$, see [GLC2, Sect. D.2.4].

Assume now that $\mathcal{A}_X$ is *holonomic*. We claim that in this case, the category $\text{Modif}(\mathcal{A})$ has an initial object, to be denoted $j_{!, \text{fact}}(\mathcal{A}|_{X-x})$.

Namely, consider $\mathcal{A}_{\text{Ran}} \in \text{D-mod}(\text{Ran})$, and

$$(C.3) \quad (j_{\text{Ran}})_! \circ (j_{\text{Ran}})^!(\mathcal{A}_{\text{Ran}}) \in \text{D-mod}(\text{Ran}),$$

where:

- j denotes the open embedding $X-x \hookrightarrow X$;
- j_{Ran} denotes the (open embedding) of the Ran space of $X-x$ into the Ran space of X .

It is easy to see that the factorization structure on $\mathcal{A}_{\text{Ran}}$ uniquely extends to one on (C.3), thereby making (C.3) into an object of $\text{Modif}(\mathcal{A})$. This is our sought-for $j_{!, \text{fact}}(\mathcal{A}|_{X-x})$.

The fact that it is the initial object in this category also follows from the construction.

If $\mathcal{A}$ is unital, then $j_{!, \text{fact}}(\mathcal{A}|_{X-x})$ is automatically unital: this follows from the fact that the map

$$\omega_{\text{Ran}} \rightarrow (j_{\text{Ran}})_! \circ j_{\text{Ran}}^!(\omega_{\text{Ran}})$$

is an isomorphism⁸⁰ and [GLC2, Proposition C.7.13].

C.3.2. Let $j_{!, \text{fact}}(\mathcal{A}|_{X-x})_x$ denote the $!$ -fiber of $j_{!, \text{fact}}(\mathcal{A}|_{X-x})$, viewed as an object of $\mathcal{A}\text{-mod}_x^{\text{fact}}$. By [GLC2, Proposition D.2.7], we obtain that $j_{!, \text{fact}}(\mathcal{A}|_{X-x})_x$ is the object that corepresents the forgetful functor

$$\mathbf{oblv}_{\mathcal{A}} : \mathcal{A}\text{-mod}_x^{\text{fact}} \rightarrow \mathbf{Vect}.$$

I.e.,

$$j_{!, \text{fact}}(\mathcal{A}|_{X-x})_x \simeq P_{\mathcal{A}, x},$$

where $P_{\mathcal{A}, x}$ is as in [GLC2, Sect. D.2.5].

⁸⁰This can be seen by reduction to the case of $X = \mathbb{A}^1$ and the contraction principle.

C.3.3. Assume now that $\mathcal{A}$ is commutative, i.e., of the form $\text{Fact}(A)$, where

$$A \in \text{ComAlg}(\text{D-mod}(X)),$$

see [GLC2, Sect. B.10.1].

By assumption, A is holonomic as a D-module on X . In this case, the construction of $j_{!,\text{fact}}(\mathcal{A}|_{X-x})$ given by (C.3) implies that it has a natural structure of commutative algebra object in $\text{FactAlg}^{\text{untl}}(X)$, and hence by [GLC2, Proposition C.8.6], is of the form $\text{Fact}(j_{!,\text{ComAlg}} \circ j^!(A))$, where $j_{!,\text{ComAlg}} \circ j^!(A)$ is the initial object in the category

$$\text{Modif}(A) := \{A' \in \text{ComAlg}(\text{D-mod}(X)), A'|_{X-x} \simeq A|_{X-x}\}.$$

C.3.4. Note that

$$(C.4) \quad j_{!,\text{fact}}(\mathcal{A}|_{X-x})_x \simeq (j_{!,\text{ComAlg}} \circ j^!(A))_x$$

as objects of

$$\mathcal{A}\text{-mod}_x^{\text{fact}} \simeq j_{!,\text{fact}}(\mathcal{A}|_{X-x})\text{-mod}_x^{\text{fact}},$$

where we view $(j_{!,\text{ComAlg}} \circ j^!(A))_x$ as an object of $j_{!,\text{fact}}(\mathcal{A}|_{X-x})\text{-mod}_x^{\text{fact}}$ via

$$(j_{!,\text{ComAlg}} \circ j^!(A))_x\text{-mod} = j_{!,\text{fact}}(\mathcal{A}|_{X-x})\text{-mod}_x^{\text{com}} \rightarrow j_{!,\text{fact}}(\mathcal{A}|_{X-x})\text{-mod}_x^{\text{fact}},$$

see [GLC2, Sect. B.10.5-6].

C.4. Proof of Proposition C.2.4.

C.4.1. We apply the discussion in Sect. C.3.3 to $\mathcal{A} := \Omega_{\mathbb{H}}^{\text{spec}}$, so that $A = \text{inv}_{\mathbb{H}}(k) \otimes \mathcal{O}_{X_{\text{dR}}}$. Consider the corresponding objects

$$j_{!,\text{fact}}(\Omega_{\mathbb{H}}^{\text{spec}}|_{X-x})$$

and

$$j_{!,\text{ComAlg}} \circ j^!(\text{inv}_{\mathbb{H}}(k) \otimes \mathcal{O}_{X_{\text{dR}}}) \in \text{ComAlg}(\text{D-mod}(X)).$$

We have:

$$(C.5) \quad j_{!,\text{fact}}(\Omega_{\mathbb{H}}^{\text{spec}}|_{X-x})_x \simeq \mathbf{P}_{\Omega_{\mathbb{H}}^{\text{spec}},x}$$

as objects of $\Omega_{\mathbb{H}}^{\text{spec}}\text{-mod}_x^{\text{fact}}$.

In particular, we obtain that $\mathbf{P}_{\Omega_{\mathbb{H}}^{\text{spec}},x}$ is an *object* (rather than a pro-object) of $\Omega_{\mathbb{H}}^{\text{spec}}\text{-mod}_x^{\text{fact}}$.

C.4.2. In order to prove Proposition C.2.4, it suffices to show that the map (C.2) becomes an isomorphism after applying the forgetful functor $\mathbf{oblv}_{\Omega_{\mathbb{H}}^{\text{spec}}}$.

Thus, using the identifications (C.4) and (C.5), our goal is to show that the map

$$(C.6) \quad (j_{!,\text{ComAlg}} \circ j^!(\text{inv}_{\mathbb{H}}(k) \otimes \mathcal{O}_{X_{\text{dR}}}))_x \rightarrow \Gamma(\text{LS}_{H,x}^{\text{mer}}, \mathcal{O}_{\text{LS}_{H,x}^{\text{mer}}})$$

is an isomorphism.

C.4.3. Recall the construction

$$(R \in \text{ComAlg}(\text{Vect})) \rightsquigarrow (\text{"Spec}(R)" \in \text{PreStk}),$$

see [GLC2, Sect. 12.7.6]. Namely, for an affine test-scheme $\text{Spec}(R')$,

$$\text{Maps}_{\text{PreStk}}(\text{Spec}(R'), \text{"Spec}(R)") := \text{Maps}_{\text{ComAlg}(\text{Vect})}(R, R').$$

We have a tautologically defined map

$$(C.7) \quad R \rightarrow \Gamma(\text{"Spec}(R)", \mathcal{O}_{\text{"Spec}(R)"}),$$

see [GLC2, Equation (12.46)].

Recall also (see [GLC2, Sect. 12.7.6]) that R is said to be “as good as connective” if:

- The map (C.7) is an isomorphism;
- The induced functor $\text{QCoh}(\text{"Spec}(R)") \rightarrow R\text{-mod}$ is an equivalence.

This construction makes sense in the relative situation: for a prestack $\mathcal{Y}$ and $R \in \text{ComAlg}(\text{QCoh}(\mathcal{Y}))$ we can consider

$$\text{“Spec}_{\mathcal{Y}}(R)” \in \text{PreStk}/\mathcal{Y},$$

so that for $(S \xrightarrow{\mathcal{Y}} \mathcal{Y}) \in \text{Sch}_{/\mathcal{Y}}^{\text{aff}}$, we have

$$S \times_{\mathcal{Y}} \text{“Spec}_{\mathcal{Y}}(R)” = \text{“Spec}(\Gamma(S, y^*(R)))”.$$

C.4.4. We will deduce the fact that (C.6) from

Proposition C.4.5. *There is a canonical isomorphism*

$$\text{LS}_{H,x}^{\text{mer}} \simeq \text{“Spec} \left((j_{!, \text{ComAlg}} \circ j^!(\text{inv}_H(k) \otimes \mathcal{O}_{X_{\text{dR}}}))_x \right)” ,$$

so that the map (C.6) identifies with the map (C.7).

Proposition C.4.6. *The commutative algebra*

$$(j_{!, \text{ComAlg}} \circ j^!(\text{inv}_H(k) \otimes \omega_X))_x$$

is as good as connective.

C.5. Proof of Proposition C.4.5.

C.5.1. Consider $\text{pt}/H \times X_{\text{dR}}$ as a constant prestack over X_{dR} . We can identify

$$\text{LS}_{H,x}^{\text{mer}} \simeq \mathfrak{L}_x(\text{pt}/H).$$

Explicitly, for $R' \in \text{ComAlg}(\text{Vect}^{\leq 0})$, we have

$$(C.8) \quad \text{Maps}(\text{Spec}(R'), \text{LS}_{H,x}^{\text{mer}}) \simeq \text{Maps}_{\text{PreStk}/X_{\text{dR}}}(\text{Spec}_{X_{\text{dR}}}(R'((t))), \text{pt}/H \times X_{\text{dR}}),$$

where:

- t is a local coordinate at x ;
- We regard $R'((t))$ as an object of $\text{ComAlg}(\text{QCoh}(X_{\text{dR}})^{\leq 0})$.

C.5.2. Since H is unipotent, we have a canonical identification

$$\text{pt}/H \simeq \text{“Spec}(\text{inv}_H(k))”.$$

Hence, we can rewrite the right-hand side in (C.8) as

$$\text{Maps}_{\text{ComAlg}(\text{QCoh}(X_{\text{dR}}))}(\text{inv}_H(k) \otimes \mathcal{O}_{X_{\text{dR}}}, R'((t))),$$

which identifies by adjunction with

$$\text{Maps}_{\text{ComAlg}(\text{QCoh}(X_{\text{dR}}))}(j_{!, \text{ComAlg}} \circ j^!(\text{inv}_H(k) \otimes \mathcal{O}_{X_{\text{dR}}}), R'[[t]]),$$

and which we rewrite further as

$$\text{Maps}_{\text{ComAlg}(\text{Vect})}((j_{!, \text{ComAlg}} \circ j^!(\text{inv}_H(k) \otimes \mathcal{O}_{X_{\text{dR}}}))_x, R'),$$

which is by definition the same as

$$\text{Maps}_{\text{PreStk}}(\text{Spec}(R'), \text{“Spec} \left((j_{!, \text{ComAlg}} \circ j^!(\text{inv}_H(k) \otimes \mathcal{O}_{X_{\text{dR}}}))_x \right)”).$$

The identification of the maps (C.6) and (C.7) follows from the construction.

□[Proposition C.4.5]

C.6. Proof of Proposition C.4.6.

C.6.1. We identify tautologically

$$(j_!, \text{ComAlg} \circ j^!(\text{inv}_H(k) \otimes \omega_X))_x \simeq ((j_{\text{Ran}})_! \circ (j_{\text{Ran}})^!(\Omega_H^{\text{spec}}))_{\{x\}},$$

where $\{x\}$ is viewed as a point of Ran .

The assertion of the proposition étale-local, so we can assume that

$$(x \in X) = (0 \in \mathbb{A}^1).$$

In this case, by the contraction principle, we can replace the operation of taking the $!$ -fiber at the point $\{0\} \in \text{Ran}(\mathbb{A}^1)$ by $C_c(\text{Ran}(\mathbb{A}^1), -)$.

Further,

$$C_c(\text{Ran}(\mathbb{A}^1), (j_{\text{Ran}})_! \circ (j_{\text{Ran}})^!(\Omega_H^{\text{spec}})) \simeq C_c(\text{Ran}(\mathbb{A}^1 - 0), (j_{\text{Ran}})^!(\Omega_H^{\text{spec}})).$$

The rest of the proof repeats the proof of the assertion in [GLC2, Sect. 12.7.7] given in *loc. cit.* Sects. 12.7.8-12.7.14.

C.6.2. Namely, using [GLC2, Lemma 12.7.10] we reduce to the case when $\mathfrak{h}$ is abelian. In the latter case

$$\text{inv}_H(k) \simeq \text{Sym}(\mathfrak{h}^*[-1]),$$

and

$$C_c(\text{Ran}(\mathbb{A}^1 - 0), j_{\text{Ran}}^!(\Omega_H^{\text{spec}})) \simeq \text{Sym}(\mathfrak{h}^*[-1]) \otimes \text{Sym}(\mathfrak{h}^*).$$

(The last isomorphism is due to the fact that for any connected Y , the "constant D-algebra" functor $\text{ComAlg} \rightarrow \text{ComAlg}(\text{D-mod}(Y))$ admits a (partially defined) left adjoint calculated as $C_c(\text{Ran}(Y), \text{Fact}(-))$, while this left adjoint sends $\text{Sym}(\mathcal{M})$, $\mathcal{M} \in \text{D-mod}(Y)$ to $\text{Sym}(C_c(Y, \mathcal{M}))$.)

□[Proposition C.4.6]

Remark C.6.3. Note that the factorization algebra that we have denoted by Ω_H^{spec} is *not* the same as the factorization algebra $\Omega(\mathfrak{h})$ in [GLC2, Sect. 12.5.4].

Namely, the latter is $\text{Fact}(C_{\text{chev}}(\mathfrak{h} \otimes D_X))$, while the former is $\text{Fact}(C_{\text{chev}}(\mathfrak{h} \otimes k_X))$, where $k_X \in \text{D-mod}(X)$ is "the constant sheaf" and C_{chev} denotes the cohomological Chevalley complex of a Lie- $*$ algebra, see [BD2, Sect. 1.4.10].

APPENDIX D. COMPACTNESS OF Ω^{spec}

The goal of this section is to prove Theorem B.6.2.

D.1. The shearing trick.

D.1.1. Consider the cocharacter $2\check{\rho}_P$ of $Z_{\check{M}}$. Note that the symmetric monoidal category $\text{Rep}(\check{M})$ admits an automorphism, given by sending an object V with central character λ to

$$V[2\langle 2\check{\rho}_P, \lambda \rangle].$$

Denote this automorphism by $\text{Shear}_{\check{M}}$. We refer it to as the *shearing* automorphism of $\text{Rep}(\check{M})$.

D.1.2. Denote

$$\text{inv}_{\check{N}_P^-}(k)^{\Rightarrow} := \text{Shear}_{\check{M}}(\text{inv}_{\check{N}_P^-}(k)) \in \text{ComAlg}(\text{Rep}(\check{M}))$$

and

$$\Omega^{\text{spec}, \Rightarrow} := \text{Shear}_{\check{M}}(\Omega^{\text{spec}}) \in \text{FactAlg}^{\text{untl}}(X, \text{Rep}(\check{M})),$$

so that

$$\Omega^{\text{spec}, \Rightarrow} \simeq \text{Fact}(\text{inv}_{\check{N}_P^-}(k)^{\Rightarrow}).$$

Clearly, Theorem B.6.2 is equivalent to the assertion that $\Omega^{\text{spec}, \Rightarrow}$ is *ULA* as an object of

$$\Omega^{\text{spec}, \Rightarrow}\text{-mod}^{\text{fact}}(\text{Rep}(\check{M})).$$

D.1.3. We observe:

Lemma D.1.4. *The object*

$$\mathrm{inv}_{\check{N}_P^-}(k)^{\Rightarrow} \in \mathrm{Rep}(\check{M})$$

is connective.

Proof. Note that $H^i(\mathrm{inv}_{\check{N}_P^-}(k))$ is a subquotient of $\Lambda^i((\check{\mathfrak{n}}_P^-)^*)$. The result follows now from the fact that the weights of $2\check{\rho}_P$ on $(\check{\mathfrak{n}}_P^-)^*$ are strictly positive. $\square$

By [GLC2, Proposition B.9.18], from Lemma D.1.4 we obtain that the lax factorization category $\Omega^{\mathrm{spec}, \Rightarrow}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{M}))$ carries a t-structure, such that the forgetful functor

$$\mathbf{oblv}_{\Omega^{\mathrm{spec}, \Rightarrow}} : \Omega^{\mathrm{spec}, \Rightarrow}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{M})) \rightarrow \mathrm{Rep}(\check{M})$$

is t-exact.

Moreover, $\Omega^{\mathrm{spec}, \Rightarrow}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{M}))$ is left-complete in this t-structure.

D.1.5. Let $\mathbf{D}$ be a DG category equipped with a t-structure, compatible with filtered colimits (i.e., the subcategory $\mathbf{D}^{\geq 0}$ is preserved by filtered colimits). Recall that an object $\mathbf{d} \in \mathbf{D}$ is said to be *almost compact* if the functor

$$\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, -) : \mathbf{D} \rightarrow \mathbf{Vect}$$

is preserved by filtered colimits *when restricted to $\mathbf{D}^{\geq -n}$ for every individual $n \in \mathbb{N}$.*

D.1.6. Let now $\mathbf{A}$ be a lax factorization category equipped with a t-structure⁸¹, and let $\mathcal{A}$ be a factorization algebra object in it.

We shall say that $\mathcal{A}$ is *almost ULA* in $\mathcal{A}\text{-mod}^{\mathrm{fact}}(\mathbf{A})$ if for every affine scheme $S \rightarrow \mathrm{Ran}$, the object $\mathcal{A}_S \in \mathcal{A}\text{-mod}^{\mathrm{fact}}(\mathbf{A})_S$ is almost compact.

D.1.7. We claim:

Proposition D.1.8. *The factorization algebra $\Omega^{\mathrm{spec}, \Rightarrow}$ is almost ULA in $\Omega^{\mathrm{spec}, \Rightarrow}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{M}))$.*

Proof. Consider the forgetful functor

$$\mathbf{oblv}_{\check{M}} : \mathrm{Rep}(\check{M}) \rightarrow \mathbf{Vect}.$$

It is easy to see that it suffices to show that the factorization algebra $\mathbf{oblv}_{\check{M}}(\Omega^{\mathrm{spec}, \Rightarrow})$ is almost ULA in $\mathbf{oblv}_{\check{M}}(\Omega^{\mathrm{spec}, \Rightarrow}\text{-mod}^{\mathrm{fact}})$.

Since $\mathbf{oblv}_{\check{M}}(\mathrm{inv}_{\check{N}_P^-}(k)^{\Rightarrow})$ is finite-dimensional, it is almost of finite type. Now the required assertion follows from [CR, Theorem 4.6.1]. $\square$

D.2. From almost compactness to compactness. In the previous subsection we proved that for every affine scheme $S \rightarrow \mathrm{Ran}$, the object

$$\Omega_S^{\mathrm{spec}, \Rightarrow} \in \Omega^{\mathrm{spec}, \Rightarrow}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{M}))_S$$

is almost compact.

In this subsection, we will show that it is actually compact, implying Theorem B.6.2.

⁸¹See [GLC2, Sect. B.11.11] for what this means

D.2.1. Let $\mathbf{D}$ be as in Sect. D.1.5. We shall say that an object $\mathbf{d} \in \mathbf{D}$ has a *finite cohomological dimension* if there exist an integer k , such that

$$\mathrm{Hom}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}'[k']) = 0 \text{ for } \mathbf{d}' \in \mathbf{D}^{\heartsuit} \text{ and } k' > k,$$

where

$$\mathrm{Hom}_{\mathbf{D}}(-, -) := H^0(\mathcal{H}om_{\mathbf{D}}(-, -)).$$

Such an integer k is called an upper bound on the cohomological dimension.

The following lemma is well-known:

Lemma D.2.2. *Assume that $\mathbf{D}$ is left-complete in its t -structure. Let $\mathbf{d} \in \mathbf{D}$ be almost compact. Then $\mathbf{d}$ is compact if and only if it has a finite cohomological dimension of $\mathbf{d}$.*

We include the proof for completeness.

Proof. Suppose that $\mathbf{d}$ has a finite cohomological dimension; let k be an upper bound on its cohomological dimension. Note that by the left-completeness assumption, we have

$$(D.1) \quad \mathrm{Hom}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}') = 0 \text{ for any } \mathbf{d}' \in \mathbf{D}^{<-(k+1)}.$$

Indeed,

$$\mathrm{Hom}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}') = H^0(\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \mathbf{d}')) \simeq H^0(\lim_n \mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \tau^{\geq -n}(\mathbf{d}'))),$$

while each $\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \tau^{\geq -n}(\mathbf{d}'))$ has a vanishing H^0 and H^{-1} (the latter is needed to kill the term arising from $R^1\mathrm{lim}$).

Let

$$\mathrm{colim}_i \mathbf{d}'_i \rightarrow \mathbf{d}'$$

be a filtered colimit diagram in $\mathbf{D}$. We need to show that

$$\mathrm{colim}_i \mathrm{Hom}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}'_i) \rightarrow \mathrm{Hom}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}')$$

is an isomorphism, where the colimit in the left-hand side is taken in Vect .

Consider the map between the long exact cohomology sequences corresponding to the fiber sequences

$$i \mapsto \tau^{<-(k+1)}(\mathbf{d}'_i) \rightarrow \mathbf{d}'_i \rightarrow \tau^{\geq -(k+1)}(\mathbf{d}'_i)$$

and

$$\tau^{<-(k+1)}(\mathbf{d}') \rightarrow \mathbf{d}' \rightarrow \tau^{\geq -(k+1)}(\mathbf{d}').$$

We have

$$\mathrm{colim}_i H^n(\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \tau^{<-(k+1)}(\mathbf{d}'_i))) = 0 \text{ and } H^n(\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \tau^{<-(k+1)}(\mathbf{d}'))) = 0 \text{ for } n \geq 0,$$

by (D.1).

The map

$$\mathrm{colim}_i H^n(\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \tau^{\geq -(k+1)}(\mathbf{d}'_i))) \rightarrow H^n(\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \tau^{\geq -(k+1)}(\mathbf{d}'))) \text{ for all } n$$

is an isomorphism, by the assumption of almost compactness.

Hence, the map

$$\mathrm{colim}_i H^0(\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \mathbf{d}'_i)) \rightarrow H^0(\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \mathbf{d}'))$$

is an isomorphism, as required.

Assume now that the cohomological dimension of $\mathbf{d}$ is infinite. Let $\mathbf{d}'_k \in \mathbf{D}^{\heartsuit}$ be a sequence of objects and n_k an increasing sequence of natural numbers such that

$$\mathrm{Hom}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}'_k[n_k]) \neq 0.$$

Let $\alpha_k \in \mathrm{Hom}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}'_k[n_k])$ be non-zero elements.

Set

$$\mathbf{d}' := \bigoplus_{k \in \mathbb{N}} \mathbf{d}'_k[n_k] \simeq \operatorname{colim}_k \left(\bigoplus_{1, \dots, k} \mathbf{d}'_k[n_k] \right).$$

By the left completeness assumption, the map

$$\mathbf{d}' \rightarrow \lim_k \tau^{\geq -n_k}(\mathbf{d}') \simeq \lim_k \left(\bigoplus_{1, \dots, k} \mathbf{d}'_k[n_k] \right) \simeq \lim_k \left(\prod_{1, \dots, k} \mathbf{d}'_k[n_k] \right) = \prod_k \mathbf{d}'_k[n_k]$$

is an isomorphism. Hence, the elements $\{\alpha_k\}$ give rise to an element $\alpha \in \operatorname{Hom}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}')$, which does not lie in the image of

$$\operatorname{colim}_k \operatorname{Hom}_{\mathbf{D}} \left(\mathbf{d}, \left(\bigoplus_{1, \dots, k} \mathbf{d}'_k[n_k] \right) \right).$$

Hence $\mathbf{d}$ is not compact. $\square$

D.2.3. Thus, we obtain that in order to prove that $\Omega_S^{\operatorname{spec}, \Rightarrow}$ is compact in $\Omega^{\operatorname{spec}, \Rightarrow} \text{-mod}^{\operatorname{fact}}(\operatorname{Rep}(\check{M}))_S$, it suffices to show that it has a finite cohomological dimension.

Since Ran is the colimit of prestacks of the form X_{dR}^I , $I \in \mathbf{fSet}$, with no restriction of generality we can assume that S is smooth.

Consider the following general paradigm. Let $\mathbf{D}$ be an $\operatorname{QCoh}(S)$ -linear category (i.e., $\mathcal{O}_S$ maps to the Hochschild homology⁸² of $\mathbf{D}$).

Let $\mathbf{D}$ be equipped with a t-structure, compatible with filtered colimits. For every $f : S' \rightarrow S$, denote

$$\mathbf{D}' := \operatorname{QCoh}(S') \otimes_{\operatorname{QCoh}(S)} \mathbf{D}.$$

We equip $\mathbf{D}'$ with the tensor product t-structure.

D.2.4. We claim:

Proposition D.2.5. *Assume that $\mathbf{d}$ is almost compact and that S is smooth. Then $\mathbf{d}$ is compact if and only if there exists an integer n such that for every geometric point $s : \operatorname{Spec}(k') \rightarrow S$, the object*

$$s^*(\mathbf{d}) \in \mathbf{D}_s := \operatorname{Vect}_{k'} \otimes_{\operatorname{QCoh}(S)} \mathbf{D}$$

has cohomological dimension bounded by n .

D.2.6. Before we prove Proposition D.2.5, let us finish the proof of Theorem B.6.2.

By Proposition D.2.5, it is sufficient to show that for a fixed $\underline{x} \in \operatorname{Ran}$, the object

$$\Omega_{\underline{x}}^{\operatorname{spec}, \Rightarrow} \in \Omega^{\operatorname{spec}, \Rightarrow} \text{-mod}^{\operatorname{fact}}(\operatorname{Rep}(\check{M}))_{\underline{x}}$$

has cohomological dimension uniformly bounded by some integer n .

By Theorem B.4.9 (which has been proved independently),

$$\Omega^{\operatorname{spec}} \text{-mod}^{\operatorname{fact}}(\operatorname{Rep}(\check{M}))_{\underline{x}} \simeq \operatorname{IndCoh}(\operatorname{LS}_{\check{P}^-, \underline{x}}^{\operatorname{mon-free} \check{M}}),$$

and under this equivalence, the object

$$\Omega_{\underline{x}}^{\operatorname{spec}} \in \Omega^{\operatorname{spec}} \text{-mod}^{\operatorname{fact}}(\operatorname{Rep}(\check{M}))_{\underline{x}}$$

corresponds to

$$i_*^{\operatorname{IndCoh}}(\mathcal{O}_{\operatorname{LS}_{\check{P}^-, \underline{x}}^{\operatorname{reg}}}) \in \operatorname{IndCoh}(\operatorname{LS}_{\check{P}^-, \underline{x}}^{\operatorname{mon-free} \check{M}}).$$

The object $i_*^{\operatorname{IndCoh}}(\mathcal{O}_{\operatorname{LS}_{\check{P}^-, \underline{x}}^{\operatorname{reg}}})$ is compact.

Hence, $\Omega_{\underline{x}}^{\operatorname{spec}, \Rightarrow}$ is compact as an object of $\Omega^{\operatorname{spec}, \Rightarrow} \text{-mod}^{\operatorname{fact}}(\operatorname{Rep}(\check{M}))_{\underline{x}}$. By Lemma D.2.2, it has a finite cohomological dimension. Let n_1 be an integer that bounds its cohomological dimension when $\underline{x}$ is a singleton x (it is easy to see that this category does not depend on x , up to extending scalars).

⁸²A.k.a., derived Bernstein center.

Let $S \rightarrow \text{Ran}$ factor through X^I . We take

$$n := |I| \cdot n_1.$$

By the above, this choice of n has the required property.

□[Theorem B.6.2]

D.3. Proof of Proposition D.2.5.

D.3.1. For any $\mathbf{d}, \mathbf{d}' \in \mathbf{D}$, let $\underline{\mathcal{H}om}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}') \in \text{QCoh}(S)$ be their inner Hom object, i.e.,

$$\mathcal{H}om_{\text{QCoh}(S)}(\mathcal{F}, \underline{\mathcal{H}om}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}')) := \mathcal{H}om_{\mathbf{D}}(\mathcal{F} \otimes \mathbf{d}, \mathbf{d}').$$

For $\mathcal{F} \in \text{QCoh}(S)$, we have a naturally defined map

$$(D.2) \quad \mathcal{F} \otimes \underline{\mathcal{H}om}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}') \rightarrow \underline{\mathcal{H}om}_{\mathbf{D}}(\mathbf{d}, \mathcal{F} \otimes \mathbf{d}')$$

D.3.2. We claim:

Lemma D.3.3. *Assume that $\mathbf{d}$ is almost compact and $\mathbf{d}' \in \mathbf{D}^{>-\infty}$ and $\mathcal{F}$ is of finite Tor-dimension. Then the map (D.2) is an isomorphism.*

Proof. The assumption that $\mathcal{F}$ is of finite Tor-dimension implies that it can be written as a finite colimit of flat objects. So, we can assume that $\mathcal{F}$ is flat.

In this case, by Lazard's lemma, we can write $\mathcal{F}$ as a filtered colimit of objects $\mathcal{F}_i$, where each $\mathcal{F}_i$ is finitely generated and free. It suffices to show that the map

$$\text{colim}_i \underline{\mathcal{H}om}_{\mathbf{D}}(\mathbf{d}, \mathcal{F}_i \otimes \mathbf{d}') \rightarrow \underline{\mathcal{H}om}_{\mathbf{D}}(\mathbf{d}, \mathcal{F} \otimes \mathbf{d}')$$

is an isomorphism, which is equivalent to

$$\text{colim}_i \mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \mathcal{F}_i \otimes \mathbf{d}') \rightarrow \mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \mathcal{F} \otimes \mathbf{d}')$$

being an isomorphism.

However, the latter holds by the assumption that $\mathbf{d}$ is almost compact.

□

Corollary D.3.4. *Let $f : S' \rightarrow S$ be a morphism of finite Tor-dimension between affine schemes. Then for $\mathbf{d}$ almost compact and $\mathbf{d}' \in \mathbf{D}^{>-\infty}$, the natural map*

$$f^*(\underline{\mathcal{H}om}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}')) \rightarrow \underline{\mathcal{H}om}_{\mathbf{D}'}(f^*(\mathbf{d}), f^*(\mathbf{d}'))$$

is an isomorphism.

Proof. It is enough to show that the induced map

$$(D.3) \quad f_* \circ f^*(\underline{\mathcal{H}om}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}')) \rightarrow f_* (\underline{\mathcal{H}om}_{\mathbf{D}'}(f^*(\mathbf{d}), f^*(\mathbf{d}')))$$

is an isomorphism.

By the projection formula

$$f_* \circ f^*(\underline{\mathcal{H}om}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}')) \simeq f_*(\mathcal{O}_{S'}) \otimes \underline{\mathcal{H}om}_{\mathbf{D}}(\mathbf{d}, \mathbf{d}').$$

Furthermore, for any $\tilde{\mathbf{d}}' \in \mathbf{D}'$, we have tautologically

$$f_* (\underline{\mathcal{H}om}_{\mathbf{D}'}(f^*(\mathbf{d}), \tilde{\mathbf{d}}')) \simeq \underline{\mathcal{H}om}_{\mathbf{D}}(\mathbf{d}, f_*(\tilde{\mathbf{d}}')).$$

In particular,

$$f_* (\underline{\mathcal{H}om}_{\mathbf{D}'}(f^*(\mathbf{d}), f^*(\mathbf{d}')) \simeq \underline{\mathcal{H}om}_{\mathbf{D}}(\mathbf{d}, f_* \circ f^*(\mathbf{d})) \simeq \underline{\mathcal{H}om}_{\mathbf{D}}(\mathbf{d}, f_*(\mathcal{O}_{S'} \otimes \mathbf{d})).$$

By construction, the following diagram commutes

$$\begin{array}{ccc} f_* \circ f^*(\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \mathbf{d}')) & \xrightarrow{(D.3)} & f_*(\mathcal{H}om_{\mathbf{D}'}(f^*(\mathbf{d}), f^*(\mathbf{d}')))) \\ \sim \downarrow & & \downarrow \sim \\ f_*(\mathcal{O}_{S'}) \otimes \mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \mathbf{d}') & \xrightarrow{(D.2)} & \mathcal{H}om_{\mathbf{D}}(\mathbf{d}, f_*(\mathcal{O}_{S'}) \otimes \mathbf{d}). \end{array}$$

Hence, the required assertion follows from Lemma D.3.3. $\square$

D.3.5. We are now ready to prove Proposition D.2.5. The “only if” direction is evident. We now prove the “if” direction.

Set $m := \dim(S)$. Let n be as in the statement of Proposition D.2.5. We will show that the cohomological dimension of $\mathbf{d}$ is bounded by $n + 2m$.

Fix an object $\mathbf{d}' \in \mathbf{D}^\vee$. We wish to show that $\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \mathbf{d}') \in \mathbf{Vect}$ lives in degrees $\leq (k + 2n)$. Consider the object

$$\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \mathbf{d}') \in \mathbf{QCoh}(S).$$

It suffices to show that $\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \mathbf{d}')$ belongs to $\mathbf{QCoh}(S)^{\leq (n+2m)}$. By [GLC1, Proposition 2.3.2], it suffices to show that for every geometric point $s : \mathrm{Spec}(k') \rightarrow S$,

$$s^*(\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \mathbf{d}')) \in \mathbf{Vect}_{k'}^{\leq n+m}.$$

By Corollary D.3.4, we have

$$s^*(\mathcal{H}om_{\mathbf{D}}(\mathbf{d}, \mathbf{d}')) \simeq \mathrm{Hom}_{\mathbf{D}_s}(s^*(\mathbf{d}), s^*(\mathbf{d}')).$$

Hence, it suffices to show that the functor

$$s^* : \mathbf{D} \rightarrow \mathbf{D}_s$$

has cohomological amplitude bounded on the left by m .

We factor the morphism s as

$$\mathrm{Spec}(k') \xrightarrow{s'} S' := \mathrm{Spec}(k') \times_{\mathrm{Spec}(k)} S \rightarrow S.$$

The pullback functor

$$\mathbf{D} \rightarrow \mathbf{D}' := \mathbf{Vect}_{k'} \otimes_{\mathbf{Vect}} \mathbf{D} \simeq \mathbf{QCoh}(S') \otimes_{\mathbf{QCoh}(S)} \mathbf{D}$$

is t-exact. Hence, it suffices to show that the functor

$$(s')^* : \mathbf{D}' \rightarrow \mathbf{D}_s$$

has cohomological amplitude bounded on the left by m .

The morphism s' is a regular closed embedding of codimension m . Hence, we have

$$(s')^* \simeq (s')^! [m]$$

(up to tensoring by a line) where $(s')^!$ is the *right* adjoint to the functor

$$s'_* : \mathbf{D}' \rightarrow \mathbf{D}_s.$$

Since s'_* is t-exact, the functor $(s')^!$ is left t-exact, implying the desired assertion.

$\square$ [Proposition D.2.5]

APPENDIX E. PROOF OF SEMI-INFINITE GEOMETRIC SATAKE

The goal of this section is to prove Theorem 2.4.3.

Like Theorem 2.7.6, in order to prove Theorem 2.4.3 we need to cross the Langlands bridge, and the overall strategy of the proof will be similar to that of Theorem 2.7.6: we will interpret both sides as the category of factorization modules over the same factorization algebra, but in the factorization category $\mathrm{Rep}(\tilde{G}) \otimes \mathrm{Rep}(\tilde{M})$, rather than just $\mathrm{Rep}(\tilde{M})$.

E.1. Another orientation session. As with Sect. B, the material here has a close relationship to existing literature.

The first such theorem is [ABG], which proves a pointwise assertion in the Iwahori context. A factorization theorem was recently obtained in [CR], which used coinvariants and $\mathfrak{T}$ instead of invariants and Ω .

The argument given here is similar to the one in [CR], making the necessary modifications and reducing to Theorem 2.7.6 instead of [Ra4].

There is one additional result that was not previously considered: the identification of the semi-infinite IC sheaf Proposition 2.5.9, which is proved in Sect. E.9.

E.2. The factorization algebra $\Omega(R)^{\text{spec}}$. In this subsection we will interpret the spectral side in Theorem 2.4.3 as the category of factorization modules over a factorization algebra in $\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})$.

E.2.1. Consider the symmetric monoidal categories

$$\text{Rep}(\check{G}) \otimes \text{Rep}(\check{G}), \text{Rep}(\check{G}) \otimes \text{Rep}(\check{P}^-) \text{ and } \text{Rep}(\check{G}) \otimes \text{Rep}(\check{M}),$$

which we turn into factorization categories by the recipe of [GLC2, Sect. B.11.8].

Let $R_{\check{G}}$ be the regular representation of $\check{G}$, viewed as a (commutative) factorization algebra in $\text{Rep}(\check{G}) \otimes \text{Rep}(\check{G})$.

By a slight abuse of notation, we will denote by the same character $R_{\check{G}}$ its image under the restriction functor

$$\text{Rep}(\check{G}) \otimes \text{Rep}(\check{G}) \rightarrow \text{Rep}(\check{G}) \otimes \text{Rep}(\check{P}^-).$$

E.2.2. Consider the (lax unital) factorization functor

$$\text{inv}_{\check{N}_P^-} : \text{Rep}(\check{P}^-) \rightarrow \text{Rep}(\check{M}).$$

Denote

$$\Omega(R)^{\text{spec}} := (\text{Id} \otimes \text{inv}_{\check{N}_P^-})(R_{\check{G}}) \in \text{FactAlg}^{\text{unlt}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})).$$

E.2.3. Our current goal is to construct a (strictly unital) factorization functor

$$\mathbf{F}^{\text{spec}} : \mathbf{I}(\check{G}, \check{P}^-)^{\text{spec, loc}} \rightarrow \Omega(R)^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M}))$$

We first construct a (lax unital) factorization functor

$$\text{pre-}\mathbf{F}^{\text{spec}} : \mathbf{I}(\check{G}, \check{P}^-)^{\text{spec, loc}} \rightarrow \text{Rep}(\check{G}) \otimes \text{Rep}(\check{M}).$$

E.2.4. We define the functor $\text{pre-}\mathbf{F}^{\text{spec}}$ as the $(\text{IndCoh}, *)$ -pushforward functor along the morphism

$$\text{LS}_G^{\text{reg}} \times_{\text{LS}_G^{\text{mer}}} \text{LS}_{\check{P}^-}^{\text{mer}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_M^{\text{reg}} \rightarrow \text{LS}_G^{\text{reg}} \times \text{LS}_M^{\text{reg}}.$$

Explicitly, in terms of (A.2), the functor $\text{pre-}\mathbf{F}^{\text{spec}}$ is given by

$$\text{IndCoh}^*(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-))_{\mathfrak{L}_{\nabla}^+(\check{G}) \times \mathfrak{L}_{\nabla}^+(\check{M})} \xrightarrow{\Gamma(\mathfrak{L}_{\nabla}(\check{G}/\check{N}_P^-), -)^{\text{IndCoh}}} \text{Vect}_{\mathfrak{L}_{\nabla}^+(\check{G}) \times \mathfrak{L}_{\nabla}^+(\check{M})} \simeq \text{Rep}(\check{G}) \otimes \text{Rep}(\check{M}).$$

By construction, the functor $\text{pre-}\mathbf{F}^{\text{spec}}$ is linear with respect to actions of

$$\text{Sph}_G^{\text{spec}} \otimes \text{Sph}_M^{\text{spec}}$$

on the two sides.

E.2.5. Note that we can view the functor $\text{pre-}\mathbf{F}^{\text{spec}}$ also as follows: it corresponds to the functor $J^{-,\text{spec},\text{pre-enh}}$ of (A.13) via the self-duality of $\text{Rep}(\check{G})$.

Consider the composition

$$(E.1) \quad \text{Rep}(\check{G}) \otimes_{\text{Sph}_G^{\text{spec}}} \mathbf{I}(\check{G}, \check{P}^-)^{\text{spec},\text{loc}} \xrightarrow{\text{Id} \otimes \text{pre-}\mathbf{F}^{\text{spec}}} \text{Rep}(\check{G}) \otimes_{\text{Sph}_G^{\text{spec}}} (\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})) \rightarrow \text{Rep}(\check{M}),$$

where the second arrow, i.e.,

$$(E.2) \quad \text{Rep}(\check{G}) \otimes_{\text{Sph}_G^{\text{spec}}} (\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})) \rightarrow \text{Rep}(\check{M}),$$

is induced by the duality pairing

$$\text{Rep}(\check{G}) \otimes_{\text{Sph}_G^{\text{spec}}} \text{Rep}(\check{G}) \rightarrow \text{Vect}.$$

By construction, the functor (E.1) identifies with the functor $J^{-,\text{spec},\text{enh}}$.

Remark E.2.6. Let us view $\mathbf{I}(\check{G}, \check{P}^-)^{\text{spec},\text{loc}}$ as acted on by $\text{Rep}(\check{M})$ via pullback along

$$\text{LS}_G^{\text{reg}} \times_{\text{LS}_G^{\text{mer}}} \text{LS}_{\check{P}^-}^{\text{mer}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_M^{\text{reg}} \rightarrow \text{LS}_M^{\text{reg}}.$$

When we view $\text{pre-}\mathbf{F}^{\text{spec}}$ as a $\text{Sph}_G^{\text{spec}} \otimes \text{Rep}(\check{M})$ -linear functor, it can be uniquely recovered from the $\text{Sph}_G^{\text{spec}}$ -linear functor

$$\text{pre-pre-}\mathbf{F}^{\text{spec}} : \mathbf{I}(\check{G}, \check{P}^-)^{\text{spec},\text{loc}} \rightarrow \text{Rep}(\check{G}),$$

equal to the composition of $\text{pre-}\mathbf{F}^{\text{spec}}$ and

$$(\text{Id} \otimes \text{inv}_M) : \text{Rep}(\check{G}) \otimes \text{Rep}(\check{M}) \rightarrow \text{Rep}(\check{G}).$$

E.2.7. Recall that the functor

$$\iota_*^{\text{IndCoh}} : \text{Rep}(\check{P}^-) \rightarrow \mathbf{I}(\check{G}, \check{P}^-)^{\text{spec},\text{loc}}$$

is strictly unital.

The composite functor

$$\text{pre-}\mathbf{F}^{\text{spec}} \circ \iota_*^{\text{IndCoh}} : \text{Rep}(\check{P}^-) \rightarrow \text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})$$

is the functor of direct image along

$$\text{LS}_{\check{P}^-}^{\text{reg}} \rightarrow \text{LS}_G^{\text{reg}} \times \text{LS}_M^{\text{reg}},$$

i.e., the functor of induction along

$$\check{P}^- \rightarrow \check{G} \times \check{M}.$$

In particular, we obtain that the functor $\text{pre-}\mathbf{F}^{\text{spec}}$ sends the factorization unit to $\Omega(R)^{\text{spec}}$. Hence, the functor $\text{pre-}\mathbf{F}^{\text{spec}}$ upgrades to a functor

$$\mathbf{I}(\check{G}, \check{P}^-)^{\text{spec},\text{loc}} \rightarrow \Omega(R)^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})),$$

which is the sought-for functor $\mathbf{F}^{\text{spec}}$.

E.2.8. As in Sect. B.3.4, the action of $\text{Sph}_G^{\text{spec}} \otimes \text{Sph}_M^{\text{spec}}$ on $\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})$ automatically extends to an action on

$$\Omega(R)^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})).$$

Furthermore, as in Sect. B.3.5, the $\text{Sph}_G^{\text{spec}} \otimes \text{Sph}_M^{\text{spec}}$ -linear structure on $\text{pre-}\mathbf{F}^{\text{spec}}$ automatically extends to a $\text{Sph}_G^{\text{spec}} \otimes \text{Sph}_M^{\text{spec}}$ -linear structure on the functor $\mathbf{F}^{\text{spec}}$.

E.2.9. Note that the image of

$$\mathbf{1}_{\mathrm{Rep}(\check{G})} \otimes \Omega(R)^{\mathrm{spec}} \in \mathrm{FactAlg}^{\mathrm{unl}} \left(\mathrm{Rep}(\check{G}) \underset{\mathrm{Sph}_G^{\mathrm{spec}}}{\otimes} (\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M})) \right)$$

along the functor (E.2) identifies with

$$\Omega^{\mathrm{spec}} \in \mathrm{FactAlg}^{\mathrm{unl}}(\mathrm{Rep}(\check{M})).$$

Hence, the functor (E.1) upgrades to a functor

$$(E.3) \quad \mathrm{Rep}(\check{G}) \underset{\mathrm{Sph}_G^{\mathrm{spec}}}{\otimes} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \rightarrow \Omega^{\mathrm{spec}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{M})).$$

Unwinding the definitions, we obtain that the functor (E.3) identifies with the composition

$$\begin{aligned} \mathrm{Rep}(\check{G}) \underset{\mathrm{Sph}_G^{\mathrm{spec}}}{\otimes} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} &\xrightarrow{(A.19)} \\ &\rightarrow \mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}} \hat{\mathcal{M}})^{\wedge}_{\mathrm{mon-free}})^{\mathrm{Q}_{\mathrm{mon-free}}} \Omega^{\mathrm{spec}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{M})). \end{aligned}$$

E.3. Fully faithfulness on the eventually coconnective part.

E.3.1. Recall that the factorization category $\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$ carries a naturally defined t-structure, Sect. A.7.3

The goal of this subsection is to prove the following key assertion:

Proposition E.3.2. *The functor $\mathbf{F}^{\mathrm{spec}}$ is fully faithful when restricted to the eventually coconnective subcategory of $\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$, i.e., for every affine scheme S , the corresponding functor*

$$(\mathrm{I}(\check{G}, \check{P}^-)_S^{\mathrm{spec}, \mathrm{loc}})^{>-\infty} \rightarrow (\Omega(R)^{\mathrm{spec}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M}))_S)^{>-\infty}$$

is fully faithful.

The rest of this subsection is devoted to the proof of this proposition.

E.3.3. Recall the functor

$$(E.4) \quad \mathrm{Sph}_G^{\mathrm{spec}} \rightarrow R_{\check{G}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{G}))$$

of [GLC2, Equation (E.30)], and recall that it induces an equivalence between the eventually coconnective subcategories of the two sides (by [GLC2, Proposition E.9.4]).

Applying to (E.4) the tensoring-up procedure $- \underset{\mathrm{Rep}(\check{G})}{\otimes} \mathrm{Rep}(\check{P}^-)$, we obtain a functor

$$(E.5) \quad \mathrm{Sph}_G^{\mathrm{spec}} \underset{\mathrm{Rep}(\check{G})}{\otimes} \mathrm{Rep}(\check{P}^-) \rightarrow R_{\check{G}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{P}^-)),$$

which also induces an equivalence between the eventually coconnective subcategories of the two sides, which follows formally from the corresponding property of (E.4) (or, alternatively, by the same argument as in [GLC2, Proposition E.9.4]).

E.3.4. We will think of $\mathrm{Rep}(\check{G}) \underset{\mathrm{Sph}_G^{\mathrm{spec}}}{\otimes} \mathrm{Rep}(\check{P}^-)$ as

$$\mathrm{IndCoh}^*(\mathrm{LS}_{\check{G}}^{\mathrm{reg}} \underset{\mathrm{LS}_G^{\mathrm{mer}}}{\times} \mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}).$$

Denote by ι the map

$$\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}} \rightarrow \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \underset{\mathrm{LS}_M^{\mathrm{mer}}}{\times} \mathrm{LS}_M^{\mathrm{reg}},$$

and also the base-changed map

$$\mathrm{LS}_G^{\mathrm{reg}} \underset{\mathrm{LS}_G^{\mathrm{mer}}}{\times} \mathrm{LS}_{\check{P}^-}^{\mathrm{reg}} \rightarrow \mathrm{LS}_G^{\mathrm{reg}} \underset{\mathrm{LS}_G^{\mathrm{mer}}}{\times} \mathrm{LS}_{\check{P}^-}^{\mathrm{mer}} \underset{\mathrm{LS}_M^{\mathrm{mer}}}{\times} \mathrm{LS}_M^{\mathrm{reg}}.$$

Consider the resulting functor

$$(E.6) \quad \iota_*^{\text{IndCoh}} : \text{IndCoh}^*(\text{LS}_G^{\text{reg}} \times_{\text{LS}_G^{\text{mer}}} \text{LS}_{\check{P}^-}^{\text{reg}}) \rightarrow \text{IndCoh}^*(\text{LS}_G^{\text{reg}} \times_{\text{LS}_G^{\text{mer}}} \text{LS}_{\check{P}^-}^{\text{mer}} \times_{\text{LS}_M^{\text{mer}}} \text{LS}_M^{\text{reg}}).$$

Similar to Sect. A.5.3, one shows that the functor ι_*^{IndCoh} of (E.6) admits a continuous right adjoint, giving rise to a monadic adjunction $(\iota_*^{\text{IndCoh}}, \iota^!)$.

E.3.5. Note that by construction, the following diagram commutes:

$$(E.7) \quad \begin{array}{ccc} \text{Sph}_G^{\text{spec}} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{P}^-) & \xrightarrow{\iota_*^{\text{IndCoh}}} & \text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}} \\ (E.5) \downarrow & & \downarrow \mathbf{F}^{\text{spec}} \\ R_{\check{G}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{P})) & \xrightarrow{\text{Id} \otimes \text{inv}_{\check{N}_{\check{P}}^-}} & \Omega(R)^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})). \end{array}$$

We will prove:

Proposition E.3.6.

- (a) *The bottom horizontal arrow in (E.7) admits a continuous right adjoint, so that the resulting monad on $R_{\check{G}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{P}))$ is left t-exact.*
- (b) *The natural transformation in the diagram*

$$(E.8) \quad \begin{array}{ccc} \text{Sph}_G^{\text{spec}} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{P}^-) & \xleftarrow{\iota^!} & \text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}} \\ \downarrow (E.5) & \searrow & \downarrow \mathbf{F}^{\text{spec}} \\ R_{\check{G}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{P})) & \xleftarrow{\quad} & \Omega(R)^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})), \end{array}$$

obtained from (E.7) by passing to right adjoints along the horizontal arrows, is an isomorphism.

E.3.7. Before we prove Proposition E.3.6, let us show how it implies the statement of Proposition E.3.2.

Point (a) of Proposition E.3.6 implies that the category $\Omega(R)^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M}))$ carries a t-structure, uniquely characterized by the fact that the functor

$$\text{Id} \otimes \text{inv}_{\check{N}_{\check{P}}^-} : R_{\check{G}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{P})) \rightarrow \Omega(R)^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M}))$$

is t-exact.

We will show that the functor $\mathbf{F}^{\text{spec}}$ induces an equivalence between the eventually coconnective subcategories of the two sides.

The functor ι_*^{IndCoh} is t-exact. Hence, it identifies $(\text{I}(\check{G}, \check{P}^-)^{\text{spec, loc}})^{>-\infty}$ with

$$(\iota^! \circ \iota_*^{\text{IndCoh}})\text{-mod} \left(\left(\text{Sph}_G^{\text{spec}} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{P}^-) \right)^{>-\infty} \right).$$

Similarly, the functor $\text{Id} \otimes \text{inv}_{\check{N}_{\check{P}}^-}$ identifies $(\Omega(R)^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})))^{>-\infty}$ with the category of modules over the corresponding monad acting on $(R_{\check{G}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{P})))^{>-\infty}$.

Now, point (b) of Proposition E.3.6 implies that the functor (E.5) intertwines the two monads, i.e., the equivalence of (E.5) extends to an equivalence between the corresponding categories of modules.

□[Proposition E.3.2]

E.4. Proof of Proposition E.3.6.

E.4.1. Let $\Phi : \mathbf{C} \rightarrow \mathbf{D}$ be a (strictly unital) functor between unital lax factorization categories. Assume that Φ admits a right adjoint.⁸³ Then this right adjoint, to be denoted Φ^R , carries a natural (lax unital) factorization structure (see [GLC2, Sect. C.11.21]).

Let $\mathcal{A}$ be a (unital) factorization algebra in $\mathbf{C}$. Then functor Φ induces a (strictly unital) factorization functor between lax factorization categories

$$(E.9) \quad \Phi_{\mathcal{A}} : \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{C}) \rightarrow \Phi(\mathcal{A})\text{-mod}^{\text{fact}}(\mathbf{D}),$$

which makes the diagram

$$(E.10) \quad \begin{array}{ccc} \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{C}) & \xrightarrow{\Phi_{\mathcal{A}}} & \Phi(\mathcal{A})\text{-mod}^{\text{fact}}(\mathbf{D}) \\ \text{oblv}_{\mathcal{A}} \downarrow & & \downarrow \text{oblv}_{\Phi(\mathcal{A})} \\ \mathbf{C} & \xrightarrow{\Phi} & \mathbf{D} \end{array}$$

commute.

We claim that the functor $\Phi_{\mathcal{A}}$ of (E.9) admits a continuous right adjoint, and that the natural transformation in the diagram

$$\begin{array}{ccc} \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{C}) & \xleftarrow{(\Phi_{\mathcal{A}})^R} & \Phi(\mathcal{A})\text{-mod}^{\text{fact}}(\mathbf{D}) \\ \text{oblv}_{\mathcal{A}} \downarrow & \searrow & \downarrow \text{oblv}_{\Phi(\mathcal{A})} \\ \mathbf{C} & \xleftarrow{\Phi^R} & \mathbf{D}, \end{array}$$

obtained from (E.10) by passing to right adjoints along the horizontal arrows, is an isomorphism.

Indeed, it is easy to see that the right adjoint $(\Phi_{\mathcal{A}})^R$ of $\Phi_{\mathcal{A}}$ is the composition of the functor

$$(\Phi^R)_{\Phi(\mathcal{A})} : \Phi(\mathcal{A})\text{-mod}^{\text{fact}}(\mathbf{D}) \rightarrow \Phi^R(\Phi(\mathcal{A}))\text{-mod}^{\text{fact}}(\mathbf{C}),$$

followed by the functor of restriction

$$\Phi^R(\Phi(\mathcal{A}))\text{-mod}^{\text{fact}}(\mathbf{C}) \rightarrow \mathcal{A}\text{-mod}^{\text{fact}}(\mathbf{C})$$

along the homomorphism of factorization algebras

$$\mathcal{A} \rightarrow \Phi^R(\Phi(\mathcal{A})).$$

E.4.2. We will apply the above paradigm as follows. We will start with a (lax unital) factorization functor between unital factorization categories:

$$\Phi' : \mathbf{C} \rightarrow \mathbf{D}',$$

and we will set $\mathbf{D} := \Phi'(\mathbf{1}_{\mathbf{C}})\text{-mod}^{\text{fact}}(\mathbf{D}')$. Let Φ denote the resulting functor $\mathbf{C} \rightarrow \mathbf{D}$.

Note that for any $\mathcal{A} \in \text{FactAlg}^{\text{unl}}(\mathbf{C})$, we have

$$\Phi(\mathcal{A})\text{-mod}^{\text{fact}}(\mathbf{D}) \simeq \Phi'(\mathcal{A})\text{-mod}^{\text{fact}}(\mathbf{D}').$$

E.4.3. We take

$$\mathbf{C} := \text{Rep}(\check{G}) \otimes \text{Rep}(\check{P}^-), \quad \mathbf{D}' := \text{Rep}(\check{G}) \otimes \text{Rep}(\check{M}), \quad \Phi' := \text{Id} \otimes \text{inv}_{\check{N}_P^-} \quad \text{and} \quad \mathcal{A} := R_{\check{G}},$$

so that

$$\Phi'(\mathbf{1}_{\mathbf{C}}) = \mathbf{1}_{\text{Rep}(\check{G})} \otimes \Omega^{\text{spec}}.$$

It follows from Theorem B.6.2 that the functor $\text{Id} \otimes \text{inv}_{\check{N}_P^-}$ preserves compactness, and hence admits a right adjoint.

This implies the existence of a continuous right adjoint in point (a) of Proposition E.3.6.

⁸³Of course, this right adjoint is supposed to be continuous when evaluated on every $S \rightarrow \text{Ran}$.

E.4.4. We now show that the monad in point (a) of Proposition E.3.6 is left t-exact. By Sect. E.4.1, it suffices to show that the monad corresponding to the functor

$$\mathrm{Id} \otimes \mathrm{inv}_{\tilde{N}_P^-} : \mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{P}^-) \rightarrow (\mathbf{1}_{\mathrm{Rep}(\check{G})} \otimes \Omega^{\mathrm{spec}})\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M}))$$

is left t-exact.

It suffices to show that the monad corresponding to the functor

$$\mathrm{inv}_{\tilde{N}_P^-} : \mathrm{Rep}(\check{P}^-) \rightarrow \Omega^{\mathrm{spec}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{M}))$$

is left t-exact.

However, by Theorem B.6.7, the latter adjunction identifies with

$$(\iota)_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}^*(\mathrm{LS}_{\check{P}^-}^{\mathrm{reg}}) \rightleftarrows \mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}\check{M}})^{\wedge}_{\mathrm{mon-free}}) : (\iota)^!,$$

and $(\iota)^! \circ (\iota)_*^{\mathrm{IndCoh}}$ is left t-exact by Proposition A.7.2.

E.4.5. We now prove point (b) of Proposition E.3.6. By Sect. E.4.1, it suffices to show that the natural transformation in the diagram

$$(E.11) \quad \begin{array}{ccc} \mathrm{Sph}_{\check{G}}^{\mathrm{spec}} \otimes_{\mathrm{Rep}(\check{G})} \mathrm{Rep}(\check{P}^-) & \xleftarrow{\iota^!} & \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \\ \downarrow & \searrow & \downarrow \mathrm{pre-F}^{\mathrm{spec}} \\ \mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{P}^-) & \xleftarrow{(\mathbf{1}_{\mathrm{Rep}(\check{G})} \otimes \Omega^{\mathrm{spec}})\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M}))} & \end{array}$$

obtained by passing to right adjoints along the horizontal arrows in the commutative diagram

$$(E.12) \quad \begin{array}{ccc} \mathrm{Sph}_{\check{G}}^{\mathrm{spec}} \otimes_{\mathrm{Rep}(\check{G})} \mathrm{Rep}(\check{P}^-) & \xrightarrow{\iota_*^{\mathrm{IndCoh}}} & \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \\ \downarrow & & \downarrow \mathrm{pre-F}^{\mathrm{spec}} \\ \mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{P}^-) & \xrightarrow{(\mathbf{1}_{\mathrm{Rep}(\check{G})} \otimes \Omega^{\mathrm{spec}})\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M}))} & \end{array}$$

is an isomorphism.

Still equivalently, it suffices to show that the natural transformation in the diagram

$$(E.13) \quad \begin{array}{ccc} \mathrm{Sph}_{\check{G}}^{\mathrm{spec}} \otimes_{\mathrm{Rep}(\check{G})} \mathrm{Rep}(\check{P}^-) & \xleftarrow{\iota^!} & \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \\ \downarrow & \searrow & \downarrow \\ \mathrm{Rep}(\check{P}^-) & \xleftarrow{\Omega^{\mathrm{spec}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{M}))} & \end{array}$$

obtained by passing to right adjoints along the horizontal arrows in the commutative diagram

$$(E.14) \quad \begin{array}{ccc} \mathrm{Sph}_{\check{G}}^{\mathrm{spec}} \otimes_{\mathrm{Rep}(\check{G})} \mathrm{Rep}(\check{P}^-) & \xrightarrow{\iota_*^{\mathrm{IndCoh}}} & \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \\ \downarrow & & \downarrow \\ \mathrm{Rep}(\check{P}^-) & \xrightarrow{\Omega^{\mathrm{spec}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{M}))} & \end{array}$$

is an isomorphism.

E.4.6. Using Theorem B.6.7, we rewrite diagram (E.14) as

$$\begin{array}{ccc}
 \mathrm{IndCoh}^*(\mathrm{LS}_G^{\mathrm{reg}} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{LS}_{P^-}^{\mathrm{reg}}) & \xrightarrow{\iota_*^{\mathrm{IndCoh}}} & \mathrm{IndCoh}^*(\mathrm{LS}_G^{\mathrm{reg}} \times_{\mathrm{LS}_G^{\mathrm{mer}}} \mathrm{LS}_{P^-}^{\mathrm{mer}} \times_{\mathrm{LS}_M^{\mathrm{mer}}} \mathrm{LS}_M^{\mathrm{reg}}) \\
 \downarrow & & \downarrow \\
 \mathrm{IndCoh}^*(\mathrm{LS}_{P^-}^{\mathrm{reg}}) & \xrightarrow{(\iota')_*^{\mathrm{IndCoh}}} & \mathrm{IndCoh}^*((\mathrm{LS}_{P^-}^{\mathrm{mon-free}})^{\wedge}_{\check{M}})_{\mathrm{mon-free}},
 \end{array}$$

in which the vertical arrows are the $(\mathrm{IndCoh}, *)$ -pushforward functors. The required assertion follows now by base change.

E.5. **The functor $\mathbf{F}^{\mathrm{geom}}$.** In this subsection we will map the geometric side in Theorem 2.4.3 to the category of modules over a factorization algebra in $\mathrm{Whit}_*(G) \otimes \mathrm{Whit}^!(M)$.

E.5.1. Recall the functors

$$J_{\mathrm{Whit}}^{-, \mathrm{pre-enh}} : \mathrm{Whit}^!(G) \otimes \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} \rightarrow \mathrm{Whit}^!(M)$$

and

$$J_{\mathrm{Whit}}^{-, \mathrm{enh}} : \mathrm{Whit}^!(G) \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} \rightarrow \mathrm{Whit}^!(M).$$

from (1.45).

By duality, the functor $J_{\mathrm{Whit}}^{-, \mathrm{pre-enh}}$ gives rise to a functor

$$(E.15) \quad \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} \rightarrow \mathrm{Whit}_*(G) \otimes \mathrm{Whit}^!(M),$$

which we will denote by $\mathrm{pre-F}^{\mathrm{geom}}$.

By construction, the functor $\mathrm{pre-F}^{\mathrm{geom}}$ is linear with respect to the actions of $\mathrm{Sph}_G \otimes \mathrm{Sph}_M$ on the two sides.

E.5.2. Consider the composition

$$\begin{aligned}
 (E.16) \quad \mathrm{Whit}^!(G) \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} & \xrightarrow{\mathrm{Id} \otimes \mathrm{pre-F}^{\mathrm{geom}}} \\
 & \rightarrow \mathrm{Whit}^!(G) \otimes_{\mathrm{Sph}_G} (\mathrm{Whit}_*(G) \otimes \mathrm{Whit}^!(M)) \rightarrow \mathrm{Whit}^!(M),
 \end{aligned}$$

where the second arrow, i.e.,

$$(E.17) \quad \mathrm{Whit}^!(G) \otimes_{\mathrm{Sph}_G} (\mathrm{Whit}_*(G) \otimes \mathrm{Whit}^!(M)) \rightarrow \mathrm{Whit}^!(M)$$

is induced by the duality pairing

$$\mathrm{Whit}^!(G) \otimes_{\mathrm{Sph}_G} \mathrm{Whit}_*(G) \rightarrow \mathrm{Vect}.$$

By construction, the functor (E.16) identifies with the functor $J_{\mathrm{Whit}}^{-, \mathrm{enh}}$.

E.5.3. Let us view the two sides of (E.15) as acted on by $\mathrm{Sph}_G \otimes \mathrm{Rep}(\check{M})$ via

$$\mathrm{Sat}_M^{\mathrm{nv}} : \mathrm{Rep}(\check{M}) \rightarrow \mathrm{Sph}_M.$$

The functor $\mathrm{pre-F}^{\mathrm{geom}}$ is recovered uniquely from the Sph_G -linear functor

$$\mathrm{pre-pre-F}^{\mathrm{geom}} : \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} \rightarrow \mathrm{Whit}_*(G),$$

equal to the composition of $\mathrm{pre-F}^{\mathrm{geom}}$ followed by the functor

$$\mathrm{Whit}^!(M) \rightarrow \mathrm{Vect},$$

given by taking the $!$ -fiber at the unit point of the affine Grassmannian.

Note that the functor $\text{pre-pre-F}^{\text{geom}}$ equals the composition

$$(E.18) \quad I(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \xrightarrow{\alpha_{\rho_M(\omega_X), \text{taut}}} I(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X)}) \rightarrow \text{Whit}_*(G).$$

E.5.4. Let

$$\Omega(R) \in \text{FactAlg}^{\text{untl}}(\text{Whit}_*(G) \otimes \text{Whit}^!(M))$$

be the image of the factorization unit in $I(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}}$ along the functor $\text{pre-F}^{\text{geom}}$.

Thus, we obtain that the functor $\text{pre-F}^{\text{geom}}$ lifts to a functor

$$I(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \rightarrow \Omega(R)\text{-mod}^{\text{fact}}(\text{Whit}_*(G) \otimes \text{Whit}^!(M)),$$

which we denote by F^{geom} .

E.5.5. As in Sect. B.3.4, the $\text{Sph}_G \otimes \text{Sph}_M$ -action on $\text{Whit}_*(G) \otimes \text{Whit}^!(M)$ canonically lifts to an action of $\text{Sph}_G \otimes \text{Sph}_M$ on $\Omega(R)\text{-mod}^{\text{fact}}(\text{Whit}_*(G) \otimes \text{Whit}^!(M))$.

Furthermore, as in Sect. B.3.5, the $\text{Sph}_G \otimes \text{Sph}_M$ -linear structure on $\text{pre-F}^{\text{geom}}$ canonically lifts to a $\text{Sph}_G \otimes \text{Sph}_M$ -linear structure on F^{geom} .

E.5.6. Note that the image of

$$\mathbf{1}_{\text{Whit}^!(G)} \otimes \Omega(R) \in \text{FactAlg}^{\text{untl}}\left(\text{Whit}^!(G) \otimes_{\text{Sph}_G} (\text{Whit}_*(G) \otimes \text{Whit}^!(M))\right)$$

along the functor (E.17) identifies canonically with

$$\Omega \in \text{FactAlg}^{\text{untl}}(\text{Whit}^!(M)).$$

Hence, the functor (E.16) upgrades to a functor

$$(E.19) \quad \text{Whit}^!(G) \otimes_{\text{Sph}_G} I(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \rightarrow \Omega\text{-mod}^{\text{fact}}(\text{Whit}^!(M)).$$

Unwinding the construction, we obtain that the functor (E.19) identifies with the functor

$$\text{Whit}^!(G) \otimes_{\text{Sph}_G} I(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \simeq \text{Whit}^!(\mathfrak{L}(G))^{-, \frac{\infty}{2}\text{-Accs}} \xrightarrow{J_{\text{Whit}, \Omega}^{-, \frac{\infty}{2}}} \Omega\text{-mod}^{\text{fact}}(\text{Whit}^!(M)).$$

E.6. Connecting the geometric and spectral sides. We now come to the key point in the proof of Theorem 2.4.3.

E.6.1. Denote

$$\text{FLE}_{\check{G}, \infty, \tau} := \tau_G \circ \text{FLE}_{\check{G}, \infty}, \quad \text{Rep}(\check{G}) \rightarrow \text{Whit}_*(G).$$

Consider the functor

$$\text{FLE}_{\check{G}, \infty, \tau}^{-1} \otimes \text{CS}_{M, \tau} : \text{Whit}_*(G) \otimes \text{Whit}^!(M) \rightarrow \text{Rep}(\check{G}) \otimes \text{Rep}(\check{M}).$$

From Theorem B.5.4 we obtain:

Corollary E.6.2. *The functor $\text{FLE}_{\check{G}, \infty, \tau}^{-1} \otimes \text{CS}_{M, \tau}$ sends*

$$\Omega(R) \mapsto \Omega(R)^{\text{spec}}.$$

E.6.3. Set

$$\begin{aligned} \text{pre-pre-F} &:= \text{FLE}_{\check{G}, \infty, \tau}^{-1} \circ \text{pre-pre-F}^{\text{geom}}, \quad I(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \rightarrow \text{Rep}(\check{G}), \\ \text{pre-F} &:= (\text{FLE}_{\check{G}, \infty, \tau}^{-1} \otimes \text{CS}_{M, \tau}) \circ \text{pre-F}^{\text{geom}}, \quad I(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \rightarrow \text{Rep}(\check{G}) \otimes \text{Rep}(\check{M}). \end{aligned}$$

From Corollary E.6.2 we obtain that the functor pre-F upgrades to a functor

$$\text{F} : I(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \rightarrow \Omega(R)^{\text{spec}}\text{-mod}^{\text{fact}}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})).$$

E.6.4. The functor F is $\text{Sph}_{\check{G}}^{\text{spec}} \otimes \text{Sph}_{\check{M}}^{\text{spec}}$ -linear, where the action on $I(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}}$ is given by

$$\text{Sph}_{\check{G}}^{\text{spec}} \otimes \text{Sph}_{\check{M}}^{\text{spec}} \xrightarrow{\text{Sat}_{\check{G}}^{-1} \otimes \text{Sat}_{\check{M}, \tau}^{-1}} \text{Sph}_G \otimes \text{Sph}_M.$$

E.6.5. Consider the composition

$$(E.20) \quad \mathrm{Whit}^!(G) \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} \xrightarrow{\mathrm{CS}_{G,\tau} \otimes \mathrm{pre-F}} \mathrm{Rep}(\check{G}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} (\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M})) \rightarrow \mathrm{Rep}(\check{M}),$$

where the second arrow is as in (E.2).

From Sect. E.5.2 we obtain that the functor (E.20) identifies with

$$\mathrm{Sat}_{M,\tau} \circ J_{\mathrm{Whit}}^{-,\mathrm{enh}}.$$

E.7. Construction of the functor. In this subsection we will construct the functor $\mathrm{Sat}^{-,\frac{\infty}{2}}$. We will reduce the assertion that it is an equivalence to a pointwise statement.

E.7.1. Recall that the category $\mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}}$ is compactly generated⁸⁴ by objects of the form

$$\mathcal{F} \star_M \mathbf{1}_{\mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}}}, \quad \mathcal{F} \in (\mathrm{Sph}_M)^c.$$

Similarly, the category $\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}$ is compactly generated by objects of the form

$$\mathcal{F} \star_{\check{M}} \mathbf{1}_{\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}}}, \quad \mathcal{F} \in (\mathrm{Sph}_{\check{M}}^{\mathrm{spec}})^c.$$

Moreover, the latter objects are eventually coconnective. In particular, we obtain that

$$(\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}})^c \subset (\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}})^{>-\infty}.$$

E.7.2. Since the functors $\mathbf{F}$ and $\mathbf{F}^{\mathrm{spec}}$ are $\mathrm{Sph}_M^{\mathrm{spec}}$ -linear, we obtain that the essential image under $\mathbf{F}$ of compact objects of $\mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}}$ is contained in the essential image under $\mathbf{F}^{\mathrm{spec}}$ of $(\mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}})^{>-\infty}$.

Using Proposition E.3.2, we obtain that the functor $\mathbf{F}$ canonically lifts to a functor

$$\mathrm{Sat}^{-,\frac{\infty}{2}} : \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} \rightarrow \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}},$$

which makes the diagram

$$(E.21) \quad \begin{array}{ccc} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} & \xrightarrow{\mathrm{Sat}^{-,\frac{\infty}{2}}} & \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \\ & \searrow \mathbf{F} & \swarrow \mathbf{F}^{\mathrm{spec}} \\ & \Omega(R)^{\mathrm{spec}\text{-mod}} \mathrm{fact}(\mathrm{Rep}(\check{G}) \otimes \mathrm{Rep}(\check{M})) & \end{array}$$

commute.

Furthermore, by construction, the functor $\mathrm{Sat}^{-,\frac{\infty}{2}}$ preserves compactness.

E.7.3. The linearity of the functors $\mathbf{F}$ and $\mathbf{F}^{\mathrm{spec}}$ with respect to

$$(\mathrm{Sph}_G^{\mathrm{spec}} \otimes \mathrm{Sph}_M^{\mathrm{spec}})^c$$

implies that the functor $\mathrm{Sat}^{-,\frac{\infty}{2}}$ is also linear with respect to $(\mathrm{Sph}_G^{\mathrm{spec}} \otimes \mathrm{Sph}_M^{\mathrm{spec}})^c$ on compacts.

Hence, we obtain that the functor $\mathrm{Sat}^{-,\frac{\infty}{2}}$ is linear with respect to $\mathrm{Sph}_G^{\mathrm{spec}} \otimes \mathrm{Sph}_M^{\mathrm{spec}}$.

E.7.4. Combining Sects. E.2.5 and E.6.5, we obtain that the functor $\mathrm{Sat}^{-,\frac{\infty}{2}}$ makes the diagram (2.21) commute.

⁸⁴I.e., this holds when evaluated on any affine scheme S mapping to Ran .

E.7.5. Furthermore, it follows from Sects. E.5.6 and E.2.9 that we have a commutative diagram

$$\begin{array}{ccc}
\mathrm{Whit}^!(G) \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} & \xrightarrow{\mathrm{CS}_{G,\tau} \otimes \mathrm{Sat}^{-,\frac{\infty}{2}}} & \mathrm{Rep}(\check{G}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}} \\
\sim \downarrow & & \sim \downarrow \text{(A.19)} \\
\mathrm{Whit}^!(\mathfrak{L}(G))^{-,\frac{\infty}{2}\text{-Accs}} & & \mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}\check{M}})_{\mathrm{mon-free}}^\wedge) \\
J_{\mathrm{Whit},\Omega}^{-,\frac{\infty}{2}} \downarrow & & \downarrow \mathrm{Q}_{\mathrm{mon-free}}^\wedge \\
\Omega\text{-mod}^{\mathrm{fact}}(\mathrm{Whit}^!(M)) & \xrightarrow{\mathrm{Sat}_{M,\tau}} & \Omega^{\mathrm{spec}}\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{M})).
\end{array}$$

Combining with Theorems B.7.6 and B.6.7, we obtain a commutative diagram

$$\begin{array}{ccc}
\mathrm{Whit}^!(G) \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)_{\rho_P(\omega_X)}^{\mathrm{loc}} & \xrightarrow{\mathrm{CS}_{G,\tau} \otimes \mathrm{Sat}^{-,\frac{\infty}{2}}} & \mathrm{Rep}(\check{G}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec},\mathrm{loc}} \\
\sim \downarrow & & \text{(A.19)} \downarrow \sim \\
\mathrm{Whit}^!(\mathfrak{L}(G))^{-,\frac{\infty}{2}\text{-Accs}} & \xrightarrow{\mathrm{CS}_\tau^{-,\frac{\infty}{2}}} & \mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}\check{M}})_{\mathrm{mon-free}}^\wedge).
\end{array}
\tag{E.22}$$

E.7.6. In order to prove Theorem 2.4.3, it remains to show that the functor $\mathrm{Sat}^{-,\frac{\infty}{2}}$ is an equivalence.

Since it preserves compactness, by [GLC2, Proposition 6.2.6], it suffices to show that $\mathrm{Sat}^{-,\frac{\infty}{2}}$ induces an equivalence

$$\mathrm{I}(G, P^-)_{\rho_P(\omega_X),\underline{x}}^{\mathrm{loc}} \rightarrow \mathrm{I}(\check{G}, \check{P}^-)_{\underline{x}}^{\mathrm{spec},\mathrm{loc}}
\tag{E.23}$$

for every geometric point $\underline{x} \in \mathrm{Ran}$.

We will prove this in the next subsection.

E.8. Proof of the pointwise equivalence.

E.8.1. The functor (E.23) preserves compactness, and is essentially surjective on compact generators. Hence, it suffices to show that it is fully faithful on compacts.

E.8.2. Recall the paradigm of [GLC2, Sect. 7.1]. Let

$$(\mathrm{I}(G, P^-)_{\rho_P(\omega_X),\underline{x}}^{\mathrm{loc}})_{\mathrm{temp}} \text{ and } (\mathrm{I}(\check{G}, \check{P}^-)_{\underline{x}}^{\mathrm{spec},\mathrm{loc}})_{\mathrm{temp}}$$

denote the tempered localizations of the two sides, where the temperedness is considered with respect to G (and not M).

First, we claim that the functor

$$(\mathrm{I}(G, P^-)_{\rho_P(\omega_X),\underline{x}}^{\mathrm{loc}})_{\mathrm{temp}} \rightarrow (\mathrm{I}(\check{G}, \check{P}^-)_{\underline{x}}^{\mathrm{spec},\mathrm{loc}})_{\mathrm{temp}},
\tag{E.24}$$

induced by (E.23), is an equivalence.

In order to prove this, by [GLC2, Corollary 7.1.12], it suffices to show that the induced functor

$$\mathrm{Rep}(\check{G}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(G, P^-)_{\rho_P(\omega_X),\underline{x}}^{\mathrm{loc}} \rightarrow \mathrm{Rep}(\check{G}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)_{\underline{x}}^{\mathrm{spec},\mathrm{loc}}
\tag{E.25}$$

is an equivalence.

However, by (E.22), the functor (E.25) identifies with the functor

$$\mathrm{CS}_\tau^{-,\frac{\infty}{2}} : \mathrm{Whit}^!(\mathfrak{L}(G))_{\underline{x}}^{-,\frac{\infty}{2}\text{-Accs}} \rightarrow \mathrm{IndCoh}^*((\mathrm{LS}_{\check{P}^-}^{\mathrm{mon-free}\check{M}})_{\mathrm{mon-free}}^\wedge)_{\underline{x}},$$

and hence is an equivalence by (the pointwise version of) Theorem 2.7.6.

E.8.3. Given that (E.24) is an equivalence, in order to prove that (E.23) is fully faithful on compacts, it suffices to show that both functors

$$(E.26) \quad I(G, P^-)_{\rho_P(\omega_X), \underline{x}}^{\text{loc}} \rightarrow (I(G, P^-)_{\rho_P(\omega_X), \underline{x}}^{\text{loc}})_{\text{temp}}$$

and

$$(E.27) \quad I(\check{G}, \check{P}^-)_{\underline{x}}^{\text{spec, loc}} \rightarrow (I(\check{G}, \check{P}^-)_{\underline{x}}^{\text{spec, loc}})_{\text{temp}}$$

are fully faithful on compacts.

E.8.4. We first prove the assertion concerning (E.27).

At a fixed $\underline{x} \in \text{Ran}$, the category $I(\check{G}, \check{P}^-)_{\underline{x}}^{\text{spec, loc}}$ is

$$\text{IndCoh}(\text{LS}_{\check{G}, \underline{x}}^{\text{reg}} \times_{\text{LS}_{\check{G}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{mer}} \times_{\text{LS}_{\check{M}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{M}, \underline{x}}^{\text{reg}}),$$

where $\text{LS}_{\check{G}, \underline{x}}^{\text{reg}} \times_{\text{LS}_{\check{G}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{mer}} \times_{\text{LS}_{\check{M}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{M}, \underline{x}}^{\text{reg}}$ is an algebraic stack locally almost of finite type.

We claim that the localization (E.27) identifies with

$$\Psi : \text{IndCoh}(\text{LS}_{\check{G}, \underline{x}}^{\text{reg}} \times_{\text{LS}_{\check{G}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{mer}} \times_{\text{LS}_{\check{M}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{M}, \underline{x}}^{\text{reg}}) \rightarrow \text{QCoh}(\text{LS}_{\check{G}, \underline{x}}^{\text{reg}} \times_{\text{LS}_{\check{G}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{mer}} \times_{\text{LS}_{\check{M}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{M}, \underline{x}}^{\text{reg}}).$$

This would imply the fully faithfulness on compacts: indeed $\text{Coh}(-)$ maps fully faithfully to $\text{QCoh}(-)$.

E.8.5. By [GLC2, Corollary 7.1.8], we can identify $(I(\check{G}, \check{P}^-)_{\underline{x}}^{\text{spec, loc}})_{\text{temp}}$ with

$$\text{QCoh}(\text{LS}_{\check{G}, \underline{x}}^{\text{reg}})_{\text{QCoh}((\text{LS}_{\check{G}, \underline{x}}^{\text{mer}})^{\wedge}_{\text{reg}})} \otimes_{\text{Sph}_{\check{G}}^{\text{spec}}} \left(\text{QCoh}(\text{LS}_{\check{G}, \underline{x}}^{\text{reg}}) \otimes_{\text{Sph}_{\check{G}}^{\text{spec}}} I(\check{G}, \check{P}^-)_{\underline{x}}^{\text{spec, loc}} \right),$$

which we further identify with

$$\text{QCoh}(\text{LS}_{\check{G}, \underline{x}}^{\text{reg}})_{\text{QCoh}((\text{LS}_{\check{G}, \underline{x}}^{\text{mer}})^{\wedge}_{\text{reg}})} \otimes_{\text{QCoh}((\text{LS}_{\check{P}^-, \underline{x}}^{\text{mer}})^{\wedge}_{\text{reg}})} \text{IndCoh}((\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}})^{\wedge}_{\text{mon-free}}).$$

Since $(\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}})^{\wedge}_{\text{mon-free}}$ is the formal completion of a smooth algebraic stack, namely, $\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}} \hat{M}$ along $\text{LS}_{\check{P}^-, \underline{x}}^{\text{reg}}$, and

$$\text{LS}_{\check{G}, \underline{x}}^{\text{reg}} \times_{\text{LS}_{\check{G}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}} \hat{M} \quad \text{and} \quad \text{LS}_{\check{P}^-, \underline{x}}^{\text{reg}}$$

coincide as *subsets* of $\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}} \hat{M}$, we obtain that

$$\text{QCoh}(\text{LS}_{\check{G}, \underline{x}}^{\text{reg}})_{\text{QCoh}((\text{LS}_{\check{G}, \underline{x}}^{\text{mer}})^{\wedge}_{\text{reg}})} \otimes_{\text{QCoh}((\text{LS}_{\check{P}^-, \underline{x}}^{\text{mer}})^{\wedge}_{\text{reg}})} \text{IndCoh}((\text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}} \hat{M})^{\wedge}_{\text{mon-free}}) \simeq \text{QCoh}(\text{LS}_{\check{G}, \underline{x}}^{\text{reg}} \times_{\text{LS}_{\check{G}, \underline{x}}^{\text{mer}}} \text{LS}_{\check{P}^-, \underline{x}}^{\text{mon-free}} \hat{M}),$$

as required.

E.8.6. We now prove that (E.26) is fully faithful on compacts. Since we are working at a fixed $\underline{x} \in \text{Ran}$, the category $I(G, P^-)_{\rho_P(\omega_X), \underline{x}}^{\text{loc}}$ is equivalent to

$$\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X), \underline{x}})^{I_P^-, \text{ren}}.$$

We claim that the functor

$$\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X), \underline{x}})^{I_P^-, \text{ren}} \rightarrow \left(\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X), \underline{x}})^{I_P^-, \text{ren}} \right)_{\text{temp}}$$

is fully faithful on the eventually coconnective subcategory (with respect to the usual t-structure).

This would imply the desired assertion, since compact objects in $\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\omega_X), \underline{x}})^{I_P^-, \text{ren}}$ are eventually coconnective.

E.8.7. To prove the claim, it suffices to show that the functor

$$\text{temp} : \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X)\underline{x}}) \rightarrow \left(\text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X)\underline{x}}) \right)_{\text{temp}}$$

is fully faithful on the eventually coconnective subcategory.

To show this, it suffices to prove that for any $\mathcal{F} \in \text{D-mod}_{\frac{1}{2}}(\text{Gr}_{G,\rho(\omega_X)\underline{x}})^c$, the object

$$\text{coFib}(\text{temp}^L \circ \text{temp}(\mathcal{F}) \rightarrow \mathcal{F})$$

is infinitely connective.

We have

$$\text{coFib}(\text{temp}^L \circ \text{temp}(\mathcal{F}) \rightarrow \mathcal{F}) \simeq \mathcal{F} \star \text{coFib}(\text{temp}^L \circ \text{temp}(\delta_{1,\text{Gr}_G}) \rightarrow \delta_{1,\text{Gr}_G}).$$

Since $\mathcal{F}$ is compact, the functor $\mathcal{F} \star -$ has a bounded cohomological amplitude. Hence, it suffices to show that

$$\text{coFib}(\text{temp}^L \circ \text{temp}(\delta_{1,\text{Gr}_G}) \rightarrow \delta_{1,\text{Gr}_G})$$

is infinitely connective.

E.8.8. Since Sat_G is a t-exact equivalence, it suffices to show that

$$\text{coFib}(\text{temp}^L \circ \text{temp}(\mathcal{F}') \rightarrow \mathcal{F}')$$

is infinitely connective for any $\mathcal{F}' \in \text{Sph}_G^{\text{spec}}$.

However,

$$\text{temp}^L \circ \text{temp}(\mathcal{F}') = \Xi \circ \Psi(\mathcal{F}'),$$

for the stack $\text{Hecke}_{G,\underline{x}}^{\text{spec,loc}}$, and the assertion follows.

□[Theorem 2.4.3]

E.9. Identification of the semi-infinite IC sheaf. The goal of this subsection is to prove Proposition 2.5.9.

E.9.1. Consider $\text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}}$ as acted on by $\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})$ via

$$\text{Sat}_G^{\text{nv}} : \text{Rep}(\check{G}) \rightarrow \text{Sph}_G \text{ and } \text{Sat}_{M,\tau}^{\text{nv}} : \text{Rep}(\check{M}) \rightarrow \text{Sph}_M.$$

Let $\overline{\check{G}/\check{N}_P^-}$ be the affine closure of $\check{G}/\check{N}_P^-$ and consider the algebra $\mathcal{O}_{\overline{\check{G}/\check{N}_P^-}}$ of global functions on $\overline{\check{G}/\check{N}_P^-}$ as a commutative algebra equipped with an action of $\check{G} \times \check{M}$. Denote the resulting commutative factorization algebra in $\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})$ by $R_{\overline{\check{G}/\check{N}_P^-}}$.

According to [Ga5, Sect. 5.5.2] (adapted to a general parabolic⁸⁵), we have a canonical action of $R_{\overline{\check{G}/\check{N}_P^-}}$ on

$$\Delta^{-, \frac{\infty}{2}} \in \text{FactAlg} \left(\text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}} \right).$$

Furthermore, the object $\text{IC}_{\text{Ran}}^{-, \frac{\infty}{2}} \in \text{I}(G, P^-)_{\rho_P(\omega_X)}^{\text{loc}}$ identifies with

$$(E.28) \quad R_{\check{G}} \otimes_{R_{\overline{\check{G}/\check{N}_P^-}}} \Delta^{-, \frac{\infty}{2}},$$

where:

- We view $R_{\check{G}}$ as a commutative factorization algebra in $\text{ComAlg}(\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M}))$ via restriction along the homomorphism $\check{M} \rightarrow \check{P}^- \rightarrow \check{G}$;
- The homomorphism $R_{\overline{\check{G}/\check{N}_P^-}} \rightarrow R_{\check{G}}$ is induced by the map

$$\check{G} \rightarrow \check{G}/\check{N}_P^- \rightarrow \overline{\check{G}/\check{N}_P^-}.$$

⁸⁵The case of a general parabolic is considered in [FH].

E.9.2. We consider $I(\check{G}, \check{P}^-)^{\text{spec, loc}}$ as equipped with an action of $\text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})$ via

$$\text{nv} : \text{Rep}(\check{G}) \rightarrow \text{Sph}_{\check{G}}^{\text{spec}} \quad \text{and} \quad \text{nv} : \text{Rep}(\check{M}) \rightarrow \text{Sph}_{\check{M}}^{\text{spec}}.$$

Consider the object

$$\Delta^{-, \text{spec}, \frac{\infty}{2}} \simeq \iota_*^{\text{IndCoh}}(\mathcal{O}_{\text{LS}_{\check{P}^-}^{\text{reg}}}) \in I(\check{G}, \check{P}^-)^{\text{spec, loc}}.$$

It is equipped with an action of $R_{\check{G}/\check{N}_P^-}$, obtained from the homomorphism

$$\mathcal{O}_{\check{G}/\check{N}_P^-} \rightarrow \mathcal{O}_{\check{G}/\check{N}_P^-},$$

where we identify $\mathcal{O}_{\check{G}/\check{N}_P^-} \in \text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})$ with the direct image of $\mathcal{O}_{\text{pt}/\check{P}^-}$ along the projection

$$\text{pt}/\check{P}^- \rightarrow \text{pt}/\check{G} \times \text{pt}/\check{M}.$$

Note that the object $\text{IC}^{-, \text{spec}, \frac{\infty}{2}}$ identifies tautologically with

$$(E.29) \quad R_{\check{G}} \otimes_{R_{\check{G}/\check{N}_P^-}} \Delta^{-, \text{spec}, \frac{\infty}{2}}.$$

E.9.3. Under the equivalence $\text{Sat}^{-, \frac{\infty}{2}}$, the object $\Delta^{-, \frac{\infty}{2}} \simeq \mathbf{1}_{I(\check{G}, \check{P}^-)^{\text{loc}}_{\rho_P(\omega_X)}}$ corresponds to

$$\Delta^{-, \text{spec}, \frac{\infty}{2}} \simeq \mathbf{1}_{I(\check{G}, \check{P}^-)^{\text{spec, loc}}}.$$

Comparing (E.28) and (E.29), we obtain that in order to prove Proposition 2.5.9, it suffices to show that under the above identification, the actions of $R_{\check{G}/\check{N}_P^-}$ on $\Delta^{-, \frac{\infty}{2}}$ and $\Delta^{-, \text{spec}, \frac{\infty}{2}}$ correspond to one another, up to an automorphism of $R_{\check{G}/\check{N}_P^-} \in \text{Rep}(\check{G}) \otimes \text{Rep}(\check{M})$, induced by the action on $\check{G}/\check{N}_P^-$ of an element $z \in Z_{\check{M}}$.

Indeed, the required isomorphism between (E.28) and (E.29) will then be induced by the action of z on $R_{\check{G}}$.

E.9.4. We note that $\Delta^{-, \text{spec}, \frac{\infty}{2}}$ belongs to the heart of the natural t-structure on $I(\check{G}, \check{P}^-)^{\text{spec, loc}}$, i.e., the object $\Delta_{X_{\text{dR}}}^{-, \text{spec}, \frac{\infty}{2}}[-1]$ belongs to heart of the structure on $I(\check{G}, \check{P}^-)^{\text{spec, loc}}_{X_{\text{dR}}}$ (see [GLC2, Sect. B.8.13]).

Hence, the question of two actions of a *classical* commutative factorization algebra being equal can be checked at the level of fibers at closed points $\underline{x} \in \text{Ran}$. By factorization, we can assume that $\underline{x}$ is a singleton $\{x\}$.

I.e., we need to show that the two actions of $\mathcal{O}_{\check{G}/\check{N}_P^-}$ on

$$\mathcal{O}_{\text{pt}/\check{P}^-} \in \text{QCoh}(\text{pt}/\check{P}^-) \simeq \text{Rep}(\check{P}^-)$$

coincide.

E.9.5. Note that for a (classical) commutative algebra A equipped with an action of $\check{G} \times \check{M}$, the datum of an action of A on $\mathcal{O}_{\text{pt}/\check{P}^-}$ is equivalent to the datum of a $\check{G} \times \check{M}$ -equivariant map

$$\check{G}/\check{N}_P^- \rightarrow \text{Spec}(A).$$

We take $A = \mathcal{O}_{\check{G}/\check{N}_P^-}$. We need to show that the map

$$(E.30) \quad \check{G}/\check{N}_P^- \rightarrow \overline{\check{G}/\check{N}_P^-}$$

corresponding to the action of $R_{\check{G}/\check{N}_P^-, x}$ on $\Delta_x^{-, \frac{\infty}{2}}$ is the tautological embedding

$$\check{G}/\check{N}_P^- \hookrightarrow \overline{\check{G}/\check{N}_P^-}.$$

E.9.6. We claim that it is enough to show that the image of (E.30) is contained in $\check{G}/\check{N}_P^- \subset \overline{\check{G}/\check{N}_P^-}$.

Indeed, the set of $\check{G} \times \check{M}$ -equivariant self-maps of $\check{G}/\check{N}_P^-$ is in bijection with $Z_{\check{M}}$.

E.9.7. Suppose, by contradiction that the image of (E.30) is *not* contained in $\check{G}/\check{N}_P^- \subset \overline{\check{G}/\check{N}_P^-}$. Then it is contained in the boundary

$$\overline{\check{G}/\check{N}_P^-} - \check{G}/\check{N}_P^-.$$

However, this means that there exists a highest weight $\lambda \in \Lambda_G^+$ such that the corresponding piece of the Drinfeld-Plücker datum on $\Delta_x^{-, \frac{\infty}{2}}$ (see [Ga5, Sect. 5.4.2])

$$(E.31) \quad V_G^\lambda \star_{\check{G}} \Delta_x^{-, \frac{\infty}{2}} \rightarrow V_M^\lambda \star_{\check{M}} \Delta_x^{-, \frac{\infty}{2}}$$

(here V_G^λ and V_M^λ are irreducible representations of $\check{G}$ and $\check{M}$, respectively) is 0.

However, it follows from the construction of the Drinfeld-Plücker datum on $\Delta_x^{-, \frac{\infty}{2}}$ in [Ga4, Sect.6.3] that the maps (E.31) are all non-zero.

□[Proposition 2.5.9]

E.9.8. *Proof of Proposition 2.6.8.* The proof will proceed along the lines parallel to those of Proposition 2.5.9:

The object $\Delta_{\text{co}}^{-, \text{spec}, \frac{\infty}{2}}$ carries a natural action on $R_{\check{G}/\check{N}_P^-}$, and we have

$$\text{IC}_{\text{co}}^{-, \text{spec}, \frac{\infty}{2}} \simeq R_{\check{G}} \otimes_{R_{\check{G}/\check{N}_P^-}} \Delta_{\text{co}}^{-, \text{spec}, \frac{\infty}{2}}.$$

Hence, in order to prove Proposition 2.6.8, it suffices to show that with respect to the identification

$$(2.29)(\Delta_{\text{co}}^{-, \text{spec}, \frac{\infty}{2}}) \simeq \Delta^{-, \text{spec}, \frac{\infty}{2}}$$

of Sect. 2.6.4, the above action of $R_{\check{G}/\check{N}_P^-}$ on $\Delta_{\text{co}}^{-, \text{spec}, \frac{\infty}{2}}$ corresponds to the action of $R_{\check{G}/\check{N}_P^-}$ on $\Delta^{-, \text{spec}, \frac{\infty}{2}}$ from Sect. E.9.2.

However, this follows from the (essential) uniqueness of such an action proved above.

□[Proposition 2.5.9]

APPENDIX F. PROOF OF THEOREM 6.9.3

Throughout this section, unless otherwise stated, we work at the pointwise level.

F.1. Regular Kac-Moody representations.

F.1.1. Recall that we have a DG category $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}$ of Kac-Moody representations at critical level constructed in [FG2].

This DG category has a t-structure with heart $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^\heartsuit$ the subcategory of $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G}^\heartsuit$ consisting of Kac-Moody representations on which the center $\mathfrak{Z}$ of the (completed, critically-twisted) enveloping algebra acts through its quotient $\mathfrak{z} = \text{Image}(\mathfrak{Z} \rightarrow \text{End}(\mathbb{V}_{\mathfrak{g}, \text{crit}_G}))$ as in [GLC2, Sect. 4.6.1]. There is a canonical action of $\text{QCoh}(\text{Op}_G^{\text{reg}})$ on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}$, where we are using the Feigin-Frenkel isomorphism $\mathfrak{z} \simeq \mathcal{O}_{\text{Op}_G^{\text{reg}}}$.

According to [RY], there is an action of $\mathfrak{L}(G)$ on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}$ compatible with the forgetful functor to $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G}$ and commuting with the action of $\text{QCoh}(\text{Op}_G^{\text{reg}})$.

Remark F.1.2. The existence of this action is subtle and is related to the localization theorem of [RY] (cf. below). However, in practice, we only will be interested in the bounded below category $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^+$, where one can use a substitute as in [RY].

Moreover, we also will only be interested in Iwahori invariants in $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}$, where the localization theorem itself is more classical, giving another way to circumvent these issues.

This is to say: we are referencing [RY] for the existence of this action because the reference exists and the resulting formalism is convenient, not because we actually need the full strength of these results.

F.1.3. *The localization theorem.* We now recall:

Theorem F.1.4 ([RY]). *There is a canonical t -exact equivalence*

$$\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_G) \otimes_{\mathrm{Rep}(\check{G})} \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{reg}}) \simeq \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \mathrm{reg}}$$

of $\mathfrak{L}(G)$ -categories.

We will only need the following consequence due to [FG3].

Corollary F.1.5. *There is a canonical t -exact equivalence*

$$\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_G)^{I_P} \otimes_{\mathrm{Rep}(\check{G})} \mathrm{QCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{reg}}) \simeq \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \mathrm{reg}}^{I_P}$$

of categories acted on by the $(\mathfrak{L}(G), I_P)$ -Hecke category.

F.2. Transverse Kac-Moody representations.

F.2.1. Below, we will define a certain subcategory

$$(\widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \mathrm{reg}}^{I_P})_{\mathrm{trans}} \subset \widehat{\mathfrak{g}}\text{-mod}_{\mathrm{crit}_G, \mathrm{reg}}^{I_P}.$$

We will denote the embedding of this subcategory by j_* . The functor j_* will admit a left adjoint j^* . Geometrically, one is meant to picture j as the embedding of the open Bruhat cell into the flag variety.

First, we define a related category

$$\left(\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_G)^{I_P} \otimes_{\mathrm{Rep}(\check{G})} \mathrm{Rep}(\check{B}) \right)_{\mathrm{trans}} \subset \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_G)^{I_P} \otimes_{\mathrm{Rep}(\check{G})} \mathrm{Rep}(\check{B}).$$

Here $\mathrm{Rep}(\check{G})$ acts on $\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_G)$ via naive Satake.

F.2.2. Namely, note that the $*$ -averaging functor

$$\mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)} \rightarrow \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_G)^{I_P}$$

is an equivalence, similar to the proof of Lemma 3.7.4. At the pointwise level, we can apply Theorem 2.4.3 to identify

$$(F.1) \quad \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M), \mathrm{ren}} \simeq \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \simeq \mathrm{IndCoh}(\mathrm{pt} / \check{G} \times_{\check{\mathfrak{g}}/\check{G}} \check{\mathfrak{n}}_P^- / \check{P}^-).$$

Here we implicitly trivialize ω_X in a neighborhood of our point, allowing us to omit the $\rho_P(\omega_X)$ -twist on the first term.

As in [AG, Sect. 12], we obtain a similar equivalence

$$(F.2) \quad \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_G)^{I_P} \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{pt} / \check{G} \times_{\check{\mathfrak{g}}/\check{G}} \check{\mathfrak{n}}_P^- / \check{P}^-)$$

from the corresponding equivalence with $\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)$ -invariants instead of I_P .

We then obtain

$$(F.3) \quad \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_G)^{I_P} \otimes_{\mathrm{Rep}(\check{G})} \mathrm{Rep}(\check{B}) \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\check{\mathfrak{n}}_P^- / \check{P}^- \times \mathrm{pt} / \check{B}).$$

F.2.3. There is a canonical map from $\check{\mathfrak{n}}_P^-/\check{P}^- \times_{\check{\mathfrak{g}}/\check{G}} \text{pt}/\check{B}$ to $\check{P}^- \setminus \check{G}/\check{B}$. We let

$$(\check{\mathfrak{n}}_P^-/\check{P}^- \times_{\check{\mathfrak{g}}/\check{G}} \text{pt}/\check{B})_{\text{trans}} = (0 \times_{\check{\mathfrak{g}}} \check{\mathfrak{n}}_P^-)/\check{B}_{\check{M}}$$

denote the preimage of the open Bruhat cell (where $\check{B}(M) \subset \check{M} \subset \check{P}^-$ is the Borel subgroup of $\check{M}$).

We then define

$$\left(\text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}) \right)_{\text{trans}}$$

as the subcategory corresponding to

$$\text{IndCoh}_{\text{Nilp}}((\check{\mathfrak{n}}_P^-/\check{P}^- \times_{\check{\mathfrak{g}}/\check{G}} \text{pt}/\check{B})_{\text{trans}})$$

under (F.3).

F.2.4. We clearly have the desired functors j_* and j^* in this case – we will use the same notation in the setting of either

$$\text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}) \text{ or } \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}.$$

F.2.5. Next, note that the canonical map $\text{Op}_G^{\text{reg}} \rightarrow \text{pt}/\check{G}$ factors canonically through $\text{pt}/\check{B}$. Therefore, we can write

$$\begin{aligned} \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P} &\simeq \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P} \otimes_{\text{Rep}(\check{G})} \text{QCoh}(\text{Op}_G^{\text{reg}}) \simeq \\ &\simeq \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}) \otimes_{\text{Rep}(\check{B})} \text{QCoh}(\text{Op}_G^{\text{reg}}). \end{aligned}$$

We then define $(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P})_{\text{trans}}$ as the subcategory corresponding to

$$\left(\text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}) \right)_{\text{trans}} \otimes_{\text{Rep}(\check{B})} \text{QCoh}(\text{Op}_G^{\text{reg}})$$

under the above equivalence.

F.3. Wakimoto modules as objects of the transverse subcategory. We now reduce the proof of Theorem 6.9.3 to a series of intermediate results.

F.3.1. In what follows, we let⁸⁶

$$\mathbb{W}_{\mathfrak{g}, \text{crit}} := \text{Wak}(\text{Vac}(M)_{\text{crit}_M - \check{\rho}_P}).$$

In a similar spirit, we let

$$\mathbb{V}_{\mathfrak{g}, \text{crit}} := \text{Vac}(G)_{\text{crit}_G}.$$

We remark that $\mathbb{W}_{\mathfrak{g}, \text{crit}}$ receives a map of factorization algebras from $\mathbb{V}_{\mathfrak{g}, \text{crit}}$; it follows formally that $\mathbb{W}_{\mathfrak{g}, \text{crit}}$ lies in $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P, \heartsuit} \subset \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G}^{I_P, \heartsuit}$.

F.3.2. First, we claim:

Theorem F.3.3. $\mathbb{W}_{\mathfrak{g}, \text{crit}}$ lies in $(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P})_{\text{trans}}$.

F.3.4. It follows from Theorem F.3.3 that the canonical map

$$\mathbb{V}_{\mathfrak{g}, \text{crit}} \rightarrow \mathbb{W}_{\mathfrak{g}, \text{crit}}$$

induces a map

$$(F.4) \quad j^*(\mathbb{V}_{\mathfrak{g}, \text{crit}}) \rightarrow \mathbb{W}_{\mathfrak{g}, \text{crit}} \in (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P})_{\text{trans}}$$

or equivalently

$$j_* \circ j^*(\mathbb{V}_{\mathfrak{g}, \text{crit}}) \rightarrow \mathbb{W}_{\mathfrak{g}, \text{crit}} \in \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P}.$$

⁸⁶Unlike in Sect. 3.6.3, we do not distinguish between the object of $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P, \heartsuit}$ and the underlying object in $\text{Vect}^{\heartsuit}$.

F.3.5. Next, we claim:

Theorem F.3.6. *The map (F.4) is an isomorphism.*

Remark F.3.7. In the Borel case, Theorem F.3.6 amounts to the assertion of [FG1, Theorem 4.11]. The proof we give below is similar in spirit, but has some changes to more easily adapt to the general parabolic case.

F.3.8. Let us now deduce Theorem 6.9.3 from Theorem F.3.6.

At the pointwise level, there are canonical equivalences

$$(F.5) \quad \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \simeq \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mer}} \times_{\mathrm{LS}_G^{\mathrm{mer}}} (\check{\mathfrak{n}}_P^-)^\wedge / \check{P}^-) \simeq \\ \simeq \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mer}} \times_{\mathrm{LS}_G^{\mathrm{mer}}} (\check{\mathfrak{n}}_P^-)^\wedge / \check{P}^-).$$

Here the last isomorphism comes as in [GLC2, Proposition 3.8.7].

Under the equivalence

$$\mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)^{\mathrm{loc}} \stackrel{(5.18)}{\simeq} \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}}) \otimes_{\mathrm{Sph}_G^{\mathrm{spec}}} \mathrm{I}(\check{G}, \check{P}^-)^{\mathrm{spec}, \mathrm{loc}} \stackrel{(F.5)}{\simeq} \\ \simeq \mathrm{QCoh}(\mathrm{Op}_G^{\mathrm{mer}} \times_{\mathrm{LS}_G^{\mathrm{mer}}} (\check{\mathfrak{n}}_P^-)^\wedge / \check{P}^-)$$

the image of the $\mathbb{V}_{\mathfrak{g}, \mathrm{crit}}$ under

$$\mathrm{KL}(G)_{\mathrm{crit}_G} \rightarrow \mathrm{KL}(G)_{\mathrm{crit}_G} \otimes_{\mathrm{Sph}_G} \mathrm{I}(G, P^-)^{\mathrm{loc}}$$

maps to the structure sheaf of

$$\mathrm{Op}_G^{\mathrm{reg}} \times_{\mathrm{LS}_G^{\mathrm{reg}}} \mathrm{pt} / \check{P}^-.$$

It follows by functoriality that $j_* \circ j^*(\mathbb{V}_{\mathfrak{g}, \mathrm{crit}})$ maps to the structure sheaf of

$$\mathrm{Op}_G^{\mathrm{reg}} \times_{\mathrm{LS}_G^{\mathrm{reg}}} \mathrm{pt} / \check{P}^- \times_{\check{B} \backslash \check{G} / \check{P}^-} \mathrm{pt} / \check{B}(M) = \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{reg}}.$$

This is the same as $\mathrm{co}J_{\mathrm{Op}^{\mathrm{reg}}_{\check{M}, \check{P}^-}}(\mathcal{O}_{\mathrm{Op}^{\mathrm{reg}}_{\check{M}, \check{P}^-}})$ by construction. Therefore, the existence of an isomorphism as in Theorem 6.9.3 follows from Theorem F.3.6.

The fact that this isomorphism satisfies (i) from Theorem 6.9.3 is obvious from the constructions above. It remains to see that this isomorphism satisfies (ii) from Theorem 6.9.3. We do so below.

F.3.9. We consider the following two objects of $\mathrm{Rep}(\check{M})$.

On the one hand, we consider the tautological $\check{M}$ -torsor $\mathcal{P}_{\check{M}, \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{reg}}}$ on $\mathrm{MOp}_{\check{G}, \check{P}^-}$, and we let $\mathcal{O}_{\mathcal{P}_{\check{M}, \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{reg}}}} \in \mathrm{Rep}(\check{M})$ be the space of functions on it, viewed as an $\check{M}$ -representation.

On the other hand, we consider the object $\mathcal{O}'_{\mathcal{P}_{\check{M}, \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{reg}}}}$ equal to the image of $\mathrm{Wak}^{\mathrm{enh}}(\mathrm{Vac}_{\mathrm{crit}_M - \check{\rho}_P})$ along the composition (6.42) $\circ$ (5.18). Note that we can rewrite $\mathcal{O}'_{\mathcal{P}_{\check{M}, \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{reg}}}}$ as

$$\mathrm{DS} \circ \mathrm{Wak}(\mathrm{Sat}_M^{\mathrm{nv}}(R_M) \star_M \mathrm{Vac}_{\mathrm{crit}_M - \check{\rho}_P}).$$

We need to compare two isomorphisms between $\mathcal{O}_{\mathcal{P}_{\check{M}, \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{reg}}}}$ and $\mathcal{O}'_{\mathcal{P}_{\check{M}, \mathrm{MOp}_{\check{G}, \check{P}^-}^{\mathrm{reg}}}}$.

The first isomorphism η_1 comes from the commutativity used in Theorem 6.9.3 (ii), i.e., it traces back to (6.24).

The second isomorphism η_2 comes using the calculation of the image of $\mathbb{W}_{\mathfrak{g}, \mathrm{crit}}$ under (5.18) given in Sect. F.3.8.

It suffices to show that $\mathbf{oblv}_{\check{M}}(\eta_1) = \mathbf{oblv}_{\check{M}}(\eta_2)$ as morphism in $\mathbf{Vect}$, where is the forgetful functor $\mathbf{oblv}_{\check{M}} : \mathbf{Rep}(\check{M}) \rightarrow \mathbf{Vect}$.

F.3.10. Unwinding the construction, we obtain that the two induced maps

$$\mathbf{oblv}_{\check{M}}(\mathcal{O}_{\check{M}, \mathbf{MOP}_{\check{G}, \check{P}^-}^{\text{reg}}}^{\mathcal{P}}) \rightrightarrows \mathbf{oblv}_{\check{M}}(\mathcal{O}_{\check{M}, \mathbf{MOP}_{\check{G}, \check{P}^-}^{\text{reg}}}^{\mathcal{P}'})$$

coincide when precomposed with

$$\mathcal{O}_{\mathbf{Op}_G^{\text{reg}}} \rightarrow \mathbf{oblv}_{\check{M}}(\mathcal{O}_{\check{M}, \mathbf{MOP}_{\check{G}, \check{P}^-}^{\text{reg}}}^{\mathcal{P}}).$$

In particular, we obtain that $\mathbf{oblv}_{\check{M}}(\eta_1)$ and $\mathbf{oblv}_{\check{M}}(\eta_2)$ are equal evaluated on the vacuum vector $1 \in \mathcal{O}_{\check{M}, \mathbf{MOP}_{\check{G}, \check{P}^-}^{\text{reg}}}^{\mathcal{P}}$.

In addition, both $\mathbf{oblv}_{\check{M}}(\eta_1)$ and $\mathbf{oblv}_{\check{M}}(\eta_2)$ are $\mathbf{oblv}_{\check{M}}(\mathcal{O}_{\check{M}, \mathbf{MOP}_{\check{G}, \check{P}^-}^{\text{reg}}}^{\mathcal{P}})$ -linear, where:

- $\mathcal{O}_{\check{M}, \mathbf{MOP}_{\check{G}, \check{P}^-}^{\text{reg}}}^{\mathcal{P}}$ acts on itself naturally;
- $\mathcal{O}_{\check{M}, \mathbf{MOP}_{\check{G}, \check{P}^-}^{\text{reg}}}^{\mathcal{P}}$ acts on $\mathcal{O}_{\check{M}, \mathbf{MOP}_{\check{G}, \check{P}^-}^{\text{reg}}}^{\mathcal{P}'}$ via:
 - The identification $\mathbf{MOP}_{\check{G}, \check{P}^-}^{\text{reg}} \simeq \mathbf{Op}_{\check{M}, \check{\rho}_P}^{\text{reg}}$ as schemes mapping to $\text{pt}/\check{M}$;
 - The action of ${}^{\text{st}}\mathcal{O}_{\check{M}, \mathbf{Op}_{\check{M}, \check{\rho}_P}^{\text{reg}}}$ on $\text{Sat}_{\check{M}}^{\text{bv}}(R_{\check{M}}) \star \text{Vac}_{\text{crit}_M - \check{\rho}_P}$ (as an object of $\text{KL}(M)_{\text{crit}_M - \check{\rho}_P}$, equipped with an action of $\check{M}$).

The above two properties combined imply that $\mathbf{oblv}_{\check{M}}(\eta_1) = \mathbf{oblv}_{\check{M}}(\eta_2)$.

F.4. Proof of Theorem F.3.3.

F.4.1. Note that Sph_M acts on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{\mathfrak{L}(N_{P^-}) \cdot \mathfrak{L}^+(M)}$, and hence on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P}$. This induces an action of $\mathbf{Rep}(\check{M})$ on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P}$ via naive Satake (for M). We will denote this action as

$$(\mathcal{M} \in \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P}, W \in \mathbf{Rep}(\check{M})) \mapsto \mathcal{M} \star_M W.$$

F.4.2. For $V \in \mathbf{Rep}(\check{G})^\vee$, let $\mathcal{V}_{\mathbf{Op}_G^{\text{reg}}}$ denote the corresponding vector bundle on $\mathbf{Op}_G^{\text{reg}}$.

By (F.2), there is a canonical action of $\mathbf{Rep}(\check{P}^-)$ on $\text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P}$, and hence on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P}$. The underlying action of $\mathbf{Rep}(\check{G})$ is given by the action of $\text{QCoh}(\mathbf{Op}_G^{\text{reg}})$, while the underlying action of $\mathbf{Rep}(\check{M})$ is given by Sect. F.4.1.

It follows that there are natural maps

$$(F.6) \quad \mathcal{V}_{\mathbf{Op}_G^{\text{reg}}}^\lambda \otimes_{\mathcal{O}_{\mathbf{Op}_G^{\text{reg}}}} \mathcal{M} \rightarrow \mathcal{M} \star_M W^\lambda, \quad \mathcal{M} \in \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P}.$$

where V^λ is the highest weight representation of $\check{G}$ and $W^\lambda = (V^\lambda)_{\check{N}_{P^-}}$ is the highest weight representation of $\check{M}$ for a given dominant coweight λ . The above maps satisfy Plücker relations in the natural sense.

F.4.3. Since the $\check{G}$ -bundle on $\mathbf{Op}_G^{\text{reg}}$ is equipped with a reduction to $\check{B}$, there are canonical line bundles $\mathcal{L}_{\mathbf{Op}_G^{\text{reg}}}^\lambda$ ($\lambda \in \Lambda$) on $\mathbf{Op}_G^{\text{reg}}$ with maps

$$(F.7) \quad \mathcal{L}_{\mathbf{Op}_G^{\text{reg}}}^\lambda \rightarrow \mathcal{V}_{\mathbf{Op}_G^{\text{reg}}}^\lambda, \quad \lambda \in \Lambda_G^+$$

satisfying Plücker relations.

⁸⁷Here $\mathcal{P}_{\check{M}, \mathbf{Op}_{\check{M}, \check{\rho}_P}^{\text{reg}}}$ denotes the tautological $\check{M}$ -torsor on $\mathbf{Op}_{\check{M}, \check{\rho}_P}^{\text{reg}}$.

F.4.4. Unwinding the definitions, we obtain that an object $\mathcal{M} \in \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P}$ lies in the transverse subcategory if and only if the following conditions hold:

For every $\lambda \in \Lambda_G^+ \cap \Lambda_{G,P}$ (so that W^λ is 1-dimensional), the composition

$$\mathcal{L}_{\text{Op}_G^{\text{reg}}}^\lambda \otimes_{\mathcal{O}_{\text{Op}_G^{\text{reg}}}} \mathcal{M} \rightarrow \mathcal{V}_{\text{Op}_G^{\text{reg}}}^\lambda \otimes_{\mathcal{O}_{\text{Op}_G^{\text{reg}}}} \mathcal{M} \rightarrow \mathcal{M} \star_M W^\lambda$$

is an isomorphism, where:

- The first arrow comes from (F.7);
- The second arrow comes from (F.6).

F.4.5. Recall now that the action of $\mathcal{O}_{\text{Op}_G^{\text{reg}}}$ on $\mathbb{W}_{\mathfrak{g}, \text{crit}}$ extends to an action of $\mathcal{O}_{\text{MOp}_{G, \check{P}^-}^{\text{reg}}}$.

For $W \in \text{Rep}(\check{P}^-)^\heartsuit$, let $\mathcal{W}_{\text{MOp}_{G, \check{P}^-}^{\text{reg}}}$ denote the induced vector bundle on $\text{MOp}_{G, \check{P}^-}^{\text{reg}}$.

By “birth of opers” for $\check{M}$ (see [GLC2, Sect. 5.2]), we have canonical isomorphisms

$$\begin{aligned} \text{(F.8)} \quad \mathbb{W}_{\mathfrak{g}, \text{crit}} \star_M W &:= \text{Wak}(\text{Vac}(M)_{\text{crit}_M - \check{\rho}_P}) \star_M W := \text{Wak} \left(\text{Vac}(M)_{\text{crit}_M - \check{\rho}_P} \star_M \text{Sat}_M^{\text{nv}}(W) \right) \simeq \\ &\simeq \text{Wak} \left(\text{Vac}(M)_{\text{crit}_M - \check{\rho}_P} \otimes_{\mathcal{O}_{\text{Op}_M^{\text{reg}}}} \mathcal{W}_{\text{Op}_M^{\text{reg}}} \right) \simeq \mathcal{W}_{\text{MOp}_{G, \check{P}^-}^{\text{reg}}} \otimes_{\mathcal{O}_{\text{MOp}_{G, \check{P}^-}^{\text{reg}}}} \mathbb{W}_{\mathfrak{g}, \text{crit}}, \quad W \in \text{Rep}(\check{M}). \end{aligned}$$

F.4.6. Note that there are natural maps

$$\text{(F.9)} \quad \mathcal{V}_{\text{Op}_G^{\text{reg}}}^\lambda|_{\text{MOp}_{G, \check{P}^-}^{\text{reg}}} \rightarrow \mathcal{W}_{\text{MOp}_{G, \check{P}^-}^{\text{reg}}}^\lambda,$$

where $\mathcal{W}_{\text{MOp}_{G, \check{P}^-}^{\text{reg}}}^\lambda$ is the vector bundle corresponding to $W^\lambda \in \text{Rep}(\check{M}) \rightarrow \text{Rep}(\check{P}^-)$, $\lambda \in \Lambda_G^+$.

Now [FG1, Proposition 2.6.1] asserts that these isomorphisms are compatible with the natural Drinfeld-Plücker structures on both sides, i.e., the diagrams

$$\begin{array}{ccc} \mathcal{V}_{\text{Op}_G^{\text{reg}}}^\lambda \otimes_{\mathcal{O}_{\text{Op}_G^{\text{reg}}}} \mathbb{W}_{\mathfrak{g}, \text{crit}} & \xrightarrow{\text{(F.6)}} & \mathbb{W}_{\mathfrak{g}, \text{crit}} \star_M W^\lambda \\ \sim \downarrow & & \sim \downarrow \text{(F.8)} \\ \mathcal{V}_{\text{Op}_G^{\text{reg}}}^\lambda|_{\text{MOp}_{G, \check{P}^-}^{\text{reg}}} \otimes_{\mathcal{O}_{\text{MOp}_{G, \check{P}^-}^{\text{reg}}}} \mathbb{W}_{\mathfrak{g}, \text{crit}} & \xrightarrow{\text{(F.9)}} & \mathcal{W}_{\text{MOp}_{G, \check{P}^-}^{\text{reg}}} \otimes_{\mathcal{O}_{\text{MOp}_{G, \check{P}^-}^{\text{reg}}}} \mathbb{W}_{\mathfrak{g}, \text{crit}} \end{array}$$

commute.

This makes it manifest that $\mathbb{W}_{\mathfrak{g}, \text{crit}}$ satisfies the condition from Sect. F.4.4, since the maps (F.9) have the required property.

Remark F.4.7. Formally, [FG1, Proposition 2.6.1] only deals with the Borel case. However, it is shown in [FG1, Sect. 2.8] that Proposition 2.6.1 implies its parallel version for the parabolic being the whole group; similar reasoning shows that the Borel case implies the general parabolic case.

F.5. Proof of Theorem F.3.6.

F.5.1. We will prove the following two results.

Lemma F.5.2. *The canonical map $j^*(\mathbb{V}_{\mathfrak{g}, \text{crit}}) \rightarrow \mathbb{W}_{\mathfrak{g}, \text{crit}}$ is an isomorphism after applying the functor*

$$(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P})_{\text{trans}} \xrightarrow{j^*} \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P} \rightarrow \widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G} \xrightarrow{\text{DS}} \text{Vect}.$$

Lemma F.5.3. *The functor DS is conservative when restricted to the eventually coconnective transverse subcategory*

$$(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P})_{\text{trans}}^{>-\infty} := (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P})_{\text{trans}} \cap (\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P})^{>-\infty}.$$

F.5.4. Clearly the combination of these two lemmas yields Theorem F.3.6.

F.6. Proof of Lemma F.5.2.

F.6.1. *More on the transverse subcategory.* We now need to look closer at the transverse subcategory and the functor j^* .

Note that the open cell $(\check{B} \backslash \check{G} / \check{P}^-)_{\text{trans}}$ is isomorphic to $\check{B}(M)$. Therefore, the functor

$$j^* : \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{IP} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}) \rightarrow \left(\text{D-mod}_{\text{crit}}(\text{Gr}_G)^{IP} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}) \right)_{\text{trans}}$$

factors as

$$\begin{array}{ccc} \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{IP} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}) & \xrightarrow{\pi^*} & \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{IP} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}(M)) \\ & & \downarrow \tilde{j}^* \\ & & \left(\text{D-mod}_{\text{crit}}(\text{Gr}_G)^{IP} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}) \right)_{\text{trans}}. \end{array}$$

Each of the functors π^* and $\tilde{j}^*$ admit right adjoints, which we denote as π_* and $\tilde{j}_*$ respectively.

We obtain a canonically defined natural transformation

$$(F.10) \quad \pi_* \circ \pi^* \rightarrow \pi_* \circ \tilde{j}_* \circ \tilde{j}^* \circ \pi^* = j_* \circ j^*.$$

Remark F.6.2. On the spectral side, π^* and $\tilde{j}^*$ correspond to the maps

$$\text{pt} / \check{B}(M) \xrightarrow{\tilde{j}} \check{P}^- \backslash \check{G} / \check{B}(M) \xrightarrow{\pi} \check{P}^- \backslash \check{G} / \check{B}.$$

F.6.3. Note that we have a Cartesian square

$$(F.11) \quad \begin{array}{ccc} \text{MOP}_{\check{G}, \check{P}^-}^{\text{reg}} & \longrightarrow & \text{pt} / \check{B}(M) \\ \text{p}^{\text{Miu}} \downarrow & & \downarrow \\ \text{OP}_{\check{G}}^{\text{reg}} & \longrightarrow & \text{pt} / \check{B}. \end{array}$$

From here, we obtain that the endofunctor $\pi_* \circ \pi^*$ (on either side of (F.11)) corresponds to

$$(-) \otimes_{\mathcal{O}_{\text{OP}_{\check{G}}^{\text{reg}}}} \mathcal{O}_{\text{MOP}_{\check{G}, \check{P}^-}^{\text{reg}}}.$$

Therefore, the natural transformation (F.10) corresponds to a map

$$(F.12) \quad \mathbb{V}_{\mathfrak{g}, \text{crit}} \otimes_{\mathcal{O}_{\text{OP}_{\check{G}}^{\text{reg}}}} \mathcal{O}_{\text{MOP}_{\check{G}, \check{P}^-}^{\text{reg}}} \rightarrow j_* \circ j^*(\mathbb{V}_{\mathfrak{g}, \text{crit}})$$

whose composition with

$$\mathbb{V}_{\mathfrak{g}, \text{crit}} \rightarrow \mathbb{V}_{\mathfrak{g}, \text{crit}} \otimes_{\mathcal{O}_{\text{OP}_{\check{G}}^{\text{reg}}}} \mathcal{O}_{\text{MOP}_{\check{G}, \check{P}^-}^{\text{reg}}}$$

is the unit map for the adjunction (j^*, j_*) .

Lemma F.6.4. *The composition*

$$\mathbb{V}_{\mathfrak{g}, \text{crit}} \otimes_{\mathcal{O}_{\text{OP}_{\check{G}}^{\text{reg}}}} \mathcal{O}_{\text{MOP}_{\check{G}, \check{P}^-}^{\text{reg}}} \rightarrow j_* \circ j^*(\mathbb{V}_{\mathfrak{g}, \text{crit}}) \rightarrow \mathbb{W}_{\mathfrak{g}, \text{crit}}$$

is the canonical map coming from the map $\mathbb{V}_{\mathfrak{g}, \text{crit}} \rightarrow \mathbb{W}_{\mathfrak{g}, \text{crit}}$ and the canonical action of $\mathcal{O}_{\text{MOP}_{\check{G}, \check{P}^-}^{\text{reg}}}$ on $\mathbb{W}_{\mathfrak{g}, \text{crit}}$ extending the action of $\mathcal{O}_{\text{OP}_{\check{G}}^{\text{reg}}}$.

This follows again from [FG1, Proposition 2.6.1].

F.6.5. Recall now that the map

$$\mathbb{V}_{\mathfrak{g},\text{crit}} \otimes_{\mathcal{O}_{\text{OP}_G^{\text{reg}}}} \mathcal{O}_{\text{MOP}_{\check{G},\check{P}^-}^{\text{reg}}} \rightarrow \mathbb{W}_{\mathfrak{g},\text{crit}}$$

yields an isomorphism after applying the Drinfeld-Sokolov functor DS.

Therefore, by Lemma F.6.4, it suffices to show that

$$(F.13) \quad \mathbb{V}_{\mathfrak{g},\text{crit}} \otimes_{\mathcal{O}_{\text{OP}_G^{\text{reg}}}} \mathcal{O}_{\text{MOP}_{\check{G},\check{P}^-}^{\text{reg}}} \rightarrow j_* \circ j^*(\mathbb{V}_{\mathfrak{g},\text{crit}})$$

yields an isomorphism after applying DS.

To prove the latter, it suffices to show that the image of (F.13) in $\text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G,\text{reg}})$ becomes an isomorphism.

Hence, by the definition of the $j_* \circ j^*$ functor on $\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G,\text{reg}}^{IP}$, it suffices to prove the following:

Proposition F.6.6. *For any $\mathcal{F}_0 \in \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{\mathfrak{L}^+(G)}$ with image*

$$\mathcal{F} \in \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{IP} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}),$$

the map

$$\pi_* \circ \pi^*(\mathcal{F}) \xrightarrow{(F.10)} j_* \circ j^*(\mathcal{F})$$

yields an isomorphism after applying the functor

$$(F.14) \quad \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{IP} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}) \rightarrow \text{D-mod}_{\text{crit}}(\text{Gr}_G) \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}) \rightarrow \text{Whit}_*(G) \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}).$$

The rest of this subsection is devoted to the proof of Proposition F.6.6.

F.6.7. *Initial reduction step.* Since $\text{D-mod}_{\text{crit}}(\text{Gr}_G)^{\mathfrak{L}^+(G)}$ is compactly generated by eventually coconnective objects, we are reduced to the case where $\mathcal{F}_0$ is in the heart of the t-structure. By naive Satake, it corresponds to a $\check{G}$ -representation V .

By $\text{Rep}(\check{G})$ -equivariance of the functors appearing in this statement, we can assume V is trivial, i.e., that $\mathcal{F}_0$ is the unit object δ_{1,Gr_G} .

F.6.8. We now translate the statement to the spectral side. Note that by the construction of the equivalence $\text{Sat}^{-,\frac{\infty}{2}}$, we have a commutative diagram of factorization categories⁸⁸

$$\begin{array}{ccc} \text{I}(G, P^-)^{\text{loc}} & \xrightarrow{\text{Sat}^{-,\frac{\infty}{2}}} & \text{I}(\check{G}, \check{P}^-)^{\text{spec,loc}} \\ \downarrow & & \downarrow \text{pre-F}^{\text{spec}} \\ & & \text{Rep}(\check{G}) \otimes \text{Rep}(\check{M}) \\ & & \downarrow \text{id} \otimes \text{inv}_{\check{M}} \\ \text{Whit}_*(G) & \xrightarrow{\text{FLE}_{\check{G},\infty,\tau}} & \text{Rep}(\check{G}) \end{array}$$

where:

- The left vertical arrow is the composition

$$\begin{aligned} \text{I}(G, P^-)^{\text{loc}} &:= \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M), \text{ren}} \rightarrow \\ &\rightarrow \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{\mathfrak{L}(N_P^-) \cdot \mathfrak{L}^+(M)} \rightarrow \text{D-mod}_{\text{crit}}(\text{Gr}_G) \rightarrow \text{Whit}_*(G); \end{aligned}$$

- The right vertical arrow above is the $(\text{IndCoh}, *)$ -pushforward to LS_G^{reg} .

⁸⁸We are omitting the translation by $\rho_P(\omega_X)$.

Now at the *pointwise* level, observe that the left vertical arrow above factors as

$$(F.15) \quad I(G, P^-)^{\text{loc}} \simeq \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P, \text{ren}} \rightarrow \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P} \rightarrow \text{D-mod}_{\text{crit}}(\text{Gr}_G) \rightarrow \text{Whit}_*(G)$$

with first arrow being an equivalence.

Therefore, we obtain a commutative diagram

$$\begin{array}{ccc} \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P, \text{ren}} & \xrightarrow{(F.1)} & \text{IndCoh}(\text{pt}/\check{G} \times_{\check{\mathfrak{g}}/\check{G}} \check{\mathfrak{n}}_P^-/\check{P}^-) \\ \downarrow & & \downarrow (p_1)_*^{\text{IndCoh}} \\ \text{Whit}_*(G) & \xrightarrow{\text{FLE}_{\check{G}, \infty, \tau}} & \text{QCoh}(\text{pt}/\check{G}) = \text{Rep}(\check{G}), \end{array}$$

where:

- p_1 denotes the projection $\text{pt}/\check{G} \times_{\check{\mathfrak{g}}/\check{G}} \check{\mathfrak{n}}_P^-/\check{P}^- \rightarrow \text{pt}/\check{G}$;
- The left vertical arrow is the composition of the last three arrows in (F.15).

In turn, this yields

$$(F.16) \quad \begin{array}{ccc} \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P, \text{ren}} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}) & \longrightarrow & \text{IndCoh}(\check{\mathfrak{n}}_P^-/\check{P}^- \times_{\check{\mathfrak{g}}/\check{G}} \text{pt}/\check{B}) \\ \downarrow & & \downarrow (p_2)_*^{\text{IndCoh}} \\ \text{Whit}_*(G) \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}) & \xrightarrow{\text{FLE}_{\check{G}, \infty, \tau} \otimes \text{Id}} & \text{QCoh}(\text{pt}/\check{B}) = \text{Rep}(\check{B}), \end{array}$$

where:

- p_2 denotes the projection $\check{\mathfrak{n}}_P^-/\check{P}^- \times_{\check{\mathfrak{g}}/\check{G}} \text{pt}/\check{B} \rightarrow \text{pt}/\check{B}$;
- The left vertical arrow is the functor (F.14).

F.6.9. The image of $\delta_{1, \text{Gr}_G} \otimes \mathbf{1}_{\text{Rep}(\check{B})}$ along the top horizontal arrow in (F.16) is $\iota_*^{\text{IndCoh}}(\mathcal{O}_{\check{P}^- \setminus \check{G}/\check{B}})$ for ι being the map

$$\check{P}^- \setminus \check{G}/\check{B} = \text{pt}/\check{P}^- \times_{\text{pt}/\check{G}} \text{pt}/\check{B} \rightarrow \check{\mathfrak{n}}_P^-/\check{P}^- \times_{\check{\mathfrak{g}}/\check{G}} \text{pt}/\check{B}.$$

Thus, we have reduced the assertion of the proposition to the following:

Lemma F.6.10. *The morphism*

$$\pi_* \circ \pi^*(\mathcal{O}_{\check{P}^- \setminus \check{G}/\check{B}}) \rightarrow j_* \circ j^*(\mathcal{O}_{\check{P}^- \setminus \check{G}/\check{B}})$$

induces an isomorphism after taking direct image along

$$\check{P}^- \setminus \check{G}/\check{B} \rightarrow \text{pt}/\check{B}.$$

Proof. It suffices to show that the map

$$\mathcal{O}_{\check{P}^- \setminus \check{G}/\check{B}(M)} \rightarrow \tilde{j}_* \circ \tilde{j}^*(\mathcal{O}_{\check{P}^- \setminus \check{G}/\check{B}(M)})$$

induces an isomorphism after taking direct image along

$$\check{P}^- \setminus \check{G}/\check{B}(M) \rightarrow \text{pt}/\check{B}(M).$$

However, this follows from the fact that

$$k \rightarrow \Gamma(\check{P}^- \setminus \check{G}, \mathcal{O}_{\check{P}^- \setminus \check{G}})$$

is an isomorphism. □

□[Proposition F.6.6]

F.7. Proof of Lemma F.5.3.

F.7.1. We follow the same train of ideas as in Sect. F.6.

F.7.2. We have the following commutative diagram.

$$\begin{array}{ccccc}
(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P})_{\text{trans}} & \simeq & \left(\text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P} \otimes_{\text{Rep}(\check{G})} \text{QCoh}(\text{Op}_{\check{G}}^{\text{reg}}) \right)_{\text{trans}} & \rightarrow & \left(\text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}) \right)_{\text{trans}} \\
\downarrow J_* & & \downarrow J_* & & \downarrow J_* \\
\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P} & \simeq & \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P} \otimes_{\text{Rep}(\check{G})} \text{QCoh}(\text{Op}_{\check{G}}^{\text{reg}}) & \longrightarrow & \text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P} \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}) \\
\downarrow & & \downarrow & & \downarrow \\
\text{Whit}_*(\widehat{\mathfrak{g}}\text{-mod}_{\text{crit}_G, \text{reg}}^{I_P}) & \simeq & \text{Whit}_*(G) \otimes_{\text{Rep}(\check{G})} \text{QCoh}(\text{Op}_{\check{G}}^{\text{reg}}) & \longrightarrow & \text{Whit}_*(G) \otimes_{\text{Rep}(\check{G})} \text{Rep}(\check{B}).
\end{array}$$

In the above diagram, the right horizontal arrows are conservative because $\text{Op}_{\check{G}}^{\text{reg}} \rightarrow \text{pt}/\check{B}$ is affine. Moreover, the equivalences and rightward arrows are t-exact. Therefore, it is enough to prove that the right vertical composition is conservative when restricted to

$$\left(\text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P} \otimes_{\text{Rep}(\check{G})} \text{QCoh}(\text{Op}_{\check{G}}^{\text{reg}}) \right)_{\text{trans}} \cap \left(\text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P} \otimes_{\text{Rep}(\check{G})} \text{QCoh}(\text{Op}_{\check{G}}^{\text{reg}}) \right)^{>-\infty}.$$

F.7.3. We now pass to the spectral side. We record the following (well-known) assertion.

Lemma F.7.4. *The equivalence (F.2) has bounded cohomological amplitude for the natural t-structures on both sides.*

Proof. The heart of the t-structure on $\text{IndCoh}_{\text{Nilp}}(\text{pt}/\check{G} \times \check{\mathfrak{n}}_P^-/\check{P}^-)$ is $\text{Rep}(\check{P}^-)^{\vee}$. Therefore, it suffices to see that for each $V \in \text{Rep}(\check{P}^-)^{\vee}$ maps to an object in degrees $[-N, N]$ in $\text{D-mod}_{\text{crit}}(\text{Gr}_G)^{I_P}$ for some N independent of V .

First, when V comes from a representation of $\check{M}_{\text{ab}}$, the assertion is standard, cf. [AB, Theorem 5] for a perversity statement at the level of sheaves on $I_P \backslash \mathcal{L}(G)/I_P$ and then note that further pushforward to $I_P \backslash \text{Gr}_G$ has bounded amplitude.

Next, we deduce the claim for any V of the form $V_1 \otimes V_2$ for $V_1 \in \text{Rep}(\check{G})^{\vee}$ and $V_2 \in \text{Rep}(\check{M}_{\text{ab}})^{\vee}$ from the t-exactness of the action of $\text{Rep}(\check{G})$ on $\text{D-mod}_{\text{crit}}(\text{Gr}_G)$.

Finally, every $V \in \text{Rep}(\check{P})^{\vee}$ has a bounded resolution by terms of the form $V_1 \otimes V_2$ as above – this follows by a standard argument from quasi-affineness of $\check{G}/[\check{P}, \check{P}]$. $\square$

F.7.5. Applying Lemma F.7.4 and using (F.16), we obtain that it suffices to show that the functor

$$\text{IndCoh}_{\text{Nilp}}((\check{\mathfrak{n}}_P^-/\check{P}^- \times \text{pt}/\check{B})_{\text{trans}}) \xrightarrow{(p_2)_*^{\text{IndCoh}}} \text{IndCoh}(\text{pt}/\check{B}) \simeq \text{Rep}(\check{B})$$

is conservative on the eventually coconnective subcategory.

Recall that

$$(\check{\mathfrak{n}}_P^-/\check{P}^- \times \text{pt}/\check{B})_{\text{trans}} = (0 \times_{\check{\mathfrak{g}}} \check{\mathfrak{n}}_P^-)/\check{B}_{\check{M}}$$

We factor the above map p_2 as

$$(0 \times_{\check{\mathfrak{g}}} \check{\mathfrak{n}}_P^-)/\check{B}_{\check{M}} \rightarrow \text{pt}/\check{B}(M) \rightarrow \text{pt}/\check{B}.$$

The $(\text{IndCoh}, *)$ -pushforward functor with respect to the first arrow is conservative on the eventually coconnective subcategory, since this morphism is affine.

Finally, the pushforward functor along $\mathrm{pt}/\check{B}(M) \rightarrow \mathrm{pt}/\check{B}$ is conservative, since the above map is affine (indeed, $\check{B}/\check{B}_{\check{M}} \simeq \check{N}_P$ is affine).

□[Lemma F.5.3]

REFERENCES

- [AB] S. Arkhipov and R. Bezrukavnikov, *Perverse sheaves on affine flags and Langlands dual group*, Israel Journal of Mathematics 170 (2009): 135–183.
- [ABG] S. Arkhipov, R. Bezrukavnikov and V. Ginzburg, *Quantum groups, the loop Grassmannian, and the Springer resolution*, Journal of the American Mathematical Society 17.3 (2004): 595–678.
- [AG] D. Arinkin and D. Gaitsgory, *Singular support of coherent sheaves, and the Geometric Langlands Conjecture*, Selecta Math. N.S. **21** (2015), 1–199.
- [AGKRRV] D. Arinkin, D. Gaitsgory, D. Kazhdan, S. Raskin, N. Rozenblyum and Y. Varshavsky, *The stack of local systems with restricted variation and geometric Langlands theory with nilpotent singular support*, arXiv:2010.01906.
- [Bar] J. Barlev, *D-modules on spaces of rational maps*, Compositio Mathematica **150** (214), 835–876.
- [BD1] A. Beilinson and V. Drinfeld, *Quantization of Hitchin's integrable system and Hecke eigensheaves*, available at: https://people.mpim-bonn.mpg.de/gaitsgde/grad_2009/
- [BD2] A. Beilinson and V. Drinfeld, *Chiral algebras*, American Mathematical Soc. **51** (2004).
- [BFGM] A. Braverman, M. Finkelberg, D. Gaitsgory and I. Mirković, *Intersection cohomology of Drinfeld's compactifications*, Selecta Mathematica 8.3 (2002), 381–418.
- [BG1] A. Braverman and D. Gaitsgory, *Geometric Eisenstein series*, Invent. Math. **150** (2002), 287–84.
- [BG2] A. Braverman and D. Gaitsgory, *Deformations of local systems and Eisenstein series*, GAFA **17** (2008), 1788–1850.
- [CR] J. Campbell and S. Raskin, *Langlands duality on the Beilinson-Drinfeld Grassmannian*, arXiv:2310.19734.
- [Ch1] L. Chen, *Nearby cycles on Drinfeld-Gaitsgory-Vinberg interpolation Grassmannian and long intertwining functor*, Duke Math. Jour. **172** (2023), 447–553.
- [Ch2] L. Chen, *Nearby cycles in dualities in geometric Langlands program*, PhD Thesis (Harvard University 2021), available at: <https://windshower.github.io/linchen/papers/dissertation.pdf>
- [CF] L. Chen and Y. Fu, *An extension of Kazhdan-Lusztig equivalence*, arXiv:2111.14606.
- [CFGY] L. Chen, Y. Fu, D. Gaitsgory and D. Yang, *Loop group actions via factorization*, forthcoming.
- [Dr] V. Drinfeld, *Two-dimensional ℓ -adic representations of the fundamental group of a curve over a finite field and automorphic forms on $GL(2)$* , Amer. Jour. of Math. **105** (1983), 85–114.
- [DG1] V. Drinfeld and D. Gaitsgory, *On some finiteness questions for algebraic stacks*, Geometric and Functional Analysis 23.1 (2013): 149–294.
- [DG2] V. Drinfeld and D. Gaitsgory, *Compact generation of the category of D-modules on the stack of G -bundles on a curve*, Cambridge Journal of Mathematics 3.1 (2015): 19–125.
- [DG3] V. Drinfeld and D. Gaitsgory, *Geometric constant term functor(s)*, Selecta Mathematica 22 (2016): 1881–1951.
- [FH] J. Faergeman, A. Hayash, *Parabolic geometric Eisenstein series and constant term functors*, forthcoming.
- [FR] J. Faergeman and S. Raskin, *Non-vanishing of geometric Whittaker coefficients for reductive groups*, arXiv:2207.02955.
- [Fr] E. Frenkel, *Lectures on Wakimoto modules, opers and the center at the critical level*, arXiv:math/0210029.
- [FG1] E. Frenkel and D. Gaitsgory, *Geometric realizations of Wakimoto modules at the critical level*, Duke Math. J. **143** (2008), 117–203.
- [FG2] E. Frenkel and D. Gaitsgory, *D-modules on the affine flag variety and representations of affine Kac-Moody algebras*, Represent. Theory **13** (2009), 470–608.
- [FG3] E. Frenkel and D. Gaitsgory, *Localization of D-modules on the affine Grassmannian*, Annals of Mathematics (2009): 1339–1381.
- [FGV] E. Frenkel, D. Gaitsgory and K. Vilonen, *On the geometric Langlands conjecture*, Jour. Amer. Math. Soc. **15** (2002), 367–417.
- [Ga1] D. Gaitsgory, *Outline of the proof of the Geometric Langlands Conjecture for $GL(2)$* , Astérisque **370** (2015), 1–112.
- [Ga2] D. Gaitsgory, *A conjectural extension of the Kazhdan-Lusztig equivalence*, Publications of RIMS **57** (2021), 1227–1376.
- [Ga3] D. Gaitsgory, *A "strange" functional equation for Eisenstein series and Verdier duality on the moduli stack of bundles*, Annales Scientifiques de l'ENS **50** (2017), 1123–1162.
- [Ga4] D. Gaitsgory, *The semi-infinite intersection cohomology sheaf*, Adv. in Math. **327** (2018), 789–868.
- [Ga5] D. Gaitsgory, *The semi-infinite intersection cohomology sheaf-II: the Ran space version*, in: Representation theory and algebraic geometry, Trends in Mathematics, Birkhäuser (2022).
- [Ga6] D. Gaitsgory, *The local and global versions of the Whittaker category*, PAMQ **16**, (2020), 775–904.
- [GLC1] D. Gaitsgory and S. Raskin, *Proof of the geometric Langlands conjecture I: construction of the functor*, arXiv:2405.03599
- [GLC2] D. Arinkin, D. Beraldo, L. Chen, J. Faergeman, D. Gaitsgory, K. Lin, S. Raskin and N. Rozenblyum, *Proof of the geometric Langlands conjecture II: Kac-Moody localization and the FLE*, arXiv:2405.03648
- [GLC4] D. Arinkin, D. Beraldo, L. Chen, J. Faergeman, D. Gaitsgory, K. Lin, S. Raskin and N. Rozenblyum, *Proof of the geometric Langlands conjecture IV: Ambidexterity*, available at: <https://people.mpim-bonn.mpg.de/gaitsgde/GLC/>
- [GLC5] D. Gaitsgory and S. Raskin, *Proof of the geometric Langlands conjecture V: The multiplicity one theorem*, available at: <https://people.mpim-bonn.mpg.de/gaitsgde/GLC/>
- [GLys] D. Gaitsgory and S. Lysenko, *Parameters and duality for the metaplectic geometric Langlands theory*, Selecta Math. New Ser. **24** (2018), 227–301. Also arXiv: 1608.00284.
- [GaRo3] D. Gaitsgory and N. Rozenblyum, *A study in derived algebraic geometry, Vol. 1: Correspondences and Duality*, Mathematical surveys and monographs **221** (2017), AMS, Providence, RI.
- [GaRo4] D. Gaitsgory and N. Rozenblyum, *A study in derived algebraic geometry, Vol. 2: Deformations, Lie Theory and Formal Geometry*, Mathematical surveys and monographs **221** (2017), AMS, Providence, RI.
- [Laum] G. Laumon, *Correspondance de Langlands géométrique pour les corps de fonctions*, Duke Math. J. 54.1 (1987): 309–359.
- [Lin] K. Lin, *Poincaré series and miraculous duality*, arXiv:2211.05282.

- [Ra1] S. Raskin, *A geometric proof of the Feigin-Frenkel theorem*, Representation Theory **16** (2012), 489–512.
- [Ra2] S. Raskin, *W-algebras and Whittaker categories*, Selecta Mathematica **27** (2021).
- [Ra3] S. Raskin, *Chiral principal series categories I: Finite dimensional calculations*, Adv. in Math. **388** (2021).
- [Ra4] S. Raskin, *Chiral principal series categories II: the factorizable Whittaker category*, available at: <https://gauss.math.yale.edu/~sr2532/cpsii.pdf>
- [RY] S. Raskin and D. Yang, *Affine Beilinson-Bernstein localization at the critical level*, Annals of Mathematics **200.2** (2024): 487–527.
- [Ro] N. Rozenblyum, *Connections on moduli spaces and infinitesimal Hecke modifications*, arXiv:2108.07745.
- [Ya-D] D. Yang, *Categorical Moy-Prasad theory*, arXiv:2104.12917.
- [Ya-R] R. Yang, *Twisted Whittaker category on affine flags and the category of representations of the mixed quantum group*, Compositio Mathematica **160** (2024), 1349–1417.

PROOF OF THE GEOMETRIC LANGLANDS CONJECTURE IV: AMBIDEXTERITY

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ABSTRACT. This paper performs the following steps toward the proof of GLC in the de Rham setting:

- (i) We deduce GLC for $G = GL_n$;
- (ii) We prove that the Langlands functor $\mathbb{L}_G$ constructed in [GLC1], when restricted to the cuspidal category, is *ambidextrous*;
- (iii) We reduce GLC to the study of a *classical vector bundle with connection*, denoted $\mathcal{A}_{G,\text{irred}}$, on the stack $\text{LS}_G^{\text{irred}}$ of irreducible local systems;
- (iv) We prove that GLC is equivalent to the contractibility of the space of generic oper structures on irreducible local systems;
- (v) Using [BKS], we deduce GLC for classical groups.

CONTENTS

Introduction	2
0.1. What is done in this paper?	2
0.2. How is ambidexterity proved?	3
0.3. Relation to opers	5
0.4. Contents	7
0.5. Acknowledgements	7
1. Summary of the Langlands functor	8
1.1. The functor $\mathbb{L}_G$ via the Whittaker model	8
1.2. Langlands functor and Eisenstein series	9
1.3. The Langlands functor on the cuspidal part	10
1.4. Spectral action	11
1.5. Conservativity	13
1.6. The algebra $\mathcal{A}_{G,\text{irred}}$	13
1.7. Proof of GLC for $G = GL_n$	14
2. Left adjoint as the dual	15
2.1. The dual automorphic category	15
2.2. The dual of the cuspidal category	16
2.3. Duality and the Poincaré functors	17
2.4. Self-duality on the spectral side	18
2.5. Left adjoint vs dual	18
3. Right adjoint as the dual	21
3.1. The ambidexterity statement	21
3.2. Critical localization	22
3.3. The spectral Poincaré functor	23
3.4. Proof of Theorem 3.1.2	26
3.5. An addendum: ambidexterity for eigensheaves	27
4. The expression for $\mathcal{A}_{G,\text{irred}}$ via opers	29

4.1. Statement of the result	29
4.2. Combining with ambidexterity	30
4.3. Proof of Proposition 4.2.8	31
4.4. Proof of Theorem 3.1.8	32
4.5. Contractibility of opers	32
4.6. Proof of Theorem 4.1.5	33
5. Proof of Theorem 4.6.3	35
5.1. Strategy of the proof	35
5.2. Framework for the proof of Proposition 5.1.2	36
5.3. An abstract version of Proposition 5.2.4	37
5.4. A digression: the category $\mathrm{QCoh}_{\mathrm{co}}$	38
5.5. Abstract version of Proposition 5.1.5: the absolute case	40
5.6. Abstract version of Proposition 5.1.5: the relative case	41
Appendix A. Proof of Proposition 5.3.4	43
A.1. The unital Ran space	43
A.2. IndCoh on categorical prestacks	44
A.3. A reformulation	45
A.4. A description of relative D-modules	46
A.5. Proof of Proposition A.3.7	48
A.6. Proof of Lemma A.5.3	49
Appendix B. Proof of Proposition 5.5.5	49
B.1. A reformulation in terms of unital structure	49
B.2. The local unital structure	51
B.3. An expression for the global sections functor	53
B.4. Inputting the unitality structure	55
Appendix C. Proof of Proposition 5.6.8	57
C.1. Localization functor in the abstract setting	57
C.2. Proof of Proposition 5.6.8	58
C.3. Proof of Proposition C.1.5	61
C.4. Proof of Proposition C.1.7	61
References	64

INTRODUCTION

This paper is the fourth in the series of five papers, whose combined content will prove the geometric Langlands conjecture (GLC), as it was formulated in [GLC1, Conjecture 1.6.7].

0.1. What is done in this paper?

0.1.1. In the papers [GLC1, GLC2, GLC3] we constructed the *Langlands functor*

$$(0.1) \quad \mathbb{L}_G : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}),$$

and GLC says that (0.1) is an equivalence.

0.1.2. The main result of [GLC3] says that (0.1) induces an equivalence

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}},$$

where:

- $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ is the full subcategory generated by Eisenstein series from proper Levi subgroups;
- $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$ is the full subcategory consisting of objects, set-theoretically supported on the locus of reducible local systems.

0.1.3. As one of the first steps in this paper we will show that GLC is equivalent to the statement that the induced functor

$$(0.2) \quad \mathbb{L}_{G,\text{cusp}} : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}} \rightarrow \text{IndCohNilp}(\text{LS}_G^{\text{irred}}),$$

is an equivalence (see Corollary 1.3.10). (Note also that $\text{IndCohNilp}(\text{LS}_G^{\text{irred}})$ is the same as the usual $\text{QCoh}(\text{LS}_G^{\text{irred}})$ category).

Thus, the proof of GLC amounts to the study of the functor $\mathbb{L}_{G,\text{cusp}}$.

0.1.4. Before we even begin the discussion of the main results of this paper, we observe (see Sect. 1.7) that the above considerations already allow us to deduce GLC for $G = GL_n$.

Namely, the fact that $\mathbb{L}_{G,\text{cusp}}$ is fully faithful for GL_n follows from [Ga2] (or, in a more modern language, from [Be1]).

We then show that its essential surjectivity is equivalent to the existence of (non-zero) Hecke eigen-sheaves attached to irreducible local systems, which was established in [FGV] using geometric methods (or, alternatively, in [BD1] using localization at the critical level).

0.1.5. The main result of this paper, Theorem 3.1.4, which we call the Ambidexterity Theorem, says that the left and right adjoints of the functor $\mathbb{L}_{G,\text{cusp}}$ are isomorphic.

This already gets us pretty close to the statement that $\mathbb{L}_{G,\text{cusp}}$ is an equivalence. Yet, we will need to “milk” the ambidexterity statement some more in order to obtain the actual proof. Some of this milking will be preformed in this paper, and some will be delegated to its sequel.

0.1.6. An additional crucial input comes from the paper [FR] (combined with [Be2]), which says that the functor $\mathbb{L}_{G,\text{cusp}}$ is conservative. This implies that in order to prove GLC, it is sufficient to show that the monad

$$(0.3) \quad \mathbb{L}_{G,\text{cusp}} \circ \mathbb{L}_{G,\text{cusp}}^L$$

acting on $\text{QCoh}(\text{LS}_G^{\text{irred}})$ is isomorphic to the identity functor.

We observe (see Sect. 1.6.2) that the monad (0.3) is given by tensor product with an associative algebra object

$$(0.4) \quad \mathcal{A}_{G,\text{irred}} \in \text{QCoh}(\text{LS}_G^{\text{irred}}).$$

The monad (0.3) is an equivalence if and only if the unit map

$$(0.5) \quad \mathcal{O}_{\text{LS}_G^{\text{irred}}} \rightarrow \mathcal{A}_{G,\text{irred}}$$

is an isomorphism in $\text{QCoh}(\text{LS}_G^{\text{irred}})$.

0.1.7. Now, the Ambidexterity Theorem tells us something about the structure of $\mathcal{A}_{G,\text{irred}}$. Namely, it implies that $\mathcal{A}_{G,\text{irred}}$ is self-dual as an object of $\text{QCoh}(\text{LS}_G^{\text{irred}})$. In particular, it is perfect, and hence compact.

However, we prove more: we show (assuming that G is semi-simple) that $\mathcal{A}_{G,\text{irred}}$ is a *classical vector bundle*, equipped with a flat connection (see Theorem 3.1.8).

Thus, we can view $\mathcal{A}_{G,\text{irred}}$ as a classical local system on $\text{LS}_G^{\text{irred}}$. We also show that this local system has a finite monodromy (see Proposition 4.2.8); this latter statement will play a role in the final step of the proof of GLC in the next paper in this series.

0.1.8. The above additional pieces of information concerning $\mathcal{A}_{G,\text{irred}}$ result from Corollary 4.2.5, which says that the fiber of $\mathcal{A}_{G,\text{irred}}$ at a given irreducible local system σ is isomorphic to the homology of the space of *generic oper structures* on σ .

We will explain the mechanism for this in Sect. 0.3.

0.2. How is ambidexterity proved?

0.2.1. The proof of the Ambidexterity Theorem is obtained by essentially staring at what we call the Fundamental Commutative Diagram (see [GLC2, Diagram (18.14)]):

$$(0.6) \quad \begin{array}{ccc} \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xrightarrow[\sim]{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{coeff}_G \uparrow & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\ \mathrm{Loc}_G \uparrow & & \uparrow \mathrm{Poinc}_{\check{G},*}^{\mathrm{spec}} \\ \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} & \xrightarrow[\sim]{\mathrm{FLE}_{G, \mathrm{crit}}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}}, \end{array}$$

where we ignore some cohomological shifts and twists by constant lines.

Remark 0.2.2. In fact, (0.6) is a special case at levels (crit for G , ∞ for $\check{G}$) of an analogous diagram that is expected to exist in the quantum case:

$$(0.7) \quad \begin{array}{ccc} \mathrm{Whit}^!(G)_{\kappa, \mathrm{Ran}} & \xrightarrow[\sim]{\mathrm{FLE}_{\check{G}, \check{\kappa}}^\vee} & \mathrm{KL}(\check{G})_{-\check{\kappa}, \mathrm{Ran}} \\ \mathrm{coeff}_G \uparrow & & \uparrow \Gamma_{\check{G}, -\check{\kappa}} \\ \mathrm{D-mod}_\kappa(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_{G, \kappa}} & \mathrm{D-mod}_{-\check{\kappa}}(\mathrm{Bun}_{\check{G}})_{\mathrm{co}} \\ \mathrm{Loc}_{G, \kappa} \uparrow & & \uparrow \mathrm{Poinc}_{\check{G},*} \\ \mathrm{KL}(G)_{\kappa, \mathrm{Ran}} & \xrightarrow[\sim]{\mathrm{FLE}_{G, \kappa}} & \mathrm{Whit}_*(\check{G})_{-\check{\kappa}, \mathrm{Ran}}. \end{array}$$

A remarkable feature of the quantum diagram is that it is *self-dual*: i.e., if we dualize all categories and arrows in (0.7) we obtain a similar diagram, but for $((G, \kappa), (\check{G}, -\check{\kappa}))$ replaced by $((\check{G}, \check{\kappa}), (G, -\kappa))$.

0.2.3. We break (0.6) into the upper and lower portions, i.e.,

$$(0.8) \quad \begin{array}{ccc} \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xrightarrow[\sim]{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{coeff}_G \uparrow & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \end{array}$$

and

$$(0.9) \quad \begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\ \mathrm{Loc}_G \uparrow & & \uparrow \mathrm{Poinc}_{\check{G},*}^{\mathrm{spec}} \\ \mathrm{KL}(G)_{\mathrm{crit}, \mathrm{Ran}} & \xrightarrow[\sim]{\mathrm{FLE}_{G, \mathrm{crit}}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}}, \end{array}$$

and we combine (0.8) (resp., (0.9)) with the inclusion of (resp., projection to) the cuspidal subcategory:

$$\begin{array}{ccc} \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xrightarrow[\sim]{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{coeff}_G \uparrow & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\ \mathbf{e} \uparrow & & \uparrow j_* \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xrightarrow{\mathbb{L}_{G, \mathrm{cusp}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \end{array}$$

and

$$\begin{array}{ccc}
\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xrightarrow{\mathbb{L}_{G,\mathrm{cusp}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{irred}}) \\
\uparrow \mathbf{e}^L & & \uparrow J^* \\
\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\
\uparrow \mathrm{Loc}_G & & \uparrow \mathrm{Poinc}_{\check{G},*}^{\mathrm{spec}} \\
\mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & \xrightarrow[\sim]{\mathrm{FLE}_{G,\mathrm{crit}}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}},
\end{array}$$

respectively.

0.2.4. Thus, we obtain the diagrams

$$\begin{array}{ccc}
\mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xrightarrow[\sim]{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\
\uparrow \mathrm{coeff}_G \circ \mathbf{e} & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh} \circ J_*} \\
\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xrightarrow{\mathbb{L}_{G,\mathrm{cusp}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{irred}})
\end{array}
\tag{0.10}$$

and

$$\begin{array}{ccc}
\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xrightarrow{\mathbb{L}_{G,\mathrm{cusp}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{irred}}) \\
\uparrow \mathbf{e}^L \circ \mathrm{Loc}_G & & \uparrow J^* \circ \mathrm{Poinc}_{\check{G},*}^{\mathrm{spec}} \\
\mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & \xrightarrow[\sim]{\mathrm{FLE}_{G,\mathrm{crit}}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}},
\end{array}
\tag{0.11}$$

respectively.

The key feature of the latter diagrams is that in (0.10) the right vertical arrow is fully faithful, and in (0.11) the left vertical arrow is a *Verdier quotient* (a.k.a., is a localization).

0.2.5. Starting from diagrams (0.10) and (0.11), the ambidexterity assertion is proved as follows.

Consider the functor *dual* to $\mathbb{L}_{G,\mathrm{cusp}}$ (with respect to the natural self-dualities of the two sides, see Sect. 2).

The point now is that the vertical arrows in (0.10) admit *left* adjoints, and these left adjoints are *essentially*¹ isomorphic to the duals of the original functors. Combined with the fact that the right vertical arrow is fully faithful, this implies that the dual of $\mathbb{L}_{G,\mathrm{cusp}}$ is isomorphic to the left adjoint of $\mathbb{L}_{G,\mathrm{cusp}}$.

Similarly, the vertical arrows in (0.11) admit *right* adjoints, and these right adjoints are *essentially* isomorphic to the duals of the original functors. Combined with the fact that the left vertical arrow is a Verdier quotient, this implies that the dual of $\mathbb{L}_{G,\mathrm{cusp}}$ is isomorphic to the right adjoint of $\mathbb{L}_{G,\mathrm{cusp}}$.

Thus, we have identified both the left and right adjoints of $\mathbb{L}_{G,\mathrm{cusp}}$ with its dual.

0.3. Relation to opers. We now turn to the statements announced in Sect. 0.1.8 that relate the fiber of the object $\mathcal{A}_{G,\mathrm{irred}}$ at a given $\sigma \in \mathrm{LS}_G^{\mathrm{irred}}$ to the homology of the space $\mathrm{Op}_{\check{G},\sigma}^{\mathrm{gen}}$ of generic oper structures on σ .

¹Essentially:=up to some twists.

0.3.1. Let

$$\mathcal{B}_{G,\text{irred}}^{\text{Op}} \in \text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}}),$$

be the object obtained by applying the *left* forgetful functor

$$\text{oblv}^l : \text{D-mod}(\text{LS}_{\check{G}}^{\text{irred}}) \rightarrow \text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$$

to the object

$$(\pi_{\text{Ran}}^{\text{irred}})!(\omega_{\text{Op}_{\check{G}}^{\text{mon-free,irred}}(X^{\text{gen}})_{\text{Ran}}}),$$

where:

- $\text{Op}_{\check{G}}^{\text{mon-free}}(X^{\text{gen}})_{\text{Ran}}$ is the space of pairs (σ, o) , where $\sigma \in \text{LS}_{\check{G}}$, and o is a generic oper structure on it;
- $\pi_{\text{Ran}} : \text{Op}_{\check{G}}^{\text{mon-free}}(X^{\text{gen}})_{\text{Ran}} \rightarrow \text{LS}_{\check{G}}$ is the tautological map $(\sigma, o) \mapsto \sigma$;
- $\text{Op}_{\check{G}}^{\text{mon-free,irred}}(X^{\text{gen}})_{\text{Ran}}$ and $\pi_{\text{Ran}}^{\text{irred}}$ is the base change of the above objects along the inclusion $\text{LS}_{\check{G}}^{\text{irred}} \rightarrow \text{LS}_{\check{G}}$.

By construction, $\mathcal{B}_{G,\text{irred}}^{\text{Op}}$ is naturally a co-commutative coalgebra in $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$.

Note that since the map $\pi_{\text{Ran}}^{\text{irred}}$ is pseudo-proper (see Sect. 3.3.11), the fiber of $\mathcal{B}_{G,\text{irred}}^{\text{Op}}$ at a given $\sigma \in \text{LS}_{\check{G}}^{\text{irred}}$ is indeed given by the homology of the space

$$\text{Op}_{G,\sigma}^{\text{gen}} := \{\sigma\} \times_{\text{LS}_{\check{G}}} \text{Op}_{\check{G}}^{\text{mon-free}}(X^{\text{gen}})$$

of generic oper structures on σ .

0.3.2. The point of departure is Theorem 4.6.3, which says that the comonad on

$$\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}^{\text{irred}}) \simeq \text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$$

given by

$$\text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}} \circ (\text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}})^R, \quad \text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}} := j^* \circ \text{Poinc}_{\check{G},*}^{\text{spec}}$$

is isomorphic to the comonad given by tensoring with $\mathcal{B}_{G,\text{irred}}^{\text{Op}}$.

0.3.3. Consider the comonad

$$\mathbb{L}_G \circ \mathbb{L}_G^R$$

acting on $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$. Since it is $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$ -linear, it is given by tensor product with a co-associative coalgebra object, denoted

$$\mathcal{B}_{G,\text{irred}} \in \text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}}).$$

The fact that the left vertical arrow in (0.11) is a Verdier quotient implies that we have an isomorphism of comonads

$$\mathbb{L}_G \circ \mathbb{L}_G^R \simeq \text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}} \circ (\text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}})^R.$$

Combining with Theorem 4.6.3, we obtain an isomorphism²

$$\mathcal{B}_{G,\text{irred}} \simeq \mathcal{B}_{G,\text{irred}}^{\text{Op}}.$$

However, the Ambidexterity Theorem implies that $\mathcal{A}_{G,\text{irred}} \simeq \mathcal{B}_{G,\text{irred}}$ (as plain objects of $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$). Combining, we obtain an isomorphism

$$(0.12) \quad \mathcal{A}_{G,\text{irred}} \simeq \mathcal{B}_{G,\text{irred}}^{\text{Op}},$$

also as plain objects of $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$.

From here we obtain the desired statements relating the fiber of $\mathcal{A}_{G,\text{irred}}$ at a given irreducible local system σ with the homology of $\text{Op}_{G,\sigma}^{\text{gen}}$.

²One can show that this isomorphism respects the co-associative coalgebra structures, but we will neither prove nor use this fact.

0.3.4. Note, however, that the isomorphism (0.12) gives us more. Namely, since we already know that $\mathcal{A}_{G,\text{irred}}$ is concentrated in cohomological degree 0, we obtain that the connected components of $\text{Op}_{G,\sigma}^{\text{gen}}$ are *homologically contractible*.

And since GLC is equivalent to the fact that the map (0.5) is an isomorphism, we obtain that it is equivalent to either of the following:

- For every irreducible σ , the space $\text{Op}_{G,\sigma}^{\text{gen}}$ is homologically contractible;
- For every irreducible σ , the space $\text{Op}_{G,\sigma}^{\text{gen}}$ is connected.

0.3.5. Recall now that a recent result of [BKS] proves the homological contractibility of the spaces $\text{Op}_{G,\sigma}^{\text{gen}}$, whenever G (and hence $\check{G}$) is classical.

Hence, we obtain that GLC is a theorem for classical G .

0.4. **Contents.** We now briefly review the contents of this paper section-by-section.

0.4.1. In Sect. 1 we review the contents of [GLC1, GLC2, GLC3] relevant for this paper, and draw some consequences. In particular:

- We show that the functor $\mathbb{L}_G$ is an equivalence if and only if the corresponding functor $\mathbb{L}_{G,\text{cusp}}$ is;
- We show that $\mathbb{L}_{G,\text{cusp}}$ is an equivalence if and only if the object $\mathcal{A}_{G,\text{irred}}$ is isomorphic to the structure sheaf;
- We deduce GLC for GL_n .

0.4.2. In Sect. 2 we review the self-duality identifications on the two sides of (0.2), and we show that the *left* adjoint of $\mathbb{L}_{G,\text{cusp}}$ identifies with its dual, up to a twist. This uses the compatibility of the functor $\mathbb{L}_G$ with the Whittaker model, i.e., the upper portion of (0.6).

In Sect. 3, we show that the *right* adjoint of $\mathbb{L}_{G,\text{cusp}}$ also identifies with its dual (up to the same twist). This uses the compatibility of the functor $\mathbb{L}_G$ with localization at the critical level, i.e., the lower portion of (0.6).

Combining, we deduce the Ambidexterity Theorem, which says that the left and right adjoints of $\mathbb{L}_{G,\text{cusp}}$ are isomorphic.

From here, we deduce that the object $\mathcal{A}_{G,\text{irred}} \in \text{QCoh}(\text{LS}_G^{\text{irred}})$ is self-dual, and hence compact.

0.4.3. In Sect. 4 we express $\mathcal{A}_{G,\text{irred}}$ via the space of generic oper structures.

As a result, we prove that $\mathcal{A}_{G,\text{irred}}$ is a classical vector bundle (when G is semi-simple).

And we deduce GLC for classical groups.

0.4.4. In Sect. 5 we reduce Theorem 4.1.5 stated in the previous section to the combination of two general assertions about the space of rational maps. These assertions are proved in Sect. A and Sects. B+C, respectively.

0.4.5. *Conventions and notation.* The conventions and notation in this paper follow those in [GLC3].

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1. SUMMARY OF THE LANGLANDS FUNCTOR

In the papers [GLC1], a functor

$$\mathbb{L}_G : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$$

was constructed.

The geometric Langlands conjecture, a.k.a. GLC ([GLC1, Conjecture 1.6.7]), says that the functor $\mathbb{L}_G$ is an equivalence. For the duration of this paper, we will assume the validity of GLC for proper Levi subgroups of G .

In this section we will summarize the properties of $\mathbb{L}_G$ relevant for this paper, and draw some consequences.

1.1. The functor $\mathbb{L}_G$ via the Whittaker model. In this subsection we will recall the “main” feature of the functor $\mathbb{L}_G$; its compatibility with the Whittaker model.

1.1.1. Recall (see [GLC2, Sects. 9.3 and 9.4]) that the category $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ is related to the *Whittaker category* by a pair of adjoint functors

$$\mathrm{Poinc}_{G,!} : \mathrm{Whit}^!(G)_{\mathrm{Ran}} \rightleftarrows \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) : \mathrm{coeff}_G.$$

1.1.2. Recall also that the category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$ is related to the category $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}$ by a pair of adjoint functors

$$\mathrm{Loc}_{\check{G}}^{\mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightleftarrows \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) : \Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}},$$

see [GLC2, Sect. 17.6].

Note, however, that the functor $\mathrm{Loc}_{\check{G}}^{\mathrm{spec}}$ factors as

$$\mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{\Xi_{\{0\}, \mathrm{Nilp}}} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}),$$

and the functor $\Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}}$ factors as

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \xrightarrow{\Psi_{\{0\}, \mathrm{Nilp}}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{\Gamma_{\check{G}}^{\mathrm{spec}}} \mathrm{Rep}(\check{G})_{\mathrm{Ran}},$$

where

$$\Xi_{\{0\}, \mathrm{Nilp}} : \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \rightleftarrows \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) : \Psi_{\{0\}, \mathrm{Nilp}}$$

are the natural embedding and projection, respectively.

1.1.3. We record the following (see Sect. C.1.9 for the proof):

Proposition 1.1.4. *The functor*

$$\Gamma_{\check{G}}^{\mathrm{spec}} : \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \rightarrow \mathrm{Rep}(\check{G})_{\mathrm{Ran}}$$

is fully faithful.

1.1.5. The Langlands functor $\mathbb{L}_G$ is *essentially*³ determined by the property that it makes the diagram

$$(1.1) \quad \begin{array}{ccc} \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xrightarrow[\sim]{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{coeff}_G[2\delta_{N_{\rho}(\omega_X)}] \uparrow & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}, \mathrm{IndCoh}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \end{array}$$

commute, where CS_G is the geometric Casselman-Shalika equivalence (see [GLC2, Sect. 1.4]), and

$$\delta_{N_{\rho}(\omega_X)} = \dim(\mathrm{Bun}_{N_{\rho}(\omega_X)}).$$

Remark 1.1.6. The commutation of (1.1) is the point of departure for any of the constructions of the Langlands functor.

³See [GLC1, Sects. 1.4 and 1.6] for what the word “essentially” refers to.

1.1.7. It is shown in [GLC3, Theorem 16.1.2] that the functor $\mathbb{L}_G$ admits a left adjoint, to be denoted $\mathbb{L}_G^L$. Passing to the left adjoints in (1.1), we obtain a commutative diagram

$$(1.2) \quad \begin{array}{ccc} \mathrm{Whit}^!(G)_{\mathrm{Ran}} & \xleftarrow[\sim]{\mathrm{CS}_G^{-1}} & \mathrm{Rep}(\tilde{G})_{\mathrm{Ran}} \\ \mathrm{Poinc}_{G,!}[-2\delta_{N_{\rho}(\omega_X)}] \downarrow & & \downarrow \mathrm{Loc}_G^{\mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xleftarrow{\mathbb{L}_G^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}). \end{array}$$

1.2. **Langlands functor and Eisenstein series.** In this subsection we will summarize the properties of $\mathbb{L}_G$ relevant for this paper that have to do with the Eisenstein series and constant term functors.

1.2.1. A key property of the functor $\mathbb{L}_G$ is that it commutes with the Eisenstein functors, i.e., for a parabolic P with Levi quotient M , the diagram

$$(1.3) \quad \begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}}) \\ \mathrm{Eis}_{!,\mathrm{twk}} \downarrow & & \downarrow \mathrm{Eis}^{\mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}) \end{array}$$

commutes, where $\mathrm{Eis}_{!,\mathrm{twk}}$ is a “tweaked” $!$ -Eisenstein functor, where the tweak involves a translation along Bun_M and a cohomological shift (the precise details of the tweak are irrelevant for this paper), see [GLC3, Theorem 14.2.2].

1.2.2. Passing to the right adjoints along the vertical arrows in (1.3), we obtain a diagram

$$(1.4) \quad \begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xrightarrow{\mathbb{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}}) \\ \mathrm{CT}_{*,\mathrm{twk}} \uparrow & & \uparrow \mathrm{CT}^{\mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}), \end{array}$$

which *a priori* commutes up to a natural transformation (in the above diagram, $\mathrm{CT}_{*,\mathrm{twk}}$ is a “tweaked” Constant Term functor, set to be the right adjoint of $\mathrm{Eis}_{!,\mathrm{twk}}$).

However, one of the main results of the paper [GLC3], namely, Theorem 15.1.2 in *loc. cit.*, says that the natural transformation in (1.4) is an isomorphism. I.e., the diagram (1.4) commutes.

1.2.3. Note that by passing to left adjoints along all arrows in (1.4), we obtain a commutative diagram

$$(1.5) \quad \begin{array}{ccc} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) & \xleftarrow{\mathbb{L}_M^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{M}}) \\ \mathrm{Eis}_{!,\mathrm{twk}} \downarrow & & \downarrow \mathrm{Eis}^{\mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xleftarrow{\mathbb{L}_G^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}). \end{array}$$

1.2.4. Let

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \subset \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

be the full subcategory generated by the essential images of the functors $\mathrm{Eis}_!$ (equivalently, $\mathrm{Eis}_{!,\mathrm{twk}}$) for proper parabolics,

Let

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})_{\mathrm{red}} \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$$

be the full subcategory consisting of objects, set-theoretically supported on the locus

$$\mathrm{LS}_{\tilde{G}}^{\mathrm{red}} \subset \mathrm{LS}_{\tilde{G}}$$

consisting of *reducible* local systems, i.e., the union of the images of the (proper) maps

$$\mathrm{LS}_{\tilde{P}} \rightarrow \mathrm{LS}_{\tilde{G}}$$

for proper parabolic subgroups.

Combining diagrams (1.3) and (1.5) we obtain that the functors $\mathbb{L}_G$ and $\mathbb{L}_G^L$ send the subcategories

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \text{ and } \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})_{\mathrm{red}}$$

to one another, thereby inducing a pair of adjoint functors

$$(1.6) \quad \mathbb{L}_{G,\mathrm{Eis}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \rightleftarrows \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})_{\mathrm{red}} : \mathbb{L}_{G,\mathrm{Eis}}^L.$$

The main result of [GLC3], namely, Theorem 17.1.2 in *loc. cit.* says:

Theorem 1.2.5. *The adjoint functors in (1.6) are (mutually inverse) equivalences.*

1.3. The Langlands functor on the cuspidal part. In this subsection we will study the restriction of $\mathbb{L}_G$ to the cuspidal subcategory. We will show that GLC is equivalent to the statement that the resulting functor

$$\mathbb{L}_{G,\mathrm{cusp}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}}) \simeq \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}})$$

is an equivalence.

1.3.1. Let

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} := \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \right)^\perp$$

be the cuspidal subcategory.

Tautologically, $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}$ is the intersection of the kernels of the functors CT_* (equivalently, $\mathrm{CT}_{*,\mathrm{twk}}$) for all proper parabolics.

1.3.2. Denote by $\mathbf{e}$ the tautological embedding

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} \hookrightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G).$$

Since $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}}$ is generated by objects that are compact in $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$, the functor $\mathbf{e}$ admits a left adjoint, to be denoted $\mathbf{e}^L$.

1.3.3. Let

$$\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}} \xrightarrow{j} \mathrm{LS}_{\tilde{G}}$$

be the embedding of the irreducible locus. We can regard $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}})$ as a full subcategory of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$ of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})$ via j_* , and as such it identifies with

$$(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})_{\mathrm{red}})^\perp.$$

Tautologically,

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}})_{\mathrm{red}} = \ker(j^*).$$

1.3.4. From the commutation of (1.4), we obtain that the functor $\mathbb{L}_G$ sends $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}$ to $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}})$. Denote the resulting functor by

$$(1.7) \quad \mathbb{L}_{G,\mathrm{cusp}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}}).$$

I.e., we have a commutative diagram

$$(1.8) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}) \\ \mathbf{e} \uparrow & & \uparrow j_* \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xrightarrow{\mathbb{L}_{G,\mathrm{cusp}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}}). \end{array}$$

Note that from the commutation of (1.3) we obtain a commutative diagram

$$(1.9) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\ \mathbf{e}^L \downarrow & & \downarrow j^* \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xrightarrow{\mathbb{L}_{G,\mathrm{cusp}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}). \end{array}$$

1.3.5. Note also that when we view $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}$ and $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$ as quotient categories of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ and $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$ via $\mathbf{e}^L$ and j^* , respectively, from the commutation of (1.5), we obtain that there exists a well-defined functor

$$\mathbb{L}_{G,\mathrm{cusp}}^L : \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}$$

that makes the diagram

$$(1.10) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xleftarrow{\mathbb{L}_G^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\ \mathbf{e}^L \downarrow & & \downarrow j^* \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xleftarrow{\mathbb{L}_{G,\mathrm{cusp}}^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}). \end{array}$$

commute.

Taking into account (1.9), we obtain that the functors

$$(1.11) \quad \mathbb{L}_{G,\mathrm{cusp}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} \rightleftarrows \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})_{\mathrm{red}} : \mathbb{L}_{G,\mathrm{cusp}}^L$$

are mutually adjoint, i.e.,

$$\mathbb{L}_{G,\mathrm{cusp}}^L \simeq (\mathbb{L}_{G,\mathrm{cusp}})^L.$$

1.3.6. We claim, however:

Proposition 1.3.7. *The functor $\mathbb{L}_G^L$ sends $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$ to $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}$.*

Proof. Follows from [GLC2, Proposition 17.2.2]. We will supply an alternative argument in Sect. 1.4.8. $\square$

Corollary 1.3.8. *The following diagram commutes:*

$$(1.12) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & \xleftarrow{\mathbb{L}_G^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \\ \mathbf{e} \uparrow & & \uparrow j_* \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xleftarrow{\mathbb{L}_{G,\mathrm{cusp}}^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}). \end{array}$$

1.3.9. Taking into account Theorem 1.2.5, we obtain:

Corollary 1.3.10. *The functor $\mathbb{L}_G$ is an equivalence if and only if so is the functor $\mathbb{L}_{G,\mathrm{cusp}}$.*

1.4. **Spectral action.** In this subsection we will recall another crucial feature of the functor $\mathbb{L}_G$: its compatibility with the $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -actions on the two sides.

1.4.1. Recall (see e.g., [GLC1, Theorem 1.2.4]) that the Hecke action gives rise to an action of the monoidal category $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$.

1.4.2. We have:

Proposition 1.4.3. *With respect to the $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -action on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$, the full subcategory*

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

is set-theoretically supported on $\mathrm{LS}_{\check{G}}^{\mathrm{red}}$, i.e.,

$$(1.13) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{Eis}} \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) = 0.$$

This proposition is probably well-known. We will supply a proof for completeness in Sect. 1.4.9.

1.4.4. As a formal consequence of Proposition 1.4.3, we obtain:

Corollary 1.4.5. *We have an inclusion*

$$(1.14) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}$$

as full subcategories of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$.

In fact, a stronger assertion is true (to be proved in Sect. 1.5.6):

Theorem 1.4.6. *The inclusion (1.14) is an equality.*

1.4.7. By the construction of the functor $\mathbb{L}_G$, it is $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -linear, where $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ acts on $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$ naturally.

Since the symmetric monoidal category $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ is rigid, we obtain that the functor $\mathbb{L}_G^L$ is also equipped with a natural $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -linear structure.

Moreover, the monad

$$\mathbb{L}_G^L \circ \mathbb{L}_G,$$

acting on $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$, is $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -linear.

1.4.8. *Second proof of Proposition 1.3.7.* By $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -linearity, the functor $\mathbb{L}_G^L$ sends

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}),$$

viewed as a full subcategory of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$, to

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}),$$

viewed as a full subcategory of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$.

However, the former is tautologically the same as $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$, while the latter is contained in $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}$ by Corollary 1.4.5.

□[Proposition 1.3.7]

1.4.9. *Proof of Proposition 1.4.3.* According to [BG]⁴, the functor $\mathrm{Eis}_!$ can be factored as a composition

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{M}})} \mathrm{QCoh}(\mathrm{LS}_{\check{P}}) \xrightarrow{\mathrm{Eis}_!^{\mathrm{part.enh}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G),$$

where the functor $\mathrm{Eis}_!^{\mathrm{part.enh}}$ is $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -linear.

This implies the required assertion.

□[Proposition 1.4.3]

⁴The paper [BG] only treats the case of $P = B$. The general case will be treated in a forthcoming paper of J. Fægervan and A. Hayash.

1.5. Conservativity. In this subsection we recall a crucial result from [FR], which says that the functor $\mathbb{L}_{G,\text{cusp}}$ is conservative.

In a sense, this unveils the main reason why GLC holds: that the functor $\mathbb{L}_G$ does not lose information (in a very coarse sense, by sending some objects to zero).

1.5.1. We now import the following result from [FR, Theorem A] (which is a combination of [FR, Theorem B] and [Be2, Theorem A]):

Theorem 1.5.2. *The functor $\mathbb{L}_{G,\text{cusp}}$ is conservative.*

Remark 1.5.3. Note that in the case when $G = GL_n$, the assertion of Theorem 1.5.2 follows immediately from (the much more elementary) Theorem 1.7.2 below.

1.5.4. We now claim:

Theorem 1.5.5. *The functor $\mathbb{L}_G$ is conservative.*

Proof. The follows immediately by combining Theorems 1.5.2 and 1.2.5. □

1.5.6. *Proof of Theorem 1.4.6.* The assertion of the theorem is equivalent to

$$(1.15) \quad \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}} \otimes_{\text{QCoh}(\text{LS}_{\check{G}})} \text{QCoh}(\text{LS}_{\check{G}})_{\text{red}} = 0,$$

where $\text{QCoh}(\text{LS}_{\check{G}})_{\text{red}} = \ker(j^*)$.

By Theorem 1.5.2, it suffices to show that the functor $\mathbb{L}_{G,\text{cusp}}$ annihilates the subcategory (1.15). However, $\mathbb{L}_{G,\text{cusp}}$ sends this category to

$$\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}^{\text{irred}}) \otimes_{\text{QCoh}(\text{LS}_{\check{G}})} \text{QCoh}(\text{LS}_{\check{G}})_{\text{red}},$$

and the latter is obviously 0. □[Theorem 1.4.6]

1.6. The algebra $\mathcal{A}_{G,\text{irred}}$. In this subsection we will introduce an object

$$\mathcal{A}_{G,\text{irred}} \in \text{AssocAlg}(\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})),$$

which encodes the monad $\mathbb{L}_{G,\text{cusp}} \circ \mathbb{L}_{G,\text{cusp}}^L$.

We will show that the validity of GLC is equivalent to the fact that the unit map (1.16) is an isomorphism.

The proof of GLC that will be presented in the sequel to this paper will amount to the showing that the algebraic geometry and topology of $\text{LS}_{\check{G}}^{\text{irred}}$ essentially force this map to be an isomorphism (modulo a certain computation on the automorphic side).

1.6.1. Consider the monad $\mathbb{L}_{G,\text{cusp}} \circ \mathbb{L}_{G,\text{cusp}}^L$ on $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}^{\text{irred}})$ corresponding to the adjoint functors (1.11). By Sect. 1.4.7, this monad is $\text{QCoh}(\text{LS}_{\check{G}})$ -linear.

Since the action of $\text{QCoh}(\text{LS}_{\check{G}})$ on $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}^{\text{irred}})$ factors through

$$j^* : \text{QCoh}(\text{LS}_{\check{G}}) \rightarrow \text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}}),$$

we obtain that $\mathbb{L}_{G,\text{cusp}} \circ \mathbb{L}_{G,\text{cusp}}^L$ is $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$ -linear.

1.6.2. Note that $\text{Nilp}|_{\text{LS}_{\check{G}}^{\text{irred}}} = \{0\}$, so

$$\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}^{\text{irred}}) = \text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}}).$$

Hence, the monad $\mathbb{L}_{G,\text{cusp}} \circ \mathbb{L}_{G,\text{cusp}}^L$ is a $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$ -linear monad on $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$ itself, and thus corresponds to a unital associative algebra, to be denoted

$$\mathcal{A}_{G,\text{irred}} \in \text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}}).$$

1.6.3. The unit of the $(\mathbb{L}_{G,\text{cusp}}^L, \mathbb{L}_{G,\text{cusp}})$ -adjunction corresponds to the unit map

$$(1.16) \quad \mathcal{O}_{\text{LS}_{\check{G}}^{\text{irred}}} \rightarrow \mathcal{A}_{G,\text{irred}}.$$

Tautologically, the functor $\mathbb{L}_{G,\text{cusp}}^L$ is fully faithful if and only if the map (1.16) is an isomorphism.

1.6.4. Given Theorem 1.5.2 and Corollary 1.3.10, we obtain:

Corollary 1.6.5. *The functor $\mathbb{L}_G$ is an equivalence if and only if the map (1.16) is an isomorphism in $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$.*

1.7. **Proof of GLC for $G = GL_n$.** In this subsection, we will show how Theorem 1.2.5 allows us to prove GLC in the case when $G = GL_n$.

1.7.1. The point of departure is the following result, established in [Be1] (or, in a slightly different language, in [Ga2]):

Theorem 1.7.2. *The restriction of the functor coeff_G to the subcategory*

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}} \subset \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$$

is fully faithful.

1.7.3. From Theorem 1.7.2 we will now deduce:

Corollary 1.7.4. *The functor $\mathbb{L}_{G,\text{cusp}}$ is fully faithful.*

Proof. From (1.1) and (1.8), we obtain a commutative diagram

$$\begin{array}{ccc} \text{Whit}^!(G)_{\text{Ran}} & \xrightarrow[\sim]{\text{CS}_G} & \text{Rep}(\check{G})_{\text{Ran}} \\ \text{coeff}_G[2\delta_{N_{\rho(\omega_X)}}] \uparrow & & \uparrow \Gamma_{\check{G}}^{\text{spec}, \text{IndCoh}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \\ \mathbf{e} \uparrow & & \uparrow j_* \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}} & \xrightarrow{\mathbb{L}_{G,\text{cusp}}} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}^{\text{irred}}). \end{array}$$

It is sufficient to show that the composite right vertical arrow in the above diagram is fully faithful. Indeed, this would imply that the fully-faithfulness of the functors

$$\text{coeff}_G[2\delta_{N_{\rho(\omega_X)}}] \circ \mathbf{e} \text{ and } \mathbb{L}_{G,\text{cusp}}$$

are logically equivalent.

Note that since

$$\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}}) = \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}^{\text{irred}}),$$

the right vertical arrow in the above diagram can be identified with

$$(1.17) \quad \text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}}) \xrightarrow{j_*} \text{QCoh}(\text{LS}_{\check{G}}) \xrightarrow{\Gamma_{\check{G}}^{\text{spec}}} \text{Rep}(\check{G})_{\text{Ran}}.$$

Now, in the composition (1.17) both arrows are fully faithful: this is obvious for j_* , and for $\Gamma_{\check{G}}^{\text{spec}}$ this is the content of Proposition 1.1.4. $\square$

Remark 1.7.5. Note that the proof of Corollary 1.7.4 shows that it is actually logically equivalent to Theorem 1.7.2. Hence, once we establish GLC, we will know that Theorem 1.7.2 also holds for any G .

1.7.6. By Corollary 1.6.5, in order to prove GLC, we need to show that the map (1.16) is an isomorphism in $\mathrm{QCoh}(\mathrm{LS}_G^{\mathrm{irred}})$. Since $\mathrm{LS}_G^{\mathrm{irred}}$ is eventually coconnective, it is sufficient to show that for any field-valued point

$$\sigma : \mathrm{Spec}(K) \rightarrow \mathrm{LS}_G^{\mathrm{irred}},$$

the resulting map

$$(1.18) \quad K \rightarrow \mathcal{A}_{G,\sigma}$$

is an isomorphism, where $\mathcal{A}_{G,\sigma}$ denotes the fiber of $\mathcal{A}_{G,\mathrm{irred}}$ at σ .

Applying base change to the functor $\mathbb{L}_{G,\mathrm{cusp}}$ along σ , we obtain a functor

$$\mathbb{L}_{G,\sigma} : \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} \otimes_{\mathrm{QCoh}(\mathrm{LS}_G^{\mathrm{irred}})} \mathrm{Vect}_K \rightarrow \mathrm{Vect}_K.$$

Since the functor $\mathbb{L}_{G,\mathrm{cusp}}$ is fully faithful and admits a left adjoint, we obtain that $\mathbb{L}_{G,\sigma}$ is also fully faithful. In particular, $\mathbb{L}_{G,\sigma}$ is conservative. Hence, by Barr-Beck, it can be identified with the forgetful functor

$$\mathcal{A}_{G,\sigma}\text{-mod} \rightarrow \mathrm{Vect}_K.$$

Such a functor can be fully faithful either when (1.18) is an isomorphism (which is what we want to show), or when $\mathcal{A}_{G,\sigma} = 0$. Thus, it remains to rule out the latter possibility.

1.7.7. We need to show that the category

$$\mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_\sigma := \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} \otimes_{\mathrm{QCoh}(\mathrm{LS}_G^{\mathrm{irred}})} \mathrm{Vect}_K$$

is non-zero. For this, we can further replace K by its algebraic closure.

Performing base change $k \rightsquigarrow K$, we can assume that $K = k$. Then the category $\mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_\sigma$ is, by definition, the category of *Hecke eigen-sheaves* with respect to σ .

However, it was shown in [FGV] that for $G = GL_n$ and σ irreducible, the category $\mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_\sigma$ contains a non-zero object.

Remark 1.7.8. Alternatively, the proof of the existence of a non-zero Hecke eigensheaf for a given irreducible local system, valid for any G , follows by combining the [BD1] construction of Hecke eigensheaves via localization at the critical level and the result of [Ari], which says that any irreducible local system carries a generic oper structure.

□[GLC for $G = GL_n$]

2. LEFT ADJOINT AS THE DUAL

In this section we will establish the "first half" of the Ambidexterity Theorem, namely that the functor *left adjoint* to $\mathbb{L}_{G,\mathrm{cusp}}$ is, up to a certain twist, is canonically isomorphic to its dual.

In order to do so, we will first have to show that the source and the target of $\mathbb{L}_{G,\mathrm{cusp}}$ are canonically self-dual.

2.1. The dual automorphic category. In this subsection we recall, following [DG] or [Ga1], the description of the dual of the category $\mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$.

2.1.1. Recall the category $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}$. It is defined as the *colimit*

$$\mathrm{colim}_U \mathrm{D}\text{-mod}_{\frac{1}{2}}(U),$$

where U runs over the poset of quasi-compact open substacks of Bun_G , and for $U_1 \xrightarrow{j_{1,2}} U_2$ the corresponding functor

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(U_1) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(U_2)$$

is $(j_{1,2})_*$.

For a given quasi-compact open

$$(2.1) \quad U \xhookrightarrow{j} \mathrm{Bun}_G,$$

we let

$$j_{\mathrm{co},*} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(U) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}$$

denote the tautological functor.

2.1.2. The category $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}$ is endowed with a tautologically defined functor

$$\mathrm{Ps}\text{-Id}^{\mathrm{nv}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G),$$

characterized by the following property:

For a quasi-compact open as in (2.1), we have

$$(2.2) \quad \mathrm{Ps}\text{-Id}^{\mathrm{nv}} \circ j_{\mathrm{co},*} \simeq j_*,$$

as functors

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(U) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G).$$

2.1.3. Note that Verdier duality on Bun_G gives rise to a canonical identification

$$(2.3) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)^\vee \xrightarrow{\mathbf{D}^{\mathrm{Verdier}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}.$$

It is characterized by the requirement that for (2.1), we have

$$j_{\mathrm{co},*} \simeq (j^*)^\vee,$$

where we identify

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(U)^\vee \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(U)$$

via usual Verdier duality, also denoted $\mathbf{D}^{\mathrm{Verdier}}$.

2.2. The dual of the cuspidal category. In this subsection we will use Sect. 2.1 to show that the cuspidal automorphic category is canonically self-dual.

2.2.1. Let

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co},\mathrm{Eis}} \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}$$

be the full subcategory, generated by the essential images of the functors

$$\mathrm{Eis}_{\mathrm{co},*} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_M)_{\mathrm{co}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}$$

(see [Gal, Sect. 1.4]) for *proper* parabolic subgroups.

Set

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co},\mathrm{cusp}} := \left(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co},\mathrm{Eis}} \right)^\perp.$$

Let

$$\mathbf{e}_{\mathrm{co}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co},\mathrm{cusp}} \hookrightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}$$

denote the tautological embedding. It admits a *left adjoint*, making $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co},\mathrm{cusp}}$ into a *localization* of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}$.

2.2.2. The identification (2.3) gives rise to an identification

$$(2.4) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}^{\vee} \xrightarrow{\mathbf{D}_{\mathrm{cusp}}^{\mathrm{Verdier}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co},\mathrm{cusp}},$$

so that

$$\mathbf{e}_{\mathrm{co}} \simeq (\mathbf{e}^L)^{\vee} \text{ and } \mathbf{e}_{\mathrm{co}}^L \simeq \mathbf{e}^{\vee}.$$

2.2.3. Recall now that the category $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co},\mathrm{cusp}}$ has the following property (see [Ga1, Proposition 2.3.4]): there exists a quasi-compact open substack

$$U_0 \xrightarrow{j_0} \mathrm{Bun}_G,$$

such that the functor

$$\mathbf{e}_{\mathrm{co}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co},\mathrm{cusp}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}$$

factors as

$$(j_0)_{\mathrm{co},*} \circ \mathbf{e}_{U_0,\mathrm{co}},$$

for (an automatically uniquely defined fully faithful functor)

$$\mathbf{e}_{U_0,\mathrm{co}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co},\mathrm{cusp}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(U_0).$$

2.2.4. Furthermore, according to [Ga1, Theorem 2.2.7], the functor $\mathrm{Ps}\text{-Id}^{\mathrm{nv}}$ sends $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co},\mathrm{cusp}}$ to $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}$, and the resulting functor

$$\mathrm{Ps}\text{-Id}_{\mathrm{cusp}}^{\mathrm{nv}} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co},\mathrm{cusp}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}$$

is an equivalence.

We automatically have

$$(2.5) \quad \mathbf{e} \circ \mathrm{Ps}\text{-Id}_{\mathrm{cusp}}^{\mathrm{nv}} \simeq (j_0)_* \circ \mathbf{e}_{U_0,\mathrm{co}}.$$

2.2.5. Thus, combining (2.4) with the equivalence $\mathrm{Ps}\text{-Id}_{\mathrm{cusp}}^{\mathrm{nv}}$ we obtain a self-duality

$$(2.6) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}^{\vee} \xrightarrow{\mathrm{Ps}\text{-Id}_{\mathrm{cusp}}^{\mathrm{nv}} \circ \mathbf{D}_{\mathrm{cusp}}^{\mathrm{Verdier}}} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}.$$

2.2.6. For later use we introduce the following notation:

Let

$$(2.7) \quad \mathbf{e}_{U_0} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(U_0)$$

be the uniquely defined functor, so that

$$\mathbf{e}_{U_0,\mathrm{co}} \simeq \mathbf{e}_{U_0} \circ \mathrm{Ps}\text{-Id}_{\mathrm{cusp}}^{\mathrm{nv}}.$$

The functor $\mathbf{e}_{U_0}$ is automatically fully faithful and

$$\mathbf{e} \simeq (j_0)_* \circ \mathbf{e}_{U_0}.$$

Let $\mathbf{e}_{U_0}^L$ denote the left adjoint of $\mathbf{e}_{U_0}$. We have:

$$(2.8) \quad \mathbf{e}^L \simeq \mathbf{e}_{U_0}^L \circ j_0^*.$$

2.3. Duality and the Poincaré functors. In this section we will recall the result of [Lin] that says that the $!$ - and $*$ -versions of the geometric Poincaré functor become isomorphic, once we project to the cuspidal automorphic category.

2.3.1. Recall (see [GLC2, Sect. 9.5]) that in addition to the functor

$$\mathrm{Poinc}_{G,!} : \mathrm{Whit}^!(G)_{\mathrm{Ran}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G),$$

there exists a naturally defined functor

$$\mathrm{Poinc}_{G,*} : \mathrm{Whit}_*(G)_{\mathrm{Ran}} \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}.$$

2.3.2. By construction, with respect the duality (2.3) and the canonical duality

$$(2.9) \quad (\mathrm{Whit}^!(G)_{\mathrm{Ran}})^\vee \simeq \mathrm{Whit}_*(G)_{\mathrm{Ran}},$$

we have

$$(2.10) \quad (\mathrm{coeff}_G)^\vee \simeq \mathrm{Poinc}_{G,*}.$$

2.3.3. Recall now (see [GLC2, Theorem 1.3.13]) that, in addition to the duality (2.9), there exists a canonical equivalence

$$(2.11) \quad \Theta_{\mathrm{Whit}(G)} : \mathrm{Whit}_*(G)_{\mathrm{Ran}} \simeq \mathrm{Whit}^!(G)_{\mathrm{Ran}}.$$

The following assertion is a “baby” version of the main result of [Lin] (see Lemma 1.3.4 in *loc. cit.*):

Theorem 2.3.4. *The functors*

$$\mathbf{e}^L \circ \mathrm{Poinc}_{G,!} \circ \Theta_{\mathrm{Whit}(G)} \text{ and } \mathbf{e}^L \circ \mathrm{Ps}\text{-}\mathrm{Id}^{\mathrm{nv}} \circ \mathrm{Poinc}_{G,*} [2\delta_{N_\rho(\omega_X)}],$$

$$\mathrm{Whit}_*(G)_{\mathrm{Ran}} \rightrightarrows \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}$$

are canonically isomorphic.

2.4. Self-duality on the spectral side. In this short subsection we set up our conventions regarding the self-duality of the category $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$.

2.4.1. Let us identify $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ with its own dual via the *naive duality*

$$(2.12) \quad \mathrm{QCoh}(\mathrm{LS}_{\check{G}})^\vee \xrightarrow{\mathbf{D}^{\mathrm{naive}}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}).$$

I.e., the corresponding anti-self equivalence of

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}})^c = \mathrm{QCoh}(\mathrm{LS}_{\check{G}})^{\mathrm{perf}}$$

is given by monoidal dualization.

2.4.2. The self-duality (2.12) induces a self-duality

$$(2.13) \quad \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})^\vee \xrightarrow{\mathbf{D}^{\mathrm{naive}}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}).$$

2.4.3. Recall now (see, e.g., [AGKRRV1, Sect. 11.3]) that the canonical self-duality on $\mathrm{Rep}(\check{G})$ gives rise to a self-duality of the category $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}$.

2.4.4. We have:

Lemma 2.4.5. *With respect to the above self-dualities of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ and $\mathrm{Rep}(\check{G})_{\mathrm{Ran}}$, the functors*

$$\mathrm{Loc}_{\check{G}}^{\mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \leftrightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) : \Gamma_{\check{G}}^{\mathrm{spec}}$$

identify with each other’s duals:

$$(\mathrm{Loc}_{\check{G}}^{\mathrm{spec}})^\vee \simeq \Gamma_{\check{G}}^{\mathrm{spec}} \text{ and } (\Gamma_{\check{G}}^{\mathrm{spec}})^\vee \simeq \mathrm{Loc}_{\check{G}}^{\mathrm{spec}}.$$

2.5. Left adjoint vs dual. In this subsection we finally formulate and prove the main result of this section, Theorem 2.5.3, which says that the left adjoint and the dual of $\mathbb{L}_{G,\mathrm{cusp}}$ are isomorphic, up to a twist.

2.5.1. Consider the functor *dual* to $\mathbb{L}_{G,\text{cusp}}$

$$\mathbb{L}_{G,\text{cusp}}^\vee : \text{QCoh}(\text{LS}_G^{\text{irred}})^\vee \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}}^\vee.$$

Using the identifications (2.13) and (2.6), we can think of $\mathbb{L}_{G,\text{cusp}}^\vee$ as a functor

$$(2.14) \quad \text{QCoh}(\text{LS}_G^{\text{irred}}) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}}.$$

Let

$$\Phi_{G,\text{cusp}} : \text{QCoh}(\text{LS}_G^{\text{irred}}) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}}$$

denote the composition of the functor (2.14) with the Chevalley involution and the shift $[-2\delta_{N_{\rho(\omega_X)}}]$, i.e.,

$$\Phi_{G,\text{cusp}} := \tau_G \circ \mathbb{L}_{G,\text{cusp}}^\vee[-2\delta_{N_{\rho(\omega_X)}}].$$

2.5.2. We are going to prove:

Theorem 2.5.3. *The functor $\Phi_{G,\text{cusp}}$ identifies canonically with $\mathbb{L}_{G,\text{cusp}}^L$.*

This theorem is a particular case of [GLC3, Theorem 16.2.3], and its proof is much simpler in that it only uses Theorem 2.3.4, rather than the full force of the result from [Lin]. We will supply a proof for the sake of completeness, and it occupies the rest of this subsection.

2.5.4. By Proposition 1.1.4, it suffices to establish an isomorphism

$$\mathbb{L}_{G,\text{cusp}}^L \circ j^* \circ \text{Loc}_G^{\text{spec}} \simeq \Phi_{G,\text{cusp}} \circ j^* \circ \text{Loc}_G^{\text{spec}}$$

as functors

$$\text{Rep}(\check{G}) \rightrightarrows \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}}.$$

We will do so by showing that both diagrams

$$(2.15) \quad \begin{array}{ccc} \text{Whit}^!(G)_{\text{Ran}} & \xleftarrow[\sim]{\text{CS}_G^{-1}} & \text{Rep}(\check{G})_{\text{Ran}} \\ \text{Poinc}_{G,!}[-2\delta_{N_{\rho(\omega_X)}}] \downarrow & & \downarrow \text{Loc}_G^{\text{spec}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & & \text{QCoh}(\text{LS}_{\check{G}}) \\ \mathbf{e}^L \downarrow & & \downarrow j^* \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}} & \xleftarrow{\mathbb{L}_{G,\text{cusp}}^L} & \text{QCoh}(\text{LS}_G^{\text{irred}}) \end{array}$$

and

$$(2.16) \quad \begin{array}{ccc} \text{Whit}^!(G)_{\text{Ran}} & \xleftarrow{\text{CS}_G^{-1}} & \text{Rep}(\check{G})_{\text{Ran}} \\ \text{Poinc}_{G,!}[-2\delta_{N_{\rho(\omega_X)}}] \downarrow & & \downarrow \text{Loc}_G^{\text{spec}} \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & & \text{QCoh}(\text{LS}_{\check{G}}) \\ \mathbf{e}^L \downarrow & & \downarrow j^* \\ \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}} & \xleftarrow{\Phi_{G,\text{cusp}}} & \text{QCoh}(\text{LS}_G^{\text{irred}}) \end{array}$$

commute.

2.5.5. The commutation of (2.15) is immediate from (1.2) and (1.10). Thus, it remains to deal with (2.16).

First, according to [GLC2, Lemma 1.4.12], we have

$$\tau_G \circ \mathrm{CS}_G^{-1} \simeq \Theta_{\mathrm{Whit}(G)} \circ \mathrm{CS}_G^\vee.$$

Combining with Theorem 2.3.4, this allows us to replace (2.16) by

$$(2.17) \quad \begin{array}{ccc} \mathrm{Whit}_*(G)_{\mathrm{Ran}} & \xleftarrow{\mathrm{CS}_G^\vee} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{Poinc}_{G,*} \downarrow & & \downarrow \mathrm{Loc}_G^{\mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{co} & & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \\ \mathbf{e}^L \circ \mathrm{Ps-Id}^{\mathrm{nv}} \downarrow & & \downarrow j^* \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xleftarrow{\mathbb{L}_{G,\mathrm{irred}}^\vee[-2\delta_{N_{\rho(\omega_X)}}]} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}). \end{array}$$

2.5.6. Using

$$\mathbf{e}^L \circ \mathrm{Ps-Id}^{\mathrm{nv}} \simeq \mathrm{Ps-Id}_{\mathrm{cusp}}^{\mathrm{nv}} \circ \mathbf{e}^\vee,$$

we can rewrite (2.17) as

$$\begin{array}{ccc} \mathrm{Whit}_*(G)_{\mathrm{Ran}} & \xleftarrow{\mathrm{CS}_G^\vee} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{Poinc}_{G,*} \downarrow & & \downarrow \mathrm{Loc}_G^{\mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{co} & & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \\ \mathrm{Ps-Id}_{\mathrm{cusp}}^{\mathrm{nv}} \circ \mathbf{e}^\vee \downarrow & & \downarrow j^* \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xleftarrow{\mathbb{L}_{G,\mathrm{irred}}^\vee[-2\delta_{N_{\rho(\omega_X)}}]} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}), \end{array}$$

and further by

$$(2.18) \quad \begin{array}{ccc} \mathrm{Whit}_*(G)_{\mathrm{Ran}} & \xleftarrow{\mathrm{CS}_G^\vee} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\ \mathrm{Poinc}_{G,*} \downarrow & & \downarrow \mathrm{Loc}_G^{\mathrm{spec}} \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{co} & & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \\ \mathbf{e}^\vee \downarrow & & \downarrow j^* \\ \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{co,\mathrm{cusp}} & \xleftarrow{\mathbb{L}_{G,\mathrm{irred}}^\vee[-2\delta_{N_{\rho(\omega_X)}}]} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}), \end{array}$$

where we now think of $\mathbb{L}_{G,\mathrm{irred}}^\vee$ as a functor

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{co,\mathrm{cusp}}$$

via (2.13) and (2.4).

2.5.7. Passing to the dual functors in (2.18), we obtain that it is equivalent to

$$\begin{array}{ccc}
 \mathrm{Whit}_*(G)_{\mathrm{Ran}} & \xrightarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\
 \mathrm{coeff}_G \uparrow & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}} \\
 \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G) & & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \\
 \mathrm{e} \uparrow & & \uparrow j_* \\
 \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xrightarrow{\mathbb{L}_{G,\mathrm{irred}}[-2\delta_{N_{\rho(\omega_X)}}]} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}).
 \end{array}$$

However, the commutativity of the latter diagram follows from (1.1) and (1.8).

□[Theorem 2.5.3]

3. RIGHT ADJOINT AS THE DUAL

In this section we will assume that G is semi-simple⁵.

We will prove the “second half” of the ambidexterity theorem, namely, that the functor *right adjoint* to $\mathbb{L}_{G,\mathrm{cusp}}$ is isomorphic to the (twist of) the dual of $\mathbb{L}_{G,\mathrm{cusp}}$.

The full Ambidexterity Theorem says that the left and right adjoints of $\mathbb{L}_{G,\mathrm{cusp}}$ are isomorphic. Of course, this statement *a posteriori* follows from GLC, but in the current strategy, it constitutes a step in its proof.

3.1. The ambidexterity statement.

3.1.1. We continue to regard the categories $\mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}$ and $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$ as self-dual via the identifications (2.13) and (2.6), respectively.

Recall the functor $\Phi_{G,\mathrm{cusp}}$, see Sect. 2.5.1. We will prove:

Theorem 3.1.2. *The functor $\Phi_{G,\mathrm{cusp}}$ identifies canonically with the right adjoint of $\mathbb{L}_{G,\mathrm{cusp}}$.*

3.1.3. Before we prove the theorem, let us draw some consequences. First, by combining Theorems 2.5.3 and 3.1.2, we obtain what we call the Ambidexterity Theorem:

Main Theorem 3.1.4. *The left and right adjoints of $\mathbb{L}_{G,\mathrm{cusp}}$ are isomorphic.*

Corollary 3.1.5. *The endofunctor*

$$\mathbb{L}_{G,\mathrm{cusp}} \circ \mathbb{L}_{G,\mathrm{cusp}}^L$$

of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$ is isomorphic to its own left and right adjoint.

3.1.6. Recall (see Sect. 1.6.2) that the functor $\mathbb{L}_{G,\mathrm{cusp}} \circ \mathbb{L}_{G,\mathrm{cusp}}^L$ is given by tensoring with the associative algebra object

$$\mathcal{A}_{G,\mathrm{irred}} \in \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}).$$

From Corollary 3.1.5 we obtain:

Corollary 3.1.7. *The object $\mathcal{A}_{G,\mathrm{irred}} \in \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$ is self-dual. In particular, it belongs to $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})^{\mathrm{perf}}$, i.e., it is compact.*

Eventually we will prove an even more precise version of the second part of the above corollary (see Sect. 4.4):

Main Theorem 3.1.8. *The object $\mathcal{A}_{G,\mathrm{irred}} \in \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$ is a classical vector bundle, which is equipped with a naturally defined flat connection⁶.*

Remark 3.1.9. Note that it makes sense to talk about classical vector bundles on $\mathrm{LS}_{\check{G}}^{\mathrm{irred}}$, since, under the assumption that $\check{G}$ is semi-simple, the stack $\mathrm{LS}_{\check{G}}^{\mathrm{irred}}$ is classical and smooth.

⁵This assumption is just a convenience. The statement holds for any reductive G , just the proof would involve slightly more notation.

⁶Note that Corollary 4.2.5 and Proposition 4.2.8 imply that the resulting local system on $\mathrm{LS}_{\check{G}}^{\mathrm{irred}}$ has a finite monodromy

3.1.10. The rest of this section is devoted to the proof of Theorem 3.1.2.

3.2. Critical localization. In this subsection, we will show that the right adjoint of the critical localization functor (functor Loc_G below) is *essentially* isomorphic to its dual, once we restrict to the cuspidal automorphic category.

3.2.1. Let Loc_G be the functor

$$\text{KL}(G)_{\text{crit}, \text{Ran}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$$

of [GLC2, Sect. 14.1.4].

3.2.2. Denote by $\text{Loc}_{G, \text{cusp}}$ the composite functor

$$\text{KL}(G)_{\text{crit}, \text{Ran}} \xrightarrow{\text{Loc}_G} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \xrightarrow{\mathbf{e}_L} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}}.$$

We have the following counterpart of Proposition 1.1.4:

Proposition 3.2.3. *The functor $\text{Loc}_{G, \text{cusp}}$ is Verdier quotient.*

Proof. Let U_0 be as in Sect. 2.2.3. Denote

$$\text{Loc}_{G, U_0} := j_0^* \circ \text{Loc}_G, \quad \text{KL}(G)_{\text{crit}, \text{Ran}} \rightarrow \text{D-mod}_{\frac{1}{2}}(U_0),$$

so that

$$(3.1) \quad \text{Loc}_{G, \text{cusp}} \simeq \mathbf{e}_{U_0}^L \circ \text{Loc}_{G, U_0}$$

where $\mathbf{e}_{U_0}^L$ is the left adjoint of the functor $\mathbf{e}_{U_0}$ of (2.7).

It is known that for any quasi-compact U , the corresponding functor $\text{Loc}_{G, U}$ is a Verdier quotient (see [GLC2, Theorem 13.4.2]). Now, the assertion follows from (3.1), since $\mathbf{e}_{U_0}^L$ is also a Verdier quotient. $\square$

3.2.4. Recall (see [GLC2, Sect. 2.2.4]) that the category $\text{KL}(G)_{\text{crit}, \text{Ran}}$ is also canonically self-dual. Thus, using the self-duality of $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}}$ given by (2.6), we can regard the dual of $\text{Loc}_{G, \text{cusp}}$ as a functor

$$(3.2) \quad \text{Loc}_{G, \text{cusp}}^\vee : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}} \rightarrow \text{KL}(G)_{\text{crit}, \text{Ran}}.$$

The following assertion is a counterpart of Lemma 2.4.5:

Proposition 3.2.5. *We have a canonical identification between $\text{Loc}_{G, \text{cusp}}^\vee$ and*

$$\text{Loc}_{G, \text{cusp}}^R \otimes \det(\Gamma(X, \mathcal{O}_X) \otimes \mathfrak{g})[\delta_G],$$

where $\delta_G = \dim(\text{Bun}_G)$.

Proof. Let Loc_{G, U_0} be as in the proof of Proposition 3.2.3. It follows formally that we have the identifications

$$\text{Loc}_{G, \text{cusp}}^\vee \simeq \text{Loc}_{G, U_0}^\vee \circ \mathbf{e}_{U_0} \quad \text{and} \quad \text{Loc}_{G, \text{cusp}}^R \simeq \text{Loc}_{G, U_0}^R \circ \mathbf{e}_{U_0}.$$

Thus, in order to prove Proposition 3.2.5, it suffices to show that for a quasi-compact $U \subset \text{Bun}_G$, with respect to the Verdier self-duality of $\text{D-mod}_{\frac{1}{2}}(U)$, we have

$$(3.3) \quad \text{Loc}_{G, U}^\vee \simeq \text{Loc}_{G, U}^R \otimes \det(\Gamma(X, \mathcal{O}_X) \otimes \mathfrak{g})[\delta_G]$$

as functors

$$\text{D-mod}_{\frac{1}{2}}(U) \rightarrow \text{KL}(G)_{\text{crit}, \text{Ran}}.$$

Denote by $\text{Loc}_{G, \text{crit}}$ the composition of Loc_G with the identification

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \simeq \text{D-mod}_{\text{crit}}(\text{Bun}_G)$$

of [GLC2, Equation (10.2)], and similarly for Bun_G replaced by U .

In terms of the self-duality of $\mathrm{D}\text{-mod}_{\mathrm{crit}}(U)$ specified in [GLC2, Sect. 10.5.2], we can reformulate (3.3) as

$$(3.4) \quad \mathrm{Loc}_{G,\mathrm{crit},U}^{\vee} \simeq \mathrm{Loc}_{G,\mathrm{crit},U}^R.$$

Note that by the definition of the functor $\mathrm{Loc}_{G,\mathrm{crit}}$ in [GLC2, Sect. 11.3.3-11.3.4], the functor $\mathrm{Loc}_{G,\mathrm{crit},U}^R$ is the functor $\Gamma_{G,\mathrm{crit}} \circ j_{\mathrm{co},*}$, where

$$\Gamma_{G,\mathrm{crit}} : \mathrm{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Bun}_G)_{\mathrm{co}} \rightarrow \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}}$$

is the functor of [GLC2, Sect. 10.2.5].

We have:

$$\mathrm{Loc}_{G,\mathrm{crit},U}^{\vee} \simeq \mathrm{Loc}_{G,\mathrm{crit}}^{\vee} \circ (j_*)^{\vee} = \mathrm{Loc}_{G,\mathrm{crit}}^{\vee} \circ j_{\mathrm{co},*}.$$

Hence, in order to prove (3.4) it remains to identify

$$\mathrm{Loc}_{G,\mathrm{crit}}^{\vee} \simeq \Gamma_{G,\mathrm{crit}}.$$

However, the latter is the statement of [GLC2, Proposition 10.5.7(b)]. $\square$

3.3. The spectral Poincaré functor. In this section we will show that the right adjoint of the spectral Poincaré functor (functor $\mathrm{Poinc}_{G,*}^{\mathrm{spec}}$ below) is *essentially* isomorphic to its dual, once we restrict to the locus of irreducible local systems.

NB: it is in this subsection that the assumption that G (rather $\check{G}$) is semi-simple is used⁷.

3.3.1. Let

$$\mathrm{Poinc}_{G,*}^{\mathrm{spec}} : \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$$

be the functor of [GLC2, Sect. 17.4.2].

Denote by $\mathrm{Poinc}_{G,*}^{\mathrm{spec}}|_{\mathrm{irred}}$ the composite functor

$$\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \xrightarrow{\mathrm{Poinc}_{G,*}^{\mathrm{spec}}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \xrightarrow{j^*} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}).$$

3.3.2. Recall now (see [GLC2, Sect. 3.2]) that in addition to $\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}}$, we can consider the category

$$\mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}},$$

and we have a canonical identification

$$(3.5) \quad (\mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}})^{\vee} \xrightarrow{\mathrm{D}^{\mathrm{Serre}}} \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}}.$$

In addition, we have an identification

$$\Theta_{\mathrm{Op}^{\mathrm{mon-free}}(\check{G})} : \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}},$$

see [GLC2, Equation (3.21)].

Composing we obtain a datum of self-duality:

$$(3.6) \quad \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}}^{\vee} \xrightarrow{\Theta_{\mathrm{Op}^{\mathrm{mon-free}}(\check{G})} \circ \mathrm{D}^{\mathrm{Serre}}} \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}}.$$

⁷In fact, it is used twice, and one can show that the two usages cancel each other out. We just chose not to go through this exercise.

3.3.3. We are going to prove:

Proposition 3.3.4. *With respect to the self-dualities (3.6) and (2.13), we have a canonical identification between the functor dual to $\mathrm{Poinc}_{\check{G},*,\mathrm{irred}}^{\mathrm{spec}}$ and*

$$(\mathrm{Poinc}_{\check{G},*,\mathrm{irred}}^{\mathrm{spec}})^R \otimes \mathbb{I}_{\mathrm{Kost}}[-\delta_G],$$

where $\mathbb{I}_{\mathrm{Kost}}$ is the line of [GLC2, Sect. 17.2.2].

The rest of this subsection is devoted to the proof of Proposition 3.3.4.

3.3.5. Recall (see [GLC2, Sect. 17.4.1]) that in addition to the functor $\mathrm{Poinc}_{\check{G},*}^{\mathrm{spec}}$, there exists a functor

$$\mathrm{Poinc}_{\check{G},!}^{\mathrm{spec}} : \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}} \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}).$$

Denote

$$\mathrm{Poinc}_{\check{G},!,\mathrm{irred}}^{\mathrm{spec}} := j^* \circ \mathrm{Poinc}_{\check{G},!}^{\mathrm{spec}}.$$

According to [GLC2, Theorem 17.4.7], we have:

$$\mathrm{Poinc}_{\check{G},!}^{\mathrm{spec}} \otimes \mathbb{I}_{\mathrm{Kost}}[-\delta_G] \simeq \mathrm{Poinc}_{\check{G},*}^{\mathrm{spec}} \circ \Theta_{\mathrm{Op}^{\mathrm{mon-free}}(\check{G})}.$$

Hence, the assertion of the proposition can be reformulated as an isomorphism

$$(3.7) \quad (\mathrm{Poinc}_{\check{G},!,\mathrm{irred}}^{\mathrm{spec}})^\vee \simeq (\mathrm{Poinc}_{\check{G},*,\mathrm{irred}}^{\mathrm{spec}})^R[2\delta_G],$$

as functors

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \Rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}},$$

where we regard $(\mathrm{Poinc}_{\check{G},!,\mathrm{irred}}^{\mathrm{spec}})^\vee$ as a functor

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \xrightarrow{(2.13)} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})^\vee \rightarrow \mathrm{IndCoh}^!(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}}^\vee \simeq \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}}.$$

To simplify the notation, we will prove a variant of (3.7), where instead of the entire Ran we worked at a fixed point $\underline{x} \in \mathrm{Ran}$. I.e., we will prove

$$(3.8) \quad (\mathrm{Poinc}_{\check{G},!,\underline{x},\mathrm{irred}}^{\mathrm{spec}})^\vee \simeq (\mathrm{Poinc}_{\check{G},*,\underline{x},\mathrm{irred}}^{\mathrm{spec}})^R[2\delta_G]$$

as functors

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \Rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}),$$

3.3.6. Recall that

$$\mathrm{Poinc}_{\check{G},*,\underline{x}}^{\mathrm{spec}} \quad \text{and} \quad \mathrm{Poinc}_{\check{G},!,\underline{x}}^{\mathrm{spec}}$$

are given by

$$(\pi_{\underline{x}})_*^{\mathrm{IndCoh}} \circ (s_{\underline{x}})^{\mathrm{IndCoh},*} \quad \text{and} \quad (\pi_{\underline{x}})_*^{\mathrm{IndCoh}} \circ (s_{\underline{x}})^!,$$

respectively, for the morphisms

$$\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}} \xleftarrow{s_{\underline{x}}} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(X - \underline{x}) \xrightarrow{\pi_{\underline{x}}} \mathrm{LS}_{\check{G}},$$

see [GLC2, Sects. 17.4.1 and 17.4.2].

Let

$$\mathrm{coeff}_{\check{G},\underline{x}}^{\mathrm{spec}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}})$$

denote the functor

$$(s_{\underline{x}})_*^{\mathrm{IndCoh}} \circ (\pi_{\underline{x}})^!.$$

For future use, we also introduce the notation for the Ran version of this functor

$$\mathrm{coeff}_{\check{G}}^{\mathrm{spec}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}) \rightarrow \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}})_{\mathrm{Ran}}.$$

3.3.7. Let

$$(3.9) \quad \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})^\vee \xrightarrow{\mathbf{D}^{\mathrm{Serre}}} \mathrm{IndCoh}(\mathrm{LS}_{\check{G}})$$

be the identification, given by *Serre duality*.

Note that with respect to identifications (3.9) and (3.5), we have

$$(3.10) \quad (\mathrm{Poinc}_{\check{G},!,\underline{x}}^{\mathrm{spec}})^\vee \simeq \mathrm{coeff}_{\check{G},\underline{x}}^{\mathrm{spec}}.$$

3.3.8. Due to the assumption that G (and hence $\check{G}$) is semi-simple, the stack $\mathrm{LS}_{\check{G}}^{\mathrm{irred}}$ is smooth, so the natural embedding

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \hookrightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$$

is an equivalence.

In particular, the identification (3.9) induces an identification

$$(3.11) \quad \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})^\vee \xrightarrow{\mathbf{D}^{\mathrm{Serre}}} \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}).$$

From (3.10) we obtain

$$(j^* \circ \mathrm{Poinc}_{\check{G},!,\underline{x}}^{\mathrm{spec}})^\vee \simeq \mathrm{coeff}_{\check{G},\underline{x}}^{\mathrm{spec}} \circ j_* =: \mathrm{coeff}_{\check{G},\underline{x},\mathrm{irred}}^{\mathrm{spec}},$$

as functors

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \rightrightarrows \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G},\underline{x}}^{\mathrm{mon-free}}),$$

where we use (3.11) to identify $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$ with its own dual.

3.3.9. Note now that since $\check{G}$ is semi-simple, the Killing form on $\check{\mathfrak{g}}$ defines a canonical symplectic structure on $\mathrm{LS}_{\check{G}}^{\mathrm{irred}}$. Hence,

$$\mathbf{D}^{\mathrm{Serre}} \simeq \mathbf{D}^{\mathrm{naive}}[\dim(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}(X))] = \mathbf{D}^{\mathrm{naive}}[2\delta_G].$$

Hence, (3.8) becomes equivalent to an isomorphism

$$(3.12) \quad \mathrm{coeff}_{\check{G},\underline{x},\mathrm{irred}}^{\mathrm{spec}} \simeq (\mathrm{Poinc}_{\check{G},*,\underline{x},\mathrm{irred}}^{\mathrm{spec}})^R.$$

3.3.10. Thus, we have to establish an adjunction between

$$j^* \circ (\pi_{\underline{x}})_*^{\mathrm{IndCoh}} \circ (s_{\underline{x}})^* \text{ and } (s_{\underline{x}})_*^{\mathrm{IndCoh}} \circ (\pi_{\underline{x}})^! \circ j_*.$$

Since the functors $((s_{\underline{x}})_*^{\mathrm{IndCoh}}, (s_{\underline{x}})^*^{\mathrm{IndCoh}})$ form an adjoint pair, it suffices to establish an adjunction between

$$j^* \circ (\pi_{\underline{x}})_*^{\mathrm{IndCoh}} \text{ and } (\pi_{\underline{x}})^! \circ j_*.$$

3.3.11. Set

$$\mathrm{Op}_{\check{G}}^{\mathrm{mon-free,irred}}(X - \underline{x}) := \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(X - \underline{x}) \times_{\mathrm{LS}_{\check{G}}} \mathrm{LS}_{\check{G}}^{\mathrm{irred}}.$$

Let $\pi_{\underline{x}}^{\mathrm{irred}}$ denote the resulting morphism

$$\mathrm{Op}_{\check{G}}^{\mathrm{mon-free,irred}}(X - \underline{x}) \rightarrow \mathrm{LS}_{\check{G}}^{\mathrm{irred}}.$$

By base change, the required adjunction is equivalent to an adjunction between

$$\mathrm{IndCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free,irred}}(X - \underline{x})) \xrightarrow{(\pi_{\underline{x}}^{\mathrm{irred}})_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$$

and

$$\mathrm{IndCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \xrightarrow{(\pi_{\underline{x}}^{\mathrm{irred}})^!} \mathrm{IndCoh}(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free,irred}}(X - \underline{x})).$$

However, this follows from the fact that, under the assumption that $\check{G}$ is semi-simple, the morphism $\pi_{\underline{x}}^{\mathrm{irred}}$ is ind-proper. Indeed, the generic non-degeneracy condition for opers is automatic, once the underlying local system is irreducible.

□[Proposition 3.3.4]

3.4. Proof of Theorem 3.1.2. This proof will amount to comparing the commutative diagrams obtained from the diagram expressing the compatibility of $\mathbb{L}_G$ with critical localization (diagram (3.14) below) by passage to right adjoint and dual functors, respectively.

3.4.1. By Proposition 3.2.3, in order to construct an isomorphism

$$(\mathbb{L}_{G,\text{cusp}})^R \simeq \Phi_{G,\text{cusp}},$$

it suffices to establish an isomorphism

$$(3.13) \quad (\text{Loc}_{G,\text{cusp}})^\vee \circ (\mathbb{L}_{G,\text{cusp}})^R \simeq (\text{Loc}_{G,\text{cusp}})^\vee \circ \Phi_{G,\text{cusp}}.$$

3.4.2. Recall that according to [GLC2, Theorem 18.5.2], we have the following commutative diagram:

$$(3.14) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \\ \text{Loc}_G \otimes \mathfrak{l} \uparrow & & \uparrow \text{Poinc}_{G,*}^{\text{spec}} \\ \text{KL}(G)_{\text{crit,Ran}} & \xrightarrow{\text{FLE}_{G,\text{crit}}} & \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})_{\text{Ran}}, \end{array}$$

where :

- $\text{FLE}_{G,\text{crit}}$ is the *critical FLE functor* of [GLC2, Equation (6.7)];
- $\mathfrak{l}$ is the comologically graded line

$$\mathfrak{l}_{G,N_{\rho(\omega_X)}}^{\otimes \frac{1}{2}} \otimes \mathfrak{l}_{N_{\rho(\omega_X)}}^{\otimes -1}[-\delta_{N_{\rho(\omega_X)}}],$$

where:

- $\mathfrak{l}_{G,N_{\rho(\omega_X)}}^{\otimes \frac{1}{2}}$ is the (non-graded) line of [GLC2, Equation (9.7)];
- $\mathfrak{l}_{N_{\rho(\omega_X)}}$ is the (non-graded) line of [GLC2, Equation (14.2)];
- $\delta_{N_{\rho(\omega_X)}} = \dim(\text{Bun}_{N_{\rho(\omega_X)}})$.

3.4.3. Concatenating diagrams (3.14) and (1.9), we obtain a commutative diagram

$$(3.15) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}} & \xrightarrow{\mathbb{L}_{G,\text{cusp}}} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}^{\text{irred}}) \\ \text{Loc}_{G,\text{cusp}} \otimes \mathfrak{l} \uparrow & & \uparrow \text{Poinc}_{G,*}^{\text{spec}} \\ \text{KL}(G)_{\text{crit,Ran}} & \xrightarrow{\text{FLE}_{G,\text{crit}}} & \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})_{\text{Ran}}. \end{array}$$

Passing to the right adjoints in (3.15) we obtain a diagram

$$(3.16) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}} & \xleftarrow{(\mathbb{L}_{G,\text{cusp}})^R} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}^{\text{irred}}) \\ \text{Loc}_{G,\text{cusp}}^R \downarrow & & \downarrow (\text{Poinc}_{G,*}^{\text{spec}})^{R \otimes \mathfrak{l}} \\ \text{KL}(G)_{\text{crit,Ran}} & \xleftarrow{\text{FLE}_{G,\text{crit}}^{-1}} & \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})_{\text{Ran}}. \end{array}$$

We will establish (3.13) by showing that the diagram

$$(3.17) \quad \begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}} & \xleftarrow{\Phi_{G,\text{cusp}}} & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}^{\text{irred}}) \\ \text{Loc}_{G,\text{cusp}}^R \downarrow & & \downarrow (\text{Poinc}_{G,*}^{\text{spec}})^{R \otimes \mathfrak{l}} \\ \text{KL}(G)_{\text{crit,Ran}} & \xleftarrow{\text{FLE}_{G,\text{crit}}^{-1}} & \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})_{\text{Ran}} \end{array}$$

commutes as well.

3.4.4. Consider the diagram obtained by passing to the duals in (3.15):

$$(3.18) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xleftarrow{(\mathbb{L}_{G,\mathrm{cusp}})^\vee} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{irred}}) \\ \mathrm{Loc}_{G,\mathrm{cusp}}^\vee \otimes \mathbb{I} \downarrow & & \downarrow (\mathrm{Poinc}_{G,*,\mathrm{irred}}^{\mathrm{spec}})^\vee \\ \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & \xleftarrow{\mathrm{FLE}_{G,\mathrm{crit}}^\vee} & \mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}} \end{array}$$

Recall now that according to [GLC2, Theorem 8.1.4], we have a canonical identification

$$\mathrm{FLE}_{G,\mathrm{crit}}^\vee \simeq \tau_G \circ \mathrm{FLE}_{G,\mathrm{crit}}^{-1} \circ \Theta_{\mathrm{Op}^{\mathrm{mon-free}}(\check{G})}$$

as functors

$$\mathrm{IndCoh}^!(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}} \rightarrow \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}}.$$

Combining with Proposition 3.3.4, we can rewrite (3.18) as

$$(3.19) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xleftarrow{\tau_G \circ (\mathbb{L}_{G,\mathrm{cusp}})^\vee} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{irred}}) \\ \mathrm{Loc}_{G,\mathrm{cusp}}^\vee \otimes \mathbb{I} \downarrow & & \downarrow (\mathrm{Poinc}_{G,*,\mathrm{irred}}^{\mathrm{spec}})^R \otimes \mathbb{I}_{\mathrm{Kost}}^{\otimes -1}[\delta_G] \\ \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & \xleftarrow{\mathrm{FLE}_{G,\mathrm{crit}}^{-1}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}}, \end{array}$$

and further as

$$(3.20) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xleftarrow{\Phi_{G,\mathrm{cusp}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{irred}}) \\ \mathrm{Loc}_{G,\mathrm{cusp}}^\vee[2\delta_{N_{\rho(\omega_X)}}] \downarrow & & \downarrow (\mathrm{Poinc}_{G,*,\mathrm{irred}}^{\mathrm{spec}})^R \otimes \mathbb{I}^{\otimes -1} \otimes \mathbb{I}_{\mathrm{Kost}}^{\otimes -1}[\delta_G] \\ \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & \xleftarrow{\mathrm{FLE}_{G,\mathrm{crit}}^{-1}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}}. \end{array}$$

Taking into account Proposition 3.2.5, we can further rewrite (3.20) as

$$(3.21) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xleftarrow{\Phi_{G,\mathrm{cusp}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_G^{\mathrm{irred}}) \\ \mathrm{Loc}_{G,\mathrm{cusp}}^R[2\delta_{N_{\rho(\omega_X)}}] \downarrow & & \downarrow (\mathrm{Poinc}_{G,*,\mathrm{irred}}^{\mathrm{spec}})^R \otimes \mathbb{I}^{\otimes -1} \otimes \mathbb{I}_{\mathrm{Kost}}^{\otimes -1} \otimes \det(\Gamma(X, \mathcal{O}_X) \otimes \mathfrak{g})^{\otimes -1} \\ \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & \xleftarrow{\mathrm{FLE}_{G,\mathrm{crit}}^{-1}} & \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mon-free}})_{\mathrm{Ran}}. \end{array}$$

3.4.5. Comparing (3.21) with the desired diagram (3.17), we conclude that it establish the isomorphism

$$\mathbb{I}_{\mathrm{Kost}} \otimes \det(\Gamma(X, \mathcal{O}_X) \otimes \mathfrak{g})^{\otimes -1} \simeq (\mathbb{I}_{G,N_{\rho(\omega_X)}}^{\otimes \frac{1}{2}})^{\otimes 2} \otimes \mathbb{I}_{N_{\rho(\omega_X)}}^{\otimes -2}.$$

However, the latter is given by [GLC3, Proposition 15.1.10].

□[Theorem 3.1.2]

3.5. An addendum: ambidexterity for eigensheaves. The contents of this subsection will not be used elsewhere in the paper. Here we will explain another approach to ambidexterity, albeit so far working only for Hecke eigensheaves (or more generally D-modules with nilpotent singular support, see Remark 3.5.9).

3.5.1. Fix a point $\sigma \in \mathrm{LS}_G^{\mathrm{irred}}$, and let

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_\sigma := \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{QCoh}(\mathrm{LS}_G)} \mathrm{Vect}$$

be the corresponding category of Hecke eigensheaves, where

$$\mathrm{QCoh}(\mathrm{LS}_G) \rightarrow \mathrm{Vect}$$

is the functor of $*$ -fiber of σ , to be denoted $(i_\sigma)^*$.

Note that since σ was assumed irreducible, the forgetful functor

$$(3.22) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_\sigma \xrightarrow{\mathrm{oblv}^\sigma} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$$

lands in $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}$, see Corollary 1.4.5.

3.5.2. The functor $\mathbb{L}_G$ induces a functor

$$\mathbb{L}_{G,\sigma} : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_\sigma \rightarrow \mathrm{Vect}.$$

According to Theorem 3.1.4, the left and rights adjoints of $\mathbb{L}_{G,\sigma}$ are (canonically) isomorphic. In this subsection we will exhibit another way of constructing such an isomorphism⁸.

3.5.3. Note that by construction, the functor $\mathbb{L}_{G,\sigma}$ is isomorphic to the composition

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_\sigma \xrightarrow{\mathbf{oblv}_\sigma} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{\mathrm{coeff}_G^{\mathrm{Vac, glob}}} \mathrm{Vect},$$

where $\mathrm{coeff}_G^{\mathrm{Vac, glob}}$ is as in [GLC2, Sect. 9.6.3].

The left adjoint of $\mathbb{L}_{G,\sigma}$, denoted $\mathbb{L}_{G,\sigma}^L$ sends the generator $k \in \mathrm{Vect}$ to the object

$$(i_\sigma)^*(\mathrm{Poinc}_{G,!}^{\mathrm{Vac, glob}}),$$

where:

- $(i_\sigma)^*$ denotes the functor

$$\begin{aligned} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) &\simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})} \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}) \xrightarrow{\mathrm{id} \otimes (i_\sigma)^*} \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})} \mathrm{Vect} = \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_\sigma, \end{aligned}$$

left adjoint to the forgetful functor $\mathbf{oblv}_\sigma$;

- $\mathrm{Poinc}_{G,!}^{\mathrm{Vac, glob}} \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ is the object from [GLC1, Sect. 1.3].

3.5.4. Thus, we wish to construct a canonical isomorphism

$$(3.23) \quad \mathcal{H}om_{\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_\sigma}(\mathcal{F}, \mathbb{L}_{G,\sigma}^L(\mathbf{V})) \simeq \mathcal{H}om_{\mathrm{Vect}}(\mathrm{coeff}_G^{\mathrm{Vac, glob}} \circ \mathbf{oblv}_\sigma(\mathcal{F}), \mathbf{V})$$

for $\mathcal{F} \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_\sigma$ and $\mathbf{V} \in \mathrm{Vect}$.

We will rewrite both sides of (3.23) and show that they are canonically isomorphic.

3.5.5. Using Theorem 2.3.4, we rewrite $\mathbb{L}_{G,\sigma}^L(\mathbf{V})$ as

$$(3.24) \quad (i_\sigma)^*(\mathrm{Ps}\text{-Id}^{\mathrm{nv}}(\mathrm{Poinc}_{G,*}^{\mathrm{Vac, glob}})) \otimes \mathbf{V},$$

where

$$\mathrm{Poinc}_{G,*}^{\mathrm{Vac, glob}} := \mathbb{D}^{\mathrm{Verdier}}(\mathrm{Poinc}_{G,!}^{\mathrm{Vac, glob}})[-2\delta_{N_\rho(\omega_X)}] \in \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{co}}.$$

Since σ is a smooth point of $\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}}$, we have

$$(i_\sigma)^* \simeq (i_\sigma)^![\dim(\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}})] \otimes \det(T_\sigma^*(\mathrm{LS}_{\tilde{G}})),$$

where $i_\sigma : \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}}) \rightarrow \mathrm{Vect}$ is the functor of !-pullback, which is defined for maps of finite Tor-dimension, and is the *right* adjoint of $(i_\sigma)_*$, since the map $i_\sigma : \mathrm{pt} \rightarrow \mathrm{LS}_{\tilde{G}}^{\mathrm{irred}}$ is proper.

In particular, the functor

$$\begin{aligned} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) &\simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})} \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}) \xrightarrow{\mathrm{id} \otimes (i_\sigma)^!} \\ &\rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})} \mathrm{Vect} = \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_\sigma \end{aligned}$$

is the right adjoint of $\mathbf{oblv}_\sigma$.

⁸However, it is not obvious that the isomorphism we will construct in this subsection is the same as one from Theorem 3.1.4.

Note also that using the symplectic structure on LS_G^{irred} , we can trivialize the line $\det(T_\sigma^*(LS_G))$, and we note that $\dim(LS_G^{\text{irred}}) = 2 \dim(\text{Bun}_G)$.

Combining, we obtain that the left-hand side in (3.23) identifies with

$$(3.25) \quad \mathcal{H}om_{\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)}(\mathbf{oblv}_\sigma(\mathcal{F}), \text{Ps-Id}^{\text{bv}}(\text{Poinc}_{G,*}^{\text{Vac, glob}}) \otimes \mathbf{V})[2 \dim(\text{Bun}_G)].$$

3.5.6. Denote

$$\mathcal{F}' := \mathbf{oblv}_\sigma(\mathcal{F}).$$

We rewrite (3.25) using Verdier duality as

$$(3.26) \quad \mathcal{H}om_{\text{Vect}}(C_c(\text{Bun}_G, \mathcal{F}' \otimes^* \text{Poinc}_{G,!}^{\text{Vac, glob}}), \mathbf{V})[2 \dim(\text{Bun}_G) + 2\delta_{N_{\rho(\omega_X)}}].$$

And we rewrite the right-hand side of (3.23) as

$$(3.27) \quad \mathcal{H}om_{\text{Vect}}(C(\text{Bun}_G, \mathcal{F}' \otimes^! \text{Poinc}_{G,*}^{\text{Vac, glob}}), \mathbf{V})[2\delta_{N_{\rho(\omega_X)}}].$$

Hence, in order to establish (3.23), we need to construct an isomorphism

$$(3.28) \quad C_c(\text{Bun}_G, \mathcal{F}' \otimes^* \text{Poinc}_{G,!}^{\text{Vac, glob}})[-2 \dim(\text{Bun}_G)] \simeq C(\text{Bun}_G, \mathcal{F}' \otimes^! \text{Poinc}_{G,*}^{\text{Vac, glob}}).$$

3.5.7. By the main theorem of [Lin], we have

$$\text{Poinc}_{G,!}^{\text{Vac, glob}}[-2 \dim(\text{Bun}_G)] \simeq \text{Mir}_{\text{Bun}_G}(\text{Poinc}_{G,*}^{\text{Vac, glob}}).$$

Now (3.28) follows from the fact that $\mathcal{F}'$ has nilpotent singular support (see [AGKRRV1, Corollary 14.4.10]) combined with the next general assertion from [AGKRRV2, Theorem 3.4.2]:

Theorem 3.5.8. *For any $\mathcal{F}' \in \text{D-mod}_{\frac{1}{2}, \text{Nilp}}(\text{Bun}_G)$ and any $\mathcal{F}'' \in \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{co}}$, there is a canonical isomorphism*

$$C_c(\text{Bun}_G, \mathcal{F}' \otimes^* \text{Mir}_{\text{Bun}_G}(\mathcal{F}'')) \simeq C(\text{Bun}_G, \mathcal{F}' \otimes^! \mathcal{F}'').$$

Remark 3.5.9. The above argument can be generalized so that it proves ambidexterity for the functor induced by $\mathbb{L}_G$

$$\text{D-mod}_{\frac{1}{2}, \text{Nilp}}(\text{Bun}_G)_{\text{cusp}} \rightarrow \text{IndCoh}(LS_G^{\text{irred, restr}}),$$

where

$$\text{D-mod}_{\frac{1}{2}, \text{Nilp}}(\text{Bun}_G)_{\text{cusp}} := \text{D-mod}_{\frac{1}{2}, \text{Nilp}}(\text{Bun}_G) \cap \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\text{cusp}}$$

and

$$LS_G^{\text{irred, restr}} := LS_G^{\text{restr}} \cap LS_G^{\text{irred}}.$$

Note that $LS_G^{\text{irred, restr}}$ is a disjoint union of formal schemes, each of which is isomorphic to the formal completion of a point in a smooth symplectic scheme of dimension $2 \dim(\text{Bun}_G)$.

4. THE EXPRESSION FOR $\mathcal{A}_{G, \text{irred}}$ VIA OPERS

In this section we will prove that the object $\mathcal{A}_{G, \text{irred}} \in \text{QCoh}(LS_G^{\text{irred}})$ can be expressed via opers.

This will lead to a number of structural results concerning $\mathcal{A}_{G, \text{irred}}$, as well as the space of generic oper structures on irreducible local systems.

Furthermore, given the recent result of [BKS], we will deduce GLC for classical groups.

4.1. Statement of the result.

4.1.1. Consider the space $\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(X^{\mathrm{gen}})_{\mathrm{Ran}}$ fibered over Ran , whose fiber over $\underline{x} \in \mathrm{Ran}$ is

$$\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(X - \underline{x}).$$

Let π_{Ran} denote the resulting map

$$\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(X^{\mathrm{gen}})_{\mathrm{Ran}} \rightarrow \mathrm{LS}_{\check{G}}.$$

Set

$$\mathrm{Op}_{\check{G}}^{\mathrm{mon-free,irred}}(X^{\mathrm{gen}})_{\mathrm{Ran}} := \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}(X^{\mathrm{gen}})_{\mathrm{Ran}} \times_{\mathrm{LS}_{\check{G}}} \mathrm{LS}_{\check{G}}^{\mathrm{irred}}.$$

Let $\pi_{\mathrm{Ran}}^{\mathrm{irred}}$ denote the resulting map

$$\mathrm{Op}_{\check{G}}^{\mathrm{mon-free,irred}}(X^{\mathrm{gen}})_{\mathrm{Ran}} \rightarrow \mathrm{LS}_{\check{G}}^{\mathrm{irred}}.$$

Note that the morphism $\pi_{\mathrm{Ran}}^{\mathrm{irred}}$ is pseudo-proper, i.e., a (not necessarily filtered) colimit of proper maps, see Sect. 3.3.11.

4.1.2. Consider the object⁹

$$\mathcal{B}_{G,\mathrm{irred}}^{\mathrm{Op}} := \mathbf{oblv}^l \left((\pi_{\mathrm{Ran}}^{\mathrm{irred}})^! (\omega_{\mathrm{Op}_{\check{G}}^{\mathrm{mon-free,irred}}(X^{\mathrm{gen}})_{\mathrm{Ran}}}) \right) \in \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}),$$

where

$$\mathbf{oblv}^l : \mathrm{D-mod}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$$

is the “left” forgetful functor, see [GaRo1, Equation (5.3)].

By construction, $\mathcal{B}_{G,\mathrm{irred}}^{\mathrm{Op}}$ is a co-commutative coalgebra in $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$.

4.1.3. Denote by

$$\mathbb{L}_{G,\mathrm{cusp}}^R : \mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}}$$

the functor *right adjoint* to $\mathbb{L}_{G,\mathrm{cusp}}$.

Since the monoidal category $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$ is rigid and the functor $\mathbb{L}_{G,\mathrm{cusp}}$ is $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$ -linear, the functor $\mathbb{L}_{G,\mathrm{cusp}}^R$ is also naturally $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$ -linear.

Hence, the comonad

$$\mathbb{L}_{G,\mathrm{cusp}} \circ \mathbb{L}_{G,\mathrm{cusp}}^R$$

acting on $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$ is given by tensoring by a co-associative coalgebra object, to be denoted $\mathcal{B}_{G,\mathrm{irred}}$.

4.1.4. The main result of this section reads:

Theorem 4.1.5. *There exists a canonical isomorphism between*

$$\mathcal{B}_{G,\mathrm{irred}} \simeq \mathcal{B}_{G,\mathrm{irred}}^{\mathrm{Op}}$$

as plain objects of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}}^{\mathrm{irred}})$.

Remark 4.1.6. One can show that the isomorphism of Theorem 4.1.5 respects the co-associative coalgebra structures on the two sides. However, we will neither prove¹⁰ nor use this.

4.2. Combining with ambidexterity. Prior to proving Theorem 4.1.5 we will derive some consequences.

4.2.1. Note that *a priori*, the comonad

$$\mathbb{L}_{G,\mathrm{cusp}} \circ \mathbb{L}_{G,\mathrm{cusp}}^R$$

is the right adjoint of the monad

$$\mathbb{L}_{G,\mathrm{cusp}} \circ \mathbb{L}_{G,\mathrm{cusp}}^L.$$

Hence, the *coalgebra* $\mathcal{B}_{G,\mathrm{irred}}$ identifies a priori with the monoidal dual of the *algebra* $\mathcal{A}_{G,\mathrm{irred}}$.

⁹The superscript “Op” in the notation below refers to opers and *not* to the opposite algebra structure.

¹⁰See, however, Remark 4.6.7.

4.2.2. Combining with Corollary 3.1.7 we obtain:

Corollary 4.2.3. *There is a canonical isomorphism*

$$\mathcal{A}_{G,\text{irred}} \simeq \mathcal{B}_{G,\text{irred}}$$

as objects of $\text{QCoh}(\text{LS}_G^{\text{irred}})$.

4.2.4. Combing further with Theorem 4.1.5, we obtain:

Corollary 4.2.5. *There is a canonical isomorphism*

$$\mathcal{A}_{G,\text{irred}} \simeq \mathcal{B}_{G,\text{irred}}^{\text{Op}}$$

as objects of $\text{QCoh}(\text{LS}_G^{\text{irred}})$.

And as a result:

Corollary 4.2.6. *The object $\mathcal{B}_{G,\text{irred}}^{\text{Op}} \in \text{QCoh}(\text{LS}_G^{\text{irred}})$ is compact.*

4.2.7. Consider the object

$$(4.1) \quad (\pi_{\text{Ran}}^{\text{irred}})_!(\omega_{\text{Op}_G^{\text{mon-free,irred}}(X^{\text{gen}})_{\text{Ran}}}) \in \text{D-mod}(\text{LS}_G^{\text{irred}}).$$

Assuming Theorem 3.1.8 for a moment, we obtain that the object (4.1) has the form

$$\omega_{\text{LS}_G^{\text{irred}}} \otimes \underline{\mathcal{B}}_{G,\text{irred}}^{\text{Op}},$$

where $\underline{\mathcal{B}}_{G,\text{irred}}^{\text{Op}}$ is a *classical local system of finite rank* on $\text{LS}_G^{\text{irred}}$.

We will prove the following assertion, which would be needed for the final step in the proof of GLC:

Proposition 4.2.8. *The local system $\underline{\mathcal{B}}_{G,\text{irred}}^{\text{Op}}$ has a finite monodromy, i.e., it trivializes over a finite étale cover of $\text{LS}_G^{\text{irred}}$.*

4.3. **Proof of Proposition 4.2.8.** We will deduce Proposition 4.2.8 from Theorem 4.1.5.

4.3.1. Denote

$$\mathcal{F} := (\pi_{\text{Ran}}^{\text{irred}})_!(\omega_{\text{Op}_G^{\text{mon-free,irred}}(X^{\text{gen}})_{\text{Ran}}})[-n],$$

where¹¹ $n = \dim(\text{LS}_G)$.

Since the map $\pi_{\text{Ran}}^{\text{irred}}$ is pseudo-proper, this object can be written as

$$\text{colim}_{i \in I} \mathcal{F}_i \quad \mathcal{F}_i \in \text{D-mod}(\text{LS}_G^{\text{irred}}),$$

over some diagram I , where each $\mathcal{F}_i$ is of the form

$$(f_i)_{*,\text{dR}}(\omega_{\mathcal{Y}_i})[-n],$$

where $f_i : \mathcal{Y}_i \rightarrow \text{LS}_G^{\text{irred}}$ is a proper map of algebraic stacks.

For each index i , consider the Stein factorization of the map f_i

$$\mathcal{Y}_i \rightarrow \mathcal{Y}_i^0 \xrightarrow{f_i^0} \text{LS}_G^{\text{irred}},$$

so that f_i^0 is a finite map.

Denote

$$\mathcal{F}_i^0 := (f_i^0)_{*,\text{dR}}(\omega_{\mathcal{Y}_i^0})[-n].$$

¹¹The cohomological shift is introduced for the sake of perverse normalization.

4.3.2. Let η be the generic point of a connected component of $\mathrm{LS}_G^{\mathrm{irred}}$. It is enough to show that $\mathcal{F}|_\eta$ has a finite monodromy.

We have

$$\mathcal{F}_i|_\eta \in \mathrm{D}\text{-mod}(\eta)^{\leq 0}, \quad \mathcal{F}_i^0|_\eta \in \mathrm{D}\text{-mod}(\eta)^\heartsuit$$

and the map

$$\mathcal{F}_i \rightarrow \mathcal{F}_i^0$$

induces an isomorphism

$$H^0(\mathcal{F}_i|_\eta) \rightarrow \mathcal{F}_i^0|_\eta.$$

By Theorem 3.1.8 combined with Corollary 4.2.5, the object $\mathcal{F}|_\eta$ is concentrated in cohomological degree 0. Hence, we obtain that

$$\mathcal{F}|_\eta \simeq \operatorname{colim}_{i \in I} \mathcal{F}_i^0|_\eta.$$

Since $\mathcal{F}|_\eta$ is finite-dimensional, we obtain that I contains a finite sub-diagram $I^f \subset I$ such that the map

$$\bigoplus_{i \in I^f} \mathcal{F}_i^0|_\eta \rightarrow \mathcal{F}|_\eta$$

is surjective.

Since each $\mathcal{F}_i^0|_\eta$ has a finite monodromy, we obtain that so does $\mathcal{F}|_\eta$.

□[Proposition 4.2.8]

4.4. Proof of Theorem 3.1.8. In this subsection we will continue to assume Theorem 4.1.5, and deduce Theorem 3.1.8.

4.4.1. Since $\mathcal{A}_{G,\mathrm{irred}}$ is perfect, in order to prove that it is a classical vector bundle, it suffices to show that the $*$ -fibers of $\mathcal{A}_{G,\mathrm{irred}}$ at k -points of $\mathrm{LS}_G^{\mathrm{irred}}$ are concentrated in cohomological degree 0.

By the self-duality assertion in Corollary 3.1.7, it suffices to show that the $*$ -fibers of $\mathcal{A}_{G,\mathrm{irred}}$ are concentrated in *non-positive* cohomological degrees.

4.4.2. By Corollary 4.2.5, it suffices to show that the $*$ -fibers of $\mathcal{B}_{G,\mathrm{irred}}^{\mathrm{Op}}$ are concentrated in *non-positive* cohomological degrees.

However, the $*$ -fiber of $\mathcal{B}_{G,\mathrm{irred}}^{\mathrm{Op}}$ at a k -point $\sigma \in \mathrm{LS}_G^{\mathrm{irred}}$ is isomorphic to $\mathrm{C}(\mathrm{Op}_{G,\sigma}^{\mathrm{gen}})$, where

$$\mathrm{Op}_{G,\sigma}^{\mathrm{gen}} := \{\sigma\} \times_{\mathrm{LS}_G^{\mathrm{irred}}} \mathrm{Op}_G^{\mathrm{mon-free,irred}}(X^{\mathrm{gen}})_{\mathrm{Ran}}.$$

It is automatically concentrated in *non-positive* cohomological degrees, being the *homology* of a prestack.

4.4.3. The D-module structure on $\mathcal{A}_{G,\mathrm{irred}}$ comes from the isomorphism of Corollary 4.2.5.

□[Theorem 3.1.8]

4.5. Contractibility of opers. In this subsection we will continue to assume Theorem 4.1.5. We will show that the validity of GLC is equivalent to the contractibility (and, in fact, just connectedness) of the space of generic oper structures on irreducible local systems.

4.5.1. Note that in the course of the proof of Theorem 3.1.8 above we have established:

Corollary 4.5.2. *The homology of the fibers of the map $\pi_{\mathrm{Ran}}^{\mathrm{irred}}$ is acyclic off degree 0.*

This can be equivalently reformulated as follows:

Corollary 4.5.3. *The connected components of the fibers of the map $\pi_{\mathrm{Ran}}^{\mathrm{irred}}$ are homologically contractible.*

4.5.4. Applying Corollary 1.6.5, we obtain:

Corollary 4.5.5. *The following assertions are equivalent:*

- (i) *The functor $\mathbb{L}_G$ is equivalence.*
- (ii) *The fibers of the map $\pi_{\text{Ran}}^{\text{irred}}$ are connected.*
- (iii) *The fibers of the map $\pi_{\text{Ran}}^{\text{irred}}$ are homologically contractible.*

4.5.6. In particular, we obtain that GLC is equivalent to the following conjecture:

Conjecture 4.5.7. *The space of generic oper structures on a given irreducible local system is homologically contractible.*

Remark 4.5.8. Note that the “bottom” layer of Conjecture 4.5.7 says that the space of generic oper structures on a given irreducible local system is *non-empty*. This statement is actually a theorem, thanks to [Ari].

Remark 4.5.9. The assertion of Conjecture 4.5.7 is easy for $G = GL_n$. In particular, in this way we obtain another proof of GLC in this case (i.e., one that is logically different from that in Sect. 1.7)¹².

4.5.10. Recall now that thanks to [BKS], Conjecture 4.5.7 is actually a theorem whenever G is a classical group¹³. Hence, we obtain:

Main Theorem 4.5.11. *The geometric Langlands conjecture holds when G is a classical group.*

Remark 4.5.12. Formally speaking, the main theorem of [BKS] establishes Conjecture 4.5.7 for a slightly different notion of oper, namely, for $\mathfrak{g}$ -opers, rather than $\check{G}$ -opers (and it is the latter that appears in Conjecture 4.5.7). In other words, [BKS] implies Conjecture 4.5.7 not for $\check{G}$ itself but rather for its adjoint quotient.

However, as we shall see in the sequel to this paper, the statement of GLC for a given pair $(G, \check{G})$ formally follows from the case when $\check{G}$ is replaced by its adjoint quotient (resp., G is replaced by the simply-connected cover of its derived group).

4.6. **Proof of Theorem 4.1.5.** As we shall presently see, the proof of the theorem follows almost immediately from diagram (3.15), once we combine the following pieces of information:

- The functor $\text{FLE}_{G, \text{crit}}$ is an equivalence;
- The functor $\mathbb{L}_{G, \text{cusp}}$ is a Verdier quotient.

4.6.1. Consider the adjoint pair

$$(4.2) \quad \text{Poinc}_{G, *, \text{irred}}^{\text{spec}} : \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})_{\text{Ran}} \rightleftarrows \text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}}) : \text{coeff}_{\check{G}, \text{irred}}^{\text{spec}},$$

see (3.12), where $\text{coeff}_{\check{G}, \text{irred}}^{\text{spec}}$ is the version of the functor $\text{coeff}_{\check{G}, \underline{x}, \text{irred}}^{\text{spec}}$ when $\underline{x}$ varies in families along Ran .

4.6.2. We will deduce Theorem 4.1.5 from the following assertion, which takes place purely on the spectral side.

Theorem 4.6.3. *The comonad on $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$ corresponding to (4.2) is given by tensor product with $\mathcal{B}_{G, \text{irred}}^{\text{Op}}$.*

This theorem will be proved in Sect. 5. Let us assume it, and proceed with the proof of Theorem 4.1.5.

¹²The difference between the two arguments is that one uses a fully faithfulness assertion on the automorphic side, another on the spectral side.

¹³Here by a classical group we mean a reductive group whose root datum is of type A, B, C or D.

4.6.4. Since we only want to identify

$$(4.3) \quad \mathcal{B}_{G,\text{irred}} \simeq \mathcal{B}_{G,\text{irred}}^{\text{Op}}$$

as objects of $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$ (and not as co-algebras), it suffices to construct an isomorphism of comonads

$$(4.4) \quad \text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}} \circ (\text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}})^R \simeq \mathbb{L}_{G,\text{cusp}} \circ (\mathbb{L}_{G,\text{cusp}})^R$$

acting on $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$. Indeed, each side of (4.3) is obtained by applying the corresponding side of (4.4) to $\mathcal{O}_{\text{LS}_{\check{G}}^{\text{irred}}}$.

4.6.5. Since $\text{FLE}_{G,\text{crit}}$ is an equivalence, the left-hand side in (4.4) identifies with

$$\text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}} \circ \text{FLE}_{G,\text{crit}} \circ \text{FLE}_{G,\text{crit}}^R \circ (\text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}})^R.$$

Since $\text{Loc}_{G,\text{cusp}}$ is a Verdier quotient, the right-hand side in (4.4) identifies with

$$\mathbb{L}_{G,\text{irred}} \circ (\text{Loc}_{G,\text{cusp}} \otimes \mathbb{I}) \circ ((\text{Loc}_{G,\text{cusp}} \otimes \mathbb{I})^R \circ (\mathbb{L}_{G,\text{cusp}})^R).$$

4.6.6. Hence, it suffices to establish an isomorphism between the comonads

$$\begin{aligned} \text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}} \circ \text{FLE}_{G,\text{crit}} \circ \text{FLE}_{G,\text{crit}}^R \circ (\text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}})^R &\simeq \\ &\simeq \mathbb{L}_{G,\text{irred}} \circ (\text{Loc}_{G,\text{cusp}} \otimes \mathbb{I}) \circ ((\text{Loc}_{G,\text{cusp}} \otimes \mathbb{I})^R \circ (\mathbb{L}_{G,\text{cusp}})^R), \end{aligned}$$

which is the same as

$$\begin{aligned} (\text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}} \circ \text{FLE}_{G,\text{crit}}) \circ (\text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}} \circ \text{FLE}_{G,\text{crit}})^R &\simeq \\ &\simeq (\mathbb{L}_{G,\text{irred}} \circ (\text{Loc}_{G,\text{cusp}} \otimes \mathbb{I})) \circ (\mathbb{L}_{G,\text{irred}} \circ (\text{Loc}_{G,\text{cusp}} \otimes \mathbb{I}))^R. \end{aligned}$$

However, this follows formally from the commutativity of (3.15).

□[Theorem 4.1.5]

Remark 4.6.7. Note that Theorem 4.1.5 only says that $\mathcal{B}_{G,\text{irred}}$ and $\mathcal{B}_{G,\text{irred}}^{\text{Op}}$ are isomorphic as objects of $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$, but not as co-associative co-algebras. One can upgrade the proof given above to an isomorphism of coalgebras along the following lines:

It follows from the construction that both comonads in (4.4) are linear with respect to the action of $\text{Rep}(\check{G})_{\text{Ran}}$ on $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$ via the functor $j^* \circ \text{Loc}_{\check{G}}^{\text{spec}}$, and the isomorphism between them constructed above respects this structure.

It also follows from the construction that the $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$ -linear structure on $\mathbb{L}_{G,\text{cusp}} \circ (\mathbb{L}_{G,\text{cusp}})^R$ agrees with the above $\text{Rep}(\check{G})_{\text{Ran}}$ -linear structure.

The comonad given by tensor product with $\mathcal{B}_{G,\text{irred}}^{\text{Op}}$ has a tautological linear structure with respect to $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$, and hence also with respect to $\text{Rep}(\check{G})_{\text{Ran}}$. It follows from the proof of Theorem 4.6.3 given in the next section that the above $\text{Rep}(\check{G})_{\text{Ran}}$ -linear structure on $- \otimes \mathcal{B}_{G,\text{irred}}^{\text{Op}}$ agrees with the $\text{Rep}(\check{G})_{\text{Ran}}$ -linear structure on $\text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}} \circ (\text{Poinc}_{\check{G},*,\text{irred}}^{\text{spec}})^R$.

Thus, we obtain that the two comonads

$$- \otimes \mathcal{B}_{G,\text{irred}} \text{ and } - \otimes \mathcal{B}_{G,\text{irred}}^{\text{Op}}$$

are identified as $\text{Rep}(\check{G})_{\text{Ran}}$ -linear comonads. Since the functor $j^* \circ \text{Loc}_{\check{G}}^{\text{spec}}$ is a Verdier quotient, this implies that the above identification is automatically $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$ -linear. The latter is equivalent to the identification of $\mathcal{B}_{G,\text{irred}}$ and $\mathcal{B}_{G,\text{irred}}^{\text{Op}}$ as co-associative coalgebras in $\text{QCoh}(\text{LS}_{\check{G}}^{\text{irred}})$.

5. PROOF OF THEOREM 4.6.3

The rest of the paper is devoted to the proof of Theorem 4.6.3. In particular, it takes place purely on the spectral side.

We will break up Theorem 4.6.3 into two assertions: Propositions 5.1.2 and 5.1.5. The former can be informally phrased as “the Ran integral equates the quasi-coherent and de Rham direct images”. The latter can be informally phrased as “the Ran integral erases the difference between the local and the global”.

It turns out that both these assertions are quite general, i.e., have nothing to do with the specifics of opers or local systems.

5.1. Strategy of the proof.

5.1.1. Consider the tautological natural transformation

$$(5.1) \quad (\pi_{\text{Ran}})_*^{\text{IndCoh}} \circ \text{oblv}_{\text{Op}_G^{\text{mon-free}}(X^{\text{gen}})_{\text{Ran}}}^r \rightarrow \text{oblv}_{\text{LS}_G}^r \circ (\pi_{\text{Ran}})_*^{\text{dR}},$$

as functors

$$\text{D-mod}(\text{Op}_G^{\text{mon-free}}(X^{\text{gen}})_{\text{Ran}}) \rightrightarrows \text{IndCoh}(\text{LS}_G).$$

We will prove

Proposition 5.1.2. *The natural transformation (5.1) is an isomorphism when evaluated on objects in the essential image of the functor*

$$\pi_{\text{Ran}}^! : \text{D-mod}(\text{LS}_G) \rightarrow \text{D-mod}(\text{Op}_G^{\text{mon-free}}(X^{\text{gen}})_{\text{Ran}}).$$

Remark 5.1.3. For the validity of Proposition 5.1.2, it is essential that we work with the entire Ran and not a fixed $\underline{x} \in \text{Ran}$.

5.1.4. Let $(s_{\text{Ran}})_*^{\text{IndCoh}}$ denote the functor

$$\text{IndCoh}(\text{Op}_G^{\text{mon-free}}(X^{\text{gen}})_{\text{Ran}}) \rightarrow \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}})_{\text{Ran}},$$

and let $(s_{\text{Ran}})^{\text{IndCoh},*}$ denote its left adjoint.

The counit of the $((s_{\text{Ran}})^{\text{IndCoh},*}, (s_{\text{Ran}})_*^{\text{IndCoh}})$ -adjunction defines a natural transformation

$$(5.2) \quad (\pi_{\text{Ran}})_*^{\text{IndCoh}} \circ (s_{\text{Ran}})^{\text{IndCoh},*} \circ (s_{\text{Ran}})_*^{\text{IndCoh}} \rightarrow (\pi_{\text{Ran}})_*^{\text{IndCoh}}.$$

We will prove:

Proposition 5.1.5. *The natural transformation (5.2), composed with the coarsening functor*

$$\Psi_{\text{LS}_G} : \text{IndCoh}(\text{LS}_G) \rightarrow \text{QCoh}(\text{LS}_G),$$

is an isomorphism, when evaluated on objects in the essential image of the functor

$$\pi_{\text{Ran}}^! : \text{IndCoh}(\text{LS}_G) \rightarrow \text{IndCoh}(\text{Op}_G^{\text{mon-free}}(X^{\text{gen}})_{\text{Ran}}).$$

5.1.6. We claim that the combination of Propositions 5.1.2 and 5.1.5 implies the assertion of Theorem 4.6.3.

Recall that the morphism $\pi_{\text{Ran}}^{\text{irred}}$ is pseudo-proper, so we can identify $(\pi_{\text{Ran}}^{\text{irred}})^! \simeq ((\pi_{\text{Ran}}^{\text{irred}})_*)^R$. Hence, the comonad

$$\text{Poinc}_{G,*,\text{irred}}^{\text{spec}} \circ (\text{Poinc}_{G,*,\text{irred}}^{\text{spec}})^R$$

identifies with

$$j^* \circ (\pi_{\text{Ran}})_*^{\text{IndCoh}} \circ (s_{\text{Ran}})^{\text{IndCoh},*} \circ (s_{\text{Ran}})_*^{\text{IndCoh}} \circ (\pi_{\text{Ran}})^! \circ j_*.$$

According to Proposition 5.1.5, this comonad maps isomorphically to the comonad

$$j^* \circ (\pi_{\text{Ran}})_*^{\text{IndCoh}} \circ (\pi_{\text{Ran}})^! \circ j_* \simeq (\pi_{\text{Ran}}^{\text{irred}})_*^{\text{IndCoh}} \circ (\pi_{\text{Ran}}^{\text{irred}})^!.$$

In particular, we obtain that this comonad is obtained by the $!$ -tensor product with the coalgebra object

$$(5.3) \quad (\pi_{\text{Ran}}^{\text{irred}})_*^{\text{IndCoh}} \circ (\pi_{\text{Ran}}^{\text{irred}})^! (\omega_{\text{LS}_G^{\text{irred}}}) \simeq (\pi_{\text{Ran}}^{\text{irred}})_*^{\text{IndCoh}} (\omega_{\text{Op}_G^{\text{mon-free,irred}}(X^{\text{gen}})_{\text{Ran}}}).$$

5.1.7. Restricting along the horizontal arrows in the Cartesian diagram

$$\begin{array}{ccc} \mathrm{Op}_G^{\mathrm{mon-free, irred}}(X^{\mathrm{gen}})_{\mathrm{Ran}} & \xrightarrow{j} & \mathrm{Op}_G^{\mathrm{mon-free}}(X^{\mathrm{gen}})_{\mathrm{Ran}} \\ \pi_{\mathrm{Ran}}^{\mathrm{irred}} \downarrow & & \downarrow \pi_{\mathrm{Ran}} \\ \mathrm{LS}_G^{\mathrm{irred}} & \xrightarrow{j} & \mathrm{LS}_{\tilde{G}}, \end{array}$$

from Proposition 5.1.2 we obtain that the natural transformation

$$(5.4) \quad (\pi_{\mathrm{Ran}}^{\mathrm{irred}})_*^{\mathrm{IndCoh}} \circ \mathbf{oblv}_{\mathrm{Op}_G^{\mathrm{mon-free, irred}}(X^{\mathrm{gen}})_{\mathrm{Ran}}}^r \rightarrow \mathbf{oblv}_{\mathrm{LS}_G^{\mathrm{irred}}}^r \circ (\pi_{\mathrm{Ran}}^{\mathrm{irred}})_*^{\mathrm{dR}}$$

is an isomorphism, when evaluated on objects lying in the essential image of

$$(\pi_{\mathrm{Ran}}^{\mathrm{irred}})^! : \mathrm{D-mod}(\mathrm{LS}_G^{\mathrm{irred}}) \rightarrow \mathrm{D-mod}(\mathrm{Op}_G^{\mathrm{mon-free, irred}}(X^{\mathrm{gen}})_{\mathrm{Ran}}).$$

Hence, we obtain that the coalgebra (5.3) maps isomorphically to

$$\mathbf{oblv}_{\mathrm{LS}_G^{\mathrm{irred}}}^r \left((\pi_{\mathrm{Ran}}^{\mathrm{irred}})_*^{\mathrm{dR}} \circ (\pi_{\mathrm{Ran}}^{\mathrm{irred}})^! (\omega_{\mathrm{LS}_G^{\mathrm{irred}}}) \right) \simeq \mathbf{oblv}_{\mathrm{LS}_G^{\mathrm{irred}}}^r \circ (\pi_{\mathrm{Ran}}^{\mathrm{irred}})_*^{\mathrm{dR}} (\omega_{\mathrm{Op}_G^{\mathrm{mon-free, irred}}(X^{\mathrm{gen}})_{\mathrm{Ran}}}).$$

Finally, we note that

$$\mathbf{oblv}_{\mathrm{LS}_G^{\mathrm{irred}}}^r(-) \simeq \mathbf{oblv}_{\mathrm{LS}_G^{\mathrm{irred}}}^l(-) \otimes \omega_{\mathrm{LS}_G^{\mathrm{irred}}}.$$

□[Theorem 4.6.3]

5.2. Framework for the proof of Proposition 5.1.2. In this subsection we will explain a general framework for the proof of Proposition 5.1.2: it has to do with a morphism between D-prestacks over X .

5.2.1. Consider the prestack

$$(\mathrm{Op}_G^{\mathrm{mon-free}}(X^{\mathrm{gen}})_{\mathrm{Ran}})_{\mathrm{dR}^{\mathrm{rel}}} := (\mathrm{Op}_G^{\mathrm{mon-free}}(X^{\mathrm{gen}})_{\mathrm{Ran}})_{\mathrm{dR}} \times_{(\mathrm{LS}_{\tilde{G}})_{\mathrm{dR}}} \mathrm{LS}_{\tilde{G}},$$

so that

$$\mathrm{IndCoh}((\mathrm{Op}_G^{\mathrm{mon-free}}(X^{\mathrm{gen}})_{\mathrm{Ran}})_{\mathrm{dR}^{\mathrm{rel}}}),$$

is the category of *relative* D-modules on $\mathrm{Op}_G^{\mathrm{mon-free}}(X^{\mathrm{gen}})_{\mathrm{Ran}}$ with respect to the projection π_{Ran} .

Denote by

$$\mathbf{ind}^{\mathrm{rel}} : \mathrm{IndCoh}(\mathrm{Op}_G^{\mathrm{mon-free}}(X^{\mathrm{gen}})_{\mathrm{Ran}}) \rightleftarrows \mathrm{IndCoh}((\mathrm{Op}_G^{\mathrm{mon-free}}(X^{\mathrm{gen}})_{\mathrm{Ran}})_{\mathrm{dR}^{\mathrm{rel}}}) : \mathbf{oblv}^{\mathrm{rel}}$$

the resulting pair of adjoint functors.

Consider also the functors

$$(5.5) \quad (\pi_{\mathrm{Ran}})_{*, \mathrm{dR}^{\mathrm{rel}}}^{\mathrm{IndCoh}} : \mathrm{IndCoh}((\mathrm{Op}_G^{\mathrm{mon-free}}(X^{\mathrm{gen}})_{\mathrm{Ran}})_{\mathrm{dR}^{\mathrm{rel}}}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}})$$

and

$$(5.6) \quad (\pi_{\mathrm{Ran}})_{\mathrm{dR}^{\mathrm{rel}}}^! : \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}) \rightarrow \mathrm{IndCoh}((\mathrm{Op}_G^{\mathrm{mon-free}}(X^{\mathrm{gen}})_{\mathrm{Ran}})_{\mathrm{dR}^{\mathrm{rel}}}).$$

5.2.2. As in Sect. 5.1.1 we have a natural transformation

$$(5.7) \quad (\pi_{\mathrm{Ran}})_*^{\mathrm{IndCoh}} \circ \mathbf{oblv}^{\mathrm{rel}} \rightarrow (\pi_{\mathrm{Ran}})_{*, \mathrm{dR}^{\mathrm{rel}}},$$

as functors

$$\mathrm{IndCoh}((\mathrm{Op}_G^{\mathrm{mon-free}}(X^{\mathrm{gen}})_{\mathrm{Ran}})_{\mathrm{dR}^{\mathrm{rel}}}) \rightleftarrows \mathrm{IndCoh}(\mathrm{LS}_{\tilde{G}}).$$

The assertion of Proposition 5.1.2 follows immediately from the one:

Proposition 5.2.3. *The natural transformation (5.7) is an isomorphism when evaluated on objects in the essential image of the functor (5.6).*

In its turn, Proposition 5.2.3 follows from the next assertion:

Proposition 5.2.4. *The counit of the adjunction*

$$\mathbf{ind}^{\text{rel}} \circ \mathbf{oblv}^{\text{rel}} \rightarrow \text{Id}$$

is an isomorphism, when evaluated on objects in the essential image of the functor (5.6).

5.3. An abstract version of Proposition 5.2.4. In this subsection we will show that Proposition 5.2.4 is a particular case of a general assertion that has to do with a morphism $f : \mathcal{Z} \rightarrow \mathcal{Y}$ of D-prestacks over X .

5.3.1. Consider the prestack $\text{Sect}_{\nabla}(X, \mathcal{Y})$ of horizontal sections of $\mathcal{Y}$, as well as

$$\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Y})_{\text{Ran}} \text{ and } \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}}$$

that associate to a point $\underline{x} \in \text{Ran}$ the spaces of horizontal sections of $\mathcal{Y}$ and $\mathcal{Z}$ over $X - \underline{x}$, respectively.

Note that we have a tautological map

$$\text{Sect}_{\nabla}(X, \mathcal{Y}) \times \text{Ran} \rightarrow \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Y})_{\text{Ran}}$$

(and similarly for $\mathcal{Z}$).

Denote

$$\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}} := \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}} \times_{\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Y})_{\text{Ran}}} (\text{Sect}_{\nabla}(X, \mathcal{Y}) \times \text{Ran}).$$

Denote by π_{Ran} the natural projection

$$\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}} \rightarrow \text{Sect}_{\nabla}(X, \mathcal{Y})$$

and by $\pi_{\text{Ran}, \text{dR}^{\text{rel}}}$ the map

$$(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}})_{\text{dR}^{\text{rel}}} \rightarrow \text{Sect}_{\nabla}(X, \mathcal{Y})$$

5.3.2. We will impose the following finiteness conditions on $\mathcal{Y}$ and $\mathcal{Z}$:

- $\mathcal{Y}$ is *sectionally left* in the sense of [Ro, Sect. 3.1.3(ii)], i.e.,
 - The prestack $\text{Sect}_{\nabla}(X, \mathcal{Y})$ is locally almost of finite type;
 - The condition of [Ro, Sect. 3.1.3(ii)] is satisfied for points of $\text{Sect}_{\nabla}(X, \mathcal{Y})$;
- $\mathcal{Z}$ is *meromorphically sectionally left* relative to $\mathcal{Y}$, i.e.,
 - The prestack $\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}}$ is locally almost of finite type;
 - The condition of [Ro, Sect. 3.1.3(ii)] is satisfied for points of $\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}}$.

5.3.3. Denote

$$(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}})_{\text{dR}^{\text{rel}}} := (\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}})_{\text{dR}} \times_{\text{Sect}_{\nabla}(X, \mathcal{Y})_{\text{dR}}} \text{Sect}_{\nabla}(X, \mathcal{Y}).$$

Let

$$(5.8) \quad \mathbf{ind}^{\text{rel}} : \text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}}) \rightleftarrows \text{IndCoh}((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}})_{\text{dR}^{\text{rel}}}) : \mathbf{oblv}^{\text{rel}}$$

the resulting pair of adjoint functors.

We have:

Proposition 5.3.4. *The counit of the adjunction*

$$\mathbf{ind}^{\text{rel}} \circ \mathbf{oblv}^{\text{rel}} \rightarrow \text{Id}$$

is an isomorphism, when evaluated on objects in the essential image of the pullback functor

$$(\pi_{\text{Ran}, \text{dR}^{\text{rel}}})^! : \text{IndCoh}(\text{Sect}_{\nabla}(X, \mathcal{Y})) \rightarrow \text{IndCoh}((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}})_{\text{dR}^{\text{rel}}}).$$

The proof will be given in Sect. A.

5.3.5. Note that Proposition 5.2.4 is indeed a particular case of Proposition 5.3.4: we take $\mathcal{Y}$ to be the constant D-stack with fiber $\text{pt}/\check{G}$ and $\mathcal{Z} := \text{Op}_{\check{G}}$.

□[Proposition 5.2.4]

5.4. A digression: the category $\mathrm{QCoh}_{\mathrm{co}}$. In order to formulate (an abstract version of) Proposition 5.1.5, it will be convenient to introduce a general construction of a certain variant of the category of quasi-coherent sheaves on a prestack, denoted $\mathrm{QCoh}_{\mathrm{co}}(-)$ (see [GLC2, Sect. A.1] for a more detailed discussion).

5.4.1. Let $\mathcal{W}$ be a prestack. We define the category $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{W})$ by

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}) := \operatorname{colim}_{S \rightarrow \mathcal{W}, S \in \mathrm{Sch}^{\mathrm{aff}}} \mathrm{QCoh}(S),$$

where the colimit is taken with respect to the *direct image functors*¹⁴.

5.4.2. If $\mathcal{W}$ has an affine diagonal, we have a functor

$$\Omega_{\mathcal{W}} : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}) \rightarrow \mathrm{QCoh}(\mathcal{W}).$$

Namely, it corresponds to the compatible family of direct image functors

$$\mathrm{QCoh}(S) \rightarrow \mathrm{QCoh}(\mathcal{W}), \quad S \in \mathrm{Sch}_{/\mathcal{W}}^{\mathrm{aff}},$$

which are well-defined since the morphisms $S \rightarrow \mathcal{W}$ are affine.

5.4.3. *Example.* Suppose that $\mathcal{W}$ is a *scheme*. Then it is easy to see that in this case the functor $\Omega_{\mathcal{W}}$ is an equivalence.

In fact, according to [Ga4, Proposition 6.2.7 and Theorem 2.2.6], the same is true when $\mathcal{W}$ is an eventually coconnective quasi-compact algebraic stack of finite type with an affine diagonal.

Note that $\mathcal{W} = \mathrm{LS}_{\bar{G}}$ is an example of such an algebraic stack.

Remark 5.4.4. We do not know whether $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{W})$ is dualizable. However, $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{W})$ is tautologically the pre-dual of $\mathrm{QCoh}(\mathcal{W})$, i.e.,

$$\mathrm{QCoh}(\mathcal{W}) \simeq \mathrm{Funct}(\mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}), \mathrm{Vect}),$$

where $\mathrm{Funct}(-, -)$ is the category of colimit-preserving functors.

In particular, if $\mathcal{W}$ is such that $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{W})$ is dualizable, then so is $\mathrm{QCoh}(\mathcal{W})$.

5.4.5. *Example.* Let $\mathcal{W}$ be an ind-scheme, written as

$$\mathcal{W} = \operatorname{colim}_i W^i, \quad W^i \in \mathrm{Sch},$$

where the transition maps $W^i \rightarrow W^j$ are closed embeddings.

In this case,

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}) \simeq \operatorname{colim}_i \mathrm{QCoh}(W^i)$$

where the colimit is taken with respect to the direct image functors.

Note that if $\mathcal{W}$ is of ind-finite type, we have a naturally defined functor

$$(5.9) \quad \Psi_{\mathcal{W}} : \mathrm{IndCoh}(\mathcal{W}) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}).$$

Indeed, we can write

$$\mathrm{IndCoh}(\mathcal{W}) \simeq \operatorname{colim}_i \mathrm{IndCoh}(W^i)$$

(under direct image functors) and (5.9) is given by the compatible family of functors

$$\Psi_{W^i} : \mathrm{IndCoh}(W^i) \rightarrow \mathrm{QCoh}(W^i).$$

One can show that (5.9) is an equivalence if $\mathcal{W}$ is formally smooth.

¹⁴The reason for the notation “ $\mathrm{QCoh}_{\mathrm{co}}$ ” is that it is a version of the QCoh category, i.e., we take the colimit with respect to the *-direct image maps, instead of the limit with respect to the *-pullback maps.

5.4.6. The assignment

$$\mathcal{W} \rightsquigarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W})$$

has the following functoriality properties for maps $f : \mathcal{W}_1 \rightarrow \mathcal{W}_2$:

- We have the direct image functor

$$f_* : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}_1) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}_2).$$

- If f is *schematic*, we also have the pullback functor

$$f^* : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}_2) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}_1),$$

which is a left adjoint of f_* .

- For a pullback square

$$\begin{array}{ccc} \mathcal{W}_1 & \xrightarrow{f} & \mathcal{W}_2 \\ \downarrow & & \downarrow \\ S_1 & \xrightarrow{f} & S_2, \end{array}$$

where S_1 and S_2 are affine schemes, the functor

$$\mathrm{QCoh}(S_1) \otimes_{\mathrm{QCoh}(S_2)} \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}_2) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}_1),$$

defined by f^* , is an equivalence;

- If f is *schematic and of finite Tor dimension*, we also have the $!$ -pullback functor

$$f^! : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}_2) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}_1).$$

Note that if f is also proper, then the functors $(f_*, f^!)$ are mutually adjoint.

5.4.7. Let $\mathcal{W}_{\mathrm{Ran}}$ be a prestack over Ran . Set

$$(5.10) \quad \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W})_{\mathrm{Ran}} := \lim_{S \in \mathrm{Sch}_{\mathrm{aff}}^{\mathrm{aff}}, S \rightarrow \mathrm{Ran}} \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}_{\mathrm{Ran}} \times_{\mathrm{Ran}} S) \otimes_{\mathrm{QCoh}(S)} \mathrm{IndCoh}(S),$$

where the limit is formed using the $*$ -pullback functors along the $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}_{\mathrm{Ran}} \times_{\mathrm{Ran}} S)$ -factors and $!$ -pullback functors along the $\mathrm{IndCoh}(S)$ -factors.

Thus, an object $\mathcal{F} \in \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W})_{\mathrm{Ran}}$ gives rise to an object

$$\mathcal{F}_{S, \underline{x}} \in \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}_{\mathrm{Ran}} \times_{\mathrm{Ran}} S) \otimes_{\mathrm{QCoh}(S)} \mathrm{IndCoh}(S)$$

for every $\underline{x} \in \mathrm{Ran}(S)$.

In the case when $\mathcal{W}_{\mathrm{Ran}} \rightarrow \mathrm{Ran}$ is schematic, so that for every $(S, \underline{x})$ as above we have

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}_{\mathrm{Ran}} \times_{\mathrm{Ran}} S) \simeq \mathrm{QCoh}(\mathcal{W}_{\mathrm{Ran}} \times_{\mathrm{Ran}} S),$$

we will simply write $\mathrm{QCoh}(\mathcal{W})_{\mathrm{Ran}}$ instead of $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{W})_{\mathrm{Ran}}$.

Remark 5.4.8. The assignment

$$S \rightsquigarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}_{\mathrm{Ran}} \times_{\mathrm{Ran}} S)$$

naturally forms a sheaf of categories over Ran , to be denoted

$$\underline{\mathrm{QCoh}}_{\mathrm{co}}(\mathcal{W})_{\mathrm{Ran}}.$$

The above definition of $\mathrm{QCoh}_{\mathrm{co}}(\mathcal{W})_{\mathrm{Ran}}$ is a particular case of the following construction: for any sheaf of categories $\underline{\mathbf{C}}_{\mathrm{Ran}}$ over Ran , we can assign the category

$$\mathbf{C}_{\mathrm{Ran}} := \lim_{S \in \mathrm{Sch}_{\mathrm{aff}}^{\mathrm{aff}}, S \rightarrow \mathrm{Ran}} \mathbf{C}(S) \otimes_{\mathrm{QCoh}(S)} \mathrm{IndCoh}(S).$$

Note that since Ran is 1-affine, we have

$$\mathbf{C}_{\mathrm{Ran}} \simeq \Gamma(\mathrm{Ran}, \underline{\mathbf{C}}_{\mathrm{Ran}}) \otimes_{\mathrm{QCoh}(\mathrm{Ran})} \mathrm{IndCoh}(\mathrm{Ran}).$$

In particular, since the functor

$$\mathrm{QCoh}(\mathrm{Ran}) \rightarrow \mathrm{IndCoh}(\mathrm{Ran}), \quad \mathcal{F} \mapsto \mathcal{F} \otimes \omega_{\mathrm{Ran}}$$

is an equivalence, we have an equivalence

$$\mathbf{C}_{\mathrm{Ran}} \simeq \Gamma(\mathrm{Ran}, \underline{\mathbf{C}}_{\mathrm{Ran}}).$$

5.4.9. Let $p_{\mathcal{W}_{\mathrm{Ran}}}$ denote the projection

$$\mathcal{W}_{\mathrm{Ran}} \rightarrow \mathrm{Ran}.$$

Note that we have a well-defined functor

$$(p_{\mathcal{W}_{\mathrm{Ran}}})_* : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W})_{\mathrm{Ran}} \rightarrow \mathrm{IndCoh}(\mathrm{Ran}) \simeq \mathrm{D-mod}(\mathrm{Ran}).$$

Let us denote by

$$\Gamma^{\mathrm{IndCoh}_{\mathrm{Ran}}}(\mathcal{W}_{\mathrm{Ran}}, -) : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W})_{\mathrm{Ran}} \rightarrow \mathrm{Vect}$$

the functor equal to the composition

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{W})_{\mathrm{Ran}} \xrightarrow{(p_{\mathcal{W}_{\mathrm{Ran}}})_*} \mathrm{D-mod}(\mathrm{Ran}) \xrightarrow{\Gamma^{\mathrm{IndCoh}}(\mathrm{Ran}, -)} \mathrm{Vect},$$

where we can alternatively think of $\Gamma^{\mathrm{IndCoh}}(\mathrm{Ran}, -)$ as the functor

$$\mathrm{C}_c(\mathrm{Ran}, -) : \mathrm{D-mod}(\mathrm{Ran}) \rightarrow \mathrm{Vect},$$

left adjoint to $k \mapsto \omega_{\mathrm{Ran}}$.

5.4.10. Assume now that $\mathcal{W}_{\mathrm{Ran}}$ is locally almost of finite type (so that $\mathrm{IndCoh}(\mathcal{W}_{\mathrm{Ran}})$ is defined) and assume that $\mathcal{W}_{\mathrm{Ran}} \rightarrow \mathrm{Ran}$ is a relative ind-scheme.

We claim that in this case, there exists a well-defined functor

$$(5.11) \quad \Psi_{\mathcal{W}_{\mathrm{Ran}}} : \mathrm{IndCoh}(\mathcal{W}_{\mathrm{Ran}}) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W})_{\mathrm{Ran}},$$

which is a variant of (5.9).

Indeed, we can write

$$\mathrm{IndCoh}(\mathcal{W}_{\mathrm{Ran}}) \simeq \lim_{S \in \mathrm{Sch}_{\mathrm{aff}}^{\mathrm{aff}}, S \rightarrow \mathrm{Ran}} \mathrm{IndCoh}(\mathcal{W}_{\mathrm{Ran}} \times_{\mathrm{Ran}} S),$$

so it is enough to define a compatible family of functors

$$(5.12) \quad \mathrm{IndCoh}(\mathcal{W}_{\mathrm{Ran}} \times_{\mathrm{Ran}} S) \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathcal{W}_{\mathrm{Ran}} \times_{\mathrm{Ran}} S).$$

Write

$$\mathcal{W}_{\mathrm{Ran}} \times_{\mathrm{Ran}} S \simeq \mathrm{colim}_i \mathcal{W}_S^i,$$

where $\mathcal{W}_S^i$ are schemes, and the transition maps $\mathcal{W}_S^i \rightarrow \mathcal{W}_S^j$ are closed embeddings.

The functors (5.12) are given by the compatible family of functors

$$\mathrm{IndCoh}(\mathcal{W}_S^i) \rightarrow \mathrm{QCoh}(\mathcal{W}_S^i) \otimes_{\mathrm{QCoh}(S)} \mathrm{IndCoh}(S),$$

Serre-dual to the tautological functors

$$\mathrm{QCoh}(\mathcal{W}_S^i) \otimes_{\mathrm{QCoh}(S)} \mathrm{IndCoh}(S) \rightarrow \mathrm{IndCoh}(\mathcal{W}_S^i),$$

given by $!$ -pullback along $\mathcal{W}_S^i \rightarrow S$.

5.5. Abstract version of Proposition 5.1.5: the absolute case. As with Proposition 5.1.2, we will prove an abstract statement, of which Proposition 5.1.5 is a particular case. The general set-up involves a morphism

$$\mathcal{Z} \rightarrow \mathcal{Y}$$

of D-prestacks as in Sect. 5.3.1. For expository purposes, we will first consider the absolute situation, i.e., one when $\mathcal{Y} = \mathrm{pt}$.

5.5.1. Let $\mathcal{Z}$ be an affine D-scheme over X . For $\underline{x} \in \text{Ran}$ we will denote by $\mathfrak{L}_{\nabla}^+(\mathcal{Z})_{\underline{x}}$ (resp., $\mathfrak{L}_{\nabla}(\mathcal{Z})_{\underline{x}}$) the scheme (resp., ind-scheme) $\text{Sect}_{\nabla}(\mathcal{D}_{\underline{x}}, \mathcal{Z})$ (resp., $\text{Sect}_{\nabla}(\mathcal{D}_{\underline{x}} - \underline{x}, \mathcal{Z})$).

Consider the corresponding categories

$$\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Z})_{\underline{x}}) \text{ and } \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z})_{\underline{x}}).$$

In addition, we can consider the ind-scheme $\text{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z})$, and the categories

$$\text{IndCoh}(\text{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z})) \text{ and } \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z})).$$

5.5.2. Letting $\underline{x} \in \text{Ran}$ move in a family over Ran , we obtain the spaces $\mathfrak{L}_{\nabla}^+(\mathcal{Z})_{\text{Ran}}$ and $\mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}}$, where

$$\mathfrak{L}_{\nabla}^+(\mathcal{Z})_{\text{Ran}} \rightarrow \text{Ran}$$

is a relative scheme, and

$$\mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}} \rightarrow \text{Ran}$$

is a relative ind-scheme. Consider also the relative ind-scheme $\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}}$.

We define the categories $\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Z}))_{\text{Ran}}$, $\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}}$ and $\text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}))_{\text{Ran}}$ by the recipe of Sect. 5.4.7.

5.5.3. Consider the map

$$s_{\mathcal{Z}, \text{Ran}} : \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}} \rightarrow \mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}},$$

obtained by restricting horizontal sections along

$$\mathcal{D}_{\underline{x}} - \underline{x} \rightarrow X - \underline{x}.$$

When $\mathcal{Z}$ is unambiguous, we will simply write s_{Ran} instead of $s_{\mathcal{Z}, \text{Ran}}$.

We have an adjoint pair of functors

$$(s_{\mathcal{Z}, \text{Ran}})^* : \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}} \rightleftarrows \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}))_{\text{Ran}} : (s_{\mathcal{Z}, \text{Ran}})_*.$$

5.5.4. An abstract version of Proposition 5.1.5 (in the absolute case) case reads:

Proposition 5.5.5. *The natural transformation*

$$\Gamma^{\text{IndCohRan}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}}, -) \circ (s_{\mathcal{Z}, \text{Ran}})^* \circ (s_{\mathcal{Z}, \text{Ran}})_* \rightarrow \Gamma^{\text{IndCohRan}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}}, -)$$

is an isomorphism, when evaluated on the image of $\omega_{\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}}}$ along

$$\text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}}) \xrightarrow{\Psi_{\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}}}} \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}))_{\text{Ran}}.$$

The proof will be given in Sect. B.

5.6. **Abstract version of Proposition 5.1.5: the relative case.** In this section we will introduce a relative version of the set-up of Sect. 5.5.

5.6.1. Let $\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}}$ have the same meaning as in Sect. 5.3.1. We will view it as a relative ind-scheme over

$$\text{Sect}_{\nabla}(X, \mathcal{Y}) \times \text{Ran}.$$

Consider the corresponding category

$$\text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y}))_{\text{Ran}}.$$

Let π_{Ran} denote the projection

$$\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}} \rightarrow \text{Sect}_{\nabla}(X, \mathcal{Y}).$$

Combined with the projection

$$p_{\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}}} : \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}} \rightarrow \text{Ran},$$

we obtain a map

$$\pi_{\text{Ran}} \times p_{\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}}} : \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}} \rightarrow \text{Sect}_{\nabla}(X, \mathcal{Y}) \times \text{Ran}.$$

5.6.2. Let us denote by

$$(\pi_{\text{Ran}})_*^{\text{IndCohRan}} : \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y}))_{\text{Ran}} \rightarrow \text{QCoh}(\text{Sect}_{\nabla}(X, \mathcal{Y}))$$

the composite functor

$$\begin{aligned} \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y}))_{\text{Ran}} &\xrightarrow{(\pi_{\text{Ran}} \times p_{\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}}})^*} \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X, \mathcal{Y})) \otimes \text{IndCoh}(\text{Ran}) \rightarrow \\ &\xrightarrow{\Omega_{\text{Sect}_{\nabla}(X, \mathcal{Y})} \otimes \text{Id}} \text{QCoh}(\text{Sect}_{\nabla}(X, \mathcal{Y})) \otimes \text{IndCoh}(\text{Ran}) \xrightarrow{\text{Id} \otimes \Gamma^{\text{IndCoh}}(\text{Ran}, -)} \text{QCoh}(\text{Sect}_{\nabla}(X, \mathcal{Y})). \end{aligned}$$

Note that we have a commutative diagram

$$(5.13) \quad \begin{array}{ccc} \text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y}))_{\text{Ran}} & \xrightarrow{\Psi_{\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}}}} & \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y}))_{\text{Ran}} \\ (\pi_{\text{Ran}})_*^{\text{IndCoh}} \downarrow & & \downarrow (\pi_{\text{Ran}})_*^{\text{IndCohRan}} \\ \text{IndCoh}(\text{Sect}_{\nabla}(X, \mathcal{Y})) & \xrightarrow{\Psi_{\text{Sect}_{\nabla}(X, \mathcal{Y})}} & \text{QCoh}(\text{Sect}_{\nabla}(X, \mathcal{Y})), \end{array}$$

where the top horizontal arrow is the functor of (5.11).

5.6.3. Let us be in the situation described in [GLC2, Sects. 4.5.1-4.5.2.], with the following change of notations: what was denoted by $\mathcal{Y}$ (resp., $\mathcal{Y}_0$) in *loc. cit.* we denote by $\mathcal{Z}$ (resp., $\mathcal{Y}$).

Let us be given an affine map

$$s_{\mathcal{T}, \text{Ran}} : \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}} \rightarrow \mathcal{T}_{\text{Ran}}$$

that fits into a commutative (but not necessarily Cartesian) diagram

$$(5.14) \quad \begin{array}{ccc} \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}} & \xrightarrow{s_{\mathcal{T}, \text{Ran}}} & \mathcal{T}_{\text{Ran}} \\ \pi_{\text{Ran}} \times p_{\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}}} \downarrow & & \downarrow \mathfrak{L}(f) \\ \text{Sect}_{\nabla}(X, \mathcal{Y}) \times \text{Ran} & \xrightarrow{s_{\mathcal{Y}, \text{Ran}}} & \mathfrak{L}^+(\mathcal{Y})_{\text{Ran}}. \end{array}$$

5.6.4. We will now make the following additional structural assumption:

Consider the fiber product

$$\mathcal{T}_{\text{Sect}_{\nabla}(X, \mathcal{Y}), \text{Ran}} := (\text{Sect}_{\nabla}(X, \mathcal{Y}) \times \text{Ran}) \times_{\mathfrak{L}^+(\mathcal{Y})_{\text{Ran}}} \mathcal{T}_{\text{Ran}}.$$

We require that there exist a $\text{Sect}_{\nabla}(X, \mathcal{Y})$ -family of affine D-schemes

$$\mathcal{Z}_{\text{Sect}_{\nabla}(X, \mathcal{Y})} \rightarrow \text{Sect}_{\nabla}(X, \mathcal{Y}) \times X_{\text{dR}},$$

such that:

- $\mathcal{T}_{\text{Sect}_{\nabla}(X, \mathcal{Y}), \text{Ran}}$ identifies with the (relative over $\text{Sect}_{\nabla}(X, \mathcal{Y})$) factorization space of horizontal loops $\mathfrak{L}_{\nabla}(\mathcal{Z}_{\text{Sect}_{\nabla}(X, \mathcal{Y})})_{\text{Ran}} / \text{Sect}_{\nabla}(X, \mathcal{Y})$;
- The map

$$s'_{\mathcal{T}, \text{Ran}} : \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}} \rightarrow \mathcal{T}_{\text{Sect}_{\nabla}(X, \mathcal{Y}), \text{Ran}},$$

arising from (5.14), identifies with the evaluation map

$$s_{\mathcal{Z}_{\text{Sect}_{\nabla}(X, \mathcal{Y})}} : \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}_{\text{Sect}_{\nabla}(X, \mathcal{Y})} / \text{Sect}_{\nabla}(X, \mathcal{Y}))_{\text{Ran}} \rightarrow \mathfrak{L}_{\nabla}(\mathcal{Z}_{\text{Sect}_{\nabla}(X, \mathcal{Y})})_{\text{Ran}} / \text{Sect}_{\nabla}(X, \mathcal{Y}).$$

Finally, we require that the above data be compatible with the unital structures in the natural sense.

Remark 5.6.5. For our applications, we will take $\mathcal{Z} = \text{Op}_{\check{G}}$ and $\mathcal{Y} = \text{pt}/\check{G}$ and $\mathcal{T} := \text{Op}_{\check{G}}^{\text{mon-free}}$. In this case $\text{Sect}_{\nabla}(X, \mathcal{Y}) = \text{LS}_{\check{G}}$, and $\mathcal{Z}_{\text{Sect}_{\nabla}(X, \mathcal{Y})}$ is the D-scheme parameterized by $\text{LS}_{\check{G}}$ that classifies oper structures on a given local system.

5.6.6. We have an adjoint pair of functors

$$(s_{\mathcal{T}, \text{Ran}})^* : \text{QCoh}_{\text{co}}(\mathcal{T})_{\text{Ran}} \rightleftarrows \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y}))_{\text{Ran}} : (s_{\mathcal{T}, \text{Ran}})_*.$$

5.6.7. We claim:

Proposition 5.6.8. *The natural transformation*

$$(\pi_{\text{Ran}})_*^{\text{IndCoh}_{\text{Ran}}} \circ (s_{\mathcal{T}, \text{Ran}})^* \circ (s_{\mathcal{T}, \text{Ran}})_* \rightarrow (\pi_{\text{Ran}})_*^{\text{IndCoh}_{\text{Ran}}}$$

is an isomorphism, when evaluated on objects that lie in the essential image of the functor

$$(5.15) \quad \text{IndCoh}(\text{Sect}_{\nabla}(X, \mathcal{Y})) \xrightarrow{\pi_{\text{Ran}}^!} \text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}}) \rightarrow \\ \xrightarrow{\Psi_{\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}}}} \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}}).$$

The proof will be given in Sect. C.

5.6.9. Let us show how Proposition 5.6.8 implies Proposition 5.1.5.

We apply Proposition 5.6.8 to the spaces specified in Remark 5.6.5, so that

$$\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}} = \text{Op}_{\tilde{G}}^{\text{mon-free}}(X^{\text{gen}})_{\text{Ran}}.$$

We have a commutative diagram

$$(5.16) \quad \begin{array}{ccc} \text{IndCoh}(\text{Op}_{\tilde{G}}^{\text{mon-free}}(X^{\text{gen}})_{\text{Ran}}) & \xrightarrow{\Psi_{\text{Op}_{\tilde{G}}^{\text{mon-free}}(X^{\text{gen}})_{\text{Ran}}}} & \text{QCoh}_{\text{co}}(\text{Op}_{\tilde{G}}^{\text{mon-free}}(X^{\text{gen}})_{\text{Ran}}) \\ (s_{\text{Ran}})_*^{\text{IndCoh}} \downarrow & & \downarrow (s_{\text{Ran}})_* \\ \text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}})_{\text{Ran}} & \xrightarrow{\Psi_{(\text{Op}_{\tilde{G}}^{\text{mon-free}})_{\text{Ran}}}} & \text{QCoh}_{\text{co}}(\text{Op}_{\tilde{G}}^{\text{mon-free}})_{\text{Ran}}. \end{array}$$

Moreover, the diagram

$$(5.17) \quad \begin{array}{ccc} \text{IndCoh}(\text{Op}_{\tilde{G}}^{\text{mon-free}}(X^{\text{gen}})_{\text{Ran}}) & \xrightarrow{\Psi_{\text{Op}_{\tilde{G}}^{\text{mon-free}}(X^{\text{gen}})_{\text{Ran}}}} & \text{QCoh}_{\text{co}}(\text{Op}_{\tilde{G}}^{\text{mon-free}}(X^{\text{gen}})_{\text{Ran}}) \\ (s_{\text{Ran}})^*, \text{IndCoh} \uparrow & & \uparrow (s_{\text{Ran}})^* \\ \text{IndCoh}^*(\text{Op}_{\tilde{G}}^{\text{mon-free}})_{\text{Ran}} & \xrightarrow{\Psi_{(\text{Op}_{\tilde{G}}^{\text{mon-free}})_{\text{Ran}}}} & \text{QCoh}_{\text{co}}(\text{Op}_{\tilde{G}}^{\text{mon-free}})_{\text{Ran}}. \end{array}$$

obtained from (5.16) by passing to left adjoints along the vertical arrows, commutes as well.

The conclusion of Proposition 5.1.5 follows now from Proposition 5.6.8, by juxtaposing the commutative diagrams (5.13), (5.16) and (5.17).

□[Proposition 5.1.5]

APPENDIX A. PROOF OF PROPOSITION 5.3.4

The idea of the proof of Proposition 5.3.4 can be summarized by the following slogan: the unital version of the space of *rational* horizontal sections maps isomorphically to its own de Rham prestack.

We will deduce it from the main theorem of [Ro] by a rather formal manipulation.

A.1. The unital Ran space. In order to prove Proposition 5.3.4 we will need to work with the *unital Ran space*, which is no longer a prestack (i.e., a functor from affine schemes to ∞ -groupoids) but rather a *categorical prestack*, i.e., a functor from affine schemes to ∞ -categories (see [GLC2, Sect. C.5] for a more detailed discussion).

A.1.1. Recall the notion of *categorical prestack*, see [Ro, Appendix C]. By definition, this is a functor

$$(\mathrm{Sch}^{\mathrm{aff}})^{\mathrm{op}} \rightarrow 1\text{-Cat},$$

where 1-Cat denotes the $(\infty, 1)$ -category of $(\infty, 1)$ -categories.

Thus, a categorical prestack $\mathcal{X}$ assigns to an affine scheme S a category, to be denoted $\mathcal{X}(S)$, and to a map $f : S_1 \rightarrow S_2$ a functor

$$\mathcal{X}(f) : \mathcal{X}(S_2) \rightarrow \mathcal{X}(S_1),$$

equipped with a datum of compatibility for compositions.

A.1.2. Let $\mathrm{Ran}^{\mathrm{unl}}$ be the unital version of the Ran space, see [Ga4, Sect. 4.2] or [Ro, Sect. 2.1]. I.e., $\mathrm{Ran}^{\mathrm{unl}}$ associates to an affine scheme S the *category* of finite subsets of $\mathrm{Hom}(S, X_{\mathrm{dR}})$, where the morphisms are given by inclusion.

Let

$$\mathbf{t} : \mathrm{Ran} \rightarrow \mathrm{Ran}^{\mathrm{unl}}$$

denote the tautological map.

A.1.3. Along with the prestacks

$$\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z})_{\mathrm{Ran}}, \mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}}, (\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}})_{\mathrm{dR}^{\mathrm{rel}}}, \text{ etc}$$

one can consider their unital versions, which are now *categorical prestacks*, denoted

$$(A.1) \quad \mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z})_{\mathrm{Ran}^{\mathrm{unl}}}, \mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{unl}}}, (\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{unl}}})_{\mathrm{dR}^{\mathrm{rel}}},$$

respectively, see [Ro, Sect. 3.3.1].

Explicitly, for an affine scheme S , the category $\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z})_{\mathrm{Ran}^{\mathrm{unl}}}(S)$ consists of pairs $(\underline{x}, z)$, where $\underline{x} \in \mathrm{Ran}^{\mathrm{unl}}(S)$ and z is a horizontal section of $\mathcal{Z}$ on $X \times S - \mathrm{Graph}_{\underline{x}}$.

A morphism $(\underline{x}_1, z_1) \rightarrow (\underline{x}_2, z_2)$ is an inclusion $\underline{x}_1 \subseteq \underline{x}_2$ and an identification

$$z_1|_{X \times S - \mathrm{Graph}_{\underline{x}_2}} \simeq z_2.$$

And similarly for the other two categorical prestacks in (A.1).

A.1.4. By definition, the projections from the categorical prestacks in (A.1) to $\mathrm{Ran}^{\mathrm{unl}}$ are *value-wise co-Cartesian fibrations in groupoids*.

Denote by $\pi_{\mathrm{Ran}^{\mathrm{unl}}}$ the projection from

$$(\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{unl}}})_{\mathrm{dR}^{\mathrm{rel}}} \rightarrow \mathrm{Sect}_{\nabla}(X, \mathcal{Y}).$$

A.1.5. We will denote by $\mathbf{t}$ the maps from the non-unital to the unital versions. We have

$$\pi_{\mathrm{Ran}^{\mathrm{unl}}} \circ \mathbf{t} = \pi_{\mathrm{Ran}}.$$

A.2. IndCoh on categorical prestacks.

A.2.1. Let $\mathcal{X}$ be a categorical prestack *locally almost of finite type*, see [Ro, Sect. C.1.3] for what this means. In this case, it makes sense to talk about the category $\mathrm{IndCoh}(\mathcal{X})$ (see [Ga4, Sect. 2.2] or [Ro, Sect. C.3]).

Namely, an object $\mathcal{F} \in \mathrm{IndCoh}(\mathcal{X})$ associates to an affine scheme S (assumed almost of finite type) a functor

$$\mathcal{X}(S) \rightarrow \mathrm{IndCoh}(S),$$

in a way compatible with $!$ -pullback for morphisms between affine schemes.

We will denote this data as follows:

- For an object $x \in \mathcal{X}(S)$, we have an object

$$x^!(\mathcal{F}) \in \mathrm{IndCoh}(S);$$

- For a morphism $x_1 \xrightarrow{\alpha} x_2$ in $\mathcal{X}(S)$ a morphism

$$x_1^!(\mathcal{F}) \rightarrow x_2^!(\mathcal{F})$$

in $\text{IndCoh}(S)$.

A.2.2. We let

$$\text{IndCoh}(\mathcal{X})_{\text{str}} \subset \text{IndCoh}(\mathcal{X})$$

be the full subcategory, consisting of objects $\mathcal{F} \in \text{IndCoh}(\mathcal{X})$ such that for every affine test-scheme S and an arrow

$$x_1 \xrightarrow{\alpha} x_2, \quad x_1, x_2 \in \mathcal{X}(S),$$

the resulting map

$$x_1^!(\mathcal{F}) \rightarrow x_2^!(\mathcal{F})$$

is an isomorphism.

In other words, if we denote by

$$\mathcal{X} \xrightarrow{\text{str}} \mathcal{X}_{\text{str}}$$

the prestack, obtained from $\mathcal{X}$ by inverting all arrows, the pullback functor

$$\text{IndCoh}(\mathcal{X}_{\text{str}}) \xrightarrow{\text{str}^!} \text{IndCoh}(\mathcal{X})$$

defines an equivalence

$$\text{IndCoh}(\mathcal{X}_{\text{str}}) \xrightarrow{\sim} \text{IndCoh}(\mathcal{X})_{\text{str}}.$$

A.2.3. We claim:

Lemma A.2.4. *The natural diagram of categories*

$$\begin{array}{ccc} \text{IndCoh}(\mathcal{X}_{\text{dR}})_{\text{str}} & \longrightarrow & \text{IndCoh}(\mathcal{X})_{\text{str}} \\ \downarrow & & \downarrow \\ \text{IndCoh}(\mathcal{X}_{\text{dR}}) & \longrightarrow & \text{IndCoh}(\mathcal{X}) \end{array}$$

is a pullback square.

Proof. Follows from the fact that for an affine scheme S almost of finite type, the $!$ -pullback functor with respect to $S_{\text{red}} \rightarrow S$ is conservative. $\square$

A.3. A reformulation.

A.3.1. Note that the projection

$$(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}})_{\text{dRrel}} \xrightarrow{\pi_{\text{Ran}, \text{dRrel}}} \text{Sect}_{\nabla}(X, \mathcal{Y})$$

factors as

$$\begin{aligned} (\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}})_{\text{dRrel}} &\xrightarrow{\text{t}} (\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dRrel}} \xrightarrow{\text{str}} \\ &\rightarrow \left(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}_{\text{str}}} \right)_{\text{dRrel}} \xrightarrow{(\pi_{\text{Ran}^{\text{untl}}, \text{dRrel}})^{\text{str}}} \text{Sect}_{\nabla}(X, \mathcal{Y}), \end{aligned}$$

where

$$\left(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}_{\text{str}}} \right)_{\text{dRrel}} := ((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dRrel}})_{\text{str}}.$$

Hence, the pullback functor

$$\pi_{\text{Ran}^{\text{untl}}, \text{dRrel}}^! : \text{IndCoh}(\text{Sect}_{\nabla}(X, \mathcal{Y})) \rightarrow \text{IndCoh}((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dRrel}})$$

maps to

$$\text{IndCoh}((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dRrel}})_{\text{str}} \subset \text{IndCoh}((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dRrel}}).$$

A.3.2. We obtain that Proposition 5.3.4 follows from the next more precise statement:

Proposition A.3.3. *The counit of the adjunction*

$$\mathbf{ind}^{\text{rel}} \circ \mathbf{oblv}^{\text{rel}} \rightarrow \text{Id}$$

is an isomorphism, when evaluated on objects in the essential image along $\mathbf{t}^!$ of

$$\text{IndCoh}((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dR}^{\text{rel}}})_{\text{str}} \subset \text{IndCoh}((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dR}^{\text{rel}}}).$$

A.3.4. Consider the commutative diagram

$$(A.2) \quad \begin{array}{ccc} \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}} & \xrightarrow{\mathbf{t}} & \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}} \\ \downarrow & & \downarrow \\ (\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}})_{\text{dR}^{\text{rel}}} & \xrightarrow{\mathbf{t}} & (\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dR}^{\text{rel}}} \end{array}$$

This diagram is value-wise Cartesian. Hence, we have a well-defined pair of adjoint functors

$$(A.3) \quad \mathbf{ind}_{\text{untl}}^{\text{rel}} : \text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}}) \rightleftarrows \text{IndCoh}((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dR}^{\text{rel}}}) : \mathbf{oblv}_{\text{untl}}^{\text{rel}},$$

and both functors are compatible with their non-unital counterparts (5.8) via $\mathbf{t}^!$.

A.3.5. We also have a commutative diagram

$$\begin{array}{ccc} \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}} & \xrightarrow{\text{str}} & \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}_{\text{str}}^{\text{untl}}} \\ \downarrow & & \downarrow \\ (\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dR}^{\text{rel}}} & \xrightarrow{\text{str}} & (\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}_{\text{str}}^{\text{untl}}})_{\text{dR}^{\text{rel}}}, \end{array}$$

where

$$\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}_{\text{str}}^{\text{untl}}} := (\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{str}},$$

which is value-wise Cartesian. Hence, we have another pair of adjoint functors

$$(A.4) \quad \mathbf{ind}_{\text{untl}, \text{str}}^{\text{rel}} : \text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}_{\text{str}}^{\text{untl}}}) \rightleftarrows \text{IndCoh}((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}_{\text{str}}^{\text{untl}}})_{\text{dR}^{\text{rel}}}) : \mathbf{oblv}_{\text{untl}, \text{str}}^{\text{rel}},$$

where both functors are compatible with their non-strict counterparts (A.3) via $\text{str}^!$.

We can equivalently think of (A.4) as an adjunction

$$(A.5) \quad \mathbf{ind}_{\text{untl}, \text{str}}^{\text{rel}} : (\text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}}))_{\text{str}} \rightleftarrows (\text{IndCoh}((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dR}^{\text{rel}}}))_{\text{str}} : \mathbf{oblv}_{\text{untl}, \text{str}}^{\text{rel}}.$$

A.3.6. The assertion of Proposition A.3.3 follows from the following even more precise statement:

Proposition A.3.7. *The functor*

$$\mathbf{oblv}_{\text{untl}, \text{str}}^{\text{rel}} : (\text{IndCoh}((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dR}^{\text{rel}}}))_{\text{str}} \rightarrow (\text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}}))_{\text{str}}$$

is an equivalence.

A.4. A description of relative D-modules. In order to prove Proposition A.3.7, we will describe the category

$$\text{IndCoh}((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dR}^{\text{rel}}})$$

à la [Ro, Corollary 4.6.10].

A.4.1. As a warm-up, let us fix a point $\underline{x}$, and consider the prestack

$$\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}/\mathcal{Y}) := \mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}) \times_{\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y})} \mathrm{Sect}_{\nabla}(X, \mathcal{Y})$$

along with its variant

$$\begin{aligned} \mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}/\mathcal{Y})_{\mathrm{dR}^{\mathrm{rel}}} &:= \mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z})_{\mathrm{dR}} \times_{\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Y})_{\mathrm{dR}}} \mathrm{Sect}_{\nabla}(X, \mathcal{Y}) \simeq \\ &\simeq (\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}/\mathcal{Y}))_{\mathrm{dR}} \times_{\mathrm{Sect}_{\nabla}(X, \mathcal{Y})_{\mathrm{dR}}} \mathrm{Sect}_{\nabla}(X, \mathcal{Y}). \end{aligned}$$

We will describe the category

$$\mathrm{IndCoh}(\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}/\mathcal{Y})_{\mathrm{dR}^{\mathrm{rel}}})$$

along with its forgetful (i.e., pullback) functor to $\mathrm{IndCoh}(\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}/\mathcal{Y}))$.

A.4.2. Consider the map

$$\mathrm{add}_{\underline{x}} : \mathrm{Ran}^{\mathrm{untl}} \rightarrow \mathrm{Ran}^{\mathrm{untl}},$$

given by

$$\underline{y} \mapsto \underline{y} \cup \underline{x}.$$

Set

$$\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{untl}}, \underline{x}} := \mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{untl}}} \times_{\mathrm{Ran}^{\mathrm{untl}}, \mathrm{add}_{\underline{x}}} \mathrm{Ran}^{\mathrm{untl}}.$$

Restriction along $X - (\underline{y} \cup \underline{x}) \subset X - \underline{x}$ gives rise to a map

$$(A.6) \quad \mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}/\mathcal{Y}) \times \mathrm{Ran}^{\mathrm{untl}} \rightarrow \mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{untl}}, \underline{x}}.$$

A.4.3. Denote by

$$\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{untl}}, \underline{x}}^{\wedge}$$

the formal completion of $\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{untl}}, \underline{x}}$ along (A.6).

The projection

$$\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}/\mathcal{Y}) \times \mathrm{Ran}^{\mathrm{untl}} \rightarrow \mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}/\mathcal{Y})_{\mathrm{dR}^{\mathrm{rel}}}$$

tautologically extends to map

$$(A.7) \quad \mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{untl}}, \underline{x}}^{\wedge} \rightarrow \mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}/\mathcal{Y})_{\mathrm{dR}^{\mathrm{rel}}}.$$

A.4.4. The following is a version of [Ro, Corollary 4.6.10], where we allow poles at $\underline{x}$:

Theorem A.4.5. *The functor*

$$\begin{aligned} &\mathrm{IndCoh}(\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}/\mathcal{Y})_{\mathrm{dR}^{\mathrm{rel}}}) \rightarrow \\ &\rightarrow \mathrm{IndCoh}(\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}/\mathcal{Y})) \times_{\mathrm{IndCoh}(\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}/\mathcal{Y}) \times \mathrm{Ran}^{\mathrm{untl}})} \mathrm{IndCoh}\left(\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{untl}}, \underline{x}}^{\wedge}\right), \end{aligned}$$

given by pullback along the maps $\mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}/\mathcal{Y}) \rightarrow \mathrm{Sect}_{\nabla}(X - \underline{x}, \mathcal{Z}/\mathcal{Y})_{\mathrm{dR}^{\mathrm{rel}}}$ and (A.7), is an equivalence.

Remark A.4.6. In fact, this theorem is a particular case of [Ro, Corollary 4.6.10]: replace the original $\mathcal{Z}$ by its restriction of scalars along $X - \underline{x} \rightarrow X$.

A.4.7. We will now state a version of Theorem A.4.5, where we let $\underline{x}$ vary along Ran^{untl} . Consider the map

$$\text{add} : \text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}} \rightarrow \text{Ran}^{\text{untl}}, \quad \underline{x}_1, \underline{x}_2 \mapsto \underline{x}_1 \cup \underline{x}_2.$$

Set

$$\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{add}} := \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}} \times_{\text{Ran}^{\text{untl}, \text{add}}} (\text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}),$$

and

$$\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{pr}_1} := \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}} \times_{\text{Ran}^{\text{untl}, \text{pr}_1}} (\text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}}),$$

where

$$\text{pr}_1 : \text{Ran}^{\text{untl}} \times \text{Ran}^{\text{untl}} \rightarrow \text{Ran}^{\text{untl}}$$

is the projection on the first factor. In other words,

$$\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{pr}_1} \simeq \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}} \times \text{Ran}^{\text{untl}}.$$

Restriction along $X - (\underline{x}_1 \cup \underline{x}_2) \subset X - \underline{x}_1$ gives rise to a map

$$(A.8) \quad \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{pr}_1} \rightarrow \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{add}}.$$

A.4.8. Denote by

$$\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{add}}^{\wedge}$$

the formal completion of $\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{add}}$ along (A.8).

The projection

$$\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{pr}_1} \rightarrow (\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dR}^{\text{rel}}}$$

tautologically extends to a map

$$(A.9) \quad \text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{add}}^{\wedge} \rightarrow (\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dR}^{\text{rel}}}.$$

The following is a version of Theorem A.4.5 in families:

Theorem A.4.9. *The functor*

$$(A.10) \quad \text{IndCoh}((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dR}^{\text{rel}}}) \rightarrow \\ \rightarrow \text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}}) \times_{\text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{pr}_1})} \text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{add}}^{\wedge}),$$

given by pullback along the maps $\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}} \rightarrow (\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dR}^{\text{rel}}}$ and (A.9), is an equivalence.

A.5. Proof of Proposition A.3.7.

A.5.1. The functor (A.10) induces a functor

$$(A.11) \quad \text{IndCoh}((\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{dR}^{\text{rel}}})_{\text{str}} \rightarrow \\ \rightarrow \text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{Ran}^{\text{untl}}})_{\text{str}} \times_{\text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{pr}_1})_{\text{str}}} \text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{add}}^{\wedge})_{\text{str}},$$

where the two sides in (A.11) are full subcategories in the corresponding sides in (A.10).

Since the functor (A.10) is an equivalence, we obtain that (A.11) is fully faithful.

A.5.2. We will prove:

Lemma A.5.3. *The functor*

$$\text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{add}}^{\wedge})_{\text{str}} \rightarrow \text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}/\mathcal{Y})_{\text{pr}_1})_{\text{str}}$$

is an equivalence.

Let us assume this lemma for a moment and finish the proof of Proposition A.3.7.

A.5.4. By Lemma A.5.3, we obtain that the right-hand side in (A.11) projects isomorphically onto the first factor. Hence, we obtain that the pullback functor

$$(A.12) \quad \mathrm{IndCoh}((\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{untl}}})_{\mathrm{dR}^{\mathrm{rel}}})_{\mathrm{str}} \rightarrow \mathrm{IndCoh}(\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{untl}}})_{\mathrm{str}},$$

which is the functor $\mathbf{oblv}_{\mathrm{untl}, \mathrm{str}}^{\mathrm{rel}}$ of Proposition A.3.7, is fully faithful.

It remains to show that the functor (A.12) is essentially surjective.

A.5.5. Let $\mathcal{F}$ be an object in $\mathrm{IndCoh}(\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{untl}}})_{\mathrm{str}}$, which, by Lemma A.5.3, we interpret as an object in the right-hand side of (A.11).

By Theorem A.4.9 it corresponds to an object $\mathcal{F}_{\mathrm{dR}} \in \mathrm{IndCoh}((\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{untl}}})_{\mathrm{dR}^{\mathrm{rel}}})$, and we only need to show that $\mathcal{F}_{\mathrm{dR}}$ is strict. However, this follows from (a relative version of) Lemma A.2.4. $\square$ [Proposition A.3.7]

A.6. Proof of Lemma A.5.3.

A.6.1. We will prove that the map

$$(\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{pr}_1})_{\mathrm{str}} \rightarrow (\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{add}}^{\wedge})_{\mathrm{str}}$$

is an isomorphism of prestacks.

A.6.2. We claim:

Lemma A.6.3. *Let $\mathcal{W} \rightarrow \mathrm{Ran}^{\mathrm{untl}} \times \mathrm{Ran}^{\mathrm{untl}}$ be a map of categorical prestacks, which is a value-wise co-Cartesian fibration. Then the induced map*

$$\mathcal{W} \times_{\mathrm{Ran}^{\mathrm{untl}} \times \mathrm{Ran}^{\mathrm{untl}}, \Delta} \mathrm{Ran}^{\mathrm{untl}} \rightarrow \mathcal{W}$$

induces an isomorphism

$$\left(\mathcal{W} \times_{\mathrm{Ran}^{\mathrm{untl}} \times \mathrm{Ran}^{\mathrm{untl}}, \Delta} \mathrm{Ran}^{\mathrm{untl}} \right)_{\mathrm{str}} \rightarrow \mathcal{W}_{\mathrm{str}}.$$

Proof. This follows from the fact that the diagonal map $\Delta : \mathrm{Ran}^{\mathrm{untl}} \rightarrow \mathrm{Ran}^{\mathrm{untl}} \times \mathrm{Ran}^{\mathrm{untl}}$ is value-wise cofinal. $\square$

A.6.4. Applying Lemma A.6.3, it suffices to show that the map

$$\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{pr}_1} \times_{\mathrm{Ran}^{\mathrm{untl}} \times \mathrm{Ran}^{\mathrm{untl}}, \Delta} \mathrm{Ran}^{\mathrm{untl}} \rightarrow \mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{add}}^{\wedge} \times_{\mathrm{Ran}^{\mathrm{untl}} \times \mathrm{Ran}^{\mathrm{untl}}, \Delta} \mathrm{Ran}^{\mathrm{untl}}$$

induces an isomorphism on strictifications.

We claim that the above map is actually an isomorphism as-is.

A.6.5. Since the operation of formal completion commutes with fiber products, it suffices to show that the map

$$\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{pr}_1} \times_{\mathrm{Ran}^{\mathrm{untl}} \times \mathrm{Ran}^{\mathrm{untl}}, \Delta} \mathrm{Ran}^{\mathrm{untl}} \rightarrow \mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{add}} \times_{\mathrm{Ran}^{\mathrm{untl}} \times \mathrm{Ran}^{\mathrm{untl}}, \Delta} \mathrm{Ran}^{\mathrm{untl}}$$

is an isomorphism.

However, the latter is evident on the nose.

$\square$ [Lemma A.5.3]

APPENDIX B. PROOF OF PROPOSITION 5.5.5

In this section we let $\mathcal{Z}$ be an arbitrary affine D-scheme over X .

We will show that the assertion of Proposition 5.5.5 essentially amounts to [BD1, Proposition 4.6.5], combined with some *unitality* considerations.

B.1. A reformulation in terms of unital structure.

B.1.1. Let us return to the setting of Sect. 5.4.7. Suppose that $\mathcal{W}_{\text{Ran}}$ extends to a categorical prestack $\mathcal{W}_{\text{Ran}^{\text{unl}}}$ over Ran^{unl} , so that

$$\mathcal{W}_{\text{Ran}} \simeq \mathcal{W}_{\text{Ran}^{\text{unl}}} \times_{\text{Ran}^{\text{unl}}} \text{Ran},$$

and

$$\mathcal{W}_{\text{Ran}^{\text{unl}}} \rightarrow \text{Ran}^{\text{unl}}$$

is a value-wise co-Cartesian fibration in groupoids.

Effectively, this means that for $S \in \text{Sch}^{\text{aff}}$ and a map $\alpha : \underline{x}_1 \rightarrow \underline{x}_2$ in $\text{Ran}^{\text{unl}}(S)$ we have a map of prestacks

$$\alpha_{\mathcal{W}} : \mathcal{W}_{\text{Ran}} \times_{\text{Ran}, \underline{x}_1} S \rightarrow \mathcal{W}_{\text{Ran}} \times_{\text{Ran}, \underline{x}_2} S.$$

We will refer to $\mathcal{W}_{\text{Ran}^{\text{unl}}}$ as the unital structure on $\mathcal{W}_{\text{Ran}}$.

Let $\mathfrak{t}$ denote the tautological map

$$\mathcal{W}_{\text{Ran}} \rightarrow \mathcal{W}_{\text{Ran}^{\text{unl}}}.$$

B.1.2. To the data as above we can attach a category $\text{QCoh}_{\text{co}}(\mathcal{W})_{\text{Ran}^{\text{unl}}}$. Namely, the data of an object of $\text{QCoh}_{\text{co}}(\mathcal{W})_{\text{Ran}^{\text{unl}}}$ consists of an object $\mathcal{F} \in \text{QCoh}_{\text{co}}(\mathcal{W})_{\text{Ran}}$, and for every α as above of a map

$$((\alpha_{\mathcal{W}})_* \otimes \text{Id}_{\text{IndCoh}(S)}) (\mathcal{F}_{S, \underline{x}_1}) \rightarrow \mathcal{F}_{S, \underline{x}_2}$$

in

$$\text{QCoh}_{\text{co}}(\mathcal{W}_{\text{Ran}} \times_{\text{Ran}, \underline{x}_2} S) \otimes_{\text{QCoh}(S)} \text{IndCoh}(S).$$

Let $\mathfrak{t}^!$ denote the tautological forgetful functor

$$\text{QCoh}_{\text{co}}(\mathcal{W})_{\text{Ran}^{\text{unl}}} \rightarrow \text{QCoh}_{\text{co}}(\mathcal{W})_{\text{Ran}}.$$

B.1.3. Note that the space $\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}^{\text{unl}}}$ from Sect. A.1.3 provides a unital structure on $\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}}$.

We will deduce Proposition 5.5.5 from the following more precise statement:

Proposition B.1.4. *The natural transformation*

$$\Gamma^{\text{IndCoh}_{\text{Ran}}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}}, -) \circ (s_{\mathcal{Z}, \text{Ran}})^* \circ (s_{\mathcal{Z}, \text{Ran}})_* \rightarrow \Gamma^{\text{IndCoh}_{\text{Ran}}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}}, -)$$

is an isomorphism, when evaluated on the essential image of the functor

$$\mathfrak{t}^! : \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}))_{\text{Ran}^{\text{unl}}} \rightarrow \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}))_{\text{Ran}}.$$

In the rest of this subsection we will show how Proposition B.1.4 implies Proposition 5.5.5.

B.1.5. Assume that $\mathcal{W}_{\text{Ran}^{\text{unl}}}$ is locally almost of finite type (in particular, $\mathcal{W}_{\text{Ran}}$ is locally almost of finite type, so that the category $\text{IndCoh}(\mathcal{W}_{\text{Ran}})$ is well-ddefined). We claim that we have a well-defined category $\text{IndCoh}(\mathcal{W}_{\text{Ran}^{\text{unl}}})$.

By definition, the data of an object of $\text{IndCoh}(\mathcal{W}_{\text{Ran}^{\text{unl}}})$ consists of an object $\mathcal{F} \in \text{QCoh}_{\text{co}}(\mathcal{W}_{\text{Ran}})$, i.e., for every $\underline{x} \in \text{Ran}(S)$ we have an object $\mathcal{F}_{S, \underline{x}} \in \text{IndCoh}(\mathcal{W}_{\text{Ran}} \times_{\text{Ran}, \underline{x}} S)$, and for every α as above of a map

$$(\alpha_{\mathcal{W}})_*(\mathcal{F}_{S, \underline{x}_1}) \rightarrow \mathcal{F}_{S, \underline{x}_2}$$

in $\text{IndCoh}(S)$.

Let $\mathfrak{t}^!$ denote the tautological forgetful functor

$$\text{IndCoh}(\mathcal{W}_{\text{Ran}^{\text{unl}}}) \rightarrow \text{IndCoh}(\mathcal{W}_{\text{Ran}}).$$

B.1.6. Assume now that $\mathcal{W}_{\text{Ran}} \rightarrow \text{Ran}$ is a relative ind-scheme. Then as in Sect. 5.4.10 we have a functor

$$(B.1) \quad \Psi_{\mathcal{W}_{\text{Ran}^{\text{untl}}}} : \text{IndCoh}(\mathcal{W}_{\text{Ran}^{\text{untl}}}) \rightarrow \text{QCoh}_{\text{co}}(\mathcal{W})_{\text{Ran}^{\text{untl}}},$$

which makes the diagram

$$\begin{array}{ccc} \text{IndCoh}(\mathcal{W}_{\text{Ran}^{\text{untl}}}) & \xrightarrow{\Psi_{\mathcal{W}_{\text{Ran}^{\text{untl}}}}} & \text{QCoh}_{\text{co}}(\mathcal{W})_{\text{Ran}^{\text{untl}}} \\ \downarrow \text{t}^! & & \downarrow \text{t}^! \\ \text{IndCoh}(\mathcal{W}_{\text{Ran}}) & \xrightarrow{\Psi_{\mathcal{W}_{\text{Ran}}}} & \text{QCoh}_{\text{co}}(\mathcal{W})_{\text{Ran}} \end{array}$$

commutes.

B.1.7. Thus, we obtain that in order to prove Proposition 5.5.5, it suffices to show that the object

$$\omega_{\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}}} \in \text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}})$$

lies in the essential image of the functor

$$\text{t}^! : \text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}^{\text{untl}}}) \rightarrow \text{IndCoh}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}}).$$

However, this is true for any $\mathcal{W}_{\text{Ran}^{\text{untl}}}$ for which the maps $\alpha_{\mathcal{W}}$ are proper, which is the case for $\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}^{\text{untl}}}$.

□[Proposition 5.5.5]

B.2. The local unital structure.

B.2.1. Recall the categories $\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Z}))_{\text{Ran}}$ and $\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}}$, see Sect. 5.4.7. We will now introduce their variants, to be denoted

$$\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Z}))_{\text{Ran}^{\text{untl}}} \text{ and } \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}^{\text{untl}}},$$

respectively.

In order to do so, as in Sect. B.1.2, we must attach to a map $\alpha : \underline{x}_1 \rightarrow \underline{x}_2$ in $\text{Ran}^{\text{untl}}(S)$ functors

$$(B.2) \quad \text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Z}))_{\text{Ran}} \times_{\text{Ran}, \underline{x}_1} S \rightarrow \text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Z}))_{\text{Ran}} \times_{\text{Ran}, \underline{x}_2} S$$

and

$$(B.3) \quad \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}} \times_{\text{Ran}, \underline{x}_1} S \rightarrow \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}} \times_{\text{Ran}, \underline{x}_2} S,$$

respectively.

B.2.2. Recall that for a point $\underline{x}$ of Ran we have

$$\mathfrak{L}_{\nabla}^+(\mathcal{Z})_{\underline{x}} \simeq \text{Sect}_{\nabla}(\mathcal{D}_{\underline{x}}, \mathcal{Z}) \text{ and } \mathfrak{L}_{\nabla}(\mathcal{Z})_{\underline{x}} \simeq \text{Sect}_{\nabla}(\mathcal{D}_{\underline{x}} - \underline{x}, \mathcal{Z}).$$

For $\underline{x}_1 \subseteq \underline{x}_2$, set

$$\mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Z})_{\underline{x}_1 \subseteq \underline{x}_2} := \text{Sect}_{\nabla}(\mathcal{D}_{\underline{x}_2} - \underline{x}_1, \mathcal{Z}).$$

Restriction along

$$\mathcal{D}_{\underline{x}_1} \subseteq \mathcal{D}_{\underline{x}_2}$$

gives rise to a map

$$(B.4) \quad \mathfrak{L}_{\nabla}^+(\mathcal{Z})_{\underline{x}_2} \rightarrow \mathfrak{L}_{\nabla}^+(\mathcal{Z})_{\underline{x}_1}.$$

We define the functor

$$(B.5) \quad \text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Z})_{\underline{x}_1}) \rightarrow \text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Z})_{\underline{x}_2})$$

to be given by pullback along (B.4).

B.2.3. Restriction along

$$\mathcal{D}_{\underline{x}_1} - \underline{x}_1 \subseteq \mathcal{D}_{\underline{x}_2} - \underline{x}_1 \supseteq \mathcal{D}_{\underline{x}_2} - \underline{x}_2$$

defines maps

$$(B.6) \quad \mathfrak{L}_{\nabla}(\mathcal{Z})_{\underline{x}_1} \leftarrow \mathfrak{L}_{\nabla}^{\text{mer} \rightsquigarrow \text{reg}}(\mathcal{Z})_{\underline{x}_1 \subseteq \underline{x}_2} \rightarrow \mathfrak{L}_{\nabla}(\mathcal{Z})_{\underline{x}_2}.$$

The operation of $*$ -pull and $*$ -push along (B.6) gives rise to a functor

$$(B.7) \quad \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z})_{\underline{x}_1}) \rightarrow \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z})_{\underline{x}_2}).$$

B.2.4. The operations in (B.5) and (B.7) make sense when $\underline{x}_1$ and $\underline{x}_2$ are S -points of Ran , and give rise to the sought-for functors (B.2) and (B.3), respectively.

We will denote by $\mathfrak{t}^!$ the corresponding forgetful functors

$$\text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Z}))_{\text{Ran}^{\text{untl}}} \rightarrow \text{QCoh}(\mathfrak{L}_{\nabla}^+(\mathcal{Z}))_{\text{Ran}} \text{ and } \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}^{\text{untl}}} \rightarrow \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}},$$

respectively.

B.2.5. We will deduce Proposition B.1.4 from the following even more precise assertion:

Proposition B.2.6. *The natural transformation*

$$\begin{aligned} \Gamma^{\text{IndCohRan}}(\mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}}, -) &\rightarrow \Gamma^{\text{IndCohRan}}(\mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}}, -) \circ (s_{\mathcal{Z}, \text{Ran}})_* \circ (s_{\mathcal{Z}, \text{Ran}})^* \simeq \\ &\simeq \Gamma^{\text{IndCohRan}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z})_{\text{Ran}}, -) \circ (s_{\mathcal{Z}, \text{Ran}})^*, \end{aligned}$$

arising from the unit of the $((s_{\mathcal{Z}, \text{Ran}})^*, (s_{\mathcal{Z}, \text{Ran}})_*)$ -adjunction, is an isomorphism, when evaluated on objects lying in the essential image of the functor

$$\mathfrak{t}^! : \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}^{\text{untl}}} \rightarrow \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}}.$$

In the rest of this subsection we will show how Proposition B.2.6 implies Proposition B.1.4.

B.2.7. Let $\mathcal{F}_{\text{glob}}$ be an object of $\text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}))_{\text{Ran}}$, and assume that it lies in the essential image of

$$\mathfrak{t}^! : \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}))_{\text{Ran}^{\text{untl}}} \rightarrow \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, \mathcal{Z}))_{\text{Ran}}.$$

We wish to show that the map

$$\begin{aligned} \Gamma^{\text{IndCohRan}}(\mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}}, (s_{\mathcal{Z}, \text{Ran}})_* \circ (s_{\mathcal{Z}, \text{Ran}})^* \circ (s_{\mathcal{Z}, \text{Ran}})_*(\mathcal{F}_{\text{glob}})) &\rightarrow \\ &\rightarrow \Gamma^{\text{IndCohRan}}(\mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}}, (s_{\mathcal{Z}, \text{Ran}})_*(\mathcal{F}_{\text{glob}})), \end{aligned}$$

induced by the counit of the $((s_{\mathcal{Z}, \text{Ran}})^*, (s_{\mathcal{Z}, \text{Ran}})_*)$ -adjunction, is an isomorphism.

It is sufficient to show that the map

$$\begin{aligned} \Gamma^{\text{IndCohRan}}(\mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}}, (s_{\mathcal{Z}, \text{Ran}})_*(\mathcal{F}_{\text{glob}})) &\rightarrow \\ &\rightarrow \Gamma^{\text{IndCohRan}}(\mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}}, (s_{\mathcal{Z}, \text{Ran}})_* \circ (s_{\mathcal{Z}, \text{Ran}})^* \circ (s_{\mathcal{Z}, \text{Ran}})_*(\mathcal{F}_{\text{glob}})), \end{aligned}$$

induced by the unit of the adjunction, is an isomorphism.

B.2.8. Denote

$$\mathcal{F}_{\text{loc}} := (s_{\mathcal{Z}, \text{Ran}})_*(\mathcal{F}_{\text{glob}}) \in \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}}.$$

Thus, we have to show that the map

$$(B.8) \quad \Gamma^{\text{IndCohRan}}(\mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}}, \mathcal{F}_{\text{loc}}) \rightarrow \Gamma^{\text{IndCohRan}}(\mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}}, (s_{\mathcal{Z}, \text{Ran}})_* \circ (s_{\mathcal{Z}, \text{Ran}})^*(\mathcal{F}_{\text{loc}})),$$

induced by the unit of the adjunction, is an isomorphism.

B.2.9. Note that the functor

$$(s_{Z, \text{Ran}})_* : \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, Z))_{\text{Ran}} \rightarrow \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran}}$$

gives rise to a functor

$$(s_{Z, \text{Ran}}^{\text{untl}})_* : \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, Z))_{\text{Ran}^{\text{untl}}} \rightarrow \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran}^{\text{untl}}},$$

so that the diagram

$$\begin{array}{ccc} \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, Z))_{\text{Ran}^{\text{untl}}} & \xrightarrow{(s_{Z, \text{Ran}}^{\text{untl}})_*} & \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran}^{\text{untl}}} \\ \downarrow \mathfrak{t}^! & & \downarrow \mathfrak{t}^! \\ \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, Z))_{\text{Ran}} & \xrightarrow{(s_{Z, \text{Ran}})_*} & \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran}} \end{array}$$

commutes.

B.2.10. Hence, the assumption on $\mathcal{F}_{\text{glob}}$ implies that $\mathcal{F}_{\text{loc}}$ lies in the essential image of

$$\mathfrak{t}^! : \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran}^{\text{untl}}} \rightarrow \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran}}.$$

Hence, the isomorphism (B.8) follows from Proposition B.2.6.

□[Proposition B.1.4]

B.3. An expression for the global sections functor. In this subsection we will recall the expression for the functor

$$\Gamma^{\text{IndCohRan}}(\text{Sect}_{\nabla}(X^{\text{gen}}, Z)_{\text{Ran}}, -) \circ (s_{Z, \text{Ran}})^*$$

in terms of *factorization homology* à la [BD1, Sect. 4.6].

B.3.1. To any categorical prestack $\mathcal{W}$ we can attach the prestack $\mathcal{W}^{\rightarrow}$ that classifies arrows in $\mathcal{W}$. I.e., for a test affine scheme S , the groupoid $\mathcal{W}^{\rightarrow}(S)$ classifies triples

$$(w_1 \in \mathcal{W}(S), w_2 \in \mathcal{W}(S), \alpha : w_1 \rightarrow w_2).$$

B.3.2. Denote

$$\text{Ran}^{\subseteq} := (\text{Ran}^{\text{untl}})^{\rightarrow}.$$

Denote by

$$\text{pr}_{\text{small}}, \text{pr}_{\text{big}} : \text{Ran}^{\subseteq} \rightarrow \text{Ran}$$

the maps that correspond to the source and the target of the arrow, respectively.

Explicitly, the groupoid $\text{Ran}^{\subseteq}(S)$ consists of

$$\{\underline{x}_1, \underline{x}_2 \in \text{Ran}(S) \mid \underline{x}_1 \subseteq \underline{x}_2\},$$

and the maps pr_{small} and pr_{big} send a point as above to $\underline{x}_1$ and $\underline{x}_2$, respectively.

B.3.3. Denote

$$\mathfrak{L}_{\nabla}^+(Z)_{\text{Ran}^{\subseteq}, \text{small}} := \mathfrak{L}_{\nabla}^+(Z)_{\text{Ran}} \times_{\text{Ran}, \text{pr}_{\text{small}}} \text{Ran}^{\subseteq}, \quad \mathfrak{L}_{\nabla}^+(Z)_{\text{Ran}^{\subseteq}, \text{big}} := \mathfrak{L}_{\nabla}^+(Z)_{\text{Ran}} \times_{\text{Ran}, \text{pr}_{\text{big}}} \text{Ran}^{\subseteq}$$

and

$$\mathfrak{L}_{\nabla}(Z)_{\text{Ran}^{\subseteq}, \text{small}} := \mathfrak{L}_{\nabla}(Z)_{\text{Ran}} \times_{\text{Ran}, \text{pr}_{\text{small}}} \text{Ran}^{\subseteq}, \quad \mathfrak{L}_{\nabla}(Z)_{\text{Ran}^{\subseteq}, \text{big}} := \mathfrak{L}_{\nabla}(Z)_{\text{Ran}} \times_{\text{Ran}, \text{pr}_{\text{big}}} \text{Ran}^{\subseteq}.$$

Proceeding as in Sect. 5.4.7, one can define the corresponding categories

$$\text{QCoh}(\mathfrak{L}_{\nabla}^+(Z))_{\text{Ran}^{\subseteq}, \text{small}}, \quad \text{QCoh}(\mathfrak{L}_{\nabla}^+(Z))_{\text{Ran}^{\subseteq}, \text{big}},$$

and

$$\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran}^{\subseteq}, \text{small}} \quad \text{and} \quad \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran}^{\subseteq}, \text{big}}$$

respectively.

B.3.4. Denote by

$$p_{\mathfrak{L}_{\nabla}^+(Z)_{\text{Ran}}} : \mathfrak{L}_{\nabla}^+(Z)_{\text{Ran}} \rightarrow \text{Ran}$$

and

$$p_{\mathfrak{L}_{\nabla}^+(Z)_{\text{Ran} \subseteq, \text{small}}} : \mathfrak{L}_{\nabla}^+(Z)_{\text{Ran} \subseteq, \text{small}} \rightarrow \text{Ran}^{\subseteq} \text{ and } p_{\mathfrak{L}_{\nabla}^+(Z)_{\text{Ran} \subseteq, \text{big}}} : \mathfrak{L}_{\nabla}^+(Z)_{\text{Ran} \subseteq, \text{big}} \rightarrow \text{Ran}^{\subseteq},$$

as well as

$$p_{\mathfrak{L}_{\nabla}(Z)_{\text{Ran}}} : \mathfrak{L}_{\nabla}(Z)_{\text{Ran}} \rightarrow \text{Ran}$$

and

$$p_{\mathfrak{L}_{\nabla}(Z)_{\text{Ran} \subseteq, \text{small}}} : \mathfrak{L}_{\nabla}(Z)_{\text{Ran} \subseteq, \text{small}} \rightarrow \text{Ran}^{\subseteq} \text{ and } p_{\mathfrak{L}_{\nabla}(Z)_{\text{Ran} \subseteq, \text{big}}} : \mathfrak{L}_{\nabla}(Z)_{\text{Ran} \subseteq, \text{big}} \rightarrow \text{Ran}^{\subseteq},$$

the resulting maps.

We will consider the corresponding functors

$$(p_{\mathfrak{L}_{\nabla}^+(Z)_{\text{Ran}}})_* : \text{QCoh}(\mathfrak{L}_{\nabla}^+(Z))_{\text{Ran}} \rightarrow \text{IndCoh}(\text{Ran})$$

and

$$(p_{\mathfrak{L}_{\nabla}^+(Z)_{\text{Ran} \subseteq, \text{big}}})_* : \text{QCoh}(\mathfrak{L}_{\nabla}^+(Z))_{\text{Ran} \subseteq, \text{big}} \rightarrow \text{IndCoh}(\text{Ran}^{\subseteq})$$

as well

$$(p_{\mathfrak{L}_{\nabla}(Z)_{\text{Ran}}})_* : \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran}} \rightarrow \text{IndCoh}(\text{Ran})$$

and

$$(p_{\mathfrak{L}_{\nabla}(Z)_{\text{Ran} \subseteq, \text{big}}})_* : \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran} \subseteq, \text{big}} \rightarrow \text{IndCoh}(\text{Ran}^{\subseteq}).$$

B.3.5. Note now that the unital structures on the categories $\text{QCoh}(\mathfrak{L}_{\nabla}^+(Z))_{\text{Ran}}$ and $\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran}}$, determined by $\text{QCoh}(\mathfrak{L}_{\nabla}^+(Z))_{\text{Ran}^{\text{unl}}}$, and $\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran}^{\text{unl}}}$, respectively, give rise to functors

$$(B.9) \quad \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}^+(Z))_{\text{Ran} \subseteq, \text{small}} \rightarrow \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}^+(Z))_{\text{Ran} \subseteq, \text{big}}$$

and

$$(B.10) \quad \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran} \subseteq, \text{small}} \rightarrow \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran} \subseteq, \text{big}}.$$

Denote the compositions

$$\text{QCoh}(\mathfrak{L}_{\nabla}^+(Z))_{\text{Ran}} \xrightarrow{\text{pr}_{\text{small}}^!} \text{QCoh}(\mathfrak{L}_{\nabla}^+(Z))_{\text{Ran} \subseteq, \text{small}} \xrightarrow{(B.9)} \text{QCoh}(\mathfrak{L}_{\nabla}^+(Z))_{\text{Ran} \subseteq, \text{big}}$$

and

$$\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran}} \xrightarrow{\text{pr}_{\text{small}}^!} \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran} \subseteq, \text{small}} \xrightarrow{(B.10)} \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran} \subseteq, \text{big}}$$

in both instances by ins. unit; we will refer to this functor as the “insertion of the unit”.

B.3.6. Let

$$\text{diag} : \text{Ran} \rightarrow \text{Ran}^{\subseteq}$$

denote the diagonal map,

$$\underline{x} \mapsto (\underline{x} \subseteq \underline{x}).$$

In terms of Sect. B.3.1, it corresponds to the identity morphisms on objects of $\mathcal{W}_{\text{Ran}^{\text{unl}}}(S)$.

Note that we have pullback squares

$$(B.11) \quad \begin{array}{ccc} \mathfrak{L}_{\nabla}^+(Z)_{\text{Ran}} & \xrightarrow{\text{diag}} & \mathfrak{L}_{\nabla}^+(Z)_{\text{Ran} \subseteq, \text{big}} \\ p_{\mathfrak{L}_{\nabla}^+(Z)_{\text{Ran}}} \downarrow & & \downarrow p_{\mathfrak{L}_{\nabla}^+(Z)_{\text{Ran} \subseteq, \text{big}}} \\ \text{Ran} & \xrightarrow{\text{diag}} & \text{Ran}^{\subseteq}. \end{array}$$

and

$$(B.12) \quad \begin{array}{ccc} \mathfrak{L}_{\nabla}(Z)_{\text{Ran}} & \xrightarrow{\text{diag}} & \mathfrak{L}_{\nabla}(Z)_{\text{Ran} \subseteq, \text{big}} \\ p_{\mathfrak{L}_{\nabla}(Z)_{\text{Ran}}} \downarrow & & \downarrow p_{\mathfrak{L}_{\nabla}(Z)_{\text{Ran} \subseteq, \text{big}}} \\ \text{Ran} & \xrightarrow{\text{diag}} & \text{Ran}^{\subseteq}. \end{array}$$

Note also that we have a canonical identification

$$\text{diag}^! \circ \text{ins. unit} \simeq \text{Id},$$

in both instances.

B.3.7. Recall that $s_{Z, \text{Ran}}$ (or simply $s_{Z, \text{Ran}}$) denotes the map

$$\text{Sect}_{\nabla}(X^{\text{gen}}, Z)_{\text{Ran}} \rightarrow \mathfrak{L}_{\nabla}(Z)_{\text{Ran}}.$$

We will use the same symbol $s_{Z, \text{Ran}}$ (or simply $s_{Z, \text{Ran}}$) to denote the map

$$\text{Sect}_{\nabla}(X, Z) \times \text{Ran} \rightarrow \mathfrak{L}_{\nabla}^+(Z)_{\text{Ran}}.$$

The following assertion is a variant with parameters of [BD2, Proposition 4.6.5]:

Proposition B.3.8.

(a) *The functor*

$$\text{QCoh}(\mathfrak{L}_{\nabla}^+(Z))_{\text{Ran}} \xrightarrow{(s_{Z, \text{Ran}})^*} \text{QCoh}(\text{Sect}_{\nabla}(X, Z)) \otimes \text{IndCoh}(\text{Ran}) \xrightarrow{\Gamma(\text{Sect}_{\nabla}(X, Z), -) \otimes \text{Id}} \text{IndCoh}(\text{Ran})$$

identifies canonically with

$$\begin{aligned} \text{QCoh}(\mathfrak{L}_{\nabla}^+(Z))_{\text{Ran}} &\xrightarrow{\text{ins. unit}} \text{QCoh}(\mathfrak{L}_{\nabla}^+(Z))_{\text{Ran} \subseteq, \text{big}} \xrightarrow{(p_{\mathfrak{L}_{\nabla}^+(Z)}^+)^*_{\text{Ran} \subseteq, \text{big}}} \text{IndCoh}(\text{Ran}^{\subseteq}) \rightarrow \\ &\xrightarrow{(\text{pr}_{\text{small}})^*} \text{IndCoh}(\text{Ran}). \end{aligned}$$

Under the above identification, the map

$$(p_{\mathfrak{L}_{\nabla}^+(Z)}^+)^*_{\text{Ran}} \rightarrow (p_{\mathfrak{L}_{\nabla}^+(Z)}^+)^*_{\text{Ran}} \circ (s_{Z, \text{Ran}})^* \circ (s_{Z, \text{Ran}})^*,$$

given by the unit of the $((s_{Z, \text{Ran}})^*, (s_{Z, \text{Ran}})_*)$ -adjunction, corresponds to the map

$$\begin{aligned} (p_{\mathfrak{L}_{\nabla}^+(Z)}^+)^*_{\text{Ran}} &\simeq (\text{pr}_{\text{small}})^* \circ (p_{\mathfrak{L}_{\nabla}^+(Z)}^+)^*_{\text{Ran} \subseteq, \text{big}} \circ \text{diag}_* \simeq \\ &\simeq (\text{pr}_{\text{small}})^* \circ (p_{\mathfrak{L}_{\nabla}^+(Z)}^+)^*_{\text{Ran} \subseteq, \text{big}} \circ \text{diag}_* \circ \text{diag}^! \circ \text{ins. unit} \rightarrow (\text{pr}_{\text{small}})^* \circ (p_{\mathfrak{L}_{\nabla}^+(Z)}^+)^*_{\text{Ran} \subseteq, \text{big}} \circ \text{ins. unit}. \end{aligned}$$

(b) *The functor*

$$\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran}} \xrightarrow{(s_{Z, \text{Ran}})^*} \text{QCoh}_{\text{co}}(\text{Sect}_{\nabla}(X^{\text{gen}}, Z))_{\text{Ran}} \xrightarrow{(p_{\text{Sect}_{\nabla}(X^{\text{gen}}, Z)})^*_{\text{Ran}}} \text{IndCoh}(\text{Ran})$$

identifies canonically with

$$\begin{aligned} \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran}} &\xrightarrow{\text{ins. unit}} \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(Z))_{\text{Ran} \subseteq, \text{big}} \xrightarrow{(p_{\mathfrak{L}_{\nabla}(Z)}}^*_{\text{Ran} \subseteq, \text{big}}} \text{IndCoh}(\text{Ran}^{\subseteq}) \\ &\rightarrow (\text{pr}_{\text{small}})^* \text{IndCoh}(\text{Ran}). \end{aligned}$$

Under the above identification, the map

$$(p_{\mathfrak{L}_{\nabla}(Z)}}^*_{\text{Ran}} \rightarrow (p_{\mathfrak{L}_{\nabla}(Z)}}^*_{\text{Ran}} \circ (s_{Z, \text{Ran}})^* \circ (s_{Z, \text{Ran}})^*,$$

given by the unit of the $((s_{Z, \text{Ran}})^*, (s_{Z, \text{Ran}})_*)$ -adjunction, corresponds to the map

$$\begin{aligned} (p_{\mathfrak{L}_{\nabla}(Z)}}^*_{\text{Ran}} &\simeq (\text{pr}_{\text{small}})^* \circ (p_{\mathfrak{L}_{\nabla}(Z)}}^*_{\text{Ran} \subseteq, \text{big}} \circ \text{diag}_* \simeq \\ &\simeq (\text{pr}_{\text{small}})^* \circ (p_{\mathfrak{L}_{\nabla}(Z)}}^*_{\text{Ran} \subseteq, \text{big}} \circ \text{diag}_* \circ \text{diag}^! \circ \text{ins. unit} \rightarrow (\text{pr}_{\text{small}})^* \circ (p_{\mathfrak{L}_{\nabla}(Z)}}^*_{\text{Ran} \subseteq, \text{big}} \circ \text{ins. unit}. \end{aligned}$$

B.4. Inputting the unitality structure. In this subsection we will prove Proposition B.2.6 by combining Proposition B.3.8(b) with a cofinality argument.

B.4.1. Note that in the situation of Sect. B.3.1, the prestack $\mathcal{W}^\rightarrow$ itself can also be extended to a categorical prestack.

Applying this to Ran^{untl} , we obtain a categorical prestack, denoted $\text{Ran}^{\subseteq, \text{untl}}$. Explicitly, the space of morphisms

$$(\underline{x}_1 \subseteq \underline{x}_2) \rightarrow (\underline{x}'_1 \subseteq \underline{x}'_2)$$

is

$$\begin{cases} \{*\} & \text{if } \underline{x}_1 \subseteq \underline{x}'_1 \text{ and } \underline{x}_2 \subseteq \underline{x}'_2, \\ \emptyset, & \text{otherwise.} \end{cases}$$

Denote by

$$\mathbf{t}^{\subseteq} : \text{Ran}^{\text{untl}} \rightarrow \text{Ran}^{\subseteq, \text{untl}}$$

the corresponding map.

B.4.2. The following assertion is obtained by unwinding the constructions:

Lemma B.4.3. *The composite functor*

$$\begin{aligned} \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}^{\text{untl}}} &\xrightarrow{\mathbf{t}^!} \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}} \xrightarrow{\text{ins.unit}} \\ &\rightarrow \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}^{\subseteq, \text{big}}} \xrightarrow{(p_{\mathfrak{L}_{\nabla}^+(\mathcal{Z})_{\text{Ran}^{\subseteq, \text{big}}}})^*} \text{IndCoh}(\text{Ran}^{\subseteq}) \end{aligned}$$

factors via a functor

$$\text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}^{\text{untl}}} \rightarrow \text{IndCoh}(\text{Ran}^{\subseteq, \text{untl}}),$$

followed by $(\mathbf{t}^{\subseteq})^!$.

B.4.4. Note that from (B.12), we obtain commutative diagrams

$$\begin{array}{ccc} \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}^{\subseteq, \text{big}}} & \xrightarrow{\text{diag}^!} & \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}} \\ \downarrow (p_{\mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}^{\subseteq, \text{big}}}})^* & & \downarrow (p_{\mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}}})^* \\ \text{IndCoh}(\text{Ran}^{\subseteq}) & \xrightarrow{\text{diag}^!} & \text{IndCoh}(\text{Ran}) \end{array}$$

and

$$\begin{array}{ccc} \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}} & \xrightarrow{\text{diag}_*} & \text{QCoh}_{\text{co}}(\mathfrak{L}_{\nabla}(\mathcal{Z}))_{\text{Ran}^{\subseteq, \text{big}}} \\ \downarrow (p_{\mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}}})^* & & \downarrow (p_{\mathfrak{L}_{\nabla}(\mathcal{Z})_{\text{Ran}^{\subseteq, \text{big}}}})^* \\ \text{IndCoh}(\text{Ran}) & \xrightarrow{\text{diag}_*} & \text{IndCoh}(\text{Ran}^{\subseteq}). \end{array}$$

Hence, combining Proposition B.3.8(b) and Lemma B.4.3, we obtain that in order to deduce Proposition B.2.6, it suffices to prove the following assertion:

Proposition B.4.5. *The natural transformation*

$$\begin{aligned} \Gamma^{\text{IndCoh}}(\text{Ran}, -) \circ \text{diag}^! &\simeq \Gamma^{\text{IndCoh}}(\text{Ran}, -) \circ (\text{pr}_{\text{small}})_* \circ \text{diag}_* \circ \text{diag}^! \rightarrow \\ &\rightarrow \Gamma^{\text{IndCoh}}(\text{Ran}, -) \circ (\text{pr}_{\text{small}})_* \simeq \Gamma^{\text{IndCoh}}(\text{Ran}^{\subseteq}, -) \end{aligned}$$

of functors $\text{IndCoh}(\text{Ran}^{\subseteq}) \rightrightarrows \text{Vect}$, is an isomorphism, when evaluated on the essential image of the functor

$$(\mathbf{t}^{\subseteq})^! : \text{IndCoh}(\text{Ran}^{\subseteq, \text{untl}}) \rightarrow \text{IndCoh}(\text{Ran}^{\subseteq}).$$

□[Proposition B.2.6]

B.4.6. *Proof of Proposition B.4.5.* We need to show that the natural transformation

$$(B.13) \quad C_c(\mathrm{Ran}, -) \circ \mathrm{diag}^! \circ (\mathbf{t}^\subseteq)^! \rightarrow C_c(\mathrm{Ran}^\subseteq, -) \circ (\mathbf{t}^\subseteq)^!,$$

as functors

$$\mathrm{IndCoh}(\mathrm{Ran}^{\subseteq, \mathrm{untl}}) \Rightarrow \mathrm{Vect},$$

is an isomorphism.

First, as in [Ga4, Theorem 4.6.2], one shows that the map $\mathbf{t}^\subseteq$ is *universally homologically cofinal*. Hence, the natural transformation

$$C_c(\mathrm{Ran}^\subseteq, -) \circ (\mathbf{t}^\subseteq)^! \rightarrow C_c(\mathrm{Ran}^{\subseteq, \mathrm{untl}}, -),$$

as functors

$$\mathrm{IndCoh}(\mathrm{Ran}^{\subseteq, \mathrm{untl}}) \Rightarrow \mathrm{Vect},$$

is an isomorphism.

Consider now the composition

$$\mathbf{t}^\subseteq \circ \mathrm{diag} : \mathrm{Ran} \rightarrow \mathrm{Ran}^{\subseteq, \mathrm{untl}}.$$

It is easily seen to be value-wise cofinal. Hence, the natural transformation

$$C_c(\mathrm{Ran}, -) \circ \mathrm{diag}^! \circ (\mathbf{t}^\subseteq)^! \rightarrow C_c(\mathrm{Ran}^{\subseteq, \mathrm{untl}}, -),$$

as functors

$$\mathrm{IndCoh}(\mathrm{Ran}^{\subseteq, \mathrm{untl}}) \Rightarrow \mathrm{Vect},$$

is an isomorphism.

Combining, we obtain that (B.13) is also an isomorphism, as desired.

□[Proposition B.4.5]

APPENDIX C. PROOF OF PROPOSITION 5.6.8

We will show that Proposition 5.6.8 amounts to a parameterized version of Proposition 5.5.5, combined with a fully-faithfulness assertion regarding the *localization functor* Loc_Y .

C.1. Localization functor in the abstract setting.

C.1.1. Let $\mathcal{Y}_{\mathrm{Ran}}$ satisfy the following conditions:

- The diagonal map $\mathcal{Y}_{\mathrm{Ran}} \rightarrow \mathcal{Y}_{\mathrm{Ran}} \times_{\mathrm{Ran}} \mathcal{Y}_{\mathrm{Ran}}$ is affine. Note that this formally implies that the diagonal map of $\mathrm{Sect}(X, \mathcal{Y})$ is affine;
- For every $S \rightarrow \mathrm{Ran}$, the prestack $\mathcal{Y}_{\mathrm{Ran}} \times_{\mathrm{Ran}} S$ is *passable* (see [GaRo2, Chapter 3, Sect. 3.5.1] for what this means);
- The prestack $\mathrm{Sect}(X, \mathcal{Y})$ is passable.

C.1.2. Note that the above conditions imply that the morphism

$$s_{\mathcal{Y}, \mathrm{Ran}} : \mathrm{Sect}(X, \mathcal{Y}) \times \mathrm{Ran} \rightarrow \mathcal{Y}_{\mathrm{Ran}}$$

behaves nicely with respect to push-forwards:

For any prestack $\mathcal{W}$ mapping to $\mathcal{Y}_{\mathrm{Ran}}$, and the base-changed map

$$s'_{\mathcal{Y}, \mathrm{Ran}} : (\mathrm{Sect}(X, \mathcal{Y}) \times \mathrm{Ran}) \times_{\mathcal{Y}_{\mathrm{Ran}}} \mathcal{W} \rightarrow \mathcal{W},$$

the functor

$$(s'_{\mathcal{Y}, \mathrm{Ran}})_* : \mathrm{QCoh}((\mathrm{Sect}(X, \mathcal{Y}) \times \mathrm{Ran}) \times_{\mathcal{Y}_{\mathrm{Ran}}} \mathcal{W}) \rightarrow \mathrm{QCoh}(\mathcal{W}),$$

right adjoint to $(s'_{\mathcal{Y}, \mathrm{Ran}})^*$, commutes with colimits, and satisfies the base change formula. This follows from [GaRo2, Chapter 3, Proposition 3.5.3].

C.1.3. Consider the resulting pair of adjoint functors

$$(s_{\mathcal{Y}, \text{Ran}})^* : \text{QCoh}(\mathcal{Y})_{\text{Ran}} \rightleftarrows \text{QCoh}(\text{Sect}(X, \mathcal{Y})) \otimes \text{IndCoh}(\text{Ran}) : (s_{\mathcal{Y}, \text{Ran}})_*$$

Denote

$$\text{Loc}_{\mathcal{Y}} := (\text{Id} \otimes \Gamma^{\text{IndCoh}}(\text{Ran}, -)) \circ (s_{\mathcal{Y}, \text{Ran}})^*, \quad \text{QCoh}(\mathcal{Y})_{\text{Ran}} \rightarrow \text{QCoh}(\text{Sect}(X, \mathcal{Y})).$$

The right adjoint $\text{Loc}_{\mathcal{Y}}^R$ of $\text{Loc}_{\mathcal{Y}}$ is thus given by

$$(s_{\mathcal{Y}, \text{Ran}})_* \circ (\text{Id} \otimes (\omega_{\text{Ran}} \otimes -)), \quad \text{QCoh}(\text{Sect}(X, \mathcal{Y})) \rightarrow \text{QCoh}(\mathcal{Y})_{\text{Ran}}.$$

C.1.4. We will prove:

Proposition C.1.5. *Let $\mathcal{Y}$ be a D -prestack with an affine diagonal, satisfying the finiteness assumptions of Sect. C.1.1 above. Then the natural transformation*

$$\text{Loc}_{\mathcal{Y}} \circ (s_{\mathcal{Y}, \text{Ran}})_* \simeq (\text{Id} \otimes \Gamma^{\text{IndCoh}}(\text{Ran}, -)) \circ (s_{\mathcal{Y}, \text{Ran}})^* \circ (s_{\mathcal{Y}, \text{Ran}})_* \rightarrow (\text{Id} \otimes \Gamma^{\text{IndCoh}}(\text{Ran}, -)),$$

arising from the counit of the $((s_{\mathcal{Y}, \text{Ran}})^, (s_{\mathcal{Y}, \text{Ran}})_*)$ -adjunction, is an isomorphism, when evaluated on the essential image of the functor*

$$t^! \otimes \text{Id} : \text{QCoh}(\text{Sect}(X, \mathcal{Y})) \otimes \text{IndCoh}(\text{Ran})^{\text{untl}} \rightarrow \text{QCoh}(\text{Sect}(X, \mathcal{Y})) \otimes \text{IndCoh}(\text{Ran}).$$

This proposition will be proved in Sect. C.3.

C.1.6. Combined with the contractibility of the Ran space, i.e., the fact that the map

$$\Gamma^{\text{IndCoh}}(\text{Ran}, \omega_{\text{Ran}}) \rightarrow k,$$

given by the counit of the $(\Gamma^{\text{IndCoh}}(\text{Ran}, -), \omega_{\text{Ran}} \otimes -)$ -adjunction, is an isomorphism, from Proposition C.1.5 we obtain:

Proposition C.1.7. *Let $\mathcal{Y}$ be a D -prestack with an affine diagonal, satisfying the finiteness assumptions of Sect. C.1.1 above. Then the counit of the adjunction*

$$\text{Loc}_{\mathcal{Y}} \circ \text{Loc}_{\mathcal{Y}}^R \rightarrow \text{Id},$$

is an isomorphism.

Note that Proposition C.1.7 can be restated as:

Corollary C.1.8. *Under the above assumptions on $\mathcal{Y}$, the functor*

$$\text{Loc}_{\mathcal{Y}}^R : \text{QCoh}(\text{Sect}(X, \mathcal{Y})) \rightarrow \text{QCoh}(\mathcal{Y})_{\text{Ran}}$$

is fully faithful.

C.1.9. Note that for $\mathcal{Y}$ being the constant D -stack with fiber $\text{pt}/\check{G}$, we have

$$\text{Sect}(X, \mathcal{Y}) \simeq \text{LS}_{\check{G}}.$$

Furthermore, we have a tautological identification

$$\text{QCoh}(\mathcal{Y})_{\text{Ran}} \simeq \text{Rep}(\check{G})_{\text{Ran}}.$$

Under this identification, we have

$$\text{Loc}_{\mathcal{Y}} \simeq \text{Loc}_{\check{G}}^{\text{spec}}$$

and

$$\text{Loc}_{\mathcal{Y}}^R \simeq \Gamma_{\check{G}}^{\text{spec}}.$$

Hence, Corollary C.1.8 contains Proposition 1.1.4 as a particular case.

Remark C.1.10. As far as the actual proof of Proposition C.1.5 is concerned, we will first establish Proposition C.1.7, and then deduce the general case stated in Proposition C.1.5.

C.2. Proof of Proposition 5.6.8. In this subsection we will assume Proposition C.1.5 and deduce Proposition 5.6.8.

C.2.1. Along with the category $\mathrm{QCoh}_{\mathrm{co}}(\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y}))_{\mathrm{Ran}}$, we will consider its unital version

$$\mathrm{QCoh}_{\mathrm{co}}(\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y}))_{\mathrm{Ran}^{\mathrm{unl}}},$$

equipped with a forgetful functor

$$(C.1) \quad \mathbf{t}^! : \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y}))_{\mathrm{Ran}^{\mathrm{unl}}} \rightarrow \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y}))_{\mathrm{Ran}}.$$

We will show that the natural transformation

$$(C.2) \quad (\pi_{\mathrm{Ran}})_*^{\mathrm{IndCoh}_{\mathrm{Ran}}} \circ (s_{\mathcal{T}, \mathrm{Ran}})^* \circ (s_{\mathcal{T}, \mathrm{Ran}})_* \rightarrow (\pi_{\mathrm{Ran}})_*^{\mathrm{IndCoh}_{\mathrm{Ran}}},$$

induced by the counit of the $((s_{\mathcal{T}, \mathrm{Ran}})^*, (s_{\mathcal{T}, \mathrm{Ran}})_*)$ -adjunction, is an isomorphism, when evaluated on objects lying in the essential image of the forgetful functor (C.1).

This will imply the assertion of Proposition 5.6.8, since the functor (5.15) factors as

$$\begin{aligned} \mathrm{IndCoh}(\mathrm{Sect}_{\nabla}(X, \mathcal{Y})) &\xrightarrow{\pi_{\mathrm{Ran}}^!} \mathrm{IndCoh}(\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y}))_{\mathrm{Ran}^{\mathrm{unl}}} \rightarrow \\ &\xrightarrow{\Psi_{\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}^{\mathrm{unl}}}}} \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y}))_{\mathrm{Ran}^{\mathrm{unl}}} \xrightarrow{\mathbf{t}^!} \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y}))_{\mathrm{Ran}}. \end{aligned}$$

C.2.2. Recall the space

$$\mathcal{T}_{\mathrm{Sect}_{\nabla}(X, \mathcal{Y}), \mathrm{Ran}} := (\mathrm{Sect}_{\nabla}(X, \mathcal{Y}) \times \mathrm{Ran})_{\mathfrak{L}^+(\mathcal{Y})_{\mathrm{Ran}}} \times \mathcal{T}_{\mathrm{Ran}}.$$

Consider the corresponding category

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{T}_{\mathrm{Sect}_{\nabla}(X, \mathcal{Y})})_{\mathrm{Ran}}.$$

Let us denote by π'_{Ran} the projection

$$\mathcal{T}_{\mathrm{Sect}_{\nabla}(X, \mathcal{Y}), \mathrm{Ran}} \rightarrow \mathrm{Sect}(X, \mathcal{Y}).$$

A procedure similar to that defining the functor $(\pi_{\mathrm{Ran}})_*^{\mathrm{IndCoh}_{\mathrm{Ran}}}$ gives rise to a functor

$$(\pi'_{\mathrm{Ran}})_*^{\mathrm{IndCoh}_{\mathrm{Ran}}} : \mathrm{QCoh}_{\mathrm{co}}(\mathcal{T}_{\mathrm{Sect}_{\nabla}(X, \mathcal{Y})})_{\mathrm{Ran}} \rightarrow \mathrm{QCoh}(\mathrm{Sect}(X, \mathcal{Y})).$$

C.2.3. Note that the morphism

$$s_{\mathcal{T}, \mathrm{Ran}} : \mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}} \rightarrow \mathcal{T}_{\mathrm{Ran}}$$

factors as

$$\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y})_{\mathrm{Ran}} \xrightarrow{s'_{\mathcal{T}, \mathrm{Ran}}} \mathcal{T}_{\mathrm{Sect}_{\nabla}(X, \mathcal{Y}), \mathrm{Ran}} \xrightarrow{s'_{\mathcal{Y}, \mathrm{Ran}}} \mathcal{T}_{\mathrm{Ran}},$$

where $s'_{\mathcal{Y}, \mathrm{Ran}}$ is a base change of the map

$$s_{\mathcal{Y}, \mathrm{Ran}} : \mathrm{Sect}(X, \mathcal{Y}) \times \mathrm{Ran} \rightarrow \mathfrak{L}_{\nabla}^+(\mathcal{Y})_{\mathrm{Ran}},$$

which appears in Proposition C.1.5.

Thus, we can factor $(s_{\mathcal{T}, \mathrm{Ran}})_*$ as

$$\mathrm{QCoh}_{\mathrm{co}}(\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y}))_{\mathrm{Ran}} \xrightarrow{(s'_{\mathcal{T}, \mathrm{Ran}})^*} \mathrm{QCoh}_{\mathrm{co}}(\mathcal{T}_{\mathrm{Sect}_{\nabla}(X, \mathcal{Y})})_{\mathrm{Ran}} \xrightarrow{(s'_{\mathcal{Y}, \mathrm{Ran}})^*} \mathrm{QCoh}_{\mathrm{co}}(\mathcal{T}_{\mathrm{Ran}})$$

and $(s_{\mathcal{T}, \mathrm{Ran}})^*$ as

$$\mathrm{QCoh}_{\mathrm{co}}(\mathcal{T}_{\mathrm{Ran}}) \xrightarrow{(s'_{\mathcal{Y}, \mathrm{Ran}})^*} \mathrm{QCoh}_{\mathrm{co}}(\mathcal{T}_{\mathrm{Sect}_{\nabla}(X, \mathcal{Y})})_{\mathrm{Ran}} \xrightarrow{(s'_{\mathcal{T}, \mathrm{Ran}})^*} \mathrm{QCoh}_{\mathrm{co}}(\mathrm{Sect}_{\nabla}(X^{\mathrm{gen}}, \mathcal{Z}/\mathcal{Y}))_{\mathrm{Ran}}.$$

C.2.4. Consider the natural transformation

$$(C.3) \quad (\pi'_{\text{Ran}})_*^{\text{IndCohRan}} \circ (s'_{\mathcal{Y}, \text{Ran}})^* \circ (s'_{\mathcal{Y}, \text{Ran}})_* \circ (s'_{\mathcal{T}, \text{Ran}})_* \rightarrow \\ \rightarrow (\pi_{\text{Ran}})_*^{\text{IndCohRan}} \circ (s'_{\mathcal{T}, \text{Ran}})^* \circ (s'_{\mathcal{Y}, \text{Ran}})^* \circ (s'_{\mathcal{Y}, \text{Ran}})_* \circ (s'_{\mathcal{T}, \text{Ran}})_*,$$

arising from the *unit* of the $((s'_{\mathcal{T}, \text{Ran}})^*, (s'_{\mathcal{T}, \text{Ran}})_*)$ -adjunction.

Its composition with (C.2) is the natural transformation

$$(C.4) \quad (\pi'_{\text{Ran}})_* \circ (s'_{\mathcal{Y}, \text{Ran}})^* \circ (s'_{\mathcal{Y}, \text{Ran}})_* \circ (s'_{\mathcal{T}, \text{Ran}})_* \rightarrow (\pi'_{\text{Ran}})_* \circ (s'_{\mathcal{T}, \text{Ran}})_* \simeq (\pi_{\text{Ran}})_*,$$

arising from the *counit* of the $((s'_{\mathcal{Y}, \text{Ran}})^*, (s'_{\mathcal{Y}, \text{Ran}})_*)$ -adjunction.

We will show that both (C.3) and (C.4) are isomorphisms when evaluated on objects lying in the essential image of the functor (C.1). This will imply that (C.2) is also an isomorphism on such objects.

C.2.5. *Verification that (C.3) is an isomorphism.* Along with $\text{QCoh}_{\text{co}}(\mathcal{T}_{\text{Sect}_{\nabla}(X, \mathcal{Y})})_{\text{Ran}}$ we can consider its unital version

$$\text{QCoh}_{\text{co}}(\mathcal{T}_{\text{Sect}_{\nabla}(X, \mathcal{Y})})_{\text{Ran}^{\text{unital}}}.$$

Note that the functors

$$(s'_{\mathcal{Y}, \text{Ran}})^*, (s'_{\mathcal{Y}, \text{Ran}})_*, (s'_{\mathcal{T}, \text{Ran}})^*, (s'_{\mathcal{T}, \text{Ran}})_*$$

upgrade to functors

$$(s'_{\mathcal{Y}, \text{Ran}^{\text{unital}}})^*, (s'_{\mathcal{Y}, \text{Ran}^{\text{unital}}})_*, (s'_{\mathcal{T}, \text{Ran}^{\text{unital}}})^*, (s'_{\mathcal{T}, \text{Ran}^{\text{unital}}})_*$$

between the corresponding unital categories.

Hence, the functor

$$(s'_{\mathcal{Y}, \text{Ran}})^* \circ (s'_{\mathcal{Y}, \text{Ran}})_* \circ (s'_{\mathcal{T}, \text{Ran}})_*$$

sends objects that lie in the essential image of the functor (C.1) to objects that lie in the essential image of the corresponding functor

$$(C.5) \quad \mathbf{t}^! : \text{QCoh}_{\text{co}}(\mathcal{T}_{\text{Sect}_{\nabla}(X, \mathcal{Y})})_{\text{Ran}^{\text{unital}}} \rightarrow \text{QCoh}_{\text{co}}(\mathcal{T}_{\text{Sect}_{\nabla}(X, \mathcal{Y})})_{\text{Ran}}.$$

We obtain that it is enough to show that the natural transformation

$$(C.6) \quad (\pi'_{\text{Ran}})_*^{\text{IndCohRan}} \circ (s'_{\mathcal{T}, \text{Ran}})^* \rightarrow (\pi_{\text{Ran}})_*^{\text{IndCohRan}},$$

arising from the *counit* of the $((s'_{\mathcal{T}, \text{Ran}})^*, (s'_{\mathcal{T}, \text{Ran}})_*)$ -adjunction, is an isomorphism when evaluated on objects lying in the essential image of the functor (C.5).

However, thanks to the identification

$$s'_{\mathcal{T}, \text{Ran}} \simeq s_{\mathcal{Z}_{\text{Sect}_{\nabla}(X, \mathcal{Y})}}$$

(see Sect. 5.6.4), the latter statement is a parameterized (by $\text{Sect}(X, \mathcal{Y})$) version of Proposition B.2.6 for the relative affine scheme $\mathcal{Z}_{\text{Sect}_{\nabla}(X, \mathcal{Y})}$.

C.2.6. *Verification that (C.4) is an isomorphism.* As above, it is enough to show that the natural transformation

$$(C.7) \quad (\pi'_{\text{Ran}})_*^{\text{IndCohRan}} \circ (s'_{\mathcal{Y}, \text{Ran}})^* \circ (s'_{\mathcal{Y}, \text{Ran}})_* \rightarrow (\pi'_{\text{Ran}})_*^{\text{IndCohRan}}$$

is an isomorphism when evaluated on objects lying in the essential image of the functor (C.5).

However, by base change along the Cartesian diagram

$$\begin{array}{ccc} \mathcal{T}_{\text{Sect}_{\nabla}(X, \mathcal{Y}), \text{Ran}} & \xrightarrow{s'_{\mathcal{Y}, \text{Ran}}} & \mathcal{T}_{\text{Ran}} \\ \downarrow & & \downarrow \\ \text{Sect}(X, \mathcal{Y}) \times \text{Ran} & \xrightarrow{s_{\mathcal{Y}, \text{Ran}}} & \mathcal{Y}_{\text{Ran}}, \end{array}$$

this reduces to the assertion of Proposition C.1.5.

□[Proposition 5.6.8]

C.3. Proof of Proposition C.1.5.

C.3.1. We will first reduce the assertion of Proposition C.1.5 to that of Proposition C.1.7, and then prove Proposition C.1.7.

We need to show that the natural transformation

$$(\mathrm{Id} \otimes \Gamma^{\mathrm{IndCoh}}(\mathrm{Ran}^{\mathrm{untl}}, -)) \circ (s_{\mathrm{Ran}^{\mathrm{untl}}})^* \circ (s_{\mathrm{Ran}^{\mathrm{untl}}})_* \rightarrow (\mathrm{Id} \otimes \Gamma^{\mathrm{IndCoh}}(\mathrm{Ran}^{\mathrm{untl}}, -))$$

arising from the counit of the $((s_{\mathrm{Ran}^{\mathrm{untl}}})^*, (s_{\mathrm{Ran}^{\mathrm{untl}}})_*)$ -adjunction, is an isomorphism.

C.3.2. First, note that the left-lax symmetric monoidal structure on the functor

$$\Gamma^{\mathrm{IndCoh}}(\mathrm{Ran}^{\mathrm{untl}}, -) : \mathrm{IndCoh}(\mathrm{Ran}^{\mathrm{untl}}) \rightarrow \mathrm{Vect},$$

arising by adjunction from the monoidal structure on the functor $\omega_{\mathrm{Ran}^{\mathrm{untl}}} \otimes -$, is actually strictly symmetric monoidal structure. Indeed, this follows from the fact that the diagonal morphism

$$\mathrm{Ran}^{\mathrm{untl}} \rightarrow \mathrm{Ran}^{\mathrm{untl}} \times \mathrm{Ran}^{\mathrm{untl}}$$

is value-wise cofinal.

C.3.3. Similarly, we obtain that the functor

$$(\mathrm{Id} \otimes \Gamma^{\mathrm{IndCoh}}(\mathrm{Ran}^{\mathrm{untl}}, -)) \circ (s_{\mathrm{Ran}^{\mathrm{untl}}})^* : \mathrm{QCoh}(\mathcal{Y})_{\mathrm{Ran}} \rightarrow \mathrm{QCoh}(\mathrm{Sect}(X, \mathcal{Y}))$$

is $\mathrm{IndCoh}(\mathrm{Ran}^{\mathrm{untl}})$ -linear, where $\mathrm{IndCoh}(\mathrm{Ran}^{\mathrm{untl}})$ acts on $\mathrm{QCoh}(\mathrm{Sect}(X, \mathcal{Y}))$ via the symmetric monoidal functor $\Gamma^{\mathrm{IndCoh}}(\mathrm{Ran}^{\mathrm{untl}}, -)$.

C.3.4. This implies that we have a canonical isomorphism between the functor

$$(\mathrm{Id} \otimes \Gamma^{\mathrm{IndCoh}}(\mathrm{Ran}^{\mathrm{untl}}, -)) \circ (s_{\mathrm{Ran}^{\mathrm{untl}}})^* \circ (s_{\mathrm{Ran}^{\mathrm{untl}}})_*$$

and

$$\left((\mathrm{Id} \otimes \Gamma^{\mathrm{IndCoh}}(\mathrm{Ran}^{\mathrm{untl}}, -)) \circ (s_{\mathrm{Ran}^{\mathrm{untl}}})^* \circ (s_{\mathrm{Ran}^{\mathrm{untl}}})_* \circ (\mathrm{Id} \otimes (\omega_{\mathrm{Ran}^{\mathrm{untl}}} \otimes -)) \right) \otimes \Gamma^{\mathrm{IndCoh}}(\mathrm{Ran}^{\mathrm{untl}}, -),$$

and this isomorphism is compatible with the map of both to

$$(\mathrm{Id} \otimes \Gamma^{\mathrm{IndCoh}}(\mathrm{Ran}^{\mathrm{untl}}, -)) \simeq \left((\mathrm{Id} \otimes \Gamma^{\mathrm{IndCoh}}(\mathrm{Ran}^{\mathrm{untl}}, -)) \circ (\mathrm{Id} \otimes (\omega_{\mathrm{Ran}^{\mathrm{untl}}} \otimes -)) \right) \otimes \Gamma^{\mathrm{IndCoh}}(\mathrm{Ran}^{\mathrm{untl}}, -).$$

However, the latter map is an isomorphism, by Proposition C.1.7.

□[Proposition C.1.5]

C.4. Proof of Proposition C.1.7.

C.4.1. Due to the assumption that $\mathrm{Sect}(X, \mathcal{Y})$ is passable, self-functors on $\mathrm{QCoh}(\mathrm{Sect}(X, \mathcal{Y}))$ are in bijection with objects of $\mathrm{QCoh}(\mathrm{Sect}(X, \mathcal{Y}) \times \mathrm{Sect}(X, \mathcal{Y}))$, and the identity endofunctor is given by

$$(\Delta_{\mathrm{Sect}(X, \mathcal{Y})})_*(\mathcal{O}_{\mathrm{Sect}(X, \mathcal{Y})}).$$

Thus, we need to show that the map

$$(C.8) \quad (\mathrm{Id} \otimes (\mathrm{Loc}_{\mathcal{Y}} \circ \mathrm{Loc}_{\mathcal{Y}}^R))((\Delta_{\mathrm{Sect}(X, \mathcal{Y})})_*(\mathcal{O}_{\mathrm{Sect}(X, \mathcal{Y})})) \rightarrow (\Delta_{\mathrm{Sect}(X, \mathcal{Y})})_*(\mathcal{O}_{\mathrm{Sect}(X, \mathcal{Y})})$$

is an isomorphism.

C.4.2. We rewrite the left-hand side in (C.8) as the image of $\mathcal{O}_{\text{Sect}(X, \mathcal{Y})}$ along the push-pull along the diagram

$$\begin{array}{ccc}
 \text{Sect}(X, \mathcal{Y}) & & \\
 \Delta_{\text{Sect}(X, \mathcal{Y})} \downarrow & & \\
 \text{Sect}(X, \mathcal{Y}) \times \text{Sect}(X, \mathcal{Y}) & \longleftarrow & \text{Sect}(X, \mathcal{Y}) \times \text{Sect}(X, \mathcal{Y}) \times \text{Ran} \\
 & \downarrow \text{id} \times s_{\mathcal{Y}, \text{Ran}} & \\
 & \text{Sect}(X, \mathcal{Y}) \times \mathcal{Y}_{\text{Ran}} & \xleftarrow{\text{id} \times s_{\mathcal{Y}, \text{Ran}}} \text{Sect}(X, \mathcal{Y}) \times \text{Sect}(X, \mathcal{Y}) \times \text{Ran} \\
 & & \downarrow \\
 & & \text{Sect}(X, \mathcal{Y}) \times \text{Sect}(X, \mathcal{Y}).
 \end{array}$$

By base change, we rewrite this as push-pull along

$$\begin{array}{ccc}
 \text{Sect}(X, \mathcal{Y}) \times \text{Sect}(X, \mathcal{Y}) \times \text{Ran} & \longrightarrow & \text{Sect}(X, \mathcal{Y}) \times \text{Sect}(X, \mathcal{Y}) \\
 \downarrow s_{\mathcal{Y}, \text{Ran}} \times s_{\mathcal{Y}, \text{Ran}} & & \\
 \mathcal{Y}_{\text{Ran}} & \xrightarrow{\Delta_{\mathcal{Y}_{\text{Ran}}/\text{Ran}}} & \mathcal{Y}_{\text{Ran}} \times_{\text{Ran}} \mathcal{Y}_{\text{Ran}}
 \end{array}$$

of the object

$$(p_{\mathcal{Y}_{\text{Ran}}})^*(\omega_{\text{Ran}}) \in \text{QCoh}(\mathcal{Y})_{\text{Ran}}.$$

C.4.3. Consider the following version of the set-up of Sect. B.3.

Let $f : \mathcal{Z}_1 \rightarrow \mathcal{Z}_2$ is an affine morphism of D-prestacks. Consider the following commutative (but non-Cartesian) diagram

$$\begin{array}{ccc}
 \text{Sect}(X, \mathcal{Z}_1) \times \text{Ran} & \xrightarrow{s_{\mathcal{Z}_1, \text{Ran}}} & \mathcal{Z}_{1, \text{Ran}} \\
 \text{Sect}(f) \times \text{id} \downarrow & & \downarrow f_{\text{Ran}} \\
 \text{Sect}(X, \mathcal{Z}_2) \times \text{Ran} & \xrightarrow{s_{\mathcal{Z}_2, \text{Ran}}} & \mathcal{Z}_{2, \text{Ran}}.
 \end{array}$$

The $((s_{\mathcal{Z}_1, \text{Ran}})^*, (s_{\mathcal{Z}_1, \text{Ran}})_*)$ - and $((s_{\mathcal{Z}_2, \text{Ran}})^*, (s_{\mathcal{Z}_2, \text{Ran}})_*)$ -adjunctions give rise to natural transformation

$$s_{\mathcal{Z}_2, \text{Ran}}^* \circ (f_{\text{Ran}})_* \rightarrow (\text{Sect}(f)_* \otimes \text{Id})_* \circ s_{\mathcal{Z}_1, \text{Ran}}^*$$

as functors

$$\text{QCoh}(\mathcal{Z}_1)_{\text{Ran}} \rightrightarrows \text{QCoh}(\text{Sect}(X, \mathcal{Z}_2)) \otimes \text{IndCoh}(\text{Ran}).$$

Consider the induced natural transformation

$$\begin{aligned}
 \text{(C.9)} \quad & (\text{Id} \otimes \Gamma^{\text{IndCoh}}(\text{Ran}, -)) \circ s_{\mathcal{Z}_2, \text{Ran}}^* \circ (f_{\text{Ran}})_* \rightarrow \\
 & \rightarrow (\text{Id} \otimes \Gamma^{\text{IndCoh}}(\text{Ran}, -)) \circ (\text{Sect}(f)_* \otimes \text{Id})_* \circ s_{\mathcal{Z}_1, \text{Ran}}^* \simeq \text{Sect}(f)_* \circ (\text{Id} \otimes \Gamma^{\text{IndCoh}}(\text{Ran}, -)) \circ s_{\mathcal{Z}_1, \text{Ran}}^*
 \end{aligned}$$

as functors

$$\text{QCoh}(\mathcal{Z}_1)_{\text{Ran}} \rightrightarrows \text{QCoh}(\text{Sect}(X, \mathcal{Z}_2)).$$

The following is a parametrized version of Proposition B.3.8(a):

Proposition C.4.4. *The natural transformation (C.9) is an isomorphism, when evaluated on objects lying in the essential image of the forgetful functor*

$$\mathbf{t}^! : \text{QCoh}(\mathcal{Z}_1)_{\text{Ran}^{\text{until}}} \rightarrow \text{QCoh}(\mathcal{Z}_1)_{\text{Ran}}.$$

Corollary C.4.5. *The natural transformation (C.9) is an isomorphism, when evaluated on the object*

$$(p_{\mathcal{Z}_1, \text{Ran}})^*(\omega_{\text{Ran}}) \in \text{QCoh}(\mathcal{Z}_1)_{\text{Ran}}.$$

C.4.6. We will apply the above to

$$\mathcal{Z}_1 = \mathcal{Y}, \mathcal{Z}_2 = \mathcal{Y} \times \mathcal{Y}$$

and f being the diagonal map.

Unwinding the definitions, we obtain that the map

$$\begin{aligned} \text{(C.10)} \quad & (\text{Id} \otimes \Gamma^{\text{IndCoh}}(\text{Ran}, -)) \circ (s_{\mathcal{Y}, \text{Ran}} \times_{\text{Ran}} s_{\mathcal{Y}, \text{Ran}})^* \circ (\Delta_{\mathcal{Y}_{\text{Ran}}})_* ((p_{\mathcal{Y}_{\text{Ran}}})^*(\omega_{\text{Ran}})) \xrightarrow{\text{(C.9)}} \\ & \rightarrow (\Delta_{\text{Sect}(X, \mathcal{Y})})_* \circ (\text{Id} \otimes \Gamma^{\text{IndCoh}}(\text{Ran}, -)) \circ (s_{\mathcal{Y}, \text{Ran}})^* ((p_{\mathcal{Y}_{\text{Ran}}})^*(\omega_{\text{Ran}})) \xrightarrow{\sim} (\Delta_{\text{Sect}(X, \mathcal{Y})})_* (\mathcal{O}_{\text{Sect}(X, \mathcal{Y})}) \end{aligned}$$

identifies with the map

$$(\text{Id} \otimes \Gamma^{\text{IndCoh}}(\text{Ran}, -)) \circ (s_{\mathcal{Y}, \text{Ran}} \times_{\text{Ran}} s_{\mathcal{Y}, \text{Ran}})^* \circ (\Delta_{\mathcal{Y}_{\text{Ran}}})_* ((p_{\mathcal{Y}_{\text{Ran}}})^*(\omega_{\text{Ran}})) \rightarrow (\Delta_{\text{Sect}(X, \mathcal{Y})})_* (\mathcal{O}_{\text{Sect}(X, \mathcal{Y})})$$

of (C.8).

Hence, the latter map is an isomorphism by Corollary C.4.5.

□[Proposition C.1.7]

REFERENCES

- [Ari] D. Arinkin, *Irreducible connections admit generic oper structures*, arXiv:1602.08989.
- [AGKRRV1] D. Arinkin, D. Gaitsgory, D. Kazhdan, S. Raskin, N. Rozenblyum and Y. Varshavsky, *The stack of local systems with restricted variation and geometric Langlands theory with nilpotent singular support*, arXiv:2010.01906.
- [AGKRRV2] D. Arinkin, D. Gaitsgory, D. Kazhdan, S. Raskin, N. Rozenblyum and Y. Varshavsky, *Duality for automorphic sheaves with nilpotent singular support*, arXiv:2012.07665.
- [BD1] A. Beilinson and V. Drinfeld, *Quantization of Hitchin's integrable system and Hecke eigensheaves*, available at http://people.math.harvard.edu/~gaitsgde/grad_2009/
- [BD2] A. Beilinson and V. Drinfeld, *Chiral algebras*, AMS Colloquium Publications **51**, AMS (2004).
- [BKS] D. Beraldo, D. Kazhdan and T. Schlank, *Contractibility of the space of generic opers for classical groups*, arXiv:1808.05801.
- [Be1] D. Beraldo, *On the extended Whittaker category*, Selecta Mathematica **25** (2019).
- [Be2] D. Beraldo, *On the geometric Ramanujan conjecture*, arXiv:2103.17211.
- [BG] A. Braverman and D. Gaitsgory, *Deformations of local systems and Eisenstein series*, joint with A. Braverman, GAFA **17** (2008), 1788–1850.
- [DG] V. Drinfeld and D. Gaitsgory, *Compact generation of the category of D -modules on the stack of G -bundles on a curve* Cambridge Math Journal, **3** (2015), 19–125.
- [FR] J. Faergeman and S. Raskin, *Non-vanishing of geometric Whittaker coefficients for reductive groups*, arXiv:2207.02955.
- [FGV] E. Frenkel, D. Gaitsgory and K. Vilonen, *On the geometric Langlands conjecture*, Jour. Amer. Math. Soc. **15** (2002), 367–417.
- [Ga1] D. Gaitsgory, *A “strange” functional equation for Eisenstein series and Verdier duality on the moduli stack of bundles*, Annales Scientifiques de l’ENS **50** (2017), 1123–1162.
- [Ga2] D. Gaitsgory, *On a vanishing conjecture appearing in the geometric Langlands correspondence*, Ann. Math. **160** (2004), 617–682.
- [Ga3] D. Gaitsgory, *Sheaves of categories and the notion of 1-affineness*, Contemporary Mathematics **643** (2015), 1–99.
- [Ga4] D. Gaitsgory, *The Atiyah-Bott formula for the cohomology of Bun_G* , arXiv:1505.02331.
- [GaRo1] D. Gaitsgory and N. Rozenblyum, *Crystals and D -modules*, PAMQ **10**, (2014), 57–155.
- [GaRo2] D. Gaitsgory and N. Rozenblyum, *A study in derived algebraic geometry, Vol. 1: Correspondences and Duality*, Mathematical surveys and monographs **221** (2017), AMS, Providence, RI.
- [GLC1] D. Gaitsgory and S. Raskin, *Proof of the geometric Langlands conjecture I: construction of the functor*, arXiv:2405.03599
- [GLC2] D. Arinkin, D. Beraldo, D. Gaitsgory, J. Faergeman, L. Chen, K. Lin, S. Raskin and N. Rozenblyum, *Proof of the geometric Langlands conjecture II: Kac-Moody localization and the FLE*, arXiv:2405.03648
- [GLC3] J. Campbell, L. Chen, D. Gaitsgory and S. Raskin, *Proof of the geometric Langlands conjecture III: compatibility with parabolic induction*, arXiv:2409.07051
- [Lin] K. Lin, *Poincaré series and miraculous duality*, arXiv:2211.05282.
- [Ro] N. Rozenblyum, *Connections on moduli spaces and infinitesimal Hecke modifications*, arXiv:2108.07745.

“Fear will keep the local systems in line.”
 (Grand Moff Tarkin, *Star Wars: Episode IV – A New Hope*)

PROOF OF THE GEOMETRIC LANGLANDS CONJECTURE V: THE MULTIPLICITY ONE THEOREM

DENNIS GAITSGORY AND SAM RASKIN

To V. Drinfeld

CONTENTS

Introduction	1
0.1. What was done in previous papers?	1
0.2. What is done in this paper?	2
0.3. Complications stemming from a non-trivial center	4
0.4. What is not done in this paper?	5
0.5. Contents	7
0.6. Acknowledgements	7
1. Summary of the preceding results	8
1.1. The Langlands functor and its categorical properties	8
1.2. Spectral properties	8
1.3. Restriction to the irreducible locus	8
1.4. Properties of $\mathcal{A}_{G,\text{irred}}$	9
2. Reduction to the case when G is almost simple and simply-connected	10
2.1. Compatibility between Langlands functors	10
2.2. Proof of Proposition 2.1.6	10
2.3. Changing the group	13
2.4. Proof of Theorem 2.3.4	14
3. Low genus cases	17
3.1. What do we need to prove?	17
3.2. The case of $g = 0$	17
3.3. The case $g = 1$	17
4. Calculation of the fundamental group	18
4.1. The fundamental groupoid of $\text{Bun}_{\tilde{G}}$	18
4.2. Line bundles on $\text{Bun}_{\tilde{G}}$	19
4.3. The fundamental group of $\text{LS}_{\tilde{G}}^{\text{irred}}$	20
5. The core of the proof	22
5.1. Action of the center	22
5.2. Endomorphisms of the vacuum Poincaré object	22
5.3. Algebraic geometry of $\text{LS}_{\tilde{G}}$	23
5.4. Structure of $\mathcal{A}_{G,\text{irred}}$	23
5.5. Proof of GLC	24
5.6. Additional remarks	26
6. The vacuum Poincaré object	27
6.1. How to calculate those endomorphisms?	27
6.2. Proof of Theorems 5.2.3 and 5.2.4	27

Date: September 13, 2024.

6.3. Proof of Theorems 5.2.7 and 5.2.8	28
7. Geometry of Bun_G	29
7.1. Proof of Proposition 4.1.6	29
7.2. Proof of Proposition 4.3.5	30
7.3. Proof of Proposition 5.3.5	31
7.4. Proof of Proposition 4.3.10	32
8. 2-Fourier-Mukai transform of the automorphic category	34
8.1. The notion of 2-Fourier-Mukai transform	34
8.2. 2-Fourier-Mukai transform and Poincaré duality	36
8.3. Example: the usual Fourier-Mukai transform	37
8.4. Some further compatibilities	38
8.5. Two sheaves associated with automorphic category	39
8.6. Proof of Theorem 8.5.8	41
8.7. Proof of Proposition 8.6.7(a)	43
8.8. Proof of Proposition 8.6.7(b)	45
8.9. Geometric Langlands for non-pure inner forms	48
8.10. Arithmetic consequences	49
Appendix A. Review of (semi-)stability for G -bundles	50
A.1. Definition of (semi-)stability	50
A.2. A characterization of (semi-)stability	51
References	51

INTRODUCTION

This paper is a conclusion of the series of papers [GLC1, GLC2, GLC3, GLC4], written jointly with D. Arinkin, D. Beraldo, J. Campbell, L. Chen, J. Færgeman, K. Lin, and N. Rozenblyum.

In the preceding papers, we have constructed the Langlands functor

$$(0.1) \quad \mathbb{L}_G : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}),$$

and established many of its properties.

In this paper we will prove the geometric Langlands conjecture (GLC) by showing that $\mathbb{L}_G$ is an equivalence.

This theorem confirms the original vision of Beilinson-Drinfeld, which was circulated by them in discussions of their work [BD]. A precise version of the conjecture was originally given in [AG, Conjecture 11.2.3]. The functor $\mathbb{L}_G$ was constructed in [GLC1], and a version of GLC with this precise functor included appears as [GLC1, Conjecture 1.6.7]. As discussed in [GLC1], the functor $\mathbb{L}_G$ is the unique possible equivalence that is compatible with the *Whittaker* and *Eisenstein* compatibilities of [Ga4].

0.1. What was done in previous papers? We begin with a short summary of the key points of [GLC1, GLC2, GLC3, GLC4]. This material is discussed in greater detail in Sect. 1.

0.1.1. Recall from [GLC1, Sect. 1.1] that $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ denotes the category of *half-twisted* D -modules on Bun_G . This is the category that appears on the automorphic side of the (de Rham) version of the geometric Langlands conjecture (GLC).

In [GLC1], we used the functor of Whittaker coefficient to construct the *geometric Langlands functor*

$$(0.2) \quad \mathbb{L}_G : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}),$$

where the right hand side was defined in [AG]. This functor is linear with respect to the *spectral action* of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ (see [GLC1, Sect. 1.2]). The geometric Langlands conjecture states that $\mathbb{L}_G$ is an equivalence of categories.

0.1.2. We proved in [GLC4, Theorem 1.6.2] (building on [FR] and [GLC3]) that the functor $\mathbb{L}_G$ is conservative. We also proved that $\mathbb{L}_G$ admits a left adjoint, to be denoted $\mathbb{L}_G^L$. Moreover, we proved that the composition of the functor $\mathbb{L}_G$ and its left adjoint $\mathbb{L}_G^L$, which is an endofunctor of $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}})$, is given by tensoring by an object

$$\mathcal{A}_G \in \text{QCoh}(\text{LS}_{\check{G}}).$$

This object $\mathcal{A}_G$ is naturally an associative algebra object in $\text{QCoh}(\text{LS}_{\check{G}})$.

By Barr-Beck, we obtain an equivalence

$$\begin{array}{ccc} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) & \xrightarrow{\sim} & \mathcal{A}_G\text{-mod}(\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}})) \\ & \searrow \mathbb{L}_G & \downarrow \\ & & \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \end{array}$$

between $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ and the category of $\mathcal{A}_G$ -modules in $\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}})$.

Therefore, the geometric Langlands conjecture amounts to the assertion that the unit map

$$(0.3) \quad \mathcal{O}_{\text{LS}_{\check{G}}} \rightarrow \mathcal{A}_G,$$

is an isomorphism in $\text{QCoh}(\text{LS}_{\check{G}})$.

0.1.3. The main theorem of [GLC3] asserts that the restriction of the map (0.3) to the locus of *reducible* local systems is an isomorphism.

Therefore, it remains to show that the map

$$(0.4) \quad \mathcal{O}_{\text{LS}_{\check{G}}^{\text{irred}}} \rightarrow \mathcal{A}_G|_{\text{LS}_{\check{G}}^{\text{irred}}},$$

induced by (0.3), is an isomorphism.

0.1.4. The paper [GLC4] developed some structural features of $\mathcal{A}_{G,\text{irred}}$.

Namely, this work proves that

$$\mathcal{A}_{G,\text{irred}} := \mathcal{A}_G|_{\text{LS}_{\check{G}}^{\text{irred}}}$$

is a *classical* vector bundle¹ that carries a flat connection. Moreover, this connection has finite monodromy.

0.1.5. The above observations mark the starting point of the present paper.

0.2. What is done in this paper?

0.2.1. First, let us observe what needs to be done given the preliminaries above.

Let σ be an *irreducible* $\check{G}$ -local system. We need to show that the fiber $\mathcal{A}_{G,\sigma}$ of $\mathcal{A}_G$ at σ is the unit associative algebra.

As $\mathcal{A}_{G,\sigma}\text{-mod}$ is the category $\text{Hecke}_{\sigma}(\text{D-mod}(\text{Bun}_G))$ of Hecke eigensheaves with eigenvalue σ , the above amounts to showing that there is a *unique* (up to tensoring with a vector space) Hecke eigensheaf for each such σ .

By the construction of the Langlands functor $\mathbb{L}_G$, this amounts to two assertions, (i) the existence of a Hecke eigensheaf for an irreducible spectral parameter σ , and (ii) a multiplicity one theorem: we need to show that cuspidal objects of $\text{D-mod}(\text{Bun}_G)$ can be (uniquely) reconstructed from their Whittaker coefficients.

¹Here we are for simplicity assuming that G is semi-simple, i.e., that $Z_G^0 = \{1\}$.

0.2.2. We now outline the argument presented in the paper.

For simplicity, we assume that G is simple (in particular, of adjoint type) and that the genus g is at least 2; we also exclude the case when $g = 2$ and $G = PGL_2$. The proof is based on the three observations concerning the topology of $LS_{\tilde{G}}^{\text{irred}}$, the algebraic geometry of $LS_{\tilde{G}}$, and the sheaf theory of Bun_G :

- $LS_{\tilde{G}}^{\text{irred}}$ is simply-connected (Theorem 4.3.2);
- $LS_{\tilde{G}}$ is Cohen-Macaulay and the complement of $LS_{\tilde{G}}^{\text{irred}}$ has codimension ≥ 2 (Corollary 5.3.3, Proposition 5.3.5);
- Endomorphisms of the vacuum Poincaré sheaf $\text{Poinc}_{G,!}^{\text{Vac, glob}} \in \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ are just scalars (Theorem 5.2.3).

We remind here that the vacuum Poincaré sheaf appears in characterizing the Langlands functor $\mathbb{L}_G$, cf. [GLC1, Sect. 1.4].

The last of these three results is the simplest to prove. But as we will see below, it is ultimately the point most responsible for addressing the *multiplicity one* problem mentioned in Sect. 0.2.1.

0.2.3. The above three observations combine as follows.

The first observation combined with the features of $\mathcal{A}_{G, \text{irred}}$ mentioned in Sect. 0.1.4 implies that $\mathcal{A}_{G, \text{irred}}$ is isomorphic to the direct sum of several copies of the structure sheaf, i.e.,

$$\mathcal{A}_{G, \text{irred}} \simeq \mathcal{O}_{LS_{\tilde{G}}^{\text{irred}}}^{\oplus n}.$$

As (0.4) is a morphism of algebras, it suffices to show that $n = 1$.

0.2.4. The second observation, combined with the fact that (0.3) is an isomorphism on the *reducible* locus, implies that

$$H^0(\Gamma(LS_{\tilde{G}}, \mathcal{A}_G)) \rightarrow H^0(\Gamma(LS_{\tilde{G}}^{\text{irred}}, \mathcal{A}_{G, \text{irred}}))$$

is an isomorphism (see Sect. 5.5.2 for more details).

So, we obtain that

$$\begin{aligned} (0.5) \quad \dim(H^0(\Gamma(LS_{\tilde{G}}, \mathcal{A}_G))) &= \dim(H^0(\Gamma(LS_{\tilde{G}}^{\text{irred}}, \mathcal{A}_{G, \text{irred}}))) = \\ &= \dim(H^0(\Gamma(LS_{\tilde{G}}^{\text{irred}}, \mathcal{O}_{LS_{\tilde{G}}^{\text{irred}}}^{\oplus n}))) \geq n. \end{aligned}$$

0.2.5. Finally, by the construction of $\mathbb{L}_G$, we have

$$\Gamma(LS_{\tilde{G}}, \mathcal{A}_G) \simeq \mathcal{E}nd_{\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)}(\text{Poinc}_{G,!}^{\text{Vac}}).$$

The third observation implies that H^0 of the right-hand side is one-dimensional. Combined with (0.5), this implies that $n = 1$, as desired.

Remark 0.2.6. The outline above uses a special feature of the de Rham setting. By contrast, Betti or étale moduli stacks of local systems have infinite dimensional algebras of global functions, while the de Rham moduli stack has very few global functions. For this reason, our strategy does not adapt to either Betti or étale settings. In particular, the dimension count above carries no meaning in those settings, and our overall strategy for controlling multiplicities does not adapt.

With that said, we remind that it was shown in [GLC1, Theorem 3.5.2] that the de Rham version of GLC proved in this paper implies the Betti version.

0.2.7. *Deficiencies.* An awkward aspect of this paper is that the argument outlined above does not apply in low genus. Namely, the three features mentioned in Sect. 0.2.2 break down when $g \leq 1$ (and the first two also for $g = 2$ when G is of type A_1).

Thus, the proof of GLC we give is *not* uniform across all reductive groups and genera.

0.2.8. We treat the outlying cases as follows (still assuming that G is simple):

First, when $g = 0$, there is nothing to prove, as $\mathrm{LS}_G^{\mathrm{irred}}$ is empty in this case.

Second, we recall from [GLC4, Theorem 1.8.2] that multiplicity one problems can be settled for arbitrary genus and groups of type A_n using features of the geometry of opers, which is much easier to understand in this case.

Thus, it remains to treat the case of $g = 1$ and a simple group of type different than A_n . However, in this case a theorem of [KS, BFM] asserts that $\mathrm{LS}_G^{\mathrm{irred}}$ is again empty.

0.3. Complications stemming from a non-trivial center. This subsection can be skipped on the first pass. That said, here we will point to a fun part of this paper: the 2-categorical Fourier-Mukai transform.

0.3.1. In the outline in Sect. 0.2, we have assumed that G is of adjoint type. However, this case is *insufficient* in order to deduce GLC for any reductive group G .

In fact, we prove (see Corollary 2.3.10) that GLC reduces to *almost simple simply-connected* groups.

In this subsection we indicate how the outline in Sect. 0.2 needs to be modified in order to treat this case.

0.3.2. Assume that G is semi-simple. Then if $\check{G}$ is not simply-connected, $\mathrm{LS}_{\check{G}}^{\mathrm{irred}}$ is not connected, nor are its connected components simply-connected. In fact

$$\pi_0(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}) \simeq \pi_0(\mathrm{Bun}_{\check{G}}) \simeq (Z_G)^\vee.$$

We denote this bijection by

$$\alpha \in (Z_G)^\vee \rightsquigarrow \mathrm{LS}_{G,\alpha}^{\mathrm{irred}},$$

where $\mathrm{LS}_{G,\alpha}^{\mathrm{irred}}$ is a connected component of $\mathrm{LS}_G^{\mathrm{irred}}$.

Moreover, each connected component of $\mathrm{LS}_G^{\mathrm{irred}}$ has abelian fundamental group, and its characters are in bijection with Bun_{Z_G} . We denote this bijection by

$$(\mathcal{P}_{Z_G} \in \mathrm{Bun}_{Z_G}) \rightsquigarrow \mathcal{L}_{\mathcal{P}_{Z_G}},$$

where $\mathcal{L}_{\mathcal{P}_{Z_G}}$ is a 1-dimensional local system.

0.3.3. With this understood, it is not difficult to adapt the proof explained in Sect. 0.2, modulo the following issue:

We need to know that:

- The functor $\mathbb{L}_G$ sends the direct summand of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ on which the action of Z_G by automorphisms of the identity functor is given by α to the direct summand of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$ consisting of sheaves supported on $\mathrm{LS}_{G,\alpha}^{\mathrm{irred}}$.
- The functor $\mathbb{L}_G$ intertwines the automorphism of $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ given by translation by $\mathcal{P}_{Z_G}$ with the automorphism of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$ given by tensor product with $\mathcal{L}_{\mathcal{P}_{Z_G}}$.

0.3.4. The above two properties can be formulated purely on the geometric side, in terms of the spectral action of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$, see Theorems 5.1.5 and 5.1.7, respectively.

The proofs of these two theorems involve a fun manipulation (that appears to be new) with the *2-categorical Fourier-Mukai transform*.

0.3.5. The 2-categorical Fourier-Mukai transform in the case at hand is an equivalence of 2-categories between *sheaves of categories* over the 2-stacks $\mathrm{Ge}_{Z_G}(X)$ and $\mathrm{Ge}_{\pi_1(\check{G})}(X)$, which classify gerbes on X with respect to Z_G and $\pi_1(\check{G})$, respectively.

It turns out that one can upgrade $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ to a sheaf of categories over $\mathrm{Ge}_{Z_G}(X)$ and over $\mathrm{Ge}_{\pi_1(\check{G})}(X)$, and the assertion is that the resulting two sheaves of categories map to one another under the 2-categorical Fourier-Mukai transform. Per Remark 8.1.5, the prestacks $\mathrm{Ge}_{Z_G}(X)$ and $\mathrm{Ge}_{\pi_1(\check{G})}(X)$ are *not* 1-affine, so it is essential to work with sheaves of categories here.

The above assertion implies the two properties mentioned in Sect. 0.3.3. But in fact it carries much more information. For example, it contains the answer to the following question:

What the spectral side of GLC, when on the automorphic side instead of the usual Bun_G , we consider the stack of bundles with respect to a non-pure inner form of G (i.e., a twist by a G_{ad} -torsor, which does not come from a G -torsor)?

It turns out that the answer is the twist of (the usual) $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$ by a gerbe on $\mathrm{LS}_{\check{G}}$ that is attached to the G_{ad} -torsor in question.

0.4. What is not done in this paper?

0.4.1. First, there are a number of nearby problems that we do not consider and do not know how to solve. Here are some:

- Geometric Langlands with Iwahori ramification.
- Quantum geometric Langlands.
- Local geometric Langlands with wild ramification.
- Global geometric Langlands with wild ramification.
- Restricted geometric Langlands for ℓ -adic sheaves (for curves in positive characteristic).
- Geometric Langlands for Fargues-Fontaine curves.

0.4.2. Moreover, *even in the (global, untwisted, unramified, de Rham) setting of the present paper*, we feel there is more to understand. The tricks indicated in Sect. 0.2 analyze automorphic sheaves via their shadows (specifically, the core part of the argument uses a dimension count). In the remainder of this subsection, we indicate questions and projects that might let us observe them more directly.

0.4.3. As in Sect. 0.2.1, we need to be able to recover cuspidal D-modules – or equivalently: eigen-sheaves with irreducible spectral parameters – from their Whittaker coefficients. This can be stated more formally either as showing that the composition

$$(0.6) \quad \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)_{\mathrm{cusp}} \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{\mathrm{coeff}_G^{\mathrm{Vac, glob}}} \mathrm{Whit}(\mathrm{Gr}_{G, \mathrm{Ran}})$$

is fully faithful, or showing that the functor of vacuum Whittaker coefficient

$$\mathrm{Hecke}_{\sigma}(\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \xrightarrow{\mathrm{coeff}_G^{\mathrm{Vac, glob}}} \mathrm{Vect}$$

is an equivalence for any irreducible local system σ .

For $G = GL_n$ (or a quotient thereof), the *mirabolic trick* suffices to prove this result; see [Be1] for a strong version of this assertion. We remind that a similar assertion holds at the level of automorphic functions, although in the arithmetic setup, such assertions are *not* valid for more general reductive groups.

A posteriori, the same statement for general G follows from the results of the present paper. But one can imagine tackling this statement more directly.

Below we record some possible strategies that we know have been discussed previously in the geometric Langlands community. We allow ourselves more informality in this discussion than elsewhere in this text.

0.4.4. *Proof via contractibility of opers?* Recall that $\mathcal{A}_G \in \mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ was defined using sheaves on Bun_G , i.e., it crosses the barrier between G and $\check{G}$ – as should be no surprise.

However, one key point of [GLC4] is that there is an alternative construction of $\mathcal{A}_{G,\mathrm{irred}}$ purely in $\check{G}$ (i.e., spectral) terms: this sheaf is the relative homology of the space of rational opers (with irreducible underlying local system). Equivalently, for irreducible σ , the fiber $\mathcal{A}_{G,\sigma}$ can be identified with homology of the space of generic oper structures on σ .

Therefore, as recorded in [GLC4, Conjecture 4.5.7], it suffices to show that this space of generic oper structures is *contractible*. Note that one piece of this assertion – namely, that this space is non-empty – is a theorem of D. Arinkin, [Ari].

The contractibility assertion is easy for $G = GL_n$, and has recently been established in [BKS] for all classical groups. So this provides a uniform proof of GLC for classical groups, see [GLC4, Theorem 4.5.11].

The contractibility assertion for general G follows a posteriori from the results of this paper. But conceivably, one could find an a priori proof of this assertion for a general G that would yield a more satisfying conclusion to the geometric Langlands conjecture.

We remark that in [GLC4, Corollary 4.5.5], we showed that the space of generic oper structures on σ has vanishing homology in positive degrees, so it only remains to show that it is *connected*. In fact, as $\mathcal{A}_{G,\mathrm{irred}}$ is a vector bundle, it suffices to check this for a single irreducible local system σ (per connected component of $\mathrm{LS}_G^{\mathrm{irred}}$).

0.4.5. *Proof via microlocal sheaf theory?* To date, the most successful geometric approach to studying Whittaker coefficients of automorphic sheaves for general reductive groups is [FR], where it was shown that the functor (0.6) is *conservative*. One could try extending the techniques of [FR] to obtain a geometric proof of fully faithfulness. Here we briefly fantasize about one possible form this idea might take.

First, [FR] works primarily in the setting of sheaves *with nilpotent singular support* as in [AGKRRV1]. One key point of [FR] is the picture that the vacuum Whittaker coefficient is the microstalk at the basepoint of the global nilpotent cone Nilp , a picture that was then realized in a better and more precise form in [NT].

On the one hand, by [Wa], there is a category $\mu\mathrm{Perv}(T^*(\mathrm{Bun}_G))$ of *microsheaves supported on* $T^*(\mathrm{Bun}_G)$ and a subcategory $\mu\mathrm{Perv}_{\mathrm{Nilp}}(T^*(\mathrm{Bun}_G))$ of *microsheaves supported on the nilpotent cone*. Moreover, there is a fully faithful *microlocalization* functor $\mathrm{Perv}_{\mathrm{Nilp}}(\mathrm{Bun}_G) \rightarrow \mu\mathrm{Perv}_{\mathrm{Nilp}}(T^*(\mathrm{Bun}_G))$.

For $\overset{\circ}{\mathrm{Nilp}} \subset \mathrm{Nilp}$ the open of *generically regular* nilpotent Higgs fields, [FR, Theorem G] morally says that the composition

$$\mathrm{Perv}_{\mathrm{Nilp}}(\mathrm{Bun}_G)_{\mathrm{temp}} \rightarrow \mathrm{Perv}_{\mathrm{Nilp}}(\mathrm{Bun}_G) \rightarrow \mu\mathrm{Perv}_{\mathrm{Nilp}}(T^*(\mathrm{Bun}_G)) \rightarrow \mu\mathrm{Perv}_{\overset{\circ}{\mathrm{Nilp}}}(T^*(\mathrm{Bun}_G))$$

remains fully faithful.

On the other hand, one can dream of a strengthening of [NT] that expresses *all* Whittaker coefficients from the microlocalization to $\mu\mathrm{Perv}_{\overset{\circ}{\mathrm{Nilp}}}(T^*(\mathrm{Bun}_G))$, with this microlocal functor being fully faithful for some natural geometric reasons.

This would also yield another resolution to the geometric Langlands conjecture, and one that might teach us more than the present paper does.

0.4.6. *Proof via Verlinde gluing?* By [GLC1, Theorem 3.5.6], we can deduce the de Rham geometric Langlands conjecture from its Betti counterpart [BZN].

As in [BZN] Sect. 4.6, a better understanding of the automorphic side of Betti Langlands could allow one to glue $\mathrm{Shv}_{\frac{1}{2},\mathrm{Nilp}}^{\mathrm{Betti}}(\mathrm{Bun}_G)$ via degeneration to a nodal curve, using Bezrukavnikov-style equivalences as local input. As we understand, there is active and on-going work of D. Nadler and Z. Yun in this direction.

0.4.7. *Proof via arithmetic?* Suppose for simplicity that G is adjoint and the genus is at least 2, so that $\mathrm{LS}_G^{\mathrm{irred}}$ is connected.

Recall that $\mathcal{A}_G|_{\mathrm{LS}_G^{\mathrm{irred}}}$ is a vector bundle. As mentioned in Sect. 0.4.4, it follows that $\mathcal{A}_G|_{\mathrm{LS}_G^{\mathrm{irred}}}$ has one-dimensional fibers if it has a one-dimensional fiber at *any* point. This in turn should reduce by specialization to a statement in characteristic p . Using the intimate relationship between geometric and arithmetic Langlands established in [AGKRRV2], one should be able to reduce de Rham geometric Langlands to a suitable multiplicity one statement for essentially *any* class of unramified cusp forms (equipped with the action of V. Lafforgue's excursion operators).

With that said, we are not aware of any such arithmetic results for general reductive groups.

0.4.8. *Proof via... something else?* We do not intend to represent the above as an exhaustive summary of discussions that have occurred. More seriously, we anticipate future innovations whose form we do not yet know.

0.5. Contents.

0.5.1. Let us briefly review the actual contents of this paper.

In Sect. 1 we summarize the results of [GLC1, GLC2, GLC3, GLC4] that will be used in this paper.

In Sect. 2 we show that it is sufficient to prove GLC in the special case when the group G is almost simple and simply-connected.

In Sect. 3 we prove GLC when the genus g of our curve is either 0 or 1.

In Sect. 4 we express the fundamental groupoid of $\mathrm{LS}_G^{\mathrm{irred}}$ in terms of Z_G . (In particular, we show that if G is simple, then $\mathrm{LS}_G^{\mathrm{irred}}$ is connected and simply-connected, under the assumption that $g \geq 2$, and excluding the case $g = 2$ and $G = \mathrm{PGL}_2$.)

In Sect. 5 we prove GLC for curves of genus ≥ 2 , essentially elaborating on the outline in Sect. 0.2. Here we also state some further results whose proofs are given in later sections.

In Sect. 6 we calculate (in the case $g \geq 2$) endomorphisms of the vacuum Poincaré object, and show that they consist essentially only of scalars.

In Sect. 7 we prove the properties concerning algebraic geometry and topology of the stacks Bun_G and LS_G .

In Sect. 8 we introduce the 2-categorical Fourier-Mukai transform and use it to establish the compatibility between the action of Bun_{Z_G} on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ and the spectral action. This material is used to adapt the discussion of Sect. 0.2 to the case when G is not of adjoint type. As a byproduct, we obtain a version of geometric Langlands for non-pure inner twists of G .

0.5.2. *Conventions and notation.* This paper follows the conventions and notation of the [GLC1, GLC2, GLC3, GLC4] series.

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1. SUMMARY OF THE PRECEDING RESULTS

In this section we summarize the results from [GLC1, GLC2, GLC3, GLC4] that will be used in the present paper for the proof of GLC.

1.1. The Langlands functor and its categorical properties.

1.1.1. In [GLC1, Sect. 1], we have constructed a functor

$$\mathbb{L}_G : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}).$$

It satisfies the following properties:

- The functor $\mathbb{L}_G$ admits a left adjoint $\mathbb{L}_G^L$, see [GLC3, Theorem 16.1.2];
- The functor $\mathbb{L}_G$ is conservative, see [GLC4, Theorem 1.6.2].

1.1.2. The geometric Langlands conjecture (GLC) says:

Conjecture 1.1.3. *The functor $\mathbb{L}_G$ is an equivalence.*

1.1.4. By the above, Conjecture 1.1.3 is equivalent to the fact that the unit of the adjunction

$$(1.1) \quad \mathrm{Id}_{\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})} \rightarrow \mathbb{L}_G \circ \mathbb{L}_G^L$$

is an isomorphism.

1.2. Spectral properties.

1.2.1. Recall that the Hecke action gives rise to an action of the monoidal category $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$, see [GLC1, Sect. 1.2].

The functor $\mathbb{L}_G$ has the following features with respect to this action:

- The functor $\mathbb{L}_G$ is $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -linear, i.e., it intertwines the actions of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ on the two sides, see [GLC1, Sect. 1.7]. As $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ is rigid monoidal, this implies that the functor $\mathbb{L}_G^L$ is also $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -linear;
- The $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -linear monad $\mathbb{L}_G \circ \mathbb{L}_G^L$ on $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}})$ is given by a (uniquely defined) associative algebra object

$$\mathcal{A}_G \in \mathrm{QCoh}(\mathrm{LS}_{\check{G}}),$$

see [GLC3, Theorem 16.4.2].

1.2.2. The unit map

$$(1.2) \quad \mathcal{O}_{\mathrm{LS}_{\check{G}}} \rightarrow \mathcal{A}_G$$

for the associative algebra $\mathcal{A}_G$ in $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ gives rise to the map (1.1). Therefore, to see that (1.1) is an isomorphism of functors, it suffices to show that (1.2) is an isomorphism of quasi-coherent sheaves.

1.3. Restriction to the irreducible locus.

1.3.1. Let

$$\mathrm{LS}_{\tilde{G}}^{\mathrm{red}} \subset \mathrm{LS}_{\tilde{G}}$$

be the closed substack consisting of *reducible* local systems, i.e., the union of the images of the (proper) maps

$$\mathrm{LS}_{\tilde{P}} \rightarrow \mathrm{LS}_{\tilde{G}},$$

where $\tilde{P} \subsetneq \tilde{G}$ are proper parabolics of $\tilde{G}$.

Let

$$\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}} \xrightarrow{j} \mathrm{LS}_{\tilde{G}}$$

denote the embedding of the complement to $\mathrm{LS}_{\tilde{G}}^{\mathrm{red}}$, i.e., $\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}}$ is the open substack of *irreducible* local systems.

1.3.2. Let

$$(1.3) \quad \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})_{\mathrm{red}} \subset \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})$$

be the full subcategory consisting of objects set-theoretically supported on $\mathrm{LS}_{\tilde{G}}^{\mathrm{red}}$, i.e.,

$$\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}})_{\mathrm{red}} = \ker \left(j^* : \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}}) \right).$$

The above faithful embedding (1.3) admits a right adjoint, denoted $\hat{\tau}^!$. Explicitly,

$$\hat{\tau}^! \simeq \mathrm{Fib}(\mathrm{Id} \rightarrow j_* \circ j^*(-)).$$

1.3.3. The following is the main result of [GLC3] (it is equivalent to Theorem 17.1.2 in *loc. cit.*, see Sect. 17.3.3):

- The map $\hat{\tau}^!(\mathcal{O}_{\mathrm{LS}_{\tilde{G}}}) \rightarrow \hat{\tau}^!(\mathcal{A}_G)$ induced by (1.2) is an isomorphism.

1.3.4. Set

$$\mathcal{A}_{G,\mathrm{irred}} := j^*(\mathcal{A}_G).$$

Given the isomorphism in Sect. 1.3.3, we obtain that the fact that (1.2) is an isomorphism (and hence, the statement of GLC) is equivalent to the fact that the map

$$(1.4) \quad \mathcal{O}_{\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}}} \rightarrow \mathcal{A}_{G,\mathrm{irred}},$$

induced by (1.2) is an isomorphism.

1.4. Properties of $\mathcal{A}_{G,\mathrm{irred}}$.

1.4.1. In this subsection we will assume that G is semi-simple. In this case $\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}}$ is a classical smooth algebraic stack.

1.4.2. The following property of $\mathcal{A}_{G,\mathrm{irred}}$ is one of the two main results of the paper [GLC4], see Theorem 3.1.8 in *loc. cit.*:

- The object $\mathcal{A}_{G,\mathrm{irred}} \in \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}})$ is a vector bundle (in particular, it is concentrated in cohomological degree 0);

1.4.3. In addition, we have:

- The object $\mathcal{A}_{G,\mathrm{irred}}$ is canonically of the form $\mathbf{oblv}^l(\mathcal{F})$, where $\mathcal{F} \in \mathrm{D-mod}(\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}})$ and

$$\mathbf{oblv}^l : \mathrm{D-mod}(\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}}^{\mathrm{irred}})$$

denotes the “left” forgetful functor. This is [GLC4, Corollary 4.2.5].

- The above object $\mathcal{F}$ is a local system with a finite monodromy (in particular, it is concentrated in cohomological degree 0). This is [GLC4, Proposition 4.2.8].

2. REDUCTION TO THE CASE WHEN G IS ALMOST SIMPLE AND SIMPLY-CONNECTED

For a number of technical reasons,² in the main argument in the proof of GLC, we will need to assume that the group G is almost simple and simply-connected. In this section we will perform a reduction to this case.

2.1. Compatibility between Langlands functors.

2.1.1. Let G_1 and G_2 be a pair of reductive groups, and let

$$\phi : G_1 \rightarrow G_2$$

be an *almost isogeny*, i.e., a map that induces an isogeny of their derived groups.

By a slight abuse of notation we will denote by the same symbol ϕ the induced map

$$\mathrm{Bun}_{G_1} \rightarrow \mathrm{Bun}_{G_2}.$$

2.1.2. The map ϕ induces a map

$$\phi^\vee : \check{G}_2 \rightarrow \check{G}_1.$$

By a slight abuse of notation, we will denote by the same symbol $\phi^\vee$ the induced map

$$\mathrm{LS}_{\check{G}_2} \rightarrow \mathrm{LS}_{\check{G}_1}.$$

2.1.3. Note that the functor

$$(2.1) \quad \phi^! : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})$$

is linear with respect to the action of $\mathrm{Rep}(\check{G}_1)_{\mathrm{Ran}}$, where the action on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2})$ is via the functor

$$\mathrm{Rep}(\check{G}_1)_{\mathrm{Ran}} \rightarrow \mathrm{Rep}(\check{G}_2)_{\mathrm{Ran}},$$

given by restriction along $\phi^\vee$.

Hence, (2.1) is $\mathrm{QCoh}(\mathrm{LS}_{\check{G}_1})$ -linear, where the action on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2})$ is via

$$(\phi^\vee)^* : \mathrm{QCoh}(\mathrm{LS}_{\check{G}_1}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}_2}).$$

2.1.4. Note also that the functor

$$(\phi^\vee)_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}(\mathrm{LS}_{\check{G}_2}) \rightarrow \mathrm{IndCoh}(\mathrm{LS}_{\check{G}_1})$$

sends

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}_2}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}_1}).$$

This follows, e.g., from [AG, Proposition 7.1.3(b)].

2.1.5. We will prove:

Proposition 2.1.6. *The following diagram of functors commutes*

$$(2.2) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}) & \xrightarrow{\mathbb{L}_{G_1}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}_1}) \\ \phi^! \uparrow & & (\phi^\vee)_*^{\mathrm{IndCoh}} \uparrow \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}) & \xrightarrow{\mathbb{L}_{G_2}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}_2}). \end{array}$$

Moreover, this datum of commutativity is compatible with the action of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}_1})$.

2.2. Proof of Proposition 2.1.6.

²One reason is not very serious: if G has a center of positive dimension, the stack $\mathrm{LS}_{\check{G}}^{\mathrm{irred}}$ is not smooth, which is a silly annoyance. The more serious reason is that when $g = 2$ we will need to assume that G has no factors of A_1 in its Dynkin diagram, see the preamble to Sect. 5.5.

2.2.1. First, we can factor ϕ as

$$G_1 \xrightarrow{\phi'} G_1 \times Z_{G_2}^0 \xrightarrow{\phi''} G_2.$$

The proposition evidently holds for ϕ' . Hence, it is sufficient to prove it for ϕ'' . Thus, we can assume that our ϕ is surjective.

2.2.2. When ϕ is surjective, we factor it as

$$G_1 \xrightarrow{\phi'} G_1 / \ker(\phi)^0 \xrightarrow{\phi''} G_2,$$

so that the kernel of ϕ' is a connected torus, and ϕ'' is an actual isogeny.

The case of a surjection with kernel being a connected torus will be analyzed explicitly in Sect. 2.4.2.

Hence, it is enough to prove Proposition 2.1.6 for ϕ being an isogeny (this assumption will be used in one place in the course of the proof, namely, in Sect. 2.2.4).

2.2.3. First, we claim that it is sufficient show that the outer square in the following diagram commutes:

$$(2.3) \quad \begin{array}{ccccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}) & \xrightarrow{\mathbb{L}_{G_1}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}_1}) & \longrightarrow & \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}_1}) \\ \phi^! \uparrow & & (\phi^\vee)_* \uparrow & & (\phi^\vee)_* \uparrow \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}) & \xrightarrow{\mathbb{L}_{G_2}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}_2}) & \longrightarrow & \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}_2}), \end{array}$$

where the horizontal composite functors are

$$\mathbb{L}_{G_1, \mathrm{coarse}} \text{ and } \mathbb{L}_{G_2, \mathrm{coarse}},$$

respectively (see [GLC1, Sect. 1.4]).

Indeed, suppose that (2.3) commutes, and let us show that this uniquely upgrades to the commutation of (2.2).

2.2.4. It is enough to show that the two circuits in (2.2) are isomorphic when restricted to the subcategory of compact objects in $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2})$. Since the functor

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}_1}) \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}_1})$$

is fully faithful on

$$\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}_1})^{>-\infty} \subset \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}_1}),$$

it suffices to show that both circuits in (2.2), when restricted to

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2})^c \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2})$$

map to $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}_1})^{>-\infty}$.

By [GLC1, Theorem 1.6.2], the functors $\mathbb{L}_{G_i, \mathrm{coarse}}$ send

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_i})^c \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\tilde{G}_i})^{>-\infty},$$

$i = 1, 2$.

The statement about the counterclockwise circuit follows now because the functor $(\phi^\vee)_*$ has finite cohomological amplitude.

The statement for the clockwise circuit follows similarly: indeed since $\phi : G_1 \rightarrow G_2$ is an isogeny, the map

$$\phi : \mathrm{Bun}_{G_1} \rightarrow \mathrm{Bun}_{G_2}$$

is étale, and hence the functor

$$\phi^! : \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2})$$

preserves compactness.

2.2.5. The same argument implies the following:

Once we prove that the outer circuit in (2.3) commutes as a diagram of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}_1})$ -module categories, it would follow that its restriction to compact objects commutes as a diagram of $\mathrm{Perf}(\mathrm{LS}_{\check{G}_1})$ -module categories, so the resulting left commuting square in (2.3) is one of $\mathrm{Perf}(\mathrm{LS}_{\check{G}_1})$ -module categories, and hence of $\mathrm{Ind}(\mathrm{Perf}(\mathrm{LS}_{\check{G}_1})) \simeq \mathrm{QCoh}(\mathrm{LS}_{\check{G}_1})$ -module categories.

2.2.6. Let us now prove the commutativity of

$$(2.4) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}) & \xrightarrow{\mathbb{L}_{G_1, \text{coarse}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}_1}) \\ \phi^! \uparrow & & (\phi^\vee)_* \uparrow \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}) & \xrightarrow{\mathbb{L}_{G_2, \text{coarse}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}_2}). \end{array}$$

It will follow from the construction that the data of commutativity is compatible with the Hecke action of $\mathrm{Rep}(\check{G}_1)_{\mathrm{Ran}}$, and hence also of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}_1})$.

2.2.7. Since the functor

$$\Gamma_{\check{G}_1}^{\mathrm{spec}} : \mathrm{QCoh}(\mathrm{LS}_{\check{G}_1}) \rightarrow \mathrm{Rep}(\check{G}_1)_{\mathrm{Ran}}$$

is fully faithful, it suffices to show that the outer square in the concatenation of (2.4) with the commutative diagram

$$(2.5) \quad \begin{array}{ccc} \mathrm{QCoh}(\mathrm{LS}_{\check{G}_1}) & \longrightarrow & \mathrm{Rep}(\check{G}_1)_{\mathrm{Ran}} \\ (\phi^\vee)_* \uparrow & & \uparrow \mathrm{coInd}^{\phi^\vee} \\ \mathrm{QCoh}(\mathrm{LS}_{\check{G}_2}) & \longrightarrow & \mathrm{Rep}(\check{G}_2)_{\mathrm{Ran}} \end{array}$$

commutes, where

$$\mathrm{coInd}^{\phi^\vee} : \mathrm{Rep}(\check{G}_2) \rightarrow \mathrm{Rep}(\check{G}_1)$$

is the functor of co-induction, i.e., the right adjoint to the functor

$$\mathrm{Res}^{\phi^\vee} : \mathrm{Rep}(\check{G}_1) \rightarrow \mathrm{Rep}(\check{G}_2).$$

2.2.8. Note that pullback with respect to the map

$$\phi : \mathrm{Gr}_{G_1} \rightarrow \mathrm{Gr}_{G_2}$$

induces a functor

$$\phi^! : \mathrm{Whit}(G_2) \rightarrow \mathrm{Whit}(G_1).$$

It is easy to see that the following diagram commutes:

$$\begin{array}{ccc} \mathrm{Whit}(G_1) & \xrightarrow{\mathrm{CS}_{G_1}} & \mathrm{Rep}(\check{G}_1) \\ \phi^! \uparrow & & \uparrow \mathrm{coInd}^{\phi^\vee} \\ \mathrm{Whit}(G_2) & \xrightarrow{\mathrm{CS}_{G_2}} & \mathrm{Rep}(\check{G}_2). \end{array}$$

Combining this with the commutative diagrams [GLC2, Equation (18.4)]

$$\begin{array}{ccc} \mathrm{Whit}(G_i)_{\mathrm{Ran}} & \longrightarrow & \mathrm{Rep}(\check{G}_i)_{\mathrm{Ran}} \\ \mathrm{coeff}_{G_i}[2\delta_{N_{\rho(\omega_X)}}] \uparrow & & \uparrow \Gamma_{\check{G}_i}^{\mathrm{spec}} \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_i}) & \xrightarrow{\mathbb{L}_{G_i, \text{coarse}}} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}_i}) \end{array}$$

for $i = 1, 2$, we obtain that the required commutativity is equivalent to the commutativity of

$$(2.6) \quad \begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}) & \xrightarrow{\mathrm{coeff}_{G_1}} & \mathrm{Whit}(G_1)_{\mathrm{Ran}} \\ \phi^! \uparrow & & \uparrow \phi^! \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}) & \xrightarrow{\mathrm{coeff}_{G_2}} & \mathrm{Whit}(G_2)_{\mathrm{Ran}}. \end{array}$$

2.2.9. Passing to left adjoints, the commutativity of (2.6) is equivalent to the commutativity of

$$\begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}) & \xleftarrow{\mathrm{Poinc}_{G_1,!}} & \mathrm{Whit}(G_1)_{\mathrm{Ran}} \\ \phi_! \downarrow & & \downarrow \phi_! \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}) & \xleftarrow{\mathrm{Poinc}_{G_2,!}} & \mathrm{Whit}(G_2)_{\mathrm{Ran}}, \end{array}$$

while the latter is tautological. $\square$ [Proposition 2.1.6]

2.3. Changing the group.

2.3.1. Passing to left adjoint functors in (2.2), we obtain a commutative diagram

$$\begin{array}{ccc} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}) & \xleftarrow{\mathbb{L}_{G_1}^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}_1}) \\ \phi_! \downarrow & & \downarrow (\phi^\vee)^* \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}) & \xleftarrow{\mathbb{L}_{G_2}^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}_2}), \end{array}$$

compatible with the action of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}_1})$. Tensoring up, we obtain a commutative diagram

$$(2.7) \quad \begin{array}{ccc} \mathrm{QCoh}(\mathrm{LS}_{\check{G}_2}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}_1})} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}) & \xleftarrow{\mathbb{L}_{G_1}^L} & \mathrm{QCoh}(\mathrm{LS}_{\check{G}_2}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}_1})} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}_1}) \\ \downarrow & & \downarrow \\ \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}) & \xleftarrow{\mathbb{L}_{G_2}^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}_2}). \end{array}$$

2.3.2. We claim:

Proposition 2.3.3. *The functor*

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}_2}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}_1})} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}_1}) \rightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}_2})$$

is an equivalence.

Indeed, this is particular case (extended to stacks in a straightforward way) of [AG, Corollary 7.6.2]. In Sect. 2.4, we will prove:

Theorem 2.3.4. *The functor*

$$(2.8) \quad \mathrm{QCoh}(\mathrm{LS}_{\check{G}_2}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}_1})} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2})$$

is an equivalence.

2.3.5. Combining Proposition 2.3.3 and Theorem 2.3.4, we obtain:

Corollary 2.3.6. *Suppose that the functor $\mathbb{L}_{G_1}$ is an equivalence. Then so is $\mathbb{L}_{G_2}$.*

2.3.7. For a given reductive group G , take $G_2 = G$, and let $G_1 := G_{\text{sc}}$ be the simply-connected cover of its derived group. Let ϕ be the canonical map

$$G_{\text{sc}} \rightarrow G.$$

As a particular case of Corollary 2.3.6, we obtain:

Corollary 2.3.8. *If GLC holds for G_{sc} , then it also holds for G .*

2.3.9. Let $G = G_1 \times G_2$. We have

$$\text{Bun}_G \simeq \text{Bun}_{G_1} \times \text{Bun}_{G_2},$$

and hence

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \simeq \text{D-mod}_{\frac{1}{2}}(\text{Bun}_{G_1}) \otimes \text{D-mod}_{\frac{1}{2}}(\text{Bun}_{G_2}).$$

We also have:

$$\check{G} \simeq \check{G}_1 \times \check{G}_2,$$

and hence

$$\text{LS}_{\check{G}} \simeq \text{LS}_{\check{G}_1} \times \text{LS}_{\check{G}_2},$$

so that

$$\text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}}) \simeq \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}_1}) \otimes \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}_2}).$$

It is clear that under the above identifications,

$$\mathbb{L}_G \simeq \mathbb{L}_{G_1} \otimes \mathbb{L}_{G_2}.$$

Combining with Corollary 2.3.8, we obtain:

Corollary 2.3.10. *If GLC holds for all G that are almost simple and simply-connected, then it holds for any reductive G .*

2.4. Proof of Theorem 2.3.4.

2.4.1. We will distinguish two special types of homomorphisms ϕ :

Type A: ϕ is injective;

Type B: ϕ is surjective with a connected kernel.

Note that any ϕ can be factored as a composition

$$G_1 \rightarrow G'_1 \rightarrow G'_2 \rightarrow G_2,$$

where:

- The homomorphisms $G_1 \rightarrow G'_1$ and $G'_2 \rightarrow G_2$ are of type B;
- $G'_1 \rightarrow G'_2$ is of type A.

So, it is enough to prove Theorem 2.3.4 for ϕ of each of the above two types separately.

2.4.2. *Proof for type B.* Set

$$T := \ker(\phi).$$

(In this subsection T is just a torus, i.e., it is *not* the Cartan subgroup of either G_1 or G_2 .)

We have an action of Bun_T on Bun_{G_1} , and

$$(2.9) \quad \text{Bun}_{G_2} \simeq \text{Bun}_{G_1} / \text{Bun}_T.$$

We also have a projection

$$\text{LS}_{\check{G}_1} \rightarrow \text{LS}_{\check{T}},$$

and

$$(2.10) \quad \text{LS}_{\check{G}_2} \simeq \text{pt} \times_{\text{LS}_{\check{T}}} \text{LS}_{\check{G}_1}$$

where $\text{pt} \rightarrow \text{LS}_{\check{T}}$ is the unit point.

From (2.9) we obtain that the naturally defined functor

$$(2.11) \quad \mathrm{Vect}_{\mathrm{D-mod}(\mathrm{Bun}_T)} \otimes \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2})$$

is an equivalence, where:

- $\mathrm{D-mod}(\mathrm{Bun}_T)$ acts on $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})$ by $!$ -convolution;
- The functor $\mathrm{D-mod}(\mathrm{Bun}_T) \rightarrow \mathrm{Vect}$ is cohomology with compact supports.

From (2.10) we obtain:

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}_2}) \simeq \mathrm{Vect}_{\mathrm{QCoh}(\mathrm{LS}_{\check{T}})} \otimes \mathrm{QCoh}(\mathrm{LS}_{\check{G}_1}),$$

and hence

$$\mathrm{QCoh}(\mathrm{LS}_{\check{G}_2})_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}_1})} \otimes \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})$$

can be rewritten as

$$\mathrm{Vect}_{\mathrm{QCoh}(\mathrm{LS}_{\check{T}})} \otimes \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}).$$

However, by Fourier-Mukai (i.e., GLC for tori)

$$\mathrm{QCoh}(\mathrm{LS}_{\check{T}}) \stackrel{\mathrm{FM}}{\simeq} \mathrm{D-mod}(\mathrm{Bun}_T).$$

Hence, we obtain

$$\begin{aligned} \mathrm{QCoh}(\mathrm{LS}_{\check{G}_2})_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}_1})} \otimes \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}) &\simeq \mathrm{Vect}_{\mathrm{QCoh}(\mathrm{LS}_{\check{T}})} \otimes \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}) \stackrel{\mathrm{FM}}{\simeq} \\ &\simeq \mathrm{Vect}_{\mathrm{D-mod}(\mathrm{Bun}_T)} \otimes \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}) \stackrel{(2.11)}{\simeq} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}). \end{aligned}$$

This is the desired equivalence (2.8).

2.4.3. Proof for type A. First, replacing G_1 by its derived group, we obtain that it is enough to consider the case when G_1 is semi-simple, which we will from now on assume.

Denote by T the cokernel of ϕ , and denote by ψ the projection

$$\mathrm{Bun}_{G_2} \rightarrow \mathrm{Bun}_T.$$

Consider $\mathrm{D-mod}(\mathrm{Bun}_T)$ as a (symmetric) monoidal category with respect to the pointwise $\overset{!}{\otimes}$ tensor product, and let it act on $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2})$ via $\psi^!(-) \overset{!}{\otimes} (-)$.

Denote

$$(2.12) \quad \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})' := \mathrm{Vect}_{\mathrm{D-mod}(\mathrm{Bun}_T)} \otimes \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}),$$

where $\mathrm{D-mod}(\mathrm{Bun}_T) \rightarrow \mathrm{Vect}$ is the functor of $!$ -fiber at the unit point.

The functor $\phi^!$ naturally factors as

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}) \rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})' \xrightarrow{(\phi^!)'} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}),$$

and it is easy to see that the functor $(\phi^!)'$ is fully faithful. In fact, its essential image is a *direct summand* in $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})$, described as follows:

Note that the group Z_{G_1} (which is finite, due to the assumption that G_1 is semi-simple) acts by automorphisms of the identity functor of $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})$. Hence, the category $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})$ splits as a direct sum according to characters of Z_{G_1} :

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}) = \bigoplus_{\alpha \in (Z_{G_1})^\vee} \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})_\alpha.$$

Now,

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})' = \bigoplus_{\alpha} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})_{\alpha},$$

where α runs over the subset consisting of those characters that vanish on the subgroup

$$\ker(Z_{G_1} \rightarrow Z_{G_2}/Z_{G_2}^0).$$

The functor $\phi_!$ factors as

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}) \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})' \xrightarrow{(\phi_!)'} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}),$$

where the first arrow is the corresponding orthogonal projection.

2.4.4. Consider the map

$$\phi^{\vee} : \mathrm{LS}_{\tilde{G}_2} \rightarrow \mathrm{LS}_{\tilde{G}_1}.$$

Let

$$(2.13) \quad \mathrm{LS}'_{\tilde{G}_1} \subset \mathrm{LS}_{\tilde{G}_1}$$

be the union of connected components that lie in the essential image of $\phi^{\vee}$. The following assertion is a particular case of Theorem 5.1.5 below:

Proposition 2.4.5. *The full subcategory*

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})' \subset \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})$$

equals

$$\mathrm{QCoh}(\mathrm{LS}'_{\tilde{G}_1}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}_1})} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1}).$$

Let us assume this proposition and proceed with the proof of Case A of Theorem 2.3.4. We obtain that the functor (2.8) is an equivalence if and only if the functor

$$(2.14) \quad \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}_2}) \otimes_{\mathrm{QCoh}(\mathrm{LS}'_{\tilde{G}_1})} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})' \rightarrow \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}),$$

induced by $(\phi_!)'$, is an equivalence.

2.4.6. Consider $\mathrm{QCoh}(\mathrm{LS}_{\tilde{T}})$ as a monoidal category with respect to *convolution*. As such, it acts on $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}_2})$ via the map

$$\psi^{\vee} : \mathrm{LS}_{\tilde{T}} \rightarrow \mathrm{LS}_{\tilde{G}_2}.$$

Note that using the Fourier-Mukai equivalence (i.e., GLC for tori)

$$\mathrm{QCoh}(\mathrm{LS}_{\tilde{T}}) \xrightarrow{\mathrm{FM}} \mathrm{D}\text{-mod}(\mathrm{Bun}_T),$$

we can rewrite $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_1})'$ as

$$\mathrm{Vect} \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{T}})} \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2}),$$

where the functor $\mathrm{QCoh}(\mathrm{LS}_{\tilde{T}}) \rightarrow \mathrm{Vect}$ is $\Gamma(\mathrm{LS}_{\tilde{T}}, -)$.

2.4.7. Note the action of $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}_2})$ on $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2})$ is compatible with the actions of $\mathrm{QCoh}(\mathrm{LS}_{\tilde{T}})$ on both. Thus, (2.14) can be rewritten as the special case (for $\mathbf{C} := \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_2})$) of the functor

$$(2.15) \quad \mathrm{QCoh}(\mathrm{LS}_{\tilde{G}_2}) \otimes_{\mathrm{QCoh}(\mathrm{LS}'_{\tilde{G}_1})} \left(\mathrm{Vect} \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\tilde{T}})} \mathbf{C} \right) \rightarrow \mathbf{C},$$

defined for a DG category $\mathbf{C}$, equipped with an action of $\mathrm{QCoh}(\mathrm{LS}_{\tilde{G}_2})$ and a compatible action of $\mathrm{QCoh}(\mathrm{LS}_{\tilde{T}})$.

2.4.8. We claim that (2.15) is an equivalence for any such $\mathbf{C}$. Indeed, let $\mathcal{Y}$ be an algebraic stack with an affine diagonal, and let

$$\tilde{\mathcal{Y}} \rightarrow \mathcal{Y}$$

be a torsor with respect to a group-stack $\mathcal{T}$, also with an affine diagonal.

Assume that both $\mathcal{Y}$ and $\mathcal{T}$ are quasi-compact, locally almost of finite type and eventually coconnective (so that [Ga3, Theorem 2.2.6] is applicable).

Then the 2-category of DG categories tensored over $\mathrm{QCoh}(\tilde{\mathcal{Y}})$ and equipped with a compatible action of $\mathrm{QCoh}(\mathcal{T})$ is equivalent to the 2-category of DG categories tensored over $\mathrm{QCoh}(\mathcal{Y})$, with the mutually inverse equivalences being

$$\mathbf{C} \mapsto \mathrm{Vect}_{\mathrm{QCoh}(\mathcal{T})} \otimes_{\mathrm{QCoh}(\mathcal{T})} \mathbf{C}$$

and

$$\mathbf{D} \mapsto \mathrm{QCoh}(\tilde{\mathcal{Y}}) \otimes_{\mathrm{QCoh}(\mathcal{Y})} \mathbf{D}.$$

□[Theorem 2.3.4]

3. LOW GENUS CASES

The device that we use to prove the GLC breaks down when X has genus 0 or 1. In this section, we treat these cases separately. We highlight the key role played by the main results of [GLC3] and [GLC4] in this material.

3.1. What do we need to prove? According to Sect. 1.3.4, in order to prove GLC, we need to prove that the map (1.4) is an isomorphism.

We will show that this is automatic when X has low genus.

3.2. The case of $g = 0$. Note that for a curve of genus 0, we have $\mathrm{LS}^{\mathrm{irred}}(X) = \emptyset$, so that (1.4) holds trivially.

3.3. The case $g = 1$.

3.3.1. According to Corollary 2.3.10, we can assume that G is almost simple and simply-connected. We will separate two cases:

- (a) $G = SL_n$;
- (b) $G \neq SL_n$.

3.3.2. In case (a), the dual group $\tilde{G}$ is isomorphic to PGL_n . In this case, [GLC4, Conjecture 4.5.7] is known (in fact, it is a trivial particular case of [BKS]).³

This implies GLC by [GLC4, Corollary 4.5.5].

3.3.3. Note that this proof covers the case of $G = SL_n$ for *any* genus.

3.3.4. We now consider case (b).

Proposition 3.3.5 ([KS], [BFM]). *Let $\mathbf{G}$ be an adjoint group different from PGL_n . Then for a curve of genus 1, we have $\mathrm{LS}_{\mathbf{G}}^{\mathrm{irred}} = \emptyset$.*

From the proposition, we obtain that (1.4) holds trivially in this case.

³Fix an irreducible PGL_n local system σ , and choose its generic lift to an SL_n -local system; denote the underlying vector bundle by $\mathcal{E}$. Then the space of generic oper structures on σ is isomorphic to the space of generically defined line subbundles in $\mathcal{E}$, and this space is known to be homologically contractible by [Ga5].

3.3.6. *Proof of Proposition 3.3.5.* Appealing to Riemann-Hilbert, it is enough to show that a Riemann surface X of genus 1 does not admit irreducible (Betti) $\mathbf{G}_{\mathbb{C}}$ -local systems.

A $\mathbf{G}_{\mathbb{C}}$ -local system σ on X is given by a pair of commuting elements

$$g_1, g_2 \in \mathbf{G}_{\mathbb{C}}.$$

Consider the subgroup

$$Z_{\mathbf{G}}(g_1, g_2) \simeq \text{Aut}_{\sigma}.$$

A standard argument shows that if σ is irreducible then (in any genus) the Lie algebra of Aut_{σ} is zero. Hence, $Z_{\mathbf{G}}(g_1, g_2)$ is finite.

Since g_1 and g_2 commute, the subgroup that they generate is contained in $Z_{\mathbf{G}}(g_1, g_2)$, and hence is itself finite. Hence, the pair (g_1, g_2) is contained in a compact form K of $\mathbf{G}_{\mathbb{C}}$.

However, now [BFM, Proposition 4.1.1]⁴ implies that $K \simeq PSU_n$ for some n ; hence $\mathbf{G} \simeq PGL_n$. $\square$ [Proposition 3.3.5]

4. CALCULATION OF THE FUNDAMENTAL GROUP

In this section we let G be a semi-simple group.

One of the key parts of the argument in the proof of GLC is that⁵ the fundamental group of the stack $\text{LS}_{\check{G}}^{\text{irred}}$ is small (outside a few exceptional cases).

For example, if G is adjoint (in which case $\check{G}$ is simply-connected), the stack $\text{LS}_{\check{G}}^{\text{irred}}$ is also simply-connected. The reader may choose to focus on this case on the first pass.

For a general G , we will show that the fundamental group of $\text{LS}_{\check{G}}^{\text{irred}}$ is controlled by the finite group Z_G .

We remark that the arguments in this section are of de Rham nature. It would be nice to also have a direct topological proof of Theorem 4.3.2 in its Betti incarnation.

4.1. The fundamental groupoid of $\text{Bun}_{\check{G}}$.

4.1.1. Let $\mathcal{S}$ be a connective spectrum. We can regard it as a constant prestack, and we let $\mathcal{S}_{\text{et}}$ be its étale sheafification.

For example, if $\mathcal{S} = B(\Gamma)$, where Γ is a finite abelian group, then $B(\Gamma)_{\text{et}}$ is the étale stack pt/Γ .

4.1.2. Consider the tautological map

$$\check{G} \rightarrow \text{pt}/\pi_1(\check{G}),$$

where $\pi_1(\check{G})$ denotes the étale fundamental group of $\check{G}$.

The above map induces a map

$$\text{pt}/\check{G} \rightarrow B^2(\pi_1(\check{G}))_{\text{et}},$$

and hence to a map

$$(4.1) \quad \text{Bun}_{\check{G}} = \mathbf{Maps}(X, \text{pt}/\check{G}) \rightarrow \mathbf{Maps}(X, B^2(\pi_1(\check{G}))_{\text{et}}) =: \text{Ge}_{\pi_1(\check{G})}(X),$$

where $\mathbf{Maps}(-, -)$ denoted the prestack of maps.

Remark 4.1.3. The map (4.1) means that to a $\check{G}$ -bundle we can canonically associate an étale $\pi_1(\check{G})$ -gerbe. Namely, this is the gerbe of étale-local lifts of our bundle to the simply-connected cover of $\check{G}$.

⁴A related result is established also in [KS].

⁵Under the assumption that $g \geq 2$ (and if $g = 2$, the Dynkin diagram of G has no A_1 factors).

4.1.4. Note that we can think of $\mathrm{Ge}_{\pi_1(\check{G})}(X)$ also as

$$B^2(C(X, \pi_1(\check{G})))_{\mathrm{et}},$$

where $C(X, \pi_1(\check{G}))$ is the spectrum of étale cochains on X with coefficients in $\pi_1(\check{G})$.

Accordingly, the (2)-stack $\mathrm{Ge}_{\pi_1(\check{G})}(X)$ splits into connected components indexed by $H^2(X, \pi_1(\check{G}))$. The neutral connected component is canonically isomorphic to

$$B(\mathrm{Bun}_{\pi_1(\check{G})})_{\mathrm{et}}.$$

4.1.5. We will prove:

Proposition 4.1.6. *The map (4.1) defines an isomorphism of $\tau_{\leq 1}$ truncations of étale homotopy types.*

The concrete meaning of this proposition is that the map (4.1) defines a bijection on the sets of connected components, and on each connected component an isomorphism of étale fundamental groups.

The proof will be given in Sect. 7.1.

4.2. **Line bundles on $\mathrm{Bun}_{\check{G}}$.** We will now use the map (4.1) to construct line bundles on $\mathrm{Bun}_{\check{G}}$ starting from Z_G -torsors. This will be part of a more general construction, which will be extensively used in Sect. 8.

4.2.1. Note that we have a canonical identification

$$(4.2) \quad \pi_1(\check{G}) \simeq (Z_G)^\vee(1),$$

where $(-)^\vee$ denotes Cartier duality and (1) denotes the Tate twist.

4.2.2. Combining (4.2) with Verdier duality

$$C(X, Z_G)^\vee \simeq B^2(C(X, (Z_G)^\vee(1))),$$

we obtain an identification

$$(4.3) \quad C(X, Z_G)^\vee \simeq B^2(C(X, \pi_1(\check{G}))),$$

and in particular a bilinear pairing

$$(4.4) \quad B^2(C(X, Z_G)) \times B^2(C(X, \pi_1(\check{G}))) \rightarrow B^2(\mu_\infty).$$

4.2.3. After étale sheafification, from (4.4) we obtain a bilinear pairing

$$(4.5) \quad \mathrm{Ge}_{Z_G}(X) \times \mathrm{Ge}_{\pi_1(\check{G})}(X) \rightarrow \mathrm{Ge}_{\mu_\infty}(\mathrm{pt}),$$

where

$$\mathrm{Ge}_{\mu_\infty}(\mathrm{pt}) := B^2(\mu_\infty)_{\mathrm{et}}.$$

4.2.4. Looping (4.5) along the first factor, we obtain a pairing⁶

$$(4.6) \quad \mathrm{Bun}_{Z_G} \times \mathrm{Ge}_{\pi_1(\check{G})}(X) \rightarrow B(\mu_\infty)_{\mathrm{et}} \rightarrow \mathrm{pt} / \mathbb{G}_m.$$

In particular, we obtain that a point

$$\mathcal{P}_{Z_G} \in \mathrm{Bun}_{Z_G}$$

gives rise to a canonically defined μ_∞ -torsor, to be denoted

$$\mathcal{L}_{\mathcal{P}_{Z_G}},$$

on $\mathrm{Ge}_{\pi_1(\check{G})}(X)$.

The fact that (4.4) is a perfect pairing implies that every μ_∞ -torsor on every connected component of $\mathrm{Ge}_{\pi_1(\check{G})}(X)$ is the restriction of $\mathcal{L}_{\mathcal{P}_{Z_G}}$ for some $\mathcal{P}_{Z_G}$.

⁶We will return to the untruncated pairing (4.5) in Sect. 8, where it will play a fundamental role.

4.2.5. We will denote by the same symbol $\mathcal{L}_{\mathcal{P}_{Z_G}}$ the pullback of the above μ_∞ -torsor along the map (4.1). By a slight abuse of notation, we will continue to use the same symbol $\mathcal{L}_{\mathcal{P}_{Z_G}}$ the corresponding étale local system of k -vector spaces on $\text{Bun}_{\tilde{G}}$.

By Proposition 4.1.6, every étale local system of k -vector spaces on a given component of $\text{Bun}_{\tilde{G}}$ splits as a direct sum of 1-dimensional ones, and each of the latter is isomorphic to the restriction of $\mathcal{L}_{\mathcal{P}_{Z_G}}$ for some $\mathcal{P}_{Z_G}$.

4.2.6. Since $\mathcal{L}_{\mathcal{P}_{Z_G}}$, viewed as an étale local system of k -vector spaces comes from a μ_∞ -torsor, we can canonically associate to it a de Rham local system, which we will still denote by the same character $\mathcal{L}_{\mathcal{P}_{Z_G}}$.

We will also use the same symbol $\mathcal{L}_{\mathcal{P}_{Z_G}}$ to denote the corresponding line bundle (viewed either as the line bundle underlying the corresponding de Rham local system, or equivalently, as a $\mathcal{O}^\times$ -torsor induced by the map $\mu_\infty \rightarrow \mathcal{O}^\times$).

By a further abuse of notation, we will use the same symbol $\mathcal{L}_{\mathcal{P}_{Z_G}}$ to denote its pullback (in all three incarnations: étale, de Rham, coherent) along the map

$$\text{LS}_{\tilde{G}} \rightarrow \text{Bun}_{\tilde{G}}.$$

4.3. The fundamental group of $\text{LS}_{\tilde{G}}^{\text{irred}}$.

4.3.1. Consider the map

$$(4.7) \quad \text{LS}_{\tilde{G}}^{\text{irred}} \rightarrow \text{LS}_{\tilde{G}} \rightarrow \text{Bun}_{\tilde{G}}.$$

The main result of this subsection is the following assertion:

Theorem 4.3.2. *Assume that $g \geq 2$, and if $g = 2$, then its root system does not have A_1 factors. Then the map (4.7) induces an isomorphism on the $\tau_{\leq 1}$ truncations of étale homotopy types.*

Before we proceed to the proof, we record the following corollary, obtained by combining Theorem 4.3.2 and Proposition 4.1.6:

Corollary 4.3.3. *For every connected component of $\text{LS}_{\tilde{G}}^{\text{irred}}$, every étale local system of k -vector spaces on it splits as a direct sum of 1-dimensional ones. Each of the latter is isomorphic to the restriction of $\mathcal{L}_{\mathcal{P}_{Z_G}}$ for some $\mathcal{P}_{Z_G} \in \text{Bun}_{Z_G}$.*

The rest of this subsection is devoted to the proof of Theorem 4.3.2.

4.3.4. Let

$$\text{Bun}_{\tilde{G}}^{\text{stbl}} \subset \text{Bun}_{\tilde{G}}$$

be the stable locus. We will prove:

Proposition 4.3.5. *Under the assumptions on G and g specified in Theorem 4.3.2, the complement of $\text{Bun}_{\tilde{G}}^{\text{stbl}}$ in $\text{Bun}_{\tilde{G}}$ has codimension ≥ 2 .*

Remark 4.3.6. Statements of this type are classical in the literature of Bun_G ; they begin with [NR, Sect. 9]. The literature we found concerned coarse moduli spaces instead of moduli stacks, so we include the argument for Proposition 4.3.5 in Sect. 7.2. There are no significant differences between our argument and those in the existing literature.

4.3.7. Combining Propositions 4.3.5 and 4.1.6, we obtain:

Corollary 4.3.8. *Under the assumptions on G and g specified in Theorem 4.3.2, for every connected component of $\text{Bun}_{\tilde{G}}^{\text{stbl}}$, every étale local system of k -vector spaces on it splits as a direct sum of 1-dimensional ones. Each of the latter is isomorphic to the restriction of $\mathcal{L}_{\mathcal{P}_{Z_G}}$ for some $\mathcal{P}_{Z_G} \in \text{Bun}_{Z_G}$.*

4.3.9. Denote

$$\mathrm{LS}_{\check{G}}^{\mathrm{stbl}} := \mathrm{LS}_{\check{G}} \times_{\mathrm{Bun}_{\check{G}}} \mathrm{Bun}_{\check{G}}^{\mathrm{stbl}}.$$

The following is well-known:

Proposition 4.3.10. *The map*

$$\mathrm{LS}_{\check{G}}^{\mathrm{stbl}} \rightarrow \mathrm{Bun}_{\check{G}}^{\mathrm{stbl}}$$

is smooth and surjective. The fibers of this map are affine spaces.

For the sake of completeness, we will supply a proof in Sect. 7.4. Combining Proposition 4.3.10 and Corollary 4.3.8, we obtain:

Corollary 4.3.11. *Under the assumptions on G and g specified in Theorem 4.3.2, for every connected component of $\mathrm{LS}_{\check{G}}^{\mathrm{stbl}}$, every étale local system of k -vector spaces on it splits as a direct sum of 1-dimensional ones. Each of the latter is isomorphic to the restriction of $\mathcal{L}_{\mathcal{P}_{Z_G}}$ for some $\mathcal{P}_{Z_G} \in \mathrm{Bun}_{Z_G}$.*

4.3.12. *Proof of Theorem 4.3.2.* First, we claim that there is an inclusion

$$\mathrm{LS}_{\check{G}}^{\mathrm{stbl}} \subset \mathrm{LS}_{\check{G}}^{\mathrm{irred}}$$

with both spaces being smooth Deligne-Mumford stacks, and both are open inside $\mathrm{LS}_{\check{G}}$.

It is enough to establish the inclusion at the level of k -points. Suppose $\sigma \in \mathrm{LS}_{\check{G}}^{\mathrm{stbl}}$; we need to show it does not admit a reduction $\sigma_{\check{P}}$ to any proper parabolic $\check{P} \subset \check{G}$. Suppose we had such a reduction. Then, by the definition of stability (see Appendix A), we would have $\langle 2\check{\rho}_{\check{P}}, \deg(\sigma_{\check{P}}) \rangle < 0$. However, the above integer is the degree of the line bundle induced from $\sigma_{\check{P}}$ using $2\check{\rho}_{\check{P}}$, viewed as a homomorphism $\check{P} \rightarrow \check{M} \rightarrow \mathbb{G}_m$. Since the line bundle is endowed with a connection, its degree must be zero, which is a contradiction.

Thus, given Corollary 4.3.11, in order to prove Theorem 4.3.2, it suffices to show that $\mathrm{LS}_{\check{G}}^{\mathrm{stbl}}$ is dense in $\mathrm{LS}_{\check{G}}^{\mathrm{irred}}$. This is equivalent to saying that $\mathrm{LS}_{\check{G}}^{\mathrm{stbl}}$ has a non-empty intersection with every irreducible component of $\mathrm{LS}_{\check{G}}^{\mathrm{irred}}$. Since $\mathrm{LS}_{\check{G}}^{\mathrm{irred}}$ is smooth, its irreducible components are the same as connected components.

According to Corollary 5.3.7 below, the embedding

$$\mathrm{LS}_{\check{G}}^{\mathrm{irred}} \hookrightarrow \mathrm{LS}_{\check{G}}$$

induces a bijection on the sets of connected components.

Hence, we obtain that it is sufficient to show that the embedding

$$\mathrm{LS}_{\check{G}}^{\mathrm{stbl}} \hookrightarrow \mathrm{LS}_{\check{G}}$$

induces a bijection on the sets of connected components.

We have a commutative square

$$\begin{array}{ccc} \mathrm{LS}_{\check{G}}^{\mathrm{stbl}} & \longrightarrow & \mathrm{LS}_{\check{G}} \\ \downarrow & & \downarrow \\ \mathrm{Bun}_{\check{G}}^{\mathrm{stbl}} & \longrightarrow & \mathrm{Bun}_{\check{G}}. \end{array}$$

In it, the lower horizontal arrow and the left vertical arrow induce bijections on the sets of connected components, by Propositions 4.3.5 and 4.3.10, respectively.

Hence, the desired assertion follows from the fact that the map

$$\mathrm{LS}_{\check{G}} \rightarrow \mathrm{Bun}_{\check{G}}$$

induces a bijection on π_0 (see [BD, Proposition 2.11.4]).

□[Theorem 4.3.2]

5. THE CORE OF THE PROOF

Throughout this section, we will assume that G is semi-simple.

After some preparations, in this section we will give a proof of GLC for curves of genus ≥ 2 .

We note that the proof is particularly simple when G is adjoint, so that $\check{G}$ is simply-connected (in this case, one needs neither Sect. 5.1 nor Theorem 5.2.7). The reader may choose to focus on this case on the first pass.

5.1. Action of the center.

5.1.1. Note that the (abelian) group-stack Bun_{Z_G} acts on Bun_G , and this action lifts to an action of Bun_{Z_G} on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$.

For $\mathcal{P}_{Z_G} \in \text{Bun}_{Z_G}$, we will denote the corresponding automorphism of $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ by

$$(5.1) \quad \mathcal{P}_{Z_G} \cdot -.$$

5.1.2. Note that the group Z_G acts on the unit point $\mathcal{P}_{Z_G}^0 \in \text{Bun}_{Z_G}$. Since

$$\mathcal{P}_{Z_G}^0 \cdot (-)$$

is the identity functor, we obtain that Z_G acts by automorphisms of the identity endofunctor of $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$.

Let

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \simeq \bigoplus_{\alpha} \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\alpha}, \quad \alpha \in (Z_G)^{\vee}$$

denote the corresponding decomposition. Denote by $\mathbf{P}_{\alpha}$ the idempotent on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$, corresponding to $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\alpha}$.

5.1.3. Let $\pi_{1,\text{alg}}(\check{G})$ denote the *algebraic fundamental group* of $\check{G}$, i.e., the quotient of the coweight lattice by the coroot lattice. We have

$$\pi_{1,\text{alg}}(\check{G}) \simeq \pi_0(\text{Bun}_{\check{G}}) \simeq \pi_0(\text{LS}_{\check{G}}).$$

Note that we have

$$\pi_{1,\text{alg}}(\check{G}) \simeq \pi_1(\check{G})(-1),$$

so that we have a canonical identification

$$(Z_G)^{\vee} \simeq \pi_{1,\text{alg}}(\check{G}).$$

5.1.4. For a given $\alpha \in (Z_G)^{\vee}$, let $\text{LS}_{\check{G},\alpha}$ denote the corresponding connected component of $\text{LS}_{\check{G}}$. Consider the corresponding idempotent

$$\mathcal{O}_{\text{LS}_{\check{G},\alpha}} \in \text{QCoh}(\text{LS}_{\check{G}}).$$

We will prove (see Sect. 8.5.10):

Theorem 5.1.5. *For $\alpha \in (Z_G)^{\vee}$ as above, the idempotent*

$$\mathcal{O}_{\text{LS}_{\check{G},\alpha}} \otimes (-) : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G),$$

where $\otimes$ denotes the spectral action of $\text{QCoh}(\text{LS}_{\check{G}})$ on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$, identifies canonically with $\mathbf{P}_{\alpha}$.

5.1.6. We will also prove (see Sect. 8.5.10):

Theorem 5.1.7. *For $\mathcal{P}_{Z_G} \in \text{Bun}_{Z_G}$ and the corresponding $\mathcal{L}_{\mathcal{P}_{Z_G}} \in \text{QCoh}(\text{LS}_{\check{G}})$, the functor*

$$\mathcal{L}_{\mathcal{P}_{Z_G}}^{\otimes -1} \otimes (-) : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G),$$

where $\otimes$ denotes the spectral action of $\text{QCoh}(\text{LS}_{\check{G}})$ on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$, identifies canonically with (5.1).

5.2. Endomorphisms of the vacuum Poincaré object. In this subsection, we will assume that $g \geq 2$.

5.2.1. Recall the object

$$\mathrm{Poinc}_{G,!}^{\mathrm{Vac},\mathrm{glob}} \in \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G),$$

see [GLC2, Sect. 9.6.4]. For $\alpha \in (Z_G)^\vee$, let $\mathrm{Poinc}_{G,!,\alpha}^{\mathrm{Vac}}$ denote the corresponding direct summand.

5.2.2. We will prove (see Sect. 6.2):

Theorem 5.2.3. *For every α , the map*

$$k \rightarrow H^0(\mathrm{End}(\mathrm{Poinc}_{G,!,\alpha}^{\mathrm{Vac}}))$$

is an isomorphism.

In fact, we will prove a more precise result, but only Theorem 5.2.3 will be needed for the proof of GLC:

Theorem 5.2.4. *$H^i(\mathrm{End}(\mathrm{Poinc}_{G,!}^{\mathrm{Vac},\mathrm{glob}})) = 0$ for $i \neq 0$.*

As a corollary of Theorem 5.2.3, we obtain:

Corollary 5.2.5. $\dim \left(H^0(\mathrm{End}(\mathrm{Poinc}_{G,!}^{\mathrm{Vac},\mathrm{glob}})) \right) = |Z_G|.$

5.2.6. We will also prove (see Sect. 6.3):

Theorem 5.2.7. *For a non-trivial $\mathcal{P}_{Z_G} \in \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$,*

$$H^0 \left(\mathcal{H}om(\mathrm{Poinc}_{G,!}^{\mathrm{Vac},\mathrm{glob}}, \mathcal{P}_{Z_G} \cdot \mathrm{Poinc}_{G,!}^{\mathrm{Vac},\mathrm{glob}}) \right) = 0.$$

As in the case of Theorem 5.2.3, we will actually prove a more precise result (but only Theorem 5.2.7 will be needed for the proof of GLC):

Theorem 5.2.8. *For a non-trivial $\mathcal{P}_{Z_G} \in \mathrm{D}\text{-}\mathrm{mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$,*

$$\mathcal{H}om(\mathrm{Poinc}_{G,!}^{\mathrm{Vac},\mathrm{glob}}, \mathcal{P}_{Z_G} \cdot \mathrm{Poinc}_{G,!}^{\mathrm{Vac},\mathrm{glob}}) = 0.$$

5.3. **Algebraic geometry of $\mathrm{LS}_{\check{G}}$.** In this subsection, we continue to assume that $g \geq 2$.

5.3.1. First, we recall (see [BD, Proposition 2.11.2]):

Theorem 5.3.2. *The stack $\mathrm{LS}_{\check{G}}$ is a classical locally complete intersection of dimension*

$$\dim(\mathfrak{g}) \cdot 2(g-1).$$

Corollary 5.3.3. *The stack $\mathrm{LS}_{\check{G}}$ is Cohen-Macaulay of dimension $\dim(\mathfrak{g}) \cdot 2(g-1)$.*

5.3.4. Next we claim:

Proposition 5.3.5. *Excluding the case of $g = 2$ with the root system of G containing an A_1 factor, the complement to $\mathrm{LS}_{\check{G}}^{\mathrm{irred}}$ in $\mathrm{LS}_{\check{G}}$ has codimension ≥ 2 .*

This proposition will be proved in Sect. 7.3.

5.3.6. Note that from Corollary 5.3.3 and Proposition 5.3.5, we obtain:

Corollary 5.3.7. *The embedding $\mathrm{LS}_{\check{G}}^{\mathrm{irred}} \hookrightarrow \mathrm{LS}_{\check{G}}$ induces a bijection between the sets of connected components.*

5.4. **Structure of $\mathcal{A}_{G,\mathrm{irred}}$.** In this subsection we continue to assume that $g \geq 2$, and we will exclude the case that $g = 2$ and the root system of G contains a factor of A_1 (as in Proposition 5.3.5).

5.4.1. Recall that $\mathcal{A}_{G,\mathrm{irred}}$ is a vector bundle on $\mathrm{LS}_{\check{G}}^{\mathrm{irred}}$ (see Sect. 1.4.2). The next proposition provides an explicit description of the shape that $\mathcal{A}_{G,\mathrm{irred}}$ can have. This description will play a crucial role in the proof of GLC given below.

Proposition 5.4.2. *The restriction of $\mathcal{A}_{G,\mathrm{irred}}$ to every connected component is isomorphic to a direct sum of line bundles, each of which is a restriction of some $\mathcal{L}_{\mathcal{P}_{Z_G}}$ (see Sect. 4.2.6).*

5.4.3. *Proof of Proposition 5.4.2.* According to Sect. 1.4, the object $\mathcal{A}_{G,\text{irred}}$ is the vector bundle underlying a local system $\mathcal{F}$ with finite monodromy (in particular, it has regular singularities).

Thanks to the finite monodromy property, we can think of $\mathcal{F}$ as an étale local system of k -vector spaces. The assertion of the proposition follows now from Corollary 4.3.3.

□[Proposition 5.4.2]

5.5. **Proof of GLC.** Let g and G be as in Sect. 5.4. Note that it is sufficient to prove GLC under these assumptions:

Indeed, by Sect. 3, we can assume that $g \geq 2$. By Corollary 2.3.10, we may assume that G is almost simple and simply-connected. By Sect. 3.3.3 we can assume that it is not isomorphic to SL_2 .

5.5.1. *Step 0.* Fix a connected component $\text{LS}_{\check{G},\alpha}$ of $\text{LS}_{\check{G}}$. Denote

$$\text{LS}_{\check{G},\alpha}^{\text{irred}} := \text{LS}_{\check{G},\alpha} \cap \text{LS}_{\check{G}}^{\text{irred}}.$$

Using Proposition 5.4.2, we can write

$$(5.2) \quad \mathcal{A}_{G,\text{irred}}|_{\text{LS}_{\check{G},\alpha}^{\text{irred}}} \simeq \bigoplus_{\mathcal{P}_{Z_G} \in \text{Bun}_{Z_G}} \mathcal{L}_{\mathcal{P}_{Z_G}}^{\oplus n_{\mathcal{P}_{Z_G},\alpha}}|_{\text{LS}_{\check{G},\alpha}^{\text{irred}}}$$

for some integers $n_{\mathcal{P}_{Z_G},\alpha}$.

It is sufficient to show that for every α

$$n_{\mathcal{P}_{Z_G},\alpha} = \begin{cases} 1 & \text{if } \mathcal{P}_{Z_G} \text{ is trivial;} \\ 0 & \text{if } \mathcal{P}_{Z_G} \text{ is non-trivial.} \end{cases}$$

Indeed, since each $\mathcal{A}_{G,\text{irred}}|_{\text{LS}_{\check{G},\alpha}^{\text{irred}}}$ is a unital associative algebra in $\text{QCoh}(\text{LS}_{\check{G},\alpha}^{\text{irred}})$, the latter will automatically imply (see [GLC3, Lemma 17.3.7]) that the unit map

$$(5.3) \quad \mathcal{O}_{\text{LS}_{\check{G},\alpha}^{\text{irred}}} \rightarrow \mathcal{A}_{G,\text{irred}}|_{\text{LS}_{\check{G},\alpha}^{\text{irred}}}$$

is an isomorphism, i.e., (1.4) is an isomorphism.

5.5.2. *Step 1.* Fix a particular $\mathcal{P}_{Z_G}$. We are going to prove that the map

$$\mathcal{A}_G \otimes \mathcal{L}_{\mathcal{P}_{Z_G}} \rightarrow j_* \circ j^*(\mathcal{A}_G \otimes \mathcal{L}_{\mathcal{P}_{Z_G}}) = j_*(\mathcal{A}_{G,\text{irred}}) \otimes \mathcal{L}_{\mathcal{P}_{Z_G}}$$

induces an isomorphism at the level of $H^0(\Gamma(\text{LS}_{\check{G}}, -))$.

Recall that $\hat{\iota}^!$ denotes the right adjoint to the embedding

$$\text{QCoh}(\text{LS}_{\check{G}})_{\text{red}} \hookrightarrow \text{QCoh}(\text{LS}_{\check{G}}),$$

so that for every $\mathcal{F} \in \text{QCoh}(\text{LS}_{\check{G}})$ we have a fiber sequence

$$\hat{\iota}^!(\mathcal{F}) \rightarrow \mathcal{F} \rightarrow j \circ j^*(\mathcal{F}).$$

Thus, it is sufficient to prove that

$$\hat{\iota}^!(\mathcal{A}_G \otimes \mathcal{L}_{\mathcal{P}_{Z_G}})$$

is concentrated in cohomological degrees ≥ 2 .

We have

$$\hat{\iota}^!(\mathcal{A}_G \otimes \mathcal{L}_{\mathcal{P}_{Z_G}}) \simeq \hat{\iota}^!(\mathcal{A}_G) \otimes \mathcal{L}_{\mathcal{P}_{Z_G}},$$

where $\mathcal{L}_{\mathcal{P}_{Z_G}}$ is a line bundle. So it is enough to show that

$$\hat{\iota}^!(\mathcal{A}_G)$$

is concentrated in cohomological degrees ≥ 2 .

Recall (see Sect. 1.3.3) that the unit map

$$\mathcal{O}_{\text{LS}_{\check{G}}} \rightarrow \mathcal{A}_G$$

induces an isomorphism

$$\hat{t}^!(\mathcal{O}_{\mathrm{LS}_{\check{G}}}) \rightarrow \hat{t}^!(\mathcal{A}_G).$$

Thus, it remains to show that

$$\hat{t}^!(\mathcal{O}_{\mathrm{LS}_{\check{G}}})$$

is concentrated in cohomological degrees ≥ 2 .

However, this follows from Proposition 5.3.5 and Corollary 5.3.3.

Remark 5.5.3. Note that the last step in the above argument shows that the map

$$\mathcal{O}_{\mathrm{LS}_{\check{G}}} \rightarrow j_*(\mathcal{O}_{\mathrm{LS}_{\check{G}}^{\mathrm{irred}}})$$

also induces an isomorphism at the level of $H^0(\Gamma(\mathrm{LS}_{\check{G}}, -))$.

5.5.4. *Step 2.* We will now show that $n_{\mathcal{P}_{Z_G}, \alpha} = 0$ if $\mathcal{P}_{Z_G}$ is non-trivial.

We begin by showing that

$$(5.4) \quad \Gamma(\mathrm{LS}_{\check{G}}, \mathcal{L}_{\mathcal{P}_{Z_G}}^{\otimes -1} \otimes \mathcal{A}_G) = 0.$$

Recall that

$$\mathcal{A}_G = \mathbb{L}_G \circ \mathbb{L}_G^L(\mathcal{O}_{\mathrm{LS}_{\check{G}}}).$$

Therefore, we have

$$(5.5) \quad \begin{aligned} \Gamma(\mathrm{LS}_{\check{G}}, \mathcal{L}_{\mathcal{P}_{Z_G}}^{\otimes -1} \otimes \mathcal{A}_G) &= \Gamma(\mathrm{LS}_{\check{G}}, \mathcal{L}_{\mathcal{P}_{Z_G}}^{\otimes -1} \otimes \mathbb{L}_G \circ \mathbb{L}_G^L(\mathcal{O}_{\mathrm{LS}_{\check{G}}})) \stackrel{\text{QCoh}(\mathrm{LS}_{\check{G}})\text{-linearity of } \mathbb{L}_G^L \text{ and } \mathbb{L}_G}{\simeq} \\ &\simeq \Gamma(\mathrm{LS}_{\check{G}}, \mathbb{L}_G \circ \mathbb{L}_G^L(\mathcal{L}_{\mathcal{P}_{Z_G}}^{\otimes -1} \otimes \mathcal{O}_{\mathrm{LS}_{\check{G}}})) = \mathcal{H}om_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})}(\mathcal{O}_{\mathrm{LS}_{\check{G}}}, \mathbb{L}_G \circ \mathbb{L}_G^L(\mathcal{L}_{\mathcal{P}_{Z_G}}^{\otimes -1} \otimes \mathcal{O}_{\mathrm{LS}_{\check{G}}})) \simeq \\ &\stackrel{\text{adjunction}}{\simeq} \mathcal{H}om_{\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)}(\mathbb{L}_G^L(\mathcal{O}_{\mathrm{LS}_{\check{G}}}), \mathbb{L}_G^L(\mathcal{L}_{\mathcal{P}_{Z_G}}^{\otimes -1} \otimes \mathcal{O}_{\mathrm{LS}_{\check{G}}})). \end{aligned}$$

Recall that

$$\mathbb{L}_G^L(\mathcal{O}_{\mathrm{LS}_{\check{G}}}) \simeq \mathrm{Poinc}_{G,!}^{\mathrm{Vac, glob}}.$$

Hence, since the functor $\mathbb{L}_G^L$ is $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ -linear, and taking into account Theorem 5.1.7, we can rewrite the right-hand side in (5.5) as

$$\mathcal{H}om(\mathrm{Poinc}_{G,!}^{\mathrm{Vac, glob}}, \mathcal{P}_{Z_G} \cdot \mathrm{Poinc}_{G,!}^{\mathrm{Vac, glob}}).$$

By Theorem 5.2.7, this expression vanishes, so the same is true of (5.4).

Applying Step 1 (with $\mathcal{L}_{\mathcal{P}_{Z_G}}^{\otimes -1}$ instead of $\mathcal{L}_{\mathcal{P}_{Z_G}}$, we find that

$$H^0\left(\Gamma(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}, \mathcal{L}_{\mathcal{P}_{Z_G}}^{\otimes -1} \otimes \mathcal{A}_{G, \mathrm{irred}})\right) = H^0\left(\Gamma(\mathrm{LS}_{\check{G}}, \mathcal{L}_{\mathcal{P}_{Z_G}}^{\otimes -1} \otimes \mathcal{A}_G)\right) = 0.$$

Hence, we obtain

$$H^0\left(\Gamma(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}, \mathcal{L}_{\mathcal{P}_{Z_G}}^{\otimes -1} \otimes \mathcal{A}_{G, \mathrm{irred}})\right) = 0.$$

However, by definition, $\mathcal{L}_{\mathcal{P}_{Z_G}}^{\otimes -1} \otimes \mathcal{A}_{G, \mathrm{irred}}$ carries $n_{\mathcal{P}_{Z_G}, \alpha}$ direct summands isomorphic to $\mathcal{O}_{\mathrm{LS}_{\check{G}, \alpha}^{\mathrm{irred}}}$. Therefore, we obtain that

$$0 = \dim H^0\left(\Gamma(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}, \mathcal{L}_{\mathcal{P}_{Z_G}}^{\otimes -1} \otimes \mathcal{A}_{G, \mathrm{irred}})\right) \geq H^0\left(\Gamma(\mathrm{LS}_{\check{G}}^{\mathrm{irred}}, \mathcal{O}_{\mathrm{LS}_{\check{G}, \alpha}^{\mathrm{irred}}}^{\oplus n_{\mathcal{P}_{Z_G}, \alpha}})\right) \geq n_{\mathcal{P}_{Z_G}, \alpha}$$

meaning that $n_{\mathcal{P}_{Z_G}, \alpha} = 0$, as was desired.

5.5.5. *Step 3.* Thus, we obtain that the decomposition (5.2) is in fact of the form

$$(5.6) \quad \mathcal{A}_{G,\text{irred}}|_{\text{LS}_{\check{G},\alpha}^{\text{irred}}} \simeq \mathcal{O}_{\text{LS}_{\check{G},\alpha}^{\text{irred}}}^{\oplus n_\alpha},$$

for some integers n_α , and we wish to show that they are all equal to 1.

By Step 1, for every α , we have

$$H^0 \left(\Gamma(\text{LS}_{\check{G},\alpha}, \mathcal{A}_G|_{\text{LS}_{\check{G},\alpha}}) \right) \simeq H^0 \left(\Gamma(\text{LS}_{\check{G},\alpha}^{\text{irred}}, \mathcal{A}_{G,\text{irred}}|_{\text{LS}_{\check{G},\alpha}^{\text{irred}}}) \right).$$

Hence,

$$\dim H^0 \left(\Gamma(\text{LS}_{\check{G},\alpha}, \mathcal{A}_G|_{\text{LS}_{\check{G},\alpha}}) \right) = \dim H^0 \left(\Gamma(\text{LS}_{\check{G},\alpha}^{\text{irred}}, \mathcal{O}_{\text{LS}_{\check{G},\alpha}^{\text{irred}}}^{\oplus n_\alpha}) \right) \geq n_\alpha.$$

Therefore,

$$\dim H^0 \left(\Gamma(\text{LS}_{\check{G}}, \mathcal{A}_G) \right) \geq \sum_{\alpha} n_\alpha.$$

Now,

$$(5.7) \quad \Gamma(\text{LS}_{\check{G}}, \mathcal{A}_G) \simeq \mathcal{H}om_{\text{D-mod } \frac{1}{2}(\text{Bun}_G)}(\mathbb{L}_G^L(\mathcal{O}_{\text{LS}_{\check{G}}}), \mathbb{L}_G^L(\mathcal{O}_{\text{LS}_{\check{G}}})) \simeq \mathcal{H}om_{\text{D-mod } \frac{1}{2}(\text{Bun}_G)}(\text{Poinc}_{G,!}^{\text{Vac, glob}}, \text{Poinc}_{G,!}^{\text{Vac, glob}}).$$

Applying Corollary 5.2.5, we obtain

$$\sum_{\alpha} 1 = |Z_G| = |\pi_0(\text{LS}_{\check{G}})| \geq \sum_{\alpha} n_\alpha.$$

Hence, in order to prove the desired equality, it suffices to show that $n_\alpha \neq 0$ for all α . I.e., we have to show that $\mathcal{A}_{G,\text{irred}}$ does not vanish on any connected component $\text{LS}_{\check{G},\alpha}^{\text{irred}}$.

5.5.6. *Step 4.* By Theorem 5.1.5, we have:

$$\Gamma(\text{LS}_{\check{G},\alpha}, \mathcal{A}_G|_{\text{LS}_{\check{G},\alpha}}) \simeq \mathcal{E}nd_{\text{D-mod } \frac{1}{2}(\text{Bun}_G)}(\text{Poinc}_{G,!,\alpha}^{\text{Vac}}).$$

As $\text{Poinc}_{G,!,\alpha}^{\text{Vac}} \neq 0$ by Theorem 5.2.3, we must have:

$$0 \neq \text{id} \in H^0 \left(\mathcal{E}nd(\text{Poinc}_{G,!,\alpha}^{\text{Vac}}) \right) \simeq H^0 \left(\Gamma(\text{LS}_{\check{G},\alpha}, \mathcal{A}_G|_{\text{LS}_{\check{G},\alpha}}) \right) \simeq H^0 \left(\Gamma(\text{LS}_{\check{G},\alpha}^{\text{irred}}, \mathcal{A}_{G,\text{irred}}|_{\text{LS}_{\check{G},\alpha}^{\text{irred}}}) \right).$$

□[GLC]

5.6. Additional remarks.

5.6.1. For G simply-connected, the assertion that all $\mathcal{A}_{G,\text{irred}}|_{\text{LS}_{\check{G},\alpha}^{\text{irred}}}$ are non-zero can be also deduced from the main result of [Ari]:

Let σ be a point of $\text{LS}_{\check{G},\alpha}^{\text{irred}}$. Recall that according to [GLC4, Theorem 3.1.5], the fiber of $\mathcal{A}_G$ at σ is isomorphic to the homology of the space of generic oper structures on σ .

Thus, it is sufficient to know that the latter space is non-empty. However, this is precisely the result⁷ of [Ari].

⁷Note that the result of [Ari] is about $\mathfrak{g}$ -opers, which are different from $\check{G}$ -opers, unless $\check{G}$ is adjoint. However, as we have seen earlier, it is sufficient to prove GLC in the latter case.

5.6.2. Note that taking into account Theorem 5.2.4, and knowing that $\mathcal{O}_{\mathrm{LS}_{\check{G}}} \simeq \mathcal{A}_G$, from (5.7) we obtain:

Corollary 5.6.3. *For $g \geq 2$, for every connected component α of $\mathrm{LS}_{\check{G}}$, the map*

$$k \rightarrow \Gamma(\mathrm{LS}_{\check{G},\alpha}, \mathcal{O}_{\mathrm{LS}_{\check{G},\alpha}})$$

is an isomorphism.

Remark 5.6.4. One can prove Corollary 5.6.3 directly by a deformation argument using [FT, Theorem 4.2].

Remark 5.6.5. We remark that Corollary 5.6.3 is a special feature of the de Rham setting; there are many more global functions on the Betti moduli stack.

5.6.6. Similarly, from Theorem 5.2.8, we obtain:

Corollary 5.6.7. *For $g \geq 2$, for a non-trivial $\mathcal{P}_{Z_G}$, we have $\Gamma(\mathrm{LS}_{\check{G}}, \mathcal{L}_{\mathcal{P}_{Z_G}}) = 0$.*

6. THE VACUUM POINCARÉ OBJECT

In this section we will prove Theorems 5.2.3 and 5.2.7, along with their strengthenings, given by Theorems 5.2.4 and 5.2.8, respectively.

6.1. How to calculate those endomorphisms?

6.1.1. Recall that the object $\mathrm{Poinc}_{G,!}^{\mathrm{vac},\mathrm{glob}}$ is the $!$ -direct image along the map

$$\mathfrak{p} : \mathrm{Bun}_{N,\rho(\omega_X)} \rightarrow \mathrm{Bun}_G$$

of $(\chi^{\mathrm{glob}})^*(\exp)$, see [GLC1, Sect. 1.3.6].

We factor the above map $\mathfrak{p}$ as a composition

$$(6.1) \quad \mathrm{Bun}_{N,\rho(\omega_X)} \xrightarrow{f} \mathrm{Bun}_{N,\rho(\omega_X)} / T \hookrightarrow \mathrm{Bun}_B^{(g-1) \cdot 2\rho} \rightarrow \mathrm{Bun}_G.$$

6.1.2. Here is the crucial observation:

Since $g \geq 2$, the coweight $(g-1) \cdot 2\rho$ belongs to Λ_G^{++} , and hence the map

$$\mathrm{Bun}_B^{(g-1) \cdot 2\rho} \rightarrow \mathrm{Bun}_G$$

is a locally closed embedding (see [DG, Theorem 7.4.3(1')]). Hence, the $!$ -direct image with respect to it is fully faithful.

Hence, for the proofs of the theorems of in this section, we can perform the calculations on $\mathrm{Bun}_B^{(g-1) \cdot 2\rho}$. Here we remark that because $Z_G \subset B$, the action of Bun_{Z_G} on Bun_G lifts to an action on $\mathrm{Bun}_B^{(g-1) \cdot 2\rho}$.

6.2. Proof of Theorems 5.2.3 and 5.2.4.

6.2.1. The map

$$(6.2) \quad \mathrm{Bun}_{N,\rho(\omega_X)} / T \hookrightarrow \mathrm{Bun}_B^{(g-1) \cdot 2\rho}$$

is a closed embedding as it comes via base-change from the closed embedding $\mathrm{pt} / T \xrightarrow{\rho(\omega_X)} \mathrm{Bun}_T$.

Note that the action of

$$B(Z_G) \subset \mathrm{Bun}_{Z_G}$$

on $\mathrm{Bun}_B^{(g-1) \cdot 2\rho}$ preserves the above locally closed embedding.

Hence, in order to prove the theorems in this subsection, we can perform the calculations on $\mathrm{Bun}_{N,\rho(\omega_X)} / T$.

6.2.2. Let r be the semi-simple rank of G . For each vertex of the Dynkin diagram we have a canonically defined map

$$\mathrm{Bun}_{N,\rho(\omega_X)} \rightarrow \mathbb{A}^1.$$

The resulting map

$$\mathrm{Bun}_{N,\rho(\omega_X)} \rightarrow (\mathbb{A}^1)^r = \mathbb{A}^r \xrightarrow{\mathrm{sum}} \mathbb{A}^1$$

is the map χ^{glob} .

We have a Cartesian square

$$(6.3) \quad \begin{array}{ccc} \mathrm{Bun}_{N,\rho(\omega_X)} & \xrightarrow{\chi^{\mathrm{glob}}} & \mathbb{A}^r \\ \mathfrak{f} \downarrow & & \downarrow \mathfrak{f}' \\ \mathrm{Bun}_{N,\rho(\omega_X)}/T & \xrightarrow{\chi^{\mathrm{glob}}/T} & \mathbb{A}^r/T, \end{array}$$

where T acts on $\mathbb{A}^r$ via

$$T \rightarrow T_{\mathrm{ad}} \simeq (\mathbb{G}_m)^r.$$

6.2.3. Since

$$Z_G = \ker(T \rightarrow T_{\mathrm{ad}}),$$

we have an action of $B(Z_G)$ on $\mathbb{A}^r/T$, and the above map

$$\chi^{\mathrm{glob}}/T : \mathrm{Bun}_{N,\rho(\omega_X)}/T \rightarrow \mathbb{A}^r/T$$

is $B(Z_G)$ -equivariant.

Since the horizontal arrows in (6.3) are smooth with affine space fibers (and hence, the corresponding pullback functors are fully faithful), we obtain that in order to prove Theorems 5.2.3 and 5.2.4, it suffices to perform the corresponding calculation on $\mathbb{A}^r/T$.

6.2.4. Note that the map

$$\mathbb{A}^r \rightarrow \mathbb{A}^r/T$$

naturally factors as

$$\mathbb{A}^r = \mathbb{A}^r \times \mathrm{pt} \rightarrow \mathbb{A}^r \times \mathrm{pt}/Z_G \simeq \mathbb{A}^r/Z_G \rightarrow \mathbb{A}^r/T,$$

and we have a Cartesian square

$$\begin{array}{ccc} \mathbb{A}^r \times \mathrm{pt}/Z_G & \longrightarrow & \mathbb{A}^r \\ \mathfrak{f}' \downarrow & & \downarrow \mathfrak{f}'' \\ \mathbb{A}^r/T & \longrightarrow & \mathbb{A}^r/T_{\mathrm{ad}} \simeq (\mathbb{A}^1/\mathbb{G}_m)^r. \end{array}$$

This reduces the assertions of Theorems 5.2.3 and 5.2.4 to the claim that the map

$$k \rightarrow \mathcal{E}nd(\mathfrak{f}_!'' \circ \mathrm{sum}^*(\mathrm{exp}))$$

is an isomorphism.

6.2.5. Since

$$\mathrm{sum}^*(\mathrm{exp}) \simeq \mathrm{exp}^{\boxtimes r},$$

by Künneth formula, we are reduced to showing that the map

$$k \rightarrow \mathcal{E}nd(\mathfrak{f}_!'''(\mathrm{exp}))$$

is an isomorphism, where

$$\mathfrak{f}_!''' : \mathbb{A}^1 \rightarrow \mathbb{A}^1/\mathbb{G}_m.$$

However, the latter is well-known: the object

$$\mathfrak{f}_!'''(\mathrm{exp}) \in \mathrm{D}\text{-mod}(\mathbb{A}^1/\mathbb{G}_m)$$

is the $*$ -direct image of $k \in \mathrm{D}\text{-mod}(\mathrm{pt})$ along

$$\mathrm{pt} \simeq \mathbb{A}^1 - 0/\mathbb{G}_m \hookrightarrow \mathbb{A}^1/\mathbb{G}_m.$$

□[Theorems 5.2.3 and 5.2.4]

6.3. Proof of Theorems 5.2.7 and 5.2.8.

6.3.1. To prove Theorems 5.2.7 and 5.2.8, it suffices to show that the image of the closed embedding (6.2) and its translate by means of a non-trivial $\mathcal{P}_{Z_G}$ are disjoint.

For that, it suffices to show that their images under the projection

$$\mathrm{Bun}_B \rightarrow \mathrm{Bun}_T$$

are disjoint.

6.3.2. For the latter it is sufficient to show that under the further projection

$$\mathrm{Bun}_T \rightarrow \mathrm{Bun}_T / B(T)$$

(where $\mathrm{Bun}_T / B(T)$ is the coarse moduli scheme), the above two images correspond to two distinct points:

$$\rho(\omega_X) \text{ and } \mathcal{P}_{Z_G} \cdot \rho(\omega_X).$$

6.3.3. However, the latter follows from the fact that $\mathcal{P}_{Z_G}$ is *non-trivial* as a T -bundle.

□[Theorems 5.2.7 and 5.2.8]

7. GEOMETRY OF Bun_G

The goal of this section is to prove Propositions 4.1.6, 4.3.5, 5.3.5 and 4.3.10.

For the duration of this section, we will change the notation from $\check{G}$ to G .

7.1. Proof of Proposition 4.1.6.

7.1.1. Let G' be a reductive group equipped with a surjective map

$$\phi : G' \rightarrow G,$$

such that

- The kernel T_0 of ϕ is a (connected) torus (which automatically lies in the center of G');
- The derived group of G' is simply-connected.

7.1.2. Denote

$$T_1 := G'_{\mathrm{ab}}.$$

We obtain an isogeny

$$\phi' := T_0 \rightarrow T_1,$$

and it is easy to see that we have a canonical isomorphism

$$(7.1) \quad \ker(\phi') \simeq \pi_1(G).$$

7.1.3. *Example.* One can take G' be the *dual* group of

$$\check{G} \times \check{T} / Z_{\check{G}},$$

where $\check{T}$ is the (abstract) Cartan of $\check{G}$.

Then

$$T_0 = (\check{T} / Z_{\check{G}})^\vee \simeq T_{\mathrm{sc}},$$

where T_{sc} is the (abstract) Cartan of the simply-connected cover G_{sc} of G . We also have $T_1 = T$, so (7.1) becomes the isomorphism

$$\pi_1(G) \simeq \ker(G_{\mathrm{sc}} \rightarrow G) \simeq \ker(T_{\mathrm{sc}} \rightarrow T).$$

7.1.4. *Example.* Let $G = PGL_n$. In this case we can take $G' := GL_n$. We have

$$T_0 \simeq \mathbb{G}_m \text{ and } T_0 \simeq \mathbb{G}_m,$$

and the map ϕ' is raising to the power n .

Then (7.1) becomes the identification

$$\pi_1(PGL_n) \simeq \mu_n.$$

7.1.5. Note that the map

$$(7.2) \quad \text{Bun}_{G'} \rightarrow \text{Bun}_{T_1}$$

is smooth with fibers that are connected and simply-connected:

Indeed, the fibers are isomorphic to Bun_{G_1} , where G_1 is a twisted form of $[G', G']$, the derived group of G' , and the moduli stack of bundles for a simply-connected group is connected and simply-connected.

Hence, we obtain that the map (7.2) induces an isomorphism of the $\tau_{\leq 1}$ -truncations of étale homotopy types.

7.1.6. The map ϕ induces an isomorphism

$$\text{Bun}_{G'} / \text{Bun}_{T_0} \simeq \text{Bun}_G.$$

Hence, we obtain that the map

$$\text{Bun}_G \simeq \text{Bun}_{G'} / \text{Bun}_{T_0} \rightarrow \text{Bun}_{T_1} / \text{Bun}_{T_0}$$

induces an isomorphism of the $\tau_{\leq 1}$ -truncations of étale homotopy types.

7.1.7. Finally, we note that the isomorphism (7.1) induces an identification

$$\text{Bun}_{T_1} / \text{Bun}_{T_0} \simeq B^2(C(X, \pi_1(G)))_{\text{et}},$$

and the resulting map

$$\text{Bun}_G \rightarrow B^2(C(X, \pi_1(G)))_{\text{et}}$$

is the same as (4.1).

□[Proposition 4.1.6]

7.2. Proof of Proposition 4.3.5.

7.2.1. At this point, we refer the reader to Appendix A for background on stable bundles and some relevant notation.

Let $\text{Bun}_G^{\text{unstbl}} \subset \text{Bun}_G$ be the closed substack of unstable⁸ bundles. We need to show $\text{Bun}_G^{\text{unstbl}}$ has codimension ≥ 2 under our hypotheses on X .

By definition of stability for G -bundles, every point of $\text{Bun}_G^{\text{unstbl}}$ is in the image of some map Bun_P^λ for $P \subsetneq G$ a maximal parabolic with Levi quotient M and $\lambda \in \pi_{1, \text{alg}}(M)$ satisfying $\langle 2\check{\rho}_P, \lambda \rangle \geq 0$.

Therefore, it suffices to show

$$\dim(\text{Bun}_P^\lambda) \leq \dim(\text{Bun}_G) - 2$$

for such λ .

⁸Here “unstable” means “not stable”, rather than “not semi-stable”.

7.2.2. By Riemann-Roch,

$$\dim(\mathrm{Bun}_G) = \dim(\mathfrak{g}) \cdot (g-1) = \dim(\mathfrak{m}) \cdot (g-1) + 2 \dim(\mathfrak{n}(P)) \cdot (g-1)$$

and

$$\dim(\mathrm{Bun}_P^\lambda) = \dim(\mathfrak{m}) \cdot (g-1) + \dim(\mathfrak{n}(P)) \cdot (g-1) - \langle 2\check{\rho}_P, \lambda \rangle,$$

where $2\check{\rho}_P$ is as in Appendix A.

Therefore, we have to show

$$2 - \langle 2\check{\rho}_P, \lambda \rangle \leq \dim(\mathfrak{n}(P)) \cdot (g-1).$$

By assumption on λ , the left hand side is at most 2. As P is a proper parabolic and $g \geq 2$, this inequality obviously holds outside the exceptional case where $\dim(\mathfrak{n}(P)) = (g-1) = 1$, which only happens if $g = 2$ and G_{ad} contains a PGL_2 factor.

□[Proposition 4.3.5]

Remark 7.2.3. Note that the assertion of Proposition 4.3.5 is false for $G = SL_2$ and $g = 2$: in this case the dimension of the semi-stable but unstable locus is 2, which is $>$ than

$$1 = 3 - 2 = \dim(\mathrm{Bun}_G) - 2.$$

7.3. Proof of Proposition 5.3.5. It is enough to show that for every maximal parabolic subgroup $P \subsetneq G$, we have

$$(7.3) \quad \dim(\mathrm{LS}_P) \leq \dim(\mathrm{LS}_G) - 2 = \dim(\mathfrak{g}) \cdot (2g-2) - 2.$$

7.3.1. Consider the stack LS_M . It is quasi-smooth of virtual dimension

$$\dim(\mathfrak{m}) \cdot (2g-2),$$

and if $g \geq 2$, by Theorem 5.3.2, its underlying classical stack is a locally complete intersection of dimension

$$\dim(\mathfrak{m}) \cdot (2g-2) + \dim(\mathfrak{z}_M),$$

where $\mathfrak{z}_M := \mathrm{Lie}(Z_M)$.

Indeed, this follows by considering the fibration $\mathrm{LS}_M \rightarrow \mathrm{LS}_{M/[M,M]}$, applying Theorem 5.3.2 for the derived group $[M, M]$, and noting that LS_{G_m} has dimension one more than its virtual dimension by explicit analysis.

7.3.2. Consider the map

$$(7.4) \quad \mathfrak{q} : \mathrm{LS}_P \rightarrow \mathrm{LS}_M.$$

It is quasi-smooth of virtual relative dimension

$$\dim(\mathfrak{n}(P)) \cdot (2g-2).$$

Lemma 7.3.3.

- (a) Each fiber of the map $\mathfrak{q}$ has dimension $\leq \dim(\mathfrak{n}(P)) \cdot (2g-1)$.
- (b) There exists a dense open substack of LS_M over which $\mathfrak{q}$ is smooth.

Let us assume this lemma for a moment and proceed with the proof of (7.3).

It follows from point (b) of the lemma that the generic fiber of $\mathfrak{q}$ has dimension $\dim(\mathfrak{n}(P)) \cdot (2g-2)$, so the substack of LS_M over which $\mathfrak{q}$ has fibers of larger dimension has codimension at least one. We obtain:

Corollary 7.3.4. $\dim(\mathrm{LS}_P) \leq \dim(\mathrm{LS}_M) + \dim(\mathfrak{n}(P)) \cdot (2g-1) - 1$.

7.3.5. From Corollary 7.3.4, we obtain:

$$\dim(\mathrm{LS}_P) \leq \dim(\mathfrak{m}) \cdot (2g - 2) + \dim(\mathfrak{z}_M) + \dim(\mathfrak{n}(P)) \cdot (2g - 1) - 1.$$

Thus it remains to show that, under the assumptions of Proposition 5.3.5,

$$(7.5) \quad \begin{aligned} \dim(\mathfrak{m}) \cdot (2g - 2) + \dim(\mathfrak{z}_M) + \dim(\mathfrak{n}(P)) \cdot (2g - 1) - 1 &\leq \dim(\mathfrak{g}) \cdot (2g - 2) - 2 \\ &= \dim(\mathfrak{m}) \cdot (2g - 2) + 2 \dim(\mathfrak{n}(P)) \cdot (2g - 2) - 2. \end{aligned}$$

This is equivalent to

$$(7.6) \quad \dim(\mathfrak{z}_M) + 1 \leq \dim(\mathfrak{n}(P)) \cdot (2g - 3).$$

7.3.6. We now use the assumption that G is semi-simple and that the corank of P is one, so that $\dim(\mathfrak{z}_M) = 1$, i.e., (7.6) becomes

$$(7.7) \quad 2 \leq \dim(\mathfrak{n}(P)) \cdot (2g - 3).$$

This holds automatically if $g \geq 3$. If $g = 2$, the above inequality can only be violated if $\dim(\mathfrak{n}(P)) = 1$, but this only happens if the Dynkin diagram of G has an A_1 factor.

□[Proposition 5.3.5]

Remark 7.3.7. Note that the assertion of Proposition 5.3.5 is false for $G = SL_2$ and $g = 2$: in this case the dimension of the reducible locus is 5, which is greater than

$$4 = 6 - 2 = \dim(\mathrm{LS}_G) - 2.$$

7.3.8. *Proof of Lemma 7.3.3(b).* It suffices to show that for every $\sigma_M \in \mathrm{LS}_M$, there exists a point σ'_M that lies in the same irreducible component, over which the fiber of (7.4) is smooth.

Note that for $\sigma_M \in \mathrm{LS}_M$ and

$$\sigma_P \in \mathfrak{q}^{-1}(\sigma_M) \simeq \mathrm{LS}_{N(P)\sigma_M},$$

the obstruction to the smoothness of the fiber is

$$H^2(X, \mathfrak{n}(P)_{\sigma_P}).$$

The latter is non-zero if for some subquotient V of $\mathfrak{n}(P)$ as a M -representation, the local system V_{σ_M} admits a trivial quotient.

Let Z_M^0 denote the neutral connected component of Z_M and consider its action on $\mathfrak{n}(P)$. It acts on every V as above by a *non-trivial* character. Hence, for a generic point $\sigma_Z \in \mathrm{LS}_{Z_M}$ and

$$\sigma'_M := \sigma_Z \otimes \sigma_M,$$

the local system $V_{\sigma'_M}$ will not have trivial quotients.

Since $\mathrm{LS}_{Z_M^0}$ is irreducible, its action on LS_M preserves irreducible components, i.e., σ' lies in the same irreducible component as σ .

□[Lemma 7.3.3(b)]

7.3.9. *Proof of Lemma 7.3.3(a).* We will use the following assertion:

Lemma 7.3.10. *Let Y be a quasi-smooth scheme of virtual dimension d . Suppose that m is an integer such that for all field-valued points $y \in Y$ we have*

$$\dim(H^{-1}(T_y^*(Y))) \leq m.$$

Then $\dim(Y) \leq d + m$.

Proof. It is enough to show that for every field-valued point $y \in Y$, the dimension of the classical cotangent space to Y at y is $\leq d + m$. However, the classical cotangent space is just $H^0(T_y^*(Y))$. We have

$$\dim(H^0(T_y^*(Y))) = \dim(H^0(T_y^*(Y)) - \dim(H^1(T_y^*(Y))) + \dim(H^1(T_y^*(Y))),$$

where $\dim(H^0(T_y^*(Y)) - \dim(H^1(T_y^*(Y)))$ is the virtual dimension of Y .

□

The assertion of the lemma automatically extends to algebraic stacks. We apply it to the fibers of the map (7.4), i.e., to the stacks

$$\mathrm{LS}_{N(P)\sigma_M}, \quad \sigma_M \in \mathrm{LS}_M.$$

It remains to show that

$$\dim(H^{-1}(T_{\sigma_P}^*(\mathrm{LS}_{N(P)\sigma_M}))) \leq \dim(\mathfrak{n}(P)), \quad \sigma_P \in \mathrm{LS}_{N(P)\sigma}.$$

We have:

$$T_{\sigma_P}^*(\mathrm{LS}_{N(P)\sigma_M}) \simeq C(X, \mathfrak{n}(P)_{\sigma_P}[1])^\vee,$$

so

$$H^{-1}(T_{\sigma_P}^*(\mathrm{LS}_{N(P)\sigma_M})) \simeq H^2(X, \mathfrak{n}(P)_{\sigma_P})^\vee,$$

which identifies with

$$H^0(X, (\mathfrak{n}(P))_{\sigma_P}^\vee)$$

by Verdier duality.

We clearly have

$$\dim(H^0(X, (\mathfrak{n}(P))_{\sigma_P}^\vee)) \leq \dim(\mathfrak{n}(P)).$$

□[Lemma 7.3.3(a)]

7.4. Proof of Proposition 4.3.10.

7.4.1. The fact that the non-empty fibers of the map

$$\mathrm{LS}_G \rightarrow \mathrm{Bun}_G$$

are affine spaces is completely general:

For a given $\mathcal{P}_G \in \mathrm{Bun}_G$, the fiber in question is a torsor for the (derived) vector space

$$(7.8) \quad \Gamma(X, \mathfrak{g}_{\mathcal{P}_G} \otimes \omega_X).$$

Warning: In the above formula ω_X stands for the canonical line bundle on X , and *not* the dualizing sheaf of X , which is the $[1]$ shift of that. This deviates from the conventions adopted in this series, according to which for a prestack $\mathcal{Y}$, we denote by $\omega_{\mathcal{Y}}$ the dualizing sheaf on $\mathcal{Y}$. So, the curve X itself is the only exception for this convention.

7.4.2. Let us show that the map in question is smooth over the stable locus. This is equivalent to the fibers being smooth as derived schemes.

By the above torsor property, it suffices to show that if $\mathcal{P}_G \in \mathrm{Bun}_G$ is stable, then the derived vector space (7.8) is classical, i.e., that

$$H^1(X, \mathfrak{g}_{\mathcal{P}_G} \otimes \omega_X) = 0.$$

By Serre duality (and using the Killing form on $\mathfrak{g}$), this is equivalent to

$$H^0(X, \mathfrak{g}_{\mathcal{P}_G}) = 0.$$

I.e., we need to show that stable bundles do not admit infinitesimal automorphisms. This is standard; we supply a proof for completeness.

7.4.3. Suppose the contrary. Let A be an infinitesimal automorphism of $\mathcal{P}_G$. First, we show that A is nilpotent.

Consider the characteristic polynomial of the $\mathcal{O}_X$ -valued Higgs field A , i.e., the map

$$X \rightarrow \mathfrak{t} // W =: \mathfrak{a}$$

coming from A .

This map is necessarily constant; denote its image by a .

Let $t \in \mathfrak{t}$ be a semi-simple element that maps under $\mathfrak{t} \rightarrow \mathfrak{a}$ to a . In this case, $\mathcal{P}_G$ admits a reduction to $Z_G(t)$, which is a Levi subgroup.

If a were not nilpotent, we would have $t \neq 0$, and $Z_G(t)$ is a *proper* Levi subgroup. However, the existence of such a reduction contradicts the assumption that $\mathcal{P}_G$ is stable.

7.4.4. Thus, A is (non-zero) nilpotent. The Jacobson-Morosov theory supplies a (decreasing) filtration on the vector bundle $\mathfrak{g}_{\mathcal{P}_G}$ so that

$$(\mathfrak{g}_{\mathcal{P}_G})^{\geq 1} \subset \mathfrak{g}_{\mathcal{P}_G}$$

is the unipotent radical of a parabolic reduction canonically associated to A , and the operator $(\mathrm{ad}_A)^n$ defines a map

$$(7.9) \quad \mathrm{gr}^{-n}(\mathfrak{g}_{\mathcal{P}_G}) \rightarrow \mathrm{gr}^n(\mathfrak{g}_{\mathcal{P}_G}),$$

which is an isomorphism at the generic point of X .

Since $\mathcal{P}_G$ was assumed stable, $\deg((\mathfrak{g}_{\mathcal{P}_G})^{\geq 1}) < 0$. Hence, for some $n \geq 1$, we have $\deg(\mathrm{gr}^n(\mathfrak{g}_{\mathcal{P}_G})) < 0$. However,

$$\deg(\mathrm{gr}^{-n}(\mathfrak{g}_{\mathcal{P}_G})) = -\deg(\mathrm{gr}^n(\mathfrak{g}_{\mathcal{P}_G})),$$

and this contradicts the existence of (7.9).

7.4.5. It remains show that every stable G -bundle *admits* a connection. For a general $\mathcal{P}_G \in \mathrm{Bun}_G$, the obstruction to having a connection is given by its *Atiyah class*, which is an element of

$$H^1(X, \mathfrak{g}_{\mathcal{P}_G} \otimes \omega_X).$$

However, we have just proved that this group is zero for stable $\mathcal{P}_G$.

□[Proposition 4.3.10]

Remark 7.4.6. The above argument can be refined to prove the following criterion (originally due to A. Weil) for a G -bundle $\mathcal{P}_G$ to admit a connection:

This happens if and only if, for every reduction of $\mathcal{P}_G$ to a *Levi subgroup* M , this reduction, viewed as an M -bundle, has degree 0. See [AB] for more details.

8. 2-FOURIER-MUKAI TRANSFORM OF THE AUTOMORPHIC CATEGORY

The goal of this section is to prove Theorems 5.1.5 and 5.1.7. We will do so by considering a more general picture that involves twisting the constant group-scheme with fiber G by Z_G -gerbes.

8.1. The notion of 2-Fourier-Mukai transform. Recall that the usual Fourier-Mukai transform is a functor between categories of quasi-coherent sheaves on a pair of prestacks.

In this subsection we introduce the notion of *2-Fourier-Mukai transform*, which is a (2-)functor between 2-categories of *sheaves of categories* on a pair of prestacks.

8.1.1. Let $\mathcal{Y}_1$ and $\mathcal{Y}_2$ be a pair of prestacks equipped with a map

$$(8.1) \quad \mathcal{Y}_1 \times \mathcal{Y}_2 \rightarrow \mathrm{pt}/\mathbb{G}_m,$$

i.e., a line bundle, denoted $\mathcal{L}_{1,2}$ on $\mathcal{Y}_1 \times \mathcal{Y}_2$.

Assume that the functor $(p_2)_* : \mathrm{QCoh}(\mathcal{Y}_1 \times \mathcal{Y}_2) \rightarrow \mathrm{QCoh}(\mathcal{Y}_2)$ is continuous (this happens, e.g., when $\mathcal{Y}_1$ is quasi-compact with an affine diagonal).

Consider the functor

$$(8.2) \quad \mathrm{FM}_{\mathcal{Y}_1 \rightarrow \mathcal{Y}_2} : \mathrm{QCoh}(\mathcal{Y}_1) \rightarrow \mathrm{QCoh}(\mathcal{Y}_2), \quad \mathcal{F} \mapsto (p_2)_*(\mathcal{L}_{1,2} \otimes p_1^*(\mathcal{F})).$$

We shall say that the map (8.1) is of Fourier-Mukai type if the functor (8.2) is an equivalence.

8.1.2. Let us now be given a map

$$(8.3) \quad \mathcal{Y}_1 \times \mathcal{Y}_2 \rightarrow \mathrm{Ge}_{\mathbb{G}_m}(\mathrm{pt}),$$

where $\mathrm{Ge}_{\mathbb{G}_m}(\mathrm{pt}) = B^2(\mathbb{G}_m)_{\mathrm{et}}$ is the (2-algebraic) stack classifying $\mathbb{G}_m$ -gerbes. Let $\mathcal{G}_{1,2}$ denote the corresponding gerbe on $\mathcal{Y}_1 \times \mathcal{Y}_2$.

Recall the notion of a *sheaf of categories*, see [Ga3, Sect. 1.1]. Consider the 2-functor

$$(8.4) \quad 2\text{-FM}_{\mathcal{Y}_1 \rightarrow \mathcal{Y}_2} : \mathrm{ShvCat}(\mathcal{Y}_1) \rightarrow \mathrm{ShvCat}(\mathcal{Y}_2), \quad \mathbf{C} \mapsto (p_2)_*((p_1^*(\mathbf{C}))_{\mathcal{G}_{1,2}}),$$

where:

- For a morphism $f : \mathcal{Y}' \rightarrow \mathcal{Y}''$ between prestacks, f^* denotes the pullback functor $\mathrm{ShvCat}(\mathcal{Y}'') \rightarrow \mathrm{ShvCat}(\mathcal{Y}')$, see [Ga3, Sect. 3.1.2] (in *loc. cit.* it is denoted $\mathbf{cores}_f$);
- For a morphism $f_* : \mathcal{Y}' \rightarrow \mathcal{Y}''$ between prestacks, f_* denotes the pushforward functor $\mathrm{ShvCat}(\mathcal{Y}') \rightarrow \mathrm{ShvCat}(\mathcal{Y}'')$, see [Ga3, Sect. 3.1.3] (in *loc. cit.* it is denoted $\mathbf{ind}_f$);
- For a prestack $\mathcal{Y}$, and $\mathbf{C} \in \mathrm{ShvCat}(\mathcal{Y})$ and a $\mathbb{G}_m$ -gerbe $\mathcal{G}$ on $\mathcal{Y}$, we denote by $\mathbf{C}_{\mathcal{G}}$ the twist of $\mathbf{C}$ by $\mathcal{G}$, see [GLys, Sect. 1.7.2].

We shall say that the map (8.3) is of 2-Fourier-Mukai type if the functor (8.4) is an equivalence.

Note that the notion of being of 2-Fourier-Mukai type is in general asymmetric.

8.1.3. *Example.* Let Γ be a finite abelian group, and let $\Gamma^\vee$ be its Cartier dual. Take

$$\mathcal{Y}_1 = B^2(\Gamma)_{\mathrm{et}} =: \mathrm{Ger}_\Gamma(\mathrm{pt})$$

and

$$\mathcal{Y}_2 = \Gamma^\vee.$$

Then $\mathrm{ShvCat}(\mathcal{Y}_1)$ is the 2-category of DG categories acted on by pt/Γ . In other words, these are categories $\mathbf{C}$ equipped with an action of Γ on $\mathrm{Id}_{\mathbf{C}}$. Decomposing with respect to the characters of Γ , we obtain that a datum of such $\mathbf{C}$ is equivalent to the datum of a category graded by $\Gamma^\vee$

$$(8.5) \quad \mathbf{C} \mapsto \{\mathbf{C}_\chi, \chi \in \Gamma^\vee\}.$$

Evaluation defines a map

$$(8.6) \quad \mathrm{Ger}_\Gamma(\mathrm{pt}) \times \Gamma^\vee \rightarrow \mathrm{Ge}_{\mathbb{G}_m}(\mathrm{pt}).$$

We claim that (8.6) is of 2-Fourier-Mukai type.

Indeed, unwinding the definitions, we obtain that the functor $2\text{-FM}_{\mathrm{Ger}_\Gamma \rightarrow \Gamma^\vee}$ is given exactly by (8.5), and hence is an equivalence.

8.1.4. Swapping the factors in (8.6) we obtain a pairing

$$(8.7) \quad \Gamma^\vee \times \mathrm{Ger}_\Gamma(\mathrm{pt}) \rightarrow \mathrm{Ge}_{\mathbb{G}_m}(\mathrm{pt}),$$

and it is easy to see that it is also of 2-Fourier-Mukai type.

Indeed, the corresponding functor $2\text{-FM}_{\Gamma^\vee \rightarrow \mathrm{Ger}_\Gamma(\mathrm{pt})}$ is the inverse of $2\text{-FM}_{\mathrm{Ger}_\Gamma(\mathrm{pt}) \rightarrow \Gamma^\vee}$ up to the inversion on Γ .

Remark 8.1.5. The central players in the paper [Ga3] are prestacks that are *1-affine*, i.e., those for each the functor of *enhanced* global sections

$$(8.8) \quad \mathrm{ShvCat}(\mathcal{Y}) \xrightarrow{\Gamma^{\mathrm{enh}}(\mathcal{Y}, -)} \mathrm{QCoh}(\mathcal{Y})\text{-}\mathbf{mod}$$

is an equivalence.

Note that the prestack $\mathrm{Ger}(\mathrm{pt})$ is *not* 1-affine. Namely $\mathrm{QCoh}(\mathrm{Ger}(\mathrm{pt})) \simeq \mathrm{Vect}$, and the functor (8.8) sends $\mathbf{C}$ as above to $\mathbf{C}_0$, i.e., the fiber of $2\text{-FM}_{\mathrm{Ger}(\mathrm{pt}) \rightarrow \Gamma^\vee}(\mathbf{C})$ at the point $0 \in \Gamma^\vee$.

8.1.6. *Example.* For Γ as above, take

$$\mathcal{Y}_1 := \mathrm{pt}/\Gamma \text{ and } \mathcal{Y}_2 := \mathrm{pt}/\Gamma^\vee.$$

Cup product defines a map

$$(8.9) \quad \mathrm{pt}/\Gamma \times \mathrm{pt}/\Gamma^\vee \rightarrow \mathrm{Ge}_{\mathbb{G}_m}(\mathrm{pt}).$$

We claim that (8.9) is of 2-Fourier-Mukai type.

Note that $\mathrm{ShvCat}(\mathrm{pt}/\Gamma)$ (resp., $\mathrm{ShvCat}(\mathrm{pt}/\Gamma^\vee)$) identifies with the 2-category of DG categories equipped with an action of $\mathrm{QCoh}(\Gamma)$ (resp., $\mathrm{QCoh}(\Gamma^\vee)$), viewed as a monoidal category with respect to *convolution*. Note also that pt/Γ is 1-affine, and

$$\mathrm{QCoh}(\mathrm{pt}/\Gamma) \simeq \mathrm{Rep}(\Gamma).$$

Unwinding the definitions, we obtain that the functor $2\text{-FM}_{\mathrm{pt}/\Gamma \rightarrow \mathrm{pt}/\Gamma^\vee}$ identifies with

$$\mathrm{ShvCat}(\mathrm{pt}/\Gamma) \xrightarrow{\Gamma^{\mathrm{enh}}(\mathrm{pt}/\Gamma, -)} \mathrm{Rep}(\Gamma)\text{-}\mathbf{mod} \simeq \mathrm{QCoh}(\Gamma^\vee)\text{-}\mathbf{mod},$$

where we identify

$$\mathrm{Rep}(\Gamma) \simeq \mathrm{QCoh}(\Gamma^\vee)$$

by Fourier transform.

Note also that the composition

$$2\text{-FM}_{\mathrm{pt}/\Gamma^\vee \rightarrow \mathrm{pt}/\Gamma} \circ 2\text{-FM}_{\mathrm{pt}/\Gamma \rightarrow \mathrm{pt}/\Gamma^\vee}$$

is the identity endofunctor of $\mathrm{ShvCat}(\mathrm{pt}/\Gamma)$ up to the inversion involution on Γ .

8.2. 2-Fourier-Mukai transform and Poincaré duality. In this subsection we consider a particular pair of prestacks that are 2-Fourier-Mukai dual to each other.

Both sides have to do with gerbes for a finite abelian group on a smooth and complete curve X .

8.2.1. Let Γ be a finite abelian group as above. Take

$$\mathcal{Y}_1 := \mathrm{Ge}_\Gamma(X) \text{ and } \mathcal{Y}_2 := \mathrm{Ge}_{\Gamma^\vee(1)}(X),$$

where $(-)(1)$ denotes the Tate twist, so that

$$\Gamma^\vee \simeq \mathrm{Hom}(\Gamma, \mathbb{Z}/n\mathbb{Z})(1)$$

for $n \cdot \Gamma = 0$.

Verdier duality defines a pairing

$$B^2(C(X, \Gamma)) \times B^2(C(X, \Gamma^\vee(1))) \rightarrow B^2(\mu_\infty).$$

Applying étale sheafification, we obtain a pairing

$$(8.10) \quad \mathrm{Ge}_\Gamma(X) \times \mathrm{Ge}_{\Gamma^\vee(1)}(X) \rightarrow \mathrm{Ge}_{\mathbb{G}_m}(\mathrm{pt}).$$

8.2.2. We claim:

Theorem 8.2.3.

(a) *The pairing (8.10) is of 2-Fourier-Mukai type.*

(b) *The composition*

$$2\text{-FM}_{\mathrm{Ge}_{\Gamma^\vee(1)}(X) \rightarrow \mathrm{Ge}_\Gamma(X)} \circ 2\text{-FM}_{\mathrm{Ge}_\Gamma(X) \rightarrow \mathrm{Ge}_{\Gamma^\vee(1)}(X)}$$

is the involution of $\mathrm{ShvCat}(\mathrm{Ge}_\Gamma(X))$ coming from the inversion on Γ ,

$$(\Gamma^\vee(1))^\vee(1) \simeq \Gamma.$$

8.2.4. *Proof of Theorem 8.2.3.* Choose a point $x \in X$. Then we can split

$$\mathrm{Ge}_\Gamma(X) \simeq \mathrm{Ge}_\Gamma(\mathrm{pt}) \times \mathrm{pt} / H_{\mathrm{et}}^1(X, \Gamma) \times H_{\mathrm{et}}^2(X, \Gamma)$$

and

$$\mathrm{Ge}_{\Gamma^\vee(1)}(X) \simeq \mathrm{Ge}_{\Gamma^\vee(1)}(\mathrm{pt}) \times \mathrm{pt} / H_{\mathrm{et}}^1(X, \Gamma^\vee(1)) \times H_{\mathrm{et}}^2(X, \Gamma^\vee(1)).$$

The pairing (8.10) splits as a product of:

- The pairing (8.6), where we identify $H^2(X, \Gamma^\vee(1)) \simeq \Gamma^\vee$;
- The pairing (8.6) with the two sides swapped, where we identify $H_{\mathrm{et}}^2(X, \Gamma) \simeq (\Gamma^\vee(1))^\vee$;
- The pairing (8.9), where we identify $H_{\mathrm{et}}^1(X, \Gamma^\vee(1)) \simeq H_{\mathrm{et}}^1(X, \Gamma)^\vee$.

Now the assertion of the theorem follows by combining the examples from Sects. 8.1.3, 8.1.4 and 8.1.6.

□[Theorem 8.2.3]

8.2.5. For a $\Gamma^\vee(1)$ -gerbe $\mathfrak{G}_{\Gamma^\vee(1)}$ on X , let $\mathcal{G}_{\mathfrak{G}_{\Gamma^\vee(1)}}$ be the $\mathbb{G}_m$ -gerbe on $\mathrm{Ge}_\Gamma(X)$, corresponding to the restriction of the map (8.10) along

$$\mathrm{Ge}_\Gamma(X) \times \{\mathfrak{G}_{\Gamma^\vee(1)}\} \rightarrow \mathrm{Ge}_\Gamma(X) \times \mathrm{Ge}_{\Gamma^\vee(1)}(X).$$

We obtain that for an object

$$\underline{\mathbf{C}}_\Gamma \in \mathrm{ShvCat}(\mathrm{Ge}_\Gamma(X))$$

and the corresponding object

$$\underline{\mathbf{C}}_{\Gamma^\vee(1)} := 2\text{-FM}_{\mathrm{Ge}_\Gamma(X) \rightarrow \mathrm{Ge}_{\Gamma^\vee(1)}(X)}(\underline{\mathbf{C}}_\Gamma) \in \mathrm{ShvCat}(\mathrm{Ge}_{\Gamma^\vee(1)}(X)),$$

we have

$$(8.11) \quad \underline{\mathbf{C}}_{\Gamma^\vee(1)}|_{\mathfrak{G}_{\Gamma^\vee(1)}} \simeq \Gamma\left(\mathrm{Ge}_\Gamma(X), (\underline{\mathbf{C}}_\Gamma)_{\mathcal{G}_{\mathfrak{G}_{\Gamma^\vee(1)}}}\right),$$

where:

- $(-)|_{\mathfrak{G}_{\Gamma^\vee(1)}}$ denotes the fiber of a given sheaf of categories at the point $\mathfrak{G}_{\Gamma^\vee(1)} \in \mathrm{Ge}_{\Gamma^\vee(1)}(X)$;
- $(-)_{\mathcal{G}}$ denotes the twist of a given sheaf of categories over some prestack by a $\mathbb{G}_m$ -gerbe $\mathcal{G}$ on that prestack.

By the involutivity assertion in Theorem 8.2.3, we also obtain that for a Γ -gerbe $\mathfrak{G}_\Gamma$ on X , and the corresponding $\mathbb{G}_m$ -gerbe $\mathcal{G}_{\mathfrak{G}_\Gamma}$ on $\mathrm{Ge}_{\Gamma^\vee(1)}(X)$, we have

$$(8.12) \quad \underline{\mathbf{C}}_\Gamma|_{\mathfrak{G}_\Gamma^{-1}} \simeq \Gamma\left(\Gamma^\vee(1), (\underline{\mathbf{C}}_{\Gamma^\vee(1)})_{\mathcal{G}_{\mathfrak{G}_\Gamma}}\right).$$

8.2.6. For $\underline{\mathbf{C}}_\Gamma$ as above, denote

$$\mathbf{C}_\Gamma := \Gamma(\mathrm{Ge}_\Gamma(X), \underline{\mathbf{C}}_\Gamma) \text{ and } \mathbf{C}_{\Gamma^\vee(1)} := \Gamma(\mathrm{Ge}_{\Gamma^\vee(1)}(X), \underline{\mathbf{C}}_{\Gamma^\vee(1)}).$$

Let $\mathfrak{G}_\Gamma^0$ (resp., $\mathfrak{G}_{\Gamma^\vee(1)}^0$) denote the trivial Γ -gerbe (resp., $\Gamma^\vee(1)$ -gerbe) on X . As a particular case of (8.11), we obtain an equivalence

$$(8.13) \quad \underline{\mathbf{C}}_{\Gamma^\vee(1)}|_{\mathfrak{G}_{\Gamma^\vee(1)}^0} \simeq \mathbf{C}_\Gamma,$$

and as a particular case of (8.12) we obtain an equivalence

$$(8.14) \quad \underline{\mathbf{C}}_\Gamma|_{\mathfrak{G}_\Gamma^0} \simeq \mathbf{C}_{\Gamma^\vee(1)}.$$

8.2.7. Note that $\mathbf{C}_\Gamma$ is a category acted on by $\mathrm{QCoh}(\mathrm{Ger}_\Gamma(X))$. For

$$\alpha \in \Gamma(-1) \simeq H_{\mathrm{et}}^2(X, \Gamma) \simeq \pi_0(\mathrm{Ger}_\Gamma(X)),$$

consider the corresponding idempotent

$$\mathcal{O}_{\mathrm{Ger}_\Gamma(X), \alpha} \in \mathrm{QCoh}(\mathrm{Ger}_\Gamma(X))$$

as acting on $\mathbf{C}_\Gamma$.

Note also that $\Gamma^\vee(1)$ acts by automorphisms of the identity functor on $\mathfrak{G}_{\Gamma^\vee(1)}^0$, and hence also by the automorphisms of the identity functor of $\underline{\mathbf{C}}_{\Gamma^\vee(1)}|_{\mathfrak{G}_{\Gamma^\vee(1)}^0}$. For

$$\alpha \in \Gamma(-1) \simeq (\Gamma^\vee(1))^\vee$$

let $\mathbf{P}_\alpha$ denote the corresponding idempotent on $\underline{\mathbf{C}}_{\Gamma^\vee(1)}|_{\mathfrak{G}_{\Gamma^\vee(1)}^0}$.

Unwinding the construction, we obtain:

Lemma 8.2.8. *Under the identification (8.13), the action of $\mathcal{O}_{\mathrm{Ger}_\Gamma(X), \alpha}$ on $\mathbf{C}_\Gamma$ corresponds to the action of $\mathbf{P}_\alpha$ on $\underline{\mathbf{C}}_{\Gamma^\vee(1)}|_{\mathfrak{G}_{\Gamma^\vee(1)}^0}$.*

8.2.9. As in Sect. 4.2.4, a point

$$\mathcal{P}_{\Gamma^\vee(1)} \in \mathrm{Bun}_{\Gamma^\vee(1)}$$

gives rise to a line bundle denoted $\mathcal{L}_{\mathcal{P}_{\Gamma^\vee(1)}}$ on $\mathrm{Ger}_\Gamma(X)$.

In particular, we consider the endofunctor

$$\mathcal{L}_{\mathcal{P}_{\Gamma^\vee(1)}} \otimes (-)$$

of $\mathbf{C}_\Gamma$.

We can view $\mathcal{P}_{\Gamma^\vee(1)}$ itself as an automorphism of $\mathfrak{G}_{\Gamma^\vee(1)}^0$. And as such, it induces an autoequivalence of $\underline{\mathbf{C}}_{\Gamma^\vee(1)}|_{\mathfrak{G}_{\Gamma^\vee(1)}^0}$.

Unwinding the construction, we obtain:

Lemma 8.2.10. *Under the identification (8.13), the action of $\mathcal{L}_{\mathcal{P}_{\Gamma^\vee(1)}}^{\otimes -1}$ on $\mathbf{C}_\Gamma$ corresponds to the action of $\mathcal{P}_{\Gamma^\vee(1)}$ on $\underline{\mathbf{C}}_{\Gamma^\vee(1)}|_{\mathfrak{G}_{\Gamma^\vee(1)}^0}$.*

8.3. Example: the usual Fourier-Mukai transform.

8.3.1. Let

$$1 \rightarrow \Gamma \rightarrow T_1 \rightarrow T \rightarrow 1$$

be an isogeny of tori. Consider the dual isogeny

$$1 \rightarrow \Gamma^\vee(1) \rightarrow T^\vee \rightarrow T_1^\vee \rightarrow 1.$$

Consider the corresponding maps

$$\mathbf{p}_\Gamma : \mathrm{Bun}_T \rightarrow \mathrm{Ger}_\Gamma(X) \text{ and } \mathbf{p}_{\Gamma^\vee(1)} : \mathrm{Bun}_{T_1^\vee} \rightarrow \mathrm{Ger}_{\Gamma^\vee(1)}(X).$$

8.3.2. Consider the unit sheaf of categories

$$\underline{\mathrm{QCoh}}(\mathrm{Bun}_T)$$

over Bun_T , and let

$$\underline{\mathrm{QCoh}}^\Gamma(\mathrm{Bun}_T) := (\mathbf{p}_\Gamma)_*(\underline{\mathrm{QCoh}}(\mathrm{Bun}_T))$$

be its direct image along $\mathbf{p}_\Gamma$, viewed as a sheaf of categories over $\mathrm{Ger}_\Gamma(X)$.

Similarly, consider

$$\underline{\mathrm{QCoh}}^{\Gamma^\vee(1)}(\mathrm{Bun}_{T_1^\vee}) := (\mathbf{p}_{\Gamma^\vee(1)})_*(\underline{\mathrm{QCoh}}(\mathrm{Bun}_{T_1^\vee}))$$

as a sheaf of categories over $\mathrm{Ger}_{\Gamma^\vee(1)}(X)$.

8.3.3. We claim:

Lemma 8.3.4. *With respect to the equivalence $2\text{-FM}_{\text{Ge}_\Gamma(X) \rightarrow \text{Ge}_{\Gamma^\vee(1)}(X)}$, the objects*

$$\underline{\text{QCoh}}^\Gamma(\text{Bun}_T) \text{ and } \underline{\text{QCoh}}^{\Gamma^\vee(1)}(\text{Bun}_{T_1^\vee})$$

correspond to one another.

The proof follows from the usual properties of the usual Fourier-Mukai equivalences

$$\text{QCoh}(\text{Bun}_T) \xrightarrow{\text{FM}} \text{QCoh}(\text{Bun}_{T^\vee}) \text{ and } \text{QCoh}(\text{Bun}_{T_1}) \xrightarrow{\text{FM}} \text{QCoh}(\text{Bun}_{T_1^\vee}).$$

8.4. Some further compatibilities. In this subsection we review some further properties of the equivalence of Theorem 8.2.3 that will be needed for the proof of Theorem 8.5.8 below.

This subsection could be skipped on the first pass and returned to when necessary.

8.4.1. Note that for any $\mathfrak{G}_{\Gamma^\vee(1)} \in \text{Ge}_{\Gamma^\vee(1)}(X)$ we have the evaluation functor

$$\text{ev}|_{\mathfrak{G}_{\Gamma^\vee(1)}} : \mathbf{C}_{\Gamma^\vee(1)} \rightarrow (\underline{\mathbf{C}}_{\Gamma^\vee(1)})|_{\mathfrak{G}_{\Gamma^\vee(1)}}.$$

More generally, for a $\mathbb{G}_m$ -gerbe $\mathcal{G}$ on $\text{Ge}_{\Gamma^\vee(1)}(X)$, we have the evaluation functor

$$\text{ev}_{\mathcal{G}}|_{\mathfrak{G}_{\Gamma^\vee(1)}} : \Gamma(\text{Ge}_{\Gamma^\vee(1)}(X), (\underline{\mathbf{C}}_{\Gamma^\vee(1)})_{\mathcal{G}}) \rightarrow (\underline{\mathbf{C}}_{\Gamma^\vee(1)})_{\mathcal{G}}|_{\mathfrak{G}_{\Gamma^\vee(1)}}.$$

8.4.2. For a given $\mathfrak{G}_{\Gamma^\vee(1)} \in \text{Ge}_{\Gamma^\vee(1)}(X)$, let

$$(8.15) \quad (\text{ev}_{\mathcal{G}_{\Gamma^\vee(1)}}|_{\mathfrak{G}_\Gamma^0})^L : \underline{\mathbf{C}}_\Gamma|_{\mathfrak{G}_\Gamma^0} \rightarrow \Gamma(\text{Ge}_\Gamma(X), (\underline{\mathbf{C}}_\Gamma)_{\mathcal{G}_{\Gamma^\vee(1)}})$$

be the left adjoint of the evaluation functor

$$\Gamma(\text{Ge}_\Gamma(X), (\underline{\mathbf{C}}_\Gamma)_{\mathcal{G}_{\Gamma^\vee(1)}}) \xrightarrow{\text{ev}_{\mathcal{G}_{\Gamma^\vee(1)}}|_{\mathfrak{G}_\Gamma^0}} (\underline{\mathbf{C}}_\Gamma)_{\mathcal{G}_{\Gamma^\vee(1)}}|_{\mathfrak{G}_\Gamma^0} \simeq \underline{\mathbf{C}}_\Gamma|_{\mathfrak{G}_\Gamma^0}.$$

8.4.3. Unwinding the constructions we obtain:

Lemma 8.4.4. *The following diagram commutes:*

$$(8.14) \quad \begin{array}{ccc} \mathbf{C}_{\Gamma^\vee(1)} & \xrightarrow{\text{ev}|_{\mathfrak{G}_{\Gamma^\vee(1)}}} & (\underline{\mathbf{C}}_{\Gamma^\vee(1)})|_{\mathfrak{G}_{\Gamma^\vee(1)}} \\ \uparrow \sim & & \sim \uparrow (8.11) \\ \underline{\mathbf{C}}_\Gamma|_{\mathfrak{G}_\Gamma^0} & \xrightarrow{(8.15)} & \Gamma(\text{Ge}_\Gamma(X), (\underline{\mathbf{C}}_\Gamma)_{\mathcal{G}_{\Gamma^\vee(1)}}). \end{array}$$

8.4.5. Let now

$$0 \rightarrow \Gamma_1 \rightarrow \Gamma \rightarrow \Gamma_2 \rightarrow 0$$

be a short exact sequence of finite abelian groups, and let

$$0 \rightarrow \Gamma_2^\vee \rightarrow \Gamma^\vee \rightarrow \Gamma_1^\vee \rightarrow 0$$

be the dual short exact sequence.

Fix $\mathfrak{G}_{\Gamma_1^\vee(1)} \in \text{Ge}_{\Gamma_1^\vee(1)}(X)$, and let $\mathcal{G}_{\mathfrak{G}_{\Gamma_1^\vee(1)}}$ be the corresponding $\mathbb{G}_m$ -gerbe on $\text{Ge}_{\Gamma_1}(X)$.

Generalizing (8.11), we have:

Lemma 8.4.6. *There is a canonical equivalence*

$$\Gamma(\text{Ge}_{\Gamma_1}(X), (\underline{\mathbf{C}}_\Gamma|_{\text{Ge}_{\Gamma_1}(X)})_{\mathcal{G}_{\mathfrak{G}_{\Gamma_1^\vee(1)}}}) \simeq \Gamma(\text{Ge}_{\Gamma^\vee(1)}(X) \times_{\text{Ge}_{\Gamma_1^\vee(1)}(X)} \{\mathfrak{G}_{\Gamma_1^\vee(1)}\}, \underline{\mathbf{C}}_{\Gamma^\vee(1)}).$$

8.5. Two sheaves associated with automorphic category. Let G be a semi-simple group and consider the category

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G).$$

We will upgrade it to two objects

$$\underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{Z_G}(\mathrm{Bun}_{G_{\mathrm{ad}}}) \in \mathrm{ShvCat}(\mathrm{Ge}_{Z_G}(X)) \text{ and } \underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{\pi_1(\check{G})}(\mathrm{Bun}_G) \in \mathrm{ShvCat}(\mathrm{Ge}_{\pi_1(G)}(X)).$$

We will state Theorem 8.5.8 that says that the above two objects correspond to one another under the 2-Fourier-Mukai transform. The nature of the two constructions will then immediately yield Theorems 5.1.5 and 5.1.7.

8.5.1. The short exact sequence of groups

$$1 \rightarrow Z_G \rightarrow G \rightarrow G_{\mathrm{ad}} \rightarrow 1$$

gives rise to a map

$$(8.16) \quad \mathfrak{p}_{Z_G} : \mathrm{Bun}_{G_{\mathrm{ad}}} \rightarrow \mathrm{Ge}_{Z_G}(X).$$

Consider the induced map

$$(\mathfrak{p}_{Z_G})_{\mathrm{dR}} : (\mathrm{Bun}_{G_{\mathrm{ad}}})_{\mathrm{dR}} \rightarrow (\mathrm{Ge}_{Z_G}(X))_{\mathrm{dR}}.$$

8.5.2. Note, however, that (by nil-invariance of étale cohomology) for a finite abelian group Γ , the map of prestacks

$$\mathrm{Ge}_{\Gamma}(X) \rightarrow (\mathrm{Ge}_{\Gamma}(X))_{\mathrm{dR}}$$

is an isomorphism.

Hence, we can regard $(\mathfrak{p}_{Z_G})_{\mathrm{dR}}$ as a map

$$(8.17) \quad (\mathrm{Bun}_{G_{\mathrm{ad}}})_{\mathrm{dR}} \rightarrow \mathrm{Ge}_{Z_G}(X).$$

8.5.3. The category $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_{\mathrm{ad}}})$ is (tautologically) the category of global sections of a sheaf of categories, denoted

$$\underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{G_{\mathrm{ad}}}(\mathrm{Bun}_{G_{\mathrm{ad}}})$$

over $(\mathrm{Bun}_{G_{\mathrm{ad}}})_{\mathrm{dR}}$.

Set

$$\underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{Z_G}(\mathrm{Bun}_{G_{\mathrm{ad}}}) := ((\mathfrak{p}_{Z_G})_{\mathrm{dR}})_* (\underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{G_{\mathrm{ad}}}(\mathrm{Bun}_{G_{\mathrm{ad}}})) \in \mathrm{ShvCat}(\mathrm{Ge}_{Z_G}(X)).$$

8.5.4. Tautologically, we have

$$(8.18) \quad \Gamma \left(\mathrm{Ge}_{Z_G}(X), \underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{Z_G}(\mathrm{Bun}_{G_{\mathrm{ad}}}) \right) \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_{\mathrm{ad}}}).$$

In addition,

$$(8.19) \quad \left(\underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{Z_G}(\mathrm{Bun}_{G_{\mathrm{ad}}}) \right) |_{\mathfrak{G}_{Z_G}^0} \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G),$$

see the notations in Sect. 8.2.5.

More generally, for a given $\mathfrak{G}_{Z_G} \in \mathrm{Ge}_{Z_G}(X)$, we have

$$(8.20) \quad \left(\underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{Z_G}(\mathrm{Bun}_{G_{\mathrm{ad}}}) \right) |_{\mathfrak{G}_{Z_G}} \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G, \mathfrak{G}_{Z_G}}),$$

where

$$\mathrm{Bun}_{G, \mathfrak{G}_{Z_G}} := \mathrm{Bun}_G \times_{\mathrm{Ge}_{Z_G}(X)} \{\mathfrak{G}_{Z_G}\}.$$

Remark 8.5.5. We can think $\mathrm{Bun}_{G, \mathfrak{S}_{Z_G}}$ as follows: pick a G_{ad} -torsor $\mathcal{P}_{G_{\mathrm{ad}}}$ that maps to $\mathfrak{S}_{Z_G}$, and let

$$G_{\mathcal{P}_{G_{\mathrm{ad}}}}$$

be the corresponding (non-pure) inner form of the constant group-scheme with fiber G .

Then

$$\mathrm{Bun}_{G, \mathfrak{S}_{Z_G}} \simeq \mathrm{Bun}_{G_{\mathcal{P}_{G_{\mathrm{ad}}}}}.$$

(Note that different choices for $\mathcal{P}_{G_{\mathrm{ad}}}$ differ by G -torsors, and hence the corresponding moduli spaces $\mathrm{Bun}_{G_{\mathcal{P}_{G_{\mathrm{ad}}}}}$ are a priori canonically isomorphic.)

8.5.6. We now consider $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ as equipped with the spectral action of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$. Since the stack $\mathrm{LS}_{\check{G}}$ is 1-affine, we can canonically attach to $\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ an object

$$(8.21) \quad \underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{\check{G}}(\mathrm{Bun}_G) \in \mathrm{ShvCat}(\mathrm{LS}_{\check{G}}),$$

The short exact sequence of groups⁹

$$1 \rightarrow \pi_1(\check{G}) \rightarrow \check{G}_{\mathrm{sc}} \rightarrow \check{G} \rightarrow 1$$

give rise to a map

$$(8.22) \quad \mathfrak{p}_{\pi_1(\check{G})} : \mathrm{LS}_{\check{G}} \rightarrow \mathrm{Ge}_{\pi_1(\check{G})}(X).$$

Denote

$$\underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{\pi_1(\check{G})}(\mathrm{Bun}_G) := (\mathfrak{p}_{\pi_1(\check{G})})_* (\underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{\check{G}}(\mathrm{Bun}_G)) \in \mathrm{ShvCat}(\mathrm{Ge}_{\pi_1(\check{G})}(X)).$$

Note that tautologically,

$$(8.23) \quad \Gamma \left(\mathrm{Ge}_{\pi_1(\check{G})}(X), \underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{\pi_1(\check{G})}(\mathrm{Bun}_G) \right) \simeq \mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Bun}_G).$$

8.5.7. We will prove:

Theorem 8.5.8. *Under the identification $\pi_1(\check{G}) \simeq (Z_G)^\vee(1)$, we have*

$$2\text{-FM}_{\mathrm{Ge}_{Z_G}(X) \rightarrow \mathrm{Ge}_{\pi_1(\check{G})}(X)} \left(\underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{Z_G}(\mathrm{Bun}_{G_{\mathrm{ad}}}) \right) \simeq \underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{\pi_1(\check{G})}(\mathrm{Bun}_G),$$

up to the inversion involution¹⁰ on Z_G .

Combining Theorem 8.5.8 with the involutivity assertion in Theorem 8.2.3, we obtain:

Corollary 8.5.9. *Under the identification $Z_G \simeq (\pi_1(\check{G}))^\vee(1)$, we have:*

$$2\text{-FM}_{\mathrm{Ge}_{\pi_1(\check{G})}(X) \rightarrow \mathrm{Ge}_{Z_G}(X)} \left(\underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{\pi_1(\check{G})}(\mathrm{Bun}_G) \right) \simeq \underline{\mathrm{D}\text{-mod}}_{\frac{1}{2}}^{Z_G}(\mathrm{Bun}_{G_{\mathrm{ad}}}).$$

8.5.10. Let us show how Corollary 8.5.9 implies Theorems 5.1.5 and 5.1.7.

Indeed, the two theorems follow immediately from Lemma 8.2.8 and 8.2.10, respectively.

□[Theorems 5.1.5 and 5.1.7]

8.6. Proof of Theorem 8.5.8.

⁹Note that $\check{G}_{\mathrm{sc}}$ is the Langlands dual of G_{ad} .

¹⁰The inversion involution has to do with our normalization of the geometric Satake equivalence.

8.6.1. We start by constructing a functor

$$(8.24) \quad \underline{\mathrm{D-mod}}_{\frac{1}{2}}^{\pi_1(\check{G})}(\mathrm{Bun}_G) \rightarrow 2\text{-FM}_{\mathrm{Ge}_{Z_G}(X) \rightarrow \mathrm{Ge}_{\pi_1(\check{G})}(X)} \left(\underline{\mathrm{D-mod}}_{\frac{1}{2}}^{Z_G}(\mathrm{Bun}_{G_{\mathrm{ad}}}) \right),$$

up to the inversion involution.

Since $\mathrm{Ge}_{\pi_1(\check{G})}(X)$ is algebro-geometrically discrete, the datum of (8.24) consists of the data of functors

$$(8.25) \quad \underline{\mathrm{D-mod}}_{\frac{1}{2}}^{\pi_1(\check{G})}(\mathrm{Bun}_G)|_{\mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}} \rightarrow 2\text{-FM}_{\mathrm{Ge}_{Z_G}(X) \rightarrow \mathrm{Ge}_{\pi_1(\check{G})}(X)} \left(\underline{\mathrm{D-mod}}_{\frac{1}{2}}^{Z_G}(\mathrm{Bun}_{G_{\mathrm{ad}}}) \right)|_{\mathfrak{G}_{\pi_1(\check{G})}}$$

that depend functorially on $\mathfrak{G}_{\pi_1(\check{G})} \in \mathrm{Ge}_{\pi_1(\check{G})}(X)$.

Applying (8.11), the datum of (8.25) is equivalent to that of a functor

$$(8.26) \quad \underline{\mathrm{D-mod}}_{\frac{1}{2}}^{\pi_1(\check{G})}(\mathrm{Bun}_G)|_{\mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}} \rightarrow \Gamma \left(\mathrm{Ge}_{Z_G}(X), \underline{\mathrm{D-mod}}_{\frac{1}{2}}^{Z_G}(\mathrm{Bun}_{G_{\mathrm{ad}}})_{\mathfrak{G}_{\pi_1(\check{G})}} \right).$$

We rewrite the right-hand side in (8.26) as

$$(8.27) \quad \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_{\mathrm{ad}}})_{\mathfrak{G}_{\pi_1(\check{G})}},$$

where by a slight abuse of notation we regard $\mathfrak{G}_{\pi_1(\check{G})}$ as a $\mathbb{G}_m$ -gerbe on $(\mathrm{Bun}_{G_{\mathrm{ad}}})_{\mathrm{dR}}$.

8.6.2. We rewrite the left-hand side in (8.26) as

$$(8.28) \quad \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})} \mathrm{QCoh}(\mathrm{LS}_{\check{G}_{\mathrm{sc}}, \mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}}),$$

where

$$\mathrm{LS}_{\check{G}_{\mathrm{sc}}, \mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}} := \mathrm{LS}_{\check{G}} \times_{\mathrm{Ge}_{\pi_1(\check{G})}(X)} \{\mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}\}.$$

The notation $\mathrm{LS}_{\check{G}_{\mathrm{sc}}, \mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}}$ expresses the fact that this stack is a twisted form of $\mathrm{LS}_{\check{G}_{\mathrm{sc}}}$. Namely, for $\mathfrak{G}_{\pi_1(\check{G})} = \mathfrak{G}_{\pi_1(\check{G})}^0$ we have

$$\mathrm{LS}_{\check{G}_{\mathrm{sc}}, \mathfrak{G}_{\pi_1(\check{G})}^0} = \mathrm{LS}_{\check{G}_{\mathrm{sc}}}.$$

8.6.3. Let $\mathrm{Rep}(\check{G}_{\mathrm{sc}})_{\mathrm{Ran}, \mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}}$ be the $\mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}$ -twist of $\mathrm{Rep}(\check{G}_{\mathrm{sc}})$, i.e., this is the factorization category that associates to a point $\underline{x} \in \mathrm{Ran}$ the category

$$\mathrm{Rep}(\check{G}_{\mathrm{sc}})_{\underline{x}, \mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}} := \mathrm{QCoh}(\mathrm{LS}_{\check{G}_{\mathrm{sc}}, \mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}}^{\mathrm{reg}}|_{\underline{x}}),$$

where:

- $\mathrm{LS}_{\check{G}_{\mathrm{sc}}, \mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}}^{\mathrm{reg}}|_{\underline{x}} := \mathrm{LS}_{\check{G}, \underline{x}} \times_{\mathrm{Ge}_{\pi_1(\check{G})}(\mathcal{D}_{\underline{x}})} \{\mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}|_{\mathcal{D}_{\underline{x}}}\};$
- $\mathrm{Ge}_{\pi_1(\check{G})}(\mathcal{D}_{\underline{x}})$ denotes the space of $\pi_1(\check{G})$ -gerbes on the formal disc $\mathcal{D}_{\underline{x}}$ around $\underline{x}$.

8.6.4. Recall now that we have the (symmetric) monoidal localization functors

$$\mathrm{Loc}_{\check{G}}^{\mathrm{spec}} : \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}}) \text{ and } \mathrm{Loc}_{\check{G}_{\mathrm{sc}}}^{\mathrm{spec}} : \mathrm{Rep}(\check{G}_{\mathrm{sc}})_{\mathrm{Ran}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}_{\mathrm{sc}}}).$$

We have the corresponding twisted version:

$$\mathrm{Loc}_{\check{G}_{\mathrm{sc}}, \mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}}^{\mathrm{spec}} : \mathrm{Rep}(\check{G}_{\mathrm{sc}})_{\mathrm{Ran}, \mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}} \rightarrow \mathrm{QCoh}(\mathrm{LS}_{\check{G}_{\mathrm{sc}}, \mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}}),$$

and as in the case of $\mathrm{Loc}_{\check{G}_{\mathrm{sc}}}^{\mathrm{spec}}$, one shows that the functor $\mathrm{Loc}_{\check{G}_{\mathrm{sc}}, \mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}}^{\mathrm{spec}}$ is a *localization*, i.e., its right adjoint is fully faithful. See [GLC4, Prop. C.1.7] for a general result encompassing these statements.

8.6.5. Let ϕ denote the tautological map $G \rightarrow G_{\text{ad}}$, and also the map

$$\text{Bun}_G \rightarrow \text{Bun}_{G_{\text{ad}}}.$$

Note that the pullback along ϕ of $\mathcal{G}_{\mathfrak{S}_{\pi_1(\check{G})}}$, viewed as a gerbe on $(\text{Bun}_{G_{\text{ad}}})_{\text{dR}}$ canonically trivializes. Hence, we obtain a functor

$$(8.29) \quad \phi_! : \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}})_{\mathcal{G}_{\mathfrak{S}_{\pi_1(\check{G})}}}.$$

8.6.6. We will construct the following gerbe-twisted version of the Hecke action:

Proposition 8.6.7. *There is a canonically defined action of the monoidal category $\text{Rep}(\check{G}_{\text{sc}})_{\text{Ran}, \mathfrak{S}_{\pi_1(\check{G})}^{\otimes -1}}$ on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}})_{\mathcal{G}_{\mathfrak{S}_{\pi_1(\check{G})}}}$. Moreover, the following properties hold:*

(a) *The functor $\phi_!$ of (8.29) is equivariant with respect to the $\text{Rep}(\check{G})_{\text{Ran}}$ -action on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$ via the tautological functor*

$$\text{Rep}(\check{G})_{\text{Ran}} \rightarrow \text{Rep}(\check{G}_{\text{sc}})_{\text{Ran}, \mathfrak{S}_{\pi_1(\check{G})}^{\otimes -1}}.$$

(b) *The resulting $\text{Rep}(\check{G}_{\text{sc}})_{\text{Ran}, \mathfrak{S}_{\pi_1(\check{G})}^{\otimes -1}}$ -action on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}})_{\mathcal{G}_{\mathfrak{S}_{\pi_1(\check{G})}}}$ factors canonically via the localization functor*

$$\text{Loc}_{\check{G}_{\text{sc}}, \mathfrak{S}_{\pi_1(\check{G})}^{\otimes -1}}^{\text{spec}} : \text{Rep}(\check{G}_{\text{sc}})_{\text{Ran}, \mathfrak{S}_{\pi_1(\check{G})}^{\otimes -1}} \rightarrow \text{QCoh}(\text{LS}_{\check{G}_{\text{sc}}, \mathfrak{S}_{\pi_1(\check{G})}^{\otimes -1}}).$$

The proposition will be proved in Sects. 8.7 and 8.8.

8.6.8. We now return to the sought-for functor (8.26), thought of as a functor

$$(8.30) \quad \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G) \otimes_{\text{QCoh}(\text{LS}_{\check{G}})} \text{QCoh}(\text{LS}_{\check{G}_{\text{sc}}, \mathfrak{S}_{\pi_1(\check{G})}^{\otimes -1}}) \rightarrow (\text{Bun}_{G_{\text{ad}}})_{\mathcal{G}_{\mathfrak{S}_{\pi_1(\check{G})}}}.$$

Its construction follows from Proposition 8.6.7 by considering the adjoint pair

$$\text{QCoh}(\text{LS}_{\check{G}})\text{-mod} \rightleftarrows \text{QCoh}(\text{LS}_{\check{G}_{\text{sc}}, \mathfrak{S}_{\pi_1(\check{G})}^{\otimes -1}})\text{-mod},$$

where the right adjoint is the forgetful functor, and the left adjoint is the tensoring up functor with respect to

$$\text{QCoh}(\text{LS}_{\check{G}}) \rightarrow \text{QCoh}(\text{LS}_{\check{G}_{\text{sc}}, \mathfrak{S}_{\pi_1(\check{G})}^{\otimes -1}}).$$

8.6.9. The next assertion results from Lemma 8.4.4:

Lemma 8.6.10. *The functor*

$$(8.31) \quad \Gamma \left(\text{Ge}_{\pi_1(\check{G})}(X), \underline{\text{D-mod}}_{\frac{1}{2}}^{\pi_1(\check{G})}(\text{Bun}_G) \right) \rightarrow \\ \rightarrow \Gamma \left(\text{Ge}_{\pi_1(\check{G})}(X), 2\text{-FM}_{\text{Ge}_{Z_G}(X) \rightarrow \text{Ge}_{\pi_1(\check{G})}(X)} \left(\underline{\text{D-mod}}_{\frac{1}{2}}^{Z_G}(\text{Bun}_{G_{\text{ad}}}) \right) \right)$$

induced by (8.24), identifies via

$$\Gamma \left(\text{Ge}_{\pi_1(\check{G})}(X), \underline{\text{D-mod}}_{\frac{1}{2}}^{\pi_1(\check{G})}(\text{Bun}_G) \right) \simeq \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$$

and

$$(8.32) \quad \Gamma \left(\text{Ge}_{\pi_1(\check{G})}(X), 2\text{-FM}_{\text{Ge}_{Z_G}(X) \rightarrow \text{Ge}_{\pi_1(\check{G})}(X)} \left(\underline{\text{D-mod}}_{\frac{1}{2}}^{Z_G}(\text{Bun}_{G_{\text{ad}}}) \right) \right) \stackrel{(8.12)}{\simeq} \\ \simeq \left(\underline{\text{D-mod}}_{\frac{1}{2}}^{Z_G}(\text{Bun}_{G_{\text{ad}}}) \right) |_{\mathfrak{S}_{Z_G}^0} \simeq \text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$$

with the identity endofunctor on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)$.

8.6.11. We are now ready to prove that the map (8.24) is an equivalence.

Note that by point (b) of Proposition 8.6.7, for every $\mathfrak{G}_{\pi_1(\tilde{G})} \in \text{Ge}_{\pi_1(\tilde{G})}(X)$, the category

$$2\text{-FM}_{\text{Ge}_{Z_G}(X) \rightarrow \text{Ge}_{\pi_1(\tilde{G})}(X)} \left(\underline{\text{D-mod}}_{\frac{1}{2}}^{Z_G}(\text{Bun}_{G_{\text{ad}}}) \right) |_{\mathfrak{G}_{\pi_1(\tilde{G})}^{\otimes -1}} \simeq \text{D-mod}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}})_{\mathfrak{G}_{\pi_1(\tilde{G})}}$$

is a module over $\text{QCoh}(\text{LS}_{\tilde{G}_{\text{sc}}, \mathfrak{G}_{\pi_1(\tilde{G})}^{\otimes -1}})$, while $\text{LS}_{\tilde{G}_{\text{sc}}, \mathfrak{G}_{\pi_1(\tilde{G})}^{\otimes -1}}$ is 1-affine. Hence, the object

$$2\text{-FM}_{\text{Ge}_{Z_G}(X) \rightarrow \text{Ge}_{\pi_1(\tilde{G})}(X)} \left(\underline{\text{D-mod}}_{\frac{1}{2}}^{Z_G}(\text{Bun}_{G_{\text{ad}}}) \right) \in \text{ShvCat}(\text{Ge}_{\pi_1(\tilde{G})}(X))$$

is canonically of the form

$$(\mathfrak{p}_{\pi_1(\tilde{G})})_* \left(\underline{\text{D-mod}}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}})_{\mathfrak{G}_{\text{univ}}} \right),$$

where

$$\underline{\text{D-mod}}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}})_{\mathfrak{G}_{\text{univ}}} \in \text{ShvCat}(\text{LS}_{\tilde{G}}).$$

Moreover, by construction, the map (8.24) comes from a map

$$(8.33) \quad \underline{\text{D-mod}}_{\frac{1}{2}}^{\tilde{G}}(\text{Bun}_G) \rightarrow \underline{\text{D-mod}}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}})_{\mathfrak{G}_{\text{univ}}}$$

in $\text{ShvCat}(\text{LS}_{\tilde{G}})$, where $\underline{\text{D-mod}}_{\frac{1}{2}}^{\tilde{G}}(\text{Bun}_G)$ is as in (8.21).

It is sufficient to show that the map (8.33) is an equivalence. However, the stack $\text{LS}_{\tilde{G}}$ is 1-affine, and hence, the functor

$$\Gamma(\text{LS}_{\tilde{G}}, -) : \text{ShvCat}(\text{LS}_{\tilde{G}}) \rightarrow \text{DGCat}$$

is conservative.

Hence, it is sufficient to show that the resulting functor

$$\Gamma(\text{LS}_{\tilde{G}}, \underline{\text{D-mod}}_{\frac{1}{2}}^{\tilde{G}}(\text{Bun}_G)) \rightarrow \Gamma(\text{LS}_{\tilde{G}}, \underline{\text{D-mod}}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}})_{\mathfrak{G}_{\text{univ}}})$$

is an equivalence.

However, the latter functor identifies with the functor (8.31), and hence is an equivalence by Lemma 8.6.10.

□[Theorem 8.5.8]

8.7. Proof of Proposition 8.6.7(a).

8.7.1. Let us return to the setting of Sect. 8.2. Let $\underline{x}$ be a point of Ran . Consider the spaces

$$\text{Ge}_{\Gamma}(X)_{\underline{x}} := \text{Fib}(\text{Ge}_{\Gamma}(X) \rightarrow \text{Ge}_{\Gamma}(X - \underline{x}))$$

and

$$\text{Ge}_{\Gamma}(\mathcal{D}_{\underline{x}})_{\underline{x}} := \text{Fib}(\text{Ge}_{\Gamma}(\mathcal{D}_{\underline{x}}) \rightarrow \text{Ge}_{\Gamma}(\mathcal{D}_{\underline{x}}^{\times})),$$

where

$$\mathcal{D}_{\underline{x}}^{\times} := \mathcal{D}_{\underline{x}} - \underline{x}.$$

Restriction along $\mathcal{D}_{\underline{x}} \rightarrow X$ defines an *isomorphism*

$$\text{Ge}_{\Gamma}(X)_{\underline{x}} \rightarrow \text{Ge}_{\Gamma}(\mathcal{D}_{\underline{x}})_{\underline{x}}.$$

Remark 8.7.2. Here is an explicit description of the spaces $\text{Ge}_{\Gamma}(\mathcal{D}_{\underline{x}})$, $\text{Ge}_{\Gamma}(\mathcal{D}_{\underline{x}}^{\times})$ and $\text{Ge}_{\Gamma}(\mathcal{D}_{\underline{x}})_{\underline{x}}$:

Write Γ as the kernel of a homomorphism of two tori $T_0 \rightarrow T_1$. Then

$$\text{Ge}_{\Gamma}(\mathcal{D}_{\underline{x}}) = B^2(\ker(\mathfrak{L}^+(T_0)_{\underline{x}} \rightarrow \mathfrak{L}^+(T_1)_{\underline{x}}))_{\text{et}}.$$

When $\underline{x}$ is a singleton, the above space is just $\text{Ge}_{\Gamma}(\text{pt})$.

Further,

$$\text{Ge}_{\Gamma}(\mathcal{D}_{\underline{x}}^{\times}) \simeq B^2(\text{Fib}(\mathfrak{L}(T_0)_{\underline{x}} \rightarrow \mathfrak{L}(T_1)_{\underline{x}}))_{\text{et}} \simeq B^1(\text{coFib}(\mathfrak{L}(T_0)_{\underline{x}} \rightarrow \mathfrak{L}(T_1)_{\underline{x}}))_{\text{et}}.$$

Finally,

$$\text{Ge}_{\Gamma}(\mathcal{D}_{\underline{x}})_{\underline{x}} \simeq \text{Gr}_{T_1, \underline{x}} / \text{Gr}_{T_0, \underline{x}}.$$

8.7.3. A local variant of (8.10) is a pairing

$$(8.34) \quad \mathrm{Ge}_\Gamma(\mathcal{D}_{\underline{x}})_{\underline{x}} \times \mathrm{Ge}_{\Gamma^\vee(1)}(\mathcal{D}_{\underline{x}}) \rightarrow \mathrm{Ge}_{\mathbb{G}_m}(\mathrm{pt}),$$

and it is easy to show that (8.34) is also of Fourier-Mukai type.

In particular, to a $\Gamma^\vee(1)$ -gerbe $\mathfrak{G}_{\Gamma^\vee(1)}^{\mathrm{loc}}$ on $\mathcal{D}_{\underline{x}}$ we can canonically associate a $\mathbb{G}_m$ -gerbe $\mathcal{G}_{\mathfrak{G}_{\Gamma^\vee(1)}^{\mathrm{loc}}}$ on $\mathrm{Ge}_\Gamma(\mathcal{D}_{\underline{x}})_{\underline{x}}$.

8.7.4. Note that we have the following commutative diagram of pairings

$$(8.35) \quad \begin{array}{ccc} \mathrm{Ge}_\Gamma(X)_{\underline{x}} \times \mathrm{Ge}_{\Gamma^\vee(1)}(X) & \longrightarrow & \mathrm{Ge}_\Gamma(X) \times \mathrm{Ge}_{\Gamma^\vee(1)}(X) \\ \simeq \uparrow & & \\ \mathrm{Ge}_\Gamma(\mathcal{D}_{\underline{x}})_{\underline{x}} \times \mathrm{Ge}_{\Gamma^\vee(1)}(X) & & \downarrow (8.10) \\ \downarrow & & \\ \mathrm{Ge}_\Gamma(\mathcal{D}_{\underline{x}})_{\underline{x}} \times \mathrm{Ge}_{\Gamma^\vee(1)}(\mathcal{D}_{\underline{x}}) & \xrightarrow{(8.34)} & \mathrm{Ge}_{\mathbb{G}_m}(\mathrm{pt}). \end{array}$$

In particular, for a $\Gamma^\vee(1)$ -gerbe $\mathfrak{G}_{\Gamma^\vee(1)}$ on X and

$$\mathfrak{G}_{\Gamma^\vee(1)}^{\mathrm{loc}} := \mathfrak{G}_{\Gamma^\vee(1)}|_{\mathcal{D}_{\underline{x}}}$$

we have

$$\mathcal{G}_{\mathfrak{G}_{\Gamma^\vee(1)}^{\mathrm{loc}}} |_{\mathrm{Ge}_\Gamma(X)_{\underline{x}}} \simeq \mathcal{G}_{\mathfrak{G}_{\Gamma^\vee(1)}} |_{\mathrm{Ge}_\Gamma(X)_{\underline{x}}}.$$

8.7.5. Let

$$\mathrm{Hecke}_{G_{\mathrm{ad}}, \underline{x}}^{\mathrm{loc}} := \mathfrak{L}^+(G_{\mathrm{ad}})_{\underline{x}} \backslash \mathfrak{L}(G_{\mathrm{ad}})_{\underline{x}} / \mathfrak{L}^+(G_{\mathrm{ad}})_{\underline{x}}$$

be the local Hecke stack for G_{ad} at x .

We have a natural projection

$$\mathrm{Hecke}_{G_{\mathrm{ad}}, \underline{x}}^{\mathrm{loc}} \rightarrow \mathrm{Ge}_{Z_G}(\mathcal{D}_{\underline{x}}) \times_{\mathrm{Ge}_{Z_G}(\mathcal{D}_{\underline{x}}^\times)} \mathrm{Ge}_{Z_G}(\mathcal{D}_{\underline{x}}).$$

The commutative group structure on the space of gerbes gives rise to a map

$$\mathrm{Ge}_{Z_G}(\mathcal{D}_{\underline{x}}) \times_{\mathrm{Ge}_{Z_G}(\mathcal{D}_{\underline{x}}^\times)} \mathrm{Ge}_{Z_G}(\mathcal{D}_{\underline{x}}) \rightarrow \mathrm{Ge}_{Z_G}(\mathcal{D}_{\underline{x}})_{\underline{x}}.$$

Composing we obtain a map

$$(8.36) \quad \mathrm{Hecke}_{G_{\mathrm{ad}}, \underline{x}}^{\mathrm{loc}} \rightarrow \mathrm{Ge}_{Z_G}(\mathcal{D}_{\underline{x}})_{\underline{x}}.$$

Remark 8.7.6. Note that the map (8.36) induces a bijection on the sets of connected components when G is simply-connected.

8.7.7. Let $\mathcal{G}^{\mathrm{loc}}$ be a $\mathbb{G}_m$ -gerbe on $\mathrm{Ge}_{Z_G}(\mathcal{D}_{\underline{x}})_{\underline{x}}$. By a slight abuse of notation, we will denote by

$$\mathrm{Sph}(G_{\mathrm{ad}})_{\underline{x}, \mathcal{G}^{\mathrm{loc}}}$$

the corresponding twisted version of the category

$$\mathrm{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Hecke}_{G_{\mathrm{ad}}, \underline{x}}^{\mathrm{loc}}),$$

obtained by pulling back the gerbe $\mathcal{G}^{\mathrm{loc}}$ along (8.36).

Assume now that $\mathcal{G}^{\mathrm{loc}}$ is *multiplicative* (with respect to the group structure on $\mathrm{Ge}_{Z_G}(\mathcal{D}_{\underline{x}})_{\underline{x}}$). Then the category $\mathrm{Sph}(G_{\mathrm{ad}})_{\underline{x}, \mathcal{G}^{\mathrm{loc}}}$ acquires a natural monoidal structure.

8.7.8. Let $\mathfrak{G}_{\pi_1(\check{G})}^{\text{loc}}$ be a $\pi_1(\check{G})$ -gerbe on $\mathcal{D}_{\underline{x}}$. Note that since the pairing (8.34) is bilinear, the corresponding $\mathbb{G}_m$ -gerbe $\mathcal{G}_{\mathfrak{G}_{\pi_1(\check{G})}^{\text{loc}}}$ on $\text{Ge}_{Z_G}(\mathcal{D}_{\underline{x}})_{\underline{x}}$ has a natural multiplicative structure.

The following is a twisted version of the (naive) geometric Satake functor:

Lemma 8.7.9. *There exists a symmetric monoidal functor*

$$\text{Sat}_{G_{\text{ad}}, \mathfrak{G}_{\pi_1(\check{G})}^{\text{loc}}}^{\text{nv}} : \text{Rep}(\check{G}_{\text{sc}})_{\underline{x}, \mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}} \rightarrow \text{Sph}(G_{\text{ad}})_{\underline{x}, \mathcal{G}_{\mathfrak{G}_{\pi_1(\check{G})}^{\text{loc}}}}.$$

Proof. The pairing (8.34) induces a bijection between:

- The set of characters of the (finite) group $\text{Maps}(\mathcal{D}_{\underline{x}}, \pi_1(\check{G}))$, which is a subgroup of $\text{Maps}(\mathcal{D}_{\underline{x}}, \pi_1(\check{G}_{\text{ad}})) \simeq \text{Maps}(\mathcal{D}_{\underline{x}}, Z_{\check{G}_{\text{sc}}}) \simeq Z_{(\check{G}_{\text{sc}})_{\underline{x}}}$;
- The set $\pi_0(\text{Ge}_{Z_G}(\mathcal{D}_{\underline{x}})_{\underline{x}})$, which is a quotient of the set

$$\pi_0(\text{Hecke}_{G_{\text{ad}}, \underline{x}}^{\text{loc}}) \simeq \pi_0(\text{Ge}_{Z_{G_{\text{sc}}}}(\mathcal{D}_{\underline{x}})_{\underline{x}}).$$

The assertion of the lemma follows from the fact that under the usual (naive) geometric Satake functor

$$\text{Sat}_{G_{\text{ad}}}^{\text{nv}} : \text{Rep}(\check{G}_{\text{sc}})_{\underline{x}} \rightarrow \text{Sph}(G_{\text{ad}})_{\underline{x}},$$

the decomposition of $\text{Rep}(\check{G}_{\text{sc}})_{\underline{x}}$ according to central characters corresponds to the decomposition of $\text{Sph}(G_{\text{ad}})_{\underline{x}}$ along the connected components of $\text{Hecke}_{G_{\text{ad}}, \underline{x}}^{\text{loc}}$, according to support. Indeed, this observation implies that $\text{Sat}_{G_{\text{ad}}}^{\text{nv}} : \text{Rep}(\check{G}_{\text{sc}})_{\underline{x}} \rightarrow \text{Sph}(G_{\text{ad}})_{\underline{x}}$ is equivariant for the action of $B(Z_{(\check{G}_{\text{sc}})_{\underline{x}}})$ on both sides, and therefore we can twist $\text{Sat}_{G_{\text{ad}}}^{\text{nv}}$ by $Z_{(\check{G}_{\text{sc}})_{\underline{x}}}$ -gerbes. $\square$

8.7.10. The constructions in Sect. 8.7.5 have immediate counterparts for the global Hecke stack

$$\text{Hecke}_{G_{\text{ad}}, \underline{x}}^{\text{glob}} := \text{Bun}_{G_{\text{ad}}} \times_{\text{Bun}_{G_{\text{ad}}}(X - \underline{x})} \text{Bun}_{G_{\text{ad}}}.$$

In particular, a multiplicative $\mathbb{G}_m$ -gerbe $\mathcal{G}^{\text{loc}}$ on $\text{Ge}_{Z_G}(X)_{\underline{x}}$ gives rise to a monoidal category

$$(8.37) \quad \text{D-mod}_{\frac{1}{2}}(\text{Hecke}_{G_{\text{ad}}, \underline{x}}^{\text{glob}})_{\mathcal{G}^{\text{loc}}}.$$

Note that pullback defines a monoidal functor

$$\text{Sph}(G_{\text{ad}})_{\underline{x}, \mathcal{G}^{\text{loc}}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Hecke}_{G_{\text{ad}}, \underline{x}}^{\text{glob}})_{\mathcal{G}^{\text{loc}}}.$$

Assume now that $\mathcal{G}^{\text{loc}}$ is obtained by restriction along

$$\text{Ge}_{Z_G}(X)_{\underline{x}} \rightarrow \text{Ge}_{Z_G}(X)$$

of a multiplicative $\mathbb{G}_m$ -gerbe $\mathcal{G}$ on $\text{Ge}_{Z_G}(X)$.

Then we have a natural monoidal action of (8.37) on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}})_{\mathcal{G}}$.

8.7.11. Combining the above ingredients, we obtain that for a $\pi_1(\check{G})$ -gerbe $\mathfrak{G}_{\pi_1(\check{G})}$ on X , we have a monoidal action of $\text{Rep}(\check{G}_{\text{sc}})_{\underline{x}, \mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}}$ on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}})_{\mathcal{G}_{\mathfrak{G}_{\pi_1(\check{G})}^{\text{loc}}}}$.

This construction makes sense in families as $\underline{x}$ moves over the Ran space, thereby giving rise to the sought-for action of $\text{Rep}(\check{G}_{\text{sc}})_{\text{Ran}, \mathfrak{G}_{\pi_1(\check{G})}^{\otimes -1}}$ on $\text{D-mod}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}})_{\mathcal{G}_{\mathfrak{G}_{\pi_1(\check{G})}^{\text{loc}}}}$.

The compatibility in point (a) of Proposition 8.6.7 follows by construction.

$\square$ [Proposition 8.6.7(a)]

8.8. **Proof of Proposition 8.6.7(b).** We will show how to adapt the proof of [Ga4, Theorem 4.5.2] to apply in the current gerbe-twisted situation.

8.8.1. Choose a point $x \in X$. Since $H_{\text{et}}^2(X - x, \pi_1(\check{G})) = 0$, we can choose a trivialization of $\mathfrak{G}_{\pi_1(\check{G})}$ over $X - x$. I.e., we can assume that $\mathfrak{G}_{\pi_1(\check{G})}$ comes from an object

$$\mathfrak{G}_{\pi_1(\check{G})}^{\text{loc}} \in \text{Ge}_{\pi_1(\check{G})}(X)_x.$$

Note that the space $\text{Ge}_{\pi_1(\check{G})}(X)_x$ is canonically the discrete set $\pi_1(\check{G})(-1)$, which is in bijection with the set of characters of Z_G . Denote the element corresponding to our gerbe by χ .

8.8.2. We claim now that the proof of the spectral decomposition theorem in [Ga4, Sect. 11.1] applies in the current context. Recall that in *loc. cit.* the proof was based on considering the localization functor

$$\text{Loc}_{G_{\text{ad}}} : \text{KL}(G_{\text{ad}})_{\text{crit}, \text{Ran}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}}).$$

In the present twisted situation we will need to make the following modifications.

8.8.3. We replace Ran by its relative version Ran_x that classifies finite subsets of X that contain the point x .

For further discussion, in order to simplify the notation, we will work with a fixed element $\underline{x} \in \text{Ran}_x$. Write $\underline{x} = \underline{x}' \sqcup \{x\}$.

8.8.4. We replace the category

$$\text{KL}(G_{\text{ad}})_{\text{crit}, \underline{x}} \simeq \text{KL}(G_{\text{ad}})_{\text{crit}, \underline{x}'} \otimes \text{KL}(G_{\text{ad}})_{\text{crit}, x}$$

by

$$\text{KL}(G_{\text{ad}})_{\text{crit}, \underline{x}, \chi} := \text{KL}(G_{\text{ad}})_{\text{crit}, \underline{x}'} \otimes \text{KL}(G_{\text{ad}})_{\text{crit}, x, \chi},$$

where $\text{KL}(G_{\text{ad}})_{\text{crit}, x, \chi}$ is the full subcategory of $\text{KL}(G)_{\text{crit}, x}$, consisting of objects, on which

$$Z_G \subset \mathfrak{L}^+(G)_x$$

acts by the character χ .

8.8.5. Note that by (8.35), the $\mathbb{G}_m$ -gerbe $\mathfrak{G}_{\pi_1(\check{G})}$ on $\text{Ge}_{Z_G}(X)$ is obtained by pullback via the map

$$\text{Ge}_{Z_G}(X) \rightarrow \text{Ge}_{Z_G}(\mathcal{D}_x) \simeq \text{Ge}_{Z_G}(\text{pt}) \simeq B^2(Z_G)_{\text{et}}$$

from the $\mathbb{G}_m$ -gerbe on $B^2(Z_G)_{\text{et}}$ corresponding to the character χ .

The pullback of $\mathfrak{G}_{\pi_1(\check{G})}$ to $\text{Bun}_{G_{\text{ad}}}$ trivializes over the cover

$$\text{Bun}_{G_{\text{ad}}} \times_{\text{pt}/\mathfrak{L}^+(G_{\text{ad}})_x} \text{pt}/\mathfrak{L}^+(G)_x,$$

and corresponds to the multiplicative line bundle on pt/Z_G given by χ .

From here we obtain that we have a well-defined localization functor

$$\text{Loc}_{G_{\text{ad}}, \underline{x}, \chi} : \text{KL}(G_{\text{ad}})_{\text{crit}, \underline{x}, \chi} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}})_{\mathfrak{G}_{\pi_1(\check{G})}}.$$

8.8.6. Recall that the category $\text{KL}(G_{\text{ad}})_{\text{crit}, \underline{x}}$ is acted on by

$$\text{QCoh}(\text{Op}_{G_{\text{sc}}, \underline{x}}^{\text{mon-free}}).$$

The key point in the proof of [Ga4, Theorem 4.5.2] is the fact (going back to [BD] and reviewed in [GLC2, Sects. 15-16]) that the functor

$$\text{Loc}_{G_{\text{ad}}, \underline{x}} : \text{KL}(G_{\text{ad}})_{\text{crit}, \underline{x}} \rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}})$$

factors as

$$\begin{aligned} \text{KL}(G_{\text{ad}})_{\text{crit}, \underline{x}} &\rightarrow \text{KL}(G_{\text{ad}})_{\text{crit}, \underline{x}} \otimes_{\text{QCoh}(\text{Op}_{G_{\text{sc}}, \underline{x}}^{\text{mon-free}})} \text{QCoh}(\text{Op}_{G_{\text{sc}}}^{\text{mon-free}}(X - x)) \xrightarrow{\text{Loc}_{G_{\text{ad}}, \underline{x}}^{\text{glob}}} \\ &\rightarrow \text{D-mod}_{\frac{1}{2}}(\text{Bun}_{G_{\text{ad}}}), \end{aligned}$$

where the functor $\mathrm{Loc}_{\check{G}_{\mathrm{ad}}, x}^{\mathrm{glob}}$ is $\mathrm{QCoh}(\mathrm{LS}_{\check{G}_{\mathrm{sc}}})$ -linear with respect to the tautological projection $\mathrm{Op}_{\check{G}_{\mathrm{sc}}}^{\mathrm{mon-free}}(X - x) \rightarrow \mathrm{LS}_{\check{G}_{\mathrm{sc}}}$.

We will now explain the modification of this construction.

8.8.7. Consider the space $\mathrm{Op}_{\check{G}, x}^{\mathrm{mon-free}}$. We claim that there is a canonically defined map

$$(8.38) \quad \mathrm{Op}_{\check{G}, x}^{\mathrm{mon-free}} \rightarrow \mathrm{Ge}_{\pi_1(\check{G})}(\mathcal{D}_x)_x.$$

To construct it, it suffices to show that the composition

$$(8.39) \quad \mathrm{Op}_{\check{G}, x}^{\mathrm{mer}} \rightarrow \mathrm{LS}_{\check{G}, x}^{\mathrm{mer}} \rightarrow \mathrm{Ge}_{\pi_1(\check{G})}(\mathcal{D}_x^\times)$$

factors through the point of $\mathrm{Ge}_{\pi_1(\check{G})}(\mathcal{D}_x^\times)$, corresponding to the trivial $\pi_1(\check{G})$ -gerbe,

Note that the map (8.39) factors as

$$\mathrm{Op}_{\check{G}, x}^{\mathrm{mer}} \rightarrow \mathrm{LS}_{\check{G}, x}^{\mathrm{mer}} \rightarrow \mathrm{Bun}_{\check{G}}(\mathcal{D}_x^\times) \rightarrow \mathrm{Ge}_{\pi_1(\check{G})}(\mathcal{D}_x^\times),$$

while the map

$$\mathrm{Op}_{\check{G}, x}^{\mathrm{mer}} \rightarrow \mathrm{LS}_{\check{G}, x}^{\mathrm{mer}} \rightarrow \mathrm{Bun}_{\check{G}}(\mathcal{D}_x^\times)$$

in turn factors as

$$\mathrm{Op}_{\check{G}, x}^{\mathrm{mer}} \rightarrow \mathrm{Bun}_{\check{B}}(\mathcal{D}_x^\times) \rightarrow \mathrm{Bun}_{\check{G}}(\mathcal{D}_x^\times).$$

Hence, (8.39) factors as

$$(8.40) \quad \mathrm{Op}_{\check{G}, x}^{\mathrm{mer}} \rightarrow \mathrm{Bun}_{\check{B}}(\mathcal{D}_x^\times) \rightarrow \mathrm{Bun}_{\check{T}}(\mathcal{D}_x^\times) \rightarrow \mathrm{Ge}_{\pi_1(\check{G})}(\mathcal{D}_x^\times),$$

where we think of $\pi_1(\check{G})$ as $\ker(\check{T} \rightarrow \check{T}_{\mathrm{sc}})$.

However, the map

$$\mathrm{Op}_{\check{G}, x}^{\mathrm{mer}} \rightarrow \mathrm{Bun}_{\check{B}}(\mathcal{D}_x^\times) \rightarrow \mathrm{Bun}_{\check{T}}(\mathcal{D}_x^\times)$$

corresponds to the point $2\check{\rho}(\omega^{\otimes \frac{1}{2}}) \in \mathrm{Bun}_{\check{T}}(\mathcal{D}_x^\times)$, and that point lifts to a point of $\mathrm{Bun}_{\check{T}_{\mathrm{sc}}}(\mathcal{D}_x^\times)$. This implies that the map (8.40) factors via the trivial gerbe.

Remark 8.8.8. We claim that $\mathrm{Op}_{\check{G}, x}^{\mathrm{mon-free}}$ maps in fact to a twisted form of the affine Grassmannian of the group $\check{G}$, so that (8.38) factors via this map¹¹.

Indeed, recall that for a fixed curve X , we can think of $\check{G}$ -opers as connections of the standard form on a fixed $\check{G}$ -bundle $\mathcal{P}_{\check{G}}^{\mathrm{Op}}$, induced from a particular Borel bundle for a principal SL_2 -triple, see [GLC2, Sect. 3.1.4].

Consider the twisted affine Grassmannian $\mathrm{Gr}_{\check{G}, \mathcal{P}_{\check{G}}^{\mathrm{Op}}, x}$, i.e., the moduli space of pairs

$$(\mathcal{P}_{\check{G}}, \alpha),$$

where $\mathcal{P}_{\check{G}}$ is a $\check{G}$ -bundle on $\mathcal{D}_x$, and α is an isomorphism $\mathcal{P}_{\check{G}} \simeq \mathcal{P}_{\check{G}}^{\mathrm{Op}}$ over $\mathcal{D}_x^\times$.

Note that we can think of a point of $\mathrm{Op}_{\check{G}, x}^{\mathrm{mon-free}}$ as a triple

$$(A, \mathcal{P}_{\check{G}}, \alpha),$$

where:

- A is a connection of the standard oper form on $\mathcal{P}_{\check{G}}^{\mathrm{Op}}$ over $\mathcal{D}_x^\times$;
- $\mathcal{P}_{\check{G}}$ is a $\check{G}$ -bundle on $\mathcal{D}_x$;
- α is an isomorphism $\mathcal{P}_{\check{G}} \simeq \mathcal{P}_{\check{G}}^{\mathrm{Op}}$ over $\mathcal{D}_x^\times$, so that the *a priori meromorphic* connection on $\mathcal{P}_{\check{G}}$, induced by A via α is *regular*.

The assignment

$$(A, \mathcal{P}_{\check{G}}, \alpha) \mapsto (\mathcal{P}_{\check{G}}, \alpha)$$

is the sought-for map

$$\mathrm{Op}_{\check{G}, x}^{\mathrm{mon-free}} \rightarrow \mathrm{Gr}_{\check{G}, \mathcal{P}_{\check{G}}^{\mathrm{Op}}, x}.$$

¹¹This remark is inessential for the sequel and the reader may choose to skip it.

8.8.9. In the twisted situation we replace

$$\mathrm{Op}_{\check{G}_{\mathrm{sc}}, \underline{x}}^{\mathrm{mon-free}} \simeq \mathrm{Op}_{\check{G}_{\mathrm{sc}}, \underline{x}'}^{\mathrm{mon-free}} \times \mathrm{Op}_{\check{G}_{\mathrm{sc}}, x}^{\mathrm{mon-free}}$$

by

$$\mathrm{Op}_{\check{G}_{\mathrm{sc}}, \underline{x}, \chi}^{\mathrm{mon-free}} := \mathrm{Op}_{\check{G}_{\mathrm{sc}}, \underline{x}'}^{\mathrm{mon-free}} \times \mathrm{Op}_{\check{G}_{\mathrm{sc}}, x, \chi}^{\mathrm{mon-free}},$$

where $\mathrm{Op}_{\check{G}_{\mathrm{sc}}, x, \chi}^{\mathrm{mon-free}}$ is the preimage of the point χ under the projection (8.38).

Note that with respect to the $\mathrm{QCoh}(\mathrm{Op}_{\check{G}, x}^{\mathrm{mon-free}})$ -action on $\mathrm{KL}(G)_{\mathrm{crit}, x}$, we can identify the subcategory $\mathrm{KL}(G_{\mathrm{ad}})_{\mathrm{crit}, x, \chi}$ with the direct summand that is supported over

$$\mathrm{Op}_{\check{G}_{\mathrm{sc}}, x, \chi}^{\mathrm{mon-free}} \subset \mathrm{Op}_{\check{G}, x}^{\mathrm{mon-free}}.$$

8.8.10. Consider the space

$$\mathrm{Op}_{\check{G}_{\mathrm{sc}}}^{\mathrm{mon-free}}(X - x)_{\chi} := \mathrm{Op}_{\check{G}_{\mathrm{sc}}}(X - x) \times_{\mathrm{LS}_{\check{G}_{\mathrm{sc}}}(X - x)} \mathrm{LS}_{\check{G}_{\mathrm{sc}}, \mathfrak{G}_{\pi_1(\check{G})}}.$$

Note that we have a naturally defined map

$$\mathrm{Op}_{\check{G}_{\mathrm{sc}}}^{\mathrm{mon-free}}(X - x)_{\chi} \rightarrow \mathrm{Op}_{\check{G}_{\mathrm{sc}}, \underline{x}, \chi}^{\mathrm{mon-free}}.$$

Now, by the same principle as in [Ga4, Theorem 10.3.4] (see also [GLC2, Sects. 15-16]), we obtain that the functor $\mathrm{Loc}_{G_{\mathrm{ad}}, \underline{x}, \chi}$ factors as

$$\begin{aligned} \mathrm{KL}(G_{\mathrm{ad}})_{\mathrm{crit}, \underline{x}, \chi} &\rightarrow \mathrm{KL}(G_{\mathrm{ad}})_{\mathrm{crit}, \underline{x}, \chi} \otimes_{\mathrm{QCoh}(\mathrm{Op}_{\check{G}_{\mathrm{sc}}, \underline{x}, \chi}^{\mathrm{mon-free}})} \mathrm{QCoh}(\mathrm{Op}_{\check{G}_{\mathrm{sc}}}^{\mathrm{mon-free}}(X - x)_{\chi}) \xrightarrow{\mathrm{Loc}_{G_{\mathrm{ad}}, \underline{x}, \chi}^{\mathrm{glob}}} \\ &\rightarrow \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G_{\mathrm{ad}}})_{\mathfrak{G}_{\pi_1(\check{G})}}, \end{aligned}$$

where the functor $\mathrm{Loc}_{G_{\mathrm{ad}}, \underline{x}, \chi}^{\mathrm{glob}}$ is linear with respect to $\mathrm{QCoh}(\mathrm{LS}_{\check{G}_{\mathrm{sc}}, \mathfrak{G}_{\pi_1(\check{G})}})$.

Now the argument parallel to that in [Ga4, Sect. 11.1] establishes the factorization of the action stated in Proposition 8.6.7(b).

□[Proposition 8.6.7(b)]

8.9. Geometric Langlands for non-pure inner forms.

8.9.1. Note that, in view of Remark 8.5.5, from Corollary 8.5.9 we obtain an expression of the twisted categories $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G, \mathfrak{G}_{Z_G})$ in terms of the usual $\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G)$ and the spectral action.

Namely, using (8.11) and the fact that $\mathrm{LS}_{\check{G}}$ is 1-affine, we obtain:

Corollary 8.9.2. *For a Z_G -gerbe $\mathfrak{G}_{Z_G}$ on X , we have a canonical equivalence:*

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G, \mathfrak{G}_{Z_G}^{-1}}) \simeq \mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_G) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})} \mathrm{QCoh}(\mathrm{LS}_{\check{G}})_{\mathfrak{G}_{Z_G}},$$

where $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})_{\mathfrak{G}_{Z_G}}$ is the twist of $\mathrm{QCoh}(\mathrm{LS}_{\check{G}})$ by the pullback of the gerbe $\mathfrak{G}_{Z_G}$ on $\mathrm{Ge}_{\pi_1(\check{G})}(X)$ along the map $\mathfrak{p}_{\pi_1(\check{G})}$ of (8.22).

Combining with GLC for G , we obtain a form of GLC for non-pure inner twists:

Corollary 8.9.3. *There is a canonical equivalence:*

$$\mathrm{D-mod}_{\frac{1}{2}}(\mathrm{Bun}_{G, \mathfrak{G}_{Z_G}^{-1}}) \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{\check{G}}) \otimes_{\mathrm{QCoh}(\mathrm{LS}_{\check{G}})} \mathrm{QCoh}(\mathrm{LS}_{\check{G}})_{\mathfrak{G}_{Z_G}}.$$

8.9.4. Let G_{sc} be the simply-connected cover of G ; consider the short exact sequence

$$1 \rightarrow \pi_1(G) \rightarrow G_{\mathrm{sc}} \rightarrow G \rightarrow 1$$

and the resulting map

$$\mathfrak{p}_{\pi_1(G)} : \mathrm{Bun}_G \rightarrow \mathrm{Ge}_{\pi_1(G)}(X).$$

For a $\mathbb{G}_m$ -gerbe $\mathfrak{G}$ on $\mathrm{Ge}_{\pi_1(G)}(X)$, let us denote by $\mathrm{D-mod}_{\frac{1}{2}, \mathfrak{G}}(\mathrm{Bun}_G)$ the corresponding category of gerbe-twisted D-modules on Bun_G .

8.9.5. On the dual side we have the short exact sequence

$$1 \rightarrow Z_{\check{G}} \rightarrow \check{G} \rightarrow \check{G}_{\text{ad}} \rightarrow 1,$$

and a map

$$\mathfrak{p}_{Z_{\check{G}}} : \text{LS}_{\check{G}_{\text{ad}}} \rightarrow \text{Ge}_{Z_{\check{G}}}(X).$$

To a point $\mathfrak{G}_{Z_{\check{G}}} \in \text{Ge}_{Z_{\check{G}}}(X)$ we can associate a gives rise to a (non-pure) inner twist of $\text{LS}_{\check{G}}$:

$$\text{LS}_{\check{G}, \mathfrak{G}_{Z_{\check{G}}}} := \text{LS}_{\check{G}_{\text{ad}}} \times_{\text{Ge}_{Z_{\check{G}}}(X)} \{\mathfrak{G}_{Z_{\check{G}}}\}.$$

8.9.6. Consider the short exact sequence

$$0 \rightarrow \pi_1(G) \rightarrow Z_{G_{\text{sc}}} \rightarrow Z_G \rightarrow 0$$

and its dual

$$0 \rightarrow \pi_1(\check{G}) \rightarrow \pi_{\check{G}_{\text{ad}}} \rightarrow Z_{\check{G}} \rightarrow 0.$$

Combining Theorem 8.5.8 and Lemma 8.4.6 we obtain:

Corollary 8.9.7. *For a $Z_{\check{G}}$ -gerbe $\mathfrak{G}_{Z_{\check{G}}}$ on X , there is a canonical equivalence*

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\mathfrak{G}_{Z_{\check{G}}}} \simeq \text{D-mod}_{\frac{1}{2}}(\text{Bun}_{G_{\text{sc}}}) \otimes_{\text{QCoh}(\text{LS}_{\check{G}_{\text{ad}}})} \text{QCoh}(\text{LS}_{\check{G}, \mathfrak{G}_{Z_{\check{G}}}}).$$

Combining with GLC for G_{sc} we obtain:

Corollary 8.9.8. *For $\mathfrak{G}_{Z_{\check{G}}} \in \text{Ge}_{Z_{\check{G}}}(X)$ there is a canonical equivalence*

$$\text{D-mod}_{\frac{1}{2}}(\text{Bun}_G)_{\mathfrak{G}_{Z_{\check{G}}}} \simeq \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}, \mathfrak{G}_{Z_{\check{G}}}}).$$

Remark 8.9.9. We expect that equivalences parallel to Corollaries 8.9.3 and 8.9.8 also take place in the framework of Fargues-Scholze theory of [FS].

8.10. Arithmetic consequences. In this subsection we will appeal to the notations introduced in [AGKRRV1, Sect. 24]. We will work over the ground field $k = \overline{\mathbb{F}}_p$, and we will assume that Z_G and $Z_{\check{G}}$ have orders prime to p .

8.10.1. In this subsection we will assume that GLC holds (for constant group-schemes) in the context of ℓ -adic sheaves over the ground field $\overline{\mathbb{F}}_p$, see [AGKRRV1, Conjecture 21.2.7]:

$$(8.41) \quad \text{Shv}_{\text{Nilp}, \frac{1}{2}}(\text{Bun}_G) \simeq \text{IndCoh}_{\text{Nilp}}(\text{LS}_G^{\text{restr}}).$$

Then by the same principle as in Corollary 8.9.3, from (8.41) one can derive a GLC-type equivalence for non-pure inner forms of G :

For a Z_G -gerbe $\mathfrak{G}_{Z_G}$ on X , we have

$$(8.42) \quad \text{Shv}_{\text{Nilp}, \frac{1}{2}}(\text{Bun}_{G, \mathfrak{G}_{Z_G}^{-1}}) \simeq \text{IndCoh}_{\text{Nilp}}(\text{LS}_G^{\text{restr}}) \otimes_{\text{QCoh}(\text{LS}_G^{\text{restr}})} \text{QCoh}(\text{LS}_G^{\text{restr}})_{\mathfrak{G}_{Z_G}},$$

where we view $\mathfrak{G}_{Z_G}$ as a $\mu_{\infty}(\overline{\mathbb{F}}_q)$ -gerbe on $\text{Ge}_{\pi_1(\check{G})}$, and we turn it into a $\mathbb{G}_m$ -gerbe via

$$\mu_{\infty}(\overline{\mathbb{F}}_q) \hookrightarrow \mu_{\infty}(\overline{\mathbb{Q}}_{\ell}) \subset \mathbb{G}_m.$$

8.10.2. Similarly, for a $Z_{\check{G}}$ -gerbe $\mathfrak{G}_{Z_{\check{G}}}$ on X , combining Corollary 8.9.8 and (8.41), we obtain:

$$(8.43) \quad \text{Shv}_{\text{Nilp}, \frac{1}{2}}(\text{Bun}_G)_{\mathfrak{G}_{Z_{\check{G}}}} \simeq \text{IndCoh}_{\text{Nilp}}(\text{LS}_{\check{G}, \mathfrak{G}_{Z_{\check{G}}}}^{\text{restr}}),$$

where we view $\mathfrak{G}_{Z_{\check{G}}}$ as a $\mu_{\infty}(\overline{\mathbb{F}}_q)$ -gerbe on $\text{Ge}_{\pi_1(G)}$, and we turn it into a $\overline{\mathbb{Q}}_{\ell}^{\times}$ -gerbe via

$$(8.44) \quad \mu_{\infty}(\overline{\mathbb{F}}_q) \hookrightarrow \mu_{\infty}(\overline{\mathbb{Q}}_{\ell}) \subset \overline{\mathbb{Q}}_{\ell}^{\times}.$$

8.10.3. Assume now that X and G are defined over $\mathbb{F}_q$. Let $\mathfrak{G}_{Z_G}$ be a Z_G -gerbe that is also defined over $\mathbb{F}_q$, and hence so is the stack $\mathrm{Bun}_{G, \mathfrak{G}_{Z_G}^{-1}}$. Thus, we can consider the corresponding space of automorphic functions

$$\mathrm{Funct}_c(\mathrm{Bun}_{G, \mathfrak{G}_{Z_G}^{-1}}(\mathbb{F}_q), \overline{\mathbb{Q}}_\ell).$$

The Frobenius-equivariant structure on $\mathfrak{G}_{Z_G}$ gives rise to a Frobenius-equivariant structure on the $\mathbb{G}_m$ -gerbe $\mathfrak{p}_{\pi_1(\tilde{G})}^*(\mathcal{G}_{\mathfrak{G}_{Z_G}})$ over $\mathrm{LS}_{\tilde{G}}^{\mathrm{restr}}$. Hence, the restriction of $\mathfrak{p}_{\pi_1(\tilde{G})}^*(\mathcal{G}_{\mathfrak{G}_{Z_G}})$ to

$$\mathrm{LS}_{\tilde{G}}^{\mathrm{arithm}} := (\mathrm{LS}_{\tilde{G}}^{\mathrm{restr}})^{\mathrm{Frob}}$$

gives rise to a line bundle on $\mathrm{LS}_{\tilde{G}}^{\mathrm{arithm}}$, to be denoted $\mathcal{L}_{\mathfrak{G}_{Z_G}}$.

As in [AGKRRV1, Conjecture 24.8.6], applying the categorical trace of Frobenius to the two sides of (8.42), we obtain an isomorphism of vector spaces

$$(8.45) \quad \mathrm{Funct}_c(\mathrm{Bun}_{G, \mathfrak{G}_{Z_G}^{-1}}(\mathbb{F}_q), \overline{\mathbb{Q}}_\ell) \simeq \Gamma(\mathrm{LS}_{\tilde{G}}^{\mathrm{arithm}}, \omega_{\mathrm{LS}_{\tilde{G}}^{\mathrm{arithm}}} \otimes \mathcal{L}_{\mathfrak{G}_{Z_G}}).$$

Thus, (8.45) is an expression for the (spherical) automorphic category for a (non-pure) inner form of G .

8.10.4. Let now $\mathfrak{G}_{Z_{\tilde{G}}}$ be a $Z_{\tilde{G}}$ -gerbe on X defined over $\mathbb{F}_q$. Then the stack $\mathrm{LS}_{\tilde{G}, \mathfrak{G}_{Z_{\tilde{G}}}}^{\mathrm{restr}}$ acquires an action of the Frobenius automorphism. Denote

$$\mathrm{LS}_{\tilde{G}, \mathfrak{G}_{Z_{\tilde{G}}}}^{\mathrm{arithm}} := \left(\mathrm{LS}_{\tilde{G}, \mathfrak{G}_{Z_{\tilde{G}}}}^{\mathrm{restr}} \right)^{\mathrm{Frob}}.$$

The $\overline{\mathbb{Q}}_\ell^\times$ -gerbe $\mathfrak{p}_{\pi_1(\tilde{G})}^*(\mathcal{G}_{\mathfrak{G}_{Z_{\tilde{G}}}})$ (see (8.44)) on Bun_G gives rise to a $\overline{\mathbb{Q}}_\ell^\times$ -torsor over $\mathrm{Bun}_G(\mathbb{F}_q)$, to be denoted $\mathcal{P}_{\mathfrak{G}_{Z_{\tilde{G}}}}$. Consider the space

$$\mathrm{Sect}_c(\mathrm{Bun}_G(\mathbb{F}_q), \mathcal{P}_{\mathfrak{G}_{Z_{\tilde{G}}}})$$

of its compactly supported sections.

As in [AGKRRV1, Conjecture 24.8.6], applying the categorical trace of Frobenius to the two sides of (8.43), we obtain an isomorphism of vector spaces

$$(8.46) \quad \mathrm{Sect}_c(\mathrm{Bun}_G(\mathbb{F}_q), \mathcal{P}_{\mathfrak{G}_{Z_{\tilde{G}}}}) \simeq \Gamma(\mathrm{LS}_{\tilde{G}, \mathfrak{G}_{Z_{\tilde{G}}}}^{\mathrm{arithm}}, \omega_{\mathrm{LS}_{\tilde{G}, \mathfrak{G}_{Z_{\tilde{G}}}}^{\mathrm{arithm}}}).$$

Thus, (8.46) is an expression for the *metaplectic* (spherical) automorphic category of G , which is given in terms of the (non-pure) inner twist of $\tilde{G}$.

APPENDIX A. REVIEW OF (SEMI-)STABILITY FOR G -BUNDLES

We briefly review the basic notions from the theory of (semi-)stable G -bundles following the original source [Ra].

A.1. Definition of (semi-)stability. In what follows, we only consider parabolic subgroups P containing our fixed Borel.

A.1.1. Notation related to root data. Recall that Λ denotes the coweight lattice of G and $\tilde{\Lambda}$ denotes the weight lattice. As is standard, $2\check{\rho}_G \in \tilde{\Lambda}$ denotes the sum of the positive roots.

Let P be a parabolic subgroup of G with Levi quotient M . As the weight lattices of M and G coincide, we also have the weight $2\check{\rho}_M \in \tilde{\Lambda}$. We set $2\check{\rho}_P := 2\check{\rho}_G - 2\check{\rho}_M$, which is the sum of the roots occurring in $\mathfrak{n}(P)$.

We remark that $\langle 2\check{\rho}_P, \alpha_i \rangle = 0$ for every vertex i in the Dynkin diagram I_M of M , i.e., for each simple coroot α_i whose $\mathfrak{sl}_2$ maps into $\mathfrak{m}$. Indeed, we have $\langle 2\check{\rho}_G, \alpha_i \rangle = \langle 2\check{\rho}_M, \alpha_i \rangle = 2$ for such α_i .

It follows that for a coweight $\lambda \in \Lambda$, the value of $\langle 2\check{\rho}_P, \lambda \rangle$ only depends on the class of λ in $\Lambda / \mathrm{Span}\{\alpha_i\}_{i \in I_M} =: \pi_{1, \mathrm{alg}}(M)$.

A.1.2. We have the following definition (cf. [Ra]):

Definition A.1.3. A G -bundle $\mathcal{P}_G$ on X is *semi-stable* (resp. *stable*) if for every maximal (proper) parabolic subgroup $P \subsetneq G$ and every reduction $\mathcal{P}_P$ of $\mathcal{P}_G$ to P , we have

$$\langle 2\check{\rho}_P, \deg(\mathcal{P}_P) \rangle \leq 0 \quad (\text{resp. } < 0).$$

Here we remind that $\deg(\mathcal{P}_P)$ is an element of $\pi_{1,\text{alg}}(M)$.

We remark that the integer $\langle 2\check{\rho}_P, \deg(\mathcal{P}_P) \rangle$ appearing above is the degree of the vector bundle $\mathfrak{n}(P)_{\mathcal{P}_P}$ on X .

Example A.1.4. This definition is rigged to recover the usual one for $G = GL_n$.

Indeed, suppose $\mathcal{E}$ has rank n and P is the maximal parabolic whose reductions correspond to subbundles $\mathcal{E}_0 \subset \mathcal{E}$ of rank m . Then a straightforward calculation yields

$$\langle 2\check{\rho}_P, \deg(\mathcal{P}_P) \rangle = \text{rank}(\mathcal{E}) \cdot \deg(\mathcal{E}_0) - \text{rank}(\mathcal{E}_0) \cdot \deg(\mathcal{E}).$$

A.2. A characterization of (semi-)stability.

A.2.1. We have the following basic result.

Proposition A.2.2. *For a G -bundle $\mathcal{P}_G$ on X , the following conditions are equivalent.*

- (a) $\mathcal{P}_G$ is semi-stable (resp. stable).
- (b) For every proper parabolic subgroup $P \subsetneq G$ (possibly not of corank 1) and every reduction $\mathcal{P}_P$ of $\mathcal{P}_G$ to P , we have

$$\langle 2\check{\rho}_P, \deg(\mathcal{P}_P) \rangle \leq 0 \quad (\text{resp. } < 0)$$

- (c) For every reduction $\mathcal{P}_B$ of $\mathcal{P}_G$ to the Borel, we have:

$$(A.1) \quad \deg(\mathcal{P}_B) = \sum_{i \in I_G} n_i \alpha_i + \varepsilon, \quad n_i \in \mathbb{Q}^{\leq 0}, \quad \varepsilon \in \mathbb{Q} \cdot \Lambda_{Z_G} \quad (\text{resp. } n_i \in \mathbb{Q}^{< 0}).$$

Here Λ_{Z_G} is the set of coweights mapping into the center of G .

A.2.3. *Proof of Proposition A.2.2.* The key point is to observe

$$(A.2) \quad \begin{cases} \langle 2\check{\rho}_P, \varepsilon \rangle = 0, \\ \langle 2\check{\rho}_P, \alpha_i \rangle = 0 & \text{if } i \in I_M \\ \langle 2\check{\rho}_P, \alpha_i \rangle \geq 2 & \text{if } i \notin I_M \end{cases}$$

where the last expression follows as $\langle 2\check{\rho}_G, \alpha_i \rangle = 2$ and $2\check{\rho}_M$ is a sum of roots $\check{\alpha}_j$ with $j \in I_M$ (so $j \neq i$). Then (b) tautologically implies (a), (c) implies (b) by (A.2), and (a) implies (c) again by noting that for P_i the maximal parabolic corresponding to $i \in I_G$, we have $\langle 2\check{\rho}_{P_i}, \deg(\mathcal{P}_B) \rangle = n_i \langle 2\check{\rho}_P, \alpha_i \rangle \in n_i \cdot \mathbb{Z}^{> 0}$ by (A.2).

□[Proposition A.2.2]

A.2.4. In the above, the condition (c) immediately matches the notion of semi-stability used in [DG] (see *loc. cit.* Lemma 7.3.2), and it matches the notion of stability implicitly suggested in *loc. cit.*

REFERENCES

- [AB] A. Hassan and I. Biswas, *On holomorphic principal bundles over a compact Riemann surface admitting a flat connection*, Mathematische Annalen 322, **2** (2002), 333-346.
- [Ari] D. Arinkin, *Irreducible connections admit generic oper structures*, arXiv:1602.08989.
- [AG] D. Arinkin and D. Gaitsgory, *Singular support of coherent sheaves, and the Geometric Langlands Conjecture*, Selecta Math. N.S. **21** (2015), 1–199.
- [AGKRRV1] D. Arinkin, D. Gaitsgory, D. Kazhdan, S. Raskin, N. Rozenblyum and Y. Varshavsky, *The stack of local systems with restricted variation and geometric Langlands theory with nilpotent singular support*, arXiv:2010.01906.
- [AGKRRV2] D. Arinkin, D. Gaitsgory, D. Kazhdan, S. Raskin, N. Rozenblyum and Y. Varshavsky, *Automorphic functions as the trace of Frobenius*, arXiv:2102.07906.
- [Bel1] D. Beraldo, *On the extended Whittaker category*, Selecta Mathematica **25** (2019), 1–55.
- [Be2] D. Beraldo, *On the geometric Ramanujan conjecture*, arXiv:2103.17211.
- [BD] A. Beilinson and V. Drinfeld, *Quantization of Hitchin's integrable system and Hecke eigensheaves*, available at http://people.math.harvard.edu/~gaitsgde/grad_2009/

- [BFM] A. Borel, R. Friedman and J. Morgan, *Almos commuting elements in compact Lie groups*, *Memoirs of AMS* **157** (2002).
- [BKS] D. Beraldo, D. Kazhdan and T. Schlank, *Contractibility of the space of generic opers for classical groups*, arXiv:1808.05801.
- [BZN] D. Ben-Zvi and D. Nadler, *Betti geometric Langlands*, *Algebraic geometry: Salt Lake City, 2015* **97** (2018): 3–41.
- [DG] V. Drinfeld and D. Gaitsgory, *Compact generation of the category of D -modules on the stack of G -bundles on a curve* *Cambridge Math Journal*, **3** (2015), 19–125.
- [FR] J. Faergeman and S. Raskin, *Non-vanishing of geometric Whittaker coefficients for reductive groups*, arXiv:2207.02955.
- [FS] L. Fargues and P. Scholze, *Geometrization of the local Langlands correspondence*, arXiv:2102.13459 (2021).
- [FT] E. Frenkel and C. Teleman, *Geometric Langlands correspondence near opers*, *Journal of the Ramanujan Mathematical Society* **28A** (2013), 123–147.
- [Ga1] D. Gaitsgory, *A "strange" functional equation for Eisenstein series and Verdier duality on the moduli stack of bundles*, *Annales Scientifiques de l'ENS* **50** (2017), 1123–1162.
- [Ga2] D. Gaitsgory, *The Atiyah-Bott formula for the cohomology of Bun_G* , arXiv:1505.02331.
- [Ga3] D. Gaitsgory, *Sheaves of categories and the notion of 1-affineness*, *Contemporary Mathematics* **643** (2015), 1–99.
- [Ga4] D. Gaitsgory, *Outline of the proof of the Geometric Langlands Conjecture for $GL(2)$* , *Astérisque* **370** (2015), 1–112.
- [Ga5] D. Gaitsgory, *Contractibility of the space of rational maps*, *Inventiones mathematicae*, **191** no. 1 (2013), 91–196.
- [GLys] D. Gaitsgory and S. Lysenko, *Parameters and duality for the metaplectic geometric Langlands theory*, *Selecta Math. New Ser.* **24** (2018), 227–301. Also arXiv: 1608.00284
- [GLC1] D. Gaitsgory and S. Raskin, *Proof of the geometric Langlands conjecture I: construction of the functor*, arXiv:2405.03599
- [GLC2] D. Arinkin, D. Beraldo, L. Chen, J. Faergeman, D. Gaitsgory, K. Lin, S. Raskin and N. Rozenblyum, *Proof of the geometric Langlands conjecture II: Kac-Moody localization and the FLE*, arXiv:2405.03648
- [GLC3] J. Campbell, L. Chen, D. Gaitsgory and S. Raskin, *Proof of the geometric Langlands conjecture III: compatibility with parabolic induction*, arXiv:2409.07051
- [GLC4] D. Arinkin, D. Beraldo, L. Chen, J. Faergeman, D. Gaitsgory, K. Lin, S. Raskin and N. Rozenblyum, *Proof of the geometric Langlands conjecture IV: ambidexterity*, available at <https://people.mpim-bonn.mpg.de/gaitsgde/GLC/>
- [KS] V. Kac and A. V. Smigla, *Vacuum structure in supersymmetric Yang-Mills theories with any gauge group*, in: *The many faces of the Superworld*, World Scientific (2000), 185–234.
- [NR] M. Narasimhan and S. Ramanan, *Moduli of vector bundles on a compact Riemann surface*, *Annals of Mathematics* (1969), 14–51.
- [NT] D. Nadler and J. Taylor, *The Whittaker functional is a shifted microstalk*, arXiv:2206.09216 (2022).
- [Ra] A. Ramanathan, *Stable principal bundles on a compact Riemann surface*, *Mathematische Annalen* **213** (1975): 129–152.
- [Ro] N. Rozenblyum, *Connections on moduli spaces and infinitesimal Hecke modifications*, arXiv:2108.07745.
- [Wa] I. Waschkies, *The stack of microlocal perverse sheaves*, *Bulletin de la société mathématique de France* **132**, **3** (2004): 397–462.